

SINAMICS S120/S150

List Manual · 11/2009

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SINAMICS S120/S150

List Manual

Valid for

Drive

SINAMICS

Firmware Version

4.3 SP1

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Safety Notices

These Operating Instructions contain information which you must observe to ensure your own personal safety, as well as to prevent damage to the product and connected equipment. The notices referring to your personal safety are highlighted in the manual by a warning triangle. Notices that relate only to material damage have no warning triangle. The notices shown below are graded according to the level of hazard (from most to least hazardous):



Danger

Indicates that death or serious injury **will** result if proper precautions are not taken.



Alarm

Indicates that death or serious injury **may** result if proper precautions are not taken.



Caution

With a warning triangle, Caution indicates that minor injury **may** result if proper precautions are not taken.

Caution

Without a warning triangle, Caution indicates that material damage may result if proper precautions are not taken.

Notice

Indicates that an undesirable result or condition may occur if the corresponding instructions are not observed.

If more than one level of hazard exists, the warning for the highest hazard level is always used. A warning on a warning triangle indicating possible personal injury may also include a warning relating to material damage.

Qualified Personnel

The associated device/system may only be installed and operated in conjunction with this documentation. The equipment/system may only be commissioned and operated by **qualified personnel**. For the purpose of the safety information in this documentation, a “qualified person” is someone who is authorized to start up, ground, and tag equipment, systems, and circuits in accordance with established safety procedures.

Proper Use of Siemens Products

Please observe the following:



Alarm

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Preface

SINAMICS documentation

The SINAMICS documentation is organized in 2 parts:

- General documentation/catalogs
- Manufacturer/service documentation

At <http://www.siemens.com/motioncontrol/docu>, information is available on the following topics:

- Ordering documentation

Here you can find an up-to-date overview of publications

- Downloading documentation

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- Researching documentation online

Information on DOConCD and direct access to the publications in DOCon-Web.

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My Documentation Manager provides you with a range of features for creating your own machine documentation

- Training and FAQs

Information on the range of training courses and FAQs (frequently asked questions) are available via the page navigation.

Usage phases and their tools/documents (as an example)

Table Preface-1 Usage phases and the available tools/documents

Usage phase	Tools/documents
Orientation	SINAMICS S Sales Documentation
Planning/configuration	SIZER configuration tool Configuration Manuals, Motors
Decision making/ordering	SINAMICS S Catalogs
Installation/assembly	• SINAMICS S120 Equipment Manual for Control Units and Additional System Components • SINAMICS S120 Equipment Manual for Booksize Power Units • SINAMICS S120 Equipment Manual for Built-In Power Units • SINAMICS S150 Operating Instructions
Commissioning	• STARTER parameterization and commissioning tool • SINAMICS S120 Getting Started • SINAMICS S120 Commissioning Manual • SINAMICS S120 CANopen Commissioning Manual • SINAMICS S120 Function Manual • SINAMICS S20/S150 List Manual • SINAMICS S150 Operating Instructions
Usage/operation	• SINAMICS S120 Commissioning Manual • SINAMICS S20/S150 List Manual • SINAMICS S150 Operating Instructions
Maintenance/servicing	• SINAMICS S120 Commissioning Manual • SINAMICS S20/S150 List Manual • SINAMICS S150 Operating Instructions

Target group

This documentation is aimed at machine manufacturers, commissioning engineers, and service personnel who use SINAMICS.

Benefits

This documentation contains the comprehensive information about parameters, function block diagrams, and faults and alarms required to commission and service the system.

This manual should be used in addition to the other manuals and tools provided for the product.

Standard version

The scope of the functionality described in this document can differ from the scope of the functionality of the drive system that is actually supplied.

- Other functions not described in this documentation might be able to be executed in the drive system. This does not, however, represent an obligation to supply such functions with a new control or when servicing.
- The documentation can also contain descriptions of functions that are not available in a particular product version of the drive system. The functionalities of the supplied drive system should only be taken from the ordering documentation.
- Extensions or changes made by the machine manufacturer must be documented by the machine manufacturer.

To aid ease of comprehension, this documentation does not contain all detailed information on all product types. This documentation cannot take into consideration every conceivable type of installation, operation and service/maintenance.

Search tools

The following guides are provided to help you locate information in this manual:

1. Table of contents
 - General table of contents for the complete manual (after the preface).
 - Table of contents for function block diagrams (Section 2.1).
2. List of abbreviations
3. List of References
4. Index

Technical support

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Questions regarding this manual

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You can find a fax form is at the end of this document.

Internet address for SINAMICS

<http://www.siemens.com/sinamics>

EC Declaration of Conformity

The EC Declaration of Conformity for the EMC Directive can be obtained from:

- Internet
<http://support.automation.siemens.com>
Product/Order No.: 15257461

- Branch offices
At the relevant regional office of the I DT MC Business Unit of Siemens AG.

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Parameters

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1.1 Overview of parameters

1.1.1 Explanation of list of parameters

Basic structure of parameter descriptions

The data in the following example has been chosen at random. The table below contains all the information that can be included in a parameter description. Some of the information is optional.

The parameter list (See Chapter 1.2) is structured as follows:

----- **Start of example** -----

pxxxx[0 to n]	BICO: Full parameter name/Abbreviated name				
Drive object (function module)	Can be changed: C1(x), C2(x), U, T	Calculated: CALC_MOD_REG		Access level: 2	
	Data type: Unsigned32/Integer16	Dynamic index: CDS, p0170		Function block diagram: 2080	
	P group: Closed loop control	Unit group: 7_1		Unit selection: p0505	
	Not for motor type: FEM	Normalizing: p2000		Expert list: 1	
	Min 0.00 [Nm]	Max 10.00 [Nm]		Factory setting 0.00 [Arms]	
Description:	Text				
Values:	0: Name and meaning of value 0 1: Name and meaning of value 1 2: Name and meaning of value 2 etc.				
Recommendation:	Text				
Index:	[0] = Name and meaning of index 0 [1] = Name and meaning of index 1 [2] = Name and meaning of index 2 etc.				
Bit array:	Bit	Signal name	1 signal	0 signal	FP
	00	Name and meaning of bit 0	Yes	No	8010
	01	Name and meaning of bit 1	Yes	No	-
	02	Name and meaning of bit 2	Yes	No	8012
		etc.			
Dependency:	Text See also: pxxxx, rxxxx See also: Fxxxxx, Axxxxx				
Danger:	Alarm:	Caution:	Safety notices with a warning triangle		
					
Caution:	Notice:	Safety notices without a warning triangle			
Note:	Information which might be useful.				

pxxxx[0 to n] Parameter number

The parameter number is made up of a "p" or "r", followed by the parameter number and the index (optional).

Examples of number representation in the parameter list:

- p... Adjustable parameter (read and write parameter)
- r... Display parameters (read-only)
- p0918 Adjustable parameter 918
- p0099[0 to 3] Adjustable parameter 99, indices 0 to 3
- p1001[0 to n] Adjustable parameter 1001, indices 0 to n (n = configurable)
- r0944 Display parameter 944

Other examples of notation in the documentation:

- p1070[1] Adjustable parameter 1070, index 1
- p2098[1].3 Adjustable parameter 2098, index 1 bit 3
- r0945[2](3) Display parameter 945, index 2 of drive object 3
- p0795.4 Adjustable parameter 795, bit 4
- r2129.0 to 15 Display parameter 2129 with bit array (maximum 16 bits)

The following applies to adjustable parameters:

The parameter value "when shipped" is specified under "Factory setting" with the relevant unit in square parentheses. The value can be adjusted within the range defined by "Min" and "Max".

The term "linked parameterization" is used in cases where changes to adjustable parameters affect the settings of other parameters.

Linked parameterization can, for example, occur as a result of the following actions or parameters:

- Execution of macros
p0015, p0700, p1000, p1500
- Setting PROFIBUS telegram (BICO interconnections)
p0922
- Setting component lists
p0230, p0300, p0301, p0400
- Automatic calculation and pre-assignment
p0112, p0340, p0578, p3900
- Restoring factory settings
p0970

The following applies to display parameters:

The fields "Min", "Max" and "Factory setting" are specified with a dash "-" and the relevant unit in square parentheses.

Note:

The parameter list can contain parameters that are not visible in the expert lists of the respective commissioning software (e.g. parameters for trace functions).

BICO: Full parameter name/Abbreviated name

The following abbreviations can appear in front of the parameter name:

- **BI:** Binector Input
This parameter is used for selecting the source of a digital signal.
- **BO:** Binector Output
This parameter is available as a digital signal for interconnection with other parameters.
- **CI:** Connector Input
This parameter is used for selecting the source of an "analog" signal.
- **CO:** Connector Output
This parameter is available as an "analog" signal for interconnection with other parameters.
- **CO/BO:** Connector/Binector Output
This parameter is available as an "analog" and digital signal for interconnection with other parameters.

Note:

A connector input (CI) cannot be interconnected with just any connector output (CO, signal source).

When interconnecting a connector input using the commissioning software, only those signal sources which are possible are listed.

Drive Object (function module)

A drive object (DO) is an independent, "self-contained" functional unit that has its own parameters and, in some cases, also faults and alarms.

When carrying out commissioning using the commissioning software, you can select/deselect additional functions and their parameters by activating/deactivating function modules accordingly.

Note:

References: /FH1/ SINAMICS S120 Function Manual Drive Functions

The parameter list specifies the associated drive object and function module for each individual parameter.

Examples:

- p1070 CI: Main setpoint
SERVO (extended setpoint), VECTOR
The parameter is available only in association with drive object SERVO and the "Extended setpoint channel" function module or with drive object VECTOR irrespective of activated function modules.
- p1055 BI: Jog bit 0
SERVO, VECTOR
The parameter is available in association with drive objects SERVO and VECTOR irrespective of activated function modules, i.e. it is also available with every activated function module belonging to the respective drive object.

A parameter can belong to a single, multiple, or all drive objects.

The following information relating to "Drive object" and "Function module" can be displayed under the parameter number:

Table 1-1 Data in the "Drive object (function module)" field

Drive object (function module)	Type	Meaning
All objects	-	All drive objects have this parameter.
A_INF	10	Active Infeed Control Closed-loop controlled, self-commutated infeed/regenerative feedback unit for generating a constant DC-link voltage
A_INF (line transformer)	-	Active Infeed with "Line transformer" function module (r0108.4).
A_INF (dynamic line buffering)	-	Active Infeed with "Dynamic line buffering" function module (r0108.7).
A_INF (system droop control)	-	Active Infeed with "System droop control" function module (r0108.12).
A_INF (parallel)	-	Active Infeed with "Parallel connection" function module (r0108.15).
A_INF (master/slave)	-	Active Infeed with "Master/Slave" function module (r0108.16).
A_INF (brk mod ext)	-	Active Infeed with "Braking Module External" function module (r0108.26).
A_INF (cooling unit)	-	Active Infeed with "Cooling unit" function module (r0108.28)
A_INF (PROFINET)	-	Active Infeed with "PROFINET" function module (r0108.31).
B_INF	30	Basic Infeed closed-loop control Unregulated line infeed unit (without feedback) for rectifying the line voltage of the DC link
B_INF (parallel)	-	Basic Infeed with "Parallel connection" function module (r0108.15).
B_INF (brk mod ext)	-	Basic Infeed with "Braking Module External" function module (r0108.26).
B_INF (cooling unit)	-	Basic Infeed with "Cooling unit" function module (r0108.28)
B_INF (PROFINET)	-	Basic Infeed with "PROFINET" function module (r0108.31).
CU	-	Control Unit, all versions.

Table 1-1 Data in the "Drive object (function module)" field, continued

Drive object (function module)	Type	Meaning
CU (CAN)	-	Control Unit with "CAN" function module.
CU (COMM BOARD)		Control Unit with "COMM BOARD" function module.
CU (PROFINET)		Control Unit with "PROFINET" function module.
CU_S	1	Control Unit SINAMICS S120.
CU_S_S150	-	Control Unit SINAMICS S150.
CU_S_CU310DP	-	Control Unit SINAMICS S120 AC Drive with PROFIBUS interface.
CU_S_CU310PN	-	Control Unit SINAMICS S120 AC Drive with PROFINET interface.
CU_CX32	-	Controller Extension for boosting the processing performance
CU_LINK	254	Object for Controller Extension 32 (CX32)
CU_I	-	Control Unit SINAMICS Integrated (only SIMOTION D4x5).
ENCODER	300	Object for a DRIVE-CLiQ encoder.
HUB	150	DRIVE-CLiQ Hub Module.
S_INF	20	Smart Infeed control Unregulated line infeed/feedback unit for generating the DC link voltage.
S_INF (parallel)	-	Smart Infeed with "Parallel connection" function module (r0108.15).
S_INF (brk mod ext)	-	Smart Infeed with "Braking Module External" function module (r0108.26).
S_INF (cooling unit)	-	Smart Infeed with "Cooling unit" function module (r0108.28).
S_INF (PROFINET)	-	Smart Infeed with "PROFINET" function module (r0108.31).
SERVO	11	Servo drive.
SERVO (extended M_ctrl)	-	Servo drive with "Extended torque control" function module (r0108.1).
SERVO (cl. loop position ctrl)	-	Servo drive with "Closed loop position control" function module (r0108.3).
SERVO (EPOS)	-	Servo drive with "Basic positioner" function module (r0108.4).
SERVO (APC)	-	Servo drive with "Advanced Positioning Control (APC)" function module (r0108.7).
SERVO (extended setpoint)	-	Servo drive with "Extended setpoint channel" function module (r0108.8).
SERVO (Lin)	-	Servo drive with "Linear motor" function module (r0108.12).
SERVO (Safety red)	-	Servo drive with "Safety rotary axis" function module (r0108.13).
SERVO (ext. brake)	-	Servo drive with "Extended brake control" function module (r0108.14).
SERVO (Tech_ctrl)	-	Servo drive with "Technology controller" function module (r0108.16)
SERVO (ext. msg)	-	Servo drive with "Extended messages/monitoring functions" function module (r0108.17)
SERVO (cooling unit)	-	Servo drive with "Cooling unit" function module (r0108.28).

Table 1-1 Data in the "Drive object (function module)" field, continued

Drive object (function module)	Type	Meaning
SERVO (CAN)	-	Servo drive with "CAN" function module (r0108.29).
SERVO (PROFINET)	-	Servo drive with "PROFINET" function module (r0108.31).
TB30	100	Terminal Board 30
TM120	207	Terminal Module 120
TM15	203	Terminal Module 15 (SIMOTION D4xx only).
TM15DI_DO	204	Terminal Module 15 (for SINAMICS)
TM17	202	Terminal Module 17 (SIMOTION D4xx only).
TM31	200	Terminal Module 31.
TM41	201	Terminal Module 41.
VECTOR	12	Vector drive.
VECTOR (n/M)	-	Vector drive with "Closed-loop speed/torque control" function module (r0108.2).
VECTOR (pos ctrl)	-	Vector drive with "Position control" function module (r0108.3).
VECTOR (EPOS)	-	Vector drive with "Basic positioner" function module (r0108.4).
VECTOR (Safety red)	-	Vector drive with "Safety rotary axis" function module (r0108.13).
VECTOR (ext. brake)	-	Vector drive with "Extended brake control" function module (r0108.14).
VECTOR (parallel)	-	Vector drive with "Parallel connection" function module (r0108.15).
VECTOR (tech_ctrl)	-	Vector drive with "Technology controller" function module (r0108.16).
VECTOR (ext. mess.)	-	Vector drive with "Extended messages/monitoring functions" function module (r0108.17).
VECTOR (cooling unit)	-	Vector drive with "Cooling unit" function module (r0108.28).
VECTOR (CAN)	-	Vector drive with "CAN" function module (r0108.29).
VECTOR (PROFINET)	-	Vector drive with "PROFINET" function module (r0108.31).

Note:

The drive object type is used to identify the drive objects in the drive system (e.g. r0107, r0975[1]).

Can be changed

The "-" sign indicates that the parameter can be changed in any object state and that the change will be effective immediately.

The letters "C1(x), C2(x), T, U" ((x): optional) mean that the parameter can be changed only in the specified drive object state and that the change will not take effect until the object switches to another state. This can be a single state or multiple states.

The following states may be specified:

- C1(x) Device commissioning C1: **Commissioning 1**
 Device commissioning is in progress (p0009 > 0).
 Pulses cannot be released.
 The parameter can only be changed when the device commissioning settings (p0009 > 0) are as follows:
 - C1: Can be changed for all settings p0009 > 0.
 - C1(x): Can only be changed when the settings are p0009 = x.
 A modified parameter value does not take effect until device commissioning mode is exited with p0009 = 0.
- C2(x) Drive object commissioning C2: **Commissioning 2**
 Drive commissioning is in progress (p0009 = 0 and p0010 > 0).
 Pulses cannot be released.
 The parameter can only be changed when the drive commissioning settings (p0010 > 0) are as follows:
 - C2: Can be changed for all settings p0010 > 0.
 - C2(x): Can only be changed when p0010 = x.
 A modified parameter value does not take effect until drive commissioning mode is exited with p0010 = 0.
- U Operation U: **Run**
 Pulses are released.
- T Ready T: **Ready to run**
 The pulses are not released and the status "C1(x)" or "C2(x)" is not active.

Note:

Parameter p0009 is CU-specific (present on Control Unit).

Parameter p0010 is drive-specific (present for each drive object).

The operating state of the individual drive objects is displayed in r0002.

Calculated

Specifies whether the parameter is influenced by automatic calculations.

The calculation attribute defines which activities influence the parameter.

The following attributes apply:

- CALC_MOD_ALL
 - p0340 = 1
 - Project download with commissioning software and send from p0340 = 3
- CALC_MOD_CON
 - p0340 = 1, 3, 4
- CALC_MOD_EQU
 - p0340 = 1, 2
- CALC_MOD_LIM_REF
 - p0340 = 1, 3, 5
 - p0578 = 1
- CALC_MOD_REG
 - p0340 = 1, 3

Note:

For p3900 > 0, p0340 = 1 is also called automatically.

After p1910 = 1, p0340 = 3 is automatically called.

Access level

Specifies the access level required to be able to display and change the respective parameter. The required access level can be set via p0003.

The system uses the following access levels:

1. Standard
2. Extended
3. Expert
4. Service
 - parameters with this access level are password protected.
5. Macro (the parameter can only be changed via macro)

Note:

Parameter p0003 is CU-specific (present on Control Unit).

Data type

The information on the data type can consist of the following two items (separated by a slash):

- First item
Data type of the parameter
- Second item (for binector or connector input only)
Data type of the signal source to be interconnected (binector/connector output)

Parameters can have the following data types:

- I8 Integer8 8-bit integer
- I16 Integer16 16-bit integer
- I32 Integer32 32-bit integer
- U8 Unsigned8 8 bits without sign
- U16 Unsigned16 16 bits without sign
- U32 Unsigned32 32 bits without sign
- Float FloatingPoint32 32-bit floating point number

Depending on the data type of the BICO input parameters (signal sink) and BICO output parameters (signal source), the following combinations are possible when BICO interconnections are established:

Table 1-2 Possible combinations of BICO interconnections

	BICO input parameter			
	CI parameter			BI parameter
BICO output parameter	Unsigned32 / Integer16	Unsigned32 / Integer32	Unsigned32 / FloatingPoint32	Unsigned32 / Binary
CO: Unsigned8	x	x	–	–
CO: Unsigned16	x	x	–	–
CO: Integer16	x	x	–	–
CO: Unsigned32	x	x	–	–
CO: Integer32	x	x	–	–
CO: FloatingPoint32	x	x	x ¹	–
BO: Unsigned8	–	–	–	x
BO: Unsigned16	–	–	–	x
Legend: x: BICO interconnection permitted –: BICO interconnection not permitted				
1 Exception: BICO input parameters with data type "Unsigned32/FloatingPoint32" can also be interconnected with the following BICO output parameters, despite the fact that these are not of the "FloatingPoint32" data type: CO: r8850, CO: r8860, CO: r2050, CO: r2060				

Table 1-2 Possible combinations of BICO interconnections, continued

	BICO input parameter			
	CI parameter			BI parameter
BICO output parameter	Unsigned32 / Integer16	Unsigned32 / Integer32	Unsigned32 / FloatingPoint32	Unsigned32 / Binary
BO: Integer16	–	–	–	x
BO: Unsigned32	–	–	–	x
BO: Integer32	–	–	–	x
BO: FloatingPoint32	–	–	–	–
Legend: x: BICO interconnection permitted –: BICO interconnection not permitted				
1 Exception: BICO input parameters with data type "Unsigned32/FloatingPoint32" can also be interconnected with the following BICO output parameters, despite the fact that these are not of the "FloatingPoint32" data type: CO: r8850, CO: r8860, CO: r2050, CO: r2060				

Dynamic index

For parameters with a dynamic index [0 to n], the following information is specified here:

- Data set (if available).
- Parameter for the number of indices ($n = \text{number} - 1$).

This field can contain the following information:

- "CDS, p0170" (Command Data Set, CDS count)

Example:

p1070[0] → main setpoint [command data set 0]

p1070[1] → main setpoint [command data set 1], etc.

- "DDS, p0180" (Drive Data Set, DDS count)
- "EDS, p0140" (Encoder Data Set, EDS count)
- "MDS, p0130" (Motor Data Set, MDS count)
- "PDS, p0120" (Power unit Data Set, PDS count)
- "p2615" (traversing blocks count)

Note:

Information on the data sets can be taken from the following references:

References: /FH1/ SINAMICS S120 Function Manual Drive Functions
Section "Data Sets"

Function block diagram

The parameter is included in this function block diagram. The structure of the parameter function and its relationship with other parameters is shown in the specified function block diagram.

Example:

Function block diagram: 3060: Function block diagram number 3060.3

3: Signal path (optional)

P group (refers only to access via BOP (Basic Operator Panel))

Specifies the function group to which the parameter belongs. The required parameter group can be set via p0004.

Note:

Parameter p0004 is CU-specific (present on Control Unit).

"Unit", "Unit group", and "Unit selection"

The standard unit of a parameter is specified in square parentheses after the values for "Min", "Max", and "Factory setting".

For parameters where the unit can be switched, the specifications for "Unit group" and "Unit selection" determine the group to which this parameter belongs and with which parameter the unit can be switched over.

Example:

Unit group: 7_1, unit selection: p0505

The parameter belongs to unit group 7_1 and the unit can be switched over using p0505.

Note:

Detailed information on switching over units can be found in the following references:

References: /FH1/ SINAMICS S120 Function Manual Drive Functions

References: /BA3/ SINAMICS S150 Operating Instructions

All the potential unit groups and possible unit selections are listed below.

Table 1-3 Unit groups (p0100)

Unit group	Unit selection for p0100 =		Reference value at %
	0	1	
7_4	Nm	lbf ft	-
8_4	N	lbf	-

Table 1-3 Unit groups (p0100), continued

Unit group	Unit selection for p0100 =		Reference value at %
	0	1	
14_2	W	HP	-
14_6	kW	HP	-
25_1	kgm ²	lb ft ²	-
27_1	kg	lb	-
28_1	Nm/A	lbf ft/A	-
29_1	N/Arms	lbf/Arms	-
30_1	m	ft	-

Table 1-4 Unit groups (p0349)

Unit group	Unit selection for p0349 =		Reference value at %
	1	2	
15_1	mH	%	$\frac{1000 \cdot p0304}{2 \cdot \pi \cdot \sqrt{3} \cdot p0305 \cdot p0310}$
16_1	Ohm	%	$\frac{p0304}{\sqrt{3} \cdot p0305}$

Table 1-5 Unit groups (p0505)

Unit group	Unit selection for p0505 =				Reference value at %
	1	2	3	4	
2_1	Hz	%	Hz	%	p2000
2_2	kHz	%	kHz	%	p2000
3_1	rpm	%	rpm	%	p2000
4_1	m/min	%	ft/min	%	p2000
4_2	m/min	m/min	ft/min	ft/min	-
5_1	Vrms	%	Vrms	%	p2001
5_2	V	%	V	%	p2001
5_3	V	%	V	%	p2001
6_1	mA rms	%	mA rms	%	p2002
6_2	A rms	%	A rms	%	p2002
6_3	mA	%	mA	%	p2002
6_4	A	%	A	%	p2002
6_5	A	%	A	%	p2002
7_1	Nm	%	lbf ft	%	p2003

Table 1-5 Unit groups (p0505), continued

Unit group	Unit selection for p0505 =				Reference value at %
	1	2	3	4	
7_2	Nm	Nm	lbf ft	lbf ft	-
7_3	Nm	%	lbf ft	%	1.0
8_1	N	%	lbf	%	p2003
8_2	N	N	lbf	lbf	-
8_3	N	%	lbf	%	1.0
14_1	W	%	HP	%	r2004
14_3	W	%	HP	%	r2004
14_4	W	%	HP	%	r2004
14_5	kW	%	HP	%	r2004
14_7	kW	%	HP	%	r2004
14_8	kW	%	HP	%	r2004
14_9	W	W	HP	HP	-
14_10	kW	kW	HP	HP	-
14_11	var	%	var	%	r2004
14_12	kvar	%	kvar	%	r2004
17_1	Nms/rad	%	lbf ft s/rad	%	p2000/p2003
18_1	V/A	%	V/A	%	p2002/p2001
19_1	A/V	%	A/V	%	p2001/p2002
21_1	°C	°C	°F	°F	-
21_2	K	K	°F	°F	-
22_1	m/s ²	m/s ²	ft/s ²	ft/s ²	-
22_2	m/s ²	%	ft/s ²	%	p2007
23_1	Vrms s/m	Vrms s/m	Vrms s/ft	Vrms s/ft	-
24_1	Ns/m	Ns/m	lbf s/ft	lbf s/ft	-
24_2	Ns/m	%	lbf s/ft	%	p2000/p2003
26_1	m/s ³	m/s ³	ft/s ³	ft/s ³	-
39_1	1/s ²	%	1/s ²	%	p2007

Table 1-6 Unit group (p0595)

Unit group	Unit selection for p0595 =		Reference value at %
	Value	Unit	
9_1	The values that can be set and the technological units are shown in p0595 (See Chapter 1.2).		

Parameter values

Min	Minimum value of the parameter [unit]
Max	Maximum value of the parameter [unit]
Factory setting	Shipped value (default) [unit]
	A different value may be displayed for certain parameters (e.g. p1800) at the initial commissioning stage. Reason: The setting of these parameters is determined by the operating environment of the Control Unit (e.g. depending on converter type, macro, power module).

Note:

For SINAMICS G150/G130/S150, the macros and their settings are provided in the following documentation:

References: /BAx/ x = 1, 2, 3
SINAMICS G150/G130/S150 Operating Instructions

Not for motor type

Specifies for which motor type this parameter has no significance

ASM: Induction motor

FEM: Separately excited synchronous motor

PEM: Permanent-magnet synchronous motor

REL: Reluctance motor/SIEMOSYN motor

Scaling

Specification of the reference value with which a signal value is automatically converted for a BICO interconnection.

The following reference values are possible:

- p2000 to p2007: Reference speed, reference voltage, etc.
- TEMP: 100 = 100%
- PERCENT: 1.0 = 100 %
- 4000h: 4000 hex = 100%

Expert list

Specifies whether this parameter is available in the expert list of the specified drive objects in the commissioning software.

1: Parameter does exist in the expert list.

0: Parameter exists in the expert list.

Notice:

Users assume full responsibility for using parameters marked "Expert list: 0" (parameter does not exist in the expert list).

These parameters and their functionalities have not been tested and no further user documentation is available for them (e.g. description of functions). Moreover, no support is provided for these parameters by "Technical Support" (hotline).

Description

Explanation of the function of a parameter

Values

Lists the possible values of a parameter

Recommendation

Information on recommended settings

Index

The name and meaning of each individual index is specified for indexed parameters.

The following applies to the values (Min, Max, Factory setting) of indexed adjustable parameters:

- Min, Max:

The adjustment range and unit apply to all indices.

- Factory setting:

When all indices have the same factory setting, index 0 is specified with the unit to represent all indices.

When the indices have different factory settings, they are all listed individually with the unit.

Bit array

For parameters with bit arrays, the following information is provided about each bit:

- Bit number and signal name
- Meaning with signal states 0 and 1
- Function block diagram (optional)

The signal is shown in this function block diagram.

Dependency

Conditions that must be fulfilled in connection with this parameter. Also includes special effects that can occur between this parameter and others.

See also: List of other relevant parameters to be considered

Safety notices

Important information that must be observed to avoid the risk of physical injury or material damage.

Information that must be observed to avoid any problems.

Information that the user or operator may find useful.

Danger

The description of this safety notice can be found at the beginning of this manual (see **Safety Notices**).



Alarm

The description of this safety notice can be found at the beginning of this manual (see **Safety Notices**).



Caution

The description of this safety notice can be found at the beginning of this manual (see **Safety Notices**).



Caution

The description of this safety notice can be found at the beginning of this manual (see **Safety Notices**).

Notice

The description of this safety notice can be found at the beginning of this manual (see **Safety Notices**).

Note

Information that the user or operator may find useful.

1.1.2 Number ranges of parameters

Note:

The following number ranges represent an overview for all parameters in SINAMICS.

The parameters for the product described in this List Manual are described in detail in Chapter 1.2.

Parameters are grouped into the following number ranges:

Table 1-7 Number ranges of parameters

Range		Description
From	To	
0000	0099	Operation and visualization
0100	0199	Commissioning
0200	0299	Power unit
0300	0399	Motor
0400	0499	Encoder
0500	0599	Technology and units
0600	0699	Thermal motor monitoring and motor model, maximum current
0700	0799	Command sources and terminals on the Control Unit, measuring sockets
0800	0839	CDS, DDS data sets (e.g. switch over, copy)
0840	0879	Sequence control (e.g. source for ON/OFF1)
0880	0899	Control and status words
0900	0999	PROFIBUS/PROFIdrive
1000	1199	Setpoint channel
1200	1299	Functions (e.g. motor holding brake)
1300	1399	V/f control
1400	1799	Closed-loop control
1800	1899	Gating unit
1900	1999	Power unit and motor identification
2000	2099	Communication (PROFIBUS)
2100	2199	Faults and alarms, monitoring functions
2200	2399	Technology controller
2900	2930	Fixed values (e.g. percentage, torque)
3400	3699	Infeed control (Active Line Module)
3800	3899	Friction characteristic

Table 1-7 Number ranges of parameters, continued

Range		Description
From	To	
3900	3999	Administration parameters
4000	4199	Terminal Board, Terminal Module (e.g. TB30, TM31)
4200	4399	Terminal Module (e.g. TM15, TM17)
6000	6999	SINAMICS GM/SM/GL
7000	7499	Parallel connection of power units
7800	7899	EEPROM read/write parameters
8500	8599	Data and macro management
8600	8799	CAN bus
8800	8899	Communication Board
9300	9399	Safety Integrated
9400	9499	Parameter consistency and storage
9500	9899	Safety Integrated
9900	9949	Topology
9950	9999	Diagnostics (internal)
10000	10099	Safety Integrated
11000	11299	Free technology controller 1, 2, 3
20000	20999	Free function blocks (FBLOCKS)
21000	25999	Drive Control Chart (DCC)
50000	53999	SINAMICS DC MASTER (DC closed loop control)
61000	61001	PROFINET

1.2 List of parameters

Product: SINAMICS S120/S150, Version: 4301400, Language: eng
 Objects: A_INF, B_INF, CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR

r0002	Infeed operating display / INF op_display		
A_INF, S_INF	Can be changed: - Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 250	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Operating display for the infeed.		
Value:	0: Operation - everything enabled 21: Ready for operation - set "Operation enable" = "1" (p0852) 31: Rdy for sw on - pre-chrg running (p0857) 32: Ready for switching on - set "ON/OFF1" = "0/1" (p0840) 35: Switching on inhibited - carry out first commissioning (p0010) 41: Switching on inhibited - set "ON/OFF1" = "0" (p0840) 42: Switching on inhibited - set "OC/OFF2" = "1" (p0844, p0845) 44: Switching on inhibited - connect 24 V to terminal EP (hardware) 45: Switching on inhibited - remove fault cause, acknowledge fault 46: Switching on inhibited - exit comm mode (p0009, p0010) 60: Infeed deactivated/not operational 200: Wait for booting/partial booting 250: Device signals a topology error		
Dependency:	Refer to: r0046		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		
Note:	OC: Operating condition EP: Enable Pulses (pulse enable) COMM: Commissioning		

r0002	Infeed operating display / INF op_display		
B_INF	Can be changed: - Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 250	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Operating display for the infeed.		
Value:	0: Operation - everything enabled 31: Rdy for sw on - pre-chrg running (p0857) 32: Ready for switching on - set "ON/OFF1" = "0/1" (p0840) 35: Switching on inhibited - carry out first commissioning (p0010) 41: Switching on inhibited - set "ON/OFF1" = "0" (p0840) 42: Switching on inhibited - set "OC/OFF2" = "1" (p0844, p0845) 44: Switching on inhibited - connect 24 V to terminal EP (hardware) 45: Switching on inhibited - remove fault cause, acknowledge fault 46: Switching on inhibited - exit comm mode (p0009, p0010) 60: Infeed deactivated/not operational 200: Wait for booting/partial booting 250: Device signals a topology error		
Dependency:	Refer to: r0046		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		
Note:	OC: Operating condition COMM: Commissioning		

r0002	Control Unit operating display / CU op_display		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 117	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Operating display for the Control Unit (CU).		
Value:	0: Operation 10: Ready 20: Wait for run-up 25: Wait for automatic FW update of DRIVE-CLiQ components 31: Commissioning software download active 33: Remove/acknowledge topology error 34: Exit commissioning mode 35: Carry out first commissioning 70: Initialization 80: Reset active 99: Internal software error 101: Specify topology 111: Insert drive object 112: Delete drive object 113: Change drive object number 114: Change component number 115: Run parameter download 117: Delete component		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		

r0002	Encoder DO operating display / Enc DO op_display		
ENCODER	Can be changed: - Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 250	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Operating display for encoder drive object		
Value:	0: Encoder in cyclic operation 35: Carry out first commissioning (p0010) 45: Remove fault cause, acknowledge fault 46: Exit commissioning mode (p0009, p0010) 60: Encoder deactivated 200: Wait for booting/partial booting 250: Device signals a topology error		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		

r0002	DRIVE-CLiQ Hub Module operating display / Hub op_display		
HUB	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	250	-
Description:	Operating display for the DRIVE-CLiQ Hub Module.		
Value:	0: Module in cyclic operation 40: Module not in cyclic operation 50: Alarm 60: Fault 70: Initialization 120: Module deactivated 200: Wait for booting/partial booting 250: Device signals a topology error		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		

r0002	Drive operating display / Drv op_display		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	250	-
Description:	Operating display for the drive.		
Value:	0: Operation - everything enabled 10: Operation - set "enable setpoint" = "1" (p1142, p1152) 11: Operation - set "enable speed controller" = "1" (p0856) 12: Operation - RFG frozen, set "RFG start" = "1" (p1141) 13: Operation - set "enable RFG" = "1" (p1140) 14: Oper. - MotID, excit. running and/or brake opens, SS2, SOS 15: Operation - open brake (p1215) 16: Operation - withdraw braking with OFF1 using "ON/OFF1" = "1" 17: Operation - braking with OFF3 can only be interrupted with OFF2 18: Operation - brake on fault, remove fault, acknowledge 19: Operation - armature short-circ./DC brake act. (p1230, p1231) 21: Ready for operation - set "Operation enable" = "1" (p0852) 22: Ready for operation - demagnetizing running (p0347) 23: Ready for operation - set "Infeed operation" = "1" (p0864) 31: Ready for switching on - set "ON/OFF1" = "0/1" (p0840) 35: Switching on inhibited - carry out first commissioning (p0010) 41: Switching on inhibited - set "ON/OFF1" = "0" (p0840) 42: Switching on inhibited - set "OC/OFF2" = "1" (p0844, p0845) 43: Switching on inhibited - set "OC/OFF3" = "1" (p0848, p0849) 44: Switching on inhibited - connect 24 V to terminal EP (hardware) 45: Switching on inhibited - rectify fault, acknowledge fault, STO 46: Switching on inhibited - exit comm mode (p0009, p0010) 60: Drive object deactivated/not operational 200: Wait for booting/partial booting 250: Device signals a topology error		
Dependency:	Refer to: r0046		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		

Note:

OC: Operating condition
 EP: Enable Pulses (pulse enable)
 RFG: Ramp-function generator
 COMM: Commissioning
 MotID: Motor data identification
 SS2: Safe Stop 2
 SOS: Safe Operating Stop
 STO: Safe Torque Off

r0002	TB30 operating display / TB30 op_display		
TB30	Can be changed: - Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 250	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Operating display for Terminal Board 30 (TB30).		
Value:	0: Module in cyclic operation 40: Module not in cyclic operation 60: Fault 70: Initialization 80: Reset active 120: Module deactivated 200: Wait for run-up 250: Device signals a topology error		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		

r0002	TM120 operating display / TM120 op_display		
TM120	Can be changed: - Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 250	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Operating display for Terminal Module 120 (TM120)		
Value:	0: Module in cyclic operation 40: Module not in cyclic operation 50: Alarm 60: Fault 70: Initialization 120: Module deactivated 200: Wait for booting/partial booting 250: Device signals a topology error		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		

r0002			
TM15 operating display / TM15 op_display			
TM15	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	250	-
Description:	Operating display for Terminal Module 15 (TM15).		
Value:	0: Module in cyclic operation 40: Module not in cyclic operation 50: Alarm 60: Fault 70: Initialization 120: Module deactivated 200: Wait for booting/partial booting 250: Device signals a topology error		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		
r0002			
TM15DI/DO operating display / TM15D op_display			
TM15DI_DO	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	250	-
Description:	Operating display for Terminal Module 15 (TM15).		
Value:	0: Module in cyclic operation 40: Module not in cyclic operation 50: Alarm 60: Fault 70: Initialization 120: Module deactivated 200: Wait for booting/partial booting 250: Device signals a topology error		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		
r0002			
TM17 operating display / TM17 op_display			
TM17	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	250	-
Description:	Operating display for Terminal Module 17 (TM17).		
Value:	0: Module in cyclic operation 40: Module not in cyclic operation 50: Alarm 60: Fault 70: Initialization 120: Module deactivated 200: Wait for booting/partial booting 250: Device signals a topology error		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		

r0002	TM31 operating display / TM31 op_display		
TM31	Can be changed: - Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 250	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Operating display for Terminal Module 31 (TM31).		
Value:	0: Module in cyclic operation 40: Module not in cyclic operation 50: Alarm 60: Fault 70: Initialization 120: Module deactivated 200: Wait for booting/partial booting 250: Device signals a topology error		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		
r0002	TM41 operating display / TM41 op_display		
TM41	Can be changed: - Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 250	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Operating display for Terminal Module 41 (TM41).		
Value:	0: Operation - everything enabled 10: Operation - set "enable setpoint" = "1" (p1142) 12: Operation - RFG frozen, set "RFG start" = "1" (p1141) 13: Operation - set "enable RFG" = "1" (p1140) 18: Operation - brake on fault, remove fault, acknowledge 21: Ready for operation - set "Operation enable" = "1" (p0852) 31: Ready for switching on - set "ON/OFF1" = "0/1" (p0840) 41: Switching on inhibited - set "ON/OFF1" = "1/0" (p0840) 42: Switching on inhibited - set "OC/OFF2" = "1" (p0844) 43: Switching on inhibited - set "OC/OFF3" = "1" (p0848) 45: Switching on inhibited - remove fault cause, acknowledge fault 46: Switching on inhibited - exit comm mode (p0009, p0010) 120: Module deactivated 200: Wait for booting/partial booting 250: Device signals a topology error		
Notice:	For several missing enable signals, the corresponding value with the highest number is displayed.		
Note:	OC: Operating condition RFG: Ramp-function generator COMM: Commissioning		

r0002 TM54F operating display / TM54F op_displayTM54F_MA,
TM54F_SL**Can be changed:** -**Data type:** Integer16**P-Group:** -**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

250

Access level: 1**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Operating display for Terminal Module 54F (TM54F).**Value:**

0: Module in cyclic operation
 40: Module not in cyclic operation
 50: Alarm
 60: Fault
 70: Initialization
 120: Module deactivated
 200: Wait for booting/partial booting
 250: Device signals a topology error

p0003 BOP access level / BOP acc_levelCU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** C1, U, T**Data type:** Integer16**P-Group:** -**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

4

Access level: 1**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

1

Description: Sets the access level for reading and writing parameters via the Basic Operator Panel (BOP).**Value:**

0: User-defined
 1: Standard
 2: Extended
 3: Expert
 4: Service

Note:

Access level 0 (user-defined):
 Parameters from the user-defined list (p0013). Not used as of firmware version 2.6 (p0016).
 Access level 1 (standard):
 Parameters for the simplest operator control possibility (e.g. p1120 = ramp-function generator, ramp-up time).
 Access level 2 (extended):
 Parameters to operate the basic functions of the drive unit.
 Access level 3 (experts):
 Expert know-how is required for these parameters (e.g. BICO parameterization).
 Access level 4 (service):
 For these parameters, it is necessary that authorized service personnel enter the appropriate password (p3950).

p0004	BOP display filter / BOP disp_filter		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C2(1), U, T Data type: Integer16 P-Group: - Not for motor type: ASM Min 0 Max 99	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the display filter for parameters with the Basic Operator Panel (BOP).		
Value:	0: All parameters 1: Displays, signals 2: Power unit 3: Motor 4: Encoder/pos enc 5: Technology/units 7: Digital inputs/outputs commands sequence control 8: Analog inputs/outputs 10: Setpoint channel/ramp-fct generator 12: Functions 13: V/f control 14: Control 15: Data sets 17: Basic positioner 18: Gating unit 19: Motor identification 20: Communication 21: Faults, alarms, monitoring functions 25: Closed loop position control 28: Free function blocks 47: Trace and function generator 50: OA parameters 90: Topology 95: Safety Integrated 98: Command Data Sets (CDS) 99: Drive Data Sets (DDS)		
Dependency:	Refer to: p0003		
Notice:	The display filter via p0004 provides precise filtering and displays the corresponding parameters only when p0009 and p0010 = 0.		
Note:	The set access level via p0003 is also relevant for the display filter via p0004. Examples (assumption: p0009 = p0010 = 0): p0003 = 1, p0004 = 3 --> Only the parameters for the motor are displayed with access level 1. p0003 = 2, p0004 = 3 --> Only the parameters for the motor are displayed with access levels 1 and 2.		

p0005[0...1]	BOP operating display selection / BOP op_disp sel		
A_INF, B_INF, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VEC- TOR	Can be changed: U, T Data type: Unsigned16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 65535	Factory setting [0] 2 [1] 0
Description:	Sets the parameter number and parameter index for display for p0006 = 2, 4 for the Basic Operator Panel (BOP). Examples for the SERVO drive object: p0005[0] = 21, p0005[1] = 0: Actual speed smoothed (r0021) p0005[0] = 25, p0005[1] = 0: Output voltage smoothed (r0025) p0005[0] = 27, p0005[1] = 0: Absolute current actual value, smoothed (r0027)		
Index:	[0] = Parameter number [1] = Parameter index		
Dependency:	Refer to: p0006		
Note:	Procedure: 1. The parameter number to be displayed should be set in index 0. Only the monitoring parameters (read-only parameters) can be set that actually exist for the actual drive object. If the set parameter number is not indexed, or if there is an index in index 1 that lies outside the valid range of the set parameter, then index 1 is automatically set to 0. 2. The index that belongs to the parameter set in index 0 should be set in index 1. The permissible changes in index 1 always depend on the parameter number set in index 0.		
p0006	BOP operating display mode / BOP op_disp mode		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 4	Factory setting 4
Description:	Sets the mode of the operating display for the Basic Operator Panel (BOP) in the operating states "ready for operation" and "operation".		
Value:	0: Operation --> r0021, otherwise r0020 <--> r0021 1: Operation --> r0021, otherwise r0020 2: Operation --> p0005, otherwise p0005 <--> r0020 3: Operation --> r0002, otherwise r0002 <--> r0020 4: p0005		
Dependency:	Refer to: p0005		
Note:	Mode 0 ... 3 can only be selected if also r0020, r0021 are available on the drive object. Mode 4 is available for all drive objects.		

p0006			
BOP operating display mode / BOP op_disp mode			
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min 4	Max 4	Factory setting 4
Description:	Sets the mode of the operating display for the Basic Operator Panel (BOP) in the operating states "ready for operation" and "operation".		
Value:	4: p0005		
Dependency:	Refer to: p0005		
Note:	Mode 0 ... 3 can only be selected if also r0020, r0021 are available on the drive object. Mode 4 is available for all drive objects.		

p0007			
BOP background lighting / BOP lighting			
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min 0 [s]	Max 2000 [s]	Factory setting 0 [s]
Description:	Sets the delay time until the background lighting of the Basic Operator Panel (BOP) is switched off. If no keys are actuated, then the background lighting automatically switches itself off after this time has expired.		
Note:	p0007 = 0: Background lighting is always switched on (factory setting).		

p0008			
BOP drive object after booting / BOP DO after boot			
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min 1	Max 65535	Factory setting 1
Description:	Sets the required drive object that is active at the Basic Operator Panel (BOP) after booting.		
Note:	The value from p0008 initializes the display on the Basic Operator Panel (BOP) at the top left after booting. The drive object Control Unit is selected using the value 1.		

p0009			
Device commissioning parameter filter / Dev comm par_filt			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1, T Data type: Integer16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 10000	Factory setting 1
Description:	Sets the device and basic drive commissioning. By appropriately setting this parameter, those parameters are filtered that can be written into in the various commissioning steps.		

Value:	0: Ready 1: Device configuration 2: Defining the drive type/function module 3: Drive base configuration 4: Data set base configuration 29: Device download 30: Parameter reset 50: OA application configuration 55: OA application installation 101: Topology input 111: Insert drive object 112: Delete drive object 113: Change drive object number 114: Change component number 115: Parameter download 117: Delete component 10000: Ready (asynchronous)
Notice:	For p0009 = 10000 the following applies: After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.
Note:	The drives can only be powered up outside the device commissioning (the inverter enabled). In this case, p0009 must be 0 (Ready) and the individual drive objects must have already gone into operation (p0010). p0009 = 1: Device configuration At the first commissioning of the device, after booting, the device is in the "device configuration" state. To start the internal automatic first commissioning of the drive unit, p0009 should be set to 0 (Ready) after the ID for the actual topology (r0098) was transferred into the ID for the target topology (p0099). To do this, it is sufficient to set a single index value of p0099[x] the same as r0098[x]. Before the device has been completely commissioned, no other parameter can be changed. After the first commissioning was carried out, in this state, when required, other basic device configuration parameters can be adapted (e.g. the basic sampling time in p0110). p0009 = 2: Defines the drive type / function module In this state, the drive object types and/or the function modules can be changed or selected for the individual drive objects. To do this, the drive object type can be set using p0107[0...15] and the function can be set using p0108[0...15] (refer to p0101[0...15]). p0009 = 3: Drive basis configuration In this state, after the device has been commissioned for the first time, basic changes can be made for the individual drive objects (e.g. sampling times in p0111, p0112, p0115 and the number of data sets in p0120, p0130, p0140, p0170, p0180). p0009 = 4: Data set basis configuration In this state, after the device has been commissioned for the first time, for the individual drive objects changes can be made regarding the assignment of the components (p0121, p0131, p0141, p0151, p0161) to the individual data sets and the assignment of the power unit, motor and encoder to the drive data sets (p0185, ...). p0009 = 29: Device download If a download is made using the commissioning software, the device is automatically brought into this state. After the download has been completed, p0009 is automatically set to 0 (ready). It is not possible to manually set p0009 to this value. p0009 = 30: Parameter reset In order to bring the complete unit into the "first commissioning" state or to load the parameters saved using p0977, to start, p0009 must be set to this value. p0976 can then be changed to the required value. p0009 = 50: OA application configuration In this state, after the device has been commissioned for the first time, changes can be made for the individual drive objects regarding the activity (p4956) of the OA applications. p0009 = 55: OA application installation OA applications can be installed and/or uninstalled in this state. p0009 = 101: Topology input In this state, the DRIVE-CLiQ target topology can be entered using p9902 and p9903. p0009 = 111: Insert drive object This state allows a new drive object to be inserted using p9911.

p0009 = 112: Delete drive object

This state allows existing drive objects to be deleted using p9912 after the device has been commissioned for the first time.

p0009 = 113: Change drive object number

This state allows the drive object number of existing drive objects to be changed using p9913 after the device has been commissioned for the first time.

p0009 = 114: Change component number

This state allows the component number of existing components to be changed using p9914 after the device has been commissioned for the first time.

p0009 = 115: Parameter download

This state allows the complete device and drive commissioning using the parameter services.

p0009 = 117: Delete component

This state allows components to be deleted using p9917 after the device has been commissioned for the first time.

p0010	Infeed commissioning parameter filter / INF comm par_filt		
A_INF, B_INF, S_INF	Can be changed: C2(1), T	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	30	1
Description:	Sets the parameter filter to commission an infeed unit. Setting this parameter filters out the parameters that can be written into in the various commissioning steps.		
Value:	0: Ready 1: Quick commissioning 2: Power unit commissioning 5: Technological application/units 29: Only Siemens int 30: Parameter reset		
Note:	The drive can only be powered up outside the drive commissioning (inverter enable). To realize this, this parameter must be set to 0. For p3900 not equal to 0, at the end of the quick commissioning, this parameter is automatically reset to 0. Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1.		

p0010	Encoder DO commissioning parameter filter / EncDO com par_filt		
ENCODER	Can be changed: C2(1), T	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	30	0
Description:	Sets the parameter filter to commission an encoder drive object. Setting this parameter filters out the parameters that can be written into in the various commissioning steps. For the BOP, this setting also causes the read access operations to be filtered.		
Value:	0: Ready 4: Encoder commissioning 5: Technological application/units 29: Only Siemens int 30: Parameter reset		
Note:	Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1.		

p0010	Drive commissioning parameter filter / Drv comm. par_filt		
SERVO, VECTOR	Can be changed: C2(1), T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000	Access level: 1 Func. diagram: 2800, 2846 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the parameter filter to commission a drive. Setting this parameter filters out the parameters that can be written into in the various commissioning steps.		
Value:	0: Ready 1: Quick commissioning 2: Power unit commissioning 3: Motor commissioning 4: Encoder commissioning 5: Technological application/units 15: Data sets 17: Basic positioner commissioning 25: Position control commissioning 29: Only Siemens int 30: Parameter reset 95: Safety Integrated commissioning 10000: Ready with immediate feedback signal		
Notice:	For p0010 = 10000 the following applies: After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	The drive can only be powered up outside the drive commissioning (inverter enable). To realize this, this parameter must be set to 0. For p3900 not equal to 0, at the end of the quick commissioning, this parameter is automatically reset to 0. Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1. p0010 = 10000 corresponds to p0010 = 0. Unlike with p0010 = 0, the parameter modification is applied immediately and the calculations are made in the background. Further parameter modifications cannot be made while the calculations are being performed.		

p0010	TB30 commissioning parameter filter / TB30 comm.par_filt		
TB30	Can be changed: C2(1), T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 30	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the parameter filter for commissioning a Terminal Board 30 (TB30). Setting this parameter filters out the parameters that can be written into in the various commissioning steps. For the BOP, this setting also causes the read access operations to be filtered.		
Value:	0: Ready 29: Only Siemens int 30: Parameter reset		
Dependency:	Refer to: p0970		
Note:	Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1.		

p0010 TM120 commissioning parameter filter / TM120 com par_filt			
TM120	Can be changed: C2(1), T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 30	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the parameter filter for commissioning a Terminal Module 120 (TM120). Setting this parameter filters out the parameters that can be written into in the various commissioning steps. For the BOP, this setting also causes the read access operations to be filtered.		
Value:	0: Ready 29: Only Siemens int 30: Parameter reset		
Dependency:	Refer to: p0970		
Note:	Only the following values are possible: p0010 = 0, 30 Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1.		
p0010 TM15 commissioning parameter filter / TM15 comm par_filt			
TM15	Can be changed: C2(1), T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 30	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the parameter filter for commissioning a Terminal Module 15 (TM15). Setting this parameter filters out the parameters that can be written into in the various commissioning steps. For the BOP, this setting also causes the read access operations to be filtered.		
Value:	0: Ready 29: Only Siemens int 30: Parameter reset		
Dependency:	Refer to: p0970		
Note:	Only the following values are possible: p0010 = 0, 30 Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1.		
p0010 TM15DI/DO commissioning the parameterizing filter / TM15D com par_filt			
TM15DI_DO	Can be changed: C2(1), T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 30	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the parameter filter for commissioning a Terminal Module 15 (TM15). Setting this parameter filters out the parameters that can be written into in the various commissioning steps. For the BOP, this setting also causes the read access operations to be filtered.		
Value:	0: Ready 29: Only Siemens int 30: Parameter reset		
Dependency:	Refer to: p0970		
Note:	Only the following values are possible: p0010 = 0, 30 Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1.		

p0010	TM17 commissioning parameter filter / TM17 comm par_filt		
TM17	Can be changed: C2(1), T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 30	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the parameter filter for commissioning a Terminal Module 17 (TM17). Setting this parameter filters out the parameters that can be written into in the various commissioning steps. For the BOP, this setting also causes the read access operations to be filtered.		
Value:	0: Ready 29: Only Siemens int 30: Parameter reset		
Dependency:	Refer to: p0970		
Note:	Only the following values are possible: p0010 = 0, 30 Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1.		
p0010	TM31 commissioning parameter filter / TM31 comm par_filt		
TM31	Can be changed: C2(1), T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 30	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the parameter filter for commissioning a Terminal Module 31 (TM31). Setting this parameter filters out the parameters that can be written into in the various commissioning steps. For the BOP, this setting also causes the read access operations to be filtered.		
Value:	0: Ready 29: Only Siemens int 30: Parameter reset		
Dependency:	Refer to: p0970		
Note:	Only the following values are possible: p0010 = 0, 30 Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1.		
p0010	TM41 commissioning parameter filter / TM41 comm par_filt		
TM41	Can be changed: C2(1), T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 30	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the parameter filter for commissioning a Terminal Module 41 (TM41). Setting this parameter filters out the parameters that can be written into in the various commissioning steps. For the BOP, this setting also causes the read access operations to be filtered.		
Value:	0: Ready 4: Encoder commissioning 5: Technological application/units 29: Only Siemens int 30: Parameter reset		
Dependency:	Refer to: p0970		
Note:	Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1.		

p0010	TM54F commissioning parameter filter / TM54F com par_filt		
TM54F_MA	Can be changed: C2(1), T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 95	Access level: 1 Func. diagram: 2847 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the parameter filter for commissioning a Terminal Module 54F (TM54F). Setting this parameter filters out the parameters that can be written into in the various commissioning steps. For the BOP, this setting also causes the read access operations to be filtered.		
Value:	0: Ready 29: Only Siemens int 30: Parameter reset 95: Safety Integrated commissioning		
Dependency:	Refer to: p0970		
Note:	Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1.		
p0011	BOP password entry (p0013) / BOP passw ent p13		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the password for the Basic Operator Panel (BOP).		
Dependency:	Refer to: p0012, p0013		
p0012	BOP password acknowledgement (p0013) / BOP passw ackn p13		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Acknowledges the password for the Basic Operator Panel (BOP).		
Dependency:	Refer to: p0011, p0013		

p0013[0...49]	BOP user-defined list / BOP list		
A_INF, B_INF, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VEC- TOR	Can be changed: U, T Data type: Unsigned16 P-Group: Functions Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 65535	Factory setting 0
Description:	Sets the required parameters to read and write via the Basic Operator Panel (BOP). Activation: 1. p0003 = 3 (expert). 2. p0013[0...49] = requested parameter number 3. If required, enter p0011 = password in order to prevent non-authorized deactivation. 4. p0016 = 1 --> activates the selected user-defined list. Deactivation/change: 1. p0003 = 3 (expert). 2. If required, p0012 = p0011, in order to be authorized to change or deactivate the list. 3. If required p0013[0...49] = required parameter number. 4. p0016 = 1 --> activates the modified user-defined list. 5. p0003 = 0 --> deactivates the user-defined list.		
Dependency:	Refer to: p0009, p0011, p0012, p0976		
Note:	The following parameters can be read and written on the Control Unit drive object: - p0003 (access stage) - p0009 (device commissioning, parameter filter) - p0012 (BOP password acknowledgement (p0013)) The following applies for the user-defined list: - password protection is only available on the drive object Control Unit and is valid for all of the drive objects. - p0013 cannot be included in the user-defined list for all drive objects. - p0003, p0009, p0011, p0012, p0976 cannot, for the drive object Control Unit, be included in the user-defined list. - the user-defined list can be cleared and deactivated "restore factory setting". A value of 0 means: Entry is empty.		
p0015	Macro drive object / Macro DO		
A_INF, B_INF, S_INF, SERVO, TM120, TM15DI_DO, TM31, VECTOR	Can be changed: C2(1) Data type: Unsigned32 P-Group: Commands Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 999999	Factory setting 0
Description:	Runs the corresponding macro files. The selected macro file must be available on the memory card/device memory. Example: p0015 = 6 --> the macro file PM000006.ACX is run.		
Dependency:	Refer to: p0700, p1000, p1500, r8570		

- Notice:** After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.
No errors were issued during quick commissioning (p3900 = 1) when writing to parameters of the QUICK_IBN group!
- Note:** The macros in the specified directory are displayed in r8570. r8570 is not in the expert list of the commissioning software.
Macros available as standard are described in the technical documentation of the particular product.
The parameter is not influenced by setting the factory setting.

p0015**Macro drive unit / Macro drv unit**

CU_CX32, CU_I,	Can be changed: C1	Calculated: -	Access level: 1
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: -	Units group: -	Unit selection: -
CU_S_CU310PN	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	999999	0

- Description:** Runs the corresponding macro files.
The selected macro file must be available on the memory card/device memory.
Example:
p0015 = 6 --> the macro file PM000006.ACX is run.
- Dependency:** Refer to: p0700, p1000, p1500, r8570
- Notice:** After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.
- Note:** The macros in the specified directory are displayed in r8570. r8570 is not in the expert list of the commissioning software.
Macros available as standard are described in the technical documentation of the particular product.
The parameter is not influenced by setting the factory setting.

p0015**Macro drive unit / Macro drv unit**

CU_S_S150	Can be changed: C1	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	999999	1

- Description:** Runs the corresponding macro files.
The selected macro file must be available on the memory card/device memory.
Example:
p0015 = 6 --> the macro file PM000006.ACX is run.
- Dependency:** Refer to: p0700, p1000, p1500, r8570
- Notice:** After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.
- Note:** The macros in the specified directory are displayed in r8570. r8570 is not in the expert list of the commissioning software.
Macros available as standard are described in the technical documentation of the particular product.
The parameter is not influenced by setting the factory setting.

p0016 Activate BOP user-defined list / BOP user list actCU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** C1, U, T**Data type:** Integer16**P-Group:** -**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

1

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description: Setting for activating/deactivating the user-defined list for the Basic Operator Panel (BOP).
If p0016 = 1, then it is only possible to access parameters in the parameter list (p0013).**Value:** 0: BOP user-defined list deactivated
1: BOP user-defined list activated**Dependency:** Refer to: p0011, p0012, p0013**Note:** The user-defined list can only be deactivated with p0011 = p0012**r0018 Control Unit Firmware-Version / CU FW version**CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** -**Data type:** Unsigned32**P-Group:** -**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 1**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Displays the firmware version of the Control Unit.**Dependency:** Refer to: r0128, r0148, r0158, r0197, r0198**Note:** Example:
The value 1010100 should be interpreted as V01.01.01.00.**r0019.0...14 CO/BO: Control word BOP / STW BOP**CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** -**Data type:** Unsigned16**P-Group:** Displays, signals**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Displays the control word for the Basic Operator Panel (BOP).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	ON / OFF (OFF1)	ON	OFF (OFF1)	-
	01	No coast-down / coast-down (OFF2)	No coast down	Coast down (OFF2)	-
	02	No Quick Stop / Quick Stop (OFF3)	No Quick Stop	Quick Stop (OFF3)	-
	07	Acknowledge fault (0 -> 1)	Yes	No	-
	13	Motorized potentiometer raise	Yes	No	-
	14	Motorized potentiometer lower	Yes	No	-

r0020 Speed setpoint smoothed / n_set smth

SERVO, VECTOR

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [rpm]

Calculated: -**Dynamic index:** -**Units group:** 3_1**Scaling:** p2000**Max**

- [rpm]

Access level: 2**Func. diagram:** 5020, 6799**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [rpm]

Description: Displays the currently smoothed speed setpoint at the input of the speed controller or V/f characteristic (after the interpolator).**Dependency:** Refer to: r0060

Note: Smoothing time constant = 100 ms
The signal is not suitable as a process quantity and may only be used as a display quantity.
The speed setpoint is available smoothed (r0020) and unsmoothed (r0060).

r0020	Velocity setpoint smoothed / v_set smth		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 2 Func. diagram: 5020, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the currently smoothed velocity setpoint at the input of the velocity controller or V/f characteristic (after the interpolator).		
Dependency:	Refer to: r0060		
Note:	Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The velocity setpoint is available smoothed (r0020) and unsmoothed (r0060).		

r0021	CO: Actual speed smoothed / n_act smooth		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 2 Func. diagram: 1580, 1680, 4710, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the smoothed actual value of the motor speed.		
Dependency:	Refer to: r0063		
Note:	Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The value displayed in r0021 is the smoothed value of r0063.		

r0021	CO: Actual velocity smoothed / v_act smooth		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 2 Func. diagram: 1580, 1680, 4710, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the smoothed actual value of the motor velocity.		
Dependency:	Refer to: r0063		
Note:	Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The value displayed in r0021 is the smoothed value of r0063.		

r0022 Speed actual value rpm smoothed / n_ist rpm smooth

SERVO, VECTOR

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 1580, 1680, 4710, 6799**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** p2000**Expert list:** 1**Min****Max****Factory setting**

- [rpm]

- [rpm]

- [rpm]

Description:

Displays the smoothed actual value of the motor speed.

r0022 is identical to r0021, however, it always has units of rpm and contrary to r0021 cannot be changed over.

Dependency:

Refer to: r0063

Note:

Smoothing time constant = 100 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The value displayed in r0022 is the smoothed value of r0063.

r0022 Actual velocity smoothed / v_act smooth

SERVO (Lin)

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 1580, 1680, 4710, 6799**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** p2000**Expert list:** 1**Min****Max****Factory setting**

- [m/min]

- [m/min]

- [m/min]

Description:

Displays the smoothed actual value of the motor velocity.

Dependency:

Refer to: r0063

Note:

Smoothing time constant = 100 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The value displayed in r0022 is the smoothed value of r0063.

r0024 CO: Line supply frequency smoothed / f_line smooth

A_INF, S_INF

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 8850, 8950**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** p2000**Expert list:** 1**Min****Max****Factory setting**

- [Hz]

- [Hz]

- [Hz]

Description:

Displays the smoothed line supply frequency.

Dependency:

Refer to: r0066

Note:

Smoothing time constant = 300 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The line frequency is available smoothed (r0024) and unsmoothed (r0066).

A positive sign of the frequency is obtained when the line supply phases U, V and W are connected with the correct phase sequence.

A negative sign of the frequency is obtained when the 3 line phases are interchanged therefore designating a negative direction of the rotating field of the 3-phase line supply voltage.

r0024	Output frequency smoothed / f_outp smooth		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: p2000 Max - [Hz]	Access level: 3 Func. diagram: 1690, 5300, 5730, 6799 Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the smoothed converter frequency.		
Dependency:	Refer to: r0066		
Note:	Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The output frequency is available smoothed (r0024) and unsmoothed (r0066).		
r0025[0...3]	CO: Input voltage smoothed / V_inp smooth		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [Vrms]	Access level: 2 Func. diagram: 8850, 8950 Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the smoothed actual value of the input voltage.		
Index:	[0] = Voltage at input terminals of power unit from line supply model [1] = Voltage at VSM or at input terminals of the line filter [2] = Voltage of the voltage source from the line supply model [3] = Smoothed voltage of voltage source from line supply model		
Dependency:	Refer to: r0072		
Note:	Smoothing time constant = 300 ms The signals are not suitable as process quantity and may only be used as display quantities. The input voltages are available smoothed (r0025) and unsmoothed (r0072). Re r0025[0]: Pulsed voltage at the line supply input terminals of the power unit. The value is calculated from the modulation depth r0074 and is therefore only correct in the closed-loop controlled mode and when the pulses are enabled. Re r0025[1]: Absolute voltage at the input terminals of the line filter or the connection point of a VSM. The value is calculated from the VSM measured values r3661 and r3662 and is therefore equal to 0 if a VSM is not connected. Re r0025[2]: Estimated value for the voltage of the voltage source that is calculated in the voltage model of the line supply PLL. Re r0025[3]: Smoothed display value of the filtered source voltage from r0072[3].		
r0025	CO: Output voltage smoothed / V_outp smooth		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [Vrms]	Access level: 2 Func. diagram: 1690, 5730, 6799 Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the smoothed output voltage of the power unit.		
Dependency:	Refer to: r0072		

Note: Smoothing time constant = 100 ms
The signal is not suitable as a process quantity and may only be used as a display quantity.
The output voltage is available smoothed (r0025) and unsmoothed (r0072).

r0026			
CO: DC link voltage smoothed / Vdc smooth			
A_INF, B_INF, S_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5730, 6799, 8750, 8850, 8950
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min - [V]	Max - [V]	Factory setting - [V]
Description:	Displays the smoothed actual value of the DC link voltage.		
Dependency:	Refer to: r0070		
Notice:	This smoothed signal is not suitable for diagnostics or evaluation of dynamic operations. In this case, the unsmoothed value should be used.		
Note:	A_INF, B_INF, S_INF: smoothing time constant = 300 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The DC link voltage is available smoothed (r0026) and unsmoothed (r0070).		

r0026			
CO: DC link voltage smoothed / Vdc smooth			
SERVO	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5730, 8750, 8850, 8950
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min - [V]	Max - [V]	Factory setting - [V]
Description:	Displays the smoothed actual value of the DC link voltage.		
Dependency:	Refer to: r0070		
Notice:	For SINAMICS S120 AC Drive (AC/AC) the following applies: When measuring a DC link voltage < 200 V, for the Power Module (e.g. PM340) a valid measured value is not supplied. In this case, when an external 24 V power supply is connected, a value of approx. 24 V is displayed in the display parameter.		
Note:	SERVO, VECTOR: Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The DC link voltage is available smoothed (r0026) and unsmoothed (r0070).		

r0026			
CO: DC link voltage smoothed / Vdc smooth			
VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 6799, 8750, 8850, 8950
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min - [V]	Max - [V]	Factory setting - [V]
Description:	Displays the smoothed actual value of the DC link voltage.		
Dependency:	Refer to: r0070		
Notice:	For SINAMICS S120 AC Drive (AC/AC) the following applies: When measuring a DC link voltage < 200 V, for the Power Module (e.g. PM340) a valid measured value is not supplied. In this case, when an external 24V power supply is connected, a value of approx. 24 V is displayed in the display parameter.		

Note: SERVO, VECTOR: Smoothing time constant = 100 ms
 The signal is not suitable as a process quantity and may only be used as a display quantity.
 The DC link voltage is available smoothed (r0026) and unsmoothed (r0070).

r0027	CO: Absolute actual current smoothed / I_act abs val smth		
A_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32	Calculated: - Dynamic index: -	Access level: 2 Func. diagram: 5730, 6799, 8850, 8950
	P-Group: Displays, signals Not for motor type: -	Units group: - Scaling: p2002	Unit selection: - Expert list: 1
	Min - [Arms]	Max - [Arms]	Factory setting - [Arms]
Description:	Displays the smoothed absolute actual current value.		
Dependency:	Refer to: r0068		
Notice:	This smoothed signal is not suitable for diagnostics or evaluation of dynamic operations. In this case, the unsmoothed value should be used.		
Note:	A_INF, S_INF, VECTOR: Smoothing time constant = 300 ms SERVO: Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The absolute current actual value is available smoothed (r0027) and unsmoothed (r0068).		

r0027	CO: Absolute actual current smoothed / I_act abs val smth		
B_INF	Can be changed: - Data type: FloatingPoint32	Calculated: - Dynamic index: -	Access level: 2 Func. diagram: 8750
	P-Group: Displays, signals Not for motor type: -	Units group: 6_4 Scaling: p2002	Unit selection: p0505 Expert list: 1
	Min - [A]	Max - [A]	Factory setting - [A]
Description:	Displays the smoothed absolute actual current value.		
Dependency:	Refer to: r0068		
Notice:	This smoothed signal is not suitable for diagnostics or evaluation of dynamic operations. In this case, the unsmoothed value should be used. For Basic Line Modules of chassis format, the displayed value is invalid as these units do not have any current sensing.		
Note:	Smoothing time constant = 300 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The absolute current actual value is available smoothed (r0027) and unsmoothed (r0068).		

r0028	Modulation depth smoothed / Mod_depth smth		
A_INF, SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32	Calculated: - Dynamic index: -	Access level: 3 Func. diagram: 5730, 6799, 8950
	P-Group: Displays, signals Not for motor type: -	Units group: - Scaling: p2002	Unit selection: - Expert list: 1
	Min - [%]	Max - [%]	Factory setting - [%]
Description:	Displays the smoothed actual value of the modulation depth.		
Dependency:	Refer to: r0074		
Note:	A_INF: Smoothing time constant = 300 ms SERVO, VECTOR: Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The modulation depth is available smoothed (r0028) and unsmoothed (r0074).		

r0029 Reactive current actual value smoothed / I_{react} smooth

A_INF, S_INF

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 8850, 8950**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** p2002**Expert list:** 1**Min****Max****Factory setting**

- [Arms]

- [Arms]

- [Arms]

Description: Displays the smoothed actual value of the reactive current component.**Note:** Smoothing time constant = 300 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The reactive current actual value is available smoothed (r0029) and unsmoothed (r0076).

r0029 Current actual value field-generating smoothed / I_d_act smooth

SERVO, VECTOR

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 5730, 6799**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** p2002**Expert list:** 1**Min****Max****Factory setting**

- [Arms]

- [Arms]

- [Arms]

Description: Displays the smoothed field-generating actual current.**Dependency:** Refer to: r0076**Note:** SERVO: Smoothing time constant = 100 ms

VECTOR: Smoothing time constant = 300 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The field-generating current actual value is available smoothed (r0029) and unsmoothed (r0076).

r0030 Active current actual value smoothed / I_{active} smooth

A_INF, S_INF

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 8850, 8950**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** p2002**Expert list:** 1**Min****Max****Factory setting**

- [Arms]

- [Arms]

- [Arms]

Description: Displays the smoothed actual value of the active current components.**Dependency:** Refer to: r0078**Note:** Smoothing time constant = 300 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The active current actual value is available smoothed (r0030) and unsmoothed (r0078).

r0030 Current actual value torque-generating smoothed / I_q_act smooth

SERVO, VECTOR

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 5730, 6799**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** p2002**Expert list:** 1**Min****Max****Factory setting**

- [Arms]

- [Arms]

- [Arms]

Description: Displays the smoothed torque-generating actual current.**Dependency:** Refer to: r0078

Note: SERVO: Smoothing time constant = 100 ms
 VECTOR: Smoothing time constant = 300 ms
 The signal is not suitable as a process quantity and may only be used as a display quantity.
 The following applies for SERVO:
 The torque-generating current actual value is available smoothed (r0030 with 100 ms, r0078[1] with p0045) and unsmoothed (r0078[0]).
 The following applies for VECTOR:
 The torque-generating current actual value is available smoothed (r0030 with 300 ms) and unsmoothed (r0078).

r0030			
Current actual value force generating smoothed / Iq_act smooth			
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 5730, 6799 Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the smoothed force-generating actual current.		
Dependency:	Refer to: r0078		
Note:	SERVO: Smoothing time constant = 100 ms VECTOR: Smoothing time constant = 300 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The following applies for SERVO: The force-generating current actual value is available smoothed (r0030 with 100 ms, r0078[1] with p0045) and unsmoothed (r0078[0]). The following applies for VECTOR: The torque-generating current actual value is available smoothed (r0030 with 300 ms) and unsmoothed (r0078).		
r0031			
Actual torque smoothed / M_act smooth			
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 2 Func. diagram: 5730, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the smoothed torque actual value.		
Dependency:	Refer to: r0080		
Note:	Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The active current actual value is available smoothed (r0031) and unsmoothed (r0080).		
r0031			
Force actual value smoothed / F_act smooth			
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 2 Func. diagram: 5730, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the smoothed force setpoint.		
Dependency:	Refer to: r0080		
Note:	Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The active current actual value is available smoothed (r0031) and unsmoothed (r0080).		

r0032	CO: Active power actual value smoothed / P_actv_act smth		
A_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: 14_10 Scaling: r2004 Max - [kW]	Access level: 2 Func. diagram: 5730, 6799, 8750, 8850, 8950 Unit selection: p0505 Expert list: 1 Factory setting - [kW]
Description:	Displays the smoothed actual value of the active power.		
Dependency:	Refer to: r0082		
Notice:	This smoothed signal is not suitable for diagnostics or evaluation of dynamic operations. In this case, the unsmoothed value should be used.		
Note:	Significance for the drive: Power output at the motor shaft Significance for the infeed: Line power drawn For A_INF, B_INF and S_INF the following applies: The active power is available smoothed (r0032 with 300 ms) and unsmoothed (r0082). The following applies for SERVO: The active power is available smoothed (r0032 with 100 ms, r0082[1] with p0045) and unsmoothed (r0082[0]). For VECTOR and VECTORMV, the following applies: The active power is available smoothed (r0032 with 100 ms) and unsmoothed (r0082).		

r0032	CO: Active power actual value smoothed / P_actv_act smth		
B_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: 14_10 Scaling: r2004 Max - [kW]	Access level: 2 Func. diagram: 5730, 6799, 8750, 8850, 8950 Unit selection: p0505 Expert list: 1 Factory setting - [kW]
Description:	Displays the smoothed actual value of the active power.		
Dependency:	Refer to: r0082		
Notice:	This smoothed signal is not suitable for diagnostics or evaluation of dynamic operations. In this case, the unsmoothed value should be used. For Basic Line Modules of chassis format, the displayed value is invalid as these units do not have any current sensing.		
Note:	Significance for the drive: Power output at the motor shaft Significance for the infeed: Line power drawn For A_INF, B_INF and S_INF the following applies: The active power is available smoothed (r0032 with 300 ms) and unsmoothed (r0082). The following applies for SERVO: The active power is available smoothed (r0032 with 100 ms, r0082[1] with p0045) and unsmoothed (r0082[0]). For VECTOR and VECTORMV, the following applies: The active power is available smoothed (r0032 with 100 ms) and unsmoothed (r0082).		

r0033	Torque utilization smoothed / M_util smooth		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 8012 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the smoothed torque utilization as a percentage. The torque utilization is obtained from the required smoothed torque in reference to the torque limit, scaled using p2196.		
Note:	Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The torque utilization is available smoothed (r0033) and unsmoothed (r0081). For M_set total (r0079) > M_max offset (p1532), the following applies: - demanded torque = M_set total - M_max offset - actual torque limit = M_max upper effective (r1538) - M_max offset For M_set total (r0079) <= M_max offset (p1532), the following applies: - demanded torque = M_max offset - M_set total - actual torque limit = M_max offset - M_max lower effective (r1539) For the actual torque limit = 0, the following applies: r0033 = 100 % For the actual torque limit < 0, the following applies: r0033 = 0 %		
r0033	Force utilization smoothed / F_util smooth		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 8012 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the smoothed force utilization as a percentage. The force utilization is obtained from the required smoothed force referred to the force limit.		
Note:	Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The force utilization is available smoothed (r0033) and unsmoothed (r0081). For F_set total (r0079) > F_max offset (p1532), the following applies: - demanded torque = F_set total - F_max offset - actual force limit = F_max upper effective (r1538) - F_max offset For F_set total (r0079) <= F_max offset (p1532), the following applies: - demanded force = F_max offset - F_set total - actual force limit = F_max offset - F_max lower effective (r1539) For the actual force limit = 0, the following applies: r0033 = 100 % For the actual force limit < 0, the following applies: r0033 = 0 %		

r0033 Torque utilization smoothed / M_util smooth

VECTOR

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 8012**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** PERCENT**Expert list:** 1**Min****Max****Factory setting**

- [%]

- [%]

- [%]

Description:

Displays the smoothed torque utilization as a percentage.

The torque utilization is obtained from the required smoothed torque in reference to the torque limit, scaled using p2196.

Note:

Smoothing time constant = 100 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The torque utilization is available smoothed (r0033) and unsmoothed (r0081).

For M_set total (r0079) > 0, the following applies:

- Required torque = M_set total

- Actual torque limit = M_max upper effective (r1538)

For M_set total (r0079) ≤ 0, the following applies:

- Required torque = - M_set total

- Actual torque limit = - M_max lower effective (r1539)

For the actual torque limit = 0, the following applies: r0033 = 100 %

For the actual torque limit < 0, the following applies: r0033 = 0 %

r0034 Motor utilization / Motor utilization

SERVO, VECTOR

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 8017**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** ASM, REL, FEM**Scaling:** PERCENT**Expert list:** 1**Min****Max****Factory setting**

- [%]

- [%]

- [%]

Description:

Displays the motor utilization from the thermal I2t motor model.

Dependency:

The motor utilization is only determined for permanent-magnet synchronous motors and if the I2t motor model is activated.

The motor utilization is formed from the ratio between the I2t motor model temperature (minus 40 Kelvin) and the reference value p0605 (motor overtemperature, fault threshold) - 40 Kelvin. If p0605 is reduced, r0034 increases and the motor temperature remains the same.

Refer to: p0611, p0612, p0615

Note:

Smoothing time constant = 100 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

A value of r0034 = -200.0 % indicates an invalid display, for example, because the thermal I2t motor model was not activated or was incorrectly parameterized.

r0035	CO: Temperature input / Temp_input		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: - Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 2 Func. diagram: 8850, 8950 Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the temperature currently measured at X21 (booksize) or X41 (chassis). When using an Active Interface Module (p0220 = 41 ... 45) a bimetallic sensor must be connected up to monitor the temperature. The temperature sensor type is indicated using p0601 and cannot be changed when an Active Interface Module is being used. Temperature within permissible limit values: r0035 = -50°C Temperature outside the permissible limit values: r0035 = -250°C		
Dependency:	Refer to: A06260, F06261, F06262		
Notice:	The function in r0192.11 must be available in order to obtain a correct display.		
Note:	For r0035 equal to -200.0 °C, the following applies: - "no sensor" selected in p0601! For r0035 equal to -300.0 °C, the following applies: - a KTY84 is selected in p0601 but is not connected! - the temperature display is not valid (temperature sensor error)!		

r0035	CO: Temperature input / Temp_input		
B_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: - Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 2 Func. diagram: 8750 Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the temperature currently measured at X21 (booksize) or X41 (chassis). For a BLM with internal Braking Module, a bimetallic sensor must be connected up to monitor the temperature of the braking resistor. The temperature sensor type is indicated using p0601 and cannot be changed for the existing internal Braking Module. Temperature within permissible limit values: r0035 = -50°C Temperature outside the permissible limit values: r0035 = -250°C		
Dependency:	Refer to: F06907, F06908		
Notice:	The function in r0192.11 must be available in order to obtain a correct display.		
Note:	For r0035 equal to -200.0 °C, the following applies: - "no sensor" selected in p0601! For r0035 equal to -300.0 °C, the following applies: - a KTY84 is selected in p0601 but is not connected! - the temperature display is not valid (temperature sensor error)!		

r0035	CO: Motor temperature / Mot_temp		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: - Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 2 Func. diagram: 7008, 8016, 8017 Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the actual temperature in the motor.		
Note:	For r0035 not equal to -200.0 °C, the following applies: - this temperature display is valid. - a KTY sensor is connected. - the thermal model for the induction motor is activated (p0612 bit 1 = 1 and temperature sensor deactivated: p0600 = 0 or p0601 = 0). For r0035 equal to -200.0 °C, the following applies: - this temperature display is not valid (temperature sensor error). - A PTC sensor or bimetallic NC contact is connected. - the temperature sensor of the synchronous motor is deactivated (p0600 = 0 or p0601 = 0).		
r0036	CO: Power unit overload I2t / PU overload I2t		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 8014 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the power unit overload determined using the I2t calculation. A current reference value is defined for the I2t monitoring of the power unit. It represents the current that can be conducted by the power unit without any influence of the switching losses (e.g. the continuously permissible current of the capacitors, inductances, busbars, etc.). If the I2t reference current of the power unit is not exceeded, then an overload (0 %) is not displayed. In the other case, the degree of thermal overload is calculated, whereby 100% results in a trip.		
Dependency:	Refer to: p0290, p0294 Refer to: F30005		
r0037[0...19]	CO: Power unit temperatures / PU temperatures		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: - Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: 8014 Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the temperatures in the power unit.		

Index:

- [0] = Inverter, maximum value
- [1] = Depletion layer maximum value
- [2] = Rectifier maximum value
- [3] = Air intake
- [4] = Interior of power unit
- [5] = Inverter 1
- [6] = Inverter 2
- [7] = Inverter 3
- [8] = Inverter 4
- [9] = Inverter 5
- [10] = Inverter 6
- [11] = Rectifier 1
- [12] = Rectifier 2
- [13] = Depletion layer 1
- [14] = Depletion layer 2
- [15] = Depletion layer 3
- [16] = Depletion layer 4
- [17] = Depletion layer 5
- [18] = Depletion layer 6
- [19] = Cooling system liquid intake

Note: The value of -200 indicates that there is no measuring signal.
r0037[0]: Maximum value of the inverter temperatures (r0037[5...10]).
r0037[1]: Maximum value of the depletion layer temperatures (r0037[13...18]).
r0037[2]: Maximum value of the rectifier temperatures (r0037[11...12]).
The maximum value is the temperature of the hottest inverter, depletion layer, or rectifier.

r0037[0...1] Control Unit temperature / CU temperature			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min - [°C]	Max - [°C]	Factory setting - [°C]
Description:	Displays the Control Unit temperature. An appropriate message is output when 87 °C is exceeded.		
Index:	[0] = Control Unit temperature actual [1] = Control Unit temperature maximum		
Dependency:	Refer to: A01009		
Note:	The value of -200 indicates that there is no measuring signal.		

r0038 Power factor smoothed / Cos phi smooth			
A_INF, S_INF, VEC- TOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 6799, 8850, 8950
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the smoothed actual power factor.		
Note:	Smoothing time constant = 300 ms The signal is not suitable as a process quantity and may only be used as a display quantity. Significance for the motor: Motor power factor Significance for the infeed: Power factor at the connection point (p3470, p3471)		

r0039 Energy consumption / Energy consumption				
VECTOR	Can be changed: -		Calculated: -	Access level: 2
	Data type: FloatingPoint32		Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals		Units group: -	Unit selection: -
	Not for motor type: -		Scaling: -	Expert list: 1
	Min	Max	Factory setting	
	- [kWh]	- [kWh]	- [kWh]	
Description:	Displays the electrical energy used since the last reset.			
Dependency:	Refer to: p0040			
p0040 Reset energy consumption display / Energy usage reset				
A_INF, S_INF, VECTOR	Can be changed: U, T		Calculated: -	Access level: 2
	Data type: Unsigned8		Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals		Units group: -	Unit selection: -
	Not for motor type: -		Scaling: -	Expert list: 1
	Min	Max	Factory setting	
	0	1	0	
Description:	Setting to reset the energy consumption display (r0039). Procedure: Set p0040 = 0 --> 1. The display is reset and the parameter is automatically set to zero.			
Dependency:	Refer to: r0039			
p0045 Smoothing time constant, display values / T_smth display				
SERVO, VECTOR	Can be changed: U, T		Calculated: -	Access level: 2
	Data type: FloatingPoint32		Dynamic index: -	Func. diagram: 4715, 5610, 5730, 6714, 8012
	P-Group: -		Units group: -	Unit selection: -
	Not for motor type: -		Scaling: -	Expert list: 1
	Min	Max	Factory setting	
	0.00 [ms]	1000.00 [ms]	1.00 [ms]	
Description:	Sets the smoothing time constant for the following display values: SERVO: r0078[1], r0079[1], r0081 (calculated from the quantities smoothed with p0045), r0082[1]. VECTOR: r0063[1], r0068[1], r0080[1], r0082[1].			
r0046.0...29 CO/BO: Missing enable sig / Missing enable sig				
A_INF, S_INF	Can be changed: -		Calculated: -	Access level: 1
	Data type: Unsigned32		Dynamic index: -	Func. diagram: 8834, 8934
	P-Group: Displays, signals		Units group: -	Unit selection: -
	Not for motor type: -		Scaling: -	Expert list: 1
	Min	Max	Factory setting	
	-	-	-	
Description:	Displays missing enable signals that are preventing the closed-loop infeed control from being commissioned.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	OFF1 enable missing	Yes	No
	01	OFF2 enable missing	Yes	No
	03	Operation enable missing	Yes	No
	08	EP terminals enable missing	Yes	No
	16	OFF1 enable internal missing	Yes	No
	17	OFF2 enable internal missing	Yes	No
	19	Pulse enable internal missing	Yes	No
	26	Infeed inactive or not operational	Yes	No
	29	Cooling system ready signal missing	Yes	No

Dependency: Refer to: r0002

Note: The value r0046 = 0 indicates that all enable signals for the infeed are present.

Bit 00 = 1 (enable signal missing), if:

- the signal source in p0840 is a 0 signal.
- there is a "switching on inhibited".

Bit 01 = 1 (enable signal missing), if:

- the signal source in p0844 or p0845 is a 0 signal.

Bit 03 = 1 (enable signal missing), if:

- the signal source in p0852 is a 0 signal.

Bit 08 = 1 (enable signal missing), if:

- the pulse enable via terminal EP is missing (booksize: X21, chassis: X41).

Bit 16 = 1 (enable signal missing), if:

- there is an OFF1 fault response. The system is only enabled if the fault is removed and was acknowledged and the "switching on inhibited" withdrawn with OFF1 = 0.

Bit 17 = 1 (enable signal missing), if:

- the commissioning mode is selected (p0009 > 0 or p0010 > 0) or there is an OFF2 fault response or the OFF1 signal source (p0840) is changed.

Bit 19 = 1 (internal pulse enable missing), if:

- synchronization is running between the basic clock cycle, DRIVE-CLiQ clock cycle and application clock cycle.

Bit 26 = 1 (enable signal missing), if:

- the infeed is inactive (p0105 = 0) or is not operational (r7850[DO-Index]=0).

Bit 29 = 1 (enable signal missing), if:

- the cooling system ready signal via BI: p0266[1] missing.

r0046.0...29		CO/BO: Missing enable sig / Missing enable sig				
B_INF	Can be changed: -		Calculated: -		Access level: 1	
	Data type: Unsigned32		Dynamic index: -		Func. diagram: 8734	
	P-Group: Displays, signals		Units group: -		Unit selection: -	
	Not for motor type: -		Scaling: -		Expert list: 1	
	Min		Max		Factory setting	
	-		-		-	
Description:		Displays missing enable signals that are preventing the closed-loop infeed control from being commissioned.				
Bit field:		Bit	Signal name	1 signal	0 signal	FP
	00	OFF1 enable missing	Yes	No	-	
	01	OFF2 enable missing	Yes	No	-	
	08	EP terminals enable missing	Yes	No	-	
	16	OFF1 enable internal missing	Yes	No	-	
	17	OFF2 enable internal missing	Yes	No	-	
	26	Infeed inactive or not operational	Yes	No	-	
	29	Cooling system ready signal missing	Yes	No	-	
Dependency:		Refer to: r0002				

Note: The value r0046 = 0 indicates that all enable signals for the infeed are present.

Bit 00 = 1 (enable signal missing), if:

- the signal source in p0840 is a 0 signal.
- there is a "switching on inhibited".

Bit 01 = 1 (enable signal missing), if:

- the signal source in p0844 or p0845 is a 0 signal.

Bit 16 = 1 (enable signal missing), if:

- there is an OFF1 fault response. The system is only enabled if the fault is removed and was acknowledged and the "switching on inhibited" withdrawn with OFF1 = 0.

Bit 17 = 1 (enable signal missing), if:

- the commissioning mode is selected (p0009 > 0 or p0010 > 0) or there is an OFF2 fault response or the OFF1 signal source (p0840) is changed.

Bit 26 = 1 (enable signal missing), if:

- the infeed is inactive (p0105 = 0) or is not operational (r7850[DO-Index]=0).

Bit 29 = 1 (enable signal missing), if:

- the cooling system ready signal via BI: p0266[1] missing.

r0046.0...31**CO/BO: Missing enable sig / Missing enable sig**

SERVO, VECTOR

Can be changed: -**Calculated:** -**Access level:** 1**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 2634**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays missing enable signals that are preventing the closed-loop drive control from being commissioned.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	OFF1 enable missing	Yes	No	-
01	OFF2 enable missing	Yes	No	-
02	OFF3 enable missing	Yes	No	-
03	Operation enable missing	Yes	No	-
04	Armature short-circuit / DC current brake, enable missing	Yes	No	7014, 7016
05	STOP2 enable missing	Yes	No	-
06	STOP1 enable missing	Yes	No	-
08	EP terminals enable missing	Yes	No	-
09	Infeed enable missing	Yes	No	-
10	Ramp-function generator enable missing	Yes	No	-
11	Ramp-function generator start missing	Yes	No	-
12	Setpoint enable missing	Yes	No	-
16	OFF1 enable internal missing	Yes	No	-
17	OFF2 enable internal missing	Yes	No	-
18	OFF3 enable internal missing	Yes	No	-
19	Pulse enable internal missing	Yes	No	-
20	Armature short-circuit/DC current brake internal enable missing	Yes	No	7014, 7016
21	STOP2 enable internal missing	Yes	No	-
22	STOP1 enable internal missing	Yes	No	-
25	Function bypass active	Yes	No	-
26	Drive inactive or not operational	Yes	No	-
27	Demagnetizing not completed	Yes	No	-
28	Brake open missing	Yes	No	-
29	Cooling system ready signal missing	Yes	No	-
30	Speed controller inhibited	Yes	No	-
31	Jog setpoint active	Yes	No	-

Dependency:

Refer to: r0002

Note: The value r0046 = 0 indicates that all enable signals for this drive are present.

Bit 00 = 1 (enable signal missing), if:

- the signal source in p0840 is a 0 signal.
- there is a "switching on inhibited".

Bit 01 = 1 (enable signal missing), if:

- the signal source in p0844 or p0845 is a 0 signal.

Bit 02 = 1 (enable signal missing), if:

- the signal source in p0848 or p0849 is a 0 signal.

Bit 03 = 1 (enable signal missing), if:

- the signal source in p0852 is a 0 signal.

Bit 04 = 1 (armature short-circuit active), if:

- the signal source in p1230 has a 1 signal

Bit 05, Bit 06: Being prepared

Bit 08 = 1 (enable signal missing), if:

- the pulse enable via terminal EP is missing (booksize: X21, chassis: X41).

Bit 09 = 1 (enable signal missing), if:

- the signal source in p0864 is a 0 signal.

Bit 10 = 1 (enable signal missing), if:

- the signal source in p1140 is a 0 signal.

Bit 11 = 1 (enable signal missing) if the speed setpoint is frozen, because:

- the signal source in p1141 is a 0 signal.
- the speed setpoint is entered from jogging and the two signal sources for jogging, bit 0 (p1055) and bit 1 (p1056) have a 1 signal.

Bit 12 = 1 (enable signal missing), if:

- the signal source in p1142 is a 0 signal.
- When activating the function module "basic positioner" (r0108.4 = 1), the signal source in p1142 is set to a 0 signal.

Bit 16 = 1 (enable signal missing), if:

- there is an OFF1 fault response. The system is only enabled if the fault is removed and was acknowledged and the "switching on inhibited" withdrawn with OFF1 = 0.

Bit 17 = 1 (enable signal missing), if:

- commissioning mode is selected (p0009 > 0 or p0010 > 0).
- there is an OFF2 fault response.
- the drive is inactive (p0105 = 0) or is not operational (r7850[DO-Index]=0).

Bit 18 = 1 (enable signal missing), if:

- OFF3 has still not been completed or an OFF3 fault response is present.

Bit 19 = 1 (internal pulse enable missing), if:

- synchronization is running between the basic clock cycle, DRIVE-CLiQ clock cycle and application clock cycle.

Bit 20 = 1 (internal armature short-circuit active), if:

- the drive is not in the state "S4: Operation" or "S5x" (refer to function diagram 2610).
- the internal pulse enable is missing (r0046.19 = 0).

Bit 21 = 1 (enable signal missing), if:

The pulses have been enabled and the speed setpoint has still not been enabled, because:

- the holding brake opening time (p1216) has still not expired.
- the motor has still not been magnetized (induction motor).
- the encoder has not been calibrated (V/f vector and synchronous motor)

Bit 22: Being prepared

Bit 26 = 1 (enable signal missing), if:

- the drive is inactive (p0105 = 0) or is not operational (r7850[DO-Index]=0).

Bit 27 = 1 (enable signal missing), if:

- demagnetizing has still not been completed (only for vector).

Bit 28 = 1 (enable signal missing), if:

- the holding brake is closed or has still not been opened.

Bit 29 = 1 (enable signal missing), if:

- the cooling system ready signal via BI: p0266[1] missing.

Bit 30 = 1 (speed controller inhibited), if one of the following reasons is present:

- A 0 signal is available via BI: p0856.

- the function generator with current input is active.

- the measuring function "current controller reference frequency characteristic" is active.

- the pole position identification is active.

- motor data identification is active (only certain steps).

Bit 31 = 1 (enable signal missing), if:

- the speed setpoint from jog 1 or 2 is entered.

r0046.0...31**CO/BO: Missing enable sig / Missing enable sig**

SERVO (Lin)

Can be changed: -

Calculated: -

Access level: 1

Data type: Unsigned32

Dynamic index: -

Func. diagram: 2634

P-Group: Displays, signals

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

-

-

-

Description:

Displays missing enable signals that are preventing the closed-loop drive control from being commissioned.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	OFF1 enable missing	Yes	No	-
01	OFF2 enable missing	Yes	No	-
02	OFF3 enable missing	Yes	No	-
03	Operation enable missing	Yes	No	-
04	Armature short-circuit / DC current brake, enable missing	Yes	No	7014, 7016
05	STOP2 enable missing	Yes	No	-
06	STOP1 enable missing	Yes	No	-
08	EP terminals enable missing	Yes	No	-
09	Infeed enable missing	Yes	No	-
10	Ramp-function generator enable missing	Yes	No	-
11	Ramp-function generator start missing	Yes	No	-
12	Setpoint enable missing	Yes	No	-
16	OFF1 enable internal missing	Yes	No	-
17	OFF2 enable internal missing	Yes	No	-
18	OFF3 enable internal missing	Yes	No	-
19	Pulse enable internal missing	Yes	No	-
20	Armature short-circuit/DC current brake internal enable missing	Yes	No	7014, 7016
21	STOP2 enable internal missing	Yes	No	-
22	STOP1 enable internal missing	Yes	No	-
25	Function bypass active	Yes	No	-
26	Drive inactive or not operational	Yes	No	-
27	Demagnetizing not completed	Yes	No	-
28	Brake open missing	Yes	No	-
29	Cooling system ready signal missing	Yes	No	-
30	Velocity controller inhibited	Yes	No	-
31	Jog setpoint active	Yes	No	-

Dependency:

Refer to: r0002

Note:

The value r0046 = 0 indicates that all enable signals for this drive are present.

Bit 00 = 1 (enable signal missing), if:

- the signal source in p0840 is a 0 signal.

- there is a "switching on inhibited".

Bit 01 = 1 (enable signal missing), if:

- the signal source in p0844 or p0845 is a 0 signal.

Bit 02 = 1 (enable signal missing), if:

- the signal source in p0848 or p0849 is a 0 signal.

- Bit 03 = 1 (enable signal missing), if:
- the signal source in p0852 is a 0 signal.
- Bit 04 = 1 (armature short-circuit active), if:
- the signal source in p1230 has a 1 signal
- Bit 05, Bit 06: Being prepared
- Bit 08 = 1 (enable signal missing), if:
- the pulse enable via terminal EP is missing (booksize: X21, chassis: X41).
- Bit 09 = 1 (enable signal missing), if:
- the signal source in p0864 is a 0 signal.
- Bit 10 = 1 (enable signal missing), if:
- the signal source in p1140 is a 0 signal.
- Bit 11 = 1 (enable signal missing) if the velocity setpoint is frozen, because:
- the signal source in p1141 is a 0 signal.
 - the velocity setpoint is entered from jogging and the two signal sources for jogging, bit 0 (p1055) and bit 1 (p1056) have a 1 signal.
- Bit 12 = 1 (enable signal missing), if:
- the signal source in p1142 is a 0 signal.
 - When activating the function module "basic positioner" (r0108.4 = 1), the signal source in p1142 is set to a 0 signal.
- Bit 16 = 1 (enable signal missing), if:
- there is an OFF1 fault response. The system is only enabled if the fault is removed and was acknowledged and the "switching on inhibited" withdrawn with OFF1 = 0.
- Bit 17 = 1 (enable signal missing), if:
- commissioning mode is selected (p0009 > 0 or p0010 > 0).
 - there is an OFF2 fault response.
 - the drive is inactive (p0105 = 0) or is not operational (r7850[DO-Index]=0).
- Bit 18 = 1 (enable signal missing), if:
- OFF3 has still not been completed or an OFF3 fault response is present.
- Bit 19 = 1 (internal pulse enable missing), if:
- synchronization is running between the basic clock cycle, DRIVE-CLiQ clock cycle and application clock cycle.
- Bit 20 = 1 (internal armature short-circuit active), if:
- the drive is not in the state "S4: Operation" or "S5x" (refer to function diagram 2610).
 - the internal pulse enable is missing (r0046.19 = 0).
- Bit 21 = 1 (enable signal missing), if:
- The pulses have been enabled and the velocity setpoint has still not been enabled, because:
- the holding brake opening time (p1216) has still not expired.
 - the motor has still not been magnetized (induction motor).
- Bit 22: Being prepared
- Bit 26 = 1 (enable signal missing), if:
- the drive is inactive (p0105 = 0) or is not operational (r7850[DO-Index]=0).
- Bit 27 = 1 (enable signal missing), if:
- demagnetizing has still not been completed (only for vector).
- Bit 28 = 1 (enable signal missing), if:
- the holding brake is closed or has still not been opened.
- Bit 29 = 1 (enable signal missing), if:
- the cooling system ready signal via BI: p0266[1] missing.
- Bit 30 = 1 (velocity controller inhibited), if one of the following reasons is present:
- A 0 signal is available via BI: p0856.
 - the function generator with current input is active.
 - the measuring function "current controller reference frequency characteristic" is active.
 - the pole position identification is active.
 - motor data identification is active (only certain steps).
- Bit 31 = 1 (enable signal missing), if:
- the velocity setpoint from jog 1 or 2 is entered.

r0047	Status, identification / Status ident		
SERVO	Can be changed: - Data type: Integer16 P-Group: Displays, signals Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 104	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the currently executed steps or the first step after the enable for the motor identification and pole position identification routines.		
Value:	0: No measurement 1: PolID: Wait for brake closing time 2: PolID: Measurement, step 1 3: PolID: Measurement, step 2 4: PolID: Measurement, step 3 5: PolID: Measurement, step 4 6: PolID: Measurement, stage 2 7: PolID: Measurement evaluation 8: PolID: Measurement end 11: MotID: Inductance measurement, step 1 12: MotID: Inductance measurement, step 2 13: MotID: Inductance measurement evaluation 14: MotID: Resistance measurement evaluation 15: MotID: Fine synchronization, step 1 16: MotID: Fine synchronization, step 2 17: MotID: Fine synchronization, step 3 18: MotID: Fine synchronization, end 20: MotID: Rotating inductance measurement, step 1 21: MotID: Rotating inductance measurement, step 2 22: MotID: Rotating inductance measurement, step 3 23: MotID: Rotating inductance measurement, step 4 24: MotID: Rotating Inductance measurement evaluation 25: MotID: Rotating Inductance measurement end 30: MotID: Induction motor measurement, step 1 31: MotID: Induction motor measurement, step 2 32: MotID: Induction motor measurement, step 3 33: MotID: Induction motor measurement, step 4 34: MotID: Induction motor measurement, step 5 35: MotID: Induction motor measurement, step 6 36: MotID: Induction motor measurement, step 7 37: MotID: Induction motor measurement, step 8 38: MotID: Induction motor measurement, step 9 40: MotID: Commutating angle, step 1 41: MotID: Commutating angle, step 2 42: MotID: Commutating angle, step 3 43: MotID: Commutating angle, step 4 45: MotID: Commutating angle rotating, step 1 46: MotID: Commutating angle rotating, step 2 47: MotID: Commutating angle rotating, step 3 48: MotID: Commutating angle rotating complete 50: MotID: kT determination, step 1 51: MotID: kT determination, step 2 52: MotID: kT determination, step 3 53: MotID: kT determination evaluation 54: MotID: kT determination end 60: MotID: Reluctance constant measurement, step 1 61: MotID: Reluctance constant measurement, step 2 62: MotID: Reluctance constant measurement, step 3 63: MotID: Reluctance constant measurement end 70: MotID: Moment of inertia measurement, step 1 71: MotID: Moment of inertia measurement, step 2 72: MotID: Moment of inertia measurement, step 3		

- 73: MotID: Moment of inertia measurement end
- 80: MotID: Magnetizing inductance measurement, step 1
- 81: MotID: Magnetizing inductance measurement, step 2
- 82: MotID: Magnetizing inductance measurement, step 3
- 83: MotID: Magnetizing inductance measurement evaluation
- 84: MotID: Magnetizing inductance measurement end
- 90: MotID: Saturation characteristic. step 1
- 91: MotID: Saturation characteristic. step 2
- 92: MotID: Saturation characteristic. step 3
- 93: MotID: Saturation characteristic evaluation 1
- 94: MotID: Saturation characteristic evaluation 2
- 95: MotID: Saturation characteristic end
- 96: MotID: Converter model, step 1
- 97: MotID: Converter model, step 2
- 98: MotID: Converter model, step 3
- 99: MotID: Converter model, step 4
- 100: PolID: Motion-based, step 1
- 101: PolID: Motion-based, step 2
- 102: PolID: Motion-based, step 3
- 103: PolID: Motion-based, step 4
- 104: PolID: Motion-based, step 5

r0047 Motor data ident. routine and speed controller optimization / MotID and n_opt

VECTOR	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	300	-

Description: Displays the actual status for the motor data identification (standstill measurement) and the speed/velocity controller optimization (rotating measurement).

Value:

- 0: No measurement
- 115: Measurement q leakage inductance (part 2)
- 120: Speed controller optimization (vibration test)
- 140: Calculate speed controller setting
- 150: Measurement, moment of inertia
- 170: Measurement, magnetizing current and saturation characteristic
- 190: Speed encoder test
- 195: Measurement q leakage inductance (part 1)
- 200: Rotating measurement selected
- 210: Pole position identification selected
- 220: identification, leakage inductance
- 230: Identification, rotor time constant
- 240: Identification, stator inductance
- 250: Identification, stator inductance LQLD
- 270: Identification, stator resistance
- 290: Identification, valve lockout time
- 300: Standstill measurement selected

Note: r0047 = 300 is also displayed if encoder calibration p1990 is selected.

r0049[0...3] Motor data set/encoder data set effective / MDS/EDS effective

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned8	Dynamic index: -	Func. diagram: 8565
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the effective Motor Data Set (MDS) and the effective Encoder Data Sets (EDS).

Index:
 [0] = Motor Data Set MDS effective
 [1] = Encoder 1 Encoder Data Set EDS effective
 [2] = Encoder 2 Encoder Data Set EDS effective
 [3] = Encoder 3 Encoder Data Set EDS effective

Dependency: Refer to: p0186, p0187, p0188, p0189, r0838

Note: Value 99 means the following: No encoder assigned (not configured).

r0050.0...3 CO/BO: Command Data Set CDS effective / CDS effective

A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned8	Dynamic index: -	Func. diagram: 8560
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the effective Command Data Set (CDS).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	CDS eff., bit 0	On	Off	-
	01	CDS eff., bit 1	On	Off	-
	02	CDS eff., bit 2	On	Off	-
	03	CDS eff., bit 3	On	Off	-

Dependency: Refer to: p0810, p0811, r0836

Note: The Command Data Set selected using a binector input (e.g. p0810) is displayed using r0836.

r0051.0...4 CO/BO: Drive Data Set DDS effective / DDS effective

ENCODER, SERVO, TM41, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned8	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the effective Drive Data Set (DDS).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DDS eff., bit 0	On	Off	-
	01	DDS eff., bit 1	On	Off	-
	02	DDS eff., bit 2	On	Off	-
	03	DDS eff., bit 3	On	Off	-
	04	DDS eff., bit 4	On	Off	-

Dependency: Refer to: p0820, p0821, p0822, p0823, p0824, r0837

Note: The drive data set changeover is suppressed when selecting the motor identification, during the rotating measurement, the encoder calibration and the friction characteristic record.

r0056.1...15 CO/BO: Status word, closed-loop control / ZSW cl-loop ctrl

SERVO	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2526
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word of the closed-loop control.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	01	Demagnetizing completed	Yes	No	-
	04	Magnetizing completed	Yes	No	2701
	08	Field weakening active	Yes	No	-
	14	Vdc_max controller active	Yes	No	-
	15	Vdc_min controller active	Yes	No	-

Note: Re bit 04:
The bit is immediately set after power-on
Exception:
For an induction motor with brake (except for p1215 = 2), the bit is only set when 60% of the reference flux is reached.

r0056.0...15 CO/BO: Status word, closed-loop control / ZSW cl-loop ctrl

VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2526
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word of the closed-loop control.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Initialization completed	Yes	No	-
	01	Demagnetizing completed	Yes	No	-
	02	Pulse enable present	Yes	No	-
	03	Soft starting present	Yes	No	-
	04	Magnetizing completed	Yes	No	-
	05	Voltage boost when starting	Active	Inactive	6300
	06	Acceleration voltage	Active	Inactive	6300
	07	Frequency negative	Yes	No	6719
	08	Field weakening active	Yes	No	-
	09	Voltage limit active	Yes	No	6714
	10	Slip limit active	Yes	No	6310
	11	Frequency limit active	Yes	No	6719
	12	Current limiting controller voltage output active	Yes	No	-
	13	Current/torque limiting	Active	Inactive	6060
	14	Vdc_max controller active	Yes	No	6220, 6320
	15	Vdc_min controller active	Yes	No	6220, 6320

r0060	CO: Speed setpoint before the setpoint filter / n_set before filt.		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 2701, 2704, 5020, 6030, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the actual speed setpoint at the input of the speed controller or V/f characteristic (after the interpolator).		
Dependency:	Refer to: r0020		
Note:	The speed setpoint is available smoothed (r0020) and unsmoothed (r0060).		
r0060	CO: Velocity setpoint before the setpoint filter / v_set before filt		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 2701, 2704, 5020, 6030, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the actual velocity setpoint at the input of the velocity controller or V/f characteristic (after the interpolator).		
Dependency:	Refer to: r0020		
Note:	The velocity setpoint is available smoothed (r0020) and unsmoothed (r0060).		
r0061	CO: Actual speed unsmoothed / n_act unsmoothed		
ENCODER	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 2 Func. diagram: 1580, 4710, 4715 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the actual speed values sensed by the encoders (unsmoothed).		
r0061	CO: Actual velocity unsmoothed / v_act unsmoothed		
ENCODER (Lin enc)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 2 Func. diagram: 1580, 4710, 4715 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the actual velocities sensed by the encoders (unsmoothed).		

r0061[0...1]	CO: Actual speed unsmoothed / n_act unsmoothed		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 2 Func. diagram: 1580, 4710, 4715 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the actual speed values sensed by the encoders (unsmoothed).		
Index:	[0] = Encoder 1 [1] = Encoder 2		

r0061[0...1]	CO: Actual velocity unsmoothed / v_act unsmoothed		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 2 Func. diagram: 1580, 4710, 4715 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the actual velocities sensed by the encoders (unsmoothed).		
Index:	[0] = Encoder 1 [1] = Encoder 2		

r0061[0...2]	CO: Actual speed unsmoothed / n_act unsmoothed		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 2 Func. diagram: 1580, 4710, 4715 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the actual speed values sensed by the encoders (unsmoothed).		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Note:	For V/f operating modes, the speeds from encoder 2 and encoder 3 are not displayed.		

r0062	CO: Speed setpoint after the filter / n_set after filter		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 1590, 1750, 5020, 5030, 5210, 6030 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the actual speed setpoint after the setpoint filters.		

r0062	CO: Velocity setpoint after the filter / v_set after filter		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 1590, 1750, 5020, 5030, 5210, 6030 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the actual velocity setpoint after the setpoint filters.		
r0063	CO: Actual speed smoothed / n_act smooth		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 1580, 1590, 4710, 5300 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the current smoothed actual speed for speed control.		
Dependency:	Refer to: r0021, r0061, p1441		
Note:	The speed actual value is calculated in encoderless operation. For operation with encoder, r0063 is smoothed with p1441. The actual speed is available as a display quantity with additional smoothing in r0021.		
r0063	CO: Actual velocity smoothed / v_act smooth		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 1580, 1590, 4710, 5300 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the actual smoothed velocity actual value for velocity control.		
Dependency:	Refer to: r0021, r0061, p1441		
Note:	The velocity actual value is calculated in encoderless operation. For operation with encoder, r0063 is smoothed with p1441. The actual velocity is available as a display quantity with additional smoothing in r0021.		
r0063	CO: Actual speed value / n_act		
TM41	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the current smoothed actual speed.		
Note:	For Terminal Module 41 (TM41), this value is used to interconnect with standard telegram 3 and is always zero.		

r0063[0...2]	CO: Actual speed value / n_act		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 1680, 4715 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the actual speed of the closed-loop speed control and the V/f control.		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045 [2] = Calculated from f_set - f_slip		
Dependency:	Refer to: r0021		
Note:	The speed actual value is calculated in encoderless operation and for V/f control. For operation with encoder, r0063[0] is smoothed with p1441. The speed actual value (r0063[0]) is additionally displayed - smoothed with p0045 - in r0063[1]. The speed (r0063[2]) calculated from the output frequency and slip can only be compared with the speed actual value (r0063[0]) in the steady-state. The actual speed (r0063[0]) is available as a display quantity with additional smoothing in r0021.		
r0064	CO: Speed controller system deviation / n_ctrl system dev		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 5040, 6040 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the actual system deviation of the speed controller.		
Note:	In servo control mode with active reference model, the system deviation to the P component of the speed controller is displayed.		
r0064	CO: Velocity controller system deviation / v_ctrl system dev		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 5040, 6040 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the actual system deviation of the velocity controller.		
Note:	In servo control mode with active reference model, the system deviation to the P component of the velocity controller is displayed.		
r0065	Slip frequency / f_Slip		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Hz]	Calculated: - Dynamic index: - Units group: 2_1 Scaling: p2000 Max - [Hz]	Access level: 3 Func. diagram: 1710, 6310, 6727, 6730, 6732 Unit selection: p0505 Expert list: 1 Factory setting - [Hz]
Description:	Displays the slip frequency for induction motors (ASM).		

r0066 CO: Line frequency / f_line

A_INF, S_INF

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [Hz]

Calculated: -**Dynamic index:** -**Units group:** 2_1**Scaling:** p2000**Max**

- [Hz]

Access level: 3**Func. diagram:** 8850, 8864, 8950, 8964**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [Hz]

Description:

Displays the line frequency.

Dependency:

Refer to: r0024

Note:

The line frequency is available smoothed (r0024) and unsmoothed (r0066).

A positive sign of the frequency is obtained when the line supply phases U, V and W are connected with the correct phase sequence.

A negative sign of the frequency is obtained when the 3 line phases are interchanged therefore designating a negative direction of the rotating field of the 3-phase line supply voltage.

r0066 CO: Output frequency / f_outp

SERVO, VECTOR

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [Hz]

Calculated: -**Dynamic index:** -**Units group:** 2_1**Scaling:** p2000**Max**

- [Hz]

Access level: 3**Func. diagram:** 1690, 5300, 5730, 6310, 6730, 6731, 6799**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [Hz]

Description:

Displays the Motor Module output frequency.

Dependency:

Refer to: r0024

Note:

The output frequency is available smoothed (r0024) and unsmoothed (r0066).

r0067[0...1] Absolute current value permissible / I_abs val perm

A_INF, S_INF

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [Arms]

Calculated: -**Dynamic index:** -**Units group:** 6_2**Scaling:** p2002**Max**

- [Arms]

Access level: 3**Func. diagram:** -**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [Arms]

Description:

Displays the actual permissible absolute line-side current.

Index:

[0] = Motor mode

[1] = Regenerative mode

Dependency:

The permissible current is the minimum from the maximum converter current (r0209), the parameterized current limits (p3530 to p3533) as well as the maximum permissible current of line filter (r3534).

Refer to: p3530, p3531, r3534

r0067	CO: Output current, maximum / I_outp max		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 5722, 6300, 6640, 6724 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the maximum output current of the Motor Module.		
Dependency:	The maximum output current is determined by the parameterized current limit and the motor and converter thermal protection. Refer to: p0290, p0640		
r0068	CO: Absolute current actual value / I_act abs val		
A_INF, S_INF, SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 5730, 7017, 8014, 8017, 8850, 8950 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays actual absolute current.		
Dependency:	Refer to: r0027		
Notice:	Bei A_INF, S_INF the following applies: The value is updated with the current controller sampling time. The following applies for SERVO: The value is updated with a sampling time of 1 ms.		
Note:	Absolute current value = $\sqrt{I_q^2 + I_d^2}$ The absolute current actual value is available smoothed (r0027) and unsmoothed (r0068).		
r0068	CO: DC current in the DC link / Idc DC link		
B_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: - Units group: 6_4 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: 8014, 8750 Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the DC current in the DC link.		
Dependency:	Refer to: r0027		
Notice:	For Basic Line Modules of chassis format, the displayed value is invalid as these units do not have any current sensing.		
Note:	The DC current in the DC link is available smoothed (r0027) and unsmoothed (r0068).		

r0068[0...1]	CO: Absolute current actual value / I_act abs val		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1690, 6714, 6799, 7017, 8014, 8017, 8018 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays actual absolute current.		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		
Dependency:	Refer to: r0027		
Notice:	The value is updated with the current controller sampling time.		
Note:	Absolute current value = $\sqrt{I_q^2 + I_d^2}$ The absolute value of the current actual value is available smoothed (r0027 with 300 ms, r0068[1] with p0045) and unsmoothed (r0068[0]).		
r0069[0...6]	Phase current actual value / I_phase act value		
A_INF, S_INF, SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: - Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: 1630, 5730, 6714, 6730, 6731, 8850, 8950 Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the measured actual phase currents as peak value.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W [3] = Phase U offset [4] = Phase V offset [5] = Phase W offset [6] = Total U, V, W		
Note:	In indices 3 ... 5, the offset currents of the 3 phases, which are added to correct the phase currents, are displayed. The sum of the 3 corrected phase currents is displayed in index 6.		
r0069[0...6]	CO: Phase current actual value / I_phase act value		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: - Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: 1630, 5730, 6714, 6730, 6731, 8850, 8950 Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the measured actual phase currents as peak value.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W [3] = Phase U offset [4] = Phase V offset [5] = Phase W offset [6] = Total U, V, W		
Note:	In indices 3 ... 5, the offset currents of the 3 phases, which are added to correct the phase currents, are displayed. The sum of the 3 corrected phase currents is displayed in index 6.		

r0070	CO: Actual DC link voltage / Vdc act val		
A_INF, B_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: 5_2 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 1774, 8750, 8850, 8864, 8940, 8950, 8964 Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the measured actual value of the DC link voltage.		
Dependency:	Refer to: r0026		
Note:	The DC link voltage is available smoothed (r0026) and unsmoothed (r0070).		
r0070	CO: Actual DC link voltage / Vdc act val		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: 5_2 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 5730 Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the measured actual value of the DC link voltage.		
Dependency:	Refer to: r0026		
Notice:	For SINAMICS S120 AC Drive (AC/AC) the following applies: When measuring a DC link voltage < 200 V, for the Power Module (e.g. PM340) a valid measured value is not supplied. In this case, when an external 24V power supply is connected, a value of approx. 24 V is displayed.		
Note:	The DC link voltage is available smoothed (r0026) and unsmoothed (r0070).		
r0070	CO: Actual DC link voltage / Vdc act val		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: 5_2 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 6723, 6724, 6730, 6731, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the measured actual value of the DC link voltage.		
Dependency:	Refer to: r0026		
Notice:	For SINAMICS S120 AC Drive (AC/AC) the following applies: When measuring a DC link voltage < 200 V, for the Power Module (e.g. PM340) a valid measured value is not supplied. In this case, when an external 24V power supply is connected, a value of approx. 24 V is displayed.		
Note:	The DC link voltage is available smoothed (r0026) and unsmoothed (r0070).		

r0071	Maximum output voltage / V_output max		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 1710, 6300, 6640, 6722, 6723, 6724, 6725, 6727 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the maximum output voltage.		
Dependency:	The maximum output voltage depends on the actual DC link voltage (r0070) and the maximum modulation depth (p1803).		
Note:	As the (driven) motor load increases, the maximum output voltage drops as a result of the reduction in DC link voltage.		
r0072[0...3]	CO: Input voltage / V_input		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 8850, 8950 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the current power unit input voltage (Line Module).		
Index:	[0] = Voltage at input terminals of power unit from line supply model [1] = Voltage at VSM or at input terminals of the line filter [2] = Voltage of the voltage source from the line supply model [3] = Smoothed voltage of voltage source from line supply model		
Note:	The input voltages are available smoothed (r0025) and unsmoothed (r0072). r0072[0]: Displays the pulsed voltage at the line supply input terminals of the power unit. The value is calculated from the modulation depth (r0074) and is therefore only correct in the closed-loop controlled mode and when the pulses are enabled. r0072[1]: Displays the absolute voltage at the input terminals of the line filter or the connection point of a Voltage Sensing Module (VSM). The value is calculated from the VSM measured values r3661 and r3662 and is therefore equal to 0 if a VSM is not connected. r0072[2]: Displays the estimated value for the voltage of the voltage source that is calculated in the voltage model of the line supply PLL. (input quantities of the model are the measured values of the line currents and the DC link voltage as well as the characteristics of the line filter p0225, p0226 as well as the line inductance p3424). r0072[3]: Displays the smoothed value for the source voltage in r0072[2]. The PT1 smoothing time constant is set in p3472.		
r0072	CO: Output voltage / V_output		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 1630, 5730, 6730, 6731, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the actual power unit output voltage (Motor Module).		
Dependency:	Refer to: r0025		
Note:	The output voltage is available smoothed (r0025) and unsmoothed (r0072).		

r0073	Maximum modulation depth / Modulat_depth max		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Modulation Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 6724 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the maximum modulation depth.		
Dependency:	Refer to: p1803		
r0074	CO: Modulat_depth / Modulat_depth		
A_INF, SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 5730, 6730, 6731, 6799, 8940, 8950 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the actual modulation depth.		
Dependency:	Refer to: r0028		
Note:	For space vector modulation, 100% corresponds to the maximum output voltage without overcontrol. Values above 100 % indicate an overcontrol condition - values below 100% have no overcontrol. The phase voltage (phase-to-phase, rms) is calculated as follows: $(r0074 * r0070) / (\sqrt{2} * 100 \%)$. The modulation depth is available smoothed (r0028) and unsmoothed (r0074).		
r0075	CO: Reactive current setpoint / I_react_set		
A_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 8946 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the reactive current setpoint.		
Dependency:	Refer to: r3471, p3610		
Note:	The reactive current requirement of a line filter should be covered by the controlled infeed/regenerative feedback so that the converter always operates with a power factor of 1 compared to the line. Setpoint r0075 includes the reactive current for a line filter that depends on the current operating point (r3471). If the line phases are reversed and the line voltage therefore has a negative orientation ($r0066 < 0$), it should be noted that the sign of the reactive current is reversed.		
r0075	CO: Current setpoint field-generating / Id_set		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1630, 5714, 5722, 6714 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the field-generating current setpoint (Id_set).		
Note:	This value is irrelevant for the V/f control mode.		

r0076	CO: Reactive current actual value / I_reactive_act		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1774, 1775, 8850, 8946, 8950 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the reactive current actual value.		
Dependency:	Refer to: r0029, r0075		
Note:	The reactive current actual value is available smoothed (r0029) and unsmoothed (r0076).		
r0076	CO: Current actual value field-generating / Id_act		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1630, 1710, 5714, 5730, 6714, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the field-generating current actual value (Id_act).		
Dependency:	Refer to: r0029		
Note:	This value is irrelevant for the V/f control mode. The field-generating current actual value is available smoothed (r0029) and unsmoothed (r0076).		
r0077	CO: Active current setpoint / I_active_set		
A_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1774, 8940, 8946 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the active current setpoint (Iq_set).		
r0077	CO: Current setpoint torque-generating / Iq_set		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1630, 1774, 5714, 6710, 6714, 6719 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the torque/force generating current setpoint.		
Note:	This value is irrelevant for the V/f control mode.		

r0077	CO: Current setpoint force-generating / Iq_set		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1630, 1774, 5714, 6710, 6714, 6719 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the torque/force generating current setpoint.		
Note:	This value is irrelevant for the V/f control mode.		
r0078	CO: Active current actual value / I_active_act		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1774, 1775, 8850, 8946, 8950 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the actual value for the active current.		
Dependency:	Refer to: r0030		
Note:	The active current actual value is available smoothed (r0030) and unsmoothed (r0078).		
r0078[0...1]	CO: Current actual value torque-generating / Iq_act		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1630, 5714, 5730 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the torque-generating current actual value (Iq_act).		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		
Dependency:	Refer to: r0030, p0045		
Note:	These values are irrelevant for the V/f control mode. The torque-generating current actual value is available smoothed (r0030 with 100 ms, r0078[1] with p0045) and unsmoothed (r0078[0]).		
r0078[0...1]	CO: Current actual value force-generating / Iq_act		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1630, 5714, 5730 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the force-generating current actual value (Iq_act).		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		
Dependency:	Refer to: r0030, p0045		

Note: These values are irrelevant for the V/f control mode.
The force-generating current actual value is available smoothed (r0030 with 100 ms, r0078[1] with p0045) and unsmoothed (r0078[0]).

r0078	CO: Current actual value torque-generating / Iq_act		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 1710, 6310, 6714, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the torque-generating current actual value (Iq_act).		
Dependency:	Refer to: r0030		
Note:	This value is irrelevant for the V/f control mode. The torque-generating current actual value is available smoothed (r0030 with 300 ms) and unsmoothed (r0078).		

r0079[0...1]	CO: Torque setpoint total / M_set total		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 5610, 8012 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the torque setpoint at the output of the speed controller (before clock cycle interpolation).		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		

r0079[0...1]	CO: Total force setpoint / F_set total		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: 5610, 8012 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the force setpoint at the output of the velocity controller (before clock cycle interpolation).		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		

r0079	CO: Torque setpoint total / M_set total		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 1700, 1710, 6030, 6060, 6710, 8012 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the torque setpoint at the output of the speed controller (before clock cycle interpolation).		

r0080 CO: Torque actual value / M_act

SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 5730 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the actual torque value.		
Dependency:	Refer to: r0031		
Note:	The torque actual value is available smoothed (r0031) and unsmoothed (r0080).		

r0080 CO: Force actual value / F_act

SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: 5730 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the actual force value.		
Dependency:	Refer to: r0031		
Note:	The force actual value is available smoothed (r0031) and unsmoothed (r0080).		

r0080[0...1] CO: Torque actual value / M_act

VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 6714, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the actual torque value.		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		
Dependency:	Refer to: r0031		
Note:	The torque actual value is available smoothed (r0031 with 100 ms, r0080[1] with p0045) and unsmoothed (r0080[0]).		

r0081 CO: Torque utilization / M_Utilization

SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 8012 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the torque utilization as a percentage. The torque utilization is obtained from the required smoothed torque referred to the torque limit.		
Dependency:	Refer to: r0033		
Note:	The torque utilization is available smoothed (r0033) and unsmoothed (r0081). The torque utilization is obtained from the required torque referred to the torque limit as follows: - Positive torque: $r0081 = ((r0079 + p1532) / (r1538 - p1532)) * 100 \%$ - Negative torque: $r0081 = ((-r0079 + p1532) / (-r1538 + p1532)) * 100 \%$ For SERVO, the following applies: The calculation of the torque utilization depends on the selected smoothing time constant (p0045).		

r0081 CO: Force utilization / F_utilization

SERVO (Lin)

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [%]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** PERCENT**Max**

- [%]

Access level: 3**Func. diagram:** 8012**Unit selection:** -**Expert list:** 1**Factory setting**

- [%]

Description:

Displays the force utilization as a percentage.

The force utilization is obtained from the required smoothed force referred to the force limit.

Dependency:

Refer to: r0033

Note:

The force utilization is available smoothed (r0033) and unsmoothed (r0081).

The calculation of the force utilization depends on the selected smoothing time constant (p0045).

r0081 CO: Torque utilization / M_Utilization

VECTOR

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [%]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** PERCENT**Max**

- [%]

Access level: 3**Func. diagram:** 8012**Unit selection:** -**Expert list:** 1**Factory setting**

- [%]

Description:

Displays the torque utilization as a percentage.

The torque utilization is obtained from the required smoothed torque referred to the torque limit.

Dependency:

Refer to: r0033

Note:

The torque utilization is available smoothed (r0033) and unsmoothed (r0081).

The torque utilization is obtained from the required torque referred to the torque limit as follows:

- Positive torque: $r0081 = (r0079 / r1538) * 100 \%$ - Negative torque: $r0081 = (-r0079 / -r1539) * 100 \%$

For SERVO, the following applies:

The calculation of the torque utilization depends on the selected smoothing time constant (p0045).

r0082 CO: Active power actual value / P_act

A_INF, S_INF

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [kW]

Calculated: -**Dynamic index:** -**Units group:** 14_7**Scaling:** r2004**Max**

- [kW]

Access level: 3**Func. diagram:** 8750, 8850, 8950**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [kW]

Description:

Displays the instantaneous active power.

Dependency:

Refer to: r0032

Note:

The active power is available smoothed (r0032) and unsmoothed (r0082).

r0082	CO: Active power actual value / P_act		
B_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: 14_7 Scaling: r2004 Max - [kW]	Access level: 3 Func. diagram: 8750, 8850, 8950 Unit selection: p0505 Expert list: 1 Factory setting - [kW]
Description:	Displays the instantaneous active power.		
Dependency:	Refer to: r0032		
Notice:	For Basic Line Modules of chassis format, the displayed value is invalid as these units do not have any current sensing.		
Note:	The active power is available smoothed (r0032) and unsmoothed (r0082).		
r0082[0...2]	CO: Active power actual value / P_act		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: 14_5 Scaling: r2004 Max - [kW]	Access level: 3 Func. diagram: 5730 Unit selection: p0505 Expert list: 1 Factory setting - [kW]
Description:	Displays the instantaneous active power.		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045 [2] = Electric power		
Dependency:	Refer to: r0032		
Note:	The mechanical active power is available smoothed (r0032 with 100 ms, r0082[1] with p0045) and unsmoothed (r0082[0]).		
r0082[0...2]	CO: Active power actual value / P_act		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: 14_8 Scaling: r2004 Max - [kW]	Access level: 3 Func. diagram: 5730 Unit selection: p0505 Expert list: 1 Factory setting - [kW]
Description:	Displays the instantaneous active power.		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045 [2] = Electric power		
Dependency:	Refer to: r0032		
Note:	The mechanical active power is available smoothed (r0032 with 100 ms, r0082[1] with p0045) and unsmoothed (r0082[0]).		

r0082[0...2]	CO: Active power actual value / P_act		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: 14_5 Scaling: r2004 Max - [kW]	Access level: 3 Func. diagram: 6714, 6799 Unit selection: p0505 Expert list: 1 Factory setting - [kW]
Description:	Displays the instantaneous active power.		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045 [2] = Electric power		
Dependency:	Refer to: r0032		
Note:	The mechanical active power is available smoothed (r0032 with 100 ms, r0082[1] with p0045) and unsmoothed (r0082[0]).		
r0083	CO: Flux setpoint / Flux setpoint		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the flux setpoint.		
r0084	CO: Flux actual value / Flux act val		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the flux actual value.		
r0084[0...1]	CO: Flux actual value / Flux act val		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 6726, 6730, 6731 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the flux actual value.		
Index:	[0] = Unsmoothed [1] = Smoothed		
Note:	The flux actual value (index 1) smoothed with p1585 is only displayed for separately-excited synchronous motors. In the following cases, the unsmoothed flux actual value is also displayed: - in the range of the current model. - during the pole position identification. - for I/f control. - for a stalled drive.		

r0087	CO: Actual power factor / Cos phi act		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 6714, 6730, 6732, 6799 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the actual active power factor.		
r0088	CO: DC link voltage setpoint / Vdc setpoint		
A_INF, SERVO (Tech_ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: 5_2 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 8940, 8964 Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the setpoint for the DC link voltage.		
r0088	CO: DC link voltage setpoint / Vdc setpoint		
VECTOR (Tech_ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: 5_2 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the setpoint for the DC link voltage.		
r0089[0...2]	Actual phase voltage / U_phase act val		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: 5_3 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 6719 Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the actual phase voltage.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		
Note:	The values are determined from the transistor power-on duration.		

p0092 Clock synchronous operation pre-assignment/check / Clock sync op			
CU_CX32, CU_I	Can be changed: C1(1)	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 1	Factory setting 1
Description:	Setting to pre-assign/check the sampling times for the internal controller clock cycles for clock-synchronous PROFIBUS operation. p0092 = 1: The controller clock cycles are set so that clock synchronous PROFIBUS operation is possible. If it is not possible to change the controller clock cycles of the clock-cycle synchronous PROFIBUS operation, then an appropriate message is output. The presetting of the controller clock cycles can result in a derating of the Motor Module (e.g. p0115[0] = 400 µs --> 375 µs). When the drive unit utilization (r9976) is calculated, its maximum computing time load has already been calculated during ramp-up for clock-cycle synchronous operation and taken into account in r9976 (V4.3 and higher), if fixed DCC runtime groups "Receive AFTER IF1 PROFIdrive PZD" and "Send BEFORE IF1 PROFIdrive PZD" are used. p0092 = 0: The controller clock cycles are set without any restrictions by the clock-cycle PROFIBUS operation (as for up to version V2.3). When the drive unit utilization (r9976) is calculated, its maximum computing time load has already been calculated during ramp-up for non-clock-cycle-synchronous operation and taken into account in r9976 (V4.3 and higher), if fixed DCC runtime groups "Receive AFTER IF1 PROFIdrive PZD" and "Send BEFORE IF1 PROFIdrive PZD" are used.		
Value:	0: No isochronous PROFIBUS 1: Isochronous PROFIBUS		
Dependency:	Refer to: r0110, p0115 Refer to: A01223, A01224		
Caution:	Only current controller clock cycles (p0115 index 0) which are integers of 125 µs are permitted for isochronous mode.		
			
Notice:	p0092 only affects the automatic default for the clock cycles (p0115) in the drive. If the clock cycles are modified subsequently in expert mode (p0112 = 0), p0092 = 0 should be set so that the new values are not overwritten again by the automatic default when the parameters are downloaded. Current controller clock cycle values must continue to be integers of 125 µs for synchronous mode.		

p0092	Clock synchronous operation pre-assignment/check / Clock sync op		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Setting to pre-assign/check the sampling times for the internal controller clock cycles for clock-synchronous PROFIBUS operation. p0092 = 1: The controller clock cycles are set so that clock synchronous PROFIBUS operation is possible. If it is not possible to change the controller clock cycles of the clock-cycle synchronous PROFIBUS operation, then an appropriate message is output. The presetting of the controller clock cycles can result in a derating of the Motor Module (e.g. p0115[0] = 400 µs --> 375 µs). When the drive unit utilization (r9976) is calculated, its maximum computing time load has already been calculated during ramp-up for clock-cycle synchronous operation and taken into account in r9976 (V4.3 and higher), if fixed DCC runtime groups "Receive AFTER IF1 PROFIdrive PZD" and "Send BEFORE IF1 PROFIdrive PZD" are used. p0092 = 0: The controller clock cycles are set without any restrictions by the clock-cycle PROFIBUS operation (as for up to version V2.3). When the drive unit utilization (r9976) is calculated, its maximum computing time load has already been calculated during ramp-up for non-clock-cycle-synchronous operation and taken into account in r9976 (V4.3 and higher), if fixed DCC runtime groups "Receive AFTER IF1 PROFIdrive PZD" and "Send BEFORE IF1 PROFIdrive PZD" are used.		
Value:	0: No isochronous PROFIBUS 1: Isochronous PROFIBUS		
Dependency:	Refer to: r0110, p0115 Refer to: A01223, A01224		
Caution:	Only current controller clock cycles (p0115 index 0) which are integers of 125 µs are permitted for isochronous mode.		
			
Notice:	p0092 only affects the automatic default for the clock cycles (p0115) in the drive. If the clock cycles are modified subsequently in expert mode (p0112 = 0), p0092 = 0 should be set so that the new values are not overwritten again by the automatic default when the parameters are downloaded. Current controller clock cycle values must continue to be integers of 125 µs for synchronous mode.		
r0093	CO: Pole position angle electrically normalized / Pole pos el norm		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°]	Calculated: - Dynamic index: - Units group: - Scaling: p2005 Max - [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [°]
Description:	Displays the normalized electrical pole position angle.		
Dependency:	Refer to: r0094, p0431, r1778		

Notice: When the pole position angle (r0093) is output via test socket Tx (x = 0, 1, 2) to adjust the encoder (to determine the angular commutation offset) the test socket being used must be parameterized as follows:

p0771[x] = r0093
 p0777[x] = 0 %
 p0778[x] = 0 V
 p0779[x] = 400 %
 p0780[x] = 4 V
 p0783[x] = 0 V
 p0784[x] = 0

For p1821 = 1 (counter-clockwise direction of rotation) the following applies:

In order to adjust the encoder using the EMF method, the value, determined using the oscilloscope, must be inverted and then entered in p0431.

Note: For operation with encoder and pulse suppression, the following applies:

- the value is generated from r0094 + 180 °.
- this angle can be used to adjust the encoders of synchronous motors.

For pulse enable, the following applies:

- the value indicates the transformation angle used by the control + 180 °.
- this value is, contrary to r0094, also applicable (provides information) for encoderless operation and after a pole position identification routine.

r0094		CO: Transformation angle / Transformat_angle	
A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8850, 8950
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2005	Expert list: 1
	Min - [°]	Max - [°]	Factory setting - [°]
Description:	Displays the transformation angle.		
Note:	The transformation angle corresponds to the line supply angle.		

r0094	CO: Transformation angle / Transformat_angle		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 1580, 1680, 1690, 4710, 6714, 6730, 6731, 6732
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2005	Expert list: 1
	Min - [°]	Max - [°]	Factory setting - [°]
Description:	Displays the transformation angle.		
Dependency:	Refer to: r0093, p0431, r1778		
Note:	The transformation angle corresponds to the electrical commutation angle.		
	If no pole position identification is carried out (p1982), and the encoder is adjusted, the following applies: The encoder supplies the value and indicates the electrical angle of the flux position (d axis).		

p0097	Select drive object type / Select DO type		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Integer16 P-Group: Topology Not for motor type: - Min 0 Max 17	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Executes an automatic device configuration. In so doing, p0099, p0107 and p0108 are appropriately set.		
Value:	0: No selection 1: Drive object type SERVO 2: Drive object type VECTOR 3: SINAMICS GM (DFEMV & VECTORMV) 4: SINAMICS SM (AFEMV & VECTORMV) 5: SINAMICS GL (VECTORGL) 6: SINAMICS SL (VECTORSL) 12: Drive object type VECTOR parallel circuit 13: Drive object type VECTORMV - GM parallel circuit 14: Drive object type VECTORMV - SM parallel circuit 15: Drive object type DC_CTRL 16: Drive object type SERVO HMI 17: Drive object type VECTOR HMI		
Dependency:	Refer to: r0098, p0099 Refer to: A01330		
Note:	For p0097 = 0, p0099 is automatically set to the factor setting. The possible settings are dependent upon the device type.		

r0098[0...5]	Actual device topology / Device_act topo		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Topology Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the automatically detected actual device topology in coded form.		
Index:	[0] = DRIVE-CLiQ socket X100 [1] = DRIVE-CLiQ socket X101 [2] = DRIVE-CLiQ socket X102 [3] = DRIVE-CLiQ socket X103 [4] = DRIVE-CLiQ socket X104 [5] = DRIVE-CLiQ socket X105		
Dependency:	Refer to: p0097, p0099		
Note:	Topology coding: abcd efgh hex a = number of Active Line Modules b = number of Motor Modules c = number of motors d = number of encoders (or the line supply voltage sensing for Active Line Modules) e = number of additional encoders (or the line supply voltage sensing for Active Line Modules) f = number of Terminal Modules g = number of Terminal Boards h = reserved if the value 0 is displayed in all indices, then components are not detected via DRIVE-CLiQ. If a value F hex occurs at a position of the coding (abcd efgh hex), then an overflow has occurred.		

p0099[0...5]	Device target topology / Device_target topo		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Unsigned32 P-Group: Topology Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the device target topology in coded form (refer to r0098). The setting is made during commissioning. Deactivated or non-available components are also counted		
Index:	[0] = DRIVE-CLiQ socket X100 [1] = DRIVE-CLiQ socket X101 [2] = DRIVE-CLiQ socket X102 [3] = DRIVE-CLiQ socket X103 [4] = DRIVE-CLiQ socket X104 [5] = DRIVE-CLiQ socket X105		
Dependency:	The parameter can only be written into for p0097 = 0. To perform an automatic device configuration run, an index of the device target topology must be set to the value of the device actual topology in r0098 for acknowledgement. An index of the device actual topology with a value other than 0 must be selected. Refer to: p0097, r0098 Refer to: A01330		
Note:	The parameter can only be set to the values 0, the value of the actual device topology, the value of the actual device target topology and FFFFFFFF hex. If the value 0 is displayed in all of the indices, then the system has still not been commissioned. The value FFFFFFFF hex indicates that the topology was not generated by the automatic device configuration but was commissioned using the commissioning software (e.g. using parameter download).		
p0100	IEC/NEMA mot stds / IEC/NEMA mot stds		
SERVO	Can be changed: C2(1) Data type: Integer16 P-Group: Converter Not for motor type: FEM Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Defines whether the motor and drive converter power settings (e.g. rated motor power - p0307) are expressed in [kW] or [hp]. Depending on the selection, the rated motor frequency (p0310) is either set to 50 Hz or 60 Hz. The following applies for IEC drives: The power factor (p0308) should be parameterized. The following applies for NEMA drives: The efficiency (p0309) should be parameterized.		
Value:	0: IEC-Motor (50 Hz, SI units) 1: NEMA motor (60 Hz, US units)		
Dependency:	If p0100 is changed, all of the rated motor parameters are reset. Only then are possible unit changeovers made. The units of all motor parameters are changed that are involved in the selection of IEC or NEMA (e.g. r0206, p0307, p0316, r0333, r0334, p0341, p0344, r1493, r1969). Refer to: r0206, p0210, p0300, p0304, p0305, p0307, p0308, p0309, p0310, p0311, p0312, p0314, p0320, p0322, p0323, p0335, r0336, r0337, p0338, p1800		
Note:	The parameter can only be changed for vector control (p0107). The parameter value is not reset when the factory setting is restored (p0010 = 30, p0970).		

p0100	IEC/NEMA mot stds / IEC/NEMA mot stds		
VECTOR	Can be changed: C2(1) Data type: Integer16 P-Group: Converter Not for motor type: FEM Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Defines whether the motor and drive converter power settings (e.g. rated motor power - p0307) are expressed in [kW] or [hp]. Depending on the selection, the rated motor frequency (p0310) is either set to 50 Hz or 60 Hz. The following applies for IEC drives: The power factor (p0308) should be parameterized. The following applies for NEMA drives: The efficiency (p0309) should be parameterized.		
Value:	0: IEC-Motor (50 Hz, SI units) 1: NEMA motor (60 Hz, US units)		
Dependency:	If p0100 is changed, all of the rated motor parameters are reset. Only then are possible unit changeovers made. The units of all motor parameters are changed that are involved in the selection of IEC or NEMA (e.g. r0206, p0307, p0316, r0333, r0334, p0341, p0344, r1493, r1969). Refer to: r0206, p0210, p0300, p0304, p0305, p0307, p0308, p0309, p0310, p0311, p0312, p0314, p0320, p0322, p0323, p0335, r0336, r0337, p0338, p1800		
Note:	The parameter can only be changed for vector control (p0107). The parameter value is not reset when the factory setting is restored (p0010 = 30, p0970).		
p0101[0...23]	Drive object numbers / DO numbers		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Unsigned16 P-Group: Topology Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 62	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter contains the object number via which every drive object can be addressed. The number of an existing drive object is entered into each index. The numbers are automatically assigned once and can no longer be changed as long as the object has not been deleted. In the commissioning software, this object number cannot be entered using the expert list, but is automatically assigned when inserting an object.		

Index:	[0] = Drive object number Control Unit
	[1] = Drive object number object 1
	[2] = Drive object number object 2
	[3] = Drive object number object 3
	[4] = Drive object number object 4
	[5] = Drive object number object 5
	[6] = Drive object number object 6
	[7] = Drive object number object 7
	[8] = Drive object number object 8
	[9] = Drive object number object 9
	[10] = Drive object number object 10
	[11] = Drive object number object 11
	[12] = Drive object number object 12
	[13] = Drive object number object 13
	[14] = Drive object number object 14
	[15] = Drive object number object 15
	[16] = Drive object number object 16
	[17] = Drive object number object 17
	[18] = Drive object number object 18
	[19] = Drive object number object 19
	[20] = Drive object number object 20
	[21] = Drive object number object 21
	[22] = Drive object number object 22
	[23] = Drive object number object 23

Note: Value = 0: No drive object is defined.

r0102[0...1] Number of drive objects / DO count

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Topology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the number of existing or existing and prepared drive objects.

Index: [0] = Existing drive objects
[1] = Existing and prepared drive objects

Dependency: Refer to: p0101

Note: The numbers of the drive objects are in p0101.

Index 0:

Displays the number of drive objects that have already been set up.

Index 1:

Displays the number of drive objects that have already been set up and, in addition, the drive objects that still have to be set up.

r0103 Application-specific view / Appl_spec view

A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	65535	-

Description: Displays the application-specific view of the individual drive object.

Dependency: Refer to: p0107, r0107

p0103[0...23]	Application-specific view / Appl_spec view		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(2) Data type: Unsigned16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The application-specific view of an existing drive object is entered into each index. The parameter cannot be changed.		
Dependency:	Refer to: p0107, r0107		
Note:	In the non-volatile memory, the application-specific views are defined in files with the following structure: PDxxxyyy.ACX xxx: Application-specific view (p0103) yyy: Type of drive object (p0107) Example: PD052011.ACX --> "011" stands for the drive object, type SERVO --> "052" is the number of the view for this drive object		
p0105	Activate/deactivate drive object / DO act/deact		
A_INF, B_INF, CU_LINK, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15DI_DO, TM17, VECTOR	Can be changed: T Data type: Integer16 P-Group: Closed-loop control Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Setting to activate/deactivate a drive object.		
Value:	0: Deactivate drive object 1: Activate drive object 2: Drive object, deactivate and not present		
Recommend.:	After inserting all of the components of a drive object, before activating, first wait for Alarm A01316.		
Dependency:	Refer to: r0106 Refer to: A01316		
Caution:	It is not permissible to deactivate drive objects with safety functions enabled.		
Notice:	The following applies when activating: If components are inserted for the first time and the appropriate drive object is activated, then the drive system is automatically booted. To do this, the pulses of all of the drive objects must be suppressed.		
Note:	Setting a drive object to deactivate principally corresponds to the "parking axis" function. however, here, all of the DRIVE-CLiQ components, assigned to the drive object, are involved.		
p0105	Activate/deactivate drive object / DO act/deact		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: Closed-loop control Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Setting to activate/deactivate a drive object.		
Value:	0: Deactivate drive object 1: Activate drive object		

Dependency: Refer to: r0106

Notice: The following applies when activating:

If components are inserted for the first time and the appropriate drive object is activated, then the drive system is automatically booted. To do this, the pulses of all of the drive objects must be suppressed.

p0105 Activate/deactivate drive object / DO act/deact

TM15, TM31, TM41	Can be changed: T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	2	1

Description: Setting to activate/deactivate a drive object.

Value: 0: Deactivate drive object
1: Activate drive object
2: Drive object, deactivate and not present

Recommend.: After inserting all of the components of a drive object, before activating, first wait for Alarm A01316.

Dependency: Refer to: r0106
Refer to: A01316

Warning: A drive that is moved by simulating the inputs of a Terminal Module is brought to a standstill while this parameter is being changed over.



Caution: It is not permissible to deactivate drive objects with safety functions enabled.

Notice: The following applies when activating:

If components are inserted for the first time and the appropriate drive object is activated, then the drive system is automatically booted. To do this, the pulses of all of the drive objects must be suppressed.

Note: Setting a drive object to deactivate principally corresponds to the "parking axis" function. however, here, all of the DRIVE-CLiQ components, assigned to the drive object, are involved.

p0105 Activate/deactivate drive object / DO act/deact

TM54F_MA, TM54F_SL	Can be changed: T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	2	1

Description: Setting to activate/deactivate a drive object.

Value: 0: Deactivate drive object
1: Activate drive object
2: Drive object, deactivate and not present

Recommend.: After inserting all of the components of a drive object, before activating, first wait for Alarm A01316.

Dependency: Refer to: r0106
Refer to: A01316

Caution: It is not permissible to deactivate drive objects with safety functions enabled:
TM54F can only be deactivated if all of the axes connected to it via P10010 have been deactivated or are not enabled on the connected safety axes.

Notice: The following applies when activating:

If components are inserted for the first time and the appropriate drive object is activated, then the drive system is automatically booted. To do this, the pulses of all of the drive objects must be suppressed.

Note: Setting a drive object to deactivate principally corresponds to the "parking axis" function. however, here, all of the DRIVE-CLiQ components, assigned to the drive object, are involved.

r0106 Drive object active/inactive / DO act/inact

A_INF, B_INF,
CU_CX32, CU_I,
CU_LINK, CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, HUB,
S_INF, SERVO,
TB30, TM120, TM15,
TM15DI_DO, TM17,
TM31, TM41, VEC-
TOR

Can be changed: -
Data type: Integer16
P-Group: Closed-loop control
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 2
Func. diagram: -
Unit selection: -
Expert list: 1

	Min	Max	Factory setting
	0	1	-

Description: Displays the "active/inactive" state of a drive object.

Value:
0: Drive object inactive
1: Drive object active

Dependency: Refer to: p0105

r0107 Drive object type / DO type

A_INF, B_INF,
CU_LINK,
ENCODER, HUB,
S_INF, SERVO,
TB30, TM120, TM15,
TM15DI_DO, TM17,
TM31, TM41,
TM54F_MA,
TM54F_SL, VEC-
TOR

Can be changed: -
Data type: Integer16
P-Group: Closed-loop control
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 2
Func. diagram: -
Unit selection: -
Expert list: 1

	Min	Max	Factory setting
	0	300	-

Description: Displays the type of each drive object.

Value:
0: -
1: SINAMICS S
2: SINAMICS G
3: SINAMICS I
4: SINAMICS CX32
5: SINAMICS GM
6: SINAMICS DC
7: SINAMICS GL
9: SINAMICS S110
10: ACTIVE INFEED CONTROL
11: SERVO
12: VECTOR
13: VECTORMV
14: VECTORGL
16: VECTORSL
17: DC_CTRL
18: VECTORM2C
19: VECTORDM
20: SMART INFEED CONTROL
30: BASIC INFEED CONTROL
40: ACTIVE INFEED CONTROLMV
41: BASIC INFEED CONTROLMV
42: ACTIVE INFEED CONTROLM2C
51: SINAMICS G120 230 (SingleDO-Drive which combines Device+Vector)
52: SINAMICSG120 240_2 (SingleDO-Drive which combines Device+Vector)
100: TB30 (Terminal Board)

101: SINAMICS SL
 102: SINAMICS SM2
 104: SINAMICS SM2I
 150: DRIVE-CLiQ Hub Module
 200: TM31 (Terminal Module)
 201: TM41 (Terminal Module)
 202: TM17 High Feature (Terminal Module)
 203: TM15 (Terminal Module)
 204: TM15 (Terminal Module for SINAMICS)
 205: TM54F - Master (Terminal Module)
 206: TM54F - Slave (Terminal Module)
 207: TM120 (Terminal Module)
 254: CU-LINK
 300: ENCODER

Dependency: Refer to: p0103, r0103

p0107[0...23] Drive object type / DO type

CU_CX32, CU_I,	Can be changed: C1(2)	Calculated: -	Access level: 2
CU_S,	Data type: Integer16	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: -	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 1
CU_S_S150	Min	Max	Factory setting
	0	300	0

Description: The type of an existing drive object is entered into each index.

Value:

0: -
 1: SINAMICS S
 2: SINAMICS G
 3: SINAMICS I
 4: SINAMICS CX32
 5: SINAMICS GM
 6: SINAMICS DC
 7: SINAMICS GL
 9: SINAMICS S110
 10: ACTIVE INFEED CONTROL
 11: SERVO
 12: VECTOR
 13: VECTORMV
 14: VECTORGL
 16: VECTORSL
 17: DC_CTRL
 18: VECTORM2C
 19: VECTORDM
 20: SMART INFEED CONTROL
 30: BASIC INFEED CONTROL
 40: ACTIVE INFEED CONTROLMV
 41: BASIC INFEED CONTROLMV
 42: ACTIVE INFEED CONTROLM2C
 51: SINAMICS G120 230 (SingleDO-Drive which combines Device+Vector)
 52: SINAMICSG120 240_2 (SingleDO-Drive which combines Device+Vector)
 100: TB30 (Terminal Board)
 101: SINAMICS SL
 102: SINAMICS SM2
 104: SINAMICS SM2I
 150: DRIVE-CLiQ Hub Module
 200: TM31 (Terminal Module)
 201: TM41 (Terminal Module)
 202: TM17 High Feature (Terminal Module)
 203: TM15 (Terminal Module)

204: TM15 (Terminal Module for SINAMICS)
 205: TM54F - Master (Terminal Module)
 206: TM54F - Slave (Terminal Module)
 207: TM120 (Terminal Module)
 254: CU-LINK
 300: ENCODER

Index:

[0] = Drive object type, Control Unit
 [1] = Drive object type, object 1
 [2] = Drive object type, object 2
 [3] = Drive object type, object 3
 [4] = Drive object type, object 4
 [5] = Drive object type, object 5
 [6] = Drive object type, object 6
 [7] = Drive object type, object 7
 [8] = Drive object type, object 8
 [9] = Drive object type, object 9
 [10] = Drive object type, object 10
 [11] = Drive object type, object 11
 [12] = Drive object type, object 12
 [13] = Drive object type, object 13
 [14] = Drive object type, object 14
 [15] = Drive object type, object 15
 [16] = Drive object type, object 16
 [17] = Drive object type, object 17
 [18] = Drive object type, object 18
 [19] = Drive object type, object 19
 [20] = Drive object type, object 20
 [21] = Drive object type, object 21
 [22] = Drive object type, object 22
 [23] = Drive object type, object 23

Dependency:

Refer to: p0103, r0103

Caution:

If you change this parameter and exit the device commissioning mode, then the complete software will be set up again and all of the previous drive parameter settings are deleted.



Note:

The number (p0101) and the associated drive object type are in the same index.

For SINAMICS S a drive object type can only be changed between SERVO and VECTOR. If you change the parameter and exit drive start-up (p0009 from 2 to 0) the drive parameters are set up again.

r0108

Drive object, function module / DO function module

A_INF, B_INF,
S_INF

Can be changed: -

Calculated: -

Access level: 2

Data type: Unsigned32

Dynamic index: -

Func. diagram: -

P-Group: Closed-loop control

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

-

-

-

Description:

Displays the activated function module for the particular drive object.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
04	Line transformer	Activated	Not activated	-
07	Dynamic line support	Activated	Not activated	-
12	Line droop reg	Activated	Not activated	-
15	Parallel cct. config.	Activated	Not activated	-
18	Free function blocks	Activated	Not activated	-
19	Master/Slave	Activated	Not activated	-
26	Braking Module external	Activated	Not activated	-
27	Fan	Activated	Not activated	-
28	Cooling system	Activated	Not activated	-
31	PROFINET	Activated	Not activated	-

Note:

A "function module" is a functional expansion of a drive object that can be activated when commissioning.

19	Bit 19	On	Off	-
20	Bit 20	On	Off	-
21	Bit 21	On	Off	-
22	Bit 22	On	Off	-
23	Bit 23	On	Off	-
24	Bit 24	On	Off	-
25	Bit 25	On	Off	-
26	Bit 26	On	Off	-
27	Bit 27	On	Off	-
28	Bit 28	On	Off	-
29	Bit 29	On	Off	-
30	Bit 30	On	Off	-
31	Bit 31	On	Off	-

Note: A "function module" is a functional expansion of a drive object that can be activated when commissioning.

r0108 Drive object, function module / DO function module

ENCODER	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the activated function module for the particular drive object.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	12	Linear encoder	Activated	Not activated	-
	18	Free function blocks	Activated	Not activated	-
	31	PROFINET	Activated	Not activated	-

Note: A "function module" is a functional expansion of a drive object that can be activated when commissioning.

r0108 Drive object, function module / DO function module

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the activated function module for the particular drive object.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	01	Extended closed-loop torque control	Activated	Not activated	-
	02	Closed-loop speed/torque control	Activated	Not activated	-
	03	Closed loop position control	Activated	Not activated	-
	04	Basic positioner	Activated	Not activated	-
	07	Advanced Positioning Control (APC)	Activated	Not activated	-
	08	Extended setpoint channel	Activated	Not activated	-
	12	Linear motor	Activated	Not activated	-
	13	Safety rotary axis	Activated	Not activated	-
	14	Extended brake control	Activated	Not activated	-
	15	Parallel cct. config.	Activated	Not activated	-
	16	Technology controller	Activated	Not activated	-
	17	Extended signals/monitoring	Activated	Not activated	-
	18	Free function blocks	Activated	Not activated	-
	20	Software gating unit	Not activated	Activated	-
	25	Fail-safe inputs/outputs of the CU	Activated	Not activated	-
	27	Fan	Activated	Not activated	-
	28	Cooling system	Activated	Not activated	-
	29	CAN	Activated	Not activated	-
	31	PROFINET	Activated	Not activated	-

Note: A "function module" is a functional expansion of a drive object that can be activated when commissioning.

r0108 Drive object, function module / DO function moduleTB30, TM120,
TM15DI_DO, TM31,
TM41**Can be changed:** -**Data type:** Unsigned32**P-Group:** Closed-loop control**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 2**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Displays the activated function module for the particular drive object.**Bit field:****Bit Signal name**

18 Free function blocks

31 PROFINET

1 signal

Activated

Activated

0 signal

Not activated

Not activated

FP

-

-

Note: A "function module" is a functional expansion of a drive object that can be activated when commissioning.**r0110[0...2] Basic sampling times / t_basis**CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** -**Data type:** FloatingPoint32**P-Group:** Closed-loop control**Not for motor type:** -**Min**

0.00 [µs]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

10000.00 [µs]

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

- [µs]

Description: Displays the basic sampling times.

The sampling times are set using p0112 and p0115. The values for the basic sampling times are determined as a result of these settings.

Index:

[0] = Basic sampling time 0

[1] = Basic sampling time 1

[2] = Basic sampling time 2

r0111 Basic sampling time selection / t_basis selA_INF, B_INF,
CU_CX32, CU_I,
CU_LINK, CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150, HUB,
S_INF, SERVO,
VECTOR**Can be changed:** -**Data type:** Integer16**P-Group:** Closed-loop control**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Min**

0

Max

2

Factory setting

-

Description: Displays the selected basic sampling time for this drive object.**Dependency:** Refer to: r0110**r0111 Basic sampling time selection / t_basis sel**TB30, TM15,
TM15DI_DO, TM17,
TM31, TM41**Can be changed:** -**Data type:** Integer16**P-Group:** Closed-loop control**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

2

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Displays the selected basic sampling time for this drive object.**Dependency:** Refer to: r0110**Note:**

For TB30 and the Terminal Module, this parameter has no significance.

For TB30 and certain Terminal Modules, the sampling times can be set using p4099 (see description of p4099 for the Module in question).

p0112 Sampling times presetting p0115 / t_sample for p0115			
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: C1(3)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 5	Factory setting 3
Description:	Pre-assignment of the sampling times in p0115. The clock cycles for the current controller / speed controller / flux controller / setpoint channel / position controller / positioning / technology controller are pre-assigned as follows: SINAMICS S, servo drive: p0112 = 1: 250 / 250 / 250 / 4000 / 2000 / 8000 / 4000 µs (for chassis units) p0112 = 2: 125 / 250 / 250 / 4000 / 2000 / 8000 / 4000 µs p0112 = 3: 125 / 125 / 125 / 4000 / 1000 / 4000 / 4000 µs p0112 = 4: 62.5 / 62.5 / 62.5 / 1000 / 1000 / 2000 / 1000 µs p0112 = 5: Not possible SINAMICS S, Active Infeed (p0112 = 1 not for p0092 = 1): p0112 = 1: 400 / - / - / 1600 µs (presetting for the rated pulse frequency = 2.5 kHz) p0112 = 2: 250 / - / - / 2000 µs (presetting for the rated pulse frequency = 4.0 kHz) p0112 = 3: 125 / - / - / 2000 µs p0112 = 4: 125 / - / - / 1000 µs p0112 = 5: 125 / - / - / 500 µs SINAMICS S, Smart Infeed (p0112 = 1 not for p0092 = 1): p0112 = 1: 400 / - / - / 1600 µs (presetting for the rated pulse frequency = 2.5 kHz) p0112 = 2: 250 / - / - / 2000 µs (presetting for the rated pulse frequency = 4.0 kHz) p0112 = 3: 250 / - / - / 2000 µs p0112 = 4: 250 / - / - / 1000 µs p0112 = 5: Not possible SINAMICS S, Basic Infeed, booksize: p0112 = 4: 250 / - / - / 2000 µs SINAMICS S, Basic Infeed, chassis: p0112 = 1: 2000 / - / - / 2000 µs p0112 = 2: 2000 / - / - / 2000 µs (presetting) p0112 = 3: 2000 / - / - / 2000 µs p0112 = 4: Not possible p0112 = 5: Not possible SINAMICS S/G, vector drive (p0112 = 1 not for p0092 = 1 and not for PM340): p0112 = 1: 400 / 1600 / 1600 / 1600 / 3200 / 3200 / 3200 µs (for rated pulse frequency = 1.25, 2.5 kHz) p0112 = 2: 250 / 1000 / 2000 / 1000 / 2000 / 4000 / 4000 µs p0112 = 3: 250 / 1000 / 1000 / 1000 / 2000 / 4000 / 4000 µs (for rated pulse frequency = 2.0, 4.0 kHz) SINAMICS S, vector drive: p0112 = 4: 250 / 500 / 1000 / 500 / 1000 / 2000 / 2000 µs p0112 = 5: 250 / 250 / 1000 / 250 / 1000 / 2000 / 1000 µs		
Value:	0: Expert 1: xLow 2: Low 3: Standard 4: High 5: xHigh		
Recommend.:	When changing the sampling times of the current and speed controllers (also refer to p0115), then we recommend that after exiting commissioning (p0009 = 0) the controller settings are recalculated using p0340 = 4.		

Dependency: It is prohibited to select a parameter value from p0112 if the associated current controller clock cycle cannot set (e.g. p0112 = 1 is not possible for a vector drive and PM340 power unit).
Refer to: p0092

Note: For p0112 = 0 (expert) the individual sampling times in p0115 can be adjusted.
p0112 = 1 cannot be set for a power unit type PM340 (refer to r0203) for vector drives.

p0113 Minimum pulse frequency, selection / f_puls min sel

SERVO	Can be changed: C1(3)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	2.000 [kHz]	4.000 [kHz]	4.000 [kHz]

Description: The current controller sampling time (p0115[0]) is pre-assigned by selecting the minimum pulse frequency.

Dependency: The parameter can only be changed with p0112 = 0 (expert). For isochronous operation (p0092 = 1) the parameter can only be set so that a current controller clock cycle of 125 µs is obtained as an integer number.
The required pulse frequency can be set in p1800 after commissioning (p0009 = p0010 = 0).
Refer to: p0112, r0114, p0115, p1800

Note: The current controller sampling time (p0115[0]) is set to the inverse value of twice the minimum pulse frequency.
For p0113 = 2.0 kHz, p0115[0] = 250 µs is set, for p0113 = 4.0 kHz, p0115[0] = 125 µs is set. The current controller sampling time (p0115[0]), calculated from the pulse frequency, is set in a grid of 1.25 µs.
For a power unit type PM340 (refer to r0203), only the values 2.0 and 4.0 kHz can be set.

p0113 Minimum pulse frequency, selection / f_puls min sel

VECTOR	Can be changed: C1(3)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	1.000 [kHz]	2.000 [kHz]	2.000 [kHz]

Description: The current controller sampling time (p0115[0]) is pre-assigned by selecting the minimum pulse frequency.

Dependency: The parameter can only be changed with p0112 = 0 (expert). For isochronous operation (p0092 = 1) the parameter can only be set so that a current controller clock cycle of 125 µs is obtained as an integer number.
The required pulse frequency can be set in p1800 after commissioning (p0009 = p0010 = 0).
Refer to: p0112, r0114, p0115, p1800

Note: The current controller sampling time (p0115[0]) is set to the inverse value of twice the minimum pulse frequency.
For p0113 = 1.0 kHz, p0115[0] = 500 µs is set, for p0113 = 2.0 kHz, p0115[0] = 250 µs is set. The current controller sampling time (p0115[0]), calculated from the pulse frequency, is set in a grid of 1.25 µs.
For a power unit type PM340 (refer to r0203), only the values 1.0 and 2.0 kHz can be set. 1.0 kHz can be set in order to achieve a current controller clock cycle of 500µs. However, in this case, the minimum pulse frequency p1800 is limited to 2 kHz.

r0114[0...9] Minimum pulse frequency, recommended / f_puls min recom

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [kHz]	- [kHz]	- [kHz]

Description: Displays the recommended values (indices 0 and 1) for the minimum pulse frequency (p0113).
If the system rejects a change to p0113 because the value to be used lies outside the permitted value range, then instead the recommended value from r0114 can be used.

Index:	[0] = If only the actual drive is changed [1] = If all drives connected to the DRIVE-CLiQ line are changed [2] = 2. possible pulse frequency [3] = 3. possible pulse frequency [4] = 4. possible pulse frequency [5] = 5. possible pulse frequency [6] = 6. possible pulse frequency [7] = 7. possible pulse frequency [8] = 8. possible pulse frequency [9] = 9. possible pulse frequency
Dependency:	Refer to: p0113
Note:	After exiting commissioning (p0009 = p0010 = 0), the pulse frequencies calculated from the sampling time p115[0] are displayed in indices 1 to 9. If additional restrictions do not apply (e.g. due to having selected an output filter), these can be entered into p1800. The maximum pulse frequency of the power units was already taken into account in r0114. A value of 0 kHz does not define a recommended pulse frequency.

p0115[0...6]	Sampling times for internal control loops / t_sample int ctrl		
A_INF, B_INF, S_INF, SERVO	Can be changed: C1(3)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [µs]	16000.00 [µs]	[0] 125.00 [µs]
			[1] 125.00 [µs]
			[2] 125.00 [µs]
			[3] 4000.00 [µs]
			[4] 1000.00 [µs]
			[5] 4000.00 [µs]
			[6] 4000.00 [µs]

Description:	Sets the sampling times for the control loops. The default setting is made using p0112 and can only be individually changed for p0112 = 0 (expert).
Recommend.:	When changing the sampling times of the current and speed controllers (also refer to p0115), then we recommend that after exiting commissioning (p0009 = 0) the controller settings are recalculated using p0340 = 4.
Index:	[0] = Current controller [1] = Speed controller [2] = Flux controller [3] = Setpoint channel [4] = Position controller [5] = Positioning [6] = Technology controller
Dependency:	The sampling times can only be separately set if p0112 is 0 (expert). If a sampling time is modified in the expert mode, then all of the sampling times with higher indices are automatically changed in the same ratio as the sampling time itself was changed. Slower time slices are only taken if the calculated sampling time is also permitted. Upper limit is 8 ms. Higher-level controls must be calculated in integral ratios to lower-level controls (e.g. $p0115[1] = N * p0115[0]$; where N is an integer number). The sampling time of the speed controller (p0115[1]) can have as a maximum a value of 800% of the current controller sampling time (p0115[0]). For servo drives, the maximum sampling time of the current controller is 250 µs and for vector drives, 500 µs. The sampling times for setpoint channel (p0115[3]), position controller (p0115[4]), positioning (p0115[5]) and technology controller (p0115[6]) must have at least 2x the value of the current controller sampling time (p0115[0]). Refer to: r0110, r0111, p0112

Note: For function modules that can be activated (e.g. technology controller), the parameters values are pre-assigned.

For the Active Line Module (ALM) and Smart Line Module (SLM), the current and DC link voltage controllers operate with the same sampling time. For ALM/SLM the maximum current controller clock cycle is 400 µs.

For the Basic Line Module (BLM), the DC link voltage measurement operates in the current controller sampling time.

For BLM booksize, only the current controller sampling time of 250 µs is permitted. For BLM chassis, only the current controller sampling time of 2000 µs is permitted.

For power unit type PM340 (r0203), only current controller sampling times of 62.5 µs, 125 µs, 250 µs and 500 µs can be set. The maximum current controller clock cycle for servo drives and the minimum current controller clock cycle for vector drives is 250 µs.

If sampling times in p0115 are individually changed for p0112 = 0 (expert) then it must always be observed that the selected sampling times of the setpoint channel (p0115[3]), position controller (p0115[4]), positioning (p0115[5]) and technology controller (p0115[6]) are always greater than or equal to twice the current controller sampling time (p0115[0]).

p0115[0]			
Sampling time for supplementary functions / t_samp suppl_fct			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16000.00 [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 4000.00 [µs]
Description:	Sets the basic sampling time for supplementary functions (DCC, free function blocks) on this object. Only setting values that are an integer multiple of 125 µs are permissible.		
Index:	[0] = Basic sampl. time		
p0115[0]			
Sampling time for speed detection / t_sample n_det			
ENCODER	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 125.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 500.00 [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 125.00 [µs]
Description:	Sets the sampling times for speed detection.		
Index:	[0] = Basic sampl. time		
p0115[0...6]			
Sampling times for internal control loops / t_sample int ctrl			
SERVO (Lin)	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16000.00 [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 125.00 [µs] [1] 125.00 [µs] [2] 125.00 [µs] [3] 4000.00 [µs] [4] 1000.00 [µs] [5] 4000.00 [µs] [6] 4000.00 [µs]
Description:	Sets the sampling times for the control loops. The default setting is made using p0112 and can only be individually changed for p0112 = 0 (expert).		
Recommend.:	When changing the sampling times of the current and velocity controllers (p0115), then we recommend that after exiting commissioning (p0009 = 0) the controller settings are recalculated using p0340 = 4.		

Index:	[0] = Current controller [1] = Velocity controller [2] = Flux controller [3] = Setpoint channel [4] = Position controller [5] = Positioning [6] = Technology controller
Dependency:	The sampling times can only be separately set if p0112 is 0 (expert). If a sampling time is modified in the expert mode, then all of the sampling times with higher indices are automatically changed in the same ratio as the sampling time itself was changed. Slower time slices are only taken if the calculated sampling time is also permitted. Upper limit is 8 ms. Higher-level controls must be calculated in integral ratios to lower-level controls (e.g. $p0115[1] = N * p0115[0]$; where N is an integer number). The sampling time of the velocity controller (p0115[1]) can have as a maximum a value of 800% of the current controller sampling time (p0115[0]). For servo drives, the maximum sampling time of the current controller is 250 µs and for vector drives, 500 µs. Refer to: r0110, r0111, p0112
Note:	For function modules that can be activated (e.g. technology controller), the parameters values are pre-assigned. For the Active Line Module (ALM) and Smart Line Module (SLM), the current and DC link voltage controllers operate with the same sampling time. For ALM/SLM the maximum current controller clock cycle is 400 µs. For the Basic Line Module (BLM), the DC link voltage measurement operates in the current controller sampling time. For BLM booksize, only the current controller sampling time of 250 µs is permitted. For BLM chassis, only the current controller sampling time of 2000 µs is permitted. For power unit type PM340 (r0203), only current controller sampling times of 62.5 µs, 125 µs, 250 µs and 500 µs can be set. The maximum current controller clock cycle for servo drives and the minimum current controller clock cycle for vector drives is 250 µs. If sampling times in p0115 are individually changed for p0112 = 0 (expert) then it must always be observed that the selected sampling times of the setpoint channel (p0115[3]), position controller (p0115[4]), positioning (p0115[5]) and technology controller (p0115[6]) are always greater than or equal to twice the current controller sampling time (p0115[0]).

p0115[0]	Sampling time for supplementary functions / t_samp suppl_fct		
TB30, TM120, TM15DI_DO, TM31	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16000.00 [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 4000.00 [µs]
Description:	Sets the sampling times for supplementary functions (DCC, free function blocks) on this object. Only setting values that are an integer multiple of 125 µs are permissible.		
Index:	[0] = Basic sampl. time		
Note:	This parameter only applies to set the sampling times of possible supplementary functions. The sampling times for inputs/outputs must be set in p4099.		

p0115[0]	Sampling time for supplementary functions / t_samp suppl_fct		
TM41	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16000.00 [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 4000.00 [µs]
Description:	Sets the sampling times for supplementary functions (DCC, free function blocks) on this object. Only setting values that are an integer multiple of 125 µs are permissible.		
Index:	[0] = Basic sampl. time		
Note:	This parameter only applies to set the sampling times of possible supplementary functions. The sampling times for inputs/outputs or encoder emulation must be set in p4099.		

p0115[0...6] Sampling times for internal control loops / t_sample int ctrl			
VECTOR	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16000.00 [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 125.00 [µs] [1] 125.00 [µs] [2] 125.00 [µs] [3] 4000.00 [µs] [4] 1000.00 [µs] [5] 4000.00 [µs] [6] 4000.00 [µs]
Description:	Sets the sampling times for the control loops. The default setting is made using p0112 and can only be individually changed for p0112 = 0 (expert).		
Recommend.:	When changing the sampling times of the current and speed controllers (also refer to p0115), then we recommend that after exiting commissioning (p0009 = 0) the controller settings are recalculated using p0340 = 4.		
Index:	[0] = Current controller [1] = Speed controller [2] = Flux controller [3] = Setpoint channel [4] = Position controller [5] = Positioning [6] = Technology controller		
Dependency:	The sampling times can only be separately set if p0112 is 0 (expert). If a sampling time is modified in the expert mode, then all of the sampling times with higher indices are automatically changed in the same ratio as the sampling time itself was changed. Slower time slices are only taken if the calculated sampling time is also permitted. Upper limit is 8 ms. Higher-level controls must be calculated in integral ratios to lower-level controls (e.g. $p0115[1] = N * p0115[0]$; where N is an integer number). The sampling time of the speed controller (p0115[1]) can have as a maximum a value of 800% of the current controller sampling time (p0115[0]). The sampling times for setpoint channel (p0115[3]), position controller (p0115[4]), positioning (p0115[5]) and technology controller (p0115[6]) must have at least 2x the value of the current controller sampling time (p0115[0]). If p0092 = 1 the sampling times p0115 and the pulse frequency p1800 are checked every time the parameters are downloaded, and reset to the initial values if necessary. This check can be deactivated by setting p0092 = 0 (making this setting does not affect isochronous PROFIBUS operation). Refer to: r0110, r0111, p0112		
Note:	For function modules that can be activated (e.g. technology controller), the parameters values are pre-assigned. The minimum current controller clock cycle is 250 µs, the maximum current controller clock cycle is 500 µs. For power unit type PM340 (r0203), only these current controller sampling times of 250 µs and 500 µs can be set.		

r0116[0...1]			
Drive object clock cycle recommended / DO_clock recom			
A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, S_INF, SERVO, TB30, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min - [µs]	Max - [µs]	Factory setting - [µs]
Description:	Displays the recommended sampling time for the drive objects. r00116[0] = recommended sampling time: Recommended value which would then make the complete system operational. r00116[1] = recommended sampling time: Recommended value, which after changing other clock cycles on the DRIVE-CLiQ line, would result in an operational system.		
Index:	[0] = Change only for the actual drive object [1] = Changing all objects on the DRIVE-CLiQ line		
Dependency:	Refer to: p0115		
p0117			
Current controller computing dead time mode / I_ctrl t_dead mode			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: Closed-loop control Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 6	Factory setting 6
Description:	Sets the mode for the computing dead time of the current controller. 0: Offset (shifted) clocking, minimum computing dead time of each drive, automatic setting 1: Clocking at the same time, the dead time aligns itself to the dead time of the latest drive, automatic setting 2: Manual setting of the computing dead time, early transfer 3: Manual setting of the computing dead time, late transfer 4-6: As for 0-2, however, no early transfers are set for vectors		
Dependency:	Refer to: p0118 Refer to: A02100		
Note:	Re p0117 = 0: The times when the setpoints become effective for the individual controls is automatically and individually determined. Another computing dead time is set for each control (closed-loop) (p0118). Current is impressed for the individual controls without any offset with respect to time (improved EMC compatibility). Re p0117 = 1: The latest closed-loop control determines when the setpoints for each of the individual controls become active. The same computing dead time is set for each control (p0118). Current is impressed (flows) for the individual controls without any offset with respect to time. Re p0117 = 2: The computing dead time is manually set. The user must optimize the value in p0118. Re p0117 = 3: Only for internal Siemens use. Re p0117 = 4 ... 6: Behavior as for p0117 = 0 ... 2, however for vectors, the earliest times are not determined. The modified computing dead time mode is not effective until the drive unit is powered up again.		

p0118	Current controller computing dead time / I_ctrl t_dead		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2000.00 [µs]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [µs]
Description:	This parameter is preset as a function of the current controller sampling time (p0115[0]) and normally does not have to be changed.		
Dependency:	Refer to: p0117 Refer to: A02100		
Note:	For p0118 <= 0.005 µs, the current controller output is delayed by a complete current controller clock cycle (p0115[0]). After p0118 has been changed, we recommend that the current controller is adapted (p1715).		
p0120	Number of Power unit Data Sets (PDS) / PDS count		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: C1(3) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of Power unit Data Sets (PDS). The value corresponds to the number of power units connected together for a parallel circuit configuration.		
Dependency:	Refer to: p0107, r0107		
Note:	This parameter is only significant for drive objects A_INFEED and VECTOR with a parallel circuit configuration.		
p0121[0...n]	Power unit component number / PU comp_no		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: C1(4) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 199	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The power unit data set is assigned to a power unit using this parameter. This unique component number is assigned when parameterizing the topology. Only component numbers can be entered into this parameter that correspond to a power unit.		
Dependency:	Refer to: p0107, r0107		
Note:	For parallel circuit configurations, the parameter index is assigned to a power unit.		
p0124[0...n]	Power unit detection via LED / PU detection LED		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: U, T Data type: Unsigned8 P-Group: Converter Not for motor type: - Min 0	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Detects the power unit assigned to this drive and data set.		
Note:	While p0124 = 1, the READY LED flashes green/orange or red/orange with 2 Hz at the appropriate power unit. For parallel circuit configurations, the parameter index is assigned to a power unit.		

p0124[0...23]	Detection of main components using LED / Detection LED		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned8 P-Group: Converter Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Detects the main components of the drive object selected via the index.		
p0125[0...n]	Activate/deactivate power unit components / PU_comp act/deact		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: C1(4), T Data type: Integer16 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 2	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Setting to activate/deactivate a power unit component.		
Value:	0: Deactivate component 1: Activate component 2: Component, deactivate and not present		
Recommend.:	After inserting a component, before activating, first wait for Alarm A01317.		
Dependency:	Refer to: r0126 Refer to: A01317		
Caution:	It is not permissible to deactivate drive objects with safety functions enabled.		
Note:	The activation of a component can be rejected if the component was inserted for the first time. In this case, it is only possible to activate the component when the pulses for all of the drive objects are inhibited. For units connected in parallel, when one of the power units is deactivated, then the enable in p7001 is withdrawn.		
r0126[0...n]	Power unit components active/inactive / PU comp act/inact		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: Integer16 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the "active/inactive" state of a power unit component.		
Value:	0: Component inactive 1: Component active		
Dependency:	Refer to: p0105, p0125, p0897		
r0127[0...n]	Power unit version EPROM data / PU EPROM version		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Converter Not for motor type: - Min -	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the version of the EPROM data of the power unit.		
Dependency:	Refer to: r0147, r0157		
Note:	For parallel circuit configurations, the parameter index is assigned to a power unit.		

r0128[0...n]	Power unit, firmware version / PU FW version		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Converter Not for motor type: - Min -	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the firmware version of the power unit.		
Dependency:	Refer to: r0018, r0148, r0158, r0197, r0198		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00. For parallel circuit configurations, the parameter index is assigned to a power unit.		
p0130	Number of Motor Data Sets (MDS) / MDS count		
SERVO, VECTOR	Can be changed: C1(3) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16	Access level: 2 Func. diagram: 8575 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of Motor Data Sets (MDS).		
p0131[0...n]	Motor component number / Mot comp_no		
SERVO, VECTOR	Can be changed: C1(4) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 199	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The motor data set is assigned to a motor using this parameter. This unique component number is assigned when parameterizing the topology. Only component numbers can be entered into this parameter that correspond to a motor.		
p0139[0...2]	Copy Motor Data Set MDS / Copy MDS		
SERVO, VECTOR	Can be changed: C2(15) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 31	Access level: 2 Func. diagram: 8575 Unit selection: - Expert list: 1 Factory setting 0
Description:	Copying a Motor Data Set (MDS) into another.		
Index:	[0] = Source motor data set [1] = Target motor data set [2] = Start copying procedure		
Note:	Procedure: 1. In Index 0, enter which motor data set should be copied. 2. In Index 1, enter the motor data set data that is to be copied into. 3. Start copying: Set index 2 from 0 to 1. p0139[2] is automatically set to 0 when copying is completed. When copying, p0131 is not taken into account.		

p0140	Number of VSM data sets / VSM count		
A_INF, S_INF	Can be changed: C1(3) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of VSM data sets.		
Note:	The value cannot be changed for infeed units; it corresponds to the number of power units connected in parallel.		
p0140	Number of Encoder Data Sets (EDS) / EDS count		
ENCODER	Can be changed: C1(3) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of Encoder Data Sets (EDS).		
Note:	When parameterizing the drive with "no encoder" there must be at least one encoder data set (p0140 >= 1).		
p0140	Number of Encoder Data Sets (EDS) / EDS count		
SERVO, VECTOR	Can be changed: C1(3) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of Encoder Data Sets (EDS).		
Note:	When parameterizing the drive with "no encoder" there must be at least one encoder data set (p0140 >= 1).		
p0141[0...n]	VSM component number / VSM comp_no0		
A_INF, S_INF	Can be changed: C1(4) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: p0140 Units group: - Scaling: - Max 199	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The VSM data set is assigned to a VSM evaluation using this parameter. This unique component number is assigned when parameterizing the topology. Only component numbers that correspond to a VSM evaluation can be entered in this parameter.		
p0141[0...n]	Encoder interface (Sensor Module) component number / Enc_interf comp_no		
ENCODER, SERVO, VECTOR	Can be changed: C1(4) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 199	Access level: 3 Func. diagram: 4704, 8570 Unit selection: - Expert list: 1 Factory setting 0
Description:	This parameter is used to assign the encoder data set to an encoder evaluation (e.g. SMC). This unique component number is assigned when parameterizing the topology. Only component numbers can be entered into this parameter that correspond to an encoder evaluation.		

Note: If the encoder evaluation and encoder are integrated (motor with DRIVE-CLiQ), then their component numbers are identical.
For an SMC, different component numbers are assigned for the SMC (p0141) and the (actual) encoder (p0142).

p0142[0...n]	Encoder component number / Encoder comp_no		
ENCODER, SERVO, VECTOR	Can be changed: C1(4) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 199	Access level: 3 Func. diagram: 4704 Unit selection: - Expert list: 1 Factory setting 0
Description:	This parameter is used to assign the encoder data set to an encoder. This assignment is made using the unique component number that was assigned when parameterizing the topology. Only component numbers can be entered into this parameter that correspond to an encoder.		
Note:	If the encoder evaluation and encoder are integrated (motor with DRIVE-CLiQ), then their component numbers are identical. For an SMC, different component numbers are assigned for the SMC (p0141) and the (actual) encoder (p0142).		

p0144[0...n]	Voltage Sensing Module detection via LED / VSM detection LED		
A_INF, S_INF	Can be changed: U, T Data type: Unsigned8 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: p0140 Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Detects the Voltage Sensing Module (VSM) module assigned to this infeed.		

p0144[0...n]	Sensor Module detection via LED / SM detection LED		
ENCODER, SERVO, VECTOR	Can be changed: U, T Data type: Unsigned8 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Detects the Sensor Module assigned to this drive and data set.		
Note:	While p0144 = 1, the READY LED flashes green/orange or red/orange with 2 Hz at the appropriate Sensor Module.		

p0145[0...n]	Voltage Sensing Module, activate/deactivate / VSM act/deact		
A_INF, S_INF	Can be changed: C1(4), T Data type: Integer16 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: p0140 Units group: - Scaling: - Max 2	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Setting to activate/deactivate a Voltage Sensing Module (VSM).		
Value:	0: Deactivate component 1: Activate component 2: Component, deactivate and not present		
Recommend.:	After inserting a component, before activating, first wait for Alarm A01317.		
Dependency:	Refer to: r0146 Refer to: A01317		

Note: For chassis infeeds, it is not possible to activate/deactivate the Voltage Sensing Module (VSM) via p0145. The VSM can only be activated/deactivated in the group with the appropriate infeed via p0125[0...n].
The activation of a component can be rejected if the component was inserted for the first time.
In this case, it is only possible to activate the component when the pulses for all of the drive objects are inhibited.

p0145[0...n]		Activate/deactivate encoder interface / Enc_intf act/deact		
ENCODER, SERVO, VECTOR	Can be changed:	C1(4), U, T	Calculated: -	Access level: 2
	Data type:	Integer16	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group:	Data sets	Units group: -	Unit selection: -
	Not for motor type:	-	Scaling: -	Expert list: 1
	Min	0	Max	Factory setting
			2	1
Description:	Setting to activate/deactivate an encoder interface (Sensor Module).			
Value:	0: Deactivate component 1: Activate component 2: Component, deactivate and not present			
Recommend.:	After inserting a component, before activating, first wait for Alarm A01317.			
Dependency:	Refer to: r0146 Refer to: A01317			
Note:	The deactivation of an encoder interface corresponds to the "parking encoder" function and has the same effect. The activation of a component can be rejected if the component was inserted for the first time. In this case, it is only possible to activate the component when the pulses for all of the drive objects are inhibited. With the encoder interface for encoder 1 (motor encoder), the relevant drive object for writing the parameter must be in the "Ready for operation" state. With the encoder interface for encoders 2 and 3, the parameter can also be written during operation.			

r0146[0...n]		Voltage Sensing Module, active/inactive / VSM act/inact		
A_INF, S_INF	Can be changed:	-	Calculated: -	Access level: 2
	Data type:	Integer16	Dynamic index: p0140	Func. diagram: -
	P-Group:	Data sets	Units group: -	Unit selection: -
	Not for motor type:	-	Scaling: -	Expert list: 1
	Min	0	Max	Factory setting
			1	-
Description:	Displays the "active" or "inactive" state of a Voltage Sensing Module (VSM).			
Value:	0: Component inactive 1: Component active			
Dependency:	Refer to: p0105, p0145			

r0146[0...n]		Encoder interface active/inactive / Enc_intf act/inact		
ENCODER, SERVO, VECTOR	Can be changed:	-	Calculated: -	Access level: 2
	Data type:	Integer16	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group:	Data sets	Units group: -	Unit selection: -
	Not for motor type:	-	Scaling: -	Expert list: 1
	Min	0	Max	Factory setting
			1	-
Description:	Displays the "active" or "inactive" state of an encoder interface (Sensor Module).			
Value:	0: Component inactive 1: Component active			
Dependency:	Refer to: p0105, p0145, p0480, p0897			

r0147[0...n]	Voltage Sensing Module, EPROM data version / VSM EEPROM version		
A_INF, S_INF	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: p0140 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the version of the EPROM data of the Voltage Sensing Module (VSM).		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		

r0147[0...n]	Sensor Module EPROM data version / SM EEPROM version		
ENCODER, SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the version of the EPROM data of the Sensor Module.		
Dependency:	Refer to: r0127, r0157		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		

r0148[0...n]	Voltage Sensing Module firmware version / VSM FW version		
A_INF, S_INF	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: p0140 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the firmware version of the Voltage Sensing Module (VSM).		
Dependency:	Refer to: r0018, r0128, r0158, r0197, r0198		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		

r0148[0...n]	Sensor Module firmware version / SM FW version		
ENCODER, SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the firmware version of the Sensor Module.		
Dependency:	Refer to: r0018, r0128, r0158, r0197, r0198		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		

p0150	VSM2 data sets selection / VSM2 dat_sets qty		
A_INF, S_INF	Can be changed: C1(3) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of VSM2 data sets.		
Dependency:	The Voltage Sensing Module 2 (VSM2) can only be used if the "line transformer" function module has been activated (r0108.4 = 1).		
Note:	For the VSM2, parameters p5460 and following are significant. The Voltage Sensing Module 2 (VSM2) should always be connected to the primary side of the line transformer if at all possible.		
p0150	Number of VSM data sets / VSM dat_sets qty.		
VECTOR	Can be changed: C1(3) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of VSM data sets.		
p0151[0...n]	Voltage Sensing Module 2 component number / VSM2 comp_num		
A_INF, S_INF	Can be changed: C1(4) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: p0150 Units group: - Scaling: - Max 199	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The VSM2 data set is assigned to a VSM2 evaluation using this parameter.		
p0151[0...1]	DRIVE-CLiQ Hub Module component number / Hub comp_no		
HUB	Can be changed: C1(4) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 199	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	This parameter is used to assign the data set to a DRIVE-CLiQ Hub Module. This unique component number is assigned when parameterizing the topology. Only the numbers of components operated as hubs can be entered in these parameters. [0] = DRIVE-CLiQ node 1 [1] = DRIVE-CLiQ node 2		

p0151	Terminal Module component number / TM comp_no		
TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL	Can be changed: C1(4) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0 Max 199	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the component number for the Terminal Module. This unique component number is assigned when parameterizing the topology. Only component numbers can be entered into this parameter that correspond to a Terminal Module.		
p0151[0...n]	Voltage Sensing Module component number / VSM comp_no		
VECTOR	Can be changed: C1(4) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0 Max 199	Calculated: - Dynamic index: p0150 Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The VSM data set is assigned to a VSM evaluation using this parameter.		
p0154[0...n]	Voltage Sensing Module 2 detection via LED / VSM2 detection LED		
A_INF, S_INF	Can be changed: U, T Data type: Unsigned8 P-Group: Encoder Not for motor type: - Min 0 Max 1	Calculated: - Dynamic index: p0150 Units group: - Scaling: -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Detects the Voltage Sensing Module 2 (VSM2) assigned to this infeed.		
p0154	DRIVE-CLiQ Hub Module detection via LED / Hub detection LED		
HUB	Can be changed: U, T Data type: Unsigned8 P-Group: Encoder Not for motor type: - Min 0 Max 1	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Detects any DRIVE-CLiQ Hub Module that has been assigned.		
p0154	Terminal Module detection via LED / TM detection LED		
TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL	Can be changed: U, T Data type: Unsigned8 P-Group: Terminals Not for motor type: - Min 0 Max 1	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Detects the Terminal Module assigned to this drive and data set.		
Note:	While p0154 = 1, the READY LED flashes green/orange or red/orange with 2 Hz at the appropriate Terminal Module.		

p0155[0...n]	Voltage Sensing Module 2, activate/deactivate / VSM2 act/deact		
A_INF, S_INF	Can be changed: C1(4), T	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: p0150	Func. diagram: -
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	2	1
Description:	Setting to activate/deactivate a Voltage Sensing Module 2 (VSM2).		
Value:	0: Deactivate component 1: Activate component 2: Component, deactivate and not present		
Recommend.:	After inserting a component, before activating, first wait for Alarm A01317.		
Dependency:	Refer to: r0156 Refer to: A01317		

p0155[0...n]	Voltage Sensing Module, activate/deactivate / VSM act/deact		
VECTOR	Can be changed: C1(4), T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: p0150	Func. diagram: -
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	2	1
Description:	Setting to activate/deactivate a Voltage Sensing Module (VSM).		
Value:	0: Deactivate component 1: Activate component 2: Component, deactivate and not present		
Recommend.:	After inserting a component, before activating, first wait for Alarm A01317.		
Dependency:	Refer to: r0156 Refer to: A01317		

r0156[0...n]	Voltage Sensing Module 2, active/inactive / VSM2 act/inact		
A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: p0150	Func. diagram: -
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	-
Description:	Displays the "active" or "inactive" state of a Voltage Sensing Module 2 (VSM2).		
Value:	0: Component inactive 1: Component active		
Dependency:	Refer to: p0155		

r0156[0...n]	Voltage Sensing Module, active/inactive / VSM act/inact		
VECTOR	Can be changed: - Data type: Integer16 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: p0150 Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the "active" or "inactive" state of a Voltage Sensing Module (VSM).		
Value:	0: Component inactive 1: Component active		
Dependency:	Refer to: p0155		
r0157[0...n]	Voltage Sensing Module 2, EPROM data version / VSM2 EPROM version		
A_INF, S_INF	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: p0150 Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the version of the EPROM data of the Voltage Sensing Module 2 (VSM2).		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		
r0157	DRIVE-CLiQ Hub Module EPROM data version / Hub EPROM version		
HUB	Can be changed: - Data type: Unsigned32 P-Group: Terminals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the version of the EPROM data for the DRIVE-CLiQ Hub Module.		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		
r0157	Terminal Module EPROM data version / TM EPROM version		
TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL	Can be changed: - Data type: Unsigned32 P-Group: Terminals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the version of the EPROM data of the Terminal Module.		
Dependency:	Refer to: r0127, r0147		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		

r0157[0...n]	Voltage Sensing Module, EPROM data version / VSM EPROM version		
VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: p0150	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the version of the EPROM data of the Voltage Sensing Module (VSM).		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		

r0158[0...n]	Voltage Sensing Module 2 firmware version / VSM2 FW version		
A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: p0150	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the firmware version of the Voltage Sensing Module 2 (VSM2).		
Dependency:	Refer to: r0018, r0197, r0198		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		

r0158	DRIVE-CLiQ Hub Module firmware version / Hub FW version		
HUB	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the firmware version of the DRIVE-CLiQ Hub Module.		

r0158	Terminal Module Firmware Version / TM FW version		
TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the firmware version of the Terminal Module.		
Dependency:	Refer to: r0018, r0128, r0148, r0197, r0198		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		

r0158[0...n]	Voltage Sensing Module firmware version / VSM FW version		
VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: p0150 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the firmware version of the Voltage Sensing Module (VSM).		
Dependency:	Refer to: r0018, r0128, r0197, r0198		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		
p0161	Option board, component number / Opt board comp_num		
TB30	Can be changed: C1(4) Data type: Unsigned8 P-Group: Data sets Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 199	Access level: 4 Func. diagram: 9100 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the component number for the option board (e.g. Terminal Board 30). This unique component number is assigned when parameterizing the topology. Only component numbers can be entered into this parameter that correspond to an option board.		
p0162	CU-Link slave component number / CX32 comp_no		
CU_LINK	Can be changed: C1(4) Data type: Unsigned8 P-Group: Terminals Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 199	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
p0170	Number of Command Data Sets (CDS) / CDS count		
A_INF, B_INF, S_INF, SERVO	Can be changed: C1(3) Data type: Unsigned8 P-Group: Commands Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of Command Data Sets (CDS).		
Note:	It is possible to toggle between command parameters (BICO parameters) using this data set changeover.		
p0170	Number of Command Data Sets (CDS) / CDS count		
TM41	Can be changed: C1(3) Data type: Unsigned8 P-Group: Commands Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of Command Data Sets (CDS).		
Note:	It is possible to toggle between command parameters (BICO parameters) using this data set changeover.		

p0170 Number of Command Data Sets (CDS) / CDS count

VECTOR

Can be changed: C1(3)**Calculated:** -**Access level:** 2**Data type:** Unsigned8**Dynamic index:** -**Func. diagram:** -**P-Group:** Commands**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

2

4

2

Description: Sets the number of Command Data Sets (CDS).**Note:** It is possible to toggle between command parameters (BICO parameters) using this data set changeover.**p0180 Number of Drive Data Sets (DDS) / DDS count**

SERVO, TM41, VECTOR

Can be changed: C1(3)**Calculated:** -**Access level:** 2**Data type:** Unsigned8**Dynamic index:** -**Func. diagram:** 8565**P-Group:** Data sets**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

1

32

1

Description: Sets the number of Drive Data Sets (DDS).**p0186[0...n] Motor Data Sets (MDS) number / MDS number**

SERVO, VECTOR

Can be changed: C1(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned8**Dynamic index:** DDS, p0180**Func. diagram:** 8575**P-Group:** Data sets**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0

15

0

Description: Using the parameter, each Drive Data Set (= index) is assigned the associated Motor Data Set (MDS).
The parameter value therefore corresponds to the number of the assigned motor data set.**p0187[0...n] Encoder 1 encoder data set number / Enc 1 EDS number**

SERVO, VECTOR

Can be changed: C1(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned8**Dynamic index:** DDS, p0180**Func. diagram:** 1580, 8570**P-Group:** Data sets**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0

99

99

Description: Using the parameter, each Drive Data Set (= index) is assigned the associated Encoder Data Set (EDS) for encoder 1.

The parameter value therefore corresponds to the number of the assigned encoder data set.

Example:

Encoder data set 0 should be assigned to encoder 1 in drive data set 2.

--> p0187[2] = 0

Note: A value of 99 means that no encoder has been assigned to this drive data set (not configured).

p0188[0...n]	Encoder 2 encoder data set number / Enc 2 EDS number		
SERVO, VECTOR	Can be changed: C1(4)	Calculated: -	Access level: 3
	Data type: Unsigned8	Dynamic index: DDS, p0180	Func. diagram: 1580, 8570
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 99	Factory setting 99
Description:	Using the parameter, each Drive Data Set (= index) is assigned the associated Encoder Data Set (EDS) for encoder 2. The parameter value therefore corresponds to the number of the assigned encoder data set.		
Note:	A value of 99 means that no encoder has been assigned to this drive data set (not configured).		

p0189[0...n]	Encoder 3 encoder data set number / Enc 3 EDS number		
SERVO, VECTOR	Can be changed: C1(4)	Calculated: -	Access level: 3
	Data type: Unsigned8	Dynamic index: DDS, p0180	Func. diagram: 1580, 8570
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 99	Factory setting 99
Description:	Using the parameter, each Drive Data Set (= index) is assigned the associated Encoder Data Set (EDS) for encoder 3. The parameter value therefore corresponds to the number of the assigned encoder data set.		
Note:	A value of 99 means that no encoder has been assigned to this drive data set (not configured).		

r0192	Power unit firmware properties / PU FW property				
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -		
	P-Group: Converter	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the properties supported by the power unit firmware.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Edge modulation possible	Yes	No	-
	01	Free telegram can be selected	Yes	No	-
	02	Smart mode possible for Active Line Module	Yes	No	-
	03	Safety Integrated possible for VECTOR	Yes	No	-
	06	Liquid cooling	Yes	No	-
	07	SERVO pulse frequency changeover, DDS-dependent	Yes	No	-
	08	Simulation mode possible	Yes	No	-
	09	Internal armature short-circuit possible	Yes	No	-
	10	Autonomous internal armature short-circuit possible	Yes	No	-
	11	Infeed temperature inputs X21.1/2	Yes	No	-
	12	Integral normalized to half the gating unit clock cycle freq.	Yes	No	-
	13	Filtering thermal power unit current limit possible	Yes	No	-
	14	DC link compensation possible in power unit	Yes	No	-
	15	PT100 temperature evaluation possible	Yes	No	-
	16	Gating unit with pulse frequency wobulation possible	Yes	No	-
	17	Compound brake possible	Yes	No	-
	18	Extended voltage range possible	Yes	No	-

19	Gating unit available with current limitation control	Yes	No	-
20	Component status possible	Yes	No	-
21	Temperature evaluation via Motor Module / CU terminals possible	Yes	No	-
22	Reduced device supply voltage possible	Yes	no	-
23	Current measurement oversampling available	Yes	No	-
25	Internal fan operating hours counter available	Yes	No	-
26	Software gating unit in the CU is supported	Yes	No	-

Notice: This information represents the characteristics/features of the power unit firmware. It does not provide information/data about the characteristics/features of the hardware (e.g. bit 06 = 1 means that although the firmware supports "liquid cooling", a power unit with liquid cooling does not have to be used).

Note: Re bit 09:
The Motor Module supports the internal armature short-circuit. The function is internally required for voltage protection (p1231 = 3).
Re bit 10:
The Motor Module supports the autonomous internal voltage protection. If the voltage protection function is internally activated (p1231 = 3) the Motor Module decides autonomously - using the DC link voltage - as to whether the short-circuit is activated.
Re bit 23:
The component supports the detection of current actual values (and the detection of valve close durations) with double clocking and phase shift.

r0194[0...n]		VSM properties / VSM properties			
A_INF, S_INF	Can be changed:	-	Calculated:	-	Access level: 4
	Data type:	Unsigned32	Dynamic index:	p0140	Func. diagram: -
	P-Group:	Encoder	Units group:	-	Unit selection: -
	Not for motor type:	-	Scaling:	-	Expert list: 1
	Min	-	Max	-	Factory setting
Description: Displays the properties supported by the Voltage Sensing Module (VSM).					
Bit field:		Bit	Signal name	1 signal	0 signal
		00	Reserved	Yes	No
					FP
					-

r0194[0...n]		VSM properties / VSM properties			
VECTOR	Can be changed:	-	Calculated:	-	Access level: 4
	Data type:	Unsigned32	Dynamic index:	p0150	Func. diagram: -
	P-Group:	Encoder	Units group:	-	Unit selection: -
	Not for motor type:	-	Scaling:	-	Expert list: 1
	Min	-	Max	-	Factory setting
Description: Displays the properties supported by the Voltage Sensing Module (VSM).					
Bit field:		Bit	Signal name	1 signal	0 signal
		00	Reserved	Yes	No
					FP
					-

r0196[0...255]	DRIVE-CLiQ component status / DLQ comp status		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the status of DRIVE-CLiQ components. r0196[0...1]: Not used r0196[2]: Status of DRIVE-CLiQ component with component number 2 ... r0196[255]: Status of DRIVE-CLiQ component with component number 255		
Note:	Structure of status value: Bits 31 ... 08, 07, 06 ... 04, 03 ... 00 Re Bit 31 ... 08: Reserved Re Bit 07: 1: Part of target topology, 0: Only in actual topology Re Bit 06 ... 04: 1: Active, 0: Inactive or parked Re bit 03 ... 00: 0: Component data not available. 1: Power-up, acyclic DRIVE-CLiQ communication (LED = orange). 2: Ready for operation, cyclic DRIVE-CLiQ communication (LED = green). 3: Alarm (LED = green). 4: Fault (LED = red). 5: Detection via LED and ready for operation (LED = green/orange). 6: Detection via LED and alarm (LED = green/orange). 7: Detection via LED and fault (LED = red/orange). 8: Downloading firmware (LED = green/red at 0.5 Hz). 9: Firmware downloading completed, Waiting for POWER ON (LED = green/red at 2.0 Hz).		
r0197	Bootloader version / Bootloader vers		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the bootloader version.		
Dependency:	Refer to: r0018, r0128, r0148, r0158, r0198		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		

r0198[0...1]			
BIOS and EEPROM data version / BIOS/EEPROM vers			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the BIOS and EEPROM data version. Re r0198[0] Displays the BIOS version. Re r0198[1] Displays the EEPROM data version.		
Dependency:	Refer to: r0018, r0128, r0148, r0158, r0197		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		
p0199[0...24]			
Drive object name / DO name			
All objects	Can be changed: C1 Data type: Unsigned16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Freely assignable name for a drive object. In the commissioning software, this name cannot be entered using the expert list, but is specified in the configuration assistant. The object name can be subsequently modified in the Project Navigator using standard Windows resources.		
Note:	The parameter is not influenced by setting the factory setting.		
r0200[0...n]			
Power unit code number actual / PU code no. act			
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Converter Not for motor type: - Min -	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the unique code number of the power unit.		
Note:	No power unit found: r0200 = p0201. For parallel circuit configurations, the parameter index is assigned to a power unit.		
p0201[0...n]			
Power unit code number / PU code no			
A_INF, B_INF, S_INF	Can be changed: C2(2) Data type: Unsigned16 P-Group: Converter Not for motor type: - Min 0	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the actual code number from r0200 to acknowledge the power unit being used. When commissioned for the first time, the code number is automatically transferred from r0200 into p0201.		
Dependency:	Refer to: F07815		

Note: The parameter is used to identify when the drive is being commissioned for the first time.
The power unit commissioning can only be exited (p0201 = r0200), if the actual and acknowledged code numbers are identical (p0010 = 2).
For parallel circuit configurations, the parameter index is assigned to a power unit.

p0201[0...n]	Power unit code number / PU code no		
SERVO, VECTOR	Can be changed: C2(2) Data type: Unsigned16 P-Group: Converter Not for motor type: - Min 0	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the actual code number from r0200 to acknowledge the power unit being used. When commissioned for the first time, the code number is automatically transferred from r0200 into p0201.		
Dependency:	Refer to: F07815		
Notice:	When p0201 = 10000, the rated power unit data is reloaded and dependent parameters are set (e.g. p0205, p0210, p0230, p0857, p1800). p0201 is then automatically assigned the value of r0200 if the code number of the power unit could be read. A warm start must be performed after this procedure (automatically if necessary).		
Note:	The parameter is used to identify when the drive is being commissioned for the first time. The power unit commissioning can only be exited (p0201 = r0200), if the actual and acknowledged code numbers are identical (p0010 = 2). However, if the comparator in p9906 or p9908 is at 2 (low) or 3 (minimum), the power unit commissioning is automatically set to p0201 = r0200 upon exiting. When the code number is changed, the connection voltage (p0210) is checked and, if necessary, adjusted. For parallel circuit configurations, the parameter index is assigned to a power unit.		

r0203[0...n]	Actual power unit type / PU actual type		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: Integer16 P-Group: Converter Not for motor type: - Min 2	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 400	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the type of power unit found.		
Value:	2: MICROMASTER 440 3: MICROMASTER 411 4: MICROMASTER 410 5: MICROMASTER 436 6: MICROMASTER 440 PX 7: MICROMASTER 430 100: SINAMICS S 101: SINAMICS S (value) 102: SINAMICS S (combi) 112: PM220 (SINAMICS G120) 113: PM230 (SINAMICS G120) 114: PM240 (SINAMICS G120) 115: PM250 (SINAMICS G120) 116: PM260 (SINAMICS G120) 118: SINAMICS G120 Px 120: PM340 (SINAMICS S120) 150: SINAMICS G 200: SINAMICS GM 250: SINAMICS SM 260: SINAMICS SM120 300: SINAMICS GL 350: SINAMICS SL 400: SINAMICS DCM		
Note:	For parallel circuit configurations, the parameter index is assigned to a power unit.		

r0203[0...15]	Memory card name / Sp_card name																																																																									
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -																																																																							
Description:	Displays the name of the memory card in ASCII code. r0203[0]: Name character 1 ... r0203[15]: Name character 16 For the commissioning software, the ASCII characters are displayed uncoded.																																																																									
Notice:	An ASCII table (excerpt) can be found, for example, in the Appendix of the List Manual:																																																																									
r0204[0...n]	Power unit hardware properties / PU HW property																																																																									
A_INF, B_INF, S_INF	Can be changed: - Data type: Unsigned32 P-Group: Converter Not for motor type: - Min -	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -																																																																							
Description:	Displays the properties supported by the power unit hardware.																																																																									
Bit field:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr> <tr> <td>00</td><td>Device type</td><td>DC/AC device</td><td>AC/DC device</td><td>-</td></tr> <tr> <td>01</td><td>RFI filter available</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>02</td><td>Active Line Module available</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>03</td><td>Smart Line Module available</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>04</td><td>Basic Line Module available with thyristor bridge</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>05</td><td>Basic Line Module available with diode bridge</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>06</td><td>Liquid cooling with cooling system (chassis PU)</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>07</td><td>F3E regenerative feedback into the line supply</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>08</td><td>Internal Braking Module</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>09</td><td>Different cooling type supported</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>12</td><td>Safe Brake Control (SBC) supported</td><td>No</td><td>Yes</td><td>-</td></tr> <tr> <td>13</td><td>Safety Integrated supported</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>14</td><td>Internal LC output filter</td><td>Yes</td><td>No</td><td>-</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FP	00	Device type	DC/AC device	AC/DC device	-	01	RFI filter available	Yes	No	-	02	Active Line Module available	Yes	No	-	03	Smart Line Module available	Yes	No	-	04	Basic Line Module available with thyristor bridge	Yes	No	-	05	Basic Line Module available with diode bridge	Yes	No	-	06	Liquid cooling with cooling system (chassis PU)	Yes	No	-	07	F3E regenerative feedback into the line supply	Yes	No	-	08	Internal Braking Module	Yes	No	-	09	Different cooling type supported	Yes	No	-	12	Safe Brake Control (SBC) supported	No	Yes	-	13	Safety Integrated supported	Yes	No	-	14	Internal LC output filter	Yes	No	-			
Bit	Signal name	1 signal	0 signal	FP																																																																						
00	Device type	DC/AC device	AC/DC device	-																																																																						
01	RFI filter available	Yes	No	-																																																																						
02	Active Line Module available	Yes	No	-																																																																						
03	Smart Line Module available	Yes	No	-																																																																						
04	Basic Line Module available with thyristor bridge	Yes	No	-																																																																						
05	Basic Line Module available with diode bridge	Yes	No	-																																																																						
06	Liquid cooling with cooling system (chassis PU)	Yes	No	-																																																																						
07	F3E regenerative feedback into the line supply	Yes	No	-																																																																						
08	Internal Braking Module	Yes	No	-																																																																						
09	Different cooling type supported	Yes	No	-																																																																						
12	Safe Brake Control (SBC) supported	No	Yes	-																																																																						
13	Safety Integrated supported	Yes	No	-																																																																						
14	Internal LC output filter	Yes	No	-																																																																						
Note:	For parallel circuit configurations, the parameter index is assigned to a power unit.																																																																									

r0204[0...n]	Power unit hardware properties / PU HW property			
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Converter Not for motor type: - Min -	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -	
Description:	Displays the properties supported by the power unit hardware.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Device type	DC/AC device	AC/AC device
	01	RFI filter available	Yes	No
	02	Active Line Module available	Yes	No
	03	Smart Line Module available	Yes	No
	04	Basic Line Module available with thyristor bridge	Yes	No
	05	Basic Line Module available with diode bridge	Yes	No
	06	Liquid cooling with cooling system (chassis PU)	Yes	No
	07	F3E regenerative feedback into the line supply	Yes	No
	08	Internal Braking Module	Yes	No
	09	Different cooling type supported	Yes	No
	12	Safe Brake Control (SBC) supported	No	Yes
	13	Safety Integrated supported	Yes	No
	14	Internal LC output filter	Yes	No
Note:	For parallel circuit configurations, the parameter index is assigned to a power unit.			
p0205	Power unit application / PU application			
VECTOR	Can be changed: C2(1, 2) Data type: Integer16 P-Group: Converter Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 7	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 6	
Description:	Overloading the load duty cycles applies under the prerequisite that before and after the overload, the drive converter is operated with its base load current - in this case, a load duty cycle of 300 s is used as basis. For booksize drive units, the following applies: Only the setting p0205 = 0 can be selected. In this particular case, the base load current has a load duty cycle of 150 % for 60 s and 176 % for 30 s. For chassis units, the following applies: The base load current for a slight overload condition is based on a load duty cycle 110 % for 60 s and 150 % for 10 s. The base load current for a high overload condition is based on a load duty cycle 150 % for 60 s and 160 % for 10 s.			
Value:	0: Load duty cycle with high overload for vector drives 1: Load duty cycle with low overload for vector drives 6: S1 duty cycle for servo drives (feed drive) 7: S6 duty cycle for servo drives (spindle drive)			
Note:	When the parameter is changed, all of the motor parameters and the control mode are pre-assigned according to the selected application. The parameter has not influence when calculating the thermal overload. p0205 can only be changed to the settings that are saved in the power unit EEPROM. The parameter value is not reset when the factory setting is restored (see p0010 = 30, p0970).			

r0206[0...4]	Rated power unit power / PU P_{rated}		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: 14_6 Scaling: - Max - [kW]	Access level: 2 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [kW]
Description:	Displays the rated power unit power for various load duty cycles.		
Index:	[0] = Rating plate [1] = Load duty cycle with low overload [2] = Load duty cycle with high overload [3] = S1 cont duty cyc [4] = S6 load duty cycle		
Dependency:	IECdrives (p0100 = 0): Units kW NEMA drives (p0100 = 1): Units hp Refer to: p0100, p0205		

r0207[0...4]	Rated power unit current / PU PI_{rated}		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Arms]	Access level: 2 Func. diagram: 8014 Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the rated power unit power for various load duty cycles.		
Index:	[0] = Rating plate [1] = Load duty cycle with low overload [2] = Load duty cycle with high overload [3] = S1 cont duty cyc [4] = S6 load duty cycle		
Dependency:	Refer to: p0205		

r0208	Rated power unit line supply voltage / PU V_{rated}		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Vrms]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the rated line supply voltage of the power unit. r0208 = 400 : 380 - 480 V +/-10 % r0208 = 500 : 500 - 600 V +/-10 % r0208 = 690 : 660 - 690 V +/-10 % For the Basic Line Module (BLM) the following applies: r0208 = 690 : 500 - 690 V +/-10 %		

r0209[0...4]	Power unit, maximum current / PU I_{max}		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Arms]	Access level: 2 Func. diagram: 8750, 8850, 8950 Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the maximum output current of the power unit.		
Index:	[0] = Catalog [1] = Load duty cycle with high overload [2] = Load duty cycle with low overload [3] = S1 load duty cycle [4] = S6 load duty cycle		
Dependency:	Refer to: p0205		
p0210	Drive unit line supply voltage / Supply voltage		
A_INF, S_INF	Can be changed: C2(1) Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 100 [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000 [Vrms]	Access level: 1 Func. diagram: 8860, 8960 Unit selection: - Expert list: 1 Factory setting 400 [Vrms]
Description:	Sets the drive unit supply voltage (3-ph. AC). The value corresponds to the rms value of the phase-to-phase rated line supply voltage.		
Dependency:	Refer to: p3400		
Warning:	If the infeed is continually in the controlled mode with high DC link voltages (p3510 > 660 V), depending on the particular application, this can damage the connected motors that have not been specified for these high voltages. Before an active infeed with a line supply voltage p0210 > 415 V goes into pulsed operation it must be ensured that all of the components connected to the DC link can be permanently operated with DC link voltages exceeding 660 V.		
			
Notice:	Controlled operation of booksize power units for p0210 > 415 V is possible if the maximum steady-state DC link voltage (p0280) is increased as follows: p0280 ≥ 1.5 * p0210 and p0280 > 660 V. In this case, the setpoint of the DC link voltage p3510 is not automatically adapted. We recommend p3510 = 1.5 * p0210. Closed-loop voltage controlled operation is active with p3400.0 = 0 and p3400.3 = 1. For p0210 > 415 V for booksize power units with a supply voltage of 3-ph. 380 ... 480 V, the Smart Mode is automatically activated (p3400.0 = 1). In this case, the Smart Mode cannot be deactivated. This is because in the voltage controlled mode, the maximum steady-state DC link voltage (p0280) would be exceeded. For booksize power units with supply voltage of 3-ph. 380 ... 480 V AC, the following applies: 380 V ≤ p0210 ≤ 400 V → Pre-assignment, setpoint for the DC link voltage: p3510 = 600 V 401 V ≤ p0210 ≤ 415 V → Pre-assignment, setpoint for the DC link voltage: p3510 = 625 V 416 V ≤ p0210 ≤ 480 V → Smart Mode with non-regulated DC link voltage: p3510 = 1.35 * p0210		
Note:	When pre-assigning the setpoint for the DC link voltage (p3510), the following is generally valid: p3510 = 1.5 * p0210 The voltage range for the supply voltage depends on the type and the voltage class of the power unit. For booksize drive units, the following applies: Active Line Module, 400 V unit: 180 V ≤ p0210 ≤ 480 V Smart Line Module, 400 V unit: 180 V ≤ p0210 ≤ 480 V For chassis units, the following applies: Active Line Module, 400 V unit: 180 V ≤ p0210 ≤ 480 V Active Line Module, 690 V unit: 660 V ≤ p0210 ≤ 690 V Active Line Module, 500/690 V unit: 380 V ≤ p0210 ≤ 690 V Smart Line Module, 400 V unit: 380 V ≤ p0210 ≤ 480 V Smart Line Module, 690 V unit: 500 V ≤ p0210 ≤ 690 V		

p0210	Drive unit line supply voltage / Supply voltage		
B_INF	Can be changed: C2(1) Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 70 [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000 [Vrms]	Access level: 1 Func. diagram: 8760 Unit selection: - Expert list: 1 Factory setting 400 [Vrms]
Description:	Sets the drive unit supply voltage (3-ph. AC). The value corresponds to the rms value of the phase-to-phase rated line supply voltage.		
Dependency:	The parameter can be reduced to p0210 = 70 V if p0212 bit 0 has been set.		
Caution:	If the line supply voltage is higher than the entered value, the Vdc controller may be automatically deactivated in some cases to prevent the motor from accelerating. In this case, an appropriate alarm is output.		
Notice:	When connected to 3-ph. 230 V AC (only booksize units) the following must be observed: - the undervoltage and overvoltage limits change (r0296, r0297). - when using the internal braking chopper of Basic Line Modules (20 or 40 kW) the threshold when the braking chopper becomes active is reduced to 385 V. When using an external braking chopper, it must be ensured that a suitable activation threshold is used. - all of the components connected to this DC link must also be adapted to the low line supply voltage. It is especially important that the rated DC voltage of all of the drives connected to this DC link is set with p0210 (e.g. $p0210(\text{SERVO}) = 1.35 * p0210(\text{B_INF}) = 310 \text{ V}$). - it is not possible to use a Control Supply Module (CSM) to generate a 24 V supply from the DC link, as the minimum continuous DC link voltage should not be below 430 V.		
Note:	The supply voltage range depends on the voltage class of the power unit. 400 V chassis units: $380 \text{ V} \leq p0210 \leq 480 \text{ V}$ 690 V chassis units: $500 \text{ V} \leq p0210 \leq 690 \text{ V}$ 400 V booksize units can also be connected to 3-ph. 230 V AC: 400 V booksize units: $180 \text{ V} \leq p0210 \leq 480 \text{ V}$ A reduced supply voltage up to 70 V is possible if p0212 bit 0 = 1 has been set.		
p0210	Drive unit line supply voltage / Supply voltage		
SERVO	Can be changed: C2(2), T Data type: Unsigned16 P-Group: Converter Not for motor type: - Min 1 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 63000 [V]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 600 [V]
Description:	Sets the drive unit supply voltage. AC/AC unit: The rms value of the phase-to-phase line supply voltage should be entered. DC/AC unit: The rated DC voltage of the connection busbar should be entered.		
Dependency:	Set p1254, p1294 (automatic detection of the Vdc switch-on levels) = 0. The switch-in thresholds of the Vdc_max controller are then directly determined using p0210.		
Caution:	If the line supply voltage is higher than the entered value, the Vdc controller may be automatically deactivated in some cases to prevent the motor from accelerating. In this case, an appropriate alarm is output.		

Note: Setting ranges for p0210 as a function of the rated power unit voltage:

V_{rated} = 400 V:
 - p0210 = 380 ... 480 V (AC/AC), 510 ... 720 V (DC/AC)

V_{rated} = 500 V:
 - p0210 = 500 ... 600 V (AC/AC), 675 ... 900 V (DC/AC)

V_{rated} = 660 V ... 690 V:
 - p0210 = 660 ... 690 V (AC/AC), 890 ... 1035 V (DC/AC)

V_{rated} = 500 V ... 690 V:
 - p0210 = 500 ... 690 V (AC/AC), 675 ... 1035 V (DC/AC)

The pre-charging switch-in threshold for the DC link voltage (V_{dc}) is calculated from p0210:
 V_{dc_pre} = p0210 * 0.82 * 1.35 (AC/AC)
 V_{dc_pre} = p0210 * 0.82 (DC/AC)

The undervoltage thresholds for the DC link voltage (V_{dc}) are calculated from p0210 as a function of the rated power unit voltage:

V_{rated} = 400 V:
 - V_{min} = p0210 * 0.78 (AC/AC) > 330 V, p0210 * 0.60 (DC/AC) > 380 V

V_{rated} = 500 V:
 - V_{min} = p0210 * 0.76 (AC/AC) > 410 V

V_{rated} = 660 V ... 690 V:
 - V_{min} = p0210 * 0.82 (AC/AC) > 565 V, p0210 * 0.63 (DC/AC) > 650 V

V_{rated} = 500 V ... 690 V:
 - V_{min} = p0210 * 0.82 (AC/AC) > 420 V, p0210 * 0.63 (DC/AC) > 480 V

p0210			
Drive unit line supply voltage / Supply voltage			
VECTOR	Can be changed: C2(2), T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1 [V]	Max 63000 [V]	Factory setting 600 [V]
Description:	Sets the drive unit supply voltage. AC/AC unit: The rms value of the phase-to-phase line supply voltage should be entered. DC/AC unit: The rated DC voltage of the connection busbar should be entered.		
Dependency:	Set p1254, p1294 (automatic detection of the V _{dc} switch-on levels) = 0. The switch-in thresholds of the V _{dc_max} controller are then directly determined using p0210. The parameter can be reduced to p0210 = 100 V if p0212 bit 0 has been set. Refer to: p0212		
Caution:	If the line supply voltage is higher than the entered value, the V _{dc} controller may be automatically deactivated in some cases to prevent the motor from accelerating. In this case, an appropriate alarm is output.		

Note: Setting ranges for p0210 as a function of the rated power unit voltage:
 $V_{rated} = 400 \text{ V}$:
 - p0210 = 380 ... 480 V (AC/AC), 510 ... 720 V (DC/AC)
 $V_{rated} = 500 \text{ V}$:
 - p0210 = 500 ... 600 V (AC/AC), 675 ... 900 V (DC/AC)
 $V_{rated} = 660 \text{ V} \dots 690 \text{ V}$:
 - p0210 = 660 ... 690 V (AC/AC), 890 ... 1035 V (DC/AC)
 $V_{rated} = 500 \text{ V} \dots 690 \text{ V}$:
 - p0210 = 500 ... 690 V (AC/AC), 675 ... 1035 V (DC/AC)
 The pre-charging switch-in threshold for the DC link voltage (Vdc) is calculated from p0210:
 $V_{dc_pre} = p0210 * 0.82 * 1.35 \text{ (AC/AC)}$
 $V_{dc_pre} = p0210 * 0.82 \text{ (DC/AC)}$
 The undervoltage thresholds for the DC link voltage (Vdc) are calculated from p0210 as a function of the rated power unit voltage:
 $V_{rated} = 400 \text{ V}$:
 - $V_{min} = p0210 * 0.78 \text{ (AC/AC)} > 330 \text{ V}$, $p0210 * 0.60 \text{ (DC/AC)} > 380 \text{ V}$
 $V_{rated} = 500 \text{ V}$:
 - $V_{min} = p0210 * 0.76 \text{ (AC/AC)} > 410 \text{ V}$
 $V_{rated} = 660 \text{ V} \dots 690 \text{ V}$:
 - $V_{min} = p0210 * 0.82 \text{ (AC/AC)} > 565 \text{ V}$, $p0210 * 0.63 \text{ (DC/AC)} > 650 \text{ V}$
 $V_{rated} = 500 \text{ V} \dots 690 \text{ V}$:
 - $V_{min} = p0210 * 0.82 \text{ (AC/AC)} > 420 \text{ V}$, $p0210 * 0.63 \text{ (DC/AC)} > 480 \text{ V}$

p0211	Rated line freq / Rated line freq		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 10 [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100 [Hz]	Access level: 3 Func. diagram: 8864, 8964 Unit selection: - Expert list: 1 Factory setting 50 [Hz]
Description:	Sets the rated line frequency for the infeed.		
Dependency:	Refer to: p3409		
Notice:	For p3409 = 1, the following applies: After operation has been enabled, the rated line supply frequency (p0211) is automatically set to a value of 50 Hz or 60 Hz corresponding to the currently measured frequency. This means that the parameter value of p0211 is, under certain circumstances, changed. For p3409 = 0, the following applies: The system does not change parameter p0211.		

p0212	Power unit configuration / PU configuration				
B_INF	Can be changed: C2(2)		Calculated: -		Access level: 4
	Data type: Unsigned16		Dynamic index: -		Func. diagram: -
	P-Group: Converter		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		0000 bin
Description:	Configuration of the power unit.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Reduced device supply voltage	Yes	No	-
Dependency:	Reduced supply voltages (bit 0 = 1) are only possible on booksize power units. Bit 0 = 1 can only be set if r0192 bit 22 = 1. Refer to: r0192, p0210				

Caution:

Re bit 00:

Working with reduced input voltages deactivates undervoltage detection.

Note:

Re bit 0 = 0:

It is not possible to reduce the supply voltage in p0210.

Re bit 0 = 1:

With this setting the supply voltage in p0210 can be reduced to 70 V. Bit 0 = 1 can only be set for booksize power units with a rated power of up to 40 kW.

p0212**Power unit configuration / PU configuration**

VECTOR

Can be changed: C2(2)**Calculated:** -**Access level:** 4**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** -**P-Group:** Converter**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

0000 bin

Description:

Configuration of the power unit.

Bit field:**Bit** **Signal name****1 signal****0 signal****FP**

00 Reduced device supply voltage

Yes

No

-

Dependency:

Reduced supply voltages (bit 0 = 1) are only possible on booksize power units.

Bit 0 = 1 can only be set if r0192 bit 22 = 1.

Refer to: r0192, p0210

Caution:

Re bit 00:

Working with reduced input voltages deactivates undervoltage detection.

Note:

Re bit 00 = 0:

It is not possible to reduce the supply voltage in p0210.

Re bit 00 = 1:

With this setting the supply voltage in p0210 can be reduced to 100 V.

Only operating mode p1300 = 19 is possible.

p0220[0...1]**Infeed line filter type / INF line filt type**

A_INF

Can be changed: C2(1)**Calculated:** -**Access level:** 3**Data type:** Integer16**Dynamic index:** -**Func. diagram:** 8950**P-Group:** Converter**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0

45

0

Description:

Sets the line filter type for the Active Line Module (ALM).

Using the line filter type, filter capacitance (p0221), filter resistance (p0222) and inductance (p0223) and resistance (p0224) of the reactor are pre-assigned.

For an Active Line Module (ALM), the power is automatically selected corresponding to the Active Interface Module (AIM) and the line filter type (p0220) preset as follows:

- "booksize" format: p0220 = 41 ... 45

- "chassis" format: p0220 = 10 ... 19

Value:	0: No line filter 1: Wideband Line Filter booksize 400 V 16 kW (6SL3000-0BE21-6AA0) 2: Wideband Line Filter booksize 400 V 36 kW (6SL3000-0BE23-6AA0) 3: Wideband Line Filter booksize 400 V 55 kW (6SL3000-0BE25-5AA0) 4: Wideband Line Filter booksize 400 V 80 kW (6SL3000-0BE28-0AA0) 5: Wideband Line Filter booksize 400 V 120 kW (6SL3000-0BE31-2AA0) 10: AIM F 400 V 132 kW 160 kW (6SL3300-7TE32-6Ax0) 11: AIM G 400 V 235 kW (6SL3300-7TE33-8Ax0) 12: AIM G 400 V 300 kW (6SL3300-7TE35-0Ax0) 13: AIM H 400 V 380 kW 500 kW (6SL3300-7TE38-4Ax0) 14: AIM J 400 V 630 kW 900 kW (6SL3300-7TE41-4Ax0) 15: AIM F 690 V 150 kW (6SL3300-7Tx31-4Ax0) 16: AIM G 690 V 330 kW (6SL3300-7Tx33-1Ax0) 17: AIM H 690 V 560 kW (6SL3300-7Tx35-8Ax0) 18: AIM J 690 V 800 kW (6SL3300-7Tx37-4Ax0) 19: AIM J 690 V 1100 kW 1400 kW (6SL3300-7Tx41-3Ax0) 31: Basic Line Filter booksize 400 V 16 kW (6SL3000-0BE21-6DA0) 32: Basic Line Filter booksize 400 V 36 kW (6SL3000-0BE23-6DA0) 33: Basic Line Filter booksize 400 V 55 kW (6SL3000-0BE25-5DA0) 34: Basic Line Filter Booksize 400 V 80 kW (6SL3000-0BE28-0DAx) 35: Basic Line Filter Booksize 400 V 120 kW (6SL3000-0BE31-2DAx) 41: AIM 400 V 16 kW (6SL3100-0BE21-6AB0) 42: AIM 400 V 36 kW (6SL3100-0BE23-6AB0) 43: AIM 400 V 55 kW (6SL3100-0BE25-5AB0) 44: AIM 400 V 80 kW (6SL3100-0BE28-0AB0) 45: AIM 400 V 120 kW (6SL3100-0BE31-2AB0)
Index:	[0] = Line filter [1] = Line filter, optional
Notice:	"Booksize" format: When using an Active Interface Module (AIM), it is absolutely necessary that the terminals for the temperature switch between the Active Interface Module (X121.1/2) and the Active Line Module (X21.1/2) are connected.
Note:	For booksize units, when using an Active Interface Module in p0220[0] it is also possible to use a Basic Filter that is parameterized in p0220[1]. The setting of the filter capacitance (p0221) and filter resistance (p0222) - derived from p0220[0, 1] - are required in the closed-loop voltage controlled mode to automatically compensate the filter reactive current. For two power ratings, the same line filter is used for both power ratings. AIM: Active Interface Module

p0221[0...1]	Infeed filter capacitance / INF C_filter		
A_INF	Can be changed: C2(1) Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 0.00 [μF]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000.00 [μF]	Access level: 3 Func. diagram: 8950 Unit selection: - Expert list: 1 Factory setting 0.00 [μF]
Description:	Sets the filter capacitance of the line filter (connected in a delta configuration).		
Index:	[0] = Line filter [1] = Line filter, optional		
Note:	When a Siemens line filter is used (p0220) this parameter is automatically preset with the correct value. For a parallel circuit, the value corresponds to the capacitance of a power unit. Index 0 refers to the first line filter from p0220[0]. Index 1 refers to the optional second line filter from p0220[1].		

p0222[0...1] Infeed filter resistance / INF R_filter

A_INF	Can be changed: C2(1)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00000 [Ohm]	Max 100.00000 [Ohm]	Factory setting 0.00000 [Ohm]
Description:	Sets the filter resistance in series with the filter capacitance.		
Index:	[0] = Line filter [1] = Line filter, optional		
Note:	When a Siemens line filter is used (p0220) this parameter is automatically preset with the correct value. For a parallel circuit, the value corresponds to the resistance of a power unit. Index 0 refers to the first line filter from p0220[0]. Index 1 refers to the optional second line filter from p0220[1].		

p0223 Infeed inductance between filter and power unit / INF L filter/PU

A_INF, S_INF	Can be changed: C2(1)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8850, 8950
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.001 [mH]	Max 1000.000 [mH]	Factory setting 2.100 [mH]
Description:	Sets the inductance between the filter and power unit.		
Note:	The parameter is automatically pre-assigned depending on the power unit being used and matches the specified Siemens line reactors. For a parallel circuit, the value corresponds to the inductance of a power unit.		

p0224 Infeed resistance between filter and power unit / INF R filter/PU

A_INF, S_INF	Can be changed: C2(1)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8850, 8950
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00000 [Ohm]	Max 100.00000 [Ohm]	Factory setting 0.00100 [Ohm]
Description:	Sets the resistance between the filter and power unit		
Note:	The parameter is automatically pre-assigned depending on the power unit being used and matches the specified Siemens line reactors. For a parallel circuit, the value corresponds to the resistance of a power unit.		

p0225 Infeed inductance between line supply and filter / INF L line/filter

A_INF, S_INF	Can be changed: C2(1)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8850, 8950
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.001 [mH]	Max 1000.000 [mH]	Factory setting 0.001 [mH]
Description:	Sets the inductance between line supply and filter.		
Note:	The value must be, for example, appropriately increased if an additional inductance (reactor or transformer is installed in front of the filter).		

p0226	Infeed resistance between line supply and filter / INF R line/filter		
A_INF, S_INF	Can be changed: C2(1) Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 0.00 [Ohm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [Ohm]	Access level: 3 Func. diagram: 8850, 8950 Unit selection: - Expert list: 1 Factory setting 0.00 [Ohm]
Description:	Sets the resistance between the line supply and filter.		
Note:	The value must be, for example, appropriately increased if an additional resistor is installed in front of the filter.		
p0227	Infeed, DC link capacitance, power unit / INF C		
A_INF, S_INF	Can be changed: C2(1) Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 0.20 [mF]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [mF]	Access level: 3 Func. diagram: 8850, 8950 Unit selection: - Expert list: 1 Factory setting 1.00 [mF]
Description:	Sets the total DC link capacitance.		
Note:	The total DC link capacitance of a DC link group comprises the sum of the sub-capacitances of all motor/infeed modules and the additional DC link capacitors.		
p0230	Drive filter type, motor side / Drv filt type mot		
VECTOR	Can be changed: C2(1, 2) Data type: Integer16 P-Group: Converter Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the type of the filter at the motor side.		
Value:	0: No filter 1: Motor reactor 2: dv/dt filter 3: Sine-wave filter, Siemens 4: Sine-wave filter, third-party		
Dependency:	The following parameters are influenced using p0230: p0230 = 1: --> p0233 (power unit, motor reactor) = filter inductance p0230 = 3: --> p0233 (power unit, motor reactor) = filter inductance --> p0234 (power unit sine-wave filter capacitance) = filter capacitance --> p0290 (power unit overload response) = inhibit pulse frequency reduction --> p1082 (maximum speed) = Fmax filter / pole pair number --> p1800 (pulse frequency) >= nominal pulse frequency of the filter --> p1802 (modulator modes) = space vector modulation without overcontrol --> p1811 (modulator configuration) = wobulation amplitude --> p1909 (motor data identification, control word) = only Rs measurement p0230 = 4: --> p0290 (power unit overload response) = inhibit pulse frequency reduction --> p1802 (modulator modes) = space vector modulation without overcontrol --> p1811 (modulator configuration) = wobulation amplitude --> p1909 (motor data identification, control word) = only Rs measurement		

The user must set the following parameters according to the data sheet of the sine-wave filter and also the user must check whether they are permitted.

--> p0233 (power unit, motor reactor) = filter inductance

--> p0234 (power unit sine-wave filter capacitance) = filter capacitance

--> p1082 (maximum speed) = Fmax filter / pole pair number

--> p1800 (pulse frequency) >= nominal pulse frequency of the filter

Refer to: p0233, p0234, p0290, p1082, p1800, p1802

Note: if a filter type cannot be selected, then this filter type is not permitted for the Motor Module.

p0230 = 1:

The output frequency of booksize/blocksize power units with output reactors is restricted to 120 Hz.

p0230 = 2:

Chassis-type power units with dv/dt filter, depending on the rated pulse frequency, may only be operated with a maximum pulse frequency of p1800 = 2.5 kHz or 4 kHz. The output frequency is limited to 150 Hz.

p0230 = 3:

Sine-wave filters with a rated pulse frequency of 1.25 or 2.5 kHz should only be operated with a current controller sampling rate p0115[0] = 400 µs, sine-wave filters with a rated pulse frequency of 2 or 4 kHz with p0115[0] = 250 µs.

The sine-wave filter cannot be selected if the current controller sampling rate has not been appropriately set.

Chassis power units with sine-wave filter are limited to output frequencies of 115 Hz or 150 Hz.

p0233		Power unit motor reactor / PU mot reactor		
SERVO	Can be changed:	C2(2), U, T	Calculated: -	Access level: 4
	Data type:	FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group:	Converter	Units group: -	Unit selection: -
	Not for motor type:	-	Scaling: -	Expert list: 1
	Min		Max	Factory setting
	0.000 [mH]		1000.000 [mH]	0.000 [mH]
Description:	Enter the inductance of a filter connected at the power unit output.			
Dependency:	This parameter is automatically preset when you select a filter via p0230 if a SIEMENS filter is defined for the power unit.			
	Refer to: p0230			
Note:	The parameter cannot be changed if the power unit has an internal sine-wave filter.			

p0233		Power unit motor reactor / PU mot reactor		
VECTOR	Can be changed:	C2(2), U, T	Calculated: -	Access level: 2
	Data type:	FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group:	Converter	Units group: -	Unit selection: -
	Not for motor type:	-	Scaling: -	Expert list: 1
	Min		Max	Factory setting
	0.000 [mH]		1000.000 [mH]	0.000 [mH]
Description:	Enter the inductance of a filter connected at the power unit output.			
Dependency:	This parameter is automatically preset when you select a filter via p0230 if a SIEMENS filter is defined for the power unit.			
	Refer to: p0230			
Note:	When exiting the quick commissioning using p3900 = 1, the parameter value is set to the value of the defined SIEMENS filter or to zero. For this reason, the parameter value of a third-party filter only has to be entered outside the commissioning phase (p0010 = 0) and then the controller calculation (p0340 = 3) is carried out.			

p0234	Power unit sine-wave filter capacitance / PU sine filter C		
SERVO	Can be changed: C2(2), U, T Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 0.000 [μF]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.000 [μF]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [μF]
Description:	Enters the capacitance of a sine-wave filter connected at the power unit output.		
Dependency:	This parameter is automatically preset when you select a filter via p0230 if a SIEMENS filter is defined for the power unit. Refer to: p0230		
Note:	The parameter value includes the sum of all of the capacitances of a phase connected in series (phase - ground). The parameter cannot be changed if the power unit has an internal sine-wave filter.		
p0234	Power unit sine-wave filter capacitance / PU sine filter C		
VECTOR	Can be changed: C2(2), U, T Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 0.000 [μF]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.000 [μF]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [μF]
Description:	Enters the capacitance of a sine-wave filter connected at the power unit output.		
Dependency:	This parameter is automatically preset when you select a filter via p0230 if a SIEMENS filter is defined for the power unit. Refer to: p0230		
Note:	The parameter value includes the sum of all of the capacitances of a phase connected in series (phase - ground). When exiting the quick commissioning using p3900 = 1, the parameter value is set to the value of the defined SIEMENS filter or to zero. For this reason, the parameter value of a third-party filter only has to be entered outside the commissioning phase (p0010 = 0).		
p0235	Number of reactors connected in series / Qty L in series		
VECTOR	Can be changed: C2(1, 2) Data type: Unsigned8 P-Group: Converter Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Number of reactors connected at the power unit output.		
Dependency:	Refer to: p0230		
Caution:	If the number of motor reactors connected in series does not correspond to the parameter value, then this can result in an unfavorable control behavior.		
Note:	The parameter cannot be changed for chassis units and for p0230 = 1.		

r0238	Power unit internal resistance / PU R_int		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min - [Ohm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Ohm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Ohm]
Description:	Displays the internal resistance of the power unit (IGBT and line resistance).		
Note:	For a parallel circuit, the value corresponds to the resistance of a power unit. Index 0 refers to the first line filter from p0220[0]. Index 1 refers to the optional second line filter from p0220[1].		
p0249	Power unit cooling type / PU cool type		
S_INF, SERVO, VECTOR	Can be changed: C2(1, 2) Data type: Integer16 P-Group: Converter Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the cooling type for booksize compact power units. This therefore defines whether for these power units, the internal air cooling is shut down and instead, the "Cold-Plate" cooling type is used.		
Value:	0: Air cooling int 1: Cold-Plate		
Note:	For booksize compact power units, there is a 4 at the 5th position in the Order No. The parameter is irrelevant for all other power unit types.		
p0251[0...n]	Operating hours counter power unit fan / PU fan t_oper		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T Data type: Unsigned32 P-Group: Modulation Not for motor type: - Min 0 [h]	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 4294967295 [h]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [h]
Description:	Displays the power unit fan operating hours. The number of hours operated can only be reset to 0 in this parameter (e.g. after a fan has been replaced).		
Dependency:	Refer to: p0252		
p0252	Maximum operating time power unit fan / PU fan t_oper max		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T Data type: Unsigned32 P-Group: Modulation Not for motor type: - Min 0 [h]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000 [h]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 40000 [h]
Description:	Sets the maximum operating time of the power unit fan. The pre-alarm (warning) is output 500 hours before this set value. The monitoring is deactivated with p0252 = 0.		
Dependency:	Refer to: p0251		
Note:	For chassis units, the maximum operating time in the power unit parameter is set to 50000 via the factory setting.		

p0254[0...n]	Power unit internal fan operating hours counter / PU int fan t_oper		
A_INF, S_INF	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0 [h]	Max 4294967295 [h]	Factory setting 0 [h]
Description:	Displays the power unit internal fan operating hours. The number of hours operated can only be reset to 0 in this parameter (e.g. after a fan has been replaced).		
Dependency:	Refer to: p0252		

p0255[0...1]	Power unit contactor monitoring time / PU contactor t_mon		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 6500 [ms]	Factory setting 0 [ms]
Description:	Sets the monitoring time for internal monitoring of the contactor feedback contacts.		
Index:	[0] = Pre-charge contactor [1] = Bridging contactor		
Dependency:	Refer to: F30060, F30061		
Note:	This parameter is only effective for chassis power units with 3 AC line connection and line contactors. A value of 0 deactivates the associated line contactor monitoring.		

p0260	Cooling system, starting time 1 / RKA start time 1		
A_INF (Cooling system), B_INF (Cooling system), S_INF (Cooling system), SERVO (Cooling system), VECTOR (Cooling system)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9795
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [s]	Max 60.0 [s]	Factory setting 5.0 [s]
Description:	Sets starting time 1 to monitor the cooling system after power-on command. After powering up, the following signals must be present within starting time 1: - "RKA powered up" - "RKA liquid flow OK" When a fault occurs, an appropriate message is output.		
Dependency:	Refer to: F49152, F49153		
Note:	RKA: Cooling system		

p0261 Cooling system, starting time 2 / RKA start time 2

A_INF (Cooling system), B_INF (Cooling system), S_INF (Cooling system), SERVO (Cooling system), VECTOR (Cooling system)

Can be changed: U, T

Data type: FloatingPoint32

P-Group: Converter

Not for motor type: -

Calculated: -

Dynamic index: -

Units group: -

Scaling: -

Access level: 3

Func. diagram: 9795

Unit selection: -

Expert list: 1

Min

0.0 [s]

Max

1200.0 [s]

Factory setting

180.0 [s]

Description:

Sets starting time 2 to monitor the cooling system after power-on command.

After powering up, the following signals must be present within starting time 2:

- "RKA conductivity, no fault"

- "RKA conductivity, no alarm"

When a fault occurs, an appropriate message is output.

Dependency:

Refer to: p0266

Refer to: F49151, A49171

p0262 Cooling system, fault conductivity delay time / RKA cond t_del

A_INF (Cooling system), B_INF (Cooling system), S_INF (Cooling system), SERVO (Cooling system), VECTOR (Cooling system)

Can be changed: U, T

Data type: FloatingPoint32

P-Group: Converter

Not for motor type: -

Calculated: -

Dynamic index: -

Units group: -

Scaling: -

Access level: 3

Func. diagram: 9795

Unit selection: -

Expert list: 1

Min

0.0 [s]

Max

30.0 [s]

Factory setting

0.0 [s]

Description:

Sets the delay time for the fault "RKA: Conductive limit value exceeded" during operation.

The fault is only output if, during operation, the conductivity exceeds the permissible fault value and the value remains for a longer time than is set in this parameter.

Dependency:

Refer to: F49151

p0263 Cooling system fault liquid flow, delay time / RKA flow t_del

A_INF (Cooling system), B_INF (Cooling system), S_INF (Cooling system), SERVO (Cooling system), VECTOR (Cooling system)

Can be changed: U, T

Data type: FloatingPoint32

P-Group: Converter

Not for motor type: -

Calculated: -

Dynamic index: -

Units group: -

Scaling: -

Access level: 3

Func. diagram: 9795

Unit selection: -

Expert list: 1

Min

0.0 [s]

Max

20.0 [s]

Factory setting

3.0 [s]

Description:

Sets the delay time for the fault "RKA: Liquid flow too low".

The fault is only output if the cause is present for a time longer than is set in this parameter.

Dependency:

Refer to: F49153

p0264					Cooling system, run-on time / RKA run-on time						
A_INF (Cooling system), B_INF (Cooling system), S_INF (Cooling system), SERVO (Cooling system), VECTOR (Cooling system)					Can be changed: U, T		Calculated: -		Access level: 3		
					Data type: FloatingPoint32		Dynamic index: -		Func. diagram: 9795		
					P-Group: Converter		Units group: -		Unit selection: -		
					Not for motor type: -		Scaling: -		Expert list: 1		
					Min		Max		Factory setting		
					0.0 [s]		180.0 [s]		30.0 [s]		
Description:					Sets the run-up time of the cooling system after a power-off command.						
r0265.0...3					BO: Cooling system, control word / RKA CTW						
A_INF (Cooling system), B_INF (Cooling system), S_INF (Cooling system), SERVO (Cooling system), VECTOR (Cooling system)					Can be changed: -		Calculated: -		Access level: 3		
					Data type: Unsigned8		Dynamic index: -		Func. diagram: -		
					P-Group: Commands		Units group: -		Unit selection: -		
					Not for motor type: -		Scaling: -		Expert list: 1		
					Min		Max		Factory setting		
					-		-		-		
Description:					Displays the control word for the cooling system.						
Bit field:					Bit	Signal name	1 signal		0 signal	FP	
					00	Power up cooling system	Activating		Deactivating	-	
					01	Message converter off	Off		On	-	
					02	Acknowledge faults	Acknowledgement		No acknowledgement	-	
					03	Leakage sensing OK	No leaked liquid		Leaked liquid	-	
p0266[0...7]					BI: Cooling system, feedback signals, signal source / RKA fdbk S_src						
A_INF (Cooling system), B_INF (Cooling system), S_INF (Cooling system), SERVO (Cooling system), VECTOR (Cooling system)					Can be changed: U, T		Calculated: -		Access level: 3		
					Data type: Unsigned32 / Binary		Dynamic index: -		Func. diagram: -		
					P-Group: Communications		Units group: -		Unit selection: -		
					Not for motor type: -		Scaling: -		Expert list: 1		
					Min		Max		Factory setting		
					-		-		1		
Description:					Sets the signal sources for the feedback signals from the cooling system.						
Index:					[0] = Cooling system powered up [1] = Cooling system ready to be powered up [2] = Cooling system, no alarm present [3] = Cooling system, no fault present [4] = Cooling system, no leaked liquid [5] = Cooling system, liquid flow OK [6] = Cooling system, conductivity < fault threshold [7] = Cooling system, conductivity < alarm threshold						

r0267.0...7 BO: Cooling system status word / RKA ZSW

A_INF (Cooling system), B_INF (Cooling system), S_INF (Cooling system), SERVO (Cooling system), VECTOR (Cooling system)

Can be changed: -
Data type: Unsigned16
P-Group: Commands
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: -
Unit selection: -
Expert list: 1

Min	Max	Factory setting
-	-	-

Description: Displays the status word of the cooling system.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	RKA powered up	Yes	No	-
	01	RKA ready to be powered up	Yes	No	-
	02	RKA no alarm present	Yes	No	-
	03	RKA no fault present	Yes	No	-
	04	RKA no leaked fluid	Yes	No	-
	05	RKA liquid flow OK	Yes	No	-
	06	RKA conductivity, no fault	Yes	No	9974
	07	RKA conductivity, no alarm	Yes	No	9974

Dependency: Refer to: p0266

p0278 DC link voltage undervoltage threshold reduction / Vdc V_under red

SERVO, VECTOR

Can be changed: T
Data type: FloatingPoint32
P-Group: Converter
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: -
Unit selection: -
Expert list: 1

Min	Max	Factory setting
-80 [V]	0 [V]	0 [V]

Description: Sets the absolute value by which the threshold to initiate the undervoltage fault (F30003) is reduced.

Dependency: Refer to: p0210, r0296
Refer to: F30003

Notice: When using a Control Supply Module (CSM) for 24 V supply from the DC link, the minimum continuous DC link voltage may not lie below 430 V. DC link voltages in the range 300 ... 430 V are permissible up to a duration of 1 min.

Note: The resulting shutdown threshold can be read in r0296 and is dependent on the selected rated voltage (p0210) and the power unit being used.

p0279 DC link voltage offset alarm threshold / Vdc offs A thresh

A_INF, B_INF, S_INF

Can be changed: T
Data type: FloatingPoint32
P-Group: Converter
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: 8760, 8864, 8964
Unit selection: -
Expert list: 1

Min	Max	Factory setting
0 [V]	500 [V]	0 [V]

Description: Sets the voltage threshold to initiate alarm A06810.

The value represents an offset so that the alarm threshold is obtained from the total of r0296 and p0279.

Dependency: Refer to: p0210, r0296
Refer to: A06810

Note: The absolute value of the undervoltage threshold r0296 depends on the selected unit supply voltage (p0210).

p0280			
DC link voltage maximum steady-state / Vdc_max stat			
A_INF	Can be changed: C2(1), T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8940, 8964
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	270 [V]	1500 [V]	660 [V]
Description:	Sets the maximum steady-state DC link voltage. When the DC link voltage setpoint reaches the threshold, alarm A06800 is output. The setpoint for the DC link voltage in p3510 is limited to the value in p0280. The voltage can be increased (boosted) using the modulation depth reserve controller. The modulation depth reserve (p3480) can be too low if p0210 (drive unit supply voltage) was incorrectly parameterized, a line overvoltage condition is present or a high reactive current is required.		
Dependency:	Refer to: p0210 Refer to: A06800		
Warning:	Before increasing the voltage limit for pulsed operation of a controlled booksize infeed with line supply voltages p0210 > 415 V it should be checked whether the motors connected to the DC link are specified for the higher motor voltages. The warning information associated with p0210 must be carefully observed.		
			
Caution:	All motors connected to the DC link must be rated for the maximum DC link voltage set in this parameter.		
			
Notice:	For chassis power units, for the extended line supply voltage range from 500 V to 690 V, the value in p0280 is automatically adapted if the line supply voltage in p0210 is changed. The individual parameter setting for p0280 is then lost and if necessary must be re-entered.		
Note:	A brief dynamic increase of the DC link voltage does not result in an alarm. Presetting values: 380 ... 480 V booksize units: 660 V 380 ... 480 V chassis units: 750 V 500 ... 690 V chassis units: 0.875 * p0210 + 502 V Maximum values: 380 ... 480 V booksize units: 785 V 380 ... 480 V chassis units: 785 V 500 ... 690 V chassis units: 1130 V		

p0281			
Line supply overvoltage, alarm threshold / V_I_over A thresh			
A_INF, S_INF	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8860, 8960
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	100 [%]	200 [%]	110 [%]
Description:	Sets the alarm threshold for a line supply overvoltage condition. The setting is made as a percentage of the drive unit supply voltage (p0210).		
Dependency:	Refer to: p0211, p0221, p0222, p0223, p0224, p0225, p0226		
Note:	If synchronizing voltages are not detected, the line supply voltage is estimated using a model. It is therefore important to ensure that drive unit data is correctly specified.		

p0282	Line supply undervoltage, alarm threshold / V_I_under A thresh		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 10 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100 [%]	Access level: 3 Func. diagram: 8860, 8960 Unit selection: - Expert list: 1 Factory setting 85 [%]
Description:	Sets the alarm threshold for a line undervoltage condition. The setting is made as a percentage of the drive unit supply voltage (p0210).		
Dependency:	Refer to: p0222, p0224, p0225, p0226, p3421, p3422 Refer to: A06105		
Note:	If synchronizing voltages are not detected, the line supply voltage is estimated using a model. It is therefore important to ensure that drive unit data is correctly specified.		
p0283	Line supply undervoltage, shutdown (trip) threshold / V_I_under tr_thrs		
A_INF, S_INF	Can be changed: C2(1), T Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 10 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100 [%]	Access level: 3 Func. diagram: 8860, 8960 Unit selection: - Expert list: 1 Factory setting 75 [%]
Description:	Sets the shutdown threshold for the line supply undervoltage. The setting is made as a percentage of the drive unit supply voltage (p0210).		
Dependency:	Refer to: p0282 Refer to: F06100		
Notice:	For booksize Active Line Modules, the following applies: When operated without Active Interface Module (p0220 = 41 ... 45), the minimum shutdown threshold is 75 %.		
p0284	Line supply frequency exceeded, alarm threshold / f_I_exc A thresh		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 100.0 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 300.0 [%]	Access level: 3 Func. diagram: 8864, 8964 Unit selection: - Expert list: 1 Factory setting 110.0 [%]
Description:	Sets the alarm threshold for an excessively high line frequency.		
Dependency:	Set as a percentage of the rated line frequency. Refer to: p0211		
p0285	Line supply frequency undershot, alarm threshold / f_I_under A thresh		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 0.0 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.0 [%]	Access level: 3 Func. diagram: 8864, 8964 Unit selection: - Expert list: 1 Factory setting 90.0 [%]
Description:	Sets the alarm threshold for an excessively low line frequency.		
Dependency:	Set as a percentage of the rated line frequency. Refer to: p0211		

p0287[0...1]	Ground fault monitoring thresholds / Gnd flt threshold		
A_INF, S_INF, SERVO, VECTOR	Can be changed: T Data type: FloatingPoint32 P-Group: - Not for motor type: - Min 0.0 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 6.0 [%] [1] 16.0 [%]
Description:	Sets the shutdown thresholds for the ground fault monitoring. The setting is made as a percentage of the maximum power unit current (r0209).		
Index:	[0] = Threshold for pulse inhibit [1] = Threshold for pulse enable		
Dependency:	Refer to: F30021		
Note:	The parameter only applies to booksize and chassis power units. Deactivating the ground fault monitoring: - Sequence: --> p0287[1] = 0 --> p0287[0] = 0 - irrespective of the firmware version of the power unit. Sets the thresholds: - the prerequisite is at least firmware version 2.2 of the power unit.		
r0289	CO: Maximum power unit output current / PU I_outp max		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the actual maximum output current of the power unit taking into account derating factors.		
p0290	Power unit overload response / PU overld response		
SERVO, VECTOR	Can be changed: T Data type: Integer16 P-Group: Converter Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 3 Func. diagram: 8014 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the response to a thermal overload condition of the power unit. The following quantities can result in a response to thermal overload: - heat sink temperature (r0037.0) - chip temperature (r0037.1) - power unit overload I2T (r0036) Possible measures to avoid thermal overload: - reduce the output current limit r0289 and r0067 (for closed-loop speed/velocity or torque/force control) or the output frequency (for V/f control) indirectly via the output current limit and the intervention of the current limiting controller). - reduce the pulse frequency (only for vector control). A reduction, if parameterized, is always realized after an appropriate alarm is output.		
Value:	0: Reduce output current or output frequency 1: No reduction, shutdown when overload threshold is reached 2: Reduce I_output or f_output and f_pulse (not using I2t) 3: Reduce the pulse frequency (not using I2t)		

- Dependency:** If a sine-wave filter is parameterized as output filter (p0230 = 3, 4), then only responses can be selected without pulse frequency reduction (p0290 = 0, 1).
If a fault or alarm is present, then r2135.13 or r2135.15 is set.
Refer to: r0036, r0037, p0108, r0108, p0230, r2135
Refer to: A05000, A05001, A07805
- Caution:** If the thermal overload of the power unit is not sufficiently reduced by the actions taken, the drive is always shut down. This means that the power unit is always protected irrespective of the setting of this parameter.
- Note:** The setting p0290 = 0, 2 is only practical if the load decreases with decreasing speed (e.g. for applications with variable torque such as for pumps and fans).
Under overload conditions, the current and torque limit are reduced, and therefore the motor is braked and forbidden speed ranges (e.g. minimum speed p1080 and suppression [skip] speeds p1091 ... p1094) can be passed through.
For p0290 = 2, 3, the I2t overload detection of the power unit does not influence the responses.
When the motor data identification routine is selected, p290 cannot be changed.

r0293	CO: Power unit alarm threshold model temperature / PU Tmodel_A_thresh		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Converter	Units group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min - [°C]	Max - [°C]	Factory setting - [°C]
Description:	Temperature alarm threshold for the difference from the chip and heat sink temperature in the thermal model.		
Dependency:	Refer to: r0037 Refer to: F30024		
Note:	The parameter is only relevant for chassis power units.		

p0294	Power unit alarm with I2t overload / PU I2t alarm thresh		
A_INF, S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8014
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 10.0 [%]	Max 100.0 [%]	Factory setting 95.0 [%]
Description:	Sets the alarm threshold for the I2t power unit overload. Drive: If this threshold is exceeded, an overload alarm is generated and the system responds as parameterized in p0290. Infeed: When the threshold value is exceeded, only an overload alarm is output.		
Dependency:	Refer to: r0036, p0290 Refer to: A07805		
Note:	The I2t fault threshold is 100 %. If this value is exceeded, fault F30005 is output.		

p0294	Power unit alarm with I2t overload / PU I2t alm thresh		
B_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 10.0 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.0 [%]	Access level: 3 Func. diagram: 8014 Unit selection: - Expert list: 1 Factory setting 95.0 [%]
Description:	Sets the alarm threshold for the I2t power unit overload.		
Dependency:	Refer to: r0036 Refer to: A07805		
Note:	The parameter is only relevant for booksize units!		
p0295	Fan run-on time / Fan run-on time		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 0 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 600 [s]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [s]
Description:	Sets the fan run-on time after the pulses for the power unit have been canceled.		
Note:	Under certain circumstances, the fan can continue to run for longer than was set (e.g. as a result of the excessively high heat sink temperature). For values less than 1 s, a 1 s run on time for the fan is effective.		
r0296	DC link voltage undervoltage threshold / Vdc V_lower_thresh		
A_INF, B_INF, S_INF	Can be changed: - Data type: Unsigned16 P-Group: Converter Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [V]	Access level: 2 Func. diagram: 8750, 8760, 8850, 8864, 8950, 8964 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	If the DC link voltage falls below the threshold specified here, the infeed is tripped due to a DC link undervoltage condition.		
Dependency:	Refer to: F30003		
r0296	DC link voltage undervoltage threshold / Vdc V_lower_thresh		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Converter Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [V]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [V]
Description:	If the DC link voltage falls below this threshold, the Motor Module is shut down due to a DC link undervoltage condition (F30003).		
Dependency:	Refer to: p0278 Refer to: F30003		
Note:	For booksize units, the following applies: The undervoltage threshold can be reduced with p0278.		

r0297	DC link voltage overvoltage threshold / Vdc V_upper_thresh		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Converter Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [V]	Access level: 2 Func. diagram: 8750, 8760, 8850, 8864, 8950, 8964 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	If the DC link voltage exceeds the threshold specified here, the drive unit is tripped due to DC link overvoltage.		
Dependency:	Refer to: F30002		
p0300[0...n]	Motor type selection / Mot type sel		
SERVO	Can be changed: C2(1, 3) Data type: Integer16 P-Group: Motor Not for motor type: - Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10001	Access level: 1 Func. diagram: 6310 Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the motor type or start to read in the motor parameters for a motor with DRIVE-CLiQ (p0300 = 10000). The following applies for p0300 < 10000: The first digit of the parameter value always defines the general motor type and corresponds to the third-party motor belonging to a motor list: 1 = Rotating induction motor 2 = Rotating synchronous motor 3 = Linear induction motor (reserved) 4 = Linear synchronous motor The type information must be entered to filter motor-specific parameters and to optimize the operating characteristics and behavior. For example, for synchronous motors, power factor (p0308) is neither used nor displayed (in the BOP/AOP).		
Value:	0: No motor 1: Induction motor (rotating) 2: Synchronous motor (rotating, permanent-magnet) 4: Synchronous motor (linear, permanent-magnet) 102: 1PH2 induction motor 104: 1PH4 induction motor 107: 1PH7 induction motor 108: 1PH8 induction motor 134: 1PM4 induction motor 136: 1PM6 induction motor 166: 1PL6 induction motor 200: 1PH8 synchronous motor 206: 1FT6 synchronous motor 207: 1FT7 synchronous motor 236: 1FK6 synchronous motor 237: 1FK7 synchronous motor 261: 1FE1 synchronous motor 276: 1FS6 synchronous motor 283: 1FW3 synchronous motor 286: 1FW6 synchronous motor 291: 2SP1 synchronous motor 401: 1FN1 synchronous motor (linear) 403: 1FN3 synchronous motor (linear) 406: 1FN6 synchronous motor (linear) 10000: Motor with DRIVE-CLiQ 10001: Motor with DRIVE-CLiQ 2nd data set		

- Dependency:** When the motor type is changed, the code number in p0301 may be reset to 0.
If p0300 is changed during quick commissioning (p0010 = 1), then the matching technological application (p0500) is automatically pre-assigned. This does not occur when commissioning the motor (p0010 = 3). If 10000 is written to p0300 for a parameter download, then p0500 is pre-assigned with DRIVE-CLiQ corresponding to the motor type.
Refer to: p0301
- Caution:** If a catalog motor is selected (p0300 ≥ 100) and an associated motor code number (p0301), then the parameters, that are associated with this list cannot be changed (write protection). The write protection is canceled if the motor type p0300 is set to a non-Siemens motor that matches p0301 (e.g. p0300 = 2 for p0301 = 2xxxx). Write protection is automatically canceled when the results of motor data identification are copied to the motor parameters.
- Notice:** If required, the list of motor codes/encoder codes can be found in the Appendix of the List Manual.
- Note:** With p0300 = 10000, for a motor with DRIVE-CLiQ, the motor parameters are automatically downloaded, with p0300 = 10001, the motor parameters of a second data set (if available).
If a motor type has not been selected (p0300 = 0), then the drive commissioning routine cannot be exited.
A motor type with a value above p0300 ≥ 100 describes motors for which a motor parameter list exists.
Motor types with a value below p0300 < 100 correspond to the selection of a third-party motor. When appropriately selected, this means that the motor parameters are pre-assigned the settings for a third-party motor.
This also applies for parameters for a motor with DRIVE-CLiQ. In this case p0300 can only be set to p0300 = 10000 or 10001 (read motor parameters) or to the corresponding non-Siemens motor (first digit of the motor code number) in order to be able to cancel the write protection.

p0300[0...n]		Motor type selection / Mot type sel	
VECTOR	Can be changed: C2(1, 3)	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: 6310
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	10001	0
Description:	Selects the motor type or starts to read in the motor parameters for a motor with DRIVE-CLiQ (p0300 = 10000 or 10001, if there is a second data set).		
	The following applies for p0300 < 10000: The first digit of the parameter value always defines the general motor type and corresponds to the third-party motor belonging to a motor list:		
	1 = Rotating induction motor		
	2 = Rotating synchronous motor		
	3 = Linear induction motor (reserved)		
	4 = Linear synchronous motor		
	5 = Synchronous motor separately-excited		
	7 = SIEMOSYN motor		
	8 = Reluctance motor		
	The type information must be entered to filter motor-specific parameters and to optimize the operating characteristics and behavior. For example, for synchronous motors, power factor (p0308) is neither used nor displayed (in the BOP/AOP).		

Value:	0: No motor 1: Induction motor (rotating) 2: Synchronous motor (rotating, permanent-magnet) 5: Synchronous motor (separately excited) 7: SIEMOSYN motor 8: Reluctance motor 11: 1LA1 standard induction motor 12: 1LE2 standard induction motor (NEMA) 15: 1LA5 standard induction motor 16: 1LA6 standard induction motor 17: 1LA7 standard induction motor 18: 1LA8 / 1PQ8 standard induction motor 102: 1PH2 induction motor 104: 1PH4 induction motor 107: 1PH7 induction motor 108: 1PH8 induction motor 134: 1PM4 induction motor 136: 1PM6 induction motor 166: 1PL6 induction motor 283: 1FW3 synchronous motor 10000: Motor with DRIVE-CLiQ 10001: Motor with DRIVE-CLiQ 2nd data set
Dependency:	Motors, selection 206, 236, 237 cannot be operated (also not as motor with DRIVE-CLiQ). p0300 = 5 cannot be selected with SINAMICS G. When the motor type is changed, the code number in p0301 may be reset to 0. p0300 = 12 can only be selected for p0100 = 1 (NEMA). When selecting a motor type from the 1LA5 and 1LA7 series, parameters p0335, p0626, p0627 and p0628 of the thermal motor model are pre-assigned as a function of p0307 and p0311. Refer to: p0301
Caution:	A permanent-magnet synchronous motor cannot be operated with an SSI encoder without HTL/TTL track. A separately-excited synchronous motor can only be operated with an SSI encoder if this is used as the second encoder and an HTL/TTL encoder is used as the first encoder.
	
Caution:	If a catalog motor is selected (p0300 >= 100) and an associated motor code number (p0301), then the parameters, that are associated with this list cannot be changed (write protection). The write protection is canceled if the motor type p0300 is set to a non-Siemens motor that matches p0301 (e.g. p0300 = 2 for p0301 = 2xxxx). Write protection is automatically canceled when the results of motor data identification are copied to the motor parameters.
Notice:	If required, the list of motor codes/encoder codes can be found in the Appendix of the List Manual. For 1PQ8 motors (p0300 = 18) the fan type p0335 should be set to 5.
Note:	With p0300 = 10000, for a motor with DRIVE-CLiQ, the motor parameters are automatically downloaded, with p0300 = 10001, the motor parameters of a second data set (if available). If a motor type has not been selected (p0300 = 0), then the drive commissioning routine cannot be exited. A motor type with a value above p0300 >= 100 describes motors for which a motor parameter list exists. Motor types with a value below p0300 < 100 correspond to the selection of a third-party motor. When appropriately selected, this means that the motor parameters are pre-assigned the settings for a third-party motor. This also applies for parameters for a motor with DRIVE-CLiQ. In this case p0300 can only be set to p0300 = 10000 or 10001 (read motor parameters) or to the corresponding non-Siemens motor (first digit of the motor code number) in order to be able to cancel the write protection.

p0301[0...n]	Motor code number selection / Mot code No. sel		
SERVO	Can be changed: C2(1, 3) Data type: Unsigned16 P-Group: Motor Not for motor type: FEM Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 65535	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to select a motor from a motor parameter list. When changing the code number (with the exception to the value 0), all of the motor parameters are pre-assigned from the internally available parameter lists.		

- Dependency:** Code numbers can only be selected for motor types that correspond to the motor type selected in p0300. For 1PH2, 1PH4, 1PH7, 1PM4, 1PM6, 1FT6 motors, code numbers are also possible, whose fourth decimal position is greater by a value of 1 or 2 than the matching motor type in p0300. For 1FE1 motors, the third decimal position can be higher by a value of 1.
Refer to: p0300
- Notice:** If required, the list of motor codes/encoder codes can be found in the Appendix of the List Manual.
- Note:** The motor code number can only be changed if the matching catalog motor was first selected in p0300.
For a motor with DRIVE-CLiQ, p0301 cannot be changed. In this case, p0301 is automatically written to the code number of the motor parameter read in (r0302) if p0300 is set to 10000.
When selecting a catalog motor (p0300 >= 100), drive commissioning can only be exited if a code number is selected.
If, for direct drives, the motor code number (p0301) is changed, this does not automatically result in the angular commutation offset being determined (p0431).

p0301[0...n]	Motor code number selection / Mot code No. sel		
SERVO (Lin)	Can be changed: C2(1, 3) Data type: Unsigned16 P-Group: Motor Not for motor type: FEM Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 65535	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to select a motor from a motor parameter list. When changing the code number (with the exception to the value 0), all of the motor parameters are pre-assigned from the internally available parameter lists.		
Dependency:	Code numbers can only be selected for motor types that correspond to the motor type selected in p0300. Refer to: p0300		
Notice:	If required, the list of motor codes/encoder codes can be found in the Appendix of the List Manual.		
Note:	The motor code number can only be changed if the matching catalog motor was first selected in p0300. When selecting a catalog motor (p0300 >= 100), drive commissioning can only be exited if a code number is selected.		

p0301[0...n]	Motor code number selection / Mot code No. sel		
VECTOR	Can be changed: C2(1, 3) Data type: Unsigned16 P-Group: Motor Not for motor type: FEM Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 65535	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to select a motor from a motor parameter list. When changing the code number (with the exception to the value 0), all of the motor parameters are pre-assigned from the internally available parameter lists.		
Dependency:	Code numbers can only be selected for motor types that correspond to the motor type selected in p0300. For 1PH2, 1PH4, 1PH7, 1PM4, 1PM6, 1FT6 motors, code numbers are also possible, whose fourth decimal position is greater by a value of 1 or 2 than the matching motor type in p0300. For 1FE1 motors, the third decimal position can be higher by a value of 1. Refer to: p0300		
Notice:	If required, the list of motor codes/encoder codes can be found in the Appendix of the List Manual.		
Note:	The motor code number can only be changed if the matching catalog motor was first selected in p0300. For a motor with DRIVE-CLiQ, p0301 cannot be changed. In this case, p0301 is automatically written to the code number of the motor parameter read in (r0302) if p0300 is set to 10000. When selecting a catalog motor (p0300 >= 100), drive commissioning can only be exited if a code number is selected.		

r0302[0...n]	Motor code number of motor with DRIVE-CLiQ / Motor code Mot DLQ																												
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Motor Not for motor type: - Min -	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -																										
Description:	Displays the motor code number from the saved motor data from a motor with DRIVE-CLiQ.																												
Note:	Drive commissioning can only be exited if the code number that was downloaded (r0302) matches the stored code number (p0301). If the numbers differ, then the motor data set should be re-loaded using p0300 = 10000. The motor data are always expected from the first encoder that is assigned to the drive data sets (refer to p0187 = encoder 1) data set number. The value is not updated cyclically but only on specific events (e.g. update DRIVE-CLiQ device). r0302 = 0: No motor with DRIVE-CLiQ found																												
r0303[0...n]	Motor status word of motor with DRIVE-CLiQ / Mot ZSW Mot DLQ																												
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Motor Not for motor type: - Min -	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -																										
Description:	Displays the status word of the automatic motor parameter sensing of a motor with DRIVE-CLiQ. Motor parameter sensing takes place during warm restarts, following downloading, and when powering up, if the SMI is connected to the Motor Module and the encoder is activated (see p0145). If the motor data is reloaded (p0300 = 10000, 10001), r0303 is also updated.																												
Bit field:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr> <tr> <td>00</td><td>Motor data sets selection</td><td>Two</td><td>One</td><td>-</td></tr> <tr> <td>01</td><td>Motor connection type</td><td>Delta</td><td>Star</td><td>-</td></tr> <tr> <td>02</td><td>Windings are switchable</td><td>can be chng over</td><td>cannot be chg over</td><td>-</td></tr> <tr> <td>03</td><td>Number of switchable windings bit 0</td><td>2</td><td>0</td><td>-</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FP	00	Motor data sets selection	Two	One	-	01	Motor connection type	Delta	Star	-	02	Windings are switchable	can be chng over	cannot be chg over	-	03	Number of switchable windings bit 0	2	0	-			
Bit	Signal name	1 signal	0 signal	FP																									
00	Motor data sets selection	Two	One	-																									
01	Motor connection type	Delta	Star	-																									
02	Windings are switchable	can be chng over	cannot be chg over	-																									
03	Number of switchable windings bit 0	2	0	-																									
Dependency:	Refer to: p0145, p0300																												
Note:	Re bit 0: Value 0 indicates that the first SMI data set is selected, value 1 indicates that the second is selected [MDS]. Re bit 1: Displays the connection type for the winding [MDS]. Re bit 2: Displays whether the windings can be changed over. Re bit 3: Value 0 indicates that no windings that can be changed over are present, value 1 indicates that two are present.																												
p0304[0...n]	Rated motor voltage / Mot V_{rated}																												
SERVO, VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0 [Vrms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 20000 [Vrms]	Access level: 1 Func. diagram: 6300, 6724 Unit selection: - Expert list: 1 Factory setting 0 [Vrms]																										
Description:	Sets the rated motor voltage (rating plate).																												
Dependency:	Refer to: p0349																												
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.																												
Note:	When the parameter value is entered the connection type of the motor (star-delta) must be taken into account.																												

p0305[0...n]	Rated motor current / Mot I_{rated}		
SERVO	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.00 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10000.00 [Arms]	Access level: 1 Func. diagram: 6300 Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the rated motor current (rating plate).		
Dependency:	Refer to: p0349		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0305 is changed during quick commissioning (p0010 = 1), then the maximum current p0640 is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).		
Note:	When the parameter value is entered the connection type of the motor (star-delta) must be taken into account.		
p0305[0...n]	Rated motor current / Mot I_{rated}		
VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.00 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10000.00 [Arms]	Access level: 1 Func. diagram: 6300 Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the rated motor current (rating plate).		
Dependency:	Refer to: p0349		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0305 is changed during quick commissioning (p0010 = 1), then the maximum current p0640 is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3). If the rated motor current exceeds twice the maximum drive converter current (r0209), then the maximum current is reduced due to the current harmonics that increase overproportionally (r0067).		
Note:	When the parameter value is entered the connection type of the motor (star-delta) must be taken into account.		
p0306[0...n]	Number of motors connected in parallel / Motor qty		
SERVO	Can be changed: C2(1, 3) Data type: Unsigned8 P-Group: Motor Not for motor type: - Min 1	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Number of motors that can be operated in parallel using one motor data set. Depending on the motor number entered, internally an equivalent motor is calculated. The following should be carefully observed for motors connected in series: The following rating plate data should only be entered for one motor: - resistances and inductances: p0350, p0352, p0353, p0354, p0356, p0357, p0358, p0360 - currents: p0305, p0318, p0320, p0323, p0325, p0329, p0338, p0391, p0392 - torques/forces: p0312, p0319 - power ratings: p0307 - masses/moments of inertia: p0341, p0344 All other parameters take into account the replacement/equivalent motor (e.g. r0331, r0370, r0373, r0374).		
Dependency:	Refer to: r0331, r0370, r0373, r0374, r0376, r0377, r0382		

Caution:

The motors to be connected in parallel must be of the same type and size (same order no. (MLFB)).
The mounting regulations when connecting motors in parallel must be carefully maintained! Especially for synchronous motors, the pole position of motors that are rigidly coupled with one another (mechanically) must be identical.
The number of motors set must correspond to the number of motors that are actually connected in parallel.
After changing p0306, it is imperative that the control parameters are adapted (e.g. using automatic calculation with p0340 = 1).

Notice:

If p0306 is changed during quick commissioning (p0010 = 1), then the maximum current p0640 is appropriately pre-assigned. This is not the case when commissioning the motor (p0010 = 3).

p0306[0...n]**Number of motors connected in parallel / Motor qty**

VECTOR

Can be changed: C2(1, 3)**Calculated:** -**Access level:** 1**Data type:** Unsigned8**Dynamic index:** MDS, p0130**Func. diagram:** -**P-Group:** Motor**Units group:** -**Unit selection:** -**Not for motor type:** FEM**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

1

50

1

Description:

Number of motors that can be operated in parallel using one motor data set.
Depending on the motor number entered, internally an equivalent motor is calculated.
The following should be carefully observed for motors connected in series:
The following rating plate data should only be entered for one motor:
- resistances and inductances: p0350 ... p0361
- currents: p0305, p0320, p0323, p0325, p0329, p0389, p0390, p0391, p0392
- power ratings: p0307
- masses/moments of inertia: p0341, p0344

All other parameters take into account the replacement/equivalent motor (e.g. r0331, r0333).

Dependency:

Refer to: r0331

Caution:

The motors to be connected in parallel must be of the same type and size (same order no. (MLFB)).
The mounting regulations when connecting motors in parallel must be carefully maintained! Especially for synchronous motors, the pole position of motors that are rigidly coupled with one another (mechanically) must be identical.
The number of motors set must correspond to the number of motors that are actually connected in parallel.
After changing p0306, it is imperative that the control parameters are adapted (e.g. using automatic calculation with p0340 = 1).

Notice:

If p0306 is changed during quick commissioning (p0010 = 1), then the maximum current p0640 is appropriately pre-assigned. This is not the case when commissioning the motor (p0010 = 3).

Note:

Only operation with V/f characteristic makes sense if more than 10 identical motors are connected in parallel.
Separately-excited synchronous motors must not be connected in parallel. Synchronous and reluctance motors that are not coupled with one another align themselves when the pulses are switched in. If the motors have different load levels, then equalization currents flow between the motors.

p0307[0...n]**Rated motor power / Mot P_{rated}**

SERVO

Can be changed: C2(1, 3)**Calculated:** -**Access level:** 1**Data type:** FloatingPoint32**Dynamic index:** MDS, p0130**Func. diagram:** -**P-Group:** Motor**Units group:** 14_6**Unit selection:** p0100**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0.00 [kW]

100000.00 [kW]

0.00 [kW]

Description:

Sets the rated motor power (rating plate).

Dependency:

IECdrives (p0100 = 0): Units kW
NEMA drives (p0100 = 1): Units hp
Refer to: p0100

Caution:

This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.

Note:

This parameter is automatically preset for motors from the motor list (p0301).

p0307[0...n]	Rated motor power / Mot P_{rated}		
VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min -100000.00 [kW]	Calculated: - Dynamic index: MDS, p0130 Units group: 14_6 Scaling: - Max 100000.00 [kW]	Access level: 1 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.00 [kW]
Description:	Sets the rated motor power (rating plate).		
Dependency:	IECdrives (p0100 = 0): Units kW NEMA drives (p0100 = 1): Units hp Refer to: p0100		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	This parameter is automatically preset for motors from the motor list (p0301). For generators, a negative rated power should be entered.		
p0308[0...n]	Rated motor power factor / Mot cos_{phi}_{rated}		
SERVO, VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL Min 0.000	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 1.000	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the rated motor power factor (cos phi, rating plate). For a parameter value of 0.000, the power factor is internally calculated and displayed in r0332.		
Dependency:	This parameter is only available for IEC motors (p0100 = 0). Refer to: p0100, p0309, r0332		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx).		
p0309[0...n]	Rated motor efficiency / Mot eta_{rated}		
VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL Min 0.0 [%]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 99.9 [%]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [%]
Description:	Sets the rated motor efficiency (rating plate). For a parameter value of 0.0, the power factor is internally calculated and displayed in r0332.		
Dependency:	This parameter is only available for NEMA motors (p0100 = 1). Refer to: p0100, p0308, r0332		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx).		

p0310[0...n]	Rated motor frequency / Mot f_{rated}		
SERVO	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.00 [Hz]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 3000.00 [Hz]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [Hz]
Description:	Sets the rated motor frequency (rating plate).		
Dependency:	The number of pole pairs (r0313) is automatically recalculated when the parameter is changed (together with p0311), if p0314 = 0. If p0310 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. Refer to: p0311, r0313, p0314		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0310 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).		
Note:	The parameter is automatically pre-assigned for induction motors from the motor list (p0301). For synchronous motors, the parameter is not required and must therefore be pre-assigned zero. For p0310 = 0, it is not possible to calculate the pole pair; instead, it must be entered in p0314.		

p0310[0...n]	Rated motor frequency / Mot f_{rated}		
VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.00 [Hz]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 3000.00 [Hz]	Access level: 1 Func. diagram: 6300 Unit selection: - Expert list: 1 Factory setting 0.00 [Hz]
Description:	Sets the rated motor frequency (rating plate).		
Dependency:	The number of pole pairs (r0313) is automatically recalculated when the parameter is changed (together with p0311), if p0314 = 0. The rated frequency is restricted to values between 1.00 Hz and 650.00 Hz. Refer to: p0311, r0313, p0314		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0310 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).		

p0311[0...n]	Rated motor speed / Mot n_{rated}		
SERVO, VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [rpm]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 210000.0 [rpm]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [rpm]
Description:	Sets the rated motor speed (rating plate). For VECTOR the following applies (p0107): For p0311 = 0, the rated motor slip of induction motors is internally calculated and displayed in r0330. It is especially important to correctly enter the rated motor speed for vector control and slip compensation for V/f control.		

- Dependency:** If p0311 is changed and for p0314 = 0, the pole pair (r0313) is recalculated automatically.
Refer to: p0310, r0313, p0314
- Caution:** This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.
- Notice:** If p0311 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).

p0311[0...n]		Rated motor velocity / Mot v _{rated}	
SERVO (Lin)	Can be changed: C2(1, 3)	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [m/min]	Max 6000.0 [m/min]	Factory setting 0.0 [m/min]
Description:	Sets the rated motor velocity (rating plate).		
Dependency:	The pole pair width is set in p0315. Refer to: p0310, r0313, p0314		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0311 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).		

p0312[0...n]		Rated motor torque / Mot M _{rated}	
SERVO	Can be changed: C2(3)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: 7_4	Unit selection: p0100
	Not for motor type: ASM, REL, FEM	Scaling: -	Expert list: 1
	Min 0.00 [Nm]	Max 1000000.00 [Nm]	Factory setting 0.00 [Nm]
Description:	Sets the rated motor torque (rating plate).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		

p0312[0...n]		Rated motor force / Mot F_rated	
SERVO (Lin)	Can be changed: C2(3)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: 8_4	Unit selection: p0100
	Not for motor type: ASM, REL, FEM	Scaling: -	Expert list: 1
	Min 0.00 [N]	Max 1000000.00 [N]	Factory setting 0.00 [N]
Description:	Sets the rated motor force (rating plate).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		

r0313[0...n]	Motor pole pair number, actual (or calculated) / Mot PolePairNo act		
SERVO	Can be changed: - Data type: Unsigned16 P-Group: Motor Not for motor type: - Min -	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 5300 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of motor pole pairs. The value is used for internal calculations. Values: r0313 = 1: 2-pole motor r0313 = 2: 4-pole motor etc.		
Dependency:	For p0314 > 0, the entered value is displayed in r0313. For p0314 = 0, the pole pair number (r0313) is automatically calculated from the rated frequency (p0310) and the rated speed (p0311). Refer to: p0310, p0311, p0314		
Note:	For the automatic calculation, the pole pair number is set to the value of 2 if the rated speed or the rated frequency is zero.		
r0313[0...n]	Motor pole pair number, actual (or calculated) / Mot PolePairNo act		
VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Motor Not for motor type: - Min -	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 5300 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of motor pole pairs. The value is used for internal calculations. Values: r0313 = 1: 2-pole motor r0313 = 2: 4-pole motor etc.		
Dependency:	For p0314 > 0, the entered value is displayed in r0313. For p0314 = 0, the pole pair number (r0313) is automatically calculated from the rated power (p0307), rated frequency (p0310) and rated speed (p0311). Refer to: p0307, p0310, p0311, p0314		
Note:	For the automatic calculation, the pole pair number is set to the value of 2 if the rated speed or the rated frequency is zero.		
p0314[0...n]	Motor pole pair number / Mot pole pair No.		
SERVO	Can be changed: C2(1, 3) Data type: Unsigned16 P-Group: Motor Not for motor type: - Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 4000	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the motor pole pair number. Values: p0314 = 1: 2-pole motor p0314 = 2: 4-pole motor etc.		
Dependency:	For p0314 = 0, the pole pair number is automatically calculated from the rated frequency (p0310) and the rated speed (p0311) and displayed in r0313.		

Notice: If p0314 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is appropriately pre-assigned. This is not the case when commissioning the motor (p0010 = 3).

For induction motors, the value need only be input if the rated data of a generator is entered therefore resulting in a negative rated slip. In this case, the number of pole pairs in r0313 is too low by 1 and must be manually corrected.

p0314[0...n]		Motor pole pair number / Mot pole pair No.		
VECTOR	Can be changed: C2(1, 3) Data type: Unsigned16 P-Group: Motor Not for motor type: - Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 255	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0	
Description:	Sets the motor pole pair number. Values: p0314 = 1: 2-pole motor p0314 = 2: 4-pole motor etc.			
Dependency:	For p0314 = 0, the pole pair number is automatically calculated from the rated frequency (p0310) and the rated speed (p0311) and displayed in r0313.			
Notice:	If p0314 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is appropriately pre-assigned. This is not the case when commissioning the motor (p0010 = 3). For induction motors, the value need only be input if the rated data of a generator is entered therefore resulting in a negative rated slip. In this case, the number of pole pairs in r0313 is too low by 1 and must be manually corrected.			

p0315[0...n]		Motor pole pair width / MotPolePair width		
SERVO (Lin)	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 1.00 [mm]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 1000.00 [mm]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 30.00 [mm]	
Description:	Sets the pole pair width of the linear motor.			
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.			
Note:	This parameter is automatically preset for motors from the motor list (p0301).			

p0316[0...n]		Motor torque constant / Mot kT		
SERVO	Can be changed: C2(1, 3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.00 [Nm/A]	Calculated: - Dynamic index: MDS, p0130 Units group: 28_1 Scaling: - Max 400.00 [Nm/A]	Access level: 1 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.00 [Nm/A]	
Description:	Sets the torque constant of the synchronous motor. p0316 = 0: The torque constant is calculated from the motor data. p0316 > 0: The selected value is used as torque constant.			
Dependency:	Refer to: r0334, r1937			
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.			
Note:	This parameter is not used for induction motors (p0300 = 1xx).			

p0316[0...n]	Motor force constant / Mot kT		
SERVO (Lin)	Can be changed: C2(1, 3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.00 [N/Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: 29_1 Scaling: - Max 15000.00 [N/Arms]	Access level: 1 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.00 [N/Arms]
Description:	Sets the force constant of the synchronous motor. p0316 = 0: The force constant is calculated from the motor data. p0316 > 0: The selected value is used as force constant.		
Dependency:	Refer to: r0334, r1937		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	This parameter is automatically preset for motors from the motor list (p0301).		
p0316[0...n]	Motor torque constant / Mot kT		
VECTOR	Can be changed: C2(1, 3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.00 [Nm/A]	Calculated: - Dynamic index: MDS, p0130 Units group: 28_1 Scaling: - Max 400.00 [Nm/A]	Access level: 2 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.00 [Nm/A]
Description:	Sets the torque constant of the synchronous motor. p0316 = 0: The torque constant is calculated from the motor data. p0316 > 0: The selected value is used as torque constant.		
Dependency:	Refer to: r0334		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	This parameter is not used for induction motors (p0300 = 1xx).		
p0317[0...n]	Motor voltage constant / Mot kE		
SERVO	Can be changed: C2(3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.0 [Vrms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 24000.0 [Vrms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [Vrms]
Description:	Sets the voltage constant for synchronous motors. Units for rotating synchronous motors: Vrms/(1000 rpm), phase-to-phase		
Dependency:	Refer to: r1938		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	This parameter is not used for induction motors (p0300 = 1xx).		

p0317[0...n]	Motor voltage constant / Mot kE		
SERVO (Lin)	Can be changed: C2(3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.0 [Vrms s/m]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 5000.0 [Vrms s/m]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [Vrms s/m]
Description:	Sets the voltage constant for synchronous motors. Units for linear synchronous motors: Vrms s/m, phase		
Dependency:	Refer to: r1938		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	This parameter is automatically preset for motors from the motor list (p0301).		
p0318[0...n]	Motor stall current / Mot I_standstill		
SERVO	Can be changed: C2(3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.00 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10000.00 [Arms]	Access level: 3 Func. diagram: 8017 Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the stall current for synchronous motors (p0300 = 2xx).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The parameter is used for the I2t monitoring of the motor (refer to p0611). This parameter is not used for induction motors (p0300 = 1xx).		
p0318[0...n]	Motor stall current / Mot I_standstill		
VECTOR	Can be changed: C2(3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.00 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10000.00 [Arms]	Access level: 4 Func. diagram: 8017 Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the stall current for synchronous motors (p0300 = 2xx).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The parameter is used for the I2t monitoring of the motor (refer to p0611). This parameter is not used for induction motors (p0300 = 1xx).		

p0319[0...n]	Motor stall torque / Mot M_standstill		
SERVO	Can be changed: C2(3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.00 [Nm]	Calculated: - Dynamic index: MDS, p0130 Units group: 7_4 Scaling: - Max 100000.00 [Nm]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.00 [Nm]
Description:	Sets the standstill (stall) torque for rotating synchronous motors (p0300 = 2xx).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	This parameter is not used for induction motors (p0300 = 1xx). This parameter value is not evaluated from a control-related perspective.		
p0319[0...n]	Motor stall force / Mot F_standstill		
SERVO (Lin)	Can be changed: C2(3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.00 [N]	Calculated: - Dynamic index: MDS, p0130 Units group: 8_4 Scaling: - Max 100000.00 [N]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.00 [N]
Description:	Sets the standstill (stall) force for linear synchronous motors (p0300 = 4xx).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	This parameter is automatically preset for motors from the motor list (p0301). This parameter value is not evaluated from a control-related perspective.		
p0320[0...n]	Motor rated magnetizing current/short-circuit current / Mot I_mag_rated		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: REL, FEM Min 0.000 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 5000.000 [Arms]	Access level: 3 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting 0.000 [Arms]
Description:	Induction motors: Sets the rated motor magnetizing current. For p0320 = 0.000 the magnetizing current is internally calculated and displayed in r0331. Synchronous motors: Sets the rated motor short-circuit current.		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The magnetization current p0320 for induction motors (not for catalog motors) is reset when quick commissioning is exited with p3900 > 0. VECTOR: If, for induction motors, the magnetizing current p0320 is changed outside the commissioning phase (p0010 > 0), then the magnetizing inductance p0360 is changed so that the EMF r0337 remains constant.		

p0322[0...n]	Maximum motor speed / Mot n_max		
SERVO	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [rpm]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 210000.0 [rpm]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [rpm]
Description:	Sets the maximum motor speed.		
Dependency:	Refer to: p1082		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0322 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).		
p0322[0...n]	Motor maximum velocity / Mot v_max		
SERVO (Lin)	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [m/min]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 1000.0 [m/min]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [m/min]
Description:	Sets the maximum motor velocity.		
Dependency:	Refer to: p1082		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0322 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).		
p0322[0...n]	Maximum motor speed / Mot n_max		
VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [rpm]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 210000.0 [rpm]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [rpm]
Description:	Sets the maximum motor speed.		
Dependency:	Refer to: p1082		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0322 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).		

p0323[0...n]	Maximum motor current / Mot I_max		
SERVO	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, FEM Min 0.00 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 20000.00 [Arms]	Access level: 1 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the maximum permissible motor current (e.g. demagnetizing current for synchronous motors).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0323 is changed during quick commissioning (p0010 = 1), then the maximum current p0640 is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).		
Note:	The parameter has no effect for induction motors. For synchronous motors, a value must always be entered for the maximum motor current. p0323 is a motor data. The user-selectable current limit is entered into p0640.		
p0323[0...n]	Maximum motor current / Mot I_max		
VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, FEM Min 0.00 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 20000.00 [Arms]	Access level: 3 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the maximum permissible motor current (e.g. demagnetizing current for synchronous motors).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0323 is changed during quick commissioning (p0010 = 1), then the maximum current p0640 is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).		
Note:	This parameter is automatically preset for motors from the motor list (p0301). The parameter has no effect for induction motors. The parameter has not effect for synchronous motors if a value of 0.0 is entered. The user-selectable current limit is entered into p0640.		
p0324[0...n]	Winding maximum speed / Winding n_max		
SERVO	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [rpm]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 210000.0 [rpm]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [rpm]
Description:	Sets the electrical maximum speed of the winding.		
Dependency:	Refer to: p1082		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0324 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned appropriately. This is not the case when commissioning the motor (p0010 = 3).		

p0324[0...n]	Winding maximum velocity / Winding v_max		
SERVO (Lin)	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [m/min]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 1000.0 [m/min]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [m/min]
Description:	Sets the electrical maximum velocity of the winding.		
Dependency:	Refer to: p1082		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0324 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned appropriately. This is not the case when commissioning the motor (p0010 = 3).		
p0324[0...n]	Winding maximum speed / Winding n_max		
VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [rpm]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 210000.0 [rpm]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [rpm]
Description:	Sets the electrical maximum speed of the winding.		
Dependency:	Refer to: p1082		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0324 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned appropriately. This is not the case when commissioning the motor (p0010 = 3).		
p0325[0...n]	Motor pole position identification current, 1st phase / Mot PolID I 1st ph		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.000 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10000.000 [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [Arms]
Description:	Sets the current for the 1st phase of the two-stage technique for pole position identification routine. The current of the 2nd phase is set in p0329. The two-stage technique is selected with p1980 = 4.		
Dependency:	Refer to: p0329, p1980, p1981, p1982, p1983, r1984, r1985, r1987, p1990 Refer to: F07995		
Notice:	When the motor code (p0301) is changed, it is possible that p0325 is not pre-assigned. p0325 can be pre-assigned using p0340 = 3.		
Note:	The value is automatically pre-assigned for the following events: - For p0325 = 0 and automatic calculation of the closed-loop control parameters (p0340 = 1, 2, 3). - for quick commissioning (p3900 = 1, 2, 3).		

p0326[0...n]	Motor stall torque correction factor / Mot M_stall_corr		
SERVO	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: REL, FEM Min 5 [%]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 60 [%]
Description:	Sets the correction factor for the stall torque/force at a 600 V DC link voltage.		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300). The reference value for this parameter is inversely proportional to the leakage inductance of the motor (p0353, p0354, p0356). The following applies for firmware version 2.6 SP2 and higher: If leakage inductances are changed for motor data identification, the value in p0326 is automatically adapted to maintain the stall torque.		
p0326[0...n]	Motor stall force correction factor / Mot F_stall_corr		
SERVO (Lin)	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: REL, FEM Min 5 [%]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 60 [%]
Description:	Sets the correction factor for the stall force at a 600 V DC link voltage.		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300). The reference value for this parameter is inversely proportional to the leakage inductance of the motor (p0353, p0354, p0356). The following applies for firmware version 2.6 SP2 and higher: If leakage inductances are changed for motor data identification, the value in p0326 is automatically adapted to maintain the stall torque.		
p0327[0...n]	Optimum motor load angle / Mot phi_load opt		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.0 [°]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 135.0 [°]	Access level: 3 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting 90.0 [°]
Description:	Sets the optimum load angle for synchronous motors with reluctance torque (e.g. 1FE ... motors). This parameter has no significance for induction motors. SERVO: The load angle is measured at 1.5 x rated motor current. VECTOR: The load angle is measured at the rated motor current.		
Dependency:	Refer to: r1947		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		

Note: For synchronous motors without reluctance torque, a angle of 90 degrees must be set.
When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).

p0328[0...n] Motor reluctance torque constant / Mot kT_reluctance			
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: ASM, REL, FEM	Scaling: -	Expert list: 1
	Min -1000.00 [mH]	Max 1000.00 [mH]	Factory setting 0.00 [mH]
Description:	Sets the reluctance torque constant for synchronous motors with reluctance torque (e.g. 1FE ... motors). This parameter has no significance for induction motors.		
Dependency:	Refer to: r1939		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	For synchronous motors without reluctance torque, the value 0 must be set.		

p0328[0...n] Motor reluctance force constant / Mot kT_reluctance			
SERVO (Lin)	Can be changed: C2(3), U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: ASM, REL, FEM	Scaling: -	Expert list: 1
	Min -1000.00 [mH]	Max 1000.00 [mH]	Factory setting 0.00 [mH]
Description:	Sets the reluctance force constant for synchronous motors with reluctance force (e.g. 1FE ... motors). This parameter has no significance for induction motors.		
Dependency:	Refer to: r1939		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	For synchronous motors without reluctance torque, the value 0 must be set.		

p0329[0...n] Motor pole position identification current / Mot PolID current			
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: ASM, REL, FEM	Scaling: -	Expert list: 1
	Min 0.00 [Arms]	Max 10000.00 [Arms]	Factory setting 0.00 [Arms]
Description:	Sets the current for the pole position identification routine. For a two-stage technique, the current is set for the second phase.		
Dependency:	Refer to: p0325, p1980, p1981, p1982, p1983, r1984, r1985, r1987, p1990 Refer to: F07995		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		

r0330[0...n] Rated motor slip / Mot slip Rated

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
	Min - [Hz]	Max - [Hz]	Factory setting - [Hz]

Description: Displays the rated motor slip.

Dependency: The rated slip is calculated from the rated frequency, rated speed and number of pole pairs.
Refer to: p0310, p0311, r0313

Note: The parameter is not used for synchronous motors (p0300 = 2xx).

r0331[0...n] Actual motor magnetizing current/short-circuit current / Mot I_mag_rtd act

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 5722, 6722, 6724
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: REL, FEM	Scaling: -	Expert list: 1
	Min - [Arms]	Max - [Arms]	Factory setting - [Arms]

Description: Induction motor:
Displays the rated magnetizing current from p0320.
For p0320 = 0, the internally calculated magnetizing current is displayed.
Synchronous motor:
Displays the rated short-circuit current from p0320.

Dependency: If p0320 was not entered, then the parameter is calculated from the rating plate parameters.

Note: In the case of multi-motor operation r0331 is increased by the factor p0306 compared to p0320.

r0332[0...n] Rated motor power factor / Mot cos_phi Rated

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: PEM, REL	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

Description: Displays the rated power factor for induction motors.
For IEC motors, the following applies (p0100 = 0):
For p0308 = 0, the internally-calculated power factor is displayed.
For p0308 > 0, this value is displayed.
For NEMA motors, the following applies (p0100 = 1):
For p0309 = 0, the internally-calculated power factor is displayed.
For p0309 > 0, this value is converted into the power factor and displayed.

Dependency: If p0308 is not entered, the parameter is calculated from the rating plate parameters.

Note: The parameter is not used for synchronous motors (p0300 = 2xx).

r0333[0...n]	Rated motor torque / Mot M_{rated}		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min - [Nm]	Calculated: - Dynamic index: MDS, p0130 Units group: 7_4 Scaling: - Max - [Nm]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [Nm]
Description:	Displays the rated motor torque.		
Dependency:	IEC drives (p0100 = 0): unit Nm NEMA drives (p0100 = 1): unit lbf ft		
Note:	For induction and reluctance motors, r0333 is calculated from p0307 and p0311. For synchronous motors, r0333 is calculated from p0305, p0316, p0327 and p0328. The result can deviate from the input in p0312. If p0316 = 0, then r0333 = p0312 is displayed. In the case of multi-motor operation r0333 is increased by the factor p0306 compared to the rated torque of an individual motor.		
r0333[0...n]	Rated motor force / Mot F_{rated}		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min - [N]	Calculated: - Dynamic index: MDS, p0130 Units group: 8_4 Scaling: - Max - [N]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [N]
Description:	Displays the rated motor force.		
Dependency:	IEC drives (p0100 = 0): unit Nm NEMA drives (p0100 = 1): unit lbf ft		
Note:	For synchronous motors, r0333 is calculated from p0305, p0316, p0327 and p0328. The result can deviate from the input in p0312. If p0316 = 0, then r0333 = p0312 is displayed.		
r0334[0...n]	Actual motor-torque constant / Mot kT_{act}		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min - [Nm/A]	Calculated: - Dynamic index: MDS, p0130 Units group: 28_1 Scaling: - Max - [Nm/A]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [Nm/A]
Description:	Displays the torque constant of the synchronous motor used.		
Dependency:	IEC drives (p0100 = 0): unit Nm / A NEMA drives (p0100 = 1): unit lbf ft / A Refer to: p0316		
Note:	This parameter is not used for induction motors (p0300 = 1xx). For synchronous motors, parameter r0334 = p0316 is displayed. If p0316 = 0, r0334 is calculated from p0305 and p0312.		

r0334[0...n]	Actual motor force constant / Mot kT act		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min - [N/Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: 29_1 Scaling: - Max - [N/Arms]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [N/Arms]
Description:	Displays the force constant of the synchronous motor used.		
Dependency:	IEC drives (p0100 = 0): unit Nm / A NEMA drives (p0100 = 1): unit lbf ft / A Refer to: p0316		
Note:	For synchronous motors, parameter r0334 = p0316 is displayed. if p0316 = 0, r0334 is calculated from p0305 and p0312.		
r0334[0...n]	Actual motor-torque constant / Mot kT act		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min - [Nm/A]	Calculated: - Dynamic index: MDS, p0130 Units group: 28_1 Scaling: - Max - [Nm/A]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [Nm/A]
Description:	Displays the torque constant of the synchronous motor used.		
Dependency:	IEC drives (p0100 = 0): unit Nm / A NEMA drives (p0100 = 1): unit lbf ft / A Refer to: p0316		
Note:	This parameter is not used for induction motors (p0300 = 1xx). For synchronous motors, parameter r0334 = p0316 is displayed. When p0316 = 0, r0334 is calculated from p0305 and p0312 or p0305, p0307, and p0311.		
p0335[0...n]	Motor cooling type / Motor cooling type		
SERVO, VECTOR	Can be changed: C2(1, 3), T Data type: Integer16 P-Group: Motor Not for motor type: PEM, REL, FEM Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 128	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the motor cooling system used.		
Value:	0: Non-ventilated 1: Forced cooling 2: Liquid cooling 4: Non-ventilated and internal fan 5: Forced cooling and internal fan 6: Liquid cooling and internal fan 128: No fan		
Dependency:	For 1LA5 and 1LA7 motors (refer to p0300), the parameter is preset as a function of p0307 and p0311.		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The parameter influences the thermal 3-mass motor model. 1LA1 and 1LA8 motors are characterized by the fact that they have an internal rotor fan. This "internal cooling" lies within the motor frame and is not visible. Air is not directly exchanged with the motor ambient air. For 1PQ8 motors, p0335 should be set to 5 as these motors are force-ventilated motors. 1LA7 motors, frame size 56 are operated without fan.		

r0336[0...n]	Actual rated motor frequency / Mot f_{rated act}		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min - [Hz]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max - [Hz]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the rated frequency of the motor. For p0310 > 0, this value is displayed.		
Dependency:	Refer to: p0311, p0314		
Note:	For p0310 = 0 or for synchronous motors, the rated motor frequency r0336 is calculated from the rated speed and the pole pair number. For p0310 > 0, this value is displayed (not for synchronous motors).		
r0337[0...n]	Rated motor EMF / Mot EMF_{rated}		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max - [Vrms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the rated EMF of the motor.		
Note:	EMF: Electromagnetic force		
r0337[0...n]	Rated motor EMF / Mot EMF_{rated}		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: REL Min - [Vrms s/m]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max - [Vrms s/m]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Vrms s/m]
Description:	Displays the rated EMF of the motor.		
Note:	EMF: Electromagnetic force		
p0338[0...n]	Motor limit current / Mot I_{limit}		
SERVO	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.00 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10000.00 [Arms]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the motor limit current for synchronous motors (for a 600 V DC link voltage). Using this current, the maximum torque is achieved at the rated speed (voltage limit characteristic).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Notice:	If p0338 is changed during quick commissioning (p0010 = 1), then the maximum current p0640 is appropriately pre-assigned. This is not the case when commissioning the motor (p0010 = 3).		

r0339[0...n]	Rated motor voltage / Mot V_{rated}		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max - [Vrms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the rated motor voltage		
Note:	For induction motors (p0300 = 1xx) the parameter is set to p0304. For synchronous motors, parameter r0339 = p0304 is displayed. If p0304 = 0, then r0339 is calculated from p0305 and p0316.		
p0340	Automatic calculation, control parameters / Calc auto par		
A_INF, S_INF	Can be changed: T Data type: Integer16 P-Group: Closed-loop control Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Setting to reset and automatically calculate filter and control (closed-loop) parameters.		
Value:	0: No calculation 1: Complete re-calculation. of control parameters with COMM data 2: Reset control parameters		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0. The following parameters are influenced using p0340: p0340 = 1: --> All of the parameters influenced for p0340 = 2 --> p3421 = p0223, p0225 --> p3422 = p0227 --> p3424 = p0225 --> p3415, p3425, p3555, p3614, p3620, p3622 are reset to the factory setting dependent on the particular unit. p0340 = 2: --> p3560, p3562, p3564, p3603, p3615 and p3617 are reset to the factory setting. For S_INF, these control parameters are not available		
Note:	When quick commissioning is exited using p3900 > 0, p0340 is automatically set to 1. At the end of the calculations, p0340 is automatically set to 0.		
p0340[0...n]	Automatic calculation, motor/control parameters / Calc auto par		
SERVO, VECTOR	Can be changed: C2(3), T Data type: Integer16 P-Group: Motor Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 5	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Setting to automatically calculate motor parameters and V/f open-loop and closed-loop control parameters from the rating plate data.		
Value:	0: No calculation 1: Complete calculation 2: Calculation of equivalent circuit diagram parameters 3: Calculation of closed-loop control parameters 4: Calculation of controller parameters 5: Calculation of technological limits and threshold values		

Notice:

After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.

The following parameters are influenced using p0340:

The parameters designated with (*) are not overwritten for catalog motors (p0300 > 100).

SERVO:

p0340 = 1:

--> All of the parameters influenced for p0340 = 2, 3, 4, 5

--> p0341 (*)

--> p0342, p0344, p0600, p0640, p1082, p2000, p2001, p2002, p2003, p2005, p2007

p0340 = 2:

--> p0350 (*), p0354 (*), p0356 (*), p0358 (*), p0360 (*)

--> p0625 (matching p0350)

p0340 = 3:

--> All of the parameters influenced for p0340 = 4, 5

--> p0325 (is only calculated for p0325 = 0)

--> p0348 (*) (is only calculated for p0348 = 0)

--> p0441, p0442, p0443, p0444, p0445 (only for 1FT6, 1FK6, 1FK7 motors)

--> p0492, p1082, p1980, p1319, p1326, p1327, p1612, p1752, p1755

p0340 = 4:

--> p1441, p1460, p1462, p1463, p1464, p1465, p1470, p1472, p1590, p1592, p1596, p1656, p1657, p1658, p1659, p1715, p1717

--> p1461 (for p0348 > p0322, p1461 is set to 100 %)

--> p1463 (for p0348 > p0322, p1463 is set to 400 %)

p0340 = 5:

--> p1037, p1038, p1520, p1521, p1530, p1531, p2140 ... p2142, p2148, p2150, p2155, p2161, p2162, p2163, p2164, p2175, p2177, p2194, p3820 ... p3829

VECTOR:

p0340 = 1:

--> All of the parameters influenced for p0340 = 2, 3, 4, 5

--> p0341 (*)

--> p0342, p0344, p0600, p0640, p1082, p1231, p1232, p1333, p1349, p1441, p1442, p1576, p1577, p1609, p1619, p1620, p1621, p1654, p1726, p1825, p1828 ... p1832, p1909, p1959, p2000, p2001, p2002, p2003, p2005, p2007, p3927, p3928

p0340 = 2:

--> p0350 (*), p0354 ... p0361 (*), p0652 ... p0660

--> p0625 (matching p0350)

p0340 = 3:

--> All of the parameters influenced for p0340 = 4, 5

--> p0346, p0347, p0492, p0622, p1262, p1320 ... p1327, p1582, p1584, p1612, p1616, p1744, p1748, p1749, p1755, p1756, p2178

p0340 = 4:

--> p1290, p1292, p1293, p1299, p1338, p1339, p1340, p1341, p1345, p1346, p1460, p1461, p1462, p1463, p1464, p1465, p1470, p1472, p1590, p1592, p1600, p1628, p1629, p1630, p1643, p1703, p1715, p1717, p1740, p1756, p1757, p1760, p1761, p1764, p1767, p1781, p1783, p1785, p1786, p1795, p7036, p7037, p7038

p0340 = 5:

--> p260 ... p264, p1037, p1038, p1520, p1521, p1530, p1531, p1574, p1750, p1802, p1803, p2140, p2142, p2148, p2150, p2161, p2162, p2163, p2164, p2175, p2177, p2194, p3207, p3208, p3815, p3820 ... p3829

Note:

p0340 = 1 contains the calculations of p0340 = 2, 3, 4, 5 without overwriting the motor parameters from the Siemens motor lists (p0301 > 0).

p0340 = 2 calculates the motor parameters (p0350 ... p0360), but only if it does involve a Siemens catalog motor (p0301 = 0).

p0340 = 3 contains the calculations of p0340 = 4, 5.

p0340 = 4 only calculates the controller parameters.

p0340 = 5 only calculates the controller limits.

When quick commissioning is exited using p3900 > 0, p0340 is automatically set to 1.

At the end of the calculations, p0340 is automatically set to 0.

If the STARTER commissioning software writes a 3 into p0340 when "downloading to target device", then this corresponds to a "complete calculation of the motor/control parameters without equivalent circuit diagram data". The same calculations are carried out as for p0340 = 1, however, without the equivalent circuit diagram parameters of the motor (p0340 = 2), the motor moment of inertia (p0341) and the motor weight (p0344).

For third-party linear synchronous motors (p0300 = 4) equivalent circuit diagram data are not calculated (p0340 = 2).

p0341[0...n]		Motor moment of inertia / Mot M_mom of inert	
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 1700, 5042, 5210, 6030, 6031
	P-Group: Motor	Units group: 25_1	Unit selection: p0100
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.000000 [kgm²]	Max 100000.000000 [kgm²]	Factory setting 0.000000 [kgm²]
Description:	Sets the motor moment of inertia (without load).		
Dependency:	IEC drives (p0100 = 0): unit kg m^2 NEMA drives (p0100 = 1): unit lb ft^2 The parameter value is included, together with p0342, in the rated starting time of the motor. Refer to: p0342, r0345		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	SERVO: p0341 * p0342 + p1498 influence the speed/torque pre-control in encoderless operation. VECTOR: The product of p0341 * p0342 is used when the speed controller (p0340 = 4) is calculated automatically.		

p0341[0...n]	Motor weight / Mot weight		
SERVO (Lin)	Can be changed: C2(3), U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 5042, 5210
	P-Group: Motor	Units group: 27_1	Unit selection: p0100
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.000000 [kg]	Max 10000.000000 [kg]	Factory setting 0.000000 [kg]
Description:	Sets the high moments of inertia (without load).		
Dependency:	IEC drives (p0100 = 0): unit kg m^2		
	NEMA drives (p0100 = 1): unit lb ft^2		
	The parameter value is included, together with p0342, in the rated starting time of the motor.		
	Refer to: p0342, r0345		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	This parameter is automatically preset for motors from the motor list (p0301).		
	SERVO:		
	p0341 * p0342 + p1498 influence the speed/torque pre-control in encoderless operation.		

p0342[0...n]	Ratio between the total and motor moment of inertia / Mot MomInert Ratio		
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 1700, 5042, 5210, 6030, 6031
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 1.000	Max 10000.000	Factory setting 1.000
Description:	Sets the ratio between the total moment of inertia/mass (load + motor) and the intrinsic motor moment of inertia/mass (no load).		
Dependency:	This means that together with p0341, the rated starting (accelerating time) of the motor is calculated for a vector drive. Refer to: p0341, r0345, p1498		
Note:	SERVO: p0341 * p0342 + p1498 influence the speed/torque pre-control in encoderless operation. VECTOR: The product of p0341 * p0342 is used when the speed controller (p0340 = 4) is calculated automatically.		
p0342[0...n]	Ratio between the total and motor force of inertia / Mot MomInert Ratio		
SERVO (Lin)	Can be changed: C2(3), U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 5042, 5210
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 1.000	Max 10000.000	Factory setting 1.000
Description:	Sets the ratio between the total force of inertia/mass (load + motor) and the intrinsic motor force of inertia/mass (no load).		
Dependency:	This means that together with p0341, the rated starting (accelerating time) of the motor is calculated for a vector drive. Refer to: p0341, r0345, p1498		
Note:	SERVO: p0341 * p0342 + p1498 influence the velocity/force pre-control in encoderless operation. VECTOR: The product of p0341 * p0342 is used when the speed controller (p0340 = 4) is calculated automatically.		
r0343[0...n]	Identified motor rated current / Mot I_{rated} ident		
VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [Arms]	Max 10000.00 [Arms]	Factory setting - [Arms]
Description:	Identified motor rated current		

p0344[0...n]	Motor weight (for the thermal motor model) / Mot weight th mod		
SERVO, VECTOR	Can be changed: C2(3), T Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL, FEM Min 0.0 [kg]	Calculated: CALC_MOD_ALL Dynamic index: MDS, p0130 Units group: 27_1 Scaling: - Max 50000.0 [kg]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.0 [kg]
Description:	Sets the motor weight.		
Dependency:	IEC drives (p0100 = 0): unit kg NEMA drives (p0100 = 1): unit lb		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The parameter influences the thermal 3 mass model of the induction motor. The parameter is not used for synchronous motors (p0300 = 2xx).		
r0345[0...n]	Nominal motor starting time / Mot t_start_rated		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: REL Min - [s]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max - [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [s]
Description:	Displays the rated motor starting time. This time corresponds to the time from standstill up to reaching the motor rated speed and the acceleration with motor rated torque (r0333).		
Dependency:	Refer to: r0313, r0333, r0336, p0341, p0342		
p0346[0...n]	Motor excitation build-up time / Mot t_excitation		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.000 [s]	Calculated: CALC_MOD_REG Dynamic index: MDS, p0130 Units group: - Scaling: - Max 20.000 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the excitation build-up time of the motor. This involves the delay time between enabling the pulses and enabling the ramp-function generator. The induction motor is magnetized during this time.		
Caution:	If there is insufficient magnetization under load or if the acceleration rate is too high, then an induction motor can stall (refer to the note). This is especially true for encoderless vector control or V/f control.		
			
Notice:	If the parameter is set to 0 s for separately-excited synchronous motors (p0300 = 5), then an excitation current setpoint is generated even if the drive is powered down. In the base speed range, this is the no-load excitation current (p0389). In the field-weakening range, the value is reduced with the inverse value of the actual speed. An excitation current setpoint is not generated during demagnetizing (p0347) and if an encoder fault is detected.		
Note:	The parameter is calculated using p0340 = 1, 3. For induction motors, the result depends on the rotor time constant (r0384). If this time is excessively reduced, this can result in an inadequate magnetizing of the induction motor. This is the case if the current limit is reached while building up magnetizing. For induction motors, the parameter cannot be set to 0 s (internal limit: 0.1 * r0384). For permanent-magnet synchronous motors and vector control, the value depends on the stator time constant (r0386). Here, it defines the time to establish the current for encoderless operation immediately after the pulses have been enabled.		

p0347[0...n]	Motor de-excitation time / Mot t_de-excitat.		
SERVO	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.000 [s]	Calculated: CALC_MOD_REG Dynamic index: MDS, p0130 Units group: - Scaling: - Max 20.000 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the demagnetizing time (for induction motors) after the inverter pulses have been canceled. The inverter pulses cannot be switched in (enabled) within this delay time. For SERVO, the de-excitation time is only used for the DC current brake.		
Note:	The parameter is calculated using p0340 = 1, 3. For induction motors, the result depends on the rotor time constant (r0384). if this time is shortened too much, then this can result in an inadequate demagnetizing of the induction motor and in an overcurrent condition when the pulses are subsequently enabled (only when the flying restart function is activated and the motor is rotating).		
p0347[0...n]	Motor de-excitation time / Mot t_de-excitat.		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.000 [s]	Calculated: CALC_MOD_REG Dynamic index: MDS, p0130 Units group: - Scaling: - Max 20.000 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the demagnetizing time (for induction motors) after the inverter pulses have been canceled. The inverter pulses cannot be switched in (enabled) within this delay time.		
Note:	The parameter is calculated using p0340 = 1, 3. For induction motors, the result depends on the rotor time constant (r0384). if this time is shortened too much, then this can result in an inadequate demagnetizing of the induction motor and in an overcurrent condition when the pulses are subsequently enabled (only when the flying restart function is activated and the motor is rotating).		
p0348[0...n]	Speed at the start of field weakening Vdc = 600 V / Mot n_field weaken		
SERVO	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [rpm]	Calculated: CALC_MOD_REG Dynamic index: MDS, p0130 Units group: - Scaling: - Max 210000.0 [rpm]	Access level: 3 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting 0.0 [rpm]
Description:	Sets the speed at the start of field weakening for a DC link voltage of 600 V.		
Dependency:	Refer to: p0320, r0331		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		

p0348[0...n]	Velocity at the start of field weakening Vdc = 600 V / Mot v_field weaken		
SERVO (Lin)	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [m/min]	Calculated: CALC_MOD_REG Dynamic index: MDS, p0130 Units group: - Scaling: - Max 1000.0 [m/min]	Access level: 3 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting 0.0 [m/min]
Description:	Sets the velocity at the start of field weakening for a DC link voltage of 600 V.		
Dependency:	Refer to: p0320, r0331		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
p0349	System of units, motor equivalent circuit diagram data / Unit_sys mot ESB		
SERVO, VECTOR	Can be changed: C2(3) Data type: Integer16 P-Group: Motor Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the actual system of units for motor equivalent circuit diagram data.		
Value:	1: System of units, physical 2: System of units, referred		
Dependency:	Refer to: p0304, p0305, p0310		
Note:	The reference parameter for resistances of the rated motor impedance $Z = p0304 / (1.732 * p0305)$ is in the % units system. Inductances are converted into a resistance using the factor $2 * \pi * p0310$. If a reference parameter (p0304, p305, p0310) is zero, then it is not possible to make a changeover to "referred" values (per unit values).		
p0350[0...n]	Motor stator resistance, cold / Mot R_stator cold		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.00000 [Ohm]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max 2000.00000 [Ohm]	Access level: 2 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [Ohm]
Description:	Sets the stator resistance of the motor at ambient temperature p0625.		
Dependency:	Refer to: p0625, r1912		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The motor identification routine determines the stator resistance from the total stator resistance minus the cable resistance (p0352).		

p0352[0...n]	Cable resistance / Mot R_cable cold		
SERVO	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.00000 [Ohm]	Calculated: - Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max 120.00000 [Ohm]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [Ohm]
Description:	Resistance of the power cable between the Motor Module and motor.		
Caution:	The cable resistance should be entered prior to motor data identification. If it is used subsequently, the difference by which p0352 was changed must be subtracted from the stator resistance p0350 or motor data identification must be repeated.		
			
Note:	The parameter influences the temperature adaptation of the stator resistance. The motor identification routine does not change the cable resistance. This is subtracted from the total measured stator resistance in order to calculate the stator resistance (p0350, p0352). The cable resistance is reset when quick commissioning is exited with p3900 > 0.		

p0352[0...n]	Cable resistance / Mot R_cable cold		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.00000 [Ohm]	Calculated: - Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max 120.00000 [Ohm]	Access level: 2 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [Ohm]
Description:	Resistance of the power cable between the Motor Module and motor.		
Dependency:	Refer to: p7003		
Caution:	The cable resistance should be entered prior to motor data identification. If it is used subsequently, the difference by which p0352 was changed must be subtracted from the stator resistance p0350 or motor data identification must be repeated.		
			
Notice:	Parallel circuits with one winding system (p7003 = 0): p0352 includes the feeder cable resistance of an individual Motor Module. The total feeder cable resistance is obtained from p0352 divided by the number of activated Motor Modules (refer to r0395). Parallel circuits with multi-winding system (p7003 = 1): p0352 includes the complete feeder cable resistance and is directly added to the stator resistance (refer to r0395).		
Note:	The parameter influences the temperature adaptation of the stator resistance. The motor identification sets the cable resistance to 20% of the measured total resistance if p0352 is zero at the time that the measurement is made. If p0352 is not zero, then the value is subtracted from the measured total stator resistance to calculate stator resistance p0350. In this case, p0350 is a minimum of 10% of the measured value. Exception: For parallel circuit configurations with one winding system (p07003 = 0), the cable resistance is directly measured. It is important to note that only the component of an individual Motor Module is entered into p0352. The cable resistance is reset when quick commissioning is exited with p3900 > 0.		

p0353[0...n]	Motor series inductance / Mot L_series		
SERVO	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.000 [mH]	Calculated: - Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 1000000.000 [mH]	Access level: 2 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.000 [mH]
Description:	Sets the series inductance.		
Note:	For the automatic calculation with p0340 = 1 or 3, the calculation of p0348 is influenced by p0353 if p0348 was 0. For the automatic calculation with p0340 = 1, 3 or 4, the calculation of p1715 is influenced by p0353. The series inductance is reset when quick commissioning is exited with p3900 > 0. The reference value for p0326 is inversely proportional to the leakage inductance of the motor (p0353, p0354, p0356).		

p0353[0...n]	Motor series inductance / Mot L_series		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.000 [mH]	Calculated: - Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 1000000.000 [mH]	Access level: 2 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.000 [mH]
Description:	Sets the series inductance.		
Note:	For the automatic calculation with p0340 = 1, 3 or 4, the calculation of p1715 is influenced by p0353. The series inductance is reset when quick commissioning is exited with p3900 > 0.		

p0354[0...n]	Motor rotor resistance cold / damping resistance d axis / Mot R_r cold / RDd		
SERVO	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL Min 0.00000 [Ohm]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max 300.00000 [Ohm]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [Ohm]
Description:	Sets the rotor/secondary section resistance of the motor at the ambient temperature p0625. For separately-excited synchronous motors: Sets the damping resistance in the rotor direction (d-axis). This parameter value is automatically calculated using the motor model (p0340 = 1, 2) or using the motor identification routine (p1910) (not for separately-excited synchronous motors).		
Dependency:	Refer to: p0625		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx). The reference value for p0326 is inversely proportional to the leakage inductance of the motor (p0353, p0354, p0356).		

p0354[0...n]	Motor rotor resistance cold / damping resistance d axis / Mot R_r cold / RDd		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL Min 0.00000 [Ohm]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max 300.00000 [Ohm]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [Ohm]
Description:	Sets the rotor/secondary section resistance of the motor at the ambient temperature p0625. For separately-excited synchronous motors: Sets the damping resistance in the rotor direction (d-axis). This parameter value is automatically calculated using the motor model (p0340 = 1, 2) or using the motor identification routine (p1910) (not for separately-excited synchronous motors).		
Dependency:	Refer to: p0625		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx).		
p0355[0...n]	Motor damping resistance, q axis / Mot R_{damp} q		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 0.00000 [Ohm]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max 300.00000 [Ohm]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [Ohm]
Description:	Sets the damping resistance of the separately-excited synchronous motor quadrature to the rotor direction (q axis). This parameter value is automatically calculated using the motor model (p0340 = 1, 2).		
p0356[0...n]	Motor stator leakage inductance / Mot L_{stator} leak.		
SERVO	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.00000 [mH]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 1000.00000 [mH]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [mH]
Description:	Induction motor, separately-excited synchronous motor: Sets the rotor leakage inductance of the motor. Synchronous motor: Sets the stator quadrature axis inductance of the motor. This parameter value is automatically calculated using the motor model (p0340 = 1, 2) or using the motor identification routine (p1910).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	This parameter is automatically preset for motors from the motor list (p0301). The reference value for p0326 is inversely proportional to the leakage inductance of the motor (p0353, p0354, p0356).		

p0356[0...n]	Motor stator leakage inductance / Mot L_stator leak.		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.00000 [mH]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 1000.00000 [mH]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [mH]
Description:	Induction motor, separately-excited synchronous motor: Sets the rotor leakage inductance of the motor. Synchronous motor: Sets the stator quadrature axis inductance of the motor. This parameter value is automatically calculated using the motor model (p0340 = 1, 2) or using the motor identification routine (p1910).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	This parameter is automatically preset for motors from the motor list (p0301). If the stator leakage inductance (p0356) for induction motors is changed outside the commissioning phase (p0010 > 0), the magnetizing inductance (p0360) is automatically adapted to the new EMF (r0337). You are then advised to repeat the measurement for the saturation characteristic (p1960). For permanent-magnet synchronous motors (p0300 = 2), this is the non-saturated value and is, therefore, ideal for a low current.		
p0357[0...n]	Motor stator inductance, d axis / Mot L_stator d		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.00000 [mH]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 1000.00000 [mH]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [mH]
Description:	Sets the stator direct-axis inductance of the synchronous motor. This parameter value is automatically calculated using the motor model (p0340 = 1, 2) or using the motor identification routine (p1910).		
Note:	The parameter is not used for separately-excited synchronous motors (p0300 = 5). For permanent-magnet synchronous motors (p0300 = 2), this is the non-saturated value and is ideal for a low current.		
p0358[0...n]	Motor rotor leakage inductance / damping inductance, d axis / Mot L_r leak / LDd		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL Min 0.00000 [mH]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 1000.00000 [mH]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [mH]
Description:	Sets the rotor/secondary section leakage inductance of the motor. For separately-excited synchronous motors: Sets the damping inductance in the rotor direction (d-axis). This value is automatically calculated using the motor model (p0340 = 1, 2) or using the motor identification routine (p1910) (not for separately-excited synchronous motors).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx). VECTOR: If the rotor leakage inductance (p0358) for induction motors is changed outside the commissioning phase (p0010 > 0), then the magnetizing inductance (p0360) is automatically adapted to the new EMF (r0337). You are then advised to repeat the measurement for the saturation characteristic (p1960).		

p0359[0...n]	Motor damping inductance, q axis / Mot L_damp q		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 0.00000 [mH]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 1000.00000 [mH]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [mH]
Description:	Sets the damping inductance of the separately-excited synchronous motor quadrature to the rotor direction (q axis). This parameter value is automatically calculated using the motor model (p0340 = 1, 2).		
p0360[0...n]	Motor magnetizing inductance/magn. inductance, d axis saturated / Mot Lh/Lh d sat		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL Min 0.00000 [mH]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 10000.00000 [mH]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [mH]
Description:	Sets the magnetizing inductance of the motor. For separately-excited synchronous motors: Sets the saturated magnetizing inductance in the rotor direction (d-axis). This parameter value is automatically calculated using the motor model (p0340 = 1, 2) or using the motor identification routine (p1910) (not for separately-excited synchronous motors).		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx).		
p0361[0...n]	Motor magnetizing inductance q axis, saturated / Mot L_magn q sat		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 0.00000 [mH]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 10000.00000 [mH]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [mH]
Description:	Sets the saturated magnetizing inductance of the separately-excited synchronous motor quadrature to the rotor direction (q axis). This parameter value is automatically calculated using the motor model (p0340 = 1, 2).		

p0362[0...n] Saturation characteristic flux 1 / Mot saturat.flux 1			
VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 10.0 [%]	Max 300.0 [%]	Factory setting 60.0 [%]
Description:	The saturation characteristic (flux as mapping of current) is defined using 4 points.		
	This parameter specifies the y coordinate (flux) for the 1st value pair of the characteristic.		
	Induction motors (ASM) and separately-excited synchronous motors (SESM):		
	The saturation characteristic describes the mapping of the magnetizing current onto the motor flux.		
	The parameter sets the first motor flux as a [%] referred to the rated motor flux.		
	Permanent-magnet synchronous motors (PESM):		
	The saturation characteristic describes the mapping of the stator quadrature axis current onto the stator quadrature axis flux.		
	The parameter sets the first stator quadrature axis flux as a [%] referred to the product of the unsaturated quadrature inductance (p0356) and the rated motor current.		
	The following applies for the flux values:		
	p0362 < p0363 < p0364 < p0365		
Dependency:	The following applies for the stator quadrature axis flux values (PESM):		
	20% < p0362 < p0363 < p0364 < p0365		
	Refer to: p0366		
	Caution:		
Note:	For permanent-magnet synchronous motors (PESM):		
	If the parameters are not set as specified, i.e. not in ascending order and to more than 20%, the unsaturated quadrature inductance (p0356) is always used for quadrature axis flux calculation purposes.		
	For induction motors, p0362 = 100 % corresponds to the rated motor flux.		
	For separately-excited synchronous motors p0362 = 100% corresponds to an induced terminal voltage with the magnitude of the rated motor voltage (under no-load conditions at the synchronous speed).		
	With permanent-magnet synchronous motors, p0362 = 100% corresponds to the product of the unsaturated quadrature inductance (p0356) and the rated motor current (p0305).		
	When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		
p0363[0...n] Saturation characteristic flux 2 / Mot saturat.flux 2			
VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 10.0 [%]	Max 300.0 [%]	Factory setting 85.0 [%]
Description:	The saturation characteristic (flux as mapping of current) is defined using 4 points.		
	This parameter specifies the y coordinate (flux) for the 1st value pair of the characteristic.		
	Induction motors (ASM) and separately-excited synchronous motors (SESM):		
	The saturation characteristic describes the mapping of the magnetizing current onto the motor flux.		
	The parameter sets the second motor flux as a [%] referred to the rated motor flux.		
	Permanent-magnet synchronous motors (PESM):		
	The saturation characteristic describes the mapping of the stator quadrature axis current onto the stator quadrature axis flux.		
The parameter sets the second stator quadrature axis flux as a [%] referred to the product of the unsaturated quadrature inductance (p0356) and the rated motor current.			

Dependency: The following applies for the flux values:
 $p0362 < p0363 < p0364 < p0365$
The following applies for the stator quadrature axis flux values (PESM):
 $20\% < p0362 < p0363 < p0364 < p0365$
Refer to: p0367

Caution: For permanent-magnet synchronous motors (PESM):
If the parameters are not set as specified, i.e. not in ascending order and to more than 20%, the unsaturated quadrature inductance (p0356) is always used for quadrature axis flux calculation purposes.

Note: For induction motors, $p0363 = 100\%$ corresponds to the rated motor flux.
For separately-excited synchronous motors $p0363 = 100\%$ corresponds to an induced terminal voltage with the magnitude of the rated motor voltage (under no-load conditions at the synchronous speed).
With permanent-magnet synchronous motors, $p0362 = 100\%$ corresponds to the product of the unsaturated quadrature inductance (p0356) and the rated motor current (p0305).
When quick commissioning is exited with $p3900 > 0$, then the parameter is reset if a catalog motor has not been selected (refer to p0300).

p0364[0...n]		Saturation characteristic flux 3 / Mot saturat.flux 3	
VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 10.0 [%]	Max 300.0 [%]	Factory setting 115.0 [%]
Description:	The saturation characteristic (flux as mapping of current) is defined using 4 points. This parameter specifies the y coordinate (flux) for the 1st value pair of the characteristic. Induction motors (ASM) and separately-excited synchronous motors (SESM): The saturation characteristic describes the mapping of the magnetizing current onto the motor flux. The parameter sets the third motor flux as a [%] referred to the rated motor flux. Permanent-magnet synchronous motors (PESM): The saturation characteristic describes the mapping of the stator quadrature axis current onto the stator quadrature axis flux. The parameter sets the third stator quadrature axis flux as a [%] referred to the product of the unsaturated quadrature inductance (p0356) and the rated motor current.		
Dependency:	The following applies for the flux values: p0362 < p0363 < p0364 < p0365 The following applies for the stator quadrature axis flux values (PESM): 20% < p0362 < p0363 < p0364 < p0365 Refer to: p0368		
Caution:	For permanent-magnet synchronous motors (PESM): If the parameters are not set as specified, i.e. not in ascending order and to more than 20%, the unsaturated quadrature inductance (p0356) is always used for quadrature axis flux calculation purposes.		
Note:	For induction motors, p0364 = 100 % corresponds to the rated motor flux. For separately-excited synchronous motors p0364 = 100% corresponds to an induced terminal voltage with the magnitude of the rated motor voltage (under no-load conditions at the synchronous speed). With permanent-magnet synchronous motors, p0362 = 100% corresponds to the product of the unsaturated quadrature inductance (p0356) and the rated motor current (p0305). When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		

p0365[0...n]	Saturation characteristic flux 4 / Mot saturat.flux 4		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 10.0 [%]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 125.0 [%]
Description:	The saturation characteristic (flux as mapping of current) is defined using 4 points. This parameter specifies the y coordinate (flux) for the 1st value pair of the characteristic. Induction motors (ASM) and separately-excited synchronous motors (SESM): The saturation characteristic describes the mapping of the magnetizing current onto the motor flux. The parameter sets the fourth motor flux as a [%] referred to the rated motor flux. Permanent-magnet synchronous motors (PESM): The saturation characteristic describes the mapping of the stator quadrature axis current onto the stator quadrature axis flux. The parameter sets the fourth stator quadrature axis flux as a [%] referred to the product of the unsaturated quadrature inductance (p0356) and the rated motor current.		
Dependency:	The following applies for the flux values: p0362 < p0363 < p0364 < p0365 The following applies for the stator quadrature axis flux values (PESM): 20% < p0362 < p0363 < p0364 < p0365 Refer to: p0369		
Caution:	For permanent-magnet synchronous motors (PESM): If the parameters are not set as specified, i.e. not in ascending order and to more than 20%, the unsaturated quadrature inductance (p0356) is always used for quadrature axis flux calculation purposes.		
Note:	For induction motors, p0365 = 100 % corresponds to the rated motor flux. For separately-excited synchronous motors p0365 = 100% corresponds to an induced terminal voltage with the magnitude of the rated motor voltage (under no-load conditions at the synchronous speed). With permanent-magnet synchronous motors, p0362 = 100% corresponds to the product of the unsaturated quadrature inductance (p0356) and the rated motor current (p0305). When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		
p0366[0...n]	Saturation characteristic I_mag 1 / Mot sat. I_mag 1		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 5.0 [%]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 800.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 50.0 [%]
Description:	The saturation characteristic (flux as mapping of current) is defined using 4 points. This parameter specifies the x coordinate for the 1st value pair of the characteristic. Induction motors (ASM) and separately-excited synchronous motors (SESM): The saturation characteristic describes the mapping of the magnetizing current onto the motor flux. The parameter sets the first magnetizing current as a [%] referred to the rated magnetizing current r0331 (ASM), which in turn is referred to the no-load excitation current (SESM). Permanent-magnet synchronous motors (PESM): The saturation characteristic describes the mapping of the stator quadrature axis current onto the stator quadrature axis flux. The parameter sets the first stator quadrature axis current as a [%] referred to the rated motor current (p0305).		

Dependency:	The following applies for the magnetizing currents: $p0366 < p0367 < p0368 < p0369$ The following applies for the stator quadrature axis current values (PESM): $20\% < p0366 < p0367 < p0368 < p0369$ Refer to: p0362
Caution:	For permanent-magnet synchronous motors (PESM): If the parameters are not set as specified, i.e. not in ascending order and to more than 20%, the unsaturated quadrature inductance (p0356) is always used for quadrature axis flux calculation purposes.
Note:	When quick commissioning is exited with $p3900 > 0$, then the parameter is reset if a catalog motor has not been selected (refer to p0300).

p0367[0...n]	Saturation characteristic I_mag 2 / Mot sat. I_mag 2		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 5.0 [%]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 800.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 75.0 [%]
Description:	The saturation characteristic (flux as mapping of current) is defined using 4 points. This parameter specifies the x coordinate for the 1st value pair of the characteristic. Induction motors (ASM) and separately-excited synchronous motors (SESM): The saturation characteristic describes the mapping of the magnetizing current onto the motor flux. The parameter sets the second magnetizing current as a [%] referred to the rated magnetizing current r0331 (ASM), which in turn is referred to the no-load excitation current (SESM). Permanent-magnet synchronous motors (PESM): The saturation characteristic describes the mapping of the stator quadrature axis current onto the stator quadrature axis flux. The parameter sets the second stator quadrature axis current as a [%] referred to the rated motor current (p0305).		
Dependency:	The following applies for the magnetizing currents: $p0366 < p0367 < p0368 < p0369$ The following applies for the stator quadrature axis current values (PESM): $20\% < p0366 < p0367 < p0368 < p0369$ Refer to: p0363		
Caution:	For permanent-magnet synchronous motors (PESM): If the parameters are not set as specified, i.e. not in ascending order and to more than 20%, the unsaturated quadrature inductance (p0356) is always used for quadrature axis flux calculation purposes.		
Note:	When quick commissioning is exited with $p3900 > 0$, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		

p0368[0...n]	Saturation characteristic I_mag 3 / Mot sat. I_mag 3		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 5.0 [%]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 800.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 150.0 [%]
Description:	The saturation characteristic (flux as mapping of current) is defined using 4 points. This parameter specifies the x coordinate for the 1st value pair of the characteristic. Induction motors (ASM) and separately-excited synchronous motors (SESM): The saturation characteristic describes the mapping of the magnetizing current onto the motor flux. The parameter sets the third magnetizing current as a [%] referred to the rated magnetizing current r0331 (ASM), which in turn is referred to the no-load excitation current (SESM). Permanent-magnet synchronous motors (PESM): The saturation characteristic describes the mapping of the stator quadrature axis current onto the stator quadrature axis flux. The parameter sets the third stator quadrature axis current as a [%] referred to the rated motor current (p0305).		
Dependency:	The following applies for the magnetizing currents: p0366 < p0367 < p0368 < p0369 The following applies for the stator quadrature axis current values (PESM): 20% < p0366 < p0367 < p0368 < p0369 Refer to: p0364		
Caution:	For permanent-magnet synchronous motors (PESM): If the parameters are not set as specified, i.e. not in ascending order and to more than 20%, the unsaturated quadrature inductance (p0356) is always used for quadrature axis flux calculation purposes.		
Note:	When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		
p0369[0...n]	Saturation characteristic I_mag 4 / Mot sat. I_mag 4		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 5.0 [%]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 800.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 210.0 [%]
Description:	The saturation characteristic (flux as mapping of current) is defined using 4 points. This parameter specifies the x coordinate for the 1st value pair of the characteristic. Induction motors (ASM) and separately-excited synchronous motors (SESM): The saturation characteristic describes the mapping of the magnetizing current onto the motor flux. The parameter sets the fourth magnetizing current as a [%] referred to the rated magnetizing current r0331 (ASM), which in turn is referred to the no-load excitation current (SESM). Permanent-magnet synchronous motors (PESM): The saturation characteristic describes the mapping of the stator quadrature axis current onto the stator quadrature axis flux. The parameter sets the fourth stator quadrature axis current as a [%] referred to the rated motor current (p0305).		
Dependency:	The following applies for the magnetizing currents: p0366 < p0367 < p0368 < p0369 The following applies for the stator quadrature axis current values (PESM): 20% < p0366 < p0367 < p0368 < p0369 Refer to: p0365		

Caution: For permanent-magnet synchronous motors (PESM):

If the parameters are not set as specified, i.e. not in ascending order and to more than 20%, the unsaturated quadrature inductance (p0356) is always used for quadrature axis flux calculation purposes.

Note: When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).

r0370[0...n] Motor stator resistance, cold / Mot R_stator cold			
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min - [Ohm]	Calculated: - Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max - [Ohm]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [Ohm]
Description:	Displays the motor stator resistance at an ambient temperature p0625. The value does not include the cable resistance.		
Dependency:	Refer to: p0625		
r0372[0...n] Cable resistance / Mot R_cable			
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min - [Ohm]	Calculated: - Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max - [Ohm]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [Ohm]
Description:	Displays the total cable resistance between Motor Module and motor, as well as the internal converter resistance.		
Dependency:	Refer to: r0238, p0352		
r0373[0...n] Motor rated stator resistance / Mot R_stator rated			
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL, FEM Min - [Ohm]	Calculated: - Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max - [Ohm]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [Ohm]
Description:	Displays the rated motor stator resistance at rated temperature (total of p0625 and p0627).		
Dependency:	Refer to: p0627		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx).		
r0374[0...n] Motor rotor resistance cold / damping resistance d axis / Mot R_r cold / RDd			
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL Min - [Ohm]	Calculated: - Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max - [Ohm]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [Ohm]
Description:	Displays the rotor/secondary section resistance of the motor for the ambient temperature p0625. For separately-excited synchronous motors: Displays the damping resistance in the rotor direction (d-axis).		
Dependency:	Refer to: p0625		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx).		

r0375[0...n]	Motor damping resistance, q axis / Mot R_damp q		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min - [Ohm]	Calculated: - Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max - [Ohm]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [Ohm]
Description:	Displays the damping resistance of the separately-excited synchronous motor quadrature to the rotor direction (q axis).		
r0376[0...n]	Rated motor rotor resistance / Mot R_rotor rated		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL, FEM Min - [Ohm]	Calculated: - Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max - [Ohm]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [Ohm]
Description:	Displays the rated (nominal) rotor/secondary section resistance of the motor at the rated temperature (total of p0625 and p0628).		
Dependency:	Refer to: p0628		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx).		
r0377[0...n]	Motor leakage inductance, total / Mot L_leak total		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min - [mH]	Calculated: - Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max - [mH]	Access level: 4 Func. diagram: 6640 Unit selection: p0349 Expert list: 1 Factory setting - [mH]
Description:	Induction motor: Displays the stator leakage inductance of the motor including the series inductance (p0353). Synchronous motor: Displays the stator quadrature axis inductance of the motor including the series inductance (p0353).		
r0377[0...n]	Motor leakage inductance, total / Mot L_leak total		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min - [mH]	Calculated: - Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max - [mH]	Access level: 4 Func. diagram: 6640 Unit selection: p0349 Expert list: 1 Factory setting - [mH]
Description:	Induction motor, separately-excited synchronous motor: Displays the stator leakage inductance of the motor, including the series inductance (p0353) and the motor reactor (p0233). Synchronous motor: Displays the stator quadrature inductance, including the series inductance (p0353) and the motor reactor (p0233).		

r0378[0...n]	Motor stator inductance, d axis / Mot L_stator_d		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min - [mH]	Calculated: - Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [mH]
Description:	Displays the stator longitudinal inductance of the synchronous motor including the series inductance (p0353) and the motor reactor (p0233).		
Note:	The parameter is not used for separately-excited synchronous motors (p0300 = 5).		
r0380[0...n]	Motor damping inductance, d axis / Mot L_damping_d		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min - [mH]	Calculated: - Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [mH]
Description:	Displays the damping inductance of the separately-excited synchronous motor in the rotor direction (d-axis).		
r0381[0...n]	Motor damping inductance, q axis / Mot L_damping_q		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min - [mH]	Calculated: - Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [mH]
Description:	Displays the damping inductance of a separately-excited synchronous motor quadrature to the rotor direction (q axis).		
r0382[0...n]	Motor magnetizing inductance transformed / Lh d axis saturated / Mot L_m tr/Lhd sat		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL Min - [mH]	Calculated: - Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [mH]
Description:	Displays the magnetizing inductance of the motor. For separately-excited synchronous motors: Displays the saturated magnetizing inductance in the rotor direction (d-axis).		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx).		
r0383[0...n]	Motor magnetizing inductance q axis, saturated / Mot L_magn q sat		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min - [mH]	Calculated: - Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting - [mH]
Description:	Displays the saturated magnetizing inductance of a separately-excited synchronous motor quadrature to the rotor direction (q axis).		

r0384[0...n]	Motor rotor time constant / damping time constant d axis / Mot T_rotor/T_Dd		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL Min - [ms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max - [ms]	Access level: 4 Func. diagram: 6722 Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the rotor time constant. For separately-excited synchronous motors: Displays the damping time constant in the rotor direction (d-axis).		
Note:	The parameter is not used for synchronous motors. The value is calculated from the total of the inductances on the rotor side (p0358, p0360) divided by the rotor/damping resistance (p0354). The temperature adaptation of the rotor resistance for induction motors is not taken into account.		
r0385[0...n]	Motor damping time constant, q axis / Mot T_Dq		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min - [ms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max - [ms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the damping time constant of a separately-excited synchronous motor quadrature to the rotor direction (q axis).		
Note:	The value is calculated from the total of the inductances on the damping side (p0359, p0361) divided by the damping resistance (p0355).		
r0386[0...n]	Motor stator leakage time constant / Mot T_stator leak		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min - [ms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max - [ms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the stator leakage time constant.		
Note:	The value is calculated from the total of all leakage inductances (p0233*, p0353, p0356, p0358) divided by the total of all motor resistances (p0350, p0352, p0354). The temperature adaptation of the resistances is not taken into account. * only applies for VECTOR (r0107).		
r0387[0...n]	Motor stator leakage time constant, q axis / Mot T_Sleak /T_Sq		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min - [ms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max - [ms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the stator leakage time constant quadrature to the rotor direction (q axis).		
Note:	The value is calculated from the total of all leakage inductances (p0233, p0356, p0359) divided by the total of all motor resistances (p0350, p0352, p0355). The temperature adaptation of the resistances is not taken into account.		

p0389[0...n]			
Excitation rated no-load current / Exc I_noload_rated			
VECTOR	Can be changed: C2(1, 3)	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 6727
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [A]	10000.00 [A]	0.00 [A]
Description:	Sets the rated no-load current (I_F0) for the excitation.		
p0390[0...n]			
Rated excitation current / Exc I_rated			
VECTOR	Can be changed: C2(1, 3)	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 6727
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [A]	10000.00 [A]	0.00 [A]
Description:	Setting the rated current (I_F) of the controlled excitation rectifier (DC master).		
p0391[0...n]			
Current controller adaptation, starting point KP / I_adapt pt KP			
SERVO	Can be changed: C2(3), U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 5714
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [Arms]	6000.00 [Arms]	0.00 [Arms]
Description:	Sets the starting point of the current-dependent current controller adaptation where the current controller gain p1715 is effective.		
Dependency:	Refer to: p0392, p0393, p1402, p1715		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	For p0393 = 100 % or p1402 bit 2 = 0, the current controller adaptation is disabled and p1715 is effective over the entire range.		
p0391[0...n]			
Current controller adaptation, starting point KP / I_adapt pt KP			
VECTOR	Can be changed: C2(3), U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 6714
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [Arms]	6000.00 [Arms]	0.00 [Arms]
Description:	Sets the starting point of the current-dependent current controller adaptation where the current controller gain p1715 is effective.		
Dependency:	Refer to: p0392, p0393, p1402, p1715		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		

p0392[0...n] Current controller adaptation, starting point KP adapted / I_adapt pt KP adap			
SERVO	Can be changed: C2(3), U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 5714
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [Arms]	Max 6000.00 [Arms]	Factory setting 0.00 [Arms]
Description:	Sets the starting point of the current-dependent current controller adaptation where the adapted current controller gain p1715 * p0393 is effective.		
Dependency:	Refer to: p0391, p0393, p1402, p1715		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	For p0393 = 100 % or p1402 bit 2 = 0, the current controller adaptation is disabled and p1715 is effective over the entire range.		
p0392[0...n] Current controller adaptation, starting point KP adapted / I_adapt pt KP adap			
VECTOR	Can be changed: C2(3), U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 6714
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [Arms]	Max 6000.00 [Arms]	Factory setting 0.00 [Arms]
Description:	Sets the starting point of the current-dependent current controller adaptation where the adapted current controller gain p1715 * p0393 is effective.		
Dependency:	Refer to: p0391, p0393, p1402, p1715		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		
p0393[0...n] Current controller adaptation p gain adaptation / I_adapt Kp adapt			
SERVO	Can be changed: C2(3), U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 5714
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 1000.00 [%]	Factory setting 100.00 [%]
Description:	Sets the factor for the current controller P gain in the adaptation range (current greater than p0392). The value is referred to p1715.		
Dependency:	Refer to: p0391, p0392, p1402, p1715		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	For p0393 = 100 % or p1402 bit 2 = 0, the current controller adaptation is disabled and p1715 is effective over the entire range.		

p0393[0...n]	Current controller adaptation P gain scaling / I_adapt Kp scal		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: REL Min 0.00 [%]	Calculated: CALC_MOD_REG Dynamic index: MDS, p0130 Units group: - Scaling: - Max 1000.00 [%]	Access level: 3 Func. diagram: 6714 Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the factor for the current controller P gain in the adaptation range (e.g. r0078 > p0392, if p0392 > p0391). The value is referred to p1715.		
Dependency:	Refer to: p0391, p0392, p1402, p1715		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	For p0393 = 100 % or p1402 bit 2 = 0, the current controller adaptation is disabled and p1715 is effective over the entire range. When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		
r0395[0...n]	Actual stator resistance / R_stator act		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min - [Ohm]	Calculated: - Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max - [Ohm]	Access level: 3 Func. diagram: 6300, 6730, 6731, 6732 Unit selection: p0349 Expert list: 1 Factory setting - [Ohm]
Description:	Displays the actual stator resistance (phase value). The parameter value also contains the temperature-independent cable resistance.		
Dependency:	In the case of induction motors the parameter is also affected by the motor temperature model. Refer to: p0350, p0352, p0620		
Note:	In each case, only the stator resistance of the active Motor Data Set is included with the stator temperature of the thermal motor model.		
r0396[0...n]	Actual rotor resistance / R_rotor act		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL, FEM Min - [Ohm]	Calculated: - Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max - [Ohm]	Access level: 3 Func. diagram: 6730 Unit selection: p0349 Expert list: 1 Factory setting - [Ohm]
Description:	Displays the actual rotor/secondary section resistance (phase value). The parameter is affected by the motor temperature model.		
Dependency:	Refer to: p0354, p0620		
Note:	In each case, only the rotor resistance of the active Motor Data Set is included with the rotor temperature of the thermal motor model. This parameter is not used for synchronous motors (p0300 = 2xx).		

p0398[0...n]	Angle magnet. decoupling (cross saturation) coeff 1 / Magn decoupl C1		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, FEM Min -10.000000	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10.000000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000000
Description:	The magnetic cross coupling of the motor's d and q axes caused by saturation (current-dependent) leads to an angle offset affecting the axis system d'q'; this decouples the magnetic quantities. The angle offset can be described as a third-order polynomial function of the load current consumed: $\text{phiOffset} = f(C1 \cdot i_q + C3 \cdot i_q^3)$ This parameter is the coefficient C1; it describes the linear load impact effect.		
p0399[0...n]	Angle magnet. decoupling (cross saturation) coeff 3 / Magn decoupl C3		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, FEM Min -10.000000	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10.000000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000000
Description:	The magnetic cross coupling of the motor's d and q axes caused by saturation (current-dependent) leads to an angle offset affecting the axis system d'q'; this decouples the magnetic quantities. The angle offset can be described as a third-order polynomial function of the load current consumed: $\text{phiOffset} = f(C1 \cdot i_q + C3 \cdot i_q^3)$ This parameter is the coefficient C3; it describes the cubic load impact effect.		
p0400[0...n]	Encoder type selection / Enc_typ sel		
ENCODER, SERVO, VECTOR	Can be changed: C2(1, 4) Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 10100	Access level: 1 Func. diagram: 1580, 4704 Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the encoder from the list of encoder types supported.		
Value:	0: No encoder 202: DRIVE-CLiQ encoder AS20, singleturn 204: DRIVE-CLiQ encoder AM20, multiturn 4096 242: DRIVE-CLiQ encoder AS24, singleturn 244: DRIVE-CLiQ encoder AM24, multiturn 4096 1001: Resolver 1 speed 1002: Resolver 2 speed 1003: Resolver 3 speed 1004: Resolver 4 speed 2001: 2048, 1 Vpp, A/B C/D R 2002: 2048, 1 Vpp, A/B R 2003: 256, 1 Vpp, A/B R 2004: 400, 1 Vpp, A/B R 2005: 512, 1 Vpp, A/B R 2006: 192, 1 Vpp, A/B R 2007: 480, 1 Vpp, A/B R 2008: 800, 1 Vpp, A/B R 2010: 18000, 1 Vpp, A/B R distance-coded 2051: 2048, 1 Vpp, A/B, EnDat, Multiturn 4096 2052: 32, 1 Vpp, A/B, EnDat, Multiturn 4096 2053: 512, 1 Vpp, A/B, EnDat, Multiturn 4096		

2054: 16, 1 Vpp, A/B, EnDat, Multiturn 4096
 2055: 2048, 1 Vpp, A/B, EnDat, Singleturn
 2081: 2048, 1 Vpp, A/B, SSI, Singleturn
 2082: 2048, 1 Vpp, A/B, SSI, Multiturn 4096
 2083: 2048, 1 Vpp, A/B, SSI, singleturn, error bit
 2084: 2048, 1 Vpp, A/B, SSI, multiturn 4096, error bit
 2110: 4000 nm, 1 Vpp, A/B R distance-coded
 2111: 20000 nm, 1 Vpp, A/B R distance-coded
 2112: 40000 nm, 1 Vpp, A/B R distance-coded
 2151: 16000 nm, 1 Vpp, A/B, EnDat, resolution 100 nm
 3001: 1024 HTL A/B R
 3002: 1024 TTL A/B R
 3003: 2048 HTL A/B R
 3005: 1024 HTL A/B
 3006: 1024 TTL A/B
 3007: 2048 HTL A/B
 3008: 2048 TTL A/B
 3009: 1024 HTL A/B unipolar
 3011: 2048 HTL A/B unipolar
 3020: 2048 TTL A/B R, with sense
 3081: SSI, Singleturn, 24 V
 3082: SSI, Multiturn 4096, 24 V
 3090: 4096, HTL, A/B, SSI, Singleturn
 3109: 2000 nm, TTL, A/B R distance-coded
 9999: User-defined
 10000: Identify encoder
 10050: Encoder with EnDat interface identified
 10051: DRIVE-CLiQ encoder identified
 10058: Digital encoder (absolute) identified
 10059: Digital encoder (incremental) identified
 10100: Identify encoder (waiting)

Caution: An encoder type with p0400 < 9999 defines an encoder for which there is an encoder parameter list. When selecting a catalog encoder (p0400 < 9999) the parameters from the encoder parameter list cannot be changed (write protection). To remove write protection, the encoder type should be set to a third-party encoder (p0400 = 9999).

Notice: If required, the list of motor codes/encoder codes can be found in the Appendix of the List Manual.

Note: The connected encoder can be identified by setting p0400 to 10000 or 10100. This assumes that the encoder supports this method, which is possible in the following cases: Motor with DRIVE-CLiQ, encoder with EnDat interface, DRIVE-CLiQ encoder.

The encoder data (e.g. pulse number p0408) can only be changed when p0400 = 9999.

When using an encoder with track A/B and zero pulse, as standard, fine synchronization is not set using a zero mark. If, for a synchronous motor, fine synchronization is to be realized using a zero mark, then the following must be executed:

- set p0400 to 9999
- set p0404.15 to 1

Prerequisite:

Coarse synchronization must be selected (e.g. pole position identification) and the zero pulse of the encoder must be either mechanically or electronically (p0431) adjusted to the pole position.

For p0400 = 10000 the following applies:

If an identification is not possible, then p0400 is set to 0.

For p0400 = 10100 the following applies:

If an identification is not possible, p0400 remains set to 10100 until it becomes possible.

p0401[0...n]	Encoder type, OEM selection / Enc type OEM sel		
ENCODER, SERVO, VECTOR	Can be changed: C2(1, 4) Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 32767	Access level: 2 Func. diagram: 1580, 4704 Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the encoder from the list of encoder types that the OEM supports.		
Note:	The connected encoder can be identified by p0400 = 10000. This means that the encoder must support this and is possible in the following cases: Motor with DRIVE-CLiQ, encoder with EnDat interface. If an identification is not possible, then p0400 is set to 0. The encoder data (e.g. pulse number p0408) can only be changed when p0400 = 9999. Using p0400 = 20000, the encoder type can be selected from the list of OEM encoders using p0401.		
p0402[0...n]	Gearbox type selection / Gearbox type sel		
ENCODER, SERVO, VECTOR	Can be changed: C2(1, 4) Data type: Integer16 P-Group: Encoder Not for motor type: - Min 1	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 10100	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 9999
Description:	Selects the gearbox type to preset the inversion and the gearbox factor. Measuring gear factor = motor or load revolutions / encoder revolutions.		
Value:	1: Gearbox 1:1 not inverted 2: Gearbox 2:7 inverted 3: Gearbox 4:17 inverted 4: Gearbox 2:10 inverted 9999: Gearbox, user-defined 10000: Identify gearbox 10100: Identify gearbox		
Dependency:	Refer to: p0410, p0432, p0433		
Note:	Re p0402 = 1: Automatic setting of p0410 = 0000 bin, p0432 = 1, p0433 = 1. Re p0402 = 2: Automatic setting of p0410 = 0011 bin, p0432 = 7, p0433 = 2. Re p0402 = 3: Automatic setting of p0410 = 0011 bin, p0432 = 17, p0433 = 4. Re p0402 = 4: Automatic setting of p0410 = 0011 bin, p0432 = 10, p0433 = 2. Re p0402 = 9999: No automatic setting of p0410, p0432, p0433. The parameters should be manually set. Re p0402 = 10000: It is only possible to identify the gearbox type for a motor with DRIVE-CLiQ. Parameters p0410, p0432 and p0433 are set corresponding to the identified gearbox. If an identification is not possible, then p0402 is set to 9999.		
p0404[0...n]	Encoder configuration effective / Enc_config eff		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 4010, 4704 Unit selection: - Expert list: 1 Factory setting 0000 bin
Description:	Settings for the basic encoder properties.		

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Linear encoder	Yes	No	-
	01	Abs value encoder	Yes	No	-
	02	Multiturn encoder	Yes	No	-
	03	Track A/B sq-wave	Yes	No	-
	04	Track A/B sinus	Yes	No	-
	05	Track C/D	Yes	No	-
	06	Hall sensor	Yes	No	-
	08	EnDat encoder	Yes	No	-
	09	SSI encoder	Yes	No	-
	10	DRIVE-CLiQ encoder	Yes	No	-
	11	Digital encoder	Yes	no	-
	12	Equidistant zero mark	Yes	No	-
	13	Irregular zero mark	Yes	No	-
	14	Distance-coded zero mark	Yes	No	-
	15	Commutation with zero mark (not ASM)	Yes	No	-
	16	Acceleration	Yes	No	-
	17	Track A/B analog	Yes	No	-
	20	Voltage level 5 V	Yes	No	-
	21	Voltage level 24 V	Yes	No	-
	22	Remote sense (only SMC30)	Yes	No	-
	23	Resolver excit.	Yes	No	-

Caution: This parameter is automatically preset for encoders from the encoder list (p0400).

When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.

Notice: If an SSI encoder (bit 9 = 1) is used as motor encoder for permanent-magnet synchronous motors, then this is only permissible in conjunction with an additional A/B track (bit 4 = 1 or bit 5 = 1).

Note: ZM: Zero mark

SMC: Sensor Module Cabinet

If a technique to determine the commutation information/data has not been selected (e.g. track C/D, Hall sensor), and the encoder pulse number is an integer multiple of the pole number, then the following applies:

The track A/B is adjusted to match the magnetic position of the motor.

Re bit 01, 02 (absolute encoder, multiturn encoder):

These bits can only be selected for EnDat encoders, SSI encoders or DRIVE-CLiQ encoders.

Re bit 10 (DRIVE-CLiQ encoder):

This bit is only used for the large-scale integrated DRIVE-CLiQ encoders that provide their encoder data directly in DRIVE-CLiQ format without converting this data. This bit is not, therefore, set for first-generation DRIVE-CLiQ encoders.

Re bit 12 (equidistant zero mark):

The zero marks occur at regular intervals (e.g. rotary encoder with 1 zero mark per revolution or linear encoder with constant zero mark distance).

The bit activates monitoring of the zero mark distance (p0424/p0425, linear/rotary) or in the case of the linear encoder with 1 zero mark and p0424 = 0 zero mark monitoring is activated.

Re bit 13 (irregular zero mark):

The zero marks occur at irregular intervals (e.g. a linear scale with only 1 zero mark in the traversing range). The zero mark distance is not monitored.

Re bit 14 (distance-coded zero mark):

The distance (clearance) between two or several consecutive zero marks allows the absolute position to be calculated.

Re bit 15 (commutation with zero mark):

Only applicable for synchronous motors. For distance-coded zero marks, the following applies:

The phase sequence of the C/D track (if available) must be the same as the phase sequence of the encoder (A/B track).

The phase sequence of the Hall signal (if available) must be the same as the phase sequence of the motor. Further, the position of the Hall sensor must be mechanically adjusted to the motor EMF.

The fine synchronization is only started after two zero marks have been passed.

p0405[0...n] Square-wave encoder track A/B / Sq-wave enc A/BENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** EDS, p0140**Func. diagram:** 4704**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**
1111 bin**Description:** Settings for the track A/B of a square-wave encoder. For square-wave encoders, p0404.3 must also be 1.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Signal	Bipolar	Unipolar	-
	01	Level	TTL	HTL	-
	02	Track monitoring	A/B <> -A/B	None	-
	03	Zero pulse	Same as A/B track	24 V unipolar	-
	04	Switching thresh	High	Low	-
	05	Pulse/direction	Yes	No	-

Caution: This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.**Note:** Re bit 02:

When the function is activated, track monitoring can be deactivated by setting p437.26.

p0407[0...n] Linear encoder grid division / Enc grid divENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** EDS, p0140**Func. diagram:** 4010, 4704**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**
16000 [nm]**Description:** Sets the grid division for a linear encoder.**Caution:** This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.**Note:** The lowest permissible value is 250 nm.**p0408[0...n] Rotary encoder pulse No. / Rot enc pulse No.**ENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** EDS, p0140**Func. diagram:** 4010, 4704**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**
2048**Description:** Sets the number of pulses for a rotary encoder.**Caution:** This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.**Note:** The number of pole pairs for a resolver is entered here.

The smallest permissible value is 1 pulse.

p0408**Rotary encoder pulse No. / Rot enc pulse No.**

TM41

Can be changed: C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 9674, 9676**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

1000

16384

2048

Description:

Sets the number of pulses for a rotary encoder.

Danger:

In the SINAMICS operating mode (p4400 = 1) the following applies:

The number of pulses per encoder revolution of the TM41 (p0408) must always precisely correspond to the number of pulses per encoder revolution of the encoder interconnected at connector input p4420. If this condition is not taken into account, then the TM41 zero mark is not in synchronism with the interconnected encoder.

Note:

TM41 SAC: order no. = 6SL3055-0AA00-3PA0

TM41 DAC: order no. = 6SL3055-0AA00-3PA1

The following applies for TM41 SAC:

- Every time the pulse number is changed, a POWER ON must be carried out for CU and TM41.

- Minimum/maximum value: 1000/8192

The following applies for TM41 DAC:

- Minimum/maximum value: 1000/16384

p0410[0...n]**Encoder inversion actual value / Enc inv act value**ENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** EDS, p0140**Func. diagram:** -**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

0000 bin

Description:

Setting to invert actual values.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Invert speed actual value	Yes	No	4710, 4715
01	Invert position actual value	Yes	No	4704

Note:

The inversion influences the following parameters:

Bit 00: r0061, r0063 (exception: encoderless control), r0094

Bit 01: r0482, r0483

p0410[0...n]**Encoder inversion actual value / Enc inv act value**ENCODER (Lin enc),
SERVO (Lin)**Can be changed:** C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** EDS, p0140**Func. diagram:** -**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

0000 bin

Description:

Setting to invert actual values.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Invert velocity actual value	Yes	No	4710, 4715
01	Invert position actual value	Yes	No	4704

Note:

The inversion influences the following parameters:

Bit 00: r0061, r0063 (exception: encoderless control), r0094

Bit 01: r0482, r0483

p0411[0...n]	Measuring gear, configuration / Meas gear config				
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: EDS, p0140	Func. diagram: -		
	P-Group: Encoder	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Sets the configuration for position tracking of a measuring gear.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Measuring gear, activate position tracking	Yes	No	-
	01	Axis type	Linear axis	Rotary axis	-
	02	Measuring gear, reset position	Yes	No	-
	03	Meas. gearbox, activate pos. tracking for incremental encoders	Yes	No	-
Notice:	For p0411.3 = 1 the following applies: If position tracking is activated for incremental encoders, only the position actual value is stored. Axis or encoder motion is not detected when deactivated! Any tolerance window entered in p0413 has no effect.				
Note:	For the following events, the non-volatile, saved position values are automatically reset: - when an encoder replacement has been identified. - when changing the configuration of the Encoder Data Set (EDS).				

p0412[0...n]	Measuring gear, absolute encoder, rotary, revolutions, virtual / Abs rot rev				
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: EDS, p0140	Func. diagram: -		
	P-Group: Encoder	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	0	4194303	0		
Description:	Sets the number of rotations that can be resolved for a rotary encoder with activated position tracking of the measuring gear.				
Dependency:	This parameter is only of significance for an absolute encoder (p0404.1 = 1) with activated position tracking (p0411.0 = 1) and for an incremental encoder with activated position tracking (p0411.3 = 1).				
Note:	The resolution that is set must be able to be represented using r0483. For rotary axes/modulo axes, the following applies: p0411.0 = 1: This parameter is preset with p0421 and can be changed. p0411.3 = 1: The parameter value is preset to the highest possible value. The highest possible value depends on the pulse number (p0408) and the fine resolution (p0419). For linear axes, the following applies: p0411.0 = 1: This parameter is pre-assigned with p0421, expanded by 6 bits for multiturn information (maximum number of overflows) and cannot be changed. p0411.3 = 1: The parameter value is preset to the highest possible value. The highest possible value depends on the pulse number (p0408) and the fine resolution (p0419).				

p0413[0...n] Measuring gear, position tracking tolerance window / Pos track window			
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: FloatingPoint32 P-Group: Encoder Not for motor type: - Min 0.00	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 4294967300.00	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00
Description:	Sets a tolerance window for position tracking. After the system is powered up, the difference between the saved position and the actual position is determined, and depending on this, the following is initiated: Difference within the tolerance window --> The position is reproduced as a result of the encoder actual value. Difference outside the tolerance window --> An appropriate message is output.		
Dependency:	Refer to: F31501, F32501, F33501		
Caution:	Rotation, e.g. through a complete encoder range is not detected.		
			
Note:	The value is entered in integer (complete) encoder pulses. For p0411.0 = 1, the value is automatically pre-assigned quarter of the encoder range. Example: Quarter of the encoder range = (p0408 * p0421) / 4 It is possible that the tolerance window may not be able to be precisely set due to the data type (floating point number with 23 bit mantissa).		

p0414[0...n] Redundant coarse position value relevant bits (identified) / Relevant bits			
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 16	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 16
Description:	Sets the number of relevant bits for the redundant coarse position value.		

p0415[0...n] Gx_XIST1 Coarse position safe most significant bit (identified) / Gx_XIST1 safe MSB			
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 31	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 14
Description:	Sets the bit number for the safe most significant bit (MSB) of the Gx_XIST1 coarse position.		
Note:	MSB: Most Significant Bit		

p0418[0...n]	Fine resolution Gx_XIST1 (in bits) / Enc fine Gx_XIST1		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned8 P-Group: Encoder Not for motor type: - Min 2	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 18	Access level: 3 Func. diagram: 4010, 4704 Unit selection: - Expert list: 1 Factory setting 11
Description:	Sets the fine resolution in bits of the incremental position actual values.		
Note:	The parameter applies for the following process data: - Gx_XIST1 - Gx_XIST2 for reference mark or flying measurement The fine resolution specifies the fraction between two encoder pulses. Depending on the physical measurement principle, an encoder pulse can be broken down into a different number of fractions (e.g. squarewave encoder: 2 bit = resolution 4, sin/cos encoder: Typical 11 bit = resolution 2048). For a squarewave encoder, with the factory setting, the least significant bits have the value zero, i.e. they do not supply any useful information. For especially high quality measuring systems, the fine resolution must be increased corresponding to the available accuracy.		
p0418	Fine resolution Gx_XIST1 (in bits) / Enc fine Gx_XIST1		
TM41	Can be changed: C2(4) Data type: Unsigned8 P-Group: Encoder Not for motor type: - Min 2	Calculated: - Dynamic index: - Units group: - Scaling: - Max 18	Access level: 3 Func. diagram: 9674, 9676 Unit selection: - Expert list: 1 Factory setting 11
Description:	Sets the fine resolution in bits of the incremental position actual values.		
Danger:	In the SINAMICS operating mode (p4400 = 1) the following applies: The fine resolution of the TM41 (p0418) must always precisely correspond to the fine resolution of the encoder interconnected at connector input p4420. If this condition is not taken into account, then the TM41 zero mark is not in synchronism with the interconnected encoder. Further, in this case, the frequency of the pulses for the encoder emulation differs from the pulse frequency of the leading encoder. For TM41 SAC (order no. = 6SL3055-0AA00-3PA0), the following applies: Every time the fine resolution has been changed, a POWER ON must be performed for the CU and TM41.		
			
p0419[0...n]	Fine resolution absolute value Gx_XIST2 (in bits) / Enc fine Gx_XIST2		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned8 P-Group: Encoder Not for motor type: - Min 2	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 18	Access level: 3 Func. diagram: 4704, 4710 Unit selection: - Expert list: 1 Factory setting 9
Description:	Sets the fine resolution in bits of the absolute position actual values.		
Dependency:	Refer to: p0418		
Note:	This parameter applies to process data Gx_XIST2 when reading the absolute value.		

p0420[0...n]					Encoder connection / Enc_connection				
ENCODER, SERVO, VECTOR		Can be changed: C2(4)		Calculated: -		Access level: 4			
		Data type: Unsigned16		Dynamic index: EDS, p0140		Func. diagram: -			
		P-Group: Encoder		Units group: -		Unit selection: -			
		Not for motor type: -		Scaling: -		Expert list: 1			
		Min		Max		Factory setting			
		-		-		0000 bin			
Description:					Selecting the encoder connection.				
Bit field:		Bit	Signal name	1 signal	0 signal	FP			
		00	SUB-D	Yes	No	-			
		01	Terminal	Yes	No	-			

p0421[0...n]					Absolute encoder rotary multiturn resolution / Enc abs multiturn				
ENCODER, SERVO, VECTOR		Can be changed: C2(4)		Calculated: -		Access level: 3			
		Data type: Unsigned16		Dynamic index: EDS, p0140		Func. diagram: 4704			
		P-Group: Encoder		Units group: -		Unit selection: -			
		Not for motor type: -		Scaling: -		Expert list: 1			
		Min		Max		Factory setting			
		0		65535		4096			
Description:					Sets the number of rotations that can be resolved for a rotary absolute encoder.				
Caution:					This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.				

p0422[0...n]					Absolute encoder linear measuring step resolution / Enc abs meas step				
ENCODER, SERVO, VECTOR		Can be changed: C2(4)		Calculated: -		Access level: 3			
		Data type: Unsigned32		Dynamic index: EDS, p0140		Func. diagram: 4704			
		P-Group: Encoder		Units group: -		Unit selection: -			
		Not for motor type: -		Scaling: -		Expert list: 1			
		Min		Max		Factory setting			
		0 [nm]		4294967295 [nm]		100 [nm]			
Description:					Sets the resolution of the absolute position for a linear absolute encoder.				
Caution:					This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.				
Note:					The serial protocol of an absolute encoder provides the position with a certain resolution , e.g. 100 nm. This value must be entered here.				

p0423[0...n]					Absolute encoder rotary singleturn resolution / Enc abs singleturn				
ENCODER, SERVO, VECTOR		Can be changed: C2(4)		Calculated: -		Access level: 3			
		Data type: Unsigned32		Dynamic index: EDS, p0140		Func. diagram: 4704			
		P-Group: Encoder		Units group: -		Unit selection: -			
		Not for motor type: -		Scaling: -		Expert list: 1			
		Min		Max		Factory setting			
		0		1073741823		8192			
Description:					Sets the number of measuring steps per revolution for a rotary absolute encoder. The resolution refers to the absolute position.				
Caution:					This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.				

p0424[0...n] Encoder, linear zero mark distance / Enc lin ZM_distENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Data type:** Unsigned16**P-Group:** Encoder**Not for motor type:** -**Min**

0 [mm]

Calculated: -**Dynamic index:** EDS, p0140**Units group:** -**Scaling:** -**Max**

65535 [mm]

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

20 [mm]

Description:

Sets the distance between two zero marks for a linear encoder. This information is used for zero mark monitoring.

Caution:

This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.

Note:

For distance-coded zero marks, this means the basic distance.

p0425[0...n] Encoder, rotary zero mark distance / Enc rot dist ZMENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Data type:** Unsigned32**P-Group:** Encoder**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** EDS, p0140**Units group:** -**Scaling:** -**Max**

16777215

Access level: 3**Func. diagram:** 4704, 8570**Unit selection:** -**Expert list:** 1**Factory setting**

2048

Description:

Sets the distance in pulses between two zero marks for a rotary encoder. This information is used for zero mark monitoring.

Caution:

This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.

Note:

For distance-coded zero marks, this means the basic distance.

p0426[0...n] Encoder zero mark differential distance / Enc diff distanceENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Data type:** Unsigned16**P-Group:** Encoder**Not for motor type:** -**Min**

1

Calculated: -**Dynamic index:** EDS, p0140**Units group:** -**Scaling:** -**Max**

65535

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

1

Description:

Differential distance with distance-coded zero marks [signal periods]. Corresponds to jump displacement of "zero mark with interference".

Caution:

This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.

p0427[0...n] Encoder SSI baud rate / Enc SSI baud rateENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Data type:** FloatingPoint32**P-Group:** Encoder**Not for motor type:** -**Min**

0 [kHz]

Calculated: -**Dynamic index:** EDS, p0140**Units group:** -**Scaling:** -**Max**

65535 [kHz]

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

100 [kHz]

Description:

Sets the baud rate for an SSI encoder.

Caution:

This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.

Note:

SSI: Synchronous Serial Interface

p0428[0...n] Encoder SSI monoflop time / Enc SSI t_monoflopENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** EDS, p0140**Func. diagram:** -**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0 [µs]

65535 [µs]

30 [µs]

Description: Sets the minimum delay time between two data transfers of the absolute value for an SSI encoder.**Caution:** This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.**p0429[0...n] Encoder SSI configuration / Enc SSI config**ENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** EDS, p0140**Func. diagram:** -**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

0000 bin

Description: Sets the configuration for an SSI encoder.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Transfer code	Binary code	Gray code	-
	02	Transfer absolute value twice	Yes	No	-
	06	Data line during the monoflop time	High level	Low level	-

Caution: This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.**Note:** Re bit 06:

The quiescent signal level of the data line corresponds to the inverted, set level.

p0430[0...n] Sensor Module configuration / SM configENCODER, SERVO,
VECTOR**Can be changed:** C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** EDS, p0140**Func. diagram:** -**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

1110 0000 0000 1000 0000
0000 0000 0000 bin**Description:** Sets the configuration of the Sensor Module.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	17	Burst oversampling	Yes	No	-
	18	Continuous oversampling (reserved)	Yes	No	-
	19	Safety position actual value sensing	Yes	No	-
	20	Speed calculation mode (only SMC30)	Incremental diff	Flank time meas	-
	21	Zero mark tolerance	Yes	No	-
	22	Rot pos adapt	Yes	No	-
	23	Deselection commutation with zero mark	Yes	No	-
	24	Commutation with selected zero mark	Yes	No	-
	25	Disconnection of encoder power supply on parking	Yes	No	-
	27	SSI position value extrapolation	Yes	No	-
	28	Cubic correction	Yes	No	-
	29	Phase correction	Yes	No	-
	30	Amplitude correction	Yes	No	-
	31	Offset correction	Yes	No	-

Note:

Re bit 17 (burst oversampling):

Displays whether burst oversampling has been activated.

Re bit 18 (continuous oversampling):

Displays whether continuous oversampling has been activated.

Re bit 19 (safety position actual value sensing):

Displays whether the safety encoder position is sent in the cyclic telegram.

Re bit 20 (speed calculation mode):

If bit = 0 the speed is calculated via edge time measurement with extrapolation between edges. In this mode the speed actual value 0 measuring time can be set in p0453.

If bit = 1 the speed is calculated via incremental difference without extrapolation. In this mode p0453 is irrelevant.

Re bit 21 (zero mark tolerance):

If bit = 1 there is no fault response (F3x100) if the number of track pulses between two zero pulses does not match the configured number on one occasion. If the next zero mark is correct, no fault occurs.

Re bit 22 (rotor position adaptation):

If bit = 1 correction might take several steps (+/- 1 quadrant per revolution) if the zero mark arrived in a tolerance bandwidth (+/- 1/12 of the electrical revolution).

Re bit 23 (deselection commutation with zero mark):

Only possible with the Sensor Module property "Commutation with zero mark can be deselected" (p0458.23 = yes).

This should only be set for encoders that have not been adjusted.

Re bit 24 (commutation with selected zero mark):

Sets commutation to selected zero mark.

Re bit 25 (disconnection of encoder power supply on parking):

If activated, the encoder power supply is disconnected completely on parking (0 V), regardless of whether the operating voltage is 5 V or 24 V. If not activated, the encoder power supply is not disconnected on parking (although the 24 V operating voltage is reduced to 5 V).

Re bit 27 (extrapolation SSI position value):

Sets the interpolation of the position values from the SSI encoder.

Re bit 28 (cubic correction):

Sets whether cubic correction is activated for the A-B analog signals.

Re bit 29 (phase correction):

Sets whether phase correction is activated for the A-B analog signals.

Re bit 30 (amplitude correction):

Sets whether amplitude correction is activated for the A-B analog signals.

Re bit 31 (offset correction):

Sets whether offset correction is activated for the A-B analog signals.

p0430[0...n]**Sensor Module configuration / SM config**

ENCODER (Lin enc),
SERVO (Lin)

Can be changed: C2(4)

Data type: Unsigned32

P-Group: Encoder

Not for motor type: -

Min

-

Calculated: -

Dynamic index: EDS, p0140

Units group: -

Scaling: -

Max

-

Access level: 3

Func. diagram: -

Unit selection: -

Expert list: 1

Factory setting

1110 0000 0000 1000 0000
0000 0000 0000 bin

Description:

Sets the configuration of the Sensor Module.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	17	Burst oversampling	Yes	No	-
	18	Continuous oversampling (reserved)	Yes	No	-
	19	Safety position actual value sensing	Yes	No	-
	20	Velocity calculation mode (only SMC30)	Incremental diff	Flank time meas	-
	21	Zero mark tolerance	Yes	No	-
	22	Rot pos adapt	Yes	No	-
	23	Deselection commutation with zero mark	Yes	No	-
	24	Commutation with selected zero mark	Yes	No	-
	25	Disconnection of encoder power supply on parking	Yes	No	-
	27	SSI position value extrapolation	Yes	No	-
	28	Cubic correction	Yes	No	-
	29	Phase correction	Yes	No	-
	30	Amplitude correction	Yes	No	-
	31	Offset correction	Yes	No	-

Note:

Re bit 17 (burst oversampling):
Displays whether burst oversampling has been activated.

Re bit 18 (continuous oversampling):
Displays whether continuous oversampling has been activated.

Re bit 19 (safety position actual value sensing):
Displays whether the safety encoder position is sent in the cyclic telegram.

Re bit 20 (speed calculation mode):
If bit = 0 the speed is calculated via edge time measurement with extrapolation between edges. In this mode the speed actual value 0 measuring time can be set in p0453.
If bit = 1 the speed is calculated via incremental difference without extrapolation. In this mode p0453 is irrelevant.

Re bit 21 (zero mark tolerance):
If bit = 1 there is no fault response (F3x100) if the number of track pulses between two zero pulses does not match the configured number on one occasion. If the next zero mark is correct, no fault occurs.

Re bit 22 (rotor position adaptation):
If bit = 1 correction might take several steps (+/- 1 quadrant per revolution) if the zero mark arrived in a tolerance bandwidth (+/- 1/12 of the electrical revolution).

Re bit 23 (deselection commutation with zero mark):
Only possible with the Sensor Module property "Commutation with zero mark can be deselected" (p0458.23 = yes).
This should only be set for encoders that have not been adjusted.

Re bit 24 (commutation with selected zero mark):
Sets commutation to selected zero mark.

Re bit 25 (disconnection of encoder power supply on parking):
If activated, the encoder power supply is disconnected completely on parking (0 V), regardless of whether the operating voltage is 5 V or 24 V. If not activated, the encoder power supply is not disconnected on parking (although the 24 V operating voltage is reduced to 5 V).

Re bit 27 (extrapolation SSI position value):
Sets the interpolation of the position values from the SSI encoder.

Re bit 28 (cubic correction):
Sets whether cubic correction is activated for the A-B analog signals.

Re bit 29 (phase correction):
Sets whether phase correction is activated for the A-B analog signals.

Re bit 30 (amplitude correction):
Sets whether amplitude correction is activated for the A-B analog signals.

Re bit 31 (offset correction):
Sets whether offset correction is activated for the A-B analog signals.

p0431[0...n]	Angular commutation offset / Ang_com offset		
SERVO, VECTOR	Can be changed: C2(4) Data type: FloatingPoint32 P-Group: Encoder Not for motor type: - Min -180.00 [°]	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 180.00 [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [°]
Description:	Sets the angular commutation offset.		
Dependency:	The value is taken into account in r0094. Refer to: r0094, r1778		
Caution:	When the firmware is upgraded from V2.3 to V2.4 or higher, the value must be reduced by 60° if all the following conditions are fulfilled: - The motor is a synchronous motor (p0300 = 2, 2xx, 4, 4xx). - The encoder is a resolver (p0404.23 = 1). - The actual speed value is inverted (p0410.0 = 1).		
Notice:	The angular commutation offset cannot be generally taken from other drive systems. As a minimum - the sign of the offset determined for SIMODRIVE 611 digital and SIMODRIVE 611 universal must be reversed for SINAMICS (p0431 (SINAMICS) = -p1016 (SIMODRIVE)).		
Note:	Angular commutation offset, angular difference between electrical position of encoder and flux position. For p0404.5 = 1 (track C/D) the following applies: The angular offset in p0431 acts on track A/B, the zero mark on track C/D. For p0404.6 = 1 (Hall sensor) the following applies: The angular offset in p0431 acts on track A/B and the zero mark.		
p0432[0...n]	Gearbox factor, encoder revolutions / Grbx_fact enc_rev		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Integer16 P-Group: Encoder Not for motor type: - Min 1	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 10000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the encoder revolutions for the gearbox factor of the encoder evaluation. The gearbox factor specifies the ratio between the encoder shaft and motor shaft (for motor encoders) or between the encoder shaft and the load.		
Dependency:	This parameter can only be set for p0402 = 9999. Refer to: p0402, p0410, p0433		
Note:	Negative gearbox factors should be implemented with p0410.		
p0433[0...n]	Gearbox factor, motor/load revolutions / Grbx_fact mot_rev		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Integer16 P-Group: Encoder Not for motor type: - Min 1	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 10000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the motor and load revolutions for the gearbox factor of the encoder evaluation. The gearbox factor specifies the ratio between the encoder shaft and motor shaft (for motor encoders) or between the encoder shaft and the load.		
Dependency:	This parameter can only be set for p0402 = 9999. Refer to: p0402, p0410, p0432		
Note:	Negative gearbox factors should be implemented with p0410.		

p0434[0...n]	Encoder SSI error bit / Enc SSI error bit		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the position and level of the error bit in the SSI protocol.		
Notice:	The bit may only be positioned before (p0446) or after (p0448) the absolute value in the SSI protocol.		
Note:	Value = dcba ba: Position of the error bit in the protocol (0 ... 63). c: Level (0: Low level, 1: High level). d: Status of the evaluation (0: Off, 1: On with 1 error bit, 2: On with 2 error bits ... 9: On with 9 error bits). For several error error bits, the following applies: - the position specified under ba and the additional bits are assigned increasing consecutively. - the level set under c applies to all error bits. Example: p0434 = 1013 --> The evaluation is switched in and the error bit is at position 13 with a low level. p0434 = 1113 --> The evaluation is switched in and the error bit is at position 13 with a high level.		

p0435[0...n]	Encoder SSI alarm bit / Enc SSI alarm bit		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the position and level of the alarm bit in the SSI protocol.		
Notice:	The bit may only be positioned before (p0446) or after (p0448) the absolute value in the SSI protocol.		
Note:	Value = dcba ba: Position of the alarm bit in protocol (0 ... 63). c: Level (0: Low level, 1: High level). d: State of the evaluation (0: Off, 1: On). Example: p0435 = 1014 --> The evaluation is switched in and the alarm bit is at position 14 with a low level. p0435 = 1114 --> The evaluation is switched in and the alarm bit is at position 14 with a high level.		

p0436[0...n]	Encoder SSI parity bit / Enc SSI parity bit		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the position and parity of the parity bit in the SSI protocol.		
Notice:	The bit may only be positioned before (p0446) or after (p0448) the absolute value in the SSI protocol.		

Note: Value = dcba
ba: Position of the parity bit in the protocol (0 ... 63).
c: Parity (0: even, 1: uneven).
d: State of the evaluation (0: Off, 1: On).
Example:
p0436 = 1015
--> The evaluation is switched in and the parity bit is at position 15 with even parity.
p0436 = 1115
--> The evaluation is switched in and the parity bit is at position 15 with uneven parity.

p0437[0...n] Sensor Module configuration extended / SM config ext			
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0011 0000 0000 0000 0000 1000 0000 0000 bin

Description: Sets the extended configuration of the Sensor Module.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activating the data logger	Yes	No	-
	01	Zero mark edge detection	Yes	No	-
	02	Correction position actual value XIST1	Yes	No	-
	04	Edge evaluation	Yes	No	-
	05	Edge evaluation	Yes	No	-
	06	Freeze the speed actual value for dn/dt errors	Yes	No	-
	07	Accumulate uncorrected encoder pulses	Yes	No	-
	11	Automatic alarm acknowledgment	Yes	No	-
	26	Track monitoring deselection	Yes	No	-
	28	EnDat linear encoder monitoring incremental/absolute	Yes	No	-
	29	EnDat encoder initialization with high accuracy	Yes	No	-
	31	Analog unipolar track monitoring	Yes	No	-

Dependency: Refer to: p0430, r0459

Note: A value of zero is displayed if an encoder is not present.

Re bit 00:
When the data logger (trace) is activated, in the case of a fault, data before and after the event are recorded (traced) and saved in files on the non-volatile memory medium. Experts can then evaluate this data.

Re bit 01:
If bit = 0, the zero mark is evaluated by ANDing tracks A and B and the zero mark.
For bit = 1, the zero mark is evaluated depending on the direction of rotation detected. For a positive direction of rotation, the positive edge of the zero mark is considered and for a negative direction of rotation, the negative edge of the zero mark.

Re bit 02:
If the bit is set, in the event of a deviation less than the tolerance window for the zero mark (p4681, p4682), the pulses per revolution are corrected. If the bit is not set, encoder fault F3x131 is triggered.

Re Bit 04 and Bit 05:
The current hardware only supports 1x or 4x signal evaluation.
Bit 5/4 = 0/0: Signal evaluation per period, 4x.
Bit 5/4 = 1/0: Signal evaluation per period, 4x.
Bit 5/4 = 0/1: Signal evaluation per period, 1x.
Bit 5/4 = 1/1: Illegal setting.

Re bit 06:

If the function is active, when dn/dt monitoring responds, the speed actual value is internally frozen for a time equivalent to two current controller clock cycles. The rotor position continues to be integrated. The actual value is then re-enabled after this time has expired.

Re bit 07:

If the bit is set, the encoder pulses which have not been corrected are added to r4688 at the zero mark.

Re bit 11:

If the bit is set, within a certain time grid the Sensor Module checks whether the fault cause is still present. This enables the Module to switch from the fault state to the operating state and provide valid actual values automatically. The faults are displayed until the user acknowledges them.

Re bit 20:

If the bit is set, the bandwidth of the analog filter for SMx10 (resolver) and SMx20 (sin/cos encoder) can be set via parameter p4660.

Re bit 26:

Track monitoring is deactivated for the square-wave encoders when the bit is set, even if the monitoring function is selected in p405.2.

Re bit 28:

Monitoring of the difference between incremental and absolute position in the case of linear encoders.

Re bit 29:

When the bit is set, the EnDat encoder is initialized under a certain speed and, therefore, with high accuracy. If initialization at a higher speed is requested, fault F31151, F32151, or F33151 is output.

Re bit 31:

When monitoring is active, the levels of the individual track signals and the corresponding inverted track signals are monitored separately.

p0437[0...n]		Sensor Module configuration extended / SM config ext			
ENCODER (Lin enc), SERVO (Lin)	Can be changed: C2(4)		Calculated: -	Access level: 3	
	Data type: Unsigned32		Dynamic index: EDS, p0140	Func. diagram: -	
	P-Group: Encoder		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min -		Max -	Factory setting 0011 0000 0000 0000 0000 1000 0000 0000 bin	
Description:		Sets the extended configuration of the Sensor Module.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activating the data logger	Yes	No	-
	01	Zero mark edge detection	Yes	No	-
	02	Correction position actual value XIST1	Yes	No	-
	04	Edge evaluation	Yes	No	-
	05	Edge evaluation	Yes	No	-
	06	Freeze the speed actual value for dn/dt errors	Yes	No	-
	07	Accumulate uncorrected encoder pulses	Yes	No	-
	11	Automatic alarm acknowledgment	Yes	No	-
	26	Track monitoring deselection	Yes	No	-
	28	EnDat linear encoder monitoring incremental/absolute	Yes	No	-
	29	EnDat encoder initialization with high accuracy	Yes	No	-
	31	Analog unipolar track monitoring	Yes	No	-
Dependency:		Refer to: p0430, r0459			

Note: A value of zero is displayed if an encoder is not present.

Re bit 00:
When the data logger (trace) is activated, in the case of a fault, data before and after the event are recorded (traced) and saved in files on the non-volatile memory medium. Experts can then evaluate this data.

Re bit 01:
If bit = 0, the zero mark is evaluated by ANDing tracks A and B and the zero mark.
For bit = 1, the zero mark is evaluated depending on the direction detected. For a positive direction, the positive edge of the zero mark is considered and for a negative direction, the negative edge of the zero mark.

Re bit 02:
If the bit is set, in the event of a deviation less than the tolerance window for the zero mark (p4681, p4682), the pulses per revolution are corrected. If the bit is not set, encoder fault F3x131 is triggered.

Re Bit 04 and Bit 05:
Bit 5/4 = 0/0: Signal evaluation per period, 4x.
Bit 5/4 = 1/0: Signal evaluation per period, 4x.
Bit 5/4 = 0/1: Signal evaluation per period, 1x.
Bit 5/4 = 1/1: Illegal setting.

Re bit 06:
If the function is active, when dn/dt monitoring responds, the velocity actual value is internally frozen for a time equivalent to two current controller clock cycles. The rotor position continues to be integrated. The actual value is then re-enabled after this time has expired.

Re bit 07:
If the bit is set, the encoder pulses detected as faulty between two zero marks are accumulated (r4688).

Re bit 29:
When the bit is set, the EnDat encoder is initialized under a certain velocity and, therefore, with high accuracy. If initialization at a higher velocity is requested, fault F31151, F32151, or F33151 is output.

Re bit 31:
When monitoring is active, the levels of the individual track signals and the corresponding inverted track signals are monitored separately.

p0438[0...n]		Squarewave encoder filter time / Enc t_filt	
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [µs]	100.00 [µs]	0.64 [µs]
Description:	Sets the filter time for a squarewave encoder.		
	The hardware of the squarewave encoder only supports the following values:		
	No filtering		
	0.04 µs		
	0.64 µs		
	2.56 µs		
	10.24 µs		
	20.48 µs		
Dependency:	Refer to: r0452		
Notice:	If the filter time is too long, the track signals A/B/R may be suppressed and the appropriate messages output.		
Note:	The most suitable filter time depends on the number of pulses and maximum speed of the square-wave encoder.		
	The filter time is automatically corrected to the next value when entering a non-specified value. In this case, no message is output.		
	The effective filter time is displayed in r0452.		

p0439[0...n]	Encoder ramp-up time / Enc ramp-up time		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned16 P-Group: Encoder Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 65535 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Ramp-up time of encoder. The encoder supplies stable track signals once this time has elapsed.		
Caution:	This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.		
p0440[0...n]	Copy encoder serial number / Copy enc ser_no		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Copies the actual serial number of the encoder belonging to this Encoder Data Set (EDS) to p0441 ... p0445. Example: For p0440[0] = 1, the serial number of the encoder belonging EDS0 is copied to p0441[0] ... p0445[0].		
Value:	0: No action 1: Transfer serial number		
Dependency:	Refer to: p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0462, r0463, r0464, p1990		
Note:	For encoders with serial number, encoder replacement is monitored in order to request angular commutation calibration (adjustment) for motor encoders and absolute calibration for direct measuring systems with absolute value data. The serial number, which from then onwards is used for monitoring purposes, can be transferred using p0440. In the following cases, copying is automatically started in the following cases: 1.) When commissioning 1FT6, 1FK6, 1FK7 motors. 2.) When writing into p0431. 3.) For p1990 = 1. p0440 is automatically set to 0 when the copying has been completed. In order to permanently accept the copied values, it is necessary to save in a non-volatile fashion (p0977).		
p0441[0...n]	Encoder commissioning serial number part 1 / Enc comm ser_no 1		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min 0000 hex	Calculated: CALC_MOD_ALL Dynamic index: EDS, p0140 Units group: - Scaling: - Max FFFF FFFF hex	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Serial number part 1 of the encoder for the commissioning.		
Dependency:	Refer to: p0440, p0442, p0443, p0444, p0445, r0460, r0461, r0462, r0463, r0464 Refer to: F07414		
Note:	A value of zero is displayed if an encoder is not present.		

p0442[0...n]	Encoder commissioning serial number part 2 / Enc comm ser_no 2		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: CALC_MOD_ALL	Access level: 4
	Data type: Unsigned32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Serial number part 2 of the encoder for the commissioning.		
Dependency:	Refer to: p0440, p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0462, r0463, r0464 Refer to: F07414		
Note:	A value of zero is displayed if an encoder is not present.		
p0443[0...n]	Encoder commissioning serial number part 3 / Enc comm ser_no 3		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: CALC_MOD_ALL	Access level: 4
	Data type: Unsigned32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Serial number part 3 of the encoder for the commissioning.		
Dependency:	Refer to: p0440, p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0462, r0463, r0464 Refer to: F07414		
Note:	A value of zero is displayed if an encoder is not present.		
p0444[0...n]	Encoder commissioning serial number part 4 / Enc comm ser_no 4		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: CALC_MOD_ALL	Access level: 4
	Data type: Unsigned32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Serial number part 4 of the encoder for the commissioning.		
Dependency:	Refer to: p0440, p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0462, r0463, r0464 Refer to: F07414		
Note:	A value of zero is displayed if an encoder is not present.		
p0445[0...n]	Encoder commissioning serial number part 5 / Enc comm ser_no 5		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: CALC_MOD_ALL	Access level: 4
	Data type: Unsigned32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Serial number part 5 of the encoder for the commissioning.		
Dependency:	Refer to: p0440, p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0462, r0463, r0464 Refer to: F07414		
Note:	A value of zero is displayed if an encoder is not present.		

p0446[0...n]	Encoder SSI number of bits before the absolute value / Enc SSI bit before		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 65535	Factory setting 0
Description:	Sets the number of bits before the absolute value in the SSI protocol.		
Caution:	This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.		
Note:	For example, error bit, alarm bit or parity bit can be positioned at these bits.		
p0447[0...n]	Encoder SSI number of bits absolute value / Enc SSI bit val		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 65535	Factory setting 25
Description:	Sets the number of bits for the absolute value in the SSI protocol.		
Caution:	This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.		
p0448[0...n]	Encoder SSI number of bits after the absolute value / Enc SSI bit after		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 65535	Factory setting 0
Description:	Sets the number of bits after the absolute value in the SSI protocol.		
Caution:	This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.		
Note:	For example, error bit, alarm bit or parity bit can be positioned at these bits.		
p0449[0...n]	Encoder SSI number of bits, filler bits / Enc SSI fill bits		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 65535	Factory setting 1
Description:	Sets the number of filler bits for double absolute value transfer in the SSI protocol.		
Dependency:	Refer to: p0429		
Caution:	This parameter is automatically preset for encoders from the encoder list (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.		
Note:	This parameter is only of significance for p0429.2 = 1.		

r0451[0...2] Commutation angle factor / Enc commut_factor

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 4710
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the relationship between the electrical and mechanical pole positions.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Note:	A value of zero is displayed if an encoder is not present.		

r0452 Squarewave encoder filter time display / Enc t_filt displ

ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [µs]	Max - [µs]	Factory setting - [µs]
Description:	Displays the effective filter time for a squarewave encoder. The filter time is set using p0438.		
Dependency:	Refer to: p0438		
Note:	A value of zero is displayed if an encoder is not present.		

r0452[0...2] Squarewave encoder filter time display / Enc t_filt displ

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [µs]	Max - [µs]	Factory setting - [µs]
Description:	Displays the effective filter time for a squarewave encoder. The filter time is set using p0438.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p0438		
Note:	A value of zero is displayed if an encoder is not present.		

p0453[0...n] Measuring time pulse encoder evaluation zero speed / t_meas enc_eva n_0

ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.10 [ms]	Max 10000.00 [ms]	Factory setting 1000.00 [ms]
Description:	Sets the measuring time for evaluating zero speed. If no further encoder pulses are detected during this time, a speed actual value of zero is output internally.		
Dependency:	Refer to: r0452		
Note:	This function is required for slow-running motors so that actual speeds close to zero can be output correctly.		

r0455 Encoder configuration recognized / Enc config act

ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the detected encoder configuration.
In this case, the encoder must automatically support the function (e.g. encoder with EnDat interface).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Linear encoder	Yes	No	-
	01	Abs value encoder	Yes	No	-
	02	Multiturn encoder	Yes	No	-
	03	Track A/B sq-wave	Yes	No	-
	04	Track A/B sinus	Yes	No	-
	05	Track C/D	Yes	No	-
	06	Hall sensor	Yes	No	-
	08	EnDat encoder	Yes	No	-
	09	SSI encoder	Yes	No	-
	10	DRIVE-CLiQ encoder	Yes	No	-
	11	Digital encoder	Yes	no	-
	12	Equidistant zero mark	Yes	No	-
	13	Irregular zero mark	Yes	No	-
	14	Distance-coded zero mark	Yes	No	-
	15	Commutation with zero mark (not ASM)	Yes	No	-
	16	Acceleration	Yes	No	-
	17	Track A/B analog	Yes	No	-
	20	Voltage level 5 V	Yes	No	-
	21	Voltage level 24 V	Yes	No	-
	22	Remote sense (only SMC30)	Yes	No	-
	23	Resolver excit.	Yes	No	-

Dependency: Refer to: p0404

Note: ZM: Zero mark
This parameter is only used for diagnostics.
A value of zero is displayed if an encoder is not present.
Re bit 20, 21 (voltage level 5 V, voltage level 24 V):
The voltage level cannot be detected. Therefore, these bits are always set to 0.

r0455[0...2] Encoder configuration recognized / Enc config act

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the detected encoder configuration.
In this case, the encoder must automatically support the function (e.g. encoder with EnDat interface).

Index: [0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Linear encoder	Yes	No	-
	01	Abs value encoder	Yes	No	-
	02	Multiturn encoder	Yes	No	-
	03	Track A/B sq-wave	Yes	No	-
	04	Track A/B sinus	Yes	No	-
	05	Track C/D	Yes	No	-
	06	Hall sensor	Yes	No	-
	08	EnDat encoder	Yes	No	-
	09	SSI encoder	Yes	No	-
	10	DRIVE-CLiQ encoder	Yes	No	-
	11	Digital encoder	Yes	no	-
	12	Equidistant zero mark	Yes	No	-
	13	Irregular zero mark	Yes	No	-
	14	Distance-coded zero mark	Yes	No	-
	15	Commutation with zero mark (not ASM)	Yes	No	-
	16	Acceleration	Yes	No	-
	17	Track A/B analog	Yes	No	-
	20	Voltage level 5 V	Yes	No	-
	21	Voltage level 24 V	Yes	No	-
	22	Remote sense (only SMC30)	Yes	No	-
	23	Resolver excit.	Yes	No	-

Dependency: Refer to: p0404

Note: ZM: Zero mark

This parameter is only used for diagnostics.

A value of zero is displayed if an encoder is not present.

Re bit 20, 21 (voltage level 5 V, voltage level 24 V):

The voltage level cannot be detected. Therefore, these bits are always set to 0.

r0456 Encoder configuration supported / Enc config supp

ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Contains the encoder configuration supported by the Sensor Module.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Linear encoder	Yes	No	-
	01	Abs value encoder	Yes	No	-
	02	Multiturn encoder	Yes	No	-
	03	Track A/B sq-wave	Yes	No	-
	04	Track A/B sinus	Yes	No	-
	05	Track C/D	Yes	No	-
	06	Hall sensor	Yes	No	-
	08	EnDat encoder	Yes	No	-
	09	SSI encoder	Yes	No	-
	10	DRIVE-CLiQ encoder	Yes	No	-
	11	Digital encoder	Yes	no	-
	12	Equidistant zero mark	Yes	No	-
	13	Irregular zero mark	Yes	No	-
	14	Distance-coded zero mark	Yes	No	-
	15	Commutation with zero mark (not ASM)	Yes	No	-
	16	Acceleration	Yes	No	-
	17	Track A/B analog	Yes	No	-
	20	Voltage level 5 V	Yes	No	-
	21	Voltage level 24 V	Yes	No	-
	22	Remote sense (only SMC30)	Yes	No	-
	23	Resolver excit.	Yes	No	-

Dependency: Refer to: p0404
Note: ZM: Zero mark
 This parameter is only used for diagnostics.
 A value of zero is displayed if an encoder is not present.

r0456[0...2] Encoder configuration supported / Enc config supp

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
-	-	-	

Description: Contains the encoder configuration supported by the Sensor Module.

Index:
 [0] = Encoder 1
 [1] = Encoder 2
 [2] = Encoder 3

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Linear encoder	Yes	No	-
	01	Abs value encoder	Yes	No	-
	02	Multiturn encoder	Yes	No	-
	03	Track A/B sq-wave	Yes	No	-
	04	Track A/B sinus	Yes	No	-
	05	Track C/D	Yes	No	-
	06	Hall sensor	Yes	No	-
	08	EnDat encoder	Yes	No	-
	09	SSI encoder	Yes	No	-
	10	DRIVE-CLiQ encoder	Yes	No	-
	11	Digital encoder	Yes	no	-
	12	Equidistant zero mark	Yes	No	-
	13	Irregular zero mark	Yes	No	-
	14	Distance-coded zero mark	Yes	No	-
	15	Commutation with zero mark (not ASM)	Yes	No	-
	16	Acceleration	Yes	No	-
	17	Track A/B analog	Yes	No	-
	20	Voltage level 5 V	Yes	No	-
	21	Voltage level 24 V	Yes	No	-
	22	Remote sense (only SMC30)	Yes	No	-
	23	Resolver excit.	Yes	No	-

Dependency: Refer to: p0404
Note: ZM: Zero mark
 This parameter is only used for diagnostics.
 A value of zero is displayed if an encoder is not present.

r0458 Sensor Module properties / SM properties					
ENCODER	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned32		Dynamic index: -		Func. diagram: 4704
	P-Group: Encoder		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		-
Description: Sets the Sensor Module configuration.					
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Encoder data available	Yes	No	-
	01	Motor data available	Yes	No	-
	02	Temperature sensor connection available	Yes	No	-
	03	Connection for PTC for motor with DRIVE-CLiQ also available	Yes	No	-
	04	Module temperature available	Yes	No	-
	05	Absolute encoder p0408/p0421, no power of 2	Yes	No	-
	06	Sensor Module permits parking/unparking	Yes	No	-
	07	Hall sensor can be combined with actual value inversion	Yes	No	-
	08	Evaluation through several temperature channels possible	Yes	No	-
	09	Encoder fault and its associated information available	Yes	No	-
	10	Speed diagnostics in the Sensor Module	Yes	No	-
	11	Configuring without park state possible	Yes	No	-
	12	Extended functions available	Yes	No	-
	13	Extended encoder fault handling	Yes	No	-
	14	Extended singleturn/multiturn information available	Yes	No	-
	15	Valuation figures available	Yes	No	-
	16	Pole position identification	Yes	No	-
	17	Burst oversampling	Yes	No	-
	19	Safety position actual value sensing	Yes	No	-
	20	Extended speed calculation being used (only SMC30)	Yes	No	-
	21	Zero mark tolerance	Yes	No	-
	22	Rot pos adapt	Yes	No	-
	23	Commutation with zero mark can be deselected	Yes	No	-
	24	Commutation with selected zero mark	Yes	No	-
	25	Disconnection of encoder power supply on parking supported	Yes	No	-
	27	SSI position value extrapolation	Yes	No	-
	28	Cubic correction	Yes	No	-
	29	Phase correction	Yes	No	-
	30	Amplitude correction	Yes	No	-
	31	Offset correction	Yes	No	-
Dependency: Refer to: p0437, p0600, p0601					

Note: A value of zero is displayed if an encoder is not present.

Re bit 11:
When the property is set, the following parameters can be changed without the actual value in the encoder interface becoming invalid (state r0481.14 = 1 "parking encoder active"):
p0314, p0315, p0430, p0431, p0441, p0442, p0443, p0444, p0445

Re bit 12:
The extended functions can be configured using p0437.

Re bit 13:
Encoder faults can be acknowledged via Gn_STW.15.

Re bit 14:
Only for internal Siemens use.

Re bit 23:
When the property is set, commutation with zero mark can be deselected using p0430.23.

Re bit 24:
If the property is set, commutation to the selected zero mark can be carried out.

r0458 Sensor Module properties / SM properties

ENCODER (Lin enc)	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 4704 Unit selection: - Expert list: 1 Factory setting -
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Description: Sets the Sensor Module configuration.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Encoder data available	Yes	No	-
	01	Motor data available	Yes	No	-
	02	Temperature sensor connection available	Yes	No	-
	03	Connection for PTC for motor with DRIVE-CLiQ also available	Yes	No	-
	04	Module temperature available	Yes	No	-
	05	Absolute encoder p0408/p0421, no power of 2	Yes	No	-
	06	Sensor Module permits parking/unparking	Yes	No	-
	07	Hall sensor can be combined with actual value inversion	Yes	No	-
	08	Evaluation through several temperature channels possible	Yes	No	-
	09	Encoder fault and its associated information available	Yes	No	-
	10	Velocity diagnostics in the Sensor Module	Yes	No	-
	11	Configuring without park state possible	Yes	No	-
	12	Extended functions available	Yes	No	-
	13	Extended encoder fault handling	Yes	No	-
	14	Extended singleturn/multiturn information available	Yes	No	-
	15	Valuation figures available	Yes	No	-
	16	Pole position identification	Yes	No	-
	17	Burst oversampling	Yes	No	-
	19	Safety position actual value sensing	Yes	No	-
	20	Extended velocity calculation available (only SMC30)	Yes	No	-
	21	Zero mark tolerance	Yes	No	-
	22	Rot pos adapt	Yes	No	-
	23	Commutation with zero mark can be deselected	Yes	No	-
	24	Commutation with selected zero mark	Yes	No	-

25	Disconnection of encoder power supply on parking supported	Yes	No	-
27	SSI position value extrapolation	Yes	No	-
28	Cubic correction	Yes	No	-
29	Phase correction	Yes	No	-
30	Amplitude correction	Yes	No	-
31	Offset correction	Yes	No	-

Dependency: Refer to: p0437, p0600, p0601

Note: A value of zero is displayed if an encoder is not present.

Re bit 11:

When the property is set, the following parameters can be changed without the actual value in the encoder interface becoming invalid (state r0481.14 = 1 "parking encoder active"):

p0314, p0315, p0430, p0431, p0441, p0442, p0443, p0444, p0445

Re bit 12:

The extended functions can be configured using p0437.

Re bit 13:

Encoder faults can be acknowledged via Gn_STW.15.

Re bit 14:

Only for internal Siemens use.

Re bit 23:

When the property is set, commutation with zero mark can be deselected using p0430.23.

Re bit 24:

If the property is set, commutation to the selected zero mark can be carried out.

r0458[0...2] Sensor Module properties / SM properties

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 4704
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Sets the Sensor Module configuration.

Index:
 [0] = Encoder 1
 [1] = Encoder 2
 [2] = Encoder 3

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Encoder data available	Yes	No	-
	01	Motor data available	Yes	No	-
	02	Temperature sensor connection available	Yes	No	-
	03	Connection for PTC for motor with DRIVE-CLiQ also available	Yes	No	-
	04	Module temperature available	Yes	No	-
	05	Absolute encoder p0408/p0421, no power of 2	Yes	No	-
	06	Sensor Module permits parking/unparking	Yes	No	-
	07	Hall sensor can be combined with actual value inversion	Yes	No	-
	08	Evaluation through several temperature channels possible	Yes	No	-
	09	Encoder fault and its associated information available	Yes	No	-
	10	Speed diagnostics in the Sensor Module	Yes	No	-
	11	Configuring without park state possible	Yes	No	-
	12	Extended functions available	Yes	No	-
	13	Extended encoder fault handling	Yes	No	-
	14	Extended singleturn/multiturn information available	Yes	No	-
	15	Valuation figures available	Yes	No	-

16	Pole position identification	Yes	No	-
17	Burst oversampling	Yes	No	-
19	Safety position actual value sensing	Yes	No	-
20	Extended speed calculation being used (only SMC30)	Yes	No	-
21	Zero mark tolerance	Yes	No	-
22	Rot pos adapt	Yes	No	-
23	Commutation with zero mark can be deselected	Yes	No	-
24	Commutation with selected zero mark	Yes	No	-
25	Disconnection of encoder power supply on parking supported	Yes	No	-
27	SSI position value extrapolation	Yes	No	-
28	Cubic correction	Yes	No	-
29	Phase correction	Yes	No	-
30	Amplitude correction	Yes	No	-
31	Offset correction	Yes	No	-

Dependency: Refer to: p0437, p0600, p0601

Note: A value of zero is displayed if an encoder is not present.

Re bit 11:

When the property is set, the following parameters can be changed without the actual value in the encoder interface becoming invalid (state r0481.14 = 1 "parking encoder active"):

p0314, p0315, p0430, p0431, p0441, p0442, p0443, p0444, p0445

Re bit 12:

The extended functions can be configured using p0437.

Re bit 13:

Encoder faults can be acknowledged via Gn_STW.15.

Re bit 14:

Only for internal Siemens use.

Re bit 23:

When the property is set, commutation with zero mark can be deselected using p0430.23.

Re bit 24:

If the property is set, commutation to the selected zero mark can be carried out.

r0458[0...2] Sensor Module properties / SM properties

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 4704
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Sets the Sensor Module configuration.

Index:
 [0] = Encoder 1
 [1] = Encoder 2
 [2] = Encoder 3

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Encoder data available	Yes	No	-
	01	Motor data available	Yes	No	-
	02	Temperature sensor connection available	Yes	No	-
	03	Connection for PTC for motor with DRIVE-CLiQ also available	Yes	No	-
	04	Module temperature available	Yes	No	-
	05	Absolute encoder p0408/p0421, no power of 2	Yes	No	-
	06	Sensor Module permits parking/unparking	Yes	No	-
	07	Hall sensor can be combined with actual value inversion	Yes	No	-

List of parameters

08	Evaluation through several temperature channels possible	Yes	No	-
09	Encoder fault and its associated information available	Yes	No	-
10	Velocity diagnostics in the Sensor Module	Yes	No	-
11	Configuring without park state possible	Yes	No	-
12	Extended functions available	Yes	No	-
13	Extended encoder fault handling	Yes	No	-
14	Extended singleturn/multiturn information available	Yes	No	-
15	Valuation figures available	Yes	No	-
16	Pole position identification	Yes	No	-
17	Burst oversampling	Yes	No	-
19	Safety position actual value sensing	Yes	No	-
20	Extended velocity calculation available (only SMC30)	Yes	No	-
21	Zero mark tolerance	Yes	No	-
22	Rot pos adapt	Yes	No	-
23	Commutation with zero mark can be deselected	Yes	No	-
24	Commutation with selected zero mark	Yes	No	-
25	Disconnection of encoder power supply on parking supported	Yes	No	-
27	SSI position value extrapolation	Yes	No	-
28	Cubic correction	Yes	No	-
29	Phase correction	Yes	No	-
30	Amplitude correction	Yes	No	-
31	Offset correction	Yes	No	-

Dependency:

Refer to: p0437, p0600, p0601

Note:

A value of zero is displayed if an encoder is not present.

Re bit 11:

When the property is set, the following parameters can be changed without the actual value in the encoder interface becoming invalid (state r0481.14 = 1 "parking encoder active"):

p0314, p0315, p0430, p0431, p0441, p0442, p0443, p0444, p0445

Re bit 12:

The extended functions can be configured using p0437.

Re bit 13:

Encoder faults can be acknowledged via Gn_STW.15.

Re bit 14:

Only for internal Siemens use.

Re bit 23:

When the property is set, commutation with zero mark can be deselected using p0430.23.

Re bit 24:

If the property is set, commutation to the selected zero mark can be carried out.

r0459	Sensor Module properties extended / SM prop ext				
ENCODER	Can be changed: -		Calculated: -	Access level: 3	
	Data type: Unsigned32		Dynamic index: -	Func. diagram: -	
	P-Group: Encoder		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the extended properties supported by the Sensor Module.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activating the data logger	Yes	No	-
	01	Zero mark edge detection	Yes	No	-
	02	Correction position actual value XIST1	Yes	No	-
	04	Edge evaluation	Yes	No	-
	05	Edge evaluation	Yes	No	-
	06	Freeze the speed actual value for dn/dt errors	Yes	No	-
	07	Accumulate uncorrected encoder pulses	Yes	No	-
	09	Support function p0426, p0439	Yes	No	-
	10	Pulse/direction interface	Yes	No	-
	11	Automatic alarm acknowledgment	Yes	No	-
	26	Track monitoring deselection	Yes	No	-
	28	EnDat linear encoder monitoring incremental/absolute	Yes	No	-
	29	EnDat encoder initialization with high accuracy	Yes	No	-
	31	Analog unipolar track monitoring	Yes	No	-
Dependency:	Refer to: p0437				
Note:	A value of zero is displayed if an encoder is not present.				
	Re bit 09:				
	Parameter p0426 or p0439 has been modified. These functions are not supported by the connected Sensor Module.				

r0459	Sensor Module properties extended / SM prop ext				
ENCODER (Lin enc)	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -		
	P-Group: Encoder	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the extended properties supported by the Sensor Module.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activating the data logger	Yes	No	-
	01	Zero mark edge detection	Yes	No	-
	02	Correction position actual value XIST1	Yes	No	-
	04	Edge evaluation	Yes	No	-
	05	Edge evaluation	Yes	No	-
	06	Freeze actual velocity for dn/dt errors	Yes	No	-
	07	Accumulate uncorrected encoder pulses	Yes	No	-
	09	Support function p0426, p0439	Yes	No	-
	10	Pulse/direction interface	Yes	No	-
	11	Automatic alarm acknowledgment	Yes	No	-
	26	Track monitoring deselection	Yes	No	-
	28	EnDat linear encoder monitoring incremental/absolute	Yes	No	-
	29	EnDat encoder initialization with high accuracy	Yes	No	-
	31	Analog unipolar track monitoring	Yes	No	-

Dependency: Refer to: p0437

Note: A value of zero is displayed if an encoder is not present.

Re bit 09:

Parameter p0426 or p0439 has been modified. These functions are not supported by the connected Sensor Module.

r0459[0...2] Sensor Module properties extended / SM prop ext

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the extended properties supported by the Sensor Module.

Index:
 [0] = Encoder 1
 [1] = Encoder 2
 [2] = Encoder 3

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activating the data logger	Yes	No	-
	01	Zero mark edge detection	Yes	No	-
	02	Correction position actual value XIST1	Yes	No	-
	04	Edge evaluation	Yes	No	-
	05	Edge evaluation	Yes	No	-
	06	Freeze the speed actual value for dn/dt errors	Yes	No	-
	07	Accumulate uncorrected encoder pulses	Yes	No	-
	09	Support function p0426, p0439	Yes	No	-
	10	Pulse/direction interface	Yes	No	-
	11	Automatic alarm acknowledgment	Yes	No	-
	26	Track monitoring deselection	Yes	No	-
	28	EnDat linear encoder monitoring incremental/absolute	Yes	No	-
	29	EnDat encoder initialization with high accuracy	Yes	No	-
	31	Analog unipolar track monitoring	Yes	No	-

Dependency: Refer to: p0437

Note: A value of zero is displayed if an encoder is not present.

Re bit 09:

Parameter p0426 or p0439 has been modified. These functions are not supported by the connected Sensor Module.

r0459[0...2] Sensor Module properties extended / SM prop ext

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the extended properties supported by the Sensor Module.

Index:
 [0] = Encoder 1
 [1] = Encoder 2
 [2] = Encoder 3

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activating the data logger	Yes	No	-
	01	Zero mark edge detection	Yes	No	-
	02	Correction position actual value XIST1	Yes	No	-
	04	Edge evaluation	Yes	No	-
	05	Edge evaluation	Yes	No	-
	06	Freeze actual velocity for dn/dt errors	Yes	No	-
	07	Accumulate uncorrected encoder pulses	Yes	No	-
	09	Support function p0426, p0439	Yes	No	-
	10	Pulse/direction interface	Yes	No	-
	11	Automatic alarm acknowledgment	Yes	No	-
	26	Track monitoring deselection	Yes	No	-
	28	EnDat linear encoder monitoring incremental/absolute	Yes	No	-
	29	EnDat encoder velocity monitoring on initialization	Yes	No	-
	31	Analog unipolar track monitoring	Yes	No	-

Dependency: Refer to: p0437

Note: A value of zero is displayed if an encoder is not present.

Re bit 09:

Parameter p0426 or p0439 has been modified. These functions are not supported by the connected Sensor Module.

r0460 Encoder serial number part 1 / Enc ser_no 1

ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the actual serial number part 1 of the appropriate encoder.

Dependency: Refer to: p0441, p0442, p0443, p0444, p0445, r0461, r0462, r0463, r0464

r0460[0...2] Encoder serial number part 1 / Enc ser_no 1

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the actual serial number part 1 of the appropriate encoder.

Index:
[0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

Dependency: Refer to: p0441, p0442, p0443, p0444, p0445, r0461, r0462, r0463, r0464

r0461 Encoder serial number part 2 / Enc ser_no 2

ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the actual serial number part 2 of the appropriate encoder.

Dependency: Refer to: p0441, p0442, p0443, p0444, p0445, r0460, r0462, r0463, r0464

r0461[0...2]	Encoder serial number part 2 / Enc ser_no 2		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the actual serial number part 2 of the appropriate encoder.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p0441, p0442, p0443, p0444, p0445, r0460, r0462, r0463, r0464		
r0462	Encoder serial number part 3 / Enc ser_no 3		
ENCODER	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the actual serial number part 3 of the appropriate encoder.		
Dependency:	Refer to: p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0463, r0464		
r0462[0...2]	Encoder serial number part 3 / Enc ser_no 3		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the actual serial number part 3 of the appropriate encoder.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0463, r0464		
r0463	Encoder serial number part 4 / Enc ser_no 4		
ENCODER	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the actual serial number part 4 of the appropriate encoder.		
Dependency:	Refer to: p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0462, r0464		

r0463[0...2] Encoder serial number part 4 / Enc ser_no 4			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the actual serial number part 4 of the appropriate encoder.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0462, r0464		

r0464 Encoder serial number part 5 / Enc ser_no 5			
ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the actual serial number part 5 of the appropriate encoder.		
Dependency:	Refer to: p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0462, r0463		

r0464[0...2] Encoder serial number part 5 / Enc ser_no 5			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the actual serial number part 5 of the appropriate encoder.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p0441, p0442, p0443, p0444, p0445, r0460, r0461, r0462, r0463		

r0465[0...27] Encoder 1 identification number/serial number / Enc1 ID_no/Ser_no			
ENCODER, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned8	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the serial number of encoder 1. Index 0 = first character of the identification number ... Index x = 20 hex (blank) --> separation between the identification number of serial number Index x + 1 = 2F hex (slash) --> separation between the identification number of serial number Index x + 2 = 20 hex (blank) --> separation between the identification number of serial number Index x + 3 = first character of the serial number ... Index y with contents = last character of the serial number		

Dependency: Refer to: r0460, r0461, r0462, r0463, r0464
Notice: An ASCII table (excerpt) can be found, for example, in the Appendix of the List Manual:
Note: The individual characters of the identification number/serial number are available coded as ASCII characters.

r0466[0...27]	Encoder 2 identification number/serial number / Enc2 ID_no/Ser_no		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned8	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the serial number of encoder 2.
Index 0 = first character of the identification number
...
Index x = 20 hex (blank) --> separation between the identification number of serial number
Index x + 1 = 2F hex (slash) --> separation between the identification number of serial number
Index x + 2 = 20 hex (blank) --> separation between the identification number of serial number
Index x + 3 = first character of the serial number
...
Index y with contents = last character of the serial number

Dependency: Refer to: r0460, r0461, r0462, r0463, r0464
Notice: An ASCII table (excerpt) can be found, for example, in the Appendix of the List Manual:
Note: The individual characters of the identification number/serial number are available coded as ASCII characters.

r0467[0...27]	Encoder 3 identification number/serial number / Enc3 ID_no/Ser_no		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned8	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the serial number of encoder 3.
Index 0 = first character of the identification number
...
Index x = 20 hex (blank) --> separation between the identification number of serial number
Index x + 1 = 2F hex (slash) --> separation between the identification number of serial number
Index x + 2 = 20 hex (blank) --> separation between the identification number of serial number
Index x + 3 = first character of the serial number
...
Index y with contents = last character of the serial number

Dependency: Refer to: r0460, r0461, r0462, r0463, r0464
Notice: An ASCII table (excerpt) can be found, for example, in the Appendix of the List Manual:
Note: The individual characters of the identification number/serial number are available coded as ASCII characters.

r0470	Redundant coarse value valid bits / Valid bits		
ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the valid bits of the redundant coarse position value.

Dependency: Refer to: p9323, p9523

r0470[0...2] Redundant coarse value valid bits / Valid bits			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the valid bits of the redundant coarse position value.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p9323, p9523		

r0471 Redundant coarse value fine resolution bits / Fine bit			
ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the number of valid bits for the fine resolution of the redundant coarse position value.		
Dependency:	Refer to: p9324, p9524		

r0471[0...2] Redundant coarse value fine resolution bits / Fine bit			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the number of valid bits for the fine resolution of the redundant coarse position value.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p9324, p9524		

r0472 Redundant coarse position value relevant bits / Relevant bits			
ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the number of relevant bits for the redundant coarse position value.		

r0472[0...2]	Redundant coarse position value relevant bits / Relevant bits		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the number of relevant bits for the redundant coarse position value.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		

r0474	Redundant coarse position value configuration / Red pos config				
ENCODER	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned32		Dynamic index: -		Func. diagram: -
	P-Group: Encoder		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		-
Description:	Displays the encoder configuration for the redundant coarse position value.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Incrementer	Yes	No	-
	01	Encoder CRC least significant byte first	Yes	No	-
	02	Redundant coarse position val. most significant bit left-aligned	Yes	No	-
Dependency:	Refer to: p9315, p9515				

r0474[0...2]		Redundant coarse position value configuration / Red pos config					
SERVO, VECTOR	Can be changed: -		Calculated: -		Access level: 3		
	Data type: Unsigned32		Dynamic index: -		Func. diagram: -		
	P-Group: Encoder		Units group: -		Unit selection: -		
	Not for motor type: -		Scaling: -		Expert list: 1		
	Min		Max		Factory setting		
	-		-		-		
Description:		Displays the encoder configuration for the redundant coarse position value.					
Index:		[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3					
Bit field:	Bit	Signal name	1 signal		0 signal		FP
	00	Incrementer	Yes		No		-
	01	Encoder CRC least significant byte first	Yes		No		-
	02	Redundant coarse position val. most significant bit left-aligned	Yes		No		-
Dependency:		Refer to: p9315, p9515					

r0475	Gx_XIST1 coarse position safe most significant bit / Gx_XIST1 safe MSB		
ENCODER	Can be changed: - Data type: Unsigned16 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the bit number for the safe most significant bit (MSB) of the Gx_XIST1 coarse position.		
Note:	MSB: Most Significant Bit		
r0475[0...2]	Gx_XIST1 coarse position safe most significant bit / Gx_XIST1 safe MSB		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the bit number for the safe most significant bit (MSB) of the Gx_XIST1 coarse position.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Note:	MSB: Most Significant Bit		
r0477	CO: Measuring gear, position difference / Meas gear pos diff		
ENCODER	Can be changed: - Data type: Integer32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the position difference before the measuring gear between powering down and powering up.		
Dependency:	Refer to: F31501, F32501, F33501		
Note:	The increments are displayed in the format the same as r0483. The position difference should be read in encoder increments.		
r0477[0...2]	CO: Measuring gear, position difference / Meas gear pos diff		
SERVO, VECTOR	Can be changed: - Data type: Integer32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the position difference before the measuring gear between powering down and powering up.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: F31501, F32501, F33501		
Note:	The increments are displayed in the format the same as r0483. The position difference should be read in encoder increments.		

r0479	CO: Diagnostics encoder position actual value Gn_XIST1 / Diag Gn_XIST1		
ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: 4704
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the encoder actual position value Gn_XIST1 according to PROFIdrive for diagnostics.
In contrast to p0482, the value is updated in each DRIVE-CLiQ basic clock cycle and displayed with sign.

Caution:



Following ramping-up or after a data set changeover, the new value is present at connector inputs which are inter-connected to connector output r0479 and under certain circumstances take 100 ms to become available. Reason: These interconnections are updated in the background, unlike interconnections involving other connector outputs (e.g. CO: r0482).

The value is immediately available when non-cyclically reading r0479 (e.g. via the expert list).

r0479[0...2]	CO: Diagnostics encoder position actual value Gn_XIST1 / Diag Gn_XIST1		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: 4704
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the encoder actual position value Gn_XIST1 according to PROFIdrive for diagnostics.
In contrast to p0482, the value is updated in each DRIVE-CLiQ basic clock cycle and displayed with sign.

Index:

[0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

Caution:



Following ramping-up or after a data set changeover, the new value is present at connector inputs which are inter-connected to connector output r0479 and under certain circumstances take 100 ms to become available. Reason: These interconnections are updated in the background, unlike interconnections involving other connector outputs (e.g. CO: r0482).

The value is immediately available when non-cyclically reading r0479 (e.g. via the expert list).

r0479	CO: Diagnostics encoder position actual value Gn_XIST1 / Diag Gn_XIST1		
TM41	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: 9674, 9676
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the encoder actual position value Gn_XIST1 according to PROFIdrive for diagnostics.
In contrast to p0482, the value is updated in each DRIVE-CLiQ basic clock cycle and displayed with sign.

p0480				
CI: Signal source for encoder control word Gn_STW / Enc S_src Gn_STW				
ENCODER	Can be changed: T	Calculated: -	Access level: 3	
	Data type: Unsigned32 / Integer16	Dynamic index: -	Func. diagram: 1580, 4720	
	P-Group: Encoder	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description: Sets the signal source for the encoder control word Gn_STW according to PROFIdrive.				
Note: When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: CI: p0480[0] = r2520[0], CI: p0480[1] = r2520[1] and CI: p0480[2] = r2520[2]				
p0480[0...2]				
CI: Signal source for encoder control word Gn_STW / Enc S_src Gn_STW				
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3	
	Data type: Unsigned32 / Integer16	Dynamic index: -	Func. diagram: 1580, 4720	
	P-Group: Encoder	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description: Sets the signal source for the encoder control word Gn_STW according to PROFIdrive.				
Index: [0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3				
Note: When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: CI: p0480[0] = r2520[0], CI: p0480[1] = r2520[1] and CI: p0480[2] = r2520[2]				
r0481				
CO: Encoder status word Gn_ZSW / Enc Gn_ZSW				
ENCODER	Can be changed: -	Calculated: -	Access level: 3	
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 4010, 4704, 4730	
	P-Group: Encoder	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	-	
Description: Displays the encoder status word Gn_ZSW according to PROFIdrive.				
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Function 1 active	Yes	No
	01	Function 2 active	Yes	No
	02	Function 3 active	Yes	No
	03	Function 4 active	Yes	No
	04	Value 1	Displayed in r0483	Not present
	05	Value 2	Displayed in r0483	Not present
	06	Value 3	Displayed in r0483	Not present
	07	Value 4	Displayed in r0483	Not present
	08	Measuring probe 1 deflected	Yes	No
	09	Measuring probe 2 deflected	Yes	No
	11	Encoder fault acknowledge active	Yes	No
	13	Absolute value cyclically	Displayed in r0483	No
	14	Parking encoder active	Yes	No
	15	Encoder fault	Displayed in r0483	None
Notice: Information on Gn_STW/Gn_ZSW can, e.g. be found in the following literature: SINAMICS S120 Function Manual Drive Functions				

Note: Re bit 14:
Displays the acknowledgement for "activate parking encoder" (Gn_STW.14 = 1) or encoder position actual value (Gn_XIST1) invalid.
Re bit 14, 15:
r0481.14 = 1 and r0481.15 = 0 can have one of the following causes:
- the encoder is parked.
- the encoder is deactivated.
- the encoder is being commissioned.
- no parameterized encoder available.
- encoder data set is being changed over.
r0481.14 = 1 and r0481.15 = 1 has the following significance:
An encoder error has occurred and the encoder position actual value (Gn_XIST1) is invalid.

r0481[0...2] CO: Encoder status word Gn_ZSW / Enc Gn_ZSW

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 4010, 4704, 4730
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the encoder status word Gn_ZSW according to PROFIdrive.

Index:
[0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Function 1 active	Yes	No	-
	01	Function 2 active	Yes	No	-
	02	Function 3 active	Yes	No	-
	03	Function 4 active	Yes	No	-
	04	Value 1	Displayed in r0483	Not present	-
	05	Value 2	Displayed in r0483	Not present	-
	06	Value 3	Displayed in r0483	Not present	-
	07	Value 4	Displayed in r0483	Not present	-
	08	Measuring probe 1 deflected	Yes	No	-
	09	Measuring probe 2 deflected	Yes	No	-
	11	Encoder fault acknowledge active	Yes	No	9676
	13	Absolute value cyclically	Displayed in r0483	No	-
	14	Parking encoder active	Yes	No	-
	15	Encoder fault	Displayed in r0483	None	-

Notice: Information on Gn_STW/Gn_ZSW can, e.g. be found in the following literature:
SINAMICS S120 Function Manual Drive Functions

Note: Re bit 14:
Displays the acknowledgement for "activate parking encoder" (Gn_STW.14 = 1) or encoder position actual value (Gn_XIST1) invalid.
Re bit 14, 15:
r0481.14 = 1 and r0481.15 = 0 can have one of the following causes:
- the encoder is parked.
- the encoder is deactivated.
- the encoder is being commissioned.
- no parameterized encoder available.
- encoder data set is being changed over.
r0481.14 = 1 and r0481.15 = 1 has the following significance:
An encoder error has occurred and the encoder position actual value (Gn_XIST1) is invalid.

r0481 CO: Encoder status word Gn_ZSW / Enc Gn_ZSW

TM41	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the encoder status word Gn_ZSW according to PROFIdrive.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Function 1 active	Yes	No	-
	01	Function 2 active	Yes	No	-
	02	Function 3 active	Yes	No	-
	03	Function 4 active	Yes	No	-
	04	Value 1	Displayed in r0483	Not present	-
	05	Value 2	Displayed in r0483	Not present	-
	06	Value 3	Displayed in r0483	Not present	-
	07	Value 4	Displayed in r0483	Not present	-
	08	Measuring probe 1 deflected	Yes	No	-
	09	Measuring probe 2 deflected	Yes	No	-
	11	Encoder fault acknowledge active	Yes	No	9676
	13	Absolute value cyclically	Displayed in r0483	No	-
	14	Parking encoder active	Yes	No	-
	15	Encoder fault	Displayed in r0483	None	-

Notice: Information on Gn_STW/Gn_ZSW can, e.g. be found in the following literature:
SINAMICS S120 Function Manual Drive Functions

Note: For p4401 = 0, the following applies:
For Terminal Module 41 (TM41), this value is used to interconnect with standard telegram 3 and is always zero.
For p4401 = 1, the following applies:
r0481.0 indicates as to whether the zero mark synchronization is active.
r0481.4 indicates whether the zero mark of the incremental encoder was found.
r0481.14 indicates whether the output of track A/B is activated.

r0482 CO: Encoder actual position value Gn_XIST1 / Enc Gn_XIST1

ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1580, 2450, 4010, 4704, 4735, 4740
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the encoder actual position value Gn_XIST1 according to PROFIdrive.

Note: - this value is reset if necessary when the "parking axis" function is deselected.
- in this value, the measuring gear (p0432, p0433) is only taken into account when the position tracking is activated (p0411.0 = 1).

r0482[0...2]	CO: Encoder actual position value Gn_XIST1 / Enc Gn_XIST1		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 1580, 2450, 4010, 4704, 4735, 4740 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the encoder actual position value Gn_XIST1 according to PROFIdrive.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Note:	- this value is reset if necessary when the "parking axis" function is deselected. - in this value, the measuring gear (p0432, p0433) is only taken into account when the position tracking is activated (p0411.0 = 1).		
r0482	CO: Encoder actual position value Gn_XIST1 / Enc Gn_XIST1		
TM41	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 9674 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the encoder actual position value Gn_XIST1 according to PROFIdrive.		
r0483	CO: Encoder actual position value Gn_XIST2 / Enc Gn_XIST2		
ENCODER	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 1580, 2450, 4010, 4704 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the encoder actual position value Gn_XIST2 according to PROFIdrive.		
Recommend.:	Possible causes of the error codes: Error code 4097 and 4098: Defective Control Unit hardware. Error codes 4099 and 4100: Too many measuring pulses have occurred.		
Notice:	The encoder position actual value must be requested using the encoder control word Gn_STW.13.		

Note:

- in this value, the measuring gear (p0432, p0433) is only taken into account when the position tracking is activated (p0411.0 = 1).
- if GxZSW.15 = 1 (r0481), then an error code with the following significance is located in Gx_XIST2 (r0483):

- 1: Encoder fault.
- 2: Possible position shift in Gx_XIST1.
- 3: Reserved.
- 4: Abort, reference mark search.
- 5: Abort, retrieve reference value.
- 6: Abort, flying measurement.
- 7: Abort, retrieve measured value.
- 8: Abort, absolute value transfer.
- 3841: Function not supported.
- 4097: Abort, reference mark search due to an initialization error.
- 4098: Abort, flying measurement due to an initialization error.
- 4099: Abort, reference mark search due to a measuring error.
- 4100: Abort, flying measurement due to a measuring error.

r0483[0...2] CO: Encoder actual position value Gn_XIST2 / Enc Gn_XIST2			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1580, 2450, 4010, 4704
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the encoder actual position value Gn_XIST2 according to PROFIdrive.		
Recommend.:	Possible causes of the error codes: Error code 4097 and 4098: Defective Control Unit hardware. Error codes 4099 and 4100: Too many measuring pulses have occurred.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Notice:	The encoder position actual value must be requested using the encoder control word Gn_STW.13.		
Note:	- in this value, the measuring gear (p0432, p0433) is only taken into account when the position tracking is activated (p0411.0 = 1). - if GxZSW.15 = 1 (r0481), then an error code with the following significance is located in Gx_XIST2 (r0483): 1: Encoder fault. 2: Possible position shift in Gx_XIST1. 3: Reserved. 4: Abort, reference mark search. 5: Abort, retrieve reference value. 6: Abort, flying measurement. 7: Abort, retrieve measured value. 8: Abort, absolute value transfer. 3841: Function not supported. 4097: Abort, reference mark search due to an initialization error. 4098: Abort, flying measurement due to an initialization error. 4099: Abort, reference mark search due to a measuring error. 4100: Abort, flying measurement due to a measuring error.		

r0483	CO: Encoder actual position value Gn_XIST2 / Enc Gn_XIST2		
TM41	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the encoder actual position value Gn_XIST2 according to PROFIdrive.		
Notice:	The encoder position actual value must be requested using the encoder control word Gn_STW.13.		
Note:	SIMOTION (p4400 = 0) operating mode: This value is used for interconnection with standard telegram 3 and is always zero. SINAMICS (p4400 = 1) operating mode: Once automatic zero mark synchronization is complete, the position of the zero mark of the leading encoder is displayed in this parameter. The leading encoder is interconnected via connector input p4420.		
r0484	CO: Redundant coarse encoder position + CRC Gn_XIST1 / Enc red pos+CRC		
ENCODER	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the redundant coarse encoder position including CRC (Cyclic Redundancy Check). Upper 16 bits: CRC over the redundant coarse encoder position. Lower 16 bits: Redundant coarse encoder position. On SMx encoder modules, the encoder coarse position count direction is opposite to r0482 (encoder actual value Gn_XIST1). The value contains 2 bit fine resolution. On DRIVE-CLiQ encoders, the encoder coarse position count direction corresponds to r0482 and the encoder coarse position contains 9 valid bits and no fine resolution bits.		
Dependency:	The values are valid when the safety position actual value sensing is activated (p0430.19 = 1). Refer to: p0430		
Note:	This absolute value does not change, contrary to r0482, when deselecting the function "parking axis".		
r0484[0...2]	CO: Redundant coarse encoder position + CRC Gn_XIST1 / Enc red pos+CRC		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the redundant coarse encoder position including CRC (Cyclic Redundancy Check). Upper 16 bits: CRC over the redundant coarse encoder position. Lower 16 bits: Redundant coarse encoder position. On SMx encoder modules, the encoder coarse position count direction is opposite to r0482 (encoder actual value Gn_XIST1). The value contains 2 bit fine resolution. On DRIVE-CLiQ encoders, the encoder coarse position count direction corresponds to r0482 and the encoder coarse position contains 9 valid bits and no fine resolution bits.		

Index: [0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

Dependency: The values are valid when the safety position actual value sensing is activated (p0430.19 = 1).
Refer to: p0430

Note: This absolute value does not change, contrary to r0482, when deselecting the function "parking axis".

r0485 **CO: Measuring gear, encoder raw value incremental / Enc raw val incr**

ENCODER

Can be changed: -	Calculated: -	Access level: 1
Data type: Unsigned32	Dynamic index: -	Func. diagram: -
P-Group: Encoder	Units group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min	Max	Factory setting
-	-	-

Description: Displays the raw value of the incremental encoder actual value before the measuring gear.

r0485[0...2] **CO: Measuring gear, encoder raw value incremental / Enc raw val incr**

SERVO, VECTOR

Can be changed: -	Calculated: -	Access level: 1
Data type: Unsigned32	Dynamic index: -	Func. diagram: -
P-Group: Encoder	Units group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min	Max	Factory setting
-	-	-

Description: Displays the raw value of the incremental encoder actual value before the measuring gear.

Index: [0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

r0486 **CO: Measuring gear, encoder raw value absolute / Enc raw val abs**

ENCODER

Can be changed: -	Calculated: -	Access level: 1
Data type: Unsigned32	Dynamic index: -	Func. diagram: -
P-Group: Encoder	Units group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min	Max	Factory setting
-	-	-

Description: Displays the raw value of the absolute encoder actual value before the measuring gear.

r0486[0...2] **CO: Measuring gear, encoder raw value absolute / Enc raw val abs**

SERVO, VECTOR

Can be changed: -	Calculated: -	Access level: 1
Data type: Unsigned32	Dynamic index: -	Func. diagram: -
P-Group: Encoder	Units group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min	Max	Factory setting
-	-	-

Description: Displays the raw value of the absolute encoder actual value before the measuring gear.

Index: [0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

r0487 Diagnostic encoder control word Gn_STW / Enc Gn_STW

ENCODER

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1580, 4704, 4720, 4740**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the encoder control word Gn_STW according to PROFIdrive for diagnostics.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Request function 1	Yes	No	-
01	Request function 2	Yes	No	-
02	Request function 3	Yes	No	-
03	Request function 4	Yes	No	-
04	Request command bit 0	Yes	No	-
05	Request command bit 1	Yes	No	-
06	Request command bit 2	Yes	No	-
07	Flying measurement mode/search for reference mark	Flying measurement	Reference marks	-
13	Request absolute value cyclic	Yes	No	-
14	Request parking encoder	Yes	No	-
15	Request acknowledge encoder fault	Yes	No	-

Notice:

Information on Gn_STW/Gn_ZSW should be taken from the corresponding product documentation.

Note:

The signal source for the encoder control word is set with p0480.

r0487[0...2] Diagnostic encoder control word Gn_STW / Enc Gn_STW

SERVO, VECTOR

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1580, 4704, 4720, 4740**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the encoder control word Gn_STW according to PROFIdrive for diagnostics.

Index:

[0] = Encoder 1
 [1] = Encoder 2
 [2] = Encoder 3

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Request function 1	Yes	No	-
01	Request function 2	Yes	No	-
02	Request function 3	Yes	No	-
03	Request function 4	Yes	No	-
04	Request command bit 0	Yes	No	-
05	Request command bit 1	Yes	No	-
06	Request command bit 2	Yes	No	-
07	Flying measurement mode/search for reference mark	Flying measurement	Reference marks	-
13	Request absolute value cyclic	Yes	No	-
14	Request parking encoder	Yes	No	-
15	Request acknowledge encoder fault	Yes	No	-

Notice:

Information on Gn_STW/Gn_ZSW should be taken from the corresponding product documentation.

Note:

The signal source for the encoder control word is set with p0480.

p0488	Measuring probe 1 input terminal / Meas probe 1 inp		
ENCODER	Can be changed: U, T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 3 Func. diagram: 4740 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal to connect probe 1.		
Value:	0: No measuring probe 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9)		
Dependency:	Refer to: p0489, p0490, p0728		
Notice:	To select the values: For Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).		
Note:	DI/DO: Bidirectional digital input/output The terminal must be set as input (p0728). Refer to the encoder interface for PROFIdrive. If parameterization is rejected, check whether the terminal is already being used in p0580, p0680, p2517 or p2518.		

p0488[0...2]	Measuring probe 1 input terminal / Meas probe 1 inp		
SERVO, VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 3 Func. diagram: 4740 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal to connect probe 1.		
Value:	0: No measuring probe 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9)		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p0489, p0490, p0728		
Notice:	To select the values: For Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).		
Note:	DI/DO: Bidirectional digital input/output The terminal must be set as input (p0728). Refer to the encoder interface for PROFIdrive. If parameterization is rejected, check whether the terminal is already being used in p0580, p0680, p2517 or p2518.		

p0489	Measuring probe 2 input terminal / Meas probe 2 inp		
ENCODER	Can be changed: U, T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 3 Func. diagram: 4740 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal to connect probe 2.		
Value:	0: No measuring probe 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9)		
Dependency:	Refer to: p0488, p0490, p0728		
Notice:	To select the values: For Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).		
Note:	DI/DO: Bidirectional digital input/output The terminal must be set as input (p0728). Refer to the encoder interface for PROFIdrive. If parameterization is rejected, check whether the terminal is already being used in p0580, p0680, p2517 or p2518.		

p0489[0...2]	Measuring probe 2 input terminal / Meas probe 2 inp		
SERVO, VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 3 Func. diagram: 4740 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal to connect probe 2.		
Value:	0: No measuring probe 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9)		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p0488, p0490, p0728		
Notice:	To select the values: For Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).		
Note:	DI/DO: Bidirectional digital input/output The terminal must be set as input (p0728). Refer to the encoder interface for PROFIdrive. If parameterization is rejected, check whether the terminal is already being used in p0580, p0680, p2517 or p2518.		

p0490	Invert measuring probe or equivalent zero mark / Meas. probe invert				
CU_CX32, CU_I, CU_S, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 3		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 4740		
	P-Group: Encoder	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Setting to invert the digital input signals to connect a measuring probe or an equivalent zero mark.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	Inverted	Not inverted	-
	09	DI/DO 9 (X122.10)	Inverted	Not inverted	-
	10	DI/DO 10 (X122.12)	Inverted	Not inverted	-
	11	DI/DO 11 (X122.13)	Inverted	Not inverted	-
	12	DI/DO 12 (X132.9)	Inverted	Not inverted	-
	13	DI/DO 13 (X132.10)	Inverted	Not inverted	-
	14	DI/DO 14 (X132.12)	Inverted	Not inverted	-
	15	DI/DO 15 (X132.13)	Inverted	Not inverted	-
Dependency:	Refer to: p0488, p0489, p0493, p0495, p0728				
Notice:	To select the values: For Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).				
Note:	The terminal must be set as input. When the measuring probe or the equivalent zero mark is inverted, this has no effect on the status displays of the digital inputs (r0721, r0722, r0723). DI: Digital input, DO: Digital output				

p0490	Invert measuring probe or equivalent zero mark / Meas. probe invert				
CU_S_CU310DP, CU_S_CU310PN	Can be changed: U, T	Calculated: -	Access level: 3		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 4740		
	P-Group: Encoder	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Setting to invert the digital input signals to connect a measuring probe or an equivalent zero mark.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	09	DI/DO 9 (X122.10)	Inverted	Not inverted	-
	10	DI/DO 10 (X122.12)	Inverted	Not inverted	-
	11	DI/DO 11 (X122.13)	Inverted	Not inverted	-
Dependency:	Refer to: p0488, p0489, p0493, p0495, p0728				
Notice:	To select the values: For Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).				
Note:	The terminal must be set as input. When the measuring probe or the equivalent zero mark is inverted, this has no effect on the status displays of the digital inputs (r0721, r0722, r0723). DI: Digital input, DO: Digital output				

p0491 Motor encoder fault response ENCODER / Fault resp ENCODER			
SERVO	Can be changed: T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the behavior for the ENCODER fault response (motor encoder). This means, for example, if an encoder fault occurs, encoderless operation can be automatically selected with a shutdown behavior that can be selected.		
Value:	0: Encoder fault results in OFF2 1: Enc fault results in encoderless oper. and oper. continues 2: Encoder fault results in encoderless operation and OFF1 3: Encoder fault results in encoderless operation and OFF3 4: Encoder fault results in an armature short-cct internal/DC brake		
Dependency:	The following parameters are relevant for encoderless operation. Refer to: p0341, p0342, p1470, p1472, p1517, p1612, p1755 Refer to: F07575		
Caution:	 For a value = 1, 2, 3, the following applies: - encoderless operation must have been started. For a value = 1, the following applies: - in spite of the motor encoder fault that has occurred, the motor continues to operate.		
Note:	For a value = 1, 2, 3, the following applies: - for encoderless operation the following condition must be fulfilled: $p1800 \geq n / (2 * p0115[0])$, $n = 1, 2$, etc. - Refer to the status signal "encoderless operation due to a fault" (BO: r1407.13). - If, when setting r1407.13, a different drive data set is selected (e.g. interconnection from p0820), then the open-loop or closed-loop control type p1300 of this data set must match that of the original data set (e.g. p1300 = 21). Encoderless closed-loop controlled operation is kept when changing over. For a value = 4, the following applies: - The value can only be set for all motor data sets when p1231 = 3, 4. - For synchronous motors, an armature short circuit is initiated on an encoder fault. - For induction motors, DC braking is initiated on an encoder fault. The DC brake must be commissioned (p1232, p1233, p1234).		

p0491 Motor encoder fault response ENCODER / Fault resp ENCODER			
TM41	Can be changed: T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the behavior for the ENCODER fault response (motor encoder). This means, for example, if an encoder fault occurs, encoderless operation can be automatically selected with a shutdown behavior that can be selected.		
Value:	0: Encoder fault results in OFF2 1: Enc fault results in encoderless oper. and oper. continues 2: Encoder fault results in encoderless operation and OFF1 3: Encoder fault results in encoderless operation and OFF3 4: Encoder fault results in an armature short-cct internal/DC brake		
Dependency:	The following parameters are relevant for encoderless operation. Refer to: p0341, p0342, p1470, p1472, p1517, p1612, p1755 Refer to: F07575		

Caution:



For a value = 1, 2, 3, the following applies:

- encoderless operation must have been started.

For a value = 1, the following applies:

- in spite of the motor encoder fault that has occurred, the motor continues to operate.

Note:

For a value = 1, 2, 3, the following applies:

- Refer to the status signal "encoderless operation due to a fault" (BO: r1407.13).

- If, when setting r1407.13, a different drive data set is selected (e.g. interconnection from p0820), then the open-loop or closed-loop control type p1300 of this data set must match that of the original data set (e.g. p1300 = 21). Encoderless closed-loop controlled operation is kept when changing over.

For a value = 4, the following applies:

- The value can only be set for all motor data sets when p1231 = 3, 4.
- For synchronous motors, an armature short circuit is initiated on an encoder fault.
- For induction motors, DC braking is initiated on an encoder fault. The DC brake must be commissioned (p1232, p1233, p1234).

p0491 Motor encoder fault response ENCODER / Fault resp ENCODER

VECTOR

Can be changed: T

Calculated: -

Access level: 3

Data type: Integer16

Dynamic index: -

Func. diagram: -

P-Group: Encoder

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

0

4

0

Description:

Sets the behavior for the ENCODER fault response (motor encoder).

This means, for example, if an encoder fault occurs, encoderless operation can be automatically selected with a shutdown behavior that can be selected.

Value:

- 0: Encoder fault results in OFF2
- 1: Enc fault results in encoderless oper. and oper. continues
- 2: Encoder fault results in encoderless operation and OFF1
- 3: Encoder fault results in encoderless operation and OFF3
- 4: Encoder fault results in an armature short-cct internal/DC brake

Dependency:

The following parameters are relevant for encoderless operation.

Refer to: p0341, p0342, p1470, p1472, p1517, p1612, p1755

Refer to: F07575

Caution:



For a value = 1, 2, 3, the following applies:

- encoderless operation must have been started.

For a value = 1, the following applies:

- in spite of the motor encoder fault that has occurred, the motor continues to operate.

Note:

For a value = 1, 2, 3, the following applies:

- Refer to the status signal "encoderless operation due to a fault" (BO: r1407.13).

- Not possible for separately excited synchronous motors (p0300 = 5).

p0492 Square-wave encoder, maximum speed difference per sampling cycle / n_dif max/samp_cyc

ENCODER

Can be changed: U, T

Calculated: CALC_MOD_REG

Access level: 3

Data type: FloatingPoint32

Dynamic index: -

Func. diagram: -

P-Group: Encoder

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

0.00 [rpm]

210000.00 [rpm]

0.00 [rpm]

Description:

Sets the maximum permissible speed difference within the current controller sampling time for squarewave encoders.

Dependency:

Refer to: F31118, A31418

Note: For a value of 0.0, the speed change monitoring is disabled.
if the set maximum speed difference is only exceeded for one sampling time of the current controller, then an appropriate alarm is output. However, if the maximum speed difference is exceeded over several sampling times, then a corresponding fault is output.

p0492 Square-wave encoder, max. velocity difference per sampling cycle / v_dif max/samp_cyc

ENCODER (Lin enc)	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [m/min]	1000.00 [m/min]	0.00 [m/min]

Description: Sets the maximum permissible velocity difference within the current controller sampling time for square-wave encoders.

Dependency: Refer to: F31118, A31418

Note: For a value of 0.0, velocity change monitoring is disabled.
if the set maximum velocity difference is only exceeded for one sampling time of the current controller, then an appropriate alarm is output. However, if the maximum speed difference is exceeded over several sampling times, then a corresponding fault is output.

p0492 Square-wave encoder, maximum speed difference per sampling cycle / n_dif max/samp_cyc

SERVO, VECTOR	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [rpm]	210000.00 [rpm]	0.00 [rpm]

Description: Sets the maximum permissible speed difference within the current controller sampling time for squarewave encoders.

When the value is exceeded, depending on p0491, either encoderless closed-loop speed/torque control is selected or the drive is powered down.

Dependency: Refer to: F31118, A31418, F32118, A32418, F33118, A33418

Note: For a value of 0.0, the speed change monitoring is disabled.
if the set maximum speed difference is only exceeded for one sampling time of the current controller, then an appropriate alarm is output. However, if the maximum speed difference is exceeded over several sampling times, then a corresponding fault is output.

The following applies for VECTOR:

The parameter is only pre-assigned when selecting p0340 = 1, 3.

The following applies for SERVO, VECTORMV:

The speed actual value used for the monitoring is a floating average between p0115[0] and p0115[1].

p0492 Square-wave encoder, max. velocity difference per sampling cycle / v_dif max/samp_cyc			
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Encoder Not for motor type: - Min 0.00 [m/min]	Calculated: CALC_MOD_REG Dynamic index: - Units group: - Scaling: - Max 1000.00 [m/min]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [m/min]
Description:	Sets the maximum permissible speed difference between two computing cycles when square-wave encoders are evaluated. When the value is exceeded, depending on p0491, either encoderless closed-loop velocity/force control is selected or the drive is powered down.		
Dependency:	Refer to: F31118, A31418, F32118, A32418, F33118, A33418		
Note:	For a value of 0.0, velocity change monitoring is disabled. When half of the parameter value is exceeded, an alarm is already generated and the velocity change is limited to this.		

p0493 Zero mark selection, input terminal / ZM_sel inp_term			
ENCODER	Can be changed: U, T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal for selecting the reference mark via BERO/switching signal when performing referencing with several zero marks. The encoder interface supplies the position of the reference mark, which was detected immediately after the positive edge of the BERO signal.		
Value:	0: No selection via BERO 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9)		
Dependency:	Refer to: p0490		
Notice:	For CU310, CX32, NX10 and NX15, only DI/DO 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).		
Note:	Refer to the encoder interface for PROFIdrive. The terminal must be set as input (p0728). For p0493 = 0 (factory setting) the following applies: - there is no logic operation between the reference mark search and an input signal. For p0493 > 0, the following applies: - the positive edge of the input signal is evaluated. If the negative edge is to be evaluated, signal inversion must be parameterized via p0490. - if a parameter change is rejected, a check should be performed as to whether the input terminal is already being used in p0580, p0680, p2517, or p2518.		

p0493[0...n]	Zero mark selection, input terminal / ZM_sel inp_term		
SERVO, VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 8	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal for selecting the reference mark via BERO/switching signal when performing referencing with several zero marks. The encoder interface supplies the position of the reference mark, which was detected immediately after the positive edge of the BERO signal.		
Value:	0: No selection via BERO 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9)		
Dependency:	Refer to: p0490		
Notice:	For CU310, CX32, NX10 and NX15, only DI/DO 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).		
Note:	Refer to the encoder interface for PROFIdrive. The terminal must be set as input (p0728). For p0493 = 0 (factory setting) the following applies: - there is no logic operation between the reference mark search and an input signal. For p0493 > 0, the following applies: - the positive edge of the input signal is evaluated. If the negative edge is to be evaluated, signal inversion must be parameterized via p0490. - if a parameter change is rejected, a check should be performed as to whether the input terminal is already being used in p0580, p0680, p2517, or p2518.		

p0495	Equivalent zero mark, input terminal / Zero mark inp		
ENCODER	Can be changed: U, T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 3 Func. diagram: 4735 Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the input terminal for connecting an equivalent zero mark (external encoder zero mark).		
Value:	0: No equivalent zero mark (evaluation of the encoder zero mark) 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9)		
Dependency:	Refer to: p0490		
Notice:	For CU310, CX32, NX10 and NX15, only DI/DO 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).		

Note: Refer to the encoder interface for PROFIdrive.
The terminal must be set as input.
For p0495 = 0 (factory setting), the encoder zero mark is evaluated as zero mark.
For p0495 > 0, the following applies:
Depending on the direction of motion, the positive or negative edge at the appropriate input is evaluated.
- increasing position actual values (r0482) --> the 0/1 edge is evaluated.
- decreasing position actual values (r0482) --> the 1/0 edge is evaluated.
Only one zero mark is supported. If function 2, 3 or 4 is selected, this results in a fault message in Gn_ZSW.
The inversion of the inputs via p0490 affects the function "referencing with equivalent zero mark". This is the reason that the edge evaluation is interchanged as a function of the direction of motion.
An input can only be assigned to one encoder as measuring probe 1, 2 or equivalent zero mark. Exception: The same encoder can be simultaneously used as measuring probe and equivalent zero mark as both functions cannot be simultaneously requested.

p0495[0...2] Equivalent zero mark, input terminal / Zero mark inp

SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 4735
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	8	0

Description: Selects the input terminal for connecting an equivalent zero mark (external encoder zero mark).

Value:

- 0: No equivalent zero mark (evaluation of the encoder zero mark)
- 1: DI/DO 9 (X122.10)
- 2: DI/DO 10 (X122.12)
- 3: DI/DO 11 (X122.13)
- 4: DI/DO 13 (X132.10)
- 5: DI/DO 14 (X132.12)
- 6: DI/DO 15 (X132.13)
- 7: DI/DO 8 (X122.9)
- 8: DI/DO 12 (X132.9)

Index:

- [0] = Encoder 1
- [1] = Encoder 2
- [2] = Encoder 3

Dependency: Refer to: p0490

Notice: For CU310, CX32, NX10 and NX15, only DI/DO 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).

Note: Refer to the encoder interface for PROFIdrive.
The terminal must be set as input.
For p0495 = 0 (factory setting), the encoder zero mark is evaluated as zero mark.
For p0495 > 0, the following applies:
Depending on the direction of motion, the positive or negative edge at the appropriate input is evaluated.
- increasing position actual values (r0482) --> the 0/1 edge is evaluated.
- decreasing position actual values (r0482) --> the 1/0 edge is evaluated.
Only one zero mark is supported. If function 2, 3 or 4 is selected, this results in a fault message in Gn_ZSW.
The inversion of the inputs via p0490 affects the function "referencing with equivalent zero mark". This is the reason that the edge evaluation is interchanged as a function of the direction of motion.
An input can only be assigned to one encoder as measuring probe 1, 2 or equivalent zero mark. Exception: The same encoder can be simultaneously used as measuring probe and equivalent zero mark as both functions cannot be simultaneously requested.

p0496	Encoder diagnostic signal selection / Enc diag select		
ENCODER	Can be changed: U, T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 70	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the trace signal to be output in r0497, r0498 and r0499 for encoder diagnostics.		
Value:	0: Inactive 1: r0497: Mechanical revolution 10: r0498: Raw value, track A, r0499: Raw value, track B 11: r0498: Fine position X (-A/2), r0499: Fine position Y (-B/2) 12: r0498: Fine position Phi, r0499: - 13: r0498: Offset correction X, r0499: Offset correction Y 14: r0498: Phase correction X, r0499: Amplitude correction Y 15: r0498: Cubic correction X, r0499: Fine position X 16: r0498: oversampling channel A, r0499: oversampling channel B 17: r0498: fan-out, amount, r0499: fan-out, number 18: r0498: Oversampling angle, r0499: Oversampling amount 20: r0498: Raw value, track C, r0499: Raw value, track D 21: r0498: CD position X (-D/2), r0499: CD position Y (C/2) 22: r0498: CD position Phi, r0499: CD pos. Phi - mech. revolution 23: r0497: Zero mark status 24: r0498: Raw value, track R, r0499: Zero mark status 25: r0498: Raw value, track A, r0499: Raw value, track R 30: r0497: Absolute position serial 31: r0497: Absolute position, incremental 32: r0497: Zero mark position 33: r0497: Correction absolute position difference 40: r0498: Raw temperature, r0499: Temperature in 0.1 °C 41: r0498: Resistance in 0.1 Ohm, r0499: Temperature in 0.1 °C 42: r0497: Resistance 2500 Ohm 51: r0497: Absolute speed difference (dn/dt) 52: r0497: Xact1 corrected quadrants 60: Analog sensor: r0498: raw val chann. A, r0499: raw val chann. B 61: Analog sensor: r0498: fine pos chann. A, r0499: fine pos chann. B 70: Resolver: r0498: Transformation ratio, r0499: phase		
Dependency:	Refer to: r0497, r0498, r0499		
Note:	Re p0496 = 1: 360 ° <--> 2 ³² Re p0496 = 10 (resolver): 2900 mV <--> 26214 dec Re p0496 = 10, 20 (sin/cos 1 Vpp, EnDat): 500 mV <--> 21299 dec Re p0496 = 11 (resolver): 2900 mV <--> 13107 dec, internal processor offset is corrected Re p0496 = 11, 21 (sin/cos 1 Vpp, EnDat): 500 mV <--> 10650 dec, internal processor offset is corrected Re p0496 = 12: 180 ° fine position <--> 32768 dec Re p0496 = 13 (resolver): 2900 mV <--> 13107 dec Re p0496 = 13 (sin/cos 1 Vpp, EnDat): 500 mV <--> 10650 dec Re p0496 = 14: 100 % <--> 16384 dec Re p0496 = 15: 100 % <--> 16384 dec Re p0496 = 16: (resolver): channel A: 2900 mV <--> 26214 dec, channel B: 2900 mV <--> 26214 dec Re p0496 = 16: (sin/cos 1 Vpp, EnDat) channel A: 500 mV <--> 21299 dec, channel B: 500 mV <--> 21299 dec Re p0496 = 17 (resolver): absolute value: 2900 mV <--> 13107 dec, number: 1 ... 8 Re p0496 = 17 (sin/cos 1 Vpp, EnDat): absolute value 500 mV <--> 10650 dec, number: 1 ... 8 Re p0496 = 18 (resolver): angle: signal period <--> 2 ¹⁶ , absolute value: 2900 mV <--> 13107 dec Re p0496 = 18 (sin/cos 1 Vpp, EnDat): angle: signal period <--> 2 ¹⁶ , absolute value: 500 mV <--> 10650 dec Re p0496 = 22: 180 ° <--> 32768 dec Re p0496 = 23, 24: r0497.31 (r0499.15) set for at least 1 current controller cycle when encoder zero mark detected Re p0496 = 24, 25: 500 mV <--> 21299 dec		

Re p0496 = 30: Rotary: 1 singleturn measuring step <--> 1 dec, linear: 1 measuring step <--> 1 dec
 Re p0496 = 31: Absolute position, incremental in 1/4 encoder pulses
 Re p0496 = 32: Zero mark position in 1/4 encoder pulses
 Re p0496 = 33: counter offset absolute value in 1/4 encoder pulses
 Re p0496 = 40: $r0498 <--> (R_KTY/1 \text{ kOhm} - 0.9) * 32768$
 Re p0496 = 42: 2500 Ohm <--> 2^{32}
 Re p0496 = 51: 1 rpm <--> 1000 dec
 Re p0496 = 52: In 1/4 encoder pulses
 Re p0496 = 70: r: 100% <--> 32768 dec, phase: 180 ° <--> 18000 dec

p0496[0...2]		Encoder diagnostic signal selection / Enc diag select	
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	70	0
Description:	Selects the trace signal to be output in r0497, r0498 and r0499 for encoder diagnostics.		
Value:	0: Inactive		
	1: r0497: Mechanical revolution		
	10: r0498: Raw value, track A, r0499: Raw value, track B		
	11: r0498: Fine position X (-A/2), r0499: Fine position Y (-B/2)		
	12: r0498: Fine position Phi, r0499: -		
	13: r0498: Offset correction X, r0499: Offset correction Y		
	14: r0498: Phase correction X, r0499: Amplitude correction Y		
	15: r0498: Cubic correction X, r0499: Fine position X		
	16: r0498: oversampling channel A, r0499: oversampling channel B		
	17: r0498: fan-out, amount, r0499: fan-out, number		
	18: r0498: Oversampling angle, r0499: Oversampling amount		
	20: r0498: Raw value, track C, r0499: Raw value, track D		
	21: r0498: CD position X (-D/2), r0499: CD position Y (C/2)		
	22: r0498: CD position Phi, r0499: CD pos. Phi - mech. revolution		
	23: r0497: Zero mark status		
	24: r0498: Raw value, track R, r0499: Zero mark status		
	25: r0498: Raw value, track A, r0499: Raw value, track R		
	30: r0497: Absolute position serial		
	31: r0497: Absolute position, incremental		
	32: r0497: Zero mark position		
	33: r0497: Correction absolute position difference		
	40: r0498: Raw temperature, r0499: Temperature in 0.1 °C		
	41: r0498: Resistance in 0.1 Ohm, r0499: Temperature in 0.1 °C		
	42: r0497: Resistance 2500 Ohm		
	51: r0497: Absolute speed difference (dn/dt)		
	52: r0497: Xact1 corrected quadrants		
	60: Analog sensor: r0498: raw val chann. A, r0499: raw val chann. B		
	61: Analog sensor: r0498: fine pos chann. A, r0499: fine pos chann. B		
	70: Resolver: r0498: Transformation ratio, r0499: phase		
Index:	[0] = Encoder 1		
	[1] = Encoder 2		
	[2] = Encoder 3		
Dependency:	Refer to: r0497, r0498, r0499		

Note:

Re p0496 = 1: 360 ° <--> 2³²

Re p0496 = 10 (resolver): 2900 mV <--> 26214 dec

Re p0496 = 10, 20 (sin/cos 1 Vpp, EnDat): 500 mV <--> 21299 dec

Re p0496 = 11 (resolver): 2900 mV <--> 13107 dec, internal processor offset is corrected

Re p0496 = 11, 21 (sin/cos 1 Vpp, EnDat): 500 mV <--> 10650 dec, internal processor offset is corrected

Re p0496 = 12: 180 ° fine position <--> 32768 dec

Re p0496 = 13 (resolver): 2900 mV <--> 13107 dec

Re p0496 = 13 (sin/cos 1 Vpp, EnDat): 500 mV <--> 10650 dec

Re p0496 = 14: 100 % <--> 16384 dec

Re p0496 = 15: 100 % <--> 16384 dec

Re p0496 = 16: (resolver): channel A: 2900 mV <--> 26214 dec, channel B: 2900 mV <--> 26214 dec

Re p0496 = 16: (sin/cos 1 Vpp, EnDat) channel A: 500 mV <--> 21299 dec, channel B: 500 mV <--> 21299 dec

Re p0496 = 17 (resolver): absolute value: 2900 mV <--> 13107 dec, number: 1 ... 8

Re p0496 = 17 (sin/cos 1 Vpp, EnDat): absolute value 500 mV <--> 10650 dec, number: 1 ... 8

Re p0496 = 18 (resolver): angle: signal period <--> 2¹⁶, absolute value: 2900 mV <--> 13107 dec

Re p0496 = 18 (sin/cos 1 Vpp, EnDat): angle: signal period <--> 2¹⁶, absolute value: 500 mV <--> 10650 dec

Re p0496 = 22: 180 ° <--> 32768 dec

Re p0496 = 23, 24: r0497.31 (r0499.15) set for at least 1 current controller cycle when encoder zero mark detected

Re p0496 = 24, 25: 500 mV <--> 21299 dec

Re p0496 = 30: Rotary: 1 singleturn measuring step <--> 1 dec, linear: 1 measuring step <--> 1 dec

Re p0496 = 31: Absolute position, incremental in 1/4 encoder pulses

Re p0496 = 32: Zero mark position in 1/4 encoder pulses

Re p0496 = 33: counter offset absolute value in 1/4 encoder pulses

Re p0496 = 40: r0498 <--> (R_KTY/1 kOhm - 0.9) * 32768

Re p0496 = 42: 2500 Ohm <--> 2³²

Re p0496 = 51: 1 rpm <--> 1000 dec

Re p0496 = 52: In 1/4 encoder pulses

Re p0496 = 70: r: 100% <--> 32768 dec, phase: 180 ° <--> 18000 dec

r0497 Encoder diagnostic signal double word / Enc diag DW

ENCODER	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the trace signal for encoder diagnostics (double word).
The signal to be output is selected in p0496.

Dependency: Refer to: p0496, r0498, r0499

r0497[0...2] CO: Encoder diagnostic signal double word / Enc diag DW

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the trace signal for encoder diagnostics (double word).
The signal to be output is selected in p0496.

Index:
[0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

Dependency: Refer to: p0496, r0498, r0499

r0498	Encoder diagnostic signal low word / Enc diag low word		
ENCODER	Can be changed: - Data type: Integer16 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the trace signal for encoder diagnostics (low component). The signal to be output is selected in p0496.		
Dependency:	Refer to: p0496, r0497, r0499		

r0498[0...2]	CO: Encoder diagnostic signal low word / Enc diag low word		
SERVO, VECTOR	Can be changed: - Data type: Integer16 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the trace signal for encoder diagnostics (low component). The signal to be output is selected in p0496.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p0496, r0497, r0499		

r0499	Encoder diagnostic signal high word / Enc diag high word		
ENCODER	Can be changed: - Data type: Integer16 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the trace signal for encoder diagnostics (high component). The signal to be output is selected in p0496.		
Dependency:	Refer to: p0496, r0497, r0498		

r0499[0...2]	CO: Encoder diagnostic signal high word / Enc diag high word		
SERVO, VECTOR	Can be changed: - Data type: Integer16 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the trace signal for encoder diagnostics (high component). The signal to be output is selected in p0496.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p0496, r0497, r0498		

p0500 Technology application / Tec application			
SERVO	Can be changed: C2(1, 5), T Data type: Integer16 P-Group: Applications Not for motor type: - Min 100	Calculated: - Dynamic index: - Units group: - Scaling: - Max 102	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100
Description:	Sets the technology application. The parameter influences the calculation of open-loop and closed-loop control parameters that is e.g. initiated using p0578.		
Value:	100: Standard drive (SERVO) 101: Feed drive (limit current limitation) 102: Spindle drive (rated current limitation)		
Dependency:	Refer to: p1520, p1521, p1530, p1531, p2000, p2175, p2177		
Caution: 	After changing over the technological application and then calculating the open-loop and closed-loop parameters, the behavior of the motor can have changed very significantly (e.g. the same setpoint results in a higher speed due to a different reference speed). For this reason extreme caution must be taken when the motor is started for the first time.		
Note:	The calculation of parameters dependent on the technology application can be called up as follows: - when exiting quick commissioning using p3900 > 0 - when writing p0340 = 1, 3, 5 - when writing p0578 = 1 For p0500 = 100 and when the calculation is initiated, the following parameters are set: - p1520/p1521 = rated motor torque (r0333) - p1530/p1531= 2*pi*r0333*p0311 (rotary) or r0333*p0311 (linear) - p2000 = rated motor speed (p0311) - p2175 = factory setting - p2177 = factory setting For p0500 = 101 and when the calculation is initiated, the following parameters are set: - p1520/p1521 = torque at the maximum motor current (p0323) - p1530/p1531= power at the maximum motor current (p0323) and rated motor speed (p0311) - p2000 = rated motor speed (p0311) - p2175 = maximum value - p2177 = 0.2 s For p0500 = 102 and when the calculation is initiated, the following parameters are set: - p1520/p1521 = rated motor torque (r0333) - p1530/p1531= 2*pi*r0333*p0311 (rotary) or r0333*p0311 (linear) - p2000 = maximum motor speed (p0322) if p0322 is not equal to 0, otherwise rated motor speed (p0311) - p2175 = factory setting - p2177 = factory setting		
p0500 Technology application / Tec application			
VECTOR	Can be changed: C2(1, 5), T Data type: Integer16 P-Group: Applications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the technology application. The parameter influences the calculation of open-loop and closed-loop control parameters that is e.g. initiated using p0578.		

Value: 0: Standard drive (VECTOR)
1: Pumps and fans
2: Sensorless closed-loop control down to $f = 0$ (passive loads)

Dependency: Refer to: p2175, p2177

Note: The calculation of parameters dependent on the technology application can be called up as follows:
- when exiting quick commissioning using p3900 > 0
- when writing p0340 = 1, 3, 5
- when writing p0578 = 1
For p0500 = 0 and when the calculation is initiated, the following parameters are set:
p1574 = 10 V (for separately-excited synchronous motors: 20 V)
p1750 bit 2 = 0
p1802 = 4 (SVM/FLB without overcontrol)
p1803 = 106 %
For p0500 = 1 and when the calculation is initiated, the following parameters are set:
p1574 = 2 V (for separately-excited synchronous motors: 4 V)
p1750 bit 2 = 0
p1802 = 9 (edge modulation), if r0192 bit 0 = 1
p1802 = 4, if r0192 bit 0 = 0 or parallel circuit configuration with single-winding system (p7003)
p1803 = 106 %
For p0500 = 2 and when the calculation is initiated, the following parameters are set:
p1574 = 2 V (for separately-excited synchronous motors: 4 V)
p1750 bit 2 = 1: Sensorless closed-loop control of induction motors effective up to a frequency of zero.
This operating mode is possible for passive loads. These include applications where the load does not generate regenerative torque when breaking away and the motor comes to a standstill (zero speed) itself when the pulses are inhibited.
p1802 = 4 (SVM/FLB without overcontrol)
p1803 = 106 %
The setting of p1750 is only relevant for induction motors.
p1802 and p1803 are only changed, in all cases, if a sine-wave output filter (p0230 = 3, 4) has not been selected.

p0505 Selecting the system of units / Select unit sys

A_INF, B_INF,
ENCODER, S_INF,
SERVO, TM41, VEC-
TOR

Can be changed: C2(5)
Data type: Integer16
P-Group: Applications
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 1
Func. diagram: -
Unit selection: -
Expert list: 1
Factory setting
1

Min
1

Max
4

Description: Adjustable parameter of the actual system of units.

Value: 1: System of units SI
2: System of units, referred/SI
3: US system of units
4: System of units, referred/US

Dependency: The parameter cannot be changed if the master control was fetched.

Caution: If a per unit representation is selected and if reference parameters (e.g. p2000) are subsequently changed, then the physical significance of some closed-loop control parameters will also be adapted where as a result, the closed-loop control behavior can change (refer to p1576, p1621, p1744, p1752, p1755 and p1609, p1612, p1619, p1620).



Note: Reference parameter for the unit system % are, for example, p2000 ... p2004. Depending on what has been selected, these are displayed using either SI or US units.

p0528 Controller gain, system of units / Ctrl_gain unit_sys

ENCODER, SERVO, TM41	Can be changed: C2(5)	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Applications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0

Description: Sets the system of units for the controller gains.

Value: 0: Representation, physical/% (p0505)
1: Representation, no dimensions (referred)

Note: For p0528 = 0 (physical/%), the following applies:
Using p0505, the dependent parameters can be changed over between physical and % representation.
For SERVO (r0107) the following applies:
The parameter is pre-assigned a value of 0 and cannot be changed.

p0528 Controller gain, system of units / Ctrl_gain unit_sys

VECTOR	Can be changed: C2(5)	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Applications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	1

Description: Sets the system of units for the controller gains.

Value: 0: Representation, physical/% (p0505)
1: Representation, no dimensions (referred)

Note: For VECTOR (r0107) the following applies:
The parameter is pre-assigned a value of 1 and cannot be changed.

p0530[0...n] Bearing type selection / Bearing type sel

SERVO, VECTOR	Can be changed: C2(1, 3)	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: FEM	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	104	0

Description: The parameter is used to select a bearing type.

0 = No selection

1 = Manual entry

101 = STANDARD

102 = PERFORMANCE

103 = HIGH PERFORMANCE

104 = ADVANCED LIFETIME

If the bearing type changes the bearing code number (p0531) is pre-assigned accordingly.

Dependency: Refer to: p0301, p0531

Caution: If a valid bearing type (p0530) is entered, the parameters in the bearing list cannot be changed (write protection).
Write protection is canceled if bearing type 1 is entered.

Note: p0530 cannot be changed on a motor with DRIVE-CLiQ.

p0531[0...n]	Bearing code number selection / Bear. code num sel		
SERVO, VECTOR	Can be changed: C2(3) Data type: Unsigned16 P-Group: Motor Not for motor type: FEM Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 65535	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to display and enter the bearing code number. It is pre-assigned automatically (from the parameter lists which are available internally) when p0301 and p0530 are entered and cannot be changed (write protection). The information in p0530 should be observed when removing write protection. When changing the bearing code number (except in the case of changing the value to 0), all of the bearing parameters are pre-assigned from the parameter lists which are available internally.		
Dependency:	Refer to: p0301, p0530		
Note:	p0531 cannot be changed on a motor with DRIVE-CLiQ.		
p0532[0...n]	Bearing maximum speed / Bearing n_max		
SERVO	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: FEM Min 0.0 [rpm]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 210000.0 [rpm]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [rpm]
Description:	Sets the maximum motor bearing speed.		
Dependency:	Refer to: p1082		
Caution:	This parameter is pre-assigned in the case of motors from the motor list (p0301) if a bearing type (p0530) or a bearing code (p0531) is selected. When selecting a catalog motor, this parameter cannot be changed (write protection). The information in p0530 should be observed when removing write protection.		
Notice:	If p0532 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned appropriately. This is not the case when commissioning the motor (p0010 = 3).		
p0532[0...n]	Bearing maximum velocity / Bearing v_max		
SERVO (Lin)	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: FEM Min 0.0 [m/min]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 1000.0 [m/min]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [m/min]
Description:	Sets the maximum bearing velocity.		
Dependency:	Refer to: p1082		
Caution:	This parameter is pre-assigned in the case of motors from the motor list (p0301) if a bearing type (p0530) or a bearing code (p0531) is selected. When selecting a catalog motor, this parameter cannot be changed (write protection). The information in p0530 should be observed when removing write protection.		
Notice:	If p0532 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned appropriately. This is not the case when commissioning the motor (p0010 = 3).		

p0532[0...n]	Bearing maximum speed / Bearing n_max		
VECTOR	Can be changed: C2(1, 3) Data type: FloatingPoint32 P-Group: Motor Not for motor type: FEM Min 0.0 [rpm]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 210000.0 [rpm]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [rpm]
Description:	Sets the maximum motor bearing speed.		
Dependency:	Refer to: p1082		
Caution:	This parameter is pre-assigned in the case of motors from the motor list (p0301) if a bearing type (p0530) or a bearing code (p0531) is selected. When selecting a catalog motor, this parameter cannot be changed (write protection). The information in p0530 should be observed when removing write protection.		
Notice:	If p0532 is changed during quick commissioning (p0010 = 1), then the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned appropriately. This is not the case when commissioning the motor (p0010 = 3).		
p0570	Inhibit list values effective number / Inhib list no		
SERVO, VECTOR	Can be changed: U, T Data type: Unsigned8 P-Group: Applications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 50	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the number of parameters in the inhibit list p0571 that should be withdrawn from the automatic motor and closed-loop control parameter calculation (refer to p0340, p0578), starting from index 0.		
Note:	Defines the number of entries in p0571 that should be taken into account. This means that a value of 0 deactivates the complete list.		
p0571[0...49]	Inhibit list, motor/closed-loop control parameter calculation / Inhib list calc		
SERVO	Can be changed: U, T Data type: Integer16 P-Group: Applications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2142	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The inhibit list contains parameters that should be withdrawn from the automatic motor and closed-loop control parameter calculation (p0340, p0578).		
Value:	0: No parameter 348: Speed at the start of field weakening Vdc = 600 V 600: Motor temperature sensor 640: Current limit 1082: Maximum speed 1441: Actual speed smoothing time 1460: Speed controller P gain 1462: Speed controller integral time 1470: Speed controller P gain, encoderless 1472: Speed controller integral time, encoderless 1520: Torque limit upper/motoring 1521: Torque limit lower/regenerative 1530: Power limit motoring 1531: Power limit regenerative 1590: Flux controller P gain 1592: Flux controller integral time 1656: Activates current setpoint filter 2141: Speed threshold 1 2142: Hysteresis speed 1		

Note: p0570 defines the number of entries (starting at index 0) for which the inhibit should apply. p0572 can be used to define for which drive data sets the inhibit list should apply.
If a motor data set is entered into a parameter number, then this is not overwritten as soon as only one drive data set refers to the motor data set (p0186).

p0571[0...49] Inhibit list, motor/closed-loop control parameter calculation / Inhib list calc

VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Applications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	2142	0

Description: The inhibit list contains parameters that should be withdrawn from the automatic motor and closed-loop control parameter calculation (p0340, p0578).

Value:

- 0: No parameter
- 600: Motor temperature sensor
- 640: Current limit
- 1082: Maximum speed
- 1460: Speed controller P gain
- 1462: Speed controller integral time
- 1470: Speed controller P gain, encoderless
- 1472: Speed controller integral time, encoderless
- 1520: Torque limit upper/motoring
- 1521: Torque limit lower/regenerative
- 1530: Power limit motoring
- 1531: Power limit regenerative
- 1590: Flux controller P gain
- 1592: Flux controller integral time
- 2141: Speed threshold 1
- 2142: Hysteresis speed 1

Note: p0570 defines the number of entries (starting at index 0) for which the inhibit should apply. p0572 can be used to define for which drive data sets the inhibit list should apply.
If a motor data set is entered into a parameter number, then this is not overwritten as soon as only one drive data set refers to the motor data set (p0186).

p0572[0...n] Activate inhibit list / Act inhib list

SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Applications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0

Description: Enable ID as to whether the parameters of the inhibit list p0571 should be overwritten when calculating the motor and closed-loop control parameters for the particular drive data set (index = DDS).

Value:

- 0: No
- 1: Yes

Note:

- 0: The automatic calculation (p0340, p0578) also overwrites the parameters of list p0571.
- 1: The automatic calculation (p0340, p0578) does not overwrite the parameters of list p0571.

p0573	Inhibit automatic reference value calculation / Inhibit calc		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Applications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Inhibits the calculation of reference parameters (e.g. p2000) when automatically calculating the motor and closed-loop control parameters (p0340, p3900).		
Value:	0: No 1: Yes		
Notice:	The inhibit for the reference value calculation is canceled when new motor parameters (e.f. p0305) are entered and only one drive data set exists (p0180 = 1). This is the case during initial commissioning. Once the motor and control parameters have been calculated (see p3900, p0340), the inhibit for the reference value calculation is automatically reactivated.		
Note:	0: The automatic calculation (p0340, p3900) overwrites the reference parameters. 1: The automatic calculation (p0340, p3900) does not overwrite the reference parameters.		
p0578[0...n]	Calculate parameters that are dependent on the technology/units / Calc tec par		
SERVO, VECTOR	Can be changed: C2(5), T Data type: Integer16 P-Group: Applications Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	This parameter is used to calculate all parameters that depend on the technology of the application (p0500). All of the parameters are calculated that can also be determined using p0340 = 5.		
Value:	0: No calculation 1: Complete parameterization		
Note:	At the end of the calculations, p0578 is automatically set to 0.		
p0580	Measuring probe, input terminal / MT input terminal		
SERVO	Can be changed: U, T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal for the measuring probe for speed actual value measurement.		
Value:	0: No measuring probe 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9)		
Dependency:	Refer to: p0581, p0728 Refer to: A07498		
Notice:	To select the values: For CU310, Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).		

Note: DI/DO: Bidirectional digital input/output
The terminal must be set as input (p0728).
If a parameter change is rejected, a check should be performed as to whether the input terminal is already being used in p0488, p0489, p0493, p0495, p0680, p2517, or p2518.

p0581	Meas probe, edge / MT edge		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0
Description:	Sets the edge to evaluate the measuring probe signal for speed actual value measurement. 0: 0/1 edge 1: 1/0 edge		
Dependency:	Refer to: p0580		

p0582	Measuring probe, pulses per revolution / MT pulses per rev		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	1	12	1
Description:	Sets the number of pulses per revolution (e.g. for disks with holes).		

p0583	Measuring probe, maximum measuring time / MT t_meas max		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.040 [s]	10.000 [s]	10.000 [s]
Description:	Sets the maximum measuring time for the measuring probe. If a new pulse is not received before the maximum measuring time has expired, then the speed actual value in r0586 is set to zero. This timer is re-started with the next pulse.		
Dependency:	Refer to: r0586		

r0586	CO: Measuring probe, speed actual value / MT n_act		
SERVO	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min	Max	Factory setting
	- [rpm]	- [rpm]	- [rpm]
Description:	Displays the speed actual value measured using the BERO.		
Dependency:	Refer to: p0580, p0583		
Note:	For p0580 = 0 (no measuring probe), a value of zero is displayed here.		

r0586	CO: Measuring probe, velocity actual value / MT v_act		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocity actual value measured using the BERO.		
Dependency:	Refer to: p0580, p0583		
Note:	For p0580 = 0 (no measuring probe), a value of zero is displayed here.		
r0587	CO: Measuring probe, measuring time measured / MT t_meas measured		
SERVO	Can be changed: - Data type: Unsigned32 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time between the last two BERO pulses. The measuring time is specified as 32-bit value with a resolution of 1/48 µs. If a new pulse is not received before the maximum measured time in p0583 expires, then r0587 is set to the maximum measuring time.		
Dependency:	Refer to: p0580		
Note:	For p0580 = 0 (no measuring probe), a value of zero is displayed here.		
r0588	CO: Measuring probe, pulse counter / MT pulse counter		
SERVO	Can be changed: - Data type: Unsigned32 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of measuring pulses that have occurred (been received) up until now.		
Dependency:	Refer to: p0580		
Note:	After reaching 4294967295 ($2^{32} - 1$), the counter starts again at 0.		
r0589	Measuring probe, delay time / MT t_delay		
SERVO	Can be changed: - Data type: Unsigned32 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time since the last measuring pulse was detected. The delay time is specified as 32-bit value with a resolution of 1/48 µs. When a measuring pulse occurs (is received) the delay time is reset and is limited to the maximum measuring time in p0583.		
Dependency:	Refer to: p0580		
Note:	For p0580 = 0 (no measuring probe), a value of zero is displayed here.		

p0595 Selecting technological units / Select tech units

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: C2(5)	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Applications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1	Max 32	Factory setting 1

Description: Selects the units for the parameters of the technology controller.

Value:

- 1: %
- 2: 1 referred, no dimensions
- 3: Bar
- 4: °C
- 5: Pa
- 6: ltr/s
- 7: m³/s
- 8: ltr/min
- 9: m³/min
- 10: ltr/h
- 11: m³/h
- 12: kg/s
- 13: kg/min
- 14: kg/h
- 15: t/min
- 16: t/h
- 17: N
- 18: kN
- 19: Nm
- 20: psi
- 21: °F
- 22: gallon/s
- 23: inch³/s
- 24: gallon/min
- 25: inch³/min
- 26: gallon/h
- 27: inch³/h
- 28: lb/s
- 29: lb/min
- 30: lb/h
- 31: lbf
- 32: lbf ft

Dependency: Only units of parameters with unit group 9_1 can be changed over using this parameter.
Refer to: p0596

p0596 Reference quantity, technological units / Ref tech units

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.01	Max 340.28235E36	Factory setting 1.00

Description: Sets the reference quantity for the technological units. When changing over using changeover parameter 595 to absolute units, all of the parameters involved refer to the reference quantity.

Dependency: Refer to: p0595

p0600[0...n] Motor temperature sensor for monitoring / Mot temp_sensor			
SERVO	Can be changed: C2(3), U, T Data type: Integer16 P-Group: Motor Not for motor type: - Min 0	Calculated: CALC_MOD_ALL Dynamic index: MDS, p0130 Units group: - Scaling: - Max 21	Access level: 2 Func. diagram: 8016 Unit selection: - Expert list: 1 Factory setting 1
Description: Sets the sensor to monitor the motor temperature.			
Value: 0: No sensor 1: Temperature sensor via encoder 1 2: Temperature sensor via encoder 2 3: Temperature sensor via encoder 3 10: Temperature sensor via a BICO interconnection 11: Temperature sensor via Motor Module / CU terminals 20: Temperature sensor via a BICO interconnection p608 21: Temperature sensor via a BICO interconnection p609			
Dependency: Refer to: r0458, p0601, p0603			
Caution: If, for a selected temperature sensor (p0600 > 0), the motor temperature sensor is not connected but another encoder, then the temperature adaptation of the motor resistances must be switched out. Otherwise, in controlled-loop operation, torque errors will occur that will mean that the drive will not be able to be stopped.			
			
Notice: The parameter is calculated in the drive using p0340 and is inhibited for p0340 > 0.			
Note: Re p0600 = 0: With induction motors, the motor temperature is calculated using the motor temperature model (see also p0612, bit 1). Re p0600 = 1, 2, 3: Bimetallic switch (p0601 = 4) and PT100 temperature sensor (p0601 = 5) are not supported. Re p0600 = 10: The BICO interconnection should be executed via connector input CI: p0603. Re p0600 = 11: For SINAMICS S120 AC Drive (AC/AC) and using the Control Unit Adapter CUA31, the temperature sensor is connected at the adapter (X210). Re p0600 = 20, 21: The BICO interconnection should be executed via connector input CI: p0608 or p0609. Associated parameters: p0601, p4600..4603, p4610..p4613			

p0600[0...n] Motor temperature sensor for monitoring / Mot temp_sensor			
VECTOR	Can be changed: C2(3), U, T Data type: Integer16 P-Group: Motor Not for motor type: - Min 0	Calculated: CALC_MOD_ALL Dynamic index: MDS, p0130 Units group: - Scaling: - Max 21	Access level: 2 Func. diagram: 8016 Unit selection: - Expert list: 1 Factory setting 0
Description: Sets the sensor to monitor the motor temperature.			
Value: 0: No sensor 1: Temperature sensor via encoder 1 2: Temperature sensor via encoder 2 3: Temperature sensor via encoder 3 10: Temperature sensor via a BICO interconnection 11: Temperature sensor via Motor Module / CU terminals 20: Temperature sensor via a BICO interconnection p608 21: Temperature sensor via a BICO interconnection p609			
Dependency: Refer to: r0458, p0601, p0603			

Caution:



Notice:

Note:

If, for a selected temperature sensor (p0600 > 0), the motor temperature sensor is not connected but another encoder, then the temperature adaptation of the motor resistances must be switched out. Otherwise, in controlled-loop operation, torque errors will occur that will mean that the drive will not be able to be stopped.

The parameter is calculated in the drive using p0340 and is inhibited for p0340 > 0.

Re p0600 = 0:

With induction motors, the motor temperature is calculated using the motor temperature model (see also p0612, bit 1).

Re p0600 = 1, 2, 3:

Bimetallic switch (p0601 = 4) and PT100 temperature sensor (p0601 = 5) are not supported.

Re p0600 = 10:

The BICO interconnection should be executed via connector input CI: p0603.

Re p0600 = 11:

For SINAMICS S120 AC Drive (AC/AC) and using the Control Unit Adapter CUA31, the temperature sensor is connected at the adapter (X210).

Re p0600 = 20, 21:

The BICO interconnection should be executed via connector input CI: p0608 or p0609.

Associated parameters: p0601, p4600..4603, p4610..p4613

p0601

Temperature sensor, sensor type / Temp_sens type

A_INF, B_INF,
S_INF

Can be changed: C2(3), U, T

Calculated: -

Access level: 2

Data type: Integer16

Dynamic index: -

Func. diagram: -

P-Group: Motor

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

0

4

0

Description:

Sets the sensor type for the temperature measurement at input X21 (booksize) or X41 (chassis).

The measured value is displayed in r0035.

Value:

- 0: No sensor
- 1: PTC alarm & timer
- 2: KTY84
- 4: Bimetallic NC contact alarm & timer

Dependency:

Refer to: r0035

Note:

The measured value display depends on the selected sensor type.

Re p0601 = 0 (no sensor):

--> r0035 = -200 °C

Re p0601 = 1 (PTC alarm & timer):

Tripping resistance = 1650 Ohm (lower resistance --> r0035 = -50 °C, higher resistance --> r0035 = 250 °C).

Re p0601 = 2 (KTY84):

Displays the temperature in °C.

Re p0601 = 4 (bimetallic NC contact alarm & timer):

r0035 = -50 °C

--> The tripping resistance is less than 100 Ohm (bimetallic NC contact is closed or has a short-circuit).

r0035 = 250 °C

--> The tripping resistance is greater than 100 Ohm (bimetallic NC contact is open, not connected or has a wire breakage).

When using the following components, a value of 4 is set as the factory setting and can no longer be changed:

- Basic Line Module (BLM) with internal Braking Module.
- Active Line Module (ALM) with line filter Active Interface Module (AIM, p0220[0] = 41 ... 45).

In these cases, in addition to the temperature display, the temperature is also monitored.

p0601[0...n]	Motor temperature sensor type / Mot_temp_sens type		
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	11	2
Description:	Sets the sensor type for the motor temperature monitoring.		
Value:	0: No sensor 1: PTC alarm & timer 2: KTY84 3: KTY84 and PTC (only for motors with DRIVE-CLiQ): 4: Bimetallic NC contact alarm & timer (only for temp_eval via MM) 5: PT100 10: Evaluation via several temperature channels SME12x 11: Evaluation via several temperature channels BICO		
Dependency:	The thermal motor model is only calculated for p0612.1 = 1. Refer to: r0458, p0600, p0612		
Note:	The temperature sensor for the temperature evaluation is set in p0600. For p0600 = 10 (temperature sensor via a BICO interconnection), the setting in p0601 has no significance. Information on using temperature sensors is provided in the following literature: - hardware description of the appropriate components - SINAMICS S120 Commissioning Manual Re p0601 = 1 (PTC alarm & timer): Tripping resistance = 1650 Ohm. After the tripping resistance has been exceeded, an appropriate alarm is output and after the delay time set in p0606 has expired, an appropriate fault is output. Re p0601 = 3 (KTY84 and PTC (only for motors with DRIVE-CLiQ)): For motors with DRIVE-CLiQ and 2 temperature sensors, the value is automatically set. Re p0601 = 4 (bimetallic NC contact alarm & timer (only for temperature evaluation via the Motor Module)): Tripping resistance = 100 Ohm. After tripping, an appropriate alarm is output and after the delay time set in p0606 has expired, an appropriate fault is output. Re p0601 = 5 (PT100): It is only possible to evaluate a PT100 for p0600 = 11 and r0192 bit 15 = 1. Re p0601 = 10 (evaluation through several temperature channels (SME12x)): Not permitted for p0600 = 0, 10, 11. Associated parameters: p4600 ... p4603 (can be switched via EDS) For r0458.8 = 1, a temperature evaluation is supported through several temperature channels. Examples: When evaluating using SME120 or SME125, 4 temperature channels are available (parameterized using p4600, p4601, p4602, p4603). When evaluating using CU310 and CUA32, 2 temperature channels are available (encoder interface, parameterization via p4600 / terminal block, parameterization via p4601). Re p0601 = 11 (evaluation via several temperature channels (BICO)): Not permitted for p0600 = 0, 10, 11. Associated parameters: p4610 ... p4613 (can be switched via MDS)		

p0602	Par_circuit power unit number, temperature sensor / PU_No temp_sensor		
VECTOR (Parallel)	Can be changed: C2(3), U, T Data type: Unsigned16 P-Group: Motor Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the power unit number to which the temperature sensor is connected. The value corresponds to the Power unit Data Set number (PDS) of the power unit. The number of power unit data sets is defined in p0120.		
p0603	CI: Motor temperature signal source / Mot temp S_src		
SERVO, VECTOR	Can be changed: C2(3), T Data type: Unsigned32 / FloatingPoint32 P-Group: Motor Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: TEMP Max -	Access level: 2 Func. diagram: 8016 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to evaluate the motor temperature via a BICO interconnection.		
Dependency:	Refer to: p0600		
Note:	Temperature sensor KTY: Valid temperature range -48 °C ... 248 °C. PTC temperature sensor: For a value = -50 °C, the following applies: Motor temperature < nominal response temperature of the PTC. For a value = 250 °C, the following applies: Motor temperature >= nominal response temperature of the PTC. Note: When using a Terminal Module 31 (TM31), the following applies: - the sensor type used is set using p4100. - the temperature signal is interconnected using CO: r4105.		
p0604[0...n]	Motor temperature alarm threshold / Mot_temp al thr		
SERVO	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [°C]	Calculated: - Dynamic index: MDS, p0130 Units group: 21_1 Scaling: - Max 200.0 [°C]	Access level: 2 Func. diagram: 8016 Unit selection: p0505 Expert list: 1 Factory setting 120.0 [°C]
Description:	Sets the alarm threshold for monitoring the motor temperature.		
Dependency:	Refer to: p0606		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The hysteresis for canceling the alarm is 2 Kelvin. When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		

p0604[0...n]	Motor temperature alarm threshold / Mot_temp al thr		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [°C]	Calculated: - Dynamic index: MDS, p0130 Units group: 21_1 Scaling: - Max 200.0 [°C]	Access level: 2 Func. diagram: 8016 Unit selection: p0505 Expert list: 1 Factory setting 130.0 [°C]
Description:	Sets the alarm threshold for monitoring the motor temperature.		
Dependency:	Refer to: p0606		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The hysteresis for canceling the alarm is 2 Kelvin. When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		
p0605[0...n]	Motor temperature fault threshold / Mot_temp flt thr		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [°C]	Calculated: - Dynamic index: MDS, p0130 Units group: 21_1 Scaling: - Max 200.0 [°C]	Access level: 2 Func. diagram: 8016, 8017 Unit selection: p0505 Expert list: 1 Factory setting 145.0 [°C]
Description:	Sets the fault threshold to monitor the motor temperature.		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.		
Note:	The hysteresis for canceling the fault is 2 Kelvin. When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		
p0606[0...n]	Motor temperature timer / Mot_temp timer		
SERVO	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 600.000 [s]	Access level: 2 Func. diagram: 8016 Unit selection: - Expert list: 1 Factory setting 240.000 [s]
Description:	Sets the timer for the alarm threshold for the motor temperature monitoring function. This timer is started when the temperature alarm threshold (p0604) is exceeded. If the timer expires before the temperature in the meantime falls below the alarm threshold, the fault F07011 is output. If the temperature fault threshold (p0605) is prematurely exceeded before the timer has expired, then fault F07011 is immediately output. As long as the motor temperature has still not exceeded the fault threshold and the alarm thresholds have again been undershot, the fault can be acknowledged.		
Dependency:	Refer to: p0604, p0605 Refer to: F07011, A07910		
Note:	With p0606 = 0 s, the timer is deactivated and only the fault threshold is effective. KTY sensor: When setting the minimum value, the timer is disabled and a fault is not output until p0605 is exceeded. PTC sensor, bimetallic NC contact: The timer minimum value has no special significance.		

p0606[0...n]	Motor temperature timer / Mot_temp timer		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 600.000 [s]	Access level: 2 Func. diagram: 8016 Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the timer for the alarm threshold for the motor temperature monitoring function. This timer is started when the temperature alarm threshold (p0604) is exceeded. If the timer expires before the temperature in the meantime falls below the alarm threshold, the fault F07011 is output. If the temperature fault threshold (p0605) is prematurely exceeded before the timer has expired, then fault F07011 is immediately output. As long as the motor temperature has still not exceeded the fault threshold and the alarm thresholds have again been undershot, the fault can be acknowledged.		
Dependency:	Refer to: p0604, p0605 Refer to: F07011, A07910		
Note:	With p0606 = 0 s, the timer is deactivated and only the fault threshold is effective. KTY sensor: When setting the minimum value, the timer is disabled and a fault is not output until p0605 is exceeded. PTC sensor, bimetallic NC contact: The timer minimum value has no special significance.		
p0607[0...n]	Temperature sensor fault timer / Sensor fault time		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 600.000 [s]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.100 [s]
Description:	Sets the timer between the output of alarm and fault for a temperature sensor fault. If there is a sensor fault, this timer is started. If the sensor fault is still present after the timer has expired, a corresponding fault message is output.		
Note:	If the motor is an induction motor, the timer is switched off when setting the minimum value and no alarm is output. Temperature monitoring is then based on the thermal model.		
p0608[0...3]	CI: Motor temperature signal source 2 / MotTempSignal2		
SERVO, VECTOR	Can be changed: C2(3), T Data type: Unsigned32 / FloatingPoint32 P-Group: Motor Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: TEMP Max -	Access level: 2 Func. diagram: 8016 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets signal source 2 to evaluate the motor temperature via a BICO interconnection.		
Index:	[0] = Motor temperature channel 1 [1] = Motor temperature channel 2 [2] = Motor temperature channel 3 [3] = Motor temperature channel 4		
Dependency:	Refer to: p0600		

Note: Temperature sensor KTY: Valid temperature range -48 °C ... 248 °C.
 PTC/bimetal temperature sensor:
 For a value of -50 °C, the following applies: Motor temperature < nominal response temperature of the PTC (bimetal contact closed).
 For a value of 250 °C, the following applies: Motor temperature >= nominal response temperature of the PTC (bimetal contact open).
Note:
 When using a Terminal Module 120 (TM120), the following applies:
 - the sensor type used is set using p4100.
 - the temperature signal is interconnected using CO: r4105.

p0609[0...3] CI: Motor temperature signal source 3 / MotTempSignal3

SERVO, VECTOR	Can be changed: C2(3), T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets signal source 3 to evaluate the motor temperature via a BICO interconnection.

Index:
 [0] = Motor temperature channel 1
 [1] = Motor temperature channel 2
 [2] = Motor temperature channel 3
 [3] = Motor temperature channel 4

Dependency: Refer to: p0600

Note: Temperature sensor KTY: Valid temperature range -48 °C ... 248 °C.
 PTC/bimetal temperature sensor:
 For a value of -50 °C, the following applies: Motor temperature < nominal response temperature of the PTC (bimetal contact closed).
 For a value of 250 °C, the following applies: Motor temperature >= nominal response temperature of the PTC (bimetal contact open).
Note:
 When using a Terminal Module 120 (TM120), the following applies:
 - the sensor type used is set using p4100.
 - the temperature signal is interconnected using CO: r4105.

p0610[0...n] Motor overtemperature response / Mot temp response

VECTOR	Can be changed: C2(3), T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	2	1

Description: Sets the system response when the motor temperature reaches the alarm threshold.

Value:
 0: No response only alarm no reduction of I_max
 1: Alarm with reduction of I_max and fault
 2: Alarm and fault no reduction of I_max

Dependency: Refer to: p0601, p0604, p0605
 Refer to: F07011, A07910

Note: The I_max reduction is not executed for PTC (p0601 = 1) or bimetallic NC contact (p0601 = 4).
 The I_max reduction results in a lower output frequency.

p0611[0...n]	I2t motor model thermal time constant / I2t mot_mod T																		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0 [s]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 20000 [s]	Access level: 3 Func. diagram: 8017 Unit selection: - Expert list: 1 Factory setting 0 [s]																
Description:	Sets the winding time constant. The time constant specifies the warm-up time of the cold stator winding when loaded with the motor standstill current up until a temperature rise of 63 % of the continuously permissible winding temperature has been reached.																		
Dependency:	This parameter is only used for synchronous motors (p0300 = 2xx). Refer to: r0034, p0612, p0615 Refer to: F07011, A07012, A07910																		
Caution:	This parameter is automatically preset from the motor database for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.																		
Note:	When parameter p0611 is reset to 0, then this switches out the thermal I2t motor model (also refer to p0612). If no temperature sensor is parameterized, then the ambient temperature for the thermal motor model is referred to p0625.																		
p0612[0...n]	Thermal motor model configuration / Therm Mot_mod conf																		
SERVO, VECTOR	Can be changed: U, T Data type: Unsigned16 P-Group: - Not for motor type: REL, FEM Min -	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 8017 Unit selection: - Expert list: 1 Factory setting 0010 bin																
Description:	Sets the configuration for the thermal motor model.																		
Bit field:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr> <tr> <td>00</td><td>Activate I2t motor model</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>01</td><td>Activate motor temperature model</td><td>Yes</td><td>No</td><td>-</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FP	00	Activate I2t motor model	Yes	No	-	01	Activate motor temperature model	Yes	No	-			
Bit	Signal name	1 signal	0 signal	FP															
00	Activate I2t motor model	Yes	No	-															
01	Activate motor temperature model	Yes	No	-															
Dependency:	Refer to: r0034, p0611, p0615																		
Note:	Re bit 00: This bit is only used for permanent-magnet synchronous motors (p0300 = 2xx). It is only possible to switch in thermal I2t monitoring with a time constant greater than zero (p0611 > 0). Re bit 01: This bit is used to activate/deactivate the thermal motor model for induction motors.																		
p0615[0...n]	I2t motor model fault threshold / I2t mot_mod thresh																		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.0 [°C]	Calculated: - Dynamic index: MDS, p0130 Units group: 21_1 Scaling: - Max 220.0 [°C]	Access level: 2 Func. diagram: 8017 Unit selection: p0505 Expert list: 1 Factory setting 180.0 [°C]																
Description:	Sets the fault threshold for monitoring using the thermal I2t motor model.																		
Dependency:	The parameter is only used for permanent-magnet synchronous motors (p0300 = 2xx). Refer to: r0034, p0611, p0612 Refer to: F07011, A07012																		
Caution:	This parameter is automatically preset for motors from the motor list (p0301). When selecting a catalog motor, this parameter cannot be changed (write protection). Information in p0300 should be carefully observed when removing write protection.																		

p0616[0...n]	Motor overtemperature alarm threshold 1 / Mot temp alarm 1		
SERVO	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [°C]	Calculated: - Dynamic index: MDS, p0130 Units group: 21_1 Scaling: - Max 200.0 [°C]	Access level: 2 Func. diagram: 8016 Unit selection: p0505 Expert list: 1 Factory setting 195.0 [°C]
Description:	Sets the alarm threshold 1 for monitoring the motor temperature.		
Note:	The alarm threshold is not, as for p0604, coupled to the timer p0606. The hysteresis for canceling the fault is 2 Kelvin.		

p0616[0...n]	Motor overtemperature alarm threshold 1 / Mot temp alarm 1		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.0 [°C]	Calculated: - Dynamic index: MDS, p0130 Units group: 21_1 Scaling: - Max 200.0 [°C]	Access level: 2 Func. diagram: 8016 Unit selection: p0505 Expert list: 1 Factory setting 130.0 [°C]
Description:	Sets the alarm threshold 1 for monitoring the motor temperature.		
Note:	The alarm threshold is not, as for p0604, coupled to the timer p0606. The hysteresis for canceling the fault is 2 Kelvin.		

p0620[0...n]	Thermal adaptation, stator and rotor resistance / Mot therm_adapt R		
SERVO	Can be changed: C2(3), U, T Data type: Integer16 P-Group: Motor Not for motor type: - Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 2	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2
Description:	Sets the thermal adaptation of the stator/primary section resistance and rotor/secondary section resistance according to r0395 and r0396.		
Value:	0: No thermal adaptation of stator and rotor resistances 1: Resistances adapted to the temperatures of the thermal model 2: Resistances adapted to the measured stator winding temperature		
Note:	For p0620 = 1, the following applies: The stator resistance is adapted using the temperature in r0035 and the rotor resistance together with the model temperature in r0633. For p0620 = 2, the following applies: The stator resistance is adapted using the temperature in r0035. If applicable, the rotor temperature for adapting the rotor resistance is calculated from the stator temperature (r0035) as follows: $\text{theta_R} = (\text{r0628} + \text{r0625}) / (\text{r0627} + \text{r0625}) * \text{r0035}$		

p0620[0...n]	Thermal adaptation, stator and rotor resistance / Mot therm_adapt R		
VECTOR	Can be changed: C2(3), U, T Data type: Integer16 P-Group: Motor Not for motor type: - Min 0	Calculated: CALC_MOD_ALL Dynamic index: MDS, p0130 Units group: - Scaling: - Max 2	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the thermal adaptation of the stator/primary section resistance and rotor/secondary section resistance according to r0395 and r0396.		

Value: 0: No thermal adaptation of stator and rotor resistances
1: Resistances adapted to the temperatures of the thermal model
2: Resistances adapted to the measured stator winding temperature

Note: For p0620 = 1, the following applies:
The stator resistance is adapted using the temperature in r0035 and the rotor resistance together with the model temperature in r0633.
For p0620 = 2, the following applies:
The stator resistance is adapted using the temperature in r0035. If applicable, the rotor temperature for adapting the rotor resistance is calculated from the stator temperature (r0035) as follows:
$$\theta_{R} = (r0628 + r0625) / (r0627 + r0625) * r0035$$

p0621[0...n] Identification stator resistance after restart / Rst_ident Restart			
VECTOR (n/M)	Can be changed: C2(3), T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	2	0
Description:	Selects the identification of the stator resistance after booting the Control Unit (only for vector control). The identification is used to measure the actual stator resistance and from the ratio of the result of motor data identification (p0350) to the matching ambient temperature (p0625) the actual mean temperature of the stator winding is calculated. The result is used to initialize the thermal motor model. p0621 = 1: Identification of the stator resistance only when the drive is powered up for the first time (pulse enable) after booting the Control Unit. p0621 = 2: Identification of the stator resistance every time the drive is powered up (pulse enable).		
Value:	0: No temperature identification 1: Temperature identification after restart 2: Temperature identification after each power-up		
Dependency:	- perform motor data identification (see p1910) with cold motor. - enter ambient temperature at time of motor data identification in p0625. Refer to: p0622, r0623		
Notice:	The calculated stator temperature can only be compared with the measured value of a temperature sensor (KTY) to a certain extent, as the sensor is usually the warmest point of the stator winding, whereas the measured value of identification reflects the mean value of the stator winding. Furthermore this is a short-time measurement with limited accuracy that is performed during the magnetizing phase of the induction motor.		
Note:	The measurement is carried out: - For induction motors - When vector control is active (see p1300) - If a temperature sensor (KTY) has not been connected - When the motor is at a standstill when switched on When a flying restart is performed on a rotating motor, the temperatures of the thermal motor model are set to a third of the overtemperatures. This occurs only once, however, when the CU is booted (e.g. after a power failure). If identification is activated, the magnetizing time is determined via p0622 and not via p0346. Quick magnetizing (p1401.6) is de-energized internally and alarm A07416 is displayed. The speed is enabled after completion of the measurement.		

p0622[0...n]	Motor excitation time for Rs_ident after powering up again / t_excit Rs_id		
VECTOR (n/M)	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL, FEM Min 0.000 [s]	Calculated: CALC_MOD_REG Dynamic index: MDS, p0130 Units group: - Scaling: - Max 20.000 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the excitation time of the motor for the stator resistance identification after powering up again (restart).		
Dependency:	Refer to: p0621, r0623		
Note:	For p0622 < p0346 the following applies: If identification is activated, the magnetizing time is influenced by p0622. The speed is enabled after measurement is complete, but not before the time in p0346 has elapsed (see r0056 bit 4). The time taken for measurement also depends on the settling time of the measured current. For p0622 >= p0346 the following applies: Parameter p0622 is internally limited to the magnetizing time p0346, so that p0346 represents the maximum possible magnetizing time during identification. The entire measurement period (magnetizing plus measurement settling time plus measuring time) will always be greater than p0346.		
r0623	Stator resistance of Rs identification after powering up again / R_Stator Reset_Id		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM, REL, FEM Min - [Ohm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Ohm]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Ohm]
Description:	Displays the identified stator resistance after the Rs identification after powering up again.		
Dependency:	Refer to: p0621, p0622		
p0624[0...n]	Motor Temperature Offset PT100 / Mot T_offset PT100		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min -100.0 [K]	Calculated: - Dynamic index: MDS, p0130 Units group: 21_2 Scaling: - Max 100.0 [K]	Access level: 3 Func. diagram: 8016 Unit selection: p0505 Expert list: 1 Factory setting 0.0 [K]
Description:	Sets the temperature offset for the PT100 measured value. If there is a difference between the motor temperature displayed in r0035 and the actual motor temperature, this offset can be entered in this parameter, thereby compensating for the difference.		
Dependency:	Refer to: p0600, p0601, p0602		
Note:	The parameter only takes effect with the following settings: - Temperature sensor of the power unit detected (p0600 = 11). - Sensor type PT100 selected (p0601 = 5). If the resistance in series with the PT100 (e.g. the cable resistance of the feeder cable) is known, the following conversion formula must be used: Offset in p0624 = Measured resistance in ohms x 2.5 K/Ohm Example: Measured cable resistance = 2 Ohm --> 2 Ohm x 2.5 K / Ohm = 5.0 K		

p0625[0...n]	Motor ambient temperature / Mot T_ambient		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min -40 [°C]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: 21_1 Scaling: - Max 80 [°C]	Access level: 3 Func. diagram: 8016 Unit selection: p0505 Expert list: 1 Factory setting 20 [°C]
Description:	Defines the ambient temperature of the motor for calculating the motor temperature model.		
Note:	The parameters for stator and rotor resistance (p0350, p0354) refer to this temperature. If the thermal I2t motor model is activated for permanent-magnet synchronous motors (refer to p0611), p0625 is included in the model calculation if a temperature sensor is not being used (see p0601).		
p0626[0...n]	Motor overtemperature, stator core / Mot T_over core		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL, FEM Min 20 [K]	Calculated: - Dynamic index: MDS, p0130 Units group: 21_2 Scaling: - Max 200 [K]	Access level: 3 Func. diagram: 8016 Unit selection: p0505 Expert list: 1 Factory setting 50 [K]
Description:	Defines the rated overtemperature of the stator core referred to the ambient temperature.		
Dependency:	For 1LA5 and 1LA7 motors (refer to p0300), the parameter is preset as a function of p0307 and p0311.		
Note:	Refer to: p0625		
Note:	When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		
p0627[0...n]	Motor overtemperature, stator winding / Mot T_over stator		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL, FEM Min 20 [K]	Calculated: - Dynamic index: MDS, p0130 Units group: 21_2 Scaling: - Max 200 [K]	Access level: 3 Func. diagram: 8016 Unit selection: p0505 Expert list: 1 Factory setting 80 [K]
Description:	Defines the rated overtemperature of the stator winding referred to the ambient temperature.		
Dependency:	For 1LA5 and 1LA7 motors (refer to p0300), the parameter is preset as a function of p0307 and p0311.		
Note:	Refer to: p0625		
Note:	When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		
p0628[0...n]	Motor overtemperature rotor winding / Mot T_over rotor		
SERVO, VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: PEM, REL, FEM Min 20 [K]	Calculated: - Dynamic index: MDS, p0130 Units group: 21_2 Scaling: - Max 200 [K]	Access level: 3 Func. diagram: 8016 Unit selection: p0505 Expert list: 1 Factory setting 100 [K]
Description:	Defines the rated overtemperature of the squirrel cage rotor referred to ambient temperature.		
Dependency:	For 1LA5 and 1LA7 motors (refer to p0300), the parameter is preset as a function of p0307 and p0311.		
Note:	Refer to: p0625		
Note:	When quick commissioning is exited with p3900 > 0, then the parameter is reset if a catalog motor has not been selected (refer to p0300).		

r0630[0...n]	Motor temperature model ambient temperature / MotTMod T_amb.		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 8016
	P-Group: Motor	Units group: 21_1	Unit selection: p0505
	Not for motor type: PEM, REL, FEM	Scaling: TEMP	Expert list: 1
	Min	Max	Factory setting
	- [°C]	- [°C]	- [°C]
Description:	Displays the ambient temperature of the motor temperature model.		

r0631[0...n]	Motor temperature model, stator core temperature / MotTMod T_core		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 8016
	P-Group: Motor	Units group: 21_1	Unit selection: p0505
	Not for motor type: PEM, REL, FEM	Scaling: TEMP	Expert list: 1
	Min	Max	Factory setting
	- [°C]	- [°C]	- [°C]
Description:	Displays the stator core temperature of the motor temperature model.		

r0632[0...n]	Motor temperature model, stator winding temperature / MotTMod T_copper		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 8016
	P-Group: Motor	Units group: 21_1	Unit selection: p0505
	Not for motor type: PEM, REL, FEM	Scaling: TEMP	Expert list: 1
	Min	Max	Factory setting
	- [°C]	- [°C]	- [°C]
Description:	Displays the stator winding temperature of the motor temperature model.		

r0633[0...n]	Motor temperature model, rotor temperature / MotTMod T_rotor		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 8016
	P-Group: Motor	Units group: 21_1	Unit selection: p0505
	Not for motor type: PEM, REL, FEM	Scaling: TEMP	Expert list: 1
	Min	Max	Factory setting
	- [°C]	- [°C]	- [°C]
Description:	Displays the rotor temperature of the motor temperature model.		

p0634[0...n]	Q flux flux constant unsaturated / PSIQ KPSI UNSAT		
VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: ASM, FEM	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.000 [Vs rms]	100.000 [Vs rms]	0.000 [Vs rms]
Description:	The non-linear and cross-coupled quadrature axis flux functions are defined using 4 coefficients. The parameter weights the unsaturated component of the quadrature axis flux function.		

p0635[0...n]	Q flux quadrature axis current constant unsaturated / PSIQ KIQ UNSAT		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, FEM Min 0.00 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10000.00 [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	The non-linear and cross-coupled quadrature axis flux functions are defined using 4 coefficients. This parameter describes the interdependency of the unsaturated component of the quadrature axis current.		
Dependency:	Refer to: p0634		
p0636[0...n]	Q flux direct axis current constant unsaturated / PSIQ KID UNSAT		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, FEM Min 0.00 [Arms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10000.00 [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	The non-linear and cross-coupled quadrature axis flux functions are defined using 4 coefficients. This parameter describes the interdependency of the unsaturated component of the direct axis current.		
Dependency:	Refer to: p0634		
p0637[0...n]	Q flux flux gradient saturated / PSIQ Grad SAT		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, FEM Min 0.00 [mH]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 10000.00 [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [mH]
Description:	The non-linear and cross-coupled quadrature axis flux functions are defined using 4 coefficients. This parameter describes the gradients of the saturated component over the quadrature axis current.		
Dependency:	Refer to: p0634, p0635, p0636		
p0640[0...n]	Current limit / Current limit		
SERVO, VECTOR	Can be changed: C2(1, 3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.00 [Arms]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000.00 [Arms]	Access level: 2 Func. diagram: 5722, 6640 Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the current limit.		
Dependency:	Refer to: r0209, p0323		

Note: The parameter is part of the quick commissioning (p0010 = 1); this means that it is appropriately pre-assigned when changing p0305, p0323 and p0338.

The current limit p0640 is limited to r0209 and p0323. The limit to p0323 is not realized if a value of zero is entered there.

The resulting current limit is displayed in r0067 and if required, r0067 is reduced by the thermal model of the Motor Module.

The torque and power limits (p1520, p1521, p1530, p1531) matching the current limit are automatically calculated when exiting the quick commissioning using p3900 > 0 or using the automatic parameterization with p0340 = 3, 5. For VECTOR the following applies (p0107):

p0640 is limited to $4.0 \cdot p0305$.

p0640 is pre-assigned for the automatic self commissioning routine (e.g. to $1.5 \cdot p0305$, with p0305 = r0207[1]).

p0640 must be entered when commissioning the system. This is the reason that p0640 is not calculated by the automatic parameterization when exiting the quick commissioning (p3900 > 0).

For SERVO the following applies (p0107):

p0640 is pre-assigned as follows using the automatic parameterization (p0340 = 1, p3900 > 0) taking into account the limits r0209 and r0323:

- for induction motors: $p0640 = 1.5 \cdot p0305$
- for synchronous motors: $p0640 = p0338$

p0641[0...n]		CI: Current limit, variable / Curr lim var		
VECTOR	Can be changed:	T	Calculated: -	Access level: 3
	Data type:	Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 6640
	P-Group:	Closed-loop control	Units group: -	Unit selection: -
	Not for motor type:	-	Scaling: PERCENT	Expert list: 1
	Min	-	Max	Factory setting

Description: Sets the signal source for the variable current limit.
The value is referred to p0640.

p0642[0...n]		Encoderless operation current reduction / Encoderl op I_red		
SERVO	Can be changed:	C2(1, 3), U, T	Calculated: -	Access level: 1
	Data type:	FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group:	Motor	Units group: -	Unit selection: -
	Not for motor type:	-	Scaling: -	Expert list: 1
	Min	0.00 [%]	Max	Factory setting

Description: Sets the reduction for the current limit in encoderless operation.
The value is referred to p0640.

Dependency: Refer to: r0209, p0323, p0491, p0640, p1300, p1404

Note: If the motor is operated both with encoder as well as without encoder (e.g. p0491 is not equal to 0 or p1404 < p1082) then the maximum current can be reduced in encoderless operation. This reduces disturbing saturation-related motor data changes in encoderless operation.

p0643[0...n]		Overvoltage protection for synchronous motors / Overvolt_protect		
SERVO	Can be changed:	T	Calculated: -	Access level: 3
	Data type:	Integer16	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group:	Motor	Units group: -	Unit selection: -
	Not for motor type:	-	Scaling: -	Expert list: 1
	Min	0	Max	Factory setting

Description: Sets the overvoltage protection for synchronous motors in the field-weakening range.

Value:

- 0: No measure
- 1: Voltage Protection Module (VPM)

Dependency:	Refer to: p0316, p1082, p1231, p9601, p9801 Refer to: F07432, F07906, F07907
Notice:	When the speed limiting is removed, the user is responsible for implementing a suitable overvoltage protection.
Note:	In the field-weakening range, synchronous motors can, when a fault condition exists, generate high DC link voltages. The following possibilities exist to protect the drive system from being destroyed due to overvoltage: - limit the maximum speed (p1082) without any additional protection. The maximum speed without protection is calculated as follows: Rotary motors: $p1082 \text{ [rpm]} \leq 11.695 * p0297/p0316 \text{ [Nm/A]}$ Linear motors: $p1082 \text{ [m/min]} \leq 73.484 * p0297/0316 \text{ [N/A]}$ - use a Voltage Protection Module (VPM) in conjunction with the function "Safe Torque Off" (p9601, p9801). When a fault condition exists, the VPM short-circuits the motors. During the short-circuit, the pulses must be suppressed - this means that the terminals for the function "Safe Torque Off" must be connected to the VPM. - activating the internal voltage protection (IVP) with p1231 = 3.

p0643[0...n]	Overvoltage protection for synchronous motors / Overvolt_protect		
VECTOR (n/M)	Can be changed: C2(3)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0
Description:	Sets the overvoltage protection for synchronous motors in the field-weakening range.		
Value:	0: No measure 1: Voltage Protection Module (VPM)		
Dependency:	Refer to: p0316, p1082, p1231, p9601, p9801 Refer to: F07432, F07906, F07907		
Notice:	When the speed limiting is removed, the user is responsible for implementing a suitable overvoltage protection.		
Note:	In the field-weakening range, synchronous motors can, when a fault condition exists, generate high DC link voltages. The following possibilities exist to protect the drive system from being destroyed due to overvoltage: - limit the maximum speed (p1082) without any additional protection. The maximum speed without protection is calculated as follows: Rotary motors: $p1082 \text{ [rpm]} \leq 11.695 * p0297/p0316 \text{ [Nm/A]}$ Linear motors: $p1082 \text{ [m/min]} \leq 73.484 * p0297/0316 \text{ [N/A]}$ - use a Voltage Protection Module (VPM) in conjunction with the function "Safe Torque Off" (p9601, p9801). When a fault condition exists, the VPM short-circuits the motors. During the short-circuit, the pulses must be suppressed - this means that the terminals for the function "Safe Torque Off" must be connected to the VPM. - activating the internal voltage protection (IVP) with p1231 = 3.		

p0645[0...n]	Motor kT characteristic kT1 / Mot kT char kT1		
SERVO (Ext M_ctrl, Lin)	Can be changed: C2(1, 3), U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: ASM, REL, FEM	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [N/Arms]	200.00 [N/Arms]	0.00 [N/Arms]
Description:	Sets the constant kT1 for the kT characteristic. $kT(iq) = kT1 + kT3 * iq^2 + kT5 * iq^4 + kT7 * iq^6$		
Dependency:	Refer to: p0316, p0646, p0647, p0648, p1780		
Note:	For the standard setting, the value in p0316 is effective. The value in p0316 is ignored and the kT characteristic is effective, if the following conditions are fulfilled: - the function module "expanded torque control" has been activated (r0108 = 1). - the kT characteristic has been activated (p1780.9 = 1).		

p0645[0...n]	Motor kT characteristic kT1 / Mot kT char kT1		
SERVO (Ext M_ctrl)	Can be changed: C2(1, 3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min 0.00 [Nm/A]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 200.00 [Nm/A]	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [Nm/A]
Description:	Sets the constant kT1 for the kT characteristic. $kT(iq) = kT1 + kT3 * iq^2 + kT5 * iq^4 + kT7 * iq^6$		
Dependency:	Refer to: p0316, p0646, p0647, p0648, p1780		
Note:	For the standard setting, the value in p0316 is effective. The value in p0316 is ignored and the kT characteristic is effective, if the following conditions are fulfilled: - the function module "expanded torque control" has been activated (r0108 = 1). - the kT characteristic has been activated (p1780.9 = 1).		
p0646[0...n]	Motor kT characteristic kT3 / Mot kT char kT3		
SERVO (Ext M_ctrl)	Can be changed: C2(1, 3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min -	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Sets the constant kT3 for the kT characteristic. $kT(iq) = kT1 + kT3 * iq^2 + kT5 * iq^4 + kT7 * iq^6$		
Dependency:	Refer to: p0316, p0645, p0647, p0648, p1780		
Note:	For the standard setting, the value in p0316 is effective. The value in p0316 is ignored and the kT characteristic is effective, if the following conditions are fulfilled: - the function module "expanded torque control" has been activated (r0108 = 1). - the kT characteristic has been activated (p1780.9 = 1).		
p0647[0...n]	Motor kT characteristic kT5 / Mot kT char kT5		
SERVO (Ext M_ctrl)	Can be changed: C2(1, 3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min -	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Sets the constant kT5 for the kT characteristic. $kT(iq) = kT1 + kT3 * iq^2 + kT5 * iq^4 + kT7 * iq^6$		
Dependency:	Refer to: p0316, p0645, p0646, p0648, p1780		
Note:	For the standard setting, the value in p0316 is effective. The value in p0316 is ignored and the kT characteristic is effective, if the following conditions are fulfilled: - the function module "expanded torque control" has been activated (r0108 = 1). - the kT characteristic has been activated (p1780.9 = 1).		

p0648[0...n]	Motor kT characteristic kT7 / Mot kT char kT7		
SERVO (Ext M_ctrl)	Can be changed: C2(1, 3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, REL, FEM Min -	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Sets the constant kT7 for the kT characteristic. $kT(iq) = kT1 + kT3 * iq^2 + kT5 * iq^4 + kT7 * iq^6$		
Dependency:	Refer to: p0316, p0645, p0646, p0647, p1780		
Note:	For the standard setting, the value in p0316 is effective. The value in p0316 is ignored and the kT characteristic is effective, if the following conditions are fulfilled: - the function module "expanded torque control" has been activated (r0108 = 1). - the kT characteristic has been activated (p1780.9 = 1).		
p0650[0...n]	Actual motor operating hours / Mot t_oper act		
SERVO, VECTOR	Can be changed: T Data type: Unsigned32 P-Group: Motor Not for motor type: - Min 0 [h]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 4294967295 [h]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [h]
Description:	Displays the operating hours for the corresponding motor. The motor operating time counter continues to run when the pulses are enabled. When the pulse enable is withdrawn, the counter is held and the value saved.		
Dependency:	The following prerequisites must be fulfilled in order to be able to save the operating hours counter in a non-volatile fashion: - firmware with V2.2 or higher. - Control Unit 320 (CU320) with hardware version C or higher (module with NVRAM). Refer to: p0651 Refer to: A01590		
Note:	The operating hours counter in p0650 can only be reset to 0. In this case, p0651 is automatically set to 0. For p0651 = 0, the operating hours counter is disabled. The operating hours counter only runs for MDS0 and MDS1 (Motor Data Set).		
p0651[0...n]	Motor operating hours maintenance interval / Mot t_op maint		
SERVO, VECTOR	Can be changed: T Data type: Unsigned32 P-Group: Motor Not for motor type: - Min 0 [h]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 150000 [h]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [h]
Description:	Sets the service/maintenance intervals in hours for the appropriate motor. An appropriate fault is output when the operating hours set here are reached.		
Dependency:	Refer to: p0650 Refer to: A01590		
Note:	For p0651 = 0, the operating hours counter is disabled. The operating hours counter only runs for MDS0 and MDS1 (Motor Data Set).		

p0652[0...n]	Motor stator resistance, scaling / Mot R_stator scal		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 10.0 [%]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300.0 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Factor to evaluate the stator resistance.		
Dependency:	Refer to: p0350, r0370		
p0653[0...n]	Motor stator leakage inductance, scaling / Mot L_S_leak scal		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 10.0 [%]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300.0 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Factor to evaluate the stator leakage inductance.		
Dependency:	Refer to: p0356, r0377		
p0655[0...n]	Motor magnetizing inductance, d axis saturated scaling / Mot L_m d sat scal		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 10.0 [%]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300.0 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Factor to evaluate the magnetizing inductance in the direction of the rotor axis (d axis).		
Dependency:	Refer to: p0360, r0382		
p0656[0...n]	Motor magnetizing inductance, q axis, saturated scaling / Mot L_m q sat scal		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 10.0 [%]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300.0 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Factor to evaluate the magnetizing inductance quadrature to the rotor axis (q axis).		
Dependency:	Refer to: p0361, r0383		
p0657[0...n]	Motor damping inductance, d axis scaling / Mot L_damp d scal		
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 10.0 [%]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300.0 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Factor to evaluate the damping inductance in the direction of the rotor axis (d axis).		
Dependency:	Refer to: p0358, r0380		

p0658[0...n] Motor damping inductance, q axis scaling / Mot L_damp q scal			
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 10.0 [%]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300.0 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Factor to evaluate the damping inductance quadrature to the rotor axis (q axis).		
Dependency:	Refer to: p0359, r0381		
p0659[0...n] Motor damping resistance, d axis scaling / Mot R_damp d scal			
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 10.0 [%]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300.0 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Factor to evaluate the damping resistance in the direction of the rotor axis (d axis).		
Dependency:	Refer to: p0354, r0374		
p0660[0...n] Motor damping resistance, q axis scaling / Mot R_damp q scal			
VECTOR	Can be changed: C2(3), U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: ASM, PEM, REL Min 10.0 [%]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: - Scaling: - Max 300.0 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Factor to evaluate the damping resistance quadrature to the rotor axis (q axis).		
Dependency:	Refer to: p0355, r0375		
p0680[0...5] Central measuring probe, input terminal / Cen meas inp			
CU_CX32, CU_I, CU_S, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the digital input used for the function "central measuring probe evaluation". p0680[0]: Digital input, measuring probe 1 p0680[1]: Digital input, measuring probe 2 p0680[2]: Digital input, measuring probe 3 p0680[3]: Digital input, measuring probe 4 p0680[4]: Digital input, measuring probe 5 p0680[5]: Digital input, measuring probe 6		
Value:	0: No measuring probe 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9)		

Dependency:	Refer to: p0728
Notice:	To select the values: For CU310, Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).
Note:	Prerequisite: The DI/DO must be set as input (p0728.x = 0). DI/DO: Bidirectional digital input/output If a parameter change is rejected, a check should be performed as to whether the input terminal is already being used in p0488, p0489, p0493, p0495, p0580, p2517, or p2518.

p0680[0...5] Central measuring probe, input terminal / Cen meas inp

CU_S_CU310DP, CU_S_CU310PN	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	3	0

Description: Sets the digital input used for the function "central measuring probe evaluation".

p0680[0]: Digital input, measuring probe 1
p0680[1]: Digital input, measuring probe 2
p0680[2]: Digital input, measuring probe 3
p0680[3]: Digital input, measuring probe 4
p0680[4]: Digital input, measuring probe 5
p0680[5]: Digital input, measuring probe 6

Value:
0: No measuring probe
1: DI/DO 9 (X122.10)
2: DI/DO 10 (X122.12)
3: DI/DO 11 (X122.13)

Dependency: Refer to: p0728

Notice: To select the values:
For CU310, Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).

Note: Prerequisite: The DI/DO must be set as input (p0728.x = 0).

DI/DO: Bidirectional digital input/output

If a parameter change is rejected, a check should be performed as to whether the input terminal is already being used in p0488, p0489, p0493, p0495, p0580, p2517, or p2518.

p0681 BI: Central measuring probe, synchronizing signal signal source / Cen meas sync_sig

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the synchronizing signal (SYN) of the function "central measuring probe evaluation".
The signal is used to synchronize the common system time between the master and slave.

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p0682 CI: Central measuring probe, control word signal source / Cen meas STW S_src

CU_CX32, CU_I,	Can be changed: T	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32 / Integer16	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Displays, signals	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 1
CU_S_S150	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the control word of the function "central measuring probe evaluation".

p0684 Central measuring probe evaluation technique / Cen meas eval_tech

CU_CX32, CU_I,	Can be changed: U, T	Calculated: -	Access level: 3
CU_S,	Data type: Integer16	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Encoder	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 1
CU_S_S150	Min	Max	Factory setting
	0	1	0

Description: Sets the evaluation technique for the "central measuring probe evaluation" function.

Value:
 0: Measurement with handshake
 1: Measurement without handshake

Note: During measurement without a handshake, the probe may have a higher evaluation frequency.
 The setting "Measurement without handshake" must be supported by the higher-level control.

r0685 Central measuring probe, control word display / Cen meas STW disp

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Commands	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 1
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the control word for the function "central measuring probe evaluation".

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Falling edge, measuring probe 1	Yes	No	-
	01	Falling edge, measuring probe 2	Yes	No	-
	02	Falling edge, measuring probe 3	Yes	No	-
	03	Falling edge, measuring probe 4	Yes	No	-
	04	Falling edge, measuring probe 5	Yes	No	-
	05	Falling edge, measuring probe 6	Yes	No	-
	08	Rising edge, measuring probe 1	Yes	No	-
	09	Rising edge, measuring probe 2	Yes	No	-
	10	Rising edge, measuring probe 3	Yes	No	-
	11	Rising edge, measuring probe 4	Yes	No	-
	12	Rising edge, measuring probe 5	Yes	No	-
	13	Rising edge, measuring probe 6	Yes	No	-

r0686[0...5]	CO: Central measuring probe, measuring time rising edge / CenMeas t_meas 0/1			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -	
Description:	Displays the measuring time for a rising edge at the digital input for the "central measuring probe evaluation" function. The measuring time is specified as 16-bit value with a resolution of 0.25 µs. r0686[0]: Measuring time, rising edge measuring probe 1 r0686[1]: Measuring time, rising edge measuring probe 2 r0686[2]: Measuring time, rising edge measuring probe 3 r0686[3]: Measuring time, rising edge measuring probe 4 r0686[4]: Measuring time, rising edge measuring probe 5 r0686[5]: Measuring time, rising edge measuring probe 6			

r0687[0...5]	CO: Central measuring probe, measuring time falling edge / CenMeas t_meas 1/0			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -	
Description:	Displays the measuring time for a falling edge at the digital input for the "central measuring probe evaluation" function. The measuring time is specified as 16-bit value with a resolution of 0.25 µs. r0687[0]: Measuring time, falling edge measuring probe 1 r0687[1]: Measuring time, falling edge measuring probe 2 r0687[2]: Measuring time, falling edge measuring probe 3 r0687[3]: Measuring time, falling edge measuring probe 4 r0687[4]: Measuring time, falling edge measuring probe 5 r0687[5]: Measuring time, falling edge measuring probe 6			

r0688	CO: Central measuring probe, status word display / Cen meas ZSW disp				
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -		
Description:	Displays the status word for the function "central measuring probe evaluation".				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Digital input, measuring probe 1	High	Low	-
	01	Digital input, measuring probe 2	High	Low	-
	02	Digital input, measuring probe 3	High	Low	-
	03	Digital input, measuring probe 4	High	Low	-
	04	Digital input, measuring probe 5	High	Low	-
	05	Digital input, measuring probe 6	High	Low	-
	08	Sub-sampling, measuring probe 1	High	Low	-

09	Sub-sampling, measuring probe 2	High	Low	-
10	Sub-sampling, measuring probe 3	High	Low	-
11	Sub-sampling, measuring probe 4	High	Low	-
12	Sub-sampling, measuring probe 5	High	Low	-
13	Sub-sampling, measuring probe 6	High	Low	-

p0700[0...n] Macro Binector Input (BI) / Macro BIA_INF, B_INF,
S_INF, SERVO,
VECTOR**Can be changed:** C2(1), T**Data type:** Unsigned32**P-Group:** Commands**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** CDS, p0170**Units group:** -**Scaling:** -**Max**

999999

Access level: 1**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description:

Runs the corresponding macro files.

The binector inputs (BI) of the appropriate Command Data Set (CDS) are appropriately interconnected.

The selected macro file must be available on the memory card/device memory.

Example:

p0700 = 6 --> the macro file PM000006.ACX is run.

Dependency:

Refer to: p0015, p1000, p1500, r8571

Notice:

No errors were issued during quick commissioning (p3900 = 1) when writing to parameters of the QUICK_IBN group!

Note:

The macros in the specified directory are displayed in r8571. r8571 is not in the expert list of the commissioning software.

Macros available as standard are described in the technical documentation of the particular product.

BI: Binector input

r0721 CX digital inputs, terminal actual value / CX DI actual value

CU_CX32

Can be changed: -**Data type:** Unsigned32**P-Group:** Commands**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 2**Func. diagram:** 2220, 2230,
2231**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description:

Displays the actual value at the digital inputs.

This means that the actual input signal can be checked at terminal DI x or DI/DO x prior to switching from the simulation mode (p0795.x = 1) to terminal mode (p0795.x = 0). The input signal at terminal DI x is displayed in bit x of r0721.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	DI 0 (X122.1)	High	Low	-
01	DI 1 (X122.2)	High	Low	-
02	DI 2 (X122.3)	High	Low	-
03	DI 3 (X122.4)	High	Low	-
08	DI/DO 8 (X122.9)	High	Low	-
09	DI/DO 9 (X122.10)	High	Low	-
10	DI/DO 10 (X122.12)	High	Low	-
11	DI/DO 11 (X122.13)	High	Low	-
16	DI 16 (X122.5)	High	Low	-
17	DI 17 (X122.6)	High	Low	-

Note:

If a DI/DO is parameterized as output (p0728.x = 1), then r0721.x = 0 is displayed.

DI: Digital input

DI/DO: Bidirectional digital input/output

r0721 CU digital inputs, terminal actual value / CU DI actual value					
CU_I, CU_S, CU_S_S150	Can be changed: -		Calculated: -	Access level: 2	
	Data type: Unsigned32		Dynamic index: -	Func. diagram: 1510, 2020, 2030, 2031, 2100, 2120, 2130, 2131, 2132, 2133	
	P-Group: Commands		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min	Max	Factory setting		
-	-	-			
Description:					
Displays the actual value at the digital inputs. This means that the actual input signal can be checked at terminal DI x or DI/DO x prior to switching from the simulation mode (p0795.x = 1) to terminal mode (p0795.x = 0). The input signal at terminal DI x is displayed in bit x of r0721.					
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	High	Low	-
	01	DI 1 (X122.2)	High	Low	-
	02	DI 2 (X122.3)	High	Low	-
	03	DI 3 (X122.4)	High	Low	-
	04	DI 4 (X132.1)	High	Low	-
	05	DI 5 (X132.2)	High	Low	-
	06	DI 6 (X132.3)	High	Low	-
	07	DI 7 (X132.4)	High	Low	-
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-
	12	DI/DO 12 (X132.9)	High	Low	-
	13	DI/DO 13 (X132.10)	High	Low	-
	14	DI/DO 14 (X132.12)	High	Low	-
	15	DI/DO 15 (X132.13)	High	Low	-
	16	DI 16 (X122.5)	High	Low	-
	17	DI 17 (X122.6)	High	Low	-
	20	DI 20 (X132.5)	High	Low	-
	21	DI 21 (X132.6)	High	Low	-
Notice:					
To the terminal designation: The first designation is valid for CU320, the second for CU310.					
Note:					
If a DI/DO is parameterized as output (p0728.x = 1), then r0721.x = 0 is displayed. DI: Digital input DI/DO: Bidirectional digital input/output					

r0721 CU digital inputs, terminal actual value / CU DI actual value			
CU_S_CU310DP, CU_S_CU310PN	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1510, 2020, 2030, 2031, 2100, 2120, 2130, 2131, 2132, 2133
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the actual value at the digital inputs. This means that the actual input signal can be checked at terminal DI x or DI/DO x prior to switching from the simulation mode (p0795.x = 1) to terminal mode (p0795.x = 0). The input signal at terminal DI x is displayed in bit x of r0721.		

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	High	Low	-
	01	DI 1 (X122.2)	High	Low	-
	02	DI 2 (X122.3)	High	Low	-
	03	DI 3 (X122.4)	High	Low	-
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-

Notice: To the terminal designation:

The first designation is valid for CU320, the second for CU310.

Note: If a DI/DO is parameterized as output (p0728.x = 1), then r0721.x = 0 is displayed.

DI: Digital input

DI/DO: Bidirectional digital input/output

r0722.0...17 CO/BO: CX digital inputs, status / CX DI status

CU_CX32	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2220, 2230, 2231
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the digital inputs.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	High	Low	-
	01	DI 1 (X122.2)	High	Low	-
	02	DI 2 (X122.3)	High	Low	-
	03	DI 3 (X122.4)	High	Low	-
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-
	16	DI 16 (X122.5)	High	Low	-
	17	DI 17 (X122.6)	High	Low	-

Dependency: Refer to: r0723

Note: DI: Digital input

DI/DO: Bidirectional digital input/output

r0722.0...21	CO/BO: CU digital inputs, status / CU DI status				
CU_I, CU_S, CU_S_S150	Can be changed: -	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1510, 2020, 2030, 2031, 2100, 2120, 2130, 2131, 2132, 2133		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the status of the digital inputs.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	High	Low	-
	01	DI 1 (X122.2)	High	Low	-
	02	DI 2 (X122.3)	High	Low	-
	03	DI 3 (X122.4)	High	Low	-
	04	DI 4 (X132.1)	High	Low	-
	05	DI 5 (X132.2)	High	Low	-
	06	DI 6 (X132.3)	High	Low	-
	07	DI 7 (X132.4)	High	Low	-
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-
	12	DI/DO 12 (X132.9)	High	Low	-
	13	DI/DO 13 (X132.10)	High	Low	-
	14	DI/DO 14 (X132.12)	High	Low	-
	15	DI/DO 15 (X132.13)	High	Low	-
	16	DI 16 (X122.5)	High	Low	-
	17	DI 17 (X122.6)	High	Low	-
	20	DI 20 (X132.5)	High	Low	-
	21	DI 21 (X132.6)	High	Low	-
Dependency:	Refer to: r0723				
Note:	DI: Digital input DI/DO: Bidirectional digital input/output				

r0722.0...11	CO/BO: CU digital inputs, status / CU DI status				
CU_S_CU310DP, CU_S_CU310PN	Can be changed: -	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1510, 2020, 2030, 2031, 2100, 2120, 2130, 2131, 2132, 2133		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the status of the digital inputs.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	High	Low	-
	01	DI 1 (X122.2)	High	Low	-
	02	DI 2 (X122.3)	High	Low	-
	03	DI 3 (X122.4)	High	Low	-
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-
Dependency:	Refer to: r0723				
Note:	DI: Digital input				
	DI/DO: Bidirectional digital input/output				

r0723.0...17 CO/BO: CX digital inputs, status inverted / CX DI status inv

CU_CX32

Can be changed: -**Calculated:** -**Access level:** 1**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 2220, 2230, 2231**P-Group:** Commands**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the inverted status of the digital inputs.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	DI 0 (X122.1)	High	Low	-
01	DI 1 (X122.2)	High	Low	-
02	DI 2 (X122.3)	High	Low	-
03	DI 3 (X122.4)	High	Low	-
08	DI/DO 8 (X122.9)	High	Low	-
09	DI/DO 9 (X122.10)	High	Low	-
10	DI/DO 10 (X122.12)	High	Low	-
11	DI/DO 11 (X122.13)	High	Low	-
16	DI 16 (X122.5)	High	Low	-
17	DI 17 (X122.6)	High	Low	-

Dependency:

Refer to: r0722

Note:

DI: Digital input

DI/DO: Bidirectional digital input/output

r0723.0...21 CO/BO: CU digital inputs, status inverted / CU DI status invCU_I, CU_S,
CU_S_S150**Can be changed:** -**Calculated:** -**Access level:** 1**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 1510, 2020, 2030, 2031, 2100, 2120, 2130, 2131 2132, 2133**P-Group:** Commands**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the inverted status of the digital inputs.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	DI 0 (X122.1)	High	Low	-
01	DI 1 (X122.2)	High	Low	-
02	DI 2 (X122.3)	High	Low	-
03	DI 3 (X122.4)	High	Low	-
04	DI 4 (X132.1)	High	Low	-
05	DI 5 (X132.2)	High	Low	-
06	DI 6 (X132.3)	High	Low	-
07	DI 7 (X132.4)	High	Low	-
08	DI/DO 8 (X122.9)	High	Low	-
09	DI/DO 9 (X122.10)	High	Low	-
10	DI/DO 10 (X122.12)	High	Low	-
11	DI/DO 11 (X122.13)	High	Low	-
12	DI/DO 12 (X132.9)	High	Low	-
13	DI/DO 13 (X132.10)	High	Low	-
14	DI/DO 14 (X132.12)	High	Low	-
15	DI/DO 15 (X132.13)	High	Low	-
16	DI 16 (X122.5)	High	Low	-
17	DI 17 (X122.6)	High	Low	-
20	DI 20 (X132.5)	High	Low	-
21	DI 21 (X132.6)	High	Low	-

Dependency: Refer to: r0722
Note: DI: Digital input
 DI/DO: Bidirectional digital input/output

r0723.0...11	CO/BO: CU digital inputs, status inverted / CU DI status inv				
CU_S_CU310DP, CU_S_CU310PN	Can be changed: -	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1510, 2020, 2030, 2031, 2100, 2120, 2130, 2131 2132, 2133		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the inverted status of the digital inputs.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	High	Low	-
	01	DI 1 (X122.2)	High	Low	-
	02	DI 2 (X122.3)	High	Low	-
	03	DI 3 (X122.4)	High	Low	-
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-
Dependency:	Refer to: r0722				
Note:	DI: Digital input				
	DI/DO: Bidirectional digital input/output				

p0728	CX set input or output / CX DI or DO				
CU_CX32	Can be changed: T		Calculated: -		Access level: 1
	Data type: Unsigned32		Dynamic index: -		Func. diagram: 2230, 2231
	P-Group: Commands		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		0000 bin
Description:	Sets the bidirectional digital inputs/outputs as an input or output.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	Output	Input	-
	09	DI/DO 9 (X122.10)	Output	Input	-
	10	DI/DO 10 (X122.12)	Output	Input	-
	11	DI/DO 11 (X122.13)	Output	Input	-
Note:	DI/DO: Bidirectional digital input/output				

p0728	CU, set input or output / CU DI or DO				
CU_I, CU_S, CU_S_S150	Can be changed: T	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1510, 2030, 2031, 2130, 2131, 2132, 2133		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Sets the bidirectional digital inputs/outputs as an input or output.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	Output	Input	-
	09	DI/DO 9 (X122.10)	Output	Input	-
	10	DI/DO 10 (X122.12)	Output	Input	-
	11	DI/DO 11 (X122.13)	Output	Input	-
	12	DI/DO 12 (X132.9)	Output	Input	-
	13	DI/DO 13 (X132.10)	Output	Input	-
	14	DI/DO 14 (X132.12)	Output	Input	-
	15	DI/DO 15 (X132.13)	Output	Input	-
Notice:	To the terminal designation: The first designation is valid for CU320, the second for CU310.				
Note:	DI/DO: Bidirectional digital input/output				

p0728	CU, set input or output / CU DI or DO				
CU_S_CU310DP, CU_S_CU310PN	Can be changed: T	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1510, 2030, 2031, 2130, 2131, 2132, 2133		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Sets the bidirectional digital inputs/outputs as an input or output.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	Output	Input	-
	09	DI/DO 9 (X122.10)	Output	Input	-
	10	DI/DO 10 (X122.12)	Output	Input	-
	11	DI/DO 11 (X122.13)	Output	Input	-
Notice:	To the terminal designation: The first designation is valid for CU320, the second for CU310.				
Note:	DI/DO: Bidirectional digital input/output				

r0729	CX digital outputs access authority / CX DO access			
CU_CX32	Can be changed: -	Calculated: -	Access level: 1	
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2030, 2031, 2130, 2131, 2132, 2133	
	P-Group: Commands	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	-	
Description:	Displays the access authority at the digital outputs. Bit = 1: The control has access authority to the digital output via PROFIBUS or direct access. Bit = 0: The drive has access authority to the digital output or the digital input/output is not set as digital output or is not available.			

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-

Dependency: Refer to: p0728, p0738, p0739, p0740, p0741, p0742, p0743, p0744, p0745, r0747, p0748

Note: The DI/DO must be connected as output (p0728).

DI/DO: Bidirectional digital input/output

r0729 CU digital outputs access authority / CU DO access

CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2030, 2031, 2130, 2131, 2132, 2133
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the access authority at the digital outputs.

Bit = 1:

The control has access authority to the digital output via PROFIBUS or direct access.

Bit = 0:

The drive has access authority to the digital output or the digital input/output is not set as digital output or is not available.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-
	12	DI/DO 12 (X132.9)	High	Low	-
	13	DI/DO 13 (X132.10)	High	Low	-
	14	DI/DO 14 (X132.12)	High	Low	-
	15	DI/DO 15 (X132.13)	High	Low	-

Dependency: Refer to: p0728, p0738, p0739, p0740, p0741, p0742, p0743, p0744, p0745, r0747, p0748

Note: The DI/DO must be connected as output (p0728).

DI/DO: Bidirectional digital input/output

p0738 BI: CX signal source for terminal DI/DO 8 / CX S_src DI/DO 8

CU_CX32	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2230
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for terminal DI/DO 8 (X122.7).

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: Prerequisite: The DI/DO must be set as an output (p0728.8 = 1).

DI/DO: Bidirectional digital input/output

p0738	BI: CU, signal source for terminal DI/DO 8 / CU S_src DI/DO 8		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 1510, 2030, 2130
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for terminal DI/DO 8 (X122.7).

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: Prerequisite: The DI/DO must be set as an output (p0728.8 = 1).
DI/DO: Bidirectional digital input/output

p0739	BI: CX signal source for terminal DI/DO 9 / CX S_src DI/DO 9		
CU_CX32	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2230
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for terminal DI/DO 9 (X122.8).

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: Prerequisite: The DI/DO must be set as an output (p0728.9 = 1).
DI/DO: Bidirectional digital input/output

p0739	BI: CU, signal source for terminal DI/DO 9 / CU S_src DI/DO 9		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2030, 2130
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for terminal DI/DO 9 (X122.8).

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: Prerequisite: The DI/DO must be set as an output (p0728.9 = 1).
DI/DO: Bidirectional digital input/output

p0740	BI: CX signal source for terminal DI/DO 10 / CX S_src DI/DO 10		
CU_CX32	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2231
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for terminal DI/DO 10 (X122.10).

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: Prerequisite: The DI/DO must be set as an output (p0728.10 = 1).
DI/DO: Bidirectional digital input/output

p0740	BI: CU, signal source for terminal DI/DO 10 / CU S_src DI/DO 10		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 2031, 2131 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 10 (X122.10).		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Prerequisite: The DI/DO must be set as an output (p0728.10 = 1). DI/DO: Bidirectional digital input/output		
p0741	BI: CX signal source for terminal DI/DO 11 / CX S_src DI/DO 11		
CU_CX32	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 2231 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 11 (X122.11).		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Prerequisite: The DI/DO must be set as an output (p0728.11 = 1). DI/DO: Bidirectional digital input/output		
p0741	BI: CU, signal source for terminal DI/DO 11 / CU S_src DI/DO 11		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 1510, 2031, 2131 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 11 (X122.11).		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Prerequisite: The DI/DO must be set as an output (p0728.11 = 1). DI/DO: Bidirectional digital input/output		
p0742	BI: CU, signal source for terminal DI/DO 12 / CU S_src DI/DO 12		
CU_I, CU_S, CU_S_S150	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 1510, 2132 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 12 (X132.7).		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Prerequisite: The DI/DO must be set as an output (p0728.12 = 1). DI/DO: Bidirectional digital input/output		

p0743	BI: CU, signal source for terminal DI/DO 13 / CU S_src DI/DO 13			
CU_I, CU_S, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 1	
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2132	
	P-Group: Commands	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description:	Sets the signal source for terminal DI/DO 13 (X132.8).			
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.			
Note:	Prerequisite: The DI/DO must be set as an output (p0728.13 = 1). DI/DO: Bidirectional digital input/output			

p0744	BI: CU, signal source for terminal DI/DO 14 / CU S_src DI/DO 14			
CU_I, CU_S, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 1	
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2133	
	P-Group: Commands	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description:	Sets the signal source for terminal DI/DO 14 (X132.10).			
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.			
Note:	Prerequisite: The DI/DO must be set as an output (p0728.14 = 1). DI/DO: Bidirectional digital input/output			

p0745	BI: CU, signal source for terminal DI/DO 15 / CU S_src DI/DO 15			
CU_I, CU_S, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 1	
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 1510, 2133	
	P-Group: Commands	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description:	Sets the signal source for terminal DI/DO 15 (X132.11).			
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.			
Note:	Prerequisite: The DI/DO must be set as an output (p0728.15 = 1). DI/DO: Bidirectional digital input/output			

r0747	CX digital outputs, status / CX DO status				
CU_CX32	Can be changed: -	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the status of digital outputs.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-
Note:	Inversion using p0748 has been taken into account. DI/DO: Bidirectional digital input/output				

r0747 CU, digital outputs status / CU DO statusCU_I, CU_S,
CU_S_S150**Can be changed:** -**Calculated:** -**Access level:** 1**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 2130, 2131,
2132, 2133**P-Group:** Commands**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description: Displays the status of digital outputs.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-
	12	DI/DO 12 (X132.9)	High	Low	-
	13	DI/DO 13 (X132.10)	High	Low	-
	14	DI/DO 14 (X132.12)	High	Low	-
	15	DI/DO 15 (X132.13)	High	Low	-

Note: Inversion using p0748 has been taken into account.
DI/DO: Bidirectional digital input/output**r0747 CU, digital outputs status / CU DO status**CU_S_CU310DP,
CU_S_CU310PN**Can be changed:** -**Calculated:** -**Access level:** 1**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 2130, 2131,
2132, 2133**P-Group:** Commands**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description: Displays the status of digital outputs.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-

Note: Inversion using p0748 has been taken into account.
DI/DO: Bidirectional digital input/output**p0748 CX invert digital outputs / CX DO invert**

CU_CX32

Can be changed: U, T**Calculated:** -**Access level:** 1**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 2230, 2231**P-Group:** Commands**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description: Setting to invert the signals at the digital outputs.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	Inverted	Not inverted	-
	09	DI/DO 9 (X122.10)	Inverted	Not inverted	-
	10	DI/DO 10 (X122.12)	Inverted	Not inverted	-
	11	DI/DO 11 (X122.13)	Inverted	Not inverted	-

Notice: If telegram 39x is set via p0922 in SINAMICS Integrated, the inversion of the output has no effect.**Note:** DI/DO: Bidirectional digital input/output

p0748	CU, invert digital outputs / CU DO invert				
CU_I, CU_S, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2030, 2031, 2130, 2131, 2132, 2133		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Setting to invert the signals at the digital outputs.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	Inverted	Not inverted	-
	09	DI/DO 9 (X122.10)	Inverted	Not inverted	-
	10	DI/DO 10 (X122.12)	Inverted	Not inverted	-
	11	DI/DO 11 (X122.13)	Inverted	Not inverted	-
	12	DI/DO 12 (X132.9)	Inverted	Not inverted	-
	13	DI/DO 13 (X132.10)	Inverted	Not inverted	-
	14	DI/DO 14 (X132.12)	Inverted	Not inverted	-
	15	DI/DO 15 (X132.13)	Inverted	Not inverted	-
Notice:	If telegram 39x is set via p0922 in SINAMICS Integrated, the inversion of the output has no effect.				
Note:	DI/DO: Bidirectional digital input/output				

p0748	CU, invert digital outputs / CU DO invert				
CU_S_CU310DP, CU_S_CU310PN	Can be changed: U, T	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2030, 2031, 2130, 2131, 2132, 2133		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Setting to invert the signals at the digital outputs.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X122.9)	Inverted	Not inverted	-
	09	DI/DO 9 (X122.10)	Inverted	Not inverted	-
	10	DI/DO 10 (X122.12)	Inverted	Not inverted	-
	11	DI/DO 11 (X122.13)	Inverted	Not inverted	-
Notice:	If telegram 39x is set via p0922 in SINAMICS Integrated, the inversion of the output has no effect.				
Note:	DI/DO: Bidirectional digital input/output				

p0771[0...2]	CI: Test sockets signal source / TestSktSigSrce		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 / Integer16 P-Group: Terminals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max -	Access level: 2 Func. diagram: 8134 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the signal to be output at the test sockets.		
Index:	[0] = T0 [1] = T1 [2] = T2		
Dependency:	Can only be set when p0776 = 99. Refer to: r0772, r0774, p0776, p0777, p0778, p0779, p0780, p0783, p0784, r0786		

r0772[0...2]	Test sockets output signal / TestSksSignalVal		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min - [%] Description: Displays the actual value of the signal to be output. Index: [0] = T0 [1] = T1 [2] = T2 Dependency: Refer to: p0771, r0774, p0776, p0777, p0778, p0779, p0780, p0783, p0784, r0786	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%] 	Access level: 2 Func. diagram: 8134 Unit selection: - Expert list: 1 Factory setting - [%]
r0774[0...2]	Test sockets output voltage / TestSks V_output		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min - [V] Description: Displays the actual output voltage for the test sockets. Index: [0] = T0 [1] = T1 [2] = T2 Dependency: Refer to: p0771, r0772, p0776, p0777, p0778, p0779, p0780, p0783, p0784, r0786	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [V] 	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [V]
p0776[0...2]	Test socket mode / Test skt mode		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: Terminals Not for motor type: - Min 96 Description: Sets the mode for the test sockets. Value: 96: Physical address (32-bit integer signal unsigned) 97: Physical address (32-bit integer signal) 98: Physical address (32-bit floating-point signal) 99: BICO signal Index: [0] = T0 [1] = T1 [2] = T2 Dependency: Refer to: p0771, r0772, r0774, p0777, p0778, p0779, p0780, p0783, p0784, r0786, p0788, p0789, r0790	Calculated: - Dynamic index: - Units group: - Scaling: - Max 99 	Access level: 4 Func. diagram: 8134 Unit selection: - Expert list: 1 Factory setting 99
p0777[0...2]	Test socket characteristic value x1 / Test skt char x1		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -100000.00 [%] Description: The scaling characteristic for the test sockets is defined using two points. This parameter specifies the x coordinate (percentage) of the first point on the characteristic.	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000.00 [%]	Access level: 2 Func. diagram: 8134 Unit selection: - Expert list: 1 Factory setting 0.00 [%]

Index: [0] = T0
[1] = T1
[2] = T2

Dependency: Can only be set when p0776 = 99.
Refer to: p0778, p0779, p0780, r0786

Note: The value 0.00 % corresponds to 2.49 V.

p0778[0...2] Test socket characteristic value y1 / Test skt char y1

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min 0.00 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4.98 [V]	Access level: 2 Func. diagram: 8134 Unit selection: - Expert list: 1 Factory setting 2.49 [V]
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Description: The scaling characteristic for the test sockets is defined using two points. This parameter specifies the y coordinate (output voltage) of the first point on the characteristic.

Index: [0] = T0
[1] = T1
[2] = T2

Dependency: Can only be set when p0776 = 99.
Refer to: p0777, p0779, p0780, r0786

p0779[0...2] Test socket characteristic value x2 / Test skt char x2

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -100000.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 427.9E9 [%]	Access level: 2 Func. diagram: 8134 Unit selection: - Expert list: 1 Factory setting 100.00 [%]
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Description: The scaling characteristic for the test sockets is defined using two points. This parameter specifies the x coordinate (percentage) of the second point on the characteristic.

Index: [0] = T0
[1] = T1
[2] = T2

Dependency: Can only be set when p0776 = 99.
Refer to: p0777, p0778, p0780, r0786

Note: The value 100.00 % corresponds to 4.98 V.

p0780[0...2] Test socket characteristic value y2 / Test skt char y2

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min 0.00 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4.98 [V]	Access level: 2 Func. diagram: 8134 Unit selection: - Expert list: 1 Factory setting 4.98 [V]
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Description: The scaling characteristic for the test sockets is defined using two points. This parameter specifies the y coordinate (output voltage) of the second point on the characteristic.

Index: [0] = T0
[1] = T1
[2] = T2

Dependency: Can only be set when p0776 = 99.
Refer to: p0777, p0778, p0779, r0786

p0783[0...2]	Test sockets offset / Test skt offset		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -4.98 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4.98 [V]	Access level: 2 Func. diagram: 8134 Unit selection: - Expert list: 1 Factory setting 0.00 [V]
Description:	Sets an additional offset for the test sockets.		
Index:	[0] = T0 [1] = T1 [2] = T2		
p0784[0...2]	Test socket limit on/off / TestSktLim on/off		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: Terminals Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: 8134 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the limit for a signal to be output via test sockets.		
Value:	0: Limiting off 1: Limiting on		
Index:	[0] = T0 [1] = T1 [2] = T2		
Note:	Limiting on: If signals are output outside the permissible measuring range, the signal is limited to 4.98 V or to 0 V. Limiting off: If signals are output outside the permissible measuring range, this causes signal overflow. In the case of signal overflow, the signal jumps from 0 V to 4.98 V or from 4.98 V to 0 V.		
r0786[0...2]	Test socket scaling per volt / TestSktScale/Volt		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 8134 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the scaling of the signal to be output. A change in the output voltage by 1 volt corresponds to the value in this parameter. The units are determined by the interconnected test signal.		
Index:	[0] = T0 [1] = T1 [2] = T2		
Dependency:	Refer to: p0771, r0772, r0774, p0777, p0778, p0779, p0780, p0783, p0784		
Note:	Example: r0786[0] = 1500.0 and the measuring signal is r0063 (CO: Actual speed smoothed [rpm]). A change of 1 V at the output of test socket T0 corresponds to 1500.0 [rpm].		

p0788[0...2]	Test sockets physical address / Test skt PhyAddr		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Terminals Not for motor type: - Min 0000 bin	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1111 1111 1111 1111 1111 1111 1111 1111 bin	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 bin
Description:	Sets the physical address to output signals via the test sockets.		
Index:	[0] = T0 [1] = T1 [2] = T2		
Dependency:	Changes only become effective if p0776 does not equal 99. Refer to: p0789, r0790		

p0789[0...2]	Test sockets physical address gain / TestSktPhyAddrGain		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -340.28235E36	Calculated: - Dynamic index: - Units group: - Scaling: - Max 340.28235E36	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.00000
Description:	Sets the gain of a signal output of a physical address via test sockets.		
Index:	[0] = T0 [1] = T1 [2] = T2		
Dependency:	Changes only become effective if p0776 does not equal 99. Refer to: p0788		

r0790[0...2]	Test sockets physical address signal value / TestSktPhyAddrVal		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Terminals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the actual value of a signal determined via a physical address.		
Index:	[0] = T0 [1] = T1 [2] = T2		
Dependency:	Only effective when p0776 = 97 or p0776 = 96. Refer to: p0788		

p0795 CX digital inputs, simulation mode / CX DI simulation

CU_CX32

Can be changed: U, T**Calculated:** -**Access level:** 2**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 2220, 2230, 2231**P-Group:** Commands**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**
0000 bin**Description:** Sets the simulation mode for digital inputs.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	Simulation	Terminal eval.	-
	01	DI 1 (X122.2)	Simulation	Terminal eval.	-
	02	DI 2 (X122.3)	Simulation	Terminal eval.	-
	03	DI 3 (X122.4)	Simulation	Terminal eval.	-
	08	DI/DO 8 (X122.9)	Simulation	Terminal eval.	-
	09	DI/DO 9 (X122.10)	Simulation	Terminal eval.	-
	10	DI/DO 10 (X122.12)	Simulation	Terminal eval.	-
	11	DI/DO 11 (X122.13)	Simulation	Terminal eval.	-
	16	DI 16 (X122.5)	Simulation	Terminal eval.	-
	17	DI 17 (X122.6)	Simulation	Terminal eval.	-

Dependency: The setpoint for the input signals is specified using p0796.

Refer to: p0796, p9620

Notice: If a digital input is used as signal source for the function "STO" (BI: p9620) then it is not permissible to select the simulation mode and this is rejected.**Note:** This parameter is not saved when data is backed-up (p0971, p0977).

DI: Digital input

DI/DO: Bidirectional digital input/output

p0795 CU digital inputs simulation mode / CU DI simulationCU_I, CU_S,
CU_S_S150**Can be changed:** U, T**Calculated:** -**Access level:** 2**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 1510, 2020, 2030, 2031, 2100, 2120, 2130, 2131, 2132, 2133**P-Group:** Commands**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**
0000 bin**Description:** Sets the simulation mode for digital inputs.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	Simulation	Terminal eval.	-
	01	DI 1 (X122.2)	Simulation	Terminal eval.	-
	02	DI 2 (X122.3)	Simulation	Terminal eval.	-
	03	DI 3 (X122.4)	Simulation	Terminal eval.	-
	04	DI 4 (X132.1)	Simulation	Terminal eval.	-
	05	DI 5 (X132.2)	Simulation	Terminal eval.	-
	06	DI 6 (X132.3)	Simulation	Terminal eval.	-
	07	DI 7 (X132.4)	Simulation	Terminal eval.	-
	08	DI/DO 8 (X122.9)	Simulation	Terminal eval.	-
	09	DI/DO 9 (X122.10)	Simulation	Terminal eval.	-
	10	DI/DO 10 (X122.12)	Simulation	Terminal eval.	-
	11	DI/DO 11 (X122.13)	Simulation	Terminal eval.	-
	12	DI/DO 12 (X132.9)	Simulation	Terminal eval.	-
	13	DI/DO 13 (X132.10)	Simulation	Terminal eval.	-
	14	DI/DO 14 (X132.12)	Simulation	Terminal eval.	-

15	DI/DO 15 (X132.13)	Simulation	Terminal eval.	-
16	DI 16 (X122.5)	Simulation	Terminal eval.	-
17	DI 16 (X122.6)	Simulation	Terminal eval.	-
20	DI 16 (X132.5)	Simulation	Terminal eval.	-
21	DI 16 (X132.6)	Simulation	Terminal eval.	-

Dependency: The setpoint for the input signals is specified using p0796.

Refer to: p0796, p9620

Notice: If a digital input is used as signal source for the function "STO" (BI: p9620) then it is not permissible to select the simulation mode and this is rejected.

Note: This parameter is not saved when data is backed-up (p0971, p0977).

DI: Digital input

DI/DO: Bidirectional digital input/output

p0795

CU digital inputs simulation mode / CU DI simulation

CU_S_CU310DP,
CU_S_CU310PN

Can be changed: U, T

Calculated: -

Access level: 2

Data type: Unsigned32

Dynamic index: -

Func. diagram: 1510, 2020,
2030, 2031, 2100, 2120, 2130,
2131, 2132, 2133

P-Group: Commands

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting
0000 bin

Description: Sets the simulation mode for digital inputs.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	Simulation	Terminal eval.	-
	01	DI 1 (X122.2)	Simulation	Terminal eval.	-
	02	DI 2 (X122.3)	Simulation	Terminal eval.	-
	03	DI 3 (X122.4)	Simulation	Terminal eval.	-
	08	DI/DO 8 (X122.9)	Simulation	Terminal eval.	-
	09	DI/DO 9 (X122.10)	Simulation	Terminal eval.	-
	10	DI/DO 10 (X122.12)	Simulation	Terminal eval.	-
	11	DI/DO 11 (X122.13)	Simulation	Terminal eval.	-

Dependency: The setpoint for the input signals is specified using p0796.

Refer to: p0796, p9620

Notice: If a digital input is used as signal source for the function "STO" (BI: p9620) then it is not permissible to select the simulation mode and this is rejected.

Note: This parameter is not saved when data is backed-up (p0971, p0977).

DI: Digital input

DI/DO: Bidirectional digital input/output

p0796	CX digital inputs, simulation mode, setpoint / CX DI simul setp				
CU_CX32	Can be changed: U, T	Calculated: -	Access level: 2		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2020, 2030, 2031		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Sets the setpoint for the input signals in the digital input simulation mode.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	High	Low	-
	01	DI 1 (X122.2)	High	Low	-
	02	DI 2 (X122.3)	High	Low	-
	03	DI 3 (X122.4)	High	Low	-
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-
	16	DI 16 (X122.5)	High	Low	-
	17	DI 17 (X122.6)	High	Low	-
Dependency:	The simulation of a digital input is selected using p0795.				
	Refer to: p0795				
Note:	DI: Digital input				
	DI/DO: Bidirectional digital input/output				
	This parameter is not saved when data is backed-up (p0971, p0977).				

p0796	CU digital inputs simulation mode setpoint / CU DI simul setp				
CU_I, CU_S, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 2		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1510, 2020, 2030, 2031, 2100, 2120, 2130, 2131, 2132, 2133		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Sets the setpoint for the input signals in the digital input simulation mode.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	High	Low	-
	01	DI 1 (X122.2)	High	Low	-
	02	DI 2 (X122.3)	High	Low	-
	03	DI 3 (X122.4)	High	Low	-
	04	DI 4 (X132.1)	High	Low	-
	05	DI 5 (X132.2)	High	Low	-
	06	DI 6 (X132.3)	High	Low	-
	07	DI 7 (X132.4)	High	Low	-
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-
	12	DI/DO 12 (X132.9)	High	Low	-
	13	DI/DO 13 (X132.10)	High	Low	-
	14	DI/DO 14 (X132.12)	High	Low	-
	15	DI/DO 15 (X132.13)	High	Low	-
	16	DI 16 (X122.5)	High	Low	-
	17	DI 17 (X122.6)	High	Low	-
	20	DI 20 (X132.5)	High	Low	-
	21	DI 21 (X132.6)	High	Low	-

Dependency: The simulation of a digital input is selected using p0795.
Refer to: p0795

Note: DI: Digital input
DI/DO: Bidirectional digital input/output
This parameter is not saved when data is backed-up (p0971, p0977).

p0796 CU digital inputs simulation mode setpoint / CU DI simul setp

CU_S_CU310DP, CU_S_CU310PN	Can be changed: U, T Data type: Unsigned32	Calculated: - Dynamic index: -	Access level: 2 Func. diagram: 1510, 2020, 2030, 2031, 2100, 2120, 2130, 2131, 2132, 2133 Unit selection: - Expert list: 1 Factory setting 0000 bin
	P-Group: Commands Not for motor type: - Min -	Units group: - Scaling: - Max -	

Description: Sets the setpoint for the input signals in the digital input simulation mode.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X122.1)	High	Low	-
	01	DI 1 (X122.2)	High	Low	-
	02	DI 2 (X122.3)	High	Low	-
	03	DI 3 (X122.4)	High	Low	-
	08	DI/DO 8 (X122.9)	High	Low	-
	09	DI/DO 9 (X122.10)	High	Low	-
	10	DI/DO 10 (X122.12)	High	Low	-
	11	DI/DO 11 (X122.13)	High	Low	-

Dependency: The simulation of a digital input is selected using p0795.
Refer to: p0795

Note: DI: Digital input
DI/DO: Bidirectional digital input/output
This parameter is not saved when data is backed-up (p0971, p0977).

p0799[0...2] CX inputs/outputs, sampling time / CX I/O t_sampl

CU_CX32	Can be changed: C1(3) Data type: FloatingPoint32	Calculated: - Dynamic index: -	Access level: 3 Func. diagram: 2020, 2030, 2031 Unit selection: - Expert list: 1 Factory setting 4000.00 [µs]
	P-Group: Commands Not for motor type: - Min 0.00 [µs]	Units group: - Scaling: - Max 5000.00 [µs]	

Description: Sets the sampling time for the inputs and outputs.

Index: [0] = Digital inputs/outputs (DI/DO)
[1] = Not available - analog inputs (AI)
[2] = Not available - analog outputs (AO)

Dependency: The parameter can only be modified for p0009 = 3, 29.
Refer to: p0009

Note: The modified sampling time is not effective until the drive unit is powered up again.

p0799[0...2]	CU inputs/outputs, sampling time / CU I/O t_sampl			
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Commands Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 5000.00 [µs]	Access level: 3 Func. diagram: 2020, 2030, 2031 Unit selection: - Expert list: 1 Factory setting 4000.00 [µs]	
Description:	Sets the sampling time for the inputs and outputs.			
Index:	[0] = Digital inputs/outputs (DI/DO) [1] = Not available - analog inputs (AI) [2] = Not available - analog outputs (AO)			
Dependency:	The parameter can only be modified for p0009 = 3, 29. Refer to: p0009			
Note:	The modified sampling time is not effective until the drive unit is powered up again.			
p0806	BI: Inhibit master control / PcCtrl inhibit			
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0	
Description:	Sets the signal source to block the master control.			
Dependency:	Refer to: r0807			
Note:	The master control is used from the commissioning software (drive control panel) and from the Advanced Operator Panel (AOP, LOCAL mode).			
r0807.0	BO: Master control active / PcCtrl active			
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: Unsigned8 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -	
Description:	Displays what has the master control. The drive can be controlled via the BICO interconnection or from external (e.g. the commissioning software).			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Master control active	Yes	No
				FP 5030, 6031
Dependency:	Refer to: p0806			
Notice:	The master control only influences control word 1 and speed setpoint 1. Other control words/setpoints can be transferred from another automation device.			
Note:	Bit 0 = 0: BICO interconnection active Bit 0 = 1: Master control for PC/AOP The master control is used from the commissioning software (drive control panel) and from the Advanced Operator Panel (AOP, LOCAL mode).			

p0809[0...2] Copy Command Data Set CDS / Copy CDS

SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned8	Dynamic index: -	Func. diagram: 8560
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	15	0

Description: Copies one Command Data Set (CDS) into another.

Index:
 [0] = Source Command Data Set
 [1] = Target Command Data Set
 [2] = Start copying procedure

Note:
 Procedure:
 1. In Index 0, enter which command data set should be copied.
 2. In Index 1, enter the command data set that is to be copied into.
 3. Start copying: Set index 2 from 0 to 1.
 p0809[2] is automatically set to 0 when copying is completed.

p0810 BI: Command Data Set selection CDS bit 0 / CDS select., bit 0

SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 8560
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source to select the Command Data Set bit 0 (CDS bit 0).

Dependency: Refer to: r0050, p0811, r0836

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note:
 The Command Data Set selected using the binector inputs is displayed in r0836.
 The currently effective command data set is displayed in r0050.
 A Command Data Set can be copied using p0809.

p0811 BI: Command data set selection CDS bit 1 / CDS select., bit 1

VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 8560
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source to select the Command Data Set bit 1 (CDS bit 1).

Dependency: Refer to: r0050, p0810, r0836

Note:
 The Command Data Set selected using the binector inputs is displayed in r0836.
 The currently effective command data set is displayed in r0050.
 A Command Data Set can be copied using p0809.

p0819[0...2] Copy Drive Data Set DDS / Copy DDS

SERVO, TM41, VEC-TOR	Can be changed: C2(15)	Calculated: -	Access level: 2
	Data type: Unsigned8	Dynamic index: -	Func. diagram: 8565
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 31	Factory setting 0

Description: Copies one Drive Data Set (DDS) into another.

Index:
[0] = Source Drive Data Set
[1] = Target Drive Data Set
[2] = Start copying procedure

Note: Procedure:
1. In Index 0, enter which drive data set is to be copied.
2. In Index 1, enter the drive data set data that is to be copied into.
3. Start copying: Set index 2 from 0 to 1.
p0819[2] is automatically set to 0 when copying is completed.

p0820[0...n] BI: Drive Data Set selection DDS bit 0 / DDS select., bit 0

SERVO, TM41, VEC-TOR	Can be changed: C2(15), T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 8565, 8575
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0

Description: Sets the signal source to select the Drive Data Set, bit 0 (DDS, bit 0).

Dependency: Refer to: r0051, r0837

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p0821[0...n] BI: Drive Data Set selection DDS bit 1 / DDS select., bit 1

SERVO, TM41, VEC-TOR	Can be changed: C2(15), T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 8565
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0

Description: Sets the signal source to select the Drive Data Set, bit 1 (DDS, bit 1).

Dependency: Refer to: r0051, r0837

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p0822[0...n] BI: Drive Data Set selection DDS bit 2 / DDS select., bit 2

SERVO, TM41, VEC-TOR	Can be changed: C2(15), T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 8565
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0

Description: Sets the signal source to select the Drive Data Set, bit 2 (DDS, bit 2).

Dependency: Refer to: r0051, r0837

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p0823[0...n] BI: Drive Data Set selection DDS bit 3 / DDS select., bit 3

SERVO, TM41, VEC-TOR	Can be changed: C2(15), T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 8565
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source to select the Drive Data Set, bit 3 (DDS, bit 3).

Dependency: Refer to: r0051, r0837

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p0824[0...n] BI: Drive Data Set selection DDS bit 4 / DDS select., bit 4

SERVO, TM41, VEC-TOR	Can be changed: C2(15), T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 8565, 8575
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source to select the Drive Data Set, bit 4 (DDS, bit 4).

Dependency: Refer to: r0051, r0837

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p0826[0...n] Motor changeover, motor number / Mot_chng mot No.

SERVO	Can be changed: C2(3)	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: MDS, p0130	Func. diagram: 8575
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	15	0

Description: Sets the freely-assignable motor number for the motor changeover.

Dependency: Refer to: p0827

Caution: When changing over motor data sets with the same motor number (e.g. star-delta changeover) and for a motor with brake, the motor brake remains open during the changeover.

Note: When the motor data sets are changed over, the following applies:
The same motor number signifies the same thermal model.

p0826[0...n] Motor changeover, motor number / Mot_chng mot No.

VECTOR	Can be changed: C2(3)	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: MDS, p0130	Func. diagram: 8575
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	15	0

Description: Sets the freely-assignable motor number for the motor changeover.

Dependency: Refer to: p0827

Caution: When changing over motor data sets with the same motor number (e.g. star-delta changeover) and for a motor with brake, the motor brake remains open during the changeover.

Note: When the motor data sets are changed over, the following applies:
The same motor number signifies the same thermal model.
For the same motor number, the correction values of the Rs, Lh or kT adaptation are applied for the data set changeover (refer to r1782, r1787, r1797).

p0827[0...n]	Motor changeover status word bit number / Mot_chg ZSW bitNo.		
SERVO, VECTOR	Can be changed: C2(3)	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: MDS, p0130	Func. diagram: 8575
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 15	Factory setting 0
Description:	Sets the bit number for every motor data set. Example: p0827[0] = 0: For MDS0, r0830.0 is switched. p0827[1] = 5: For MDS1, r0830.5 is switched.		
Dependency:	Refer to: p0826, r0830		
Note:	A motor is only changed over (a new motor selected) after the pulses have been suppressed. When the motor data sets are changed over, the following applies: Bit numbers that are not identical, signify that the motor must be changed over.		

p0828[0...n]	BI: Motor changeover, feedback signal / Mot_chng fdbk sig		
SERVO, VECTOR	Can be changed: C2(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 8575
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source for the feedback signal when changing over the motor. For p0833.0 = 1 the following applies: This feedback signal (0/1 edge) is required after a motor changeover to enable the pulses.		
Dependency:	Refer to: p0833		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		

r0830.0...15	CO/BO: Motor changeover, status word / Mot_chngov ZSW		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 8575
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

Description: Displays the status word of the motor changeover.
These signals can be connected to digital outputs to change over the motor.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Motor selection, bit 0	High	Low	-
	01	Motor selection, bit 1	High	Low	-
	02	Motor selection, bit 2	High	Low	-
	03	Motor selection, bit 3	High	Low	-
	04	Motor selection, bit 4	High	Low	-
	05	Motor selection, bit 5	High	Low	-
	06	Motor selection, bit 6	High	Low	-
	07	Motor selection, bit 7	High	Low	-
	08	Motor selection, bit 8	High	Low	-
	09	Motor selection, bit 9	High	Low	-
	10	Motor selection, bit 10	High	Low	-

11	Motor selection, bit 11	High	Low	-
12	Motor selection, bit 12	High	Low	-
13	Motor selection, bit 13	High	Low	-
14	Motor selection, bit 14	High	Low	-
15	Motor selection, bit 15	High	Low	-

Dependency: Refer to: p0827

p0831[0...15] **BI: Motor changeover, contactor feedback / Mot_chg cont fdbk**

SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 8575
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the feedback signal of the contactors when changing over motors. There is a fixed inter-relationship between energizing the contactor and the feedback signal.

Example:

A changeover is to be made between MDS0 (motor 0) and MDS1 (motor 1). The contactors should be switched using bit 4 (contactor 0) and 5 (contactor 1). The changeover should be made with an interconnection of the feedback signal.

Implementation:

MDS0: p0827[0] = 4, interconnect output to switch contactor 0 to r0830.4, p0831[4] = "input, feedback signal, contactor 0"

MDS1: p0827[1] = 5, interconnect output to switch contactor 1 to r0830.5, p0831[5] = "input, feedback signal, contactor 1"

The following sequence applies when changing over from MDS0 to MDS1:

1. The status bit r0830.4 is deleted. When the feedback signal (p0831[4]) is connected, the system waits until the feedback signal "contactor open" is displayed. If the feedback signal is not connected, then the system waits for the switch-off interlocking time of 320 ms.
2. The status bit r0830.5 is set. If the feedback signal (p0831[5]) is connected, the system waits until the feedback signal "contactor closed" is displayed. If the feedback signal is not connected, then the system waits for the switch-on interlocking time of 160 ms.

Index:

- [0] = Feedback signal contactor 0
- [1] = Feedback signal contactor 1
- [2] = Feedback signal contactor 2
- [3] = Feedback signal contactor 3
- [4] = Feedback signal contactor 4
- [5] = Feedback signal contactor 5
- [6] = Feedback signal contactor 6
- [7] = Feedback signal contactor 7
- [8] = Feedback signal contactor 8
- [9] = Feedback signal contactor 9
- [10] = Feedback signal contactor 10
- [11] = Feedback signal contactor 11
- [12] = Feedback signal contactor 12
- [13] = Feedback signal contactor 13
- [14] = Feedback signal contactor 14
- [15] = Feedback signal contactor 15

r0832.0...15 CO/BO: Mot. changeover, contactor feedback sig. status word / Mot_chng fdbk ZSW

SERVO, VECTOR

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 8575**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the status word of the contactor feedback signals when changing over a motor.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Feedback signal contactor 0	Closed	Opened	-
01	Feedback signal contactor 1	Closed	Opened	-
02	Feedback signal contactor 2	Closed	Opened	-
03	Feedback signal contactor 3	Closed	Opened	-
04	Feedback signal contactor 4	Closed	Opened	-
05	Feedback signal contactor 5	Closed	Opened	-
06	Feedback signal contactor 6	Closed	Opened	-
07	Feedback signal contactor 7	Closed	Opened	-
08	Feedback signal contactor 8	Closed	Opened	-
09	Feedback signal contactor 9	Closed	Opened	-
10	Feedback signal contactor 10	Closed	Opened	-
11	Feedback signal contactor 11	Closed	Opened	-
12	Feedback signal contactor 12	Closed	Opened	-
13	Feedback signal contactor 13	Closed	Opened	-
14	Feedback signal contactor 14	Closed	Opened	-
15	Feedback signal contactor 15	Closed	Opened	-

Dependency:

Refer to: p0831

p0833 Data set changeover configuration / DS_chng config

SERVO

Can be changed: C2(15)**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 8575**P-Group:** Data sets**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

0000 bin

Description:

Sets the configuration for the motor and encoder changeover.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Contactor changeover from the application/drive	application	Drive	-
01	Pulse suppression by application/drive	application	Drive	-
02	Suppress drive parking for EDS changeover	Yes	No	-

Note:

Re bit 00:

When the bit is set and the motor has to be changed over, then p0827 must be set differently in the appropriate motor data sets.

Re bit 02:

The bit defines whether, for an EDS changeover, the status signal Gn_ZSW.14 is suppressed (parking encoder active).

p0833 Data set changeover configuration / DS_chng config				
VECTOR	Can be changed: C2(15)		Calculated: -	Access level: 2
	Data type: Unsigned16		Dynamic index: -	Func. diagram: 8575
	P-Group: Data sets		Units group: -	Unit selection: -
	Not for motor type: -		Scaling: -	Expert list: 1
	Min		Max	Factory setting
	-		-	0010 bin
Description: Sets the configuration for the motor and encoder changeover.				
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Contactor changeover from the applica- tion/drive	application	Drive
	01	Pulse suppression by application/drive	application	Drive
	02	Suppress drive parking for EDS changeover	Yes	No
Note:	Re bit 00: When the bit is set and the motor has to be changed over, then p0827 must be set differently in the appropriate motor data sets. Re bit 02: The bit defines whether, for an EDS changeover, the status signal Gn_ZSW.14 is suppressed (parking encoder active). Re motor changeover to running motor: The "flying restart" function should also be activated (p1200) when changing over to a motor that is already running.			

r0835.2 CO/BO: Data set changeover status word / DDS_ZSW				
ENCODER, TM41	Can be changed: -		Calculated: -	Access level: 2
	Data type: Unsigned16		Dynamic index: -	Func. diagram: 8575
	P-Group: Displays, signals		Units group: -	Unit selection: -
	Not for motor type: -		Scaling: -	Expert list: 1
	Min		Max	Factory setting
	-		-	-
Description: Displays the status word for the drive data set changeover.				
Bit field:	Bit	Signal name	1 signal	0 signal
	02	Internal parameter calculation active	Yes	No
Note:	Re bit 02: A data set changeover is delayed by the time required for the internal parameter calculation.			

r0835.0...11 CO/BO: Data set changeover status word / DDS_ZSW				
SERVO, VECTOR	Can be changed: -		Calculated: -	Access level: 2
	Data type: Unsigned16		Dynamic index: -	Func. diagram: 8575
	P-Group: Displays, signals		Units group: -	Unit selection: -
	Not for motor type: -		Scaling: -	Expert list: 1
	Min		Max	Factory setting
	-		-	-
Description: Displays the status word for the drive data set changeover.				
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Motor changeover active	Yes	No
	01	Encoder changeover active	Yes	No
	02	Internal parameter calculation active	Yes	No
	04	Armature short circuit active	Yes	No
	05	Identification running	Yes	No
	06	Friction characteristic record running	Yes	No
	07	Rotating measurement running	Yes	No
	08	Motor data identification running	Yes	No
	10	Wait for pulse suppression	Yes	No
	11	Wait for motor changeover feedback signal	Yes	No

Note: This parameter is only supplied with up-to-date values if data set changeover has been selected or is running.

Re bit 00:
The signal is only influenced when a motor changeover is set via p0827 (unequal bit numbers).

Re bit 01:
The signal is only influenced when an encoder changeover is set via p0187, p0188, or p0189.

Re bit 02:
A data set changeover is delayed by the time required for the internal parameter calculation.

Re bit 04:
A data set changeover is only carried out when the armature short circuit is not activated.

Re bit 05:
The following applies for SERVO:
A data set changeover is only carried out when pole position identification, encoder adjustment, motor data identification, and rotating measurement are not running.
The following applies for VECTOR:
A data set changeover is only carried out when pole position identification is not running.

Re bit 06:
A data set changeover is only carried out when the friction characteristic record is not running.

Re bit 07 (VECTOR only):
A data set changeover is only carried out when rotating measurement is not running.

Re bit 08 (VECTOR only):
A data set changeover is only carried out when motor data identification is not running.

Re bit 10:
A motor changeover is set with p0833.1 = 1. It can only be carried out when the application performs pulse suppression.

Re bit 11:
A motor changeover is set with p0833.0 = 1. The pulses are only enabled when the "Motor changeover feedback" signal is detected.

r0836.0...3		CO/BO: Command Data Set CDS selected / CDS selected				
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: -		Calculated: -		Access level: 2	
	Data type: Unsigned8		Dynamic index: -		Func. diagram: 1530, 8560	
	P-Group: Displays, signals		Units group: -		Unit selection: -	
	Not for motor type: -		Scaling: -		Expert list: 1	
	Min -		Max -		Factory setting -	
Description:		Displays the command data set (CDS) selected via the binector input.				
Bit field:		Bit	Signal name	1 signal	0 signal	FP
		00	CDS select. bit 0	On	Off	-
		01	CDS select. bit 1	On	Off	-
		02	CDS select. bit 2	On	Off	-
		03	CDS select. bit 3	On	Off	-
Dependency:		Refer to: r0050, p0810, p0811				
Note:		Command data sets are selected via binector input p0810 and following. The currently effective command data set is displayed in r0050.				

r0837.0...4 CO/BO: Drive Data Set DDS selected / DDS selected

ENCODER, SERVO, TM41, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned8	Dynamic index: -	Func. diagram: 8565
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the drive data set (DDS) selected via the binector input.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DDS select. bit 0	On	Off	-
	01	DDS select. bit 1	On	Off	-
	02	DDS select. bit 2	On	Off	-
	03	DDS select. bit 3	On	Off	-
	04	DDS select. bit 4	On	Off	-

Dependency: Refer to: r0051, p0820, p0821, p0822, p0823, p0824

Note: Drive data sets are selected via binector input p0820 and following.
The currently effective drive data set is displayed in r0051.
If there is only one data set, then a value of 0 is displayed in this parameter and not the selection via binector inputs.

r0838[0...3] Motor/Encoder Data Set selected / MDS/EDS selected

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned8	Dynamic index: -	Func. diagram: 8565
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the selected Motor Data Set (MDS) and the selected Encoder Data Sets (EDS).

Index:
[0] = Motor Data Set MDS selected
[1] = Encoder 1 Encoder Data Set EDS selected
[2] = Encoder 2 Encoder Data Set EDS selected
[3] = Encoder 3 Encoder Data Set EDS selected

Dependency: Refer to: r0049, p0186, p0187, p0188, p0189

Note: Value 99 means the following: No encoder assigned (not configured).

p0839 Motor changeover contactor control delay time / Mot_chg ctrl t_del

SERVO, VECTOR	Can be changed: C2(3)	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0 [ms]	500 [ms]	0 [ms]

Description: Sets the delay time for the contactor control for the motor changeover.

The delay time is taken into account in the following cases:

- for feedback signal, previous contactor "Open". The new motor contactor is controlled (energized) after the delay time has expired.
- for the feedback signal, new motor contactor "Closed". The pulses are enabled after the delay time has expired.

p0840[0...n]	BI: ON/OFF1 / ON/OFF1		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2501, 2610, 8720, 8820, 8920
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for control word 1 bit 0 (ON/OFF1).		
Recommend.:	When the setting for this binector input is changed, the motor can only be switched on by means of an appropriate signal change of the source.		
Dependency:	Refer to: p1055, p1056		
Notice:	For BI: p0840 = 0 signal, the motor can be moved, jogging using BI: p1055 or BI: p1056. The command "ON/OFF1" can be issued using BI: p0840 or using BI: p1055/p1056. Only the signal source that originally powered up can also power down again. The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For drives with closed-loop speed control (p1300 = 20, 21), the following applies: Bit 0 = 0: OFF1 (braking with the ramp-function generator, then pulse suppression and switching on inhibited) For drives with closed-loop torque control (p1300 = 22, 23), the following applies: Bit 0 = 0: Immediate pulse suppression For drives with closed-loop torque control (activated using p1501), the following applies: Bit 0 = 0: No dedicated braking response, but pulse suppression when standstill is detected (p1226, p1227) For drives with closed-loop speed/torque control, the following applies: Bit 0 = 0/1: ON (pulses can be enabled) For active infeeds (Active Line Module and Smart Line Module) the following applies: Bit 0 = 0: OFF1 (reduce Vdc along the ramp, then pulse suppression and pre-charging contactor/line contactor open) Bit 0 = 0/1: ON (pre-charging contactor/line contactor closed, pulses can be enabled) For passive infeeds (Basic Line Module) the following applies: Bit 0 = 0: OFF1 (pre-charging contactor/line contactor open) Bit 0 = 0/1: ON (pre-charging contactor/line contactor closed) r0863.1 of a drive can also be selected as signal source.		

p0840	BI: ON/OFF1 / ON/OFF1		
TM41	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9677
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for control word 1 bit 0 (ON/OFF1).		
Recommend.:	When the setting for this binector input is changed, the motor can only be switched on by means of an appropriate signal change of the source.		
Dependency:	Refer to: p1055, p1056		
Notice:	Only the signal source that originally powered up can also power down again. The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Bit 0 = 0: OFF1 (pulse suppression and switching on inhibited) Bit 0 = 0/1: ON (pulses can be enabled) This parameter has no function in the "SINAMICS" operating mode (p4400 = 1).		

p0844[0...n]	BI: 1. OFF2 / 1. OFF2		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2501, 8720, 8820, 8920 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the 1st OC/OFF2. The AND logic operation of the 1st OC/OFF2 and 2nd OC/OFF2 results in control word 1, bit 1 (OC/OFF2).		
Caution:	When "master control from PC" is activated, this binector input is ineffective.		
			
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For drives, the following applies: Bit 1 = 0: OFF2 (immediate pulse suppression and switching on inhibited) Bit 1 = 1: No OFF2 (enable is possible) For infeed units, the following applies: Bit 1 = 0: OFF2 (immediate pulse suppression for Active Infeed Modules and Smart Line Modules, pre-charging contactor/line contactor open and switching on inhibited) Bit 1 = 1: No OFF2 (enable is possible) OC: Operating condition		

p0844	BI: OFF2 / OFF2		
TM41	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 9677 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for control word 1, bit 1 (OC/OFF2).		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Bit 1 = 0: OFF2 (immediate pulse suppression and switching on inhibited) Bit 1 = 1: No OFF2 (enable is possible) OC: Operating condition This parameter has no function in the "SINAMICS" operating mode (p4400 = 1).		

p0845[0...n]	BI: 2. OFF2 / 2. OFF2		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2501, 8720, 8820, 8920 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the 2nd OC/OFF2. The AND logic operation of the 1st OC/OFF2 and 2nd OC/OFF2 results in control word 1, bit 1 (OC/OFF2).		

Note: For drives, the following applies:
 Bit 1 = 0: OFF2 (immediate pulse suppression and switching on inhibited)
 Bit 1 = 1: No OFF2 (enable is possible)
 For infeed units, the following applies:
 Bit 1 = 0: OFF2 (immediate pulse suppression for Active Infeed Modules and Smart Line Modules, pre-charging contactor/line contactor open and switching on inhibited)
 Bit 1 = 1: No OFF2 (enable is possible)
 OC: Operating condition

p0848[0...n]	BI: 1. OFF3 / 1. OFF3		
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2501
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1

Description: Sets the signal source for the 1st OC/OFF3.
 The AND logic operation of the 1st OC/OFF3 and 2nd OC/OFF3 results in control word 1, bit 2 (OC/OFF3).
Caution: When "master control from PC" is activated, this binector input is ineffective.



Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.
Note: Bit 2 = 0: OFF3 (braking along the OFF3 ramp (p1135), then pulse suppression and switching on inhibited)
 Bit 2 = 1: No OFF3 (enable is possible)
 OC: Operating condition

p0848	BI: OFF3 / OFF3		
TM41	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9677
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1

Description: Sets the signal source for control word 1, bit 2 (OC/OFF3).
Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.
Note: Bit 2 = 0: OFF3 (pulse suppression and switching on inhibited)
 Bit 2 = 1: No OFF3 (enable is possible)
 OC: Operating condition
 This parameter has no function in the "SINAMICS" operating mode (p4400 = 1).

p0849[0...n]	BI: 2. OFF3 / 2. OFF3		
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2501
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1

Description: Sets the signal source for the 2nd OC/OFF3.
 The AND logic operation of the 1st OC/OFF3 and 2nd OC/OFF3 results in control word 1, bit 2 (OC/OFF3).
Note: Bit 2 = 0: OFF3 (braking along the OFF3 ramp (p1135), then pulse suppression and switching on inhibited)
 Bit 2 = 1: No OFF3 (enable is possible)
 OC: Operating condition

p0852[0...n]	BI: Operation enable / Operation enable		
A_INF, S_INF, SERVO, VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2501, 8820, 8920 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for control word 1 bit 3 (enable operation)		
Caution:	When "master control from PC" is activated, this binector input is ineffective.		
			
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Bit 3 = 0: Inhibit operation (cancel pulses) Bit 3 = 1: Enable operation (pulses can be enabled)		
p0852	BI: Operation enable / Operation enable		
TM41	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 9677 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for control word 1 bit 3 (enable operation)		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Bit 3 = 0: Inhibit operation (cancel pulses) Bit 3 = 1: Enable operation (pulses can be enabled) This parameter has no function in the "SINAMICS" operating mode (p4400 = 1).		
p0854[0...n]	BI: Master ctrl by PLC / Master ctrl by PLC		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2501, 8720, 8820, 8920 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for control word 1 bit 10 (master control by PLC).		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Bit 10 = 0: No master control by PLC Bit 10 = 1: Master control by PLC This bit is used to initiate a response for the drives when the control fails (F07220). If there is no control available, then BI: p0854 should be set to a 1 signal. If a control is available, control word 1 bit 10 = 1 (STW1.10, PZD1) must be set so that the received data is updated. This applies regardless of the setting in p0854 and even in the case of free telegram configuration (p0922 = 999).		

p0854	BI: Master ctrl by PLC / Master ctrl by PLC		
ENCODER	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2501, 8720, 8820, 8920 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for control word 1 bit 10 (master control by PLC).		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Bit 10 = 0: No master control by PLC Bit 10 = 1: Master control by PLC This bit is used to initiate a response for the drives when the control fails (F07220). If there is no control available, then BI: p0854 should be set to a 1 signal. If a control is available, control word 1 bit 10 = 1 (STW1.10, PZD1) must be set so that the received data is updated. This applies regardless of the setting in p0854 and even in the case of free telegram configuration (p0922 = 999).		
p0854	BI: Master ctrl by PLC / Master ctrl by PLC		
TM41	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 9677, 9678 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for control word 1 bit 10 (master control by PLC).		
Dependency:	Refer to: p1155		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Bit 10 = 0: No master control by PLC Bit 10 = 1: Master control by PLC For the TM41, a response can be initiated using this bit if the control fails. The parameter is only effective in the "SIMOTION" operating mode (p4400 = 0). In the "SINAMICS" operating mode, the setpoints at CI: p4420 are evaluated independently of p0854. Further, the setting of p2037 should be observed.		
p0855[0...n]	BI: Unconditionally release holding brake / Uncond open brake		
SERVO, VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2501, 2701, 2707 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the command "unconditionally open holding brake".		
Dependency:	Refer to: p0858		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	The signal via BI: p0858 (unconditionally close holding brake) has a higher priority than via BI: p0855 (unconditionally open holding brake).		

p0856[0...n]	BI: Speed controller enable / n_ctrl enable		
SERVO, VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2501, 2701, 2707 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the command "enable speed controller" (r0898.12). 0 signal: Set the I component and speed controller output to zero. 1 signal: Enable speed controller.		
Dependency:	Refer to: r0898		
Note:	If "enable speed controller" is withdrawn, then an existing brake will be closed. If "speed controller enable" is withdrawn, the pulses are not suppressed.		
p0856[0...n]	BI: Velocity controller enable / v_ctrl enable		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2501, 2701, 2707 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the command "enable speed controller" (r0898.12). 0 signal: Set the I component and speed controller output to zero. 1 signal: Enable speed controller.		
Dependency:	Refer to: r0898		
Note:	If "enable speed controller" is withdrawn, then an existing brake will be closed. If "speed controller enable" is withdrawn, the pulses are not suppressed.		
p0857	Power unit monitoring time / PU t_monit		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T Data type: FloatingPoint32 P-Group: Commands Not for motor type: - Min 100.0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 60000.0 [ms]	Access level: 2 Func. diagram: 8760, 8864, 8964 Unit selection: - Expert list: 1 Factory setting 6000.0 [ms]
Description:	Sets the monitoring time for the power unit. The following applies for infeeds and drives: The monitoring time is started after an 0/1 edge of the ON/OFF1 command. If the power unit does not return a READY signal within the monitoring time, then fault F06000 (infeeds) or F07802 (drives) is output. For drives, the following also applies: After the pulse enable (operation enabled, p0852), the monitoring time is re-started. If the infeed does not signal ready to the drive within the monitoring time (using BI: p0864 of the drive), fault F07840 is initiated.		
Dependency:	Refer to: F06000, F07802, F07840, F30027		
Notice:	The maximum time to pre-charge the DC link is monitored in the power unit and cannot be changed. The maximum duration of the pre-charging depends on the power class and the power unit design. The monitoring time for the pre-charging is started after the ON command (BI: p0840 = 0/1 signal). Fault F30027 is output when the maximum pre-charging duration is exceeded.		

Note: The factory setting for p0857 depends on the power class and the design of the power unit.
The monitoring time for the ready signal of the power unit includes the time to pre-charge the DC link and, if relevant, the de-bounce time of the contactors.
If an excessively low value is entered into p0857, then after enable, this results in the corresponding fault.

p0858[0...n]	BI: Unconditionally close holding brake / Uncond close brake		
SERVO	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2501, 2701, 2707
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	9719.13
Description:	Sets the signal source for the command "unconditionally close holding brake".		
Dependency:	Refer to: p0855		
Note:	The signal via BI: p0858 (unconditionally close holding brake) has a higher priority than via BI: p0855 (unconditionally open holding brake). For a 1 signal via BI: p0858, the command "unconditionally close the holding brake" is executed and internally a zero setpoint is entered.		

p0858[0...n]	BI: Unconditionally close holding brake / Uncond close brake		
VECTOR	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2501, 2701, 2707
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the command "unconditionally close holding brake".		
Dependency:	Refer to: p0855		
Note:	The signal via BI: p0858 (unconditionally close holding brake) has a higher priority than via BI: p0855 (unconditionally open holding brake). For a 1 signal via BI: p0858, the command "unconditionally close the holding brake" is executed and internally a zero setpoint is entered.		

p0860	BI: Line cont. fdbk sig / Line contact feedb		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2634, 8734, 8834, 8934
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	863.1
Description:	Sets the signal source for the feedback signal from the line contactor.		
Recommend.:	When the monitoring is activated (BI: p0860 not equal to r0863.1), then to control the line contactor, signal BO: r0863.1 of its own drive object should be used.		
Dependency:	Refer to: p0861, r0863 Refer to: F07300		
Notice:	The line contactor monitoring is deactivated if the control signal of the particular drive object is set as the signal source for the feedback signal of the line contactor (BI: p0860 = r0863.1).		
Note:	The state of the line contactor is monitored depending on signal BO: r0863.1. When the monitoring is activated (BI: p0860 not equal to r0863.1), fault F07300 is then also output if the contactor is closed before it is controlled using r0863.1.		

p0861				
Line contactor monitoring time / LineContact t_mon				
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T		Calculated: -	
	Data type: FloatingPoint32		Dynamic index: -	
	P-Group: Commands		Units group: -	
	Not for motor type: -		Scaling: -	
Description:	Min		Max	
	0 [ms]		5000 [ms]	
	Sets the monitoring time of the line contactor.			
	This time starts each time that the line contactor switches (r0863.1). If a feedback signal is not received from the line contactor within the time, a message is output.			
Dependency:		Refer to: p0860, r0863 Refer to: F07300		
Note:		The monitoring function is disabled for the factory setting of p0860.		
p0862				
Power unit ON delay / PU t_on				
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T		Calculated: -	
	Data type: FloatingPoint32		Dynamic index: -	
	P-Group: Commands		Units group: -	
	Not for motor type: -		Scaling: -	
Description:	Min		Max	
	0 [ms]		65000 [ms]	
	Sets the delay time for the control command of the power unit and a line contactor, if used.			
	This means that it is possible to realize a shifted (delayed) pre-charging or power-on using a single ON command.			
Note:		When the infeed units are active, before the line contactor is closed, an offset adjustment of the current measurement is carried out for a duration of 120 ms (p3491).		
r0863.0...2				
CO/BO: Drive coupling status word/control word / CoupleZSW/STW				
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: -		Calculated: -	
	Data type: Unsigned16		Dynamic index: -	
	P-Group: Commands		Units group: -	
	Not for motor type: -		Scaling: -	
Description:	Min		Max	
	-		-	
	Displays the status and control words of the drive coupling.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Closed-loop control operation	Yes	No
				2610, 6495, 8732, 8832, 8932, 9794
	01	Energize contactor	Yes	No
				2610, 2634, 8732, 8734, 8832, 8834, 8932, 8934
	02	Infeed line supply failure	Yes	No
				-
Dependency:		Refer to: p0864		

Note: Re bit 00:
 Bit 0 signals that the infeed is ready.
 When the operating signal is transferred via BO: r0863.0 this allows several drives to start (run-up) staggered over time when they are simultaneously powered up.
 To realize this, the following connections/interconnections are required:
 Drive 1: Internconnect BI: p0864 with BO: r0863.0 of the infeed
 Drive 2: Internconnect BI: p0864 with BO: r0863.0 of drive 1
 Drive 3: Internconnect BI: p0864 with BO: r0863.0 of drive 2, etc.
 The first drive only transfers the operating signal to the next drive after it has reached its ready condition.
 Re bit 01:
 Bit 1 is used to control an external line contactor.
 Re bit 02:
 This bit only signals line supply failure for Active Infeed (A_INF) and Smart Infeed (S_INF).

p0864	BI: Infeed operation / INF operation		
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 1773, 1774, 2610
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the operating signal of the infeed (e.g. BO: r0863.0).		
Dependency:	Refer to: r0863		
Note:	The sequence control of a servo/vector drive requires this signal. The following applies for an infeed without DRIVE-CLiQ: For these infeeds, the "ready" message is available via an output terminal. This signal must be connected to a digital input. The drives supplied from this infeed must use this signal as ready signal (BI: p0864 = digital input).		

p0868	Power unit DC switch debounce time / PU DC sw t_deboun		
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0 [ms]	65000 [ms]	65000 [ms]
Description:	Sets the debounce time for the DC circuit breaker for Motor Modules in "chassis" format.		
Note:	The following applies if p0868 = 65000 ms: The debounce time defined internally in the power unit's EEPROM is implemented.		

r0873	CO/BO: Infeed, total operation / INF total oper		
B_INF, S_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 8732, 8832
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the operational readiness of the infeeds when using Smart Line Module (SLM) and Basic Line Module (BLM) together (mixed operation). In order that signal BO: r0873 is available at one of the infeeds, then BI: p0874 of the one infeed must be interconnected to BO: r0863.0 of the other infeed.		
Dependency:	Refer to: r0863, p0874		
Note:	Mixed operation is not possible with the Active Line Module (ALM)!		

p0874				
BI: Smart/ Basic Line Module operation / SLM/BLM operation				
B_INF, S_INF	Can be changed: T	Calculated: -	Access level: 2	
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 8732, 8832	
	P-Group: Commands	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description:	Setting to interconnect the ready signal for mixed operation of Smart Line Module (SLM) and Basic Line Module (BLM). In order that signal BO: r0873 is available at one of the infeeds, then BI: p0874 of the one infeed must be interconnected to BO: r0863.0 of the other infeed.			
Dependency:	Refer to: r0863, r0873			
Note:	Mixed operation is not possible with the Active Line Module (ALM)!			
p0895[0...n]				
BI: Activate/deactivate power unit components / PU_comp act/deact				
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 1	
	Data type: Unsigned32 / Binary	Dynamic index: PDS, p0120	Func. diagram: -	
	P-Group: Displays, signals	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	1	
Description:	Sets the signal source to activate/deactivate a power unit component.			
Dependency:	BI: p0895 = 0 signal Deactivating power unit components BI: p0895 = 1 signal Activating power unit components Refer to: p0125, r0126 Refer to: A05054			
Caution:	It is not permissible to deactivate drive objects with safety functions enabled.			
Note:	The power unit is only deactivated when the pulses are suppressed. For units connected in parallel, when one of the power units is deactivated, then the enable in p7001 is withdrawn.			
r0896.0				
BO: Parking axis, status word / Parking axis, ZSW				
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2	
	Data type: Unsigned8	Dynamic index: -	Func. diagram: -	
	P-Group: Displays, signals	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	-	
Description:	Displays the status word for the "parking axis" function.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Parking axis active	Yes	No
Dependency:	Refer to: p0897			

p0897 BI: Parking axis selection / Parking axis sel

SERVO, VECTOR

Can be changed: T**Calculated:** -**Access level:** 2**Data type:** Unsigned32 / Binary**Dynamic index:** -**Func. diagram:** -**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

0

Description: Sets the signal source to select the "parking axis" function.**Dependency:** BI: p0897 = 0 signal

The function "parking axis" is not selected.

BI: p0897 = 1 signal

The function "parking axis" is selected.

Refer to: r0896

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.**Note:** After it has been selected the "parking axis" function only becomes active when the pulses are suppressed.**r0898.0...10 CO/BO: Control word sequence control infeed / STW seq_ctrl INF**

A_INF, S_INF

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1530, 8820, 8920**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description: Displays control word 1 of the infeed.**Bit field:****Bit Signal name****1 signal****0 signal****FP**

00 ON/OFF1

Yes

No

-

01 OC / OFF2

Yes

No

-

03 Operation enable

Yes

No

-

05 Inhibit motoring operation

Yes

No

-

06 Inhibit regenerative

Yes

No

-

10 Master ctrl by PLC

Yes

No

-

Note: OC: Operating condition**r0898.0...10 CO/BO: Control word sequence control infeed / STW seq_ctrl INF**

B_INF

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 8720**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description: Displays control word 1 of the infeed.**Bit field:****Bit Signal name****1 signal****0 signal****FP**

00 ON/OFF1

Yes

No

-

01 OC / OFF2

Yes

No

-

10 Master ctrl by PLC

Yes

No

-

Note: OC: Operating condition

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r0898.0...14 CO/BO: Control word sequence control / STW seq_ctrl

SERVO (Lin)

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1530, 2501**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the control word of the sequence control.

Bit field:**Bit Signal name****1 signal****0 signal****FP**

00 ON/OFF1

Yes

No

-

01 OC / OFF2

Yes

No

-

02 OC / OFF3

Yes

No

-

03 Operation enable

Yes

No

-

04 Ramp-function generator enable

Yes

No

-

05 Continue ramp-function generator

Yes

No

-

06 Velocity setpoint enable

Yes

No

-

07 Command open brake

Yes

No

-

08 Jog 1

Yes

No

-

09 Jog 2

Yes

No

-

10 Master ctrl by PLC

Yes

No

-

12 Velocity controller enable

Yes

No

-

14 Command close brake

Yes

No

-

Note:

OC: Operating condition

r0898.0...13 CO/BO: Control word sequence control / STW seq_ctrl

TM41

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 9678**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the control word of the sequence control.

Bit field:**Bit Signal name****1 signal****0 signal****FP**

00 ON/OFF1

Yes

No

-

01 OC / OFF2

Yes

No

-

02 OC / OFF3

Yes

No

-

03 Operation enable

Yes

No

-

04 Ramp-function generator enable

Yes

No

-

05 Start ramp-function generator

Yes

No

-

06 Speed setpoint enable

Yes

No

-

07 Acknowledge fault

Yes

No

-

10 Master ctrl by PLC

Yes

No

-

13 Zero mark enable

Yes

No

-

Note:

OC: Operating condition

r0899.0...12 CO/BO: Status word sequence control infeed / ZSW seq_ctrl INF

A_INF, S_INF

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1530, 8826, 8926**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the status word of the infeed sequence control.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Ready for sw on	Yes	No	-
01	Ready	Yes	No	-
02	Operation enabled	Yes	No	-
04	No OFF2 active	OFF2 inactive	OFF2 active	-
06	Sw on inhibit	Yes	No	-
08	Power-up active	Yes	No	-
09	Control request	Yes	No	-
11	Pre-charging compl	Yes	No	-
12	Line contactor closed	Yes	No	8934

Note:

Re bit 12:

The feedback signal of a line contactor (auxiliary contact) can be interconnected via BI: p0860.

r0899.0...12 CO/BO: Status word sequence control infeed / ZSW seq_ctrl INF

B_INF

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 8726**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the status word of the infeed sequence control.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Ready for sw on	Yes	No	-
01	Ready	Yes	No	-
02	Operation enabled	Yes	No	-
04	No OFF2 active	OFF2 inactive	OFF2 active	-
06	Sw on inhibit	Yes	No	-
09	Control request	Yes	No	-
11	Pre-charging compl	Yes	No	-
12	Line contactor closed	Yes	No	-

Note:

Re bits 00, 01, 02, 04, 06, 09:

For PROFIdrive, these signals are used for status word 1.

r0899.0...15 CO/BO: Status word drive object 1 / ZSW DO1

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 2
CU_S,	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Displays, signals	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 1
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the status word from drive object 1 (Control Unit).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	reserved	Yes	No	-
	03	Fault present	Yes	No	-
	07	Alarm present	Yes	No	-
	08	System time synchronized	Yes	No	-
	12	Slave sign-of-life bit 0	Yes	No	-
	13	Slave sign-of-life bit 1	Yes	No	-
	14	Slave sign-of-life bit 2	Yes	No	-
	15	Slave sign-of-life bit 3	Yes	No	-

r0899.9 CO/BO: Status word sequence control encoder DO / ZSW seq_ctrl encDO

ENCODER	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word for sequence control on the encoder drive object.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	09	Control request	Yes	No	-

Note: For PROFIdrive, this signal is used for status word ZSW2_ENC.

r0899.0...15 CO/BO: Status word sequence control / ZSW seq_ctrl

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2503
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word of the sequence control.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Ready for sw on	Yes	No	-
	01	Ready	Yes	No	-
	02	Operation enabled	Yes	No	-
	03	Jog active	Yes	No	-
	04	No coasting active	OFF2 inactive	OFF2 active	-
	05	No Quick Stop active	OFF3 inactive	OFF3 active	-
	06	Switching on inhibited active	Yes	No	-
	07	Drive ready	Yes	No	-
	08	Controller enable	Yes	No	-
	09	Control request	Yes	No	-
	11	Pulses enabled	Yes	No	-
	12	Holding brake open	Yes	No	-
	13	Command close holding brake	Yes	No	-
	14	Pulse enable from the brake control	Yes	No	-
	15	Setpoint enable from the brake control	Yes	No	-

Note: Re bits 00, 01, 02, 04, 05, 06, 09:
For PROFIdrive, these signals are used for status word 1.
Re bit 13:
When the "Safe Brake Control" (SBC) is activated and selected, the brake is no longer controlled using this signal.
Re bit 14, 15:
These signals are only of significance when the "extended brake control" function module is activated (r0108.14 = 1).

r0899.0...15 CO/BO: Status word sequence control / ZSW seq_ctrl

TM41	Can be changed: - Data type: Unsigned16 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 9680 Unit selection: - Expert list: 1 Factory setting -
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Description: Displays the status word of the sequence control.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Ready for sw on	Yes	No	-
	01	Ready	Yes	No	-
	02	Operation enabled	Yes	No	-
	04	Coast down active	No	Yes	-
	05	Quick Stop active	No	Yes	-
	06	Sw on inhibit	Yes	No	-
	09	Control request	Yes	No	-
	13	Zero mark enabled	Yes	No	-
	14	Track A/B enabled	Yes	No	-
	15	Interface encoder emulation enabled	Yes	No	-

Note: Re bit 00, 01, 02, 06:
For PROFIdrive, these signals are used for status word 1.

p0915[0...29] TM15 PROFIdrive PZD setpoint assignment / TM15 PD PZD setp			
TM15	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	4273	[0] 4201
			[1] 4204
			[2] 4205
			[3] 4211
			[4] 4212
			[5] 4213
			[6] 0
			[7] 0
			[8] 0
			[9] 0
			[10] 0
			[11] 0
			[12] 0
			[13] 0
			[14] 0
			[15] 0
			[16] 0
			[17] 0
			[18] 0
			[19] 0
			[20] 0
			[21] 0
			[22] 0
			[23] 0
			[24] 0
			[25] 0
			[26] 0
			[27] 0
			[28] 0
			[29] 0
Description:	Is used to assign the process data received from the master (PZD, setpoints).		
Value:	0: ZERO 4201: r4201 (system time for synchronization) 4204: r4204 (control digital output 0 ... 15) 4205: r4205 (control digital output 16 ... 23) 4211: r4211 (edge mode digital input 0 ... 7) 4212: r4212 (edge mode digital input 8 ... 15) 4213: r4213 (edge mode digital input 16 ... 23) 4250: r4250 (set/resetting time digital output 0) 4251: r4251 (set/resetting time digital output 1) 4252: r4252 (set/resetting time digital output 2) 4253: r4253 (set/resetting time digital output 3) 4254: r4254 (set/resetting time digital output 4) 4255: r4255 (set/resetting time digital output 5) 4256: r4256 (set/resetting time digital output 6) 4257: r4257 (set/resetting time digital output 7) 4258: r4258 (set/resetting time digital output 8) 4259: r4259 (set/resetting time digital output 9) 4260: r4260 (set/resetting time digital output 10)		

4261: r4261 (set/resetting time digital output 11)
4262: r4262 (set/resetting time digital output 12)
4263: r4263 (set/resetting time digital output 13)
4264: r4264 (set/resetting time digital output 14)
4265: r4265 (set/resetting time digital output 15)
4266: r4266 (set/resetting time digital output 16)
4267: r4267 (set/resetting time digital output 17)
4268: r4268 (set/resetting time digital output 18)
4269: r4269 (set/resetting time digital output 19)
4270: r4270 (set/resetting time digital output 20)
4271: r4271 (set/resetting time digital output 21)
4272: r4272 (set/resetting time digital output 22)
4273: r4273 (set/resetting time digital output 23)

Index:

[0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5
[5] = PZD 6
[6] = PZD 7
[7] = PZD 8
[8] = PZD 9
[9] = PZD 10
[10] = PZD 11
[11] = PZD 12
[12] = PZD 13
[13] = PZD 14
[14] = PZD 15
[15] = PZD 16
[16] = PZD 17
[17] = PZD 18
[18] = PZD 19
[19] = PZD 20
[20] = PZD 21
[21] = PZD 22
[22] = PZD 23
[23] = PZD 24
[24] = PZD 25
[25] = PZD 26
[26] = PZD 27
[27] = PZD 28
[28] = PZD 29
[29] = PZD 30

Note:

Example:

The telegram for the setpoints should have the following process data (PZD) and assignments:

PZD 1 (r4201), PZD 2 (r4204), PZD 3 (r4250)

The setpoint assignment must be realized as follows:

p0915[0] = 4201 - 16 bit

p0915[1] = 4204 - 16 bit

p0915[2] = 4250 - 16 bit

p0915[3] = 0

...

p0915[29] = 0

p0915[0...35] TM17 PROFIdrive PZD setpoint assignment / TM17 PD PZD setp			
TM17	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	4265	[0] 4201
			[1] 4204
			[2] 4211
			[3] 4212
			[4] 0
			[5] 0
			[6] 0
			[7] 0
			[8] 0
			[9] 0
			[10] 0
			[11] 0
			[12] 0
			[13] 0
			[14] 0
			[15] 0
			[16] 0
			[17] 0
			[18] 0
			[19] 0
			[20] 0
			[21] 0
			[22] 0
			[23] 0
			[24] 0
			[25] 0
			[26] 0
			[27] 0
			[28] 0
			[29] 0
			[30] 0
			[31] 0
			[32] 0
			[33] 0
			[34] 0
			[35] 0

Description: Is used to assign the process data received from the master (PZD, setpoints).

Value: 0: ZERO
 4201: r4201 (system time for synchronization)
 4204: r4204 (control digital output 0 ... 15)
 4211: r4211 (edge mode digital input 0 ... 7)
 4212: r4212 (edge mode digital input 8 ... 15)
 4250: r4250 (set/resetting time digital output 0)
 4251: r4251 (set/resetting time digital output 1)
 4252: r4252 (set/resetting time digital output 2)
 4253: r4253 (set/resetting time digital output 3)
 4254: r4254 (set/resetting time digital output 4)
 4255: r4255 (set/resetting time digital output 5)

4256: r4256 (set/resetting time digital output 6)
4257: r4257 (set/resetting time digital output 7)
4258: r4258 (set/resetting time digital output 8)
4259: r4259 (set/resetting time digital output 9)
4260: r4260 (set/resetting time digital output 10)
4261: r4261 (set/resetting time digital output 11)
4262: r4262 (set/resetting time digital output 12)
4263: r4263 (set/resetting time digital output 13)
4264: r4264 (set/resetting time digital output 14)
4265: r4265 (set/resetting time digital output 15)

Index:

[0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5
[5] = PZD 6
[6] = PZD 7
[7] = PZD 8
[8] = PZD 9
[9] = PZD 10
[10] = PZD 11
[11] = PZD 12
[12] = PZD 13
[13] = PZD 14
[14] = PZD 15
[15] = PZD 16
[16] = PZD 17
[17] = PZD 18
[18] = PZD 19
[19] = PZD 20
[20] = PZD 21
[21] = PZD 22
[22] = PZD 23
[23] = PZD 24
[24] = PZD 25
[25] = PZD 26
[26] = PZD 27
[27] = PZD 28
[28] = PZD 29
[29] = PZD 30
[30] = PZD 31
[31] = PZD 32
[32] = PZD 33
[33] = PZD 34
[34] = PZD 35
[35] = PZD 36

Note:

Example:

The telegram for the setpoints should have the following process data (PZD) and assignments:

PZD 1 (r4201), PZD 2 (r4204), PZD 3 (r4250), PZD 4 (r4250)

The setpoint assignment must be realized as follows:

p0915[0] = 4201 - 16 bit

p0915[1] = 4204 - 16 bit

p0915[2] = 4250 - 32 bit - specified twice one after the other

p0915[3] = 4250 - 32 bit

p0915[4] = 0

...

p0915[35] = 0

p0916[0...29]	TM15 PROFIdrive PZD actual value assignment / TM15 PD PZD actVal		
TM15	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	4373	[0] 4301
			[1] 4304
			[2] 4305
			[3] 4311
			[4] 4312
			[5] 4313
			[6] 0
			[7] 0
			[8] 0
			[9] 0
			[10] 0
			[11] 0
			[12] 0
			[13] 0
			[14] 0
			[15] 0
			[16] 0
			[17] 0
			[18] 0
			[19] 0
			[20] 0
			[21] 0
			[22] 0
			[23] 0
			[24] 0
			[25] 0
			[26] 0
			[27] 0
			[28] 0
			[29] 0
Description:	Is used to assign the process data to be sent to the master (PZD, actual values).		
Value:	0: ZERO 4301: r4301 (module synchronization) 4304: r4304 (status digital input 0 ... 15) 4305: r4305 (status digital input 16 ... 23) 4311: r4311 (edge status digital input 0 ... 7) 4312: r4312 (edge status digital input 8 ... 15) 4313: r4313 (edge status digital input 16 ... 23) 4350: r4350 (edge times digital input 0) 4351: r4351 (edge times digital input 1) 4352: r4352 (edge times digital input 2) 4353: r4353 (edge times digital input 3) 4354: r4354 (edge times digital input 4) 4355: r4355 (edge times digital input 5) 4356: r4356 (edge times digital input 6) 4357: r4357 (edge times digital input 7) 4358: r4358 (edge times digital input 8) 4359: r4359 (edge times digital input 9) 4360: r4360 (edge times digital input 10)		

4361: r4361 (edge times digital input 11)
4362: r4362 (edge times digital input 12)
4363: r4363 (edge times digital input 13)
4364: r4364 (edge times digital input 14)
4365: r4365 (edge times digital input 15)
4366: r4366 (edge times digital input 16)
4367: r4367 (edge times digital input 17)
4368: r4368 (edge times digital input 18)
4369: r4369 (edge times digital input 19)
4370: r4370 (edge times digital input 20)
4371: r4371 (edge times digital input 21)
4372: r4372 (edge times digital input 22)
4373: r4373 (edge times digital input 23)

Index:

[0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5
[5] = PZD 6
[6] = PZD 7
[7] = PZD 8
[8] = PZD 9
[9] = PZD 10
[10] = PZD 11
[11] = PZD 12
[12] = PZD 13
[13] = PZD 14
[14] = PZD 15
[15] = PZD 16
[16] = PZD 17
[17] = PZD 18
[18] = PZD 19
[19] = PZD 20
[20] = PZD 21
[21] = PZD 22
[22] = PZD 23
[23] = PZD 24
[24] = PZD 25
[25] = PZD 26
[26] = PZD 27
[27] = PZD 28
[28] = PZD 29
[29] = PZD 30

Note:

Example:

The telegram for the actual values should have the following process data (PZD) and assignments:

PZD 1 (r4301), PZD 2 (r4304), PZD 3 (r4350)

The actual value assignment must be implemented as follows:

p0916[0] = 4301 - 16 bit

p0916[1] = 4304 - 16 bit

p0916[2] = 4350 - 16 bit

p0916[3] = 0

...

p0916[29] = 0

p0916[0...35] TM17 PROFIdrive PZD actual value assignment / TM17 PD PZD actVal			
TM17	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	4365	[0] 4301
			[1] 4304
			[2] 4311
			[3] 4312
			[4] 0
			[5] 0
			[6] 0
			[7] 0
			[8] 0
			[9] 0
			[10] 0
			[11] 0
			[12] 0
			[13] 0
			[14] 0
			[15] 0
			[16] 0
			[17] 0
			[18] 0
			[19] 0
			[20] 0
			[21] 0
			[22] 0
			[23] 0
			[24] 0
			[25] 0
			[26] 0
			[27] 0
			[28] 0
			[29] 0
			[30] 0
			[31] 0
			[32] 0
			[33] 0
			[34] 0
			[35] 0
Description:	Is used to assign the process data to be sent to the master (PZD, actual values).		
Value:	0: ZERO 4301: r4301 (module synchronization) 4304: r4304 (status digital input 0 ... 15) 4311: r4311 (edge status digital input 0 ... 7) 4312: r4312 (edge status digital input 8 ... 15) 4350: r4350 (edge times digital input 0) 4351: r4351 (edge times digital input 1) 4352: r4352 (edge times digital input 2) 4353: r4353 (edge times digital input 3) 4354: r4354 (edge times digital input 4) 4355: r4355 (edge times digital input 5)		

4356: r4356 (edge times digital input 6)
 4357: r4357 (edge times digital input 7)
 4358: r4358 (edge times digital input 8)
 4359: r4359 (edge times digital input 9)
 4360: r4360 (edge times digital input 10)
 4361: r4361 (edge times digital input 11)
 4362: r4362 (edge times digital input 12)
 4363: r4363 (edge times digital input 13)
 4364: r4364 (edge times digital input 14)
 4365: r4365 (edge times digital input 15)

Index:

[0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5
 [5] = PZD 6
 [6] = PZD 7
 [7] = PZD 8
 [8] = PZD 9
 [9] = PZD 10
 [10] = PZD 11
 [11] = PZD 12
 [12] = PZD 13
 [13] = PZD 14
 [14] = PZD 15
 [15] = PZD 16
 [16] = PZD 17
 [17] = PZD 18
 [18] = PZD 19
 [19] = PZD 20
 [20] = PZD 21
 [21] = PZD 22
 [22] = PZD 23
 [23] = PZD 24
 [24] = PZD 25
 [25] = PZD 26
 [26] = PZD 27
 [27] = PZD 28
 [28] = PZD 29
 [29] = PZD 30
 [30] = PZD 31
 [31] = PZD 32
 [32] = PZD 33
 [33] = PZD 34
 [34] = PZD 35
 [35] = PZD 36

Note:

Example:

The telegram for the actual values should have the following process data (PZD) and assignments:

PZD 1 (r4301), PZD 2 (r4304), PZD 3 (r4350), PZD 4 (r4350)

The setpoint assignment must be realized as follows:

p0916[0] = 4301 - 16 bit

p0916[1] = 4304 - 16 bit

p0916[2] = 4350 - 32 bit - specified twice one after the other

p0916[3] = 4350 - 32 bit

p0916[4] = 0

...

p0916[35] = 0

p0918	PROFIBUS address / PB address		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 1 Max 126	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 126	Access level: 2 Func. diagram: 1520, 2410 Unit selection: - Expert list: 1
Description:	Displays or sets the PROFIBUS address for PROFIBUS interface on the Control Unit. The address can be set as follows: 1) Using the address switch on the Control Unit. --> p0918 can then only be read and displays the selected address. --> A change only becomes effective after a POWER ON. 2) Using p0918 --> Only if the address 00 hex, 7F hex, 80 hex, or FF hex has been set using the address switch. --> The address is saved in a non-volatile fashion using the function "copy from RAM to ROM". --> A change only becomes effective after a POWER ON.		
Note:	Permissible PROFIBUS addresses: 1 ... 126 (01 hex ... 7E hex) Address 126 is used for commissioning. Every PROFIBUS address change only becomes effective after a POWER ON. The parameter is not influenced by setting the factory setting.		
p0922	PROFIdrive telegram selection / PD Telegr_sel		
A_INF, B_INF, S_INF	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 370 Max 999	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 999	Access level: 1 Func. diagram: 1520, 2420, 2423, 2447, 2457, 2481, 2483 Unit selection: - Expert list: 1
Description:	Sets the send and receive telegram.		
Value:	370: SIEMENS telegram 370, PZD-1/1 371: SIEMENS telegram 371, PZD-5/8 999: Free telegram configuration with BICO		
Dependency:	Refer to: F01505, F01506		
Note:	If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999.		
p0922	PROFIdrive telegram selection / PD Telegr_sel		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 390 Max 999	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 999	Access level: 1 Func. diagram: 1520, 2420, 2423, 2481, 2483 Unit selection: - Expert list: 1
Description:	Sets the send and receive telegram.		
Value:	390: SIEMENS telegram 390, PZD-2/2 391: SIEMENS telegram 391, PZD-3/7 392: SIEMENS telegram 392, PZD-3/15 999: Free telegram configuration with BICO		

p0922	PROFIdrive telegram selection / PD Telegr_sel		
ENCODER	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 81 Max 999	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 1 Func. diagram: 1520, 2420, 2422, 2423, 2468, 2470 Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram.		
Value:	81: SIEMENS telegram 81, PZD-2/6 82: SIEMENS telegram 82, PZD-2/7 83: SIEMENS telegram 83, PZD-2/8 999: Free telegram configuration with BICO		
Dependency:	Refer to: p2038 Refer to: F01505, F01506		
Note:	If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999.		
p0922	PROFIdrive telegram selection / PD Telegr_sel		
SERVO	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 1 Max 999	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 1 Func. diagram: 1520, 2420, 2422, 2423, 2468, 2470 Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram.		
Value:	1: Standard telegram 1, PZD-2/2 2: Standard telegram 2, PZD-4/4 3: Standard telegram 3, PZD-5/9 4: Standard telegram 4, PZD-6/14 5: Standard telegram 5, PZD-9/9 6: Standard telegram 6, PZD-10/14 102: SIEMENS telegram 102, PZD-6/10 103: SIEMENS telegram 103, PZD-7/15 105: SIEMENS telegram 105, PZD-10/10 106: SIEMENS telegram 106, PZD-11/15 116: SIEMENS telegram 116, PZD-11/19 118: SIEMENS telegram 118, PZD-11/19 125: SIEMENS telegram 125, PZD-14/10 126: SIEMENS telegram 126, PZD-15/15 136: SIEMENS telegram 136, PZD-15/19 220: SIEMENS telegram 220, PZD-10/10 999: Free telegram configuration with BICO		
Dependency:	Refer to: p2038 Refer to: F01505, F01506		
Note:	For p0922 = 100 ... 199, p2038 is automatically set to 1 and p2038 can no longer be changed. This means that for these telegrams, the "SIMODRIVE 611 universal" interface mode is set and cannot be changed. If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999.		

p0922	PROFIdrive telegram selection / PD Telegr_sel		
SERVO (EPOS, Pos ctrl)	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 7	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 1 Func. diagram: 1520, 2420, 2422, 2423, 2468, 2470 Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram.		
Value:	7: Standard telegram 7, PZD-2/2 9: Standard telegram 9, PZD-10/5 110: SIEMENS telegram 110, PZD-12/7 111: SIEMENS telegram 111, PZD-12/12 999: Free telegram configuration with BICO		
Dependency:	Refer to: p2038 Refer to: F01505, F01506		
Note:	For p0922 = 100 ... 199, p2038 is automatically set to 1 and p2038 can no longer be changed. This means that for these telegrams, the "SIMODRIVE 611 universal" interface mode is set and cannot be changed. If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999.		
p0922	PROFIdrive telegram selection / PD Telegr_sel		
SERVO (Pos ctrl)	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 999	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 1 Func. diagram: 1520, 2420, 2422, 2423, 2468, 2470 Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram.		
Value:	999: Free telegram configuration with BICO		
Dependency:	Refer to: p2038 Refer to: F01505, F01506		
Note:	For p0922 = 100 ... 199, p2038 is automatically set to 1 and p2038 can no longer be changed. This means that for these telegrams, the "SIMODRIVE 611 universal" interface mode is set and cannot be changed. If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999.		
p0922	PROFIdrive telegram selection / PD Telegr_sel		
TM15, TM17	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 0	Access level: 1 Func. diagram: 2481, 2483 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the send and receive telegram.		
Value:	0: Free telegram configuring with p0915/p0916		

p0922	PROFIdrive telegram selection / PD Telegr_sel		
TM41	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 3	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 1 Func. diagram: 1520, 9677, 9679, 9681, 9683 Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram.		
Value:	3: Standard telegram 3, PZD-5/9 999: Free telegram configuration with BICO		
Dependency:	Refer to: F01505, F01506		
Note:	If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999.		
p0922	PROFIdrive telegram selection / PD Telegr_sel		
VECTOR	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 1 Func. diagram: 1520, 2420, 2422, 2423, 2468, 2470 Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram.		
Value:	1: Standard telegram 1, PZD-2/2 2: Standard telegram 2, PZD-4/4 20: Standard telegram 20, PZD-2/6 220: SIEMENS telegram 220, PZD-10/10 352: SIEMENS telegram 352, PZD-6/6 999: Free telegram configuration with BICO		
Dependency:	Refer to: F01505, F01506		
Caution:	Telegrams 2, 3 and 4 are not suitable for sensorless vector control (p1300 = 20). For sensorless vector control, it is necessary that the setpoint speed is entered at the ramp-function generator input (e.g. p1070) and not after the ramp-function generator (p1155).		
Note:	If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999.		
p0922	PROFIdrive telegram selection / PD Telegr_sel		
VECTOR (EPOS, n/M, Pos ctrl)	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 7	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 1 Func. diagram: 1520, 2420, 2422, 2423, 2468, 2470 Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram.		
Value:	7: Standard telegram 7, PZD-2/2 9: Standard telegram 9, PZD-10/5 110: SIEMENS telegram 110, PZD-12/7 111: SIEMENS telegram 111, PZD-12/12 999: Free telegram configuration with BICO		
Dependency:	Refer to: F01505, F01506		

Caution: Telegrams 2, 3 and 4 are not suitable for sensorless vector control (p1300 = 20). For sensorless vector control, it is necessary that the setpoint speed is entered at the ramp-function generator input (e.g. p1070) and not after the ramp-function generator (p1155).

Note: If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited.

The inhibited interconnections can only be changed again after setting value 999.

p0922	PROFIdrive telegram selection / PD Telegr_sel		
VECTOR (n/M, Pos ctrl)	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 999	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 1 Func. diagram: 1520, 2420, 2422, 2423, 2468, 2470 Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram.		
Value:	999: Free telegram configuration with BICO		
Dependency:	Refer to: F01505, F01506		
Caution:	Telegram 2, 3 and 4 are not suitable for sensorless vector control (p1300 = 20). For sensorless vector control, it is necessary that the setpoint speed is entered at the ramp-function generator input (e.g. p1070) and not after the ramp-function generator (p1155).		
Note:	If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited.		
	The inhibited interconnections can only be changed again after setting value 999.		

p0922	PROFIdrive telegram selection / PD Telegr_sel		
VECTOR (n/M)	Can be changed: C2(1), T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 1 Func. diagram: 1520, 2420, 2422, 2423, 2468, 2470 Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram.		
Value:	1: Standard telegram 1, PZD-2/2 2: Standard telegram 2, PZD-4/4 3: Standard telegram 3, PZD-5/9 4: Standard telegram 4, PZD-6/14 20: Standard telegram 20, PZD-2/6 220: SIEMENS telegram 220, PZD-10/10 352: SIEMENS telegram 352, PZD-6/6 999: Free telegram configuration with BICO		
Dependency:	Refer to: F01505, F01506		
Caution:	Telegram 2, 3 and 4 are not suitable for sensorless vector control (p1300 = 20). For sensorless vector control, it is necessary that the setpoint speed is entered at the ramp-function generator input (e.g. p1070) and not after the ramp-function generator (p1155).		
Note:	If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited.		
	The inhibited interconnections can only be changed again after setting value 999.		

r0924[0...1]	ZSW bit pulses enabled / ZSW pulses enab		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2454, 2456 Unit selection: - Expert list: 1 Factory setting -
Description:	Display of the position of the "Pulses enabled" status word bit in the PROFIdrive telegram		
Index:	[0] = Signal number [1] = Bit position		
p0925	PROFIdrive clock synchronous sign-of-life tolerance / PD SoL_tol		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, SERVO, TM41, VECTOR (n/M)	Can be changed: U, T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: 2410 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of tolerated consecutive sign-of-life errors of the clock-cycle synchronous master. The sign-of-life signal is normally received in PZD4 (control word 2) from the master.		
Dependency:	Refer to: p2045, r2065 Refer to: F01912		
Note:	The sign-of-life monitoring is disabled for p0925 = 65535.		
r0930	PROFIdrive operating mode / PD operating mode		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the operating mode. 1: Closed-loop speed controlled operation with ramp-function generator 2: Closed-loop position controlled operation 3: Closed-loop speed controlled operation without ramp-function generator		
r0944	CO: Counter for fault buffer changes / Fault buff change		
All objects	Can be changed: - Data type: Unsigned16 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 8060 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays fault buffer changes. This counter is incremented every time the fault buffer changes.		
Recommend.:	Used to check whether the fault buffer has been read out consistently.		
Dependency:	Refer to: r0945, r0947, r0948, r0949, r2109		

r0945[0...63] Fault code / Fault code

All objects	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1750, 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the numbers of faults that have occurred.

Dependency: Refer to: r0947, r0948, r0949, r2109, r2130, r2133, r2136, r3120, r3122

Note: The buffer parameters are cyclically updated in the background (refer to status signal in r2139).

Fault buffer structure (general principle):

r0945[0], r0949[0], r0948[0], r2109[0], r3115[0] --> actual fault case, fault 1

...

r0945[7], r0949[7], r0948[7], r2109[7], r3115[7] --> actual fault case, fault 8

r0945[8], r0949[8], r0948[8], r2109[8], r3115[8] --> 1st acknowledged fault case, fault 1

...

r0945[15], r0949[15], r0948[15], r2109[15], r3115[15] --> 1st acknowledged fault case, fault 8

...

r0945[56], r0949[56], r0948[56], r2109[56], r3115[56] --> 7th acknowledged fault case, fault 1

...

r0945[63], r0949[63], r0948[63], r2109[63], r3115[63] --> 7th acknowledged fault case, fault 8

r0946[0...65534] Fault code list / Fault code list

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min	Max	Factory setting
	-	-	-

Description: Lists the fault codes stored in the drive unit.

The indices can only be accessed with a valid fault code.

Dependency: The parameter assigned to the fault code is entered in r0951 under the same index.

r0947[0...63] Fault number / Fault number

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1750, 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: This parameter is identical to r0945.

r0948[0...63] Fault time received in milliseconds / t_fault rcv ms

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1750, 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [ms]	- [ms]	- [ms]

Description: Displays the system runtime in milliseconds when the fault occurred.

Dependency: Refer to: r0945, r0947, r0949, r2109, r2114, r2130, r2133, r2136, r3115, r3120, r3122

Notice: The time comprises r2130 (days) and r0948 (milliseconds).

Note: The buffer parameters are cyclically updated in the background (refer to status signal in r2139).
The structure of the fault buffer and the assignment of the indices is shown in r0945.
When the parameter is read via PROFIdrive, the TimeDifference data type applies.

r0949[0...63] Fault value / Fault value

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: 1750, 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
-	-	-	

Description: Displays additional information about the fault that occurred (as integer number).

Dependency: Refer to: r0945, r0947, r0948, r2109, r2130, r2133, r2136, r3115, r3120, r3122

Note: The buffer parameters are cyclically updated in the background (refer to status signal in r2139).
The structure of the fault buffer and the assignment of the indices is shown in r0945.

p0952 Fault cases, counter / Fault cases qty

All objects	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1710, 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
0	65535	0	

Description: Number of fault situations that have occurred since the last reset.

Dependency: The fault buffer is deleted (cleared) by setting p0952 to 0.

Refer to: r0945, r0947, r0948, r0949, r2109, r2130, r2133, r2136

r0963 PROFIBUS baud rate / PB baud rate

CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
0	255	-	

Description: Displays the corresponding value for the PROFIBUS baud rate.

Value:

0:	9.6 kbit/s
1:	19.2 kbit/s
2:	93.75 kbit/s
3:	187.5 kbit/s
4:	500 kbit/s
6:	1.5 Mbit/s
7:	3 Mbit/s
8:	6 Mbit/s
9:	12 Mbit/s
10:	31.25 kbit/s
11:	45.45 kbit/s
255:	Baud rate unknown

r0964[0...6]	Device identification / Device ident.		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the device identification.		
Index:	[0] = Company (Siemens = 42) [1] = Device type [2] = Firmware version [3] = Firmware date (year) [4] = Firmware date (day/month) [5] = Number of drive objects [6] = Firmware patch/hot fix		
Note:	Example: r0964[0] = 42 --> SIEMENS r0964[1] = 5000 --> SINAMICS S CU320 r0964[1] = 5200 --> SINAMICS G CU320 r0964[2] = 102 --> first part of the firmware version V01.02 (second part, refer to index 6) r0964[3] = 2003 --> year 2003 r0964[4] = 1401 --> 14th of January r0964[5] = 4 --> 4 drive objects r0964[6] = 600 --> second part, firmware version (complete version: V01.02.06.00)		
r0965	PROFIdrive profile number / PD profile number		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the PROFIdrive profile number and profile version. Constant value = 0329 hex. Byte 1: Profile number = 03 hex = PROFIdrive profile Byte 2: Profile version = 29 hex = Version 4.1		
Note:	When the parameter is read via PROFIdrive, the Octet String 2 data type applies.		
p0969	System runtime relative / t_System relative		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Unsigned32 P-Group: Displays, signals Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4294967295 [ms]	Access level: 3 Func. diagram: 1750, 8060 Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Displays the system runtime in ms since the last POWER ON.		
Note:	The value in p0969 can only be reset to 0. The value overflows after approx. 49 days. When the parameter is read via PROFIdrive, the TimeDifference data type applies.		

p0970	Reset infeed parameter / INF par reset		
A_INF, B_INF, S_INF	Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to initiate a reset of the parameters of an individual infeed unit. The parameters of the basic commissioning (refer to p0009) are in this case not reset (p0107, p0108, p0121, p0170). These can only be reset using the factory setting of the complete drive unit (p0976). The sampling times (p0111, p0112, p0115) are only not reset if this results in a conflict with the basic clock cycle (p0110).		
Value:	0: Inactive 1: Start a parameter reset 100: Start a BICO interconnection reset		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	A factory setting run can only be started if p0010 was first set to 30 (parameter reset). At the end of the calculations, p0970 is automatically set to 0. Parameter reset has been completed if p0970 and p0010 have been set to 0.		
p0970	ENCODER reset parameters / ENC par reset		
ENCODER	Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to initiate the reset of the parameters on the ENCODER drive object. Parameter p0141 is not reset. It is only reset if the entire drive unit is reset to the factory settings (p0976).		
Value:	0: Inactive 1: Start a parameter reset 100: Start a BICO interconnection reset		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	A factory setting run can only be started if p0010 was first set to 30 (parameter reset). At the end of the calculations, p0970 is automatically set to 0. Parameter reset has been completed if p0970 and p0010 have been set to 0.		
p0970	Reset drive parameters / Drive par reset		
SERVO, VECTOR	Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to initiate the reset of the parameters of an individual drive unit. Parameters p0100, p0205 (only for VECTOR) and the parameters of the basic drive commissioning (p0009) are not reset (p0107, p0108, p0111, p0112, p0115, p0121, p0130, p0131, p0140, p0141, p0142, p0170, p0186 ... p0189). These can only be reset using the factory setting of the complete drive unit (p0976).		
Value:	0: Inactive 1: Start a parameter reset 5: Starts a safety parameter reset 100: Start a BICO interconnection reset		

Dependency: Refer to: F01659

Notice: After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.

Note: A factory setting run can only be started if p0010 was first set to 30 (parameter reset).
At the end of the calculations, p0970 is automatically set to 0. Parameter reset has been completed if p0970 and p0010 have been set to 0.
If a Safety Integrated function is parameterized (p9601), the safety parameters will not be reset if p0970 = 1. In this case, an error message (F1659) is output with fault value 2.

p0970 TB30 reset parameters / TB30 par reset

Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
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Description: The parameter is used to initiate a reset of the parameters on Terminal Board 30 (TB30).
The sampling time p4099 is not reset if in so doing a conflict occurs with the basic clock cycle.
Parameter p0161 is not reset. It is only reset if the entire drive unit is reset to the factory settings (p0976).

Value: 0: Inactive
1: Start a parameter reset
100: Start a BICO interconnection reset

Notice: After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.

Note: A factory setting run can only be started if p0010 was first set to 30 (parameter reset).
At the end of the calculations, p0970 is automatically set to 0. Parameter reset has been completed if p0970 and p0010 have been set to 0.

p0970 TM120 reset parameters / TM120 par reset

Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
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Description: The parameter is used to initiate a reset of the parameters on Terminal Module 120 (TM120).

Value: 0: Inactive
1: Start a parameter reset
100: Start a BICO interconnection reset

Dependency: Refer to: p0010

Notice: After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.

Note: A factory setting run can only be started if p0010 was first set to 30 (parameter reset).
At the end of the calculations, p0970 is automatically set to 0.

p0970 TM15 reset parameter / TM15 par reset

Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
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Description: The parameter is used to initiate a reset of the parameters on Terminal Module 15 (TM15).
Parameter p0151 is not reset. It is only reset if the entire drive unit is reset to the factory settings (p0976).

Value:	0: Inactive 1: Start a parameter reset 100: Start a BICO interconnection reset
Dependency:	Refer to: p0010
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.
Note:	A factory setting run can only be started if p0010 was first set to 30 (parameter reset). At the end of the calculations, p0970 is automatically set to 0.

p0970 TM15DI/DO reset parameter / TM15D par reset

TM15DI_DO	Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to initiate a reset of the parameters on Terminal Module 15 (TM15). The sampling time p4099 is not reset if in so doing a conflict occurs with the basic clock cycle. Parameter p0151 is not reset. It is only reset if the entire drive unit is reset to the factory settings (p0976).		
Value:	0: Inactive 1: Start a parameter reset 100: Start a BICO interconnection reset		
Dependency:	Refer to: p0010		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	A factory setting run can only be started if p0010 was first set to 30 (parameter reset). At the end of the calculations, p0970 is automatically set to 0.		

p0970 TM17 reset parameter / TM17 par reset

TM17	Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to initiate a reset of the parameters on Terminal Module 17 (TM17). Parameter p0151 is not reset. It is only reset if the entire drive unit is reset to the factory settings (p0976).		
Value:	0: Inactive 1: Start a parameter reset 100: Start a BICO interconnection reset		
Dependency:	Refer to: p0010		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	A factory setting run can only be started if p0010 was first set to 30 (parameter reset). At the end of the calculations, p0970 is automatically set to 0.		

p0970 TM31 reset parameters / TM31 par reset

TM31	Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to initiate a reset of the parameters on Terminal Module 31 (TM31). The sampling time p4099 is not reset if in so doing a conflict occurs with the basic clock cycle. Parameter p0151 is not reset. It is only reset if the entire drive unit is reset to the factory settings (p0976).		
Value:	0: Inactive 1: Start a parameter reset 100: Start a BICO interconnection reset		
Dependency:	Refer to: p0010		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	A factory setting run can only be started if p0010 was first set to 30 (parameter reset). At the end of the calculations, p0970 is automatically set to 0.		

p0970 TM41 reset parameters / TM41 par reset

TM41	Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to initiate a reset of the parameters on Terminal Module 41 (TM41). Parameter p0151 is not reset. It is only reset if the entire drive unit is reset to the factory settings (p0976).		
Value:	0: Inactive 1: Start a parameter reset 100: Start a BICO interconnection reset		
Dependency:	Refer to: p0010		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	A factory setting run can only be started if p0010 was first set to 30 (parameter reset). At the end of the calculations, p0970 is automatically set to 0.		

p0970 TM54F reset parameters / TM54F par reset

TM54F_MA	Can be changed: C2(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The parameter is used to initiate a reset of the parameters on Terminal Module 54F (TM54F). Parameter p0151 is not reset. It is only reset if the entire drive unit is reset to the factory settings (p0976).		
Value:	0: Inactive 1: Start a parameter reset 100: Start a BICO interconnection reset		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		

Note: A factory setting run can only be started if p0010 was first set to 30 (parameter reset).
At the end of the calculations, p0970 is automatically set to 0. Parameter reset has been completed if p0970 and p0010 have been set to 0.

p0971			
Save drive object parameters / Drv_obj par save			
All objects	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Factory settings	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0
Description:	Setting to save the parameter of the particular drive object in the non-volatile memory.		
Value:	0: Inactive 1: Save drive object		
Dependency:	Refer to: p0977, p1960, p3845, r3996		
Caution:	The Control Unit power supply may only be powered down after data has been saved (i.e. after data save has been started, wait until the parameter again has the value 0).		
Notice:	Writing to parameters is inhibited while saving. The progress while saving is displayed in r3996.		
Note:	Starting from the particular drive object, the following parameters are saved: CU3xx: Device-specific parameters and PROFIBUS device parameters. Other objects: Parameters of the actual object and PROFIBUS device parameters. Prerequisite: In order that the parameter of a drive object, saved with p0971 = 1, is read the next time that the Control Unit is booted, then all parameters must, as a minimum, have first been saved once with p0977 = 1.		

p0972			
Drive unit reset / Drv_unit reset			
CU_CX32, CU_I,	Can be changed: U, T	Calculated: -	Access level: 1
CU_S,	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: -	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 1
CU_S_S150	Min	Max	Factory setting
	0	3	0
Description:	Sets the required procedure to execute a hardware reset for the drive unit.		
Value:	0: Inactive 1: Hardware-Reset immediate 2: Hardware reset preparation 3: Hardware reset after cyclic communication has failed		
Danger:	It must be absolutely ensured that the system is in a safe condition. The memory card of the Control Unit must not be accessed.		
			
Notice:	For SIMOTION or SINUMERIK with integrated SINAMICS, the hardware reset acts on the complete system and depends on the state of the control.		

Note:

Re value = 1:
Reset is immediately executed and communications interrupted.
After communications have been established, check the reset operation (refer below).
Re value = 2:
Help to check the reset operation.
Firstly, set p0972 = 2 and then read back. Secondly, set p0972 = 1 (it is possible that this request is possibly no longer acknowledged). The communication is then interrupted.
After communications have been established, check the reset operation (refer below).
Re value = 3:
The reset is executed after interrupting cyclic communication. This setting is used to implement a synchronized reset by a control for several drive units.
If the cyclic communication is active for both PROFIdrive interfaces, then the reset is executed after completing both cycle communications.
After communications have been established, check the reset operation (refer below).
To check the reset operation:
After the drive unit has been restarted and communications have been established, read p0972 and check the following:
p0972 = 0? --> The reset was successfully executed.
p0972 > 0? --> The reset was not executed.

r0975[0...10] Drive object identification / DO identification			
All objects	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the identification of the drive object.		
Index:	[0] = Company (Siemens = 42) [1] = Drive object type [2] = Firmware version [3] = Firmware date (year) [4] = Firmware date (day/month) [5] = PROFIdrive drive object, type class [6] = PROFIdrive drive object, sub-type Class 1 [7] = Drive object number [8] = Reserved [9] = Reserved [10] = Firmware patch/hot fix		
Note:	Example: r0975[0] = 42 --> SIEMENS r0975[1] = 11 --> SERVO drive object type r0975[2] = 102 --> first part, firmware version V01.02 (second part, refer to index 10) r0975[3] = 2003 --> year 2003 r0975[4] = 1401 --> 14th of January r0975[5] = 1 --> PROFIdrive drive object, type class r0975[6] = 9 --> PROFIdrive drive object sub-type class 1 r0975[7] = 2 --> drive object number = 2 r0975[8] = 0 (reserved) r0975[9] = 0 (reserved) r0975[10] = 600 --> second part, firmware version (complete version: V01.02.06.00)		

p0976	Reset and load all parameters / Reset load all par		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(30) Data type: Unsigned16 P-Group: Factory settings Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1013	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Resets or downloads all parameters of the drive system.		
Value:	0: Inactive 1: Start reset of all parameters to factory setting 2: Start dnlod of param. saved in non-volatile mem w/ p0977=1 3: Start dnlod of volatile parameters from RAM 10: Start dnlod of param. saved in non-volatile mem w/ p0977=10 11: Start dnlod of param. saved in non-volatile mem w/ p0977=11 12: Start dnlod of param. saved in non-volatile mem w/ p0977=12 20: Start download of Siemens internal setting 20 21: Start download of Siemens internal setting 21 22: Start download of Siemens internal setting 22 23: Start download Siemens internal setting 23 24: Start download Siemens internal setting 24 25: Start download Siemens internal setting 25 26: Start download Siemens internal setting 26 100: Start resetting of all BICO interconnections 1011: Start dnlod of param. saved in volatile mem w/ p0977=1011 1012: Start dnlod of param. saved in volatile mem w/ p0977=1012 1013: Start dnlod of param. saved in volatile mem w/ p0977=1013		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	After all of the parameters have been reset to their factory setting, the system must be commissioned for the first time again. Resetting or loading is realized in the non-volatile memory. Procedure: 1. Set p0009 = 30 (parameter reset). 2. Set p0976 to "required value". The system is rebooted. p0976 is automatically set to 0 and p0009 is automatically set to 1 after this has been carried out.		

p0977**Save all parameters / Save all par**

CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150

Can be changed: U, T**Data type:** Unsigned16**P-Group:** Factory settings**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

1013

Access level: 1**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description:

Saves all parameters of the drive system to the non-volatile memory.

Value:

0: Inactive

1: Save in non-volatile memory - downloaded at POWER ON

10: Save as opt. in non-vol. memory - downloaded w/ p0976=10

11: Save as opt. in non-vol. memory - downloaded w/ p0976=11

12: Save as opt. in non-vol. memory - downloaded w/ p0976=12

20: Save in non-volatile memory as setting 20 (reserved)

21: Save in non-volatile memory as setting 21 (reserved)

22: Save in non-volatile memory as setting 22 (reserved)

23: Save in non-volatile memory as setting 23 (reserved)

24: Save in non-volatile memory as setting 24 (reserved)

25: Save in non-volatile memory as setting 25 (reserved)

26: Save in non-volatile memory as setting 26 (reserved)

1011: Save in volatile memory, downloaded with p0976=1011

1012: Save in volatile memory, downloaded with p0976=1012

1013: Save in volatile memory, downloaded with p0976=1013

Dependency:

Refer to: p0976, p1960, p3845, r3996

Caution:

The Control Unit power supply may only be powered down after data has been saved (i.e. after data save has been started, wait until the parameter again has the value 0).

Notice:

Writing to parameters is inhibited while saving.

The progress while saving is displayed in r3996.

Note:

Parameters saved with p0977 = 10, 11 or 12 can be downloaded again with p0976 = 10, 11 or 12.

p0978[0...24] List of drive objects / List of the DO			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1)	Calculated: -	Access level: 2
	Data type: Unsigned8	Dynamic index: -	Func. diagram: -
	P-Group: Topology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	255	[0] 1
			[1] 0
			[2] 0
			[3] 0
			[4] 0
			[5] 0
			[6] 0
			[7] 0
			[8] 0
			[9] 0
			[10] 0
			[11] 0
			[12] 0
			[13] 0
			[14] 0
			[15] 0
			[16] 0
			[17] 0
			[18] 0
			[19] 0
			[20] 0
			[21] 0
			[22] 0
			[23] 0
			[24] 0
Description:	This parameter is an image of p0101 in conformance with PROFIdrive. Parameters p0101 and p0978 contain the following information: 1) The same number of drive objects 2) The same drive objects In this sense, they are consistent. Difference between p0101 and p0978: p0978 can be re-sorted and a zero inserted in order to identify those drive objects that participate in the process data exchange and to define their sequence in the process data exchange. Drive objects that are listed after the first zero, are excluded from the process data exchange. For p0978, in addition, the value 255 can be inserted a multiple number of times. p0978[n] = 255 means: The drive object is visible for the PROFIBUS master and is empty (without any actual process data exchange). This allows cyclic communications of a PROFIBUS master with unchanged configuring to the drive units with a lower number of drive objects.		
Dependency:	Refer to: p0101, p0971, p0977		
Note:	p0978 cannot be changed when the drive system is first commissioned. The reason for this is that at this time the actual topology has still not been acknowledged (p0099 is still not equal to r0098 and p0009 is set to 0).		

r0979[0...10] PROFIdrive encoder format / PD encoder format

ENCODER, TM41

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 4010, 4704**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the actual position encoder used according to PROFIdrive.

Index:

[0] = Header
 [1] = Type, encoder 1
 [2] = Resolution enc 1
 [3] = Shift factor G1_XIST1
 [4] = Shift factor G1_XIST2
 [5] = Distinguishable revolutions encoder 1
 [6] = Reserved
 [7] = Reserved
 [8] = Reserved
 [9] = Reserved
 [10] = Reserved

Note:

Information about the individual indices can be taken from the following literature:
 PROFIdrive Profile Drive Technology

r0979[0...10] PROFIdrive encoder format / PD encoder format

ENCODER (Lin enc)

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 4010, 4704**P-Group:** Encoder**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the actual position encoder used according to PROFIdrive.

Index:

[0] = Header
 [1] = Type, encoder 1
 [2] = Resolution enc 1
 [3] = Shift factor G1_XIST1
 [4] = Shift factor G1_XIST2
 [5] = Distinguishable distance encoder 1
 [6] = Reserved
 [7] = Reserved
 [8] = Reserved
 [9] = Reserved
 [10] = Reserved

Note:

Information about the individual indices can be taken from the following literature:
 PROFIdrive Profile Drive Technology

r0979[0...30] PROFIdrive encoder format / PD encoder format			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 4010, 4704
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the actual position encoder used according to PROFIdrive.		
Index:	[0] = Header [1] = Type, encoder 1 [2] = Resolution enc 1 [3] = Shift factor G1_XIST1 [4] = Shift factor G1_XIST2 [5] = Distinguishable revolutions encoder 1 [6] = Reserved [7] = Reserved [8] = Reserved [9] = Reserved [10] = Reserved [11] = Type, encoder 2 [12] = Resolution enc 2 [13] = Shift factor G2_XIST1 [14] = Shift factor G2_XIST2 [15] = Distinguishable revolutions encoder 2 [16] = Reserved [17] = Reserved [18] = Reserved [19] = Reserved [20] = Reserved [21] = Type, encoder 3 [22] = Resolution enc 3 [23] = Shift factor G3_XIST1 [24] = Shift factor G3_XIST2 [25] = Distinguishable revolutions encoder 3 [26] = Reserved [27] = Reserved [28] = Reserved [29] = Reserved [30] = Reserved		
Note:	Information about the individual indices can be taken from the following literature: PROFIdrive Profile Drive Technology		

r0979[0...30] PROFIdrive encoder format / PD encoder format

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 4010, 4704
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the actual position encoder used according to PROFIdrive.

Index:

- [0] = Header
- [1] = Type, encoder 1
- [2] = Resolution enc 1
- [3] = Shift factor G1_XIST1
- [4] = Shift factor G1_XIST2
- [5] = Distinguishable distance encoder 1
- [6] = Reserved
- [7] = Reserved
- [8] = Reserved
- [9] = Reserved
- [10] = Reserved
- [11] = Type, encoder 2
- [12] = Resolution enc 2
- [13] = Shift factor G2_XIST1
- [14] = Shift factor G2_XIST2
- [15] = Distinguishable distance encoder 2
- [16] = Reserved
- [17] = Reserved
- [18] = Reserved
- [19] = Reserved
- [20] = Reserved
- [21] = Type, encoder 3
- [22] = Resolution enc 3
- [23] = Shift factor G3_XIST1
- [24] = Shift factor G3_XIST2
- [25] = Distinguishable distance encoder 3
- [26] = Reserved
- [27] = Reserved
- [28] = Reserved
- [29] = Reserved
- [30] = Reserved

Note: Information about the individual indices can be taken from the following literature:
PROFIdrive Profile Drive Technology

r0980[0...299] List of existing parameters 1 / List avail par 1

All objects	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min	Max	Factory setting
	-	-	-

Description: Displays the parameters that exist for this drive.

Dependency: Refer to: r0981, r0989

Note: The existing parameters are displayed in indices 0 to 298. If an index contains the value 0, then the list ends here. In a long list, index 299 contains the parameter number at which position the list continues.

This list consists solely of the following parameters:

r0980[0...299], r0981[0...299] ... r0989[0...299]

The parameters in this list are not displayed in the expert list of the commissioning software. However, they can be read from a higher-level control system (e.g. PROFIBUS master).

r0981[0...299]	List of existing parameters 2 / List avail par 2		
All objects	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the parameters that exist for this drive.		
Dependency:	Refer to: r0980, r0989		
Note:	The existing parameters are displayed in indices 0 to 298. If an index contains the value 0, then the list ends here. In a long list, index 299 contains the parameter number at which position the list continues. This list consists solely of the following parameters: r0980[0...299], r0981[0...299] ... r0989[0...299] The parameters in this list are not displayed in the expert list of the commissioning software. However, they can be read from a higher-level control system (e.g. PROFIBUS master).		
r0989[0...299]	List of existing parameters 10 / List avail par 10		
All objects	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the parameters that exist for this drive.		
Dependency:	Refer to: r0980, r0981		
Note:	The existing parameters are displayed in indices 0 to 298. If an index contains the value 0, then the list ends here. This list consists solely of the following parameters: r0980[0...299], r0981[0...299] ... r0989[0...299] The parameters in this list are not displayed in the expert list of the commissioning software. However, they can be read from a higher-level control system (e.g. PROFIBUS master).		
r0990[0...99]	List of modified parameters 1 / List chang. par 1		
All objects	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays those parameters with a value other than the factory setting for this drive.		
Dependency:	Refer to: r0991, r0999		
Note:	Modified parameters are displayed in indices 0 to 98. If an index contains the value 0, then the list ends here. In a long list, index 99 contains the parameter number at which position the list continues. This list consists solely of the following parameters: r0990[0...99], r0991[0...99] ... r0999[0...99] The parameters in this list are not displayed in the expert list of the commissioning software. However, they can be read from a higher-level control system (e.g. PROFIBUS master).		

r0991[0...99]	List of modified parameters 2 / List chang. par 2		
All objects	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays those parameters with a value other than the factory setting for this drive.		
Dependency:	Refer to: r0990, r0999		
Note:	Modified parameters are displayed in indices 0 to 98. If an index contains the value 0, then the list ends here. In a long list, index 99 contains the parameter number at which position the list continues. This list consists solely of the following parameters: r0990[0...99], r0991[0...99] ... r0999[0...99] The parameters in this list are not displayed in the expert list of the commissioning software. However, they can be read from a higher-level control system (e.g. PROFIBUS master).		
r0999[0...99]	List of modified parameters 10 / List chang. par 10		
All objects	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays those parameters with a value other than the factory setting for this drive.		
Dependency:	Refer to: r0990, r0991		
Note:	Modified parameters are displayed in indices 0 to 98. If an index contains the value 0, then the list ends here. This list consists solely of the following parameters: r0990[0...99], r0991[0...99] ... r0999[0...99] The parameters in this list are not displayed in the expert list of the commissioning software. However, they can be read from a higher-level control system (e.g. PROFIBUS master).		
p1000[0...n]	Macro Connector Inputs (CI) for speed setpoints / Macro CI n_set		
SERVO, VECTOR	Can be changed: C2(1), T Data type: Unsigned32 P-Group: Commands Not for motor type: - Min 0	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max 999999	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Runs the corresponding macro files. The Connector Inputs (CI) for the speed setpoints of the appropriate Command Data Set (CDS) are appropriately interconnected. The selected macro file must be available on the memory card/device memory. Example: p1000 = 6 --> the macro file PM000006.ACX is run.		
Dependency:	Refer to: p0015, p0700, p1500, r8572		
Notice:	No errors were issued during quick commissioning (p3900 = 1) when writing to parameters of the QUICK_IBN group!		
Note:	The macros in the specified directory are displayed in r8572. r8572 is not in the expert list of the commissioning software. Macros available as standard are described in the technical documentation of the particular product. CI: Connector Input		

p1000[0...n]	Macro Connector Inputs (CI) for velocity setpoints / Macro CI v_set		
SERVO (Lin)	Can be changed: C2(1), T Data type: Unsigned32 P-Group: Commands Not for motor type: - Min 0	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max 999999	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Runs the corresponding macro files. The Connector Inputs (CI) for the velocity setpoints of the appropriate Command Data Set (CDS) are appropriately interconnected. The selected macro file must be available on the memory card/device memory. Example: p1000 = 6 --> the macro file PM000006.ACX is run.		
Dependency:	Refer to: p0015, p0700, p1500, r8572		
Notice:	No errors were issued during quick commissioning (p3900 = 1) when writing to parameters of the QUICK_IBN group!		
Note:	The macros in the specified directory are displayed in r8572. r8572 is not in the expert list of the commissioning software. Macros available as standard are described in the technical documentation of the particular product. CI: Connector Input		
p1001[0...n]	CO: Fixed velocity setpoint 1 / n_set_fixed 1		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 1021, 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 1.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1001[0...n]	CO: Fixed speed setpoint 1 / n_set_fixed 1		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 1021, 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 1.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1002[0...n]	CO: Fixed velocity setpoint 2 / n_set_fixed 2		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 2.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1002[0...n]	CO: Fixed speed setpoint 2 / n_set_fixed 2		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 2.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1003[0...n]	CO: Fixed velocity setpoint 3 / n_set_fixed 3		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 3.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1003[0...n]	CO: Fixed speed setpoint 3 / n_set_fixed 3		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 3.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1004[0...n]	CO: Fixed velocity setpoint 4 / n_set_fixed 4		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 4.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1004[0...n]	CO: Fixed speed setpoint 4 / n_set_fixed 4		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 4.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1005[0...n]	CO: Fixed velocity setpoint 5 / n_set_fixed 5		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 5.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1005[0...n]	CO: Fixed speed setpoint 5 / n_set_fixed 5		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 5.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1006[0...n]	CO: Fixed velocity setpoint 6 / n_set_fixed 6		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 6.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1006[0...n]	CO: Fixed speed setpoint 6 / n_set_fixed 6		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 6.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1007[0...n]	CO: Fixed velocity setpoint 7 / n_set_fixed 7		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 7.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1007[0...n]	CO: Fixed speed setpoint 7 / n_set_fixed 7		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 7.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1008[0...n]	CO: Fixed velocity setpoint 8 / n_set_fixed 8		
SERVO (Extended setp., Lin)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3010
	P-Group: Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min -1000.000 [m/min]	Max 1000.000 [m/min]	Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 8.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1008[0...n]	CO: Fixed speed setpoint 8 / n_set_fixed 8		
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3010
	P-Group: Setpoints	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min -210000.000 [rpm]	Max 210000.000 [rpm]	Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 8.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1009[0...n]	CO: Fixed velocity setpoint 9 / n_set_fixed 9		
SERVO (Extended setp., Lin)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3010
	P-Group: Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min -1000.000 [m/min]	Max 1000.000 [m/min]	Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 9.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1009[0...n]	CO: Fixed speed setpoint 9 / n_set_fixed 9		
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3010
	P-Group: Setpoints	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min -210000.000 [rpm]	Max 210000.000 [rpm]	Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 9.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1010[0...n]	CO: Fixed velocity setpoint 10 / n_set_fixed 10		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 10.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1010[0...n]	CO: Fixed speed setpoint 10 / n_set_fixed 10		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 10.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1011[0...n]	CO: Fixed velocity setpoint 11 / n_set_fixed 11		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 11.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1011[0...n]	CO: Fixed speed setpoint 11 / n_set_fixed 11		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 11.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1012[0...n]	CO: Fixed velocity setpoint 12 / n_set_fixed 12		
SERVO (Extended setp., Lin)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3010
	P-Group: Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min -1000.000 [m/min]	Max 1000.000 [m/min]	Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 12.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1012[0...n]	CO: Fixed speed setpoint 12 / n_set_fixed 12		
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3010
	P-Group: Setpoints	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min -210000.000 [rpm]	Max 210000.000 [rpm]	Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 12.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1013[0...n]	CO: Fixed velocity setpoint 13 / n_set_fixed 13		
SERVO (Extended setp., Lin)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3010
	P-Group: Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min -1000.000 [m/min]	Max 1000.000 [m/min]	Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 13.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1013[0...n]	CO: Fixed speed setpoint 13 / n_set_fixed 13		
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3010
	P-Group: Setpoints	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min -210000.000 [rpm]	Max 210000.000 [rpm]	Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 13.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1014[0...n]	CO: Fixed velocity setpoint 14 / n_set_fixed 14		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 14.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1014[0...n]	CO: Fixed speed setpoint 14 / n_set_fixed 14		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 14.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1015[0...n]	CO: Fixed velocity setpoint 15 / n_set_fixed 15		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 2 Func. diagram: 1021, 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets a value for the fixed speed / velocity setpoint 15.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1015[0...n]	CO: Fixed speed setpoint 15 / n_set_fixed 15		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 1021, 3010 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets a value for the fixed speed / velocity setpoint 15.		
Dependency:	Refer to: p1020, p1021, p1022, p1023, r1024, r1197		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p1020[0...n]	BI: Fixed velocity setpoint selection Bit 0 / v_set_fixed Bit 0		
SERVO (Extended setp., Lin)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2505 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select the fixed velocity setpoint.		
Dependency:	Selects the required fixed velocity setpoint using p1020 ... p1023. Displays the number of the actual fixed velocity setpoint in r1197. Sets values for the fixed velocity setpoints 1 ... 15 using p1001 ... p1015. Refer to: p1021, p1022, p1023, r1197		
Note:	If a fixed velocity setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).		
p1020[0...n]	BI: Fixed speed setpoint selection Bit 0 / n_set_fixed Bit 0		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2505 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for selecting the fixed speed setpoint.		
Dependency:	Selects the required fixed speed setpoint using p1020 ... p1023. Displays the number of the actual fixed speed setpoint in r1197. Sets the values for the fixed speed setpoints 1 ... 15 using p1001 ... p1015. Refer to: p1021, p1022, p1023, r1197		
Note:	If a fixed speed setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).		
p1021[0...n]	BI: Fixed velocity setpoint selection Bit 1 / v_set_fixed Bit 1		
SERVO (Extended setp., Lin)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2505 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select the fixed velocity setpoint.		
Dependency:	Selects the required fixed velocity setpoint using p1020 ... p1023. Displays the number of the actual fixed velocity setpoint in r1197. Sets values for the fixed velocity setpoints 1 ... 15 using p1001 ... p1015. Refer to: p1020, p1022, p1023, r1197		
Note:	If a fixed velocity setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).		

p1021[0...n] BI: Fixed speed setpoint selection Bit 1 / n_set_fixed Bit 1

SERVO (Extended setp.), VECTOR

Can be changed: T**Data type:** Unsigned32 / Binary**P-Group:** Setpoints**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** CDS, p0170**Units group:** -**Scaling:** -**Max**

-

Access level: 3**Func. diagram:** 2505**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description: Sets the signal source for selecting the fixed speed setpoint.**Dependency:** Selects the required fixed speed setpoint using p1020 ... p1023.

Displays the number of the actual fixed speed setpoint in r1197.

Sets the values for the fixed speed setpoints 1 ... 15 using p1001 ... p1015.

Refer to: p1020, p1022, p1023, r1197

Note: If a fixed speed setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).**p1022[0...n] BI: Fixed velocity setpoint selection Bit 2 / v_set_fixed Bit 2**

SERVO (Extended setp., Lin)

Can be changed: T**Data type:** Unsigned32 / Binary**P-Group:** Setpoints**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** CDS, p0170**Units group:** -**Scaling:** -**Max**

-

Access level: 3**Func. diagram:** 2505**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description: Sets the signal source to select the fixed velocity setpoint.**Dependency:** Selects the required fixed velocity setpoint using p1020 ... p1023.

Displays the number of the actual fixed velocity setpoint in r1197.

Sets values for the fixed velocity setpoints 1 ... 15 using p1001 ... p1015.

Refer to: p1020, p1021, p1023, r1197

Note: If a fixed velocity setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).**p1022[0...n] BI: Fixed speed setpoint selection Bit 2 / n_set_fixed Bit 2**

SERVO (Extended setp.), VECTOR

Can be changed: T**Data type:** Unsigned32 / Binary**P-Group:** Setpoints**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** CDS, p0170**Units group:** -**Scaling:** -**Max**

-

Access level: 3**Func. diagram:** 2505**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description: Sets the signal source for selecting the fixed speed setpoint.**Dependency:** Selects the required fixed speed setpoint using p1020 ... p1023.

Displays the number of the actual fixed speed setpoint in r1197.

Sets the values for the fixed speed setpoints 1 ... 15 using p1001 ... p1015.

Refer to: p1020, p1021, p1023, r1197

Note: If a fixed speed setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).

p1023[0...n]	BI: Fixed velocity setpoint selection Bit 3 / v_set_fixed Bit 3		
SERVO (Extended setp., Lin)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2505 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select the fixed velocity setpoint.		
Dependency:	Selects the required fixed velocity setpoint using p1020 ... p1023. Displays the number of the actual fixed velocity setpoint in r1197. Sets values for the fixed velocity setpoints 1 ... 15 using p1001 ... p1015. Refer to: p1020, p1021, p1022, r1197		
Note:	If a fixed velocity setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).		
p1023[0...n]	BI: Fixed speed setpoint selection Bit 3 / n_set_fixed Bit 3		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2505 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for selecting the fixed speed setpoint.		
Dependency:	Selects the required fixed speed setpoint using p1020 ... p1023. Displays the number of the actual fixed speed setpoint in r1197. Sets the values for the fixed speed setpoints 1 ... 15 using p1001 ... p1015. Refer to: p1020, p1021, p1022, r1197		
Note:	If a fixed speed setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).		
r1024	CO: Fixed velocity setpoint effective / v_set_fixed eff		
SERVO (Extended setp., Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 1550, 3010 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the selected and effective fixed velocity setpoint. This setpoint is the output value for the fixed velocity setpoints and must be appropriately interconnected (e.g. with the main setpoint).		
Recommend.:	Interconnect the signal with main setpoint (p1070).		
Dependency:	Selects the required fixed velocity setpoint using p1020 ... p1023. Displays the number of the actual fixed velocity setpoint in r1197. Sets values for the fixed velocity setpoints 1 ... 15 using p1001 ... p1015. Refer to: p1070, r1197		
Note:	If a fixed velocity setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).		

r1024	CO: Fixed speed setpoint effective / n_set_fixed eff																												
SERVO (Extended setp.), VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 1550, 3010 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]																										
Description:	Displays the selected and effective fixed speed setpoint. This setpoint is the output value for the fixed speed setpoints and must be appropriately interconnected (e.g. with the main setpoint).																												
Recommend.:	Interconnect the signal with main setpoint (p1070).																												
Dependency:	Selects the required fixed speed setpoint using p1020 ... p1023. Displays the number of the actual fixed speed setpoint in r1197. Sets the values for the fixed speed setpoints 1 ... 15 using p1001 ... p1015. Refer to: p1070, r1197																												
Note:	If a fixed speed setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).																												
p1030[0...n]	Motorized potentiometer configuration / Mop configuration																												
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: Unsigned16 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3020 Unit selection: - Expert list: 1 Factory setting 0110 bin																										
Description:	Sets the configuration for the motorized potentiometer.																												
Bit field:	<table><tr><th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr><tr><td>00</td><td>Data save active</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>01</td><td>Automatic mode, ramp-function generator active</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>02</td><td>Initial rounding-off active</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>03</td><td>Save in NVRAM active</td><td>Yes</td><td>No</td><td>-</td></tr></table>	Bit	Signal name	1 signal	0 signal	FP	00	Data save active	Yes	No	-	01	Automatic mode, ramp-function generator active	Yes	No	-	02	Initial rounding-off active	Yes	No	-	03	Save in NVRAM active	Yes	No	-			
Bit	Signal name	1 signal	0 signal	FP																									
00	Data save active	Yes	No	-																									
01	Automatic mode, ramp-function generator active	Yes	No	-																									
02	Initial rounding-off active	Yes	No	-																									
03	Save in NVRAM active	Yes	No	-																									
Notice:	The following prerequisites must be fulfilled in order to be able to save the setpoint (Bit 03 = 1) in a non-volatile fashion: - Firmware with V2.3 or higher. - Control Unit 320 (CU320) with hardware version C or higher (module with NVRAM).																												
Note:	Re bit 00: 0: The setpoint for the motorized potentiometer is not saved and after ON is entered using p1040. 1: The setpoint for the motorized potentiometer is saved after OFF and after ON set to the saved value. In order to save in a non-volatile fashion, bit 03 should be set to 1. Re bit 01: 0: Without ramp-function generator in the automatic mode (ramp-up/ramp-down time = 0). 1: With ramp-function generator in the automatic mode. For manual operation (0 signal via BI: p1041), the ramp-function generator is always active. Re bit 02: 0: Without initial rounding-off 1: With initial rounding-off. The selected ramp-up/down time is correspondingly exceeded. The initial rounding-off is a sensitive way of specifying small changes (progressive reaction when keys are pressed). The jerk for the initial rounding-off is independent of the ramp-up time and only depends on the selected maximum speed (p1082). It is calculated as follows: $r = 0.01 \% * p1082 [1/s] / 0.13^2 [s^2]$ The jerk acts up until the maximum acceleration is reached ($a_{max} = p1082 [1/s] / p1047 [s]$), and then the drive continues to run linearly with a constant rate of acceleration. The higher the maximum acceleration (the lower that p1047 is), the longer the ramp-up time increases with respect to the set ramp-up time.																												

Re bit 03:

0: Non-volatile data save deactivated.

1: The setpoint for the motorized potentiometer is saved in a non-volatile fashion (for bit 00 = 1).

p1035[0...n]	BI: Motorized potentiometer setpoint raise / Mop raise		
SERVO (Extended setp.), VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2442, 2505, 3020
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to increase the setpoint for the motorized potentiometer		
Dependency:	Refer to: p1036		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
p1035	BI: Zero marks enable / ZM enable		
TM41	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9678
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to enable the zero marks.		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For TM41, this parameter has no function. The zero mark can only be switched in or switched out using p4401.		
p1036[0...n]	BI: Motorized potentiometer lower setpoint / Mop lower		
SERVO (Extended setp.), VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2442, 2505
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to reduce the setpoint for the motorized potentiometer.		
Dependency:	Refer to: p1035		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
p1037[0...n]	Motorized potentiometer maximum velocity / Mop n_max		
SERVO (Extended setp., Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3020
	P-Group: Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-1000.000 [m/min]	1000.000 [m/min]	0.000 [m/min]
Description:	Sets the maximum speed/velocity for the motorized potentiometer.		
Note:	This parameter is automatically pre-assigned in the commissioning phase. The setpoint output from the motorized potentiometer is limited to this value.		

p1037[0...n]	Motorized potentiometer maximum speed / Mop n_max		
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3020
	P-Group: Setpoints	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -210000.000 [rpm]	Max 210000.000 [rpm]	Factory setting 0.000 [rpm]
Description:	Sets the maximum speed/velocity for the motorized potentiometer.		
Note:	This parameter is automatically pre-assigned in the commissioning phase. The setpoint output from the motorized potentiometer is limited to this value.		
p1038[0...n]	Motorized potentiometer minimum velocity / Mop n_min		
SERVO (Extended setp., Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3020
	P-Group: Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -1000.000 [m/min]	Max 1000.000 [m/min]	Factory setting 0.000 [m/min]
Description:	Sets the minimum speed/velocity for the motorized potentiometer.		
Note:	This parameter is automatically pre-assigned in the commissioning phase. The setpoint output from the motorized potentiometer is limited to this value.		
p1038[0...n]	Motorized potentiometer minimum speed / Mop n_min		
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3020
	P-Group: Setpoints	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -210000.000 [rpm]	Max 210000.000 [rpm]	Factory setting 0.000 [rpm]
Description:	Sets the minimum speed/velocity for the motorized potentiometer.		
Note:	This parameter is automatically pre-assigned in the commissioning phase. The setpoint output from the motorized potentiometer is limited to this value.		
p1039[0...n]	BI: Motorized potentiometer inversion / Mop inversion		
SERVO (Extended setp.), VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 3020
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source to invert the minimum speed/velocity or the maximum speed/velocity for the motorized potentiometer.		
Dependency:	Refer to: p1037, p1038		
Note:	The inversion is only active during "motorized potentiometer raise" or "motorized potentiometer lower".		

p1040[0...n]	Motorized potentiometer starting value / Mop start value		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: - Max 1000.000 [m/min]	Access level: 2 Func. diagram: 3020 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets the starting value for the motorized potentiometer. This starting value becomes effective after the drive has been powered up.		
Dependency:	Only effective if p1030.0 = 0. Refer to: p1030		
p1040[0...n]	Motorized potentiometer starting value / Mop start value		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3020 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets the starting value for the motorized potentiometer. This starting value becomes effective after the drive has been powered up.		
Dependency:	Only effective if p1030.0 = 0. Refer to: p1030		
p1041[0...n]	BI: Motorized potentiometer manual/automatic / Mop manual/auto		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3020 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to change over from manual to automatic when using a motorized potentiometer. In the manual mode, the setpoint is changed using two signals - raise and lower. In the automatic mode, the setpoint must be interconnected via a connector input.		
Dependency:	Refer to: p1030, p1035, p1036, p1042		
Note:	The effectiveness of the internal ramp-function generator can be set in automatic mode.		
p1042[0...n]	CI: Motorized potentiometer automatic setpoint / Mop auto setpoint		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3020 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the setpoint of the motorized potentiometer in the automatic mode.		
Dependency:	Refer to: p1041		

p1043[0...n]	BI: Motorized potentiometer accept setpoint / Mop accept set val		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3020 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to accept the setting value for the motorized potentiometer.		
Dependency:	Refer to: p1044		
Note:	The setting value (CI: p1044) becomes effective for a 0/1 edge of the setting command (BI: p1043).		
p1044[0...n]	CI: Motorized potentiometer setting value / Mop set val		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3020 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the setting value for the motorized potentiometer.		
Dependency:	Refer to: p1043		
Note:	The setting value (CI: p1044) becomes effective for a 0/1 edge of the setting command (BI: p1043).		
r1045	CO: Mot. potentiom. velocity setp. in front of ramp-fct. gen. / Mop n_set bef RFG		
SERVO (Extended setp., Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 3020 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Sets the effective setpoint in front of the internal motorized potentiometer ramp-function generator.		
r1045	CO: Mot. potentiometer speed setp. in front of ramp-fct. gen. / Mop n_set bef RFG		
SERVO (Extended setp.), VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 3020 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Sets the effective setpoint in front of the internal motorized potentiometer ramp-function generator.		
p1047[0...n]	Motorized potentiometer ramp-up time / Mop ramp-up time		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000.000 [s]	Access level: 2 Func. diagram: 3020 Unit selection: - Expert list: 1 Factory setting 10.000 [s]
Description:	Sets the ramp-up time for the internal ramp-function generator for the motorized potentiometer. The setpoint is changed from zero up to the speed/velocity limit (p1082) within this time (if no initial rounding-off has been activated).		

Dependency: Refer to: p1030, p1048, p1082

Note: When the initial rounding-off is activated (p1030.2) the ramp-up time is correspondingly extended.

p1048[0...n] Motorized potentiometer ramp-down time / Mop ramp-down time

SERVO (Extended setp.), VECTOR

Can be changed: U, T

Calculated: -

Access level: 2

Data type: FloatingPoint32

Dynamic index: DDS, p0180

Func. diagram: 3020

P-Group: Setpoints

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

0.000 [s]

1000.000 [s]

10.000 [s]

Description: Sets the ramp-down time for the internal ramp-function generator for the motorized potentiometer. The setpoint is changed from the speed/velocity limit (p1082) to zero within this time (if no initial rounding-off has been activated).

Dependency: Refer to: p1030, p1047, p1082

Note: The deceleration time is extended corresponding to the activated initial rounding-off (p1030.2).

r1050 CO: Motor. potentiometer setpoint after the ramp-function generator / Mop setp after RFG

SERVO (Extended setp., Lin)

Can be changed: -

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dynamic index: -

Func. diagram: 1550, 3020

P-Group: Setpoints

Units group: 4_1

Unit selection: p0505

Not for motor type: -

Scaling: p2000

Expert list: 1

Min

Max

Factory setting

- [m/min]

- [m/min]

- [m/min]

Description: Sets the effective setpoint after the internal motorized potentiometer ramp-function generator. This setpoint is the output value of the motorized potentiometer and must be appropriately interconnected onwards (e.g. with the main setpoint).

Recommend.: Interconnect the signal with main setpoint (p1070).

Dependency: Refer to: p1070

Note: For "With ramp-function generator", after an OFF1, OFF2, OFF3 or for a 0 signal via BI: p0852 (inhibit operation, cancel pulses) the ramp-function generator output (r1050) is set to the starting value (configuration via p1030.0).

r1050 CO: Motor. potentiometer setpoint after the ramp-function generator / Mop setp after RFG

SERVO (Extended setp.), VECTOR

Can be changed: -

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dynamic index: -

Func. diagram: 1550, 3020

P-Group: Setpoints

Units group: 3_1

Unit selection: p0505

Not for motor type: -

Scaling: p2000

Expert list: 1

Min

Max

Factory setting

- [rpm]

- [rpm]

- [rpm]

Description: Sets the effective setpoint after the internal motorized potentiometer ramp-function generator. This setpoint is the output value of the motorized potentiometer and must be appropriately interconnected onwards (e.g. with the main setpoint).

Recommend.: Interconnect the signal with main setpoint (p1070).

Dependency: Refer to: p1070

Note: For "With ramp-function generator", after an OFF1, OFF2, OFF3 or for a 0 signal via BI: p0852 (inhibit operation, cancel pulses) the ramp-function generator output (r1050) is set to the starting value (configuration via p1030.0).

p1051[0...n]	CI: Velocity limit RFG positive direction / v_limit RFG pos		
SERVO (Extended setp., Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3050 Unit selection: - Expert list: 1 Factory setting 1083[0]
Description:	Sets the signal source for the velocity limit of the positive direction on the ramp-function generator input.		
p1051[0...n]	CI: Speed limit RFG positive direction of rotation / n_limit RFG pos		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3050 Unit selection: - Expert list: 1 Factory setting 1083[0]
Description:	Sets the signal source for the speed limit of the positive direction on the ramp-function generator input.		
p1052[0...n]	CI: Velocity limit RFG negative direction / v_limit RFG neg		
SERVO (Extended setp., Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3050 Unit selection: - Expert list: 1 Factory setting 1086[0]
Description:	Sets the signal source for the velocity limit of the negative direction on the ramp-function generator input.		
p1052[0...n]	CI: Speed limit RFG negative direction of rotation / n_limit RFG neg		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3050 Unit selection: - Expert list: 1 Factory setting 1086[0]
Description:	Sets the signal source for the speed limit of the negative direction on the ramp-function generator input.		
p1055[0...n]	BI: Jog bit 0 / Jog bit 0		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2501, 3030 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for jog 1.		
Recommend.:	When the setting for this binector input is changed, the motor can only be switched on by means of an appropriate signal change of the source.		
Dependency:	Refer to: p0840, p1058		
Notice:	The drive is enabled for jogging using BI: p1055 or BI: p1056. The command "ON/OFF1" can be issued using BI: p0840 or using BI: p1055/p1056. Only the signal source that was used to power up can also be used to power down again.		

p1056[0...n]	BI: Jog bit 1 / Jog bit 1		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2501, 3030 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for jog 2.		
Recommend.:	When the setting for this binector input is changed, the motor can only be switched on by means of an appropriate signal change of the source.		
Dependency:	Refer to: p0840, p1059		
Notice:	The drive is enabled for jogging using BI: p1055 or BI: p1056. The command "ON/OFF1" can be issued using BI: p0840 or using BI: p1055/p1056. Only the signal source that was used to power up can also be used to power down again.		
p1058[0...n]	Jog 1 velocity setpoint / Jog 1 n_set		
SERVO (Extended setp., Lin)	Can be changed: T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: - Max 1000.000 [m/min]	Access level: 2 Func. diagram: 1550, 3030 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets the speed/velocity for jog 1. Jogging is level-triggered and allows the motor to be incrementally moved.		
Dependency:	Refer to: p1055, p1056		
p1058[0...n]	Jog 1 speed setpoint / Jog 1 n_set		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.000 [rpm]	Access level: 2 Func. diagram: 1550, 3030 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets the speed/velocity for jog 1. Jogging is level-triggered and allows the motor to be incrementally moved.		
Dependency:	Refer to: p1055, p1056		
p1059[0...n]	Jog 2 velocity setpoint / Jog 2 n_set		
SERVO (Extended setp., Lin)	Can be changed: T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: - Max 1000.000 [m/min]	Access level: 2 Func. diagram: 1550, 3030 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets the speed/velocity for jog 2. Jogging is level-triggered and allows the motor to be incrementally moved.		
Dependency:	Refer to: p1055, p1056		

p1059[0...n]	Jog 2 speed setpoint / Jog 2 n_set		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.000 [rpm]	Access level: 2 Func. diagram: 1550, 3030 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets the speed/velocity for jog 2. Jogging is level-triggered and allows the motor to be incrementally moved.		
Dependency:	Refer to: p1055, p1056		
p1063[0...n]	Velocity limit setpoint channel / v_limit setp		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: - Max 1000.000 [m/min]	Access level: 1 Func. diagram: 3040 Unit selection: p0505 Expert list: 1 Factory setting 1000.000 [m/min]
Description:	Sets the speed limit/velocity limit effective in the setpoint channel.		
Dependency:	Refer to: p1082, p1083, p1085, p1086, p1088		
p1063[0...n]	Speed limit setpoint channel / n_limit setp		
SERVO (Extended setp.)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.000 [rpm]	Access level: 1 Func. diagram: 3040 Unit selection: p0505 Expert list: 1 Factory setting 210000.000 [rpm]
Description:	Sets the speed limit/velocity limit effective in the setpoint channel.		
Dependency:	Refer to: p1082, p1083, p1085, p1086, p1088		
p1063[0...n]	Speed limit setpoint channel / n_limit setp		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.000 [rpm]	Access level: 1 Func. diagram: 3040 Unit selection: p0505 Expert list: 1 Factory setting 40000.000 [rpm]
Description:	Sets the speed limit/velocity limit effective in the setpoint channel.		
Dependency:	Refer to: p1082, p1083, p1085, p1086, p1088		
p1070[0...n]	CI: Main setpoint / Main setpoint		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 1550, 3030 Unit selection: - Expert list: 1 Factory setting 1024[0]
Description:	Sets the signal source for the main setpoint. Examples: r1024: Fixed speed setpoint effective r1050: Motor. potentiometer setpoint after the ramp-function generator		

Dependency: Refer to: p1071, r1073, r1078

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p1071[0...n] CI: Main setpoint scaling / Main setp scal

SERVO (Extended setp.), VECTOR

Can be changed: T

Data type: Unsigned32 / FloatingPoint32

P-Group: Setpoints

Not for motor type: -

Min

-

Calculated: -

Dynamic index: CDS, p0170

Units group: -

Scaling: PERCENT

Max

-

Access level: 3

Func. diagram: 1550, 3030

Unit selection: -

Expert list: 1

Factory setting

1

Description: Sets the signal source for scaling the main setpoint.

r1073 CO: Main setpoint effective / Main setpoint eff

SERVO (Extended setp., Lin)

Can be changed: -

Data type: FloatingPoint32

P-Group: Setpoints

Not for motor type: -

Min

- [m/min]

Calculated: -

Dynamic index: -

Units group: 4_1

Scaling: p2000

Max

- [m/min]

Access level: 3

Func. diagram: 3030

Unit selection: p0505

Expert list: 1

Factory setting

- [m/min]

Description: Displays the effective main setpoint. The value shown is the main setpoint after scaling.

r1073 CO: Main setpoint effective / Main setpoint eff

SERVO (Extended setp.), VECTOR

Can be changed: -

Data type: FloatingPoint32

P-Group: Setpoints

Not for motor type: -

Min

- [rpm]

Calculated: -

Dynamic index: -

Units group: 3_1

Scaling: p2000

Max

- [rpm]

Access level: 3

Func. diagram: 3030

Unit selection: p0505

Expert list: 1

Factory setting

- [rpm]

Description: Displays the effective main setpoint. The value shown is the main setpoint after scaling.

p1075[0...n] CI: Supplementary setpoint / Suppl setpoint

SERVO (Extended setp.), VECTOR

Can be changed: T

Data type: Unsigned32 / FloatingPoint32

P-Group: Setpoints

Not for motor type: -

Min

-

Calculated: -

Dynamic index: CDS, p0170

Units group: -

Scaling: p2000

Max

-

Access level: 3

Func. diagram: 1550, 3030

Unit selection: -

Expert list: 1

Factory setting

0

Description: Sets the signal source for the supplementary setpoint.

Dependency: Refer to: p1076, r1077, r1078

p1076[0...n] CI: Supplementary setpoint scaling / Suppl setp scal

SERVO (Extended setp.), VECTOR

Can be changed: T

Data type: Unsigned32 / FloatingPoint32

P-Group: Setpoints

Not for motor type: -

Min

-

Calculated: -

Dynamic index: CDS, p0170

Units group: -

Scaling: PERCENT

Max

-

Access level: 3

Func. diagram: 1550, 3030

Unit selection: -

Expert list: 1

Factory setting

1

Description: Sets the signal source for scaling the supplementary setpoint.

r1077	CO: Supplementary setpoint effective / Suppl setpoint eff		
SERVO (Extended setp., Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 3030 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the effective supplementary setpoint. The value shown is the additional setpoint after scaling.		
r1077	CO: Supplementary setpoint effective / Suppl setpoint eff		
SERVO (Extended setp.), VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 3030 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the effective supplementary setpoint. The value shown is the additional setpoint after scaling.		
r1078	CO: Total setpoint effective / Total setpoint eff		
SERVO (Extended setp., Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 3030 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the total effective setpoint. The value indicates the sum of the effective main setpoint and supplementary setpoint.		
r1078	CO: Total setpoint effective / Total setpoint eff		
SERVO (Extended setp.), VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 3030 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the total effective setpoint. The value indicates the sum of the effective main setpoint and supplementary setpoint.		
p1080[0...n]	Minimum velocity / Minimum speed		
SERVO (Extended setp., Lin)	Can be changed: C2(1), T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: - Max 1000.000 [m/min]	Access level: 1 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets the lowest possible speed/velocity. This value is not undershot in operation.		
Note:	The parameter value applies for both motor directions of rotation. In exceptional cases, the motor can operate below this value (e.g. when reversing).		

p1080[0...n]	Minimum speed / Minimum speed		
SERVO (Extended setp.), VECTOR	Can be changed: C2(1), T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 19500.000 [rpm]	Access level: 1 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets the lowest possible speed/velocity. This value is not undershot in operation.		
Note:	The parameter value applies for both motor directions of rotation. In exceptional cases, the motor can operate below this value (e.g. when reversing).		
p1082[0...n]	Maximum speed / n_Max		
SERVO	Can be changed: C2(1), T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.000 [rpm]	Access level: 1 Func. diagram: 3020, 3050, 3060, 3070, 3095, 5300 Unit selection: p0505 Expert list: 1 Factory setting 1500.000 [rpm]
Description:	Sets the highest possible speed.		
Dependency:	Refer to: p0115, p0322		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	The parameter applies for both motor directions. The parameter has a limiting effect and is the reference quantity for all ramp-up and ramp-down times (e.g. down ramps, ramp-function generator, motor potentiometer). Since the parameter is part of quick commissioning (p0010 = 1), it is defined appropriately when p0310, p0311, and p0322 are changed. The following limits are always effective for p1082: $p1082 \leq p0322$, if $p0322 > 0$ $p1082 \leq 60 / (10.0 * p0115[0] * r0313)$ $p1082 \leq 60 * \text{Maximum power unit pulse frequency} / (5.0 * r0313)$ For the automatic calculation ($p0340 = 1$) the value of the parameter is pre-assigned the maximum motor speed ($p0322$). If $p0322 = 0$, the rated motor speed ($p0311$) is used as default (pre-assignment) value. For induction motors that are not catalog motors ($p0301 = 0$), the synchronous no-load speed is used as default (pre-assignment) value ($p0310 * 60 / r0313$). For synchronous motors, the following additionally applies: In the automatic calculation ($p0340 = 1$), p1082 is limited to speeds for which the rated current of the power unit (S1 continuous operation $r0207[3]$) is not sufficient as field current: $p1082 < p0348 / (1 - r0207 / r0331)$, valid for $r0207[3] < r0331$. On the other hand, an additional limit is effective, which prevents the EMF from exceeding the maximum DC link voltage (see p0643 and p1231). The effective assignment of the motor data set parameter (e.g. p0311) to the drive data set parameter p1082 when pre-assigning should be taken from p0186. p1082 is also available in the quick commissioning ($p0010 = 1$); this means that when exiting via $p3900 > 0$, the value is not changed.		

p1082[0...n]	Maximum velocity / v_max		
SERVO (Lin)	Can be changed: C2(1), T	Calculated: CALC_MOD_ALL	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3020, 3050, 3060, 3070, 3095, 5300
	P-Group: Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [m/min]	Max 1000.000 [m/min]	Factory setting 1000.000 [m/min]
Description:	Sets the highest possible velocity.		
Dependency:	Refer to: p0115, p0322		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	The parameter applies for both motor directions. The parameter has a limiting effect and is the reference quantity for all ramp-up and ramp-down times (e.g. down ramps, ramp-function generator, motor potentiometer). Since the parameter is part of quick commissioning (p0010 = 1), it is defined appropriately when p0310, p0311, and p0322 are changed. The following limits are always effective for p1082: $p1082 \leq p0322, \text{ if } p0322 > 0$ $p1082 \leq 60 / (10.0 * p0115[0] * r0313)$ $p1082 \leq 60 * \text{Maximum power unit pulse frequency} / (5.0 * r0313)$ For the automatic calculation (p0340 = 1) the value of the parameter is pre-assigned the maximum motor speed (p0322). If p0322 = 0, the rated motor speed (p0311) is used as default (pre-assignment) value. For induction motors that are not catalog motors (p0301 = 0), the synchronous no-load speed is used as default (pre-assignment) value (p0310 * 60 / r0313). For synchronous motors, the following additionally applies: In the automatic calculation (p0340 = 1), p1082 is limited to speeds for which the rated current of the power unit (S1 continuous operation r0207[3]) is not sufficient as field current: $p1082 < p0348 / (1 - r0207 / r0331)$, valid for $r0207[3] < r0331$. On the other hand, an additional limit is effective, which prevents the EMF from exceeding the maximum DC link voltage (see p0643 and p1231). The effective assignment of the motor data set parameter (e.g. p0311) to the drive data set parameter p1082 when pre-assigning should be taken from p0186. p1082 is also available in the quick commissioning (p0010 = 1); this means that when exiting via p3900 > 0, the value is not changed.		

p1082[0...n]	Maximum speed / n_Max		
VECTOR	Can be changed: C2(1), T	Calculated: CALC_MOD_ALL	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3020, 3050, 3060, 3070, 3095, 5300
	P-Group: Setpoints	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [rpm]	Max 210000.000 [rpm]	Factory setting 1500.000 [rpm]
Description:	Sets the highest possible speed.		
Dependency:	Refer to: p0115, p0230, r0313, p0322, r0336		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		

Note: The parameter applies for both motor directions. The parameter has a limiting effect and is the reference quantity for all ramp-up and ramp-down times (e.g. down ramps, ramp-function generator, motor potentiometer). Since the parameter is part of quick commissioning (p0010 = 1), it is defined appropriately when p0310, p0311, and p0322 are changed.

The following limits are always effective for p1082:

$$p1082 \leq p0322, \text{ if } p0322 > 0$$

$$p1082 \leq 60 * \text{Minimum} (15 * r0336, 650 \text{ Hz}) / r0313$$

$$p1082 \leq 60 * \text{Maximum power unit pulse frequency} / (k * r0313)$$

$$k = 12 \text{ for vector control } (r0108.2 = 1), k = 6.5 \text{ for V/f control } (r0108.2 = 0)$$

If a sine-wave filter (p0230 = 3) is parameterized as output filter, then the maximum speed is limited corresponding to the maximum permissible filter output frequency (refer to the filter data sheet). For reactors and dv/dt filters, it is limited to 150 Hz / r0313 (for chassis power units) or 120 Hz / r0313 (for booksize power units).

For the automatic calculation (p0340 = 1) the value of the parameter is pre-assigned the maximum motor speed (p0322). If p0322 = 0, the rated motor speed (p0311) is used as default (pre-assignment) value. For induction motors that are not catalog motors (p0301 = 0), the synchronous no-load speed is used as default (pre-assignment) value (p0310 * 60 / r0313).

For synchronous motors, the following additionally applies:

The maximum speed p1082 is restricted to speeds (r1084) where the EMF does not exceed the DC link voltage.

The effective assignment of the motor data set parameter (e.g. p0311) to the drive data set parameter p1082 when pre-assigning should be taken from p0186.

p1082 is also available in the quick commissioning (p0010 = 1); this means that when exiting via p3900 > 0, the value is not changed.

For vector control (p1300 = 20 ... 23) the maximum speed is limited to $60.0 / (8.333 * p0115[0] * r0313)$. This can be identified by a reduction in r1084. p1082 is not changed in this process due to the fact that the operating mode p1300 can be changed over.

When using sine-wave filters (p0230 = 3, 4), the maximum speed r1084 is limited to 70% of the resonant frequency of the filter capacitance and the motor leakage inductance.

p1083[0...n]		CO: Speed limit in positive direction of rotation / n_limit pos		
SERVO	Can be changed:	U, T	Calculated: -	Access level: 2
	Data type:	FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3050, 3095
	P-Group:	Setpoints	Units group: 3_1	Unit selection: p0505
	Not for motor type:	-	Scaling: p2000	Expert list: 1
	Min	0.000 [rpm]	Max	Factory setting
			210000.000 [rpm]	210000.000 [rpm]
Description:	Sets the maximum speed for the positive direction.			
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.			
p1083[0...n]		CO: Velocity limit positive direction / v_limit pos		
SERVO (Lin)	Can be changed:	U, T	Calculated: -	Access level: 2
	Data type:	FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3050, 3095
	P-Group:	Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type:	-	Scaling: p2000	Expert list: 1
	Min	0.000 [m/min]	Max	Factory setting
			1000.000 [m/min]	1000.000 [m/min]
Description:	Sets the maximum velocity for the positive direction.			
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.			

p1083[0...n]	CO: Speed limit in positive direction of rotation / n_limit pos		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 2 Func. diagram: 3050, 6732 Unit selection: p0505 Expert list: 1 Factory setting 40000.000 [rpm]
Description:	Sets the maximum speed for the positive direction.		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
r1084	CO: Speed limit positive effective / n_limit pos eff		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 3050, 3095 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the effective positive speed limit.		
Dependency:	Refer to: p1082, p1083, p1085		
r1084	CO: Velocity limit positive effective / v_limit pos eff		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 3050, 3095 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the effective positive velocity limit.		
Dependency:	Refer to: p1082, p1083, p1085		
p1085[0...n]	CI: Velocity limit positive direction / v_limit pos		
SERVO (Extended setp., Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3050 Unit selection: - Expert list: 1 Factory setting 1083[0]
Description:	Sets the signal source for the velocity limit of the positive direction.		
p1085[0...n]	CI: Speed limit in positive direction of rotation / n_limit pos		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3050 Unit selection: - Expert list: 1 Factory setting 1083[0]
Description:	Sets the signal source for the speed limit of the positive direction.		

p1086[0...n]	CO: Speed limit in negative direction of rotation / n_limit neg		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 0.000 [rpm]	Access level: 2 Func. diagram: 3050, 3095 Unit selection: p0505 Expert list: 1 Factory setting -210000.000 [rpm]
Description:	Sets the speed limit for the negative direction.		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1086[0...n]	CO: Velocity limit negative direction / v_limit neg		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -1000.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 0.000 [m/min]	Access level: 2 Func. diagram: 3050, 3095 Unit selection: p0505 Expert list: 1 Factory setting -1000.000 [m/min]
Description:	Sets the velocity limit for the negative direction.		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1086[0...n]	CO: Speed limit in negative direction of rotation / n_limit neg		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -210000.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 0.000 [rpm]	Access level: 2 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting -40000.000 [rpm]
Description:	Sets the speed limit for the negative direction.		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
r1087	CO: Speed limit negative effective / n_limit neg eff		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 3050, 3095 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the effective negative speed limit.		
Dependency:	Refer to: p1082, p1086, p1088		
r1087	CO: Velocity limit negative effective / v_limit neg eff		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 3050, 3095 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the effective negative velocity limit.		
Dependency:	Refer to: p1082, p1086, p1088		

p1088[0...n]	CI: Velocity limit negative direction / n_limit neg		
SERVO (Extended setp., Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3050 Unit selection: - Expert list: 1 Factory setting 1086[0]
Description:	Sets the signal source for the speed/velocity limit of the negative direction.		
p1088[0...n]	CI: Speed limit in negative direction of rotation / n_limit neg		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3050 Unit selection: - Expert list: 1 Factory setting 1086[0]
Description:	Sets the signal source for the speed/velocity limit of the negative direction.		
p1091[0...n]	Skip velocity 1 / v_skip 1		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets skip velocity 1.		
Dependency:	Refer to: p1092, p1093, p1094, p1101		
Note:	The skip (suppression) velocities can be used to prevent the effects of mechanical resonance.		
p1091[0...n]	Skip speed 1 / n_skip 1		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets skip speed 1.		
Dependency:	Refer to: p1092, p1093, p1094, p1101		
Note:	The skip (suppression) speeds can be used to prevent the effects of mechanical resonance.		
p1092[0...n]	Skip velocity 2 / v_skip 2		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets skip velocity 2.		
Dependency:	Refer to: p1091, p1093, p1094, p1101		

p1092[0...n]	Skip speed 2 / n_skip 2		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets skip speed 2.		
Dependency:	Refer to: p1091, p1093, p1094, p1101		
p1093[0...n]	Skip velocity 3 / v_skip 3		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets skip velocity 3.		
Dependency:	Refer to: p1091, p1092, p1094, p1101		
p1093[0...n]	Skip speed 3 / n_skip 3		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets skip speed 3.		
Dependency:	Refer to: p1091, p1092, p1094, p1101		
p1094[0...n]	Skip velocity 4 / v_skip 4		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets skip velocity 4.		
Dependency:	Refer to: p1091, p1092, p1093, p1101		
p1094[0...n]	Skip speed 4 / n_skip 4		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets skip speed 4.		
Dependency:	Refer to: p1091, p1092, p1093, p1101		

p1101[0...n]	Skip velocity bandwidth / v_skip bandwidth		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [m/min]
Description:	Sets the bandwidth for the skip velocities 1 to 4.		
Dependency:	Refer to: p1091, p1092, p1093, p1094		
Note:	The setpoint velocities are skipped (suppressed) in the range of the skip velocity +/-p1101. Steady-state operation is not possible in the skipped (suppressed) velocity range. The skip (suppression) range is skipped. Example: p1091 = 600 and p1101 = 20 --> setpoint velocities between 580 and 620 [rpm] are skipped. For the skip bandwidths, the following hysteresis behavior applies: For a setpoint velocity coming from below, the following applies: r1170 < 580 [m/min] and 580 [m/min] <= r1114 <= 620 [m/min] --> r1119 = 580 [m/min] For a setpoint velocity coming from above, the following applies: r1170 > 620 [m/min] and 580 [m/min] <= r1114 <= 620 [m/min] --> r1119 = 620 [m/min]		
p1101[0...n]	Skip speed bandwidth / n_skip bandwidth		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [rpm]
Description:	Sets the bandwidth for the skip speeds/velocities 1 to 4.		
Dependency:	Refer to: p1091, p1092, p1093, p1094		
Note:	The setpoint (reference) speeds are skipped (suppressed) in the range of the skip speed +/-p1101. Steady-state operation is not possible in the skipped (suppressed) speed range. The skip (suppression) range is skipped. Example: p1091 = 600 and p1101 = 20 --> setpoint speeds between 580 and 620 [rpm] are skipped. For the skip bandwidths, the following hysteresis behavior applies: For a setpoint speed coming from below, the following applies: r1170 < 580 [rpm] and 580 [rpm] <= r1114 <= 620 [rpm] --> r1119 = 580 [rpm] For a setpoint speed coming from above, the following applies: r1170 > 620 [rpm] and 580 [rpm] <= r1114 <= 620 [rpm] --> r1119 = 620 [rpm]		
p1110[0...n]	BI: Inhibit negative direction / Inhib neg dir		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2505, 3040 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to disable the negative direction.		
Dependency:	Refer to: p1111		

p1111[0...n]	BI: Inhibit positive direction / Inhib pos dir		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2505, 3040 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to disable the positive direction.		
Dependency:	Refer to: p1110		
r1112	CO: Velocity setpoint after minimum limiting / v_set n. min_lim		
SERVO (Extended setp., Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocity setpoint after the minimum limiting.		
Dependency:	Refer to: p1091, p1092, p1093, p1094, p1101		
r1112	CO: Speed setpoint after minimum limiting / n_set n. min_lim		
SERVO (Extended setp.), VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 3050 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed setpoint after the minimum limiting.		
Dependency:	Refer to: p1091, p1092, p1093, p1094, p1101		
p1113[0...n]	BI: Setpoint inversion / Setp inv		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2441, 2442, 2505, 3040 Unit selection: - Expert list: 1 Factory setting 0
Dependency:	Refer to: r1198		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
r1114	CO: Setpoint after the direction limiting / Setp after limit		
SERVO (Extended setp., Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 1550, 3040, 3050 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the speed/velocity setpoint after the changeover and limiting the direction.		

r1114	CO: Setpoint after the direction limiting / Setp after limit		
SERVO (Extended setp.), VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 1550, 3040, 3050 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed/velocity setpoint after the changeover and limiting the direction.		
p1115	Ramp-function generator selection / RFG selection		
SERVO (Extended setp.), VECTOR	Can be changed: T Data type: Integer16 P-Group: Setpoints Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: 1550, 3080 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the ramp-function generator type.		
Value:	0: Basic ramp-function generator 1: Extended ramp-function generator		
Note:	Another ramp-function generator type can only be selected when the motor is at a standstill.		
r1119	CO: Ramp-function generator setpoint at the input / RFG setp at inp		
SERVO (Extended setp., Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 1550, 1690, 3050, 3060, 3070 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the setpoint at the input of the ramp-function generator.		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	The setpoint is influenced by other functions, e.g. skip (suppressed) speeds, minimum and maximum limits.		
r1119	CO: Ramp-function generator setpoint at the input / RFG setp at inp		
SERVO (Extended setp.), VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 1550, 1690, 3050, 3060, 3070 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the setpoint at the input of the ramp-function generator.		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	The setpoint is influenced by other functions, e.g. skip (suppressed) speeds, minimum and maximum limits.		

p1120[0...n]	Ramp-function generator ramp-up time / RFG ramp-up time		
SERVO (Extended setp., Lin)	Can be changed: C2(1), U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 999999.000 [s]	Access level: 1 Func. diagram: 3060, 3070 Unit selection: - Expert list: 1 Factory setting 10.000 [s]
Description:	The drive is accelerated from standstill (setpoint = 0) up to the maximum velocity (p1082) in this time.		
Dependency:	Refer to: p1082		
p1120[0...n]	Ramp-function generator ramp-up time / RFG ramp-up time		
SERVO (Extended setp.)	Can be changed: C2(1), U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 999999.000 [s]	Access level: 1 Func. diagram: 3060, 3070 Unit selection: - Expert list: 1 Factory setting 10.000 [s]
Description:	The ramp-function generator ramps-up the speed setpoint from standstill (setpoint = 0) up to the maximum speed (p1082) in this time.		
Dependency:	Refer to: p1082		
p1120[0...n]	Ramp-function generator ramp-up time / RFG ramp-up time		
VECTOR	Can be changed: C2(1), U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 999999.000 [s]	Access level: 1 Func. diagram: 3060, 3070 Unit selection: - Expert list: 1 Factory setting 10.000 [s]
Description:	The ramp-function generator ramps-up the speed setpoint from standstill (setpoint = 0) up to the maximum speed (p1082) in this time.		
Dependency:	Refer to: p1082		
Note:	The parameter is adapted during the rotating measurement (p1960 > 0). This is the reason that during the rotating measurement, the motor can accelerate faster than was originally parameterized. For V/f control and encoderless vector control (see p1300), ramp-up times of 0 s are not expedient. The setting should be based on the startup times (r0345) of the motor.		
p1121[0...n]	Ramp-function generator ramp-down time / RFG ramp-down time		
SERVO	Can be changed: C2(1), U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 999999.000 [s]	Access level: 1 Func. diagram: 3060, 3070 Unit selection: - Expert list: 1 Factory setting 10.000 [s]
Description:	The ramp-function generator ramps-down the speed setpoint from the maximum speed (p1082) down to standstill (setpoint = 0) in this time. Further, the ramp-down time is always effective for OFF1.		
Dependency:	Refer to: p1082		
Note:	The following applies for SERVO: The ramp-function generator is only available when the function module "extended setpoint channel" is active (r0108.8 = 1).		

p1121[0...n]	Ramp-function generator ramp-down time / RFG ramp-down time		
SERVO (Lin)	Can be changed: C2(1), U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3060, 3070
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 999999.000 [s]	Factory setting 0.000 [s]
Description:	The drive is decelerated from the maximum velocity (p1082) down to standstill (setpoint = 0) in this time.		
Dependency:	Refer to: p1082		
Note:	The following applies for SERVO: The ramp-function generator is only available when the function module "extended setpoint channel" is active (r0108.8 = 1).		
p1121[0...n]	Ramp-function generator ramp-down time / RFG ramp-down time		
VECTOR	Can be changed: C2(1), U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3060, 3070
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 999999.000 [s]	Factory setting 10.000 [s]
Description:	The ramp-function generator ramps-down the speed setpoint from the maximum speed (p1082) down to standstill (setpoint = 0) in this time. Further, the ramp-down time is always effective for OFF1.		
Dependency:	Refer to: p1082		
Note:	For V/f control and encoderless vector control (see p1300), ramp-down times of 0 s are not recommended. The setting should be based on the startup times (r0345) of the motor.		
p1122[0...n]	BI: Bypass ramp-function generator / Bypass RFG		
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2505
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source for bypassing the ramp generator (ramp-up and ramp-down times = 0).		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For VECTOR in encoderless operation, it is not permissible that the ramp-function generator is bypassed.		
p1130[0...n]	Ramp-function generator initial rounding-off time / RFG t_start_round		
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3070
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 30.000 [s]	Factory setting 0.000 [s]
Description:	Sets the initial rounding-off time for the extended ramp generator. The value applies to ramp-up and ramp-down.		
Note:	Rounding-off times avoid an abrupt response and prevent damage to the mechanical system.		

p1131[0...n]	Ramp-function generator final rounding-off time / RFG t_end_delay		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 30.000 [s]	Access level: 2 Func. diagram: 3070 Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the final rounding-off time for the extended ramp generator. The value applies to ramp-up and ramp-down.		
Note:	Rounding-off times avoid an abrupt response and prevent damage to the mechanical system.		
p1134[0...n]	Ramp-function generator rounding-off type / RFG round-off type		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Setpoints Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: 3070 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the smoothed response to the OFF1 command or the reduced setpoint for the extended ramp-function generator.		
Value:	0: Cont. smoothing 1: Discont smoothing		
Dependency:	No effect up to initial rounding-off time (p1130) > 0 s.		
Note:	p1134 = 0 (continuous smoothing) If the setpoint is reduced while ramping-up, initially a final rounding-off is carried out and then the ramp-up completed. During the final rounding-off, the output of the ramp-function generator continues to go in the direction of the previous setpoint (overshoot). After the final rounding-off has been completed, the output goes toward the new setpoint. p1134 = 1 (discontinuous smoothing) If the setpoint is reduced while ramping-up, then the output goes immediately in the direction of the new setpoint. For the setpoint change there is no rounding-off.		
p1135[0...n]	OFF3 ramp-down time / OFF3 t_RD		
SERVO	Can be changed: C2(1), U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 600.000 [s]	Access level: 2 Func. diagram: 3060, 3070 Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the ramp-down time from the maximum speed down to zero speed for the OFF3 command.		
Note:	This time can be exceeded if the DC link voltage reaches its maximum value.		
p1135[0...n]	OFF3 ramp-down time / OFF3 t_RD		
SERVO (Lin)	Can be changed: C2(1), U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 600.000 [s]	Access level: 2 Func. diagram: 3060, 3070 Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the ramp-down time from the maximum velocity down to zero speed for the OFF3 command.		
Note:	This time can be exceeded if the DC link voltage reaches its maximum value.		

p1135[0...n]	OFF3 ramp-down time / OFF3 t_RD		
VECTOR	Can be changed: C2(1), U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3060, 3070
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 600.000 [s]	Factory setting 3.000 [s]
Description:	Sets the ramp-down time from the maximum speed down to zero speed for the OFF3 command.		
Note:	This time can be exceeded if the DC link voltage reaches its maximum value.		
p1136[0...n]	OFF3 initial rounding-off time / RFGOFF3 t_strt_rnd		
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3070
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 30.000 [s]	Factory setting 0.000 [s]
Description:	Sets the initial rounding-off time for OFF3 for the extended ramp generator.		
p1137[0...n]	OFF3 final rounding-off time / RFG OFF3 t_end_del		
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 3070
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 30.000 [s]	Factory setting 0.000 [s]
Description:	Sets the final rounding-off time for OFF3 for the extended ramp generator.		
p1140[0...n]	BI: Ramp-function generator enable / RFG enable		
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2501
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 1
Description:	Sets the signal source for control word 1 bit 4 (operating condition/disable ramp-function generator).		
Dependency:	Refer to: p1141, p1142		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Bit 4 = 0: Inhibit ramp-function generator (the ramp-function generator output is set to zero) Bit 4 = 1: Operating condition (the ramp-function generator can be enabled)		
p1140	BI: Ramp-function generator enable / RFG enable		
TM41	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9678
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 1
Description:	Sets the signal source for control word 1 bit 4 (operating condition/disable ramp-function generator).		
Dependency:	Refer to: p1141, p1142		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		

Note: Bit 4 = 0: Inhibit ramp-function generator (the ramp-function generator output is set to zero)
 Bit 4 = 1: Operating condition (the ramp-function generator can be enabled)
 This parameter has no function in the "SINAMICS" operating mode (p4400 = 1).

p1141[0...n] BI: Continue ramp-function generator / Continue RFG			
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2501
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 1
Description: Sets the signal source for control word 1 bit 5 (continue ramp-function generator/freeze ramp-function generator).			
Dependency: Refer to: p1140, p1142			
Notice: The ramp-function generator is, independent of the state of the signal source, active in the following cases: - OFF1/OFF3. - ramp-function generator output within the suppression bandwidth. - ramp-function generator output below the minimum speed.			
Note: Bit 5 = 0: Freeze ramp-function generator Bit 5 = 1: Continue ramp-function generator			

p1141 BI: Continue ramp-function generator / Continue RFG			
TM41	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9678
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 1
Description: Sets the signal source for control word 1 bit 5 (continue ramp-function generator/freeze ramp-function generator).			
Dependency: Refer to: p1140, p1142			
Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.			
Note: Bit 5 = 0: Freeze ramp-function generator Bit 5 = 1: Continue ramp-function generator This parameter has no function in the "SINAMICS" operating mode (p4400 = 1).			

p1142[0...n] BI: Speed setpoint enable / n_set enable			
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2501
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 1
Description: Sets the signal source for control word 1 bit 6 (enable setpoint/disable setpoint).			
Dependency: Refer to: p1140, p1141			
Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.			
Note: Bit 6 = 0: Inhibit setpoint (the ramp-function generator input is set to zero) Bit 6 = 1: Enable setpoint			

p1142[0...n]	BI: Velocity setpoint enable / v_set enable		
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2501
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source for control word 1 bit 6 (enable setpoint/disable setpoint).		
Dependency:	Refer to: p1140, p1141		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Bit 6 = 0: Inhibit setpoint (the ramp-function generator input is set to zero) Bit 6 = 1: Enable setpoint		
p1142	BI: Speed setpoint enable / n_set enable		
TM41	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9674, 9678
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source for control word 1 bit 6 (enable setpoint/disable setpoint).		
Dependency:	Refer to: p1140, p1141		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Bit 6 = 0: Inhibit setpoint (the ramp-function generator input is set to zero) Bit 6 = 1: Enable setpoint This parameter has no function in the "SINAMICS" operating mode (p4400 = 1).		
p1143[0...n]	BI: Ramp-function generator, accept setting value / Accept RFG set val		
SERVO (Extended setp.), VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 3060, 3070
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for accepting the setting value of the ramp-function generator.		
Dependency:	The signal source for the ramp-function generator setting value is set using parameters. Refer to: p1144		
Note:	0/1 signal: The ramp-function generator output is immediately (without delay) set to the setting value of the ramp-function generator. 1 signal: The setting value of the ramp-function generator is effective. 1/0 signal: The input value of the ramp-function generator is effective. The ramp-function generator output is adapted to the input value using the ramp-up time or the ramp-down time. 0 signal: The input value of the ramp-function generator is effective.		

p1144[0...n]	CI: Ramp-function generator setting value / RFG setting value		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 3060, 3070 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the ramp-function generator setting value.		
Dependency:	The signal source for accepting the setting value is set using parameters. Refer to: p1143		
p1145[0...n]	Ramp-function generator tracking intensity. / RFG track intens		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 50.0	Access level: 3 Func. diagram: 3080 Unit selection: - Expert list: 1 Factory setting 1.3
Description:	Sets the ramp-function generator tracking. The output value of the ramp-function generator is tracked (corrected) corresponding to the maximum possible drive acceleration. The reference value is the deviation at the speed/velocity controller input that is necessary to ensure that the motor accelerates at the torque/force limit.		
Recommend.:	p1145 = 0.0: This value deactivates the ramp-function generator tracking. p1145 = 0.0 ... 1.0: Generally, these values are not practical. They cause the motor to accelerate below its torque limit. The lower the selected value, the greater the margin between the controller and torque limit when accelerating. p1145 > 1.0: The greater the value, the higher the permissible deviation between the speed setpoint and speed actual value.		
Notice:	If ramp-function generator tracking is activated and the ramp time is set too short, this can cause unsteady acceleration. Remedy: - switch off ramp-function generator tracking (p1145 = 0). - increase the ramp-up/ramp-down time (p1120, p1121).		
Note:	In the V/f mode, ramp-function generator tracking is not active. For SERVO with V/f operation, the following applies: The complete ramp-function generator is not active, i.e. ramp-up and ramp-down time = 0.		
p1148[0...n]	Ramp-function gen., tolerance for ramp-up and ramp-down active / RFG tol HL/RL act		
SERVO (Extended setp., Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: - Max 10.000 [m/min]	Access level: 3 Func. diagram: 3060, 3070 Unit selection: p0505 Expert list: 1 Factory setting 0.200 [m/min]
Description:	Sets the tolerance value for the status of the ramp-function generator (ramp-up active, ramp-down active). If the input of the ramp-function generator does not change in comparison to the output by more than the entered tolerance time, then the status bits "ramp-up active" and "ramp-down active" are not influenced.		
Dependency:	Refer to: r1199		

p1148[0...n]	Ramp-function gen., tolerance for ramp-up and ramp-down active / RFG tol HL/RL act		
SERVO (Extended setp.), VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 1000.000 [rpm]	Access level: 3 Func. diagram: 3060, 3070 Unit selection: p0505 Expert list: 1 Factory setting 19.800 [rpm]
Description:	Sets the tolerance value for the status of the ramp-function generator (ramp-up active, ramp-down active). If the input of the ramp-function generator does not change in comparison to the output by more than the entered tolerance time, then the status bits "ramp-up active" and "ramp-down active" are not influenced.		
Dependency:	Refer to: r1199		
r1149	CO: Ramp-function generator, acceleration / RFG acceleration		
SERVO (Extended setp., Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/s ²]	Calculated: - Dynamic index: - Units group: 22_2 Scaling: p2007 Max - [m/s ²]	Access level: 3 Func. diagram: 3060, 3070 Unit selection: p0505 Expert list: 1 Factory setting - [m/s ²]
Description:	Displays the acceleration of the ramp-function generator.		
Dependency:	Refer to: p1145		
r1149	CO: Ramp-function generator, acceleration / RFG acceleration		
SERVO (Extended setp.), VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rev/s ²]	Calculated: - Dynamic index: - Units group: 39_1 Scaling: p2007 Max - [rev/s ²]	Access level: 3 Func. diagram: 3060, 3070 Unit selection: p0505 Expert list: 1 Factory setting - [rev/s ²]
Description:	Displays the acceleration of the ramp-function generator.		
Dependency:	Refer to: p1145		
r1150	CO: Ramp-function generator velocity setpoint at the output / RFG n_set at outp		
SERVO (Extended setp., Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 1550, 3080 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the setpoint at the output of the ramp-function generator.		
r1150	CO: Ramp-function generator speed setpoint at the output / RFG n_set at outp		
SERVO (Extended setp.), VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 1550, 3080 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the setpoint at the output of the ramp-function generator.		

p1151[0...n]	Ramp-function generator configuration / RFG config			
SERVO (Extended setp.), VECTOR	Can be changed: U, T	Calculated: -	Access level: 2	
	Data type: Unsigned16	Dynamic index: DDS, p0180	Func. diagram: 3070	
	P-Group: Setpoints	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0000 bin	
Description:	Sets the configuration for the extended ramp-function generator.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Disable rounding-off at the zero cross-over	Yes	No
Caution:	Re bit 00 = 1: If the ramp-up time is longer than the ramp-down time (p1120 > p1121), then there is an acceleration step at the zero crossover. This can have a negative impact on the mechanical system.			
Note:	Re bit 00 = 1: When the direction change is changed there is no rounding-off before and after the zero crossover.			
p1152	BI: Setpoint 2 enable / Setp 2 enab			
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: T	Calculated: -	Access level: 3	
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2711, 4015	
	P-Group: Commands	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	899.15	
Description:	Sets the signal source for "setpoint 2 enable".			
p1155[0...n]	CI: Speed controller speed setpoint 1 / n_ctrl n_set 1			
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3	
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 1550, 3080, 5030, 6031	
	P-Group: Setpoints	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: p2000	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description:	Sets the signal source for speed setpoint 1 of the speed controller.			
Dependency:	The effectiveness of this setpoint depends on, e.g. STW1.4 and STW1.6. Refer to: r0002, p0840, p0844, p0848, p0852, p0854, r0898, p1140, p1142, p1160, r1170, p1189, p1412, p1414, p1417, p1418			
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.			
p1155[0...n]	CI: Velocity controller, velocity setpoint 1 / v_ctrl n_set 1			
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 3	
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 1550, 3080, 5030, 6031	
	P-Group: Setpoints	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: p2000	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description:	Sets the signal source for velocity setpoint 1 of the velocity controller.			
Dependency:	The effectiveness of this setpoint depends on, e.g. STW1.4 and STW1.6. Refer to: r0002, p0840, p0844, p0848, p0852, p0854, r0898, p1140, p1142, p1160, r1170, p1189, p1412, p1414, p1417, p1418			
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.			

p1155	CI: TM41 incremental encoder emulation speed setpoint 1 / Enc_emulat n_set 1		
TM41	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 9674 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for speed setpoint 1 of the incremental encoder emulation. The speed setpoint is processed corresponding to the sequencer of the TM41.		
Dependency:	The effectiveness of this setpoint depends on control word 1 (STW1). Refer to: r0898		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
p1160[0...n]	CI: Speed controller speed setpoint 2 / n_ctrl n_set 2		
SERVO, VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 1550, 3080 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for speed setpoint 2 of the speed controller.		
Dependency:	Refer to: p1155, r1170		
Note:	For OFF1/OFF3, the ramp-function generator ramp is effective. The ramp-function generator is set (SERVO: to the actual value, VECTOR: To the setpoint (r1170)) and stops the drive corresponding to the ramp-downtime (p1121 or p1135). While stopping via the ramp-function generator, STW1.4 is effective (enable ramp-function generator). When the function module "position control" (r0108.3 = 1) is activated, this connector input is interconnected as follows as standard: CI: p1160 = r2562		
p1160[0...n]	CI: Velocity controller, velocity setpoint 2 / v_ctrl n_set 2		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 1550, 3080 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for velocity setpoint 2 of the velocity controller.		
Dependency:	Refer to: p1155, r1170		
Note:	For OFF1/OFF3, the ramp-function generator ramp is effective. The ramp-function generator is set to the actual value and stops the drive corresponding to the ramp-downtime (p1121 or p1135). While stopping via the ramp-function generator, STW1.4 is effective (enable ramp-function generator). For the function module "position control" (r0108.3 = 1), this connector input is interconnected as follows as standard: CI: p1160 = r2562		

r1169	CO: Speed controller, speed setpoints 1 and 2 / n_ctrl n_set 1/2		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 3080 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed setpoint after the addition of the speed setpoint 1 (p1155) and speed setpoint 2 (p1160).		
Dependency:	Refer to: p1155, p1160		
Note:	The value is only correctly displayed at r0899.2 = 1 (operation enabled).		
r1169	CO: Velocity controller, velocity setpoints 1 and 2 / v_ctrl n_set 1/2		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 3080 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocity setpoint after the addition of the velocity setpoint 1 (p1155) and velocity setpoint 2 (p1160).		
Dependency:	Refer to: p1155, p1160		
Note:	The value is only correctly displayed at r0899.2 = 1 (operation enabled).		
r1170	CO: Speed controller, setpoint sum / n_ctrl setp sum		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 1550, 1590, 1690, 1700, 1750, 3080, 5020, 6030 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed setpoint after selecting the ramp-function generator and adding the speed setpoint 1 (p1155) and speed setpoint 2 (p1160).		
Dependency:	Refer to: r1150, p1155, p1160		
r1170	CO: Velocity controller, setpoint sum / v_ctrl setp sum		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 1550, 1590, 1750, 3080, 5020 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocity setpoint after selecting the ramp-function generator and adding the velocity setpoint 1 (p1155) and velocity setpoint 2 (p1160).		
Dependency:	Refer to: r1150, p1155, p1160		

p1189[0...n]		Speed setpoint configuration / n_ctrl config			
SERVO, VECTOR	Can be changed: U, T		Calculated: -		Access level: 2
	Data type: Unsigned16		Dynamic index: DDS, p0180		Func. diagram: 3080
	P-Group: Setpoints		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		0011 bin
Description:		Sets the configuration for the speed setpoint.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Interpolation ramp-fct gen/speed controller active	Yes	No	3080
	01	Interpol. op-loop ctrl /speed controller active	Yes	No	3080
Note:		Re bit 01: The interpolator is only effective for clock-cycle synchronous PROFIBUS operation and when the master receives a sign-of-life (STW 2.12 ... STW 2.15).			

p1189[0...n]		Velocity setpoint configuration / v_ctrl config			
SERVO (Lin)	Can be changed: U, T		Calculated: -		Access level: 2
	Data type: Unsigned16		Dynamic index: DDS, p0180		Func. diagram: 3080
	P-Group: Setpoints		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		0011 bin
Description:		Sets the configuration for the velocity setpoint.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Interpolation ramp-fct gen/velocity controller active	Yes	No	3080
	01	Interpol. op-loop ctrl /velocity controller active	Yes	No	3080
Note:		Re bit 01: The interpolator is only effective for clock-cycle synchronous PROFIBUS operation and when the master receives a sign-of-life (STW 2.12 ... STW 2.15).			

p1189		Incremental encoder emulation configuration / Enc_emulat config			
TM41	Can be changed: U, T		Calculated: -		Access level: 2
	Data type: Unsigned16		Dynamic index: -		Func. diagram: 9674
	P-Group: Setpoints		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		0010 bin
Description:		Sets the configuration for the incremental encoder emulation.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	01	Interpol. op-loop ctrl /speed controller active	Yes	No	9674
Note:		The parameter is not effective in the SINAMICS operating mode (p4400 = 1). Re bit 01: The interpolator is only effective for clock-cycle synchronous PROFIBUS operation and when the master receives a sign-of-life (STW 2.12 ... STW 2.15).			

p1190			
CI: DSC position deviation XERR / DSC XERR			
SERVO	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 1550, 3090
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the position deviation XERR for DSC (position controller output of the higher-level control).		
Dependency:	Clock cycle synchronous operation must be activated for DSC. The position controller gain factor (KPC), the position deviation (XERR) and the speed setpoint (N_SOLL_B) must be included in the setpoint telegram. At least the encoder interface (Gx_XIST1) must be included in the actual value telegram. The position actual value used for the internal position controller can be selected using p1192. Refer to: p1191, p1192		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	DSC: Dynamic Servo Control		
p1191			
CI: DSC position controller gain KPC / DSC KPC			
SERVO	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 1550, 3090
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the position controller gain KPC for DSC.		
Dependency:	Clock cycle synchronous operation must be activated for DSC. Refer to: p1190		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	DSC: Dynamic Servo Control		
p1192[0...n]			
DSC enc selection / DSC enc selection			
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: 3090
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	1	3	1
Description:	Sets the number of the encoder used for DSC.		
Value:	1: Encoder 1 (motor encoder) 2: Encoder 2 3: Encoder 3		
Note:	DSC: Dynamic Servo Control Value 1 corresponds to encoder 1 (motor encoder); the encoder data set is assigned via p0187. Value 2 corresponds to encoder 2; the encoder data set is assigned via p0188. Value 3 corresponds to encoder 3; the encoder data set is assigned via p0189.		

p1193[0...n]	DSC encoder adaptation factor / DSC encodAdaptFact		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000000.000	Access level: 3 Func. diagram: 3090 Unit selection: - Expert list: 1 Factory setting 1.000
Description:	Sets the factor to adapt the encoder when using either encoder 2 or 3 for DSC. The factor sets the ratio of the pulse difference between the motor encoder and the selected encoder for the same distance moved through. This factor takes into account e.g. gear ratios, differences in the number of encoder pulses.		
Dependency:	Refer to: p1192		
Note:	Example: Encoder 1: Motor encoder with 2048 pulses/revolution, ballscrew with 10 mm/revolution pitch Encoder 2: Linear scale with 20 µm grid division as direct measuring system p1193 = number of pulses, encoder 1 per motor revolution / number of pulses, encoder 2 per motor revolution p1193 = 2048 / (10 mm / 20 µm) = 4.096		
r1197	Fixed velocity setpoint number actual / n_set_fixed No act		
SERVO (Extended setp., Lin)	Can be changed: - Data type: Unsigned32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3010 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of the selected fixed speed/velocity setpoint.		
Dependency:	Refer to: p1020, p1021, p1022, p1023		
Note:	If a fixed speed setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).		
r1197	Fixed speed setpoint number actual / n_set_fixed No act		
SERVO (Extended setp.), VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Setpoints Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3010 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of the selected fixed speed/velocity setpoint.		
Dependency:	Refer to: p1020, p1021, p1022, p1023		
Note:	If a fixed speed setpoint has not been selected (p1020 ... p1023 = 0, r1197 = 0), then r1024 = 0 (setpoint = 0).		

r1198.0...15**CO/BO: Control word setpoint channel / STW setpoint chan**

SERVO (Extended setp.), VECTOR

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1530, 2505**P-Group:** Setpoints**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the control word for the setpoint channel.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Fixed setpoint bit 0	Yes	No	3010
01	Fixed setpoint bit 1	Yes	No	3010
02	Fixed setpoint bit 2	Yes	No	3010
03	Fixed setpoint bit 3	Yes	No	3010
05	Inhibit negative direction	Yes	No	3040
06	Inhibit positive direction	Yes	No	3040
11	Setpoint inversion	Yes	No	3040
13	Motorized potentiometer raise	Yes	No	3020
14	Motorized potentiometer lower	Yes	No	3020
15	Bypass ramp-function generator	Yes	No	3060, 3070

r1199.0...6**CO/BO: Ramp-function generator status word / RFG ZSW**

SERVO (Extended setp.), VECTOR

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1550, 3080, 8010**P-Group:** Setpoints**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

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Description:

Displays the status word for the ramp-function generator (RFG).

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Ramp-up active	Yes	No	-
01	Ramp-down active	Yes	No	-
02	Ramp-function generator active	Yes	No	-
03	Ramp-function generator set	Yes	No	-
04	Ramp-function generator held	Yes	No	-
05	Ramp-function generator tracking active	Yes	No	-
06	Maximum limit active	Yes	No	-

Note:

Re bit 02:

The bit is an OR logic operation - bit 00 and bit 01.

p1200[0...n]	FlyRest oper mode / FlyRest op_mode		
VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: 1690
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0	Max 6	Factory setting 0
Description:	Sets the operating mode for flying restart. The flying restart allows the drive converter to be powered up while the motor is still rotating. In so doing, the drive converter output frequency is changed until the actual motor speed/velocity is found. The motor then accelerates up to the setpoint at the ramp-function generator setting.		
Value:	0: Flying restart inactive 1: Flying restart always active (start in setpoint direction) 2: FlyRestart active after on, fault, OFF2 (start in setp. dir.) 3: FlyRestart active after fault, OFF2 (start in setp. direction) 4: Flying restart always active (start only in setpoint direction) 5: FlyRestart active after on, fault, OFF2 (start only in setp_dir) 6: FlyRestart active after fault, OFF2 (start only in setp. dir.)		
Dependency:	For induction motors, the following applies: A differentiation is made between flying restart for V/f control and for vector control (p1300). Flying restart, V/f control: p1202, p1203, r1204 Flying restart, vector control: p1202, p1203, r1205 For synchronous motors, the following applies: Flying restart is not possible with V/f control or if, in the case of sensorless vector control, a Voltage Sensing Module has not been connected and parameterized. Refer to: p1201 Refer to: F07330, F07331		
Notice:	The "flying restart" function must be used in cases where the motor may still be running (e.g. after a brief line supply interruption) or is being driven by the load. The system might otherwise shut down as a result of overcurrent. It does not make sense to use "flying restart" together with the "motor holding brake function" (p1215 > 0) because then the flying restart will always be realized with the motor stationary.		
Note:	For p1200 = 1, 4, the following applies: Flying restart is active after faults, OFF1, OFF2, OFF3. For p1200 = 2, 5, the following applies: The "power-on" is the first power-on operation after the drive system has been booted. This is practical for motors with a high-inertia load. For p1200 = 1, 2, 3, the following applies: The search is made in both directions. For p1200 = 4, 5, 6, the following applies: The search is only made in the setpoint direction. For operation with encoder, the following applies: p1200 = 1, 4 as well as p1200 = 2, 5 and p1200 = 3, 6 have the same meaning. For V/f control (p1300 < 20), the following applies: The speed can only be sensed for values above approx. 5 % of the rated motor speed. For lower speeds, it is assumed that the motor is at a standstill. If p1200 is changed while commissioning (p0009, p0010 > 0), then it is possible that the old value will no longer be able to be set. The reason for this is that the dynamic limits of p1200 have been changed by a parameter that was set when the drive was commissioned (e.g. p0300).		

p1201[0...n]	BI: Flying restart enable signal source / Fly_res enab S_src		
VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: -
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1

Description: Sets the signal source to enable the "flying restart" function.

Dependency: Refer to: p1200

Note: Withdrawing the enable signal has the same effect as setting p1200 = 0.

p1202[0...n]	FlyRest srch curr / FlyRest I_src		
VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: PEM, REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	10 [%]	400 [%]	100 [%]

Description: Sets the search current for the "flying restart" function.

The value is referred to the motor magnetizing current.

Dependency: Refer to: r0331

Caution: An unfavorable parameter value can result in the motor behaving in an uncontrollable fashion.



Note:

In V/f control mode, the parameter serves as a threshold value for establishing the current at the beginning of the flying restart function. When the threshold value is reached, the prevailing search current is set dependent upon the frequency on the basis of voltage inputs.

Reducing the search current can also improve flying restart performance (if the system moment of inertia is not very high, for example).

p1203[0...n]	Flying restart search rate factor / FlyRst v_Srch Fact		
VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: PEM, REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	10 [%]	4000 [%]	100 [%]

Description: The value influences the rate at which the output frequency is changed during a flying restart . A higher value results in a longer search time.

Caution: An unfavorable parameter value can result in the motor behaving in an uncontrollable fashion.

For vector control, a value that is too low or too high can cause flying restart to become unstable.



Note:

The parameter factory setting is selected so that standard induction motors that are rotating can be found and restarted as quickly as possible (fast flying restart).

With this presetting, if the motor is not found (e.g. for motors that are accelerated as a result of active loads) we recommend that the search rate is reduced (by increasing p1203).

r1204.0...13	CO/BO: Flying restart, V/f control status / FlyRest Vf st			
VECTOR	Can be changed: -		Calculated: -	Access level: 4
	Data type: Unsigned16		Dynamic index: -	Func. diagram: -
	P-Group: Functions		Units group: -	Unit selection: -
	Not for motor type: PEM, REL		Scaling: -	Expert list: 1
	Min	Max	Factory setting	
	-	-	-	
Description:	Displays the status for checking and monitoring flying restart states in the V/f control mode.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Current impressed	Yes	No
	01	No current flow	Yes	No
	02	Voltage input	Yes	No
	03	Voltage reduced	Yes	No
	04	Start ramp-function generator	Yes	No
	05	Wait for execution	Yes	No
	06	Slope filter act	Yes	No
	07	Positive gradient	Yes	No
	08	Current < thresh	Yes	No
	09	Current minimum	Yes	No
	10	Search in the positive direction	Yes	No
	11	Stop after positive direction	Yes	No
	12	Stop after negative direction	Yes	No
	13	No result	Yes	No

r1205.0...15	CO/BO: Flying restart, vector control status / FlyRest vector st				
VECTOR	Can be changed: -		Calculated: -	Access level: 4	
	Data type: Unsigned16		Dynamic index: -	Func. diagram: -	
	P-Group: Functions		Units group: -	Unit selection: -	
	Not for motor type: PEM, REL		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	-	
Description:	Displays the status for checking and monitoring flying restart states in the vector control mode.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Speed adaptation circuit record angle	Yes	No	-
	01	Speed adaptation circuit set gain to 0	Yes	No	-
	02	Isd channel enable	Yes	No	-
	03	Speed control switched out	Yes	No	-
	04	Quadrature arm switched in	Yes	No	-
	05	Special transformation active	Yes	No	-
	06	Speed adaptation circuit set I comp to 0	Yes	No	-
	07	Current control on	Yes	No	-
	08	Isd_set = 0 A	Yes	No	-
	09	Frequency held	Yes	No	-
	10	Search in the positive direction	Yes	No	-
	11	Search Started	Yes	No	-
	12	Current impressed	Yes	No	-
	13	Search interrupted	Yes	No	-
	14	Speed adaptation circuit deviation = 0	Yes	No	-
	15	Speed control activated	Yes	No	-

Note: Bits 0..9: Used by the control in internal sequences during the flying restart.
Depending on the motor type (p0300), the number of active bits differs.
Bits 10..15: Are used to monitor the flying restart sequence.
For PEM, only bits 10, 11 and 15 are supported.

p1206[0...9]	Set the fault number without automatic restart / Fault_No w/o AR		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: U, T Data type: Unsigned16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the faults for which automatic restart should not be effective.		
Dependency:	The setting is only effective for p1210 = 6, 16. Refer to: p1210		
p1207	BI: AR connection following drive object / AR connection DO		
A_INF, B_INF, S_INF	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Modifies the pre-charging monitoring of the infeed. The active automatic restart (AR) of the following drive object can be interconnected using this binector input (BI: p1207 = r1214.2). This means that when the automatic restart is operational, the pre-charging monitoring of the infeed is deactivated and is only reactivated under the following conditions: - the absolute current in the DC link is greater than 2 % of the maximum current (r0209) of the infeed to provide protection against short-circuit in the DC link. - if a Voltage Sensing Module (VSM) is being used, the line supply voltage amplitude is greater than 3 % of the parameterized unit supply voltage (p0210) to protect the pre-charging resistors against continuous filter current when the line supply partially returns.		
Dependency:	Refer to: r0209, p0210, r1214		
p1208[0...1]	BI: AR modification infeed / AR modification		
SERVO, VECTOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to modify the automatic restart (AR). Interconnections between the automatic restart and infeed: With the following interconnection in the mode p1210 = 6, the automatic restart can respond to infeed faults: BI: p1208[0] = r2139.3 With the following interconnection, in the mode p1210 = 4, the automatic restart can respond to line supply failure of the infeed: BI: p1208[1] = r0863.2		
Index:	[0] = Infeed fault [1] = Infeed line supply failure		
Dependency:	Refer to: r0863, r2139		

p1210 Automatic restart, mode / AR mode			
A_INF, B_INF, S_INF	Can be changed: U, T Data type: Integer16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 6	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the automatic restart mode (AR).		
Value:	0: Inhibit automatic restart 1: Acknowledge all faults without restarting 4: Restart after line supply failure w/o additional start attempts 6: Restart after fault with additional start attempts		
Dependency:	The automatic restart requires an active ON command, e.g. that is available at a digital input. If, for p1210 > 1, there is no active ON command, then the automatic restart is interrupted. When using an Advanced Operator Panel (AOP) in the LOCAL Mode, then there is no automatic restart. Refer to: p0840, p0857, p1267 Refer to: F30003		
Danger: 	If the automatic restart is activated (p1210 > 1) if there is an ON command (refer to p0840), the drive is powered up as soon as any fault messages that are present can be acknowledged. This also occurs after the line supply returns or the Control Unit boots if the DC link voltage is again present or the feedback of the line supply infeed (refer to p0864) is again available. This automatic power-up sequence can only be interrupted by withdrawing the ON command.		
Caution:	A change is only accepted and made in the state "initialization" (r1214.0) and "wait for alarm" (r1214.1). For p1210 > 1, the infeed is automatically started.		
Note:	When automatic restart mode is activated, the supply voltage must remain connected (e.g. backed up by UPS). Re p1210 = 1: Faults that are present are automatically acknowledged. If new faults occur after a successful fault acknowledgment, then these are also automatically acknowledged again. A minimum time of p1212 + 1 s must expire between a successful fault acknowledgement and a fault re-occurring if the signal ON/OFF1 (control word 1 bit 0) is at a HIGH signal level. If the signal ON/OFF1 is at a LOW signal level, then the time between a successful fault acknowledgement and a new fault must be at least 1 s. For p1210 = 1, fault F07320 is not generated if the acknowledgement attempt was not successful, for example, due to frequently occurring faults. Re p1210 = 4: An automatic restart is only executed if fault F06200 has occurred. If additional faults are present, then these faults are also acknowledged and when successful, starting continues. If the 24 V Control Unit power supply fails, then this is interpreted as a line supply failure. Re p1210 = 6: An automatic restart is carried out if any fault has occurred.		

p1210 Automatic restart, mode / AR mode			
SERVO	Can be changed: U, T Data type: Integer16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the automatic restart mode (AR).		
Value:	0: Inhibit automatic restart 1: Acknowledge all faults without restarting 4: Restart after line supply failure w/o additional start attempts 6: Restart after fault with additional start attempts 14: Restart after line supply failure following man. acknowledgment 16: Restart after fault following manual acknowledgment		

- Dependency:** The automatic restart requires an active ON command, e.g. that is available at a digital input.
If, for $p1210 > 1$, there is no active ON command, then the automatic restart is interrupted.
When using an Operator Panel in the LOCAL mode, then there is no automatic start.
For $p1210 = 14, 16$, a manual acknowledgement is required for an automatic restart.
Refer to: p0840, p0857, p1267
Refer to: F30003
- Danger:**  If the automatic restart is activated ($p1210 > 1$) if there is an ON command (refer to p0840), the drive is powered up as soon as any fault messages that are present can be acknowledged. This also occurs after the line supply returns or the Control Unit boots if the DC link voltage is again present or the feedback of the line supply infeed (refer to p0864) is again available. This automatic power-up sequence can only be interrupted by withdrawing the ON command.
- Caution:** A change is only accepted and made in the state "initialization" (r1214.0) and "wait for alarm" (r1214.1). When faults are present, therefore, the parameter cannot be changed.
For $p1210 > 1$, the motor is automatically started.
- Note:** When automatic restart mode is activated, the supply voltage must remain connected (e.g. backed up by UPS).
Re $p1210 = 1$:
Faults that are present are automatically acknowledged. If new faults occur after a successful fault acknowledgment, then these are also automatically acknowledged again. For $p1210 = 1$, fault F07320 is not generated if the acknowledgment attempt was unsuccessful, for example, because the monitoring time p1213 index 0 was exceeded.
Re $p1210 = 4$:
An automatic restart is only carried out if fault F30003 occurred at the Motor Module or a high signal is present at binector input p1208[1]. If additional faults are present, then these faults are also acknowledged and when successful, starting continues. If the 24 V Control Unit power supply fails, then this is interpreted as a line supply failure.
Re $p1210 = 6$:
An automatic restart is carried out if any fault has occurred or there is a high signal at binector input p1208[0].
Re $p1210 = 14$:
As for $p1210 = 4$. However, faults that are present must be manually acknowledged.
Re $p1210 = 16$:
As for $p1210 = 6$. However, faults that are present must be manually acknowledged.

p1210		Automatic restart, mode / AR mode	
VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 16	Factory setting 0
Description:	Sets the automatic restart mode (AR).		
Value:	0: Inhibit automatic restart 1: Acknowledge all faults without restarting 4: Restart after line supply failure w/o additional start attempts 6: Restart after fault with additional start attempts 14: Restart after line supply failure following man. acknowledgment 16: Restart after fault following manual acknowledgment		
Recommend.:	For brief line supply failures, the motor shaft may still be rotating when restarting. The "flying restart" function (p1200) might need to be activated to restart while the motor shaft is still rotating.		
Dependency:	The automatic restart requires an active ON command, e.g. that is available at a digital input. If, for p1210 > 1, there is no active ON command, then the automatic restart is interrupted. When using an Operator Panel in the LOCAL mode, then there is no automatic start. For p1210 = 14, 16, a manual acknowledgement is required for an automatic restart. Refer to: p0840, p0857, p1267 Refer to: F30003		

Danger:

If the automatic restart is activated ($p1210 > 1$) if there is an ON command (refer to p0840), the drive is powered up as soon as any fault messages that are present can be acknowledged. This also occurs after the line supply returns or the Control Unit boots if the DC link voltage is again present or the feedback of the line supply infeed (refer to p0864) is again available. This automatic power-up sequence can only be interrupted by withdrawing the ON command.

Caution:

A change is only accepted and made in the state "initialization" (r1214.0) and "wait for alarm" (r1214.1). When faults are present, therefore, the parameter cannot be changed.

For $p1210 > 1$, the motor is automatically started.

Note:

When automatic restart mode is activated, the supply voltage must remain connected (e.g. backed up by UPS).

Re $p1210 = 1$:

Faults that are present are automatically acknowledged. If new faults occur after a successful fault acknowledgment, then these are also automatically acknowledged again. For $p1210 = 1$, fault F07320 is not generated if the acknowledgment attempt was unsuccessful, for example, because the monitoring time p1213 index 0 was exceeded.

Re $p1210 = 4$:

An automatic restart is only carried out if fault F30003 occurred at the Motor Module or a high signal is present at binector input p1208[1]. If additional faults are present, then these faults are also acknowledged and when successful, starting continues. If the 24 V Control Unit power supply fails, then this is interpreted as a line supply failure.

Re $p1210 = 6$:

An automatic restart is carried out if any fault has occurred or there is a high signal at binector input p1208[0].

Re $p1210 = 14$:

As for $p1210 = 4$. However, faults that are present must be manually acknowledged.

Re $p1210 = 16$:

As for $p1210 = 6$. However, faults that are present must be manually acknowledged.

p1211**Automatic restart, start attempts / AR start attempts**

A_INF, B_INF,
S_INF

Can be changed: U, T

Calculated: -

Access level: 3

Data type: Unsigned16

Dynamic index: -

Func. diagram: -

P-Group: Functions

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

0

10

3

Description:

Sets the start attempts of the automatic restart function for $p1210 = 4, 6$.

Dependency:

The setting of this parameter is always effective for $p1210 = 6$. For $p1210 = 4$, the parameter only has an influence if an additional line phase failure (F6200) occurs at the start attempt.

Refer to: p1210, r1214

Refer to: F07320

Caution:

A change is only accepted and made in the state "initialization" (r1214.0) and "wait for alarm" (r1214.1).

Notice:

After fault F07320 occurs, the power-on command must be withdrawn and all of the faults acknowledged so that the automatic restart function is reactivated.

After a complete blackout the start counter always starts with the counter value that applied before the blackout, and decrements this startup attempt by 1. If a further attempt to acknowledge is started by the automatic restart function prior to blackout, e.g. when the CU remains active on blackout longer than the time $p1212 / 2$, the fault counter will already have been decremented by 1. In this case, the fault counter is thus decreased by the value 2.

Note:

A start attempt starts immediately when a fault occurs. The restart attempt is considered to have been completed if the infeed is powered up and an additional delay time of 1 s has expired.

As long as a fault is present, an acknowledge command is generated in the time intervals of $p1212 / 2$. When successfully acknowledged, the start counter is decremented. If, after this, a fault re-occurs before a restart has been completed, then acknowledgement starts again from the beginning.

Fault F07320 is output if, after several faults occur, the number of parameterized start attempts has been reached. After a successful start attempt (i.e. a fault/error has no longer occurred up to the end of the power-up operation) the start counter is again reset to the parameter value after 1 s. If faults re-occur, the parameterized number of start attempts is again available.

At least one start attempt is always carried out.

After a line supply failure, acknowledgement is immediate and when the line supply returns, the system is powered up. If, between successfully acknowledging the line fault and the line supply returning, another fault occurs, then its acknowledgement also causes the start counter to be decremented.

p1211	Automatic restart, start attempts / AR start attempts		
SERVO, VECTOR	Can be changed: U, T Data type: Unsigned16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 3
Description:	Sets the start attempts of the automatic restart function for p1210 = 4, 6.		
Dependency:	Refer to: p1210, r1214 Refer to: F07320		
Caution:	A change is only accepted and made in the state "initialization" (r1214.0) and "wait for alarm" (r1214.1).		
Notice:	After fault F07320 occurs, the power-on command must be withdrawn and all of the faults acknowledged so that the automatic restart function is reactivated. After a complete blackout the start counter always starts with the counter value that applied before the blackout, and decrements this startup attempt by 1. If a further attempt to acknowledge is started by the automatic restart function prior to blackout, e.g. when the CU remains active on blackout longer than the time p1212 / 2, the fault counter will already have been decremented by 1. In this case, the fault counter is thus decreased by the value 2.		
Note:	A start attempt starts immediately when a fault occurs. The start attempt is considered to be completed if the motor was magnetized (r0056.4 = 1) and an additional delay time of 1 s has expired. As long as a fault is present, an acknowledge command is generated in the time intervals of p1212 / 2. When successfully acknowledged, the start counter is decremented. If, after this, a fault re-occurs before a restart has been completed, then acknowledgement starts again from the beginning. Fault F07320 is output if, after several faults occur, the number of parameterized start attempts has been reached. After a successful start attempt, i.e. a fault/error has no longer occurred up to the end of the magnetizing phase, the start counter is again reset to the parameter value after 1 s. If a fault re-occurs - the parameterized number of start attempts is again available. At least one start attempt is always carried out. After a line supply failure, acknowledgement is immediate and when the line supply returns, the system is powered up. If, between successfully acknowledging the line fault and the line supply returning, another fault occurs, then its acknowledgement also causes the start counter to be decremented.		
p1212	Automatic restart, delay time start attempts / AR t_wait start		
A_INF, B_INF, S_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.1 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.0 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.0 [s]
Description:	Sets the delay time up to restart.		
Dependency:	This parameter setting is active for p1210 = 4, 6. For p1210 = 1, the following applies: Faults are only acknowledged, no restart. Refer to: p1210, r1214		
Caution:	A change is only accepted and made in the state "initialization" (r1214.0) and "wait for alarm" (r1214.1).		
Note:	The faults are automatically acknowledged and the system is powered up again after half of the waiting time has expired and after the full waiting time has expired.		

p1212	Automatic restart, delay time start attempts / AR t_wait start		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.1 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.0 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.0 [s]
Description:	Sets the delay time up to restart.		
Dependency:	This parameter setting is active for p1210 = 4, 6. For p1210 = 1, the following applies: Faults are only automatically acknowledged in half of the waiting time, no restart. Refer to: p1210, r1214		
Caution:	A change is only accepted and made in the state "initialization" (r1214.0) and "wait for alarm" (r1214.1).		
Note:	The faults are automatically acknowledged after half of the waiting time has expired and the full waiting time. If the cause of a fault is not removed in the first half of the delay time, then it is no longer possible to acknowledge in the waiting time.		
p1213[0...1]	Automatic restart, monitoring time / AR t_monitoring		
A_INF, B_INF, S_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.0 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000.0 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [s]
Description:	Sets the monitoring time of the automatic restart (AR).		
Index:	[0] = For restart [1] = To reset the fault counter		
Dependency:	Refer to: p1210, r1214		
Caution:	A change is only accepted and made in the state "initialization" (r1214.0) and "wait for alarm" (r1214.1).		
Notice:	After fault F07320 occurs, the power-on command must be withdrawn and all of the faults acknowledged so that the automatic restart function is reactivated.		
Note:	Index 0: The monitoring time starts when the faults are detected. If the automatic acknowledgements are not successful, the monitoring time runs again. If the drive has not restarted at the end of the monitoring time, fault F07320 is signaled. The monitoring is deactivated with p1213 = 0. If p1213 is set to a value which is lower than in p1212, fault F07320 is generated at each restart. If, for p1210 = 1, the time in p1213 is set lower than in p1212, then fault F07320 is also generated at each restart. The monitoring time must be extended if the faults that occur cannot be immediately and successfully acknowledged (e.g. for faults that are permanently present). Index 1: The fault counter (refer to r1214) is only set back to the starting value p1211 if, after successful restart, the time in p1213 index 1 has expired. The delay time is not effective for fault acknowledgement without automatic restart (p1210 = 1). After a power failure (blackout) the delay time only starts after the line supply returns and the Control Unit boots. The fault counter is set to p1211, if F07320 occurred, the power-on command is withdrawn and the fault is acknowledged. The fault counter is immediately updated if the starting value p1211 or the mode p1210 is changed.		

p1213[0...1] Automatic restart, monitoring time / AR t_monitoring					
SERVO, VECTOR	Can be changed: U, T		Calculated: -	Access level: 3	
	Data type: FloatingPoint32		Dynamic index: -	Func. diagram: -	
	P-Group: Functions		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min	Max	Factory setting		
	0.0 [s]	10000.0 [s]	0.0 [s]		
Description:	Sets the monitoring time of the automatic restart (AR).				
Index:	[0] = For restart [1] = To reset the fault counter				
Dependency:	Refer to: p1210, r1214				
Caution:	A change is only accepted and made in the state "initialization" (r1214.0) and "wait for alarm" (r1214.1).				
Notice:	After fault F07320 occurs, the power-on command must be withdrawn and all of the faults acknowledged so that the automatic restart function is reactivated.				
Note:	Index 0: The monitoring time starts when the faults are detected. If the automatic acknowledgements are not successful, the monitoring time runs again. If, after the monitoring time has expired, the drive has still not successfully started again (flying restart and magnetizing of the motor must have been completed: r0056.4 = 1), then fault F07320 is output. The monitoring is deactivated with p1213 = 0. If p1213 is set lower than the sum of p1212, the magnetizing time p0346 and the additional delay time due to the flying restart, then fault F07320 is generated at each restart. If, for p1210 = 1, the time in p1213 is set lower than in p1212, then fault F07320 is also generated at each restart. The monitoring time must be extended if the faults that occur cannot be immediately and successfully acknowledged (e.g. for faults that are permanently present). Index 1: The fault counter (refer to r1214) is only set back to the starting value p1211 if, after successful restart, the time in p1213 index 1 has expired. The delay time is not effective for fault acknowledgement without automatic restart (p1210 = 1). After a power failure (blackout) the delay time only starts after the line supply returns and the Control Unit boots. The fault counter is set to p1211, if F07320 occurred, the power-on command is withdrawn and the fault is acknowledged. The fault counter is immediately updated if the starting value p1211 or the mode p1210 is changed.				
r1214.0...15 CO/BO: Automatic restart, status / AR status					
A_INF, B_INF, S_INF	Can be changed: -		Calculated: -	Access level: 3	
	Data type: Unsigned16		Dynamic index: -	Func. diagram: -	
	P-Group: Functions		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the status of the automatic restart (AR).				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Initialization	Yes	No	-
	01	Wait for alarm	Yes	No	-
	02	Auto restart act	Yes	No	-
	03	Setting the acknowledgement command	Yes	No	-
	04	Acknowledge alarms	Yes	No	-
	05	Restart	Yes	No	-
	06	Delay time running after automatic power-up	Yes	No	-
	07	Fault	Yes	No	-
	10	Effective fault	Yes	No	-
	12	Start count. bit 0	On	Off	-
	13	Start count. bit 1	On	Off	-
	14	Start count. bit 2	On	Off	-
	15	Start count. bit 3	On	Off	-

Note:

Re bit 00:
State to display the single initialization after POWER ON.

Re bit 01:
State in which the automatic restart function waits for faults (initial state).

Re bit 02:
General display that a fault has been identified and that the restart or acknowledgement has been initiated.

Re bit 03:
Displays the acknowledge command within the "acknowledge alarms" state (bit 4 = 1). For bit 5 = 1 or bit 6 = 1, the acknowledge command is continually displayed.

Re bit 04:
State in which the faults that are present are acknowledged. The state is exited again after successful acknowledgement. A change is only made into the next state if it is signaled that a fault is no longer present after an acknowledgement command (bit 3 = 1).

Re bit 05:
State in which the drive is automatically powered up (only for p1210 = 4, 6).

Re bit 06:
State in which the system waits after having been powered up, to the end of the start attempt.
For p1210 = 1, this signal is directly set after the faults have been successfully acknowledged.

Re bit 07:
State which is assumed after a fault occurs within the automatic restart function.

Re bits 12 ... 15:
Actual state of the start counter (binary coded).

r1214.0...15 CO/BO: Automatic restart, status / AR status

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the automatic restart (AR).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Initialization	Yes	No	-
	01	Wait for alarm	Yes	No	-
	02	Auto restart act	Yes	No	-
	03	Setting the acknowledgement command	Yes	No	-
	04	Acknowledge alarms	Yes	No	-
	05	Restart	Yes	No	-
	06	Delay time running after automatic power-up	Yes	No	-
	07	Fault	Yes	No	-
	10	Effective fault	Yes	No	-
	12	Start count. bit 0	On	Off	-
	13	Start count. bit 1	On	Off	-
	14	Start count. bit 2	On	Off	-
	15	Start count. bit 3	On	Off	-

Note:

Re bit 00:
State to display the single initialization after POWER ON.

Re bit 01:
State in which the automatic restart function waits for faults (initial state).

Re bit 02:
General display that a fault has been identified and that the restart or acknowledgement has been initiated.

Re bit 03:
Displays the acknowledge command within the "acknowledge alarms" state (bit 4 = 1). For bit 5 = 1 or bit 6 = 1, the acknowledge command is continually displayed.

Re bit 04:

State in which the faults that are present are acknowledged. The state is exited again after successful acknowledgement. A change is only made into the next state if it is signaled that a fault is no longer present after an acknowledgement command (bit 3 = 1).

Re bit 05:

State in which the drive is automatically powered up (only for p1210 = 4, 6).

Re bit 06:

State in which the system waits after having been powered up, to the end of the start attempt (to the end of the magnetizing process).

For p1210 = 1, this signal is directly set after the faults have been successfully acknowledged.

Re bit 07:

State which is assumed after a fault occurs within the automatic restart function. This is only reset after acknowledging the fault and withdrawing the power-on command.

Re bit 10:

When the automatic restart function is active, r1214 bit 7 is displayed, otherwise the effective fault r2139 bit 3.

Re bits 12 ... 15:

Actual state of the start counter (binary coded).

p1215 Motor holding brake configuration / Brake config			
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: -	Func. diagram: 2701, 2707, 2711
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	3	0
Description:	Sets the holding brake configuration.		
Value:	0: No motor holding brake being used 1: Motor holding brake acc. to sequence control 2: Motor holding brake always open 3: Motor holding brake like sequence control, connection via BICO		
Dependency:	Refer to: p1216, p1217, p1226, p1227, p1228, p1278		
Caution:	For the setting p1215 = 0, if a brake is used, it remains closed. If the motor moves, this will destroy the brake.		
Notice:	If p1215 was set to 1 or if p1215 was set to 3, then when the pulses are suppressed, the brake is closed even if the motor is still rotating. Pulse suppression can either be caused by a 0 signal at p0844, p0845 or p0852 or as a result of a fault with OFF2 response. If this is not desirable (e.g. for a flying restart), then the brake can be kept open using a 1 signal at p0855.		
Note:	If the configuration is set to "no holding brake present" when booting, then the motor holding brake will be automatically identified. If a motor holding brake is detected, the configuration is set to "motor holding brake as for sequence control". If a holding brake integrated in the motor is used, then it is not permissible that p1215 is set to 3. if an external motor holding brake is being used, then p1215 should be set to 3 and r0899.12 should be interconnected as control signal. When the function module "extended brake control" is activated (r0108.14 = 1), r1229.1 should be interconnected as control signal. The parameter can only be set to zero when the pulses are inhibited. The parameterization "no motor holding brake available" and "Safe Brake Control" enabled (p1215 = 0, p9602 = 1, p9802 = 1) is not practical if there is no motor holding brake. The parameterization "motor holding brake the same as sequence control, connection via BICO" and "Safe Brake Control" enabled (p1215 = 3, p9602 = 1, p9802 = 1) is not practical.		

p1216 Motor holding brake, opening time / Brake t_{open}

SERVO, VECTOR

Can be changed: U, T**Calculated:** -**Access level:** 2**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 2701, 2711**P-Group:** Functions**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0 [ms]

10000 [ms]

100 [ms]

Description:

Sets the time to open the motor holding brake.

After controlling the holding brake (opens), the speed/velocity setpoint remains at zero for this time. After this, the speed/velocity setpoint is enabled.

Recommend.:

This time should be set longer than the actual opening time of the brake. This ensures that the drive cannot accelerate when the brake is applied.

Dependency:

Refer to: p1215, p1217

p1217 Motor holding brake closing time / Brake t_{close}

SERVO, VECTOR

Can be changed: U, T**Calculated:** -**Access level:** 2**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 2701, 2711**P-Group:** Functions**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0 [ms]

10000 [ms]

100 [ms]

Description:

Sets the time to apply the motor holding brake.

After OFF1 or OFF3 and the holding brake is controlled (the brake closes), then the drive remains closed-loop controlled for this time stationary with a speed setpoint/velocity setpoint of zero. The pulses are suppressed when the time expires.

Recommend.:

This time should be set longer than the actual closing time of the brake. This ensures that the pulses are only suppressed after the brake has closed.

Dependency:

Refer to: p1215, p1216

Notice:

If the selected closing time is too short with respect to the actual closing time of the brake, then the load can sag. If the closing time is selected to be too long with respect to the actual closing time of the brake, the control works against the brake and therefore reduces its lifetime.

p1218[0...1] BI: Open motor holding brake / Open brakeSERVO (Extended
brk), VECTOR
(Extended brk)**Can be changed:** T**Calculated:** -**Access level:** 2**Data type:** Unsigned32 / Binary**Dynamic index:** -**Func. diagram:** 2707**P-Group:** Functions**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

1

Description:

Sets the signal source for a conditional opening of the motor holding brake.

Dependency:

Refer to: p1215

Note:

[0]: Signal, open brake, AND logic operation, input 1

[1]: Signal, open brake, AND logic operation, input 2

p1219[0...3]	BI: Immediately close motor holding brake / Close brake		
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 2707 Unit selection: - Expert list: 1 Factory setting [0] 0 [1] 0 [2] 0 [3] 1229.9
Description:	Sets the signal source for an unconditional (immediate) closing of the motor holding brake.		
Dependency:	Refer to: p1215, p1275		
Note:	[0]: Signal, immediately close brake, inversion via p1275.0 [1]: Signal, immediately close brake, inversion via p1275.1 [2]: Signal, immediately close brake [3]: Signal, immediately close brake - refer to the factory setting These four signals form an OR logic operation.		
p1220	CI: Open motor holding brake, signal source, threshold / Open brake thresh		
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max -	Access level: 2 Func. diagram: 2707 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the command "open brake".		
Dependency:	Refer to: p1215, p1221, r1229, p1277		
p1221	Open motor holding brake, threshold / Open brake thresh		
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 200.00 [%]	Access level: 2 Func. diagram: 2707 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the threshold value for the command "open brake".		
Dependency:	Refer to: p1220, r1229, p1277		
p1222	BI: Motor holding brake feedback signal brake closed / Brake feedb closed		
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 2711 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the feedback signal "brake closed". For motor holding brakes with feedback signal, the signal "brake closed" can be activated using p1275.5 = 1.		
Dependency:	Refer to: p1223, p1275		

Note: 1 signal: Brake closed.
 When braking with 1 feedback signal, the inverted feedback signal is connected to the BICO input for the second feedback signal (p1223).
 For r1229.5 = 1, OFF/OFF3 are suppressed to prevent the drive accelerating by a load that drives the motor - whereby OFF2 remains effective.

p1223	BI: Motor holding brake feedback signal brake open / Brake feedb open		
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 2711 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the feedback signal "brake open". For motor holding brakes with feedback signal, the signal "brake open" can be activated using p1275.5 = 1.		
Dependency:	Refer to: p1222, p1275		
Note:	1 signal: Brake open. When braking with 1 feedback signal, the inverted feedback signal is connected to the BICO input for the second feedback signal (p1222).		

p1224[0...3]	BI: Close motor holding brake at standstill / Brk close standst		
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 2704 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for close brake at standstill.		
Dependency:	Refer to: p1275		
Note:	[0]: Signal, close brake at standstill, inversion via p1275.2 [1]: Signal, close brake at standstill, inversion via p1275.3 [2]: Signal, close brake at standstill [3]: Signal, close brake at standstill These four signals form an OR logic operation.		

p1225	CI: Standstill detection, threshold value / Standstill thresh		
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2000 Max -	Access level: 2 Func. diagram: 2704 Unit selection: - Expert list: 1 Factory setting 63[0]
Description:	Sets the signal source "threshold value" for the standstill identification.		
Dependency:	Refer to: p1226, p1228, r1229		

p1226[0...n]				Threshold for zero speed detection / n_standst n_thresh			
SERVO, VECTOR	Can be changed: U, T		Calculated: -		Access level: 2		
	Data type: FloatingPoint32		Dynamic index: DDS, p0180		Func. diagram: 2701, 2704		
	P-Group: Functions		Units group: 3_1		Unit selection: p0505		
	Not for motor type: -		Scaling: -		Expert list: 1		
	Min 0.00 [rpm]		Max 210000.00 [rpm]		Factory setting 20.00 [rpm]		
Description:				Sets the speed threshold for the standstill identification. Acts on the actual value and setpoint monitoring. When braking with OFF1 or OFF3, when the threshold is undershot, standstill is identified. The following applies when the brake control is activated: When the threshold is undershot, the brake control is started and the system waits for the brake closing time in p1217. The pulses are then suppressed. if the brake control is not activated, the following applies: When the threshold is undershot, the pulses are suppressed and the drive coasts down.			
Dependency:				Refer to: p1215, p1216, p1217, p1227			
Notice:				For reasons relating to the compatibility to earlier software versions, a parameter value of 0 in indices 1 to 31 is overwritten with the parameter value in index 0 when the Control Unit boots.			
Note:				Standstill is detected if the actual speed drops below the speed threshold in p1226 or if the monitoring time (p1227) - started when speed setpoint <= speed threshold (p1226) - has expired.			
				The actual value sensing is subject to measuring noise. For this reason, standstill cannot be detected if the speed threshold is too low.			

p1226[0...n]				Standstill detection, velocity threshold / v_standst v_thresh			
SERVO (Lin)	Can be changed: U, T		Calculated: -		Access level: 2		
	Data type: FloatingPoint32		Dynamic index: DDS, p0180		Func. diagram: 2701, 2704		
	P-Group: Functions		Units group: 4_1		Unit selection: p0505		
	Not for motor type: -		Scaling: -		Expert list: 1		
	Min 0.00 [m/min]		Max 1000.00 [m/min]		Factory setting 0.20 [m/min]		
Description:				Sets the velocity threshold for the standstill identification. Acts on the actual value and setpoint monitoring. When braking with OFF1 or OFF3, when the threshold is undershot, standstill is identified. The following applies when the brake control is activated: When the threshold is undershot, the brake control is started and the system waits for the brake closing time in p1217. The pulses are then suppressed. if the brake control is not activated, the following applies: When the threshold is undershot, the pulses are suppressed and the drive coasts down.			
Dependency:				Refer to: p1215, p1216, p1217, p1227			
Notice:				For reasons relating to the compatibility to earlier software versions, a parameter value of 0 in indices 1 to 31 is overwritten with the parameter value in index 0 when the Control Unit boots.			
Note:				Standstill is detected if the actual velocity drops below the velocity threshold in p1226 or if the monitoring time (p1227) - started when the velocity setpoint <= velocity threshold (p1226) - has expired.			
				The actual value sensing is subject to measuring noise. For this reason, standstill cannot be detected if the velocity threshold is too low.			

p1227	Zero speed detection monitoring time / n_standst t_monit		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 300.000 [s]	Access level: 2 Func. diagram: 2701, 2704 Unit selection: - Expert list: 1 Factory setting 4.000 [s]
Description:	Sets the monitoring time for the standstill identification. When braking with OFF1 or OFF3, standstill is identified after this time has expired, after the setpoint speed has fallen below p1226 (also refer to p1145). After this, the brake control is started, the system waits for the closing time in p1217 and then the pulses are suppressed.		
Dependency:	Refer to: p1215, p1216, p1217, p1226		
Notice:	For p1145 > 0.0 (RFG tracking) the setpoint is not equal to zero dependent on the selected value. This can therefore cause the monitoring time in p1227 to be exceeded. In this case, for a driven motor, the pulses are not suppressed.		
Note:	Standstill (zero speed) is detected if, during the complete monitoring time (p1227), the speed setpoint falls below the speed threshold (p1226). For p1227 = 300.000 s, the following applies: The monitoring is deactivated. For p1227 = 0.000 s, the following applies: With OFF1 or OFF3 and a ramp-down time = 0, the pulses are immediately suppressed and the motor "coasts" down.		
p1228	Pulse suppression delay time / Pulse suppr t_del		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10.000 [s]	Access level: 2 Func. diagram: 2701, 2704 Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the delay time for pulse suppression. After OFF1 or OFF3 and zero speed detection, the system waits for this time to expire and the pulses are then suppressed.		
Dependency:	Refer to: p1226, p1227		
Note:	Standstill (zero speed) is detected if, during the complete delay time (p1228), the speed actual value falls below the speed threshold (p1226).		

r1229.1...11 CO/BO: Motor holding brake status word / Brake ZSW

SERVO (Extended brk), VECTOR (Extended brk)

Can be changed: -

Calculated: -

Access level: 2

Data type: Unsigned32

Dynamic index: -

Func. diagram: -

P-Group: Functions

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

-

-

-

Description: Displays the status word for the motor holding brake.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	01	Command open brake (continuous signal)	Yes	No	2711
	03	Pulse enable, extended brake control	Yes	No	2711
	04	Brake does not open	Yes	No	2711
	05	Brake does not close	Yes	No	2711
	06	Brake threshold exceeded	Yes	No	2707
	07	Brake threshold undershot	Yes	No	2704
	08	Brake monitoring time expired	Yes	No	2704
	09	Pulse enable request missing/n_ctrl inhibited	Yes	No	2707
	10	Brake OR logic operation result	Yes	No	2707
	11	Brake AND logic operation result	Yes	No	2707

p1230[0...n] BI: Armature short-circuit / DC brake activation / ASC act

SERVO, VECTOR

Can be changed: U, T

Calculated: -

Access level: 1

Data type: Unsigned32 / Binary

Dynamic index: CDS, p0170

Func. diagram: 7014, 7016, 7017

P-Group: Functions

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

-

-

0

Description: Sets the signal source to activate the armature short-circuit or DC brake.

Dependency: Refer to: p1231, p1232, p1233, p1234, p1235, p1236, p1237, r1238, r1239, p1345, p1346

Note: 1 signal: Armature short-circuit/DC brake is deactivated.

0 signal: Armature short-circuit/DC brake is deactivated.

p1231[0...n] Armature short-circuit / DC brake configuration / ASC config

SERVO, VECTOR

Can be changed: U, T

Calculated: -

Access level: 1

Data type: Integer16

Dynamic index: MDS, p0130

Func. diagram: 7014, 7016, 7017

P-Group: Functions

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

0

14

0

Description: Setting to activate the various types for armature short-circuit / DC brake.

Value:

- 0: No function
- 1: External armature short-circuit with contactor feedback signal
- 2: Ext. armature short circuit without contactor feedback signal
- 3: Internal voltage protection
- 4: Internal armature short-circuit / DC brake
- 14: DC brake under starting speed

Dependency: Refer to: p0300, p1230, p1232, p1233, p1234, p1235, p1236, p1237, r1238, r1239, p1345, p1346

Danger:

Re p1231 = 1, 2:

- only short-circuit-proof motors may be used, or suitable resistors must be used to short-circuit the motor

Re p1231 = 3:

- when the internal voltage protection is active, after pulse suppression, all of the motor terminals are at half of the DC link voltage (without an internal voltage protection, the motor terminals are at zero potential)!

- it is only permissible to use motors that are short-circuit proof (p0320 < p0323).

- The Motor Module must be able to conduct 180% short-circuit current (r0320) of the motor (r0209).

- the internal voltage protection cannot be interrupted due to a fault response. If an overcurrent condition occurs during the active, internal voltage protection, then this can destroy the Motor Module and/or the motor.

- if the Motor Module does not support the autonomous, internal voltage protection (r0192.10 = 0), in order to ensure safe, reliable functioning when the line supply fails, an external 24 V power supply (UPS) must be used for the components.

- if the Motor Module does support the autonomous, internal voltage protection (r0192.10 = 1), in order to ensure safe, reliable functioning when the line supply fails, the 24 V power supply for the components must be provided through a Control Supply Module.

- if the internal voltage protection is active, it is not permissible that the motor is driven by the load for a longer period of time (e.g. as a result of loads that move the motor or another coupled motor).

Re p1231 = 4 and synchronous motor:

- when armature short-circuit is active, all of the motor terminals are at half of the DC link potential.

- it is only permissible to use motors that are short-circuit proof (p0320 < p0323).

- The Motor Module must be able to conduct 180% short-circuit current (r0320) of the motor (r0209).

Note:

Re p1231 = 1, 2:

The external armature short circuit can only be selected for synchronous motors (p0300). In this case, control bit BO: r1239.0 must be interconnected (e.g. to a digital input) to control the external contactor.

The external armature short circuit cannot be set as a fault response. It can be triggered via binector input p1230. It is also always activated in the case of pulse suppression.

Re p1231 = 3:

Internal voltage protection (using an internal armature short circuit) can only be selected for synchronous motors (p0300) and Motor Modules in booksize or chassis format. Further, it is not permissible for Safety to be active on blocksize Motor Modules (i.e. p9501 = 0 and p9601 = 0). The internal voltage protection prevents the DC link capacitance from being charged if there is no possibility of regenerating the EMF of a motor operated in the field-weakening mode. The Motor Module must support this function (r0192.9 = 1).

a) If the Motor Module does not support the autonomous, internal armature short-circuit (r0192.10 = 0), the armature short-circuit is activated as soon as the activation criterion is fulfilled (refer below):

b) If the Motor Module supports the autonomous internal voltage protection (r0192.10 = 1), then the Motor Module itself decides - using the DC link voltage - as to whether the short-circuit should be activated. In this case, protection is also provided even if the DRIVE-CLiQ connection between the Control Unit and Motor Module was interrupted. The short circuit is activated if the DC link voltage exceeds 800 V. If the DC link voltage falls below 450 V, then the short-circuit is withdrawn. This therefore ensures that the required input voltage for the Control Supply Module is maintained.

Re p1231 = 4:

The function is activated as soon as the activation criterion is fulfilled.

- the function can be initiated by OFF2

a) For synchronous motors (p0300 = 2xx, 4xx), the internal armature short-circuit is initiated.

- the Motor Module must support this function (r0192.9 = 1).

b) For induction motors (p0300 = 1xx), the DC brake is initiated.

Activation criterion (one of the following criteria is fulfilled):

- 1 signal via binector input p1230 (DC brake activation)

- the drive is not in the state "S4: Operation" or in S5x (refer to function diagram 2610).

- the internal pulse enable is missing (r0046.19 = 0).

Re p1231 = 14:

This DC brake can only be set on induction motors.

The DC brake is triggered if a 1 signal is pending at binector input p1230 during operation and the current speed is below the starting speed p1234. Then, following upstream demagnetization (see p0347), the braking current p1232 is injected for the duration of time set in p1233. This is followed by automatic switch-off. During operation the command for DC braking can be withdrawn at any time.

DC braking by means of fault response continues to be possible.

Re p1231 = 3, 4, 14:

The value can only be changed to values not equal to 3 or 4 or 14 if p0491 is not equal to 4 and p2101 is not equal to 6 (armature short-circuit/DC brake not set).

In the case of SERVO, the DC brake does not function in V/f operation (p1317).

Note:

ASC: Armature short circuit

IVP: Internal Voltage Protection

UPS: Uninterruptible Power Supply

CSM: Control Supply Module

DC brake

p1232[0...n]		DC braking, braking current / DCBRK I_brake		
SERVO		Can be changed: U, T	Calculated: -	Access level: 1
		Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 7017
		P-Group: Functions	Units group: -	Unit selection: -
		Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
		Min 0.00 [Arms]	Max 10000.00 [Arms]	Factory setting 0.00 [Arms]
Description:		Sets the braking current for DC braking.		
Dependency:		Refer to: p1230, p1231, p1233, p1234, r1239, p1345, p1346		
Note:		A change to the braking current becomes effective the next time that the DC brake is powered up. The value for p1232 is specified as an rms value in the 3-phase system. The magnitude of the braking current is the same as that of an identical output current at frequency zero (see r0067, r0068, p0640).		
p1232[0...n]		DC braking, braking current / DCBRK I_brake		
VECTOR		Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 1
		Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 7017
		P-Group: Functions	Units group: -	Unit selection: -
		Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
		Min 0.00 [Arms]	Max 10000.00 [Arms]	Factory setting 0.00 [Arms]
Description:		Sets the braking current for DC braking.		
Dependency:		Refer to: p1230, p1231, p1233, p1234, r1239, p1345, p1346		
Note:		A change to the braking current becomes effective the next time that the DC brake is powered up. The value for p1232 is specified as an rms value in the 3-phase system. The magnitude of the braking current is the same as that of an identical output current at frequency zero (see r0067, r0068, p0640). For the current controller, the settings of parameters p1345 and p1346 (I_max limiting controller) are used.		
p1233[0...n]		DC braking time / DCBRK time		
SERVO, VECTOR		Can be changed: U, T	Calculated: -	Access level: 1
		Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: 7017
		P-Group: Functions	Units group: -	Unit selection: -
		Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
		Min 0.0 [s]	Max 3600.0 [s]	Factory setting 1.0 [s]
Description:		Sets the DC braking time (as fault response).		
Dependency:		Refer to: p1230, p1231, p1232, p1234, r1239		
Note:		If a speed encoder is being used, DC braking is ended as soon as the drive falls below the standstill threshold p1226.		

p1234[0...n]	Speed at the start of DC braking / DCBRK n_start		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: PEM, REL, FEM Min 0.00 [rpm]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 210000.00 [rpm]	Access level: 1 Func. diagram: 7017 Unit selection: - Expert list: 1 Factory setting 210000.00 [rpm]
Description:	Sets the starting speed for DC braking. If the actual speed falls below this threshold, then DC braking is activated.		
Dependency:	Refer to: p1230, p1231, p1232, p1233, r1239		
Caution:	If an encoder fault occurs during closed-loop operation with encoder, controlled deceleration of the drive down to the start speed p1234 of the DC current brake. In this case, the DC brake is activated immediately and injects the braking current p1232 for the braking current time p1233 after demagnetization. The braking current and braking duration must, therefore, be dimensioned accordingly for this situation so that the drive can be decelerated to standstill.		

p1234[0...n]	DC braking, starting velocity / DCBRK v_start		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: PEM, REL, FEM Min 0.00 [m/min]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 1000.00 [m/min]	Access level: 1 Func. diagram: 7017 Unit selection: - Expert list: 1 Factory setting 1000.00 [m/min]
Description:	Sets the starting speed for DC braking. If the actual velocity falls below this threshold, then DC braking is activated.		
Dependency:	Refer to: p1230, p1231, p1232, p1233, r1239		
Caution:	If an encoder fault occurs during closed-loop operation with encoder, controlled deceleration of the drive down to the start speed p1234 of the DC current brake. In this case, the DC brake is activated immediately and injects the braking current p1232 for the braking current time p1233 after demagnetization. The braking current and braking duration must, therefore, be dimensioned accordingly for this situation so that the drive can be decelerated to standstill.		

p1234[0...n]	Speed at the start of DC braking / DCBRK n_start		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: PEM, REL, FEM Min 0.00 [rpm]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 210000.00 [rpm]	Access level: 1 Func. diagram: 7017 Unit selection: - Expert list: 1 Factory setting 40000.00 [rpm]
Description:	Sets the starting speed for DC braking. If the actual speed falls below this threshold, then DC braking is activated.		
Dependency:	Refer to: p1230, p1231, p1232, p1233, r1239		
Caution:	If an encoder fault occurs during closed-loop operation with encoder, controlled deceleration of the drive down to the start speed p1234 of the DC current brake. In this case, the DC brake is activated immediately and injects the braking current p1232 for the braking current time p1233 after demagnetization. The braking current and braking duration must, therefore, be dimensioned accordingly for this situation so that the drive can be decelerated to standstill.		

p1235[0...n]	BI: External armature short-circuit, contactor feedback signal / ASC ext feedback		
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: -
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the contactor feedback signal for external armature short-circuit.		
Dependency:	Refer to: p1230, p1231, p1236, p1237, r1239		
Notice:	In order that the pulses are not enabled when the contactor is closed, the contactor feedback signal must lag by a sufficiently long time when opening the contactor.		
Note:	1 signal: The contactor is closed. 0 signal: The contactor is open.		
p1236[0...n]	Ext. armature short-cct., contactor feedback signal monit. time / ASC ext t_monit		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0 [ms]	1000 [ms]	200 [ms]
Description:	Sets the monitoring time of the contactor feedback signal for the external armature short-circuit configuration. If the contactor feedback signal (p1235) is parameterized, then the appropriate feedback signal (r1239.1) is expected within this monitoring time after either opening or closing the contactor.		
Dependency:	Refer to: p1230, p1231, p1235, p1237, r1239 Refer to: A07904, F07905		
p1237[0...n]	External armature short-circuit, waiting time when opening / ASC ext t_wait		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0 [ms]	1000 [ms]	200 [ms]
Description:	Sets the delay time when opening the contactor of the external armature short-circuit. If no contactor feedback signal has been selected (p1235), then the system waits for this time before the pulses are switched in.		
Dependency:	Refer to: p1230, p1231, p1235, p1236, r1239		
Notice:	This delay time must be at least long enough so that the contactor contacts reliably open before the pulses are switched in. The delay time must be greater than the contactor response time. The Motor Module can be damaged if the delay time is too short.		

r1238	CO: Armature short-circuit, external state / EASC state		
SERVO, VECTOR	Can be changed: - Data type: Integer16 P-Group: Functions Not for motor type: ASM Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 6	Access level: 1 Func. diagram: 2610 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the state for the external armature short-circuit.		
Value:	0: Powered down 1: Ready 2: Active 3: Active - feedback signal "Closed" OK 4: Active - feedback signal "Closed" missing 5: Prompt to remove the armature short-circuit 6: Active - feedback signal "Open" missing		
Dependency:	Refer to: p1230, p1231, p1235, p1236, p1237, r1239 Refer to: A07904, F07905		
Note:	Activation criterion (one of the following criteria is fulfilled): - the signal at BI: p1230 (armature short-circuit activation) is 0. - the drive is not in the state "S4: Operation" or in S5x (refer to function diagram 2610). - the internal pulse enable is missing (r0046.19 = 0). Re state "switched out" (r1238 = 0): - the external armature short-circuit can be selected with p1231 = 1. Re state "ready" (r1238 = 1): - as soon as the activation criterion is fulfilled, then a transition is made into the state "active" (r1238 = 2). Regarding the state "active" (r1238 = 2), "active - feedback signal "Closed" OK" (r1238 = 3), "active - feedback signal "Closed" missing" (r1238 = 4): - the control signal to close contactor r1239.0 is set to "1" (closed) and the pulses are suppressed. - if a contactor feedback signal is not connected (BI: p1235 = 0 signal), then a transition is immediately made into state 3. - if a contactor feedback signal is connected, then a transition is made into state 3 if the feedback signal at BI: p1235 goes to "1" (closed) within the monitoring time (p1236). - otherwise, a transition is made into state 4. Re state "prompt to remove the armature short-circuit" (r1238 = 5): - the activation criterion is no longer fulfilled. An attempt is made to again remove the armature short circuit. - the control signal to close the contactor r1239.0 is set to "0" (open) and the pulses remain suppressed. - if a contactor feedback signal is not connected (BI: p1235 = 0 signal), the system waits for the delay time (p1237) to expire until a transition is made into state 1. - if a contactor feedback signal is connected, the system waits until the feedback signal at BI: p1235 goes to "0" (open) until a transition is made into state 1. If this does not occur within the monitoring time (p1236), then a transition is made into state 6. Re state "active - feedback signal "Open" missing" (r1238 = 6): - this error state can be exited by deselecting the external armature short-circuit (p1231 = 0).		

r1239.0...11 CO/BO: Armature short-circuit / DC brake status word / ASC/DCBRK ZSW

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word for armature short-circuit.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	External armature short-circuit	Active	Inactive	-
	01	External armature short-circuit, contactor feedback signal	Closed	Open	-
	02	External armature short-circuit ready	Yes	No	-
	03	External armature short-circuit with contactor feedback signal	Yes	No	-
	04	Internal armature short-circuit	Active	Inactive	-
	05	Internal armature short circuit, feedback signal from power unit	Active	Inactive	-
	06	Internal armature short-circuit ready	Yes	No	-
	08	DC brake active	Yes	No	7017
	10	DC brake ready	Yes	No	7017
	11	Armature short circuit/DC brake selected	Yes	No	-

Dependency: Refer to: p1230, p1231, p1232, p1233, p1234, p1235, p1236, p1237

Note: External armature short-circuit (bits 0 ... 3):

Re bit 00:

Using this signal, the motor is short-circuited through an external contactor circuit. This means that this BO: p1239.0 must be interconnected e.g. to a digital output.

Re bit 01:

This signal indicates the state of the contactor to establish the armature short-circuit. To do this, BI: p1235 must be interconnected to a digital input.

Re bit 02:

The external armature short-circuit configuration is ready and is activated as soon as the activation criterion is fulfilled.

Re bit 03:

1: A feedback signal from an external contactor was parameterized in BI: p1235.

Internal voltage protection / internal armature short-circuit (bits 4 ... 6):

Re bit 04:

a) Internal voltage protection (p1231 = 3) was selected and the Motor Module does not support the autonomous internal voltage protection (r0192.10 = 0).

The Control Unit issues the command to the Motor Module to short-circuit the motor through the power semiconductors.

a) Internal voltage protection (p1231 = 3) was selected and the Motor Module supports the autonomous internal voltage protection (r0192.10 = 1).

The Motor Module decides autonomously whether the armature short-circuit is activated. In this case, the following applies: r1239.4 = r1239.5.

c) Internal armature short-circuit (p1231 = 4) was selected.

The Control Unit issues the command to the Motor Module to short-circuit the motor through the power semiconductors.

Re bit 05:

The Motor Module signals that the motor is short-circuited in the Motor Module through the power semiconductors.

Re bit 06:

a) Internal voltage protection (p1231 = 3) was selected and the Motor Module does not support the autonomous internal voltage protection (r0192.10 = 0).

The internal voltage protection is ready and is activated as soon as the activation criterion is fulfilled.

a) Internal voltage protection (p1231 = 3) was selected and the Motor Module supports the autonomous internal voltage protection (r0192.10 = 1).

The internal voltage protection is ready and the Motor Module decides autonomously - using the DC link voltage - whether the short-circuit is activated. In this case, protection is also provided even if the DRIVE-CLiQ connection between the Control Unit and Motor Module was interrupted. The short-circuit is activated if the DC link voltage exceeds 800 V. If the DC link voltage falls below 450 V, then the short-circuit is withdrawn.

c) Internal armature short-circuit (p1231 = 4) was selected.

The internal armature short-circuit is ready and is activated as soon as the activation criterion is fulfilled.

Activation criterion (one of the following criteria is fulfilled):

- the signal at BI: p1230 (armature short-circuit activation) is 1.
- the drive is not in the state "S4: Operation" or in S5x (refer to function diagram 2610).
- the internal pulse enable is missing (r0046.19 = 0).

p1240[0...n]		Vdc controller or Vdc monitoring configuration / Vdc_ctrl config	
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: 5650
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0	Max 9	Factory setting 0
Description:	Sets the configuration of the controller or monitoring for the DC link voltage (Vdc).		
Value:	0: Inhib Vdc ctrl 1: Vdc_max controller enable 2: Vdc_min controller (kinetic buffering) enable 3: Vdc_min controller and Vdc_max controller enable 4: Activates Vdc_max monitoring 5: Activates Vdc_min monitoring 6: Activates Vdc_min monitoring and Vdc_max monitoring 7: Vdc_max controller without accelerating enable 8: Vdc_min controller without braking enable 9: Vdc_min and Vdc_max controller w/o braking/accelerating enable		
Dependency:	Refer to: p1244, p1248, p1250, p1532		
Notice:	During a few steps of the rotating measurement (p1960 = 1) the Vdc_min controller and/or Vdc_max controller is disabled.		

Note: p1240 = 1, 3:
When the upper DC link voltage threshold is reached (p1244), then the following applies:
- the Vdc_max controller limits the regenerative energy in order that the DC link voltage is kept below the maximum DC link voltage when braking.
- when other drives regenerate into the DC link, then the Vdc_max controller causes the motor to accelerate.

p1240 = 2, 3:
When the lower DC link voltage threshold is reached (p1248), the following applies:
- the Vdc_min controller limits the energy taken from the DC link in order to keep the DC link voltage above the minimum DC link voltage when accelerating.
- the motor is braked in order to use its kinetic energy to buffer the DC link.

p1240 = 4, 5, 6:
When the threshold in p1244 or p1248 is reached, the DC link voltage monitoring initiates a fault with a response and therefore reduces additional negative effects on the DC link voltage.

p1240 = 7, 9:
As for p1240 = 1, 3. However, the motor is prevented from accelerating due to the fact that other drives are regenerating. The effective lower torque limit cannot exceed the offset of the torque limit (p1532).

p1240 = 8, 9:
As for p1240 = 2, 3. However, the motor is prevented from braking due to the fact that the DC link voltage has been lowered. The effective upper torque limit cannot be less than the offset of the torque limit (p1532).

p1240[0...n]	Vdc controller or Vdc monitoring configuration / Vdc_ctrl config		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: 6220
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	6	1

Description: Sets the controller configuration of the DC link voltage (Vdc controller) in the closed-loop control mode.

Value:

- 0: Inhib Vdc ctrl
- 1: Vdc_max controller enable
- 2: Vdc_min controller (kinetic buffering) enable
- 3: Vdc_min controller and Vdc_max controller enable
- 4: Activates Vdc_max monitoring
- 5: Activates Vdc_min monitoring
- 6: Activates Vdc_min monitoring and Vdc_max monitoring

Dependency: Refer to: p1245
Refer to: A07400, A07401, A07402, F07403, F07404, F07405, F07406

Notice: An excessively high value in p1245 can possibly negatively influence the normal operation of the drive.

Note: p1240 = 1, 3:
When the DC link voltage limit specified for the Motor Module is reached the following applies:
- the Vdc_max controller limits the regenerative energy in order that the DC link voltage is kept below the maximum DC link voltage when braking.
- the ramp-down times are automatically increased.

p1240 = 2, 3:
When the switch-in threshold of the Vdc_min controller is reached (p1245), the following applies:
- the Vdc_min controller limits the energy taken from the DC link in order to keep the DC link voltage above the minimum DC link voltage when accelerating.
- the motor is braked in order to use its kinetic energy to buffer the DC link.

p1240 = 4, 5, 6:
When the threshold in r1242 or r1246 is reached, the DC link voltage monitoring initiates a fault (F07403 or F07404) with a response and therefore reduces additional negative effects on the DC link voltage.

r1242	Vdc_max controller switch-in level / Vdc_max on_level		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 6220 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Displays the switch-in level for the Vdc_max controller. If p1254 = 0 (automatic sensing of the switch-in level = off), then the following applies: AC/AC device: $r1242 = 1.15 * \sqrt{2} * V_{\text{mains}} = 1.15 * \sqrt{2} * p0210$ (supply voltage) DC/AC device: $r1242 = 1.15 * U_{\text{dc}} = 1.15 * p0210$ (supply voltage) If p1254 = 1 (automatic sensing of the switch-in level = on), then the following applies: $r1242 = V_{\text{dc_max}} - 50.0 \text{ V}$ (Vdc_max: Overvoltage threshold of the power unit)		
Note:	The Vdc_max controller is not switched back off until the DC-link voltage falls below the threshold $0.95 * p1242$ and the controller output is zero.		
p1243[0...n]	Vdc_max controller dynamic factor / Vdc_max dyn_factor		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 1 [%]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [%]	Access level: 3 Func. diagram: 6220 Unit selection: - Expert list: 1 Factory setting 100 [%]
Description:	Sets the dynamic factor for the DC link voltage controller (Vdc_max controller). 100% means that p1250, p1251, and p1252 (gain, integral time, and rate time) are used corresponding to their basic settings and based on a theoretical controller optimization. If subsequent optimization is required, this can be carried out using the dynamic factor. In this case p1250, p1251, p1252 are weighted with the dynamic factor p1243. If several modules are connected to the DC link, then the dynamic factor must be increased corresponding to the ratio of the additional capacitances to the capacitance of the module involved.		
Note:	The presetting of the dynamic factor is based on the power units connected to DRIVE-CLiQ. It is assumed that the power units connected via DRIVE-CLiQ are also electrically connected to the DC link. If this is not the case, then the dynamic factor must be optimized manually.		
p1244[0...n]	DC link voltage threshold upper / Vdc upper thresh		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 165 [V]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1200 [V]	Access level: 3 Func. diagram: 5650 Unit selection: - Expert list: 1 Factory setting 750 [V]
Description:	Sets the upper threshold for the DC link voltage. For p1240 = 1, 3, 7, 9, this threshold is used as limit setpoint for the Vdc_max controller. For p1240 = 4, 6, for DC link voltages above this threshold, an appropriate fault is output.		
Dependency:	Refer to: p1240, p1248, p1250		
Note:	For $p1244 < 1.07 * \text{"parameterized DC link voltage"}$ input of values is rejected. For p0204.0 = 1, the following applies: "Parameterized DC link voltage" = p0210 For p0204.0 = 0, the following applies: "Parameterized DC link voltage" = $p0210 * 1.4142$		

p1245[0...n]	Vdc_min controller switch-in level (kinetic buffering) / Vdc_min on_level		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 65 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 150 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 76 [%]
Description:	Sets the switch-in level for the Vdc-min controller (kinetic buffering). The value is obtained as follows: AC/AC device: $r1246[V] = p1245[\%] * \sqrt{2} * p0210$ DC/AC device: $r1246[V] = p1245[\%] * p0210$		
Dependency:	Refer to: p0210		
Warning:	An excessively high value may adversely affect normal drive operation. The values up to 150 % are intended for operating modes p1240 = 5, 6.		
			
Note:	For SINAMICS GM/SM, the following applies: Minimum value = 0.75, Maximum value = 0.90		
r1246	Vdc_min controller switch-in level (kinetic buffering) / Vdc_min on_level		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 6220 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Displays the switch-in level for the Vdc_min controller (kinetic buffering).		
Note:	The Vdc_min controller is not switched back off until the DC-link voltage rises above the threshold $1.05 * p1246$ and the controller output is zero.		
p1247[0...n]	Vdc_min controller dynamic factor (kinetic buffering) / Vdc_min dyn_factor		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 1 [%]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [%]	Access level: 3 Func. diagram: 6220 Unit selection: - Expert list: 1 Factory setting 100 [%]
Description:	Sets the dynamic factor for the Vdc_min controller (kinetic buffering). 100% means that p1250, p1251, and p1252 (gain, integral time, and rate time) are used corresponding to their basic settings and based on a theoretical controller optimization. If subsequent optimization is required, this can be carried out using the dynamic factor. In this case p1250, p1251, p1252 are weighted with the dynamic factor p1247. If several modules are connected to the DC link, then the dynamic factor must be increased corresponding to the ratio of the additional capacitances to the capacitance of the module involved.		

p1248[0...n]	DC link voltage threshold lower / Vdc lower thresh		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 100 [V]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000 [V]	Access level: 3 Func. diagram: 5650 Unit selection: - Expert list: 1 Factory setting 450 [V]
Description:	Sets the lower threshold for the DC link voltage. For p1240 = 2, 3, 8, 9, this threshold is used as limit setpoint for the Vdc_min controller. For p1240 = 5, 6, for DC link voltages below this threshold, an appropriate fault is output.		
Dependency:	Refer to: p1240, p1244, p1250		
Note:	For p1248 > 0.93 * "parameterized DC link voltage" input of values is rejected. For p0204.0 = 1, the following applies: "Parameterized DC link voltage" = p0210 For p0204.0 = 0, the following applies: "Parameterized DC link voltage" = p0210 * 1.4142		
p1249[0...n]	Vdc_max controller speed threshold / Vdc_max n_thresh		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 0.00 [rpm]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 10.00 [rpm]
Description:	Sets the lower speed threshold for the Vdc_max controller. When this speed threshold is undershot, the Vdc_max control is switched out and the speed is controlled using the ramp-function generator.		
Note:	For fast braking where the ramp-function generator tracking was active, it is possible to prevent the drive rotating in the opposite direction by increasing the speed threshold and setting a final rounding-off time in the ramp-function generator (p1131). This is supported using a dynamic setting of the speed controller.		
p1250[0...n]	Vdc controller proportional gain / Vdc_ctrl Kp		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 0.00 [A/V]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: 19_1 Scaling: - Max 10.00 [A/V]	Access level: 3 Func. diagram: 5650 Unit selection: p0505 Expert list: 1 Factory setting 1.00 [A/V]
Description:	Sets the proportional gain for the DC-link voltage controller (Vdc_min controller, Vdc_max controller).		
Dependency:	Refer to: p1240, p1244, p1248		
p1250[0...n]	Vdc controller proportional gain / Vdc_ctrl Kp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 0.00	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.00	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.00
Description:	Sets the proportional gain for the DC-link voltage controller (Vdc_min controller, Vdc_max controller).		
Dependency:	The effective proportional gain is obtained taking into account p1243 (Vdc_max controller dynamic factor).		
Note:	The gain factor is proportional to the capacitance of the DC link. The parameter is preset to a value that is optimally adapted to the capacitance of the individual Motor Module. The capacitances of the other power units, which are connected to the DC link, can be taken into account using the dynamic factor (p1247 or p1243).		

p1251[0...n]	Vdc controller integral time / Vdc_ctrl Tn		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 0 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [ms]	Access level: 3 Func. diagram: 6220 Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Sets the integral time for the DC-link voltage controller (Vdc_min controller, Vdc_max controller).		
Dependency:	The effective integral time is obtained taking into account p1243 (Vdc_max controller dynamic factor).		
Note:	An integral time is normally not required for single axis drives. For multi-axis drives on the other hand, it may be possible to compensate for interference from other axes using the integral time (integral component) . An integral time of 0 (default) deactivates the integral component of the controller.		
p1252[0...n]	Vdc controller rate time / Vdc_ctrl t_rate		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 0 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000 [ms]	Access level: 3 Func. diagram: 6220 Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Sets the rate time constant for the DC-link voltage controller (Vdc_min controller, Vdc_max controller).		
Dependency:	The effective rate time is obtained taking into account p1243 (Vdc_max controller dynamic factor).		
p1254	Vdc_max controller automatic ON level detection / Vdc_max SenseOnLev		
VECTOR (n/M)	Can be changed: U, T Data type: Integer16 P-Group: Functions Not for motor type: REL Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Activates/deactivates the automatic sensing of the switch-in level for the Vdc_max controller.		
Value:	0: Automatic detection inhibited 1: Automatic detection enabled		
p1255[0...n]	Vdc_min controller time threshold / Vdc_min t_thresh		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000.000 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the time threshold for the Vdc_min controller (kinetic buffering). If this value is exceeded a fault is output; the required response can be parameterized. Prerequisite: p1256 = 1.		
Dependency:	Refer to: F07406		
Notice:	If a time threshold has been parameterized, the Vdc_max controller should also be activated (p1240 = 3) so that the drive does not shut down with overvoltage when Vdc_min control is exited (due to the time violation) and in the event of fault response OFF3. It is also possible to increase the OFF3 ramp-down time p1135.		

p1256[0...n]	Vdc_min controller response (kinetic buffering) / Vdc_min response		
VECTOR (n/M)	Can be changed: U, T Data type: Integer16 P-Group: Functions Not for motor type: REL Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the response for the Vdc_min controller (kinetic buffering).		
Value:	0: Buffer Vdc until undervoltage, n<p1257 -> F07405 1: Buff. Vdc until undervolt., n<p1257 -> F07405, t>p1255 -> F07406		
Dependency:	Refer to: F07405, F07406		
p1257[0...n]	Vdc_min controller speed threshold / Vdc_min n_thresh		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 0.00 [rpm]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 50.00 [rpm]
Description:	Sets the speed threshold for the Vdc-min controller (kinetic buffering). If this value is exceeded a fault is output; the required response can be parameterized .		
r1258	CO: Vdc controller output / Vdc_ctrl output		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 6220 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the actual output of the Vdc controller (DC link voltage controller)		
Note:	The regenerative power limit p1531 is used for vector control to pre-control the Vdc_max controller. The lower the power limit is set, the lower the correction signals of the controller when the voltage limit is reached.		
p1260	Bypass configuration / Bypass config		
VECTOR (Tech_ctrl)	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Value:	0: Bypass deactivated 1: Bypass with synchronization and overlap 2: Bypass with synchronization without overlap 3: Bypass without synchronization		
Note:	If the bypass function is selected ((p1260 > 0), then when the power unit restarts after POWER OFF, the state of the bypass switch is evaluated. This means that after the ramp-up, it is possible to directly change into the standby mode. This is only possible for p1267 = 0 (bypass using the control signal) and if the control command after the system has been booted is still available (p1266). This function has a higher priority than the automatic restart function (p1210). The "bypass" function can only be switched out again (p1260 = 0) if the bypass is not active or the bypass function has a fault.		

r1261.0...9 CO/BO: Bypass control/status word / Bypass STW / ZSW

VECTOR (Tech_ctrl)	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Control and feedback signals of the bypass switch.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Command switch motor - drive	Close	Open	-
	01	Command switch motor - line supply	Close	Open	-
	02	Synchronization requested	Yes	No	-
	03	Staging status	Active	Not active	-
	05	Feedback signal switch motor - drive	Closed	Opened	-
	06	Feedback signal switch motor - line supply	Closed	Opened	-
	07	Bypass command (from p1266)	Yes	No	-
	08	Feedback signal synchronization completed (from p1268)	Yes	No	-
	09	Staging requested (from p2369)	Yes	No	-

Dependency: Refer to: p2369

Note: Control bits 0 and 1 should be interconnected to the signal outputs via which the switches in the motor feeder cables should be controlled. These should be selected/dimensioned for switching under load.

p1262[0...n] Bypass dead time / Bypass t_dead

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.000 [s]	20.000 [s]	1.000 [s]

Description: Sets the dead time for non-synchronized bypass.

Note: This parameter is used to define the changeover time of the contactors. It should not be shorter than the demagnetizing time of the motor (p0347). The total changeover time for the bypass is based on the total of p1262 plus the OFF time for the relevant switch (p1274[x]).

p1263 Debypass delay time / Debypass t_del

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.000 [s]	300.000 [s]	1.000 [s]

Description: Sets the delay time to switch back to converter operation for a non-synchronized bypass.

p1264 Bypass delay time / Bypass t_del

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.000 [s]	300.000 [s]	1.000 [s]

Description: Sets the delay time for switching to line operation for a non-synchronized bypass.

p1265 Bypass speed threshold / Bypass n_thresh

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2000	Expert list: 1
	Min	Max	Factory setting
	0.00 [rpm]	210000.00 [rpm]	1480.00 [rpm]

Description: Sets the speed threshold to activate the bypass.

Note: When selecting p1260 = 3 and p1267.1 = 1, the bypass is automatically activated when this speed is reached.

p1266 BI: Bypass, control command / Bypass command

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the command to bypass.

p1267 Bypass changeover source configuration / Chngov_src config

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned8	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the cause that should initiate the bypass.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bypass via signal (BI: p1266)	Yes	No	-
	01	Bypass via reaching the speed threshold	Yes	No	-

Note: The parameter only has an effect for a non-synchronized bypass.

p1267 bit 0 = 1:

The bypass is initiated by setting a binary signal. When the command is reset, after the debypass delay time (p1263) has expired, operation at the Motor Module is re-selected.

p1267 bit 1 = 1:

When the speed threshold entered in p1265 is reached, the bypass is switched in. The system only switches back when the speed setpoint again falls below the threshold value.

p1268 BI: Bypass, feedback synchronization completed / FdbkSig sync compl

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	3819.2

Description: Input for the feedback signal that synchronization was successfully completed.

Dependency: Refer to: r3819

p1269[0...1]	BI: Bypass switch feedback signal / Bypass FS		
VECTOR (Tech_ctrl)	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the feedback signal of the bypass switch.		
Index:	[0] = Switch motor - drive [1] = Switch motor - line supply		
Note:	In the case of switches without a feedback signal, connect the corresponding control bit as the signal source: p1269 index 0 = r1261 bit 0 p1269 index 1 = r1261 bit 1		
p1272	Simulation mode / Simulation mode		
VECTOR	Can be changed: T Data type: Integer16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	In the simulation mode, the closed-loop control or V/f control can be operated without motor. The simulation mode is used to test the power unit. Even though the DC link voltage is missing, the pulses are enabled when powering up. The DC link pre-charging is bypassed and the undervoltage detection is disabled. Closed-loop speed control with an encoder is possible if the torque setpoint (r0079) is used in order to operate a second drive in the closed-loop torque controlled mode.		
Value:	0: Off 1: On		
Dependency:	The following functions are deactivated in the simulation mode: - motor data identification routine - motor data identification routine, rotating without encoder - pole position identification routine For V/f control and encoderless vector control, flying restart is not carried out (refer to p1200). Refer to: r0192, p1900, p1910, p1960, p1990 Refer to: A07825, F07826		
Notice:	In simulation mode, binector output r0863.1 = 1 is set. This is why you need to check whether other devices are powered up via this signal before activating simulation mode. You might need to disconnect the corresponding BICO interconnection temporarily.		
Note:	Simulation mode is only possible for DC link voltages below 40 V. In order that the closed-loop control can be calculated, the displayed DC link voltage (r0026, r0070) is set to the rated DC link voltage (refer to p0210). Closed-loop current control and motor model are switched out (disabled) - the same is true for the speed controller for encoderless closed-loop speed control. When fault messages occur, the parameter is not automatically reset. This function is not implemented for SINAMICS GM.		

p1274[0...1]	Bypass switch monitoring time / Switch t_monit			
VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2	
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -	
	P-Group: -	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	0 [ms]	5000 [ms]	1000 [ms]	
Description:	Sets the bypass switch monitoring time.			
Index:	[0] = Switch motor - drive [1] = Switch motor - line supply			
Note:	The monitoring is deactivated with p1274 = 0 ms. The changeover time for the bypass (p1262) is extended by the time set in this parameter.			
p1275	Motor holding brake control word / Brake STW			
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: U, T	Calculated: -	Access level: 2	
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -	
	P-Group: Functions	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0000 bin	
Description:	Sets the control word for the motor holding brake.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Inverting BI: 1219[0]	Yes	No
	01	Inverting BI: 1219[1]	Yes	No
	02	Inverting BI: 1224[0]	Yes	No
	03	Inverting BI: 1224[1]	Yes	No
	05	Brake with feedback	Yes	No
				FP
				2707
				2707
				2704
				2704
				2711
p1276	Motor holding brake, standstill detection, bypass / Brk standst bypass			
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: U, T	Calculated: -	Access level: 2	
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2704	
	P-Group: Functions	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	0.000 [s]	300.000 [s]	300.000 [s]	
Description:	Sets the delay time for closing the brake at standstill. After this time has expired, if the "close brake at standstill" or OFF1/OFF3 is present, the brake is closed and the pulses are suppressed. For p1276 = 300.000 s, the timer is deactivated - this means that the timer output is always zero.			
p1277	Motor holding brake, braking threshold delay exceeded / Del thresh exceed.			
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: U, T	Calculated: -	Access level: 2	
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2707	
	P-Group: Functions	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	0.000 [s]	300.000 [s]	0.000 [s]	
Description:	Sets the delay time for the signal "braking threshold exceeded" (BO: r1229.6).			
Dependency:	Refer to: p1220, p1221, r1229			

p1278	Brake control, diagnostics evaluation / Brake diagnostics		
SERVO, VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the brake control type (with or without diagnostics evaluation). Example for brake control with diagnostics evaluation. - brake control in the Motor Modules in booksize format - Safe Brake Relay for AC Drive Example for brake control without diagnostics evaluation. - Brake Relay for AC Drive		
Value:	0: Brake control with diagnostics evaluation 1: Brake control without diagnostics evaluation		
Note:	If the configuration of the motor holding brake (p1215) is set to "no holding brake present" when booting, then an automatic identification of the motor holding brake will be carried out. If a brake control is detected without diagnostics evaluation (e.g. Brake Relay for AC Drive), then the parameter is set to "brake control without diagnostics evaluation". It is not permissible to parameterize "brake control without diagnostics evaluation" and also enable "safe brake control" (p1278 = 1, p9602 = 1, p9802 = 1).		
p1279[0...3]	BI: Motor holding brake, OR/AND logic operation / Brake OR AND		
SERVO (Extended brk), VECTOR (Extended brk)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 2707 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the OR/AND logic operation.		
Dependency:	Refer to: r1229		
Note:	[0]: OR logic operation, input 1 --> the result is displayed in r1229.10. [1]: OR logic operation, input 2 --> the result is displayed in r1229.10. [2]: AND logic operation, input 1 --> the result is displayed in r1229.11. [3]: AND logic operation, input 2 --> the result is displayed in r1229.11.		
p1280[0...n]	Vdc controller or Vdc monitoring configuration (V/f) / Vdc_ctr config V/f		
VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 6	Access level: 3 Func. diagram: 1690, 6320 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the configuration of the controller for the DC link voltage (Vdc controller) in the V/f operating mode.		
Value:	0: Inhib Vdc ctrl 1: Vdc_max controller enable 2: Vdc_min controller (kinetic buffering) enable 3: Vdc_min controller and Vdc_max controller enable 4: Activates Vdc_max monitoring 5: Activates Vdc_min monitoring 6: Activates Vdc_min monitoring and Vdc_max monitoring		
Note:	p1240 = 4, 5, 6: When the threshold in r1282 or r1286 is reached, the DC link voltage monitoring initiates a fault (F07403 or F07404) with a response and therefore reduces additional negative effects on the DC link voltage.		

r1282	Vdc_max controller switch-in level (V/f) / Vdc_max on_level		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 6320 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Displays the switch-in level for the Vdc_max controller. If p1294 = 0 (automatic sensing of the switch-in level = off), then the following applies: AC/AC device: $r1282 = 1.15 * \sqrt{2} * V_{\text{mains}} = 1.15 * \sqrt{2} * p0210$ (supply voltage) DC/AC device: $r1282 = 1.15 * V_{\text{dc}} = 1.15 * p0210$ (supply voltage) If p1294 = 1 (automatic sensing of the switch-in level = on), then the following applies: $r1282 = V_{\text{dc_max}} - 50.0 \text{ V}$ (Vdc_max: Overvoltage threshold of the power unit)		
Note:	The Vdc_max controller is not switched back off until the DC-link voltage falls below the threshold $0.95 * p1282$ and the controller output is zero.		
p1283[0...n]	Vdc_max controller dynamic factor (V/f) / Vdc_max dyn_factor		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 1 [%]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [%]	Access level: 3 Func. diagram: 6320 Unit selection: - Expert list: 1 Factory setting 100 [%]
Description:	Sets the dynamic factor for the DC link voltage controller (Vdc_max controller). 100% means that p1290, p1291, and p1292 (gain, integral time, and rate time) are used in accordance with their basic settings and on the basis of a theoretical controller optimization. If subsequent optimization is required, this can be carried out using the dynamic factor. In this case, p1290, p1291, and p1292 are weighted with the dynamic factor p1283. If several modules are connected to the DC link, then the dynamic factor must be increased corresponding to the ratio of the additional capacitances to the capacitance of the module involved.		
Note:	The presetting of the dynamic factor is based on the power units connected to DRIVE-CLiQ. It is assumed that the power units connected via DRIVE-CLiQ are also electrically connected to the DC link. If this is not the case, then the dynamic factor must be optimized manually.		
p1285[0...n]	Vdc_min controller switch-in level (kinetic buffering) (V/f) / Vdc_min on_level		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 65 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 150 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 76 [%]
Description:	Sets the switch-in level for the Vdc-min controller (kinetic buffering). The value is obtained as follows: AC/AC unit: $p1286[\text{V}] = p1285[\%] * \sqrt{2} * p0210$ DC/AC unit: $p1286[\text{V}] = p1285[\%] * p0210$		
Warning:	An excessively high value may adversely affect normal drive operation. The values up to 150 % are intended for operating modes p1240 = 5, 6.		



r1286	Vdc_min controller switch-in level (kinetic buffering) (V/f) / Vdc_min on_level		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 6320 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Displays the switch-in level for the Vdc_min controller (kinetic buffering).		
Note:	The Vdc_min controller is not switched back off until the DC-link voltage rises above the threshold $1.05 \cdot p1286$ and the controller output is zero.		
p1287[0...n]	Vdc_min controller dynamic factor (kinetic buffering) (V/f) / Vdc_min dyn_factor		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 1 [%]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [%]	Access level: 3 Func. diagram: 6320 Unit selection: - Expert list: 1 Factory setting 100 [%]
Description:	Sets the dynamic factor for the Vdc_min controller (kinetic buffering). 100% means that p1290, p1291, and p1292 (gain, integral time, and rate time) are used corresponding to their basic settings and based on a theoretical controller optimization. If subsequent optimization is required, this can be carried out using the dynamic factor. In this case, p1290, p1291, and p1292 are weighted with the dynamic factor p1287. If several modules are connected to the DC link, then the dynamic factor must be increased corresponding to the ratio of the additional capacitances to the capacitance of the module involved.		
Note:	The presetting of the dynamic factor is based on the power units connected to DRIVE-CLiQ. It is assumed that the power units connected via DRIVE-CLiQ are also electrically connected to the DC link. If this is not the case, then the dynamic factor must be optimized manually.		
p1288[0...n]	Vdc_max controller feedback coupling factor ramp-fct. gen. (V/f) / Vdc_max factor RFG		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.000	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.500
Description:	Sets the feedback factor for the ramp-function generator. Its ramp times are decelerated relative to the output signal of the Vdc_max controller.		
Note:	For values p1288 = 0.0 to 0.5, the controller dynamics are automatically adapted internally.		
p1289[0...n]	Vdc_max controller speed threshold (V/f) / Vdc_max n_thresh		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.00 [rpm]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 10.00 [rpm]
Description:	Sets the lower speed threshold for the Vdc_max controller. When this speed threshold is undershot, the Vdc_max control is switched out and the speed is controlled using the ramp-function generator.		

p1290[0...n]	Vdc controller proportional gain (V/f) / Vdc_ctrl Kp		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.00	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.00	Access level: 3 Func. diagram: 6320 Unit selection: - Expert list: 1 Factory setting 1.00
Description:	Sets the proportional gain for the Vdc controller (DC link voltage controller).		
Note:	The gain factor is proportional to the capacitance of the DC link. The parameter is preset to a value that is optimally adapted to the capacitance of the individual Motor Module. The capacitances of the other power units which are connected to the DC link can be taken into account using the dynamic factor (p1287 or p1283).		
p1291[0...n]	Vdc controller integral time (V/f) / Vdc_ctrl Tn		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [ms]	Access level: 3 Func. diagram: 6320 Unit selection: - Expert list: 1 Factory setting 40 [ms]
Description:	Sets the integral time for the Vdc controller (DC link voltage controller).		
p1292[0...n]	Vdc controller rate time (V/f) / Vdc_ctrl t_rate		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000 [ms]	Access level: 3 Func. diagram: 6320 Unit selection: - Expert list: 1 Factory setting 10 [ms]
Description:	Sets the rate time constant for the Vdc controller (DC link voltage controller).		
p1293[0...n]	Vdc min controller output limit (V/f) / Vdc_min outp_lim		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.00 [Hz]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 600.00 [Hz]	Access level: 3 Func. diagram: 6320 Unit selection: - Expert list: 1 Factory setting 10.00 [Hz]
Description:	Sets the output limit for the Vdc min controller (DC link undervoltage controller).		
p1294	Vdc_max controller automatic detection ON signal level (V/f) / Vdc_max SenseOnLev		
VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Activates/deactivates the automatic sensing of the switch-in level for the Vdc_max controller. When the sensing function is deactivated, the activation threshold r1282 for the Vdc_max controller is determined from the parameterized connection voltage p0210.		
Value:	0: Automatic detection inhibited 1: Automatic detection enabled		

p1295[0...n]	Vdc_min controller time threshold (V/f) / Vdc_min t_thresh		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000.000 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the time threshold for the Vdc_min controller (kinetic buffering). If this value is exceeded a fault is output; the required response can be parameterized. Prerequisite: p1296 = 1.		
Notice:	If a time threshold has been parameterized, the Vdc_max controller should also be activated (p1280 = 3) so that the drive does not shut down with overvoltage when Vdc_min control is exited (due to the time violation) and in the event of fault response OFF3. It is also possible to increase the OFF3 ramp-down time p1135.		
p1296[0...n]	Vdc_min controller response (kinetic buffering) (V/f / Vdc_min response		
VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the response for the Vdc_min controller (kinetic buffering).		
Value:	0: Buffer Vdc until undervoltage, n<p1297 -> F07405 1: Buff. Vdc until undervolt., n<p1297 -> F07405, t>p1295 -> F07406		
Note:	Re p1296 = 1: The quick stop ramp entered in p1135 must not be equal to zero, to prevent overcurrent shutdown if F07406 is triggered.		
p1297[0...n]	Vdc_min controller speed threshold (V/f) / Vdc_min n_thresh		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.00 [rpm]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 50.00 [rpm]
Description:	Sets the speed threshold for the Vdc-min controller (kinetic buffering). If this value is exceeded a fault is output; the required response can be parameterized .		
r1298	CO: Vdc controller output (V/f) / Vdc_ctrl output		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 6320 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the actual output of the Vdc controller (DC link voltage controller)		

p1300[0...n]	Open-loop/closed-loop control operating mode / Op/cl-lp ctrl_mode		
SERVO	Can be changed: C2(1), T Data type: Integer16 P-Group: V/f open-loop control Not for motor type: - Min 20 Max 23	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 23	Access level: 2 Func. diagram: 1590, 1690, 5060, 6300 Unit selection: - Expert list: 1 Factory setting 21
Description:	Sets the open and closed loop control mode of a drive.		
Value:	20: Speed control (encoderless) 21: Speed control (with encoder) 23: Torque control (with encoder)		
Dependency:	Closed-loop speed or torque control (with encoder) cannot be selected if the encoder type is not entered (p0400). Refer to: p0108, r0108, p0300, p0311, p0400, p1501		
Note:	The closed-loop torque control can only be changed over in operation (p1300 = 20, 21) by selecting the closed-loop speed control (p1501). At the changeover, the setting of p1300 does not change. In this case, the actual state is displayed in r1407, bit 2 and bit 3. For encoderless operation (p1404 = 0 or p1300 = 20), the following applies: - The following condition must be fulfilled: $p1800 \geq n / (2 * p0115[0])$, $n = 1, 2, \dots$ - For motors with a small power rating (< 300 W) we recommend to set $n \geq 2$.		

p1300[0...n]	Open-loop/closed-loop control operating mode / Op/cl-lp ctrl_mode		
SERVO (Lin)	Can be changed: C2(1), T Data type: Integer16 P-Group: V/f open-loop control Not for motor type: - Min 20 Max 23	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 23	Access level: 2 Func. diagram: 1590, 1690, 5060, 6300 Unit selection: - Expert list: 1 Factory setting 21
Description:	Sets the open and closed loop control mode of a drive.		
Value:	20: Velocity control (encoderless) 21: Velocity control (with encoder) 23: Force control (with encoder)		
Dependency:	Closed-loop velocity or force control (with encoder) cannot be selected if the encoder type is not entered (p0400). Refer to: p0108, r0108, p0300, p0311, p0400, p1501		
Note:	It is only possible to change over to force control during operation (p1501) if velocity control is selected (p1300 = 20, 21). At the changeover, the setting of p1300 does not change. In this case, the actual state is displayed in r1407, bit 2 and bit 3. For encoderless operation (p1404 = 0 or p1300 = 20), the following applies: - The following condition must be fulfilled: $p1800 \geq n / (2 * p0115[0])$, $n = 1, 2, \dots$ - For motors with a small power rating (< 300 W) we recommend to set $n \geq 2$.		

p1300[0...n]	Open-loop/closed-loop control operating mode / Op/cl-lp ctrl_mode		
VECTOR	Can be changed: C2(1), T Data type: Integer16 P-Group: V/f open-loop control Not for motor type: - Min 0 Description: Sets the V/f control mode of the drive. Value: 0: V/f control with linear characteristic 1: V/f control with linear characteristic and FCC 2: V/f control with parabolic characteristic 3: V/f control with parameterizable characteristic 4: V/f control with linear characteristic and ECO 5: V/f control for drives requiring a precise freq. (e.g. textiles) 6: V/f control for drives requiring a precise frequency and FCC 7: V/f control for a parabolic characteristic and ECO 19: V/f control with independent voltage setpoint	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 19	Access level: 2 Func. diagram: 1690, 6300, 6310, 6320 Unit selection: - Expert list: 1 Factory setting 0
Recommend.:	The use of the vector control operating modes is recommended for synchronous motors.		
Dependency:	If you are working with reduced supply voltages (p0212, bit 0 = 1), only V/f control with independent voltage setpoint (p1300 = 19) can be set as the operating mode.		
Notice:	Active slip compensation is required in the V/f control types with Eco mode (p1300 = 4, 7). The scaling of the slip compensation (p1335) should be set so that the slip is completely compensated (generally 100%). The Eco mode is only effective in steady-state operation and when the ramp-function generator is not bypassed. In the case of analog setpoints, if required the tolerance for ramp-up and ramp-down should be actively increased for the ramp-function generator using p1148 in order to reliably signal a steady-state condition.		
Note:	For the open-loop control modes p1300 = 5 and 6, the slip compensation p1335 and the resonance damping p1338 are internally switched out (disabled) in order to be able to precisely set the output frequency. For the open-loop control modes p1300 = 4 and 7 (Eco mode), the efficiency can be optimized by varying the voltage (when the operating point is constant). During operation (the pulses enabled) the open-loop control mode cannot be changed by changing over drive data sets.		

p1300[0...n]	Open-loop/closed-loop control operating mode / Op/cl-lp ctrl_mode		
VECTOR (n/M)	Can be changed: C2(1), T Data type: Integer16 P-Group: V/f open-loop control Not for motor type: - Min 0 Description: Sets the open and closed loop control mode of a drive. Value: 0: V/f control with linear characteristic 1: V/f control with linear characteristic and FCC 2: V/f control with parabolic characteristic 3: V/f control with parameterizable characteristic 4: V/f control with linear characteristic and ECO 5: V/f control for drives requiring a precise freq. (e.g. textiles) 6: V/f control for drives requiring a precise frequency and FCC 7: V/f control for a parabolic characteristic and ECO 18: I/f control with fixed current 19: V/f control with independent voltage setpoint 20: Speed control (encoderless) 21: Speed control (with encoder) 22: Torque control (encoderless) 23: Torque control (with encoder)	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 23	Access level: 2 Func. diagram: 1690, 1700, 6300, 8012 Unit selection: - Expert list: 1 Factory setting 20

- Recommend.:** The use of the vector control operating modes is recommended for synchronous motors.
- Dependency:** Closed-loop speed or torque control (with encoder) cannot be selected if the encoder type is not entered (p0400).
Closed-loop speed or torque control can be selected if the closed-loop speed/torque control was selected as operating mode (p0108.2).
Only operation with V/f characteristic is possible if the rated motor speed is not entered (p0311).
A reluctance motor can only be operated in a V/f control mode (p1300 < 20).
Sensorless control on separately excited synchronous motors is only possible with a VSM module (see p0150, p0151).
Refer to: p0108, r0108, p0300, p0311, p0400, p1501
- Notice:** Active slip compensation is required in the V/f control types with Eco mode (p1300 = 4, 7). The scaling of the slip compensation (p1335) should be set so that the slip is completely compensated (generally 100%).
The Eco mode is only effective in steady-state operation and when the ramp-function generator is not bypassed. In the case of analog setpoints, if required the tolerance for ramp-up and ramp-down should be actively increased for the ramp-function generator using p1148 in order to reliably signal a steady-state condition.
- Note:** The closed-loop torque control can only be changed over in operation (p1300 = 20, 21) by selecting the closed-loop speed control (p1501). At the changeover, the setting of p1300 does not change. In this case, the actual state is displayed in r1407, bit 2 and bit 3.
For the open-loop control modes p1300 = 5 and 6 (textile sector), slip compensation p1335, resonance damping p1338, and the I_{max} frequency controller are switched off internally so that the output frequency can be set precisely. The I_{max} voltage controller remains active.
For the open-loop control modes p1300 = 4 and 7 (Eco mode), the efficiency can be optimized by varying the voltage (when the operating point is constant).
Separately-excited synchronous motors can only be operated in modes p1300 = 20, 21 and 23 - or for diagnostic purposes in modes p1300 = 0, 3 and 18. For I/f control (p1300 = 18), the current amplitude can be set using p1609. Both for V/f as well as for I/f control only a small load may be applied to the separately-excited synchronous motor because the excitation current is not calculated as a function of the load.
During operation (the pulses enabled) the open-loop/closed-loop control mode cannot be changed by changing over drive data sets.
p1300 is pre-assigned depending on r0108.2 and p0187.

p1302[0...n]**V/f control configuration / V/f configuration**

VECTOR

Can be changed: T**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** DDS, p0180**Func. diagram:** -**P-Group:** V/f open-loop control**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

0000 bin

Description:

Sets the configuration for V/f control.

Bit field:**Bit** **Signal name****1 signal****0 signal****FP**

00 U_output starting angle zero

Yes

No

-

01 Take the sign of the setpoint voltage into account

Yes

No

-

Note:

Re bit 00:

If the bit is set the device will always start up with setpoint angle zero on pulse enable. This also affects the setpoint angle for DC braking (p1231).

Re bit 01:

If the bit is set, in the case of V/f control with independent voltage setpoint (p1300 = 19) and negative setpoint voltages at the input of p1330, the setpoint angle is rotated through 180 degrees, thereby achieving a negative output voltage. This renders the voltage boost p1310, p1311 ineffective.

p1310[0...n]	Voltage boost permanent / V_boost perm		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 250.0 [%]	Access level: 2 Func. diagram: 1690, 6300 Unit selection: - Expert list: 1 Factory setting 50.0 [%]
Description:	Defines the voltage boost as a [%] referred to the rated motor current (p0305). The magnitude of the permanent voltage boost is reduced with increasing frequency so that at the rated motor frequency, the rated motor voltage is present. The magnitude of the boost in Volt at a frequency of zero is defined as follows: $\text{Voltage boost [V]} = 1.732 * p0305 (\text{rated motor current [A]}) \times r0395 (\text{stator/primary section resistance [ohm]}) \times p1310 (\text{permanent voltage boost [\%]}) / 100 \%$ At low output frequencies, there is only a low output voltage in order to maintain the motor flux. However, the output voltage can be too low in order to achieve the following: - magnetize the induction motor. - hold the load. - compensate for losses in the system. This is the reason that the output voltage can be increased using p1310. The voltage boost can be used for both linear as well as square-law V/f characteristics and is calculated as follows: $\text{Voltage boost} = 1.732 * p0305 (\text{rated motor current}) \times r0395 (\text{stator/primary section resistance}) \times p1310 (\text{permanent voltage boost})$		
Dependency:	The current limit p0640 limits the boost. For vector control, the permanent voltage boost (p1310) has no effect as the drive converter automatically sets the optimum operating conditions. Refer to: p1300, p1311, p1312, r1315		
Notice:	The voltage boost increases the motor temperature (particularly at zero speed).		
Note:	The voltage boost is only effective for V/f control (p1300). The boost values are combined with one another if the permanent voltage boost (p1310) is used in conjunction with other boost parameters (acceleration boost (p1311), voltage boost for starting (p1312)). However, these parameters are assigned the following priorities: p1310 > p1311, p1312		
p1311[0...n]	Voltage boost at acceleration / V_boost accelerate		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 250.0 [%]	Access level: 2 Func. diagram: 1690, 6300 Unit selection: - Expert list: 1 Factory setting 0.0 [%]
Description:	p1311 only results in a voltage boost when accelerating and generates a supplementary torque to accelerate the load. The voltage boost becomes effective for a positive setpoint increase and disappears as soon as the setpoint has been reached. The build-up and withdrawal of the voltage boost are smoothed. The magnitude of the boost in Volt at a frequency of zero is defined as follows: $\text{Voltage boost [V]} = 1.732 * p0305 (\text{rated motor current [A]}) \times r0395 (\text{stator/primary section resistance [ohm]}) \times p1311 (\text{voltage boost when accelerating [\%]}) / 100 \%$		
Dependency:	The current limit p0640 limits the boost. Refer to: p1300, p1310, p1312, r1315		
Notice:	The voltage boost results in a higher motor temperature increase.		
Note:	The voltage boost when accelerating can improve the response to small, positive setpoint changes. Assigning priorities for the voltage boosts: refer to p1310		

p1312[0...n]	Voltage boost when starting / V_boost starting		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 250.0 [%]	Access level: 2 Func. diagram: 1690, 6300 Unit selection: - Expert list: 1 Factory setting 0.0 [%]
Description:	p1312 causes an additional voltage boost when starting, however, only for the first acceleration phase. The voltage boost becomes effective for a positive setpoint increase and disappears as soon as the setpoint has been reached. The build-up and withdrawal of the voltage boost are smoothed.		
Dependency:	The current limit p0640 limits the boost. Refer to: p1300, p1310, p1311, r1315		
Notice:	The voltage boost results in a higher motor temperature increase.		
Note:	The voltage boost when accelerating can improve the response to small, positive setpoint changes. Assigning priorities for the voltage boosts: refer to p1310		
r1315	Voltage boost total / V_boost total		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 6300 Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the total resulting voltage boost in volt (p1310 + p1311 + p1312).		
Dependency:	Refer to: p1310, p1311, p1312		
p1317[0...n]	V/f control activation / Vf act		
SERVO	Can be changed: T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: 5718 Unit selection: - Expert list: 1 Factory setting 0
Description:	Activates the V/f control with linear characteristic. 0: Operation as set in p1300. 1: Activates the V/f control.		
Value:	0: Off (p1300 eff) 1: On		
Dependency:	Refer to: p1318, p1319, p1326, p1327		
p1318[0...n]	V/f control ramp-up/ramp-down time / Uf t_rmp-up_rmp-dn		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 999999.000 [s]	Access level: 3 Func. diagram: 5300 Unit selection: - Expert list: 1 Factory setting 10.000 [s]
Description:	Sets the ramp-up and ramp-down time for the V/f control. The ramp-function generator requires this time to reach the maximum speed (p1082) from zero.		
Dependency:	Refer to: p1317, p1319, p1326, p1327		
Note:	This ramp is used for stall protection and operates independently of any ramp-function generator that might have been configured.		

p1319[0...n] V/f control voltage at zero frequency / Uf V at f=0 Hz			
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.0 [Vrms]	Calculated: CALC_MOD_REG Dynamic index: DDS, p0180 Units group: - Scaling: - Max 50.0 [Vrms]	Access level: 3 Func. diagram: 5300 Unit selection: - Expert list: 1 Factory setting 0.0 [Vrms]
Description:	The linear characteristic for the V/f control is defined by 0 Hz / p1319 and p1326 / p1327. This parameter specifies the voltage for a frequency of 0 Hz.		
Dependency:	Activates the V/f control using p1317. Refer to: p1317, p1326, p1327		
Note:	Linear interpolation is carried out between the points 0 Hz/p1319 and p1326/p1327.		
p1320[0...n] V/f control programmable characteristic frequency 1 / Uf char f1			
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.00 [Hz]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: - Scaling: - Max 3000.00 [Hz]	Access level: 3 Func. diagram: 6300 Unit selection: - Expert list: 1 Factory setting 0.00 [Hz]
Description:	The programmable characteristic for the V/f control is defined using 4 points and 0 Hz/p1310. This parameter specifies the voltage of the first point along the characteristic.		
Dependency:	Selects the freely programmable characteristic using p1300 = 3. The following applies to the frequency values: p1320 <= p1322 <= p1324 <= p1326. Otherwise, a standard characteristic is used that contains the rated motor operating point. Refer to: p1300, p1310, p1311, p1321, p1322, p1323, p1324, p1325, p1326, p1327		
Note:	Linear interpolation is carried out between the points 0 Hz/p1310, p1320/p1321 ... p1326/p1327. The voltage boost when accelerating (p1311) is also applied to the freely programmable V/f characteristic.		
p1321[0...n] V/f control programmable characteristic voltage 1 / Vf char U1			
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.0 [Vrms]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000.0 [Vrms]	Access level: 3 Func. diagram: 6300 Unit selection: - Expert list: 1 Factory setting 0.0 [Vrms]
Description:	The programmable characteristic for the V/f control is defined using 4 points and 0 Hz/p1310. This parameter specifies the voltage of the first point along the characteristic.		
Dependency:	Selects the freely programmable characteristic using p1300 = 3. Refer to: p1310, p1311, p1320, p1322, p1323, p1324, p1325, p1326, p1327		
Note:	Linear interpolation is carried out between the points 0 Hz/p1310, p1320/p1321 ... p1326/p1327. The voltage boost when accelerating (p1311) is also applied to the freely programmable V/f characteristic.		
p1322[0...n] V/f control programmable characteristic frequency 2 / Vf char f2			
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.00 [Hz]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: - Scaling: - Max 3000.00 [Hz]	Access level: 3 Func. diagram: 6300 Unit selection: - Expert list: 1 Factory setting 0.00 [Hz]
Description:	The programmable characteristic for the V/f control is defined using 4 points and 0 Hz/p1310. This parameter specifies the voltage of the second point along the characteristic.		

Dependency: The following applies to the frequency values: p1320 <= p1322 <= p1324 <= p1326. Otherwise, a standard characteristic is used that contains the rated motor operating point.
Refer to: p1310, p1311, p1320, p1321, p1323, p1324, p1325, p1326, p1327

p1323[0...n]	V/f control programmable characteristic voltage 2 / Vf char U2		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6300
	P-Group: V/f open-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [Vrms]	Max 10000.0 [Vrms]	Factory setting 0.0 [Vrms]
Description:	The programmable characteristic for the V/f control is defined using 4 points and 0 Hz/p1310. This parameter specifies the voltage of the second point along the characteristic.		
Dependency:	Refer to: p1310, p1311, p1320, p1321, p1322, p1324, p1325, p1326, p1327		

p1324[0...n]	V/f control programmable characteristic frequency 3 / Vf char f3		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6300
	P-Group: V/f open-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [Hz]	Max 3000.00 [Hz]	Factory setting 0.00 [Hz]
Description:	The programmable characteristic for the V/f control is defined using 4 points and 0 Hz/p1310. This parameter specifies the voltage of the third point along the characteristic.		
Dependency:	The following applies to the frequency values: p1320 <= p1322 <= p1324 <= p1326. Otherwise, a standard characteristic is used that contains the rated motor operating point. Refer to: p1310, p1311, p1320, p1321, p1322, p1323, p1325, p1326, p1327		

p1325[0...n]	V/f control programmable characteristic voltage 3 / Vf char U3		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6300
	P-Group: V/f open-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [Vrms]	Max 10000.0 [Vrms]	Factory setting 0.0 [Vrms]
Description:	The programmable characteristic for the V/f control is defined using 4 points and 0 Hz/p1310. This parameter specifies the voltage of the third point along the characteristic.		
Dependency:	Refer to: p1310, p1311, p1320, p1321, p1322, p1323, p1324, p1326, p1327		

p1326[0...n]	V/f control programmable characteristic frequency 4 / Vf char f4		
SERVO, VECTOR	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5300, 6300
	P-Group: V/f open-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [Hz]	Max 10000.00 [Hz]	Factory setting 0.00 [Hz]
Description:	In the servo control mode the following applies: The linear characteristic for the V/f control is defined by 0 Hz / p1319 and p1326 / p1327. For vector control, the following applies: The programmable characteristic for the V/f control is defined using 4 points and 0 Hz/p1310. This parameter specifies the frequency of the fourth point along the characteristic.		

Dependency:	In the servo control mode the following applies: Activates the V/f control using p1317. For vector control, the following applies: Selects the freely programmable characteristic using p1300 = 3. The following applies to the frequency values: p1320 ≤ p1322 ≤ p1324 ≤ p1326. Otherwise, a standard characteristic is used that contains the rated motor operating point. Refer to: p1310, p1311, p1317, p1319, p1320, p1321, p1322, p1323, p1324, p1325, p1327
Note:	In the servo control mode the following applies: Linear interpolation is carried out between the points 0 Hz / p1319 and p1326 / p1327. For vector control, the following applies: Linear interpolation is carried out between the points 0 Hz/p1310, p1320/p1321 ... p1326/p1327. For output frequencies above p1326, the characteristic is extrapolated with the gradient between the characteristic points p1324/p1325 and p1326/p1327. The voltage boost when accelerating (p1311) is also applied to the freely programmable V/f characteristic.

p1327[0...n] V/f control programmable characteristic voltage 4 / Vf char U4

SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.0 [Vrms]	Calculated: CALC_MOD_REG Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000.0 [Vrms]	Access level: 3 Func. diagram: 5300, 6300 Unit selection: - Expert list: 1 Factory setting 0.0 [Vrms]
Description:	In the servo control mode the following applies: The linear characteristic for the V/f control is defined by 0 Hz / p1319 and p1326 / p1327. For vector control, the following applies: The programmable characteristic for the V/f control is defined using 4 points and 0 Hz/p1310. This parameter specifies the voltage of the fourth point along the characteristic.		
Dependency:	In the servo control mode the following applies: Activates the V/f control using p1317. For vector control, the following applies: Selects the freely programmable characteristic using p1300 = 3. Refer to: p1310, p1311, p1317, p1319, p1320, p1321, p1322, p1323, p1324, p1325, p1326		
Note:	In the servo control mode the following applies: Linear interpolation is carried out between the points 0 Hz / p1319 and p1326 / p1327. For vector control, the following applies: Linear interpolation is carried out between the points 0 Hz/p1310, p1320/p1321 ... p1326/p1327. The voltage boost when accelerating (p1311) is also applied to the freely programmable V/f characteristic.		

p1330[0...n] CI: V/f control independent voltage setpoint / Vf V_set independ.

VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2001 Max -	Access level: 3 Func. diagram: 6300 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the voltage setpoint for V/f control with an independent voltage setpoint (p1300 = 19).		
Dependency:	Selects the V/f control with independent voltage setpoint via p1300 = 19. Refer to: p1300		

p1333[0...n] V/f control FCC starting frequency / V/f FCC f_start

VECTOR

Can be changed: U, T**Calculated:** CALC_MOD_ALL**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** 6310**P-Group:** V/f open-loop control**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0.00 [Hz]

3000.00 [Hz]

0.00 [Hz]

Description:

Sets the starting frequency at which FCC (Flux Current Control) is activated.

Dependency:

The correct operating mode must be set (p1300 = 1, 6).

Warning:

An excessively low value can result in instability.

**Note:**

For p1333 = 0 Hz, the FCC starting frequency is automatically set to 6 % of the rated motor frequency.

p1334[0...n] V/f control slip compensation starting frequency / Slip comp start

VECTOR

Can be changed: U, T**Calculated:** CALC_MOD_ALL**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** 6310**P-Group:** V/f open-loop control**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0.00 [Hz]

3000.00 [Hz]

0.00 [Hz]

Description:

Sets the starting frequency of the slip compensation.

Note:

For p1334 = 0, the starting frequency of the slip compensation is automatically set to 6 % of the rated motor frequency.

p1335[0...n] Slip compensation, scaling / Slip comp scal

VECTOR

Can be changed: U, T**Calculated:** -**Access level:** 2**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** 1690, 6310**P-Group:** V/f open-loop control**Units group:** -**Unit selection:** -**Not for motor type:** PEM, REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0.0 [%]

600.0 [%]

0.0 [%]

Description:

Sets the setpoint for slip compensation in [%] referred to r0330 (motor rated slip).

p1335 = 0.0 %: Slip compensation deactivated.

p1335 = 100.0 %: The slip is completely compensated.

Dependency:

Prerequisite for a precise slip compensation for p1335 = 100 % are the precise motor parameters (p0350 ... p0360).

If the parameters are not precisely known, a precise compensation can be achieved by varying p1335.

For V/f control types with Eco optimization (4 and 7), the slip compensation must be activated in order to guarantee correct operation.

Note:

The purpose of slip compensation is to maintain a constant motor speed regardless of the applied load. The fact that the motor speed decreases with increasing load is a typical characteristic of induction motors.

For synchronous motors, this effect does not occur and the parameter has no effect in this case.

For the open-loop control modes p1300 = 5 and 6 (textile sector), the slip compensation is internally disabled in order to be able to precisely set the output frequency.

If p1335 is changed during commissioning (p0009, p0010 > 0), then it is possible that the old value will no longer be able to be set. The reason for this is that the dynamic limits of p1335 have been changed by a parameter that was set when the drive was commissioned (e.g. p0300).

p1336[0...n]	Slip compensation limit value / Slip comp lim val		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: PEM, REL Min 0.00 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 600.00 [%]	Access level: 2 Func. diagram: 6310 Unit selection: - Expert list: 1 Factory setting 250.00 [%]
Description:	Sets the limit value for slip compensation in [%] referred to r0330 (motor rated slip).		
r1337	CO: Actual slip compensation / Slip comp act val		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: PEM, REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 6310 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the actual compensated slip [%] referred to r0330 (rated motor slip).		
Dependency:	p1335 > 0 %: Slip compensation active. Refer to: p1335		
p1338[0...n]	V/f mode resonance damping gain / Vf Res_damp gain		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.00	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.00	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.00
Description:	Sets the gain for resonance damping for V/f control. In V/f mode, the resonance damping function dampens oscillations that are frequently experienced by induction motors in certain speed ranges and by synchronous motors above even low speeds.		
Dependency:	Refer to: p1317, p1339, p1349		
Note:	Resonance damping is active in the following ranges: - Active: 3.1 Hz ... p1349 - Build-up (linear): 3.1 ... 4.77 Hz - Reduction (linear): 0.95 * p1349 ... p1349 Where the value = 1 and at the oscillation amplitude of the rated current, the rated slip frequency is switched in for induction motors, while a frequency of 10 Hz is switched in for synchronous motors.		
p1338[0...n]	V/f mode resonance damping gain / Vf Res_damp gain		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.00	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.00	Access level: 3 Func. diagram: 1690, 6310 Unit selection: - Expert list: 1 Factory setting 0.00
Description:	Sets the gain for resonance damping for V/f control.		
Dependency:	Refer to: p1300, p1339, p1349		

Note: The resonance damping function dampens active current oscillations that frequency occur under no-load conditions.

The resonance damping is active in a range of approximately 5 ... 90 % of the rated motor frequency (p0310), but up to a maximum of 45 Hz.

For the open-loop control modes p1300 = 5 and 6 (textile sectors), the resonance damping is internally disabled in order that the output frequency can be precisely set.

p1339[0...n] V/f mode resonance damping filter time constant / Vf Res_damp T			
SERVO	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: V/f open-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1.00 [ms]	Max 1000.00 [ms]	Factory setting 20.00 [ms]
Description:	Sets the filter time constant for resonance damping for V/f control.		
Dependency:	Refer to: p1317, p1338, p1349		
Note:	The filter time constant must be greater than the oscillation period of the oscillation to be dampened.		

p1339[0...n] V/f mode resonance damping filter time constant / Vf Res_damp T			
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6310
	P-Group: V/f open-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1.00 [ms]	Max 1000.00 [ms]	Factory setting 20.00 [ms]
Description:	Sets the filter time constant for resonance damping for V/f control.		
Dependency:	Refer to: p1300, p1338, p1349		

p1340[0...n] I_max frequency controller proportional gain / I_max_ctrl Kp			
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 1690
	P-Group: V/f open-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000	Max 0.500	Factory setting 0.000
Description:	Sets the proportional gain of the I_max voltage controller. The I_max controller reduces the drive converter output current if the maximum current (r0067) is exceeded. In the V/f operating modes (p1300) for the I_max control, one controller is used that acts on the output frequency and one controller that acts on the output voltage. The frequency controller reduces the current by decreasing the converter output frequency. The frequency is reduced down to a minimum value (equaling twice rated slip). If the overcurrent condition cannot be successfully resolved using this measure, then the drive converter output voltage is reduced using the I_max voltage controller. Once the overcurrent condition has been resolved, the drive is accelerated along the ramp set in p1120 (ramp-up time).		
Dependency:	In the V/f modes (p1300) for textile applications and for external voltage setpoints, only the I_max voltage controller is used.		
Notice:	When deactivating the I_max controller, the following must be carefully observed: When the maximum current (r0067) is exceeded, the output current is no longer reduced, however, overcurrent alarm messages are generated. The drive is shut down if the overcurrent limit (r0209) is exceeded.		
Note:	The I_max limiting controller becomes ineffective if the ramp-function generator is deactivated with p1122 = 1. p1341 = 0: I_max frequency controller deactivated and I_max voltage controller activated over the complete speed range.		

p1341[0...n]	I_max frequency controller integral time / I_max_ctrl Tn		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.000 [s]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 50.000 [s]	Access level: 3 Func. diagram: 1690 Unit selection: - Expert list: 1 Factory setting 0.300 [s]
Description:	Sets the integral time for the I_max frequency controller.		
Dependency:	Refer to: p1340		
Note:	When p1341 = 0, the current limiting controller influencing the frequency is deactivated and only the current limiting controller influencing the output voltage remains active (p1345, p1346).		
r1343	CO: I_max controller frequency output / I_max_ctrl f_outp		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 1690 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the effective frequency limit.		
Dependency:	Refer to: p1340		
r1344	I_max controller voltage output / I_max_ctrl V_outp		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 1690 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the amount by which the converter output voltage is reduced.		
Dependency:	Refer to: p1340		
p1345[0...n]	DC brake proportional gain / DCBRK Kp		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.000	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100000.000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the proportional gain for the DC brake (p1230, p1231).		
Dependency:	Refer to: p1346		
Note:	Current controller adaptation is not effective with the DC brake.		

p1345[0...n]	I_max voltage controller proportional gain / I_max_V_ctrl Kp		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.000	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100000.000	Access level: 3 Func. diagram: 1690 Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the proportional gain for the I_max voltage controller.		
Dependency:	Refer to: p1340		
Note:	The controller settings are also used in the current controller of the DC brake (refer to p1232).		
p1346[0...n]	DC brake integral time / DCBRK Tn		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.000 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 50.000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.030 [ms]
Description:	Sets the integral time for the DC brake (p1230, p1231).		
Dependency:	Refer to: p1345		
Note:	For p1346 = 0, the following applies: The integral time of the DC brake is deactivated.		
p1346[0...n]	I_max voltage controller integral time / I_max_V_ctrl Tn		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.000 [s]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 50.000 [s]	Access level: 3 Func. diagram: 1690 Unit selection: - Expert list: 1 Factory setting 0.030 [s]
Description:	Sets the integral time for the I_max voltage controller.		
Dependency:	Refer to: p1340		
Note:	p1346 = 0: Integral time of the I_max voltage controller deactivated. The controller settings are also used in the current controller of the DC brake (refer to p1232).		
p1349[0...n]	V/f mode resonance damping maximum frequency / Vf res_damp F_max		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.00 [Hz]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: - Scaling: - Max 3000.00 [Hz]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 3000.00 [Hz]
Description:	Sets the maximum output frequency for resonance damping for V/f control. Resonance damping is inactive above this output frequency.		
Dependency:	Refer to: p1338, p1339		
Note:	Resonance damping is active in the following ranges: - Active: 3.1 Hz ... p1349 - Build-up (linear): 3.1 ... 4.77 Hz - Reduction (linear): 0.95 * p1349 ... p1349		

p1349[0...n]	V/f mode resonance damping maximum frequency / Vf res_damp F_max		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: - Min 0.00 [Hz]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: - Scaling: - Max 3000.00 [Hz]	Access level: 3 Func. diagram: 6310 Unit selection: - Expert list: 1 Factory setting 0.00 [Hz]
Description:	Sets the maximum output frequency for resonance damping for V/f control. Resonance damping is inactive above this output frequency.		
Dependency:	Refer to: p1338, p1339		
Note:	For p1349 = 0, the changeover limit is automatically set to 95 % of the rated motor frequency - however, to a max. of 45 Hz.		
p1350[0...n]	Soft starting / Soft starting		
VECTOR	Can be changed: U, T Data type: Integer16 P-Group: V/f open-loop control Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: 1690 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets whether the voltage is continuously increased during the magnetizing phase (p1350 = 1, On) or whether it jumps directly to the voltage boost (p1350 = 0, Off).		
Value:	0: Off 1: On		
Note:	The settings for this parameter have the following advantages and disadvantages: 0 = off (jump directly to voltage boost) Advantage: Flux is established quickly -> torque is quickly available Disadvantage: The motor can move while it is being magnetized 1 = on (voltage is continually established) Advantage: The motor is unlikely to rotate Disadvantage: The flux is established slower -> torque is available later		
p1351[0...n]	CO: Motor holding brake starting frequency / Brake f_start		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: V/f open-loop control Not for motor type: REL Min -300.00 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: PERCENT Max 300.00 [%]	Access level: 3 Func. diagram: 6310 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the frequency setting value at the slip compensation output for starting up with motor holding brake.		
Dependency:	Setting parameter p1351 > 0 activates slip compensation automatically (p1335 = 100%) if it has been deactivated up to this point.		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	Connected with p1352 a value of 100% corresponds to the motor rated slip r0330.		

p1356[0...n]				
CI: V/f control, angular setpoint / Vf ang setpoint				
VECTOR	Can be changed: U, T	Calculated: -	Access level: 3	
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: -	
	P-Group: V/f open-loop control	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: p2005	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description: Sets the signal source for the differential angular generation for V/f control.				
p1358[0...n]				
Angular difference, symmetrizing, actual angle / Sym act angle				
VECTOR	Can be changed: U, T	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -	
	P-Group: V/f open-loop control	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	0	1	0	
Description: Sets the dead time for the symmetrizing of the actual angle value for the differential angular generation. The selected multiplier refers to the current controller clock cycle (dead time= p1358 * p0115[0]).				
r1359				
CO: Angular difference / Angular difference				
VECTOR	Can be changed: -	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -	
	P-Group: V/f open-loop control	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: p2005	Expert list: 1	
	Min	Max	Factory setting	
	- [°]	- [°]	- [°]	
Description: Displays the output of the differential angular generation.				
Note: The difference between the setpoint angle, read-in in p1356 and the actual value of the V/f control delayed with p1358 is displayed.				
p1400[0...n]				
Speed control configuration / n_ctrl config				
SERVO	Can be changed: U, T	Calculated: -	Access level: 2	
	Data type: Unsigned32	Dynamic index: DDS, p0180	Func. diagram: 1590, 5490	
	P-Group: Closed-loop control	Units group: -	Unit selection: -	
	Not for motor type: REL	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0000 0011 1010 0000 bin	
Description: Sets the configuration for the closed-loop speed control.				
Bit field:				
Bit	Signal name	1 signal	0 signal	FP
03	Reference model speed setpoint, I component	On	Off	5030
04	Torque limiting active in motoring/regenerative mode	Yes	No	-
05	Kp/Tn adaptation active	Yes	No	-
07	Interpolation speed pre-control active	Yes	No	-
08	Interpolation torque setpoint active	Yes	No	-
09	Damping for encoderless open-loop controlled oper.	Yes	No	-
10	Speed pre-control	For balancing	For setp_filter 2	-
11	Encoderless oper. speed actual value starting value	Setpoint	0.0	-
12	Encoderless operation changeover	Steady-state	When accelerating	-
13	Motoring/regenerative depending on	Speed setpoint	Actual speed value	-
16	I component for limiting	Enable	Hold	-

Note:

Re bit 07:

The interpolator is only effective for clock-cycle synchronous PROFIBUS operation and when the master receives a sign-of-life (STW 2.12 ... STW 2.15). Further, for active Dynamic Servo Control (DSC) an additional dead time of one speed controller clock cycle is obtained.

Re bit 11:

If the motor rotates when the pulses are enabled, then we recommend p1400.11 = 1 (starting value = setpoint) with the matching sign.

If the motor remains stationary (zero speed) when the pulses are enabled, then we recommend p1400.11 = 0 (starting value = 0.0).

Re bit 12:

If a changeover is made from operation with encoder to encoderless operation while accelerating (with the threshold from p1404), then we recommend p1400.12 = 0.

If the changeover is made from operation with encoder to encoderless at constant speed/velocity (e.g. with a DDS changeover or if there is an encoder fault via p0491) then we recommend p1400.12 = 1.

p1400[0...n]**Velocity control, configuration / v_ctrl config**

SERVO (Lin)

Can be changed: U, T**Calculated:** -**Access level:** 2**Data type:** Unsigned32**Dynamic index:** DDS, p0180**Func. diagram:** 1590, 5490**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

0000 0011 1010 0000 bin

Description:

Sets the configuration for the closed-loop velocity control.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
03	Reference model velocity setpoint I component	On	Off	5030
04	Force limiting active in motoring/regenerative mode	Yes	No	-
05	Kp/Tn adaptation active	Yes	No	-
07	Interpolation velocity controller pre-control active	Yes	No	-
08	Interpolation force setpoint active	Yes	No	-
09	Damping for encoderless open-loop controlled oper.	Yes	No	-
10	Velocity pre-control	For balancing	For setp_filter 2	-
11	Encoderless oper. velocity actual value starting value	Setpoint	0.0	-
12	Encoderless operation changeover	Steady-state	When accelerating	-
13	Motoring/regenerative depending on	Speed setpoint	Actual speed value	-
16	I component for limiting	Enable	Hold	-

Note:

Re bit 07:

The interpolator is only effective for clock-cycle synchronous PROFIBUS operation and when the master receives a sign-of-life (STW 2.12 ... STW 2.15). Further, for active Dynamic Servo Control (DSC) an additional dead time of one velocity controller clock cycle is obtained.

Re bit 11:

If the motor rotates when the pulses are enabled, then we recommend p1400.11 = 1 (starting value = setpoint) with the matching sign.

If the motor remains stationary (zero speed) when the pulses are enabled, then we recommend p1400.11 = 0 (starting value = 0.0).

Re bit 12:

If a changeover is made from operation with encoder to encoderless operation while accelerating (with the threshold from p1404), then we recommend p1400.12 = 0.

If the changeover is made from operation with encoder to encoderless at constant speed/velocity (e.g. with a DDS changeover or if there is an encoder fault via p0491) then we recommend p1400.12 = 1.

p1400[0...n] Speed control configuration / n_ctrl config

VECTOR (n/M)

Can be changed: U, T**Calculated:** -**Access level:** 2**Data type:** Unsigned32**Dynamic index:** DDS, p0180**Func. diagram:** 6490**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

1000 0000 0010 0001 bin

Description: Sets the configuration for the closed-loop speed control.**Bit field:****Bit Signal name****1 signal****0 signal****FP**

00 Automatic Kp/Tn adaptation active

Yes

No

6040

01 Encoderless vector control freeze I comp

Yes

No

6040

02 Acceleration pre-control source

External (p1495)

Internal (n_set)

6031

03 Reference model speed setpoint, I component

On

Off

6031

05 Kp/Tn adaptation active

Yes

No

-

06 Free Tn adaptation active

Yes

No

-

14 Torque pre-control

Always active

For n_ctrl enab

-

15 Encoderless vector control, speed pre-control

Yes

No

6030

Note:

Re bit 01:

When the bit is set, the I component of the speed controller is kept when changing into the open-loop controlled mode.

p1401[0...n] Flux control configuration / Flux ctrl config

VECTOR (n/M)

Can be changed: U, T**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** DDS, p0180**Func. diagram:** 6491**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** PEM, REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

1110 bin

Description: Sets the configuration for flux setpoint control**Bit field:****Bit Signal name****1 signal****0 signal****FP**

00 Flux setpoint soft starting active

Yes

No

6722,

6725

01 Flux setpoint differentiation active

Yes

No

6723,

6726

02 Flux build-up control active

Yes

No

6722,

6723,

6725,

6726

03 Flux characteristic, load-dependent

Yes

No

6725

04 Flux controller (ASM with encoder)

Yes

No

-

05 Flux impression (ASM with encoder)

with model chngov

From 30 % n Rated

-

06 Quick magnetizing

Yes

No

6722

07 Pre-control speed limitation

Yes

No

6640

08 Speed limiting controller

With M_limits

With I_limits

6640

Note:

Re bit 00 (not for permanent-magnet synchronous motors):

Initially, the flux is only established with a low rate of rise when magnetizing the induction motor. The flux setpoint p1570 is reached again at the end of the magnetizing time p0346.

Re bit 01 (not for permanent-magnet synchronous motors):

Initially, the flux is only established with a low rate of rise when magnetizing the induction motor. The flux setpoint p1570 is reached again at the end of the magnetizing time p0346. When quick magnetizing (p1401.6 = 1) is selected, soft starting is internally deactivated and alarm A07416 is displayed.

The flux differentiation can be switched out if a significant ripple occurs in the field-generating current setpoint (r0075) when entering the field weakening range. However, this is not suitable for fast acceleration operations because then, the flux decays more slowly and the voltage limiting responds.

Re bit 02 (not for permanent-magnet synchronous motors):

The flux build-up control operates during the magnetizing phase p0346 of the induction motor. If it is switched out, a constant current setpoint is injected and the flux is built up corresponding to the rotor time constant. When quick magnetizing (p1401.6 = 1) is selected and when flux build-up control is de-energized alarm A07416 is displayed.

Re bit 03:

The load-dependent calculation of the flux characteristic is only available for separately-excited synchronous motors.

Re bit 04 (only for vector control with sensor):

The flux controller does not operate in the range of the current model and not in the range of the flux impression (refer to p1750.4).

Re bit 05 (only for vector control with sensor):

Extremely rugged control operation is possible by directly toggling between the current model and flux impression. We therefore recommend that, in addition, the time-controlled model change is switched in (p1750.4 = 1) or the model changeover limits are significantly increased (p1752 > 0.35 * p0311; p1753 = 5 %).

Re bit 06 (not for induction motors):

Magnetizing is performed with maximum current (0.9 * r0067). With active identification of the stator resistance (see p0621) quick magnetizing is internally deactivated and alarm A07416 is displayed. During a flying restart of a rotating motor (see p1200) no quick magnetizing takes place.

Re bit 07:

if the speed of the drive exceeds the effective speed limit of the speed limiting controller, the torque limit is reduced linearly to zero as the deviation becomes greater. This reduces the integral component of the speed controller and, in turn, the overshoot during load shedding (see also F07901 and p2162).

Re bit 08:

The speed limiting controller sets the speed to maximum by opening the torque limits as far as the current limits (bit 08 = 0) or taking the torque limits into account (bit 08 = 1).

p1402[0...n]				
Closed-loop current control and motor model configuration / I_ctrl config				
SERVO	Can be changed: T		Calculated: -	Access level: 3
	Data type: Unsigned16		Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control		Units group: -	Unit selection: -
	Not for motor type: -		Scaling: -	Expert list: 1
	Min		Max	Factory setting
	-		-	0100 bin
Description: Sets the configuration for the closed-loop control and the motor model.				
Bit field:	Bit	Signal name	1 signal	0 signal
	01	Park encoder for n_ist > p1404	Yes	No
	02	Current controller adaptation active	Yes	No
	04	Torque-speed pre-control with encoder	Yes	No
Note:				
	Re bit 01:			
	When the bit is set, the encoder is parked as soon as the actual speed is greater than the changeover speed (p1404). The encoder state is displayed in r0487.14.			
	Re bit 02:			
	The current controller adaptation (p0391 ... p0393) is only calculated when the bit is set.			

p1402[0...n]	Closed-loop current control and motor model configuration / I_ctrl config				
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 3		
	Data type: Unsigned16	Dynamic index: DDS, p0180	Func. diagram: -		
	P-Group: Closed-loop control	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0100 bin		
Description:	Sets the configuration for the closed-loop control and the motor model.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	01	Park encoder for v_act > p1404	Yes	No	-
	02	Current controller adaptation active	Yes	No	-
	04	Force-velocity pre-control with encoder	Yes	No	-
Note:	Re bit 01: When the bit is set, the encoder is parked as soon as the actual velocity is greater than the changeover velocity (p1404). The encoder state is displayed in r0487.14. Re bit 02: The current controller adaptation (p0391 ... p0393) is only calculated when the bit is set.				

p1402[0...n]	Closed-loop current control and motor model configuration / I_ctrl config				
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3		
	Data type: Unsigned16	Dynamic index: DDS, p0180	Func. diagram: -		
	P-Group: Closed-loop control	Units group: -	Unit selection: -		
	Not for motor type: REL	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0001 bin		
Description:	Sets the configuration for the closed-loop control and the motor model.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Speed-following error correction active	Yes	No	-
	02	Current controller adaptation active	Yes	No	-
	08	Changeover current model/voltage model with speed setpoint	Yes	No	-
Note:	Re bit 00: When the bit is set, the speed following error is compensated that is obtained as a result of the smoothing time constant in p1441. Re bit 02: The current controller adaptation (p0391 ... p0393) is only calculated when the bit is set. Re bit 08: Only with encoderless control of separately excited synchronous motors.				

p1404[0...n]	Encoderless operation changeover speed / Encoderl op n_chg				
SERVO	Can be changed: T	Calculated: -	Access level: 3		
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 1590, 5060		
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	0.00 [rpm]	210000.00 [rpm]	210000.00 [rpm]		
Description:	Sets the speed to change over between operation with and without encoder. Above this speed, the drive system is automatically operated in encoderless mode.				

Note: The changeover speed applies when changing over between operation with and without encoder.
 Separate speed controllers should be set when operating with and without encoder.
 Operation with encoder: p1460 (Kp), p1462 (Tn), p1461, p1463, p1457, p1458 (parameters for speed controller adaptation)
 Operation without encoder: p1470 (Kp), p1472 (Tn)
 For encoderless operation (p1404 = 0 or p1300 = 20), the following applies:
 - The condition must be fulfilled: $p1800 \geq n / (2 * p0115[0])$, $n = 1, 2, \dots$
 - For motors with a small power rating (< 300 W) we recommend to set $n \geq 2$.

p1404[0...n]		Encoderless operation changeover velocity / Encoderl op v_chg	
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 1590, 5060
	P-Group: Closed-loop control	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [m/min]	1000.00 [m/min]	1000.00 [m/min]
Description:	Sets the velocity to change over between operation with and without encoder. Above this velocity, the drive system is automatically operated in encoderless mode.		
Note:	The changeover speed applies when changing over between operation with and without encoder. Separate speed controllers should be set when operating with and without encoder. Operation with encoder: p1460 (Kp), p1462 (Tn), p1461, p1463, p1457, p1458 (parameters for speed controller adaptation) Operation without encoder: p1470 (Kp), p1472 (Tn) For encoderless operation (p1404 = 0 or p1300 = 20), the following applies: - The condition must be fulfilled: $p1800 \geq n / (2 * p0115[0])$, $n = 1, 2, \dots$ - For motors with a small power rating (< 300 W) we recommend to set $n \geq 2$.		
	For encoderless operation (p1404 = 0 or p1300 = 20), the following applies: - The condition must be fulfilled: $p1800 \geq n / (2 * p0115[0])$, $n = 1, 2, \dots$ - For motors with a small power rating (< 300 W) we recommend to set $n \geq 2$.		

r1406.8...12		CO/BO: Control word speed controller / STW n_ctrl			
SERVO	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned16		Dynamic index: -		Func. diagram: 1530, 2520
	P-Group: Closed-loop control		Units group: -		Unit selection: -
	Not for motor type: REL		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		-
Description:		Displays the control word of the speed controller.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	Travel to fixed stop active	Yes	No	-
	12	Torque control active	Yes	No	-

r1406.8...12		CO/BO: Control word velocity controller / STW v_ctrl			
SERVO (Lin)	Can be changed: -		Calculated: -	Access level: 3	
	Data type: Unsigned16		Dynamic index: -	Func. diagram: 1530, 2520	
	P-Group: Closed-loop control		Units group: -	Unit selection: -	
	Not for motor type: REL		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	-	
Description:		Displays the control word of the velocity controller.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	Travel to fixed stop active	Yes	No	-
	12	Force control active	Yes	No	-

r1406.4...15 CO/BO: Control word speed controller / STW n_ctrl

VECTOR (n/M)

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** -**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the control word of the speed controller.

Bit field:**Bit Signal name****1 signal****0 signal****FP**

04 Hold speed controller I component

Yes

No

6040

05 Set speed controller I component

Yes

No

6040

08 Travel to fixed stop

Yes

No

8012

11 Droop enable

Yes

No

6030

12 Torque control active

Yes

No

6060

15 Set speed adaptation controller I component

Yes

No

-

r1407.0...13 CO/BO: Status word speed controller / ZSW n_ctrl

SERVO

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1530, 2522**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the status word of the speed controller.

Bit field:**Bit Signal name****1 signal****0 signal****FP**

00 V/f control active

Yes

No

-

01 Encoderless operation active

Yes

No

-

02 Torque control active

Yes

No

8010

04 Speed setpoint from DSC

Yes

No

2522

05 Speed controller I component frozen

Yes

No

-

06 Speed controller I component set

Yes

No

-

07 Torque limit reached

Yes

No

5610

08 Upper torque limit active

Yes

No

5610

09 Lower torque limit active

Yes

No

5610

11 Speed setpoint limited

Yes

No

-

13 Encoderless operation due to a fault

Yes

No

-

Note:

Re bit 04:

The following conditions must be fulfilled to set to 1:

- CI: p1190 and CI: p1191 must be interconnected with a signal source that is not equal to zero.
- it is not permissible that OFF1, OFF3 or STOP2 are active.
- it is not permissible that the motor data identification is active.
- Master control must not be active.

The following conditions can mean that the DSC function is not active in spite of the fact that the bit is set:

- clock-cycle synchronous operation is not selected (r2054 not equal to 4).
- the PROFIBUS is not clock-cycle synchronous (r2064[0] not equal to 1).
- DSC is not switched on at the control side; this means that KPC = 0 is transferred as the value at CI: p1191.

r1407.0...13 CO/BO: Status word velocity controller / ZSW v_ctrl

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2522
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word of the velocity controller.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	V/f control active	Yes	No	-
	01	Encoderless operation active	Yes	No	-
	02	Force control active	Yes	No	8010
	04	Velocity setpoint from DSC	Yes	No	2522
	05	Velocity controller I component frozen	Yes	No	-
	06	Velocity controller I component set	Yes	No	-
	07	Force limit reached	Yes	No	5610
	08	Upper force limit active	Yes	No	5610
	09	Lower force limit active	Yes	No	5610
	11	Velocity setpoint limited	Yes	No	-
	13	Encoderless operation due to a fault	Yes	No	-

Note: Re bit 04:

The following conditions must be fulfilled to set to 1:

- Cl: p1190 and Cl: p1191 must be interconnected with a signal source that is not equal to zero.
- it is not permissible that OFF1, OFF3 or STOP2 are active.
- it is not permissible that the motor data identification is active.
- Master control must not be active.

The following conditions can mean that the DSC function is not active in spite of the fact that the bit is set:

- clock-cycle synchronous operation is not selected (r2054 not equal to 4).
- the PROFIBUS is not clock-cycle synchronous (r2064[0] not equal to 1).
- DSC is not switched on at the control side; this means that KPC = 0 is transferred as the value at Cl: p1191.

r1407.0...15 CO/BO: Status word speed controller / ZSW n_ctrl

VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2522
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word of the speed controller.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	V/f control active	Yes	No	-
	01	Encoderless operation active	Yes	No	-
	02	Torque control active	Yes	No	6030, 6060, 8010
	03	Speed control active	Yes	No	6040
	05	Speed controller I component frozen	Yes	No	6040
	06	Speed controller I component set	Yes	No	6040
	07	Torque limit reached	Yes	No	6060
	08	Upper torque limit active	Yes	No	6060
	09	Lower torque limit active	Yes	No	6060
	10	Droop enabled	Yes	No	6030

11	Speed setpoint limited	Yes	No	6030
12	Ramp-function generator set	Yes	No	-
13	Encoderless operation due to a fault	Yes	No	-
14	I/f control active	Yes	No	-
15	Torque limit reached (without pre-control)	Yes	No	6060

r1408.0...9 CO/BO: Status word current controller / ZSW I_ctrl

SERVO	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2530, 5040
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the current controller status word.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	CI-loop curr ctrl	Active	Not active	-
	04	Limit Vd	Active	Not active	-
	05	Limit Vq	Active	Not active	-
	06	Positive limiting Iq	Active	Not active	-
	07	Negative limiting Iq	Active	Not active	-
	08	Limit iq_set	Active	Not active	-
	09	Limit id_set	Active	Not active	-

Note: The set current limit is taken into account in upstream torque limiting. Bits 6, 7, and 8 are, therefore, only set in the event of overshoots on account of the current setpoint filter.

r1408.0...15 CO/BO: Status word current controller / ZSW I_ctrl

VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2530
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the current controller status word.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Current ctrl act	Active	Not active	-
	01	Id control, I component limiting	Active	Not active	6714
	03	Voltage limiting	Active	Not active	6714
	10	Speed adaptation, limiting	Active	Not active	-
	11	Speed adaptation, speed deviation	Out tolerance	In tolerance	6719
	12	Motor stalled	Yes	No	6719, 8018
	15	Excitation current differential exceeded	Yes	No	-

p1412[0...n] Speed setpoint filter, dead time / n_set dead time

TM41	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 9674
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.000 [ms]	1.000 [ms]	0.000 [ms]

Description: Sets the delay of the speed setpoint for the incremental encoder emulation.

Note: The parameter is not effective in the SINAMICS operating mode (p4400 = 1).

p1414[0...n]	Speed setpoint filter activation / n_set_filt act				
SERVO	Can be changed: U, T		Calculated: -	Access level: 3	
	Data type: Unsigned16		Dynamic index: DDS, p0180	Func. diagram: 5020	
	P-Group: Closed-loop control		Units group: -	Unit selection: -	
	Not for motor type: REL		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0000 bin	
Description:	Setting for activating/deactivating the speed setpoint filter.				
Recommend.:	If only one filter is required, filter 1 should be activated and filter 2 deactivated, to avoid excessive processing time.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activate filter 1	Yes	No	-
	01	Activate filter 2	Yes	No	-
Dependency:	The individual speed setpoint filters are parameterized as of p1415.				

p1414[0...n]	Velocity setpoint filter activation / v_set_filt act				
SERVO (Lin)	Can be changed: U, T		Calculated: -	Access level: 3	
	Data type: Unsigned16		Dynamic index: DDS, p0180	Func. diagram: 5020	
	P-Group: Closed-loop control		Units group: -	Unit selection: -	
	Not for motor type: REL		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0000 bin	
Description:	Setting for activating/deactivating the velocity setpoint filter.				
Recommend.:	If only one filter is required, filter 1 should be activated and filter 2 deactivated, to avoid excessive processing time.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activate filter 1	Yes	No	-
	01	Activate filter 2	Yes	No	-
Dependency:	The individual velocity setpoint filters are parameterized as of p1415.				

p1414[0...n]	Speed setpoint filter activation / n_set_filt act				
TM41	Can be changed: U, T		Calculated: -	Access level: 3	
	Data type: Unsigned16		Dynamic index: DDS, p0180	Func. diagram: 9674	
	P-Group: Closed-loop control		Units group: -	Unit selection: -	
	Not for motor type: REL		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0000 bin	
Description:	Setting for activating/deactivating speed setpoint filter 1 for the incremental encoder emulation.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activate filter 1	Yes	No	9674
Dependency:	The speed setpoint filter can be parameterized using p1417 and p1418. Refer to: p1417, p1418				
Note:	The parameter is not effective in the SINAMICS operating mode (p4400 = 1).				

p1415[0...n]	Speed setpoint filter 1 type / n_set_filt 1 typ		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0	Max 2	Factory setting 0
Description:	Sets the type for speed setpoint filter 1.		
Value:	0: Low pass: PT1 1: Low pass: PT2 2: General 2nd-order filter		
Dependency:	PT1 low pass: p1416 PT2 low pass: p1417, p1418 General filter: p1417 ... p1420		

p1415[0...n]	Velocity setpoint filter 1 type / v_setp_filt 1 typ		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0	Max 2	Factory setting 0
Description:	Sets the type for speed setpoint filter 1.		
Value:	0: Low pass: PT1 1: Low pass: PT2 2: General 2nd-order filter		
Dependency:	PT1 low pass: p1416 PT2 low pass: p1417, p1418 General filter: p1417 ... p1420		

p1416[0...n]	Speed setpoint filter 1 time constant / n_set_filt 1 T		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 5000.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the time constant for the speed setpoint filter 1 (PT1).		
Dependency:	Refer to: p1414, p1415		
Note:	For SERVO (p0107) the following applies: This parameter is only effective if the speed filter is set as a PT1 low pass.		

p1416[0...n]	Velocity setpoint filter 1 time constant / v_set_filt 1 T		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 5000.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the time constant for the velocity setpoint filter 1 (PT1).		
Dependency:	Refer to: p1414, p1415		
Note:	This parameter is only effective if the velocity filter is set as a PT1 low pass.		

p1416[0...n]	Speed setpoint filter 1 time constant / n_set_filt 1 T		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6030
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 5000.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the time constant for the speed setpoint filter 1 (PT1).		
Note:	For SERVO (p0107) the following applies: This parameter is only effective if the speed filter is set as a PT1 low pass.		
p1417[0...n]	Speed setpoint filter 1 denominator natural frequency / n_set_filt 1 fn_d		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 2000.0 [Hz]
Description:	Sets the denominator natural frequency for speed setpoint filter 1 (PT2, general filter).		
Dependency:	Refer to: p1414, p1415		
Note:	This parameter is only effective if the speed filter is parameterized as a PT2 low pass or as general filter. The filter is only effective if the natural frequency is less than half of the sampling frequency.		
p1417[0...n]	Velocity setpoint filter 1 denominator natural frequency / v_set_filt 1 fn_d		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 2000.0 [Hz]
Description:	Sets the denominator natural frequency for velocity setpoint filter 1 (PT2, general filter).		
Dependency:	Refer to: p1414, p1415		
Note:	This parameter is only effective if the velocity filter is parameterized as a PT2 low pass or as general filter. The filter is only effective if the natural frequency is less than half of the sampling frequency.		
p1417[0...n]	Speed setpoint filter 1 denominator natural frequency / n_set_filt 1 fn_d		
TM41	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 9674
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 2000.0 [Hz]
Description:	Sets the denominator natural frequency for the speed setpoint filter 1 (PT2) of the incremental encoder emulation.		
Dependency:	Refer to: p1414		
Note:	The parameter is not effective in the SINAMICS operating mode (p4400 = 1). This parameter is only effective if the speed setpoint filter in p1414 is activated. The filter is only effective if the natural frequency is less than half of the sampling frequency.		

p1418[0...n]	Speed setpoint filter 1 denominator damping / n_set_filt 1 D_d		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.001	Max 10.000	Factory setting 0.700
Description:	Sets the denominator damping for speed setpoint filter 1 (PT2, general filter).		
Dependency:	Refer to: p1414, p1415		
Note:	This parameter is only effective if the speed filter is parameterized as a PT2 low pass or as general filter.		

p1418[0...n]	Velocity setpoint filter 1 denominator damping / v_set_filt 1 D_d		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.001	Max 10.000	Factory setting 0.700
Description:	Sets the denominator damping for velocity setpoint filter 1 (PT2, general filter).		
Dependency:	Refer to: p1414, p1415		
Note:	This parameter is only effective if the velocity filter is parameterized as a PT2 low pass or as general filter.		

p1418[0...n]	Speed setpoint filter 1 denominator damping / n_set_filt 1 D_d		
TM41	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 9674
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.001	Max 1.000	Factory setting 0.700
Description:	Sets the denominator damping for the speed setpoint filter 1 (PT2) of the incremental encoder emulation.		
Dependency:	Refer to: p1414		
Note:	The parameter is not effective in the SINAMICS operating mode (p4400 = 1). This parameter is only effective if the speed setpoint filter in p1414 is activated.		

p1419[0...n]	Speed setpoint filter 1 numerator natural frequency / n_set_filt 1 fn_n		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 2000.0 [Hz]
Description:	Sets the numerator natural frequency for speed setpoint filter 1 (general filter).		
Dependency:	Refer to: p1414, p1415		
Note:	This parameter is only effective if the speed filter is set as a general filter. The filter is only effective if the natural frequency is less than half of the sampling frequency.		

p1419[0...n]	Velocity setpoint filter 1 numerator natural frequency / v_set_filt 1 fn_n		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 2000.0 [Hz]
Description:	Sets the numerator natural frequency for velocity setpoint filter 1 (general filter).		
Dependency:	Refer to: p1414, p1415		
Note:	This parameter is only effective if the velocity filter is set as a general filter. The filter is only effective if the natural frequency is less than half of the sampling frequency.		

p1420[0...n]	Speed setpoint filter 1 numerator damping / n_set_filt 1 D_n		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.000	Max 10.000	Factory setting 0.700
Description:	Sets the numerator damping for speed setpoint filter 1 (general filter).		
Dependency:	Refer to: p1414, p1415		
Note:	This parameter is only effective if the speed filter is set as a general filter.		

p1420[0...n]	Velocity setpoint filter 1 numerator damping / v_set_filt 1 D_n		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.000	Max 10.000	Factory setting 0.700
Description:	Sets the numerator damping for velocity setpoint filter 1 (general filter).		
Dependency:	Refer to: p1414, p1415		
Note:	This parameter is only effective if the velocity filter is set as a general filter.		

p1421[0...n]	Speed setpoint filter 2 type / n_set_filt 2 typ		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0	Max 2	Factory setting 0
Description:	Sets the type for speed setpoint filter 2.		
Value:	0: Low pass: PT1 1: Low pass: PT2 2: General 2nd-order filter		
Dependency:	PT1 low pass: p1422 PT2 low pass: p1423, p1424 General filter: p1423 ... p1426		

p1421[0...n]	Velocity setpoint filter 2 type / v_setp_filt 2 typ		
SERVO (Lin)	Can be changed: U, T Data type: Integer16 P-Group: Closed-loop control Not for motor type: REL Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: 5020 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the type for speed setpoint filter 2.		
Value:	0: Low pass: PT1 1: Low pass: PT2 2: General 2nd-order filter		
Dependency:	PT1 low pass: p1422 PT2 low pass: p1423, p1424 General filter: p1423 ... p1426		
p1422[0...n]	Speed setpoint filter 2 time constant / n_set_filt 2 T		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 5000.00 [ms]	Access level: 3 Func. diagram: 5020 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the time constant for the speed setpoint filter 2 (PT1).		
Dependency:	Refer to: p1414, p1421		
Note:	This parameter is only effective if the speed filter is set as a PT1 low pass.		
p1422[0...n]	Velocity setpoint filter 2 time constant / v_set_filt 2 T		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 5000.00 [ms]	Access level: 3 Func. diagram: 5020 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the time constant for the velocity setpoint filter 2 (PT1).		
Dependency:	Refer to: p1414, p1421		
Note:	This parameter is only effective if the velocity filter is set as a PT1 low pass.		
p1423[0...n]	Speed setpoint filter 2 denominator natural frequency / n_set_filt 2 fn_d		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 5020 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the denominator natural frequency for speed setpoint filter 2 (PT2, general filter).		
Dependency:	Refer to: p1414, p1421		
Note:	This parameter is only effective if the speed filter is parameterized as a PT2 low pass or as general filter. The filter is only effective if the natural frequency is less than half of the sampling frequency.		

p1423[0...n]	Velocity setpoint filter 2 denominator natural frequency / v_set_filt 2 fn_d		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 2000.0 [Hz]
Description:	Sets the denominator natural frequency for velocity setpoint filter 2 (PT2, general filter).		
Dependency:	Refer to: p1414, p1421		
Note:	This parameter is only effective if the velocity filter is parameterized as a PT2 low pass or as general filter. The filter is only effective if the natural frequency is less than half of the sampling frequency.		
p1424[0...n]	Speed setpoint filter 2 denominator damping / n_set_filt 2 D_d		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.001	Max 10.000	Factory setting 0.700
Description:	Sets the denominator damping for speed setpoint filter 2 (PT2, general filter).		
Dependency:	Refer to: p1414, p1421		
Note:	This parameter is only effective if the speed filter is parameterized as a PT2 low pass or as general filter.		
p1424[0...n]	Velocity setpoint filter 2 denominator damping / v_set_filt 2 D_d		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.001	Max 10.000	Factory setting 0.700
Description:	Sets the denominator damping for velocity setpoint filter 2 (PT2, general filter).		
Dependency:	Refer to: p1414, p1421		
Note:	This parameter is only effective if the velocity filter is parameterized as a PT2 low pass or as general filter.		
p1425[0...n]	Speed setpoint filter 2 numerator natural frequency / n_set_filt 2 fn_n		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5020
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 2000.0 [Hz]
Description:	Sets the numerator natural frequency for speed setpoint filter 2 (general filter).		
Dependency:	Refer to: p1414, p1421		
Note:	This parameter is only effective if the speed filter is set as a general filter. The filter is only effective if the natural frequency is less than half of the sampling frequency.		

p1425[0...n]	Velocity setpoint filter 2 numerator natural frequency / v_set_filt 2 fn_n		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 5020 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the numerator natural frequency for velocity setpoint filter 2 (general filter).		
Dependency:	Refer to: p1414, p1421		
Note:	This parameter is only effective if the velocity filter is set as a general filter. The filter is only effective if the natural frequency is less than half of the sampling frequency.		
p1426[0...n]	Speed setpoint filter 2 numerator damping / n_set_filt 2 D_n		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 5020 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the numerator damping for speed setpoint filter 2 (general filter).		
Dependency:	Refer to: p1414, p1421		
Note:	This parameter is only effective if the speed filter is set as a general filter.		
p1426[0...n]	Velocity setpoint filter 2 numerator damping / v_set_filt 2 D_n		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 5020 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the numerator damping for velocity setpoint filter 2 (general filter).		
Dependency:	Refer to: p1414, p1421		
Note:	This parameter is only effective if the velocity filter is set as a general filter.		
p1428[0...n]	Speed pre-control balancing dead time / n_prectrBal t_dead		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 2.0	Access level: 3 Func. diagram: 5030, 5042, 5210 Unit selection: - Expert list: 1 Factory setting 0.0
Description:	Sets the dead time to symmetrize the speed setpoint for active torque pre-control. The selected multiplier refers to the speed controller clock cycle (dead time= p1428 * p0115[1]).		
Dependency:	In conjunction with p1429, this parameter can emulate the characteristics of how the torque is established (dynamic response of closed current control loop). Refer to: p1429, p1511		

p1428[0...n]	Velocity pre-control balancing dead time / n_prectrBal t_dead		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5030, 5042, 5210
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0	Max 2.0	Factory setting 0.0
Description:	Sets the dead time to symmetrize the velocity setpoint for active force pre-control. The selected multiplier refers to the velocity controller clock cycle (dead time= p1428 * p0115[1]).		
Dependency:	In conjunction with p1429, this parameter can emulate the characteristics of how the force is established (dynamic response of closed current control loop). Refer to: p1429, p1511		
p1428[0...n]	Speed pre-control balancing dead time / n_prectrBal t_dead		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6031
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0	Max 3.0	Factory setting 0.0
Description:	Sets the dead time to symmetrize the speed setpoint for active torque pre-control. The selected multiplier refers to the speed controller clock cycle (dead time= p1428 * p0115[1]).		
Dependency:	In conjunction with p1429, this parameter can emulate the characteristics of how the torque is established (dynamic response of closed current control loop). The parameter is only effective if the acceleration model is supplied using external acceleration signals (p1400.2 = 1). For p1400.2 = 0, a fixed dead time is used. Refer to: p1429, p1511		
p1429[0...n]	Speed pre-control balancing time constant / n_prectr bal T		
SERVO, VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5030, 5042, 5210, 6031
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 10000.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the time constant (PT1) for symmetrizing the speed setpoint for active torque pre-control.		
Dependency:	In conjunction with p1428, this parameter can emulate the characteristics of how torque is established (dynamic response of the closed current control loop). For VECTOR (r0107) the following applies: The parameter is only effective if the acceleration model is supplied using external acceleration signals (p1400.2 = 1). For p1400.2 = 0, time constant p1442 (or p1452 for encoderless vector control) is used. Refer to: p1428, p1511		

p1429[0...n]	Velocity pre-control balancing time constant / n_prectr bal T		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000.00 [ms]	Access level: 3 Func. diagram: 5030, 5042, 5210 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the time constant (PT1) for symmetrizing the velocity setpoint for active force pre-control.		
Dependency:	In conjunction with p1428, this parameter can emulate the characteristics of how the force is established (dynamic response of closed current control loop). Refer to: p1428, p1511		
p1430[0...n]	CI: Speed pre-control / n_prectrl		
SERVO	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 1550, 1590, 5020 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for speed pre-control channel (speed pre-control or torque pre-control).		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
p1430[0...n]	CI: Velocity pre-control / v_prectrl		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 1550, 1590, 5020 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the velocity pre-control channel (velocity pre-control or force pre-control).		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
r1431	CO: Speed pre-control to motor model / n_pctrl mot_mod		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 4 Func. diagram: 6030 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed setpoint for pre-controlling the motor model with sensorless vector control.		
Note:	With p1400 bit 15 = 0 or encoderless torque control, the pre-control signal is kept continuously in the range of the voltage model.		

r1432 CO: Speed pre-control after symmetrizing / n_prectr after sym			
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 5030 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed pre-control value after symmetrizing for the torque build-up (emulates the closed current control loop).		
Dependency:	Symmetrizing can be parameterized with p1428 and/or p1429.		
r1432 CO: Velocity pre-control after symmetrizing / n_prectr after sym			
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 5030 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocity pre-control value after symmetrizing for the force build-up (emulates the closed current control loop).		
Dependency:	Symmetrizing can be parameterized with p1428 and/or p1429.		
p1433[0...n] Speed controller reference model natural frequency / n_ctrl RefMod fn			
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 8000.0 [Hz]	Access level: 3 Func. diagram: 5030, 6031 Unit selection: - Expert list: 1 Factory setting 0.0 [Hz]
Description:	Sets the natural frequency of a PT2 element for the reference model of the speed controller.		
Recommend.:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (actual speed value) are virtually identical when the I component of the speed controller is disabled.		
Dependency:	Together with p1434 and p1435, the characteristics (in the time domain) of the closed-loop speed control (P) can be emulated. For VECTOR (r0107) the following applies: The reference model is activated with p1400.3 = 1. For encoderless vector control (p1300 = 20) the reference model is disabled in open-loop speed controlled operation (refer to p1755). Refer to: p1434, p1435		
p1433[0...n] Velocity controller reference model natural frequency / v_ctrl RefMod fn			
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 8000.0 [Hz]	Access level: 3 Func. diagram: 5030, 6031 Unit selection: - Expert list: 1 Factory setting 0.0 [Hz]
Description:	Sets the natural frequency of a PT2 element for the reference model of the velocity controller.		
Recommend.:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (velocity actual value) are virtually identical when the I component of the velocity controller is disabled.		
Dependency:	Together with p1434 and p1435, the characteristics (in the time domain) of the closed-loop velocity control (P) can be emulated. Refer to: p1434, p1435		

p1434[0...n]	Speed controller reference model damping / n_ctrl RefMod D		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 5.000	Access level: 3 Func. diagram: 5030, 6031 Unit selection: - Expert list: 1 Factory setting 1.000
Description:	Sets the damping of a PT2 element for the reference model of the speed controller.		
Recommend.:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (actual speed value) are virtually identical when the I component of the speed controller is disabled.		
Dependency:	In conjunction with p1433 and p1435, the characteristics (in time) of the P-controlled speed control loop can be emulated. For VECTOR (r0107) the following applies: The reference model is activated with p1400.3 = 1. Refer to: p1433, p1435		
p1434[0...n]	Velocity controller reference model damping / v_ctrl RefMod D		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 5.000	Access level: 3 Func. diagram: 5030, 6031 Unit selection: - Expert list: 1 Factory setting 1.000
Description:	Sets the damping of a PT2 element for the reference model of the velocity controller.		
Recommend.:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (velocity actual value) are virtually identical when the I component of the velocity controller is disabled.		
Dependency:	Together with p1433 and p1435, the characteristics (in the time domain) of the P-controlled velocity control loop can be emulated. Refer to: p1433, p1435		
p1435[0...n]	Speed controller reference model dead time / n_ctrRefMod t_dead		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 2.00	Access level: 2 Func. diagram: 5030, 6031 Unit selection: - Expert list: 1 Factory setting 0.00
Description:	Sets the "fractional" dead time for the reference model of the speed controller. This parameter emulates the computing dead time of the proportionally controlled speed control loop. The selected multiplier refers to the speed controller clock cycle (dead time= p1435 * p0115[1]).		
Recommend.:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (actual speed value) are virtually identical when the I component of the speed controller is disabled.		
Dependency:	In conjunction with p1433 and p1434, the characteristics (in time) of the P-controlled speed control loop can be emulated. For VECTOR (r0107) the following applies: The reference model is activated with p1400.3 = 1. Refer to: p0115, p1433, p1434		

p1435[0...n]	Velocity controller reference model dead time / v_ctrRefMod t_dead		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 2.00	Access level: 2 Func. diagram: 5030, 6031 Unit selection: - Expert list: 1 Factory setting 0.00
Description:	Sets the "fractional" dead time for the reference model of the velocity controller. This parameter emulates the computing dead time of the proportionally controlled velocity control loop. The selected multiplier refers to the velocity controller clock cycle (dead time= p1435 * p0115[1]).		
Recommend.:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (velocity actual value) are virtually identical when the I component of the velocity controller is disabled.		
Dependency:	Together with p1433 and p1434, the characteristics (in the time domain) of the P-controlled velocity control loop can be emulated. Refer to: p0115, p1433, p1434		
p1435[0...n]	Speed controller reference model dead time / n_ctrRefMod t_dead		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 3.00	Access level: 2 Func. diagram: 5030, 6031 Unit selection: - Expert list: 1 Factory setting 0.00
Description:	Sets the "fractional" dead time for the reference model of the speed controller. This parameter emulates the computing dead time of the proportionally controlled speed control loop. The selected multiplier refers to the speed controller clock cycle (dead time= p1435 * p0115[1]).		
Recommend.:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (actual speed value) are virtually identical when the I component of the speed controller is disabled.		
Dependency:	In conjunction with p1433 and p1434, the characteristics (in time) of the P-controlled speed control loop can be emulated. For VECTOR (r0107) the following applies: The reference model is activated with p1400.3 = 1. Refer to: p0115, p1433, p1434		
r1436	CO: Speed controller reference model speed setpoint output / RefMod n_set outp		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 5030, 6031 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed setpoint at the output of the reference model.		
Dependency:	For VECTOR (r0107) the following applies: The reference model is activated with p1400.3 = 1.		

r1436	CO: Velocity controller, reference model velocity_setpoint output / RefMod v_set outp		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 5030, 6031 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocity setpoint at the output of the reference model.		
Dependency:	For VECTOR (r0107) the following applies: The reference model is activated with p1400.3 = 1.		
p1437[0...n]	CI: Speed controller, reference model I component input / n_ctrRefMod I_comp		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 6031 Unit selection: - Expert list: 1 Factory setting 1436[0]
Description:	Sets the signal source for speed setpoint for the integral component of the speed controller.		
Dependency:	The reference model is activated with p1400.3 = 1. Refer to: p1400		
Caution:	It should be ensured that a speed setpoint is selected as signal source that corresponds to the setpoint for the P component of the speed controller.		
r1438	CO: Speed controller, speed setpoint / n_ctrl n_set		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 1550, 1590, 1700, 5030, 5040, 5042, 5210, 5300, 5620, 6031, 6040 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed setpoint after setpoint limiting for the P component of the speed controller. For V/f operation, the value that is displayed is of no relevance.		
Dependency:	Refer to: r1439		
Note:	In the standard state (the reference model is deactivated), r1438 = r1439.		
r1438	CO: Velocity controller, velocity setpoint / v_ctrl v_set		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 1550, 1590, 5030, 5040, 5042, 5210, 5300, 5620 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocity setpoint after setpoint limiting for the P component of the velocity controller. For V/f operation, the value that is displayed is of no relevance.		
Dependency:	Refer to: r1439		
Note:	In the standard state (the reference model is deactivated), r1438 = r1439.		

r1439	Speed setpoint, I component / n_set I_comp		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 5030, 5040, 6031 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed setpoint for the I component of the speed controller (output of the reference model after the setpoint limiting).		
Dependency:	Refer to: r1438		
Note:	In the standard state (the reference model is deactivated), r1438 = r1439.		
r1439	Velocity setpoint, I component / v_set I_comp		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 5030, 5040, 6031 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocity setpoint for the I component of the velocity controller (output of the reference model after the setpoint limiting).		
Dependency:	Refer to: r1438		
Note:	In the standard state (the reference model is deactivated), r1438 = r1439.		
p1440[0...n]	CI: Speed controller speed actual value / n_ctrl n_act		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 6040 Unit selection: - Expert list: 1 Factory setting 63[0]
Description:	Sets the signal source for the speed actual value of the speed controller.		
Dependency:	Refer to: r1443		
Caution:	Speed control with encoder (p1300 = 21):		
	For the speed or position signal of the motor model there must always be a motor encoder available (evaluation via SMC/SMI; see also p0400). The actual speed of the motor r0061 and the position data for synchronous motors continue to come from this motor encoder and are not affected by the setting of parameter p1440.		
Notice:	Speed control without encoder (p1300 = 20):		
	Dependent upon the transmission path of the external speed signal there will be dead times which have to be taken into account when setting the speed controller parameters (p1470, p1472) and can lead to dynamic losses accordingly. It is for this reason that signal transmission times have to be kept as low as possible.		
	So that the speed controller can also work at standstill, set p1750 bit 2 = 1 (controlled operation as of zero speed for passive loads). If you do not make this setting, operation will switch to open-loop speed control in the low speed range, switching the closed-loop speed controller off and rendering the measured actual speed ineffective.		

p1441[0...n]	Actual speed smoothing time / n_ist T_smooth		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 50.00 [ms]	Access level: 3 Func. diagram: 4710, 4715 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the smoothing time constant (PT1) for the speed actual value.		
Dependency:	Refer to: r0063, p1451		
Note:	The speed actual value should be smoothed for encoders with a low pulse number or for resolvers. After this parameter has been changed, we recommend that the speed controller is adapted and/or the speed controller settings checked Kp (p1460) and Tn (p1462).		
p1441[0...n]	Actual velocity smoothing time / v_act T_smooth		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 50.00 [ms]	Access level: 3 Func. diagram: 4710, 4715 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the smoothing time constant (PT1) for the velocity actual value.		
Dependency:	Refer to: r0063, p1451		
Note:	The speed actual value should be smoothed for encoders with a low pulse number or for resolvers. After this parameter has been changed, we recommend that the velocity controller is adapted and/or the velocity controller settings checked Kp (p1460) and Tn (p1462).		
p1441[0...n]	Actual speed smoothing time / n_ist T_smooth		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000.00 [ms]	Access level: 3 Func. diagram: 4710, 4715 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the smoothing time constant (PT1) for the speed actual value.		
Dependency:	Refer to: r0063		
Notice:	Smoothing times above 20 ms are only possible if the drive is accelerated or braked with the appropriately long ramp-up/ramp-down times. Otherwise, significant torque errors can occur and there is the danger that the drive is powered down (tripped) with F07902 (motor stalled).		
Note:	The speed actual value should be smoothed for encoders with a low pulse number or for resolvers. After this parameter has been changed, we recommend that the speed controller is adapted and/or the speed controller settings checked Kp (p1460) and Tn (p1462).		
p1442[0...n]	Speed controller speed actual value smoothing time / n_ctr n_act T_smth		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: - Scaling: - Max 32000.00 [ms]	Access level: 2 Func. diagram: 1700, 6040 Unit selection: - Expert list: 1 Factory setting 4.00 [ms]
Description:	Sets the smoothing time for the actual speed value of the speed controller for closed-loop control with encoder.		
Note:	The smoothing must be increased if there is gear backlash. For longer smoothing times, the integral time of the speed controller must also be increased (e.g. using p0340 = 4).		

r1443 CO: Speed actual value at actual value input speed controller / n_act n_ctrl inp

VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 6040
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2000	Expert list: 1
	Min	Max	Factory setting
	- [rpm]	- [rpm]	- [rpm]
Description:	Displays the speed actual value at the speed controller's free-wiring actual value input p1440.		
Dependency:	Refer to: p1440		
Note:	This speed signal is only used by the speed controller and not by the motor model.		

r1444 Speed controller, speed setpoint steady-state (static) / n_ctrl n_set stat

SERVO, VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5030
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2000	Expert list: 1
	Min	Max	Factory setting
	- [rpm]	- [rpm]	- [rpm]
Description:	Displays the sum of all speed setpoints that are present. The following sources are available for the displayed setpoint: - setpoint at the ramp-function generator input (r1119). - speed setpoint 1 (p1155). - speed setpoint 2 (p1160). - speed setpoint for the speed pre-control (p1430). - setpoint from DSC (for DSC active). - setpoint via PC (for master control active).		
Dependency:	Refer to: r1119, p1155, p1160, p1430		

r1444 Velocity controller, velocity setpoint, total / v_ctrl v_set stat

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5030
	P-Group: Closed-loop control	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2000	Expert list: 1
	Min	Max	Factory setting
	- [m/min]	- [m/min]	- [m/min]
Description:	Displays the sum of all velocity setpoints that are present. The following sources are available for the displayed setpoint: - setpoint at the ramp-function generator input (r1119). - velocity setpoint 1 (p1155). - velocity setpoint 2 (p1160). - velocity setpoint for the velocity pre-control (p1430). - setpoint from DSC (for DSC active). - setpoint via PC (for master control active, p3983).		
Dependency:	Refer to: r1119, p1155, p1160, p1430		

r1445 CO: Actual speed smoothed / n_act smooth

VECTOR (n/M)

Can be changed: -**Calculated:** -**Access level:** 4**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 6040**P-Group:** Closed-loop control**Units group:** 3_1**Unit selection:** p0505**Not for motor type:** REL**Scaling:** p2000**Expert list:** 1**Min****Max****Factory setting**

- [rpm]

- [rpm]

- [rpm]

Description:

Displays the actual smoothed actual speed for speed control.

p1451[0...n] Speed actual value smoothing time sensorless / n_act t_sm SL

SERVO

Can be changed: U, T**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** -**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0 [ms]

100 [ms]

0 [ms]

Description:

Sets the smoothing time for the calculated speed actual value in sensorless operation.

Dependency:

Refer to: p1441

p1451[0...n] Velocity actual value smoothing time sensorless / v_act t_sm SL

SERVO (Lin)

Can be changed: U, T**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** -**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0 [ms]

100 [ms]

0 [ms]

Description:

Sets the smoothing time for the calculated velocity actual value in sensorless operation.

Dependency:

Refer to: p1441

p1451[0...n] Motor model speed actual value smoothing time SLVC / Mot_mod n_act t_sm

VECTOR (n/M)

Can be changed: U, T**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** -**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** ASM, PEM, REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0 [ms]

100 [ms]

4 [ms]

Description:

Sets the smoothing time for the speed actual value calculated by the motor model in sensorless operation.

p1452[0...n] Speed controller speed actual value smoothing time (SLVC) / n_C n_act T_s SLVC

VECTOR (n/M)

Can be changed: U, T**Calculated:** -**Access level:** 2**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** 1700, 6040**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0.00 [ms]

32000.00 [ms]

10.00 [ms]

Description:

Sets the smoothing time for the actual speed of the speed controller for encoderless closed-loop speed control.

Note:

The smoothing must be increased if there is gear backlash. For longer smoothing times, the integral time of the speed controller must also be increased (e.g. using p0340 = 4).

r1454	CO: Speed controller system deviation I component / n_ctrl sys dev Tn		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 5040 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the system deviation of the I component of the speed controller. When the reference model is inactive (p1433 = 0 Hz), this parameter corresponds to the system deviation of the complete PI controller (r1454 = r0064).		
r1454	CO: Velocity controller system deviation I component / v_ctrl sys dev Tn		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 5040 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the system deviation of the I component of the velocity controller. When the reference model is inactive (p1433 = 0 Hz), this parameter corresponds to the system deviation of the complete PI controller (r1454 = r0064).		
r1454	CO: Speed controller system deviation I component / n_ctrl sys dev Tn		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 6040 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the system deviation of the I component of the speed controller. When the reference model is inactive (p1433 = 0 Hz), this parameter corresponds to the system deviation of the complete PI controller (r1454 = r0064).		
p1455[0...n]	CI: Speed controller P gain adaptation signal / n_ctrl Adpt_sig Kp		
SERVO	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 5050 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the source for the adaptation signal to additionally adapt the P gain of the speed controller.		
Dependency:	Refer to: p1456, p1457, p1458, p1459		
p1455[0...n]	CI: Velocity controller, P gain adaptation signal / v_ctrl Adpt_sig Kp		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 5050 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the source for the adaptation signal to additionally adapt the P gain of the velocity controller.		
Dependency:	Refer to: p1456, p1457, p1458, p1459		

p1455[0...n]	CI: Speed controller P gain adaptation signal / n_ctrl Adpt_sig Kp		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 6050 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the source for the adaptation signal to additionally adapt the P gain of the speed controller.		
Dependency:	Refer to: p1456, p1457, p1458, p1459		
p1456[0...n]	Speed controller P gain adaptation lower starting point / n_ctrl AdaptKpLow		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 400.00 [%]	Access level: 3 Func. diagram: 5050 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the lower starting point of the adaptation range for the additional adaptation of the P gain of the speed controller. The values are in % and refer to the set source of the adaptation signal.		
Dependency:	Refer to: p1455, p1457, p1458, p1459		
p1456[0...n]	Velocity controller P gain adaptation, lower starting point / v_ctrl AdaptKpLow		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 400.00 [%]	Access level: 3 Func. diagram: 5050 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the lower starting point of the adaptation range for the additional adaptation of the P gain of the velocity controller. The values are in % and refer to the set source of the adaptation signal.		
Dependency:	Refer to: p1455, p1457, p1458, p1459		
p1456[0...n]	Speed controller P gain adaptation lower starting point / n_ctrl AdaptKpLow		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 400.00 [%]	Access level: 3 Func. diagram: 6050 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the lower starting point of the adaptation range for the additional adaptation of the P gain of the speed controller. The values are in % and refer to the set source of the adaptation signal.		
Dependency:	Refer to: p1455, p1457, p1458, p1459		
Note:	If the upper transition point p1457 of the speed controller adaptation is set to lower values than the lower transition p1456, then the controller gain below p1457 is adapted with p1459 and above p1456, with p1458.		

p1457[0...n]	Speed controller P gain adaptation upper starting point / n_ctrl AdaptKp up		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 400.00 [%]	Factory setting 0.00 [%]
Description:	Sets the upper starting point of the adaptation range for the additional adaptation of the P gain of the speed controller. The values are in % and refer to the set source of the adaptation signal.		
Dependency:	Refer to: p1455, p1456, p1458, p1459		
p1457[0...n]	Velocity controller P gain adaptation upper starting point / v_ctrl AdaptKp up		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 400.00 [%]	Factory setting 0.00 [%]
Description:	Sets the upper starting point of the adaptation range for the additional adaptation of the P gain of the velocity controller. The values are in % and refer to the set source of the adaptation signal.		
Dependency:	Refer to: p1455, p1456, p1458, p1459		
p1457[0...n]	Speed controller P gain adaptation upper starting point / n_ctrl AdaptKp up		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 400.00 [%]	Factory setting 0.00 [%]
Description:	Sets the upper starting point of the adaptation range for the additional adaptation of the P gain of the speed controller. The values are in % and refer to the set source of the adaptation signal.		
Dependency:	Refer to: p1455, p1456, p1458, p1459		
Note:	If the upper transition point p1457 of the speed controller adaptation is set to lower values than the lower transition p1456, then the controller gain below p1457 is adapted with p1459 and above p1456, with p1458.		
p1458[0...n]	Adaptation factor, lower / Adapt_factor lower		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 200000.0 [%]	Factory setting 100.0 [%]
Description:	Sets the adaptation factor before the adaptation range (0 % ... p1456) to additionally adapt the P gain of the speed/velocity controller.		
Dependency:	Refer to: p1455, p1456, p1457, p1459		

p1458[0...n]	Adaptation factor, lower / Adapt_factor lower		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 200000.0 [%]	Factory setting 100.0 [%]
Description:	Sets the adaptation factor before the adaptation range (0 % ... p1456) to additionally adapt the P gain of the speed/velocity controller.		
Dependency:	Refer to: p1455, p1456, p1457, p1459		
Note:	If the upper transition point p1457 of the speed controller adaptation is set to lower values than the lower transition p1456, then the controller gain below p1457 is adapted with p1459 and above p1456, with p1458.		
p1459[0...n]	Adaptation factor, upper / Adapt_factor upper		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 200000.0 [%]	Factory setting 100.0 [%]
Description:	Sets the adaptation factor after the adaptation range (> p1457) to additionally adapt the P gain of the speed/velocity controller.		
Dependency:	Refer to: p1455, p1456, p1457, p1458		
p1459[0...n]	Adaptation factor, upper / Adapt_factor upper		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 200000.0 [%]	Factory setting 100.0 [%]
Description:	Sets the adaptation factor after the adaptation range (> p1457) to additionally adapt the P gain of the speed/velocity controller.		
Dependency:	Refer to: p1455, p1456, p1457, p1458		
Note:	If the upper transition point p1457 of the speed controller adaptation is set to lower values than the lower transition p1456, then the controller gain below p1457 is adapted with p1459 and above p1456, with p1458.		
p1460[0...n]	Speed controller P gain adaptation speed, lower / n_ctrl Kp n lower		
SERVO	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5040, 5042
	P-Group: Closed-loop control	Units group: 17_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.000 [Nms/rad]	Max 999999.000 [Nms/rad]	Factory setting 0.300 [Nms/rad]
Description:	Sets the P gain of the speed controller before the adaptation speed range (0 ... p1464). This value corresponds to the basic setting of the P gain of the speed controller without adaptation (p1461 = 100 %).		
Dependency:	Refer to: p1461, p1464, p1465		
Note:	When automatically calculating the speed controller, only the motor moment of inertia is taken into account (p0341). For higher load moments of inertia (p0342 > 1 or p1498 > 0), you are advised to check the speed controller gain.		

p1460[0...n]	Velocity controller, P gain adaptation velocity, lower / v_ctrl Kp n lower		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5040, 5042
	P-Group: Closed-loop control	Units group: 24_2	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.000 [Ns/m]	Max 999999.000 [Ns/m]	Factory setting 10.000 [Ns/m]
Description:	Sets the P gain of the velocity controller before the adaptation velocity range (0 ... p1464). This value corresponds to the basic setting of the P gain of the velocity controller without adaptation (p1461 = 100 %).		
Dependency:	Refer to: p1461, p1464, p1465		
Note:	When automatically calculating the velocity controller, only the motor inertia is taken into account (p0341). For higher inertias (p0342 > 1 or p1498 > 0), you are advised to check the velocity controller gain.		
p1460[0...n]	Speed controller P gain adaptation speed, lower / n_ctrl Kp n lower		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 1700, 6040
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.000	Max 999999.000	Factory setting 0.300
Description:	Sets the P gain of the speed controller before the adaptation speed range (0 ... p1464). This value corresponds to the basic setting of the P gain of the speed controller without adaptation (p1461 = 100 %).		
Dependency:	For p0528 = 1, the speed controller gain is represented without any dimensions. Refer to: p1461, p1464, p1465		
p1461[0...n]	Speed controller Kp adaptation speed, upper scaling / n_ctrl Kp n upper		
SERVO	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 200000.0 [%]	Factory setting 100.0 [%]
Description:	Sets the P gain of the speed controller for the upper adaptation speed range (> p1465). The entry is made referred to the P gain for the lower adaptation speed range of the speed controller (% referred to p1460).		
Dependency:	Refer to: p1460, p1464, p1465		
Note:	When automatically calculating the speed controller, only the motor moment of inertia is taken into account (p0341). For higher load moments of inertia (p0342 > 1 or p1498 > 0), you are advised to check the speed controller gain.		
p1461[0...n]	Velocity controller, P gain adaptation velocity, upper / v_ctrl Kp n upper		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 200000.0 [%]	Factory setting 100.0 [%]
Description:	Sets the P gain of the velocity controller for the upper adaptation velocity range (> p1465). The entry is made referred to the P gain for the lower adaptation velocity range of the velocity controller (% referred to p1460).		
Dependency:	Refer to: p1460, p1464, p1465		
Note:	When automatically calculating the velocity controller, only the motor inertia is taken into account (p0341). For higher inertias (p0342 > 1 or p1498 > 0), you are advised to check the velocity controller gain.		

p1461[0...n]	Speed controller Kp adaptation speed, upper scaling / n_ctrl Kp n upper		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200000.0 [%]	Access level: 3 Func. diagram: 6050 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the P gain of the speed controller for the upper adaptation speed range (> p1465). The entry is made referred to the P gain for the lower adaptation speed range of the speed controller (% referred to p1460).		
Dependency:	Refer to: p1460, p1464, p1465		
Note:	If the upper transition point p1465 of the speed controller adaptation is set to lower values than the lower transition p1464, then the controller gain below p1465 is adapted with p1461. This means that an adaptation can be implemented for low speeds without having to change the controller parameters.		
p1462[0...n]	Speed controller integral time adaptation speed lower / n_ctrl Tn n lower		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100000.00 [ms]	Access level: 2 Func. diagram: 1700, 5040, 5042, 6040 Unit selection: - Expert list: 1 Factory setting 20.00 [ms]
Description:	Sets the integration time of the speed controller before the adaptation speed range (0 ... p1464). This value corresponds to the basic setting of the integral time of the speed controller without adaptation (p1461 = 100 %).		
Dependency:	Refer to: p1463, p1464, p1465		
p1462[0...n]	Velocity contr. integral act. time adaptation velocity lower / v_ctrl Tn n lower		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100000.00 [ms]	Access level: 2 Func. diagram: 5040, 5042 Unit selection: - Expert list: 1 Factory setting 20.00 [ms]
Description:	Sets the integration time of the velocity controller before the adaptation velocity range (0 ... p1464). This value corresponds to the basic setting of the integral time of the velocity controller without adaptation (p1461 = 100 %).		
Dependency:	Refer to: p1463, p1464, p1465		
p1463[0...n]	Speed controller Tn adaptation speed, upper scaling / n_ctrl Tn n upper		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200000.0 [%]	Access level: 3 Func. diagram: 5050 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the integral time of the speed controller after the adaptation speed range (> p1465). The entry is made referred to the integral time for the lower adaptation speed range of the speed controller (% referred to p1462).		
Dependency:	Refer to: p1462, p1464, p1465		

p1463[0...n]	Velocity contr. integral act. time adaptation velocity upper / v_ctrl Tn n upper		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200000.0 [%]	Access level: 3 Func. diagram: 5050 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the integral time of the velocity controller after the adaptation velocity range (> p1465). The entry is made referred to the integral time for the lower adaptation velocity range of the velocity controller (% referred to p1462).		
Dependency:	Refer to: p1462, p1464, p1465		
p1463[0...n]	Speed controller Tn adaptation speed, upper scaling / n_ctrl Tn n upper		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200000.0 [%]	Access level: 3 Func. diagram: 6050 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the integral time of the speed controller after the adaptation speed range (> p1465). The entry is made referred to the integral time for the lower adaptation speed range of the speed controller (% referred to p1462).		
Dependency:	Refer to: p1462, p1464, p1465		
Note:	If the upper transition point p1465 of the speed controller adaptation is set to lower values than the lower transition point p1464, then the controller integral time below p1465 is adapted with p1463. This means that an adaptation can be implemented for low speeds without having to change the controller parameters.		
p1464[0...n]	Speed controller adaptation speed, lower / n_ctrl n lower		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [rpm]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: 5050 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [rpm]
Description:	Sets the lower adaptation speed of the speed controller. No adaptation is effective below this speed.		
Dependency:	Refer to: p1460, p1461, p1462, p1463, p1465		
p1464[0...n]	Velocity controller adaptation velocity, lower / v_ctrl n lower		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [m/min]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: 4_1 Scaling: - Max 1000.00 [m/min]	Access level: 3 Func. diagram: 5050 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [m/min]
Description:	Sets the lower adaptation velocity of the velocity controller. No adaptation is effective below this velocity.		
Dependency:	Refer to: p1460, p1461, p1462, p1463, p1465		

p1464[0...n]	Speed controller adaptation speed, lower / n_ctrl n lower		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6050
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 0.00 [rpm]
Description:	Sets the lower adaptation speed of the speed controller. No adaptation is effective below this speed.		
Dependency:	Refer to: p1460, p1461, p1462, p1463, p1465		
Note:	If the upper transition point p1465 of the speed controller adaptation is set to lower values than the lower transition point p1464, then the controller below p1465 is adapted with p1461 or p1463. This means that an adaptation can be implemented for low speeds without having to change the controller parameters.		
p1465[0...n]	Speed controller adaptation speed, upper / n_ctrl n upper		
SERVO	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5050
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 210000.00 [rpm]
Description:	Sets the upper adaptation speed of the speed controller. No adaptation is effective above this speed. For P gain, p1460 * p1461 is effective. For the integral time, p1462 * p1463 is effective.		
Dependency:	Refer to: p1460, p1461, p1462, p1463, p1464		
p1465[0...n]	Velocity controller adaptation velocity, upper / v_ctrl n upper		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5050
	P-Group: Closed-loop control	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 1000.00 [m/min]	Factory setting 1000.00 [m/min]
Description:	Sets the upper adaptation velocity of the velocity controller. No adaptation is effective above this velocity. For P gain, p1460 * p1461 is effective. For the integral time, p1462 * p1463 is effective.		
Dependency:	Refer to: p1460, p1461, p1462, p1463, p1464		
p1465[0...n]	Speed controller adaptation speed, upper / n_ctrl n upper		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6050
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 210000.00 [rpm]
Description:	Sets the upper adaptation speed of the speed controller. No adaptation is effective above this speed. For P gain, p1460 * p1461 is effective. For the integral time, p1462 * p1463 is effective.		
Dependency:	Refer to: p1460, p1461, p1462, p1463, p1464		
Note:	If the upper transition point p1465 of the speed controller adaptation is set to lower values than the lower transition point p1464, then the controller below p1465 is adapted with p1461 or p1463. This means that an adaptation can be implemented for low speeds without having to change the controller parameters.		

p1466[0...n]	CI: Speed controller P-gain scaling / n_ctrl Kp scal		
SERVO	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 5050 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the scaling of the P gain of the speed controller. This also makes the effective P gain (including adaptations) scalable.		
p1466[0...n]	CI: Velocity controller P gain scaling / v_ctrl Kp scal		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 5050 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the scaling of the P gain of the velocity controller. This also makes the effective P gain (including adaptations) scalable.		
p1466[0...n]	CI: Speed controller P-gain scaling / n_ctrl Kp scal		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 6050 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the scaling of the P gain of the speed controller. This also makes the effective P gain (including adaptations) scalable.		
r1468	Speed controller P-gain effective / n_ctrl Kp eff		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nms/rad]	Calculated: - Dynamic index: - Units group: 17_1 Scaling: - Max - [Nms/rad]	Access level: 3 Func. diagram: 5040, 5042, 5210 Unit selection: p0505 Expert list: 1 Factory setting - [Nms/rad]
Description:	Displays the effective P gain of the speed controller.		
r1468	Velocity controller P gain effective / v_ctrl Kp eff		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Ns/m]	Calculated: - Dynamic index: - Units group: 24_2 Scaling: - Max - [Ns/m]	Access level: 3 Func. diagram: 5040, 5042, 5210 Unit selection: p0505 Expert list: 1 Factory setting - [Ns/m]
Description:	Displays the effective P gain of the velocity controller.		

r1468	CO: Speed controller P-gain effective / n_ctrl Kp eff		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 6040 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the effective P gain of the speed controller.		
Dependency:	For p0528 = 1, the speed controller gain is represented without any dimensions. In this case, connector output signal r1468 is increased by a factor of 100 in order to improve the resolution.		
r1469	Speed controller integral time effective / n_ctrl Tn eff		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [ms]	Access level: 3 Func. diagram: 5040, 5042, 6040 Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the effective integral time of the speed controller.		
r1469	Velocity controller integral time effective / v_ctrl Tn eff		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [ms]	Access level: 3 Func. diagram: 5040, 5042 Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the effective integral time of the velocity controller.		
p1470[0...n]	Speed controller encoderless operation P-gain / n_ctrl SLVC Kp		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000 [Nms/rad]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: 17_1 Scaling: - Max 999999.000 [Nms/rad]	Access level: 2 Func. diagram: 5210 Unit selection: p0505 Expert list: 1 Factory setting 0.300 [Nms/rad]
Description:	Sets the P gain for encoderless operation for the speed controller.		
Note:	When automatically calculating the speed controller, only the motor moment of inertia is taken into account (p0341). For higher load moments of inertia (p0342 > 1 or p1498 > 0), you are advised to check the speed controller gain.		
p1470[0...n]	Velocity controller encoderless operation P-gain / v_ctrl SLVC Kp		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000 [Ns/m]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: 24_2 Scaling: - Max 999999.000 [Ns/m]	Access level: 2 Func. diagram: 5210 Unit selection: p0505 Expert list: 1 Factory setting 10.000 [Ns/m]
Description:	Sets the P gain for encoderless operation for the velocity controller.		
Note:	When automatically calculating the velocity controller, only the motor inertia is taken into account (p0341). For higher inertias (p0342 > 1 or p1498 > 0), you are advised to check the velocity controller gain.		

p1470[0...n]	Speed controller encoderless operation P-gain / n_ctrl SLVC Kp		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6040, 6050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.000	Max 999999.000	Factory setting 0.300
Description:	Sets the P gain for encoderless operation for the speed controller.		
Dependency:	For p0528 = 1, the speed controller gain is represented without any dimensions.		
Note:	The product p0341 * p0342 is taken into account when automatically calculating the speed controller (p0340 = 1, 3, 4).		
p1472[0...n]	Speed controller encoderless operation integral time / n_ctrl SLVC Tn		
SERVO	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5210
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 100000.0 [ms]	Factory setting 20.0 [ms]
Description:	Set the integral time for encoderless operation for the speed controller.		
p1472[0...n]	Velocity controller encoderless operation integral time / v_ctrl SLVC Tn		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5210
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 100000.0 [ms]	Factory setting 20.0 [ms]
Description:	Set the integral time for encoderless operation for the velocity controller.		
p1472[0...n]	Speed controller encoderless operation integral time / n_ctrl SLVC Tn		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6040, 6050
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 100000.0 [ms]	Factory setting 20.0 [ms]
Description:	Set the integral time for encoderless operation for the speed controller.		
p1475[0...n]	CI: Speed controller torque setting value for motor holding brake / n_ctrl M_sv MHB		
VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 6040
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source for the torque setting value when starting up with motor holding brake.		
Dependency:	The switching in of the torque setting value for the motor holding brake has a higher priority than the setting of the integrator value using p1477 and p1478.		
Note:	The setting of the integral output of the speed controller begins after magnetizing (see p0346, r0056 bit 4) and ends at the end of the brake control opening time p1216. A setting value of zero means that no setting procedure will take place.		

p1476[0...n]	BI: Speed controller hold integrator / n_ctrl integ stop		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2520, 5040, 5042, 5210, 6040
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to hold the integrator for the speed controller.		
p1476[0...n]	BI: Velocity controller hold integrator / v_ctrl integ stop		
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2520, 5040, 5042, 5210
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to hold the integrator for the velocity controller.		
p1477[0...n]	BI: Speed controller set integrator value / n_ctrl integ set		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2520, 5040, 5042, 5210, 6040
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to set the integrator setting value (p1478).		
Dependency:	Refer to: p1478, p1479		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For the interface mode "SIMODRIVE 611 universal" (p2038 = 1), p1477 and p1478 are used for the signal STW2.6 (integrator inhibit, speed controller).		
p1477[0...n]	BI: Velocity controller set integrator value / v_ctrl integ set		
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2520, 5040, 5042, 5210
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to set the integrator setting value (p1478).		
Dependency:	Refer to: p1478, p1479		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For the interface mode "SIMODRIVE 611 universal" (p2038 = 1), p1477 and p1478 are used for the signal STW2.6 (integrator inhibit, speed controller).		

p1478[0...n]	CI: Speed controller integrator setting value / n_ctr integ_setVal		
SERVO	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 5040, 5042, 5210 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the integrator setting value for the velocity controller. The signal to set this integrator setting value is interconnected via p1477.		
Dependency:	Refer to: p1477, p1479		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For the interface mode "SIMODRIVE 611 universal" (p2038 = 1), p1477 and p1478 are used for the signal STW2.6 (integrator inhibit, speed controller).		
p1478[0...n]	CI: Velocity controller integrator value / v_ctr integ_setVal		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 5040, 5042, 5210 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the integrator setting value for the velocity controller. The signal to set this integrator setting value is interconnected via p1477.		
Dependency:	Refer to: p1477, p1479		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For the interface mode "SIMODRIVE 611 universal" (p2038 = 1), p1477 and p1478 are used for the signal STW2.6 (integrator inhibit, speed controller).		
p1478[0...n]	CI: Speed controller integrator setting value / n_ctr integ_setVal		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 6040 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the integrator setting value for the velocity controller. The signal to set this integrator setting value is interconnected via p1477.		
Dependency:	The setting value of the speed controller integrator is weighted with the scaling factor of the signal source in p1479. If p1478 is interconnected to the integral output of the speed controller (r1482), then after the magnetizing time (r0346) and if the speed controller is enabled, the integral component of the controller is set to the last value before the pulse inhibit. This value is set if no setting command (p1477) is interconnected or, at the instant that the pulses were inhibited, a setting command is available, which is not deactivated up to the next time that the pulses are inhibited. For encoderless vector control, in addition p1400.1 should be set to 1 so that when the drive is stopped, the integral component of the speed controller is not controlled down to zero. In order that when setting the integrator output, only the static torque is detected, we recommend that the accelerating torque is completely pre-controlled (e.g. p1496). If p1478 is interconnected to another output other than r1482, then after magnetizing and speed controller enable, the integral output is set once if the setting command is not interconnected (p1477 = 0). Refer to: p1477, p1479		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		

p1479[0...n]	CI: Speed controller integrator setting value scaling / n_ctrl I_val scal		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 6040 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for scaling the integrator setting value (p1478) of the speed controller.		
Dependency:	Refer to: p1477, p1478		
r1480	CO: Speed controller PI torque output / n_ctrl PI-M_output		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 1590, 5040, 5042, 5060, 5210, 6060 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the torque setpoint at the output of the PI speed controller.		
r1480	CO: Velocity controller PI force output / v_ctrl PI-F_output		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: 1590, 5040, 5042, 5060, 5210 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the force setpoint at the output of the PI velocity controller.		
r1481	CO: Speed controller P torque output / n_ctrl P-M_output		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 5040, 5042, 5210, 6040 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the torque setpoint at the output of the P speed controller.		
r1481	CO: Velocity controller P force output / v_ctrl P-F_output		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: 5040, 5042, 5210 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the force setpoint at the output of the P velocity controller.		

r1482 CO: Speed controller I torque output / n_ctrl I-M_outputSERVO, VECTOR
(n/M)**Can be changed:** -**Data type:** FloatingPoint32**Calculated:** -**Dynamic index:** -**Access level:** 3**Func. diagram:** 5040, 5042, 5210, 6030, 6040**P-Group:** Closed-loop control**Units group:** 7_1**Unit selection:** p0505**Not for motor type:** REL**Scaling:** p2003**Expert list:** 1**Min**
- [Nm]**Max**
- [Nm]**Factory setting**
- [Nm]**Description:** Displays the torque setpoint at the output of the I speed controller.**r1482 CO: Velocity controller I force output / v_ctrl I-F_output**

SERVO (Lin)

Can be changed: -**Data type:** FloatingPoint32**Calculated:** -**Dynamic index:** -**Access level:** 3**Func. diagram:** 5040, 5042, 5210**P-Group:** Closed-loop control**Units group:** 8_1**Unit selection:** p0505**Not for motor type:** REL**Scaling:** p2003**Expert list:** 1**Min**
- [N]**Max**
- [N]**Factory setting**
- [N]**Description:** Displays the force setpoint at the output of the I velocity controller.**p1486[0...n] CI: Droop compensation torque / Droop M_comp**

VECTOR (n/M)

Can be changed: T**Data type:** Unsigned32 / FloatingPoint32**Calculated:** -**Dynamic index:** CDS, p0170**Access level:** 3**Func. diagram:** 6030**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** p2003**Expert list:** 1**Min**
-**Max**
-**Factory setting**
0**Description:** Sets the signal source for the compensation torque to be output within the droop calculation.**p1487[0...n] Droop compensation torque scaling / Droop M_comp scal**

VECTOR (n/M)

Can be changed: U, T**Data type:** FloatingPoint32**Calculated:** -**Dynamic index:** DDS, p0180**Access level:** 3**Func. diagram:** 6030**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** PERCENT**Expert list:** 1**Min**
-2000.0 [%]**Max**
2000.0 [%]**Factory setting**
100.0 [%]**Description:** Sets the scaling for the compensation torque within the droop calculation.**p1488[0...n] Droop input source / Droop input source**

VECTOR (n/M)

Can be changed: U, T**Data type:** Integer16**Calculated:** -**Dynamic index:** DDS, p0180**Access level:** 3**Func. diagram:** 6030**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min**
0**Max**
3**Factory setting**
0**Description:** Sets the source for droop feedback.**Value:**
0: Droop feedback not connected
1: Droop from torque setpoint
2: Droop from speed controller output
3: Droop from integral output, speed controller**Dependency:** Refer to: p1489, r1490, p1492

p1489[0...n]	Droop feedback scaling / Droop scaling		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 0.500	Access level: 3 Func. diagram: 6030 Unit selection: - Expert list: 1 Factory setting 0.050
Description:	Sets the scaling for the droop feedback		
Dependency:	Refer to: p1488, r1490, p1492		
Note:	Example: A value of 0.05 means that for a torque equal to the rated motor torque, the rated motor speed is reduced by 5 %.		
r1490	CO: Droop feedback speed reduction / Droop n_reduction		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 6030 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the output signal of the droop calculation. The droop feedback result is subtracted from the speed setpoint when activated (p1492).		
Dependency:	Refer to: p1488, p1489, p1492		
p1492[0...n]	BI: Droop feedback enable / Droop enable		
VECTOR (n/M)	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2520, 6030 Unit selection: - Expert list: 1 Factory setting 0
Description:	Enables the droop to be applied to the speed/velocity setpoint.		
Dependency:	Refer to: p1488, p1489, r1490		
Note:	Even when not enabled, the droop speed is calculated but not subtracted from the setpoint speed. This makes it possible to subtract the result of this calculation from the speed of another drive.		
r1493	CO: Moment of inertia, total / M_inertia total		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [kgm²]	Calculated: - Dynamic index: - Units group: 25_1 Scaling: - Max - [kgm²]	Access level: 3 Func. diagram: 5042, 5210 Unit selection: p0100 Expert list: 1 Factory setting - [kgm²]
Description:	Displays the parameterized total moment of inertia ((p0341 * p0342) + p1498) without evaluation by the scaling via p1497.		
Dependency:	Refer to: p1300, p1402, p1404, p1497		
Note:	The parameterized total moment of inertia, taking into account p1497, influences the torque pre-control. In encoderless operation or when the torque-speed pre-control with encoder (p1402.4 = 1) is activated, then torque-speed pre-control is activated.		

r1493	CO: Load mass / Load mass		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [kg]	Calculated: - Dynamic index: - Units group: 27_1 Scaling: - Max - [kg]	Access level: 3 Func. diagram: 5042, 5210 Unit selection: p0100 Expert list: 1 Factory setting - [kg]
Description:	Displays the parameterized total weight ((p0341 * p0342) + p1498) without evaluation by the scaling via p1497.		
Dependency:	Refer to: p1300, p1402, p1404, p1497		
Note:	The parameterized total weight, taking into account p1497, influences the force-velocity pre-control in encoderless operation or when activated, force-velocity pre-control with encoder (p1402.4).		
r1493	CO: Moment of inertia, total / M_inertia total		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [kgm ²]	Calculated: - Dynamic index: - Units group: 25_1 Scaling: - Max - [kgm ²]	Access level: 3 Func. diagram: 6031 Unit selection: p0100 Expert list: 1 Factory setting - [kgm ²]
Description:	Displays the parameterized total moment of inertia ((p0341 * p0342) + p1496) without evaluation by the scaling via p1497.		
p1494[0...n]	Speed controller integrator feedback time constant / n_ctr integ_fdbk T		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000.00 [ms]	Access level: 2 Func. diagram: 5040, 5042, 5210 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the time constant of the PT1 filter for integrator feedback. The integrator of the speed/velocity controller is re-parameterized to become a PT1 filter through a feedback element (1st Order low pass filter characteristics). The following applies: p1494 < 0.25 (2 * p0115[1]) --> the PT1 filter is not active - the pure integrator is effective. p1494 >= 0.25 (2 * p0115[1]) --> the PT1 filter is active and has replaced the pure integrator.		
Note:	Applications: Motion at zero setpoint and dominant stiction can be suppressed but this has a negative impact on the remaining setpoint-actual value difference. This can be used, for example, to avoid oscillation of a position-controlled axis at standstill (stick-slip effect) or overshoot when traversing (moving) in micrometer steps. Also prevents tension/stressing for axes that are mechanically and rigidly coupled with one another (e.g. for synchronous spindles, master - slave axes).		

p1494[0...n]	Velocity controller integrator feedback time constant / v_ctr integ_fdbk T		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000.00 [ms]	Access level: 2 Func. diagram: 5040, 5042, 5210 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the time constant of the PT1 filter for integrator feedback. The integrator of the speed/velocity controller is re-parameterized to become a PT1 filter through a feedback element (1st Order low pass filter characteristics). The following applies: p1494 < 0.25 (2 * p0115[1]) --> the PT1 filter is not active - the pure integrator is effective. p1494 >= 0.25 (2 * p0115[1]) --> the PT1 filter is active and has replaced the pure integrator.		
Note:	Applications: Motion at zero setpoint and dominant stiction can be suppressed but this has a negative impact on the remaining setpoint-actual value difference. This can be used, for example, to avoid oscillation of a position-controlled axis at standstill (stick-slip effect) or overshoot when traversing (moving) in micrometer steps. Also prevents tension/stressing for axes that are mechanically and rigidly coupled with one another (e.g. for synchronous spindles, master - slave axes).		
p1495[0...n]	CI: Acceleration pre-control / a_prectrl		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2007 Max -	Access level: 3 Func. diagram: 6031 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the acceleration pre-control.		
Dependency:	The signal source for the acceleration is activated with p1400.2 = 1. For p1400.2 = 0, the acceleration pre-control is calculated from the speed setpoint change from r0062. For p1400.2 = 0 and activate reference model (p1400.3 = 1) the acceleration pre-control is switched out. Refer to: p1400, p1496		
Note:	If the acceleration is entered as external signal, then the accelerating torque is calculated as follows (r1518): $r1518 = \text{acceleration (\% of p2007)} / 100 \% * (p2007 * 60 \text{ s}) / p0311 * r0345 / 1 \text{ s} * r0333$		
p1496[0...n]	Acceleration pre-control scaling / a_before scaling		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000.0 [%]	Access level: 3 Func. diagram: 1700, 6031 Unit selection: - Expert list: 1 Factory setting 0.0 [%]
Description:	Sets the scaling for the acceleration pre-control of the speed/velocity controller.		
Dependency:	When the reference model is activated (p1400.3 = 1) and for an internal acceleration pre-control (p1400.2 = 0), the acceleration pre-control is switched out (disabled). The reference model (p1400.3 = 1) and external acceleration pre-control (p1400.2 = 1) can be operated together. Refer to: p0341, p0342		
Note:	The parameter is set to 100% by the rotating measurement (refer to p1960). The acceleration pre-control may not be used if the speed setpoint manifests significant ripple (e.g. analog setpoint) and the rounding-off in the speed ramp-function generator is disabled. We also recommend that the pre-control mode is not used if there is gearbox backlash.		

p1497[0...n]	CI: Moment of inertia, scaling / M_mom inert scal		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 5042, 5210, 6030, 6031 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for scaling the moment of inertia.		
p1497[0...n]	CI: Motor weight scaling / Motor weight scal		
SERVO (Lin)	Can be changed: U, T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 5042, 5210 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for scaling the motor weight.		
p1498[0...n]	Load moment of inertia / Load mom of inert		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00000 [kgm²]	Calculated: - Dynamic index: DDS, p0180 Units group: 25_1 Scaling: - Max 100000.00000 [kgm²]	Access level: 3 Func. diagram: 5042, 5210 Unit selection: p0100 Expert list: 1 Factory setting 0.00000 [kgm²]
Description:	Sets the load moment of inertia.		
Note:	(p0341 * p0342) + p1498 influence the speed/torque pre-control in encoderless operation.		
p1498[0...n]	Load mass / Load mass		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00000 [kg]	Calculated: - Dynamic index: DDS, p0180 Units group: 27_1 Scaling: - Max 10000.00000 [kg]	Access level: 3 Func. diagram: 5042, 5210 Unit selection: p0100 Expert list: 1 Factory setting 0.00000 [kg]
Description:	Sets the load mass.		
Note:	(p0341 * p0342) + p1498 influence the velocity/force pre-control in encoderless operation.		
p1499[0...n]	Accelerating for torque control, scaling / a for M_ctrl scal		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 400.0 [%]	Access level: 3 Func. diagram: 6030 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the scaling for the acceleration integrator at low speeds (only for encoderless torque control).		
Dependency:	Refer to: p0341, p0342		

p1500[0...n]	Macro Connector Inputs (CI) for torque setpoints / Macro CI M_set		
SERVO, VECTOR	Can be changed: C2(1), T Data type: Unsigned32 P-Group: Commands Not for motor type: REL Min 0	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max 999999	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Runs the corresponding macro files. The Connector Inputs (CI) for the torque setpoints of the appropriate Command Data Set (CDS) are appropriately interconnected. The selected macro file must be available on the memory card/device memory. Example: p1500 = 6 --> the macro file PM000006.ACX is run.		
Dependency:	Refer to: p0015, p0700, p1000, r8573		
Notice:	No errors were issued during quick commissioning (p3900 = 1) when writing to parameters of the QUICK_IBN group!		
Note:	The macros in the specified directory are displayed in r8573. r8573 is not in the expert list of the commissioning software. Macros available as standard are described in the technical documentation of the particular product. CI: Connector Input		
p1500[0...n]	Macro Connector Inputs (CI) for force setpoints / Macro CI F_set		
SERVO (Lin)	Can be changed: C2(1), T Data type: Unsigned32 P-Group: Commands Not for motor type: REL Min 0	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max 999999	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Runs the corresponding macro files. The connector inputs (CI) for the force setpoints of the appropriate command data set (CDS) are appropriately interconnected. The selected macro file must be available on the memory card/device memory. Example: p1500 = 6 --> the macro file PM000006.ACX is run.		
Dependency:	Refer to: p0015, p0700, p1000, r8573		
Notice:	No errors were issued during quick commissioning (p3900 = 1) when writing to parameters of the QUICK_IBN group!		
Note:	The macros in the specified directory are displayed in r8573. r8573 is not in the expert list of the commissioning software. Macros available as standard are described in the technical documentation of the particular product. CI: Connector Input		
p1501[0...n]	BI: Change over between closed-loop speed/torque control / Changeov n/M_ctrl		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 1700, 2520, 5060, 6060 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for toggling between speed and torque control.		
Dependency:	The input connectors to enter the torque are provided using p1511, p1512 and p1513. Refer to: p1300		

- Caution:** If the closed-loop torque control is not activated (p1300) and a change is made to closed-loop torque control (p1501), OFF1 (p0840) does not have its own braking response but pulse suppression when standstill is detected (p1226, p1227).
- Notice:** The parameter may be protected as a result of p0922 or p2079 and cannot be changed.
- Note:** 0 signal: Closed-loop speed control
1 signal: Closed-loop torque control

p1501[0...n] BI: Change over velocity/force control / Changeov n/F_ctrl

SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2520, 5060
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for toggling between velocity and force control.

Dependency: The input connectors to enter the force are provided using p1511, p1512 and p1513.
Refer to: p1300

Caution: If the closed-loop force control is not activated (p1300) and a change is made to closed-loop force control (p1501), OFF1 (p0840) does not have its own braking response but pulse suppression when standstill is detected (p1226, p1227).

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: 0 signal: Velocity control
1 signal: Force control

p1503[0...n] CI: Torque setpoint / M_set

VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 1700, 6060
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the torque setpoint for torque control.

Note: A change is made to closed-loop torque control if, in p1300, closed-loop torque control was selected or if the selection was made using the changeover source in p1501.
it is also possible to change over in operation using p1501.

r1508 CO: Torque setpoint before supplementary torque / M_set bef. M_suppl

VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 6030, 6060, 6722
	P-Group: Closed-loop control	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min	Max	Factory setting
	- [Nm]	- [Nm]	- [Nm]

Description: Displays the torque setpoint before entering the supplementary torque.

For closed-loop speed control, r1508 corresponds to the speed controller output; for closed-loop torque control, r1508 corresponds to the torque setpoint of the signal source assigned in p1503.

r1509	CO: Torque setpoint before torque limiting / M_set before M_lim		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 1590, 5060, 5610 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the total torque setpoint before the torque limiting (total of the controller output, supplementary torque and if required, the pre-control torque, encoderless operation). In the closed-loop speed controlled mode, r1509 = p1480 + r1515 + pre-controlled torque, encoderless operation. r1509 and r1515 are identical for the closed-loop torque control.		
r1509	CO: Force setpoint before force limiting / F_set before F_lim		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: 1590, 5060, 5610 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the total torque setpoint before the force limiting (total of the controller output, supplementary force and if required, the pre-control force, encoderless operation). In the closed-loop speed controlled mode, r1509 = p1480 + r1515 + pre-control force, encoderless operation. r1509 and r1515 are identical for the closed-loop force control.		
p1511[0...n]	CI: Supplementary torque 1 / M_suppl 1		
SERVO, VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 5060, 6060 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for supplementary torque 1.		
p1511[0...n]	CI: Supplementary force 1 / F_suppl 1		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 5060, 6060 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for supplementary force 1.		
p1512[0...n]	CI: Supplementary torque 1 scaling / M_suppl 1 scal		
SERVO, VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 5060, 6060 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for scaling the supplementary torque 1.		

p1512[0...n]	CI: Supplementary force 1 scaling / F_suppl 1 scal		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 5060, 6060 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for scaling the supplementary force 1.		
p1513[0...n]	CI: Supplementary torque 2 / M_suppl 2		
SERVO, VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 5060, 6060 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for supplementary torque 2.		
p1513[0...n]	CI: Supplementary force 2 / F_suppl 2		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 5060, 6060 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for supplementary force 2.		
p1514[0...n]	Supplementary torque 2 scaling / M_suppl 2 scal		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -2000.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: PERCENT Max 2000.0 [%]	Access level: 3 Func. diagram: 1700, 6060 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the scaling for supplementary torque 2.		
r1515	Supplementary torque total / M_suppl total		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 2 Func. diagram: 5040, 5060 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the total supplementary torque. The displayed value is the total of supplementary torque values 1 and 2 (p1511, p1512, p1513, p1514).		

r1515 Supplementary force total / F_suppl total

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5040, 5060
	P-Group: Closed-loop control	Units group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min - [N]	Max - [N]	Factory setting - [N]
Description: Displays the total supplementary force. The displayed value is the total of supplementary forces 1 and 2 (p1511, p1512, p1513, p1514).			

r1516 CO: Supplementary torque and acceleration torque / M_suppl + M_accel

VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 6060
	P-Group: Closed-loop control	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min - [Nm]	Max - [Nm]	Factory setting - [Nm]
Description: Displays the total supplementary torque and the accelerating torque. The displayed value is the total of the smoothed supplementary torque and the accelerating torque (p1516 = p1518[1] + r1515).			

p1517[0...n] Accelerating torque smoothing time constant / M_accel T_smooth

SERVO, VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5042, 5210, 6060
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 100.00 [ms]	Factory setting 4.00 [ms]
Description: Sets the smoothing time constant of the accelerating torque.			
Note: For servo drives, the parameter is only effective in encoderless operation. For vector drives, the acceleration pre-control is inhibited if the smoothing is set to the maximum value.			

p1517[0...n] Acceleration force smoothing time constant / F_accel T_smooth

SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5042, 5210
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 100.00 [ms]	Factory setting 4.00 [ms]
Description: Sets the smoothing time constant of the accelerating force.			
Note: For servo drives, the parameter is only effective in encoderless operation.			

r1518[0...1] CO: Accelerating torque / M_accel

SERVO	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5042, 5210
	P-Group: Closed-loop control	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min - [Nm]	Max - [Nm]	Factory setting - [Nm]
Description: Displays the accelerating torque to pre-control the speed controller for torque-speed pre-control (p1402.4 = 1 or in encoderless operation).			

Index: [0] = Unsmoothed
[1] = Smoothed
Dependency: Refer to: p0341, p0342, p1300, p1402, r1493, p1497, p1498

r1518[0...1]	CO: Accelerating force / F_accel		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: 5042, 5210 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the accelerating torque to pre-control the velocity controller for force- velocity pre-control (p1402.4 = 1 or in encoderless operation).		
Index:	[0] = Unsmoothed [1] = Smoothed		
Dependency:	Refer to: p0341, p0342, p1300, p1402, r1493, p1497, p1498		

r1518[0...1]	CO: Accelerating torque / M_accel		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 6060 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the accelerating torque for pre-control of the speed controller.		
Index:	[0] = Unsmoothed [1] = Smoothed		
Dependency:	Refer to: p0341, p0342, p1496		

p1520[0...n]	CO: Torque limit upper/motoring / M_max upper/mot		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -1000000.00 [Nm]	Calculated: CALC_MOD_LIM_REF Dynamic index: DDS, p0180 Units group: 7_1 Scaling: p2003 Max 20000000.00 [Nm]	Access level: 2 Func. diagram: 5620, 5630, 6630 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Nm]
Description:	Sets the fixed upper torque limit or the torque limit when motoring.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p0500, p1521, p1522, p1523, p1532, r1538, r1539		
Danger:	For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Negative values when setting the upper torque limit (p1520 < 0) can result in the motor accelerating in an uncontrollable fashion.		
			
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	For VECTOR the following applies (p0107): The torque limit is limited to 400% of the rated motor torque. When automatically calculating the motor/closed-loop control parameters (p0340), the torque limit is set to match the current limit (p0640).		

p1520[0...n]	CO: Force limit upper/motoring / F_max upper/mot		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5620, 5630, 6630
	P-Group: Closed-loop control	Units group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min -1000000.00 [N]	Max 20000000.00 [N]	Factory setting 0.00 [N]
Description:	Sets the fixed upper or force limit when motoring.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p0500, p1521, p1522, p1523, p1532, r1538, r1539		
Danger:	For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Negative values when setting the upper torque limit (p1520 < 0) can result in the motor accelerating in an uncontrollable fashion.		
			
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	For VECTOR the following applies (p0107): The torque limit is limited to 400% of the rated motor torque. When automatically calculating the motor/closed-loop control parameters (p0340), the torque limit is set to match the current limit (p0640).		
p1520[0...n]	CO: Torque limit upper / M_max upper		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6630
	P-Group: Closed-loop control	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min -1000000.00 [Nm]	Max 20000000.00 [Nm]	Factory setting 0.00 [Nm]
Description:	Sets the fixed, upper torque limit.		
Dependency:	Refer to: p1521, p1522, p1523, r1538, r1539		
Danger:	Negative values when setting the upper torque limit (p1520 < 0) can result in the motor accelerating in an uncontrollable fashion.		
			
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	The torque limit is limited to 400% of the rated motor torque. When automatically calculating the motor/closed-loop control parameters (p0340), the torque limit is set to match the current limit (p0640).		
p1521[0...n]	CO: Torque limit lower/regenerative / M_max lower/regen		
SERVO	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5620, 5630, 6630
	P-Group: Closed-loop control	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min -20000000.00 [Nm]	Max 1000000.00 [Nm]	Factory setting 0.00 [Nm]
Description:	Sets the fixed lower torque limit or the torque limit when regenerating.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p0500, p1520, p1522, p1523, p1532		

Danger:



Notice:

Note:

For p1400.4 = 0 (torque limiting, upper/lower) the following applies:

Positive values when setting the lower torque limit (p1521 > 0) can result in the motor accelerating in an uncontrollable fashion.

A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

For VECTOR the following applies (p0107):

The torque limit is limited to 400% of the rated motor torque. When automatically calculating the motor/closed-loop control parameters (p0340), the torque limit is set to match the current limit (p0640).

p1521[0...n]

SERVO (Lin)

CO: Force limit lower/regenerative / F_max lower/regen

Can be changed: U, T

Calculated:
CALC_MOD_LIM_REF

Access level: 2

Data type: FloatingPoint32

Dynamic index: DDS, p0180

Func. diagram: 5620, 5630, 6630

P-Group: Closed-loop control

Units group: 8_1

Unit selection: p0505

Not for motor type: REL

Scaling: p2003

Expert list: 1

Min
-20000000.00 [N]

Max
1000000.00 [N]

Factory setting
0.00 [N]

Description:

Sets the fixed lower or force limit when regenerating.

Dependency:

p1400 bit 4 = 0: Upper / lower

p1400 bit 4 = 1: Motoring / regenerative

Refer to: p0500, p1520, p1522, p1523, p1532

Danger:



Notice:

Note:

For p1400.4 = 0 (torque limiting, upper/lower) the following applies:

Positive values when setting the lower torque limit (p1521 > 0) can result in the motor accelerating in an uncontrollable fashion.

A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

For VECTOR the following applies (p0107):

The torque limit is limited to 400% of the rated motor torque. When automatically calculating the motor/closed-loop control parameters (p0340), the torque limit is set to match the current limit (p0640).

p1521[0...n]

VECTOR (n/M)

CO: Torque limit lower / M_max lower

Can be changed: U, T

Calculated:
CALC_MOD_LIM_REF

Access level: 2

Data type: FloatingPoint32

Dynamic index: DDS, p0180

Func. diagram: 5620, 5630, 6630

P-Group: Closed-loop control

Units group: 7_1

Unit selection: p0505

Not for motor type: REL

Scaling: p2003

Expert list: 1

Min
-20000000.00 [Nm]

Max
1000000.00 [Nm]

Factory setting
0.00 [Nm]

Description:

Sets the fixed, lower torque limit.

Dependency:

Refer to: p1520, p1522, p1523, p1532

Danger:



Notice:

Note:

A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

The torque limit is limited to 400% of the rated motor torque. When automatically calculating the motor/closed-loop control parameters (p0340), the torque limit is set to match the current limit (p0640).

p1522[0...n]	CI: Torque limit upper/motoring / M_max upper/mot		
SERVO	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 1610, 5620, 5630, 6630 Unit selection: - Expert list: 1 Factory setting 1520[0]
Description:	Sets the signal source for the upper or torque/force limit when motoring.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p1520, p1521, p1523, p1532		
Danger:	For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Negative values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.		
			
p1522[0...n]	CI: Force limit upper/motoring / F_max upper/mot		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 1610, 5620, 5630, 6630 Unit selection: - Expert list: 1 Factory setting 1520[0]
Description:	Sets the signal source for the upper or torque/force limit when motoring.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p1520, p1521, p1523, p1532		
Danger:	For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Negative values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.		
			
p1522[0...n]	CI: Torque limit upper / M_max upper		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 6630 Unit selection: - Expert list: 1 Factory setting 1520[0]
Description:	Sets the signal source for the upper torque limit.		
Dependency:	Refer to: p1520, p1521, p1523		
Danger:	Negative values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.		
			

p1523[0...n]	CI: Torque limit lower/regenerative / M_max lower/regen		
SERVO	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 1610, 5620, 5630, 6630 Unit selection: - Expert list: 1 Factory setting 1521[0]
Description:	Sets the signal source for the lower or torque/force limit when regenerating.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p1520, p1521, p1522, p1532		
Danger:	For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Positive values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.		
			
p1523[0...n]	CI: Force limit lower/regenerative / F_max lower/regen		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 1610, 5620, 5630, 6630 Unit selection: - Expert list: 1 Factory setting 1521[0]
Description:	Sets the signal source for the lower or torque/force limit when regenerating.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p1520, p1521, p1522, p1532		
Danger:	For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Positive values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.		
			
p1523[0...n]	CI: Torque limit lower / M_max lower		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 3 Func. diagram: 6630 Unit selection: - Expert list: 1 Factory setting 1521[0]
Description:	Sets the signal source for the lower torque limit.		
Dependency:	Refer to: p1520, p1521, p1522		
Danger:	Positive values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.		
			

p1524[0...n]	CO: Torque limit upper/motoring scaling / M_max up/mot scal		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -2000.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: PERCENT Max 2000.0 [%]	Access level: 3 Func. diagram: 5620, 5630, 6630 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the scaling for the upper torque limit or the torque limit when motoring.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	This parameter can be freely interconnected. The value has the meaning stated above if it is interconnected from connector input p1528.		
p1524[0...n]	CO: Force limit upper/motoring scaling / F_max up/mot scal		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -2000.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: PERCENT Max 2000.0 [%]	Access level: 3 Func. diagram: 5620, 5630, 6630 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the scaling for the upper force limit or the force limit when motoring.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	This parameter can be freely interconnected. The value has the meaning stated above if it is interconnected from connector input p1528.		
p1524[0...n]	CO: Torque limit upper scaling / M_max upper scal		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -2000.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: PERCENT Max 2000.0 [%]	Access level: 3 Func. diagram: 5620, 5630, 6630 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the scaling for the upper torque limit.		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	This parameter can be freely interconnected. The value has the meaning stated above if it is interconnected from connector input p1528.		

p1525[0...n]	CO: Torque limit lower/regenerative scaling / M_max low/gen scal		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5620, 5630, 6630
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min -2000.0 [%]	Max 2000.0 [%]	Factory setting 100.0 [%]
Description:	Sets the scaling for the lower torque limit or the torque limit when regenerating.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	This parameter can be freely interconnected. The value has the meaning stated above if it is interconnected from connector input p1528.		
p1525[0...n]	CO: Force limit lower/regenerative scaling / F_max lo/reg scal		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5620, 5630, 6630
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min -2000.0 [%]	Max 2000.0 [%]	Factory setting 100.0 [%]
Description:	Sets the scaling for the lower force limit or the force limit when regenerating.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	This parameter can be freely interconnected. The value has the meaning stated above if it is interconnected from connector input p1528.		
p1525[0...n]	CO: Torque limit lower scaling / M_max lower scal		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6630
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min -2000.0 [%]	Max 2000.0 [%]	Factory setting 100.0 [%]
Description:	Sets the scaling for the lower torque limit.		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	This parameter can be freely interconnected. The value has the meaning stated above if it is interconnected from connector input p1528.		

r1526	CO: Torque limit upper/motoring without offset / M_max up w/o offs		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 5620, 5630, 6630, 6640 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the upper torque limit of all torque limits without offset.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p1520, p1521, p1522, p1523, p1528, p1529		
r1526	CO: Force limit upper/motoring without offset / F_max up w/o offs		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: 5620, 5630, 6630, 6640 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the upper force limit of all force limits without offset.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p1520, p1521, p1522, p1523, p1528, p1529		
r1526	CO: Torque limit upper without offset / M_max up w/o offs		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 5620, 5630, 6630, 6640 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the upper torque limit of all torque limits without offset.		
Dependency:	Refer to: p1520, p1521, p1522, p1523, p1528, p1529		
r1527	CO: Torque limit lower/regenerative without offset / M_max low w/o offs		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 5620, 5630, 6630, 6640 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the lower torque limit of all torque limits without offset.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p1520, p1521, p1522, p1523, p1528, p1529		

r1527	CO: Force limit lower/regenerative without offset / F_max low w/o offs		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: 5620, 5630, 6630, 6640 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the lower force limit of all force limits without offset.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative Refer to: p1520, p1521, p1522, p1523, p1528, p1529		
r1527	CO: Torque limit lower without offset / M_max low w/o offs		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 5620, 5630, 6630, 6640 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the lower torque limit of all torque limits without offset.		
Dependency:	Refer to: p1520, p1521, p1522, p1523, p1528, p1529		
p1528[0...n]	CI: Torque limit upper/motoring scaling / M_max up/mot scal		
SERVO	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 1610, 3617, 5620, 5630 Unit selection: - Expert list: 1 Factory setting 1524[0]
Description:	Sets the signal source for the scaling of the upper or motoring torque limit in p1522.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative		
Danger:	For p1400.4 = 0 (torque limiting, upper/lower) the following applies:		
	Negative values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
p1528[0...n]	CI: Force limit upper/motoring scaling / F_max up/mot scal		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 1610, 3617, 5620, 5630 Unit selection: - Expert list: 1 Factory setting 1524[0]
Description:	Sets the signal source for the scaling of the upper or motoring force limit in p1522.		
Dependency:	p1400 bit 4 = 0: Upper / lower p1400 bit 4 = 1: Motoring / regenerative		

Danger:

For p1400.4 = 0 (torque limiting, upper/lower) the following applies:

Negative values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.

Notice:

The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p1528[0...n]**CI: Torque limit upper scaling / M_max upper scal**

VECTOR (n/M)

Can be changed: T**Calculated:** -**Access level:** 3**Data type:** Unsigned32 / FloatingPoint32**Dynamic index:** CDS, p0170**Func. diagram:** 6630**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** PERCENT**Expert list:** 1**Min****Max****Factory setting**

-

-

1524[0]

Description:

Sets the signal source for the scaling of the upper torque limit in p1522.

Danger:

For p1400.4 = 0 (torque limiting, upper/lower) the following applies:

Negative values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.

Notice:

The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p1529[0...n]**CI: Torque limit lower/regenerative scaling / M_max low/gen scal**

SERVO

Can be changed: T**Calculated:** -**Access level:** 3**Data type:** Unsigned32 / FloatingPoint32**Dynamic index:** CDS, p0170**Func. diagram:** 1610, 3617, 5620, 5630**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** PERCENT**Expert list:** 1**Min****Max****Factory setting**

-

-

1525[0]

Description:

Sets the signal source for the scaling of the lower torque limit or the regenerative torque limit in p1523.

Dependency:

p1400 bit 4 = 0: Upper / lower

p1400 bit 4 = 1: Motoring / regenerative

Danger:

For p1400.4 = 0 (torque limiting, upper/lower) the following applies:

Positive values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.

Notice:

The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p1529[0...n]**CI: Force limit lower/regenerative scaling / F_max lo/reg scal**

SERVO (Lin)

Can be changed: T**Calculated:** -**Access level:** 3**Data type:** Unsigned32 / FloatingPoint32**Dynamic index:** CDS, p0170**Func. diagram:** 1610, 3617, 5620, 5630**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** PERCENT**Expert list:** 1**Min****Max****Factory setting**

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1525[0]

Description:

Sets the signal source for the scaling of the lower force limit or the regenerative force limit in p1523.

Dependency:

p1400 bit 4 = 0: Upper / lower

p1400 bit 4 = 1: Motoring / regenerative

Danger:

For p1400.4 = 0 (torque limiting, upper/lower) the following applies:

Positive values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.

Notice:

The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p1529[0...n]			
VECTOR (n/M)	Cl: Torque limit lower scaling / M_max lower scal		
	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 6630
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	1525[0]
Description:	Sets the signal source for the scaling of the lower torque limit in p1523.		
Danger:	For p1400.4 = 0 (torque limiting, upper/lower) the following applies:		
	Positive values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		

p1530[0...n]			
SERVO, VECTOR (n/M)	Power limit motoring / P_max mot		
	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5640, 6640
	P-Group: Closed-loop control	Units group: 14_5	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [kW]	100000.00 [kW]	0.00 [kW]
Description:	Sets the power limit when motoring.		
Dependency:	Refer to: p0500, p1531		
Note:	For VECTOR the following applies (p0107):		
	The power limit is limited to 300% of the rated motor power.		

p1530[0...n]			
SERVO (Lin)	Power limit motoring / P_max mot		
	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5640, 6640
	P-Group: Closed-loop control	Units group: 14_8	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [kW]	100000.00 [kW]	0.00 [kW]
Description:	Sets the power limit when motoring.		
Dependency:	Refer to: p0500, p1531		
Note:	For VECTOR the following applies (p0107):		
	The power limit is limited to 300% of the rated motor power.		

p1531[0...n]			
SERVO	Power limit regenerative / P_max gen		
	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5640, 6640
	P-Group: Closed-loop control	Units group: 14_5	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-100000.00 [kW]	-0.01 [kW]	-0.01 [kW]
Description:	Sets the regenerative power limit.		
Dependency:	Refer to: p0500, p1530		

p1531[0...n]	Power limit regenerative / P_max gen		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5640, 6640
	P-Group: Closed-loop control	Units group: 14_8	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -100000.00 [kW]	Max -0.01 [kW]	Factory setting -0.01 [kW]
Description:	Sets the regenerative power limit.		
Dependency:	Refer to: p0500, p1530		
p1531[0...n]	Power limit regenerative / P_max gen		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6640
	P-Group: Closed-loop control	Units group: 14_5	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -100000.00 [kW]	Max -0.01 [kW]	Factory setting -0.01 [kW]
Description:	Sets the regenerative power limit.		
Dependency:	Refer to: p0500, p1530		
Note:	The power limit is limited to 300% of the rated motor power. For power units without regenerative feedback into the line supply, the regenerative power limit is preset to 30% of the motoring power limit p1530 and in the ratio rated drive converter power to rated motor power.		
p1532[0...n]	CO: Torque limit offset / M_max offset		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5620, 5630, 5650, 7010, 8012
	P-Group: Closed-loop control	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min -100000.00 [Nm]	Max 100000.00 [Nm]	Factory setting 0.00 [Nm]
Description:	Sets the torque offset for the torque limit.		
Dependency:	Refer to: p1520, p1521, p1522, p1523, p1528, p1529		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p1532[0...n]	CO: Force offset, force limit / F_max offset		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5620, 5630, 5650, 7010, 8012
	P-Group: Closed-loop control	Units group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min -100000.00 [N]	Max 100000.00 [N]	Factory setting 0.00 [N]
Description:	Sets the force offset for the force limit.		
Dependency:	Refer to: p1520, p1521, p1522, p1523, p1528, p1529		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

r1533	Current limit torque-generating total / Iq_max total		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 5640, 5722, 6640 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the maximum torque/force generating current as a result if all current limits.		
r1533	Current limit force-generating total / Iq_max total		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 5640, 5722, 6640 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the maximum torque/force generating current as a result if all current limits.		
r1534	CO: Torque limit upper total / M_max upper total		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 1610, 5620, 5630, 5640 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the upper torque limit of all torque limits.		
Dependency:	Refer to: p1520, p1521, p1522, p1523, p1528, p1529, p1532		
r1534	CO: Force limit upper total / F_max upper total		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: 1610, 5620, 5630, 5640 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the upper force limit of all force limits.		
Dependency:	Refer to: p1520, p1521, p1522, p1523, p1528, p1529, p1532		
r1535	CO: Torque limit lower total / M_max lower total		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: 1610, 5620, 5630, 5640 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the lower torque limit of all torque limits.		
Dependency:	Refer to: p1520, p1521, p1522, p1523, p1528, p1529, p1532		

r1535	CO: Force limit lower total / F_max lower total		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: 1610, 5620, 5630, 5640 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the lower force limit of all force limits.		
Dependency:	Refer to: p1520, p1521, p1522, p1523, p1528, p1529, p1532		
r1536	Current limit maximum torque-generating current / Isq_max		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 4 Func. diagram: 6640, 6710 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the maximum limit for the torque-generating current component.		
r1537	Current limit minimum torque-generating current / Isq_min		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 4 Func. diagram: 6640, 6710 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the minimum limit for the torque-generating current component.		
r1538	CO: Upper effective torque limit / M_max upper eff		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 2 Func. diagram: 1610, 1700, 5610, 5650, 6060, 6640 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the currently effective upper torque limit.		
Note:	The effective upper torque limit is reduced with respect to the selected upper torque limit p1520, if the current limit p0640 is reduced or the rated magnetizing current of the induction motor p0320 is increased. The following applies in the case of VECTOR: This may be the case for rotating measurements (see p1960). The following applies in the case of VECTOR: Further variable torque limiting is possible (e.g. binector input p1540). The torque limit p1520 can be recalculated using p0340 = 1, 3 or 5.		

r1538	CO: Upper force limit effective / F_max upper eff		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 2 Func. diagram: 1610, 5610, 5650 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the currently effective upper force limit.		
Note:	The effective, upper force limit is reduced with respect to the selected upper force limit p1520 if the current limit p0640 is reduced. The force limit p1520 can be recalculated using p0340 = 1, 3 or 5.		
r1539	CO: Lower effective torque limit / M_max lower eff		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 2 Func. diagram: 1610, 1700, 5610, 5650, 6060, 6640 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the currently effective lower torque limit.		
Note:	The effective lower torque limit is reduced with respect to the selected lower torque limit p1521, if the current limit p0640 is reduced or the rated magnetizing current of the induction motor p0320 is increased. The following applies in the case of VECTOR: This may be the case for rotating measurements (see p1960). The following applies in the case of VECTOR: Further variable torque limiting is possible (e.g. binector input p1541). The torque limit p1520 can be recalculated using p0340 = 1, 3 or 5.		
r1539	CO: Lower force limit effective / F_max lower eff		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 2 Func. diagram: 1610, 5610, 5650 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the currently effective lower force limit.		
Note:	The effective lower force limit is reduced with respect to the selected lower force limit p1521 if the current limit p0640 is reduced. The force limit p1520 can be recalculated using p0340 = 1, 3 or 5.		
p1540[0...n]	CI: Torque limit speed controller upper scaling / M_max n-ctr upScal		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 1700, 6060 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the scaling of the upper torque limiting to limit the speed controller output.		

p1541[0...n]	CI: Torque limit. speed controller lower scaling / M_max nctr lowScal		
VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 1700, 6060
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source for the scaling of the lower torque limiting to limit the speed controller output.		
p1542[0...n]	CI: Travel to fixed stop torque reduction / TfS M_red		
SERVO	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 5610
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the torque/force reduction when traversing to a fixed stop. This value is converted into a factor and is interconnected to the scaling of the torque/force limits.		
Dependency:	Refer to: p1528, p1529, r1543, p1544, p1545		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
p1542[0...n]	CI: Travel to fixed stop force reduction / TfS F_red		
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 5610
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the torque/force reduction when traversing to a fixed stop. This value is converted into a factor and is interconnected to the scaling of the torque/force limits.		
Dependency:	Refer to: p1528, p1529, r1543, p1544, p1545		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
r1543	CO: Travel to fixed stop torque scaling / TfS M scal		
SERVO	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5610
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]
Description:	Displays the internally converted factor to interconnect to the scaling of the torque/force limits.		
Dependency:	Refer to: p1528, p1529, p1542, p1544, p1545		
r1543	CO: Travel to fixed stop force scaling / TfS F scal		
SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5610
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]
Description:	Displays the internally converted factor to interconnect to the scaling of the torque/force limits.		
Dependency:	Refer to: p1528, p1529, p1542, p1544, p1545		

p1544	Travel to fixed stop evaluation torque reduction / TfS M_red eval		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535 [%]	Access level: 3 Func. diagram: 5610 Unit selection: - Expert list: 1 Factory setting 100 [%]
Description:	Sets the evaluation for the torque/force reduction when traversing to a fixed stop.		
Dependency:	Refer to: p1528, p1529, p1542, r1543, p1545		
Note:	4000 hex (16384 dec) in the MOMRED control word corresponds to a reduction by the percentage specified in this parameter.		
p1544	Travel to fixed stop evaluation force reduction / TfS F_red eval		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535 [%]	Access level: 3 Func. diagram: 5610 Unit selection: - Expert list: 1 Factory setting 100 [%]
Description:	Sets the evaluation for the torque/force reduction when traversing to a fixed stop.		
Dependency:	Refer to: p1528, p1529, p1542, r1543, p1545		
Note:	4000 hex (16384 dec) in the MOMRED control word corresponds to a reduction by the percentage specified in this parameter.		
p1545[0...n]	BI: Activates travel to a fixed stop / TfS activation		
SERVO	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2520, 3617, 8012 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to activate/deactivate the "travel to fixed stop" function 1: Travel to fixed stop is active 0: Travel to fixed stop is inactive		
Dependency:	Refer to: p1542, r1543, p1544		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
p1545[0...n]	BI: Activates travel to a fixed stop / TfS activation		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2520, 3617, 8012 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to activate/deactivate the "travel to fixed stop" function 1: Travel to fixed stop is active 0: Travel to fixed stop is inactive		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	EPOS uses the parameter (refer to p2686). When traveling to fixed stop, the fault F07900 "motor locked" is suppressed.		

p1546	Speed threshold motoring/regenerating / n_thresh mot/regen		
SERVO	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 20.00 [rpm]
Description:	Sets the speed threshold for the motoring/regenerative limit. For speeds where the absolute value is less than p1546, then the following applies: - For p1400.13 = 0: Motoring limit (speed threshold is compared to the speed actual value). - For p1400.13 = 1: Regenerative limiting (speed threshold is compared to the speed setpoint).		

p1546	Velocity threshold motoring/regenerative / v_thresh mot/regen		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 1000.00 [m/min]	Factory setting 0.20 [m/min]
Description:	Sets the velocity threshold for the motoring/regenerative limit. For velocities where the absolute value is less than p1546, then the following applies: - For p1400.13 = 0: Motoring limit (velocity threshold is compared to the velocity actual value). - For p1400.13 = 1: Regenerative limiting (velocity threshold is compared to the velocity setpoint).		

r1547[0...1]	CO: Torque limit for speed controller output / M_max outp n_ctrl		
VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 6060
	P-Group: Closed-loop control	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min - [Nm]	Max - [Nm]	Factory setting - [Nm]
Description:	Displays the torque limit to limit the speed controller output.		
Index:	[0] = Upper limit [1] = Lower limit		

r1548[0...1]	CO: Stall current limit torque-generating maximum / Isq_max stall		
VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: 6_2	Unit selection: p0505
	Not for motor type: REL	Scaling: p2002	Expert list: 1
	Min - [Arms]	Max - [Arms]	Factory setting - [Arms]
Description:	Displays the limit for the torque-generating current component using the stall calculation, the current limit of the Motor Module as well as the parameterization in p0640.		
Index:	[0] = Upper limit [1] = Lower limit		

r1549	CO: Stall power actual value / P_stall		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: 14_5 Scaling: r2004 Max - [kW]	Access level: 3 Func. diagram: 5640 Unit selection: p0505 Expert list: 1 Factory setting - [kW]
Description:	Displays the instantaneous stall power.		
Dependency:	Refer to: p0326		
Note:	The stall power is influenced by p0326, p0353, p0354 and p0356.		
r1549	CO: Stall power actual value / P_stall		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: 14_8 Scaling: r2004 Max - [kW]	Access level: 3 Func. diagram: 5640 Unit selection: p0505 Expert list: 1 Factory setting - [kW]
Description:	Displays the instantaneous stall power.		
Dependency:	Refer to: p0326		
Note:	The stall power is influenced by p0326, p0353, p0354 and p0356.		
p1550[0...n]	BI: Transfer actual torque as torque offset / Accept act torque		
SERVO	Can be changed: T Data type: Unsigned32 / Binary P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 9718.23
Description:	For a positive edge, the actual torque (r0079[0]) at this instant in time is used instead of the torque offset from p1532 as long as p1550 remains at 1.		
p1550[0...n]	BI: Transfer actual force as force offset / Accept act force		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 9718.23
Description:	For a positive edge, the actual force (r0079[0]) at this instant in time is used instead of the force offset from p1532 as long as p1550 remains at 1.		

p1551[0...n]	BI: Torque limit variable/fixed signal source / M_lim var/fixS_src		
SERVO, VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 5620, 5630, 6060, 6630
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source to change over the torque limits between variable and fixed torque limit. 1 signal from BI: p1551: The variable torque limit applies (fixed torque limit + scaling). 0 signal from BI: p1551: The fixed torque limit applies. Example: In order that for a Quick Stop (OFF3) the fixed torque limit is effective, BI: p1551 must be interconnected to r0899.5.		

p1551[0...n]	BI: Force limit variable/fixed signal source / F_lim var/fixS_src		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 5620, 5630, 6060, 6630
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source to change over the force limits between variable and fixed force limit. 1 signal from BI: p1551: The variable force limit applies (fixed force limit + scaling). 0 signal from BI: p1551: The fixed force limit applies. Example: In order that for a Quick Stop (OFF3) the fixed force limit is effective, BI: p1551 must be interconnected to r0899.5.		

p1552[0...n]	CI: Torque limit upper scaling without offset / M_max up w/o offs		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 5060, 6060
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source for the scaling of the upper torque limiting to limit the speed controller output without taking into account the current and power limits.		

p1552[0...n]	CI: Force limit upper scaling without offset / F_max up w/o offs		
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 5060, 6060
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source for the scaling of the upper force limiting to limit the velocity controller output without taking into account the current and power limits.		

p1554[0...n]	CI: Torque limit lower scaling without offset / M_max low w/o offs		
SERVO, VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 5060, 6060 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the scaling of the lower torque limiting to limit the speed controller output without taking into account the current and power limits.		
p1554[0...n]	CI: Force limit lower scaling without offset / F_max low w/o offs		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 5060, 6060 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the scaling of the lower force limiting to limit the velocity controller output without taking into account the current and power limits.		
p1555[0...n]	CI: Power limit / P_max		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: r2004 Max -	Access level: 3 Func. diagram: 6640 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the motoring and negative regenerative power limit.		
Dependency:	Refer to: p1530, p1531		
Note:	The resulting power limit when motoring is the minimum from p1530 and the signal that is read in, the resulting regenerative power limit is the maximum from p1531 and the negative signal that is read in.		
p1556[0...n]	Power limit scaling / P_max_scale		
VECTOR (n/M)	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 340.28235E36	Access level: 2 Func. diagram: 6640 Unit selection: - Expert list: 1 Factory setting 0.00
Description:	Sets the scaling of the signal source for the motoring and negative regenerative power limit. 0 signifies no power limiting.		
p1569[0...n]	CI: Supplementary torque 3 / M_suppl 3		
SERVO, VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Functions Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2003 Max -	Access level: 2 Func. diagram: 7010 Unit selection: - Expert list: 1 Factory setting 3841[0]
Description:	Sets the signal source for supplementary torque 3.		
Dependency:	Refer to: p3842		

Notice: The signal input is after the torque limit (r1538, r1539). For vector drives, the signals that are entered are only limited by the current and power limits.

Note: The signal input is preferably used to enter the friction characteristic. The friction compensation is also effective if the speed controller output reaches its torque limits, but the current limits have still not been reached (this only applies to vector drives).

p1569[0...n] CI: Supplementary force 3 / F_suppl 3

SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 7010
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min	Max	Factory setting
	-	-	3841[0]

Description: Sets the signal source for supplementary force 3.

Dependency: Refer to: p3842

Notice: The signal input is after the torque limit (r1538, r1539). For vector drives, the signals that are entered are only limited by the current and power limits.

Note: The signal input is preferably used to enter the friction characteristic. The friction compensation is also effective if the velocity controller output reaches its force limits, but the current limits have still not been reached (this only applies to vector drives).

p1570[0...n] CO: Flux setpoint / Flux setpoint

VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6722
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	50.0 [%]	200.0 [%]	100.0 [%]

Description: Sets the flux setpoint referred to rated motor flux.

Notice: A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

Note: For p1570 > 100%, the flux setpoint increases as a function of the load from 100% (no-load operation) to the setting in p1570 (above rated motor torque), if p1580 > 0% has been set.

p1571[0...n] CI: Supplementary flux setpoint / Suppl flux setp

VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 6725
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: ASM, PEM, REL	Scaling: p2003	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the supplementary flux setpoint.

Notice: Low flux setpoints can cause the drive to stall at higher loads. This is the reason that the flux setpoint should only be adapted for slow load changes.

Note: The supplementary flux setpoint is limited to +/- 50 %.

p1572[0...n]	Supplementary flux setpoint / Suppl flux setp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: PERCENT Max 100.0 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [%]
Description:	Sets the supplementary flux setpoint for the flux controller. The value is referred to the rated motor flux.		
Notice:	The parameter should be set back to 0% again for normal closed-loop control operation.		
Note:	The parameter is used to optimize the flux controller. The current model is not influenced by the setting.		
p1573[0...n]	Flux threshold value magnetizing / Flux thresh magnet		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL Min 10.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: PERCENT Max 200.0 [%]	Access level: 3 Func. diagram: 6722 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the flux threshold value for enabling the speed setpoint and the end of magnetizing (r0056.4).		
Note:	The parameter only has an influence if the flux actual value reaches the threshold value p1573 more quickly during magnetizing than the time set in p0346. The parameter has no influence for flying restart (see p1200) and after DC braking (see p1231).		
p1574[0...n]	Voltage reserve dynamic / V_reserve dyn		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [Vrms]	Calculated: CALC_MOD_LIM_REF Dynamic index: DDS, p0180 Units group: 5_1 Scaling: - Max 150.0 [Vrms]	Access level: 3 Func. diagram: 6723, 6724 Unit selection: p0505 Expert list: 1 Factory setting 10.0 [Vrms]
Description:	Sets a dynamic voltage reserve.		
Note:	In the field weakening range, it must be expected that the control dynamic performance is somewhat restricted due to the limited possibilities of controlling/adjusting the voltage. This can be improved by increasing the voltage margin (reserve). Increasing the reserve reduces the steady-state maximum output voltage (r0071).		
p1576[0...n]	Flux boost, adaptation speed, lower / Flux boost n lower		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.00 [rpm]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: 6725 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [rpm]
Description:	Sets the lower adaptation speed of the flux boost. Below this speed, p1570 is set as reference (setpoint) flux.		

p1577[0...n]	Flux boost adaptation speed, upper / Flux boost n upper		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6725
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1
	Min 1.0 [%]	Max 10000.0 [%]	Factory setting 200.0 [%]
Description:	Sets the upper adaptation speed of the flux boost. Above this speed, the rated motor flux (100 %) is set as reference (setpoint) flux.		
Dependency:	The parameter value refers to the lower adaptation speed of the flux boost. Refer to: p1576		
p1578[0...n]	Flux reduction flux decrease smoothing time / Flux red dec t_sm		
SERVO	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5722
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
	Min 20 [ms]	Max 5000 [ms]	Factory setting 200 [ms]
Description:	Sets the smoothing time for the flux setpoint when decreasing the flux due to flux reduction (p1581 < 100 %).		
Dependency:	Refer to: p1579, p1581		
p1579[0...n]	Flux reduction flux build-up smoothing time / Flux red up t_sm		
SERVO	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5722
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 5000 [ms]	Factory setting 4 [ms]
Description:	Sets the smoothing time for the flux setpoint for the flux build-up due to flux reduction (p1581 < 100 %).		
Dependency:	Refer to: p1578, p1581		
Note:	An excessively long smoothing time extends the time until the maximum torque is reached from the no-load phase.		
p1580[0...n]	Efficiency optimization / Efficiency opt.		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6722
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
	Min 0 [%]	Max 100 [%]	Factory setting 0 [%]
Description:	Sets the efficiency optimization. When optimizing the efficiency, the flux setpoint of the closed-loop control is adapted as a function of the load. For p1580 = 100 %, under no-load operating conditions, the flux setpoint is reduced to 50 % of the rated motor flux.		
Note:	It only makes sense to activate this function if the dynamic response requirements of the speed controller are low. In order to avoid oscillations, if required, the speed controller parameters should be adapted (increase Tn, reduce Kp). Further, the smoothing time of the flux setpoint filter (p1582) should be increased.		

p1581[0...n]	Flux reduction factor / Flux red factor		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min 20 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100 [%]	Access level: 2 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting 100 [%]
Description:	Sets the factor to which the flux is reduced under no-load conditions. For a value of 100%, the flux reduction is switched out. This parameter refers to the flux saved in the field weakening characteristic. By reducing the flux, the losses in induction motors can be reduced under no-load conditions or at low torques. However, the time it takes to reach the maximum torque is extended.		
Recommend.:	For induction motors with closed rotor slots, we recommend that the integral time of the current controller (p1717) is e.g. increased to three times the value. For stable operation, the maximum field-weakening factor in operation with an encoder must be less than 16 and in operation without an encoder must be less than 4. Lower field weakening factors are recommended. The field weakening factor is calculated as follows: $(p1082 * 100 \% * 600 \text{ V}) / (p0348 * p1581 * p0070)$ In order to reduce losses due to magnetizing and demagnetizing, we recommend that the smoothing times are adapted for flux decrease (p1578) and flux build-up (p1579). In order to reduce the losses as a result of building-up and reducing the torque, we recommend that the torque setpoint is smoothed (current setpoint filter (p1656 ...) or speed actual value filter (p1441)).		
Dependency:	Refer to: p1578, p1579		
Note:	It only makes sense to activate this function if there are low dynamic requirements placed on the speed controller and there are frequent phases with a low load. In order to avoid oscillations, if required, the speed controller parameters should be adapted (decrease Kp (p1460, p1470), increase Tn (p1462, p1472)). When used without an encoder, flux reduction is not possible for induction motors with closed rotor slots.		
p1582[0...n]	Flux setpoint smoothing time / Flux setp T_smth		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 4 [ms]	Calculated: CALC_MOD_REG Dynamic index: DDS, p0180 Units group: - Scaling: - Max 5000 [ms]	Access level: 3 Func. diagram: 6722 Unit selection: - Expert list: 1 Factory setting 15 [ms]
Description:	Sets the smoothing time for the flux setpoint.		
r1583	Flux setpoint smoothed / Flux setp smooth		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 4 Func. diagram: 6722, 6723 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the smoothed flux setpoint. The value is referred to the rated motor flux.		

p1584[0...n]	Field weakening operation, flux setpoint smoothing time / Field weak T_smth		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6722
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 20000 [ms]	Factory setting 0 [ms]
Description:	Sets the smoothing time for the flux setpoint in the field-weakening range		
Recommend.:	Smoothing should be especially used if there is no regenerative feedback into the line supply. This means that the DC link voltage can quickly increase in regenerative operation		
Note:	Only the flux setpoint rise is smoothed		
p1585[0...n]	Flux actual value, smoothing time / Flux actVal T_smth		
SERVO	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 1000 [ms]	Factory setting 0 [ms]
Description:	Sets the smoothing time for the flux actual value.		
p1585[0...n]	Flux actual value, smoothing time / Flux actVal T_smth		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 1000 [ms]	Factory setting 0 [ms]
Description:	Sets the smoothing time for the flux actual value.		
p1586[0...n]	Field weakening characteristic, scaling / Field weak scal		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL, FEM	Scaling: -	Expert list: 1
	Min 80.0 [%]	Max 120.0 [%]	Factory setting 100.0 [%]
Description:	Sets the scaling of the pre-control characteristic for the start of field weakening. For values above 100% and for partial load situations, the field weakening starts at higher speeds.		
Note:	If the start of field weakening is shifted to lower speeds, then the voltage reserve is increased for partial load situations. If the start of field weakening is shifted to higher speeds, the voltage reserve is appropriately reduced so that for fast load changes, it can be expected that this will have a negative impact on the dynamic performance.		
r1589	Field-weakening current, pre-control value / FieldWkCurrPrectrl		
VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 6724
	P-Group: Closed-loop control	Units group: 6_2	Unit selection: p0505
	Not for motor type: ASM, REL, FEM	Scaling: p2002	Expert list: 1
	Min - [Arms]	Max - [Arms]	Factory setting - [Arms]
Description:	Displays the pre-control value for the field weakening current.		

p1590[0...n]	Flux controller P gain / Flux controller Kp		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL Min 0.0 [A/Vs]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 999999.0 [A/Vs]	Access level: 3 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting 10.0 [A/Vs]
Description:	Sets the proportional gain of the flux controller.		
Note:	For synchronous motors, this parameters has no effect. The value is automatically pre-assigned dependent on the motor when the drive system is first commissioned. When calculating controller parameters (p0340 = 4), this value is recalculated.		
p1590[0...n]	Flux controller P gain / Flux controller Kp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL Min 0.0	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 999999.0	Access level: 4 Func. diagram: 6723 Unit selection: - Expert list: 1 Factory setting 10.0
Description:	Sets the proportional gain of the flux controller.		
Note:	The value is automatically pre-assigned dependent on the motor when the drive system is first commissioned. When calculating controller parameters (p0340 = 4), this value is recalculated.		
p1592[0...n]	Flux controller integral time / Flux controller Tn		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL Min 0 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [ms]	Access level: 3 Func. diagram: 5722 Unit selection: - Expert list: 1 Factory setting 30 [ms]
Description:	Sets the integral time of the flux controller.		
Note:	For synchronous motors, this parameters has no effect. The value is automatically pre-assigned dependent on the motor when the drive system is first commissioned. When calculating controller parameters (p0340 = 4), this value is recalculated.		
p1592[0...n]	Flux controller integral time / Flux controller Tn		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL Min 0 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [ms]	Access level: 4 Func. diagram: 6723 Unit selection: - Expert list: 1 Factory setting 30 [ms]
Description:	Sets the integral time of the flux controller.		
Note:	The value is automatically pre-assigned dependent on the motor when the drive system is first commissioned. When calculating controller parameters (p0340 = 4), this value is recalculated.		

r1593[0...1]	CO: Field weakening controller / flux controller output / Field/FI_ctrl outp		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 4 Func. diagram: 6723, 6724, 6726 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the output of the field weakening controller (synchronous motor) or the output of the flux controller (separately-excited synchronous motor, induction motor).		
Index:	[0] = PI output [1] = I output		
p1594[0...n]	Field-weakening controller, P gain / Field_ctrl Kp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, REL, FEM Min 0.00	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000.00	Access level: 4 Func. diagram: 6724 Unit selection: - Expert list: 1 Factory setting 0.00
Description:	Sets the P gain of the field-weakening controller.		
p1596[0...n]	Field weakening controller integral-action time / Field_ctrl Tn		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 10 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [ms]	Access level: 3 Func. diagram: 6723, 6724 Unit selection: - Expert list: 1 Factory setting 50 [ms]
Description:	Sets the integral-action time of the field-weakening controller.		
r1597	CO: Field weakening controller output / Field_ctrl output		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 4 Func. diagram: 6723 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the output of the field weakening controller. The value is referred to the rated motor flux.		
r1598	CO: Total flux setpoint / Flux setp total		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 6714, 6723, 6724, 6725, 6726, 8018 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the effective flux setpoint. The value is referred to the rated motor flux.		

p1599[0...n]	Flux controller, excitation current difference / Flux ctr I_exc_dif		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 3.0 [%]
Description:	Sets the permissible difference between the actual excitation current and the excitation current setpoint. The excitation current flux controller is active within this difference. If the difference lies outside the specified limit value, then the I component of the excitation current flux controller is kept. Instead of this, for the flux controller of the field-generating current, an additional I controller is switched in (integral time according to p1592). If the difference again lies within the bandwidth, the I component of the excitation current flux controller is reactivated and the I component of the flux controller of the field-generating current is reduced as an exponential function with respect to time. The reduction of the I component over time depends on the rotor time constant (r0384).		
p1600[0...n]	P flux controller, P gain / P flux ctrl Kp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.0	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 999999.0	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10.0
Description:	Sets the proportional gain of the P flux controller for separately-excited synchronous motors.		
Note:	The value is automatically pre-assigned dependent on the motor when the drive system is first commissioned. When calculating controller parameters (p0340 = 4), this value is recalculated.		
r1602	CO: Flux controller P output / Flux ctrl P outp		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 4 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the output of the P flux controller for separately-excited synchronous motors (FEM).		
p1605[0...n]	Pulse technique pattern configuration / Pulse config		
VECTOR (n/M)	Can be changed: U, T Data type: Integer16 P-Group: Closed-loop control Not for motor type: ASM, REL, FEM Min 1	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2
Description:	Sets the applied pulse patterns for estimating the continuous rotor position. Remark: See p1750 for the activation of the pulse-pattern technique.		
Value:	1: pm 2: ppm		
Dependency:	Refer to: p1750		
Note:	When commissioning a catalog motor, the technique is automatically selected depending on the motor type being used.		

r1606	CO: Pulse technique pattern current / Pulse config		
VECTOR (n/M)	Can be changed: - Data type: Integer16 P-Group: Closed-loop control Not for motor type: ASM, REL, FEM Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the currently applied pulse patterns for estimating the continuous rotor position.		
Value:	0: None 1: pm 2: ppmm		
Dependency:	Refer to: p1605, p1750		
p1607[0...n]	Pulse technique stimulus / Pulse stimu		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, REL, FEM Min 0.000 [mVs]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 20000.000 [mVs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 32.000 [mVs]
Description:	Sets the excitation amplitude (voltage-time pulse) for the pulse technique for estimating the continuous rotor position.		
Dependency:	Refer to: p1605, p1750		
r1608[0...6]	CO: Pulse technique response / Pulse config		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: ASM, REL, FEM Min - [A]	Calculated: - Dynamic index: - Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the signal responses to the excitation of the pulse technique.		
Index:	[0] = Phase R [1] = Phase S [2] = D estimated [3] = Q estimated [4] = D estimated AC [5] = Q estimated AC [6] = Pointer length AC		
Dependency:	Refer to: p1605, p1607, p1750		
p1609[0...n]	Current setpoint for I/f operation / I_set I/f oper		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.00 [Arms]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 6_2 Scaling: - Max 10000.00 [Arms]	Access level: 3 Func. diagram: 6727 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the stator current setpoint for operation of separately-excited synchronous motors (FEM) in the operating mode I/f (p1300 = 18).		

p1610[0...n]	Torque setpoint static (SLVC) / M_set static		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -200.0 [%] Description: Sets the static torque setpoint for encoderless vector control (SLVC). This parameter is entered as a percentage referred to the rated motor torque (r0333). For encoderless vector control, when the motor model is shut down, an absolute current is impressed. p1610 represents the maximum load that occurs at a constant setpoint speed. Notice: p1610 should always be set to at least 10 % higher than the maximum steady-state load that can occur. Note: For p1610 = 0%, a current setpoint is calculated that corresponds to the no-load case (ASM: rated magnetizing current). For p1610 = 100 %, a current setpoint is calculated that corresponds to the rated motor torque. Negative values are converted into positive setpoints in the case of induction and permanent-magnet synchronous motors.	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200.0 [%] Description: Sets the static torque setpoint for encoderless vector control (SLVC). This parameter is entered as a percentage referred to the rated motor torque (r0333). For encoderless vector control, when the motor model is shut down, an absolute current is impressed. p1610 represents the maximum load that occurs at a constant setpoint speed. Notice: p1610 should always be set to at least 10 % higher than the maximum steady-state load that can occur. Note: For p1610 = 0%, a current setpoint is calculated that corresponds to the no-load case (ASM: rated magnetizing current). For p1610 = 100 %, a current setpoint is calculated that corresponds to the rated motor torque. Negative values are converted into positive setpoints in the case of induction and permanent-magnet synchronous motors.	Access level: 2 Func. diagram: 1710, 6721, 6722, 6726 Unit selection: - Expert list: 1 Factory setting 50.0 [%]
p1611[0...n]	Supplementary accelerating torque (SLVC) / M_suppl_accel		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%] Description: Enters the dynamic torque setpoint for the low-speed range for encoderless vector control (SLVC). This parameter is entered as a percentage referred to the rated motor torque (r0333). Note: When accelerating and braking p1611 is added to p1610 and the resulting total torque is converted into an appropriate current setpoint and controlled. For pure accelerating torques, it is always favorable to use the torque pre-control of the speed controller (p1496).	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200.0 [%] Description: Enters the dynamic torque setpoint for the low-speed range for encoderless vector control (SLVC). This parameter is entered as a percentage referred to the rated motor torque (r0333). Note: When accelerating and braking p1611 is added to p1610 and the resulting total torque is converted into an appropriate current setpoint and controlled. For pure accelerating torques, it is always favorable to use the torque pre-control of the speed controller (p1496).	Access level: 2 Func. diagram: 1710, 6721, 6722, 6726 Unit selection: - Expert list: 1 Factory setting 0.0 [%]
p1612[0...n]	Current setpoint, open-loop control, encoderless / I_setCtrEncoderl		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [Arms] Description: Sets the current setpoint for controlled (open-loop) encoderless operation. Note: The value is effective at speeds less than p1755 and represents a reserve for a possibly existing load torque or torque error in the moment of inertia.	Calculated: CALC_MOD_REG Dynamic index: DDS, p0180 Units group: 6_2 Scaling: - Max 10000.00 [Arms] Description: Sets the current setpoint for controlled (open-loop) encoderless operation. Note: The value is effective at speeds less than p1755 and represents a reserve for a possibly existing load torque or torque error in the moment of inertia.	Access level: 2 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Arms]
p1612[0...n]	Current setpoint magnetizing open-loop controlled / Id_set ctrl		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.00 [Arms] Description: Sets the magnetizing current setpoint in the open-loop controlled encoderless operation. The value is only valid during the current model orientation.	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 6_2 Scaling: - Max 10000.00 [Arms] Description: Sets the magnetizing current setpoint in the open-loop controlled encoderless operation. The value is only valid during the current model orientation.	Access level: 2 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Arms]

Dependency: Refer to: p1610, p1611

Note: The value is effective at speeds less than p1755 and represents a reserve for a possibly existing load torque or torque error in the moment of inertia.

p1616[0...n]	Current setpoint smoothing time / I_set T_smooth		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 4 [ms]	Calculated: CALC_MOD_REG Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [ms]	Access level: 3 Func. diagram: 6721, 6722, 6726 Unit selection: - Expert list: 1 Factory setting 40 [ms]
Description:	Sets the smoothing time for the current/torque setpoint in the open-loop-controlled operating range in the case of sensorless vector control.		
Note:	This parameter is only effective in the range where current is injected for encoderless vector control. In the case of induction motors the current setpoint is calculated from parameter value p1610 and in the case of separately excited synchronous motors the torque setpoint is calculated from parameter value p1611.		
r1617	CO: Torque setpoint (controlled) / M_setp sv SLVC		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min - [Nm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Nm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Nm]
Description:	Torque setpoint for sensorless control of the separately excited synchronous motor in the open-loop-controlled operating range (under p1755 * p1756).		
r1618	Current model controller, pre-control / I_mod_ctrl prectrl		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 4 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the pre-control value of the current model controller. It involves a magnetizing current in the d-direction.		
p1619[0...n]	Setpoint/actual value tracking threshold / SetAct track thrsh		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.00 [Arms]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 6_2 Scaling: p2002 Max 10000.00 [Arms]	Access level: 3 Func. diagram: 6727 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Arms]
Description:	Threshold for setpoint/actual value tracking of the stator current in the q direction of the current model.		

p1620[0...n]	Stator current, minimum / I_{stator} min		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min -10000.00 [Arms]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 6_2 Scaling: - Max 10000.00 [Arms]	Access level: 3 Func. diagram: 6727 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the minimum stator current for separately-excited synchronous motors (FEM). A negative value means that the field-generating stator current (d-axis) has a negative sign. The valid value is internally limited to 50% of the rated motor current (p0305).		
p1621[0...n]	Changeover speed, inner cos phi = 1 / n_chngov cos phi=1		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.00 [rpm]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: 6727 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [rpm]
Description:	Sets the speed where a change is made from the inner to the outer cos phi = 1. If the value that is entered exceeds the rated speed, then a change is made to the inner cos phi = 1 over the complete speed range.		
p1622[0...n]	Field-generating current setpoint smoothing time constant / Id_setp T_{smth}		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.1 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200.0 [ms]	Access level: 3 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting 20.0 [ms]
Description:	Sets the smoothing time constant for the setpoint of the field-generating current components. The current filtered in this way is included in the calculation of the cos phi.		
r1623[0...1]	Field-generating current setpoint (steady-state) / Id_set stationary		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: PEM, REL Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 4 Func. diagram: 6723, 6726, 6727 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the steady-state field generating current setpoint (Id_set).		
Note:	Index 1 shows the stationary field-generating current on the stator side in the case of separately excited synchronous motors without the excitation current monitoring component (r1644).		

r1624	Field-generating current setpoint, total / Id_setp total		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: REL Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 4 Func. diagram: 6640, 6721, 6723, 6727 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the limited field-generating current setpoint (Id_set). This value comprises the steady-state field-generating current setpoint r1623 and a dynamic component that is only set when changes are made to the flux setpoint.		
p1625[0...n]	Excitation current setpoint calibration / I_exc_setp cal		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 10.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200.0 [%]	Access level: 3 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Gain factor to weight the excitation current setpoint.		
r1626	CO: Excitation current setpoint / I_exc_setp		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the calculated excitation current setpoint.		
Dependency:	Refer to: p0390		
r1627	CO: Current model load angle / I_mod load angle		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min - [°]	Calculated: - Dynamic index: - Units group: - Scaling: p2005 Max - [°]	Access level: 3 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting - [°]
Description:	Displays the load angle of the current model.		
p1628[0...n]	Current model controller, dynamic factor / I_mod_ctr dyn_fact		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: ASM, PEM, REL Min 1 [%]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 400 [%]	Access level: 3 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting 50 [%]
Description:	Dynamic factor of the model controller in the current model		

p1629[0...n]	Current model controller P gain / I_mod_ctrl Kp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.000	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100000.000	Access level: 4 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the proportional gain of the current model controller. This value is automatically preset using p3900 or p0340 when commissioning has been completed.		
p1630[0...n]	Current model controller integral time / I_ctrl Tn		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.00 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000.00 [ms]	Access level: 4 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the proportional gain of the current model controller. This value is automatically preset using p3900 or p0340 when commissioning has been completed.		
r1631	Current model controller, P gain effective / I_mod ctrl Kp eff		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the effective P gain of the current model controller.		
r1632	Current model controller integral time effective / I_mod_ctrl Tn eff		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min - [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [ms]	Access level: 3 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the effective integral time of the current model controller.		
r1633	Current model, flux setpoint / I_mod flux setp		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the effective flux setpoint of the current model. The value is referred to the rated motor flux.		

r1634 Current model, flux actual value / I_mod flux act val

VECTOR (n/M)

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Closed-loop control**Not for motor type:** ASM, PEM, REL**Min**

- [%]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** PERCENT**Max**

- [%]

Access level: 3**Func. diagram:** 6727**Unit selection:** -**Expert list:** 1**Factory setting**

- [%]

Description: Displays the effective flux actual value of the current model. The value is referred to the rated motor flux.**r1635 Current model controller, I component / I_mod_ctrl I_comp**

VECTOR (n/M)

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Closed-loop control**Not for motor type:** ASM, PEM, REL**Min**

- [Arms]

Calculated: -**Dynamic index:** -**Units group:** 6_2**Scaling:** p2002**Max**

- [Arms]

Access level: 4**Func. diagram:** 6727**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [Arms]

Description: Displays the I component of the current model controller.**r1636 Current model controller output / I_mod_ctrl outp**

VECTOR (n/M)

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Closed-loop control**Not for motor type:** ASM, PEM, REL**Min**

- [Arms]

Calculated: -**Dynamic index:** -**Units group:** 6_2**Scaling:** p2002**Max**

- [Arms]

Access level: 4**Func. diagram:** 6727**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [Arms]

Description: Displays the output of the current model controller.**r1637 Current model, magnetizing current, d axis / I_mod I_mag d-ax**

VECTOR (n/M)

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Closed-loop control**Not for motor type:** ASM, PEM, REL**Min**

- [Arms]

Calculated: -**Dynamic index:** -**Units group:** 6_2**Scaling:** p2002**Max**

- [Arms]

Access level: 4**Func. diagram:** 6727**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [Arms]

Description: Displays the magnetizing current of the current model in the d-axis.**r1638 Current model, magnetizing current, q axis / I_mod I_mag q-ax**

VECTOR (n/M)

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Closed-loop control**Not for motor type:** ASM, PEM, REL**Min**

- [Arms]

Calculated: -**Dynamic index:** -**Units group:** 6_2**Scaling:** p2002**Max**

- [Arms]

Access level: 4**Func. diagram:** 6727**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [Arms]

Description: Displays the magnetizing current of the current model in the q-axis.

r1639	CO: Current model Isq after actual value tracking / I_mod Isq track		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 4 Func. diagram: 6727 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the stator current in the q axis after current actual value tracking.		
p1640[0...n]	CI: Excitation current actual value / I_exc_act val		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the excitation current actual value		
r1641	Excitation current actual value / I_exc_act val		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 6727, 6497 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the excitation current actual value that is read in.		
Dependency:	Refer to: p0390		
p1642[0...n]	Minimum excitation current / Min I_exc		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.1 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 50.0 [%]	Access level: 4 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting 5.0 [%]
Description:	Sets the minimum excitation current. This means that negative excitation currents can be avoided.		
p1643[0...n]	Gain factor, minimum excitation current closed-loop control / Min I_exc Kp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.00	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 5.00	Access level: 4 Func. diagram: 6727 Unit selection: - Expert list: 1 Factory setting 0.40
Description:	Sets the gain factor for the minimum excitation current, closed-loop control. This is active if the excitation current is below 75% of p1642.		
Dependency:	Refer to: p1642		

r1644	Excitation current monitoring output / I_exc_monit outp		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 4 Func. diagram: 6727 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the output of the excitation current monitoring for separately excited synchronous motors.		
p1645[0...6]	BI: Excitation feedback signals signal source / Exc FS S_src		
VECTOR (n/M)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: ASM, PEM, REL Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 6495 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the individual feedback signals from the excitation.		
Index:	[0] = Excitation ready to be powered up [1] = Excitation ready [2] = Excitation operational [3] = Excitation group signal fault [4] = Excitation group signal alarm [5] = Not used [6] = Not used		
Dependency:	Refer to: r1649		
p1646	Excitation monitoring time / Excit t_monit		
VECTOR (n/M)	Can be changed: T Data type: FloatingPoint32 P-Group: Commands Not for motor type: ASM, PEM, REL Min 2.0 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1300.0 [s]	Access level: 2 Func. diagram: 6495 Unit selection: - Expert list: 1 Factory setting 20.0 [s]
Description:	Sets the monitoring time of the excitation.		
Note:	After an ON command, the feedback signal must be received within this monitoring time. After the on command for the excitation (r1648.0 = 1), its feedback signal must be available at r1649.1 within this monitoring time (BI: p1645[1]). The same monitoring time is effective after the excitation is enabled for operation (r1648.3 = 1) up to the feedback signal "excitation operational" (r1649.2 = 1, BI: p1645[2]).		
p1647	Excitation switch-off delay time / Exc t_off		
VECTOR (n/M)	Can be changed: T Data type: FloatingPoint32 P-Group: Commands Not for motor type: ASM, PEM, REL Min 0.5 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 5.0 [s]	Access level: 2 Func. diagram: 6495 Unit selection: - Expert list: 1 Factory setting 0.8 [s]
Description:	Sets the switch-off delay time to shut down the excitation equipment.		
Note:	The delay time starts if, when powering down, r0863.0 = 0. r1648.0 and r1648.3 are reset at the end of the delay time.		

r1648.0...11 CO/BO: Excitation, control word / Excitation STW

VECTOR (n/M)

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 6495**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** ASM, PEM, REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the control word for the excitation equipment.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Power up excitation	Yes	No	-
01	Excitation no OFF2	Yes	No	-
02	Excitation no OFF3	Yes	No	-
03	Excitation operation enable	Yes	No	-
07	Excitation acknowledge fault	Yes	No	-
10	Master control by excitation equipment	Yes	No	-
11	ccw rotating field excitation invert excitation current setpoint	Yes	No	-

r1649.0...7 CO/BO: Excitation status word / Excitation ZSW

VECTOR (n/M)

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 6495**P-Group:** Displays, signals**Units group:** -**Unit selection:** -**Not for motor type:** ASM, PEM, REL**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the status word of the excitation equipment.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Excitation ready to be powered up feedback signal	Yes	No	-
01	Excitation ready feedback signal	Yes	No	-
02	Excitation operational feedback signal	Yes	No	-
03	Excitation group signal fault	Yes	No	-
07	Excitation group signal alarm	Yes	No	-

Dependency:

Refer to: p1645

r1650 Current setpoint torque-generating before filter / Iq_set before filt

SERVO, VECTOR (n/M)

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 5710**P-Group:** Closed-loop control**Units group:** 6_2**Unit selection:** p0505**Not for motor type:** REL**Scaling:** p2002**Expert list:** 1**Min****Max****Factory setting**

- [Arms]

- [Arms]

- [Arms]

Description:

Displays the torque generating current setpoint Iqset after the torque limits and the clock cycle interpolation is ahead of the current setpoint filters.

r1650 Current setpoint force-generating before filter / Iq_set before filt

SERVO (Lin)

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** 5710**P-Group:** Closed-loop control**Units group:** 6_2**Unit selection:** p0505**Not for motor type:** REL**Scaling:** p2002**Expert list:** 1**Min****Max****Factory setting**

- [Arms]

- [Arms]

- [Arms]

Description:

Displays the force generating current setpoint Iqset after the force limits and the clock cycle interpolation is ahead of the current setpoint filters.

r1651	CO: Torque setpoint, function generator / M_set FG		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the torque setpoint of the function generator.		
r1651	CO: Force setpoint, function generator / F_set FG		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the force setpoint of the function generator.		
p1653[0...n]	Current setpoint torque-generating smoothing time minimum / Isq_s T_smth min		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL Min 0.1 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 20.0 [ms]	Access level: 4 Func. diagram: 6710 Unit selection: - Expert list: 1 Factory setting 0.1 [ms]
Description:	Sets the minimum smoothing time constant for the setpoint of the torque-generating current components.		
p1654[0...n]	Curr. setpoint torque-gen. smoothing time field weakening range / Isq_s T_smth FW		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL Min 0.1 [ms]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: - Scaling: - Max 50.0 [ms]	Access level: 4 Func. diagram: 6710 Unit selection: - Expert list: 1 Factory setting 4.8 [ms]
Description:	Sets the smoothing time constant for the setpoint of the torque-generating current components.		
Note:	The smoothing time does not become effective until the field-weakening range is reached.		
p1655[0...1]	CI: Current setpoint filter natural frequency tuning / I_set_filt f_n		
VECTOR (n/M)	Can be changed: U, T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: 1710, 6710 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for tuning the natural frequency of the current setpoint filter.		
Index:	[0] = Filter 1 [1] = Filter 2		

p1656[0...n]	Activates current setpoint filter / I_setp_filt act				
SERVO	Can be changed: U, T		Calculated: CALC_MOD_CON	Access level: 3	
	Data type: Unsigned16		Dynamic index: DDS, p0180	Func. diagram: 5710	
	P-Group: Closed-loop control		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0001 bin	
Description:	Setting for activating/deactivating the current setpoint filter.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Filter 1	Active	Inactive	-
	01	Filter 2	Active	Inactive	-
	02	Filter 3	Active	Inactive	-
	03	Filter 4	Active	Inactive	-
Dependency:	The individual current setpoint filters are parameterized as of p1657.				
Note:	If not all of the filters are required, then the filters should be used consecutively starting from filter 1.				

p1656[0...n]	Activates current setpoint filter / I_setp_filt act				
VECTOR (n/M)	Can be changed: U, T		Calculated: -	Access level: 3	
	Data type: Unsigned16		Dynamic index: DDS, p0180	Func. diagram: 6710	
	P-Group: Closed-loop control		Units group: -	Unit selection: -	
	Not for motor type: REL		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0001 bin	
Description:	Setting for activating/deactivating the current setpoint filter.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Filter 1	Active	Inactive	-
	01	Filter 2	Active	Inactive	-
Dependency:	The individual current setpoint filters are parameterized as of p1657.				
Note:	If not all of the filters are required, then the filters should be used consecutively starting from filter 1.				

p1657[0...n]	Current setpoint filter 1 type / I_set_filt 1 Typ				
SERVO, VECTOR (n/M)	Can be changed: U, T		Calculated: CALC_MOD_CON	Access level: 3	
	Data type: Integer16		Dynamic index: DDS, p0180	Func. diagram: 5710, 6710	
	P-Group: Closed-loop control		Units group: -	Unit selection: -	
	Not for motor type: REL		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	1		2	1	
Description:	Sets the current setpoint filter 1 as low pass (PT2) or as extended general 2nd-order filter.				
Value:	1: Low pass: PT2 2: General 2nd-order filter				
Dependency:	Current setpoint filter 1 is activated via p1656.0 and parameterized via p1657 ... p1661.				
Note:	For an extended general 2nd-order filter, by inserting the same natural frequency in both the numerator and in the denominator, i.e. bandstop frequency, a bandstop filter is implemented. If the numerator damping of zero is selected, the bandstop frequency is completely suppressed. The denominator damping can be determined from the equation for the 3 dB bandwidth: $f_{3dB} \text{ bandwidth} = 2 * D_{denominator} * f_{bandstop} \text{ frequency}$				

p1658[0...n]	Current setpoint filter 1 denominator natural frequency / I_set_filt 1 fn_n		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting 1999.0 [Hz]
Description:	Sets the denominator natural frequency for current setpoint filter 1 (PT2, general filter).		
Dependency:	Current setpoint filter 1 is activated via p1656.0 and parameterized via p1657 ... p1661.		
p1659[0...n]	Current setpoint filter 1 denominator damping / I_set_filt 1 D_n		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.001	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the denominator damping for current setpoint filter 1.		
Dependency:	Current setpoint filter 1 is activated via p1656.0 and parameterized via p1657 ... p1661.		
p1660[0...n]	Current setpoint filter 1 numerator natural frequency / I_set_filt 1 fn_z		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting 1999.0 [Hz]
Description:	Sets the numerator natural frequency for current setpoint filter 1 (general filter).		
Dependency:	Current setpoint filter 1 is activated via p1656.0 and parameterized via p1657 ... p1661.		
p1661[0...n]	Current setpoint filter 1 numerator damping / I_set_filt 1 D_z		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the numerator damping for current setpoint filter 1.		
Dependency:	Current setpoint filter 1 is activated via p1656.0 and parameterized via p1657 ... p1661.		
p1662[0...n]	Current setpoint filter 2 type / I_set_filt 2 Typ		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: Integer16 P-Group: Closed-loop control Not for motor type: REL Min 1	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the current setpoint filter 2 as low pass (PT2) or as extended general 2nd-order filter.		
Value:	1: Low pass: PT2 2: General 2nd-order filter		
Dependency:	Current setpoint filter 2 is activated via p1656.1 and parameterized via p1662 ... p1666.		

Note: For an extended general 2nd-order filter, by inserting the same natural frequency in both the numerator and in the denominator, i.e. bandstop frequency, a bandstop filter is implemented. If the numerator damping of zero is selected, the bandstop frequency is completely suppressed.

The denominator damping can be determined from the equation for the 3 dB bandwidth:

$$f_{3dB} \text{ bandwidth} = 2 * D_{\text{denominator}} * f_{\text{bandstop frequency}}$$

p1663[0...n]	Current setpoint filter 2 denominator natural frequency / I_set_filt 2 fn_n		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting 1999.0 [Hz]
Description:	Sets the denominator natural frequency for current setpoint filter 2 (PT2, general filter).		
Dependency:	Current setpoint filter 2 is activated via p1656.1 and parameterized via p1662 ... p1666.		
p1664[0...n]	Current setpoint filter 2 denominator damping / I_set_filt 2 D_n		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.001	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the denominator damping for current setpoint filter 2.		
Dependency:	Current setpoint filter 2 is activated via p1656.1 and parameterized via p1662 ... p1666.		
p1665[0...n]	Current setpoint filter 2 numerator natural frequency / I_set_filt 2 fn_z		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting 1999.0 [Hz]
Description:	Sets the numerator natural frequency for current setpoint filter 2 (general filter).		
Dependency:	Current setpoint filter 2 is activated via p1656.1 and parameterized via p1662 ... p1666.		
p1666[0...n]	Current setpoint filter 2 numerator damping / I_set_filt 2 D_z		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the numerator damping for current setpoint filter 2.		
Dependency:	Current setpoint filter 2 is activated via p1656.1 and parameterized via p1662 ... p1666.		

p1667[0...n]	Current setpoint filter 3 type / I_set_filt 3 Typ		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: 5710
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1	Max 2	Factory setting 1
Description:	Sets the current setpoint filter 3 as low pass (PT2) or as extended general 2nd-order filter.		
Value:	1: Low pass: PT2 2: General 2nd-order filter		
Dependency:	Current setpoint filter 3 is activated via p1656.2 and parameterized via p1667 ... p1671.		
p1668[0...n]	Current setpoint filter 3 denominator natural frequency / I_set_filt 3 fn_n		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5710
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 1999.0 [Hz]
Description:	Sets the denominator natural frequency for current setpoint filter 3 (PT2, general filter).		
Dependency:	Current setpoint filter 3 is activated via p1656.2 and parameterized via p1667 ... p1671.		
p1669[0...n]	Current setpoint filter 3 denominator damping / I_set_filt 3 D_n		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5710
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.001	Max 10.000	Factory setting 0.700
Description:	Sets the denominator damping for current setpoint filter 3.		
Dependency:	Current setpoint filter 3 is activated via p1656.2 and parameterized via p1667 ... p1671.		
p1670[0...n]	Current setpoint filter 3 numerator natural frequency / I_set_filt 3 fn_z		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5710
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 1999.0 [Hz]
Description:	Sets the numerator natural frequency for current setpoint filter 3 (general filter).		
Dependency:	Current setpoint filter 3 is activated via p1656.2 and parameterized via p1667 ... p1671.		
p1671[0...n]	Current setpoint filter 3 numerator damping / I_set_filt 3 D_z		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5710
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000	Max 10.000	Factory setting 0.700
Description:	Sets the numerator damping for current setpoint filter 3.		
Dependency:	Current setpoint filter 3 is activated via p1656.2 and parameterized via p1667 ... p1671.		

p1672[0...n]	Current setpoint filter 4 type / I_set_filt 4 Typ		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: 5710
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1	Max 2	Factory setting 1
Description:	Sets the current setpoint filter 4 as low pass (PT2) or as extended general 2nd-order filter.		
Value:	1: Low pass: PT2 2: General 2nd-order filter		
Dependency:	Current setpoint filter 4 is activated via p1656.3 and parameterized via p1672 ... p1676.		
p1673[0...n]	Current setpoint filter 4 denominator natural frequency / I_set_filt 4 fn_n		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5710
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 1999.0 [Hz]
Description:	Sets the denominator natural frequency for current setpoint filter 4 (PT2, general filter).		
Dependency:	Current setpoint filter 4 is activated via p1656.3 and parameterized via p1672 ... p1676.		
p1674[0...n]	Current setpoint filter 4 denominator damping / I_set_filt 4 D_n		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5710
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.001	Max 10.000	Factory setting 0.700
Description:	Sets the denominator damping for current setpoint filter 4.		
Dependency:	Current setpoint filter 4 is activated via p1656.3 and parameterized via p1672 ... p1676.		
p1675[0...n]	Current setpoint filter 4 numerator natural frequency / I_set_filt 4 fn_n		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5710
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.5 [Hz]	Max 16000.0 [Hz]	Factory setting 1999.0 [Hz]
Description:	Sets the numerator natural frequency for current setpoint filter 4 (general filter).		
Dependency:	Current setpoint filter 4 is activated via p1656.3 and parameterized via p1672 ... p1676.		
p1676[0...n]	Current setpoint filter 4 numerator damping / I_set_filt 4 D_z		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5710
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000	Max 10.000	Factory setting 0.700
Description:	Sets the numerator damping for current setpoint filter 4.		
Dependency:	Current setpoint filter 4 is activated via p1656.3 and parameterized via p1672 ... p1676.		

p1699	Filter data acceptance / Filt data accept		
SERVO, VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0	Max 1	Factory setting 0
Description:	Activates data acceptance for parameter changes for the filter. p1699 = 0: The new filter data are immediately accepted. p1699 = 1: The new filter data are only accepted when this parameter is reset.		
Dependency:	Refer to: p1414, p1415, p1416, p1417, p1418, p1419, p1420, p1421, p1422, p1423, p1424, p1425, p1426, p1656, p1657, p1658, p1659, p1660, p1661, p1662, p1663, p1664, p1665, p1666, p1667, p1668, p1669, p1670, p1671, p1672, p1673, p1674, p1675, p1676		

p1701[0...n]	Current controller reference model dead time / I_ctrRefMod t_dead		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 5714
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0	Max 1.0	Factory setting 1.0
Description:	Sets the fractional dead time for the current controller reference model. This parameter emulates the computing dead time of the proportionally controlled current control loop.		
Note:	Dead time = p1701 * p0115[0]		

p1702[0...n]	Isd current controller pre-control scaling / Isd_ctr_prectrScal		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6714
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: ASM, REL	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 200.0 [%]	Factory setting 70.0 [%]
Description:	Sets the scaling of the dynamic current controller pre-control for the flux-generating current component Isd.		
Note:	The parameter is effective for permanent and separately-excited synchronous motors.		

p1703[0...n]	Isq current controller pre-control scaling / Isq_ctr_prectrScal		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 4
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6714
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 200.0 [%]	Factory setting 70.0 [%]
Description:	Sets the scaling of the dynamic current controller pre-control for the torque/force-generating current component Isq.		

p1704[0...n]	Isq current controller pre-control EMF scaling / Isq_ctrl EMF scal		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200.0 [%]	Access level: 4 Func. diagram: 6714, 6726 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the scaling of the EMF pre-control for the Isq current controller.		
p1705[0...n]	Flux setpoint/actual value tracking threshold / Flux track thresh		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, PEM, REL Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: PERCENT Max 100.0 [%]	Access level: 4 Func. diagram: 6714, 6726 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Threshold for the setpoint - actual value tracking of the EMF pre-control of the Isq current controller.		
p1715[0...n]	Current controller P gain / I_ctrl Kp		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000 [V/A]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: 18_1 Scaling: - Max 100000.000 [V/A]	Access level: 3 Func. diagram: 5714, 7017 Unit selection: p0505 Expert list: 1 Factory setting 0.000 [V/A]
Description:	Sets the proportional gain of the current controller for the lower adaptation current range. This value is automatically preset using p3900 or p0340 when commissioning has been completed.		
Dependency:	Refer to: p0391, p0392, p0393		
Note:	For p0393 = 100 %, the current controller adaptation is disabled and p1715 is effective over the entire range.		
p1715[0...n]	Current controller P gain / I_ctrl Kp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100000.000	Access level: 3 Func. diagram: 1710, 6714, 7017 Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the proportional gain of the current controller for the lower adaptation current range. This value is automatically preset using p3900 or p0340 when commissioning has been completed.		
Dependency:	Refer to: p0391, p0392, p0393		
Note:	For p0393 = 100 %, the current controller adaptation is disabled and p1715 is effective over the entire range.		

p1717[0...n]	Current controller integral-action time / I_ctrl Tn		
SERVO, VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000.00 [ms]	Access level: 3 Func. diagram: 1710, 5714, 6714, 7017 Unit selection: - Expert list: 1 Factory setting 2.00 [ms]
Description:	Sets the integral-action time of the current controller.		
Dependency:	Refer to: p1715		
r1718	CO: Isq controller output / Isq_ctrl outp		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 4 Func. diagram: 6714 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the actual output of the Isq current controller (torque/force generating current, PI controller). The value contains the proportional and integral components of the PI controller.		
r1719	Isq controller integral component / Isq_ctrl I_comp		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 4 Func. diagram: 6714 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the integral component of the Isq current controller (torque/force-generating current, PI controller).		
r1723	CO: Isd controller output / Isd_ctrl outp		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 4 Func. diagram: 6714 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the actual output of the Isd current controller (flux-generating current, PI controller). The value contains the proportional and integral components of the PI controller.		
r1724	Isd controller integral component / Isd_ctrl I_comp		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 4 Func. diagram: 6714 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the integral component of the Isd current controller (flux-generating current, PI controller).		

r1725	Isd controller integral component limit / Isd_ctrl I_limit		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 4 Func. diagram: 6714 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the limit value for the integral component of the Isd current controller.		
p1726[0...n]	Quadrature arm decoupling, scaling / Transv_decpl scal		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200.0 [%]	Access level: 4 Func. diagram: 6714 Unit selection: - Expert list: 1 Factory setting 75.0 [%]
Description:	Sets the scaling of the quadrature arm decoupling		
Note:	This parameter is ineffective for encoderless vector control. In this case, p1727 is always used. If p1726 is set to 0, then the quadrature de-coupling is deactivated. The integral component of the Isd current controller remains effective in the complete speed control range. For the closed-loop control of synchronous motors, this parameter is used to scale the current controller de-coupling.		
p1727[0...n]	Quadrature arm decoupling at voltage limit scaling / TrnsvDecplVmaxScal		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200.0 [%]	Access level: 4 Func. diagram: 6714 Unit selection: - Expert list: 1 Factory setting 50.0 [%]
Description:	Sets the scaling of quadrature arm decoupling when the voltage limit is reached.		
r1728	De-coupling voltage, in-line axis / V_dir-axis_decoupl		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 6714 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the actual output of the quadrature channel de-coupling for the d axis.		
r1729	De-coupling voltage, quadrature axis / V_quad_decoupl		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 6714 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the actual output of the quadrature channel de-coupling for the q axis.		

r1732	CO: Direct-axis voltage setpoint / Direct V set		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 1630, 5714, 6714, 5718 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the direct-axis voltage setpoint Ud.		
r1732[0...1]	CO: Direct-axis voltage setpoint / Direct V set		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 1630, 5714, 6714, 5718 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the direct-axis voltage setpoint Ud.		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		
r1733	CO: Quadrature-axis voltage setpoint / Quad V set		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 1630, 5714, 5718, 6714, 6719 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the quadrature-axis component of voltage setpoint Uq.		
r1733[0...1]	CO: Quadrature-axis voltage setpoint / Quad V set		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 3 Func. diagram: 1630, 5714, 5718, 6714, 6719 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the quadrature-axis component of voltage setpoint Uq.		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		
p1740[0...n]	Gain resonance damping for encoderless closed loop control / Gain res_damp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.025
Description:	Defines the gain of the controller for resonance damping for operation with encoderless vector control in the range that current is impressed.		

p1744[0...n]	Motor model speed threshold stall detection / MotMod n_thr stall		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL, FEM	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 100.00 [rpm]
Description:	Sets the speed threshold value to detect a stalled motor. If the adaptation controller output exceeds the parameterized speed difference, then bit 11 in status word p1408 is set.		
Dependency:	If a stalled drive is detected (p1408.11 set), fault 7902 is output after the delay time in p2178. Refer to: p2178		
Note:	Speed monitoring is only effective in operation with a speed encoder (refer to p1300).		
p1745[0...n]	Motor model error threshold stall detection / MotMod ThreshStall		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 1000.0 [%]	Factory setting 5.0 [%]
Description:	Sets the fault threshold in order to detect a motor that has stalled. If the error signal (r1746) exceeds the parameterized error threshold, then bit 12 in status word p1408 is set.		
Dependency:	If a stalled drive is detected (p1408.12 set), fault F07902 is output after the delay time set in p2178. Refer to: p2178		
Note:	Monitoring is only effective in the low-speed range (below p1755 * (100% - p1756)).		
r1746	Motor model error signal stall detection / MotMod sig stall		
VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min - [%]	Max - [%]	Factory setting - [%]
Description:	Signal to initiate stall detection		
Note:	The signal is not calculated while magnetizing and only in the low speed range (below p1755 * (100 % - p1756)).		
p1748[0...n]	Motor model lower changeover speed n_set -> n_act / Lower n_chngov		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 90.0 [%]	Factory setting 50.0 [%]
Description:	Sets the lower speed for the transition "n_set -> n_act" in encoderless operation. This value is entered as a percentage referred to p1749.		
Dependency:	Refer to: p1749, p1752		

p1749[0...n]					
Motor model upper changeover speed n_set -> n_act / Upper n_chgov					
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3		
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -		
	P-Group: Closed-loop control	Units group: -	Unit selection: -		
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1		
	Min 0.0 [%]	Max 99.0 [%]	Factory setting 50.0 [%]		
Description:	Sets the upper speed for the transition "n_set -> n_act" in sensorless operation. This value is entered as a percentage of p1755.				
Dependency:	Refer to: p1748, p1752				
p1750[0...n]					
Motor model configuration / MotMod config					
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3		
	Data type: Unsigned8	Dynamic index: DDS, p0180	Func. diagram: -		
	P-Group: Closed-loop control	Units group: -	Unit selection: -		
	Not for motor type: REL, FEM	Scaling: -	Expert list: 1		
	Min -	Max -	Factory setting 0000 bin		
Description:	Sets the configuration for the motor model. Bit 0 = 1: Forces open-loop speed-controlled starting (ASM). Bit 1 = 1: Forces the system to pass through frequency zero, open-loop-controlled (ASM). Bit 2 = 1: Drive remains in full closed-loop control mode, even at zero frequency (ASM). Bit 3 = 1: Motor model evaluates the saturation characteristic (ASM). Bit 4 = 1: Time-controlled change between current and observer models (ASM). Bit 5 = 1: HF signal injection to estimate the continuous rotor position (PESM). Bit 6 = 1: If the motor is blocked, encoderless vector control remains speed-controlled (ASM).				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Controlled start	Yes	No	-
	01	Controlled through 0 Hz	Yes	No	-
	02	Closed-loop ctrl oper. down to zero freq. for passive loads	Yes	No	-
	03	Motor model Lh_pre = f(PsiEst)	Yes	No	-
	04	Model changeover	Time controlled	Freq. controlled	-
	05	Cl.-loop control mode PESM up to f=0 with HF signal injection	Yes	No	-
	06	Closed-loop control when motor is blocked	Yes	No	-
Caution:	Do not use bit 6 = 1 if the motor can be reversed by the load. Long wait times due to blocking (p2177) can cause the motor to stall. In this case you should deactivate the function or use closed-loop control throughout the speed range (note the information re bit 2 = 1).				
					
Note:	Bit 0 ... Bit 2 only have influence for sensorless vector control, bit 4 only for vector control with sensor. Bit 2 is pre-assigned depending on p0500. Re bit 02 = 1: The sensorless vector control is effective down to zero frequency. A change is not made into the open-loop speed controlled mode. This operating mode is possible for passive loads. These include applications where the load itself does not generate any active torque and therefore only acts reactively to the drive torque of the induction motor. If bit 2 is set to 1, then bit 3 is also automatically activated. Manual deselection is possible and can make sense if, for third-party motors, the saturation characteristic (p1960) was not carried out. Generally, for standard SIEMENS motors, the already pre-assigned (default value) saturation characteristic is adequate. For bit 2 = 1, the selection of bits 0 and 1 is ignored. Re bit 02 = 0: Bit 3 is deactivated automatically.				

Re bit 05 = 1:

The selection of HF signal injection is only relevant for permanent-magnet synchronous motors (PESM). Bits 0..4 are not relevant for PESM.

For the purpose of component configuration, an automatic system run-up is triggered on first-time activation. Activation is, therefore, only possible outside of motor commissioning (p0010 = 0).

Re bit 06 = 1:

The following applies for encoderless vector control of induction motors only: If the motor is blocked (see p2175, p2177), the time condition p1758 is bypassed and there is no changeover to open-loop control.

r1751 Motor model status / MotMod status

VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL, FEM	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the motor model.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Controlled operation	Active	Inactive	6721
	01	Set ramp-function generator	Active	Inactive	-
	02	Stop RsLh adaptation	Yes	No	-
	03	Feedback	Active	Inactive	-
	04	Encoder operation	Active	Inactive	-
	05	Holding angle	Yes	No	-
	06	Acceleration criterion	Active	Inactive	-
	07	Set angular integrator PEM	No	Yes	-
	08	Stop Kt adaptation PEM	No	Yes	-
	09	PolID active PEM SLVC	No	Yes	-
	10	I injection PEM	No	Yes	-
	11	Speed controller output cannot be set to zero	Yes	No	-
	12	Rs adapt waits	Yes	No	-
	13	Motor operation	Yes	No	-
	14	Stator frequency sign	Positive	Negative	-
	15	Torque sign	Motor mode	Regenerative mode	-
	16	Pulse injection active PEM	Yes	No	-

p1752[0...n] Motor model changeover speed operation with encoder / MotMod n_chgov enc

SERVO	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [rpm]	210000.00 [rpm]	210000.00 [rpm]

Description: Sets the speed to change over the motor model for operation with encoder.

Dependency: Refer to: p1756

Note: Induction motor (ASM):

The motor model is influenced for speeds/velocities greater than p1752.

Synchronous motor (SRM):

A monitoring (F07412) is activated for speeds/velocities greater than p1752.

The motor model is additionally influenced when kT adaptation is activated (p1780.3 = 1).

p1752[0...n]	Motor model with encoder changeover velocity / MotMod v_chgov enc		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [m/min]	Calculated: CALC_MOD_REG Dynamic index: DDS, p0180 Units group: 4_1 Scaling: - Max 1000.00 [m/min]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 1000.00 [m/min]
Description:	Sets the velocity to change over the motor model for operation with encoder.		
Dependency:	Refer to: p1756		
Note:	Induction motor (ASM): The motor model is influenced for speeds/velocities greater than p1752. Synchronous motor (SRM): A monitoring (F07412) is activated for speeds/velocities greater than p1752. The motor model is additionally influenced when kT adaptation is activated (p1780.3 = 1).		
p1752[0...n]	Motor model changeover speed operation with encoder / MotMod n_chgov enc		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [rpm]	Calculated: CALC_MOD_REG Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 210000.00 [rpm]
Description:	Sets the speed to change over the motor model for operation with encoder.		
Dependency:	In V/f characteristic mode the parameter is of no significance. Refer to: p1756		
p1753[0...n]	Motor model changeover speed hysteresis operation with encoder / MotMod n_chgovHysE		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: CALC_MOD_REG Dynamic index: DDS, p0180 Units group: - Scaling: - Max 90.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [%]
Description:	Sets the hysteresis for the changeover speed of the motor model for operation with speed encoder.		
Dependency:	Refer to: p1752		
Note:	The value refers to p1752. In the case of separately excited synchronous motors, the lower hysteresis value is calculated with $p1752 * p1753$; in the case of all other types of motor, $p1752 * (1 - p1753)$ is used.		
p1754[0...n]	Flux angle difference smoothing time / Angle diff T_smth		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, REL Min 0.1 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000.0 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 5.0 [ms]
Description:	Sets the smoothing time constant to filter the main flux angle difference from the voltage and current models. The filtered value is included in the calculation of the total flux angle. PESM: Sets the smoothing time constant for the angular difference display between motor model and encoder.		
Note:	In the case of a separately excited synchronous motor and sensorless vector control, the parameter must be set to the minimum value to improve motor model changeover.		

p1755[0...n] Motor model changeover speed encoderless operation / MotMod n_chgSnsorl

SERVO	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 210000.00 [rpm]
Description:	Sets the speed to change over the motor model to encoderless operation.		
Dependency:	Refer to: p1756		
Note:	The changeover speed applies for the changeover between open-loop and closed-loop control mode.		

p1755[0...n] Motor model changeover velocity encoderless operation / MotMod v_chgSnsorl

SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 1000.00 [m/min]	Factory setting 1000.00 [m/min]
Description:	Sets the velocity to change over the motor model to encoderless operation.		
Dependency:	Refer to: p1756		
Note:	The changeover velocity applies for the changeover between open-loop and closed-loop control mode.		

p1755[0...n] Motor model changeover speed encoderless operation / MotMod n_chgSnsorl

VECTOR	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 210000.00 [rpm]
Description:	Sets the speed to change over the motor model to encoderless operation.		
Dependency:	In V/f characteristic mode the parameter is of no significance. Refer to: p1756		
Notice:	The changeover speed represents the steady-state minimum speed up to which the motor model can be used in sensorless steady-state operation. If the stability is not adequate close to the changeover speed, it may make sense to increase the parameter value.		
Note:	The changeover speed applies for the changeover between open-loop and closed-loop control mode.		

p1756 Motor model changeover speed hysteresis / MotMod n_chgov hys

SERVO	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 90.0 [%]	Factory setting 5.0 [%]
Description:	Sets the hysteresis for the changeover speed/velocity of the motor model.		
Dependency:	Refer to: p1752, p1755		
Note:	The value is entered relative to p1404, p1752 or p1755.		

p1756	Motor model changeover velocity hysteresis / MotMod v_chgov hys		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: CALC_MOD_CON Dynamic index: - Units group: - Scaling: - Max 90.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 5.0 [%]
Description:	Sets the hysteresis for the changeover speed/velocity of the motor model.		
Dependency:	Refer to: p1752, p1755		
Note:	The value is entered relative to p1404, p1752 or p1755.		
p1756	Motor model changeover speed hysteresis encoderless operation / MotMod n_chgov hys		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.0 [%]	Calculated: CALC_MOD_REG Dynamic index: - Units group: - Scaling: - Max 95.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 50.0 [%]
Description:	Sets the hysteresis for the changeover speed of the motor model for encoderless operation.		
Dependency:	In V/f characteristic mode the parameter is of no significance. Refer to: p1755		
Note:	The parameter value refers to p1755. In the case of separately excited synchronous motors, the lower hysteresis value is calculated with $p1755 * p1756$; in the case of all other types of motor, $p1755 * (1 - p1756)$ is used.		
p1757[0...n]	Motor model w/o enc. op./cl.-loop controlled stab. controller Kp / MotMod w/o enc Kp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL, FEM Min 0.01	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.00	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.70
Description:	Sets the gain of the transient response controller when the motor model changes over from open-loop controlled operation to closed-loop controlled operation.		
Note:	Only for ASM and PSM in encoderless operation: The settling range starts at $0.5 * p1755 * p1756$. For ASM it ends at $p1755 * p1756$ or at p1755, if p1759 is at the maximum value. For PSM it always ends at $p1755 * p1756$.		
p1758[0...n]	Motor model changeover delay time closed/open-loop control / MotMod t_cl_op		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min 100 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1000 [ms]
Description:	Sets the minimum time for falling below the changeover speed when changing from closed-loop controlled operation to open-loop controlled operation.		
Dependency:	Refer to: p1755, p1756		

p1759[0...n]	Motor model changeover delay time open/closed loop control / MotMod t op_cl		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 2000 [ms]	Factory setting 0 [ms]
Description:	Sets the minimum time for exceeding the changeover speed when changing from open-loop controlled operation to closed-loop controlled operation.		
Dependency:	Refer to: p1755, p1756		
Note:	When p1759 = 2000 ms, the delay time becomes ineffective and the model changeover is determined by the output frequency only.		
p1760[0...n]	Motor model with encoder speed adaptation Kp / MotMod wE n_ada Kp		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL, FEM	Scaling: -	Expert list: 1
	Min 0.000	Max 100000.000	Factory setting 1000.000
Description:	Sets the proportional gain of the controller for speed adaptation with encoder		
p1761[0...n]	Motor model with encoder speed adaptation Tn / MotMod wE n_ada Tn		
VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL, FEM	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 1000 [ms]	Factory setting 4 [ms]
Description:	Sets the integral-action time of the controller for speed adaptation with encoder		
r1762[0...1]	Motor model deviation component 1 / MotMod dev comp 1		
VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 6721, 6730, 6731
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL, FEM	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Induction motor (ASM): Displays the referred imaginary system deviation for the adaptation circuit of the motor model. Permanent-magnet synchronous motor (PESM): Displays the system deviation for speed adaptation. r1762.0: Angular deviation [rad-el] of the estimated EMF. r1762.1: Angular deviation [electr. deg.] of the low-level signal response for pulse technique.		
Index:	[0] = Deviation Model1 [1] = Deviation Model2		

r1763	Motor model deviation component 2 / MotMod dev comp 2		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL, FEM Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Induction motor (ASM): Displays the referred real system deviation for the adaptation circuit of the motor model. Permanent-magnet synchronous motor (PESM): Not used.		
p1764[0...n]	Motor model without encoder speed adaptation Kp / MotMod woE n_adaKp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL, FEM Min 0.000	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100000.000	Access level: 3 Func. diagram: 6730 Unit selection: - Expert list: 1 Factory setting 1000.000
Description:	Sets the proportional gain of the controller for speed adaptation without encoder.		
r1765	Motor model, speed adaptation Kp effective / MotM n_ada Kp act		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL, FEM Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the effective proportional gain of the controller for the speed adaptation.		
p1766[0...n]	Motor model voltage model calculation enable / V_mod calc enab		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, REL Min 0.0 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 90.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 50.0 [%]
Description:	Sets the speed to enable the voltage model to calculate the speed actual value. This value is entered as a percentage referred to p1748.		
Dependency:	Refer to: p1748, p1752		
p1767[0...n]	Motor model without encoder speed adaptation Tn / MotMod woE n_adaTn		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL, FEM Min 1 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 200 [ms]	Access level: 3 Func. diagram: 6730 Unit selection: - Expert list: 1 Factory setting 4 [ms]
Description:	Sets the integral time of the controller for speed adaptation without encoder		

r1768	Motor model, speed adaptation Vi effective / MotM n_ada Vi act		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL, FEM Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the effective gain of the integral component of the controller for speed adaptation.		
r1770	CO: Motor model speed adaptation proportional component / MotMod n_adapt Kp		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL, FEM Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 6730 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the P component of the controller for speed adaptation.		
r1771	CO: Motor model speed adaptation I comp. / MotMod n_adapt Tn		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL, FEM Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 6730 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the I component of the controller for speed adaptation.		
r1773[0...1]	Motor model slip speed / MotMod slip		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL, FEM Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: - Max - [rpm]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays estimated (speed) signals of the motor model: r1773.0: Displays the estimated (mechanical) slip of the motor model. r1773.1: Displays the estimated input speed of the motor model.		
Index:	[0] = Estimated slip speed [1] = Estimated speed		
p1774[0...n]	Motor model, offset voltage compensation alpha / MotMod offs comp A		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min -5.000 [V]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 5.000 [V]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Sets the offset voltage in the alpha direction; this compensates the offset voltages of the drive converter/inverter at low speeds. The value is valid for the rated (nominal) pulse frequency of the Motor Module.		
Note:	The value is preset during the rotating measurement.		

p1775[0...n]	Motor model, offset voltage compensation beta / MotMod offs comp B		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min -5.000 [V]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 5.000 [V]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Sets the offset voltage in the beta direction; this compensates the offset voltages of the drive converter/inverter at low speeds. The value is valid for the rated (nominal) pulse frequency of the Motor Module.		
Note:	The value is preset during the rotating measurement.		
r1776[0...6]	Motor model status signals / MotMod status sig		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the internal status signals of the motor model: Index 0: Changeover ramp between current and voltage models Index 1: Changeover ramp for model tracking (encoderless induction motors only) Index 2: Changeover ramp for zero frequency range (encoderless induction motors only) Index 3: Transition ramp actual speed from speed setpoint to model value (encoderless FEM) Index 4: Speed controller enable (encoderless FEM) Index 5: Transition ramp between current and voltage models (encoderless FEM) Index 6: Transition ramp for EMF deviation at PLL input (encoderless PESM)		
Index:	[0] = Changeover ramp motor model [1] = Changeover ramp model tracking [2] = Changeover ramp zero frequency encoderless ASM [3] = Changeover ramp actual speed encoderless FEM [4] = Enable speed controller encoderless FEM [5] = Changeover ramp motor model encoderless FEM [6] = Changeover ramp motor model encoderless PESM		
Note:	Indices 3 through 5 are only relevant in the case of encoderless control of separately excited synchronous motors.		
r1778	Motor model flux angle difference / MotMod ang. diff.		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min - [°]	Calculated: - Dynamic index: - Units group: - Scaling: p2005 Max - [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [°]
Description:	Induction motor (ASM): Displays the difference between the motor model flux angle and the transformation angle. Permanent-magnet synchronous motor (PESM): Displays the angular difference between motor model and encoder.		
Dependency:	A setting for smoothing the display can be made using p1754.		
Notice:	The display only makes sense for corrected actual value inversion, encoder pulse number and pole pair number. Example: Moving in encoderless operation at a speed not equal to zero and without load. -> Check the sign of r0061 and r0063. If the sign is not equal, then change p0410.0. -> Check the stationary value of r0061 and r0063. If the value is not equal, change the encoder pulse number (p0408) or pole pair number (p0314).		

r1778 Motor model flux angle difference / MotMod ang. diff.

VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: p2005	Expert list: 1
	Min	Max	Factory setting
	- [°]	- [°]	- [°]

Description: Induction motor (ASM):
Displays the difference between the motor model flux angle and the transformation angle.
Permanent-magnet synchronous motor (PESM):
Displays the angular difference between motor model and encoder.

Dependency: A setting for smoothing the display can be made using p1754.

r1779 Motor model absolute flux / MotMod abs flux

VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL, FEM	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]

Description: Displays the absolute value of the flux of the motor model.

p1780[0...n] Motor model adaptation configuration / MotMod adapt conf

SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0010 0000 bin

Description: Sets the configuration for the adaptation circuit of the motor model.
Induction motor (ASM): Rs, Rr (only for operation with encoder), Lh and offset compensation.
Permanent magnet synchronous motor (PEM): kT

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	03	Select motor model PEM kT adaptation	Yes	No	-
	05	Select ASM Rr adaptation (only with encoder)	Yes	No	-
	07	Select T(valve) with Rs adaptation	Yes	No	-
	08	Compensation voltage emulation error in the drive converter	Yes	No	-
	09	kT(iq) characteristic active	Yes	No	-

Caution: For the PEM kT adaptation (p1780.3) as well as the compensation of the voltage emulation error (p1780.8) and for the kT(iq) characteristic (p1780.9), the function module "Extended torque control" (r0108.1) should be activated.



Note: ASM: Induction motor
PEM: Permanent magnet synchronous motor
The kT adaptation is only active at a speed greater than the changeover speed with encoder (p1752). Near the current limit when strongly saturating motors are operated.

p1780[0...n] Motor/converter model adaptation configuration / MotMod adapt conf						
SERVO (Ext M_ctrl)	Can be changed:	U, T	Calculated:	-	Access level:	3
	Data type:	Unsigned16	Dynamic index:	DDS, p0180	Func. diagram:	-
	P-Group:	Closed-loop control	Units group:	-	Unit selection:	-
	Not for motor type:	REL	Scaling:	-	Expert list:	1
	Min	-	Max	-	Factory setting	0000 0001 0010 1000 bin
Description: Sets the configuration for the adaptation circuit of the motor model. Induction motor (ASM): Rs, Rr (only for operation with encoder), Lh and offset compensation. Permanent magnet synchronous motor (PEM): kT						
Bit field:	Bit	Signal name	1 signal	0 signal	FP	
	03	Select motor model PEM kT adaptation	Yes	No	-	
	05	Select ASM Rr adaptation (only with encoder)	Yes	No	-	
	07	Select T(valve) with Rs adaptation	Yes	No	-	
	08	Compensation voltage emulation error in the drive converter	Yes	No	-	
	09	kT(iq) characteristic active	Yes	No	-	
 Caution: For the PEM kT adaptation (p1780.3) as well as the compensation of the voltage emulation error (p1780.8) and for the kT(iq) characteristic (p1780.9), the function module "Extended torque control" (r0108.1) should be activated.						
 Note: ASM: Induction motor PEM: Permanent magnet synchronous motor The kT adaptation and the kT characteristic can be simultaneously selected. Re kT adaptation (p1780.3 = 1): - the kT adaptation is only active at a speed greater than the changeover speed with encoder (p1752). - beforehand, an identification of the voltage emulation error must be started (p1909.14 = 1). - If the electrical configuration (e.g. Motor Module, cable routing) or the pulse frequency (p1800) changes, then a new identification run must be carried out. - To identify the voltage emulation error the Motor Module should still be warm. - the motor temperature (r0035) should not change significantly (i.e. it should not be identified immediately after a load duty cycle). Re kT(iq) characteristic (p1780.9 = 1): - for the kT(iq) characteristic $kT(iq) = kT + kT3 * iq^2 + kT5 * iq^4 + kT7 * iq^6$ the parameters must first be identified (p1959.6 = 1) (kT: p0316, kT3: p0646, kT5: p0647, kT7: p0647).						
p1780[0...n] Motor model adaptation configuration / MotMod adapt conf						
VECTOR	Can be changed:	U, T	Calculated:	-	Access level:	3
	Data type:	Unsigned16	Dynamic index:	DDS, p0180	Func. diagram:	-
	P-Group:	Closed-loop control	Units group:	-	Unit selection:	-
	Not for motor type:	REL	Scaling:	-	Expert list:	1
	Min	-	Max	-	Factory setting	0111 1100 bin
Description: Sets the configuration for the adaptation circuit of the motor model. Induction motor (ASM): Rs, Rr (only for operation with encoder), Lh and offset compensation. Permanent magnet synchronous motor (PEM): kT						

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	01	Select motor model ASM Rs adaptation	Yes	No	-
	02	Select motor model ASM Lh adaptation	Yes	No	-
	03	Select motor model PEM kT adaptation	Yes	No	-
	04	Select motor model offset adaptation	Yes	No	-
	05	Select ASM Rr adaptation (only with encoder)	Yes	No	-
	06	Select pole position identification PEM encoderless	Yes	No	-
	07	Select T(valve) with Rs adaptation	Yes	No	-

Dependency: In V/f characteristic operating mode only bit 7 is relevant.

Note: ASM: Induction motor

PEM: Permanent magnet synchronous motor

When selecting the compensation of the valve interlocking via Rs (bit 7), the compensation in the gating unit is deactivated and is instead taken into account in the motor model.

In order that the correction values of the Rs, Lh and kT adaptation (selected using Bit 0 ... Bit 2) are correctly accepted when changing over the drive data set, a dedicated motor number must be entered into p0826 for each different motor.

p1781[0...n] Motor model Rs adaptation integral time / MotMod Rs Tn

VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
	Min 10 [ms]	Max 10000 [ms]	Factory setting 100 [ms]

Description: Sets the integral time for the Rs adaptation of the motor model for an induction motor (ASM).

r1782[0...n] Motor model Rs adaptation corrective value / MotMod Rs corr

VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
	Min - [Ohm]	Max - [Ohm]	Factory setting - [Ohm]

Description: Displays the corrective value for the Rs adaptation of the motor model for an induction motor (ASM).

Dependency: Refer to: p0826, p1780

Note: The display of the inactive data sets is only updated when changing over the data set.

p1783[0...n] Motor model Rs adaptation Kp / MotMod Rs Kp

VECTOR (n/M)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: PEM, REL, FEM	Scaling: -	Expert list: 1
	Min 0.000	Max 1.000	Factory setting 0.100

Description: Sets the proportional gain for the Rs adaptation of the motor model for an induction motor (ASM).

p1785[0...n]	Motor model Lh adaptation Kp / MotMod Lh Kp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min 0.000	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1.000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.100
Description:	Sets the proportional gain for the Lh adaptation of the motor model for an induction motor (ASM).		
p1786[0...n]	Motor model Lh adaptation integral time / MotMod Lh Tn		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min 10 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100 [ms]
Description:	Sets the integral time for the Lh adaptation of the motor model for an induction motor (ASM).		
r1787[0...n]	Motor model Lh adaptation corrective value / MotMod Lh corr		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min - [mH]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max - [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the corrective value for the Lh adaptation of the motor model for an induction motor (ASM).		
Dependency:	Refer to: p0826, p1780		
Note:	The adaptation result is reset if the magnetizing inductance of the induction motor is changed (p0360, r0382). This also happens when changing over the data set if a different motor is not being used (p0826). The display of the inactive data sets is only updated when changing over the data set.		
r1789	Motor model Rs adaptation switch-in frequency / MotMod Rs f_on		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Hz]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the power-on stator frequency for the Rs adaptation for the induction motor (ASM).		
r1790	Motor model Rs adaptation power-on slip / MotMod Rs fslip		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Hz]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the power-on slip frequency for the Rs adaptation for the induction motor (ASM).		

r1791	Motor model Lh adaptation power-on frequency / MotMod Lh f_on		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Hz]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the power-on stator frequency/ primary section frequency for the Lh adaptation for the induction motor (ASM).		
r1792	Motor model Lh adaptation power-on slip / MotMod Lh fslip		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: PEM, REL, FEM Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Hz]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the power-on slip frequency for the Lh adaptation for the induction motor (ASM).		
p1795[0...n]	Motor model kT adaptation smoothing time / MotMod kT T_smth		
SERVO (Ext M_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, REL, FEM Min 1 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100 [ms]
Description:	Sets the smoothing time of the kT adaptation of the motor model for a permanent-magnet synchronous motor (PEM).		
Dependency:	Refer to: p1780, r1797		
p1795[0...n]	Motor model kT adaptation integral time / MotMod kT Tn		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, REL, FEM Min 10 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000 [ms]	Access level: 3 Func. diagram: 6731 Unit selection: - Expert list: 1 Factory setting 100 [ms]
Description:	Sets the integral time of the kT adaptation of the motor model for a permanent-magnet synchronous motor (PEM).		
r1797	Motor model kT adaptation corrective value / MotMod kT corr		
SERVO (Ext M_ctrl, Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, REL, FEM Min - [N/Arms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [N/Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [N/Arms]
Description:	Displays the corrective value of the kT adaptation of the motor model for a permanent-magnet synchronous motor (PEM).		
Dependency:	Refer to: p1780, p1795		

r1797	Motor model kT adaptation corrective value / MotMod kT corr		
SERVO (Ext M_ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, REL, FEM Min - [Nm/A]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Nm/A]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Nm/A]
Description:	Displays the corrective value of the kT adaptation of the motor model for a permanent-magnet synchronous motor (PEM).		
Dependency:	Refer to: p1780, p1795		
r1797[0...n]	Motor model kT adaptation corrective value / MotMod kT corr		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, REL, FEM Min - [Nm/A]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max - [Nm/A]	Access level: 3 Func. diagram: 6731 Unit selection: - Expert list: 1 Factory setting - [Nm/A]
Description:	Displays the corrective value of the kT adaptation of the motor model for a permanent-magnet synchronous motor (PEM).		
Dependency:	Refer to: p0826, p1780		
Note:	The display of the inactive data sets is only updated when changing over the data set.		
p1798[0...n]	Motor model pulse technique speed adaptation Kp / MotMod pulses Kp		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: ASM, REL, FEM Min 0.000	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000.000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.000
Description:	Sets the proportional gain Kp for speed adaptation with active pulse technique for the estimation of the continuous rotor position.		
p1800[0...n]	Pulse frequency setpoint / Pulse freq setp		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Modulation Not for motor type: - Min 1.000 [kHz]	Calculated: CALC_MOD_ALL Dynamic index: DDS, p0180 Units group: - Scaling: - Max 32.000 [kHz]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 4.000 [kHz]
Description:	Sets the drive converter switching frequency. This parameter is preset to the rated converter value when the drive is first commissioned.		
Dependency:	The pulse frequency can, depending on the current controller sampling time (p0115[0]) assume the following values: a) $p1800 = 1000 / (p0115[0] * n)$ with $n = 2, 3, 4$ b) $p1800 = 1000 * n / p0115[0]$ with $n = 1, 2, 3, 4, \dots$ Example: $p0115[0] = 125 \mu s \rightarrow p1800 = 2, 2.6, 4 \text{ kHz}$ (from equation a) $p0115[0] = 125 \mu s \rightarrow p1800 = 8, 16 \text{ kHz}$ (from equation b) Possible setting values can be taken from r0114 (if p0009 = p0010 = 0). Refer to: r0110, r0111, p0112, p0113, r0114, p0115, p0230, p1817		

Note: The maximum possible pulse frequency is also determined by the power unit being used.
When the pulse frequency is increased, depending on the particular power unit, the maximum output current can be reduced (derating, refer to r0067).
If p1800 is changed while commissioning (p0009, p0010 > 0), then it is possible that the old value will no longer be able to be set. The reason for this is that the dynamic limits of p1800 have been changed by a parameter that was set when the drive was commissioned (e.g. p1082).
For encoderless operation (p1404 = 0 or p1300 = 20), the following conditions apply:
 $p1800 = 1 / (2 * p0115[0])$
or
 $p1800 \geq n / p0115[0]$, $n = 1, 2, \dots$
For motors with a low power rating (< 300 W) we recommend that p1800 is set acc. to the second condition.

p1800[0...n]		Pulse frequency setpoint / Pulse freq setp	
VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1.000 [kHz]	Max 16.000 [kHz]	Factory setting 4.000 [kHz]
Description:	Sets the drive converter switching frequency. This parameter is preset to the rated converter value when the drive is first commissioned.		
Dependency:	The pulse frequency can, depending on the current controller sampling time (p0115[0]) assume the following values: p1800 = 1 / (p0115[0] * 2) or p1800 = n / p0115[0] where n = 1, 2, 3, ... Example: p0115[0] = 250 μs --> p1800 = 2, 4, 8, 12, 16 kHz Possible setting values can be taken from r0114 (if p0009 = p0010 = 0). If p0092 = 1 the sampling times p0115 and the pulse frequency p1800 are checked every time the parameters are downloaded, and reset to the initial values if necessary. This check can be deactivated by setting p0092 = 0 (making this setting does not affect isochronous PROFIBUS operation). If wobulation is selected (p1810.2), the pulse frequency can only be changed as part of pulse enabling to values with the following ratio: a) p1800 <= 1000 / p0115[0] for p1811 > 0 % b) p1800 <= 1000 * 2 / p0115[0] for p 1811 = 0 % Under pulse inhibit p1800 > 1000 / p0115[0] -> p1811 = 0 p1800 > 1000 * 2 / p0115[0] -> 1810.2 = 0 and p1811 = 0 (applicable for all indices) Refer to: r0110, r0111, p0112, p0113, r0114, p0115, p0230, p1817		
Note:	The maximum possible pulse frequency is also determined by the power unit being used. When the pulse frequency is increased, depending on the particular power unit, the maximum output current can be reduced (derating, refer to r0067). If a sine-wave filter is parameterized as output filter (p0230 = 3), then the pulse frequency cannot be changed below the minimum value required for the filter. If p1800 is changed while commissioning (p0009, p0010 > 0), then it is possible that the old value will no longer be able to be set. The reason for this is that the dynamic limits of p1800 have been changed by a parameter that was set when the drive was commissioned (e.g. p1082).		

r1801[0...1]	CO: Pulse frequency / Pulse frequency		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kHz]	Calculated: - Dynamic index: - Units group: - Scaling: p2000 Max - [kHz]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [kHz]
Description:	Displays the actual converter switching frequency.		
Index:	[0] = Actual [1] = Minimum value of the modulator		
Note:	The selected pulse frequency (p1800) may be reduced if the drive converter has an overload condition (p0290). The following applies for vector drives (p0107): The pulse frequency can also be reduced when changing over the modulator to an optimized pulse pattern. This is used to avoid overcontrol. In the case of chassis power units, two-thirds of the setpoint pulse frequency is displayed in the FLB modulation range.		

p1802[0...n]	Modulator mode / Modulator mode		
VECTOR	Can be changed: T Data type: Integer16 P-Group: Modulation Not for motor type: - Min 0	Calculated: CALC_MOD_LIM_REF Dynamic index: DDS, p0180 Units group: - Scaling: - Max 9	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the modulator mode.		
Value:	0: Automatic changeover SVM/FLB 1: Flat top modulation (FLB) 2: Space vector modulation (SVM) 3: SVM without overcontrol 4: SVM/FLB without overcontrol 5: SVM with pulse frequency reduction 6: SVM/FLB with pulse frequency reduction 7: No edge modulation up to 100 Hz 8: No edge modulation up to 60 Hz 9: Edge modulation		
Dependency:	If a sine-wave filter is parameterized as output filter (p0230 = 3, 4), or if the power unit firmware is not able to calculate edge modulation (r0192 bit0 = 0), then only space vector modulation without overcontrol can be set as modulation type (p1802 = 3). p1802 > 6: Wobulation is deactivated, p1810.2 = 0 and p1811 = 0 (valid for all indices) Refer to: r0192, p0230, p7003		
Notice:	If the edge modulation is enabled (p1802 > 6), then the current actual value correction should be activated (p1840.0 = 0), if the Motor Module is connected to a controlled (regulated) DC link (Active Infeed).		
Note:	When modulation modes are enabled that could lead to overmodulation (p1802 = 0, 1, 2, 5, 6), the modulation depth must be limited using p1803 (pre-assignment, p1803 = 98%). The higher the overmodulation, the greater the current ripple and torque ripple. When changing p1802[x], the values for all of the other existing indices are also changed. p1802 = 7, 8 should be used if the drive is operated below 100 Hz or 60 Hz, and it is necessary to avoid changing over to edge modulation. Above these output frequencies, the modulation depth remains limited so that there the full output voltage of the edge modulation is not reached.		

p1803[0...n]	Maximum modulation depth / Modulat depth max		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 6723
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 20.0 [%]	Max 150.0 [%]	Factory setting 100.0 [%]
Description:	Defines the maximum modulation depth.		
Note:	p1803 = 100% is the overcontrol limit for space vector modulation (for an ideal drive converter without any switching delay). If an optimized pulse pattern is enabled (edge modulation), then the modulation depth is limited to below the output frequency of 28 Hz as there is no optimized pulse pattern in this range.		
p1804[0...n]	Filter time constant smoothed modulation index / T_filt mod_idxSmth		
VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 10000.0 [ms]	Factory setting 10.0 [ms]
Description:	Filter time constant for the smoothed modulation index to change over the modulator mode.		
p1806[0...n]	Filter time constant Vdc correction / T_filt Vdc_corr		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 4
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 10000.0 [ms]	Factory setting 0.0 [ms]
Description:	Sets the filter time constant of the DC link voltage used to calculate the modulation depth.		
r1807	Actual DC link voltage to calculate the modulation depth / VdcActValMod_depth		
VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: 5_2	Unit selection: p0505
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min - [V]	Max - [V]	Factory setting - [V]
Description:	DC link voltage that is used to convert the setpoint voltage into an equivalent modulation depth.		
r1808	DC link voltage actual value for V_max calculation / Vdc act val V_max		
VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: 5_2	Unit selection: p0505
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min - [V]	Max - [V]	Factory setting - [V]
Description:	DC link voltage used to determine the maximum possible output voltage.		

r1809	CO: Modulator mode actual / Modulator mode act			
VECTOR	Can be changed: - Data type: Integer16 P-Group: Modulation Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -	
Description:	Displays the effective modulator mode.			
Value:	1: Flat top modulation (FLB) 2: Space vector modulation (SVM) 3: Edge modulation from 28 Hz; 23:3 4: Edge modulation from 28 Hz; 19:1 5: Edge modulation from 60 Hz; 17:3 6: Edge modulation from 60 Hz; 17:1 7: Edge modulation from 100 Hz; 9:2 8: Edge modulation from 100 Hz; 9:1			

p1810	Modulator configuration / Modulator config																																						
A_INF, S_INF	Can be changed: U, T Data type: Unsigned16 P-Group: Modulation Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1000 0000 0001 0000 bin																																				
Description:	Sets the configuration for the modulator.																																						
Bit field:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr> <tr> <td>02</td><td>Activate wobulation</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>04</td><td>Disable wobulation amplitude</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>05</td><td>Activate extended current limitation control</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>06</td><td>Activate isochronous current limitation</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>07</td><td>Activate voltage impression with dynamic current limits</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>15</td><td>Activate flat-top modulation</td><td>Yes</td><td>No</td><td>-</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FP	02	Activate wobulation	Yes	No	-	04	Disable wobulation amplitude	Yes	No	-	05	Activate extended current limitation control	Yes	No	-	06	Activate isochronous current limitation	Yes	No	-	07	Activate voltage impression with dynamic current limits	Yes	No	-	15	Activate flat-top modulation	Yes	No	-			
Bit	Signal name	1 signal	0 signal	FP																																			
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06	Activate isochronous current limitation	Yes	No	-																																			
07	Activate voltage impression with dynamic current limits	Yes	No	-																																			
15	Activate flat-top modulation	Yes	No	-																																			
Dependency:	If bit 2 is set from 1 to 0, p1811 = 0 is set.																																						
Notice:	Bit 02 = 1 can only be set subject to the following prerequisites: - Pulse inhibit - r0192.16 = 1 - $p1800 < 2 * 1000/p0115[0]$ Bit 15 can only be changed subject to the following prerequisites: - Pulse inhibit																																						
Note:	Re bit 02 = 0: A gating unit that does not permit wobulation is used. Re bit 02 = 1: A gating unit that permits wobulation is used. For a pulse frequency wobulation amplitude equal to zero ($p1811 = 0$), the maximum possible pulse frequency in $p1800 = 2 * 1/\text{current controller clock cycle } (p0115[0])$. For a wobulation amplitude greater than zero ($p1811 > 0$), the maximum possible pulse frequency in $p1800 = 1/\text{current controller clock cycle } (p0115[0])$. Re bit 04 = 0: The pulse frequency wobulation amplitude ($p1811$) is enabled. Only applies if bit 02 = 1. Re bit 04 = 1: The pulse frequency wobulation amplitude ($p1811$) is disabled. Only applies if bit 02 = 1.																																						

Re bit 15 = 0:
To deactivate flat-top control mode, p3400.1 also needs to be set to 0.
Re bit 15 = 1:
Flat-top control mode is active regardless of the setting for p3400.1.

p1810 Modulator configuration / Modulator config					
VECTOR	Can be changed: U, T		Calculated: -	Access level: 3	
	Data type: Unsigned16		Dynamic index: -	Func. diagram: -	
	P-Group: Modulation		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0010 bin	
Description:	Sets the configuration for the modulator.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Avg value filter for V_lim (only for Vdc_comp. in modulator)	Yes	No	-
	01	DC link voltage compensation in the current control	Yes	No	-
	02	Wobulation activated	Yes	No	-
	03	Current measurem. oversampling activated (for PESM pulse techn.)	Yes	No	-
Dependency:	If bit 2 is set from 1 to 0, p1811 = 0 is set.				
Notice:	Bit 1 = 0 can only be set under a pulse inhibit and for r0192.14 = 1. Bit 2 can only be set to 1 subject to the following prerequisites: - Pulse inhibit - r0192.16 = 1 - p1800 < 2 * 1000/p0115[0]				
Note:	Re bit 00 = 0: Voltage limitation from the minimum of the DC link voltage (lower ripple in the output current, reduced output voltage). Re bit 00 = 1: Voltage limitation from averaged DC link voltage (higher output voltage with increased ripple in the output current). The selection is only valid if the DC link compensation is not performed in the Control Unit (bit 1 = 0). Re bit 01 = 0: DC link voltage compensation in the modulator. Re bit 01 = 1: DC link voltage compensation in the current control. Re bit 02 = 0: A gating unit that does not permit wobulation is used. Edge modulation is not possible with a parallel connection. Re bit 02 = 1: A gating unit that permits wobulation is used. For a wobulation amplitude equal to zero (p1811 = 0), the maximum possible pulse frequency in p1800 = 2 * 1/current controller clock cycle (p0115[0]). For a wobulation amplitude greater than zero (p1811 > 0), the maximum possible pulse frequency in p1800 = 1/current controller clock cycle (p0115[0]). Re bit 03 = 1: The component supports current actual value sensing (and the detection of valve close durations) with double clocking and phase shift. Activation must be supported by the component (r0192.23) and the system has to be restarted in order for it to take effect.				

p1811	Pulse frequency wobulation amplitude / Puls wobb ampli				
A_INF, S_INF	Can be changed: T	Calculated: -	Access level: 4		
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -		
	P-Group: Modulation	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	0 [%]	20 [%]	0 [%]		
Description:	Sets the amplitude of the statistical wobulation signal.				
	This signal is used to vary the pulse frequency to create a more pleasant sound.				
Note:	It is only possible to change the parameter if wobulation is activated with bit 2 in parameter P1810.				
p1811[0...n]	Pulse frequency wobulation amplitude / Puls wobb ampli				
VECTOR	Can be changed: U, T	Calculated: -	Access level: 3		
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -		
	P-Group: Modulation	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	0 [%]	20 [%]	0 [%]		
Description:	Sets the amplitude of the statistical wobulation signal.				
	This signal is used to vary the pulse frequency to create a more pleasant sound.				
Note:	p1811 > 0 is possible, if the following applies: - p1810.2 (modulator configuration) = 1 (wobulation activated) - p1800 (pulse frequency) <= 1/p115[0] - p0230 (output filter) < 3 (no sine-wave filter)				
p1812	BI: Offset calibration output current measurement / Off_calibr I_outp				
VECTOR	Can be changed: T	Calculated: -	Access level: 4		
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -		
	P-Group: Modulation	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	1		
Description:	Sets the signal source to activate/deactivate offset calibration for output current measurement.				
Caution:	The absence of offset calibration can have a negative effect on control properties. Offset calibration must be performed before switching on the power unit for the first time after POWER ON.				
					
Note:	Offset calibration is only performed with pulses suppressed and can take up to one second.				
p1815	Phase for PWM generation subgroup / Ph for PWM subgr				
A_INF, S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3		
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -		
	P-Group: Modulation	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0001 bin		
Description:	Sets bit 0 for recording the power unit in the subgroup for the "offset clocking".				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Recording in subgroup for offset clocking	Yes	No	-
Dependency:	Refer to: p1818, p1819				

Note: A change only becomes effective after booting.

If any of the following secondary conditions are not fulfilled, then "offset clocking" is deactivated for all power units in the subgroup. This means that all power units of the subgroup are not clocked with offset.

Secondary conditions for clocking with an offset:

- the PWM frequency (p1800[D]) of all power units in the subgroup must be the same.
- the PWM frequency (p1800[D]) must be the same in all drive data sets in the subgroup.
- the following must apply for the ratio between the PWM cycle (1/p1800[D]) and the current controller cycle (p0115[0]):

The ratio (1/p1800[D]) / (p0115[0]) must be an even integer number (2, 4, 6, ...) for all power units in the subgroup.

or

The ratio (p0115[0]) / (1/p1800[D]) must be an integer number (1, 2, 3, ...) for all power units in the subgroup.

p1816 Set phase for PWM generation manually / Set Ph for PWM			
A_INF, S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-1	16	-1
Description: Sets manual setting and overwriting of automatically determined phase shift for "offset clocking".			
For p1816 = 1, the following applies:			
Automatic mode. The phase shift value is automatically determined.			
For p1816 = 0 ... 16, the following applies:			
Manual mode. The user should define the phase shift value as follows:			
1. PWM cycle (1/p1800) > current controller clock cycle (p0115[0])			
The power unit executes a phase shift from Tshift = current controller cycle (p0115[0]) * p1816.			
2. PWM cycle (1/p1800) ≤ current controller clock cycle (p0115[0])			
For p1816 ≥ 1, the power unit executes a phase shift from Tshift = PWM cycle/2.			
Dependency: Refer to: r0116, p1800, p1819			
p1817 Minimum ratio, pulse frequency to the output frequency / Min f_puls / f_max			
VECTOR	Can be changed: C2(2)	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	8.3	15.0	12.0
Description: Sets the minimum ratio between the pulse frequency and the output frequency.			
Notice: If the ratio between the pulse frequency and the output frequency is reduced, then oscillations can occur in the output current that can result in significant levels of current ripple with the appropriate negative effects.			
Note: When the maximum speed is changed, the pulse frequency p1800 is automatically limited to this minimum ratio. It is not permissible to reduce the pulse frequency if this would result in this ratio being undershot.			
p1818 Phase for PWM generation configuration / Ph for PWM config			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	1
Description: Sets the phase shift for offset clocking.			
For the first active power unit, it is specified whether clocking is to start at 0° (value = 0) or 180° (value = 1). All other active power units are clocked alternately according to the setting made here.			

Dependency: Refer to: p1819
Note: A change only becomes effective after a POWER ON.
The parameter is not influenced by setting the factory setting.

p1819 Phase for PWM generation / Ph for PWM

A_INF	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-1	16	0

Description: Display for "offset clocking". Depending on the particular case, the value is interpreted differently:

- Case:
The PWM clock cycle (1/p1800[D]) is greater than the current controller clock cycle (p115[0]) and is an integer and even multiple of it (e.g. p0115[0] = 125 µs, p1800[D] = 4 kHz, 2 kHz, 1 kHz).
The value displayed refers to the phase shift in the current-controlled cycles to be applied by the power unit.
- Case:
The current controller clock cycle (p0115[0]) is greater than or equal to the PWM clock cycle (1/p1800[D]) and is an integer multiple (e.g. p0115[0] = 125 µs, p1800[D] = 8 kHz, 16 kHz).
The displayed value 1 means that the power unit is to apply a phase shift of 180 ° (from the PWM cycle).
- Case:
If neither case 1 nor case 2 applies to at least one power unit, then only 0 can be entered for all of the power units.

Dependency: Refer to: p0108, r0108, p0115, p1800, p1818

Note: For reasons of compatibility, the parameter is an adjustable parameter. However, it functions solely as a read parameter. This means that factory setting -1 no longer has any significance and is only available for reasons of compatibility.

p1819 Phase for PWM generation / Ph for PWM

S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-1	16	-1

Description: Display for "offset clocking". Depending on the particular case, the value is interpreted differently:

- Case:
The PWM clock cycle (1/p1800[D]) is greater than the current controller clock cycle (p115[0]) and is an integer and even multiple of it (e.g. p0115[0] = 125 µs, p1800[D] = 4 kHz, 2 kHz, 1 kHz).
The value displayed refers to the phase shift in the current-controlled cycles to be applied by the power unit.
- Case:
The current controller clock cycle (p0115[0]) is greater than or equal to the PWM clock cycle (1/p1800[D]) and is an integer multiple (e.g. p0115[0] = 125 µs, p1800[D] = 8 kHz, 16 kHz).
The displayed value 1 means that the power unit is to apply a phase shift of 180 ° (from the PWM cycle).
- Case:
If neither case 1 nor case 2 applies to at least one power unit, then only 0 can be entered for all of the power units.

Dependency: Refer to: p0108, r0108, p0115, p1800, p1818

Note: For reasons of compatibility, the parameter is an adjustable parameter. However, it functions solely as a read parameter. This means that factory setting -1 no longer has any significance and is only available for reasons of compatibility.

p1820[0...n]	Reverse the output phase sequence / Outp_ph_seq rev		
VECTOR	Can be changed: C2(3) Data type: Integer16 P-Group: Motor Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: 6732 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the phase sequence reversal for the motor. If the motor does not rotate in the required direction, then the output phase sequence can be reversed using this parameter. This means that with the same setpoint, the motor direction is reversed without reversing the encoder actual value. When a speed encoder is being used, it may be necessary to also invert the encoder actual value (p0410).		
Value:	0: Off 1: On		
Dependency:	Refer to: p1821		
Note:	This setting can only be changed when the pulses are inhibited. p1821 can be used to reverse the phase sequence and encoder actual value.		
p1821[0...n]	Dir of rot / Dir of rot		
SERVO, VECTOR	Can be changed: C2(3) Data type: Integer16 P-Group: Motor Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: 4704, 4710, 4711, 4715, 5730, 6730, 6731, 6732 Unit selection: - Expert list: 1 Factory setting 0
Description:	Setting to change the direction of rotation. If the parameter is changed, it reverses the direction of rotation of the motor and the encoder actual value without changing the setpoint.		
Value:	0: Clockwise 1: Counter-clockwise		
Dependency:	Refer to: F07434		
Notice:	An appropriate fault is output for a drive data set changeover where the direction of rotation changes and the pulses are enabled.		
Note:	For operation with the phase sequence U/V/W, the direction of rotation is defined when viewing the face side of the motor output shaft. When changing the direction of rotation, the rotating field direction of the current controller is reversed. The speed actual value (e.g. r0063) is also reversed so that the control sense is kept and internally causing the direction of rotation to be reversed with the same setpoint. Further, the position actual values of the actual encoder are reversed (e.g. r0482[0...2]). For VECTOR, the following applies: p1820 can be used to reverse the direction of the motor without reversing the encoder actual value.		

p1821[0...n]	Direction / Direction		
SERVO (Lin)	Can be changed: C2(3) Data type: Integer16	Calculated: - Dynamic index: DDS, p0180	Access level: 3 Func. diagram: 4704, 4710, 4711, 4715, 5730, 6730, 6731, 6732
	P-Group: Motor Not for motor type: - Min 0	Units group: - Scaling: - Max 1	Unit selection: - Expert list: 1 Factory setting 0
Description:	Setting to change the direction. If the parameter is changed, it reverses the direction of rotation of the motor and the encoder actual value without changing the setpoint.		
Value:	0: Clockwise 1: Counter-clockwise		
Dependency:	Refer to: F07434		
Notice:	An appropriate fault is output for a drive data set changeover where the direction of rotation changes and the pulses are enabled.		
Note:	For operation with the phase sequence U/V/W, the direction is defined when viewing the face side of the motor output shaft. When changing the direction, the rotating field direction of the current controller is reversed. The actual velocity (e.g. r0063) is also reversed so that the control sense is kept and internally causing the direction of rotation to be reversed with the same setpoint. Further, the position actual values of the actual encoder are reversed (e.g. r0482[0...2]). For VECTOR, the following applies: p1820 can be used to reverse the direction of the motor without reversing the encoder actual value.		
p1825	Converter valve threshold voltage / Threshold voltage		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Modulation Not for motor type: - Min 0.0 [Vrms]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 100.0 [Vrms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.6 [Vrms]
Description:	Sets the threshold voltage drop of the valves (power semiconductor devices) to be compensated.		
Note:	The value is automatically calculated in the motor data identification routine.		
p1827	Infeed compensation valve lockout time operating mode / INFcomp t_lockMode		
A_INF, S_INF	Can be changed: U, T Data type: Integer16 P-Group: Modulation Not for motor type: PEM, REL Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the operating mode for the compensation of the valve lockout time.		
Value:	0: Compensation valve lockout time deactivated 1: Compensation valve lockout time activated		
Note:	The compensation is always active, independent of the value of this parameter if the closed-loop control is activated to suppress circulating currents (p7035) for power units connected in parallel.		

p1828	Compensation valve lockout time phase U / Comp t_lock ph U		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [µs]	Max 1000000.00 [µs]	Factory setting 0.00 [µs]
Description:	Sets the valve lockout time to compensate for phase U.		
Note:	The value is automatically calculated in the motor data identification routine. For type PM340 power units, the parameter is limited to 3.98 µs.		
p1829	Compensation valve lockout time phase V / Comp t_lock ph V		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [µs]	Max 1000000.00 [µs]	Factory setting 0.00 [µs]
Description:	Sets the valve lockout time to compensate for phase V.		
Note:	For type PM340 power units, the parameter is limited to 3.98 µs.		
p1830	Compensation valve lockout time phase W / Comp t_lock ph W		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [µs]	Max 1000000.00 [µs]	Factory setting 0.00 [µs]
Description:	Sets the valve lockout time to compensate for phase W.		
Note:	For type PM340 power units, the parameter is limited to 3.98 µs.		
p1832	Dead time compensation current level / t_dead_comp I_lev		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [Arms]	Max 10000.0 [Arms]	Factory setting 0.0 [Arms]
Description:	Above the current level, the dead time - resulting from the converter switching delays - is compensated by a previously calculated constant value. If the relevant phase current setpoint falls below the absolute value defined by p1832, the corrective value for this phase is continuously reduced.		
Dependency:	The factor setting of p1832 is automatically set to 0.02 * rated drive converter current (r0207).		

r1837 Gating unit configuration / Gating unit config

VECTOR

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** -**P-Group:** Modulation**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Display for the configuration of the gating unit driver.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Modulation depth for a flying restart	Limited	Not limited	-
01	Modulation depth for Vdc closed-loop control	Limited	Not limited	-
02	Vdc_min controller	Active	Not active	-
03	Motor data identification routine	Active	Not active	-
04	Current offset calculation	Active	Not active	-
05	Simulation mode	Active	Not active	-
06	Reverse the output phase sequence	Active	Not active	-
07	Counter-clockwise direction of rotation	Active	Not active	-
08	Synchronization (bypass)	Active	Not active	-
09	F7801 monitor by application	active	not active	-
10	Chassis Drive active	Yes	no	-
11	Short-circuit test active	no	Yes	-
12	FL modulation prohibited	Yes	No	-
13	F3E present	Yes	No	-
14	PS-Asic3 present	Yes	No	-
15	CU230 or CU240	Yes	No	-

p1840[0...n] Actual value correction configuration / AVC config

VECTOR

Can be changed: T**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** DDS, p0180**Func. diagram:** -**P-Group:** Modulation**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

0000 bin

Description:

Configuration of the actual value correction

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Actual value correction deactivated	Yes	No	-
01	Compares the integrals from modulator and setpoint	Yes	No	-

Dependency:

Refer to: p1802

Note:

During operation (the pulses enabled) the configuration cannot be changed by changing over drive data sets.

r1841 Actual value correction status word / AVC status

VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Status of the actual value correction

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Hardware for the actual value correction detected	Yes	No	-
	01	Automatic shutdown (too many switching instants)	Yes	No	-
	02	Integral normalized to half the gating unit clock cycle freq.	Yes	No	-
	03	Actual value correction temporarily suppressed	Yes	No	-
	15	Actual value correction active	Yes	No	-

p1845[0...n] Actual value correction evaluation factor Lsig / ActV_corr FactLsig

VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00	10.00	1.00

Description: Sets the weighting factor for the leakage inductance of the L-R element of the actual value correction.

Dependency: Refer to: p0391, p0392, p0393

Note: The load-dependent adaptation of the leakage inductance of the current actual value correction is defined using p0391 ... p0393.

p1846[0...n] Actual value correction damping factor / ActV_corr D_factor

VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00	10.00	1.00

Description: Sets the damping factor for the actual value correction.

The factor multiplies the T0/Tsig ratio in the feedback branch of the LR element

r1848[0...5]	Actual value correction phase currents / ActVal_corr I_corr		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Modulation Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays phase correction currents as well as the drive converter phase currents		
Index:	[0] = Harmonics, phase U [1] = Harmonics, phase V [2] = Harmonics, phase W [3] = Measured value phase U [4] = Measured value phase V [5] = Measured value phase W		
r1849[0...5]	Actual value correction phase voltages / ActVal_corr V_corr		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Modulation Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the phase correction voltages and the drive converter phase voltages		
Index:	[0] = Harmonics, phase U [1] = Harmonics, phase V [2] = Harmonics, phase W [3] = Measured value phase U [4] = Measured value phase V [5] = Measured value phase W		
p1900	Motor data identification and rotating measurement / MotID and rot meas		
VECTOR	Can be changed: C2(1), T Data type: Integer16 P-Group: Motor identification Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2
Description:	Sets the motor data identification and speed controller optimization. p1900 = 0: Function inhibited. p1900 = 2: Induction motors --> set p1910 = 1 and p1960 = 0 Permanent-magnet or separately-excited synchronous motors --> set p1910 = 1, p1990 = 1 and p1960 = 0 When the drive enable signals are present, a motor data identification routine is carried out at standstill with the next power-on command. Current flows through the motor which means that it can align itself by up to a quarter of a revolution. For permanent-magnet or separately-excited synchronous motors, the encoder is adjusted with the next power-on command. The motor must be free to rotate and rotates through 1.5 revolutions of the motor encoder.		
Value:	0: Inhibited 2: Identify motor data at standstill		
Dependency:	In the simulation mode, the parameter cannot be written into. When selecting the motor data identification routine, the drive data set changeover is suppressed. Refer to: p1272, p1300, p1910 Refer to: F07990, A07991		

- Notice:** If there is a motor holding brake, it must be open (p1215 = 2).
To permanently accept the determined settings they must be saved in a non-volatile fashion (p0971, p0977).
- Note:** The motor and control parameters are only optimally set when both measurements are carried out (initially at standstill, and then with the motor rotating).
An appropriate alarm is output when the parameter is set.
The power-on command must remain set during a measurement and after the measurement has been completed, the drive automatically resets it.
The duration of the measurements can lie between 0.3 s and several minutes. This time is, for example, influenced by the motor size and the mechanical conditions.
p1900 is automatically set to 0 after the motor data identification routine has been completed.

p1900		Motor data identification and rotating measurement / MotID and rot meas	
VECTOR (n/M)	Can be changed: C2(1), T	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 3	Factory setting 2
Description:	Sets the motor data identification and speed controller optimization. p1900 = 0: Function inhibited. p1900 = 1: Induction motors --> set p1910 = 1 and p1960 = 0, 1, 2 depending on p1300 Permanent-magnet or separately-excited synchronous motors --> set p1910 = 1, p1990 = 1 and p1960 = 0, 1, 2 depending on p1300 When the drive enable signals are present, a motor data identification routine is carried out at standstill with the next power-on command. Current flows through the motor which means that it can align itself by up to a quarter of a revolution. For permanent-magnet or separately-excited synchronous motors, the encoder is adjusted with the next power-on command. The motor must be free to rotate and rotates through 1.5 revolutions of the motor encoder. With the following power-on command, a rotating motor data identification routine is carried out - and in addition, a speed controller optimization by making measurements at different motor speeds. p1900 = 2: Induction motors --> set p1910 = 1 and p1960 = 0 Permanent-magnet or separately-excited synchronous motors --> set p1910 = 1, p1990 = 1 and p1960 = 0 When the drive enable signals are present, a motor data identification routine is carried out at standstill with the next power-on command. Current flows through the motor which means that it can align itself by up to a quarter of a revolution. For permanent-magnet or separately-excited synchronous motors, the encoder is adjusted with the next power-on command. The motor must be free to rotate and rotates through 1.5 revolutions of the motor encoder. p1900 = 3: Sets p1960 = 0, 1, 2 depending on p1300 This setting should only be selected if the motor data identification was already carried out at standstill. When the drive enable signals are present, with the next power-on command, a rotating motor data identification routine is carried out - and in addition, a speed controller optimization by making measurements at different motor speeds.		
Value:	0: Inhibited 1: Identify motor data at standstill and with motor rotating 2: Identify motor data at standstill 3: Identify motor data with motor rotating		
Dependency:	In the simulation mode, the parameter cannot be written into. When selecting the motor data identification routine, the drive data set changeover is suppressed. Refer to: p1272, p1300, p1910, p1960, p1990 Refer to: A07980, A07981, F07982, F07983, F07984, F07985, F07986, A07987, F07988, F07990, A07991		
Notice:	If there is a motor holding brake, it must be open (p1215 = 2). To permanently accept the determined settings they must be saved in a non-volatile fashion (p0971, p0977).		

Note: The motor and control parameters are only optimally set when both measurements are carried out (initially at stand-still, and then with the motor rotating).
 An appropriate alarm is output when the parameter is set.
 The power-on command must remain set during a measurement and after the measurement has been completed, the drive automatically resets it.
 The duration of the measurements can lie between 0.3 s and several minutes. This time is, for example, influenced by the motor size and the mechanical conditions.
 p1900 is automatically set to 0 after the motor data identification routine has been completed.

p1901 Test pulse evaluation configuration / Test puls config

VECTOR **Can be changed:** T **Calculated:** - **Access level:** 2
Data type: Unsigned32 **Dynamic index:** - **Func. diagram:** -
P-Group: Motor identification **Units group:** - **Unit selection:** -
Not for motor type: - **Scaling:** - **Expert list:** 1
Min **Max** **Factory setting**
 - - 0000 bin

Description: Sets the configuration for the test pulse evaluation.
 Settable options:
 Bit 0: Short-circuit test
 Check for conductor-to-conductor short circuit when pulse is enabled.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Test pulse for phase short-circuit active	Yes	No	-

Note: For short-circuit test:
 If the test was successful once after POWER ON (see r1902.0), it is not repeated.
 If a conductor-to-conductor short-circuit is detected during the test, this is displayed in r1902.1.

r1902 Test pulse evaluation status / Status test puls

VECTOR **Can be changed:** - **Calculated:** - **Access level:** 2
Data type: Unsigned32 **Dynamic index:** - **Func. diagram:** -
P-Group: - **Units group:** - **Unit selection:** -
Not for motor type: - **Scaling:** - **Expert list:** 1
Min **Max** **Factory setting**
 - - -

Description: Status display of test pulse evaluation.
 Short-circuit test:
 Bit0: No faults detected during short-circuit test.
 Bit1: Phase short-circuit detected.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Short-circuit test	Executed	Not executed	-
01	Phase short-circuit detected	Yes	no	-

p1909[0...n] Motor data identification control word / MotID STW

SERVO, SERVO (Lin) **Can be changed:** T **Calculated:** CALC_MOD_ALL **Access level:** 3
Data type: Unsigned16 **Dynamic index:** MDS, p0130 **Func. diagram:** -
P-Group: Motor identification **Units group:** - **Unit selection:** -
Not for motor type: - **Scaling:** - **Expert list:** 1
Min **Max** **Factory setting**
 - - 0010 0111 0000 0000 bin

Description: Sets the configuration of the motor data identification.

Recommend.: For the stationary motor data identification, if a motor holding brake is being used it should be opened and the motor finely synchronized before the measurement. This should only be done if it can be safely carried out and no external forces can act on the motor. This determines the angular commutation offset (p1909.13, p0431).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	Measure D inductance	Yes	No	-
	09	Measure Q inductance	Yes	No	-
	10	Magnetizing field inductance and measure rotor resistance	Yes	No	-
	13	Measure commutation angle and direction of rotation	Yes	No	-
	14	Determining the voltage emulation error	Yes	No	-
Dependency:	Refer to: p1910, r1912, r1913, r1915, r1925, r1927, r1932, r1933, r1934, r1935, r1936, r1950, r1951, p1952, p1953				
Note:	For an induction motor (ASM) the following bits: 8, 9, 10, 13 are effective For a synchronous motor (SRM) the following bits: 8, 9, 13, 14 are effective Re bit 14: - after successfully determining the voltage emulation error, the display of the phase voltage actual values r0089 and the active power actual value r0082 and the torque actual value r0080 are significantly more accurate. - the voltage emulation errors should be identified with the Motor Module in the warm state. - the motor temperature (r0035) should not change significantly (i.e. it should not be identified immediately after a load duty cycle).				

p1909[0...n]	Motor data identification control word / MotID STW			
SERVO (Ext M_ctrl, Lin)	Can be changed: T	Calculated: CALC_MOD_ALL	Access level: 3	
	Data type: Unsigned16	Dynamic index: MDS, p0130	Func. diagram: -	
	P-Group: Motor identification	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0110 0111 0000 0000 bin	

Description: Sets the configuration of the motor data identification.

Recommend.: For the stationary motor data identification, if a motor holding brake is being used it should be opened and the motor finely synchronized before the measurement. This should only be done if it can be safely carried out and no external forces can act on the motor. This determines the angular commutation offset (p1909.13, p0431).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	Measure D inductance	Yes	No	-
	09	Measure Q inductance	Yes	No	-
	10	Magnetizing field inductance and measure rotor resistance	Yes	No	-
	13	Measure commutation angle and direction of rotation	Yes	No	-
	14	Determining the voltage emulation error	Yes	No	-
Dependency:	Refer to: p1910, r1912, r1913, r1915, r1925, r1927, r1932, r1933, r1934, r1935, r1936, r1950, r1951, p1952, p1953				
Note:	For an induction motor (ASM) the following bits: 8, 9, 10, 13 are effective For a synchronous motor (SRM) the following bits: 8, 9, 13, 14 are effective Re bit 14: - after successfully determining the voltage emulation error, the display of the phase voltage actual values r0089 and the active power actual value r0082 and the torque actual value r0080 are significantly more accurate. - the voltage emulation errors should be identified with the Motor Module in the warm state. - the motor temperature (r0035) should not change significantly (i.e. it should not be identified immediately after a load duty cycle).				

p1909[0...n]	Motor data identification control word / MotID STW			
VECTOR	Can be changed: T Data type: Unsigned16 P-Group: Motor identification Not for motor type: - Min -	Calculated: CALC_MOD_ALL Dynamic index: MDS, p0130 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 bin	
Description:	Sets the configuration of the motor data identification.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Stator inductance estimate no measurement	Yes	No
	01	Cl.-loop current control w/ dead-beat controller	Yes	No
	02	Rotor time constant estimate no measurement	Yes	No
	03	Leakage inductance estimate no measurement	Yes	No
	04	Activates the identification dynamic leakage inductance	Yes	No
	05	Determine Tr and Lsig evaluation in the time range	Yes	No
	06	Activate vibration damping	Yes	No
	07	Deactivate vibration detection	Yes	No
	11	Deactivate pulse measurement Lq Ld	Yes	No
	12	Deactivate rotor resistance Rr measurement	Yes	No
	14	Deactivate valve interlocking time measurement	Yes	No
	15	Determine only stator resistance, valve voltage fault, dead time	Yes	No
Note:	The following applies to permanent-magnet synchronous motors: Without deselection in bit 11, in the closed-loop control mode, the direct inductance LD and the quadrature inductance Lq are measured at a low current. When deselecting with bit 11 or in the V/f mode, the stator inductance is measured at half the rated motor current. If the stator inductance is not measured but is to be estimated, then bit 0 should be set and bit 11 should be deselected.			
p1910	Motor data identification routine, stationary (standstill) / MotID standstill			
SERVO	Can be changed: T Data type: Integer16 P-Group: Motor identification Not for motor type: - Min -3	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0	
Description:	Setting to control the motor data identification with the motor stationary.			
Value:	-3: Accept identified parameters -2: Acknowledge encoder inversion actual value (F07993) -1: Start motor data identification without acceptance 0: Inactive/inhibit 1: Start motor data identification with acceptance			
Recommend.:	For motors with brakes, the brake should be opened before carrying out the stationary motor data identification routine (p1215 = 2) as long as this can be done without incurring any danger. The commutation angle and the direction of rotation are also determined.			
Dependency:	Refer to: p1909, r1912, r1913, r1915, r1925, r1927, r1932, r1933, r1934, r1935, r1936, r1950, r1951, p1952, p1953 Refer to: F07990, A07991, F07993			

Caution:



For motors without brake or with the brake open (p1215 = 2), for the stationary (zero speed) measurement, the motor may rotate slightly.

Notice:

If there is a motor holding brake, it must be open (p1215 = 2).

Note:

To permanently accept the determined settings they must be saved in a non-volatile fashion (p0971, p0977).

Motor data identification can only be selected when the pulses of all of the drive objects of the Control Unit have been suppressed. After selection, all of the other drive objects of the Control Unit are interlocked so that they cannot be powered up until the motor data identification has been completed or deselected.

After a started motor identification is ended, the parameter is automatically reset to 0.

A motor data identification that is presently being carried out can be terminated with p1910 = 0.

p1910

Motor data identification selection / MotID selection

VECTOR

Can be changed: T

Calculated: -

Access level: 2

Data type: Integer16

Dynamic index: -

Func. diagram: -

P-Group: Motor identification

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

0

25

1

Description:

Sets the motor data identification routine.

The motor data identification routine is carried out after the next power-on command.

p1910 = 1:

All motor data and the drive converter characteristics are identified and then transferred to the following parameters:

p0350, p0354, p0356, p0357, p0358, p0360, p1825, p1828, p1829, p1830

After this, the control parameter p0340 = 3 is automatically calculated.

Value:

- 0: Inhibited
- 1: Complete identification (ID) and acceptance of motor data
- 2: Complete identification (ID) of motor data without acceptance
- 3: ID of the saturation characteristic and acceptance
- 4: ID of the saturation characteristic without acceptance
- 5: ID of dynamic leakage inductance Lsig (r1920) without acceptance
- 6: ID of lockout time (r1926) without acceptance
- 7: ID of stator resistance Rs (r1912) without acceptance
- 8: ID of stator inductance Ls (r1915) Rr (r1927) without acceptance
- 9: ID of rotor time constant Tr (r1913) without acceptance
- 10: ID of static leakage inductance Lsig (r1914) without acceptance
- 20: Voltage vector input
- 21: Voltage vector input without filter
- 22: Rectangular voltage vector input without filter
- 23: Triangular voltage vector input without filter
- 24: Rectangular voltage vector input with filter
- 25: Triangular voltage vector input with filter

Dependency:

"Quick commissioning" must be carried out (p0010 = 1) before executing the motor data identification routine!

In the simulation mode, the parameter cannot be written into. When selecting the motor data identification routine, the drive data set changeover is suppressed.

Refer to: p1272, p1900

Caution:

After the motor data identification (p1910 > 0) has been selected, alarm A07991 is output and a motor data identification routine is carried out as follows at the next power-on command:

- current flows through the motor and a voltage is present at the drive converter output terminals.
- during the identification routine, the motor shaft can rotate through a maximum of half a revolution.
- however, no torque is generated.

Notice:

If there is a motor holding brake, it must be open (p1215 = 2).

To permanently accept the determined settings they must be saved in a non-volatile fashion (p0971, p0977).

Note:	When setting p1910, the following should be observed: 1. "With acceptance" means: The parameters specified in the description are overwritten with the identified values and therefore have an influence on the controller setting. 2. "Without acceptance" means: The identified parameters are only displayed in the range r1912 ... r1926. The controller settings remain unchanged. 3. p1910 = 3, 4, 5 can only be selected for induction motors.		
p1911	Number of phases to be identified / Qty ph to ident		
VECTOR	Can be changed: T Data type: Integer16 P-Group: Motor identification Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of phases to be identified.		
Value:	1: 1 phase U 2: 2 phases U, V 3: 3 phases U, V, W		
Note:	When identifying with several phases, the accuracy increases and also the time it takes to make the measurement.		
r1912	Stator resistance identified / R_stator ident		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Ohm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Ohm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Ohm]
Description:	Displays the identified stator resistance.		
Dependency:	Refer to: p1909, p1910, r1913, r1915, r1925, r1927, r1932, r1933, r1934, r1935, r1936, r1950, r1951, p1952, p1953		
r1912[0...2]	Identified stator resistance / R_stator ident		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Ohm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Ohm]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Ohm]
Description:	Displays the identified stator resistance.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		

r1913	Rotor time constant identified / T_rotor ident		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM Min - [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the identified rotor time constant.		
Dependency:	Refer to: p1909, p1910, r1912, r1915, r1925, r1927, r1932, r1933, r1934, r1935, r1936, r1950, r1951, p1952, p1953		
r1913[0...2]	Identified rotor time constant / T_rotor ident		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM Min - [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [ms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the identified rotor time constant.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		
r1914[0...2]	Identified total leakage inductance / L_total_leak ident		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the identified total leakage inductance.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		
r1915	Stator inductance identified / L_stator ident		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the identified stator inductance.		
Dependency:	Refer to: p1909, p1910, r1912, r1913, r1925, r1927, r1932, r1933, r1934, r1935, r1936, r1950, r1951, p1952, p1953		

r1915[0...2]	Identified nominal stator inductance / L_stator ident		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the nominal stator inductance identified.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		
r1916[0...2]	Identified stator inductance 1 / L_stator 1 ident		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the nominal stator inductance identified for the 1st point of the saturation characteristic.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		
r1917[0...2]	Identified stator inductance 2 / L_stator 2 ident		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the nominal stator inductance identified for the 2nd point of the saturation characteristic.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		
r1918[0...2]	Identified stator inductance 3 / L_stator 3 ident		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the nominal stator inductance identified for the 3rd point of the saturation characteristic.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		

r1919[0...2]	Identified stator inductance 4 / L_stator 4 ident		
VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [mH]	- [mH]	- [mH]
Description:	Displays the nominal stator inductance identified for the 4th point of the saturation characteristic.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		

r1920[0...2]	Identified dynamic leakage inductance / L_leak dyn ident		
VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [mH]	- [mH]	- [mH]
Description:	Displays the identified dynamic total leakage inductance.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		

r1921[0...2]	Identified dynamic leakage inductance 1 / L_leak 1 dyn id		
VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [mH]	- [mH]	- [mH]
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		

r1922[0...2]	Identified dynamic leakage inductance 2 / L_leak 2 dyn id		
VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [mH]	- [mH]	- [mH]
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		

r1923[0...2]	Identified dynamic leakage inductance 3 / L_leak 3 dyn id		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		
r1924[0...2]	Identified dynamic leakage inductance 4 / L_leak 4 dyn id		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		
r1925	Threshold voltage identified / V_threshold ident		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Vrms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the identified threshold voltage of the power unit.		
Dependency:	Refer to: p1909, p1910, r1912, r1913, r1915, r1927, r1932, r1933, r1934, r1935, r1936, r1950, r1951, p1952, p1953		
r1925[0...2]	Identified threshold voltage / V_threshold ident		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Vrms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the identified IGBT threshold voltage.		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		

r1926[0...2] Identified effective valve lockout time / t_lock_valve id

VECTOR

Can be changed: -**Calculated:** -**Access level:** 4**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** -**P-Group:** Motor identification**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

- [μs]

- [μs]

- [μs]

Description:

Displays the identified effective valve lockout time.

Index:

[0] = Phase U

[1] = Phase V

[2] = Phase W

r1927 Rotor resistance identified / R_rotor ident

SERVO

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** -**P-Group:** Motor identification**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

- [Ohm]

- [Ohm]

- [Ohm]

Description:

Displays the identified rotor resistance.

Dependency:

Refer to: p1909, p1910, r1912, r1913, r1915, r1925, r1932, r1933, r1934, r1935, r1936, r1950, r1951, p1952, p1953

r1927[0...2] Identified rotor resistance / R_rotor ident

VECTOR

Can be changed: -**Calculated:** -**Access level:** 4**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** -**P-Group:** Motor identification**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

- [Ohm]

- [Ohm]

- [Ohm]

Description:

Displays the identified rotor resistance

Index:

[0] = Phase U

[1] = Phase V

[2] = Phase W

r1929[0...2] Identified cable resistance / R_cable ident

VECTOR

Can be changed: -**Calculated:** -**Access level:** 4**Data type:** FloatingPoint32**Dynamic index:** -**Func. diagram:** -**P-Group:** Motor identification**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

- [Ohm]

- [Ohm]

- [Ohm]

Description:

Displays the identified cable resistance.

Index:

[0] = Phase U

[1] = Phase V

[2] = Phase W

r1932[0...19] d inductance identified / Ld ident

SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the identified (differential) d-inductance.		
Dependency:	Refer to: p1909, p1910, r1912, r1913, r1915, r1925, r1927, r1933, r1934, r1935, r1936, r1950, r1951, p1952, p1953		
Note:	The Ld characteristic consists of the value pairs from p1932 and p1933 with the same index. This value corresponds to the value of the total leakage inductance (r0377).		

r1933[0...19] d inductance identification current / Ld I_ident

SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the identification current of the d inductance.		
Dependency:	Refer to: p1909, p1910, r1912, r1913, r1915, r1925, r1927, r1932, r1934, r1935, r1936, r1950, r1951, p1952, p1953		
Note:	The Ld characteristic consists of the value pairs from p1932 and p1933 with the same index.		

r1934[0...9] q inductance identified / Lq ident

SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the identified (differential) q-inductance.		
Dependency:	Refer to: p1909, p1910, r1932, r1933		
Note:	The Lq characteristic consists of the value pairs from p1934 and p1935 with the same index. This value corresponds to the value of the total leakage inductance (r0377).		

r1934[0...9] q inductance identified / Lq ident

VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the identified (differential) q-inductance.		
Dependency:	Refer to: r1935, p1959, p1960		
Note:	The Lq characteristic consists of the value pairs from p1934 and p1935 with the same index. This value corresponds to the value of the total leakage inductance (r0377).		

r1935[0...20] Identification current / I_ident			
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the identification current for the identification of the q-inductance ([0...9]) as well as the torque constant ([10]) and the torque characteristic ([11...20]).		
Index:	[0] = q inductance identification current measuring point 1 [1] = q inductance identification current measuring point 2 [2] = q inductance identification current measuring point 3 [3] = q inductance identification current measuring point 4 [4] = q inductance identification current measuring point 5 [5] = q inductance identification current measuring point 6 [6] = q inductance identification current measuring point 7 [7] = q inductance identification current measuring point 8 [8] = q inductance identification current measuring point 9 [9] = q inductance identification current measuring point 10 [10] = Torque constant identification current [11] = Torque characteristic identification current measuring point 1 [12] = Torque characteristic identification current measuring point 2 [13] = Torque characteristic identification current measuring point 3 [14] = Torque characteristic identification current measuring point 4 [15] = Torque characteristic identification current measuring point 5 [16] = Torque characteristic identification current measuring point 6 [17] = Torque characteristic identification current measuring point 7 [18] = Torque characteristic identification current measuring point 8 [19] = Torque characteristic identification current measuring point 9 [20] = Torque characteristic identification current measuring point 10		
Dependency:	Refer to: p1909, p1910, r1934, p1959, p1960		
Note:	- the Lq characteristic consists of the value pairs from r1934 and r1935 with the same index. - the torque constant is identified with the current r1935[10] and displayed in r1937[0]. If the reluctance torque constant is identified (p1959.7 = 1), the torque constant is identified with 150% rated current (p0305), otherwise with 100% rated current. - the torque characteristic (r1937[1...10]) is identified in the range between the rated current (p0305) and the maximum current (p0640) (r1935[11...20]).		

r1935[0...20] Identification current / I_ident			
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the identification current for the identification of the q-inductance ([0...9]) as well as the force constant ([10]) and the force characteristic ([11...20]).		

Index:	[0] = q inductance identification current measuring point 1 [1] = q inductance identification current measuring point 2 [2] = q inductance identification current measuring point 3 [3] = q inductance identification current measuring point 4 [4] = q inductance identification current measuring point 5 [5] = q inductance identification current measuring point 6 [6] = q inductance identification current measuring point 7 [7] = q inductance identification current measuring point 8 [8] = q inductance identification current measuring point 9 [9] = q inductance identification current measuring point 10 [10] = Force constant identification current [11] = Force characteristic identification current measuring point 1 [12] = Force characteristic identification current measuring point 2 [13] = Force characteristic identification current measuring point 3 [14] = Force characteristic identification current measuring point 4 [15] = Force characteristic identification current measuring point 5 [16] = Force characteristic identification current measuring point 6 [17] = Force characteristic identification current measuring point 7 [18] = Force characteristic identification current measuring point 8 [19] = Force characteristic identification current measuring point 9 [20] = Force characteristic identification current measuring point 10
Dependency:	Refer to: p1909, p1910, r1934, p1959, p1960
Note:	- the Lq characteristic consists of the value pairs from r1934 and r1935 with the same index. - the force constant is identified with the current r1935[10] and displayed in r1937[0]. If the reluctance force constant is identified (p1959.7 = 1), the force constant is identified with 150% rated current (p0305), otherwise with 100% rated current. - the force characteristic (r1937[1...10]) is identified in the range between the rated current (p0305) and the maximum current (p0640) (r1935[11...20]).

r1935[0...9]	q inductance identification current / Lq I_ident		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Arms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the identification current to identify the q inductance ([0...9]).		
Dependency:	Refer to: r1934, p1959, p1960		
Note:	The Lq characteristic consists of the value pairs from r1934 and r1935 with the same index.		

r1936	Magnetizing inductance identified / L_H ident		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the identified magnetizing inductance(gamma equivalent circuit diagram).		
Dependency:	Refer to: p1909, p1910, r1913, r1915, r1927, p1959, p1960, r1962, r1963		
Note:	This value corresponds to the value of the transformed magnetizing inductance (r0382).		

r1937[0...10] Torque constant identified / kT ident			
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Nm/A]	Calculated: - Dynamic index: - Units group: 28_1 Scaling: - Max - [Nm/A]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [Nm/A]
Description:	Displays the identified torque constant/torque characteristic over the q current.		
Index:	[0] = Torque constant identified [1] = Torque characteristic identified measuring point 1 [2] = Torque characteristic identified measuring point 2 [3] = Torque characteristic identified measuring point 3 [4] = Torque characteristic identified measuring point 4 [5] = Torque characteristic identified measuring point 5 [6] = Torque characteristic identified measuring point 6 [7] = Torque characteristic identified measuring point 7 [8] = Torque characteristic identified measuring point 8 [9] = Torque characteristic identified measuring point 9 [10] = Torque characteristic identified measuring point 10		
Dependency:	Refer to: r1938, r1939, p1959, p1960, r1969		
Note:	- the value in r1937[0] corresponds to the torque constant (p0316) and was identified with the current in r1935[10]. If the reluctance torque is identified (p1959.7 = 1), the torque constant is identified with 150% rated current (p0305), otherwise with 100% rated current. - if indices r1937[1...10] are not equal to zero, they show the values of the torque characteristic identified for the current in r1935[11...20]. The torque characteristic is identified in the range between rated current (p0305) and maximum current (p0640).		

r1937[0...10] Force constant identified / kT ident			
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [N/Arms]	Calculated: - Dynamic index: - Units group: 29_1 Scaling: - Max - [N/Arms]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [N/Arms]
Description:	Displays the identified force constant.		
Index:	[0] = Force constant identified [1] = Force characteristic identified measuring point 1 [2] = Force characteristic identified measuring point 2 [3] = Force characteristic identified measuring point 3 [4] = Force characteristic identified measuring point 4 [5] = Force characteristic identified measuring point 5 [6] = Force characteristic identified measuring point 6 [7] = Force characteristic identified measuring point 7 [8] = Force characteristic identified measuring point 8 [9] = Force characteristic identified measuring point 9 [10] = Force characteristic identified measuring point 10		
Dependency:	Refer to: r1938, r1939, p1959, p1960, r1969		
Note:	- the value in r1937[0] corresponds to the force constant (p0316) and was identified with the current in r1935[10]. If the reluctance force constant is identified (p1959.7 = 1), the force constant is identified with 150% rated current (p0305), otherwise with 100% rated current.		

r1938	Voltage constant identified / kE ident		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Vrms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the identified voltage constant.		
Dependency:	Refer to: r1937, r1939, p1959, p1960, r1969		
Note:	This value corresponds to the voltage constant (p0317).		

r1938	Voltage constant identified / kE ident		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Vrms s/m]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Vrms s/m]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Vrms s/m]
Description:	Displays the identified voltage constant.		
Dependency:	Refer to: r1937, r1939, p1959, p1960, r1969		
Note:	This value corresponds to the voltage constant (p0317).		

r1939	Reluctance torque constant identified / kT_reluct ident		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the identified reluctance torque constant.		
Dependency:	Refer to: r1937, r1938, p1959, p1960, r1969		
Note:	This value corresponds to the reluctance torque constant (p0328).		

r1939	Reluctance force constant identified / kT_reluct ident		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the identified reluctance force constant.		
Dependency:	Refer to: r1937, r1938, p1959, p1960, r1969		
Note:	This value corresponds to the reluctance force constant (p0328).		

r1947	Optimum load angle identified / phi_load ident		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [°]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [°]
Description:	Displays the identified, optimum load angle.		
Note:	This value corresponds to the optimum load angle (p0327).		
r1948	Magnetizing current identified / I_mag ident		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the identified magnetizing current.		
Dependency:	Refer to: r1936, p1959, p1960		
Note:	This value corresponds to the magnetizing current (p0320 / r0331).		
r1950[0...19]	Voltage emulation error voltage values / V_error V_values		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [V]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [V]
Description:	The identified characteristic of the voltage emulation error is displayed r1950[0...19] and r1951[0...19].		
Dependency:	Refer to: r1951		
r1951[0...19]	Voltage emulation error current values / V_error I_error		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [A]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [A]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [A]
Description:	The identified characteristic of the voltage emulation error is displayed r1950[0...19] and r1951[0...19].		
Dependency:	Refer to: r1950		
p1952[0...n]	Voltage emulation error final value / V_error final val		
SERVO (Ext M_ctrl)	Can be changed: T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.000 [V]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.000 [V]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Sets the final value to compensate the voltage emulation error.		
Dependency:	Refer to: p1953		

Note: The voltage emulation error is calculated and compensated for every phase according to the following formula:

$$u_error = u0 * i / (abs(i) + i0)$$

 u0: This is set in p1952.
 i0: This is set in p1953.
 i: Phase current to which the emulation error u_error belongs.

p1953[0...n] Voltage emulation error current offset / V_error I_offset

SERVO (Ext M_ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [A]	Max 100.000 [A]	Factory setting 0.000 [A]

Description: Sets the current offset to compensate the voltage emulation error.

Dependency: Refer to: p1952

Note: The voltage emulation error is calculated and compensated for every phase according to the following formula:

$$u_error = u0 * i / (abs(i) + i0)$$

 u0: This is set in p1952.
 i0: This is set in p1953.
 i: Phase current to which the emulation error u_error belongs.

p1958[0...n] Rotating measurement ramp-up/ramp-down time / Rot meas t_r up/dn

SERVO	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -1.00 [s]	Max 999999.00 [s]	Factory setting -1.00 [s]

Description: Sets the ramp-up/ramp-down time for the rotating measurement.

The following applies for negative values:

When the function module "extended setpoint channel" is activated (r0108.8 = 1), the maximum of the ramp-up/ramp-down time of the setpoint channel becomes effective. When this function module is inactive, then no ramp-up/ramp-down time is effective.

The following applies for positive values:

The selected ramp-up/ramp-down time becomes effective.

Recommend.: A ramp-up/ramp-down time should not be activated for the motor data identification (p1958 = 0) as long as this can be safely done without incurring any danger. This means that the identification is complete and more accurate. When the ramp-up/ramp-down time is activated, the following steps of the rotating motor data identification are not executed:

- p1959.5 (identifying the q inductance)
- p1959.7 (identifying the reluctance torque constant)

Dependency: Refer to: p1959, p1960

p1958[0...n]	Moving measurement ramp-up/ramp-down time / Mov meas t_r up/dn				
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 2		
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -		
	P-Group: Motor identification	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-1.00 [s]	999999.00 [s]	-1.00 [s]		
Description:	Sets the ramp-up/ramp-down time for the moving measurement. The following applies for negative values: When the function module "extended setpoint channel" is activated (r0108.8 = 1), the maximum of the ramp-up/ramp-down time of the setpoint channel becomes effective. When this function module is inactive, then no ramp-up/ramp-down time is effective. The following applies for positive values: The selected ramp-up/ramp-down time becomes effective.				
Recommend.:	A ramp-up/ramp-down time should not be activated for the motor data identification (p1958 = 0) as long as this can be safely done without incurring any danger. This means that the identification is complete and more accurate. When the ramp-up/ramp-down time is activated, the following steps of the moving motor data identification are not executed: - p1959.5 (identifying the q inductance) - p1959.7 (identifying the reluctance force constant)				
Dependency:	Refer to: p1959, p1960				
p1959[0...n]	Rotating measurement configuration / Rot meas config				
SERVO	Can be changed: T	Calculated: CALC_MOD_ALL	Access level: 3		
	Data type: Unsigned16	Dynamic index: MDS, p0130	Func. diagram: -		
	P-Group: Motor identification	Units group: -	Unit selection: -		
	Not for motor type: REL	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 1110 1110 0111 bin		
Description:	Sets the configuration of the rotating measurement.				
Recommend.:	A direction inhibit should not be activated for the rotating measurement (p1959.14 = 1 and p1959.15 = 1) as long as this can be done without incurring any danger. This means that the identification is complete and more accurate. When the direction inhibit is activated, the reluctance torque constant (p1959.7) is not identified and the angular commutation offset (p1959.10, p0431) is inaccurately determined. The reluctance torque constant (p1959.7) is also not identified in encoderless operation.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Reserved	Yes	No	-
	01	Identify the saturation characteristic	Yes	No	-
	02	Identify the moment of inertia	Yes	No	-
	05	Identify the q inductance	Yes	No	-
	06	Identify the torque constant	Yes	No	-
	07	Identify the reluctance torque constant	Yes	No	-
	08	Identify the q inductance at the test stand	Yes	No	-
	09	Identify the magnetizing current / magnetizing inductance	Yes	No	-
	10	Identify the commutation angle and direction of rotation	Yes	No	-
	11	Identify rotor resistance	Yes	No	-
	14	Positive direction permitted	Yes	No	-
	15	Negative direction permitted	Yes	No	-
Dependency:	Refer to: p1958, p1960				

- Notice:** The step p1959.8 (identify q inductance on the test stand) may only be selected if the drive can be kept at zero speed or at a fixed speed either using a test stand or other mechanical measures.
During steps p1959.2 (identifying the moment of inertia) and p1959.6 (identifying the torque constant) the Vdc_min controller is disabled (p1240).
During step p1959.7 (identifying the reluctance torque constant) the Vdc_min controller and Vdc_max controller are disabled (p1240).
- Note:** For an induction motor (ASM), the following bits 1, 2, 5, 8, 9, 10, 14, 15 are effective
For a synchronous motor (SRM), the following bits 2, 5, 6, 7, 8, 10, 14, 15 are effective
Re bit 05:
For "motor holding brake the same as sequence control" (p1215 = 1 or 3), the Lq characteristic is only measured up to approximately the rated motor current (p0305) instead of up to the current limit (p0640). Before carrying out the rotation measurement for motors with brake, the brake should be opened (p1215 = 2) - as long as this can be done without incurring any danger.
Re bit 10:
If the motor holding brake is set just the same as the sequence control (p1215 = 1 or 3), the commutation angle and the direction of rotation are not measured. Before carrying out the rotation measurement for motors with brake, the brake should be opened (p1215 = 2) - as long as this can be done without incurring any danger.
Re bit 14, 15:
The following applies for bit 14 and 15 = 0:
When the function module "extended setpoint channel" is activated (r0108.8 = 1), the direction inhibit of the setpoint channel becomes effective. No direction of inhibit is effective if the function module is inactive.
The following applies for minimum bit 14 = 1 or bit 15 = 1:
The direction inhibit set in p1959 becomes effective.

p1959[0...n]		Moving measurement configuration / Mov meas config				
SERVO (Lin)	Can be changed:	T	Calculated:	CALC_MOD_ALL	Access level:	3
	Data type:	Unsigned16	Dynamic index:	MDS, p0130	Func. diagram:	-
	P-Group:	Motor identification	Units group:	-	Unit selection:	-
	Not for motor type:	REL	Scaling:	-	Expert list:	1
	Min	-	Max	-	Factory setting	0000 1110 1110 0111 bin
Description:		Sets the configuration of the moving measurement.				
Recommend.:		A direction inhibit should not be activated for the moving measurement (p1959.14 = 1 and p1959.15 = 1) as long as this can be done without incurring any danger. This means that the identification is complete and more accurate. When the direction inhibit is activated, the reluctance force constant (p1959.7) is not identified and the angular commutation offset (p1959.10, p0431) is inaccurately determined. The reluctance force constant (p1959.7) is also not identified in encoderless operation.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP	
	00	Reserved	Yes	No	-	
	01	Identify the saturation characteristic	Yes	No	-	
	02	Identify the moment of inertia	Yes	No	-	
	05	Identify the q inductance	Yes	No	-	
	06	Identify the force constant	Yes	No	-	
	07	Identify the reluctance force constant	Yes	No	-	
	08	Identify the q inductance at the test stand	Yes	No	-	
	09	Identify the magnetizing current / magnetizing inductance	Yes	No	-	
	10	Identify commutation angle and direction	Yes	No	-	
	11	Identify rotor resistance	Yes	No	-	
	14	Positive direction permitted	Yes	No	-	
	15	Negative direction permitted	Yes	No	-	
Dependency:		Refer to: p1958, p1960				

- Notice:** The step p1959.8 (identify q inductance on the test stand) may only be selected if the drive can be kept at zero speed or at a fixed speed either using a test stand or other mechanical measures.
During steps p1959.2 (identifying the moment of inertia) and p1959.6 (identifying the torque constant) the Vdc_min controller is disabled (p1240).
During step p1959.7 (identifying the reluctance torque constant) the Vdc_min controller and Vdc_max controller are disabled (p1240).
- Note:** For an induction motor (ASM), the following bits 1, 2, 5, 8, 9, 10, 14, 15 are effective
For a synchronous motor (SRM), the following bits 2, 5, 6, 7, 8, 10, 14, 15 are effective
Re bit 05:
For "motor holding brake the same as sequence control" (p1215 = 1 or 3), the Lq characteristic is only measured up to approximately the rated motor current (p0305) instead of up to the current limit (p0640). Before carrying out the rotation measurement for motors with brake, the brake should be opened (p1215 = 2) - as long as this can be done without incurring any danger.
Re bit 10:
If the motor holding brake is set just the same as the sequence control (p1215 = 1 or 3), the commutation angle and the direction of rotation are not measured. Before carrying out the rotation measurement for motors with brake, the brake should be opened (p1215 = 2) - as long as this can be done without incurring any danger.
Re bit 14, 15:
The following applies for bit 14 and 15 = 0:
When the function module "extended setpoint channel" is activated (r0108.8 = 1), the direction inhibit of the setpoint channel becomes effective. No direction of inhibit is effective if the function module is inactive.
The following applies for minimum bit 14 = 1 or bit 15 = 1:
The direction inhibit set in p1959 becomes effective.

p1959[0...n]		Rotating measurement configuration / Rot meas config			
VECTOR (n/M)	Can be changed: T		Calculated: CALC_MOD_ALL		Access level: 2
	Data type: Unsigned16		Dynamic index: DDS, p0180		Func. diagram: -
	P-Group: Motor identification		Units group: -		Unit selection: -
	Not for motor type: REL		Scaling: -		Expert list: 1
	Min -		Max -		Factory setting 0001 1111 bin
Description:		Sets the configuration of the rotating measurement.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Enc test active	Yes	No	-
	01	Saturation characteristic identification	Yes	No	-
	02	Moment of inertia identification	Yes	No	-
	03	Recalculates the speed controller parameters	Yes	No	-
	04	Speed controller optimization (vibration test)	Yes	No	-
	05	q leakage inductance ident. (for current controller adaptation)	Yes	No	-
Dependency:		Refer to: F07988			
Note:		The encoder is only tested if the rotating measurement with encoder is selected (p1960 = 2). The following parameters are influenced for the individual optimization steps: Bit 00: None Bit 01: p0320, p0360, p0362 ... p0369 Bit 02: p0341, p0342 Bit 03: p1400.0, p1458, p1459, p1460, p1462, p1463, p1470, p1472, p1496 Bit 04: Dependent on p1960 Bit 05: p0391, p0392, p0393, p1402.2 only for induction motors p1960 = 1, 3: p1400.0, p1458, p1459, p1470, p1472, p1496 p1960 = 2, 4: p1458, p1459, p1460, p1461, p1462, p1463, p1496 The identification of the q leakage inductance can only be carried out for unloaded motors or motors with a low (load approx. 30% below the rated motor torque). Only then is a current controller adaptation (p0391 ... p0393) parameterized if the q-leakage inductance under no-load conditions is at least 30 % higher than the total leakage inductance (p0356, p0358).			

p1960	Rotating measurement selection / Rot meas sel		
SERVO	Can be changed: T Data type: Integer16 P-Group: Motor identification Not for motor type: REL Min -3	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Activates the rotating measurement.		
Value:	-3: Accept identified parameters -2: Acknowledge encoder inversion actual value (F07993) -1: Start motor data identification without acceptance 0: Inactive/inhibit 1: Start motor data identification with acceptance		
Recommend.:	Before carrying out the rotation measurement for motors with brake, the brake should be opened (p1215 = 2) - as long as this can be done without incurring any danger. The commutation angle and the direction are also determined.		
Dependency:	Refer to: r1934, r1935, r1936, r1937, r1938, r1939, r1947, r1948, p1958, p1959, r1962, r1963, r1969 Refer to: F07990, A07991, F07993		
Danger:	For the rotating measurement, the motor is accelerated up to the maximum speed. Only the parameterized current limit (p0640) and the maximum speed (p1082) are effective. The behavior of the motor can be influenced using the direction inhibit (p1959.14, p1959.15) and the ramp-up/ramp-down time (p1958).		
			
Notice:	If there is a motor holding brake, it must be open (p1215 = 2).		
	To permanently accept the determined settings they must be saved in a non-volatile fashion (p0971, p0977).		
Note:	The rotating measurement can only be selected when the pulses of all of the drive objects of the Control Unit have been suppressed. After selection, all of the other drive objects of the Control Unit are interlocked so that they cannot be powered up until the rotating measurement has been completed or deselected. When the rotating measurement is activated (p1960 = 1), it is not possible to save the parameters (p0971, p0977).		
p1960	Moving measurement selection / Mov meas sel		
SERVO (Lin)	Can be changed: T Data type: Integer16 P-Group: Motor identification Not for motor type: REL Min -3	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Activates the rotating measurement.		
Value:	-3: Accept identified parameters -2: Acknowledge encoder inversion actual value (F07993) -1: Start motor data identification without acceptance 0: Inactive/inhibit 1: Start motor data identification with acceptance		
Recommend.:	Before carrying out the rotation measurement for motors with brake, the brake should be opened (p1215 = 2) - as long as this can be done without incurring any danger. The commutation angle and the direction are also determined.		
Dependency:	Refer to: r1934, r1935, r1936, r1937, r1938, r1939, r1947, r1948, p1958, p1959, r1962, r1963, r1969 Refer to: F07990, A07991, F07993		
Danger:	For the moving measurement, the motor is accelerated up to the maximum velocity. Only the parameterized current limit (p0640) and the maximum velocity (p1082) are effective. The behavior of the motor can be influenced using the direction inhibit (p1959.14, p1959.15) and the ramp-up/ramp-down time (p1958).		
			
Notice:	If there is a motor holding brake, it must be open (p1215 = 2). To permanently accept the determined settings they must be saved in a non-volatile fashion (p0971, p0977).		

Note: The moving measurement can only be selected when the pulses of all of the drive objects of the Control Unit have been suppressed. After selection, all of the other drive objects of the Control Unit are interlocked so that they cannot be powered up until the moving measurement has been completed or deselected.
When the moving measurement is activated (p1960 = 1), it is not possible to save the parameters (p0971, p0977).

p1960		Rotating measurement selection / Rot meas sel	
VECTOR	Can be changed: T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0	Max 4	Factory setting 0
Description:	Sets the rotating measurement.		
	The rotating measurement is carried out after the next power-on command.		
	The setting possibilities of the parameter depend on the open-loop/closed-loop control mode (p1300).		
	p1300 < 20 (V/f open-loop control):		
	It is not possible to select rotating measurement or speed controller optimization.		
	p1300 = 20, 22 (encoderless operation):		
	Only rotating measurement or speed controller optimization can be selected in the encoderless mode.		
	p1300 = 21, 23 (operation with encoder):		
	Both versions (encoderless and with encoder) of the rotating measurement and speed controller optimization can be selected.		
	Value:	0: Inhibited	
1: Rotating measurement in encoderless operation			
2: Rotating measurement with encoder			
3: Speed controller optimization in encoderless operation			
4: Speed controller optimization with encoder			
Dependency:	Before the rotating measurement is carried out, the motor data identification routine (p1900, p1910, r3925) should have already been done.		
	In the simulation mode, a value of 1 cannot be written into the parameter.		
	When selecting the rotating measurement, the drive data set changeover is suppressed.		
	Refer to: p1272, p1300, p1900, p1959		
	Refer to: A07987		
Danger:	For drives with a mechanical system that limits the distance moved, it must be ensured that this is not reached during the rotating measurement. If this is not the case, then it is not permissible that the measurement is carried out.		
			
Notice:	If there is a motor holding brake, it must be open (p1215 = 2).		
	To permanently accept the determined settings they must be saved in a non-volatile fashion (p0971, p0977).		
Note:	When the rotating measurement is activated, it is not possible to save the parameters (p0971, p0977).		
	Parameter changes are automatically made for the rotating measurement (e.g. p1120); this is the reason that up to the end of the measurement, and if no faults are present, no manual changes should be made.		
	The ramp-up and ramp-down times (p1120, p1121) are limited, for the rotating measurement, to 900 s.		
	For speed controller optimization with encoder (p1960 = 2, 4), the speed controller for encoderless operation is also pre-assigned (p1470, p1472).		
	Depending on whether the speed controller optimization is carried out with or without encoder, different Kp/Tn adaptations of the speed controller are set (p1464, p1465). If the drive should be controlled with as well as without speed encoder, then we recommend the use of two drive data sets (p0180). These can then be executed with different speed controller adaptations.		



p1961	Saturation characteristic speed to determine / Sat_char n determ		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min 26 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 75 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 40 [%]
Description:	Sets the speed to determine the saturation characteristic and the encoder test. The percentage value is referred to p0310 (rated motor frequency).		
Dependency:	Refer to: p0310, p1959 Refer to: F07983		
Note:	The saturation characteristics should be determined at an operating point with the lowest possible load.		
r1962[0...9]	Saturation characteristic magnetizing current identified / Sat_char I_mag		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM, REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the magnetizing currents of the identified saturation characteristic. The values are referred to r0331.		
Dependency:	Refer to: p1959, p1960, r1963		
Note:	The saturation characteristic consists of the value pairs from p1962 and p1963 with the same index.		
r1962[0...4]	Saturation characteristic magnetizing current / Sat_char I_mag		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM, REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the magnetizing currents of the identified saturation characteristic. The values are referred to r0331. After they have been determined, the values are transferred to p0366 ... p0369.		
Index:	[0] = Value 1 [1] = Value 2 [2] = Value 3 [3] = Value 4 [4] = Value 5		
Dependency:	Refer to: r0331		
r1963[0...9]	Saturation characteristic stator flux identified / Sat_char flux		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM, REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the stator flux of the identified saturation characteristic. The values are referred to the stator flux at the magnetizing current (r0331).		

Dependency: Refer to: p1959, p1960, r1962

Note: The saturation characteristic consists of the value pairs from p1962 and p1963 with the same index.

r1963[0...4]	Saturation characteristic magnetizing inductance / Sat_char L_main		
VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: PEM, REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]
Description:	Displays the magnetizing inductances of the identified saturation characteristic. The values are referred to r0382.		
Index:	[0] = Value 1 [1] = Value 2 [2] = Value 3 [3] = Value 4 [4] = Value 5		
Dependency:	Refer to: r0382		

r1964[0...4]	Saturation characteristic rotor flux / Sat_char rot flux		
VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: PEM, REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]
Description:	Displays the rotor flux values of the identified saturation characteristic. After they have been determined, the values are transferred to p0362 ... p0365.		
Index:	[0] = Value 1 [1] = Value 2 [2] = Value 3 [3] = Value 4 [4] = Value 5		

p1965	Speed_ctrl_opt speed / n_opt speed		
VECTOR (n/M)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	10 [%]	75 [%]	40 [%]
Description:	Sets the speed for the identification of the moment of inertia and the vibration test. Induction motor: The percentage value is referred to p0310 (rated motor frequency). Synchronous motor: The percentage value is referred to the minimum from p0310 (rated motor frequency) and p1082 (maximum speed).		
Dependency:	Refer to: p0310, p1959 Refer to: F07984, F07985		
Note:	In order to calculate the inertia, sudden speed changes are carried out - the specified value corresponds to the lower speed setpoint. This value is increased by 20 % for the upper speed value. The q leakage inductance (refer to p1959 bit 5) is determined at zero speed and at 50% of p1965 - however, with a maximum output frequency of 15 Hz and at a minimum of 10% of the rated motor speed.		

p1967	Speed_ctrl_opt dynamic factor / n_opt dyn_factor		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min 1 [%]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 400 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100 [%]
Description:	Sets the dynamic response factor for speed controller optimization.		
Dependency:	Refer to: p1959 Refer to: F07985		
Note:	For a rotating measurement, this parameter can be used to optimize the speed controller. p1967 = 100 % --> speed controller optimization according to a symmetric optimum. p1967 > 100 % --> optimization with a higher dynamic response (Kp higher, Tn lower).		
r1968	Speed_ctrl_opt dynamic factor actual / n_opt dyn_fact act		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the dynamic factor which is actually achieved for the vibration test		
Dependency:	Refer to: p1959, p1967 Refer to: F07985		
Note:	This dynamic factor only refers to the control mode of the speed controller set in p1960.		
r1969	Moment of inertia identified / M_inertia ident		
SERVO	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min - [kgm²]	Calculated: - Dynamic index: - Units group: 25_1 Scaling: - Max - [kgm²]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [kgm²]
Description:	Displays the identified moment of inertia.		
Dependency:	IEC drives (p0100 = 0): unit kg m² NEMA drives (p0100 = 1): unit lb ft² Refer to: p0341, p0342, p1498, p1959, p1960		
r1969	High load inertia identified / High load inert id		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min - [kg]	Calculated: - Dynamic index: - Units group: 27_1 Scaling: - Max - [kg]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [kg]
Description:	Displays the identified high load inertia.		
Dependency:	IEC drives (p0100 = 0): unit kg m² NEMA drives (p0100 = 1): unit lb ft² Refer to: p0341, p0342, p1498, p1959, p1960		

r1969	Speed_ctrl_opt moment of inertia determined / n_opt M_inert det		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min - [kgm²]	Calculated: - Dynamic index: - Units group: 25_1 Scaling: - Max - [kgm²]	Access level: 4 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting - [kgm²]
Description:	Displays the determined moment of inertia of the drive. After it has been determined, the value is transferred to p0341, p0342.		
Dependency:	IEC drives (p0100 = 0): unit kg m² NEMA drives (p0100 = 1): unit lb ft² Refer to: p0341, p0342, p1959 Refer to: F07984		
r1970[0...1]	Speed_ctrl_opt vibration test vibration frequency determined / n_opt f_vibration		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Hz]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the vibration frequencies determined by the vibration test.		
Index:	[0] = Frequency low [1] = Frequency high		
Dependency:	Refer to: p1959 Refer to: F07985		
r1971[0...1]	Speed_ctrl_opt vibration test standard deviation determined / n_opt std. deviat.		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Hz]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the standard deviations of the vibration frequencies determined by the vibration test		
Index:	[0] = Standard deviation of low frequency [1] = Standard deviation of high frequency		
Dependency:	Refer to: p1959 Refer to: F07985		
r1972[0...1]	Speed_ctrl_opt vibration test number of periods determined / n_opt period qty		
VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of periods determined by the vibration test.		
Index:	[0] = No. of periods of the low frequency [1] = No. of periods of the high frequency		
Dependency:	Refer to: p1959 Refer to: F07985		

r1973[0...1]	Encoder, pulse number identified / Pulse No. ident																																																															
SERVO	Can be changed: - Data type: Integer32 P-Group: Motor identification Not for motor type: REL Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -																																																													
Description:	Index 0: Rotating motors: Displays the identified encoder pulse number (per revolution). Linear motors: Encoder pulse number per meter. Grid division = 1/p1973 [meter]. Index 1: Rotating motors: No significance. Linear motors: Identified grid division in nm.																																																															
Index:	[0] = Rotating motor encoder pulse number [1] = Linear motor, grid division in nm																																																															
Notice:	Due to the measuring accuracy (approx. 5 %) only the approximate value is shown in p1973 and may not be directly transferred into p0407 or p0408. An incorrect pole pair number (r0313, p0314) or pole pair width (p0315) results in an incorrect value in p1973.																																																															
Note:	A negative signal indicates an incorrect polarity of the encoder signal.																																																															
r1973	Rotating measurement, encoder test pulse number determined / n_opt pulse No.																																																															
VECTOR (n/M)	Can be changed: - Data type: Integer32 P-Group: Motor identification Not for motor type: REL Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -																																																													
Description:	Displays the number of pulses determined during the vibration test.																																																															
Note:	A negative signal indicates an incorrect polarity of the encoder signal.																																																															
r1979.0...11	BO: Speed_ctrl_opt status / n_opt status																																																															
VECTOR (n/M)	Can be changed: - Data type: Unsigned16 P-Group: Motor identification Not for motor type: REL Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -																																																													
Description:	Displays the status to check and monitor the states of speed controller optimization.																																																															
Bit field:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr> <tr> <td>00</td><td>Speed controller optimization activated</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>01</td><td>Speed controller optimization completed</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>02</td><td>Speed controller optimization interrupted</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>04</td><td>Enc test active</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>05</td><td>Saturation char. identification active</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>06</td><td>Moment of inertia identification active</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>07</td><td>Recalc. speed controller parameters active</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>08</td><td>Speed controller vibration test active</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>09</td><td>Magnetizing inductance adapt. active</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>10</td><td>Operation with encoder after encoderless operation</td><td>Yes</td><td>No</td><td>-</td></tr> <tr> <td>11</td><td>q-leakage inductance identification</td><td>Yes</td><td>No</td><td>-</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FP	00	Speed controller optimization activated	Yes	No	-	01	Speed controller optimization completed	Yes	No	-	02	Speed controller optimization interrupted	Yes	No	-	04	Enc test active	Yes	No	-	05	Saturation char. identification active	Yes	No	-	06	Moment of inertia identification active	Yes	No	-	07	Recalc. speed controller parameters active	Yes	No	-	08	Speed controller vibration test active	Yes	No	-	09	Magnetizing inductance adapt. active	Yes	No	-	10	Operation with encoder after encoderless operation	Yes	No	-	11	q-leakage inductance identification	Yes	No	-			
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p1980[0...n] Pole position identification technique / PolID technique			
SERVO	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 99	Factory setting 99
Description:	Sets the pole position identification technique.		
Value:	0: Saturation-based 1st + 2nd harmonics 1: Saturation-based 1st harmonics 4: Saturation-based, 2-stage 10: Motion-based 99: No technique selected		
Dependency:	Refer to: p0325, p0329, p1981, p1982, p1983, r1984, r1985, r1987		
Notice:	If the incorrect technique is applied, this can cause the motor to accelerate in an uncontrolled fashion. Under the following conditions, the integral time must be disabled (p1996 = 0): - p1980 = 10 (motion-based) - motor encoder with track A/B sq-wave (p0404.3 = 1) - p0430.20 = 0 (flank time measurement) Once the integral time has been disabled, the motion is increased during the identification (a minimum of 90 ° electrical). As a result of this, the maximum distance (p1981) must also be increased.		
Note:	When commissioning a catalog motor, the technique is automatically selected depending on the motor type being used. The following applies for 1FN3 motors: A technique with 2nd harmonic may not be used (do not use p1980 = 0, 4). For 1FN7 motors, the following applies: A two-stage technique may not be used (do not use p1980 = 4). The automatically set value in p0329 may not be changed.		

p1980[0...n] Pole position identification technique / PolID technique			
VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: ASM	Scaling: -	Expert list: 1
	Min 1	Max 10	Factory setting 4
Description:	Sets the pole position identification technique.		
Value:	1: Voltage pulsing, first harmonic 4: Voltage pulsing, 2-stage 6: Voltage pulsing, 2-stage invert 10: DC current injection		
Dependency:	In the simulation mode, the parameter cannot be written into. Refer to: p1272, p1780		
Note:	Voltage pulse technique (p1980 = 1, 4) cannot be applied to separately-excited synchronous motors (p0300 = 5) and for operation with sine-wave output filters (p0230).		

p1981[0...n]	Pole position identification maximum distance / PolID distance max		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0 [°]	Calculated: CALC_MOD_ALL Dynamic index: MDS, p0130 Units group: - Scaling: - Max 180 [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10 [°]
Description:	Sets the maximum distance (electrical angle) when carrying out the pole position identification routine. If this distance (travel) is exceeded, an appropriate fault is output.		
Dependency:	Refer to: p0325, p0329, p1980, p1982, p1983, r1984, r1985, r1987, p1990 Refer to: F07995		
Notice:	Value = 180 °: Monitoring is deactivated.		
p1982[0...n]	Pole position identification selection / PolID selection		
SERVO	Can be changed: T Data type: Integer16 P-Group: Motor identification Not for motor type: - Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Activates the pole position identification routine to determine the commutation angle and to carry out a plausibility check.		
Value:	0: Pole position identification off 1: Pole position identification for commutation 2: Pole position identification for plausibility check		
Recommend.:	Re p1982 = 1: This is used for synchronous motors with motor encoder without absolute data. The information/data regarding the absolute commutation angle is supplied via a track C/D, Hall sensors, an absolute encoder or from the pole position identification routine. Re p1982 = 2: This is used for synchronous motor with motor encoder with absolute data to check this data.		
Dependency:	Refer to: p0325, p0329, p1980, p1981, p1983, r1984, r1985, r1987, p1990		
p1982[0...n]	Pole position identification selection / PolID selection		
VECTOR	Can be changed: T Data type: Integer16 P-Group: Motor identification Not for motor type: - Min 0	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Activates the pole position identification routine to determine the commutation angle and to carry out a plausibility check.		
Value:	0: Pole position identification off 1: Pole position identification for commutation 2: Pole position identification for plausibility check		

Recommend.: Re p1982 = 1:
This is used for synchronous motors with motor encoder without absolute data.
The information/data regarding the absolute commutation angle is supplied via a track C/D, Hall sensors, an absolute encoder or from the pole position identification routine.
Re p1982 = 2:
This is used for synchronous motor with motor encoder with absolute data to check this data.
For VECTOR, the following applies:
With p1982 = 2, each time the pulses are enabled it is checked whether the absolute position supplied from the encoder does not exceed a deviation of 45 degrees to the identified pole wheel position.
With separately-excited synchronous motors (p0300 = 5), pole position identification cannot be selected if an encoder with position data is used (e.g. SSI encoder).

Dependency: Refer to: p0325, p0329, p1980, p1981, p1983, r1984, r1985, r1987, p1990

Note: For encoderless operation, the pole position identification routine is selected with p1780.6

p1983 **Pole position identification, test / PolID test**

SERVO **Can be changed:** U, T **Calculated:** - **Access level:** 3
Data type: Unsigned16 **Dynamic index:** - **Func. diagram:** -
P-Group: Motor identification **Units group:** - **Unit selection:** -
Not for motor type: - **Scaling:** - **Expert list:** 1
Min **Max** **Factory setting**
0 1 0

Description: Starts the pole position identification routine for test purposes.
p1983 = 1:
Start of pole position identification. The parameter is set to zero automatically on completion of the identification process.

Dependency: Refer to: p0325, p0329, p1980, p1981, p1982, r1984, r1985, r1987, p1990

Notice: For p1983 = 1 and if the pulses are not enabled, then the function is only executed the next time that the pulses are enabled.

Note: When this test is executed, it does not influence the commutation angle.

r1984 **Pole position identification, angular difference / PolID ang diff**

SERVO, VECTOR **Can be changed:** - **Calculated:** - **Access level:** 3
(n/M) **Data type:** FloatingPoint32 **Dynamic index:** - **Func. diagram:** -
P-Group: Motor identification **Units group:** - **Unit selection:** -
Not for motor type: - **Scaling:** - **Expert list:** 1
Min **Max** **Factory setting**
- [°] - [°] - [°]

Description: Displays the angular difference between the actual electrical commutation angle and the angle determined by the pole position identification.

Dependency: Refer to: p0325, p0329, p1980, p1981, p1982, p1983, r1985, r1987, p1990

Note: When the pole position identification routine is executed several times using p1983, the spread of the measured values can be determined using this value. At the same position, the spread should be less than 2 degrees electrical.

r1985	Pole position identification, saturation characteristic / PolID sat_char		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the saturation characteristic of the pole position identification routine. The values for the characteristic of the last saturation-based pole position identification routine are output every 1 ms in order to record signals (e.g. trace).		
Dependency:	Refer to: p0325, p0329, p1980, p1981, p1982, p1983, r1984, r1987, p1990		
r1987	Pole position identification trigger characteristic / PolID trig_char		
SERVO, VECTOR (n/M)	Can be changed: - Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the trigger characteristic of the pole position identification routine. The values for the characteristic of the last pole position identification routine are output every 1 ms in order to record signals (e.g. trace). The values for trigger characteristic and saturation characteristic are always output in synchronism from a time perspective.		
Dependency:	Refer to: p0325, p0329, p1980, p1981, p1982, p1983, r1984, r1985		
Note:	The following information and data can be taken from the trigger characteristic. - the value -100% marks the angle at the start of the measurement. - the value +100 % marks the commutation angle determined from the pole position identification routine.		
p1990	Encoder adjustment, determine angular commutation offset / Enc_adj det ang		
SERVO	Can be changed: U, T Data type: Integer16 P-Group: Motor identification Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	This function is only required for synchronous motors and can be started when commissioning for the first time or after replacing an encoder. The function acts on the active motor data set. When adjusting the encoder, the angular commutation offset is determined and transferred into p0431. Alarm A07971 is output while the angular commutation offset is being determined. p1990 is automatically set to 0 after the angular commutation offset has been determined. p1990 = 0: Deactivated p1990 = 1: Activated with transfer		
Dependency:	Refer to: p0325, p0329, p0431, p1980, p1981, p1982, p1983, r1984, r1985, r1987, p1999 Refer to: A07971		

- Caution:** In order to prevent an incorrect orientation of the electrical pole position (uncontrolled motor movement), the automatically determined angular commutation offset (p0431) should, for reasons of safety, be checked using one of the following recommendations:
Recommendation 1:
Set encoderless operation (p1300 = 20 or p1404 = 0), deselect pole position identification (p1982 = 0), operate under no-load conditions with a speed > p1755, correct the actual value inversion (p0410.0) (e.g. r0061 = r0063), read the angular error in r1778; the result in r1778 should be approximately 0, for |r1778| > 2 degrees, add the value to p0431 - taking into account the sign - and enter in p0431.
Recommendation 2:
Set the current limit to 0 (p0640 = 0), activate travel to fixed stop (p1545 = 1), record r0089[0] (phase voltage) and r0093 (electrically normalized pole position) (e.g. trace) while the motor is externally moved; in this case, the rising zero crossover of the phase voltage must coincide with the 360 ° --> 0 ° step (jump) from r0093.
Recommendation 3:
Measure the phase voltage V (measure phase U with respect to the virtual star point using 3 resistors) and r0093 (electrically normalized pole position); the rising zero crossover of the phase voltage must coincide with the 360 ° --> 0 ° step (jump) of r0093.
Recommendation 4:
Determine the average value from several results of a pole position identification routine executed as test (p1983) at various electrical angles and add the value to p0431 - taking into account the sign and enter into p0431.
- Notice:** For p1990 = 1 and with the pulses not enabled, the function is only executed the next time that the pulses are enabled.
- Note:** If fault F07414 is present, the following applies:
First set p1990 to 1, then acknowledge the fault and then issue the enable signals.

p1990		Encoder adjustment, determine angular commutation offset / Enc_adj det ang	
VECTOR	Can be changed: T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: ASM	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	2	0
Description:	This function is only required for synchronous motors and can be started when commissioning for the first time or after replacing an encoder. The function acts on the active motor data set. Alarm A07971 is output while the angular commutation offset is being determined. p1990 is automatically set to 0 after the angular commutation offset has been determined. For p1990 = 1 (encoder adjustment with transfer), the following applies: The angular commutation offset is determined and transferred into p0431. For p1990 = 2 (encoder adjustment for checking), the following applies: The angular commutation offset is determined and is not transferred into p0431. For a deviation of more than 6 ° electrical, fault F07413 is output.		
Value:	0: Deactivated 1: Activated with transfer 2: Activated for checking		
Dependency:	In the simulation mode, the parameter cannot be written into. When selecting the encoder adjustment, the changeover of the drive data sets is suppressed. Encoder adjustment is only carried out if the function module for speed/torque control is available (r0108 bit2 = 1). Refer to: p0325, p0329, p0431, p1272, p1900		
Caution:	When the encoder is being adjusted, the motor must be operated without a load - and if a motor holding brake is being used, this must be opened.		



p1991[0...n]		Motor changeover, angular commutation correction / Ang_com corr	
SERVO, VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-180 [°]	180 [°]	0 [°]
Description:		Sets the angle that is added to the commutating angle.	
Caution:		If the angular correction is not correctly set, when changing over and with closed-loop torque control, the motor can accelerate to high speeds in spite of the fact that a setpoint of zero has been entered.	
			

r1992	Pole position identification diagnostics / PolID diag				
SERVO	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -		
	P-Group: -	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays diagnostics information for the pole position identification routine.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Critical encoder fault occurred	Yes	No	-
	02	Enc parking active	Yes	No	-
	05	Encoder fault Class 1	Yes	No	-
	06	Encoder fault Class 2	Yes	No	-
	07	Pole position identification for encoder carried out	Yes	No	-
	08	Fine synchronization carried out	Yes	No	-
	09	Coarse synchronization carried out	Yes	No	-
	10	Commutation information available	Yes	No	-
	11	Speed information available	Yes	No	-
	12	Position information available	Yes	No	-
	15	Zero mark passed	Yes	No	-

r1992	Pole position identification diagnostics / PolID diag				
SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -		
	P-Group: -	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
-	-	-			
Description:	Displays diagnostics information for the pole position identification routine.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Critical encoder fault occurred	Yes	No	-
	02	Enc parking active	Yes	No	-
	05	Encoder fault Class 1	Yes	No	-
	06	Encoder fault Class 2	Yes	No	-
	07	Pole position identification for encoder car- ried out	Yes	No	-
	08	Fine synchronization carried out	Yes	No	-
	09	Coarse synchronization carried out	Yes	No	-
	10	Commutation information available	Yes	No	-
	11	Velocity information available	Yes	No	-
	12	Position information available	Yes	No	-
	15	Zero mark passed	Yes	No	-

p1993[0...n]	Pole position identification current, motion-based / PolID I mot_bas		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.00 [Arms]	Calculated: CALC_MOD_EQU Dynamic index: MDS, p0130 Units group: - Scaling: - Max 20000.00 [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [Arms]
Description:	Sets the current when executing the motion-based pole position identification.		
Dependency:	Refer to: p1980, p1982, p1994		
p1994[0...n]	Pole position identification rise time motion-based / PolID T mot_bas		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 2500 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100 [ms]
Description:	Sets the rise time of the current when executing the motion-based pole position identification.		
Dependency:	Refer to: p1980, p1982, p1993		
p1995[0...n]	Pole position identification gain, motion-based / PolID kp mot_bas		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.000 [Nms/rad]	Calculated: CALC_MOD_CON Dynamic index: MDS, p0130 Units group: 17_1 Scaling: - Max 999999.000 [Nms/rad]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 0.300 [Nms/rad]
Description:	Sets the gain when executing the motion-based pole position identification.		
p1995[0...n]	Pole position identification gain, motion-based / PolID kp mot_bas		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.000 [Ns/m]	Calculated: CALC_MOD_CON Dynamic index: MDS, p0130 Units group: 24_2 Scaling: - Max 999999.000 [Ns/m]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 10.000 [Ns/m]
Description:	Sets the gain when executing the motion-based pole position identification.		
p1996[0...n]	Pole position identification, integral time motion-based / PolID Tn mot_bas		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.0 [ms]	Calculated: CALC_MOD_CON Dynamic index: MDS, p0130 Units group: - Scaling: - Max 500.0 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2.0 [ms]
Description:	Sets the integral time when executing the motion-based pole position identification.		
Note:	The value 0 deactivates the I component. Once the integral time has been disabled, the motion is increased during the identification (a minimum of 90 ° electrical).		

p1997[0...n]	Pole position identification, smoothing time motion-based / PolID t_sm mot_bas		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.0 [ms]	Calculated: CALC_MOD_CON Dynamic index: MDS, p0130 Units group: - Scaling: - Max 50.0 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [ms]
Description:	Sets the smoothing time when executing the motion-based pole position identification.		
p1999[0...n]	Ang. commutation offset calibr. and pole position ID - scaling / ComOffsCalib scal		
VECTOR (n/M)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 10 [%]	Calculated: - Dynamic index: MDS, p0130 Units group: - Scaling: - Max 5000 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100 [%]
Description:	Sets the scaling for the runtime of the automatic encoder calibration and the current-impressing technique for the pole position identification routine.		
Dependency:	Refer to: p0341, p0342		
Caution:	For P1999 > 100% (setting, large moments of inertia):		
	There is no locked rotor monitoring (A7970.2).		
	The plausibility check of the encoder signal (A7970.4) only checks the sign.		
Note:	For high moments of inertia, it is practical to scale the runtime of the calibration higher.		
p2000	Reference frequency / Ref freq		
A_INF, B_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 0.10 [Hz]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 1000.00 [Hz]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 50.00 [Hz]
Description:	Sets the reference quantity for the frequency. All frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The following applies: Reference frequency (in Hz)		
p2000	Reference speed reference frequency / Ref_n Ref_f		
ENCODER	Can be changed: T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 6.00 [rpm]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 210000.00 [rpm]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 3000.00 [rpm]
Description:	Sets the reference quantity for speed and frequency. All speeds or frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The following applies: Reference frequency (in Hz) = reference speed (in rpm) / 60)		
Note:	If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.		

p2000 Reference velocity, reference frequency / Ref_v Ref_f			
ENCODER (Lin enc)	Can be changed: T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 0.60 [m/min]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 600.00 [m/min]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 120.00 [m/min]
Description:	Sets the reference quantity for velocity and frequency. All velocities or frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The following applies: Reference frequency (in Hz) = reference velocity (in (m/min) / 60)		
Note:	If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.		

p2000 Reference speed reference frequency / Ref_n Ref_f			
SERVO	Can be changed: T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 6.00 [rpm]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 210000.00 [rpm]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 3000.00 [rpm]
Description:	Sets the reference quantity for speed and frequency. All speeds or frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The following applies: Reference frequency (in Hz) = reference speed (in (rpm) / 60)		
Dependency:	Refer to: p0500, p2001, p2002, p2003, r2004		
Note:	For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using p0573 = 1. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. Example 1: The signal of an analog input (e.g. r4055[0]) is connected to a speed setpoint (e.g. p1070[0]). The actual percentage input value is cyclically converted into the absolute speed setpoint using the reference speed (p2000). Example 2: The setpoint from PROFIBUS (r2050[1]) is connected to a speed setpoint (e.g. p1070[0]). The actual input value is cyclically converted into a percentage value via the pre-specified scaling 4000 hex. This percentage value is converted to the absolute speed setpoint via reference speed (p2000).		

p2000 Reference velocity, reference frequency / Ref_v Ref_f			
SERVO (Lin)	Can be changed: T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 0.60 [m/min]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 600.00 [m/min]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 120.00 [m/min]
Description:	Sets the reference quantity for velocity and frequency. All velocities or frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The following applies: Reference frequency (in Hz) = reference velocity (in (m/min) / 60)		
Dependency:	Refer to: p0500, p2001, p2002, p2003, r2004		

Note: For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is at the factory setting.

If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.

Example 1:
The signal of an analog input (e.g. r4055[0]) is connected to a velocity setpoint (e.g. p1070[0]). The actual percentage input value is cyclically converted into the absolute velocity setpoint using the reference velocity (p2000).

Example 2:
The setpoint from PROFIBUS (r2050[1]) is connected to a velocity setpoint (e.g. p1070[0]). The actual input value is cyclically converted into a percentage value via the pre-specified scaling 4000 hex. This percentage value is converted to the absolute velocity setpoint via reference velocity (p2000).

p2000	Reference speed reference frequency / Ref_n Ref_f		
TM41	Can be changed: T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 6.00 [rpm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 210000.00 [rpm]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 3000.00 [rpm]
Description:	Sets the reference quantity for speed and frequency. All speeds or frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The following applies: Reference frequency (in Hz) = reference speed (in rpm) / 60		
Dependency:	Refer to: p2001, p2002, p2003, r2004		
Note:	For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using p0573 = 1. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. Example 1: The signal of an analog input (e.g. r4055[0]) is connected to a speed setpoint (e.g. p1070[0]). The actual percentage input value is cyclically converted into the absolute speed setpoint using the reference speed (p2000). Example 2: The setpoint from PROFIBUS (r2050[1]) is connected to a speed setpoint (e.g. p1070[0]). The actual input value is cyclically converted into a percentage value via the pre-specified scaling 4000 hex. This percentage value is converted to the absolute speed setpoint via reference speed (p2000).		

p2000	Reference speed reference frequency / Ref_n Ref_f		
VECTOR	Can be changed: T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 6.00 [rpm]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 210000.00 [rpm]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 3000.00 [rpm]
Description:	Sets the reference quantity for speed and frequency. All speeds or frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The following applies: Reference frequency (in Hz) = reference speed (in rpm) / 60		
Dependency:	Refer to: p2001, p2002, p2003, r2004		

Note: For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using p0573 = 1.

If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.

Example 1:
The signal of an analog input (e.g. r4055[0]) is connected to a speed setpoint (e.g. p1070[0]). The actual percentage input value is cyclically converted into the absolute speed setpoint using the reference speed (p2000).

Example 2:
The setpoint from PROFIBUS (r2050[1]) is connected to a speed setpoint (e.g. p1070[0]). The actual input value is cyclically converted into a percentage value via the pre-specified scaling 4000 hex. This percentage value is converted to the absolute speed setpoint via reference speed (p2000).

p2001		Reference voltage / Reference voltage	
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	10 [Vrms]	100000 [Vrms]	1000 [Vrms]
Description:	Sets the reference quantity for voltages. All voltages specified as relative value are referred to this reference quantity. This also applies for direct voltage values (= rms value) like the DC-link voltage. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.		
Note:	For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using p0573 = 1. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. For infeed units, the parameterized device supply voltage (p0210) is pre-assigned as the reference quantity. Example: The actual value of the DC link voltage (r0070) is connected to a test socket (e.g. p0771[0]). The actual voltage value is cyclically converted into a percentage of the reference voltage (p2001) and output according to the parameterized scaling.		
p2002		Reference current / Reference current	
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.10 [Arms]	100000.00 [Arms]	100.00 [Arms]
Description:	Sets the reference quantity for currents. All currents specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.		
Notice:	If various DDS are used with different motor data, then the reference quantities remain the same as these are not changed over with the DDS. The resulting conversion factor should be taken into account (e.g. for trace records). Example: p2002 = 100 A Reference quantity 100 A corresponds to 100 % p305[0] = 100 A Rated motor current 100 A for MDS0 in DDS0 --> 100 % corresponds to 100 % of the rated motor current p305[1] = 50 A Rated motor current 50 A for MDS1 in DDS1 --> 100 % corresponds to 200 % of the rated motor current		

Note: For the automatic calculation ($p0340 = 1$, $p3900 > 0$) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using $p0573 = 1$.

If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.

For infeed units, the rated line current, which is obtained from the rated power and parameterized rated line supply voltage ($p2002 = r0206 / p0210 / 1.73$) is pre-assigned as the reference quantity.

Example:

The actual value of a phase current ($r0069[0]$) is connected to a test socket (e.g. $p0771[0]$). The actual current value is cyclically converted into a percentage of the reference current ($p2002$) and output according to the parameterized scaling.

p2003 Reference torque / Reference torque			
SERVO, TM41, VEC-TOR	Can be changed: T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: 7_2	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.01 [Nm]	Max 20000000.00 [Nm]	Factory setting 1.00 [Nm]
Description:	Sets the reference quantity for torques. All torques specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.		
Note:	For the automatic calculation ($p0340 = 1$, $p3900 > 0$) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using $p0573 = 1$. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. Example: The actual value of the total torque ($r0079$) is connected to a test socket (e.g. $p0771[0]$). The actual torque is cyclically converted into a percentage of the reference torque ($p2003$) and output according to the parameterized scaling.		

p2003 Reference force / Reference force			
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: 8_2	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.01 [N]	Max 20000000.00 [N]	Factory setting 100.00 [N]
Description:	Sets the reference quantity for forces. All forces specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.		
Note:	For the automatic calculation ($p0340 = 1$, $p3900 > 0$) an appropriate pre-assignment is only made if the parameter is at the factory setting. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. Example: The actual value of the total force ($r0079[0]$) is connected to a test socket (e.g. $p0771[0]$). The actual force is cyclically converted into a percentage of the reference force ($p2003$) and output according to the parameterized scaling.		

r2004	Reference power / Reference power		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: 14_10 Scaling: - Max - [kW]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [kW]
Description:	Displays the reference quantity for power ratings. All power ratings specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.		
Dependency:	This value is calculated as follows: Infeed: Calculated from voltage times current. Closed-loop control: Calculated from torque times speed. Refer to: p2000, p2001, p2002, p2003		
Note:	If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. The reference power is calculated as follows: - $2 * \pi * \text{reference speed} / 60 * \text{reference torque (motor)}$ - $\text{reference voltage} * \text{reference current} * \text{root}(3) \text{ (infeed)}$		
p2005	Reference angle / Reference angle		
SERVO, TM41, VEC- TOR	Can be changed: T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 90.00 [°]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 180.00 [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 90.00 [°]
Description:	Sets the reference quantity for angle. All angles specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.		
Note:	For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using p0573 = 1. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.		
p2007	Reference acceleration / Ref accel		
SERVO, TM41, VEC- TOR	Can be changed: T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 0.01 [rev/s²]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 500000.00 [rev/s²]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.01 [rev/s²]
Description:	Sets the reference quantity for acceleration rates. All acceleration rates specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.		
Note:	For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using p0573 = 1. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. The reference acceleration is calculated as follows: Reference speed (p2000) converted from 1/min to 1/s divided by 1 s --> $p2007 = p2000 [\text{rpm}] / (60 [\text{s/min}] * 1 [\text{s}])$		

p2007	Reference acceleration / Ref accel		
SERVO (Lin)	Can be changed: T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 0.01 [m/s²]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: 22_1 Scaling: - Max 10000.00 [m/s²]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 0.01 [m/s²]
Description:	Sets the reference quantity for acceleration rates. All acceleration rates specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.		
Note:	For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using p0573 = 1. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. The reference acceleration is calculated as follows: Reference speed (p2000) converted from 1/min to 1/s divided by 1 s --> p2007 = p2000 [rpm] / (60 [s/min] * 1 [s])		
r2019[0...7]	Comm int error statistics / Comm err		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the receive errors at the commissioning interface (RS232).		
Index:	[0] = Number of error-free telegrams [1] = Number of rejected telegrams [2] = Number of framing errors [3] = Number of overrun errors [4] = Number of parity errors [5] = Number of starting character errors [6] = Number of checksum errors [7] = Number of length errors		
p2020	Field bus interface baud rate / Field bus baud		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 4	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 8
Description:	Sets the baud rate for the fieldbus interface USS.		
Value:	4: 2400 Baud 5: 4800 Baud 6: 9600 Baud 7: 19200 Baud 8: 38400 Baud		
Note:	Field bus interface. Changes only become effective after POWER ON. The parameter is not influenced by setting the factory setting.		

p2021	Field bus interface address / Field bus address		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 30	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Displays or sets the address for the fieldbus interface. The address can be set as follows: 1) Using the address switch on the Control Unit. --> p2021 displays the address setting. --> A change only becomes effective after a POWER ON. 2) Using p2021 --> Only if an address of 0 or an address which is invalid for the fieldbus selected in p2030 has been set using the address switch. --> The address is saved in a non-volatile fashion using the function "copy from RAM to ROM". --> A change only becomes effective after a POWER ON.		
Dependency:	Refer to: p2030		
Note:	Changes only become effective after POWER ON. The parameter is not influenced by setting the factory setting.		
p2022	Field bus int USS PZD no. / Field bus USS PZD		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2
Description:	Sets the number of 16-bit words in the PZD part of the USS telegram for the field bus interface.		
Dependency:	Refer to: p2030		
Note:	The parameter is not influenced by setting the factory setting.		
p2023	Field bus int USS PKW no. / Field bus USS PKW		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 127	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 127
Description:	Sets the number of 16-bit words in the PKW part of the USS telegram for the field bus interface.		
Value:	0: PKW 0 words 3: PKW 3 words 4: PKW 4 words 127: PKW variable		
Dependency:	Refer to: p2030		
Note:	The parameter is not influenced by setting the factory setting.		

r2029[0...7] Field bus int error statistics / Field bus error																																			
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -																																
Description:	Displays the receive errors on the field bus interface (USS).																																		
Index:	[0] = Number of error-free telegrams [1] = Number of rejected telegrams [2] = Number of framing errors [3] = Number of overrun errors [4] = Number of parity errors [5] = Number of starting character errors [6] = Number of checksum errors [7] = Number of length errors																																		
p2030 Field bus int protocol selection / Field bus protocol																																			
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 3	Calculated: - Dynamic index: - Units group: - Scaling: - Max 6	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 3																																
Description:	Sets the communication protocol for the field bus interface.																																		
Value:	3: PROFIBUS 6: USS(RS232) at X140																																		
Note:	Changes only become effective after POWER ON. The parameter is not influenced by setting the factory setting.																																		
r2032 Master control, control word effective / PcCtrl STW eff																																			
A_INF, B_INF, S_INF	Can be changed: - Data type: Unsigned16 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -																																
Description:	Displays the effective control word 1 (STW1) of the drive for the master control.																																		
Bit field:	<table><tr><td>Bit</td><td>Signal name</td><td>1 signal</td><td>0 signal</td><td>FP</td></tr><tr><td>00</td><td>ON/OFF1</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>01</td><td>OC / OFF2</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>03</td><td>Operation enable</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>07</td><td>Acknowledge fault</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>10</td><td>Master ctrl by PLC</td><td>Yes</td><td>No</td><td>-</td></tr></table>	Bit	Signal name	1 signal	0 signal	FP	00	ON/OFF1	Yes	No	-	01	OC / OFF2	Yes	No	-	03	Operation enable	Yes	No	-	07	Acknowledge fault	Yes	No	-	10	Master ctrl by PLC	Yes	No	-				
Bit	Signal name	1 signal	0 signal	FP																															
00	ON/OFF1	Yes	No	-																															
01	OC / OFF2	Yes	No	-																															
03	Operation enable	Yes	No	-																															
07	Acknowledge fault	Yes	No	-																															
10	Master ctrl by PLC	Yes	No	-																															
Notice:	The master control only influences control word 1 and speed setpoint 1. Other control words/setpoints can be transferred from another automation device.																																		
Note:	OC: Operating condition																																		

r2032 Master control, control word effective / PcCtrl STW eff

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the effective control word 1 (STW1) of the drive for the master control.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	ON/OFF1	Yes	No	-
	01	OC / OFF2	Yes	No	-
	02	OC / OFF3	Yes	No	-
	03	Operation enable	Yes	No	-
	04	Ramp-function generator enable	Yes	No	-
	05	Start ramp-function generator	Yes	No	-
	06	Speed setpoint enable	Yes	No	-
	07	Acknowledge fault	Yes	No	-
	08	Jog bit 0	Yes	No	3030
	09	Jog bit 1	Yes	No	3030
	10	Master ctrl by PLC	Yes	No	-

Notice: The master control only influences control word 1 and speed setpoint 1. Other control words/setpoints can be transferred from another automation device.

Note: OC: Operating condition

r2032 Master control, control word effective / PcCtrl STW eff

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the effective control word 1 (STW1) of the drive for the master control.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	ON/OFF1	Yes	No	-
	01	OC / OFF2	Yes	No	-
	02	OC / OFF3	Yes	No	-
	03	Operation enable	Yes	No	-
	04	Ramp-function generator enable	Yes	No	-
	05	Start ramp-function generator	Yes	No	-
	06	Velocity setpoint enable	Yes	No	-
	07	Acknowledge fault	Yes	No	-
	08	Jog bit 0	Yes	No	3030
	09	Jog bit 1	Yes	No	3030
	10	Master ctrl by PLC	Yes	No	-

Notice: The master control only influences control word 1 and speed setpoint 1. Other control words/setpoints can be transferred from another automation device.

Note: OC: Operating condition

p2035	Fieldbus interface USS PIV drive object number / Fieldbus USS DO_no		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 62	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2
Description:	Sets the drive object number for communication via the field bus interface (USS).		
Dependency:	Refer to: p0978		
Note:	p2035 defines the destination for USS parameter requests (PIV). p0978[0] defines the destination for USS process data (PZD). The parameter is available globally on all drive objects. The parameter is not influenced by setting the factory setting.		
p2037	PROFIdrive STW1.10 = 0 mode / PD STW1.10=0 mode		
A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VEC- TOR	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the processing mode for PROFIdrive STW1.10 "master control by PLC". Generally, control word 1 is received with the first receive word (PZD1) (this is in conformance to the PROFIdrive profile). The behavior of STW1.10 = 0 corresponds to that of the PROFIdrive profile. For other applications that deviate from this, the behavior can be adapted using this particular parameter.		
Value:	0: Freeze setpoints and continue to process sign-of-life 1: Freeze setpoints and sign-of-life 2: Setpoints are not frozen		
Recommend.:	Do not change the setting p2037 = 0.		
Note:	If the STW1 is not transferred according to the PROFIdrive with PZD1 (with bit 10 "master control by PLC"), then p2037 should be set to 2.		
p2038	PROFIdrive STW/ZSW interface mode / PD STW/ZSW IF mode		
SERVO, VECTOR	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the interface mode of the PROFIdrive control words and status words. When selecting a telegram via p0922 (p2079), this parameter influences the device-specific assignment of the bits in the control and status words.		
Value:	0: SINAMICS 1: SIMODRIVE 611 universal 2: VIK-NAMUR		
Dependency:	Refer to: p0922, p2079		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For p0922 (p2079) = 100 ... 199, p2038 is automatically set to 1 and p2038 can no longer be changed. This means that for these telegrams, the "SIMODRIVE 611 universal" interface mode is set and cannot be changed.		

p2038	PROFIdrive STW/ZSW interface mode / PD STW/ZSW IF mode		
SERVO (EPOS, Pos ctrl), VECTOR (EPOS, Pos ctrl)	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 0	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Displays the interface mode of the PROFIdrive control words and status words.		
Value:	0: SINAMICS		
Dependency:	Refer to: p0922, p2079		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For p0922 (p2079) = 7, 9, 110, 111, p2038 is automatically set to 0 and cannot be changed.		

p2039	Select debug monitor interface / Sel. debug monitor		
CU_I	Can be changed: U, T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	The serial interface for the debug monitor is COM1 (X140) or COM2 (internal). Value = 0: COM2 (internal) Value = 1: COM1 (X140), commissioning protocol is deactivated Value = 2: COM2 (internal) Value = 3: Reserved		

p2039	Select debug monitor interface / Sel. debug monitor		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The serial interface for the debug monitor is COM1 (X140) or COM2 (internal). Value = 0: COM2 (internal) Value = 1: COM1 (X140), commissioning protocol is deactivated Value = 2: COM2 (internal) Value = 3: Reserved		

p2040	COMM INT monitoring time / C INT t_monit		
CU_CX32, CU_I, CU_LINK	Can be changed: U, T Data type: FloatingPoint32 P-Group: Communications Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 20 [ms]
Description:	Sets the monitoring time to monitor the process data received via the internal communications interface. If no process data is received within this time, an appropriate message is output.		
Dependency:	Refer to: F01910		
Note:	0: The monitoring is deactivated.		

p2040	Fieldbus interface monitoring time / Fieldbus t_monit			
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -	
	P-Group: Communications	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min 0 [ms]	Max 65535000 [ms]	Factory setting 100 [ms]	
Description:	Sets the monitoring time to monitor the process data received via the fieldbus interface. If no process data is received within this time, an appropriate message is output.			
Dependency:	Refer to: F01910			
Note:	0: The monitoring is deactivated.			

p2042	PROFIBUS Ident Number / PB Ident No.			
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T	Calculated: -	Access level: 3	
	Data type: Integer16	Dynamic index: -	Func. diagram: -	
	P-Group: Communications	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min 0	Max 1	Factory setting 0	
Description:	Sets the PROFIBUS Ident Number (PNO-ID). SINAMICS can be operated with various identities on PROFIBUS. This allows the use of a PROFIBUS GSD that is independent of the device (e.g. PROFIdrive VIK-NAMUR with Ident Number 3AA0 hex).			
Value:	0: SINAMICS S/G 1: VIK-NAMUR			
Note:	Every change only becomes effective after a POWER ON.			

r2043.0...2	BO: PROFIdrive PZD state / PD PZD state				
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned8	Dynamic index: -	Func. diagram: 2410		
	P-Group: Communications	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min -	Max -	Factory setting -		
Description:	Displays the PROFIdrive PZD state.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Setpoint failure	Yes	No	-
	01	Clock cycle synchronous operation active	Yes	No	-
	02	Fieldbus running	Yes	No	-
Dependency:	Refer to: p2044				
Note:	When using the "setpoint failure" signal, the bus can be monitored and an application-specific response triggered when the setpoint fails.				

p2044 PROFIdrive fault delay / PD fault delayA_INF, B_INF,
ENCODER, S_INF,
SERVO, TM41, VEC-
TOR**Can be changed:** U, T
Data type: FloatingPoint32
P-Group: Communications
Not for motor type: -**Calculated:** -
Dynamic index: -
Units group: -
Scaling: -**Access level:** 3
Func. diagram: 2410
Unit selection: -
Expert list: 1
Factory setting
0 [s]**Description:** Sets the delay time to initiate fault F01910 after a setpoint failure.
The time until the fault is initiated can be used by the application. This means that it is possible to respond to the failure while the drive is still operational (e.g. emergency retraction).**Dependency:** Refer to: r2043
Refer to: F01910**p2045 CI: PROFIdrive clock-cyc. synchr. master sign-of-life, signal source / PD mast-SoL S_src**CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, SERVO,
TM41, VECTOR
(n/M)**Can be changed:** T
Data type: Unsigned32 / Integer16
P-Group: Communications
Not for motor type: -**Calculated:** -
Dynamic index: -
Units group: -
Scaling: -**Access level:** 3
Func. diagram: 2410
Unit selection: -
Expert list: 1**Min**
-
Max
-
Factory setting
0**Description:** Connector input for the sign-of-life of the clock synchronous PROFIBUS/PROFINET master.
The sign-of-life is expected at bits 12 to 15. Bits 0 to 11 are not evaluated.
The sign-of-life signal is normally received in PZD4 (control word 2) from the PROFIBUS/PROFINET master.**Dependency:** Refer to: p0925, r2065**Notice:** The parameter may be protected as a result of p0922 or p2079 and cannot be changed.**p2047 PROFIBUS additional monitoring time / PB suppl t_monit**CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** U, T
Data type: FloatingPoint32
P-Group: Communications
Not for motor type: -**Calculated:** -
Dynamic index: -
Units group: -
Scaling: -**Access level:** 3
Func. diagram: 2410
Unit selection: -
Expert list: 1
Factory setting
0 [ms]**Min**
0 [ms]
Max
20000 [ms]**Description:** Sets the additional monitoring time to monitor the process data received via PROFIBUS.
The additional monitoring time enables short bus faults to be compensated.
If no process data is received within this time, an appropriate message is output.**Recommend.:** Do not set the additional monitoring time for clock-synchronous operation.**Dependency:** Refer to: F01910**p2048 IF1 PROFIdrive PZD sampling time / IF1 PZD t_sample**CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** C1(3)
Data type: FloatingPoint32
P-Group: Communications
Not for motor type: -**Calculated:** -
Dynamic index: -
Units group: -
Scaling: -**Access level:** 3
Func. diagram: -
Unit selection: -
Expert list: 1
Factory setting
4.00 [ms]**Min**
1.00 [ms]
Max
16.00 [ms]**Description:** Sets the sampling time for the cyclic interface 1 (IF1).**Note:** For clock cycle synchronous operation, the specified bus cycle time applies (Tdp).

r2050[0...4] CO: IF1 PROFIdrive PZD receive word / IF1 PZD recv word

A_INF, B_INF,
CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150, S_INF,
TB30, TM120,
TM15DI_DO, TM31

Can be changed: -**Data type:** Integer16**P-Group:** Communications**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** 4000H**Access level:** 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Min**

-

Max

-

Factory setting

-

Description:

Connector output to interconnect PZD (setpoints) with word format received from the PROFIBUS master.

Index:

[0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5

Note:

IF1: Interface 1

r2050[0...3] CO: IF1 PROFIdrive PZD receive word / IF1 PZD recv word

ENCODER

Can be changed: -**Data type:** Integer16**P-Group:** Communications**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** 4000H**Access level:** 3**Func. diagram:** 2440, 2468**Unit selection:** -**Expert list:** 1**Min**

-

Max

-

Factory setting

-

Description:

Connector output to interconnect PZD (setpoints) with word format received from the PROFIBUS master.

Index:

[0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4

Dependency:

Refer to: r2060

Note:

IF1: Interface 1

r2050[0...19] CO: IF1 PROFIdrive PZD receive word / IF1 PZD recv word

SERVO, TM41

Can be changed: -**Data type:** Integer16**P-Group:** Communications**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** 4000H**Access level:** 3**Func. diagram:** 2440, 2468**Unit selection:** -**Expert list:** 1**Min**

-

Max

-

Factory setting

-

Description:

Connector output to interconnect PZD (setpoints) with word format received from the PROFIBUS master.

Index:

[0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5
[5] = PZD 6
[6] = PZD 7
[7] = PZD 8
[8] = PZD 9
[9] = PZD 10
[10] = PZD 11
[11] = PZD 12
[12] = PZD 13
[13] = PZD 14

[14] = PZD 15
[15] = PZD 16
[16] = PZD 17
[17] = PZD 18
[18] = PZD 19
[19] = PZD 20

Dependency: Refer to: r2060
Note: IF1: Interface 1

r2050[0...31] CO: IF1 PROFIdrive PZD receive word / IF1 PZD recv word

VECTOR	Can be changed: - Data type: Integer16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: 4000H Max -	Access level: 3 Func. diagram: 2440, 2468 Unit selection: - Expert list: 1 Factory setting -
--------	--	--	--

Description: Connector output to interconnect PZD (setpoints) with word format received from the PROFIBUS master.

Index:

[0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5
[5] = PZD 6
[6] = PZD 7
[7] = PZD 8
[8] = PZD 9
[9] = PZD 10
[10] = PZD 11
[11] = PZD 12
[12] = PZD 13
[13] = PZD 14
[14] = PZD 15
[15] = PZD 16
[16] = PZD 17
[17] = PZD 18
[18] = PZD 19
[19] = PZD 20
[20] = PZD 21
[21] = PZD 22
[22] = PZD 23
[23] = PZD 24
[24] = PZD 25
[25] = PZD 26
[26] = PZD 27
[27] = PZD 28
[28] = PZD 29
[29] = PZD 30
[30] = PZD 31
[31] = PZD 32

Dependency: Refer to: r2060
Note: IF1: Interface 1

p2051[0...7]	CI: IF1 PROFIdrive PZD send word / IF1 PZD send word		
A_INF, B_INF, S_INF	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Selects the PZD (actual values) with word format to be sent to the PROFIBUS master.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	IF1: Interface 1		

p2051[0...14]	CI: IF1 PROFIdrive PZD send word / IF1 PZD send word		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Selects the PZD (actual values) with word format to be sent to the PROFIBUS master.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12 [12] = PZD 13 [13] = PZD 14 [14] = PZD 15		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	IF1: Interface 1		

p2051[0...11]	CI: IF1 PROFIdrive PZD send word / IF1 PZD send word		
ENCODER	Can be changed: U, T Data type: Unsigned32 / Integer16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: 4000H Max -	Access level: 3 Func. diagram: 2470 Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the PZD (actual values) with word format to be sent to the PROFIBUS master.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12		
Dependency:	Refer to: p2061		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	IF1: Interface 1		

p2051[0...27]	CI: IF1 PROFIdrive PZD send word / IF1 PZD send word		
SERVO, TM41	Can be changed: U, T Data type: Unsigned32 / Integer16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: 4000H Max -	Access level: 3 Func. diagram: 2470 Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the PZD (actual values) with word format to be sent to the PROFIBUS master.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12 [12] = PZD 13 [13] = PZD 14 [14] = PZD 15 [15] = PZD 16 [16] = PZD 17 [17] = PZD 18 [18] = PZD 19 [19] = PZD 20 [20] = PZD 21 [21] = PZD 22 [22] = PZD 23		

[23] = PZD 24
 [24] = PZD 25
 [25] = PZD 26
 [26] = PZD 27
 [27] = PZD 28

Dependency: Refer to: p2061

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: IF1: Interface 1

p2051[0...4] CI: IF1 PROFIdrive PZD send word / IF1 PZD send word

TB30, TM120,
 TM15DI_DO, TM31

Can be changed: U, T

Calculated: -

Access level: 3

Data type: Unsigned32 / Integer16

Dynamic index: -

Func. diagram: -

P-Group: Communications

Units group: -

Unit selection: -

Not for motor type: -

Scaling: 4000H

Expert list: 1

Min

Max

Factory setting

-

-

0

Description: Selects the PZD (actual values) with word format to be sent to the PROFIBUS master.

Index:
 [0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: IF1: Interface 1

p2051[0...31] CI: IF1 PROFIdrive PZD send word / IF1 PZD send word

VECTOR

Can be changed: U, T

Calculated: -

Access level: 3

Data type: Unsigned32 / Integer16

Dynamic index: -

Func. diagram: 2470

P-Group: Communications

Units group: -

Unit selection: -

Not for motor type: -

Scaling: 4000H

Expert list: 1

Min

Max

Factory setting

-

-

0

Description: Selects the PZD (actual values) with word format to be sent to the PROFIBUS master.

Index:
 [0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5
 [5] = PZD 6
 [6] = PZD 7
 [7] = PZD 8
 [8] = PZD 9
 [9] = PZD 10
 [10] = PZD 11
 [11] = PZD 12
 [12] = PZD 13
 [13] = PZD 14
 [14] = PZD 15
 [15] = PZD 16
 [16] = PZD 17
 [17] = PZD 18
 [18] = PZD 19
 [19] = PZD 20
 [20] = PZD 21
 [21] = PZD 22
 [22] = PZD 23
 [23] = PZD 24
 [24] = PZD 25

[25] = PZD 26
[26] = PZD 27
[27] = PZD 28
[28] = PZD 29
[29] = PZD 30
[30] = PZD 31
[31] = PZD 32

Dependency: Refer to: p2061

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: IF1: Interface 1

r2053[0...7] IF1 PROFIdrive diagnostics PZD send word / IF1 diag send word

A_INF, B_INF, S_INF	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
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Description: Displays the PZD (actual values) with word format sent to the PROFIBUS master.

Index:
[0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5
[5] = PZD 6
[6] = PZD 7
[7] = PZD 8

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Note: IF1: Interface 1

r2053[0...14] IF1 PROFIdrive diagnostics PZD send word / IF1 diag send word					
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned16		Dynamic index: -		Func. diagram: -
	P-Group: Communications		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		-
Description: Displays the PZD (actual values) with word format sent to the PROFIBUS master.					
Index:					
[0] = PZD 1					
[1] = PZD 2					
[2] = PZD 3					
[3] = PZD 4					
[4] = PZD 5					
[5] = PZD 6					
[6] = PZD 7					
[7] = PZD 8					
[8] = PZD 9					
[9] = PZD 10					
[10] = PZD 11					
[11] = PZD 12					
[12] = PZD 13					
[13] = PZD 14					
[14] = PZD 15					
Bit field:					
	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-
Note: IF1: Interface 1					

r2053[0...11] IF1 PROFIdrive diagnostics PZD send word / IF1 diag send word				
ENCODER	Can be changed: - Calculated: - Access level: 3 Data type: Unsigned16 Dynamic index: - Func. diagram: 2450, 2470 P-Group: Communications Units group: - Unit selection: - Not for motor type: - Scaling: - Expert list: 1 Min Max Factory setting - - -			
Description:	Displays the PZD (actual values) with word format sent to the PROFIBUS master.			
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Bit 0	On	Off
	01	Bit 1	On	Off
	02	Bit 2	On	Off
	03	Bit 3	On	Off
	04	Bit 4	On	Off
	05	Bit 5	On	Off
	06	Bit 6	On	Off
	07	Bit 7	On	Off
	08	Bit 8	On	Off
	09	Bit 9	On	Off
	10	Bit 10	On	Off
	11	Bit 11	On	Off
	12	Bit 12	On	Off
	13	Bit 13	On	Off
	14	Bit 14	On	Off
	15	Bit 15	On	Off
Dependency:	Refer to: p2051, p2061			
Note:	IF1: Interface 1			

r2053[0...27] IF1 PROFIdrive diagnostics PZD send word / IF1 diag send word				
SERVO, TM41	Can be changed: - Calculated: - Access level: 3 Data type: Unsigned16 Dynamic index: - Func. diagram: 2450, 2470 P-Group: Communications Units group: - Unit selection: - Not for motor type: - Scaling: - Expert list: 1 Min Max Factory setting - - -			
Description:	Displays the PZD (actual values) with word format sent to the PROFIBUS master.			
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9			

[9] = PZD 10
 [10] = PZD 11
 [11] = PZD 12
 [12] = PZD 13
 [13] = PZD 14
 [14] = PZD 15
 [15] = PZD 16
 [16] = PZD 17
 [17] = PZD 18
 [18] = PZD 19
 [19] = PZD 20
 [20] = PZD 21
 [21] = PZD 22
 [22] = PZD 23
 [23] = PZD 24
 [24] = PZD 25
 [25] = PZD 26
 [26] = PZD 27
 [27] = PZD 28

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-
Dependency:	Refer to: p2051, p2061				
Note:	IF1: Interface 1				

r2053[0...4]	IF1 PROFIdrive diagnostics PZD send word / IF1 diag send word		
TB30, TM120, TM15DI_DO, TM31	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the PZD (actual values) with word format sent to the PROFIBUS master.

Index:
 [0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Note: IF1: Interface 1

r2053[0...31] IF1 PROFIdrive diagnostics PZD send word / IF1 diag send word

VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2450, 2470
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the PZD (actual values) with word format sent to the PROFIBUS master.

Index:

- [0] = PZD 1
- [1] = PZD 2
- [2] = PZD 3
- [3] = PZD 4
- [4] = PZD 5
- [5] = PZD 6
- [6] = PZD 7
- [7] = PZD 8
- [8] = PZD 9
- [9] = PZD 10
- [10] = PZD 11
- [11] = PZD 12
- [12] = PZD 13
- [13] = PZD 14
- [14] = PZD 15
- [15] = PZD 16
- [16] = PZD 17
- [17] = PZD 18
- [18] = PZD 19
- [19] = PZD 20
- [20] = PZD 21
- [21] = PZD 22
- [22] = PZD 23
- [23] = PZD 24
- [24] = PZD 25
- [25] = PZD 26
- [26] = PZD 27
- [27] = PZD 28
- [28] = PZD 29
- [29] = PZD 30
- [30] = PZD 31
- [31] = PZD 32

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Dependency: Refer to: p2051, p2061

Note: IF1: Interface 1

r2054 COMM INT state / C INT state

CU_CX32, CU_I, CU_LINK	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	255	-

Description: Status display for the internal communications interface.

Value:

- 0: No initialization
- 1: Fatal fault
- 2: Initialization
- 3: Send configuration
- 4: Receive configuration
- 5: Non-cyclic communication
- 6: Cyclic communications but no setpoints (stop/no clock cycle)
- 255: Cyclic communication

r2054 PROFIBUS status / PB status

CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 2410
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	4	-

Description: Status display for the PROFIBUS interface.

Value:

- 0: Off
- 1: No connection (search for baud rate)
- 2: Connection OK (baud rate found)
- 3: Cyclic connection with master (data exchange)
- 4: Cyclic data OK

Note:

Re r2054 = 3:

In state 3 (the LED flashes green), a cyclic connection has been established to the PROFIBUS master; however, one of the following prerequisites is missing for cyclic operation:

- No setpoints are being received as the PROFIBUS master is in the STOP condition.

Only for clock-cycle synchronous operation, the following applies:

- The drive is not in synchronism as the global control (GC) has an error.

Re r2054 = 4:

In the status 4 (LED green), the cyclic connection to the PROFIBUS master has been established and setpoints are being received. The clock cycle synchronization is OK, the global control (GC) is error-free.

This state does not provide any statement regarding the quality of the clock cycle synchronous sign-of-life characters on the drive objects.

r2055[0...2]		PROFIBUS diagnostics standard / PB diag standard		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed:	-	Calculated:	-
	Data type:	Unsigned16	Dynamic index:	-
	P-Group:	Communications	Units group:	-
	Not for motor type:	-	Scaling:	-
	Min	-	Max	-
Description:	Diagnostics display for the PROFIBUS interface.			
Index:	[0] = Master bus address [1] = Master input total length bytes [2] = Master output total length bytes			

r2057		PROFIBUS address switch diagnostics / PB addr diagn		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed:	-	Calculated:	-
	Data type:	Unsigned16	Dynamic index:	-
	P-Group:	Communications	Units group:	-
	Not for motor type:	-	Scaling:	-
	Min	-	Max	-
Description:	Displays the setting of the PROFIBUS address switch "DP ADDRESS" on the Control Unit.			
Dependency:	Refer to: p0918			

r2058[0...139]		COMM INT receive configuration data / C INT E_config_dat		
CU_CX32, CU_I, CU_LINK	Can be changed:	-	Calculated:	-
	Data type:	Unsigned16	Dynamic index:	-
	P-Group:	Communications	Units group:	-
	Not for motor type:	-	Scaling:	-
	Min	-	Max	-
Description:	Displays the configuration data received via COMM BOARD.			

r2059[0...7]	COMM INT identification data / C INT ident_dat		
CU_CX32, CU_I, CU_LINK	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the identification data of the COMM BOARD.		
Note:	Index 0: CB data structure version (e.g.: 100 = V1.00). Index 1: CB driver version (e.g.: 100 = V1.00). Index 2: Company, (e.g.: 42 = Siemens). Index 3: Device type Index 4: Firmware version. Index 5: Firmware date (year). Index 6: Firmware date (day/month).		
r2060[0...2]	CO: IF1 PROFIdrive PZD receive double word / IF1 PZD recv DW		
ENCODER	Can be changed: - Data type: Integer32 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: 4000H Max -	Access level: 3 Func. diagram: 2440, 2468 Unit selection: - Expert list: 1 Factory setting -
Description:	Connector output to interconnect PZD (setpoints) with double word format received from the PROFIBUS master.		
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4		
Dependency:	Refer to: r2050		
Note:	IF1: Interface 1		
r2060[0...18]	CO: IF1 PROFIdrive PZD receive double word / IF1 PZD recv DW		
SERVO, TM41	Can be changed: - Data type: Integer32 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: 4000H Max -	Access level: 3 Func. diagram: 2440, 2468 Unit selection: - Expert list: 1 Factory setting -
Description:	Connector output to interconnect PZD (setpoints) with double word format received from the PROFIBUS master.		
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4 [3] = PZD 4 + 5 [4] = PZD 5 + 6 [5] = PZD 6 + 7 [6] = PZD 7 + 8 [7] = PZD 8 + 9 [8] = PZD 9 + 10 [9] = PZD 10 + 11 [10] = PZD 11 + 12 [11] = PZD 12 + 13 [12] = PZD 13 + 14		

[13] = PZD 14 + 15
[14] = PZD 15 + 16
[15] = PZD 16 + 17
[16] = PZD 17 + 18
[17] = PZD 18 + 19
[18] = PZD 19 + 20

Dependency: Refer to: r2050
Notice: A maximum of 4 indices of the "trace" function can be used.
Note: IF1: Interface 1

r2060[0...30] CO: IF1 PROFIdrive PZD receive double word / IF1 PZD recv DW

VECTOR	Can be changed: - Data type: Integer32 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: 4000H Max -	Access level: 3 Func. diagram: 2440, 2468 Unit selection: - Expert list: 1 Factory setting -
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Description: Connector output to interconnect PZD (setpoints) with double word format received from the PROFIBUS master.

Index:

[0] = PZD 1 + 2
[1] = PZD 2 + 3
[2] = PZD 3 + 4
[3] = PZD 4 + 5
[4] = PZD 5 + 6
[5] = PZD 6 + 7
[6] = PZD 7 + 8
[7] = PZD 8 + 9
[8] = PZD 9 + 10
[9] = PZD 10 + 11
[10] = PZD 11 + 12
[11] = PZD 12 + 13
[12] = PZD 13 + 14
[13] = PZD 14 + 15
[14] = PZD 15 + 16
[15] = PZD 16 + 17
[16] = PZD 17 + 18
[17] = PZD 18 + 19
[18] = PZD 19 + 20
[19] = PZD 20 + 21
[20] = PZD 21 + 22
[21] = PZD 22 + 23
[22] = PZD 23 + 24
[23] = PZD 24 + 25
[24] = PZD 25 + 26
[25] = PZD 26 + 27
[26] = PZD 27 + 28
[27] = PZD 28 + 29
[28] = PZD 29 + 30
[29] = PZD 30 + 31
[30] = PZD 31 + 32

Dependency: Refer to: r2050
Notice: A maximum of 4 indices of the "trace" function can be used.
Note: IF1: Interface 1

p2061[0...10]	CI: IF1 PROFIdrive PZD send double word / IF1 PZD send DW		
ENCODER	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 2470
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Selects the PZD (actual values) with double word format to be sent to the PROFIBUS master.		
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4 [3] = PZD 4 + 5 [4] = PZD 5 + 6 [5] = PZD 6 + 7 [6] = PZD 7 + 8 [7] = PZD 8 + 9 [8] = PZD 9 + 10 [9] = PZD 10 + 11 [10] = PZD 11 + 12		
Dependency:	Refer to: p2051		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	IF1: Interface 1		

p2061[0...26]	CI: IF1 PROFIdrive PZD send double word / IF1 PZD send DW		
SERVO, TM41	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 2470
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Selects the PZD (actual values) with double word format to be sent to the PROFIBUS master.		
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4 [3] = PZD 4 + 5 [4] = PZD 5 + 6 [5] = PZD 6 + 7 [6] = PZD 7 + 8 [7] = PZD 8 + 9 [8] = PZD 9 + 10 [9] = PZD 10 + 11 [10] = PZD 11 + 12 [11] = PZD 12 + 13 [12] = PZD 13 + 14 [13] = PZD 14 + 15 [14] = PZD 15 + 16 [15] = PZD 16 + 17 [16] = PZD 17 + 18 [17] = PZD 18 + 19 [18] = PZD 19 + 20 [19] = PZD 20 + 21 [20] = PZD 21 + 22 [21] = PZD 22 + 23 [22] = PZD 23 + 24 [23] = PZD 24 + 25 [24] = PZD 25 + 26 [25] = PZD 26 + 27 [26] = PZD 27 + 28		

Dependency: Refer to: p2051
Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.
Note: IF1: Interface 1

p2061[0...30] CI: IF1 PROFIdrive PZD send double word / IF1 PZD send DW

VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 2470
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Selects the PZD (actual values) with double word format to be sent to the PROFIBUS master.

Index:

- [0] = PZD 1 + 2
- [1] = PZD 2 + 3
- [2] = PZD 3 + 4
- [3] = PZD 4 + 5
- [4] = PZD 5 + 6
- [5] = PZD 6 + 7
- [6] = PZD 7 + 8
- [7] = PZD 8 + 9
- [8] = PZD 9 + 10
- [9] = PZD 10 + 11
- [10] = PZD 11 + 12
- [11] = PZD 12 + 13
- [12] = PZD 13 + 14
- [13] = PZD 14 + 15
- [14] = PZD 15 + 16
- [15] = PZD 16 + 17
- [16] = PZD 17 + 18
- [17] = PZD 18 + 19
- [18] = PZD 19 + 20
- [19] = PZD 20 + 21
- [20] = PZD 21 + 22
- [21] = PZD 22 + 23
- [22] = PZD 23 + 24
- [23] = PZD 24 + 25
- [24] = PZD 25 + 26
- [25] = PZD 26 + 27
- [26] = PZD 27 + 28
- [27] = PZD 28 + 29
- [28] = PZD 29 + 30
- [29] = PZD 30 + 31
- [30] = PZD 31 + 32

Dependency: Refer to: p2051
Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.
Note: IF1: Interface 1

r2063[0...10] IF1 PROFIdrive diagnostics PZD send double word / IF1 diag send DW				
ENCODER	Can be changed: -		Calculated: -	Access level: 3
	Data type: Unsigned32		Dynamic index: -	Func. diagram: 2450, 2470
	P-Group: Communications		Units group: -	Unit selection: -
	Not for motor type: -		Scaling: -	Expert list: 1
	Min	Max	Factory setting	
	-	-	-	
Description:	Displays the PZD (actual values) with double word format sent to the PROFIBUS/PROFINET master.			
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4 [3] = PZD 4 + 5 [4] = PZD 5 + 6 [5] = PZD 6 + 7 [6] = PZD 7 + 8 [7] = PZD 8 + 9 [8] = PZD 9 + 10 [9] = PZD 10 + 11 [10] = PZD 11 + 12			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Bit 0	On	Off
	01	Bit 1	On	Off
	02	Bit 2	On	Off
	03	Bit 3	On	Off
	04	Bit 4	On	Off
	05	Bit 5	On	Off
	06	Bit 6	On	Off
	07	Bit 7	On	Off
	08	Bit 8	On	Off
	09	Bit 9	On	Off
	10	Bit 10	On	Off
	11	Bit 11	On	Off
	12	Bit 12	On	Off
	13	Bit 13	On	Off
	14	Bit 14	On	Off
	15	Bit 15	On	Off
	16	Bit 16	On	Off
	17	Bit 17	On	Off
	18	Bit 18	On	Off
	19	Bit 19	On	Off
	20	Bit 20	On	Off
	21	Bit 21	On	Off
	22	Bit 22	On	Off
	23	Bit 23	On	Off
	24	Bit 24	On	Off
	25	Bit 25	On	Off
	26	Bit 26	On	Off
	27	Bit 27	On	Off
	28	Bit 28	On	Off
	29	Bit 29	On	Off
	30	Bit 30	On	Off
	31	Bit 31	On	Off
Notice:	A maximum of 4 indices of the "trace" function can be used.			
Note:	IF1: Interface 1			

r2063[0...26] IF1 PROFIdrive diagnostics PZD send double word / IF1 diag send DW					
SERVO, TM41	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned32		Dynamic index: -		Func. diagram: 2450, 2470
	P-Group: Communications		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		-
Description:	Displays the PZD (actual values) with double word format sent to the PROFIBUS/PROFINET master.				
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4 [3] = PZD 4 + 5 [4] = PZD 5 + 6 [5] = PZD 6 + 7 [6] = PZD 7 + 8 [7] = PZD 8 + 9 [8] = PZD 9 + 10 [9] = PZD 10 + 11 [10] = PZD 11 + 12 [11] = PZD 12 + 13 [12] = PZD 13 + 14 [13] = PZD 14 + 15 [14] = PZD 15 + 16 [15] = PZD 16 + 17 [16] = PZD 17 + 18 [17] = PZD 18 + 19 [18] = PZD 19 + 20 [19] = PZD 20 + 21 [20] = PZD 21 + 22 [21] = PZD 22 + 23 [22] = PZD 23 + 24 [23] = PZD 24 + 25 [24] = PZD 25 + 26 [25] = PZD 26 + 27 [26] = PZD 27 + 28				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-
	16	Bit 16	On	Off	-
	17	Bit 17	On	Off	-
	18	Bit 18	On	Off	-
	19	Bit 19	On	Off	-
	20	Bit 20	On	Off	-
	21	Bit 21	On	Off	-
	22	Bit 22	On	Off	-
	23	Bit 23	On	Off	-

24	Bit 24	On	Off	-
25	Bit 25	On	Off	-
26	Bit 26	On	Off	-
27	Bit 27	On	Off	-
28	Bit 28	On	Off	-
29	Bit 29	On	Off	-
30	Bit 30	On	Off	-
31	Bit 31	On	Off	-

Notice: A maximum of 4 indices of the "trace" function can be used.

Note: IF1: Interface 1

r2063[0...30] IF1 PROFIdrive diagnostics PZD send double word / IF1 diag send DW

VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2450, 2470
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the PZD (actual values) with double word format sent to the PROFIBUS/PROFINET master.

Index:

- [0] = PZD 1 + 2
- [1] = PZD 2 + 3
- [2] = PZD 3 + 4
- [3] = PZD 4 + 5
- [4] = PZD 5 + 6
- [5] = PZD 6 + 7
- [6] = PZD 7 + 8
- [7] = PZD 8 + 9
- [8] = PZD 9 + 10
- [9] = PZD 10 + 11
- [10] = PZD 11 + 12
- [11] = PZD 12 + 13
- [12] = PZD 13 + 14
- [13] = PZD 14 + 15
- [14] = PZD 15 + 16
- [15] = PZD 16 + 17
- [16] = PZD 17 + 18
- [17] = PZD 18 + 19
- [18] = PZD 19 + 20
- [19] = PZD 20 + 21
- [20] = PZD 21 + 22
- [21] = PZD 22 + 23
- [22] = PZD 23 + 24
- [23] = PZD 24 + 25
- [24] = PZD 25 + 26
- [25] = PZD 26 + 27
- [26] = PZD 27 + 28
- [27] = PZD 28 + 29
- [28] = PZD 29 + 30
- [29] = PZD 30 + 31
- [30] = PZD 31 + 32

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-

09	Bit 9	On	Off	-
10	Bit 10	On	Off	-
11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-
16	Bit 16	On	Off	-
17	Bit 17	On	Off	-
18	Bit 18	On	Off	-
19	Bit 19	On	Off	-
20	Bit 20	On	Off	-
21	Bit 21	On	Off	-
22	Bit 22	On	Off	-
23	Bit 23	On	Off	-
24	Bit 24	On	Off	-
25	Bit 25	On	Off	-
26	Bit 26	On	Off	-
27	Bit 27	On	Off	-
28	Bit 28	On	Off	-
29	Bit 29	On	Off	-
30	Bit 30	On	Off	-
31	Bit 31	On	Off	-

Notice: A maximum of 4 indices of the "trace" function can be used.

Note: IF1: Interface 1

r2064[0...7] PROFIdrive diagnostics clock synchronous mode / PD diag clock sync

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: 2410
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the last parameter received from the fieldbus master for clock synchronism.
The parameters for clock synchronism are created when configuring the bus and are transferred at the start of cyclic operation from the master to the slave.

Index: [0] = Clock synchronous mode activated
[1] = Bus cycle time (Tdp) [μs]
[2] = Master cycle time (Tmapc) [μs]
[3] = Instant of actual value acquisition (Ti) [μs]
[4] = Instant of setpoint acquisition (To) [μs]
[5] = Data exchange interval (Tdx) [μs]
[6] = PLL window (Tpll-w) [1/12 μs]
[7] = PLL delay time (Tpll-d) [1/12 μs]

r2065 PROFIdrive master sign-of-life, diagnostics / PD mast-SoL diag

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, SERVO, TM41, VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2410
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays how often the sign-of-life from the clock synchronous fieldbus master failed.
An appropriate fault is output when the tolerance, specified in p0925, is exceeded.

Dependency: Refer to: F01912

r2074[0...4]	IF1 PROFIdrive diagnostics bus address PZD receive / IF1diag addr recv		
A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, S_INF, TB30, TM120, TM15DI_DO, TM31	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the PROFIBUS address of the sender from which the process data (PZD) is received.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5		
Note:	IF1: Interface 1 Value range: 0 - 125: Bus address of the sender 65535: Not assigned		

r2074[0...3]	IF1 PROFIdrive diagnostics bus address PZD receive / IF1diag addr recv		
ENCODER	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the PROFIBUS address of the sender from which the process data (PZD) is received.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4		
Note:	IF1: Interface 1 Value range: 0 - 125: Bus address of the sender 65535: Not assigned		

r2074[0...19]	IF1 PROFIdrive diagnostics bus address PZD receive / IF1diag addr recv		
SERVO, TM41	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the PROFIBUS address of the sender from which the process data (PZD) is received.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8		

[8] = PZD 9
[9] = PZD 10
[10] = PZD 11
[11] = PZD 12
[12] = PZD 13
[13] = PZD 14
[14] = PZD 15
[15] = PZD 16
[16] = PZD 17
[17] = PZD 18
[18] = PZD 19
[19] = PZD 20

Note: IF1: Interface 1
Value range:
0 - 125: Bus address of the sender
65535: Not assigned

r2074[0...31] IF1 PROFIdrive diagnostics bus address PZD receive / IF1diag addr recv

VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
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Description: Displays the PROFIBUS address of the sender from which the process data (PZD) is received.

Index:

[0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5
[5] = PZD 6
[6] = PZD 7
[7] = PZD 8
[8] = PZD 9
[9] = PZD 10
[10] = PZD 11
[11] = PZD 12
[12] = PZD 13
[13] = PZD 14
[14] = PZD 15
[15] = PZD 16
[16] = PZD 17
[17] = PZD 18
[18] = PZD 19
[19] = PZD 20
[20] = PZD 21
[21] = PZD 22
[22] = PZD 23
[23] = PZD 24
[24] = PZD 25
[25] = PZD 26
[26] = PZD 27
[27] = PZD 28
[28] = PZD 29
[29] = PZD 30
[30] = PZD 31
[31] = PZD 32

Note: IF1: Interface 1
Value range:
0 - 125: Bus address of the sender
65535: Not assigned

r2075[0...4] IF1 PROFIdrive diagnostics telegram offset PZD receive / IF1 diag offs recv

A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, S_INF, TB30, TM120, TM15DI_DO, TM31	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
--	--	---	--

Min	Max	Factory setting
-	-	-

Description: Displays the PZD byte offset in the PROFIdrive receive telegram (master output).

Index: [0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5

Note: IF1: Interface 1
Value range:
0 - 242: Byte offset
65535: Not assigned

r2075[0...3] IF1 PROFIdrive diagnostics telegram offset PZD receive / IF1 diag offs recv

ENCODER	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the PZD byte offset in the PROFIdrive receive telegram (master output).

Index: [0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4

Note: IF1: Interface 1
Value range:
0 - 242: Byte offset
65535: Not assigned

r2075[0...19] IF1 PROFIdrive diagnostics telegram offset PZD receive / IF1 diag offs recv

SERVO, TM41

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** -**P-Group:** Communications**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the PZD byte offset in the PROFIdrive receive telegram (master output).

Index:

[0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5
 [5] = PZD 6
 [6] = PZD 7
 [7] = PZD 8
 [8] = PZD 9
 [9] = PZD 10
 [10] = PZD 11
 [11] = PZD 12
 [12] = PZD 13
 [13] = PZD 14
 [14] = PZD 15
 [15] = PZD 16
 [16] = PZD 17
 [17] = PZD 18
 [18] = PZD 19
 [19] = PZD 20

Note:

IF1: Interface 1

Value range:

0 - 242: Byte offset

65535: Not assigned

r2075[0...31] IF1 PROFIdrive diagnostics telegram offset PZD receive / IF1 diag offs recv

VECTOR

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** -**P-Group:** Communications**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the PZD byte offset in the PROFIdrive receive telegram (master output).

Index:

[0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5
 [5] = PZD 6
 [6] = PZD 7
 [7] = PZD 8
 [8] = PZD 9
 [9] = PZD 10
 [10] = PZD 11
 [11] = PZD 12
 [12] = PZD 13
 [13] = PZD 14
 [14] = PZD 15
 [15] = PZD 16
 [16] = PZD 17

[17] = PZD 18
 [18] = PZD 19
 [19] = PZD 20
 [20] = PZD 21
 [21] = PZD 22
 [22] = PZD 23
 [23] = PZD 24
 [24] = PZD 25
 [25] = PZD 26
 [26] = PZD 27
 [27] = PZD 28
 [28] = PZD 29
 [29] = PZD 30
 [30] = PZD 31
 [31] = PZD 32

Note: IF1: Interface 1
 Value range:
 0 - 242: Byte offset
 65535: Not assigned

r2076[0...7] IF1 PROFdrive diagnostics telegram offset PZD send / IF1 diag offs send

A_INF, B_INF,
 S_INF

Can be changed: -

Calculated: -

Access level: 3

Data type: Unsigned16

Dynamic index: -

Func. diagram: -

P-Group: Communications

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

-

-

-

Description:

Displays the PZD byte offset in the PROFdrive send telegram (master output).

Index:

[0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5
 [5] = PZD 6
 [6] = PZD 7
 [7] = PZD 8

Note:

IF1: Interface 1
 Value range:
 0 - 242: Byte offset
 65535: Not assigned

r2076[0...14] IF1 PROFIdrive diagnostics telegram offset PZD send / IF1 diag offs send

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the PZD byte offset in the PROFIdrive send telegram (master output).

Index:
 [0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5
 [5] = PZD 6
 [6] = PZD 7
 [7] = PZD 8
 [8] = PZD 9
 [9] = PZD 10
 [10] = PZD 11
 [11] = PZD 12
 [12] = PZD 13
 [13] = PZD 14
 [14] = PZD 15

Note:
 IF1: Interface 1
 Value range:
 0 - 242: Byte offset
 65535: Not assigned

r2076[0...11] IF1 PROFIdrive diagnostics telegram offset PZD send / IF1 diag offs send

ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the PZD byte offset in the PROFIdrive send telegram (master output).

Index:
 [0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5
 [5] = PZD 6
 [6] = PZD 7
 [7] = PZD 8
 [8] = PZD 9
 [9] = PZD 10
 [10] = PZD 11
 [11] = PZD 12

Note:
 IF1: Interface 1
 Value range:
 0 - 242: Byte offset
 65535: Not assigned

r2076[0...27] IF1 PROFIdrive diagnostics telegram offset PZD send / IF1 diag offs send

SERVO, TM41	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the PZD byte offset in the PROFIdrive send telegram (master output).

Index:

- [0] = PZD 1
- [1] = PZD 2
- [2] = PZD 3
- [3] = PZD 4
- [4] = PZD 5
- [5] = PZD 6
- [6] = PZD 7
- [7] = PZD 8
- [8] = PZD 9
- [9] = PZD 10
- [10] = PZD 11
- [11] = PZD 12
- [12] = PZD 13
- [13] = PZD 14
- [14] = PZD 15
- [15] = PZD 16
- [16] = PZD 17
- [17] = PZD 18
- [18] = PZD 19
- [19] = PZD 20
- [20] = PZD 21
- [21] = PZD 22
- [22] = PZD 23
- [23] = PZD 24
- [24] = PZD 25
- [25] = PZD 26
- [26] = PZD 27
- [27] = PZD 28

Note: IF1: Interface 1
Value range:
0 - 242: Byte offset
65535: Not assigned

r2076[0...4] IF1 PROFIdrive diagnostics telegram offset PZD send / IF1 diag offs send

TB30, TM120,	Can be changed: -	Calculated: -	Access level: 3
TM15DI_DO, TM31	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the PZD byte offset in the PROFIdrive send telegram (master output).

Index:

- [0] = PZD 1
- [1] = PZD 2
- [2] = PZD 3
- [3] = PZD 4
- [4] = PZD 5

Note: IF1: Interface 1
Value range:
0 - 242: Byte offset
65535: Not assigned

r2076[0...31] IF1 PROFIdrive diagnostics telegram offset PZD send / IF1 diag offs send

VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the PZD byte offset in the PROFIdrive send telegram (master output).

Index:

- [0] = PZD 1
- [1] = PZD 2
- [2] = PZD 3
- [3] = PZD 4
- [4] = PZD 5
- [5] = PZD 6
- [6] = PZD 7
- [7] = PZD 8
- [8] = PZD 9
- [9] = PZD 10
- [10] = PZD 11
- [11] = PZD 12
- [12] = PZD 13
- [13] = PZD 14
- [14] = PZD 15
- [15] = PZD 16
- [16] = PZD 17
- [17] = PZD 18
- [18] = PZD 19
- [19] = PZD 20
- [20] = PZD 21
- [21] = PZD 22
- [22] = PZD 23
- [23] = PZD 24
- [24] = PZD 25
- [25] = PZD 26
- [26] = PZD 27
- [27] = PZD 28
- [28] = PZD 29
- [29] = PZD 30
- [30] = PZD 31
- [31] = PZD 32

Note: IF1: Interface 1
Value range:
0 - 242: Byte offset
65535: Not assigned

r2077[0...15]	PROFIBUS diagnostics peer-to-peer data transfer addresses / PB diag peer addr		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the addresses of the slaves (peers) where peer-to-peer data transfer has been configured via PROFIBUS.		
p2079	PROFIdrive PZD telegram selection extended / PD PZD telegr ext		
A_INF, B_INF, S_INF	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 370	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram. Contrary to p0922, a telegram can be selected using p2079 and subsequently expanded. For p0922 < 999 the following applies: p2079 has the same value and is inhibited. All of the interconnections and extensions contained in the telegram are inhibited. For p0922 = 999 the following applies: p2079 can be freely set. If p2079 is also set to 999, then all of the interconnections can be set. For p0922 = 999 and p2079 < 999 the following applies: The interconnections contained in the telegram are inhibited. However, the telegram can be extended.		
Value:	370: SIEMENS telegram 370, PZD-1/1 371: SIEMENS telegram 371, PZD-5/8 999: Free telegram configuration with BICO		
Dependency:	Refer to: p0922		
p2079	PROFIdrive PZD telegram selection extended / PD PZD telegr ext		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 390	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram. Contrary to p0922, a telegram can be selected using p2079 and subsequently expanded. For p0922 < 999 the following applies: p2079 has the same value and is inhibited. All of the interconnections and extensions contained in the telegram are inhibited. For p0922 = 999 the following applies: p2079 can be freely set. If p2079 is also set to 999, then all of the interconnections can be set. For p0922 = 999 and p2079 < 999 the following applies: The interconnections contained in the telegram are inhibited. However, the telegram can be extended.		
Value:	390: SIEMENS telegram 390, PZD-2/2 391: SIEMENS telegram 391, PZD-3/7 392: SIEMENS telegram 392, PZD-3/15 999: Free telegram configuration with BICO		

p2079	PROFIdrive PZD telegram selection extended / PD PZD telegr ext		
ENCODER	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 81	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram. Contrary to p0922, a telegram can be selected using p2079 and subsequently expanded. For p0922 < 999 the following applies: p2079 has the same value and is inhibited. All of the interconnections and extensions contained in the telegram are inhibited. For p0922 = 999 the following applies: p2079 can be freely set. If p2079 is also set to 999, then all of the interconnections can be set. For p0922 = 999 and p2079 < 999 the following applies: The interconnections contained in the telegram are inhibited. However, the telegram can be extended.		
Value:	81: SIEMENS telegram 81, PZD-2/6 82: SIEMENS telegram 82, PZD-2/7 83: SIEMENS telegram 83, PZD-2/8 999: Free telegram configuration with BICO		
Dependency:	Refer to: p0922		
p2079	PROFIdrive PZD telegram selection extended / PD PZD telegr ext		
SERVO	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram. Contrary to p0922, a telegram can be selected using p2079 and subsequently expanded. For p0922 < 999 the following applies: p2079 has the same value and is inhibited. All of the interconnections and extensions contained in the telegram are inhibited. For p0922 = 999 the following applies: p2079 can be freely set. If p2079 is also set to 999, then all of the interconnections can be set. For p0922 = 999 and p2079 < 999 the following applies: The interconnections contained in the telegram are inhibited. However, the telegram can be extended.		
Value:	1: Standard telegram 1, PZD-2/2 2: Standard telegram 2, PZD-4/4 3: Standard telegram 3, PZD-5/9 4: Standard telegram 4, PZD-6/14 5: Standard telegram 5, PZD-9/9 6: Standard telegram 6, PZD-10/14 102: SIEMENS telegram 102, PZD-6/10 103: SIEMENS telegram 103, PZD-7/15 105: SIEMENS telegram 105, PZD-10/10 106: SIEMENS telegram 106, PZD-11/15 116: SIEMENS telegram 116, PZD-11/19 118: SIEMENS telegram 118, PZD-11/19 125: SIEMENS telegram 125, PZD-14/10 126: SIEMENS telegram 126, PZD-15/15 136: SIEMENS telegram 136, PZD-15/19 220: SIEMENS telegram 220, PZD-10/10 999: Free telegram configuration with BICO		

Dependency: Refer to: p0922

p2079		PROFIdrive PZD telegram selection extended / PD PZD telegr ext	
SERVO (EPOS, Pos ctrl), VECTOR (EPOS, n/M, Pos ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 7	Max 999	Factory setting 999
Description:	Sets the send and receive telegram. Contrary to p0922, a telegram can be selected using p2079 and subsequently expanded. For p0922 < 999 the following applies: p2079 has the same value and is inhibited. All of the interconnections and extensions contained in the telegram are inhibited. For p0922 = 999 the following applies: p2079 can be freely set. If p2079 is also set to 999, then all of the interconnections can be set. For p0922 = 999 and p2079 < 999 the following applies: The interconnections contained in the telegram are inhibited. However, the telegram can be extended.		
Value:	7: Standard telegram 7, PZD-2/2 9: Standard telegram 9, PZD-10/5 110: SIEMENS telegram 110, PZD-12/7 111: SIEMENS telegram 111, PZD-12/12 999: Free telegram configuration with BICO		
Dependency:	Refer to: p0922		

p2079		PROFIdrive PZD telegram selection extended / PD PZD telegr ext	
SERVO (Pos ctrl), VECTOR (n/M, Pos ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	999	999	999
Description:	Sets the send and receive telegram.		
	Contrary to p0922, a telegram can be selected using p2079 and subsequently expanded.		
	For p0922 < 999 the following applies:		
	p2079 has the same value and is inhibited. All of the interconnections and extensions contained in the telegram are inhibited.		
	For p0922 = 999 the following applies:		
	p2079 can be freely set. If p2079 is also set to 999, then all of the interconnections can be set.		
Value:	For p0922 = 999 and p2079 < 999 the following applies:		
	The interconnections contained in the telegram are inhibited. However, the telegram can be extended.		
	999: Free telegram configuration with BICO		
	Dependency: Refer to: p0922		

p2079	PROFIdrive PZD telegram selection extended / PD PZD telegr ext		
TM41	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 3	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram. Contrary to p0922, a telegram can be selected using p2079 and subsequently expanded. For p0922 < 999 the following applies: p2079 has the same value and is inhibited. All of the interconnections and extensions contained in the telegram are inhibited. For p0922 = 999 the following applies: p2079 can be freely set. If p2079 is also set to 999, then all of the interconnections can be set. For p0922 = 999 and p2079 < 999 the following applies: The interconnections contained in the telegram are inhibited. However, the telegram can be extended.		
Value:	3: Standard telegram 3, PZD-5/9 999: Free telegram configuration with BICO		
Dependency:	Refer to: p0922		
p2079	PROFIdrive PZD telegram selection extended / PD PZD telegr ext		
VECTOR	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram. Contrary to p0922, a telegram can be selected using p2079 and subsequently expanded. For p0922 < 999 the following applies: p2079 has the same value and is inhibited. All of the interconnections and extensions contained in the telegram are inhibited. For p0922 = 999 the following applies: p2079 can be freely set. If p2079 is also set to 999, then all of the interconnections can be set. For p0922 = 999 and p2079 < 999 the following applies: The interconnections contained in the telegram are inhibited. However, the telegram can be extended.		
Value:	1: Standard telegram 1, PZD-2/2 2: Standard telegram 2, PZD-4/4 20: Standard telegram 20, PZD-2/6 220: SIEMENS telegram 220, PZD-10/10 352: SIEMENS telegram 352, PZD-6/6 999: Free telegram configuration with BICO		
Dependency:	Refer to: p0922		

p2079	PROFIdrive PZD telegram selection extended / PD PZD telegr ext		
VECTOR (n/M)	Can be changed: T Data type: Integer16 P-Group: Communications Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 999	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 999
Description:	Sets the send and receive telegram. Contrary to p0922, a telegram can be selected using p2079 and subsequently expanded. For p0922 < 999 the following applies: p2079 has the same value and is inhibited. All of the interconnections and extensions contained in the telegram are inhibited. For p0922 = 999 the following applies: p2079 can be freely set. If p2079 is also set to 999, then all of the interconnections can be set. For p0922 = 999 and p2079 < 999 the following applies: The interconnections contained in the telegram are inhibited. However, the telegram can be extended.		
Value:	1: Standard telegram 1, PZD-2/2 2: Standard telegram 2, PZD-4/4 3: Standard telegram 3, PZD-5/9 4: Standard telegram 4, PZD-6/14 20: Standard telegram 20, PZD-2/6 220: SIEMENS telegram 220, PZD-10/10 352: SIEMENS telegram 352, PZD-6/6 999: Free telegram configuration with BICO		
Dependency:	Refer to: p0922		
p2080[0...15]	BI: Binector-connector converter status word 1 / Bin/con ZSW1		
A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, S_INF, SERVO, TB30, TM120, TM15DI_DO, TM31, TM41, VEC- TOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: 2472 Unit selection: - Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Selects bits to be sent to the fieldbus master. The individual bits are combined to form status word 1.		

Index:

- [0] = Bit 0
- [1] = Bit 1
- [2] = Bit 2
- [3] = Bit 3
- [4] = Bit 4
- [5] = Bit 5
- [6] = Bit 6
- [7] = Bit 7
- [8] = Bit 8
- [9] = Bit 9
- [10] = Bit 10
- [11] = Bit 11
- [12] = Bit 12
- [13] = Bit 13
- [14] = Bit 14
- [15] = Bit 15

Dependency: Refer to: p2088, r2089

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p2081[0...15] BI: Binector-connector converter status word 2 / Bin/con ZSW2

A_INF, B_INF,
CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, S_INF,
SERVO, TB30,
TM120, TM15DI_DO,
TM31, TM41, VEC-
TOR

Can be changed: U, T
Data type: Unsigned32 / Binary
P-Group: Communications
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: 2472
Unit selection: -
Expert list: 1

Min	Max	Factory setting
-	-	0

Description: Selects bits to be sent to the fieldbus master.
The individual bits are combined to form status word 2.

Index:

- [0] = Bit 0
- [1] = Bit 1
- [2] = Bit 2
- [3] = Bit 3
- [4] = Bit 4
- [5] = Bit 5
- [6] = Bit 6
- [7] = Bit 7
- [8] = Bit 8
- [9] = Bit 9
- [10] = Bit 10
- [11] = Bit 11
- [12] = Bit 12
- [13] = Bit 13
- [14] = Bit 14
- [15] = Bit 15

Dependency: Refer to: p2088, r2089

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: For clock synchronous operation, bit 12 to 15 to transfer the sign-of-life are reserved in status word 2 - and may not be freely interconnected.

p2082[0...15] BI: Binector-connector converter status word 3 / Bin/con ZSW3			
A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, S_INF, SERVO, TB30, TM120, TM15DI_DO, TM31, TM41, VEC- TOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: 2472 Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Selects bits to be sent to the fieldbus master. The individual bits are combined to form free status word 3.		
Index:	[0] = Bit 0 [1] = Bit 1 [2] = Bit 2 [3] = Bit 3 [4] = Bit 4 [5] = Bit 5 [6] = Bit 6 [7] = Bit 7 [8] = Bit 8 [9] = Bit 9 [10] = Bit 10 [11] = Bit 11 [12] = Bit 12 [13] = Bit 13 [14] = Bit 14 [15] = Bit 15		
Dependency:	Refer to: p2088, r2089		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		

p2083[0...15] BI: Binector-connector converter status word 4 / Bin/con ZSW4			
A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, S_INF, SERVO, TB30, TM120, TM15DI_DO, TM31, TM41, VEC- TOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: 2472 Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Selects bits to be sent to the fieldbus master. The individual bits are combined to form free status word 4.		

Index:

- [0] = Bit 0
- [1] = Bit 1
- [2] = Bit 2
- [3] = Bit 3
- [4] = Bit 4
- [5] = Bit 5
- [6] = Bit 6
- [7] = Bit 7
- [8] = Bit 8
- [9] = Bit 9
- [10] = Bit 10
- [11] = Bit 11
- [12] = Bit 12
- [13] = Bit 13
- [14] = Bit 14
- [15] = Bit 15

Dependency: Refer to: p2088, r2089

p2084[0...15] BI: Binector-connector converter status word 5 / Bin/con ZSW5			
A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, S_INF, SERVO, TB30, TM120, TM15DI_DO, TM31, TM41, VEC- TOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2472
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Selects bits to be sent to the fieldbus master.
The individual bits are combined to form free status word 5.

Index:

- [0] = Bit 0
- [1] = Bit 1
- [2] = Bit 2
- [3] = Bit 3
- [4] = Bit 4
- [5] = Bit 5
- [6] = Bit 6
- [7] = Bit 7
- [8] = Bit 8
- [9] = Bit 9
- [10] = Bit 10
- [11] = Bit 11
- [12] = Bit 12
- [13] = Bit 13
- [14] = Bit 14
- [15] = Bit 15

Dependency: Refer to: p2088, r2089

p2088[0...4] Invert binector-connector converter status word / Bin/con ZSW inv					
A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, S_INF, SERVO, TB30, TM120, TM15DI_DO, TM31, TM41, VEC- TOR	Can be changed: U, T		Calculated: -		Access level: 3
	Data type: Unsigned16		Dynamic index: -		Func. diagram: 2472
	P-Group: Communications		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		0000 bin
Description:	Setting to invert the individual binector inputs of the binector connector converter.				
Index:	[0] = Status word 1 [1] = Status word 2 [2] = Free status word 3 [3] = Free status word 4 [4] = Free status word 5				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	Inverted	Not inverted	-
	01	Bit 1	Inverted	Not inverted	-
	02	Bit 2	Inverted	Not inverted	-
	03	Bit 3	Inverted	Not inverted	-
	04	Bit 4	Inverted	Not inverted	-
	05	Bit 5	Inverted	Not inverted	-
	06	Bit 6	Inverted	Not inverted	-
	07	Bit 7	Inverted	Not inverted	-
	08	Bit 8	Inverted	Not inverted	-
	09	Bit 9	Inverted	Not inverted	-
	10	Bit 10	Inverted	Not inverted	-
	11	Bit 11	Inverted	Not inverted	-
	12	Bit 12	Inverted	Not inverted	-
	13	Bit 13	Inverted	Not inverted	-
	14	Bit 14	Inverted	Not inverted	-
	15	Bit 15	Inverted	Not inverted	-
Dependency:	Refer to: p2080, p2081, p2082, p2083, r2089				

r2089[0...4] CO: Send binector-connector converter status word / Bin/con ZSW send

A_INF, B_INF,
CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, S_INF,
SERVO, TB30,
TM120, TM15DI_DO,
TM31, TM41, VEC-
TOR

Can be changed: -**Data type:** Unsigned16**P-Group:** Communications**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 3**Func. diagram:** 2472**Unit selection:** -**Expert list:** 1**Min**

-

Max

-

Factory setting

-

Description: Connector output to interconnect the status words to a PZD send word.

Index:
[0] = Status word 1
[1] = Status word 2
[2] = Free status word 3
[3] = Free status word 4
[4] = Free status word 5

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Dependency: Refer to: p2051, p2080, p2081, p2082, p2083**Note:** r2089 together with p2080 to p2084 forms five binector-connector converters.**r2090.0...15 BO: IF1 PROFIBUS PZD1 receive bit-serial / IF1 PZD1 recv bitw**

A_INF, B_INF,
CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, S_INF,
SERVO, TB30,
TM120, TM15DI_DO,
TM31, TM41, VEC-
TOR

Can be changed: -**Data type:** Unsigned16**P-Group:** Communications**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 3**Func. diagram:** 2468**Unit selection:** -**Expert list:** 1**Min**

-

Max

-

Factory setting

-

Description: Binector output for bit-serial interconnection of PZD1 (normally control word 1) received from the PROFIBUS master.

List of parameters

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Note: IF1: Interface 1

r2091.0...15 BO: IF1 PROFIdrive PZD2 receive bit-serial / IF1 PZD2 recv bitw

A_INF, B_INF,	Can be changed: -	Calculated: -	Access level: 3
CU_CX32, CU_I,	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2468
CU_S,	P-Group: Communications	Units group: -	Unit selection: -
CU_S_CU310DP,	Not for motor type: -	Scaling: -	Expert list: 1
CU_S_CU310PN,			
CU_S_S150,			
ENCODER, S_INF,			
SERVO, TB30,			
TM120, TM15DI_DO,			
TM31, TM41, VEC-			
TOR			

Min	Max	Factory setting
-	-	-

Description: Binector output for bit-serial interconnection of PZD2 received from the PROFIBUS master.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Note: IF1: Interface 1

r2092.0...15 BO: IF1 PROFIdrive PZD3 receive bit-serial / IF1 PZD3 recv bitwENCODER, SERVO,
TM41, VECTOR**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 2468**P-Group:** Communications**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Binector output for bit-serial interconnection of PZD3 received from the PROFIBUS master.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Bit 0	On	Off	-
01	Bit 1	On	Off	-
02	Bit 2	On	Off	-
03	Bit 3	On	Off	-
04	Bit 4	On	Off	-
05	Bit 5	On	Off	-
06	Bit 6	On	Off	-
07	Bit 7	On	Off	-
08	Bit 8	On	Off	-
09	Bit 9	On	Off	-
10	Bit 10	On	Off	-
11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-

Note:

IF1: Interface 1

r2093.0...15 BO: IF1 PROFIdrive PZD4 receive bit-serial / IF1 PZD4 recv bitwENCODER, SERVO,
TM41, VECTOR**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 2468**P-Group:** Communications**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Binector output for bit-serial interconnection of PZD4 (normally control word 2) received from the PROFIBUS master.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Bit 0	On	Off	-
01	Bit 1	On	Off	-
02	Bit 2	On	Off	-
03	Bit 3	On	Off	-
04	Bit 4	On	Off	-
05	Bit 5	On	Off	-
06	Bit 6	On	Off	-
07	Bit 7	On	Off	-
08	Bit 8	On	Off	-
09	Bit 9	On	Off	-
10	Bit 10	On	Off	-
11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-

Note:

IF1: Interface 1

r2094.0...15**BO: Connector-binector converter binector output / Con/bin outp**

A_INF, B_INF,
CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, S_INF,
SERVO, TB30,
TM120, TM15DI_DO,
TM31, TM41, VEC-
TOR

Can be changed: -**Data type:** Unsigned16**P-Group:** Communications**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 3**Func. diagram:** 2468**Unit selection:** -**Expert list:** 1**Min**

-

Max

-

Factory setting

-

Description:

Binector output for bit-serial onward interconnection of a PZD word received from the fieldbus master.
The PZD is selected via p2099[0].

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Bit 0	On	Off	-
01	Bit 1	On	Off	-
02	Bit 2	On	Off	-
03	Bit 3	On	Off	-
04	Bit 4	On	Off	-
05	Bit 5	On	Off	-
06	Bit 6	On	Off	-
07	Bit 7	On	Off	-
08	Bit 8	On	Off	-
09	Bit 9	On	Off	-
10	Bit 10	On	Off	-
11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-

Dependency:

Refer to: p2099

r2095.0...15**BO: Connector-binector converter binector output / Con/bin outp**

A_INF, B_INF,
CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, S_INF,
SERVO, TB30,
TM120, TM15DI_DO,
TM31, TM41, VEC-
TOR

Can be changed: -**Data type:** Unsigned16**P-Group:** Communications**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 3**Func. diagram:** 2468**Unit selection:** -**Expert list:** 1**Min**

-

Max

-

Factory setting

-

Description:

Binector output for bit-serial interconnection of a PZD word received from the fieldbus master.
The PZD is selected via p2099[1].

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Dependency: Refer to: p2099

p2098[0...1] Inverter connector-binector converter binector output / Con/bin outp inv

A_INF, B_INF,
CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, S_INF,
SERVO, TB30,
TM120, TM15DI_DO,
TM31, TM41, VEC-
TOR

Can be changed: U, T
Data type: Unsigned16
P-Group: Communications
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: 2468
Unit selection: -
Expert list: 1

Min	Max	Factory setting
-	-	0000 bin

Description: Setting to invert the individual binector outputs of the connector-binector converter.
Using p2098[0], the signals of CI: p2099[0] are influenced.
Using p2098[1], the signals of CI: p2099[1] are influenced.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	Inverted	Not inverted	-
	01	Bit 1	Inverted	Not inverted	-
	02	Bit 2	Inverted	Not inverted	-
	03	Bit 3	Inverted	Not inverted	-
	04	Bit 4	Inverted	Not inverted	-
	05	Bit 5	Inverted	Not inverted	-
	06	Bit 6	Inverted	Not inverted	-
	07	Bit 7	Inverted	Not inverted	-
	08	Bit 8	Inverted	Not inverted	-
	09	Bit 9	Inverted	Not inverted	-
	10	Bit 10	Inverted	Not inverted	-
	11	Bit 11	Inverted	Not inverted	-
	12	Bit 12	Inverted	Not inverted	-
	13	Bit 13	Inverted	Not inverted	-
	14	Bit 14	Inverted	Not inverted	-
	15	Bit 15	Inverted	Not inverted	-

Dependency: Refer to: r2094, r2095, p2099

p2099[0...1]	CI: Connector-binector converter signal source / Con/bin S_src		
A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, S_INF, SERVO, TB30, TM120, TM15DI_DO, TM31, TM41, VEC- TOR	Can be changed: U, T Data type: Unsigned32 / Integer16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: 2468 Unit selection: - Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source for the connector-binector converter. A PZD receive word can be selected as signal source. The signals are available to be serially passed-on (interconnection).		
Dependency:	Refer to: r2094, r2095		
Note:	From the signal source set via the connector input, the corresponding lower 16 bits are converted. p2099[0...1] together with r2094.0...15 and r2095.0...15 forms two connector-binector converters: Connector input p2099[0] to binector output in r2094.0...15 Connector input p2099[1] to binector output in r2095.0...15		
p2100[0...19]	Setting the fault number for fault response / F_no F response		
All objects	Can be changed: U, T Data type: Unsigned16 P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: 1750, 8075 Unit selection: - Expert list: 1 Factory setting
	Min 0	Max 65535	[0] 0 [1] 0 [2] 0 [3] 0 [4] 0 [5] 0 [6] 0 [7] 0 [8] 0 [9] 0 [10] 0 [11] 0 [12] 0 [13] 0 [14] 0 [15] 0 [16] 0 [17] 0 [18] 0 [19] 0
Description:	Selects the faults for which the fault response should be changed		
Dependency:	The fault is selected and the required response is set under the same index. Refer to: p2101		

Notice: For the following cases, it is not possible to re-parameterize the fault response to a fault:

- if there is no existing fault number.
- the message type is not "fault" (F).
- when a fault is present.

p2101[0...19] Setting the fault response / Fault response			
A_INF, B_INF, S_INF	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 1750, 8075
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	2	[0] 0
			[1] 0
			[2] 0
			[3] 0
			[4] 0
			[5] 0
			[6] 0
			[7] 0
			[8] 0
			[9] 0
			[10] 0
			[11] 0
			[12] 0
			[13] 0
			[14] 0
			[15] 0
			[16] 0
			[17] 0
			[18] 0
			[19] 0
Description:	Sets the fault response for the selected fault.		
Value:	0: NONE 1: OFF1 2: OFF2		
Dependency:	The fault is selected and the required response is set under the same index.		

p2101[0...19] Setting the fault response / Fault response			
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 1750, 8075
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	0	[0] 0
			[1] 0
			[2] 0
			[3] 0
			[4] 0
			[5] 0
			[6] 0
			[7] 0
			[8] 0
			[9] 0
			[10] 0
			[11] 0
			[12] 0
			[13] 0
			[14] 0
			[15] 0
			[16] 0
			[17] 0
			[18] 0
			[19] 0
Description:	Sets the fault response for the selected fault.		
Value:	0: NONE		
Dependency:	The fault is selected and the required response is set under the same index.		

p2101[0...19] Setting the fault response / Fault response

SERVO, TM41, VEC-TOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 1750, 8075
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	7	[0] 0
			[1] 0
			[2] 0
			[3] 0
			[4] 0
			[5] 0
			[6] 0
			[7] 0
			[8] 0
			[9] 0
			[10] 0
			[11] 0
			[12] 0
			[13] 0
			[14] 0
			[15] 0
			[16] 0
			[17] 0
			[18] 0
			[19] 0

Description: Sets the fault response for the selected fault.

Value:

- 0: NONE
- 1: OFF1
- 2: OFF2
- 3: OFF3
- 4: STOP1 (being developed)
- 5: STOP2
- 6: IASC/DCBRAKE
- 7: ENCODER (p0491)

Dependency: The fault is selected and the required response is set under the same index.
Refer to: p2100

Notice: It is not possible to re-parameterize the response to a specific fault for faults that are already present (queued).

Note: The fault response can only be changed for faults with the appropriate identification (see the List Manual, chapter "Faults and alarms").

Example:
 F12345 and fault response = OFF3 (OFF1, OFF2, NONE)
 --> The default fault response OFF3 can be changed to OFF1, OFF2 or NONE.

Re value = 1 (OFF1):
 Braking along the ramp-function generator down ramp followed by a pulse inhibit.

Re value = 2 (OFF2):
 Internal/external pulse inhibit.

Re value = 3 (OFF3):
 Braking along the OFF3 down ramp followed by a pulse inhibit.

Re value = 5 (STOP2):
 n_set = 0

Re value = 6 (armature short-circuit, internal/DC brake):
 The value can only be set for all motor data sets when p1231 = 3, 4.
 a) For synchronous motors (p0300 = 2xx, 4xx), an internal armature short-circuit is executed.
 b) For induction motors (p0300 = 1xx), a DC brake is initiated.

Re value = 7 (ENCODER (p0491)):
 The fault response set in p0491 is executed if applicable.

Note:
 IASC: Internal armature short circuit
 DCBRAKE: Direct current brake

p2102			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	BI: Acknowledge all faults / Ackn all faults		
	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2546, 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to acknowledge all faults at all drive objects of the drive system.		
Note:	A fault acknowledgement is triggered with a 0/1 signal.		

p2103[0...n]			
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	BI: 1. Acknowledge faults / 1. Acknowledge		
	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2441, 2442, 2443, 2447, 2475, 2546, 9220, 9677, 9678
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the first signal source to acknowledge faults.		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	A fault acknowledgement is triggered with a 0/1 signal.		

p2103			
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	BI: 1. Acknowledge faults / 1. Acknowledge		
	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the first signal source to acknowledge faults.		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	A fault acknowledgement is triggered with a 0/1 signal.		

p2104[0...n]			
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	BI: 2. Acknowledge faults / 2. Acknowledge		
	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2546, 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the second signal source to acknowledge faults.		
Note:	A fault acknowledgement is triggered with a 0/1 signal.		

p2104			
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	BI: 2. Acknowledge faults / 2. Acknowledge		
	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the second signal source to acknowledge faults.		
Note:	A fault acknowledgement is triggered with a 0/1 signal.		

p2105[0...n]			
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	BI: 3. Acknowledge faults / 3. Acknowledge		
	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2546, 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the third signal source to acknowledge faults.		
Note:	A fault acknowledgement is triggered with a 0/1 signal.		

p2105	BI: 3. Acknowledge faults / 3. Acknowledge		
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the third signal source to acknowledge faults.		
Note:	A fault acknowledgement is triggered with a 0/1 signal.		
p2106[0...n]	BI: External fault 1 / External fault 1		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: -	Access level: 3 Func. diagram: 2546 Unit selection: - Expert list: 1 Factory setting
	Min	Max	
	-	-	1
Description:	Sets the signal source for external fault 1.		
Dependency:	Refer to: F07860		
Note:	An external fault is triggered with a 1/0 signal. If this fault is output at the Control Unit, then it is transferred to all existing drive objects.		
p2106	BI: External fault 1 / External fault 1		
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source for external fault 1.		
Dependency:	Refer to: F07860		
Note:	An external fault is triggered with a 1/0 signal. If this fault is output at the Control Unit, then it is transferred to all existing drive objects.		
p2107[0...n]	BI: External fault 2 / External fault 2		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: -	Access level: 3 Func. diagram: 2546 Unit selection: - Expert list: 1 Factory setting
	Min	Max	
	-	-	1
Description:	Sets the signal source for external fault 2.		
Dependency:	Refer to: F07861		

Note: An external fault is triggered with a 1/0 signal.
If this fault is output at the Control Unit, then it is transferred to all existing drive objects.

p2107			
BI: External fault 2 / External fault 2			
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min -	Max -	Factory setting 1
Description: Sets the signal source for external fault 2.			
Dependency: Refer to: F07861			
Note: An external fault is triggered with a 1/0 signal. If this fault is output at the Control Unit, then it is transferred to all existing drive objects.			

p2108[0...n]			
BI: External fault 3 / External fault 3			
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: -	Access level: 3 Func. diagram: 2546 Unit selection: - Expert list: 1
	Min -	Max -	Factory setting 1
Description: Sets the signal source for external fault 3. External fault 3 is initiated by the following AND logic operation: - BI: p2108 negated - BI: p3111 - BI: p3112 negated			
Dependency: Refer to: p3110, p3111, p3112 Refer to: F07862			
Note: An external fault is triggered with a 1/0 signal. If this fault is output at the Control Unit, then it is transferred to all existing drive objects.			

p2108			
BI: External fault 3 / External fault 3			
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min -	Max -	Factory setting 1
Description: Sets the signal source for external fault 3. External fault 3 is initiated by the following AND logic operation: - BI: p2108 negated - BI: p3111 - BI: p3112 negated			

Dependency: Refer to: p3110, p3111, p3112

Refer to: F07862

Note: An external fault is triggered with a 1/0 signal.

If this fault is output at the Control Unit, then it is transferred to all existing drive objects.

r2109[0...63] Fault time removed in milliseconds / t_flt resolved ms

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1750, 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [ms]	- [ms]	- [ms]

Description: Displays the system runtime in milliseconds when the fault was removed.

Dependency: Refer to: r0945, r0947, r0948, r0949, r2114, r2130, r2133, r2136, r3115, r3120, r3122

Notice: The time comprises r2136 (days) and r2109 (milliseconds).

Note: The buffer parameters are cyclically updated in the background (refer to status signal in r2139).

The structure of the fault buffer and the assignment of the indices is shown in r0945.

r2110[0...63] Alarm number / Alarm number

All objects	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 8065
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: This parameter is identical to r2122.

p2111 Alarm counter / Alarm counter

All objects	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1750, 8065
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	65535	0

Description: Number of alarms that have occurred after the last reset.

Dependency: When p2111 is set to 0, the following is initiated:

- all of the alarms of the alarm buffer that have gone [0...7] are transferred into the alarm history [8...63].
- the alarm buffer [0...7] is deleted.

Refer to: r2110, r2122, r2123, r2124, r2125

Note: The parameter is reset to 0 at POWER ON.

p2112[0...n] BI: External alarm 1 / External alarm 1

A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 2546
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1

Description: Sets the signal source for external alarm 1.

Dependency: Refer to: A07850

Note: An external alarm is triggered with a 1/0 signal.

p2112			
BI: External alarm 1 / External alarm 1			
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min -	Max -	Factory setting 1
Description:	Sets the signal source for external alarm 1.		
Dependency:	Refer to: A07850		
Note:	An external alarm is triggered with a 1/0 signal.		

r2114[0...1]			
System runtime / System runtime			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the total system runtime for the drive unit. The time comprises r2114[0] (milliseconds) and r2114[1] (days). After r2114[0] has reached a value of 86.400.000 ms (24 hours) this value is reset and r2114[1] is incremented.		
Index:	[0] = Milliseconds [1] = Days		
Dependency:	Refer to: r0948, r2109, r2123, r2125, r2130, r2136, r2145, r2146		
Note:	The time in r2114 is used to display the fault and alarm times. When the electronic power supply is switched out, the counter value is saved. After the drive unit is powered up, the counter continues to run with the value that was saved the last time that the drive unit was powered down.		

p2116[0...n]			
BI: External alarm 2 / External alarm 2			
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: -	Access level: 3 Func. diagram: 2546 Unit selection: - Expert list: 1
	Min -	Max -	Factory setting 1
Description:	Sets the signal source for external alarm 2.		
Dependency:	Refer to: A07851		
Note:	An external alarm is triggered with a 1/0 signal.		

p2116	BI: External alarm 2 / External alarm 2		
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source for external alarm 2.		
Dependency:	Refer to: A07851		
Note:	An external alarm is triggered with a 1/0 signal.		
p2117[0...n]	BI: External alarm 3 / External alarm 3		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: -	Access level: 3 Func. diagram: 2546 Unit selection: - Expert list: 1 Factory setting
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source for external alarm 3.		
Dependency:	Refer to: A07852		
Note:	An external alarm is triggered with a 1/0 signal.		
p2117	BI: External alarm 3 / External alarm 3		
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	1
Description:	Sets the signal source for external alarm 3.		
Dependency:	Refer to: A07852		
Note:	An external alarm is triggered with a 1/0 signal.		
p2118[0...19]	Sets the message number for message type. / Msg_no Msg_type		
All objects	Can be changed: U, T Data type: Unsigned16 P-Group: Messages Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: 1750, 8075 Unit selection: - Expert list: 1 Factory setting
	Min	Max	Factory setting
	0	65535	0
Description:	Selects faults or alarms for which the message type should be changed.		
Dependency:	Selects the fault or alarm selection and sets the required type of message realized under the same index. Refer to: p2119		

Notice: It is not possible to re-parameterize the message type in the following cases:
- if there is no existing message number.
- if a message is present.

p2119[0...19]	Setting the message type / Message type		
All objects	Can be changed: U, T Data type: Integer16 P-Group: Messages Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 3 Func. diagram: 1750, 8075 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the message type for the selected fault or alarm.		
Value:	1: Fault (F) 2: Alarm (A) 3: No message (N)		
Dependency:	Selects the fault or alarm selection and sets the required type of message realized under the same index. Refer to: p2118		
Notice:	It is not possible to re-parameterize the message type for the existing faults or alarms.		
Note:	The message type can only be changed for messages with the appropriate identification. Example: F12345(A) --> Fault F12345 can be changed to alarm A12345. In this case, the message number that may be possibly entered in p2100[0...19] and p2126[0...19] is automatically removed.		

r2120	CO: Sum of fault and alarm buffer changes / Sum buffer changed		
All objects	Can be changed: - Data type: Unsigned16 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: 8065 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the sum of all of the fault and alarm buffer changes in the drive unit.		
Dependency:	Refer to: r0944, r2121		

r2121	CO: Counter, alarm buffer changes / Alrm buff changed		
All objects	Can be changed: - Data type: Unsigned16 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 8065 Unit selection: - Expert list: 1 Factory setting -
Description:	This counter is incremented every time the alarm buffer changes.		
Dependency:	Refer to: r2110, r2122, r2123, r2124, r2125		

r2122[0...63]	Alarm code / Alarm code		
All objects	Can be changed: - Data type: Unsigned16 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 1750, 8065 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of alarms that have occurred.		
Dependency:	Refer to: r2110, r2123, r2124, r2125, r2134, r2145, r2146, r3121, r3123		

Note: The buffer parameters are cyclically updated in the background (refer to status signal in r2139).
 Alarm buffer structure (general principle):
 r2122[0], r2124[0], r2123[0], r2125[0] --> alarm 1 (the oldest)
 ...
 r2122[7], r2124[7], r2123[7], r2125[7] --> Alarm 8 (the latest)
 When the alarm buffer is full, the alarms that have gone are entered into the alarm history:
 r2122[8], r2124[8], r2123[8], r2125[8] --> Alarm 1 (the latest)
 ...
 r2122[63], r2124[63], r2123[63], r2125[63] --> alarm 56 (the oldest)

r2123[0...63] Alarm time received in milliseconds / t_alarm recv ms

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1750, 8065
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [ms]	Max - [ms]	Factory setting - [ms]

Description: Displays the system runtime in milliseconds when the alarm occurred.
Dependency: Refer to: r2110, r2114, r2122, r2124, r2125, r2134, r2145, r2146, r3121, r3123
Notice: The time comprises r2145 (days) and r2123 (milliseconds).
Note: The buffer parameters are cyclically updated in the background (refer to status signal in r2139).
 The structure of the alarm buffer and the assignment of the indices is shown in r2122.

r2124[0...63] Alarm value / Alarm value

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: 1750, 8065
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

Description: Displays additional information about the active alarm (as integer number).
Dependency: Refer to: r2110, r2122, r2123, r2125, r2134, r2145, r2146, r3121, r3123
Note: The buffer parameters are cyclically updated in the background (refer to status signal in r2139).
 The structure of the alarm buffer and the assignment of the indices is shown in r2122.

r2125[0...63] Alarm time removed in milliseconds / t_alarm res ms

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1750, 8065
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [ms]	Max - [ms]	Factory setting - [ms]

Description: Displays the system runtime in milliseconds when the alarm was cleared.
Dependency: Refer to: r2110, r2114, r2122, r2123, r2124, r2134, r2145, r2146, r3121, r3123
Notice: The time comprises r2146 (days) and r2125 (milliseconds).
Note: The buffer parameters are cyclically updated in the background (refer to status signal in r2139).
 The structure of the alarm buffer and the assignment of the indices is shown in r2122.

p2126[0...19]	Setting fault number for acknowledge mode / Fault_no ackn_mode		
All objects	Can be changed: U, T Data type: Unsigned16 P-Group: Messages Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: 1750, 8075 Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the faults for which the acknowledge mode is to be changed		
Dependency:	Selects the faults and sets the required acknowledge mode realized under the same index Refer to: p2127		
Notice:	It is not possible to re-parameterize the acknowledge mode of a fault in the following cases: - if there is no existing fault number. - the message type is not "fault" (F). - when a fault is present.		
p2127[0...19]	Sets acknowledgement mode / Acknowledge mode		
All objects	Can be changed: U, T Data type: Integer16 P-Group: Messages Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 3 Func. diagram: 1750, 8075 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the acknowledge mode for selected fault.		
Value:	1: Acknowledgment only using POWER ON 2: Ack IMMEDIATELY after the fault cause has been removed 3: Acknowledgement only for PULSE INHIBIT		
Dependency:	Selects the faults and sets the required acknowledge mode realized under the same index Refer to: p2126		
Notice:	It is not possible to re-parameterize the acknowledge mode of a fault in the following cases: - if there is no existing fault number. - the message type is not "fault" (F). - when a fault is present.		
Note:	The acknowledge mode can only be changed for faults with the appropriate identification. Example: F12345 and acknowledge mode = POWER ON (IMMEDIATELY) --> The acknowledge mode can be changed from POWER ON to IMMEDIATELY.		
p2128[0...15]	Selecting fault/alarm code for trigger / Message trigger		
All objects	Can be changed: U, T Data type: Unsigned16 P-Group: Messages Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: 1750, 8070 Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects faults or alarms which can be used as trigger.		
Dependency:	Refer to: r2129		

r2129.0...15**CO/BO: Trigger word for faults and alarms / Trigger word**

All objects

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1530, 8070**P-Group:** Messages**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Trigger signal for the selected faults and alarms

Bit field:**Bit** **Signal name****1 signal****0 signal****FP**

00 Trigger signal p2128[0]

On

Off

-

01 Trigger signal p2128[1]

On

Off

-

02 Trigger signal p2128[2]

On

Off

-

03 Trigger signal p2128[3]

On

Off

-

04 Trigger signal p2128[4]

On

Off

-

05 Trigger signal p2128[5]

On

Off

-

06 Trigger signal p2128[6]

On

Off

-

07 Trigger signal p2128[7]

On

Off

-

08 Trigger signal p2128[8]

On

Off

-

09 Trigger signal p2128[9]

On

Off

-

10 Trigger signal p2128[10]

On

Off

-

11 Trigger signal p2128[11]

On

Off

-

12 Trigger signal p2128[12]

On

Off

-

13 Trigger signal p2128[13]

On

Off

-

14 Trigger signal p2128[14]

On

Off

-

15 Trigger signal p2128[15]

On

Off

-

Dependency:

If one of the faults or alarms selected in p2128[n] occurs, then the particular bit of this binector output is set.

Refer to: p2128

Note:

CO: r2129 = 0 --> None of the selected messages has occurred.

CO: r2129 > 0 --> At least one of the selected messages has occurred.

r2130[0...63]**Fault time received in days / t_fault recv days**

All objects

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 8060**P-Group:** Messages**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the system runtime in days when the fault occurred.

Dependency:

Refer to: r0945, r0947, r0948, r0949, r2109, r2114, r2133, r2136, r3115, r3120, r3122

Notice:

The time comprises r2130 (days) and r0948 (milliseconds).

Note:

The buffer parameters are cyclically updated in the background (refer to status signal in r2139).

r2131 CO: Actual fault code / Actual fault code

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the code of the oldest active fault.		
Note:	0: No fault present.		

r2132 CO: Actual alarm code / Actual alarm code

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 8065
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the code of the last alarm that occurred.		
Note:	0: No alarm present.		

r2133[0...63] Fault value for float values / Fault val float

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays additional information about the fault that occurred for float values.		
Dependency:	Refer to: r0945, r0947, r0948, r0949, r2109, r2130, r2136, r3115		
Note:	The buffer parameters are cyclically updated in the background (refer to status signal in r2139).		

r2134[0...63] Alarm value for float values / Alarm value float

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8065
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays additional information about the active alarm for float values.		
Dependency:	Refer to: r2110, r2122, r2123, r2124, r2125, r2145, r2146, r3121, r3123		
Note:	The buffer parameters are cyclically updated in the background (refer to status signal in r2139).		

r2135.0...15**CO/BO: Status word faults/alarms 2 / ZSW fault/alarm 2**

A_INF, B_INF,
CU_CX32, CU_I,
CU_LINK, CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, HUB,
S_INF, SERVO,
TB30, TM120, TM15,
TM15DI_DO, TM17,
TM31, TM41,
TM54F_MA,
TM54F_SL

Can be changed: -**Data type:** Unsigned16**P-Group:** Displays, signals**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 2**Func. diagram:** 1530, 2548**Unit selection:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the second status word of faults and alarms.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Fault encoder 1	Yes	No	-
01	Fault encoder 2	Yes	No	-
02	Fault encoder 3	Yes	No	-
10	Fault transformer overtemperature	Yes	No	-
11	Alarm transformer overtemperature	Yes	No	-
12	Fault motor overtemperature	Yes	No	-
13	Fault power unit thermal overload	Yes	No	-
14	Alarm motor overtemperature	Yes	No	-
15	Alarm power unit thermal overload	Yes	No	-

r2135.0...15**CO/BO: Status word faults/alarms 2 / ZSW fault/alarm 2**

VECTOR

Can be changed: -**Data type:** Unsigned16**P-Group:** Displays, signals**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 2**Func. diagram:** 1530, 2548**Unit selection:** -**Expert list:** 1**Min****Max****Factory setting**

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-

Description:

Displays the second status word of faults and alarms.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Fault encoder 1	Yes	No	-
01	Fault encoder 2	Yes	No	-
02	Fault encoder 3	Yes	No	-
12	Fault motor overtemperature	Yes	No	-
13	Fault power unit thermal overload	Yes	No	-
14	Alarm motor overtemperature	Yes	No	-
15	Alarm power unit thermal overload	Yes	No	-

r2136[0...63]**Fault time removed in days / t_fit resolv. days**

All objects

Can be changed: -**Data type:** Unsigned16**P-Group:** Messages**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 3**Func. diagram:** 8060**Unit selection:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the system runtime in days when the fault was removed.

Dependency:

Refer to: r0945, r0947, r0948, r0949, r2109, r2114, r2130, r2133, r3115, r3120, r3122

Notice:

The time comprises r2136 (days) and r2109 (milliseconds).

Note:

The buffer parameters are cyclically updated in the background (refer to status signal in r2139).

r2138.7...15 CO/BO: Control word faults/alarms / STW fault/alarm

All objects	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2546
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the control word of the faults and alarms.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	07	Acknowledge fault	Yes	No	-
	10	External alarm 1 (A07850) effective	Yes	No	-
	11	External alarm 2 (A07851) effective	Yes	No	-
	12	External alarm 3 (A07852) effective	Yes	No	-
	13	External fault 1 (F07860) effective	Yes	No	-
	14	External fault 2 (F07861) effective	Yes	No	-
	15	External fault 3 (F07862) effective	Yes	No	-

Dependency: Refer to: p2103, p2104, p2105, p2106, p2107, p2108, p2112, p2116, p2117, p3110, p3111, p3112

r2139.0...12 CO/BO: Status word faults/alarms 1 / ZSW fault/alarm 1

All objects	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2548
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the first status word of faults and alarms.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Being acknowledged	Yes	No	-
	01	Acknowledgment required	Yes	No	-
	03	Fault present	Yes	No	-
	05	Safety message present	Yes	No	-
	06	Internal message 1 present	Yes	No	-
	07	Alarm present	Yes	No	-
	08	Internal message 2 present	Yes	No	-
	11	Alarm class bit 0	High	Low	-
	12	Alarm class bit 1	High	Low	-

Note: Re bit 03, 05, 07:

These bits are set if at least one fault/alarm occurs. Data is entered into the fault/alarm buffer with delay. This is the reason that the fault/alarm buffer should only be read if, after "fault present"/"alarm present" has occurred, a change in the buffer was also detected (r0944, r9744, r2121).

Re bit 06, 08:

These status bits are used for internal diagnostic purposes only.

Re bit 11, 12:

These status bits are used for the classification of internal alarm classes and are intended for diagnostic purposes only on certain automation systems with SINAMICS functionality.

p2140[0...n] Hysteresis speed 2 / n_hysteresis 2

SERVO, VECTOR

Can be changed: U, T**Calculated:**
CALC_MOD_LIM_REF**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** 8010**P-Group:** Messages**Units group:** 3_1**Unit selection:** p0505**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0.00 [rpm]

300.00 [rpm]

90.00 [rpm]

Description: Sets the hysteresis speed (bandwidth) for the following signals:

"|n_act| <= speed threshold value 2" (BO: r2197.1)

"|n_act| > speed threshold value 2" (BO: r2197.2)

Dependency: Refer to: p2155, r2197**p2140[0...n] Hysteresis velocity 2 / v_hysteresis 2**

SERVO (Lin)

Can be changed: U, T**Calculated:**
CALC_MOD_LIM_REF**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** 8010**P-Group:** Messages**Units group:** 4_1**Unit selection:** p0505**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0.00 [m/min]

10.00 [m/min]

0.90 [m/min]

Description: Sets the hysteresis velocity (bandwidth) for the following signals:

"|n_act| <= speed threshold value 2" (BO: r2197.1)

"|n_act| > speed threshold value 2" (BO: r2197.2)

Dependency: Refer to: p2155, r2197**p2141[0...n] Speed threshold 1 / n_thresh val 1**

SERVO, VECTOR

Can be changed: U, T**Calculated:**
CALC_MOD_LIM_REF**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** 8010**P-Group:** Messages**Units group:** 3_1**Unit selection:** p0505**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0.00 [rpm]

210000.00 [rpm]

5.00 [rpm]

Description: Sets the speed threshold value for the signal "f or n comparison value reached or exceeded" (BO: r2199.1).**Dependency:** Refer to: p2142, r2199**p2141[0...n] Velocity threshold value 1 / v_thresh val 1**

SERVO (Lin)

Can be changed: U, T**Calculated:**
CALC_MOD_LIM_REF**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** DDS, p0180**Func. diagram:** 8010**P-Group:** Messages**Units group:** 4_1**Unit selection:** p0505**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0.00 [m/min]

1000.00 [m/min]

0.05 [m/min]

Description: Sets the velocity threshold value for the signal "f or n comparison value reached or exceeded" (BO: r2199.1).**Dependency:** Refer to: p2142, r2199

p2142[0...n]	Hysteresis speed 1 / n_hysteresis 1		
SERVO, VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 300.00 [rpm]	Factory setting 2.00 [rpm]
Description:	Sets the hysteresis speed (bandwidth) for the signal "f or n / v comparison value reached or exceeded" (BO: r2199.1).		
Dependency:	Refer to: p2141, r2199		
p2142[0...n]	Hysteresis velocity 1 / v_hysteresis 1		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 10.00 [m/min]	Factory setting 0.02 [m/min]
Description:	Sets the hysteresis velocity (bandwidth) for the signal "f or n / v comparison value reached or exceeded" (BO: r2199.1).		
Dependency:	Refer to: p2141, r2199		
p2144[0...n]	BI: Motor stall monitoring enable (negated) / Mot stall enab neg		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 8012
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source for the negated enable (0 = enable) of the motor stall monitoring.		
Dependency:	Refer to: p2163, p2164, p2166, r2197, r2198 Refer to: F07900		
Note:	If the enable signal is connected to r2197.7 then the stall signal is suppressed if there is no speed setpoint - actual value deviation.		
r2145[0...63]	Alarm time received in days / t_alarm recv days		
All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 8065
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the system runtime in days when the alarm occurred.		
Dependency:	Refer to: r2110, r2114, r2122, r2123, r2124, r2125, r2134, r2146, r3121, r3123		
Notice:	The time comprises r2145 (days) and r2123 (milliseconds).		
Note:	The buffer parameters are cyclically updated in the background (refer to status signal in r2139).		

r2146[0...63]	Alarm time removed in days / t_alarm res days		
All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 8065
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the system runtime in days when the alarm was cleared.		
Dependency:	Refer to: r2110, r2114, r2122, r2123, r2124, r2125, r2134, r2145, r3121, r3123		
Notice:	The time comprises r2146 (days) and r2125 (milliseconds).		
Note:	The buffer parameters are cyclically updated in the background (refer to status signal in r2139).		

p2147	Delete fault buffer of all drive objects / Del fault buffer		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: 8060
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0
Description:	Setting to delete the fault buffer of all existing drive objects.		
Value:	0: Inactive 1: Start to delete the fault buffer of all drive objects		
Dependency:	Refer to: r0945, r0947, r0948, r0949, r2109, r2130, r2133, r2136		
Note:	p2147 is automatically set to 0 after execution.		

p2148[0...n]	BI: Ramp-function generator active / HLG active		
SERVO, VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 8010
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the signal "ramp-function generator active" for the following signals/messages: "Speed setpoint - actual value deviation within tolerance t_on" (BO: r2199.4) "Ramp-up/ramp-down completed" (BO: r2199.5)		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	The binector input is automatically pre-assigned to r1199.2. The following applies for SERVO: The pre-assignment is only made when the function module "setpoint channel" is activated (r0108.8 = 1).		

p2149[0...n]					Monitoring configuration / Monit config				
SERVO	Can be changed: U, T		Calculated: -		Access level: 3				
	Data type: Unsigned16		Dynamic index: DDS, p0180		Func. diagram: 8010, 8013				
	P-Group: Messages		Units group: -		Unit selection: -				
	Not for motor type: -		Scaling: -		Expert list: 1				
	Min		Max		Factory setting				
	-		-		0000 bin				
Description:					Configuration word for signals and monitoring functions.				
Bit field:					Bit	Signal name	1 signal	0 signal	FP
					00	Enable alarm A07903	Yes	No	8010
					01	Load monitoring only in the 1st quadrant	Yes	No	8013
					03	n_act > p2155 own hysteresis	Yes	No	8010
					15	Automatic parameterization carried out (p0340 = 1, p3900 > 0)	Yes	No	-
Dependency:					Refer to: r2197				
					Refer to: A07903				
Note:					Re bit 00:				
					Alarm A07903 is output when the bit is set with r2197.7 = 0 (n_set <> n_act).				
					Re bit 01:				
					When the bit is set, load monitoring is only carried out in the 1st quadrant as a result of the positive characteristic parameters (p2182 ... p2190).				
					Re bit 03:				
					When the bit is set, r2197 bit 1 and bit 2 are determined via separate hystereses.				
					Re bit 15:				
					The bit indicates whether the automatic parameterization (p0340 = 1, p3900 > 0) for the parameters of the extended monitoring functions was carried out. If the bit is not set (e.g. when the configuration is activated (p0108.15)), the parameterization is automatically carried out during booting even if r3925.0 is already 1.				

p2149[0...n] Monitoring configuration / Monit config					
VECTOR	Can be changed: U, T		Calculated: -	Access level: 3	
	Data type: Unsigned16		Dynamic index: DDS, p0180	Func. diagram: 8010, 8013	
	P-Group: Messages		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min	Max	Factory setting		
	-	-	0001 bin		
Description:	Configuration word for signals and monitoring functions.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Enable alarm A07903	Yes	No	8010
	01	Load monitoring only in the 1st quadrant	Yes	No	8013
	03	n_act > p2155 own hysteresis	Yes	No	8010
	15	Automatic parameterization carried out	Yes	No	-
	(p0340 = 1, p3900 > 0)				
Dependency:	Refer to: r2197				
	Refer to: A07903				

Note:

Re bit 00:
Alarm A07903 is output when the bit is set with $r2197.7 = 0$ ($n_set \neq n_act$).

Re bit 01:
When the bit is set, load monitoring is only carried out in the 1st quadrant as a result of the positive characteristic parameters (p2182 ... p2190).

Re bit 03:
When the bit is set, r2197 bit 1 and bit 2 are determined via separate hystereses.

Re bit 15:
The bit indicates whether the automatic parameterization ($p0340 = 1$, $p3900 > 0$) for the parameters of the extended monitoring functions was carried out. If the bit is not set (e.g. when the configuration is activated (p0108.15)), the parameterization is automatically carried out during booting even if r3925.0 is already 1.

p2150[0...n]	Hysteresis speed 3 / n_hysteresis 3		
SERVO, VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 300.00 [rpm]	Factory setting 2.00 [rpm]
Description:	Sets the hysteresis speed (bandwidth) for the following signals: "n_act < speed threshold value 3" (BO: r2199.0) "n_set >= 0" (BO: r2198.5) "n_act >= 0" (BO: r2197.3)		
Dependency:	Refer to: p2161, r2197, r2199		

p2150[0...n]	Hysteresis velocity 3 / v_hysteresis 3		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 3.00 [m/min]	Factory setting 0.02 [m/min]
Description:	Sets the hysteresis velocity (bandwidth) for the following signals: "n_act < speed threshold value 3" (BO: r2199.0) "n_set >= 0" (BO: r2198.5) "n_act >= 0" (BO: r2197.3)		
Dependency:	Refer to: p2161, r2197, r2199		

p2151[0...n]	CI: Speed setpoint for messages/signals / n_set for msg		
SERVO	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 8010
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min -	Max -	Factory setting 1438[0]
Description:	Sets the signal source for the speed setpoint for the following messages: "Speed setpoint - actual value deviation within tolerance t_off" (BO: r2197.7) "Ramp-up/ramp-down completed" (BO: r2199.5) "n_set < p2161" (BO: r2198.4) "n_set > 0" (BO: r2198.5)		
Dependency:	Refer to: r2197, r2198, r2199		

p2151[0...n]	CI: Velocity setpoint for messages/signals / v_set for msg		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 8010 Unit selection: - Expert list: 1 Factory setting 1438[0]
Description:	Sets the signal source for the velocity setpoint for the following messages: "Velocity setpoint - actual value deviation within tolerance t_off" (BO: r2197.7) "Ramp-up/ramp-down completed" (BO: r2199.5) " v_set < p2161" (BO: r2198.4) "v_set > 0" (BO: r2198.5)		
Dependency:	Refer to: r2197, r2198, r2199		
p2151[0...n]	CI: Speed setpoint for messages/signals / n_set for msg		
VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 8010 Unit selection: - Expert list: 1 Factory setting 1170[0]
Description:	Sets the signal source for the speed setpoint for the following messages: "Speed setpoint - actual value deviation within tolerance t_off" (BO: r2197.7) "Ramp-up/ramp-down completed" (BO: r2199.5) " n_set < p2161" (BO: r2198.4) "n_set > 0" (BO: r2198.5)		
Dependency:	Refer to: r2197, r2198, r2199		
p2153[0...n]	Speed actual value filter time constant / n_act_filt T		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000000 [ms]	Access level: 3 Func. diagram: 8010 Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Sets the time constant of the PT1 element to smooth the speed / velocity actual value. The smoothed actual speed/velocity is compared with the threshold values and is only used for messages and signals.		
Dependency:	Refer to: r2169		
p2153[0...n]	Velocity actual value filter time constant / v_act_filt T		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000000 [ms]	Access level: 3 Func. diagram: 8010 Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Sets the time constant of the PT1 element to smooth the speed / velocity actual value. The smoothed actual speed/velocity is compared with the threshold values and is only used for messages and signals.		
Dependency:	Refer to: r2169		

p2154[0...n]	CI: Speed setpoint 2 / n_set 2		
SERVO, VECTOR	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 8010 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for speed setpoint 2. The sum of CI: p2151 and CI: p2154 is used for the following messages/signals: "Speed setpoint - actual value deviation within tolerance t_off" (BO: r2197.7) "Speed setpoint - actual value deviation within tolerance t_on" (BO: r2199.4) "Ramp-up/ramp-down completed" (BO: r2199.5)		
Dependency:	Refer to: p2151, r2197, r2199		
p2154[0...n]	CI: Velocity setpoint 2 / v_set 2		
SERVO (Lin)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: 8010 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the velocity setpoint 2. The sum of CI: p2151 and CI: p2154 is used for the following messages/signals: "Velocity setpoint - actual value deviation within tolerance t_off" (BO: r2197.7) "Velocity setpoint - actual value deviation within tolerance t_on" (BO: r2199.4) "Ramp-up/ramp-down completed" (BO: r2199.5)		
Dependency:	Refer to: p2151, r2197, r2199		
p2155[0...n]	Speed threshold 2 / n_thresh val 2		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [rpm]	Calculated: CALC_MOD_LIM_REF Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: 8010 Unit selection: p0505 Expert list: 1 Factory setting 900.00 [rpm]
Description:	Sets the speed threshold value for the following messages: " n_act <= speed threshold value 2" (BO: r2197.1) " n_act > speed threshold value 2" (BO: r2197.2)		
Dependency:	Refer to: p2140, r2197		

p2155[0...n]	Velocity threshold value 2 / v_thresh val 2		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 1000.00 [m/min]	Factory setting 9.00 [m/min]
Description:	Sets the velocity threshold value for the following messages: " v_act <= velocity threshold value 2" (BO: r2197.1) " v_act > velocity threshold value 2" (BO: r2197.2)		
Dependency:	Refer to: p2140, r2197		

p2156[0...n]	On delay, comparison value reached / t_on cmpr val rchd		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 10000.0 [ms]	Factory setting 0.0 [ms]
Description:	Sets the switch-in delay time for the signal "comparison value reached" (BO: r2199.1).		
Dependency:	Refer to: p2141, p2142, r2199		

p2161[0...n]	Speed threshold 3 / n_thresh val 3		
SERVO, VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 5.00 [rpm]
Description:	Sets the speed threshold value for the signal " n_act < speed threshold value 3" (BO: r2199.0).		
Dependency:	Refer to: p2150, r2199		

p2161[0...n]	Velocity threshold value 3 / v_thresh val 3		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 1000.00 [m/min]	Factory setting 0.05 [m/min]
Description:	Sets the velocity threshold value for the signal " v_act < velocity threshold value 3" (BO: r2199.0).		
Dependency:	Refer to: p2150, r2199		

p2162[0...n]	Hysteresis speed $n_act > n_max$ / Hyst $n_act > n_max$		
SERVO, VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 60000.00 [rpm]	Factory setting 0.00 [rpm]
Description:	Sets the hysteresis speed (bandwidth) for the signal "n_act > n_max" (BO: r2197.6).		
Dependency:	Refer to: r1084, r1087, r2197		
Notice:	For p0322 = 0, the following applies: $p2162 \leq 0.1 * p0311$ For p0322 > 0, the following applies: $p2162 \leq 1.02 * p0322 - p1082$ If one of the conditions is violated, p2162 is appropriately and automatically reduced when exiting the commissioning mode.		
Note:	For a negative speed limit (r1087) the hysteresis is effective below the limit value and for a positive speed limit (r1084) above the limit value. If significant overshoot occurs in the maximum speed range (e.g. due to load shedding), you are advised to increase the dynamic response of the speed controller (if possible). If this is insufficient, the hysteresis p2162 can only be increased by more than 10% of the rated speed when the maximum speed (p0322) of the motor is sufficiently greater than the speed limit p1082.		
p2162[0...n]	Hysteresis velocity $v_act > v_max$ / Hyst $v_act > v_max$		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 1000.00 [m/min]	Factory setting 6.00 [m/min]
Description:	Sets the hysteresis velocity (bandwidth) for the signal "v_act > v_max" (BO: r2197.6).		
Dependency:	Refer to: r1084, r1087, r2197		
Notice:	For p0322 = 0, the following applies: $p2162 \leq 0.1 * p0311$ For p0322 > 0, the following applies: $p2162 \leq 1.02 * p0322 - p1082$ If one of the conditions is violated, p2162 is appropriately and automatically reduced when exiting the commissioning mode.		
Note:	For a negative velocity limit (r1087) the hysteresis is effective below the limit value and for a positive velocity limit (r1084) above the limit value.		
p2163[0...n]	Speed threshold 4 / n_thresh val 4		
SERVO, VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 90.00 [rpm]
Description:	Sets the speed threshold value for the "speed setpoint - actual value deviation in tolerance t_off" signal/message (BO: r2197.7).		
Dependency:	Refer to: p2164, p2166, r2197		

p2163[0...n]	Velocity threshold value 4 / v_thresh val 4		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 1000.00 [m/min]	Factory setting 0.90 [m/min]
Description:	Sets the velocity threshold value for the "speed setpoint - actual value deviation in tolerance t_off" message (BO: r2197.7).		
Dependency:	Refer to: p2164, p2166, r2197		
p2164[0...n]	Hysteresis speed 4 / n_hysteresis 4		
SERVO, VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 200.00 [rpm]	Factory setting 2.00 [rpm]
Description:	Sets the hysteresis speed (bandwidth) for the "speed setpoint - actual value deviation in tolerance t_off" signal/message (BO: r2197.7).		
Dependency:	Refer to: p2163, p2166, r2197		
p2164[0...n]	Hysteresis velocity 4 / v_hysteresis 4		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 10.00 [m/min]	Factory setting 0.02 [m/min]
Description:	Sets the hysteresis velocity (bandwidth) for the "speed setpoint - actual value deviation in tolerance t_off" message (BO: r2197.7).		
Dependency:	Refer to: p2163, p2166, r2197		
p2166[0...n]	Off delay n_act = n_set / t_del_off n_i=n_so		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 10000.0 [ms]	Factory setting 200.0 [ms]
Description:	Sets the switch-off delay time for the "speed setpoint - actual value deviation in tolerance t_off" signal/message (BO: r2197.7).		
Dependency:	Refer to: p2163, p2164, r2197		

p2166[0...n]	Off delay $v_{act} = v_{set} / t_{del_off\ n_i=n_{so}}$		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 10000.0 [ms]	Factory setting 200.0 [ms]
Description:	Sets the switch-off delay time for the "velocity setpoint - actual value deviation in tolerance t_{off} " signal/message (BO: r2197.7).		
Dependency:	Refer to: p2163, p2164, r2197		
p2167[0...n]	Switch-on delay $n_{act} = n_{set} / t_{on\ n_{act}=n_{set}}$		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 10000.0 [ms]	Factory setting 200.0 [ms]
Description:	Sets the switch-on delay for the "speed setpoint - actual value deviation in tolerance t_{on} " signal/message (BO: r2199.4).		
p2167[0...n]	On delay $v_{act} = v_{set} / t_{on\ n_{act}=n_{set}}$		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8010
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 10000.0 [ms]	Factory setting 200.0 [ms]
Description:	Sets the switch-on delay for the "speed setpoint - actual value deviation in tolerance t_{on} " signal/message (BO: r2199.4).		
r2169	CO: Actual speed smoothed signals / n_{act} smth message		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 1750, 8010, 8012, 8013
	P-Group: Messages	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min - [rpm]	Max - [rpm]	Factory setting - [rpm]
Description:	Displays the smoothed actual speed for messages/signals.		
Dependency:	Refer to: p2153		
r2169	CO: Actual velocity smoothed signals / v_{act} smth message		
SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 1750, 8010, 8012, 8013
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min - [m/min]	Max - [m/min]	Factory setting - [m/min]
Description:	Displays the smoothed actual velocity for messages/signals.		
Dependency:	Refer to: p2153		

p2174[0...n]	Torque threshold value 1 / M_thresh val 1		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 20000000.00 [Nm]	Access level: 2 Func. diagram: 8012 Unit selection: p0505 Expert list: 1 Factory setting 5.13 [Nm]
Description:	Sets the torque threshold value for the signal "Torque setpoint < torque threshold value 1" (BO: r2198.10).		
Dependency:	Refer to: p2195, r2198		
p2174[0...n]	Force threshold value 1 / F_thresh val 1		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 20000000.00 [N]	Access level: 2 Func. diagram: 8012 Unit selection: p0505 Expert list: 1 Factory setting 1000.00 [N]
Description:	Sets the force threshold value for the signal "Torque setpoint < torque threshold value 1" (BO: r2198.10).		
Dependency:	Refer to: p2195, r2198		
p2174[0...n]	Torque threshold value 1 / M_thresh val 1		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 20000000.00 [Nm]	Access level: 2 Func. diagram: 8012 Unit selection: p0505 Expert list: 1 Factory setting 5.13 [Nm]
Description:	Sets the torque threshold value for the messages: "Torque setpoint < torque threshold value 1 and n_set reached" (BO: r2198.9) "Torque setpoint < torque threshold value 1" (BO: r2198.10) "Torque setpoint > torque threshold value 1" (BO: r2198.13)		
Dependency:	Refer to: p2195, r2198		
p2175[0...n]	Motor locked speed threshold / Mot lock n_thresh		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [rpm]	Calculated: CALC_MOD_LIM_REF Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: 8012 Unit selection: p0505 Expert list: 1 Factory setting 120.00 [rpm]
Description:	Sets the speed threshold for the message "Motor locked" (BO: r2198.6).		
Dependency:	Refer to: p0500, p2177, r2198		

p2175[0...n]	Motor locked, velocity threshold / Mot lock v_thresh		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8012
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 1000.00 [m/min]	Factory setting 1.20 [m/min]
Description:	Sets the velocity threshold for the message "Motor locked" (BO: r2198.6).		
Dependency:	Refer to: p0500, p2177, r2198		
p2175[0...n]	Motor locked speed threshold / Mot lock n_thresh		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8012
	P-Group: Messages	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 120.00 [rpm]
Description:	Sets the speed threshold for the message "Motor locked" (BO: r2198.6).		
Dependency:	Refer to: p0500, p2177, r2198		
Note:	The following applies for encoderless vector control: At low speeds in open-loop speed controlled operation (see p1755, p1756), a locked motor cannot be detected.		
p2177[0...n]	Motor locked delay time / Mot lock t_del		
SERVO, VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8012
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 65.000 [s]	Factory setting 1.000 [s]
Description:	Sets the delay time for the message "Motor locked" (BO: r2198.6). If "Motor locked" is identified within this time, then ZSW2.6 is set and an appropriate fault is output.		
Dependency:	Refer to: p0500, p2175, r2198		
p2178[0...n]	Motor stalled delay time / Mot stall t_del		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_REG	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8012
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 10.000 [s]	Factory setting 0.010 [s]
Description:	Sets the delay time for the message "Motor stalled" (BO: r2198.7). If "Motor stalled" is identified within this time, then ZSW2.7 is set and an appropriate fault is output.		
Dependency:	Refer to: r2198		

p2181[0...n]	Load monitoring response / Load monit resp		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T Data type: Integer16 P-Group: Messages Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 6	Access level: 3 Func. diagram: 8013 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the response when evaluating the load monitoring.		
Value:	0: Load monitoring disabled 1: A07920 for torque/speed too low 2: A07921 for torque/speed too high 3: A07922 for torque/speed out of tolerance 4: F07923 for torque/speed too low 5: F07924 for torque/speed too high 6: F07925 for torque/speed out of tolerance		
Dependency:	Refer to: p2182, p2183, p2184, p2185, p2186, p2187, p2188, p2189, p2190, p2192, r2198 Refer to: A07920, A07921, A07922, F07923, F07924, F07925		
Note:	The response to the faults F07923 ... F07925 can be set. F07926 is evaluated only if p2181 is not zero.		
p2182[0...n]	Load monitoring velocity threshold 1 / n_thresh 1		
SERVO (Extended sig, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [m/min]	Calculated: - Dynamic index: DDS, p0180 Units group: 4_1 Scaling: - Max 1000.00 [m/min]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 0.05 [m/min]
Description:	Sets the speed/torque envelope curve for load monitoring. The envelope curve (upper and lower envelope curve) is defined as follows based on 3 speed thresholds: p2182 (n_threshold 1) --> p2185 (M_threshold 1, upper), p2186 (M_threshold 1, lower) p2183 (n_threshold 2) --> p2187 (M_threshold 2, upper), p2188 (M_threshold 2, lower) p2184 (n_threshold 3) --> p2189 (M_threshold 3, upper), p2190 (M_threshold 3, lower)		
Dependency:	The following applies: p2182 < p2183 < p2184 Refer to: p2183, p2184, p2185, p2186 Refer to: A07926		
p2182[0...n]	Load monitoring speed threshold value 1 / n_thresh 1		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 150.00 [rpm]
Description:	Sets the speed/torque envelope curve for load monitoring. The envelope curve (upper and lower envelope curve) is defined as follows based on 3 speed thresholds: p2182 (n_threshold 1) --> p2185 (M_threshold 1, upper), p2186 (M_threshold 1, lower) p2183 (n_threshold 2) --> p2187 (M_threshold 2, upper), p2188 (M_threshold 2, lower) p2184 (n_threshold 3) --> p2189 (M_threshold 3, upper), p2190 (M_threshold 3, lower)		
Dependency:	The following applies: p2182 < p2183 < p2184 Refer to: p2183, p2184, p2185, p2186 Refer to: A07926		

p2183[0...n]	Load monitoring velocity threshold 2 / n_thresh 2		
SERVO (Extended sig, Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8013
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [m/min]	1000.00 [m/min]	0.05 [m/min]
Description:	Sets the speed/torque envelope curve for load monitoring. The envelope curve (upper and lower envelope curve) is defined as follows based on 3 speed thresholds: p2182 (n_threshold 1) --> p2185 (M_threshold 1, upper), p2186 (M_threshold 1, lower) p2183 (n_threshold 2) --> p2187 (M_threshold 2, upper), p2188 (M_threshold 2, lower) p2184 (n_threshold 3) --> p2189 (M_threshold 3, upper), p2190 (M_threshold 3, lower)		
Dependency:	The following applies: p2182 < p2183 < p2184 Refer to: p2182, p2184, p2187, p2188 Refer to: A07926		

p2183[0...n]	Load monitoring speed threshold value 2 / n_thresh 2		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8013
	P-Group: Messages	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [rpm]	210000.00 [rpm]	900.00 [rpm]
Description:	Sets the speed/torque envelope curve for load monitoring. The envelope curve (upper and lower envelope curve) is defined as follows based on 3 speed thresholds: p2182 (n_threshold 1) --> p2185 (M_threshold 1, upper), p2186 (M_threshold 1, lower) p2183 (n_threshold 2) --> p2187 (M_threshold 2, upper), p2188 (M_threshold 2, lower) p2184 (n_threshold 3) --> p2189 (M_threshold 3, upper), p2190 (M_threshold 3, lower)		
Dependency:	The following applies: p2182 < p2183 < p2184 Refer to: p2182, p2184, p2187, p2188 Refer to: A07926		

p2184[0...n]	Load monitoring velocity threshold 3 / n_thresh 3		
SERVO (Extended sig, Lin)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8013
	P-Group: Messages	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [m/min]	1000.00 [m/min]	0.05 [m/min]
Description:	Sets the speed/torque envelope curve for load monitoring. The envelope curve (upper and lower envelope curve) is defined as follows based on 3 speed thresholds: p2182 (n_threshold 1) --> p2185 (M_threshold 1, upper), p2186 (M_threshold 1, lower) p2183 (n_threshold 2) --> p2187 (M_threshold 2, upper), p2188 (M_threshold 2, lower) p2184 (n_threshold 3) --> p2189 (M_threshold 3, upper), p2190 (M_threshold 3, lower)		
Dependency:	The following applies: p2182 < p2183 < p2184 Refer to: p2182, p2183, p2189, p2190 Refer to: A07926		

p2184[0...n]	Load monitoring speed threshold value 3 / n_thresh 3		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [rpm]	Calculated: - Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 1500.00 [rpm]
Description:	Sets the speed/torque envelope curve for load monitoring. The envelope curve (upper and lower envelope curve) is defined as follows based on 3 speed thresholds: p2182 (n_threshold 1) --> p2185 (M_threshold 1, upper), p2186 (M_threshold 1, lower) p2183 (n_threshold 2) --> p2187 (M_threshold 2, upper), p2188 (M_threshold 2, lower) p2184 (n_threshold 3) --> p2189 (M_threshold 3, upper), p2190 (M_threshold 3, lower)		
Dependency:	The following applies: p2182 < p2183 < p2184 Refer to: p2182, p2183, p2189, p2190 Refer to: A07926		

p2185[0...n]	Load monitoring force threshold 1, upper / M_thresh 1 upper		
SERVO (Extended sig, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 100000.00 [N]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 100000.00 [N]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2185 > p2186 Refer to: p2182, p2186 Refer to: A07926		
Note:	The upper envelope curve is defined by p2185, p2187 and p2189.		

p2185[0...n]	Load monitoring torque threshold 1, upper / M_thresh 1 upper		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 20000000.00 [Nm]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 10000000.00 [Nm]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2185 > p2186 Refer to: p2182, p2186 Refer to: A07926		
Note:	The upper envelope curve is defined by p2185, p2187 and p2189.		

p2186[0...n]	Load monitoring force threshold 1, lower / M_thresh 1 lower		
SERVO (Extended sig, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 100000.00 [N]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [N]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2186 < p2185 Refer to: p2182, p2185 Refer to: A07926		
Note:	The lower envelope curve is defined by p2186, p2188 and p2190.		
p2186[0...n]	Load monitoring torque threshold 1, lower / M_thresh 1 lower		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 20000000.00 [Nm]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Nm]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2186 < p2185 Refer to: p2182, p2185 Refer to: A07926		
Note:	The lower envelope curve is defined by p2186, p2188 and p2190.		
p2187[0...n]	Load monitoring force threshold 2, upper / M_thresh 2 upper		
SERVO (Extended sig, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 100000.00 [N]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 100000.00 [N]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2187 > p2188 Refer to: p2183, p2188 Refer to: A07926		
Note:	The upper envelope curve is defined by p2185, p2187 and p2189.		
p2187[0...n]	Load monitoring torque threshold 2, upper / M_thresh 2 upper		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 20000000.00 [Nm]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 10000000.00 [Nm]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2187 > p2188 Refer to: p2183, p2188 Refer to: A07926		
Note:	The upper envelope curve is defined by p2185, p2187 and p2189.		

p2188[0...n]	Load monitoring force threshold 2, lower / M_thresh 2 lower		
SERVO (Extended sig, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 100000.00 [N]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [N]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2188 < p2187 Refer to: p2183, p2187 Refer to: A07926		
Note:	The lower envelope curve is defined by p2186, p2188 and p2190.		
p2188[0...n]	Load monitoring torque threshold 2, lower / M_thresh 2 lower		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 20000000.00 [Nm]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Nm]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2188 < p2187 Refer to: p2183, p2187 Refer to: A07926		
Note:	The lower envelope curve is defined by p2186, p2188 and p2190.		
p2189[0...n]	Load monitoring force threshold 3, upper / M_thresh 3 upper		
SERVO (Extended sig, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 100000.00 [N]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 100000.00 [N]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2189 > p2190 Refer to: p2184, p2190 Refer to: A07926		
Note:	The upper envelope curve is defined by p2185, p2187 and p2189.		
p2189[0...n]	Load monitoring torque threshold 3, upper / M_thresh 3 upper		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 20000000.00 [Nm]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 10000000.00 [Nm]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2189 > p2190 Refer to: p2184, p2190 Refer to: A07926		
Note:	The upper envelope curve is defined by p2185, p2187 and p2189.		

p2190[0...n]	Load monitoring force threshold 3, lower / M_thresh 3 lower		
SERVO (Extended sig, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 100000.00 [N]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [N]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2190 < p2189 Refer to: p2184, p2189 Refer to: A07926		
Note:	The lower envelope curve is defined by p2186, p2188 and p2190.		
p2190[0...n]	Load monitoring torque threshold 3, lower / M_thresh 3 lower		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 20000000.00 [Nm]	Access level: 3 Func. diagram: 8013 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Nm]
Description:	Sets the speed/torque / velocity/force envelope curve for the load monitoring.		
Dependency:	The following applies: p2190 < p2189 Refer to: p2184, p2189 Refer to: A07926		
Note:	The lower envelope curve is defined by p2186, p2188 and p2190.		
p2192[0...n]	Load monitoring delay time / Load monit t_del		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 65.00 [s]	Access level: 3 Func. diagram: 8013 Unit selection: - Expert list: 1 Factory setting 10.00 [s]
Description:	Sets the delay time to evaluate the load monitoring.		
p2194[0...n]	Torque threshold value 2 / M_thresh val 2		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min 0.00 [%]	Calculated: CALC_MOD_LIM_REF Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.00 [%]	Access level: 2 Func. diagram: 8012 Unit selection: - Expert list: 1 Factory setting 90.00 [%]
Description:	Sets the torque/force threshold value for the signal "Torque utilization < torque threshold value 2" (BO: r2199.11). The message "torque setpoint < p2174" (BO: r2198.10) and "torque utilization < p2194" (BO: r2199.11) are only evaluated after the run-up and the delay time has expired.		
Dependency:	Refer to: r0033, p2195, r2199		

p2194[0...n]	Force threshold value 2 / F_thresh val 2		
SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8012
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 100.00 [%]	Factory setting 90.00 [%]
Description:	Sets the torque/force threshold value for the signal "Torque/force utilization < torque/force threshold value 2" (BO: r2199.11). The message "torque/force setpoint < p2174" (BO: r2198.10) and "torque/force utilization < p2194" (BO: r2199.11) are only evaluated after the run-up and the delay time has expired.		
Dependency:	Refer to: r0033, p2195, r2199		
p2195[0...n]	Torque utilization switch-off delay / M_util t_off		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8012
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 1000.0 [ms]	Factory setting 800.0 [ms]
Description:	Sets the switch-off delay time for the negated signal "run-up completed". The message "torque setpoint < p2174" (BO: r2198.10) and "torque utilization < p2194" (BO: r2199.11) are only evaluated after the run-up and the delay time has expired.		
Dependency:	Refer to: p2174, p2194		
p2195[0...n]	Force utilization switch-off delay / F_util t_off		
SERVO (Lin)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8012
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 1000.0 [ms]	Factory setting 800.0 [ms]
Description:	Sets the switch-off delay time for the negated signal "run-up completed". The message "force setpoint < p2174" (BO: r2198.10) and "force utilization < p2194" (BO: r2199.11) are only evaluated after the run-up and the delay time has expired.		
Dependency:	Refer to: p2174, p2194		
p2196[0...n]	Torque utilization scaling / M_util scal		
SERVO, VECTOR	Can be changed: C2(1, 3), U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 1000.00 [%]	Factory setting 100.00 [%]
Description:	Sets the scaling factor for torque utilization (r0033).		

r2197.1...7 CO/BO: Status word monitoring 1 / ZSW monitor 1

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2534
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the first status word for monitoring functions.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	01	n_act <= speed threshold value 2	Yes	No	8010
	02	n_act > speed threshold value 2	Yes	No	8010
	03	n_act >= 0	Yes	No	8010
	06	n_act > n_max	Yes	No	8010
	07	Speed setp - act val deviation in tolerance t_off	Yes	No	8010

Note: Re bit 01, 02:
The threshold value is set in p2155 and the hysteresis in p2140.
Re bit 03:
The hysteresis is set in p2150.
Re bit 06:
The hysteresis is set in p2162.
Re bit 07:
The threshold value is set in p2163 and the hysteresis is set in p2164.

r2197.1...7 CO/BO: Status word monitoring 1 / ZSW monitor 1

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2534
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the first status word for monitoring functions.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	01	v_act <= velocity threshold value 2"	Yes	No	8010
	02	v_act > velocity threshold value 2	Yes	No	8010
	03	v_act >= 0	Yes	No	8010
	06	v_act > v_max	Yes	No	8010
	07	Velocity setpoint - actual value deviation in tolerance t_off	Yes	No	8010

Note: Re bit 01, 02:
The threshold value is set in p2155 and the hysteresis in p2140.
Re bit 03:
The hysteresis is set in p2150.
Re bit 06:
The hysteresis is set in p2162.
Re bit 07:
The threshold value is set in p2163 and the hysteresis is set in p2164.

r2198.4...12	CO/BO: Status word monitoring 2 / ZSW monitor 2				
SERVO	Can be changed: -	Calculated: -	Access level: 2		
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2536		
	P-Group: Messages	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the second status word for monitoring functions.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	04	n_set < p2161	Yes	No	8010
	05	n_set > 0	Yes	No	8010
	06	Motor locked	Yes	No	8012
	10	M_set < torque threshold value 1	Yes	No	8012
	11	Load monitoring signals an alarm	Yes	No	8013
	12	Load monitoring signals a fault condition	Yes	No	8013
Note:	Re bit 07: For servo drives, bit 07 is not used and is always inactive. Re bit 10: The torque threshold value 1 is set in p2174. Re bit 12: If the fault condition is removed, bit 12 is reset to 0. This is also the case even if the alarm message is still present.				

r2198.4...12	CO/BO: Status word monitoring 2 / ZSW monitor 2			
SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 2	
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1530, 2536	
	P-Group: Messages	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	-	
Description:	Displays the second status word for monitoring functions.			
Bit field:	Bit	Signal name	1 signal	0 signal
	04	n_set < p2161	Yes	No
	05	v_set > 0	Yes	No
	06	Motor locked	Yes	No
	10	Force setpoint < force threshold value 1	Yes	No
	11	Load monitoring signals an alarm	Yes	No
	12	Load monitoring signals a fault condition	Yes	No
Note:	Re bit 07: For servo drives, bit 07 is not used and is always inactive. Re bit 10: The force threshold value 1 is set in p2174.			

r2198.4...12 CO/BO: Status word monitoring 2 / ZSW monitor 2

VECTOR

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1530, 2536**P-Group:** Messages**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the second status word for monitoring functions.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
04	n_set < p2161	Yes	No	8010
05	n_set > 0	Yes	No	8010
06	Motor locked	Yes	No	8012
07	Motor stalled	Yes	No	8012
10	M_set < torque threshold value 1	Yes	No	8012
11	Load monitoring signals an alarm	Yes	No	8013
12	Load monitoring signals a fault condition	Yes	No	8013

Note:

Re bit 07:

For servo drives, bit 07 is not used and is always inactive.

Re bit 10:

The torque threshold value 1 is set in p2174.

Re bit 12:

If the fault condition is removed, bit 12 is reset to 0. This is also the case even if the alarm message is still present.

r2199.0...11 CO/BO: Status word monitoring 3 / ZSW monitor 3

SERVO

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1530, 2537**P-Group:** Messages**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the third status word for monitoring functions.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	n_act < speed threshold value 3	Yes	No	8010
01	f or n comparison value reached or exceeded	Yes	No	8010
04	Speed setp - act val deviation in tolerance t_on	Yes	No	8010
05	Ramp-up/ramp-down completed	Yes	No	8010
06	Current below the zero current threshold	Yes	No	-
11	Torque utilization < torque threshold value 2	Yes	No	8012

Note:

Re bit 00:

The speed threshold value 3 is set in p2161.

Re bit 01:

The comparison value is set in p2141. We recommend setting the hysteresis (p2142) for canceling the bit to a value lower than that in p2141. Otherwise, the bit will never be reset.

Re bit 11:

The torque threshold value 2 is set in p2194.

r2199.0...11 CO/BO: Status word monitoring 3 / ZSW monitor 3

SERVO (Lin)

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1530, 2537**P-Group:** Messages**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the third status word for monitoring functions.

Bit field:**Bit Signal name****1 signal****0 signal****FP**

00 |v_act| < velocity threshold value 3

Yes

No

8010

01 f or v comparison value reached or exceeded

Yes

No

8010

04 Velocity setpoint - actual value deviation in tolerance t_{on}

Yes

No

8010

05 Ramp-up/ramp-down completed

Yes

No

8010

06 Current below the zero current threshold

Yes

No

-

11 Force utilization < force threshold value 2

Yes

No

8012

Note:

Re bit 00:

The velocity threshold value 3 is set in p2161.

Re bit 01:

The comparison value is set in p2141.

Re bit 11:

The force threshold value 2 is set in p2194.

r2199.0...11 CO/BO: Status word monitoring 3 / ZSW monitor 3

VECTOR

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** 1530, 2537, 8018**P-Group:** Messages**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description:

Displays the third status word for monitoring functions.

Bit field:**Bit Signal name****1 signal****0 signal****FP**

00 |n_act| < speed threshold value 3

Yes

No

8010

01 f or n comparison value reached or exceeded

Yes

No

8010

04 Speed setp - act val deviation in tolerance t_{on}

Yes

No

8010

05 Ramp-up/ramp-down completed

Yes

No

8010

06 Current below the zero current threshold

Yes

No

-

07 Speed deviation model/external in tolerance

Yes

No

8012

11 Torque utilization < torque threshold value 2

Yes

No

8012

Note:

Re bit 00:

The speed threshold value 3 is set in p2161.

Re bit 01:

The comparison value is set in p2141. We recommend setting the hysteresis (p2142) for canceling the bit to a value lower than that in p2141. Otherwise, the bit will never be reset.

Re bit 11:

The torque threshold value 2 is set in p2194.

p2200[0...n] BI: Technology controller enable / Tec_ctrl enable

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source to switch in/switch out the technology controller.
The technology controller is switched in with a 1 signal.

p2201[0...n] CO: Technology controller, fixed value 1 / Tec_ctrl fix val 1

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-200.00 [%]	200.00 [%]	10.00 [%]

Description: Sets the value for fixed value 1 of the technology controller.

Dependency: Refer to: p2220, p2221, p2222, p2223, r2224, r2229

Notice: A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

p2202[0...n] CO: Technology controller, fixed value 2 / Tec_ctrl fix val 2

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-200.00 [%]	200.00 [%]	20.00 [%]

Description: Sets the value for fixed value 2 of the technology controller.

Dependency: Refer to: p2220, p2221, p2222, p2223, r2224, r2229

Notice: A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

p2203[0...n] CO: Technology controller, fixed value 3 / Tec_ctrl fix val 3

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-200.00 [%]	200.00 [%]	30.00 [%]

Description: Sets the value for fixed value 3 of the technology controller.

Dependency: Refer to: p2220, p2221, p2222, p2223, r2224, r2229

Notice: A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

p2204[0...n]	CO: Technology controller, fixed value 4 / Tec_ctrl fix val 4		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 40.00 [%]
Description:	Sets the value for fixed value 4 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p2205[0...n]	CO: Technology controller, fixed value 5 / Tec_ctrl fix val 5		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 50.00 [%]
Description:	Sets the value for fixed value 5 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p2206[0...n]	CO: Technology controller, fixed value 6 / Tec_ctrl fix val 6		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 60.00 [%]
Description:	Sets the value for fixed value 6 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p2207[0...n]	CO: Technology controller, fixed value 7 / Tec_ctrl fix val 7		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 70.00 [%]
Description:	Sets the value for fixed value 7 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p2208[0...n]	CO: Technology controller, fixed value 8 / Tec_ctrl fix val 8		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 80.00 [%]
Description:	Sets the value for fixed value 8 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p2209[0...n]	CO: Technology controller, fixed value 9 / Tec_ctrl fix val 9		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 90.00 [%]
Description:	Sets the value for fixed value 9 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p2210[0...n]	CO: Technology controller, fixed value 10 / Tec_ctrl fix val10		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 100.00 [%]
Description:	Sets the value for fixed value 10 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p2211[0...n]	CO: Technology controller, fixed value 11 / Tec_ctrl fix val11		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 110.00 [%]
Description:	Sets the value for fixed value 11 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p2212[0...n]	CO: Technology controller, fixed value 12 / Tec_ctrl fix val12		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 120.00 [%]
Description:	Sets the value for fixed value 12 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p2213[0...n]	CO: Technology controller, fixed value 13 / Tec_ctrl fix val13		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 130.00 [%]
Description:	Sets the value for fixed value 13 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p2214[0...n]	CO: Technology controller, fixed value 14 / Tec_ctrl fix val14		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 140.00 [%]
Description:	Sets the value for fixed value 14 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
p2215[0...n]	CO: Technology controller, fixed value 15 / Tec_ctrl fix val15		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7950
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 150.00 [%]
Description:	Sets the value for fixed value 15 of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222, p2223, r2224, r2229		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		

p2216[0...n]	Technology controller fixed value selection method / Tec_ctr FixVal sel		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1	Max 2	Factory setting 2
Description:	Selects the method that can be used to select the fixed setpoints.		
Value:	1: Fixed value selection direct 2: Fixed value selection binary		
p2220[0...n]	BI: Technology controller fixed value selection bit 0 / Tec_ctrl sel bit 0		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 7950
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source to select the fixed value of the technology controller.		
Dependency:	Refer to: p2221, p2222, p2223		
p2221[0...n]	BI: Technology controller fixed value selection bit 1 / Tec_ctrl sel bit 1		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 7950
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source to select the fixed value of the technology controller.		
Dependency:	Refer to: p2220, p2222, p2223		
p2222[0...n]	BI: Technology controller fixed value selection bit 2 / Tec_ctrl sel bit 2		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 7950
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source to select the fixed value of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2223		
p2223[0...n]	BI: Technology controller fixed value selection bit 3 / Tec_ctrl sel bit 3		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 7950
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source to select the fixed value of the technology controller.		
Dependency:	Refer to: p2220, p2221, p2222		

r2224	CO: Technology controller, fixed value effective / Tec_ctr FixVal eff				
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: -	Calculated: -	Access level: 2		
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7950		
	P-Group: Technology	Units group: 9_1	Unit selection: p0595		
	Not for motor type: -	Scaling: PERCENT	Expert list: 1		
	Min	Max	Factory setting		
	- [%]	- [%]	- [%]		
Description:	Displays the selected and effective fixed value of the technology controller.				
Dependency:	Refer to: r2229				
r2225.0	CO/BO: Technology controller fixed value selection status word / Tec_ctrl FW status				
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -		
	P-Group: Technology	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Status word of the fixed value selection of the technology controller.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Technology controller fixed value selected	Yes	No	-
r2229	Technology controller number actual / Tec_ctrl No. act				
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: -	Calculated: -	Access level: 2		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 7950		
	P-Group: Technology	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the number of the selected fixed setpoint of the technology controller.				
Dependency:	Refer to: r2224				
p2230[0...n]	Technology controller motorized potentiometer configuration / Tec_ctr mop config				
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3		
	Data type: Unsigned32	Dynamic index: DDS, p0180	Func. diagram: 7954		
	P-Group: Technology	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0100 bin		
Description:	Sets the configuration for the motorized potentiometer of the technology controller.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Data save active	Yes	No	-
	02	Initial rounding-off active	Yes	No	-
	03	Non-volatile data save active for p2230.0 = 1	Yes	No	-
Dependency:	Refer to: r2231, p2240				
Notice:	The following prerequisites must be fulfilled in order to be able to save the setpoint in a non-volatile fashion:				
	- Firmware with V2.3 or higher.				
	- Control Unit 320 (CU320) with hardware version C or higher (module with NVRAM).				

Note:

Re bit 00:
 0: The setpoint for the motorized potentiometer is not saved and after ON is entered using p2240.
 1: The setpoint for the motorized potentiometer is saved and after ON is entered using r2231. In order to save in a non-volatile fashion, bit 03 should be set to 1.

Re bit 02:
 0: Without initial rounding-off
 1: With initial rounding-off.

The selected ramp-up/down time is correspondingly exceeded. The initial rounding-off is a sensitive way of specifying small changes (progressive reaction when keys are pressed). The jerk for initial rounding is independent of the ramp-up time and only depends on the selected maximum value (p2237).

It is calculated as follows:

$$r = 0.0001 * \text{MAX}(p2237, |p2238|) [\%] / 0.13^2 [s^2]$$

The jerk is effective until the maximum acceleration is reached ($a_{\text{max}} = p2237 [\%] / p2247 [s]$ or $a_{\text{max}} = p2238 [\%] / p2248 [s]$), after which the drive continues to run linearly with constant acceleration. The higher the maximum acceleration (the lower that p2247 is), the longer the ramp-up time increases with respect to the set ramp-up time.

Re bit 03:
 0: Non-volatile data save deactivated.
 1: The setpoint for the motorized potentiometer is saved in a non-volatile fashion (for p2230.0 = 1).

r2231 Technology controller motorized potentiometer setpoint memory / Tec_ctrl mop mem

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7954
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [%]	Max - [%]	Factory setting - [%]

Description: Displays the setpoint memory for the motorized potentiometer of the technology controller.
 For p2230.0 = 1, the last setpoint that was saved is entered after ON.

Dependency: Refer to: p2230

p2235[0...n] BI: Technology controller motorized potentiometer raise setpoint / Tec_ctrl mop raise

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 7954
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0

Description: Sets the signal source to increase the setpoint for the motorized potentiometer of the technology controller.

Dependency: Refer to: p2236

p2236[0...n] BI: Technology controller motorized potentiometer lower setpoint / Tec_ctrl mop lower

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 7954
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0

Description: Sets the signal source to reduce the setpoint for the motorized potentiometer of the technology controller.

Dependency: Refer to: p2235

p2237[0...n]	Technology controller motorized potentiometer maximum value / Tec_ctrl mop max		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7954
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 100.00 [%]
Description:	Sets the maximum value for the motorized potentiometer of the technology controller.		
Dependency:	Refer to: p2238		
p2238[0...n]	Technology controller motorized potentiometer minimum value / Tec_ctrl mop min		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7954
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting -100.00 [%]
Description:	Sets the minimum value for the motorized potentiometer of the technology controller.		
Dependency:	Refer to: p2237		
p2240[0...n]	Technology controller motorized potentiometer starting value / Tec_ctrl mop start		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7954
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 0.00 [%]
Description:	Sets the starting value for the motorized potentiometer of the technology controller. For p2230.0 = 0, this setpoint is entered after ON.		
Dependency:	Refer to: p2230		
r2245	CO: Technology controller mot. potentiometer setpoint before RFG / Tec_ctr mop befRFG		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7954
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min - [%]	Max - [%]	Factory setting - [%]
Description:	Sets the effective setpoint in front of the internal motorized potentiometer ramp-function generator of the technology controller.		
Dependency:	Refer to: r2250		

p2247[0...n]		Technology controller motorized potentiometer ramp-up time / Tec_ctr mop t_r-up			
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -		Access level: 2	
	Data type: FloatingPoint32	Dynamic index: DDS, p0180		Func. diagram: 7954	
	P-Group: Technology	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min 0.0 [s]	Max 1000.0 [s]		Factory setting 10.0 [s]	
Description:	Sets the ramp-up time for the internal ramp-function generator for the motorized potentiometer of the technology controller.				
Dependency:	Refer to: p2248				
Note:	The time is referred to 100 %. When the initial rounding-off is activated (p2230.2 = 1) the ramp-up is correspondingly extended.				
p2248[0...n]		Technology controller motorized potentiometer ramp-down time / Tec_ctrMop t_rdown			
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -		Access level: 2	
	Data type: FloatingPoint32	Dynamic index: DDS, p0180		Func. diagram: 7954	
	P-Group: Technology	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min 0.0 [s]	Max 1000.0 [s]		Factory setting 10.0 [s]	
Description:	Sets the ramp-down time for the internal ramp-function generator for the motorized potentiometer of the technology controller.				
Dependency:	Refer to: p2247				
Note:	The time is referred to 100 %. When the initial rounding-off is activated (p2230.2 = 1) the ramp-down is correspondingly extended.				
r2250		CO: Technology controller motorized potentiometer setpoint after RFG / Tec_ctr mop aftRFG			
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: -	Calculated: -		Access level: 2	
	Data type: FloatingPoint32	Dynamic index: -		Func. diagram: 7954	
	P-Group: Technology	Units group: 9_1		Unit selection: p0595	
	Not for motor type: -	Scaling: PERCENT		Expert list: 1	
	Min - [%]	Max - [%]		Factory setting - [%]	
Description:	Displays the effective setpoint after the internal ramp-function generator for the motorized potentiometer of the technology controller.				
Dependency:	Refer to: r2245				
p2252		Technology controller configuration / Tec_ctrl config			
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -		Access level: 3	
	Data type: Unsigned16	Dynamic index: -		Func. diagram: -	
	P-Group: Modulation	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min -	Max -		Factory setting 0111 bin	
Description:	Sets the configuration of the technology controller.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Ramp-up/down time independent of set-point sign	Yes	No	-
	01	Integrator independent of Kp	Yes	No	-
	02	Output signal without ramp active	Yes	No	-

Dependency: Refer to: p2257, p2258, p2280, p2285

Note: Re bit 00 = 0:

The ramp-down time (p2258) switches to the ramp-up time (p2257) when the sign for the output signal r2260 changes. When the sign changes, the output signal is kept at zero for one arithmetic cycle.

Re bit 00 = 1:

When r2260 exhibits a positive gradient, the ramp-up time (p2257) is active; when it exhibits a negative gradient, the ramp-down time (p2258) is active. The sign for r2260 does not have any effect on the ramp time.

Re bit 01 = 0:

The integration time of the PID controller is evaluated with the gain factor Kp (p2280) (p2285 = integral time).

Re bit 01 = 1:

The integration time of the PID controller is independent of the gain factor (p2285 = integration time) if p2280 > 0.

Re bit 02 = 0:

When the PID controller is deactivated via p2200, the output signal r2294 is reduced to zero via the ramp-down time p2293.

Re bit 02 = 1:

When the PID controller is deactivated via p2200, the output signal r2294 is set directly to zero.

p2253[0...n] CI: Technology controller setpoint 1 / Tec_ctrl setp 1

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -	Max -	Factory setting 0

Description: Sets the signal source for the setpoint 1 of the technology controller.

Dependency: Refer to: p2254, p2255

p2254[0...n] CI: Technology controller setpoint 2 / Tec_ctrl setp 2

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -	Max -	Factory setting 0

Description: Sets the signal source for the setpoint 2 of the technology controller.

Dependency: Refer to: p2253, p2256

p2255 Technology controller setpoint 1 scaling / Tec_ctrl set1 scal

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 100.00 [%]	Factory setting 100.00 [%]

Description: Sets the scaling for the setpoint 1 of the technology controller.

Dependency: Refer to: p2253

p2256	Technology controller setpoint 2 scaling / Tec_ctrl set2 scal		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 100.00 [%]	Factory setting 100.00 [%]
Description:	Sets the scaling for the setpoint 2 of the technology controller.		
Dependency:	Refer to: p2254		
p2257	Technology controller, ramp-up time / Tec_ctrl t_ramp-up		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [s]	Max 650.00 [s]	Factory setting 1.00 [s]
Description:	Sets the ramp-up time of the technology controller.		
Dependency:	Refer to: p2252, p2258		
Note:	The ramp-up time is referred to 100 %.		
p2258	Technology controller ramp-down time / Tec_ctrl t_ramp-dn		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [s]	Max 650.00 [s]	Factory setting 1.00 [s]
Description:	Sets the ramp-down time of the technology controller.		
Dependency:	Refer to: p2252, p2257		
Note:	The ramp-down time is referred to 100 %.		
r2260	CO: Technology controller setpoint after ramp-function generator / Tec_ctr set aftRFG		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min - [%]	Max - [%]	Factory setting - [%]
Description:	Sets the setpoint after the ramp-function generator of the technology controller.		
p2261	Technology controller setpoint filter time constant / Tec_ctrl set T		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 60.000 [s]	Factory setting 0.000 [s]
Description:	Sets the time constant for the setpoint filter (PT1) of the technology controller.		

r2262	CO: Technology controller setpoint after filter / Tec_ctr set aftFlt		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Technology Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: 9_1 Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: 7958 Unit selection: p0595 Expert list: 1 Factory setting - [%]
Description:	Displays the smoothed setpoint after the setpoint filter (PT1) of the technology controller.		
p2263	Technology controller type / Tec_ctrl type		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T Data type: Integer16 P-Group: Technology Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: 7958 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the technology controller type.		
Value:	0: D component in the actual value signal 1: D component in the fault signal		
p2264[0...n]	CI: Technology controller actual value / Tec_ctrl act val		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T Data type: Unsigned32 / FloatingPoint32 P-Group: Technology Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: PERCENT Max -	Access level: 2 Func. diagram: 7958 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the actual value of the technology controller.		
p2265	Technology controller actual value filter time constant / Tec_ctrl act T		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Technology Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 60.000 [s]	Access level: 2 Func. diagram: 7958 Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the time constant for the actual value filter (PT1) of the technology controller.		
r2266	CO: Technology controller actual value after filter / Tec_ctr act aftFlt		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Technology Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: 9_1 Scaling: PERCENT Max - [%]	Access level: 2 Func. diagram: 7958 Unit selection: p0595 Expert list: 1 Factory setting - [%]
Description:	Displays the smoothed actual value after the filter (PT1) of the technology controller		

p2267	Technology controller upper limit actual value / Tec_ctrl u_lim act		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 100.00 [%]
Description:	Upper limit for the actual value signal of the technology controller.		
Dependency:	Refer to: p2264, p2265, p2271 Refer to: F07426		
Notice:	If the actual value exceeds the parameter value, this results in fault F7426.		

p2268	Technology controller lower limit actual value / Tec_ctrl l_lim act		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: 9_1	Unit selection: p0595
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting -100.00 [%]
Description:	Lower limit for the actual value signal of the technology controller.		
Dependency:	Refer to: p2264, p2265, p2271 Refer to: F07426		
Notice:	If the actual value falls below the parameter value, this results in fault F7426.		

p2269	Technology controller gain actual value / Tech_ctrl gain act		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 500.00 [%]	Factory setting 100.00 [%]
Description:	Scaling factor for the actual value of the technology controller.		
Dependency:	Refer to: p2264, p2265, p2267, p2268, p2271		
Note:	For 100%, the actual value is not changed.		

p2270	Technology controller actual value function selection / Tech_ctrl act_fct		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 3	Factory setting 0
Description:	Selects an arithmetic function that should be applied to the actual value signal of the technology controller.		
Value:	0: No function 1: Root function (root from x) 2: Square function (x * x) 3: Cube function (x * x * x)		
Dependency:	Refer to: p2264, p2265, p2267, p2268, p2269, p2271		

p2271	Technology controller actual value inversion (sensor type) / Tech_ctrl act inv		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T Data type: Integer16 P-Group: Technology Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the inversion of the actual value signal of the technology controller. This depends on the type of the actual value sensor.		
Value:	0: No inversion 1: Inversion of technology controller actual value signal		
Caution:	 If the actual value inversion is incorrectly selected, then the closed-loop control with the technology controller can become unstable and can oscillate. The correct setting can be determined as follows: - inhibit the technology controller (P2200 = 0). - increase the motor speed and in so doing, measure the actual value signal (of the technology controller). - if the actual value increases with increasing motor speed, then the inversion should be switched out. - if the actual value decreases with increasing motor frequency, then the inversion should be set.		
r2272	CO: Technology controller actual value scaled / Tech_ctrl act scal		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Technology Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: 9_1 Scaling: PERCENT Max - [%]	Access level: 2 Func. diagram: 7958 Unit selection: p0595 Expert list: 1 Factory setting - [%]
Description:	Displays the scaled actual value signal of the technology controller.		
Dependency:	Refer to: p2264, p2265, r2266, p2267, p2268, p2269, p2270, p2271		
r2273	CO: Technology controller error / Tec_ctrl error		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Technology Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: 9_1 Scaling: PERCENT Max - [%]	Access level: 2 Func. diagram: 7958 Unit selection: p0595 Expert list: 1 Factory setting - [%]
Description:	Displays the error (system deviation) between the setpoint and actual value of the technology controller.		
Dependency:	Refer to: p2263		
p2274	Technology controller differentiation, time constant / Tec_ctrl D comp T		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Technology Not for motor type: - Min 0.000 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 60.000 [s]	Access level: 2 Func. diagram: 7958 Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the time constant for the differentiation (D component) of the technology controller.		
Note:	p2274 = 0: Differentiation is disabled.		

p2280	Technology controller proportional gain / Tec_ctrl Kp		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000	Max 1000.000	Factory setting 1.000
Description:	Sets the proportional gain (P component) of the technology controller.		
Dependency:	Refer to: p2252		
Note:	p2280 = 0: The proportional gain is disabled.		
p2285	Technology controller integral time / Tec_ctrl Tn		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 60.000 [s]	Factory setting 0.000 [s]
Description:	Sets the integral time (I component, integrating time constant) of the technology controller.		
Dependency:	Refer to: p2252		
Note:	p2285 = 0: The integral time is disabled.		
p2286[0...n]	BI: Hold technology controller integrator / Tec_ctr integ stop		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: CDS, p0170	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source to hold the integrator for the technology controller.		
p2289[0...n]	CI: Technology controller pre-control signal / Tec_ctrl prectrl		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source for the pre-control signal of the technology controller.		
p2291	CO: Technology controller maximum limiting / Tec_ctrl max_limit		
SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -200.00 [%]	Max 200.00 [%]	Factory setting 100.00 [%]
Description:	Sets the maximum limit of the technology controller.		
Dependency:	Refer to: p2292		

Caution:

The maximum limit must always be greater than the minimum limit (p2291 > p2292).



p2292

CO: Technology controller minimum limiting / Tec_ctrl min_lim

SERVO (Tech_ctrl),
VECTOR (Tech_ctrl)

Can be changed: U, T

Calculated: -

Access level: 2

Data type: FloatingPoint32

Dynamic index: -

Func. diagram: 7958

P-Group: Technology

Units group: -

Unit selection: -

Not for motor type: -

Scaling: PERCENT

Expert list: 1

Min

Max

Factory setting

-200.00 [%]

200.00 [%]

0.00 [%]

Description:

Sets the minimum limit of the technology controller.

Dependency:

Refer to: p2291

Caution:

The maximum limit must always be greater than the minimum limit (p2291 > p2292).



p2293

Technology controller ramp-up/ramp-down time / Tec_ctr ramp up/dn

SERVO (Tech_ctrl),
VECTOR (Tech_ctrl)

Can be changed: U, T

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dynamic index: -

Func. diagram: 7958

P-Group: Technology

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

0.00 [s]

100.00 [s]

1.00 [s]

Description:

Sets the ramping time for the output signal of the technology controller.

Dependency:

Refer to: p2291, p2292

Note:

The time refers to the set maximum and minimum limits (p2291, p2292).

r2294

CO: Technology controller output signal / Tec_ctrl outp_sig

SERVO (Tech_ctrl),
VECTOR (Tech_ctrl)

Can be changed: -

Calculated: -

Access level: 2

Data type: FloatingPoint32

Dynamic index: -

Func. diagram: 7958

P-Group: Technology

Units group: -

Unit selection: -

Not for motor type: -

Scaling: PERCENT

Expert list: 1

Min

Max

Factory setting

- [%]

- [%]

- [%]

Description:

Displays the output signal of the technology controller.

Dependency:

Refer to: p2295

p2295

CO: Technology controller output scaling / Tec_ctrl outp_scal

SERVO (Tech_ctrl),
VECTOR (Tech_ctrl)

Can be changed: U, T

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dynamic index: -

Func. diagram: 7958

P-Group: Technology

Units group: -

Unit selection: -

Not for motor type: -

Scaling: PERCENT

Expert list: 1

Min

Max

Factory setting

-100.00 [%]

100.00 [%]

100.00 [%]

Description:

Sets the scaling for the output signal of the technology controller.

p2296[0...n] CI: Technology controller output scaling / Tec_ctrl outp scal

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	2295[0]

Description: Sets the signal source for the scaling value of the technology controller.

Dependency: Refer to: p2295

p2297[0...n] CI: Technology controller maximum limit signal source / Tec_ctrl m_lm s_sc

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	2291[0]

Description: Sets the signal source for the maximum limiting of the technology controller.

Dependency: Refer to: p2291

p2298[0...n] CI: Technology controller minimum limit signal source / Tec_ctrl min_l s_s

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	2292[0]

Description: Sets the signal source for the minimum limiting of the technology controller.

Dependency: Refer to: p2292

p2299[0...n] CI: Technology controller limit offset / Tech_ctrl lim offs

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: CDS, p0170	Func. diagram: 7958
	P-Group: Technology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the offset of the output limiting of the technology controller.

r2349.0...11 CO/BO: Technology controller status word / Tec_ctrl stat word

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: - Data type: Unsigned32 P-Group: Technology Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 7958 Unit selection: - Expert list: 1 Factory setting -
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Description: Displays the status word of the technology controller.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Technology controller deactivated	Yes	No	-
	01	Technology controller limited	Yes	No	-
	02	Technology controller, motorized potentiometer limited max.	Yes	No	-
	03	Technology controller, motorized potentiometer limited min.	Yes	No	-
	08	Technology controller actual value at the minimum	Yes	No	-
	09	Technology controller actual value at the maximum	Yes	No	-
	10	Technology controller output at the minimum	Yes	No	-
	11	Technology controller output at the maximum	Yes	No	-

p2369 BI: Staging control word / Staging STW

VECTOR (Tech_ctrl)	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
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Description: Sets the signal source to select the "staging" function.

When the function is selected, monitoring of the switches is deactivated with the "bypass" function. This means that the Motor Module can be connected to other motors via an external control without switch monitoring responding.

p2398 Hibernation operating mode / Hib op_mode

SERVO (Tech_ctrl), VECTOR (Tech_ctrl)	Can be changed: T Data type: Integer16 P-Group: Technology Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
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Description: Sets the operating mode for the hibernation function.

Value:
 0: Hibernation inhibited
 1: Hibernation active

Dependency: Refer to: p2200

Caution: When this function is active, the motor can start again automatically.



Note: When the hibernation function (p2398 = 1) is activated, its behavior is defined as to whether the technology controller is additionally switched in (closed loop) or switched out (open loop).
p2200 is used to define the technology controller enable and p2251 its operating mode.
p2200 = 0, p2251 = 0, 1:
Hibernation operates without technology controller (open loop)
p2200 = 1, p2251 = 0:
Hibernation operates with technology controller (closed loop)
p2200 = 1, p2251 = 1:
Hibernation operates without technology controller (open loop) as its output is only used as supplementary setpoint and not as main speed setpoint.

p2502[0...n] LR encoder assignment / Encoder assignment			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: C2(25) Data type: Integer16 P-Group: Closed loop position control Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 3	Access level: 1 Func. diagram: 4010 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the assigned encoder. The actual value preprocessing and the closed-loop position control are carried out using the assigned encoder.		
Value:	0: No encoder 1: Encoder 1 2: Encoder 2 3: Encoder 3		
Dependency:	Refer to: p0187, p0188, p0189		
Notice:	For the setting p2502 = 0 (no encoder), closed-loop position control is not possible. This setting is only practical as supportive measure to implement encoderless closed-loop speed control (e.g. if the motor encoder is defective).		
Note:	The assigned encoder (p2502 = 1, 2, 3) must be allocated an encoder data set (p0187, p0188, p0189).		

p2503[0...n] LR length unit LU per 10 mm / LU per 10 mm			
SERVO (APC, Pos ctrl), VECTOR (Pos ctrl)	Can be changed: C2(25) Data type: Unsigned32 P-Group: Closed loop position control Not for motor type: - Min 1 [LU]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 2147483647 [LU]	Access level: 1 Func. diagram: 4010 Unit selection: - Expert list: 1 Factory setting 10000 [LU]
Description:	Sets the neutral length units LU per 10 mm. Therefore, for a linear scale, a reference is established between the physical arrangement and the neutral length units LU used in the drive. Example: Linear scale, 10 mm should be broken down to units of μm (i.e. 1 LU = 1 μm). --> p2503 = 10000		
Note:	The assignment to the grid spacing can be achieved using this for a rotary axis with linear encoder.		

p2504[0...n] LR motor/load motor distance / Mot/load motor dis			
SERVO (APC, Lin, Pos ctrl)	Can be changed: C2(25) Data type: Unsigned32 P-Group: Closed loop position control Not for motor type: - Min 1	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1048576	Access level: 1 Func. diagram: 4010, 4704, 4711 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the motor distance for the gearbox factor between the motor shaft and load shaft. Gearbox factor = motor distance (p2504) / load path (p2505)		

Dependency: Refer to: p0432, p0433, p2505

Note: The gearbox factor between the encoder shaft and the motor shaft is set using p0432 and p0433.

p2504[0...n] LR motor/load motor revolutions / Mot/load motor rev

SERVO (APC, Pos ctrl), VECTOR (Pos ctrl)

Can be changed: C2(25)

Data type: Unsigned32

Calculated: -

Dynamic index: DDS, p0180

Access level: 1

Func. diagram: 4010, 4704, 4711

P-Group: Closed loop position control

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min
1

Max
1048576

Factory setting
1

Description: Sets the motor revolutions for the gearbox factor between the motor shaft and load shaft.

Gearbox factor = motor revolutions (p2504) / load revolutions (p2505)

Dependency: Refer to: p0432, p0433, p2505

Note: The gearbox factor between the encoder shaft and the motor shaft is set using p0432 and p0433.

p2505[0...n] LR motor/load motor revolutions / Mot/load motor rev

SERVO (APC, Pos ctrl), VECTOR (Pos ctrl)

Can be changed: C2(25)

Data type: Integer32

Calculated: -

Dynamic index: DDS, p0180

Access level: 1

Func. diagram: 4010, 4704, 4711

P-Group: Closed loop position control

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min
-1048576

Max
1048576

Factory setting
1

Description: Sets the load revolutions for the gearbox factor between the motor shaft and load shaft.

Gearbox factor = motor revolutions (p2504) / load revolutions (p2505)

Dependency: Refer to: p0432, p0433, p2504

Note: The gearbox factor between the encoder shaft and the motor shaft is set using p0432 and p0433.

p2506[0...n] LR length unit LU per load path / LU per load path

SERVO (APC, Lin, Pos ctrl)

Can be changed: C2(25)

Data type: Unsigned32

Calculated: -

Dynamic index: DDS, p0180

Access level: 1

Func. diagram: 4010

P-Group: Closed loop position control

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min
1 [LU]

Max
2147483647 [LU]

Factory setting
10000 [LU]

Description: Sets the neutral length units LU per load path.

Therefore, for a rotary encoder, a reference is established between the physical arrangement and the neutral length units LU used in the drive.

Example:

Rotary encoder, ballscrew with 10 mm/revolution, 10 mm should be broken down to units of μm (i.e. 1 LU = 1 μm).

--> One load path corresponds to 10000 LU

--> p2506 = 10000

Note: The position controller can only process position setpoints in the interpolator clock cycle (IPO clock cycle) in integer length units (LU, Length Unit). This is the reason that speed setpoints that are not a multiple integer of 1 LU per IPO clock cycle can only be realized as an average. The result speed setpoint steps are especially noticeable for a high loop gain or when the pre-control is active. Increasing p2506 counteracts this behavior.

p2506[0...n] LR length unit LU per load revolution / LU per load rev			
SERVO (APC, Pos ctrl), VECTOR (Pos ctrl)	Can be changed: C2(25)	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: DDS, p0180	Func. diagram: 4010
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1 [LU]	Max 2147483647 [LU]	Factory setting 10000 [LU]
Description:	Sets the neutral length units LU per load revolution. Therefore, for a rotary encoder, a reference is established between the physical arrangement and the neutral length units LU used in the drive. Example: Rotary encoder, ballscrew with 10 mm/revolution, 10 mm should be broken down to units of μm (i.e. 1 LU = 1 μm). --> One load revolution corresponds to 10000 LU --> p2506 = 10000		
Note:	The position controller can only process position setpoints in the interpolator clock cycle (IPO clock cycle) in integer length units (LU, Length Unit). This is the reason that speed setpoints that are not a multiple integer of 1 LU per IPO clock cycle can only be realized as an average. The result speed setpoint steps are especially noticeable for a high load gain or when the pre-control is active. Increasing p2506 counteracts this behavior.		

p2507[0...n] LR absolute encoder adjustment status / Abs_enc_adj stat			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: EDS, p0140	Func. diagram: 4010
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 3	Factory setting 1
Description:	Activating the adjustment and display of the status of the adjustment for absolute encoders.		
Value:	0: Error occurred while adjusting 1: Absolute encoder not adjusted 2: Absolute encoder not adjusted and encoder adjustment initiated 3: Absolute encoder adjusted		
Dependency:	Refer to: p2525, p2598, p2599		
Caution:	For rotating absolute encoders, when adjusting, a range is set up symmetrically around zero with half of the encoder range, within which the position must be re-established after powering down/powering up. In this range, it is only permissible that the encoder overflows. After the adjustment has been completed, it must be guaranteed that the range is not exited. The reason for this is that outside the range, there is no clear reference any longer between the encoder actual value and mechanical system. If the reference point (CI: p2598) lies in this range, then the position actual value is set when adjusting to the reference point. Otherwise, adjustment is canceled with F07443. There is no overflow for linear absolute encoders. This means that after the adjustment, the position can be re-established in the complete traversing range after powering down/powering up. When adjusting, the position actual value is set to the reference point.		
			
Note:	The encoder adjustment is initiated with p2507 = 2. The status is displayed using the other values. In order to permanently save the determined position offset (p2525) it must be saved in a non-volatile fashion (p0971, p0977). This adjustment can only be initiated for an absolute encoder.		

p2508[0...3]	BI: LR activate reference mark search / Ref_mark act		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 4010
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the function "activate reference mark search".		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p0490, p0495, p2502, p2509, r2684 Refer to: A07495		
Notice:	When activating the function "set position actual value" while the function "reference mark search" is activated, then the function "reference mark search" is automatically deactivated.		
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: BI: p2508[0] = r2684.0 The function can only be activated using a 0/1 signal if no reference function is active (r2526.2). If "reference mark search" and "measuring probe evaluation" are simultaneously activated, then no function is activated and the actual function is interrupted.		
p2509[0...3]	BI: LR activating measuring probe evaluation / MT_eval act		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 4010
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the function "activating the measuring probe evaluation". 0/1 signal: The function "activate measuring probe evaluation" is started.		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p0488, p0489, p0490, p2502, p2508, p2510, p2511, p2517, p2518 Refer to: A07495		
Notice:	When the "set position actual value" is activated while the function "measuring probe evaluation" is activated, then the function "measuring probe evaluation" is automatically deactivated.		
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: BI: p2509[0] = r2684.1 The function can only be activated using a 0/1 signal if no reference function is active (r2526.2). If "reference mark search" and "measuring probe evaluation" are simultaneously activated, then no function is activated and the actual function is interrupted.		

p2510[0...3]	BI: LR selecting measuring probe evaluation / MT_eval select		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3615, 4010
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to select the measuring probe. 1 signal = measuring probe 2 is activated for BI: p2509 = 0/1 edge. 0 signal = measuring probe 1 is activated for BI: p2509 = 0/1 edge.		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p2502, p2509, p2511		
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: BI: p2509[0] = r2684.1 The measuring probe is selected at the 0/1 signal transition at r2684.1 (flying referencing active).		
p2511[0...3]	BI: LR measuring probe evaluation edge / MT_eval edge		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3615, 4010
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the edge evaluation of the measuring probe. 1 signal = falling edge of the measuring probe (p2510) is activated for BI: p2509 = 0/1 edge. 0 signal = rising edge of the measuring probe (p2510) is activated for BI: p2509 = 0/1 edge.		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p2502, p2509, p2510		
p2512[0...3]	BI: LR pos. actual value preprocessing activate corr. value (edge) / ActVal_prepCorrAct		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 4010, 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the function "activate position actual value preprocessing, corrective value (edge)". 0/1 signal: The corrective value available through CI: p2513 is activated.		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p2502, p2513, r2684		
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: BI: p2512[0] = r2684.7		

p2513[0...3]			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	CI: LR Position actual value preprocessing, corrective value / Act val_prep corr		
	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 4010, 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the corrective value for position actual value preprocessing.		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p2502, p2512, r2521, r2685		
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: CI: p2513[0] = r2685 For BI: p2512[0] = 0/1 signal, the position actual value (CO: r2521[0]) is corrected corresponding to the value via CI: p2513[0]. In so doing, the sign of the corrective value present is taken into account.		

p2514[0...3]			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	BI: LR activate position actual value setting / s_act setting act		
	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 4010
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to activate the function "set position actual value".		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p2502, p2515 Refer to: A07495, A07497		
Warning:	As long as the position actual value is set, encoder increments that are received are not evaluated. In this state, any position difference cannot be corrected!		
			
Notice:	When the function "set position actual value" is activated while the function "reference mark search" or "measuring probe evaluation" is activated, then the corresponding function is deactivated.		
Note:	BI: p2514 = 1 signal: The position actual value is set to the setting value in CI: p2515. Alarm A07497 "position setting value activated" is output. Encoder increments that are received in the meantime, are not taken into account. BI: p2514 = 1/0 signal: The position actual value preprocessing is activated and is based on the setting value.		

p2515[0...3]	CI: LR position actual setting, setting value / s_act set setVal		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T Data type: Unsigned32 / Integer32 P-Group: Closed loop position control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 4010 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the setting value of the function "setting position actual value".		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p2502, p2514		
p2516[0...3]	CI: LR position offset / Position offset		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T Data type: Unsigned32 / Integer32 P-Group: Closed loop position control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 4010 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the position offset.		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p2502, r2667		
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: CI: p2516[0] = r2667		
p2517[0...2]	LR direct measuring probe 1 / Direct MT 1		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T Data type: Integer16 P-Group: Closed loop position control Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 18	Access level: 3 Func. diagram: 4010 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal for direct measuring probe 1. The direct measuring probe can either be parameterized as a non-cyclic (value 1 ... 8) or a cyclic (value 11 ... 18) measuring probe. After it has been activated via BI: p2509 = 0/1 signal, the non-cyclic measuring probe measures once and can be used with EPOS. After it has been activated via the p2509 = 1 signal, the cyclic measuring probe measures cyclically and cannot be used with EPOS. In order to process signals faster, the direct measuring probe bypasses the handshake technique via the encoder control word and encoder status word.		

Value:	0: No measuring probe 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9) 11: DI/DO 9 cyclic 12: DI/DO 10 cyclic 13: DI/DO 11 cyclic 14: DI/DO 13 cyclic 15: DI/DO 14 cyclic 16: DI/DO 15 cyclic 17: DI/DO 8 cyclic 18: DI/DO 12 cyclic
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3
Dependency:	Refer to: p0490, p0728, p2509, p2510, p2511
Notice:	To select the values: For CU310, Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).
Note:	DI/DO: Bidirectional digital input/output The terminal must be set as input (p0728). If a parameter change is rejected, a check should be performed as to whether the input terminal is already being used in p0488, p0489, p0493, p0495, p0580, or p0680. Direct measurement via p2517 has a higher priority than measurements via p0488. For the direct measuring probe evaluation, the DP clock cycle must be integer multiple of the position controller clock cycle.

p2518[0...2]	LR direct measuring probe 2 / Direct MT 2		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T Data type: Integer16 P-Group: Closed loop position control Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 18	Access level: 3 Func. diagram: 4010 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal for direct measuring probe 2. The direct measuring probe can either be parameterized as a non-cyclic (value 1 ... 6) or a cyclic (value 11 ... 16) measuring probe. After it has been activated via BI: p2509 = 0/1 signal, the non-cyclic measuring probe measures once and can be used with EPOS. After it has been activated via the p2509 = 1 signal, the cyclic measuring probe measures cyclically and cannot be used with EPOS. In order to process signals faster, the direct measuring probe bypasses the handshake technique via the encoder control word and encoder status word.		

Value:	0: No meas probe 1: DI/DO 9 (X122.10) 2: DI/DO 10 (X122.12) 3: DI/DO 11 (X122.13) 4: DI/DO 13 (X132.10) 5: DI/DO 14 (X132.12) 6: DI/DO 15 (X132.13) 7: DI/DO 8 (X122.9) 8: DI/DO 12 (X132.9) 11: DI/DO 9 cyclic 12: DI/DO 10 cyclic 13: DI/DO 11 cyclic 14: DI/DO 13 cyclic 15: DI/DO 14 cyclic 16: DI/DO 15 cyclic 17: DI/DO 8 cyclic 18: DI/DO 12 cyclic
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3
Dependency:	Refer to: p0490, p0728, p2509, p2510, p2511
Notice:	To select the values: For CU310, Cx32, NX10 and NX15, only DI/DO 8, 9, 10, 11 can be selected as fast inputs (refer to the Equipment Manual).
Note:	DI/DO: Bidirectional digital input/output The terminal must be set as input (p0728). If a parameter change is rejected, a check should be performed as to whether the input terminal is already being used in p0488, p0489, p0493, p0495, p0580, or p0680. Direct measurement via p2518 has a higher priority than measurements via p0489. For the direct measuring probe evaluation, the DP clock cycle must be integer multiple of the position controller clock cycle.

p2519[0...n]	LR position actual value preprocessing config. DDS changeover / s_act config DDS		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T Data type: Integer16 P-Group: Closed loop position control Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 5	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the behavior of the position actual value preprocessing for the position controller for a DDS changeover. Re p2519 = 1: In the following cases, for a DDS changeover, the actual position actual value becomes invalid and the reference point is reset: - the EDS effective for the closed-loop position control changes. - the encoder assignment changes (p2502). - the mechanical relationships change (p2503 ... p2506). - the direction of rotation changes (p1821). For absolute encoders, the status of the adjustment (p2507) is also reset if the same absolute encoder remains selected for the closed-loop position control, but the mechanical relationships or the direction of rotation have changed. In the operation state, in addition, a fault (F07494) is generated.		
Notice:	The remaining setting values are intended for expanded functionality.		
Note:	The behavior for a DDS changeover is determined using the value of p2519 in the target data set.		

r2520[0...2]	CO: LR Position actual value preprocessing, encoder control word / ActVal_prep STW				
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: -	Calculated: -	Access level: 1		
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 4010		
	P-Group: Closed loop position control	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the encoder control word generated by the position actual value preprocessing.				
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Request function 1	Yes	No	-
	01	Request function 2	Yes	No	-
	02	Request function 3	Yes	No	-
	03	Request function 4	Yes	No	-
	04	Request command bit 0	Yes	No	-
	05	Request command bit 1	Yes	No	-
	06	Request command bit 2	Yes	No	-
	07	Flying measurement mode/search for reference mark	Flying measurement	Reference marks	-
	13	Request absolute value cyclic	Yes	No	-
	14	Request parking encoder	Yes	No	-
	15	Request acknowledge encoder fault	Yes	No	-
Dependency:	Refer to: p0480				
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: CI: p0480[0] = r2520[0], CI: p0480[1] = r2520[1] and CI: p0480[2] = r2520[2]				

r2521[0...3]	CO: LR position actual value / s_act			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: -	Calculated: -	Access level: 1	
	Data type: Integer32	Dynamic index: -	Func. diagram: 4010	
	P-Group: Closed loop position control	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	- [LU]	- [LU]	- [LU]	
Description:	Displays the actual position actual value determined by the position actual value preprocessing.			
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3			
Dependency:	Refer to: p2502, r2526			
Note:	r2526.0 = 1 --> The position actual value in r2521[0] for the position control is valid. r2527.0 = 1 --> The position actual value in r2521[1] for encoder 1 is valid. r2528.0 = 1 --> The position actual value in r2521[2] for encoder 2 is valid. r2529.0 = 1 --> The position actual value in r2521[3] for encoder 3 is valid.			

r2521[0...3]	CO: LR position actual value / s_act		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer32	Dynamic index: -	Func. diagram: 4010
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [LU]	- [LU]	- [LU]
Description:	Displays the actual position actual value determined by the position actual value preprocessing.		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p2502, r2526		
Note:	r2526.0 = 1 --> The position actual value in r2521[0] for the position control is valid. r2527.0 = 1 --> The position actual value in r2521[1] for encoder 1 is valid. r2528.0 = 1 --> The position actual value in r2521[2] for encoder 2 is valid. r2529.0 = 1 --> The position actual value in r2521[3] for encoder 3 is valid.		

r2522[0...3] CO: LR velocity actual value / v_actSERVO (Pos ctrl),
VECTOR (Pos ctrl)**Can be changed:** -**Data type:** Integer32**P-Group:** Closed loop position control**Not for motor type:** -**Min**

- [1000 LU/min]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

- [1000 LU/min]

Access level: 1**Func. diagram:** 4010**Unit selection:** -**Expert list:** 1**Factory setting**

- [1000 LU/min]

Description: Displays the velocity actual value determined by the position actual value preprocessing.**Index:**
[0] = Closed loop position control
[1] = Encoder 1
[2] = Encoder 2
[3] = Encoder 3**Dependency:** Refer to: p2502, r2526**Note:**
r2526.0 = 1 --> The velocity actual value in r2522[0] for the position control is valid.
r2527.0 = 1 --> The velocity actual value in r2522[1] for encoder 1 is valid.
r2528.0 = 1 --> The velocity actual value in r2522[2] for encoder 2 is valid.
r2529.0 = 1 --> The velocity actual value in r2522[3] for encoder 3 is valid.**r2523[0...3] CO: LR measured value / Measured value**SERVO (Pos ctrl),
VECTOR (Pos ctrl)**Can be changed:** -**Data type:** Integer32**P-Group:** Closed loop position control**Not for motor type:** -**Min**

- [LU]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

- [LU]

Access level: 1**Func. diagram:** 4010**Unit selection:** -**Expert list:** 1**Factory setting**

- [LU]

Description: Displays the value determined by the function "reference mark search" and "measuring probe evaluation".**Index:**
[0] = Closed loop position control
[1] = Encoder 1
[2] = Encoder 2
[3] = Encoder 3**Dependency:** Refer to: p2502, r2526**Note:**
r2526.2 = 1 --> The measured value in r2523[0] for the position control is valid.
r2527.2 = 1 --> The measured value in r2523[1] for encoder 1 is valid.
r2528.2 = 1 --> The measured value in r2523[2] for encoder 2 is valid.
r2529.2 = 1 --> The measured value in r2523[3] for encoder 3 is valid.**r2524 CO: LR LU/mm / LU/mm**SERVO (Lin, Pos
ctrl)**Can be changed:** -**Data type:** FloatingPoint32**P-Group:** Closed loop position control**Not for motor type:** -**Min**

- [LU]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

- [LU]

Access level: 1**Func. diagram:** 4010**Unit selection:** -**Expert list:** 1**Factory setting**

- [LU]

Description: Displays the internal length units LU/mm.**Dependency:** Refer to: p0404

r2524				
CO: LR LU/revolution / LU/revolution				
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: -		Calculated: -	
	Data type: FloatingPoint32		Dynamic index: -	
	P-Group: Closed loop position control		Units group: -	
	Not for motor type: -		Scaling: -	
	Min - [LU]		Max - [LU]	
Access level: 1				
Func. diagram: 4010				
Unit selection: -				
Expert list: 1				
Factory setting - [LU]				
Description: Displays the internal length units LU/motor revolution.				
Dependency: Refer to: p0404				
p2525[0...n]				
CO: LR encoder adjustment, offset / Enc_adj offset				
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T		Calculated: -	
	Data type: Unsigned32		Dynamic index: EDS, p0140	
	P-Group: Closed loop position control		Units group: -	
	Not for motor type: -		Scaling: -	
	Min 0 [LU]		Max 4294967295 [LU]	
Access level: 4				
Func. diagram: 4010				
Unit selection: -				
Expert list: 1				
Factory setting 0 [LU]				
Description: For the absolute encoder adjustment, a drive determines the position offset.				
Dependency: Refer to: p0404				
Note: The position offset is only relevant for absolute encoders. The drive determines it when making the adjustment and the user should not change it.				
r2526.0...9				
CO/BO: LR status word / ZSW				
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: -		Calculated: -	
	Data type: Unsigned16		Dynamic index: -	
	P-Group: Closed loop position control		Units group: -	
	Not for motor type: -		Scaling: -	
	Min -		Max -	
Access level: 1				
Func. diagram: -				
Unit selection: -				
Expert list: 1				
Factory setting -				
Description: Displays the status word of the position controller.				
Bit field:				
Bit	Signal name	1 signal	0 signal	FP
00	Position actual value valid	Yes	No	4010, 4015
01	Referencing active	Yes	No	4010
02	Measured value valid	Yes	No	3615, 4010
03	Closed-loop position control active	Yes	No	4015
04	Fixed stop reached	Yes	No	3617, 4025
05	Fixed stop outside window	Yes	No	3617, 4025
06	Position controller output limited	Yes	No	4015
07	Request tracking mode	Yes	No	-
08	Clamping active when traveling to fixed stop	Yes	No	4025
09	Setting value for adjustment valid	Yes	No	-
Dependency: Refer to: r2521, r2522, r2523				
Note: Re bit 04: The signal is influenced via p2634. Re bit 05: The signal is influenced via p2635.				

r2527.0...2 CO/BO: LR actual value sensing status word encoder 1 / ActValSensZSW enc1

SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word of the position actual value sensing for encoder 1.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Position actual value valid	Yes	No	-
	01	Referencing active	Yes	No	-
	02	Measured value valid	Yes	No	-

r2528.0...2 CO/BO: LR actual value sensing status word encoder 2 / ActValSensZSW enc2

SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word of the position actual value sensing for encoder 2.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Position actual value valid	Yes	No	-
	01	Referencing active	Yes	No	-
	02	Measured value valid	Yes	No	-

r2529.0...2 CO/BO: LR actual value sensing status word encoder 3 / ActValSensZSW enc3

SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word of the position actual value sensing for encoder 3.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Position actual value valid	Yes	No	-
	01	Referencing active	Yes	No	-
	02	Measured value valid	Yes	No	-

p2530 CI: LR position setpoint / s_set

SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 4015, 4020
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the position setpoint of the position controller.

Dependency: Refer to: r2665

Note: When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: BI: p2530 = r2665

p2531			
CI: LR velocity setpoint / v_set			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description: Sets the signal source for the velocity setpoint of the position controller.			
Dependency: Refer to: r2666			
Note: When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: BI: p2531 = r2666			
p2532			
CI: LR position actual value / s_act			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 4015, 4020, 4025
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 2521[0]
Description: Sets the signal source for the position actual value of the position controller.			
Dependency: Refer to: r2521			
p2533[0...n]			
LR position setpoint filter, time constant / s_set_filt T			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 1000.00 [ms]	Factory setting 0.00 [ms]
Description: Sets the time constant for the position setpoint filter (PT1).			
Note: The effective Kv factor (position loop gain) is reduced with the filter. This allows a softer control behavior with improved tolerance with respect to noise/disturbances. Applications: - reduces the pre-control dynamic response. - jerk limiting.			
p2534[0...n]			
LR velocity pre-control factor / V_prectrl fact			
SERVO (Lin, Pos ctrl)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4015, 4025
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 200.00 [%]	Factory setting 0.00 [%]
Description: Setting to activate and weight the velocity pre-control value. Value = 0 % --> The pre-control is deactivated.			
Dependency: Refer to: p2535, p2536, r2563			
Note: When the axis control loop is optimally set as well as a precisely determined equivalent time constant of the velocity control loop, the pre-control factor is 100%.			

p2534[0...n]	LR speed pre-control factor / n_prectrI fact		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4015, 4025
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 200.00 [%]	Factory setting 0.00 [%]
Description:	Setting to activate and weight the speed pre-control value. Value = 0 % --> The pre-control is deactivated.		
Dependency:	Refer to: p2535, p2536, r2563		
Note:	When the axis control loop is optimally set as well as a precisely determined equivalent time constant of the speed control loop, the pre-control factor is 100%.		
p2535[0...n]	LR velocity pre-control balancing filter dead time / v_prectrFlt t_dead		
SERVO (Lin, Pos ctrl)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00	Max 2.00	Factory setting 0.00
Description:	Sets the "fractional" dead time to emulate the timing behavior of the velocity control loop. The selected multiplier refers to the position controller clock cycle (deadtime= p2535 * p0115[4]).		
Dependency:	Refer to: p0115, p2536		
Notice:	When speed pre-control is active (p2534 > 0 %), the following applies: In addition to the set dead time (p2535), internally two position controller clock cycles are effective. When speed pre-control is inactive (p2534 = 0 %), the following applies: No dead time is effective (p2535 and internal).		
Note:	Together with p2536, the timing behavior of the velocity control loop can be emulated.		
p2535[0...n]	LR speed pre-control balancing filter dead time / n_prectrFlt t_dead		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00	Max 2.00	Factory setting 0.00
Description:	Sets the "fractional" dead time to emulate the timing behavior of the speed control loop. The selected multiplier refers to the position controller clock cycle (deadtime= p2535 * p0115[4]).		
Dependency:	Refer to: p0115, p2536		
Notice:	When speed pre-control is active (p2534 > 0 %), the following applies: In addition to the set dead time (p2535), internally two position controller clock cycles are effective. When speed pre-control is inactive (p2534 = 0 %), the following applies: No dead time is effective (p2535 and internal).		
Note:	Together with p2536, the timing behavior of the closed-loop control loop can be emulated.		

p2536[0...n]			
SERVO (Lin, Pos ctrl)	LR velocity pre-control, balancing filter PT1 / v_prectrl filt PT1		
	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
Description:	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [ms]	100.00 [ms]	0.00 [ms]
Dependency:	Sets a PT1 filter to emulate the timing behavior of the velocity control loop.		
Notice:	Refer to: p2535		
Note:	When speed pre-control is inactive (p2534 = 0 %), the following applies:		
	If a PT1 filter has been set, it is not effective.		
Note:	Together with p2535, the timing behavior of the velocity control loop can be emulated.		
p2536[0...n]			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	LR speed pre-control, symmetrizing filter PT1 / n_prectrl filt PT1		
	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
Description:	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [ms]	100.00 [ms]	0.00 [ms]
Dependency:	Sets a PT1 filter to emulate the timing behavior of the closed-speed control loop.		
Notice:	Refer to: p2535		
Note:	When speed pre-control is inactive (p2534 = 0 %), the following applies:		
	If a PT1 filter has been set, it is not effective.		
Note:	Together with p2535, the timing behavior of the closed-loop control loop can be emulated.		
p2537			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	CI: LR position controller adaptation / Adaptation		
	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
Description:	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	1
Dependency:	Sets the signal source for the adaptation of the proportional gain of the position controller.		
Notice:	Refer to: p2538		
p2538[0...n]			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	LR proportional gain / Kp		
	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
Description:	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.000 [1000/min]	300.000 [1000/min]	1.000 [1000/min]
Dependency:	Sets the proportional gain (P gain, position loop gain, Kv factor) of the position controller.		
Notice:	Refer to: p2537, p2539, p2555, r2557, r2558		
Note:	The proportional gain is used define at which traversing velocity which following error is obtained (without pre-control)		
	Low proportional gain:		
	Slow response to a setpoint - actual value difference, the following error becomes large.		
	High proportional gain:		
Note:	Fast response to the setpoint - actual value difference, the following error becomes small.		

p2539[0...n]	LR integral time / Tn		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100000.00 [ms]	Access level: 1 Func. diagram: 4015 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Setting to activate the integral time of the position controller. Value = 0 ms --> The I component of the position controller is deactivated.		
Dependency:	Refer to: p2538, r2559		
p2540	CO: LR position controller output, velocity limit / LR_outp limit		
SERVO (Lin, Pos ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min 0.000 [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max 1000.000 [m/min]	Access level: 3 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting 1000.000 [m/min]
Description:	Sets the velocity limit of the position controller output.		
Dependency:	Refer to: p2541		
p2540	CO: LR position controller output, speed limit / LR_outp limit		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min 0.000 [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max 210000.000 [rpm]	Access level: 3 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting 210000.000 [rpm]
Description:	Sets the speed limit of the position controller output.		
Dependency:	Refer to: p2541		
p2541	CI: LR position controller output, velocity limit signal source / LR_outp lim S_src		
SERVO (Lin, Pos ctrl)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2000 Max -	Access level: 1 Func. diagram: 4015 Unit selection: - Expert list: 1 Factory setting 2540[0]
Description:	Sets the signal source for the position controller output limit.		
Dependency:	Refer to: p2540		
p2541	CI: LR position controller output, speed limit signal source / LR_outp lim S_src		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2000 Max -	Access level: 1 Func. diagram: 4015 Unit selection: - Expert list: 1 Factory setting 2540[0]
Description:	Sets the signal source for the position controller output limit.		
Dependency:	Refer to: p2540		

p2542	LR standstill window / Standstill window		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T Data type: Unsigned32 P-Group: Closed loop position control Not for motor type: - Min 0 [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147483647 [LU]	Access level: 1 Func. diagram: 4020 Unit selection: - Expert list: 1 Factory setting 200 [LU]
Description:	Sets the standstill window for the standstill monitoring function. After the standstill monitoring time expires, it is cyclically checked whether the difference between the setpoint and actual position is located within the standstill window and, if required, an appropriate fault is output. Value = 0 --> The standstill monitoring is deactivated.		
Dependency:	Refer to: p2543, p2544 Refer to: F07450		
Note:	The following applies for the setting of the standstill and positioning window: Standstill window (p2542) >= positioning window (p2544)		
p2543	LR standstill monitoring time / t_standstill monit		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000.00 [ms]	Access level: 1 Func. diagram: 4020 Unit selection: - Expert list: 1 Factory setting 200.00 [ms]
Description:	Sets the standstill monitoring time for the standstill monitoring function. After the standstill monitoring time expires, it is cyclically checked whether the difference between the setpoint and actual position is located within the standstill window and, if required, an appropriate fault is output.		
Dependency:	Refer to: p2542, p2545 Refer to: F07450		
Note:	The following applies for the setting of the standstill and positioning monitoring time: Standstill monitoring time (p2543) <= positioning monitoring time (p2545)		
p2544	LR positioning window / Pos_window		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T Data type: Unsigned32 P-Group: Closed loop position control Not for motor type: - Min 0 [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147483647 [LU]	Access level: 1 Func. diagram: 4020 Unit selection: - Expert list: 1 Factory setting 40 [LU]
Description:	Sets the positioning window for the positioning monitoring function. After the positioning monitoring time expires, it is checked once as to whether the difference between the setpoint and actual position lies within the positioning window and if required an appropriate fault is output. Value = 0 --> The positioning monitoring function is deactivated.		
Dependency:	Refer to: p2542, p2545, r2684 Refer to: F07451		
Note:	The following applies for the setting of the standstill and positioning window: Standstill window (p2542) >= positioning window (p2544)		

p2545 LR positioning monitoring time / t_pos_monitSERVO (Pos ctrl),
VECTOR (Pos ctrl)**Can be changed:** U, T**Data type:** FloatingPoint32**P-Group:** Closed loop position control**Not for motor type:** -**Min**

0.00 [ms]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

100000.00 [ms]

Access level: 1**Func. diagram:** 4020**Unit selection:** -**Expert list:** 1**Factory setting**

1000.00 [ms]

Description:

Sets the positioning monitoring time for the positioning monitoring.

After the positioning monitoring time expires, it is checked once as to whether the difference between the setpoint and actual position lies within the positioning window and if required an appropriate fault is output.

Dependency:

Refer to: p2543, p2544, r2684

Refer to: F07451

Note:

The following applies for the setting of the standstill and positioning monitoring time:

Standstill monitoring time (p2543) <= positioning monitoring time (p2545)

p2546[0...n] LR dynamic following error monitoring tolerance / s_delta_monit tolSERVO (Pos ctrl),
VECTOR (Pos ctrl)**Can be changed:** U, T**Data type:** Unsigned32**P-Group:** Closed loop position control**Not for motor type:** -**Min**

0 [LU]

Calculated: -**Dynamic index:** DDS, p0180**Units group:** -**Scaling:** -**Max**

2147483647 [LU]

Access level: 1**Func. diagram:** 4025**Unit selection:** -**Expert list:** 1**Factory setting**

1000 [LU]

Description:

Sets the tolerance for the dynamic following error monitoring.

If the dynamic following error (r2563) exceeds the selected tolerance, then an appropriate fault is output.

Value = 0 --> The dynamic following error monitoring is deactivated.

Dependency:

Refer to: r2563, r2684

Refer to: F07452

Note:

The tolerance bandwidth is intended to prevent the dynamic following error monitoring incorrectly responding due to operational control sequences (e.g. during load surges).

p2547 LR cam switching position 1 / Cam position 1SERVO (Pos ctrl),
VECTOR (Pos ctrl)**Can be changed:** U, T**Data type:** Integer32**P-Group:** Closed loop position control**Not for motor type:** -**Min**

-2147483648 [LU]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

2147483647 [LU]

Access level: 1**Func. diagram:** 4025**Unit selection:** -**Expert list:** 1**Factory setting**

0 [LU]

Description:

Sets the cam switching position 1.

Dependency:

Refer to: p2548, r2683

Caution:

Only after the axis has been referenced can it be guaranteed that the cam switching signals when output have a "true" position reference.

**Note:**

Position actual value <= cam switching position 1 --> r2683.8 = 1 signal

Position actual value > cam switching position 1 --> r2683.8 = 0 signal

p2548 LR cam switching position 2 / Cam position 2SERVO (Pos ctrl),
VECTOR (Pos ctrl)**Can be changed:** U, T**Data type:** Integer32**P-Group:** Closed loop position control**Not for motor type:** -**Min**

-2147483648 [LU]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

2147483647 [LU]

Access level: 1**Func. diagram:** 4025**Unit selection:** -**Expert list:** 1**Factory setting**

0 [LU]

Description:

Sets the cam switching position 2.

Dependency:

Refer to: p2547, r2683

Caution:

Only after the axis has been referenced can it be guaranteed that the cam switching signals when output have a "true" position reference.

**Note:**

Position actual value <= cam switching position 2 --> r2683.9 = 1 signal

Position actual value > cam switching position 2 --> r2683.9 = 0 signal

p2549 BI: LR enable 1 / Enable 1SERVO (Pos ctrl),
VECTOR (Pos ctrl)**Can be changed:** T**Data type:** Unsigned32 / Binary**P-Group:** Closed loop position control**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 1**Func. diagram:** 4015**Unit selection:** -**Expert list:** 1**Factory setting**

899.2

Description:

Sets the signal source for the position controller enable 1.

Dependency:

Refer to: r0899, p2550

Note:

The position controller is enabled by ANDing BI: p2549 and BI: p2550.

p2550 BI: LR enable 2 / Enable 2SERVO (Pos ctrl),
VECTOR (Pos ctrl)**Can be changed:** T**Data type:** Unsigned32 / Binary**P-Group:** Closed loop position control**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 1**Func. diagram:** 4015**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description:

Sets the signal source for the position controller enable 2.

Dependency:

Refer to: p2549

Note:

The position controller is enabled by ANDing BI: p2549 and BI: p2550.

When the function module "closed-loop position control" or "basic positioner" is activated, the following BICO inter-connection is established: BI: p2550 = 1

p2551	BI: LR setpoint signal present / Sig s_set pres		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Closed loop position control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 4020 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the "setpoint present" signal. BI: p2551 = 1 signal: The end of the positioning operation on the setpoint side is signaled and the positioning and standstill monitoring activated. BI: p2551 = 0 signal: The start of a positioning operation or tracking mode on the setpoint side is signaled and the positioning and standstill monitoring deactivated.		
Dependency:	Refer to: p2554, r2683		
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: BI: p2551 = r2683.2		
p2552	BI: LR signal travel to fixed stop active / Signal TfS act		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Closed loop position control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 4025 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the signal "travel to fixed stop active". BI: p2552 = 1 signal: The activity associated with travel to fixed stop is signaled and the detection of the fixed stop is started via the maximum following error (p2634).		
Dependency:	Refer to: r2683		
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: BI: p2552 = r2683.14		
p2553	BI: LR signal fixed stop reached / Signal fixed stop		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Closed loop position control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 4025 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the signal "fixed stop reached". BI: p2553 = 1 signal: When the fixed stop is reached, this is signaled and the fixed stop monitoring window is activated.		
Dependency:	Refer to: r2683		
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: BI: p2553 = r2683.12		

p2554	BI: LR signal traversing command active / Sig trav_cmnd act		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Closed loop position control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 4020 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the signal "traversing command active". BI: p2554 = 1 signal: It is signaled that positioning is active and therefore the positioning monitoring is not activated with the signal "set-point present" (p2551).		
Dependency:	Refer to: p2551, r2684		
Note:	When the function module "basic positioner" (r0108.4 = 1) is activated, the following BICO interconnection is established: BI: p2554 = r2684.15		
p2555	CI: LR LU/revolution LU/mm / LU/rev LU/mm		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 4015 Unit selection: - Expert list: 1 Factory setting 2524[0]
Description:	Sets the signal source for the reference of the internal length units LU to motor revolution for rotary encoders and to mm for linear encoders.		
Dependency:	Refer to: p0404, r2524		
Note:	The signal value is used to convert the length unit to the speed or velocity setpoint.		
r2556	CO: LR position setpoint after setpoint smoothing / s_set after interp		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: - Data type: Integer32 P-Group: Closed loop position control Not for motor type: - Min - [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [LU]	Access level: 1 Func. diagram: 4015 Unit selection: - Expert list: 1 Factory setting - [LU]
Description:	Displays the position setpoint after the setpoint smoothing.		
r2557	CO: LR position controller input, system deviation / LR_inp sys dev		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: - Data type: Integer32 P-Group: Closed loop position control Not for motor type: - Min - [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [LU]	Access level: 1 Func. diagram: 4015 Unit selection: - Expert list: 1 Factory setting - [LU]
Description:	Displays the difference between the position setpoint and the position actual value at the position controller input.		

r2558	CO: LR position controller output, P component / LR_outp P comp		
SERVO (Lin, Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the P component at the output of the position controller for the velocity setpoint.		
r2558	CO: LR position controller output, P component / LR_outp P comp		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the P component at the output of the position controller for the speed setpoint.		
r2559	CO: LR position controller output, I component / LR_outp I comp		
SERVO (Lin, Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the I component at the output of the position controller for the velocity setpoint.		
r2559	CO: LR position controller output, I component / LR_outp I comp		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the I component at the output of the position controller for the speed setpoint.		
r2560	CO: LR velocity setpoint / v_set		
SERVO (Lin, Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocity setpoint after limiting (CI: p2541).		

r2560	CO: LR speed setpoint / n_set		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed setpoint after limiting (CI: p2541).		
r2561	CO: LR velocity pre-control value / v_prectrl val		
SERVO (Lin, Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocity setpoint due to the pre-control.		
r2561	CO: LR speed pre-control value / n_prectrl val		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speed setpoint due to the pre-control.		
r2562	CO: LR velocity setpoint, total / v_set total		
SERVO (Lin, Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the total velocity setpoint.		
	This value is obtained from the sum of the velocity pre-control and position controller output.		
Dependency:	Refer to: r2560, r2561		
r2562	CO: LR total speed setpoint / n_set total		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the total speed setpoint		
	This value is obtained from the sum of the speed pre-control and position controller output.		
Dependency:	Refer to: r2560, r2561		

r2563	CO: LR following error dynamic model / Follow error dyn		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: - Data type: Integer32 P-Group: Closed loop position control Not for motor type: - Min - [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [LU]	Access level: 1 Func. diagram: 4025 Unit selection: - Expert list: 1 Factory setting - [LU]
Description:	Displays the dynamic following error. This value is the deviation, corrected by the velocity-dependent component, between the position setpoint and the position actual value.		
Note:	For p2534 >= 100 % (pre-control activated) the following applies: The dynamic following error (r2563) corresponds to the system deviation (r2557) at the position controller input. For 0 % < p2534 < 100 % (pre-control activated) or p2534 = 0 % (pre-control deactivated) the following applies: The dynamic following error (r2563) is the deviation between the measured position actual value and a value that is calculated from the position setpoint via a PT1 model. This compensates the system-related velocity-dependent system deviation for a P controller.		
r2564	CO: LR force pre-control value / F_prectrl val		
SERVO (Lin, Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [N]	Calculated: - Dynamic index: - Units group: 8_1 Scaling: p2003 Max - [N]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [N]
Description:	Displays the force pre-control value.		
Dependency:	Refer to: p1511, p1512		
Note:	The force pre-control value is the derivation over time of the velocity pre-control value and is referred to a high inertia mass of 1000.0 kg. When using the pre-control, then this should be evaluated corresponding to the actual mass.		
r2564	CO: LR torque pre-control value / M_prectrl val		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed loop position control Not for motor type: - Min - [Nm]	Calculated: - Dynamic index: - Units group: 7_1 Scaling: p2003 Max - [Nm]	Access level: 1 Func. diagram: 4015 Unit selection: p0505 Expert list: 1 Factory setting - [Nm]
Description:	Displays the torque pre-control value.		
Dependency:	Refer to: p1511, p1512		
Note:	The torque pre-control value is the derivation over time of the speed pre-control value and is referred to a moment of inertia of 1 kgm ² /2 PI. When using the pre-control, then this should be evaluated corresponding to the actual moment of inertia.		
r2565	CO: LR following error actual / Following err act		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: - Data type: Integer32 P-Group: Closed loop position control Not for motor type: - Min - [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [LU]	Access level: 1 Func. diagram: 4015 Unit selection: - Expert list: 1 Factory setting - [LU]
Description:	Displays the actual following error. This value is the deviation between the position setpoint - after fine interpolation - and the position actual value.		

Notice: When speed pre-control is active (p2534 > 0 %), the following applies:
 To calculate this value, the position setpoint is delayed by two position controller clock cycles.
 When speed pre-control is inactive (p2534 = 0 %), the following applies:
 To calculate this value, the position setpoint is delayed by two position controller clock cycles.

r2566			
LR velocity input pre-control / v_inp_prectrl			
SERVO (Lin, Pos ctrl)	Can be changed: -	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min - [m/min]	Max - [m/min]	Factory setting - [m/min]
Description:	Displays the velocity at the input of the pre-control channel.		
Note:	This display parameter is used for diagnostics even when the pre-control is inactive (p2534 = 0%).		
r2566			
LR speed input pre-control / n_inp_prectrl			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: -	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min - [rpm]	Max - [rpm]	Factory setting - [rpm]
Description:	Displays the speed at the input of the pre-control channel.		
Note:	This display parameter is used for diagnostics even when the pre-control is inactive (p2534 = 0%).		
p2567[0...n]			
LR force pre-control mass / F_prectrl mass			
SERVO (Lin, Pos ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: 27_1	Unit selection: p0100
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000000 [kg]	Max 10000.000000 [kg]	Factory setting 1.000000 [kg]
Description:	Sets the mass for the force pre-control.		
Dependency:	Refer to: p2534, r2564		
Note:	When calculating the force pre-control value (r2654), the derivation over time of the speed pre-control value is multiplied by p2567. For reasons associated with the compatibility to earlier firmware releases, the factory setting for p2567 = 1 kg. This means that CO: r2564 remains, as standard, the derivation over time of the velocity pre-control value and refers, as before, to a weight of 1 kg. For force pre-control, the mass can now be directly entered into p2567 (instead of subsequently evaluating the pre-control value).		
p2567[0...n]			
LR torque pre-control moment of inertia / M_prectrl M_inertia			
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4015
	P-Group: Closed loop position control	Units group: 25_1	Unit selection: p0100
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000000 [kgm²]	Max 100000.000000 [kgm²]	Factory setting 0.159155 [kgm²]
Description:	Sets the moment of inertia for the torque pre-control.		
Dependency:	Refer to: p2534, r2564		

Note: When calculating the torque pre-control value (r2654), the time derivation of the speed pre-control value is multiplied by 2 PI * p2567.
 For reasons associated with the compatibility to earlier firmware versions, the factory setting for p2567 = 1 kgm²/2 PI. This means that CO: r2564 remains as standard the derivation over time of the speed pre-control value and is referred, as before, to a moment of inertia of 1 kgm²/2 PI. For torque pre-control, the moment of inertia can now be directly entered into p2567 (instead of subsequently evaluating the pre-control value).

p2568	BI: EPOS STOP cam activation / STOP cam act		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3630
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source to activate the function "STOP cam".

BI: p2568 = 1 signal

--> The evaluation of the STOP cam minus (BI: p2569) and STOP cam plus (BI: p2570) is active.

Dependency: Refer to: p2569, p2570

Note: The traversing range can also be limited using software limit switches.

p2569	BI: EPOS STOP cam minus / STOP cam minus		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3630
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1

Description: Sets the signal source for the STOP cam in the negative direction of travel.

Recommend.: Set the OFF3 ramp-down time (p1135), so that the axis, after reaching the STOP cam at the maximum velocity, does not move through a higher braking travel than is actually available.

Sets message 07491 as alarm (A07491):

Set the maximum deceleration (p2573), so that the axis, after reaching the STOP cam at the maximum velocity, does not move through a higher braking travel than is actually available.

Dependency: Refer to: p1135, p2568, p2570, p2573, r2684

Refer to: F07491

Caution: The STOP cams are low active.



Sets message 07491 as fault (F07491):

For a 0 signal, the axis is stopped with the OFF3 ramp-down time (p1135), status signal r2684.13 is set to 1, saved and the appropriate fault is output. After the fault has been acknowledged, only motion moving away from the STOP cam is permitted.

For a 0/1 signal and valid travel direction, when the stop cam is exited, this is detected and the status signal r2684.13 is set to 0.

Sets message 07491 as alarm (A07491):

For a 0 signal, the axis is stopped with the maximum deceleration (p2573), status signal r2684.13 is set to 1, saved and the appropriate alarm is output. Only motion away from the STOP cam is permitted.

For a 0/1 signal and valid travel direction, when the stop cam is exited, this is detected and the status signal r2684.13 is set to 0 and the alarm is deleted.

p2570	BI: EPOS STOP cam plus / STOP cam plus		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3630 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the STOP cam in the positive direction of travel.		
Recommend.:	Set the OFF3 ramp-down time (p1135), so that the axis, after reaching the STOP cam at the maximum velocity, does not move through a higher braking travel than is actually available. Sets message 07492 as alarm (A07492): Set the maximum deceleration (p2573), so that the axis, after reaching the STOP cam at the maximum velocity, does not move through a higher braking travel than is actually available.		
Dependency:	Refer to: p1135, p2568, p2569, p2573, r2684 Refer to: F07492		
Caution: 	The STOP cams are low active. Sets message 07492 as fault (F07492): For a 0 signal, the axis is stopped with the OFF3 ramp-down time (p1135), status signal r2684.14 is set to 1, saved and the appropriate fault is output. After the fault has been acknowledged, only motion moving away from the STOP cam is permitted. For a 0/1 signal and valid travel direction, when the stop cam is exited, this is detected and the status signal r2684.14 is set to 0. Sets message 07492 as alarm (A07492): For a 0 signal, the axis is stopped with the maximum deceleration (p2573), status signal r2684.14 is set to 1, saved and the appropriate alarm is output. Only motion away from the STOP cam is permitted. For a 0/1 signal and valid travel direction, when the stop cam is exited, this is detected and the status signal r2684.14 is set to 0 and the alarm is deleted.		
p2571	EPOS maximum velocity / v_max		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min 1 [1000 LU/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 40000000 [1000 LU/min]	Access level: 1 Func. diagram: 3630 Unit selection: - Expert list: 1 Factory setting 30000 [1000 LU/min]
Description:	Sets the maximum velocity for the function module "basic positioner" (r0108.4).		
Dependency:	Refer to: r1084, r1087, p2503, p2504, p2505, p2506		
Note:	The maximum velocity is active in all of the operating modes of the basic positioner. The maximum velocity for the basic positioner should be aligned with the maximum speed/velocity of the speed/velocity controller: Rotary encoders: $p2571[1000 \text{ LU/min}] = \min(r1084 , r1087)[\text{rpm}] \times p2505/p2504 \times p2506/1000$ Linear encoders: $p2571[1000 \text{ LU/min}] = \min(r1084 , r1087)[\text{m/min}] \times p2503/10[\text{m}]$		

p2572	EPOS maximum acceleration / a_max		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min 1 [1000 LU/s²]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2000000 [1000 LU/s²]	Access level: 1 Func. diagram: 3630 Unit selection: - Expert list: 1 Factory setting 100 [1000 LU/s²]
Description:	Sets the maximum acceleration for the function module "basic positioner" (r0108.4).		
Dependency:	Refer to: p2619, p2644		
Note:	The maximum acceleration appears to exhibit jumps (without jerk). "Traversing blocks" operating mode: The programmed acceleration override (p2619) acts on the maximum acceleration. "Direct setpoint input/MDI" mode: The acceleration override is effective (p2644, 4000 hex = 100 %). "Jog" and "search for reference" modes No acceleration override is active. The axis starts with the maximum acceleration.		
p2573	EPOS maximum deceleration / -a_max		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min 1 [1000 LU/s²]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2000000 [1000 LU/s²]	Access level: 1 Func. diagram: 3630 Unit selection: - Expert list: 1 Factory setting 100 [1000 LU/s²]
Description:	Sets the maximum deceleration for the function module "basic positioner" (r0108.4).		
Dependency:	Refer to: p2620, p2645		
Note:	The maximum deceleration appears to exhibit jumps (without jerk). "Traversing blocks" operating mode: The programmed deceleration override (p2620) acts on the maximum deceleration. "Direct setpoint input/MDI" mode: The deceleration override is effective (p2645, 4000 hex = 100 %). "Jog" and "search for reference" modes No deceleration override is effective. The axis breaks with the maximum deceleration.		
p2574	EPOS jerk limiting / Jerk lim		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min 1 [1000 LU/s³]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000000 [1000 LU/s³]	Access level: 1 Func. diagram: 3635 Unit selection: - Expert list: 1 Factory setting 10000 [1000 LU/s³]
Description:	Sets the jerk limiting		
Dependency:	Refer to: p2572, p2573, p2575		

Note: The jerk limiting is internally converted into a jerk time as follows:

$$\text{Jerk time } Tr = \max(p2572, p2573) / p2574$$
The jerk time is internally limited to 1000 ms and is rounded-off to an integer multiple of the sampling time positioning (p0115[5]).
The jerk time is valid for the acceleration and deceleration phases also for unequal maximum acceleration (p2572) and maximum deceleration (p2573).
For unequal maximum acceleration and maximum deceleration, the motion is not optimal from a time perspective as the jerk limit cannot be used for the lower of the two values.
If, in the traversing profile, the acceleration time without jerk limiting is less than the jerk time Tr , then the motion with jerk limiting is not optimum from a time perspective.
For traversing motion with a direct transition between acceleration and deceleration (i.e. jerk time is greater than the constant velocity phase), jerk can increase up to twice the parameterized jerk.
CONTINUE_FLYING with direction reversal acts internally just like a CONTINUE_WITH_STOP without the "position reached" being set. Without jerk limiting, this behavior can hardly be noticed as, when reversing, the position setpoint is only kept at zero for one interpolator clock cycle.
For block change enable CONTINUE_WITH_STOP, jerk limiting results in a longer delay time.

p2575	BI: EPOS jerk limiting activation / Jerk limit act		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3635
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source to activate the jerk limiting.
Activating/deactivating:
- using BI: p2575 = 1 signal or 0 signal.
- using the command JERK in the traversing block (only for BI: p2575 = 0 signal).

Dependency: Refer to: p2574

Note: A change of the signal state at the binector input is only accepted at zero speed.

p2576	EPOS modulo correction, modulo range / Modulo corr range		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 3635
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	1 [LU]	2147482647 [LU]	360000 [LU]

Description: Sets the modulo range for axes with modulo correction.

Dependency: Refer to: p2577

p2577	BI: EPOS modulo correction activation / Modulo corr act		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3630, 3635
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source to activate modulo correction.

Dependency: Refer to: p2576

Note: When the signal state changes at the binector input, this only becomes effective in the "ready to power-up" state.
 Selecting modulo correction:
 The current position setpoint in the modulo range is corrected. The position actual value differs from the position setpoint by the following error and can also leave the modulo range.
 Deselecting modulo correction:
 It is based on the current position actual value.

p2578	CI: EPOS software limit switch minus signal source / SW limSw Min S_src		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 3630
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	2580[0]
Description:	Sets the signal source for the software limit switch minus.		
Dependency:	Refer to: p2579, p2580, p2581, p2582 Refer to: A07469, A07477, A07479, F07481		
Notice:	A change to the software limit switch becomes immediately effective. If the software limit switch is changed, then this results in the positions in the traversing blocks being checked.		
Note:	The following applies for the setting of the software limit switch: Software limit switch minus < software limit switch plus		

p2579	CI: EPOS software limit switch plus signal source / SW limSwPlus S_src		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 3630
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	2581[0]
Description:	Sets the signal source for the software limit switch plus.		
Dependency:	Refer to: p2578, p2580, p2581, p2582 Refer to: A07470, A07478, A07480, F07482		
Notice:	A change to the software limit switch becomes immediately effective. If the software limit switch is changed, then this results in the positions in the traversing blocks being checked.		
Note:	The following applies for the setting of the software limit switch: Software limit switch minus < software limit switch plus		

p2580	CO: EPOS software limit switch minus / SW limSwitch minus		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Integer32	Dynamic index: -	Func. diagram: 3630
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-2147482648 [LU]	2147482647 [LU]	-2147482648 [LU]
Description:	Sets the software limit switch in the negative direction of travel.		
Dependency:	Refer to: p2578, p2579, p2581, p2582		

p2581			
SERVO (EPOS), VECTOR (EPOS)	CO: EPOS software limit switch plus / SW lim switch plus		
	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Integer32	Dynamic index: -	Func. diagram: 3630
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-2147482648 [LU]	2147482647 [LU]	2147482647 [LU]
Description:	Sets the software limit switch in the positive direction of travel.		
Dependency:	Refer to: p2578, p2579, p2580, p2582		
p2582			
SERVO (EPOS), VECTOR (EPOS)	BI: EPOS software limit switch activation / SW lim sw act		
	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3630
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to activate the "software limit switch".		
Dependency:	Refer to: p2578, p2579, p2580, p2581		
Caution:	Software limit switch effective:		
	- axis is referenced (r2684.11 = 1) and BI: p2582 = 1 signal.		
	Software limit switch ineffective:		
	- modulo correction active (BI: p2577 = 1 signal).		
	- search for reference is executed.		
Notice:	Target position for relative positioning outside software limit switch:		
	The traversing block is started and the axis comes to a standstill at the software limit switch. An appropriate alarm is output and the traversing block is interrupted. Traversing blocks with valid position can be activated.		
	Target position for absolute positioning outside software limit switch:		
	In the "traversing blocks" mode, the traversing block is not started and an appropriate fault is output.		
	Axis outside the valid traversing range:		
	If the axis is already outside the valid traversing range, then an appropriate fault is output. The fault can be acknowledged at standstill. Traversing blocks with valid position can be activated.		
Note:	The traversing range can also be limited using STOP cams.		
p2583			
SERVO (EPOS), VECTOR (EPOS)	EPOS backlash compensation / Backlash comp		
	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Integer32	Dynamic index: -	Func. diagram: 3635
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-200000 [LU]	200000 [LU]	0 [LU]
Description:	Sets the amount of play (backlash) for positive or negative play.		
	0: The backlash compensation is deactivated.		
	> 0: Positive backlash (normal case)		
	When the direction is reversed, the encoder actual value leads the actual value.		
	< 0: Negative backlash		
	When the direction is reversed, the actual value leads the encoder actual value.		



Dependency: If a stationary axis is referenced by setting the reference point, or an adjusted with absolute encoder is powered up, then the setting of p2604 is relevant for entering the compensation value.
p2604 = 1:
Traveling in the positive direction -> A compensation value is immediately entered.
Traveling in the negative direction -> A compensation value is not entered
p2604 = 0:
Traveling in the positive direction -> A compensation value is not entered
Traveling in the negative direction -> A compensation value is immediately entered.
When again setting the reference point (a referenced axis) or for "flying referencing", p2604 is not relevant but instead the history of the axis.
Refer to: p2604, r2667

p2585	EPOS jog 1 setpoint velocity / Jog 1 v_set		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Integer32	Dynamic index: -	Func. diagram: 3610
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -40000000 [1000 LU/min]	Max 40000000 [1000 LU/min]	Factory setting -300 [1000 LU/min]
Description:	Sets the setpoint velocity for jog 1.		
Dependency:	Refer to: p2587, p2589, p2591		

p2586	EPOS jog 2 setpoint velocity / Jog 2 v_set		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Integer32	Dynamic index: -	Func. diagram: 3610
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -40000000 [1000 LU/min]	Max 40000000 [1000 LU/min]	Factory setting 300 [1000 LU/min]
Description:	Sets the setpoint velocity for jog 2.		
Dependency:	Refer to: p2588, p2590, p2591		

p2587	EPOS jog 1 traversing distance / Jog 1 distance		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 3610
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0 [LU]	Max 2147482647 [LU]	Factory setting 1000 [LU]
Description:	Sets the traversing distance for incremental jog 1.		
Dependency:	Refer to: p2585, p2589, p2591		
Note:	Incremental jog 1 is started with BI: p2591 = 1 signal and BI: p2589 = 0/1 signal. With BI: p2589 = 0 signal, incremental jog is interrupted.		

p2588	EPOS jog 2 traversing distance / Jog 2 distance		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 3610
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0 [LU]	Max 2147482647 [LU]	Factory setting 1000 [LU]
Description:	Sets the traversing distance for incremental jog 2.		
Dependency:	Refer to: p2586, p2590, p2591		

Note: Incremental jog 2 is started with BI: p2591 = 1 signal and BI: p2590 = 0/1 signal.
With BI: p2590 = 0 signal, incremental jogging is interrupted.

p2589	BI: EPOS jog 1 signal source / Jog 1 S_src		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3610, 3625
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for jog 1.		
Dependency:	When jogging, the axis is accelerated or braked with the maximum acceleration/deceleration (p2572/p2573). BI: p2591 = 0 signal The axis endlessly moves with the setpoint velocity, jog 1 (p2585). BI: p2591 = 1 signal The axis traverses through a parameterized distance (p2585) with the setpoint velocity, jog 1 (p2587). Refer to: p2572, p2573, p2585, p2587, p2591		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
p2590	BI: EPOS jog 2 signal source / Jog 2 S_src		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3610, 3625
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for jog 2.		
Dependency:	When jogging, the axis is accelerated or braked with the maximum acceleration/deceleration (p2572/p2573). BI: p2591 = 0 signal The axis endlessly moves with the setpoint velocity, jog 2 (p2586). BI: p2591 = 1 signal The axis traverses through a parameterized distance (p2586) with the setpoint velocity, jog 2 (p2588). Refer to: p2572, p2573, p2586, p2588, p2591		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
p2591	BI: EPOS jogging incremental / Jog incr		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3610
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for jogging incremental.		
Dependency:	Refer to: p2585, p2586, p2587, p2588, p2589, p2590		

p2593	CI: EPOS LU/revolution LU/mm / LU/rev LU/mm		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2524[0]
Description:	Sets the signal source for the reference of the internal length units LU to motor revolution for rotary encoders and to mm for linear encoders.		
Dependency:	Refer to: p0404, r2524, p2594		
Note:	The signal value is used to convert the length unit to the speed or velocity setpoint.		
p2594	CI: EPOS Maximum velocity externally limited / v_Max ext lim		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2000 Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the externally limited maximum velocity.		
Dependency:	Refer to: r2524, p2571, p2593		
Warning:	In order that the externally limited velocity can be effective for the EPOS operating modes, connector input p2593 must be correctly interconnected.		
			
p2595	BI: EPOS referencing start / Ref start		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3612, 3625, 3614 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to start the "search for reference" or "flying referencing". BI: p2595 = 0/1 signal Referencing is started. BI: p2595 = 1/0 signal Referencing is interrupted.		
Dependency:	Refer to: p2597, p2598, p2599, r2684		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Search for reference (BI: p2597 = 0 signal): The reference point approach can only be activated (0/1 edge) after traversing motion that is being processed has been completed. With the start, where relevant, the state signal "reference point set" (r2684.11) is reset. Flying referencing (BI: p2597 = 1 signal): With the start, the state signal "reference point set" (r2684.11) is not reset.		

p2596	BI: EPOS set reference point / Set ref_pt		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the "set reference point".		
Dependency:	Refer to: p2598, p2599, r2684		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Reference point setting is effective in the following operating states: - in the basic state. - for FIXED STOP with progress condition END (corresponds to the initial state). - for traversing block interrupted via BI: p2640 = 0 signal (intermediate stop). - for EPOS not enabled (BI: p2656 = 0 signal) and position actual value valid (BI: p2658 = 1 signal).		
p2597	BI: EPOS referencing type selection / Ref_typ select		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3612, 3614, 3625 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select referencing type. 1 signal: Flying referencing 0 signal: Search for reference		
Dependency:	Refer to: p2595		
Note:	Referencing is activated as follows: - Select the referencing type (BI: p2597) - Start referencing (BI: p2595 = 0/1 signal)		
p2598[0...3]	CI: EPOS reference point coordinate, signal source / Ref_pt coord S_src		
SERVO (EPOS, Pos ctrl), VECTOR (EPOS, Pos ctrl)	Can be changed: T Data type: Unsigned32 / Integer32 P-Group: Closed loop position control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3612, 3614 Unit selection: - Expert list: 1 Factory setting [0] 2599[0] [1] 0 [2] 0 [3] 0
Description:	Sets the signal source for the reference point coordinate. This value is used as reference for the following referencing operations: - search for reference - set reference point - flying referencing - absolute value adjustment		
Index:	[0] = Closed loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		

Dependency: Refer to: p2502, p2507, p2595, p2596, p2597, p2599

Note: When the function module "basic positioner" (r0108.4 = 1) is activated, the following applies:

Incremental measuring system:

After the reference point is reached, the drive accepts the current axis position from the position received via the connector input CI: p2598[0].

Absolute encoder:

When adjusting the encoder, the position received via the connector input is set as the current axis position. The position offset to the actual encoder value is displayed in p2525.

p2599**CO: EPOS reference point coordinate value / Ref_pt coord val**

SERVO (Pos ctrl),
VECTOR (Pos ctrl)

Can be changed: U, T

Calculated: -

Access level: 1

Data type: Integer32

Dynamic index: -

Func. diagram: 3612

P-Group: Closed loop position control

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

-2147482648 [LU]

2147482647 [LU]

0 [LU]

Description: Sets the position value for the reference point coordinate.

This value is set as the current axis position after referencing or adjustment.

Dependency: Refer to: p2507, p2525, p2595, p2596, p2597, p2598

p2600**EPOS search for reference, reference point offset / Ref_pt offset**

SERVO (EPOS),
VECTOR (EPOS)

Can be changed: U, T

Calculated: -

Access level: 1

Data type: Integer32

Dynamic index: -

Func. diagram: 3612

P-Group: Basic positioner

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

-2147482648 [LU]

2147482647 [LU]

0 [LU]

Description: Sets the reference point offset for search for reference.

Dependency: Refer to: p2598

p2601**EPOS flying referencing, inner window / Inner window**

SERVO (EPOS),
VECTOR (EPOS)

Can be changed: U, T

Calculated: -

Access level: 1

Data type: Integer32

Dynamic index: -

Func. diagram: 3614

P-Group: Basic positioner

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

0 [LU]

2147482647 [LU]

0 [LU]

Description: Sets the inner window for flying referencing.

Value = 0:

The evaluation of the inner window is deactivated.

Dependency: Refer to: p2597, p2602, r2684

Notice: The inner window must be set so that it is smaller than the outer window.

Note: If the difference between the reference point coordinate and detected actual position is less than the inner window, then no correction is executed for a referenced axis.

If the difference between the reference point coordinate and detected actual position is greater than the inner window and less than the outer window (p2602), then a correction is executed for a referenced axis.

p2602	EPOS flying referencing, outer window / Outer window		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min 0 [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147482647 [LU]	Access level: 1 Func. diagram: 3614 Unit selection: - Expert list: 1 Factory setting 0 [LU]
Description:	Sets the outer window for flying referencing. Value = 0: The evaluation of the outer window is deactivated.		
Dependency:	Refer to: p2597, r2684 Refer to: A07489		
Notice:	The inner window must be set so that it is smaller than the outer window.		
Note:	If the difference between the reference point coordinate and detected actual position is greater than the outer window, then no correction is executed for the referenced axis. Further, an appropriate message is output and r2684.3 is set to 1.		
p2603	EPOS flying referencing, positioning mode, relative / Pos_mode relative		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 1 Func. diagram: 3635 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the relative positioning mode for flying referencing. Value = 1: The corrected setpoint is not calculated into the traversing distance. Value = 0: The corrected setpoint is calculated into the traversing distance.		
Dependency:	Refer to: p2597, p2623, p2648		
Caution:	For p2603 = 0 the direction can change.		
p2604	BI: EPOS search for reference, start direction / Srch for ref dir		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal sources for the start direction of the search for reference. 1 signal: Start in the negative direction. 0 signal: Start in the positive direction.		
Dependency:	Refer to: p2583, p2595, p2597		

p2605	EPOS search for reference, approach velocity, reference cam / v_appr ref_cam		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min 1 [1000 LU/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 40000000 [1000 LU/min]	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 5000 [1000 LU/min]
Description:	Sets the approach velocity to the reference cam for the search for reference.		
Dependency:	The search for reference only starts with the approach velocity to the reference cam when there is a reference cam (p2607 = 1). Refer to: p2595, p2597, p2604, p2606, p2607		
Note:	When traversing to the reference cam, the velocity override is effective. If, at the start of the search for reference, the axis is already at the reference cam, then the axis immediately starts to traverse to the zero mark.		
p2606	EPOS search for reference, reference cam, maximum distance / Ref_cam max s		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min 0 [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147482647 [LU]	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 2147482647 [LU]
Description:	Sets the maximum distance after the start of the search for reference when traversing to the reference cam.		
Dependency:	Refer to: p2595, p2597, p2604, p2605, p2607 Refer to: F07458		
Note:	When using a reversing cam, the maximum distance must be set appropriately long.		
p2607	EPOS search for reference, reference cam present / Ref_cam pres		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned8 P-Group: Basic positioner Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets whether or not a reference cam is present for the search for reference. Value = 1: Reference cam present. Value = 0: No reference cam present.		
Dependency:	Refer to: p2595, p2597, p2604, p2605, p2606		
p2608	EPOS search for reference, approach velocity, zero mark / v_appr ref_ZM		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min 1 [1000 LU/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 40000000 [1000 LU/min]	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 300 [1000 LU/min]
Description:	Sets the approach velocity after detecting the reference cam to search for the zero mark for the the search for reference.		
Dependency:	If there is no reference cam (p2607 = 0), the search for reference immediately starts with the axis traversing to the zero mark. Refer to: p2595, p2597, p2604, p2607, p2609, p2610		

Caution:



If the reference cam is not adjusted so that at each search for reference the same zero mark for synchronization is detected, then an "incorrect" axis reference point is obtained.

After the reference cam has been left, the search for the zero mark is activated with a time delay due to internal factors. This is the reason that the reference cam should be adjusted in this center between two zero marks and the approach velocity should be adapted to the distance between two zero marks.

Note:

The velocity override is not effective when traversing to the zero mark.

p2609	EPOS search for reference, max. distance ref. cam and zero mark / Max s ref_cam ZM		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min 0 [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147482647 [LU]	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 20000 [LU]
Description:	Sets the maximum distance after leaving the reference cam when traversing to the zero mark.		
Dependency:	Refer to: p2595, p2597, p2604, p2607, p2608, p2610 Refer to: F07459		
p2610	EPOS search for ref., tol. bandwidth for distance to zero mark / Tol_band to ZM		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min 0 [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147482647 [LU]	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 2147482647 [LU]
Description:	Sets the tolerance bandwidth for the distance to the zero mark The zero mark is evaluated within the maximum distance between the reference cam and zero mark (p2609) minus the tolerance bandwidth for the distance to the zero mark (p2610).		
Dependency:	Refer to: p2609		
p2611	EPOS search for reference, approach velocity, reference point / v_appr ref_pt		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min 1 [1000 LU/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 40000000 [1000 LU/min]	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 300 [1000 LU/min]
Description:	Sets the approach velocity after detecting the zero mark to approach the reference point.		
Dependency:	Refer to: p2595, p2597, p2604, p2607, p2609, p2610		
Note:	When traversing to the reference point, the velocity override is not effective.		
p2612	BI: EPOS search for reference, reference cam / Ref_cam		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the reference cam.		
Dependency:	Refer to: p2607		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		

p2613	BI: EPOS search for reference reversing cam minus / Rev minus		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the reversing cam in the negative direction of travel. 1 signal: Reversing cam not reached. 0 signal: Reversing cam reached.		
Dependency:	Refer to: p2614		
Note:	If, during the search for reference from the reversing cam minus and plus, a 0 signal is detected, then the axis remains stationary (at standstill).		
p2614	BI: EPOS search for reference reversing cam plus / Rev plus		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the reversing cam in the negative direction of travel. 1 signal: Reversing cam not reached. 0 signal: Reversing cam reached.		
Dependency:	Refer to: p2613		
Note:	If, during the search for reference from the reversing cam minus and plus, a 0 signal is detected, then the axis remains stationary (at standstill).		
p2615	EPOS maximum number of traversing blocks / Trav_block qty max		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: C2(17) Data type: Unsigned8 P-Group: Basic positioner Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 64	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting 64
Description:	Sets the maximum number of traversing blocks that are available.		
Dependency:	Refer to: p2616, p2617, p2618, p2619, p2620, p2621, p2622, p2623, p2624		
p2616[0...n]	EPOS traversing block, block number / Trav_blk, blkNo.		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Integer16 P-Group: Basic positioner Not for motor type: - Min -1	Calculated: - Dynamic index: p2615 Units group: - Scaling: - Max 63	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting -1
Description:	Sets a block number. -1: Invalid block number. These blocks are not taken into account. 0 ... 63: Valid block number.		
Dependency:	The number of indices depends on p2615. Refer to: p2615, p2617, p2618, p2619, p2620, p2621, p2622, p2623, p2624		

p2617[0...n]	EPOS traversing block position / Trav_block pos		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min -2147482648 [LU]	Calculated: - Dynamic index: p2615 Units group: - Scaling: - Max 2147482647 [LU]	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting 0 [LU]
Description:	Sets the target position for the traversing block.		
Dependency:	The number of indices depends on p2615. Refer to: p2615, p2616, p2618, p2619, p2620, p2621, p2622, p2623, p2624		
Note:	The target position is approached in either relative or absolute terms depending on p2623.		
p2618[0...n]	EPOS traversing block velocity / Trav_block v		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min 1 [1000 LU/min]	Calculated: - Dynamic index: p2615 Units group: - Scaling: - Max 40000000 [1000 LU/min]	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting 600 [1000 LU/min]
Description:	Sets the velocity for the traversing block.		
Dependency:	The number of indices depends on p2615. Refer to: p2615, p2616, p2617, p2619, p2620, p2621, p2622, p2623, p2624, p2646		
Note:	The velocity can be influenced using the velocity override (p2646).		
p2619[0...n]	EPOS traversing block acceleration override / Trav_block a_over		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Basic positioner Not for motor type: - Min 1.0 [%]	Calculated: - Dynamic index: p2615 Units group: - Scaling: - Max 100.0 [%]	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the acceleration override for the traversing block. The override refers to the maximum acceleration (p2572).		
Dependency:	The number of indices depends on p2615. Refer to: p2572, p2615, p2616, p2617, p2618, p2620, p2621, p2622, p2623, p2624		
p2620[0...n]	EPOS traversing deceleration override / Trav_block -a_over		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Basic positioner Not for motor type: - Min 1.0 [%]	Calculated: - Dynamic index: p2615 Units group: - Scaling: - Max 100.0 [%]	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the deceleration override for the traversing block. The override refers to the maximum deceleration (p2573).		
Dependency:	The number of indices depends on p2615. Refer to: p2573, p2615, p2616, p2617, p2618, p2619, p2621, p2622, p2623, p2624		
Notice:	If, when calculating the traversing profile, it is identified that the target position of the next block with the programmed deceleration override will not be reached without direction reversal (flying block change), then the old (current) deceleration override remains effective.		

p2621[0...n]	EPOS traversing block task / Trav_block task		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Integer16 P-Group: Basic positioner Not for motor type: - Min 1	Calculated: - Dynamic index: p2615 Units group: - Scaling: - Max 9	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the required task for the traversing block.		
Value:	1: POSITIONING 2: FIXED STOP 3: ENDLESS_POS 4: ENDLESS_NEG 5: WAITING 6: GOTO 7: SET_O 8: RESET_O 9: JERK		
Dependency:	The number of indices depends on p2615. Refer to: p2615, p2616, p2617, p2618, p2619, p2620, p2622, p2623, p2624		
p2622[0...n]	EPOS traversing block task parameter / Trav_blk task_par		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min -2147483648	Calculated: - Dynamic index: p2615 Units group: - Scaling: - Max 2147483647	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets additional information/data of the appropriate task for the traversing block.		
Dependency:	The number of indices depends on p2615. Refer to: p2615, p2616, p2617, p2618, p2619, p2620, p2621, p2623, p2624		
Note:	The following should be set depending on the task: FIXED STOP: Clamping torque and clamping force (rotary 0...65536 [0.01 Nm], linear 0...65536 [N]) WAIT: Delay time [ms] GOTO: Block number SET_O: 1, 2 or 3 - set direct output 1, 2 or 3 (both) RESET_O: 1, 2 or 3 - set direct output 1, 2 or 3 (both) JERK: 0 - deactivate, 1 - activate		

p2623[0...n]	EPOS traversing block, task mode / Trav_block mode		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned16 P-Group: Basic positioner Not for motor type: - Min 0	Calculated: - Dynamic index: p2615 Units group: - Scaling: - Max 65535	Access level: 1 Func. diagram: 3515, 3616 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the influence of the task for the traversing block. Value = 0000 cccc bbbb aaaa cccc: Positioning mode cccc = 0000 --> ABSOLUTE cccc = 0001 --> RELATIVE cccc = 0010 --> ABS_POS (only for a rotary axis with modulo correction) cccc = 0011 --> ABS_NEG (only for a rotary axis with modulo correction) bbbb: Progression condition bbbb = 0000 --> END bbbb = 0001 --> CONTINUE WITH STOP bbbb = 0010 --> CONTINUE FLYING bbbb = 0011 --> CONTINUE EXTERNAL bbbb = 0100 --> CONTINUE EXTERNAL WAIT bbbb = 0101 --> CONTINUE EXTERNAL ALARM aaaa: IDs aaaa = 0001 --> Skip block		
Dependency:	The number of indices depends on p2615. Refer to: p2615, p2616, p2617, p2618, p2619, p2620, p2621, p2622, p2624		
p2624	EPOS traversing block, sorting / Trav_block sort		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T Data type: Unsigned16 P-Group: Basic positioner Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the traversing blocks for sorting corresponding to their block number. Procedure: Set p2624 = 0 --> 1. Sorting is started and the parameters are automatically reset to 0 once the operation has been completed.		
Dependency:	Refer to: p2615, p2616, p2617, p2618, p2619, p2620, p2621, p2622, p2623		
Note:	After sorting, the traversing blocks are written at the beginning of the memory in increasing sequence without any gaps.		
p2625	BI: EPOS traversing block selection, bit 0 / Trav_blk sel bit 0		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3616, 3640 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select the traversing block, bit 0.		
Dependency:	Binector inputs p2625, p2626, p2627, p2628, p2629 and p2630 are used to select one of the maximum of 64 traversing blocks. Refer to: p2626, p2627, p2628, p2629, p2630		

p2626	BI: EPOS traversing block selection, bit 1 / Trav_blk sel bit 1		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3616, 3640 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select the traversing block, bit 1.		
Dependency:	Binector inputs p2625, p2626, p2627, p2628, p2629 and p2630 are used to select one of the maximum of 64 traversing blocks. Refer to: p2625, p2627, p2628, p2629, p2630		
p2627	BI: EPOS traversing block selection, bit 2 / Trav_blk sel bit 2		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3616, 3640 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select the traversing block, bit 2.		
Dependency:	Binector inputs p2625, p2626, p2627, p2628, p2629 and p2630 are used to select one of the maximum of 64 traversing blocks. Refer to: p2625, p2626, p2628, p2629, p2630		
p2628	BI: EPOS traversing block selection, bit 3 / Trav_blk sel bit 3		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3616, 3640 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select the traversing block, bit 3.		
Dependency:	Binector inputs p2625, p2626, p2627, p2628, p2629 and p2630 are used to select one of the maximum of 64 traversing blocks. Refer to: p2625, p2626, p2627, p2629, p2630		
p2629	BI: EPOS traversing block selection, bit 4 / Trav_blk sel bit 4		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3616, 3640 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select the traversing block, bit 4.		
Dependency:	Binector inputs p2625, p2626, p2627, p2628, p2629 and p2630 are used to select one of the maximum of 64 traversing blocks. Refer to: p2625, p2626, p2627, p2628, p2630		

p2630	BI: EPOS traversing block selection, bit 5 / Trav_blk sel bit 5		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3616, 3640 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select the traversing block, bit 5.		
Dependency:	Binector inputs p2625, p2626, p2627, p2628, p2629 and p2630 are used to select one of the maximum of 64 traversing blocks. Refer to: p2625, p2626, p2627, p2628, p2629		
p2631	BI: EPOS activate traversing task (0 -> 1) / Trav_task act		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3616, 3625 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for "activating traversing task". BI: p2631 = 0/1 signal The traversing task, selected using BI: p2625 ... p2630, is started.		
Dependency:	Refer to: p2625, p2626, p2627, p2628, p2629, p2630, p2640, p2641		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	To start a traversing block, the axis must be referenced (r2684.11 = 1). The status signal r2684.12 = 0/1 signal is used for acknowledgement. A traversing task can be influenced using the following signals: - intermediate stop via BI: p2640. - reject traversing task via BI: p2641.		
p2632	EPOS external block change evaluation / Ext BlckChg eval		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Integer16 P-Group: Basic positioner Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 1 Func. diagram: 3615, 3616 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the mode to evaluate "external block change".		
Value:	0: External block change via the measuring probe 1: External block change via BI: p2633		
Dependency:	Refer to: p2623, p2633, r2677, r2678		
Note:	In the mode "external block change via measuring probe (p2632 = 0), the following applies: When starting a traversing block with the block change enable CONTINUE_EXTERNAL, CONTINUE_EXTERNAL_WAIT and CONTINUE_EXTERNAL_ALARM an activated "flying referencing" is interrupted. After ending the block, "flying referencing" must be reactivated via BI: p2595 = 0/1 signal.		

p2633	BI: EPOS external block change (0 -> 1) / Ext BlckChg (0->1)		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3615 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for "external block change". BI: p2633 = 0/1 signal		
Dependency:	The evaluation of the signal is only active p2632 = 1. Refer to: p2623, p2632, p2640, p2641, r2677, r2678		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	A 0/1 edge initiates a flying block change in the subsequent traversing block. When the external block change is identified, the actual position is saved in r2678. A traversing task can be influenced using the following signals: - intermediate stop via BI: p2640. - reject traversing task via BI: p2641.		
p2634[0...n]	EPOS fixed stop maximum following error / Following err max		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T Data type: Unsigned32 P-Group: Closed loop position control Not for motor type: - Min 0 [LU]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 2147482647 [LU]	Access level: 1 Func. diagram: 3617, 4025 Unit selection: - Expert list: 1 Factory setting 1000 [LU]
Description:	Sets the following error to detect the "fixed stop reached" state (r2526.4).		
Dependency:	Refer to: r2526, p2621, r2675		
Note:	The state "fixed stop reached" is detected if the following error exceeds the theoretically calculated following error value by p2634.		
p2635	EPOS fixed stop monitoring window / Fixed stop monit		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: U, T Data type: Unsigned32 P-Group: Closed loop position control Not for motor type: - Min 0 [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147482647 [LU]	Access level: 1 Func. diagram: 3617, 4025 Unit selection: - Expert list: 1 Factory setting 100 [LU]
Description:	Sets the monitoring window of the actual position after the fixed stop is reached.		
Dependency:	Refer to: r2526, r2683 Refer to: F07484		
Note:	If, after the fixed stop is reached, the end stop shifts in either the positive or negative direction by more than the value set here, then BO: r2526.5 is set to 1 and an appropriate message is output.		

p2637	BI: EPOS fixed stop reached / Fixed stop reached		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3616, 3617 Unit selection: - Expert list: 1 Factory setting 2526.4
Description:	Sets the signal source for the feedback signal "fixed stop reached". BI: p2637 = 1 signal Fixed stop is reached. BI: p2637 = 0 signal Fixed stop is not reached.		
Dependency:	Refer to: r2526, p2634		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	The identification of "fixed stop reached" is, for the factory setting, dependent on the signal BO: r2526.4 (fixed stop reached). This signal is influenced via p2634 (EPOS fixed stop, maximum following error).		
p2638	BI: EPOS fixed stop outside the monitoring window / Fixed stop outside		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3616, 3617 Unit selection: - Expert list: 1 Factory setting 2526.5
Description:	Sets the signal source for the feedback signal "fixed stop outside the monitoring window". BI: p2638 = 1 signal Fixed stop is located outside the monitoring window. BI: p2638 = 0 signal Fixed stop is inside the monitoring window.		
Dependency:	Refer to: r2526, p2635		
Note:	The identification of "fixed stop outside the monitoring window" is, for the factory setting, dependent on signal BO: r2526.5 (fixed stop outside window). This signal is influenced via p2635 (EPOS fixed stop monitoring window).		
p2639	BI: EPOS force limit reached / F_limit reached		
SERVO (EPOS, Lin)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting 1407.7
Description:	Sets the signal source for the feedback signal "force limit reached" when traversing to fixed stop. BI: p2639 = 1 signal Force limit has been reached. BI: p2639 = 0 signal Force limit has not been reached.		
Dependency:	Refer to: r1407		
Note:	The feedback signal from "force limit reached" is, for the factory setting, dependent on the signal BO: r1407.7 (torque limit reached).		

p2639	BI: EPOS torque limit reached / M_limit reached		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3616
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1407.7
Description:	Sets the signal source for the feedback signal "torque limit reached" when traversing to fixed stop. BI: p2639 = 1 signal Torque limit is reached. BI: p2639 = 0 signal Torque limit is not reached.		
Dependency:	Refer to: r1407		
Note:	The feedback signal from "torque limit reached" is, for the factory setting, dependent on the signal BO: r1407.7 (torque limit reached).		
p2640	BI: EPOS intermediate stop (0 signal) / Intermediate stop		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3616, 3620, 3625
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the "no intermediate stop/intermediate stop". BI: p2640 = 1 signal No intermediate stop. BI: p2640 = 0 signal Intermediate stop.		
Dependency:	Refer to: p2631, p2641, p2647, p2649		
Caution:	For BI: p2649 = 1 signal, the following applies: Motion starts without any explicit control signal.		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	This signal is only effective in the modes "traversing blocks" and "direct setpoint input/MDI". When activating the intermediate stop, the axis brakes with the parameterized deceleration (p2620 or p2645).		
p2641	BI: EPOS reject traversing task (0 signal) / Trav_task reject		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3616, 3620, 3625
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for "do not reject traversing task/reject traversing task". BI: p2641 = 1 signal Do not reject traversing task. BI: p2641 = 0 signal Reject traversing task.		
Dependency:	Refer to: p2631, p2640, p2647, p2649		
Caution:	For BI: p2649 = 1 signal, the following applies: Motion starts without any explicit control signal.		

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.
Note: This signal is only effective in the modes "traversing blocks" and "direct setpoint input/MDI".
 When activating reject traversing tasks, then the axis brakes with the maximum deceleration (p2573).

p2642	CI: EPOS direct setpoint input/MDI position setpoint / MDI s_set		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 3618
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	2690[0]
Description:	Sets the signal source for the position setpoint in the mode "direct setpoint input/MDI".		
Dependency:	Refer to: p2648, p2649, p2650, p2690		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Depending on p2649, the position setpoint is either transferred continuously or edge-triggered. The position setpoint input is interpreted as length unit [LU].		

p2643	CI: EPOS direct setpoint input/MDI velocity setpoint / MDI v_set		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 3618
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	2691[0]
Description:	Sets the signal source for the velocity setpoint in the "direct setpoint input/MDI mode".		
Dependency:	Refer to: p2649, p2650, p2691		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Depending on p2649, the velocity setpoint is either transferred continuously or edge-triggered. The velocity setpoint input is interpreted as [1000 LU/min].		

p2644	CI: EPOS direct setpoint input/MDI acceleration override / MDI a_over		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 3618
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	2692[0]
Description:	Sets the signal source for the acceleration override in the operating mode "direct setpoint input/MDI".		
Dependency:	Refer to: p2649, p2650, p2692		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	Depending on p2649, the acceleration override is either transferred continuously or edge-triggered. The signal value 4000 hex (16384 dec) corresponds to 100 %.		

p2645	CI: EPOS direct setpoint input/MDI deceleration override / MDI -a_over		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 3618
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	2693[0]
Description:	Sets the signal source for the deceleration override in the operating mode "direct setpoint input/MDI".		
Dependency:	Refer to: p2649, p2650, p2693		

Notice: If, when calculating the traversing profile, it is identified that the target position with the programmed deceleration override cannot be reached without reversing the direction, then when accepting the dynamic values, the larger deceleration override is accepted and becomes effective.

The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: Depending on p2649, the deceleration override is either transferred continuously or edge-triggered.
The signal value 4000 hex (16384 dec) corresponds to 100 %.

p2646 CI: EPOS velocity override / v_over

SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 3630
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
-	-	1	

Description: Sets the signal source for the velocity override.

This velocity override is effective in the following operating modes "direct setpoint input/MDI", "traversing blocks", "jogging" and "search for reference" (when approaching the reference cam).

Dependency: Refer to: p2571, p2585, p2586, p2605, p2618, p2643, r2681

Notice: The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note: The effective override (r2681) can differ from the specified override due to limits (e.g. maximum velocity).

p2647 BI: EPOS direct setpoint input/MDI selection / MDI selection

SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3620, 3625, 3640
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
-	-	0	

Description: Sets the signal source for selecting the operating mode "direct setpoint input/MDI".

Dependency: Refer to: p2640, p2641, p2642, p2643, p2644, p2645, p2646, p2648, p2649, p2650, p2651, p2652, p2653

Note: In this mode, using BI: p2653 it is possible to make a flying changeover between setting-up and positioning.
In this mode, even if the axis is not referenced (r2684.11 = 0) relative positioning is possible.

p2648 BI: EPOS direct setpoint input/MDI positioning type / MDI pos_type

SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3620
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
-	-	0	

Description: Sets the signal source for the positioning type in the mode "direct setpoint input/MDI".

BI: p2648 = 1 signal

Absolute positioning is selected.

BI: p2648 = 0 signal

Relative positioning is selected.

Dependency: Refer to: p2649, p2650, p2654

Refer to: A07461, F07488

Notice: Absolute positioning:

To traverse, the reference point must be set (r2684.11 = 1).

Relative positioning:

To traverse, it is not necessary that the reference point is set.

Note: Depending on p2649, the positioning type is either transferred continuously or edge-triggered.
Binector input p2648 is only evaluated when connector input p2654 = 0. If p2654 is a value other than 0, the positioning type is evaluated by means of the set signal source.

p2649	BI: EPOS direct setpoint input/MDI transfer type selection / MDI trans_type sel		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3620 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to define how values are transferred in the operating mode "direct setpoint input/MDI". BI: p2649 = 1 signal Values are continually transferred (refer to parameter under dependency). BI: p2649 = 0 signal The values are transferred for BI: p2650 = 0/1 signal.		
Dependency:	Refer to: p2642, p2643, p2644, p2645, p2648, p2650, p2651, p2652		
Caution:	For BI: p2649 = 1 signal, the following applies: Motion starts without any explicit control signal.		
Note:	Parameter p2649 can only be changed when p0922 (p2079) = 999.		
p2650	BI: EPOS direct setpoint input/MDI setpoint acceptance edge / MDI setp_accept		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3620 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to accept the values for edge-triggered selection (BI: p2649 = 0 signal) in the operating mode "direct setpoint input/MDI". BI: p2650 = 0/1 signal and BI: p2649 = 0 signal Values are accepted, edge-triggered (refer to parameter under dependency).		
Dependency:	Refer to: p2640, p2641, p2642, p2643, p2644, p2645, p2648, p2649, p2651, p2652, r2684		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	The status signal r2684.12 = 0/1 signal is used for acknowledgement. The operating mode "direct setpoint input/MDI" can be influenced via the following signals: - intermediate stop via BI: p2640. - reject traversing task via BI: p2641.		
p2651	BI: EPOS direct setpoint input/MDI direction selection, positive / MDI dir_sel pos		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3620 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the positive direction selection in the operating mode "direct setpoint input/MDI".		
Dependency:	Refer to: p2576, p2648, p2649, p2650, p2652, p2653, p2654		

Note: The following applies for "setting-up":

- the traversing direction can be entered using this binector input.
- if both directions (p2651, p2652) are selected, then the axis remains stationary (zero speed).
- if both directions (p2651, p2652) are deselected, then the axis remains stationary (zero speed).

The following applies for "positioning":

Using binector inputs p2651 and p2652, when the modulo correction (BI: p2577 = 1 signal) is activated and for absolute positioning (BI: p2648 = 1 signal), the traversing direction is specified as follows:

BI: p2651 / BI: p2652

0 signal / 0 signal: Absolute positioning through the shortest distance.

1 signal / 0 signal: Absolute positioning in the positive direction.

0 signal / 1 signal: Absolute positioning in the negative direction.

1 signal / 1 signal: Absolute positioning through the shortest distance.

p2652 BI: EPOS direct setpoint input/MDI direction selection, negative / MDI dir_sel neg

SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3620
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the negative direction selection in the operating mode "direct setpoint input/MDI".

Dependency: Refer to: p2576, p2648, p2649, p2650, p2651, p2653, p2654

Note: The following applies for "setting-up":

- the traversing direction can be entered using this binector input.
- if both directions (p2651, p2652) are selected, then the axis remains stationary (zero speed).
- if both directions (p2651, p2652) are deselected, then the axis remains stationary (zero speed).

The following applies for "positioning":

Using binector inputs p2651 and p2652, when the modulo correction (BI: p2577 = 1 signal) is activated and for absolute positioning (BI: p2648 = 1 signal), the traversing direction is specified as follows:

BI: p2651 / BI: p2652

0 signal / 0 signal: Absolute positioning through the shortest distance.

1 signal / 0 signal: Absolute positioning in the positive direction.

0 signal / 1 signal: Absolute positioning in the negative direction.

1 signal / 1 signal: Absolute positioning through the shortest distance.

p2653 BI: EPOS direct setpoint input/MDI setting-up selection / MDI setting-up sel

SERVO (EPOS), VECTOR (EPOS)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 3620
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for setting-up in the operating mode "direct setpoint input/MDI".

BI: p2653 = 1 signal

Setting-up selected.

BI: p2653 = 0 signal

Positioning selected.

Dependency: Refer to: p2651, p2652

Note: In the operating mode "direct setpoint input/MDI", it is possible to make a flying changeover between setting-up and positioning.

For "setup" (BI: p2653 = 1 signal), the following applies:

A traversing direction must be selected via binector inputs p2651 and p2652.

p2654			
CI: EPOS direct setpoint input/MDI mode adaptation / MDI mode adapt			
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Integer16 P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3620 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to interconnect the MDI mode to the operating mode "direct setpoint input MDI" via PROFIBUS telegram 110. CI: p2654 = 0 The binector inputs listed below are evaluated. CI: p2654 > 0 The following binector inputs are not evaluated: - BI: p2648 (positioning type) - BI: p2651 (direction selection, positive) - BI: p2652 (direction selection, negative) In this case, the following definitions apply: Signal via CI: p2654 = xx0x hex -> absolute Signal via CI: p2654 = xx1x hex -> relative Signal via CI: p2654 = xx2x hex -> abs_pos (only for modulo correction) Signal via CI: p2654 = xx3x hex -> abs_neg (only for modulo correction)		
Dependency:	Refer to: p2648, p2651, p2652		
p2655[0...1]			
BI: EPOS select tracking mode / Sel tracking mode			
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3635 Unit selection: - Expert list: 1 Factory setting [0] 1 [1] 2526.7
Description:	Sets the signal source to select tracking mode. BI: p2655[0] or BI: p2655[1] = 1 signal Tracking mode after withdrawing the enable signal from EPOS (BI: p2656 = 0 signal). BI: p2655[0] and BI: p2655[1] = 0 signal No tracking mode after withdrawing the enable signal from EPOS (BI: p2656 = 0 signal).		
Dependency:	Refer to: p2656		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	For the following events, independent of the signal that is present, tracking mode is selected: - after booting. - after a 0/1 signal at BI: p2658 (EPOS position actual value, valid feedback signal). - while a fault is present.		

p2656	BI: EPOS enable basic positioner / EPOS enable		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3635 Unit selection: - Expert list: 1 Factory setting 2526.3
Description:	Sets the signal source to enable the basic positioner. BI: p2656 = 1 signal The basic positioner is enabled. BI: p2656 = 0 signal The basic positioner is not enabled.		
Dependency:	Refer to: r2526, p2655		
p2657	CI: EPOS position actual value/position setting value / Pos act/set value		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Integer32 P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3610, 3616, 3620, 3635 Unit selection: - Expert list: 1 Factory setting 2521[0]
Description:	Sets the signal source for the position actual value/position setting value.		
Dependency:	Refer to: r2521, p2658		
Note:	In the tracking mode, the position setpoint is taken from this connector input.		
p2658	BI: EPOS pos. actual value valid, feedback signal / Pos valid feedback		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3635 Unit selection: - Expert list: 1 Factory setting 2526.0
Description:	Sets the signal source for the feedback signal "position actual value is valid". BI: p2658 = 1 signal The position actual value received via CI: p2657 is valid. BI: p2658 = 0 signal The position actual value received via CI: p2657 is invalid.		
Dependency:	Refer to: r2526, p2657		
Note:	While a 0 signal is present, the position setpoint (p2665) is held at the value of 0.		

p2659	BI: EPOS referencing active feedback signal / Ref act fdbk		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting 2526.1
Description:	Sets the signal source for the feedback signal "referencing active". BI: p2659 = 1 signal Referencing is active. BI: p2659 = 0 signal Referencing is not active.		
Dependency:	Refer to: r2526		
p2660	CI: EPOS measured value referencing / Meas val ref		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Integer32 P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3612, 3614 Unit selection: - Expert list: 1 Factory setting 2523[0]
Description:	Sets the signal source for the measured value for the function "referencing".		
Dependency:	Refer to: r2523		
p2661	BI: EPOS measured value valid, feedback signal / MeasVal valid fdbk		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3612, 3614, 3615 Unit selection: - Expert list: 1 Factory setting 2526.2
Description:	Sets the signal source for the feedback signal "measured value valid". BI: p2661 = 1 signal The measured value received via CI: p2660 is valid. BI: p2661 = 0 signal The measured value received via CI: p2660 is invalid.		
Dependency:	Refer to: r2526, p2660		
p2662	BI: EPOS adjustment value valid feedback signal / Adj val valid FS		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2526.9
Description:	Sets the signal source for the feedback signal "adjustment value valid". BI: p2662 = 1 signal The adjustment value received via CI: p2660 is valid. BI: p2662 = 0 signal The adjustment value received via CI: p2660 is not valid.		
Dependency:	Refer to: r2526, p2660		

p2663	BI: EPOS clamping active feedback signal / Clamping active FS		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting 2526.8
Description:	Sets the signal source for the feedback signal "clamping active for travel to fixed stop". BI: p2663 = 1 signal Clamping is active BI: p2663 = 0 signal Clamping is not active.		
Dependency:	Refer to: r2526		
Note:	The feedback signal from "terminals active" is, for the factory setting, dependent on the signal BO: r2526.8 (terminals active when moving to a fixed stop).		
r2665	CO: EPOS position setpoint / s_set		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min - [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [LU]	Access level: 1 Func. diagram: 3635 Unit selection: - Expert list: 1 Factory setting - [LU]
Description:	Displays the current absolute position setpoint.		
Dependency:	Refer to: p2530		
Note:	As standard, the following BICO interconnection is established: CI: p2530 = r2665		
r2666	CO: EPOS velocity setpoint / v_set		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min - [1000 LU/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [1000 LU/min]	Access level: 1 Func. diagram: 3635 Unit selection: - Expert list: 1 Factory setting - [1000 LU/min]
Description:	Displays the current velocity setpoint.		
Dependency:	Refer to: p2531		
Note:	As standard, the following BICO interconnection is established: CI: p2531 = r2666		
r2667	CO: EPOS backlash compensation value / Backlash value		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min - [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [LU]	Access level: 1 Func. diagram: 3635 Unit selection: - Expert list: 1 Factory setting - [LU]
Description:	Displays the currently effective value for backlash compensation.		
Dependency:	Refer to: p2516		
Note:	As standard, the following BICO interconnection is established: CI: p2516 = r2667		

r2669	CO: EPOS current operating mode / Op mode act																																												
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3625, 3630 Unit selection: - Expert list: 1 Factory setting -																																										
Description:	Displays the current active operating mode. Value = 00 hex -> no operating mode active Value = 01 hex -> jogging active Value = 02 hex -> search for reference active Value = 04 hex -> traversing blocks active Value = 08 hex -> Positioning for direct setpoint input/MDI active Value = 10 hex -> Setting-up for direct setpoint input/MDI active Value = 20 hex -> flying referencing active																																												
Dependency:	Refer to: p2589, p2590, p2595, p2631, p2647, p2653																																												
r2670.0...15	CO/BO: EPOS status word, active traversing block / ZSW act trav_block																																												
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Unsigned32 P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3615, 3625, 3650 Unit selection: - Expert list: 1 Factory setting -																																										
Description:	Displays the status word for the active traversing block. r2670.0: Active traversing block, bit 0 ... r2670.5: Active traversing block, bit 5 r2670.15: MDI active																																												
Bit field:	<table><tr><th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr><tr><td>00</td><td>Active traversing block, bit 0</td><td>Active</td><td>Not active</td><td>-</td></tr><tr><td>01</td><td>Active traversing block, bit 1</td><td>Active</td><td>Not active</td><td>-</td></tr><tr><td>02</td><td>Active traversing block, bit 2</td><td>Active</td><td>Not active</td><td>-</td></tr><tr><td>03</td><td>Active traversing block, bit 3</td><td>Active</td><td>Not active</td><td>-</td></tr><tr><td>04</td><td>Active traversing block, bit 4</td><td>Active</td><td>Not active</td><td>-</td></tr><tr><td>05</td><td>Active traversing block, bit 5</td><td>Active</td><td>Not active</td><td>-</td></tr><tr><td>15</td><td>MDI active</td><td>Active</td><td>Not active</td><td>-</td></tr></table>	Bit	Signal name	1 signal	0 signal	FP	00	Active traversing block, bit 0	Active	Not active	-	01	Active traversing block, bit 1	Active	Not active	-	02	Active traversing block, bit 2	Active	Not active	-	03	Active traversing block, bit 3	Active	Not active	-	04	Active traversing block, bit 4	Active	Not active	-	05	Active traversing block, bit 5	Active	Not active	-	15	MDI active	Active	Not active	-				
Bit	Signal name	1 signal	0 signal	FP																																									
00	Active traversing block, bit 0	Active	Not active	-																																									
01	Active traversing block, bit 1	Active	Not active	-																																									
02	Active traversing block, bit 2	Active	Not active	-																																									
03	Active traversing block, bit 3	Active	Not active	-																																									
04	Active traversing block, bit 4	Active	Not active	-																																									
05	Active traversing block, bit 5	Active	Not active	-																																									
15	MDI active	Active	Not active	-																																									
Dependency:	Refer to: p2631, p2647																																												
Note:	Re bit 00 ... 05: Displays the active traversing block in the traversing blocks operating mode. Re bit 15: For a 1 signal, the operating mode - direct setpoint input/MDI - is active																																												

r2671	CO: EPOS current position setpoint / s_set act		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32	Calculated: - Dynamic index: -	Access level: 1 Func. diagram: 3610, 3616, 3620
	P-Group: Basic positioner Not for motor type: -	Units group: - Scaling: -	Unit selection: - Expert list: 1
	Min - [LU]	Max - [LU]	Factory setting - [LU]
Description:	Displays the position setpoint presently being processed.		
Note:	A position of zero is displayed for non position-related tasks (e.g. ENDLESS_POS, ENDLESS_NEG).		
r2672	CO: EPOS current velocity setpoint / v_set act		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32	Calculated: - Dynamic index: -	Access level: 1 Func. diagram: 3610, 3612, 3616, 3620
	P-Group: Basic positioner Not for motor type: -	Units group: - Scaling: -	Unit selection: - Expert list: 1
	Min - [1000 LU/min]	Max - [1000 LU/min]	Factory setting - [1000 LU/min]
Description:	Displays the velocity setpoint presently being processed.		
r2673	CO: EPOS current acceleration override / a_over act		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: FloatingPoint32	Calculated: - Dynamic index: -	Access level: 1 Func. diagram: 3610, 3612, 3616, 3620
	P-Group: Basic positioner Not for motor type: -	Units group: - Scaling: -	Unit selection: - Expert list: 1
	Min - [%]	Max - [%]	Factory setting - [%]
Description:	Displays the acceleration override presently being processed.		
Note:	An override of 100% is effective in the "jogging" and "search for reference" operating modes.		
r2674	CO: EPOS current deceleration override / -a_over act		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: FloatingPoint32	Calculated: - Dynamic index: -	Access level: 1 Func. diagram: 3610, 3612, 3616, 3620
	P-Group: Basic positioner Not for motor type: -	Units group: - Scaling: -	Unit selection: - Expert list: 1
	Min - [%]	Max - [%]	Factory setting - [%]
Description:	Displays the deceleration override presently being processed.		
Note:	An override of 100% is effective in the "jogging" and "search for reference" operating modes.		

r2675	CO: EPOS current task / Task cur		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer16 P-Group: Basic positioner Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 9	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the task that is presently being processed.		
Value:	0: Inactive 1: POSITIONING 2: FIXED STOP 3: ENDLESS_POS 4: ENDLESS_NEG 5: WAITING 6: GOTO 7: SET_O 8: RESET_O 9: JERK		
Dependency:	Refer to: p2621		
r2676	CO: EPOS current task parameter / Task para cur		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the task parameter presently being processed in the "traversing blocks" operating mode.		
Dependency:	Refer to: p2622		
Note:	The following is displayed depending on the task: FIXED STOP: Clamping torque (0 ... 65536 [0.01 Nm]) or clamping force (0 ... 65536 [N]) WAIT: Delay time [ms] GOTO: Block number SET_O: 1, 2, 3 --> direct output 1, 2 or 3 (both) is set RESET_O: 1, 2, 3 --> direct output 1, 2 or 3 (both) is set JERK: 0 --> deactivate, 1 --> activate		
r2677	CO: EPOS current task mode / Task mode cur		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Unsigned16 P-Group: Basic positioner Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 3616 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the task mode presently being processed.		
Dependency:	Refer to: p2623		

r2678	CO: EPOS external block change / Ext BlckChg s_act		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min - [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [LU]	Access level: 1 Func. diagram: 3615, 3616, 3620 Unit selection: - Expert list: 1 Factory setting - [LU]
Description:	Displays the actual position for the following events: - external block change via measuring probe (p2632 = 0, BI: p2661 = 0/1 signal). - external block change via BI: p2633 (p2632 = 1, BI: p2633 = 0/1 signal). - activate traversing task (BI: p2631 = 0/1 signal).		
Dependency:	Refer to: p2631, p2632, p2633, p2661		
r2680	CO: EPOS clearance, reference cam and zero mark / Clearance cam/ZM		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min - [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [LU]	Access level: 1 Func. diagram: 3612 Unit selection: - Expert list: 1 Factory setting - [LU]
Description:	Displays the clearance determined between the reference cam and zero mark in the search for reference.		
r2681	CO: EPOS velocity override effective / v_over effective		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: FloatingPoint32 P-Group: Basic positioner Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 1 Func. diagram: 3630 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the currently effective velocity override.		
Dependency:	Refer to: p2571, p2646		
Note:	The effective override can differ from the specified override due to limits (e.g. p2571, maximum velocity).		
r2682	CO: EPOS residual distance to go / Residual distance		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min - [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [LU]	Access level: 1 Func. diagram: 3635 Unit selection: - Expert list: 1 Factory setting - [LU]
Description:	Displays the current residual distance. The remaining distance is the distance to still to be moved through up to the end of the current positioning task.		
Dependency:	Refer to: r2665, r2671, r2678		

r2683.0...14 CO/BO: EPOS status word 1 / POS_ZSW1

SERVO (Lin, Pos ctrl)	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 3645
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

Description: Displays status word 1 for the basic positioner (EPOS).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Tracking mode active	Yes	No	3635, 4020
	01	Velocity limiting active	Yes	No	3630
	02	Setpoint available	Yes	No	3635
	03	Target position reached	Yes	No	3635
	04	Axis moves forwards	Yes	No	3635
	05	Axis moves backwards	Yes	No	3635
	06	Software limit switch minus reached	Yes	No	3635
	07	Software limit switch plus reached	Yes	No	3635
	08	Position actual value <= cam switching position 1	Yes	No	4025
	09	Position actual value <= cam switching position 2	Yes	No	4025
	10	Direct output 1 via traversing block	Yes	No	3616
	11	Direct output 2 via traversing block	Yes	No	3616
	12	Fixed stop reached	Yes	No	3616, 3617
	13	Fixed stop clamping force reached	Yes	No	3616, 3617
	14	Travel to fixed stop active	Yes	No	3616, 3617

Dependency: Refer to: r2684

Note: Re bit 02, 04, 05, 06, 07:
This signals designate the state after jerk limiting.
Re bit 08, 09:
These signals are generated in the "closed-loop position control" function module.

r2683.0...14 CO/BO: EPOS status word 1 / POS_ZSW1

SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 3645
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

Description: Displays status word 1 for the basic positioner (EPOS).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Tracking mode active	Yes	No	3635, 4020
	01	Velocity limiting active	Yes	No	3630
	02	Setpoint available	Yes	No	3635
	03	Target position reached	Yes	No	3635
	04	Axis moves forwards	Yes	No	3635
	05	Axis moves backwards	Yes	No	3635
	06	Software limit switch minus reached	Yes	No	3635
	07	Software limit switch plus reached	Yes	No	3635
	08	Position actual value <= cam switching position 1	Yes	No	4025

09	Position actual value <= cam switching position 2	Yes	No	4025
10	Direct output 1 via traversing block	Yes	No	3616
11	Direct output 2 via traversing block	Yes	No	3616
12	Fixed stop reached	Yes	No	3616, 3617
13	Fixed stop clamping torque reached	Yes	No	3616, 3617
14	Travel to fixed stop active	Yes	No	3616, 3617

Dependency: Refer to: r2684

Note: Re bit 02, 04, 05, 06, 07:

This signals designate the state after jerk limiting.

Re bit 08, 09:

These signals are generated in the "closed-loop position control" function module.

r2684.0...15 CO/BO: EPOS status word 2 / POS_ZSW2

SERVO (Pos ctrl),
VECTOR (Pos ctrl)

Can be changed: -

Calculated: -

Access level: 1

Data type: Unsigned16

Dynamic index: -

Func. diagram: 3646

P-Group: Closed loop position control

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

-

-

-

Description: Displays status word 2 for the basic positioner (EPOS).

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Search for reference active	Active	Not active	3612
01	Flying referencing active	Active	Not active	3614
02	Referencing active	Active	Not active	-
03	Printing mark outside outer window	Yes	No	3614
04	Axis accelerating	Yes	No	3635
05	Axis decelerating	Yes	No	3635
06	Jerk limiting active	Yes	No	3635
07	Activate correction	Yes	No	3635
08	Following error in tolerance	Yes	No	4025
09	Modulo correction active	Yes	No	-
10	Target position reached	Yes	No	4020
11	Reference point set	Yes	No	3612, 3614, 3630
12	Acknowledgement, traversing block activated	Yes	No	3616, 3620
13	STOP cam minus active	Yes	No	3630
14	STOP cam plus active	Yes	No	3630
15	Traversing command active	Yes	No	3635

Note: Re bit 02:

The "referencing active" signal is an OR logic operation of "search for reference active" and "flying referencing active".

Re bit 00 ... 07 and 11 ... 14:

These signals are generated in the function module "basic positioner".

Re bit 08:

The signal is generated in the "closed-loop position control" function module.

r2685	CO: EPOS corrective value / Corrective value		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: Integer32 P-Group: Basic positioner Not for motor type: - Min - [LU]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [LU]	Access level: 1 Func. diagram: 3635 Unit selection: - Expert list: 1 Factory setting - [LU]
Description:	Displays the corrective value for the position actual value.		
Dependency:	Refer to: r2684		
Note:	As standard, the following BICO interconnection is established: CI: p2513 = r2685 Using this value, e.g. modulo corrections are carried out.		
r2686[0...1]	CO: EPOS force limit effective / F_limit eff		
SERVO (EPOS, Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Basic positioner Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: 3616, 3617 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the effective force limiting. r2686[0]: Displays the effective upper force limiting when traversing to fixed stop (referred to CI: p1522, CI: p1523). r2686[1]: Displays the effective lower force limiting when traversing to fixed stop (referred to CI: p1522, CI: p1523).		
Dependency:	Refer to: p1520, p1521, p1522, p1523, r2676		
Note:	As standard, the following BICO interconnections are established: CI: p1528 = r2686[0] CI: p1529 = r2686[1]		
r2686[0...1]	CO: EPOS torque limiting effective / M_limit eff		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: - Data type: FloatingPoint32 P-Group: Basic positioner Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: 3616, 3617 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the effective torque limiting. r2686[0]: Displays the effective upper torque limiting when traversing to fixed stop (referred to CI: p1522, CI: p1523). r2686[1]: Displays the effective lower torque limiting when traversing to fixed stop (referred to CI: p1522, CI: p1523).		
Dependency:	Refer to: p1520, p1521, p1522, p1523, r2676		
Note:	As standard, the following BICO interconnections are established: CI: p1528 = r2686[0] CI: p1529 = r2686[1]		

r2687 CO: EPOS force setpoint / F_set

SERVO (EPOS, Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 3616, 3617
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [N]	Max - [N]	Factory setting - [N]
Description:	Displays the effective force setpoint when reaching the fixed stop (referred to CI: p1522, CI: p1523).		
Dependency:	Refer to: p1520, p1521, p1522, p1523, r2676		

r2687 CO: EPOS torque setpoint / M_set

SERVO (EPOS), VECTOR (EPOS)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 3616, 3617
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [Nm]	Max - [Nm]	Factory setting - [Nm]
Description:	Displays the effective torque setpoint when reaching the fixed stop (referred to CI: p1522, CI: p1523).		
Dependency:	Refer to: p1520, p1521, p1522, p1523, r2676		

p2690 CO: EPOS position fixed setpoint / Pos fixed value

SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Integer32	Dynamic index: -	Func. diagram: 3618
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -2147482648 [LU]	Max 2147482647 [LU]	Factory setting 0 [LU]
Description:	Sets a fixed setpoint for the position.		
Dependency:	Refer to: p2642, p2648		
Note:	As standard, the following BICO interconnection is established: CI: p2642 = r2690		

p2691 CO: EPOS velocity fixed setpoint / v fixed value

SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 3618
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1 [1000 LU/min]	Max 40000000 [1000 LU/min]	Factory setting 600 [1000 LU/min]
Description:	Sets a fixed setpoint for the velocity.		
Dependency:	Refer to: p2643		
Note:	As standard, the following BICO interconnection is established: CI: p2643 = r2691		

p2692 CO: EPOS acceleration override, fixed setpoint / a_over fixed val

SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 3618
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.100 [%]	Max 100.000 [%]	Factory setting 100.000 [%]
Description:	Sets a fixed setpoint for the acceleration override.		
Dependency:	Refer to: p2572, p2644		

Note: As standard, the following BICO interconnection is established: CI: p2644 = r2692
The percentage value refers to the maximum acceleration (p2572).

p2693	CO: EPOS deceleration override, fixed setpoint / -a_over fixed val		
SERVO (EPOS), VECTOR (EPOS)	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 3618
	P-Group: Basic positioner	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.100 [%]	100.000 [%]	100.000 [%]

Description: Sets a fixed setpoint for the deceleration override.

Dependency: Refer to: p2573, p2645

Note: As standard, the following BICO interconnection is established: CI: p2645 = r2693
The percentage value refers to the maximum deceleration (p2573).

r2700	CO: Reference frequency / f_ref		
A_INF, B_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Display and connector output of the current reference quantity for the frequency (p2000).
All frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.
This parameter has the unit Hz.

Dependency: Refer to: p2000

Note: This parameter provides the numerical value of the reference quantity p2000 as a connector output for interconnection with Drive Control Chart (DCC). The numerical value can be adopted unchanged from this connector output in DCC.

r2700	CO: Reference speed/reference frequency / n_ref/f_ref		
ENCODER, SERVO, TM41, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Display and connector output for the reference quantity for speed and frequency (p2000).
All speeds or frequencies specified as relative value are referred to this reference quantity.
The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.
The following applies: Reference frequency (in Hz) = reference speed (in rpm) / 60
This parameter has the unit rpm.

Dependency: Refer to: p2000

Note: This parameter provides the numerical value of the reference quantity p2000 as a connector output for interconnection with Drive Control Chart (DCC). The numerical value can be adopted unchanged from this connector output in DCC.

r2700	CO: Reference velocity/reference frequency current / v_ref/f_ref act		
ENCODER (Lin enc)	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Display and connector output for the current reference quantity for velocity and frequency. All velocities or frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The following applies: Reference frequency (in Hz) = reference velocity (in m/min) / 60		
Dependency:	Refer to: p2000		
Note:	This parameter provides the numerical value of the reference quantity p2000 as a connector output for interconnection with Drive Control Chart (DCC). The numerical value can be adopted unchanged from this connector output in DCC.		
r2700	CO: Reference frequency, current / f_ref act		
S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Display and connector output of the current reference quantity for the frequency (p2000). All frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. This parameter has the unit Hz.		
Dependency:	Refer to: p2000		
Note:	This parameter provides the numerical value of the reference quantity p2000 as a connector output for interconnection with Drive Control Chart (DCC). The numerical value can be adopted unchanged from this connector output in DCC.		
r2700	CO: Reference velocity/reference frequency current / v_ref/f_ref act		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Display and connector output for the current reference quantity for velocity and frequency. All velocities or frequencies specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The following applies: Reference frequency (in Hz) = reference velocity (in m/min) / 60		
Dependency:	Refer to: p2000		

Note: This parameter represents the numerical value of the reference quantity in the currently selected units and is only available for interconnection with Drive Control Chart (DCC).
 If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.
 Example 1:
 The signal of an analog input (e.g. r4055[0]) is connected to a velocity setpoint (e.g. p1070[0]). The actual percentage input value is cyclically converted into the absolute velocity setpoint using the reference velocity (p2000).
 Example 2:
 The setpoint from PROFIBUS (r2050[1]) is connected to a velocity setpoint (e.g. p1070[0]). The actual input value is cyclically converted into a percentage value via the pre-specified scaling 4000 hex. This percentage value is converted to the absolute velocity setpoint via reference velocity (p2000).

r2701	CO: Reference voltage / Reference voltage		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Connector output of the reference quantity for voltages p2001. All voltages specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. This parameter has the unit Vrms.		
Dependency:	Refer to: p2001		
Note:	This parameter provides the numerical value of the reference quantity p2001 as a connector output for interconnection with Drive Control Chart (DCC). The numerical value can be adopted unchanged from this connector output in DCC.		

r2702	CO: Reference current / Reference current		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Connector output of the reference quantity for currents p2002. All currents specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. This parameter has the unit Arms.		
Dependency:	Refer to: p2002		
Note:	This parameter provides the numerical value of the reference quantity p2002 as a connector output for interconnection with Drive Control Chart (DCC). The numerical value can be adopted unchanged from this connector output in DCC.		

r2703	CO: Reference torque / Reference torque		
SERVO, TM41, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Connector output of the reference quantity p2003 for torque (r0108.12 = 0) or force (r0108.12 = 1). All torques specified as relative values (r0108.12 = 0) or forces (r0108.12 = 1) are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The unit of this parameter is the same as the unit selected for p2003.		

Dependency: p0505, r0108.12
Refer to: p2003

Note: This parameter provides the numerical value of the reference quantity p2003 in the currently selected unit as a connector output for interconnection with Drive Control Chart (DCC). The numerical value can be adopted unchanged from this connector output in DCC.

r2703 CO: Reference force current / Ref force cur

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the current reference quantity for forces.

All forces specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.

Dependency: p0505, r0108.12
Refer to: p2003

Note: This parameter represents the numerical value of the reference quantity in the currently selected units and is only available for interconnection with Drive Control Chart (DCC).
If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.
Example:
The actual value of the total force (r0079[0]) is connected to a test socket (e.g. p0771[0]). The actual force is cyclically converted into a percentage of the reference force (p2003) and output according to the parameterized scaling.

r2704 CO: Reference power / Reference power

A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Connector output of the reference quantity for powers p2004.

All power ratings specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex.

The unit of this parameter is the same as the unit selected for p2004.

Dependency: This value is calculated as voltage x current for the infeed and as torque x speed for closed-loop controls.
Refer to: r2004

Note: This parameter provides the numerical value of the reference quantity p2004 in the currently selected unit as a connector output for interconnection with Drive Control Chart (DCC). The numerical value can be adopted unchanged from this connector output in DCC.
The reference power is calculated as follows:
- $2 \cdot \pi \cdot \text{reference speed} / 60 \cdot \text{reference torque (motor)}$
- $\text{reference voltage} \cdot \text{reference current} \cdot \text{root}(3)$ (infeed)

r2705	CO: Reference angle / Reference angle		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Connector output of the reference quantity for angles p2005. All angles specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. This parameter has the unit degree.		
Dependency:	Refer to: p2005		
Note:	This parameter provides the numerical value of the reference quantity p2005 as a connector output for interconnection with Drive Control Chart (DCC). The numerical value can be adopted unchanged from this connector output in DCC.		
r2706	CO: Reference temperature / Reference temp		
A_INF, B_INF, S_INF, SERVO, TM120, TM31, TM41, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Connector output of the reference quantity for temperatures. All temperatures specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. This parameter has the unit degree Celsius.		
Note:	This parameter provides the numerical value of the reference quantity for the temperature as a connector output for interconnection with Drive Control Chart (DCC). The numerical value can be adopted unchanged from this connector output in DCC.		
r2707	CO: Reference acceleration / Ref accel		
SERVO, TM41, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Connector output of the reference quantity for accelerations p2007. All acceleration rates specified as relative value are referred to this reference quantity. The reference quantity in this parameter corresponds to 100% or 4000 hex or 4000 0000 hex. The unit of this parameter is the same as the unit selected for p2007.		
Dependency:	r0108.12, p0505 Refer to: p2007		
Note:	This parameter provides the numerical value of the reference quantity p2007 as a connector output for interconnection with Drive Control Chart (DCC). The numerical value in the currently selected unit can be adopted unchanged from this connector output in DCC.		

p2720[0...n] Load gear configuration / Load gear config

SERVO, VECTOR	Can be changed: C2(1, 4)	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the configuration for position tracking of a load gear.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Load gear, activate position tracking	Yes	No	-
	01	Axis type	Linear axis	Rotary axis	-
	02	Load gear, reset position	Yes	No	-

Note: For the following events, the non-volatile, saved position values are automatically reset:

- when an encoder replacement has been identified.
- when changing the configuration of the Encoder Data Set (EDS).
- when adjusting the absolute encoder again

p2721[0...n] Load gear, rotary absolute gearbox, revolutions, virtual / Abs rot rev

SERVO, VECTOR	Can be changed: C2(1, 4)	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	4194303	0

Description: Sets the number of rotations that can be resolved for a rotary absolute encoder with activated position tracking of the load gear.

Dependency: This parameter is only of significance for an absolute encoder (p0404.1 = 1) with activated position tracking of the load gear (p2720.0 = 1).

Note: The resolution that is set must be able to be represented using r2723.
 For rotary axes/modulo axes, the following applies:
 This parameter is preset with p0421 and can be changed.
 For linear axes, the following applies:
 This parameter is pre-assigned with p0421, expanded by 6 bits for multiturn information (maximum number of overflows) and cannot be changed.

p2722[0...n] Load gear, position tracking tolerance window / Pos track tol

SERVO, VECTOR	Can be changed: C2(1, 4)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00	4294967300.00	0.00

Description: Sets a tolerance window for position tracking.
 After the system is powered up, the difference between the saved position and the actual position is determined, and depending on this, the following is initiated:
 Difference within the tolerance window --> The position is reproduced as a result of the encoder actual value.
 Difference outside the tolerance window --> An appropriate message is output.

Dependency: Refer to: F07449

Caution: Rotation, e.g. through a complete encoder range is not detected.



Note: The value is entered in integer (complete) encoder pulses.
 For p2720.0 = 1, the value is automatically pre-assigned quarter of the encoder range.
 Example:
 Quarter of the encoder range = (p0408 * p0421) / 4
 It is possible that the tolerance window may not be able to be precisely set due to the data type (floating point number with 23 bit mantissa).

r2723[0...n]	CO: Load gear absolute value / Load gear abs_val		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: DDS, p0180	Func. diagram: 4010, 4704
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the absolute value after the load gear.		
Notice:	The encoder position actual value must be requested using the encoder control word Gn_STW.13.		
Note:	The increments are displayed in the format the same as r0483.		

r2724[0...n]	CO: Load gear position difference / Load gear pos diff		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the position difference before the load gear between powering down and powering up.		
Note:	The increments are displayed in the same format as for r0483/r2723.		
	If the measuring gear of the motor encoder is not activated, the position difference should be read in encoder increments.		
	If the measuring gear of the motor encoder is activated, the position difference is converted using the measuring gear factor.		

p2730[0...3]	BI: LR pos. actual value preprocessing activate neg. corr. (edge) / ActV_prep neg corr		
SERVO (Pos ctrl), VECTOR (Pos ctrl)	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 4010, 4015
	P-Group: Closed loop position control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the function "activate position actual value preprocessing, negative corrective value (edge)".		
	0/1 signal: The corrective value available through CI: p2513 is negated and activated.		
Index:	[0] = Closed-loop position control [1] = Encoder 1 [2] = Encoder 2 [3] = Encoder 3		
Dependency:	Refer to: p2502, p2513, r2684		

p2810[0...1]		BI: AND logic operation inputs / AND inputs			
SERVO, VECTOR	Can be changed: T	Calculated: -		Access level: 2	
	Data type: Unsigned32 / Binary	Dynamic index: -		Func. diagram: 2634	
	P-Group: Functions	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min	Max		Factory setting	
	-	-		0	
Description:		Sets the signal sources for the inputs of the AND logic operation.			
Dependency:		Refer to: r2811			
Note:		[0]: AND logic operation, input 1 --> the result is displayed in r2811.0. [1]: AND logic operation, input 2 --> the result is displayed in r2811.0.			
r2811.0		CO/BO: AND logic operation result / AND result			
SERVO, VECTOR	Can be changed: -	Calculated: -		Access level: 2	
	Data type: Unsigned32	Dynamic index: -		Func. diagram: 2634	
	P-Group: Functions	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min	Max		Factory setting	
	-	-		-	
Description:		Displays the result of the AND logic operation			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	AND logic operation result	Yes	No	-
Dependency:		Refer to: p2810			
p2816[0...1]		BI: OR logic operation inputs / OR inputs			
SERVO, VECTOR	Can be changed: T	Calculated: -		Access level: 2	
	Data type: Unsigned32 / Binary	Dynamic index: -		Func. diagram: 2634	
	P-Group: Functions	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min	Max		Factory setting	
	-	-		0	
Description:		Sets the signal sources for the inputs of the OR logic operation.			
Dependency:		Refer to: r2817			
Note:		[0]: OR logic operation, input 1 --> the result is displayed in r2817.0. [1]: OR logic operation, input 2 --> the result is displayed in r2817.0.			
r2817.0		CO/BO: OR logic operation result / OR result			
SERVO, VECTOR	Can be changed: -	Calculated: -		Access level: 2	
	Data type: Unsigned32	Dynamic index: -		Func. diagram: 2634	
	P-Group: Functions	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min	Max		Factory setting	
	-	-		-	
Description:		Displays the result of the OR logic operation.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	OR logic operation result	Yes	No	-
Dependency:		Refer to: p2816			

p2900[0...n]	CO: Fixed value 1 [%] / Fixed value 1 [%]		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Free function blocks Not for motor type: - Min -10000.00 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: PERCENT Max 10000.00 [%]	Access level: 3 Func. diagram: 1021 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets a fixed percentage.		
Dependency:	Refer to: p2901, p2930		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	The value can be used to interconnect a scaling function (e.g. scaling of the main setpoint)		
p2901[0...n]	CO: Fixed value 2 [%] / Fixed value 2 [%]		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Free function blocks Not for motor type: - Min -10000.00 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: PERCENT Max 10000.00 [%]	Access level: 3 Func. diagram: 1021 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets a fixed percentage.		
Dependency:	Refer to: p2900, p2930		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	The value can be used to interconnect a scaling function (e.g. scaling of the supplementary setpoint)		
r2902[0...14]	CO: Fixed values [%] / Fixed values [%]		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Free function blocks Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 1 Func. diagram: 1021 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Signal sources for frequently used percentage values.		
Index:	[0] = Fixed value +0 % [1] = Fixed value +5 % [2] = Fixed value +10 % [3] = Fixed value +20 % [4] = Fixed value +50 % [5] = Fixed value +100 % [6] = Fixed value +150 % [7] = Fixed value +200 % [8] = Fixed value -5 % [9] = Fixed value -10 % [10] = Fixed value -20 % [11] = Fixed value -50 % [12] = Fixed value -100 % [13] = Fixed value -150 % [14] = Fixed value -200 %		
Dependency:	Refer to: p2900, p2901, p2930		
Note:	The signal sources can, for example, be used to interconnect scalings.		

p2930[0...n]	CO: Fixed value M [Nm] / Fixed value M [Nm]		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Free function blocks Not for motor type: REL Min -100000.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: p2003 Max 100000.00 [Nm]	Access level: 3 Func. diagram: 1021 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Nm]
Description:	Sets a fixed value for torque.		
Dependency:	Refer to: p2900, p2901		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	The value can, for example, be used to interconnect a supplementary torque.		

p2930[0...n]	CO: Fixed value F [N] / Fixed value F [N]		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Free function blocks Not for motor type: REL Min -100000.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: p2003 Max 100000.00 [N]	Access level: 3 Func. diagram: 1021 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [N]
Description:	Sets a fixed value for force.		
Dependency:	Refer to: p2900, p2901		
Notice:	A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.		
Note:	The value can, for example, be used to interconnect a supplementary force.		

p3016	MotId torque constant identified / kT ident		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min 0.00 [Nm/A]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: 28_1 Scaling: - Max 100.00 [Nm/A]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.00 [Nm/A]
Description:	Torque constant for the synchronous motor determined by the motor data identification. This torque constant can be changed after the identification and accepted in p0316 with p1910/p1960 = -3.		
Dependency:	Refer to: p0316, r0334, r1937, p1960		

p3016	MotId force constant identified / kT ident		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min 0.00 [N/Arms]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: 29_1 Scaling: - Max 1000.00 [N/Arms]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.00 [N/Arms]
Description:	Force constant for a synchronous linear motor determined by the motor data identification. This force constant can be changed after the identification and is accepted in p0316 with p1910/p1960 = -3.		
Dependency:	Refer to: p0316, r0334, r1937, p1960		

p3017	Motld voltage constant identified / kE ident		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min 0.0 [Vrms]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 10000.0 [Vrms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [Vrms]
Description:	Voltage constant for a synchronous motor determined by the motor data identification. This voltage constant can be changed after the identification and accepted in p0317 with p1910/p1960 = -3. Units for rotating synchronous motors: Vrms/(1000 rpm), phase-to-phase		
Dependency:	Refer to: p0317, r1938, p1960		
p3017	Motld voltage constant identified / kE ident		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min 0.0 [Vrms s/m]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 1000.0 [Vrms s/m]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [Vrms s/m]
Description:	Voltage constant for a synchronous linear motor determined by the motor data identification. This voltage constant can be changed after the identification and accepted in p0317 with p1910/p1960 = -3. Units for linear synchronous motors: Vrms s/m, phase		
Dependency:	Refer to: p0317, r1938, p1960		
p3020	Motld magnetizing current identified / I_mag ident		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL, FEM Min 0.000 [Arms]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 5000.000 [Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [Arms]
Description:	Magnetizing current for an induction motor determined by the motor data identification. This magnetizing current can be changed after the identification and accepted in p0320 with p1910/p1960 = -3.		
Dependency:	Refer to: p0320, r0331, p1910, r1948, p1960		
p3027	Motld optimum load angle identified / phi_load opt ident		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min 0.0 [°]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 135.0 [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [°]
Description:	Optimum load angle for a synchronous motor determined by the motor data identification. This optimum load angle can be changed after the identification and accepted in p0327 with p1910/p1960 = -3.		
Dependency:	Refer to: p0327, r1947, p1960		

p3028	Motld reluctance torque constant identified / kT_reluct ident				
SERVO	Can be changed: U, T		Calculated: CALC_MOD_ALL	Access level: 3	
	Data type: FloatingPoint32		Dynamic index: -	Func. diagram: -	
	P-Group: Motor identification		Units group: -	Unit selection: -	
	Not for motor type: ASM, REL, FEM		Scaling: -	Expert list: 1	
	Min -1000.00 [mH]		Max 1000.00 [mH]	Factory setting 0.00 [mH]	
Description:	Reluctance torque constant for a synchronous motor determined by the motor data identification. This reluctance torque constant can be changed after the identification and accepted in p0328 with p1910/p1960 = -3.				
Dependency:	Refer to: p0328, r1939, p1960				
p3028	Motld reluctance force constant identified / kT_reluct ident				
SERVO (Lin)	Can be changed: U, T		Calculated: CALC_MOD_ALL	Access level: 3	
	Data type: FloatingPoint32		Dynamic index: -	Func. diagram: -	
	P-Group: Motor identification		Units group: -	Unit selection: -	
	Not for motor type: ASM, REL, FEM		Scaling: -	Expert list: 1	
	Min -1000.00 [mH]		Max 1000.00 [mH]	Factory setting 0.00 [mH]	
Description:	Reluctance force constant for a synchronous motor determined by the motor data identification. This reluctance force constant can be changed after the identification and accepted in p0328 with p1910/p1960 = -3.				
Dependency:	Refer to: p0328, r1939, p1960				
p3030	Motld angular commutation offset identified / Ang_com offset				
SERVO	Can be changed: U, T		Calculated: CALC_MOD_ALL	Access level: 3	
	Data type: FloatingPoint32		Dynamic index: -	Func. diagram: -	
	P-Group: Motor identification		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min -180.00 [°]		Max 180.00 [°]	Factory setting 0.00 [°]	
Description:	Angular commutation offset for a synchronous motor determined by the motor data identification. This angular commutation offset can be changed after the identification and accepted in p0431 with p1910/p1960 = -3.				
Dependency:	Refer to: p0431, p1910, p1960, r1984				
p3031	Motld encoder inversion actual value identified / EnclnvActVal ident				
SERVO	Can be changed: U, T		Calculated: CALC_MOD_ALL	Access level: 3	
	Data type: Unsigned16		Dynamic index: -	Func. diagram: -	
	P-Group: Motor identification		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min -		Max -	Factory setting 0000 bin	
Description:	Inversion of the encoder actual value determined by the motor data identification. This inversion can be changed after the identification and accepted in p0410 with p1910/p1960 = -3.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Invert speed actual value	Yes	No	4710, 4715
	01	Invert position actual value	Yes	No	4704
Dependency:	Refer to: p0410, p1910, p1960				

p3031	Motld encoder inversion actual value identified / EnclInvActVal ident																		
SERVO (Lin)	Can be changed: U, T Data type: Unsigned16 P-Group: Motor identification Not for motor type: - Min -	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 bin																
Description:	Inversion of the encoder actual value determined by the motor data identification. This inversion can be changed after the identification and accepted in p0410 with p1910/p1960 = -3.																		
Bit field:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr> <tr> <td>00</td><td>Invert velocity actual value</td><td>Yes</td><td>No</td><td>4710, 4715</td></tr> <tr> <td>01</td><td>Invert position actual value</td><td>Yes</td><td>No</td><td>4704</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FP	00	Invert velocity actual value	Yes	No	4710, 4715	01	Invert position actual value	Yes	No	4704			
Bit	Signal name	1 signal	0 signal	FP															
00	Invert velocity actual value	Yes	No	4710, 4715															
01	Invert position actual value	Yes	No	4704															
Dependency:	Refer to: p0410, p1910, p1960																		
p3041	Motld moment of inertia identified / M_inertia ident																		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min 0.000000 [kgm²]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: 25_1 Scaling: - Max 100000.000000 [kgm²]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.000000 [kgm²]																
Description:	Motor moment of inertia determined by the motor data identification. This motor moment of inertia can be changed after the identification and accepted in p0341 with p1910/p1960 = -3.																		
Dependency:	Refer to: p0341, p1960, r1969																		
p3041	Motld motor mass identified / Mot mass ident																		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: REL Min 0.000000 [kg]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: 27_1 Scaling: - Max 10000.000000 [kg]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.000000 [kg]																
Description:	Mass of the motor determined by the motor data identification. This mass can be changed after the identification and accepted in p0341 with p1910/p1960 = -3.																		
Dependency:	Refer to: p0341, p1960, r1969																		
p3042	Motld load moment of inertia identified / Load mom ident																		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.00000 [kgm²]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: 25_1 Scaling: - Max 100000.00000 [kgm²]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.00000 [kgm²]																
Description:	Load moment of inertia determined by the motor data identification. This load moment of inertia can be changed after the identification and accepted in p1498 with p1910/p1960 = -3.																		
Dependency:	Refer to: p0342, p1498, p1960, r1969																		
Note:	For p1910/p1960 = -3, p0342 is set to 1 (ratio between the total and motor).																		

p3042	Motld load mass identified / Load mass ident		
SERVO (Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.00000 [kg]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: 27_1 Scaling: - Max 10000.00000 [kg]	Access level: 3 Func. diagram: - Unit selection: p0100 Expert list: 1 Factory setting 0.00000 [kg]
Description:	Load mass determined by the motor data identification. This load mass can be changed after the identification and accepted in p1498 with p1910/p1960 = -3.		
Dependency:	Refer to: p0342, p1498, p1960, r1969		
Note:	For p1910/p1960 = -3, p0342 is set to 1 (ratio between the total and motor).		
p3045	Motld force characteristic kT1 identified / kT1 ident		
SERVO (Ext M_ctrl, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min -340.28235E36 [N/Arms]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 340.28235E36 [N/Arms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [N/Arms]
Description:	Coefficient kT1 for the force characteristic for a synchronous linear motor determined by the motor data identification. This coefficient can be changed after the identification and accepted in p0645 with p1910/p1960 = -3.		
Dependency:	Refer to: p0645, p0646, p0647, p0648, p1960, p3046, p3047, p3048		
p3045	Motld torque characteristic kT1 identified / kT1 ident		
SERVO (Ext M_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min -340.28235E36 [Nm/A]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 340.28235E36 [Nm/A]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [Nm/A]
Description:	Coefficient kT1 for the torque characteristic for a synchronous motor determined by the motor data identification. This coefficient can be changed after the identification and accepted in p0645 with p1910/p1960 = -3.		
Dependency:	Refer to: p0645, p0646, p0647, p0648, p1960, p3046, p3047, p3048		
p3046	Motld force characteristic kT3 identified / kT3 ident		
SERVO (Ext M_ctrl, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min -	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Coefficient kT3 for the force characteristic for a synchronous linear motor determined by the motor data identification. This coefficient can be changed after the identification and accepted in p0646 with p1910/p1960 = -3.		
Dependency:	Refer to: p0645, p0646, p0647, p0648, p1960, p3045, p3047, p3048		

p3046	Motld torque characteristic kT3 identified / kT3 ident		
SERVO (Ext M_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min -	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Coefficient kT3 for the torque characteristic for a synchronous motor determined by the motor data identification. This coefficient can be changed after the identification and accepted in p0646 with p1910/p1960 = -3.		
Dependency:	Refer to: p0645, p0646, p0647, p0648, p1960, p3045, p3047, p3048		
p3047	Motld force characteristic kT5 identified / kT5 ident		
SERVO (Ext M_ctrl, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min -	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Coefficient kT5 for the force characteristic for a synchronous linear motor determined by the motor data identification. This coefficient can be changed after the identification and accepted in p0647 with p1910/p1960 = -3.		
Dependency:	Refer to: p0645, p0646, p0647, p0648, p1960, p3045, p3046, p3048		
p3047	Motld torque characteristic kT5 identified / kT5 ident		
SERVO (Ext M_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min -	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Coefficient kT5 for the torque characteristic for a synchronous motor determined by the motor data identification. This coefficient can be changed after the identification and accepted in p0647 with p1910/p1960 = -3.		
Dependency:	Refer to: p0645, p0646, p0647, p0648, p1960, p3045, p3046, p3048		
p3048	Motld force characteristic kT7 identified / kT7 ident		
SERVO (Ext M_ctrl, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: ASM, REL, FEM Min -	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Coefficient kT7 for the force characteristic for a synchronous linear motor determined by the motor data identification. This coefficient can be changed after the identification and accepted in p0648 with p1910/p1960 = -3.		
Dependency:	Refer to: p0645, p0646, p0647, p0648, p1960, p3045, p3046, p3047		

p3048 Motld torque characteristic kT7 identified / kT7 ident

SERVO (Ext M_ctrl)	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: ASM, REL, FEM	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Coefficient kT7 for the torque characteristic for a synchronous motor determined by the motor data identification. This coefficient can be changed after the identification and accepted in p0648 with p1910/p1960 = -3.

Dependency: Refer to: p0645, p0646, p0647, p0648, p1960, p3045, p3046, p3047

p3049[0...n] Motld Speed at start of field weakening identified / ident

SERVO	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00000 [rpm]	210000.00000 [rpm]	0.00000 [rpm]

Description: Speed at the start of field weakening determined by the motor data identification. This start speed can be changed after the identification and accepted in p0348 with p1910/p1960 = -3.

Dependency: Refer to: p0348, p1910, p1960

p3049[0...n] Motld Speed at start of field weakening identified / v_Fieldweak ident

SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00000 [m/min]	1000.00000 [m/min]	0.00000 [m/min]

Description: Velocity at the start of field weakening determined by the motor data identification. This start velocity can be changed after the identification and accepted in p0348 with p1910/p1960 = -3.

Dependency: Refer to: p0348, p1910, p1960

p3050[0...n] Motorld stator resistance identified / R_stator ident

SERVO	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: MDS, p0130	Func. diagram: -
	P-Group: Motor identification	Units group: 16_1	Unit selection: p0349
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00000 [Ohm]	2000.00000 [Ohm]	0.00000 [Ohm]

Description: Stator resistance determined by the motor data identification. This stator resistance can be changed after the identification and accepted in p0350 with p1910/p1960 = -3.

Dependency: Refer to: p0350, p1910, r1912

p3054[0...n]	MotId rotor resistance identified / R_rotor ident		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM, REL Min 0.00000 [Ohm]	Calculated: CALC_MOD_ALL Dynamic index: MDS, p0130 Units group: 16_1 Scaling: - Max 300.00000 [Ohm]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [Ohm]
Description:	Rotor resistance for an induction motor determined by the motor data identification. This stator resistance can be changed after the identification and accepted in p0354 with p1910/p1960 = -3.		
Dependency:	Refer to: p0354, p0625, p1910, r1927, p1960		
Note:	The parameter is not used for synchronous motors (p0300 = 2xx).		
p3056[0...n]	MotId stator leakage inductance identified / L_stator leak		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.00000 [mH]	Calculated: CALC_MOD_ALL Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 1000.00000 [mH]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [mH]
Description:	Stator leakage inductance determined by the motor data identification. This stator leakage inductance can be changed after the identification and accepted in p0356 with p1910/p1960 = -3.		
Dependency:	Refer to: p0356, p1910, r1932		
p3058[0...n]	MotId rotor leakage inductance identified / L_rotor leak		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM, REL Min 0.00000 [mH]	Calculated: CALC_MOD_ALL Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 1000.00000 [mH]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [mH]
Description:	Rotor leakage induction for an induction motor determined by the motor data identification. This rotor leakage inductance can be changed after the identification and accepted in p0358 with p1910/p1960 = -3.		
Dependency:	Refer to: p0358, p1910, r1932		
p3060[0...n]	MotId magnetizing inductance identified / MotId Lh ident		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM, REL Min 0.00000 [mH]	Calculated: CALC_MOD_ALL Dynamic index: MDS, p0130 Units group: 15_1 Scaling: - Max 10000.00000 [mH]	Access level: 3 Func. diagram: - Unit selection: p0349 Expert list: 1 Factory setting 0.00000 [mH]
Description:	Magnetizing inductance for an induction motor determined by the motor data identification. This magnetizing inductance can be changed after the identification and accepted in p0360 with p1910/p1960 = -3.		
Dependency:	Refer to: p0360, p1910, r1936, p1960		

p3070	Motld voltage emulation error final value identified / V_err final ident		
SERVO (Ext M_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.000 [V]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 100.000 [V]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Final value of the voltage emulation error determined by the motor data identification. This final value can be changed after the identification and accepted in p1952 with p1910/p1960 = -3.		
Dependency:	Refer to: p1910, p1952, p1953, p3071		
p3071	Motld voltage emulation error current offset identified / V_error I_offset		
SERVO (Ext M_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: - Min 0.000 [A]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 100.000 [A]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [A]
Description:	Current offset of the voltage emulation error determined by the motor data identification. This current offset can be changed after the identification and accepted in p1953 with p1910/p1960 = -3.		
Dependency:	Refer to: p1910, p1952, p1953, p3070		
p3080	Motld flux controller P gain identified / Flux ctrl Kp ident		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM, REL Min 0.0 [A/Vs]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 999999.0 [A/Vs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [A/Vs]
Description:	P gain of the flux controller for an induction motor determined by the motor data identification. This P gain can be changed after the identification and accepted in p1590 with p1910/p1960 = -3.		
Dependency:	Refer to: p1590, p1910		
p3081	Motld flux controller integral time identified / Flux ctrl Tn ident		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor identification Not for motor type: PEM, REL Min 0 [ms]	Calculated: CALC_MOD_ALL Dynamic index: - Units group: - Scaling: - Max 10000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Integral time of the flux controller for an induction motor determined by the motor data identification. This integral time can be changed after the identification and accepted in p1592 with p1910/p1960 = -3.		
Dependency:	Refer to: p1592, p1910		

p3082 MotId current controller P gain identified / I_ctrl Kp ident

SERVO	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: 18_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.000 [V/A]	Max 100000.000 [V/A]	Factory setting 0.000 [V/A]

Description: P gain of the current controller determined by the motor data identification.

This P gain can be changed after the identification and accepted in p1715 with p1910/p1960 = -3.

Dependency: Refer to: p1715, p1910

p3083 MotId current controller integral time identified / I_ctrl Tn ident

SERVO	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 1000.00 [ms]	Factory setting 0.00 [ms]

Description: Integral time of the current controller determined by the motor data identification.

This integral time can be changed after the identification and accepted in p1717 with p1910/p1960 = -3.

Dependency: Refer to: p1717, p1910

p3088 MotId Motor model changeover speed operation with encoder ident. / MotMod n_chgSnsorI

SERVO	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00000 [rpm]	Max 210000.00000 [rpm]	Factory setting 0.00000 [rpm]

Description: Changeover speed for the motor model with encoder determined by the motor data identification.

This changeover speed can be changed after the identification and accepted in p1752 with p1910/p1960 = -3.

Dependency: Refer to: p1752, p1910

p3088 MotId Motor model changeover velocity operat. with encod. ident. / v_chg Ident encod

SERVO (Lin)	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00000 [m/min]	Max 1000.00000 [m/min]	Factory setting 0.00000 [m/min]

Description: Changeover velocity for the motor model with encoder determined by the motor data identification.

This changeover velocity can be changed after the identification and accepted in p1752 with p1910/p1960 = -3.

Dependency: Refer to: p1752, p1910

p3100	RTC time stamp mode / RTC t_stamp mode		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the mode for the time stamp p3100 = 0: Time stamp, operating hours p3100 = 1: Time stamp, UTC format		
Note:	RTC: Real Time Clock UTC: Universal Time Coordinates The UTC time started, according to the definition on 01.01.1970 at 00:00:00 and is output in days and milliseconds.		
p3101[0...1]	RTC set UTC time / RTC set UTC		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4294967295	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Setting the UTC time. This means that the drive system is synchronized to the time specified by the time master. To start p3101[1] must be written to followed by p3101[0]. After writing to p3101[0], the UTC time is accepted. p3101[0]: Milliseconds p3101[1]: Days		
r3102[0...1]	RTC read UTC time / RTC read UTC		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the current UTC time in the drive system. p3102[0]: Milliseconds p3102[1]: Days		
p3103	RTC synchronization source / RTC sync_source		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the synchronization source/technique.		
Value:	0: PROFIBUS 1: PROFINET 2: PPI 3: PROFINET PTP		

p3104	BI: RTC real time synchronization PING / RTC PING		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the PING event to set the UTC time.		
Notice:	The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
r3108[0...1]	RTC last synchronization deviation / RTC sync_dev		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the last synchronization deviation that was determined. r3108[0]: Milliseconds r3108[1]: Days		
p3109	RTC real time synchronization, tolerance window / RTC sync tol		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: - Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100 [ms]
Description:	Sets the tolerance window for time synchronization. When this tolerance window is exceeded, an appropriate alarm is output.		
Dependency:	Refer to: A01099		
p3110	External fault 3, power-up delay / Ext fault 3 t_on		
All objects	Can be changed: U, T Data type: Unsigned16 P-Group: Messages Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000 [ms]	Access level: 3 Func. diagram: 2546 Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Sets the delay time for external fault 3.		
Dependency:	Refer to: p2108, p3111, p3112 Refer to: F07862		

p3111[0...n]	BI: External fault 3, enable / Ext fault 3 enab		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the enable signal of external fault 3. External fault 3 is initiated by the following AND logic operation: - BI: p2108 negated - BI: p3111 - BI: p3112 negated		
Dependency:	Refer to: p2108, p3110, p3112 Refer to: F07862		
p3111	BI: External fault 3, enable / Ext fault 3 enab		
CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM54F_MA, TM54F_SL	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2546 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the enable signal of external fault 3. External fault 3 is initiated by the following AND logic operation: - BI: p2108 negated - BI: p3111 - BI: p3112 negated		
Dependency:	Refer to: p2108, p3110, p3112 Refer to: F07862		
p3112[0...n]	BI: External fault 3 enable negated / Ext flt 3 enab neg		
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the negated enable signal of external fault 3. External fault 3 is initiated by the following AND logic operation: - BI: p2108 negated - BI: p3111 - BI: p3112 negated		
Dependency:	Refer to: p2108, p3110, p3111 Refer to: F07862		

p3112 BI: External fault 3 enable negated / Ext flt 3 enab neg

CU_CX32, CU_I,
CU_LINK, CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, HUB,
TB30, TM120, TM15,
TM15DI_DO, TM17,
TM31, TM54F_MA,
TM54F_SL

Can be changed: U, T**Data type:** Unsigned32 / Binary**P-Group:** Messages**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 3**Func. diagram:** 2546**Unit selection:** -**Expert list:** 1**Min**

-

Max

-

Factory setting

0

Description:

Sets the signal source for the negated enable signal of external fault 3.
External fault 3 is initiated by the following AND logic operation:

- BI: p2108 negated

- BI: p3111

- BI: p3112 negated

Dependency:

Refer to: p2108, p3110, p3111

Refer to: F07862

r3113.0...15 CO/BO: NAMUR message bit bar / NAMUR bit bar

All objects

Can be changed: -**Data type:** Unsigned16**P-Group:** Messages**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting****Min**

-

Max

-

-

Description:

Displays the status of NAMUR signal bit bar.

The faults or alarms are assigned to the appropriate signaling/message classes and influence a specific message bit.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Fault drive converter data electronics / software error	Yes	No	-
01	Line supply fault	Yes	No	-
02	DC link overvoltage	Yes	No	-
03	Fault drive converter power electronics	Yes	No	-
04	Drive converter overtemperature	Yes	No	-
05	Ground fault	Yes	No	-
06	Motor overload	Yes	No	-
07	Bus error	Yes	No	-
08	External safety-relevant shutdown	Yes	No	-
09	Mot encoder fault	Yes	No	-
10	Error internal communications	Yes	No	-
11	Infeed fault	Yes	No	-
15	Other faults	Yes	No	-

r3114.9...11 CO/BO: Messages status word global / Msg ZSW global

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the global status word for messages.
The appropriate bit is set if at least one message is present at the drive objects.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	09	Alarm present	Yes	No	8065
	10	Fault present	Yes	No	8060
	11	Safety message present	Yes	No	-

Note: The status bits are displayed with delay.

r3115[0...63] Fault drive object initiating / F DO initiating

All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: 1750, 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the drive object number of the initiating drive object for this fault as integer number.
Value = 63:

The fault was initiated by the drive object itself.

Dependency: Refer to: r0945, r0947, r0948, r0949, r2109, r2130, r2133, r2136, r3120, r3122

Note: The buffer parameters are cyclically updated in the background (refer to status signal in r2139).
The structure of the fault buffer and the assignment of the indices is shown in r0945.

p3116 BI: Acknowledgement automatically suppressed / Ackn suppress

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the automatic acknowledgement on the device drive object.
BI: p3116 = 1 signal
Faults present are not automatically acknowledged on the device drive object. Local device faults are not forwarded.
BI: p3116 = 0 signal
Faults present are automatically acknowledged on the device drive object. Local device faults are forwarded to the first active drive object.

Dependency: Refer to: p2102, p2103, p2104, p2105, p3981

Note: When selecting a standard telegram, the BICO interconnection for control signal STW1.10 (master control by PLC) is automatically established.

p3117				
Change safety message type / Ch. SI mess type				
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Unsigned32 P-Group: Messages Not for motor type: - Min 0 Max 1	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0	
Description:	Sets the re-parameterization of all safety messages for faults and alarms. The relevant message type during changeover is selected by the firmware. 0: Safety messages are not re-parameterized 1: Safety messages are re-parameterized			
Note:	A change only becomes effective after a POWER ON.			
r3120[0...63]				
Component number fault / Comp_num flt				
All objects	Can be changed: - Data type: Unsigned32 P-Group: Messages Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: 8060 Unit selection: - Expert list: 1 Factory setting -	
Description:	Displays the component number of the fault which has occurred.			
Dependency:	Refer to: r0945, r0947, r0948, r0949, r2109, r2130, r2133, r2136, r3122			
Note:	Value = 0: Assignment to a component not possible. The buffer parameters are cyclically updated in the background (refer to status signal in r2139). The structure of the fault buffer and the assignment of the indices is shown in r0945.			
r3121[0...63]				
Component number alarm / Comp_num alarm				
All objects	Can be changed: - Data type: Unsigned32 P-Group: Messages Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: 8065 Unit selection: - Expert list: 1 Factory setting -	
Description:	Displays the component number of the alarm which has occurred.			
Dependency:	Refer to: r2110, r2123, r2124, r2125, r2134, r2145, r2146, r3123			
Note:	Value = 0: Assignment to a component not possible. The buffer parameters are cyclically updated in the background (refer to status signal in r2139). The structure of the alarm buffer and the assignment of the indices is shown in r2122.			
r3122[0...63]				
Diagnostic attribute fault / Diag_attr fault				
All objects	Can be changed: - Data type: Unsigned32 P-Group: Messages Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: 8060 Unit selection: - Expert list: 1 Factory setting -	
Description:	Displays the diagnostic attribute of the fault which has occurred.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Hardware replacement recommended	Yes	No
Dependency:	Refer to: r0945, r0947, r0948, r0949, r2109, r2130, r2133, r2136, r3120			
			FP	-

Note: The buffer parameters are cyclically updated in the background (refer to status signal in r2139).
The structure of the fault buffer and the assignment of the indices is shown in r0945.

r3123[0...63]	Diagnostic attribute alarm / Diag_attr alarm				
All objects	Can be changed: -	Calculated: -		Access level: 3	
	Data type: Unsigned32	Dynamic index: -		Func. diagram: 8065	
	P-Group: Messages	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min	Max		Factory setting	
	-	-		-	
Description:	Displays the diagnostic attribute of the alarm which has occurred.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Hardware replacement recommended	Yes	No	-
Dependency:	Refer to: r2110, r2123, r2124, r2125, r2134, r2145, r2146				
Note:	The buffer parameters are cyclically updated in the background (refer to status signal in r2139). The structure of the alarm buffer and the assignment of the indices is shown in r2122.				

r3131	CO: Current fault value / Current flt value		
All objects	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: 8060
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the fault value of the oldest active fault.		

p3135	Suppress active fault / Supp act flt				
A_INF, B_INF, S_INF, SERVO, TM41, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 8060		
	P-Group: Messages	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Sets the suppression of r2139.3 "Fault present" for certain fault responses.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	Suppression of fault response ENCODER	On	Off	-
	10	Suppression of fault response NONE	On	Off	-
Dependency:	Refer to: p0491, r2139				
Note:	Depending on the suppression of a fault reaction in this parameter, r2139.1 "Acknowledgement required" is set when at least one fault occurs.				
	Re bit 08:				
	The suppression is only effective if p0491 = 1.				

p3201[0...n]	Excitation current outside the tolerance threshold value / I_exc n Tol thresh		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: ASM, PEM, REL Min 0.1 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10.0 [%]
Description:	Sets the threshold value for the "excitation current outside tolerance" message for the excitation current monitoring. If the absolute value of the difference between the excitation current setpoint and actual value (r1641 - r1626) exceeds the threshold value and the hysteresis is longer than the selected delay time, then fault F07913 is output. This fault is withdrawn when the threshold voltage is undershot.		
Dependency:	Refer to: r1626, r1641, p3202, p3203 Refer to: F07913		
Note:	The monitoring function is only carried out for separately-excited synchronized motors (p0300 = 5).		
p3202[0...n]	Excitation current outside the tolerance hysteresis / I_exc n Tol hyst		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: ASM, PEM, REL Min 0.1 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10.0 [%]
Description:	Sets the hysteresis for the "excitation current outside tolerance" message for the excitation current monitoring.		
Dependency:	Refer to: p3201, p3203 Refer to: F07913		
Note:	The monitoring function is only carried out for separately-excited synchronized motors (p0300 = 5).		
p3203[0...n]	Excitation current outside the tolerance delay time / I_exc n Tol t_del		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: ASM, PEM, REL Min 0.0 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.0 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.0 [s]
Description:	Sets the delay time for the "excitation current outside tolerance" message for the excitation current monitoring.		
Dependency:	Refer to: p3201, p3202 Refer to: F07913		
Note:	The monitoring function is only carried out for separately-excited synchronized motors (p0300 = 5).		
p3204[0...n]	Flux outside the tolerance threshold value / Flux n tol thresh		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Messages Not for motor type: ASM, PEM, REL Min 0.1 [%]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.0 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10.0 [%]
Description:	Sets the threshold value for the "flux outside the tolerance" message for the flux monitoring. If the absolute value of the difference between the flux setpoint and actual value (r0083 - r0084) falls below the threshold value with hysteresis longer than the selected delay time, then fault F07914 is output. This fault is withdrawn when the threshold voltage is undershot.		
Dependency:	Refer to: r0083, r0084, p3205, p3206 Refer to: F07914		

Note: The monitoring function is only carried out for separately-excited synchronized motors (p0300 = 5).
The flux monitoring is only active after magnetizing (r0056.4 = 1).

p3205[0...n]	Flux outside the tolerance hysteresis / Flux n tol hyst		
VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1
	Min 0.1 [%]	Max 50.0 [%]	Factory setting 10.0 [%]
Description:	Sets the hysteresis for the "flux outside tolerance" message for the flux monitoring.		
Dependency:	Refer to: p3204, p3206 Refer to: F07914		
Note:	The monitoring function is only carried out for separately-excited synchronized motors (p0300 = 5). The flux monitoring is only active after magnetizing (r0056.4 = 1).		

p3206[0...n]	Flux outside tolerance delay time / Flux n tol t_del		
VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1
	Min 0.0 [s]	Max 10.0 [s]	Factory setting 5.0 [s]
Description:	Sets the delay time for the "flux outside tolerance" message for the flux monitoring.		
Dependency:	Refer to: p3204, p3205 Refer to: F07914		
Note:	The monitoring function is only carried out for separately-excited synchronized motors (p0300 = 5). The flux monitoring is only active after magnetizing (r0056.4 = 1).		

p3207[0...n]	Zero current signal threshold value / I_0_sig thresh		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Messages	Units group: 6_2	Unit selection: p0505
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1
	Min 0.01 [Arms]	Max 10000.00 [Arms]	Factory setting 1.00 [Arms]
Description:	Sets the threshold value for the zero current signal for the zero current monitoring. If the absolute current falls below the threshold value then r2199.6 is set to 1 after the delay time has expired. The bit is reset if the threshold value and the hysteresis are exceeded again.		
Dependency:	Refer to: r2199, p3208, p3209		
Note:	The monitoring function is only carried out for separately-excited synchronized motors (p0300 = 5). The monitoring is only carried out for speeds less than the speed threshold value in p2161 (r2199.0 = 1).		

p3208[0...n]	Zero current signal hysteresis / I_0_sig hyst		
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Messages	Units group: 6_2	Unit selection: p0505
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1
	Min 0.01 [Arms]	Max 10000.00 [Arms]	Factory setting 1.00 [Arms]
Description:	Sets the hysteresis for the zero current signal for the zero current monitoring.		
Dependency:	Refer to: p3207, p3209		
Note:	The monitoring function is only carried out for separately-excited synchronized motors (p0300 = 5). The monitoring is only carried out for speeds less than the speed threshold value in p2161 (r2199.0 = 1).		
p3209[0...n]	Zero current signal delay time / I_0_sig t_del		
VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: ASM, PEM, REL	Scaling: -	Expert list: 1
	Min 0.00 [s]	Max 10.00 [s]	Factory setting 0.02 [s]
Description:	Sets the delay time for the zero current signal for the zero current monitoring.		
Dependency:	Refer to: p3207, p3208		
Note:	The monitoring function is only carried out for separately-excited synchronized motors (p0300 = 5). The monitoring is only carried out for speeds less than the speed threshold in p2161 (r2199.0 = 1).		
p3233[0...n]	Torque actual value filter, time constant / M_act_filt T		
SERVO (Extended sig), VECTOR (Extended sig)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8013
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 1000000 [ms]	Factory setting 0 [ms]
Description:	Sets the time constant of the PT1 element to smooth the torque actual value. The smoothed actual torque is compared with the threshold values and is only used for messages and signals.		
p3235	Phase failure signal motor monitoring time / Ph_fail t_monit		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 2000 [ms]	Factory setting 320 [ms]
Description:	Sets the monitoring time for phase failure detection of the motor.		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	Monitoring is only effective for blocksize and booksize power units. The function can be switched out with p3235 = 0. For vector drives: The monitoring is automatically deactivated during the flying restart operation for a motor that is still rotating.		

p3236[0...n]		Speed threshold 7 / n_thresh val 7			
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3		
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8012		
	P-Group: Messages	Units group: 3_1	Unit selection: p0505		
	Not for motor type: REL	Scaling: -	Expert list: 1		
	Min 0.00 [rpm]	Max 3000.00 [rpm]	Factory setting 100.00 [rpm]		
Description:	Sets the speed threshold value for the signal "speed deviation model/external" (BO: r2199.7).				
Dependency:	Refer to: r1443, r2169, r2199, p3237				
p3237[0...n]		Hysteresis speed 7 / n_hysteresis 7			
VECTOR	Can be changed: U, T	Calculated: CALC_MOD_LIM_REF	Access level: 3		
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8012		
	P-Group: Messages	Units group: 3_1	Unit selection: p0505		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min 0.00 [rpm]	Max 200.00 [rpm]	Factory setting 2.00 [rpm]		
Description:	Sets the hysteresis speed for the signal "speed deviation model/external" (BO: r2199.7).				
Dependency:	Refer to: r2199, p3236				
p3238[0...n]		OFF delay n_act_motor model = n_act external / t_del n_a = n_ext			
VECTOR	Can be changed: U, T	Calculated: -	Access level: 3		
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 8012		
	P-Group: Messages	Units group: -	Unit selection: -		
	Not for motor type: REL	Scaling: -	Expert list: 1		
	Min 0.0 [s]	Max 100.0 [s]	Factory setting 3.0 [s]		
Description:	Sets the OFF delay for the signal "speed deviation model/external in tolerance" (BO: r2199.7). The smoothed actual speed of the motor model r2169 is compared with the speed measured externally r1443 (threshold value p3236).				
Dependency:	Refer to: p3236, p3237				
p3290		Variable signaling function start / Var sig start			
SERVO	Can be changed: U, T	Calculated: -	Access level: 3		
	Data type: Unsigned8	Dynamic index: -	Func. diagram: 5301		
	P-Group: -	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min -	Max -	Factory setting 0010 bin		
Description:	Settings for start/stop and the comparison type for the variable signaling function.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activate function	Active	Not active	-
	01	Comparison with sign	With sign	Without sign	-

p3291	CI: Variable signaling function signal source / Var sig S_src		
SERVO	Can be changed: U, T Data type: Unsigned32 / Integer16 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 5301 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the variable signaling function.		
Dependency:	Refer to: p3292, p3293		
Note:	Re p3291 = 1: In this case, the signal source is defined using p3292 and p3293.		

p3292	Variable signaling function signal source address / Var sig S_src addr		
SERVO	Can be changed: U, T Data type: Unsigned32 P-Group: - Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 4 Func. diagram: 5301 Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the address of the signal source for the variable signaling function.		
Dependency:	Refer to: p3291		
Caution:	If an incorrect address and data type are set, then this can cause the software to crash.		



Note: This parameter should only be set for p3291 = 1.

p3293	Variable signaling function signal source data type / Var sig S_src type		
SERVO	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 7	Access level: 4 Func. diagram: 5301 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the data type of the signal source for the variable signaling function.		
Value:	0: Unknown 1: U8, Unsigned8 2: I8, Signed8 3: U16, Unsigned16 4: I16, Signed16 5: U32, Unsigned32 6: I32, Signed32 7: Float, FloatingPoint32		

Dependency: Refer to: p3291

Caution: If an incorrect address and data type are set, then this can cause the software to crash.



Note: This parameter should only be set for p3291 = 1.

r3294	BO: Variable signaling function output signal / Var sig outp_sig		
SERVO	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned8	Dynamic index: -	Func. diagram: 5301
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the output signal for the variable signaling function.		
Dependency:	Refer to: p3290, p3291, p3295, p3296, p3297, p3298		

p3295	Variable signaling function threshold value / Var sig thresh_val		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5301
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-340.28235E36	340.28235E36	0.000
Description:	Sets the threshold value for the variable signaling function.		

p3296	Variable signaling function hysteresis / Var sig hyst		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5301
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.000	340.28235E36	0.000
Description:	Sets the hysteresis for the variable signaling function.		

p3297	Variable signaling function pickup delay / Var sig t_pickup		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 5301
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0 [ms]	10000 [ms]	0 [ms]
Description:	Sets the pickup delay for the variable signaling function.		
Notice:	Values that do not comply with the following condition are treated just like value 0: Pickup delay (p3297) >= sampling time (p3299)		
Note:	For a value of 0, the pickup delay is disabled. The output signal is set if the condition for the 1 signal is fulfilled for longer than the selected time.		

p3298	Variable signaling function dropout delay / Var sig t_dropout		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 5301
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0 [ms]	10000 [ms]	0 [ms]
Description:	Sets the dropout delay for the variable signaling function.		
Notice:	Values that do not comply with the following condition are treated just like value 0: Dropout delay (p3298) >= sampling time (p3299)		

Note: For a value of 0, the dropout delay is disabled.
The output signal is reset if the condition for the 0 signal is fulfilled for longer than the selected time.

p3299	Variable signaling function, sampling time / Var sig t_sample		
SERVO	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 5301
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	1.000 [ms]	4.000 [ms]	4.000 [ms]
Description:	Sets the sampling time for the variable signaling function.		
Notice:	The following must apply for the setting: Sampling time (p3299) ≤ pickup delay (p3297), dropout delay (p3298)		
Note:	Only the following values can be set: 1.000, 2.000, 3.000, 4.000		

p3400	Infeed configuration word / INF config_word				
A_INF, S_INF	Can be changed: T	Calculated: -		Access level: 2	
	Data type: Unsigned16	Dynamic index: -		Func. diagram: 8940	
	P-Group: Closed-loop control	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min	Max		Factory setting	
	-	-		1010 bin	
Description:	Sets the configuration word of the infeed.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Smart Mode	On	Off	-
	01	Flat-Top Mode	On	Off	-
	03	Vdc controller	On	Off	-
	05	Line supply voltage sensing with VSM	On	Off	-
Dependency:	Refer to: p0210				
Note:	Re bit 00: In the Smart Mode, the DC link voltage is not controlled - however, infeed can still regenerate. The magnitude of DC link voltage depends on the current line supply voltage and the DC link load. For drive units belonging to the 400 V voltage class, for a drive unit supply voltage (p0210) greater than 415 V, infeed is always operated in the Smart Mode. This means that the 660 V limit can be maintained for the steady-state DC link voltage (p0280) up to a line supply voltage of 480 V. Re bit 01: If the Flat Top Mode is deactivated, switching losses are higher. This means that the full power is no longer continuously available. For p3400.0 = 1 or p1810.15 =1, this bit is not effective. Re bit 03: If the Vdc controller is switched out, overvoltage or undervoltage conditions occur in the DC link if no other voltage regulating component is located in the DC link. For p3400.0 = 1, this bit is not effective. Re bit 05: If a VSM is detected when commissioning the system, this bit is automatically set. When the bit is set, the line supply voltage input of the VSM must be connected (connected at the line side of line reactor). VSM: Voltage Sensing Module				

r3402 Infeed internal state / INF state int

A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: -	Func. diagram: 8832, 8932
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	12	-
Description:	Displays the internal status of the infeed module.		
Value:	0: Initialization 1: Fault 2: No ON command 3: Offset measurement running 4: ON delay active 5: Precharg. running 6: Pulse enable missing 7: Synchronization running 8: Voltage ramp-up active 9: Operation 10: Shutdown running 11: Identification running 12: Transformer magnetization running		

r3402 Infeed status internal BIC / INF state int

B_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: -	Func. diagram: 8932
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	6	-
Description:	Displays the internal status of the infeed module.		
Value:	0: Initialization 1: Fault 2: No ON command 3: Offset measurement running 4: ON delay active 5: Precharg. running 6: Operation		

r3405.0...7 CO/BO: Status word infeed / INF ZSW

A_INF, S_INF, SERVO (Tech_ctrl)	Can be changed: -	Calculated: -	Access level: 2		
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 8828, 8928		
	P-Group: Closed-loop control	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the status word of the infeed.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Smart Mode active	Yes	No	-
	01	Vdc-ctrl active	Yes	No	-
	02	Phase failure detected	Yes	No	-
	03	Current limit reached	Yes	No	-
	04	Infeed operates in generator/motor mode	Regenerative mode	Motor mode	-
	05	Motor mode inhibited	Yes	No	-
	06	Generator mode inhibited	Yes	No	-
	07	DC link undervoltage alarm threshold undershot	Yes	No	-

Note:

Re bit 00:
Smart Mode is activated with p3400.0.

Re bit 01:
The DC link voltage closed-loop control is activated with parameters p3400.3 and p3513.

Re bit 02:
When phase failure is detected the bit is set and alarm A06205 is output.
The bit is reset for the following events:
- the infeed had reached the normal operating state again after a phase failure has been bypassed/buffered (p3402 = 9).
- the pulse enable is withdrawn due to a fault or powering down with OFF1/OFF2.

Re bit 03:
The present current limit is displayed in r0067.

Re bit 04:
An active current setting r0078 >= 0 means infeed operation in motor mode; an active current setting r0078 < 0 means regenerative operation in generator mode.

Re bit 05:
The motor mode inhibit is activated with p3532.

Re bit 06:
The generator mode inhibit is activated with p3533.

Re bit 07:
The alarm threshold is dependent on r0296 and the setting in p0279.

r3405.7 CO/BO: Status word infeed / INF ZSW

B_INF	Can be changed: - Data type: Unsigned16 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
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Description: Displays the status word of the infeed.

Bit field:	<table border="0" style="width: 100%;"> <tr> <th style="text-align: left;">Bit</th> <th style="text-align: left;">Signal name</th> <th style="text-align: left;">1 signal</th> <th style="text-align: left;">0 signal</th> <th style="text-align: left;">FP</th> </tr> <tr> <td>07</td> <td>DC link undervoltage alarm threshold undershot</td> <td>Yes</td> <td>No</td> <td>-</td> </tr> </table>	Bit	Signal name	1 signal	0 signal	FP	07	DC link undervoltage alarm threshold undershot	Yes	No	-		
Bit	Signal name	1 signal	0 signal	FP									
07	DC link undervoltage alarm threshold undershot	Yes	No	-									

r3405.1 CO/BO: Status word DC-link control / UDC ZSW

VECTOR (Tech_ctrl)	Can be changed: - Data type: Unsigned16 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
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Description: Displays the status word for DC-link voltage control.

Bit field:	<table border="0" style="width: 100%;"> <tr> <th style="text-align: left;">Bit</th> <th style="text-align: left;">Signal name</th> <th style="text-align: left;">1 signal</th> <th style="text-align: left;">0 signal</th> <th style="text-align: left;">FP</th> </tr> <tr> <td>01</td> <td>Vdc-ctrl active</td> <td>Yes</td> <td>No</td> <td>-</td> </tr> </table>	Bit	Signal name	1 signal	0 signal	FP	01	Vdc-ctrl active	Yes	No	-		
Bit	Signal name	1 signal	0 signal	FP									
01	Vdc-ctrl active	Yes	No	-									

Note:

Re bit 01:
DC-link voltage control is disabled and enabled with p3513.

p3409	Infeed line frequency setting / INF f_line_mode		
A_INF, S_INF	Can be changed: T Data type: Integer16 P-Group: Closed-loop control Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the mode to detect the line supply frequency.		
Value:	0: Line supply frequency setting 50/60 Hz off 1: Line supply frequency setting 50/60 Hz on		
Dependency:	Refer to: p0211, p0284, p0285 Refer to: A06350, A06351, F06500		
Note:	For p3409 = 1, the following applies: After operation has been enabled, the rated line supply frequency (p0211) is automatically set to a value of 50 Hz or 60 Hz corresponding to the currently measured frequency. This means that the parameter value of p0211 is, under certain circumstances, changed. For p3409 = 0, the following applies: The system does not change parameter p0211.		
p3410	Infeed identification method / INF Ident_type		
A_INF, S_INF	Can be changed: C2(1), T Data type: Integer16 P-Group: Closed-loop control Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 5	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 5
Description:	Sets the line and DC link parameter identification routine for the infeed module.		
Value:	0: Identification (Id) off 1: Activate identification (Id) 2: Set controller settings 3: Save identification and controller settings 4: Save identification and controller settings with L adaptation 5: Reset, save Id and controller setting with L adaptation		
Dependency:	Refer to: r3411, r3412, r3414, p3415, p3416, p3417, p3421, p3422, p3424, p3555, p3560, p3614 Refer to: A06400		
Notice:	For p3410 = 1, 3, 4, 5, alarm A06400 is output and designates that the selected identification will take place the next time that the pulses are enabled.		
Note:	When p3410 = 1 an identification run for the total inductance and DC link capacitance is initiated when the pulses are next enabled. The results are displayed in r3411 and r3412. If a Voltage Sensing Module (VSM) is connected, then the line inductance (r3414) is also measured. The infeed then goes into the ready for switching on state. For p3410 = 2, the data (r3411, r3412 and r3414) determined during the identification run (p3410 = 1) are transferred into p3421, p3422 and p3424. The control loop parameters are suitably scaled to achieve a rugged controller setting (p3425); the fast controller response (p3555[2]) and the current actual value smoothing (p3614) are preset. Calculations for the controller are then repeated. The user must save the new parameters in a non-volatile fashion in order to permanently select the new controller setting. When p3410 = 3 an identification run for the inductance and DC link capacitance is initiated when the pulses are next enabled. Data determined during the identification (r3411, r3412, r3414) are used, as described under p3410 = 2 for the setting of p3421, p3422, p3424, p3425, p3555 as well as p3614, and the controller is recalculated. All of the parameters for the infeed module are then automatically stored in a non-volatile memory. The infeed continues to operate without any interruption with the new controller parameters.		

When p3410 = 4 an identification run for the inductance and DC link capacitance is initiated when the pulses are next enabled. Data determined during the identification (r3411, r3412, r3414) are used, as described under p3410 = 2 for the setting of p3421, p3422, p3424, p3425, p3555 as well as p3614, and the controller is recalculated. The line inductance identification is then repeated, if p3415[1] > p3514[0]. If the inductance measured the second time is lower than the first, the parameters are written to the current controller adaptation (p3620, p3622).. All of the parameters for the infeed module are then automatically stored in a non-volatile memory. The infeed continues to operate without any interruption with the new controller parameters.

For p3410 = 5, the same measurements and write operations are always carried out as for p3410 = 4. However, for the first identification run, initially the controller setting is reset by setting p3421 = p0223 + p0225, p3424 = p0225, p3422 = p0227 and p3425[.] = 100 %. Further, before the measurements are carried out, a brief identification run is executed to coarsely set the controller.

p3410 is automatically set to 0 after an identification run has been completed.

r3411[0...1] Infeed identified inductance / INF L ident			
A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [mH]	Max - [mH]	Factory setting - [mH]
Description:	Displays the identified total inductance. The value corresponds to the total inductance between the rigid line supply and the infeed input terminals.		
Index:	[0] = Run 1 [1] = Run 2		
Dependency:	Refer to: p3410		
Note:	The value measured in the first identification run is displayed in r3411[0] (for p3410 = 1, 3, 4, 5). This value is transferred to p3421. The value measured in the second identification run (for p3410 = 4, 5) is displayed in r3411[1] - this value is used to set the current controller adaptation (p3622). For the inductance value of the line reactor, r3411 - r3414 applies.		
r3412[0...1] Infeed DC link capacitance identified / INF C_DClk ident			
A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [mF]	Max - [mF]	Factory setting - [mF]
Description:	Displays the identified total DC link capacitance.		
Index:	[0] = Run 1 [1] = Run 2		
Dependency:	Refer to: p3410		
Note:	The value measured in the first identification run (for p3410 = 1, 3, 4, 5) is displayed in r3412[0]. For p3410 = 1, 3, this value is transferred to p3422. The DC link capacitance is not measured at the second identification run. The total DC link capacitance of a DC link group comprises the sum of the sub-capacitances of all motor/infeed modules and the additional DC link capacitors.		

r3414[0...1]	Infeed, line supply inductance identified / INF t_line ident		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the identified line supply inductance. The value corresponds to the total inductance between the stiff line supply and the connection point of the Voltage Sensing Module (VSM).		
Index:	[0] = Run 1 [1] = Run 2		
Dependency:	Refer to: p3410		
Notice:	The value is only automatically determined for the line supply identification (p3410 > 0), if operation with a Voltage Sensing Module is selected (p3400.5 = 1). Otherwise, r3414 = 0 is displayed.		
Note:	The value measured in the first identification run is displayed in r3414[0] (for p3410 = 1, 3, 4, 5). This value is transferred to p3421. The value measured in the second identification run is displayed in r3414[1] (for p3410 = 4, 5). For the inductance value of the line reactor, r3411 - r3414 applies.		
p3415[0...1]	Infeed excitation current L identification / INF I_exc L_Ident		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 1.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 75.00 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 20.00 [%]
Description:	Sets the magnitude of the excitation frequency for the L identification. The setting is made as a percentage of the maximum power unit current (r0209).		
Index:	[0] = Run 1 [1] = Run 2		
Dependency:	Refer to: p3410, r3411, p3421, p3620, p3622		
Notice:	To correctly identify the current level (p3410 = 4, 5) depending on the reactor inductance, the following must apply: p3415[0] < p3415[1] For A_INF booksize units, the following applies: The interrelationship between the reactor inductance and the current magnitude should be measured. Generally, the factory setting of p3415[0] and p3415[1] should be kept. For chassis units and S_INF booksize units, the following applies: Generally, there is only a very low inter-relationship between the reactor inductance and the current magnitude. This means that for the factory setting p3415[0] = p3415[1] = 20 %, i.e. run 2 is not executed.		
Note:	The reactive current for identification run 1 is set in p3415[0] (basic controller setting). The reactive current for identification run 2 is set in p3415[1] (adaptation of the current controller when reducing the reactor inductance with increasing current magnitude).		

p3416	Infeed excitation amplitude C identification / INF exc_amp C_Id		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.10 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.00 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2.00 [%]
Description:	Sets the level of the excitation frequency for identification of the total DC link capacitance. The amplitude is indicated as a percentage of the DC voltage setpoint ($V_{dc} = p0210 * p3510$).		
Dependency:	Refer to: p3410, r3412, p3422		
p3417	Infeed excitation frequency C identification / INF f_exc C_ID		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 10.00 [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 200.00 [Hz]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 50.00 [Hz]
Description:	Sets the level of the excitation frequency for identification of the total DC link capacitance.		
Dependency:	Refer to: p3410, r3412, p3422		
p3421	Infeed inductance / INF L		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.001 [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2000.000 [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.000 [mH]
Description:	Sets the total effective inductance for the closed-loop current control from the sum of the line supply inductance and inductance of the line reactor. This parameter is preset to the sum of p0223 and p0225.		
Dependency:	Refer to: p0223, p0225, p3410, p3425, p3614, p3622		
Note:	The controller setting is derived from this value and p3425. The value can be automatically determined using the identification run (p3410). For a parallel circuit, the value corresponds to the inductance of a power unit. For the inductance value of the line reactor, p3421 - p3424 applies.		
p3422	Infeed DC link capacitance / INF C_DCL		
A_INF, S_INF, SERVO (Tech_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.20 [mF]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2000.00 [mF]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2.00 [mF]
Description:	Sets the DC link capacitance for the closed-loop voltage control. This value is preset with p0227.		
Dependency:	Refer to: p0227, p3410, p3425		
Note:	The controller setting is derived from this value and p3425. A suitable value can be automatically determined using the identification run (p3410).		

p3422 DC-link capacitance, total / C_DC tot

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.20 [mF]	Max 2000.00 [mF]	Factory setting 2.00 [mF]
Description:	Sets the total DC link capacitance for closed-loop voltage control. The capacitance of one power unit is pre-assigned to this value. The value should be adapted according to the number of power units.		
Note:	The controller setting for the DC-link voltage controller is derived from this value.		

p3424 Infeed, line supply inductance / INF L_line

A_INF, S_INF	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.001 [mH]	Max 1000.000 [mH]	Factory setting 0.001 [mH]
Description:	Sets the line supply inductance. This parameter is preset with p0225.		
Dependency:	Refer to: p0223, p0225, p3410, p3425, p3622		
Note:	The controller setting is derived from this value and p3425. The value can be automatically determined using the identification (p3410) if operation with a Voltage Sensing Module is selected. Otherwise, p3424 is set to p3421 - p0223.		

p3425[0...1] Infeed control loop parameter scaling / INF par scal

A_INF, S_INF	Can be changed: T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1.00 [%]	Max 1000.00 [%]	Factory setting 100.00 [%]
Description:	Sets the scaling factors for controller parameters p3421, p3422 and p3424.		
Index:	[0] = Scaling, inductance [1] = Scaling, capacitance		
Dependency:	Refer to: p3410, p3421, p3422, p3424, p3614		
Note:	p3425 is automatically set to the optimum value when setting the control parameters using the line supply data identification p3410 >= 2. As the line supply inductance (p3424) increases in comparison to the total inductance (p3421), lower values must be selected for p3425. This means that the control is adapted to weak line supplies with high relative short-circuit voltage u_k or high line supply inductance (also refer to p3614). The scaled control loop parameters become effective for closed-loop control, i.e. the products $p3421 * p3425[0]$ and $p3422 * p3425[1]$ represent the controller setting.		

p3440	Smart Mode configuration / Smart Mode config																							
A_INF, S_INF	Can be changed: T Data type: Unsigned16 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0001 bin																					
Description:	Sets the configuration of the Smart Mode.																							
Bit field:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr> <tr> <td>00</td><td>Soft Pulse Mode</td><td>On</td><td>Off</td><td>-</td></tr> <tr> <td>01</td><td>Feedback control</td><td>On</td><td>Off</td><td>-</td></tr> <tr> <td>02</td><td>Deselect automatic line identification after POWER ON</td><td>Yes</td><td>No</td><td>-</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FP	00	Soft Pulse Mode	On	Off	-	01	Feedback control	On	Off	-	02	Deselect automatic line identification after POWER ON	Yes	No	-			
Bit	Signal name	1 signal	0 signal	FP																				
00	Soft Pulse Mode	On	Off	-																				
01	Feedback control	On	Off	-																				
02	Deselect automatic line identification after POWER ON	Yes	No	-																				
Note:	Re bit 00: When the pulsed mode for Smart Mode is deactivated, when regenerating, higher phase current gradients occur. Re bit 01: When feedback control is activated, the transistors are only activated for feedback. This reduces switching losses and the DC-link voltage reaches higher values (higher output voltage for the motors). In Version v04.30.xx a value of 1 is permissible for this bit for S120combi units only. Re bit 02: When automatic line identification is selected, the inductance and DC-link capacitance values are calculated on the first pulse enable after each POWER ON and saved in p3448[0...1]. In the case of manual input of p3448[0...1], automatic line identification must be deselected. In Version v04.30.xx a value of 1 is permissible for this bit for S120 combi units only.																							
p3441[0...1]	Smart Mode Vdc ctrl Kp/Tn / SLM Vdc_ctrl Kp/Tn																							
A_INF, S_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.00 [%]																					
Description:	Sets the standardized proportional gain (index 0) and the integral time (index 1) for the DC-link voltage controller (Vdc controller) in Smart Mode.																							
Index:	[0] = Proportional gain [1] = Integral time																							
Note:	A value of 100% corresponds to the basic setting derived from loop control parameters (p0115, p3409, p3448[1]).																							
p3442[0...1]	Smart Mode smoothing times / SLM t_smooth																							
A_INF, S_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 0.25 [ms] [1] 1.00 [ms]																					
Description:	Sets the time constant for PT1 filtering of the DC-link voltage for the Vdc controller (index 0) and the monitored DC-link load current (index 1) in Smart Mode.																							
Index:	[0] = DC-link voltage actual value (r3445) [1] = Monitored DC-link load current (r3446[2])																							
Dependency:	Refer to: r3445, r3446																							

p3443[0...1]	Smart Mode line commutation current threshold values / SLM line com I_thr		
A_INF, S_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 100.00 [%] [1] 200.00 [%]
Description:	Sets the current threshold values for the deactivation (index 0) and activation (index 1) of line commutation in Smart Mode.		
Index:	[0] = Deactivating [1] = Activating		
Note:	A value of 100% corresponds to the minimum feedback load current derived from the loop control parameters (p0210, p0211, p3409, p3448[0], p3432) without infeed components. To avoid frequent changeovers in operation close to the changeover point, the value for activation (index 1) must be significantly higher than the value for deactivation (index 0).		
p3444[0...1]	Smart Mode voltages / SLM voltages		
A_INF, S_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 105.00 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 90.00 [%] [1] 100.50 [%]
Description:	Sets the minimum line voltage for regenerative feedback or the DC-link voltage setpoint in Smart Mode.		
Index:	[0] = Minimum line voltage for feedback [1] = DC-link voltage setpoint		
Note:	Re index 0: A value of 100% corresponds to the supply voltage set in p0210. If the minimum line voltage is undershot, feedback is deactivated to prevent the DC-link voltage decaying in the event of a significant system voltage dip. Re index 1: A value of 100% corresponds to the rectified value of the current line voltage. The value must be greater than or equal to 100%.		
r3445[0...1]	Smart Mode voltages display / SLM voltages disp		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: 5_2 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the various voltages in Smart Mode.		
Index:	[0] = DC link voltage smoothed [1] = DC-link voltage setpoint		
Dependency:	Refer to: r0070, p3442		
Note:	Re index 0: Displays the DC-link voltage actual value measured and smoothed with p3442[0]. The smoothed value is used for the DC-link voltage controller (Vdc controller) in Smart Mode. The DC-link voltage is also available unsmoothed (r0070). Re index 1: Displays the DC-link voltage setpoint for the DC-link voltage controller (Vdc controller) in Smart Mode.		

r3446[0...2]	Smart Mode currents / SLM currents		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: - Units group: 6_4 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the various current values in Smart Mode.		
Index:	[0] = DC-link current setpoint [1] = Vdc controller I component [2] = Monitored DC-link load current		
Note:	Re index 0: Displays the DC-link current setpoint requested by the DC-link voltage controller (Vdc controller) in Smart Mode. Re index 1: Displays the I component of the DC-link voltage controller (Vdc controller). Re index 2: Displays the monitored DC-link load current.		
r3447	Smart Mode OFF angle / SLM phi_OFF		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°]	Calculated: - Dynamic index: - Units group: - Scaling: p2005 Max - [°]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [°]
Description:	Displays the OFF angle requested by the DC-link voltage controller (Vdc controller) in Smart Mode.		
Note:	A value = 30 ° deactivates feedback. A value = 0 ° requests maximum feedback (line commutation).		
p3448[0...1]	Smart Mode effective inductance/DC-link capacitance / SLM L/C effect		
A_INF, S_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 10.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000.00 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 110.00 [%] [1] 100.00 [%]
Description:	Sets the effective inductance or DC-link capacitance in Smart Mode.		
Index:	[0] = Effective inductance with ref. to p0223 [1] = Effective DC-link capacitance with ref. to p0227		
Notice:	p3448[0] is used in the calculation of the reference value of p3443 and in the controller models. p3448[1] is used in the calculation of the reference value of p3441 and in the controller models. When automatic line identification is selected (p3440.2 = 0), the following applies: - the inductance and DC-link capacitance values are calculated on the first pulse enable after each POWER ON and saved in p3448[0...1]. - values entered manually are overwritten after the next POWER ON. When automatic line identification is deselected (p3440.2 = 1), the following applies: - the inductance and DC-link capacitance values have to be entered manually.		

Note:

Re index 0:
A value of 100% corresponds to the inductance parameterized in p0223.
For a parallel circuit, the value corresponds to the inductance of a power unit.

Re index 1:
A value of 100% corresponds to the capacitance parameterized in p0227.
The value includes the capacitances of all modules connected on the DC link.

r3452 Infeed PLL status / INF PLL status

A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0000 bin	0111 bin	-

Description: Displays the status of the line supply PLL.

Value:

0:	Initialization running
1:	Error when synchronizing
2:	Line analysis
3:	Calculation line data
4:	Pulse enable missing
5:	PLL calculation
6:	Final status controlled / Smart Mode
7:	Reserved

p3458[0...1] Infeed PLL smoothing time / INF PLL t_smooth

A_INF, S_INF	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	1.0 [ms]	1000.0 [ms]	[0] 23.1 [ms] [1] 9.1 [ms]

Description: Sets the smoothing time for the line supply PLL.

Index:

[0]	= Encoderless operation line supply frequency smoothing time
[1]	= VSM operation line supply frequency smoothing time

Note: It may be necessary to reduce the smoothing time for weak line supplies with high frequency fluctuations. There is otherwise a risk of brief orientation errors and the infeed could fail.

r3460 Infeed PLL system deviation / INF PLL ctrl_dev

A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [°]	- [°]	- [°]

Description: Displays the PLL system deviation.

r3461	Infeed PLL system deviation after filtering / INF PLL ctrl_devSm		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [°]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [°]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [°]
Description:	Displays the PLL system deviation after filtering.		
Dependency:	Refer to: p3458		
p3463	Infeed, line angle change, phase failure detection / INF phi ph_fail		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -180.0 [°]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 180.0 [°]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 15.0 [°]
Description:	If the line supply angle (angle between the line supply phases) suddenly changes by this value, then a phase failure is assumed. The pulses are then inhibited for 10 ms.		
Dependency:	Refer to: A06205		
r3467[0...1]	CO: Infeed current alpha/beta line filter / INF I a/b filter		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [A]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [A]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [A]
Description:	Displays the line current at the input terminals of the line filter in alpha/beta components.		
Index:	[0] = Alpha [1] = Beta		
r3468[0...1]	CO: Infeed voltage alpha/beta line filter / INF V a/b filter		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [V]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Displays the line supply voltage at the input terminals of the line filter in alpha/beta components.		
Index:	[0] = Alpha [1] = Beta		
Note:	The input terminals of the line filter form, for infeeds, the connection point of the Voltage Sensing Module (VSM) to measure the line supply voltage. For operation with VSM (p3400.5 = 1), the following applies: The voltage measured values r3661 and r3662, transformed into the alpha/beta system are displayed in r3468. For encoderless operation without VSM (p3400.5 = 0), the following applies: The estimated values of the voltages from the line supply model of the PLL transformed into the alpha/beta system are displayed in r3468.		

p3469[0...n]	Latch delay time correction, zero crossover detection / t_latch corr PLL		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -10000.0 [μs]	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 10000.0 [μs]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.0 [μs]
Description:	Calibration value for the RC filter of the zero crossover detection of the line supply voltage in the power unit. When p3469 = 0, a new calibration is performed the next time identification is carried out with p3410 = 4 or 5.		
Note:	The calibration value is stored in the EEPROM of the power unit because it is a characteristic of the power unit.		
r3470	Infeed active current filter / INF I_act filter		
A_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: - Max - [Arms]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the active current requirement due to the line filter.		
Dependency:	Refer to: r0038, p0221, p0222		
Note:	With respect to the line supply, the sum of the active currents of the power unit (p0078) and line filter (p3470) are effective. The active current demand of the line filter is taken into account when calculating the power factor (r0038). The magnitude of the line filter active current depends on the capacitance (p0221) and the resistance (p0222) of the line filter.		
r3471	Infeed reactive current filter / INF I_reactiveFilt		
A_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: - Max - [Arms]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the reactive current requirement as a result of the line filter. The reactive current requirement of a line filter is covered by the controlled infeed/regenerative feedback so that the converter always operates with a power factor of 1 compared to the line.		
Dependency:	Refer to: r0038, r0075, r0076, p0221		
Note:	With respect to the line supply, the sum of the reactive currents of the power unit (p0076) and line filter (p3471) are effective. The reactive current requirement of the line filter is taken into account when calculating the power factor (r0038). The amount of the reactive current depends on the capacitance (p0221) of the line filter that is automatically parameterized when a line filter is selected (p0220). If the line phases are reversed and the line voltage therefore has a negative orientation (r0066 < 0), it should be noted that the sign of the reactive current is reversed.		

p3472[0...4]	Line supply PLL line supply voltage smoothing time / Line PLL V_I t_{sm}		
A_INF, S_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 1.0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 30000.0 [ms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 200.0 [ms] [1] 100.0 [ms] [2] 5000.0 [ms] [3] 8.0 [ms] [4] 8.0 [ms]
Description:	Sets the smoothing time of the line supply voltage for the line supply PLL.		
Index:	[0] = Encoderless operation line supply voltage smoothing time [1] = VSM operation line supply voltage smoothing time [2] = Detection line supply undervoltage smoothing time [3] = Detection line supply overvoltage smoothing time [4] = Detection line supply voltage step smoothing time		
Dependency:	Refer to: p3400		
Note:	For the pre-control of the line supply voltage, a smoothed value of the line supply voltage is used in the closed-loop control. p3472[0]: Sets the PT1 time constant to smooth the line supply voltage for operation without VSM (p3400.5 = 0). p3472[1]: Sets the PT1 time constant to smooth the line supply voltage for operation with VSM (p3400.5 = 1). p3472[2]: Sets the smoothing time constant to slowly detect a line supply undervoltage (F6100). p3472[3]: Sets the smoothing time constant to quickly detect line supply undervoltages for phase failure (A6205). p3472[4]: Sets the smoothing time constant to quickly adapt the line supply pre-control for line supply voltage steps (p0286).		
p3480	Infeed modulation depth limit / INF mod_depth lim		
A_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 50.0 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 110.0 [%]	Access level: 3 Func. diagram: 8940 Unit selection: - Expert list: 1 Factory setting 97.0 [%]
Description:	Sets the maximum steady-state modulation depth. When this limit is reached, the DC link voltage is boosted to maintain the control margin. This means that the control reserve is maintained.		
Dependency:	Refer to: p3481, r3485		
p3481	Infeed standby controller dynamic response / INF res_ctrl dyn		
A_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.0 [ms]	Access level: 4 Func. diagram: 8940 Unit selection: - Expert list: 1 Factory setting 7.5 [ms]
Description:	Sets the dynamic response of the reserve controller for the modulation depth. As the smoothing time increases, the response of the DC link voltage tracking becomes slower.		
Dependency:	Refer to: p3480, r3485		

r3485	Infeed standby controller output / INF res_ctrl outp		
A_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 8940 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Displays the reserve controller output for the modulation depth. The DC link voltage is increased by this voltage value - the summed setpoint for the DC link voltage is output in r0088. The summed setpoint is limited to the maximum steady-stage DC link voltage (p0280).		
Dependency:	Refer to: p3480, p3481		
p3490	Infeed OFF command delay time / INF t_del OFF		
A_INF, B_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000000.0 [ms]	Access level: 2 Func. diagram: 8732, 8832, 8932 Unit selection: - Expert list: 1 Factory setting 0.0 [ms]
Description:	Sets the delay time for the ON/OFF1 command of the infeed. After ON/OFF1 = 0 the infeed remains in operation for the specified time		
Dependency:	Refer to: p0840		
Notice:	The ON/OFF1 command of the infeed can be interrupted.		
Note:	This parameter is only relevant if a Motor Module and the infeed are controlled by the same OFF command. In this case, the delay time and the stop ramp time of the motor can be coordinated with one another.		
p3491	Infeed I-offset measurement monitoring time / INF I_offs t_monit		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Commands Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65000 [ms]	Access level: 4 Func. diagram: 8832, 8932 Unit selection: - Expert list: 1 Factory setting 2000 [ms]
Description:	Sets the monitoring time for the current-offset measurement of the power unit. The time is started with the normal end of the measurement. If the measurement is invalid and if no valid measurement can be taken within the monitoring period (phase currents too high), an appropriate message is generated.		
Note:	Set this parameter to 0 to allow variations in the delay when running-up.		
p3492	Infeed, line supply undervoltage delay time / INF V_line t_del		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Commands Not for motor type: - Min 0 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 300 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [s]
Description:	Sets the delay time for shutdown due to a line supply undervoltage condition (A06100). After the fault occurs, the power unit is only tripped (shut down) after this delay has expired. If the fault is removed during this design time, then the power unit is not tripped (shut down).		
Dependency:	Refer to: p0283 Refer to: F06100		

Note: The degree of ruggedness of the infeed with respect to fluctuations in the line supply voltage can be increased by parameterizing this delay value.
However, the following should be noted:

- the infeed power decreases proportionally (linearly) with the line supply voltage.
- when other components are connected, for low line supply voltage, operating faults or damage can occur. In this case, the specifications of the connected electrical components should always be carefully observed.

p3508 Infeed step-up factor maximum / Step-up factor max

A_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 1.60	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3.00	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.60
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Description: Sets the maximum permissible step-up factor for the power unit used in conjunction with the line filter parameterized in p0220[0].

Dependency: Refer to: p0210, p0220, p3510

Note: The maximum step-up factor determines the maximum ratio between the DC link voltage setpoint (p3510) and the unit supply voltage (p0210).
The input of the DC link voltage setpoint (p3510) is limited corresponding to the permissible step-up factor (p3508):
 $p3510 \leq p0210 * p3508$.
 Presetting values:
 380 ... 480 V booksize units without Active Interface Module: 1.60
 380 ... 480 V booksize units with Active Interface Module (p0220 = 41 ... 45): 2.00
 380 ... 480 V chassis units: 2.00
 500 ... 690 V chassis units: 2.00
 Maximum values:
 380 ... 480 V booksize units without Active Interface Module: 1.60
 380 ... 480 V booksize units with Active Interface Module (p0220 = 41 ... 45): 2.00
 380 ... 480 V chassis units: 2.00
 500 ... 690 V chassis units: 2.00
 When the filter setting (p0220) is changed, then the setting of the maximum step-up factor (p3508) is also automatically adapted.

p3510 Infeed DC link voltage setpoint / INF Vdc setp

A_INF, SERVO (Tech_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 100.00 [V]	Calculated: - Dynamic index: - Units group: 5_2 Scaling: p2001 Max 1600.00 [V]	Access level: 2 Func. diagram: 1774, 8940 Unit selection: p0505 Expert list: 1 Factory setting 600.00 [V]
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Description: Sets the setpoint for the DC link voltage.

Dependency: Refer to: p0210, p0280, p3400, p3508, p3511

Warning: Before increasing the voltage limit for pulsed operation of a controlled booksize infeed with line supply voltages p0210 > 415 V it should be checked whether the motors connected to the DC link are specified for the higher motor voltages.
The warning information associated with p0210 must be carefully observed.



Note: When the Smart Mode is activated (p3400.0 = 1) the DC link voltage is not regulated, i.e. the value entered here is in this case not effective.

The permissible range of the DC link voltage depends on the parameterized unit supply voltage (p0210) and the permissible, maximum continuous DC link voltage (p0280).

In voltage-controlled operation (p3400.0 = 0) the following applies:

$p3510 \geq 1.42 \cdot p0210$ and
 $p3510 \leq p3508 \cdot p0210$ and
 $p3510 \leq p0280$.

In the Smart Mode (p3400.0 = 1) the following applies:

The setpoint p3510 for the DC link voltage control is inactive. In order to permit an adapted display, deviating from voltage-controlled operation, the lower limit p3510 is $\geq 1.2 \cdot p0210$.

p3510	DC link voltage setpoint / Vdc setp		
VECTOR (Tech_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 100.00 [V]	Calculated: - Dynamic index: - Units group: 5_2 Scaling: p2001 Max 1600.00 [V]	Access level: 2 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 600.00 [V]
Description:	Sets the setpoint for the DC-link voltage on the motor side.		
p3511	CI: Infeed DC link voltage supplementary setpoint / INF Vdc Z_set		
A_INF, SERVO (Tech_ctrl)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max -	Access level: 3 Func. diagram: 8940 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the supplementary setpoint for the DC link voltage.		
Dependency:	Refer to: p3510		
p3511	CI: DC-link voltage supplementary setpoint / Vdc Z_set		
VECTOR (Tech_ctrl)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the supplementary setpoint for the DC-link voltage on the motor side.		
Dependency:	Refer to: p3510		
p3513	BI: Inhibit voltage-controlled operation / Inhib V_ctrl mode		
A_INF, SERVO (Tech_ctrl)	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for inhibiting the voltage-controlled mode of the infeed.		
Dependency:	Refer to: p3400, r3405		
Notice:	The DC link voltage must be controlled by a different component at the DC link; otherwise this results in an overvoltage or undervoltage condition.		

Note: The current controller remains active and can be controlled by means of its setpoint inputs (p3515, p3610). This binector input is used to change over between master (0 signal) and slave (1 signal) operation and vice versa.

p3513 BI: Inhibit voltage-controlled operation / Inhib V_ctrl mode

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	1

Description: Sets the signal source for disabling DC-link voltage control on the motor side.

p3514 Infeed supplementary active current steady-state / INF I_sup_eff stat

A_INF	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8940
	P-Group: Closed-loop control	Units group: 6_2	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-1000.00 [Arms]	1000.00 [Arms]	0.00 [Arms]

Description: Sets a steady-state supplementary setpoint for the active line supply current.

Dependency: Refer to: p3515

p3515 CI: Infeed supplementary active current / INF I_suppl eff

A_INF	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 8940
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the supplementary setpoint of the active current.

Dependency: Refer to: p3514

p3516 Infeed current distribution factor / INF I_distr_factor

A_INF	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8940, 8942
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [%]	100.00 [%]	100.00 [%]

Description: Sets the factor to be multiplied by the active current setpoint for the current controller.

Dependency: Refer to: p3579

r3517 CO: Infeed active current controller unlimited setpoint / INF I_act ctrl set

A_INF, SERVO (Tech_ctrl)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8940
	P-Group: Closed-loop control	Units group: 6_2	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [Arms]	- [Arms]	- [Arms]

Description: Displays the unlimited setpoint of the active current controller.

For a master-slave infeed configuration, the master retrieves this setpoint and distributes it to all of the slaves. The slaves operate in the current-controlled mode.

r3517	CO: DC-link controller active current setpoint / UDC I_act ctrl stp		
VECTOR (Tech_ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the unlimited setpoint of the active current controller for DC-link voltage control on the motor side.		
p3520[0...3]	CI: Infeed power pre-control / INF pre-control P		
A_INF, SERVO (Tech_ctrl)	Can be changed: U, T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for power pre-control.		
Dependency:	Refer to: p3521		
Note:	Closed-loop control of the DC link voltage is improved by pre-controlling the power required for the other modules. A non-normalized quantity is expected so that the various power reference values (r2004) of the drive objects do not have to be taken into account. The scaling factors are used to adapt the scaling (p3521).		
p3520[0...3]	CI: DC link pre-control power / Vdc pre-ctrl P		
VECTOR (Tech_ctrl)	Can be changed: U, T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for power pre-control.		
Dependency:	Refer to: p3521		
Note:	Closed-loop control of the DC link voltage is improved by pre-controlling the power required for the other modules. A non-normalized quantity is expected so that the various power reference values (r2004) of the drive objects do not have to be taken into account. The scaling factors are used to adapt the scaling (p3521).		
p3521[0...3]	Infeed pre-control power scaling / INF prectrl P scal		
A_INF, SERVO (Tech_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -100000.00000 [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max 100000.00000 [%]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.00000 [%]
Description:	Sets the scaling factor for the power pre-control.		
Dependency:	Refer to: p3520		

p3521[0...3] DC link pre-control power scaling / Vdc prectrl P scal

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min -100000.00000 [%]	Max 100000.00000 [%]	Factory setting 100.00000 [%]
Description:	Sets the scaling factor for the power pre-control.		
Dependency:	Refer to: p3520		

p3523[0...3] Infeed pre-control power smoothing / INF pre-ctrl P sm

SERVO (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 1000 [ms]	Factory setting 0 [ms]
Description:	Sets the filter time for power pre-control.		
Dependency:	Refer to: p3520		

p3523[0...3] DC link pre-control power smoothing / Vdc pre-ctrl P sm

VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 1000 [ms]	Factory setting 0 [ms]
Description:	Sets the filter time for power pre-control.		
Dependency:	Refer to: p3520		

p3530 Infeed current limit motoring / INF I_limit mot

A_INF	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8940
	P-Group: Closed-loop control	Units group: 6_2	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1.00 [Arms]	Max 100000.00 [Arms]	Factory setting 10000.00 [Arms]
Description:	Sets the motoring limit for the active line current. The currently effective current limit is displayed in r0067[0].		
Dependency:	Refer to: r0067, p3532		
Caution:	If this limit is selected lower than the maximum current permissible for the power unit (r0067), the infeed can no longer provide its full controlled power. Operating faults of the infeed can occur due to the resulting DC link undervoltage.		
Notice:	For self-commutated infeeds, the DC link voltage decreases if more power is drawn from the DC link by the connected load than can be supplied by the line because of the power unit maximum current or a limit in p3530. If the DC link voltage decreases down to the rectified value, then the complete current - necessary to cover the required active power - flows, uncontrolled into the rectifier circuit via the diodes. This is the reason that, for physical reasons, the value in p3530 cannot act as current limit that is always maintained. The value forms a current threshold from which point onwards the DC link energy is used as buffer for brief power fluctuations.		

p3531 Infeed current limit regenerative / INF I_limit regen			
A_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -100000.00 [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: - Max -1.00 [Arms]	Access level: 3 Func. diagram: 8940 Unit selection: p0505 Expert list: 1 Factory setting -10000.00 [Arms]
Description:	Sets the limit for the active line supply current when regenerating. The currently effective current limit is displayed in r0067[1].		
Dependency:	Refer to: r0067, p3533		
Caution:	If this limit is selected lower than the maximum current permissible for the power unit (r0067), the infeed can no longer provide its full controlled power. This can result in an overvoltage condition in the DC link.		
Notice:	For self-commutated infeeds, the DC link voltage increases if more power is input to the DC link through the connected source than can be fed to the line because of the maximum power unit current or a limit in p3531 If the DC link voltage exceeds the permissible threshold (p0297), defined by the hardware, then the unit is tripped due to overvoltage. The value in p3531 represents a current limit that is always maintained - however, this can result in overvoltage conditions in the DC link. The value in p3531 represents a current limit from which point onwards the capacitance of the DC link can be used as buffer for brief power fluctuations.		

p3532 BI: Infeed, inhibit motoring / INF mot mode inhib			
A_INF	Can be changed: T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 8920 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for inhibiting the motor mode of the infeed.		
Dependency:	Refer to: r3405, p3530		
Notice:	If the motor mode is inhibited although power is withdrawn from the DC link, then the DC link voltage drops to the rectified value. In this state, the DC link is post-charged through the diodes and motoring power is fed to the power unit in spite of the motoring inhibit. The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	The inhibit only becomes active after operation has been enabled and the Vdc has been ramped up (r0863.0 = 1).		

p3533 BI: Infeed, inhibit generator mode / INF gen mode inhib			
A_INF, S_INF	Can be changed: T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 8820, 8920 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to inhibit the generator mode of the infeed.		
Dependency:	Refer to: r3405, p3531		
Notice:	The DC link voltage will increase if generator mode is inhibited even though power is being regenerated into the DC link. The parameter may be protected as a result of p0922 or p2079 and cannot be changed.		
Note:	The inhibit only becomes active after operation has been enabled and the Vdc has been ramped up (r0863.0 = 1).		

r3534	Infeed line filter maximum current / INF filter I_max		
A_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Arms]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Arms]
Description:	Displays the maximum permissible current for the line filter set using p0220[0].		
Dependency:	Refer to: p0220		
Note:	The currently effective maximum current for the power unit is displayed in r0067. The value in r0067 is obtained as minimum of the current limits in r0209, p3530 ... r3534.		
r3554	Infeed Vdc controller integral component / INF Vdc_ctr I_comp		
A_INF, SERVO (Tech_ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 2 Func. diagram: 8940 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the integral action component of the DC link voltage controller (Vdc controller).		
r3554	Vdc controller integral component / Vdc_ctr I_comp		
VECTOR (Tech_ctrl)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 2 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays the integral action component of the DC link voltage controller (Vdc controller).		
p3555[0...5]	Infeed Vdc controller integral component fast intervention / Vdc_ctr I-compFast		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 200.00 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 2.00 [%] [1] 102.00 [%] [2] 0.00 [%] [3] 5.00 [%] [4] 100.00 [%] [5] 0.00 [%]
Description:	Sets the fast Vdc controller intervention for a step-like decrease of the DC link voltage due to a high motor load of the infeed. The Vdc controller fast intervention is deactivated for p3555[5] = 0 % or p3560 < 100% or p0225 > 0.5 * p0223. For a line supply and DC link identification (p3410 >= 2) the level of the fast controller intervention (p3555[2]) is automatically adapted to the line supply inductance.		

Recommend.:	Precise system knowhow is required when correctly changing this parameter! - generally, the fast controller intervention is used to improve the control behavior for high-speed load changes. The function can therefore always be deactivated with p3555[5] = 0 % if no peak load duty cycles are required in the application. - using p3555[0], the calculation of the modulation depth is determined in the case of high system deviations also when the controller intervention is deactivated. This is the reason that p3555[0] should generally not be changed.
Index:	[0] = Intervention threshold 1: Vdc deviation from the setpoint [1] = Intervention threshold 2: Vdc difference to the rectified value [2] = Fast intervention automatic scaling [3] = Fast intervention pre-control [4] = Fast intervention timeout [5] = Fast intervention manual scaling
Note:	p3555[0]: Vdc system deviation as a percentage of the setpoint of the DC link voltage (first condition to initiate fast controller intervention). The threshold is also used to internally change over the modulation depth calculation for high system deviations and should therefore generally not be changed! p3555[1]: Vdc threshold as a percentage of the rectified value of the current line supply voltage (second condition to initiate the fast controller intervention). Both threshold conditions must be fulfilled to initiate the controller intervention. p3555[2]: Percentage overall level of the fast intervention (scaling factor). For a line supply identification with p3410 >= 2, the factor is automatically adapted or, for weak line supplies with a high inductance, set to 0. p3555[3]: Percentage correction of the pre-control for a fast voltage dip (dead time compensation). p3555[4]: Percentage minimum time between controller interventions (100% corresponds to 100 ms). If high load change frequencies occur with the application, the minimum time between two controller interventions can be reduced using p3555[4]. p3555[5]: Percentage overall level of the fast intervention (scaling factor). With p3555[5] = 0, the fast controller intervention is inhibited. For weak line supplies with a high inductance, it makes sense to deactivate the fast intervention.

p3560	Infeed Vdc controller proportional gain / INF Vdc_ctrl Kp		
A_INF, SERVO (Tech_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.01 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: 8940 Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the normalized proportional gain for the DC link voltage controller (Vdc controller).		
Note:	A value of 100 % corresponds to the basic setting derived from loop control parameters (p3421, p3422).		

p3560	Vdc controller proportional gain / Vdc_ctrl Kp		
VECTOR (Tech_ctrl)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.01 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the normalized proportional gain for the DC link voltage controller (Vdc controller).		
Note:	A value of 100% corresponds to the basic setting derived from the loop control parameter (p3422).		

p3562			
Infeed,Vdc controller integral time / INF Vdc_ctrl Tn			
A_INF, SERVO (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8940
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.10 [%]	Max 100000.00 [%]	Factory setting 100.00 [%]
Description: Sets the normalized integral time for the DC link voltage controller (Vdc).			
Note: A value of 100 % corresponds to the basic setting derived from loop control parameters (p3421, p3422).			
p3562			
Vdc controller integral time / Vdc_ctrl Tn			
VECTOR (Tech_ctrl)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.10 [%]	Max 100000.00 [%]	Factory setting 100.00 [%]
Description: Sets the normalized integral time for the DC link voltage controller (Vdc).			
Note: A value of 100% corresponds to the basic setting derived from the loop control parameter (p3422).			
p3564			
Infeed Vdc monitor, time constant / INF Vdc_observe T			
A_INF	Can be changed: T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 100.0 [ms]	Factory setting 0.2 [ms]
Description: Sets the filter time constant for the DC link voltage monitor (Vdc).			
p3566			
Infeed Vdc ramp duration / INF Vdc t_ramp			
A_INF	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8932
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 40 [ms]	Max 1000 [ms]	Factory setting 100 [ms]
Description: Sets the ramp time for the DC link voltage (Vdc) when powering up and powering down.			
Powering up (pulses enabled, r0898.3 = 1):			
During this time, the DC link voltage is increased from the rectifier value after pre-charging to the voltage setpoint (p3510, p3511). The voltage setpoint is increased, when necessary, so that the modulation depth reserve (p3481) is maintained. The reactive current is set to the value 0 while ramping.			
Powering down (inhibit pulses, r0898.3 = 0):			
During this time, the DC link voltage is reduced to the rectified value ($\sqrt{2}$ * line supply voltage). The reactive current value is set to the value 0 when the ramp starts.			

p3570	CI: Master/slave active current setpoint / I_act_setp		
A_INF (Master/Slave)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2002 Max -	Access level: 3 Func. diagram: 8948 Unit selection: - Expert list: 1 Factory setting 3573[0]
Description:	Sets the signal source of the active current setpoint for the closed-loop current control of the slave. The signal value is received from the master infeed (e.g. via the multiplexer or direct).		
Dependency:	Refer to: p3513, p3571, p3572, r3573		
p3571[0...3]	CI: Master/slave active current setpoint, multiplexer input / I_act multi inp		
A_INF (Master/Slave)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2002 Max -	Access level: 3 Func. diagram: 8948 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal sources for the input values of the multiplexer. The signal values are used as setpoint for the closed-loop current control of the slave infeed.		
Index:	[0] = Multiplexer input value 0 [1] = Multiplexer input value 1 [2] = Multiplexer input value 2 [3] = Multiplexer input value 3		
Dependency:	Refer to: p3570, p3572, r3573		
Note:	For a master infeed and a slave infeed, the active current setpoint can be entered without using a multiplexer. If the multiplexer for the master/slave is not required, then it can also be used for another function.		
p3572	CI: Master/slave active current setpoint, multiplexer selection / I_act multi sel		
A_INF (Master/Slave)	Can be changed: T Data type: Unsigned32 / Integer16 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 8948 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to select the required input value for the multiplexer. CI: p3572 = 0, 1, 2, 3 --> valid values Fault F06320 is output for other values.		
Dependency:	Refer to: p3570, p3571, r3573 Refer to: F06320		
Note:	For a master infeed and a slave infeed, the active current setpoint can be entered without using a multiplexer. If the multiplexer for the master/slave is not required, then it can also be used for another function.		
r3573	CO: Master/slave active current setpoint, multiplexer output / I_act multi outp		
A_INF (Master/Slave)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 8948 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Displays (connector output) the output for the multiplexer. The signal value is used as standard for the active current setpoint for the slave infeed.		

Dependency: Refer to: p3570, p3571, p3572

Note: For a master infeed and a slave infeed, the active current setpoint can be entered without using a multiplexer. If the multiplexer for the master/slave is not required, then it can also be used for another function.

p3574[0...3] Master/slave DC link voltage monitoring / Vdc monitoring

A_INF (Master/Slave)	Can be changed: C2(1), T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8948
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -60 [V]	Max 60 [V]	Factory setting [0] 20 [V] [1] -20 [V] [2] 5 [V] [3] -5 [V]

Description: Sets the upper and lower limit values and hysteresis values for the DC link voltage monitoring. The values are entered as absolute values and refer to the DC link voltage setpoint (p3510). For a slave infeed, if the limits are violated, then the closed-loop voltage control is automatically switched in.

Index:
[0] = Vdc upper limit value
[1] = Vdc lower limit value
[2] = Vdc upper hysteresis value
[3] = Vdc lower hysteresis value

Dependency: Refer to: p0210

r3575.0...2 BO: Master/slave DC link voltage monitoring status / Vdc monit status

A_INF (Master/Slave)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 8948
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

Description: Displays the status of the DC link voltage monitoring for the master/slave.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Upper limit value reached	Yes	No	-
	01	Lower limit value reached	Yes	No	-
	02	Upper/lower limit value reached	Yes	No	-

Dependency: Refer to: r0088, p3510, p3574

p3576[0...5] Master/slave current distribution factor, multiplexer input / I_dist_factor inp.

A_INF (Master/Slave)	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8948
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [%]	Max 100.00 [%]	Factory setting 100.00 [%]

Description: Sets up to 6 factors to be multiplied by the active current setpoint for the current controller. For a master slave infeed configuration, the value reduced in this way can be distributed to the slave axes. The overall gain from the perspective of the voltage controller remains the same.

Index:
[0] = Multiplexer input value 0
[1] = Multiplexer input value 1
[2] = Multiplexer input value 2
[3] = Multiplexer input value 3
[4] = Multiplexer input value 4
[5] = Multiplexer input value 5

Dependency: Refer to: p3577, r3578, p3579

Note: If the multiplexer for the master/slave is not required, then it can also be used for another function.

p3577 **CI: Master/slave current distribution factor, multiplexer selection / I_dist_factor sel.**

A_INF (Master/Slave)

Can be changed: T

Calculated: -

Access level: 3

Data type: Unsigned32 / Integer16

Dynamic index: -

Func. diagram: 8948

P-Group: Closed-loop control

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

-

-

0

Description: Sets the signal source to select the required input value for the multiplexer.

CI: p3577 = 0, 1, 2, 3, 4, 5 --> valid values

Fault F06321 is output for other values.

Dependency: Refer to: p3576, r3578, p3579

Refer to: F06321

Note: If the multiplexer for the master/slave is not required, then it can also be used for another function.

r3578 **CO: Master/slave current distribution factor, multiplexer output / I_dist_factor outp**

A_INF (Master/Slave)

Can be changed: -

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dynamic index: -

Func. diagram: 8948

P-Group: Closed-loop control

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

- [%]

- [%]

- [%]

Description: Displays (connector output) the output for the multiplexer.

The signal value is used as standard for the current distribution factor for the infeed master slave operation.

Dependency: Refer to: p3576, p3577, p3579

Note: If the multiplexer for the master/slave is not required, then it can also be used for another function.

p3579 **CI: Master/Slave current distribution factor / I_dist_factor**

A_INF (Master/Slave)

Can be changed: T

Calculated: -

Access level: 3

Data type: Unsigned32 / FloatingPoint32

Dynamic index: -

Func. diagram: 8948

P-Group: Closed-loop control

Units group: -

Unit selection: -

Not for motor type: -

Scaling: PERCENT

Expert list: 1

Min

Max

Factory setting

-

-

3578[0]

Description: Sets the factor to be multiplied by the active current setpoint for the current controller.

For a master slave infeed configuration, the value reduced in this way can be distributed to the slave axes. The overall gain from the perspective of the voltage controller remains the same.

Dependency: Refer to: p3576, p3577, r3578

r3602 Infeed control status / INF ctrl state

A_INF	Can be changed: -	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0000 bin	1000 bin	-

Description: Displays the status of the closed-loop infeed control.

Value:	0: Initialization running
	1: Pulse enable missing
	2: Ramp-up, DC link voltage
	3: Ramp-up reactive current
	4: Shutdown running
	5: Reset identification
	6: Operation
	7: Identification running
	8: Smart Mode running

p3603 Infeed current pre-control factor D component / INF I_ctrl D-comp

A_INF	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8946
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [%]	500.00 [%]	100.00 [%]

Description: The D component of the current pre-control is determined from the device data of the filter. p3603 can be used to weigh the pre-calculated D component. If no dynamic pre-control is to be used, set the factor to zero.

r3606 Infeed active current controller system deviation / INF I_act ctrl dev

A_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8946
	P-Group: Closed-loop control	Units group: 6_2	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [Arms]	- [Arms]	- [Arms]

Description: Displays the system deviation of the active current controller.

r3608 Infeed reactive current controller system deviation / INF I_reactvCtrDev

A_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8946
	P-Group: Closed-loop control	Units group: 6_2	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [Arms]	- [Arms]	- [Arms]

Description: Displays the system deviation of the reactive current controller.

p3610	Infeed reactive current fixed setpoint / INF I_reactv F_set		
A_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -10000.0 [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: - Max 10000.0 [Arms]	Access level: 2 Func. diagram: 1774, 8946 Unit selection: p0505 Expert list: 1 Factory setting 0.0 [Arms]
Description:	Sets the fixed setpoint for the reactive current. The permissible maximum absolute value for the reactive current is the device rated current r0207. The following applies: $ p3610 \leq r0207$		
Dependency:	Refer to: r0029, r0075, r0076		
Notice:	If the line phases are reversed and the line voltage therefore has a negative orientation ($r0066 < 0$), it should be noted that the sign of the reactive current is also reversed. The negated value of p3610 is effective in display parameters r0029, r0075, r0076 as appropriate.		
Note:	p3610 < 0: Inductive reactive current is produced, i. e. the current follows the voltage. p3610 > 0: Capacitive reactive current is produced, i. e. the current leads the voltage. This definition applies to 3AC voltage systems both with positive rotational orientation ($r0066 > 0$) and for negative rotational orientation ($r0066 < 0$).		
p3611	CI: Infeed reactive current supplementary setpoint / INF I_reactv Z_set		
A_INF	Can be changed: U, T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2002 Max -	Access level: 2 Func. diagram: 8946 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the supplementary setpoint of the reactive current.		
p3614	Infeed current actual value filter smoothing time / INF I_act t_sm		
A_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Commands Not for motor type: - Min 0.000 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2.000 [ms]	Access level: 3 Func. diagram: 8950 Unit selection: - Expert list: 1 Factory setting 0.000 [ms]
Description:	Sets the time constant for the PT1 filtering of the active current actual value and reactive current actual value.		
Note:	The PT1 filter with a clock cycle dead time can be used to stabilize the closed-loop current control for extremely weak line supplies with higher relative short-circuit voltage u_k . The current actual value filter is deactivated with $p3614 = 0$. For an automatic controller setting with $p3410 \geq 2$, the current actual value filter is automatically preset.		
p3615	Infeed current controller P gain / INF I_ctrl Kp		
A_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: 8946 Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the normalized P gain for closed-loop current control of the infeed.		
Note:	A value of 100 % corresponds to the basic setting derived from loop control parameters (p3421, p3422).		

p3617	Infeed current controller integral time / INF I_ctrl Tn		
A_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.10 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000.00 [%]	Access level: 2 Func. diagram: 8946 Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the normalized integral time for the infeed current controller.		
Note:	A value of 100 % corresponds to the basic setting derived from loop control parameters (p3421, p3422).		
r3618	Infeed active current controller, integral component / INF I_act_ctrl Tn		
A_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 2 Func. diagram: 8946 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the integral component of the active current controller.		
r3619	Infeed reactive current controller integral component / INF I_reactv_ctrTn		
A_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: 5_1 Scaling: p2001 Max - [Vrms]	Access level: 2 Func. diagram: 8946 Unit selection: p0505 Expert list: 1 Factory setting - [Vrms]
Description:	Displays the integral action component of the reactive current controller.		
p3620	Infeed current controller adaptation lower application threshold / INF I_adptLowThrsh		
A_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 40.00 [%]
Description:	Sets the switch-in threshold for the current controller adaptation. The value refers to the maximum power unit current (r0209). From the starting threshold onwards, the inductance value (p3421) used for current control is reduced linearly as a function of the current value. The inductance value for the maximum power unit current is therefore $p3421 * p3622$.		
Dependency:	Refer to: p3410, p3415, p3622		
Note:	The parameter can be set automatically using the line supply identification (p3410 = 4, 5) (also refer to p3622). Pre-requisite for a reliable measurement of p3622 is that the current magnitude for run 2 (p3415[1]) is at least 10 % higher than the current magnitude for run 1 of the line supply identification. Otherwise, the measurement result is rejected. In the case of a correct measurement, p3620 is set to 80% of the current magnitude for run 1 (p3415[0]). For chassis power units, it is generally not necessary to adapt p3620 and p3622 to the characteristics of the line supply. However, when required, the current controller adaptation can be optimized by selecting suitable current magnitudes for p3415. For booksize power units, p3620 and p3622 are automatically adapted with the then valid default setting of the line identification p3415.		

p3622	Infeed current controller adaptation reduction factor / INF I_adapt factor		
A_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.01 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 85.00 [%]
Description:	Sets the inductance of the line reactor at the maximum power unit current (r0209) as a percentage of the inductance (p3421) at the application threshold (p3620).		
Dependency:	Refer to: p3410, p3415, p3620		
Note:	The parameter for a line supply identification (p3410 = 4, 5) automatically optimized, if the following applies: p3415[1] - p3415[0] > 10%. Otherwise, the measurement result is rejected. For chassis power units, it is generally not necessary to adapt p3620 and p3622 to the characteristics of the line supply. However, when required, the current controller adaptation can be optimized by selecting suitable current magnitudes for p3415. For booksize power units, p3620 and p3622 are automatically adapted with the then valid default setting of the line identification p3415.		
p3624[0...1]	Infeed harmonics controller order / INF harm_ctr order		
A_INF	Can be changed: T Data type: Unsigned16 P-Group: Closed-loop control Not for motor type: - Min 5	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max 13	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 5 [1] 7
Description:	Sets the Order of the line harmonics for the current harmonics controller. p3624[0]: Order of the line harmonics for the first harmonics controller. p3624[1]: Order of the line harmonics for the second harmonics controller.		
Dependency:	Refer to: p3625, r3626		
Note:	Harmonics in the line supply voltage can cause harmonics in the converter current. These types of current harmonics can be reduced by activating additional controller modules. Example: For a 50 Hz line supply harmonics at 250 Hz in the phase currents can be reduced by activating a harmonic controller with Order 5 (p3624[0] = 5).		
p3625[0...1]	Infeed harmonics controller scaling / INF harm_ctrl scal		
A_INF	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.0 [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max 300.0 [%]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.0 [%]
Description:	Sets the gain of the harmonics controller. p3625[0]: Gain of the first harmonics controller p3625[1]: Gain of the second harmonics controller 0 %: Controller is deactivated 100 %: Controller is activated with default gain setting		
Dependency:	Refer to: p3624, r3626		
Note:	The harmonics controller corrects the power unit voltages so that the line-side current harmonics are reduced. The order of a current harmonic, that is to be dampened using a harmonics controller, is defined using p3624.		

r3626[0...1] Infeed harmonics control output / INF harm_ctrl outp

A_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: 5_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [Vrms]	- [Vrms]	- [Vrms]
Description:	Displays the output voltages of the harmonics controller. r3626[0]: RMS value of the 5th harmonic of the controller output voltage r3626[1]: RMS value of the 7th harmonic of the controller output voltage The harmonics controller corrects the power unit voltages so that the line-side current harmonics are reduced.		
Dependency:	Refer to: p3624, p3625		

r3632 Infeed input voltage Vsd (active component) / INF V_inp Vsd

A_INF	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 1774, 8946, 8950
	P-Group: Closed-loop control	Units group: 5_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [Vrms]	- [Vrms]	- [Vrms]
Description:	Displays the voltage Vsd (active component) at the 3-phase line supply input of the power unit.		

r3633 Infeed input voltage Vsq (reactive component) / INF V_inp Vsq

A_INF	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 1774, 8946, 8950
	P-Group: Closed-loop control	Units group: 5_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [Vrms]	- [Vrms]	- [Vrms]
Description:	Displays the voltage Vsq (reactive component) at the 3-phase line supply input of the power unit.		

r3635 CO: Infeed input voltage angle / INF V_inp angle

A_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8950
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [°]	- [°]	- [°]
Description:	Displays the angle of the input voltage (relative to the line angle).		

p3660	VSM input line supply voltage, voltage scaler / VSM inp V_scaler		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max 100000.00 [%]	Access level: 3 Func. diagram: 9880 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the voltage scaler for the Voltage Sensing Module (VSM).		
Note:	When the 690 V input is used (X522) without voltage scaler, 0 % should be entered. When the 100 V input (X521) is used with voltage scaler to measure medium voltages, the dividing (scaling) factor multiplied by 100% should be entered. Example: 1000 V line supply voltage, voltage scaling, 10:1 --> voltage at the VSM input is 100 V --> p3660 = 10 * 100 % = 1000 %		
p3660[0...n]	VSM input line supply voltage, voltage scaler / VSM inp V_scaler		
VECTOR	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: p0150 Units group: - Scaling: PERCENT Max 100000.00 [%]	Access level: 3 Func. diagram: 9880 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the voltage scaler for the Voltage Sensing Module (VSM).		
Note:	When the 690 V input is used (X522) without voltage scaler, 0 % should be entered. When the 100 V input (X521) is used with voltage scaler to measure medium voltages, the dividing (scaling) factor multiplied by 100% should be entered. Example: 1000 V line supply voltage, voltage scaling, 10:1 --> voltage at the VSM input is 100 V --> p3660 = 10 * 100 % = 1000 %		
r3661	CO: VSM input line supply voltage u1 - u2 / VSM inp u1-u2		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: 5_3 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 8850, 8950, 9880 Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the voltage between phases L1 and L2.		
Dependency:	Refer to: r0025, r0072, p3660		
Note:	X521.1 or X522.1: Connection of L1 X521.2 or X522.2: Connection of L2 X521.3 or X522.3: Connection of L3 The absolute voltage value (3-ph. AC) resulting from the phase voltages is displayed unsmoothed in r0072[1] and smoothed in r0025[1].		

r3661[0...n]	CO: VSM input line supply voltage u1 - u2 / VSM inp u1-u2			
VECTOR	Can be changed: -	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: -	
	P-Group: Closed-loop control	Units group: 5_3	Unit selection: p0505	
	Not for motor type: -	Scaling: p2001	Expert list: 1	
	Min	Max	Factory setting	
	- [V]	- [V]	- [V]	
Description:	Displays the voltage between phases L1 and L2.			
Dependency:	Refer to: p3660			
Note:	X521.1 or X522.1: Connection of L1 X521.2 or X522.2: Connection of L2 X521.3 or X522.3: Connection of L3			

r3662	CO: VSM input line supply voltage u2 - u3 / VSM inp u2-u3			
A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 8850, 8950, 9880	
	P-Group: Closed-loop control	Units group: 5_3	Unit selection: p0505	
	Not for motor type: -	Scaling: p2001	Expert list: 1	
	Min	Max	Factory setting	
	- [V]	- [V]	- [V]	
Description:	Displays the voltage between phases L2 and L3.			
Dependency:	Refer to: r0025, r0072, p3660			
Note:	X521.2 or X522.2: Connection of L2 X521.3 or X522.3: Connection of L3 The absolute voltage value resulting from the phase voltages is displayed in r0072[1] and smoothed in r0025[1].			

r3662[0...n]	CO: VSM input line supply voltage u2 - u3 / VSM inp u2-u3			
VECTOR	Can be changed: -	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: -	
	P-Group: Closed-loop control	Units group: 5_3	Unit selection: p0505	
	Not for motor type: -	Scaling: p2001	Expert list: 1	
	Min	Max	Factory setting	
	- [V]	- [V]	- [V]	
Description:	Displays the voltage between phases L2 and L3.			
Dependency:	Refer to: p3660			
Note:	X521.2 or X522.2: Connection of L2 X521.3 or X522.3: Connection of L3			

r3664.0...1	BO: VSM temperature evaluation, status / VSM temp status				
A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 9886		
	P-Group: Terminals	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the status of the temperature evaluation of the Voltage Sensing Module (VSM). This displays whether the temperature actual value has exceeded the fault/alarm threshold.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Temperature alarm threshold exceeded	Yes	No	-
	01	Temperature fault threshold exceeded	Yes	No	-
Dependency:	Refer to: p3665, r3666, p3667, p3668				

r3664[0...n]	CO: VSM temperature evaluation, status / VSM temp status				
VECTOR	Can be changed: -		Calculated: -	Access level: 3	
	Data type: Unsigned16		Dynamic index: p0150	Func. diagram: 9886	
	P-Group: Terminals		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	-	
Description:	Displays the status of the temperature evaluation of the Voltage Sensing Module (VSM). This displays whether the temperature actual value has exceeded the fault/alarm threshold.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Temperature alarm threshold exceeded	Yes	No	-
	01	Temperature fault threshold exceeded	Yes	No	-
Dependency:	Refer to: p3665, r3666, p3667, p3668				
p3665[0...n]	VSM temperature evaluation, sensor type / VSM TempSensorType				
A_INF, S_INF	Can be changed: T		Calculated: -	Access level: 3	
	Data type: Integer16		Dynamic index: p0140	Func. diagram: 9886	
	P-Group: Closed-loop control		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	0		2	0	
Description:	Setting of the temperature sensor for the Voltage Sensing Module (VSM). The temperature sensor is connected to terminals X520.5 and X520.6 of the VSM.				
Value:	0: No sensor 1: PTC 2: KTY84				
p3665[0...n]	VSM temperature evaluation, sensor type / VSM TempSensorType				
VECTOR	Can be changed: T		Calculated: -	Access level: 3	
	Data type: Integer16		Dynamic index: p0150	Func. diagram: 9886	
	P-Group: Closed-loop control		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	0		2	0	
Description:	Setting of the temperature sensor for the Voltage Sensing Module (VSM). The temperature sensor is connected to terminals X520.5 and X520.6 of the VSM.				
Value:	0: No sensor 1: PTC 2: KTY84				
r3666	CO: VSM temperature KTY / VSM temp KTY				
A_INF, S_INF	Can be changed: -		Calculated: -	Access level: 3	
	Data type: FloatingPoint32		Dynamic index: -	Func. diagram: 9886	
	P-Group: Closed-loop control		Units group: 21_1	Unit selection: p0505	
	Not for motor type: -		Scaling: TEMP	Expert list: 1	
	Min		Max	Factory setting	
	- [°C]		- [°C]	- [°C]	
Description:	Displays the temperature actual value of a KTY84 temperature sensor connected to the Voltage Sensing Module (VSM). Prerequisite: A KTY84 sensor is connected and p3665 is set to 2.				
Dependency:	Refer to: p3665				

Note: For sensor type PTC (p3665 = 1), the following applies:
 - below the nominal response temperature, r3666 = -50 °C.
 - above the nominal response temperature, r3666 = 199.9 °C.

r3666[0...n]	CO: VSM temperature KTY / VSM temp KTY		
VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: 9886
	P-Group: Closed-loop control	Units group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min	Max	Factory setting
	- [°C]	- [°C]	- [°C]
Description:	Displays the temperature actual value of a KTY84 temperature sensor connected to the Voltage Sensing Module (VSM). Prerequisite: A KTY84 sensor is connected and p3665 is set to 2.		
Dependency:	Refer to: p3665		
Note:	For sensor type PTC (p3665 = 1), the following applies: - below the nominal response temperature, r3666 = -50 °C. - above the nominal response temperature, r3666 = 199.9 °C.		

p3667	VSM line filter overtemperature alarm threshold / VSMfilt_T A_thresh		
A_INF, S_INF	Can be changed: T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9886
	P-Group: -	Units group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min	Max	Factory setting
	0 [°C]	301 [°C]	150 [°C]
Description:	Sets the alarm threshold for the KTY temperature sensor of the Voltage Sensing Module (VSM) to monitor the line filter temperature. Prerequisite: A KTY84 sensor is connected and p3665 is set to 2.		
Dependency:	Refer to: p3665 Refer to: A34211		

p3667[0...n]	VSM line filter overtemperature alarm threshold / VSMfilt_T A_thresh		
VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: 9886
	P-Group: -	Units group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min	Max	Factory setting
	0.00 [°C]	301.00 [°C]	150.00 [°C]
Description:	Sets the alarm threshold for the KTY temperature sensor of the Voltage Sensing Module (VSM) to monitor the line filter temperature. Prerequisite: A KTY84 sensor is connected and p3665 is set to 2.		
Dependency:	Refer to: p3665 Refer to: A34211		

p3668	VSM line filter overtemperature shutdown threshold / VSM filt_T F_thres		
A_INF, S_INF	Can be changed: T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9886
	P-Group: -	Units group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min 0 [°C]	Max 301 [°C]	Factory setting 180 [°C]
Description:	Sets the shutdown threshold for the KTY temperature sensor of the VSM to monitor the line filter temperature.		
Dependency:	Refer to: p3667 Refer to: F34207		

p3668[0...n]	VSM line filter overtemperature shutdown threshold / VSM filt_T F_thres		
VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: 9886
	P-Group: -	Units group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min 0.00 [°C]	Max 301.00 [°C]	Factory setting 180.00 [°C]
Description:	Sets the shutdown threshold for the KTY temperature sensor of the VSM to monitor the line filter temperature.		
Dependency:	Refer to: p3667 Refer to: F34207		

p3669	VSM line filter overtemperature hysteresis / VSM filt_T hyst		
A_INF, S_INF	Can be changed: T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9886
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min 1.0 [K]	Max 50.0 [K]	Factory setting 3.0 [K]
Description:	Sets the hysteresis for the alarm threshold of the VSM to monitor the line filter temperature.		
Dependency:	Refer to: p3667		

p3669[0...n]	VSM line filter overtemperature hysteresis / VSM filt_T hyst		
VECTOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: 9886
	P-Group: -	Units group: 21_2	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min 1.00 [K]	Max 50.00 [K]	Factory setting 3.00 [K]
Description:	Sets the hysteresis for the alarm threshold of the VSM to monitor the line filter temperature.		
Dependency:	Refer to: p3667		

p3670	VSM 10 V input CT gain / VSM CT_gain		
A_INF, S_INF	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.000 [A]	Calculated: - Dynamic index: - Units group: - Scaling: p2002 Max 1000.000 [A]	Access level: 3 Func. diagram: 9880 Unit selection: - Expert list: 1 Factory setting 1.000 [A]
Description:	Sets CT gain of the CT connected at the 10 V input of the Voltage Sensing Module (VSM). The parameter specifies the current magnitude in [A] referred to the input voltage at the VSM in [V]. Example: CT with 1 V per 200 A. --> p3670 = 200		
Dependency:	Refer to: r3671, r3672		
Note:	The CT for phase 1 is connected at terminals X520.1 and X520.2 of the VSM. The CT for phase 2 is connected at terminals X520.3 and X520.4 of the VSM.		
p3670[0...n]	VSM 10 V input CT gain / VSM CT_gain		
VECTOR	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.000 [A]	Calculated: - Dynamic index: p0150 Units group: - Scaling: p2002 Max 1000.000 [A]	Access level: 3 Func. diagram: 9880 Unit selection: - Expert list: 1 Factory setting 1.000 [A]
Description:	Sets CT gain of the CT connected at the 10 V input of the Voltage Sensing Module (VSM). The parameter specifies the current magnitude in [A] referred to the input voltage at the VSM in [V]. Example: CT with 1 V per 200 A. --> p3670 = 200		
Dependency:	Refer to: r3671, r3672		
Note:	The CT for phase 1 is connected at terminals X520.1 and X520.2 of the VSM. The CT for phase 2 is connected at terminals X520.3 and X520.4 of the VSM.		
r3671	CO: VSM 10 V input CT 1 actual value / VSM CT 1 I_act		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [A]	Calculated: - Dynamic index: - Units group: - Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: 9880 Unit selection: - Expert list: 1 Factory setting - [A]
Description:	Displays the current actual value from current transducer (CT) 1 at the 10 V input of the Voltage Sensing Module (VSM).		
Dependency:	Refer to: p3670		
Note:	The CT for phase 1 is connected at terminals X520.1 and X520.2 of the VSM.		

r3671[0...n]	CO: VSM 10 V input CT 1 actual value / VSM CT 1 I_act		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [A]	Calculated: - Dynamic index: p0150 Units group: - Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: 9880 Unit selection: - Expert list: 1 Factory setting - [A]
Description:	Displays the current actual value from current transducer (CT) 1 at the 10 V input of the Voltage Sensing Module (VSM).		
Dependency:	Refer to: p3670		
Note:	The CT for phase 1 is connected at terminals X520.1 and X520.2 of the VSM.		
r3672	CO: VSM 10 V input CT 2 actual value / VSM CT 2 I_act		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [A]	Calculated: - Dynamic index: - Units group: - Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: 9880 Unit selection: - Expert list: 1 Factory setting - [A]
Description:	Displays the current actual value from current transducer (CT) 2 at the 10 V input of the Voltage Sensing Module (VSM).		
Dependency:	Refer to: p3670		
Note:	The CT for phase 2 is connected at terminals X520.3 and X520.4 of the VSM.		
r3672[0...n]	CO: VSM 10 V input CT 2 actual value / VSM CT 2 I_act		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [A]	Calculated: - Dynamic index: p0150 Units group: - Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: 9880 Unit selection: - Expert list: 1 Factory setting - [A]
Description:	Displays the current actual value from current transducer (CT) 2 at the 10 V input of the Voltage Sensing Module (VSM).		
Dependency:	Refer to: p3670		
Note:	The CT for phase 2 is connected at terminals X520.3 and X520.4 of the VSM.		
r3673	CO: VSM 10 V input 1 actual value / VSM inp 1 V_act		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 9880 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Displays the actual value of the voltage measured at the 10 V input 1 of the Voltage Sensing Modules (VSM).		
Dependency:	Refer to: p3670		
Note:	10 V input 1: Terminals X520.1 and X520.2		

r3673[0...n]	CO: VSM 10 V input 1 actual value / VSM inp 1 V_act		
VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: 9880
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the actual value of the voltage measured at the 10 V input 1 of the Voltage Sensing Modules (VSM).		
Dependency:	Refer to: p3670		
Note:	10 V input 1: Terminals X520.1 and X520.2		
r3674	CO: VSM 10 V input 2 actual value / VSM inp 2 V_act		
A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9880
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the actual value of the voltage measured at the 10 V input 2 of the Voltage Sensing Modules (VSM).		
Dependency:	Refer to: p3670		
Note:	10 V input 2: Terminals X520.3 and X520.4		
r3674[0...n]	CO: VSM 10 V input 2 actual value / VSM inp 2 V_act		
VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: 9880
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the actual value of the voltage measured at the 10 V input 2 of the Voltage Sensing Modules (VSM).		
Dependency:	Refer to: p3670		
Note:	10 V input 2: Terminals X520.3 and X520.4		
p3676	VSM line filter capacitance alarm threshold / VSMfilt C A_thresh		
A_INF, S_INF	Can be changed: T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [%]	100.00 [%]	0.00 [%]
Description:	Sets the alarm threshold for the change of the capacitance of the line filter. The monitoring of the filter capacitance is deactivated with p3676 = 0.00 %.		
Dependency:	Refer to: p3670 Refer to: A06250		
Note:	Prerequisites for monitoring the filter capacitance: The phase currents must be measured at two capacitors of the line filter. To do this, CTs should be connected at the 10 V inputs of the VSM.		

r3677[0...2]	CO: VSM line filter capacitance / VSM filt C		
A_INF, S_INF	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [μF]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [μF]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [μF]
Description:	Displays the capacitance of the line filter (for a star circuit configuration).		
Index:	[0] = Phase U [1] = Phase V [2] = Phase W		
Dependency:	Refer to: p3676		
Note:	Prerequisite: The monitoring of the filter capacitance is activated.		
p3678[0...1]	Filter monitoring threshold values / Filt.monit.thresh.		
A_INF (Line transf)	Can be changed: C2(1) Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 0.00 [%] [1] 0.00 [%]
Description:	Sets the threshold values for filter monitoring. Voltage threshold value monitoring refers to p0210. Current threshold value monitoring refers to the nominal filter current. Nominal filter current = $2 * PI * p0211 * 3 * p0221[0] * p0210 * \sqrt{2} / \sqrt{3}$		
Index:	[0] = Voltage threshold value monitoring [1] = Current threshold value monitoring		
Dependency:	Refer to: r3671, r3672, r7310, r7311 Refer to: F06855		
Note:	A setting value of 0.0 deactivates filter monitoring. Recommended setting for activation: Voltage threshold value: 3.0% Current threshold value: 500%		
p3679[0...1]	Filter monitoring minimum times / Filt.monit.min.t		
A_INF (Line transf)	Can be changed: C2(1) Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 200.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 20.00 [ms] [1] 0.50 [ms]
Description:	Sets the minimum times for filter monitoring. If the set threshold values are exceeded for at least the set time, a fault is set.		
Index:	[0] = Voltage threshold value monitoring [1] = Current threshold value monitoring		

p3680	BI: Braking Module internal inhibit / BM int inhib		
B_INF	Can be changed: T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to inhibit the internal Braking Module. 1 signal: The Braking Module is inhibited. 0 signal: The Braking Module is enabled.		
Dependency:	Refer to: A06904		
Caution:	When the Braking Module is inhibited, no energy can be dissipated in the braking resistor.		
			
p3681	BI: Activating Braking Module internal DC link fast discharge / BM intDCdischg act		
B_INF	Can be changed: T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to activate the DC link fast discharge for an internal braking module. The DC link fast discharge is started later with delay time (p3682) when the following conditions apply: - BI: p3681 = 1 signal. - an external line contactor is opened via r0863.1 "energize contactor". The DC link fast discharge is interrupted when the following conditions apply: - BI: p3681 = 0 signal. - ON command for the infeed.		
Recommend.:	The DC link fast discharge should be activated if there is an external line contactor and is correctly interconnected (r0863.1, p0860). If the DC link fast discharge is not activated together with an external line contactor, then faults could occur when pre-charging (e.g. F300027).		
Dependency:	Refer to: p3682 Refer to: F30027		
p3682	Braking Module internal DC link fast discharge delay time / BM int DC dischg t		
B_INF	Can be changed: C1(3), T Data type: Unsigned32 P-Group: Communications Not for motor type: - Min 500 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4294967295 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1000 [ms]
Description:	Sets the delay time for switching in the DC link fast discharge for an internal Braking Module.		
Dependency:	Refer to: p3681		

p3683	Braking Module internal activation threshold brake chopper / BM int chop level		
B_INF	Can be changed: C2(1) Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 110.00 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 780.00 [V]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 760.00 [V]
Description:	Sets the activation threshold for the brake chopper.		
Note:	The activation threshold is only effective if the "Reduced device supply voltage" function (p212.0 = 1) has been activated!		
r3685	BO: Digital Braking Module: Pre-alarm I2t shutdown / Dig BM A I2t shutd		
B_INF	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	The binector output uses a 1 signal to indicate that 80 % of the highest permissible I2t value has been reached in the Braking Module.		
Dependency:	Refer to: A06905		
r3686	BO: Digital Braking Module Fault / Dig BM Fault		
B_INF	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	The binector output uses a 1 signal to indicate an overcurrent fault or an I2t shutdown in the Braking Module.		
Dependency:	Refer to: F06906		
r3687	BO: Digital Braking Module pre-alarm overtemperature / Dig BM A overtemp		
B_INF	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	The binector output uses a 1 signal to indicate an overtemperature condition at the connected temperature sensor of the braking resistor.		
r3688	BO: Digital Braking Module fault overtemperature / Dig BM F overtemp		
B_INF	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	The binector output uses a 1 signal to indicate that the highest permissible overtemperature at the connected temperature sensor has been reached and has caused a trip (shutdown).		
Dependency:	Refer to: F06908		

r3689 BO: Digital Braking Module Vce fault / Dig BM Uce fault

B_INF	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: The binector output uses a 1 signal to indicate that there is a Vce fault in the Digital Braking Module.

Dependency: Refer to: F06909

p3700 APC configuration / APC config

SERVO (APC)	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 7012
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the configuration for APC (Advanced Positioning Control).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activating APC	Yes	No	-
	01	APC acceleration sensor	Yes	No	-
	08	Activating pulse de-coupling	Yes	No	-

Note: Re bit 00:

For a 1 signal, the acceleration filter output is added to the speed setpoint.

For a 0 signal, the value 0 is added. This must be used to evaluate the filter frequency characteristics.

Re bit 08:

For bit 0 = 1 and activated pulse de-coupling (bit 8 = 1), the speed of the direct measuring system (p3701) selected for APC is used as the actual value for the closed-loop speed control.

p3701 APC enc sel / APC enc sel

SERVO (APC)	Can be changed: C1(4)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Data sets	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	2	3	2

Description: Sets the number of the encoder used for for APC (Advanced Positioning Control).

Value:
2: Encoder 2
3: Encoder 3

Note: Encoder 1 is the motor encoder and cannot be used for APC as APC requires a load measuring system.

An encoder used for APC is, regarding its fault messages, treated just like a motor encoder - this means that its fault messages are assigned to the drive.

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p3708[0...n]	APC velocity actual value smoothing time encoder 2 / APC v_act t_sm 2		
SERVO (APC, Lin)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4711
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 50.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the smoothing time constant (PT1) for the velocity actual value of encoder 2 with APC (Advanced Positioning Control).		
Note:	The speed actual value should be smoothed for encoders with a low pulse number or for resolvers.		
p3708[0...n]	APC speed actual value smoothing time encoder 2 / APC n_act t_sm 2		
SERVO (APC)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4711
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 50.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the smoothing time constant (PT1) for the speed actual value of encoder 2 with APC (Advanced Positioning Control).		
Note:	The speed actual value should be smoothed for encoders with a low pulse number or for resolvers.		
p3709[0...n]	APC velocity actual value smoothing time encoder 3 / APC v_act t_sm 3		
SERVO (APC, Lin)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4711
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 50.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the smoothing time constant (PT1) for the velocity actual value of encoder 3 with APC (Advanced Positioning Control).		
Note:	The speed actual value should be smoothed for encoders with a low pulse number or for resolvers.		
p3709[0...n]	APC speed actual value smoothing time encoder 3 / APC n_act t_sm 3		
SERVO (APC)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 4711
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 50.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the smoothing time constant (PT1) for the speed actual value of encoder 3 with APC (Advanced Positioning Control).		
Note:	The speed actual value should be smoothed for encoders with a low pulse number or for resolvers.		

p3711[0...n]	APC filter 1.1 denominator natural frequency / APC Filt 1.1 fn_d		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the denominator natural frequency for filter 1.1 (PT2, general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3712[0...n]	APC filter 1.1 denominator damping / APC Filt 1.1 D_d		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.050	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the denominator damping for filter 1.1 (PT2, general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3713[0...n]	APC filter 1.1 numerator natural frequency / APC Filt 1 fn_n		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the numerator natural frequency for filter 1.1 (general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3714[0...n]	APC filter 1.1 numerator damping / APC Filt 1.1 D_n		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the numerator damping for filter 1.1 (general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3721[0...n]	APC filter 2.1 denominator natural frequency / APC Filt 2.1 fn_d		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the denominator natural frequency for filter 2.1 (PT2, general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		

p3722[0...n]	APC filter 2.1 denominator damping / APC Filt 2.1 D_d		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.050	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the denominator damping for filter 2.1 (PT2, general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3723[0...n]	APC filter 2.1 numerator natural frequency / APC Filt 2.1 fn_n		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the numerator natural frequency for filter 2.1 (general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3724[0...n]	APC filter 2.1 numerator damping / APC Filt 2.1 D_n		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the numerator damping for filter 2.1 (general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3726[0...n]	APC filter 2.2 denominator natural frequency / APC Filt 2.2 fn_d		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the denominator natural frequency for filter 2.2 (PT2, general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3727[0...n]	APC filter 2.2 denominator damping / APC Filt 2.2 D_d		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.050	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the denominator damping for filter 2.2 (PT2, general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		

p3728[0...n]	APC filter 2.2 numerator natural frequency / APC Filt 2.2 fn_n		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the numerator natural frequency for filter 2.2 (general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3729[0...n]	APC filter 2.2 numerator damping / APC Filt 2.2 D_n		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 7029 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the numerator damping for filter 2.2 (general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3731[0...n]	APC filter 3.1 denominator natural frequency / APC Filt 3.1 fn_d		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the denominator natural frequency for filter 3.1 (PT2, general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3732[0...n]	APC filter 3.1 denominator damping / APC Filt 3.1 D_d		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.050	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the denominator damping for filter 3.1 (PT2, general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3733[0...n]	APC filter 3.1 numerator natural frequency / APC Filt 3.1 fn_n		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the numerator natural frequency for filter 3.1 (general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		

p3734[0...n]	APC filter 3.1 numerator damping / APC Filt 3.1 D_n		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the numerator damping for filter 3.1 (general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3736[0...n]	APC filter 3.2 denominator natural frequency / APC Filt 3.2 fn_d		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the denominator natural frequency for filter 3.2 (PT2, general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3737[0...n]	APC filter 3.2 denominator damping / APC Filt 3.2 D_d		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.050	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the denominator damping for filter 3.2 (PT2, general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3738[0...n]	APC filter 3.2 numerator natural frequency / APC Filt 3.2 fn_n		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.5 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 16000.0 [Hz]	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 2000.0 [Hz]
Description:	Sets the numerator natural frequency for filter 3.2 (general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		
p3739[0...n]	APC filter 3.2 numerator damping / APC Filt 3.2 D_n		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.000	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10.000	Access level: 3 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.700
Description:	Sets the numerator damping for filter 3.2 (general 2nd Order filter) for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3704, p3705		

p3750[0...n]	CI: APC acceleration sensor input / APC accel input		
SERVO (APC)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: p2007 Max -	Access level: 2 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the actual value of the acceleration sensor for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3700		
p3751[0...n]	APC acceleration sensor high pass time constant / APC accel DT1 T		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min 0.00 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 10000.00 [ms]	Access level: 2 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 100.00 [ms]
Description:	Sets the time constant of the high pass filter for the acceleration sensor for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3700, p3750		
p3760[0...n]	APC load velocity controller 1 P gain / APC v_load ctr1 Kp		
SERVO (APC, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -100.000	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.000	Access level: 2 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the proportional gain of the load velocity controller 1 for APC (Advanced Positioning Control). The gain acts on the difference between the velocity setpoint and load velocity in the branch for filter 2.1 and 2.2.		
p3760[0...n]	APC load speed controller 1 P gain / APC n_load ctr1 Kp		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -100.000	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 100.000	Access level: 2 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the proportional gain of the load speed controller 1 for APC (Advanced Positioning Control). The gain acts on the difference between the speed setpoint and load speed in the branch for filter 2.1 and 2.2.		
p3761[0...n]	APC load velocity controller 1 rate time / APC v_load ctr1 Tv		
SERVO (APC, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: REL Min -500.00 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 500.00 [ms]	Access level: 2 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the rate time of the load velocity controller 1 for APC (Advanced Positioning Control). The rate time acts on the load acceleration in the branch for filter 2.1 and 2.2.		

p3761[0...n]	APC load speed controller 1 rate time / APC n_load ctr1 Tv		
SERVO (APC)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7012
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -500.00 [ms]	Max 500.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the rate time of the load speed controller 1 for APC (Advanced Positioning Control). The rate time acts on the load acceleration in the branch for filter 2.1 and 2.2.		
p3765[0...n]	APC load velocity controller 2 P gain / APC v_load ctr2 Kp		
SERVO (APC, Lin)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7012
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -100.000	Max 100.000	Factory setting 0.000
Description:	Sets the proportional gain of the load velocity controller 2 for APC (Advanced Positioning Control). The gain acts on the difference between the velocity setpoint and load velocity in the branch for filter 3.1 and 3.2.		
p3765[0...n]	APC load speed controller 2 P gain / APC n_load ctr2 Kp		
SERVO (APC)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7012
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -100.000	Max 100.000	Factory setting 0.000
Description:	Sets the proportional gain of the load speed controller 2 for APC (Advanced Positioning Control). The gain acts on the difference between the speed setpoint and load speed in the branch for filter 3.1 and 3.2.		
p3766[0...n]	APC load velocity controller 2 rate time / APC v_load ctr2 Tv		
SERVO (APC, Lin)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7012
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -500.00 [ms]	Max 500.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the rate time of the load velocity controller 2 for APC (Advanced Positioning Control). The rate time acts on the load acceleration in the branch for filter 3.1 and 3.2.		
p3766[0...n]	APC load speed controller 2 rate time / APC n_load ctr2 Tv		
SERVO (APC)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7012
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -500.00 [ms]	Max 500.00 [ms]	Factory setting 0.00 [ms]
Description:	Sets the rate time of the load speed controller 2 for APC (Advanced Positioning Control). The rate time acts on the load acceleration in the branch for filter 3.1 and 3.2.		

r3770 CO: APC load velocity / APC v_load

SERVO (APC, Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7012
	P-Group: Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min - [m/min]	Max - [m/min]	Factory setting - [m/min]
Description:	Displays the load velocity for APC (Advanced Positioning Control).		
Dependency:	Refer to: r3771		

r3770 CO: APC load speed / APC n_load

SERVO (APC)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7012
	P-Group: Setpoints	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min - [rpm]	Max - [rpm]	Factory setting - [rpm]
Description:	Displays the load speed for APC (Advanced Positioning Control).		
Dependency:	Refer to: r3771		

r3771 CO: APC load velocity smoothed / APC v_load smth

SERVO (APC, Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7012
	P-Group: Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min - [m/min]	Max - [m/min]	Factory setting - [m/min]
Description:	Displays the smoothed load velocity for APC (Advanced Positioning Control).		
Dependency:	Refer to: p1441, r3770		

r3771 CO: APC load speed smoothed / APC n_load smth

SERVO (APC)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7012
	P-Group: Setpoints	Units group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min - [rpm]	Max - [rpm]	Factory setting - [rpm]
Description:	Displays the smoothed load speed for APC (Advanced Positioning Control).		
Dependency:	Refer to: p1441, r3770		

r3772[0...1] APC filter branch 2 display values / APC branch 2 val

SERVO (APC, Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7012
	P-Group: Setpoints	Units group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min - [m/min]	Max - [m/min]	Factory setting - [m/min]
Description:	Displays the velocities in filter branch 2.		
Index:	[0] = Filter 2.1 input value [1] = Filter 2.2 output value		

r3772[0...1]	APC filter branch 2 display values / APC branch 2 val		
SERVO (APC)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 7012 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speeds in filter branch 2.		
Index:	[0] = Filter 2.1 input value [1] = Filter 2.2 output value		

r3773[0...1]	APC filter branch 3 display values / APC branch 3 val		
SERVO (APC, Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 7012 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocities in filter branch 3.		
Index:	[0] = Filter 3.1 input value [1] = Filter 3.2 output value		

r3773[0...1]	APC filter branch 3 display values / APC branch 3 val		
SERVO (APC)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 7012 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speeds in filter branch 3.		
Index:	[0] = Filter 3.1 input value [1] = Filter 3.2 output value		

r3777[0...1]	CO: APC filter branch 1 display values / APC branch 1 val		
SERVO (APC, Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: 7012 Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the velocities in filter branch 1.		
Index:	[0] = Filter 1.1 input value [1] = Filter 1.1 output value		

r3777[0...1]	CO: APC filter branch 1 display values / APC branch 1 val		
SERVO (APC)	Can be changed: - Data type: FloatingPoint32 P-Group: Setpoints Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: 7012 Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the speeds in filter branch 1.		
Index:	[0] = Filter 1.1 input value [1] = Filter 1.1 output value		
p3778[0...n]	APC velocity limit / APC v_limit		
SERVO (APC, Lin)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [m/min]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: 4_1 Scaling: - Max 1000.00 [m/min]	Access level: 2 Func. diagram: 7012 Unit selection: p0505 Expert list: 1 Factory setting 1000.00 [m/min]
Description:	Sets the velocity limit for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3779		
p3778[0...n]	APC speed limit / APC n_limit		
SERVO (APC)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [rpm]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: 3_1 Scaling: - Max 210000.00 [rpm]	Access level: 2 Func. diagram: 7012 Unit selection: p0505 Expert list: 1 Factory setting 210000.00 [rpm]
Description:	Sets the speed limit for APC (Advanced Positioning Control).		
Dependency:	Refer to: p3779		
p3779[0...n]	APC velocity limit monitoring time / APC v_limit t		
SERVO (APC, Lin)	Can be changed: U, T Data type: Unsigned32 P-Group: Closed-loop control Not for motor type: REL Min 0 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000000 [ms]	Access level: 2 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Sets the monitoring time to limit the output for APC (Advanced Positioning Control). This monitoring time is started after the selected limit value (p3778) has been exceeded. A corresponding fault is output if the limit value is not undershot before this time expires.		
Dependency:	Refer to: p3778 Refer to: F07425		

p3779[0...n]	APC speed limit monitoring time / APC n_limit t		
SERVO (APC)	Can be changed: U, T Data type: Unsigned32 P-Group: Closed-loop control Not for motor type: REL Min 0 [ms]	Calculated: CALC_MOD_CON Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1000000 [ms]	Access level: 2 Func. diagram: 7012 Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Sets the monitoring time to limit the output for APC (Advanced Positioning Control). This monitoring time is started after the selected limit value (p3778) has been exceeded. A corresponding fault is output if the limit value is not undershot before this time expires.		
Dependency:	Refer to: p3778 Refer to: F07425		
p3784[0...n]	BI: Sync-line-drive external increase voltage / Sync ext V incr		
VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Sets the signal source to increase the voltage for external line-drive synchronization. BI: p3784 = 1 signal: The voltage is increased.		
Note:	Only SINAMICS GM150.		
p3785[0...n]	BI: Sync-line-drive external decrease voltage / Sync ext V decr		
VECTOR	Can be changed: T Data type: Unsigned32 / Binary P-Group: Functions Not for motor type: - Min -	Calculated: - Dynamic index: CDS, p0170 Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Sets the signal source to decrease the voltage for external line-drive synchronization. BI: p3785 = 1 signal: The voltage is lowered.		
Note:	Only SINAMICS GM150.		
p3800[0...n]	Sync-line-drive activation / Sync act		
VECTOR	Can be changed: T Data type: Integer16 P-Group: Functions Not for motor type: - Min 0	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: 7020 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the activation for the line-drive synchronization.		
Value:	0: Sync-line-drive deactivated 1: Sync-line-drive activated VSM-INT		
Dependency:	Refer to: p3801, p3802		

Note: When the ground fault monitoring initiates a fault for overlapping synchronizing the threshold value p0287[1] for the Motor Module and the associated infeed must be appropriately increased (e.g. p0287[1] = 100 %).
 For p3800 = 1, the following applies:
 The INTERNAL voltage actual values are used for synchronization. The effects that a (sine-wave) filter - that is connected between the Motor Module and motor - has on the voltage actual values are taken into account (theoretically) by appropriately selecting p0230.
 VSM: Voltage Sensing Module

p3801[0...n]	Sync-line-drive, drive object number / Sync DO_No				
VECTOR	Can be changed: T		Calculated: -	Access level: 2	
	Data type: Unsigned16		Dynamic index: DDS, p0180	Func. diagram: 7020	
	P-Group: Functions		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	1		62	1	
Description:	Sets the drive object number of the Voltage Sensing Module (VSM) used for the line-drive synchronization.				
Dependency:	Refer to: p3800, p3802				
Notice:	The current controller sampling time p0115[0] of the drive object with the VSM used for synchronization must be identical to the current controller sampling time of the drive of the drive used to perform line synchronization.				
Note:	VSM: Voltage Sensing Module The setting p3801 = 1 is always possible (no VSM selected).				

p3802[0...n]	BI: Sync-line-drive enable / Sync enable				
VECTOR	Can be changed: T		Calculated: -	Access level: 2	
	Data type: Unsigned32 / Binary		Dynamic index: CDS, p0170	Func. diagram: 7020	
	P-Group: Functions		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0	
Description:	Sets the signal source to switch in/switch out for the line-drive synchronization. BI: p3802 = 1 signal: The line-drive synchronization is switched in.				
Dependency:	Refer to: p3800, p3801				

r3803.0	CO/BO: Sync-line-drive control word / Sync STW				
VECTOR	Can be changed: -		Calculated: -	Access level: 2	
	Data type: Unsigned32		Dynamic index: -	Func. diagram: -	
	P-Group: Functions		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	-	
Description:	Displays the control word for the line-drive synchronization.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Sync-line-drive selected	Yes	No	-
Note:	Re bit 00: For a 1 signal, p3800 > 0 is set.				

r3804	CO: Sync-line-drive target frequency / Sync f_target		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: p2000 Max - [Hz]	Access level: 2 Func. diagram: 3030, 7020 Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the target frequency for the line-drive synchronization.		
Dependency:	Refer to: A07941		
r3805	CO: Sync-line-drive frequency difference / Sync f_diff		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: p2000 Max - [Hz]	Access level: 2 Func. diagram: 7020 Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the frequency difference between the measured target frequency and output frequency of the gating unit of the closed-loop control for line-drive synchronization.		
p3806[0...n]	Sync-line-drive frequency difference threshold value / Sync f_diff thresh		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.00 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1.00 [Hz]	Access level: 2 Func. diagram: 7020 Unit selection: - Expert list: 1 Factory setting 0.10 [Hz]
Description:	Sets the threshold value of the frequency difference to activate the closed-loop phase control for line-drive synchronization. The closed-loop phase control is activated (r3819.6 = 1), if the frequency difference is less than the threshold value.		
r3808	CO: Sync-line-drive phase difference / Sync phase diff		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min - [°]	Calculated: - Dynamic index: - Units group: - Scaling: p2005 Max - [°]	Access level: 2 Func. diagram: 7020 Unit selection: - Expert list: 1 Factory setting - [°]
Description:	Displays the phase difference between the measured target phase and phase of the gating unit of the closed-loop control for line-drive synchronization.		
p3809[0...n]	Sync-line-drive phase setpoint / Sync phase setp		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min -180.00 [°]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 179.90 [°]	Access level: 2 Func. diagram: 7020 Unit selection: - Expert list: 1 Factory setting 0.00 [°]
Description:	Sets the phase setpoint for the line-drive synchronization.		

p3811[0...n]	Sync-line-drive frequency limiting / Sync f_lim		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.00 [Hz]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 1.00 [Hz]	Access level: 2 Func. diagram: 7020 Unit selection: - Expert list: 1 Factory setting 0.20 [Hz]
Description:	Sets the frequency limiting of the phase controller output for the line-drive synchronization.		
r3812	CO: Sync-line-drive correction frequency / Sync f_corr		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: p2000 Max - [Hz]	Access level: 2 Func. diagram: 3080, 7020 Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the correction frequency for the line-drive synchronization.		
p3813[0...n]	Sync-line-drive phase synchronism threshold value / Sync Ph_sync thrsh		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 1.00 [°]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 20.00 [°]	Access level: 2 Func. diagram: 7020 Unit selection: - Expert list: 1 Factory setting 2.00 [°]
Description:	Sets the threshold value of the phase synchronism for the line-drive synchronization. A prerequisite for synchronism is achieved if the phase difference is lower than the threshold value.		
Note:	Synchronism is reached (r3819.2 = 1), if the AND logic operation of the results from the phase measurement (p3813) and voltage measurement (p3815) is fulfilled.		
r3814	CO: Sync-line-drive voltage difference / Sync V_diff		
VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [Vrms]	Access level: 2 Func. diagram: 7020 Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the voltage difference between the measured target voltage and output voltage of the gating unit of the closed-loop control for line-drive synchronization.		
p3815[0...n]	Sync-line-drive voltage difference threshold value / Sync V_diff thresh		
VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Functions Not for motor type: - Min 0.00 [%]	Calculated: CALC_MOD_LIM_REF Dynamic index: DDS, p0180 Units group: - Scaling: - Max 20.00 [%]	Access level: 2 Func. diagram: 7020 Unit selection: - Expert list: 1 Factory setting 10.00 [%]
Description:	Sets the threshold value of the voltage difference for the line-drive synchronization. A prerequisite for synchronism is reached if the voltage difference is less than the threshold value.		

Note: Synchronism is reached ($r3819.2 = 1$), if the AND logic operation of the results from the phase measurement (p3813) and voltage measurement (p3815) is fulfilled.
For voltage manipulated quantity margin (reserve) of the drive converter, the amplitude difference (r3814) between the setpoint and actual value is controlled (corrected) to zero.

r3819.0...7 CO/BO: Sync-line-drive status word / Sync ZSW					
VECTOR	Can be changed: -		Calculated: -	Access level: 2	
	Data type: Unsigned32		Dynamic index: -	Func. diagram: 7020	
	P-Group: Functions		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the status word for the line-drive synchronization.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Sync-line-drive enabled	Yes	No	-
	02	Sync-line-drive synchronism reached	Yes	No	-
	03	Sync-line-drive synchronizing error	Yes	No	-
	05	Sync-line-drive frequency measurement active	Yes	No	-
	06	Sync-line-drive phase control active	Yes	No	-
	07	Sync-line-drive without drive	Yes	No	-

p3820[0...n] Friction characteristic, value n0 / Friction n0			
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 15.00 [rpm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the n coordinate of the 1st value pair of the friction characteristic.		
Dependency:	Refer to: p3830, p3845		

p3820[0...n] Friction characteristic, value v0 / Friction v0			
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 21000.00 [m/min]	Factory setting 1.50 [m/min]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the v coordinate of the 1st value pair of the friction characteristic.		
Dependency:	Refer to: p3830, p3845		

p3821[0...n]	Friction characteristic, value n1 / Friction n1		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 30.00 [rpm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the n coordinate of the 2nd value pair of the friction characteristic.		
Dependency:	Refer to: p3831, p3845		

p3821[0...n]	Friction characteristic, value v1 / Friction v1		
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 21000.00 [m/min]	Factory setting 3.00 [m/min]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the v coordinate of the 2nd value pair of the friction characteristic.		
Dependency:	Refer to: p3831, p3845		

p3822[0...n]	Friction characteristic, value n2 / Friction n2		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 60.00 [rpm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the n coordinate of the 3rd value pair of the friction characteristic.		
Dependency:	Refer to: p3832, p3845		

p3822[0...n]	Friction characteristic, value v2 / Friction v2		
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 21000.00 [m/min]	Factory setting 6.00 [m/min]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the v coordinate of the 3rd value pair of the friction characteristic.		
Dependency:	Refer to: p3832, p3845		

p3823[0...n]	Friction characteristic, value n3 / Friction n3		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 120.00 [rpm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the n coordinate of the 4th value pair of the friction characteristic.		
Dependency:	Refer to: p3833, p3845		
p3823[0...n]	Friction characteristic, value v3 / Friction v3		
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 21000.00 [m/min]	Factory setting 12.00 [m/min]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the v coordinate of the 4th value pair of the friction characteristic.		
Dependency:	Refer to: p3833, p3845		
p3824[0...n]	Friction characteristic, value n4 / Friction n4		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 150.00 [rpm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the n coordinate of the 5th value pair of the friction characteristic.		
Dependency:	Refer to: p3834, p3845		
p3824[0...n]	Friction characteristic, value v4 / Friction v4		
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 21000.00 [m/min]	Factory setting 15.00 [m/min]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the v coordinate of the 5th value pair of the friction characteristic.		
Dependency:	Refer to: p3834, p3845		

p3825[0...n]	Friction characteristic, value n5 / Friction n5		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 300.00 [rpm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the n coordinate of the 6th value pair of the friction characteristic.		
Dependency:	Refer to: p3835, p3845		

p3825[0...n]	Friction characteristic, value v5 / Friction v5		
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 21000.00 [m/min]	Factory setting 30.00 [m/min]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the v coordinate of the 6th value pair of the friction characteristic.		
Dependency:	Refer to: p3835, p3845		

p3826[0...n]	Friction characteristic, value n6 / Friction n6		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 600.00 [rpm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the n coordinate of the 7th value pair of the friction characteristic.		
Dependency:	Refer to: p3836, p3845		

p3826[0...n]	Friction characteristic, value v6 / Friction v6		
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 21000.00 [m/min]	Factory setting 60.00 [m/min]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the v coordinate of the 7th value pair of the friction characteristic.		
Dependency:	Refer to: p3836, p3845		

p3827[0...n]	Friction characteristic, value n7 / Friction n7		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 1200.00 [rpm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the n coordinate of the 8th value pair of the friction characteristic.		
Dependency:	Refer to: p3837, p3845		
p3827[0...n]	Friction characteristic, value v7 / Friction v7		
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 21000.00 [m/min]	Factory setting 120.00 [m/min]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the v coordinate of the 8th value pair of the friction characteristic.		
Dependency:	Refer to: p3837, p3845		
p3828[0...n]	Friction characteristic, value n8 / Friction n8		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 1500.00 [rpm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the n coordinate of the 9th value pair of the friction characteristic.		
Dependency:	Refer to: p3838, p3845		
p3828[0...n]	Friction characteristic, value v8 / Friction v8		
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 21000.00 [m/min]	Factory setting 150.00 [m/min]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the v coordinate of the 9th value pair of the friction characteristic.		
Dependency:	Refer to: p3838, p3845		

p3829[0...n]	Friction characteristic, value n9 / Friction n9		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 210000.00 [rpm]	Factory setting 3000.00 [rpm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the n coordinate of the 10th value pair of the friction characteristic.		
Dependency:	Refer to: p3839, p3845		
p3829[0...n]	Friction characteristic, value v9 / Friction v9		
SERVO (Lin)	Can be changed: T	Calculated: CALC_MOD_LIM_REF	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min 0.00 [m/min]	Max 21000.00 [m/min]	Factory setting 300.00 [m/min]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the v coordinate of the 10th value pair of the friction characteristic.		
Dependency:	Refer to: p3839, p3845		
p3830[0...n]	Friction characteristic, value M0 / Friction M0		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [Nm]	Max 1000000.00 [Nm]	Factory setting 0.00 [Nm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the M coordinate of the 1st value pair of the friction characteristic.		
Dependency:	Refer to: p3820, p3845		
p3830[0...n]	Friction characteristic, value F0 / Friction F0		
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [N]	Max 1000000.00 [N]	Factory setting 0.00 [N]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the F coordinate of the 1st value pair of the friction characteristic.		
Dependency:	Refer to: p3820, p3845		

p3831[0...n]	Friction characteristic, value M1 / Friction M1		
SERVO, VECTOR (n/M)	Can be changed: T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min -1000000.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 1000000.00 [Nm]	Access level: 2 Func. diagram: 7010 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Nm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the M coordinate of the 2nd value pair of the friction characteristic.		
Dependency:	Refer to: p3821, p3845		
p3831[0...n]	Friction characteristic, value F1 / Friction F1		
SERVO (Lin)	Can be changed: T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min -1000000.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 1000000.00 [N]	Access level: 2 Func. diagram: 7010 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [N]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the F coordinate of the 2nd value pair of the friction characteristic.		
Dependency:	Refer to: p3821, p3845		
p3832[0...n]	Friction characteristic, value M2 / Friction M2		
SERVO, VECTOR (n/M)	Can be changed: T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min -1000000.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 1000000.00 [Nm]	Access level: 2 Func. diagram: 7010 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Nm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the M coordinate of the 3rd value pair of the friction characteristic.		
Dependency:	Refer to: p3822, p3845		
p3832[0...n]	Friction characteristic, value F2 / Friction F2		
SERVO (Lin)	Can be changed: T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min -1000000.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 1000000.00 [N]	Access level: 2 Func. diagram: 7010 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [N]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the F coordinate of the 3rd value pair of the friction characteristic.		
Dependency:	Refer to: p3822, p3845		

p3833[0...n]	Friction characteristic, value M3 / Friction M3		
SERVO, VECTOR (n/M)	Can be changed: T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min -1000000.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 1000000.00 [Nm]	Access level: 2 Func. diagram: 7010 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Nm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the M coordinate of the 4th value pair of the friction characteristic.		
Dependency:	Refer to: p3823, p3845		
p3833[0...n]	Friction characteristic, value F3 / Friction F3		
SERVO (Lin)	Can be changed: T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min -1000000.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 1000000.00 [N]	Access level: 2 Func. diagram: 7010 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [N]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the F coordinate of the 4th value pair of the friction characteristic.		
Dependency:	Refer to: p3823, p3845		
p3834[0...n]	Friction characteristic, value M4 / Friction M4		
SERVO, VECTOR (n/M)	Can be changed: T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min -1000000.00 [Nm]	Calculated: - Dynamic index: DDS, p0180 Units group: 7_1 Scaling: - Max 1000000.00 [Nm]	Access level: 2 Func. diagram: 7010 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [Nm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the M coordinate of the 5th value pair of the friction characteristic.		
Dependency:	Refer to: p3824, p3845		
p3834[0...n]	Friction characteristic, value F4 / Friction F4		
SERVO (Lin)	Can be changed: T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min -1000000.00 [N]	Calculated: - Dynamic index: DDS, p0180 Units group: 8_1 Scaling: - Max 1000000.00 [N]	Access level: 2 Func. diagram: 7010 Unit selection: p0505 Expert list: 1 Factory setting 0.00 [N]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the F coordinate of the 5th value pair of the friction characteristic.		
Dependency:	Refer to: p3824, p3845		

p3835[0...n]	Friction characteristic, value M5 / Friction M5		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [Nm]	Max 1000000.00 [Nm]	Factory setting 0.00 [Nm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the M coordinate of the 6th value pair of the friction characteristic.		
Dependency:	Refer to: p3825, p3845		

p3835[0...n]	Friction characteristic, value F5 / Friction F5		
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [N]	Max 1000000.00 [N]	Factory setting 0.00 [N]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the F coordinate of the 6th value pair of the friction characteristic.		
Dependency:	Refer to: p3825, p3845		

p3836[0...n]	Friction characteristic, value M6 / Friction M6		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [Nm]	Max 1000000.00 [Nm]	Factory setting 0.00 [Nm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the M coordinate of the 7th value pair of the friction characteristic.		
Dependency:	Refer to: p3826, p3845		

p3836[0...n]	Friction characteristic, value F6 / Friction F6		
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [N]	Max 1000000.00 [N]	Factory setting 0.00 [N]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the F coordinate of the 7th value pair of the friction characteristic.		
Dependency:	Refer to: p3826, p3845		

p3837[0...n]	Friction characteristic, value M7 / Friction M7		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [Nm]	Max 1000000.00 [Nm]	Factory setting 0.00 [Nm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the M coordinate of the 8th value pair of the friction characteristic.		
Dependency:	Refer to: p3827, p3845		

p3837[0...n]	Friction characteristic, value F7 / Friction F7		
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [N]	Max 1000000.00 [N]	Factory setting 0.00 [N]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the F coordinate of the 8th value pair of the friction characteristic.		
Dependency:	Refer to: p3827, p3845		

p3838[0...n]	Friction characteristic, value M8 / Friction M8		
SERVO, VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [Nm]	Max 1000000.00 [Nm]	Factory setting 0.00 [Nm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the M coordinate of the 9th value pair of the friction characteristic.		
Dependency:	Refer to: p3828, p3845		

p3838[0...n]	Friction characteristic, value F8 / Friction F8		
SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [N]	Max 1000000.00 [N]	Factory setting 0.00 [N]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the F coordinate of the 9th value pair of the friction characteristic.		
Dependency:	Refer to: p3828, p3845		

p3839[0...n] Friction characteristic, value M9 / Friction M9

SERVO, VECTOR (n/M)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [Nm]	Max 1000000.00 [Nm]	Factory setting 0.00 [Nm]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the M coordinate of the 10th value pair of the friction characteristic.		
Dependency:	Refer to: p3829, p3845		

p3839[0...n] Friction characteristic, value F9 / Friction F9

SERVO (Lin)	Can be changed: T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: 7010
	P-Group: Functions	Units group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min -1000000.00 [N]	Max 1000000.00 [N]	Factory setting 0.00 [N]
Description:	The friction characteristic is defined by 10 value pairs. This parameter specifies the F coordinate of the 10th value pair of the friction characteristic.		
Dependency:	Refer to: p3829, p3845		

r3840.0...8 CO/BO: Friction characteristic, status word / Friction ZSW

SERVO, VECTOR (n/M)	Can be changed: -		Calculated: -	Access level: 2
	Data type: Unsigned32		Dynamic index: -	Func. diagram: 7010
	P-Group: Functions		Units group: -	Unit selection: -
	Not for motor type: REL		Scaling: -	Expert list: 1
	Min -		Max -	Factory setting -
Description:	Displays the state of the friction characteristic.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Friction characteristic OK	Yes	No
	01	Friction characteristic record activated	Yes	No
	02	Friction characteristic record completed	Yes	No
	03	Friction characteristic record aborted	Yes	No
	08	Friction characteristic positive direction	Yes	No

r3841 CO: Friction characteristic output / Frict outp

SERVO, VECTOR (n/M)	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7010
	P-Group: Functions	Units group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min - [Nm]	Max - [Nm]	Factory setting - [Nm]
Description:	Displays the torque of the friction characteristic dependent on the speed.		
Dependency:	Refer to: p1569, p3842		

r3841 CO: Friction characteristic output / Frict outp

SERVO (Lin)

Can be changed: -**Data type:** FloatingPoint32**P-Group:** Functions**Not for motor type:** REL**Min**

- [N]

Calculated: -**Dynamic index:** -**Units group:** 8_1**Scaling:** p2003**Max**

- [N]

Access level: 2**Func. diagram:** 7010**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [N]

Description: Displays the force of the friction characteristic dependent on the velocity.**Dependency:** Refer to: p1569, p3842**p3842 Friction characteristic activation / Frict act**SERVO, VECTOR
(n/M)**Can be changed:** T**Data type:** Integer16**P-Group:** Functions**Not for motor type:** REL**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

1

Access level: 2**Func. diagram:** 7010**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description: Setting to activate and deactivate the friction characteristic.**Value:**
0: Friction characteristic deactivated
1: Friction characteristic activated**Dependency:** Refer to: p1569, r3841, p3845**p3845 Friction characteristic record activation / Frict rec act**

SERVO, VECTOR

Can be changed: T**Data type:** Integer16**P-Group:** Functions**Not for motor type:** REL**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

3

Access level: 2**Func. diagram:** 7010**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description: Setting for the friction characteristic record.

After the next power-on command, the friction characteristic is automatically recorded.

Value:
0: Friction characteristic record deactivated
1: Friction char record activated for all directions
2: Friction char record activated for positive direction
3: Friction char record activated for negative direction**Dependency:** When selecting the friction characteristic measurement, the drive data set changeover is suppressed.

For linear drives (refer to r0108 bit 12) it is not permissible to carry out the friction characteristic measurement for mechanical systems that limit travel.

Danger:

For drives with a mechanical system that limit the distance moved, it must be ensured that during recording, the friction characteristic is not reached. If this is not the case, then it is not permissible that the measurement is carried out.

Notice:

To permanently accept the determined settings they must be saved in a non-volatile fashion (p0971, p0977).

Note:

When the friction characteristic record is active, it is not possible to save the parameters (p0971, p0977).

When the friction characteristic record is active (p3845 > 0), it is not possible to change p3820 ... p3829, p3830 ... p3839 and p3842.

When recording the friction characteristic, in addition to the friction, the motor losses are also determined (e.g. iron losses, eddy current losses and remagnetizing losses). A differentiation is not made between these individual loss components. We recommend that a motor temperature sensor is used because torque deviations can also be emulated/mapped on the characteristic due to the thermal influence.

p3846[0...n]	Friction characteristic record ramp-up/ramp-down time / Frict rec t_RFG		
SERVO, VECTOR (n/M)	Can be changed: T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 999999.000 [s]	Access level: 2 Func. diagram: 7010 Unit selection: - Expert list: 1 Factory setting 10.000 [s]
Description:	Sets the ramp-up/ramp-down time of the ramp-up/ramp-down function generator to automatically record the friction characteristic. The drive is accelerated from standstill (setpoint = 0) up to the maximum speed/velocity (p1082) in this time.		
Dependency:	Refer to: p3845		
p3847[0...n]	Friction characteristic record warm-up time / Frict rec t_warm		
SERVO, VECTOR (n/M)	Can be changed: T Data type: FloatingPoint32 P-Group: Functions Not for motor type: REL Min 0.000 [s]	Calculated: - Dynamic index: DDS, p0180 Units group: - Scaling: - Max 3600.000 [s]	Access level: 2 Func. diagram: 7010 Unit selection: - Expert list: 1 Factory setting 0.000 [s]
Description:	Sets the warm-up time. For an automatic trace (record) to start, the highest selected speed (p3829) is approached and this time is held. After this, the measurement is started with the highest speed.		
Dependency:	Refer to: p3829, p3845		
p3860	Number of Braking Modules connected in parallel / BM qty par_cct		
A_INF (Brk Mod ext), B_INF (Brk Mod ext), S_INF (Brk Mod ext)	Can be changed: C2(2) Data type: Unsigned8 P-Group: Converter Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8	Access level: 3 Func. diagram: 9951 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of Braking Modules connected in parallel in a DC link.		
Note:	The parameter can only be written to if the infeed is in the commissioning mode (p0010 = 2).		
r3861.0...7	BO: Braking Module inhibit/acknowledgement / BM inhib/ackn		
A_INF (Brk Mod ext), B_INF (Brk Mod ext), S_INF (Brk Mod ext)	Can be changed: - Data type: Unsigned32 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 9951 Unit selection: - Expert list: 1 Factory setting -
Description:	Signal to energize terminal X21.1 "inhibit/acknowledgement" on the Braking Module. This binector output is used as signal source to interconnect to a digital output. For "booksize" formats the digital output must be connected to terminal X21.1 and for "chassis" formats the digital output must be connected to terminal X21.3 of the particular Braking Module.		

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Inhibit/acknowledge Braking Module 1	High	Low	-
	01	Inhibit/acknowledge Braking Module 2	High	Low	-
	02	Inhibit/acknowledge Braking Module 3	High	Low	-
	03	Inhibit/acknowledge Braking Module 4	High	Low	-
	04	Inhibit/acknowledge Braking Module 5	High	Low	-
	05	Inhibit/acknowledge Braking Module 6	High	Low	-
	06	Inhibit/acknowledge Braking Module 7	High	Low	-
	07	Inhibit/acknowledge Braking Module 8	High	Low	-

Warning:

Check that binector outputs BO: r3861.n are connected correctly and that the appropriate digital outputs are wired correctly.

If the interconnection/wiring is incorrect, the software could execute a different (incorrect) function via binector outputs BO: r3861.n if the Braking Module develops a fault.

p3862	Braking Module DC link fast discharge delay time / BM DC-dischg t_del		
A_INF (Brk Mod ext), B_INF (Brk Mod ext), S_INF (Brk Mod ext)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9951
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	500 [ms]	4294967295 [ms]	1000 [ms]
Description:	Sets the delay time for switching in the DC link fast discharge.		
Dependency:	Refer to: p3863, r3864		
Note:	The DC link fast discharge is only possible for "booksize" formats. This function is not supported for "chassis" formats.		

p3863	BI: Activating Braking Module DC link fast discharge / BM DC-dischg act		
A_INF (Brk Mod ext), B_INF (Brk Mod ext), S_INF (Brk Mod ext)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9951
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to activate the DC link fast discharge. The DC link fast discharge is started later with delay time (p3862) when the following conditions apply: - BI: p3863 = 1 signal. - an external line contactor is opened via r0863.1 "energize contactor". The DC link fast discharge is interrupted when the following conditions apply: - BI: p3863 = 0 signal. - ON command for the infeed.		
Recommend.:	The DC link fast discharge should be activated if there is an external line contactor and is correctly interconnected (r0863.1, p0860). If the DC link fast discharge is not activated together with an external line contactor, then faults could occur when pre-charging (e.g. F30027).		
Dependency:	Refer to: r3864 Refer to: F30027		
Note:	The DC link fast discharge is only possible for "booksize" formats. This function is not supported for "chassis" formats.		

r3864.0...7 BO: Braking Module DC link fast discharge / BM DC link dischg

A_INF (Brk Mod ext), B_INF (Brk Mod ext), S_INF (Brk Mod ext)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9951
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Signal to control (energize) terminal X21.2 "DC link fast discharge" on the Braking Module.
This binector output is used as signal source to interconnect to a digital output. The digital output must be connected to terminal X21.2 of the particular Braking Module.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Fast discharge Braking Module 1	High	Low	-
	01	Fast discharge Braking Module 2	High	Low	-
	02	Fast discharge Braking Module 3	High	Low	-
	03	Fast discharge Braking Module 4	High	Low	-
	04	Fast discharge Braking Module 5	High	Low	-
	05	Fast discharge Braking Module 6	High	Low	-
	06	Fast discharge Braking Module 7	High	Low	-
	07	Fast discharge Braking Module 8	High	Low	-

Dependency: Refer to: p3863
Refer to: F30027

Warning:

It must be carefully ensured that the binector outputs BO: p3864.n are correctly interconnected and also that the appropriate digital outputs are correctly connected up.
If the interconnection/connection is incorrect, in the case of an active DC link fast discharge, the software could execute another function (incorrect function) via binector outputs BO: p3864.n or could also permanently control the DC link fast discharge even if the line contactor is closed.

Note: The DC link fast discharge is only possible for "booksize" formats. This function is not supported for "chassis" formats.

p3865[0...7] BI: Braking Module pre-warning I*t shutdown / BM I*t shutdown

A_INF (Brk Mod ext), B_INF (Brk Mod ext), S_INF (Brk Mod ext)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9951
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the signal "pre-alarm I*t shutdown (X21.3) of the Braking Module.
BI: p3865[0...7] = 0 signal --> no pre-alarm, I*t shutdown
BI: p3865[0...7] = 1 signal --> pre-alarm I*t shutdown (A06901)

Dependency: Refer to: A06901

Note: The pre-alarm I*t shutdown is only possible for "booksize" formats. This function is not supported for "chassis" formats.

p3866[0...7]		BI: Braking Module fault / BM fault		
A_INF (Brk Mod ext), B_INF (Brk Mod ext), S_INF (Brk Mod ext)	Can be changed: T	Calculated: -	Access level: 3	
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9951	
	P-Group: -	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description:	Sets the signal source for the "fault" signal of the Braking Module (X21.4 for "booksize" formats and X21.3 for "chassis" formats). BI: p3866[0...7] = 0 signal --> fault (A06900) BI: p3866[0...7] = 1 signal --> No fault For a 1 signal, an acknowledgement via BO: r3861 is automatically carried out at certain time intervals.			
Dependency:	Refer to: A06900			

p3870		Long stator configuration / Long stator config			
SERVO	Can be changed: U, T	Calculated: -	Access level: 3		
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -		
	P-Group: -	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Sets the configuration when operating a long stator motor.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Activate long stator help functions	Active	Inactive	-
	01	Suppress Gx_ZSW.14	Active	Inactive	-
Dependency:	Refer to: p3871, p3872, p3873, p3874, r3875, p3876, p3878, p3879				
Notice:	The following restrictions apply to this function: - it is not permissible to change over the drive data set. - the encoder/drive may not be parked using a PROFIBUS telegram. - a maximum of 4 drives may be connected to the Control Unit. - it is not permissible to commute with the zero mark (p0404).				
Note:	Re bit 00: All of the help functions for long stator motors can be enabled/disabled using this bit. Re bit 01: When the bit is set, bit 14 (parking encoder active) is set to 0 in the encoder status word GX_ZSW independent of whether the encoder is parked or not.				

p3871		BI: Set long stator signal source commutation angle (p3872) / Set S_src com_ang		
SERVO	Can be changed: T	Calculated: -	Access level: 3	
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -	
	P-Group: -	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description:	Sets the signal source to set the commutation angle available via connector input p3872.			
Dependency:	Refer to: p3870, p3872, p3873, p3874, r3875, p3876, p3878, p3879			
Danger:	Setting an incorrect commutation angle can result in instability in the closed-loop control and in turn injure personnel or cause damage to the machine!			
				
Note:	Setting takes place for a 0/1 signal edge.			

p3872			
SERVO	CI: Long stator signal source commutation angle / S_src com_angle		
	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2005	Expert list: 1
	Min	Max	Factory setting
	-	-	3878[0]
Description: Sets the signal source for the commutation angle. This angle is set for a 0/1 signal edge via BI: p3871.			
Dependency: Refer to: p3870, p3871, p3873, r3875, p3876, p3878, p3879			
Danger: Setting an incorrect commutation angle can result in instability in the closed-loop control and in turn injure personnel or cause damage to the machine!			
			
p3873			
SERVO	BI: Long stator sig. source changeover to cl.-loop ctrl w/ enc. / S_src ctrl w/ enc		
	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: Functions	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description: Sets the signal source to change over to closed-loop control with encoder.			
Dependency: Refer to: p3870, p3871, p3872, p3874, r3875, p3876, p3878, p3879			
Danger: Setting an incorrect commutation angle can result in instability in the closed-loop control and in turn injure personnel or cause damage to the machine!			
			
Note: BI: p3873 = 1 signal --> closed-loop control with encoder BI: p3873 = 0 signal --> encoderless closed-loop control For a 0/1 edge, the commutation angle is set from CI: p3874.			
p3874			
	CI: Long stator signal source commutation angle oper. with encoder / S_src com_ang enc		
SERVO	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2005	Expert list: 1
	Min	Max	Factory setting
	-	-	3879[0]
Description: Sets the signal source for the commutation angle for operation with encoder.			
Dependency: Refer to: p3870, p3871, p3872, p3873, r3875, p3876, p3878, p3879			
Note: This angle is set for a 0/1 signal edge via BI: p3873.			

r3875.0...1 CO/BO: Long stator status word / Long stator ZSW

SERVO	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word for long stator motors.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Sensor Module is unparked	Yes	No	-
	01	Closed-loop speed control with encoder requested	Active	Inactive	-

Dependency: Refer to: p3870, p3871, p3872, p3873, p3874, p3876, p3878, p3879

Note: The display is updated with a sampling time of 1 ms.
 Re bit 00 = 1:
 The encoder is parked. Contrary to r0481.14, parking is also displayed here if the suppression of the parking bit is active in r0481.14 (p3870.1 = 1).
 Re bit 01 = 1:
 The long-stator functions requested closed-loop speed control with encoder. In r1407.2, it is indicated as to whether an encoder is actually used for the closed-loop control.

r3875.0...1 CO/BO: Long stator status word / Long stator ZSW

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word for long stator motors.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Sensor Module is unparked	Yes	No	-
	01	Closed-loop velocity control with encoder requested	Active	Inactive	-

Dependency: Refer to: p3870, p3871, p3872, p3873, p3874, p3876, p3878, p3879

Note: The display is updated with a sampling time of 1 ms.
 Re bit 00 = 1:
 The encoder is parked. Contrary to r0481.14, parking is also displayed here if the suppression of the parking bit is active in r0481.14 (p3870.1 = 1).
 Re bit 01 = 1:
 The long-stator functions requested closed-loop velocity control with encoder. In r1407.2, it is indicated as to whether an encoder is actually used for the closed-loop control.

p3876 BI: Unpark long stator signal source 1 encoder / S_src 1 enc unpark

SERVO	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source 1 to unpark the encoder.

Dependency: Refer to: p3870, p3871, p3872, p3873, p3874, r3875, p3878, p3879

Note: BI: p3876 = 1 signal --> encoder is unparked
 BI: p3876 = 0 signal --> encoder is parked

p3878	CO: Long stator commutation angle 1 / Com_angle 1		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -180 [°]	Calculated: - Dynamic index: - Units group: - Scaling: p2005 Max 180 [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [°]
Description:	Sets the commutation angle 1 for long stator motors.		
Dependency:	Refer to: p3870, p3871, p3872, p3873, p3874, r3875, p3876, p3879		
p3879	CO: Long stator commutation angle 2 / Com_angle 2		
SERVO	Can be changed: U, T Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -180 [°]	Calculated: - Dynamic index: - Units group: - Scaling: p2005 Max 180 [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [°]
Description:	Sets the commutation angle 2 for long stator motors.		
Dependency:	Refer to: p3870, p3871, p3872, p3873, p3874, r3875, p3876, p3878		
p3900	Completion of quick commissioning / Compl_quick_comm		
A_INF, B_INF, S_INF	Can be changed: C2(1) Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Exits the quick commissioning (p0010 = 1) with automatic calculation of all of the parameters that depend on the entries made during the quick commissioning. p3900 = 1 initially includes a parameter reset (factory setting, the same as p0970 = 1) for all parameters of the drive object; however, without overwriting the entries made during the quick commissioning. The interconnections of PROFIBUS PZD telegram selection (p0922) and the interconnections via p0700 are re-established and all of the dependent filter and closed-loop control parameters are calculated (corresponding to p0340 = 1). p3900 = 2 includes the restoration of the interconnections of PROFIBUS PZD telegram selection (p0922) and the interconnections via p0700 and the calculations corresponding to p0340 = 1. p3900 = 3 only includes the end of quick commissioning.		
Value:	0: No quick parameterization 1: Quick parameterization after parameter reset 2: Quick param. (only) for controller par. and reset for BICO par 3: Completion of quick commissioning		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	When the calculations have been completed, p3900 and p0010 are automatically reset to a value of 0.		

p3900 Completion of quick commissioning / Compl quick_comm			
SERVO	Can be changed: C2(1) Data type: Integer16 P-Group: Displays, signals Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Exits quick commissioning (p0010 = 1) with automatic calculation of all parameters of all existing drive data sets that depend on the entries made during quick commissioning. p3900 = 1 initially includes a parameter reset (factory setting, the same as p0970 = 1) for all parameters of the drive object; however, without overwriting the entries made during the quick commissioning. The interconnections of PROFIBUS PZD telegram selection (p0922) and the interconnections via p0700, p1000 and p1500 are re-established and all of the dependent motor, open-loop and control-loop control parameters are calculated (corresponding to p0340 = 1). p3900 = 2 includes the restoration of the interconnections of PROFIBUS PZD telegram selection (p0922) and the interconnections via p0700, p1000 and p1500 and the calculations corresponding to p0340 = 1. p3900 = 3 only includes the calculations associated with the motor, open-loop and closed-loop control parameters corresponding to p0340 = 1.		
Value:	0: No quick parameterization 1: Quick parameterization after parameter reset 2: Quick parameterization (only) for BICO and motor parameters 3: Quick parameterization for motor parameters (only)		
Notice:	After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.		
Note:	When the calculations have been completed, p3900 and p0010 are automatically reset to a value of 0. When calculating motor, open-loop and closed-loop control parameters (such as for p0340 = 1) parameters associated with a selected Siemens catalog motor are not overwritten. If a catalog motor has not been selected (see p0300), then the following parameters are reset with p3900 > 0 in order to restore the situation that applied when commissioning the drive for the first time: for induction motors p0320, p0352, p0353, p0604, p0605, p0626 ... p0628. for synchronous motors p0326, p0327, p0352, p0353, p0391 ... p0393, p0604, p0605.		

p3900 Completion of quick commissioning / Compl quick_comm			
VECTOR	Can be changed: C2(1) Data type: Integer16 P-Group: Displays, signals Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Exits quick commissioning (p0010 = 1) with automatic calculation of all parameters of all existing drive data sets that depend on the entries made during quick commissioning. p3900 = 1 initially includes a parameter reset (factory setting, the same as p0970 = 1) for all parameters of the drive object; however, without overwriting the entries made during the quick commissioning. The interconnections of PROFIBUS PZD telegram selection (p0922) and the interconnections via p0700, p1000 and p1500 are re-established and all of the dependent motor, open-loop and control-loop control parameters are calculated (corresponding to p0340 = 1). p3900 = 2 includes the restoration of the interconnections of PROFIBUS PZD telegram selection (p0922) and the interconnections via p0700, p1000 and p1500 and the calculations corresponding to p0340 = 1. p3900 = 3 only includes the calculations associated with the motor, open-loop and closed-loop control parameters corresponding to p0340 = 1.		
Value:	0: No quick parameterization 1: Quick parameterization after parameter reset 2: Quick parameterization (only) for BICO and motor parameters 3: Quick parameterization for motor parameters (only)		

- Notice:** After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.
- Note:** When the calculations have been completed, p3900 and p0010 are automatically reset to a value of 0.
When calculating motor, open-loop and closed-loop control parameters (such as for p0340 = 1) parameters associated with a selected Siemens catalog motor are not overwritten.
If a catalog motor has not been selected (see p0300), then the following parameters are reset with p3900 > 0 in order to restore the situation that applied when commissioning the drive for the first time:
for induction motors p0320, p0352, p0353, p0362 ... p0369, p0391 ... p0393, p0604, p0605, p0626 ... p0628.
for synchronous motors p0326, p0327, p0352, p0353, p0391 ... p0393, p0604, p0605.

p3901[0...n]		Power unit EEPROM Vdc offset calibration / PU EEPROM Vdc_offs			
B_INF, VECTOR	Can be changed:	C1, C2(1), T	Calculated: -	Access level: 3	
	Data type:	FloatingPoint32	Dynamic index:	PDS, p0120	Func. diagram: -
	P-Group:	-	Units group:	-	Unit selection: -
	Not for motor type:	-	Scaling:	-	Expert list: 1
	Min	-40.0 [V]	Max	40.0 [V]	Factory setting
Description:		Differential voltage for calibrating the offset for DC-link voltage measurement.			
Dependency:		Refer to: r0192, p0212			
Caution:		Incorrect use of the calibration can have a negative impact on the closed-loop control. The parameter influences the upper and lower voltage detection.			
					
Note:		Parameter entries are directly saved in the DRIVE-CLiQ component involved.			
		The parameter is only effective in the case of booksize power units if r0192 bit 22 and p0212 bit 0 are set.			

r3925[0...n]		Identification final display / Ident final_disp			
SERVO	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned32		Dynamic index: DDS, p0180		Func. diagram: -
	P-Group: Motor		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
Description:		Displays the commissioning steps that have been carried out.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Motor/control parameters calculated (p0340 = 1, p3900 > 0)	Yes	No	-
	02	Motor data identification carried out at standstill (p1910 = 1)	Yes	No	-
	03	Rotating measurement carried out (p1960 = 1, 2)	Yes	No	-
	04	Motor encoder adjustment carried out (p1960 = 1, p1990 = 1)	Yes	No	-
	05	Motor encoder manually adjusted	Yes	No	-
	15	Motor equivalent circuit diagram parameters changed	Changed	Not changed	-
Note:		The individual bits are only set if the appropriate action has been initiated and successfully completed. When motor rating plate parameters are changed, the final display is reset. When setting the individual bits, all of the most significant bits are reset.			

r3925[0...n]		Identification final display / Ident final_disp			
VECTOR	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned32	Dynamic index: DDS, p0180	Func. diagram: -		
	P-Group: Motor	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:		Displays the commissioning steps that have been carried out.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Motor/control parameters calculated (p0340 = 1, p3900 > 0)	Yes	No	-
	02	Motor data identification carried out at standstill (p1910 = 1)	Yes	No	-
	03	Rotating measurement carried out (p1960 = 1, 2)	Yes	No	-
	04	Motor encoder adjustment carried out (p1960 = 1, p1990 = 1)	Yes	No	-
	10	Automatic parameterization only for V/f control (r0108.2 = 0)	Yes	No	-
	15	Motor equivalent circuit diagram parameters changed	Changed	Not changed	-
Note:		The individual bits are only set if the appropriate action has been initiated and successfully completed. When motor rating plate parameters are changed, the final display is reset. When setting the individual bits, all of the most significant bits are reset.			

r3927[0...n]		Motor data identification induction motor data determined / MotID ASM dat det			
SERVO	Can be changed: -	Calculated: CALC_MOD_ALL	Access level: 3		
	Data type: Unsigned32	Dynamic index: DDS, p0180	Func. diagram: -		
	P-Group: Motor identification	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:		Displays the data of an induction motor determined and accepted from the stationary motor data identification or rotating measurement.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	p0350 accepted	Yes	No	-
	01	p0354 accepted	Yes	No	-
	02	p0356 accepted	Yes	No	-
	03	p0358 accepted	Yes	No	-
	04	p0360 accepted	Yes	No	-
	05	p0320 accepted	Yes	No	-
	06	p0410 accepted	Yes	No	-
	12	p1715 accepted	Yes	No	-
	13	p1717 accepted	Yes	No	-
	14	p1590 accepted	Yes	No	-
	15	p1592 accepted	Yes	No	-
	22	p0341 accepted	Yes	No	-
	24	p0348 accepted	Yes	No	-
	25	p1752 accepted	Yes	No	-
Dependency:		Refer to: r3925			

r3927[0...n] Motor data identification control word / MotID STW				
VECTOR	Can be changed: - Calculated: CALC_MOD_ALL Access level: 3 Data type: Unsigned16 Dynamic index: DDS, p0180 Func. diagram: - P-Group: Motor identification Units group: - Unit selection: - Not for motor type: - Scaling: - Expert list: 1 Min Max Factory setting - - -			
Description:	Successfully completed component of the last motor data identification carried out.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Stator inductance estimate no measurement	Yes	No
	01	Cl.-loop current control w/ dead-beat controller	Yes	No
	02	Rotor time constant estimate no measurement	Yes	No
	03	Leakage inductance estimate no measurement	Yes	No
	04	Activates the identification dynamic leakage inductance	Yes	No
	05	Determine Tr and Lsig evaluation in the time range	Yes	No
	06	Activate vibration damping	Yes	No
	07	Deactivate vibration detection	Yes	No
	11	Deactivate pulse measurement Lq Ld	Yes	No
	12	Deactivate rotor resistance Rr measurement	Yes	No
	14	Deactivate valve interlocking time measurement	Yes	No
	15	Determine only stator resistance, valve voltage fault, dead time	Yes	No
Dependency:	Refer to: r3925			
Note:	The parameter is a copy of p1909.			

r3928[0...n] Motor data identification synchronous motor data determined / MotID PEM dat det				
SERVO	Can be changed: - Calculated: CALC_MOD_ALL Access level: 3 Data type: Unsigned32 Dynamic index: DDS, p0180 Func. diagram: - P-Group: Motor identification Units group: - Unit selection: - Not for motor type: REL Scaling: - Expert list: 1 Min Max Factory setting - - -			
Description:	Successfully completed component of the last rotating measurement carried out.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	p0350 accepted	Yes	No
	02	p0356 accepted	Yes	No
	06	p0410 accepted	Yes	No
	07	p0431 accepted	Yes	No
	08	p1952 accepted	Yes	No
	09	p1953 accepted	Yes	No
	12	p1715 accepted	Yes	No
	13	p1717 accepted	Yes	No
	18	p0316 accepted	Yes	No
	19	p0317 accepted	Yes	No
	20	p0327 accepted	Yes	No
	21	p0328 accepted	Yes	No
	22	p0341 accepted	Yes	No
	23	kT characteristic parameter accepted	Yes	No
	24	p0348 accepted	Yes	No

Dependency: Refer to: r3925

r3928[0...n] Rotating measurement configuration / Rot meas config

VECTOR (n/M)	Can be changed: -	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: Unsigned16	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Motor identification	Units group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Successfully completed component of the last rotating measurement carried out.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Enc test active	Yes	No	-
	01	Saturation characteristic identification	Yes	No	-
	02	Moment of inertia identification	Yes	No	-
	03	Recalculates the speed controller parameters	Yes	No	-
	04	Speed controller optimization (vibration test)	Yes	No	-
	05	q leakage inductance ident. (for current controller adaptation)	Yes	No	-

Dependency: Refer to: r3925

Note: The parameter is a copy of p1959.

p3950 Service parameter / Serv. par.

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1, U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: For service personnel only.

r3974 Device status word / Device status word

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the device status.

Bit 0: SW reset active

Bit 1: The writing of parameters is disabled, as parameters are being saved.

Bit 2: The writing of parameters is disabled, as a macro is running.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	SW reset active	active	inactive	-
	01	Writing of parameters disabled as parameter save in progress	active	inactive	-
	02	Writing of parameters disabled as macro in progress	active	inactive	-

r3977	BICO counter, topology / BICO counter topo		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the BICO interconnections that have been parameterized in the complete (overall) topology. The counter is incremented by one for each modified BICO interconnection.		
Dependency:	Refer to: r3978, r3979		

r3978	BICO CounterDevice / BICO CounterDevice		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the counter reading for modified BICO interconnections on this device. The counter is incremented by one for each modified BICO interconnection.		

r3979	BICO counter, drive object / BICO counter DO		
A_INF, B_INF, CU_LINK, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VEC- TOR	Can be changed: - Data type: Unsigned32 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the counter reading for modified BICO interconnections on this drive object. The counter is incremented by one for each modified BICO interconnection.		

p3981	Faults, acknowledge drive object / Faults ackn DO		
All objects	Can be changed: U, T Data type: Unsigned8 P-Group: Messages Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: 8060 Unit selection: - Expert list: 1 Factory setting 0
Description:	Setting to acknowledge all active faults of a drive object.		
Note:	Parameter should be set from 0 to 1 to acknowledge. After acknowledgement, the parameter is automatically reset to 0.		

p3985 Master control mode selection / PcCtrl mode selectA_INF, B_INF,
S_INF, SERVO,
VECTOR**Can be changed:** U, T**Data type:** Integer16**P-Group:** Setpoints**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

1

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description:

Sets the mode to change over the master control / LOCAL mode.

Value:

0: Change master control for STW1.0 = 0

1: Change master control in operation

Danger:

When changing the master control in operation, the drive can manifest undesirable behavior - e.g. it can accelerate up to another setpoint.

**r3986 Parameter count / Parameter count**

All objects

Can be changed: -**Data type:** Unsigned16**P-Group:** -**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description:

Displays the number of parameters for this drive unit.

The number comprises the device-specific and the drive-specific parameters.

Dependency:

Refer to: r0980, r0981, r0989

r3988[0...1] Boot state / Boot_stateCU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** -**Data type:** Integer16**P-Group:** -**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

10800

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description:

Index 0:

Displays the boot state.

Index 1:

Displays the partial boot state

Value:

0: Not active

1: Fatal fault

10: Fault

20: Reset all parameters

30: Drive object modified

40: Download using commissioning software

50: Parameter download using commissioning software

90: Reset Control Unit and delete drive objects

100: Start initialization

101: Wait for topology input

110: Instantiate Control Unit basis

111: Insert drive object

112: Remove drive object

113: Change drive object number

114: Change component number

115: Parameter download using commissioning software

117: Remove component

150: Wait until actual topology determined

160: Evaluate topology
 170: Instantiate Control Unit reset
 180: Initialization YDB configuration information
 200: First commissioning
 210: Create drive packages
 250: Wait for topology acknowledge
 325: Wait for input of drive type
 350: Determine drive type
 360: Write into topology-dependent parameters
 370: Wait until p0009 = 0 is set
 380: Check topology
 550: Call conversion functions for parameter
 625: Wait non-cyclic starting DRIVE-CLiQ
 650: Start cyclic operation
 660: Evaluate drive commissioning status
 670: Autom. FW update DRIVE-CLiQ components
 680: Wait for CU LINK slaves
 690: Wait non-cyclic starting DRIVE-CLiQ
 700: Save parameters
 725: Wait until DRIVE-CLiQ cyclic
 740: Check the ability to operate
 745: Start of the time slices
 750: Interrupt enable
 800: Initialization finished
 10050: Wait for synchronization
 10100: Wait for CU LINK slaves
 10150: Wait until actual topology determined
 10200: Evaluation component status
 10250: Call conversion functions for parameter
 10300: Preparation cyclic operation
 10350: Autom. FW update DRIVE-CLiQ components
 10400: Wait for slave properties
 10450: Check CX/NX status
 10500: Wait until DRIVE-CLiQ cyclic
 10550: Carry out warm start
 10600: Evaluate, encoder status
 10800: Partial boot completed

Index:
 [0] = System
 [1] = Partial boot

r3996 Parameter write inhibit status / Par_write inhib st

All objects	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned8	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays whether writing to parameters is inhibited.
 r3996 = 0: Write to parameter is not inhibited.
 0 < r3996 < 100: Write to parameter is inhibited. The value shows how the calculations are progressing.

r3998 First infeed commissioning / First inf_commA_INF, B_INF,
S_INF**Can be changed:** -**Data type:** Unsigned16**P-Group:** -**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

65535

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Displays whether the infeed must be commissioned for the first time.
0 = Yes
2 = No

r3998[0...n] First drive commissioning / First drv_comm

SERVO, VECTOR

Can be changed: -**Data type:** Unsigned16**P-Group:** -**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** DDS, p0180**Units group:** -**Scaling:** -**Max**

65535

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Displays whether the drive still has to be commissioned for the first time.
0 = Yes
2 = No

r4021 TB30 digital inputs terminal actual value / TB30 DI act value

TB30

Can be changed: -**Data type:** Unsigned32**P-Group:** Commands**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 2**Func. diagram:** 9100**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Displays the actual value at the digital inputs.
This means that the actual input signal can be checked at terminal DI x prior to switching from the simulation mode (p4095.x = 1) to the terminal mode (p4095.x = 0). The input signal of terminal DI x is displayed in bit x of r4021.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X481.1)	High	Low	-
	01	DI 1 (X481.2)	High	Low	-
	02	DI 2 (X481.3)	High	Low	-
	03	DI 3 (X481.4)	High	Low	-

Note: DI: Digital input

r4021 TM15DI/DO digital inputs, terminal actual value / TM15D DI act val

TM15DI_DO

Can be changed: -**Data type:** Unsigned32**P-Group:** Commands**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 2**Func. diagram:** 9400, 9401,
9402**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Displays the actual value at the digital inputs.
This means that the actual input signal can be checked at terminal DI x or DI/DO x prior to switching from the simulation mode (p4095.x = 1) to terminal mode (p4095.x = 0). The input signal at terminal DI x or DI/DO x is displayed in bit x of r4021.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	High	Low	-
	01	DI/DO 1 (X520.3)	High	Low	-
	02	DI/DO 2 (X520.4)	High	Low	-
	03	DI/DO 3 (X520.5)	High	Low	-
	04	DI/DO 4 (X520.6)	High	Low	-
	05	DI/DO 5 (X520.7)	High	Low	-
	06	DI/DO 6 (X520.8)	High	Low	-
	07	DI/DO 7 (X520.9)	High	Low	-
	08	DI/DO 8 (X521.2)	High	Low	-
	09	DI/DO 9 (X521.3)	High	Low	-
	10	DI/DO 10 (X521.4)	High	Low	-
	11	DI/DO 11 (X521.5)	High	Low	-
	12	DI/DO 12 (X521.6)	High	Low	-
	13	DI/DO 13 (X521.7)	High	Low	-
	14	DI/DO 14 (X521.8)	High	Low	-
	15	DI/DO 15 (X521.9)	High	Low	-
	16	DI/DO 16 (X522.2)	High	Low	-
	17	DI/DO 17 (X522.3)	High	Low	-
	18	DI/DO 18 (X522.4)	High	Low	-
	19	DI/DO 19 (X522.5)	High	Low	-
	20	DI/DO 20 (X522.6)	High	Low	-
	21	DI/DO 21 (X522.7)	High	Low	-
	22	DI/DO 22 (X522.8)	High	Low	-
	23	DI/DO 23 (X522.9)	High	Low	-

Note: If a DI/DO is parameterized as output (p4028.x = 1), then r4021.x = 0 is displayed.
DI/DO: Bidirectional digital input/output

r4021 TM31 digital inputs terminal actual value / TM31 DI act value

TM31	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1840, 9550, 9552, 9560, 9562
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the actual value at the digital inputs.
This means that the actual input signal can be checked at terminal DI x or DI/DO x prior to switching from the simulation mode (p4095.x = 1) to terminal mode (p4095.x = 0). The input signal at terminal DI x or DI/DO x is displayed in bit x of r4021.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X520.1)	High	Low	-
	01	DI 1 (X520.2)	High	Low	-
	02	DI 2 (X520.3)	High	Low	-
	03	DI 3 (X520.4)	High	Low	-
	04	DI 4 (X530.1)	High	Low	-
	05	DI 5 (X530.2)	High	Low	-
	06	DI 6 (X530.3)	High	Low	-
	07	DI 7 (X530.4)	High	Low	-
	08	DI/DO 8 (X541.2)	High	Low	-
	09	DI/DO 9 (X541.3)	High	Low	-
	10	DI/DO 10 (X541.4)	High	Low	-
	11	DI/DO 11 (X541.5)	High	Low	-

Note: If a DI/DO is parameterized as output (p4028.x = 1), then r4021.x = 0 is displayed.
DI: Digital input
DI/DO: Bidirectional digital input/output

r4021 TM41 digital inputs terminal actual value / TM41 DI act val

TM41	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the actual value at the digital inputs.
This means that the actual input signal can be checked at terminal DI x or DI/DO x prior to switching from the simulation mode (p4095.x = 1) to terminal mode (p4095.x = 0). The input signal at terminal DI x or DI/DO x is displayed in bit x of r4021.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X522.1)	High	Low	9660
	01	DI 1 (X522.2)	High	Low	9660
	02	DI 2 (X522.3)	High	Low	9660
	03	DI 3 (X522.4)	High	Low	9660
	08	DI/DO 0 (X521.1)	High	Low	9661
	09	DI/DO 1 (X521.2)	High	Low	9661
	10	DI/DO 2 (X521.3)	High	Low	9662
	11	DI/DO 3 (X521.4)	High	Low	9662

Note: If a DI/DO is parameterized as output (p4028.x = 1), then r4021.x = 0 is displayed.
DI: Digital input
DI/DO: Bidirectional digital input/output

r4022.0...3 CO/BO: TB30 digital inputs, status / TB30 DI status

TB30	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1790, 9100
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the digital inputs of the Terminal Board 30 (TB30).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X481.1)	High	Low	-
	01	DI 1 (X481.2)	High	Low	-
	02	DI 2 (X481.3)	High	Low	-
	03	DI 3 (X481.4)	High	Low	-

Dependency: Refer to: r4023

Note: DI: Digital input

r4022.0...23	CO/BO: TM15DI/DO digital inputs, status / TM15D DI status			
TM15DI_DO	Can be changed: -		Calculated: -	
	Data type: Unsigned32		Dynamic index: -	
	P-Group: Commands		Units group: -	
	Not for motor type: -		Scaling: -	
	Min		Max	
	-		-	
Description:	Displays the status of the digital inputs of Terminal Module 15 (TM15).			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	DI/DO 0 (X520.2)	High	Low
	01	DI/DO 1 (X520.3)	High	Low
	02	DI/DO 2 (X520.4)	High	Low
	03	DI/DO 3 (X520.5)	High	Low
	04	DI/DO 4 (X520.6)	High	Low
	05	DI/DO 5 (X520.7)	High	Low
	06	DI/DO 6 (X520.8)	High	Low
	07	DI/DO 7 (X520.9)	High	Low
	08	DI/DO 8 (X521.2)	High	Low
	09	DI/DO 9 (X521.3)	High	Low
	10	DI/DO 10 (X521.4)	High	Low
	11	DI/DO 11 (X521.5)	High	Low
	12	DI/DO 12 (X521.6)	High	Low
	13	DI/DO 13 (X521.7)	High	Low
	14	DI/DO 14 (X521.8)	High	Low
	15	DI/DO 15 (X521.9)	High	Low
	16	DI/DO 16 (X522.2)	High	Low
	17	DI/DO 17 (X522.3)	High	Low
	18	DI/DO 18 (X522.4)	High	Low
	19	DI/DO 19 (X522.5)	High	Low
	20	DI/DO 20 (X522.6)	High	Low
	21	DI/DO 21 (X522.7)	High	Low
	22	DI/DO 22 (X522.8)	High	Low
	23	DI/DO 23 (X522.9)	High	Low
Dependency:	Refer to: r4023, r4024, r4025			
Notice:	For the BICO interconnection of the connector output (CO) only bit 00 ... bit 15 are transferred.			
Note:	DI/DO: Bidirectional digital input/output			

r4022.0...11	CO/BO: TM31 digital inputs, status / TM31 DI status				
TM31	Can be changed: -	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1840, 9550, 9552, 9560, 9562		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the status of the digital inputs of Terminal Module 31 (TM31).				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X520.1)	High	Low	-
	01	DI 1 (X520.2)	High	Low	-
	02	DI 2 (X520.3)	High	Low	-
	03	DI 3 (X520.4)	High	Low	-
	04	DI 4 (X530.1)	High	Low	-
	05	DI 5 (X530.2)	High	Low	-
	06	DI 6 (X530.3)	High	Low	-

07	DI 7 (X530.4)	High	Low	-
08	DI/DO 8 (X541.2)	High	Low	-
09	DI/DO 9 (X541.3)	High	Low	-
10	DI/DO 10 (X541.4)	High	Low	-
11	DI/DO 11 (X541.5)	High	Low	-

Dependency: Refer to: r4023

Note: DI: Digital input

DI/DO: Bidirectional digital input/output

r4022.0...11 CO/BO: TM41 digital inputs, status / TM41 DI status

TM41	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the digital inputs of Terminal Module 41 (TM41).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X522.1)	High	Low	9660
	01	DI 1 (X522.2)	High	Low	9660
	02	DI 2 (X522.3)	High	Low	9660
	03	DI 3 (X522.4)	High	Low	9660
	08	DI/DO 0 (X521.1)	High	Low	9661
	09	DI/DO 1 (X521.2)	High	Low	9661
	10	DI/DO 2 (X521.3)	High	Low	9662
	11	DI/DO 3 (X521.4)	High	Low	9662

Dependency: Refer to: r4023

Note: DI: Digital input

DI/DO: Bidirectional digital input/output

r4023.0...3 BO: TB30 digital inputs, status inverted / TB30 DI status inv

TB30	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1790, 9100
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the inverted status of the digital inputs of the Terminal Board 30 (TB30).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X481.1)	High	Low	-
	01	DI 1 (X481.2)	High	Low	-
	02	DI 2 (X481.3)	High	Low	-
	03	DI 3 (X481.4)	High	Low	-

Dependency: Refer to: r4022

Note: DI: Digital input

r4023.0...23	CO/BO: TM15DI/DO digital inputs, status inverted / TM15D DI stat inv			
TM15DI_DO	Can be changed: -		Calculated: -	
	Data type: Unsigned32		Dynamic index: -	
	P-Group: Commands		Units group: -	
	Not for motor type: -		Scaling: -	
	Min		Max	
	-		-	
Description:	Displays the inverted status of the digital inputs of Terminal Module 15 (TM15).			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	DI/DO 0 (X520.2)	High	Low
	01	DI/DO 1 (X520.3)	High	Low
	02	DI/DO 2 (X520.4)	High	Low
	03	DI/DO 3 (X520.5)	High	Low
	04	DI/DO 4 (X520.6)	High	Low
	05	DI/DO 5 (X520.7)	High	Low
	06	DI/DO 6 (X520.8)	High	Low
	07	DI/DO 7 (X520.9)	High	Low
	08	DI/DO 8 (X521.2)	High	Low
	09	DI/DO 9 (X521.3)	High	Low
	10	DI/DO 10 (X521.4)	High	Low
	11	DI/DO 11 (X521.5)	High	Low
	12	DI/DO 12 (X521.6)	High	Low
	13	DI/DO 13 (X521.7)	High	Low
	14	DI/DO 14 (X521.8)	High	Low
	15	DI/DO 15 (X521.9)	High	Low
	16	DI/DO 16 (X522.2)	High	Low
	17	DI/DO 17 (X522.3)	High	Low
	18	DI/DO 18 (X522.4)	High	Low
	19	DI/DO 19 (X522.5)	High	Low
	20	DI/DO 20 (X522.6)	High	Low
	21	DI/DO 21 (X522.7)	High	Low
	22	DI/DO 22 (X522.8)	High	Low
	23	DI/DO 23 (X522.9)	High	Low
Dependency:	Refer to: r4022, r4024, r4025			
Notice:	For the BICO interconnection of the connector output (CO) only bit 00 ... bit 15 are transferred.			
Note:	DI/DO: Bidirectional digital input/output			

r4023.0...11	CO/BO: TM31 digital inputs, status inverted / TM31 DI status inv				
TM31	Can be changed: -		Calculated: -	Access level: 1	
	Data type: Unsigned32		Dynamic index: -	Func. diagram: 1840, 9550, 9552, 9560, 9562	
	P-Group: Commands		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the inverted status of the digital inputs of Terminal Module 31 (TM31).				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X520.1)	High	Low	-
	01	DI 1 (X520.2)	High	Low	-
	02	DI 2 (X520.3)	High	Low	-
	03	DI 3 (X520.4)	High	Low	-
	04	DI 4 (X530.1)	High	Low	-
	05	DI 5 (X530.2)	High	Low	-
	06	DI 6 (X530.3)	High	Low	-

07	DI 7 (X530.4)	High	Low	-
08	DI/DO 8 (X541.2)	High	Low	-
09	DI/DO 9 (X541.3)	High	Low	-
10	DI/DO 10 (X541.4)	High	Low	-
11	DI/DO 11 (X541.5)	High	Low	-

Dependency: Refer to: r4022

Note: DI: Digital input

DI/DO: Bidirectional digital input/output

r4023.0...11 BO: TM41 digital inputs, status inverted / TM41 DI status inv

TM41	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the inverted status of the digital inputs of Terminal Module 41 (TM41).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X522.1)	High	Low	9660
	01	DI 1 (X522.2)	High	Low	9660
	02	DI 2 (X522.3)	High	Low	9660
	03	DI 3 (X522.4)	High	Low	9660
	08	DI/DO 0 (X521.1)	High	Low	9661
	09	DI/DO 1 (X521.2)	High	Low	9661
	10	DI/DO 2 (X521.3)	High	Low	9662
	11	DI/DO 3 (X521.4)	High	Low	9662

Dependency: Refer to: r4022

Note: DI: Digital input

DI/DO: Bidirectional digital input/output

r4024 CO: TM15DI/DO digital inputs 16 ... 23 status / TM15D DI 16-23 St

TM15DI_DO	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of digital inputs 16 ... 23 of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 16 (X522.2)	On	Off	-
	01	DI/DO 17 (X522.3)	On	Off	-
	02	DI/DO 18 (X522.4)	On	Off	-
	03	DI/DO 19 (X522.5)	On	Off	-
	04	DI/DO 20 (X522.6)	On	Off	-
	05	DI/DO 21 (X522.7)	On	Off	-
	06	DI/DO 22 (X522.8)	On	Off	-
	07	DI/DO 23 (X522.9)	On	Off	-

Dependency: Refer to: r4022, r4023, r4025

Note: DI: Digital input

r4025 CO: TM15DI/DO digital inputs 16 ... 23 status inverted / TM15D DI 16-23 inv

TM15DI_DO	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the inverted status of digital inputs 16 ... 23 of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 16 (X522.2)	On	Off	-
	01	DI/DO 17 (X522.3)	On	Off	-
	02	DI/DO 18 (X522.4)	On	Off	-
	03	DI/DO 19 (X522.5)	On	Off	-
	04	DI/DO 20 (X522.6)	On	Off	-
	05	DI/DO 21 (X522.7)	On	Off	-
	06	DI/DO 22 (X522.8)	On	Off	-
	07	DI/DO 23 (X522.9)	On	Off	-

Dependency: Refer to: r4022, r4023, r4024

Note: DI: Digital input

p4028 TM15 set input or output / TM15 DI or DO

TM15	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the bidirectional digital inputs/outputs on the Terminal Module 15 (TM15) as input or output.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	Output	Input	-
	01	DI/DO 1 (X520.3)	Output	Input	-
	02	DI/DO 2 (X520.4)	Output	Input	-
	03	DI/DO 3 (X520.5)	Output	Input	-
	04	DI/DO 4 (X520.6)	Output	Input	-
	05	DI/DO 5 (X520.7)	Output	Input	-
	06	DI/DO 6 (X520.8)	Output	Input	-
	07	DI/DO 7 (X520.9)	Output	Input	-
	08	DI/DO 8 (X521.2)	Output	Input	-
	09	DI/DO 9 (X521.3)	Output	Input	-
	10	DI/DO 10 (X521.4)	Output	Input	-
	11	DI/DO 11 (X521.5)	Output	Input	-
	12	DI/DO 12 (X521.6)	Output	Input	-
	13	DI/DO 13 (X521.7)	Output	Input	-
	14	DI/DO 14 (X521.8)	Output	Input	-
	15	DI/DO 15 (X521.9)	Output	Input	-
	16	DI/DO 16 (X522.2)	Output	Input	-
	17	DI/DO 17 (X522.3)	Output	Input	-
	18	DI/DO 18 (X522.4)	Output	Input	-
	19	DI/DO 19 (X522.5)	Output	Input	-
	20	DI/DO 20 (X522.6)	Output	Input	-
	21	DI/DO 21 (X522.7)	Output	Input	-
	22	DI/DO 22 (X522.8)	Output	Input	-
	23	DI/DO 23 (X522.9)	Output	Input	-

Note: DI/DO: Bidirectional digital input/output

p4028 TM15DI/DO set input or output / TM15D DI or DO

TM15DI_DO

Can be changed: T**Calculated:** -**Access level:** 1**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 9400, 9401, 9402**P-Group:** Commands**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**
0000 bin**Description:**

Sets the bidirectional digital inputs/outputs on the Terminal Module 15 (TM15) as input or output.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	DI/DO 0 (X520.2)	Output	Input	-
01	DI/DO 1 (X520.3)	Output	Input	-
02	DI/DO 2 (X520.4)	Output	Input	-
03	DI/DO 3 (X520.5)	Output	Input	-
04	DI/DO 4 (X520.6)	Output	Input	-
05	DI/DO 5 (X520.7)	Output	Input	-
06	DI/DO 6 (X520.8)	Output	Input	-
07	DI/DO 7 (X520.9)	Output	Input	-
08	DI/DO 8 (X521.2)	Output	Input	-
09	DI/DO 9 (X521.3)	Output	Input	-
10	DI/DO 10 (X521.4)	Output	Input	-
11	DI/DO 11 (X521.5)	Output	Input	-
12	DI/DO 12 (X521.6)	Output	Input	-
13	DI/DO 13 (X521.7)	Output	Input	-
14	DI/DO 14 (X521.8)	Output	Input	-
15	DI/DO 15 (X521.9)	Output	Input	-
16	DI/DO 16 (X522.2)	Output	Input	-
17	DI/DO 17 (X522.3)	Output	Input	-
18	DI/DO 18 (X522.4)	Output	Input	-
19	DI/DO 19 (X522.5)	Output	Input	-
20	DI/DO 20 (X522.6)	Output	Input	-
21	DI/DO 21 (X522.7)	Output	Input	-
22	DI/DO 22 (X522.8)	Output	Input	-
23	DI/DO 23 (X522.9)	Output	Input	-

Note:

DI/DO: Bidirectional digital input/output

p4028 TM17 set input or output / TM17 DI or DO

TM17

Can be changed: T**Calculated:** -**Access level:** 2**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** -**P-Group:** Commands**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**
0000 bin**Description:**

Sets the bidirectional digital inputs/outputs on the Terminal Module 17 (TM17) as input or output.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	DI/DO 0 (X520.2)	Output	Input	-
01	DI/DO 1 (X520.3)	Output	Input	-
02	DI/DO 2 (X520.5)	Output	Input	-
03	DI/DO 3 (X520.6)	Output	Input	-
04	DI/DO 4 (X520.8)	Output	Input	-
05	DI/DO 5 (X520.9)	Output	Input	-
06	DI/DO 6 (X521.2)	Output	Input	-
07	DI/DO 7 (X521.3)	Output	Input	-
08	DI/DO 8 (X521.8)	Output	Input	-
09	DI/DO 9 (X521.9)	Output	Input	-
10	DI/DO 10 (X522.2)	Output	Input	-

11	DI/DO 11 (X522.3)	Output	Input	-
12	DI/DO 12 (X522.5)	Output	Input	-
13	DI/DO 13 (X522.6)	Output	Input	-
14	DI/DO 14 (X522.8)	Output	Input	-
15	DI/DO 15 (X522.9)	Output	Input	-

Note: DI/DO: Bidirectional digital input/output

p4028 TM31 set input or output / TM31 DI or DO

TM31	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1840, 9560, 9562
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the bidirectional digital inputs/outputs as input or output on the Terminal Module 31 (TM31).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 8 (X541.2)	Output	Input	-
	09	DI/DO 9 (X541.3)	Output	Input	-
	10	DI/DO 10 (X541.4)	Output	Input	-
	11	DI/DO 11 (X541.5)	Output	Input	-

p4028 TM41 set input or output / TM41 DI or DO

TM41	Can be changed: T	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the bidirectional digital inputs/outputs on the Terminal Module 41 (TM41) as input or output.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 0 (X521.1)	Output	Input	9661
	09	DI/DO 1 (X521.2)	Output	Input	9661
	10	DI/DO 2 (X521.3)	Output	Input	9662
	11	DI/DO 3 (X521.4)	Output	Input	9662

p4030 BI: TB30 signal source for terminal DO 0 / TB30 S_src DO 0

TB30	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 1790, 9102
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for digital output DO 0 (X481.5) of the Terminal Board 30 (TB30).

Note: DO: Digital output

p4030	BI: TM15DI/DO signal source for terminal DI/DO 0 / TM15D S_src DI/DO0		
TM15DI_DO	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9400 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 0 (X520.2) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.0 = 1). DI/DO: Bidirectional digital input/output		
p4030	BI: TM31 signal source for terminal DO 0 / TM31 S_src DO 0		
TM31	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 1840, 9556 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the digital output DO 0 (X542.1, X542.2, X542.3) of Terminal Module 31 (TM31). Digital output 0 of TM31 is a relay output. If the signal at the binector input p4030 is low, then terminal COM 0 (X542.2) is connected to NC 0 (X542.1). This connection also matches the mechanical quiescent setting of the relay. If the signal at the binector input p4030 is high, then terminal COM 0 (X542.2) is connected to NO 0 (X542.3).		
Note:	DO: Digital output NC: Normally Closed contact NO: Normally Open contact		
p4031	BI: TB30 signal source for terminal DO 1 / TB30 S_src DO 1		
TB30	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9102 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DO 1 (X481.6) of the Terminal Board 30 (TB30).		
Note:	DO: Digital output		
p4031	BI: TM15DI/DO signal source for terminal DI/DO 1 / TM15D S_src DI/DO1		
TM15DI_DO	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9400 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 1 (X520.3) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.1 = 1). DI/DO: Bidirectional digital input/output		

p4031 BI: TM31 signal source for terminal DO 1 / TM31 S_src DO 1			
TM31	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 1840, 9556 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the digital output DO 1 (X542.4, X542.5, X542.6) of Terminal Module 31 (TM31). Digital output 1 of TM31 is a relay output. If the signal at the binector input p4031 is low, then terminal COM 1 (X542.5) is connected to NC 1 (X542.4). This connection also matches the mechanical quiescent setting of the relay. If the signal at the binector input p4031 is high, then terminal COM 1 (X542.5) is connected to NO 1 (X542.6).		
Note:	DO: Digital output NC: Normally Closed contact NO: Normally Open contact		

p4032 BI: TB30 signal source for terminal DO 2 / TB30 S_src DO 2			
TB30	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9102 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DO 2 (X481.7) of the Terminal Board 30 (TB30).		
Note:	DO: Digital output		

p4032 BI: TM15DI/DO signal source for terminal DI/DO 2 / TM15D S_src DI/DO2			
TM15DI_DO	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9400 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 2 (X520.4) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.2 = 1). DI/DO: Bidirectional digital input/output		

p4033 BI: TB30 signal source for terminal DO 3 / TB30 S_src DO 3			
TB30	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 1790, 9102 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DO 3 (X481.8) of the Terminal Board 30 (TB30).		
Note:	DO: Digital output		

p4033	BI: TM15DI/DO signal source for terminal DI/DO 3 / TM15D S_src DI/DO3		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9400
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 3 (X520.5) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.3 = 1). DI/DO: Bidirectional digital input/output		
p4034	BI: TM15DI/DO signal source for terminal DI/DO 4 / TM15D S_src DI/DO4		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9400
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 4 (X520.6) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.4 = 1). DI/DO: Bidirectional digital input/output		
p4035	BI: TM15DI/DO signal source for terminal DI/DO 5 / TM15D S_src DI/DO5		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9400
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 5 (X520.7) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.5 = 1). DI/DO: Bidirectional digital input/output		
p4036	BI: TM15DI/DO signal source for terminal DI/DO 6 / TM15D S_src DI/DO6		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9400
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 6 (X520.8) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.6 = 1). DI/DO: Bidirectional digital input/output		

p4037	BI: TM15DI/DO signal source for terminal DI/DO 7 / TM15D S_src DI/DO7		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9400
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 7 (X520.9) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.7 = 1). DI/DO: Bidirectional digital input/output		
p4038	BI: TM15DI/DO signal source for terminal DI/DO 8 / TM15D S_src DI/DO8		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9401
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 8 (X521.2) of terminal module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.8 = 1). DI/DO: Bidirectional digital input/output		
p4038	BI: TM31 signal source for terminal DI/DO 8 / TM31 S_src DI/DO8		
TM31	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 1840, 9560
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 8 (X541.2) of Terminal Module 31 (TM31).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.8 = 1). DI/DO: Bidirectional digital input/output		
p4038	BI: TM41 signal source for terminal DI/DO 0 / TM41 S_src DI/DO 0		
TM41	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9661
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 0 (X521.1) of Terminal Module 41 (TM41).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.8 = 1). DI/DO: Bidirectional digital input/output		

p4039	BI: TM15DI/DO signal source for terminal DI/DO 9 / TM15D S_src DI/DO9		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9401
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 9 (X521.3) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.9 = 1). DI/DO: Bidirectional digital input/output		

p4039	BI: TM31 signal source for terminal DI/DO 9 / TM31 S_src DI/DO9		
TM31	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9560
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 9 (X541.3) of Terminal Module 31 (TM31).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.9 = 1). DI/DO: Bidirectional digital input/output		

p4039	BI: TM41 signal source for terminal DI/DO 1 / TM41 S_src DI/DO 1		
TM41	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9661
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 1 (X541.2) of Terminal Module 41 (TM41).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.9 = 1). DI/DO: Bidirectional digital input/output		

p4040	BI: TM15DI/DO signal source for terminal DI/DO 10 / TM15D S_srcDI/DO10		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9401
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 10 (X521.4) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.10 = 1). DI/DO: Bidirectional digital input/output		

p4040	BI: TM31 signal source for terminal DI/DO 10 / TM31 S_srcDI/DO10		
TM31	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9562 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 10 (X541.4) of Terminal Module 31 (TM31).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.10 = 1). DI/DO: Bidirectional digital input/output		
p4040	BI: TM41 signal source for terminal DI/DO 2 / TM41 S_src DI/DO 2		
TM41	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9662 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 2 (X521.3) of Terminal Module 41 (TM41).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.10 = 1). DI/DO: Bidirectional digital input/output		
p4041	BI: TM15DI/DO signal source for terminal DI/DO 11 / TM15D S_srcDI/DO11		
TM15DI_DO	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9401 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 11 (X521.5) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.11 = 1). DI/DO: Bidirectional digital input/output		
p4041	BI: TM31 signal source for terminal DI/DO 11 / TM31 S_src DI/DO11		
TM31	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 1840, 9562 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 11 (X541.5) of Terminal Module 31 (TM31).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.11 = 1). DI/DO: Bidirectional digital input/output		

p4041	BI: TM41 signal source for terminal DI/DO 3 / TM41 S_src DI/DO 3		
TM41	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9662 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 3 (X521.4) of Terminal Module 41 (TM41).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.11 = 1). DI/DO: Bidirectional digital input/output		
p4042	BI: TM15DI/DO signal source for terminal DI/DO 12 / TM15D S_srcDI/DO12		
TM15DI_DO	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9401 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 12 (X521.6) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.12 = 1). DI/DO: Bidirectional digital input/output		
p4043	BI: TM15DI/DO signal source for terminal DI/DO 13 / TM15D S_srcDI/DO13		
TM15DI_DO	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9401 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 13 (X521.7) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.13 = 1). DI/DO: Bidirectional digital input/output		
p4044	BI: TM15DI/DO signal source for terminal DI/DO 14 / TM15D S_srcDI/DO14		
TM15DI_DO	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9401 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 14 (X521.8) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.14 = 1). DI/DO: Bidirectional digital input/output		

p4045				
BI: TM15DI/DO signal source for terminal DI/DO 15 / TM15D S_srcDI/DO15				
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1	
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9401	
	P-Group: Commands	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	0	
Description:	Sets the signal source for terminal DI/DO 15 (X521.9) of Terminal Module 15 (TM15).			
Note:	Prerequisite: The DI/DO must be set as an output (p4028.15 = 1). DI/DO: Bidirectional digital input/output			

p4046				
TM31 digital outputs, limit current / TM31 DO limit curr				
TM31	Can be changed: T	Calculated: -	Access level: 2	
	Data type: Integer16	Dynamic index: -	Func. diagram: 9560	
	P-Group: Commands	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	0	1	0	
Description:	Sets the limit for the total output voltage of terminals X541.1, X541.2, X541.3 and X541.4 (DI/DO 8 ... 11) of Terminal Module 31 (TM31).			
Value:	0: 0.1 A total current limit DI/DO 8 ... 11 1: 1.0 A total current limit DI/DO 8 ... 11			
Dependency:	Refer to: p4028			
Warning:	Since the sum of the output currents at terminals X541.1, X541.2, X541.3 and X541.4 is limited, an overcurrent or short circuit at one output terminal can cause a dip in the signal at the other terminals.			
				

r4047					
TB30 digital outputs status / TB30 DO status					
TB30	Can be changed: -	Calculated: -	Access level: 1		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9102		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the status of the digital outputs of the Terminal Board 30 (TB30).				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DO 0 (X481.5)	High	Low	-
	01	DO 1 (X481.6)	High	Low	-
	02	DO 2 (X481.7)	High	Low	-
	03	DO 3 (X481.8)	High	Low	-
Note:	Inversion using p4048 has been taken into account. DO: Digital output				

r4047 TM15DI/DO digital outputs, status / TM15D DO status

TM15DI_DO	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9400, 9401, 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the digital outputs of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	High	Low	-
	01	DI/DO 1 (X520.3)	High	Low	-
	02	DI/DO 2 (X520.4)	High	Low	-
	03	DI/DO 3 (X520.5)	High	Low	-
	04	DI/DO 4 (X520.6)	High	Low	-
	05	DI/DO 5 (X520.7)	High	Low	-
	06	DI/DO 6 (X520.8)	High	Low	-
	07	DI/DO 7 (X520.9)	High	Low	-
	08	DI/DO 8 (X521.2)	High	Low	-
	09	DI/DO 9 (X521.3)	High	Low	-
	10	DI/DO 10 (X521.4)	High	Low	-
	11	DI/DO 11 (X521.5)	High	Low	-
	12	DI/DO 12 (X521.6)	High	Low	-
	13	DI/DO 13 (X521.7)	High	Low	-
	14	DI/DO 14 (X521.8)	High	Low	-
	15	DI/DO 15 (X521.9)	High	Low	-
	16	DI/DO 16 (X522.2)	High	Low	-
	17	DI/DO 17 (X522.3)	High	Low	-
	18	DI/DO 18 (X522.4)	High	Low	-
	19	DI/DO 19 (X522.5)	High	Low	-
	20	DI/DO 20 (X522.6)	High	Low	-
	21	DI/DO 21 (X522.7)	High	Low	-
	22	DI/DO 22 (X522.8)	High	Low	-
	23	DI/DO 23 (X522.9)	High	Low	-

Note: Inversion using p4048 has been taken into account.
The setting of the DI/DO as either input or output is of no significance (p4028).
DI/DO: Bidirectional digital input/output

r4047 TM31 digital outputs status / TM31 DO status

TM31	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9556, 9560, 9562
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the digital outputs of Terminal Module 31 (TM31).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DO 0 (X542.1 - 3)	High	Low	-
	01	DO 1 (X542.4 - 6)	High	Low	-
	08	DI/DO 8 (X541.2)	High	Low	-
	09	DI/DO 9 (X541.3)	High	Low	-
	10	DI/DO 10 (X541.4)	High	Low	-
	11	DI/DO 11 (X541.5)	High	Low	-

Note: Inversion using p4048 has been taken into account.
The setting of the DI/DO as either input or output is of no significance (p4028).
DO: Digital output
DI/DO: Bidirectional digital input/output

r4047 TM41 digital outputs status / TM41 DO status

TM41	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the digital outputs of Terminal Module 41 (TM41).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 0 (X521.1)	High	Low	9661
	09	DI/DO 1 (X521.2)	High	Low	9661
	10	DI/DO 2 (X521.3)	High	Low	9662
	11	DI/DO 3 (X521.4)	High	Low	9662

Note: Inversion using p4048 has been taken into account.
The setting of the DI/DO as either input or output is of no significance (p4028).
DO: Digital output
DI/DO: Bidirectional digital input/output

p4048 TB30 invert digital outputs / TB30 DO invert

TB30	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9102
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Setting to invert the signals at the digital outputs of the Terminal Board 30 (TB30).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DO 0 (X481.5)	Inverted	Not inverted	-
	01	DO 1 (X481.6)	Inverted	Not inverted	-
	02	DO 2 (X481.7)	Inverted	Not inverted	-
	03	DO 3 (X481.8)	Inverted	Not inverted	-

Note: DO: Digital output

p4048 TM15 invert digital inputs/outputs / TM15 DI/DO invert

TM15	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Setting to invert the signals at the digital inputs/outputs of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	Inverted	Not inverted	-
	01	DI/DO 1 (X520.3)	Inverted	Not inverted	-
	02	DI/DO 2 (X520.4)	Inverted	Not inverted	-
	03	DI/DO 3 (X520.5)	Inverted	Not inverted	-
	04	DI/DO 4 (X520.6)	Inverted	Not inverted	-
	05	DI/DO 5 (X520.7)	Inverted	Not inverted	-
	06	DI/DO 6 (X520.8)	Inverted	Not inverted	-

07	DI/DO 7 (X520.9)	Inverted	Not inverted	-
08	DI/DO 8 (X521.2)	Inverted	Not inverted	-
09	DI/DO 9 (X521.3)	Inverted	Not inverted	-
10	DI/DO 10 (X522.4)	Inverted	Not inverted	-
11	DI/DO 11 (X521.5)	Inverted	Not inverted	-
12	DI/DO 12 (X521.6)	Inverted	Not inverted	-
13	DI/DO 13 (X521.7)	Inverted	Not inverted	-
14	DI/DO 14 (X521.8)	Inverted	Not inverted	-
15	DI/DO 15 (X521.9)	Inverted	Not inverted	-
16	DI/DO 16 (X522.2)	Inverted	Not inverted	-
17	DI/DO 17 (X522.3)	Inverted	Not inverted	-
18	DI/DO 18 (X522.4)	Inverted	Not inverted	-
19	DI/DO 19 (X522.5)	Inverted	Not inverted	-
20	DI/DO 20 (X522.6)	Inverted	Not inverted	-
21	DI/DO 21 (X522.7)	Inverted	Not inverted	-
22	DI/DO 22 (X522.8)	Inverted	Not inverted	-
23	DI/DO 23 (X522.9)	Inverted	Not inverted	-

Note: DI/DO: Bidirectional digital input/output

p4048 TM15DI/DO invert digital outputs / TM15D DO invert

TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9400, 9401, 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Setting to invert the signals at the digital outputs of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	Inverted	Not inverted	-
	01	DI/DO 1 (X520.3)	Inverted	Not inverted	-
	02	DI/DO 2 (X520.4)	Inverted	Not inverted	-
	03	DI/DO 3 (X520.5)	Inverted	Not inverted	-
	04	DI/DO 4 (X520.6)	Inverted	Not inverted	-
	05	DI/DO 5 (X520.7)	Inverted	Not inverted	-
	06	DI/DO 6 (X520.8)	Inverted	Not inverted	-
	07	DI/DO 7 (X520.9)	Inverted	Not inverted	-
	08	DI/DO 8 (X521.2)	Inverted	Not inverted	-
	09	DI/DO 9 (X521.3)	Inverted	Not inverted	-
	10	DI/DO 10 (X521.4)	Inverted	Not inverted	-
	11	DI/DO 11 (X521.5)	Inverted	Not inverted	-
	12	DI/DO 12 (X521.6)	Inverted	Not inverted	-
	13	DI/DO 13 (X521.7)	Inverted	Not inverted	-
	14	DI/DO 14 (X521.8)	Inverted	Not inverted	-
	15	DI/DO 15 (X521.9)	Inverted	Not inverted	-
	16	DI/DO 16 (X522.2)	Inverted	Not inverted	-
	17	DI/DO 17 (X522.3)	Inverted	Not inverted	-
	18	DI/DO 18 (X522.4)	Inverted	Not inverted	-
	19	DI/DO 19 (X522.5)	Inverted	Not inverted	-
	20	DI/DO 20 (X522.6)	Inverted	Not inverted	-
	21	DI/DO 21 (X522.7)	Inverted	Not inverted	-
	22	DI/DO 22 (X522.8)	Inverted	Not inverted	-
	23	DI/DO 23 (X522.9)	Inverted	Not inverted	-

Note: DI/DO: Bidirectional digital input/output

p4048 TM17 invert digital inputs/outputs / TM17 DI/DO invert

TM17	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Setting to invert the signals at the digital inputs/outputs of Terminal Module 17 (TM17).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	Inverted	Not inverted	-
	01	DI/DO 1 (X520.3)	Inverted	Not inverted	-
	02	DI/DO 2 (X520.5)	Inverted	Not inverted	-
	03	DI/DO 3 (X520.6)	Inverted	Not inverted	-
	04	DI/DO 4 (X520.8)	Inverted	Not inverted	-
	05	DI/DO 5 (X520.9)	Inverted	Not inverted	-
	06	DI/DO 6 (X521.2)	Inverted	Not inverted	-
	07	DI/DO 7 (X521.3)	Inverted	Not inverted	-
	08	DI/DO 8 (X521.8)	Inverted	Not inverted	-
	09	DI/DO 9 (X521.9)	Inverted	Not inverted	-
	10	DI/DO 10 (X522.2)	Inverted	Not inverted	-
	11	DI/DO 11 (X522.3)	Inverted	Not inverted	-
	12	DI/DO 12 (X522.5)	Inverted	Not inverted	-
	13	DI/DO 13 (X522.6)	Inverted	Not inverted	-
	14	DI/DO 14 (X522.8)	Inverted	Not inverted	-
	15	DI/DO 15 (X522.9)	Inverted	Not inverted	-

Note: DI/DO: Bidirectional digital input/output

p4048 TM31 invert digital outputs / TM31 DO invert

TM31	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9556, 9560, 9562
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Setting to invert the signals at the digital outputs of Terminal Module 31 (TM31).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DO 0 (X542.1 - 3)	Inverted	Not inverted	-
	01	DO 1 (X542.4 - 6)	Inverted	Not inverted	-
	08	DI/DO 8 (X541.2)	Inverted	Not inverted	-
	09	DI/DO 9 (X541.3)	Inverted	Not inverted	-
	10	DI/DO 10 (X541.4)	Inverted	Not inverted	-
	11	DI/DO 11 (X541.5)	Inverted	Not inverted	-

Note: DO: Digital output
DI/DO: Bidirectional digital input/output

p4048 TM41 invert digital outputs / TM41 DO invert

TM41	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Setting to invert the signals at the digital outputs of Terminal Module 41 (TM41).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	08	DI/DO 0 (X521.1)	Inverted	Not inverted	9661
	09	DI/DO 1 (X521.2)	Inverted	Not inverted	9661
	10	DI/DO 2 (X521.3)	Inverted	Not inverted	9662
	11	DI/DO 3 (X521.4)	Inverted	Not inverted	9662

Note: DO: Digital output
DI/DO: Bidirectional digital input/output

p4049 TM15 digital inputs/outputs, set the mode / TM15 DI/DO mode

TM15	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the mode of the DI/DOs of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	I/O with time	I/O	-
	01	DI/DO 1 (X520.3)	I/O with time	I/O	-
	02	DI/DO 2 (X520.4)	I/O with time	I/O	-
	03	DI/DO 3 (X520.5)	I/O with time	I/O	-
	04	DI/DO 4 (X520.6)	I/O with time	I/O	-
	05	DI/DO 5 (X520.7)	I/O with time	I/O	-
	06	DI/DO 6 (X520.8)	I/O with time	I/O	-
	07	DI/DO 7 (X520.9)	I/O with time	I/O	-
	08	DI/DO 8 (X521.2)	I/O with time	I/O	-
	09	DI/DO 9 (X521.3)	I/O with time	I/O	-
	10	DI/DO 10 (X522.4)	I/O with time	I/O	-
	11	DI/DO 11 (X521.5)	I/O with time	I/O	-
	12	DI/DO 12 (X521.6)	I/O with time	I/O	-
	13	DI/DO 13 (X521.7)	I/O with time	I/O	-
	14	DI/DO 14 (X521.8)	I/O with time	I/O	-
	15	DI/DO 15 (X521.9)	I/O with time	I/O	-
	16	DI/DO 16 (X522.2)	I/O with time	I/O	-
	17	DI/DO 17 (X522.3)	I/O with time	I/O	-
	18	DI/DO 18 (X522.4)	I/O with time	I/O	-
	19	DI/DO 19 (X522.5)	I/O with time	I/O	-
	20	DI/DO 20 (X522.6)	I/O with time	I/O	-
	21	DI/DO 21 (X522.7)	I/O with time	I/O	-
	22	DI/DO 22 (X522.8)	I/O with time	I/O	-
	23	DI/DO 23 (X522.9)	I/O with time	I/O	-

Note: DI/DO: Bidirectional digital input/output

p4049 TM17 digital inputs/outputs, set the mode / TM17 DI/DO mode

TM17	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the mode of the DI/DO of Terminal Module 17 (TM17).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	I/O with time	I/O	-
	01	DI/DO 1 (X520.3)	I/O with time	I/O	-
	02	DI/DO 2 (X520.5)	I/O with time	I/O	-
	03	DI/DO 3 (X520.6)	I/O with time	I/O	-
	04	DI/DO 4 (X520.8)	I/O with time	I/O	-
	05	DI/DO 5 (X520.9)	I/O with time	I/O	-
	06	DI/DO 6 (X521.2)	I/O with time	I/O	-
	07	DI/DO 7 (X521.3)	I/O with time	I/O	-
	08	DI/DO 8 (X521.8)	I/O with time	I/O	-
	09	DI/DO 9 (X521.9)	I/O with time	I/O	-
	10	DI/DO 10 (X522.2)	I/O with time	I/O	-
	11	DI/DO 11 (X522.3)	I/O with time	I/O	-
	12	DI/DO 12 (X522.5)	I/O with time	I/O	-
	13	DI/DO 13 (X522.6)	I/O with time	I/O	-
	14	DI/DO 14 (X522.8)	I/O with time	I/O	-
	15	DI/DO 15 (X522.9)	I/O with time	I/O	-

Note: DI/DO: Bidirectional digital input/output

r4052[0...1] CO: TB30 analog inputs, current input voltage / TB30 AI inp_V

TB30	Can be changed: -	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9104
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]

Description: Displays the actual input voltage at the analog inputs of the Terminal Board 30 (TB30).

Index: [0] = AI 0 (X482.1/X482.2)
[1] = AI 1 (X482.3/X482.4)

Note: AI: Analog Input

r4052[0...1] CO: TM31 analog inputs, current input voltage/current / TM31 AI inp_V/I

TM31	Can be changed: -	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9566, 9568
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the actual input voltage in V when set as voltage input.

Displays the actual input current in mA when set as current input and with the load resistor switched in.

Index: [0] = AI 0 (X521.1/X521.2, S5.0)
[1] = AI 1 (X521.3/X521.4, S5.1)

Dependency: The type of analog input AI x (voltage or current input) is set using p4056.
Refer to: r4056, p4056

Note: AI: Analog Input

r4052[0]			
CO: TM41 analog inputs, current input voltage / TM41 AI inp_V			
TM41	Can be changed: - Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [V]	Access level: 1 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Displays the current input voltage in V.		
Index:	[0] = AI 0 (X523.1/X523.2)		
Note:	AI: Analog Input		
p4053[0...1]			
TB30 analog inputs, smoothing time constant / TB30 AI T_smooth			
TB30	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min 0.0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.0 [ms]	Access level: 1 Func. diagram: 9104 Unit selection: - Expert list: 1 Factory setting 0.0 [ms]
Description:	Sets the smoothing time constant of the 1st-order low pass filter for the analog inputs of the Terminal Board 30 (TB30).		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		
Note:	AI: Analog Input		
p4053[0...1]			
TM31 analog inputs, smoothing time constant / TM31 AI T_smooth			
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min 0.0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.0 [ms]	Access level: 1 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 0.0 [ms]
Description:	Sets the smoothing time constant of the 1st-order low pass filter for the analog inputs of Terminal Module 31 (TM31).		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
Note:	AI: Analog Input		
p4053[0]			
TM41 analog inputs, smoothing time constant / TM41 AI T_smooth			
TM41	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min 0.0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.0 [ms]	Access level: 1 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting 0.0 [ms]
Description:	Sets the smoothing time constant of the 1st-order low pass filter for the analog inputs of Terminal Module 41 (TM41).		
Index:	[0] = AI 0 (X523.1/X523.2)		
Note:	AI: Analog Input		

r4055[0...1]	CO: TB30 analog inputs, actual value in percent / TB30 AI value in %		
TB30	Can be changed: - Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 1 Func. diagram: 1790, 9104 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the currently referred input value of the analog inputs of Terminal Board 30 (TB30). When interconnected, the signals are referred to the reference quantities p200x and p205x.		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		
Note:	AI: Analog Input		
r4055[0...1]	CO: TM31 analog inputs, actual value in percent / TM31 AI value in %		
TM31	Can be changed: - Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 1 Func. diagram: 1840, 9566, 9568 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the currently referred input value of the analog inputs of Terminal Module 31 (TM31). When interconnected, the signals are referred to the reference quantities p200x and p205x.		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
Note:	AI: Analog Input		
r4055[0]	CO: TM41 analog inputs, current value in percent / TM41 AI value in %		
TM41	Can be changed: - Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 1 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the currently referred input value of the analog inputs of Terminal Module 41 (TM41). When interconnected, the signals are referred to the reference quantities p200x and p205x.		
Index:	[0] = AI 0 (X523.1/X523.2)		
Note:	AI: Analog Input		
r4056[0...1]	TB30 analog inputs, type / TB30 AI type		
TB30	Can be changed: - Data type: Integer16 P-Group: Terminals Not for motor type: - Min 4	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the type of analog inputs.		
Value:	4: Bipolar voltage input (-10 V ... +10 V)		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		

p4056[0...1]			
TM31 analog inputs, type / TM31 AI type			
TM31	Can be changed: U, T Data type: Integer16 P-Group: Terminals Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 5	Access level: 1 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 4
Description:	Sets the type of analog inputs of Terminal Module 31 (TM31). p4056[x] = 0, 4 correspond to a voltage input (r4052, p4057, p4059 are displayed in V). p4056[x] = 2, 3, 5 correspond to a current input (r4052, p4057, p4059 are displayed in mA). In addition, the associated switch S5 must be switched. For a voltage input, S5.1 or S5.2 must be switched to setting "V". For a current input, S5.1 or S5.2 must be switched to setting "I" (load resistor = 250 ohm is switched in).		
Value:	0: Unipolar voltage input (0 V ... +10 V) 2: Unipolar current input (0 mA ... +20 mA) 3: Unipolar current input monitored (+4 mA ... +20 mA) 4: Bipolar voltage input (-10 V ... +10 V) 5: Bipolar current input (-20 mA ... +20 mA)		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
Warning:	The maximum voltage difference between the analog input terminals AI+, AI- and the ground of the TM31 (X520.6, X530.3) may not exceed 35 V. For operation with the load resistor switched in, the voltage between the differential inputs AI+ and AI- may not exceed 15 V or the impressed current of 60 mA; if this is not carefully observed, the input will be damaged.		
Notice:	For use as voltage input, switch S500 must be set to 0 for the input involved. Switch S500 is located on the front panel of the TM31 above terminal block X521.		
Note:	When changing p4056, the parameters of the scaling characteristic (p4057, p4058, p4059, p4060) are overwritten with the following default values: For p4056 = 0, 4, p4057 is set to 0.0 V, p4058 to 0.0 %, p4059 to 10.0 V and p4060 to 100.0 %. For p4056 = 2, 5, p4057 is set to 0.0 mA, p4058 to 0.0 %, p4059 to 20.0 mA and p4060 to 100.0 %. For p4056 = 3, p4057 is set to 4.0 mA, p4058 to 0.0 %, p4059 to 20.0 mA and p4060 to 100.0 %.		
r4056			
TM41 analog input, type / TM41 AI type			
TM41	Can be changed: - Data type: Integer16 P-Group: Terminals Not for motor type: - Min 4	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the type of the analog input.		
Value:	4: Bipolar voltage input (-10 V ... +10 V)		
p4057[0...1]			
TB30 analog inputs, characteristic value x1 / TB30 AI char x1			
TB30	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -11.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 11.000 [V]	Access level: 2 Func. diagram: 9104 Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Sets the scaling characteristic for the analog inputs of Terminal Board 30 (TB30). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the x coordinate (input voltage in V) of the 1st value pair of the characteristic.		

Index: [0] = AI 0 (X482.1/X482.2)
[1] = AI 1 (X482.3/X482.4)
Note: The parameters for the characteristic do not have a limiting effect.

p4057[0...1] TM31 analog inputs, characteristic value x1 / TM31 AI char x1			
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000	Access level: 2 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the scaling characteristic for the analog inputs of Terminal Module 31 (TM31). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the x coordinate (input voltage in V or input current in mA) of the 1st value pair of the characteristic.		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
Dependency:	The unit of this parameter (V or mA) depends on the analog input type. Refer to: r4056, p4056		
Notice:	This parameter is automatically overwritten when the analog input type (p4056) is modified.		
Note:	The parameters for the characteristic do not have a limiting effect.		

p4057[0] TM41 analog input, characteristic value x1 / TM41 AI char x1			
TM41	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000 [V]	Access level: 2 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Sets the scaling characteristic for the analog inputs of Terminal Module 41 (TM41). The scaling characteristic for the analog input is defined using 2 points. This parameter specifies the x coordinate (input voltage in V) of the 1st value pair of the characteristic.		
Index:	[0] = AI 0 (X523.1/X523.2)		
Note:	The parameters for the characteristic do not have a limiting effect.		

p4058[0...1] TB30 analog inputs, characteristic value y1 / TB30 AI char y1			
TB30	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -1000.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: 9104 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the scaling characteristic for the analog inputs of Terminal Board 30 (TB30). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the y coordinate (percentage) of the 1st value pair of the characteristic.		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		
Note:	The parameters for the characteristic do not have a limiting effect.		

p4058[0...1]	TM31 analog inputs, characteristic value y1 / TM31 AI char y1		
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -1000.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the scaling characteristic for the analog inputs of Terminal Module 31 (TM31). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the y coordinate (percentage) of the 1st value pair of the characteristic.		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
Notice:	This parameter is automatically overwritten when the analog input type (p4056) is modified.		
Note:	The parameters for the characteristic do not have a limiting effect.		
p4058[0]	TM41 analog input, characteristic value y1 / TM41 AI char y1		
TM41	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -1000.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the scaling characteristic for the analog input of Terminal Module 41 (TM41). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the y coordinate (percentage) of the 1st value pair of the characteristic.		
Index:	[0] = AI 0 (X523.1/X523.2)		
Note:	The parameters for the characteristic do not have a limiting effect.		
p4059[0...1]	TB30 analog inputs, characteristic value x2 / TB30 AI char x2		
TB30	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -11.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 11.000 [V]	Access level: 2 Func. diagram: 9104 Unit selection: - Expert list: 1 Factory setting 10.000 [V]
Description:	Sets the scaling characteristic for the analog inputs of Terminal Board 30 (TB30). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the x coordinate (input voltage in V) of the 2nd value pair of the characteristic.		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		
Note:	The parameters for the characteristic do not have a limiting effect.		

p4059[0...1] TM31 analog inputs, characteristic value x2 / TM31 AI char x2			
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000	Access level: 2 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 10.000
Description:	Sets the scaling characteristic for the analog inputs of Terminal Module 31 (TM31). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the x coordinate (input voltage in V or input current in mA) of the 2nd value pair of the characteristic.		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
Dependency:	The unit of this parameter (V or mA) depends on the analog input type. Refer to: r4056, p4056		
Notice:	This parameter is automatically overwritten when the analog input type (p4056) is modified.		
Note:	The parameters for the characteristic do not have a limiting effect.		
p4059[0] TM41 analog input, characteristic value x2 / TM41 AI char x2			
TM41	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000 [V]	Access level: 2 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting 10.000 [V]
Description:	Sets the scaling characteristic for the analog input of Terminal Module 41 (TM41). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the x coordinate (input voltage in V) of the 2nd value pair of the characteristic.		
Index:	[0] = AI 0 (X523.1/X523.2)		
Note:	The parameters for the characteristic do not have a limiting effect.		
p4060[0...1] TB30 analog inputs, characteristic value y2 / TB30 AI char y2			
TB30	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -1000.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: 9104 Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the scaling characteristic for the analog inputs of Terminal Board 30 (TB30). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the y coordinate (percentage) of the 2nd value pair of the characteristic.		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		
Note:	The parameters for the characteristic do not have a limiting effect.		

p4060[0...1]	TM31 analog inputs, characteristic value y2 / TM31 AI char y2		
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -1000.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the scaling characteristic for the analog inputs of Terminal Module 31 (TM31). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the y coordinate (percentage) of the 2nd value pair of the characteristic.		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
Notice:	This parameter is automatically overwritten when the analog input type (p4056) is modified.		
Note:	The parameters for the characteristic do not have a limiting effect.		
p4060[0]	TM41 analog input, characteristic value y2 / TM41 AI char y2		
TM41	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -1000.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the scaling characteristic for the analog input of Terminal Module 41 (TM41). The scaling characteristic for the analog inputs is defined using two points. This parameter specifies the y coordinate (percentage) of the 2nd value pair of the characteristic.		
Index:	[0] = AI 0 (X523.1/X523.2)		
Note:	The parameters for the characteristic do not have a limiting effect.		
p4061[0...1]	TM31 analog inputs, wire breakage monitoring response threshold / TM31 WireBrkThresh		
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min 0.00 [mA]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.00 [mA]	Access level: 2 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 2.00 [mA]
Description:	Sets the response threshold for wire-breakage monitoring of the analog inputs of Terminal Module 31 (TM31).		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
Dependency:	For the following analog input type, the wire breakage monitoring is active: p4056[x] = 3 (unipolar current input monitored (+4 mA ... +20 mA)) Refer to: r4056, p4056		

p4062[0...1]	TM31 analog inputs, wire breakage monitoring delay time / TM31 wirebrk t_del		
TM31	Can be changed: U, T Data type: Unsigned16 P-Group: Terminals Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000 [ms]	Access level: 2 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 100 [ms]
Description:	Sets the delay time for wire-breakage monitoring of the analog inputs on Terminal Module 31 (TM31).		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
p4063[0...1]	TB30 analog inputs offset / TB30 AI offset		
TB30	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000 [V]	Access level: 2 Func. diagram: 9104 Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Sets the offset for the analog inputs of Terminal Board 30 (TB30). The offset is added to the input signal before the scaling characteristic.		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		
p4063[0...1]	TM31 analog inputs offset / TM31 AI offset		
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000	Access level: 2 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the offset for the analog inputs of Terminal Module 31 (TM31). The offset is added to the input signal before the scaling characteristic.		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
p4063[0]	TM41 analog input, offset / TM41 AI offset		
TM41	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000 [V]	Access level: 2 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Sets the offset for the analog input of Terminal Module 41 (TM41). The offset is added to the input signal before the scaling characteristic.		
Index:	[0] = AI 0 (X523.1/X523.2)		

p4066[0...1] TB30 analog inputs, activate absolute value generation / TB30 AI absVal act

TB30	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 9104
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0
Description:	Activates the absolute value generation for the analog input signals of the Terminal Board 30 (TB30).		
Value:	0: No absolute value generation 1: Absolute value generation switched in		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		

p4066[0...1] TM31 analog inputs, activate absolute value generation / TM31 AI absVal act

TM31	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 9566, 9568
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0
Description:	Activates the absolute value generation for the analog input signals of Terminal Module 31 (TM31).		
Value:	0: No absolute value generation 1: Absolute value generation switched in		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		

p4066[0] TM41 analog input, activate absolute value generation / TM41 AI absVal act

TM41	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 9663
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0
Description:	Activates the absolute value generation of the analog input signal of Terminal Module 41 (TM41).		
Value:	0: No absolute value generation 1: Absolute value generation switched in		
Index:	[0] = AI 0 (X523.1/X523.2)		

p4067[0...1] BI: TB30 analog inputs invert signal source / TB30 AI inv S_src

TB30	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9104
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to invert the analog input signals of the Terminal Board 30 (TB30).		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		

p4067[0...1]			
BI: TM31 analog inputs invert signal source / TM31 AI inv S_src			
TM31	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9566, 9568
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to invert the analog inputs signals of Terminal Module 31 (TM31).		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
p4067[0]			
BI: TM41 analog input invert signal source / TM41 AI inv S_src			
TM41	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9663
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source to invert the analog input signal of Terminal Module 41 (TM41).		
Index:	[0] = AI 0 (X523.1/X523.2)		
p4068[0...1]			
TB30 analog inputs, window to suppress noise / TB30 AI window			
TB30	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9104
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [%]	20.00 [%]	0.00 [%]
Description:	Sets the noise suppression window for the analog inputs of Terminal Board 30 (TB30). Changes less than the window are suppressed.		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		
p4068[0...1]			
TM31 analog inputs, window to suppress noise / TM31 AI window			
TM31	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9566, 9568
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [%]	20.00 [%]	0.00 [%]
Description:	Sets the window for noise suppression for the analog inputs of Terminal Module 31 (TM31). Changes less than the window are suppressed.		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		

p4068[0]	TM41 analog input, window to suppress noise / TM41 AI window		
TM41	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.00 [%]	Access level: 3 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the noise suppression window for the analog input of Terminal Module 41 (TM41). Changes less than the window are suppressed.		
Index:	[0] = AI 0 (X523.1/X523.2)		
p4069[0...1]	BI: TB30 analog inputs, signal source for enable / TB30 AI enable		
TB30	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Terminals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 9104 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for enabling the analog inputs of the Terminal Board 30 (TB30).		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		
p4069[0...1]	BI: TM31 analog inputs, signal source for enable / TM31 AI enable		
TM31	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Terminals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the enable signal of the analog inputs of Terminal Module 31 (TM31).		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
p4069[0]	BI: TM41 analog input, signal source for enable / TM41 AI enable		
TM41	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Terminals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source for the enable signal of the analog input of Terminal Module 41 (TM41).		
Index:	[0] = AI 0 (X523.1/X523.2)		

p4071[0...1]			
CI: TB30 analog outputs, signal source / TB30 AO sig_source			
TB30	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 1790, 9106
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the analog outputs of the Terminal Board 30 (TB30).		
Index:	[0] = AO 0 (X482.5/X482.6) [1] = AO 1 (X482.7/X482.8)		
Note:	AO: Analog Output		

p4071[0...1]			
CI: TM31 analog outputs, signal source / TM31 AO sig_source			
TM31	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 1840, 9572
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the analog outputs of Terminal Module 31 (TM31).		
Index:	[0] = AO 0 (X522.1, X522.2, X522.3) [1] = AO 1 (X522.4, X522.5, X522.6)		
Note:	AO: Analog Output		

r4072[0...1]			
TB30 analog outputs, output value currently referred / TB30 AO outp_val			
TB30	Can be changed: -	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9106
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]
Description:	Displays the actual referred output value of the analog outputs of the Terminal Board 30 (TB30).		
Index:	[0] = AO 0 (X482.5/X482.6) [1] = AO 1 (X482.7/X482.8)		

r4072[0...1]			
TM31 analog outputs, output value currently referred / TM31 AO outp_val			
TM31	Can be changed: -	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9572
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]
Description:	Displays the actual referred output value of the analog outputs of Terminal Module 31 (TM31).		
Index:	[0] = AO 0 (X522.1, X522.2, X522.3) [1] = AO 1 (X522.4, X522.5, X522.6)		

p4073[0...1]	TB30 analog outputs, smoothing time constant / TB30 AO T_smooth		
TB30	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9106
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 1000.0 [ms]	Factory setting 0.0 [ms]
Description:	Sets the smoothing time constant of the 1st order low pass filter for the analog outputs of the Terminal Board 30 (TB30).		
Index:	[0] = AO 0 (X482.5/X482.6) [1] = AO 1 (X482.7/X482.8)		

p4073[0...1]	TM31 analog outputs, smoothing time constant / TM31 AO T_smooth		
TM31	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9572
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [ms]	Max 1000.0 [ms]	Factory setting 0.0 [ms]
Description:	Sets the smoothing time constant of the 1st-order low pass filter for the analog outputs of Terminal Module 31 (TM31).		
Index:	[0] = AO 0 (X522.1, X522.2, X522.3) [1] = AO 1 (X522.4, X522.5, X522.6)		

r4074[0...1]	TB30 analog outputs, current output voltage / TB30 AO V_outp		
TB30	Can be changed: -	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9106
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [V]	Max - [V]	Factory setting - [V]
Description:	Displays the actual output voltage at the analog outputs of the Terminal Board 30 (TB30).		
Index:	[0] = AO 0 (X482.5/X482.6) [1] = AO 1 (X482.7/X482.8)		

r4074[0...1]	TM31 analog outputs, current output voltage/current / TM31 AO V/I_outp		
TM31	Can be changed: -	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9572
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the actual output voltage in V when set as voltage output. Displays the actual output voltage in mA when set as current output.		
Index:	[0] = AO 0 (X522.1, X522.2, X522.3) [1] = AO 1 (X522.4, X522.5, X522.6)		
Dependency:	The type of the analog output AO x (voltage or current output) is set using p4076. Refer to: r4076, p4076		
Note:	AO: Analog Output		

p4075[0...1] TB30 analog outputs, activate absolute value generation / TB30 AO absVal act

TB30	Can be changed: T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 9106
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 1	Factory setting 0
Description:	Activates the absolute value generation for the analog outputs of the Terminal Board 30 (TB30).		
Value:	0: No absolute value generation 1: Absolute value generation switched in		
Index:	[0] = AO 0 (X482.5/X482.6) [1] = AO 1 (X482.7/X482.8)		

p4075[0...1] TM31 analog outputs, activate absolute value generation / TM31 AO absVal act

TM31	Can be changed: T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 9572
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 1	Factory setting 0
Description:	Activates the absolute value generation for the analog outputs of Terminal Module 31 (TM31).		
Value:	0: No absolute value generation 1: Absolute value generation switched in		
Index:	[0] = AO 0 (X522.1, X522.2, X522.3) [1] = AO 1 (X522.4, X522.5, X522.6)		

r4076[0...1] TB30 analog outputs, type / TB30 AO type

TB30	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 4	Max 4	Factory setting -
Description:	Displays the type of analog outputs of the Terminal Board 30 (TB30).		
Value:	4: Voltage output (-10 V ... +10 V)		
Index:	[0] = AO 0 (X482.5/X482.6) [1] = AO 1 (X482.7/X482.8)		

p4076[0...1] TM31 analog outputs, type / TM31 AO type

TM31	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: 9572
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 4	Factory setting 4
Description:	Sets the type of analog outputs of Terminal Module 31 (TM31). p4076[x] = 1, 4 correspond to a voltage output (p4074, p4078, p4080, p4083 are displayed in V). p4076[x] = 0, 2, 3 correspond to a current output (p4074, p4078, p4080, p4083 are displayed in mA).		
Value:	0: Current output (0 mA ... +20 mA) 1: Voltage output (0 V ... +10 V) 2: Current output (+4 mA ... +20 mA) 3: Current output (-20 mA ... +20 mA) 4: Voltage output (-10 V ... +10 V)		

Index: [0] = AO 0 (X522.1, X522.2, X522.3)
[1] = AO 1 (X522.4, X522.5, X522.6)

Dependency: Refer to: p4077, p4078, p4079, p4080

Note: When changing p4076, the parameters of the scaling characteristic (p4077, p4078, p4079, p4080) are overwritten with the following default values:

For p4076 = 0, 3, p4077 is set to 0.0 %, p4078 to 0.0 mA, p4079 to 100.0 % and p4080 to 20.0 mA.

For p4076 = 1, 4, p4077 is set to 0.0 %, p4078 to 0.0 V, p4079 to 100.0 % and p4080 to 10.0 V.

For p4076 = 2, p4077 is set to 0.0 %, p4078 to 4.0 mA, p4079 to 100.0 % and p4080 to 20.0 mA.

p4077[0...1] TB30 analog outputs, characteristic value x1 / TB30 AO char x1

TB30	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9106
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-1000.00 [%]	1000.00 [%]	0.00 [%]

Description: Sets the scaling characteristic for the analog outputs of Terminal Board 30 (TB30).

The scaling characteristic for the analog outputs is defined using 2 points.

This parameter specifies the x coordinate (percentage) of the 1st value pair of the characteristic.

Index: [0] = AO 0 (X482.5/X482.6)
[1] = AO 1 (X482.7/X482.8)

Note: The parameters for the characteristic do not have a limiting effect.

p4077[0...1] TM31 analog outputs, characteristic value x1 / TM31 AO char x1

TM31	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9572
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-1000.00 [%]	1000.00 [%]	0.00 [%]

Description: Sets the scaling characteristic for the analog outputs of Terminal Module 31 (TM31).

The scaling characteristic for the analog outputs is defined using 2 points.

This parameter specifies the x coordinate (percentage) of the 1st value pair of the characteristic.

Index: [0] = AO 0 (X522.1, X522.2, X522.3)
[1] = AO 1 (X522.4, X522.5, X522.6)

Dependency: The unit of this parameter (V or mA) depends on the analog input type.

Refer to: r4076, p4076

Notice: This parameter is automatically overwritten when the analog output type is changed (p4076).

Note: This parameter is automatically overwritten if p4076 (type of analog output) is changed.

The parameters for the characteristic do not have a limiting effect.

p4078[0...1] TB30 analog outputs, characteristic value y1 / TB30 AO char y1

TB30	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9106
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-11.000 [V]	11.000 [V]	0.000 [V]

Description: Sets the scaling characteristic for the analog outputs of Terminal Board 30 (TB30).

The scaling characteristic for the analog outputs is defined using 2 points.

This parameter specifies the y coordinate (output voltage in V) of the 1st value pair of the characteristic.

Index: [0] = AO 0 (X482.5/X482.6)
[1] = AO 1 (X482.7/X482.8)

Note: The parameters for the characteristic do not have a limiting effect.

p4078[0...1] TM31 analog outputs, characteristic value y1 / TM31 AO char y1			
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000 [V]	Access level: 2 Func. diagram: 9572 Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Sets the scaling characteristic for the analog outputs of Terminal Module 31 (TM31). The scaling characteristic for the analog outputs is defined using 2 points. This parameter specifies the y coordinate (output voltage in V or output current in mA) of the 1st value pair of the characteristic.		
Index:	[0] = AO 0 (X522.1, X522.2, X522.3) [1] = AO 1 (X522.4, X522.5, X522.6)		
Dependency:	The unit of this parameter (V or mA) depends on the analog input type. Refer to: r4076, p4076		
Notice:	This parameter is automatically overwritten when the analog output type is changed (p4076).		
Note:	This parameter is automatically overwritten if p4076 (type of analog output) is changed. The parameters for the characteristic do not have a limiting effect.		

p4079[0...1] TB30 analog outputs, characteristic value x2 / TB30 AO char x2			
TB30	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -1000.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: 9106 Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the scaling characteristic for the analog outputs of Terminal Board 30 (TB30). The scaling characteristic for the analog outputs is defined using 2 points. This parameter specifies the x coordinate (percentage) of the 2nd value pair of the characteristic.		
Index:	[0] = AO 0 (X482.5/X482.6) [1] = AO 1 (X482.7/X482.8)		
Note:	The parameters for the characteristic do not have a limiting effect.		

p4079[0...1] TM31 analog outputs, characteristic value x2 / TM31 AO char x2			
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -1000.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 2 Func. diagram: 9572 Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the scaling characteristic for the analog outputs of Terminal Module 31 (TM31). The scaling characteristic for the analog outputs is defined using 2 points. This parameter specifies the x coordinate (percentage) of the 2nd value pair of the characteristic.		
Index:	[0] = AO 0 (X522.1, X522.2, X522.3) [1] = AO 1 (X522.4, X522.5, X522.6)		
Dependency:	Refer to: r4076, p4076		
Notice:	This parameter is automatically overwritten when the analog output type is changed (p4076).		
Note:	This parameter is overwritten if p4076 (type of analog output) is changed. The parameters for the characteristic do not have a limiting effect.		

p4080[0...1]	TB30 analog outputs, characteristic value y2 / TB30 AO char y2		
TB30	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -11.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 11.000 [V]	Access level: 2 Func. diagram: 9106 Unit selection: - Expert list: 1 Factory setting 10.000 [V]
Description:	Sets the scaling characteristic for the analog outputs of Terminal Board 30 (TB30). The scaling characteristic for the analog outputs is defined using 2 points. This parameter specifies the y coordinate (output voltage in V) of the 2nd value pair of the characteristic.		
Index:	[0] = AO 0 (X482.5/X482.6) [1] = AO 1 (X482.7/X482.8)		
Note:	The parameters for the characteristic do not have a limiting effect.		
p4080[0...1]	TM31 analog outputs, characteristic value y2 / TM31 AO char y2		
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000 [V]	Access level: 2 Func. diagram: 9572 Unit selection: - Expert list: 1 Factory setting 10.000 [V]
Description:	Sets the scaling characteristic for the analog outputs of Terminal Module 31 (TM31). The scaling characteristic for the analog outputs is defined using 2 points. This parameter specifies the y coordinate (output voltage in V or output current in mA) of the 2nd value pair of the characteristic.		
Index:	[0] = AO 0 (X522.1, X522.2, X522.3) [1] = AO 1 (X522.4, X522.5, X522.6)		
Dependency:	The unit of this parameter (V or mA) depends on the analog input type. Refer to: r4076, p4076		
Notice:	This parameter is automatically overwritten when the analog output type is changed (p4076).		
Note:	This parameter is overwritten if p4076 (type of analog output) is changed. The parameters for the characteristic do not have a limiting effect.		
p4082[0...1]	BI: TB30 analog outputs invert signal source / TB30 AO inv S_src		
TB30	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Terminals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 9106 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for inverting the analog output signals of the Terminal Board 30 (TB30).		
Index:	[0] = AO 0 (X482.5/X482.6) [1] = AO 1 (X482.7/X482.8)		

p4082[0...1]	BI: TM31 analog outputs invert signal source / TM31 AO inv S_src		
TM31	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Terminals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 9572 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to invert the analog output signals of Terminal Module 31 (TM31).		
Index:	[0] = AO 0 (X522.1, X522.2, X522.3) [1] = AO 1 (X522.4, X522.5, X522.6)		
p4083[0...1]	TB30 analog outputs, offset / TB30 AO offset		
TB30	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -10.000	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10.000	Access level: 1 Func. diagram: 9106 Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the offset for the analog outputs of Terminal Board 30 (TB30). The offset is added to the output signal after the scaling characteristic.		
Index:	[0] = AO 0 (X482.5/X482.6) [1] = AO 1 (X482.7/X482.8)		
p4083[0...1]	TM31 analog outputs, offset / TM31 AO offset		
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000	Access level: 1 Func. diagram: 9572 Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the offset for the analog outputs of Terminal Module 31 (TM31). The offset is added to the output signal after the scaling characteristic.		
Index:	[0] = AO 0 (X522.1, X522.2, X522.3) [1] = AO 1 (X522.4, X522.5, X522.6)		
Dependency:	The unit of this parameter (V or mA) depends on the analog input type. Refer to: r4076, p4076		
Note:	This means, for example, the offset of a downstream isolating amplifier can be compensated.		
p4086	BI: TM15DI/DO signal source for terminal DI/DO 16 / TM15D S_srcDI/DO16		
TM15DI_DO	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: 9402 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for terminal DI/DO 16 (X522.2) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.16 = 1). DI/DO: Bidirectional digital input/output		

p4087	BI: TM15DI/DO signal source for terminal DI/DO 17 / TM15D S_srcDI/DO17		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 17 (X522.3) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.17 = 1). DI/DO: Bidirectional digital input/output		
p4088	BI: TM15DI/DO signal source for terminal DI/DO 18 / TM15D S_srcDI/DO18		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 18 (X522.4) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.18 = 1). DI/DO: Bidirectional digital input/output		
p4089	BI: TM15DI/DO signal source for terminal DI/DO 19 / TM15D S_srcDI/DO19		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 19 (X522.5) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.19 = 1). DI/DO: Bidirectional digital input/output		
p4090	BI: TM15DI/DO signal source for terminal DI/DO 20 / TM15D S_srcDI/DO20		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for terminal DI/DO 20 (X522.6) of Terminal Module 15 (TM15).		
Note:	Prerequisite: The DI/DO must be set as an output (p4028.20 = 1). DI/DO: Bidirectional digital input/output		

p4091	BI: TM15DI/DO signal source for terminal DI/DO 21 / TM15D S_srcDI/DO21		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for terminal DI/DO 21 (X522.7) of Terminal Module 15 (TM15).

Note: Prerequisite: The DI/DO must be set as an output (p4028.21 = 1).
DI/DO: Bidirectional digital input/output

p4092	BI: TM15DI/DO signal source for terminal DI/DO 22 / TM15D S_srcDI/DO22		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for terminal DI/DO 22 (X522.8) of Terminal Module 15 (TM15).

Note: Prerequisite: The DI/DO must be set as an output (p4028.22 = 1).
DI/DO: Bidirectional digital input/output

p4093	BI: TM15DI/DO signal source for terminal DI/DO 23 / TM15D S_srcDI/DO23		
TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 1
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 9402
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for terminal DI/DO 23 (X522.9) of Terminal Module 15 (TM15).

Note: Prerequisite: The DI/DO must be set as an output (p4028.23 = 1).
DI/DO: Bidirectional digital input/output

r4094.0...23	BO: TM15 digital inputs status inverted raw data internal / TM15 DI st raw dat		
TM15DI_DO	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the inverted status of the raw data of the digital inputs of the Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	High	Low	-
	01	DI/DO 1 (X520.3)	High	Low	-
	02	DI/DO 2 (X520.4)	High	Low	-
	03	DI/DO 3 (X520.5)	High	Low	-
	04	DI/DO 4 (X520.6)	High	Low	-
	05	DI/DO 5 (X520.7)	High	Low	-
	06	DI/DO 6 (X520.8)	High	Low	-
	07	DI/DO 7 (X520.9)	High	Low	-
	08	DI/DO 8 (X521.2)	High	Low	-
	09	DI/DO 9 (X521.3)	High	Low	-
	10	DI/DO 10 (X521.4)	High	Low	-

11	DI/DO 11 (X521.5)	High	Low	-
12	DI/DO 12 (X521.6)	High	Low	-
13	DI/DO 13 (X521.7)	High	Low	-
14	DI/DO 14 (X521.8)	High	Low	-
15	DI/DO 15 (X521.9)	High	Low	-
16	DI/DO 16 (X522.2)	High	Low	-
17	DI/DO 17 (X522.3)	High	Low	-
18	DI/DO 18 (X522.4)	High	Low	-
19	DI/DO 19 (X522.5)	High	Low	-
20	DI/DO 20 (X522.6)	High	Low	-
21	DI/DO 21 (X522.7)	High	Low	-
22	DI/DO 22 (X522.8)	High	Low	-
23	DI/DO 23 (X522.9)	High	Low	-

Notice: The raw data of the digital inputs is directly displayed (e.g. without any debounce).

Note: Should only used for internal Siemens purposes (alternative r4022, r4023).

p4095 TB30 digital inputs, simulation mode / TB30 DI sim_mode

TB30	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1790, 9100
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the simulation mode for the digital inputs of the Terminal Board 30 (TB30).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X481.1)	Simulation	Terminal eval.	-
	01	DI 1 (X481.2)	Simulation	Terminal eval.	-
	02	DI 2 (X481.3)	Simulation	Terminal eval.	-
	03	DI 3 (X481.4)	Simulation	Terminal eval.	-

Dependency: The setpoint for the input signals is specified using p4096.

Refer to: p4096

Note: This parameter is not saved when data is backed-up (p0971, p0977).

DI: Digital input

p4095 TM15DI/DO digital inputs, simulation mode / TM15D DI sim_mode

TM15DI_DO	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9400, 9401, 9402
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the simulation mode for the digital inputs of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	Simulation	Terminal eval.	-
	01	DI/DO 1 (X520.3)	Simulation	Terminal eval.	-
	02	DI/DO 2 (X520.4)	Simulation	Terminal eval.	-
	03	DI/DO 3 (X520.5)	Simulation	Terminal eval.	-
	04	DI/DO 4 (X520.6)	Simulation	Terminal eval.	-
	05	DI/DO 5 (X520.7)	Simulation	Terminal eval.	-
	06	DI/DO 6 (X520.8)	Simulation	Terminal eval.	-
	07	DI/DO 7 (X520.9)	Simulation	Terminal eval.	-
	08	DI/DO 8 (X521.2)	Simulation	Terminal eval.	-
	09	DI/DO 9 (X521.3)	Simulation	Terminal eval.	-
	10	DI/DO 10 (X521.4)	Simulation	Terminal eval.	-
	11	DI/DO 11 (X521.5)	Simulation	Terminal eval.	-
	12	DI/DO 12 (X521.6)	Simulation	Terminal eval.	-

13	DI/DO 13 (X521.7)	Simulation	Terminal eval.	-
14	DI/DO 14 (X521.8)	Simulation	Terminal eval.	-
15	DI/DO 15 (X521.9)	Simulation	Terminal eval.	-
16	DI/DO 16 (X522.2)	Simulation	Terminal eval.	-
17	DI/DO 17 (X522.3)	Simulation	Terminal eval.	-
18	DI/DO 18 (X522.4)	Simulation	Terminal eval.	-
19	DI/DO 19 (X522.5)	Simulation	Terminal eval.	-
20	DI/DO 20 (X522.6)	Simulation	Terminal eval.	-
21	DI/DO 21 (X522.7)	Simulation	Terminal eval.	-
22	DI/DO 22 (X522.8)	Simulation	Terminal eval.	-
23	DI/DO 23 (X522.9)	Simulation	Terminal eval.	-

Dependency: The setpoint for the input signals is specified using p4096.

Refer to: p4096

Warning:



A drive that is moved by simulating the inputs of a Terminal Module is brought to a standstill while the Terminal Module is being activated or deactivated.

Note:

This parameter is not saved when data is backed-up (p0971, p0977).

DI/DO: Bidirectional digital input/output

p4095 TM31 digital inputs, simulation mode / TM31 DI sim_mode

TM31	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1840, 9550, 9552, 9560, 9562
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the simulation mode for the digital inputs of Terminal Module 31 (TM31).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X520.1)	Simulation	Terminal eval.	-
	01	DI 1 (X520.2)	Simulation	Terminal eval.	-
	02	DI 2 (X520.3)	Simulation	Terminal eval.	-
	03	DI 3 (X520.4)	Simulation	Terminal eval.	-
	04	DI 4 (X530.1)	Simulation	Terminal eval.	-
	05	DI 5 (X530.2)	Simulation	Terminal eval.	-
	06	DI 6 (X530.3)	Simulation	Terminal eval.	-
	07	DI 7 (X530.4)	Simulation	Terminal eval.	-
	08	DI/DO 8 (X541.2)	Simulation	Terminal eval.	-
	09	DI/DO 9 (X541.3)	Simulation	Terminal eval.	-
	10	DI/DO 10 (X541.4)	Simulation	Terminal eval.	-
	11	DI/DO 11 (X541.5)	Simulation	Terminal eval.	-

Dependency: The setpoint for the input signals is specified using p4096.

Refer to: p4096

Warning:



A drive that is moved by simulating the inputs of a Terminal Module is brought to a standstill while the Terminal Module is being activated or deactivated.

Note:

This parameter is not saved when data is backed-up (p0971, p0977).

DI: Digital input

DI/DO: Bidirectional digital input/output

p4095 TM41 digital inputs, simulation mode / TM41 DI sim_mode

TM41	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the simulation mode for the digital inputs of Terminal Module 41 (TM41).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X522.1)	Simulation	Terminal eval.	9660
	01	DI 1 (X522.2)	Simulation	Terminal eval.	9660
	02	DI 2 (X522.3)	Simulation	Terminal eval.	9660
	03	DI 3 (X522.4)	Simulation	Terminal eval.	9660
	08	DI/DO 0 (X521.1)	Simulation	Terminal eval.	9661
	09	DI/DO 1 (X521.2)	Simulation	Terminal eval.	9661
	10	DI/DO 2 (X521.3)	Simulation	Terminal eval.	9662
	11	DI/DO 3 (X521.4)	Simulation	Terminal eval.	9662

Dependency: The setpoint for the input signals is specified using p4096.
Refer to: p4096

Warning: A drive that is moved by simulating the inputs of a Terminal Module is brought to a standstill while the Terminal Module is being activated or deactivated.



Note: This parameter is not saved when data is backed-up (p0971, p0977).
DI: Digital input
DI/DO: Bidirectional digital input/output

p4096 TB30 digital inputs, simulation mode setpoint / TB30 DI sim setpt

TB30	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 1790, 9100
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the setpoint for the input signals in the simulation mode of the digital inputs of the Terminal Board 30 (TB30).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X481.1)	High	Low	-
	01	DI 1 (X481.2)	High	Low	-
	02	DI 2 (X481.3)	High	Low	-
	03	DI 3 (X481.4)	High	Low	-

Dependency: The simulation of a digital input is selected using p4095.
Refer to: p4095

Note: This parameter is not saved when data is backed-up (p0971, p0977).
DI: Digital input

p4096	TM15DI/DO digital inputs, simulation mode, setpoint / TM15D DI sim setp				
TM15DI_DO	Can be changed: U, T		Calculated: -	Access level: 2	
	Data type: Unsigned32		Dynamic index: -	Func. diagram: 9400, 9401, 9402	
	P-Group: Terminals		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0000 bin	
Description:	Sets the setpoint for the input signals in the simulation mode of the digital inputs of Terminal Module 15 (TM15).				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	High	Low	-
	01	DI/DO 1 (X520.3)	High	Low	-
	02	DI/DO 2 (X520.4)	High	Low	-
	03	DI/DO 3 (X520.5)	High	Low	-
	04	DI/DO 4 (X520.6)	High	Low	-
	05	DI/DO 5 (X520.7)	High	Low	-
	06	DI/DO 6 (X520.8)	High	Low	-
	07	DI/DO 7 (X520.9)	High	Low	-
	08	DI/DO 8 (X521.2)	High	Low	-
	09	DI/DO 9 (X521.3)	High	Low	-
	10	DI/DO 10 (X521.4)	High	Low	-
	11	DI/DO 11 (X521.5)	High	Low	-
	12	DI/DO 12 (X521.6)	High	Low	-
	13	DI/DO 13 (X521.7)	High	Low	-
	14	DI/DO 14 (X521.8)	High	Low	-
	15	DI/DO 15 (X521.9)	High	Low	-
	16	DI/DO 16 (X522.2)	High	Low	-
	17	DI/DO 17 (X522.3)	High	Low	-
	18	DI/DO 18 (X522.4)	High	Low	-
	19	DI/DO 19 (X522.5)	High	Low	-
	20	DI/DO 20 (X522.6)	High	Low	-
	21	DI/DO 21 (X522.7)	High	Low	-
	22	DI/DO 22 (X522.8)	High	Low	-
	23	DI/DO 23 (X522.9)	High	Low	-
Dependency:	The simulation of a digital input is selected using p4095. Refer to: p4095				
Note:	This parameter is not saved when data is backed-up (p0971, p0977). DI/DO: Bidirectional digital input/output				

p4096	TM31 digital inputs, simulation mode setpoint / TM31 DI sim setp				
TM31	Can be changed: U, T		Calculated: -	Access level: 2	
	Data type: Unsigned32		Dynamic index: -	Func. diagram: 1840, 9550, 9552, 9560, 9562	
	P-Group: Terminals		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0000 bin	
Description:	Sets the setpoint for the input signals in the simulation mode of the digital inputs of Terminal Module 31 (TM31).				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X520.1)	High	Low	-
	01	DI 1 (X520.2)	High	Low	-
	02	DI 2 (X520.3)	High	Low	-
	03	DI 3 (X520.4)	High	Low	-
	04	DI 4 (X530.1)	High	Low	-
	05	DI 5 (X530.2)	High	Low	-
	06	DI 6 (X530.3)	High	Low	-

07	DI 7 (X530.4)	High	Low	-
08	DI/DO 8 (X541.2)	High	Low	-
09	DI/DO 9 (X541.3)	High	Low	-
10	DI/DO 10 (X541.4)	High	Low	-
11	DI/DO 11 (X541.5)	High	Low	-

Dependency: The simulation of a digital input is selected using p4095.

Refer to: p4095

Note: This parameter is not saved when data is backed-up (p0971, p0977).

DI: Digital input

DI/DO: Bidirectional digital input/output

p4096 TM41 digital inputs, simulation mode setpoint / TM41 DI sim setp

TM41	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the setpoint for the input signals in the simulation mode of the digital inputs of Terminal Module 41 (TM41).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 0 (X522.1)	High	Low	9660
	01	DI 1 (X522.2)	High	Low	9660
	02	DI 2 (X522.3)	High	Low	9660
	03	DI 3 (X522.4)	High	Low	9660
	08	DI/DO 0 (X521.1)	High	Low	9661
	09	DI/DO 1 (X521.2)	High	Low	9661
	10	DI/DO 2 (X521.3)	High	Low	9662
	11	DI/DO 3 (X521.4)	High	Low	9662

Dependency: The simulation of a digital input is selected using p4095.

Refer to: p4095

Note: This parameter is not saved when data is backed-up (p0971, p0977).

DI: Digital input

DI/DO: Bidirectional digital input/output

p4097[0...1] TB30 analog inputs simulation mode / TB30 AI sim_mode

TB30	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: -	Func. diagram: 9104
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0

Description: Sets the simulation mode for the analog inputs of the Terminal Board 30 (TB30).

Value:
 0: No simulation mode for analog input x
 1: Simulation mode for analog input x

Index:
 [0] = AI 0 (X482.1/X482.2)
 [1] = AI 1 (X482.3/X482.4)

Dependency: The setpoint for the input voltage is specified via p4098.

Refer to: p4098

Note: This parameter is not saved when data is backed-up (p0971, p0977).

AI: Analog Input

p4097[0...1] TM31 analog inputs simulation mode / TM31 AI sim_mode			
TM31	Can be changed: U, T Data type: Integer16 P-Group: Terminals Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the simulation mode for the analog inputs of Terminal Module 31 (TM31).		
Value:	0: No simulation mode for analog input x 1: Simulation mode for analog input x		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
Dependency:	The setpoint for the input voltage is specified via p4098. Refer to: p4098		
Note:	This parameter is not saved when data is backed-up (p0971, p0977). AI: Analog Input		
p4097[0] TM41 analog input, simulation mode / TM41 AI sim_mode			
TM41	Can be changed: U, T Data type: Integer16 P-Group: Terminals Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the simulation mode for the analog input of Terminal Module 41 (TM41).		
Value:	0: No simulation mode for analog input x 1: Simulation mode for analog input x		
Index:	[0] = AI 0 (X523.1/X523.2)		
Dependency:	The setpoint for the input voltage is specified via p4098. Refer to: p4098		
Note:	This parameter is not saved when data is backed-up (p0971, p0977). AI: Analog Input		
p4098[0...1] TB30 analog inputs simulation mode setpoint / TB30 AI sim setp			
TB30	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -11.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 11.000 [V]	Access level: 2 Func. diagram: 9104 Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Sets the setpoint for the input voltage in the simulation mode of the analog inputs of Terminal Board 30 (TB30).		
Index:	[0] = AI 0 (X482.1/X482.2) [1] = AI 1 (X482.3/X482.4)		
Dependency:	The simulation of an analog input is selected using p4097. Refer to: p4097		
Note:	This parameter is not saved when data is backed-up (p0971, p0977). AI: Analog Input		

p4098[0...1] TM31 analog inputs simulation mode setpoint / TM31 AI sim setp			
TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000	Access level: 2 Func. diagram: 9566, 9568 Unit selection: - Expert list: 1 Factory setting 0.000
Description:	Sets the setpoint for the input value in simulation mode of the analog inputs of Terminal Module 31 (TM31).		
Index:	[0] = AI 0 (X521.1/X521.2, S5.0) [1] = AI 1 (X521.3/X521.4, S5.1)		
Dependency:	The simulation of an analog input is selected using p4097. If AI x is parameterized as voltage input (p4056), then the setpoint is a voltage in V. If AI x is parameterized as current input (p4056), then the setpoint is a current in mA. Refer to: r4056, p4056, p4097		
Note:	This parameter is not saved when data is backed-up (p0971, p0977). AI: Analog Input		
p4098[0] TM41 analog input, simulation mode setpoint / TM41 AI sim setp			
TM41	Can be changed: U, T Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min -20.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20.000 [V]	Access level: 2 Func. diagram: 9663 Unit selection: - Expert list: 1 Factory setting 0.000 [V]
Description:	Sets the setpoint for the input value in simulation mode of the analog input of Terminal Module 41 (TM41).		
Index:	[0] = AI 0 (X523.1/X523.2)		
Dependency:	The simulation of the analog input is selected using p4097. If AI x is parameterized as voltage input (p4056), then the setpoint is a voltage in V. If AI x is parameterized as current input (p4056), then the setpoint is a current in mA. Refer to: p4097		
Note:	This parameter is not saved when data is backed-up (p0971, p0977). AI: Analog Input		
p4099[0...2] TB30 inputs/outputs, sampling time / TB30 I/O t_sample			
TB30	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Commands Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 5000.00 [µs]	Access level: 3 Func. diagram: 1790, 9100 Unit selection: - Expert list: 1 Factory setting [0] 4000.00 [µs] [1] 4000.00 [µs] [2] 4000.00 [µs]
Description:	Sets the sampling time for the inputs and outputs of the Terminal Board 30 (TB30).		
Index:	[0] = Digital inputs/outputs (DI/DO) [1] = Analog inputs (AI) [2] = Analog outputs (AO)		
Dependency:	The parameter can only be modified for p0009 = 3, 29. The sampling times can only be set as an integer multiple of the basic sampling time (r0110, r0111). Refer to: p0009, r0110, r0111		

Note: The modified sampling time is not effective until the drive unit is powered up again.
For clock cycle synchronous PROFIBUS operation, the TB30 hardware (e.g. A/D converter) is operated with the PROFIBUS clock cycle (r2064[1]). This clock cycle is also kept after the PROFIBUS connection has been exited up to the next time that the Control Unit is powered down. In this case, a faster sampling time than the PROFIBUS clock cycle is not practical in p4099[0...2].

p4099			
TM15 inputs/outputs, sampling time / TM15 I/O t_sample			
TM15	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Commands Not for motor type: - Min 31.25 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 500.00 [µs]	Access level: 3 Func. diagram: 1780 Unit selection: - Expert list: 1 Factory setting 125.00 [µs]
Description:	The sampling time of the Terminal Module 15 (TM15) is determined by the DRIVE-CLiQ clock cycle of the line to which the component is attached. An entry is not possible using p4099. At power on, p4099 is correctly set to the resulting sampling time.		
p4099			
TM15DI/DO inputs/outputs, sampling time / TM15D I/O t_sampl			
TM15DI_DO	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Commands Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 5000.00 [µs]	Access level: 3 Func. diagram: 1781, 9400 Unit selection: - Expert list: 1 Factory setting 4000.00 [µs]
Description:	Sets the sampling time for the inputs and outputs of Terminal Module 15 (TM15).		
Dependency:	The parameter can only be modified for p0009 = 3, 29. Refer to: p0009, r0110, r0111		
Note:	- the modified sampling time is not effective until the drive unit is powered up again. - parameter p4099[0] must never equal zero.		
p4099			
TM17 inputs/outputs, sampling time / TM17 I/O t_sample			
TM17	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Commands Not for motor type: - Min 31.25 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 500.00 [µs]	Access level: 3 Func. diagram: 1782 Unit selection: - Expert list: 1 Factory setting 125.00 [µs]
Description:	The sampling time of the Terminal Module 17 (TM17) is determined by the DRIVE-CLiQ clock cycle of the line to which the component is attached. An entry is not possible using p4099. At power on, p4099 is correctly set to the resulting sampling time.		
p4099[0...2]			
TM31 inputs/outputs, sampling time / TM31 I/O t_sample			
TM31	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Commands Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 5000.00 [µs]	Access level: 3 Func. diagram: 1840, 9550 Unit selection: - Expert list: 1 Factory setting 4000.00 [µs]
Description:	Sets the sampling time for the inputs and outputs of Terminal Module 31 (TM31).		
Index:	[0] = Digital inputs/outputs (DI/DO) [1] = Analog inputs (AI) [2] = Analog outputs (AO)		

- Dependency:** The parameter can only be modified for p0009 = 3, 29.
The sampling times can only be set as an integer multiple of the basic sampling time (r0110, r0111).
Refer to: p0009, r0110, r0111
- Notice:** The sampling times entered in index 0 (digital inputs/outputs) and index 2 (analog outputs) must always be greater than or equal to the sampling time in index 1 (analog inputs).
- Note:** - the modified sampling time is not effective until the drive unit is powered up again.
- parameter p4099[0] must never equal zero.

p4099[0...3] TM41 inputs/outputs, sampling time / TM41 I/O t_sample

TM41	Can be changed: C1(3)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 1842, 9660
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [µs]	5000.00 [µs]	[0] 4000.00 [µs]
			[1] 4000.00 [µs]
			[2] 0.00 [µs]
			[3] 125.00 [µs]

Description: Sets the sampling time for the inputs and outputs of Terminal Module 41 (TM41).

Index: [0] = Digital inputs/outputs (DI/DO)
[1] = Analog inputs (AI)
[2] = Not present
[3] = Incremental encoder emulation

Dependency: The parameter can only be modified for p0009 = 3, 29.
Refer to: p0009, r0110, r0111
Refer to: F35228

Note: - the value of the sampling time of the incremental encoder emulation p4099[3] can be preset in both operating modes (p4400). The next time that the system boots, the validity of the value is checked. For an invalid value, fault F35228 is output and p4099[3] is automatically set to the correct value. The modified parameter must then be saved and POWER ON carried out. If necessary, the sampling time can be checked again, taking into account any other TM41s located on the same DRIVE-CLiQ line.
- the modified sampling time is not effective until the drive unit is powered up again.
- if there are several TM41s located on a DRIVE-CLiQ line, the same sampling time in p4099[3] must be set for all components.
- the sampling time of a TM41 in SINAMICS mode (p4400 = 1) must be the same as that of the emulated encoder.
- the sampling time of a TM41 in SIMOTION mode (p4400 = 0) is determined by the topology used
- parameter p4099[0] must never equal zero.

p4100[0...3] TM120 temperature evaluation, sensor type / TM120 sensor type

TM120	Can be changed: T	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: 9606
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	4	1

Description: Sets the temperature evaluation of Terminal Module 120 (TM120).

This means that the temperature sensor type is selected and the evaluation is switched in.

Value: 0: Evaluation disabled
1: PTC thermistor
2: KTY84
4: Bimetallic NC contact

Index: [0] = Temperature sensor channel 1
[1] = Temperature sensor channel 2
[2] = Temperature sensor channel 3
[3] = Temperature sensor channel 4

Note: The temperature sensor is connected to the following terminals:

Channel 1: X521.2(+) and X521.1(-)

Channel 2: X521.4(+) and X521.3(-)

Channel 3: X521.6(+) and X521.5(-)

Channel 4: X521.8(+) and X521.7(-)

p4100 TM31 temperature evaluation, sensor type / TM31 temp sens_typ			
TM31	Can be changed: T	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: 9576, 9577
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 2	Factory setting 0
Description:	Sets the temperature evaluation of Terminal Module 31 (TM31). This means that the temperature sensor type is selected and the evaluation is switched in.		
Value:	0: Evaluation disabled 1: PTC thermistor 2: KTY84		
Note:	The temperature sensor is connected at terminals X522.7(+) and X522.8(-).		

r4101[0...3] TM120 temperature evaluation, sensor resistance / TM120 Temp R_sen			
TM120	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 9576, 9577
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min - [Ohm]	Max - [Ohm]	Factory setting - [Ohm]
Description:	Displays the actual resistance value of the temperature sensor connected at Terminal Module 120 (TM120).		
Index:	[0] = Temperature sensor channel 1 [1] = Temperature sensor channel 2 [2] = Temperature sensor channel 3 [3] = Temperature sensor channel 4		
Note:	The maximum measurable resistance value is approx. 1700 Ohm. The temperature sensor is connected to the following terminals: Channel 1: X521.2(+) and X521.1(-) . Channel 2: X521.4(+) and X521.3(-) . Channel 3: X521.6(+) and X521.5(-) . Channel 4: X521.8(+) and X521.7(-) .		

r4101 TM31 temperature evaluation, sensor resistance / TM31 temp R_sensor			
TM31	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 9576, 9577
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min - [Ohm]	Max - [Ohm]	Factory setting - [Ohm]
Description:	Displays the actual resistance value of the temperature sensor connected at Terminal Module 31 (TM31).		
Note:	The maximum measurable resistance value is approx. 1700 Ohm. The temperature sensor is connected at terminals X522.7(+) and X522.8(-).		

p4102[0...7]		TM120 temperature evaluation fault/alarm threshold / TM120 TempThresh	
TM120	Can be changed: T Data type: Integer16 P-Group: - Not for motor type: - Min -48 [°C]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 251 [°C]	Access level: 1 Func. diagram: 9576 Unit selection: - Expert list: 1 Factory setting 251 [°C]
Description:	Sets the fault/alarm threshold for the temperature evaluation of Terminal Module 120 (TM120). Temperature actual value 1 > p4102[0] --> alarm A35211 is output. Temperature actual value 1 > p4102[1] --> fault F35207 is output. Temperature actual value 2 > p4102[2] --> alarm A35212 is output. Temperature actual value 2 > p4102[3] --> fault F35208 is output. Temperature actual value 3 > p4102[4] --> alarm A35213 is output. Temperature actual value 3 > p4102[5] --> fault F35209 is output. Temperature actual value 4 > p4102[6] --> alarm A35214 is output. Temperature actual value 4 > p4102[7] --> fault F35210 is output.		
Index:	[0] = Sensor 1 alarm threshold [1] = Sensor 1 fault threshold [2] = Sensor 2 alarm threshold [3] = Sensor 2 fault threshold [4] = Sensor 3 alarm threshold [5] = Sensor 3 fault threshold [6] = Sensor 4 alarm threshold [7] = Sensor 4 fault threshold		
Dependency:	Refer to: p4103		
Warning:	Fault F35207-F352010 only causes the drive to shut down if there is at least one BICO interconnection between the drive and TM120.		
			
Note:	A value > 250 °C deactivates the alarm or fault. The temperature sensor is connected to the following terminals: Channel 1: X521.2(+) and X521.1(-) . Channel 2: X521.4(+) and X521.3(-) . Channel 3: X521.6(+) and X521.5(-) . Channel 4: X521.8(+) and X521.7(-) .		

p4102[0...1]		TM31 temperature evaluation fault/alarm threshold / TM31 temp thresh	
TM31	Can be changed: T Data type: Integer16 P-Group: - Not for motor type: - Min -48 [°C]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 251 [°C]	Access level: 1 Func. diagram: 9576 Unit selection: - Expert list: 1 Factory setting [0] 100 [°C] [1] 120 [°C]
Description:	Sets the fault/alarm threshold for the temperature evaluation of Terminal Module 31 (TM31). Temperature actual value > p4102[0] --> alarm A35211 is output. Alarm A35211 remains present until the temperature actual value (r4105) falls below the corresponding value (p4102[0] hysteresis). Temperature actual value > p4102[1] --> fault F35207 is output. Fault F35207 remains present until the temperature actual value (r4105) falls below the corresponding value (p4102[1] hysteresis) and the fault has been acknowledged. The hysteresis value is 5 °C and cannot be changed by the user.		
Index:	[0] = Alarm threshold [1] = Fault threshold		

Dependency: Refer to: r4104

Warning: Fault F35207 only causes the drive to be shut down if there is at least one BICO interconnection between the drive and TM31.



Note: A value > 250 °C deactivates the alarm or fault.
The temperature sensor is connected at terminals X522.7(+) and X522.8(-).

p4103[0...3] TM120 temperature evaluation timer / TM120 temp t_timer

TM120	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.000 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 600.000 [ms]	Access level: 1 Func. diagram: 9576 Unit selection: - Expert list: 1 Factory setting 0.000 [ms]
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Description: Timer for the fault output of the temperature evaluation of Terminal Module 120 (TM120).
This timer is started when the temperature alarm threshold (p4102[0,2,4,6]) is exceeded.
If the timer expires before the temperature in the meantime falls below the alarm threshold, the fault F35207-F35210 is output.
If the temperature fault threshold (p4102[1,3,5,7]) is prematurely exceeded before the timer has expired, then fault F35207-F35210 is immediately output.
As long as the temperature of the TM120 has still not exceeded the fault threshold and the alarm threshold has again been undershot, the fault can be acknowledged.

Index:
 [0] = Temperature sensor channel 1
 [1] = Temperature sensor channel 2
 [2] = Temperature sensor channel 3
 [3] = Temperature sensor channel 4

Dependency: Refer to: r4104

Warning: Fault F35207-F35210 only causes the drive to shut down if there is at least one BICO interconnection between the drive and TM120.



Note: With p4103 = 0 ms, the timer is deactivated and only the fault threshold is effective.

p4103 TM31 temperature evaluation timer / TM31 temp t_timer

TM31	Can be changed: U, T Data type: FloatingPoint32 P-Group: Motor Not for motor type: - Min 0.000 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 600.000 [ms]	Access level: 1 Func. diagram: 9576 Unit selection: - Expert list: 1 Factory setting 0.000 [ms]
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Description: Timer for the fault output of the temperature evaluation of Terminal Module 31 (TM31).
This timer is started when the temperature alarm threshold (p4102[0]) is exceeded.
If the timer expires before the temperature in the meantime falls below the alarm threshold, the fault F35207 is output.
If the temperature fault threshold (p4102[1]) is prematurely exceeded before the timer has expired, then fault F35207 is immediately output.
As long as the temperature of the TM31 has still not exceeded the fault threshold and the alarm threshold has again been undershot, the fault can be acknowledged.

Dependency: Refer to: r4104

Warning: Fault F35207 only causes the drive to be shut down if there is at least one BICO interconnection between the drive and TM31.



Note: With p4103 = 0 ms, the timer is deactivated and only the fault threshold is effective.

r4104.0...7 BO: TM120 temperature evaluation, status / TM120 temp status

TM120	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1840, 9576
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status for the temperature evaluation of Terminal Module 120 (TM120).
This displays whether the temperature actual value has exceeded the fault/alarm threshold.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Temperature alarm threshold sensor 1 exceeded	Yes	No	-
	01	Temperature fault threshold sensor 1 exceeded	Yes	No	-
	02	Temperature alarm threshold sensor 2 exceeded	Yes	No	-
	03	Temperature fault threshold sensor 2 exceeded	Yes	No	-
	04	Temperature alarm threshold sensor 3 exceeded	Yes	No	-
	05	Temperature fault threshold sensor 3 exceeded	Yes	No	-
	06	Temperature alarm threshold sensor 4 exceeded	Yes	No	-
	07	Temperature fault threshold sensor 4 exceeded	Yes	No	-

Dependency: Refer to: p4102

r4104.0...1 BO: TM31 temperature evaluation, status / TM31 temp status

TM31	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 1840, 9576
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status for the temperature evaluation of Terminal Module 31 (TM31).
This displays whether the temperature actual value has exceeded the fault/alarm threshold.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Temperature alarm threshold exceeded	Yes	No	-
	01	Temperature fault threshold exceeded	Yes	No	-

Dependency: Refer to: p4102

r4105[0...3] CO: TM120 temperature evaluation, actual value / TM120 temp act val

TM120	Can be changed: -	Calculated: -	Access level: 1
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 1840, 9576
	P-Group: Terminals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min	Max	Factory setting
	- [°C]	- [°C]	- [°C]

Description: Displays the actual temperature value of the temperature evaluation of Terminal Module 120 (TM120).

Index:
[0] = Temperature sensor channel 1
[1] = Temperature sensor channel 2
[2] = Temperature sensor channel 3
[3] = Temperature sensor channel 4

Dependency:	For sensor type PTC (p4100 = 1) and bimetal (p4100 = 4), the following applies: - below the nominal response temperature, r4105 = -50 °C. - above the nominal response temperature, r4105 = 250 °C. For sensor type KTY84-130 (p4100 = 2), the following applies: - the displayed value corresponds to the temperature actual value. Refer to: p4100
Note:	For an invalid temperature actual value, (e.g. a short-circuit at the sensor input), r4105 = -300 °C is displayed. The temperature sensor is connected to the following terminals: Channel 1: X521.2(+) and X521.1(-) . Channel 2: X521.4(+) and X521.3(-) . Channel 3: X521.6(+) and X521.5(-) . Channel 4: X521.8(+) and X521.7(-) .

r4105	CO: TM31 temperature evaluation, actual value / TM31 temp actValue		
TM31	Can be changed: - Data type: FloatingPoint32 P-Group: Terminals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: - Units group: - Scaling: TEMP Max - [°C]	Access level: 1 Func. diagram: 1840, 9576 Unit selection: - Expert list: 1 Factory setting - [°C]
Description:	Displays the actual temperature value of the temperature evaluation of Terminal Module 31 (TM31).		
Dependency:	For sensor type PTC (p4100 = 1), the following applies: - below the nominal response temperature, r4105 = -50 °C. - above the nominal response temperature, r4105 = 250 °C. For sensor type KTY84-130 (p4100 = 2), the following applies: - the displayed value corresponds to the temperature actual value. Refer to: p4100		
Note:	For an invalid temperature actual value, (e.g. a short-circuit at the sensor input), r4105 = -300 °C is displayed. The temperature sensor is connected at terminals X522.7(+) and X522.8(-).		

r4154	TM41 diagnostics speed setpoint non-filtered / n_set non-filt		
TM41	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [rpm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [rpm]
Description:	Displays the unfiltered speed setpoint N_SETPT in revolutions per minute for diagnostic purposes. In contrast to p1155, this value is updated in each DRIVE-CLiQ basic clock cycle and displayed with sign.		
Dependency:	Refer to: r4155		
Note:	The parameter is not effective in the SINAMICS operating mode (p4400 = 1).		

r4155	TM41 diagnostics speed setpoint / TM41 Diag n_set		
TM41	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [rpm]	Access level: 3 Func. diagram: 9674 Unit selection: - Expert list: 1 Factory setting - [rpm]
Description:	Displays the filtered speed setpoint N_SETPT in revolutions per minute for diagnostic purposes. In contrast to p1155, this value is updated in each DRIVE-CLiQ basic clock cycle and displayed with sign.		
Dependency:	Refer to: r4154		
Note:	The parameter is not effective in the SINAMICS operating mode (p4400 = 1).		

r4201 TM15 system time for synchronization / TM15 t_system sync

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Is used to synchronize the timer of Terminal Module 15 (TM15) with the system time of the DP master.
To do this, the sign-of-life of the DP master is transferred in the form of a counter in bits 12 to 15.
At each cycle of the system of the DP master, bit 0 (SYN signal) is set for the duration of a DP master clock cycle.

r4201 TM17 system time for synchronization / TM17 t_system sync

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Is used to synchronize the timer of Terminal Module 17 (TM17) with the system time of the DP master.
To do this, the sign-of-life of the DP master is transferred in the form of a counter in bits 12 to 15.
At each cycle of the system of the DP master, bit 0 (SYN signal) is set for the duration of a DP master clock cycle.

r4204 TM15 control digital output 0 ... 15 / TM15 ctrl DO 0-15

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Used to control digital output 0 ... 15 of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	On	Off	-
	01	DI/DO 1 (X520.3)	On	Off	-
	02	DI/DO 2 (X520.4)	On	Off	-
	03	DI/DO 3 (X520.5)	On	Off	-
	04	DI/DO 4 (X520.6)	On	Off	-
	05	DI/DO 5 (X520.7)	On	Off	-
	06	DI/DO 6 (X520.8)	On	Off	-
	07	DI/DO 7 (X520.9)	On	Off	-
	08	DI/DO 8 (X521.2)	On	Off	-
	09	DI/DO 9 (X521.3)	On	Off	-
	10	DI/DO 10 (X522.4)	On	Off	-
	11	DI/DO 11 (X521.5)	On	Off	-
	12	DI/DO 12 (X521.6)	On	Off	-
	13	DI/DO 13 (X521.7)	On	Off	-
	14	DI/DO 14 (X521.8)	On	Off	-
	15	DI/DO 15 (X521.9)	On	Off	-

Note: DI/DO: Bidirectional digital input/output

r4204 TM17 control digital output 0 ... 15 / TM17 ctrl DO 0-15

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Used to control digital output 0 ... 15 of Terminal Module 17 (TM17).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	On	Off	-
	01	DI/DO 1 (X520.3)	On	Off	-
	02	DI/DO 2 (X520.5)	On	Off	-
	03	DI/DO 3 (X520.6)	On	Off	-
	04	DI/DO 4 (X520.8)	On	Off	-
	05	DI/DO 5 (X520.9)	On	Off	-
	06	DI/DO 6 (X521.2)	On	Off	-
	07	DI/DO 7 (X521.3)	On	Off	-
	08	DI/DO 8 (X521.8)	On	Off	-
	09	DI/DO 9 (X521.9)	On	Off	-
	10	DI/DO 10 (X522.2)	On	Off	-
	11	DI/DO 11 (X522.3)	On	Off	-
	12	DI/DO 12 (X522.5)	On	Off	-
	13	DI/DO 13 (X522.6)	On	Off	-
	14	DI/DO 14 (X522.8)	On	Off	-
	15	DI/DO 15 (X522.9)	On	Off	-

Note: DI/DO: Bidirectional digital input/output

r4205 TM15 control digital output 16 ... 23 / TM15 ctrl DO 16-23

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Used to control digital output 16 ... 23 of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 16 (X522.2)	On	Off	-
	01	DI/DO 17 (X522.3)	On	Off	-
	02	DI/DO 18 (X522.4)	On	Off	-
	03	DI/DO 19 (X522.5)	On	Off	-
	04	DI/DO 20 (X522.6)	On	Off	-
	05	DI/DO 21 (X522.7)	On	Off	-
	06	DI/DO 22 (X522.8)	On	Off	-
	07	DI/DO 23 (X522.9)	On	Off	-

Note: DI/DO: Bidirectional digital input/output

r4211	TM15 edge mode digital input 0 ... 7 / TM15 EdgMd DI0-7		
TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the edge mode for digital input 0 ... 7 of Terminal Module 15 (TM15).
Assignment of the digital inputs to the bits:

DI 0: r4211.1 ... 0

DI 1: r4211.3 ... 2

DI 2: r4211.5 ... 4

DI 3: r4211.7 ... 6

DI 4: r4211.9 ... 8

DI 5: r4211.11 ... 10

DI 6: r4211.13 ... 12

DI 7: r4211.15 ... 14

Possible edge modes:

Bit x, y = 0, 0 --> no edge detection

Bit x, y = 0, 1 --> rising - rising edge

Bit x, y = 1, 0 --> falling - falling edge

Bit x, y = 1, 1 --> rising - falling edge or falling - rising edge

Note: DI: Digital input

r4211	TM17 edge mode digital input 0 ... 7 / TM17 EdgMd DI 0-7		
TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the edge mode for digital input 0 ... 7 of Terminal Module 17 (TM17).
Assignment of the digital inputs to the bits:

DI 0: r4211.1 ... 0

DI 1: r4211.3 ... 2

DI 2: r4211.5 ... 4

DI 3: r4211.7 ... 6

DI 4: r4211.9 ... 8

DI 5: r4211.11 ... 10

DI 6: r4211.13 ... 12

DI 7: r4211.15 ... 14

Possible edge modes:

Bit x, y = 0, 0 --> no edge detection

Bit x, y = 0, 1 --> rising - rising edge

Bit x, y = 1, 0 --> falling - falling edge

Bit x, y = 1, 1 --> rising - falling edge or falling - rising edge

Note: DI: Digital input

r4212	TM15 edge mode digital input 8 ... 15 / TM15 EdgMd DI8-15		
TM15	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -

Description: Displays the edge mode for digital input 8 ... 15 of Terminal Module 15 (TM15).
 Assignment of the digital inputs to the bits:
 DI 8: r4212.1 ... 0
 DI 9: r4212.3 ... 2
 DI 10: r4212.5 ... 4
 DI 11: r4212.7 ... 6
 DI 12: r4212.9 ... 8
 DI 13: r4212.11 ... 10
 DI 14: r4212.13 ... 12
 DI 15: r4212.15 ... 14
 Possible edge modes:
 Bit x, y = 0, 0 --> no edge detection
 Bit x, y = 0, 1 --> rising - rising edge
 Bit x, y = 1, 0 --> falling - falling edge
 Bit x, y = 1, 1 --> rising - falling edge or falling - rising edge

Note: DI: Digital input

r4212	TM17 edge mode digital input 8 ... 15 / TM17 EdgMd DI 8-15		
TM17	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -

Description: Displays the edge mode for digital input 8 ... 15 of Terminal Module 17 (TM17).
 Assignment of the digital inputs to the bits:
 DI 8: r4212.1 ... 0
 DI 9: r4212.3 ... 2
 DI 10: r4212.5 ... 4
 DI 11: r4212.7 ... 6
 DI 12: r4212.9 ... 8
 DI 13: r4212.11 ... 10
 DI 14: r4212.13 ... 12
 DI 15: r4212.15 ... 14
 Possible edge modes:
 Bit x, y = 0, 0 --> no edge detection
 Bit x, y = 0, 1 --> rising - rising edge
 Bit x, y = 1, 0 --> falling - falling edge
 Bit x, y = 1, 1 --> rising - falling edge or falling - rising edge

Note: DI: Digital input

r4213 TM15 edge mode digital input 16 ... 23 / TM15 EdgMd DI16-23

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the edge mode for digital input 16 ... 23 of Terminal Module 15 (TM15).
Assignment of the digital inputs to the bits:

DI 16: r4213.1 ... 0

DI 17: r4213.3 ... 2

DI 18: r4213.5 ... 4

DI 19: r4213.7 ... 6

DI 20: r4213.9 ... 8

DI 21: r4213.11 ... 10

DI 22: r4213.13 ... 12

DI 23: r4213.15 ... 14

Possible edge modes:

Bit x, y = 0, 0 --> no edge detection

Bit x, y = 0, 1 --> rising - rising edge

Bit x, y = 1, 0 --> falling - falling edge

Bit x, y = 1, 1 --> rising - falling edge or falling - rising edge

Note: DI: Digital input

p4220 TM17 enable DI/DO 0 ... 5 / TM17 enable 0-5

TM17	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the enable signal with bits 0 ... 5 for DI/DO 0 ... 5 of Terminal Module 17 (TM17).

Sets the triggering of the enable signal with bits 8 ... 13.

The following assignment applies:

Enable signal for DI/DO 0, 1, 2, 3, 4 or 5 via DI/DO 10, 11, 12, 13, 14 or 15.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	With enable	Without enable	-
	01	DI/DO 1 (X520.3)	With enable	Without enable	-
	02	DI/DO 2 (X520.5)	With enable	Without enable	-
	03	DI/DO 3 (X520.6)	With enable	Without enable	-
	04	DI/DO 4 (X520.8)	With enable	Without enable	-
	05	DI/DO 5 (X520.9)	With enable	Without enable	-
	08	DI/DO 10 (X522.2)	Level-triggered	Edge-triggered	-
	09	DI/DO 11 (X522.3)	Level-triggered	Edge-triggered	-
	10	DI/DO 12 (X522.5)	Level-triggered	Edge-triggered	-
	11	DI/DO 13 (X522.6)	Level-triggered	Edge-triggered	-
	12	DI/DO 14 (X522.8)	Level-triggered	Edge-triggered	-
	13	DI/DO 15 (X522.9)	Level-triggered	Edge-triggered	-

Note: DI/DO: Bidirectional digital input/output

p4221	TM17 smoothing time constant, digital input 0 ... 15 / TM17 smooth DI				
TM17	Can be changed: U, T	Calculated: -	Access level: 2		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Sets the smoothing time constant for digital input 0 ... 15 of Terminal Module 17 (TM17).				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	Smoothing 1 µs	Smoothing 125 µs	-
	01	DI/DO 1 (X520.3)	Smoothing 1 µs	Smoothing 125 µs	-
	02	DI/DO 2 (X520.5)	Smoothing 1 µs	Smoothing 125 µs	-
	03	DI/DO 3 (X520.6)	Smoothing 1 µs	Smoothing 125 µs	-
	04	DI/DO 4 (X520.8)	Smoothing 1 µs	Smoothing 125 µs	-
	05	DI/DO 5 (X520.9)	Smoothing 1 µs	Smoothing 125 µs	-
	06	DI/DO 6 (X521.2)	Smoothing 1 µs	Smoothing 125 µs	-
	07	DI/DO 7 (X521.3)	Smoothing 1 µs	Smoothing 125 µs	-
	08	DI/DO 8 (X521.8)	Smoothing 1 µs	Smoothing 125 µs	-
	09	DI/DO 9 (X521.9)	Smoothing 1 µs	Smoothing 125 µs	-
	10	DI/DO 10 (X522.2)	Smoothing 1 µs	Smoothing 125 µs	-
	11	DI/DO 11 (X522.3)	Smoothing 1 µs	Smoothing 125 µs	-
	12	DI/DO 12 (X522.5)	Smoothing 1 µs	Smoothing 125 µs	-
	13	DI/DO 13 (X522.6)	Smoothing 1 µs	Smoothing 125 µs	-
	14	DI/DO 14 (X522.8)	Smoothing 1 µs	Smoothing 125 µs	-
	15	DI/DO 15 (X522.9)	Smoothing 1 µs	Smoothing 125 µs	-
Note:	DI/DO: Bidirectional digital input/output DI: Digital input				

p4222	TM17 time absolute/relative digital output 0 ... 15 / TM17 abs/rel 0-15				
TM17	Can be changed: T	Calculated: -	Access level: 2		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -		
	P-Group: Commands	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	0000 bin		
Description:	Sets as absolute or relative timing with bit 0 ... 15 for digital output 0 ... 15 of Terminal Module 17 (TM17).				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	Relative time	Absolute time	-
	01	DI/DO 1 (X520.3)	Relative time	Absolute time	-
	02	DI/DO 2 (X520.5)	Relative time	Absolute time	-
	03	DI/DO 3 (X520.6)	Relative time	Absolute time	-
	04	DI/DO 4 (X520.8)	Relative time	Absolute time	-
	05	DI/DO 5 (X520.9)	Relative time	Absolute time	-
	06	DI/DO 6 (X521.2)	Relative time	Absolute time	-
	07	DI/DO 7 (X521.3)	Relative time	Absolute time	-
	08	DI/DO 8 (X521.8)	Relative time	Absolute time	-
	09	DI/DO 9 (X521.9)	Relative time	Absolute time	-
	10	DI/DO 10 (X522.2)	Relative time	Absolute time	-
	11	DI/DO 11 (X522.3)	Relative time	Absolute time	-
	12	DI/DO 12 (X522.5)	Relative time	Absolute time	-
	13	DI/DO 13 (X522.6)	Relative time	Absolute time	-
	14	DI/DO 14 (X522.8)	Relative time	Absolute time	-
	15	DI/DO 15 (X522.9)	Relative time	Absolute time	-
Note:	DI/DO: Bidirectional digital input/output DO: Digital output				

r4250 TM15 set/reset time digital output 0 / TM15 t_set DO 0

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 0 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4250 TM17 set/reset time digital output 0 / TM17 t_set DO 0

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 0 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4251 TM15 set/reset time digital output 1 / TM15 t_set DO 1

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 1 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4251 TM17 set/reset time digital output 1 / TM17 t_set DO 1

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 1 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4252 TM15 set/reset time digital output 2 / TM15 t_set DO 2

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 2 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4252 TM17 set/reset time digital output 2 / TM17 t_set DO 2

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 2 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4253 TM15 set/reset time digital output 3 / TM15 t_set DO 3

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 3 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4253 TM17 set/reset time digital output 3 / TM17 t_set DO 3

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 3 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4254 TM15 set/reset time digital output 4 / TM15 t_set DO 4

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 4 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4254 TM17 set/reset time digital output 4 / TM17 t_set DO 4

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 4 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4255 TM15 set/reset time digital output 5 / TM15 t_set DO 5

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 5 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4255 TM17 set/reset time digital output 5 / TM17 t_set DO 5

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 5 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4256 TM15 set/reset time digital output 6 / TM15 t_set DO 6

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 6 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4256 TM17 set/reset time digital output 6 / TM17 t_set DO 6

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 6 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4257 TM15 set/reset time digital output 7 / TM15 t_set DO 7

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 7 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4257 TM17 set/reset time digital output 7 / TM17 t_set DO 7

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 7 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4258 TM15 set/reset time digital output 8 / TM15 t_set DO 8

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 8 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4258 TM17 set/reset time digital output 8 / TM17 t_set DO 8

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 8 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4259 TM15 set/reset time digital output 9 / TM15 t_set DO 9

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 9 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4259 TM17 set/reset time digital output 9 / TM17 t_set DO 9

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 9 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4260	TM15 set/reset time digital output 10 / TM15 t_set DO 10		
TM15	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 10 of Terminal Module 15 (TM15). The two times are specified as 8 bit values with a resolution of 64 µs.		
Note:	DO: Digital output		

r4260	TM17 set/reset time digital output 10 / TM17 t_set DO 10		
TM17	Can be changed: - Data type: Unsigned32 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 10 of Terminal Module 17 (TM17). The two times are specified as 16 bit values with a resolution of 0.25 µs.		
Note:	DO: Digital output		

r4261	TM15 set/reset time digital output 11 / TM15 t_set DO 11		
TM15	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 11 of Terminal Module 15 (TM15). The two times are specified as 8 bit values with a resolution of 64 µs.		
Note:	DO: Digital output		

r4261	TM17 set/reset time digital output 11 / TM17 t_set DO 11		
TM17	Can be changed: - Data type: Unsigned32 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 11 of Terminal Module 17 (TM17). The two times are specified as 16 bit values with a resolution of 0.25 µs.		
Note:	DO: Digital output		

r4262 TM15 set/reset time digital output 12 / TM15 t_set DO 12

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 12 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4262 TM17 set/reset time digital output 12 / TM17 t_set DO 12

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 12 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4263 TM15 set/reset time digital output 13 / TM15 t_set DO 13

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 13 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4263 TM17 set/reset time digital output 13 / TM17 t_set DO 13

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 13 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DO: Digital output

r4264	TM15 set/reset time digital output 14 / TM15 t_set DO 14		
TM15	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 14 of Terminal Module 15 (TM15). The two times are specified as 8 bit values with a resolution of 64 µs.		
Note:	DO: Digital output		

r4264	TM17 set/reset time digital output 14 / TM17 t_set DO 14		
TM17	Can be changed: - Data type: Unsigned32 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 14 of Terminal Module 17 (TM17). The two times are specified as 16 bit values with a resolution of 0.25 µs.		
Note:	DO: Digital output		

r4265	TM15 set/reset time digital output 15 / TM15 t_set DO 15		
TM15	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 15 of Terminal Module 15 (TM15). The two times are specified as 8 bit values with a resolution of 64 µs.		
Note:	DO: Digital output		

r4265	TM17 set/reset time digital output 15 / TM17 t_set DO 15		
TM17	Can be changed: - Data type: Unsigned32 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 15 of Terminal Module 17 (TM17). The two times are specified as 16 bit values with a resolution of 0.25 µs.		
Note:	DO: Digital output		

r4266 TM15 set/reset time digital output 16 / TM15 t_set DO 16

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 16 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4267 TM15 set/reset time digital output 17 / TM15 t_set DO 17

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 17 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4268 TM15 set/reset time digital output 18 / TM15 t_set DO 18

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 18 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4269 TM15 set/reset time digital output 19 / TM15 t_set DO 19

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time to set and reset for digital output 19 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DO: Digital output

r4270	TM15 set/reset time digital output 20 / TM15 t_set DO 20		
TM15	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 20 of Terminal Module 15 (TM15). The two times are specified as 8 bit values with a resolution of 64 µs.		
Note:	DO: Digital output		

r4271	TM15 set/reset time digital output 21 / TM15 t_set DO 21		
TM15	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 21 of Terminal Module 15 (TM15). The two times are specified as 8 bit values with a resolution of 64 µs.		
Note:	DO: Digital output		

r4272	TM15 set/reset time digital output 22 / TM15 t_set DO 22		
TM15	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 22 of Terminal Module 15 (TM15). The two times are specified as 8 bit values with a resolution of 64 µs.		
Note:	DO: Digital output		

r4273	TM15 set/reset time digital output 23 / TM15 t_set DO 23		
TM15	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the time to set and reset for digital output 23 of Terminal Module 15 (TM15). The two times are specified as 8 bit values with a resolution of 64 µs.		
Note:	DO: Digital output		

r4301 TM15 module synchronization / TM15 module sync

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Is used to synchronize the timer of Terminal Module 15 (TM15) with the system time of the DP master.
 Bit 12 ... 15:
 After synchronization with the DP master, the module sends its sign-of-life in the form of a counter.
 Bit 0:
 This SYNC signal is set if the module has aligned its time to the system time of the DP master.
 Bit 9:
 This bit is set if a fault has occurred on the module (r0945).

r4301 TM17 module synchronization / TM17 module sync

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Is used to synchronize the timer of Terminal Module 17 (TM17) with the system time of the DP master.
 Bit 12 ... 15:
 After synchronization with the DP master, the module sends its sign-of-life in the form of a counter.
 Bit 0:
 This SYNC signal is set if the module has aligned its time to the system time of the DP master.
 Bit 9:
 This bit is set if a fault has occurred on the module (r0945).

r4304 TM15 status, digital input 0 ... 15 / TM15 St DI 0-15

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays status for digital input 0 ... 15 of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	On	Off	-
	01	DI/DO 1 (X520.3)	On	Off	-
	02	DI/DO 2 (X520.4)	On	Off	-
	03	DI/DO 3 (X520.5)	On	Off	-
	04	DI/DO 4 (X520.6)	On	Off	-
	05	DI/DO 5 (X520.7)	On	Off	-
	06	DI/DO 6 (X520.8)	On	Off	-
	07	DI/DO 7 (X520.9)	On	Off	-
	08	DI/DO 8 (X521.2)	On	Off	-
	09	DI/DO 9 (X521.3)	On	Off	-
	10	DI/DO 10 (X522.4)	On	Off	-

11	DI/DO 11 (X521.5)	On	Off	-
12	DI/DO 12 (X521.6)	On	Off	-
13	DI/DO 13 (X521.7)	On	Off	-
14	DI/DO 14 (X521.8)	On	Off	-
15	DI/DO 15 (X521.9)	On	Off	-

Note: DI/DO: Bidirectional digital input/output

r4304 TM17 status, digital input 0 ... 15 / TM17 St DI 0-15

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays status for digital input 0 ... 15 of Terminal Module 17 (TM17).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 0 (X520.2)	On	Off	-
	01	DI/DO 1 (X520.3)	On	Off	-
	02	DI/DO 2 (X520.5)	On	Off	-
	03	DI/DO 3 (X520.6)	On	Off	-
	04	DI/DO 4 (X520.8)	On	Off	-
	05	DI/DO 5 (X520.9)	On	Off	-
	06	DI/DO 6 (X521.2)	On	Off	-
	07	DI/DO 7 (X521.3)	On	Off	-
	08	DI/DO 8 (X521.8)	On	Off	-
	09	DI/DO 9 (X521.9)	On	Off	-
	10	DI/DO 10 (X522.2)	On	Off	-
	11	DI/DO 11 (X522.3)	On	Off	-
	12	DI/DO 12 (X522.5)	On	Off	-
	13	DI/DO 13 (X522.6)	On	Off	-
	14	DI/DO 14 (X522.8)	On	Off	-
	15	DI/DO 15 (X522.9)	On	Off	-

Note: DI/DO: Bidirectional digital input/output

r4305 TM15 status, digital input 16 ... 23 / TM15 St DI 16-23

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays status for digital input 16 ... 23 of Terminal Module 15 (TM15).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI/DO 16 (X522.2)	On	Off	-
	01	DI/DO 17 (X522.3)	On	Off	-
	02	DI/DO 18 (X522.4)	On	Off	-
	03	DI/DO 19 (X522.5)	On	Off	-
	04	DI/DO 20 (X522.6)	On	Off	-
	05	DI/DO 21 (X522.7)	On	Off	-
	06	DI/DO 22 (X522.8)	On	Off	-
	07	DI/DO 23 (X522.9)	On	Off	-

Note: DI/DO: Bidirectional digital input/output

r4311	TM15 edge status digital input 0 ... 7 / TM15 EdgSt DI 0-7		
TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the edge status for digital input 0 ... 7 of Terminal Module 15 (TM15).
Assignment of the digital inputs to the bits:

DI 0: r4311.1 ... 0

DI 1: r4311.3 ... 2

DI 2: r4311.5 ... 4

DI 3: r4311.7 ... 6

DI 4: r4311.9 ... 8

DI 5: r4311.11 ... 10

DI 6: r4311.13 ... 12

DI 7: r4311.15 ... 14

Possible edge states:

Bit x, y = 0, 0 --> no edge detection

Bit x, y = 0, 1 --> 1st edge detected

Bit x, y = 1, 0 --> 2nd edge detected

Bit x, y = 1, 1 --> both edges detected

Note: DI: Digital input

r4311	TM17 edge status digital input 0 ... 7 / TM17 EdgSt DI 0-7		
TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the edge status for digital input 0 ... 7 of Terminal Module 17 (TM17).
Assignment of the digital inputs to the bits:

DI 0: r4311.1 ... 0

DI 1: r4311.3 ... 2

DI 2: r4311.5 ... 4

DI 3: r4311.7 ... 6

DI 4: r4311.9 ... 8

DI 5: r4311.11 ... 10

DI 6: r4311.13 ... 12

DI 7: r4311.15 ... 14

Possible edge states:

Bit x, y = 0, 0 --> no edge detection

Bit x, y = 0, 1 --> 1st edge detected

Bit x, y = 1, 0 --> 2nd edge detected

Bit x, y = 1, 1 --> both edges detected

Note: DI: Digital input

r4312 TM15 edge status digital input 8 ... 15 / TM15 EdgSt DI 8-15

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the edge status for digital input 8 ... 15 of Terminal Module 15 (TM15).
Assignment of the digital inputs to the bits:
DI 8: r4312.1 ... 0
DI 9: r4312.3 ... 2
DI 10: r4312.5 ... 4
DI 11: r4312.7 ... 6
DI 12: r4312.9 ... 8
DI 13: r4312.11 ... 10
DI 14: r4312.13 ... 12
DI 15: r4312.15 ... 14
Possible edge states:
Bit x, y = 0, 0 --> no edge detection
Bit x, y = 0, 1 --> 1st edge detected
Bit x, y = 1, 0 --> 2nd edge detected
Bit x, y = 1, 1 --> both edges detected

Note: DI: Digital input

r4312 TM17 edge status digital input 8 ... 15 / TM17 EdgSt DI 8-15

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the edge status for digital input 8 ... 15 of Terminal Module 17 (TM17).
Assignment of the digital inputs to the bits:
DI 8: r4312.1 ... 0
DI 9: r4312.3 ... 2
DI 10: r4312.5 ... 4
DI 11: r4312.7 ... 6
DI 12: r4312.9 ... 8
DI 13: r4312.11 ... 10
DI 14: r4312.13 ... 12
DI 15: r4312.15 ... 14
Possible edge states:
Bit x, y = 0, 0 --> no edge detection
Bit x, y = 0, 1 --> 1st edge detected
Bit x, y = 1, 0 --> 2nd edge detected
Bit x, y = 1, 1 --> both edges detected

Note: DI: Digital input

r4313 TM15 edge status digital input 16 ... 23 / TM15 EdgSt DI16-23

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the edge status for digital input 16 ... 23 of Terminal Module 15 (TM15).
Assignment of the digital inputs to the bits:

DI 16: r4313.1 ... 0

DI 17: r4313.3 ... 2

DI 18: r4313.5 ... 4

DI 19: r4313.7 ... 6

DI 20: r4313.9 ... 8

DI 21: r4313.11 ... 10

DI 22: r4313.13 ... 12

DI 23: r4313.15 ... 14

Possible edge states:

Bit x, y = 0, 0 --> no edge detection

Bit x, y = 0, 1 --> 1st edge detected

Bit x, y = 1, 0 --> 2nd edge detected

Bit x, y = 1, 1 --> both edges detected

Note: DI: Digital input

r4350 TM15 edge times digital input 0 / TM15 edge_t DI 0

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 0 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4350 TM17 edge times digital input 0 / TM17 edge_t DI 0

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 0 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4351 TM15 edge times digital input 1 / TM15 edge_t DI 1

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 1 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4351 TM17 edge times digital input 1 / TM17 edge_t DI 1

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 1 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4352 TM15 edge times digital input 2 / TM15 edge_t DI 2

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 2 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4352 TM17 edge times digital input 2 / TM17 edge_t DI 2

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 2 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4353 TM15 edge times digital input 3 / TM15 edge_t DI 3

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 3 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4353 TM17 edge times digital input 3 / TM17 edge_t DI 3

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 3 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4354 TM15 edge times digital input 4 / TM15 edge_t DI 4

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 4 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4354 TM17 edge times digital input 4 / TM17 edge_t DI 4

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 4 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4355 TM15 edge times digital input 5 / TM15 edge_t DI 5

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 5 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4355 TM17 edge times digital input 5 / TM17 edge_t DI 5

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 5 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4356 TM15 edge times digital input 6 / TM15 edge_t DI 6

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 6 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4356 TM17 edge times digital input 6 / TM17 edge_t DI 6

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 6 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4357 TM15 edge times digital input 7 / TM15 edge_t DI 7

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 7 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4357 TM17 edge times digital input 7 / TM17 edge_t DI 7

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 7 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4358 TM15 edge times digital input 8 / TM15 edge_t DI 8

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 8 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4358 TM17 edge times digital input 8 / TM17 edge_t DI 8

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 8 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4359 TM15 edge times digital input 9 / TM15 edge_t DI 9

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 9 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4359 TM17 edge times digital input 9 / TM17 edge_t DI 9

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 9 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4360 TM15 edge times digital input 10 / TM15 edge_t DI 10

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 10 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4360 TM17 edge times digital input 10 / TM17 edge_t DI 10

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 10 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4361 TM15 edge times digital input 11 / TM15 edge_t DI 11

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 11 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4361 TM17 edge times digital input 11 / TM17 edge_t DI 11

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 11 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4362 TM15 edge times digital input 12 / TM15 edge_t DI 12

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 12 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4362 TM17 edge times digital input 12 / TM17 edge_t DI 12

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 12 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4363 TM15 edge times digital input 13 / TM15 edge_t DI 13

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 13 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4363 TM17 edge times digital input 13 / TM17 edge_t DI 13

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 13 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4364 TM15 edge times digital input 14 / TM15 edge_t DI 14

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 14 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4364 TM17 edge times digital input 14 / TM17 edge_t DI 14

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 14 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4365 TM15 edge times digital input 15 / TM15 edge_t DI 15

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 15 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4365 TM17 edge times digital input 15 / TM17 edge_t DI 15

TM17	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 15 of Terminal Module 17 (TM17).
The two times are specified as 16 bit values with a resolution of 0.25 µs.

Note: DI: Digital input

r4366 TM15 edge times digital input 16 / TM15 edge_t DI 16

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 16 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4367 TM15 edge times digital input 17 / TM15 edge_t DI 17

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 17 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4368 TM15 edge times digital input 18 / TM15 edge_t DI 18

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 18 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4369 TM15 edge times digital input 19 / TM15 edge_t DI 19

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 19 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4370 TM15 edge times digital input 20 / TM15 edge_t DI 20

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 20 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4371 TM15 edge times digital input 21 / TM15 edge_t DI 21

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 21 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4372 TM15 edge times digital input 22 / TM15 edge_t DI 22

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 22 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

r4373 TM15 edge times digital input 23 / TM15 edge_t DI 23

TM15	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the time when detecting the 1st and 2nd edge for digital input 23 of Terminal Module 15 (TM15).
The two times are specified as 8 bit values with a resolution of 64 µs.

Note: DI: Digital input

p4400 TM41 incremental encoder emulation operating mode / Enc_emulat mode

TM41	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 9674, 9676
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0

Description: Sets the operating mode for the incremental encoder emulation.

Value:
0: SIMOTION
1: SINAMICS

Note: A change only becomes effective after the next boot.
Re value = 0:
Incremental encoder emulation using speed setpoint (p1155).
Re value = 1:
Incremental encoder emulation using encoder position actual value (p4420).

p4401 TM41 incremental encoder emulation mode / Enc_emulat mode

TM41	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 9674, 9676
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0011 bin

Description: Sets the mode for the incremental encoder emulation.
This parameter is used to configure the zero mark output at X520.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Zero mark enable	Yes	No	9674
	01	Zero marks synchronized with zero position of absolute encoders	Yes	No	9674

Note:

When the TM41 is operated in the SINAMICS mode (p4400 = 1), the following applies:

A new zero mark search is initiated by switching in the zero mark at the TM41. The zero mark is output at the TM41 as soon as it was synchronized with the encoder interconnected at connector input p4420.

For p4401.1 = 1, the following applies:

The zero pulse is output at X520 when the absolute encoder passes the zero position of the absolute position (modulo converted) in its absolute position.

For p4401.1 = 0, the following applies:

The zero pulse is output at X520 compatible with previous firmware versions. The zero pulse is output precisely when the TM41 (modulo converted) passes the position it was in when the 24 V supply was switched on.

r4402.0...2**CO/BO: TM41 incremental encoder emulation, status / Enc_emulat status**

TM41

Can be changed: -	Calculated: -	Access level: 3
Data type: Unsigned16	Dynamic index: -	Func. diagram: 9674, 9676
P-Group: Closed-loop control	Units group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min	Max	Factory setting
-	-	-

Description:

Displays the status of the incremental encoder emulation on Terminal Module 41 (TM41).

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Zero mark enabled	Yes	No	-
01	Tracks A/B enabled	Yes	No	-
02	Interface encoder emulation enabled	Yes	No	-

r4403**Incremental encoder emulation mode active / Enc_emul mode act**

TM41

Can be changed: -	Calculated: -	Access level: 3
Data type: Unsigned16	Dynamic index: -	Func. diagram: 9674, 9676
P-Group: Closed-loop control	Units group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min	Max	Factory setting
-	-	-

Description:

Displays the current operating mode of Terminal Module 41 (TM41).

Dependency:

Refer to: p4400

p4404**TM41 incremental encoder emulation controller options / TM41 ctrl_opt**

TM41

Can be changed: C2(4)	Calculated: -	Access level: 3
Data type: Unsigned16	Dynamic index: -	Func. diagram: -
P-Group: Encoder	Units group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min	Max	Factory setting
-	-	0001 bin

Description:

Sets the controller option for incremental encoder emulation on Terminal Module 41 (TM41).

p4404.0 = 1:

Control with minimum following error (pre-control active) for synchronous position and synchronous zero-mark emulation

p4404.1 = 1:

In the case of TTL encoders, the control response improves at slow velocities.

p4404.0 = p4404.1 = 0

Control with fixed following error.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Pre-control	Active	Inactive	-
01	Pre-control with adaptation for TTL encoder	Active	Inactive	-

Danger:

The option p4404.1 = 1 is only effective if TM41 DAC is being used.

If the possibility of a TM41 DAC (new) being replaced by a TM41 SAC (old) cannot be excluded, this option should not be set.

Note:

The parameter is only effective in the "SINAMICS" operating mode (p4400 = 1).

TM41 SAC: order no. = 6SL3055-0AA00-3PA0

TM41 DAC: order no. = 6SL3055-0AA00-3PA1

p4420		CI: TM41 incremental encoder emulation position actual value / Enc_emul s_act	
TM41	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 9676
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	Sets the signal source for the incremental encoder emulation position actual value.		
Recommend.:	The following BICO interconnection should be preferably set: CI: p4420 = r0479		
Dependency:	Refer to: p4400, r4403		
Notice:	General conditions for incremental encoder emulation can be found in the following literature: SINAMICS S120 Function Manual Drive Functions		
Note:	The parameter is not effective in the SIMOTION operating mode (p4400 = 0). Prerequisites for the signal source: For the incremental encoder emulation, an encoder is required that supplies precisely one zero mark per revolution. The following BICO interconnection should be preferably set: CI: p4420 = r0479. An encoder actual value (r0479) can only be interconnected once on a TM41. The zero mark of the incremental encoder is output at the encoder interface after successful internal, automatic synchronization. If an absolute encoder is connected to the TM41, then a zero mark is always output at the zero revolution at the encoder interface of the TM41. The same behavior applies even if no encoder parameter is interconnected.		

p4421	TM41 incremental encoder emulation deadtime compensation / Enc_emul t_dead		
TM41	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 9676
	P-Group: Setpoints	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -10.00	Max 10.00	Factory setting 0.00
Description:	Sets the deadtime compensation for incremental encoder emulation. This factor defines the multiplier in which the encoder position setpoint of the incremental encoder emulation is shifted depending on the velocity.		
Dependency:	For p4421 = 0, the deadtime compensation for the position actual value is switched out. For p4421 <> 0, the deadtime compensation is taken into account as follows: New actual value = actual value via CI: p4420 + delta s * p4421 delta s: Position change per sampling time (4099[3]), internally smoothed Refer to: p4400		
Note:	The parameter is not effective in the SIMOTION operating mode (p4400 = 0).		

p4422	TM41 position actual value inversion / TM41 s_act inv		
TM41	Can be changed: C2(4) Data type: Unsigned16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: 9676 Unit selection: - Expert list: 1 Factory setting 0
Description:	Setting to invert the position actual value for Terminal Module 41 (TM41). 0 -> Position actual value (CI: p4420) is evaluated as normal. 1 -> Position actual value (CI: p4420) is processed inverted.		
Dependency:	Refer to: p4420		
p4423	TM41 standstill adaptation / TM41 ststl_adapt		
TM41	Can be changed: C2(4) Data type: Unsigned16 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2000	Access level: 3 Func. diagram: 9676 Unit selection: - Expert list: 1 Factory setting 4
Description:	Sets standstill adaptation on Terminal Module 41 (TM41). p4423 is used to specify the number of clock cycles (one clock cycle = p4099[3]) used for encoder standstill detection. Once this time has elapsed, any potential deviation is compensated when adaptation is active. Parameter value = 0: adaptation inactive Parameter value > 0: adaptation active		
Dependency:	Refer to: r4403, p4404, p4420		
Danger:	The option p4404.1 = 1 is only effective if TM41 DAC is being used. If the possibility of a TM41 DAC (new) being replaced by a TM41 SAC (old) cannot be excluded, this option should not be set. TM41 SAC: order no. = 6SL3055-0AA00-3PA0 TM41 DAC: order no. = 6SL3055-0AA00-3PA1		
			
Note:	The parameter is only effective in the SINAMICS operating mode (p4400 = 1). The parameter value must be assigned a value of 4 or more to ensure that the system functions properly. This parameter is only relevant in the following cases: - TTL encoder is available - the controller option "Pre-control with adaptation for TTL encoder" has been activated (p4404.1 = 1)		
p4426	Incremental encoder emulation, pulses for zero mark / Enc_emul pulses ZM		
TM41	Can be changed: C2(4) Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8192	Access level: 3 Func. diagram: 9674 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets pulse number to output the zero mark for the incremental encoder simulation/emulation. Example: p0408 = 2048 (encoder pulses) p4426 = 512 (pulses for the zero mark) --> Position direction: The zero mark is output after 512 pulses. --> Negative direction: The zero mark is output after 1536 pulses.		
Dependency:	Refer to: p0408		
Note:	The pulses for the zero mark (p4426) must be less than the encoder pulse number (p0408). For p4400 = 1, this parameter has no effect.		

p4600[0...n]	Motor temperature sensor 1 sensor type / Temp_sens 1 type		
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: EDS, p0140	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 32	Factory setting 0
Description:	Sets the sensor type of the first temperature sensor for the motor temperature monitoring.		
Value:	0: No sensor 10: PTC fault 11: PTC alarm 12: PTC alarm & timer 20: KTY84 30: Bimetallic NC contact fault 31: Bimetallic NC contact alarm 32: Bimetallic NC contact alarm & timer		
Dependency:	Refer to: r0458, p0600, p0601		
Note:	This parameter is effective only when p0601 = 10. PTC thermistor: Tripping resistance = 1650 Ohm Information on using temperature sensors is provided in the following literature: - hardware description of the appropriate components - SINAMICS S120 Commissioning Manual		

p4601[0...n]	Motor temperature sensor 2 sensor type / Temp_sens 2 type		
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: EDS, p0140	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 32	Factory setting 0
Description:	Sets the sensor type of the second temperature sensor for the motor temperature monitoring.		
Value:	0: No sensor 10: PTC fault 11: PTC alarm 12: PTC alarm & timer 20: KTY84 30: Bimetallic NC contact fault 31: Bimetallic NC contact alarm 32: Bimetallic NC contact alarm & timer		
Dependency:	Refer to: r0458, p0600, p0601		
Note:	This parameter is effective only when p0601 = 10. Terminals for KTY84: X200.1, X200.2 PTC thermistor: Tripping resistance = 1650 Ohm Information on using temperature sensors is provided in the following literature: - hardware description of the appropriate components - SINAMICS S120 Commissioning Manual		

p4602[0...n] Motor temperature sensor 3 sensor type / Temp_sens 3 type

SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: EDS, p0140	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	32	0

Description: Sets the sensor type of the third temperature sensor for the motor temperature monitoring.

Value:

- 0: No sensor
- 10: PTC fault
- 11: PTC alarm
- 12: PTC alarm & timer
- 20: KTY84
- 30: Bimetallic NC contact fault
- 31: Bimetallic NC contact alarm
- 32: Bimetallic NC contact alarm & timer

Dependency: Refer to: r0458, p0600, p0601

Note: This parameter is effective only when p0601 = 10.
Terminals for PTC triplet and bimetallic: X200.3, X200.4
PTC thermistor: Tripping resistance = 1650 Ohm
Information on using temperature sensors is provided in the following literature:
- hardware description of the appropriate components
- SINAMICS S120 Commissioning Manual

p4603[0...n] Motor temperature sensor 4 sensor type / Temp_sens 4 type

SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: EDS, p0140	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	32	0

Description: Sets the sensor type of the fourth temperature sensor for the motor temperature monitoring.

Value:

- 0: No sensor
- 10: PTC fault
- 11: PTC alarm
- 12: PTC alarm & timer
- 20: KTY84
- 30: Bimetallic NC contact fault
- 31: Bimetallic NC contact alarm
- 32: Bimetallic NC contact alarm & timer

Dependency: Refer to: r0458, p0600, p0601

Note: This parameter is effective only when p0601 = 10.
Terminals for PTC triplet: X200.5, X200.6
PTC thermistor: Tripping resistance = 1650 Ohm
Information on using temperature sensors is provided in the following literature:
- hardware description of the appropriate components
- SINAMICS S120 Commissioning Manual

p4610[0...n] Motor temperature sensor 1 sensor type MDS / Temp sens1 typ MDS			
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 32	Factory setting 10
Description:	Sets the sensor type of the first temperature sensor for the motor temperature monitoring.		
Value:	0: No sensor 10: PTC fault 11: PTC alarm 12: PTC alarm & timer 20: KTY84 30: Bimetallic NC contact fault 31: Bimetallic NC contact alarm 32: Bimetallic NC contact alarm & timer		
Dependency:	Refer to: r0458, p0600, p0601		
Note:	This parameter is effective only when p0601 = 11. PTC thermistor: Tripping resistance = 1650 Ohm Information on using temperature sensors is provided in the following literature: - hardware description of the appropriate components - SINAMICS S120 Commissioning Manual		

p4611[0...n] Motor temperature sensor 2 sensor type MDS / Temp sens2 typ MDS			
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 32	Factory setting 10
Description:	Sets the sensor type of the second temperature sensor for the motor temperature monitoring.		
Value:	0: No sensor 10: PTC fault 11: PTC alarm 12: PTC alarm & timer 20: KTY84 30: Bimetallic NC contact fault 31: Bimetallic NC contact alarm 32: Bimetallic NC contact alarm & timer		
Dependency:	Refer to: r0458, p0600, p0601		
Note:	This parameter is effective only when p0601 = 11. PTC thermistor: Tripping resistance = 1650 Ohm Information on using temperature sensors is provided in the following literature: - hardware description of the appropriate components - SINAMICS S120 Commissioning Manual		

p4612[0...n] Motor temperature sensor 3 sensor type MDS / Temp sens3 typ MDS			
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 32	Factory setting 10
Description:	Sets the sensor type of the third temperature sensor for the motor temperature monitoring.		
Value:	0: No sensor 10: PTC fault 11: PTC alarm 12: PTC alarm & timer 20: KTY84 30: Bimetallic NC contact fault 31: Bimetallic NC contact alarm 32: Bimetallic NC contact alarm & timer		
Dependency:	Refer to: r0458, p0600, p0601		
Note:	This parameter is effective only when p0601 = 11. PTC thermistor: Tripping resistance = 1650 Ohm Information on using temperature sensors is provided in the following literature: - hardware description of the appropriate components - SINAMICS S120 Commissioning Manual		

p4613[0...n] Motor temperature sensor 4 sensor type MDS / Temp sens4 typ MDS			
SERVO, VECTOR	Can be changed: C2(3), U, T	Calculated: -	Access level: 2
	Data type: Integer16	Dynamic index: MDS, p0130	Func. diagram: 8016
	P-Group: Motor	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 32	Factory setting 10
Description:	Sets the sensor type of the fourth temperature sensor for the motor temperature monitoring.		
Value:	0: No sensor 10: PTC fault 11: PTC alarm 12: PTC alarm & timer 20: KTY84 30: Bimetallic NC contact fault 31: Bimetallic NC contact alarm 32: Bimetallic NC contact alarm & timer		
Dependency:	Refer to: r0458, p0600, p0601		
Note:	This parameter is effective only when p0601 = 11. PTC thermistor: Tripping resistance = 1650 Ohm Information on using temperature sensors is provided in the following literature: - hardware description of the appropriate components - SINAMICS S120 Commissioning Manual		

r4620[0...3]	Motor temperature measured / Mot_temp meas		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min	Max	Factory setting
	- [°C]	- [°C]	- [°C]
Description:	Displays the actual temperature in the motor measured through temperature channels 1 ... 4.		
Index:	[0] = Temperature channel 1 [1] = Temperature channel 2 [2] = Temperature channel 3 [3] = Temperature channel 4		
Note:	An invalid temperature is displayed using the value -200°C.		

r4640[0...95]	Encoder diagnostics state machine / Enc diag SM		
ENCODER, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Encoder diagnostics for the Profidrive interface.		

p4650	Encoder functional reserve component number / Enc fct_res num		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	399	0
Description:	Sets the component number (p0141) of the encoder whose functional reserve is to be displayed (r4651).		
Dependency:	Refer to: r4651		

r4651[0...3]	Encoder functional reserve / Enc fct_reserve		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]
Description:	Displays the functional reserve of the encoder selected via p4650. 0 ... 25 %: The function limit has been reached. A service is recommended. 26 ... 100 %: The encoder is working in the specified range.		
Index:	[0] = Incremental [1] = Reserved [2] = Abs track [3] = Code conn		
Dependency:	Refer to: p4650		

Note: Value = 999 means:
 - the component specified in p4650 is not connected
 - the encoder does not support the display of the functional reserve

p4660	Sensor module filter bandwidth / SMx f_BW		
ENCODER	Can be changed: - Data type: FloatingPoint32 P-Group: Encoder Not for motor type: - Min 0.00 [kHz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20000.00 [kHz]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [kHz]
Description:	Sets the filter bandwidth for Sensor Modules SMx10 (resolver) and SMx20 (sin/cos encoder). The value set on the Sensor Module is displayed in r4661. The Sensor Module hardware only supports the following values: - 0: The Sensor Module's default is used. - 50 kHz - 170 kHz - 500 kHz - Unlimited - only the bandwidth of the Op Amp is effective.		
Dependency:	Refer to: r4661		
Note:	A value of zero is displayed if an encoder is not present.		

p4660[0...2]	Sensor module filter bandwidth / SMx f_BW		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Encoder Not for motor type: - Min 0.00 [kHz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 20000.00 [kHz]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [kHz]
Description:	Sets the filter bandwidth for Sensor Modules SMx10 (resolver) and SMx20 (sin/cos encoder). The value set on the Sensor Module is displayed in r4661. The Sensor Module hardware only supports the following values: - 0: The Sensor Module's default is used. - 50 kHz - 170 kHz - 500 kHz - Unlimited - only the bandwidth of the Op Amp is effective.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: r4661		
Note:	A value of zero is displayed if an encoder is not present.		

r4661	Sensor Module filter bandwidth display / SMx f_BW disp		
ENCODER	Can be changed: - Data type: FloatingPoint32 P-Group: Encoder Not for motor type: - Min - [kHz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [kHz]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [kHz]
Description:	Displays the effective filter bandwidth for Sensor Modules SMx10 (resolver) and SMx20 (sin/cos encoder). The bandwidth of the filter is set using p4660.		

Dependency: Refer to: p4660

Note: A value of zero is displayed if an encoder is not present.

r4661[0...2] Sensor Module filter bandwidth display / SMx f_BW disp

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [kHz]	Max - [kHz]	Factory setting - [kHz]

Description: Displays the effective filter bandwidth for Sensor Modules SMx10 (resolver) and SMx20 (sin/cos encoder).
The bandwidth of the filter is set using p4660.

Index:
[0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

Dependency: Refer to: p4660

Note: A value of zero is displayed if an encoder is not present.

p4670[0...n] Analog sensor configuration / Ana_sens config

ENCODER, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0000 bin

Description: Sets the configuration for evaluation on the analog sensor.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	06	Set velocity to 0	Yes	No	-
	09	Fault/Alarm messages	Alarm	Fault	-
	10	Chann B act	Yes	No	-
	11	Chann A act	Yes	No	-
	13	Commutation angle constant	Yes	No	-

Notice:
Re bit 06:
Setting the bit sets the velocity actual value (r0061) permanently to 0.
Re bit 13:
Setting the bit sets the commutation angle permanently to the commutation angle offset (p0431).

Note:
Re bit 09:
A setting of bit = 0 will trigger a fault for the relevant channel if the actual value is invalid.
A setting of bit = 1 will trigger an alarm for the relevant channel if the actual value is invalid.
Re bit 10, 11:
If both channels are activated, the actual value is generated from the mean value of both channels. If a channel fails (actual value invalid), it is not included when the mean value is generated.

p4671[0...n] Analog sensor input / Ana_sens inp

ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 3	Factory setting 0

Description: Sets the input circuit for the analog sensor.

Value:

- 0: Differential
- 1: Single-ended A, B
- 2: Single-ended A*, B*
- 3: Single-ended A, B sensitive

Note:

p4671 = 0:
The two signals on a track are evaluated differentially.

p4671 = 1:
Only the non-inverted signal on a track is evaluated.

p4671 = 2:
Only the inverted signal on a track is evaluated.

p4671 = 3:
Only the non-inverted signal on a track (high resolution) is evaluated.

p4672[0...n] Analog sensor channel A voltage at actual value zero / Ana_sens A U at 0

ENCODER, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -10.0000 [V]	Max 10.0000 [V]	Factory setting 0.0000 [V]

Description: Sets the voltage when the connected sensor is at actual value zero.
At this voltage channel A supplies an actual value of zero.

p4673[0...n] Analog sensor channel A voltage per encoder period / Ana_sens A U/per

ENCODER, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -10.0000 [V]	Max 10.0000 [V]	Factory setting 6.0000 [V]

Description: Sets the output voltage range to be mapped for the connected analog sensor.
The voltage range is determined by the following parameters:

- p4672 (voltage at actual value 0)
- p4673 (voltage per encoder period)

Note: The minimum actual value which can be mapped is equal to p4672 - p4673/2.
The maximum actual value which can be mapped is equal to p4672 + p4673/2.

p4674[0...n] Analog sensor channel B voltage at actual value zero / Ana_sens B U at 0

ENCODER, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -10.0000 [V]	Max 10.0000 [V]	Factory setting 0.0000 [V]

Description: Sets the voltage when the connected sensor is at actual value zero.
At this voltage channel B supplies an actual value of zero.

p4675[0...n]		Analog sensor channel B voltage per encoder period / Ana_sens B U/per			
ENCODER, SERVO, VECTOR	Can be changed: U, T	Calculated: -		Access level: 4	
	Data type: FloatingPoint32	Dynamic index: EDS, p0140		Func. diagram: -	
	P-Group: -	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min -10.0000 [V]	Max 10.0000 [V]		Factory setting 6.0000 [V]	
Description:	Sets the output voltage range to be mapped for the connected analog sensor. The voltage range is determined by the following parameters: - p4674 (voltage at actual value 0) - p4675 (voltage per encoder period)				
Note:	The minimum actual value which can be mapped is equal to p4674 - p4675/2. The maximum actual value which can be mapped is equal to p4674 + p4675/2.				

p4676[0...n]		Analog sensor range limit threshold / Ana_sens lim thr			
ENCODER, SERVO, VECTOR	Can be changed: U, T	Calculated: -		Access level: 4	
	Data type: FloatingPoint32	Dynamic index: EDS, p0140		Func. diagram: -	
	P-Group: -	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min 0.0 [%]	Max 100.0 [%]		Factory setting 100.0 [%]	
Description:	Sets the threshold for limit monitoring of the absolute actual value on the analog sensor. If this threshold is overshoot by the actual value of a channel, a corresponding fault/alarm (p4670.9) is output.				
Dependency:	Refer to: p4673, p4675				

p4677[0...n]		Analog sensor LVDT configuration / Ana_sens LVDT conf			
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -		Access level: 4	
	Data type: Unsigned32	Dynamic index: EDS, p0140		Func. diagram: -	
	P-Group: Encoder	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min -	Max -		Factory setting 0000 bin	
Description:	Sets the configuration for LVDT mode on the analog sensor.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	LVDT ON	Yes	No	-
	01	Track B excitation	Yes	No	-
	02	Fixed value amplitude	Yes	No	-
	03	Fixed value amplitude and phase	Yes	No	-

p4680[0...n]		Zero mark monitoring tolerance permissible / ZM_monit tol perm			
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -		Access level: 3	
	Data type: Unsigned32	Dynamic index: EDS, p0140		Func. diagram: -	
	P-Group: Encoder	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min 0	Max 1000		Factory setting 4	
Description:	Sets the permissible tolerance in encoder pulses for the zero mark distance in the context of zero mark monitoring. Causes fault F3x100 to appear less frequently.				
Dependency:	Refer to: p0430 Refer to: F31100				
Note:	The parameter is activated using p0430.21 = 1 (zero mark tolerance).				

p4681[0...n]	Zero mark monitoring, tolerance window limit 1 positive / ZM tol lim 1 pos		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 1000	Factory setting 2
Description:	Sets the positive tolerance window in encoder pulses for limit 1 for the zero mark monitoring. If the deviation is less than this limit, the PPR is not corrected. If it is higher than this limit, fault F3x131 is triggered. If fault F3x131 is re-parameterized to alarm (A) or no message (N), the encoder pulses which have not been corrected are added to the accumulator (r4688). The accumulator can be deactivated using p0437.7.		
Dependency:	Refer to: p0437, p4688 Refer to: F31131		
Note:	This monitoring is activated by setting p0437.2 = 1 (position actual value correction). The positive limit describes additional pulses due to EMC.		
p4682[0...n]	Zero mark monitoring, tolerance window limit 1 negative / ZM tol lim 1 neg		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -1001	Max 0	Factory setting -1001
Description:	Sets the negative tolerance window in encoder pulses for limit 1 for the zero mark monitoring. If the deviation is less than this limit, the PPR is not corrected. If it is higher than this limit, fault F3x131 is triggered. If fault F3x131 is re-parameterized to alarm (A) or no message (N), the encoder pulses which have not been corrected are added to the accumulator (r4688). The accumulator can be deactivated using p0437.7.		
Dependency:	Refer to: p0437, p4681, p4688 Refer to: F31131		
Note:	This monitoring is activated by setting p0437.2 = 1 (position actual value correction). For a set value = -1001, the negated value of p4681 is effective. The negative limit describes the pulses lost due to a covered glass panel in the incremental encoder.		
p4683[0...n]	Zero mark monitoring tolerance window alarm threshold positive / ZM tol A_thr pos		
ENCODER, SERVO, VECTOR	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: EDS, p0140	Func. diagram: -
	P-Group: Encoder	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 100000	Factory setting 0
Description:	Sets the positive tolerance window in encoder pulses for limit 2 for the zero mark monitoring. If the zero mark deviation is higher than the tolerance set in p4681 and p4682 and fault F3x131 is re-parameterized to alarm (A) or no message (N), the accumulator p4688 is compared with this parameter and, if applicable, alarm A3x422 is output for 5 seconds.		
Dependency:	Refer to: p0437, p4681, p4682, p4688 Refer to: F31131, A31422		
Note:	Zero mark monitoring is activated by setting p0437.2 = 1 (position actual value correction).		

p4684[0...n]	Zero mark monitoring tolerance window alarm threshold negative / ZM tol A_thr neg		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Integer32 P-Group: Encoder Not for motor type: - Min -100001	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 0	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -100001
Description:	Sets the negative tolerance window in encoder pulses for limit 2 for the zero mark monitoring. If the zero mark deviation is higher than the tolerance set in p4681 and p4682 and fault F3x131 is re-parameterized to alarm (A) or no message (N), the accumulator p4688 is compared with this parameter and, if applicable, alarm A3x422 is output for 5 seconds.		
Dependency:	Refer to: p0437, p4683, p4688 Refer to: F31131, A31422		
Note:	Zero mark monitoring is activated by setting p0437.2 = 1 (position actual value correction). For a set value = -100001, the negated value of p4683 is effective.		
p4685[0...n]	Speed actual value mean value generation / n_act mean val		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 20	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the number of current controller clock cycles for mean value generation of the speed actual value.		
Note:	Value = 0, 1: No mean value generation. Higher values also mean higher dead times for the speed actual value.		
p4686[0...n]	Zero mark minimum length / ZM min length		
ENCODER, SERVO, VECTOR	Can be changed: C2(4) Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min 0	Calculated: - Dynamic index: EDS, p0140 Units group: - Scaling: - Max 10	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the minimum length for the zero mark.		
Dependency:	Refer to: p0425, p0437		
Note:	The value for the minimum length of the zero mark must be set less than p0425. The parameter is activated using p0437.1 = 1 (zero mark edge detection).		
p4688	CO: Zero mark monitoring, differential pulse count / ZM diff_pulse qty		
ENCODER	Can be changed: T Data type: Integer32 P-Group: - Not for motor type: - Min -2147483648	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147483647	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Displays the number of differential pulses for the zero mark monitoring that have accumulated. If fault F3x131 is re-parameterized to alarm (A) or no message (N), the encoder pulses which have not been corrected are added to the accumulator (p4688).		
Dependency:	Refer to: p4681, p4682, p4683, p4684		
Note:	The display can only be reset to zero.		

p4688[0...2]	CO: Zero mark monitoring, differential pulse count / ZM diff_pulse qty		
SERVO, VECTOR	Can be changed: T Data type: Integer32 P-Group: - Not for motor type: - Min -2147483648	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147483647	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Displays the number of differential pulses for the zero mark monitoring that have accumulated. If fault F3x131 is re-parameterized to alarm (A) or no message (N), the encoder pulses which have not been corrected are added to the accumulator (p4688).		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: p4681, p4682, p4683, p4684		
Note:	The display can only be reset to zero.		
r4689	CO: Squarewave encoder, diagnostics / Sq-wave enc diag		
ENCODER	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the encoder status according to PROFIdrive for a squarewave encoder.		
Dependency:	Refer to: A31422		
Note:	In the case of alarm A3x422 being output, this parameter is set for 100 ms.		
r4689[0...2]	CO: Squarewave encoder, diagnostics / Sq-wave enc diag		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Encoder Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the encoder status according to PROFIdrive for a squarewave encoder.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	Refer to: A31422		
Note:	In the case of alarm A3x422 being output, this parameter is set for 100 ms.		
p4690	SMI spare part component number / SMI comp_no		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Unsigned16 P-Group: Displays, signals Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 399	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the component number for the SMI/DQI for which motor and/or encoder data should be saved, deleted or downloaded.		
Dependency:	Refer to: p4691, p4692, p4693		

Note: DQI: DRIVE-CLiQ Sensor Integrated
SMI: SINAMICS Sensor Module Integrated

p4691 Save/download SMI spare part data / Save/DL SMI data

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 38	Factory setting 0

Description: Setting for the saving/downloading/deletion of motor and/or encoder data for the component specified in p4690 (SMI/DQI).
A backup of this data can be saved to non-volatile memory. The backup procedure is performed automatically as part of the function for saving to non-volatile memory (p0977 = 1 or "Copy RAM to ROM"). If a part is replaced, the backup can be reloaded.

Procedure:

p4690 = set component number

p4691 = 1, 2, 30: Set the required procedure (save/download/delete).

p4691 = 9, 10, 36: Feedback signal on successful completion of the procedure.

p4691 = 11... 21, 37, 38: Error vales if the procedure could not be executed successfully.

Value:

- 0: Inactive
- 1: Save SMI data
- 2: Download SMI data
- 9: SMI data downloaded and POWER ON required for component
- 10: SMI data backup complete
- 11: SMI data backup for selected component not found
- 12: Selected component not available or not connected
- 13: Insufficient memory space for backup
- 14: Format of saved data is incompatible
- 15: Transfer fault during data download
- 16: Transfer fault during data backup
- 17: Data backup does not match parameterized encoder/motor
- 18: Data backup directory not permissible
- 19: Component already contains data
- 20: Component does not contain any data
- 21: Component is not an SMI or a DQI
- 30: Delete SMI data
- 35: Confirmation of SMI data delete required
- 36: SMI data deleted and POWER ON required for component
- 37: Access level not sufficient for delete
- 38: Delete SMI data not permitted for component

Dependency: Refer to: p4690, p4692, p4693

Notice: Once SMI data has been deleted or downloaded successfully, the component has to be powered off.

Note: SMI: SINAMICS Sensor Module Integrated

DLQ encoder: DRIVE-CLiQ encoder

Help for error value = 11:

- Save the data for the original SMI on the memory card.
- Use an SMI with a suitable hardware version.

Help for error value = 12:

- Set the correct component number or connect the component.

Help for error value = 13:

- Use a memory card with more memory space.

Help for error value = 14:

- Create a data backup on the memory card corresponding to the SMI type.

Help for error value = 15:

- Check the DRIVE-CLiQ wiring for the component.

- Help for error value = 16:
- Check the DRIVE-CLiQ wiring for the component.
- Help for error value = 17:
- Save the data for the original SMI on the memory card.
- Help for error value = 18:
- Set parameter p4693 to an appropriate value.
- Help for error value = 19:
- Perform an SMI delete or use a blank SMI.
- Help for error value = 20:
- Use an SMI card that is not blank.
- Help for error value = 21:
- Set the correct component number (p4690).
- Help for error value = 35:
- Reset parameter p4691 to 30.
- Help for error value = 37:
- Set the access level to Expert or higher.
- Help for error value = 38:
- Set the correct component number (p4690 >= 200).

p4692 Save SMI spare part data of all SMIs / Save SMI data			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 29	Factory setting 0
Description:	Setting to back up the data of all SMIs and DQIs featured in the target topology.		
Value:	0: Inactive 1: Save data of all SMIs and DQIs 10: Save all data successful 13: Insufficient memory space for backup 16: Transfer fault during data backup 20: Component does not contain any data 29: Not all components from target topology saved		
Note:	SMI: SINAMICS Sensor Module Integrated p4692 = 10: Automatic on successful completion of backup procedure. p4692 = 13, 16, 20, 29: Error vales if the procedure could not be executed successfully. The procedure must be repeated if the data save operation was interrupted (e.g. if the power supply voltage failed). Help for error value = 13: - Use a memory card with more memory space. Help for error value = 16: - check the DRIVE-CLiQ connection. Help for error value = 20: - Use an SMI card that is not blank. Help for error value = 29: - Check and correct the target and actual topologies for the SMIs. - Repeat the save procedure.		

p4693[0...1] SMI spare part data backup directory / SMI dat_bkup dir			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Unsigned16 P-Group: Displays, signals Not for motor type: - Min 0 Max 399	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the directory for downloading and saving data. Example: The SMI has the component number 5 and the SMI data (motor/encoder data) is to be stored in subdirectory C205. --> p4690 = 5, p4693[0] = 205, p4691 = 1		
Index:	[0] = Subdirectory selection [1] = Reserved		
Dependency:	Refer to: p4691, r4694		
Notice:	If p4693[0] is not equal to 0 and p4693[0] is not equal to p4690, the following applies: - Only a number > 200 may be selected for the subdirectory when saving. - In the case of downloads, a selection for the subdirectory may only be made for an SMI with a component number > 200 (preliminary component number) (p4690 > 200).		
Note:	SMI: SINAMICS Sensor Module Integrated Re index 0: This index is used to select the subdirectory for saving and downloading data. The motor order number (MLFB) of the corresponding data backup is displayed in r4694. For p4693[0] = 0, the following applies: The directory is determined by the setting of p4690.		

r4694[0...19] SMI spare part data backup motor order number / SMI dat_bkup MLFB			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned8 P-Group: Displays, signals Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the motor order number (MLFB) of the data backup selected with p4693.		
Dependency:	Refer to: p4691, p4692		
Caution:	If the selected subdirectory contains a number of data sets, "More Datasets" is displayed in r4694[0...19]. If there is no SMI data (motor/encoder data) in the selected subdirectory or if the selected subdirectory does not exist, the following applies: - The number of the next subdirectory located is displayed. - This subdirectory is not checked for valid SMI data. - If another subdirectory cannot be located, 0 is displayed in r4694[0...19].		
			
Note:	SMI: SINAMICS Sensor Module Integrated		

p4700[0...1] Trace control / Trace control			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: Trace and function generator Not for motor type: - Min 0 Max 1	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Value:	0: Stop trace 1: Start trace		

Index: [0] = Trace 0
[1] = Trace 1

p4701	Measuring function, control / Meas fct ctrl		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: Trace and function generator Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Value:	0: Stop measuring function 1: Start measuring function 2: Measuring function, check parameterization		

r4705[0...1]	Trace status / Trace status		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Integer16 P-Group: Trace and function generator Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the actual status of the trace.		
Value:	0: Trace inactive 1: Trace is recording presamples 2: Trace is waiting for trigger event 3: Trace is recording 4: Recording (trace) ended		
Index:	[0] = Trace 0 [1] = Trace 1		

r4706	Measuring function, status / Meas fct status		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Integer16 P-Group: Trace and function generator Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 5	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the actual status of the measuring function.		
Value:	0: Measurement function inactive 1: Measuring function, parameterization checked 2: Measuring function waits for stabilizing time 3: Measuring function recording (tracing) 4: Measuring function, trace ended with error 5: Measuring function, trace successfully completed		

r4708[0...1] Trace memory space required / Trace mem requiredCU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** -**Data type:** Unsigned32**P-Group:** Trace and function generator**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 0**Factory setting**

-

Description: Displays the required memory in bytes for the actual parameterization.**Index:**
[0] = Trace 0
[1] = Trace 1**Dependency:** Refer to: r4799**r4709[0...1] Trace memory space required for measuring functions / Trace mem required**CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** -**Data type:** Unsigned32**P-Group:** Trace and function generator**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 0**Factory setting**

-

Description: Displays the memory space required for the current parameter setting in bytes, if the trace is used for the measuring functions.**Index:**
[0] = Trace 0
[1] = Trace 1**Dependency:** Refer to: r4799**p4710[0...1] Trace trigger condition / Trace Trig_cond**CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** U, T**Data type:** Integer16**P-Group:** Trace and function generator**Not for motor type:** -**Min**

1

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

7

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 0**Factory setting**

2

Description: Sets the trigger condition for the trace.**Value:**
1: Immediate start
2: Positive edge
3: Negative edge
4: Entry to hysteresis band
5: Leaving hysteresis band
6: Trigger at bit mask
7: Start with function generator**Index:**
[0] = Trace 0
[1] = Trace 1

p4711[0...5]	Trace trigger signal / Trace trig_signal		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Selects the trigger signal for the trace.		
Index:	[0] = Trace 0 parameter in BICO format [1] = Trace 1 parameter in BICO format [2] = Trace 0 PINx with DO Id and chart Id [3] = Trace 0 PINx with block Id and PIN Id [4] = Trace 1 PINy with DO Id and chart Id [5] = Trace 1 PINy with block Id and PIN Id		
Dependency:	Only effective when p4710 does not equal 1.		
Note:	It only makes sense to trace the PINs using the commissioning software. For index 2(4) and 3(5) equal to zero, index 0(1) can only be written and vice versa. Re index 0 ... 1: Here, the trigger signal for trace 0 or 1 is entered as parameter in the BICO format. For trace with a physical address (p4789), the data type of the trigger signal is set here. Re index 2 ...3: The triggering PIN for trace 0 is entered here. Index 2 bit 31 ... 16: Number of the Drive Object (DO), bit 15 ... 0: Number of the chart Index 3 bit 31 ... 16: Number of the block, bit 15 ... 0: Number of the PIN Re index 4 ... 5: The triggering PIN for trace 1 is entered here. Index 4 bit 31 ... 16: Number of the Drive Object (DO), bit 15 ... 0: Number of the chart Index 5 bit 31 ... 16: Number of the block, bit 15 ... 0: Number of the PIN		
p4712[0...1]	Trace trigger threshold / Trace trig_thresh		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min -340.28235E36	Calculated: - Dynamic index: - Units group: - Scaling: - Max 340.28235E36	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0.00
Description:	Sets the trigger threshold for the trace.		
Index:	[0] = Trace 0 [1] = Trace 1		
Dependency:	Only effective when p4710 = 2, 3.		
p4713[0...1]	Trace tolerance band trigger threshold / Trace trig thresh		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min -340.28235E36	Calculated: - Dynamic index: - Units group: - Scaling: - Max 340.28235E36	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0.00
Description:	Sets the first trigger threshold for trigger via tolerance band.		
Index:	[0] = Trace 0 [1] = Trace 1		
Dependency:	Only effective when p4710 = 4, 5.		

p4714[0...1]	Trace tolerance band trigger threshold / Trace trig thresh		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min -340.28235E36 Max 340.28235E36	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0.00
Description:	Sets the second trigger threshold for trigger via tolerance band		
Index:	[0] = Trace 0 [1] = Trace 1		
Dependency:	Only effective when p4710 = 4, 5.		
p4715[0...1]	Trace bit mask trigger, bit mask / Trace trig mask		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0 Max 4294967295	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Sets the bit mask for the bit mask trigger.		
Index:	[0] = Trace 0 [1] = Trace 1		
Dependency:	Only effective when p4710 = 6.		
p4716[0...1]	Trace, bit mask trigger, trigger condition / Trace Trig_cond		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0 Max 4294967295	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Sets the trigger condition for bit mask trigger.		
Index:	[0] = Trace 0 [1] = Trace 1		
Dependency:	Only effective when p4710 = 6.		
p4717	Measuring function, number of averaging operations / Meas fct avg qty		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned8 P-Group: Trace and function generator Not for motor type: - Min 0 Max 255	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
p4718	Measuring function, number of stabilizing periods / MeasFct StabPerQty		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned8 P-Group: Trace and function generator Not for motor type: - Min 0 Max 255	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0

r4719[0...1]	Trace trigger index / Trace Trig_index		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the trigger index in the trace buffer. The trigger event occurred at this point.		
Index:	[0] = Trace 0 [1] = Trace 1		
Dependency:	Only valid when p4705 = 4.		

p4720[0...1]	Trace recording cycle / Trace record_cyc		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min 0.000 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 60000.000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 1.000 [ms]
Description:	Sets the recording cycle for the trace.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4721[0...1]	Trace recording time / Trace record_time		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min 0.000 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3600000.000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 1000.000 [ms]
Description:	Sets the recording time for the trace.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4722[0...1]	Trace trigger delay / Trace trig_delay		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min -3600000.000 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3600000.000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0.000 [ms]
Description:	Sets the trigger delay for the trace. Trigger delay < 0: Pretrigger: Tracing (recording) starts the selected time before the trigger event actually occurs. Trigger delay > 0: Post trigger: Tracing does not start until the set time after the trigger event.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4723[0...1]	Time slice cycle for trace / Trace cycle		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min 0.03125 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4.00000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0.12500 [ms]
Description:	Sets the time slice cycle in which the trace is called.		
Index:	[0] = Trace 0 [1] = Trace 1		
p4724[0...1]	Trace average in the time range / Trace average		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned8 P-Group: Trace and function generator Not for motor type: - Min 0000 bin	Calculated: - Dynamic index: - Units group: - Scaling: - Max 0001 bin	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 bin
Index:	[0] = Trace 0 [1] = Trace 1		
r4725[0...1]	Trace, data type 1 traced / Trace rec type 1		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Index:	[0] = Trace 0 [1] = Trace 1		
r4726[0...1]	Trace, data type 2 traced / Trace rec type 2		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Index:	[0] = Trace 0 [1] = Trace 1		
r4727[0...1]	Trace, data type 3 traced / Trace rec type 3		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Index:	[0] = Trace 0 [1] = Trace 1		

r4728[0...1]	Trace, data type 4 traced / Trace rec type 4		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Index:	[0] = Trace 0 [1] = Trace 1		

r4729[0...1]	Trace number of recorded values / Trace rec values		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the number of traced values for each signal.		
Index:	[0] = Trace 0 [1] = Trace 1		
Dependency:	Only valid when p4705 = 4.		

p4730[0...5]	Trace record signal 0 / Trace record sig 0		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Selects the first signal to be traced.		
Index:	[0] = Trace 0 parameter in BICO format [1] = Trace 1 parameter in BICO format [2] = Trace 0 PINx with DO Id and chart Id [3] = Trace 0 PINx with block Id and PIN Id [4] = Trace 1 PINy with DO Id and chart Id [5] = Trace 1 PINy with block Id and PIN Id		

p4731[0...5]	Trace record signal 1 / Trace record sig 1		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Selects the second signal to be traced.		
Index:	[0] = Trace 0 parameter in BICO format [1] = Trace 1 parameter in BICO format [2] = Trace 0 PINx with DO Id and chart Id [3] = Trace 0 PINx with block Id and PIN Id [4] = Trace 1 PINy with DO Id and chart Id [5] = Trace 1 PINy with block Id and PIN Id		

p4732[0...5]	Trace record signal 2 / Trace record sig 2		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Selects the third signal to be traced.		
Index:	[0] = Trace 0 parameter in BICO format [1] = Trace 1 parameter in BICO format [2] = Trace 0 PINx with DO Id and chart Id [3] = Trace 0 PINx with block Id and PIN Id [4] = Trace 1 PINy with DO Id and chart Id [5] = Trace 1 PINy with block Id and PIN Id		
p4733[0...5]	Trace record signal 3 / Trace record sig 3		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Selects the fourth signal to be traced.		
Index:	[0] = Trace 0 parameter in BICO format [1] = Trace 1 parameter in BICO format [2] = Trace 0 PINx with DO Id and chart Id [3] = Trace 0 PINx with block Id and PIN Id [4] = Trace 1 PINy with DO Id and chart Id [5] = Trace 1 PINy with block Id and PIN Id		
p4734[0...5]	Trace record signal 4 / Trace record sig 4		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Selects the fifth signal to be traced.		
Index:	[0] = Trace 0 parameter in BICO format [1] = Trace 1 parameter in BICO format [2] = Trace 0 PINx with DO Id and chart Id [3] = Trace 0 PINx with block Id and PIN Id [4] = Trace 1 PINy with DO Id and chart Id [5] = Trace 1 PINy with block Id and PIN Id		

p4735[0...5]			
Trace record signal 5 / Trace record sig 5			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Selects the sixth signal to be traced.		
Index:	[0] = Trace 0 parameter in BICO format [1] = Trace 1 parameter in BICO format [2] = Trace 0 PINx with DO Id and chart Id [3] = Trace 0 PINx with block Id and PIN Id [4] = Trace 1 PINy with DO Id and chart Id [5] = Trace 1 PINy with block Id and PIN Id		
p4736[0...5]			
Trace record signal 6 / Trace record sig 6			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Selects the seventh signal to be traced.		
Index:	[0] = Trace 0 parameter in BICO format [1] = Trace 1 parameter in BICO format [2] = Trace 0 PINx with DO Id and chart Id [3] = Trace 0 PINx with block Id and PIN Id [4] = Trace 1 PINy with DO Id and chart Id [5] = Trace 1 PINy with block Id and PIN Id		
p4737[0...5]			
Trace record signal 7 / Trace record sig 7			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Selects the eighth signal to be traced.		
Index:	[0] = Trace 0 parameter in BICO format [1] = Trace 1 parameter in BICO format [2] = Trace 0 PINx with DO Id and chart Id [3] = Trace 0 PINx with block Id and PIN Id [4] = Trace 1 PINy with DO Id and chart Id [5] = Trace 1 PINy with block Id and PIN Id		

r4740[0...16383] Trace 0 trace buffer signal 0 floating point / Trace 0 trace sig0

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Trace and function generator	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min -	Max -	Factory setting -

Description: Displays the trace buffer (record buffer) for trace 0 and signal 0.
The trace (record) buffer is sub-divided into memory banks, each containing 16384 values. Parameter p4795 can be used to toggle between the individual banks.
Example A:
The first 16384 values of signal 0, trace 0 are to be read out.
In this case, memory bank 0 is set with p4795 = 0. The first 16384 values can now be read out using r4740[0] to r4740[16383].
Example B:
The values 16385 to 32768 from signal 0, trace 0 are to be read out.
In this case, memory bank 1 is set with p4795 = 1. The values can now be read out in r4740[0] to r4740[16383].

Dependency: Refer to: p4795

r4741[0...16383] Trace 0 trace buffer signal 1 floating point / Trace 0 trace sig1

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Trace and function generator	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min -	Max -	Factory setting -

Description: Displays the trace buffer (record buffer) for trace 0 and signal 1.

Dependency: Refer to: r4740, p4795

r4742[0...16383] Trace 0 trace buffer signal 2 floating point / Trace 0 trace sig2

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Trace and function generator	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min -	Max -	Factory setting -

Description: Displays the trace buffer (record buffer) for trace 0 and signal 2.

Dependency: Refer to: r4740, p4795

r4743[0...16383] Trace 0 trace buffer signal 3 floating point / Trace 0 trace sig3

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Trace and function generator	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min -	Max -	Factory setting -

Description: Displays the trace buffer (record buffer) for trace 0 and signal 3.

Dependency: Refer to: r4740, p4795

r4744[0...16383] Trace 0 trace buffer signal 4 floating point / Trace 0 trace sig4

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 4.

Dependency: Refer to: r4740, p4795

r4745[0...16383] Trace 0 trace buffer signal 5 floating point / Trace 0 trace sig5

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 5.

Dependency: Refer to: r4740, p4795

r4746[0...16383] Trace 0 trace buffer signal 6 floating point / Trace 0 trace sig6

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 6.

Dependency: Refer to: r4740, p4795

r4747[0...16383] Trace 0 trace buffer signal 7 floating point / Trace 0 trace sig7

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 7.

Dependency: Refer to: r4740, p4795

r4750[0...16383] Trace 1 trace buffer signal 0 floating point / Trace 1 trace sig0

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 0.

Dependency: Refer to: r4740, p4795

r4751[0...16383] Trace 1 trace buffer signal 1 floating point / Trace 1 trace sig1

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 1.

Dependency: Refer to: r4740, p4795

r4752[0...16383] Trace 1 trace buffer signal 2 floating point / Trace 1 trace sig2

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 2.

Dependency: Refer to: r4740, p4795

r4753[0...16383] Trace 1 trace buffer signal 3 floating point / Trace 1 trace sig3

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 3.

Dependency: Refer to: r4740, p4795

r4754[0...16383] Trace 1 trace buffer signal 4 floating point / Trace 1 trace sig4

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 4.

Dependency: Refer to: r4740, p4795

r4755[0...16383] Trace 1 trace buffer signal 5 floating point / Trace 1 trace sig5

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 5.

Dependency: Refer to: r4740, p4795

r4756[0...16383] Trace 1 trace buffer signal 6 floating point / Trace 1 trace sig6

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 6.

Dependency: Refer to: r4740, p4795

r4757[0...16383] Trace 1 trace buffer signal 7 floating point / Trace 1 trace sig7

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 7.

Dependency: Refer to: r4740, p4795

r4760[0...16383] Trace 0 trace buffer signal 0 / Trace 0 trace sig0

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 0 as integer number.

Note: For signals, data type I32 or U32, the trace buffer is assigned as follows:
r4760[0] = value 0
r4760[1] = value 1
...
r4760[8191] = value 8191
For signals, data type I16 or U16, the trace buffer is assigned as follows:
r4760[0] = value 0 (bit 31 ... 16) and value 1 (bit 15 ... 0)
r4760[1] = value 2 (bit 31 ... 16) and value 3 (bit 15 ... 0)
...
r4760[8191] = value 16382 (bit 31 ... 16) and value 16383 (bit 15 ... 0)
For signals, data type I8 or U8, the trace buffer is assigned as follows:
r4760[0] = value 0 (bit 31 ... 24) value 1 (bit 23 ... 16) value 2 (bit 15 ... 8) value 3 (bit 7 ... 0)
r4760[1] = value 4 (bit 31 ... 24) value 5 (bit 23 ... 16) value 6 (bit 15 ... 8) value 7 (bit 7 ... 0)
...
r4760[8191] = value 32764 (bit 31 ... 24) value 32765 (bit 23 ... 16) value 32766 (bit 15 ... 8) value 32767 (bit 7 ... 0)

r4761[0...16383] Trace 0 trace buffer signal 1 / Trace 0 trace sig1

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 1.

Dependency: Refer to: r4760

r4762[0...16383] Trace 0 trace buffer signal 2 / Trace 0 trace sig2

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 2.

Dependency: Refer to: r4760

r4763[0...16383] Trace 0 trace buffer signal 3 / Trace 0 trace sig3

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 3.

Dependency: Refer to: r4760

r4764[0...16383] Trace 0 trace buffer signal 4 / Trace 0 trace sig4

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 4.

Dependency: Refer to: r4760

r4765[0...16383] Trace 0 trace buffer signal 5 / Trace 0 trace sig5

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 5.

Dependency: Refer to: r4760

r4766[0...16383] Trace 0 trace buffer signal 6 / Trace 0 trace sig6

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 6.

Dependency: Refer to: r4760

r4767[0...16383] Trace 0 trace buffer signal 7 / Trace 0 trace sig7

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 0 and signal 7.

Dependency: Refer to: r4760

r4770[0...16383] Trace 1 trace buffer signal 0 / Trace 1 trace sig0

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 0.

Dependency: Refer to: r4760

r4771[0...16383] Trace 1 trace buffer signal 1 / Trace 1 trace sig1

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 1.

Dependency: Refer to: r4760

r4772[0...16383] Trace 1 trace buffer signal 2 / Trace 1 trace sig2

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 2.

Dependency: Refer to: r4760

r4773[0...16383] Trace 1 trace buffer signal 3 / Trace 1 trace sig3

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 3.

Dependency: Refer to: r4760

r4774[0...16383] Trace 1 trace buffer signal 4 / Trace 1 trace sig4

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 4.

Dependency: Refer to: r4760

r4775[0...16383] Trace 1 trace buffer signal 5 / Trace 1 trace sig5

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 5.

Dependency: Refer to: r4760

r4776[0...16383] Trace 1 trace buffer signal 6 / Trace 1 trace sig6

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 6.

Dependency: Refer to: r4760

r4777[0...16383] Trace 1 trace buffer signal 7 / Trace 1 trace sig7

CU_CX32, CU_I,	Can be changed: -	Calculated: -	Access level: 3
CU_S,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: Trace and function generator	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 0
CU_S_S150	Min	Max	Factory setting
	-	-	-

Description: Displays the trace buffer (record buffer) for trace 1 and signal 7.

Dependency: Refer to: r4760

p4780[0...1]	Trace physical address signal 0 / Trace PhyAddr Sig0		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0000 bin	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1111 1111 1111 1111 1111 1111 1111 1111 bin	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 bin
Description:	Sets the physical address for the first signal to be traced. The data type is defined using p4730.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4781[0...1]	Trace physical address signal 1 / Trace PhyAddr Sig1		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0000 bin	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1111 1111 1111 1111 1111 1111 1111 1111 bin	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 bin
Description:	Sets the physical address for the second signal to be traced. The data type is defined using p4731.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4782[0...1]	Trace physical address signal 2 / Trace PhyAddr Sig2		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0000 bin	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1111 1111 1111 1111 1111 1111 1111 1111 bin	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 bin
Description:	Sets the physical address for the third signal to be traced. The data type is defined using p4732.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4783[0...1]	Trace physical address signal 3 / Trace PhyAddr Sig3		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0000 bin	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1111 1111 1111 1111 1111 1111 1111 1111 bin	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 bin
Description:	Sets the physical address for the fourth signal to be traced. The data type is defined using p4733.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4784[0...1]	Trace physical address signal 4 / Trace PhyAddr Sig4		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0000 bin	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1111 1111 1111 1111 1111 1111 1111 1111 bin	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 bin
Description:	Sets the physical address for the fifth signal to be traced. The data type is defined using p4734.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4785[0...1]	Trace physical address signal 5 / Trace PhyAddr Sig5		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0000 bin	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1111 1111 1111 1111 1111 1111 1111 1111 bin	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 bin
Description:	Sets the physical address for the sixth signal to be traced. The data type is defined using p4735.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4786[0...1]	Trace physical address signal 6 / Trace PhyAddr Sig6		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0000 bin	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1111 1111 1111 1111 1111 1111 1111 1111 bin	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 bin
Description:	Sets the physical address for the seventh signal to be traced. The data type is defined using p4736.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4787[0...1]	Trace physical address signal 7 / Trace PhyAddr Sig7		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0000 bin	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1111 1111 1111 1111 1111 1111 1111 1111 bin	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 bin
Description:	Sets the physical address for the eighth signal to be traced. The data type is defined using p4737.		
Index:	[0] = Trace 0 [1] = Trace 1		

p4789[0...1]	Trace physical address trigger signal / Trace PhyAddr Trig		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0000 hex Max FFFF FFFF hex	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 0000 hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 hex
Description:	Sets the physical address for the trigger signal. The data type is defined by making the appropriate selection in p4711.		
Index:	[0] = Trace 0 [1] = Trace 1		
r4790[0...1]	Trace, data type 5 traced / Trace rec type 5		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Index:	[0] = Trace 0 [1] = Trace 1		
r4791[0...1]	Trace, data type 6 traced / Trace rec type 6		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Index:	[0] = Trace 0 [1] = Trace 1		
r4792[0...1]	Trace, data type 7 traced / Trace rec type 7		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Index:	[0] = Trace 0 [1] = Trace 1		
r4793[0...1]	Trace, data type 8 traced / Trace rec type 8		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Index:	[0] = Trace 0 [1] = Trace 1		

p4795	Trace memory bank changeover / Trace mem changeov		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min 0 Max 500	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Changes over the memory bank to read out the contents of the trace buffer.		
Dependency:	Refer to: r4740, r4741, r4742, r4743, r4750, r4751, r4752, r4753		
r4799	Trace memory location free / Trace mem free		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Trace and function generator Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the free memory for the trace in bytes.		
Dependency:	Refer to: r4708		
p4800	Function generator control / FG control		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: Trace and function generator Not for motor type: - Min 0 Max 3	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The function generator is started with p4800 = 1. The signal is only generated for a 1 signal of binector input p4819.		
Value:	0: Stop function generator 1: Start function generator 2: Check function generator parameterization 3: Start function generator without enable signals		
Dependency:	Refer to: p4819		
r4805	Function generator status / FG status		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Integer16 P-Group: Trace and function generator Not for motor type: - Min 0 Max 6	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the actual status of the function generator.		
Value:	0: Inactive 1: Generate accelerating ramp to offset 2: Generate parameterized signal shape 3: Generate braking ramp 4: Function generator stopped due to missing enable signals 5: Function generator waits for BI: p4819 6: Function generator parameterization has been checked		
Dependency:	Refer to: p4800, p4819		

r4806.0 BO: Function generator status signal / FG status signal

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Trace and function generator	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the function generator.
0 signal: Function generator inactive
1 signal: Function generator running

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-

p4810 Function generator mode / FG operating mode

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Trace and function generator	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	99	0

Description: Sets the operating mode of the function generator.

Value:
0: Connection at connector output r4818
1: Connection at current setpoint after filter and r4818
2: Connection as disturbing torque and r4818
3: Connection at speed setpoint after filter and r4818
4: Connection at current setpoint before filter and r4818
5: Connection at speed setpoint before filter and r4818
99: Connection at physical address and r4818

p4812 Function generator physical address / FG phys address

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Trace and function generator	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	4294967295	0

Description: Sets the physical address where the function generator is to be connected.

Dependency: Only effective when p4810 = 99.

p4813 Function generator physical address reference value / FG phys addr ref

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Trace and function generator	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	1.00	1000000.00	1.00

Description: Sets the reference value for 100 % for referred inputs.

Dependency: Only effective when p4810 = 99.

p4815[0...2]	Function generator drive number / FG drive number		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: Trace and function generator Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the required drive where the function generator is to be connected.		
Index:	[0] = First drive for connection [1] = Second drive for connection [2] = Third drive for connection		
Dependency:	Only effective when p4810 = 1, 2, 3, 4 or 5.		
Note:	For the function generator, only type SERVO or VECTOR drives can be used.		
r4818	CO: Function generator output signal / FG output signal		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the output signal for the function generator.		
Dependency:	Refer to: p4810		
Note:	The value is displayed independently of the function generator mode. The signal is available as connector output for an ongoing interconnection.		
p4819	BI: Function generator control / FG control		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: Trace and function generator Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal source to control the function generator. When the function generator is running, signal generation is stopped with a 0 signal from BI: p4819 and p4800 is set to 0.		
Dependency:	Refer to: p4800		
p4820	Function generator signal shape / FG signal shape		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: Trace and function generator Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 5	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the signal to be generated for the function generator.		
Value:	1: Square-wave 2: Staircase 3: Delta 4: Binary noise - PRBS (Pseudo Random Binary Signal) 5: Sine-wave		

p4821	Function generator period / FG period duration		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 60000.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1000.00 [ms]
Description:	Sets the period of the signal to be generated for the function generator.		
Dependency:	Ineffective when p4820 = 4 (PRBS).		
p4822	Function generator pulse width / FG pulse width		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 60000.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 500.00 [ms]
Description:	Sets the pulse width for the signal to be generated for the function generator.		
Dependency:	Only effective when p4820 = 1 (square-wave).		
p4823	Function generator bandwidth / FG bandwidth		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min 0.0025 [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16000.0000 [Hz]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 4000.0000 [Hz]
Description:	Sets the bandwidth for the signal to be generated for the function generator.		
Dependency:	Only effective when p4820 = 4 (PRBS). Refer to: p4830 Refer to: A02041		
p4824	Function generator amplitude / FG amplitude		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min -1600.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1600.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 5.00 [%]
Description:	Sets the amplitude for the signal to be generated for the function generator.		
Dependency:	Units are dependent on p4810. If p4810 = 1, 2, 4: The amplitude is referred to p2002 (reference current). If p4810 = 3, 5: The amplitude is referred to p2000 (reference speed).		

p4825	Function generator 2nd amplitude / FG 2nd amplitude		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min -1600.00 [%] Max 1600.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 7.00 [%]
Description:	Sets the second amplitude for the signal to be generated for the function generator.		
Dependency:	Only effective for p4820 = 2 (staircase). Units are dependent on p4810. If p4810 = 1, 2, 4: The amplitude is referred to p2002 (reference current). If p4810 = 3, 5: The amplitude is referred to p2000 (reference speed).		
p4826	Function generator offset / FG offset		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min -1600.00 [%] Max 1600.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets the offset (DC component) of the signal to be generated for the function generator.		
Dependency:	Units are dependent on p4810. If p4810 = 1, 2, 4: The offset is referred to p2002 (reference current). If p4810 = 3, 5: The offset is referred to p2000 (reference speed). If p4810 = 2: In order to avoid the undesirable effects of play (backlash), the offset does not act on the current setpoint, but instead on the speed setpoint.		
p4827	Function generator ramp-up time to offset / FG ramp-up offset		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min 0.00 [ms] Max 100000.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 32.00 [ms]
Description:	Sets the ramp-up time to the offset for the function generator.		
p4828	Function generator lower limit / FG lower limit		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min -10000.00 [%] Max 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -100.00 [%]
Description:	Sets the lower limit for the function generator.		
Dependency:	For p4810 = 2 the limit only applies to the current setpoint, but not the speed setpoint (offset).		

p4829	Function generator upper limit / FG upper limit		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min 0.00 [%] Max 10000.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 100.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
Description:	Sets the upper limit for the function generator.		
Dependency:	For p4810 = 2 the limit only applies to the current setpoint, but not the speed setpoint (offset).		
p4830	Function generator time slice cycle / FG time slice		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min 0.03125 [ms] Max 2.00000 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 0.12500 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
Description:	Sets the time slice cycle in which the function generator is called.		
p4831	Function generator amplitude scaling / FG amplitude scal		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min 0.00000 [%] Max 200.00000 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 100.00000 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
Description:	Sets the scaling for the amplitude of the signal waveforms for all output channels. The value can be changed while the function generator is running.		
p4832[0...2]	Function generator amplitude scaling / FG amplitude scal		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min -340.28235E36 [%] Max 340.28235E36 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 100.00000 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
Description:	Sets the scaling for the amplitude of the signal waveforms separately for each output channel. The value cannot be changed while the function generator is running.		
Index:	[0] = First drive for connection [1] = Second drive for connection [2] = Third drive for connection		
p4833[0...2]	Function generator offset scaling / FG offset scal		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: FloatingPoint32 P-Group: Trace and function generator Not for motor type: - Min -340.28235E36 [%] Max 340.28235E36 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 100.00000 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
Description:	Sets the scaling for the offset of the signal waveforms separately for each output channel. The value cannot be changed while the function generator is running.		

Index: [0] = First drive for connection
 [1] = Second drive for connection
 [2] = Third drive for connection

r4899 Status word sequence control / ZSW seq_ctrl

TM41	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word of the sequence control from Terminal Module 41 (TM41).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Ready for sw on	Yes	No	-
	01	Ready	Yes	No	-
	02	Operation enabled	Yes	No	-
	03	Fault present	Yes	No	-
	04	Coast down active	No	Yes	-
	05	Quick Stop active	No	Yes	-
	06	Sw on inhibit	Yes	No	-
	07	Alarm present	Yes	No	-
	09	Control request	Yes	No	-
	14	Motor rotates forwards	Yes	No	-

r4950 OA application count / OA no

All objects	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: OEM range	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min	Max	Factory setting
	0	10	-

Description: Displays the number of OA applications installed on the memory card/device memory.

Dependency: Refer to: r4951, r4952, r4955, p4956, r4957, r4958, r4959, r4960

Note: OA: Open Architecture

r4951 OA application identifier, total length / OA ID length

All objects	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: OEM range	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min	Max	Factory setting
	0	90	-

Description: Displays the total length of the IDs of all the OA applications installed on the memory card/device memory.

Dependency: Refer to: r4950, r4952, r4955, p4956, r4957, r4958, r4959, r4960

Note: The identifier of an OA application comprises a maximum of 8 characters plus separator.

r4952	OA application GUID, total length / OA GUID length		
All objects	Can be changed: - Data type: Unsigned16 P-Group: OEM range Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 180	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the total length of the GUIDs of all the OA applications installed on the memory card/device memory.		
Dependency:	Refer to: r4950, r4951, r4955, p4956, r4957, r4958, r4959, r4960		
Note:	The GUID of an OA application comprises 16 characters plus 1 character major information plus 1 character, minor information. GUID: Globally Unique Identifier		
r4955[0...n]	OA application identifier / OA ID		
All objects	Can be changed: - Data type: Unsigned8 P-Group: OEM range Not for motor type: - Min -	Calculated: - Dynamic index: r4951 Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the IDs of all the OA applications installed on the memory card/device memory. r4955[0...8]: Identifier of OA application 1 r4955[9...17]: Identifier of OA applications 2, ...		
Dependency:	Refer to: r4950, r4951, r4952, p4956, r4957, r4958, r4959, r4960		
Notice:	If there is no OA application, then it is not possible to access an index.		
p4956[0...n]	OA application activation / OA act		
All objects	Can be changed: C1, T Data type: Integer16 P-Group: OEM range Not for motor type: - Min 0	Calculated: - Dynamic index: r4950 Units group: - Scaling: - Max 1	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Setting to activate the OA applications installed on the memory card/device memory. r4956[0]: Activates OA application 1 r4956[1]: Activates OA application 2, ...		
Value:	0: OA application inactive 1: OA application active		
Dependency:	Refer to: r4950, r4951, r4952, r4955, r4957, r4958, r4959, r4960		
Notice:	If there is no OA application, then it is not possible to access an index.		
r4957[0...n]	OA application version / OA version		
All objects	Can be changed: - Data type: Unsigned32 P-Group: OEM range Not for motor type: - Min -	Calculated: - Dynamic index: r4950 Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the versions of the OA applications installed on the memory card/device memory. r4957[0]: Version of OA application 1 r4957[1]: Version of OA application 2, ...		
Dependency:	Refer to: r4950, r4951, r4952, r4955, p4956, r4958, r4959, r4960		

Notice: If there is no OA application, then it is not possible to access an index.

Note: Example:
The value 1010100 should be interpreted as V01.01.01.00.

r4958[0...n]	OA application interface version / OA int_version		
All objects	Can be changed: - Data type: Unsigned32 P-Group: OEM range Not for motor type: - Min -	Calculated: - Dynamic index: r4950 Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the interface versions of the OA applications installed on the memory card/device memory. r4958[0]: Interface version of OA application 1 r4958[1]: Interface version of OA applications 2, ...		
Dependency:	Refer to: r4950, r4951, r4952, r4955, p4956, r4957, r4959, r4960		
Notice:	If there is no OA application, then it is not possible to access an index.		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		

r4959[0...n]	OA application GUID / OA GUID		
All objects	Can be changed: - Data type: Unsigned8 P-Group: OEM range Not for motor type: - Min -	Calculated: - Dynamic index: r4952 Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the GUIDs of the OA applications installed on the memory card/device memory. r4959[0...15]: GUID of OA application 1 r4960[16]: Major information of OA application 1 r4960[17]: Minor information of OA application 1 r4959[18...33]: GUID of OA application 2 r4960[34]: Major information of OA application 2 r4960[35]: Minor information of OA application 2, ...		
Dependency:	Refer to: r4950, r4951, r4952, r4955, p4956, r4957, r4958, r4960		
Notice:	If there is no OA application, then it is not possible to access an index.		

r4960[0...n]	OA application GUID drive object / OA GUID DO		
All objects	Can be changed: - Data type: Unsigned8 P-Group: OEM range Not for motor type: - Min -	Calculated: - Dynamic index: r4952 Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the GUIDs of the drive object of the OA applications installed on the memory card/device memory. r4960[0...15]: GUID of this drive object of OA application 1 r4960[16]: Major information of this drive object of OA application 1 r4960[17]: Minor information of this drive object of OA application 1 r4960[18...33]: GUID of this drive object of OA application 2 r4960[34]: Major information of this drive object of OA application 2 r4960[35]: Minor information of this drive object of OA application 2, ...		
Dependency:	Refer to: r4950, r4951, r4952, r4955, p4956, r4957, r4958, r4959		
Notice:	If there is no OA application, then it is not possible to access an index.		

p4961[0...n]	OA application logbook module selection / OA logbook module		
All objects	Can be changed: T Data type: Unsigned32 P-Group: OEM range Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: r4950 Units group: - Scaling: - Max FFFF FFFF hex	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 hex
Description:	Only for service purposes.		
r4975	OA application invalid number / OA inv no.		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: OEM range Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of invalid OA applications installed on the memory card/device memory.		
Dependency:	Refer to: r4976, r4978, r4979		
Note:	OA: Open Architecture		
r4976	OA application invalid identifier, total length / OA inv ID length		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: OEM range Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the total length of the IDs of all the invalid OA applications installed on the memory card/device memory.		
Dependency:	Refer to: r4975, r4978, r4979		
Note:	The identifier of an invalid OA application comprises a maximum of 8 characters plus separator.		
r4978[0...n]	OA application invalid identifier / OA inv ID		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned8 P-Group: OEM range Not for motor type: - Min -	Calculated: - Dynamic index: r4976 Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the IDs of all the invalid OA applications installed on the memory card/device memory. r4978[0...8]: Identifier of invalid OA application 1 r4978[9...17]: Identifier of invalid OA application 2, ...		
Dependency:	Refer to: r4975, r4976, r4979		
Notice:	If there is no invalid OA application, then it is not possible to access an index.		

r4979[0...n]	OA application invalid error code / OA inv error code		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: OEM range Not for motor type: - Min -	Calculated: - Dynamic index: r4975 Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the error code of the invalid OA applications installed on the memory card/device memory. r4979[0]: Fault value of OA application 1 r4979[1]: Fault value of OA application 2, ...		
Dependency:	Refer to: r4975, r4976, r4978		
Notice:	If there is no invalid OA application, then it is not possible to access an index.		
Note:	The value in the error code must be interpreted in binary form. The bits have the following meaning: Bit 0: Incompatible OA interface version. Bit 1: OA application could not be loaded. Bit 2: Incorrect description files. Bit 3: OA application does not define a CPU type. Bit 4: OA application for this device not supported (incorrect CPU type). Bit 5: OA application for this device not supported (incorrect type ID). Bit 6: Incorrect description files (Const/Startup incompatible).		
p5401	BI: Line droop control activation / Line drp act		
A_INF (Line droop reg)	Can be changed: T Data type: Unsigned32 / Binary P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source to activate line droop control for the infeed. BI: p5401 = 1 signal: Activation of line droop control and deactivation of closed-loop DC-link voltage control/closed-loop current control. BI: p5401 = 0 signal: Activation of closed-loop DC-link voltage control/closed-loop current control and deactivation of line droop control.		
Dependency:	Refer to: r5402		
Notice:	Line droop control can only be activated if the power units have a gating unit with current limitation control (r0192.19 = 1).		
Note:	So that closed-loop DC-link voltage control is activated by the VECTOR drive object's technology controller (due to closed-loop DC-link voltage control for the infeed being deactivated) the following BICO interconnection must be set: BI: p3513 (VECTOR) = r5402.0 (A_INF)		

r5402.0...3 CO/BO: Line droop control status word / Line drp ZSW

A_INF (Line droop reg)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word of line droop regulation for the infeed.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Line droop regulation not active	Yes	No	-
	01	Line droop regulation active	Yes	No	-
	02	Line droop regulation, stand-alone	Yes	No	-
	03	Line droop regulation current limitation active	Yes	No	-

p5403[0...1] CI: Line droop control current signal source / Line drp I s_src

A_INF (Line droop reg)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	-	-	[0] 3467[0] [1] 3467[1]

Description: Sets the signal source for the current to be regulated in alpha/beta coordinates.

Index: [0] = Alpha
[1] = Beta

Dependency: Refer to: p5404

Note: The following BICO interconnections are recommended:

- Droop regulation for current/voltage at the line filter: BI: p5403 = r3467 (BI: p5404 = r3468 must be set)
- Droop regulation for current/voltage at the line transformer: BI: p5403 = r5497 (BI: p5404 = r5488 or r5498 must be set)

p5404[0...1] CI: Line droop control voltage signal source / Line drp U s_src

A_INF (Line droop reg)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	-	-	[0] 3468[0] [1] 3468[1]

Description: Sets the signal source for the voltage to be regulated in alpha/beta coordinates.

Index: [0] = Alpha
[1] = Beta

Note: The following BICO interconnections are recommended:

- Droop regulation for current/voltage at the line filter: BI: p5404 = r3468 (BI: p5403 = r3467 must be set)
- Droop regulation for current/voltage at the line transformer: BI: p5404 = r5488 or r5498 (BI: p5403 = r5497 must be set)

p5405	Line droop control frequency droop no-load frequency / Line drp f_no-ld		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 30.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max 300.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the no-load frequency (as a % of p0211) for the line droop control active power frequency droop. Droop formula (without smoothing): $r5410 = (p5405 + p5406 + p5407 * r5411[0] / r0206) * p0211$		
Dependency:	Refer to: p5409		
Note:	The droop characteristic input variable is the active power r5411[0] at the selected connection point (p5403, p5404). The output frequency calculated using the above formula is filtered in accordance with the parameterized smoothing time (p5409). The smoothed output frequency is displayed in r5410.		
p5406	Cl: Line droop control frequency droop supplementary setpoint / L drp f_suppl_setp		
A_INF (Line droop reg)	Can be changed: U, T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the frequency droop supplementary setpoint (as a % of p0211).		
Dependency:	Refer to: p5405		
p5407	Line droop control frequency droop gradient / Line drp f_grad		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -100.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 5.00 [%]
Description:	Sets the frequency droop gradient (as a % of p0211).		
Dependency:	Refer to: p5405		
p5409	Line droop control frequency droop smoothing time / Line drp f_t_sm		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 150.00 [ms]
Description:	Sets the smoothing time for the output frequency of the active power frequency droop (r5410).		
Dependency:	Refer to: p5405		
Note:	When the load changes, mechanical energy generation units induce a delayed change in frequency on account of their inertia. The converter tries to emulate this response with the assistance of the smoothing time. For the line to remain stable, all the energy generation units in a separate network have to respond in a similar manner during operation.		

r5410	Line droop control frequency droop output / Line drp f outp		
A_INF (Line droop reg)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Hz]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Hz]
Description:	Displays the smoothed output frequency of the active power frequency droop. Droop formula (without smoothing): $r5410 = (p5405 + p5406 + p5407 * r5411[0] / r0206) * p0211$		
Dependency:	Refer to: p5405		
r5411[0...1]	Line droop control frequency droop active power / Line drp f P_act		
A_INF (Line droop reg)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kW]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [kW]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [kW]
Description:	Displays the input active power of the active power frequency droop. The active power is calculated for the currents and voltages defined with connector inputs p5403 and p5404.		
Index:	[0] = Unsmoothed [1] = Smoothed		
Dependency:	Refer to: p5409		
Note:	The value is smoothed by a PT1 filter (p5409).		
r5412	Line droop control line angle / Line drp angle		
A_INF (Line droop reg)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°]	Calculated: - Dynamic index: - Units group: - Scaling: p2005 Max - [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [°]
Description:	Displays the current line angle for line droop control. The value is calculated by integrating the output frequency of the active power frequency droop (r5410).		
p5415	Line droop control voltage droop no-load voltage / Line drp U_no-ld		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 30.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max 300.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the no-load voltage (as a % of p0210) for the reactive current voltage droop for line droop control. Droop formula (without smoothing): $p5420 = (p5415 + p5416 + p5417 * r5421[0] / r0207) * p0210$		
Dependency:	Refer to: p5419		
Note:	The droop characteristic input variable is the reactive power r5421[0] at the selected connection point (p5403, p5404). The output voltage calculated using the above formula is filtered in accordance with the parameterized smoothing time p5419. The smoothed output voltage is displayed in r5420.		

p5416	CI: Line droop control voltage droop supplementary setpoint / L drp U_suppl_setp		
A_INF (Line droop reg)	Can be changed: U, T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the voltage droop supplementary setpoint (as a % of p0210).		
Dependency:	Refer to: p5415		
p5417	Line droop control voltage droop gradient / Line drp U grad		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -100.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 5.00 [%]
Description:	Sets the voltage droop gradient (as a % of p0210).		
Dependency:	Refer to: p5415		
p5419	Line droop control voltage droop smoothing time / Line drp U t_sm		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 150.00 [ms]
Description:	Sets the smoothing time for the reactive current actual value (r5421[1]). The smoothed reactive current is the input quantity for the reactive current voltage droop.		
Dependency:	Refer to: p5415		
Note:	When the load changes, mechanical energy generation units induce a delayed change in voltage on account of their electrical properties. The converter tries to emulate this response with the assistance of the smoothing time. For the line to remain stable, all the energy generation units in a separate network have to respond in a similar manner during operation.		
r5420	Line droop control voltage droop output / Line drp U outp		
A_INF (Line droop reg)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Vrms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the smoothed output voltage of the reactive current voltage droop. Droop formula (without smoothing): $p5420 = p5415 + p5416 + p5417 * r5421[0] / r0207 * p0210$		
Dependency:	Refer to: p5415		

r5421[0...1]	Line droop control voltage droop reactive current / Line drp U I_reac		
A_INF (Line droop reg)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [A]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [A]
Description:	Displays the input reactive current of the reactive current voltage droop. The reactive current is calculated for the currents and voltages defined with connector inputs p5403 and p5404.		
Index:	[0] = Unsmoothed [1] = Smoothed		
Dependency:	Refer to: p5419		
Note:	The value is smoothed by a PT1 filter (p5419).		
r5422[0...1]	Line droop control voltage droop reactive power / Line drp U Q_reac		
A_INF (Line droop reg)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [kvar]	Calculated: - Dynamic index: - Units group: 14_12 Scaling: - Max - [kvar]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [kvar]
Description:	Displays the input reactive power of the reactive current voltage droop. The reactive power is calculated for the currents and voltages defined with connector inputs p5403 and p5404.		
Index:	[0] = Unsmoothed [1] = Smoothed		
Dependency:	Refer to: p5419		
Note:	The value is smoothed by a PT1 filter (p5419).		
p5423	Line droop control supplementary inductance / Line drp L_suppl		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: - Not for motor type: - Min -10.000 [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10.000 [mH]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [mH]
Description:	Sets an imaginary supplementary inductance for the infeed. The supplementary inductance supplements the effect of the voltage droop, in particular for transient operations.		
Dependency:	Refer to: p5424		
Warning:	Unsuitable supplementary inductance and smoothing time settings will make line droop regulation unstable.		
			
Note:	The supplementary inductance is deactivated with p5523 = 0. The converter's output voltage is calculated so that the converter behaves like a supplementary inductance and as such achieves an effect similar to a line reactor with higher/lower inductance. The supplementary inductance can be used for damping in the event of resonance phenomena on the line.		

p5424	Line droop control supplementary inductance smoothing time / L drp L_suppl t_sm		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [ms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.40 [ms]
Description:	Sets the smoothing time for the effect of the supplementary inductance (p5423).		
Dependency:	Refer to: p5423		
Note:	The supplementary inductance causes current change to be fed back to the converter's output voltage.		
p5425[0...1]	CI: Line droop control voltage control signal source / L drp U_ctrl s_sc		
A_INF (Line droop reg)	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 3468[0] [1] 3468[1]
Description:	Sets the signal source for the voltage to be regulated at the connection point in alpha/beta coordinates.		
Index:	[0] = Alpha [1] = Beta		
Dependency:	Refer to: p5427		
Note:	The following BICO interconnections are recommended: - Compensation of the internal voltage drop for connection point at line filter: BI: p5425 = r3468 - Compensation of the internal voltage drop for connection point at line transformer: BI: p5425 = r5488 or r5498		
p5426	Line droop control voltage control P gain / Line drp U_ctrl Kp		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.000 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.000 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.000 [%]
Description:	Sets the proportional gain for the controller for voltage control at the connection point. The controller compensates internal voltage drops under load conditions and as such precisely converts the droop characteristic at the connection point (defined in p5425).		
Dependency:	Refer to: p5427		
Note:	The proportional component of the controller is deactivated with p5426 = 0.		
p5427	Line droop control voltage control integration time / Line drp U_ctrl Ti		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.000 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000.000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 500.000 [ms]
Description:	Sets the integration time for the controller for voltage control at the connection point. The controller compensates internal voltage drops under load conditions and as such precisely converts the droop characteristic at the connection point (defined in p5425).		
Dependency:	Refer to: p5426		
Note:	The integral component of the controller is deactivated with p5427 = 0.		

p5428[0...3]	Line droop control voltage control short circuit / Line drp U_ctrl sh		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 20.00 [%] [1] 0.00 [%] [2] 90.00 [%] [3] 2.00 [%]
Description:	Sets the parameters for rapid pre-control of the output voltage in the event of a voltage dip. The values relate to the voltage setpoint from the voltage droop (r5420). The voltage output by the converter is formed from the product of the voltage setpoint and the adaptation factor, which is calculated in rapid pre-control.		
Index:	[0] = Adaptation factor lower limit [1] = Adaptation factor increment current limit [2] = Adaptation factor increment voltage difference [3] = Adaptation factor increment increase		
Dependency:	Refer to: r5452		
Note:	Rapid adaptation of the voltage setpoint to a reduced line voltage (e.g. in the event of a line short circuit) reduces the number of additional switching operations completed by the lower-level rapid current hysteresis controller. Re index 0: Minimum value for the adaptation factor, must be complied with for the calculation of the reduced output voltage. The value is dependent upon the voltage drop at the line reactor when the current limit is reached. As such, the value results from the reactor inductance, the frequency, and the current limit. A value of 100% deactivates the function. Re index 1: When the current limit is reached, the adaptation factor is weighted with this factor in each sampling cycle. A value of 0 means that the adaptation factor will jump to the minimum value (p5428[0]) if the current limit is reached. A value of 100% deactivates the function. Re index 2: If the voltage across the line reactor exceeds the minimum value (p5428[0]) for at least 2 current controller clock cycles, the adaptation factor will be weighted with this factor (p5428[2]) in every sampling cycle. Re index 3: If none of the above criteria for reducing the adaptation factor is met, the adaptation factor is increased by the percentage value p5428[3] in each current controller clock cycle (the increase is additive). If the adaptation factor reaches the maximum value of 100%, the effect of pre-control by means of multiplication by the droop setpoint r5420 is cancelled out.		
r5429	Line droop control voltage control output / L drp U_ctrl outp		
A_INF (Line droop reg)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Vrms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [Vrms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [Vrms]
Description:	Displays the controller output for voltage control at the connection point.		
Dependency:	Refer to: p5426, p5427		

p5430	Modulation depth controller setpoint / Mod_ctrl setp		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 50.000 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 110.000 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 97.000 [%]
Description:	Sets the setpoint for the maximum stationary modulation depth.		
p5431	Modulation depth controller dynamic response / Mod_ctrl dyn		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.000 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000.000 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10.000 [ms]
Description:	Sets the dynamic response for the modulation depth controller.		
p5432[0...1]	Modulation depth controller output voltage limits / Mod_ctrl lim U		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min -100.000 [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.000 [V]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 50.000 [V] [1] -50.000 [V]
Description:	Sets the limits for the output voltage on the modulation depth controller.		
Index:	[0] = Maximum value [1] = Minimum value		
r5433	CO: Modulation depth controller output / Mod_ctrl outp		
A_INF (Line droop reg)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [V]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Display and connector output for the modulation depth controller output.		
p5434	Direct component controller low pass limit frequency / I_dc_reg PT2 f		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min 0.000 [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1.000 [Hz]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.500 [Hz]
Description:	Sets the limit frequency for the 2nd order low-pass filter to suppress the direct component in the converter current.		
Dependency:	Refer to: p5435		

p5435	Direct component controller low pass damping / I_dc_reg PT2 D				
A_INF (Line droop reg)	Can be changed: U, T		Calculated: -		Access level: 3
	Data type: FloatingPoint32		Dynamic index: -		Func. diagram: -
	P-Group: Displays, signals		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min 0.001		Max 10.000		Factory setting 1.000
Description:	Sets the damping for the 2nd order low-pass filter to suppress the direct component in the converter current.				
Dependency:	Refer to: p5434				
p5436	Direct component controller P gain / I_dc_reg Kp				
A_INF (Line droop reg)	Can be changed: U, T		Calculated: -		Access level: 3
	Data type: FloatingPoint32		Dynamic index: -		Func. diagram: -
	P-Group: Displays, signals		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min 0.000 [Ohm]		Max 10.000 [Ohm]		Factory setting 0.020 [Ohm]
Description:	Sets the proportional gain for the PI controller to suppress the direct component in the converter current.				
Dependency:	Refer to: p5437				
Note:	Recommended setting: p5436 = total resistance of reactor and transformer.				
p5437	Direct component controller integration time / I_dc_reg Ti				
A_INF (Line droop reg)	Can be changed: U, T		Calculated: -		Access level: 3
	Data type: FloatingPoint32		Dynamic index: -		Func. diagram: -
	P-Group: Displays, signals		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min 0.000 [s]		Max 100.000 [s]		Factory setting 3.500 [s]
Description:	Sets the integration time for the PI controller to suppress the direct component in the converter current.				
Dependency:	Refer to: p5436				
Note:	Recommended setting: p5437 = 70% of the control loop time constant transformer magnetizing inductance/total resistance.				
p5440	Harmonics controller bandpass filter activation / Bandpass act				
A_INF (Line droop reg)	Can be changed: U, T		Calculated: -		Access level: 4
	Data type: Unsigned16		Dynamic index: -		Func. diagram: -
	P-Group: -		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min -		Max -		Factory setting 0000 bin
Description:	Setting to activate the bandpass filter for the harmonics controller.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Filter 0	On	Off	-
	01	Filter 1	On	Off	-
	02	Filter 2	On	Off	-
	03	Filter 3	On	Off	-
Dependency:	Refer to: p5441, p5442, p5443				

p5441[0...3]	Harmonics controller bandpass filter gain / Bandpass gain		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: - Not for motor type: - Min 0.00	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000.00	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 120.00
Description:	Sets the gain for the bandpass filter for the harmonics controller.		
Index:	[0] = Filter 0 [1] = Filter 1 [2] = Filter 2 [3] = Filter 3		
Dependency:	Refer to: p5440, p5442		
p5442[0...3]	Harmonics controller bandpass filter mid-frequency / Bandpass f_mid		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: - Not for motor type: - Min 0.00 [Hz]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [Hz]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 300.00 [Hz]
Description:	Sets the mid-frequency for the bandpass filter for the harmonics controller.		
Index:	[0] = Filter 0 [1] = Filter 1 [2] = Filter 2 [3] = Filter 3		
Dependency:	Refer to: p5440, p5441, p5443		
p5443	Harmonics controller bandpass filter gain total / Bandpass gain tot		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: - Not for motor type: - Min 0.00	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.00
Description:	Sets the total gain for all bandpass filters for the harmonics controller.		
Dependency:	Refer to: p5440, p5441, p5442		
Note:	The parameter should be set to a value which is greater than or equal to the fundamental frequency.		
r5444[0...1]	Line droop control line voltage absolute value / U_line abs val		
A_INF (Line droop reg)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [V]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Displays the absolute value of the line voltage. $r5444 = \sqrt{r5445[0]^2 + r5445[1]^2}$		
Index:	[0] = Droop (p5404) [1] = Voltage correction (p5425)		
Dependency:	Refer to: r5445		

r5445[0...1] Line droop control line voltage alpha/beta component / U_line A/B comp

A_INF (Line droop reg)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the alpha/beta component of the line voltage.		
Index:	[0] = Alpha [1] = Beta		
Dependency:	Refer to: r5444		

r5446[0...1] Line droop control line voltage active/reactive component / U_line P/Q comp

A_INF (Line droop reg)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the active/reactive component of the line voltage.		
Index:	[0] = Active [1] = Reactive		

r5447 CO: Line droop control line voltage absolute value / I_line abs val

A_INF (Line droop reg)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: 6_2	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [Arms]	- [Arms]	- [Arms]
Description:	Displays the line current absolute value at the connection point (p5403, p5404).		
Dependency:	Refer to: r5448		
Note:	The following applies: $r5447 = \sqrt{r5448[0]^2 + r5448[1]^2}$		

r5448[0...1] Line droop control line current alpha/beta component / I_line A/B comp

A_INF (Line droop reg)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [A]	- [A]	- [A]
Description:	Displays the alpha/beta component of the line current.		
Index:	[0] = Alpha [1] = Beta		
Dependency:	Refer to: r5447		

r5449[0...1] Line droop control line current active/reactive component / I_line P/Q comp

A_INF (Line droop reg)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [A]	- [A]	- [A]
Description:	Displays the active/reactive component of the line current.		
Index:	[0] = Active [1] = Reactive		

p5451 BI: Current hysteresis controller operating mode / I_hyst_ctrl mode

A_INF (Line droop reg)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for setting the operating mode on the current hysteresis controller. BI: p5451 = 1 signal: Combined operation (main generator and auxiliary generator on the line). BI: p5451 = 0 signal: Isolated operation (main generator only on the line).		

r5452.0...3 CO/BO: Current hysteresis controller sequence control status word / I_hyst_ctrl seqZSW

A_INF (Line droop reg)	Can be changed: -	Calculated: -	Access level: 3																							
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -																							
	P-Group: Closed-loop control	Units group: -	Unit selection: -																							
	Not for motor type: -	Scaling: -	Expert list: 1																							
	Min	Max	Factory setting																							
	-	-	-																							
Description:	Displays the sequence control status word on the current hysteresis controller.																									
Bit field:	<table><tr><td>Bit</td><td>Signal name</td><td>1 signal</td><td>0 signal</td><td>FP</td></tr><tr><td>00</td><td>Initialization</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>01</td><td>No-load</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>02</td><td>Normal</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>03</td><td>Short circuit</td><td>Yes</td><td>No</td><td>-</td></tr></table>	Bit	Signal name	1 signal	0 signal	FP	00	Initialization	Yes	No	-	01	No-load	Yes	No	-	02	Normal	Yes	No	-	03	Short circuit	Yes	No	-
Bit	Signal name	1 signal	0 signal	FP																						
00	Initialization	Yes	No	-																						
01	No-load	Yes	No	-																						
02	Normal	Yes	No	-																						
03	Short circuit	Yes	No	-																						
Dependency:	Refer to: p5457, p5458, p5459 Refer to: F06850																									

p5453[0...5]	Current hysteresis controller overcurrent limit / I_hyst_ctrl I lim		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min 50.000 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 140.000 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting [0] 85.000 [%] [1] 85.000 [%] [2] 85.000 [%] [3] 85.000 [%] [4] 85.000 [%] [5] 115.000 [%]
Description:	Sets the permissible maximum current in the event of an overload and short circuit for the modulator to intervene and apply limiting. The effective current limits are calculated from the parameterized overcurrent (p5453) and the hysteresis width (p5454).		
Index:	[0] = No-load operation (p5451 = 1 signal) [1] = Normal operation (p5451 = 1 signal) [2] = Short-circuit operation (p5451 = 1 signal) [3] = No-load operation (p5451 = 0 signal) [4] = Normal operation (p5451 = 0 signal) [5] = Short-circuit operation (p5451 = 0 signal)		
Dependency:	The overcurrent limit (p5453) cannot be set less than or equal to the hysteresis width (p5454). Refer to: p5454, p5455		
Note:	The value refers to r0209.		
p5454[0...5]	Current hysteresis controller overcurrent hysteresis width / I_hyst_ctrl I hyst		
A_INF (Line droop reg)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min 15.000 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 50.000 [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 20.000 [%]
Description:	Sets the permissible current hysteresis in the event of an overload and short circuit for the modulator to intervene and apply limiting. The effective current limits are calculated from the parameterized overcurrent (p5453) and the hysteresis width (p5454).		
Index:	[0] = No-load operation (p5451 = 1 signal) [1] = Normal operation (p5451 = 1 signal) [2] = Short-circuit operation (p5451 = 1 signal) [3] = No-load operation (p5451 = 0 signal) [4] = Normal operation (p5451 = 0 signal) [5] = Short-circuit operation (p5451 = 0 signal)		
Dependency:	The hysteresis width (p5454) cannot be set greater than or equal to the overcurrent limit (p5453). Refer to: p5453		
Note:	The value refers to r0209.		

p5455[0...5]		Current hysteresis controller overcurrent tolerance range / I_hyst_ctrl I tol	
A_INF (Line droop reg)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 10.000 [%]	Max 100.000 [%]	Factory setting 20.000 [%]
Description:	Sets the permissible tolerance range in the event of an overload and short circuit for the modulator to intervene and apply limiting. The effective current limit is calculated from the overcurrent limit (p5453) plus the overcurrent tolerance range (p5455).		
Index:	[0] = No-load operation (p5451 = 1 signal) [1] = Normal operation (p5451 = 1 signal) [2] = Short-circuit operation (p5451 = 1 signal) [3] = No-load operation (p5451 = 0 signal) [4] = Normal operation (p5451 = 0 signal) [5] = Short-circuit operation (p5451 = 0 signal)		
Dependency:	Refer to: p5453		
Note:	The value refers to r0209.		

p5456[0...2]		Current hysteresis controller configuration / I_hyst_ctrl config				
A_INF (Line droop reg)	Can be changed:	U, T	Calculated:	-	Access level:	4
	Data type:	Unsigned16	Dynamic index:	-	Func. diagram:	-
	P-Group:	Modulation	Units group:	-	Unit selection:	-
	Not for motor type:	-	Scaling:	-	Expert list:	1
	Min	-	Max	-	Factory setting	[0] 1000 0000 0010 0000 bin [1] 1000 0000 0010 0000 bin [2] 1000 0000 0011 0000 bin
Description:		Sets the configuration for the current hysteresis controller.				
Index:		[0] = No-load [1] = Normal operation [2] = Sh-cct operation				
Bit field:	Bit	Signal name	1 signal	0 signal	FP	
	04	Disable wobulation amplitude	Yes	No	-	
	05	Activate extended current limitation control	Yes	No	-	
	06	Activate isochronous current limitation	Yes	No	-	
	15	Activate flat-top modulation	Yes	No	-	
Notice:		Re bit 04 = 0: The pulse frequency wobulation amplitude (p1811) is enabled. Only applies if p1810.2 = 1. Re bit 04 = 1: The pulse frequency wobulation amplitude (p1811) is disabled. Only applies if p1810.2 = 1.				

p5457[0...2] Current hysteresis controller pulse frequency changeover / I_hyst_ctrl f_puls

A_INF (Line droop reg)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 50.000 [%]	Max 200.000 [%]	Factory setting [0] 200.000 [%] [1] 100.000 [%] [2] 100.000 [%]

Description: Sets the pulse frequency for the operating states of the current hysteresis controller.

Index:
[0] = No-load
[1] = Normal operation
[2] = Sh-cct operation

Note:
The value refers to p1800.
Only factors of the pulse frequency in p1800 are permitted.

p5458[0...1] Current hysteresis controller minimum time operating state / I_hyst_ctrl t mode

A_INF (Line droop reg)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [s]	Max 10.000 [s]	Factory setting [0] 1.000 [s] [1] 1.250 [s]

Description: Sets the minimum times for the operating states of the current hysteresis controller.

Index:
[0] = Normal operation
[1] = Sh-cct operation

Dependency: Refer to: F06850

Note:
Re p5458[0]:
Minimum time for operating state "Rated operation" for change to "No-load operation".
Re p5458[1]:
Permissible short-circuit duration. If the short circuit is not cleared within this time, the main generator will shut down with fault F06850.

p5459[0...3] Current hysteresis controller sequence control state change / I_hyst_ctrl seq

A_INF (Line droop reg)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [%]	Max 200.000 [%]	Factory setting [0] 10.000 [%] [1] 5.000 [%] [2] 70.000 [%] [3] 75.000 [%]

Description: Sets the limits for state change in the context of sequence control on the current hysteresis controller.

Index:
[0] = Lower current limit no-load/normal operation
[1] = Upper current limit normal/no-load operation
[2] = Lower voltage limit short-circuit operation
[3] = Upper voltage limit short-circuit/normal operation

Dependency: Refer to: r5452

Note:
The current value refers to r0209.
The voltage value refers to p0210.

p5460 VSM2 input line supply voltage, voltage scaler / VSM2 inp U_scaler					
A_INF (Line transf)	Can be changed: T Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min 0.00 [%]		Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max 100000.00 [%]		Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [%]
Description:	Sets a voltage scaler for Voltage Sensing Module 2 (VSM2).				
Note:	When the 690 V input is used (X522) without voltage scaler, 0 % should be entered. When the 100 V input (X521) is used with voltage scaler to measure medium voltages, the dividing (scaling) factor multiplied by 100% should be entered. Example: 1000 V line supply voltage, voltage scaling, 10:1 --> voltage at the VSM input is 100 V --> p5460 = 10 * 100 % = 1000 %				
r5461[0...n] CO: VSM2 input line supply voltage u1 - u2 / VSM2 inp u1-u2					
A_INF (Line transf)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [V]		Calculated: - Dynamic index: p0150 Units group: 5_3 Scaling: p2001 Max - [V]		Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the voltage between phases L1 and L2.				
Note:	X521.1 or X522.1: Connection of L1 X521.2 or X522.2: Connection of L2				
r5462[0...n] CO: VSM2 input line supply voltage u2 - u3 / VSM2 inp u2-u3					
A_INF (Line transf)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [V]		Calculated: - Dynamic index: p0150 Units group: 5_3 Scaling: p2001 Max - [V]		Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the voltage between phases L2 and L3.				
Note:	X521.2 or X522.2: Connection of L2 X521.3 or X522.3: Connection of L3				
r5464[0...n] CO: VSM2 temperature evaluation status / VSM2 temp status					
A_INF (Line transf)	Can be changed: - Data type: Unsigned16 P-Group: Terminals Not for motor type: - Min -		Calculated: - Dynamic index: p0150 Units group: - Scaling: - Max -		Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the status of the temperature evaluation of Voltage Sensing Module 2 (VSM2). This displays whether the temperature actual value has exceeded the fault/alarm threshold.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Temperature alarm threshold exceeded	Yes	No	-
	01	Temperature fault threshold exceeded	Yes	No	-

p5465[0...n]	VSM2 temperature evaluation sensor type / VSM2 temp sens_typ		
A_INF (Line transf)	Can be changed: T Data type: Integer16 P-Group: Closed-loop control Not for motor type: - Min 0	Calculated: - Dynamic index: p0150 Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the temperature sensor for Voltage Sensing Module 2 (VSM2). The temperature sensor is connected to terminals X520.5 and X520.6 on the VSM2.		
Value:	0: No sensor 1: PTC 2: KTY84		
r5466[0...n]	CO: VSM2 temperature KTY / VSM2 temp KTY		
A_INF (Line transf)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [°C]	Calculated: - Dynamic index: p0150 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the temperature actual value of a KTY84 temperature sensor connected to Voltage Sensing Module 2 (VSM2). Prerequisite: A KTY84 sensor is connected and p5465 = 2 is set.		
Dependency:	Refer to: p5465		
Note:	For sensor type PTC (p5465 = 1), the following applies: - Below the nominal response temperature, r5466 = -50 °C. - Above the nominal response temperature, r5466 = 199.9 °C.		
p5467[0...n]	VSM2 overtemperature alarm threshold / VSM2 temp A_thresh		
A_INF (Line transf)	Can be changed: T Data type: FloatingPoint32 P-Group: - Not for motor type: - Min 0.00 [°C]	Calculated: - Dynamic index: p0150 Units group: 21_1 Scaling: TEMP Max 301.00 [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 150.00 [°C]
Description:	Sets the alarm threshold for the KTY temperature sensor on Voltage Sensing Module 2 (VSM2). Prerequisite: A KTY84 sensor is connected and p5465 = 2 is set.		
Dependency:	Refer to: p5465 Refer to: A34211		
p5468[0...n]	VSM2 overtemperature shutdown threshold / VSM2 temp F_thresh		
A_INF (Line transf)	Can be changed: T Data type: FloatingPoint32 P-Group: - Not for motor type: - Min 0.00 [°C]	Calculated: - Dynamic index: p0150 Units group: 21_1 Scaling: TEMP Max 301.00 [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting 180.00 [°C]
Description:	Sets the shutdown threshold for the KTY temperature sensor of the VSM2 to monitor a temperature.		
Dependency:	Refer to: p5467 Refer to: F34207		

p5469[0...n] VSM2 overtemperature hysteresis / VSM2 temp hyst

A_INF (Line transf)	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: -
	P-Group: -	Units group: 21_2	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min 1.00 [K]	Max 50.00 [K]	Factory setting 3.00 [K]
Description:	Sets the hysteresis for the warning threshold of the VSM2 to monitor a temperature.		
Dependency:	Refer to: p5467		

p5470[0...n] VSM2 10 V input CT gain / VSM2 CT_gain

A_INF (Line transf)	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min 0.000 [A]	Max 1000.000 [A]	Factory setting 1.000 [A]
Description:	Sets the CT gain of the CT connected at the 10 V input of Voltage Sensing Module 2 (VSM2). The parameter specifies the current magnitude in [A] referred to the input voltage at VSM2 in [V]. Example: CT with 1 V per 200 A. --> p5470 = 200		
Note:	The CT for phase 1 is connected at terminals X520.1 and X520.2 of VSM2. The CT for phase 2 is connected at terminals X520.3 and X520.4 of VSM2.		

r5471[0...n] CO: VSM2 10 V input CT 1 actual value / VSM2 CT1 I_act

A_INF (Line transf)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min - [A]	Max - [A]	Factory setting - [A]
Description:	Displays the current actual value from current transducer (CT) 1 at the 10 V input of Voltage Sensing Module 2 (VSM2).		
Dependency:	Refer to: p5470		
Note:	The CT for phase 1 is connected at terminals X520.1 and X520.2 of VSM2.		

r5472[0...n] CO: VSM2 10 V input CT 2 actual value / VSM2 CT2 I_act

A_INF (Line transf)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min - [A]	Max - [A]	Factory setting - [A]
Description:	Displays the current actual value from current transducer (CT) 2 at the 10 V input of Voltage Sensing Module 2 (VSM2).		
Dependency:	Refer to: p5470		
Note:	The CT for phase 2 is connected at terminals X520.3 and X520.4 of VSM2.		

r5473[0...n] CO: VSM2 10 V input 1 actual value / VSM2 inp 1 U_act

A_INF (Line transf)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the actual value of the voltage measured at the 10 V input 1 of Voltage Sensing Modules 2 (VSM2).		
Dependency:	Refer to: p5470		
Note:	10 V input 1: Terminals X520.1 and X520.2		

r5474[0...n] CO: VSM2 10 V input 2 actual value / VSM2 inp 2 U_act

A_INF (Line transf)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0150	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the actual value of the voltage measured at the 10 V input 2 of Voltage Sensing Modules 2 (VSM2).		
Dependency:	Refer to: p5470		
Note:	10 V input 2: Terminals X520.3 and X520.4		

p5480 Transformer magnetization mode / Transf mag mode

A_INF (Line transf)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	102	0
Description:	Sets the mode for transformer magnetization. 11: Identification 1 for transformer data Automatic determination of the magnetizing inductance. The magnetizing inductance determined in r5491 must be transferred to p5492 in order to take effect. Transformer identification 1 requires the "Line droop control" function module to be activated. 12: Identification 2 for transformer data Automatic determination of the transformer phase shift and the gain correction. The transformer phase shift determined in r6440 must be transferred to p6420 in order to take effect. The gain correction determined in r6441 must be transferred to p6421 in order to take effect. 13: Identification 3 for transformer data Determination of the total leakage inductance of the transformer during line data identification. p3410 = 1 is set automatically and the inductance is measured on the next power-up. Once the measurement has been taken the converter shuts down automatically and p3410 is reset (p3410 = 0). The total leakage inductance of the transformer determined in r5489 must be transferred to p5490 in order to take effect. 101: Test operation 1 The infeed switches to line droop control, but the main switch/circuit breaker does not close. Test operation requires the "Line droop control" function module to be activated. 102: Test operation 2 As test operation 1, but without synchronization with the line (VSM2 measured data r5460 and following is not used); instead, the output voltage is generated corresponding to the rated data p0210, p0211, p5486.		

Value:	0:	Deactivated
	1:	Normal operation 1
	11:	Identification of the transformer magnetizing inductance
	12:	Identification of transformer phase shift and gain correction
	13:	Identification of the total transformer leakage inductance
	101:	Test operation 1 (without activation of circuit breaker)
	102:	Test oper. 2 (w/o activation of circuit breaker, without VSM2)

Note: Re value = 13:
On low-power systems it is advisable to reduce the excitation current p3415 for inductance identification (e.g. p3415[0] = p3415[1] = 5%).
To repeat a measurement p5480 must first be set to 1 and then reset to a value of 13 once the measurement has been taken.

p5481[0...2] Transformer magnetization ramp-up time/bounce time/timeout / Transf mag t_r-up

A_INF (Line transf)	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.10 [s]	100.00 [s]	[0] 2.00 [s]
			[1] 1.00 [s]
			[2] 1.00 [s]

Description: Sets the ramp-up time for the voltage ramp, the bounce time for the circuit breaker, and a maximum time for line synchronization. If the maximum time elapses without the line being synchronized, alarm 6501 is triggered. The minimum duration of line synchronization is 25% of the maximum time p5481[2].

Index:
[0] = Voltage ramp ramp-up time
[1] = Circuit breaker bounce time
[2] = Timeout line synchronization

r5482 Transf magnetization state / Transf mag state

A_INF (Line transf)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	9	-

Description: Displays the state of sequence control for transformer magnetization.

Value:
0: Initialization
1: Wait for r3402 = 12
2: Magnetization with voltage ramp running
3: Line synchronization in progress
4: Wait for CB enable (p5483 = 1 signal)
5: Wait for bounce time for circuit breaker
6: Transition to operation running (r3402 = 9)
7: Procedure complete
8: Identification of magnetizing inductance
9: Identification of transformer phase shift and gain correction

p5483 BI: Transf. magnetiz. signal source for circuit breaker activation / Trans mag CB ON

A_INF (Line transf)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for activating the circuit breaker after following voltage ramp ramp-up.

p5484[0...2]				Transf magnetization integration times / Transf mag t_integ			
A_INF (Line transf)		Can be changed: T		Calculated: -		Access level: 3	
		Data type: FloatingPoint32		Dynamic index: -		Func. diagram: -	
		P-Group: Commands		Units group: -		Unit selection: -	
		Not for motor type: -		Scaling: -		Expert list: 1	
		Min		Max		Factory setting	
		0.00 [ms]		1000.00 [ms]		[0] 50.00 [ms]	
						[1] 50.00 [ms]	
						[2] 100.00 [ms]	
Description:		Sets the various integration times for transformer magnetization.					
Index:		[0] = Angle controller [1] = Voltage controller [2] = Voltage threshold					
p5485				Transf magnetization voltage threshold / Transf mag U_thr			
A_INF (Line transf)		Can be changed: T		Calculated: -		Access level: 3	
		Data type: FloatingPoint32		Dynamic index: -		Func. diagram: -	
		P-Group: Commands		Units group: -		Unit selection: -	
		Not for motor type: -		Scaling: -		Expert list: 1	
		Min		Max		Factory setting	
		0 [V]		300 [V]		35 [V]	
Description:		Sets the permissible voltage difference between the space vectors of the line voltage and the Active Line Module (ALM).					
Dependency:		Refer to: p5484					
p5486				Transf rated voltage primary / Transf U_rated pri			
A_INF (Line transf)		Can be changed: C2(1, 2)		Calculated: -		Access level: 3	
		Data type: FloatingPoint32		Dynamic index: -		Func. diagram: -	
		P-Group: Converter		Units group: -		Unit selection: -	
		Not for motor type: -		Scaling: -		Expert list: 1	
		Min		Max		Factory setting	
		0.00 [Vrms]		63000.00 [Vrms]		400.00 [Vrms]	
Description:		Sets the primary rated voltage of the transformer.					
p5487[0...1]				Cl: Transf primary voltage / Transf U_prim			
A_INF (Line transf)		Can be changed: T		Calculated: -		Access level: 3	
		Data type: Unsigned32 / FloatingPoint32		Dynamic index: -		Func. diagram: -	
		P-Group: Closed-loop control		Units group: -		Unit selection: -	
		Not for motor type: -		Scaling: p2001		Expert list: 1	
		Min		Max		Factory setting	
		-		-		[0] 5461[0]	
						[1] 5462[0]	
Description:		Sets the signal source for the measured phase voltage (u12, u23) on the primary side.					
Index:		[0] = Phase voltage u12 [1] = Phase voltage u23					

r5488[0...2]	CO: Transf secondary voltage transformed / Transf U_sec trans		
A_INF (Line transf)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [V]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Displays the components for the transformed voltage on the secondary side of the transformer.		
Index:	[0] = Alpha [1] = Beta [2] = Amplitude		
r5489	Transf leakage inductance identified / Transf L_l ident		
A_INF (Line transf)	Can be changed: - Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the total leakage inductance of the line transformer determined by means of automatic transformer identification (p5480 = 13). The result should be entered in parameter p5490. During identification, the value previously entered in p5490 is not effective.		
Dependency:	Refer to: p5480, p5490		
p5490	Transf leakage inductance / Transf L_leak		
A_INF (Line transf)	Can be changed: C2(1) Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min 0.001 [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.000 [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.100 [mH]
Description:	Sets the total leakage inductance of the line transformer.		
r5491	Transf magnetizing inductance identified / Transf L_m ident		
A_INF (Line transf)	Can be changed: - Data type: FloatingPoint32 P-Group: Converter Not for motor type: - Min - [mH]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mH]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mH]
Description:	Displays the magnetizing inductance of the line transformer determined by means of automatic transformer identification (p5480 = 11). The result should be entered in parameter p5492. During identification, the value previously entered in p5492 is not effective.		
Dependency:	Refer to: p5480, p5492		

p5492 Transf magnetizing inductance / Transf L_mag				
A_INF (Line transf)	Can be changed: C2(1)	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -	
	P-Group: Converter	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min 0.10 [mH]	Max 10000.00 [mH]	Factory setting 250.00 [mH]	
Description:	Sets the magnetizing inductance of the line transformer.			
Dependency:	Refer to: r5491			
r5493.0 CO/BO: Transf control signals / Transf ctrl sig				
A_INF (Line transf)	Can be changed: -	Calculated: -	Access level: 3	
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -	
	P-Group: Commands	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min -	Max -	Factory setting -	
Description:	Displays the control signals for transformer magnetization.			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Control bridging contactor	Yes	No
Dependency:	Refer to: r0863, r3402			
Note:	Re bit 00: The signal is used to control the external bridging contactor in the pre-charge branch. The external bridging contactor is closed if pre-charging is complete (r3402 > 5) and the circuit breaker has not been activated (r0863.1 = 0).			
r5497[0...1] CO: Transf secondary current / Transf I_second				
A_INF (Line transf)	Can be changed: -	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -	
	P-Group: Closed-loop control	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min - [A]	Max - [A]	Factory setting - [A]	
Description:	Displays the components for the transformer's calculated secondary current.			
Index:	[0] = Alpha [1] = Beta			
r5498[0...2] CO: Transf secondary voltage / Transf U_second				
A_INF (Line transf)	Can be changed: -	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -	
	P-Group: Closed-loop control	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min - [V]	Max - [V]	Factory setting - [V]	
Description:	Displays the components for the calculated secondary voltage.			
Index:	[0] = Alpha [1] = Beta [2] = Amplitude			

p5500**Dynamic grid control configuration / Dyn grid config**

A_INF (Dyn. line support)

Can be changed: T**Calculated:** -**Access level:** 4**Data type:** Unsigned16**Dynamic index:** -**Func. diagram:** -**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**
0000 bin**Description:**

Sets the configuration for dynamic grid control.

Bit field:

Bit	Signal name	1 signal	0 signal	FP
00	Deactivate characteristic	Yes	No	-
01	Line voltage alpha/beta amplitudes	No	Yes	-
02	Grid control mode in the event of asymmetry	No	Yes	-

Dependency:

Refer to: p5507, r5510

Note:

Re bit 00:

If p5500.0 = 0 the output value of the grid control characteristic is added to the control's reactive current setpoint.
 $r0075 = p3610 + p3611 + r3471 + r5510[0]$.

If p5500.0 = 1 the dynamic reactive current setpoint p3611 is subtracted if dynamic grid control is active (p5502.1 = 1). In this case, the following applies:

 $r0075 = p3610 + r3471 + r5510[0]$.

Re bit 01:

If p5500.1 = 0 the alpha and beta amplitudes of the line voltage, which were smoothed with p5507[2], are determined. An estimated value for the current line voltage absolute value is calculated from these amplitudes and serves as the input value for the grid control characteristic.

In the event of asymmetrical line disturbance, this means that the reactive current setpoint (r5510) can be kept constant.

If p5500.1 = 1, the smoothed absolute value of the measured line voltage is used as the input value for the grid control characteristic. Smoothing is set with p5507[3].

Re bit 02:

Only effective if p5500.1 = 0.

If p5500.2 = 0, the maximum value from the alpha and beta line voltage amplitudes is used as the input value for the grid control characteristic. The reactive current setpoint (r5510) will, therefore, remain virtually constant even in the event of asymmetrical line disturbance. Power fluctuations in the DC link are reduced.

If p5500.2 = 1, the mean value from the alpha and beta line voltage amplitudes is used as the input value for the grid control characteristic. However, the resulting higher reactive current setpoint (r5510) for grid control will cause significant power fluctuations in the DC link in the event of asymmetrical line disturbance.

p5501 BI: Dynamic grid control activation / Dyn grid act

A_INF (Dyn. line sup- port)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source to activate dynamic grid control.

BI: p5501 = 1 signal:

Activates dynamic grid control.

BI: p5501 = 0 signal:

Deactivates dynamic grid control.

Notice: Dynamic grid control can only be activated if the power units have a gating unit with current limitation control (r0192.19 = 1).

Note: If grid control is deactivated:
Line disturbance will generate the standard response to phase failures on the part of the infeed for drive applications (see A06205).

If grid control is activated:

Line disturbance will trigger grid control in accordance with the set characteristic (p5505, p5506).

r5502.0...1 CO/BO: Dynamic grid control status word / Dyn grid ZSW

A_INF (Dyn. line sup- port)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status word for dynamic grid control.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Dynamic grid control not active	Yes	No	-
	01	Dynamic grid control active	Yes	No	-

p5504[0...1] CI: Dynamic grid control line voltage input / Dyn grid U inp

A_INF (Dyn. line sup- port)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	-	-	[0] 3468[0] [1] 3468[1]

Description: Sets the signal source for the line voltage in alpha/beta coordinates.

The signals are used as input values for the characteristic for dynamic grid control (p5505, p5506).

Index: [0] = Alpha
[1] = Beta

Dependency: Refer to: r0072, r3468, r5488, r5498, p5505, p5506

Note: Possible signal sources include r3468, r5488, or r5498.

If p5504[0] = 0 or p5504[1] = 0:

The internal unsmoothed model value from current control is used for the line voltage.

p5505[0...3] Dynamic grid control characteristic voltage values / Dyn grid char U			
A_INF (Dyn. line support)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7997
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.000 [%]	100.000 [%]	[0] 10.000 [%]
			[1] 50.000 [%]
		[2] 10.000 [%]	
		[3] 50.000 [%]	
Description:	Sets the voltage values for the characteristic for dynamic grid control. The characteristic points for positive and negative voltage deviation are set separately. The positive and the negative characteristic are each defined based on their starting and finishing points. Positive voltage deviation: - Starting point: p5505[0], p5506[0] - Finishing point: p5505[1], p5506[1] Negative voltage deviation: - Starting point: p5505[2], p5506[2] - Finishing point: p5505[3], p5506[3]		
Index:	[0] = Characteristic positive starting point [1] = Characteristic positive finishing point [2] = Characteristic negative starting point [3] = Characteristic negative finishing point		
Dependency:	Refer to: p5506		
Note:	The voltage values refer to p0210. Dynamic grid control is not applied in the event of voltage deviations between the starting points of the positive and the negative characteristic (p5505[0], p5505[2]). In the event of voltage deviations above the finishing points of the positive or the negative characteristic (p5505[1], p5505[3]), grid control is limited to the reactive current setpoint of the corresponding finishing point (p5506[1], p5506[3]).		
p5506[0...3] Dynamic grid control characteristic reactive current setpoint / Dyn grid char I			
A_INF (Dyn. line support)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 7997
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.000 [%]	100.000 [%]	[0] 20.000 [%]
			[1] 100.000 [%]
		[2] 20.000 [%]	
		[3] 100.000 [%]	
Description:	Sets the reactive current setpoints for the characteristic for dynamic grid control. For more information, see p5505.		
Index:	[0] = Characteristic positive starting point [1] = Characteristic positive finishing point [2] = Characteristic negative starting point [3] = Characteristic negative finishing point		
Dependency:	Refer to: p5505		
Note:	The values refer to p0209.		

p5507[0...3] Dynamic grid control times / Dyn grid timesA_INF (Dyn. line sup-
port)**Can be changed:** U, T**Data type:** FloatingPoint32**P-Group:** Closed-loop control**Not for motor type:** -**Min**

0.00 [ms]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

10000.00 [ms]

Access level: 3**Func. diagram:** 7997**Unit selection:** -**Expert list:** 1**Factory setting**

[0] 500.00 [ms]

[1] 2.00 [ms]

[2] 20.00 [ms]

[3] 4.00 [ms]

Description: Sets the time values for dynamic grid control.**Index:**
[0] = Minimum time dynamic grid control
[1] = Minimum time line disturbance
[2] = Smoothing time alpha/beta
[3] = Smoothing time instantaneous value**Dependency:** Refer to: p5500**Note:** Re index 0:

Minimum time for continuing grid control in accordance with the characteristic once the line voltage has returned to the permissible tolerance range between the two starting points (p5505[0], p5505[2]).

Re index 1:

Minimum time for line disturbance for the start of grid control in accordance with the characteristic.

If the tolerance band between the two characteristic starting points is violated for at least this period of time, voltage control in accordance with characteristic will start up.

Re index 2:

Smoothing time for the calculation of the alpha amplitude and the beta amplitude of the line voltage if p5500.1 = 0.

An estimated value for the current line voltage absolute value is calculated from the alpha and beta amplitudes and serves as the input value for the grid control characteristic.

A smoothing time of less than a line period makes no sense.

If the estimated absolute value deviates from the smoothed measured voltage absolute value by more than 25%, the smoothed measured value is used. This corresponds to a temporary automatic switchover from p5500.1 = 0 to p5500.1 = 1.

Re index 3:

Smoothing time for the measured absolute value of the line voltage if p5500.1 = 1.

The smoothed absolute value of the line voltage is used as the input value for the grid control characteristic.

Setting p5507[3] = 0 deactivates smoothing.

p5508 Dynamic grid control Vdc threshold / Dyn grid Vdc thrA_INF (Dyn. line sup-
port)**Can be changed:** U, T**Data type:** FloatingPoint32**P-Group:** Converter**Not for motor type:** -**Min**

-200 [V]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

0 [V]

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-50 [V]

Description: Sets the threshold for the DC-link voltage (Vdc) to reduce the reactive current setpoint from dynamic grid control.
The value represents an offset in relation to the DC-link maximum voltage, with the result that the intervention threshold is obtained from the total of r0297 and p5508.**Dependency:** Refer to: r0297**Note:** To avoid imminent shutdown due to a DC-link overvoltage, the reactive current setpoint is reduced for dynamic grid control. Instead of this, the available converter current is used as the active current to reduce the DC-link voltage.

p5509[0...5] Dynamic grid control scaling values / Dyn grid scalA_INF (Dyn. line sup-
port)**Can be changed:** U, T**Data type:** FloatingPoint32**P-Group:** Closed-loop control**Not for motor type:** -**Min**

0.10 [%]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

200.00 [%]

Access level: 3**Func. diagram:** 7997**Unit selection:** -**Expert list:** 1**Factory setting**

[0] 40.00 [%]

[1] 40.00 [%]

[2] 4.00 [%]

[3] 1.00 [%]

[4] 100.00 [%]

[5] 100.00 [%]

Description:

Sets the scaling values for dynamic grid control.

Index:

[0] = Ramp reactive current at the beginning/end of grid control

[1] = Ramp reactive current when Vdc threshold is overshoot

[2] = Ramp reactive current when Vdc threshold is undershoot

[3] = Hysteresis line voltage to exit grid control

[4] = Reference voltage scaling

[5] = Current limitation scaling

Dependency:

Refer to: p5505, p5506, p5508

Note:

Re index 0:

Change in the reactive current setpoint (% per ms) at the beginning and end of dynamic grid control.

This avoids sudden changes in the reactive current if the starting points for the line voltage (p5505[0], p5505[2]) are overshoot.

Re index 1:

Change in the reactive current setpoint (% per ms) when the maximum Vdc threshold (p5508) is overshoot.

To avoid beat phenomena, the following must apply: p5509[1] > p5509[2].

Re index 2:

Change in the reactive current setpoint (% per ms) when the maximum Vdc threshold (p5508) is undershoot.

To avoid beat phenomena, the following must apply: p5509[1] > p5509[2].

Re index 3:

Sets the hysteresis for the line voltage to exit grid control (as a percentage of the supply voltage p0210).

To exit grid control, the line voltage must be in the interval reduced by the hysteresis width (the interval is defined with the starting points p5505[0] and p5505[2] and the hysteresis width p5509[3]).

Re index 4:

Sets the scaling factor for the reference voltage for dynamic grid control (as a percentage of the supply voltage p0210).

As a result, the product of p0210 * p5509[4] is applied as the voltage zero.

Re index 5:

Sets the scaling factor for the permissible maximum converter current absolute value for dynamic grid control (as a percentage of the converter maximum current r0209).

Values higher than 100% will not be applied.

r5510[0...3]	CO: Dynamic grid control output / Dyn grid output		
A_INF (Dyn. line sup-port)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [Arms]	Calculated: - Dynamic index: - Units group: 6_2 Scaling: p2002 Max - [Arms]	Access level: 3 Func. diagram: 7997 Unit selection: p0505 Expert list: 1 Factory setting - [Arms]
Description:	Display and connector output for the reactive current setpoint calculated according to the characteristic for dynamic grid control (p5505, p5506).		
Index:	[0] = Reactive current setpoint not limited [1] = Reactive current setpoint Vdc threshold [2] = Reactive current setpoint ramp [3] = Reactive current setpoint characteristic		
Dependency:	Refer to: p5505, p5506		
Note:	Re index 0: Output of characteristic following addition of reactive current setpoints prior to current limitation. The reactive current setpoint applied for current control including dynamic grid control is displayed in r0075. Re index 1: Output of characteristic following correction on the basis of the Vdc threshold (p5508). Re index 2: Output of characteristic after ramp function. Re index 3: Output of characteristic for dynamic grid control.		
r5511[0...1]	CO: Dynamic grid control line voltage amplitude / Dyn grid ampl		
A_INF (Dyn. line sup-port)	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 7997 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Display and connector output for the calculated alpha and beta amplitudes for the line voltage. The amplitudes are calculated from the alpha and beta coordinates of the current voltage measured values (p5504).		
Index:	[0] = Alpha [1] = Beta		
Dependency:	Refer to: p5500, p5504		
Note:	The alpha and beta amplitudes calculated are used to calculate grid control dependent upon the selected configuration (p5500.1).		
r5512[0...1]	CO: Dynamic grid control line voltage absolute value / Dyn grid abs val		
A_INF (Dyn. line sup-port)	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [V]	Calculated: - Dynamic index: - Units group: - Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: 7997 Unit selection: - Expert list: 1 Factory setting - [V]
Description:	Display and connector output for the voltage absolute value calculated from the alpha/beta input voltages (p5504).		
Index:	[0] = Line voltage absolute value input characteristic [1] = Line voltage absolute value smoothed		
Dependency:	Refer to: p5505, p5506		

Note: Re index 0:
Displays the effective voltage absolute value for the input of the characteristic for dynamic grid control.
Re index 1:
Displays the voltage absolute value following smoothing of the actual value according to p5507[3].
If p5500.1 = 1, this value is used as the input value for the characteristic.
If p5500.1 = 0, the alpha/beta amplitudes are used to calculate the voltage absolute value (r5511).

p6420 Line transformer phase shift / Transf ph shift

A_INF (Line transf)	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-180.00 [°]	179.90 [°]	0.00 [°]

Description: Sets the phase shift between the line transformer's primary and secondary voltages.

Note: The phase shift relates to the primary side of the transformer, which is connected to the line. The secondary side is connected to the infeed. Example: A Dy5n transformer has a phase shift of $-5 \times 30^\circ = -150^\circ$. This means that the secondary voltage is shifted from the primary voltage by -150° (the primary voltage is 150° ahead).
This phase shift must be determined when commissioning the system.

p6421 Line transformer gain adaptation / Transf gain

A_INF (Line transf)	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Converter	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	50.000 [%]	200.000 [%]	100.000 [%]

Description: Sets the gain factor correction for fine calibration of the line transformer transformation ratio.

Dependency: Refer to: r6441

r6440 Line transformer phase shift identified / Trans phase ident

A_INF (Line transf)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [°]	- [°]	- [°]

Description: Displays the phase shift between the primary and secondary voltages of the line transformer identified by automatic transformer identification (p5480 = 12).

Dependency: Refer to: p5480, p6420

Note: The phase shift relates to the primary side of the transformer, which is connected to the line. The secondary side is connected to the infeed.

Example:

A Dy5n transformer has a phase shift of $-5 \times 30^\circ = -150^\circ$. This means that the secondary voltage is shifted from the primary voltage by -150° (the primary voltage is 150° ahead).

The result should be entered in parameter p6420. During identification, the value previously entered in p6420 is not effective.

r6441	Line transformer gain adaptation identified / Transf gain ident		
A_INF (Line transf)	Can be changed: - Data type: FloatingPoint32 P-Group: Closed-loop control Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the gain factor correction identified (p5480 = 12) for fine calibration of the line transformer transformation ratio.		
Dependency:	Refer to: p6421		
Note:	The result should be entered in parameter p6421. During identification, the value previously entered in p6421 is not effective.		

r7000	Par_circuit No. of active power units / Qty active PU		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: Unsigned16 P-Group: Modulation Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the active power units for a parallel circuit configuration.		
Dependency:	Refer to: p7001		

p7001[0...n]	Par_circuit power units enable / PU enable		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: T Data type: Integer16 P-Group: Modulation Not for motor type: - Min 0	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Enables the power units in the parallel circuit configuration.		
Value:	0: Deactivated 1: Activated		
Dependency:	Refer to: r7000		
Note:	For motors with separate winding systems (p7003 = 1) it is not possible to inhibit an individual power unit. p7001 is automatically reset if a power unit is deactivated via p0125 or p0895.		

r7002[0...n]	Par_circuit status power units / Status PU		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: Integer16 P-Group: Modulation Not for motor type: - Min 0	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the status of the power units in the parallel circuit configuration.		
Value:	0: Pulses inhibited 1: Pulses enabled		
Dependency:	Refer to: r7000, p7001		

p7003	Par_circuit winding system / Winding system		
VECTOR (Parallel)	Can be changed: C2(2) Data type: Integer16 P-Group: Converter Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Specifies the winding system for a parallel circuit configuration. The following limitations/restrictions are obtained depending on the setting: One winding system (p7003 = 0): - the motor data identification routine (p1910) determines the stator resistance and the cable resistance. The cable resistance of an individual Motor Module is entered into p0352. - the current symmetrizing is activated as standard after the motor data identification routine (p7035 = 1). - individual Motor Modules can be activated and deactivated (p7001). - the edge modulation is not possible (p1802). Several separate winding systems or motors (p7003 = 1): - the motor data identification routine (p1910) determines the total (overall) resistance. The cable resistance is not measured, but instead, entered as a component of the total resistance (refer to p0352). - all Motor Modules are activated. It is not possible to deactivate a motor model. - the edge modulation can be activated (p1802).		
Value:	0: One winding system 1: Several separate winding systems or motors		
Dependency:	Refer to: p1802		
p7010	Par_circuit current dissymmetry alarm threshold / i_dissym A thresh		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Modulation Not for motor type: - Min 2 [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max 100 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 20 [%]
Description:	Sets the alarm threshold to detect current dissymmetry in the parallel circuit configuration. The deviation between the measured values and average value is evaluated. The specified value is referred to the rated power unit current (p7251[0]).		
Dependency:	Refer to: r7251 Refer to: A05052		
p7011	Par_circuit DC link voltage dissymmetry alarm threshold / Vdc_dissym A thrsh		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: U, T Data type: FloatingPoint32 P-Group: Modulation Not for motor type: - Min 2 [%]	Calculated: - Dynamic index: - Units group: - Scaling: PERCENT Max 100 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10 [%]
Description:	Sets the alarm threshold to detect dissymmetry of the DC link voltages in the parallel circuit configuration. The deviation between the measured values and average value is evaluated. The specified value is referred to the rated link voltage.		
Dependency:	Refer to: A05053		

r7020[0...n]	CO: Par_circuit deviation current in phase U / Phase U curr dev		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: PDS, p0120 Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the deviation between the measured current actual value of phase U and the average value as peak value. The maximum deviation from the average value is displayed in r7025.		
Dependency:	Refer to: r7021, r7022, r7025		
r7021[0...n]	CO: Par_circuit deviation current in phase V / Phase V curr dev		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: PDS, p0120 Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the deviation between the measured current actual value of phase V and the average value as peak value. The maximum deviation from the average value is displayed in r7026.		
Dependency:	Refer to: r7020, r7022, r7026		
r7022[0...n]	CO: Par_circuit deviation current in phase W / Phase W curr dev		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: PDS, p0120 Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the deviation between the measured current actual value of phase W and the average value as peak value. The maximum deviation from the average value is displayed in r7027.		
Dependency:	Refer to: r7020, r7021, r7027		
r7025	CO: Par_circuit max. deviation currents phase U / Phase U Max i_dev		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: - Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the maximum absolute deviation of the measured current actual values of phase U from the average value as peak value. The deviation of the individual currents from the average value is displayed in r7020.		
Dependency:	Refer to: r7020, r7026, r7027 Refer to: A05052		

r7026	CO: Par_circuit max. deviation currents phase V / Phase V Max i_dev		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: 6_5	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [A]	- [A]	- [A]
Description:	Displays the maximum absolute deviation of the measured current actual values of phase V from the average value as peak value. The deviation of the individual currents from the average value is displayed in r7021.		
Dependency:	Refer to: r7021, r7025, r7027 Refer to: A05052		
r7027	CO: Par_circuit max. deviation currents phase W / Phase W Max i_dev		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: 6_5	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [A]	- [A]	- [A]
Description:	Displays the maximum absolute deviation of the measured current actual values of phase W from the average value as peak value. The deviation of the individual currents from the average value is displayed in r7022.		
Dependency:	Refer to: r7022, r7025, r7026 Refer to: A05052		
r7030[0...n]	CO: Par_circuit DC link voltage deviation / Vdc deviation		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the deviation of the measured DC link voltage from the average value. The maximum deviation from the average value is displayed in r7031.		
Dependency:	Refer to: r7031		
r7031	CO: Par_circuit DC link voltage maximum deviation / Vdc deviation max.		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the maximum absolute deviation of the measured DC link voltage from the average value. The deviation of the individual voltages from the average value is displayed in r7030.		
Dependency:	Refer to: r7030 Refer to: A05053		

p7035 Infeed par_circuit circulating current control operating mode / Circ_I mode

A_INF (Parallel), S_INF (Parallel)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	1

Description: Sets the operating mode of the circulating current control.
The circulating current control ensures symmetrical distribution of the total currents to the individual converters.

Value: 0: Circulating current control deactivated
1: Circulating control control activated

p7035[0...n] Par_circuit circulating current control operating mode / Circ_I mode

VECTOR (Parallel)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	1

Description: Sets the operating mode of the circulating current control.
The circulating current control ensures symmetrical distribution of the total currents to the individual converters.

Value: 0: Circulating current control deactivated
1: Circulating control control activated

p7036 Infeed par_cct circulating current controller proportional gain / Circ_I Kp

A_INF (Parallel), S_INF (Parallel)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00000 [%]	1000.00000 [%]	100.00000 [%]

Description: Sets the normalized proportional gain for the circulating current controller.

Note: A value of 100 % corresponds to the basic setting derived from loop control parameters (p3421, p3622).

p7036[0...n] Par_circuit circulating current control proportional gain / Circ_I Kp

VECTOR (Parallel)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00000 [Ohm]	20.00000 [Ohm]	0.00000 [Ohm]

Description: Sets the proportional gain for the circulating current controller.
The parameter is preset to the cable resistance.

p7037	Infeed par_cct circulating current control integral time / I_circ Tn		
A_INF (Parallel), S_INF (Parallel)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0 [%]	Max 100000.0 [%]	Factory setting 100.0 [%]
Description:	Sets the normalized integral time of the circulating current controller.		
Note:	A value of 100 % corresponds to the basic setting derived from the controller sampling time p0115[0]. The integral component of the controller is deactivated with p7037 = 0.		

p7037[0...n]	Par_circuit circulating current control integral time / I_circ Tn		
VECTOR (Parallel)	Can be changed: U, T	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 2.0	Max 1000.0	Factory setting 4.0
Description:	Sets the integral time of the circulating current controller. The parameter is referred to the current controller sampling time (p0115[0]).		
Dependency:	Refer to: p0115		

p7038	Infeed par_circuit circulating current control limit / I_circ limit		
A_INF (Parallel), S_INF (Parallel)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1 [%]	Max 100 [%]	Factory setting 100 [%]
Description:	Sets the limit of the circulating current controller output values. The parameter is, depending on the phase, referred to the valve lockout times (p1828, p1829, p1830).		

p7038[0...n]	Par_circuit circulating current control limit / I_circ limit		
VECTOR (Parallel)	Can be changed: U, T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dynamic index: DDS, p0180	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1 [%]	Max 100 [%]	Factory setting 50 [%]
Description:	Sets the limit of the circulating current controller output values. The parameter is, depending on the phase, referred to the valve lockout times (p1828, p1829, p1830).		

p7040[0...n]	Par_circuit correction valve lockout time phase U / Comp t_lockout U		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -1000000.00 [μs]	Max 1000000.00 [μs]	Factory setting 0.00 [μs]
Description:	For the particular Motor Module, the correction time must be added to the valve lockout time to be compensated for phase U (p1828). The corrective value is used to compensate variations/spread in the valve lockout times of Motor Modules for a parallel circuit configuration.		
Dependency:	Refer to: p1828		
p7042[0...n]	Par_circuit correction valve lockout time phase V / Comp t_lockout V		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -1000000.00 [μs]	Max 1000000.00 [μs]	Factory setting 0.00 [μs]
Description:	For the particular Motor Module, the correction time must be added to the valve lockout time to be compensated for phase V (p1829). The corrective value is used to compensate variations/spread in the valve lockout times of Motor Modules for a parallel circuit configuration.		
Dependency:	Refer to: p1829		
p7044[0...n]	Par_circuit correction valve lockout time phase W / Comp t_lockout W		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Modulation	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -1000000.00 [μs]	Max 1000000.00 [μs]	Factory setting 0.00 [μs]
Description:	For the particular Motor Module, the correction time must be added to the valve lockout time to be compensated for phase W (p1830). The corrective value is used to compensate variations/spread in the valve lockout times of Motor Modules for a parallel circuit configuration.		
Dependency:	Refer to: p1830		
r7050[0...n]	Par_circuit circulating current phase U / Circ_I_phase U		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: 6_5	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min - [A]	Max - [A]	Factory setting - [A]
Description:	Displays the circulating current of phase U as peak value.		

r7051[0...n]	Par_circuit circulating current phase V / Circ_I_phase V		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: PDS, p0120 Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the circulating current of phase V as peak value.		
r7052[0...n]	Par_circuit circulating current phase W / Circ_I_phase W		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: PDS, p0120 Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the circulating current of phase W as peak value.		
r7100[0...99]	Par_circuit ring buffer fault/alarm code / Fault/alarm code		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: Unsigned32 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Ring buffer for faults and alarms that have occurred from power units connected in parallel (Motor Module, Active Line Infeed, Voltage Sensing Module). Displays the fault/alarm code.		
Dependency:	Refer to: r7101, r7102, r7103		
Note:	The last fault case that occurred is documented in index 0. The parameter is reset to 0 at POWER ON.		
r7101[0...99]	Par_circuit ring buffer data set number / Ring buffer Ds_No		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: Unsigned32 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Ring buffer for faults and alarms that have occurred from power units connected in parallel (Motor Module, Active Line Infeed, Voltage Sensing Module). p7101 < 100: Displays the Power unit Data Set number (PDS). p7101 >= 100: Displays the Voltage Sensing Module Data Set number (VSMDS)		
Dependency:	Refer to: r7100, r7102, r7103		
Note:	The last fault case that occurred is documented in index 0. The parameter is reset to 0 at POWER ON.		

r7102[0...99]	Par_circuit ring buffer fault/alarm received / F/A received		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: Unsigned32 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Ring buffer for faults and alarms that have occurred from power units connected in parallel (Motor Module, Active Line Infeed, Voltage Sensing Module). Displays the relative system runtime when the fault or alarm occurred.		
Dependency:	Refer to: r7100, r7101, r7103		
Note:	The last fault case that occurred is documented in index 0. The parameter is reset to 0 at POWER ON.		
r7103[0...99]	Par_circuit ring buffer fault/alarm gone / F/A gone		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: Unsigned32 P-Group: Displays, signals Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Ring buffer for faults and alarms that have occurred from power units connected in parallel (Motor Module, Active Line Infeed, Voltage Sensing Module). Displays the relative system runtime when the fault or alarm was withdrawn.		
Dependency:	Refer to: r7100, r7101, r7102		
Note:	The last fault case that occurred is documented in index 0. The parameter is reset to 0 at POWER ON.		
r7200[0...n]	Par_circuit power unit overload I2T / PU overload I2T		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [%]	Calculated: - Dynamic index: PDS, p0120 Units group: - Scaling: PERCENT Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the overload of the particular power unit in a parallel circuit configuration calculated using the I2t function. The maximum value of all power units is displayed in r0036.		
r7201[0...n]	CO: Par_circuit power unit temperatures max. inverter / PU temp max inv		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the maximum inverter temperature in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[0].		

r7202[0...n] Par_circuit power unit temperatures max. depletion layer / PU TempMaxDepLayerA_INF (Parallel),
B_INF (Parallel),
S_INF (Parallel),
VECTOR (Parallel)**Can be changed:** -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [°C]

Calculated: -**Dynamic index:** PDS, p0120**Units group:** 21_1**Scaling:** TEMP**Max**

- [°C]

Access level: 3**Func. diagram:** -**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [°C]

Description:Displays the maximum depletion layer temperature in the power unit for a parallel circuit configuration.
The maximum value of all power units is displayed in r0037[1].**r7203[0...n] Par_circuit power unit temperatures max. rectifier / PU temp max rect**A_INF (Parallel),
B_INF (Parallel),
S_INF (Parallel),
VECTOR (Parallel)**Can be changed:** -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [°C]

Calculated: -**Dynamic index:** PDS, p0120**Units group:** 21_1**Scaling:** TEMP**Max**

- [°C]

Access level: 3**Func. diagram:** -**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [°C]

Description:Displays the maximum rectifier temperature in the power unit for a parallel circuit configuration.
The maximum value of all power units is displayed in r0037[2].**r7204[0...n] Par_circuit power unit temperatures air intake / PU temp air intake**A_INF (Parallel),
B_INF (Parallel),
S_INF (Parallel),
VECTOR (Parallel)**Can be changed:** -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [°C]

Calculated: -**Dynamic index:** PDS, p0120**Units group:** 21_1**Scaling:** TEMP**Max**

- [°C]

Access level: 3**Func. diagram:** -**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [°C]

Description:Displays the air intake temperature in the power unit for a parallel circuit configuration.
The maximum value of all power units is displayed in r0037[3].**r7205[0...n] Par_circuit power unit temperatures electronics / PU temp electr**A_INF (Parallel),
B_INF (Parallel),
S_INF (Parallel),
VECTOR (Parallel)**Can be changed:** -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [°C]

Calculated: -**Dynamic index:** PDS, p0120**Units group:** 21_1**Scaling:** TEMP**Max**

- [°C]

Access level: 3**Func. diagram:** -**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [°C]

Description:Displays the temperature of the electronics module in the power unit for a parallel circuit configuration.
The maximum value of all power units is displayed in r0037[4].**r7206[0...n] Par_circuit power unit temperatures inverter 1 / PU temp inv 1**A_INF (Parallel),
B_INF (Parallel),
S_INF (Parallel),
VECTOR (Parallel)**Can be changed:** -**Data type:** FloatingPoint32**P-Group:** Displays, signals**Not for motor type:** -**Min**

- [°C]

Calculated: -**Dynamic index:** PDS, p0120**Units group:** 21_1**Scaling:** TEMP**Max**

- [°C]

Access level: 3**Func. diagram:** -**Unit selection:** p0505**Expert list:** 1**Factory setting**

- [°C]

Description:Displays the inverter temperature 1 in the power unit for a parallel circuit configuration.
The maximum value of all power units is displayed in r0037[5].

r7207[0...n]	Par_circuit power unit temperatures inverter 2 / PU temp inv 2		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the inverter temperature 2 in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[6].		
r7208[0...n]	Par_circuit power unit temperatures inverter 3 / PU temp inv 3		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the inverter temperature 3 in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[7].		
r7209[0...n]	Par_circuit power unit temperatures inverter 4 / PU temp inv 4		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the inverter temperature 4 in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[8].		
r7210[0...n]	Par_circuit power unit temperatures inverter 5 / PU temp inv 5		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the inverter temperature 5 in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[9].		
r7211[0...n]	Par_circuit power unit temperatures inverter 6 / PU temp inv 6		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays the inverter temperature 6 in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[10].		

r7212[0...n]	Par_circuit power unit temperatures inverter 1 / PU temp rect 1		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays rectifier temperature 1 in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[11].		
r7213[0...n]	Par_circuit power unit temperatures inverter 2 / PU temp rect 2		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays rectifier temperature 2 in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[12].		
r7214[0...n]	Par_circuit power unit temperatures depletion layer 1 / PU temp DepLayer 1		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays depletion layer temperature 1 in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[13].		
r7215[0...n]	Par_circuit power unit temperatures depletion layer 2 / PU temp DepLayer 2		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays depletion layer temperature 2 in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[14].		
r7216[0...n]	Par_circuit power unit temperatures depletion layer 3 / PU temp DepLayer 3		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [°C]	Calculated: - Dynamic index: PDS, p0120 Units group: 21_1 Scaling: TEMP Max - [°C]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [°C]
Description:	Displays depletion layer temperature 3 in the power unit for a parallel circuit configuration. The maximum value of all power units is displayed in r0037[15].		

r7217[0...n] Par_circuit power unit temperatures depletion layer 4 / PU temp DepLayer 4

A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min - [°C]	Max - [°C]	Factory setting - [°C]

Description: Displays depletion layer temperature 4 in the power unit for a parallel circuit configuration.
The maximum value of all power units is displayed in r0037[16].

r7218[0...n] Par_circuit power unit temperatures depletion layer 5 / PU temp DepLayer 5

A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min - [°C]	Max - [°C]	Factory setting - [°C]

Description: Displays depletion layer temperature 5 in the power unit for a parallel circuit configuration.
The maximum value of all power units is displayed in r0037[17].

r7219[0...n] Par_circuit power unit temperatures depletion layer 6 / PU temp DepLayer 6

A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: TEMP	Expert list: 1
	Min - [°C]	Max - [°C]	Factory setting - [°C]

Description: Displays depletion layer temperature 6 in the power unit for a parallel circuit configuration.
The maximum value of all power units is displayed in r0037[18].

r7220[0...n] Infeed par_circuit absolute current value motoring permissible / INF I_abs mot perm

A_INF (Parallel), S_INF (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min - [Arms]	Max - [Arms]	Factory setting - [Arms]

Description: Displays the currently permissible line-side absolute current when motoring.
The minimum value of all power units multiplied by the number of Motor Modules is displayed in r0067[0].

r7220[0...n] CO: Par_circuit drive output current maximum / Drv I_outp max

VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min - [Arms]	Max - [Arms]	Factory setting - [Arms]

Description: Displays the maximum output current of the power unit.
The minimum value of all power units multiplied by the number of Motor Modules is displayed in r0067.

r7221[0...n]	Infeed par_circuit absolute current regenerating permissible / INF I_absRegenPerm		
A_INF (Parallel), S_INF (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [Arms]	- [Arms]	- [Arms]
Description:	Displays the currently permissible line-side absolute regenerative current. The minimum value of all power units multiplied by the number of Motor Modules is displayed in r0067[1].		
r7222[0...n]	CO: Par_circuit absolute current actual value / I_act abs val		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: 6_2	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [Arms]	- [Arms]	- [Arms]
Description:	Displays actual absolute current. The summed value of all power units is displayed in r0068.		
r7223[0...n]	CO: Par_circuit phase current actual value phase U / I_phase U act val		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: 6_5	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [A]	- [A]	- [A]
Description:	Displays the measured actual value of phase U as peak value. The summed value of all power units is displayed in r0069[0].		
r7224[0...n]	CO: Par_circuit phase current actual value phase V / I_phase V act val		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: 6_5	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [A]	- [A]	- [A]
Description:	Displays the measured actual value of phase V as peak value. The summed value of all power units is displayed in r0069[1].		
r7225[0...n]	CO: Par_circuit phase current actual value phase W / I_phase W act val		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: PDS, p0120	Func. diagram: -
	P-Group: Displays, signals	Units group: 6_5	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [A]	- [A]	- [A]
Description:	Displays the measured actual value of phase W as peak value. The summed value of all power units is displayed in r0069[2].		

r7226[0...n]	CO: Par_circuit phase current actual value phase U offset / I_phase U offset		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: PDS, p0120 Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the measured offset of phase U as peak value. The summed value of all power units is displayed in r0069[3].		
r7227[0...n]	CO: Par_circuit phase current actual value phase V offset / I_phase V offset		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: PDS, p0120 Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the measured offset of phase V as peak value. The summed value of all power units is displayed in r0069[4].		
r7228[0...n]	CO: Par_circuit phase current actual value phase W offset / I_phase W offset		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: PDS, p0120 Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the measured offset of phase W as peak value. The summed value of all power units is displayed in r0069[5].		
r7229[0...n]	CO: Par_circuit phase current actual value sum U, V, W / I_phase sum UVW		
A_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [A]	Calculated: - Dynamic index: PDS, p0120 Units group: 6_5 Scaling: p2002 Max - [A]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [A]
Description:	Displays the measured sum of the currents in phases U, V and W as peak value. The summed value of all power units is displayed in r0069[6].		
r7230[0...n]	CO: Par_circuit DC link voltage actual value / Vdc_act		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: PDS, p0120 Units group: 5_2 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the measured actual value of the DC link voltage. The average value of all power units is displayed in r0070.		

r7231[0...n]	CO: Par_circuit phase voltage actual value phase U / V_phase U act val		
A_INF, S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: PDS, p0120 Units group: 5_3 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the actual voltage, phase U. The average value of all power units is displayed in r0089[0].		

r7231[0...n]	CO: Par_circuit phase voltage actual value phase U / V_phase U act val		
A_INF (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: PDS, p0120 Units group: 5_3 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the actual voltage, phase U.		

r7232[0...n]	CO: Par_circuit phase voltage actual value phase V / V_phase V act val		
A_INF, S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: PDS, p0120 Units group: 5_3 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the actual voltage, phase V. The average value of all power units is displayed in r0089[1].		

r7232[0...n]	CO: Par_circuit phase voltage actual value phase V / V_phase V act val		
A_INF (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: PDS, p0120 Units group: 5_3 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the actual voltage, phase V.		

r7233[0...n]	CO: Par_circuit phase voltage actual value phase W / V_phase W act val		
A_INF, S_INF (Parallel), VECTOR (Parallel)	Can be changed: - Data type: FloatingPoint32 P-Group: Displays, signals Not for motor type: - Min - [V]	Calculated: - Dynamic index: PDS, p0120 Units group: 5_3 Scaling: p2001 Max - [V]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [V]
Description:	Displays the actual voltage, phase W. The average value of all power units is displayed in r0089[2].		

r7250[0...4]	Par_circuit power unit rated power / PU P_rated		
A_INF (Parallel), B_INF (Parallel), S_INF (Parallel), VECTOR (Parallel)	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Converter	Units group: 14_6	Unit selection: p0100
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [kW]	- [kW]	- [kW]
Description:	Displays the rated power of the individual power units connected in parallel for various load duty cycles. The sum of the rated powers of all power units connected in parallel is displayed in r0206.		
Index:	[0] = Rating plate [1] = Load duty cycle with low overload [2] = Load duty cycle with high overload [3] = S1 cont duty cyc [4] = S6 load duty cycle		
Dependency:	The value is displayed in [kW] or [hp]. Refer to: p0100, p0205		

r7251[0...4] Par_circuit power unit rated current / PU PI_ratedA_INF (Parallel),
B_INF (Parallel),
S_INF (Parallel),
VECTOR (Parallel)**Can be changed:** -**Data type:** FloatingPoint32**P-Group:** Converter**Not for motor type:** -**Min**

- [Arms]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

- [Arms]

Access level: 2**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

- [Arms]

Description: Displays the rated current of the individual power units connected in parallel for various load duty cycles.
The sum of the rated currents of all power units connected in parallel is displayed in r0207.**Index:** [0] = Rating plate
[1] = Load duty cycle with low overload
[2] = Load duty cycle with high overload
[3] = S1 cont duty cyc
[4] = S6 load duty cycle**Dependency:** Refer to: p0205**r7252[0...4] Par_circuit maximum power unit current / PU I_max**A_INF (Parallel),
B_INF (Parallel),
S_INF (Parallel),
VECTOR (Parallel)**Can be changed:** -**Data type:** FloatingPoint32**P-Group:** Converter**Not for motor type:** -**Min**

- [Arms]

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

- [Arms]

Access level: 2**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

- [Arms]

Description: Displays the maximum output current of the individual power units connected in parallel.
The sum of the maximum currents of all power units connected in parallel is displayed in r0209.**Index:** [0] = Rating plate
[1] = Load duty cycle with low overload
[2] = Load duty cycle with high overload
[3] = S1 cont duty cyc
[4] = S6 load duty cycle**Dependency:** Refer to: p0205**r7300[0...n] CO: Par_circuit VSM input line voltage u1 - u2 / VSM inp u1-u2**A_INF (Parallel),
S_INF (Parallel)**Can be changed:** -**Data type:** FloatingPoint32**P-Group:** Closed-loop control**Not for motor type:** -**Min**

- [V]

Calculated: -**Dynamic index:** p0140**Units group:** -**Scaling:** p2001**Max**

- [V]

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

- [V]

Description: Displays the voltage between phases L1 and L2 of the particular Voltage Sensing Module (VSM) for a parallel circuit configuration.

The average value of all VSMs is displayed in r3661.

Dependency: Refer to: p3660**Note:** X521.1 or X522.1: Connection of L1
X521.2 or X522.2: Connection of L2

r7301[0...n] CO: Par_circuit VSM input line voltage u2 - u3 / VSM inp u2-u3A_INF (Parallel),
S_INF (Parallel)**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** p0140**Func. diagram:** -**P-Group:** Closed-loop control**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** p2001**Expert list:** 1**Min****Max****Factory setting**

- [V]

- [V]

- [V]

Description: Displays the voltage between phases L2 and L3 of the particular Voltage Sensing Module (VSM) for a parallel circuit configuration.

The average value of all VSMS is displayed in r3662.

Dependency: Refer to: p3660**Note:** X521.2 or X522.2: Connection of L2

X521.3 or X522.3: Connection of L3

r7305[0...n] Par_circuit VSM temperature evaluation status / VSM temp statusA_INF (Parallel),
S_INF (Parallel)**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dynamic index:** p0140**Func. diagram:** -**P-Group:** Terminals**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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Description: Displays the status of the temperature evaluation of the particular Voltage Sensing Module (VSM) for a parallel circuit configuration.

This displays whether the temperature actual value has exceeded the fault/alarm threshold.

The overall status of the temperature evaluation of all VSMS is displayed in r3664.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Temperature alarm threshold exceeded	Yes	No	-
	01	Temperature fault threshold exceeded	Yes	No	-

Dependency: Refer to: p3665, r3666, p3667, p3668**r7306[0...n] CO: Par_circuit VSM temperature KTY / VSM temp KTY**A_INF (Parallel),
S_INF (Parallel)**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dynamic index:** p0140**Func. diagram:** -**P-Group:** Closed-loop control**Units group:** 21_1**Unit selection:** p0505**Not for motor type:** -**Scaling:** TEMP**Expert list:** 1**Min****Max****Factory setting**

- [°C]

- [°C]

- [°C]

Description: Displays the temperature actual value of the KTY84 temperature sensor connected to the Voltage Sensing Module (VSM) for a parallel circuit configuration.

The maximum value of all VSMS is displayed in r3666.

Prerequisite:

A KTY84 sensor is connected and p3665 is set to 2.

Dependency: Refer to: p3665

r7310[0...n]	CO: Par_circuit VSM 10 V input CT1 actual value / VSM CT 1 I_act		
A_INF (Parallel), S_INF (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0140	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [A]	- [A]	- [A]
Description:	Displays the current actual value from current transducer (CT) 1 at the 10 V input of the particular Voltage Sensing Module (VSM) for a parallel circuit configuration. The average value of all VSMs is displayed in r3671.		
Dependency:	Refer to: p3670		
Note:	The CT for phase 1 is connected at terminals X520.1 and X520.2 of the VSM.		
r7311[0...n]	CO: Par_circuit VSM 10 V input CT2 actual value / VSM CT 2 I_act		
A_INF (Parallel), S_INF (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0140	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min	Max	Factory setting
	- [A]	- [A]	- [A]
Description:	Displays the current actual value from current transducer (CT) 2 at the 10 V input of the particular Voltage Sensing Module (VSM) for a parallel circuit configuration. The average value of all VSMs is displayed in r3672.		
Dependency:	Refer to: p3670		
Note:	The CT for phase 2 is connected at terminals X520.3 and X520.4 of the VSM.		
r7315[0...n]	CO: Par_circuit VSM 10 V input 1 actual value / VSM inp 1 V_act		
A_INF (Parallel), S_INF (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0140	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the actual value of the voltage measured at the 10 V input 1 of the Voltage Sensing Modules (VSM). The average value of all VSM is displayed in r3673.		
Dependency:	Refer to: p3670		
Note:	10 V input 1: Terminals X520.1 and X520.2		
r7316[0...n]	CO: Par_circuit VSM 10 V input 2 actual value / VSM inp 2 V_act		
A_INF (Parallel), S_INF (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: p0140	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min	Max	Factory setting
	- [V]	- [V]	- [V]
Description:	Displays the actual value of the voltage measured at the 10 V input 2 of the Voltage Sensing Modules (VSM). The average value of all VSMs is displayed in r3674.		
Dependency:	Refer to: p3670		
Note:	10 V input 2: Terminals X520.3 and X520.4		

r7320[0...n]	Par_circuit VSM line filter capacitance phase U / VSM filt C phase U		
A_INF (Parallel), S_INF (Parallel)	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: p0140	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [μF]	- [μF]	- [μF]
Description:	Displays the capacitance of the line filter, phase U of the particular Voltage Sensing Module (VSM). The average value of all VSMs is displayed in r3677[0].		
Dependency:	Refer to: p3676		
Note:	Prerequisites: The monitoring of the filter capacitance is activated.		

r7321[0...n]	Par_circuit VSM line filter capacitance phase V / VSM filt C phase V		
A_INF (Parallel), S_INF (Parallel)	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: p0140	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [μF]	- [μF]	- [μF]
Description:	Displays the capacitance of the line filter, phase V of the particular Voltage Sensing Module (VSM). The average value of all VSMs is displayed in r3677[1].		
Dependency:	Refer to: p3676		
Note:	Prerequisites: The monitoring of the filter capacitance is activated.		

r7322[0...n]	Par_circuit VSM line filter capacitance phase W / VSM filt C phase W		
A_INF (Parallel), S_INF (Parallel)	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: p0140	Func. diagram: -
	P-Group: Closed-loop control	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [μF]	- [μF]	- [μF]
Description:	Displays the capacitance of the line filter, phase W of the particular Voltage Sensing Module (VSM). The average value of all VSMs is displayed in r3677[2].		
Dependency:	Refer to: p3676		
Note:	Prerequisites: The monitoring of the filter capacitance is activated.		

p7770	NVRAM action / NVRAM action		
A_INF, B_INF, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VEC- TOR	Can be changed: T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	3	0
Description:	Sets the action to be executed for NVRAM data. At the end of the action the value is automatically set to 0.		

Value:

- 0: Inactive
- 1: Load NVRAM data to parameters
- 2: Load parameters to NVRAM
- 3: Reset

Notice: After action p7770 = 1 no more pulses may be enabled.
After action p7770 = 2, it is essential that parameters are backed up (p0977 = 1) and that a warm restart is then performed (p0009 = 30, p0976 = 2, 3). This will apply the values written.

Note: Re value = 1:
This action loads the NVRAM data to the parameters.
Re value = 2:
This action loads the parameters to the NVRAM.
Re value = 3:
This action sets parameters p7771 ... p7774 to the factory setting.
It is recommended to avoid placing unnecessary load on the subsequent upload/download operation.

p7788 Sign-of-life monitoring tolerance window /

A_INF, B_INF,
S_INF, SERVO,
VECTOR

Can be changed: U, T

Data type: Unsigned16

P-Group: Converter

Not for motor type: -

Min

1

Calculated: -

Dynamic index: -

Units group: -

Scaling: -

Max

1000

Access level: 4

Func. diagram: -

Unit selection: -

Expert list: 1

Factory setting

10

Description: The size of the window for monitoring sign-of-life errors on the power unit.

Dependency: Refer to: A30853

Note: An active window is generated by means of DRIVE-CLiQ telegrams. If more than one sign-of-life error appears in the window, A30853 is issued. The lower the value in p7788, the greater the tolerance of the monitoring procedure.

p7789 Sign-of-life monitoring fault threshold /

A_INF, B_INF,
S_INF, SERVO,
VECTOR

Can be changed: U, T

Data type: Unsigned16

P-Group: Converter

Not for motor type: -

Min

0

Calculated: -

Dynamic index: -

Units group: -

Scaling: -

Max

1000

Access level: 4

Func. diagram: -

Unit selection: -

Expert list: 1

Factory setting

1

Description: Number of consecutive sign-of-life errors on the power unit that are tolerated.

Dependency: Refer to: F30008

Note: If more sign-of-life errors than the number set in p7789 occur one after the other, F30008 is issued. The higher the value in p7789, the greater the tolerance of the monitoring procedure.

p7820 DRIVE-CLiQ component component number / DLQ comp_no

CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150

Can be changed: U, T

Data type: Unsigned16

P-Group: -

Not for motor type: -

Min

0

Calculated: -

Dynamic index: -

Units group: -

Scaling: -

Max

65535

Access level: 4

Func. diagram: -

Unit selection: -

Expert list: 1

Factory setting

0

Description: Sets the component number of the DRIVE-CLiQ component whose parameters are to be accessed.

Dependency: Refer to: p7821, p7822, r7823

p7821	DRIVE-CLiQ component parameter number / DLQ para_no		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: - Not for motor type: - Min 0 Max 65535	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the parameter number to access a parameter of a DRIVE-CLiQ component.		
Dependency:	Refer to: p7820, p7822, r7823		
p7822	DRIVE-CLiQ component parameter index / DLQ para_index		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: - Not for motor type: - Min 0 Max 65535	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the parameter index to access a parameter of a DRIVE-CLiQ component.		
Dependency:	Refer to: p7820, p7821, r7823		
r7823	DRIVE-CLiQ component read parameter value / Read DLQ value		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the parameter value read from the DRIVE-CLiQ component.		
Dependency:	Refer to: p7820, p7821, p7822		
r7825[0...6]	DRIVE-CLiQ component versions / DLQ version		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min - Max -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the firmware and EPROM versions of the DRIVE-CLiQ component selected using p7828[1].		
Index:	[0] = Reference firmware version [1] = Actual firmware version [2] = EPROM0 version [3] = EPROM1 version [4] = EPROM2 version [5] = EPROM3 version [6] = EPROM4 version		
Dependency:	Refer to: p7828		
Note:	Reference firmware version: Version on the memory card/device memory. Current firmware version: Actual version of the DRIVE-CLiQ component. EPROM version: Current EPROM version of the DRIVE-CLiQ component.		

p7826	Firmware update automatic / FW update auto		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: - Min 0 Max 2	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
Description:	Sets the behavior for the automatic firmware update of the DRIVE-CLiQ components.		
Value:	0: Deactivated 1: Upgrade and downgrade 2: Upgrade		
Notice:	If this parameter is changed, it only becomes effective the next time that the drive system boots.		
Note:	The firmware is automatically updated when the system boots. The boot can take several minutes. After the update has been completed, it is necessary to carry out a new POWER ON (power-down/power-up) for the components involved. The firmware update procedure is displayed as follows: Control Unit (LED RDY): Flashes yellow with 0.5 Hz --> firmware is being updated. Flashing yellow with 2 Hz --> POWER ON is required for the components involved. Components involved: Flashing red/green with 0.5 Hz --> firmware is being updated. Flashing red/green with 2 Hz --> POWER ON of the components is required. Only components from firmware version 2.5 support the red/green flashing at 2 Hz.		
r7827	Firmware update progress display / FW update progress		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [%] Max - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
Description:	Displays the progress when updating the firmware of the DRIVE-CLiQ components.		
p7828[0...1]	Firmware download component number / FW download number		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: - Not for motor type: - Min 0 Max 399	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 0	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
Description:	Sets the component number for the required DRIVE-CLiQ component. Index 0: Component number of the DRIVE-CLiQ component for which a firmware download is to be made. Index 1: Component number of the DRIVE-CLiQ component for which the reference firmware version, saved in r7825 on the memory card/device memory, is to be displayed.		
Index:	[0] = Firmware download [1] = Reference firmware version		
Dependency:	Refer to: p0121, p0141, p0151, p7829		
Note:	For p7828[0] = 399, the firmware for all of the existing components is downloaded. The firmware download is started with p7829 = 1.		

p7829	Activate firmware download / FW download act		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: - Min 0 Max 999	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 0	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Activating the firmware download for the DRIVE-CLiQ components specified in p7828. 1: Activate download. 0: Download successfully completed. > 1: Fault code 011: DRIVE-CLiQ component has detected a checksum error. 015: The selected DRIVE-CLiQ components did not accept the contents of the firmware file. 018: Firmware version is too old and is not accepted by the component. 019: Firmware version is not suitable for the hardware release of the component. 101: After several communication attempts, no response from the DRIVE-CLiQ component. 140: Firmware file for the DRIVE-CLiQ component not available on the memory card/device memory. 143: Component has not changed to the mode for firmware download. It was not possible to delete the existing firmware. 144: When checking the firmware that was downloaded (checksum), the component detected a fault. It is possible that the file on the memory card/device memory is defective. 145: Checking the loaded firmware (checksum) was not completed by the component in the appropriate time. 156: Component with the specified component number is not available. Additional values: Only for internal Siemens troubleshooting.		
Dependency:	Refer to: p7828		
Note:	p7829 is automatically set to 0 after the firmware has been successfully downloaded. The new firmware only becomes active at the next system run-up.		
p7830	Diagnostics telegram selection / Diag telegram		
ENCODER, SERVO, VECTOR	Can be changed: T Data type: Integer16 P-Group: - Not for motor type: - Min 0 Max 3	Calculated: - Dynamic index: - Units group: - Scaling: - Factory setting 0	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects a telegram whose contents should be shown in p7831 ... p7836.		
Value:	0: Reserved 1: First cyclic receive telegram sensor 1 2: First cyclic receive telegram sensor 2 3: First cyclic receive telegram sensor 3		
Dependency:	Refer to: r7831, r7832, r7833, r7834, r7835, r7836		

r7831[0...15] Telegram diagnostics signals / Tel diag signals			
ENCODER, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	11826	-
Description:	Displays the signals contained in the selected telegram (p7830).		
Value:	0: UNUSED 1: UNKNOWN 102: SAPAR_ID_DSA_ALARM 110: SAPAR_ALARMBITS_FLOAT_0 111: SAPAR_ALARMBITS_FLOAT_1 112: SAPAR_ALARMBITS_FLOAT_2 113: SAPAR_ALARMBITS_FLOAT_3 114: SAPAR_ALARMBITS_FLOAT_4 115: SAPAR_ALARMBITS_FLOAT_5 10500: ENC_ID_TIME_PRETRIGGER 10501: ENC_ID_TIME_SEND_TELEG_1 10502: ENC_ID_TIME_CYCLE_FINISHED 10503: ENC_ID_TIME_DELTA_FUNMAN 10504: ENC_ID_SUBTRACE_CALCTIMES 10505: ENC_ID_SYNO_PERIOD 10516: ENC_ID_ADC_TRACK_A 10517: ENC_ID_ADC_TRACK_B 10518: ENC_ID_ADC_TRACK_C 10519: ENC_ID_ADC_TRACK_D 10520: ENC_ID_ADC_TRACK_A_SAFETY 10521: ENC_ID_ADC_TRACK_B_SAFETY 10523: ENC_ID_ADC_TEMP_1 10526: ENC_ID_ADC_TRACK_R 10532: ENC_ID_TRACK_AB_X 10533: ENC_ID_TRACK_AB_Y 10534: ENC_ID_OFFSET_CORR_AB_X 10535: ENC_ID_OFFSET_CORR_AB_Y 10536: ENC_ID_AB_ABS_VALUE 10537: ENC_ID_TRACK_CD_X 10538: ENC_ID_TRACK_CD_Y 10539: ENC_ID_TRACK_CD_ABS 10542: ENC_ID_AB_RAND_X 10543: ENC_ID_AB_RAND_Y 10544: ENC_ID_AB_RAND_ABS_VALUE 10545: ENC_ID_SUBTRACE_ABS_ARRAY 10546: ENC_ID_PROC_OFFSET_0 10547: ENC_ID_PROC_OFFSET_4 10564: ENC_SELFTEMP_ACT 10565: ENC_ID_MOTOR_TEMP_TOP 10566: ENC_ID_MOTOR_TEMP_1 10580: ENC_ID_RESISTANCE_1 10590: ENC_ID_ANA_CHAN_A 10591: ENC_ID_ANA_CHAN_B 10592: ENC_ID_ANA_CHAN_X 10593: ENC_ID_ANA_CHAN_Y 10596: ENC_ID_AB_ANGLE 10597: ENC_ID_CD_ANGLE 10598: ENC_ID_MECH_ANGLE_HI 10599: ENC_ID_RM_POS_PHI_COMMU 10600: ENC_ID_PHI_COMMU 10612: ENC_ID_DIFF_CD_INC 10613: ENC_ID_RM_POS_PHI_COMMU_RFG 10628: ENC_ID_MECH_ANGLE		

10629: ENC_ID_MECH_RM_POS
 10644: ENC_ID_INIT_VECTOR
 10645: FEAT_INIT_VEKTOR
 10660: ENC_ID_SENSOR_STATE
 10661: ENC_ID_BASIC_SYSTEM
 10662: ENC_ID_REFMARK_STATUS
 10663: ENC_ID_DSA_STATUS1_SENSOR
 10664: ENC_ID_DSA_RMSTAT_HANDSHAKE
 10665: ENC_ID_DSA_CONTROL1_SENSOR
 10667: ENC_ID_SAFETY
 10676: ENC_ID_COUNTCORR_SAW_VALUE
 10677: ENC_ID_COUNTCORR_ABS_VALUE
 10678: ENC_ID_SAWTOOTH_CORR
 10692: ENC_ID_RESISTANCE_CALIB_INSTANT
 10693: ENC_ID_SERPROT_POS
 10724: ENC_ID_ACT_FUNMAN_FUNCTION
 10725: ENC_ID_SAFETY_COUNTER_CRC
 10740: ENC_ID_POS_ABSOLUTE
 10741: ENC_ID_POS_REFMARK
 10742: ENC_ID_SAWTOOTH
 10743: ENC_ID_SAFETY_PULSE_COUNTER
 10756: ENC_ID_DSA_ACTUAL_SPEED
 10757: ENC_ID_SPEED_DEV_ABS
 10772: ENC_ID_DSA_POS_XIST1
 10788: ENC_ID_AB_CROSS_CORR
 10789: ENC_ID_AB_GAIN_Y_CORR
 10790: ENC_ID_AB_PEAK_CORR
 11825: ENC_ID_RES_TRANSITION_RATIO
 11826: ENC_ID_RES_PHASE_SHIFT

Index:
 [0] =
 [1] =
 [2] =
 [3] =
 [4] =
 [5] =
 [6] =
 [7] =
 [8] =
 [9] =
 [10] =
 [11] =
 [12] =
 [13] =
 [14] =
 [15] =

r7832[0...15] Telegram diagnostics numerical format / tel diag format

ENCODER, SERVO, VECTOR	Can be changed: - Data type: Integer16 P-Group: - Not for motor type: - Min -1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 14	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
---------------------------	--	---	---

Description: Indicates the original numerical format of the signals contained in the telegram.
 The associated signal number is represented at the appropriate index in r7831.

Value:	-1:	Unknown
	0:	Boolean
	1:	Signed 1 byte
	2:	Signed 2 byte
	3:	Signed 4 byte
	4:	Signed 8 byte
	5:	Unsigned 1 byte
	6:	Unsigned 2 byte
	7:	Unsigned 4 byte
	8:	Unsigned 8 byte
	9:	Float 4 byte
	10:	Double 8 byte
	11:	mm dd yy HH MM SS MS DOW
	12:	ASCII string
	13:	SIMUMERIK frame type
	14:	SIMUMERIK axis type
Index:	[0] =	
	[1] =	
	[2] =	
	[3] =	
	[4] =	
	[5] =	
	[6] =	
	[7] =	
	[8] =	
	[9] =	
	[10] =	
	[11] =	
	[12] =	
	[13] =	
	[14] =	
	[15] =	

r7833[0...15] Telegram diagnostics unsigned / Tel diag unsigned

ENCODER, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Parameter to display a DSA signal in the unsigned-integer format.
The associated signal number is represented at the appropriate index in r7831.

Index:	[0] =	
	[1] =	
	[2] =	
	[3] =	
	[4] =	
	[5] =	
	[6] =	
	[7] =	
	[8] =	
	[9] =	
	[10] =	
	[11] =	
	[12] =	
	[13] =	
	[14] =	
	[15] =	

r7834[0...15] Telegram diagnostics signed / Tel diag signed

ENCODER, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: Integer32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Parameter to display a DSA signal in the signed-integer format.
The associated signal number is represented at the appropriate index in r7831.

Index: [0] =
 [1] =
 [2] =
 [3] =
 [4] =
 [5] =
 [6] =
 [7] =
 [8] =
 [9] =
 [10] =
 [11] =
 [12] =
 [13] =
 [14] =
 [15] =

r7835[0...15] Telegram diagnostics real / Tel diag real

ENCODER, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Parameter to display a DSA signal in the float format.
The associated signal number is represented at the appropriate index in r7831.

Index: [0] =
 [1] =
 [2] =
 [3] =
 [4] =
 [5] =
 [6] =
 [7] =
 [8] =
 [9] =
 [10] =
 [11] =
 [12] =
 [13] =
 [14] =
 [15] =

r7836[0...15] Telegram diagnostics unit / Tel diag unit			
ENCODER, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -1	Max 147	Factory setting -
Description:	Parameter to display the units of a DSA signal. The associated signal number is represented at the appropriate index in r7831.		
Value:	-1: Unknown 0: None 1: Millimeter or degrees 2: Millimeter 3: Degrees 4: mm/min or RPM 5: Millimeter / min 6: Revolutions / min 7: m/sec^2 or V/sec^2 8: m/sec^2 9: V/sec^2 10: m/sec^3 or V/sec^3 11: m/sec^3 12: V/sec^3 13: Sec 14: 16.667 / sec 15: mm/revolution 16: ACX_UNIT_COMPENSATION_CORR 18: Newton 19: Kilogram 20: Kilogram meter ^2 21: Percent 22: Hertz 23: Volt peak-to-peak 24: Amps peak-to-peak 25: Degrees Celsius 26: Degrees 28: Millimeter or degrees 29: Meters / minute 30: Meters / second 31: Ohm 32: Millihenry 33: Newton meter 34: Newton meter / Amps 35: Volt / Amp 36: Newton meter second / rad 38: 31.25 microseconds 39: Microseconds 40: Milliseconds 42: Kilowatt 43: Micro amps peak-to-peak 44: Volt seconds 45: Microvolt seconds 46: Micro Newton meter 47: Amps / Volt seconds 48: Per mille 49: Hertz / second 53: Micrometer or millidegrees 54: Micrometer 55: Millidegrees 59: Nanometer		

61:	Newton/Amps
62:	Volt seconds / meter
63:	Newton seconds / meter
64:	Micronewton
65:	Liters / minute
66:	Bar
67:	Cubic centimeters
68:	Millimeters / Volt minute
69:	Newton/Volt
80:	Millivolts peak-to-peak
81:	Volt rms
82:	Millivolts rms
83:	Amps rms
84:	Micro amps rms
85:	Micrometers / revolution
90:	Tenths of a second
91:	Hundredths of a second
92:	10 microseconds
93:	Pulses
94:	256 pulses
95:	Tenth of a pulse
96:	Revolutions
97:	100 revolutions / minute
98:	10 revolutions / minute
99:	0.1 revolutions / minute
100:	Thousandth revolution / minute
101:	Pulses / second
102:	100 pulses / second
103:	10 revolutions / (minute * second)
104:	10000 pulses/second^2
105:	0.1 Hertz
106:	0.01 Hertz
107:	0.1 / seconds
108:	Factor 0.1
109:	Factor 0.01
110:	Factor 0.001
111:	Factor 0.0001
112:	0.1 Volt peak-to-peak
113:	0.1 Volt peak-to-peak
114:	0.1 amps peak-to-peak
115:	Watt
116:	100 Watt
117:	10 Watt
118:	0.01 percent
119:	1 / second ^3
120:	0.01 percent/millisecond
121:	Pulses / revolution
122:	Microfarads
123:	Milliohm
124:	0.01 Newton meter
125:	Kilogram millimeter ^2
126:	Rad / (seconds newton meter)
127:	Henry
128:	Kelvin
129:	Hours
130:	Kilohertz
131:	Milliamperes peak-to-peak
132:	Millifarads
133:	Meter
135:	Kilowatt hours
136:	Percent
137:	Amps / Volt
138:	Volt
139:	Millivolts

140: Microvolts
 141: Amps
 142: Milliampere
 143: Micro amp
 144: Milliampere rms
 145: Millimeter
 146: Nanometer
 147: Joules

Index:

[0] =
 [1] =
 [2] =
 [3] =
 [4] =
 [5] =
 [6] =
 [7] =
 [8] =
 [9] =
 [10] =
 [11] =
 [12] =
 [13] =
 [14] =
 [15] =

r7843[0...20] Memory card serial number / Mem_card ser.no

CU_I, CU_S,
 CU_S_CU310DP,
 CU_S_CU310PN,
 CU_S_S150

Can be changed: -

Calculated: -

Access level: 1

Data type: Unsigned8

Dynamic index: -

Func. diagram: -

P-Group: -

Units group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min

Max

Factory setting

-

-

-

Description:

Displays the actual serial number of the memory card.

Dependency:

The individual characters of the serial number are displayed in the ASCII code in the indices.

Notice:

Refer to: p9920, p9921

Note:

An ASCII table (excerpt) can be found, for example, in the Appendix of the List Manual:

Example: displaying the serial number for a memory card:

r7843[0] = 49 dec --> ASCII characters = "1" --> serial number, character 1
 r7843[1] = 49 dec --> ASCII characters = "1" --> serial number, character 2
 r7843[2] = 49 dec --> ASCII characters = "1" --> serial number, character 3
 r7843[3] = 57 dec --> ASCII characters = "9" --> serial number, character 4
 r7843[4] = 50 dec --> ASCII characters = "2" --> serial number, character 5
 r7843[5] = 51 dec --> ASCII characters = "3" --> serial number, character 6
 r7843[6] = 69 dec --> ASCII characters = "E" --> serial number, character 7
 r7843[7] = 0 dec --> ASCII characters = " " --> serial number, character 8

...

r7843[19] = 0 dec --> ASCII characters = " " --> serial number, character 20

r7843[20] = 0 dec

Serial number = 111923E

r7844[0...1]	Memory card/device memory firmware version / Mem_crd/dev_mem FW		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the version of the firmware stored on the memory card/device memory.		
Note:	Example: The value 1010100 should be interpreted as V01.01.01.00.		
r7850[0...23]	Drive object operational/not operational / DO ready for oper		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Integer16 P-Group: - Not for motor type: - Min -32786	Calculated: - Dynamic index: - Units group: - Scaling: - Max 32767	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays whether, for an activated drive object, all activated topology components are available or not (or whether these can be addressed). 0: Drive object not ready for operation 1: Drive object ready for operation		
p7852	Number of indices for r7853 / Qty indices r7853		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned16 P-Group: - Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 200	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Displays the number of indices for r7853[0...n]. This corresponds to the number of DRIVE-CLiQ components that are in the target topology.		
Dependency:	Refer to: r7853		
Note:	The values are valid if all available Control Units adopt the "Initialization finished" state (r3988 = 800) following power-up.		
r7853[0...n]	Component available/not available / Comp present		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: p7852 Units group: - Scaling: - Max FFFF hex	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the component and whether this component is currently present. High byte: Component number Low byte: 0/1 (not available/available)		
Dependency:	Refer to: p7852		
Note:	The values are valid if all available Control Units adopt the "Initialization finished" state (r3988 = 800) following power-up.		

p7857	Sub-boot mode / Sub-boot mode		
A_INF, B_INF, CU_LINK, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VEC- TOR	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 1	Factory setting 1
Description:	Sets the mode for the sub-boot.		
Value:	0: Sub-boot manual 1: Sub-boot automatic		
Note:	For p7857 = 0 (manual sub-boot) the following applies: The parameter should be set to 1 to start the sub-boot.		
p7859[0...199]	Component number global / Comp_nr global		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 0
	Min -32786	Max 32767	Factory setting 0
Description:	Sets the global and unique component number in a drive system with several Control Units. Each index of the parameter corresponds to a possible local component number on the corresponding Control Unit. The indices are allocated to the global component numbers as follows: p7859[0]: Not used p7859[1]: Sets the global component number for the local component number 1 p7859[2]: Sets the global component number for the local component number 2 ... p7859[199]: Sets the global component number for the local component number 199		
Notice:	This parameter is preferably set via suitable commissioning software (e.g. UpdateAgent, STARTER, SCOUT). Changing the parameter via the AOP (Advanced Operator Panel) or BOP (Basic Operator Panel) can destroy a valid unique setting.		
Note:	The parameter is not influenced by setting the factory setting.		
r7867	Status/configuration changes global / Changes global		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays status and configuration changes of all of the drive objects in the complete unit. When changing the status or the configuration of the Control Unit or a drive object, the value of this parameter is incremented.		
Dependency:	Refer to: r7868, r7869, r7870		

r7868[0...24]	Configuration changes drive object reference / Config_chng DO ref		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Reference to the drive objects whose configuration has changed. Index 0: When changing one of the following indices, then the value in this index is increased. Index 1...n: The drive object with object number in p0101[n-1] has changed its configuration. Example: r7868[3] was incremented since the last time it was read. --> the configuration of the drive object with object number in p0101[2] was changed.		
Index:	[0] = Sum of the following indices [1] = Object number in p0101[0] [2] = Object number in p0101[1] [3] = Object number in p0101[2] [4] = Object number in p0101[3] [5] = Object number in p0101[4] [6] = Object number in p0101[5] [7] = Object number in p0101[6] [8] = Object number in p0101[7] [9] = Object number in p0101[8] [10] = Object number in p0101[9] [11] = Object number in p0101[10] [12] = Object number in p0101[11] [13] = Object number in p0101[12] [14] = Object number in p0101[13] [15] = Object number in p0101[14] [16] = Object number in p0101[15] [17] = Object number in p0101[16] [18] = Object number in p0101[17] [19] = Object number in p0101[18] [20] = Object number in p0101[19] [21] = Object number in p0101[20] [22] = Object number in p0101[21] [23] = Object number in p0101[22] [24] = Object number in p0101[23]		
Dependency:	Refer to: p0101, r7867, r7871		

r7869[0...24]	Status changes drive object reference / Status_chng DO ref		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Reference to the drive objects whose status has changed. Index 0: When changing one of the following indices, then the value in this index is increased. Index 1...n: The drive object with object number in p0101[n-1] has changed its status. Example: r7868[3] was incremented since the last time it was read. --> the status of the drive object with object number in p0101[2] was changed.		
Index:	[0] = Sum of the following indices [1] = Object number in p0101[0] [2] = Object number in p0101[1] [3] = Object number in p0101[2] [4] = Object number in p0101[3] [5] = Object number in p0101[4] [6] = Object number in p0101[5] [7] = Object number in p0101[6] [8] = Object number in p0101[7] [9] = Object number in p0101[8] [10] = Object number in p0101[9] [11] = Object number in p0101[10] [12] = Object number in p0101[11] [13] = Object number in p0101[12] [14] = Object number in p0101[13] [15] = Object number in p0101[14] [16] = Object number in p0101[15] [17] = Object number in p0101[16] [18] = Object number in p0101[17] [19] = Object number in p0101[18] [20] = Object number in p0101[19] [21] = Object number in p0101[20] [22] = Object number in p0101[21] [23] = Object number in p0101[22] [24] = Object number in p0101[23]		
Dependency:	Refer to: p0101, r7867, r7872		

r7870[0...7] Configuration changes global / Config_chng global			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the configuration changes of all of the drive objects in the complete unit.		
Index:	[0] = Sum of the following indices [1] = r7871[0] of a drive object [2] = p0101 or r0102 [3] = PROFIBUS configuration (p0978) [4] = DRIVE-CLiQ actual topology (r9900 or r9901) [5] = DRIVE-CLiQ target topology (r9902 or r9903) [6] = DRIVE-CLiQ ports (p0109) [7] = OA applications		
Dependency:	Refer to: r7867, r7871		
Note:	Index 0: When changing one of the following indices, then the value in this index is incremented. Index 1: Drive object configuration. When changing r7871[0] on a drive object, the value in this index is incremented. Index 2: Drive object, configuration unit. When changing either p0101 or r0102, the value in this index is incremented. Index 3: PROFIBUS configuration unit. When changing p0978, the value in this index is incremented. Index 4: DRIVE-CLiQ actual topology. When changing either r9900 or r9901, the value in this index is incremented. Index 5: DRIVE-CLiQ target topology. When changing either p9902 or p9903, the value in this index is incremented. Index 6: DRIVE-CLiQ ports. When changing p0109, the value in this index is incremented. Index 7: OA applications. When changing OA applications, the value in this index is incremented.		

r7871[0...10] Configuration changes drive object / Config_chng DO			
A_INF, B_INF, S_INF	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the configuration changes on the drive object.		
Index:	[0] = Sum of the following indices [1] = p0010, p0107 or p0108 [2] = Drive object name (p0199) [3] = Structure-relevant parameters (e.g. p0180) [4] = BICO interconnections [5] = Activate/deactivate drive object [6] = Data backup required [7] = Activate/deactivate component [8] = Reference or changeover parameters (e.g. p2000) [9] = Parameter count through Drive Control Chart (DCC) [10] = p0107 or p0108		
Dependency:	Refer to: r7868, r7870		
Note:	Index 0: When changing one of the following indices, then the value in this index is incremented. Index 1: Drive object commissioning: When changing p0010, p0107 or p0108, the value in this index is incremented. Index 2: Drive object name. When changing p0199, the value in this index is incremented. Index 3: Drive object structure. When changing a parameter that is relevant for the structure (e.g. number of data sets), the value in this index is incremented. Index 4: Drive object BICO interconnections. When changing r3977, the value in this index is incremented. Index 5: Drive object activity: When changing p0105, the value in this index is incremented. Index 6: Drive object, data save. 0: There are no parameter changes to save. 1: There are parameter changes to save. Index 7: Drive object component activity: When changing either p0125 or p0145, the value in this index is incremented. Index 8: Drive object changeover of units. When changing reference or changeover parameters (e.g. p2000, p0304), the value in this index is incremented. Index 9: Drive object parameter count. When changing the number of parameters by loading Drive Control Chart (DCC), the value in this index is incremented. Index 10: Drive object configuration. When changing either p0107 or p0108, the value in this index is incremented.		

r7871[0...10] Configuration changes drive object / Config_chng DO			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the configuration changes on the drive object.		
Index:	[0] = Sum of the following indices [1] = p0107 or p0108 [2] = Drive object name (p0199) [3] = Structure-relevant parameters (e.g. p0180) [4] = BICO interconnections [5] = Activate/deactivate drive object [6] = Data backup required [7] = Reserved [8] = Reference or changeover parameters (e.g. p2000) [9] = Parameter count through Drive Control Chart (DCC) [10] = p0107 or p0108		
Dependency:	Refer to: r7868, r7870		
Note:	Index 0: When changing one of the following indices, then the value in this index is incremented. Index 1: Drive object commissioning: When changing either p0107 or p0108, the value in this index is incremented. Index 2: Drive object name. When changing p0199, the value in this index is incremented. Index 3: Drive object structure. When changing a parameter that is relevant for the structure (e.g. number of data sets), the value in this index is incremented. Index 4: Drive object BICO interconnections. When changing r3977, the value in this index is incremented. Index 5: Drive object activity: When changing p0105, the value in this index is incremented. Index 6: Drive object, data save. 0: There are no parameter changes to save. 1: There are parameter changes to save. Index 8: Drive object changeover of units. When changing reference or changeover parameters (e.g. p2000, p0304), the value in this index is incremented. Index 9: Drive object parameter count. When changing the number of parameters by loading Drive Control Chart (DCC), the value in this index is incremented. Index 10: Drive object configuration. When changing either p0107 or p0108, the value in this index is incremented.		

r7871[0...10] Configuration changes drive object / Config_chng DO			
CU_LINK, HUB, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the configuration changes on the drive object.		
Index:	[0] = Sum of the following indices [1] = p0010, p0107 or p0108 [2] = Drive object name (p0199) [3] = Structure-relevant parameters (e.g. p0180) [4] = BICO interconnections [5] = Activate/deactivate drive object [6] = Data backup required [7] = Reserved [8] = Reference or changeover parameters (e.g. p2000) [9] = Parameter count through Drive Control Chart (DCC) [10] = p0107 or p0108		
Dependency:	Refer to: r7868, r7870		
Note:	Index 0: When changing one of the following indices, then the value in this index is incremented. Index 1: Drive object commissioning: When changing p0010, p0107 or p0108, the value in this index is incremented. Index 2: Drive object name. When changing p0199, the value in this index is incremented. Index 3: Drive object structure. When changing a parameter that is relevant for the structure (e.g. number of data sets), the value in this index is incremented. Index 4: Drive object BICO interconnections. When changing r3977, the value in this index is incremented. Index 5: Drive object activity: When changing p0105, the value in this index is incremented. Index 6: Drive object, data save. 0: There are no parameter changes to save. 1: There are parameter changes to save. Index 8: Drive object changeover of units. When changing reference or changeover parameters (e.g. p2000, p0304), the value in this index is incremented. Index 9: Drive object parameter count. When changing the number of parameters by loading Drive Control Chart (DCC), the value in this index is incremented. Index 10: Drive object configuration. When changing either p0107 or p0108, the value in this index is incremented.		

r7871[0...15] Configuration changes drive object / Config_chng DO			
ENCODER	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the configuration changes on the drive object.		
Index:	[0] = Sum of the following indices [1] = p0010, p0107 or p0108 [2] = Drive object name (p0199) [3] = Structure-relevant parameters (e.g. p0180) [4] = BICO interconnections [5] = Activate/deactivate drive object [6] = Data backup required [7] = Activate/deactivate component [8] = Reference or changeover parameters (e.g. p2000) [9] = Parameter count through Drive Control Chart (DCC) [10] = p0107 or p0108 [11] = Reserved [12] = Reserved [13] = Reserved [14] = Reserved [15] = Encoder type (p0400)		
Dependency:	Refer to: r7868, r7870		
Note:	Index 0: When changing one of the following indices, then the value in this index is incremented. Index 1: Drive object configuration. When changing p0010, p0107 or p0108, the value in this index is incremented. Index 2: Drive object name. When changing p0199, the value in this index is incremented. Index 3: Drive object structure. When changing a parameter that is relevant for the structure (e.g. number of data sets), the value in this index is incremented. Index 4: Drive object BICO interconnections. When changing r3977, the value in this index is incremented. Index 6: Drive object, data save. 0: There are no parameter changes to save. 1: There are parameter changes to save. Index 8: Drive object changeover of units. When changing reference or changeover parameters (e.g. p2000, p0304 ...), the value in this index is incremented. Index 9: Drive object parameter count. When changing the number of parameters by loading Drive Control Chart (DCC), the value in this index is incremented. Index 15: Encoder configuration. When changing p0400, the value in this index is incremented.		

r7871[0...15]	Configuration changes drive object / Config_chng DO		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the configuration changes on the drive object.		
Index:	[0] = Sum of the following indices [1] = p0010, p0107 or p0108 [2] = Drive object name (p0199) [3] = Structure-relevant parameters (e.g. p0180) [4] = BICO interconnections [5] = Activate/deactivate drive object [6] = Data backup required [7] = Activate/deactivate component [8] = Reference or changeover parameters (e.g. p2000) [9] = Parameter count through Drive Control Chart (DCC) [10] = p0107 or p0108 [11] = Reserved [12] = Reserved [13] = Reserved [14] = Reserved [15] = SERVO or VECTOR (e.g. p0300)		
Dependency:	Refer to: r7868, r7870		
Note:	Index 0: When changing one of the following indices, then the value in this index is incremented. Index 1: Drive object commissioning: When changing p0010, p0107 or p0108, the value in this index is incremented. Index 2: Drive object name. When changing p0199, the value in this index is incremented. Index 3: Drive object structure. When changing a parameter that is relevant for the structure (e.g. number of data sets), the value in this index is incremented. Index 4: Drive object BICO interconnections. When changing r3977, the value in this index is incremented. Index 5: Drive object activity: When changing p0105, the value in this index is incremented. Index 6: Drive object, data save. 0: There are no parameter changes to save. 1: There are parameter changes to save. Index 7: Drive object component activity: When changing either p0125 or p0145, the value in this index is incremented. Index 8: Drive object changeover of units. When changing reference or changeover parameters (e.g. p2000, p0304), the value in this index is incremented. Index 9: Drive object parameter count. When changing the number of parameters by loading Drive Control Chart (DCC), the value in this index is incremented. Index 10: Drive object configuration. When changing either p0107 or p0108, the value in this index is incremented. Index 15: SERVO/VECTOR configuration. When changing p0300, p0301 or p0400, the value in this index is incremented.		

r7871[0...10] Configuration changes drive object / Config_chng DO			
TM54F_MA, TM54F_SL	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the configuration changes on the drive object.		
Index:	[0] = Sum of the following indices [1] = p0010, p0107 or p0108 [2] = Drive object name (p0199) [3] = Structure-relevant parameters (e.g. p0180) [4] = BICO interconnections [5] = Reserved [6] = Data backup required [7] = Reserved [8] = Reference or changeover parameters (e.g. p2000) [9] = Parameter count through Drive Control Chart (DCC) [10] = p0107 or p0108		
Dependency:	Refer to: r7868, r7870		
Note:	Index 0: When changing one of the following indices, then the value in this index is incremented. Index 1: Drive object commissioning: When changing p0010, p0107 or p0108, the value in this index is incremented. Index 2: Drive object name. When changing p0199, the value in this index is incremented. Index 3: Drive object structure. When changing a parameter that is relevant for the structure (e.g. number of data sets), the value in this index is incremented. Index 4: Drive object BICO interconnections. When changing r3977, the value in this index is incremented. Index 6: Drive object, data save. 0: There are no parameter changes to save. 1: There are parameter changes to save. Index 8: Drive object changeover of units. When changing reference or changeover parameters (e.g. p2000, p0304), the value in this index is incremented. Index 9: Drive object parameter count. When changing the number of parameters by loading Drive Control Chart (DCC), the value in this index is incremented. Index 10: Drive object configuration. When changing either p0107 or p0108, the value in this index is incremented.		

r7872[0...3] Status changes drive object / Status_chng DO

All objects	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status changes on the drive object.
Index 0:
When changing one of the following indices, then the value in this index is incremented.
Index 1:
Drive object faults. When changing r0944, the value in this index is incremented.
Index 2:
Drive object alarms. When changing r2121, the value in this index is incremented.
Index 3:
Drive object safety messages. When changing r9744, the value in this index is incremented.

Index: [0] = Sum of the following indices
[1] = Faults (r0944)
[2] = Alarms (r2121)
[3] = Safety messages (r9744)

Dependency: Refer to: r7869

p7900[0...23] Drive objects priority / DO priority

CU_CX32, CU_I,	Can be changed: U, T	Calculated: -	Access level: 4
CU_S,	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
CU_S_CU310DP,	P-Group: -	Units group: -	Unit selection: -
CU_S_CU310PN,	Not for motor type: -	Scaling: -	Expert list: 1
CU_S_S150	Min	Max	Factory setting
	0	65535	0

Description: Sets the priority for processing the existing drive objects in the system.
The parameter enables a free sequence to be set for processing the drive objects. For this purpose all the drive object numbers existing in the system have to be written in the desired sequence into the corresponding indices of the parameter. After re-booting this sequence will be effective without a plausibility check.
With the factory setting the following priorities regarding processing are applicable:
- The drive objects are pre-sorted according to their type as follows: CONTROL UNIT, INFEED, SERVO, VECTOR, TM, HUB, CU-LINK
- If they are of the same type, they are sorted in ascending order according to their drive object number, i.e. the lower the number, the higher the priority for processing.

Index: [0] = Drive object number Control Unit
[1] = Drive object number object 1
[2] = Drive object number object 2
[3] = Drive object number object 3
[4] = Drive object number object 4
[5] = Drive object number object 5
[6] = Drive object number object 6
[7] = Drive object number object 7
[8] = Drive object number object 8
[9] = Drive object number object 9
[10] = Drive object number object 10
[11] = Drive object number object 11
[12] = Drive object number object 12
[13] = Drive object number object 13
[14] = Drive object number object 14
[15] = Drive object number object 15
[16] = Drive object number object 16
[17] = Drive object number object 17

[18] = Drive object number object 18
 [19] = Drive object number object 19
 [20] = Drive object number object 20
 [21] = Drive object number object 21
 [22] = Drive object number object 22
 [23] = Drive object number object 23

Notice: This parameter may only be used by qualified service personnel.

Note: If the same drive object numbers are used and if the existing drive object numbers in the system are entered incompletely, the content of this parameter is ignored entirely. The behavior as with factory setting will then become effective.

r7901[0...43] Sampling times / t_sample

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [µs]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [µs]
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Description: Displays the sampling times currently present on the drive unit.
 For r7901[x] = 0, the following applies: The time slice is not active.

r7903 Hardware sampling times still cannot be assigned / HW t_samp free

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
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Description: Displays the number of hardware sampling times that can still be assigned.
 These free sampling times can be used by OA applications such as DCC (Drive Control Chart) or FBLOCKS (free function blocks).

Note: OA: Open Architecture

p8500[0...7] BI: Data transfer 0 bitwise / Transfer 0 bit

CU_CX32, CU_LINK	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 2211 Unit selection: - Expert list: 1 Factory setting 0
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Description: Sets the signal source for bitwise data transfer.
 These signals are transferred to another Control Unit and are located in BO: r8510.0 ... 7 for further interconnection.

Index:
 [0] = Send signal to BO: r8510.0
 [1] = Send signal to BO: r8510.1
 [2] = Send signal to BO: r8510.2
 [3] = Send signal to BO: r8510.3
 [4] = Send signal to BO: r8510.4
 [5] = Send signal to BO: r8510.5
 [6] = Send signal to BO: r8510.6
 [7] = Send signal to BO: r8510.7

Dependency: Refer to: r8510

Note: Example:
 Providing the operating signals calculated on this Control Unit for the infeeds on a different Control Unit.
 p8500[0] = r0863.0 (operating signal infeed 1)
 p8500[1] = r0863.0 (operating signal infeed 2) etc.

p8500[0...7]	BI: Data transfer 0 bitwise / Transfer 0 bit		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for bitwise data transfer. These signals are available in BO: r8510.0 ... 7 for further interconnection.		
Index:	[0] = Send signal to BO: r8510.0 [1] = Send signal to BO: r8510.1 [2] = Send signal to BO: r8510.2 [3] = Send signal to BO: r8510.3 [4] = Send signal to BO: r8510.4 [5] = Send signal to BO: r8510.5 [6] = Send signal to BO: r8510.6 [7] = Send signal to BO: r8510.7		
Dependency:	Refer to: r8510		
p8501[0...21]	BI: Data transfer 1 bitwise / Transfer 1 bit		
CU_CX32	Can be changed: U, T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 2211 Unit selection: - Expert list: 1 Factory setting [0] 722.0 [1] 722.1 [2] 722.2 [3] 722.3 [4] 0 [5] 0 [6] 0 [7] 0 [8] 722.8 [9] 722.9 [10] 722.10 [11] 722.11 [12] 0 [13] 0 [14] 0 [15] 0 [16] 722.16 [17] 722.17 [18] 0 [19] 0 [20] 0 [21] 0
Description:	Sets the signal source for bitwise data transfer. These signals are transferred to another Control Unit and are located in BO: r8511.0 ... 15 for further interconnection.		

Index:

- [0] = Send signal to BO: r8511.0
- [1] = Send signal to BO: r8511.1
- [2] = Send signal to BO: r8511.2
- [3] = Send signal to BO: r8511.3
- [4] = Send signal to BO: r8511.4
- [5] = Send signal to BO: r8511.5
- [6] = Send signal to BO: r8511.6
- [7] = Send signal to BO: r8511.7
- [8] = Send signal to BO: r8511.8
- [9] = Send signal to BO: r8511.9
- [10] = Send signal to BO: r8511.10
- [11] = Send signal to BO: r8511.11
- [12] = Send signal to BO: r8511.12
- [13] = Send signal to BO: r8511.13
- [14] = Send signal to BO: r8511.14
- [15] = Send signal to BO: r8511.15
- [16] = Send signal to BO: r8511.16
- [17] = Send signal to BO: r8511.17
- [18] = Send signal to BO: r8511.18
- [19] = Send signal to BO: r8511.19
- [20] = Send signal to BO: r8511.20
- [21] = Send signal to BO: r8511.21

Dependency: Refer to: r8511

p8501[0...21] CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	BI: Data transfer 1 bitwise / Transfer 1 bit Can be changed: U, T Data type: Unsigned32 / Binary P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
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Description: Sets the signal source for bitwise data transfer.
 These signals are available in BO: r8511.0 ... 15 for further interconnection.

Index:

- [0] = Send signal to BO: r8511.0
- [1] = Send signal to BO: r8511.1
- [2] = Send signal to BO: r8511.2
- [3] = Send signal to BO: r8511.3
- [4] = Send signal to BO: r8511.4
- [5] = Send signal to BO: r8511.5
- [6] = Send signal to BO: r8511.6
- [7] = Send signal to BO: r8511.7
- [8] = Send signal to BO: r8511.8
- [9] = Send signal to BO: r8511.9
- [10] = Send signal to BO: r8511.10
- [11] = Send signal to BO: r8511.11
- [12] = Send signal to BO: r8511.12
- [13] = Send signal to BO: r8511.13
- [14] = Send signal to BO: r8511.14
- [15] = Send signal to BO: r8511.15
- [16] = Send signal to BO: r8511.16
- [17] = Send signal to BO: r8511.17
- [18] = Send signal to BO: r8511.18
- [19] = Send signal to BO: r8511.19
- [20] = Send signal to BO: r8511.20
- [21] = Send signal to BO: r8511.21

Dependency: Refer to: r8511

p8501[0...21]			
CU_LINK	BI: Data transfer 1 bitwise / Transfer 1 bit		
	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	[0] 722.0
			[1] 722.1
			[2] 722.2
			[3] 722.3
			[4] 722.4
			[5] 722.5
			[6] 722.6
			[7] 722.7
			[8] 722.8
			[9] 722.9
			[10] 722.10
			[11] 722.11
			[12] 722.12
			[13] 722.13
			[14] 722.14
			[15] 722.15
			[16] 722.16
			[17] 722.17
			[18] 0
			[19] 0
			[20] 722.20
			[21] 722.21
Description:	Sets the signal source for bitwise data transfer. These signals are transferred to another Control Unit and are located in BO: r8511.0 ... 15 for further interconnection.		
Index:	[0] = Send signal to BO: r8511.0 [1] = Send signal to BO: r8511.1 [2] = Send signal to BO: r8511.2 [3] = Send signal to BO: r8511.3 [4] = Send signal to BO: r8511.4 [5] = Send signal to BO: r8511.5 [6] = Send signal to BO: r8511.6 [7] = Send signal to BO: r8511.7 [8] = Send signal to BO: r8511.8 [9] = Send signal to BO: r8511.9 [10] = Send signal to BO: r8511.10 [11] = Send signal to BO: r8511.11 [12] = Send signal to BO: r8511.12 [13] = Send signal to BO: r8511.13 [14] = Send signal to BO: r8511.14 [15] = Send signal to BO: r8511.15 [16] = Send signal to BO: r8511.16 [17] = Send signal to BO: r8511.17 [18] = Send signal to BO: r8511.18 [19] = Send signal to BO: r8511.19 [20] = Send signal to BO: r8511.20 [21] = Send signal to BO: r8511.21		
Dependency:	Refer to: r8511		

p8502	CI: Data transfer 0 wordwise / Transfer 0 word		
CU_CX32, CU_LINK	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the wordwise data transfer (process signal). This signal value is transferred to another Control Unit and is located at CO: r8512 for further interconnection.		
Dependency:	Refer to: r8512		

p8502	CI: Data transfer 0 wordwise / Transfer 0 word		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the wordwise data transfer (process signal). This signal value is available in CO: r8512 for further interconnection.		
Dependency:	Refer to: r8512		

p8503	CI: Data transfer 1 wordwise / Transfer 1 word		
CU_CX32, CU_LINK	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the wordwise data transfer (process signal). This signal value is transferred to another Control Unit and is located in CO: r8513 for further interconnection.		
Dependency:	Refer to: r8513		

p8503	CI: Data transfer 1 wordwise / Transfer 1 word		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the wordwise data transfer (process signal). This signal value is available in CO: r8513 for further interconnection.		
Dependency:	Refer to: r8513		

p8504 CI: Data transfer 2 wordwise / Transfer 2 word

CU_CX32, CU_LINK	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the wordwise data transfer (process signal). This signal value is transferred to another Control Unit and is located in CO: r8514 for further interconnection.		
Dependency:	Refer to: r8514		

p8504 CI: Data transfer 2 wordwise / Transfer 2 word

CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the wordwise data transfer (process signal). This signal value is available in CO: r8514 for further interconnection.		
Dependency:	Refer to: r8514		

p8505 CI: Data transfer 3 wordwise / Transfer 3 word

CU_CX32, CU_LINK	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / FloatingPoint32	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the wordwise data transfer (process signal). This signal value is transferred to another Control Unit and is located in CO: r8515 for further interconnection.		
Dependency:	Refer to: r8515		

p8505 CI: Data transfer 3 wordwise / Transfer 3 word

CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Sets the signal source for the wordwise data transfer (process signal). This signal value is available in CO: r8515 for further interconnection.		
Dependency:	Refer to: r8515		

r8510.0...7 BO: Data transfer 0 receive bitwise / Trans 0 recv bit

CU_CX32, CU_LINK	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the signals of the bitwise received data.
These signals were interconnected and transferred to another Control Unit via BI: p8500[0...7].

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Receive signal from BI: p8500	On	Off	-
	01	Receive signal from BI: p8500	On	Off	-
	02	Receive signal from BI: p8500	On	Off	-
	03	Receive signal from BI: p8500	On	Off	-
	04	Receive signal from BI: p8500	On	Off	-
	05	Receive signal from BI: p8500	On	Off	-
	06	Receive signal from BI: p8500	On	Off	-
	07	Receive signal from BI: p8500	On	Off	-

Dependency: Refer to: p8500

r8510.0...7 BO: Data transfer 0 receive bitwise / Trans 0 recv bit

CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the signals of the bitwise received data.
These signals were interconnected and transferred via BI: p8500[0...7].

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Receive signal from BI: p8500	On	Off	-
	01	Receive signal from BI: p8500	On	Off	-
	02	Receive signal from BI: p8500	On	Off	-
	03	Receive signal from BI: p8500	On	Off	-
	04	Receive signal from BI: p8500	On	Off	-
	05	Receive signal from BI: p8500	On	Off	-
	06	Receive signal from BI: p8500	On	Off	-
	07	Receive signal from BI: p8500	On	Off	-

Dependency: Refer to: p8500

r8511.0...21 BO: Data transfer 1 receive bitwise / Trans 1 recv bit

CU_CX32, CU_LINK	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the signals of the bitwise received data.
These signals were interconnected and transferred to another Control Unit via BI: p8501[0...15].

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Receive signal from BI: p8501	On	Off	-
	01	Receive signal from BI: p8501	On	Off	-
	02	Receive signal from BI: p8501	On	Off	-
	03	Receive signal from BI: p8501	On	Off	-
	04	Receive signal from BI: p8501	On	Off	-
	05	Receive signal from BI: p8501	On	Off	-
	06	Receive signal from BI: p8501	On	Off	-
	07	Receive signal from BI: p8501	On	Off	-
	08	Receive signal from BI: p8501	On	Off	-
	09	Receive signal from BI: p8501	On	Off	-
	10	Receive signal from BI: p8501	On	Off	-
	11	Receive signal from BI: p8501	On	Off	-
	12	Receive signal from BI: p8501	On	Off	-
	13	Receive signal from BI: p8501	On	Off	-
	14	Receive signal from BI: p8501	On	Off	-
	15	Receive signal from BI: p8501	On	Off	-
	16	Receive signal from BI: p8501	On	Off	-
	17	Receive signal from BI: p8501	On	Off	-
	18	Receive signal from BI: p8501	On	Off	-
	19	Receive signal from BI: p8501	On	Off	-
	20	Receive signal from BI: p8501	On	Off	-
	21	Receive signal from BI: p8501	On	Off	-
Dependency:	Refer to: p8501				

r8511.0...21 BO: Data transfer 1 receive bitwise / Trans 1 recv bit

CU_I, CU_S,	Can be changed: -	Calculated: -	Access level: 2
CU_S_CU310DP,	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
CU_S_CU310PN,	P-Group: -	Units group: -	Unit selection: -
CU_S_S150	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the signals of the bitwise received data.
These signals were interconnected and transferred via BI: p8501[0...15].

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Receive signal from BI: p8501	On	Off	-
	01	Receive signal from BI: p8501	On	Off	-
	02	Receive signal from BI: p8501	On	Off	-
	03	Receive signal from BI: p8501	On	Off	-
	04	Receive signal from BI: p8501	On	Off	-
	05	Receive signal from BI: p8501	On	Off	-
	06	Receive signal from BI: p8501	On	Off	-
	07	Receive signal from BI: p8501	On	Off	-
	08	Receive signal from BI: p8501	On	Off	-
	09	Receive signal from BI: p8501	On	Off	-
	10	Receive signal from BI: p8501	On	Off	-
	11	Receive signal from BI: p8501	On	Off	-
	12	Receive signal from BI: p8501	On	Off	-
	13	Receive signal from BI: p8501	On	Off	-
	14	Receive signal from BI: p8501	On	Off	-
	15	Receive signal from BI: p8501	On	Off	-
	16	Receive signal from BI: p8501	On	Off	-
	17	Receive signal from BI: p8501	On	Off	-
	18	Receive signal from BI: p8501	On	Off	-
	19	Receive signal from BI: p8501	On	Off	-
	20	Receive signal from BI: p8501	On	Off	-
	21	Receive signal from BI: p8501	On	Off	-

Dependency: Refer to: p8501

r8512	CO: Data transfer 0 receive wordwise / Trans 0 recv word		
CU_CX32, CU_LINK	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the signals of the wordwise received data (process signal). This signal value is interconnected and transferred to another Control Unit via CI: p8502.		
Dependency:	Refer to: p8502		

r8512	CO: Data transfer 0 receive wordwise / Trans 0 recv word		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]
Description:	Displays the signals of the wordwise received data (process signal). This signal value is interconnected and transferred via CI: p8502.		
Dependency:	Refer to: p8502		

r8513	CO: Data transfer 1 receive wordwise / Trans 1 recv word		
CU_CX32, CU_LINK	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the signals of the wordwise received data (process signal). This signal value is interconnected and transferred to another Control Unit via CI: p8503.		
Dependency:	Refer to: p8503		

r8513	CO: Data transfer 1 receive wordwise / Trans 1 recv word		
CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]
Description:	Displays the signals of the wordwise received data (process signal). This signal value is interconnected and transferred via CI: p8503.		
Dependency:	Refer to: p8503		

r8514 CO: Data transfer 2 receive wordwise / Trans 2 recv word

CU_CX32, CU_LINK	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the signals of the wordwise received data (process signal).
This signal value is interconnected and transferred to another Control Unit via CI: p8504.

Dependency: Refer to: p8504

r8514 CO: Data transfer 2 receive wordwise / Trans 2 recv word

CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]

Description: Displays the signals of the wordwise received data (process signal).
This signal value is interconnected and transferred via CI: p8504.

Dependency: Refer to: p8504

r8515 CO: Data transfer 3 receive wordwise / Trans 3 recv word

CU_CX32, CU_LINK	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the signals of the wordwise received data (process signal).
This signal value is interconnected and transferred to another Control Unit via CI: p8505.

Dependency: Refer to: p8505

r8515 CO: Data transfer 3 receive wordwise / Trans 3 recv word

CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min	Max	Factory setting
	- [%]	- [%]	- [%]

Description: Displays the signals of the wordwise received data (process signal).
This signal value is interconnected and transferred via CI: p8505.

Dependency: Refer to: p8505

p8520[0...3] CU_LINK signal value scaling / Sig val scal

CU_CX32, CU_LINK	Can be changed: U, T	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2211
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00010	Max 10000.00000	Factory setting 1.00000
Description:	Sets the scaling for signal value 0 ... 3 (process signals).		
Index:	[0] = Signal value 0 (Cl: p8502) scaling [1] = Signal value 1 (Cl: p8503) scaling [2] = Signal value 2 (Cl: p8504) scaling [3] = Signal value 3 (Cl: p8505) scaling		
Dependency:	Refer to: p8502, p8503, p8504, p8505		

p8550 AOP LOCAL/REMOTE / AOP LOCAL/REMOTE

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned32 P-Group: - Not for motor type: - Min - Description: Bit field:	Calculated: - Dynamic index: - Units group: - Scaling: - Max - Is used to save the actual configuration of the Advanced Operator Panels (AOP).	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1001 bin																																																																	
	<table><tr><th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr><tr><td>00</td><td>LOCAL save</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>01</td><td>Start in LOCAL</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>02</td><td>Change in oper.</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>03</td><td>OFF acts like OFF1</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>04</td><td>OFF acts like OFF2</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>05</td><td>OFF acts like OFF3</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>06</td><td>Reserved</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>07</td><td>CW/CCW active</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>08</td><td>Jog active</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>09</td><td>Save speed setpoint</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>14</td><td>Inhibit operation</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>15</td><td>Inhibit parameterization</td><td>Yes</td><td>No</td><td>-</td></tr></table>	Bit	Signal name	1 signal	0 signal	FP	00	LOCAL save	Yes	No	-	01	Start in LOCAL	Yes	No	-	02	Change in oper.	Yes	No	-	03	OFF acts like OFF1	Yes	No	-	04	OFF acts like OFF2	Yes	No	-	05	OFF acts like OFF3	Yes	No	-	06	Reserved	Yes	No	-	07	CW/CCW active	Yes	No	-	08	Jog active	Yes	No	-	09	Save speed setpoint	Yes	No	-	14	Inhibit operation	Yes	No	-	15	Inhibit parameterization	Yes	No	-		
Bit	Signal name	1 signal	0 signal	FP																																																																
00	LOCAL save	Yes	No	-																																																																
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08	Jog active	Yes	No	-																																																																
09	Save speed setpoint	Yes	No	-																																																																
14	Inhibit operation	Yes	No	-																																																																
15	Inhibit parameterization	Yes	No	-																																																																

r8570[0...39] Macro drive object / Macro DO

A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, S_INF, SERVO, TM120, TM15DI_DO, TM31, VECTOR	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min -	Max -	Factory setting -
Description:	Displays the macro file saved in the appropriate directory on the memory card/device memory.		
Dependency:	Refer to: p0015		
Note:	For a value = 9999999, the following applies: The read operation is still running.		

r8571[0...39]	Macro Binector Input (BI) / Macro BI		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the ACX file saved in the appropriate directory in the non-volatile memory.		
Dependency:	Refer to: p0700		
Note:	For a value = 9999999, the following applies: The read operation is still running.		
r8572[0...39]	Macro Connector Inputs (CI) for speed setpoints / Macro CI n_set		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the ACX file saved in the appropriate directory in the non-volatile memory.		
Dependency:	Refer to: p1000		
Note:	For a value = 9999999, the following applies: The read operation is still running.		
r8573[0...39]	Macro Connector Inputs (CI) for torque setpoints / Macro CI M_set		
A_INF, B_INF, S_INF, SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the ACX file saved in the appropriate directory in the non-volatile memory.		
Dependency:	Refer to: p1500		
Note:	For a value = 9999999, the following applies: The read operation is still running.		
r8585	Actual macro running / Actual macro		
A_INF, B_INF, CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, S_INF, SERVO, TM120, TM15DI_DO, TM31, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Indicates which macro is presently being run in the drive object.		
Dependency:	Refer to: p0015, p0700, p1000, p1500, r8570, r8571, r8572, r8573		

r8600	CAN device type / Device type		
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: - Data type: Unsigned32 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays all of the devices connected to the CAN bus after run-up. r8600 = 00000000 hex: No drive recognized. = FFFF0192 hex: Several drives - drive 1 is an Active Line Module, servo drive or vector drive = FFFF0191 hex: Several drives - 1st drive is a Terminal Module = 02010192 hex: 1 Vector drive = 00020192 hex: 1 Servo drive = 01000192 hex: 1 Active Line Module = 00080191 hex: 1 Terminal Module		
Note:	Corresponds to the CANopen object 1000 hex. For each detected drive, the device type is displayed in object 67FF hex + 800 hex * x (x: Drive number 0 ... 7).		
r8601	CAN error register / Error register		
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: - Data type: Unsigned8 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the error register for CANopen. Bit 0: Generic error 0 signal: No error present. 1 signal: Generic error present. Bit 1 ... 3: Not supported (always a 0 signal) Bit 4: Communications error 0 signal: There is no message in the range 8700 ... 8799. 1 signal: There is at least one message (fault or alarm) in the range 8700 ... 8799. Bit 5 ... 6: Not supported (always a 0 signal) Bit 7: Fault outside the range 8700 ... 8799 0 signal: There is no fault outside the range 8700 ... 8799. 1 signal: There is at least one fault outside the range 8700 ... 8799.		
Note:	Corresponds to the CANopen object 1001 hex.		
p8602	CAN SYNC object / SYNC object		
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: C1(1), T Data type: Unsigned32 P-Group: Communications Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0080 hex
Description:	Sets the SYNC object parameter for the following CANopen objects: - 1005 hex: COB-ID		
Note:	SINAMICS operates as SYNC load. COB-ID: CAN object identification		

p8603	CAN COB ID Emergency Message / COB ID EMCY Msg		
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: C1(1), T Data type: Unsigned32 P-Group: Communications Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the COB ID of the emergency message (error telegram). It corresponds to the CANopen objects: - 1014 hex: COB-ID		
Note:	If, when downloading, the preset value 0 is downloaded, then the CANopen preset value 80 hex + Node-ID is automatically set. Online, the value 0 is rejected as, according to the CANopen Standard, COB ID 0 is not permitted here. The changeover of the node ID using the hardware switch at the CU or per software has no effect on the COB-ID EMCY. The saved value remains effective.		
p8604[0...1]	CAN node guarding / Node guarding		
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: T Data type: Unsigned16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the node guarding parameter for the following CANopen objects: - 100C hex: Guard Time - 100D hex: Life Time Factor The life time is derived by multiplying guard time by the life time factor.		
Index:	[0] = Time interval [ms] for new node guarding telegram [1] = Factor for failure of the node guarding telegram		
Dependency:	Only adjustable if heartbeat time = 0 (heartbeat is disabled). Refer to: p8606		
Note:	For p8604[0] = 0 and/or p8604[1] = 0, the node guarding protocol is not used. Either node guarding or heartbeat can be used.		
p8606	CAN Producer Heartbeat Time / Prod Heartb Time		
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: T Data type: Unsigned16 P-Group: - Not for motor type: - Min 0 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [ms]
Description:	Sets the time [ms] to cyclically send heartbeat telegrams. The smallest cycle time is 100 ms. When a 0 is written, then heartbeat telegrams are not sent.		
Dependency:	Only adjustable if guard time = 0 (node guarding disabled). Refer to: p8604		
Note:	Corresponds to the CANopen object 1017 hex. Either node guarding or heartbeat can be used.		

r8607[0...3] CAN Identity Object / Identity objectCU_S (CAN),
CU_S_S150 (CAN)**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** -**P-Group:** Communications**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

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-

-

Description: General device information display.**Index:**
[0] = Vendor ID
[1] = Product code
[2] = Revision number
[3] = Serial number**Note:** Corresponds to the CANopen object 1018 hex.
Re index 3:
The SINAMICS serial number comprises 60 bits. Of these bits, the following are displayed in this index:
Bits 0 ... 19: Consecutive number
Bits 20 ... 23: Production ID
- 0 hex: Development
- 1 hex: P1 unique number
- 2 hex: P2 unique number
- 3 hex: WA unique number
- 9 hex: Pattern
- F hex: All others
Bits 24 ... 27: Month of manufacture (0 means January, B means December)
Bits 28 ... 31: Year of manufacture (0 means 2002)**p8608 CAN Clear Bus Off Error / Clear bus off err**CU_S (CAN),
CU_S_S150 (CAN)**Can be changed:** U, T**Calculated:** -**Access level:** 3**Data type:** Integer16**Dynamic index:** -**Func. diagram:** -**P-Group:** -**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0

1

0

Description: For a bus off error, the CAN bus is restarted with p8608 = 1 after the cause of the error has been removed.**Value:**
0: Inactive
1: Start CAN controller**Note:** This parameter is automatically reset to 0 after start.**p8609[0...1] CAN Error Behaviour / Error behavior**CU_S (CAN),
CU_S_S150 (CAN)**Can be changed:** T**Calculated:** -**Access level:** 3**Data type:** Integer16**Dynamic index:** -**Func. diagram:** -**P-Group:** -**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0

2

1

Description: Sets the behavior of the CAN node referred to the communications error or equipment fault.**Value:**
0: Pre-operational
1: No change
2: Stopped**Index:**
[0] = Behavior for communication errors
[1] = Behavior for device faults**Note:** Corresponds to the CANopen object 1029 hex.

r8610[0...1] CAN First Server SDO / First server SDOCU_S (CAN),
CU_S_S150 (CAN)**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** -**P-Group:** Communications**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

-

Description:

Displays the identifier (client/server and server/client) of the SDO channel.

Index:[0] = Displays the COB ID from client to server
[1] = Displays the COB ID from server to client**Note:**Corresponds to the CANopen object 1200 hex.
SDO: Service Data Object**p8611[0...82] CAN Pre-defined Error Field / Pre_def err field**CU_S (CAN),
CU_S_S150 (CAN)**Can be changed:** U, T**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** -**P-Group:** -**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0000 hex

FFFF 1000 hex

0000 hex

Description:

Displays the Pre-defined Error Field of the CAN node.

It includes the number of all errors that have occurred, the number of errors that have occurred for each drive and the errors according to their history.

The first 16 bits represent the CANopen error code and the second 16 bits the SINAMICS error code.

Index 1 has the same structure - however, the drive object ID is in the second 16 bits instead of the SINAMICS error code.

CANopen error code:

0000 hex: No error present

8110 hex: Alarm A08751 present

8120 hex: Alarm A08752 present

8130 hex: Alarm A08700(F) with alarm value = 2 present

1000 hex: Generic error 1 present (there is at least one fault outside the range 8700 ... 8799)

1001 hex: Generic error 2 present (there is at least one alarm in the range 8700 ... 8799 with the exception of A08751, A08752, A08700)

All drive objects are acknowledged by writing the value 0 to the index 0. As soon as a fault has been acknowledged or an alarm cleared, then it is also cleared from the fault list.

Index:[0] = Number of all faults in the drive unit
[1] = Most recent drive number / fault number
[2] = Number of faults drive 1
[3] = Fault 1/ drive 1
[4] = Fault 2/ drive 1
[5] = Fault 3/ drive 1
[6] = Fault 4/ drive 1
[7] = Fault 5/ drive 1
[8] = Fault 6/ drive 1
[9] = Fault 7/ drive 1
[10] = Fault 8/ drive 1
[11] = Number of faults drive 2
[12] = Fault 1/ drive 2
[13] = Fault 2/ drive 2
[14] = Fault 3/ drive 2
[15] = Fault 4/ drive 2
[16] = Fault 5/ drive 2
[17] = Fault 6/ drive 2
[18] = Fault 7/ drive 2
[19] = Fault 8/ drive 2

[20] = Number of faults drive 3
[21] = Fault 1/ drive 3
[22] = Fault 2/ drive 3
[23] = Fault 3/ drive 3
[24] = Fault 4/ drive 3
[25] = Fault 5/ drive 3
[26] = Fault 6/ drive 3
[27] = Fault 7/ drive 3
[28] = Fault 8/ drive 3
[29] = Number of faults drive 4
[30] = Fault 1/ drive 4
[31] = Fault 2/ drive 4
[32] = Fault 3/ drive 4
[33] = Fault 4/ drive 4
[34] = Fault 5/ drive 4
[35] = Fault 6/ drive 4
[36] = Fault 7/ drive 4
[37] = Fault 8/ drive 4
[38] = Number of faults drive 5
[39] = Fault 1/ drive 5
[40] = Fault 2/ drive 5
[41] = Fault 3/ drive 5
[42] = Fault 4/ drive 5
[43] = Fault 5/ drive 5
[44] = Fault 6/ drive 5
[45] = Fault 7/ drive 5
[46] = Fault 8/ drive 5
[47] = Number of faults drive 6
[48] = Fault 1/ drive 6
[49] = Fault 2/ drive 6
[50] = Fault 3/ drive 6
[51] = Fault 4/ drive 6
[52] = Fault 5/ drive 6
[53] = Fault 6/ drive 6
[54] = Fault 7/ drive 6
[55] = Fault 8/ drive 6
[56] = Number of faults drive 7
[57] = Fault 1/ drive 7
[58] = Fault 2/ drive 7
[59] = Fault 3/ drive 7
[60] = Fault 4/ drive 7
[61] = Fault 5/ drive 7
[62] = Fault 6/ drive 7
[63] = Fault 7/ drive 7
[64] = Fault 8/ drive 7
[65] = Number of faults drive 8
[66] = Fault 1/ drive 8
[67] = Fault 2/ drive 8
[68] = Fault 3/ drive 8
[69] = Fault 4/ drive 8
[70] = Fault 5/ drive 8
[71] = Fault 6/ drive 8
[72] = Fault 7/ drive 8
[73] = Fault 8/ drive 8

[74] = Number of faults Control Unit
 [75] = Fault 1/Control Unit
 [76] = Fault 2/Control Unit
 [77] = Fault 3/Control Unit
 [78] = Fault 4/Control Unit
 [79] = Fault 5/Control Unit
 [80] = Fault 6/Control Unit
 [81] = Fault 7/Control Unit
 [82] = Fault 8/Control Unit

Note: Corresponds to the CANopen object 1003 hex.

p8620 CAN Node-ID / Node ID

CU_S (CAN), CU_S_S150 (CAN)	Can be changed: T	Calculated: -	Access level: 2
	Data type: Unsigned8	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	1	126	126

Description: Display or setting of the CANopen Node ID.
 The Node ID can be set as follows:
 1) Using the address switch on the Control Unit.
 --> p8620 can then only be read and displays the selected Node ID.
 --> A change only becomes effective after a POWER ON.
 --> CANopen Node ID and PROFIBUS address are identical.
 2) Using p8620
 --> only if all of the DIP switches - from S1 to S7 - are either set to ON or OFF.
 --> the Node ID is set as standard to 126.
 --> A change only becomes effective after save and POWER ON.

Note: Every node ID change only becomes effective after a POWER ON.
 The parameter is not influenced by setting the factory setting.
 It is only possible to independently set CANopen node ID and the PROFIBUS address using p0918 and p8620 (pre-requisite: For DIP switches, all of the switches from S1 to S7 are either set to ON or OFF).

p8622 CAN baud rate / Baud rate

CU_S (CAN), CU_S_S150 (CAN)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	7	6

Description: Setting the baud rate for the CAN bus.
 The appropriate bit timings are selected that are defined in p8623 in the associated sub-index.
 Example:

Baud rate = 20 kbit/s --> p8622 = 6 --> associated bit timing is in p8623[6].

Value:

0:	1 Mbit/s
1:	800 kbit/s
2:	500 kbit/s
3:	250 kbit/s
4:	125 kbit/s
5:	50 kbit/s
6:	20 kbit/s
7:	10 kbit/s

Dependency: Refer to: p8623

Note: The parameter is not influenced by setting the factory setting.

p8623[0...7] CAN Bit Timing selection / Bit timing select			
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0000 hex	000F 7FFF hex	[0] 1405 hex
			[1] 1605 hex
			[2] 1C05 hex
			[3] 1C0B hex
			[4] 1C17 hex
			[5] 1C3B hex
			[6] 0002 1C15 hex
			[7] 0004 1C2B hex
Description:	Sets the bit timing for the C_CAN controller to the associated and selected baud rate (p8622). Bits are distributed to the following parameters of the C_CAN controller in p8623[0...7]: Bit 0 ... 5: BRP (Baud Rate Prescaler) Bit 6 ... 7: SJW (Synchronization Jump Width) Bit 8 ... 11: TSEG1 (Time Segment 1, before the sampling point) Bit 12 ... 14: TSEG2 (Time Segment 2, after the sampling point) Bit 15: Reserved Bit 16 ... 19: BRPE (Baud Rate Prescaler Extension) Bit 20 ... 31: Reserved Example: Baud rate = 20 kbit/s --> p8622 = 6 --> associated bit timing is in p8623[6] --> 0001 2FB6		
Recommend.:	Use the factory setting when setting the bit timing.		
Index:	[0] = 1 Mbit/s [1] = 800 kbit/s [2] = 500 kbit/s [3] = 250 kbit/s [4] = 125 kbit/s [5] = 50 kbit/s [6] = 20 kbit/s [7] = 10 kbit/s		
Dependency:	Refer to: p8622		
Note:	The parameter is not influenced by setting the factory setting.		

p8630[0...2]			
CAN virtual objects / Virtual objects			
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: U, T Data type: Unsigned16 P-Group: Communications Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the drive object selection (index 0), the sub-index area (index 1) and the parameter area (index 2) when using virtual objects. This means that it is possible to access all SINAMICS parameters via CAN. Index 0 (drive object selection): 0: Not possible to access virtual CANopen objects 1: Device 2 ... 65535: Drive 1 ... 8 Index 1 (sub-index area): 0: 0 ... 255 1: 256 ... 511 2: 512 ... 767 3: 768 ... 1023 Index 2 (parameter area): 0: 1 ... 9999 1: 10000 ... 19999 2: 20000 ... 29999 3: 30000 ... 39999		
Index:	[0] = Drive object selection [1] = Sub-index range [2] = Parameter range		

p8641			
CAN Abort Connection Option Code / Abort con opt code			
SERVO (CAN), VEC-TOR (CAN)	Can be changed: T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 3
Description:	Sets the drive behavior if a CAN communication error occurs.		
Value:	0: No response 1: OFF1 2: OFF2 3: OFF3		
Dependency:	Refer to: F08700		

r8680[0...36]			
CAN Diagnosis Hardware / Diagnostics HW			
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the register of the CAN controller C_CAN: Register, Message Interface Register and Message Handler Register - referred to the CAN protocol.		

Index:	[0] = Control register
	[1] = Status register
	[2] = Error counter
	[3] = Bit timing register
	[4] = Interrupt register
	[5] = Test register
	[6] = Baud rate prescaler extension register
	[7] = Interface 1 command request register
	[8] = Interface 1 command mask register
	[9] = Interface 1 mask 1 register
	[10] = Interface 1 mask 2 register
	[11] = Interface 1 arbitration 1 register
	[12] = Interface 1 arbitration 2 register
	[13] = Interface 1 message control register
	[14] = Interface 1 data A1 register
	[15] = Interface 1 data A2 register
	[16] = Interface 1 data B1 register
	[17] = Interface 1 data B2 register
	[18] = Interface 2 command request register
	[19] = Interface 2 command mask register
	[20] = Interface 2 mask 1 register
	[21] = Interface 2 mask 2 register
	[22] = Interface 2 arbitration 1 register
	[23] = Interface 2 arbitration 2 register
	[24] = Interface 2 message control register
	[25] = Interface 2 data A1 register
	[26] = Interface 2 data A2 register
	[27] = Interface 2 data B1 register
	[28] = Interface 2 data B2 register
	[29] = Transmission request 1 register
	[30] = Transmission request 2 register
	[31] = New data 1 register
	[32] = New data 2 register
	[33] = Interrupt pending 1 register
	[34] = Interrupt pending 2 register
	[35] = Message valid 1 register
	[36] = Message valid 2 register

Note: A description of the individual registers of the C_CAN controller can be taken from "C_CAN User's Manual".

p8684	CAN NMT state after booting / NMT state boot		
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 4	Max 127	Factory setting 127
Description:	Sets the CANopen NMT state that is effective after booting.		
Value:	4: Stopped 5: Operational 127: Pre-operational		
Dependency:	Refer to: p8685		
Note:	Bootling in the NMT state pre-operational corresponds to the CANopen standard		

p8685 CAN NMT states / NMT statesCU_S (CAN),
CU_S_S150 (CAN)**Can be changed:** C1(1), U, T**Data type:** Integer16**P-Group:** Communications**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

129

Access level: 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

127

Description: Sets and displays the CANopen NMT state.**Value:**

0: Initializing

4: Stopped

5: Operational

127: Pre-operational

128: Reset node

129: Reset Communication

Note: The value 0 (initialization) is only displayed and cannot be set.

p8700[0...1] CAN Receive PDO 1 / Receive PDO 1SERVO (CAN), VEC-
TOR (CAN)**Can be changed:** C1(3), T**Data type:** Unsigned32**P-Group:** Communications**Not for motor type:** -**Min**

0000 hex

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

8000 06DF hex

Access level: 3**Func. diagram:** 9204, 9206**Unit selection:** -**Expert list:** 1**Factory setting**

[0] 8000 06DF hex

[1] 00FE hex

Description: Sets the communication parameters for CANopen Receive Process Data Object 1 (RPDO 1).**Index:**

[0] = PDO COB-ID

[1] = PDO transmission type

Dependency: A valid COB-ID can only be set for the available (existing) channel.

Refer to: p8740, p8741

Note: Corresponds to the CANopen object 1400 hex + 40 hex * x (x: Drive number 0 ... 7).

Transmission types 0, 1, FE and FF can be set.

PDO: Process Data Object

p8701[0...1] CAN Receive PDO 2 / Receive PDO 2SERVO (CAN), VEC-
TOR (CAN)**Can be changed:** C1(3), T**Data type:** Unsigned32**P-Group:** Communications**Not for motor type:** -**Min**

0000 hex

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

8000 06DF hex

Access level: 3**Func. diagram:** 9204, 9206**Unit selection:** -**Expert list:** 1**Factory setting**

[0] 8000 06DF hex

[1] 00FE hex

Description: Sets the communication parameters for CANopen Receive Process Data Object 2 (RPDO 2).**Index:**

[0] = PDO COB-ID

[1] = PDO transmission type

Dependency: A valid COB-ID can only be set for the available (existing) channel.

Refer to: p8740, p8741

Note: Corresponds to the CANopen object 1401 hex + 40 hex * x (x: Drive number 0 ... 7).

Transmission types 0, 1, FE and FF can be set.

PDO: Process Data Object

p8702[0...1] CAN Receive PDO 3 / Receive PDO 3

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max 8000 06DF hex	Factory setting [0] 8000 06DF hex [1] 00FE hex
Description:	Sets the communication parameters for CANopen Receive Process Data Object 3 (RPDO 3).		
Index:	[0] = PDO COB-ID [1] = PDO transmission type		
Dependency:	A valid COB-ID can only be set for the available (existing) channel. Refer to: p8740, p8741		
Note:	Corresponds to the CANopen object 1402 hex + 40 hex * x (x: Drive number 0 ... 7). Transmission types 0, 1, FE and FF can be set. PDO: Process Data Object		

p8703[0...1] CAN Receive PDO 4 / Receive PDO 4

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max 8000 06DF hex	Factory setting [0] 8000 06DF hex [1] 00FE hex
Description:	Sets the communication parameters for CANopen Receive Process Data Object 4 (RPDO 4).		
Index:	[0] = PDO COB-ID [1] = PDO transmission type		
Dependency:	A valid COB-ID can only be set for the available (existing) channel. Refer to: p8740, p8741		
Note:	Corresponds to the CANopen object 1403 hex + 40 hex * x (x: Drive number 0 ... 7). Transmission types 0, 1, FE and FF can be set. PDO: Process Data Object		

p8704[0...1] CAN Receive PDO 5 / Receive PDO 5

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max 8000 06DF hex	Factory setting [0] 8000 06DF hex [1] 00FE hex
Description:	Sets the communication parameters for CANopen Receive Process Data Object 5 (RPDO 5).		
Index:	[0] = PDO COB-ID [1] = PDO transmission type		
Dependency:	A valid COB-ID can only be set for the available (existing) channel. Refer to: p8740, p8741		
Note:	Corresponds to the CANopen object 1404 hex + 40 hex * x (x: Drive number 0 ... 7). Transmission types 0, 1, FE and FF can be set. PDO: Process Data Object		

p8705[0...1] CAN Receive PDO 6 / Receive PDO 6SERVO (CAN), VEC-
TOR (CAN)**Can be changed:** C1(3), T**Data type:** Unsigned32**P-Group:** Communications**Not for motor type:** -**Min**
0000 hex**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**
8000 06DF hex**Access level:** 3**Func. diagram:** 9204**Unit selection:** -**Expert list:** 1**Factory setting**
[0] 8000 06DF hex
[1] 00FE hex**Description:** Sets the communication parameters for CANopen Receive Process Data Object 6 (RPDO 6).**Index:**
[0] = PDO COB-ID
[1] = PDO transmission type**Dependency:** A valid COB-ID can only be set for the available (existing) channel.
Refer to: p8740, p8741**Note:** Corresponds to the CANopen object 1405 hex + 40 hex * x (x: Drive number 0 ... 7).
Transmission types 0, 1, FE and FF can be set.
PDO: Process Data Object**p8706[0...1] CAN Receive PDO 7 / Receive PDO 7**SERVO (CAN), VEC-
TOR (CAN)**Can be changed:** C1(3), T**Data type:** Unsigned32**P-Group:** Communications**Not for motor type:** -**Min**
0000 hex**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**
8000 06DF hex**Access level:** 3**Func. diagram:** 9204**Unit selection:** -**Expert list:** 1**Factory setting**
[0] 8000 06DF hex
[1] 00FE hex**Description:** Sets the communication parameters for CANopen Receive Process Data Object 7 (RPDO 7).**Index:**
[0] = PDO COB-ID
[1] = PDO transmission type**Dependency:** A valid COB-ID can only be set for the available (existing) channel.
Refer to: p8740, p8741**Note:** Corresponds to the CANopen object 1406 hex + 40 hex * x (x: Drive number 0 ... 7).
Transmission types 0, 1, FE and FF can be set.
PDO: Process Data Object**p8707[0...1] CAN Receive PDO 8 / Receive PDO 8**SERVO (CAN), VEC-
TOR (CAN)**Can be changed:** C1(3), T**Data type:** Unsigned32**P-Group:** Communications**Not for motor type:** -**Min**
0000 hex**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**
8000 06DF hex**Access level:** 3**Func. diagram:** 9204**Unit selection:** -**Expert list:** 1**Factory setting**
[0] 8000 06DF hex
[1] 00FE hex**Description:** Sets the communication parameters for CANopen Receive Process Data Object 8 (RPDO 8).**Index:**
[0] = PDO COB-ID
[1] = PDO transmission type**Dependency:** A valid COB-ID can only be set for the available (existing) channel.
Refer to: p8740, p8741**Note:** Corresponds to the CANopen object 1407 hex + 40 hex * x (x: Drive number 0 ... 7).
Transmission types 0, 1, FE and FF can be set.
PDO: Process Data Object

p8710[0...3] CAN Receive Mapping für RPDO 1 / Mapping RPDO 1

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Receive Process Data Object 1 (RPDO 1).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1600 hex + 40 hex * x (x: Drive number 0 ... 7). Dummy mapping not supported. The parameter can only be written online when the associated COB ID in p870x is set as invalid.		

p8711[0...3] CAN Receive Mapping für RPDO 2 / Mapping RPDO 2

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Receive Process Data Object 2 (RPDO 2).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1601 hex + 40 hex * x (x: Drive number 0 ... 7). Dummy mapping not supported. The parameter can only be written online when the associated COB ID in p870x is set as invalid.		

p8712[0...3] CAN Receive Mapping für RPDO 3 / Mapping RPDO 3

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Receive Process Data Object 3 (RPDO 3).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1602 hex + 40 hex * x (x: Drive number 0 ... 7). Dummy mapping not supported. The parameter can only be written online when the associated COB ID in p870x is set as invalid.		

p8713[0...3] CAN Receive Mapping für RPDO 4 / Mapping RPDO 4

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Receive Process Data Object 4 (RPDO 4).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1603 hex + 40 hex * x (x: Drive number 0 ... 7). Dummy mapping not supported. The parameter can only be written online when the associated COB ID in p870x is set as invalid.		

p8714[0...3] CAN Receive Mapping für RPDO 5 / Mapping RPDO 5

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Receive Process Data Object 5 (RPDO 5).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1604 hex + 40 hex * x (x: Drive number 0 ... 7). Dummy mapping not supported. The parameter can only be written online when the associated COB ID in p870x is set as invalid.		

p8715[0...3] CAN Receive Mapping für RPDO 6 / Mapping RPDO 6

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Receive Process Data Object 6 (RPDO 6).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1605 hex + 40 hex * x (x: Drive number 0 ... 7). Dummy mapping not supported. The parameter can only be written online when the associated COB ID in p870x is set as invalid.		

p8716[0...3] CAN Receive Mapping für RPDO 7 / Mapping RPDO 7

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Receive Process Data Object 7 (RPDO 7).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1606 hex + 40 hex * x (x: Drive number 0 ... 7). Dummy mapping not supported. The parameter can only be written online when the associated COB ID in p870x is set as invalid.		

p8717[0...3] CAN Receive Mapping für RPDO 8 / Mapping RPDO 8

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9204
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Receive Process Data Object 8 (RPDO 8).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1607 hex + 40 hex * x (x: Drive number 0 ... 7). Dummy mapping not supported. The parameter can only be written online when the associated COB ID in p870x is set as invalid.		

p8720[0...4] CAN Transmit PDO 1 / Transmit PDO 1

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max C000 06DF hex	Factory setting [0] C000 06DF hex [1] 00FE hex [2] 0000 hex [3] 0000 hex [4] 0000 hex
Description:	Sets the communication parameters for CANopen Transmit Process Data Object 1 (TPDO 1).		
Index:	[0] = PDO COB-ID [1] = PDO transmission type [2] = Inhibit time (in 100 µs) [3] = Reserved [4] = Event timer (in ms)		
Dependency:	A valid COB-ID can only be set for the available (existing) channel. Refer to: p8740, p8741		
Notice:	For inhibit time and even timer, the following apply: A value that is not a multiple integer of CANopen (4 ms) is rounded-off.		

Note: Corresponds to the CANopen object 1800 hex + 40 hex * x (x: Drive number 0 ... 7).
Transmission types 0, 1 ... F0, FE and FF can be set.
PDO: Process Data Object

p8721[0...4] CAN Transmit PDO 2 / Transmit PDO 2

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max C000 06DF hex	Factory setting [0] C000 06DF hex [1] 00FE hex [2] 0000 hex [3] 0000 hex [4] 0000 hex

Description: Sets the communication parameters for CANopen Transmit Process Data Object 2 (TPDO 2).

Index: [0] = PDO COB-ID
[1] = PDO transmission type
[2] = Inhibit time (in 100 µs)
[3] = Reserved
[4] = Event timer (in ms)

Dependency: A valid COB-ID can only be set for the available (existing) channel.
Refer to: p8740, p8741

Notice: For inhibit time and even timer, the following apply:
A value that is not a multiple integer of CANopen (4 ms) is rounded-off.

Note: Corresponds to the CANopen object 1801 hex + 40 hex * x (x: Drive number 0 ... 7).
Transmission types 0, 1 ... F0, FE and FF can be set.
PDO: Process Data Object

p8722[0...4] CAN Transmit PDO 3 / Transmit PDO 3

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max C000 06DF hex	Factory setting [0] C000 06DF hex [1] 00FE hex [2] 0000 hex [3] 0000 hex [4] 0000 hex

Description: Sets the communication parameters for CANopen Transmit Process Data Object 3 (TPDO 3).

Index: [0] = PDO COB-ID
[1] = PDO transmission type
[2] = Inhibit time (in 100 µs)
[3] = Reserved
[4] = Event timer (in ms)

Dependency: A valid COB-ID can only be set for the available (existing) channel.
Refer to: p8740, p8741

Notice: For inhibit time and even timer, the following apply:
A value that is not a multiple integer of CANopen (4 ms) is rounded-off.

Note: Corresponds to the CANopen object 1802 hex + 40 hex * x (x: Drive number 0 ... 7).
Transmission types 0, 1 ... F0, FE and FF can be set.
PDO: Process Data Object

p8723[0...4] CAN Transmit PDO 4 / Transmit PDO 4

SERVO (CAN), VECTOR (CAN)	Can be changed: C1(3), T Data type: Unsigned32 P-Group: Communications Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max C000 06DF hex	Access level: 3 Func. diagram: 9208, 9210 Unit selection: - Expert list: 1 Factory setting [0] C000 06DF hex [1] 00FE hex [2] 0000 hex [3] 0000 hex [4] 0000 hex
Description:	Sets the communication parameters for CANopen Transmit Process Data Object 4 (TPDO 4).		
Index:	[0] = PDO COB-ID [1] = PDO transmission type [2] = Inhibit time (in 100 µs) [3] = Reserved [4] = Event timer (in ms)		
Dependency:	A valid COB-ID can only be set for the available (existing) channel. Refer to: p8740, p8741		
Notice:	For inhibit time and even timer, the following apply: A value that is not a multiple integer of CANopen (4 ms) is rounded-off.		
Note:	Corresponds to the CANopen object 1803 hex + 40 hex * x (x: Drive number 0 ... 7). Transmission types 0, 1 ... F0, FE and FF can be set. PDO: Process Data Object		

p8724[0...4] CAN Transmit PDO 5 / Transmit PDO 5

SERVO (CAN), VECTOR (CAN)	Can be changed: C1(3), T Data type: Unsigned32 P-Group: Communications Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max C000 06DF hex	Access level: 3 Func. diagram: 9208 Unit selection: - Expert list: 1 Factory setting [0] C000 06DF hex [1] 00FE hex [2] 0000 hex [3] 0000 hex [4] 0000 hex
Description:	Sets the communication parameters for CANopen Transmit Process Data Object 5 (TPDO 5).		
Index:	[0] = PDO COB-ID [1] = PDO transmission type [2] = Inhibit time (in 100 µs) [3] = Reserved [4] = Event timer (in ms)		
Dependency:	A valid COB-ID can only be set for the available (existing) channel. Refer to: p8740, p8741		
Notice:	For inhibit time and even timer, the following apply: A value that is not a multiple integer of CANopen (4 ms) is rounded-off.		
Note:	Corresponds to the CANopen object 1804 hex + 40 hex * x (x: Drive number 0 ... 7). Transmission types 0, 1 ... F0, FE and FF can be set. PDO: Process Data Object		

p8725[0...4] CAN Transmit PDO 6 / Transmit PDO 6SERVO (CAN), VEC-
TOR (CAN)**Can be changed:** C1(3), T**Data type:** Unsigned32**P-Group:** Communications**Not for motor type:** -**Min**
0000 hex**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**
C000 06DF hex**Access level:** 3**Func. diagram:** 9208**Unit selection:** -**Expert list:** 1**Factory setting**
[0] C000 06DF hex
[1] 00FE hex
[2] 0000 hex
[3] 0000 hex
[4] 0000 hex**Description:** Sets the communication parameters for CANopen Transmit Process Data Object 6 (TPDO 6).**Index:**
[0] = PDO COB-ID
[1] = PDO transmission type
[2] = Inhibit time (in 100 µs)
[3] = Reserved
[4] = Event timer (in ms)**Dependency:** A valid COB-ID can only be set for the available (existing) channel.
Refer to: p8740, p8741**Notice:** For inhibit time and even timer, the following apply:
A value that is not a multiple integer of CANopen (4 ms) is rounded-off.**Note:** Corresponds to the CANopen object 1805 hex + 40 hex * x (x: Drive number 0 ... 7).
Transmission types 0, 1 ... F0, FE and FF can be set.
PDO: Process Data Object

p8726[0...4] CAN Transmit PDO 7 / Transmit PDO 7SERVO (CAN), VEC-
TOR (CAN)**Can be changed:** C1(3), T**Data type:** Unsigned32**P-Group:** Communications**Not for motor type:** -**Min**
0000 hex**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**
C000 06DF hex**Access level:** 3**Func. diagram:** 9208**Unit selection:** -**Expert list:** 1**Factory setting**
[0] C000 06DF hex
[1] 00FE hex
[2] 0000 hex
[3] 0000 hex
[4] 0000 hex**Description:** Sets the communication parameters for CANopen Transmit Process Data Object 7 (TPDO 7).**Index:**
[0] = PDO COB-ID
[1] = PDO transmission type
[2] = Inhibit time (in 100 µs)
[3] = Reserved
[4] = Event timer (in ms)**Dependency:** A valid COB-ID can only be set for the available (existing) channel.
Refer to: p8740, p8741**Notice:** For inhibit time and even timer, the following apply:
A value that is not a multiple integer of CANopen (4 ms) is rounded-off.**Note:** Corresponds to the CANopen object 1806 hex + 40 hex * x (x: Drive number 0 ... 7).
Transmission types 0, 1 ... F0, FE and FF can be set.
PDO: Process Data Object

p8727[0...4] CAN Transmit PDO 8 / Transmit PDO 8

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max C000 06DF hex	Factory setting [0] C000 06DF hex [1] 00FE hex [2] 0000 hex [3] 0000 hex [4] 0000 hex

Description: Sets the communication parameters for CANopen Transmit Process Data Object 8 (TPDO 8).

Index:
[0] = PDO COB-ID
[1] = PDO transmission type
[2] = Inhibit time (in 100 µs)
[3] = Reserved
[4] = Event timer (in ms)

Dependency: A valid COB-ID can only be set for the available (existing) channel.
Refer to: p8740, p8741

Notice: For inhibit time and even timer, the following apply:
A value that is not a multiple integer of CANopen (4 ms) is rounded-off.

Note: Corresponds to the CANopen object 1807 hex + 40 hex * x (x: Drive number 0 ... 7).
Transmission types 0, 1 ... F0, FE and FF can be set.
PDO: Process Data Object

p8730[0...3] CAN Transmit Mapping for TPDO 1 / Mapping TPDO 1

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex

Description: Sets the mapping parameters for CANopen Transmit Process Data Object 1 (TPDO 1).

Index:
[0] = Mapped object 1
[1] = Mapped object 2
[2] = Mapped object 3
[3] = Mapped object 4

Note: Corresponds to the CANopen object 1A00 hex + 40 hex * x (x: Drive number 0 ... 7).
The parameter can only be written online when the associated COB ID in p872x is set as invalid.

p8731[0...3] CAN Transmit Mapping for TPDO 2 / Mapping TPDO 2

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex

Description: Sets the mapping parameters for CANopen Transmit Process Data Object 2 (TPDO 2).

Index:
[0] = Mapped object 1
[1] = Mapped object 2
[2] = Mapped object 3
[3] = Mapped object 4

Note: Corresponds to the CANopen object 1A01 hex + 40 hex * x (x: Drive number 0 ... 7).
The parameter can only be written online when the associated COB ID in p872x is set as invalid.

p8732[0...3] CAN Transmit Mapping for TPDO 3 / Mapping TPDO 3

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex

Description: Sets the mapping parameters for CANopen Transmit Process Data Object 3 (TPDO 3).

Index: [0] = Mapped object 1
[1] = Mapped object 2
[2] = Mapped object 3
[3] = Mapped object 4

Note: Corresponds to the CANopen object 1A02 hex + 40 hex * x (x: Drive number 0 ... 7).
The parameter can only be written online when the associated COB ID in p872x is set as invalid.

p8733[0...3] CAN Transmit Mapping for TPDO 4 / Mapping TPDO 4

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex

Description: Sets the mapping parameters for CANopen Transmit Process Data Object 4 (TPDO 4).

Index: [0] = Mapped object 1
[1] = Mapped object 2
[2] = Mapped object 3
[3] = Mapped object 4

Note: Corresponds to the CANopen object 1A03 hex + 40 hex * x (x: Drive number 0 ... 7).
The parameter can only be written online when the associated COB ID in p872x is set as invalid.

p8734[0...3] CAN Transmit Mapping for TPDO 5 / Mapping TPDO 5

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex

Description: Sets the mapping parameters for CANopen Transmit Process Data Object 5 (TPDO 5).

Index: [0] = Mapped object 1
[1] = Mapped object 2
[2] = Mapped object 3
[3] = Mapped object 4

Note: Corresponds to the CANopen object 1A04 hex + 40 hex * x (x: Drive number 0 ... 7).
The parameter can only be written online when the associated COB ID in p872x is set as invalid.

p8735[0...3] CAN Transmit Mapping for TPDO 6 / Mapping TPDO 6

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Transmit Process Data Object 6 (TPDO 6).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1A05 hex + 40 hex * x (x: Drive number 0 ... 7). The parameter can only be written online when the associated COB ID in p872x is set as invalid.		

p8736[0...3] CAN Transmit Mapping for TPDO 7 / Mapping TPDO 7

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Transmit Process Data Object 7 (TPDO 7).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1A06 hex + 40 hex * x (x: Drive number 0 ... 7). The parameter can only be written online when the associated COB ID in p872x is set as invalid.		

p8737[0...3] CAN Transmit Mapping for TPDO 8 / Mapping TPDO 8

SERVO (CAN), VEC-TOR (CAN)	Can be changed: C1(3), T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 9208
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex
Description:	Sets the mapping parameters for CANopen Transmit Process Data Object 8 (TPDO 8).		
Index:	[0] = Mapped object 1 [1] = Mapped object 2 [2] = Mapped object 3 [3] = Mapped object 4		
Note:	Corresponds to the CANopen object 1A07 hex + 40 hex * x (x: Drive number 0 ... 7). The parameter can only be written online when the associated COB ID in p872x is set as invalid.		

p8740[0...23] CAN channel distribution / Chann assign.			
CU_S (CAN), CU_S_S150 (CAN)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	8	[0] 0
			[1] 0
			[2] 0
			[3] 0
			[4] 0
			[5] 0
			[6] 0
			[7] 0
			[8] 0
			[9] 0
			[10] 0
			[11] 0
			[12] 0
			[13] 0
			[14] 0
			[15] 0
			[16] 0
			[17] 0
			[18] 0
			[19] 0
			[20] 0
			[21] 0
			[22] 0
			[23] 0
Description:	Sets the number of channels for receive PDOs and Transmit PDOs. To define the number of channels of a drive, there are 3 indices (3 * n indices, n = 8). Index 0: Number of channels for receive PDOs (1st drive) Index 1: Number of channels for transmit PDOs (1st drive) Index 2: Reserved Correspondingly, indices 3 to 5 are valid for the 2nd drive, etc.		
Index:	[0] = Number of channels for receive PDOs (drive 1) [1] = Number of channels for transmit PDOs (drive 1) [2] = Reserved [3] = Number of channels for receive PDOs (drive 2) [4] = Number of channels for transmit PDOs (drive 2) [5] = Reserved [6] = Number of channels for receive PDOs (drive 3) [7] = Number of channels for transmit PDOs (drive 3) [8] = Reserved [9] = Number of channels for receive PDOs (drive 4) [10] = Number of channels for transmit PDOs (drive 4) [11] = Reserved [12] = Number of channels for receive PDOs (drive 5) [13] = Number of channels for transmit PDOs (drive 5) [14] = Reserved [15] = Number of channels for receive PDOs (drive 6) [16] = Number of channels for transmit PDOs (drive 6) [17] = Reserved		

[18] = Number of channels for receive PDOs (drive 7)
 [19] = Number of channels for transmit PDOs (drive 7)
 [20] = Reserved
 [21] = Number of channels for receive PDOs (drive 8)
 [22] = Number of channels for transmit PDOs (drive 8)
 [23] = Reserved

Dependency: Refer to: p8741

Note: Channel assignment not yet in effect. To acknowledge set p8741 = 1.

p8741

CAN PDO configuration acknowledgement / PDO config ackn

CU_S (CAN),
CU_S_S150 (CAN)

Can be changed: T
Data type: Integer16
P-Group: -
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: -
Unit selection: -
Expert list: 1
Factory setting
 0

Description: Acknowledges the channel distribution selection made (p8740) and the setting of the predefined connection sets (p8744).

Value:
 0: Inactive
 1: Acknowledge configuration

Dependency: Refer to: p8740, p8744

r8742

CAN number of free RPDO channels / Qty free RPDO

CU_S (CAN),
CU_S_S150 (CAN)

Can be changed: -
Data type: Unsigned16
P-Group: Communications
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: -
Unit selection: -
Expert list: 1
Factory setting
 -

Description: Displays the RPDO channels that are still available.

Dependency: Refer to: p8741

Note: The display is updated after acknowledging the configuration (p8741 = 1).
 The number only includes the RPDO channels in p8740 for which an axis is present in the topology!
 RPDO: Receive Process Data Object

r8743[0...7]

CAN assignment drive/drive ID / Drive ID

CU_S (CAN),
CU_S_S150 (CAN)

Can be changed: -
Data type: Unsigned16
P-Group: -
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: -
Unit selection: -
Expert list: 1
Factory setting
 -

Description: Displays the drive ID associated with each drive.

Index:
 [0] = Drive ID for 1st drive
 [1] = Drive ID for 2nd drive
 [2] = Drive ID for 3rd drive
 [3] = Drive ID for 4th drive
 [4] = Drive ID for 5th drive
 [5] = Drive ID for 6th drive
 [6] = Drive ID for 7th drive
 [7] = Drive ID for 8th drive

p8744 CAN PDO mapping configuration / PDO Mapping conf.

SERVO (CAN), VEC-	Can be changed: C2, T	Calculated: -	Access level: 2
TOR (CAN)	Data type: Integer16	Dynamic index: -	Func. diagram: 9204, 9206, 9208, 9210
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1	Max 2	Factory setting 2
Description:	Selector switch for the PDO mapping. Sets the mapping for download or in the online mode after acknowledging with p8741.		
Value:	1: Predefined Connection Set 2: Free PDO Mapping		

r8750[0...15] CAN mapped 16-bit receive objects / RPDO 16 mapped

SERVO (CAN), VEC-	Can be changed: -	Calculated: -	Access level: 3
TOR (CAN)	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the mapped 16-bit receive CANopen objects in the process data buffer. Example: If, e.g. the control word is mapped in an RPDO, then r8750 indicates the position of the control word in the process data buffer.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12 [12] = PZD 13 [13] = PZD 14 [14] = PZD 15 [15] = PZD 16		

r8751[0...15] CAN mapped 16-bit transmit objects / TPDO 16 mapped

SERVO (CAN), VEC-	Can be changed: -	Calculated: -	Access level: 3
TOR (CAN)	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays mapped 16-bit transmit CANopen objects in the process data buffer.

Index:

- [0] = PZD 1
- [1] = PZD 2
- [2] = PZD 3
- [3] = PZD 4
- [4] = PZD 5
- [5] = PZD 6
- [6] = PZD 7
- [7] = PZD 8
- [8] = PZD 9
- [9] = PZD 10
- [10] = PZD 11
- [11] = PZD 12
- [12] = PZD 13
- [13] = PZD 14
- [14] = PZD 15
- [15] = PZD 16

Dependency: Refer to: r8750

r8760[0...14] CAN mapped 32-bit receive objects / RPDO 32 mapped

SERVO (CAN), VEC-	Can be changed: -	Calculated: -	Access level: 3
TOR (CAN)	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the mapped 32-bit receive CANopen objects in the process data buffer.

Index:

- [0] = PZD 1 + 2
- [1] = PZD 2 + 3
- [2] = PZD 3 + 4
- [3] = PZD 4 + 5
- [4] = PZD 5 + 6
- [5] = PZD 6 + 7
- [6] = PZD 7 + 8
- [7] = PZD 8 + 9
- [8] = PZD 9 + 10
- [9] = PZD 10 + 11
- [10] = PZD 11 + 12
- [11] = PZD 12 + 13
- [12] = PZD 13 + 14
- [13] = PZD 14 + 15
- [14] = PZD 15 + 16

r8761[0...14] CAN mapped 32-bit transmit objects / TPDO 32 mapped

SERVO (CAN), VECTOR (CAN)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays mapped 32-bit transmit CANopen objects in the process data buffer.

Index:

- [0] = PZD 1 + 2
- [1] = PZD 2 + 3
- [2] = PZD 3 + 4
- [3] = PZD 4 + 5
- [4] = PZD 5 + 6
- [5] = PZD 6 + 7
- [6] = PZD 7 + 8
- [7] = PZD 8 + 9
- [8] = PZD 9 + 10
- [9] = PZD 10 + 11
- [10] = PZD 11 + 12
- [11] = PZD 12 + 13
- [12] = PZD 13 + 14
- [13] = PZD 14 + 15
- [14] = PZD 15 + 16

r8784 CO: CAN status word / Status word

SERVO (CAN), VECTOR (CAN)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 8010
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the CANopen status word.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Ready for sw on	Yes	No	-
	01	Ready	Yes	No	-
	02	Operation enabled	Yes	No	-
	03	Fault present	Yes	No	-
	04	No coasting active	Yes	No	-
	05	No Quick Stop active	Yes	No	-
	06	Switching on inhibited active	Yes	No	-
	07	Alarm present	Yes	No	-
	08	Can be freely interconnected (BI: p8785)	High	Low	-
	09	Control request	Yes	No	-
	10	Target reached	Yes	No	-
	11	Torque limit reached	Yes	No	-
	12	Velocity equal to zero	Yes	No	-
	14	Can be freely interconnected (BI: p8786)	High	Low	-
	15	Can be freely interconnected (BI: p8787)	High	Low	-

Note: Corresponds to the CANopen object 6041 hex + 800 hex * x (x: Drive number 0 ... 7).

Re bit 10:

When the ramp-function generator is activated, the interconnection from CI: p2151 = r1119 can be changed, so that to evaluate bit 10, the setpoint can be retrieved (taken) from in front of the ramp-function generator.

Re bit 10, 12:

When braking, the two bits must indicate the same state. This is the reason that the following parameters must be set the same:

p2161 (speed threshold value 3, for r2199.0) = p2163 (speed threshold value 4, for r2197.7)

p2150 (hysteresis speed 3, for r2199.0) = p2164 (hysteresis speed 4, for r2197.7)

p8785 BI: CAN status word bit 8 / Status word bit 8

SERVO (CAN), VEC-	Can be changed: T	Calculated: -	Access level: 3
TOR (CAN)	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Binector input for CANopen status word bit 8.		
Dependency:	Refer to: r8784		

p8786 BI: CAN status word bit 14 / Status word bit 14

SERVO (CAN), VEC-	Can be changed: T	Calculated: -	Access level: 3
TOR (CAN)	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Binector input for CANopen status word bit 14.		
Dependency:	Refer to: r8784		

p8787 BI: CAN status word bit 15 / Status word bit 15

SERVO (CAN), VEC-	Can be changed: T	Calculated: -	Access level: 3
TOR (CAN)	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Binector input for CANopen status word bit 15.		
Dependency:	Refer to: r8784		

p8790 CAN control word - auto interconnection / STW interc auto

SERVO (CAN), VEC-	Can be changed: C1(3), T	Calculated: -	Access level: 3
TOR (CAN)	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	0
Description:	Sets the automatic BICO interconnection of the CANopen control word.		
Value:	0: No interconn 1: Interconnection		
Dependency:	Refer to: r2050, r2090, r2091, r2092, r2093, r8750, r8795, r8850, r8890, r8891, r8892, r8893		

Note: The following BICO interconnections are automatically established if the CANopen control word is mapped at one of the locations x = 0 ... 3 in the receive process data buffer.
 For SINAMICS S120 with CBC10, the PZD interface IF2 is used:
 BI: p0840.0 = r889x.0
 BI: p0844.0 = r889x.1
 BI: p0848.0 = r889x.2
 BI: p0852.0 = r889x.3
 BI: p2103.0 = r889x.7
 For SINAMICS S110, the PZD interface IF1 is used:
 BI: p0840.0 = r209x.0
 BI: p0844.0 = r209x.1
 BI: p0848.0 = r209x.2
 BI: p0852.0 = r209x.3
 BI: p2103.0 = r209x.7
 The write access is rejected if a CANopen control word is not mapped at one of these locations.
 This also causes the project download of the commissioning software to be canceled.

r8795 CAN control word / Control word

SERVO (CAN), VECTOR (CAN)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
-	-	-	

Description: Access to the CANopen control word using SDO transfer.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	ON/OFF1	Yes	No	-
	01	Do not activate coast down	Yes	No	-
	02	Do not activate a Quick Stop	Yes	No	-
	03	Operation enable	Yes	No	-
	07	Acknowledge fault	Yes	No	-
	11	Freely interconn	High	Low	-
	12	Freely interconn	High	Low	-
	13	Freely interconn	High	Low	-
	14	Freely interconn	High	Low	-
	15	Freely interconn	High	Low	-

Dependency: Refer to: p8790

Note: Corresponds to the CANopen object 6040 hex + 800 hex * x (x: Drive number 0 ... 7).

r8796 CAN Target Velocity / Target velocity

SERVO (CAN), VECTOR (CAN)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
-	-	-	

Description: Access to the CANopen object target velocity using the SDO transfer.

The value is displayed in increments/second as standard.

Note: Corresponds to the CANopen object 60FF hex + 800 hex * x (x: Drive number 0 ... 7).

The displayed value is calculated as follows:

$r8796 = n_set [RPM] / 60 \text{ s} * p0408 * 2^{p0418} * p8798[1] / p8798[0]$

r8797 CAN Target Torque / Target torque

SERVO (CAN), VEC-TOR (CAN)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Access to the CANopen object target torque using SDO transfer.
The value is displayed as per mille (1/1000) as standard.

Note: Corresponds to the CANopen object 6071 hex + 800 hex * x (x: Drive number 0 ... 7).
The displayed value is calculated as follows:
r8797 [per mille] = M_set [Nm] / p0333 [Nm] * 1000

p8798[0...1] CAN speed conversion factor / n_conv_factor

SERVO (CAN), VEC-TOR (CAN)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	1	4294967295	1

Description: The factor converts the required velocity units into the internal velocity units (V/s).
With the factor setting, for CANopen, the velocity units are increments/second.
The parameter corresponds to the CANopen object 6094 hex.
The internal velocity is calculated as follows:

$$n_set_internal = object\ 6094.1 / object\ 6094.2 * 1/(p0408 * 2^{p0418}) * n_set_bus$$

Index: [0] = Counter
[1] = Denominator

p8835 CBE20 firmware selection / CBE20 FW sel

CU_S (PROFINET), CU_S_CU310PN, CU_S_S150 (PROFI- NET)	Can be changed: C1(1)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	1	99	1

Description: Selects the firmware version for the PROFINET COMM BOARD CBE20.

Value: 1: PROFINET V2.2
2: PN gate
3: SINAMICS Link
99: Customer-specific from the OEM directory

Note: A change only becomes effective after a POWER ON.
The parameter is not influenced by setting the factory setting.

p8836	SINAMICS Link address / SINAMICS Link add		
CU_S (PROFINET), CU_S_CU310PN, CU_S_S150 (PROFI- NET)	Can be changed: C1(1)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 64	Factory setting 0
Description:	Selects the node address for the SINAMICS Link on the Communication Board Ethernet 20 (CBE20). p8836 = 0: SINAMICS Link deactivated p8836 = 1 ... 64: SINAMICS Link node address		
Dependency:	Refer to: p8835		
Note:	SINAMICS Link requires that the appropriate CBE20 firmware version is selected (p8835 = 3). A change only becomes effective after a POWER ON. The parameter is not influenced by setting the factory setting.		

p8839[0...1]	PZD interface hardware assignment / PZD IF HW assign		
CU_S, CU_S_S150	Can be changed: C1(1)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 99	Factory setting 99
Description:	Assignment of the hardware for cyclic communications via PZD interface 1 and interface 2.		
Value:	0: Inactive 1: Communication interface integrated in the Control Unit 2: Option module 99: Automatic		
Index:	[0] = IF1 [1] = IF2		
Note:	For value = 99 (automatic) the following applies: - if neither a COMM BOARD nor CAN is inserted, then PROFIBUS communicates via IF1. - if a CBE20 is inserted, then PROFINET communicates via IF1. PROFIBUS PZD are deactivated. - CAN always communicates via IF2. For a value not equal to 99 (automatic) the following applies: - both indices must be set to a number not equal to 99 (automatic). A new setting only becomes effective after POWER ON, reset or download.		

p8840	COMM BOARD monitoring time / CB t_monit		
CU_S (COMM BOARD, PROFI- NET), CU_S_CU310PN, CU_S_S150 (COMM BOARD, PROFINET)	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0 [ms]	Max 65535000 [ms]	Factory setting 20 [ms]
Description:	Sets the monitoring time to monitor the process data received via COMM BOARD. If no process data is received within this time, an appropriate message is output.		
Dependency:	Refer to: F08501		
Note:	Value = 0: Monitoring is deactivated.		

p8841[0...39]	COMM BOARD send configuration data / CB s config_dat		
CU_S (COMM BOARD, PROFINET), CU_S_CU310PN, CU_S_S150 (COMM BOARD, PROFINET)	Can be changed: U, T Data type: Unsigned16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 65535	Factory setting 0
p8842	COMM BOARD start configuration / CB config start		
CU_S (COMM BOARD, PROFINET), CU_S_CU310PN, CU_S_S150 (COMM BOARD, PROFINET)	Can be changed: U, T Data type: Unsigned16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 1	Factory setting 0
p8848	IF2 PZD sampling time / IF2 PZD t_sample		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(3) Data type: FloatingPoint32 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min 1.00 [ms]	Max 16.00 [ms]	Factory setting 4.00 [ms]
Description:	Sets the sampling time for the cyclic interface 2 (IF2).		
r8849[0...139]	COMM BOARD receive configuration data / CB r config_dat		
CU_S (COMM BOARD, PROFINET), CU_S_CU310PN, CU_S_S150 (COMM BOARD, PROFINET)	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min -	Max -	Factory setting -
r8850[0...4]	CO: IF2 PZD receive word / IF2 PZD recv word		
A_INF, B_INF, S_INF	Can be changed: - Data type: Integer16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: 4000H	Access level: 3 Func. diagram: 2491 Unit selection: - Expert list: 1
	Min -	Max -	Factory setting -
Description:	Connector output for interconnecting the PZD (setpoints) received via interface 2 in the word format.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5		
Note:	IF2: Interface 2 PZD1 to PZD2 are displayed bit-serially in r8890 to r8891.		

r8850[0...3]	CO: IF2 PZD receive word / IF2 PZD rcv word		
ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 2485, 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Connector output for interconnecting the PZD (setpoints) received via interface 2 in the word format.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4		
Dependency:	Refer to: r8860, r8890, r8891, r8892, r8893		
Note:	IF2: Interface 2 PZD1 to PZD4 are displayed bit-serially in r8890 to r8893.		

r8850[0...15]	CO: IF2 PZD receive word / IF2 PZD rcv word		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 2485, 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Connector output for interconnecting the PZD (setpoints) received via interface 2 in the word format.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12 [12] = PZD 13 [13] = PZD 14 [14] = PZD 15 [15] = PZD 16		
Dependency:	Refer to: r8860, r8890, r8891, r8892, r8893		
Note:	IF2: Interface 2 PZD1 to PZD4 are displayed bit-serially in r8890 to r8893.		

p8851[0...7]	CI: IF2 PZD send word / IF2 PZD send word		
A_INF, B_INF, S_INF	Can be changed: U, T Data type: Unsigned32 / Integer16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: 4000H Max -	Access level: 3 Func. diagram: 2493, 9210 Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the PZD (actual values) to be sent via interface 2 in the word format.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8		
Note:	IF2: Interface 2		

p8851[0...11]	CI: IF2 PZD send word / IF2 PZD send word		
ENCODER	Can be changed: U, T Data type: Unsigned32 / Integer16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: 4000H Max -	Access level: 3 Func. diagram: 2487, 9208 Unit selection: - Expert list: 1 Factory setting 0
Description:	Selects the PZD (actual values) to be sent via interface 2 in the word format.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12		
Dependency:	Refer to: p8861		
Note:	IF2: Interface 2		

p8851[0...15]	CI: IF2 PZD send word / IF2 PZD send word		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer16	Dynamic index: -	Func. diagram: 2487, 9208
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Selects the PZD (actual values) to be sent via interface 2 in the word format.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12 [12] = PZD 13 [13] = PZD 14 [14] = PZD 15 [15] = PZD 16		
Dependency:	Refer to: p8861		
Note:	IF2: Interface 2		

r8853[0...7]	IF2 diagnostics PZD send / IF2 diag PZD send				
A_INF, B_INF, S_INF	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2493		
	P-Group: Communications	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Displays the sent PZD (actual values) sent via interface 2.				
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-

11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-

Note: IF2: Interface 2

r8853[0...11] IF2 diagnostics PZD send / IF2 diag PZD send

ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2487, 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the sent PZD (actual values) sent via interface 2.

Index:
 [0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5
 [5] = PZD 6
 [6] = PZD 7
 [7] = PZD 8
 [8] = PZD 9
 [9] = PZD 10
 [10] = PZD 11
 [11] = PZD 12

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Dependency: Refer to: p8851, p8861

Note: IF2: Interface 2

r8853[0...15] IF2 diagnostics PZD send / IF2 diag PZD send

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2487, 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the sent PZD (actual values) sent via interface 2.

Index:

- [0] = PZD 1
- [1] = PZD 2
- [2] = PZD 3
- [3] = PZD 4
- [4] = PZD 5
- [5] = PZD 6
- [6] = PZD 7
- [7] = PZD 8
- [8] = PZD 9
- [9] = PZD 10
- [10] = PZD 11
- [11] = PZD 12
- [12] = PZD 13
- [13] = PZD 14
- [14] = PZD 15
- [15] = PZD 16

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Dependency: Refer to: p8851, p8861

Note: IF2: Interface 2

r8854 COMM BOARD state / CB state

CU_S (COMM BOARD, PROFINET), CU_S_CU310PN, CU_S_S150 (COMM BOARD, PROFINET)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 255	Factory setting -
Description:	Status display for COMM BOARD.		
Value:	0: No initialization 1: Fatal fault 2: Initialization 3: Send configuration 4: Receive configuration 5: Non-cyclic communication 6: Cyclic communications but no setpoints (stop/no clock cycle) 255: Cyclic communication		

r8858[0...39] COMM BOARD read diagnostics channel / CB diag_chan read

CU_S (COMM BOARD, PROFINET), CU_S_CU310PN, CU_S_S150 (COMM BOARD, PROFINET)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

r8859[0...7] COMM BOARD identification Data / CB Ident_data

CU_S (COMM BOARD, PROFINET), CU_S_CU310PN, CU_S_S150 (COMM BOARD, PROFINET)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

r8860[0...2] CO: IF2 PZD receive double word / IF2 PZD recv DW

ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: 2485, 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Connector output for interconnecting the PZD (setpoints) received via interface 2 in the double word format.		
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4		
Dependency:	Refer to: r8850		
Note:	IF2: Interface 2		

r8860[0...14]	CO: IF2 PZD receive double word / IF2 PZD recv DW		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: 2485, 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Connector output for interconnecting the PZD (setpoints) received via interface 2 in the double word format.		
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4 [3] = PZD 4 + 5 [4] = PZD 5 + 6 [5] = PZD 6 + 7 [6] = PZD 7 + 8 [7] = PZD 8 + 9 [8] = PZD 9 + 10 [9] = PZD 10 + 11 [10] = PZD 11 + 12 [11] = PZD 12 + 13 [12] = PZD 13 + 14 [13] = PZD 14 + 15 [14] = PZD 15 + 16		
Dependency:	Refer to: r8850		
Notice:	A maximum of 4 indices of the "trace" function can be used.		
Note:	IF2: Interface 2		
p8861[0...10]	CI: IF2 PZD send double word / IF2 PZD send DW		
ENCODER	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 2487, 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Selects the PZD (actual values) to be sent via interface 2 in the double word format.		
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4 [3] = PZD 4 + 5 [4] = PZD 5 + 6 [5] = PZD 6 + 7 [6] = PZD 7 + 8 [7] = PZD 8 + 9 [8] = PZD 9 + 10 [9] = PZD 10 + 11 [10] = PZD 11 + 12		
Dependency:	Refer to: p8851		
Note:	IF2: Interface 2		

p8861[0...14]	CI: IF2 PZD send double word / IF2 PZD send DW		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer32	Dynamic index: -	Func. diagram: 2487, 9208, 9210
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min	Max	Factory setting
	-	-	0
Description:	Selects the PZD (actual values) to be sent via interface 2 in the double word format.		
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4 [3] = PZD 4 + 5 [4] = PZD 5 + 6 [5] = PZD 6 + 7 [6] = PZD 7 + 8 [7] = PZD 8 + 9 [8] = PZD 9 + 10 [9] = PZD 10 + 11 [10] = PZD 11 + 12 [11] = PZD 12 + 13 [12] = PZD 13 + 14 [13] = PZD 14 + 15 [14] = PZD 15 + 16		
Dependency:	Refer to: p8851		
Note:	IF2: Interface 2		

r8863[0...10]		IF2 diagnostics PZD send double word / IF2 diag send DW			
ENCODER	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned32		Dynamic index: -		Func. diagram: 2487
	P-Group: Communications		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		-
Description:		Displays the PZD sent via interface 2 (actual values) with double word format.			
Index:		[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4 [3] = PZD 4 + 5 [4] = PZD 5 + 6 [5] = PZD 6 + 7 [6] = PZD 7 + 8 [7] = PZD 8 + 9 [8] = PZD 9 + 10 [9] = PZD 10 + 11 [10] = PZD 11 + 12			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-

10	Bit 10	On	Off	-
11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-
16	Bit 16	On	Off	-
17	Bit 17	On	Off	-
18	Bit 18	On	Off	-
19	Bit 19	On	Off	-
20	Bit 20	On	Off	-
21	Bit 21	On	Off	-
22	Bit 22	On	Off	-
23	Bit 23	On	Off	-
24	Bit 24	On	Off	-
25	Bit 25	On	Off	-
26	Bit 26	On	Off	-
27	Bit 27	On	Off	-
28	Bit 28	On	Off	-
29	Bit 29	On	Off	-
30	Bit 30	On	Off	-
31	Bit 31	On	Off	-

Notice: A maximum of 4 indices of the "trace" function can be used.

Note: IF2: Interface 2

r8863[0...14] IF2 diagnostics PZD send double word / IF2 diag send DW

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2487
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the PZD sent via interface 2 (actual values) with double word format.

Index:

- [0] = PZD 1 + 2
- [1] = PZD 2 + 3
- [2] = PZD 3 + 4
- [3] = PZD 4 + 5
- [4] = PZD 5 + 6
- [5] = PZD 6 + 7
- [6] = PZD 7 + 8
- [7] = PZD 8 + 9
- [8] = PZD 9 + 10
- [9] = PZD 10 + 11
- [10] = PZD 11 + 12
- [11] = PZD 12 + 13
- [12] = PZD 13 + 14
- [13] = PZD 14 + 15
- [14] = PZD 15 + 16

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-

11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-
16	Bit 16	On	Off	-
17	Bit 17	On	Off	-
18	Bit 18	On	Off	-
19	Bit 19	On	Off	-
20	Bit 20	On	Off	-
21	Bit 21	On	Off	-
22	Bit 22	On	Off	-
23	Bit 23	On	Off	-
24	Bit 24	On	Off	-
25	Bit 25	On	Off	-
26	Bit 26	On	Off	-
27	Bit 27	On	Off	-
28	Bit 28	On	Off	-
29	Bit 29	On	Off	-
30	Bit 30	On	Off	-
31	Bit 31	On	Off	-

Notice: A maximum of 4 indices of the "trace" function can be used.

Note: IF2: Interface 2

p8870[0...15] SINAMICS Link receive telegram word PZD / Recv link word

A_INF (PROFINET), B_INF (PROFINET), CU_S (PROFINET), CU_S_CU310PN, CU_S_S150 (PROFI- NET), ENCODER (PROFINET), S_INF (PROFINET), SERVO (PROFI- NET), TB30 (PROFI- NET), TM120 (PROFINET), TM15 (PROFINET), TM15DI_DO (PROFINET), TM17 (PROFINET), TM31 (PROFINET), TM41 (PROFINET), VEC- TOR (PROFINET)	Can be changed: T Data type: Unsigned16 P-Group: Communications Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
---	--	---	--

Min	Max	Factory setting
0	16	0

Description: Assignment of a PZD to a telegram word from a SINAMICS Link receive telegram.
PZD p2050[index] is assigned by means of p8870[index], p8872[index].

Index: [0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5
[5] = PZD 6
[6] = PZD 7
[7] = PZD 8
[8] = PZD 9
[9] = PZD 10
[10] = PZD 11

[11] = PZD 12

[12] = PZD 13

[13] = PZD 14

[14] = PZD 15

[15] = PZD 16

Dependency: Refer to: p8872**Note:** Value range:

0: Not used

1 ... 16: Telegram word

A pair of values p8870[index], p8872[index] may only be used once in single a device.

p8871[0...15] SINAMICS Link send telegram word PZD / Send link word

A_INF (PROFINET),

B_INF (PROFINET),

CU_S (PROFINET),

CU_S_CU310PN,

CU_S_S150 (PROFI-

NET), ENCODER

(PROFINET), S_INF

(PROFINET),

SERVO (PROFI-

NET), TB30 (PROFI-

NET), TM120

(PROFINET), TM15

(PROFINET),

TM15DI_DO

(PROFINET), TM17

(PROFINET), TM31

(PROFINET), TM41

(PROFINET), VEC-

TOR (PROFINET)

Can be changed: T**Data type:** Unsigned16**P-Group:** Communications**Not for motor type:** -**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Access level:** 3**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Min**

0

Max

16

Factory setting

0

Description: Assigns a PZD to a telegram word in the SINAMICS Link send telegram.
p8871[index] assigns PZD p2051[index].**Index:**

[0] = PZD 1

[1] = PZD 2

[2] = PZD 3

[3] = PZD 4

[4] = PZD 5

[5] = PZD 6

[6] = PZD 7

[7] = PZD 8

[8] = PZD 9

[9] = PZD 10

[10] = PZD 11

[11] = PZD 12

[12] = PZD 13

[13] = PZD 14

[14] = PZD 15

[15] = PZD 16

Dependency: Refer to: p2051, p8851

Refer to: A50002

Note:

Value range:

0: Not used

1 ... 16: Send telegram word

A specific telegram word send may only be used once within a single device.

p8872[0...15] SINAMICS Link address receive PZD / Link addr recv			
A_INF (PROFINET), B_INF (PROFINET), CU_S (PROFINET), CU_S_CU310PN, CU_S_S150 (PROFI- NET), ENCODER (PROFINET), S_INF (PROFINET), SERVO (PROFI- NET), TB30 (PROFI- NET), TM120 (PROFINET), TM15 (PROFINET), TM15DI_DO (PROFINET), TM17 (PROFINET), TM31 (PROFINET), TM41 (PROFINET), VEC- TOR (PROFINET)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 64	Factory setting 0
Description:	Selects the address of the SINAMICS Link sender from which the process data (PZD) is received.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12 [12] = PZD 13 [13] = PZD 14 [14] = PZD 15 [15] = PZD 16		
Dependency:	Refer to: p8870		
Note:	Value range: 0: Not used 1 ... 64: Address		

r8874[0...4] IF2 diagnostics bus address PZD receive / IF2 diag addr recv			
A_INF, B_INF, S_INF	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the bus address of sender from which the PZD is received.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5		

r8874[0...3]	IF2 diagnostics bus address PZD receive / IF2 diag addr recv		
ENCODER	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the bus address of sender from which the PZD is received.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4		
Note:	IF2: Interface 2 Value range: 0 - 125: Bus address of the sender 255: Not assigned		

r8874[0...15]	IF2 diagnostics bus address PZD receive / IF2 diag addr recv		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Communications Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the bus address of sender from which the PZD is received.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12 [12] = PZD 13 [13] = PZD 14 [14] = PZD 15 [15] = PZD 16		
Note:	IF2: Interface 2 Value range: 0 - 125: Bus address of the sender 255: Not assigned		

r8875[0...4]	IF2 diagnostics telegram offset PZD receive / IF diag offs recv		
A_INF, B_INF, S_INF	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the byte offset of the PZD in the receive telegram.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5		

r8875[0...3]	IF2 diagnostics telegram offset PZD receive / IF diag offs recv		
ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the byte offset of the PZD in the receive telegram.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4		
Note:	IF2: Interface 2 Value range: 0 - 242: Byte offset 255: Not assigned		

r8875[0...15]	IF2 diagnostics telegram offset PZD receive / IF diag offs recv		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the byte offset of the PZD in the receive telegram.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12 [12] = PZD 13 [13] = PZD 14 [14] = PZD 15 [15] = PZD 16		

Note: IF2: Interface 2
Value range:
0 - 242: Byte offset
255: Not assigned

r8876[0...7]		IF2 diagnostics telegram offset PZD send / IF2 diag offs send	
A_INF, B_INF, S_INF	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:		Displays the byte offset of the PZD in the send telegram.	
Index:			
		[0] = PZD 1	
		[1] = PZD 2	
		[2] = PZD 3	
		[3] = PZD 4	
		[4] = PZD 5	
		[5] = PZD 6	
		[6] = PZD 7	
		[7] = PZD 8	

r8876[0...11]		IF2 diagnostics telegram offset PZD send / IF2 diag offs send	
ENCODER	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -
Description:	Displays the byte offset of the PZD in the send telegram.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12		
Note:	IF2: Interface 2 Value range: 0 - 242: Byte offset 255: Not assigned		

r8876[0...15] IF2 diagnostics telegram offset PZD send / IF2 diag offs send			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

Description: Displays the byte offset of the PZD in the send telegram.

Index:
 [0] = PZD 1
 [1] = PZD 2
 [2] = PZD 3
 [3] = PZD 4
 [4] = PZD 5
 [5] = PZD 6
 [6] = PZD 7
 [7] = PZD 8
 [8] = PZD 9
 [9] = PZD 10
 [10] = PZD 11
 [11] = PZD 12
 [12] = PZD 13
 [13] = PZD 14
 [14] = PZD 15
 [15] = PZD 16

Note:
 IF2: Interface 2
 Value range:
 0 - 242: Byte offset
 255: Not assigned

p8880[0...15] BI: IF2 binector-connector converter status word 1 / Bin/con ZSW1			
A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2489
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0

Description:
 Selects bits to be sent to the fieldbus master.
 The individual bits are combined to form status word 1.

Index:
 [0] = Bit 0
 [1] = Bit 1
 [2] = Bit 2
 [3] = Bit 3
 [4] = Bit 4
 [5] = Bit 5
 [6] = Bit 6
 [7] = Bit 7
 [8] = Bit 8
 [9] = Bit 9
 [10] = Bit 10
 [11] = Bit 11
 [12] = Bit 12
 [13] = Bit 13
 [14] = Bit 14
 [15] = Bit 15

Dependency: Refer to: p8888, r8889

p8881[0...15]	BI: IF2 binector-connector converter status word 2 / Bin/con ZSW2		
A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2489
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Selects bits to be sent to the fieldbus master.
The individual bits are combined to form status word 2.

Index: [0] = Bit 0
[1] = Bit 1
[2] = Bit 2
[3] = Bit 3
[4] = Bit 4
[5] = Bit 5
[6] = Bit 6
[7] = Bit 7
[8] = Bit 8
[9] = Bit 9
[10] = Bit 10
[11] = Bit 11
[12] = Bit 12
[13] = Bit 13
[14] = Bit 14
[15] = Bit 15

Dependency: Refer to: p8888, r8889

p8882[0...15]	BI: IF2 binector-connector converter status word 3 / Bin/con ZSW3		
A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2489
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Selects bits to be sent to the fieldbus master.
The individual bits are combined to form free status word 3.

Index: [0] = Bit 0
[1] = Bit 1
[2] = Bit 2
[3] = Bit 3
[4] = Bit 4
[5] = Bit 5
[6] = Bit 6
[7] = Bit 7
[8] = Bit 8
[9] = Bit 9
[10] = Bit 10
[11] = Bit 11
[12] = Bit 12
[13] = Bit 13
[14] = Bit 14
[15] = Bit 15

Dependency: Refer to: p8888, r8889

p8883[0...15]	BI: IF2 binector-connector converter status word 4 / Bin/con ZSW4		
A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2489
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Selects bits to be sent to the fieldbus master.
The individual bits are combined to form free status word 4.

Index: [0] = Bit 0
[1] = Bit 1
[2] = Bit 2
[3] = Bit 3
[4] = Bit 4
[5] = Bit 5
[6] = Bit 6
[7] = Bit 7
[8] = Bit 8
[9] = Bit 9
[10] = Bit 10
[11] = Bit 11
[12] = Bit 12
[13] = Bit 13
[14] = Bit 14
[15] = Bit 15

Dependency: Refer to: p8888, r8889

p8884[0...15]	BI: IF2 binector-connector converter status word 5 / Bin/con ZSW5		
A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2489
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Selects bits to be sent to the fieldbus master.
The individual bits are combined to form free status word 5.

Index: [0] = Bit 0
[1] = Bit 1
[2] = Bit 2
[3] = Bit 3
[4] = Bit 4
[5] = Bit 5
[6] = Bit 6
[7] = Bit 7
[8] = Bit 8
[9] = Bit 9
[10] = Bit 10
[11] = Bit 11
[12] = Bit 12
[13] = Bit 13
[14] = Bit 14
[15] = Bit 15

Dependency: Refer to: p8888, r8889

p8888[0...4]		IF2 invert binector-connector converter status word / Bin/con ZSW inv			
A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: U, T		Calculated: -		Access level: 3
	Data type: Unsigned16		Dynamic index: -		Func. diagram: 2489
	P-Group: Communications		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min -		Max -		Factory setting 0000 bin
Description:		Setting to invert the individual binector inputs of the binector connector converter.			
Index:		[0] = Status word 1 [1] = Status word 2 [2] = Free status word 3 [3] = Free status word 4 [4] = Free status word 5			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	Inverted	Not inverted	-
	01	Bit 1	Inverted	Not inverted	-
	02	Bit 2	Inverted	Not inverted	-
	03	Bit 3	Inverted	Not inverted	-
	04	Bit 4	Inverted	Not inverted	-
	05	Bit 5	Inverted	Not inverted	-
	06	Bit 6	Inverted	Not inverted	-
	07	Bit 7	Inverted	Not inverted	-
	08	Bit 8	Inverted	Not inverted	-
	09	Bit 9	Inverted	Not inverted	-
	10	Bit 10	Inverted	Not inverted	-
	11	Bit 11	Inverted	Not inverted	-
	12	Bit 12	Inverted	Not inverted	-
	13	Bit 13	Inverted	Not inverted	-
	14	Bit 14	Inverted	Not inverted	-
	15	Bit 15	Inverted	Not inverted	-
Dependency:		Refer to: p8880, p8881, p8882, p8883, r8889			

r8889[0...4] CO: IF2 send binector-connector converter status word / Bin/con ZSW send					
A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned16		Dynamic index: -		Func. diagram: -
	P-Group: Communications		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min -		Max -		Factory setting -
Description: Connector output to interconnect the status words to a PZD send word.					
Index: [0] = Status word 1 [1] = Status word 2 [2] = Free status word 3 [3] = Free status word 4 [4] = Free status word 5					
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-

11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-

Dependency: Refer to: p8851, p8880, p8881, p8882, p8883

Note: r8889 together with p8880 to p8883 forms four binector-connector converters.

r8890.0...15 BO: IF2 PZD1 receive bit-serial / IF2 PZD1 recv bitw

A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2485, 2491, 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Binector output for bit-serial interconnection of PZD1 (normally control word 1) received via interface 2.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Dependency: Refer to: r8850

Note: IF2: Interface 2

r8891.0...15 BO: IF2 PZD2 receive bit-serial / IF2 PZD2 recv bitw

A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2485, 2491, 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Binector output for bit-serial interconnection of PZD2 received via interface 2.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-

11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-

Dependency: Refer to: r8850

Note: IF2: Interface 2

r8892.0...15 BO: IF2 PZD3 receive bit-serial / IF2 PZD3 recv bitw

ENCODER, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2485, 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Binector output for bit-serial interconnection of PZD3 received via interface 2.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Dependency: Refer to: r8850

Note: IF2: Interface 2

r8893.0...15 BO: IF2 PZD4 receive bit-serial / IF2 PZD4 recv bitw

ENCODER, SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2485, 9204, 9206
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Binector output for bit-serial interconnection of PZD4 (normally control word 2) received via interface 2.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-

11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-

Dependency: Refer to: r8850

Note: IF2: Interface 2

r8894.0...15 BO: IF2 connector-binector converter binector output / Con/bin outp

A_INF, B_INF,
ENCODER, S_INF,
SERVO, VECTOR

Can be changed: -

Data type: Unsigned16

P-Group: Communications

Not for motor type: -

Min

-

Calculated: -

Dynamic index: -

Units group: -

Scaling: -

Max

-

Access level: 3

Func. diagram: 2485, 2491

Unit selection: -

Expert list: 1

Factory setting

-

Description: Binector output for bit-serial onward interconnection of a PZD word received from the fieldbus master.
The PZD is selected via p8899[0].

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-
	11	Bit 11	On	Off	-
	12	Bit 12	On	Off	-
	13	Bit 13	On	Off	-
	14	Bit 14	On	Off	-
	15	Bit 15	On	Off	-

Dependency: Refer to: p8899

r8895.0...15 BO: IF2 connector-binector converter binector output / Con/bin outp

A_INF, B_INF,
ENCODER, S_INF,
SERVO, VECTOR

Can be changed: -

Data type: Unsigned16

P-Group: Communications

Not for motor type: -

Min

-

Calculated: -

Dynamic index: -

Units group: -

Scaling: -

Max

-

Access level: 3

Func. diagram: 2485, 2491

Unit selection: -

Expert list: 1

Factory setting

-

Description: Binector output for bit-serial interconnection of a PZD word received from the fieldbus master.
The PZD is selected via p8899[1].

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	On	Off	-
	01	Bit 1	On	Off	-
	02	Bit 2	On	Off	-
	03	Bit 3	On	Off	-
	04	Bit 4	On	Off	-
	05	Bit 5	On	Off	-
	06	Bit 6	On	Off	-
	07	Bit 7	On	Off	-
	08	Bit 8	On	Off	-
	09	Bit 9	On	Off	-
	10	Bit 10	On	Off	-

11	Bit 11	On	Off	-
12	Bit 12	On	Off	-
13	Bit 13	On	Off	-
14	Bit 14	On	Off	-
15	Bit 15	On	Off	-

Dependency: Refer to: p8899

p8898[0...1] IF2 invert connector-binector converter binector output / Con/bin outp inv

A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2485, 2491
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Setting to invert the individual binector outputs of the connector-binector converter.
Using p8898[0], the signals of CI: p8899[0] are influenced.
Using p8898[1], the signals of CI: p8899[1] are influenced.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Bit 0	Inverted	Not inverted	-
	01	Bit 1	Inverted	Not inverted	-
	02	Bit 2	Inverted	Not inverted	-
	03	Bit 3	Inverted	Not inverted	-
	04	Bit 4	Inverted	Not inverted	-
	05	Bit 5	Inverted	Not inverted	-
	06	Bit 6	Inverted	Not inverted	-
	07	Bit 7	Inverted	Not inverted	-
	08	Bit 8	Inverted	Not inverted	-
	09	Bit 9	Inverted	Not inverted	-
	10	Bit 10	Inverted	Not inverted	-
	11	Bit 11	Inverted	Not inverted	-
	12	Bit 12	Inverted	Not inverted	-
	13	Bit 13	Inverted	Not inverted	-
	14	Bit 14	Inverted	Not inverted	-
	15	Bit 15	Inverted	Not inverted	-

Dependency: Refer to: r8894, r8895, p8899

p8899[0...1] CI: IF2 connector-binector converter signal source / Con/bin S_src

A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer16	Dynamic index: -	Func. diagram: 2485, 2491
	P-Group: Communications	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source for the connector-binector converter.
A PZD receive word can be selected as signal source. The signals are available to be serially passed-on (interconnection).

Dependency: Refer to: r8894, r8895

Note: From the signal source set via the connector input, the corresponding lower 16 bits are converted.
p8899[0...1] together with r8894.0...15 and r8895.0...15 forms two connector-binector converters:
Connector input p8899[0] to binector output in r8894.0...15
Connector input p8899[1] to binector output in r8895.0...15

p8900[0...239]	IE Name of Station / IE Name Stat		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Sets the station name for the Industrial Ethernet interface (X127) on the Control Unit. The active station name is displayed in r8910.		
Dependency:	Refer to: p8905, r8910		
Note:	An ASCII table (excerpt) can be found, for example, in the Appendix of the List Manual: The interface configuration (p8900 and following) is activated with p8905 = 1. The parameter is not influenced by setting the factory setting. IE: Industrial Ethernet		
p8901[0...3]	IE IP Address of Station / IE IP of Stat		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned8 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the IP address for the Industrial Ethernet interface (X127) on the Control Unit. The active IP address is displayed in r8911.		
Dependency:	Refer to: p8905, r8911		
Note:	The interface configuration (p8900 and following) is activated with p8905 = 1. The parameter is not influenced by setting the factory setting.		
p8902[0...3]	IE Default Gateway of Station / IE Def Gateway		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned8 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the default gateway for the Industrial Ethernet interface (X127) on the Control Unit. The active default gateway is displayed in r8912.		
Dependency:	Refer to: p8905, r8912		
Note:	The interface configuration (p8900 and following) is activated with p8905 = 1. The parameter is not influenced by setting the factory setting.		
p8903[0...3]	IE Subnet Mask of Station / IE Subnet Mask		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned8 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the subnet mask for the Industrial Ethernet interface (X127) on the Control Unit. The active subnet mask is displayed in r8913.		
Dependency:	Refer to: p8905, r8913		

Note: The interface configuration (p8900 and following) is activated with p8905 = 1.
The parameter is not influenced by setting the factory setting.

p8905	IE Interface configuration / IE IF config		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Setting to control the configuration of the Industrial Ethernet interface (X127) on the Control Unit. p8905 is automatically set to 0 at the end of an operation.		
Value:	0: No function 1: Activate configuration 2: Activate and save configuration 3: Delete configuration		
Dependency:	Refer to: p8900, p8901, p8902, p8903 Refer to: A08561		
Note:	Re p8905 = 1: The interface configuration (p8900 and following) is activated. Re p8905 = 2: The interface configuration (p8900 and following) is activated and saved to non-volatile memory. Re p8905 = 3: The interface configuration is reset to the factory setting at all points. The factory settings for the interface configuration are loaded on activation (p8905 = 1) or at the next POWER ON.		

r8910[0...239]	IE Name of Station active / IE Name Stat act		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	This parameter defines the station name of the onboard Ethernet interface.		

r8911[0...3]	IE IP Address of Station active / IE IP of Stat act		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the active IP address for the Industrial Ethernet interface (X127) on the Control Unit.		

r8912[0...3]	IE Default Gateway of Station active / IE Def Gateway act		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the active default gateway for the Industrial Ethernet interface (X127) on the Control Unit.		
r8913[0...3]	IE Subnet Mask of Station active / IE Subnet Mask act		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the active subnet mask for the Industrial Ethernet interface (X127) on the Control Unit.		
r8915[0...5]	IE MAC Address of Station / IE MAC of Station		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the MAC address for the Industrial Ethernet interface (X127) on the Control Unit.		
p8940[0...239]	CBE20 Name of Station / CBE20 Name Stat		
CU_S (PROFINET), CU_S_CU310PN, CU_S_S150 (PROFI- NET)	Can be changed: U, T Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Sets the station name for the Communication Board Ethernet 20 (CBE20).		
Note:	An ASCII table (excerpt) can be found, for example, in the Appendix of the List Manual: The CBE20 configuration (p8940 and following) is activated with p8945 = 1 (becomes effective at the next POWER ON). The parameter is not influenced by setting the factory setting.		
p8941[0...3]	CBE20 IP Address of Station / CBE20 IP of Stat		
CU_S (PROFINET), CU_S_CU310PN, CU_S_S150 (PROFI- NET)	Can be changed: U, T Data type: Unsigned8 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the IP address for the Communication Board Ethernet 20 (CBE20).		
Note:	The CBE20 configuration (p8940 and following) is activated with p8945 = 1 (becomes effective at the next POWER ON). The parameter is not influenced by setting the factory setting.		

p8942[0...3] CBE20 Default Gateway of Station / CBE20 Def GatewayCU_S (PROFINET),
CU_S_CU310PN,
CU_S_S150 (PROFI-
NET)**Can be changed:** U, T**Data type:** Unsigned8**P-Group:** -**Not for motor type:** -**Min**
0**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**
255**Access level:** 1**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**
0**Description:** Sets the default gateway for the Communication Board Ethernet 20 (CBE20).**Note:** The CBE20 configuration (p8940 and following) is activated with p8945 = 1 (becomes effective at the next POWER ON).

The parameter is not influenced by setting the factory setting.

p8943[0...3] CBE20 Subnet Mask of Station / CBE20 Subnet MaskCU_S (PROFINET),
CU_S_CU310PN,
CU_S_S150 (PROFI-
NET)**Can be changed:** U, T**Data type:** Unsigned8**P-Group:** -**Not for motor type:** -**Min**
0**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**
255**Access level:** 1**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**
0**Description:** Sets the subnet mask for the Communication Board Ethernet 20 (CBE20).**Note:** The CBE20 configuration (p8940 and following) is activated with p8945 = 1 (becomes effective at the next POWER ON).

The parameter is not influenced by setting the factory setting.

p8944 CBE20 DHCP Mode / CBE20 DHCP ModeCU_S (PROFINET),
CU_S_CU310PN,
CU_S_S150 (PROFI-
NET)**Can be changed:** U, T**Data type:** Unsigned8**P-Group:** -**Not for motor type:** -**Min**
0**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**
255**Access level:** 1**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**
0**Description:** Sets the DHCP mode for the Communication Board Ethernet 20 (CBE20).**Note:** The CBE20 configuration (p8940 and following) is activated with p8945 = 1 (becomes effective at the next POWER ON).

The parameter is not influenced by setting the factory setting.

p8945 CBE20 interface configuration / CBE20 IF configCU_S (PROFINET),
CU_S_CU310PN,
CU_S_S150 (PROFI-
NET)**Can be changed:** U, T**Data type:** Integer16**P-Group:** -**Not for motor type:** -**Min**
0**Calculated:** -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**
3**Access level:** 1**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**
0**Description:** Sets the configuration control for the Communication Board Ethernet 20 (CBE20).

p8945 is automatically set to 0 at the end of an operation.

Value:
0: No function
2: Save and activate configuration
3: Delete configuration**Note:** Re p8945 = 2:

The CBE20 configuration (p8940 and following) is saved and activated at the next POWER ON.

Re p8945 = 3:

The factory settings for the CBE20 configuration are loaded at the next POWER ON.

p9300	SI Motion monitoring clock cycle (Motor Module) / SI Mtn clock MM																							
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 500.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 25000.00 [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 12000.00 [µs]																					
Description:	Sets the monitoring clock cycle for safe motion monitoring.																							
Dependency:	Refer to: p9500, p9511 Refer to: F01652																							
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.																							
Note:	The monitoring clock cycle must be a multiple of the actual value sensing clock cycle in p9311 or of the DP clock cycle. A change only becomes effective after a POWER ON.																							
p9301	SI Motion enable safety functions (Motor Module) / SI Mtn enable MM																							
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 bin																					
Description:	Sets the enable signals for the safe motion monitoring.																							
Bit field:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr> <tr> <td>00</td><td>SOS/SLS enable</td><td>Enable</td><td>Inhibit</td><td>-</td></tr> <tr> <td>03</td><td>Actual value synchronization enable</td><td>Enable</td><td>Inhibit</td><td>-</td></tr> <tr> <td>16</td><td>Enable NX Hys Fil</td><td>Enable</td><td>Inhibit</td><td>-</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FP	00	SOS/SLS enable	Enable	Inhibit	-	03	Actual value synchronization enable	Enable	Inhibit	-	16	Enable NX Hys Fil	Enable	Inhibit	-			
Bit	Signal name	1 signal	0 signal	FP																				
00	SOS/SLS enable	Enable	Inhibit	-																				
03	Actual value synchronization enable	Enable	Inhibit	-																				
16	Enable NX Hys Fil	Enable	Inhibit	-																				
Dependency:	Refer to: p9501 Refer to: F01682, F01683																							
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.																							
Note:	A change only becomes effective after a POWER ON. SLS: Safely-Limited Speed SOS: Safe Operating Stop																							
p9302	SI Motion axis type (Motor Module) / SI Mtn AxisType MM																							
SERVO, VECTOR	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0																					
Description:	Sets the axis type (linear axis or rotary axis/spindle).																							
Value:	0: Linear axis 1: Rot axis/spindle																							
Dependency:	Refer to: p9502																							
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.																							
Note:	For the commissioning software, after changing over the axis type, the units dependent on the axis type are only updated after a project upload. A change only becomes effective after a POWER ON.																							

p9306	SI Motion function specification (Motor Module) / SI Mtn fct_spc MM				
SERVO, VECTOR	Can be changed: C2(95)		Calculated: -	Access level: 3	
	Data type: Integer16		Dynamic index: -	Func. diagram: -	
	P-Group: Safety Integrated		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	0		1	0	
Description:	Sets the function specification for Safety Integrated.				
Value:	0: Safety with encoder 1: Safety without encoder				
Dependency:	Refer to: C30711				
p9307	SI Motion function configuration MM / SI Mtn config MM				
SERVO, VECTOR	Can be changed: C2(95)		Calculated: -	Access level: 3	
	Data type: Unsigned32		Dynamic index: -	Func. diagram: -	
	P-Group: Safety Integrated		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0000 bin	
Description:	Function configuration for Safe Motion Monitoring				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Extended alarm acknowledgment	Yes	No	-
	01	Setpoint speed limit for stop F	No	Yes	-
Dependency:	Refer to: C01711				
p9311	SI Motion actual value sensing clock cycle (Motor Module) / SI Mtn act clk MM				
SERVO, VECTOR	Can be changed: C2(95)		Calculated: -	Access level: 3	
	Data type: FloatingPoint32		Dynamic index: -	Func. diagram: -	
	P-Group: Safety Integrated		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	0.0000 [µs]		25000.0000 [µs]	0.0000 [µs]	
Description:	Sets the clock cycle time of the actual value sensing for safe motion monitoring. The slower clock cycle time reduces the maximum permissible velocity - however, it ensures a lower load of the Control Unit for the safe actual value sensing. The maximum permissible velocity, which when exceeded can mean that errors occur during safe actual value sensing, is displayed in r9730. Setting criteria if the motion monitoring functions are executed without an encoder: - The actual value sensing clock cycle must be set to the same value as the current controller clock cycle (p115).				
Dependency:	Refer to: p0115, p9300, p9511 Refer to: F01652				
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.				
Note:	The parameter is only active for drive-based motion monitoring functions (p9801.2 = 1). The monitoring clock cycle from p9300 must be an integer multiple of this parameter. In the case of motion monitoring functions with encoder, the clock cycle time for actual value sensing must be an integer multiple of the current controller clock cycle and at least 4 times slower than the current controller clock cycle. A factor of at least 8 is recommended. A change only becomes effective after a POWER ON.				

p9315 SI Motion coarse position value configuration (Motor Module) / SI Mtn s config MM

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the encoder configuration for the redundant coarse position value.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Incrementer	Yes	No	-
	01	Encoder CRC least significant byte first	Yes	No	-
	02	Redundant coarse position val. most significant bit left-aligned	Yes	No	-
	16	DRIVE-CLiQ encoder	Yes	No	-

Dependency: Refer to: r0474, p9515

p9316 SI Motion encoder configuration, safety functions (Motor Module) / SI Mtn enc_cfg MM

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the configuration for the encoder and position actual value.

The encoder that is used for the safe motion monitoring functions on the Motor Module must be parameterized in this parameter.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Encoder rotating/linear	Linear	Rotating:	-
	01	Position actual value, sign change	Yes	No	-

Dependency: Refer to: p0404, p0410, p9516

p9317 SI Motion linear scale grid division (Motor Module) / SI Mtn grid MM

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.00 [nm]	250000000.00 [nm]	10000.00 [nm]

Description: Sets the grid division for a linear encoder.

The encoder that is used for the safe motion monitoring functions on the Motor Module must be parameterized in this parameter.

Dependency: Refer to: p0407, p9316

p9318	SI Motion encoder pulses per revolution (Motor Module) / SI Mtn p/rev MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2048
Description:	Sets the number of encoder pulses per revolution for rotary encoders. The encoder that is used for the safe motion monitoring functions on the Motor Module must be parameterized in this parameter.		
Dependency:	Refer to: p0408, p9316		
p9319	SI Motion fine resolution G1_XIST1 (Motor Module) / SI Mtn G1_XIST1 MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 2	Calculated: - Dynamic index: - Units group: - Scaling: - Max 18	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 11
Description:	Sets the fine resolution for G1_XIST1 in bits. The encoder that is used for the safe motion monitoring functions on the Motor Module must be parameterized in this parameter.		
Dependency:	Refer to: p0418 Refer to: F01670, F01671		
Note:	For safety functions that have not been enabled (p9301 = 0), the following applies: When booting, p9319 is automatically set the same as p0418. For safety functions that are enabled (p9301 > 0), the following applies: p9319 is checked for agreement with p0418. G1_XIST1: Encoder 1 position actual value 1 (PROFIdrive)		
p9320	SI Motion spindle pitch (Motor Module) / SI Mtn sp_pitch MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.1000 [mm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8388.0000 [mm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10.0000 [mm]
Description:	Sets the gear ratio between the encoder and load in mm/revolution for a linear axis with rotary encoder. The encoder that is used for the safe motion monitoring functions on the Motor Module must be parameterized in this parameter.		
Dependency:	Refer to: p9520		
Notice:	The fourth decimal point can be rounded-off depending on the size of the entered number (from three places before the decimal point).		

p9321[0...7]	SI Motion gearbox encoder (motor)/load denom (Motor Module) / SI Mtn denom MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147000000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the denominator for the gearbox between the encoder (or motor in the case of encoderless monitoring functions) and the load. It is not possible to change over the gearbox stages. Gearbox stage 0 is always active.		
Index:	[0] = Gearbox 1 [1] = Gearbox 2 [2] = Gearbox 3 [3] = Gearbox 4 [4] = Gearbox 5 [5] = Gearbox 6 [6] = Gearbox 7 [7] = Gearbox 8		
Dependency:	Refer to: p9322		
p9322[0...7]	SI Motion gearbox encoder (motor)/load numerator (Motor Module) / SI Mtn numer MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147000000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the numerator for the gearbox between the encoder (or motor in the case of encoderless monitoring functions) and the load. It is not possible to change over the gearbox stages. Gearbox stage 0 is always active.		
Index:	[0] = Gearbox 1 [1] = Gearbox 2 [2] = Gearbox 3 [3] = Gearbox 4 [4] = Gearbox 5 [5] = Gearbox 6 [6] = Gearbox 7 [7] = Gearbox 8		
Dependency:	Refer to: p9321		
Note:	In the case of encoderless monitoring functions (p9506), the pole pair number must be multiplied by the numerator of the gearbox ratio. Example: Gearbox ratio 1:4, pole pair number (r0313) = 2 --> p9521 = 1, p9522 = 8 (4*2)		

p9323	SI Motion red. coarse position value valid bits (Motor Module) / Valid bits MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 9
Description:	Sets the number of valid bits of the redundant coarse position value. The encoder that is used for the safe motion monitoring functions on the Motor Module must be parameterized in this parameter.		
Dependency:	Refer to: r0470, p9523		
p9324	SI Motion redundant coarse pos. value fine resolution bits (MM) / SI Mtn fine bit MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min -16	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -2
Description:	Sets the number of valid bits for the fine resolution of the redundant coarse position value. The encoder that is used for the safe motion monitoring functions on the Motor Module must be parameterized in this parameter.		
Dependency:	Refer to: r0471, p9524		
p9325	SI Motion redundant coarse pos. value relevant bits (MM) / Relevant bits MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 16
Description:	Sets the number of relevant bits for the redundant coarse position value. The encoder that is used for the safe motion monitoring functions on the Motor Module must be parameterized in this parameter.		
Dependency:	Refer to: p0414, r0472, p9525		
p9326	SI Motion encoder assignment (Motor Module) / SI Mtn encoder MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the number of the encoder that the Motor Module uses for safe motion monitoring functions.		
Dependency:	For the safe motion monitoring functions, the redundant safety position actual value sensing must be activated in the appropriate encoder data set (p0430.19 = 1). Refer to: p0187, p0188, p0189, p0430, p9526		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	For p9326 = 1 the following applies: Motor Module uses an encoder for closed-loop speed control, it involves a 1-encoder system.		

p9328[0...11]	SI Motion Sensor Module Node Identifier (Motor Module) / SI Mtn SM Ident MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned8 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max 00FF hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the node identifier of the Sensor Module that is used by the Motor Module for the motion monitoring functions.		
Dependency:	Refer to: r9881		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
p9329	SI Motion Gx_XIST1 Safe most significant bit (MM) / Gx_XIST1 MSB MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 31	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 14
Description:	Sets the bit number for the safe most significant bit (MSB) of the Gx_XIST1 coarse position. The encoder that is used for the safe motion monitoring functions on the Motor Module must be parameterized in this parameter.		
Dependency:	Refer to: p0415, r0475, p9529		
Note:	MSB: Most Significant Bit		
p9330	SI Motion standstill tolerance (Motor Module) / SI Mtn SOS Tol MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.000 [mm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.000 [mm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.000 [mm]
Description:	Sets the tolerance for the function "Safe Operating Stop" (SOS).		
Dependency:	Refer to: p9530 Refer to: C01707		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SOS: Safe Operational Stop		
p9330	SI Motion standstill tolerance (Motor Module) / SI Mtn SOS Tol MM		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.000 [°]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.000 [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.000 [°]
Description:	Sets the tolerance for the function "Safe Operating Stop" (SOS).		
Dependency:	Refer to: p9530 Refer to: C01707		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SOS: Safe Operational Stop		

p9331[0...3]	SI Motion SLS limit values (Motor Module) / SI Mtn SLS lim MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [mm/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000000.00 [mm/min]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2000.00 [mm/min]
Description:	Sets the limit values for the function "Safely-Limited Speed" (SLS).		
Index:	[0] = Limit value SLS1 [1] = Limit value SLS2 [2] = Limit value SLS3 [3] = Limit value SLP4		
Dependency:	Refer to: p9363, p9531 Refer to: C01714		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SLS: Safely-Limited Speed		
p9331[0...3]	SI Motion SLS limit values (Motor Module) / SI Mtn SLS lim MM		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [rpm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000000.00 [rpm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2000.00 [rpm]
Description:	Sets the limit values for the function "Safely-Limited Speed" (SLS).		
Index:	[0] = Limit value SLS1 [1] = Limit value SLS2 [2] = Limit value SLS3 [3] = Limit value SLP4		
Dependency:	Refer to: p9363, p9531 Refer to: C01714		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SLS: Safely-Limited Speed		
p9342	SI Motion act val comparison tol (crosswise) (Motor Module) / SI Mtn act tol MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.0010 [mm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 360.0000 [mm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.1000 [mm]
Description:	Sets the tolerance for the cross-check of the actual position between the two monitoring channels. For encoderless motion monitoring functions, the tolerance must be set to a higher value (12 degrees rotary and 1 mm linear).		
Dependency:	Refer to: p9542 Refer to: C01711		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		

p9342	SI Motion act val comparison tol (crosswise) (Motor Module) / SI Mtn act tol MM		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.0010 [°]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 360.0000 [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.1000 [°]
Description:	Sets the tolerance for the cross-check of the actual position between the two monitoring channels. For encoderless motion monitoring functions, the tolerance must be set to a higher value (12 degrees rotary and 1 mm linear).		
Dependency:	Refer to: p9542 Refer to: C01711		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
p9345	SI Motion SSM filter time (Motor Module) / SI Mtn SSM filt MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000.00 [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [µs]
Description:	Sets the filter time for the SSM feedback signal to detect standstill.		
Note:	The filter time is effective only if the function is enabled (p9300/p9500 Bit 16 = 1). The parameter is included in the data cross-check of the two monitoring channels. SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring)		
p9346	SI Motion SSM velocity limit (Motor Module) / SI Mtn SSM v_limMM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [mm/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000000.00 [mm/min]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 20.00 [mm/min]
Description:	Sets the velocity limit for the SSM feedback signal to detect standstill (n < nx). When this limit value is undershot, the signal "SSM feedback signal active" is set. If a value of 0 is entered in p9368/p9568, the value of parameter p9346/p9546 applies also for the safe acceleration monitor SBR.		
Dependency:	Refer to: p9546		
Caution:	The function "Safe Acceleration Monitor" (SBR) is switched out after the selected threshold value is undershot.		
			
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SBR: Safe Acceleration Monitor SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring)		

p9346 SI Motion SSM velocity limit (Motor Module) / SI Mtn SSM v_limMM			
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 1000000.00 [rpm]	Factory setting 20.00 [rpm]
Description:	Sets the velocity limit for the SSM feedback signal to detect standstill ($n < nx$). When this limit value is undershot, the signal "SSM feedback signal active" is set. If a value of 0 is entered in p9368/p9568, the value of parameter p9346/p9546 applies also for the safe acceleration monitor SBR.		
Dependency:	Refer to: p9546		
Caution:	The function "Safe Acceleration Monitor" (SBR) is switched out after the selected threshold value is undershot.		
			
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SBR: Safe Acceleration Monitor SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring)		

p9347 SI Motion SSM velocity hysteresis (Motor Module) / SI Mtn SSM Hyst MM			
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0010 [mm/min]	Max 500.0000 [mm/min]	Factory setting 10.0000 [mm/min]
Description:	Sets the velocity hysteresis for the SSM feedback signal to detect standstill ($n < nx$).		
Dependency:	Refer to: C01711		
Note:	The velocity hysteresis is effective only if the function is enabled (p9300/p9500 Bit 16 = 1). The parameter is included in the data cross-check of the two monitoring channels. SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring)		

p9347 SI Motion SSM velocity hysteresis (Motor Module) / SI Mtn SSM Hyst MM			
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0010 [rpm]	Max 500.0000 [rpm]	Factory setting 10.0000 [rpm]
Description:	Sets the velocity hysteresis for the SSM feedback signal to detect standstill ($n < nx$).		
Dependency:	Refer to: C01711		
Note:	The velocity hysteresis is effective only if the function is enabled (p9300/p9500 Bit 16 = 1). The parameter is included in the data cross-check of the two monitoring channels. SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring)		

p9348	SI Motion SBR actual velocity tolerance (Motor Module) / SI Mtn SBR Tol MM		
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [mm/min]	Max 120000.00 [mm/min]	Factory setting 300.00 [mm/min]
Description:	Sets the velocity tolerance for the "Safe Acceleration Monitor".		
Dependency:	Refer to: p9548 Refer to: C01706		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SBR: Safe Acceleration Monitor		
p9348	SI Motion SBR actual velocity tolerance (Motor Module) / SI Mtn SBR Tol MM		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 120000.00 [rpm]	Factory setting 300.00 [rpm]
Description:	Sets the velocity tolerance for the "Safe Acceleration Monitor".		
Dependency:	Refer to: p9548 Refer to: C01706		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SBR: Safe Acceleration Monitor		
p9349	SI Motion slip velocity tolerance (Motor Module) / SI Mtn slip MM		
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [mm/min]	Max 6000.00 [mm/min]	Factory setting 6.00 [mm/min]
Description:	Sets the velocity tolerance that is used for a 2-encoder system in cross-check between the Control Unit and the Motor Module.		
Dependency:	Refer to: p9301, p9342, p9549		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	If the "actual value synchronization" is not enabled (p9301.3 = 0), then the value parameterized in p9342 is used as tolerance in the data cross-check.		
p9349	SI Motion slip velocity tolerance (Motor Module) / SI Mtn slip MM		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 6000.00 [rpm]	Factory setting 6.00 [rpm]
Description:	Sets the velocity tolerance that is used for a 2-encoder system in cross-check between the Control Unit and the Motor Module.		
Dependency:	Refer to: p9301, p9342, p9549		

Notice: This parameter is overwritten by the copy function of the safety functions integrated in the drive.
Note: If the "actual value synchronization" is not enabled (p9301.3 = 0), then the value parameterized in p9342 is used as tolerance in the data cross-check.

p9351 SI Motion SLS changeover delay time (Motor Module) / SI Mtn SLS t MM			
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2825
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [µs]	Max 600000000.00 [µs]	Factory setting 100000.00 [µs]
Description:	Sets the delay time for the SLS changeover or for the changeover from SLS to SOS for the function "Safely-Limited Speed" (SLS). When transitioning from a higher to a lower safely-limited velocity/speed stage or to the safe operating stop (SOS), within this delay time, the "old" velocity stage remains active. Even if SLS or SOS is activated from non safety-related operation, then this delay is still applied.		
Dependency:	Refer to: p9551		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SLS: Safely Limited Speed SOS: Safe Operating Stop		

p9352 SI Motion transition time STOP C to SOS (Motor Module) / SI Mtn t C->SOS MM			
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2825
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [µs]	Max 600000000.00 [µs]	Factory setting 100000.00 [µs]
Description:	Sets the transition time from STOP C to "Safe Operating Stop" (SOS).		
Dependency:	Refer to: p9552		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SOS: Safe Operational Stop		

p9353 SI Motion transition time STOP D to SOS (Motor Module) / SI Mtn t D->SOS MM			
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2825
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [µs]	Max 600000000.00 [µs]	Factory setting 100000.00 [µs]
Description:	Sets the transition time from STOP D to "Safe Operating Stop" (SOS).		
Dependency:	Refer to: p9553		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SOS: Safe Operating Stop		

p9355 SI Motion transition time STOP F to STOP B (Motor Module) / SI Mtn t F->B MM

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2825
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [μs]	Max 600000000.00 [μs]	Factory setting 0.00 [μs]
Description:	Sets the transition time from STOP F to STOP B.		
Dependency:	Refer to: C01711		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		

p9356 SI Motion pulse suppression delay time (Motor Module) / SI Mtn IL t_del MM

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2825
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [μs]	Max 600000000.00 [μs]	Factory setting 100000.00 [μs]
Description:	Sets the delay time for the safe pulse suppression after STOP B / SS1. In the case of encoderless motion monitoring functions (p9506/p9306 = 1), the parameter has no effect.		
Dependency:	Refer to: p9360, p9556 Refer to: C01701		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SS1: Safe Stop 1		

p9357 SI Motion pulse suppression test time (Motor Module) / SI Mtn IL t MM

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [μs]	Max 10000000.00 [μs]	Factory setting 100000.00 [μs]
Description:	Sets the time after which the pulses must have been suppressed when initiating the test stop.		
Dependency:	Refer to: p9557 Refer to: C01798		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		

p9358 SI Motion acceptance test mode time limit (Motor Module) / SI Mtn acc t MM

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 5000000.00 [μs]	Max 100000000.00 [μs]	Factory setting 40000000.00 [μs]
Description:	Sets the maximum time for the acceptance test mode. If the acceptance test mode takes longer than the selected time limit, then the mode is automatically terminated.		
Dependency:	Refer to: p9558 Refer to: C01799		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		

p9360	SI Motion pulse suppression shutdown velocity (Motor Module) / SI Mtn IL v_shutMM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [mm/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 6000.00 [mm/min]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [mm/min]
Description:	Sets the shutdown velocity for pulse suppression. Below this velocity "standstill" is assumed and for STOP B / SS1, the pulses are suppressed (by changing to STOP A). In the case of encoderless motion monitoring functions, the parameter must be > 0 (recommended value: 10).		
Dependency:	Refer to: p9356, p9560		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SS1: Safe Stop 1		
p9360	SI Motion pulse suppression shutdown speed (Motor Module) / SI Mtn IL n_shutMM		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [rpm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 6000.00 [rpm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [rpm]
Description:	Sets the shutdown speed for the pulse suppression. Below this speed "standstill" is assumed and for STOP B / SS1, the pulses are suppressed (by changing to STOP A).		
Dependency:	Refer to: p9356, p9560		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SS1: Safe Stop 1		
p9363[0...3]	SI Motion SLS stop response (Motor Module) / SI Mtn SLS Stop MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 14	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2
Description:	Sets the stop response for the function "safely reduced speed" (SLS). These settings apply to the individual limit values for SLS. In the case of encoderless motion monitoring (p9306 = 1), only a value of 0 or 1 is permitted.		
Value:	0: STOP A 1: STOP B 2: STOP C 3: STOP D 4: STOP E 10: STOP A with delayed pulse suppression when the bus fails 11: STOP B with delayed pulse suppression when the bus fails 12: STOP C with delayed pulse suppression when the bus fails 13: STOP D with delayed pulse suppression when the bus fails 14: STOP E with delayed pulse suppression when the bus fails		
Index:	[0] = Limit value SLS1 [1] = Limit value SLS2 [2] = Limit value SLS3 [3] = Limit value SLP4		
Dependency:	Refer to: p9331, p9380, p9563		

Notice: This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Values 10 to 14 are being prepared and are presently ineffective.

Note: SLS: Safely-Limited Speed / SG: Safely reduced speed

p9368 SI Motion SBR velocity limit (Motor Module) / SI Mtn SBR v_limMM			
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [mm/min]	Max 1000.00 [mm/min]	Factory setting 0.00 [mm/min]
Description:	Sets the velocity limit for the "SBR" function. SBR is deactivated once the set velocity limit has been undershot.		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SBR: Safe Acceleration Monitor SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring) For p9568 = p9368 = 0, the following applies: The value in p9546/p9346 (SSM) is applied as the velocity limit for SBR.		

p9368 SI Motion SBR velocity limit (Motor Module) / SI Mtn SBR v_limMM			
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 1000.00 [rpm]	Factory setting 0.00 [rpm]
Description:	Sets the velocity limit for the "SBR" function. SBR is deactivated once the set velocity limit has been undershot.		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive.		
Note:	SBR: Safe Acceleration Monitor SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring) For p9568 = p9368 = 0, the following applies: The value in p9546/p9346 (SSM) is applied as the velocity limit for SBR.		

p9370 SI Motion acceptance test mode (Motor Module) / SI Mtn acc_mod MM			
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max 00AC hex	Factory setting 0000 hex
Description:	Setting to select and deselect the acceptance test mode.		
Value:	0: [00 hex] Deselect the acceptance test mode 172: [AC hex] Select the acceptance test mode		
Dependency:	Refer to: p9358, r9371 Refer to: C01799		

p9371	SI Motion acceptance test status (Motor Module) / SI Mtn acc_stat MM		
SERVO, VECTOR	Can be changed: - Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max 00AC hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the status of the acceptance test mode.		
Value:	0: [00 hex] Acc_mode inactive 12: [0C hex] Acc_mode not possible due to POWER ON fault 13: [0D hex] Acc_mode not possible due to incorrect ID in p9370 15: [0F hex] Acc_mode not possible due to expired Acc_timer 172: [AC hex] Acc_mode active		
Dependency:	Refer to: p9358, p9370 Refer to: C01799		
p9380	SI Motion pulse suppression delay bus failure (Motor Module) / SI Mtn t to IL MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 800000.00 [µs]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [µs]
Description:	Sets the delay time after which the pulses are safely suppressed after a bus failure.		
Dependency:	Refer to: p9363		
Notice:	This parameter is overwritten by the copy function of the safety functions integrated in the drive. The parameter is being prepared and is presently ineffective.		
p9381	SI Motion brake ramp reference value (Motor Module) / SI Mtn ramp ref MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 600.0000 [mm/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 24000.0000 [mm/min]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1500.0000 [mm/min]
Description:	Sets the reference value to define the brake ramp. The rate of rise of the brake ramp depends upon p9581 (reference value) and p9583 (monitoring time).		
Dependency:	Refer to: p9382, p9383		
p9381	SI Motion brake ramp reference value (Motor Module) / SI Mtn ramp ref MM		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 600.0000 [rpm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 24000.0000 [rpm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1500.0000 [rpm]
Description:	Sets the reference value to define the brake ramp. The rate of rise of the brake ramp depends upon p9581 (reference value) and p9583 (monitoring time).		
Dependency:	Refer to: p9382, p9383		

p9382	SI Motion brake ramp delay time (Motor Module) / SI Mtn rp t_del MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 10000.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 99000000.00 [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 250000.00 [µs]
Description:	Sets the delay time for monitoring the brake ramp. Monitoring of the brake ramp starts once the delay time has elapsed.		
Dependency:	Refer to: p9381, p9383		
p9383	SI Motion brake ramp monitoring time (Motor Module) / SI Mtn rp t_mon MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 500.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000000.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10000.00 [ms]
Description:	Sets the monitoring time to define the brake ramp. The rate of rise of the brake ramp depends upon p9581 (reference value) and p9583 (monitoring time).		
Dependency:	Refer to: p9381, p9382		
p9387	SI Motion act val sensing encoderless filter time (Motor Module) / SI Mtn EL filt MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000.00 [µs]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 25000.00 [µs]
Description:	Sets the filter time for smoothing the actual value with encoderless actual value sensing.		
Note:	This parameter is only effective for encoderless actual value sensing (p9306/p9506 = 1).		
p9388	Minimum current actual value sensing encoderless (Motor Module) / SI Mtn I_min EL MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10.00 [%]
Description:	Sets the minimum current for encoderless actual value sensing. Reducing this percentage value can adversely affect actual value sensing. - The value must be increased if C30711 has occurred with message value 1042. - The value must be decreased if C30711 has occurred with message value 1041.		
Dependency:	Refer to: C30711		
Note:	This parameter is only effective for encoderless actual value sensing (p9306/p9506 = 1).		

p9389	SI Motion voltage tolerance acceleration (Motor Module) / SI Mtn V tol MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 10.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the voltage tolerance for suppressing acceleration peaks. Increasing this percentage value means that voltage peaks will need to have a higher amplitude during acceleration procedures if they are not to affect actual value sensing. - The value must be increased if C30711 has occurred with message value 1042. - The value must be decreased if acceleration procedures have led to an excessive safety actual velocity.		
Dependency:	Refer to: C30711		
Note:	This parameter is only effective for encoderless actual value sensing (p9306/p9506 = 1).		
r9390[0...3]	SI Motion version safety motion monitoring (Motor Module) / SI Mtn Version MM		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the Safety Integrated version for the safety motion monitoring functions on the Motor Module.		
Index:	[0] = Safety Version (major release) [1] = Safety Version (minor release) [2] = Safety Version (baselevel or patch) [3] = Safety Version (hotfix)		
Dependency:	Refer to: r9590, r9770, r9870, r9890		
Note:	Example: r9390[0] = 2, r9390[1] = 60, r9390[2] = 1, r9390[3] = 0 --> SI Motion version V02.60.01.00		
r9398[0...1]	SI Motion actual checksum SI parameters (Motor Module) / SI Mtn act CRC MM		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the checksum for the checked Safety Integrated parameters of the motion monitoring function (actual checksum) on the Motor Module.		
Index:	[0] = Checksum over SI parameters for motion monitoring [1] = Checksum over SI parameters with hardware reference		
Dependency:	Refer to: p9399		
Note:	SI: Safety Integrated		

p9399[0...1]	SI Motion reference checksum SI parameters (Motor Module) / SI Mtn ref CRC MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the checksum for the checked Safety Integrated parameters of the motion monitoring functions (reference checksum) on the Motor Module.		
Index:	[0] = Checksum over SI parameters for motion monitoring [1] = Checksum over SI parameters with hardware reference		
Dependency:	Refer to: r9398		
Note:	SI: Safety Integrated		
r9406[0...19]	PS file parameter number parameter not transferred / PS parameter No.		
All objects	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the parameters that were not able to be transferred when reading the parameter back-up files (PS files) from the non-volatile memory (e.g. memory card). r9406[0] = 0 --> All of the parameter values were able to be transferred error-free. r9406[0...x] > 0 --> Displays the parameter number whose value was not able to be completely transferred or for an indexed parameter, for at least 1 index, was not able to be transferred. The first index that is not transferred is displayed in r9407.		
Dependency:	Refer to: r9407, r9408		
Note:	All indices from r9406 to r9408 designate the same parameter. r9406[x] parameter number, parameter not accepted r9407[x] parameter index, parameter not accepted r9408[x] fault code, parameter not accepted		
r9407[0...19]	PS file parameter index parameter not transferred / PS parameter index		
All objects	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 1 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the first index of the parameters that could not be transferred when the parameter backup files (PS files) were read from the non-volatile memory (e.g. memory card). If, from an indexed parameter, at least one index was not able to be transferred, then the parameter number is displayed in r9406[n] and the first index that was not transferred is displayed in r9407[n]. r9406[0] = 0 --> All of the parameter values were able to be transferred error-free. r9406[n] > 0 --> Displays r9407[n] the first index of the parameter number r9406[n] that was not transferred.		
Dependency:	Refer to: r9406, r9408		

Note: All indices from r9406 to r9408 designate the same parameter.
r9406[x] parameter number, parameter not accepted
r9407[x] parameter index, parameter not accepted
r9408[x] fault code, parameter not accepted

r9408[0...19] PS file fault code parameter not transferred / PS fault code

All objects	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Only for internal Siemens service purposes.

Dependency: Refer to: r9406, r9407

Note: All indices from r9406 to r9408 designate the same parameter.
r9406[x] parameter number, parameter not accepted
r9407[x] parameter index, parameter not accepted
r9408[x] fault code, parameter not accepted

r9409 Number of parameters to be saved / Qty par to save

All objects	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the number of modified parameters and those that have still not be saved for this drive object.

Dependency: Refer to: p0971, p0977

Note: The modified parameters that still need to be saved are internally listed in r9410 ... r9419.

r9450[0...29] Reference value change parameter with unsuccessful calculation / Ref_chg par n poss

A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VEC- TOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the parameters for which the re-calculation was unsuccessful after an internal system reference value change.

Dependency: Refer to: F07086

r9451[0...29] Units changeover adapted parameters / Unit_chngov par

A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VEC- TOR	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the parameters whose parameter would have to be changed during a units changeover.

Dependency: Refer to: F07088

r9481 Number of BICO interconnections / BICO count			
A_INF, B_INF, CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VEC- TOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min	Max	Factory setting
	-	-	-
Description:	Displays the number of BICO interconnections (signal sinks). The selected BICO interconnections should be entered into r9482[0...59] and r9483[0...59].		
Dependency:	Refer to: r9482, r9483		

r9482[0...59] BICO interconnections BI/CI parameters / BICO BI/CI par			
A_INF, B_INF, CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VEC- TOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Commands	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min	Max	Factory setting
	-	-	-
Description:	Displays the signal sinks (binector/connector inputs, BI/CI parameters). The number of BICO interconnections is displayed in r9481.		
Dependency:	Refer to: r9481, r9483		
Note:	The list is sorted according to signal sources and is structured as follows: r9842[0]: Interconnection 1 (signal sink, BICO coded), r9843[0]: Interconnection 1 (signal source, BICO coded) r9842[1]: Interconnection 2 (signal sink, BICO coded), r9843[1]: Interconnection 2 (signal source, BICO coded) ...		

r9483[0...59] BICO interconnections BO/CO parameters / BICO BO/CO par

A_INF, B_INF,
CU_CX32, CU_I,
CU_LINK, CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, S_INF,
SERVO, TB30,
TM120, TM15,
TM15DI_DO, TM17,
TM31, TM41,
TM54F_MA,
TM54F_SL, VEC-
TOR

Can be changed: -
Data type: Unsigned32
P-Group: Commands
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: -
Unit selection: -
Expert list: 0

Min

-

Max

-

Factory setting

-

Description: Displays the signal sources (binector/connector outputs, BO/CO parameters).
The number of BICO interconnections is displayed in r9481.

Dependency: Refer to: r9481, r9482

Note: The list is sorted according to signal sources and is structured as follows:
r9842[0]: Interconnection 1 (signal sink, BICO coded), r9843[0]: Interconnection 1 (signal source, BICO coded)
r9842[1]: Interconnection 2 (signal sink, BICO coded), r9843[1]: Interconnection 2 (signal source, BICO coded)
...

p9484 BICO interconnections search signal source / BICO S_src srch

A_INF, B_INF,
CU_CX32, CU_I,
CU_LINK, CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, S_INF,
SERVO, TB30,
TM120, TM15,
TM15DI_DO, TM17,
TM31, TM41,
TM54F_MA,
TM54F_SL, VEC-
TOR

Can be changed: T
Data type: Unsigned32
P-Group: -
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: -
Unit selection: -
Expert list: 0

Min

0

Max

4294967295

Factory setting

0

Description: Sets the signal source (BO/CO parameter, BICO coded) to search in the signal sinks.
The question is answered:
How often is a connection made to a signal source in the drive object and from which index are these interconnections saved (r9482[0...59], r9483[0...59])?

Dependency: Refer to: r9481, r9482, r9483, r9485, r9486

r9485	BICO interconnections signal source search count / BICO S_src srchQty		
A_INF, B_INF, CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VEC- TOR	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0
	Min	Max	Factory setting
	-	-	-
Description:	Displays the number of BICO interconnections to the signal sink being searched for. The signal source to be searched is set in p9484 (BICO-coded). The search result is contained in r9482[0...59] and r9483[0...59] and is specified by the count (r9485) and the first index (r9486).		
Dependency:	Refer to: r9481, r9482, r9483, p9484, r9486		
r9486	BICO interconnections signal source search first index / BICO S_src srchIdx		
A_INF, B_INF, CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VEC- TOR	Can be changed: - Data type: Unsigned16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0
	Min	Max	Factory setting
	-	-	-
Description:	Displays the first index of the signal source being searched for. The signal source to be searched is set in p9484 (BICO-coded). The search result is contained in r9482[0...59] and r9483[0...59] and is specified by the count (r9485) and the first index (r9486).		
Dependency:	Refer to: r9481, r9482, r9483, p9484, r9485		
r9490	Number of BICO interconnections to other drives / Qty BICO to drive		
All objects	Can be changed: - Data type: Unsigned16 P-Group: Commands Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the number of signal sources from this drive to other drives/drive objects (Binector Output/Connector Output, BO/CO).		
Dependency:	Refer to: r9491, r9492, p9493		

r9491[0...9]	BI/CI of BICO interconnections to other drives / BI/CI to drive		
All objects	Can be changed: - Data type: Unsigned32 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the signal receiver list (Binector Input/Connector Input, BI/CI) for the first interconnections between this drive and other drives/drive objects.		
Dependency:	Refer to: r9490, r9492, p9493		
Notice:	A drive cannot be deleted if this list is not empty! Otherwise, another drive would continue to attempt to read a signal from a drive that no longer existed.		
Note:	All indices of r9491 to p9493 designate the same interconnection. r9491[x] contains the signal receiver and r9492[x] the matching signal source; p9493[x] can be set to modify the interconnection.		
r9492[0...9]	BO/CO of BICO interconnections to other drives / BO/CO to drive		
All objects	Can be changed: - Data type: Unsigned32 P-Group: Commands Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the signal source list (Binector Output/Connector Output, BO/CO) for the first interconnections between this drive and other drives/drive objects.		
Dependency:	Refer to: r9490, r9491, p9493		
Notice:	A drive cannot be deleted if this list is not empty! Otherwise, another drive would continue to attempt to read a signal from a drive that no longer existed.		
Note:	All indices of r9491 to p9493 designate the same interconnection. r9491[x] contains the signal receiver and r9492[x] the matching signal source; p9493[x] can be set to modify the interconnection.		
p9493[0...9]	Reset BICO interconnections to other drives / Reset BICO to drv		
All objects	Can be changed: T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 15	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 15
Description:	Setting to reset the BICO interconnections to other drives. Each interconnection can be individually reset.		
Value:	0: Set connection to 0 1: Set connection to 1 (100 %) 2: Set connection to factory setting 15: Finished		
Dependency:	Refer to: r9490, r9491, r9492		
Note:	All indices of r9491 to p9493 designate the same interconnection. r9491[x] contains the signal receiver and r9492[x] the matching signal source; p9493[x] can be set to modify the interconnection.		

p9495	BICO behavior to deactivated drive objects / Behav to deact obj		
A_INF, B_INF, CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VEC- TOR	Can be changed: T Data type: Integer16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 2	Factory setting 0
Description:	Setting for the behavior for BICO interconnections to other non-operational/deactivated drive objects. If this drive object has BICO interconnections to other non-operational or deactivated drive objects as drain, then using these parameters, all of the associated BI/CI parameters of this drive object can be marked and then set to the factory setting or only marked.		
Value:	0: Do not do anything 1: Mark connection 2: Mark connection and then set to the factory setting		
Dependency:	Refer to: p9496, p9497, p9498, p9499 Refer to: A01318, A01507		
Note:	The BI/CI parameters involved are listed in r9498[0...29] (drain). The associated BO/CO parameters are listed in r9499[0...29] (source). However, r9498 and r9499 are only then filled if p9495 is not equal to 0, otherwise they remain empty.		
p9496	Restore BICO to the drive objects that are now activated / Rest BICO act obj		
A_INF, B_INF, CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VEC- TOR	Can be changed: T Data type: Integer16 P-Group: - Not for motor type: -	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1
	Min 0	Max 2	Factory setting 0
Description:	If this drive object has BICO interconnections to other drive objects that are either not operational or have been deactivated, then using this parameter, all of the BI/CI parameters involved with this drive object can be re-established.		
Value:	0: Do not do anything 1: Restore the connections from the list 2: Delete the connection from the list		
Dependency:	Refer to: p9495, p9497, p9498, p9499 Refer to: A01318, A01507		
Note:	The BI/CI parameters involved are listed in r9498[0...29] (drain). The associated BO/CO parameters are listed in r9499[0...29] (source). After setting p9496 to 1 or 2, r9498 and r9499 are reset, r9497 is set to 0 and p9496 itself is set to 0.		

p9497 BICO number of interconnections to deactivated drive objects / Qty to deact obj

A_INF, B_INF,
CU_CX32, CU_I,
CU_LINK, CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, SERVO,
TB30, TM120, TM15,
TM15DI_DO, TM17,
TM31, TM41, VEC-
TOR

Can be changed: T
Data type: Unsigned16
P-Group: Commands
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: -
Unit selection: -
Expert list: 1

Min
0

Max
65535

Factory setting
0

Description: Displays the number of signal sinks of this drive object to other drives/drive objects that are no longer operational/deactivated (Binector Input/Connector Input, BI/CI).

Dependency: Refer to: p9495, p9496, p9498, p9499
Refer to: A01318, A01507

Note: The parameter is only used for display purposes and cannot be written into.

p9498[0...29] BICO BI/CI parameters to deactivated drive objects / BI/CI to deact obj

A_INF, B_INF,
CU_CX32, CU_I,
CU_LINK, CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, SERVO,
TB30, TM120, TM15,
TM15DI_DO, TM17,
TM31, TM41, VEC-
TOR

Can be changed: T
Data type: Unsigned32
P-Group: Commands
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: -
Unit selection: -
Expert list: 1

Min
-

Max
-

Factory setting
0

Description: List of the BI/CI parameters that represent a connection to non-operational/deactivated drive objects.

Dependency: Refer to: p9495, p9496, p9497, p9499
Refer to: A01318, A01507

Note: All indices from r9498 to r9499 designate the same BICO interconnection.
This signal sink is in r9498[x] and the associated signal source in r9499[x].

p9499[0...29] BICO BO/CO parameters to deactivated drive objects / BO/CO to deact obj

A_INF, B_INF,
CU_CX32, CU_I,
CU_LINK, CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150,
ENCODER, SERVO,
TB30, TM120, TM15,
TM15DI_DO, TM17,
TM31, TM41, VEC-
TOR

Can be changed: T
Data type: Unsigned32
P-Group: Commands
Not for motor type: -

Calculated: -
Dynamic index: -
Units group: -
Scaling: -

Access level: 3
Func. diagram: -
Unit selection: -
Expert list: 1

Min
-

Max
-

Factory setting
0

Description: List of the BO/CO parameters that represent a connection to non-operational/deactivated drive objects.

Dependency: Refer to: p9495, p9496, p9497, p9498
Refer to: A01318, A01507

Note: All indices from r9498 to r9499 designate the same BICO interconnection.
This signal sink is in r9498[x] and the associated signal source in r9499[x].

p9500 SI Motion monitoring clock cycle (Control Unit) / SI Mtn clock CU	
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.50 [ms] Max 25.00 [ms]
	Calculated: - Dynamic index: - Units group: - Scaling: - Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 12.00 [ms]
Description:	Sets the monitoring clock cycle for safe motion monitoring.
Dependency:	Refer to: r2064, p9511 Refer to: F01652
Note:	A change only becomes effective after a POWER ON. The monitoring clock cycle must be a multiple of the actual value sensing clock cycle in p9511 (dbSI) or of the DP clock cycle (ncSI).

p9501					SI Motion enable safety functions (Control Unit) / SI Mtn enable CU						
SERVO, VECTOR					Can be changed: C2(95)		Calculated: -		Access level: 3		
					Data type: Unsigned32		Dynamic index: -		Func. diagram: -		
					P-Group: Safety Integrated		Units group: -		Unit selection: -		
					Not for motor type: -		Scaling: -		Expert list: 1		
					Min		Max		Factory setting		
					-		-		0000 bin		
Description:					Sets the enable signals for the safe motion monitoring.						
Bit field:					Bit	Signal name	1 signal	0 signal	FP		
					00	SOS/SLS (SBH/SG) enable	Enable	Inhibit	-		
					03	Actual value synchronization enable	Enable	Inhibit	-		
					16	Enable NX Hys Fil	Enable	Inhibit	-		
Dependency:					Refer to: F01682, F01683						
Note:					A change only becomes effective after a POWER ON.						
					SLS: Safely-Limited Speed / SG: Safely reduced speed						
					SOS: Safe Operating Stop / SBH: Safe operating stop						

p9502 SI Motion axis type (Control Unit) / SI Mtn ax type CU	
SERVO, VECTOR	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0 Max 1
	Calculated: - Dynamic index: - Units group: - Scaling: - Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the axis type (linear axis or rotary axis/spindle).
Value:	0: Linear axis 1: Rot axis/spindle
Note:	For the commissioning software, after changing over the axis type, the units dependent on the axis type are only updated after a project upload. A change only becomes effective after a POWER ON.

p9503	SI Motion SCA (SN) enable (Control Unit) / SI Mtn SCA enab			
SERVO, VECTOR	Can be changed: U, T Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 bin	
Description:	Setting to enable the function "Safe Cam" (SCA).			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	SCA1 (SN1) enable	Enable	Inhibit
	01	SCA2 (SN2) enable	Enable	Inhibit
	02	SCA3 (SN3) enable	Enable	Inhibit
	03	SCA4 (SN4) enab	Enable	Inhibit
	04	SCA5 (SN5) enab	Enable	Inhibit
	05	SCA6 (SN6) enable	Enable	Inhibit
	06	SCA7 (SN7) enable	Enable	Inhibit
	07	SCA8 (SN8) enable	Enable	Inhibit
	08	SCA9 (SN9) enable	Enable	Inhibit
	09	SCA10 (SN10) enable	Enable	Inhibit
	10	SCA11 (SN11) enable	Enable	Inhibit
	11	SCA12 (SN12) enable	Enable	Inhibit
	12	SCA13 (SN13) enable	Enable	Inhibit
	13	SCA14 (SN14) enable	Enable	Inhibit
	14	SCA15 (SN15) enable	Enable	Inhibit
	15	SCA16 (SN16) enable	Enable	Inhibit
	16	ESCA17 (SN17) enable	Enable	Inhibit
	17	SCA18 (SN18) enable	Enable	Inhibit
	18	SCA19 (SN19) enable	Enable	Inhibit
	19	SCA20 (SN20) enable	Enable	Inhibit
	20	SCA21 (SN21) enable	Enable	Inhibit
	21	SCA22 (SN22) enable	Enable	Inhibit
	22	SCA23 (SN23) enable	Enable	Inhibit
	23	SCA24 (SN24) enable	Enable	Inhibit
	24	SCA25 (SN25) enable	Enable	Inhibit
	25	SCA26 (SN26) enable	Enable	Inhibit
	26	SCA27 (SN27) enable	Enable	Inhibit
	27	SCA28 (SN28) enable	Enable	Inhibit
	28	SCA29 (SN29) enable	Enable	Inhibit
	29	SCA30 (SN30) enable	Enable	Inhibit
Dependency:	Refer to: p9501 Refer to: F01686			
Note:	The "Safe Cam" function (SCA) can either be enabled using p9501 or p9503. SCA: Safe Cam / SN: Safe software cam			

p9505	SI Motion SCA (SN) modulo value (Control Unit) / SI Mtn SCA modulo			
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0 [°]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 737280 [°]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0 [°]	
Description:	Sets the modulo range of the safe position actual value in degrees for the function "Safe Cam" (SCA) for rotary axes.			
Dependency:	Refer to: p9536, p9537			
Note:	SCA: Safe Cam / SN: Safe software cam			

p9506	SI Motion function specification (Control Unit) / SI Mtn fct_spc CU		
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 1	Factory setting 0
Description:	Sets the function specification for Safety Integrated.		
Value:	0: Safety with encoder 1: Safety without encoder		
Dependency:	Refer to: C01711		

p9507	SI Motion function specification (Control Unit) / SI Mtn config				
SERVO, VECTOR	Can be changed: C2(95)		Calculated: -	Access level: 3	
	Data type: Unsigned32		Dynamic index: -	Func. diagram: -	
	P-Group: Safety Integrated		Units group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min		Max	Factory setting	
	-		-	0000 bin	
Description:	Function configuration for Safe Motion Monitoring				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Extended alarm acknowledgment	Yes	No	-
	01	Setpoint speed limit for stop F	No	Yes	-
Dependency:	Refer to: C01711				

p9510	SI Motion clock-cycle synchronous PROFIBUS master / SI Mtn sync master		
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 1	Factory setting 0
Description:	Setting for the clock-cycle synchronous PROFIBUS master. The parameter must be set if the safety-relevant motion monitoring functions integrated in the drive are enabled and there is a clock-cycle synchronous PROFIBUS master. This is, for example, the case when using the following controls: - clock-cycle synchronous control for the motion control (e.g. SIMOTION). - clock-cycle synchronous PROFIsafe master (e.g. SIMATIC S7-400F).		
Value:	0: No clock-cycle synchronous PROFIBUS master 1: Clock-cycle synchronous PROFIBUS master present		
Dependency:	Refer to: C01711, A01796		
Notice:	As of firmware V2.6, the parameter has no effect.		

p9511		SI Motion actual value sensing cycle clock (Control Unit) / SI Mtn act clk CU	
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0000 [ms]	Max 25.0000 [ms]	Factory setting 0.0000 [ms]
Description:	Sets the clock cycle time of the actual value sensing for safe motion monitoring. Setting criteria if the motion monitoring functions are executed with an encoder. - A slower clock cycle time reduces the maximum permissible velocity - however, it ensures a lower load of the Control Unit for safe actual value sensing. - The maximum permissible velocity which, when exceeded, can mean that errors occur during safe actual value sensing, is displayed in r9730. - The isochronous PROFIBUS clock cycle is used as a clock cycle time for actual value sensing with a setting of 0 ms; the setting is 1 ms if isochronous operation is not being used. Setting criteria if the motion monitoring functions are executed without an encoder: - The actual value sensing clock cycle must be set to the same value as the current controller clock cycle (p115).		
Dependency:	Refer to: p0115 Refer to: F01652		
Note:	The parameter is only active for drive-based motion monitoring functions (p9601.2 = 1). The monitoring clock cycle from p9500 must be an integer multiple of this parameter. In the case of motion monitoring functions with encoder, the clock cycle time for actual value sensing must be an integer multiple of the current controller clock cycle and at least 4 times slower than the current controller clock cycle. A factor of at least 8 is recommended. The clock cycle time of the actual value sensing should not be set to more than 8 ms. A change only becomes effective after a POWER ON.		

p9515		SI Motion encoder coarse position value config (Control Unit) / SI Mtn s config CU			
SERVO, VECTOR	Can be changed: C2(95)		Calculated: -		Access level: 3
	Data type: Unsigned32		Dynamic index: -		Func. diagram: -
	P-Group: Safety Integrated		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min -		Max -		Factory setting 0000 bin
Description:		Sets the encoder configuration for the redundant coarse position value.			
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Incrementer	Yes	No	-
	01	Encoder CRC least significant byte first	Yes	No	-
	02	Redundant coarse position val. most signifi- cant bit left-aligned	Yes	No	-
	16	DRIVE-CLiQ encoder	Yes	No	-
Dependency:		Refer to: r0474, p9315			
Note:		For safe functions that are not enabled (p9501 = 0), the following applies: - p9515 is automatically set the same as p0474 when the system boots. For safety functions that are enabled (p9501 > 0), the following applies: - p9515 is checked to see that it matches p0474.			

p9516	SI Motion encoder configuration safety functions (Control Unit) / SI Mtn enc_cfg CU				
SERVO, VECTOR	Can be changed: C2(95)		Calculated: -		Access level: 3
	Data type: Unsigned16		Dynamic index: -		Func. diagram: -
	P-Group: Safety Integrated		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		0000 bin
Description:	Sets the configuration for the motor encoder and position actual value.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Motor encoder, rotating/linear	Linear	Rotating:	-
	01	Position actual value, sign change	Yes	No	-
Dependency:	Refer to: p0404, p0410 Refer to: F01671				
Note:	For safe functions that are not enabled (p9501 = 0), the following applies: - p9516.0 is automatically set when booting as for p0410.0. When booting, p9516.1 is automatically set as for p0404.1. For safety functions that are enabled (p9501 > 0), the following applies: - p9516.1 is checked to identify whether it tallies with p0404.1.				
p9517	SI Motion linear scale grid division (Control Unit) / SI Mtn grid CU				
SERVO, VECTOR	Can be changed: C2(95)		Calculated: -		Access level: 3
	Data type: FloatingPoint32		Dynamic index: -		Func. diagram: -
	P-Group: Safety Integrated		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	0.00 [nm]		250000000.00 [nm]		10000.00 [nm]
Description:	Sets the grid division for a linear motor encoder.				
Dependency:	Refer to: p0407, p9516 Refer to: F01671				
Note:	For safety functions that have not been enabled (p9501 = 0), the following applies: When booting p9517 is automatically set the same as p0407. For safety functions that are enabled (p9501 > 0), the following applies: p9517 is checked whether it coincides with p0407.				
p9518	SI Motion encoder pulses per revolution (Control Unit) / SI Mtn puls/rev CU				
SERVO, VECTOR	Can be changed: C2(95)		Calculated: -		Access level: 3
	Data type: Unsigned32		Dynamic index: -		Func. diagram: -
	P-Group: Safety Integrated		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	0		100000		2048
Description:	Sets the number of encoder pulses per revolution for rotary motor encoders.				
Dependency:	Refer to: p0408, p9516 Refer to: F01671				
Note:	For safety functions that have not been enabled (p9501 = 0), the following applies: When booting, p9518 is automatically set the same as p0408. For safety functions that are enabled (p9501 > 0), the following applies: p9518 is checked whether it coincides with p0408.				

p9519	SI Motion fine resolution G1_XIST1 (Control Unit) / SI Mtn G1_XIST1 CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 2	Calculated: - Dynamic index: - Units group: - Scaling: - Max 18	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 11
Description:	Sets the fine resolution for G1_XIST1 in bits.		
Dependency:	Refer to: p0418 Refer to: F01671		
Note:	For safety functions that have not been enabled (p9501 = 0), the following applies: When booting, p9519 is automatically set the same as p0418. For safety functions that are enabled (p9501 > 0), the following applies: p9519 is checked whether it coincides with p0418. G1_XIST1: Encoder 1 position actual value 1 (PROFIdrive)		
p9520	SI Motion spindle pitch (Control Unit) / SI Mtn Sp_pitch CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.1000 [mm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8388.0000 [mm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10.0000 [mm]
Description:	Sets the gear ratio between the encoder and load in mm/revolution for a linear axis with rotary encoder.		
Notice:	The fourth decimal point can be rounded-off depending on the size of the entered number (from three places before the decimal point).		
p9521[0...7]	SI Motion gearbox enc (motor)/load denominator (Control Unit) / SI Mtn gear denom		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147000000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the denominator for the gearbox between the encoder (or motor in the case of encoderless monitoring functions) and the load. It is not possible to change over the gearbox stages. Gearbox stage 0 is always active.		
Index:	[0] = Gearbox 1 [1] = Gearbox 2 [2] = Gearbox 3 [3] = Gearbox 4 [4] = Gearbox 5 [5] = Gearbox 6 [6] = Gearbox 7 [7] = Gearbox 8		
Dependency:	Refer to: p9522		

p9522[0...7]	SI Motion gearbox encoder (motor)/load numerator (Control Unit) / SI Mtn gear numer		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2147000000	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the numerator for the gearbox between the encoder (or motor in the case of encoderless monitoring functions) and the load. It is not possible to change over the gearbox stages. Gearbox stage 0 is always active.		
Index:	[0] = Gearbox 1 [1] = Gearbox 2 [2] = Gearbox 3 [3] = Gearbox 4 [4] = Gearbox 5 [5] = Gearbox 6 [6] = Gearbox 7 [7] = Gearbox 8		
Dependency:	Refer to: p9521		
Note:	In the case of encoderless monitoring functions (p9506), the pole pair number must be multiplied by the numerator of the gearbox ratio. Example: Gearbox ratio 1:4, pole pair number (r0313) = 2 --> p9521 = 1, p9522 = 8 (4*2)		
p9523	SI Motion redundant coarse pos. value valid bits (Control Unit) / Valid bits CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 9
Description:	Sets the number of valid bits of the redundant coarse position value. The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.		
Dependency:	Refer to: r0470, p9323		
Note:	For safe functions that are not enabled (p9501 = 0), the following applies: - p9523 is automatically set the same as r0470 when the system boots. For safety functions that are enabled (p9501 > 0), the following applies: - p9523 is checked to see that it matches r0470.		
p9524	SI Motion Redundant coarse pos. value fine resolution bits (CU) / SI Mtn fine bit CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min -16	Calculated: - Dynamic index: - Units group: - Scaling: - Max 16	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -2
Description:	Sets the number of valid bits for the fine resolution of the redundant coarse position value. The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.		
Dependency:	Refer to: r0471, p9324		

Note: For safe functions that are not enabled (p9501 = 0), the following applies:
 - p9524 is automatically set the same as r0471 when the system boots.
 For safety functions that are enabled (p9501 > 0), the following applies:
 - p9524 is checked to see that it matches r0471.

p9525 SI Motion Redundant coarse pos. value relevant bits (CU) / Relevant bits CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 16	Factory setting 16

Description: Sets the number of relevant bits for the redundant coarse position value.

Dependency: Refer to: p0414, r0472, p9325

Note: For safe functions that are not enabled (p9501 = 0), the following applies:
 - p9525 is automatically set the same as r0472 when the system boots.
 For safety functions that are enabled (p9501 > 0), the following applies:
 - p9525 is checked to see that it matches r0472.

p9526 SI Motion encoder assignment second channel / SI Mtn enc chan 2

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 1	Max 3	Factory setting 1

Description: Sets the number of the encoder that the second channel (control, Motor Module) uses for safe motion monitoring functions.

Dependency: For the safe motion monitoring functions, the redundant safety position actual value sensing must be activated in the appropriate encoder data set (p0430.19 = 1).
 Refer to: p0187, p0188, p0189, p0430

Note: For p9526 = 1, the encoder for the closed-loop speed control is used for the second channel of the motion monitoring functions (1-encoder system).
 A change only becomes effective after a POWER ON.

p9529 SI Motion Gx_XIST1 Safe most significant bit (CU) / Gx_XIST1 MSB CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 31	Factory setting 14

Description: Sets the bit number for the safe most significant bit (MSB) of the Gx_XIST1 coarse position.

Dependency: Refer to: p0415, r0475, p9329

Note: For safe functions that are not enabled (p9501 = 0), the following applies:
 - p9529 is automatically set the same as r0475 when the system boots.
 For safety functions that are enabled (p9501 > 0), the following applies:
 - p9529 is checked to see that it matches r0475.
 MSB: Most Significant Bit

p9530	SI Motion standstill tolerance (Control Unit) / SI Mtn standst_tol		
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [mm]	Max 100.000 [mm]	Factory setting 1.000 [mm]
Description:	Sets the tolerance for the function "Safe Operating Stop" (SOS).		
Dependency:	Refer to: C01707		
Note:	SOS: Safe Operating Stop / SBH: Safe operating stop		

p9530	SI Motion standstill tolerance (Control Unit) / SI Mtn standst_tol		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [°]	Max 100.000 [°]	Factory setting 1.000 [°]
Description:	Sets the tolerance for the function "Safe Operating Stop" (SOS).		
Dependency:	Refer to: C01707		
Note:	SOS: Safe Operating Stop / SBH: Safe operating stop		

p9531[0...3]	SI Motion SLS (SG) limit values (Control Unit) / SI Mtn SLS lim CU		
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [mm/min]	Max 1000000.00 [mm/min]	Factory setting 2000.00 [mm/min]
Description:	Sets the limit values for the function "Safely-Limited Speed" (SLS).		
Index:	[0] = Limit value SLS1 [1] = Limit value SLS2 [2] = Limit value SLS3 [3] = Limit value SLP4		
Dependency:	Refer to: p9532, p9561, p9563 Refer to: C01714		
Note:	SLS: Safely-Limited Speed / SG: Safely reduced speed		

p9531[0...3]	SI Motion SLS (SG) limit values (Control Unit) / SI Mtn SLS lim CU		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 1000000.00 [rpm]	Factory setting 2000.00 [rpm]
Description:	Sets the limit values for the function "Safely-Limited Speed" (SLS).		
Index:	[0] = Limit value SLS1 [1] = Limit value SLS2 [2] = Limit value SLS3 [3] = Limit value SLP4		

Dependency: Refer to: p9532, p9561, p9563
Refer to: C01714

Note: SLS: Safely-Limited Speed / SG: Safely reduced speed

p9532[0...15]		SI Motion SLS (SG) override factor (Control Unit) / SI Mtn SLS over	
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [%]	Max 100.000 [%]	Factory setting 100.000 [%]
Description:	Sets the override factor for the limit value for SLS2 and SLS4 for the function "Safely-Limited Speed" (SLS).		
Index:	[0] = SLS (SG) override factor 0 [1] = SLS (SG) override factor 1 [2] = SLS (SG) override factor 2 [3] = SLS (SG) override factor 3 [4] = SLS (SG) override factor 4 [5] = SLS (SG) override factor 5 [6] = SLS (SG) override factor 6 [7] = SLS (SG) override factor 7 [8] = SLS (SG) override factor 8 [9] = SLS (SG) override factor 9 [10] = SLS (SG) override factor 10 [11] = SLS (SG) override factor 11 [12] = SLS (SG) override factor 12 [13] = SLS (SG) override factor 13 [14] = SLS (SG) override factor 14 [15] = SLS (SG) override factor 15		
Dependency:	Refer to: p9501, p9531		
Note:	The actual override factor for SLS2 and SLS4 is selected using the safety-relevant inputs (SGE). SLS: Safely-Limited Speed / SG: Safely reduced speed		

p9533		SI Motion SLS setpoint velocity limiting (Control Unit) / SI Mtn SLS set_lim	
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.000 [%]	Max 100.000 [%]	Factory setting 80.000 [%]
Description:	This is an evaluation factor to define the setpoint limit from the selected actual speed limit. The active SLS limit value is evaluated with this factor and is made available as setpoint limit in r9733: r9733 = p9533 * p9531[0...3]		
Dependency:	This parameter only has to be parameterized for the motion monitoring functions integrated in the drive (p9601.2 = 1) Refer to: p9501, p9531, p9601		
Note:	The active actual speed limit is selected via safety-relevant inputs (SGE). When selecting SOS or a STOP A ... D, setpoint 0 is specified in r9733. SLS: Safely-Limited Speed		

p9534[0...1] SI Motion SLP (SE) upper limit values (Control Unit) / SI Mtn SLP up lim			
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -2147000.000 [mm]	Max 2147000.000 [mm]	Factory setting 100000.000 [mm]
Description:	Sets the upper limit for the function "Safely-Limited Position" (SLP).		
Index:	[0] = Limit value SLP1 (SE1) [1] = Limit value SLP2 (SE2)		
Dependency:	Refer to: p9501, p9535, p9562		
Note:	For the setting of these limit values, the following applies: p9534 > p9535 A change only becomes effective after a POWER ON. SLP: Safely-Limited Position / SE: Safe software limit switches		

p9534[0...1] SI Motion SLP (SE) upper limit values (Control Unit) / SI Mtn SLP up lim			
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -2147000.000 [°]	Max 2147000.000 [°]	Factory setting 100000.000 [°]
Description:	Sets the upper limit for the function "Safely-Limited Position" (SLP).		
Index:	[0] = Limit value SLP1 (SE1) [1] = Limit value SLP2 (SE2)		
Dependency:	Refer to: p9501, p9535, p9562		
Note:	For the setting of these limit values, the following applies: p9534 > p9535 A change only becomes effective after a POWER ON. SLP: Safely-Limited Position / SE: Safe software limit switches		

p9535[0...1] SI Motion SLP (SE) lower limit values (Control Unit) / SI Mtn SLP low lim			
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -2147000.000 [mm]	Max 2147000.000 [mm]	Factory setting -100000.000 [mm]
Description:	Sets the lower limit for the function "Safely-Limited Position" (SLP).		
Index:	[0] = Limit value SLP1 (SE1) [1] = Limit value SLP2 (SE2)		
Dependency:	Refer to: p9501, p9534, p9562		
Note:	For the setting of these limit values, the following applies: p9534 > p9535 A change only becomes effective after a POWER ON. SLP: Safely-Limited Position / SE: Safe software limit switches		

p9535[0...1]	SI Motion SLP (SE) lower limit values (Control Unit) / SI Mtn SLP low lim		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -2147000.000 [°]	Max 2147000.000 [°]	Factory setting -100000.000 [°]
Description:	Sets the lower limit for the function "Safely-Limited Position" (SLP).		
Index:	[0] = Limit value SLP1 (SE1) [1] = Limit value SLP2 (SE2)		
Dependency:	Refer to: p9501, p9534, p9562		
Note:	For the setting of these limit values, the following applies: p9534 > p9535 A change only becomes effective after a POWER ON. SLP: Safely-Limited Position / SE: Safe software limit switches		

p9536[0...29]	SI Motion SCA (SN) plus cam position (Control Unit) / SI Mtn SCA+		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -2147000.000 [mm]	Max 2147000.000 [mm]	Factory setting 10.000 [mm]
Description:	Sets the plus cam position for the function "Safe Cam" (SCA).		
Index:	[0] = Cam position SCA1 (SN1) [1] = Cam position SCA2 (SN2) [2] = Cam position SCA3 (SN3) [3] = Cam position SCA4 (SN4) [4] = Cam position SCA5 (SN5) [5] = Cam position SCA6 (SN6) [6] = Cam position SCA7 (SN7) [7] = Cam position SCA8 (SN8) [8] = Cam position SCA9 (SN9) [9] = Cam position SCA10 (SN10) [10] = Cam position SCA11 (SN11) [11] = Cam position SCA12 (SN12) [12] = Cam position SCA13 (SN13) [13] = Cam position SCA14 (SN14) [14] = Cam position SCA15 (SN15) [15] = Cam position SCA16 (SN16) [16] = Cam position SCA17 (SN17) [17] = Cam position SCA18 (SN18) [18] = Cam position SCA19 (SN19) [19] = Cam position SCA20 (SN20) [20] = Cam position SCA21 (SN21) [21] = Cam position SCA22 (SN22) [22] = Cam position SCA23 (SN23) [23] = Cam position SCA24 (SN24) [24] = Cam position SCA25 (SN25) [25] = Cam position SCA26 (SN26) [26] = Cam position SCA27 (SN27) [27] = Cam position SCA28 (SN28) [28] = Cam position SCA29 (SN29) [29] = Cam position SCA30 (SN30)		
Dependency:	Refer to: p9501, p9503, p9537		
Note:	A change only becomes effective after a POWER ON. SCA: Safe Cam / SN: Safe software cam		

p9536[0...29] SI Motion SCA (SN) plus cam position (Control Unit) / SI Mtn SCA+			
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-2147000.000 [°]	2147000.000 [°]	10.000 [°]
Description:	Sets the plus cam position for the function "Safe Cam" (SCA).		
Index:	[0] = Cam position SCA1 (SN1) [1] = Cam position SCA2 (SN2) [2] = Cam position SCA3 (SN3) [3] = Cam position SCA4 (SN4) [4] = Cam position SCA5 (SN5) [5] = Cam position SCA6 (SN6) [6] = Cam position SCA7 (SN7) [7] = Cam position SCA8 (SN8) [8] = Cam position SCA9 (SN9) [9] = Cam position SCA10 (SN10) [10] = Cam position SCA11 (SN11) [11] = Cam position SCA12 (SN12) [12] = Cam position SCA13 (SN13) [13] = Cam position SCA14 (SN14) [14] = Cam position SCA15 (SN15) [15] = Cam position SCA16 (SN16) [16] = Cam position SCA17 (SN17) [17] = Cam position SCA18 (SN18) [18] = Cam position SCA19 (SN19) [19] = Cam position SCA20 (SN20) [20] = Cam position SCA21 (SN21) [21] = Cam position SCA22 (SN22) [22] = Cam position SCA23 (SN23) [23] = Cam position SCA24 (SN24) [24] = Cam position SCA25 (SN25) [25] = Cam position SCA26 (SN26) [26] = Cam position SCA27 (SN27) [27] = Cam position SCA28 (SN28) [28] = Cam position SCA29 (SN29) [29] = Cam position SCA30 (SN30)		
Dependency:	Refer to: p9501, p9503, p9537		
Note:	A change only becomes effective after a POWER ON. SCA: Safe Cam / SN: Safe software cam		

p9537[0...29] SI Motion SCA (SN) plus cam position (Control Unit) / SI Mtn SCA-			
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-2147000.000 [mm]	2147000.000 [mm]	-10.000 [mm]
Description:	Sets the minus cam position for the function "Safe Cam" (SCA).		
Index:	[0] = Cam position SCA1 (SN1) [1] = Cam position SCA2 (SN2) [2] = Cam position SCA3 (SN3) [3] = Cam position SCA4 (SN4) [4] = Cam position SCA5 (SN5) [5] = Cam position SCA6 (SN6) [6] = Cam position SCA7 (SN7) [7] = Cam position SCA8 (SN8) [8] = Cam position SCA9 (SN9) [9] = Cam position SCA10 (SN10) [10] = Cam position SCA11 (SN11) [11] = Cam position SCA12 (SN12) [12] = Cam position SCA13 (SN13) [13] = Cam position SCA14 (SN14) [14] = Cam position SCA15 (SN15) [15] = Cam position SCA16 (SN16) [16] = Cam position SCA17 (SN17) [17] = Cam position SCA18 (SN18) [18] = Cam position SCA19 (SN19) [19] = Cam position SCA20 (SN20) [20] = Cam position SCA21 (SN21) [21] = Cam position SCA22 (SN22) [22] = Cam position SCA23 (SN23) [23] = Cam position SCA24 (SN24) [24] = Cam position SCA25 (SN25) [25] = Cam position SCA26 (SN26) [26] = Cam position SCA27 (SN27) [27] = Cam position SCA28 (SN28) [28] = Cam position SCA29 (SN29) [29] = Cam position SCA30 (SN30)		
Dependency:	Refer to: p9501, p9503, p9536		
Note:	A change only becomes effective after a POWER ON. SCA: Safe Cam / SN: Safe software cam		

p9537[0...29] SI Motion SCA (SN) plus cam position (Control Unit) / SI Mtn SCA-			
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-2147000.000 [°]	2147000.000 [°]	-10.000 [°]
Description:	Sets the minus cam position for the function "Safe Cam" (SCA).		
Index:	[0] = Cam position SCA1 (SN1) [1] = Cam position SCA2 (SN2) [2] = Cam position SCA3 (SN3) [3] = Cam position SCA4 (SN4) [4] = Cam position SCA5 (SN5) [5] = Cam position SCA6 (SN6) [6] = Cam position SCA7 (SN7) [7] = Cam position SCA8 (SN8) [8] = Cam position SCA9 (SN9) [9] = Cam position SCA10 (SN10) [10] = Cam position SCA11 (SN11) [11] = Cam position SCA12 (SN12) [12] = Cam position SCA13 (SN13) [13] = Cam position SCA14 (SN14) [14] = Cam position SCA15 (SN15) [15] = Cam position SCA16 (SN16) [16] = Cam position SCA17 (SN17) [17] = Cam position SCA18 (SN18) [18] = Cam position SCA19 (SN19) [19] = Cam position SCA20 (SN20) [20] = Cam position SCA21 (SN21) [21] = Cam position SCA22 (SN22) [22] = Cam position SCA23 (SN23) [23] = Cam position SCA24 (SN24) [24] = Cam position SCA25 (SN25) [25] = Cam position SCA26 (SN26) [26] = Cam position SCA27 (SN27) [27] = Cam position SCA28 (SN28) [28] = Cam position SCA29 (SN29) [29] = Cam position SCA30 (SN30)		
Dependency:	Refer to: p9501, p9503, p9536		
Note:	A change only becomes effective after a POWER ON. SCA: Safe Cam / SN: Safe software cam		

p9538[0...29] SI Motion SCA (SN) cam track assignment (Control Unit) / SI Mtn SCA assign.

SERVO, VECTOR

Can be changed: U, T**Data type:** Unsigned32**P-Group:** Safety Integrated**Not for motor type:** -**Min**

100

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

414

Access level: 4**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

[0] 100

[1] 101

[2] 102

[3] 103

[4] 104

[5] 105

[6] 106

[7] 107

[8] 108

[9] 109

[10] 110

[11] 111

[12] 112

[13] 113

[14] 114

[15] 200

[16] 201

[17] 202

[18] 203

[19] 204

[20] 205

[21] 206

[22] 207

[23] 208

[24] 209

[25] 210

[26] 211

[27] 212

[28] 213

[29] 214

Description:

Assigns the individual cams to the maximum of 4 cam tracks and defines the numerical value for the SGA "cam range".

p9538[0...29] = CBA dec

C = Assignment of the cam to the cam track.

Valid values are 1, 2, 3, 4.

BA = Numerical value for the SGA "cam range".

If the position lies in the range of this cam, the value BA is signaled to the safety-relevant logic via the SGA "cam range" of the cam track set using C.

Valid values are 0 ... 14. Each numerical value may only be used once for each cam track.

Examples:

p9538[0] = 207

Cam 1 (index 0) is assigned cam track 2. If the position lies within the range of this cam, a value of 7 is entered in the SGA "cam range" of the second cam track.

p9538[5] = 100

Cam 6 (index 5) is assigned cam track 1. If the position lies within the range of this cam, a value of 0 is entered in the SGA "cam range" of the first cam track.

Index:	[0] = Track assignment SCA1
	[1] = Track assignment SCA2
	[2] = Track assignment SCA3
	[3] = Track assignment SCA4
	[4] = Track assignment SCA5
	[5] = Track assignment SCA6
	[6] = Track assignment SCA7
	[7] = Track assignment SCA8
	[8] = Track assignment SCA9
	[9] = Track assignment SCA10
	[10] = Track assignment SCA11
	[11] = Track assignment SCA12
	[12] = Track assignment SCA13
	[13] = Track assignment SCA14
	[14] = Track assignment SCA15
	[15] = Track assignment SCA16
	[16] = Track assignment SCA17
	[17] = Track assignment SCA18
	[18] = Track assignment SCA19
	[19] = Track assignment SCA20
	[20] = Track assignment SCA21
	[21] = Track assignment SCA22
	[22] = Track assignment SCA23
	[23] = Track assignment SCA24
	[24] = Track assignment SCA25
	[25] = Track assignment SCA26
	[26] = Track assignment SCA27
	[27] = Track assignment SCA28
	[28] = Track assignment SCA29
	[29] = Track assignment SCA30

Dependency: Refer to: p9501, p9503

Refer to: F01681

Note: A change only becomes effective after a POWER ON.
SCA: Safe Cam / SN: Safe software cam

p9540	SI Motion SCA (SN) tolerance (Control Unit) / SI Mtn SCA tol		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.0010 [mm]	10.0000 [mm]	0.1000 [mm]
Description:	Sets the tolerance for the function "Safe Cam" (SCA). Within this tolerance, both monitoring channels may signal different signal states of the same safe cam.		
Note:	A change only becomes effective after a POWER ON.		

p9540	SI Motion SCA (SN) tolerance (Control Unit) / SI Mtn SCA tol		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0.0010 [°]	10.0000 [°]	0.1000 [°]
Description:	Sets the tolerance for the function "Safe Cam" (SCA). Within this tolerance, both monitoring channels may signal different signal states of the same safe cam.		
Note:	A change only becomes effective after a POWER ON.		

p9542	SI Motion act val comparison tol (crosswise) (Control Unit) / SI Mtn act val tol		
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0010 [mm]	Max 360.0000 [mm]	Factory setting 0.1000 [mm]
Description:	Sets the tolerance for the cross-check of the actual position between the two monitoring channels. For encoderless motion monitoring functions, the tolerance must be set to a higher value (12 degrees rotary and 1 mm linear).		
Dependency:	Refer to: C01711		

p9542	SI Motion act val comparison tol (crosswise) (Control Unit) / SI Mtn act val tol		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0010 [°]	Max 360.0000 [°]	Factory setting 0.1000 [°]
Description:	Sets the tolerance for the cross-check of the actual position between the two monitoring channels. For encoderless motion monitoring functions, the tolerance must be set to a higher value (12 degrees rotary and 1 mm linear).		
Dependency:	Refer to: C01711		

p9544	SI Motion actual value comparison tolerance (referencing) (CU) / SI Mtn ref tol		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0000 [mm]	Max 36.0000 [mm]	Factory setting 0.0100 [mm]
Description:	Sets the tolerance to check the actual values after referencing (incremental encoder) or when powering up (absolute encoder).		
Dependency:	Refer to: C01711		
Note:	A change only becomes effective after a POWER ON.		

p9544	SI Motion actual value comparison tolerance (referencing) (CU) / SI Mtn ref tol		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0000 [°]	Max 36.0000 [°]	Factory setting 0.0100 [°]
Description:	Sets the tolerance to check the actual values after referencing (incremental encoder) or when powering up (absolute encoder).		
Dependency:	Refer to: C01711		
Note:	A change only becomes effective after a POWER ON.		

p9545 SI Motion SSM (SGA n < nx) filter time (Control Unit) / SI Mtn SSM filt CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 100.00 [ms]	Factory setting 0.00 [ms]

Description: Sets the filter time for the SSM feedback signal to detect standstill.

Note: The filter time is effective only if the function is enabled (p9300/p9500 Bit 16 = 1).
The parameter is included in the data cross-check of the two monitoring channels.
SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring)

p9546 SI Motion SSM (SGA n < nx) velocity limit (CU) / SI Mtn SSM v_limCU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [mm/min]	Max 1000000.00 [mm/min]	Factory setting 20.00 [mm/min]

Description: Sets the velocity limit for the SSM feedback signal to detect standstill (n < nx).
When this limit value is undershot, the signal "SSM feedback signal active" (SGA n < n_x) is set.
If a value of 0 is entered in p9368/p9568, the value of parameter p9346/p9546 also applies for the safe acceleration monitor SBR.
Caution: The function "Safe Acceleration Monitor" (SBR) is switched out after the selected threshold value is undershot.



Note: F-DO: Failsafe Digital Output / SGA: Safety-related output
SBR: Safe Acceleration Monitor
SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring) / SGA n < nx: Safety-related output n < nx

p9546 SI Motion SSM (SGA n < nx) velocity limit (CU) / SI Mtn SSM v_limCU

SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 1000000.00 [rpm]	Factory setting 20.00 [rpm]

Description: Sets the velocity limit for the SSM feedback signal to detect standstill (n < nx).
When this limit value is undershot, the signal "SSM feedback signal active" (SGA n < n_x) is set.
If a value of 0 is entered in p9368/p9568, the value of parameter p9346/p9546 also applies for the safe acceleration monitor SBR.
Caution: The function "Safe Acceleration Monitor" (SBR) is switched out after the selected threshold value is undershot.



Note: F-DO: Failsafe Digital Output / SGA: Safety-related output
SBR: Safe Acceleration Monitor
SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring) / SGA n < nx: Safety-related output n < nx

p9547 SI Motion SSM (SGA n < nx) velocity hysteresis (CU) / SI Mtn SSM hyst CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0010 [mm/min]	Max 500.0000 [mm/min]	Factory setting 10.0000 [mm/min]

Description: Sets the velocity hysteresis for the SSM feedback signal to detect standstill (n < nx).

Dependency: Refer to: C01711

Note: The velocity hysteresis is effective only if the function is enabled (p9300/p9500 Bit 16 = 1).

The parameter is included in the data cross-check of the two monitoring channels.

SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring)

p9547 SI Motion SSM (SGA n < nx) velocity hysteresis (CU) / SI Mtn SSM hyst CU

SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.0010 [rpm]	Max 500.0000 [rpm]	Factory setting 10.0000 [rpm]

Description: Sets the velocity hysteresis for the SSM feedback signal to detect standstill (n < nx).

Dependency: Refer to: C01711

Note: The velocity hysteresis is effective only if the function is enabled (p9300/p9500 Bit 16 = 1).

The parameter is included in the data cross-check of the two monitoring channels.

SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring)

p9548 SI Motion SBR actual velocity tolerance (Control Unit) / SI Mtn SBR tol

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [mm/min]	Max 120000.00 [mm/min]	Factory setting 300.00 [mm/min]

Description: Sets the velocity tolerance for the "Safe Acceleration Monitor".

Dependency: Refer to: C01706

Note: SBR: Safe Acceleration Monitor

p9548 SI Motion SBR actual velocity tolerance (Control Unit) / SI Mtn SBR tol

SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 120000.00 [rpm]	Factory setting 300.00 [rpm]

Description: Sets the velocity tolerance for the "Safe Acceleration Monitor".

Dependency: Refer to: C01706

Note: SBR: Safe Acceleration Monitor

p9549	SI Motion slip velocity tolerance (Control Unit) / SI Mtn slip tol		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [mm/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 6000.00 [mm/min]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 6.00 [mm/min]
Description:	Sets the velocity tolerance that is used for a 2-encoder system in crosswise comparison between the two monitoring channels.		
Dependency:	Refer to: p9501, p9542		
Note:	If the "actual value synchronization" is not enabled (p9501.3 = 0), then the value parameterized in p9542 is used as tolerance in the data cross-check.		
p9549	SI Motion slip velocity tolerance (Control Unit) / SI Mtn slip tol		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [rpm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 6000.00 [rpm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 6.00 [rpm]
Description:	Sets the velocity tolerance that is used for a 2-encoder system in crosswise comparison between the two monitoring channels.		
Dependency:	Refer to: p9501, p9542		
Note:	If the "actual value synchronization" is not enabled (p9501.3 = 0), then the value parameterized in p9542 is used as tolerance in the data cross-check.		
p9550	SI Motion SGE changeover tolerance time (Control Unit) / SI Mtn SGE_chg tol		
SERVO, VECTOR	Can be changed: U, T Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000.00 [ms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 500.00 [ms]
Description:	Sets the tolerance time for the changeover of the safety-related inputs (SGE).		
p9551	SI Motion SLS (SG) changeover delay time (Control Unit) / SI Mtn SLS t CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 600000.00 [ms]	Access level: 3 Func. diagram: 2825 Unit selection: - Expert list: 1 Factory setting 100.00 [ms]
Description:	Sets the delay time for the SLS changeover or for the changeover from SLS to SOS for the function "Safely-Limited Speed" (SLS). When transitioning from a higher to a lower safely-limited velocity/speed stage or to the safe operating stop (SOS), within this delay time, the "old" velocity stage remains active. Even if SLS or SOS is activated from non safety-related operation, then this delay is still applied.		
Note:	SLS: Safely-Limited Speed / SG: Safely reduced speed SOS: Safe Operating Stop / SBH: Safe operating stop		

p9552 SI Motion transition time STOP C to SOS (SBH) (Control Unit) / SI Mtn t C->SOS CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2825
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 600000.00 [ms]	Factory setting 100.00 [ms]
Description: Sets the transition time from STOP C to "Safe Operating Stop" (SOS).			
Note: SOS: Safe Operating Stop / SBH: Safe operating stop			

p9553 SI Motion transition time STOP D to SOS (SBH) (Control Unit) / SI Mtn t D->SOS CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2825
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 600000.00 [ms]	Factory setting 100.00 [ms]
Description: Sets the transition time from STOP D to "Safe Operating Stop" (SOS).			
Note: SOS: Safe Operating Stop / SBH: Safe operating stop			

p9554 SI Motion transition time STOP E to SOS (SBH) (Control Unit) / SI Mtn time E->SOS

SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 600000.00 [ms]	Factory setting 100.00 [ms]
Description: Sets the transition time from STOP E to "Safe Operating Stop" (SOS).			
Note: SOS: Safe Operating Stop / SBH: Safe operating stop			

p9555 SI Motion transition time STOP F to STOP B (Control Unit) / SI Mtn t F->B CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2825
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 600000.00 [ms]	Factory setting 0.00 [ms]
Description: Sets the transition time from STOP F to STOP B.			
Dependency: Refer to: C01711			

p9556 SI Motion pulse suppression delay time (Control Unit) / SI Mtn IL t_del CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2825
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 600000.00 [ms]	Factory setting 100.00 [ms]
Description: Sets the delay time for the safe pulse suppression after STOP B. In the case of encoderless motion monitoring functions (p9506/p9306 = 1), the parameter has no effect.			
Dependency: Refer to: p9560 Refer to: C01701			

p9557	SI Motion pulse suppression test time (Control Unit) / SI Mtn IL t_test		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 10000.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.00 [ms]
Description:	Sets the time after which the pulses must have been suppressed when initiating the test stop.		
Dependency:	Refer to: C01798		
Note:	A change only becomes effective after a POWER ON.		
p9558	SI Motion acceptance test mode time limit (Control Unit) / SI Mtn acc_test t		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 5000.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 40000.00 [ms]
Description:	Sets the maximum time for the acceptance test mode.		
	If the acceptance test mode takes longer than the selected time limit, then the mode is automatically terminated.		
Dependency:	Refer to: C01799		
p9559	SI Motion forced checking procedure timer (Control Unit) / SI Mtn dyn timer		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [h]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 9000.00 [h]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 8.00 [h]
Description:	Sets the time to carry out the forced checking procedure and test the safety motion monitoring functions integrated in the drives.		
	Within the parameterized time, the safety functions must have been tested at least once (including deselection of the "STO" function).		
	This monitoring time is reset each time the test is carried out.		
	The signal source to initiate the forced checking procedure is set in p9705.		
Dependency:	Refer to: p9705		
	Refer to: A01697, C01798		
Note:	STO: Safe Torque Off		
p9560	SI Motion pulse suppression shutdown velocity (Control Unit) / SI Mtn IL v_shutCU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [mm/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 6000.00 [mm/min]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [mm/min]
Description:	Sets the shutdown velocity for pulse suppression.		
	Below this velocity "standstill" is assumed and for STOP B / SS1, the pulses are suppressed (by changing to STOP A).		
	In the case of encoderless motion monitoring functions, the parameter must be > 0 (recommended value: 10).		
Dependency:	Refer to: p9556		

p9560	SI Motion pulse suppression shutdown speed (Control Unit) / SI Mtn IL n_shutCU		
SERVO (Lin, Safety rot), VECTOR (Safety rot)	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [rpm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 6000.00 [rpm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [rpm]
Description:	Sets the shutdown speed for the pulse suppression. Below this speed "standstill" is assumed and for STOP B, the pulses are suppressed by changing to STOP A).		
Dependency:	Refer to: p9556		
p9560	SI Motion pulse suppression shutdown velocity (Control Unit) / SI Mtn IL v_shutd		
SERVO (Lin)	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [mm/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 6000.00 [mm/min]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [mm/min]
Description:	Sets the shutdown velocity for pulse suppression. Below this velocity "standstill" is assumed and for STOP B / SS1, the pulses are suppressed (by changing to STOP A). In the case of encoderless motion monitoring functions, the parameter must be > 0 (recommended value: 10).		
Dependency:	Refer to: p9556		
p9561	SI Motion SLS (SG) stop response (Control Unit) / SI Mtn SLS resp		
SERVO, VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 14	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 5
Description:	Sets the stop response for the function "Safely-Limited Speed" (SLS). This setting applies for all SLS limit values. An input value of less than 5 signifies personnel protection, from 10 and upwards, machine protection.		
Value:	0: STOP A 1: STOP B 2: STOP C 3: STOP D 4: STOP E 5: Sets the stop response via p9563 (SLS-specific) 10: STOP A with delayed pulse suppression when the bus fails 11: STOP B with delayed pulse suppression when the bus fails 12: STOP C with delayed pulse suppression when the bus fails 13: STOP D with delayed pulse suppression when the bus fails 14: STOP E with delayed pulse suppression when the bus fails		
Dependency:	Refer to: p9531, p9563, p9580		
Note:	SLS: Safely-Limited Speed / SG: Safely reduced speed		

p9562	SI Motion SLP (SE) stop response (Control Unit) / SI Mtn SLP resp		
SERVO, VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 2	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2
Description:	Sets the stop response for the function "Safely-Limited Position" (SLP).		
Value:	2: STOP C 3: STOP D 4: STOP E		
Dependency:	Refer to: p9534, p9535		
Note:	SLP: Safely-Limited Position / SE: Safe software limit switches		
p9563[0...3]	SI Motion SLS (SG)-specific stop response (Control Unit) / SI Mtn SLS stop		
SERVO, VECTOR	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 14	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 2
Description:	Sets the SLS-specific stop response for the function "Safely-Limited Speed" (SLS). These settings apply to the individual limit values for SLS. In the case of encoderless motion monitoring (p9306 = 1), only a value of 0 or 1 is permitted.		
Value:	0: STOP A 1: STOP B 2: STOP C 3: STOP D 4: STOP E 10: STOP A with delayed pulse suppression when the bus fails 11: STOP B with delayed pulse suppression when the bus fails 12: STOP C with delayed pulse suppression when the bus fails 13: STOP D with delayed pulse suppression when the bus fails 14: STOP E with delayed pulse suppression when the bus fails		
Index:	[0] = Limit value SLS1 [1] = Limit value SLS2 [2] = Limit value SLS3 [3] = Limit value SLP4		
Dependency:	Refer to: p9531, p9561, p9580		
Notice:	Values 10 to 14 are being prepared and are presently ineffective.		
Note:	SLS: Safely-Limited Speed / SG: Safely reduced speed		
p9568	SI Motion SBR velocity limit (Control Unit) / SI MtnSBR v_lim CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [mm/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [mm/min]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [mm/min]
Description:	Sets the velocity limit for the "SBR" function. SBR is deactivated once the set velocity limit has been undershot.		

Note: SBR: Safe Acceleration Monitor
SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring)
For p9568 = p9368 = 0, the following applies:
The value in p9546/p9346 (SSM) is applied as the velocity limit for SBR.

p9568 SI Motion SBR velocity limit (Control Unit) / SI MtnSBR v_lim CU			
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [rpm]	Max 1000.00 [rpm]	Factory setting 0.00 [rpm]
Description:	Sets the velocity limit for the "SBR" function. SBR is deactivated once the set velocity limit has been undershot.		
Note:	SBR: Safe Acceleration Monitor SSM: Safe Speed Monitor (safety-relevant feedback signal from the velocity monitoring) For p9568 = p9368 = 0, the following applies: The value in p9546/p9346 (SSM) is applied as the velocity limit for SBR.		

p9570 SI Motion acceptance test mode (Control Unit) / SI Mtn Acc_mode			
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max 00AC hex	Factory setting 0000 hex
Description:	Setting to select and deselect the acceptance test mode.		
Value:	0: [00 hex] Deselect the acceptance test mode 172: [AC hex] Select the acceptance test mode		
Dependency:	Refer to: p9558, r9571, p9601 Refer to: C01799		
Note:	Acceptance test mode can only be selected if Safety Extended Functions are enabled (p9601/9801).		

r9571 SI Motion acceptance test status (Control Unit) / SI Mtn acc_status			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max 00AC hex	Factory setting -
Description:	Displays the status of the acceptance test mode.		
Value:	0: [00 hex] Acc_mode inactive 12: [0C hex] Acc_mode not possible due to POWER ON fault 13: [0D hex] Acc_mode not possible due to incorrect ID in p9570 15: [0F hex] Acc_mode not possible due to expired Acc_timer 172: [AC hex] Acc_mode active		
Dependency:	Refer to: p9558, p9570 Refer to: C01799		

p9580 SI Motion pulse suppression delay time after bus failure (CU) / SI Mtn t to IL

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 800.00 [ms]	Factory setting 0.00 [ms]

Description: Sets the delay time after which the pulses are safely suppressed after a bus failure.

Dependency: Refer to: p9561, p9563

p9581 SI Motion brake ramp reference value (Control Unit) / SI Mtn ramp ref CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 600.0000 [mm/min]	Max 24000.0000 [mm/min]	Factory setting 1500.0000 [mm/min]

Description: Sets the reference value to define the brake ramp.

The rate of rise of the brake ramp depends upon p9581 (reference value) and p9583 (monitoring time).

Dependency: Refer to: p9582, p9583

p9581 SI Motion brake ramp reference value (Control Unit) / SI Mtn ramp ref CU

SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 600.0000 [rpm]	Max 24000.0000 [rpm]	Factory setting 1500.0000 [rpm]

Description: Sets the reference value to define the brake ramp.

The rate of rise of the brake ramp depends upon p9581 (reference value) and p9583 (monitoring time).

Dependency: Refer to: p9582, p9583

p9582 SI Motion brake ramp delay time (Control Unit) / SI Mtn rp t_del CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 10.00 [ms]	Max 99000.00 [ms]	Factory setting 250.00 [ms]

Description: Sets the delay time for monitoring the brake ramp.

Monitoring of the brake ramp starts once the delay time has elapsed.

Dependency: Refer to: p9581, p9583

p9583	SI Motion brake ramp monitoring time (Control Unit) / SI Mtn rp t_mon CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.50 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10.00 [s]
Description:	Sets the monitoring time to define the brake ramp. The rate of rise of the brake ramp depends upon p9581 (reference value) and p9583 (monitoring time).		
Dependency:	Refer to: p9581, p9582		
p9587	SI Motion act val sensing encoderless filter time (Control Unit) / SI Mtn EL filt CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [ms]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 25.00 [ms]
Description:	Sets the filter time for smoothing the actual value with encoderless actual value sensing.		
Note:	This parameter is only effective for encoderless actual value sensing (p9306/p9506 = 1).		
p9588	SI Motion min current act val sensing encoderless (Control Unit) / SI Mtn I_min EL CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 10.00 [%]
Description:	Sets the minimum current for encoderless actual value sensing. Reducing this percentage value can adversely affect actual value sensing. - The value must be increased if C1711 has occurred with message value 1042. - The value must be decreased if C1711 has occurred with message value 1041.		
Dependency:	Refer to: C01711		
Note:	This parameter is only effective for encoderless actual value sensing (p9306/p9506 = 1).		
p9589	SI Motion voltage tolerance acceleration (Control Unit) / SI Mtn V tol CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 10.00 [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1000.00 [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 100.00 [%]
Description:	Sets the voltage tolerance for suppressing acceleration peaks. Increasing this percentage value means that voltage peaks will need to have a higher amplitude during acceleration procedures if they are not to affect actual value sensing. - The value must be increased if C1711 has occurred with message value 1042. - The value must be decreased if acceleration procedures have led to an excessive safety actual velocity.		
Dependency:	Refer to: C01711		
Note:	This parameter is only effective for encoderless actual value sensing (p9306/p9506 = 1).		

r9590[0...3]	SI Motion version safety motion monitoring (Control Unit) / SI Mtn version																							
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -																					
Description:	Displays the Safety Integrated version for the safety motion monitoring functions on the Control Unit.																							
Index:	[0] = Safety Version (major release) [1] = Safety Version (minor release) [2] = Safety Version (baselevel or patch) [3] = Safety Version (hotfix)																							
Dependency:	Refer to: r9770, r9870, r9890																							
Note:	Example: r9590[0] = 2, r9590[1] = 60, r9590[2] = 1, r9590[3] = 0 --> SI Motion version V02.60.01.00																							
p9601	SI enable, functions integrated in the drive (Control Unit) / SI enable fct CU																							
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 bin																					
Description:	Sets the enable signals for safety functions on the Control Unit that are integrated in the drive. Not all of the settings listed below will be permissible, depending on the Control Unit and Motor Module or Power Module being used: - p9601 = 0: Safety functions integrated in the drive disabled. - p9601 = 1: STO/SS1 enabled via terminals. Permissible if r9771.0 = 1. - p9601 = 4: Motion monitoring functions integrated in the drive enabled via Terminal Module 54F (TM54F) (SINAMICS S120) or via an integrated F-DI/F-DO (SINAMICS S110). Permissible if r9771.5 = 1. - p9601 = 5: Motion monitoring functions integrated in the drive enabled via Terminal Module 54F (TM54F) and STO/SS1 via terminals (SINAMICS S120 only). Permissible if r9771.5 = 1. - p9601 = 8: STO/SS1 enabled via PROFIsafe. Permissible if r9771.6 = 1. - p9601 = 9: STO/SS1 enabled via PROFIsafe and STO/SS1 via terminals. Permissible if r9771.6 = 1. - p9601 = 12: Motion monitoring functions integrated in the drive enabled via PROFIsafe. Permissible if r9771.4 = 1. - p9601 = 13: Motion monitoring functions integrated in the drive enabled via PROFIsafe and STO/SS1 via terminals. Permissible if r9771.4 = 1.																							
Bit field:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr> <tr> <td>00</td><td>STO (SH) via terminals (Control Unit) enable</td><td>Enable</td><td>Inhibit</td><td>2810</td></tr> <tr> <td>02</td><td>Motion monitoring functions integr. in the drive (Control Unit)</td><td>Enable</td><td>Inhibit</td><td>-</td></tr> <tr> <td>03</td><td>PROFIsafe (Control Unit) enable</td><td>Enable</td><td>Inhibit</td><td>-</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FP	00	STO (SH) via terminals (Control Unit) enable	Enable	Inhibit	2810	02	Motion monitoring functions integr. in the drive (Control Unit)	Enable	Inhibit	-	03	PROFIsafe (Control Unit) enable	Enable	Inhibit	-			
Bit	Signal name	1 signal	0 signal	FP																				
00	STO (SH) via terminals (Control Unit) enable	Enable	Inhibit	2810																				
02	Motion monitoring functions integr. in the drive (Control Unit)	Enable	Inhibit	-																				
03	PROFIsafe (Control Unit) enable	Enable	Inhibit	-																				
Dependency:	Refer to: r9771, p9801																							
Note:	CU: Control Unit. STO: Safe Torque Off/SH: Safe standstill. SS1: Safe Stop 1 (corresponds to Stop Category 1 acc. to EN 60204). SI: Safety Integrated. SMM: Safe Motion Monitoring. F-DI: Failsafe Digital Input. F-DO: Failsafe Digital Output. A change only becomes effective after a POWER ON.																							

p9602	SI enable Safe Brake Control (Control Unit) / SI enable SBC CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: 2814 Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the enable signal for the function "Safe Brake Control" (SBC) on the Control Unit.		
Value:	0: Inhibit SBC 1: Enable SBC		
Dependency:	Refer to: p9802		
Note:	The "Safe Brake Control" function is not activated until at least one safety monitoring function has been enabled (i.e. p9501 not equal to 0 and/or p9601 not equal to 0). It does not make sense to parameterize "no motor holding brake available" and enable "Safe Brake Control" (p1215 = 0, p9602 = p9802 = 1) if there is no motor holding brake. The parameterization "motor holding brake the same as sequence control, connection via BICO" and "Safe Brake Control" enabled (p1215 = 3, p9602 = 1, p9802 = 1) is not practical. It is not permissible to parameterize "motor holding brake without feedback signals" and also enable "safe brake control" (p1278 = 1, p9602 = 1, p9802 = 1). CU: Control Unit SBC: Safe Brake Control SI: Safety Integrated		
p9610	SI PROFIsafe address (Control Unit) / SI PROFIsafe CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFE hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the PROFIsafe address of the Control Unit.		
Dependency:	Refer to: p9810		
p9620[0...7]	BI: SI signal source for STO (SH)/SBC/SS1 (Control Unit) / SI S_srcSTO/SS1 CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 / Binary P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2810 Unit selection: - Expert list: 1 Factory setting [0] 0 [1] 0 [2] 0 [3] 0 [4] 0 [5] 0 [6] 0 [7] 0
Description:	Sets the signal source for the following functions on the Control Unit: STO: Safe Torque Off / SH: Safe standstill SBC: Safe Brake Control SS1: Safe Stop 1 (time monitored)		
Dependency:	Refer to: p9601		

Note: The following signal sources are permitted:

- fixed zero (standard setting).
- digital inputs DI 0 to DI 7 on the Control Unit 320 (CU320).
- digital inputs DI 0 to DI 3 on the Controller Extensions (CX32, NX10, NX15).
- digital inputs DI 0 to DI 3 on the Control Unit 310 (CU310).

It is not permitted to establish an interconnection to a digital input in the simulation mode.

For a parallel circuit configuration of n power units, the following applies:

p9620[0] = Signal source for power unit 1

...

p9620[n-1] = Signal source for power unit n

p9621		BI: SI signal source for SBA (Control Unit) / SI s_src SBA CU	
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2810
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	An entry is added, specifying whether and/or via which digital CU input the SBA feedback signal (SBA_DIAG) is to be read in; the parameter can be interconnected as a BICO drain. 0 : No SBC with SBA (default) Bico code: To be parameterized by the user SBA and no DQ CIM: p0722.x CU signal source for DIx where x = { 0,1,2...7 } SBA with DQ CIM: p9872.3 Signal source is permanently interconnected to bit 3		
Dependency:	Refer to: p9601		
Note:	No difference is tolerated for a data cross-check between p9621 and p9821.		

p9622[0...1]		SI SBA relay wait times (Control Unit) / SI SBA WT CU	
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2810
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [ms]	Max 1000.00 [ms]	Factory setting [0] 100.00 [ms] [1] 65.00 [ms]
Description:	The relay-specific minimum wait times (ms) for evaluating the feedback signal contacts have to be taken into account. They differ for the activation and deactivation of one and the same relay.		
Index:	[0] = Relay wait time activation [1] = Relay wait time deactivation		
Dependency:	Refer to: p9850		
Note:	For a data cross-check between p9622 and p9822, a difference of one Safety monitoring clock cycle is tolerated. The parameterized time is internally rounded-off to an integer multiple of the monitoring clock cycle. The relay-specific minimum wait times (ms) for evaluating the feedback signal contacts are entered. Index 0 : for the ON time (default 100 ms) Index 1 : for the OFF time (default 65 ms)		

p9650	SI SGE changeover tolerance time (Control Unit) / SI SGE_chg tol CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2000.00 [ms]	Access level: 3 Func. diagram: 2810 Unit selection: - Expert list: 1 Factory setting 500.00 [ms]
Description:	Sets the tolerance time to change over the safety-related inputs (SGE) on the Control Unit. An SGE changeover is not effective simultaneously due to the different runtimes in the two monitoring channels. After an SGE changeover, dynamic data is not subject to a data cross-check during this tolerance time.		
Dependency:	Refer to: p9850		
Note:	For a data cross-check between p9650 and p9850, a difference of one Safety monitoring clock cycle is tolerated. The parameterized time is internally rounded-off to an integer multiple of the monitoring clock cycle. SGE: Safety-related input (e.g. STO terminals)		
p9651	SI STO/SBC/SS1 debounce time (Control Unit) / SI STO t_debou CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the CU DI debounce time used to control the SH terminal (see p9620). The debounce time is accepted rounded to whole milliseconds. The debounce time indicates the maximum duration of a fault pulse on the F-DIs, so that there are no negative effects on the SGEs.		
Note:	Example: Debounce time = 1 ms: Fault pulses of 1 ms are filtered; only pulses longer than 2 ms are processed. Debounce time = 3 ms: Fault pulses of 3 ms are filtered; only pulses longer than 4 ms are processed.		
p9652	SI Safe Stop 1 delay time (Control Unit) / SI Stop 1 t_del CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [s]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 300.00 [s]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [s]
Description:	Sets the delay time of the pulse suppression for the function "Safe Stop 1" (SS1) on the Control Unit to brake along the OFF3 down ramp (p1135).		
Dependency:	Refer to: p1135, p9852		
Note:	For a data cross-check between p9652 and p9852, a difference of one Safety monitoring clock cycle is tolerated. The parameterized time is rounded internally to an integer multiple of the monitoring clock cycle. SS1: Safe Stop 1 (corresponds to Stop Category 1 acc. to EN60204)		

p9658	SI transition time STOP F to STOP A (Control Unit) / SI STOP F->A CU		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 30000.00 [ms]	Access level: 3 Func. diagram: 2802 Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the transition period from STOP F to STOP A on the Control Unit.		
Dependency:	Refer to: r9795, p9858 Refer to: F01611		
Note:	For a data cross-check between p9658 and p9858, a difference of one Safety monitoring clock cycle is tolerated. The parameterized time is rounded internally to an integer multiple of the monitoring clock cycle. STOP F: Defect in a monitoring channel (error in the data cross-check) STOP A: Pulse suppression via the safety shutdown path		
p9659	SI forced checking procedure timer / SI FrcdCkProcTimer		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [h]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 9000.00 [h]	Access level: 3 Func. diagram: 2810 Unit selection: - Expert list: 1 Factory setting 8.00 [h]
Description:	Sets the time to carry out the dynamic update and testing the safety shutdown paths (forced checking procedure). Within the parameterized time, STO must have been deselected at least once. The monitoring time is reset each time that STO is deselected.		
Dependency:	Refer to: A01699		
Note:	STO: Safe Torque Off / SH: Safe standstill		
r9660	SI forced checking procedure remaining time / SI frc chk remain		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min - [h]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [h]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [h]
Description:	Displays the time remaining before dynamization and testing of the safety shutdown paths (forced checking procedure).		
Dependency:	Refer to: A01699		
p9700	SI Motion copy function / SI Mtn copy fct		
SERVO, VECTOR	Can be changed: C2(95), U, T Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max 00D0 hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Setting to start the required copy function. After starting, the appropriate parameters are copied from the Control Unit to the Motor Module. Once copying is complete, the parameter is automatically reset to 0.		

Value: 0: [00 hex] Copy function ended
 29: [1D hex] Start copy function node identifier
 87: [57 hex] Start copy function SI parameters
 208: [D0 hex] Start copy function SI basic parameters

Note: Re value = 57 hex and D0 hex:
 The value can only be set if the safety commissioning mode is set and the Safety Integrated password was entered.
 Re value = D0 hex:
 The following parameters are copied after starting the copy function:
 p9601/p9801, p9602/p9802, p9610/9810, p9650/p9850, p9652/p9852, p9658/p9858

p9700 SI Motion copy function / SI Mtn copy fct

TM54F_MA	Can be changed: C2(95), U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0000 hex	0057 hex	0000 hex

Description: Setting to start the required copy function.
 After starting, the appropriate parameters are copied from the Control Unit to the Motor Module.
 Once copying is complete, the parameter is automatically reset to 0.

Value: 0: [00 hex] Copy function ended
 29: [1D hex] Start copy function node identifier
 87: [57 hex] Start copy function SI parameters

Note: Re value = 57 hex:
 The value can only be set if the safety commissioning mode is set and the Safety Integrated password was entered.
 SI: Safety Integrated

p9701 Acknowledge SI Motion data / Ackn SI Mtn dat

SERVO, VECTOR	Can be changed: C2(95), U, T	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0000 hex	00EC hex	0000 hex

Description: Setting to transfer the reference checksums from the associated actual checksums after changes (SI parameters, hardware).
 After transferring the reference checksums, parameters are automatically reset to 0.

Value: 0: [00 hex] Data unchanged
 172: [AC hex] Acknowledge data change complete
 220: [DC hex] Acknowledge SI basic parameter change
 236: [EC hex] Acknowledge hardware CRC

Dependency: Refer to: r9398, p9399, r9728, p9729, r9798, p9799, r9898, p9899

Note: Re value = AC and DC hex:
 These values can only be set if the safety commissioning mode is set and the Safety Integrated password was entered.

p9701	Acknowledge SI Motion data / Ackn SI Mtn dat		
TM54F_MA, TM54F_SL	Can be changed: C2(95), U, T Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max 00EC hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Setting to transfer the reference checksums from the associated actual checksums after changes (SI parameters, hardware). After transferring the reference checksums, parameters are automatically reset to 0.		
Value:	0: [00 hex] Data unchanged 172: [AC hex] Acknowledge data change complete 236: [EC hex] Acknowledge hardware CRC		
Dependency:	Refer to: r9398, p9399, r9728, p9729, r9798, p9799, r9898, p9899		
Note:	Re value = AC hex: These values can only be set if the safety commissioning mode is set and the Safety Integrated password was entered. SI: Safety Integrated		

p9705	BI: SI Motion: Test stop signal source / SI Mtn test stop		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 / Binary P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the signal source for the test stop of the safety-relevant motion monitoring functions.		
Notice:	It is not permissible to use TM54F inputs to start the test stop.		

r9710[0...1]	SI Motion diagnostics result list 1 / SI Mtn res_list 1			
SERVO, VECTOR	Can be changed: -		Calculated: -	Access level: 3
	Data type: Unsigned32		Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated		Units group: -	Unit selection: -
	Not for motor type: -		Scaling: -	Expert list: 1
	Min		Max	Factory setting
	-		-	-
Description:	Displays result list 1 that, for the data cross-check between the monitoring channels, led to the fault.			
Index:	[0] = Result list, second channel [1] = Result list, drive			
Bit field:	Bit	Signal name	1 signal	0 signal
	00	Actual value > upper limit SOS	Yes	No
	01	Actual value > lower limit SOS	Yes	No
	02	Actual value > upper limit, SE1	Yes	No
	03	Actual value > lower limit, SE1	Yes	No
	04	Actual value > upper limit, SE2	Yes	No
	05	Actual value > lower limit, SE2	Yes	No
	06	Actual value > upper limit, SG1	Yes	No
	07	Actual value > lower limit, SG1	Yes	No
	08	Actual value > upper limit, SG2	Yes	No
	09	Actual value > lower limit, SG2	Yes	No
	10	Actual value > upper limit, SG3	Yes	No

11	Actual value > lower limit, SG3	Yes	No	-
12	Actual value > upper limit, SG4	Yes	No	-
13	Actual value > lower limit, SG4	Yes	No	-
16	Actual value > upper limit, SBR	Yes	No	-
17	Actual value > lower limit, SBR	Yes	No	-

Dependency: Refer to: C01711

r9711[0...1] SI Motion diagnostics result list 2 / SI Mtn res_list 2

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting

- - -

Description: Displays result list 2 that, for the data cross-check between the monitoring channels, led to the fault.

Index: [0] = Result list, second channel
[1] = Result list, drive

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Actual value > upper limit, SN1+	Yes	No	-
	01	Actual value > lower limit, SN1+	Yes	No	-
	02	Actual value > upper limit, SN1-	Yes	No	-
	03	Actual value > lower limit, SN1-	Yes	No	-
	04	Actual value > upper limit, SN2+	Yes	No	-
	05	Actual value > lower limit, SN2+	Yes	No	-
	06	Actual value > upper limit, SN2-	Yes	No	-
	07	Actual value > lower limit, SN2-	Yes	No	-
	08	Actual value > upper limit, SN3+	Yes	No	-
	09	Actual value > lower limit, SN3+	Yes	No	-
	10	Actual value > upper limit, SN3-	Yes	No	-
	11	Actual value > lower limit, SN3-	Yes	No	-
	12	Actual value > upper limit, SN4+	Yes	No	-
	13	Actual value > lower limit, SN4+	Yes	No	-
	14	Actual value > upper limit, SN4-	Yes	No	-
	15	Actual value > lower limit, SN4-	Yes	No	-
	16	Actual value > upper limit, n_x+	Yes	No	-
	17	Actual value > lower limit, n_x+	Yes	No	-
	18	Actual value > upper limit, n_x-	Yes	No	-
	19	Actual value > lower limit, n_x-	Yes	No	-
	20	Actual value > upper limit, modulo	Yes	No	-
	21	Actual value > lower limit, modulo	Yes	No	-

Dependency: Refer to: C01711

r9712 SI Motion diagnostics position action value motor side / SI Mtn s_act mot

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting

- - -

Description: Displays the actual motor-side position actual value for the motion monitoring functions on the Control Unit.

r9713[0...3]	SI Motion diagnostics position action value load side / SI Mtn s_act load			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3	
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -	
	P-Group: Safety Integrated	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	-	
Description:	Displays the actual load-side actual values of both monitoring channels and their difference.			
Index:	[0] = Load-side actual value on the Control Unit [1] = Load-side actual value on the second channel [2] = Load-side actual value difference Control Unit - second channel [3] = Load-side max. actual value difference CU - 2nd channel			
Dependency:	Refer to: r9724			
Note:	DCC: Data cross-check Re r9713[0]: The display of the load-side position actual value on the Control Unit is updated in the monitoring clock cycle. Re r9713[1]: The display of the load-side position actual value on the second channel is updated in the DCC clock cycle (r9724) and delayed by one DCC clock cycle. Re r9713[2]: The difference between the load-side position actual value on the Control Unit and load-side position actual value in the second channel is updated in the DCC clock cycle (r9724) and delayed by one DCC clock cycle. Re r9713[3]: The maximum difference between the load-side position actual value on the Control Unit and the load-side position actual value on the second channel.			
r9714[0...1]	SI motion diagnostics velocity / SI Mtn diag v			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3	
	Data type: Integer32	Dynamic index: -	Func. diagram: -	
	P-Group: Safety Integrated	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	-	
Description:	Displays the current velocity actual values for the motion monitoring functions on the Control Unit.			
Index:	[0] = Load-side velocity actual value on the Control Unit [1] = Current SBR velocity limit on the Control Unit			
Note:	For a linear axis, the following units apply: Micrometers per monitoring clock cycle (p9500) For a rotary axis, the following units apply: Millidegrees per monitoring clock cycle (p9500)			
r9718.23	CO/BO: SI Motion control signals 1 / SI Mtn ctrl_sig 1			
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4	
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -	
	P-Group: Safety Integrated	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min	Max	Factory setting	
	-	-	-	
Description:	Control signal 1 for safety-relevant motion monitoring functions.			
Bit field:	Bit	Signal name	1 signal	0 signal
	23	Set offset for TfS to the actual torque	Set	Reset
Note:	TfS: Traverse to fixed stop			
			FP	-

r9718.23 CO/BO: SI Motion control signals 1 / SI Mtn ctrl_sig 1

SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Control signal 1 for safety-relevant motion monitoring functions.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	23	Set offset for TfS to the actual force	Set	Reset	-

Note: TfS: Traverse to fixed stop

r9719.0...31 CO/BO: SI Motion control signals 2 / SI Mtn ctrl_sig 2

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Control signal 2 for safety-relevant motion monitoring functions.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Deselect SOS/SLS (SBH/SG)	Yes	No	-
	01	Deselect SOS (SBH)	Yes	No	-
	03	Select SLS (SG) bit 0	Set	Not set	-
	04	Select SLS (SG) bit 1	Set	Not set	-
	08	Gearbox selection, bit 0	Set	Not set	-
	09	Gearbox selection, bit 1	Set	Not set	-
	10	Gearbox selection, bit 2	Set	Not set	-
	12	Select SLP (SE)	SLP2 (SE2)	SLP1 (SE1)	-
	13	Close brake from control	Yes	No	-
	15	Select test stop	Yes	No	-
	16	SGE valid	Yes	No	-
	18	Deselect external STOP A	Yes	No	-
	19	Deselect external STOP C	Yes	No	-
	20	Deselect external STOP D	Yes	No	-
	21	Deselect external STOP E	Yes	No	-
	28	SLS (SG) override bit 0	Set	Not set	-
	29	SLS (SG) override bit 1	Set	Not set	-
	30	SLS (SG) override bit 2	Set	Not set	-
	31	SLS (SG) override bit 3	Set	Not set	-

Note: Re r9719.0 and r9719.1:

These two bits must be considered together.

- if SOS/SLS (SBH/SG) is deselected using bit 0, then assignment of bit 1 is irrelevant.

- if SOS/SLS (SBH/SG) is selected using bit 0, then a changeover is made between SOS (SBH) and SLS (SG) using bit 1.

SLP: Safely-Limited Position / SE: Safe software limit switches

SLS: Safely-Limited Speed / SG: Safely reduced speed

SOS: Safe Operating Stop / SBH: Safe operating stop

r9720.0...10 CO/BO: SI Motion control signals integrated in the drive / SI Mtn integ STW

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2840, 2855
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Control signals for safety-relevant motion monitoring functions integrated in the drive.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Deselect STO	Yes	No	-
	01	Deselect SS1	Yes	No	-
	02	Deselect SS2	Yes	No	-
	03	Deselect SOS	Yes	No	-
	04	Deselect SLS	Yes	No	-
	07	Acknowledgement	Signal edge active	No	-
	09	Select SLS bit 0	Set	Not set	-
	10	Select SLS bit 1	Set	Not set	-

Note: This parameter is only supplied with actual values in the case of Safety Integrated Extended Functions. For Safety Integrated Basic Functions (SBC, SS1, STO), the value is equal to zero.

r9721.0...15 CO/BO: SI Motion status signals / SI Mtn stat_sig

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Status signal for safety-relevant motion monitoring functions.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	SOS or SLS active	Yes	No	-
	01	SOS active	Yes	No	-
	02	Pulse enable	Deleted	Enabled	-
	03	Active SLS stage bit 0	Set	Not set	-
	04	Active SLS stage bit 1	Set	Not set	-
	05	Velocity below limit value n_x	Yes	No	-
	06	Status signals valid	Yes	No	-
	07	Safely referenced	Yes	No	-
	12	STOP A or B active	Yes	No	-
	13	STOP C active	Yes	No	-
	14	STOP D active	Yes	No	-
	15	STOP E active	Yes	No	-

Note: This parameter is only supplied with actual values in the case of Safety Integrated Extended Functions. For Safety Integrated Basic Functions (SBC, SS1, STO), the value is equal to zero.

r9722.0...15	CO/BO: SI Motion status signals integrated in the drive / SI Mtn integ stat				
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2840, 2855		
	P-Group: Safety Integrated	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Status signal for safety-relevant motion monitoring functions integrated in the drive.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	STO active	Yes	No	-
	01	SS1 active	Yes	No	-
	02	SS2 active	Yes	No	-
	03	SOS active	Yes	No	-
	04	SLS active	Yes	No	-
	07	Internal event	No	Yes	-
	09	Active SLS stage bit 0	Set	Not set	-
	10	Active SLS stage bit 1	Set	Not set	-
	11	SOS selected	Yes	No	-
	15	SSM (speed below limit value)	Yes	No	-
Notice:	Re bit 07: The signal state behaves in an opposite way to the PROFIsafe Standard.				
Note:	This parameter is only supplied with actual values in the case of Safety Integrated Extended Functions. For Safety Integrated Basic Functions (SBC, SS1, STO), the value is equal to zero. Re bit 07: An internal event is displayed if a STOP A ... F is active.				

r9722.0...15	CO/BO: SI Motion status signals integrated in the drive / SI Mtn integ stat				
SERVO (Lin)	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2840, 2855		
	P-Group: Safety Integrated	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min	Max	Factory setting		
	-	-	-		
Description:	Status signal for safety-relevant motion monitoring functions integrated in the drive.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	STO active	Yes	No	-
	01	SS1 active	Yes	No	-
	02	SS2 active	Yes	No	-
	03	SOS active	Yes	No	-
	04	SLS active	Yes	No	-
	07	Internal event	No	Yes	-
	09	Active SLS stage bit 0	Set	Not set	-
	10	Active SLS stage bit 1	Set	Not set	-
	11	SOS selected	Yes	No	-
	15	SSM (velocity below limit value)	Yes	No	-
Notice:	Re bit 07: The signal state behaves in an opposite way to the PROFIsafe Standard.				
Note:	This parameter is only supplied with actual values in the case of Safety Integrated Extended Functions. For Safety Integrated Basic Functions (SBC, SS1, STO), the value is equal to zero. Re bit 07: An internal event is displayed if a STOP A ... F is active.				

r9723.0...16 CO/BO: SI Motion diagnostic signals integrated in the drive / SI Mtn integ diag

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the diagnostic signals for safety-relevant motion monitoring functions integrated in the drive.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Forced checking procedure required	Yes	No	-
	16	Status of the SBR function	active	inactive	-

r9724 SI Motion crosswise comparison clock cycle / SI Mtn DCC clk cyc

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [ms]	- [ms]	- [ms]

Description: Displays the crosswise comparison clock cycle.

The value indicates the clock cycle time with which each individual DCC value is compared between the two monitoring channels.

Dependency: Refer to: p9500

Note: Crosswise comparison clock cycle = monitoring clock cycle (p9500) * number of data to be crosswise compared
DCC: Data cross-check

r9725[0...2] SI Motion, diagnostics STOP F / SI Mtn Diag STOP F

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Re r9725[0]:
Displays the message value that resulted in the STOP F on the drive.

Value = 0 means:

The Control Unit signaled a STOP F.

Value = 1 ... 999 means:

Number of the incorrect cross-checked data between the Control Unit and second channel.

Value >= 1000 means:

Additional diagnostic values of the drive.

Re r9725[1]:

Displays the value of the Control Unit that resulted in the STOP F.

Re r9725[2]:

Displays the value of the 2nd channel that resulted in the STOP F.

Index:
[0] = DCC error number
[1] = Control Unit DCC act value
[2] = Components DCC act val

Dependency: Refer to: C01711

Note: The significance of the individual values is described in message C01711.

p9726	SI Motion, user agreement selection/deselection / SI Mtn UserAgr sel		
SERVO, VECTOR	Can be changed: U, T Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max 00AC hex	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Setting to select and deselect the user agreement.		
Value:	0: [00 hex] Deselect user agreement 172: [AC hex] Select user agreement		
Dependency:	Refer to: r9727		
r9727	SI Motion user agreement, inside the drive / SI Mtn UserAgr int		
SERVO, VECTOR	Can be changed: - Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the internal state of the user agreement. Value = 0: User agreement is not set. Value = AC hex: User agreement is set.		
Dependency:	Refer to: p9726		
r9728[0...2]	SI Motion actual checksum, SI parameters / SI Mtn act CRC		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the checksum over the checked Safety Integrated parameters of the motion monitoring functions (actual checksum).		
Index:	[0] = Checksum over SI parameters for motion monitoring [1] = Checksum over SI parameters for actual values [2] = Checksum over SI parameters for HW		
Dependency:	Refer to: p9729 Refer to: F01680		
p9729[0...2]	SI Motion reference checksum, SI parameters / SI Mtn ref CRC		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the checksum using the checksum-tested Safety Integrated parameters for motion monitoring functions (reference checksum).		
Index:	[0] = Checksum over SI parameters for motion monitoring [1] = Checksum over SI parameters for actual values [2] = Checksum over SI parameters for HW		
Dependency:	Refer to: r9728 Refer to: F01680		

r9730	SI Motion Safe maximum velocity / SI mtn safe v_Max		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min - [mm/min]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mm/min]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mm/min]
Description:	Displays the safe maximum velocity (on the load side) that is permissible for the safe motion monitoring functions as a result of the actual value sensing. If safety is not enabled, the parameter has no significance and is set to "0".		
Note:	If the value displayed is exceeded, message C01711 is output indicating relevant subsequent faults.		
r9730	SI Motion Safe maximum velocity / SI mtn safe v_Max		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: - Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [rpm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [rpm]
Description:	Displays the safe maximum velocity (on the load side) that is permissible for the safe motion monitoring functions as a result of the actual value sensing. If safety is not enabled, the parameter has no significance and is set to "0".		
Note:	If the value displayed is exceeded, message C01711 is output indicating relevant subsequent faults.		
r9731	SI Motion safe position accuracy / SI Mtn pos acc		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min - [mm]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [mm]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [mm]
Description:	Displays the safe position accuracy (on the load side) that can be achieved as a maximum for the safe motion monitoring functions as a result of the actual value sensing. If safety is not enabled, the parameter has no significance and is set to "0".		
r9731	SI Motion safe position accuracy / SI Mtn pos acc		
SERVO (Safety rot), VECTOR (Safety rot)	Can be changed: - Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min - [°]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [°]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [°]
Description:	Displays the safe position accuracy (on the load side) that can be achieved as a maximum for the safe motion monitoring functions as a result of the actual value sensing. If safety is not enabled, the parameter has no significance and is set to "0".		

r9733[0...1]	CO: SI Motion setpoint speed limit effective / SI Mtn setp_lim		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min - [rpm]	Calculated: - Dynamic index: - Units group: 3_1 Scaling: p2000 Max - [rpm]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [rpm]
Description:	Displays the necessary setpoint speed limit as a result of the selected SI Motion monitoring functions. Contrary to the parameterization of the SI limit values, this parameter specifies the motor-side limit value and not the load-side limit value.		
Index:	[0] = Setpoint limiting positive [1] = Setpoint limiting negative		
Dependency:	r9733[0] = p9531[x] * p9533 r9733[1] = - p9531[x] * p9533 x = SLS stage selected Refer to: p9531, p9533		
Notice:	If p1051 = r9733[0] is interconnected, p1052 = r9733[1] must also be interconnected. If only the absolute value of the setpoint speed limit is required, r9733[0] is sufficient.		
Note:	The unit changeover between linear and rotary axis is not implemented via the safety changeover (p9502) but by the linear motor changeover. If the "SLS" function is not selected, r9733[0] shows p1082 and r9733[1] shows -p1082. The display in r9733 can be delayed by up to one Safety monitoring clock cycle as compared to the display in r9719/r9720 and r9721/r9722.		

r9733[0...1]	CO: SI Motion setpoint speed limit effective / SI Mtn setp_lim		
SERVO (Lin)	Can be changed: - Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min - [m/min]	Calculated: - Dynamic index: - Units group: 4_1 Scaling: p2000 Max - [m/min]	Access level: 3 Func. diagram: - Unit selection: p0505 Expert list: 1 Factory setting - [m/min]
Description:	Displays the necessary setpoint speed limit as a result of the selected SI Motion monitoring functions. Contrary to the parameterization of the SI limit values, this parameter specifies the motor-side limit value and not the load-side limit value.		
Index:	[0] = Setpoint limiting positive [1] = Setpoint limiting negative		
Dependency:	r9733[0] = p9531[x] * p9533 r9733[1] = - p9531[x] * p9533 x = SLS stage selected Refer to: p9531, p9533		
Notice:	If p1051 = r9733[0] is interconnected, p1052 = r9733[1] must also be interconnected. If only the absolute value of the setpoint speed limit is required, r9733[0] is sufficient.		
Note:	The unit changeover between linear and rotary axis is not implemented via the safety changeover (p9502) but by the linear motor changeover. If the "SLS" function is not selected, r9733[0] shows p1082 and r9733[1] shows -p1082. The display in r9733 can be delayed by up to one Safety monitoring clock cycle as compared to the display in r9719/r9720 and r9721/r9722.		

r9744			
SI message buffer changes, counter / SI msg_buffer chng			
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the changes of the safety message buffer. This counter is incremented every time that the safety message buffer changes.		
Recommend.:	This is used to check whether the safety message buffer has been read out consistently.		
Dependency:	Refer to: r9747, r9748, r9749, p9752, r9753, r9754, r9755, r9756		
r9745[0...63]			
SI component number / SI comp_num			
SERVO, TM41, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the component number of the safety message which has occurred.		
Note:	Value = 0: Assignment to a component not possible.		
r9747[0...63]			
SI message code / SI msg_code			
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the numbers of safety messages that have occurred.		
Dependency:	Refer to: r9744, r9748, r9749, p9752, r9753, r9754, r9755, r9756		
Note:	The messages type "safety message" (Cxxxxx) are entered in the message fault buffer. Message buffer structure (principle): r9747[0], r9748[0], r9749[0], r9753[0], r9754[0], r9755[0], r9756[0] --> Actual message case, safety message 1 ... r9747[7], r9748[7], r9749[7], r9753[7], r9754[7], r9755[7], r9756[7] --> Actual message case, safety message 8 r9747[8], r9748[8], r9749[8], r9753[8], r9754[8], r9755[8], r9756[8] --> 1st acknowledged message case, safety message 1 ... r9747[15], r9748[15], r9749[15], r9753[15], r9754[15], r9755[15], r9756[15] --> 1st acknowledged message case, safety message 8 ... r9747[56], r9748[56], r9749[56], r9753[56], r9754[56], r9755[56], r9756[56] --> 7th acknowledged message case, safety message 1 ... r9747[63], r9748[63], r9749[63], r9753[63], r9754[63], r9755[63], r9756[63] --> 7th acknowledged message case, safety message 8		

r9748[0...63]	SI message time received in milliseconds / SI t_msg_rcv ms		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [ms]	- [ms]	- [ms]
Description:	Displays the relative system runtime in milliseconds when the safety message occurred.		
Dependency:	Refer to: r9744, r9747, r9749, p9752, r9753, r9754, r9755, r9756		

r9749[0...63]	SI message value / SI msg_value		
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dynamic index: -	Func. diagram: -
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-
Description:	Displays the additional information about the safety message that occurred (as integer number).		
Dependency:	Refer to: r9744, r9747, r9748, p9752, r9753, r9754, r9755, r9756		

r9750[0...63]	SI diagnostic attributes / SI diag_attr				
SERVO, TM41, VECTOR	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned32		Dynamic index: -		Func. diagram: -
	P-Group: Messages		Units group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min		Max		Factory setting
	-		-		-
Description:	Displays the diagnostic attributes of the safety messages that have occurred.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Hardware replacement recommended	Yes	No	-

p9752	SI message cases, counter / SI msg_cases count		
SERVO, VECTOR	Can be changed: U, T	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Messages	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	65535	0
Description:	Number of safety messages that have occurred since the last reset.		
Dependency:	The safety message buffer is cleared by resetting the parameter to 0. Refer to: r9744, r9747, r9748, r9749, r9753, r9754, r9755, r9756		
Note:	The parameter is reset to 0 at POWER ON.		

r9753[0...63]	SI message value for float values / SI msg_val float		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays additional information about the safety message that has occurred for float values.		
Dependency:	Refer to: r9744, r9747, r9748, r9749, p9752, r9754, r9755, r9756		
r9754[0...63]	SI message time received in days / SI t_msg rcv days		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the relative system runtime in days when the safety message occurred.		
Dependency:	Refer to: r9744, r9747, r9748, r9749, p9752, r9753, r9755, r9756		
r9755[0...63]	SI message time removed in milliseconds / SI t_msg rem ms		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Messages Not for motor type: - Min - [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the relative system runtime in milliseconds when the safety message was removed.		
Dependency:	Refer to: r9744, r9747, r9748, r9749, p9752, r9753, r9754, r9756		
r9756[0...63]	SI message time removed in days / SI t_msg rem days		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Messages Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the relative system runtime in days when the safety message was removed.		
Dependency:	Refer to: r9744, r9747, r9748, r9749, p9752, r9753, r9754, r9755		
p9761	SI password input / SI password inp		
SERVO, VECTOR	Can be changed: C1, T Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: 2800 Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Enters the Safety Integrated password.		
Dependency:	Refer to: F01659		
Note:	It is not permissible to change Safety Integrated parameter settings until the Safety Integrated password has been entered.		

p9762	SI password new / SI password new		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: 2800 Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Enters a new Safety Integrated password.		
Dependency:	A change made to the Safety Integrated password must be acknowledged in the following parameter: Refer to: p9763		
p9763	SI password acknowledgement / SI ackn password		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: 2800 Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Acknowledges the new Safety Integrated password.		
Dependency:	Refer to: p9762		
Note:	The new password entered into p9762 must be re-entered in order to acknowledge. p9762 = p9763 = 0 is automatically set after the new Safety Integrated password has been successfully acknowledged.		
r9765	SI Motion forced check procedure remaining time (Control Unit) / SI Mtn dyn remain		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min - [h]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [h]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [h]
Description:	Displays the time remaining until the next dynamization and testing of the safety motion monitoring functions integrated in the drives. The signal source to initiate the forced checking procedure is parameterized in p9705.		
Dependency:	Refer to: p9705 Refer to: C01798		
r9770[0...3]	SI vers. safety fcts that run indep. in the drive (Control Unit) / SI version Drv CU		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2802 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the Safety Integrated version for the safety functions that run independently in the drive on the Control Unit.		
Index:	[0] = Safety Version (major release) [1] = Safety Version (minor release) [2] = Safety Version (baselevel or patch) [3] = Safety Version (hotfix)		
Dependency:	Refer to: r9870, r9890		

Note: Example:
r9770[0] = 2, r9770[1] = 60, r9770[2] = 1, r9770[3] = 0 --> Safety version V02.60.01.00

r9771 SI common functions (Control Unit) / SI common fct CU

SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2804 Unit selection: - Expert list: 1 Factory setting -
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Description: Displays the Safety Integrated monitoring functions supported on the Control Unit and Motor Module. The Control Unit determines this display.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	STO supported via terminals	Yes	No	2804
	01	SBC supported	Yes	No	2804
	02	SI Motion supported	Yes	No	2804
	03	SS1 supported	Yes	No	2804
	04	PROFIsafe supported for Extended Functions	Yes	No	-
	05	Drive-based motion monitoring functions supported	Yes	No	-
	06	PROFIsafe supported for Basic Functions	Yes	No	-
	07	Encoderless motion monitoring supported	Yes	No	-
	08	Safe Brake Adapter supported	Yes	No	-
	09	PROFIsafe supported BasicFunctions for parall circuit connection	Yes	No	-

Dependency: Refer to: r9871

Note: CU: Control Unit
SBC: Safe Brake Control
SI: Safety Integrated
SS1: Safe Stop 1
STO: Safe Torque Off / SH: Safe standstill

r9772.0...23 CO/BO: SI status (Control Unit) / SI status CU

SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 2804 Unit selection: - Expert list: 1 Factory setting -
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Description: Displays the Safety Integrated status on the Control Unit.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	STO selected on Control Unit	Yes	No	2810
	01	STO active on Control Unit	Yes	No	2810
	02	SS1 delay time active on the Control Unit	Yes	No	2810
	04	SBC requested	Yes	No	2814
	05	SS1 selected on the Control Unit (Basic Functions)	Yes	No	-
	06	SS1 active on the Control Unit (Basic Functions)	Yes	No	-

List of parameters

09	STOP A cannot be acknowledged, active	Yes	No	2802
10	STOP A active	Yes	No	2802
15	STOP F active	Yes	No	2802
16	STO cse: Safety comm. mode	Yes	No	-
17	STO cause selection via terminal (Basic Functions)	Yes	No	-
18	STO cause: selection via SMM	Yes	No	-
19	STO cause actual value missing	Yes	No	-
20	STO cause selection PROFIsafe (Basic Functions)	Yes	No	-
22	SS1 cause selection terminal (Basic Functions)	Yes	No	-
23	SS1 cause selection PROFIsafe (Basic Functions)	Yes	No	-

Dependency: Refer to: r9872

Note: Re bit 00:
When STO is selected, the cause is displayed in bits 16 ... 20.
Re bit 05:
When SS1 is selected, the cause is displayed in bits 22 and 23.
Re bit 18:
When the bit is set, STO is selected via PROFIsafe or Terminal Module 54F (TM54F).
Re bit 19:
With SMM encoderless no actual value sensing is possible on account of OFF2.
With SMM with encoder no actual value sensing is possible on account of parking.
SMM: Safe Motion Monitoring
Re bit 22 and 23:
These bits show via which path the SS1 has been triggered, i.e. what has started the SS1 delay time.
If the SS1 delay time is not started (e.g. because an STO is triggered at the same time), neither of the two bits is set.

r9773.0...31 CO/BO: SI status (Control Unit + Motor Module) / SI status CU+MM

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2804
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the Safety Integrated status on the drive (Control Unit + Motor Module).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	STO selected in drive	Yes	No	2804
	01	STO active in drive	Yes	No	2804
	02	SS1 delay time active in the drive	Yes	No	2804
	04	SBC requested	Yes	No	2804
	05	SS1 selected in the drive (Basic Functions)	Yes	No	-
	06	SS1 active in the drive (Basic Functions)	Yes	No	-
	31	Shutdown paths must be tested	Yes	No	2810

Note: This status is formed from the AND operation of the relevant status of the two monitoring channels.

r9774.0...31 CO/BO: SI status (group STO) / SI stat group STO

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2804
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status for Safety Integrated of the group to which this drive belongs.
These signals are an AND logic operation of the individual status signals of the drives included in this group.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	STO selected in group	Yes	No	2804
	01	STO active in group	Yes	No	2804
	02	SS1 delay time active in group	Yes	No	-
	04	SBC requested in group	Yes	No	2804
	05	SS1 selected in group (Basic Functions)	Yes	No	-
	06	SS1 active in group (Basic Functions)	Yes	No	-
	31	Shutdown paths of the group must be tested	Yes	No	2804

Dependency: Refer to: p9620, r9773

Notice: If a drive belonging to a group is deactivated via p0105, then the signals in r9774 can no longer be correctly displayed (Remedy: Before deactivating, remove this drive from the group).

Note: A group is formed by appropriately grouping the terminals for the function "Safe Torque Off" (STO).
The status of a group of n drives is, for drives 1 to n - 1 displayed with a delay of one monitoring clock cycle; this is a system-related effect.

r9776 SI Motion diagnostics / SI Mtn diag

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 4
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: The parameter is used for diagnostics.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Safety parameter changed POWER ON required	Yes	No	-

Note: Re bit 00:

The bit indicates whether a change has been made to at least one Safety parameter which will only take effect after a POWER ON.

r9780 SI monitoring clock cycle (Control Unit) / SI monitor_clk CU

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2802
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	- [ms]	- [ms]	- [ms]

Description: Displays the clock cycle time for the Safety Integrated Basic Functions on the Control Unit.

Dependency: Refer to: r9880

r9781[0...1]	SI checksum to check changes (Control Unit) / SI chg chksm CU		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the checksum for tracking changes for Safety Integrated. These are additional checksums that are created to track changes (fingerprint for the "safety logbook" functionality) to safety parameters (that are relevant for checksums).		
Index:	[0] = SI checksum to track functional changes [1] = SI checksum to track hardware-specific changes		
Dependency:	Refer to: p9601, p9729, p9799 Refer to: F01690		
r9782[0...1]	SI time stamps to check changes (Control Unit) / SI chg t CU		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min - [h]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [h]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [h]
Description:	Displays the time stamps for the checksums for tracking changes for Safety Integrated. The time stamps for the checksums for tracking changes (fingerprint for the "safety logbook" functionality) made to safety parameters are saved in parameters p9781[0] and p9781[1].		
Index:	[0] = SI time stamp for checksum to track functional changes [1] = SI time stamp for checksum to track hardware-specific changes		
Dependency:	Refer to: p9601, p9729, p9799 Refer to: F01690		
r9794[0...19]	SI crosswise comparison list (Control Unit) / SI DCC_list CU		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2802 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of the data that are being presently compared crosswise on the Control Unit. Example: r9794[0] = 1 (monitoring clock cycle) r9794[1] = 2 (enable safety functions) r9794[2] = 3 (SGE changeover, tolerance time) r9794[3] = 4 (transition time, STOP F to STOP A) ... The content of the list of crosswise-compared data is dependent upon the particular application.		
Dependency:	Refer to: r9894		
Note:	A complete list of numbers for crosswise-compared data items appears in fault F01611.		

r9795 SI diagnostics STOP F (Control Unit) / SI diag STOP F CU

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2802
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

Description: Displays the number of the cross-checked data which has caused STOP F on the Control Unit.

Dependency: Refer to: r9895

Refer to: F01611

Note: A complete list of numbers for crosswise-compared data items appears in fault F01611.

r9798 SI actual checksum SI parameters (Control Unit) / SI act_checksum CU

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2800
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting -

Description: Displays the checksum over the checked Safety Integrated parameters on the Control Unit (actual checksum).

Dependency: Refer to: p9799, r9898

p9799 SI reference checksum SI parameters (Control Unit) / SI set_checksum CU

SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2800
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFF FFFF hex	Factory setting 0000 hex

Description: Sets the checksum for the checked Safety Integrated parameters on the Control Unit (reference checksum).

Dependency: Refer to: r9798, p9899

p9801	SI enable, functions integrated in the drive (Motor Module) / SI enable fct MM																							
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 bin																					
Description:	Sets the enable signals for safety functions on the Motor Module that are integrated in the drive. Not all of the settings listed below will be permissible, depending on the Control Unit and Motor Module or Power Module being used: - p9801 = 0: Safety functions integrated in the drive disabled. - p9801 = 1: STO/SS1 enabled via terminals. Permissible if r9871.0 = 1. - p9801 = 4: Motion monitoring functions integrated in the drive enabled via Terminal Module 54F (TM54F) (SINAMICS S120) or via an integrated F-DI/F-DO (SINAMICS S110). Permissible if r9871.5 = 1. - p9801 = 5: Motion monitoring functions integrated in the drive enabled via Terminal Module 54F (TM54F) and STO/SS1 via terminals (SINAMICS S120 only). Permissible if r9871.5 = 1. - p9801 = 8: STO/SS1 enabled via PROFIsafe. Permissible if r9871.6 = 1. - p9801 = 9: STO/SS1 enabled via PROFIsafe and STO/SS1 via terminals. Permissible if r9871.6 = 1. - p9801 = 12: Motion monitoring functions integrated in the drive enabled via PROFIsafe. Permissible if r9871.4 = 1. - p9801 = 13: Motion monitoring functions integrated in the drive enabled via PROFIsafe and STO/SS1 via terminals. Permissible if r9871.4 = 1.																							
Bit field:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FP</th></tr> <tr> <td>00</td><td>STO (SH) via terminals (Motor Module) enable</td><td>Enable</td><td>Inhibit</td><td>2810</td></tr> <tr> <td>02</td><td>Motion monitoring functions integr. in the drive (Motor Module)</td><td>Enable</td><td>Inhibit</td><td>-</td></tr> <tr> <td>03</td><td>PROFIsafe (Motor Module) enable</td><td>Enable</td><td>Inhibit</td><td>-</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FP	00	STO (SH) via terminals (Motor Module) enable	Enable	Inhibit	2810	02	Motion monitoring functions integr. in the drive (Motor Module)	Enable	Inhibit	-	03	PROFIsafe (Motor Module) enable	Enable	Inhibit	-			
Bit	Signal name	1 signal	0 signal	FP																				
00	STO (SH) via terminals (Motor Module) enable	Enable	Inhibit	2810																				
02	Motion monitoring functions integr. in the drive (Motor Module)	Enable	Inhibit	-																				
03	PROFIsafe (Motor Module) enable	Enable	Inhibit	-																				
Dependency:	Refer to: p9601, r9871																							
Note:	MM: Motor Module. SI: Safety Integrated. SMM: Safe Motion Monitoring. STO: Safe Torque Off/SH: Safe standstill. SS1: Safe Stop 1 (corresponds to Stop Category 1 acc. to EN 60204). F-DI: Failsafe Digital Input. F-DO: Failsafe Digital Output. A change only becomes effective after a POWER ON.																							
p9802	SI enable Safe Brake Control (Motor Module) / SI enable SBC MM																							
SERVO, VECTOR	Can be changed: C2(95) Data type: Integer32 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 3 Func. diagram: 2814 Unit selection: - Expert list: 1 Factory setting 0																					
Description:	Sets the enable signal for the "Safe Brake Control" function (SBC) on the Motor Module. 0: Inhibit SBC 1: Enable SBC																							
Dependency:	Refer to: p9602																							

Note: The "Safe Brake Control" function is not activated until at least one safety monitoring function has been enabled (i.e. p9501 not equal to 0 and/or p9801 not equal to 0).
 It does not make sense to parameterize "no motor holding brake available" and enable "Safe Brake Control" (p1215 = 0, p9602 = p9802 = 1) if there is no motor holding brake.
 The parameterization "motor holding brake the same as sequence control, connection via BICO" and "Safe Brake Control" enabled (p1215 = 3, p9602 = 1, p9802 = 1) is not practical.
 It is not permissible to parameterize "motor holding brake without feedback signals" and also enable "safe brake control" (p1278 = 1, p9602 = 1, p9802 = 1).
 MM: Motor Module
 SBC: Safe Brake Control
 SI: Safety Integrated

p9810			
SI PROFIsafe address (Motor Module) / SI PROFIsafe MM			
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0000 hex	Max FFFE hex	Factory setting 0000 hex
Description:	Sets the PROFIsafe address of the Motor Module.		
p9821			
BI: SI signal source for SBA (Motor Module) / SI s_src SBA MM			
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2810
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min -	Max -	Factory setting 0
Description:	An entry is added, specifying whether and/or via which digital CU input the SBA feedback signal (SBA_DIAG) is to be read in; the parameter can be interconnected as a BICO drain. 0 : No SBC with SBA (default) Bico code: To be parameterized by the user SBA and no DQ CIM: p0722.x CU signal source for DIx where x = { 0,1,2...7 } SBA with DQ CIM: p9872.3 Signal source is permanently interconnected to bit 3		
Dependency:	Refer to: p9601		
Note:	No difference is tolerated for a data cross-check between p9621 and p9821.		
p9822[0...1]			
SI SBA relay wait times (Motor Module) / SI SBA t_wait MM			
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0.00 [µs]	Max 1000000.00 [µs]	Factory setting [0] 100000.00 [µs] [1] 65000.00 [µs]
Description:	Sets the wait times for activating and deactivating the relay. The relay-specific minimum wait times for evaluating the feedback signal contacts have to be set.		
Index:	[0] = Relay wait time activation [1] = Relay wait time deactivation		
Dependency:	Refer to: p9850		

p9850	SI SGE changeover tolerance time (Motor Module) / SI SGE_chg tol MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2000000.00 [µs]	Access level: 3 Func. diagram: 2810 Unit selection: - Expert list: 1 Factory setting 500000.00 [µs]
Description:	Sets the tolerance time to change over the safety-related inputs (SGE) on the Motor Module. An SGE changeover is not effective simultaneously due to the different runtimes in the two monitoring channels. After an SGE changeover, dynamic data is not subject to a data cross-check during this tolerance time.		
Dependency:	Refer to: p9650		
Note:	For a data cross-check between p9650 and p9850, a difference of one Safety monitoring clock cycle is tolerated. The parameterized time is internally rounded-off to an integer multiple of the monitoring clock cycle. SGE: Safety-related input (e.g. STO terminals)		
p9851	SI STO/SBC/SS1 debounce time (Motor Module) / SI STO t_debou MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100000.00 [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [µs]
Description:	Sets the debounce time for the EP terminal of the Motor Module. The debounce time is rounded to whole milliseconds. The debounce time indicates the maximum duration of a fault pulse on the F-DIs, so that there are no negative effects on the SGEs. Example: Debounce time = 1 ms: Fault pulses of 1 ms are filtered; only pulses longer than 2 ms are processed. Debounce time = 3 ms: Fault pulses of 3 ms are filtered; only pulses longer than 4 ms are processed.		
Note:	Example: Debounce time = 1 ms: Fault pulses of 1 ms are filtered; only pulses longer than 2 ms are processed. Debounce time = 3 ms: Fault pulses of 3 ms are filtered; only pulses longer than 4 ms are processed.		
p9852	SI Safe Stop 1 delay time (Motor Module) / SI Stop 1 t_del MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 300000.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0.00 [ms]
Description:	Sets the delay time of the pulse suppression for the function "Safe Stop 1" (SS1) on the Motor Module to brake along the OFF3 down ramp (p1135).		
Dependency:	Refer to: p1135, p9652		
Note:	For a data cross-check between p9652 and p9852, a difference of one Safety monitoring clock cycle is tolerated. The parameterized time is rounded internally to an integer multiple of the monitoring clock cycle. SS1: Safe Stop 1 (corresponds to Stop Category 1 acc. to EN60204)		

p9858 SI transition time STOP F to STOP A (Control Unit) / SI STOP F->A MM				
SERVO, VECTOR	Can be changed: C2(95)	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: 2802	
	P-Group: Safety Integrated	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min 0.00 [µs]	Max 30000000.00 [µs]	Factory setting 0.00 [µs]	
Description:	Sets the transition period from STOP F to STOP A on the Motor Module.			
Dependency:	Refer to: p9658, r9895 Refer to: F30611			
Note:	For a data cross-check between p9658 and p9858, a difference of one Safety monitoring clock cycle is tolerated. The parameterized time is rounded internally to an integer multiple of the monitoring clock cycle. STOP F: Defect in a monitoring channel (error in the data cross-check) STOP A: Pulse suppression via the safety shutdown path			

r9870[0...3] SI version safety functions integrated in drive (Motor Module) / SI version MM				
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3	
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2802	
	P-Group: Safety Integrated	Units group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min -	Max -	Factory setting -	
Description:	Displays the Safety Integrated version for the safety functions integrated in the drive on the Motor Module.			
Index:	[0] = Safety Version (major release) [1] = Safety Version (minor release) [2] = Safety Version (baselevel or patch) [3] = Safety Version (hotfix)			
Dependency:	Refer to: r9770, r9890			
Note:	Example: r9870[0] = 2, r9870[1] = 60, r9870[2] = 1, r9870[3] = 0 --> Safety version V02.60.01.00			

r9871 SI common functions (Motor Module / SI general fct MM					
SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2804		
	P-Group: Safety Integrated	Units group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min -	Max -	Factory setting -		
Description:	Displays the Safety Integrated monitoring functions supported on the Control Unit and Motor Module. The Motor Module determines this display.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	STO supported via terminals	Yes	No	2804
	01	SBC supported	Yes	No	2804
	02	SI Motion supported	Yes	No	2804
	03	SS1 supported	Yes	No	2804
	04	PROFIsafe supported for Extended Functions	Yes	No	-
	05	Drive-based motion monitoring functions supported	Yes	No	-
	06	PROFIsafe supported for Basic Functions	Yes	No	-
	07	Encoderless motion monitoring supported	Yes	No	-
	08	Safe Brake Adapter supported	Yes	No	-
	09	PROFIsafe supported BasicFunctions for parall circuit connection	Yes	No	-

Dependency: Refer to: r9771
Note: MM: Motor Module
 SBC: Safe Brake Control
 SI: Safety Integrated
 SS1: Safe Stop 1
 STO: Safe Torque Off / SH: Safe standstill

r9872.0...23 CO/BO: SI status list (Motor Module) / SI status MM

SERVO, VECTOR	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2804
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
-	-	-	

Description: Displays the Safety Integrated status on the Motor Module.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	STO on Motor Module selected	Yes	No	2810
	01	STO on Motor Module active	Yes	No	2810
	02	SS1 delay time on Motor Module active	Yes	No	2810
	03	Safe Brake Adapter feedback signal	High	Low	-
	04	SBC requested	Yes	No	2814
	05	SS1 selected on the Motor Module (Basic Functions)	Yes	No	-
	06	SS1 active on the Motor Module (Basic Functions)	Yes	No	-
	09	STOP A cannot be acknowledged, active	Yes	No	2802
	10	STOP A active	Yes	No	2802
	15	STOP F active	Yes	No	2802
	16	STO cse: Safety comm. mode	Yes	No	-
	17	STO cause selection via terminal (Basic Functions)	Yes	No	-
	18	STO cause: selection via SMM	Yes	No	-
	20	STO cause selection PROFIsafe (Basic Functions)	Yes	No	-
	22	SS1 cause selection terminal (Basic Functions)	Yes	No	-
	23	SS1 cause selection PROFIsafe (Basic Functions)	Yes	No	-

Dependency: Refer to: r9772

Notice: If communication between the Control Unit and the Motor Module is interrupted (e.g. by switching off the Motor Module), this display parameter is no longer updated. The last transferred status of the Motor Module is displayed.

Note: Re bit 00:
 When STO is selected, the cause is displayed in bits 16 ... 18 and in bit 20.
 Re bit 05:
 When SS1 is selected, the cause is displayed in bits 22 and 23.
 Re bit 18:
 When the bit is set, STO is selected via PROFIsafe or Terminal Module 54F (TM54F).
 SMM: Safe Motion Monitoring
 Re bit 22 and 23:
 These bits show via which path the SS1 has been triggered, i.e. what has started the SS1 delay time.
 If the SS1 delay time is not started (e.g. because an STO is triggered at the same time), neither of the two bits is set.

r9880	SI monitoring clock cycle (Motor Module) / SI monitor_click MM		
SERVO, VECTOR	Can be changed: - Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min - [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [ms]	Access level: 3 Func. diagram: 2802 Unit selection: - Expert list: 1 Factory setting - [ms]
Description:	Displays the clock cycle time for the Safety Integrated Basic Functions on the Motor Module.		
Dependency:	Refer to: r9780		
r9881[0...11]	SI Motion Sensor Module Node Identifier second channel / SI Mtn SM Ident		
SERVO, VECTOR	Can be changed: - Data type: Unsigned8 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the Node Identifier of the Sensor Module that the second channel uses for the motion monitoring functions.		
r9890[0...2]	SI version (Sensor Module) / SI version SM		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the Safety Integrated version on the Sensor Module.		
Index:	[0] = Safety Version (major release) [1] = Safety Version (minor release) [2] = Safety Version (baselevel or patch)		
Dependency:	Refer to: r9770, r9870		
Note:	Example: r9890[0] = 2, r9890[1] = 3, r9890[2] = 1 --> Safety-Version V02.03.01		
r9894[0...19]	SI crosswise comparison list (Motor Module) / SI DCC_list MM		
SERVO, VECTOR	Can be changed: - Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2802 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of the data that are being presently compared crosswise on the Motor Module. Example: r9894[0] = 1 (monitoring clock cycle) r9894[1] = 2 (enable safety functions) r9894[2] = 3 (SGE changeover, tolerance time) r9894[3] = 4 (transition time, STOP F to STOP A) ... The content of the list of crosswise-compared data is dependent upon the particular application.		
Dependency:	Refer to: r9794		
Note:	The complete list of numbers for data cross-check is listed in Fault F30611.		

r9895	SI diagnostics STOP F (Motor Module) / SI diag STOP F MM		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: 2802 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the number of the cross-checked data which has caused STOP F on the Motor Module.		
Dependency:	Refer to: r9795 Refer to: F30611		
Note:	The complete list of numbers for data cross-check is listed in Fault F30611.		
r9898	SI actual checksum SI parameters (Motor Module) / SI act_checksum MM		
SERVO, VECTOR	Can be changed: - Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2800 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the checksum for the checked Safety Integrated parameters on the Motor Module (actual checksum).		
Dependency:	Refer to: r9798, p9899		
p9899	SI reference checksum SI parameters (Motor Module) / SI set_checksum MM		
SERVO, VECTOR	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: 2800 Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the checksum for the checked Safety Integrated parameters on the Motor Module (reference checksum).		
Dependency:	Refer to: p9799, r9898		
r9900	Actual topology number of indices / Act topo indices		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: Topology Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the number of indices of the actual topology.		
Dependency:	Refer to: r9901		
Note:	Only for internal Siemens use. The parameter is not displayed for the STARTER commissioning software.		

r9901[0...n]	Actual topology / Act topo		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: Topology Not for motor type: - Min -	Calculated: - Dynamic index: r9900 Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting -
Description:	Displays the actual topology of the drive unit. The actual topology is sub-divided into several sections. Each of the following data is saved under an index. General data on the topology: - version - attribute to compare the actual topology and target topology - number of components Data on a component: - type component of the node ID of the component - number of DRIVE-CLiQ sockets in the Node Identifier - manufacturer and version of the Node Identifier - serial number of the Node Identifier (4 indices) - index of the component - order number (8 indices) - attribute to compare the actual topology and target topology of the component - communications address - number of port types - port type - number of ports of the port type - communications address of the associated/linked component - number of the associated/linked port - communications address of the associated/linked component - number of the associated/linked port - etc. Data on the next component: - etc.		
Dependency:	Refer to: r9900		
Note:	Only for internal Siemens use. The parameter is not displayed for the STARTER commissioning software.		
p9902	Target topology number of indices / TargetTopo indices		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: Topology Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 1
Description:	Sets the number of target topology indices.		
Dependency:	Refer to: p9903		
Note:	Only for internal Siemens use. The parameter is not displayed for the STARTER commissioning software.		

p9903[0...n]	Target topology / Target topology		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned16 P-Group: Topology Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: p9902 Units group: - Scaling: - Max FFFF hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0000 hex
Description:	Sets the target topology of the drive unit. The target topology is sub-divided into several sections. Each of the following data is saved under an index. General data on the topology: - version - attribute to compare the actual topology and target topology - number of components Data on a component: - type component of the Node Identifier of the component - number of DRIVE-CLiQ sockets in the Node Identifier - manufacturer and version of the Node Identifier - serial number of the Node Identifier (4 indices) - index of the component - order number (8 indices) - attribute to compare the actual topology and target topology of the component - component number - number of port types - port type - number of ports of the port type - component number of the associated/linked component - number of the associated/linked port - component number of the associated/linked component - number of the associated/linked port - etc. Data on the next component: - etc.		
Dependency:	Refer to: p9902		
Note:	The target topology can only be modified using the commissioning software. Only for internal Siemens use. The parameter is not displayed for the STARTER commissioning software. Changes do not become effective until they have been accepted with p9428 = 1, or on change of status from p0009 = 101 to 0 or 111.		

p9904	Topology comparison, acknowledge differences / Topo_compare ackn		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Unsigned32 P-Group: Topology Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	If, when comparing the actual topology and target topology, only error has occurred, that can be acknowledged, then using this parameter, a new comparison can be started - acknowledging the error in the target topology. Differences that can be acknowledged: - topology comparison, component shifted - topology comparison, serial number of a component has been detected to be different (byte 3 = 1) - topology comparison shows one component that is connected differently The following parameter values are available: p9904 = 1 --> the procedure is started. p9904 = 0 after starting --> the procedure has been successfully completed. p9904 = 1 after starting --> the procedure has not been successfully completed. The possible causes for an unsuccessful procedure are located in bytes 4, 3, 2. Byte 2: Number of structural differences. Byte 3: Number of differences that can be acknowledged (p9904). Byte 4: Number of differences. These differences can be resolved as follows: - sets the topology comparison (p9906 or p9907/p9908). - change over the actual topology. The appropriate action should be selected corresponding to the message that is displayed/output.		
Note:	In order to permanently accept the acknowledgement of the fault that can be resolved, then it must be saved in a non-volatile fashion (p0977).		
p9905	Device specialization / Specialization		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Unsigned16 P-Group: Topology Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	With p9905 = 1, the serial numbers and the hardware versions of all of the components are transferred from the actual topology into the target topology and a new comparison is started.. For this device specialization, the components of the target topology may only differ from those of the actual topology by the serial numbers. With p9905 = 2, the serial numbers, the hardware versions and the order numbers of all of the components are transferred from the actual topology into the target topology and a new comparison is started.. For this device specialization, the components of the target topology may only differ from those of the actual topology by the serial numbers and order numbers.		
Note:	p9905 is automatically set to 0 at the end of the operation. In order to permanently accept the data, it is necessary to save in a non-volatile fashion (p0977).		

p9906	Topology comparison, comparison stage of all components / Topo_cmpr tot comp		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Integer16 P-Group: Topology Not for motor type: - Min 0 Max 99	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the type of comparison between the actual topology and target topology. The comparison is started by setting the required value.		
Value:	0: High: Compares the complete electronic rating plate 1: Average: Compares the component type and the Order number 2: Low: Compares the component type 3: Minimum: Compares the component class 99: Topology has different comparison stages		
Note:	The electronic rating plate comprises the following data: - component type (e.g. "SMC20") - Order No. (e.g. "6SL3055-0AA0-5BA0") - manufacturer (e.g. SIEMENS) - hardware version (e.g. "A") - Serial No. (e.g. "T-P30050495") When comparing the topology, the following data is compared in the target and actual topologies: p9906 = 0: Component type, Order No., Hardware version, Manufacturer, Serial No. p9906 = 1: Component type, Order No. p9906 = 2: Component type p9906 = 3: Component class (e.g. Sensor Module or Motor Module)		
p9907	Topology comparison, comparison stage of the component number / Topo_cmpr comp_no		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Unsigned8 P-Group: Topology Not for motor type: - Min 0 Max 199	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Enters the number of the component where the setting of how the actual topology should be compared to the target topology should be changed.		
Dependency:	Refer to: p9908		
p9908	Topology comparison, comparison stage of a component / Topo_cmpr 1 comp		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Integer16 P-Group: Topology Not for motor type: - Min 0 Max 99	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the type of comparison of a component in the target topology with the actual topology. The comparison is started by setting the required value.		
Value:	0: High: Compares the complete electronic rating plate 1: Average: Compares the component type and the Order number 2: Low: Compares the component type 3: Minimum: Compares the component class 99: Topology has different comparison stages		
Dependency:	Refer to: p9907		

Note: The electronic rating plate comprises the following data:

- component type (e.g. "SMC20")
- Order No. (e.g. "6SL3055-0AA0-5BA0")
- manufacturer (e.g. SIEMENS)
- hardware version (e.g. "A")
- Serial No. (e.g. "T-P30050495")

When comparing the topology, the following data is compared in the target and actual topologies:

p9908 = 0: Component type, Order No., Hardware version, Manufacturer, Serial No.

p9908 = 1: Component type, Order No.

p9908 = 2: Component type

p9908 = 3: Component class (e.g. Sensor Module or Motor Module)

p9909 Topology comparison, component replacement / Topo_cmptr replace			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1)	Calculated: -	Access level: 3
	Data type: Unsigned8	Dynamic index: -	Func. diagram: -
	P-Group: Topology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 1	Factory setting 1
Description:	For p9909 = 1, the serial number and the hardware version of the new replaced component is automatically transferred from the actual topology into the target topology and then saved in a non-volatile fashion. For the components that have been replaced, the electronic rating plate must match as far as the following data is concerned: - component type (e.g. "SMC20") - Order No. (e.g. "6SL3055-0AA0-5BA0") For p9909 = 0, serial numbers and hardware versions are not automatically transferred. In this case, the transfer must be made using p9904.		
Dependency:	Refer to: p9904, p9905		
Note:	The modified target topology is automatically saved in a non-volatile fashion when the drive object runs-up (e.g. after a POWER ON). Special case for Control Unit and option slot modules: When replacing these components, independent of p9909, the serial number and hardware version are automatically transferred and saved in a non-volatile fashion.		

p9910 Transfer additional components into the target topology / Transfer comp			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1)	Calculated: -	Access level: 1
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Topology	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 6	Factory setting 0
Description:	Transfer additional inserted DRIVE-CLiQ components into the target topology and add the appropriate drive objects to the project.		
Value:	0: No selection 1: Drive object type SERVO 2: Drive object type VECTOR 3: SINAMICS GM (DFEMV & VECTORMV) 4: SINAMICS SM (AFEMV & VECTORMV) 5: SINAMICS GL (VECTORGL) 6: SINAMICS SL (VECTORSL)		

p9911[0...3]	Insert drive object / Drv_obj insert		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Unsigned32 P-Group: - Not for motor type: - Min 0 Max 4294967295	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	New drive objects can be created using this parameter. Index 0: The values 2 ... 62 are permissible. Index 1: Number of the drive object type (e.g. 11 for type SERVO). Index 2: Function modules defined for the drive object. Index 3: = 0: Ready. = 1: Reset (only indices 0 ... 3). = 2: Reset all (indices 0 ... 3 and flagged entries). = 3: Check and flag for insertion.		
Index:	[0] = Drive object number [1] = Drive object type [2] = Drive object function module [3] = Reset or check and flag for insertion		
Note:	Only for internal Siemens use. The parameter is not displayed for the STARTER commissioning software.		

p9912[0...1]	Delete drive object / Drv_obj delete		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(3) Data type: Unsigned16 P-Group: - Not for motor type: - Min 0 Max 62	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Drive objects can be deleted using this parameter. Index 0: The values 2 ... 62 are permissible. Index 1: = 0: Ready. = 1: Reset (only indices 0 and 1) = 2: Reset all (indices 0 and 1 and flagged entries). = 3: Check and flag for deletion. = 30: Check and flag for deletion. Keep target topology.		
Index:	[0] = Drive object number [1] = Reset or check and flag for deletion		
Note:	Only for internal Siemens use. The parameter is not displayed for the STARTER commissioning software.		

p9913[0...2]	Change drive object number / Change drv_obj_no		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(4) Data type: Unsigned16 P-Group: - Not for motor type: - Min 0 Max 62	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Existing drive objects can be assigned new numbers using these parameters. Index 0: The values 2 ... 62 are permissible. Index 1: The values 2 ... 62 are permissible. Index 2: = 0: Ready. = 1: Reset (only indices 0 ... 2). = 2: Reset all (indices 0 ... 2 and flagged entries). = 3: Check and flag for modification.		
Index:	[0] = Drive object number old [1] = Drive object number new [2] = Reset or check and flag for modification		
Note:	Only for internal Siemens use. The parameter is not displayed for the STARTER commissioning software.		

p9914[0...2]	Change component number / Change comp_no		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1 Data type: Unsigned16 P-Group: - Not for motor type: - Min 0 Max 199	Calculated: - Dynamic index: - Units group: - Scaling: -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	You can change the number of topology components using this parameter. Index 0: The values 2 ... 199 are permissible. Index 1: The values 2 ... 199 are permissible. Index 2: = 0: Ready. = 1: Reset (only indices 0 ... 2). = 2: Reset all (indices 0 ... 2 and flagged entries). = 3: Check and flag for modification.		
Index:	[0] = Component number old [1] = Component number new [2] = Reset or check and flag for modification		
Note:	Only for internal Siemens use. The parameter is not displayed for the STARTER commissioning software.		

p9915	DRIVE-CLiQ data transfer error shutdown threshold master / DLQ fault master		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Unsigned32 P-Group: Topology Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max 0007 07FF hex	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0007 02FF hex
Description:	Only for internal Siemens service purposes.		
p9916	DRIVE-CLiQ data transfer error shutdown threshold slave / DLQ fault slave		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(1) Data type: Unsigned32 P-Group: Topology Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max 0007 07FF hex	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0007 02FF hex
Description:	Only for internal Siemens service purposes.		
p9917[0...1]	Delete component / Delete comp		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: C1(30) Data type: Unsigned16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 199	Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0
Description:	Excessive components that have not been assigned can be removed from the component target topology using this parameter. Index 0: The values 2 ... 199 are permissible. Index 1: = 0: Ready. = 1: Reset (only indices 0 and 1) = 2: Reset all (indices 0 and 1 and flagged entries). = 3: Check and flag for deletion.		
Index:	[0] = Component number [1] = Reset or check and flag for deletion		
Note:	Only for internal Siemens use. The parameter is not displayed for the STARTER commissioning software.		

p9920[0...99]	Licensing, enter license key / Enter license key		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Enters the license key for this drive unit. Example of the license key: EACZ-QBCA = 69 65 67 90 45 81 66 67 65 dec (ASCII characters) Index 0 = license key character 1 (e.g. 69 dec) Index 1 = license key character 2 (e.g. 65 dec) ... Index 19 = license key character 20 (e.g. 0 dec) With the STARTER commissioning software, ASCII characters are not entered coded, i.e. the characters of the license key can be entered as printed in the Certificate of License. In this case, STARTER codes the characters.		
Dependency:	Refer to: r7843, p9921 Refer to: A13000, A13001, F13010		
Notice:	An ASCII table (excerpt) can be found, for example, in the Appendix of the List Manual:		
Note:	Only the ASCII characters, contained in a license key can be entered. When changing p9920[x] to the value 0, all of the following indices are also set to 0. After entering the license key, the license key must be activated (p9921). If the licensing is not adequate, then the following alarm is displayed together with LED: - A13000 --> licensing not sufficient - LED READY --> flashes green/red with 0.5 Hz		
p9921	Licensing, activate license key / Act license key		
CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: U, T Data type: Integer16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 1	Access level: 2 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Activates the entered license key. The following is executed when activating the license key. - the checksum of the entered license key is checked. - the entered license key is saved in a non-volatile fashion on the memory card. - re-enter the license key.		
Value:	0: Inactive 1: Activate start license key		
Dependency:	Refer to: p9920 Refer to: A13000, A13001, F13010		
Note:	Before activation, the license key entered using parameter p9920 is checked. If this check identifies an error, activation is rejected. In this case, writing a 1 to p9921 is rejected. When the license key has been activated, p9921 is automatically set to 0.		

r9925[0...99] Firmware file incorrect / FW file incorrCU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** -**Data type:** Unsigned8**P-Group:** -**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 2**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Displays the directory and name of the file whose status as shipped from the factory was identified as impermissible.**Dependency:** Refer to: r9926
Refer to: A01016**Note:** The directory and name of the file is displayed in the ASCII code.**r9926 Firmware check status / FW check status**CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** -**Data type:** Unsigned8**P-Group:** -**Not for motor type:** -**Min**

-

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

-

Access level: 2**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

-

Description: Displays the status when the firmware is checked when the system is booted.
0: Firmware not yet checked.
1: Check running.
2: Check successfully completed.
3: Check indicates an error.**Dependency:** Refer to: r9925
Refer to: A01016**p9930[0...8] System logbook activation / SYSLOG activation**CU_CX32, CU_I,
CU_S,
CU_S_CU310DP,
CU_S_CU310PN,
CU_S_S150**Can be changed:** U, T**Data type:** Unsigned8**P-Group:** -**Not for motor type:** -**Min**

0

Calculated: -**Dynamic index:** -**Units group:** -**Scaling:** -**Max**

255

Access level: 4**Func. diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting**

0

Description: Only for service purposes.**Index:** [0] = System logbook stage (0: Not active)
[1] = COM2/COM1 (0: COM2, 1: COM1)
[2] = Activate file write (0: Not active)
[3] = Display time stamp (0: Not displayed)
[4] = Reserved
[5] = Reserved
[6] = Reserved
[7] = Reserved
[8] = System logbook file size (stages, each 10 kB)**Notice:** Before powering down the Control Unit, ensure that the system logbook is switched out (p9930[0] = 0).

p9931[0...99]					System logbook module selection / SYSLOG mod select.									
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150					Can be changed: U, T Data type: Unsigned32 P-Group: - Not for motor type: - Min 0000 hex		Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex		Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex					
Description:					Only for service purposes.									
p9932					Save system logbook EEPROM / SYSLOG EEPROM save									
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150					Can be changed: U, T Data type: Unsigned8 P-Group: - Not for motor type: - Min 0		Calculated: - Dynamic index: - Units group: - Scaling: - Max 255		Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0					
Description:					Only for service purposes.									
r9935.0					BO: POWER ON delay signal / POWER ON t_delay									
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150					Can be changed: - Data type: Unsigned8 P-Group: - Not for motor type: - Min -		Calculated: - Dynamic index: - Units group: - Scaling: - Max -		Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -					
Description:					After power-on, binector output r9935.0 is set with the start of the first sampling time and is again reset after approx. 100 ms.									
Bit field:					Bit	Signal name	1 signal	0 signal	FP					
					00	POWER ON delay signal	High	Low	-					
p9941					Target topology feature delete all components / Feature delete									
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150					Can be changed: C1(1) Data type: Unsigned32 P-Group: Topology Not for motor type: - Min 0		Calculated: - Dynamic index: - Units group: - Scaling: - Max 1		Access level: 3 Func. diagram: - Unit selection: - Expert list: 0 Factory setting 0					
Description:					Writing the parameter to the value 1 deletes (sets to zero) the serial numbers of all components in the target topology. Through activation and deactivation this enables the actual topology components to be newly assigned to the target topology components.									
Note:					p9941 is automatically set to 0 at the end of the operation. A warm restart is triggered automatically after p0009 = 0.									

r9975[0...7]	System utilization measured / Sys util meas		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the measured system utilization. The higher the value displayed, the higher the system utilization.		
Index:	[0] = Computing time utilization (min) [1] = Computing time utilization (averaged) [2] = Computing time utilization (max) [3] = Largest total utilization (min) [4] = Largest total utilization (averaged) [5] = Largest total utilization (max) [6] = Reserved [7] = Reserved		
Dependency:	Refer to: r9976, r9979, r9980, r9981 Refer to: F01054, F01205		
Note:	Re index 3 ... 5: The total utilizations are determined using all sampling times used. The largest total utilizations are mapped here. The sampling time with the largest total utilization is displayed in r9979. Total utilization: Computing time load of sampling time involved including load from higher-priority sampling times (interrupts).		
r9976[0...7]	System utilization / Sys util		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the system utilization. If the utilization is greater than 100%, fault F01054 is output.		
Index:	[0] = Reserved [1] = Computing time utilization [2] = Reserved [3] = Reserved [4] = Reserved [5] = Largest total utilization [6] = Reserved [7] = Reserved		
Dependency:	Refer to: r9979, r9980 Refer to: F01054, F01205		
Note:	Re index 1: The value shows the total computing time load of the system. Re index 5: The total utilization is determined using all sampling times used. The largest total utilization is mapped here. The sampling time with the largest total utilization is displayed in r9979. Total utilization: Computing time load of sampling time involved including load from higher-priority sampling times (interrupts).		

r9979 Sampling time with largest total utilization / t_sampl lg total			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [µs]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [µs]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [µs]
Description:	Displays the sampling time with the largest total utilization.		
Dependency:	Refer to: r7901, r9976 Refer to: F01054		
Note:	The largest total utilization is displayed in r9976[5]. Total utilization: Computing time load of sampling time involved including load from higher-priority sampling times (interrupts).		

r9980[0...63] Sampling times utilization calculated / t_sampl util calc			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the calculated utilizations for the active sampling times based on the existing target topology.		
Index:	[0] = Net utilization 0 [1] = Total utilization 0 [2] = Net utilization 1 [3] = Total utilization 1 [4] = Net utilization 2 [5] = Total utilization 2 [6] = Net utilization 3 [7] = Total utilization 3 [8] = Net utilization 4 [9] = Total utilization 4 [10] = Net utilization 5 [11] = Total utilization 5 [12] = Net utilization 6 [13] = Total utilization 6 [14] = Net utilization 7 [15] = Total utilization 7 [16] = Net utilization 8 [17] = Total utilization 8 [18] = Net utilization 9 [19] = Total utilization 9 [20] = Net utilization 10 [21] = Total utilization 10 [22] = Net utilization 11 [23] = Total utilization 11 [24] = Net utilization 12 [25] = Total utilization 12 [26] = Net utilization 13 [27] = Total utilization 13 [28] = Net utilization 14 [29] = Total utilization 14 [30] = Net utilization 15 [31] = Total utilization 15 [32] = Net utilization 16 [33] = Total utilization 16		

[34] = Net utilization 17
 [35] = Total utilization 17
 [36] = Net utilization 18
 [37] = Total utilization 18
 [38] = Net utilization 19
 [39] = Total utilization 19
 [40] = Net utilization 20
 [41] = Total utilization 20
 [42] = Net utilization 21
 [43] = Total utilization 21
 [44] = Net utilization 22
 [45] = Total utilization 22
 [46] = Net utilization 23
 [47] = Total utilization 23
 [48] = Net utilization 24
 [49] = Total utilization 24
 [50] = Net utilization 25
 [51] = Total utilization 25
 [52] = Net utilization 26
 [53] = Total utilization 26
 [54] = Net utilization 27
 [55] = Total utilization 27
 [56] = Net utilization 28
 [57] = Total utilization 28
 [58] = Net utilization 29
 [59] = Total utilization 29
 [60] = Net utilization 30
 [61] = Total utilization 30
 [62] = Net utilization 31
 [63] = Total utilization 31

Dependency: Refer to: r7901, r9976, r9979
Refer to: F01054

Note: The corresponding sampling times can be read out in parameter r7901.
 Net utilization:
 Computing time load that is only called by the sampling time involved.
 Total utilization:
 Computing time load of sampling time involved including load from higher-priority sampling times (interrupts).

r9981[0...63] Sampling times utilization measured / t_sampl util meas			
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: -	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dynamic index: -	Func. diagram: -
	P-Group: -	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min - [%]	Max - [%]	Factory setting - [%]

Description: Displays the utilizations measured for the active sampling times.

Index:

[0] = Net utilization 0
 [1] = Total utilization 0
 [2] = Net utilization 1
 [3] = Total utilization 1
 [4] = Net utilization 2
 [5] = Total utilization 2
 [6] = Net utilization 3
 [7] = Total utilization 3
 [8] = Net utilization 4
 [9] = Total utilization 4
 [10] = Net utilization 5
 [11] = Total utilization 5
 [12] = Net utilization 6
 [13] = Total utilization 6

[14] = Net utilization 7
[15] = Total utilization 7
[16] = Net utilization 8
[17] = Total utilization 8
[18] = Net utilization 9
[19] = Total utilization 9
[20] = Net utilization 10
[21] = Total utilization 10
[22] = Net utilization 11
[23] = Total utilization 11
[24] = Net utilization 12
[25] = Total utilization 12
[26] = Net utilization 13
[27] = Total utilization 13
[28] = Net utilization 14
[29] = Total utilization 14
[30] = Net utilization 15
[31] = Total utilization 15
[32] = Net utilization 16
[33] = Total utilization 16
[34] = Net utilization 17
[35] = Total utilization 17
[36] = Net utilization 18
[37] = Total utilization 18
[38] = Net utilization 19
[39] = Total utilization 19
[40] = Net utilization 20
[41] = Total utilization 20
[42] = Net utilization 21
[43] = Total utilization 21
[44] = Net utilization 22
[45] = Total utilization 22
[46] = Net utilization 23
[47] = Total utilization 23
[48] = Net utilization 24
[49] = Total utilization 24
[50] = Net utilization 25
[51] = Total utilization 25
[52] = Net utilization 26
[53] = Total utilization 26
[54] = Net utilization 27
[55] = Total utilization 27
[56] = Net utilization 28
[57] = Total utilization 28
[58] = Net utilization 29
[59] = Total utilization 29
[60] = Net utilization 30
[61] = Total utilization 30
[62] = Net utilization 31
[63] = Total utilization 31

Dependency: Refer to: r7901, r9975, r9980
Refer to: F01054

Note: The corresponding sampling times can be read out in parameter r7901.
Net utilization:
Computing time load that is only called by the sampling time involved.
Total utilization:
Computing time load of sampling time involved including load from higher-priority sampling times (interrupts).

r9982[0...4]	Memory utilization: Data memory /		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the calculated data memory utilization rates based on the existing target topology.		
Index:	[0] = Fast Memory 1 [1] = Fast Memory 2 [2] = Fast Memory 3 [3] = Fast Memory 4 [4] = Heap		
Dependency:	Refer to: F01068		
r9983[0...4]	Memory utilization: Data memory measurement (actual load) /		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the measured memory utilization rates based on the existing target topology.		
Index:	[0] = Fast Memory 1 [1] = Fast Memory 2 [2] = Fast Memory 3 [3] = Fast Memory 4 [4] = Heap		
Dependency:	Refer to: F01068		
r9986[0...7]	DRIVE-CLiQ system load / DQ SysLoad		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the calculated DRIVE-CLiQ system load based on the existing target topology. The values are not made available until the RUNUP READY (800) state is adopted (see p3988). Indices 0..7 correspond to DRIVE-CLiQ ports X100..X107.		
Dependency:	Refer to: F01340		

r9987[0...7]	DRIVE-CLiQ bandwidth load / DQ BandWidthLoad		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the calculated DRIVE-CLiQ bandwidth load based on the existing target topology. The values are not made available until the RUNUP READY (800) state is adopted (see p3988). Indices 0..7 correspond to DRIVE-CLiQ ports X100..X107.		
Dependency:	Refer to: F01340		

r9988[0...7]	DRIVE-CLiQ DPRAM load / DQ DPRLoad		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: FloatingPoint32 P-Group: - Not for motor type: - Min - [%]	Calculated: - Dynamic index: - Units group: - Scaling: - Max - [%]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting - [%]
Description:	Displays the calculated DRIVE-CLiQ DPRAM load based on the existing target topology. The values are not made available until the RUNUP READY (800) state is adopted (see p3988). Indices 0..7 correspond to DRIVE-CLiQ ports X100..X107.		
Dependency:	Refer to: F01340		

p9990	DO selection for memory usage actual value determination /		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: T Data type: Unsigned16 P-Group: - Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 65535	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	The meaning of the parameter differs for reading and writing: Read: Returns the number of memory areas monitored. Write: Memory usage of a drive object: Enter the drive object ID. Memory usage of the complete system: Enter value 65535.		

r9991[0...4]	Memory usage actual values per DO /		
CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Index:	[0] = Fast Memory 1 [1] = Fast Memory 2 [2] = Fast Memory 3 [3] = Fast Memory 4 [4] = Heap		

r9992[0...4] Memory usage setpoints per DO /

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min - Index: [0] = Fast Memory 1 [1] = Fast Memory 2 [2] = Fast Memory 3 [3] = Fast Memory 4 [4] = Heap	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
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r9999[0...99] Software error internal supplementary diagnostics / SW_err int diag

CU_CX32, CU_I, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150	Can be changed: - Data type: Unsigned32 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Diagnostics parameter to display additional information for internal software errors.		
Note:	Only for internal Siemens troubleshooting.		

p10000 SI sampling time / SI t_sample

TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 1.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 25.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 12.00 [ms]
Description:	Sets the sampling time for the Terminal Module 54F (TM54F).		

p10001 SI delay time for test stop at DO 0 ... DO 3 / SI t_delay DO

TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 2.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2000.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 500.00 [ms]
Description:	Sets the wait time for testing the digital outputs 0 ... 3 (DO 0 ... DO 3). Within this time, for a forced checking procedure of the digital outputs, the signal must have been detected via the corresponding digital input DI 20 ... DI 23.		
Dependency:	Refer to: p10003, p10007, p10041, p10046		
Note:	The wait time must be set to a value greater than parameter p10017. The duration of the test stop is adapted to the debounce time set in p10017 automatically if the wait time in p10001 should be less than the debounce time.		

p10002	SI discrepancy monitoring time / SI discrep t_monit		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 1.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2000.00 [ms]	Access level: 3 Func. diagram: 2850, 2851 Unit selection: - Expert list: 1 Factory setting 500.00 [ms]
Description:	Sets the monitoring time for the discrepancy for the digital inputs. The signal states at the two associated digital inputs (F-DI) must assume the same state within this monitoring time.		
Note:	F-DI: Failsafe Digital Input		
p10003	SI forced checking procedure timer / SI FrCdCkProcTimer		
TM54F_MA	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 0.00 [h]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 8760.00 [h]	Access level: 3 Func. diagram: 2848 Unit selection: - Expert list: 1 Factory setting 8.00 [h]
Description:	Sets the time to carry out the forced checking procedure (test stop). Within the parameterized time, the digital inputs/outputs must have been subject to a forced checking procedure at least once. The forced checking procedure is started with BI: p10007 = 0/1 signal.		
Dependency:	Refer to: p10001, p10007, p10046		
r10004[0...1]	SI actual checksum TM54F parameters / SI act CRC TM54F		
TM54F_MA, TM54F_SL	Can be changed: - Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: 2847 Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the actual checksum of the checksum-checked parameters for the Terminal Module 54F (TM54F).		
p10005[0...1]	SI reference checksum TM54F parameters / SI ref CRC TM54F		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: 2847 Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Displays the reference checksum of the checksum-checked parameters for the Terminal Module 54F (TM54F).		
p10006	SI acknowledgement internal event input terminal / SI ackn int event		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Select a safety-relevant digital input for the signal "acknowledge internal event" (internal fault). The signal is transferred to the corresponding control signal of all drives. The falling edge at this input resets the status "internal event" in the drives.		

Value:	0:	Statically active
	1:	F-DI 0 (X521.2/3/6)
	2:	F-DI 1 (X521.4/5/7)
	3:	F-DI 2 (X522.1/2/7)
	4:	F-DI 3 (X522.3/4/8)
	5:	F-DI 4 (X522.5/6/9)
	6:	F-DI 5 (X531.2/3/6)
	7:	F-DI 6 (X531.4/5/7)
	8:	F-DI 7 (X532.1/2/7)
	9:	F-DI 8 (X532.3/4/8)
	10:	F-DI 9 (X532.5/6/9)
	255:	Statically inactive

p10007 BI: SI forced checking procedure F-DO 0 ... 3 signal source / SI frc_chF-DO s_sc

TM54F_MA	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dynamic index: -	Func. diagram: 2848
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0

Description: Sets the signal source to trigger the test stop (e.g. digital input of the Control Unit or one of the other Terminal Modules).

The test stop is triggered on a 0/1 signal edge.

The TM54F must not be in commissioning mode at this point (p0010 = 0).

Dependency: Refer to: p10001, p10003, p10041, p10046

Notice: Digital inputs of the TM54F must not be used to trigger the test stop.

p10008 SI operating mode TM54F / SI op_mode TM54F

TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	1	1

Description: Sets the operating mode for the Terminal Module 54F (TM54F).

Value: 0: Function interface
1: Control interface

Note: Parameter being prepared. For this firmware version, the function interface is not supported.

p10010[0...5] SI drive object assignment / SI drv_obj assign

TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dynamic index: -	Func. diagram: 2847, 2848
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	62	0

Description: Sets the drive object number for the drives that are available.

Index: [0] = Drive 1
[1] = Drive 2
[2] = Drive 3
[3] = Drive 4
[4] = Drive 5
[5] = Drive 6

Notice: If, for a drive, Terminal Module 54F (TM54F) is activated (p9601 = 5), its drive object number must be set in an index.

Note: A change only becomes effective after a POWER ON.

p10011[0...5]	SI drive group assignment / SI drv_gr assign		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min 1	Calculated: - Dynamic index: - Units group: - Scaling: - Max 4	Access level: 3 Func. diagram: 2848 Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the drive group for the drives that are available. A drive group is a combination of several drives with the same types of behavior.		
Index:	[0] = Drive 1 [1] = Drive 2 [2] = Drive 3 [3] = Drive 4 [4] = Drive 5 [5] = Drive 6		
p10012[0...5]	SI Motor Module Node Identifier Word 1 / SI MM Node ID 1		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the current Node Identifier (word 1, bit 0 ... 31) for the Motor Modules.		
Index:	[0] = Drive 1 [1] = Drive 2 [2] = Drive 3 [3] = Drive 4 [4] = Drive 5 [5] = Drive 6		
Dependency:	Refer to: p10013, p10014		
Note:	The Node Identifier (96 bit) is represented in the following 3 parameters. p10012[0] word 1 (bit 0 ... 31) for Motor Module 1 ... p10012[5] word 1 (bit 0 ... 31) for Motor Module 6 p10013[0] word 2 (bit 32 ... 63) for Motor Module 1 ... p10013[5] word 2 (bit 32 ... 63) for Motor Module 6 p10014[0] word 3 (bit 64 ... 95) for Motor Module 1 ... p10014[5] word 3 (bit 64 ... 95) for Motor Module 6		

p10013[0...5]	SI Motor Module Node Identifier Word 2 / SI MM Node ID 2		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the current Node Identifier (word 2, bit 32 ... 63) for the Motor Modules.		
Index:	[0] = Drive 1 [1] = Drive 2 [2] = Drive 3 [3] = Drive 4 [4] = Drive 5 [5] = Drive 6		
Dependency:	Refer to: p10012, p10014		
Note:	The complete Node Identifier (96 bit) is represented in p10012, p10013 and p10014.		
p10014[0...5]	SI Motor Module Node Identifier Word 3 / SI MM Node ID 3		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Sets the current Node Identifier (word 3, bit 64 ... 95) for the Motor Modules.		
Index:	[0] = Drive 1 [1] = Drive 2 [2] = Drive 3 [3] = Drive 4 [4] = Drive 5 [5] = Drive 6		
Dependency:	Refer to: p10012, p10013		
Note:	The complete Node Identifier (96 bit) is represented in p10012, p10013 and p10014.		
p10017	SI digital inputs debounce time / SI DI t_debounce		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: FloatingPoint32 P-Group: Safety Integrated Not for motor type: - Min 1.00 [ms]	Calculated: - Dynamic index: - Units group: - Scaling: - Max 100.00 [ms]	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1.00 [ms]
Description:	Parameter setting for the debounce time of the F-DIs and single-channel DIs of the TM54F. The debounce time is rounded to whole mS. The debounce time indicates the maximum duration of a fault pulse on the FDIs, so that there are no negative effects on the SGEs. Example: Debounce time 1mS : Fault pulses of 1mS are filtered; only pulses longer than 2mS are processed. Debounce time 3mS : Fault pulses of 3mS are filtered; only pulses longer than 4mS are processed.		

p10020[0...3]	SI special operating mode selection / SI spec op sel		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 3	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 1
Description:	Sets the special operating mode for the operating mode "function interface". 0 = Inactive 1 = Safe Operating Stop with braking (SS2) 2 = Safe Operating Stop without braking (SOS) 3 = Safely reduced speed without standstill (SLS) 4 = Safely reduced speed with agreement (SS2 --> SLS)		
Index:	[0] = Drive group 1 [1] = Drive group 2 [2] = Drive group 3 [3] = Drive group 4		
Dependency:	Refer to: p10008		
Note:	Parameter being prepared. For this firmware version, the function interface is not supported. SLS: Safely-Limited Speed SOS: Safe Operating Stop SS2: Safe Stop 2		

p10021[0...3]	SI Emergency Stop stop response / SI Emergency Stop		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 2	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the stop response for the drive group for Emergency Stop. The input terminal for Emergency Stop is set in p10038. 0 = Stop reaction STO 1 = Stop reaction SS1 2 = Stop reaction SS2		
Index:	[0] = Drive group 1 [1] = Drive group 2 [2] = Drive group 3 [3] = Drive group 4		
Dependency:	Refer to: p10008, p10038		
Note:	Parameter being prepared. For this firmware version, the function interface is not supported.		

p10022[0...3] SI STO input terminal / SI STO DI			
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal for STO (operating mode "control interface").		
Value:	0: Statically active 1: F-DI 0 (X521.2/3/6) 2: F-DI 1 (X521.4/5/7) 3: F-DI 2 (X522.1/2/7) 4: F-DI 3 (X522.3/4/8) 5: F-DI 4 (X522.5/6/9) 6: F-DI 5 (X531.2/3/6) 7: F-DI 6 (X531.4/5/7) 8: F-DI 7 (X532.1/2/7) 9: F-DI 8 (X532.3/4/8) 10: F-DI 9 (X532.5/6/9) 255: Statically inactive		
Index:	[0] = Drive group 1 [1] = Drive group 2 [2] = Drive group 3 [3] = Drive group 4		
Note:	Re value = 0: No terminal assigned, safety function always active. Re value = 255: No terminal assigned, safety function always inactive. STO: Safe Torque Off		

p10023[0...3] SI SS1 input terminal / SI SS1 DI			
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Assignment of the input terminals for input SS1 (operating mode = control interface) Description, refer to P10022		
Value:	0: Statically active 1: F-DI 0 (X521.2/3/6) 2: F-DI 1 (X521.4/5/7) 3: F-DI 2 (X522.1/2/7) 4: F-DI 3 (X522.3/4/8) 5: F-DI 4 (X522.5/6/9) 6: F-DI 5 (X531.2/3/6) 7: F-DI 6 (X531.4/5/7) 8: F-DI 7 (X532.1/2/7) 9: F-DI 8 (X532.3/4/8) 10: F-DI 9 (X532.5/6/9) 255: Statically inactive		
Index:	[0] = Drive group 1 [1] = Drive group 2 [2] = Drive group 3 [3] = Drive group 4		

Note: SS1: Safe Stop 1
Re value = 0:
No terminal assigned, safety function always active.
Re value = 255:
No terminal assigned, safety function always inactive.

p10024[0...3]	SI SS2 input terminal / SI SS2 DI		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Assignment of the input terminals for input SS2 (operating mode = control interface). Description, refer to p10022.		
Value:	0: Statically active 1: F-DI 0 (X521.2/3/6) 2: F-DI 1 (X521.4/5/7) 3: F-DI 2 (X522.1/2/7) 4: F-DI 3 (X522.3/4/8) 5: F-DI 4 (X522.5/6/9) 6: F-DI 5 (X531.2/3/6) 7: F-DI 6 (X531.4/5/7) 8: F-DI 7 (X532.1/2/7) 9: F-DI 8 (X532.3/4/8) 10: F-DI 9 (X532.5/6/9) 255: Statically inact		
Index:	[0] = Drive group 1 [1] = Drive group 2 [2] = Drive group 3 [3] = Drive group 4		
Note:	SS2: Safe Stop 2 Re value = 0: No terminal assigned, safety function always active. Re value = 255: No terminal assigned, safety function always inactive.		

p10025[0...3]	SI SOS input terminal / SI SOS DI		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Assignment of the input terminals for input SOS (operating mode = control interface) Description, refer to P10022		

Value:	0: Statically active
	1: F-DI 0 (X521.2/3/6)
	2: F-DI 1 (X521.4/5/7)
	3: F-DI 2 (X522.1/2/7)
	4: F-DI 3 (X522.3/4/8)
	5: F-DI 4 (X522.5/6/9)
	6: F-DI 5 (X531.2/3/6)
	7: F-DI 6 (X531.4/5/7)
	8: F-DI 7 (X532.1/2/7)
	9: F-DI 8 (X532.3/4/8)
	10: F-DI 9 (X532.5/6/9)
	255: Statically inactive
Index:	[0] = Drive group 1
	[1] = Drive group 2
	[2] = Drive group 3
	[3] = Drive group 4
Note:	SOS: Safe Operating Stop
	Re value = 0:
	No terminal assigned, safety function always active.
	Re value = 255:
	No terminal assigned, safety function always inactive.

p10026[0...3]	SI SLS input terminal / SI SLS DI		
TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	255	0
Description:	Assignment of the input terminals for input SLS (operating mode = control interface) Description, refer to P10022		
Value:	0: Statically active		
	1: F-DI 0 (X521.2/3/6)		
	2: F-DI 1 (X521.4/5/7)		
	3: F-DI 2 (X522.1/2/7)		
	4: F-DI 3 (X522.3/4/8)		
	5: F-DI 4 (X522.5/6/9)		
	6: F-DI 5 (X531.2/3/6)		
	7: F-DI 6 (X531.4/5/7)		
	8: F-DI 7 (X532.1/2/7)		
	9: F-DI 8 (X532.3/4/8)		
	10: F-DI 9 (X532.5/6/9)		
	255: Statically inactive		
Index:	[0] = Drive group 1		
	[1] = Drive group 2		
	[2] = Drive group 3		
	[3] = Drive group 4		
Note:	SLS: Safely-Limited Speed		
	Re value = 0:		
	No terminal assigned, safety function always active.		
	Re value = 255:		
	No terminal assigned, safety function always inactive.		

p10027[0...3] SI SLS_Limit(1) input terminal / SI SLS_Limit(1) DI			
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal for SLS_Limit bit 0 (operating mode "control interface").		
Value:	0: Statically active 1: F-DI 0 (X521.2/3/6) 2: F-DI 1 (X521.4/5/7) 3: F-DI 2 (X522.1/2/7) 4: F-DI 3 (X522.3/4/8) 5: F-DI 4 (X522.5/6/9) 6: F-DI 5 (X531.2/3/6) 7: F-DI 6 (X531.4/5/7) 8: F-DI 7 (X532.1/2/7) 9: F-DI 8 (X532.3/4/8) 10: F-DI 9 (X532.5/6/9) 255: Statically inactive		
Index:	[0] = Drive group 1 [1] = Drive group 2 [2] = Drive group 3 [3] = Drive group 4		
Note:	SLS: Safely-Limited Speed Re value = 0: No terminal assigned, selection bit remains statically at "0". Re value = 255: No terminal assigned, selection bit remains statically at "1".		

p10028[0...3] SI SLS_Limit(2) input terminal / SI SLS_Limit(2) DI			
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
Description:	Sets the input terminal for SLS_Limit bit 1 (operating mode "control interface").		
Value:	0: Statically active 1: F-DI 0 (X521.2/3/6) 2: F-DI 1 (X521.4/5/7) 3: F-DI 2 (X522.1/2/7) 4: F-DI 3 (X522.3/4/8) 5: F-DI 4 (X522.5/6/9) 6: F-DI 5 (X531.2/3/6) 7: F-DI 6 (X531.4/5/7) 8: F-DI 7 (X532.1/2/7) 9: F-DI 8 (X532.3/4/8) 10: F-DI 9 (X532.5/6/9) 255: Statically inactive		
Index:	[0] = Drive group 1 [1] = Drive group 2 [2] = Drive group 3 [3] = Drive group 4		

Note: SLS: Safely-Limited Speed
 Re value = 0:
 No terminal assigned, selection bit remains statically at "0".
 Re value = 255:
 No terminal assigned, selection bit remains statically at "1".

p10036[0...3]		SI special operating mode input terminal / SI spec op DI	
TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	255	0
Description:	Sets the input terminal for "special operating mode" (operating mode "function interface").		
Value:	0: Statically active 1: F-DI 0 (X521.2/3/6) 2: F-DI 1 (X521.4/5/7) 3: F-DI 2 (X522.1/2/7) 4: F-DI 3 (X522.3/4/8) 5: F-DI 4 (X522.5/6/9) 6: F-DI 5 (X531.2/3/6) 7: F-DI 6 (X531.4/5/7) 8: F-DI 7 (X532.1/2/7) 9: F-DI 8 (X532.3/4/8) 10: F-DI 9 (X532.5/6/9) 255: Statically inactive		
Index:	[0] = Drive group 1 [1] = Drive group 2 [2] = Drive group 3 [3] = Drive group 4		
Note:	Parameter being prepared. For this firmware version, the function interface is not supported. Re value = 0: No terminal assigned, static special operation. Re value = 255: No terminal assigned, static normal operation.		

p10037[0...3]		SI agreement input terminal / SI agreement DI	
TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 4
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	255	0
Description:	Sets the input terminal for "agreement" (operating mode "function interface").		
Value:	0: Statically active		
	1: F-DI 0 (X521.2/3/6)		
	2: F-DI 1 (X521.4/5/7)		
	3: F-DI 2 (X522.1/2/7)		
	4: F-DI 3 (X522.3/4/8)		
	5: F-DI 4 (X522.5/6/9)		
	6: F-DI 5 (X531.2/3/6)		
	7: F-DI 6 (X531.4/5/7)		
	8: F-DI 7 (X532.1/2/7)		
	9: F-DI 8 (X532.3/4/8)		
	10: F-DI 9 (X532.5/6/9)		
	255: Statically inactive		

Index: [0] = Drive group 1
[1] = Drive group 2
[2] = Drive group 3
[3] = Drive group 4

Note: Parameter being prepared. For this firmware version, the function interface is not supported.
Re value = 0:
No terminal assigned, no static agreement.
Re value = 255:
No terminal assigned, static agreement.

p10038[0...3] SI Emergency Stop input terminal / SI Emer Stop DI

TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Integer16 P-Group: Safety Integrated Not for motor type: - Min 0	Calculated: - Dynamic index: - Units group: - Scaling: - Max 255	Access level: 4 Func. diagram: - Unit selection: - Expert list: 1 Factory setting 0
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Description: Sets the input terminal for input "Emergency Stop" (operating mode "function interface").
The behavior of this input signal is set in p10021.

Value: 0: Statically active
1: F-DI 0 (X521.2/3/6)
2: F-DI 1 (X521.4/5/7)
3: F-DI 2 (X522.1/2/7)
4: F-DI 3 (X522.3/4/8)
5: F-DI 4 (X522.5/6/9)
6: F-DI 5 (X531.2/3/6)
7: F-DI 6 (X531.4/5/7)
8: F-DI 7 (X532.1/2/7)
9: F-DI 8 (X532.3/4/8)
10: F-DI 9 (X532.5/6/9)
255: Statically inactive

Index: [0] = Drive group 1
[1] = Drive group 2
[2] = Drive group 3
[3] = Drive group 4

Dependency: Refer to: p10008, p10021

Note: Parameter being prepared. For this firmware version, the function interface is not supported.
Re value = 0:
No terminal assigned, "Emergency Stop" statically active.
Re value = 255:
No terminal assigned, no "Emergency Stop" statically active.

p10039[0...3]		SI Safe State signal selection / SI Safe State Sel			
TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -		Access level: 3	
	Data type: Unsigned32	Dynamic index: -		Func. diagram: 2856	
	P-Group: Safety Integrated	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min	Max		Factory setting	
	-	-		0001 bin	
Description:	Sets the signals for the drive group specific signal "Safe State". Bit 0 = Power_removed Bit 1 = SS1_active Bit 2 = SS2_active Bit 3 = SOS_active Bit 4 = SLS_active Bit 5 = Reserved				
Index:	[0] = Drive group 1 [1] = Drive group 2 [2] = Drive group 3 [3] = Drive group 4				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Power_removed	Selected	Not selected	-
	01	SS1_active	Selected	Not selected	-
	02	SS2_active	Selected	Not selected	-
	03	SOS_active	Selected	Not selected	-
	04	SLS_active	Selected	Not selected	-
Note:	Bit = 0 signal --> not selected Bit = 1 signal --> selected The selected signals (high-active) are OR'ed The result of the logic operation results in the status "Safe State".				

p10040		SI F-DI input mode / SI F-DI inp_mode			
TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -		Access level: 3	
	Data type: Unsigned32	Dynamic index: -		Func. diagram: -	
	P-Group: Safety Integrated	Units group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min	Max		Factory setting	
	-	-		0000 bin	
Description:	Sets the input mode for the safety-relevant input terminals of terminal series 2.				
Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	F-DI 0, DI 1+ (X521.3)	NO contact	NC contact	2850
	01	F-DI 1, DI 3+ (X521.5)	NO contact	NC contact	2850
	02	F-DI 2, DI 5+ (X522.2)	NO contact	NC contact	2850
	03	F-DI 3, DI 7+ (X522.4)	NO contact	NC contact	2850
	04	F-DI 4, DI 9+ (X522.6)	NO contact	NC contact	2850
	05	F-DI 5, DI 11+ (X531.3)	NO contact	NC contact	2851
	06	F-DI 6, DI 13+ (X531.5)	NO contact	NC contact	2851
	07	F-DI 7, DI 15+ (X532.2)	NO contact	NC contact	2851
	08	F-DI 8, DI 17+ (X532.4)	NO contact	NC contact	2851
	09	F-DI 9, DI 19+ (X532.6)	NO contact	NC contact	2851
Note:	Only an NC contact can be connected to terminals that are not listed in the selection.				

p10041 SI F-DI enable for test / SI F-DI enab testTM54F_MA,
TM54F_SL**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dynamic index:** -**Func. diagram:** 2848**P-Group:** Safety Integrated**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

-

-

0000 bin

Description: Enable signal for the integration of F-DI in the test (forced checking procedure) of the sensor power supply.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	F-DI 0, power supply L1+	Test active	No test	-
	01	F-DI 1, power supply L1+	Test active	No test	-
	02	F-DI 2, power supply L1+	Test active	No test	-
	03	F-DI 3, power supply L1+	Test active	No test	-
	04	F-DI 4, power supply L1+	Test active	No test	-
	05	F-DI 5, power supply L2+	Test active	No test	-
	06	F-DI 6, power supply L2+	Test active	No test	-
	07	F-DI 7, power supply L2+	Test active	No test	-
	08	F-DI 8, power supply L2+	Test active	No test	-
	09	F-DI 9, power supply L2+	Test active	No test	-

Note: F-DI: Failsafe Digital Input**p10042[0...5] SI F-DO 0 signal sources / SI F-DO 0 S_src**TM54F_MA,
TM54F_SL**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** Integer16**Dynamic index:** -**Func. diagram:** 2857**P-Group:** Safety Integrated**Units group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min****Max****Factory setting**

0

779

0

Description: Sets the signal sources for F-DO 0.

The 6 signal sources in p10042[0...5] are AND'ed and the result is output at F-DO 0.

Value:	0:	No function
	1:	Drive group 1 STO active
	2:	Drive group 1 SS1 active
	3:	Drive group 1 SS2 active
	4:	Drive group 1 SOS active
	5:	Drive group 1 SLS active
	6:	Drive group 1 SSM feedback signal active
	7:	Drive group 1 safe state
	8:	Drive group 1 SOS selected
	9:	Drive group 1 internal event
	10:	Drive group 1 active SLS stage bit 0
	11:	Drive group 1 active SLS stage bit 1
	257:	Drive group 2 STO active
	258:	Drive group 2 SS1 active
	259:	Drive group 2 SS2 active
	260:	Drive group 2 SOS active
	261:	Drive group 2 SLS active
	262:	Drive group 2 SSM feedback signal active
	263:	Drive group 2 safe state
	264:	Drive group 2 SOS selected
	265:	Drive group 2 internal event
	266:	Drive group 2 active SLS stage bit 0
	267:	Drive group 2 active SLS stage bit 1

513: Drive group 3 STO active
 514: Drive group 3 SS1 active
 515: Drive group 3 SS2 active
 516: Drive group 3 SOS active
 517: Drive group 3 SLS active
 518: Drive group 3 SSM feedback signal active
 519: Drive group 3 safe state
 520: Drive group 3 SOS selected
 521: Drive group 3 internal event
 522: Drive group 3 active SLS stage bit 0
 523: Drive group 3 active SLS stage bit 1
 769: Drive group 4 STO active
 770: Drive group 4 SS1 active
 771: Drive group 4 SS2 active
 772: Drive group 4 SOS active
 773: Drive group 4 SLS active
 774: Drive group 4 SSM feedback signal active
 775: Drive group 4 safe state
 776: Drive group 4 SOS selected
 777: Drive group 4 internal event
 778: Drive group 4 active SLS stage bit 0
 779: Drive group 4 active SLS stage bit 1

Index: [0] = AND logic operation input 1
 [1] = AND logic operation input 2
 [2] = AND logic operation input 3
 [3] = AND logic operation input 4
 [4] = AND logic operation input 5
 [5] = AND logic operation input 6

Note: F-DO: Failsafe Digital Output

p10043[0...5] SI F-DO 1 signal sources / SI F-DO 1 S_src

TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 2857
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	779	0

Description: Sets the signal sources for F-DO 1.

The 6 signal sources in p10043[0...5] are AND'ed and the result is output at F-DO 1.

Value: 0: No function
 1: Drive group 1 STO active
 2: Drive group 1 SS1 active
 3: Drive group 1 SS2 active
 4: Drive group 1 SOS active
 5: Drive group 1 SLS active
 6: Drive group 1 SSM feedback signal active
 7: Drive group 1 safe state
 8: Drive group 1 SOS selected
 9: Drive group 1 internal event
 10: Drive group 1 active SLS stage bit 0
 11: Drive group 1 active SLS stage bit 1

257: Drive group 2 STO active
258: Drive group 2 SS1 active
259: Drive group 2 SS2 active
260: Drive group 2 SOS active
261: Drive group 2 SLS active
262: Drive group 2 SSM feedback signal active
263: Drive group 2 safe state
264: Drive group 2 SOS selected
265: Drive group 2 internal event
266: Drive group 2 active SLS stage bit 0
267: Drive group 2 active SLS stage bit 1
513: Drive group 3 STO active
514: Drive group 3 SS1 active
515: Drive group 3 SS2 active
516: Drive group 3 SOS active
517: Drive group 3 SLS active
518: Drive group 3 SSM feedback signal active
519: Drive group 3 safe state
520: Drive group 3 SOS selected
521: Drive group 3 internal event
522: Drive group 3 active SLS stage bit 0
523: Drive group 3 active SLS stage bit 1
769: Drive group 4 STO active
770: Drive group 4 SS1 active
771: Drive group 4 SS2 active
772: Drive group 4 SOS active
773: Drive group 4 SLS active
774: Drive group 4 SSM feedback signal active
775: Drive group 4 safe state
776: Drive group 4 SOS selected
777: Drive group 4 internal event
778: Drive group 4 active SLS stage bit 0
779: Drive group 4 active SLS stage bit 1

Index: [0] = AND logic operation input 1
[1] = AND logic operation input 2
[2] = AND logic operation input 3
[3] = AND logic operation input 4
[4] = AND logic operation input 5
[5] = AND logic operation input 6

Note: F-DO: Failsafe Digital Output

p10044[0...5]	SI F-DO 2 signal sources / SI F-DO 2 S_src		
TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 2857
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min 0	Max 779	Factory setting 0
Description:	Sets the signal sources for F-DO 2. The 6 signal sources in p10044[0...5] are AND'ed and the result is output at F-DO 2.		
Value:	0: No function 1: Drive group 1 STO active 2: Drive group 1 SS1 active 3: Drive group 1 SS2 active 4: Drive group 1 SOS active 5: Drive group 1 SLS active 6: Drive group 1 SSM feedback signal active 7: Drive group 1 safe state 8: Drive group 1 SOS selected 9: Drive group 1 internal event 10: Drive group 1 active SLS stage bit 0 11: Drive group 1 active SLS stage bit 1 257: Drive group 2 STO active 258: Drive group 2 SS1 active 259: Drive group 2 SS2 active 260: Drive group 2 SOS active 261: Drive group 2 SLS active 262: Drive group 2 SSM feedback signal active 263: Drive group 2 safe state 264: Drive group 2 SOS selected 265: Drive group 2 internal event 266: Drive group 2 active SLS stage bit 0 267: Drive group 2 active SLS stage bit 1 513: Drive group 3 STO active 514: Drive group 3 SS1 active 515: Drive group 3 SS2 active 516: Drive group 3 SOS active 517: Drive group 3 SLS active 518: Drive group 3 SSM feedback signal active 519: Drive group 3 safe state 520: Drive group 3 SOS selected 521: Drive group 3 internal event 522: Drive group 3 active SLS stage bit 0 523: Drive group 3 active SLS stage bit 1 769: Drive group 4 STO active 770: Drive group 4 SS1 active 771: Drive group 4 SS2 active 772: Drive group 4 SOS active 773: Drive group 4 SLS active 774: Drive group 4 SSM feedback signal active 775: Drive group 4 safe state 776: Drive group 4 SOS selected 777: Drive group 4 internal event 778: Drive group 4 active SLS stage bit 0 779: Drive group 4 active SLS stage bit 1		

Index:
 [0] = AND logic operation input 1
 [1] = AND logic operation input 2
 [2] = AND logic operation input 3
 [3] = AND logic operation input 4
 [4] = AND logic operation input 5
 [5] = AND logic operation input 6

Note: F-DO: Failsafe Digital Output

p10045[0...5]	SI F-DO 3 signal sources / SI F-DO 3 S_src		
TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: 2857
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0	779	0

Description: Sets the signal sources for F-DO 3.
 The 6 signal sources in p10045[0...5] are AND'ed and the result is output at F-DO 3.

Value:

- 0: No function
- 1: Drive group 1 STO active
- 2: Drive group 1 SS1 active
- 3: Drive group 1 SS2 active
- 4: Drive group 1 SOS active
- 5: Drive group 1 SLS active
- 6: Drive group 1 SSM feedback signal active
- 7: Drive group 1 safe state
- 8: Drive group 1 SOS selected
- 9: Drive group 1 internal event
- 10: Drive group 1 active SLS stage bit 0
- 11: Drive group 1 active SLS stage bit 1
- 257: Drive group 2 STO active
- 258: Drive group 2 SS1 active
- 259: Drive group 2 SS2 active
- 260: Drive group 2 SOS active
- 261: Drive group 2 SLS active
- 262: Drive group 2 SSM feedback signal active
- 263: Drive group 2 safe state
- 264: Drive group 2 SOS selected
- 265: Drive group 2 internal event
- 266: Drive group 2 active SLS stage bit 0
- 267: Drive group 2 active SLS stage bit 1
- 513: Drive group 3 STO active
- 514: Drive group 3 SS1 active
- 515: Drive group 3 SS2 active
- 516: Drive group 3 SOS active
- 517: Drive group 3 SLS active
- 518: Drive group 3 SSM feedback signal active
- 519: Drive group 3 safe state
- 520: Drive group 3 SOS selected
- 521: Drive group 3 internal event
- 522: Drive group 3 active SLS stage bit 0
- 523: Drive group 3 active SLS stage bit 1

769: Drive group 4 STO active
 770: Drive group 4 SS1 active
 771: Drive group 4 SS2 active
 772: Drive group 4 SOS active
 773: Drive group 4 SLS active
 774: Drive group 4 SSM feedback signal active
 775: Drive group 4 safe state
 776: Drive group 4 SOS selected
 777: Drive group 4 internal event
 778: Drive group 4 active SLS stage bit 0
 779: Drive group 4 active SLS stage bit 1

Index: [0] = AND logic operation input 1
 [1] = AND logic operation input 2
 [2] = AND logic operation input 3
 [3] = AND logic operation input 4
 [4] = AND logic operation input 5
 [5] = AND logic operation input 6

Note: F-DO: Failsafe Digital Output

p10046 SI test sensor feedback signal input DI 20 ... 23 / SI test sens FS

TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2848
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	0000 bin

Description: Sets the test of the feedback line for forced checking procedure.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Read back F-DO 0 in DI 20	Test active	No test	-
	01	Read back F-DO 1 in DI 21	Test active	No test	-
	02	Read back F-DO 2 in DI 22	Test active	No test	-
	03	Read back F-DO 3 in DI 23	Test active	No test	-

Note: F-DO: Failsafe Digital Output

p10047[0...3] SI selection of test mode for test stop / SI test mode sel.

TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0001 bin	0011 bin	0010 bin

Value: 1: Test mode 1 evaluation of int. diagnostic signal (passive load)
 2: Test mode 2 read back F-DO in DI (relay circuit)
 3: Test mode 3 read back F-DO in DI (actuator with feedback signal)

Note: When test mode 1 is being used, too great a load resistance on the part of the load between DO+ and DO- can lead to problems during the test stop. It is, therefore, important to make sure that the load resistance on an individual FDO does not exceed 10 Kohm.

r10051.0...9 CO/BO: SI digital inputs status / SI DI status

TM54F_MA, TM54F_SL	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2850, 2851
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the single-channel, logical, and debounced status of the safety digital inputs F-DI 0 .. 9 at Terminal Module 54F (TM54F).

If a safety function is assigned to an input (e.g. via p10022), then the following applies:

- logical "0": Safety function is selected
- logical "1": Safety function is deselected

The interrelationship between the logical level and the external voltage level at the input depends on the parameterization (refer to p10040) of the input as either NC or NO contact and is aligned to the use of a safety function:

With 24 V at the input, NC contacts have a logical "1" level, for 0 V at the input, a logical "0" level.

This means that an NC/NC contact parameterization of 0 V at both inputs of the F-DI selects the safety function, for 24 V at both inputs, deselects the safety function.

With 24 V at the input, NO contacts have a logical "0" level, for 0 V at the input, a logical "1" level.

This means that for an NC/NO contact parameterization, the level 0 V/24 V selects the safety function, the level 24 V/0 V deselects the safety function.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	F-DI 0	Logical 1	Logical 0	2850
	01	F-DI 1	Logical 1	Logical 0	2850
	02	F-DI 2	Logical 1	Logical 0	2850
	03	F-DI 3	Logical 1	Logical 0	2850
	04	F-DI 4	Logical 1	Logical 0	2850
	05	F-DI 5	Logical 1	Logical 0	2851
	06	F-DI 6	Logical 1	Logical 0	2851
	07	F-DI 7	Logical 1	Logical 0	2851
	08	F-DI 8	Logical 1	Logical 0	2851
	09	F-DI 9	Logical 1	Logical 0	2851

Dependency: Refer to: p10017, p10040

Note: F-DI: Failsafe Digital Input

r10052.0...3 CO/BO: SI digital outputs status / SI DO status

TM54F_MA, TM54F_SL	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the digital outputs at the Terminal Module 54F (TM54F).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DO 0	High	Low	2853
	01	DO 1	High	Low	2853
	02	DO 2	High	Low	2853
	03	DO 3	High	Low	2853

Note: F-DO: Failsafe Digital Output

r10053.0...3 CO/BO: SI digital inputs 20 ... 23 status / SI DI 20...23 stat

TM54F_SL	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2848
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the digital inputs at the Terminal Module 54F (TM54F).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	DI 20	High	Low	2853
	01	DI 21	High	Low	2853
	02	DI 22	High	Low	2853
	03	DI 23	High	Low	2853

r10054 SI TM54F failsafe events active / SI failsafe act

TM54F_MA, TM54F_SL	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the events that lead to the transfer of failsafe signals to all drives assigned to the TM54F. If the second channel of the TM54F transmits failsafe signals, then these are synchronized to the other channel.

Possibilities of resolving the situation:

- error during test stop: correctly perform the test stop.
- internal software error: no possibility of resolving this problem, POWER ON.
- internal synchronization problem: no possibility of resolving this problem, POWER ON.
- all other causes: remove the cause of the error and carry out a safety-relevant acknowledgement (p10006).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Commissioning mode active (p0010 = 95)	Yes	No	2847
	01	Checksum error of the safety parameters	Yes	No	-
	02	Internal synchronization problem within RM54F	Yes	No	-
	03	Internal software error	Yes	No	-
	04	Overvoltage in the TM54F	Yes	No	-
	05	Undervoltage in the TM54F	Yes	No	-
	06	Error at test stop	Yes	No	-
	07	Error for data cross-check within TM54F	Yes	No	-
	08	Overvoltage in the TM54F	Yes	No	-
	31	Failsafe events active on another channel	Yes	No	-

r10055 SI TM54F communication status drive-specific / SI comm_stat drv

TM54F_MA, TM54F_SL	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2846
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the communication status of the individual drives with the the Terminal Module 54F (TM54F).
For r10055 = 0, the following applies:
All drives assigned in p10010 communicate with the TM54F.

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Communication between drive 1 and TM54F	Not configured	Configured	-
	01	Communication between drive 2 and TM54F	Not configured	Configured	-
	02	Communication between drive 3 and TM54F	Not configured	Configured	-
	03	Communication between drive 4 and TM54F	Not configured	Configured	-
	04	Communication between drive 5 and TM54F	Not configured	Configured	-
	05	Communication between drive 6 and TM54F	Not configured	Configured	-

r10056.0 CO/BO: SI Status / SI stat

TM54F_MA	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: -
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	-	-	-

Description: Displays the status of the Terminal Module 54F (TM54F).

Bit field:	Bit	Signal name	1 signal	0 signal	FP
	00	Test stop status	Active	Inactive	-

p10061 SI password input TM54F / SI password inp

TM54F_MA, TM54F_SL	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2847
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0000 hex	FFFF FFFF hex	0000 hex

Description: Enters the Safety Integrated password for the Terminal Module 54F (TM54F).
This password is required to change the safety-relevant parameters.

p10062 SI password new TM54F / SI password new

TM54F_MA, TM54F_SL	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dynamic index: -	Func. diagram: 2847
	P-Group: Safety Integrated	Units group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min	Max	Factory setting
	0000 hex	FFFF FFFF hex	0000 hex

Description: Enters the new Safety Integrated password for the Terminal Module 54F (TM54F).

Dependency: A change made to the Safety Integrated password must be acknowledged in the following parameter:
Refer to: p10063

p10063	SI password acknowledgement TM54F / SI ackn password		
TM54F_MA, TM54F_SL	Can be changed: C2(95) Data type: Unsigned32 P-Group: Safety Integrated Not for motor type: - Min 0000 hex	Calculated: - Dynamic index: - Units group: - Scaling: - Max FFFF FFFF hex	Access level: 3 Func. diagram: 2847 Unit selection: - Expert list: 1 Factory setting 0000 hex
Description:	Acknowledgement of the new Safety Integrated password for the Terminal Module 54F (TM54F).		
Dependency:	Refer to: p10062		
Note:	The new password entered into p10062 must be re-entered in order to acknowledge. p10062 = p10063 = 0 is automatically set after the new Safety Integrated password has been successfully acknowledged.		
r10090[0...3]	SI TM54F version / SI TM54F version		
TM54F_MA, TM54F_SL	Can be changed: - Data type: Unsigned16 P-Group: Safety Integrated Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays the Safety Integrated version for the Terminal Module 54F (TM54F).		
Index:	[0] = Safety Version (major release) [1] = Safety Version (minor release) [2] = Safety Version (baselevel or patch) [3] = Safety Version (hotfix)		
Dependency:	Refer to: r9390, r9590, r9770, r9870, r9890		
Note:	Example: r10090[0] = 2, r10090[1] = 60, r10090[2] = 1, r10090[3] = 0 --> SI TM54F version V02.60.01.00		
r61000[0...239]	PROFINET Name of Station / PN Name of Station		
CU_S (PROFINET), CU_S_CU310PN, CU_S_S150 (PROFI- NET)	Can be changed: - Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays PROFINET Name of Station.		
Notice:	An ASCII table (excerpt) can be found, for example, in the Appendix of the List Manual:		
r61001[0...3]	PROFINET IP of Station / PN IP of Station		
CU_S (PROFINET), CU_S_CU310PN, CU_S_S150 (PROFI- NET)	Can be changed: - Data type: Unsigned8 P-Group: - Not for motor type: - Min -	Calculated: - Dynamic index: - Units group: - Scaling: - Max -	Access level: 3 Func. diagram: - Unit selection: - Expert list: 1 Factory setting -
Description:	Displays PROFINET IP of Station.		

1.3 Parameters for data sets

1.3.1 Parameters for command data sets (CDS)

Note:

References: /FH1/ SINAMICS S120 Function Manual Drive Functions
Section "Data Sets"

The following list contains the command-data-set-dependent parameters.

Product: SINAMICS S120/S150, Version: 4301400, Language: eng, Type: CDS

p0641[0...n]	CI: Current limit, variable / Curr lim var
p0700[0...n]	Macro Binector Input (BI) / Macro BI
p0820[0...n]	BI: Drive Data Set selection DDS bit 0 / DDS select., bit 0
p0821[0...n]	BI: Drive Data Set selection DDS bit 1 / DDS select., bit 1
p0822[0...n]	BI: Drive Data Set selection DDS bit 2 / DDS select., bit 2
p0823[0...n]	BI: Drive Data Set selection DDS bit 3 / DDS select., bit 3
p0824[0...n]	BI: Drive Data Set selection DDS bit 4 / DDS select., bit 4
p0828[0...n]	BI: Motor changeover, feedback signal / Mot_chng fdbk sig
p0840[0...n]	BI: ON/OFF1 / ON/OFF1
p0844[0...n]	BI: 1. OFF2 / 1. OFF2
p0845[0...n]	BI: 2. OFF2 / 2. OFF2
p0848[0...n]	BI: 1. OFF3 / 1. OFF3
p0849[0...n]	BI: 2. OFF3 / 2. OFF3
p0852[0...n]	BI: Operation enable / Operation enable
p0854[0...n]	BI: Master ctrl by PLC / Master ctrl by PLC
p0855[0...n]	BI: Unconditionally release holding brake / Uncond open brake
p0856[0...n]	BI: Speed controller enable / n_ctrl enable
p0856[0...n]	BI: Velocity controller enable / v_ctrl enable
p0858[0...n]	BI: Unconditionally close holding brake / Uncond close brake
p1000[0...n]	Macro Connector Inputs (CI) for speed setpoints / Macro CI n_set
p1000[0...n]	Macro Connector Inputs (CI) for velocity setpoints / Macro CI v_set
p1020[0...n]	BI: Fixed velocity setpoint selection Bit 0 / v_set_fixed Bit 0
p1020[0...n]	BI: Fixed speed setpoint selection Bit 0 / n_set_fixed Bit 0
p1021[0...n]	BI: Fixed velocity setpoint selection Bit 1 / v_set_fixed Bit 1
p1021[0...n]	BI: Fixed speed setpoint selection Bit 1 / n_set_fixed Bit 1
p1022[0...n]	BI: Fixed velocity setpoint selection Bit 2 / v_set_fixed Bit 2
p1022[0...n]	BI: Fixed speed setpoint selection Bit 2 / n_set_fixed Bit 2
p1023[0...n]	BI: Fixed velocity setpoint selection Bit 3 / v_set_fixed Bit 3
p1023[0...n]	BI: Fixed speed setpoint selection Bit 3 / n_set_fixed Bit 3
p1035[0...n]	BI: Motorized potentiometer setpoint raise / Mop raise
p1036[0...n]	BI: Motorized potentiometer lower setpoint / Mop lower
p1039[0...n]	BI: Motorized potentiometer inversion / Mop inversion
p1041[0...n]	BI: Motorized potentiometer manual/automatic / Mop manual/auto
p1042[0...n]	CI: Motorized potentiometer automatic setpoint / Mop auto setpoint
p1043[0...n]	BI: Motorized potentiometer accept setpoint / Mop accept set val
p1044[0...n]	CI: Motorized potentiometer setting value / Mop set val
p1051[0...n]	CI: Velocity limit RFG positive direction / v_limit RFG pos
p1051[0...n]	CI: Speed limit RFG positive direction of rotation / n_limit RFG pos

p1052[0...n]	CI: Velocity limit RFG negative direction / v_limit RFG neg
p1052[0...n]	CI: Speed limit RFG negative direction of rotation / n_limit RFG neg
p1055[0...n]	BI: Jog bit 0 / Jog bit 0
p1056[0...n]	BI: Jog bit 1 / Jog bit 1
p1070[0...n]	CI: Main setpoint / Main setpoint
p1071[0...n]	CI: Main setpoint scaling / Main setp scal
p1075[0...n]	CI: Supplementary setpoint / Suppl setpoint
p1076[0...n]	CI: Supplementary setpoint scaling / Suppl setp scal
p1085[0...n]	CI: Velocity limit positive direction / v_limit pos
p1085[0...n]	CI: Speed limit in positive direction of rotation / n_limit pos
p1088[0...n]	CI: Velocity limit negative direction / n_limit neg
p1088[0...n]	CI: Speed limit in negative direction of rotation / n_limit neg
p1110[0...n]	BI: Inhibit negative direction / Inhib neg dir
p1111[0...n]	BI: Inhibit positive direction / Inhib pos dir
p1113[0...n]	BI: Setpoint inversion / Setp inv
p1122[0...n]	BI: Bypass ramp-function generator / Bypass RFG
p1140[0...n]	BI: Ramp-function generator enable / RFG enable
p1141[0...n]	BI: Continue ramp-function generator / Continue RFG
p1142[0...n]	BI: Speed setpoint enable / n_set enable
p1142[0...n]	BI: Velocity setpoint enable / v_set enable
p1143[0...n]	BI: Ramp-function generator, accept setting value / Accept RFG set val
p1144[0...n]	CI: Ramp-function generator setting value / RFG setting value
p1155[0...n]	CI: Speed controller speed setpoint 1 / n_ctrl n_set 1
p1155[0...n]	CI: Velocity controller, velocity setpoint 1 / v_ctrl n_set 1
p1160[0...n]	CI: Speed controller speed setpoint 2 / n_ctrl n_set 2
p1160[0...n]	CI: Velocity controller, velocity setpoint 2 / v_ctrl n_set 2
p1201[0...n]	BI: Flying restart enable signal source / Fly_res enab S_src
p1230[0...n]	BI: Armature short-circuit / DC brake activation / ASC act
p1235[0...n]	BI: External armature short-circuit, contactor feedback signal / ASC ext feedback
p1330[0...n]	CI: V/f control independent voltage setpoint / Vf V_set independ.
p1356[0...n]	CI: V/f control, angular setpoint / Vf ang setpoint
p1430[0...n]	CI: Speed pre-control / n_prectrl
p1430[0...n]	CI: Velocity pre-control / v_prectrl
p1437[0...n]	CI: Speed controller, reference model I component input / n_ctrRefMod I_comp
p1440[0...n]	CI: Speed controller speed actual value / n_ctrl n_act
p1455[0...n]	CI: Speed controller P gain adaptation signal / n_ctrl Adpt_sig Kp
p1455[0...n]	CI: Velocity controller, P gain adaptation signal / v_ctrl Adpt_sig Kp
p1466[0...n]	CI: Speed controller P-gain scaling / n_ctrl Kp scal
p1466[0...n]	CI: Velocity controller P gain scaling / v_ctrl Kp scal
p1475[0...n]	CI: Speed controller torque setting value for motor holding brake / n_ctrl M_sv MHB
p1476[0...n]	BI: Speed controller hold integrator / n_ctrl integ stop
p1476[0...n]	BI: Velocity controller hold integrator / v_ctrl integ stop
p1477[0...n]	BI: Speed controller set integrator value / n_ctrl integ set
p1477[0...n]	BI: Velocity controller set integrator value / v_ctrl integ set
p1478[0...n]	CI: Speed controller integrator setting value / n_ctr integ_setVal
p1478[0...n]	CI: Velocity controller integrator value / v_ctr integ_setVal
p1479[0...n]	CI: Speed controller integrator setting value scaling / n_ctrl I_val scal
p1486[0...n]	CI: Droop compensation torque / Droop M_comp
p1492[0...n]	BI: Droop feedback enable / Droop enable
p1495[0...n]	CI: Acceleration pre-control / a_prectrl
p1497[0...n]	CI: Moment of inertia, scaling / M_mom inert scal
p1497[0...n]	CI: Motor weight scaling / Motor weight scal
p1500[0...n]	Macro Connector Inputs (CI) for torque setpoints / Macro CI M_set

p1500[0...n]	Macro Connector Inputs (CI) for force setpoints / Macro CI F_set
p1501[0...n]	BI: Change over between closed-loop speed/torque control / Changeov n/M_ctrl
p1501[0...n]	BI: Change over velocity/force control / Changeov n/F_ctrl
p1503[0...n]	CI: Torque setpoint / M_set
p1511[0...n]	CI: Supplementary torque 1 / M_suppl 1
p1511[0...n]	CI: Supplementary force 1 / F_suppl 1
p1512[0...n]	CI: Supplementary torque 1 scaling / M_suppl 1 scal
p1512[0...n]	CI: Supplementary force 1 scaling / F_suppl 1 scal
p1513[0...n]	CI: Supplementary torque 2 / M_suppl 2
p1513[0...n]	CI: Supplementary force 2 / F_suppl 2
p1522[0...n]	CI: Torque limit upper/motoring / M_max upper/mot
p1522[0...n]	CI: Force limit upper/motoring / F_max upper/mot
p1522[0...n]	CI: Torque limit upper / M_max upper
p1523[0...n]	CI: Torque limit lower/regenerative / M_max lower/regen
p1523[0...n]	CI: Force limit lower/regenerative / F_max lower/regen
p1523[0...n]	CI: Torque limit lower / M_max lower
p1528[0...n]	CI: Torque limit upper/motoring scaling / M_max up/mot scal
p1528[0...n]	CI: Force limit upper/motoring scaling / F_max up/mot scal
p1528[0...n]	CI: Torque limit upper scaling / M_max upper scal
p1529[0...n]	CI: Torque limit lower/regenerative scaling / M_max low/gen scal
p1529[0...n]	CI: Force limit lower/regenerative scaling / F_max lo/reg scal
p1529[0...n]	CI: Torque limit lower scaling / M_max lower scal
p1540[0...n]	CI: Torque limit speed controller upper scaling / M_max n-ctr upScal
p1541[0...n]	CI: Torque limit. speed controller lower scaling / M_max nctr lowScal
p1542[0...n]	CI: Travel to fixed stop torque reduction / TfS M_red
p1542[0...n]	CI: Travel to fixed stop force reduction / TfS F_red
p1545[0...n]	BI: Activates travel to a fixed stop / TfS activation
p1550[0...n]	BI: Transfer actual torque as torque offset / Accept act torque
p1550[0...n]	BI: Transfer actual force as force offset / Accept act force
p1551[0...n]	BI: Torque limit variable/fixed signal source / M_lim var/fixS_src
p1551[0...n]	BI: Force limit variable/fixed signal source / F_lim var/fixS_src
p1552[0...n]	CI: Torque limit upper scaling without offset / M_max up w/o offs
p1552[0...n]	CI: Force limit upper scaling without offset / F_max up w/o offs
p1554[0...n]	CI: Torque limit lower scaling without offset / M_max low w/o offs
p1554[0...n]	CI: Force limit lower scaling without offset / F_max low w/o offs
p1555[0...n]	CI: Power limit / P_max
p1569[0...n]	CI: Supplementary torque 3 / M_suppl 3
p1569[0...n]	CI: Supplementary force 3 / F_suppl 3
p1571[0...n]	CI: Supplementary flux setpoint / Suppl flux setp
p1640[0...n]	CI: Excitation current actual value / I_exc_act val
p2103[0...n]	BI: 1. Acknowledge faults / 1. Acknowledge
p2104[0...n]	BI: 2. Acknowledge faults / 2. Acknowledge
p2105[0...n]	BI: 3. Acknowledge faults / 3. Acknowledge
p2106[0...n]	BI: External fault 1 / External fault 1
p2107[0...n]	BI: External fault 2 / External fault 2
p2108[0...n]	BI: External fault 3 / External fault 3
p2112[0...n]	BI: External alarm 1 / External alarm 1
p2116[0...n]	BI: External alarm 2 / External alarm 2
p2117[0...n]	BI: External alarm 3 / External alarm 3
p2144[0...n]	BI: Motor stall monitoring enable (negated) / Mot stall enab neg
p2148[0...n]	BI: Ramp-function generator active / HLG active
p2151[0...n]	CI: Speed setpoint for messages/signals / n_set for msg
p2151[0...n]	CI: Velocity setpoint for messages/signals / v_set for msg

p2154[0...n]	Cl: Speed setpoint 2 / n_set 2
p2154[0...n]	Cl: Velocity setpoint 2 / v_set 2
p2200[0...n]	Bl: Technology controller enable / Tec_ctrl enable
p2220[0...n]	Bl: Technology controller fixed value selection bit 0 / Tec_ctrl sel bit 0
p2221[0...n]	Bl: Technology controller fixed value selection bit 1 / Tec_ctrl sel bit 1
p2222[0...n]	Bl: Technology controller fixed value selection bit 2 / Tec_ctrl sel bit 2
p2223[0...n]	Bl: Technology controller fixed value selection bit 3 / Tec_ctrl sel bit 3
p2235[0...n]	Bl: Technology controller motorized potentiometer raise setpoint / Tec_ctrl mop raise
p2236[0...n]	Bl: Technology controller motorized potentiometer lower setpoint / Tec_ctrl mop lower
p2253[0...n]	Cl: Technology controller setpoint 1 / Tec_ctrl setp 1
p2254[0...n]	Cl: Technology controller setpoint 2 / Tec_ctrl setp 2
p2264[0...n]	Cl: Technology controller actual value / Tec_ctrl act val
p2286[0...n]	Bl: Hold technology controller integrator / Tec_ctr integ stop
p2289[0...n]	Cl: Technology controller pre-control signal / Tec_ctrl prectrl
p2296[0...n]	Cl: Technology controller output scaling / Tec_ctrl outp scal
p2297[0...n]	Cl: Technology controller maximum limit signal source / Tec_ctrl m_lm s_sc
p2298[0...n]	Cl: Technology controller minimum limit signal source / Tec_ctrl min_l s_s
p2299[0...n]	Cl: Technology controller limit offset / Tec_ctrl lim offs
p3111[0...n]	Bl: External fault 3, enable / Ext fault 3 enab
p3112[0...n]	Bl: External fault 3 enable negated / Ext flt 3 enab neg
p3750[0...n]	Cl: APC acceleration sensor input / APC accel input
p3784[0...n]	Bl: Sync-line-drive external increase voltage / Sync ext V incr
p3785[0...n]	Bl: Sync-line-drive external decrease voltage / Sync ext V decr
p3802[0...n]	Bl: Sync-line-drive enable / Sync enable

1.3.2 Parameters for drive data sets (DDS)

Note:

References: /FH1/ SINAMICS S120 Function Manual Drive Functions
Section "Data Sets"

The following list contains the drive-data-set-dependent parameters.

Product: SINAMICS S120/S150, Version: 4301400, Language: eng, Type: DDS

p0186[0...n]	Motor Data Sets (MDS) number / MDS number
p0187[0...n]	Encoder 1 encoder data set number / Enc 1 EDS number
p0188[0...n]	Encoder 2 encoder data set number / Enc 2 EDS number
p0189[0...n]	Encoder 3 encoder data set number / Enc 3 EDS number
p0340[0...n]	Automatic calculation, motor/control parameters / Calc auto par
p0572[0...n]	Activate inhibit list / Act inhib list
p0578[0...n]	Calculate parameters that are dependent on the technology/units / Calc tec par
p0640[0...n]	Current limit / Current limit
p0642[0...n]	Encoderless operation current reduction / Encoderl op I_red
p1001[0...n]	CO: Fixed velocity setpoint 1 / n_set_fixed 1
p1001[0...n]	CO: Fixed speed setpoint 1 / n_set_fixed 1
p1002[0...n]	CO: Fixed velocity setpoint 2 / n_set_fixed 2
p1002[0...n]	CO: Fixed speed setpoint 2 / n_set_fixed 2
p1003[0...n]	CO: Fixed velocity setpoint 3 / n_set_fixed 3
p1003[0...n]	CO: Fixed speed setpoint 3 / n_set_fixed 3
p1004[0...n]	CO: Fixed velocity setpoint 4 / n_set_fixed 4

p1004[0...n]	CO: Fixed speed setpoint 4 / n_set_fixed 4
p1005[0...n]	CO: Fixed velocity setpoint 5 / n_set_fixed 5
p1005[0...n]	CO: Fixed speed setpoint 5 / n_set_fixed 5
p1006[0...n]	CO: Fixed velocity setpoint 6 / n_set_fixed 6
p1006[0...n]	CO: Fixed speed setpoint 6 / n_set_fixed 6
p1007[0...n]	CO: Fixed velocity setpoint 7 / n_set_fixed 7
p1007[0...n]	CO: Fixed speed setpoint 7 / n_set_fixed 7
p1008[0...n]	CO: Fixed velocity setpoint 8 / n_set_fixed 8
p1008[0...n]	CO: Fixed speed setpoint 8 / n_set_fixed 8
p1009[0...n]	CO: Fixed velocity setpoint 9 / n_set_fixed 9
p1009[0...n]	CO: Fixed speed setpoint 9 / n_set_fixed 9
p1010[0...n]	CO: Fixed velocity setpoint 10 / n_set_fixed 10
p1010[0...n]	CO: Fixed speed setpoint 10 / n_set_fixed 10
p1011[0...n]	CO: Fixed velocity setpoint 11 / n_set_fixed 11
p1011[0...n]	CO: Fixed speed setpoint 11 / n_set_fixed 11
p1012[0...n]	CO: Fixed velocity setpoint 12 / n_set_fixed 12
p1012[0...n]	CO: Fixed speed setpoint 12 / n_set_fixed 12
p1013[0...n]	CO: Fixed velocity setpoint 13 / n_set_fixed 13
p1013[0...n]	CO: Fixed speed setpoint 13 / n_set_fixed 13
p1014[0...n]	CO: Fixed velocity setpoint 14 / n_set_fixed 14
p1014[0...n]	CO: Fixed speed setpoint 14 / n_set_fixed 14
p1015[0...n]	CO: Fixed velocity setpoint 15 / n_set_fixed 15
p1015[0...n]	CO: Fixed speed setpoint 15 / n_set_fixed 15
p1030[0...n]	Motorized potentiometer configuration / Mop configuration
p1037[0...n]	Motorized potentiometer maximum velocity / Mop n_max
p1037[0...n]	Motorized potentiometer maximum speed / Mop n_max
p1038[0...n]	Motorized potentiometer minimum velocity / Mop n_min
p1038[0...n]	Motorized potentiometer minimum speed / Mop n_min
p1040[0...n]	Motorized potentiometer starting value / Mop start value
p1047[0...n]	Motorized potentiometer ramp-up time / Mop ramp-up time
p1048[0...n]	Motorized potentiometer ramp-down time / Mop ramp-down time
p1058[0...n]	Jog 1 velocity setpoint / Jog 1 n_set
p1058[0...n]	Jog 1 speed setpoint / Jog 1 n_set
p1059[0...n]	Jog 2 velocity setpoint / Jog 2 n_set
p1059[0...n]	Jog 2 speed setpoint / Jog 2 n_set
p1063[0...n]	Velocity limit setpoint channel / v_limit setp
p1063[0...n]	Speed limit setpoint channel / n_limit setp
p1080[0...n]	Minimum velocity / Minimum speed
p1080[0...n]	Minimum speed / Minimum speed
p1082[0...n]	Maximum speed / n_Max
p1082[0...n]	Maximum velocity / v_max
p1083[0...n]	CO: Speed limit in positive direction of rotation / n_limit pos
p1083[0...n]	CO: Velocity limit positive direction / v_limit pos
p1086[0...n]	CO: Speed limit in negative direction of rotation / n_limit neg
p1086[0...n]	CO: Velocity limit negative direction / v_limit neg
p1091[0...n]	Skip velocity 1 / v_skip 1
p1091[0...n]	Skip speed 1 / n_skip 1
p1092[0...n]	Skip velocity 2 / v_skip 2
p1092[0...n]	Skip speed 2 / n_skip 2
p1093[0...n]	Skip velocity 3 / v_skip 3
p1093[0...n]	Skip speed 3 / n_skip 3
p1094[0...n]	Skip velocity 4 / v_skip 4
p1094[0...n]	Skip speed 4 / n_skip 4

p1101[0...n]	Skip velocity bandwidth / v_skip bandwidth
p1101[0...n]	Skip speed bandwidth / n_skip bandwidth
p1120[0...n]	Ramp-function generator ramp-up time / RFG ramp-up time
p1121[0...n]	Ramp-function generator ramp-down time / RFG ramp-down time
p1130[0...n]	Ramp-function generator initial rounding-off time / RFG t_start_round
p1131[0...n]	Ramp-function generator final rounding-off time / RFG t_end_delay
p1134[0...n]	Ramp-function generator rounding-off type / RFG round-off type
p1135[0...n]	OFF3 ramp-down time / OFF3 t_RD
p1136[0...n]	OFF3 initial rounding-off time / RFGOFF3 t_strt_rnd
p1137[0...n]	OFF3 final rounding-off time / RFG OFF3 t_end_del
p1145[0...n]	Ramp-function generator tracking intensity. / RFG track intens
p1148[0...n]	Ramp-function gen., tolerance for ramp-up and ramp-down active / RFG tol HL/RL act
p1151[0...n]	Ramp-function generator configuration / RFG config
p1189[0...n]	Speed setpoint configuration / n_ctrl config
p1189[0...n]	Velocity setpoint configuration / v_ctrl config
p1192[0...n]	DSC enc selection / DSC enc selection
p1193[0...n]	DSC encoder adaptation factor / DSC encodAdaptFact
p1200[0...n]	FlyRest oper mode / FlyRest op_mode
p1202[0...n]	FlyRest srch curr / FlyRest I_srch
p1203[0...n]	Flying restart search rate factor / FlyRst v_Srch Fact
p1226[0...n]	Threshold for zero speed detection / n_standst n_thresh
p1226[0...n]	Standstill detection, velocity threshold / v_standst v_thresh
p1240[0...n]	Vdc controller or Vdc monitoring configuration / Vdc_ctrl config
p1243[0...n]	Vdc_max controller dynamic factor / Vdc_max dyn_factor
p1244[0...n]	DC link voltage threshold upper / Vdc upper thresh
p1245[0...n]	Vdc_min controller switch-in level (kinetic buffering) / Vdc_min on_level
p1247[0...n]	Vdc_min controller dynamic factor (kinetic buffering) / Vdc_min dyn_factor
p1248[0...n]	DC link voltage threshold lower / Vdc lower thresh
p1249[0...n]	Vdc_max controller speed threshold / Vdc_max n_thresh
p1250[0...n]	Vdc controller proportional gain / Vdc_ctrl Kp
p1251[0...n]	Vdc controller integral time / Vdc_ctrl Tn
p1252[0...n]	Vdc controller rate time / Vdc_ctrl t_rate
p1255[0...n]	Vdc_min controller time threshold / Vdc_min t_thresh
p1256[0...n]	Vdc_min controller response (kinetic buffering) / Vdc_min response
p1257[0...n]	Vdc_min controller speed threshold / Vdc_min n_thresh
p1262[0...n]	Bypass dead time / Bypass t_dead
p1280[0...n]	Vdc controller or Vdc monitoring configuration (V/f) / Vdc_ctr config V/f
p1283[0...n]	Vdc_max controller dynamic factor (V/f) / Vdc_max dyn_factor
p1285[0...n]	Vdc_min controller switch-in level (kinetic buffering) (V/f) / Vdc_min on_level
p1287[0...n]	Vdc_min controller dynamic factor (kinetic buffering) (V/f) / Vdc_min dyn_factor
p1288[0...n]	Vdc_max controller feedback coupling factor ramp-fct. gen. (V/f) / Vdc_max factor RFG
p1289[0...n]	Vdc_max controller speed threshold (V/f) / Vdc_max n_thresh
p1290[0...n]	Vdc controller proportional gain (V/f) / Vdc_ctrl Kp
p1291[0...n]	Vdc controller integral time (V/f) / Vdc_ctrl Tn
p1292[0...n]	Vdc controller rate time (V/f) / Vdc_ctrl t_rate
p1293[0...n]	Vdc min controller output limit (V/f) / Vdc_min outp_lim
p1295[0...n]	Vdc_min controller time threshold (V/f) / Vdc_min t_thresh
p1296[0...n]	Vdc_min controller response (kinetic buffering) (V/f) / Vdc_min response
p1297[0...n]	Vdc_min controller speed threshold (V/f) / Vdc_min n_thresh
p1300[0...n]	Open-loop/closed-loop control operating mode / Op/cl-lp ctrl_mode
p1302[0...n]	V/f control configuration / V/f configuration
p1310[0...n]	Voltage boost permanent / V_boost perm
p1311[0...n]	Voltage boost at acceleration / V_boost accelerate

p1312[0...n]	Voltage boost when starting / V_boost starting
p1317[0...n]	V/f control activation / Vf act
p1318[0...n]	V/f control ramp-up/ramp-down time / Uf t_rmp-up_rmp-dn
p1319[0...n]	V/f control voltage at zero frequency / Uf V at f=0 Hz
p1320[0...n]	V/f control programmable characteristic frequency 1 / Uf char f1
p1321[0...n]	V/f control programmable characteristic voltage 1 / Vf char U1
p1322[0...n]	V/f control programmable characteristic frequency 2 / Vf char f2
p1323[0...n]	V/f control programmable characteristic voltage 2 / Vf char U2
p1324[0...n]	V/f control programmable characteristic frequency 3 / Vf char f3
p1325[0...n]	V/f control programmable characteristic voltage 3 / Vf char U3
p1326[0...n]	V/f control programmable characteristic frequency 4 / Vf char f4
p1327[0...n]	V/f control programmable characteristic voltage 4 / Vf char U4
p1333[0...n]	V/f control FCC starting frequency / V/f FCC f_start
p1334[0...n]	V/f control slip compensation starting frequency / Slip comp start
p1335[0...n]	Slip compensation, scaling / Slip comp scal
p1336[0...n]	Slip compensation limit value / Slip comp lim val
p1338[0...n]	V/f mode resonance damping gain / Vf Res_damp gain
p1339[0...n]	V/f mode resonance damping filter time constant / Vf Res_damp T
p1340[0...n]	I_max frequency controller proportional gain / I_max_ctrl Kp
p1341[0...n]	I_max frequency controller integral time / I_max_ctrl Tn
p1345[0...n]	DC brake proportional gain / DCBRK Kp
p1345[0...n]	I_max voltage controller proportional gain / I_max_V_ctrl Kp
p1346[0...n]	DC brake integral time / DCBRK Tn
p1346[0...n]	I_max voltage controller integral time / I_max_V_ctrl Tn
p1349[0...n]	V/f mode resonance damping maximum frequency / Vf res_damp F_max
p1350[0...n]	Soft starting / Soft starting
p1351[0...n]	CO: Motor holding brake starting frequency / Brake f_start
p1358[0...n]	Angular difference, symmetrizing, actual angle / Sym act angle
p1400[0...n]	Speed control configuration / n_ctrl config
p1400[0...n]	Velocity control, configuration / v_ctrl config
p1401[0...n]	Flux control configuration / Flux ctrl config
p1402[0...n]	Closed-loop current control and motor model configuration / I_ctrl config
p1404[0...n]	Encoderless operation changeover speed / Encoderl op n_chg
p1404[0...n]	Encoderless operation changeover velocity / Encoderl op v_chg
p1412[0...n]	Speed setpoint filter, dead time / n_set dead time
p1414[0...n]	Speed setpoint filter activation / n_set_filt act
p1414[0...n]	Velocity setpoint filter activation / v_set_filt act
p1415[0...n]	Speed setpoint filter 1 type / n_set_filt 1 typ
p1415[0...n]	Velocity setpoint filter 1 type / v_setp_filt 1 typ
p1416[0...n]	Speed setpoint filter 1 time constant / n_set_filt 1 T
p1416[0...n]	Velocity setpoint filter 1 time constant / v_set_filt 1 T
p1417[0...n]	Speed setpoint filter 1 denominator natural frequency / n_set_filt 1 fn_d
p1417[0...n]	Velocity setpoint filter 1 denominator natural frequency / v_set_filt 1 fn_d
p1418[0...n]	Speed setpoint filter 1 denominator damping / n_set_filt 1 D_d
p1418[0...n]	Velocity setpoint filter 1 denominator damping / v_set_filt 1 D_d
p1419[0...n]	Speed setpoint filter 1 numerator natural frequency / n_set_filt 1 fn_n
p1419[0...n]	Velocity setpoint filter 1 numerator natural frequency / v_set_filt 1 fn_n
p1420[0...n]	Speed setpoint filter 1 numerator damping / n_set_filt 1 D_n
p1420[0...n]	Velocity setpoint filter 1 numerator damping / v_set_filt 1 D_n
p1421[0...n]	Speed setpoint filter 2 type / n_set_filt 2 typ
p1421[0...n]	Velocity setpoint filter 2 type / v_setp_filt 2 typ
p1422[0...n]	Speed setpoint filter 2 time constant / n_set_filt 2 T
p1422[0...n]	Velocity setpoint filter 2 time constant / v_set_filt 2 T

p1423[0...n]	Speed setpoint filter 2 denominator natural frequency / $n_set_filt\ 2\ fn_d$
p1423[0...n]	Velocity setpoint filter 2 denominator natural frequency / $v_set_filt\ 2\ fn_d$
p1424[0...n]	Speed setpoint filter 2 denominator damping / $n_set_filt\ 2\ D_d$
p1424[0...n]	Velocity setpoint filter 2 denominator damping / $v_set_filt\ 2\ D_d$
p1425[0...n]	Speed setpoint filter 2 numerator natural frequency / $n_set_filt\ 2\ fn_n$
p1425[0...n]	Velocity setpoint filter 2 numerator natural frequency / $v_set_filt\ 2\ fn_n$
p1426[0...n]	Speed setpoint filter 2 numerator damping / $n_set_filt\ 2\ D_n$
p1426[0...n]	Velocity setpoint filter 2 numerator damping / $v_set_filt\ 2\ D_n$
p1428[0...n]	Speed pre-control balancing dead time / $n_prectrBal\ t_dead$
p1428[0...n]	Velocity pre-control balancing dead time / $n_prectrBal\ t_dead$
p1429[0...n]	Speed pre-control balancing time constant / $n_prectr\ bal\ T$
p1429[0...n]	Velocity pre-control balancing time constant / $n_prectr\ bal\ T$
p1433[0...n]	Speed controller reference model natural frequency / $n_ctrl\ RefMod\ fn$
p1433[0...n]	Velocity controller reference model natural frequency / $v_ctrl\ RefMod\ fn$
p1434[0...n]	Speed controller reference model damping / $n_ctrl\ RefMod\ D$
p1434[0...n]	Velocity controller reference model damping / $v_ctrl\ RefMod\ D$
p1435[0...n]	Speed controller reference model dead time / $n_ctrRefMod\ t_dead$
p1435[0...n]	Velocity controller reference model dead time / $v_ctrRefMod\ t_dead$
p1441[0...n]	Actual speed smoothing time / $n_ist\ T_smooth$
p1441[0...n]	Actual velocity smoothing time / $v_act\ T_smooth$
p1442[0...n]	Speed controller speed actual value smoothing time / $n_ctr\ n_act\ T_smth$
p1451[0...n]	Speed actual value smoothing time sensorless / $n_act\ t_sm\ SL$
p1451[0...n]	Velocity actual value smoothing time sensorless / $v_act\ t_sm\ SL$
p1451[0...n]	Motor model speed actual value smoothing time SLVC / $Mot_mod\ n_act\ t_sm$
p1452[0...n]	Speed controller speed actual value smoothing time (SLVC) / $n_C\ n_act\ T_s\ SLVC$
p1456[0...n]	Speed controller P gain adaptation lower starting point / $n_ctrl\ AdaptKpLow$
p1456[0...n]	Velocity controller P gain adaptation, lower starting point / $v_ctrl\ AdaptKpLow$
p1457[0...n]	Speed controller P gain adaptation upper starting point / $n_ctrl\ AdaptKp\ up$
p1457[0...n]	Velocity controller P gain adaptation upper starting point / $v_ctrl\ AdaptKp\ up$
p1458[0...n]	Adaptation factor, lower / $Adapt_factor\ lower$
p1459[0...n]	Adaptation factor, upper / $Adapt_factor\ upper$
p1460[0...n]	Speed controller P gain adaptation speed, lower / $n_ctrl\ Kp\ n\ lower$
p1460[0...n]	Velocity controller, P gain adaptation velocity, lower / $v_ctrl\ Kp\ n\ lower$
p1461[0...n]	Speed controller Kp adaptation speed, upper scaling / $n_ctrl\ Kp\ n\ upper$
p1461[0...n]	Velocity controller, P gain adaptation velocity, upper / $v_ctrl\ Kp\ n\ upper$
p1462[0...n]	Speed controller integral time adaptation speed lower / $n_ctrl\ Tn\ n\ lower$
p1462[0...n]	Velocity contr. integral act. time adaptation velocity lower / $v_ctrl\ Tn\ n\ lower$
p1463[0...n]	Speed controller Tn adaptation speed, upper scaling / $n_ctrl\ Tn\ n\ upper$
p1463[0...n]	Velocity contr. integral act. time adaptation velocity upper / $v_ctrl\ Tn\ n\ upper$
p1464[0...n]	Speed controller adaptation speed, lower / $n_ctrl\ n\ lower$
p1464[0...n]	Velocity controller adaptation velocity, lower / $v_ctrl\ n\ lower$
p1465[0...n]	Speed controller adaptation speed, upper / $n_ctrl\ n\ upper$
p1465[0...n]	Velocity controller adaptation velocity, upper / $v_ctrl\ n\ upper$
p1470[0...n]	Speed controller encoderless operation P-gain / $n_ctrl\ SLVC\ Kp$
p1470[0...n]	Velocity controller encoderless operation P-gain / $v_ctrl\ SLVC\ Kp$
p1472[0...n]	Speed controller encoderless operation integral time / $n_ctrl\ SLVC\ Tn$
p1472[0...n]	Velocity controller encoderless operation integral time / $v_ctrl\ SLVC\ Tn$
p1487[0...n]	Droop compensation torque scaling / $Droop\ M_comp\ scal$
p1488[0...n]	Droop input source / $Droop\ input\ source$
p1489[0...n]	Droop feedback scaling / $Droop\ scaling$
p1494[0...n]	Speed controller integrator feedback time constant / $n_ctr\ integ_fdbk\ T$
p1494[0...n]	Velocity controller integrator feedback time constant / $v_ctr\ integ_fdbk\ T$
p1496[0...n]	Acceleration pre-control scaling / $a_before\ scaling$

p1498[0...n]	Load moment of inertia / Load mom of inert
p1498[0...n]	Load mass / Load mass
p1499[0...n]	Accelerating for torque control, scaling / a for M_ctrl scal
p1514[0...n]	Supplementary torque 2 scaling / M_suppl 2 scal
p1517[0...n]	Accelerating torque smoothing time constant / M_accel T_smooth
p1517[0...n]	Acceleration force smoothing time constant / F_accel T_smooth
p1520[0...n]	CO: Torque limit upper/motoring / M_max upper/mot
p1520[0...n]	CO: Force limit upper/motoring / F_max upper/mot
p1520[0...n]	CO: Torque limit upper / M_max upper
p1521[0...n]	CO: Torque limit lower/regenerative / M_max lower/regen
p1521[0...n]	CO: Force limit lower/regenerative / F_max lower/regen
p1521[0...n]	CO: Torque limit lower / M_max lower
p1524[0...n]	CO: Torque limit upper/motoring scaling / M_max up/mot scal
p1524[0...n]	CO: Force limit upper/motoring scaling / F_max up/mot scal
p1524[0...n]	CO: Torque limit upper scaling / M_max upper scal
p1525[0...n]	CO: Torque limit lower/regenerative scaling / M_max low/gen scal
p1525[0...n]	CO: Force limit lower/regenerative scaling / F_max lo/reg scal
p1525[0...n]	CO: Torque limit lower scaling / M_max lower scal
p1530[0...n]	Power limit motoring / P_max mot
p1531[0...n]	Power limit regenerative / P_max gen
p1532[0...n]	CO: Torque limit offset / M_max offset
p1532[0...n]	CO: Force offset, force limit / F_max offset
p1556[0...n]	Power limit scaling / P_max_scale
p1570[0...n]	CO: Flux setpoint / Flux setpoint
p1572[0...n]	Supplementary flux setpoint / Suppl flux setp
p1573[0...n]	Flux threshold value magnetizing / Flux thresh magnet
p1574[0...n]	Voltage reserve dynamic / V_reserve dyn
p1576[0...n]	Flux boost, adaptation speed, lower / Flux boost n lower
p1577[0...n]	Flux boost adaptation speed, upper / Flux boost n upper
p1578[0...n]	Flux reduction flux decrease smoothing time / Flux red dec t_sm
p1579[0...n]	Flux reduction flux build-up smoothing time / Flux red up t_sm
p1580[0...n]	Efficiency optimization / Efficiency opt.
p1581[0...n]	Flux reduction factor / Flux red factor
p1582[0...n]	Flux setpoint smoothing time / Flux setp T_smth
p1584[0...n]	Field weakening operation, flux setpoint smoothing time / Field weak T_smth
p1585[0...n]	Flux actual value, smoothing time / Flux actVal T_smth
p1586[0...n]	Field weakening characteristic, scaling / Field weak scal
p1590[0...n]	Flux controller P gain / Flux controller Kp
p1592[0...n]	Flux controller integral time / Flux controller Tn
p1594[0...n]	Field-weakening controller, P gain / Field_ctrl Kp
p1596[0...n]	Field weakening controller integral-action time / Field_ctrl Tn
p1599[0...n]	Flux controller, excitation current difference / Flux ctr I_exc_dif
p1600[0...n]	P flux controller, P gain / P flux ctrl Kp
p1605[0...n]	Pulse technique pattern configuration / Pulse config
p1607[0...n]	Pulse technique stimulus / Pulse stimu
p1609[0...n]	Current setpoint for I/f operation / I_set I/f oper
p1610[0...n]	Torque setpoint static (SLVC) / M_set static
p1611[0...n]	Supplementary accelerating torque (SLVC) / M_suppl_accel
p1612[0...n]	Current setpoint, open-loop control, encoderless / I_setCtrEncoderl
p1612[0...n]	Current setpoint magnetizing open-loop controlled / Id_set ctrl
p1616[0...n]	Current setpoint smoothing time / I_set T_smooth
p1619[0...n]	Setpoint/actual value tracking threshold / SetAct track thrsh
p1620[0...n]	Stator current, minimum / I_stator min

p1621[0...n]	Changeover speed, inner $\cos \phi = 1 / n_{\text{chngov}} \cos \phi = 1$
p1622[0...n]	Field-generating current setpoint smoothing time constant / $I_{\text{d_setp}} T_{\text{smth}}$
p1625[0...n]	Excitation current setpoint calibration / $I_{\text{exc_setp}} \text{cal}$
p1628[0...n]	Current model controller, dynamic factor / $I_{\text{mod_ctr}} \text{dyn_fact}$
p1629[0...n]	Current model controller P gain / $I_{\text{mod_ctrl}} K_p$
p1630[0...n]	Current model controller integral time / $I_{\text{ctrl}} T_n$
p1642[0...n]	Minimum excitation current / $\text{Min } I_{\text{exc}}$
p1643[0...n]	Gain factor, minimum excitation current closed-loop control / $\text{Min } I_{\text{exc}} K_p$
p1653[0...n]	Current setpoint torque-generating smoothing time minimum / $I_{\text{sq_s}} T_{\text{smth min}}$
p1654[0...n]	Curr. setpoint torque-gen. smoothing time field weakening range / $I_{\text{sq_s}} T_{\text{smth FW}}$
p1656[0...n]	Activates current setpoint filter / $I_{\text{setp_filt}} \text{act}$
p1657[0...n]	Current setpoint filter 1 type / $I_{\text{set_filt}} 1 \text{ Typ}$
p1658[0...n]	Current setpoint filter 1 denominator natural frequency / $I_{\text{set_filt}} 1 f_{n_n}$
p1659[0...n]	Current setpoint filter 1 denominator damping / $I_{\text{set_filt}} 1 D_n$
p1660[0...n]	Current setpoint filter 1 numerator natural frequency / $I_{\text{set_filt}} 1 f_{n_z}$
p1661[0...n]	Current setpoint filter 1 numerator damping / $I_{\text{set_filt}} 1 D_z$
p1662[0...n]	Current setpoint filter 2 type / $I_{\text{set_filt}} 2 \text{ Typ}$
p1663[0...n]	Current setpoint filter 2 denominator natural frequency / $I_{\text{set_filt}} 2 f_{n_n}$
p1664[0...n]	Current setpoint filter 2 denominator damping / $I_{\text{set_filt}} 2 D_n$
p1665[0...n]	Current setpoint filter 2 numerator natural frequency / $I_{\text{set_filt}} 2 f_{n_z}$
p1666[0...n]	Current setpoint filter 2 numerator damping / $I_{\text{set_filt}} 2 D_z$
p1667[0...n]	Current setpoint filter 3 type / $I_{\text{set_filt}} 3 \text{ Typ}$
p1668[0...n]	Current setpoint filter 3 denominator natural frequency / $I_{\text{set_filt}} 3 f_{n_n}$
p1669[0...n]	Current setpoint filter 3 denominator damping / $I_{\text{set_filt}} 3 D_n$
p1670[0...n]	Current setpoint filter 3 numerator natural frequency / $I_{\text{set_filt}} 3 f_{n_z}$
p1671[0...n]	Current setpoint filter 3 numerator damping / $I_{\text{set_filt}} 3 D_z$
p1672[0...n]	Current setpoint filter 4 type / $I_{\text{set_filt}} 4 \text{ Typ}$
p1673[0...n]	Current setpoint filter 4 denominator natural frequency / $I_{\text{set_filt}} 4 f_{n_n}$
p1674[0...n]	Current setpoint filter 4 denominator damping / $I_{\text{set_filt}} 4 D_n$
p1675[0...n]	Current setpoint filter 4 numerator natural frequency / $I_{\text{set_filt}} 4 f_{n_n}$
p1676[0...n]	Current setpoint filter 4 numerator damping / $I_{\text{set_filt}} 4 D_z$
p1701[0...n]	Current controller reference model dead time / $I_{\text{ctrRefMod}} t_{\text{dead}}$
p1702[0...n]	Isd current controller pre-control scaling / $I_{\text{sd_ctr_prectr}} \text{Scal}$
p1703[0...n]	Isq current controller pre-control scaling / $I_{\text{sq_ctr_prectr}} \text{Scal}$
p1704[0...n]	Isq current controller pre-control EMF scaling / $I_{\text{sq_ctrl}} \text{EMF scal}$
p1705[0...n]	Flux setpoint/actual value tracking threshold / Flux track thresh
p1715[0...n]	Current controller P gain / $I_{\text{ctrl}} K_p$
p1717[0...n]	Current controller integral-action time / $I_{\text{ctrl}} T_n$
p1726[0...n]	Quadrature arm decoupling, scaling / Transv_decpl scal
p1727[0...n]	Quadrature arm decoupling at voltage limit scaling / $\text{TrnsvDecplVmaxScal}$
p1740[0...n]	Gain resonance damping for encoderless closed loop control / Gain res_damp
p1744[0...n]	Motor model speed threshold stall detection / $\text{MotMod } n_{\text{thr stall}}$
p1745[0...n]	Motor model error threshold stall detection / $\text{MotMod ThreshStall}$
p1748[0...n]	Motor model lower changeover speed $n_{\text{set}} \rightarrow n_{\text{act}}$ / $\text{Lower } n_{\text{chngov}}$
p1749[0...n]	Motor model upper changeover speed $n_{\text{set}} \rightarrow n_{\text{act}}$ / $\text{Upper } n_{\text{chngov}}$
p1750[0...n]	Motor model configuration / MotMod config
p1752[0...n]	Motor model changeover speed operation with encoder / $\text{MotMod } n_{\text{chngov enc}}$
p1752[0...n]	Motor model with encoder changeover velocity / $\text{MotMod } v_{\text{chngov enc}}$
p1753[0...n]	Motor model changeover speed hysteresis operation with encoder / $\text{MotMod } n_{\text{chngovHysE}}$
p1754[0...n]	Flux angle difference smoothing time / $\text{Angle diff } T_{\text{smth}}$
p1755[0...n]	Motor model changeover speed encoderless operation / $\text{MotMod } n_{\text{chgSnsorl}}$
p1755[0...n]	Motor model changeover velocity encoderless operation / $\text{MotMod } v_{\text{chgSnsorl}}$
p1757[0...n]	Motor model w/o enc. op./cl.-loop controlled stab. controller K_p / $\text{MotMod w/o enc } K_p$

p1758[0...n]	Motor model changeover delay time closed/open-loop control / MotMod t cl_op
p1759[0...n]	Motor model changeover delay time open/closed loop control / MotMod t op_cl
p1760[0...n]	Motor model with encoder speed adaptation Kp / MotMod wE n_ada Kp
p1761[0...n]	Motor model with encoder speed adaptation Tn / MotMod wE n_ada Tn
p1764[0...n]	Motor model without encoder speed adaptation Kp / MotMod woE n_adaKp
p1766[0...n]	Motor model voltage model calculation enable / V_mod calc enab
p1767[0...n]	Motor model without encoder speed adaptation Tn / MotMod woE n_adaTn
p1774[0...n]	Motor model, offset voltage compensation alpha / MotMod offs comp A
p1775[0...n]	Motor model, offset voltage compensation beta / MotMod offs comp B
p1780[0...n]	Motor model adaptation configuration / MotMod adapt conf
p1780[0...n]	Motor/converter model adaptation configuration / MotMod adapt conf
p1781[0...n]	Motor model Rs adaptation integral time / MotMod Rs Tn
r1782[0...n]	Motor model Rs adaptation corrective value / MotMod Rs corr
p1783[0...n]	Motor model Rs adaptation Kp / MotMod Rs Kp
p1785[0...n]	Motor model Lh adaptation Kp / MotMod Lh Kp
p1786[0...n]	Motor model Lh adaptation integral time / MotMod Lh Tn
r1787[0...n]	Motor model Lh adaptation corrective value / MotMod Lh corr
p1795[0...n]	Motor model kT adaptation smoothing time / MotMod kT T_smth
p1795[0...n]	Motor model kT adaptation integral time / MotMod kT Tn
r1797[0...n]	Motor model kT adaptation corrective value / MotMod kT corr
p1798[0...n]	Motor model pulse technique speed adaptation Kp / MotMod pulses Kp
p1800[0...n]	Pulse frequency setpoint / Pulse freq setp
p1802[0...n]	Modulator mode / Modulator mode
p1803[0...n]	Maximum modulation depth / Modulat depth max
p1804[0...n]	Filter time constant smoothed modulation index / T_filt mod_idxSmth
p1806[0...n]	Filter time constant Vdc correction / T_filt Vdc_corr
p1811[0...n]	Pulse frequency wobulation amplitude / Puls wobb ampl
p1820[0...n]	Reverse the output phase sequence / Outp_ph_seq rev
p1821[0...n]	Dir of rot / Dir of rot
p1821[0...n]	Direction / Direction
p1840[0...n]	Actual value correction configuration / AVC config
p1845[0...n]	Actual value correction evaluation factor Lsig / ActV_corr FactLsig
p1846[0...n]	Actual value correction damping factor / ActV_corr D_factor
p1952[0...n]	Voltage emulation error final value / V_error final val
p1953[0...n]	Voltage emulation error current offset / V_error I_offset
p2140[0...n]	Hysteresis speed 2 / n_hysteresis 2
p2140[0...n]	Hysteresis velocity 2 / v_hysteresis 2
p2141[0...n]	Speed threshold 1 / n_thresh val 1
p2141[0...n]	Velocity threshold value 1 / v_thresh val 1
p2142[0...n]	Hysteresis speed 1 / n_hysteresis 1
p2142[0...n]	Hysteresis velocity 1 / v_hysteresis 1
p2149[0...n]	Monitoring configuration / Monit config
p2150[0...n]	Hysteresis speed 3 / n_hysteresis 3
p2150[0...n]	Hysteresis velocity 3 / v_hysteresis 3
p2153[0...n]	Speed actual value filter time constant / n_act_filt T
p2153[0...n]	Velocity actual value filter time constant / v_act_filt T
p2155[0...n]	Speed threshold 2 / n_thresh val 2
p2155[0...n]	Velocity threshold value 2 / v_thresh val 2
p2156[0...n]	On delay, comparison value reached / t_on cmpr val rchd
p2161[0...n]	Speed threshold 3 / n_thresh val 3
p2161[0...n]	Velocity threshold value 3 / v_thresh val 3
p2162[0...n]	Hysteresis speed n_act > n_max / Hyst n_act>n_max
p2162[0...n]	Hysteresis velocity v_act > v_max / Hyst v_act>v_max

p2163[0...n]	Speed threshold 4 / n_thresh val 4
p2163[0...n]	Velocity threshold value 4 / v_thresh val 4
p2164[0...n]	Hysteresis speed 4 / n_hysteresis 4
p2164[0...n]	Hysteresis velocity 4 / v_hysteresis 4
p2166[0...n]	Off delay n_act = n_set / t_del_off n_i=n_so
p2166[0...n]	Off delay v_act = v_set / t_del_off n_i=n_so
p2167[0...n]	Switch-on delay n_act = n_set / t_on n_act=n_set
p2167[0...n]	On delay v_act = v_set / t_on n_act=n_set
p2174[0...n]	Torque threshold value 1 / M_thresh val 1
p2174[0...n]	Force threshold value 1 / F_thresh val 1
p2175[0...n]	Motor locked speed threshold / Mot lock n_thresh
p2175[0...n]	Motor locked, velocity threshold / Mot lock v_thresh
p2177[0...n]	Motor locked delay time / Mot lock t_del
p2178[0...n]	Motor stalled delay time / Mot stall t_del
p2181[0...n]	Load monitoring response / Load monit resp
p2182[0...n]	Load monitoring velocity threshold 1 / n_thresh 1
p2182[0...n]	Load monitoring speed threshold value 1 / n_thresh 1
p2183[0...n]	Load monitoring velocity threshold 2 / n_thresh 2
p2183[0...n]	Load monitoring speed threshold value 2 / n_thresh 2
p2184[0...n]	Load monitoring velocity threshold 3 / n_thresh 3
p2184[0...n]	Load monitoring speed threshold value 3 / n_thresh 3
p2185[0...n]	Load monitoring force threshold 1, upper / M_thresh 1 upper
p2185[0...n]	Load monitoring torque threshold 1, upper / M_thresh 1 upper
p2186[0...n]	Load monitoring force threshold 1, lower / M_thresh 1 lower
p2186[0...n]	Load monitoring torque threshold 1, lower / M_thresh 1 lower
p2187[0...n]	Load monitoring force threshold 2, upper / M_thresh 2 upper
p2187[0...n]	Load monitoring torque threshold 2, upper / M_thresh 2 upper
p2188[0...n]	Load monitoring force threshold 2, lower / M_thresh 2 lower
p2188[0...n]	Load monitoring torque threshold 2, lower / M_thresh 2 lower
p2189[0...n]	Load monitoring force threshold 3, upper / M_thresh 3 upper
p2189[0...n]	Load monitoring torque threshold 3, upper / M_thresh 3 upper
p2190[0...n]	Load monitoring force threshold 3, lower / M_thresh 3 lower
p2190[0...n]	Load monitoring torque threshold 3, lower / M_thresh 3 lower
p2192[0...n]	Load monitoring delay time / Load monit t_del
p2194[0...n]	Torque threshold value 2 / M_thresh val 2
p2194[0...n]	Force threshold value 2 / F_thresh val 2
p2195[0...n]	Torque utilization switch-off delay / M_util t_off
p2195[0...n]	Force utilization switch-off delay / F_util t_off
p2196[0...n]	Torque utilization scaling / M_util scal
p2201[0...n]	CO: Technology controller, fixed value 1 / Tec_ctrl fix val 1
p2202[0...n]	CO: Technology controller, fixed value 2 / Tec_ctrl fix val 2
p2203[0...n]	CO: Technology controller, fixed value 3 / Tec_ctrl fix val 3
p2204[0...n]	CO: Technology controller, fixed value 4 / Tec_ctrl fix val 4
p2205[0...n]	CO: Technology controller, fixed value 5 / Tec_ctrl fix val 5
p2206[0...n]	CO: Technology controller, fixed value 6 / Tec_ctrl fix val 6
p2207[0...n]	CO: Technology controller, fixed value 7 / Tec_ctrl fix val 7
p2208[0...n]	CO: Technology controller, fixed value 8 / Tec_ctrl fix val 8
p2209[0...n]	CO: Technology controller, fixed value 9 / Tec_ctrl fix val 9
p2210[0...n]	CO: Technology controller, fixed value 10 / Tec_ctrl fix val10
p2211[0...n]	CO: Technology controller, fixed value 11 / Tec_ctrl fix val11
p2212[0...n]	CO: Technology controller, fixed value 12 / Tec_ctrl fix val12
p2213[0...n]	CO: Technology controller, fixed value 13 / Tec_ctrl fix val13
p2214[0...n]	CO: Technology controller, fixed value 14 / Tec_ctrl fix val14

p2215[0...n]	CO: Technology controller, fixed value 15 / Tec_ctrl fix val15
p2216[0...n]	Technology controller fixed value selection method / Tec_ctr FixVal sel
p2230[0...n]	Technology controller motorized potentiometer configuration / Tec_ctr mop config
p2237[0...n]	Technology controller motorized potentiometer maximum value / Tec_ctrl mop max
p2238[0...n]	Technology controller motorized potentiometer minimum value / Tec_ctrl mop min
p2240[0...n]	Technology controller motorized potentiometer starting value / Tec_ctrl mop start
p2247[0...n]	Technology controller motorized potentiometer ramp-up time / Tec_ctr mop t_r-up
p2248[0...n]	Technology controller motorized potentiometer ramp-down time / Tec_ctrMop t_r-down
p2502[0...n]	LR encoder assignment / Encoder assignment
p2503[0...n]	LR length unit LU per 10 mm / LU per 10 mm
p2504[0...n]	LR motor/load motor distance / Mot/load motor dis
p2504[0...n]	LR motor/load motor revolutions / Mot/load motor rev
p2505[0...n]	LR motor/load motor revolutions / Mot/load motor rev
p2506[0...n]	LR length unit LU per load path / LU per load path
p2506[0...n]	LR length unit LU per load revolution / LU per load rev
p2519[0...n]	LR position actual value preprocessing config. DDS changeover / s_act config DDS
p2533[0...n]	LR position setpoint filter, time constant / s_set_filt T
p2534[0...n]	LR velocity pre-control factor / V_prectrl fact
p2534[0...n]	LR speed pre-control factor / n_prectrl fact
p2535[0...n]	LR velocity pre-control balancing filter dead time / v_prectrlFilt t_dead
p2535[0...n]	LR speed pre-control balancing filter dead time / n_prectrlFilt t_dead
p2536[0...n]	LR velocity pre-control, balancing filter PT1 / v_prectrl filt PT1
p2536[0...n]	LR speed pre-control, symmetrizing filter PT1 / n_prectrl filt PT1
p2538[0...n]	LR proportional gain / Kp
p2539[0...n]	LR integral time / Tn
p2546[0...n]	LR dynamic following error monitoring tolerance / s_delta_monit tol
p2567[0...n]	LR force pre-control mass / F_prectrl mass
p2567[0...n]	LR torque pre-control moment of inertia / M_prectrl M_inertia
p2634[0...n]	EPOS fixed stop maximum following error / Following err max
p2720[0...n]	Load gear configuration / Load gear config
p2721[0...n]	Load gear, rotary absolute gearbox, revolutions, virtual / Abs rot rev
p2722[0...n]	Load gear, position tracking tolerance window / Pos track tol
r2723[0...n]	CO: Load gear absolute value / Load gear abs_val
r2724[0...n]	CO: Load gear position difference / Load gear pos diff
p2900[0...n]	CO: Fixed value 1 [%] / Fixed value 1 [%]
p2901[0...n]	CO: Fixed value 2 [%] / Fixed value 2 [%]
p2930[0...n]	CO: Fixed value M [Nm] / Fixed value M [Nm]
p2930[0...n]	CO: Fixed value F [N] / Fixed value F [N]
p3201[0...n]	Excitation current outside the tolerance threshold value / I_exc n Tol thresh
p3202[0...n]	Excitation current outside the tolerance hysteresis / I_exc n Tol hyst
p3203[0...n]	Excitation current outside the tolerance delay time / I_exc n Tol t_del
p3204[0...n]	Flux outside the tolerance threshold value / Flux n tol thresh
p3205[0...n]	Flux outside the tolerance hysteresis / Flux n tol hyst
p3206[0...n]	Flux outside tolerance delay time / Flux n tol t_del
p3207[0...n]	Zero current signal threshold value / I_0_sig thresh
p3208[0...n]	Zero current signal hysteresis / I_0_sig hyst
p3209[0...n]	Zero current signal delay time / I_0_sig t_del
p3233[0...n]	Torque actual value filter, time constant / M_act_filt T
p3236[0...n]	Speed threshold 7 / n_thresh val 7
p3237[0...n]	Hysteresis speed 7 / n_hysteresis 7
p3238[0...n]	OFF delay n_act_motor model = n_act external / t_del n_a = n_ext
p3704[0...n]	APC filter activation / APC filter act
p3705[0...n]	APC filter type / APC filter type

p3706[0...n]	APC sub-sampling, filter 2.x / APC sub-samp. 2.x
p3707[0...n]	APC sub-sampling, filter 3.x / APC sub-samp. 3.x
p3708[0...n]	APC velocity actual value smoothing time encoder 2 / APC v_act t_sm 2
p3708[0...n]	APC speed actual value smoothing time encoder 2 / APC n_act t_sm 2
p3709[0...n]	APC velocity actual value smoothing time encoder 3 / APC v_act t_sm 3
p3709[0...n]	APC speed actual value smoothing time encoder 3 / APC n_act t_sm 3
p3711[0...n]	APC filter 1.1 denominator natural frequency / APC Filt 1.1 fn_d
p3712[0...n]	APC filter 1.1 denominator damping / APC Filt 1.1 D_d
p3713[0...n]	APC filter 1.1 numerator natural frequency / APC Filt 1 fn_n
p3714[0...n]	APC filter 1.1 numerator damping / APC Filt 1.1 D_n
p3721[0...n]	APC filter 2.1 denominator natural frequency / APC Filt 2.1 fn_d
p3722[0...n]	APC filter 2.1 denominator damping / APC Filt 2.1 D_d
p3723[0...n]	APC filter 2.1 numerator natural frequency / APC Filt 2.1 fn_n
p3724[0...n]	APC filter 2.1 numerator damping / APC Filt 2.1 D_n
p3726[0...n]	APC filter 2.2 denominator natural frequency / APC Filt 2.2 fn_d
p3727[0...n]	APC filter 2.2 denominator damping / APC Filt 2.2 D_d
p3728[0...n]	APC filter 2.2 numerator natural frequency / APC Filt 2.2 fn_n
p3729[0...n]	APC filter 2.2 numerator damping / APC Filt 2.2 D_n
p3731[0...n]	APC filter 3.1 denominator natural frequency / APC Filt 3.1 fn_d
p3732[0...n]	APC filter 3.1 denominator damping / APC Filt 3.1 D_d
p3733[0...n]	APC filter 3.1 numerator natural frequency / APC Filt 3.1 fn_n
p3734[0...n]	APC filter 3.1 numerator damping / APC Filt 3.1 D_n
p3736[0...n]	APC filter 3.2 denominator natural frequency / APC Filt 3.2 fn_d
p3737[0...n]	APC filter 3.2 denominator damping / APC Filt 3.2 D_d
p3738[0...n]	APC filter 3.2 numerator natural frequency / APC Filt 3.2 fn_n
p3739[0...n]	APC filter 3.2 numerator damping / APC Filt 3.2 D_n
p3751[0...n]	APC acceleration sensor high pass time constant / APC accel DT1 T
p3760[0...n]	APC load velocity controller 1 P gain / APC v_load ctr1 Kp
p3760[0...n]	APC load speed controller 1 P gain / APC n_load ctr1 Kp
p3761[0...n]	APC load velocity controller 1 rate time / APC v_load ctr1 Tv
p3761[0...n]	APC load speed controller 1 rate time / APC n_load ctr1 Tv
p3765[0...n]	APC load velocity controller 2 P gain / APC v_load ctr2 Kp
p3765[0...n]	APC load speed controller 2 P gain / APC n_load ctr2 Kp
p3766[0...n]	APC load velocity controller 2 rate time / APC v_load ctr2 Tv
p3766[0...n]	APC load speed controller 2 rate time / APC n_load ctr2 Tv
p3778[0...n]	APC velocity limit / APC v_limit
p3778[0...n]	APC speed limit / APC n_limit
p3779[0...n]	APC velocity limit monitoring time / APC v_limit t
p3779[0...n]	APC speed limit monitoring time / APC n_limit t
p3800[0...n]	Sync-line-drive activation / Sync act
p3801[0...n]	Sync-line-drive, drive object number / Sync DO_No
p3806[0...n]	Sync-line-drive frequency difference threshold value / Sync f_diff thresh
p3809[0...n]	Sync-line-drive phase setpoint / Sync phase setp
p3811[0...n]	Sync-line-drive frequency limiting / Sync f_lim
p3813[0...n]	Sync-line-drive phase synchronism threshold value / Sync Ph_sync thrsh
p3815[0...n]	Sync-line-drive voltage difference threshold value / Sync V_diff thresh
p3820[0...n]	Friction characteristic, value n0 / Friction n0
p3820[0...n]	Friction characteristic, value v0 / Friction v0
p3821[0...n]	Friction characteristic, value n1 / Friction n1
p3821[0...n]	Friction characteristic, value v1 / Friction v1
p3822[0...n]	Friction characteristic, value n2 / Friction n2
p3822[0...n]	Friction characteristic, value v2 / Friction v2
p3823[0...n]	Friction characteristic, value n3 / Friction n3

p3823[0...n]	Friction characteristic, value v3 / Friction v3
p3824[0...n]	Friction characteristic, value n4 / Friction n4
p3824[0...n]	Friction characteristic, value v4 / Friction v4
p3825[0...n]	Friction characteristic, value n5 / Friction n5
p3825[0...n]	Friction characteristic, value v5 / Friction v5
p3826[0...n]	Friction characteristic, value n6 / Friction n6
p3826[0...n]	Friction characteristic, value v6 / Friction v6
p3827[0...n]	Friction characteristic, value n7 / Friction n7
p3827[0...n]	Friction characteristic, value v7 / Friction v7
p3828[0...n]	Friction characteristic, value n8 / Friction n8
p3828[0...n]	Friction characteristic, value v8 / Friction v8
p3829[0...n]	Friction characteristic, value n9 / Friction n9
p3829[0...n]	Friction characteristic, value v9 / Friction v9
p3830[0...n]	Friction characteristic, value M0 / Friction M0
p3830[0...n]	Friction characteristic, value F0 / Friction F0
p3831[0...n]	Friction characteristic, value M1 / Friction M1
p3831[0...n]	Friction characteristic, value F1 / Friction F1
p3832[0...n]	Friction characteristic, value M2 / Friction M2
p3832[0...n]	Friction characteristic, value F2 / Friction F2
p3833[0...n]	Friction characteristic, value M3 / Friction M3
p3833[0...n]	Friction characteristic, value F3 / Friction F3
p3834[0...n]	Friction characteristic, value M4 / Friction M4
p3834[0...n]	Friction characteristic, value F4 / Friction F4
p3835[0...n]	Friction characteristic, value M5 / Friction M5
p3835[0...n]	Friction characteristic, value F5 / Friction F5
p3836[0...n]	Friction characteristic, value M6 / Friction M6
p3836[0...n]	Friction characteristic, value F6 / Friction F6
p3837[0...n]	Friction characteristic, value M7 / Friction M7
p3837[0...n]	Friction characteristic, value F7 / Friction F7
p3838[0...n]	Friction characteristic, value M8 / Friction M8
p3838[0...n]	Friction characteristic, value F8 / Friction F8
p3839[0...n]	Friction characteristic, value M9 / Friction M9
p3839[0...n]	Friction characteristic, value F9 / Friction F9
p3846[0...n]	Friction characteristic record ramp-up/ramp-down time / Frict rec t_RFG
p3847[0...n]	Friction characteristic record warm-up time / Frict rec t_warm
r3925[0...n]	Identification final display / Ident final_disp
r3927[0...n]	Motor data identification induction motor data determined / MotID ASM dat det
r3928[0...n]	Motor data identification synchronous motor data determined / MotID PEM dat det
r3998[0...n]	First drive commissioning / First drv_comm
p7035[0...n]	Par_circuit circulating current control operating mode / Circ_I mode
p7036[0...n]	Par_circuit circulating current control proportional gain / Circ_I Kp
p7037[0...n]	Par_circuit circulating current control integral time / I_circ Tn
p7038[0...n]	Par_circuit circulating current control limit / I_circ limit

1.3.3 Parameters for encoder data sets (EDS)

Note:

References: /FH1/ SINAMICS S120 Function Manual Drive Functions
Section "Data Sets"

The following list contains the encoder-data-set-dependent parameters.

Product: SINAMICS S120/S150, Version: 4301400, Language: eng, Type: EDS

p0141[0...n]	Encoder interface (Sensor Module) component number / Enc_interf comp_no
p0142[0...n]	Encoder component number / Encoder comp_no
p0144[0...n]	Sensor Module detection via LED / SM detection LED
p0145[0...n]	Activate/deactivate encoder interface / Enc_intf act/deact
r0146[0...n]	Encoder interface active/inactive / Enc_intf act/inact
r0147[0...n]	Sensor Module EPROM data version / SM EEPROM version
r0148[0...n]	Sensor Module firmware version / SM FW version
p0400[0...n]	Encoder type selection / Enc_typ sel
p0401[0...n]	Encoder type, OEM selection / Enc type OEM sel
p0402[0...n]	Gearbox type selection / Gearbox type sel
p0404[0...n]	Encoder configuration effective / Enc_config eff
p0405[0...n]	Square-wave encoder track A/B / Sq-wave enc A/B
p0407[0...n]	Linear encoder grid division / Enc grid div
p0408[0...n]	Rotary encoder pulse No. / Rot enc pulse No.
p0410[0...n]	Encoder inversion actual value / Enc inv act value
p0411[0...n]	Measuring gear, configuration / Meas gear config
p0412[0...n]	Measuring gear, absolute encoder, rotary, revolutions, virtual / Abs rot rev
p0413[0...n]	Measuring gear, position tracking tolerance window / Pos track window
p0414[0...n]	Redundant coarse position value relevant bits (identified) / Relevant bits
p0415[0...n]	Gx_XIST1 Coarse position safe most significant bit (identified) / Gx_XIST1 safe MSB
p0418[0...n]	Fine resolution Gx_XIST1 (in bits) / Enc fine Gx_XIST1
p0419[0...n]	Fine resolution absolute value Gx_XIST2 (in bits) / Enc fine Gx_XIST2
p0420[0...n]	Encoder connection / Enc_connection
p0421[0...n]	Absolute encoder rotary multiturn resolution / Enc abs multiturn
p0422[0...n]	Absolute encoder linear measuring step resolution / Enc abs meas step
p0423[0...n]	Absolute encoder rotary singleturn resolution / Enc abs singleturn
p0424[0...n]	Encoder, linear zero mark distance / Enc lin ZM_dist
p0425[0...n]	Encoder, rotary zero mark distance / Enc rot dist ZM
p0426[0...n]	Encoder zero mark differential distance / Enc diff distance
p0427[0...n]	Encoder SSI baud rate / Enc SSI baud rate
p0428[0...n]	Encoder SSI monoflop time / Enc SSI t_monoflop
p0429[0...n]	Encoder SSI configuration / Enc SSI config
p0430[0...n]	Sensor Module configuration / SM config
p0431[0...n]	Angular commutation offset / Ang_com offset
p0432[0...n]	Gearbox factor, encoder revolutions / Grbx_fact enc_rev
p0433[0...n]	Gearbox factor, motor/load revolutions / Grbx_fact mot_rev
p0434[0...n]	Encoder SSI error bit / Enc SSI error bit
p0435[0...n]	Encoder SSI alarm bit / Enc SSI alarm bit
p0436[0...n]	Encoder SSI parity bit / Enc SSI parity bit
p0437[0...n]	Sensor Module configuration extended / SM config ext
p0438[0...n]	Squarewave encoder filter time / Enc t_filt
p0439[0...n]	Encoder ramp-up time / Enc ramp-up time

p0440[0...n]	Copy encoder serial number / Copy enc ser_no
p0441[0...n]	Encoder commissioning serial number part 1 / Enc comm ser_no 1
p0442[0...n]	Encoder commissioning serial number part 2 / Enc comm ser_no 2
p0443[0...n]	Encoder commissioning serial number part 3 / Enc comm ser_no 3
p0444[0...n]	Encoder commissioning serial number part 4 / Enc comm ser_no 4
p0445[0...n]	Encoder commissioning serial number part 5 / Enc comm ser_no 5
p0446[0...n]	Encoder SSI number of bits before the absolute value / Enc SSI bit before
p0447[0...n]	Encoder SSI number of bits absolute value / Enc SSI bit val
p0448[0...n]	Encoder SSI number of bits after the absolute value / Enc SSI bit after
p0449[0...n]	Encoder SSI number of bits, filler bits / Enc SSI fill bits
p0453[0...n]	Measuring time pulse encoder evaluation zero speed / t_meas enc_eva n_0
p0493[0...n]	Zero mark selection, input terminal / ZM_sel inp_term
p2507[0...n]	LR absolute encoder adjustment status / Abs_enc_adj stat
p2525[0...n]	CO: LR encoder adjustment, offset / Enc_adj offset
p4600[0...n]	Motor temperature sensor 1 sensor type / Temp_sens 1 type
p4601[0...n]	Motor temperature sensor 2 sensor type / Temp_sens 2 type
p4602[0...n]	Motor temperature sensor 3 sensor type / Temp_sens 3 type
p4603[0...n]	Motor temperature sensor 4 sensor type / Temp_sens 4 type
p4670[0...n]	Analog sensor configuration / Ana_sens config
p4671[0...n]	Analog sensor input / Ana_sens inp
p4672[0...n]	Analog sensor channel A voltage at actual value zero / Ana_sens A U at 0
p4673[0...n]	Analog sensor channel A voltage per encoder period / Ana_sens A U/per
p4674[0...n]	Analog sensor channel B voltage at actual value zero / Ana_sens B U at 0
p4675[0...n]	Analog sensor channel B voltage per encoder period / Ana_sens B U/per
p4676[0...n]	Analog sensor range limit threshold / Ana_sens lim thr
p4677[0...n]	Analog sensor LVDT configuration / Ana_sens LVDT conf
p4680[0...n]	Zero mark monitoring tolerance permissible / ZM_monit tol perm
p4681[0...n]	Zero mark monitoring, tolerance window limit 1 positive / ZM tol lim 1 pos
p4682[0...n]	Zero mark monitoring, tolerance window limit 1 negative / ZM tol lim 1 neg
p4683[0...n]	Zero mark monitoring tolerance window alarm threshold positive / ZM tol A_thr pos
p4684[0...n]	Zero mark monitoring tolerance window alarm threshold negative / ZM tol A_thr neg
p4685[0...n]	Speed actual value mean value generation / n_act mean val
p4686[0...n]	Zero mark minimum length / ZM min length

1.3.4 Parameters for motor data sets (MDS)

Note:

References: /FH1/ SINAMICS S120 Function Manual Drive Functions
Section "Data Sets"

The following list contains the motor-data-set-dependent parameters.

Product: SINAMICS S120/S150, Version: 4301400, Language: eng, Type: MDS

p0131[0...n]	Motor component number / Mot comp_no
p0300[0...n]	Motor type selection / Mot type sel
p0301[0...n]	Motor code number selection / Mot code No. sel
r0302[0...n]	Motor code number of motor with DRIVE-CLiQ / Motor code Mot DLQ
r0303[0...n]	Motor status word of motor with DRIVE-CLiQ / Mot ZSW Mot DLQ
p0304[0...n]	Rated motor voltage / Mot V_rated
p0305[0...n]	Rated motor current / Mot I_rated

p0306[0...n]	Number of motors connected in parallel / Motor qty
p0307[0...n]	Rated motor power / Mot P _{rated}
p0308[0...n]	Rated motor power factor / Mot cos_phi _{rated}
p0309[0...n]	Rated motor efficiency / Mot eta _{rated}
p0310[0...n]	Rated motor frequency / Mot f _{rated}
p0311[0...n]	Rated motor speed / Mot n _{rated}
p0311[0...n]	Rated motor velocity / Mot v _{rated}
p0312[0...n]	Rated motor torque / Mot M _{rated}
p0312[0...n]	Rated motor force / Mot F _{rated}
r0313[0...n]	Motor pole pair number, actual (or calculated) / Mot PolePairNo act
p0314[0...n]	Motor pole pair number / Mot pole pair No.
p0315[0...n]	Motor pole pair width / MotPolePair width
p0316[0...n]	Motor torque constant / Mot kT
p0316[0...n]	Motor force constant / Mot kT
p0317[0...n]	Motor voltage constant / Mot kE
p0318[0...n]	Motor stall current / Mot I _{standstill}
p0319[0...n]	Motor stall torque / Mot M _{standstill}
p0319[0...n]	Motor stall force / Mot F _{standstill}
p0320[0...n]	Motor rated magnetizing current/short-circuit current / Mot I _{mag_{rated}}
p0322[0...n]	Maximum motor speed / Mot n _{max}
p0322[0...n]	Motor maximum velocity / Mot v _{max}
p0323[0...n]	Maximum motor current / Mot I _{max}
p0324[0...n]	Winding maximum speed / Winding n _{max}
p0324[0...n]	Winding maximum velocity / Winding v _{max}
p0325[0...n]	Motor pole position identification current, 1st phase / Mot PolID I 1st ph
p0326[0...n]	Motor stall torque correction factor / Mot M _{stall_corr}
p0326[0...n]	Motor stall force correction factor / Mot F _{stall_corr}
p0327[0...n]	Optimum motor load angle / Mot phi _{load opt}
p0328[0...n]	Motor reluctance torque constant / Mot kT _{reluctance}
p0328[0...n]	Motor reluctance force constant / Mot kT _{reluctance}
p0329[0...n]	Motor pole position identification current / Mot PolID current
r0330[0...n]	Rated motor slip / Mot slip _{rated}
r0331[0...n]	Actual motor magnetizing current/short-circuit current / Mot I _{mag_{rtd act}}
r0332[0...n]	Rated motor power factor / Mot cos_phi _{rated}
r0333[0...n]	Rated motor torque / Mot M _{rated}
r0334[0...n]	Actual motor-torque constant / Mot kT act
r0334[0...n]	Actual motor force constant / Mot kT act
p0335[0...n]	Motor cooling type / Motor cooling type
r0336[0...n]	Actual rated motor frequency / Mot f _{rated act}
r0337[0...n]	Rated motor EMF / Mot EMF _{rated}
p0338[0...n]	Motor limit current / Mot I _{limit}
r0339[0...n]	Rated motor voltage / Mot V _{rated}
p0341[0...n]	Motor moment of inertia / Mot M _{mom of inert}
p0341[0...n]	Motor weight / Mot weight
p0342[0...n]	Ratio between the total and motor moment of inertia / Mot MomInert Ratio
p0342[0...n]	Ratio between the total and motor force of inertia / Mot MomInert Ratio
r0343[0...n]	Identified motor rated current / Mot I _{rated ident}
p0344[0...n]	Motor weight (for the thermal motor model) / Mot weight th mod
r0345[0...n]	Nominal motor starting time / Mot t _{start_{rated}}
p0346[0...n]	Motor excitation build-up time / Mot t _{excitation}
p0347[0...n]	Motor de-excitation time / Mot t _{de-excitat.}
p0348[0...n]	Speed at the start of field weakening V _{dc} = 600 V / Mot n _{field weaken}
p0348[0...n]	Velocity at the start of field weakening V _{dc} = 600 V / Mot v _{field weaken}

p0350[0...n]	Motor stator resistance, cold / Mot R_stator cold
p0352[0...n]	Cable resistance / Mot R_cable cold
p0353[0...n]	Motor series inductance / Mot L_series
p0354[0...n]	Motor rotor resistance cold / damping resistance d axis / Mot R_r cold / RDd
p0355[0...n]	Motor damping resistance, q axis / Mot R_damp q
p0356[0...n]	Motor stator leakage inductance / Mot L_stator leak.
p0357[0...n]	Motor stator inductance, d axis / Mot L_stator d
p0358[0...n]	Motor rotor leakage inductance / damping inductance, d axis / Mot L_r leak / LDd
p0359[0...n]	Motor damping inductance, q axis / Mot L_damp q
p0360[0...n]	Motor magnetizing inductance/magn. inductance, d axis saturated / Mot Lh/Lh d sat
p0361[0...n]	Motor magnetizing inductance q axis, saturated / Mot L_magn q sat
p0362[0...n]	Saturation characteristic flux 1 / Mot saturat.flux 1
p0363[0...n]	Saturation characteristic flux 2 / Mot saturat.flux 2
p0364[0...n]	Saturation characteristic flux 3 / Mot saturat.flux 3
p0365[0...n]	Saturation characteristic flux 4 / Mot saturat.flux 4
p0366[0...n]	Saturation characteristic I_mag 1 / Mot sat. I_mag 1
p0367[0...n]	Saturation characteristic I_mag 2 / Mot sat. I_mag 2
p0368[0...n]	Saturation characteristic I_mag 3 / Mot sat. I_mag 3
p0369[0...n]	Saturation characteristic I_mag 4 / Mot sat. I_mag 4
r0370[0...n]	Motor stator resistance, cold / Mot R_stator cold
r0372[0...n]	Cable resistance / Mot R_cable
r0373[0...n]	Motor rated stator resistance / Mot R_stator rated
r0374[0...n]	Motor rotor resistance cold / damping resistance d axis / Mot R_r cold / RDd
r0375[0...n]	Motor damping resistance, q axis / Mot R_damp q
r0376[0...n]	Rated motor rotor resistance / Mot R_rotor rated
r0377[0...n]	Motor leakage inductance, total / Mot L_leak total
r0378[0...n]	Motor stator inductance, d axis / Mot L_stator_d
r0380[0...n]	Motor damping inductance, d axis / Mot L_damping_d
r0381[0...n]	Motor damping inductance, q axis / Mot L_damping_q
r0382[0...n]	Motor magnetizing inductance transformed / Lh d axis saturated / Mot L_m tr/Lhd sat
r0383[0...n]	Motor magnetizing inductance q axis, saturated / Mot L_magn q sat
r0384[0...n]	Motor rotor time constant / damping time constant d axis / Mot T_rotor/T_Dd
r0385[0...n]	Motor damping time constant, q axis / Mot T_Dq
r0386[0...n]	Motor stator leakage time constant / Mot T_stator leak
r0387[0...n]	Motor stator leakage time constant, q axis / Mot T_Sleak /T_Sq
p0389[0...n]	Excitation rated no-load current / Exc I_noload Rated
p0390[0...n]	Rated excitation current / Exc I_Rated
p0391[0...n]	Current controller adaptation, starting point KP / I_adapt pt KP
p0392[0...n]	Current controller adaptation, starting point KP adapted / I_adapt pt KP adap
p0393[0...n]	Current controller adaptation p gain adaptation / I_adapt Kp adapt
p0393[0...n]	Current controller adaptation P gain scaling / I_adapt Kp scal
r0395[0...n]	Actual stator resistance / R_stator act
r0396[0...n]	Actual rotor resistance / R_rotor act
p0398[0...n]	Angle magnet. decoupling (cross saturation) coeff 1 / Magn decoupl C1
p0399[0...n]	Angle magnet. decoupling (cross saturation) coeff 3 / Magn decoupl C3
p0530[0...n]	Bearing type selection / Bearing type sel
p0531[0...n]	Bearing code number selection / Bear. code num sel
p0532[0...n]	Bearing maximum speed / Bearing n_max
p0532[0...n]	Bearing maximum velocity / Bearing v_max
p0600[0...n]	Motor temperature sensor for monitoring / Mot temp_sensor
p0601[0...n]	Motor temperature sensor type / Mot_temp_sens type
p0604[0...n]	Motor temperature alarm threshold / Mot_temp al thr
p0605[0...n]	Motor temperature fault threshold / Mot_temp flt thr

p0606[0...n]	Motor temperature timer / Mot_temp timer
p0607[0...n]	Temperature sensor fault timer / Sensor fault time
p0610[0...n]	Motor overtemperature response / Mot temp response
p0611[0...n]	I2t motor model thermal time constant / I2t mot_mod T
p0612[0...n]	Thermal motor model configuration / Therm Mot_mod conf
p0615[0...n]	I2t motor model fault threshold / I2t mot_mod thresh
p0616[0...n]	Motor overtemperature alarm threshold 1 / Mot temp alarm 1
p0620[0...n]	Thermal adaptation, stator and rotor resistance / Mot therm_adapt R
p0621[0...n]	Identification stator resistance after restart / Rst_ident Restart
p0622[0...n]	Motor excitation time for Rs_ident after powering up again / t_excit Rs_id
p0624[0...n]	Motor Temperature Offset PT100 / Mot T_offset PT100
p0625[0...n]	Motor ambient temperature / Mot T_ambient
p0626[0...n]	Motor overtemperature, stator core / Mot T_over core
p0627[0...n]	Motor overtemperature, stator winding / Mot T_over stator
p0628[0...n]	Motor overtemperature rotor winding / Mot T_over rotor
r0630[0...n]	Motor temperature model ambient temperature / MotTMod T_amb.
r0631[0...n]	Motor temperature model, stator core temperature / MotTMod T_core
r0632[0...n]	Motor temperature model, stator winding temperature / MotTMod T_copper
r0633[0...n]	Motor temperature model, rotor temperature / MotTMod T_rotor
p0634[0...n]	Q flux flux constant unsaturated / PSIQ KPSI UNSAT
p0635[0...n]	Q flux quadrature axis current constant unsaturated / PSIQ KIQ UNSAT
p0636[0...n]	Q flux direct axis current constant unsaturated / PSIQ KID UNSAT
p0637[0...n]	Q flux flux gradient saturated / PSIQ Grad SAT
p0643[0...n]	Overvoltage protection for synchronous motors / Overvolt_protect
p0645[0...n]	Motor kT characteristic kT1 / Mot kT char kT1
p0646[0...n]	Motor kT characteristic kT3 / Mot kT char kT3
p0647[0...n]	Motor kT characteristic kT5 / Mot kT char kT5
p0648[0...n]	Motor kT characteristic kT7 / Mot kT char kT7
p0650[0...n]	Actual motor operating hours / Mot t_oper act
p0651[0...n]	Motor operating hours maintenance interval / Mot t_op maint
p0652[0...n]	Motor stator resistance, scaling / Mot R_stator scal
p0653[0...n]	Motor stator leakage inductance, scaling / Mot L_S_leak scal
p0655[0...n]	Motor magnetizing inductance, d axis saturated scaling / Mot L_m d sat scal
p0656[0...n]	Motor magnetizing inductance, q axis, saturated scaling / Mot L_m q sat scal
p0657[0...n]	Motor damping inductance, d axis scaling / Mot L_damp d scal
p0658[0...n]	Motor damping inductance, q axis scaling / Mot L_damp q scal
p0659[0...n]	Motor damping resistance, d axis scaling / Mot R_damp d scal
p0660[0...n]	Motor damping resistance, q axis scaling / Mot R_damp q scal
p0826[0...n]	Motor changeover, motor number / Mot_chng mot No.
p0827[0...n]	Motor changeover status word bit number / Mot_chg ZSW bitNo.
p1231[0...n]	Armature short-circuit / DC brake configuration / ASC config
p1232[0...n]	DC braking, braking current / DCBRK I_brake
p1233[0...n]	DC braking time / DCBRK time
p1234[0...n]	Speed at the start of DC braking / DCBRK n_start
p1234[0...n]	DC braking, starting velocity / DCBRK v_start
p1236[0...n]	Ext. armature short-cct., contactor feedback signal monit. time / ASC ext t_monit
p1237[0...n]	External armature short-circuit, waiting time when opening / ASC ext t_wait
p1909[0...n]	Motor data identification control word / MotID STW
p1958[0...n]	Rotating measurement ramp-up/ramp-down time / Rot meas t_r up/dn
p1958[0...n]	Moving measurement ramp-up/ramp-down time / Mov meas t_r up/dn
p1959[0...n]	Rotating measurement configuration / Rot meas config
p1959[0...n]	Moving measurement configuration / Mov meas config
p1980[0...n]	Pole position identification technique / PolID technique

p1981[0...n]	Pole position identification maximum distance / PolID distance max
p1982[0...n]	Pole position identification selection / PolID selection
p1991[0...n]	Motor changeover, angular commutation correction / Ang_com corr
p1993[0...n]	Pole position identification current, motion-based / PolID I mot_bas
p1994[0...n]	Pole position identification rise time motion-based / PolID T mot_bas
p1995[0...n]	Pole position identification gain, motion-based / PolID kp mot_bas
p1996[0...n]	Pole position identification, integral time motion-based / PolID Tn mot_bas
p1997[0...n]	Pole position identification, smoothing time motion-based / PolID t_sm mot_bas
p1999[0...n]	Ang. commutation offset calibr. and pole position ID - scaling / ComOffsCalib scal
p3049[0...n]	MotId Speed at start of field weakening identified / ident
p3049[0...n]	MotId Speed at start of field weakening identified / v_Fieldweak ident
p3050[0...n]	MotId stator resistance identified / R_stator ident
p3054[0...n]	MotId rotor resistance identified / R_rotor ident
p3056[0...n]	MotId stator leakage inductance identified / L_stator leak
p3058[0...n]	MotId rotor leakage inductance identified / L_rotor leak
p3060[0...n]	MotId magnetizing inductance identified / MotId Lh ident
p4610[0...n]	Motor temperature sensor 1 sensor type MDS / Temp sens1 typ MDS
p4611[0...n]	Motor temperature sensor 2 sensor type MDS / Temp sens2 typ MDS
p4612[0...n]	Motor temperature sensor 3 sensor type MDS / Temp sens3 typ MDS
p4613[0...n]	Motor temperature sensor 4 sensor type MDS / Temp sens4 typ MDS

1.3.5 Parameters for power unit data sets (PDS)

Note:

References: /FH1/ SINAMICS S120 Function Manual Drive Functions
Section "Data Sets"

The following list contains the parameters that are dependent on the power unit data sets.

Product: SINAMICS S120/S150, Version: 4301400, Language: eng, Type: PDS

p0121[0...n]	Power unit component number / PU comp_no
p0124[0...n]	Power unit detection via LED / PU detection LED
p0125[0...n]	Activate/deactivate power unit components / PU_comp act/deact
r0126[0...n]	Power unit components active/inactive / PU comp act/inact
r0127[0...n]	Power unit version EPROM data / PU EPROM version
r0128[0...n]	Power unit, firmware version / PU FW version
r0200[0...n]	Power unit code number actual / PU code no. act
p0201[0...n]	Power unit code number / PU code no
r0203[0...n]	Actual power unit type / PU actual type
r0204[0...n]	Power unit hardware properties / PU HW property
p0251[0...n]	Operating hours counter power unit fan / PU fan t_oper
p0254[0...n]	Power unit internal fan operating hours counter / PU int fan t_oper
p0895[0...n]	BI: Activate/deactivate power unit components / PU_comp act/deact
p3469[0...n]	Latch delay time correction, zero crossover detection / t_latch corr PLL
p3901[0...n]	Power unit EEPROM Vdc offset calibration / PU EEPROM Vdc_offs
p7001[0...n]	Par_circuit power units enable / PU enable
r7002[0...n]	Par_circuit status power units / Status PU
r7020[0...n]	CO: Par_circuit deviation current in phase U / Phase U curr dev
r7021[0...n]	CO: Par_circuit deviation current in phase V / Phase V curr dev
r7022[0...n]	CO: Par_circuit deviation current in phase W / Phase W curr dev

r7030[0...n]	CO: Par_circuit DC link voltage deviation / Vdc deviation
p7040[0...n]	Par_circuit correction valve lockout time phase U / Comp t_lockout U
p7042[0...n]	Par_circuit correction valve lockout time phase V / Comp t_lockout V
p7044[0...n]	Par_circuit correction valve lockout time phase W / Comp t_lockout W
r7050[0...n]	Par_circuit circulating current phase U / Circ_I_phase U
r7051[0...n]	Par_circuit circulating current phase V / Circ_I_phase V
r7052[0...n]	Par_circuit circulating current phase W / Circ_I_phase W
r7200[0...n]	Par_circuit power unit overload I2T / PU overload I2T
r7201[0...n]	CO: Par_circuit power unit temperatures max. inverter / PU temp max inv
r7202[0...n]	Par_circuit power unit temperatures max. depletion layer / PU TempMaxDepLayer
r7203[0...n]	Par_circuit power unit temperatures max. rectifier / PU temp max rect
r7204[0...n]	Par_circuit power unit temperatures air intake / PU temp air intake
r7205[0...n]	Par_circuit power unit temperatures electronics / PU temp electr
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Function diagrams

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2.2 Explanations for the function diagrams

Function diagrams

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Explanations for the function diagrams - Explanation of the symbols (Part 1)						27.10.08 V04.03.01		SINAMICS																																															

Fig. 2-1

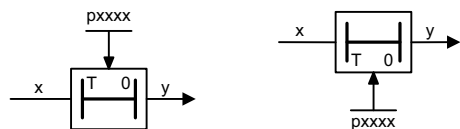
1020 – Explanations of the symbols (Part 1)

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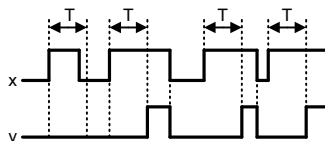
Fig. 2-2 1021 – Explanations of the symbols (Part 2)

1022 – Explanations of the symbols (Part 3)

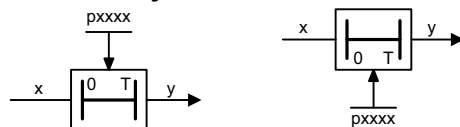
Switch-on delay



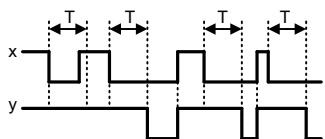
The digital signal x must have the value "1" without any interruption during the time T before output y changes to "1".



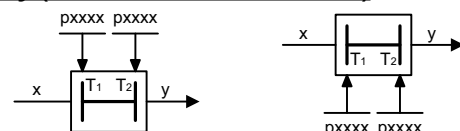
Switch-off delay



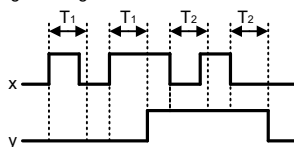
The digital signal x must have the value "0" without interruption during the time T before output y changes to "0".



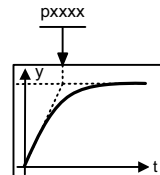
Delay (switch-on and switch-off)



The digital signal x must have the value "1" without interruption during time T_1 or must have the value "0" during time T_2 before output y changes its signal state.



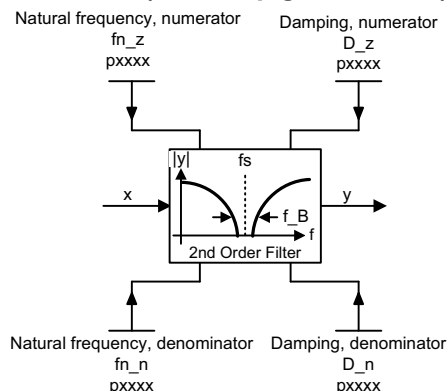
PT1 element



Delay element, first order.

ρ_{xxx} = time constant

2nd-order filter (bandstop/general filter)



Used as bandstop filter - center frequency f_s :

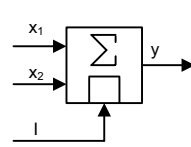
- bandwidth f_B :

$$\begin{aligned} f_{n_z} &= f_s \\ f_{n_n} &= f_s \\ D_{z} &= 0 \\ D_{n} &= \frac{f_B}{2 \cdot f_s} \end{aligned}$$

Transfer function when used as general filter

$$H(s) = \frac{\left(\frac{s}{2\pi f_{n_z}}\right)^2 + \frac{2 \cdot D_z}{2\pi f_{n_z}} \cdot s + 1}{\left(\frac{s}{2\pi f_{n_n}}\right)^2 + \frac{2 \cdot D_n}{2\pi f_{n_n}} \cdot s + 1}$$

Analog adder can be activated

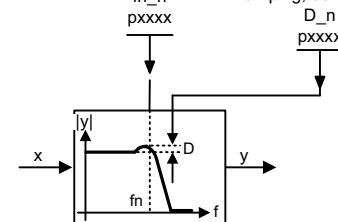


The following applies to $l = 1$ signal: $y = x_1 + x_2$

The following applies to $l = 0$ signal: $y = x_1$

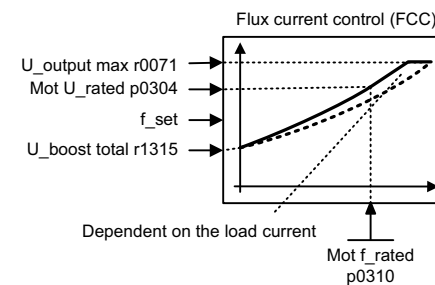
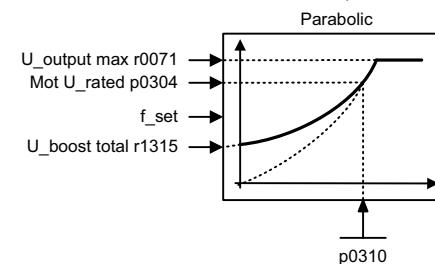
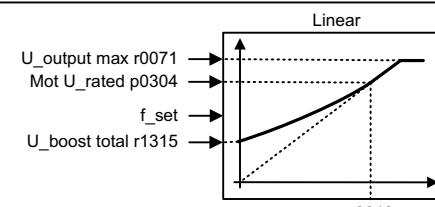
PT2 low pass

S Natural frequency, denominator
 f_n Damping, denominator



Transfer function

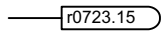
$$H(s) = \frac{1}{\left(\frac{s}{2\pi f_{n_n}}\right)^2 + \frac{2 \cdot D_{-n}}{2\pi f_{n_n}} \cdot s + 1}$$



1	2	3	4	5	6	7	8
DO: All objects					fp_1022_51_eng.vsd	Function diagram	- 1022 -
Explanations for the function diagrams - Explanation of the symbols (Part 3)					19.01.09 V04.03.01	SINAMICS	

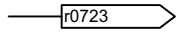
Handling BICO technology

Binector:



Binectors are binary signals that can be freely interconnected (BO = Binector Output). They represent a bit of a "BO:" display parameter (e.g. bit 15 from r0723).

Connector:



Connectors are "analog signals" that can be freely interconnected (e.g. percentage variables, speeds or torques). Connectors are also "CO:" display parameters (CO = Connector Output).

Parameterization:

At the signal destination, the required binector or connector is selected using appropriate parameters:

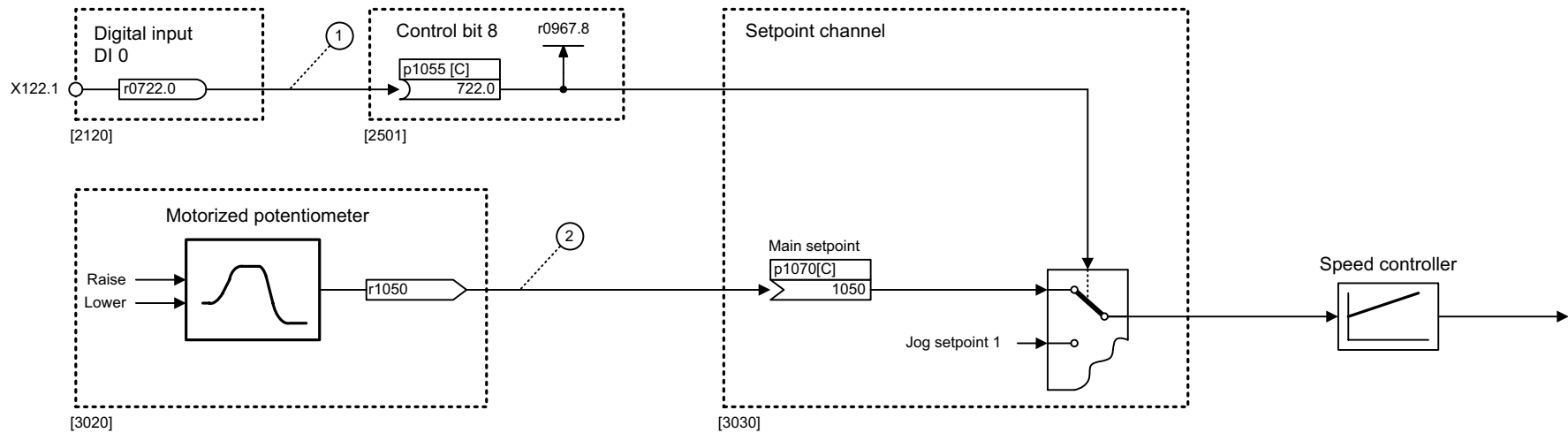
"BI:" parameter for binectors (BI = Binector Input)

or

"CI:" parameter for connectors (CI = Connector Input)

Example:

The main setpoint for the speed controller (CI: p1070) should be received from the output of the motorized potentiometer (CO: r1050) and the "jog" command (BI: p1055) from digital input DI 0 (BO: r0722.0, X122.1 terminal) on the CU320.



Parameterizing steps:

- ① p1055[0] = 722.0 Terminal X122.1 acts as "Jog bit 0".
- ② p1070[0] = 1050 The output of the motorized potentiometer acts as main setpoint for the speed controller.

1	2	3	4	5	6	7	8
DO: All objects					fp_1030_51_eng.vsd	Function diagram	- 1030 -
Explanations for the function diagrams - Handling BICO technology					19.01.09 V04.03.01	SINAMICS	

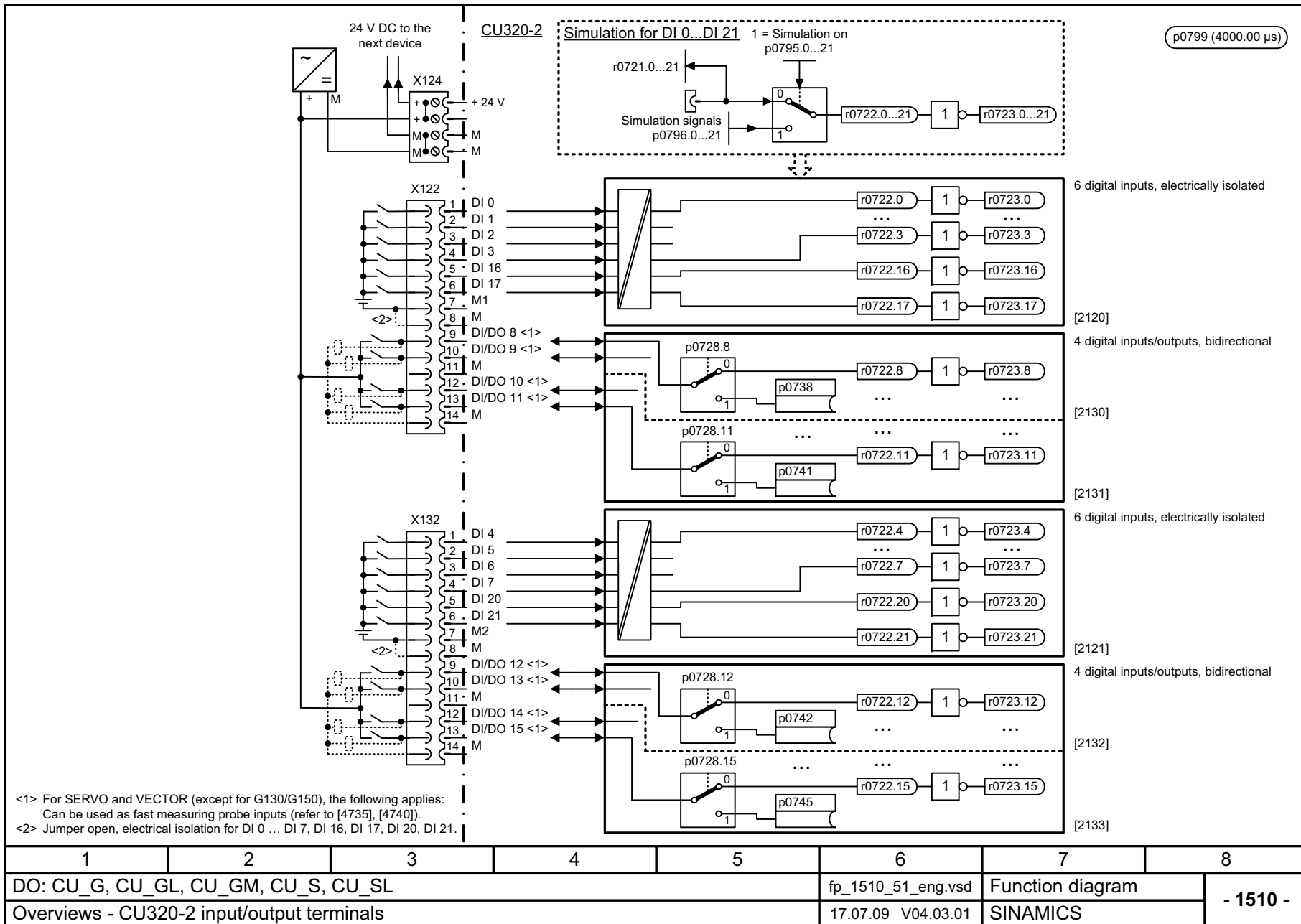
Fig. 2-4

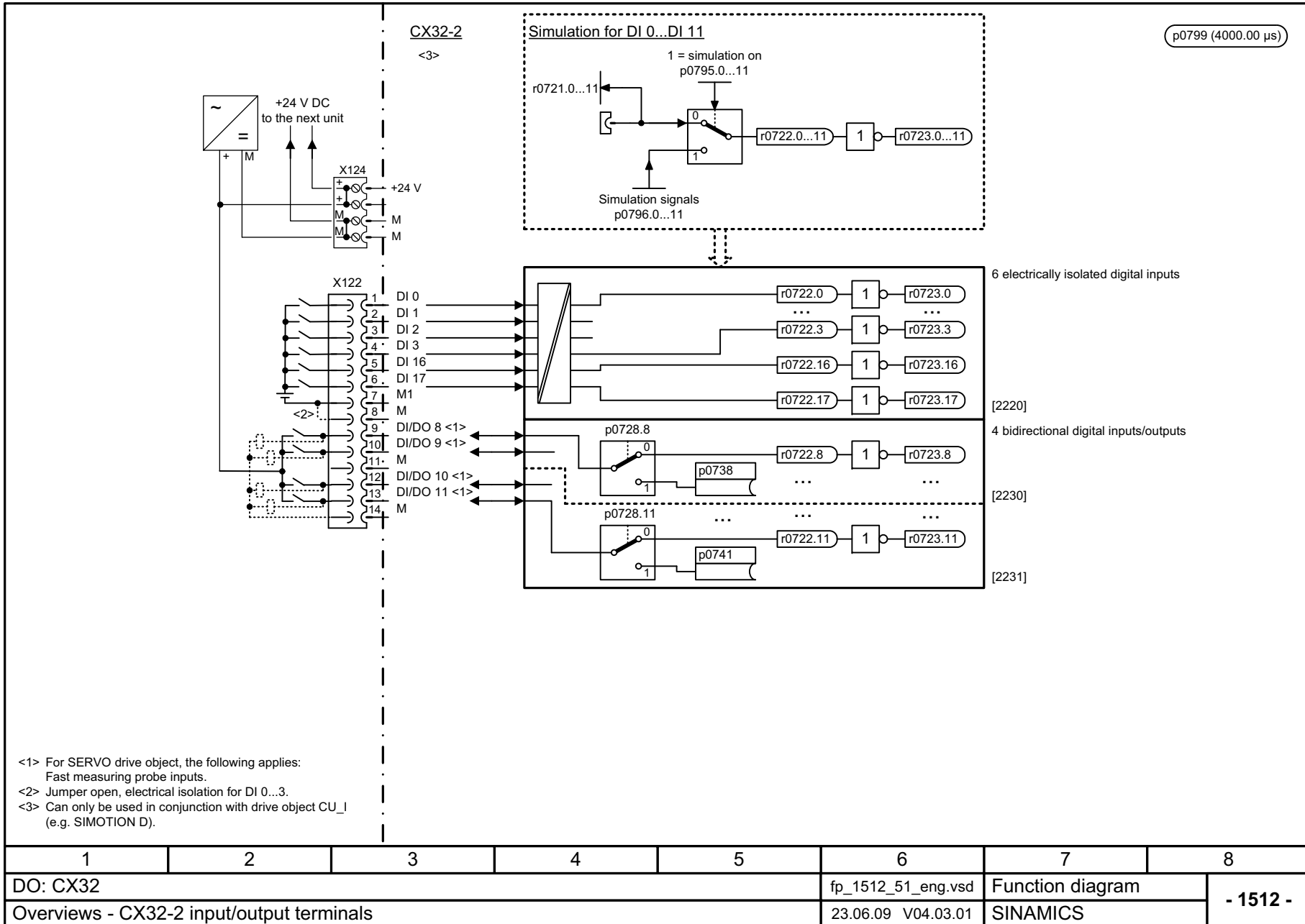
1030 – Handling BICO technology

2.3 Overviews

Function diagrams

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1840 – Terminal Module 31 (TM31)	2-1411
1842 – Terminal Module 41 (TM41)	2-1412
1850 – Terminal Module 54F (TM54F)	2-1413





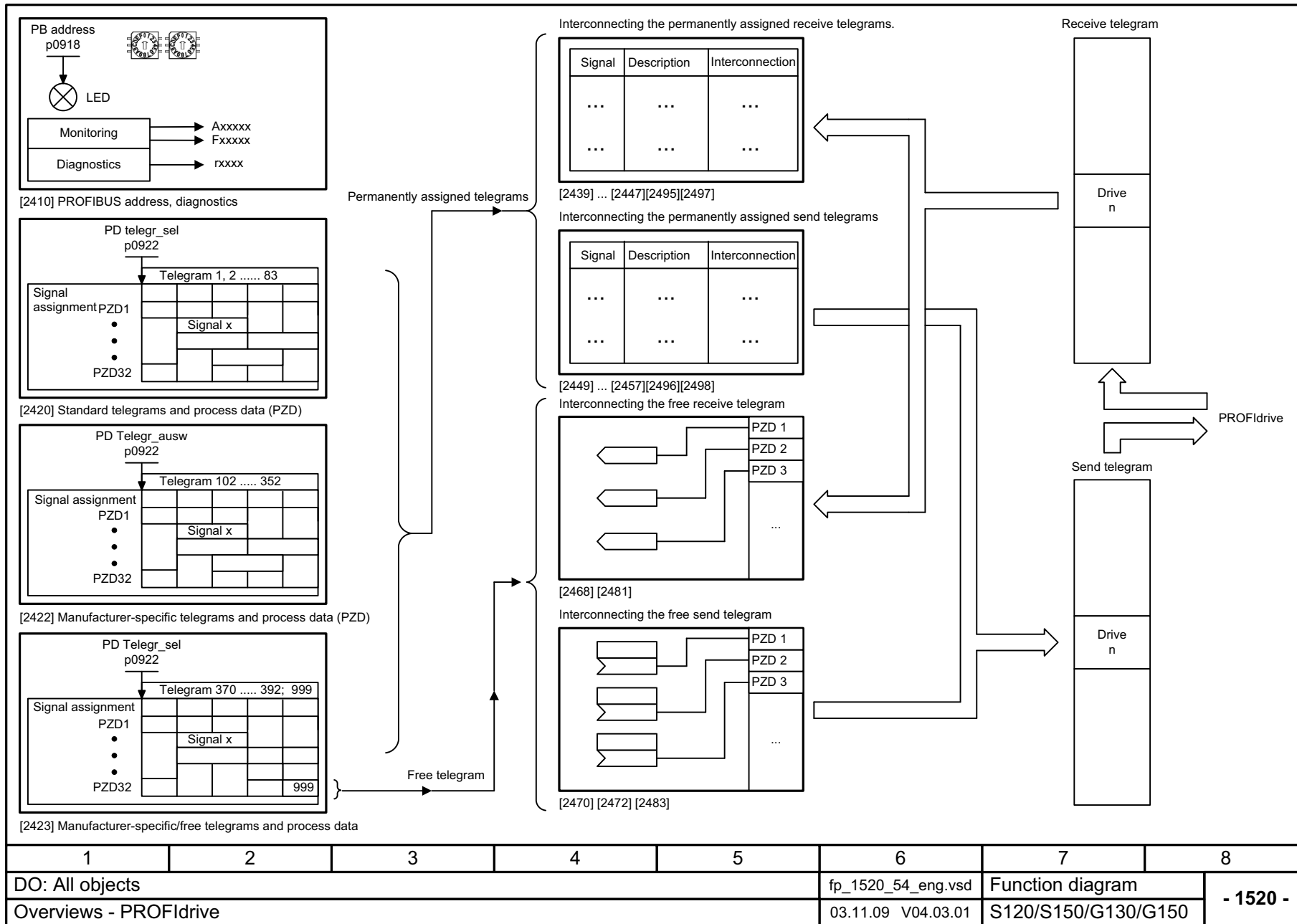


Fig. 2-7

1520 – PROFIdrive

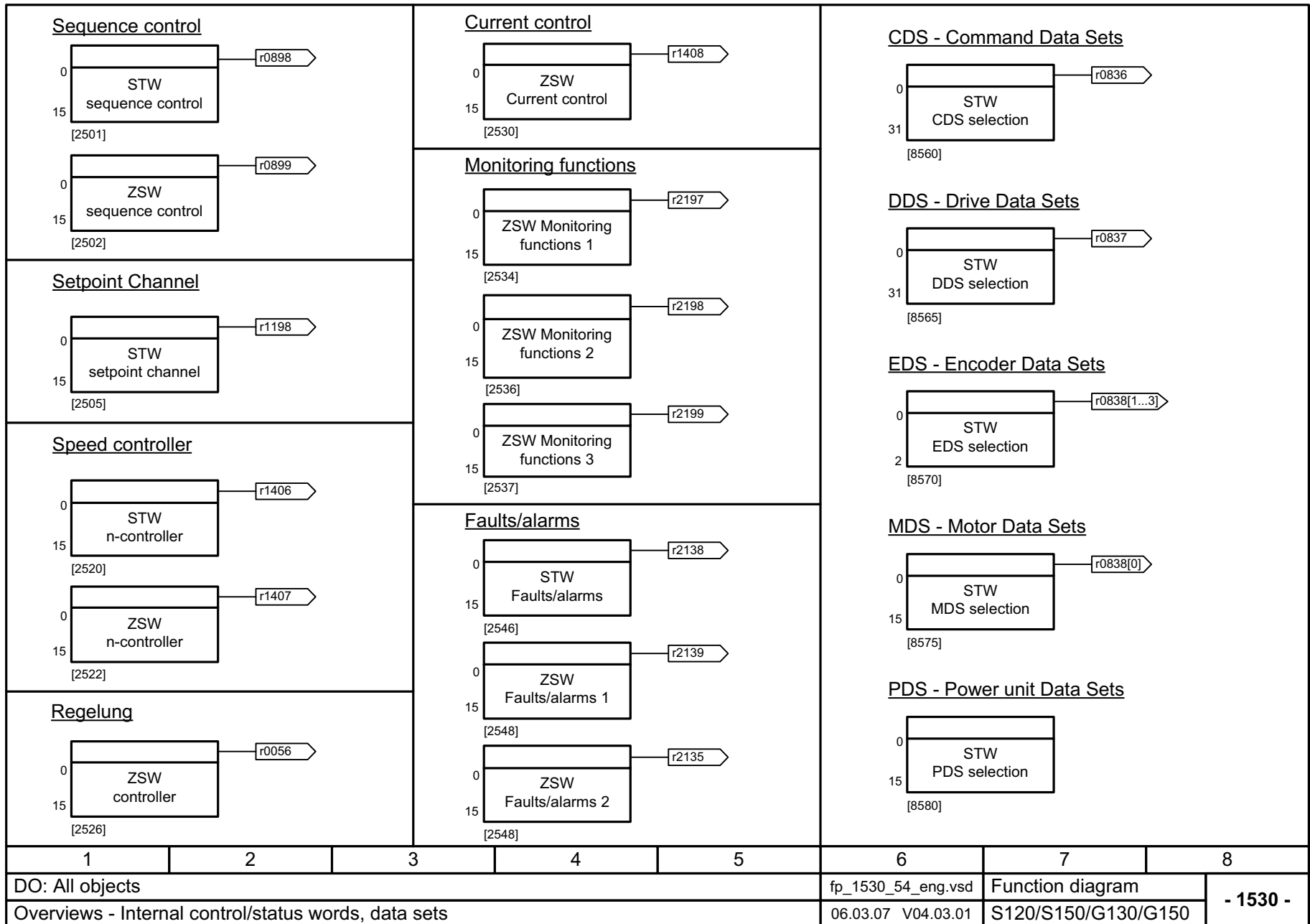
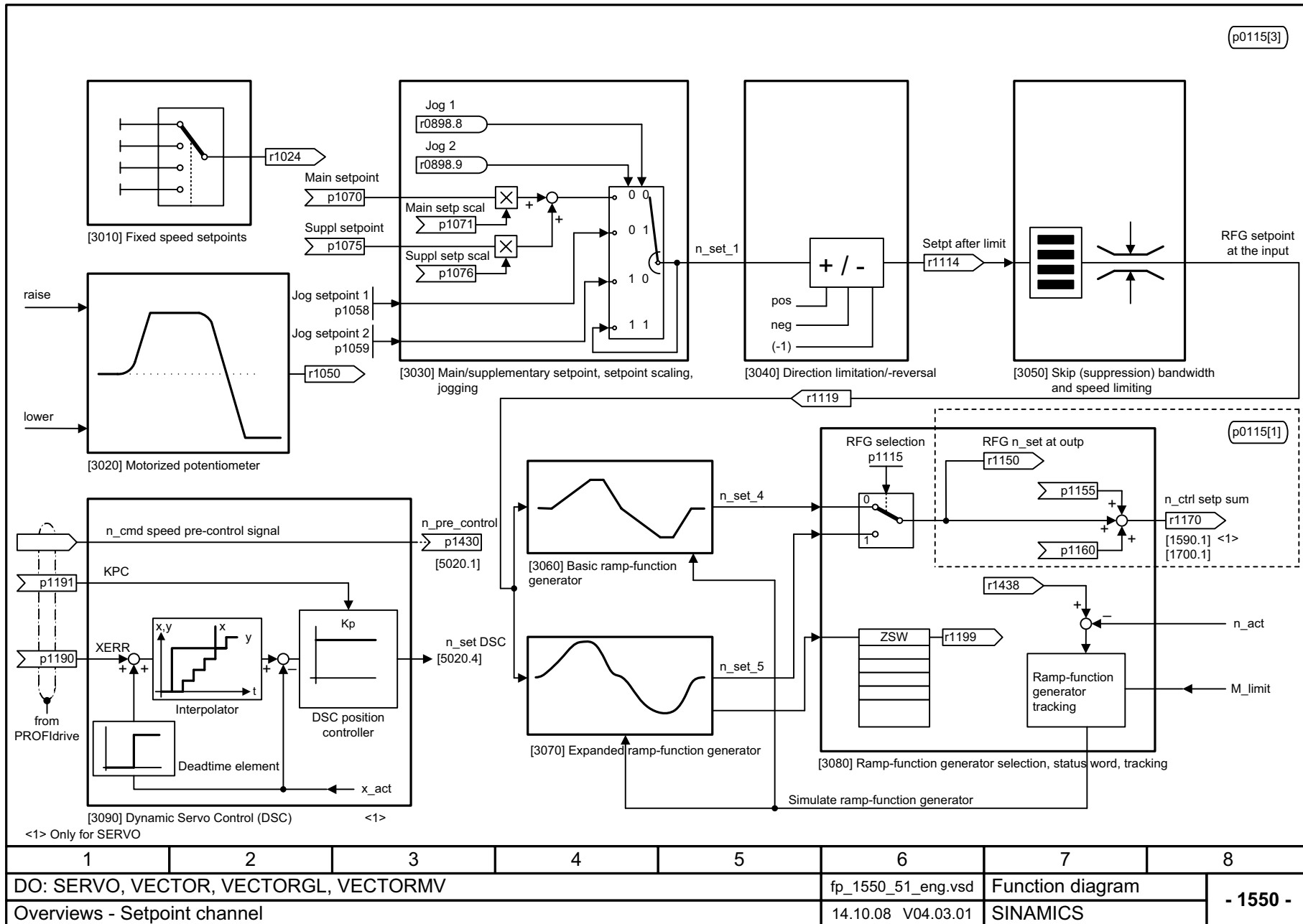
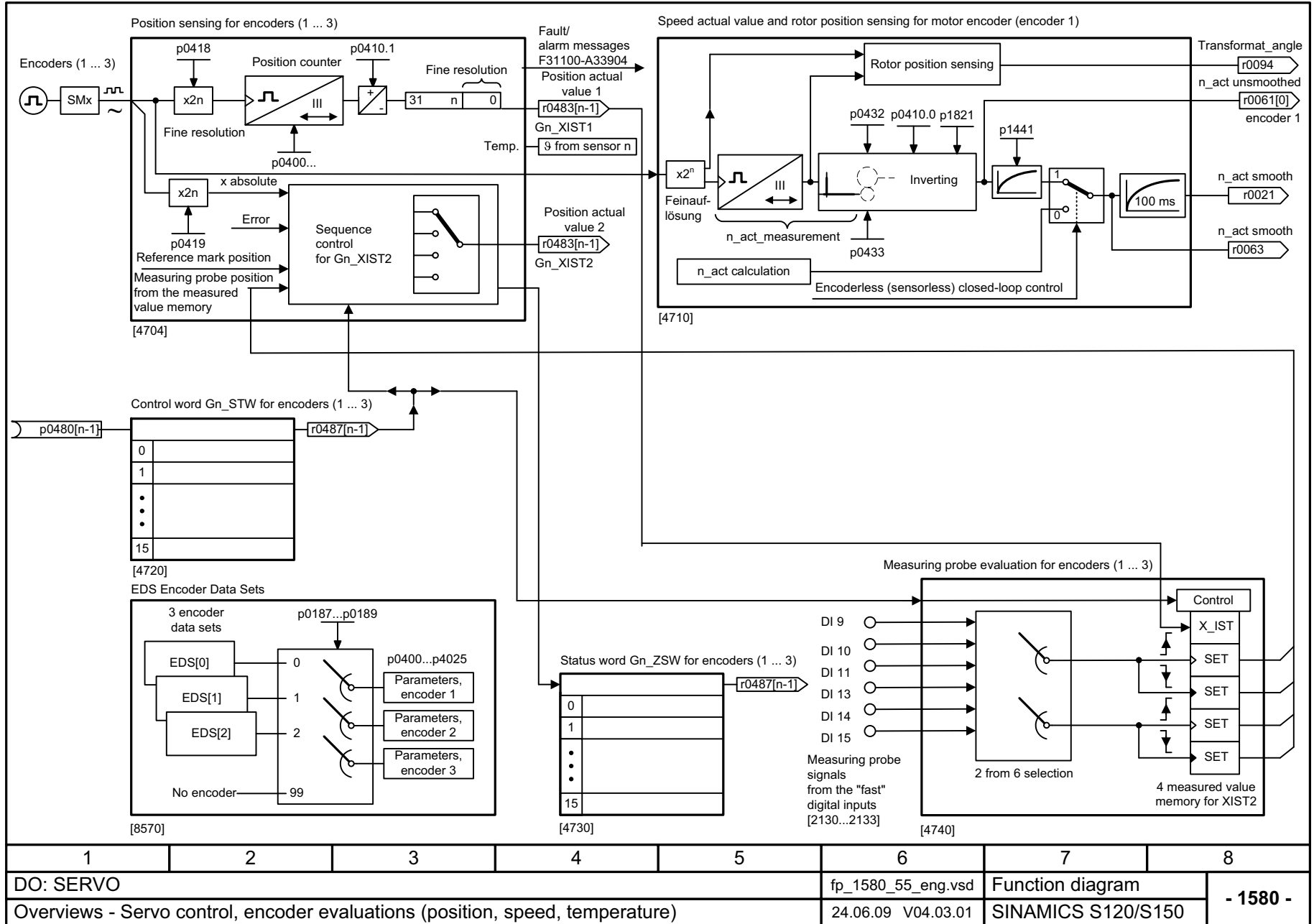
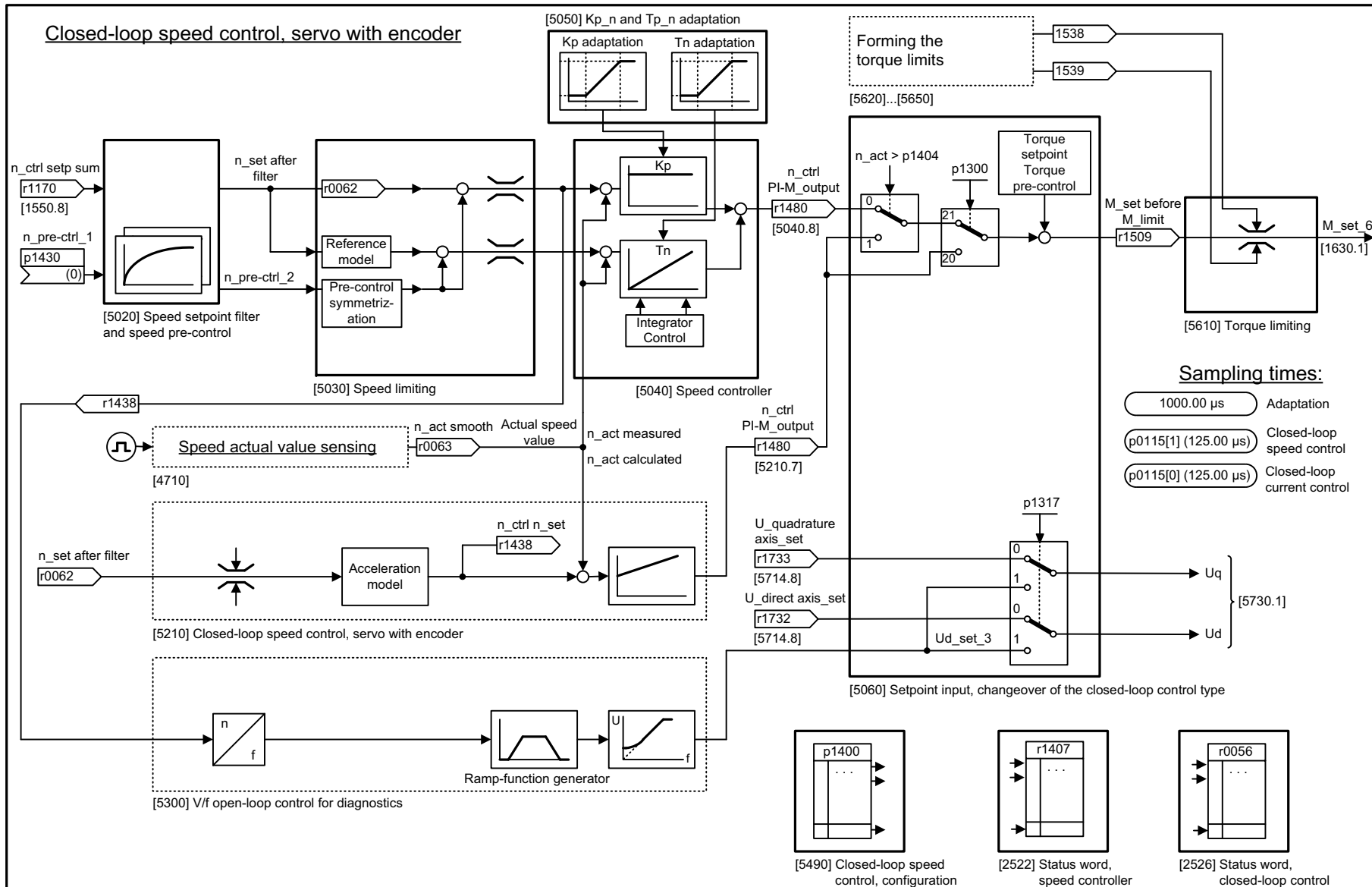


Fig. 2-8 1530 – Internal control/status words, data sets







1	2	3	4	5	6	7	8
DO: SERVO					fp_1590_01_eng.vsd	Function diagram	
Overviews - Servo control, speed control and V/f control					14.04.08 V04.03.01	SINAMICS S120	

- 1590 -

Fig. 2-11

1590 – Servo control, speed control and V/f control

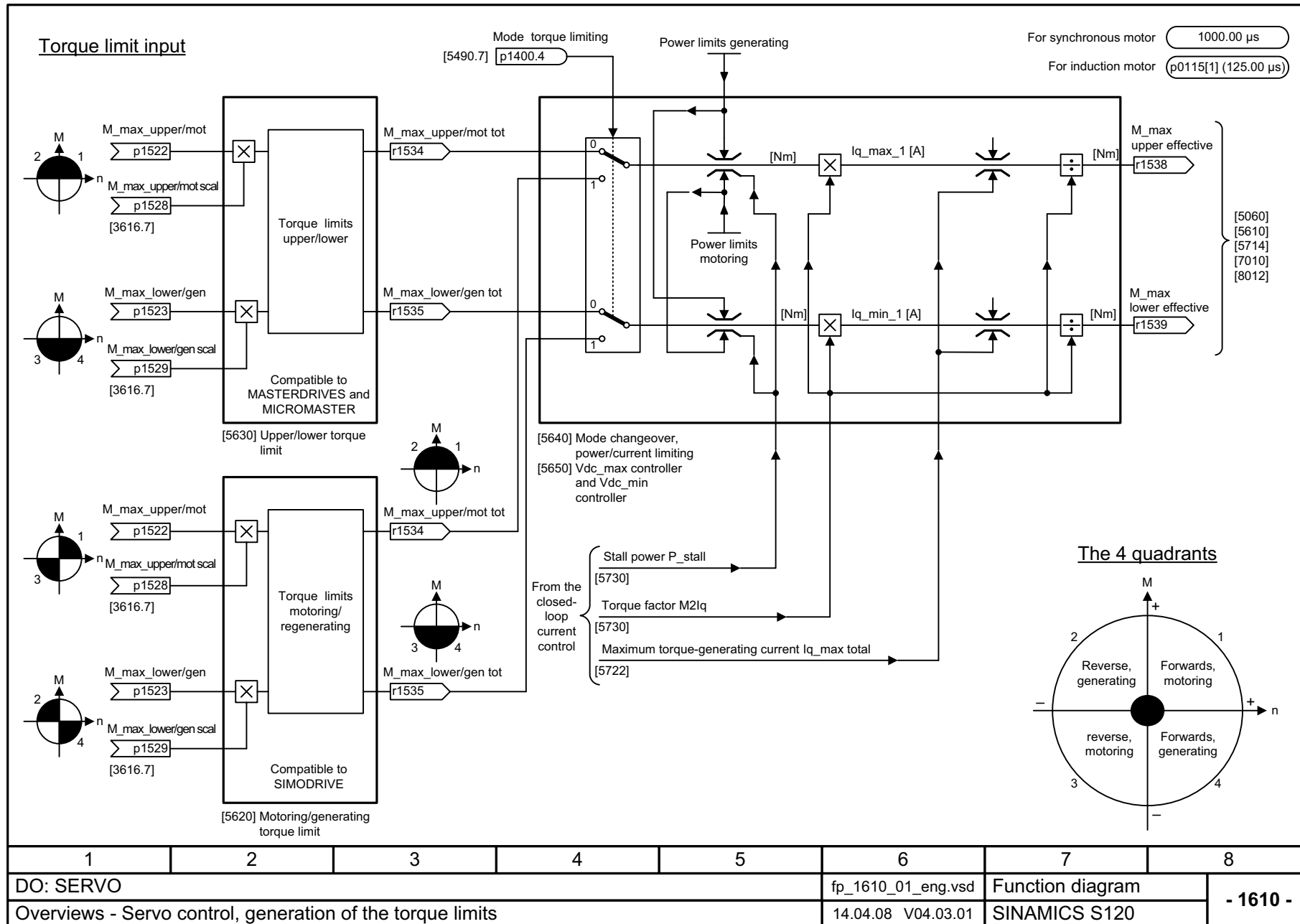


Fig. 2-12 1610 – Servo control, formation of the torque limits

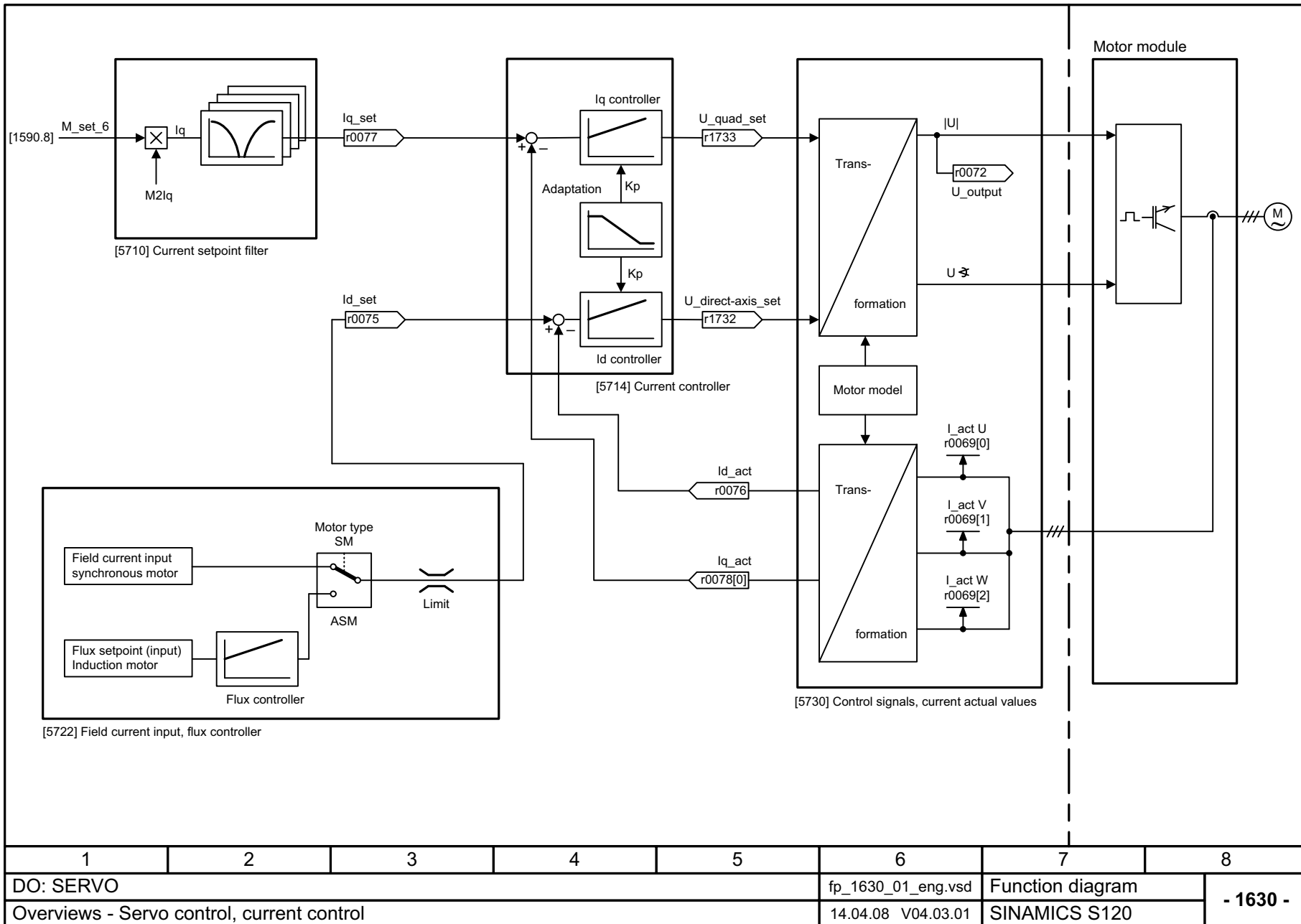
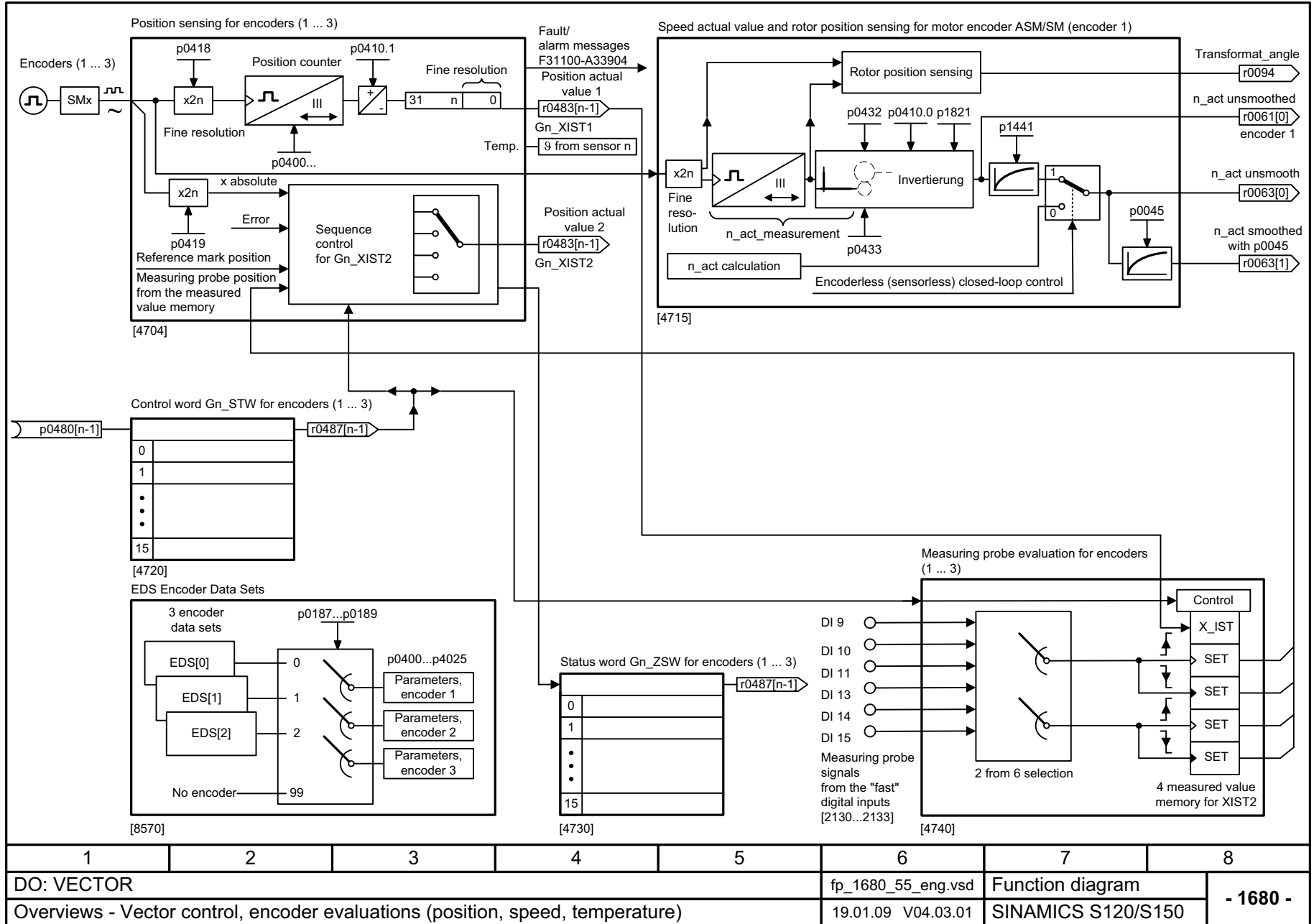
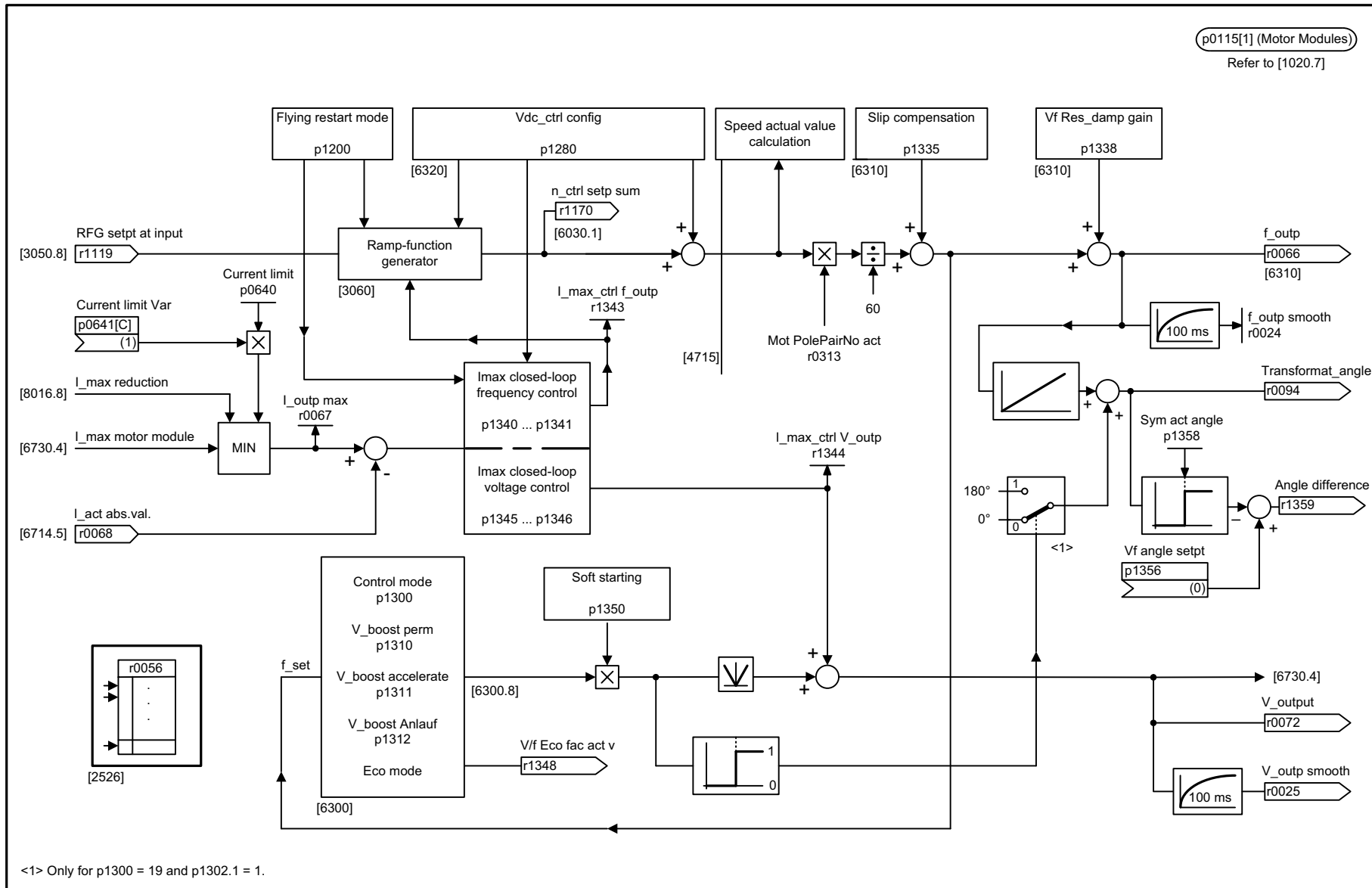


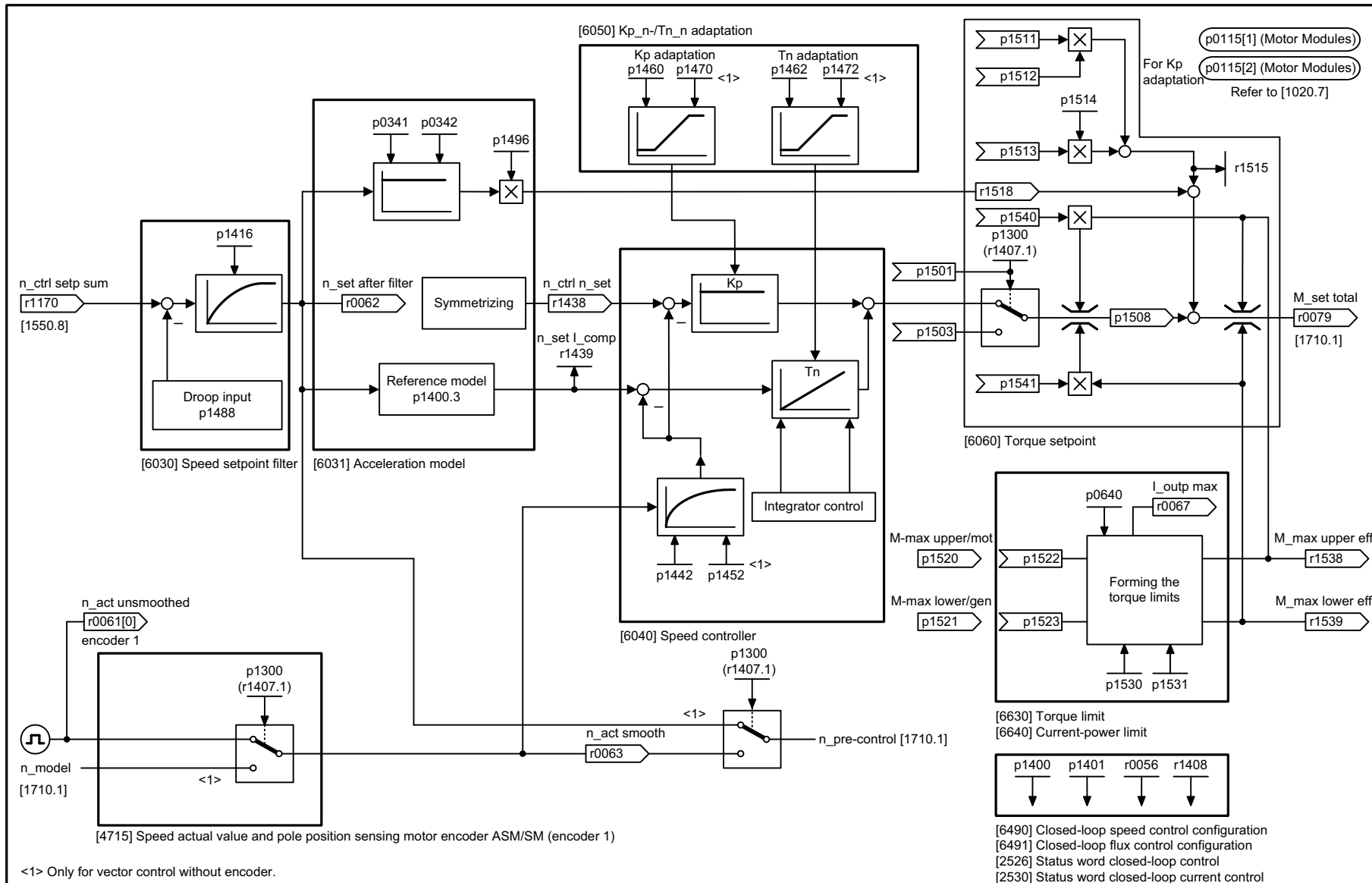
Fig. 2-13 1630 – Servo control, current control



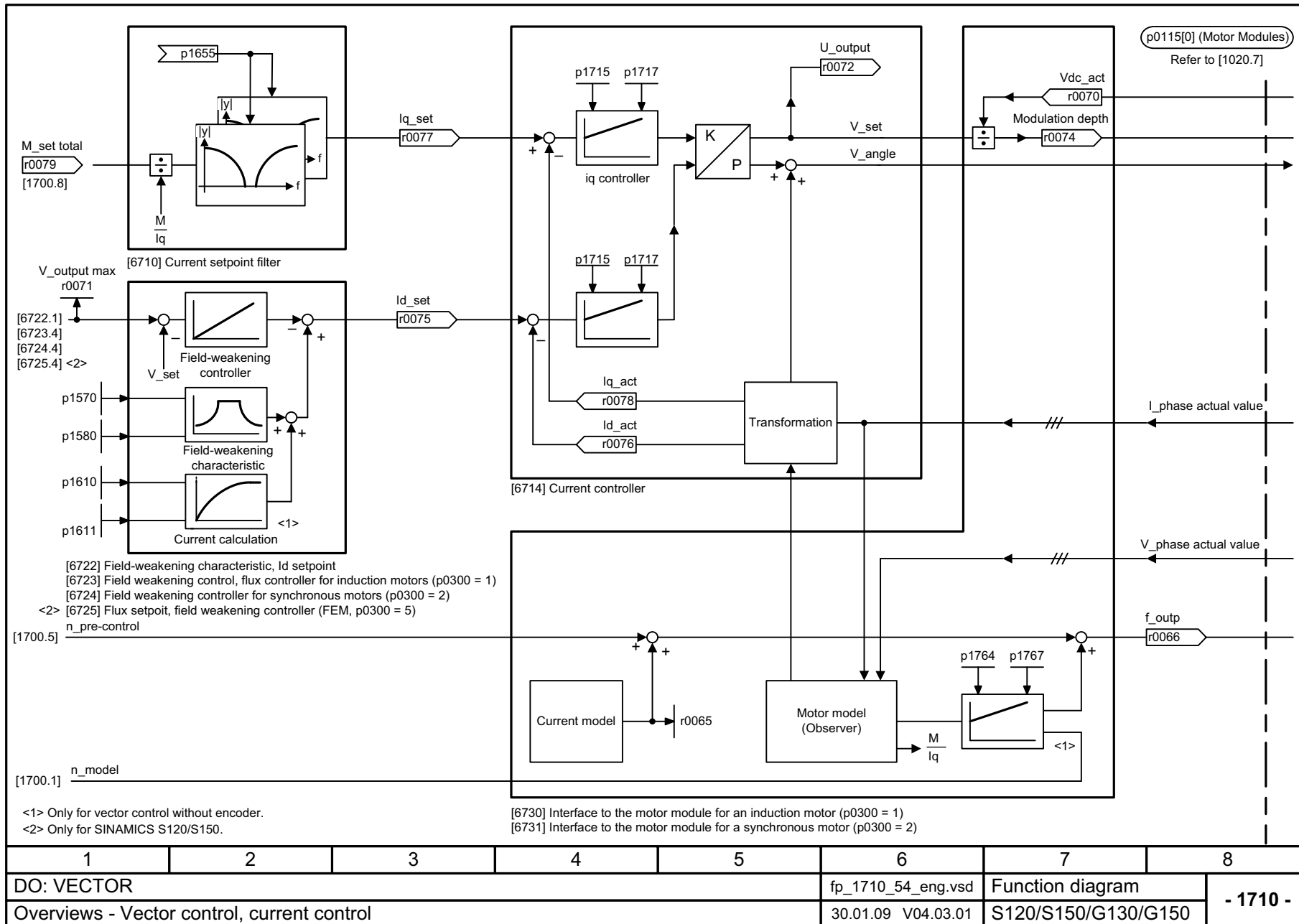
1	2	3	4	5	6	7	8	
DO: VECTOR					fp_1680_55_eng.vsd	Function diagram		- 1680 -
Overviews - Vector control, encoder evaluations (position, speed, temperature)					19.01.09 V04.03.01	SINAMICS S120/S150		

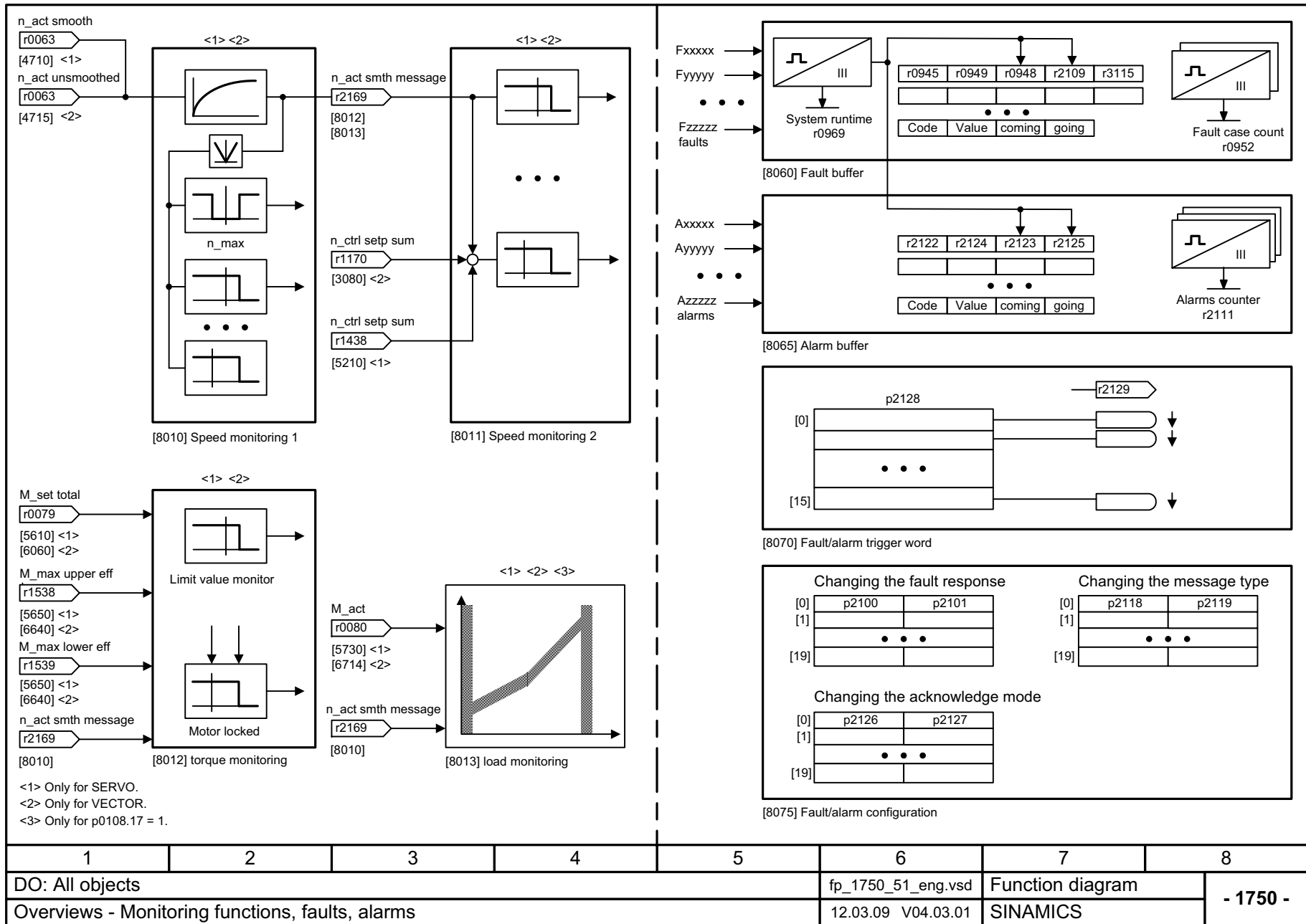


<1> Only for p1300 = 19 and p1302.1 = 1.

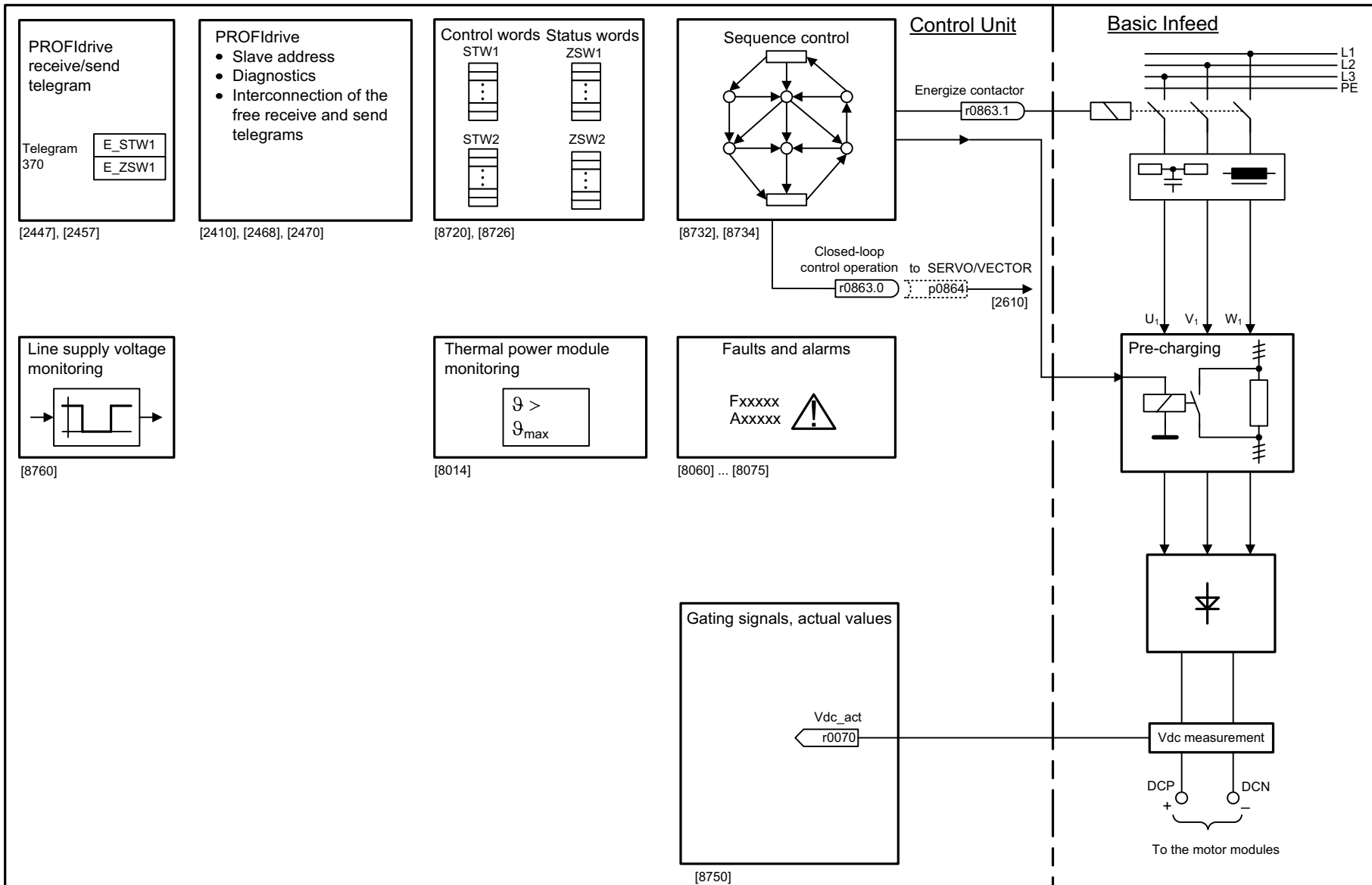


1	2	3	4	5	6	7	8
DO: VECTOR					fp_1700_54_eng.vsd	Function diagram	
Overviews - Vector control, speed control and generation of the torque limits					19.01.09 V04.03.01	S120/S150/G130/G150	
							- 1700 -



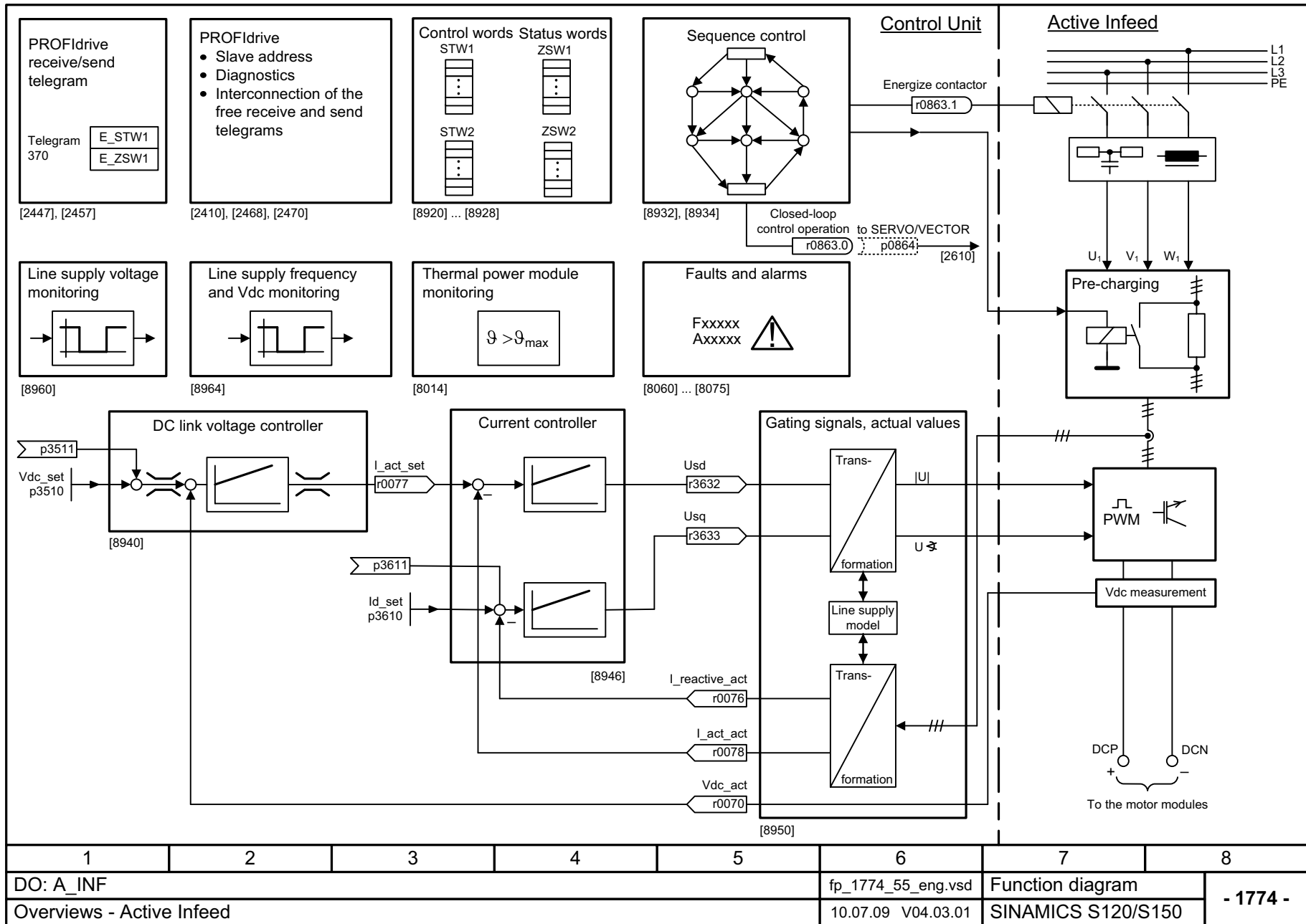


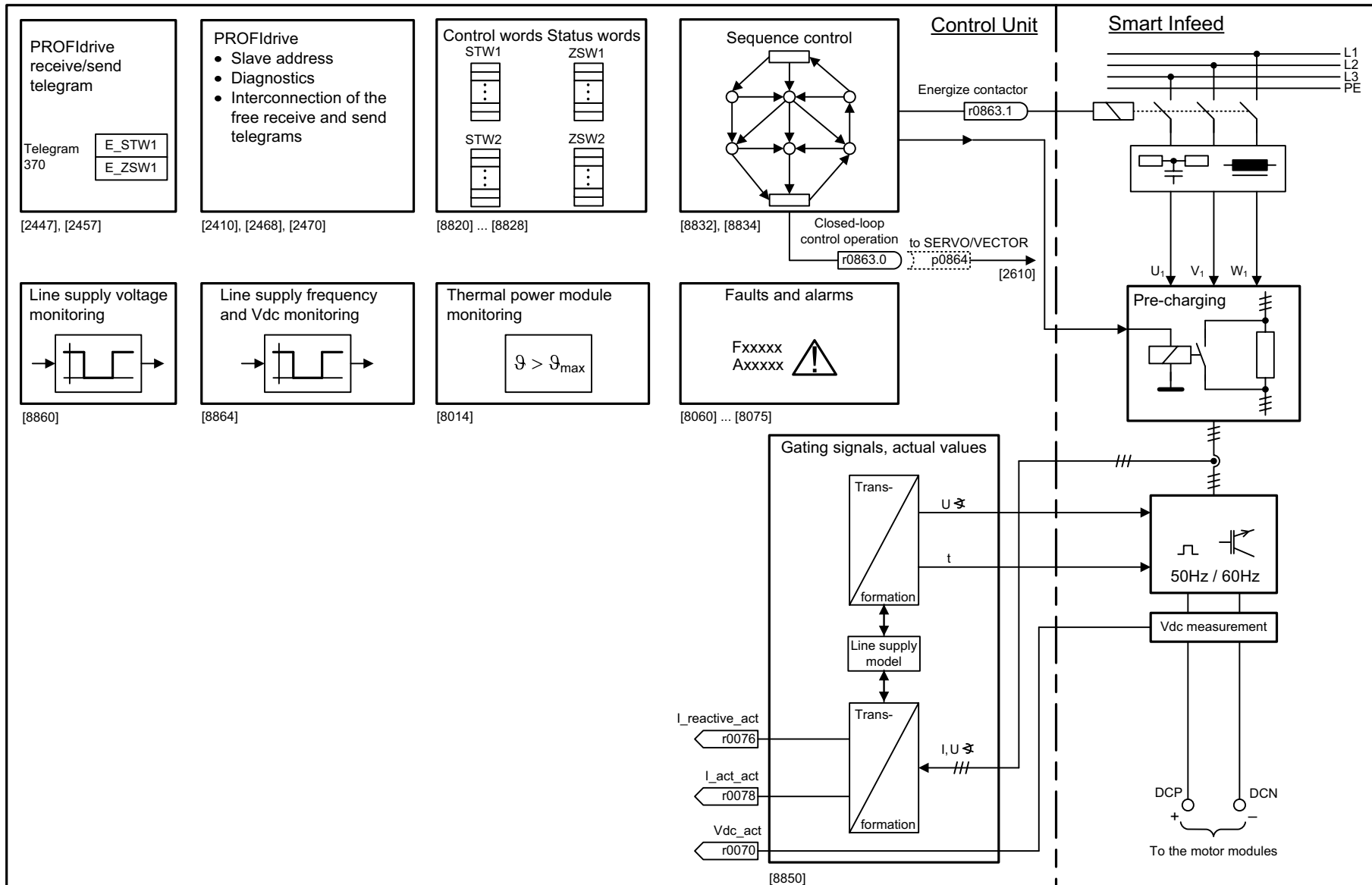
1	2	3	4	5	6	7	8	
DO: All objects					fp_1750_51_eng.vsd	Function diagram		- 1750 -
Overviews - Monitoring functions, faults, alarms					12.03.09 V04.03.01	SINAMICS		



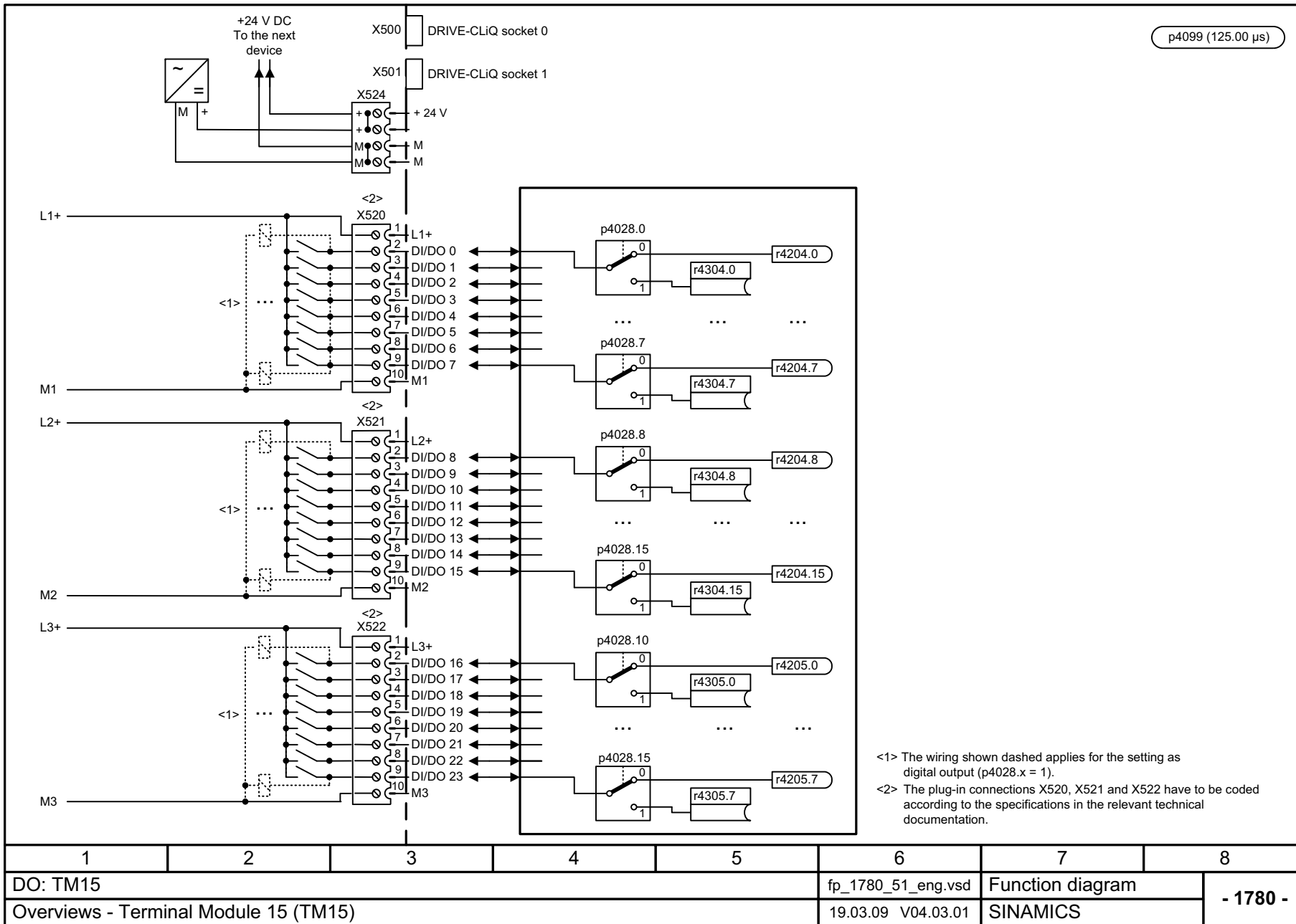
1	2	3	4	5	6	7	8
DO: B_INF					fp_1773_01_eng.vsd	Function diagram	
Overviews - Basic Infeed					14.08.08 V04.03.01	SINAMICS S120	
							- 1773 -

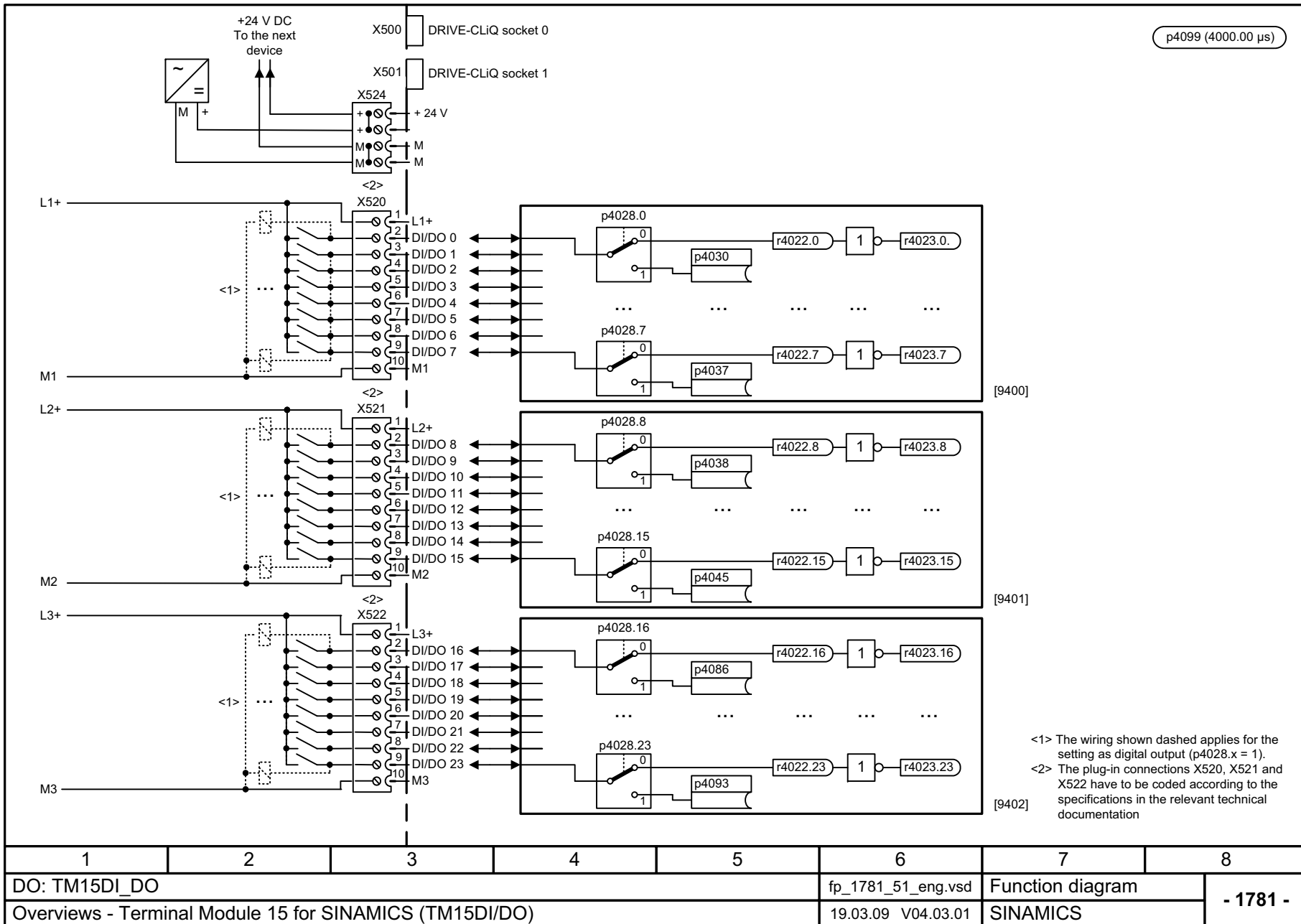
Fig. 2-19 1773 – Basic Infeed





1	2	3	4	5	6	7	8
DO: S_INF					fp_1775_01_eng.vsd	Function diagram	
Overviews - Smart Infeed					10.07.09 V04.03.01	SINAMICS S120	





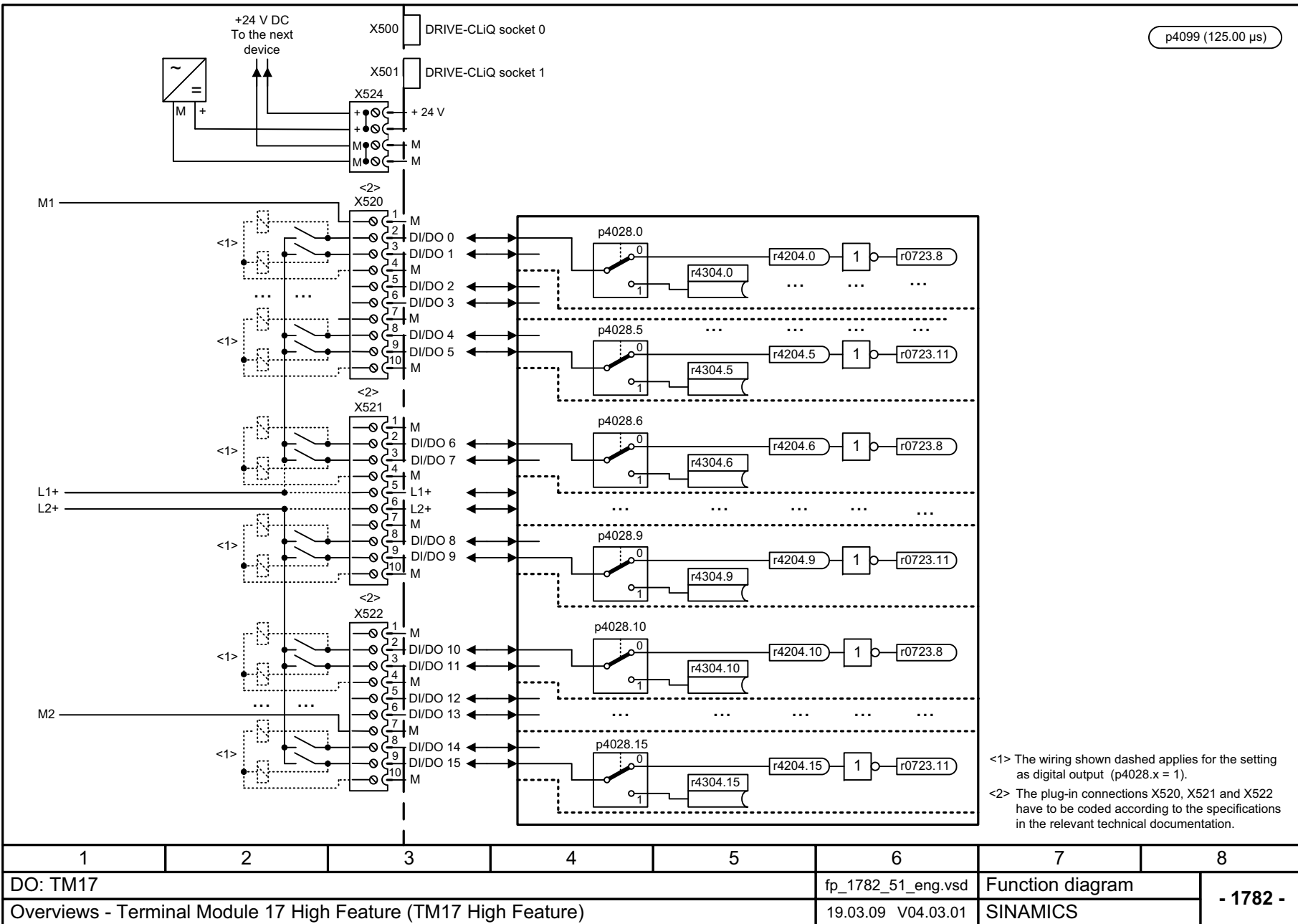
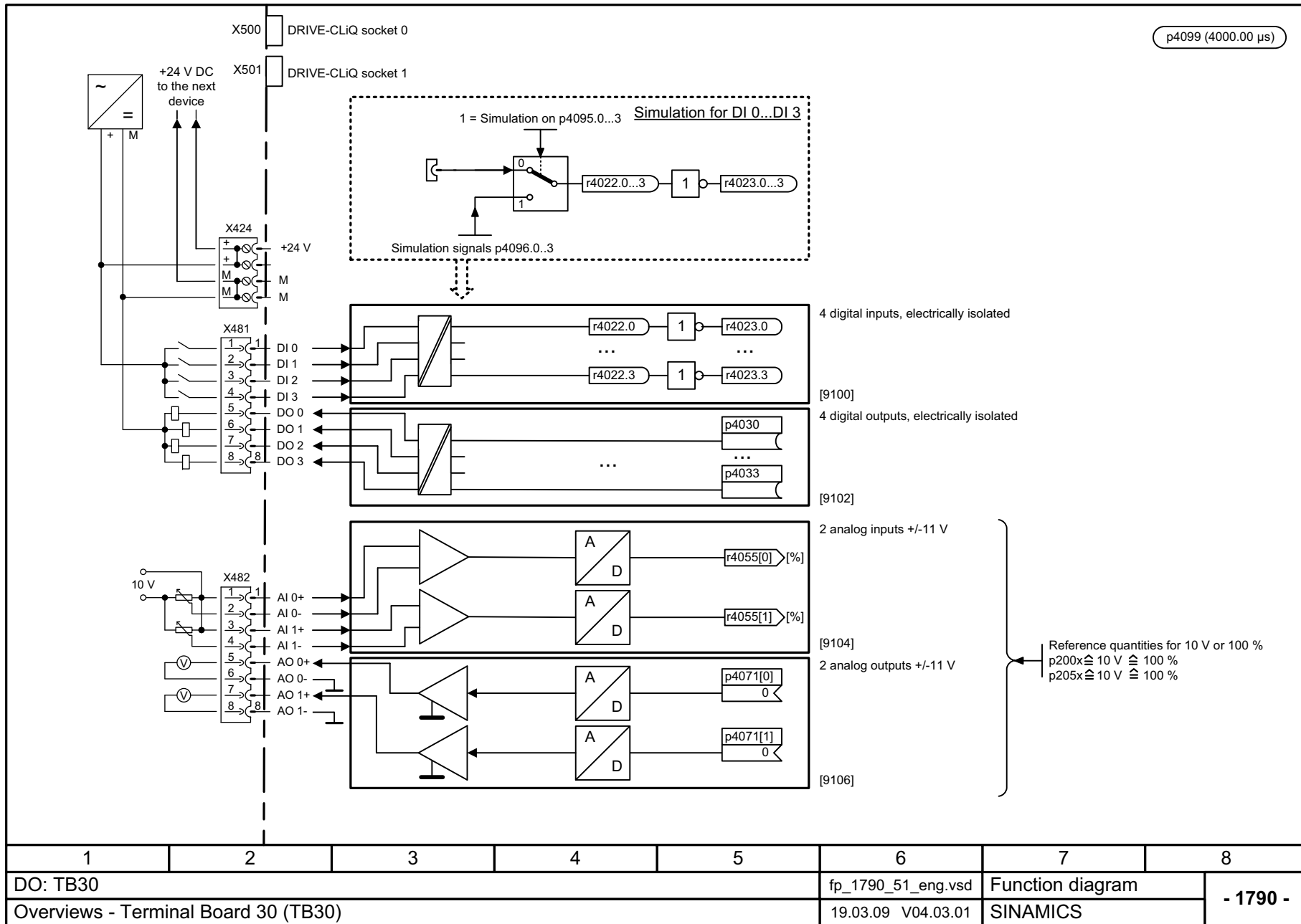
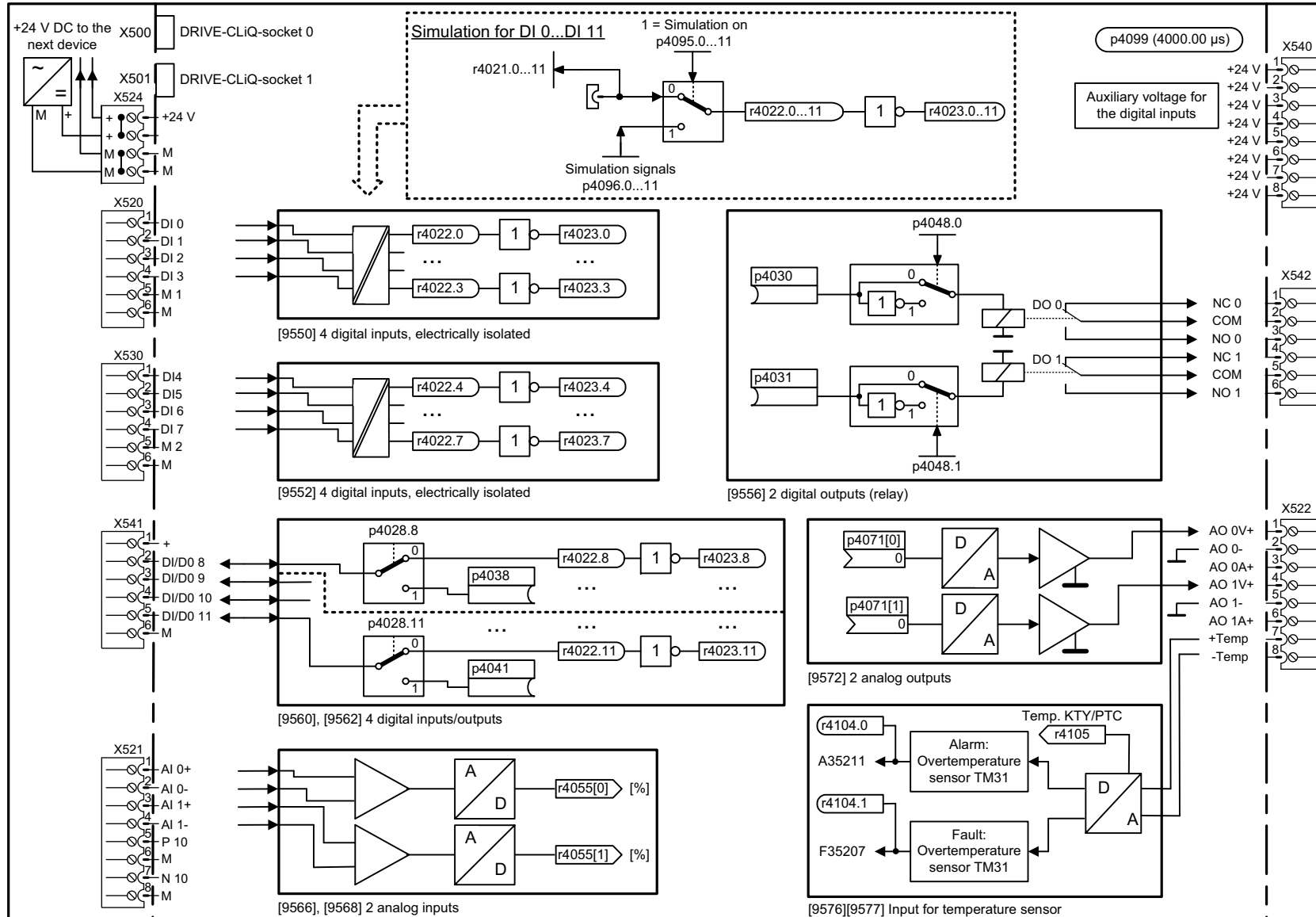
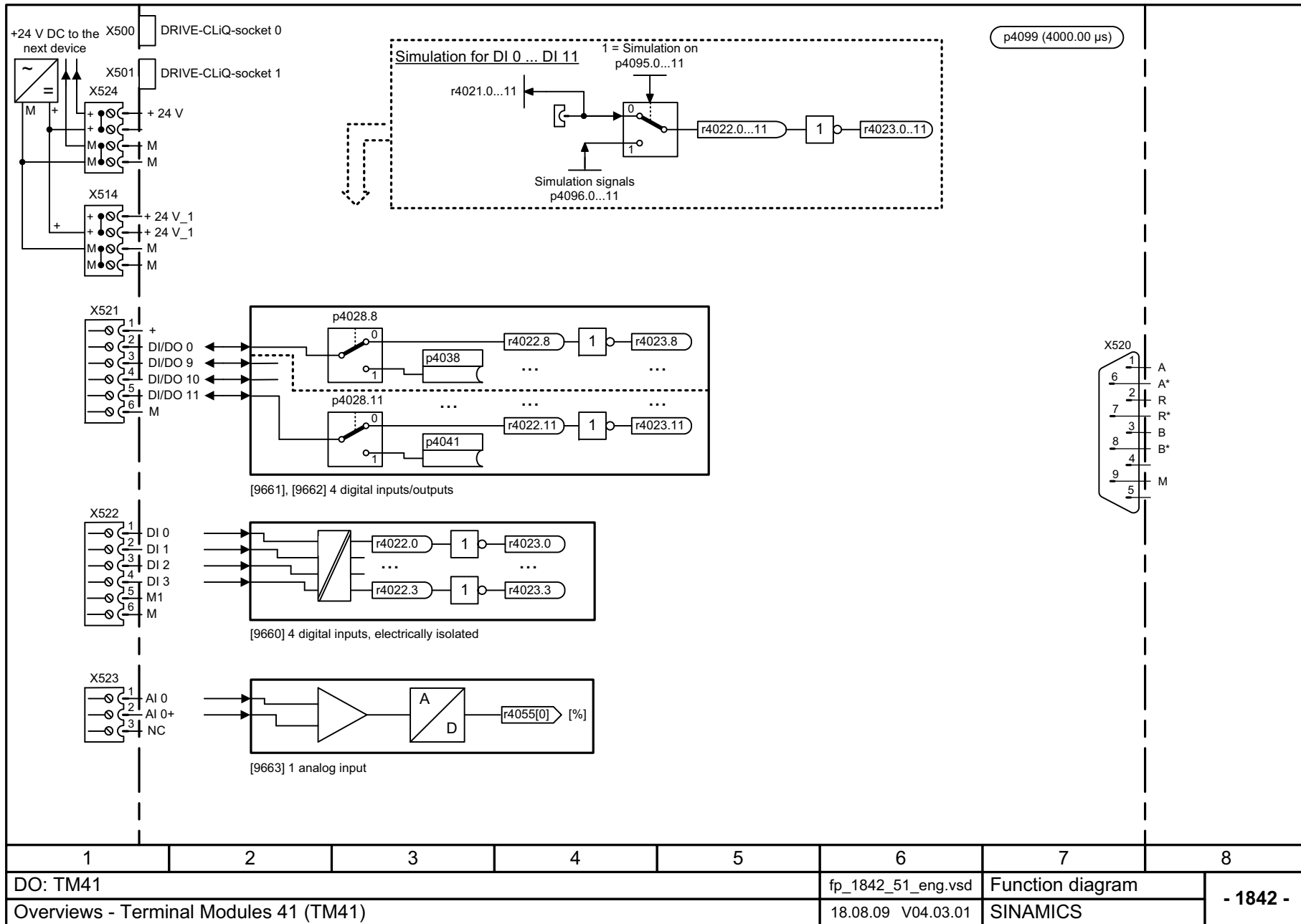


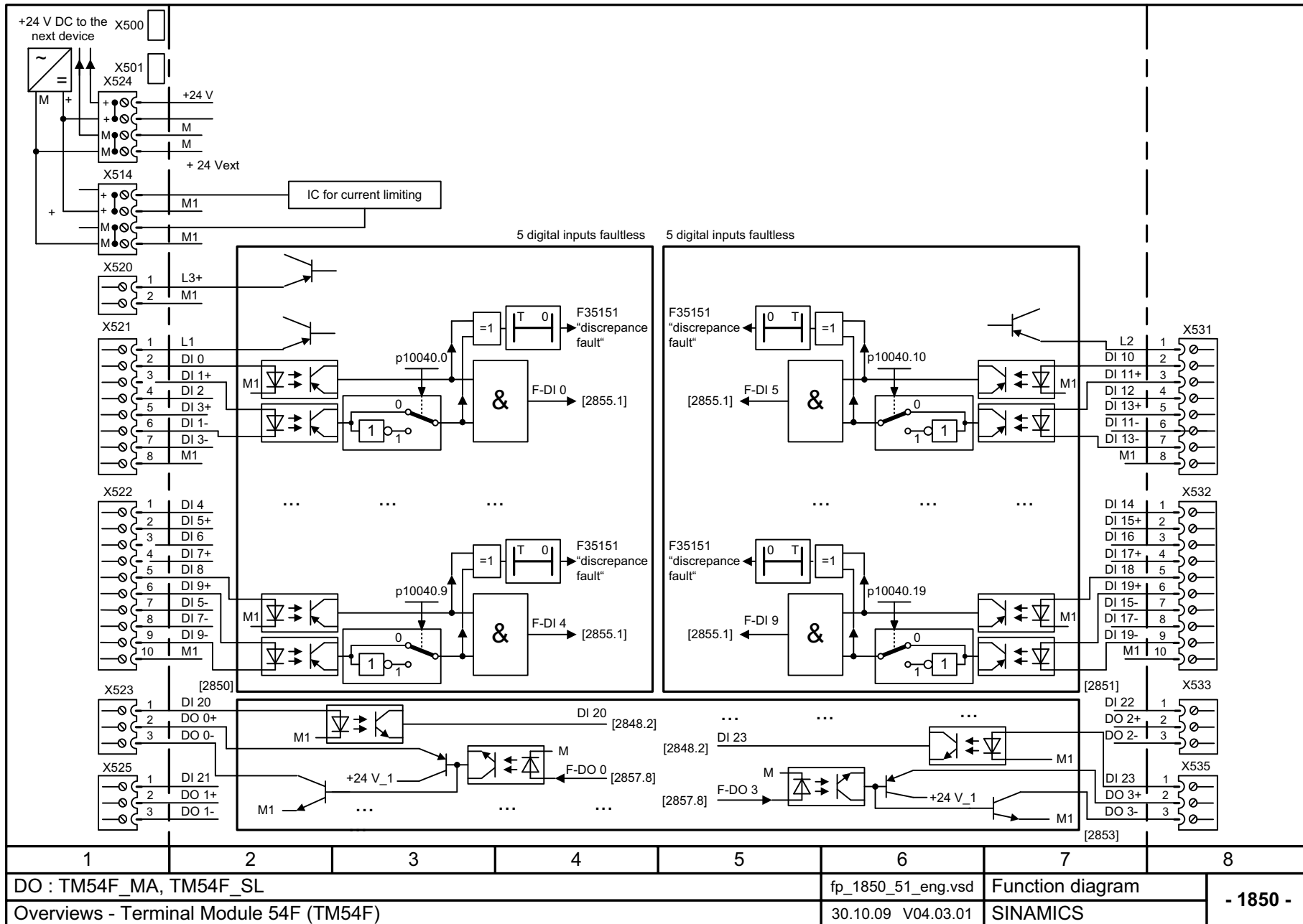
Fig. 2-24





1	2	3	4	5	6	7	8
DO: TM31					fp_1840_51_eng.vsd	Function diagram	
Overviews - Terminal Module 31 (TM31)					19.03.09 V04.03.01	SINAMICS	
							- 1840 -

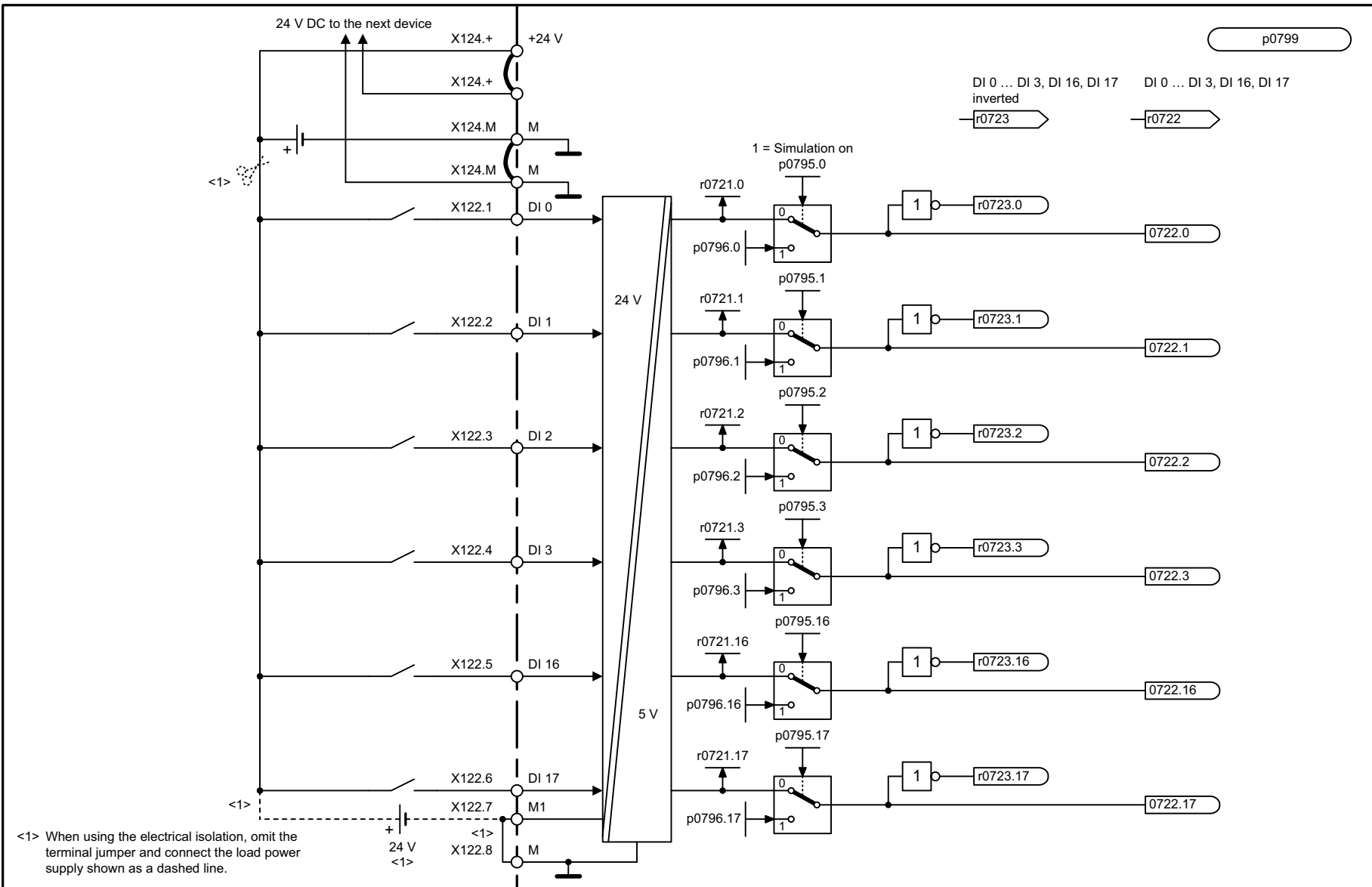




2.4 CU320-2 input/output terminals

Function diagrams

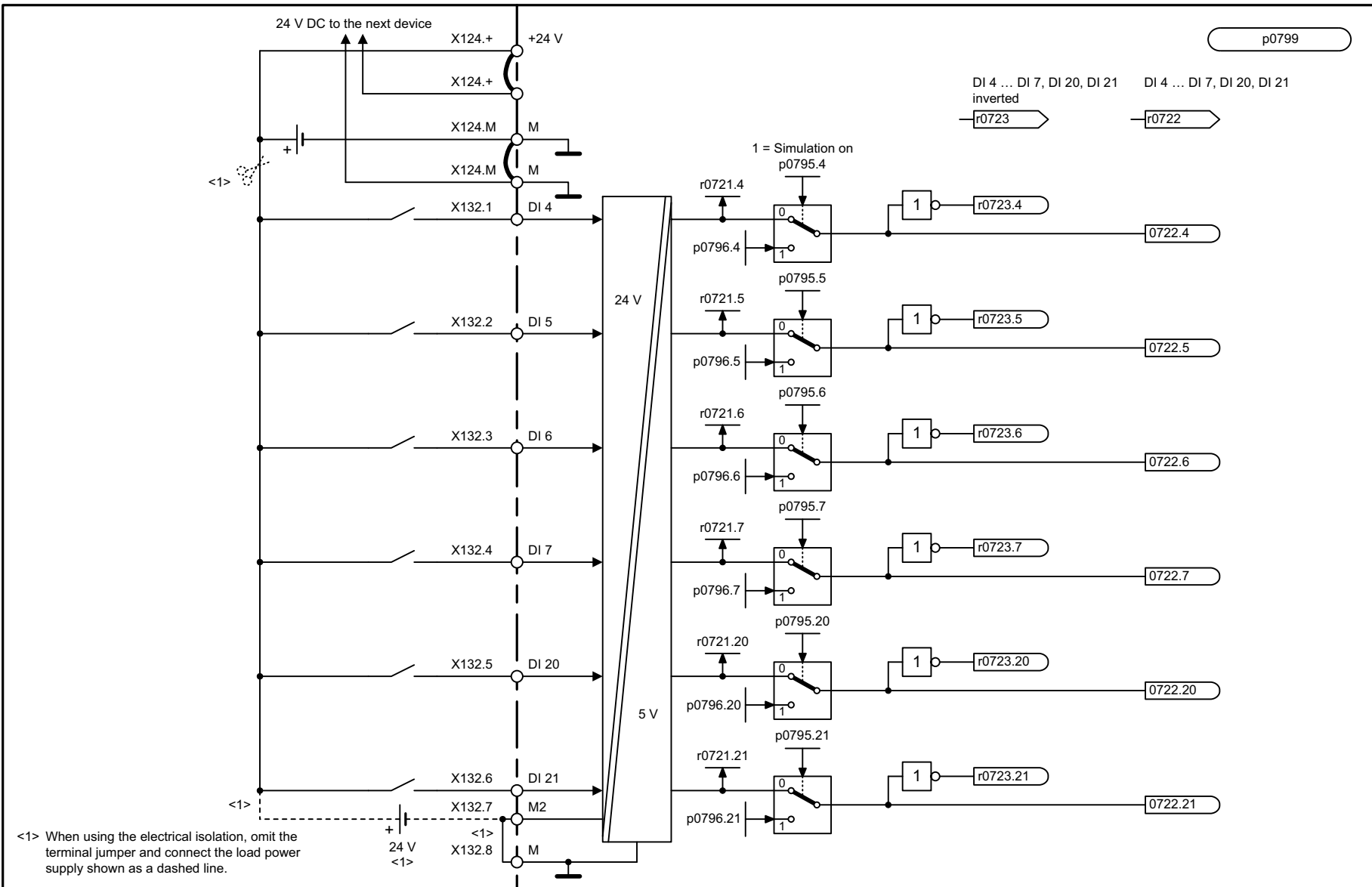
2120 – Digital inputs, isolated (DI 0 to DI 3, DI 16, DI 17)	2-1415
2121 – Digital inputs, isolated (DI 4 to DI 7, DI 20, DI 21)	2-1416
2130 – Digital inputs/outputs, bidirectional (DI/DO 8 to DI/DO 9)	2-1417
2131 – Digital inputs/outputs, bidirectional (DI/DO 10 to DI/DO 11)	2-1418
2132 – Digital inputs/outputs, bidirectional (DI/DO 12 to DI/DO 13)	2-1419
2133 – Digital inputs/outputs, bidirectional (DI/DO 14 to DI/DO 15)	2-1420



1	2	3	4	5	6	7	8
DO: CU_G, CU_GL, CU_GM, CU_S, CU_SL					fp_2120_51_eng.vsd	Function diagram	
CU320-2 input/output terminals - Digital inputs, electrically isolated (DI 0 ... DI 3, DI 16, DI 17)					24.07.09 V04.03.01	SINAMICS	

- 2120 -

Fig. 2-29 2120 – Digital inputs, isolated (DI 0 to DI 3, DI 16, DI 17)



1	2	3	4	5	6	7	8
DO: CU_G, CU_GL, CU_GM, CU_S, CU_SL					fp_2121_51_eng.vsd	Function diagram	
CU320-2 input/output terminals - Digital inputs, electrically isolated (DI 4 ... DI 7, DI 20, DI 21)					24.07.09 V04.03.01	SINAMICS	

Fig. 2-30 2121 – Digital inputs, isolated (DI 4 to DI 7, DI 20, DI 21)

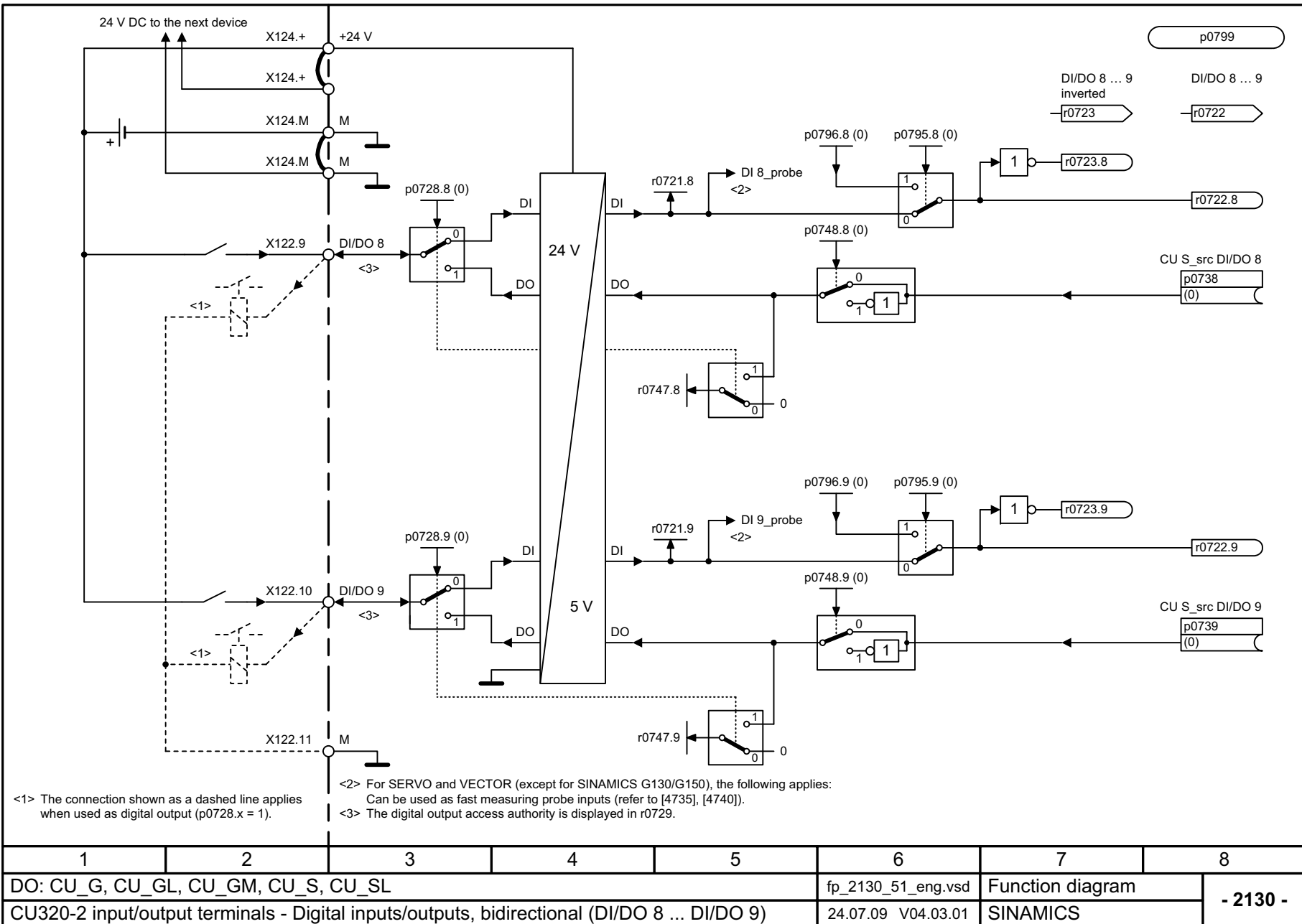


Fig. 2-31 2130 – Digital inputs/outputs, bidirectional (DI/DO 8 to DI/DO 9)

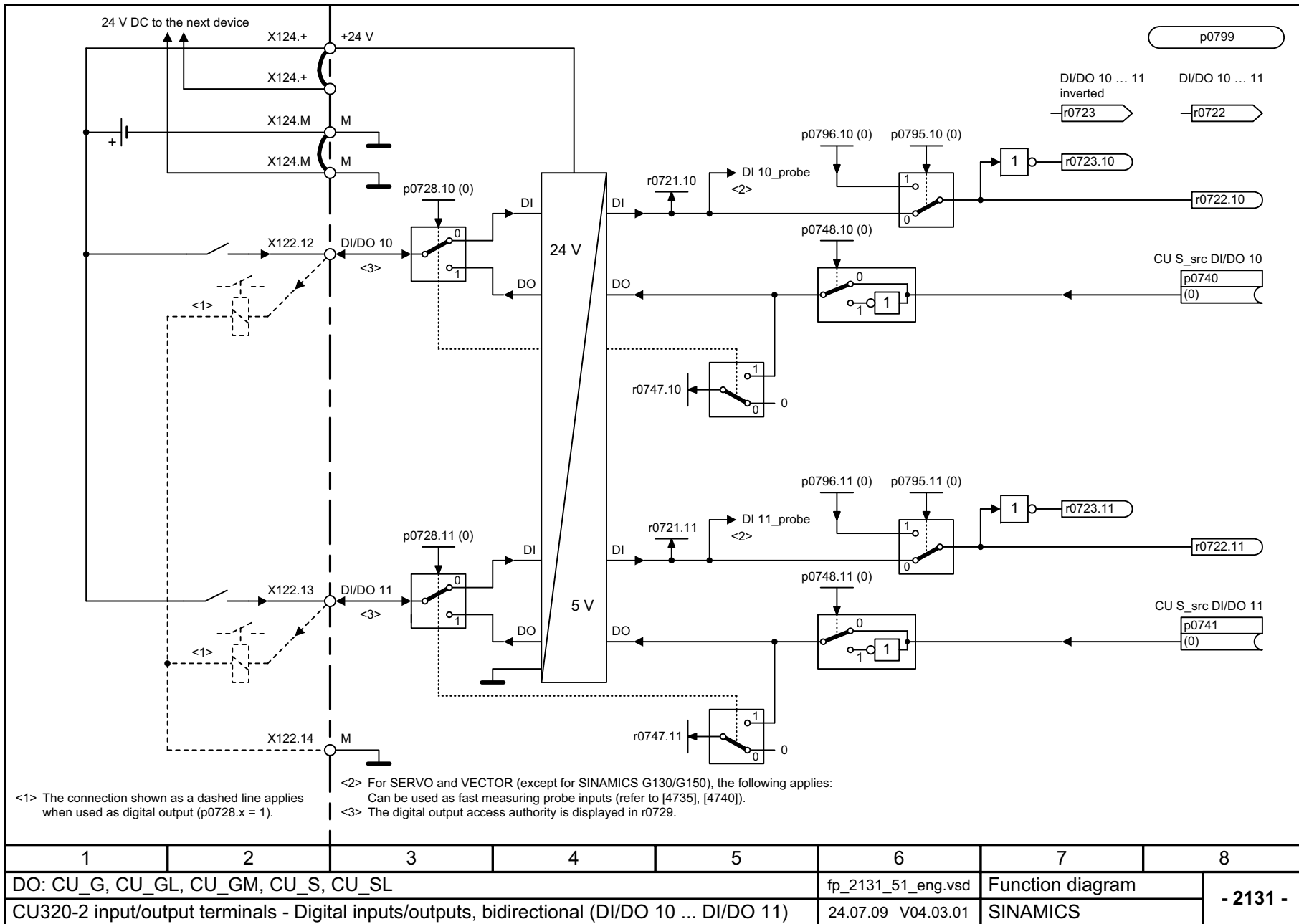


Fig. 2-32 2131 – Digital inputs/outputs, bidirectional (DI/DO 10 to DI/DO 11)

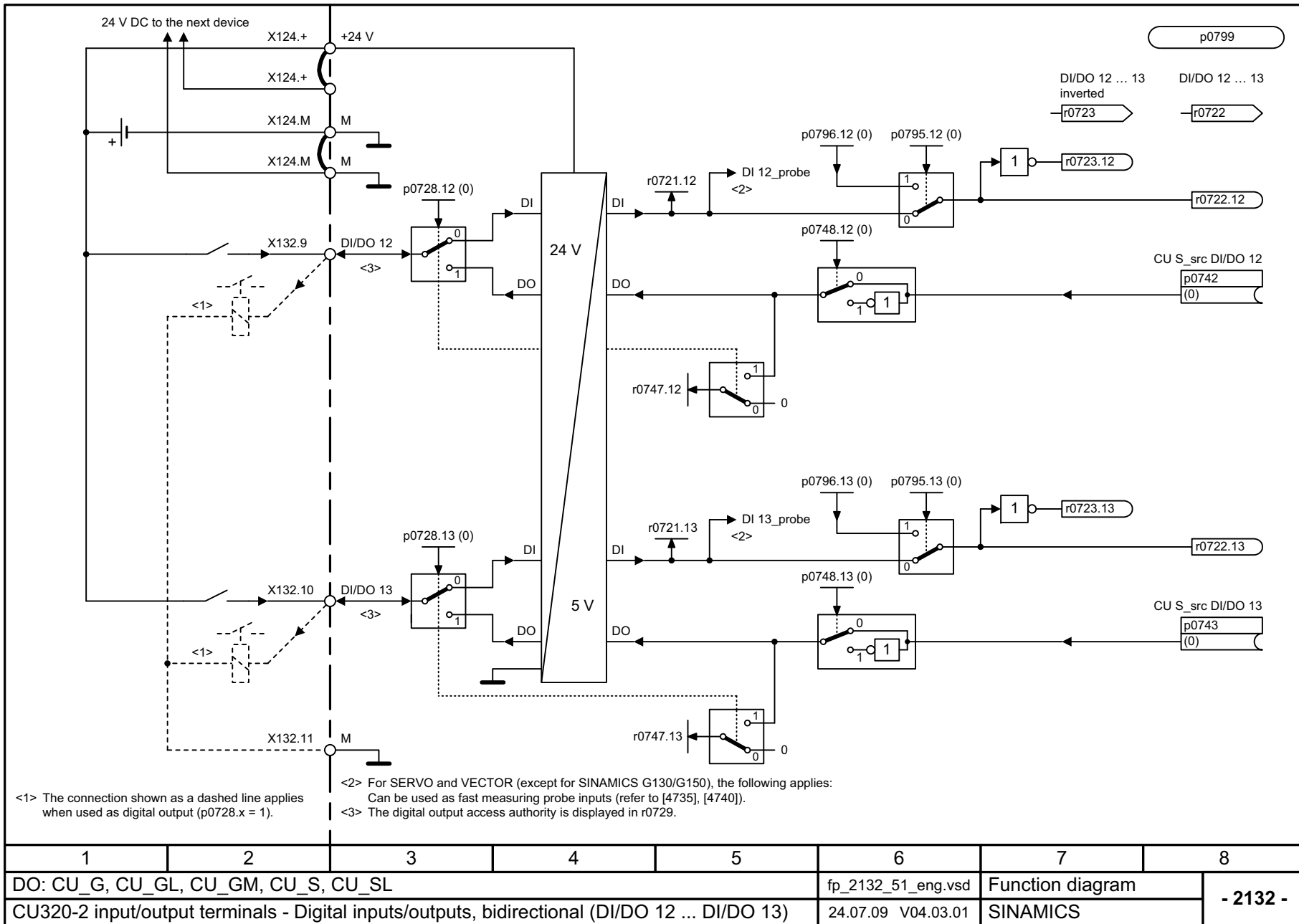


Fig. 2-33 2132 – Digital inputs/outputs, bidirectional (DI/DO 12 to DI/DO 13)

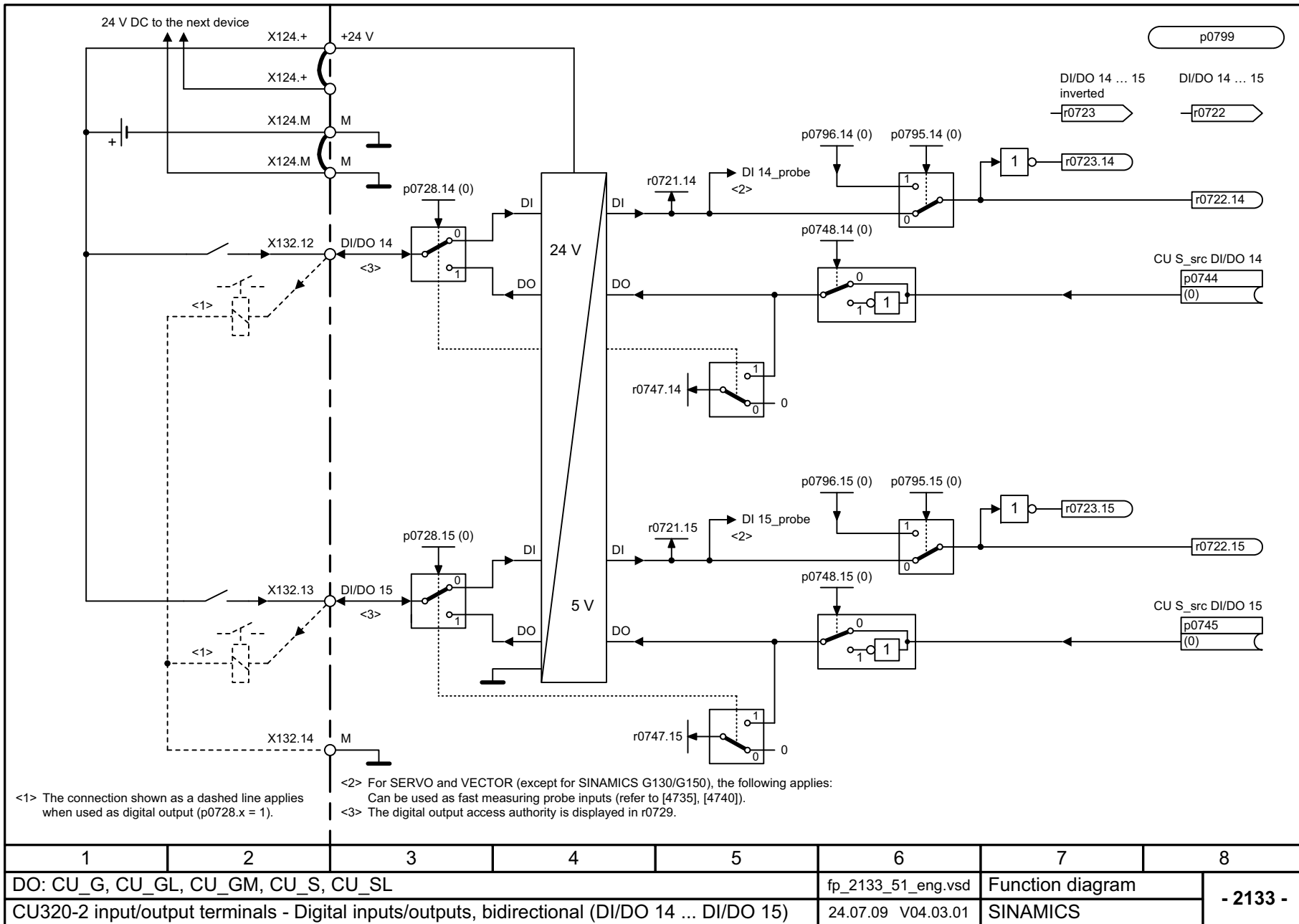


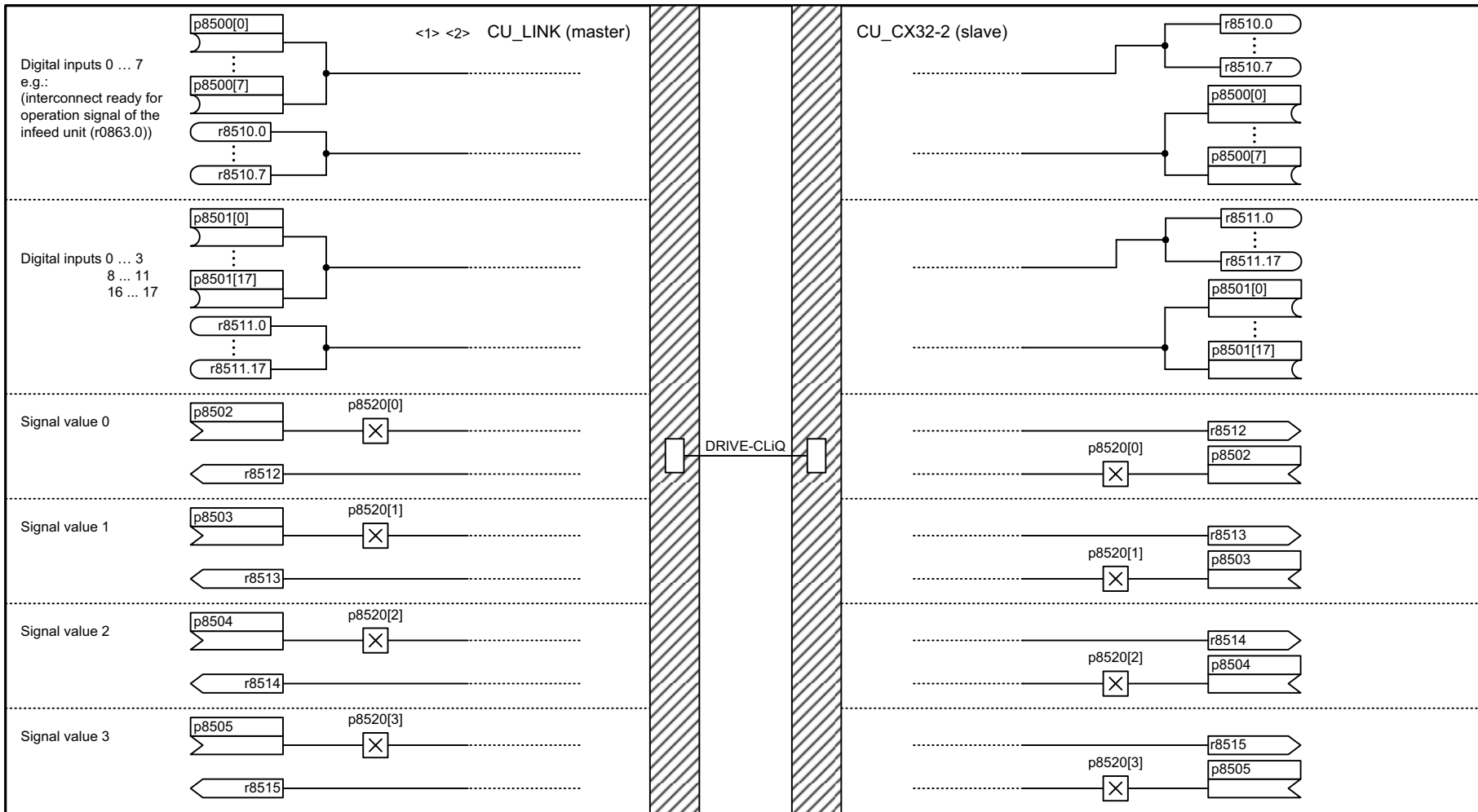
Fig. 2-34 2133 – Digital inputs/outputs, bidirectional (DI/DO 14 to DI/DO 15)

2.5 CU_LINK

Function diagrams

2211 – Data transfer

2-1422



<1> The drive object CU_LINK only exists for automation systems with integrated SINAMICS functionality (e.g. SINUMERIK NCU) and the corresponding expansion components (e.g. CX32, NX10).
The master functionality is available on the automation system, the slave functionality is available on the expansion components.
<2> p8800: CU_LINK address.
Address of the CX32 resp. NX10/15, representing the DO CU_LINK.
The address correlates with the DRIVE-CLiQ port of the CU connected to the CX/NX configured.
Value range: See p0918 (PROFIBUS address)
Only available within the unit, not visible in the STARTER, can be read, for example, through acyclic communication with DPV1 services.

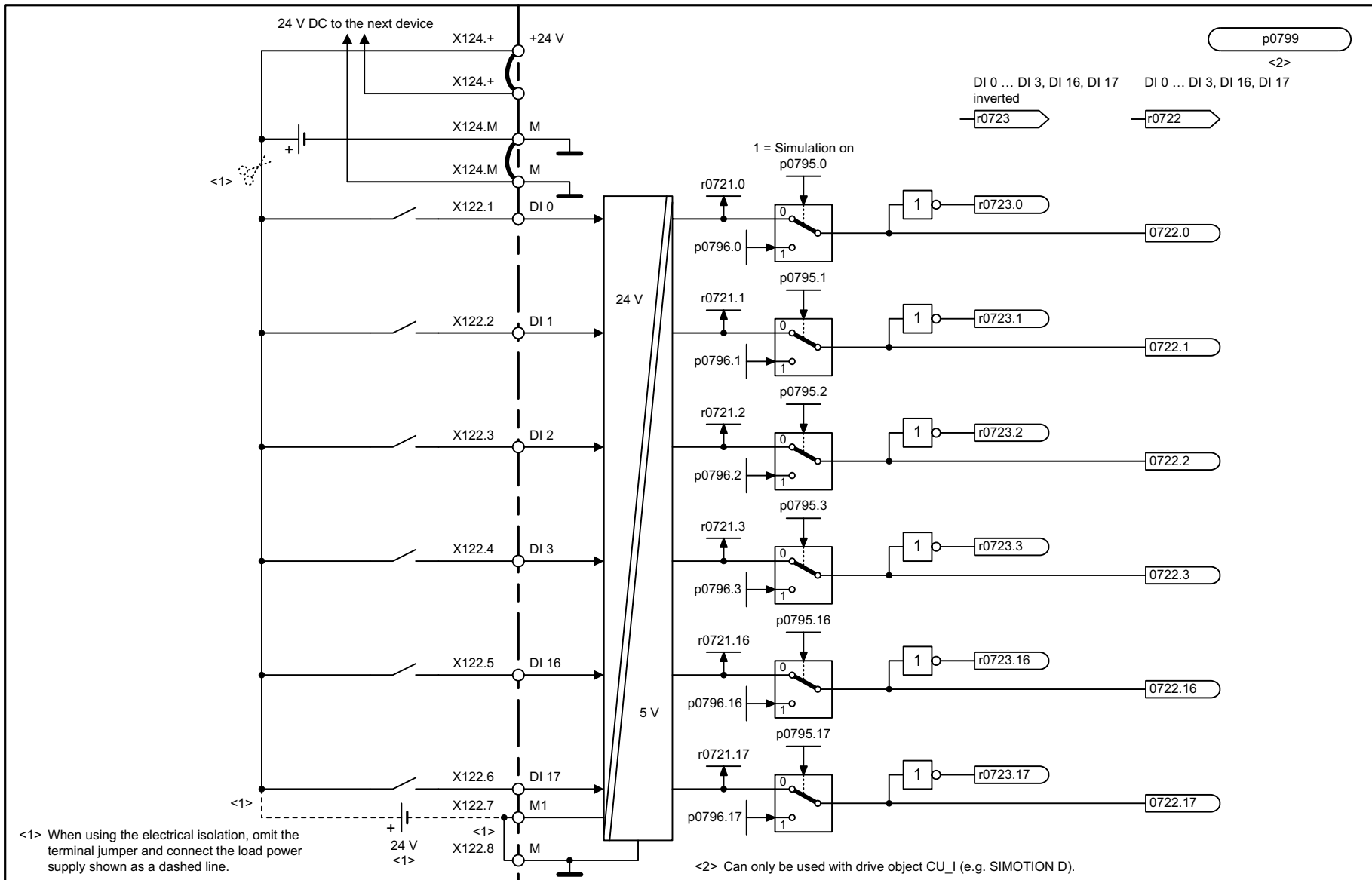
1	2	3	4	5	6	7	8
DO: CU_LINK, CU_CX32					fp_2211_51_eng.vsd	Function diagram	
CU_LINK - Data transfer					23.06.09 V04.03.01	SINAMICS	

Fig. 2-35 2211 – Data transfer

2.6 CX32-2 input/output terminals

Function diagrams

2220 – Digital inputs, isolated (DI 0 to DI 3, DI 16, DI 17)	2-1424
2230 – Digital inputs/outputs, bidirectional (DI/DO 8 to DI/DO 9)	2-1425
2231 – Digital inputs/outputs, bidirectional (DI/DO 10 to DI/DO 11)	2-1426



1	2	3	4	5	6	7	8
DO: CU_CX32					fp_2220_51_eng.vsd	Function diagram	
CX32-2 input/output terminals - Digital inputs, electrically isolated (DI 0 ... DI 3, DI 16, DI 17)					24.07.09 V04.03.01	SINAMICS	
							- 2220 -

Fig. 2-36 2220 – Digital inputs, isolated (DI 0 to DI 3, DI 16, DI 17)

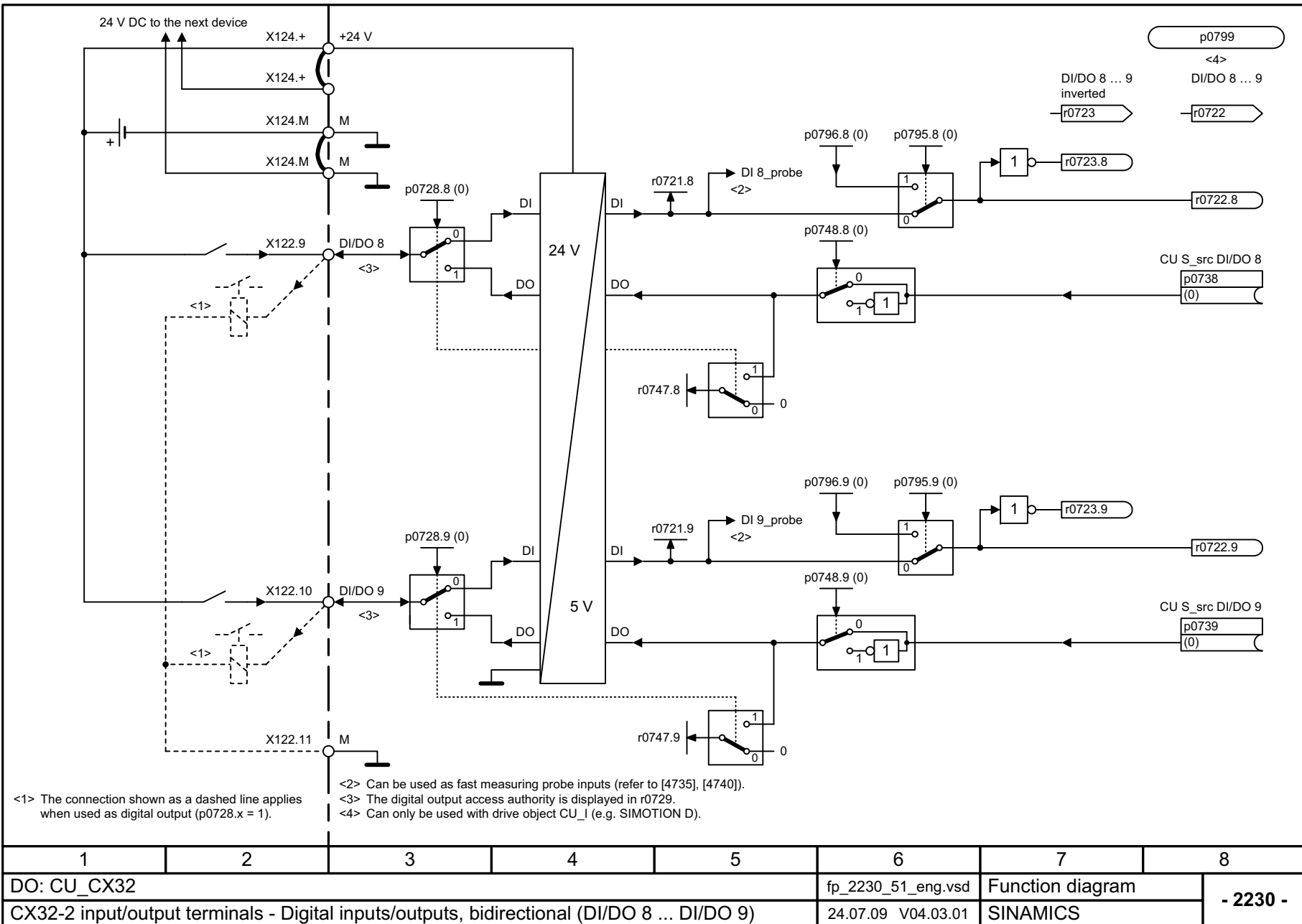


Fig. 2-37 2230 – Digital inputs/outputs, bidirectional (DI/DO 8 to DI/DO 9)

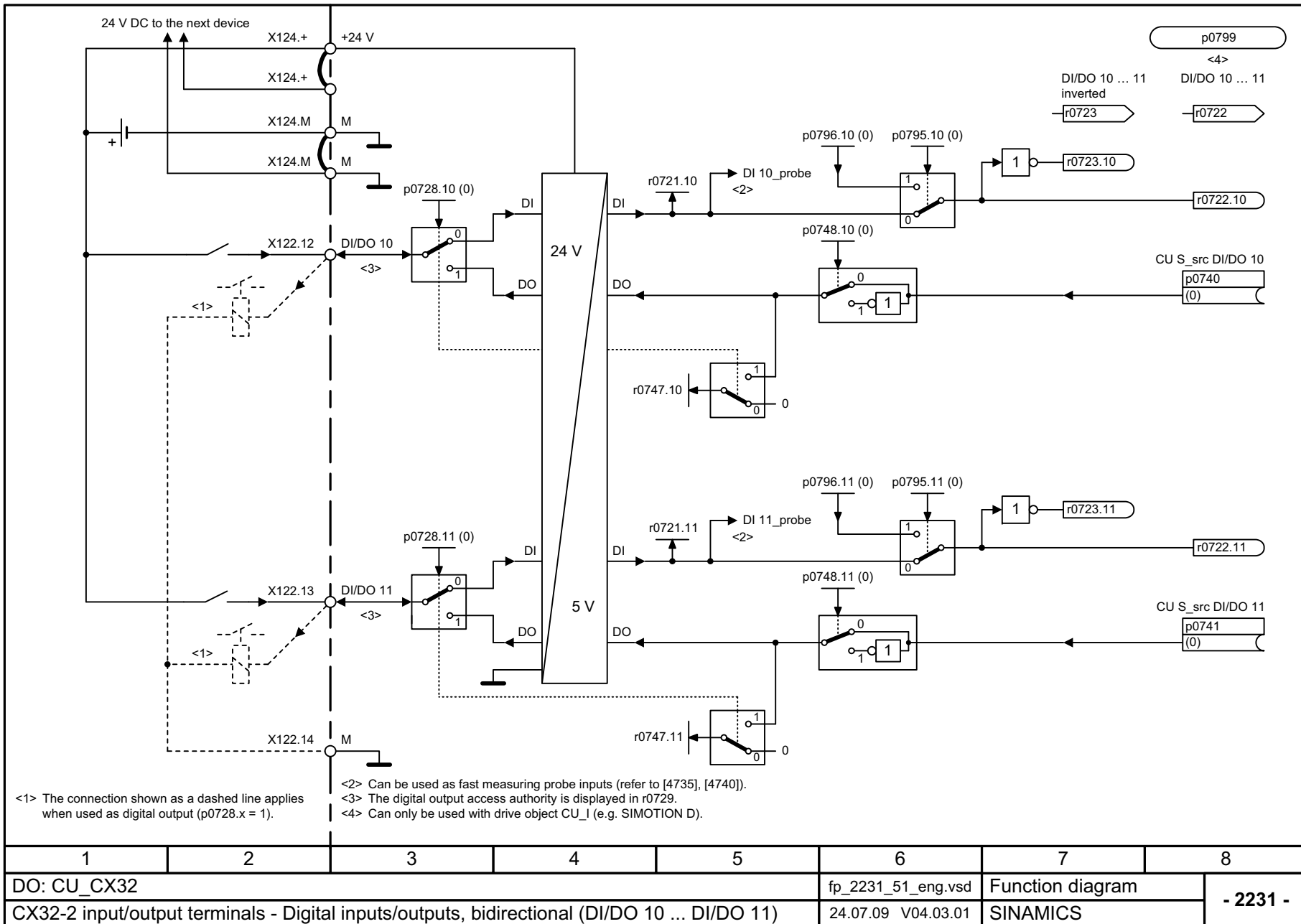


Fig. 2-38 2231 – Digital inputs/outputs, bidirectional (DI/DO 10 to DI/DO 11)

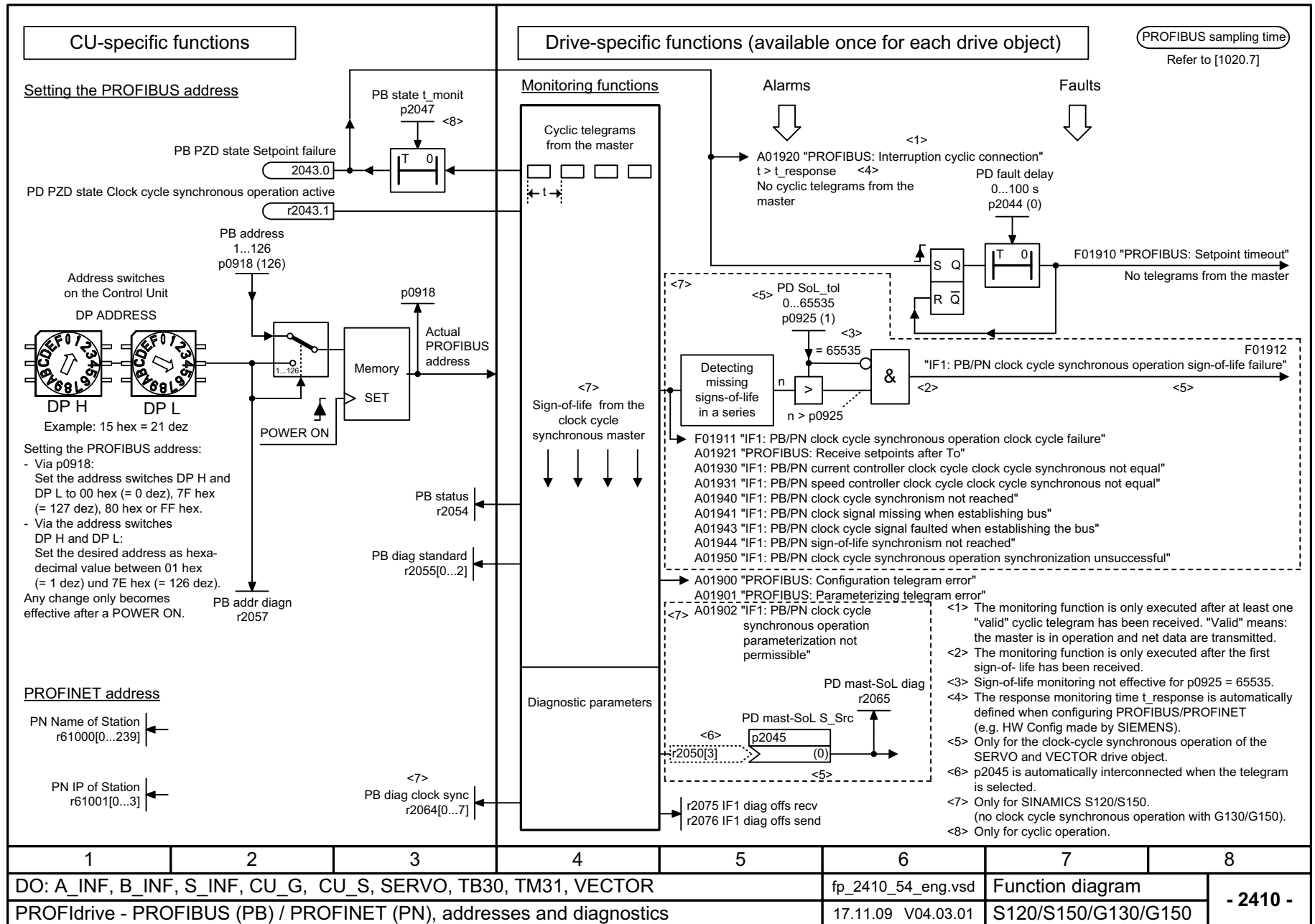
2.7 PROFIdrive

Function diagrams

2410 – PROFIBUS (PB) / PROFINET (PN), addresses and diagnostics	2-1429
2420 – Standard telegrams and process data (PZD)	2-1430
2422 – Manufacturer-specific telegrams and process data (PZD)	2-1431
2423 – Manufacturer-specific/free telegrams and process data (PZD)	2-1432
2425 – STW1_BM control word for metals sector interconnection (p2038 = 0)	2-1433
2426 – STW2_BM control word for metals sector interconnection (p2038 = 0)	2-1434
2427 – E_STW1_BM control word infeed for metals sector interconnection	2-1435
2428 – ZSW1_BM control word for metals sector interconnection (p2038 = 0)	2-1436
2429 – ZSW2_BM status word for metals sector interconnection (p2038 = 0)	2-1437
2430 – E_ZSW1_BM control word infeed for metals sector interconnection	2-1438
2433 – STW2_ENC control word for ENCODER interconnection	2-1439
2434 – ZSW2_ENC status word for ENCODER interconnection	2-1440
2439 – PZD receive signals interconnection, profile-specific	2-1441
2440 – PZD incoming signals interconnection, manufacturer-specific	2-1442
2441 – STW1 control word interconnection (p2038 = 2)	2-1443
2442 – STW1 control word interconnection (p2038 = 0)	2-1444
2443 – STW1 control word interconnection (p2038 = 1)	2-1445
2444 – STW2 control word interconnection (p2038 = 0)	2-1446
2445 – STW2 control word interconnection (p2038 = 1)	2-1447
2447 – E_STW1 control word infeed interconnection	2-1448
2449 – PZD outgoing signals interconnection, profile-specific	2-1449
2450 – PZD outgoing signals interconnection, manufacturer-specific	2-1450
2451 – ZSW1 status word interconnection (p2038 = 2)	2-1451
2452 – ZSW1 status word interconnection (p2038 = 0)	2-1452
2453 – ZSW1 status word interconnection (p2038 = 1)	2-1453
2454 – ZSW2 status word interconnection (p2038 = 0)	2-1454
2455 – ZSW2 status word interconnection (p2038 = 1)	2-1455

2456 – MELDW status word interconnection	2-1456
2457 – E_ZSW1 status word infeed interconnection	2-1457
2462 – POS_STW positioning control word interconnection (r0108.4 = 1)	2-1458
2463 – POS_STW1 positioning control word 1 interconnection (r0108.4 = 1)	2-1459
2464 – POS_STW2 positioning control word 2 interconnection (r0108.4 = 1)	2-1460
2466 – POS_ZSW1 positioning status word 1 interconnection (r0108.4 = 1)	2-1461
2467 – POS_ZSW2 positioning status word 2 interconnection (r0108.4 = 1)	2-1462
2468 – IF1 incoming telegram, free interconnection via BICO (p0922 = 999)	2-1463
2470 – IF1 outgoing telegram, free interconnection via BICO (p0922 = 999)	2-1464
2472 – IF1 status words, free interconnection	2-1465
2475 – STW1 control word 1 interconnection (r0108.4 = 1)	2-1466
2476 – SATZANW-Pos block selection interconnection (r0108.4 = 1)	2-1467
2479 – ZSW1 status word 1 interconnection (r0108.4 = 1)	2-1468
2480 – MDI_MOD-MDI mode interconnection (r0108.4 = 1)	2-1469
2481 – IF1 receive telegram, free interconnection via BICO (p0922 = 999)	2-1470
2483 – IF1 send telegram, free interconnection via BICO (p0922 = 999)	2-1471
2485 – IF2 receive telegram, free interconnection via BICO (p0922 = 999)	2-1472
2487 – IF2 send telegram, free interconnection via BICO (p0922 = 999)	2-1473
2489 – IF2 status words, free interconnection	2-1474
2491 – IF2 receive telegram, free interconnection via BICO (p0922 = 999)	2-1475
2493 – IF2 send telegram, free interconnection via BICO (p0922 = 999)	2-1476
2495 – CU_STW1 control word 1, Control Unit interconnection	2-1477
2496 – CU_ZSW1 status word 1, Control Unit interconnection	2-1478
2497 – A_DIGITAL interconnection	2-1479
2498 – E_DIGITAL interconnection	2-1480

Fig. 2-39 2410 – PROFIBUS (PB) / PROFINET (PN), addresses and diagnostics



<1> <2> <4>														PROFdrive sampling time Refer to [1020.7]													
PD Telegram select p0922 (999)																											
Interconnection is made according to														[2440] [2450] automatically													
Not suitably for sensorless vector control																											
Telegram														Appl.-Class													
1														1													
1														1, 4													
1, 4														4 DSC													
4 DSC														3													
3														3													
1														1													
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PROFdrive sampling time

Refer to [1020.7]

<1> <2> <4>
PD Telegr. select
p0922 (999)

Intercon- nection is made according to	[2440] [2450] automatically																			
	<5>		<5>		<5>		<5>		<5>		<5>		<5>		<5>		<5>		<5>	
Telegramm	102		103		105		106		110		111		116		118		125		126	
Appl.- Class	1, 4		1, 4		4 DSC		4 DSC		3		3		4 DSC		4 DSC		4 DSC		4 DSC	
PZD 1	STW1	ZSW1	STW1	ZSW1	STW1	ZSW1	STW1	ZSW1	STW1	ZSW1	STW1	ZSW1	STW1	ZSW1	STW1	ZSW1	STW1	ZSW1	STW1	ZSW1
PZD 2	NSOLL_B	NIST_B	NSOLL_B	NIST_B	NSOLL_B	NIST_B	NSOLL_B	NIST_B	SATZANW	AKTSATZ	POS_STW1	POS_ZSW1	NSOLL_B	NIST_B	NSOLL_B	NIST_B	NSOLL_B	NIST_B	NSOLL_B	NIST_B
PZD 3									POS_STW	POS_ZSW	POS_STW2	POS_ZSW2								
PZD 4	STW2	ZSW2	STW2	ZSW2	STW2	ZSW2	STW2	ZSW2	STW2	ZSW2	STW2	ZSW2	STW2	ZSW2	STW2	ZSW2	STW2	ZSW2	STW2	ZSW2
PZD 5	MOMRED	MELDW	MOMRED	MELDW	MOMRED	MELDW	MOMRED	MELDW	VERRIDE	MELDW	VERRIDE	MELDW	MOMRED	MELDW	MOMRED	MELDW	MOMRED	MELDW	MOMRED	MELDW
PZD 6	G1_STW	G1_ZSW	G1_STW	G1_ZSW	G1_STW	G1_ZSW	G1_STW	G1_ZSW	MDI_TAR	POS	MDI_TAR	POS	G1_STW	G1_ZSW	G2_STW	G2_ZSW	G1_STW	G1_ZSW	G1_STW	G1_ZSW
PZD 7		G1_XIST1	G2_STW	G1_XIST1	XERR	G1_XIST1	G2_STW	G1_XIST1	MDI_VELO	CITY	MDI_VELO	CITY	NIST_B	XERR	G1_XIST1	G3_STW	G2_XIST1	XERR	G1_XIST1	G2_STW
PZD 8				G1_XIST2	KPC	G1_XIST2	G1_XIST2	MDI_ACC												
PZD 9		G1_XIST2																		
PZD 10																				
PZD 11				G2_ZSW																
PZD 12				G2_XIST1					G2_XIST1	MDI_MODE		<3>	<3>			G2_XIST1		G3_XIST1	Reseved	
PZD 13																				
PZD 14																				
PZD 15				G2_XIST2					G2_XIST2							G2_XIST2		G3_XIST2		
PZD 16																AIST_GLATT		AIST_GLATT		
PZD 17																MSOLL_GLATT		MSOLL_GLATT		
PZD 18																PIST_GLATT		PIST_GLATT		
PZD 19																ITIST_GLATT		ITIST_GLATT		
PZD 20																				
PZD 21																				
PZD 22																				
PZD 23																				
PZD 24																				
PZD 25																				
PZD 26																				
PZD 27																				
PZD 28																				
PZD 29																				
PZD 30																				
PZD 31																				
PZD 32																				

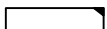
<1> Depending on the drive object, only specific telegrams can be used.

<2> If p0922 = 999 is changed to another value, the telegram is automatically assigned as specified in [2420] - [2423].

If p0922 ≠ 999 is changed to p0922 = 999, the "old" telegram assignment is maintained as specified in [2420] - [2423]!

<3> Can be freely connected.

<4> The maximum number of PZD words depends on the drive object type.

 = Position encoder signal

<5> Only for SINAMICS S120/S150.

1	2	3	4	5	6	7	8	
DO: SERVO, VECTOR					fp_2422_54_eng.vsd	Function diagram		- 2422 -
PROFIdrive – Manufacturer-specific/telegrams and Process Data (PZD)					25.11.09 V04.03.01	S120/S150/G130/G150		

Fig. 2-41 2422 – Manufacturer-specific telegrams and process data (PZD)

Fig. 2-42 2423 – Manufacturer-specific telegrams and process data (PZD)

<1*1*2*2> <4> PD Telegram select p0922 (999)																	PROFdrive sampling time Refer to [1020.7]									
Interconnection is made according to	<5>																		[2481] [2483]							
Telegram	136		220		352		370		371		390		391		392		999									
Appl.- Class	4 DSC		1		1		-		-		-		-		-		<3> - <3>									
PZD 1	STW1	ZSW1	STW1_BM	ZSW1_BM	STW1	ZSW1	E_STW1	E_ZSW1	E_STW1_BM	E_ZSW1_BM	CU_STW1	CU_ZSW1	CU_STW1	CU_ZSW1	CU_STW1	CU_ZSW1	STW1	ZSW1								
PZD 2	NSOLL_B	NIST_B	NSOLL_B	NIST_A	NSOLL_A	NIST_A_GLATT			<6>		IAIST	A_DIGITAL	E_DIGITAL	A_DIGITAL	E_DIGITAL	A_DIGITAL	E_DIGITAL	Receive telegram length freely selectable via central PROFdrive configuration in the master Transmit telegram length freely selectable via central PROFdrive configuration in the master								
PZD 3				IAIST	<3>	IAIST_GLATT			<6>		WARN_CODE			MT_STW	MT_ZSW	MT_STW	MT_ZSW									
PZD 4	STW2	ZSW2	STW2_BM	MIST <6>	<3>	MIST_GLATT			<6>		FAULT_CODE			MT1_ZS_F		MT1_ZS_F										
PZD 5	MOMRED	MELDW	M_ADD <8>	WARN_CODE	<3>	WARN_CODE			<6>		<6>			MT1_ZS_S		MT1_ZS_S										
PZD 6	G1_STW	G1_ZSW	M_LIM <7>	FAULT_CODE	<3>	FAULT_CODE			<6>				MT2_ZS_F		MT2_ZS_F											
PZD 7	G2_STW		<3>	ZSW2_BM					<6>				MT2_ZS_S		MT2_ZS_S											
PZD 8	XERR	G1_XIST1	<3>	<3>					<6>						MT3_ZS_F											
PZD 9		G1_XIST2	<3>	<3>											MT3_ZS_S											
PZD 10	KPC		<3>	<3>											MT4_ZS_F											
PZD 11		G2_ZSW													MT4_ZS_S											
PZD 12	M_VST	G2_XIST1													MT5_ZS_F											
PZD 13	Reserved														MT5_ZS_S											
PZD 14		G2_XIST2													MT6_ZS_F											
PZD 15															MT6_ZS_S											
PZD 16		AIST_GLATT																								
PZD 17		MSoll_GLATT																								
PZD 18		PIST_GLATT																								
PZD 19		ITIST_GLATT																								
PZD 20																										
PZD 21																										
PZD 22																										
PZD 23																										
PZD 24																										
PZD 25																										
PZD 26																										
PZD 27																										
PZD 28																										
PZD 29																										
PZD 30																										
PZD 31																										
PZD 32																										

<1> Depending on the drive object, only specific telegrams can be used.

<2> If p0922 = 999 is changed to another value, the telegram is automatically assigned as specified in [2420] - [2423].
If p0922 ≠ 999 is changed to p0922 = 999, the "old" telegram assignment is maintained as specified in [2420] - [2423]!

<3> In order to comply with the PROFdrive profile, PZD1 must be used as control word 1 (STW1) or status word 1 (ZSW1).
p2037 = 2 should be set if STW1 is not transferred with PZD1 as specified in the PROFdrive profile.

<4> The maximum number of PZD words depends on the drive object type.

<5> Only for S120/S150.

<6> Freely interconn.

<7> Not for V/f Steuerung.

<8> Preassignment, not disabled.

1	2	3	4	5	6	7	8
DO: A_INF, B_INF, CU_G, CU_S, S_INF, SERVO, VECTOR					fp_2423_54_eng.vsd	Function diagram	
PROFdrive - Manufacturer-specific/free telegrams and Process Data (PZD)					25.11.09 V04.03.01	S120/S150/G130/G150	

- 2423 -

Fig. 2-43 2425 – STW1_BM control word for metals sector interconnection (p2038 = 0)

Signal targets for STW1_BM					
Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW1.0	 = ON (pulses can be enabled) 0 = OFF1 (braking with ramp-function generator, then pulse cancellation and ready-to-power-up)	p0840[0] = r2090.0	[2501.3]	[2610]	-
STW1.1	1 = No OFF2 (enable is possible) 0 = OFF2 (immediate pulse cancellation and power-on inhibit)	p0844[0] = r2090.1	[2501.3]	[2610]	-
STW1.2	1 = No OFF3 (enable possible) 0 = OFF3 (braking with the OFF3 ramp p1135, then pulse cancellation and power-on inhibit)	p0848[0] = r2090.2	[2501.3]	[2610]	-
STW1.3	1 = Enable operation (pulses can be enabled) 0 = Inhibit operation (cancel pulses)	p2816[0] = r2090.3	[2501.3]	[2634.3]	-
STW1.4	1 = Operating condition (the ramp-function generator can be enabled) 0 = inhibit ramp-function generator (set the ramp-function generator output to zero)	p1140[0] = r2090.4	[2501.3]	[3060] [3070] [3080]	-
STW1.5	1 = Enable the ramp-function generator 0 = stop the ramp-function generator (freeze the ramp-function generator output)	p1141[0] = r2090.5	[2501.3]	[3060] [3070]	-
STW1.6	1 = Enable setpoint 0 = inhibit setpoint (set the ramp-function generator input to zero)	p1142[0] = r2090.6	[2501.3]	[3060] [3070] [3080]	-
STW1.7	 = Acknowledge faults	p2103[0] = r2090.7	[2546.1]	[8060]	-
STW1.8	Reserved	-	-	-	-
STW1.9	Reserved	-	-	-	-
STW1.10	1 = Control via PLC <2>	p0854[0] = r2090.10	[2501.3]	[2501]	-
STW1.11	Reserved	-	-	-	-
STW1.12	Reserved <3>	<3>	-	-	-
STW1.13	Reserved <3>	<3>	-	-	-
STW1.14	Reserved <3>	<3>	-	-	-
STW1.15	Reserved <3>	<3>	-	-	-

<1> Used in telegrams 220.

<2> STW1.10 must be set to ensure that the drive object accepts the process data (PZD).

<3> Interconnection is not disabled.

PROFIdrive sampling time
Refer to [1020.7]

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2425_54_eng.vsd	Function diagram	- 2425 -
PROFIdrive - STW1_BM-control word metal industry interconnection					06.08.09 V04.03.01	S120/S150/G130/G150	

PROFdrive sampling time
Refer to [1020.7]

Signal targets for STW2_BM					
Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW2.0	Command data set selection CDS, bit 0	p0810 = r2093.0	-	[8560]	-
STW2.1	Command data set selection CDS, bit 1	p0811 = r2093.1	-	[8560]	-
STW2.2	Drive data set selection DDS, bit 0	p0820[0] = r2093.2 <2>	-	[8565]	-
STW2.3	Drive data set selection DDS, bit 1	p0821[0] = r2093.3 <2>	-	[8565]	-
STW2.4	Drive data set selection DDS, bit 2	p0822[0] = r2093.4 <2>	-	[8565]	-
STW2.5	1 = Bypass ramp-function generator	p1122[0] = r2093.5	-	-	-
STW2.6	Reserved	-	-	-	-
STW2.7	1 = Load compensation	p1477[0] = r2093.7	-	-	-
STW2.8	1 = Droop enabled	p1492[0] = r2093.8	-	[6030]	-
STW2.9	1 = Speed controller enabled	p0856[0] = r2093.9 <2>	-	-	-
STW2.10	Reserved <2>	<2>	-	-	-
STW2.11	1 = Torque controlled operation 0 = Speed controlled operation	p1501[0] = r2093.11	-	-	-
STW2.12	Reserved <2>	<2>	-	-	-
STW2.13	Reserved <2>	<2>	-	-	-
STW2.14	Reserved <2>	<2>	-	-	-
STW2.15	Controller slave sign-of-life Toggle bit	p2081[15] = r2093.15	-	-	-

<1> Used in telegrams 220.
<2> Interconnection is not disabled.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2426_54_eng.vsd	Function diagram	- 2426 -
PROFdrive - STW2_BM-control word metal industry interconnection					06.08.09 V04.03.01	S120/S150/G130/G150	

Fig. 2-44 2426 – STW2_BM control word for metals sector interconnection (p2038 = 0)

PROFIdrive sampling time

Refer to [1020.7]

Signal targets for E_STW1_BM

Signal	Meaning	Interconnection parameters	[Function diagram] internal control word			[Function diagram] signal target			Inverted
			A_INF	B_INF <5>	S_INF <5>	A_INF	B_INF <5>	S_INF <5>	
STW1.0	ON (close pre-charging/line contactor, pulses can be enabled) 0 = OFF1 (reduce Vdc along a ramp, pulse cancel and open pre-charging/line contactor)	p0840[0] = r2090.0	[8920.3]	[8720.3]	[8820.3]	[8932]	[8732]	[8832]	-
STW1.1	1 = No OFF2 (enable is possible) 0 = OFF2 (immediate pulse cancellation and power-on inhibit)	p0844[0] = r2090.1	[8920.3]	[8720.3]	[8820.3]	[8932]	[8732]	[8832]	-
STW1.2	Reserved	-	-	-	-	-	-	-	-
STW1.3	1 = Enable operation (pulses can be enabled) <3> 0 = Inhibit operation (cancel pulses)	p0852[0] = r2090.3	[8920.3]	-	[8820.3]	[8932]	-	[8832]	-
STW1.4	Reserved	-	-	-	-	-	-	-	-
STW1.5	1 = Inhibit motoring operation <4>	p3532= r2090.5	[8920.3]	-	-	[8920]	-	-	-
STW1.6	1 = Inhibit regenerative operation <3>	p3533= r2090.6	[8920.3]	-	[8820.3]	[8920]	-	[8820]	-
STW1.7	Acknowledge faults	p2103[0] = r2090.7	[2546.3]			[8060]			-
STW1.8	Reserved <6>	<6>	-	-	-	-	-	-	-
STW1.9	Reserved <6>	<6>	-	-	-	-	-	-	-
STW1.10	1 = Control via PLC <2>	p0854[0] = r2090.10	[8920.3]	[8720.3]	[8820.3]	[8920]	[8720]	[8820]	-
STW1.11	Reserved <6>	<6>	-	-	-	-	-	-	-
STW1.12	Reserved <6>	<6>	-	-	-	-	-	-	-
STW1.13	Reserved <6>	<6>	-	-	-	-	-	-	-
STW1.14	Reserved <6>	<6>	-	-	-	-	-	-	-
STW1.15	Controller-sign-of-life Toggle Bit	p2080[15] = r2090.15	-	-	-	-	-	-	-

<1> Used in telegram 371.

<2> STW1.10 must be set to ensure that the drive object accepts the process data (PZD).

<3> Only for A_INF, S_INF

<4> Only for A_INF

<5> B_INF and S_INF only for S120.

<6> Interconnection is not disabled.

1	2	3	4	5	6	7	8
DO: A_INF, B_INF, S_INF					fp_2427_55_eng.vsd	Function diagram	- 2427 -
PROFIdrive - E_STW1-control word infeed metal industry interconnection					06.08.09 V04.03.01	SINAMICS S120/S150	

Fig. 2-45 2427 – E_STW1_BM control word infeed for metals sector interconnection

PROFdrive Abtastzeit

siehe [1020.7]

Signal sources for ZSW1_BM

Signal	Meaning	Interconnection parameters	[Function diagram] Internal status word	[Function diagram] signal source	Inverted <2>
ZSW1.0	1 = Ready to power-up	p2080[0] = r0899.0	[2503.7]	[2610]	-
ZSW1.1	1 = Ready to operate	p2080[1] = r0899.1	[2503.7]	[2610]	-
ZSW1.2	1 = Operation enabled	p2080[2] = r2811.0	[2503.7]	[2610]	-
ZSW1.3	1 = Fault present	p2080[3] = r2139.3	[2548.7]	[8060]	-
ZSW1.4	1 = No coast down active (OFF2 inactive)	p2080[4] = r0899.4	[2503.7]	[2610]	-
ZSW1.5	1 = No fast stop active (OFF3 inactive)	p2080[5] = r0899.5	[2503.7]	[2610]	-
ZSW1.6	1 = Power-on inhibit active	p2080[6] = r0899.6	[2503.7]	[2610]	-
ZSW1.7	1 = Alarm present	p2080[7] = r2139.7	[2548.7]	[8065]	-
ZSW1.8	1 = Speed setpoint - actual value deviation within tolerance t_off	p2080[8] = r2197.7	[2534.7]	[8010]	-
ZSW1.9	1 = Control requested <3>	p2080[9] = r0899.9	[2503.7]	[2503]	-
ZSW1.10	1 = f or n comparison value reached/exceeded	p2080[10] = r2199.1	[2536.7]	[8010]	-
ZSW1.11	1 = I, M, or P limit not reached <4>	p2080[11] = r1407.7	[2522.7]	[5610] [6060] <5>	✓
ZSW1.12	1 = Open holding brake	p2080[12] = r0899.12	[2503.7]	[2701]	-
ZSW1.13	Reserved <6>	<6>	-	-	-
ZSW1.14	Reserved <6>	<6>	-	-	-
ZSW1.15	Reserved <6>	<6>	-	-	-

<1> Used in telegram 220.

<2> The ZSW1 is generated using the binector-connector converter (BI: p2080[0...15], inversion: p2088[0].0...p2088[0].15)

<3> The drive object is ready to accept data.

<4> Not for VECTOR V/f.

<5> Only for SINAMICS S120.

<6> Interconnection is not disabled.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2428_54_eng.vsd	Function diagram	- 2428 -
PROFdrive - ZSW1_BM-status word metal industry interconnection					06.08.09 V04.03.01	S120/S150/G130/G150	

Fig. 2-46 2428 – ZSW1_BM control word for metals sector interconnection (p2038 = 0)

PROFIdrive sampling time
Refer to [1020.7]

Signal sources for ZSW2_BM <1>					
Signal	Meaning	Interconnection parameters	[Function diagram] internal status word	[Function diagram] signal source	Inverted
ZSW2.0	Reserved <3>	<3>	-	-	-
ZSW2.1	Reserved <3>	<3>	-	-	-
ZSW2.2	Reserved <3>	<3>	-	-	-
ZSW2.3	Reserved <3>	<3>	-	-	-
ZSW2.4	Reserved <3>	<3>	-	-	-
ZSW2.5	1 = Alarm class bit 0	p2081[5] = r2139.11	-	[2548]	-
ZSW2.6	1 = Alarm class bit 1	p2081[6] = r2139.12	-	[2548]	-
ZSW2.7	Reserved	-	-	-	-
ZSW2.8	Reserved	-	-	-	-
ZSW2.9	1 = Speed setpoint controll activ <2>	p2081[9] = r1407.11	-	-	-
ZSW2.10	1 = Upper torque limit <2>	p2081[10] = r1407.8	-	-	-
ZSW2.11	1 = Lower torque limit <2>	p2081[11] = r1407.9	-	-	-
ZSW2.12	1 = Encoderless operation due to fault	p2081[12] = r1407.13	-	-	-
ZSW2.13	SS1_ACTIVE	p2081[13] = r9773.2	-	-	-
ZSW2.14	POWER-REMOVED	p2081[14] = r9773.1	-	-	-
ZSW2.15	Controller slave sign-of-life Toggle bit	p2081[15] = r2093.15	-	-	-

<1> Used in telegrams 220.

<2> Not for VECTOR V/f.

<3> Interconnection is not disabled.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2429_54_eng.vsd	Function diagram	- 2429 -
PROFIdrive - ZSW2_BM status word metal industry interconnection					06.08.09 V04.03.01	S120/S150/G130/G150	

Fig. 2-47 2429 – ZSW2_BM status word for metals sector interconnection (p2038 = 0)

PROFdrive sampling time

Refer to [1020.7]

Signal sources for E_ZSW1_BM									
Signal	Meaning	Interconnection parameters	[Function diagram] signal source			[Function diagram] internal status word			Inverted
			A_INF	B_INF <3>	S_INF <3>	A_INF	B_INF <3>	S_INF <3>	
ZSW1.0	1 = Ready to power-up	p2080[0] = r0899.0	[8932]	[8732]	[8832]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.1	1 = Ready to operate	p2080[1] = r0899.1	[8932]	[8732]	[8832]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.2	1 = Operation enabled	p2080[2] = r0899.2	[8932]	[8732]	[8832]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.3	1 = Fault present	p2080[3] = r2139.3	[8060]			[2548.7]			-
ZSW1.4	1 = No OFF2 effective	p2080[4] = r0899.4	[8932]	[8732]	[8832]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.5	Reserved	-	-	-	-	-	-	-	-
ZSW1.6	1 = Power-on inhibit	p2080[6] = r0899.6	[8932]	[8732]	[8832]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.7	1 = Alarm present	p2080[7] = r2139.7	[8065]			[2548.7]			-
ZSW1.8	Reserved <4>	<4>	-	-	-	-	-	-	-
ZSW1.9	1 = PLC requests control <2>	p2080[9] = r0899.9	[8926]	[8726]	[8826]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.10	Reserved <4>	<4>	-	-	-	-	-	-	-
ZSW1.11	1 = Pre-charging completed	p2080[11] = r0899.11	[8950]	[8750]	[8850]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.12	1 = Line contactor closed	p2080[12] = r0899.12	[8934]	[8734]	[8834]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.13	Reserved <4>	<4>	-	-	-	-	-	-	-
ZSW1.14	Reserved <4>	<4>	-	-	-	-	-	-	-
ZSW1.15	DO-sign-of-life Toggle Bit	r2080[15] = r2090.15	-	-	-	-	-	-	-

<1> Used in telegram 371.
 <2> The drive object is ready to accept data.
 <3> Only for S120.
 <4> Interconnection is not disabled.

1	2	3	4	5	6	7	8
DO: A_INF, B_INF, S_INF					fp_2430_55_eng.vsd	Function diagram	
PROFdrive - E_ZSW1_BM-status word infeed metal industry interconnection					06.08.09 V04.03.01	SINAMICS S120/S150	
							- 2430 -

Fig. 2-48 2430 – E_ZSW1_BM control word infeed for metals sector interconnection

PROFIdrive sampling time
Refer to [1020.7]

Signal targets for STW2_ENC

<1>

Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW1.0	Reserved	-	-	-	-
STW1.1	Reserved	-	-	-	-
STW1.2	Reserved	-	-	-	-
STW1.3	Reserved	-	-	-	-
STW1.4	Reserved	-	-	-	-
STW1.5	Reserved	-	-	-	-
STW1.6	Reserved	-	-	-	-
STW1.7	 = Acknowledge faults	p2103[0] = r2090.7	[2546.1]	[8060]	-
STW1.8	Reserved	-	-	-	-
STW1.9	Reserved	-	-	-	-
STW1.10	1 = Control via PLC <2>	p0854[0] = r2090.10	[2501.3]	[2501]	-
STW1.11	Reserved	-	-	-	-
STW1.12	Master-sign-of-life, bit 0	p2045= r2050	-	[2410]	-
STW1.13	Master-sign-of-life, bit 1				
STW1.14	Master-sign-of-life, bit 2				
STW1.15	Master-sign-of-life, bit 3				

<1> Used in telegrams 81, 82, 83.

<2> STW1.10 must be set to ensure that the drive object accepts the process data (PZD).

1	2	3	4	5	6	7	8
DO: ENCODER			fp_2433_54_eng.vsd			Function diagram	- 2433 -
PROFIdrive - STW2_ENC-control word ENCODER interconnection			27.11.09 V04.03.01			SINAMICS S120/S150	

Fig. 2-49 2433 – STW2_ENC control word for ENCODER interconnection

PROFdrive sampling time
Refer to [1020.7]

Signal sources for ZSW2_ENC					
Signal	Meaning	Interconnection parameters	[Function diagram] signal source	[Function diagram] internal status word	Inverted
ZSW1.0	Reserved	-	-	-	-
ZSW1.1	Reserved	-	-	-	-
ZSW1.2	Reserved	-	-	-	-
ZSW1.3	1 = Fault present	p2080[3] = r2139.3	[2548.7]	[8060]	-
ZSW1.4	Reserved	-	-	-	-
ZSW1.5	Reserved	-	-	-	-
ZSW1.6	Reserved	-	-	-	-
ZSW1.7	1 = Alarm present	p2080[7] = r2139.7	[2548.7]	[8065]	-
ZSW1.8	Reserved	-	-	-	-
ZSW1.9	1 = Control requested <2>	p2080[9] = r0899.9	[2503.7]	[2503]	-
ZSW1.10	Reserved	-	-	-	-
ZSW1.11	Reserved	-	-	-	-
ZSW1.12	Slave-sign-of-life bit 0	p2045 = r2050[3]	-	[2410]	-
ZSW1.13	Slave-sign-of-life bit 1				
ZSW1.14	Slave-sign-of-life bit 2				
ZSW1.15	Slave-sign-of-life bit 3				

<1> Used in telegram 81, 82, 83.
<2> The drive object is ready to accept data.

1	2	3	4	5	6	7	8	
DO: ENCODER					fp_2434_54_eng.vsd	Function diagram		- 2434 -
PROFIdrive - ZSW2-ENC-status word ENCODER interconnection					27.11.09 V04.03.01	S120/S150/G130/G150		

Fig. 2-50 2434 – ZSW2_ENC status word for ENCODER interconnection

PROFdrive sampling time

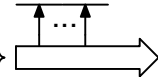
Refer to [1020.7]

Signal receivers for PZD receive signals

Signal	Meaning	PROFdrive Signal No.	<1>		<2>	
			Interconnection parameter	Function diagram	Data type	Normalization
STW1	Control word 1	1	(bitwise)	[2442][2443] <3> [2475] <3>	U16	-
STW2	Control word 2	3	(bitwise)	[2444] [2445]	U16	-
NSOLL_A	Speed setpoint A (16-bit)	5	p1070 (Erw. Soll.) p1155	[3030.2] [3080.4] <3>	I16	4000 hex $\hat{=}$ p2000
NSOLL_B	Speed setpoint B (32-bit)	7	p1070 (Erw. Soll.) p1155 p1430 (DSC) <3>	[3030.2] [3080.4] [3090.8] <3>	I32	4000 0000 hex $\hat{=}$ p2000
<3> G1_STW	Encoder 1 control word	9	p0480[0]	[4720]	U16	-
<3> G2_STW	Encoder 2 control word	13	p0480[1]	[4720]	U16	-
<3> G3_STW	Encoder 3 control word	17	p0480[2]	[4720]	U16	-
A_DIGITAL	Digital output (16-bit)	22	(bitwise)	[2497]	U16	-
<3> XERR	Position deviation	25	p1190	[3090.5]	I32	-
<3> KPC	Position controller gain factor	26	p1191	[3090.5]	I32	-
SATZANW	Pos block selection	32	(bitwise)	[2476]	U16	-
MDI_TARPOS	MDI position	34	p2642	[3618]	I32	1 hex $\hat{=}$ 1 LU
MDI_VELOCITY	MDI velocity	35	p2643	[3618]	I32	1 hex $\hat{=}$ 1000 LU/min
MDI_ACC	MDI acceleration override	36	p2644	[3618]	I16	4000 hex $\hat{=}$ 100%
MDI_DEC	MDI deceleration override	37	p2645	[3618]	I16	4000 hex $\hat{=}$ 100%
MDI_MOD	MDI mode	38	(bitwise)	[2480]	U16	-
<4> STW2_ENC	Status word 2 ENCODER	80	(bitwise)	[2433]	U16	-

PROFIBUS
PROFINETPROFdrive
receive telegram

Header
Drive object 1
Drive object 2
.
.
.
Drive object n
.
.
.
Drive object m
Trailer

[2468], [2481]
r2090...r2095 bit
r2050[0...n] WORD
r2060[0...30] DWORDTelegram assignment
according to p0922
[2420]

<1> When selecting a standard telegram or a manufacturer-specific telegram via p0922, these interconnection parameters of the command data set CDS0 are automatically set.
<2> Data type according to the PROFdrive profile: I16 = Integer16, I32 = Integer32, U16 = Unsigned16, U32 = Unsigned32.
<3> Only for SINAMICS S120.
<4> Only for ENCODER.

1	2	3	4	5	6	7	8
DO: A_INF, B_INF, CU_CX32, CU_G, CU_S, ENCODER, S_INF, SERVO, VECTOR					fp_2439_54_eng.vsd	Function diagram	- 2439 -
PROFdrive - PZD receive signals, connection of profile-specific					11.11.09 V04.03.01	S120/S150/G130/G150	

Fig. 2-51

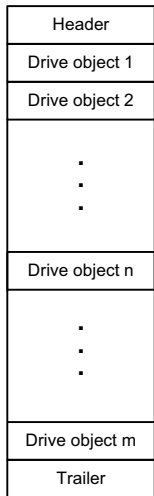
2439 – PZD receive signals interconnection, profile-specific

PROFdrive sampling time
Refer to [1020.7]

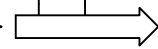
PROFIBUS
PROFINET



PROFdrive
receive telegram



[2468], [2481]
r2090...r2095 bit
r2050[0...n] WORD
r2060[0...30] DWORD



Telegram assignment
according to p0922
[2420]

Signal receivers for PZD receive signals

Signal	Meaning	PROFdrive Signal No.	<1>		<2>	
			Interconnection parameter	Function diagram	Data type	Normalization
MOMRED	Torque reduction	101	p1542	[5610.2]	I16	4000 hex $\hat{=}$ p2003
MT_STW	Measuring probe control word	130	p0682	-	U16	
POS_STW	Pos control word	203	(bitwise)	[2462]	U16	
OVERRIDE	Pos velocity override	205	p2646	[3630]	I16	4000 hex $\hat{=}$ 100%
POS_STW1	Pos control word 1	220	(bitwise)	[2463]	U16	
POS_STW2	Pos control word 2	222	(bitwise)	[2464]	U16	
MDI_MOD	Pos MDI mode	229	p2654	[3620]	U16	
M_LIM	Torque limit	310	p1503/p1552/p1554	-	U16	
M_ADD	Additional torque	311	p1495	-	U16	
E_STW1	Control word 1 for Active Infeed (ALM, SMART)	320	(bitwise)	[2447]	U16	
STW1_BM	Control word 1, variant for BM	322	(bitwise)	[2425]	U16	
STW2_BM	Control word 2, variant for BM	324	(bitwise)	[2426]	U16	
E_STW1_BM	Control word 1 for Active Infeed	326	(bitwise)	[2427]	U16	
CU_STW1	Control word 1 for Control Unit	500	(bitwise)	[2495]	U16	

<1> When selecting a standard telegram or a manufacturer-specific telegram via p0922, these interconnection parameters of the command data set CDS0 are automatically set.
 <2> Data type according to the PROFdrive profile: I16 = Integer16, I32 = Integer32, U16 = Unsigned16, U32 = Unsigned32.
 <3> Only for SINAMICS S120.

1	2	3	4	5	6	7	8
DO: A_INF, B_INF, CU_CX32, CU_G, CU_S, S_INF, SERVO, VECTOR					fp_2440_54_eng.vsd	Function diagram	- 2440 -
PROFdrive - PZD receive signals, connection of manufacturer-specific					04.11.09 V04.03.01	S120/S150/G130/G150	

Fig. 2-52 2440 – PZD incoming signals interconnection, manufacturer-specific

PROFdrive sampling time
Refer to [1020.7]

Signal targets for STW1 in Interface Mode VIK-NAMUR (p2038 = 2)					
Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW1.0	 = ON (pulses can be enabled) 0 = OFF1 (braking with ramp-function generator, then pulse cancellation & ready-to-power-up)	p0840[0] = r2090.0	[2501.3]	[2610]	-
STW1.1	1 = No OFF2 (enable is possible) <3> 0 = OFF2 (immediate pulse cancellation and power-on inhibit)	p0844[0] = r2090.1	[2501.3]	[2610]	-
STW1.2	1 = No OFF3 (enable possible) <3> 0 = OFF3 (braking with the OFF3 ramp p1135, then pulse cancellation and power-on inhibit)	p0848[0] = r2090.2	[2501.3]	[2610]	-
STW1.3	1 = Enable operation (pulses can be enabled) 0 = Inhibit operation (cancel pulses)	p0852[0] = r2090.3	[2501.3]	[2610]	-
STW1.4	1 = Operating condition (the ramp-function generator can be enabled) 0 = Inhibit ramp-function generator (set the ramp-function generator output to zero)	p1140[0] = r2090.4	[2501.3]	[3060] [3070] [3080]	-
STW1.5	1 = Enable the ramp-function generator 0 = Stop the ramp-function generator (freeze the ramp-function generator output)	p1141[0] = r2090.5	[2501.3]	[3060] [3070]	-
STW1.6	1 = Enable setpoint 0 = Inhibit setpoint (set the ramp-function generator input to zero)	p1142[0] = r2090.6	[2501.3]	[3060] [3070] [3080]	-
STW1.7	 = Acknowledge faults	p2103[0] = r2090.7	[2546.1]	[8060]	-
STW1.8	Reserved	-	-	-	-
STW1.9	Reserved	-	-	-	-
STW1.10	1 = Control via PLC <2>	p0854[0] = r2090.10	[2501.3]	[2501]	-
STW1.11	1 = Direction reversal	p1113[0] = r2090.11	[2505.3]	[3040]	-
STW1.12	Reserved	-	-	-	-
STW1.13	Reserved	-	-	-	-
STW1.14	Reserved	-	-	-	-
STW1.15	1 = CDS selection	<4> p0810[0] = 2090.15	-	[8560]	-

<1> Used in telegram 20.

<2> STW1.10 must be set to ensure that the drive object accepts the process data (PZD).

<3> OC = Operating condition.

<4> Interconnection is not disabled.

1	2	3	4	5	6	7	8
DO: VECTOR					fp_2441_54_eng.vsd	Function diagram	- 2441 -
PROFdrive - STW1 control word interconnection (p2038 = 2)					14.08.08 V04.03.01	S120/S150/G130/G150	

Fig. 2-53 2441 – STW1 control word interconnection (p2038 = 2)

Fig. 2-54 2442 – STW1 control word interconnection (p2038 = 0)

Signal targets for STW1 in Interface Mode SINAMICS (p2038 = 0)						<1>
Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted	
STW1.0	 = ON (pulses can be enabled) 0 = OFF1 (braking with ramp-function generator, then pulse cancellation and ready-to-power-up)	p0840[0] = r2090.0	[2501.3]	[2610]	-	
STW1.1	1 = No OFF2 (enable is possible) 0 = OFF2 (immediate pulse cancellation and power-on inhibit)	p0844[0] = r2090.1	[2501.3]	[2610]	-	
STW1.2	1 = No OFF3 (enable possible) 0 = OFF3 (braking with the OFF3 ramp p1135, then pulse cancellation and power-on inhibit)	p0848[0] = r2090.2	[2501.3]	[2610]	-	
STW1.3	1 = Enable operation (pulses can be enabled) 0 = Inhibit operation (cancel pulses)	p0852[0] = r2090.3	[2501.3]	[2610]	-	
STW1.4	1 = Operating condition (the ramp-function generator can be enabled) 0 = inhibit ramp-function generator (set the ramp-function generator output to zero)	p1140[0] = r2090.4	[2501.3]	[3060] [3070] [3080]	-	
STW1.5	1 = Enable the ramp-function generator 0 = stop the ramp-function generator (freeze the ramp-function generator output)	p1141[0] = r2090.5	[2501.3]	[3060] [3070]	-	
STW1.6	1 = Enable setpoint 0 = inhibit setpoint (set the ramp-function generator input to zero)	p1142[0] = r2090.6	[2501.3]	[3060] [3070] [3080]	-	
STW1.7	 = Acknowledge faults	p2103[0] = r2090.7	[2546.1]	[8060]	-	
STW1.8	Reserved	-	-	-	-	
STW1.9	Reserved	-	-	-	-	
STW1.10	1 = Control via PLC <2>	p0854[0] = r2090.10	[2501.3]	[2501]	-	
STW1.11	1 = Direction reversal <3>	p1113[0] = r2090.11	[2505.3]	[3040]	-	
STW1.12	Reserved	-	-	-	-	
STW1.13	1 = Motorized potentiometer, setpoint, raise <3>	p1035[0] = r2090.13	[2505.3]	[3020]	-	
STW1.14	1 = Motorized potentiometer, setpoint, lower <3>	p1036[0] = r2090.14	[2505.3]	[3020]	-	
STW1.15	Reserved	-	-	-	-	

PROFdrive sampling time
Refer to [1020.7]

<1> Used in telegrams 1, 2, 3, 4, 5, 6, 352.

<2> STW1.10 must be set to ensure that the drive object accepts the process data (PZD).

<3> Only for "expanded setpoint channel" and "extended ramp-function generator".

<4> OC = Operating condition.

<5> Only for SINAMICS S120.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2442_54_eng.vsd	Function diagram	- 2442 -
PROFdrive - STW1 control word interconnection (p2038 = 0)					27.11.08 V04.03.01	S120/S150/G130/G150	

Fig. 2-55 2443 – STW1 control word interconnection (p2038 = 1)

Signal targets for STW1 in Interface Mode SIMODRIVE 611 universal (p2038 = 1)					
Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW1.0	 = ON (pulses can be enabled) 0 = OFF1 (braking with ramp-function generator, then pulse cancellation, ready-to-power-up)	p0840[0] = r2090.0	[2501.3]	[2610]	-
STW1.1	1 = No OFF2 (enable is possible) <3> 0 = OFF2 (immediate pulse cancellation and power-on inhibit)	p0844[0] = r2090.1	[2501.3]	[2610]	-
STW1.2	1 = No OFF3 (enable possible) <3> 0 = OFF3 (braking with the OFF3 ramp p1135, then pulse cancellation and power-on inhibit)	p0848[0] = r2090.2	[2501.3]	[2610]	-
STW1.3	1 = Enable operation (pulses can be enabled) 0 = Inhibit operation (cancel pulses)	p0852[0] = r2090.3	[2501.3]	[2610]	-
STW1.4	1 = Operating condition (the ramp-function generator can be enabled) 0 = Inhibit ramp-function generator (set the ramp-function generator output to zero)	p1140[0] = r2090.4	[2501.3]	[3060] [3070] [3080]	-
STW1.5	1 = Enable the ramp-function generator 0 = Stop the ramp-function generator (freeze the ramp-function generator output)	p1141[0] = r2090.5	[2501.3]	[3060] [3070]	-
STW1.6	1 = Enable setpoint 0 = Inhibit setpoint (set the ramp-function generator input to zero)	p1142[0] = r2090.6	[2501.3]	[3060] [3070] [3080]	-
STW1.7	 = Acknowledge faults	p2103[0] = r2090.7	[2546.1]	[8060]	-
STW1.8	Reserved	-	-	-	-
STW1.9	Reserved	-	-	-	-
STW1.10	1 = Control via PLC <2>	p0854[0] = r2090.10	[2501.3]	[2501]	-
STW1.11	1 = Ramp-function generator active	p2148[0] = r2090.11	-	[8010]	-
STW1.12	1 = Unconditionally open the holding brake	p0855[0] = r2090.12	[2501.3]	[2701]	-
STW1.13	Reserved	-	-	-	-
STW1.14	1 = Closed-loop torque control active 0 = Closed-loop speed control active	p1501[0] = r2090.14	[2520.3]	[5060] [6060]	-
STW1.15	Reserved	-	-	-	-

<1> Used in telegrams 1, 2, 3, 4, 5, 6, 102, 103, 105, 106, 116, 118.

<2> STW1.10 must be set to ensure that the drive object accepts the process data ((PZD)).

<3> OC -> Operating condition.

PROFIdrive sampling time
Refer to [1020.7]

1	2	3	4	5	6	7	8
DO: SERVO					fp_2443_55_eng.vsd	Function diagram	- 2443 -
PROFIdrive - STW1 control word interconnection (p2038 = 1)					06.10.08 V04.03.01	SINAMICS S120/S150	

PROFdrive sampling time
Refer to [1020.7]

Signal targets for STW2 in Interface Mode SINAMICS (p2038 = 0)

Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW2.0	Drive data set selection DDS, bit 0	p0820[0] = r2093.0 <4> r2092.0	-	[8565]	-
STW2.1	Drive data set selection DDS, bit 1	p0821[0] = r2093.1 <4> r2092.1	-	[8565]	-
STW2.2	Drive data set selection DDS, bit 2	p0822[0] = r2093.2 <4> r2092.2	-	[8565]	-
STW2.3	Drive data set selection DDS, bit 3	p0823[0] = r2093.3 <4> r2092.3	-	[8565]	-
STW2.4	Drive data set selection DDS, bit 4	p0824[0] = r2093.4 <4> r2092.4	-	[8565]	-
STW2.5	Reserved	-	-	-	-
STW2.6	Reserved	-	-	-	-
STW2.7	1 = Parking axis	p0897 = r2093.7 <4> r2092.7	-	-	-
STW2.8	1 = Traverse to fixed endstop <2> <3> <5>	p1545[0] = r2093.8	[2520.2]	[8012]	-
STW2.9	Reserved	-	-	-	-
STW2.10	Reserved	-	-	-	-
STW2.11	 1 = Motor changeover, feedback Signal	p0828 = r2093.11 <4> r2092.11	-	-	-
STW2.12	Master sign-of-life, bit 0 <5>	p2045 = r2050[3] <4> r2050[2]	-	[2410]	-
STW2.13	Master sign-of-life, bit 1 <5>				
STW2.14	Master sign-of-life, bit 2 <5>				
STW2.15	Master sign-of-life, bit 3 <5>				

<1> Used in telegrams 2, 3, 4, 5, 6, 9, 110 and 111. <2> Not for telegrams 9, 110 and 111. <3> Only for SINAMICS S120. <4> Only for Telegram 9. <5> Not for Vector V/f.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2444_54_eng.vsd	Function diagram	- 2444 -
PROFdrive - STW2 control word interconnection (p2038 = 0)					26.09.08 V04.03.01	S120/S150/G130/G150	

Fig. 2-56 2444 – STW2 control word interconnection (p2038 = 0)

PROFdrive sampling time
Refer to [1020.7]

Signal targets for STW2 in Interface Mode SIMODRIVE 611 universal (p2038 = 1)					
Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW2.0	Drive data set selection DDS, bit 0	p0820[0] = r2093.0	-	[8565]	-
STW2.1	Drive data set selection DDS, bit 1	p0821[0] = r2093.1	-	[8565]	-
STW2.2	Drive data set selection DDS, bit 2	p0822[0] = r2093.2	-	[8565]	-
STW2.3	Reserved	-	-	-	-
STW2.4	1 = Bypass ramp-function generator <3>	p1122[0] = r2093.4	-	[3060] [3070]	-
STW2.5	Reserved	-	-	-	-
STW2.6	1 = Integrator inhibit, speed controller <2>	p1477[0] = r2093.6	-	[5040] [5210]	-
STW2.7	1 = Parking axis selection	p0897 = r2093.7	-	-	-
STW2.8	1 = Traverse to fixed endstop	p1545[0] = r2093.8	[2520.2]	[8012]	-
STW2.9	Drive data set selection DDS, bit 3	p0823[0] = r2093.9	-	[8565]	-
STW2.10	Drive data set selection DDS, bit 4	p0824[0] = r2093.10	-	[8565]	-
STW2.11	 Motor changeover, feedback signal	p0828 = r2093.11	-	-	-
STW2.12	Master sign-of-life, bit 0	p2045 = r2050[3]	-	[2410]	-
STW2.13	Master sign-of-life, bit 1				
STW2.14	Master sign-of-life, bit 2				
STW2.15	Master sign-of-life, bit 3				

<1> Used in telegrams 2, 3, 4, 5, 6, 102, 103, 105, 106, 116, 118.

<2> For a 1 signal, the integral component of the speed controller is cleared and the integrator is inhibited.

<3> Only if the function module "extended setpoint channel" is active (r0108.8 = 1).

1	2	3	4	5	6	7	8
DO: SERVO					fp_2445_55_eng.vsd	Function diagram	- 2445 -
PROFdrive - STW2 control word interconnection (p2038 = 1)					10.06.08 V04.03.01	SINAMICS S120/S150	

Fig. 2-57 2445 – STW2 control word interconnection (p2038 = 1)

PROFdrive sampling time

Refer to [1020.7]

Signal targets for E_STW1									
Signal	Meaning	Interconnection parameters	[Function diagram] internal control word			[Function diagram] signal target			Inverted
			A_INF	B_INF<5>	S_INF<5>	A_INF	B_INF<5>	S_INF<5>	
STW1.0	ON (close pre-charging/line contactor, pulses can be enabled) 0 = OFF1 (reduce Vdc along a ramp, pulse cancel and open pre-charging/line contactor)	p0840[0] = r2090.0	[8920.3]	[8720.3]	[8820.3]	[8932]	[8732]	[8832]	-
STW1.1	1 = No OFF2 (enable is possible) 0 = OFF2 (immediate pulse cancellation and power-on inhibit)	p0844[0] = r2090.1	[8920.3]	[8720.3]	[8820.3]	[8932]	[8732]	[8832]	-
STW1.2	Reserved		-	-	-	-	-	-	-
STW1.3	1 = Enable operation (pulses can be enabled) 0 = Inhibit operation (cancel pulses)	<3> p0852[0] = r2090.3	[8920.3]	-	[8820.3]	[8932]	-	[8832]	-
STW1.4	Reserved	-	-	-	-	-	-	-	-
STW1.5	1 = Inhibit motoring operation	<4> p3532= r2090.5	[8920.3]	-	-	[8920]	-	-	-
STW1.6	1 = Inhibit regenerative operation	<3> p3533= r2090.6	[8920.3]	-	[8820.3]	[8920]	-	[8820]	-
STW1.7	Acknowledge faults	p2103[0] = r2090.7	[2546.3]			[8060]			-
STW1.8	Reserved	-	-	-	-	-	-	-	-
STW1.9	Reserved	-	-	-	-	-	-	-	-
STW1.10	1 = Control via PLC	<2> p0854[0] = r2090.10	[8920.3]	[8720.3]	[8820.3]	[8920]	[8720]	[8820]	-
STW1.11	Reserved	-	-	-	-	-	-	-	-
STW1.12	Reserved	-	-	-	-	-	-	-	-
STW1.13	Reserved	-	-	-	-	-	-	-	-
STW1.14	Reserved	-	-	-	-	-	-	-	-
STW1.15	Reserved	-	-	-	-	-	-	-	-

<1> Used in telegram 370.

<2> STW1.10 must be set to ensure that the drive object accepts the process data (PZD).

<3> Only for A_INF, S_INF

<4> Only for A_INF

<5> B_INF and S_INF only for S120.

1	2	3	4	5	6	7	8
DO: A_INF, B_INF, S_INF					fp_2447_55_eng.vsd	Function diagram	
PROFdrive - E_STW1 control word infeed interconnection					14.04.08 V04.03.01	SINAMICS S120/S150	
							- 2447 -

Fig. 2-58 2447 – E_STW1 control word infeed interconnection

Fig. 2-59 2449 – PZD outgoing signals interconnection, profile-specific

Signal sources for PZD send signals						
Signal	Description	PROFIdrive Signal No.	Interconnection parameter	Function diagram	Data type	Normalization
ZSW1	Status word 1	2	r2089[0]	[2452][2453][2479] <2>	U16	-
ZSW2	Status word 2	4	r2089[1]	[2454][2455] <2>	U16	-
NIST_A	Speed setpoint A (16 bit)	6	r0063 SERVO r0063[0] VECTOR r0061 ENCODER	[4710] <2> [4715] [4710]	I16	4000 hex $\hat{=}$ p2000
NIST_B	Speed setpoint B (32 bit)	8	r0063 SERVO r0063[0] VECTOR r0061 ENCODER	[4710] <2> [4715] [4710]	I32	4000 0000 hex $\hat{=}$ p2000
G1_ZSW	Encoder 1 status word	10	r0481[0]	[4730]	U16	-
G1_XIST1	Encoder 1 actual position 1	11	r0482[0]	[4704]	U32	-
G1_XIST2	Encoder 1 actual position 2	12	r0483[0]	[4704]	U32	-
G2_ZSW	Encoder 2 status word	14	r0481[1]	[4730]	U16	-
G2_XIST1	Encoder 2 actual position 1	15	r0482[1]	[4704]	U32	-
G2_XIST2	Encoder 2 actual position 2	16	r0483[1]	[4704]	U32	-
G3_ZSW	Encoder 3 status word	18	r0481[2]	[4730]	U16	-
G3_XIST1	Encoder 3 actual position 1	19	r0482[2]	[4704]	U32	-
G3_XIST2	Encoder 3 actual position 2	20	r0483[2]	[4704]	U32	-
E_DIGITAL	Digital inputs	21	r2089[2]	[2459]	U16	-
XIST_A	Position actual value A	28	r2521[0]	[4010]	I32	1 hex $\hat{=}$ 1 LU
AKTSATZ	Pos selected block	33	r2670	[3650]	U16	-
IAIST_GLATT	Absolute actual current, smoothed	51	r0068[1]	[6714]	I16	4000 hex $\hat{=}$ p2002
ITIST_GLATT	Current actual value, torque-generating	52	r0078[1]	[6714]	I16	4000 hex $\hat{=}$ p2002
MIST_GLATT	Actual torque smoothed	53	r0080[1]	[6714]	I16	4000 hex $\hat{=}$ p2003
PIST_GLATT	Power factor, smoothed	54	r0082[1]	[6714]	I16	4000 hex $\hat{=}$ p2004
NIST_A_GLATT	Actual speed, smoothed	57	r0063[1]	[4715]	I16	4000 hex $\hat{=}$ p2000
MELD_NAMUR	VIK-NAMUR message bit bar	58	r3113	-	U16	-
IAIST	Absolute actual current	59	r0068[0]	[6714]	I16	4000 hex $\hat{=}$ p2002
MIST	Actual torque	60	r0080[0]	[6714]	I16	4000 hex $\hat{=}$ p2003
ZSW2_ENC	Status word 2 ENCODER	81	(bitwise)	[2434]	U16	-

<1> Data type according to the PROFIdrive profile: I16 = Integer16, I32 = Integer32, U16 = Unsigned16, U32 = Unsigned32.
<2> Only for SINAMICS S120.
<3> Only for ENCODER.

PROFIdrive sampling time
Refer to [1020.7]

PROFIdrive send telegram

Header
Drive object 1
Drive object 2
...
Drive object n
...
Drive object m
Trailer

Send words 1...16
p2051[0...15] WORD
r2053[0...15] WORD
p2061[0...14] DWORD
r2063[0...14] DWORD

Telegram assignment according to p0922 [2420]

PROFIBUS PROFINET

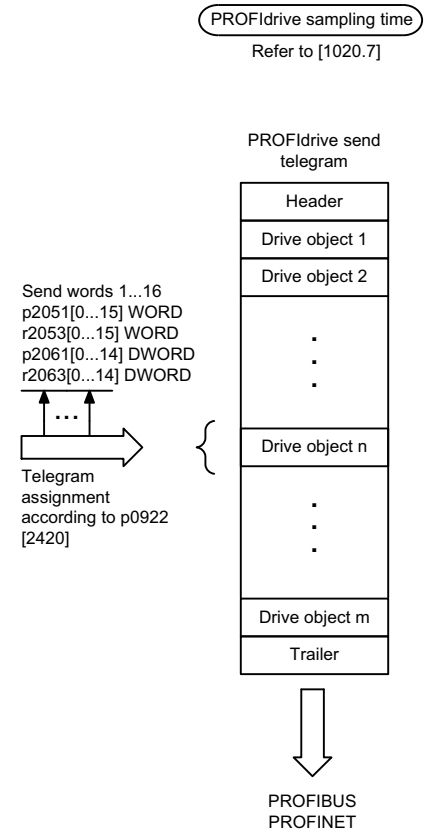
1	2	3	4	5	6	7	8
DO: A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR					fp_2449_54_eng.vsd	Function diagram	- 2449 -
PROFIdrive - PZD send signals connection of profile-specific					11.11.09 V04.03.01	S120/S150/G130/G150	

Fig. 2-60 2450 – PZD outgoing signals interconnection, manufacturer-specific

Signal sources for PZD send signals						
Signal	Description	PROFdrive Signal No.	Interconnection parameter	Function diagram	Data type	Normalization
MELDW	Message word	102	r2089[2]	[2456]	U16	-
MSOLL_GLATT	Total speed setpoint smoothed	120	r0079[1]	[5610] <2>	I16	4000 hex $\pm$ p2003
AIST_GLATT	Torque utilization smoothed	121	r0081	[8012]	I16	4000 hex $\pm$ 100%
MT_ZSW	Measuring probe status word	131	r0688	-	U16	-
MT1_ZS_F	Measuring probe 1 measuring time, falling edge	132	r0687[0]	-	U16	-
MT1_ZS_S	Measuring probe 1 measuring time, rising edge	133	r0686[0]	-	U16	-
MT2_ZS_F	Measuring probe 2 measuring time, falling edge	134	r0687[1]	-	U16	-
MT2_ZS_S	Measuring probe 2 measuring time, rising edge	135	r0686[1]	-	U16	-
MT3_ZS_F	Measuring probe 2 measuring time, falling edge	136	r0687[2]	-	U16	-
MT3_ZS_S	Measuring probe 2 measuring time, rising edge	137	r0686[2]	-	U16	-
MT4_ZS_F	Measuring probe 2 measuring time, falling edge	138	r0687[3]	-	U16	-
MT4_ZS_S	Measuring probe 2 measuring time, rising edge	139	r0686[3]	-	U16	-
MT5_ZS_F	Measuring probe 2 measuring time, falling edge	140	r0687[4]	-	U16	-
MT5_ZS_S	Measuring probe 2 measuring time, rising edge	141	r0686[4]	-	U16	-
MT6_ZS_F	Measuring probe 2 measuring time, falling edge	142	r0687[5]	-	U16	-
MT6_ZS_S	Measuring probe 2 measuring time, rising edge	143	r0686[5]	-	U16	-
POS_ZSW	Pos status word	204	r2683	[3645]	U16	-
POS_ZSW1	Pos status word 1	221	r2089[3]	[2466]	U16	-
POS_ZSW2	Pos status word 2	223	r2089[4]	[2467]	U16	-
FAULT_CODE	Fault code	301	r2131	[8060]	U16	-
WARN_CODE	Alarm code	303	r2132	[8065]	U16	-
E_ZSW1	Status word 1 for Active Infeed	321	r2089[1]	[2457]	U16	-
ZSW1_BM	Status word 1, variant for BM	323	r2089[0]	[2428]	U16	-
ZSW2_BM	Status word 2, variant for BM	325	r2089[1]	[2429]	U16	-
E_ZSW1_BM	Status word 1 for Infeed, variant for BM (ALM, BLM, SLM)	327	r2080	[2430]	U16	-
CU_ZSW1	Status word 1 for Control Unit	501	r2089[1]	[2496]	U16	-
M_VST	Torque pre-control value	-	p1513	-	I16	4000 hex $\pm$ p2003

<1> Data type according to the PROFdrive profile: I16 = Integer16, I32 = Integer32, U16 = Unsigned16, U32 = Unsigned32

<2> Only for SINAMICS S120.



1	2	3	4	5	6	7	8
DO: A_INF, B_INF, S_INF, SERVO, TM41, VECTOR					fp_2450_54_eng.vsd	Function diagram	- 2450 -
PROFdrive - PZD send signals, connection of manufacture-specific					26.11.09 V04.03.01	S120/S150/G130/G150	

PROFIdrive sampling time
Refer to [1020.7]

Signal sources for ZSW1 in Interface Mode VIK-NAMUR (p2038 = 2)

Signal	Meaning	Interconnection parameters	[Function diagram] Internal status word	[Function diagram] Signal source	Inverted <2>
ZSW1.0	1 = Ready to power-up	p2080[0] = r0899.0	[2503.7]	[2610]	-
ZSW1.1	1 = Ready to operate (DC link loaded, pulses blocked)	p2080[1] = r0899.1	[2503.7]	[2610]	-
ZSW1.2	1 = Operation enabled (drive follows n_set)	p2080[2] = r0899.2	[2503.7]	[2610]	-
ZSW1.3	1 = Fault present	p2080[3] = r2139.3	[2548.7]	[8060]	-
ZSW1.4	1 = No coast down active (OFF2 inactive)	p2080[4] = r0899.4	[2503.7]	[2610]	-
ZSW1.5	1 = No fast stop active (OFF3 inactive)	p2080[5] = r0899.5	[2503.7]	[2610]	-
ZSW1.6	1 = Power-on inhibit active	p2080[6] = r0899.6	[2503.7]	[2610]	-
ZSW1.7	1 = Alarm present	p2080[7] = r2139.7	[2548.7]	[8065]	-
ZSW1.8	1 = Speed setpoint - actual value deviation within tolerance t_off	p2080[8] = r2197.7	[2534.7]	[8010]	-
ZSW1.9	1 = Control requested <3>	p2080[9] = r0899.9	[2503.7]	[2503]	-
ZSW1.10	1 = f or n comparison value reached/exceeded	p2080[10] = r2199.1	[2536.7]	[8010]	-
ZSW1.11	1 = I, M, or P limit not reached	p2080[11] = r0056.13	[2522.7]	[6060]	✓
ZSW1.12	Reserved	-	-	-	-
ZSW1.13	1 = No motor overtemperature alarm	p2080[13] = r2135.14	[2548.7]	[8016]	✓
ZSW1.14	1 = Motor rotates forwards (n_act ≥ 0) 0 = Motor rotates backwards (n_act < 0)	p2080[14] = r2197.3	[2534.7]	[8010]	-
ZSW1.15	1 = Display CDS	<4> p2080[15] = r0836.0	-	-	-

<1> Used in telegram 20.

<2> The ZSW1 is generated using the binector-connector converter (BI: p2080[0...15], inversion: p2088[0].0...p2088[0].15)

<3> The drive object is ready to accept data.

<4> Interconnection is not disabled.

1	2	3	4	5	6	7	8
DO: VECTOR					fp_2451_54_eng.vsd	Function diagram	- 2451 -
PROFIdrive - ZSW1 status word interconnection (p2038 = 2)					14.08.08 V04.03.01	S120/S150/G130/G150	

Fig. 2-61 2451 – ZSW1 status word interconnection (p2038 = 2)

PROFdrive Abtastzeit

siehe [1020.7]

Signal sources for ZSW1 im Interface Mode SINAMICS (p2038 = 0)

<1>

Signal	Meaning	Interconnection parameters	[Function diagram] Internal status word	[Function diagram] signal source	Inverted <2>
ZSW1.0	1 = Ready to power-up	p2080[0] = r0899.0	[2503.7]	[2610]	-
ZSW1.1	1 = Ready to operate	p2080[1] = r0899.1	[2503.7]	[2610]	-
ZSW1.2	1 = Operation enabled	p2080[2] = r0899.2	[2503.7]	[2610]	-
ZSW1.3	1 = Fault present	p2080[3] = r2139.3	[2548.7]	[8060]	-
ZSW1.4	1 = No coast down active (OFF2 inactive)	p2080[4] = r0899.4	[2503.7]	[2610]	-
ZSW1.5	1 = No fast stop active (OFF3 inactive)	p2080[5] = r0899.5	[2503.7]	[2610]	-
ZSW1.6	1 = Power-on inhibit active	p2080[6] = r0899.6	[2503.7]	[2610]	-
ZSW1.7	1 = Alarm present	p2080[7] = r2139.7	[2548.7]	[8065]	-
ZSW1.8	1 = Speed setpoint - actual value deviation within tolerance t_off	p2080[8] = r2197.7	[2534.7]	[8010]	-
ZSW1.9	1 = Control requested <3>	p2080[9] = r0899.9	[2503.7]	[2503]	-
ZSW1.10	1 = f or n comparison value reached/exceeded	p2080[10] = r2199.1	[2536.7]	[8010]	-
ZSW1.11	1 = I, M, or P limit not reached <4>	p2080[11] = r1407.7	[2522.7]	[5610] [6060] <5>	✓
ZSW1.12	1 = Open holding brake	p2080[12] = r0899.12	[2503.7]	[2701]	-
ZSW1.13	1 = No motor overtemperature alarm	p2080[13] = r2135.14	[2548.7]	[8016]	✓
ZSW1.14	1 = Motor rotates forwards (n_act ≥ 0) 0 = Motor rotates backwards (n_act < 0)	p2080[14] = r2197.3	[2534.7]	[8010]	-
ZSW1.15	1 = No alarm, thermal overload, power unit	p2080[15] = r2135.15	[2548.7]	[8014]	✓

<1> Used in telegrams 1, 2, 3, 4, 5, 6, 352.

<2> The ZSW1 is generated using the binector-connector converter (BI: p2080[0...15], inversion: p2088[0].0...p2088[0].15)

<3> The drive object is ready to accept data.

<4> Not for VECTOR V/f.

<5> Only for SINAMICS S120.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2452_54_eng.vsd	Function diagram	- 2452 -
PROFdrive - ZSW1 status word interconnection (p2038 = 0)					27.11.08 V04.03.01	S120/S150/G130/G150	

Fig. 2-62 2452 – ZSW1 status word interconnection (p2038 = 0)

PROFdrive sampling time
Refer to [1020.7]

Signal sources for ZSW1 in Interface Mode SIMODRIVE 611 universal (p2038 = 1)					
Signal	Meaning	Interconnection parameters	[Function diagram] Internal status word	[Function diagram] Signal source	Inverted
ZSW1.0	1 = Ready to power-up	p2080[0] = r0899.0	[2503.7]	[2610]	-
ZSW1.1	1 = Ready to operate	p2080[1] = r0899.1	[2503.7]	[2610]	-
ZSW1.2	1 = Operation enabled	p2080[2] = r0899.2	[2503.7]	[2610]	-
ZSW1.3	1 = Fault present	p2080[3] = r2139.3	[2548.7]	[8060]	-
ZSW1.4	1 = No coast down active	p2080[4] = r0899.4	[2503.7]	[2610]	-
ZSW1.5	1 = No fast stop active	p2080[5] = r0899.5	[2503.7]	[2610]	-
ZSW1.6	1 = Power-on inhibit active	p2080[6] = r0899.6	[2503.7]	[2610]	-
ZSW1.7	1 = Alarm present	p2080[7] = r2139.7	[2548.7]	[8065]	-
ZSW1.8	1 = Speed setpoint - actual value deviation within tolerance t _{off}	p2080[8] = r2197.7	[2534.7]	[8010]	-
ZSW1.9	1 = Control requested <2>	p2080[9] = r0899.9	[2503.7]	[2503]	-
ZSW1.10	1 = f or n comparison value reached/exceeded	p2080[10] = r2199.1	[2536.7]	[8010]	-
ZSW1.11	1 = Alarm class bit 0	p2080[11] = r2139.11	[2548.7]	-	-
ZSW1.12	1 = Alarm class bit 1	p2080[12] = r2139.12	[2548.7]	-	-
ZSW1.13	Reserved	-	-	-	-
ZSW1.14	1 = Closed-loop torque control active	p2080[14] = r1407.2	[2522.7]	[2522]	-
ZSW1.15	Reserved	-	-	-	-

<1> Used in telegrams 1, 2, 3, 4, 5, 6, 102, 103, 105, 106, 116, 118.

<2> The drive object is ready to accept data.

1	2	3	4	5	6	7	8
DO: SERVO			fp_2453_01_eng.vsd			Function diagram	- 2453 -
PROFdrive - ZSW1 status word interconnection (p2038 = 1)			06.10.08 V04.03.01			SINAMICS S120	

Fig. 2-63 2453 – ZSW1 status word interconnection (p2038 = 1)

PROFdrive sampling time
Refer to [1020.7]

Signal sources for ZSW2 in Interface Mode SINAMICS (p2038 = 0)

Signal	Meaning	Interconnection parameters	[Function diagram] internal status word	[Function diagram] signal source	Inverted
ZSW2.0	1 = Drive data set DDS effective, bit 0	p2081[0] = r0051.0	-	[8565]	-
ZSW2.1	1 = Drive data set DDS effective, bit 1	p2081[1] = r0051.1	-	[8565]	-
ZSW2.2	1 = Drive data set DDS effective, bit 2	p2081[2] = r0051.2	-	[8565]	-
ZSW2.3	1 = Drive data set DDS effective, bit 3	p2081[3] = r0051.3	-	[8565]	-
ZSW2.4	1 = Drive data set DDS effective, bit 4	p2081[4] = r0051.4	-	[8565]	-
ZSW2.5	1 = Alarm class bit 0	p2081[5] = r2139.11	-	[2548]	-
ZSW2.6	1 = Alarm class bit 1	p2081[6] = r2139.12	-	[2548]	-
ZSW2.7	1 = Parking axis active	p2081[7] = r0896.0	-	-	-
ZSW2.8	1 = Traverse to fixed endstop	p2081[8] = r1406.8	-	[2520]	-
ZSW2.9	Reserved	-	-	-	-
ZSW2.10	1 = Pulses enabled	p2082[13] = r0899.11	[2503.7]	[2610]	-
ZSW2.11	1 = Data set changeover active	p2081[11] = r0835.0	-	-	-
ZSW2.12	Slave sign-of-life bit 0	Implicitly interconnected	-	-	-
ZSW2.13	Slave sign-of-life bit 1				
ZSW2.14	Slave sign-of-life bit 2				
ZSW2.15	Slave sign-of-life bit 3				

<1> Used in telegrams 2, 3, 4, 5, 6, 9, 110, 111.

<2> These signals are automatically interconnected for clock-cycle synchronous operation.

<3> Only for SINAMICS S120.

<4> Not for Vector V/f.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2454_54_eng.vsd	Function diagram	- 2454 -
PROFdrive - ZSW2 status word interconnection (p2038 = 0)					26.09.08 V04.03.01	S120/S150/G130/G150	

Fig. 2-64

2454 – ZSW2 status word interconnection (p2038 = 0)

PROFIdrive sampling time
Refer to [1020.7]

Signal sources for ZSW2 in Interface Mode SIMODRIVE 611 universal (p2038 = 1)

<1>

Signal	Meaning	Interconnection parameters	[Function diagram] Internal status word	[Function diagram] Signal source	Inverted
ZSW2.0	1 = Drive data set DDS effective, bit 0	p2081[0] = r0051.0	-	[8565]	-
ZSW2.1	1 = Drive data set DDS effective, bit 1	p2081[1] = r0051.1	-	[8565]	-
ZSW2.2	1 = Drive data set DDS effective, bit 2	p2081[2] = r0051.2	-	[8565]	-
ZSW2.3	Reserved	-	-	-	-
ZSW2.4	1 = Ramp-function generator inactive <3>	p2081[4] = r1199.2	-	[3060] [3080]	✓
ZSW2.5	1 = Holding brake open	p2081[5] = r0899.12	[2503.7]	[2701]	-
ZSW2.6	1 = Integrator inhibit, speed controller	p2081[6] = r2093.6	-	[5040] [5210]	-
ZSW2.7	1 = Parking axis active	p2081[7] = r0896.0	-	-	-
ZSW2.8	1 = Traverse to fixed endstop	p2081[8] = r1406.8	-	[2520]	-
ZSW2.9	1 = Drive data set DDS effective, bit 3	p2081[9] = r0051.3	-	[8565]	-
ZSW2.10	1 = Drive data set DDS effective, bit 4	p2081[10] = r0051.4	-	[8565]	-
ZSW2.11	1 = Data set changeover active	p2081 [11] = r0835.0	-	-	-
ZSW2.12	Slave sign-of-life bit 0	Implicitly interconnected	-	-	-
ZSW2.13	Slave sign-of-life bit 1				
ZSW2.14	Slave sign-of-life bit 2				
ZSW2.15	Slave sign-of-life bit 3				

<2>

<1> Used in telegrams 2, 3, 4, 5, 6, 102, 103, 105, 106, 116, 118.

<2> These signals are automatically interconnected for clock-cycle synchronous operation.

<3> Only if the function module "extended setpoint channel" is active (r0108.8 = 1).

1	2	3	4	5	6	7	8
DO: SERVO					fp_2455_01_eng.vsd	Function diagram	- 2455 -
PROFIdrive - ZSW2 status word interconnection (p2038 = 1)					06.10.08 V04.03.01	SINAMICS S120	

Fig. 2-65 2455 – ZSW2 status word interconnection (p2038 = 1)

PROFdrive sampling time

Refer to [1020.7]

Signal sources for MELDW

Signal	Meaning	Interconnection parameters	[Function diagram] Internal status word	[Function diagram] signal source	Inverted <2>
MELDW.0	1 = Ramp-up/ramp-down completed 0 = Ramp-function generator active	p2082[0] = r2199.5	[2537.7]	[8010]	-
MELDW.1	1 = Torque utilization [%] < torque threshold value 2 (p2194)	p2082[1] = r2199.11	[2537.7]	[8012]	-
MELDW.2	1 = n_act < speed threshold value 3 (p2161)	p2082[2] = r2199.0	[2537.7]	[8010]	-
MELDW.3	1 = n_act ≤ speed threshold value 2 (p2155)	p2082[3] = r2197.1	[2534.7]	[8010]	-
MELDW.4	Reserved	-	-	-	-
MELDW.5	Variable signaling function	p2082[5] = r3294	-	[5301]	-
MELDW.6	1 = No motor overtemperature alarm	p2082[6] = r2135.14	[2548.7]	[8016]	✓
MELDW.7	1 = No alarm, thermal overload, power unit	p2082[7] = r2135.15	[2548.7]	[8014]	✓
MELDW.8	1 = Speed setpoint - actual value deviation within tolerance t_on	p2082[8] = r2199.4	[2537.7]	[8010]	-
MELDW.9	Reserved	-	-	-	-
MELDW.10	Reserved	-	-	-	-
MELDW.11	1 = Controller enable	p2082[11] = r0899.8	[2503.7]	[2610]	-
MELDW.12	1 = Drive ready	p2082[12] = r0899.7	[2503.7]	[2610]	-
MELDW.13	1 = Pulses enabled	p2082[13] = r0899.11	[2503.7]	[2610]	-
MELDW.14	Reserved	-	-	-	-
MELDW.15	Reserved	-	-	-	-

<1> Used in telegrams 102, 103, 105, 106, 110, 111, 116, 118.

<2> The status word is generated using the binector-connector converter p2088[2].

1	2	3	4	5	6	7	8
DO: SERVO					fp_2456_01_eng.vsd	Function diagram	- 2456 -
PROFdrive - MELDW status word interconnection					23.10.08 V04.03.01	SINAMICS S120	

Fig. 2-66 2456 – MELDW status word interconnection

PROFIdrive sampling time

Refer to [1020.7]

Signal sources for E_ZSW1

<1>

Signal	Meaning	Interconnection parameters	[Function diagram] signal source			[Function diagram] internal status word			Inverted
			A_INF	B_INF <3>	S_INF <3>	A_INF	B_INF <3>	S_INF <3>	
ZSW1.0	1 = Ready to power-up	p2080[0] = r0899.0	[8932]	[8732]	[8832]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.1	1 = Ready to operate	p2080[1] = r0899.1	[8932]	[8732]	[8832]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.2	1 = Operation enabled	p2080[2] = r0899.2	[8932]	[8732]	[8832]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.3	1 = Fault present	p2080[3] = r2139.3	[8060]			[2548.7]			-
ZSW1.4	1 = No OFF2 effective	p2080[4] = r0899.4	[8932]	[8732]	[8832]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.5	Reserved	-	-	-	-	-	-	-	-
ZSW1.6	1 = Power-on inhibit	p2080[6] = r0899.6	[8932]	[8732]	[8832]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.7	1 = Alarm present	p2080[7] = r2139.7	[8065]			[2548.7]			-
ZSW1.8	Reserved	-	-	-	-	-	-	-	-
ZSW1.9	1 = PLC requests control <2>	p2080[9] = r0899.9	[8926]	[8726]	[8826]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.10	Reserved	-	-	-	-	-	-	-	-
ZSW1.11	1 = Pre-charging completed	p2080[11] = r0899.11	[8950]	[8750]	[8850]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.12	1 = Line contactor closed	p2080[12] = r0899.12	[8934]	[8734]	[8834]	[8926.7]	[8726.7]	[8826.7]	-
ZSW1.13	Reserved	-	-	-	-	-	-	-	-
ZSW1.14	Reserved	-	-	-	-	-	-	-	-
ZSW1.15	Reserved	-	-	-	-	-	-	-	-

<1> Used in telegram 370.

<2> The drive object is ready to accept data.

<3> Only for S120.

1	2	3	4	5	6	7	8
DO: A_INF, B_INF, S_INF					fp_2457_55_eng.vsd	Function diagram	- 2457 -
PROFIdrive - E_ZSW1 status word infeed interconnection					15.04.08 V04.03.01	SINAMICS S120/S150	

Fig. 2-67 2457 – E_ZSW1 status word infeed interconnection

PROFdrive sampling time
Refer to [1020.7]

Signal targets for POS_STW (positioning mode, r0108.4 = 1)

Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
POS_STW.0	1 = Tracking mode active 0 = No tracking mode active	p2655[0] = r2092.0	-	[3635]	-
POS_STW.1	1 = Set home position 0 = Do not set home position	p2596 = r2092.1	-	[3612]	-
POS_STW.2	1 = Reference cam active	p2612 = r2092.2	-	[3612]	-
POS_STW.3	Reserved	-	-	-	-
POS_STW.4	Reserved	-	-	-	-
POS_STW.5	1 = Jogging, incremental active 0 = Jogging, velocity active	p2591 = r2092.5	-	[3610]	-
POS_STW.6	Reserved	-	-	-	-
POS_STW.7	Reserved	-	-	-	-
POS_STW.8	Reserved	-	-	-	-
POS_STW.9	Reserved	-	-	-	-
POS_STW.10	Reserved	-	-	-	-
POS_STW.11	Reserved	-	-	-	-
POS_STW.12	Reserved	-	-	-	-
POS_STW.13	Reserved	-	-	-	-
POS_STW.14	Reserved	-	-	-	-
POS_STW.15	Reserved	-	-	-	-

<1> Used in telegram 110, 999.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2462_55_eng.vsd	Function diagram	- 2462 -
PROFdrive - POS_STW-Pos control word interconnection (r0108.4 = 1)					09.11.09 V04.03.01	SINAMICS S120/S150	

Fig. 2-68 2462 – POS_STW positioning control word interconnection (r0108.4 = 1)

PROFIdrive Abtastzeit

siehe [1020.7]

Signal targets for POS_STW1 (positioning mode, r0108.4 = 1)

<1>

Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
POS_STW1.0	Traversing block selection, bit 0	p2625 = r2091.0	-	-	-
POS_STW1.1	Traversing block selection, bit 1	p2626 = r2091.1	-	-	-
POS_STW1.2	Traversing block selection, bit 2	p2627 = r2091.2	-	-	-
POS_STW1.3	Traversing block selection, bit 3	p2628 = r2091.3	-	-	-
POS_STW1.4	Traversing block selection, bit 4	p2629 = r2091.4	-	-	-
POS_STW1.5	Traversing block selection, bit 5	p2630 = r2091.5	-	-	-
POS_STW1.6	Reserved	-	-	-	-
POS_STW1.7	Reserved	-	-	-	-
POS_STW1.8	1 = Absolute positioning is selected. 0 = Relative positioning is selected.	p2648 = r2091.8	-	-	-
POS_STW1.9	1 = Absolute positioning/MDI direction selection, positive. 2 = Absolute positioning/MDI direction selection, negative. 3 = Absolute positioning through the shortest distance. 0 = Absolute positioning through the shortest distance.	p2651 = r2091.9	-	-	-
POS_STW1.10		p2652 = r2091.10	-	-	-
POS_STW1.11	Reserved	-	-	-	-
POS_STW1.12	1 = Stetige Übernahme 0 = MDI-Satzwechsel mit  von Fahrauftrag aktivieren (POS_STW1.12)	p2649 = r2091.12	-	-	-
POS_STW1.13	Reserved	-	-	-	-
POS_STW1.14	1 = signal setting-up selected 0 = signal positioning selected.	p2653 = r2091.14	-	-	-
POS_STW1.15	1 = MDI selection	p2647 = r2091.15	-	-	-

<1> Used in telegram 111.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2463_55_eng.vsd	Function diagram	- 2463 -
PROFIdrive – POS_STW1-Pos control word 1 interconnection (r0108.4 = 1)					09.11.09 V04.03.01	SINAMICS S120/S150	

Fig. 2-69 2463 – POS_STW1 positioning control word 1 interconnection (r0108.4 = 1)

PROFdrive sampling time
Refer to [1020.7]

Signal targets for POS_STW2 (positioning mode, r0108.4 = 1)

Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
POS_STW2.0	1 = Tracking mode active	p2655[0] = r2092.0	-	[3635]	-
POS_STW2.1	1 = Set reference point	p2596 = r2092.1	-	[3612]	-
POS_STW2.2	1 = Reference cam active	p2612 = r2092.2	-	[3612]	-
POS_STW2.3	Reserved	-	-	-	-
POS_STW2.4	Reserved	-	-	-	-
POS_STW2.5	1 = Jogging, incremental active 0 = Jogging, velocity active	p2591 = r2092.5	-	[3610]	-
POS_STW2.6	Reserved	-	-	-	-
POS_STW2.7	Reserved	-	-	-	-
POS_STW2.8	1 = Referencing type selection for flying referencing 0 = Referencing type selection for search for reference	p2597 = r2092.8	-	-	-
POS_STW2.9	1 = Start the search for reference in the negative direction 0 = Start the search for reference in the positive direction.	p2604 = r2092.9	-	-	-
POS_STW2.10	1 = Measuring probe 2 is activated 0 = Measuring probe 1 is activated	p2510[0] = r2092.10	-	-	-
POS_STW2.11	1 = Falling edge of the measuring probe 0 = Rising edge of the measuring probe	p2511[0] = r2092.11	-	-	-
POS_STW2.12	Reserved	-	-	-	-
POS_STW2.13	Reserved	-	-	-	-
POS_STW2.14	1 = Software limit switch activation	p2582 = r2092.14	-	-	-
POS_STW2.15	1 = STOP cam active	p2568 = r2092.15	-	-	-

<1> Used in telegram 111.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2464_55_eng.vsd	Function diagram	- 2464 -
PROFdrive - POS_STW2-POS control word 2 interconnection (r0108.4 = 1)					09.11.09 V04.03.01	SINAMICS S120/S150	

Fig. 2-70 2464 – POS_STW2 positioning control word 2 interconnection (r0108.4 = 1)

PROFdrive sampling time
Refer to [1020.7]

Signal targets for POS_ZSW1 (positioning mode, r0108.4 = 1)						<1>
Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted	
POS_ZSW1.0	Active Traversing Block Bit 0 (2 ⁰)	p2083[0] = r2670[0]	-	-	-	
POS_ZSW1.1	Active Traversing Block Bit 1 (2 ¹)	p2083[1] = r2670[1]	-	-	-	
POS_ZSW1.2	Active Traversing Block Bit 2 (2 ²)	p2083[2] = r2670[2]	-	-	-	
POS_ZSW1.3	Active Traversing Block Bit 3 (2 ³)	p2083[3] = r2670[3]	-	-	-	
POS_ZSW1.4	Active Traversing Block Bit 4 (2 ⁴)	p2083[4] = r2670[4]	-	-	-	
POS_ZSW1.5	Active Traversing Block Bit 5 (2 ⁵)	p2083[5] = r2670[5]	-	-	-	
POS_ZSW1.6	Reserved	-	-	-	-	
POS_ZSW1.7	Reserved	-	-	-	-	
POS_ZSW1.8	1 = STOP cam minus active	p2083[08] = r2684[13]	-	-	-	
POS_ZSW1.9	1 = STOP cam plus active	p2083[09] = r2684[14]	-	-	-	
POS_ZSW1.10	1 = Jogging active	p2083[10] = r2094[0]	-	-	-	
POS_ZSW1.11	1 = Reference point approach active	p2083[11] = r2094[1]	-	-	-	
POS_ZSW1.12	1 = Flying referencing active	p2083[12] = r2684[1]	-	-	-	
POS_ZSW1.13	1 = Traversing Block active	p2083[13] = r2094[2]	-	-	-	
POS_ZSW1.14	1 = Set-up active	p2083[14] = r2094[4]	-	-	-	
POS_ZSW1.15	1 = MDI active 0 = MDI inactive	p2083[15] = r2670[15]	-	-	-	

<1> Used in telegram 111.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2466_55_eng.vsd	Function diagram	- 2466 -
PROFdrive – POS_ZSW1-Pos status word 1 interconnection (r0108.4 = 1)					09.11.09 V04.03.01	SINAMICS S120/S150	

Fig. 2-71 2466 – POS_ZSW1 positioning status word 1 interconnection (r0108.4 = 1)

PROFdrive Abtastzeit
siehe [1020.7]

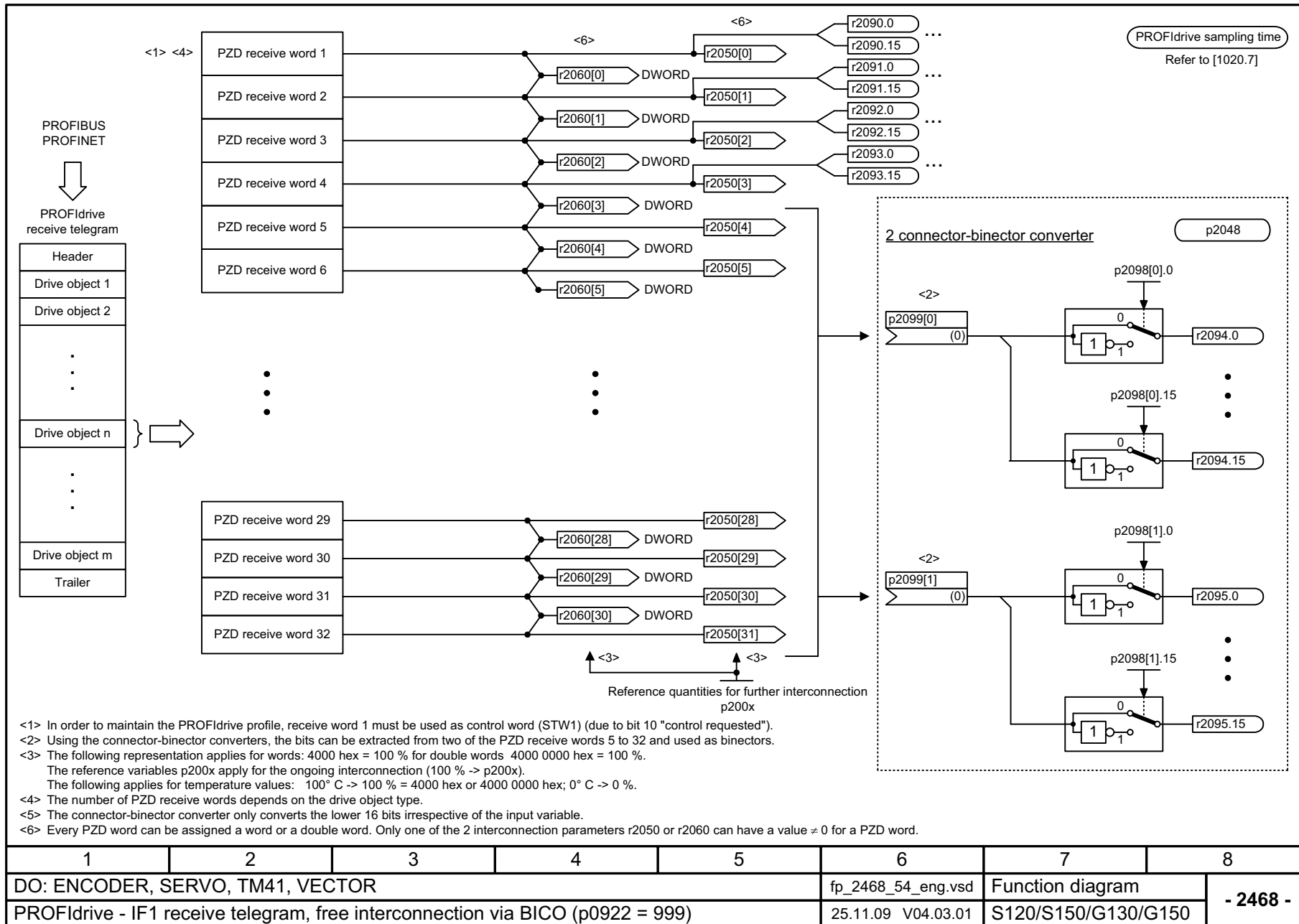
Signal targets for POS_ZSW2 (positioning mode, r0108.4 = 1)

Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
POS_ZSW2.0	1 = Tracking mode active	p2084[0] = r2683.0	-	-	-
POS_ZSW2.1	1 = Velocity limiting active	p2084[1] = r2683.1	-	-	-
POS_ZSW2.2	1 = Setpoint available	p2084[2] = r2683.2	-	-	-
POS_ZSW2.3	1 = Printing mark outside outer window	p2084[3] = r2684.3	-	-	-
POS_ZSW2.4	1 = Axis moves forward	p2084[4] = r2683.4	-	-	-
POS_ZSW2.5	1 = Axis moves backwards	p2084[5] = r2683.5	-	-	-
POS_ZSW2.6	1 = Software limit switch minus reached	p2084[6] = r2683.6	-	-	-
POS_ZSW2.7	1 = Software limit switch plus reached	p2084[7] = r2683.7	-	-	-
POS_ZSW2.8	1 = Position actual value <= cam switching position 1	p2084[8] = r2683.8	-	-	-
POS_ZSW2.9	1 = Position actual value <= cam switching position 2	p2084[9] = r2683.9	-	-	-
POS_ZSW2.10	1 = Direct output 1 via traversing block	p2084[10] = r2683.10	-	-	-
POS_ZSW2.11	1 = Direct output 2 via traversing block	p2084[11] = r2683.11	-	-	-
POS_ZSW2.12	1 = Fixed stop reached	p2084[12] = r2683.12	-	-	-
POS_ZSW2.13	1 = Fixed stop clamping torque reached	p2084[13] = r2683.13	-	-	-
POS_ZSW2.14	1 = Travel to fixed stop active	p2084[14] = r2683.14	-	-	-
POS_ZSW2.15	1 = Traversing command active	p2084[15] = r2684.15	-	-	-

<1> Verwendung in Telegramm 111.

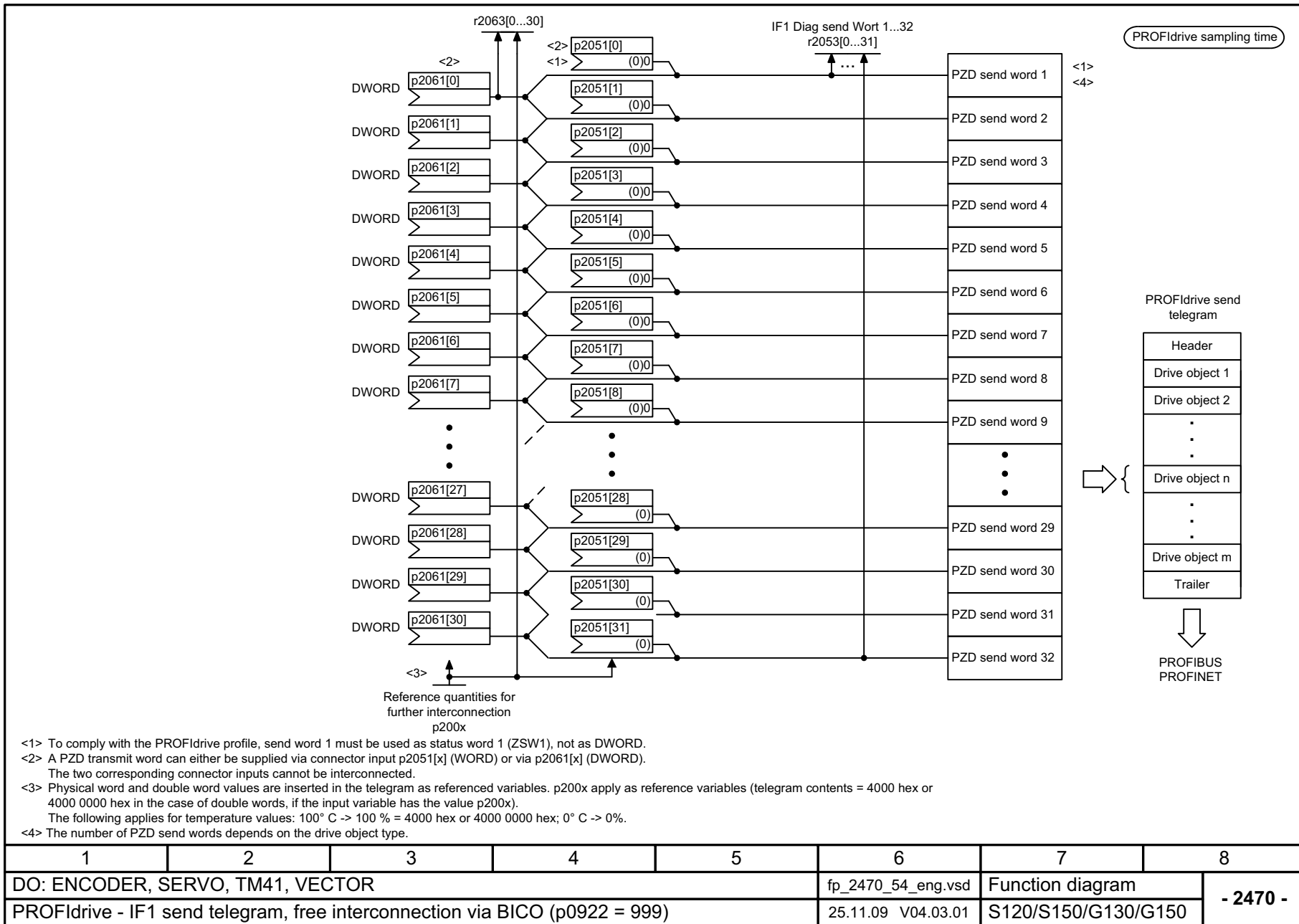
1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2467_55_eng.vsd	Function diagram	- 2467 -
PROFdrive – POS_ZSW2-Pos status word 2 interconnection (r0108.4 = 1)					09.11.09 V04.03.01	SINAMICS S120/S150	

Fig. 2-72 2467 – POS_ZSW2 positioning status word 2 interconnection (r0108.4 = 1)



1	2	3	4	5	6	7	8
DO: ENCODER, SERVO, TM41, VECTOR					fp_2468_54_eng.vsd	Function diagram	
PROFdrive - IF1 receive telegram, free interconnection via BICO (p0922 = 999)					25.11.09 V04.03.01	S120/S150/G130/G150	
							- 2468 -

Fig. 2-73 2468 – IF1 incoming telegram, free interconnection via BICO (p0922 = 999)



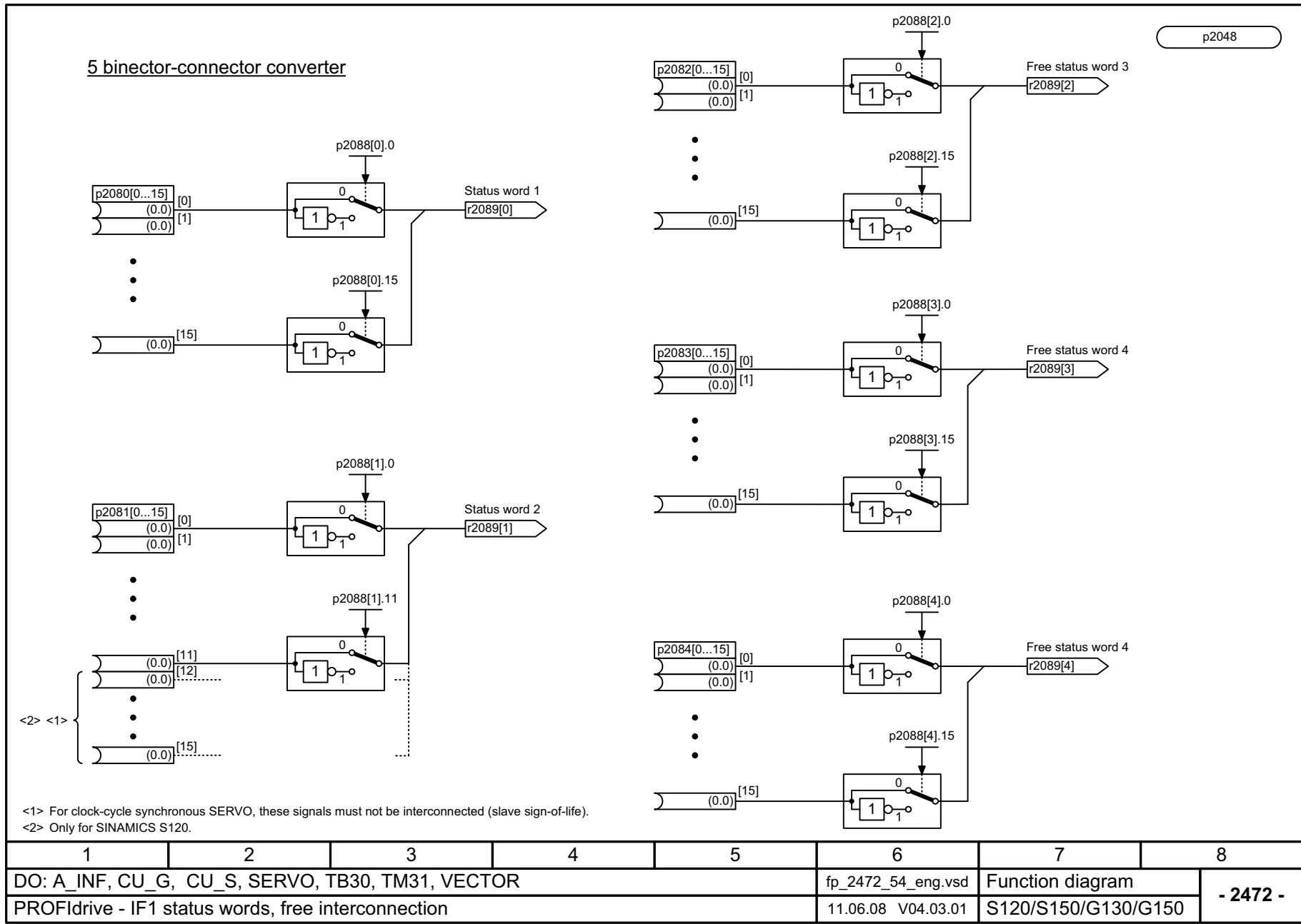


Fig. 2-75 2472 – IF1 status words, free interconnection

PROFdrive sampling time

Refer to [1020.7]

Signal targets for STW1 (positioning mode, r0108.4 = 1)

Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW1.0	= ON (pulses can be enabled) 0 = OFF1 (braking with ramp-funct. generator, then pulse cancellation and ready-to-power-up)	p0840[0] = r2090.0	[2501.3]	[2610]	-
STW1.1	1 = No OFF2 (enable is possible) <4> 0 = OFF2 (immediate pulse cancellation and power-on inhibit)	p0844[0] = r2090.1	[2501.3]	[2610]	-
STW1.2	1 = No OFF3 (enable possible) <4> 0 = OFF3 (braking with the OFF3 ramp p1135, then pulse cancellation and power-on inhibit)	p0848[0] = r2090.2	[2501.3]	[2610]	-
STW1.3	1 = Enable operation (pulses can be enabled) 0 = Inhibit operation (cancel pulses)	p0852[0] = r2090.3	[2501.3]	[2610]	-
STW1.4	1 = Do not reject traversing task 0 = Reject traversing task (ramp-down with the maximum deceleration)	p2641 = r2090.4	-	[3616.5] [3625]	-
STW1.5	1 = No intermediate stop 0 = Intermediate stop	p2640 = r2090.5	-	[3616.5] [3625]	-
STW1.6	= Activate traversing task	<3> p2631 = r2090.6 p2650 = r2090.6	-	[3620.1] [3625]	-
STW1.7	= Acknowledge faults	p2103[0] = r2090.7	[2546.1]	[8060]	-
STW1.8	1 = Jog 1 ON 0 = Jog 1 OFF	p2589 = r2090.8	-	[3610.1] [3625]	-
STW1.9	1 = Jog 2 ON 0 = Jog 2 OFF	p2590 = r2090.9	-	[3610.1] [3625]	-
STW1.10	1 = Control via PLC <2>	p0854[0] = r2090.10	[2501.3]	[2501]	-
STW1.11	1 = Start homing 0 = Stop homing	p2595 = r2090.11	-	[3612.1] [3625]	-
STW1.12	Reserved	-	-	-	-
STW1.13	= External block change	p2633 = r2090.13	-	[3615]	-
STW1.14	Reserved	-	-	-	-
STW1.15	Reserved	-	-	-	-

<1> Used in telegrams 7, 9, 110, 111.

<3> The interconnection p2649 = 0 is made additionally only in Telegram 7,9 and 110.

<2> STW1.10 must be set to ensure that the drive object accepts the process data (PZD).

<4> OC → Operating condition

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2475_55_eng.vsd	Function diagram	- 2475 -
PROFdrive - STW1 control word 1 interconnection (r0108.4 = 1)					30.09.08 V04.03.01	SINAMICS S120/S150	

Fig. 2-76 2475 – STW1 control word 1 interconnection (r0108.4 = 1)

PROFdrive sampling time
Refer to [1020.7]

Signal targets for SATZANW (positioning mode, r0108.4 = 1)						<1>
Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted	
SATZANW.0	1 = Block selection, bit 0	p2625 = r2091.0	-	[3640]	-	
SATZANW.1	1 = Block selection, bit 1	p2626 = r2091.1	-	[3640]	-	
SATZANW.2	1 = Block selection, bit 2	p2627 = r2091.2	-	[3640]	-	
SATZANW.3	1 = Block selection, bit 3	p2628 = r2091.3	-	[3640]	-	
SATZANW.4	1 = Block selection, bit 4	p2629 = r2091.4	-	[3640]	-	
SATZANW.5	1 = Block selection, bit 5	p2630 = r2091.5	-	[3640]	-	
SATZANW.6	Reserved	-	-	-	-	
SATZANW.7	Reserved	-	-	-	-	
SATZANW.8	Reserved	-	-	-	-	
SATZANW.9	Reserved	-	-	-	-	
SATZANW.10	Reserved	-	-	-	-	
SATZANW.11	Reserved	-	-	-	-	
SATZANW.12	Reserved	-	-	-	-	
SATZANW.13	Reserved	-	-	-	-	
SATZANW.14	Reserved	-	-	-	-	
SATZANW.15	1 = Activate MDI 0 = De-activate MDI	p2647 = r2091.15	-	[3625] [3640]	-	

<1> Used in telegrams 7, 9, 110.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2476_55_eng.vsd	Function diagram	- 2476 -
PROFdrive - SATZANW-Block Selection interconnection (r0108.4 = 1)					30.09.08 V04.03.01	SINAMICS S120/S150	

Fig. 2-77 2476 – SATZANW-Pos block selection interconnection (r0108.4 = 1)

PROFdrive sampling time
Refer to [1020.7]

Signal sources for ZSW1 (positioning mode, r0108.4 = 1)					
Signal	Meaning	Interconnection parameters	[Function diagram] Internal status word	[Function diagram] signal source	Inverted <2>
ZSW1.0	1 = Ready to power-up	p2080[0] = r0899.0	[2503.7]	[2610]	-
ZSW1.1	1 = Ready to operate (DC link loaded, pulses blocked)	p2080[1] = r0899.1	[2503.7]	[2610]	-
ZSW1.2	1 = Operation enabled (drive follows n_set)	p2080[2] = r0899.2	[2503.7]	[2610]	-
ZSW1.3	1 = Fault present	p2080[3] = r2139.3	[2548.7]	[8060]	-
ZSW1.4	1 = No coast down active (OFF2 inactive)	p2080[4] = r0899.4	[2503.7]	[2610]	-
ZSW1.5	1 = No fast stop active (OFF3 inactive)	p2080[5] = r0899.5	[2503.7]	[2610]	-
ZSW1.6	1 = Power-on inhibit active	p2080[6] = r0899.6	[2503.7]	[2610]	-
ZSW1.7	1 = Alarm present	p2080[7] = r2139.7	[2548.7]	[8065]	-
ZSW1.8	1 = Following error within tolerance	p2080[8] = r2684.8	[3646.7]	[4025]	-
ZSW1.9	1 = Control requested <3>	p2080[9] = r0899.9	[2503.7]	[2503]	-
ZSW1.10	1 = Target position reached	p2080[10] = r2684.10	[3646.7]	[4020] [3625]	-
ZSW1.11	1 = Home position set	p2080[11] = r2684.11	[3646.7]	[3612] [3614]	-
ZSW1.12	 = Acknowledgement traversing block activated	p2080[12] = r2684.12	[3646.7]	[3616] [3620]	-
ZSW1.13	1 = Drive at standstill	p2080[13] = r2199.0	[2537.7]	[8010] [3625]	-
ZSW1.14	1 = Axis accelerated <4>	p2080[14] = r2684.4	[3646.7]	[3635]	-
ZSW1.15	1 = Axis decelerated <4>	p2080[15] = r2684.5	[3646.7]	[3635]	-

<1> Used in telegrams 7, 9, 110, 111.

<2> The status word is generated using the binector-connector converter p2088[0].

<3> The drive object is ready to accept data.

<4> Only for telegram 111.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2479_55_eng.vsd	Function diagram	
PROFIdrive - ZSW1-Status Word 1 interconnection (r0108.4 = 1)					29.07.08 V04.03.01	SINAMICS S120/S150	
							- 2479 -

Fig. 2-78 2479 – ZSW1 status word 1 interconnection (r0108.4 = 1)

PROFdrive sampling time
Refer to [1020.7]

Signal targets for MDI_MOD (positioning mode, r0108.4 = 1)						<1>		
Signal	Meaning				Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
MDI_MOD.0	1 = Absolute positioning is selected. 0 = Relative positioning is selected.				p2648 = r2094.0	-	-	-
MDI_MOD.1	0 = Absolute positioning through the shortest distance.	1 = Absolute positioning in the positive direction.	2 = Absolute positioning in the negative direction.	3 = Absolute positioning through the shortest distance.	p2651 = r2094.1	-	-	-
MDI_MOD.2					p2652 = r2094.2	-	-	-
MDI_MOD.3	Reserved				-	-	-	-
MDI_MOD.4	Reserved				-	-	-	-
MDI_MOD.5	Reserved				-	-	-	-
MDI_MOD.6	Reserved				-	-	-	-
MDI_MOD.7	Reserved				-	-	-	-
MDI_MOD.8	Reserved				-	-	-	-
MDI_MOD.9	Reserved				-	-	-	-
MDI_MOD.10	Reserved				-	-	-	-
MDI_MOD.11	Reserved				-	-	-	-
MDI_MOD.12	Reserved				-	-	-	-
MDI_MOD.13	Reserved				-	-	-	-
MDI_MOD.14	Reserved				-	-	-	-
MDI_MOD.15	Reserved				-	-	-	-

<1> Used in telegram 9.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2480_55_eng.vsd	Function diagram	- 2480 -
PROFdrive – MDI_MOD-MDI Mode interconnection (r0108.4 = 1)					30.04.09 V04.03.01	SINAMICS S120/S150	

Fig. 2-79 2480 – MDI_MOD-MDI mode interconnection (r0108.4 = 1)

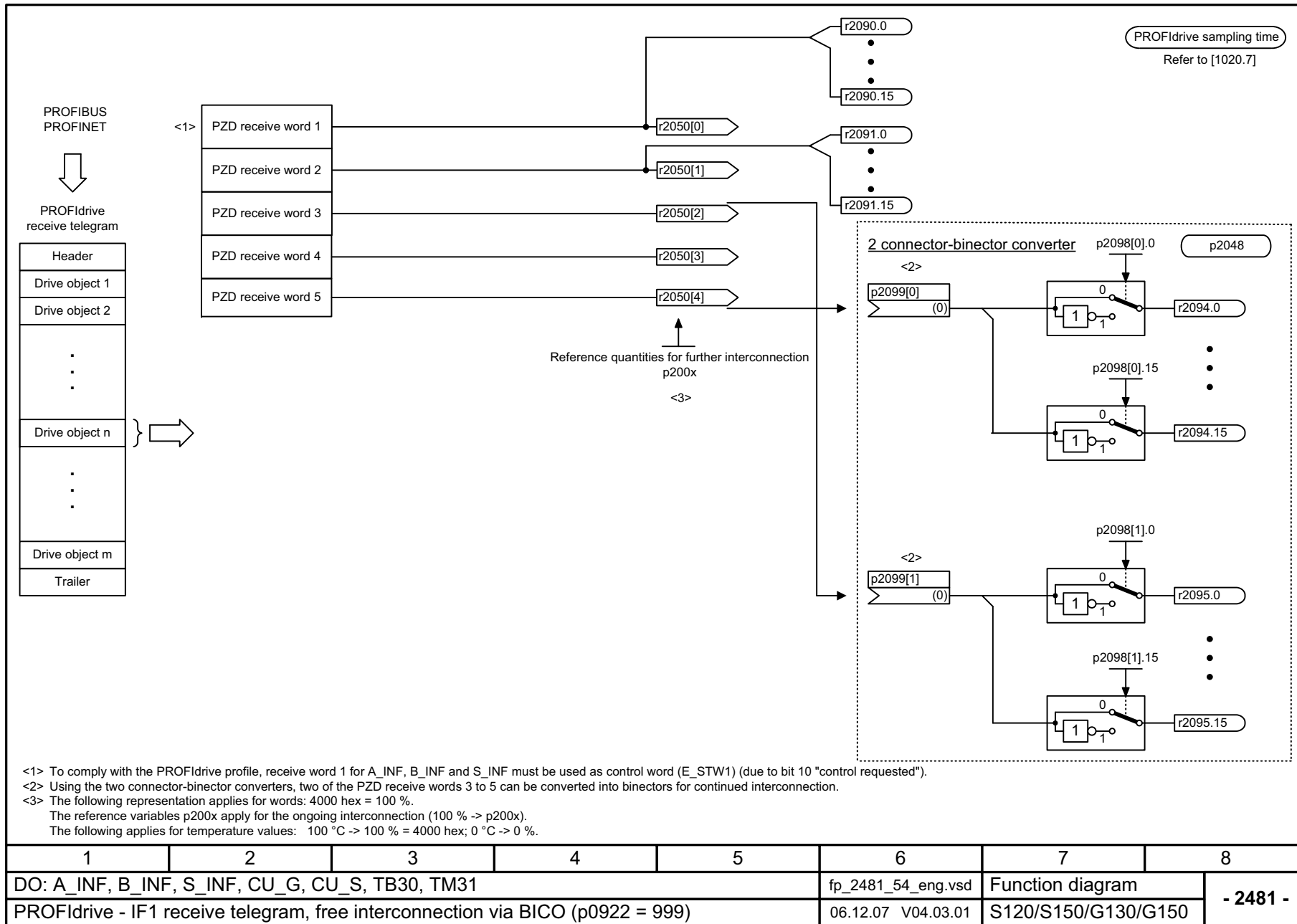
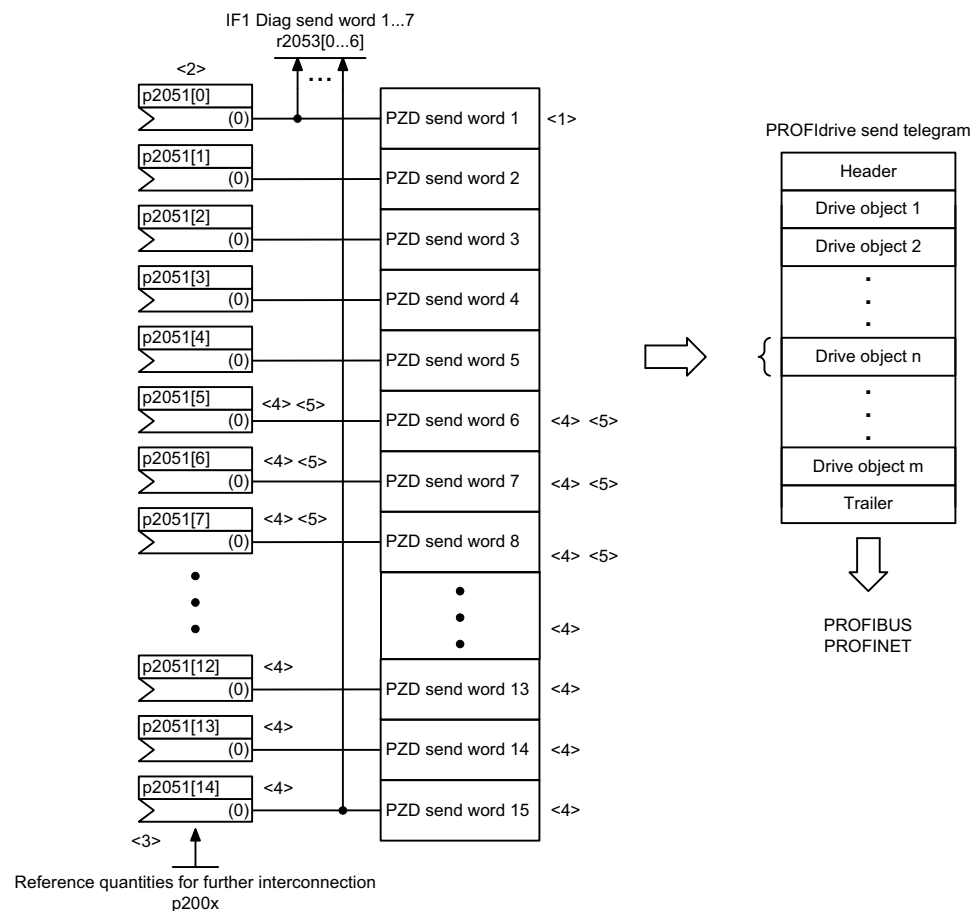


Fig. 2-80 2481 – IF1 receive telegram, free interconnection via BICO (p0922 = 999)

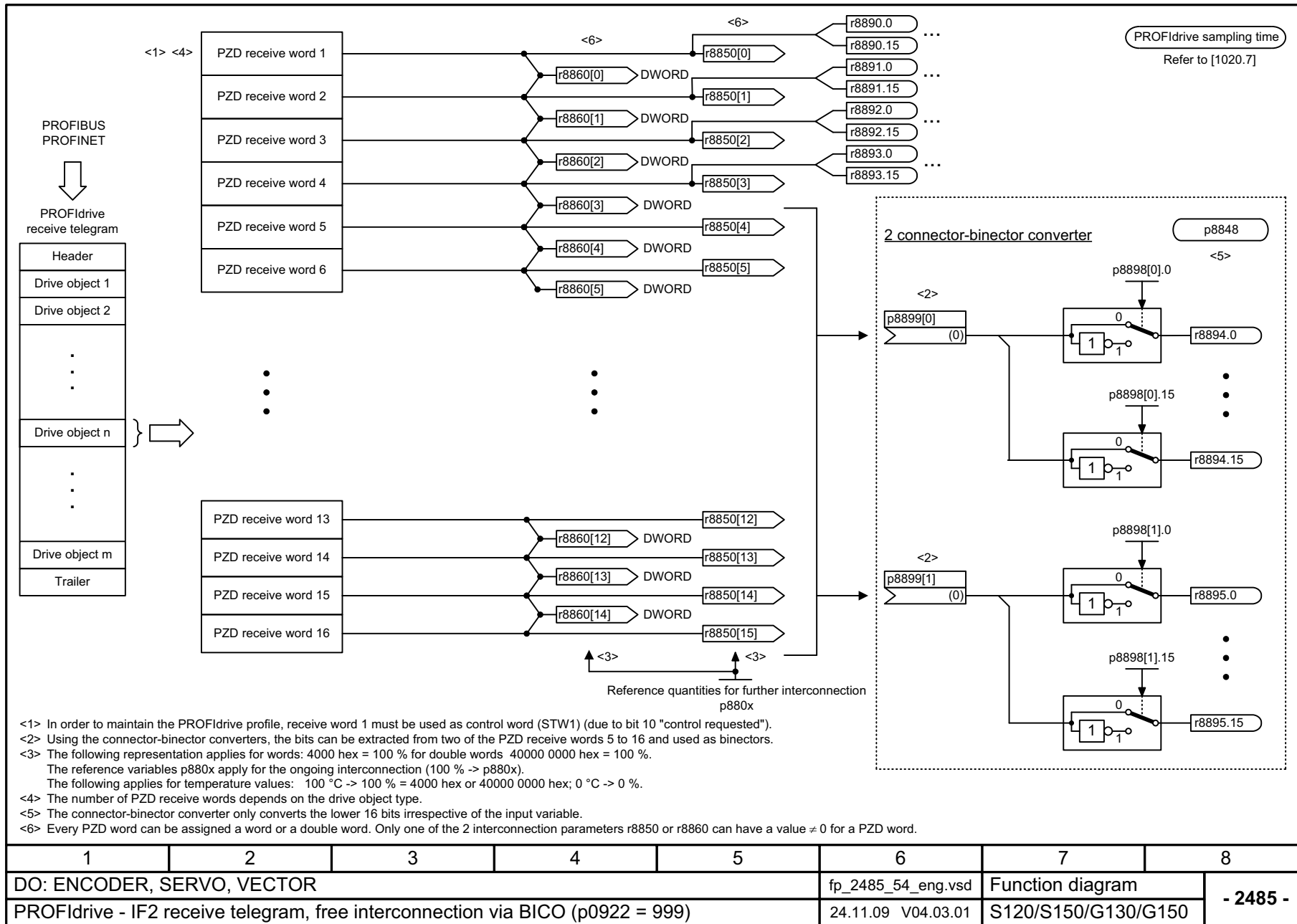
PROFdrive Abtastzeit
Refer to [1020.7]

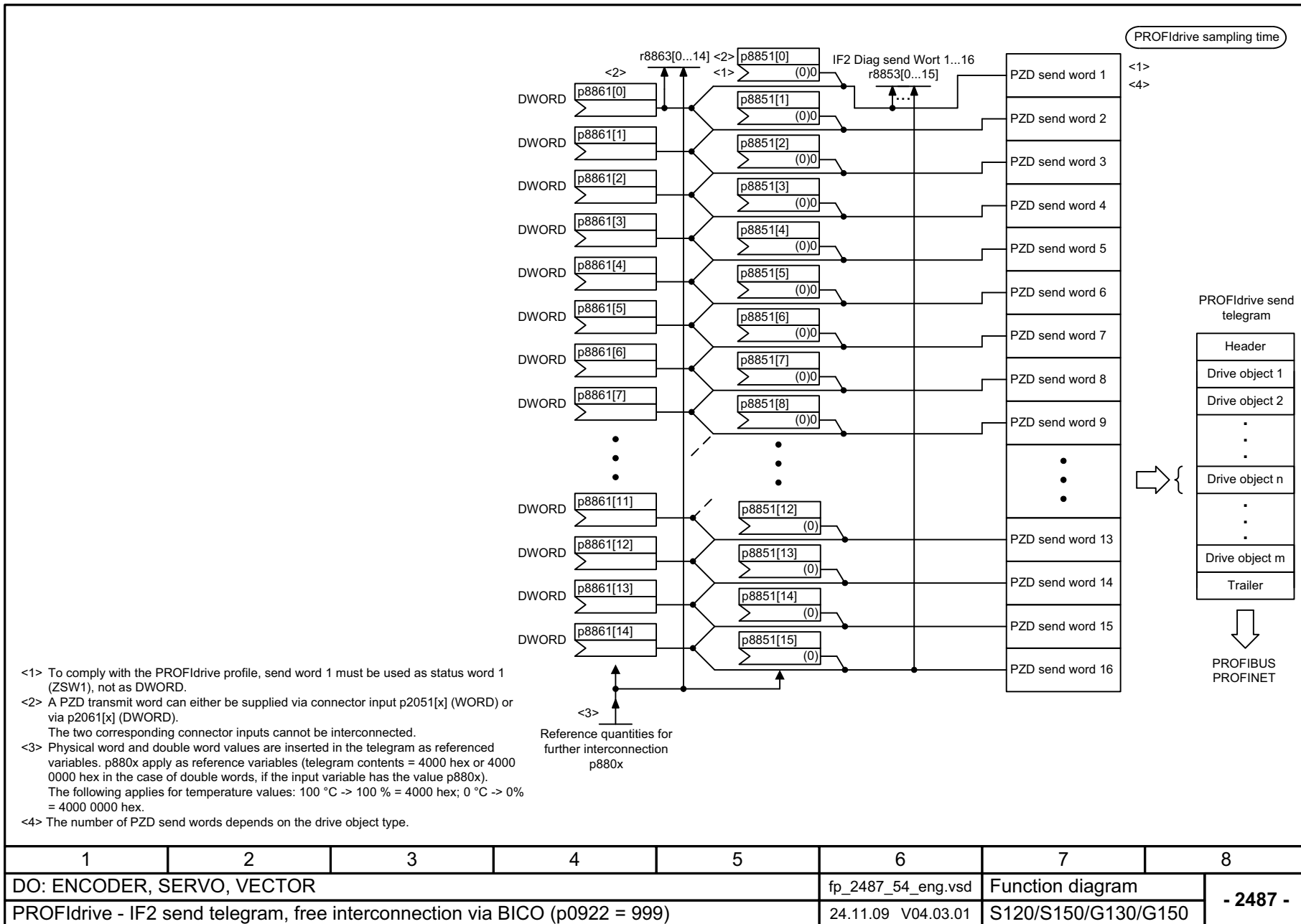


- <1> In order to maintain the PROFdrive profile, send word 1 must be used as status word 1 (E_ZSW1) for A_INF, B_INF and S_INF.
- <2> Using the binector/connector converters at [2472], bits of 4 send words can be interconnected with any binectors.
- <3> The following representation applies for words: 4000 hex = 100 %.
The reference variables p200x apply for the ongoing interconnection (100 % -> p200x).
The following applies for temperature values: 100 °C -> 100 % = 4000 hex; 0 °C -> 0 %.
- <4> Valid for CU_S.
- <5> Valid for A_INF, B_INF, S_INF.

1	2	3	4	5	6	7	8
DO: A_INF, B_INF, CU_G, CU_S, S_INF, TB30, TM15DI_DO, TM31					fp_2483_54_eng.vsd	Function diagram	
PROFdrive - IF1 send telegram, free interconnection via BICO (p0922 = 999)					06.12.07 V04.03.01	S120/S150/G130/G150	
							- 2483 -

Fig. 2-81 2483 – IF1 send telegram, free interconnection via BICO (p0922 = 999)





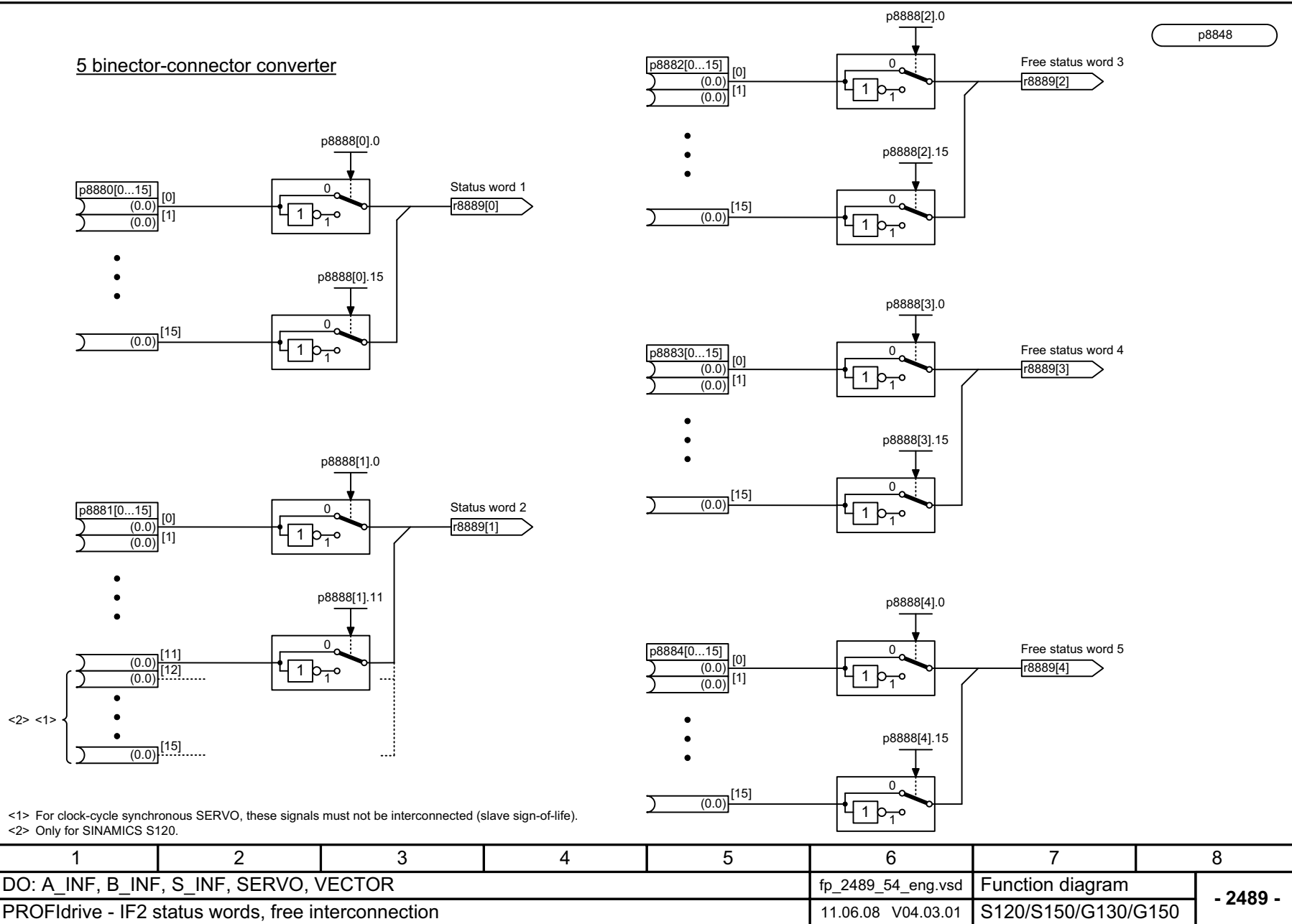
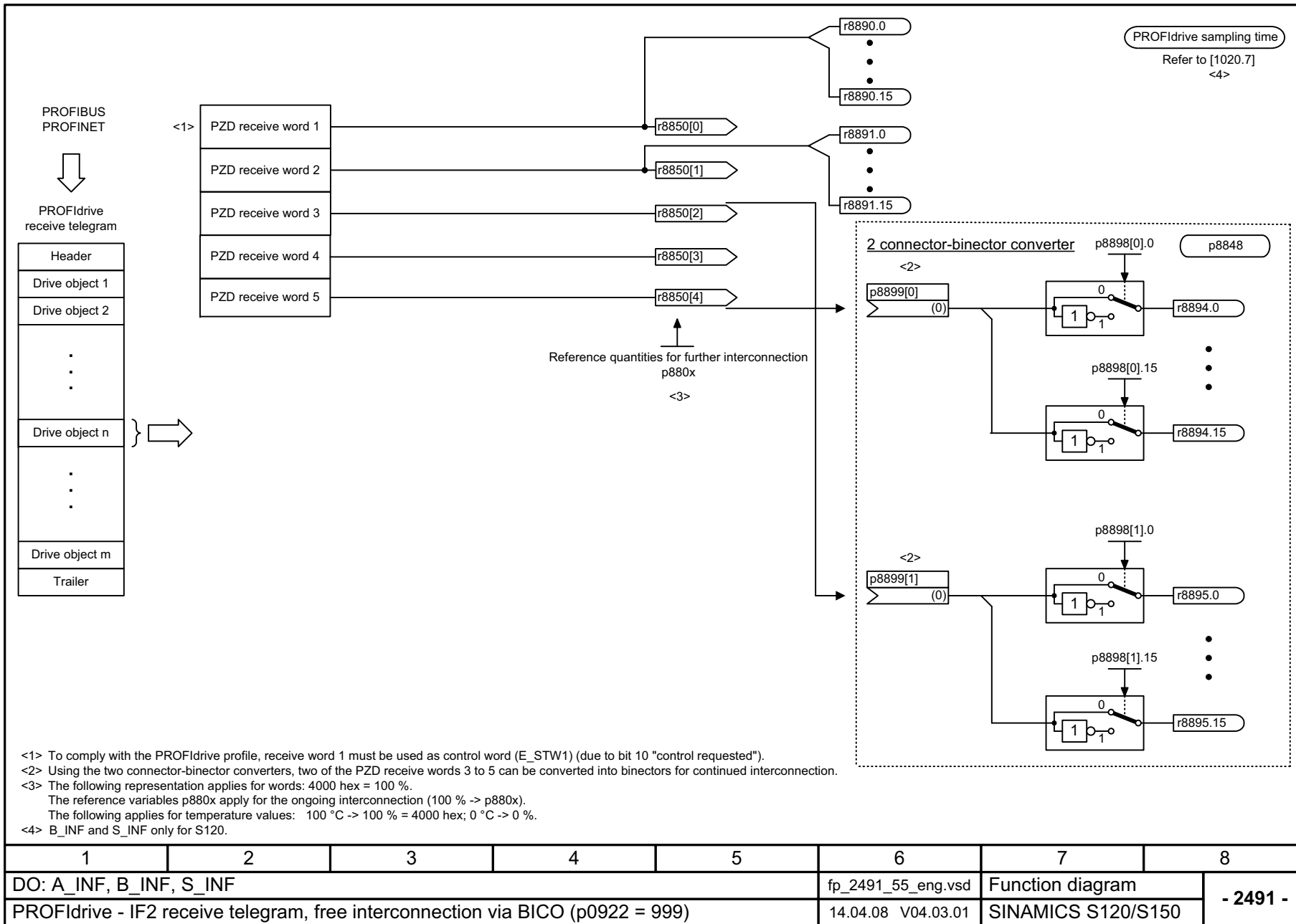
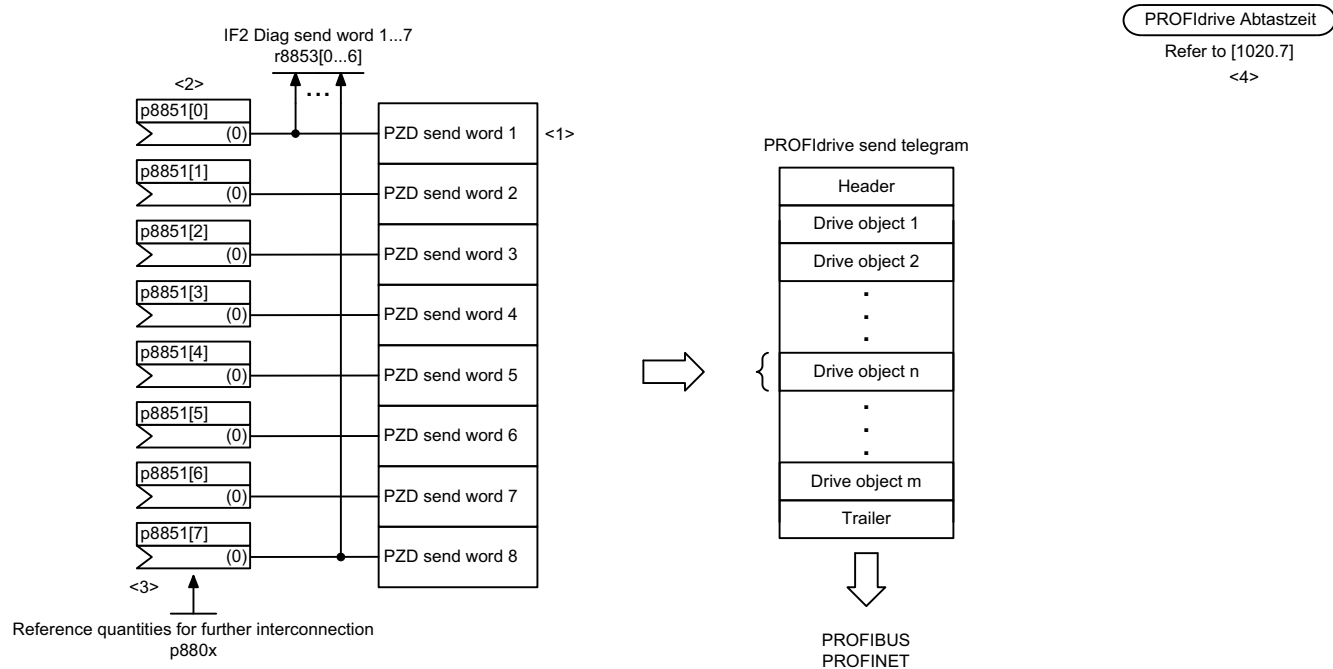


Fig. 2-84

2489 – IF2 status words, free interconnection





- <1> In order to maintain the PROFIdrive profile, send word 1 must be used as status word 1 (E_ZSW1).
- <2> Using the binector/connector converters at [2472], bits of 4 send words can be interconnected with any binectors.
- <3> The following representation applies for words: 4000 hex = 100 %.
The reference variables p880x apply for the ongoing interconnection (100 % -> p880x).
The following applies for temperature values: 100 °C -> 100 % = 4000 hex; 0 °C -> 0 %.
- <4> B_INF und S_INF only for S120.

1	2	3	4	5	6	7	8
DO: A_INF, B_INF, S_INF					fp_2493_55_eng.vsd	Function diagram	
PROFIdrive - IF2 send telegram, free interconnection via BICO (p0922 = 999)					14.04.08 V04.03.01	SINAMICS S120/S150	

Fig. 2-86 2493 – IF2 send telegram, free interconnection via BICO (p0922 = 999)

PROFdrive sampling time
Refer to [1020.7]

Signal targets for CU_STW1

Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
CU_STW1.0	Synchronization	p0681[0] = r2090.0	-	-	-
CU_STW1.1	RTC PING	p3104 = r2090.1	-	-	-
CU_STW1.2	Reserved	-	-	-	-
CU_STW1.3	Reserved	-	-	-	-
CU_STW1.4	Reserved	-	-	-	-
CU_STW1.5	Reserved	-	-	-	-
CU_STW1.6	Reserved	-	-	-	-
CU_STW1.7	Acknowledge faults	p2103[0] = r2090.7	-	-	-
CU_STW1.8	Reserved	-	-	-	-
CU_STW1.9	Reserved	-	-	-	-
CU_STW1.10	To assume control	p3116 = r2090.10	-	-	-
CU_STW1.11	Reserved	-	-	-	-
CU_STW1.12	Master sign-of-life bit 0	p2045 = r2050[0]	-	-	-
CU_STW1.13	Master sign-of-life bit 1				
CU_STW1.14	Master sign-of-life bit 2				
CU_STW1.15	Master sign-of-life bit 3				

<1> Used in telegrams 390, 391 and 392.

1	2	3	4	5	6	7	8
DO: CU_G, CU_S					fp_2495_54_eng.vsd	Function diagram	- 2495 -
PROFdrive - CU_STW1 control word 1 Control Unit interconnection					10.09.08 V04.03.01	S120/S150/G130/G150	

Fig. 2-87 2495 – CU_STW1 control word 1, Control Unit interconnection

PROFdrive sampling time
Refer to [1020.7]

Signal sources for CU_ZSW1					
Signal	Meaning	Interconnection parameters	[Function diagram] Internal status word	[Function diagram] signal source	Inverted
CU_ZSW1.0	Reserved	-	-	-	-
CU_ZSW1.1	Reserved	-	-	-	-
CU_ZSW1.2	Reserved	-	-	-	-
CU_ZSW1.3	1 = Fault present	p2081[3] = r2139.3	-	-	-
CU_ZSW1.4	Reserved	-	-	-	-
CU_ZSW1.5	Reserved	-	-	-	-
CU_ZSW1.6	Reserved	-	-	-	-
CU_ZSW1.7	1 = Alarm present	p2081[7] = r2139.7	-	-	-
CU_ZSW1.8	Synchronization (SYNC)	p2081[8] = r0899.8	-	-	-
CU_ZSW1.9	1 = No alarm present	p2081[9] = r3114.9	-	-	✓
CU_ZSW1.10	1 = No fault present	p2081[10] = r3114.10	-	-	✓
CU_ZSW1.11	1 = No safety message present	p2081[11] = r3114.11	-	-	✓
CU_ZSW1.12	Slave sign-of-life bit 0	Implicitly interconnected	-	-	-
CU_ZSW1.13	Slave sign-of-life bit 1				
CU_ZSW1.14	Slave sign-of-life bit 2				
CU_ZSW1.15	Slave sign-of-life bit 3				

<1> Used in telegrams 390, 391 and 392.

1	2	3	4	5	6	7	8	
DO: CU_G, CU_S					fp_2496_54_eng.vsd	Function diagram		- 2496 -
PROFIdrive - CU_ZSW1 status word 1 Control Unit interconnection					30.11.09 V04.03.01	S120/S150/G130/G150		

Fig. 2-88 2496 – CU_ZSW1 status word 1, Control Unit interconnection

PROFdrive sampling time
Refer to [1020.7]

Signal targets for A_DIGITAL

Signal	Meaning	Interconnection parameters <3>	[Function diagram] internal status word	[Function diagram] signal target	Inverted
A_DIGITAL.0	Digital output 8 (DI/DO 8) <2>	p0738[0] = r2091[0]	-	-	-
A_DIGITAL.1	Digital output 9 (DI/DO 9) <2>	p0739[0] = r2091[1]	-	-	-
A_DIGITAL.2	Digital output 10 (DI/DO 10) <2>	p0740[0] = r2091[2]	-	-	-
A_DIGITAL.3	Digital output 11 (DI/DO 11) <2>	p0741[0] = r2091[3]	-	-	-
A_DIGITAL.4	Digital output 12 (DI/DO 12) <2>	p0742[0] = r2091[4]	-	-	-
A_DIGITAL.5	Digital output 13 (DI/DO 13) <2>	p0743[0] = r2091[5]	-	-	-
A_DIGITAL.6	Digital output 14 (DI/DO 14) <2>	p0744[0] = r2091[6]	-	-	-
A_DIGITAL.7	Digital output 15 (DI/DO 15) <2>	p0745[0] = r2091[7]	-	-	-
A_DIGITAL.8	Reserved	-	-	-	-
A_DIGITAL.9	Reserved	-	-	-	-
A_DIGITAL.10	Reserved	-	-	-	-
A_DIGITAL.11	Reserved	-	-	-	-
A_DIGITAL.12	Reserved	-	-	-	-
A_DIGITAL.13	Reserved	-	-	-	-
A_DIGITAL.14	Reserved	-	-	-	-
A_DIGITAL.15	Reserved	-	-	-	-

<1> Used in telegrams 390, 391 and 392.

<2> Can be set via p0728 as input (DI) or output (DO).

<3> Pre-assignment, can be freely changed.

1	2	3	4	5	6	7	8
DO: CU_G, CU_S					fp_2497_54_eng.vsd	Function diagram	
PROFdrive - A_DIGITAL interconnection					06.06.08 V04.03.01	S120/S150/G130/G150	
							- 2497 -

Fig. 2-89 2497 – A_DIGITAL interconnection

PROFdrive sampling time
Refer to [1020.7]

Signal targets for E_DIGITAL

Signal	Meaning	Interconnection parameters <3>	[Function diagram] Internal status word	[Function diagram] signal target	Inverted
E_DIGITAL.0	Digital input 8 (DI/DO 8) <2>	p2082[0] = r0722[8]	-	-	-
E_DIGITAL.1	Digital input 9 (DI/DO 9) <2>	p2082[1] = r0722[9]	-	-	-
E_DIGITAL.2	Digital input 10 (DI/DO 10) <2>	p2082[2] = r0722[10]	-	-	-
E_DIGITAL.3	Digital input 11 (DI/DO 11) <2>	p2082[3] = r0722[11]	-	-	-
E_DIGITAL.4	Digital input 12 (DI/DO 12) <2>	p2082[4] = r0722[12]	-	-	-
E_DIGITAL.5	Digital input 13 (DI/DO 13) <2>	p2082[5] = r0722[13]	-	-	-
E_DIGITAL.6	Digital input 14 (DI/DO 14) <2>	p2082[6] = r0722[14]	-	-	-
E_DIGITAL.7	Digital input 15 (DI/DO 15) <2>	p2082[7] = r0722[15]	-	-	-
E_DIGITAL.8	Digital input 0 (DI 0)	p2082[8] = r0722[0]	-	-	-
E_DIGITAL.9	Digital input 1 (DI 1)	p2082[9] = r0722[1]	-	-	-
E_DIGITAL.10	Digital input 2 (DI 2)	p2082[10] = r0722[2]	-	-	-
E_DIGITAL.11	Digital input 3 (DI 3)	p2082[11] = r0722[3]	-	-	-
E_DIGITAL.12	Digital input 4 (DI 4)	p2082[12] = r0722[4]	-	-	-
E_DIGITAL.13	Digital input 5 (DI 5)	p2082[13] = r0722[5]	-	-	-
E_DIGITAL.14	Digital input 6 (DI 6)	p2082[14] = r0722[6]	-	-	-
E_DIGITAL.15	Digital input 7 (DI 7)	p2082[15] = r0722[7]	-	-	-

<1> Used in telegrams 390, 391 and 392.

<2> Can be set via p0728 as input (DI) or output (DO).

<3> Pre-assignment, can be freely changed.

1	2	3	4	5	6	7	8
DO: CU_G, CU_S					fp_2498_54_eng.vsd	Function diagram	- 2498 -
PROFdrive - E_DIGITAL interconnection					06.06.08 V04.03.01	S120/S150/G130/G150	

Fig. 2-90 2498 – E_DIGITAL interconnection

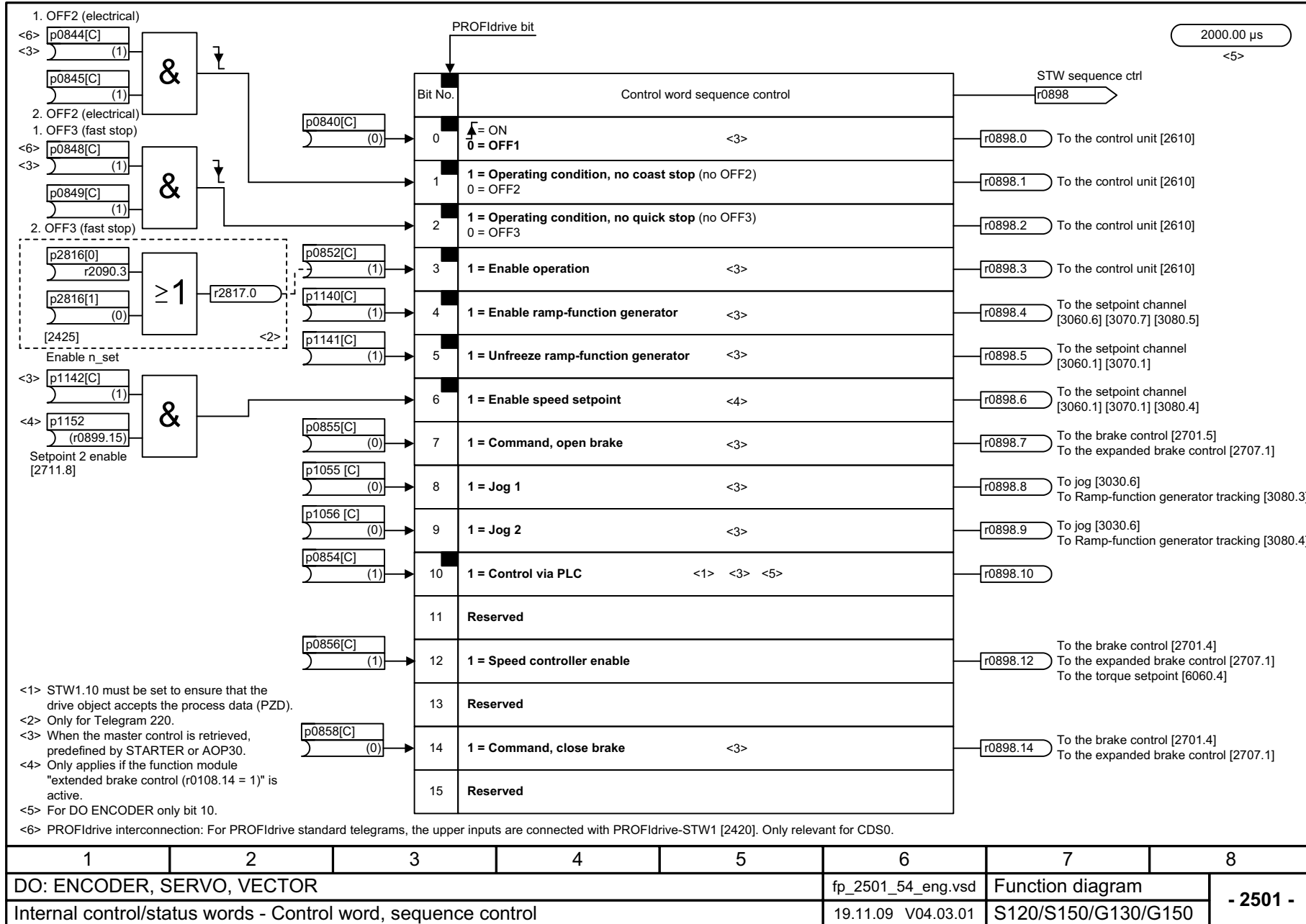
2.8 Internal control/status words

Function diagrams

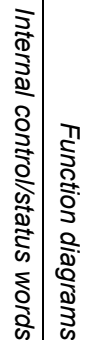
2501 – Control word sequence control	2-1482
2503 – Status word sequence control	2-1483
2505 – Control word setpoint channel	2-1484
2520 – Control word, speed controller	2-1485
2522 – Status word, speed controller	2-1486
2526 – Status word, closed loop control	2-1487
2530 – Status word, current control	2-1488
2534 – Status word, monitoring functions 1	2-1489
2536 – Status word, monitoring functions 2	2-1490
2537 – Status word, monitoring functions 3	2-1491
2546 – Control word faults/alarms	2-1492
2548 – Status word faults/alarms 1 and 2	2-1493

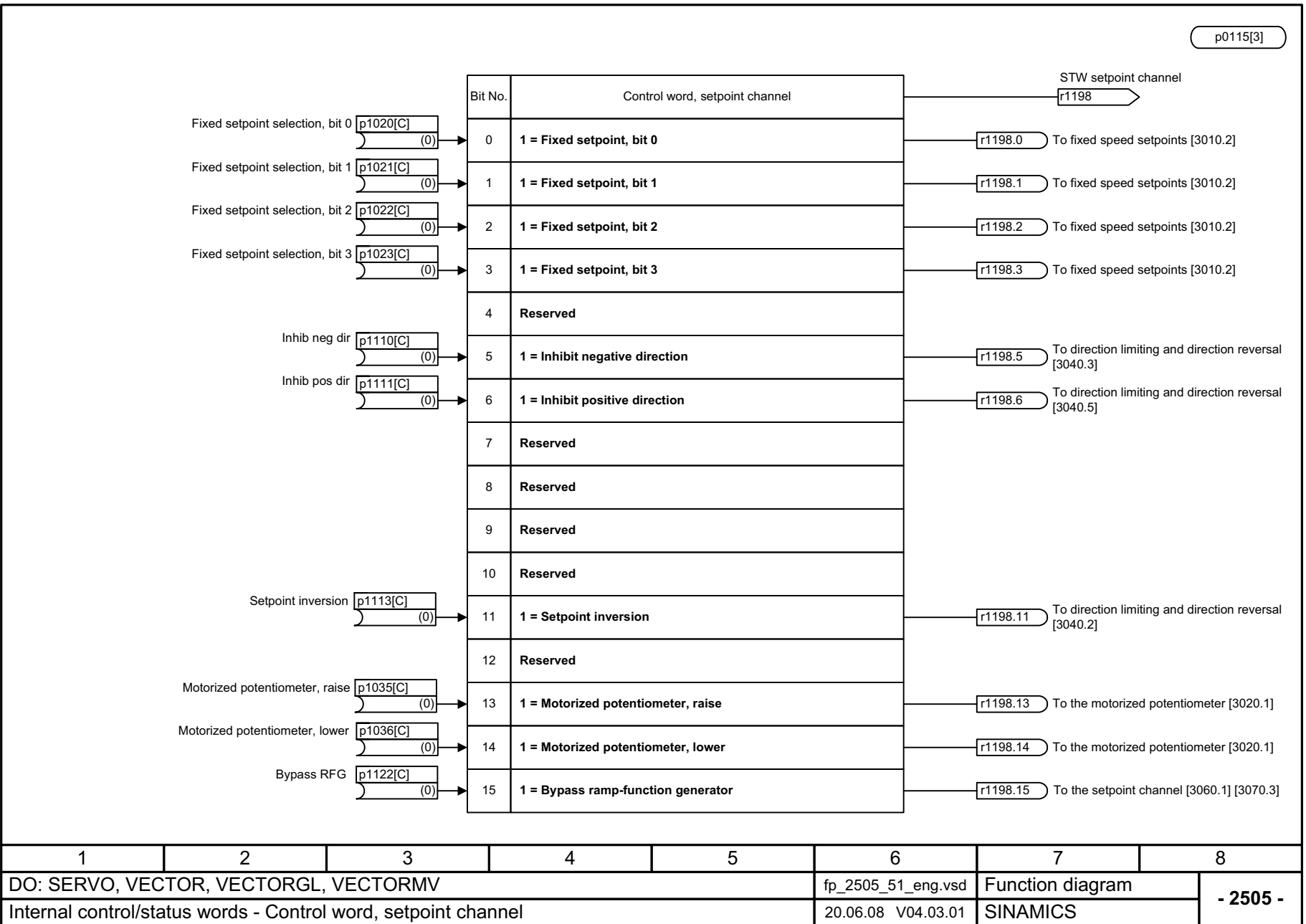
Fig. 2-91

2501 – Control word sequence control



2503 – Status word sequence control





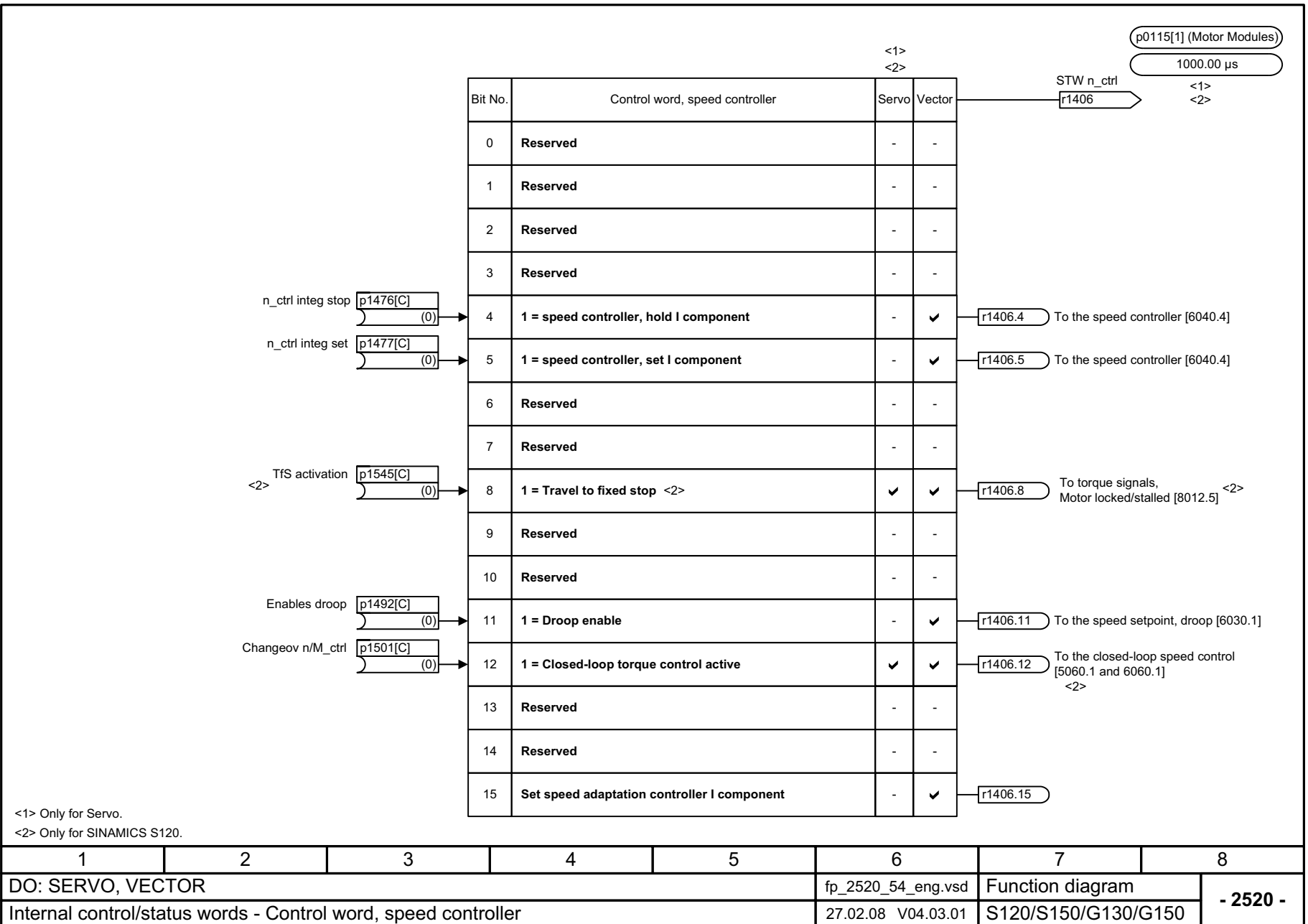
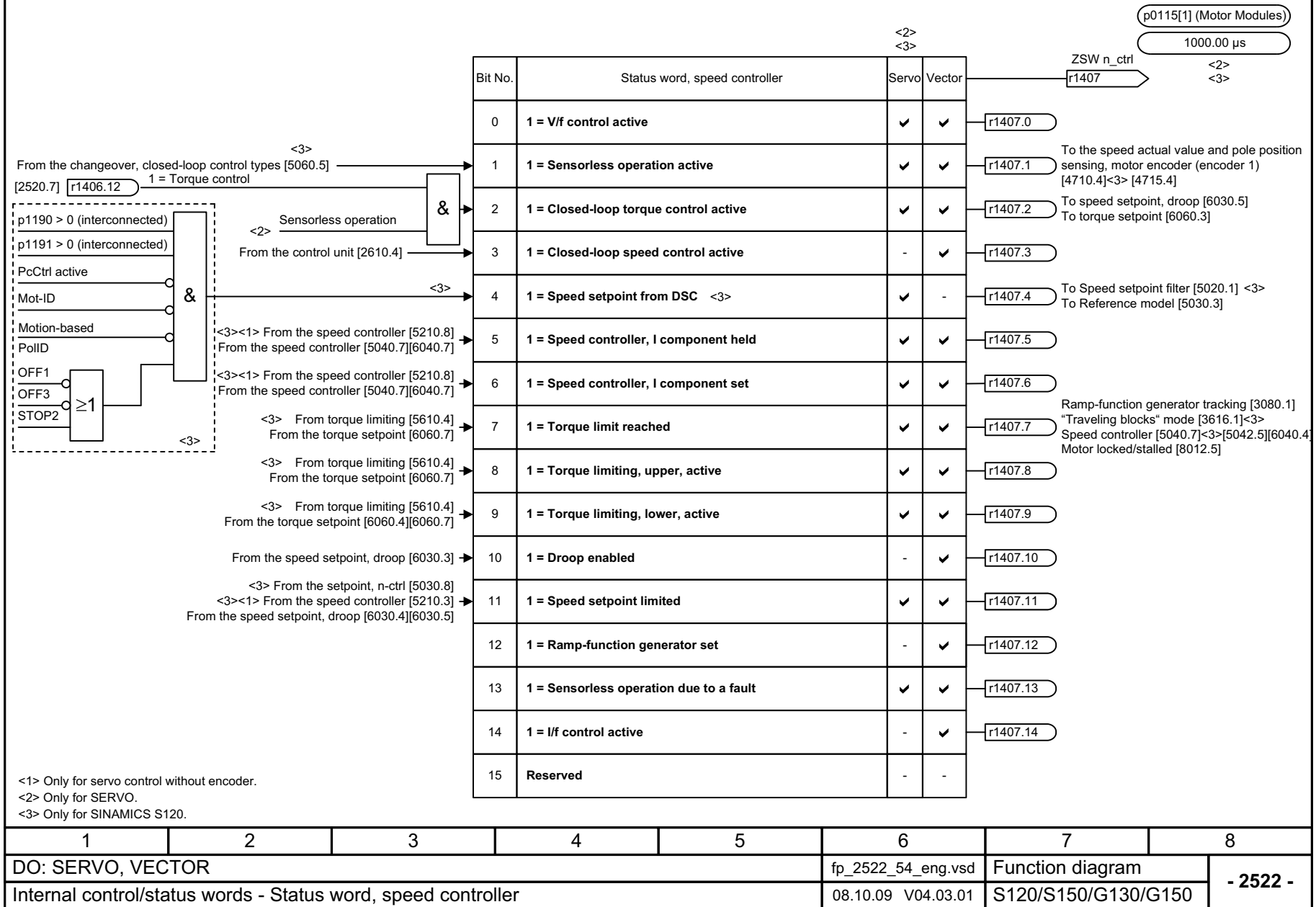


Fig. 2-94 2520 – Control word, speed controller

Fig. 2-95 2522 – Status word, speed controller



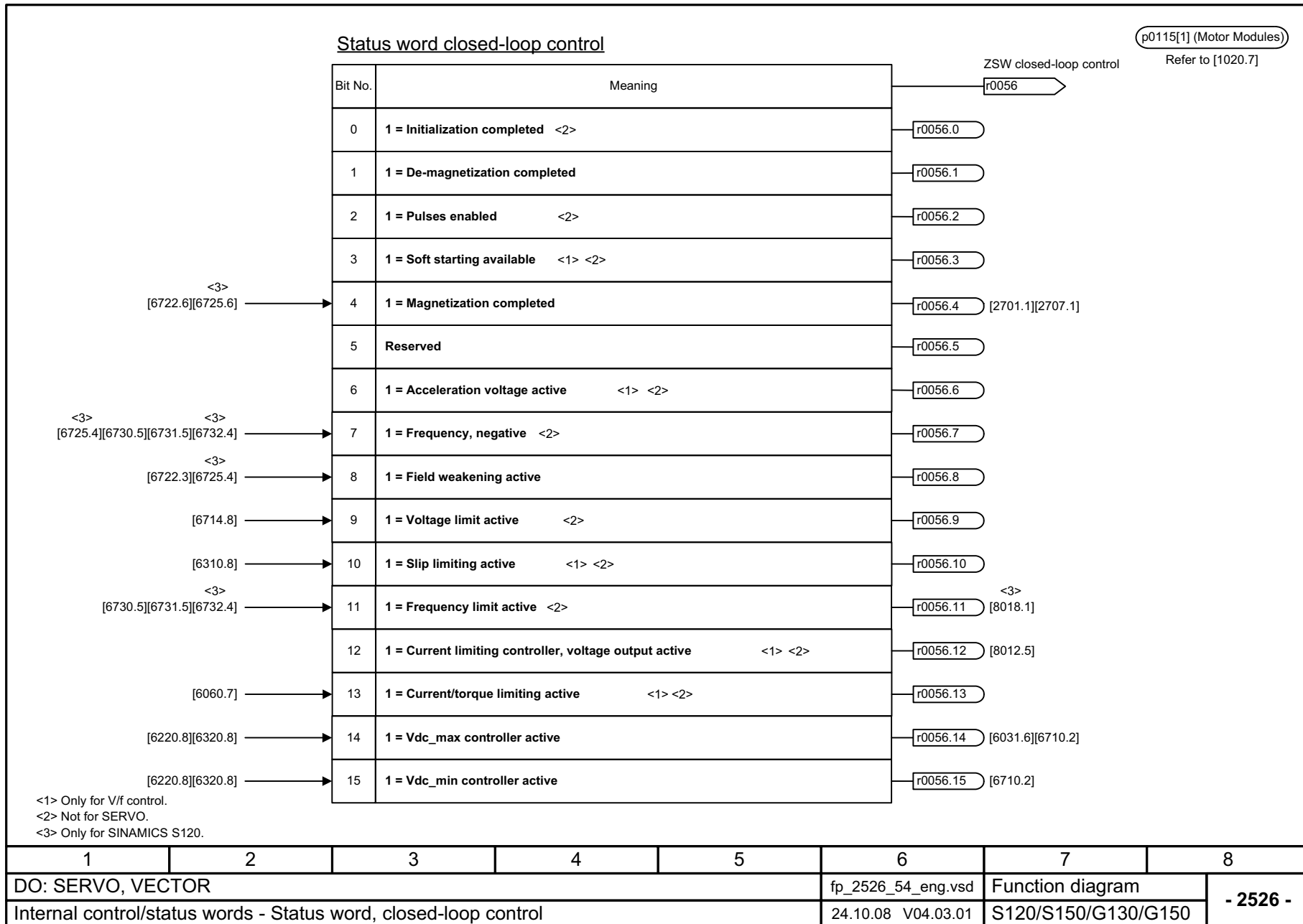


Fig. 2-96 2526 – Status word, closed loop control

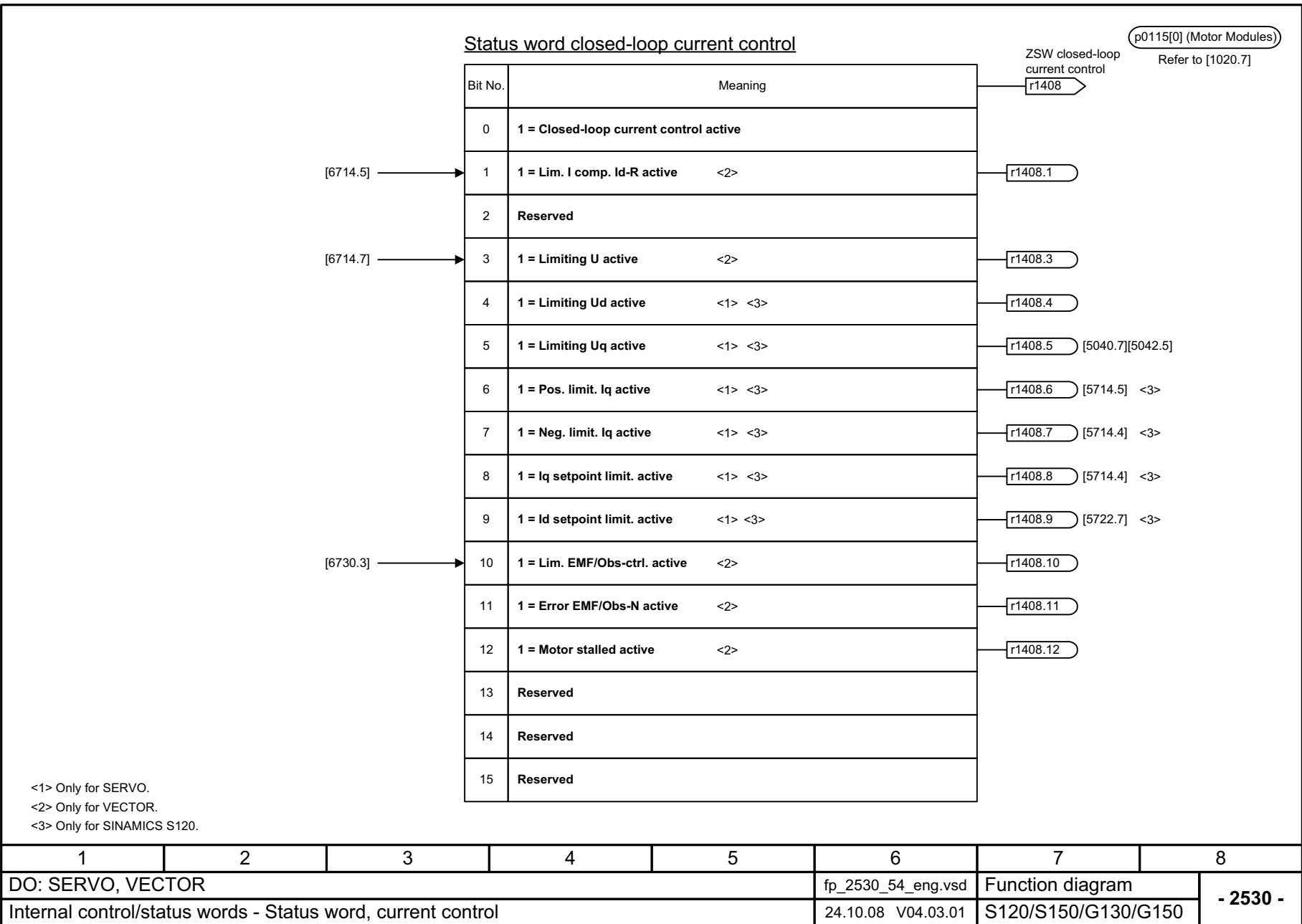


Fig. 2-97 2530 – Status word, current control

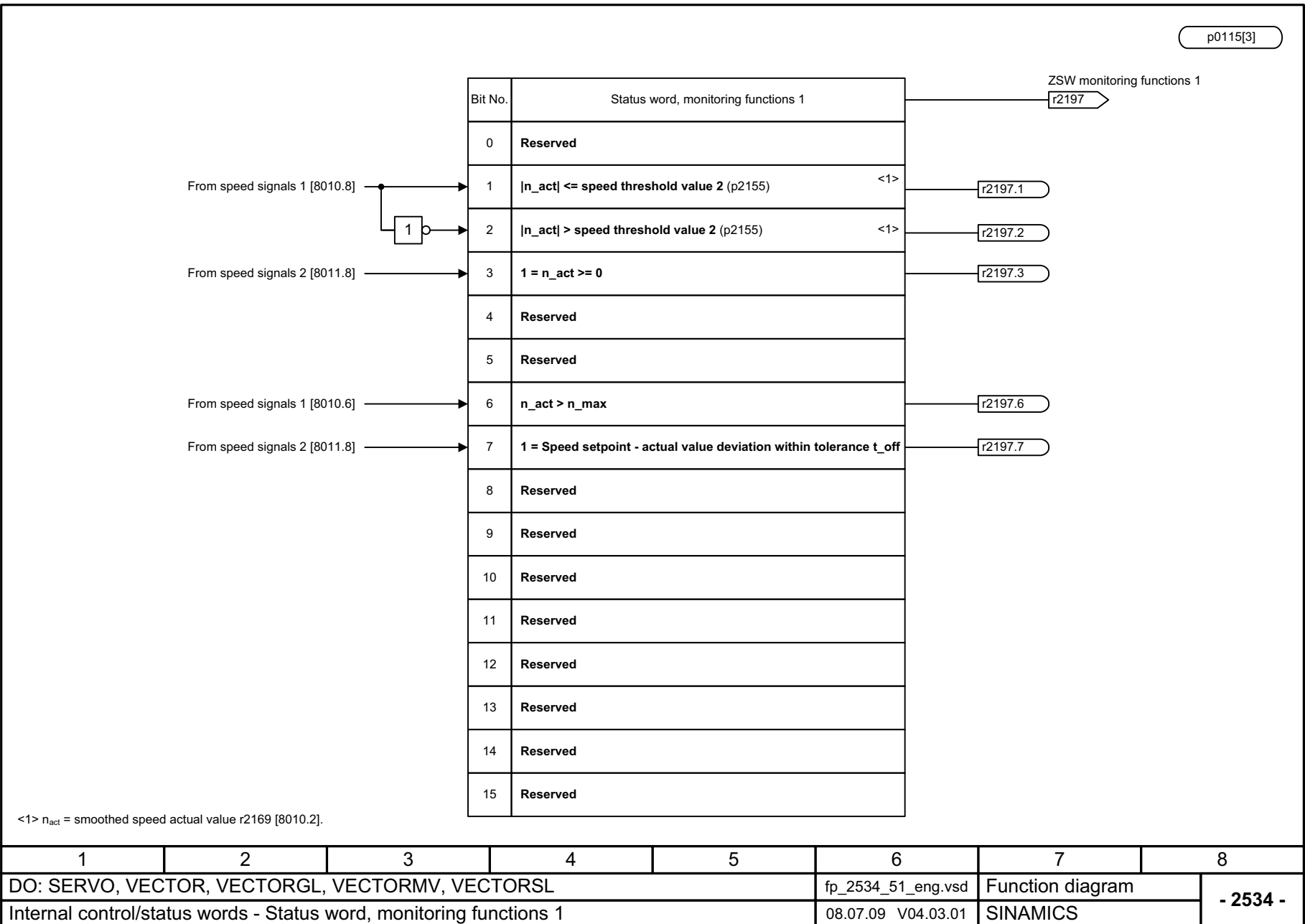


Fig. 2-98 2534 – Status word, monitoring functions 1

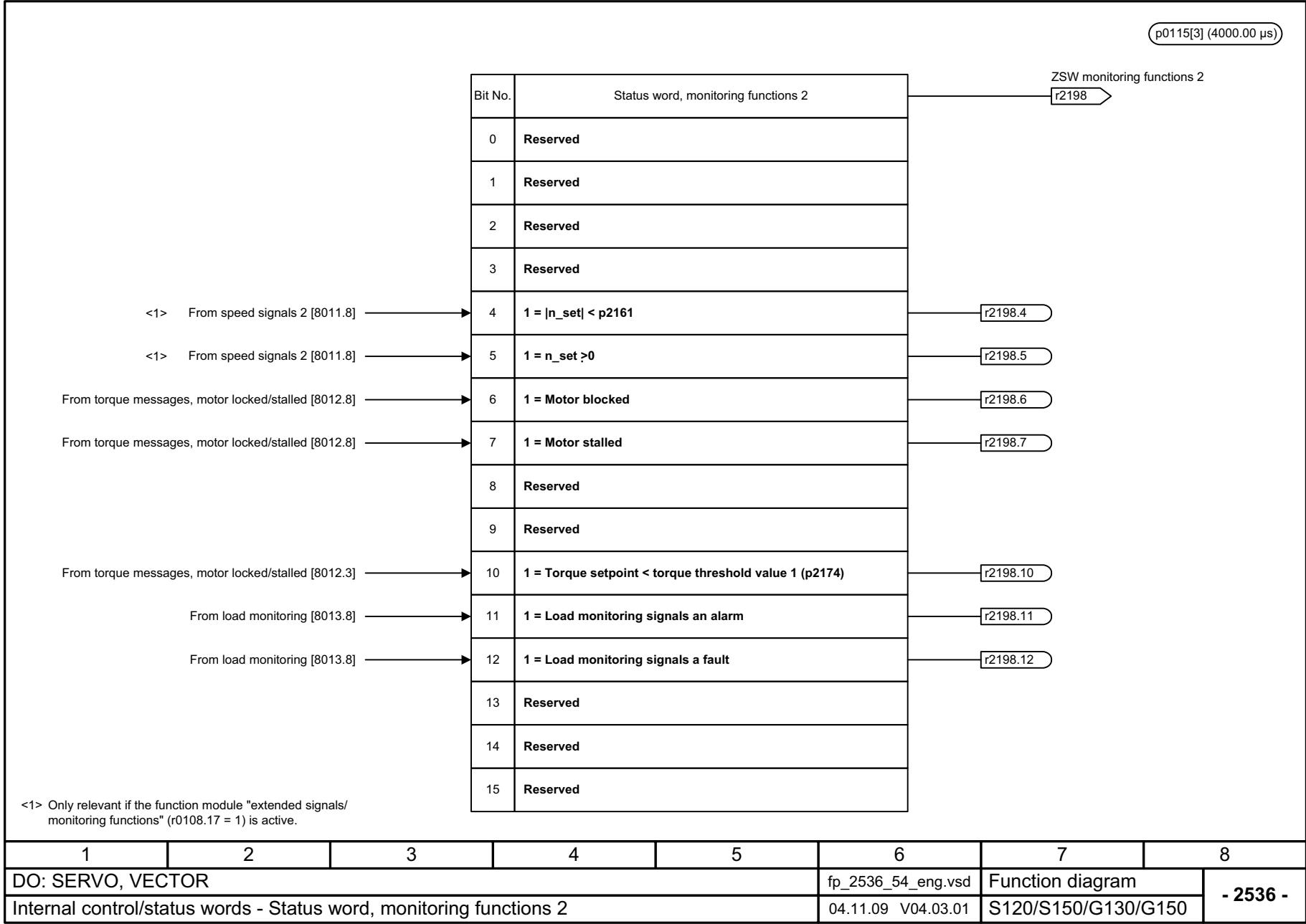


Fig. 2-99 2536 – Status word, monitoring functions 2

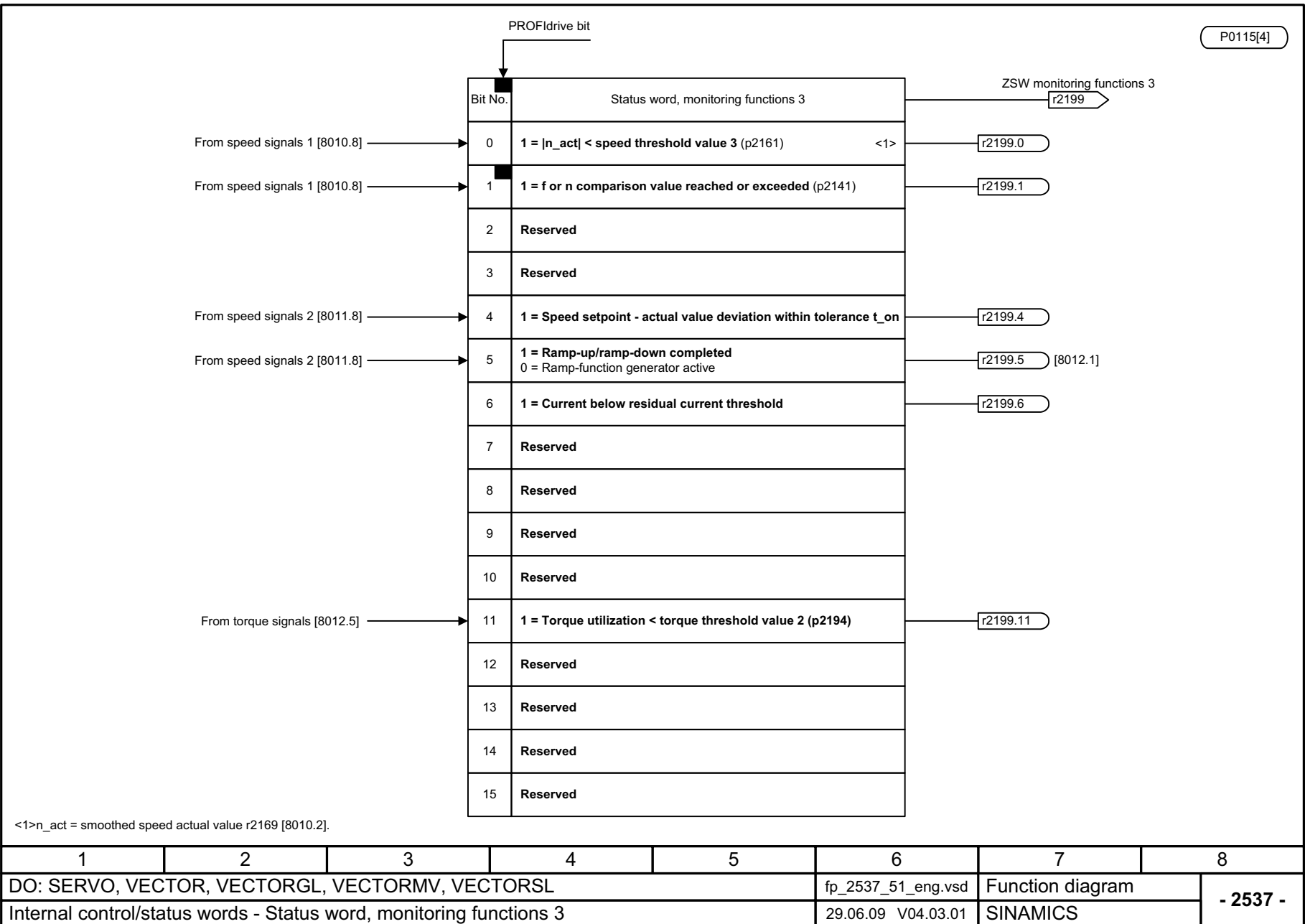
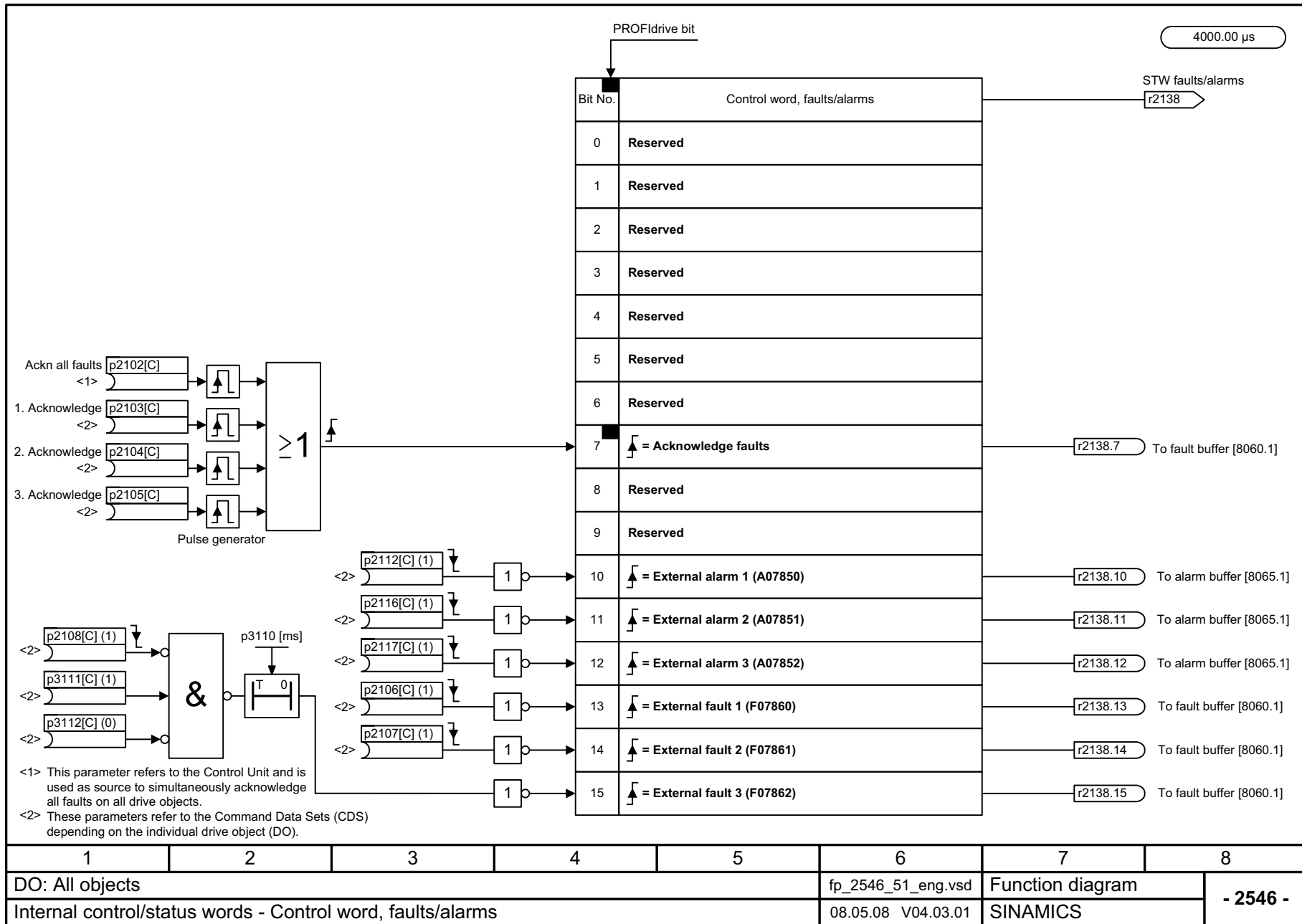
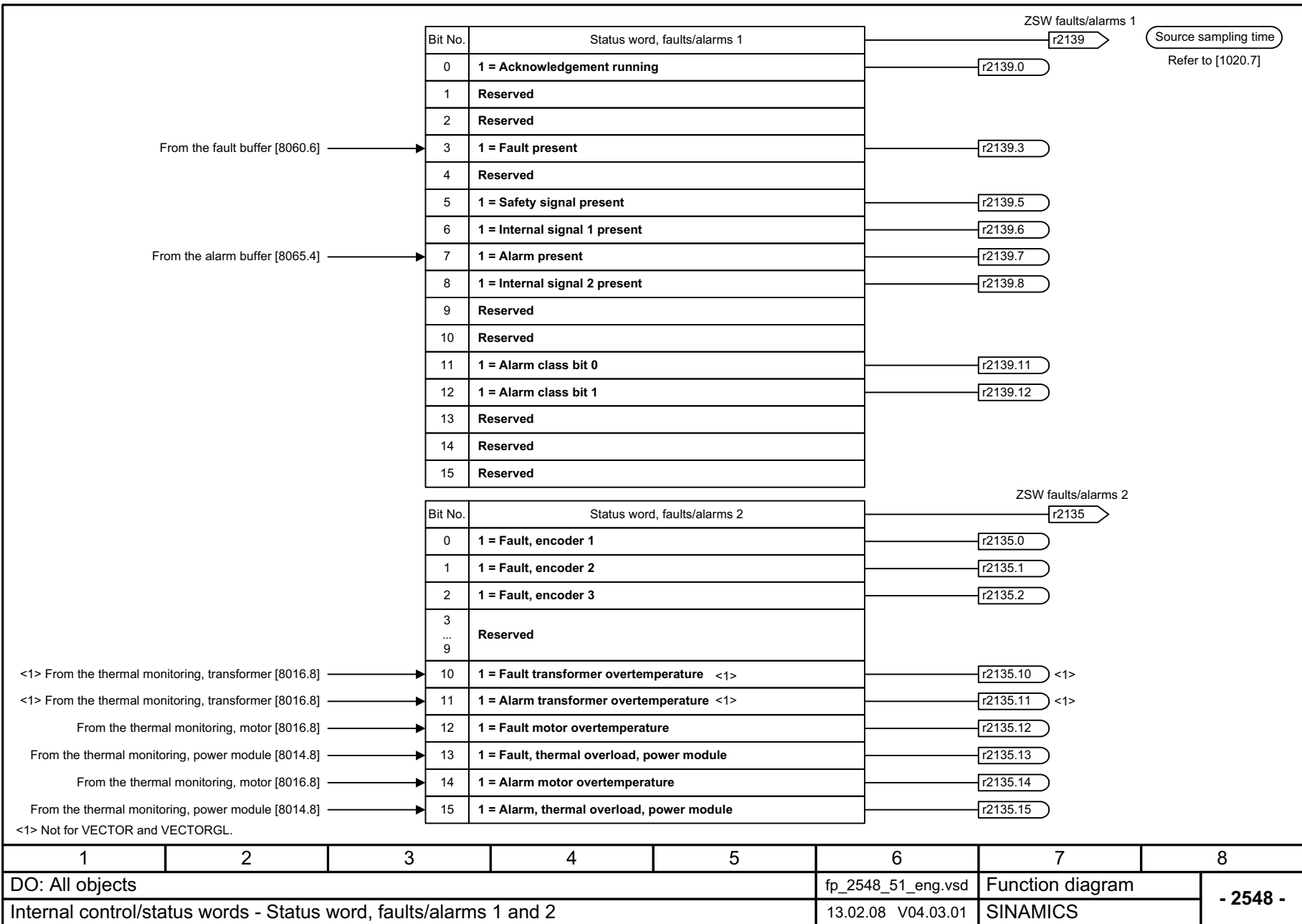


Fig. 2-100 2537 – Status word, monitoring functions 3





2.9 Sequence control

Function diagrams

2610 – Sequencer	2-1495
2634 – Missing enable signals, line contactor control, logic operation	2-1496

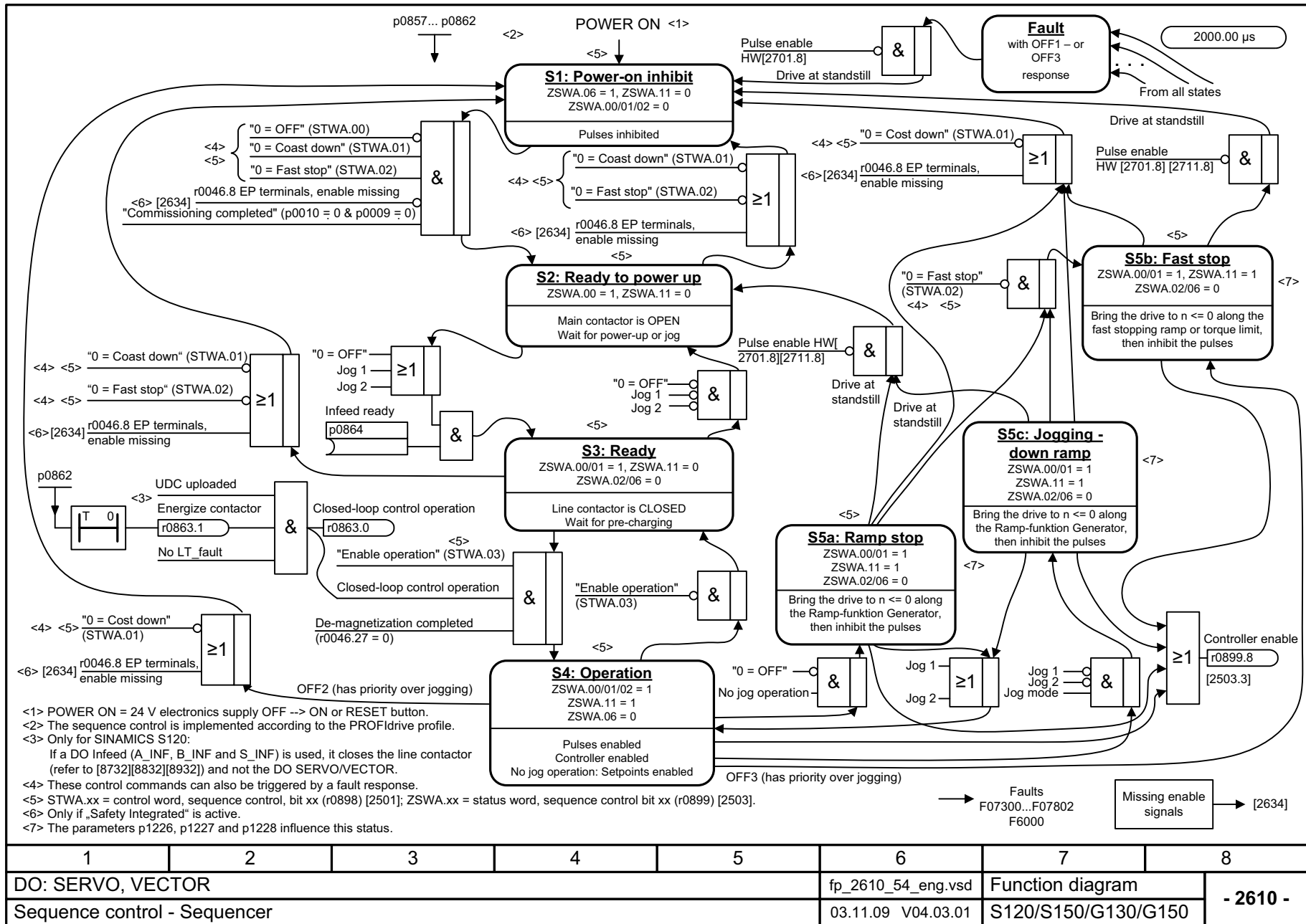


Fig. 2-103 2610 – Sequencer

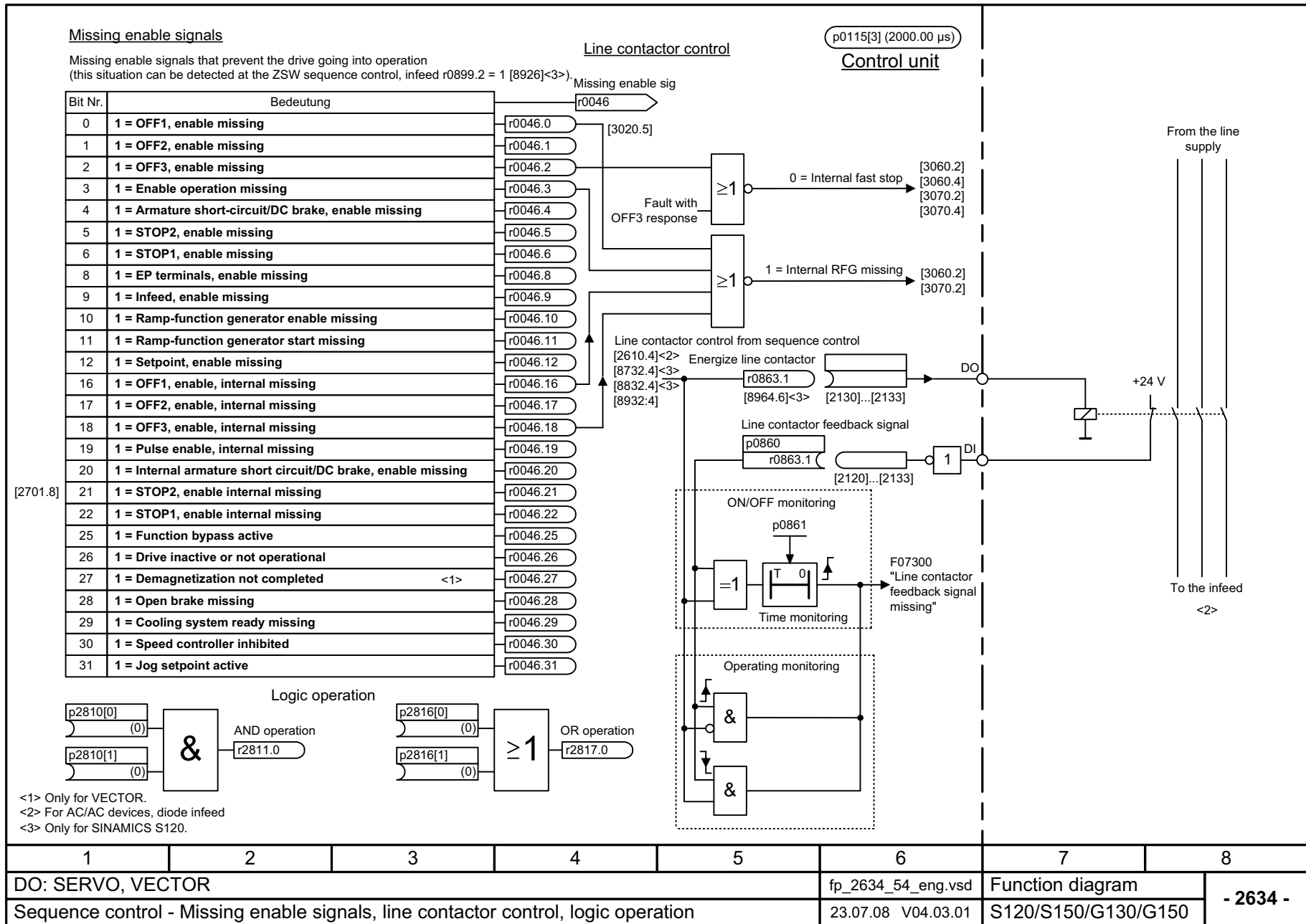
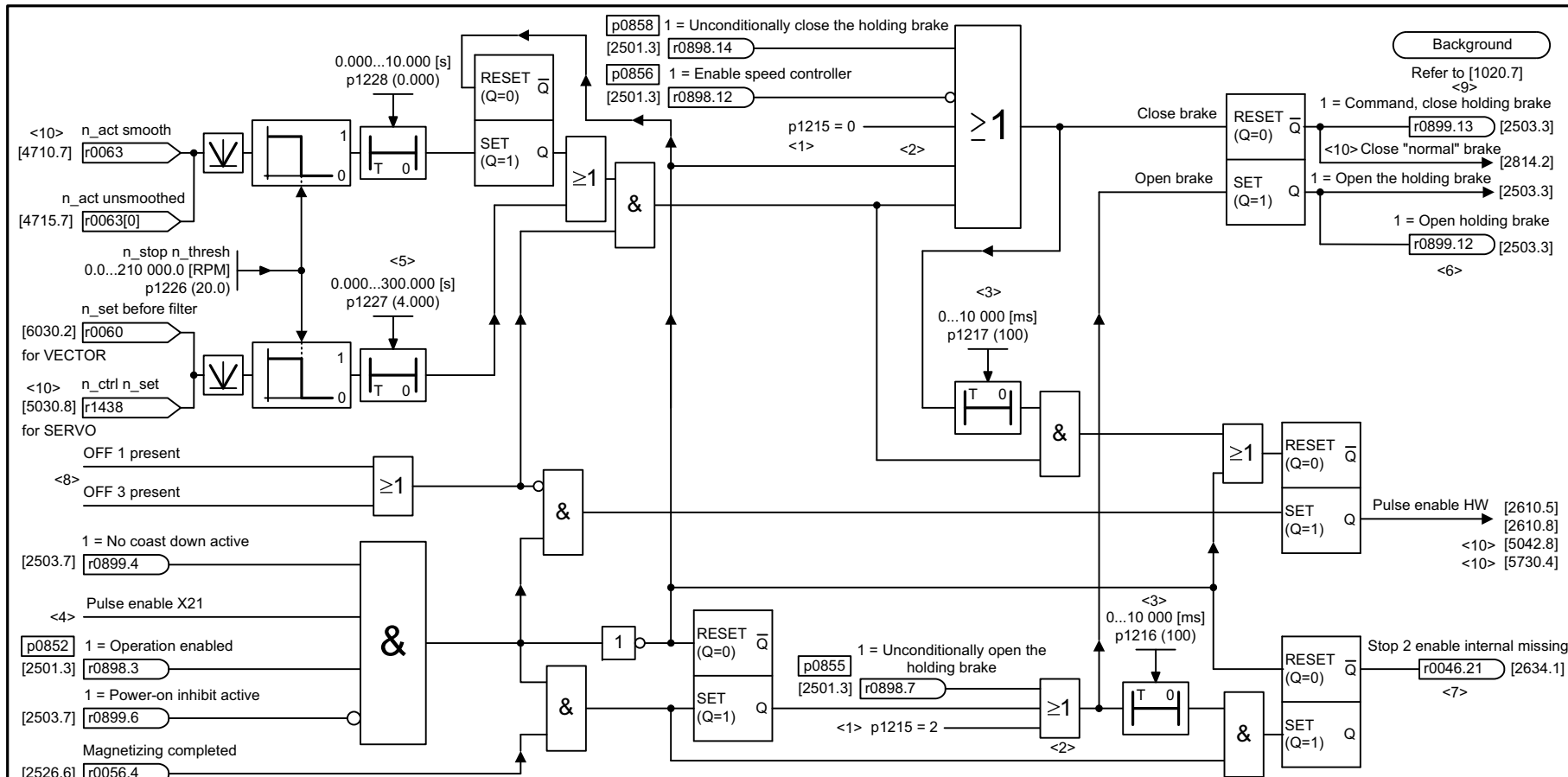


Fig. 2-104 2634 – Missing enable signals, line contactor control, logic operation

2.10 Brake control

Function diagrams

2701 – Basic brake control (r0108.14 = 0)	2-1498
2704 – Extended brake control, stationary state detection (r0108.14 = 1)	2-1499
2707 – Extended brake control, open/close brake (r0108.14 = 1)	2-1500
2711 – Extended brake control, signal outputs (r0108.14 = 1)	2-1501



- <1> Motor holding brake configuration (p1215)
0 = No motor holding brake being used.
1 = Motor holding brake acc. to sequence control.
2 = Motor holding brake always released.
3 = Motor holding brake like sequence control, connection via BICO.
- <2> Priority assignment (high -> low): p1215, p0858, p0855, p0856, sequence control.
- <3> If p1215 = 0, 2 -> t = 0 ms. <10>
- <4> Only if Safety Integrated is active (Double Motor Module: X22, Chassis: X41).
- <5> For p1227 = 300 s, the monitoring function is deactivated.
- <6> If an external motor holding brake is used, p1215 should be set to 3 and r0899.12 should be interconnected as control signal.
- <7> r0046.21 = 0, as long as r0046.0 = 1 (OFF1 enable missing or power-on inhibit).
r0046.21 = 1, if p0858 = 1 or p0856 = 0.
The signal generation is shown simplified.
- <8> The internal signal includes signals that lead to OFF1 or OFF3 (e. g. BICO or fault response).
- <9> If the brake is permanently applied or released (p0855, p0858 or p1215), the drive does not wait while the brake is released or applied.
- <10> Only for SINAMICS S120.

Note: Braking signal diagnostic evaluation (p1278) only applies for SBC (Safe Brake Control) (controls the Safe Brake Relay). <10>

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2701_54_eng.vsd	Function diagram	
Braking control - Basic braking control (r0108.14 = 0)					07.05.08 V04.03.01	S120/S150/G130/G150	
					- 2701 -		

Fig. 2-105 2701 – Basic brake control (r0108.14 = 0)

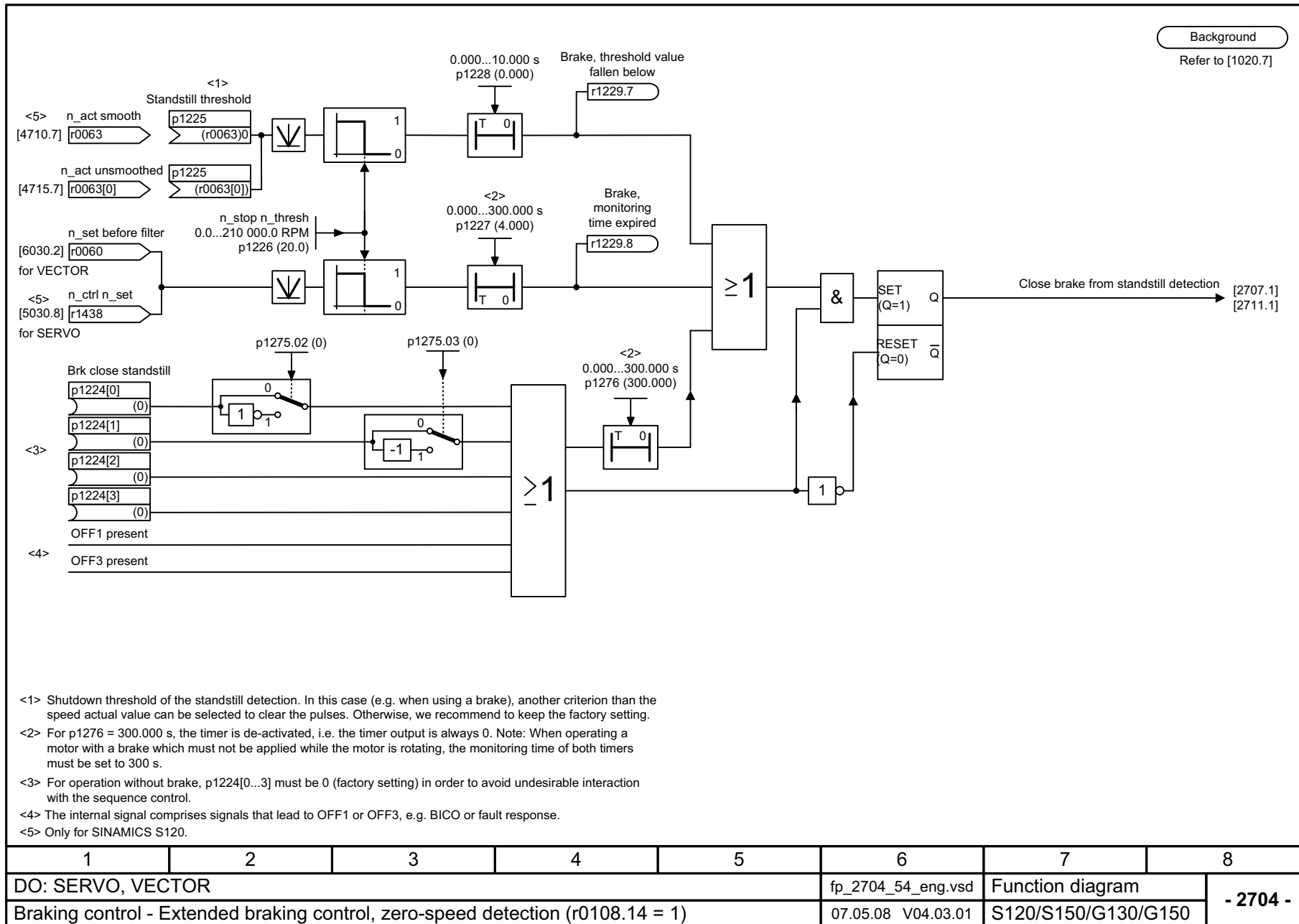
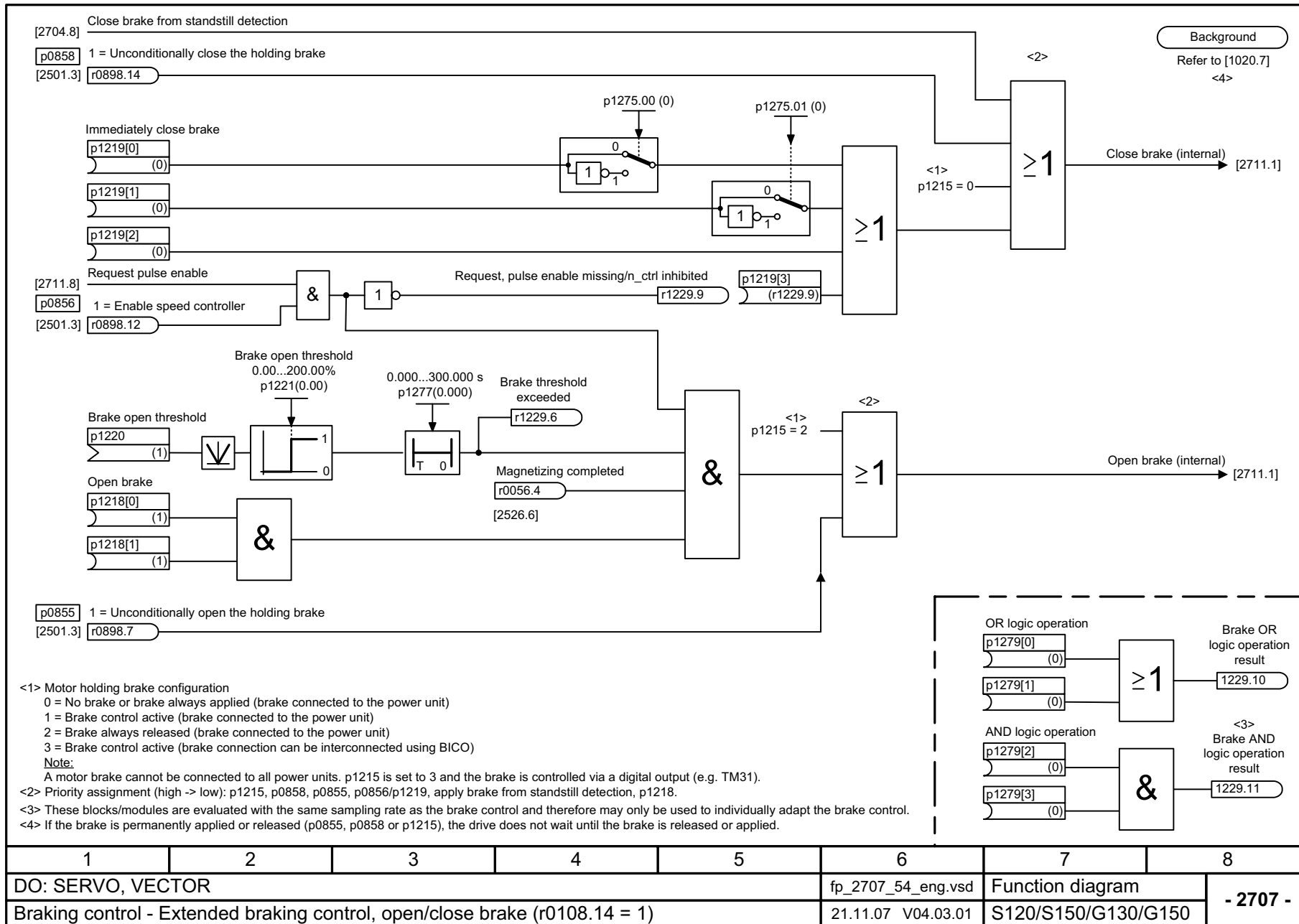
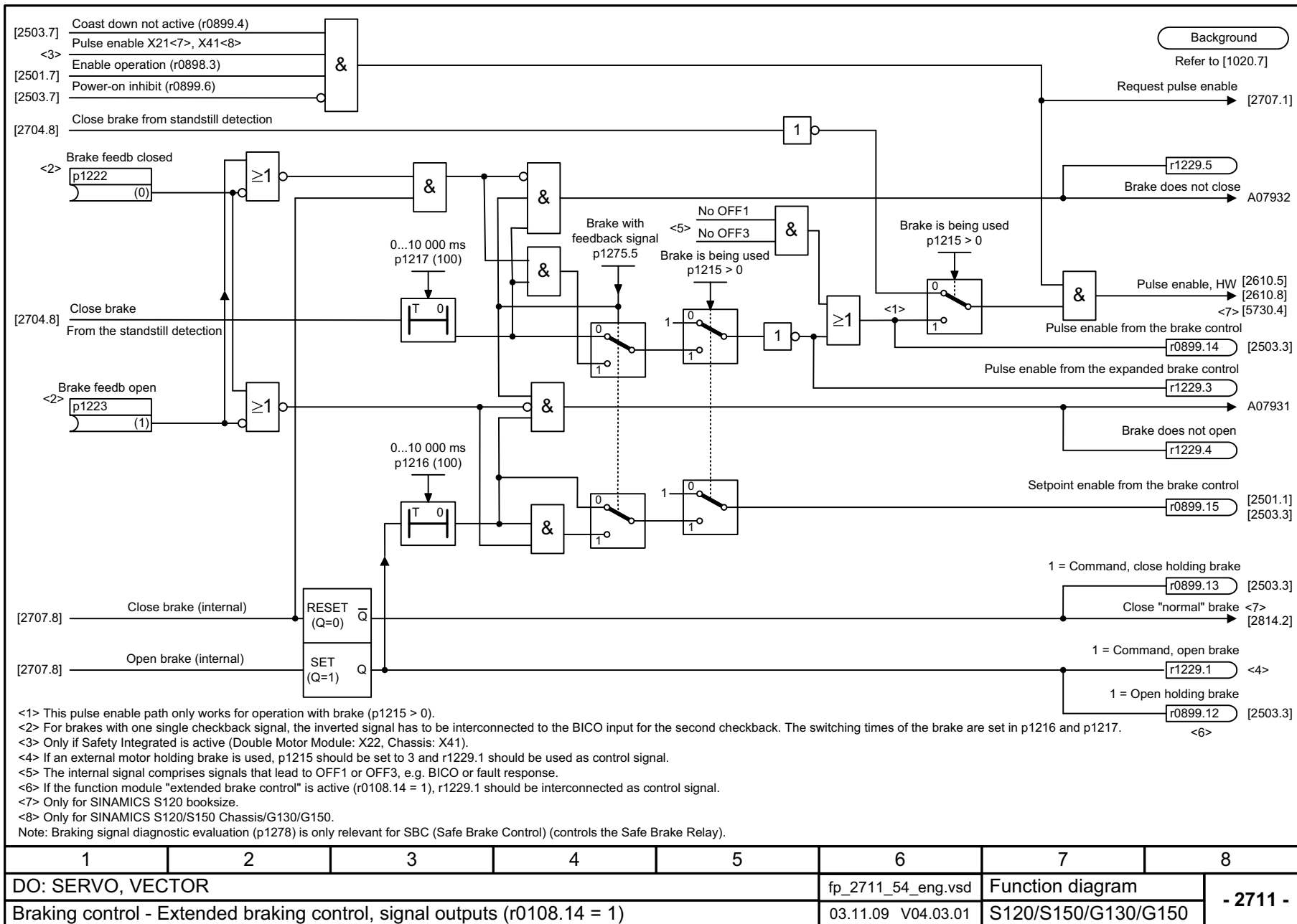


Fig. 2-106 2704 – Extended brake control, stationary state detection (r0108.14 = 1)



Function diagrams

Brake control

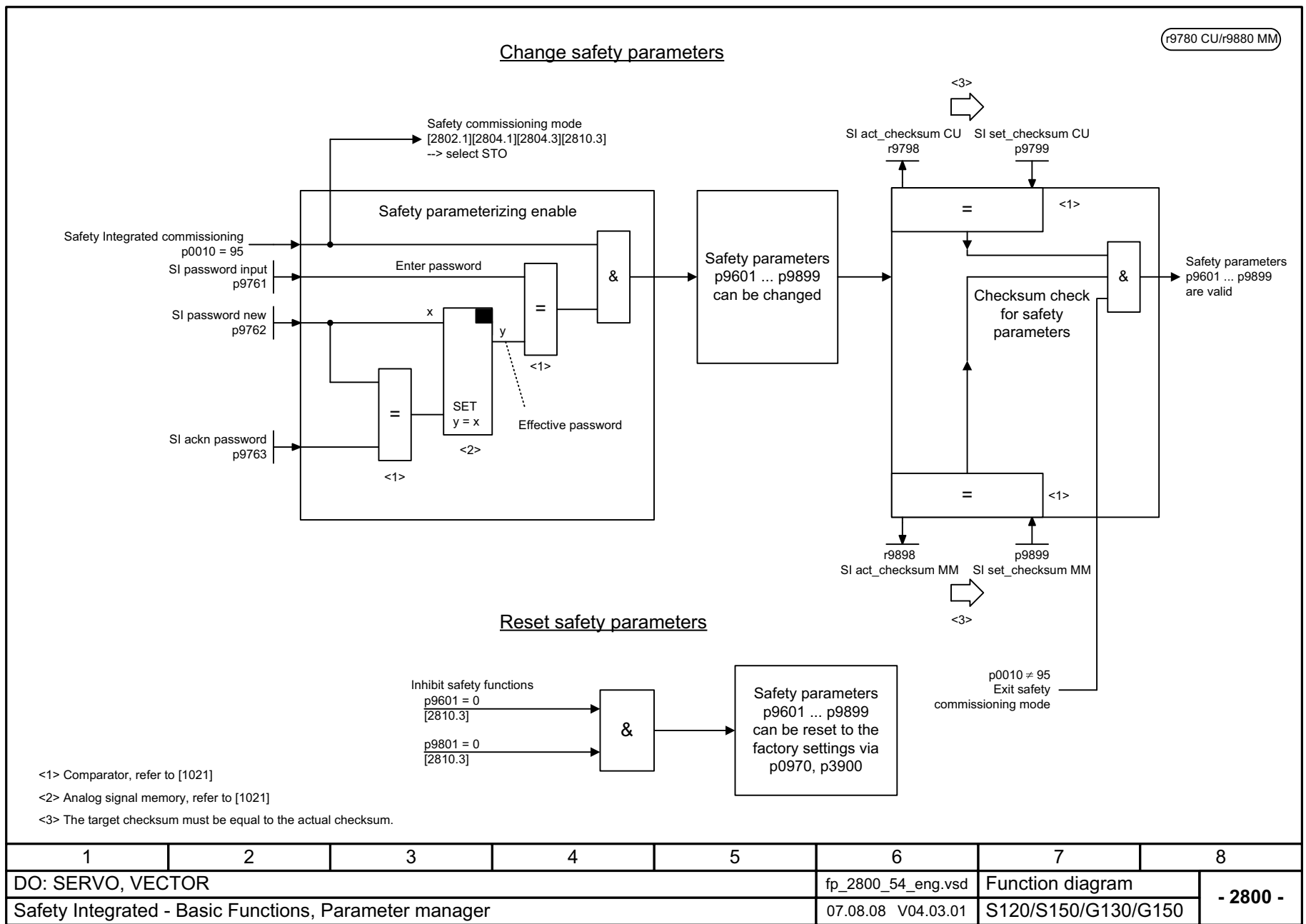


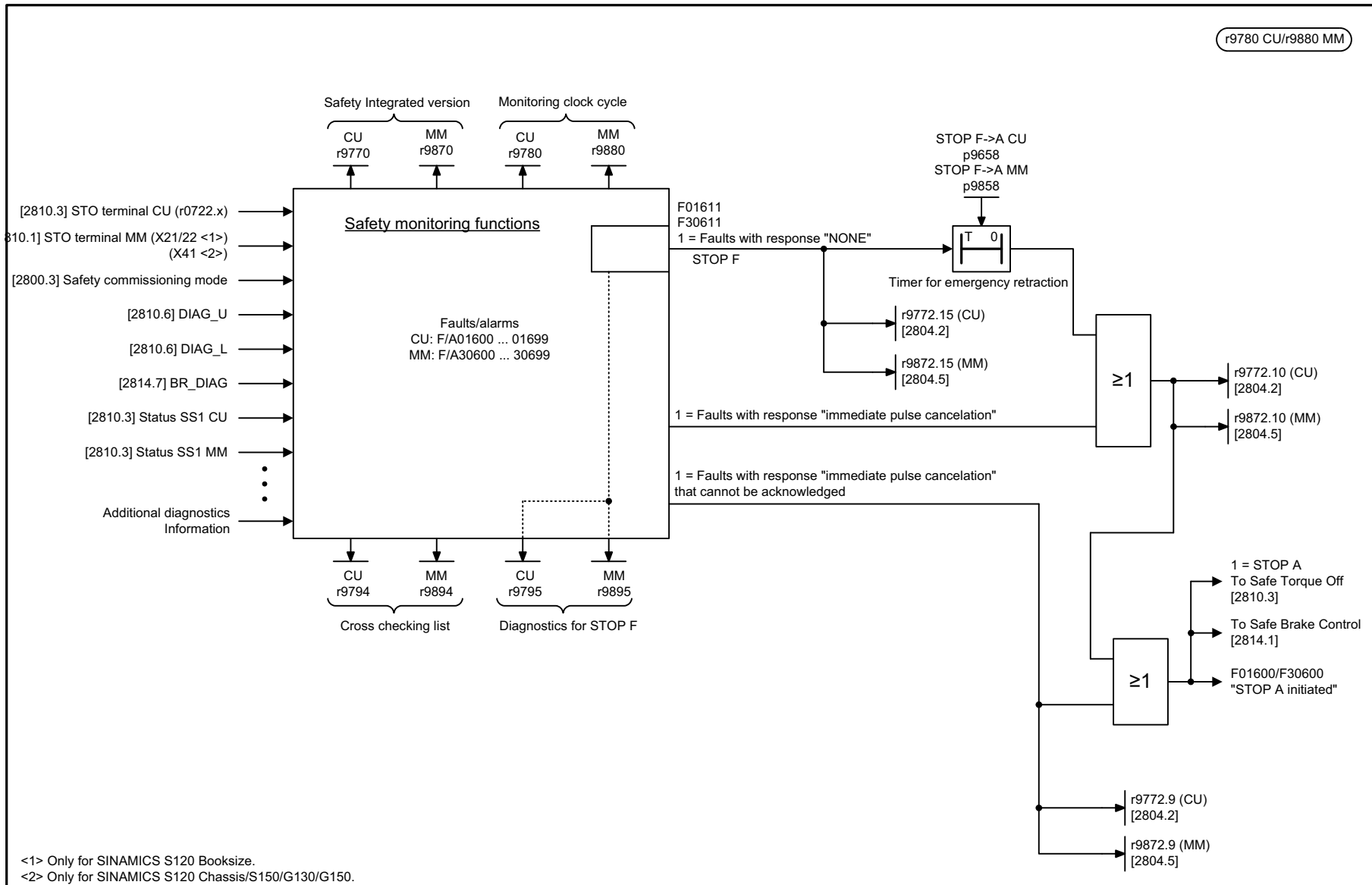
2.11 Safety Integrated

Function diagrams

2800 – Basic functions, parameter manager	2-1503
2802 – Basic functions, monitoring and faults/alarms	2-1504
2804 – Basic functions, status words	2-1505
2810 – Basic functions, STO (Safe Torque Off)/SS1 (Safe Stop 1)	2-1506
2814 – Basic functions, SBC (Safe Brake Control)	2-1507
2820 – Extended functions, SLS	2-1508
2825 – Extended functions, SS1, SS2, SOS, internal STOP B, C, D, F	2-1509
2840 – Extended functions, control word and status word	2-1510
2846 – Extended functions, parameter manager	2-1511
2847 – Extended functions, TM54F parameter manager	2-1512
2848 – Extended functions, TM54F configuration, F-DI/F-DO test	2-1513
2850 – Extended functions, TM54F (F-DI 0 to F-DI 4)	2-1514
2851 – Extended functions, TM54F (F-DI 5 to F-DI 9)	2-1515
2853 – Extended functions, TM54F (F-DO 0 to F-DO 3, DI 20 to DI 23)	2-1516
2855 – Extended functions, TM54F control interface	2-1517
2856 – Extended functions, TM54F safe state selection	2-1518
2857 – Extended functions, TM54F assignment (F-DO 0 to F-DO 3)	2-1519

Fig. 2-109

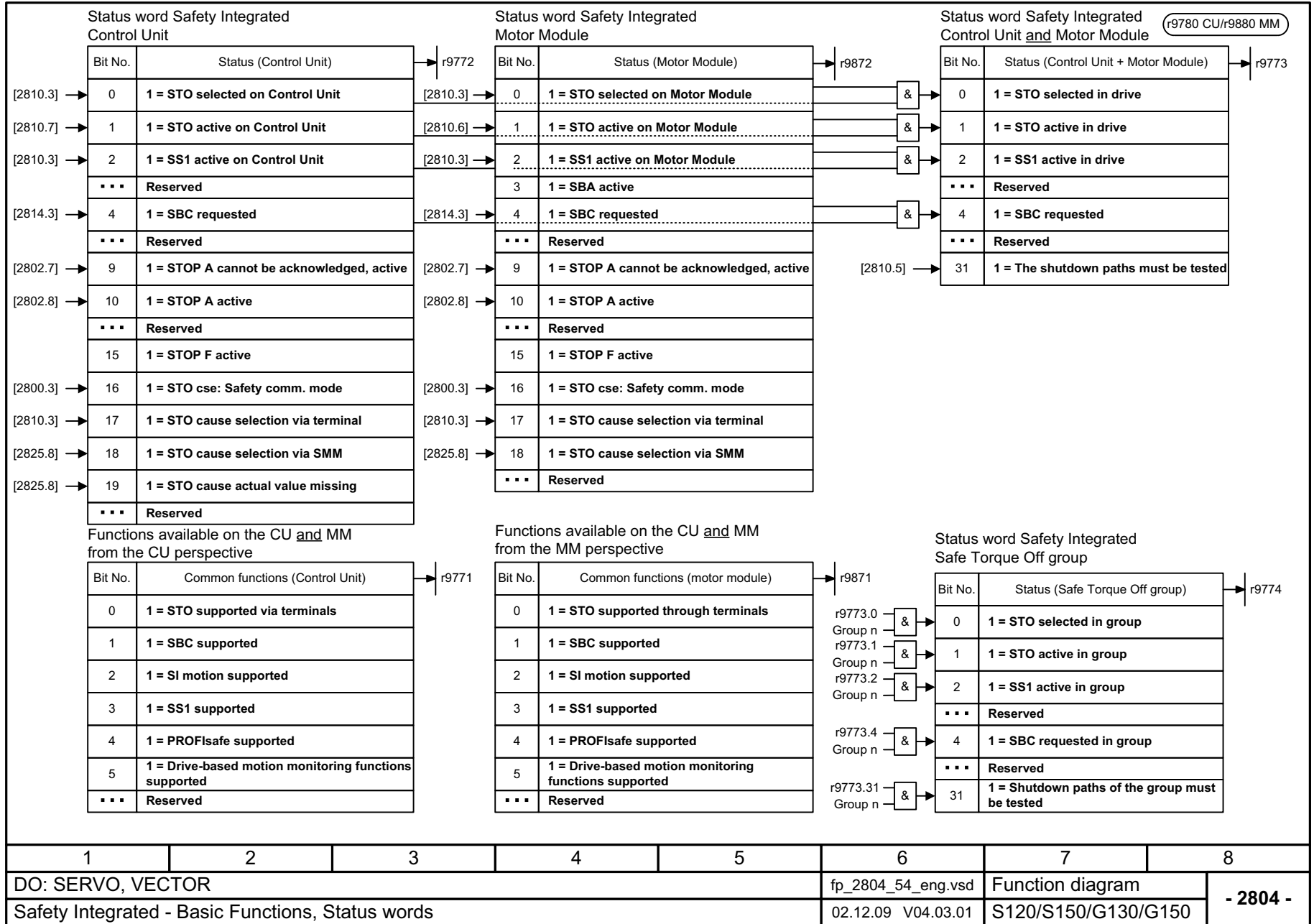




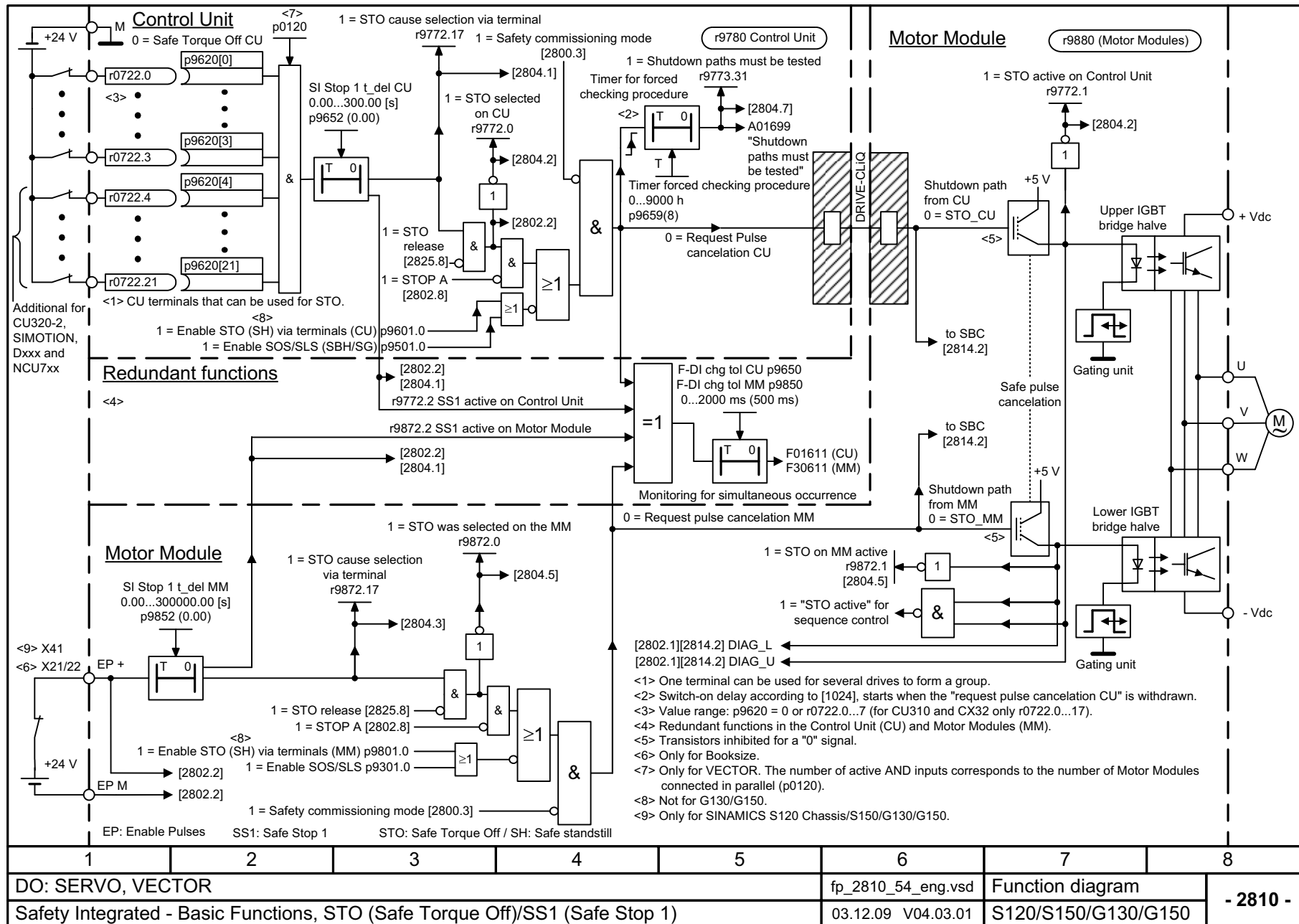
1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2802_54_eng.vsd	Function diagram	
Safety Integrated - Basic Functions, Monitoring functions and faults/alarms					02.12.09 V04.03.01	S120/S150/G130/G150	
							- 2802 -

Fig. 2-110 2802 – Basic functions, monitoring and faults/alarms

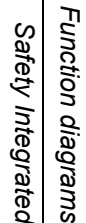
Fig. 2-111 2804 – Basic functions, status words

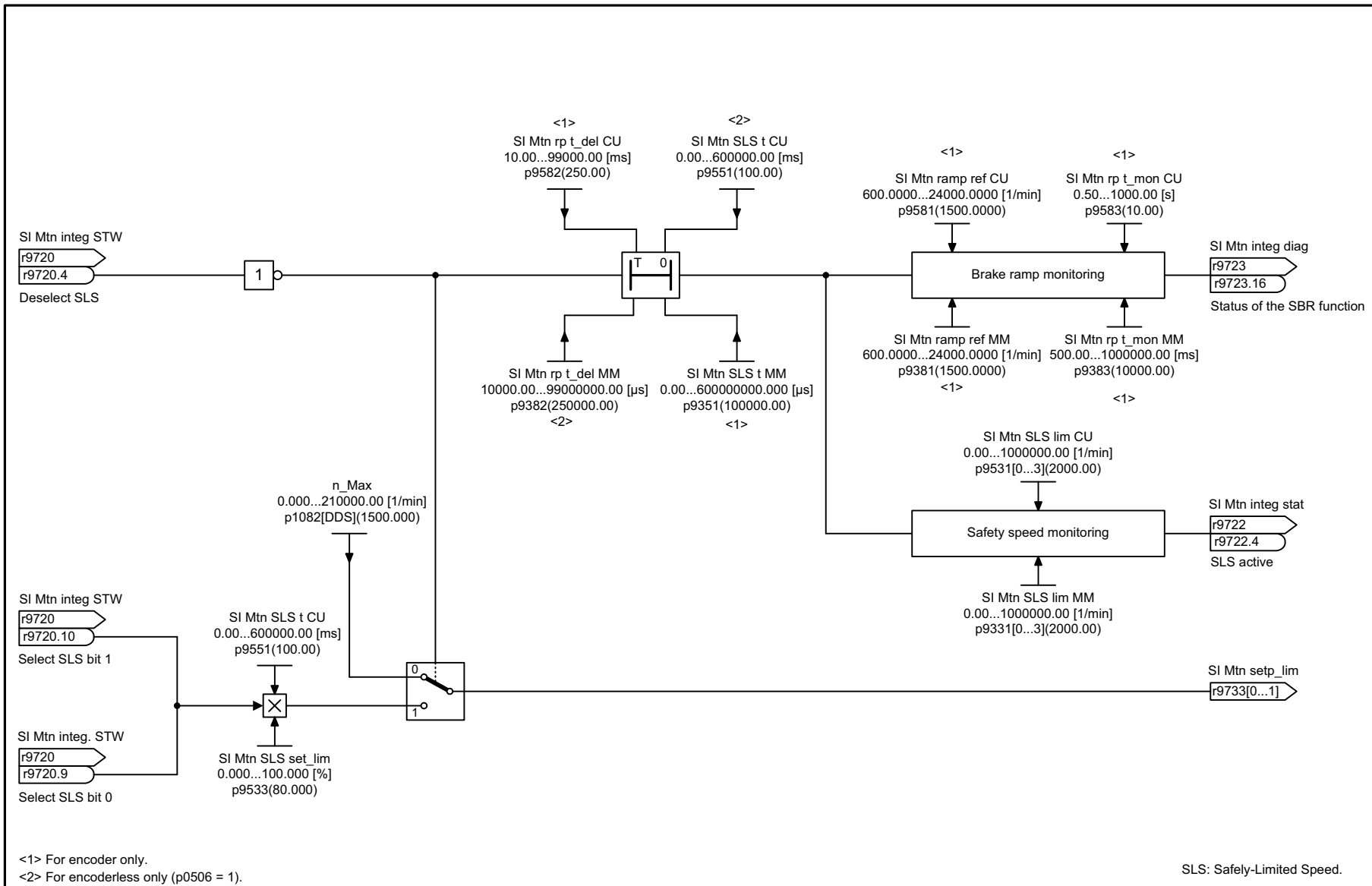


2810 – Basic functions, STO (Safe Torque Off)/SS1 (Safe Stop 1)



2814 – Basic functions, SBC (Safe Brake Control)





1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_2820_55_eng.vsd	Function diagram	
Safety Integrated - Extended Functions, SLS (Safely-Limited Speed)					19.11.09 V04.03.01	SINAMICS S120/S150	
							- 2820 -

Fig. 2-114 2820 – Extended functions, SLS

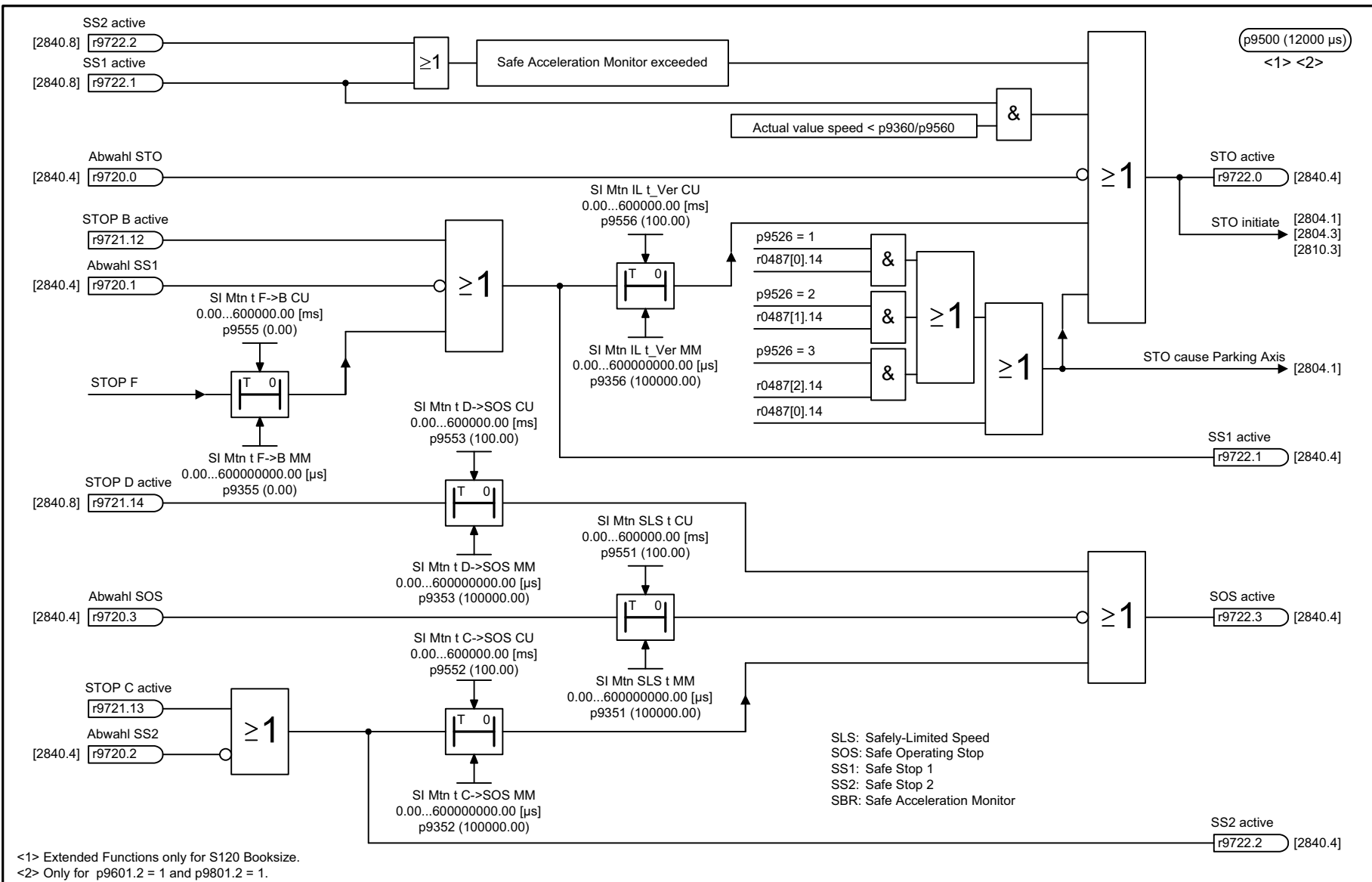
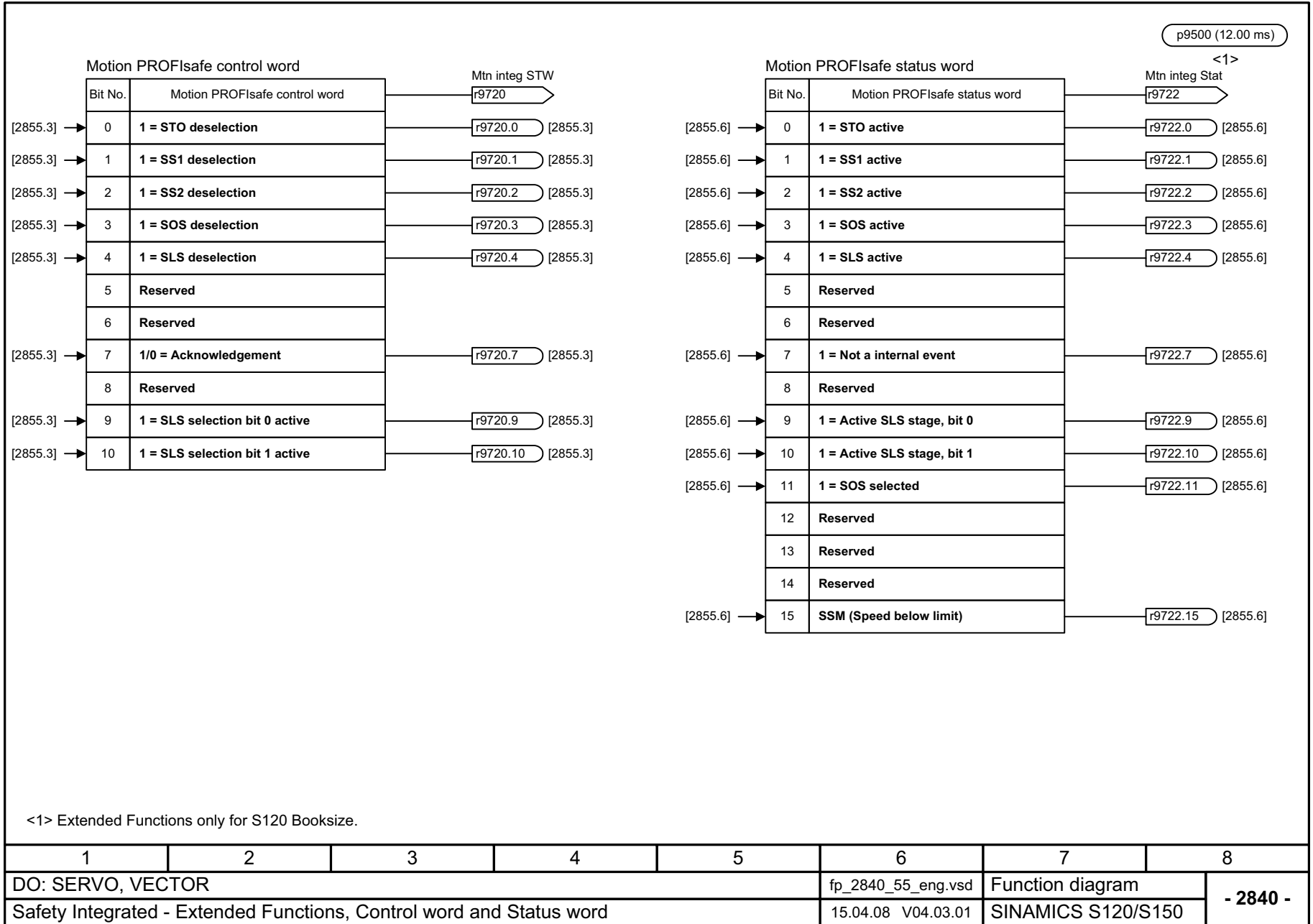


Fig. 2-115 2825 – Extended functions, SS1, SS2, SOS, internal STOP B, C, D, F

Fig. 2-116 2840 – Extended functions, control word and status word



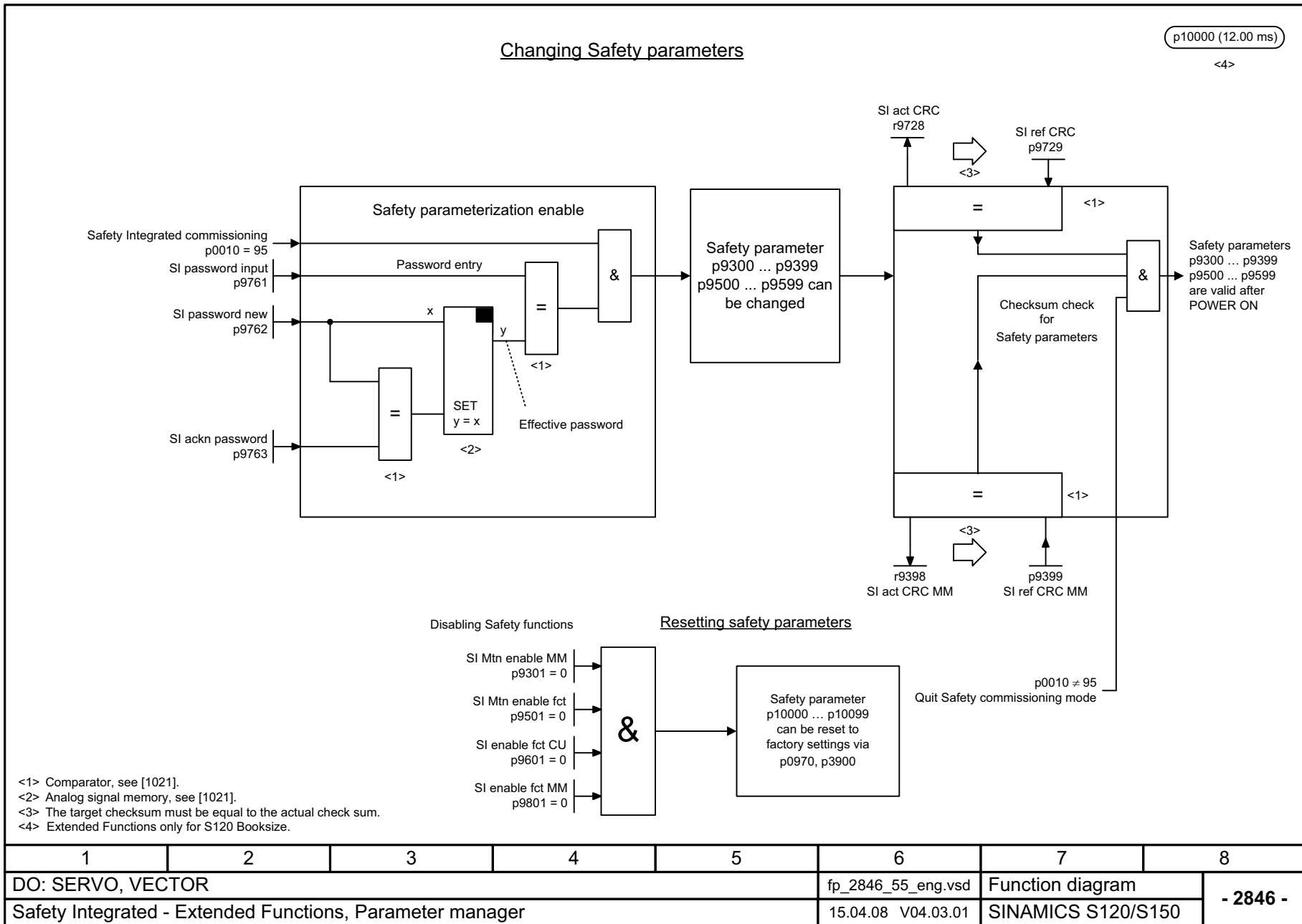


Fig. 2-117 2846 – Extended functions, parameter manager

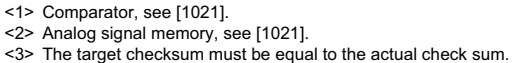


Fig. 2-118

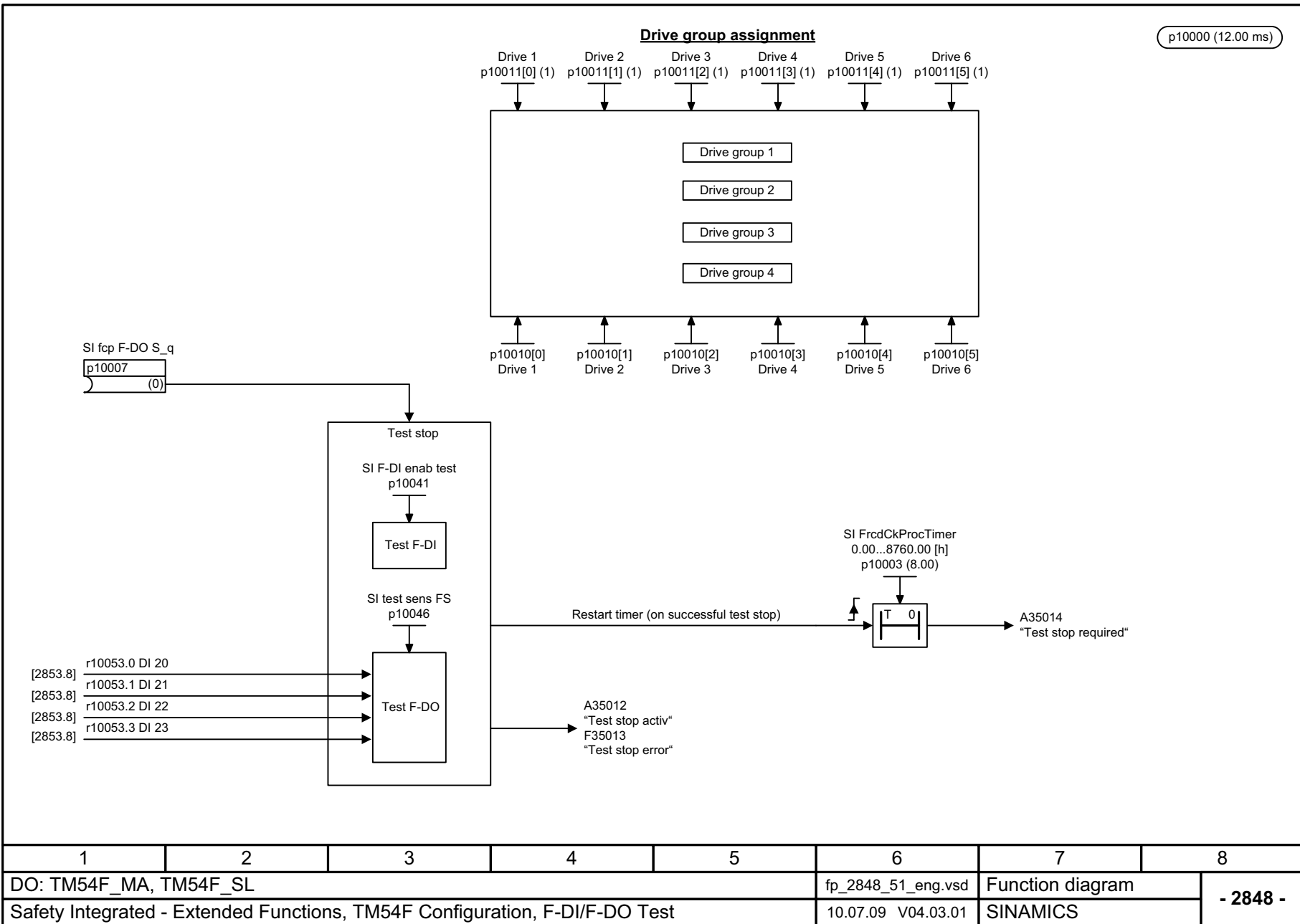


Fig. 2-119 2848 – Extended functions, TM54F configuration, F-DI/F-DO test

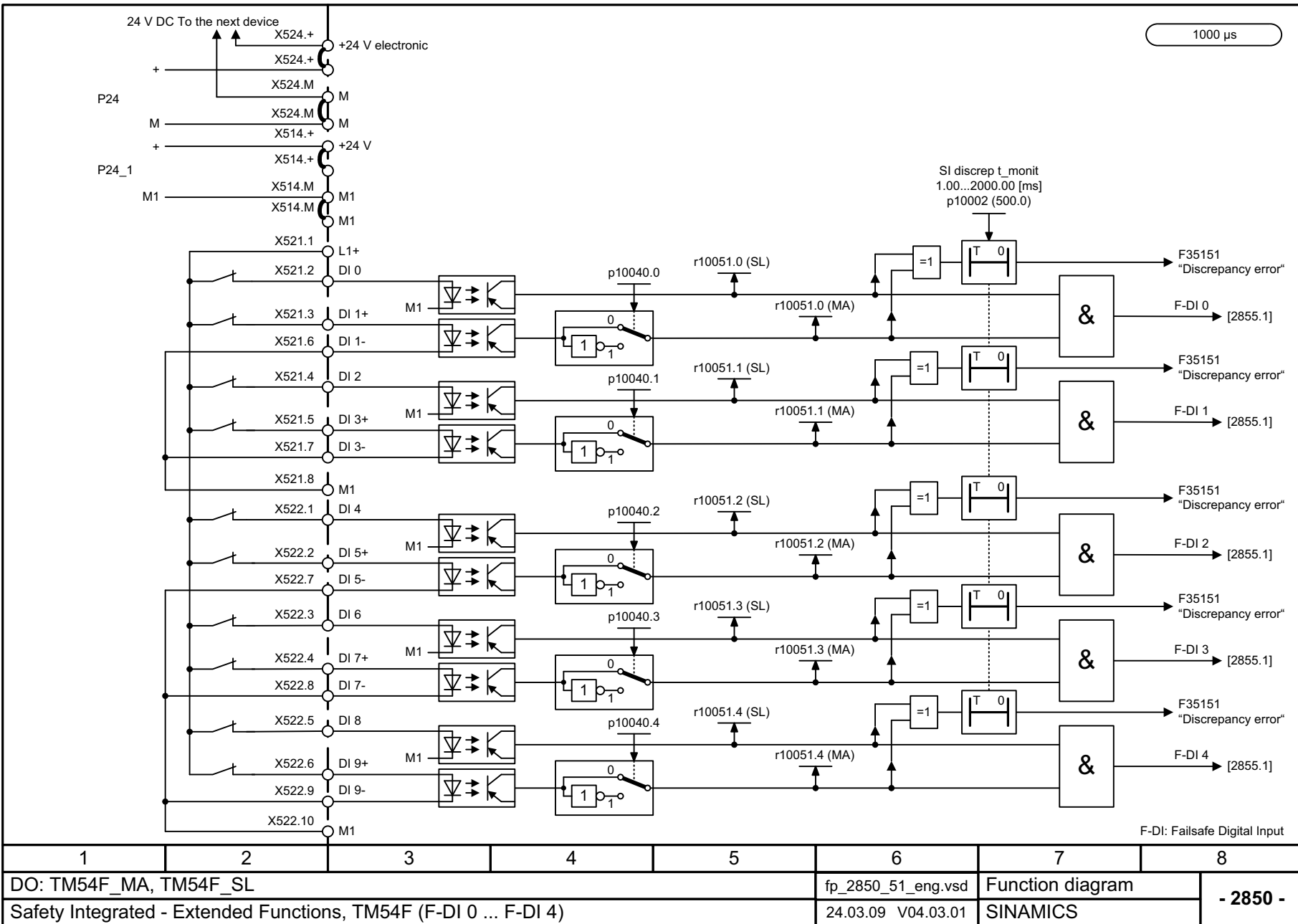


Fig. 2-120 2850 – Extended functions, TM54F (F-DI 0 to F-DI 4)

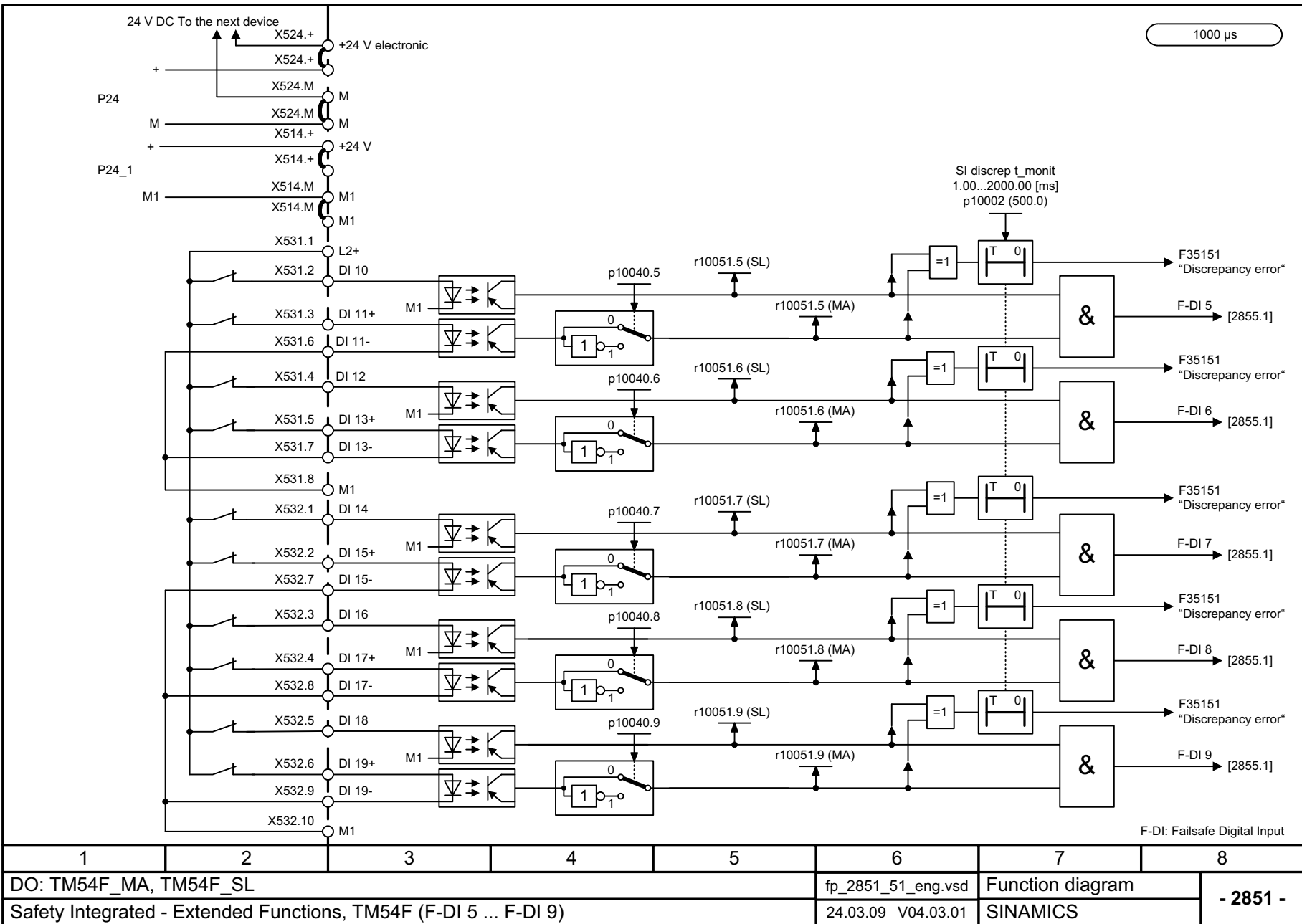


Fig. 2-121 2851 – Extended functions, TM54F (F-DI 5 to F-DI 9)

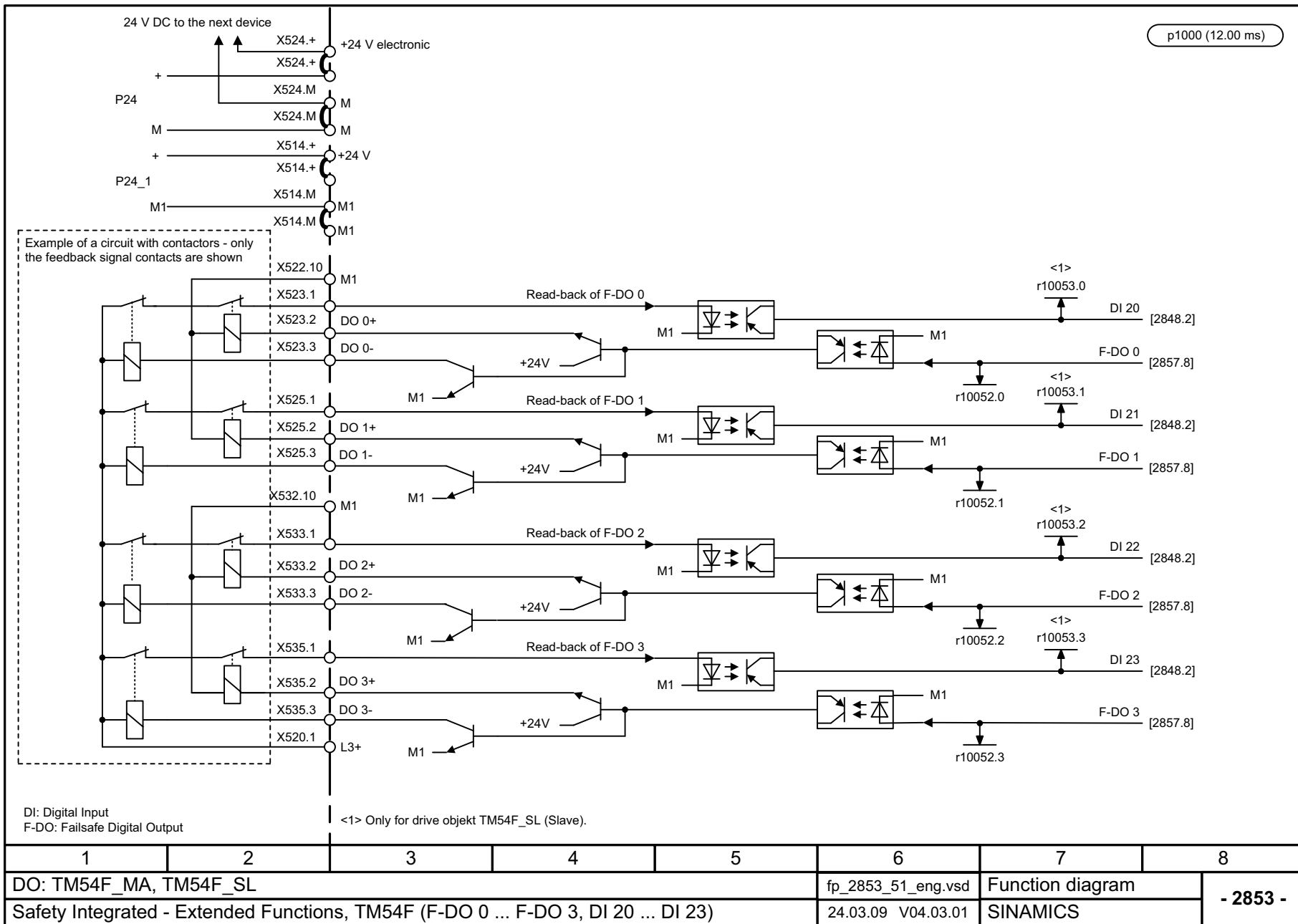


Fig. 2-122 2853 – Extended functions, TM54F (F-DO 0 to F-DO 3, DI 20 to DI 23)

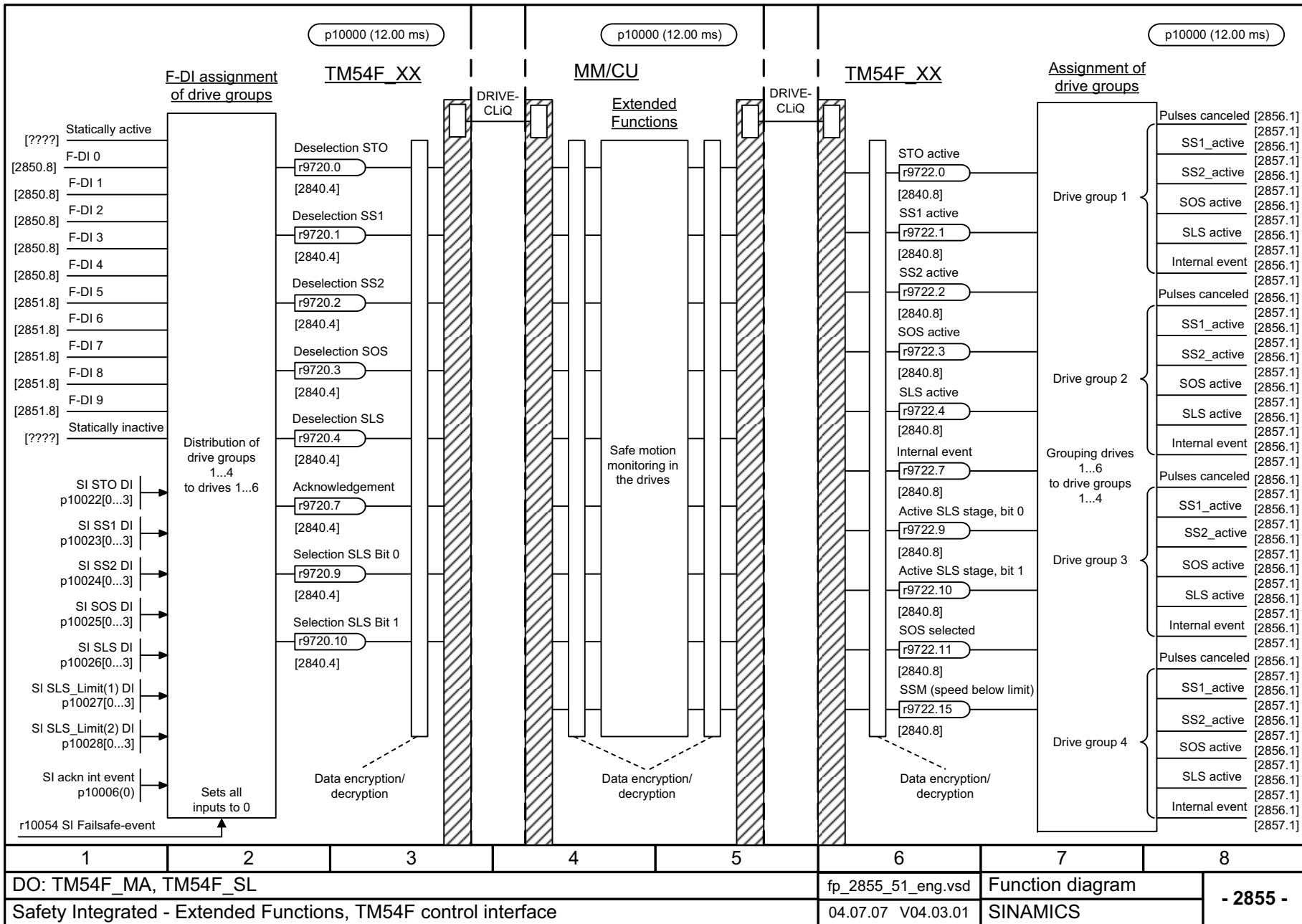


Fig. 2-123 2855 – Extended functions, TM54F control interface

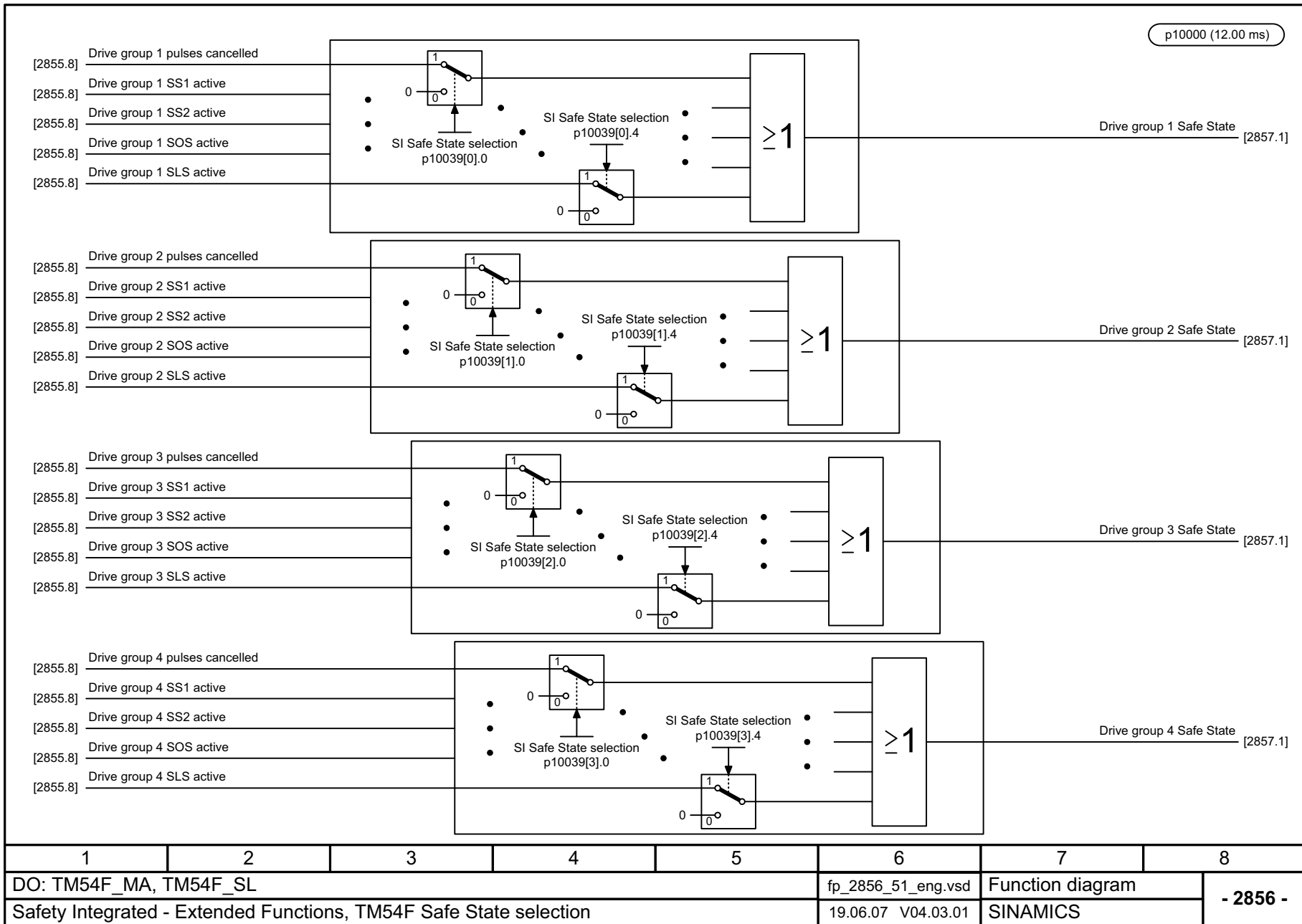


Fig. 2-124 2856 – Extended functions, TM54F safe state selection

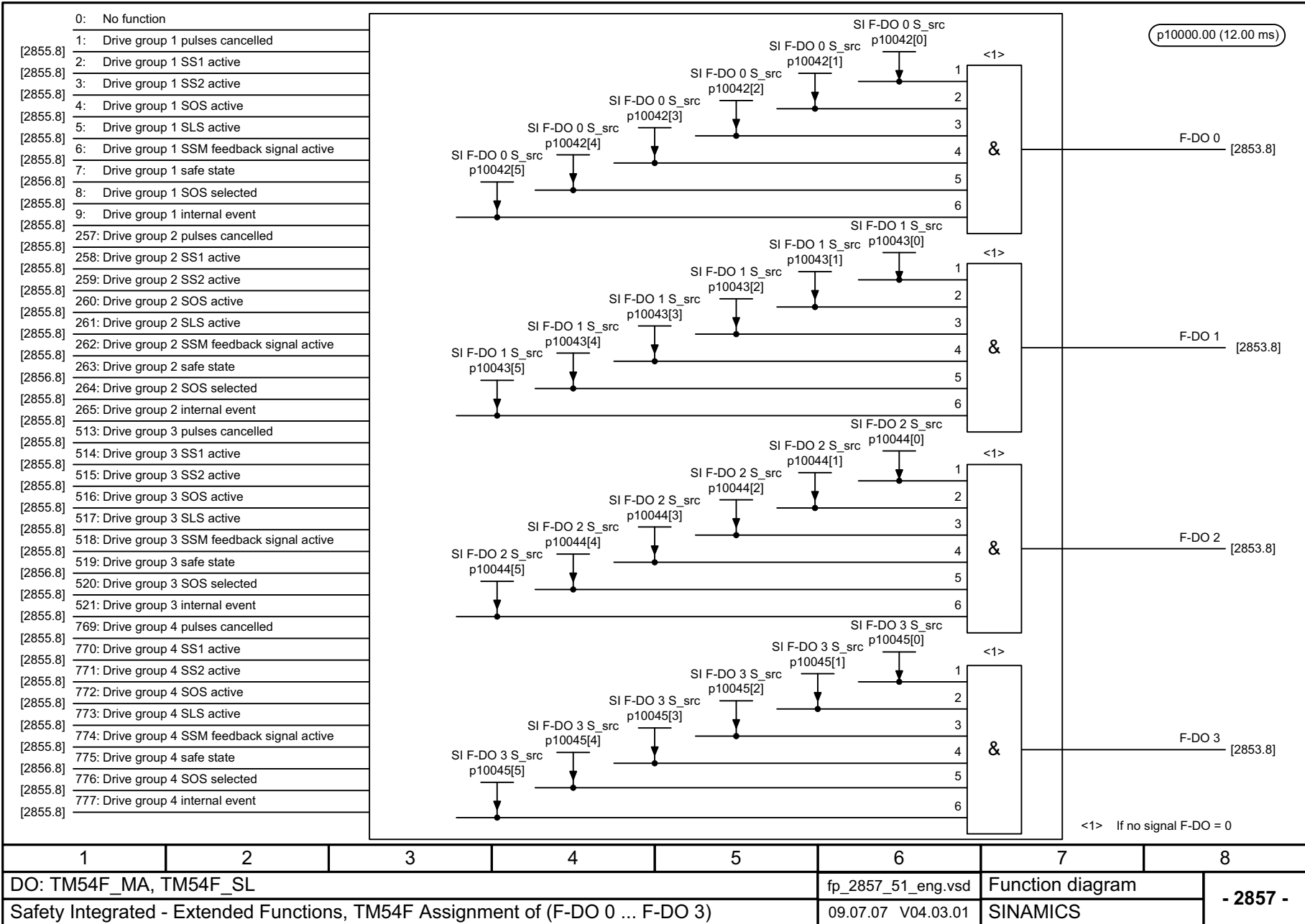
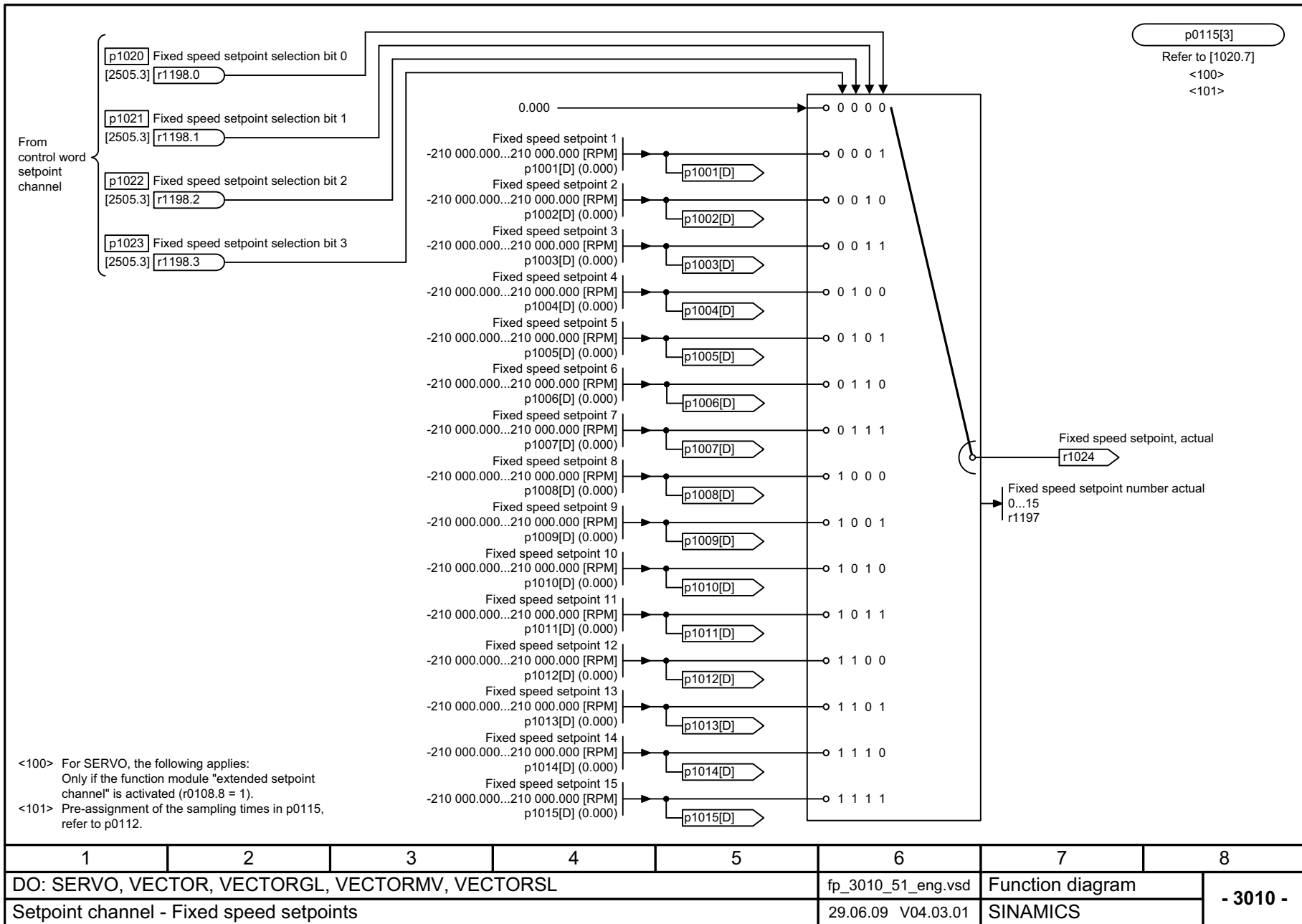


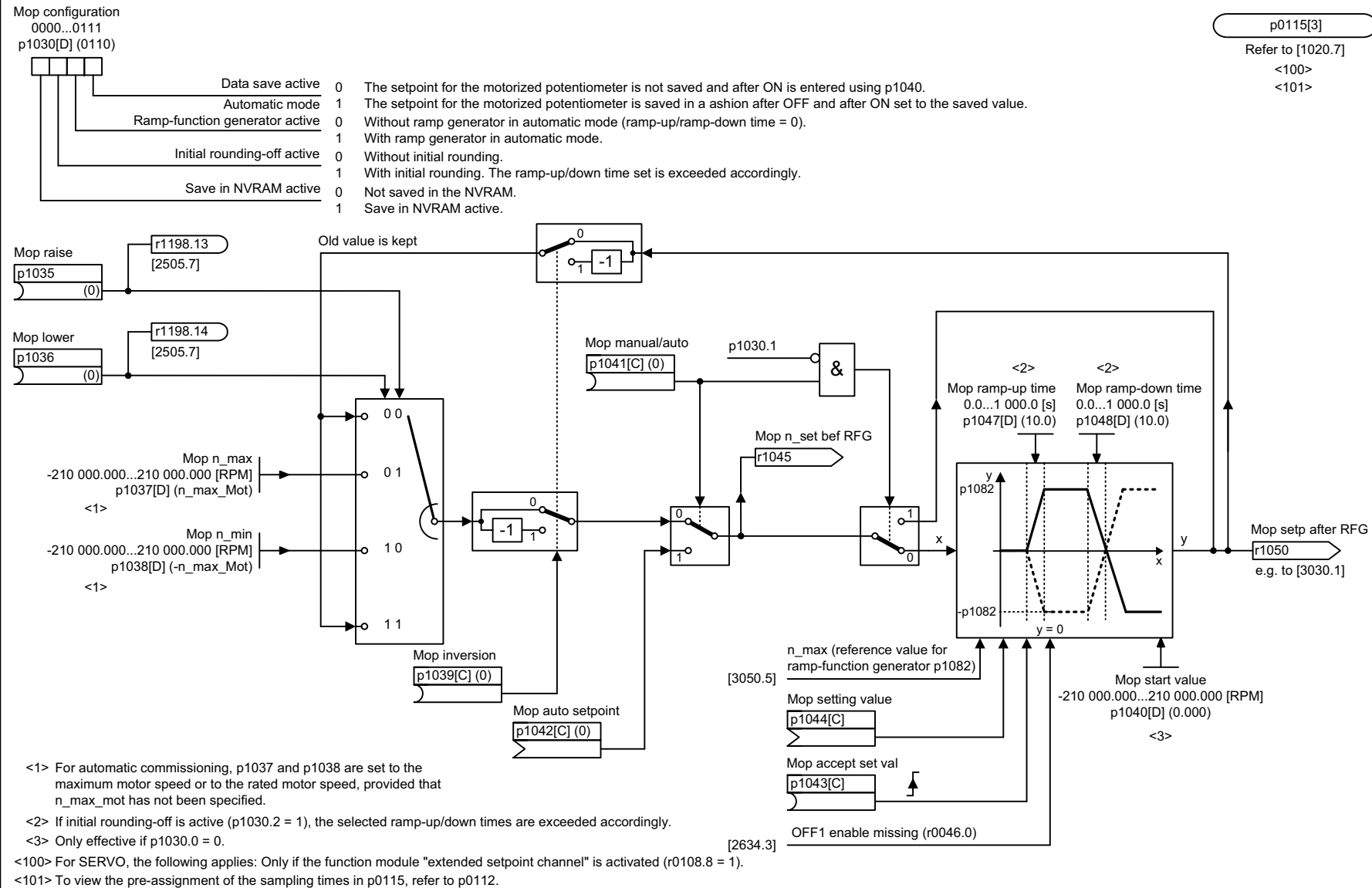
Fig. 2-125 2857 – Extended functions, TM54F assignment (F-DO 0 to F-DO 3)

2.12 Setpoint channel

Function diagrams

3010 – Fixed speed setpoints	2-1521
3020 – Motorized potentiometer	2-1522
3030 – Main/additional setpoint, setpoint scaling, jogging	2-1523
3040 – Direction of rotation limitation and reversal	2-1524
3050 – Skip frequency bands and speed limiting	2-1525
3060 – Basic ramp-function generator	2-1526
3070 – Extended ramp-function generator	2-1527
3080 – Ramp-function generator selection, status word, tracking	2-1528
3090 – Dynamic Servo Control (DSC)	2-1529

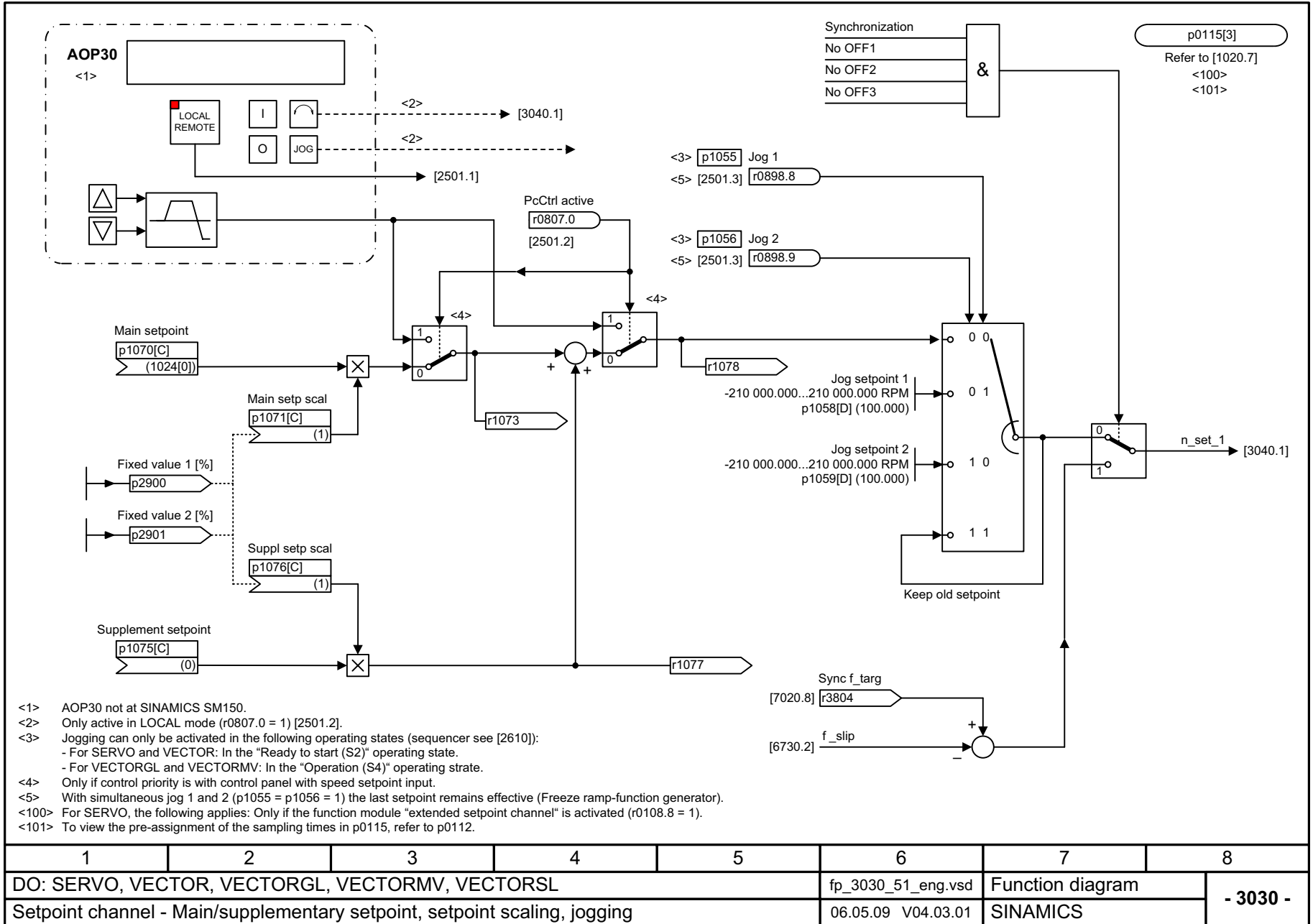




1	2	3	4	5	6	7	8
DO: SERVO, VECTOR, VECTORGL, VECTORMV, VECTORSL					fp_3020_51_eng.vsd	Function diagram	- 3020 -
Setpoint channel - Motorized potentiometer					22.10.09 V04.03.01	SINAMICS	

Fig. 2-127 3020 – Motorized potentiometer

Fig. 2-128 3030 – Main/additional setpoint, setpoint scaling, jogging



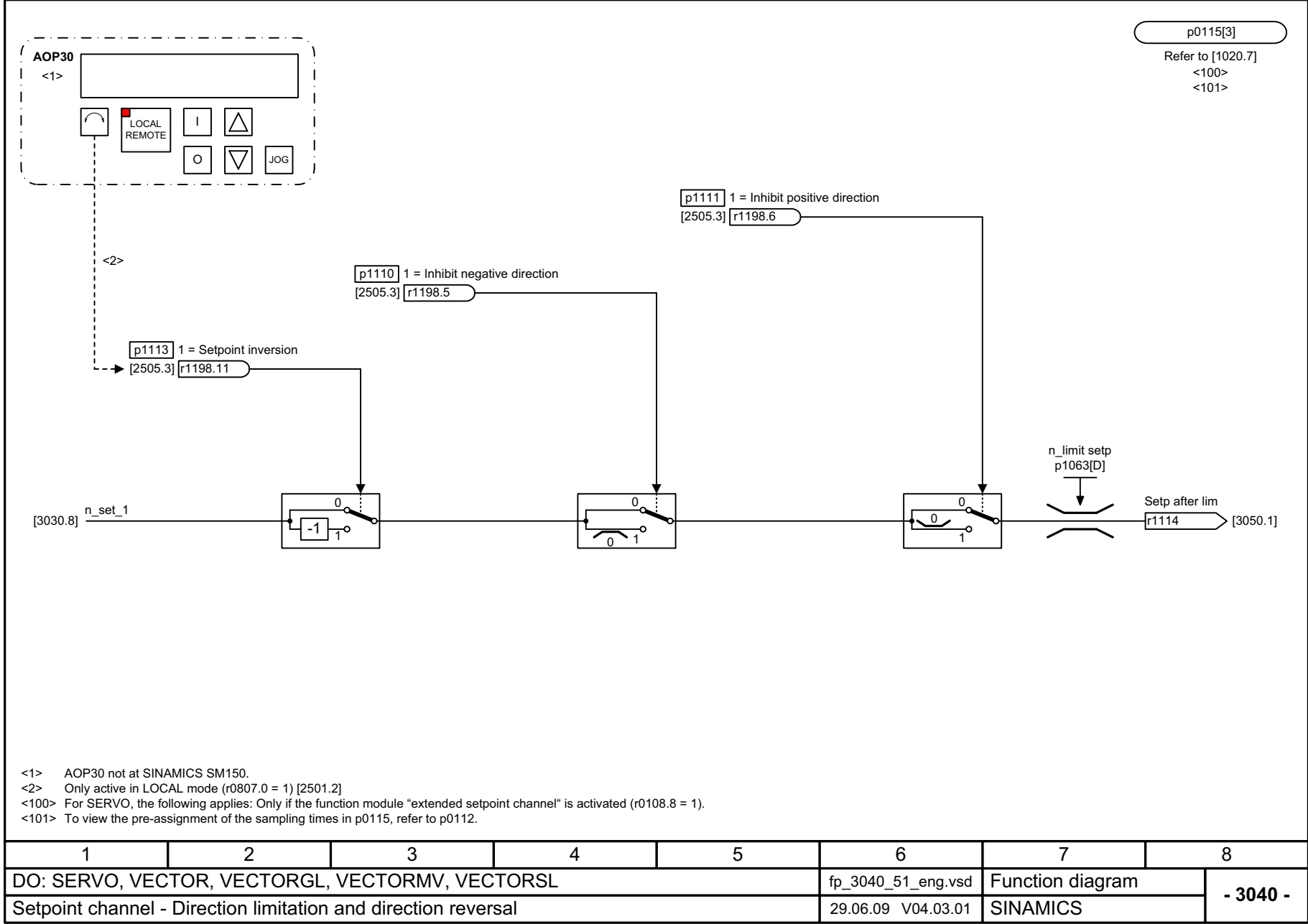
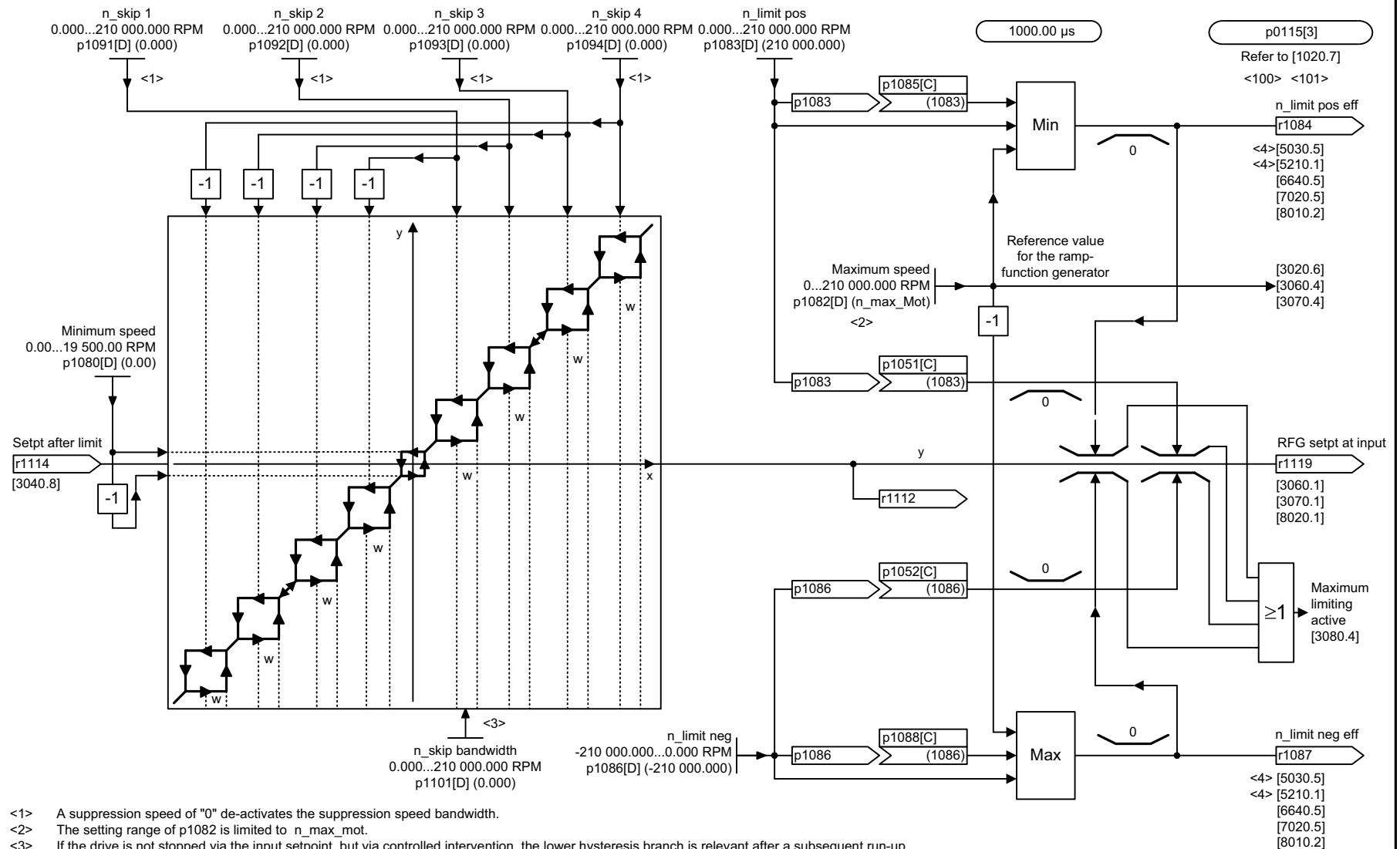


Fig. 2-130 3050 – Skip frequency bands and speed limiting



1	2	3	4	5	6	7	8
DO: SERVO, VECTOR, VECTORGL, VECTORMV, VECTORSL					fp_3050_51_eng.vsd	Function diagram	
Setpoint channel - Skip frequency bands and speed limitations					19.03.09 V04.03.01	SINAMICS	
							- 3050 -

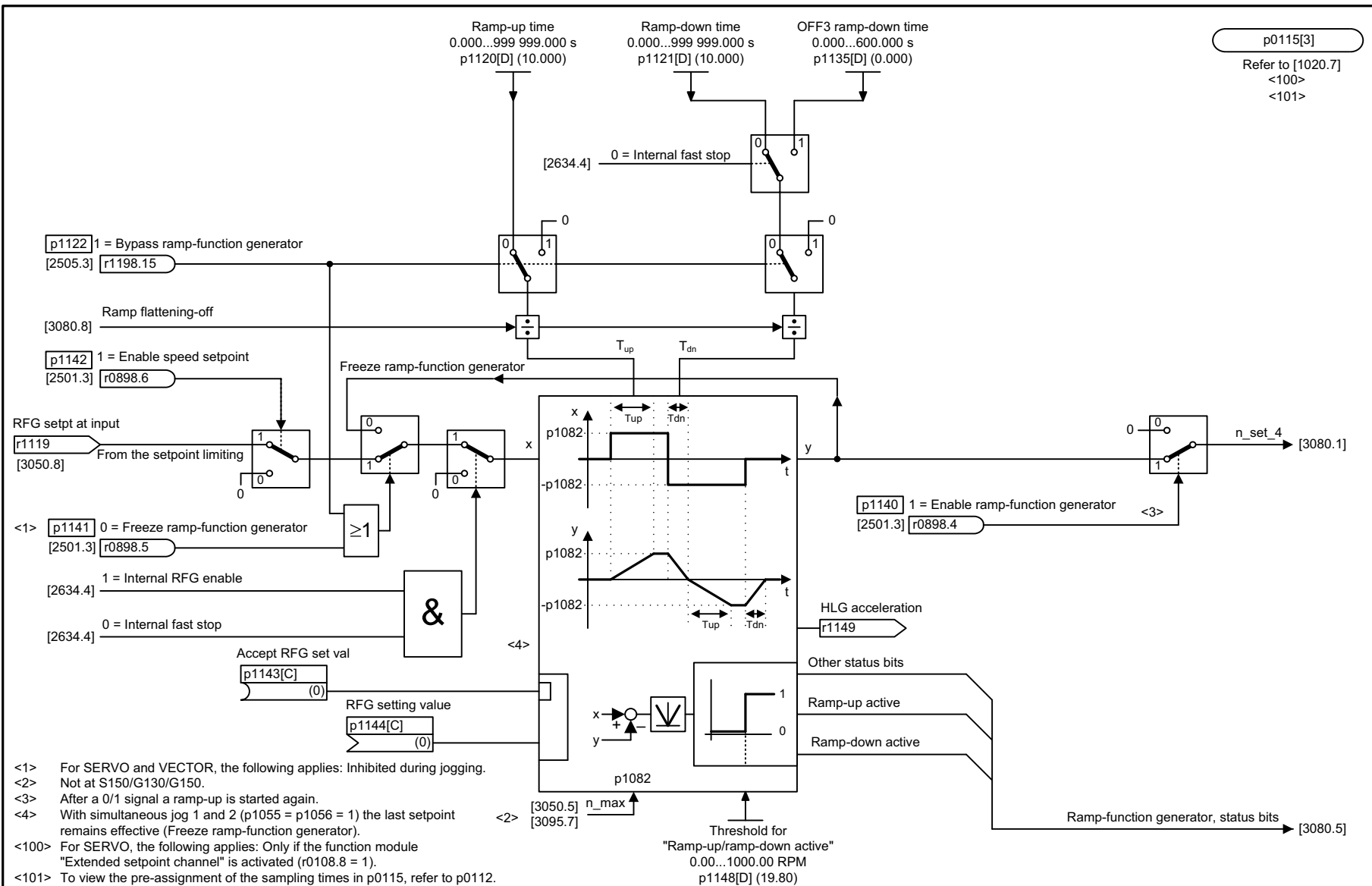
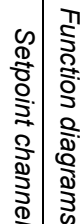


Fig. 2-131 3060 – Basic ramp-function generator

Fig. 2-132



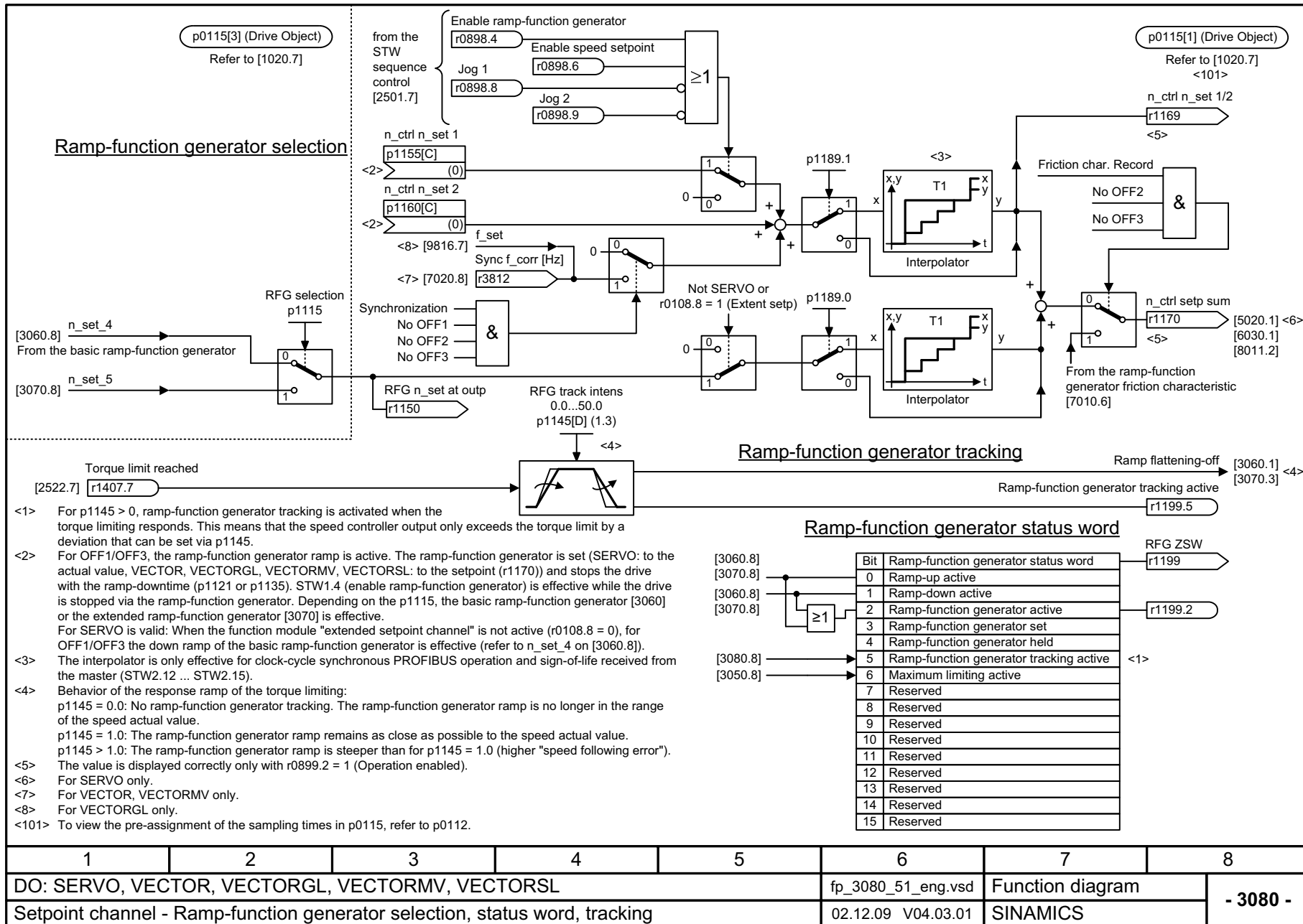
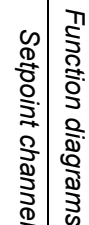


Fig. 2-133

3080 – Ramp-function generator selection, status word, tracking

Fig. 2-134

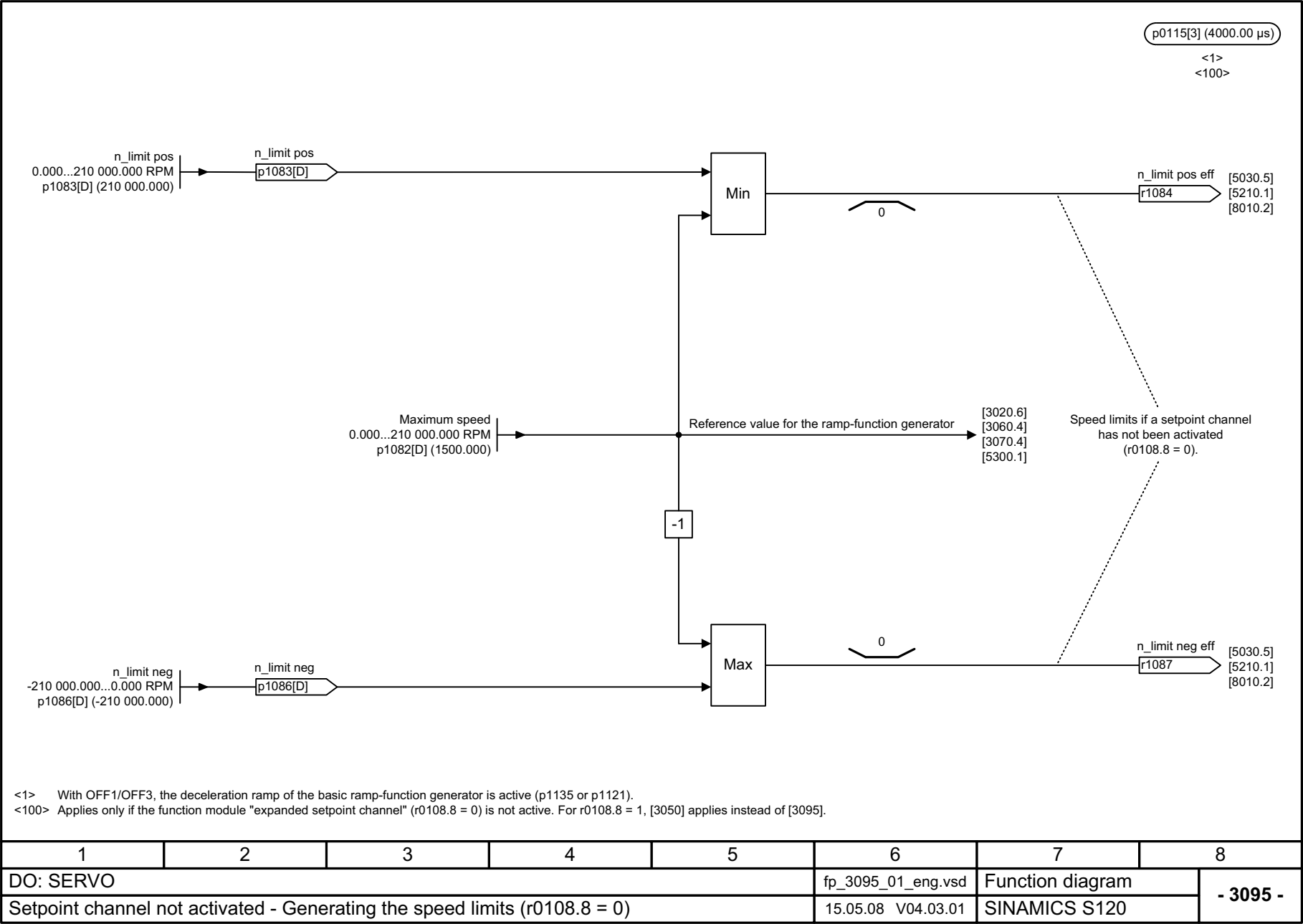


2.13 Setpoint channel not activated

Function diagrams

3095 – Formation of the speed limits ($r0108.8 = 0$)

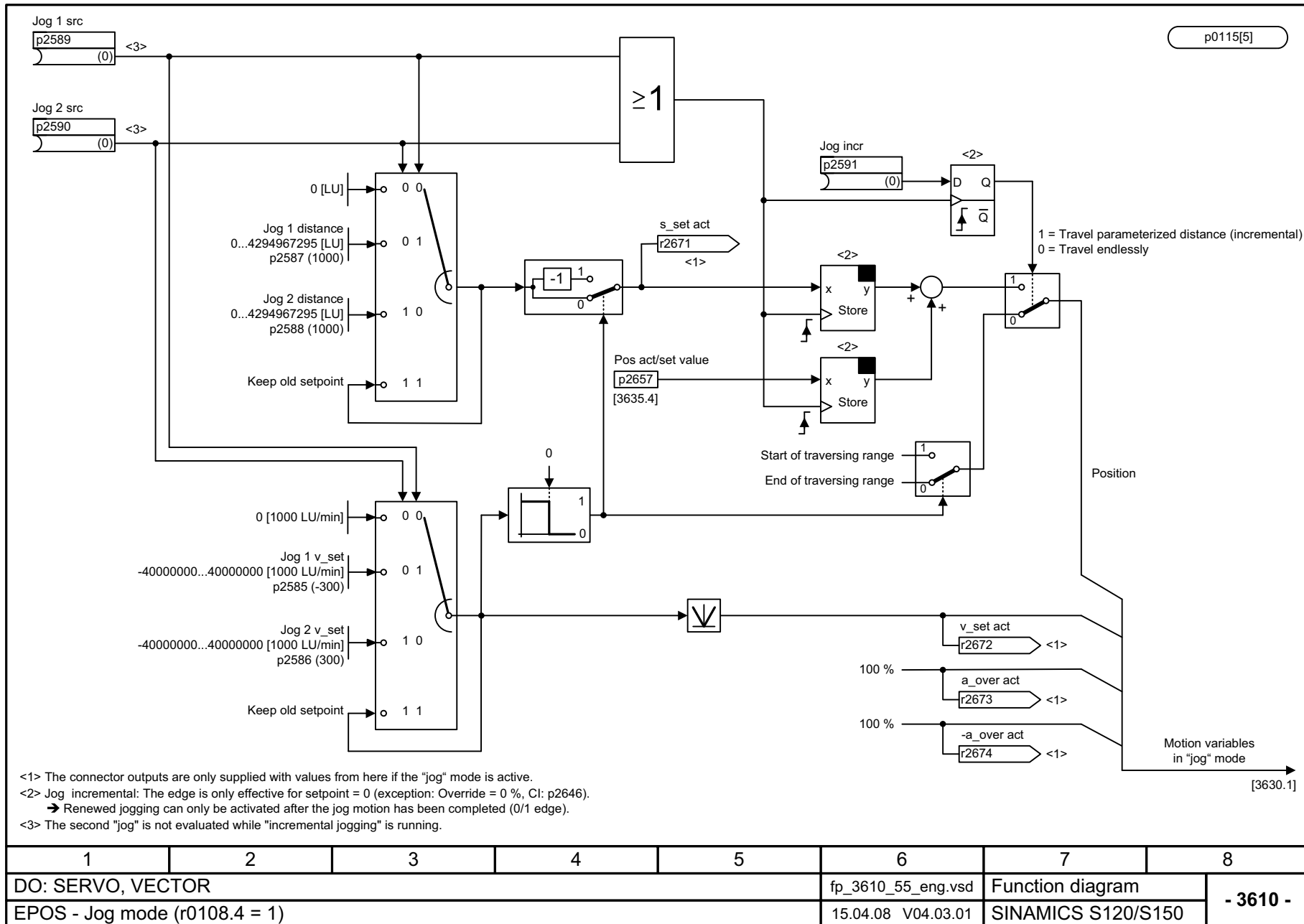
2-1531

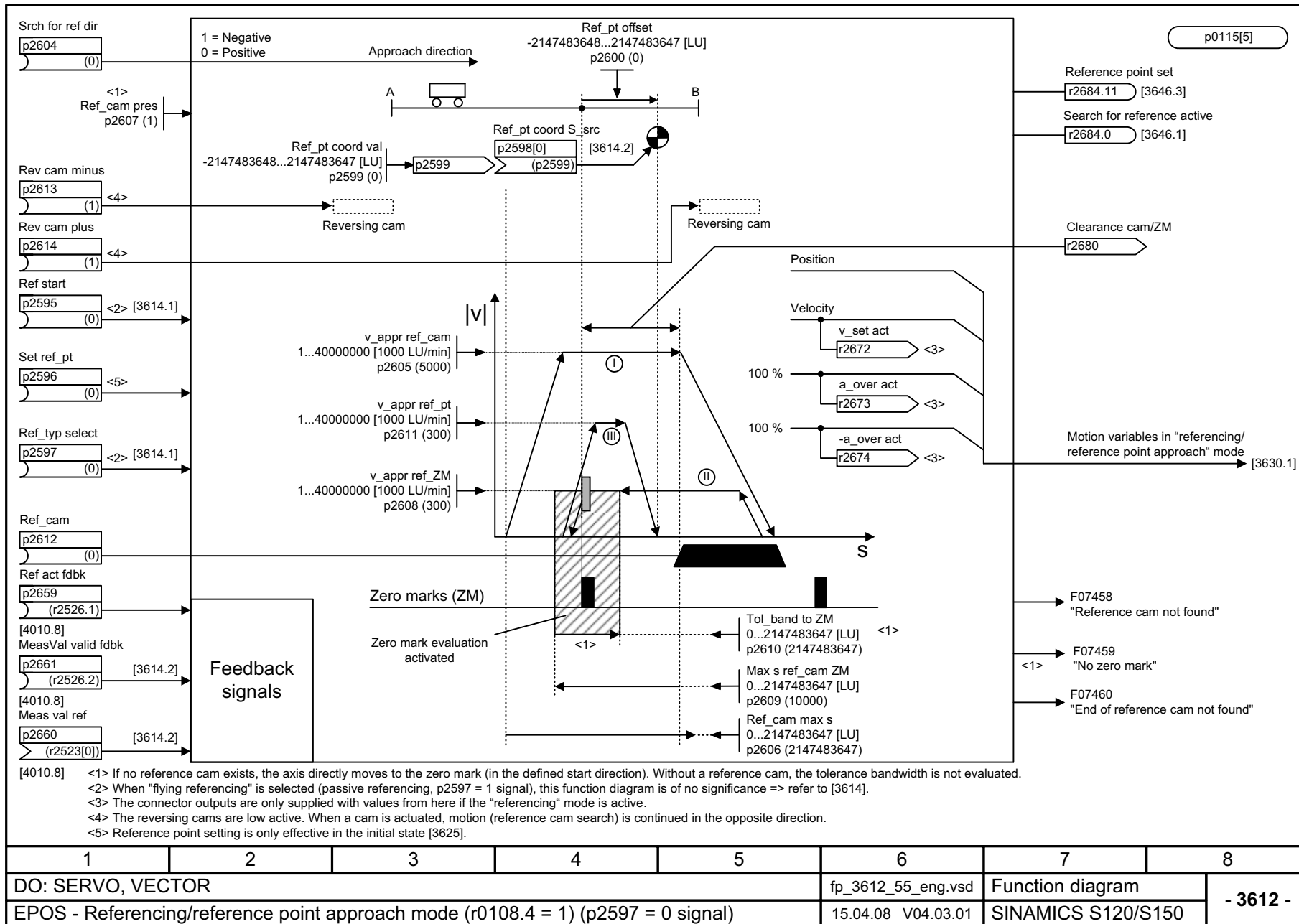


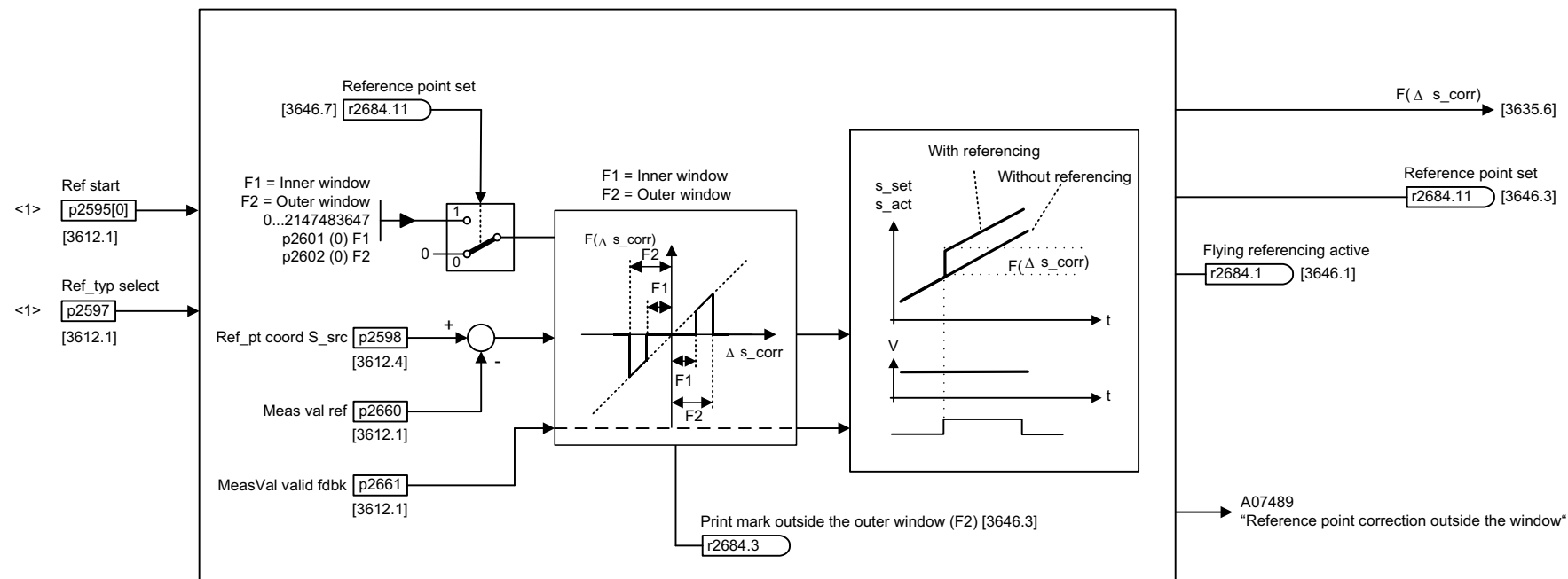
2.14 Basic positioner (EPOS)

Function diagrams

3610 – Jog mode (r0108.4 = 1)	2-1533
3612 – Referencing/reference point approach mode (r0108.4 = 1) (p2597 = 0 signal)	2-1534
3614 – Flying referencing mode (r0108.4 = 1) (p2597 = 1 signal)	2-1535
3615 – Traversing blocks, external block change mode (r0108.4 = 1)	2-1536
3616 – Traversing blocks mode (r0108.4 = 1)	2-1537
3617 – Travel to fixed stop	2-1538
3618 – Direct setpoint entry/MDI mode, dynamic values (r0108.4 = 1)	2-1539
3620 – Direct setpoint entry/MDI (r0108.4 = 1)	2-1540
3625 – Mode control (r0108.4 = 1)	2-1541
3630 – Traversing range limits (r0108.4 = 1)	2-1542
3635 – Interpolator (r0108.4 = 1)	2-1543
3640 – Control word block selection/MDI selection (r0108.4 = 1)	2-1544
3645 – Status word 1 (r0108.3 = 1, r0108.4 = 1)	2-1545
3646 – Status word 2 (r0108.3 = 1, r0108.4 = 1)	2-1546
3650 – Status word active traversing block/MDI active (r0108.4 = 1)	2-1547







<1> When "reference point approach" is selected (active referencing p2597 = 0 signal), this function diagram is of no significance => refer to [3612].
Active traversing is not directly associated with the "flying referencing" mode (passive referencing, p2597 = 1 signal).
The mode can be superimposed on the "jog" [3610], "traversing blocks" [3614] and "direct setpoint input/MDI" [3618] modes!

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_3614_55_eng.vsd	Function diagram	
EPOS - Flying referencing mode (r0108.4 = 1) (p2597 = 1 signal)					15.04.08 V04.03.01	SINAMICS S120/S150	
							- 3614 -

Fig. 2-138 3614 – Flying Referencing mode (r0108.4 = 1) (p2597 = 1 signal)

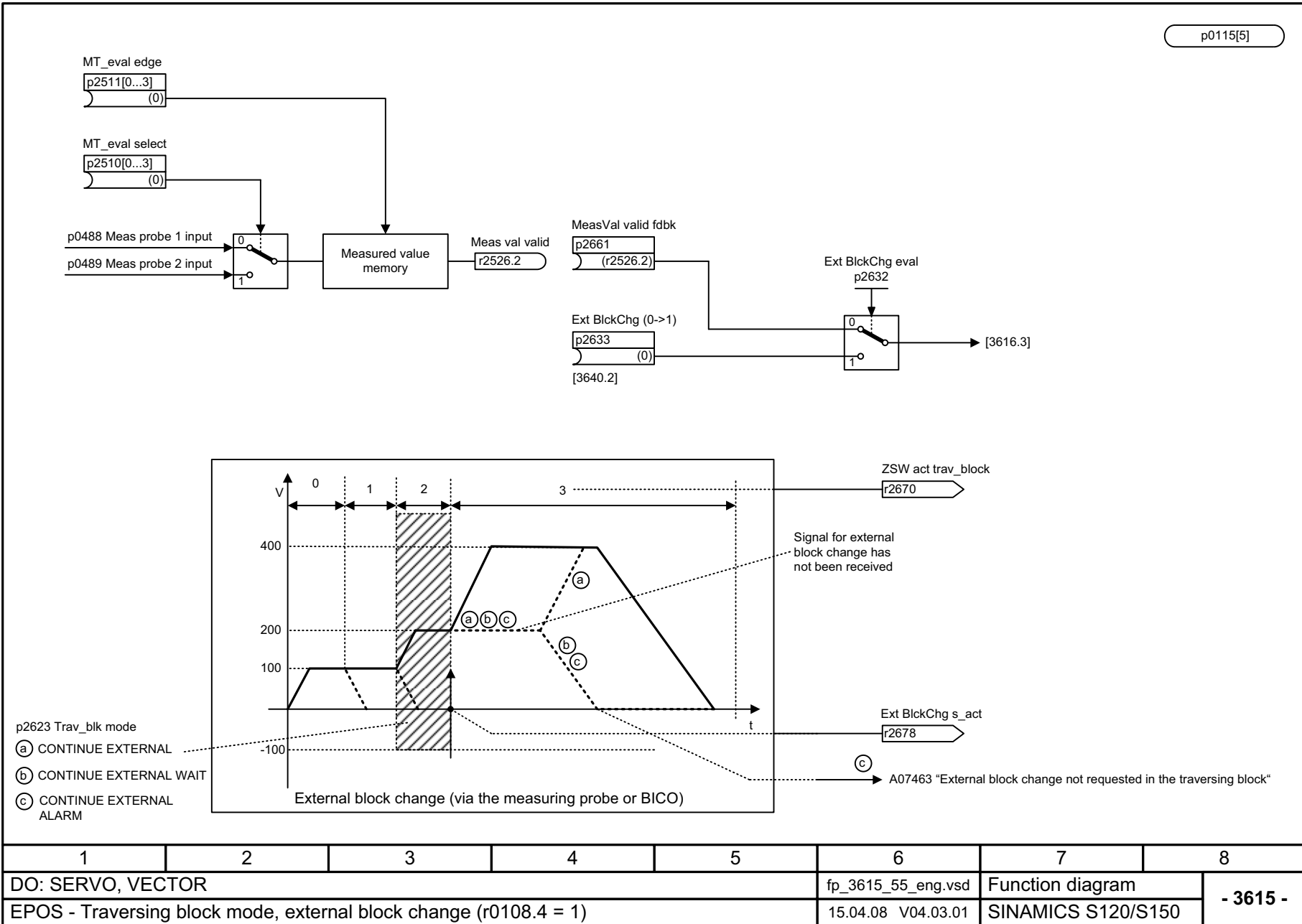
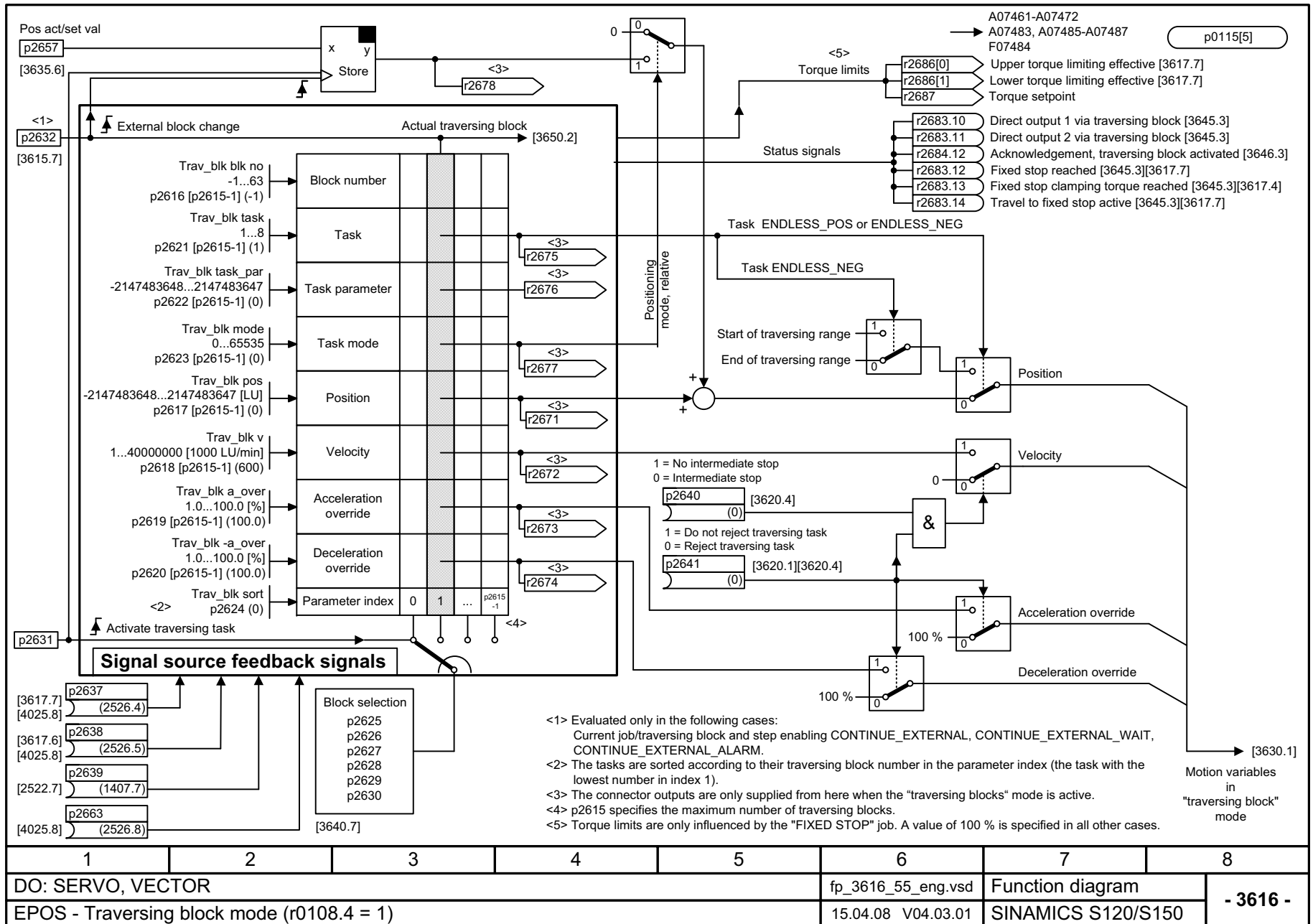


Fig. 2-140 3616 – Traversing blocks mode (r0108.4 = 1)



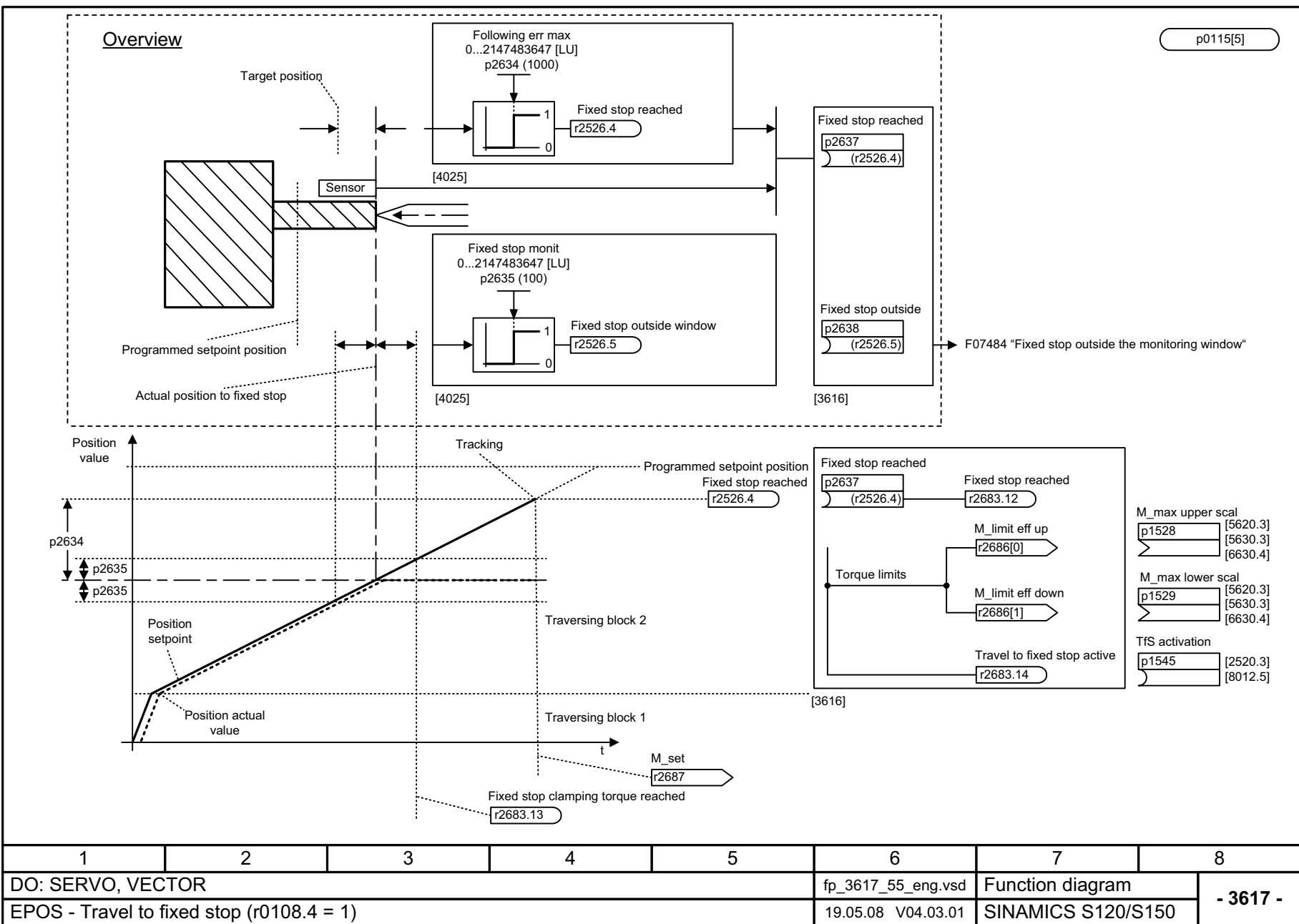


Fig. 2-141 3617 – Travel to fixed stop

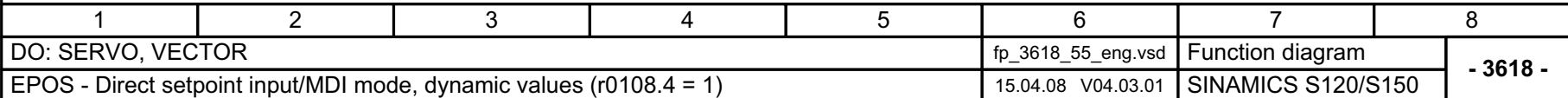
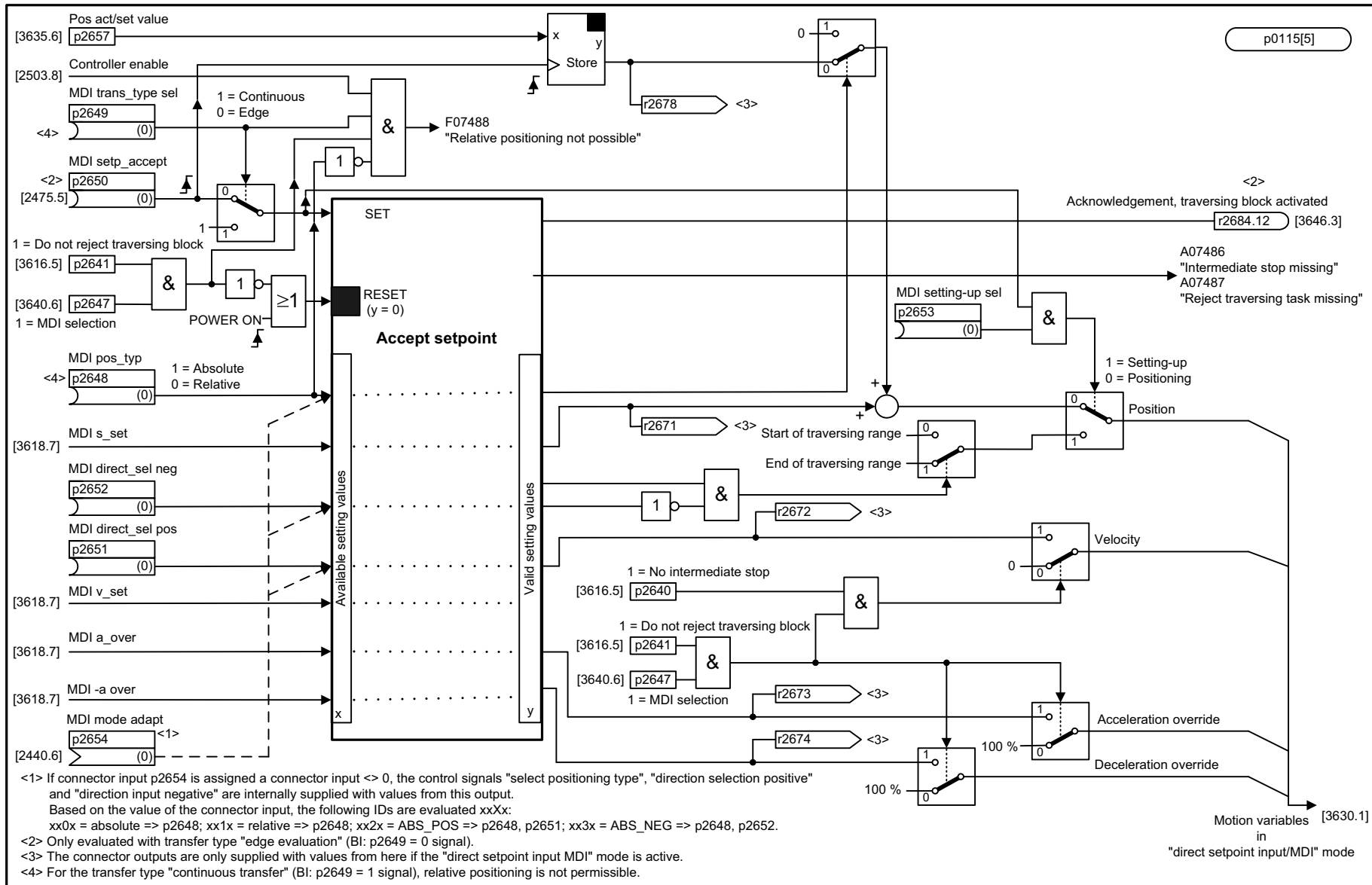


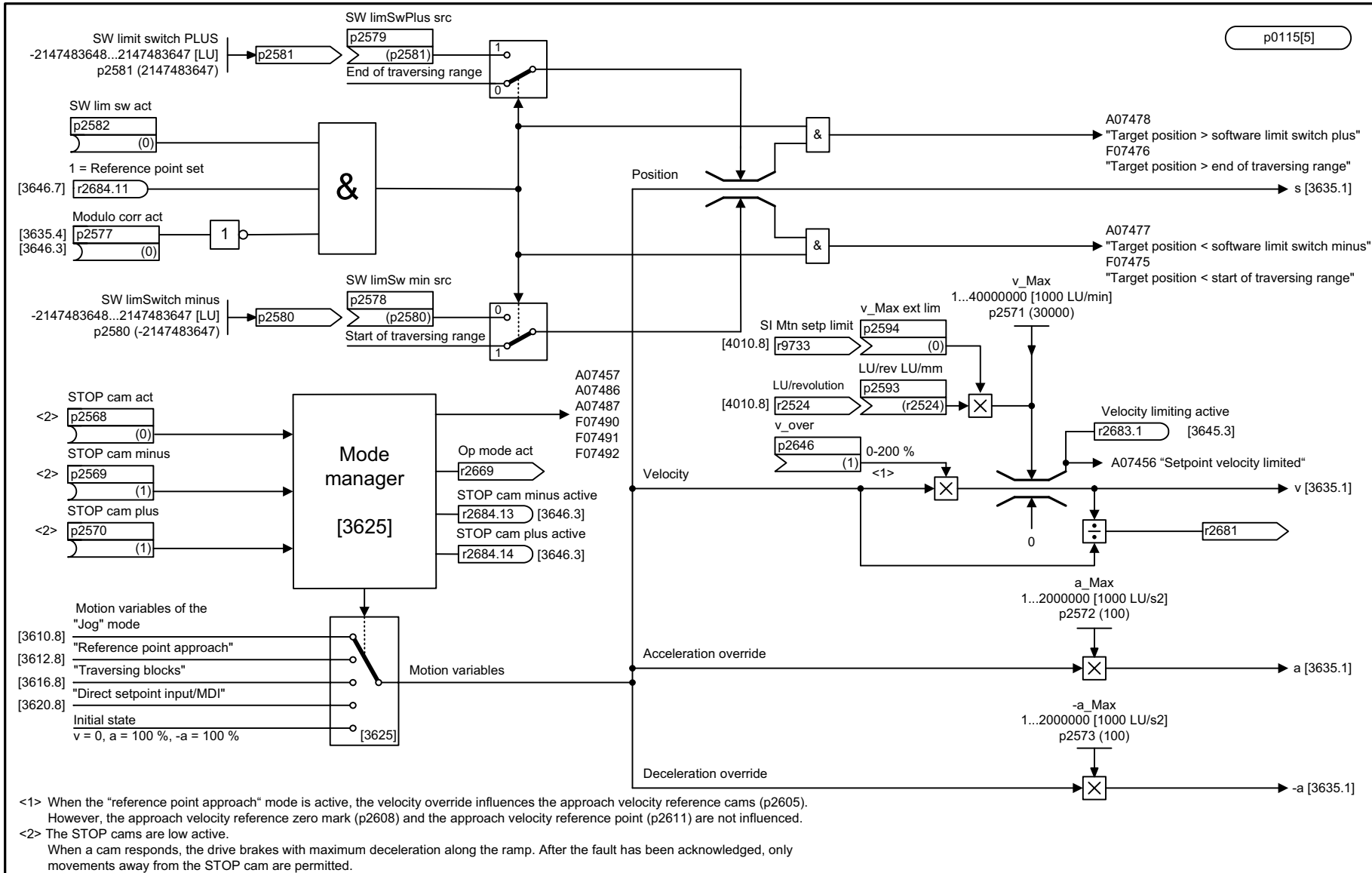
Fig. 2-142



1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_3620_55_eng.vsd	Function diagram	
EPOS - Direct setpoint input/MDI mode (r0108.4 = 1)					15.04.08 V04.03.01	SINAMICS S120/S150	
					- 3620 -		

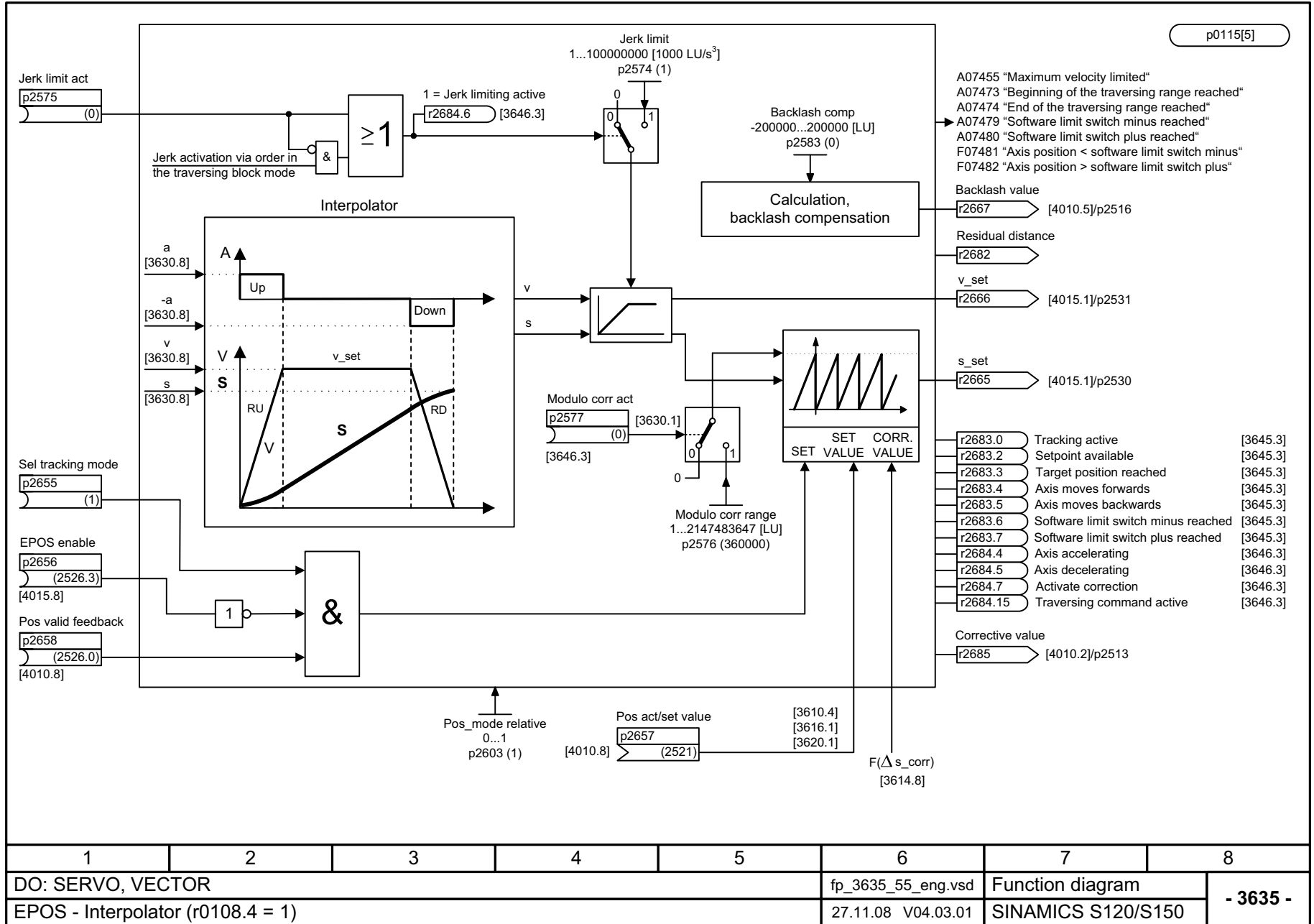
Fig. 2-144





1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_3630_55_eng.vsd	Function diagram	
EPOS - Traversing range limits (r0108.4 = 1)					08.09.09 V04.03.01	SINAMICS S120/S150	
							- 3630 -

Fig. 2-145 3630 – Traversing range limits (r0108.4 = 1)



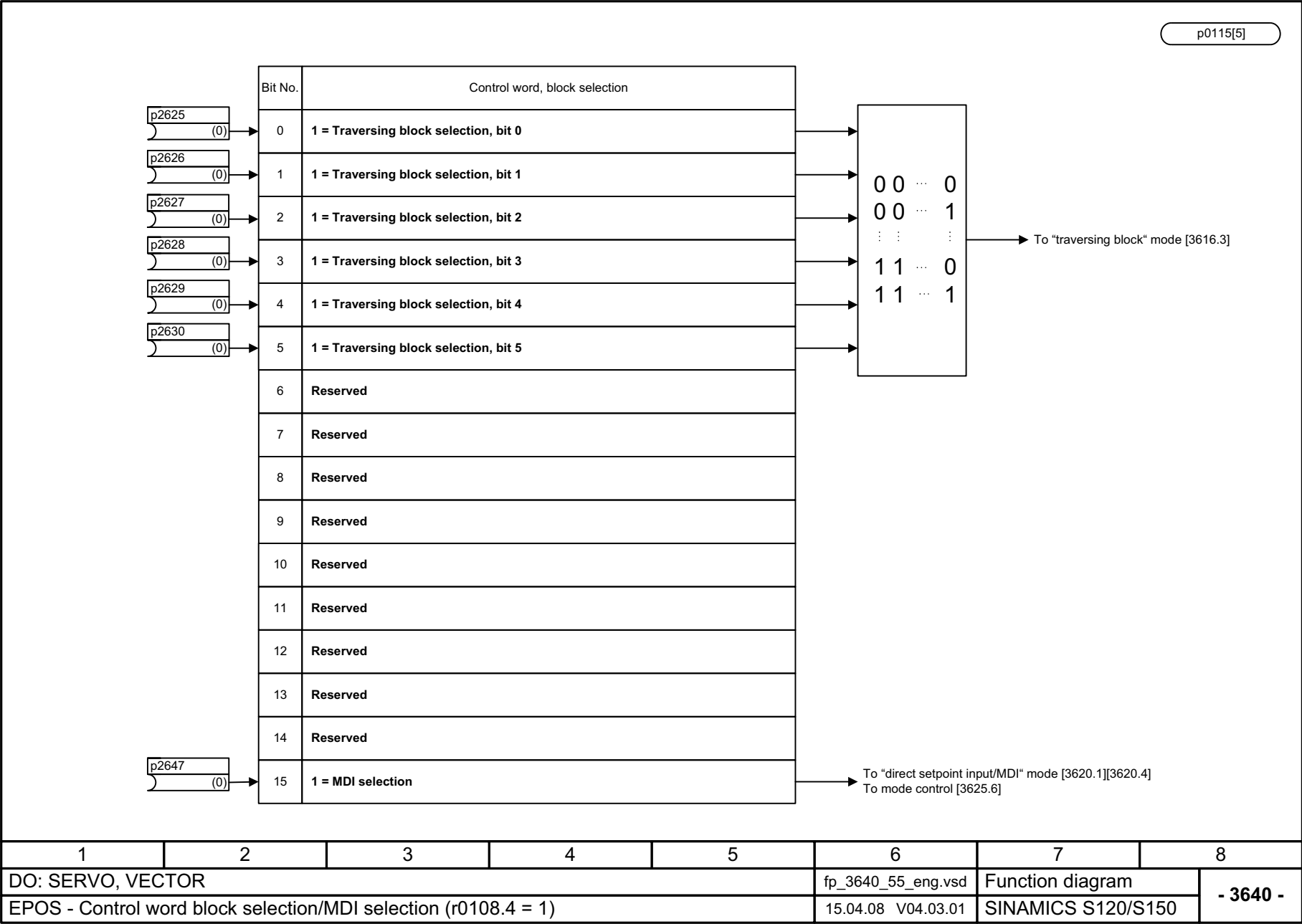


Fig. 2-147 3640 – Control word block selection/MDI selection (r0108.4 = 1)

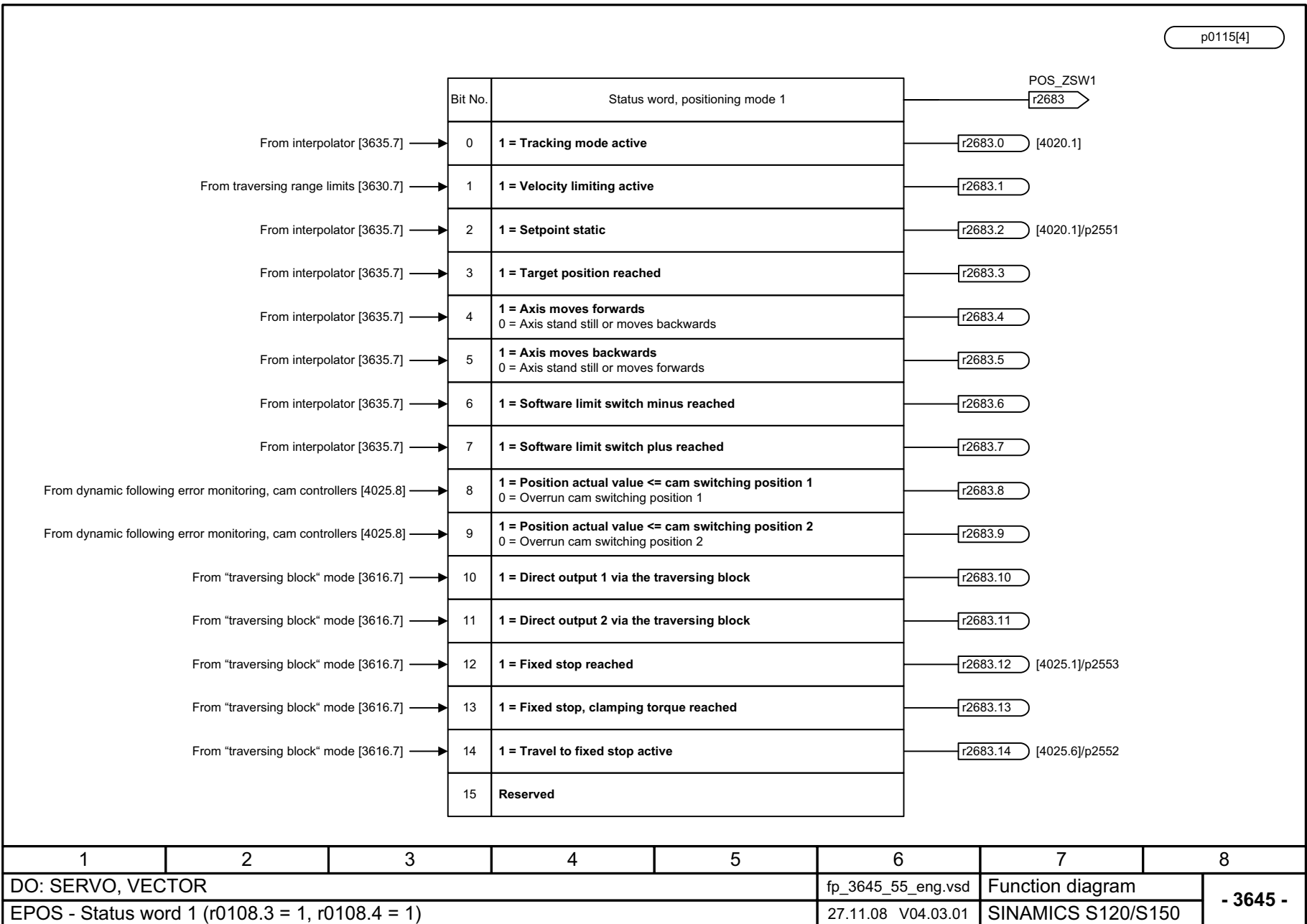
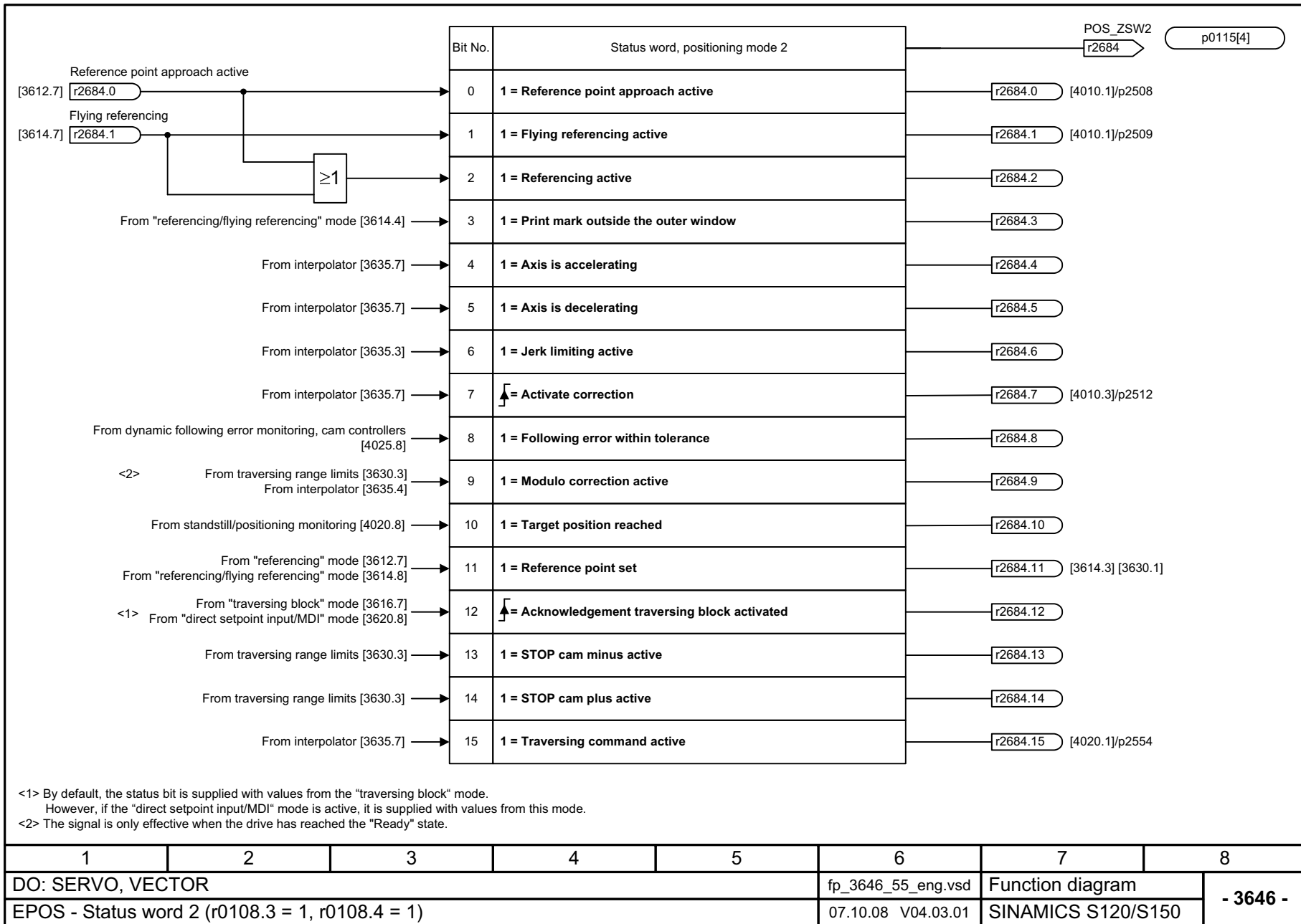


Fig. 2-148 3645 – Status word 1 (r0108.3 = 1, r0108.4 = 1)



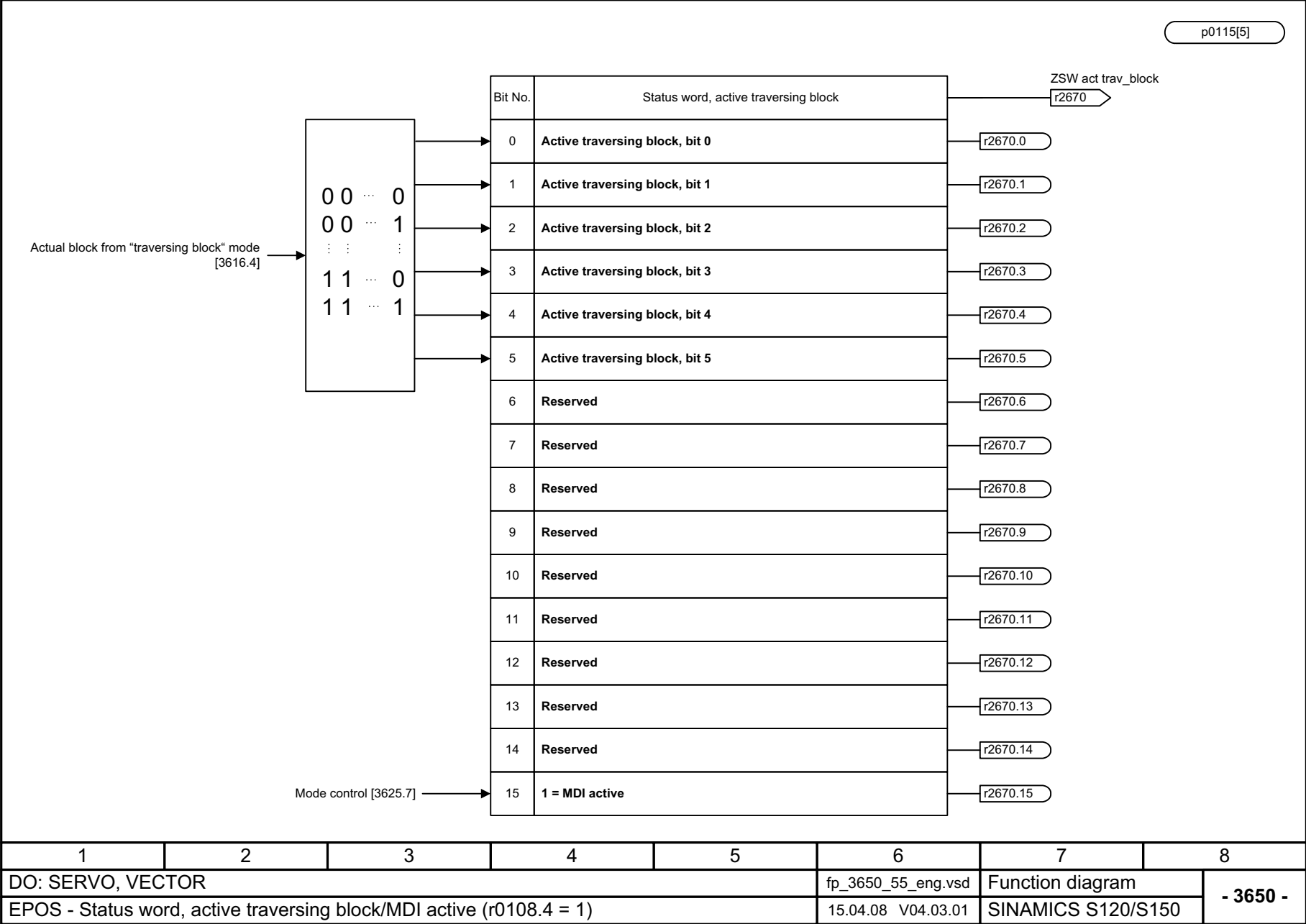


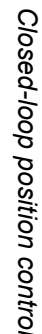
Fig. 2-150 3650 – Status word active traversing block/MDI active (r0108.4 = 1)

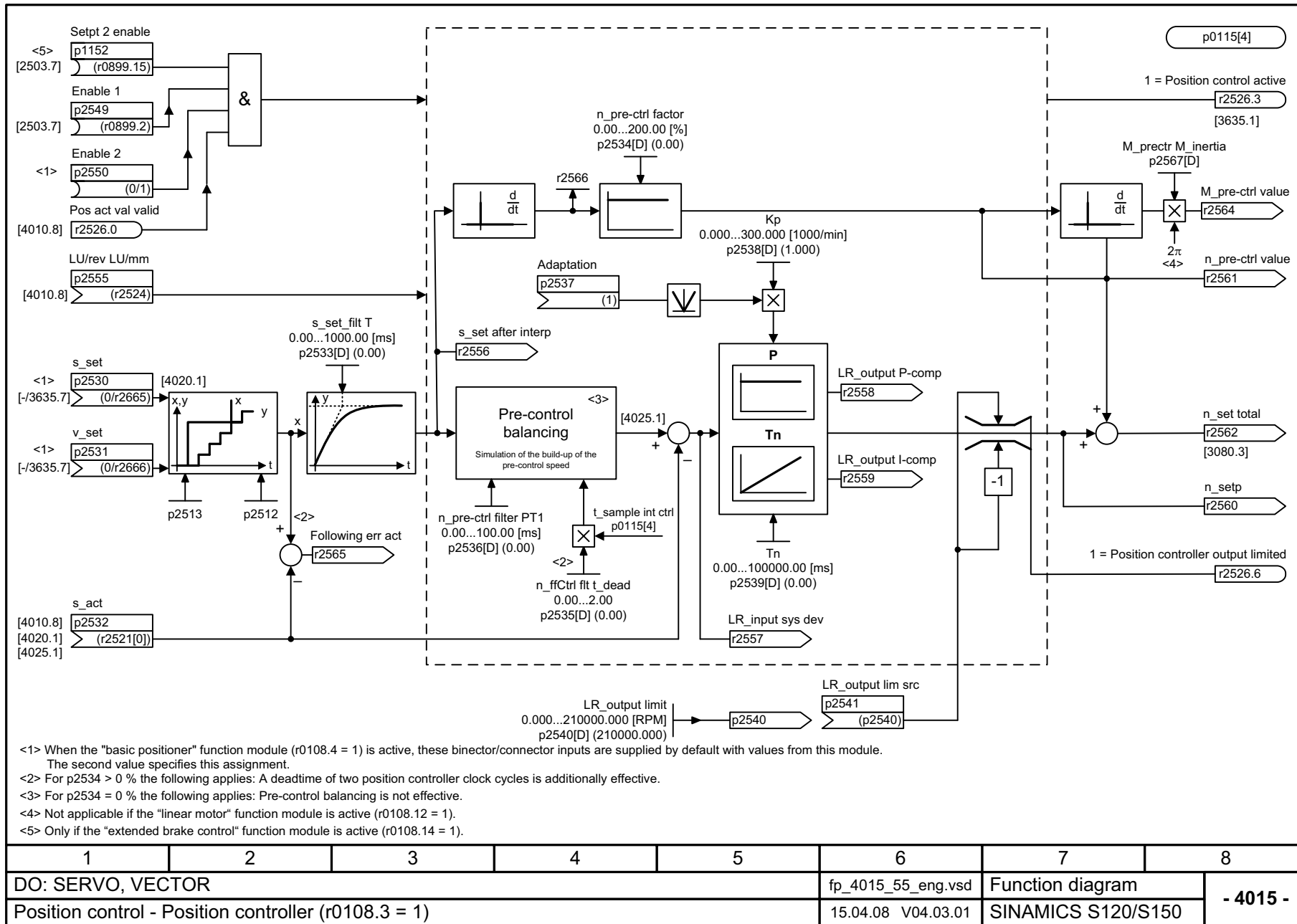
2.15 Closed-loop position control

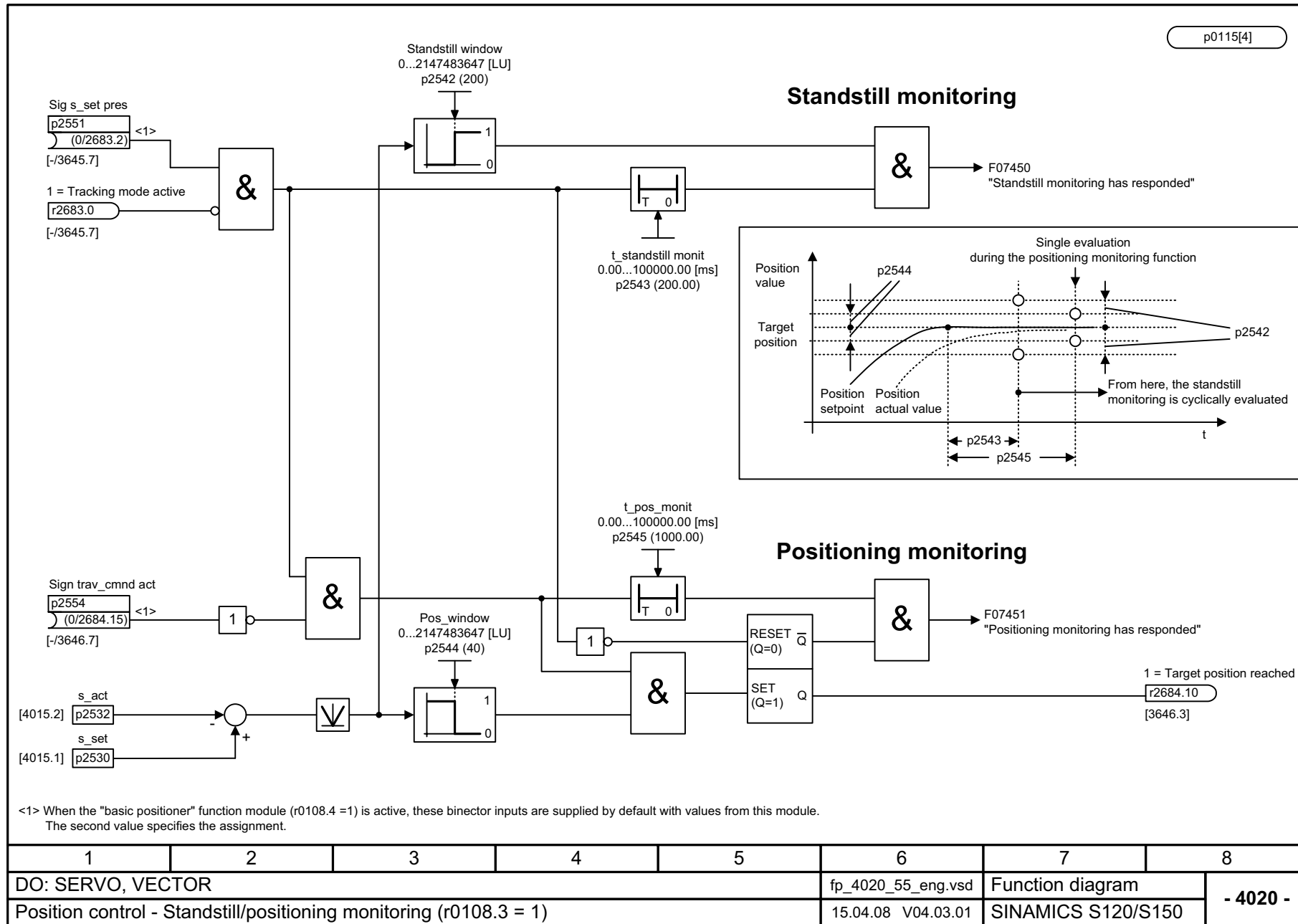
Function diagrams

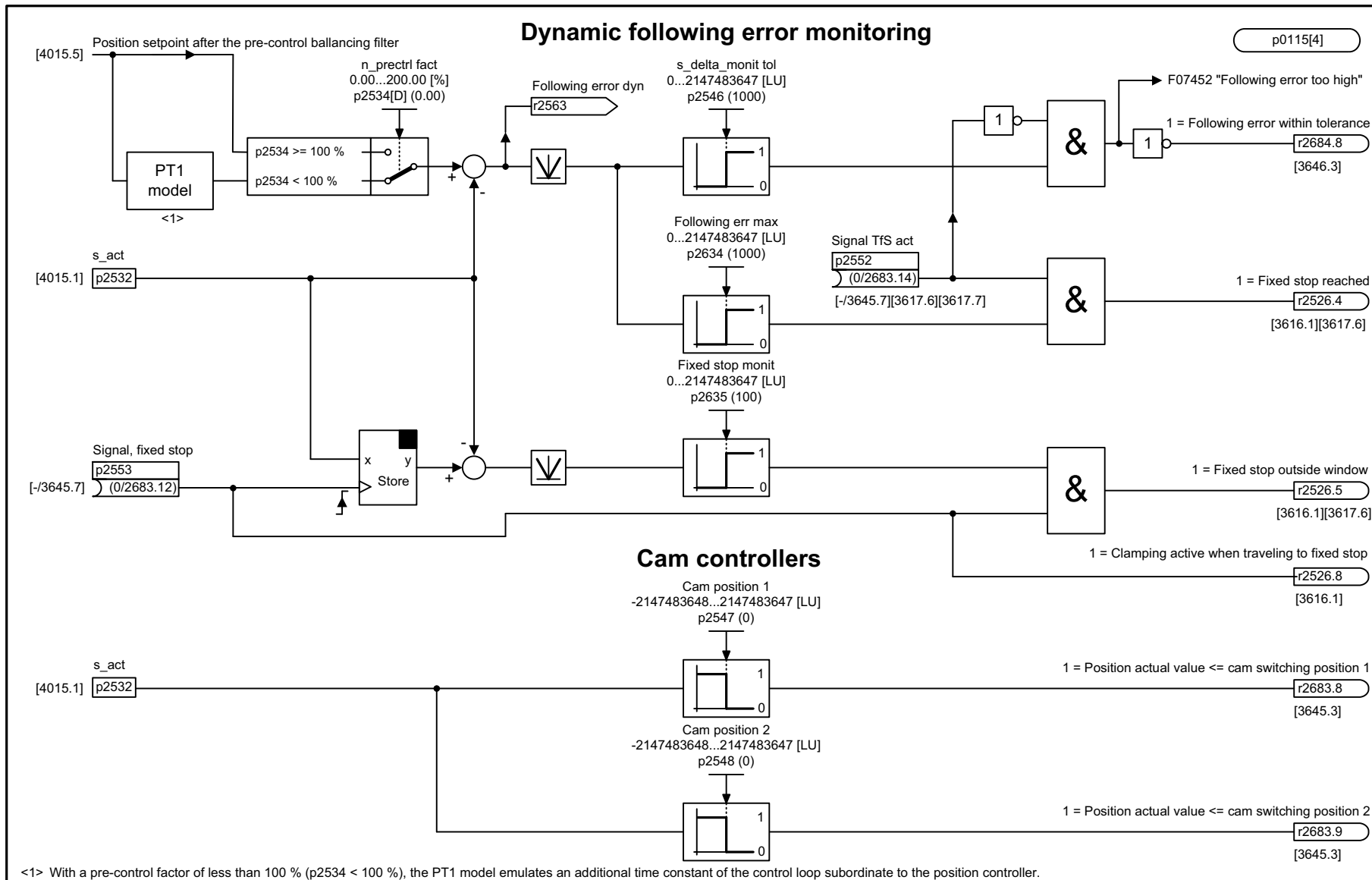
4010 – Actual position value preprocessing (r0108.3 = 1)	2-1549
4015 – Position controller (r0108.3 = 1)	2-1550
4020 – Standstill/positioning monitoring (r0108.3 = 1)	2-1551
4025 – Dynamic following error monitoring, cam controllers (r0108.3 = 1)	2-1552

Fig. 2-151









1	2	3	4	5	6	7	8	
DO: SERVO, VECTOR					fp_4025_55_eng.vsd	Function diagram		- 4025 -
Position control - Dynamic following error monitoring, cam controllers (r0108.3 = 1)					15.04.08 V04.03.01	SINAMICS S120/S150		

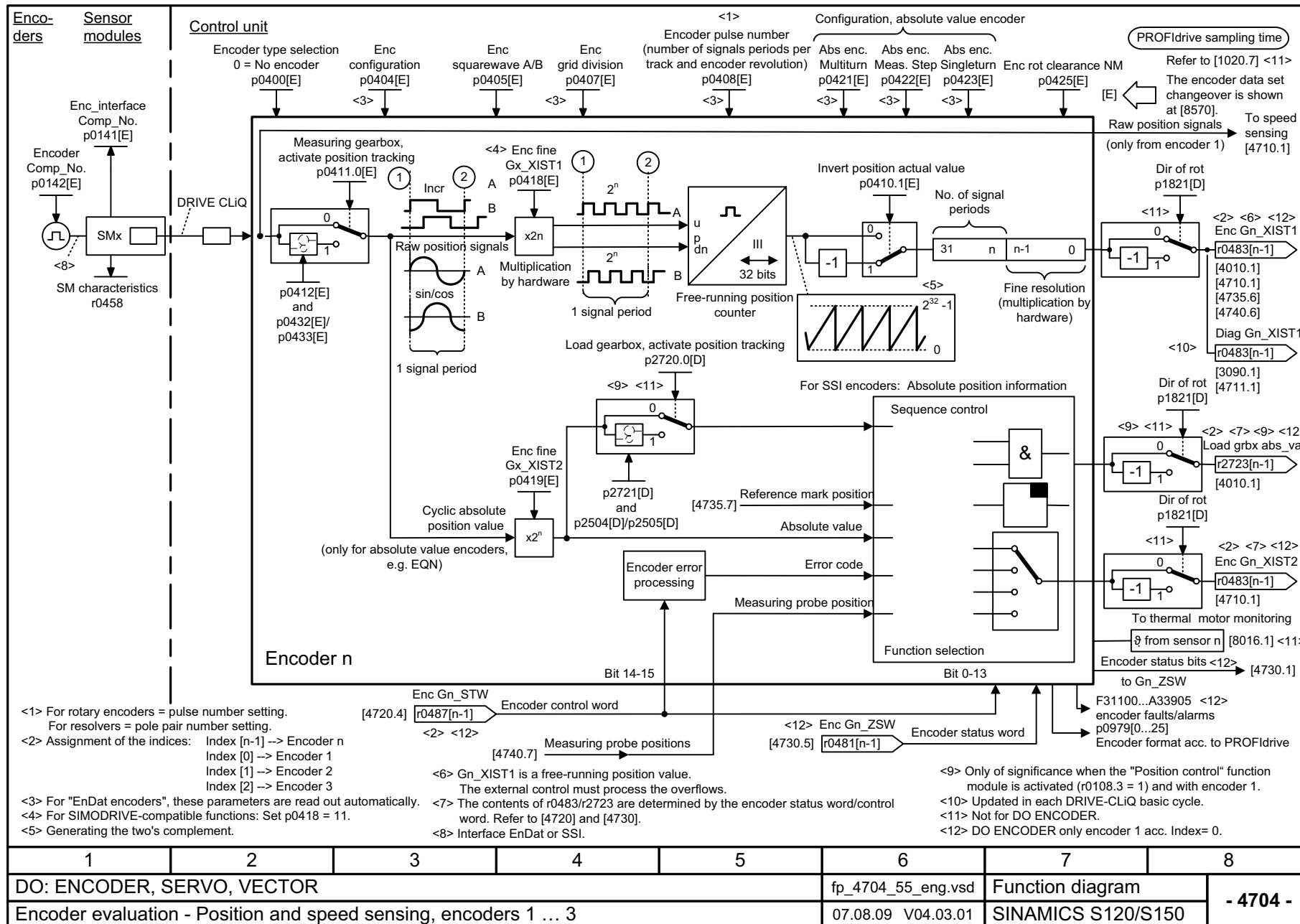
Fig. 2-154 4025 – Dynamic following error monitoring, cam controllers (r0108.3 = 1)

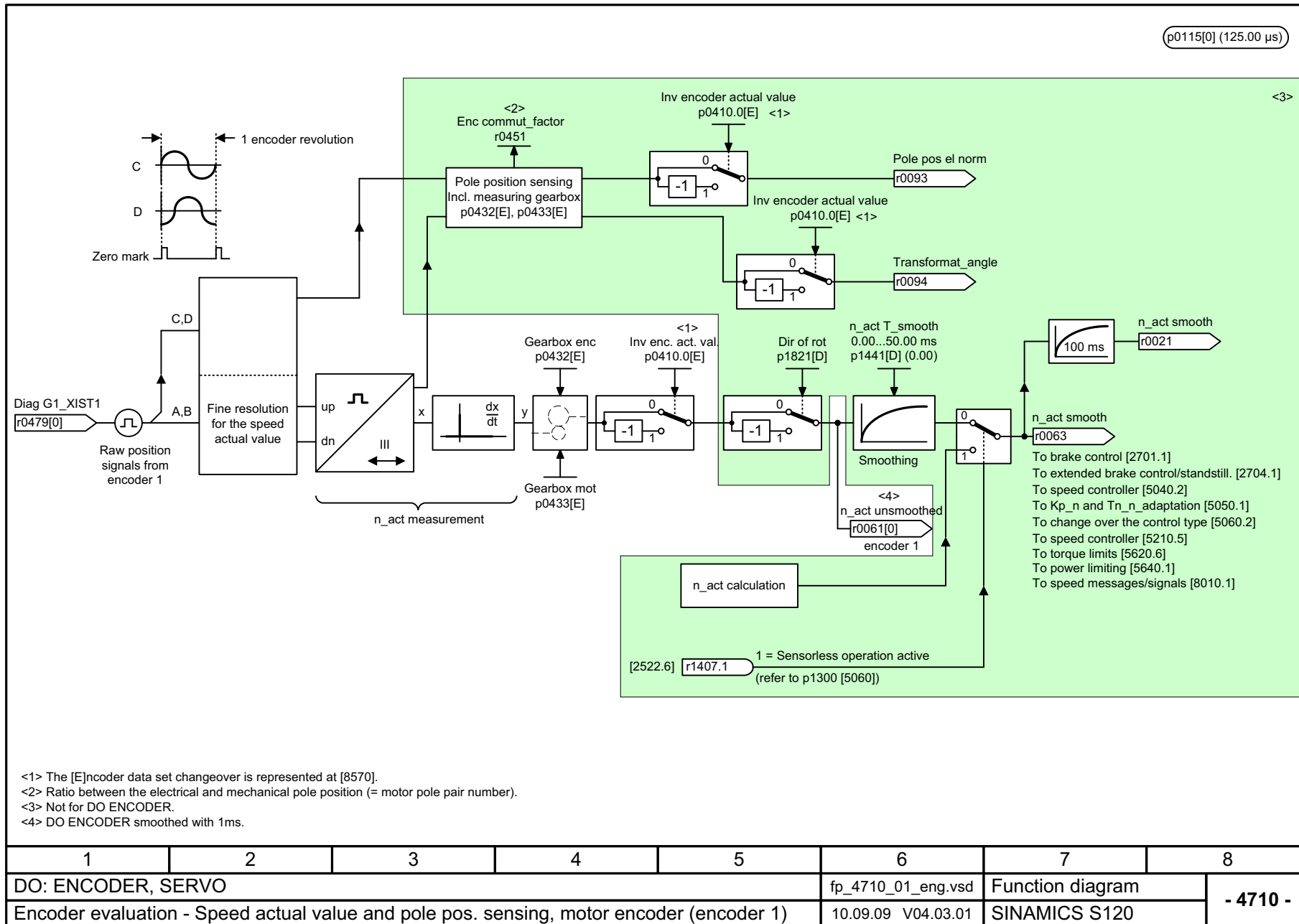
2.16 Encoder evaluation

Function diagrams

4704 – Position and temperature sensing, encoder 1 to 3	2-1554
4710 – Actual speed value and pole position sensing, motor encoder (encoder 1)	2-1555
4711 – Actual speed value sensing, encoder 2, 3 (r0108.7 = 1, APC activated)	2-1556
4715 – Actual speed value and pole position sensing, ASM/SM motor encoder (encoder 1)	2-1557
4720 – Encoder interface, incoming signals, encoder 1 to 3	2-1558
4730 – Encoder interface, outgoing signals, encoder 1 to 3	2-1559
4735 – Reference mark search with external zero mark, encoder 1 to 3	2-1560
4740 – Measuring probe evaluation, measured value memory, encoder 1 to 3	2-1561

Fig. 2-155 4704 – Position and temperature sensing, encoder 1 to 3





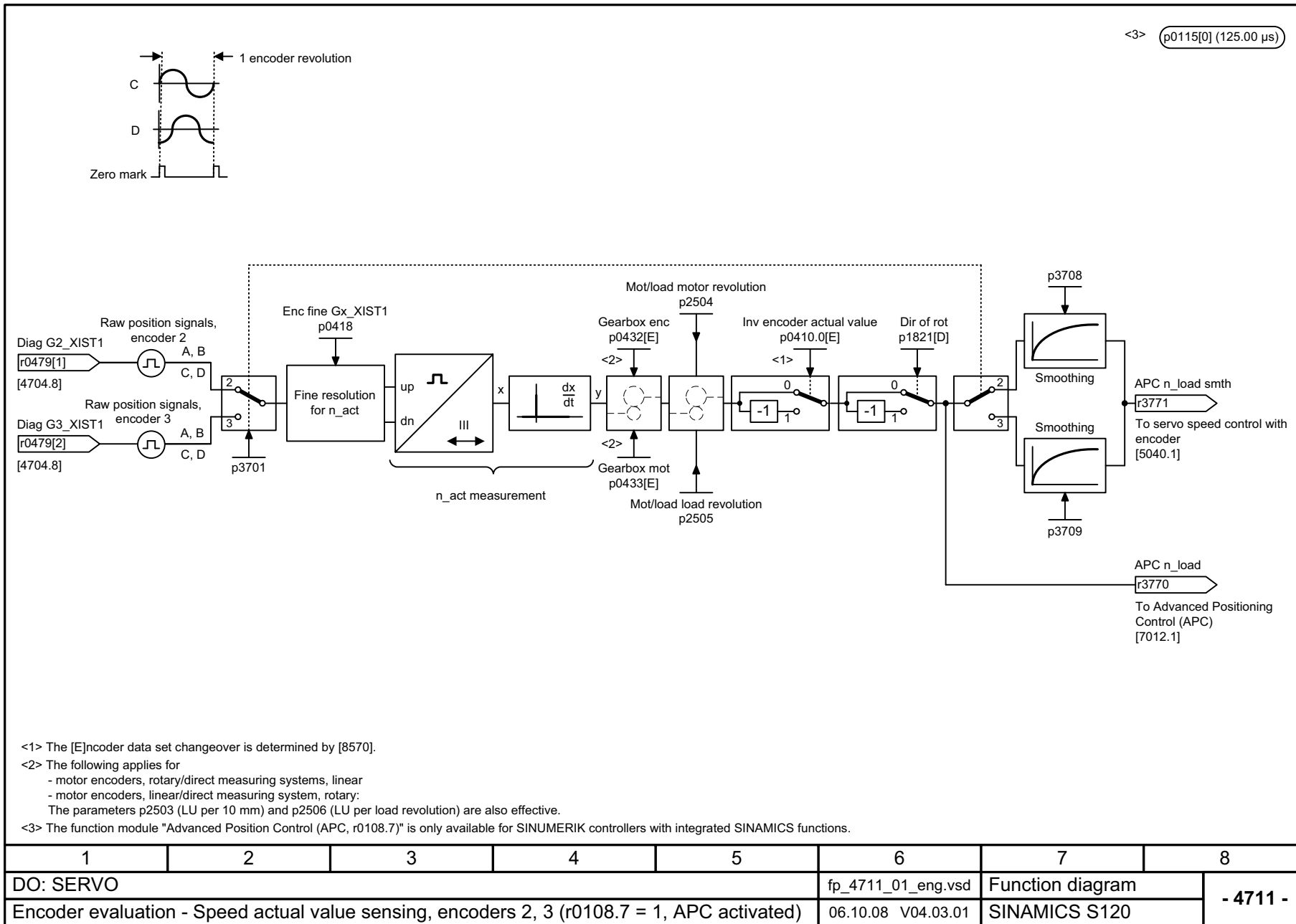
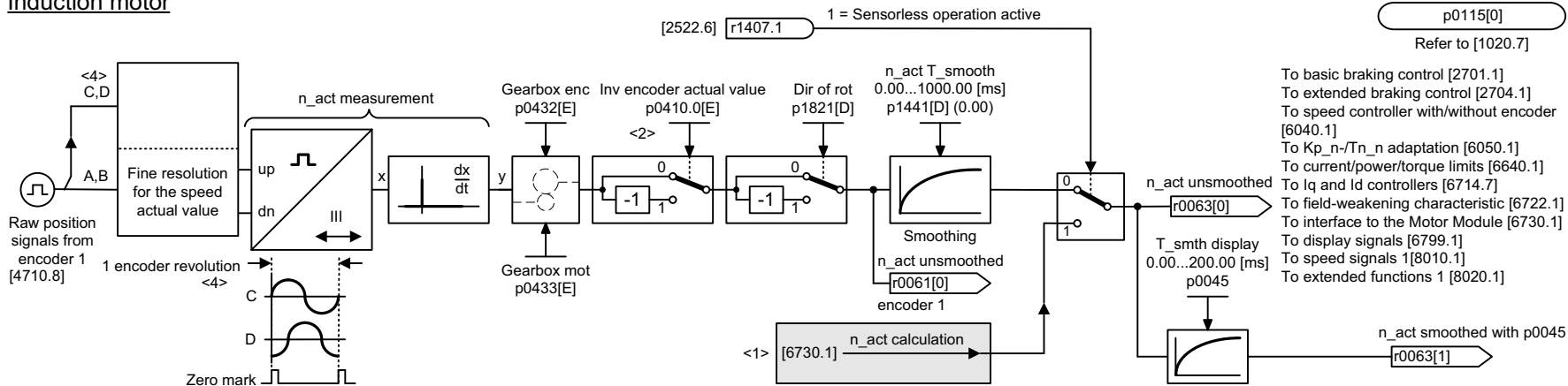


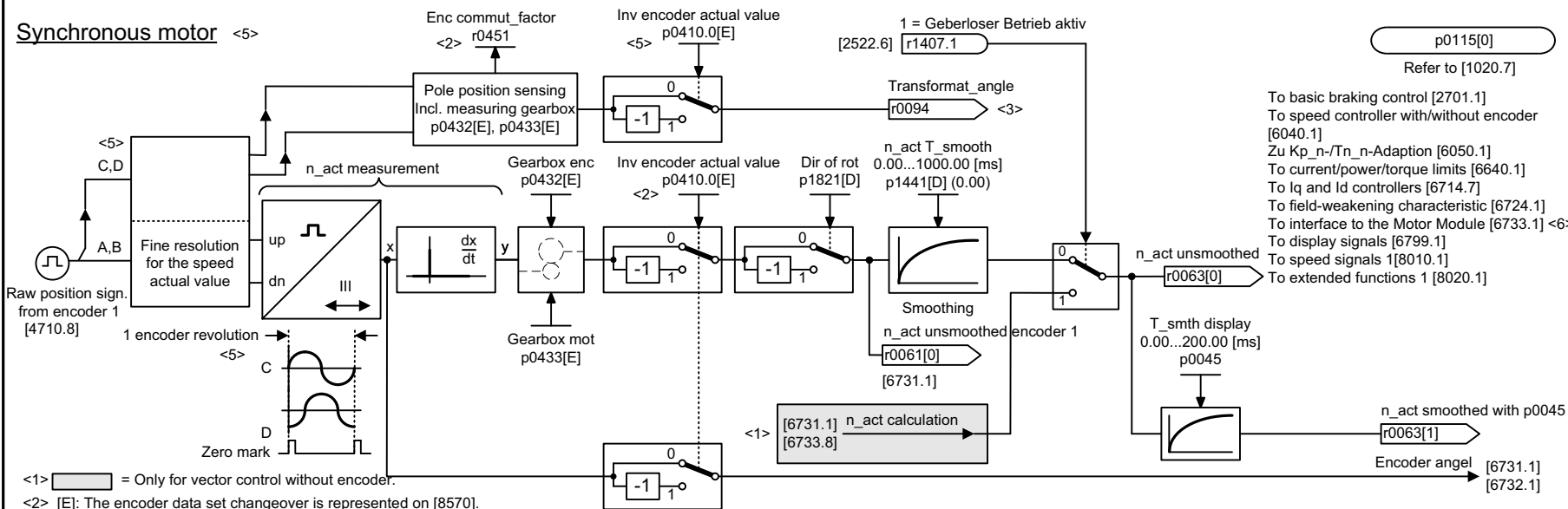
Fig. 2-157 4711 – Actual speed value sensing, encoder 2, 3 (r0108.7 = 1, APC activated)

Fig. 2-158 4715 – Actual speed value and pole position sensing, ASM/SM motor encoder (encoder 1)

Induction motor

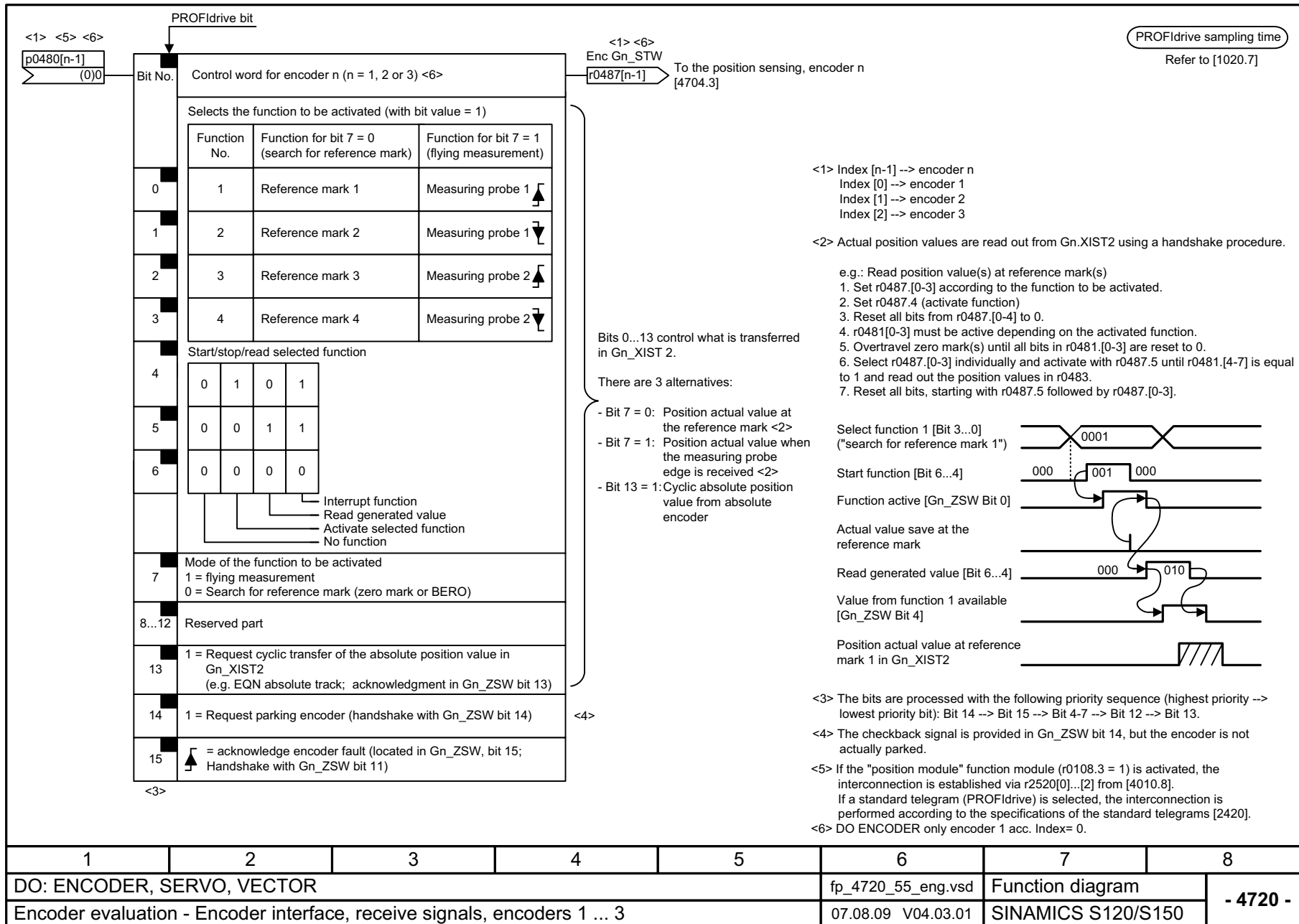


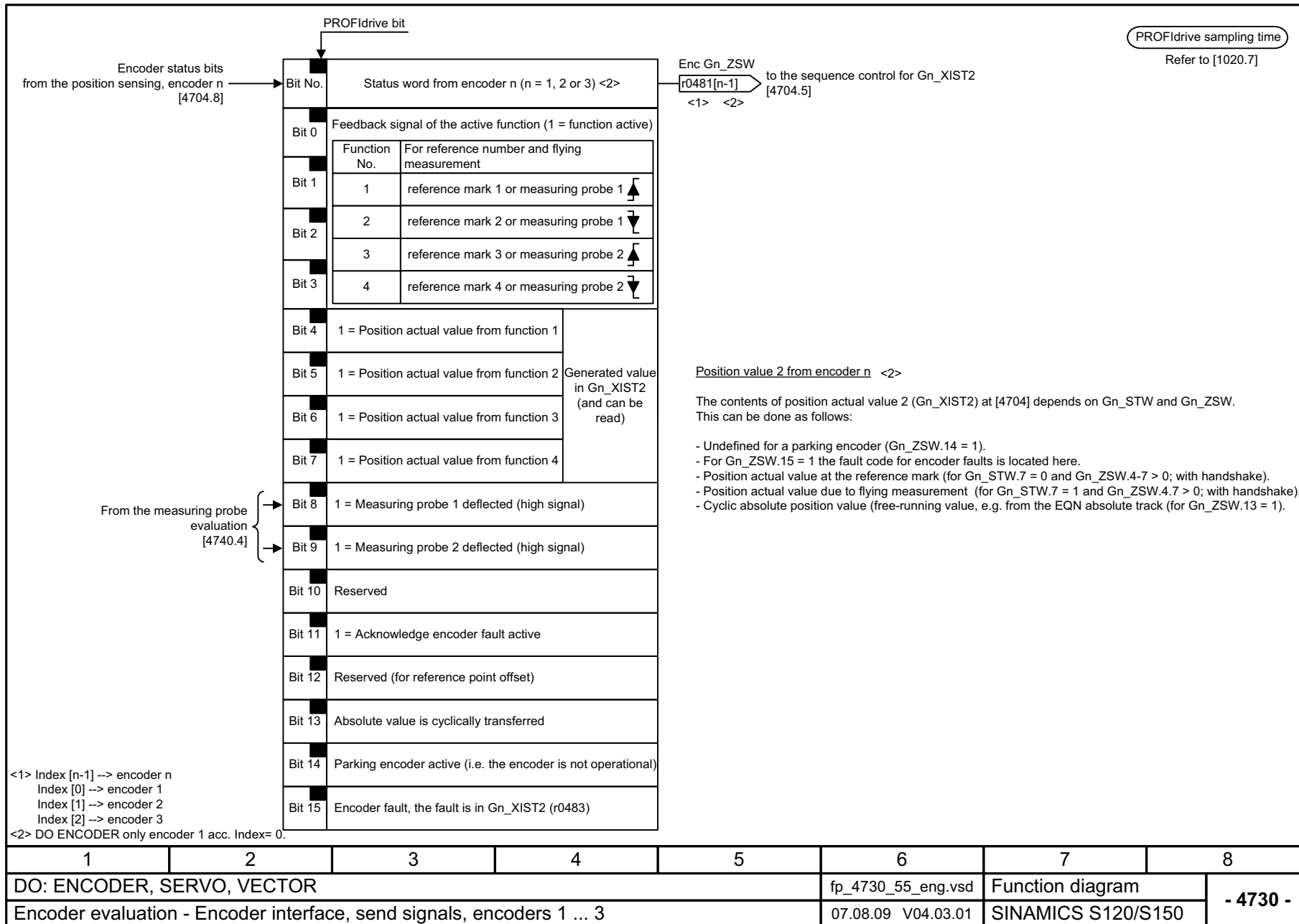
Synchronous motor <5>



- <1> [] = Only for vector control without encoder.
<2> [E]: The encoder data set changeover is represented on [8570].
<3> Only for operation with encoder and p0300 = 2.
<4> Only for S120/S150.
<5> Only for S120/S150 and p0300 = 2, 5
<6> Only for FEM.

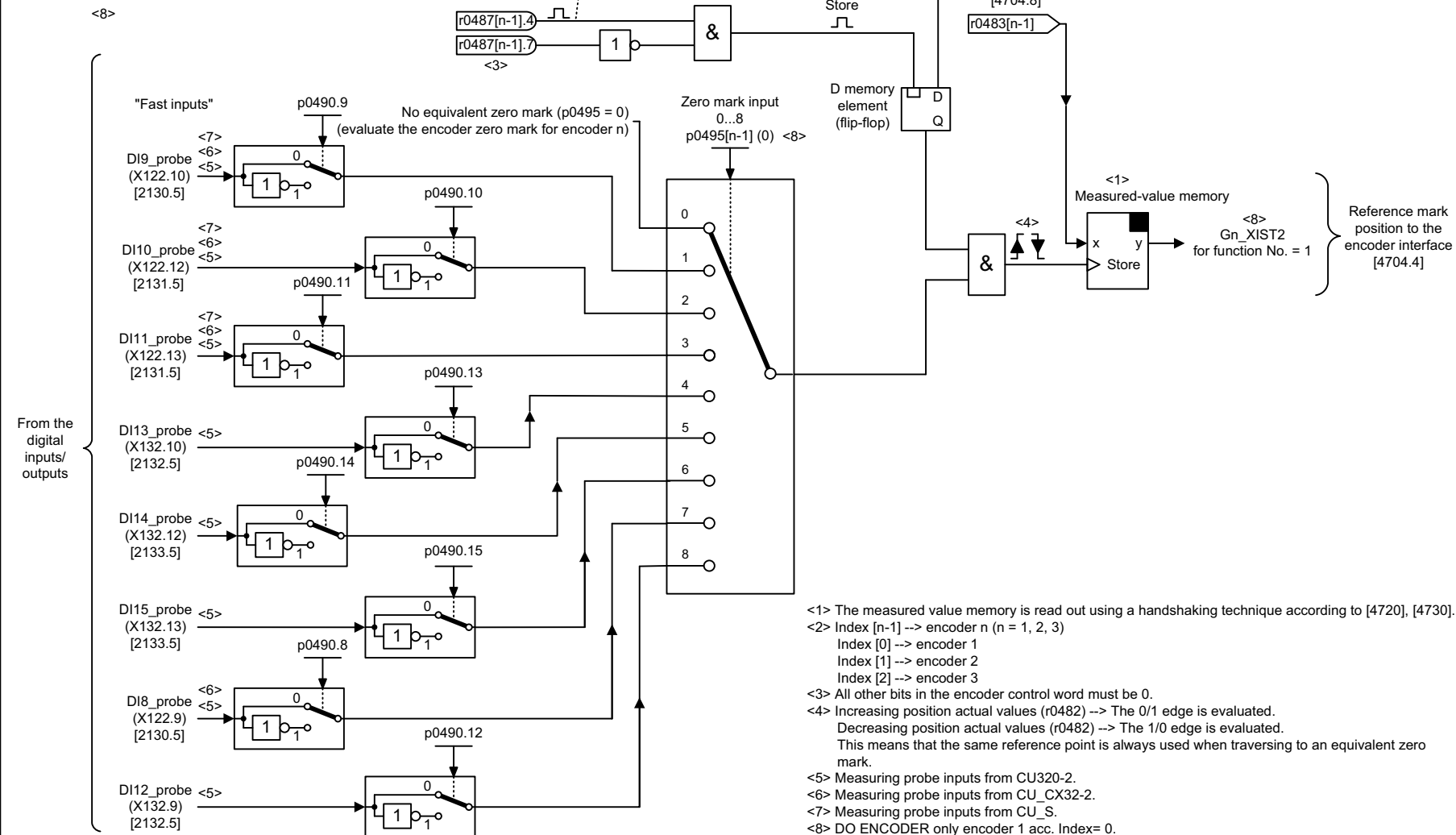
1	2	3	4	5	6	7	8
DO: VECTOR					fp_4715_54_eng.vsd	Function diagram	
Encoder evaluation - Speed act. value and pole pos. sensing, motor enc. ASM/SM (encoder 1)					29.07.09 V04.03.01	S120/S150/G130/G150	
							- 4715 -





Reference mark search

This function is provided once for each of the maximum 3 encoders.



1	2	3	4	5	6	7	8
DO: ENCODER, SERVO, VECTOR					fp_4735_55_eng.vsd	Function diagram	
Encoder evaluation - Reference mark search with equivalent zero mark, encoders 1 ... 3					07.08.09 V04.03.01	SINAMICS S120/S150	

- 4735 -

Fig. 2-161

4735 – Reference mark search with external zero mark, encoder 1 to 3

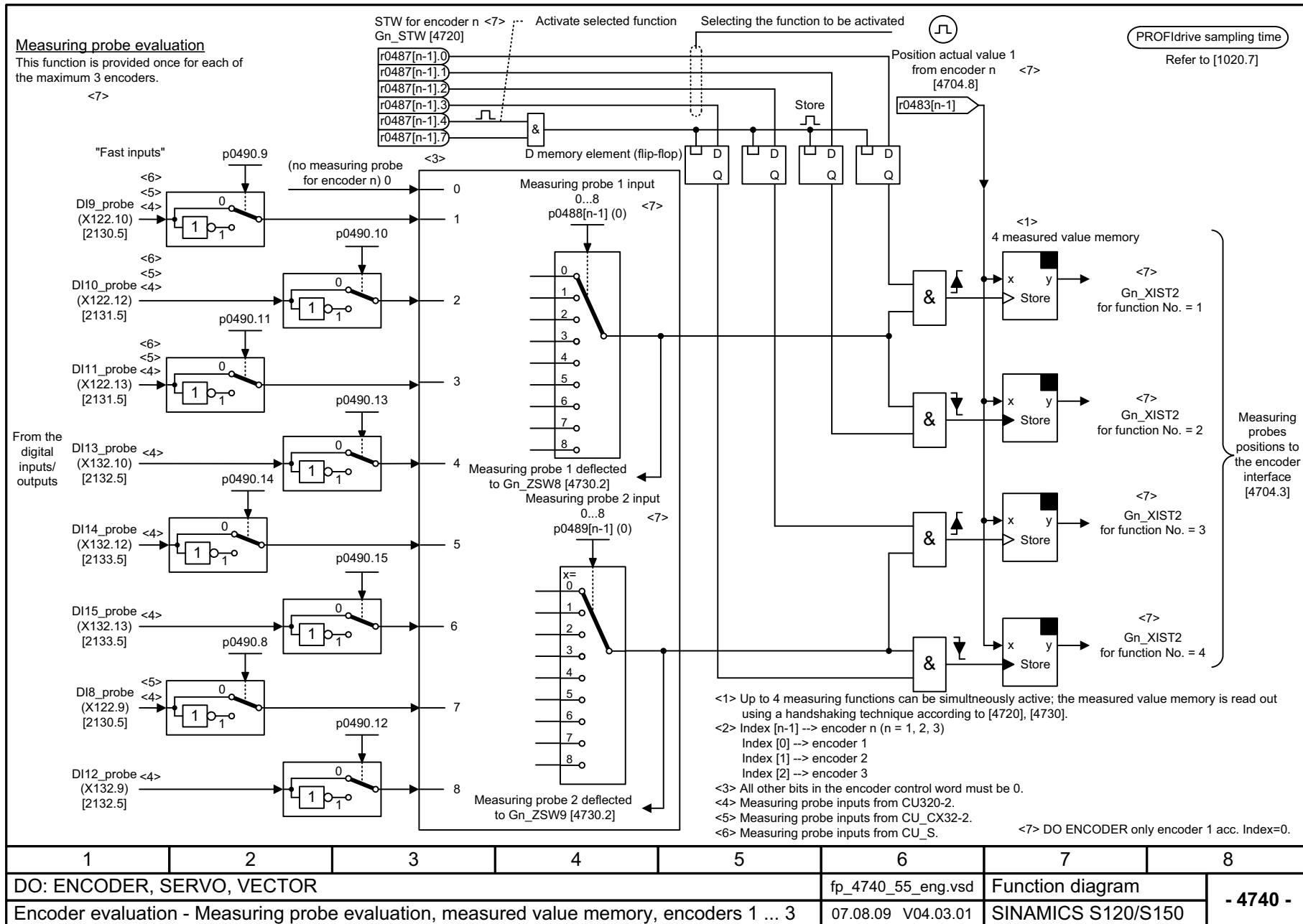


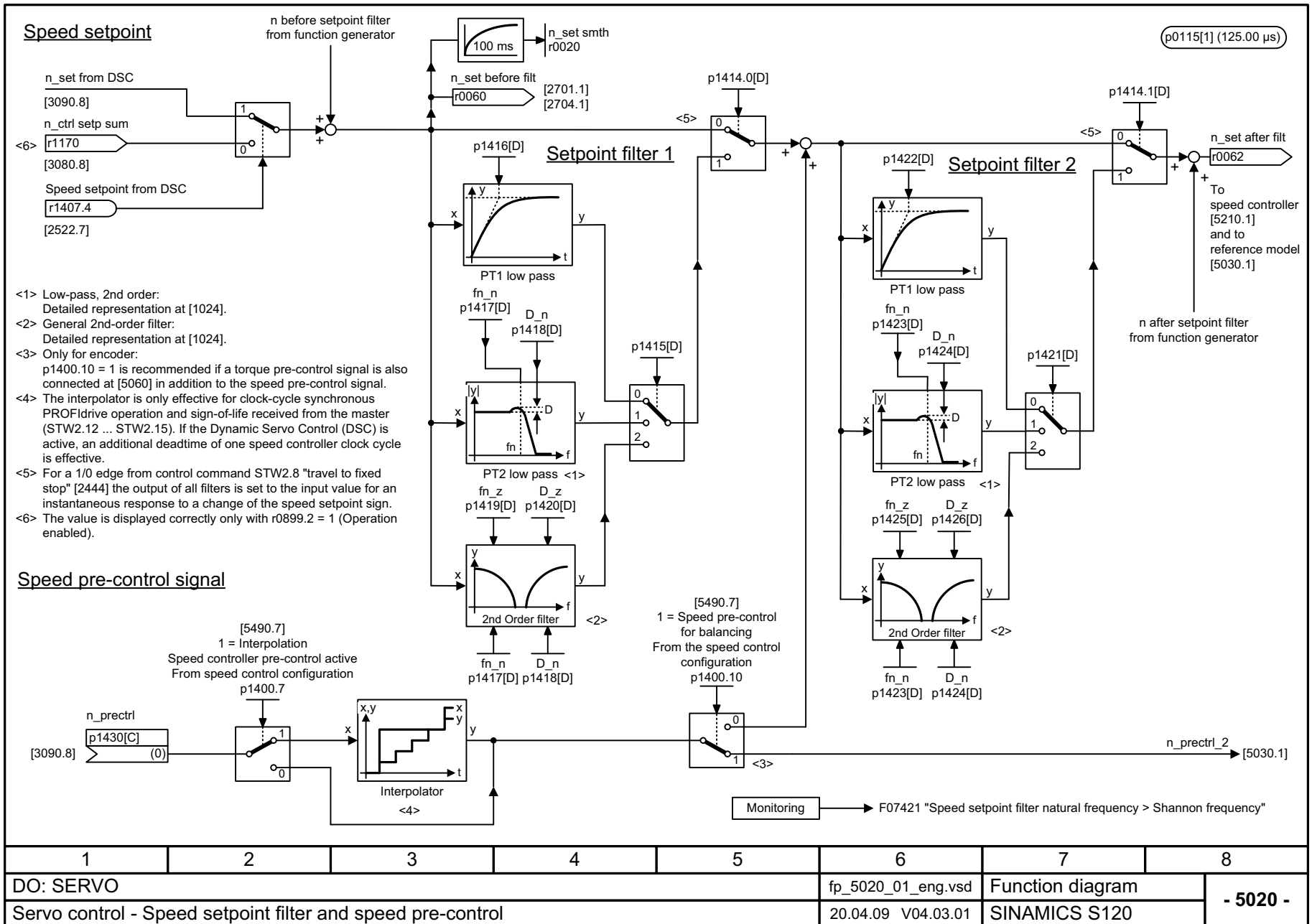
Fig. 2-162 4740 – Measuring probe evaluation, measured value memory, encoder 1 to 3

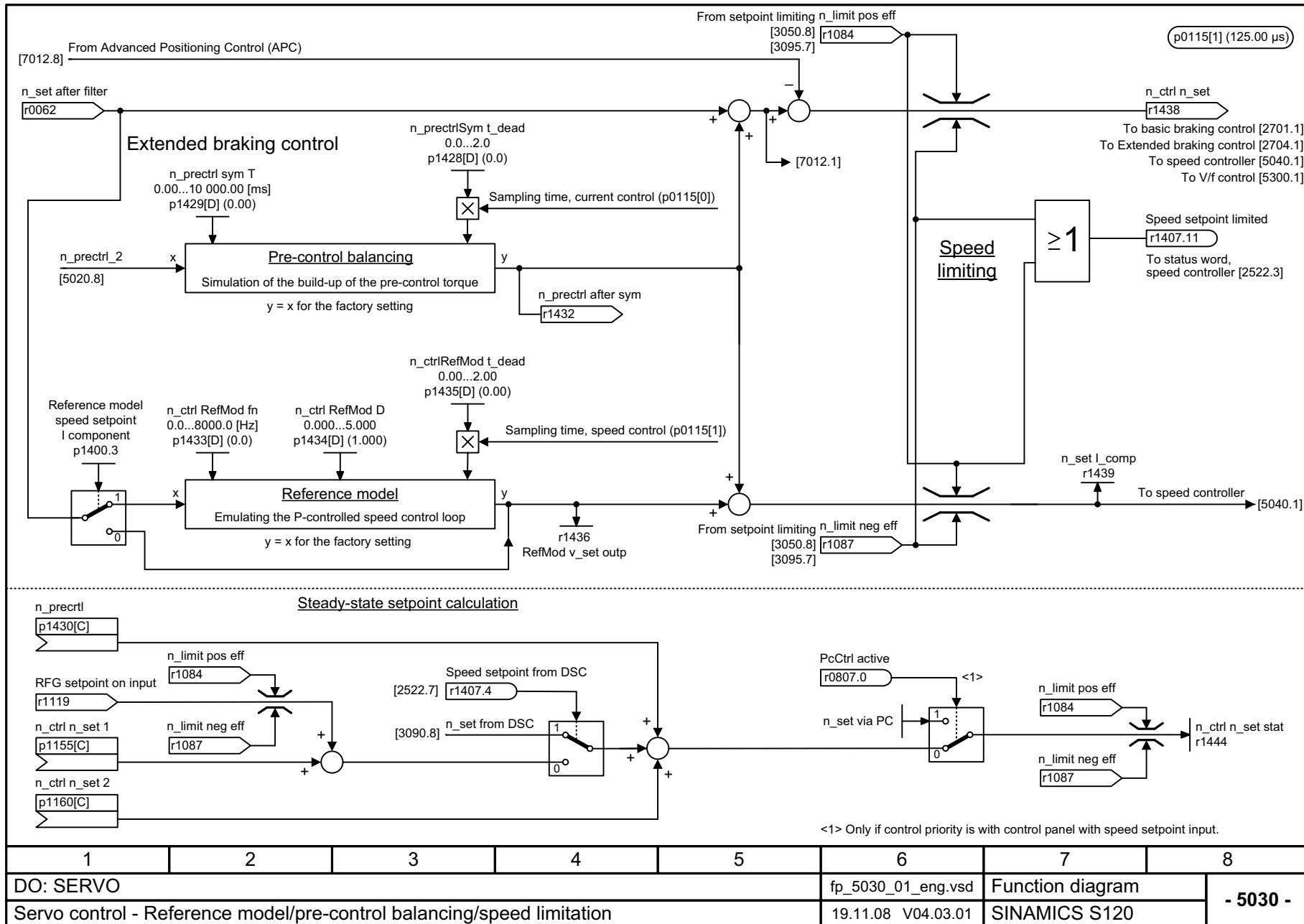
2.17 Servo control

Function diagrams

5020 – Speed setpoint filter and speed pre-control	2-1563
5030 – Reference model/pre-control balancing/speed limitation	2-1564
5040 – Speed controller with encoder	2-1565
5042 – Speed controller, torque-speed pre-control with encoder (p1402.4 = 1)	2-1566
5050 – Kp_n-/Tn_n adaptation	2-1567
5060 – Torque setpoint, control type switchover	2-1568
5210 – Speed controller without encoder	2-1569
5300 – V/f control for diagnostics	2-1570
5301 – Variable signaling function	2-1571
5490 – Speed control configuration	2-1572
5610 – Torque limitation/reduction/interpolator	2-1573
5620 – Motor/generator torque limit	2-1574
5630 – Upper/lower torque limit	2-1575
5640 – Mode switchover, power/current limitation	2-1576
5650 – Vdc_max controller and Vdc_min controller	2-1577
5710 – Current setpoint filter	2-1578
5714 – Iq and Id controller	2-1579
5722 – Field current / flux specification, flux reduction, flux controller	2-1580
5730 – Interface to the Motor Module (gating signals, actual current values)	2-1581

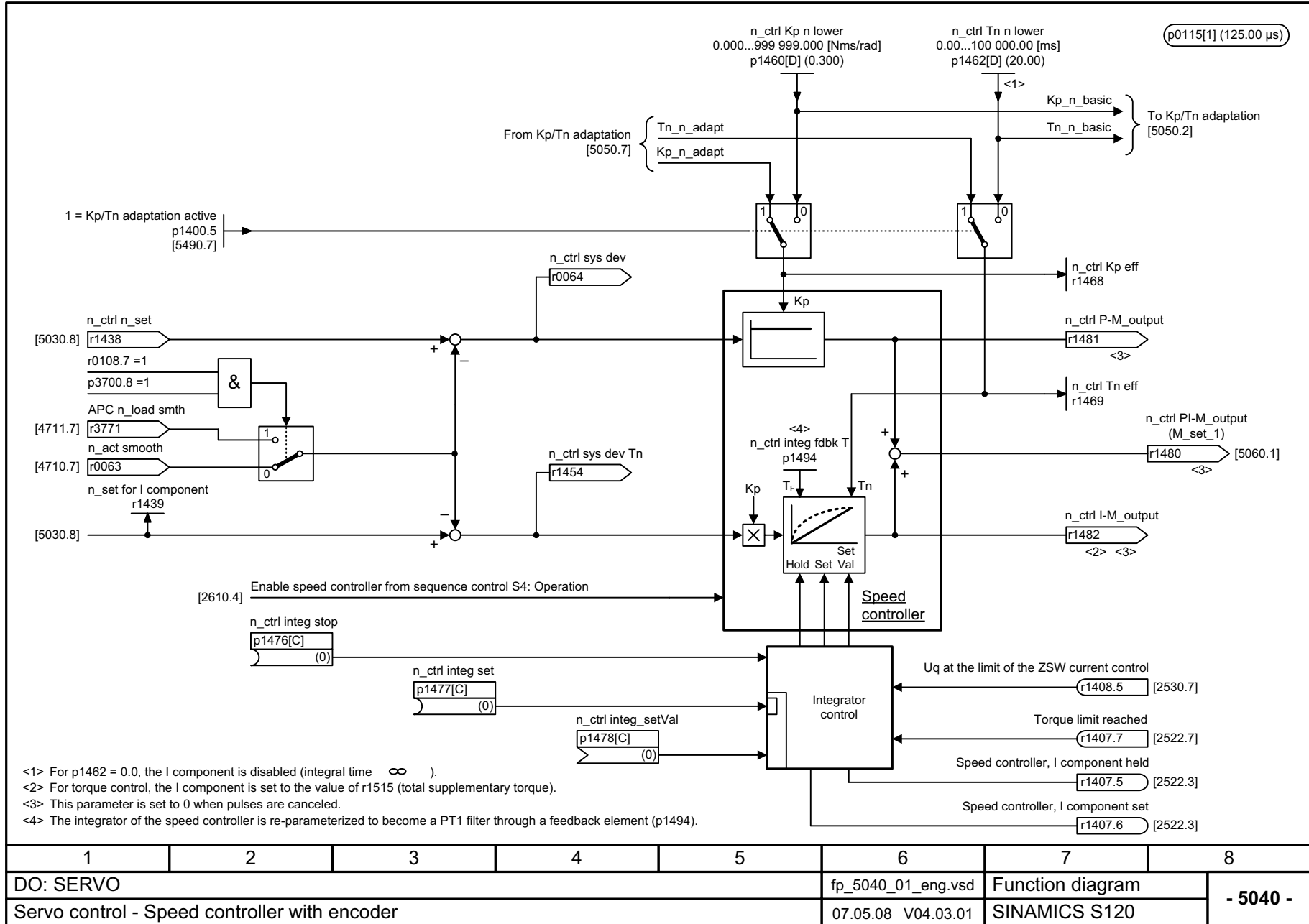
Fig. 2-163 5020 – Speed setpoint filter and speed pre-control

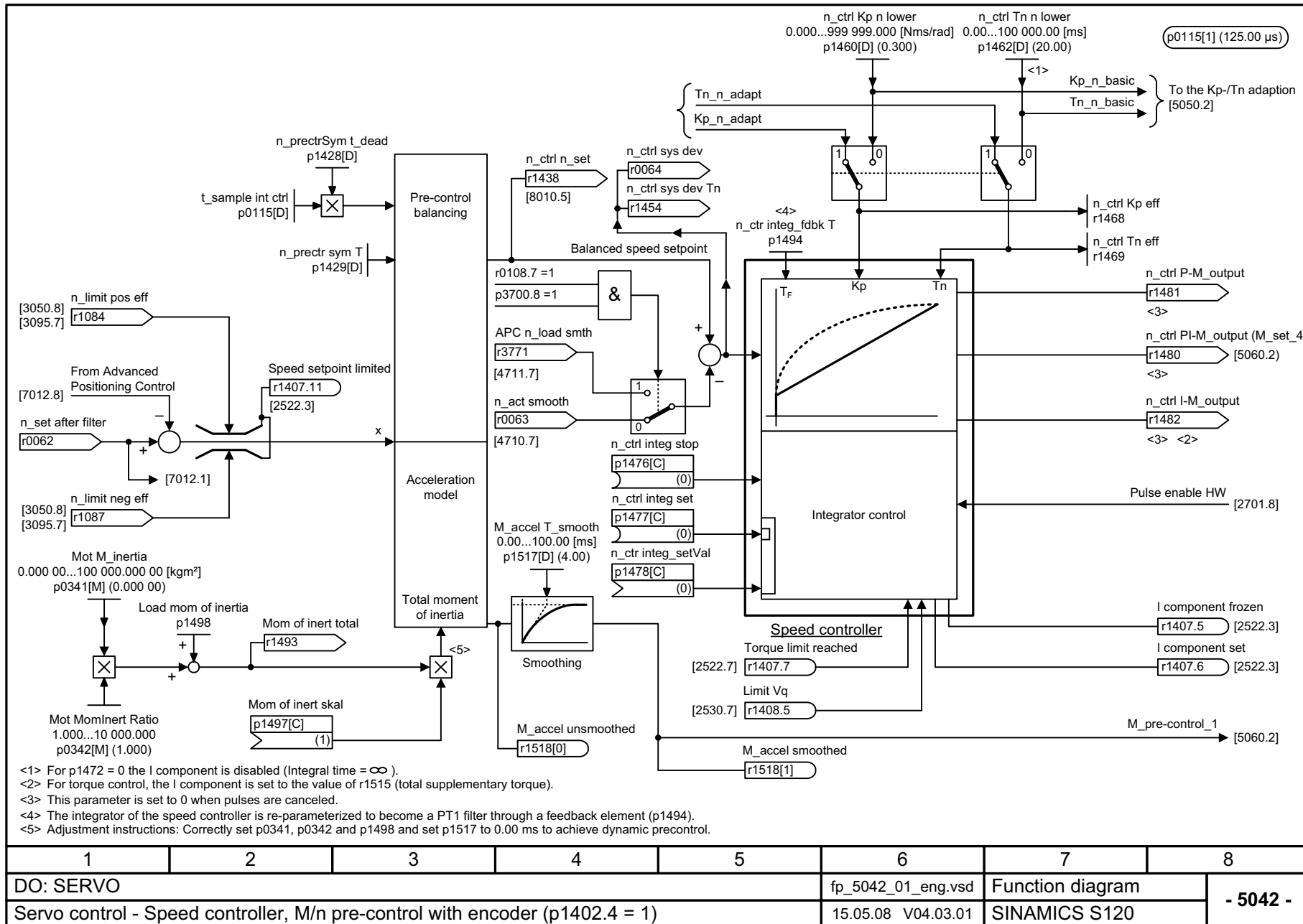




1	2	3	4	5	6	7	8
DO: SERVO					fp_5030_01_eng.vsd	Function diagram	
Servo control - Reference model/pre-control balancing/speed limitation					19.11.08 V04.03.01	SINAMICS S120	
					- 5030 -		

Fig. 2-164 5030 – Reference model/pre-control balancing/speed limitation





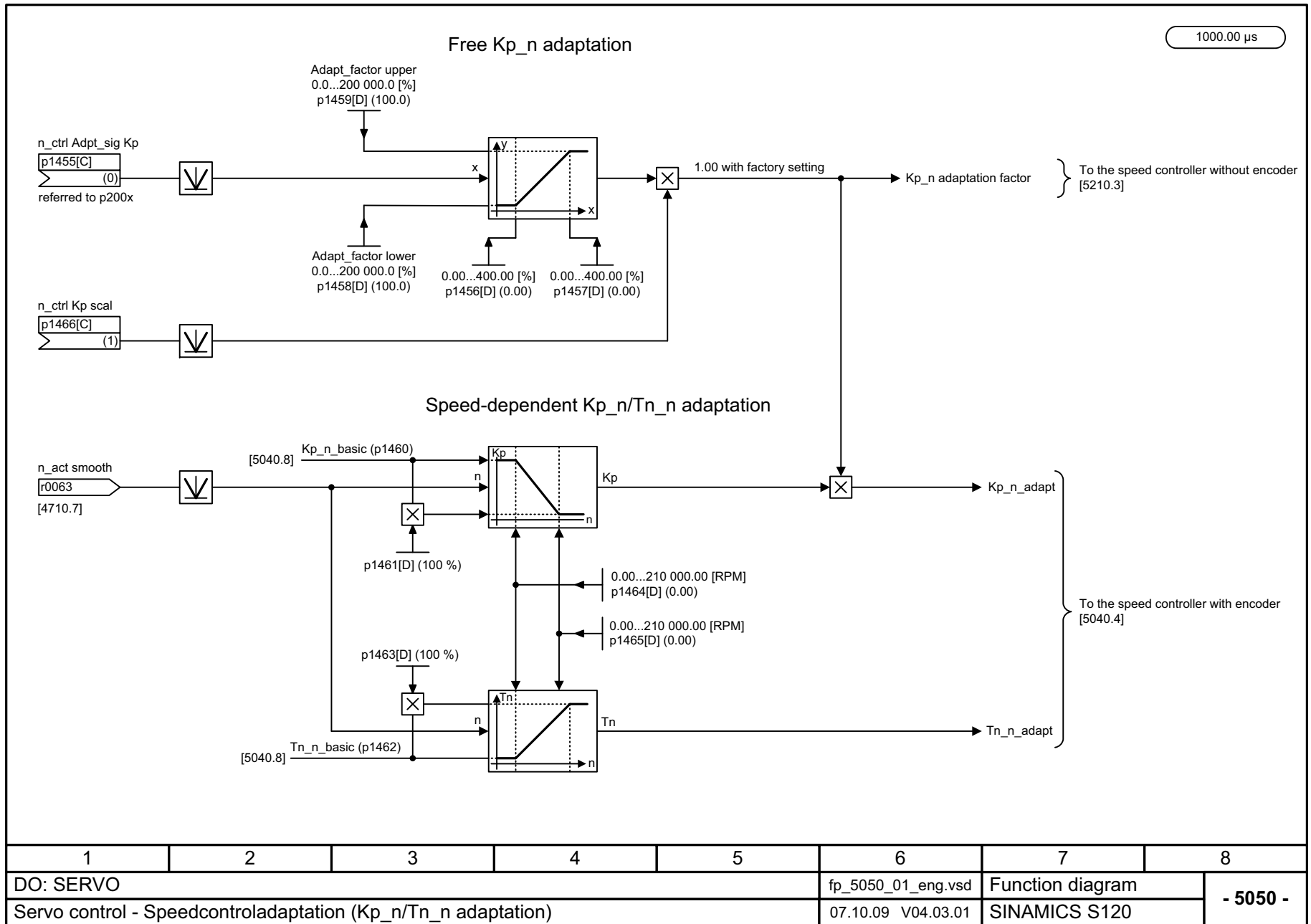
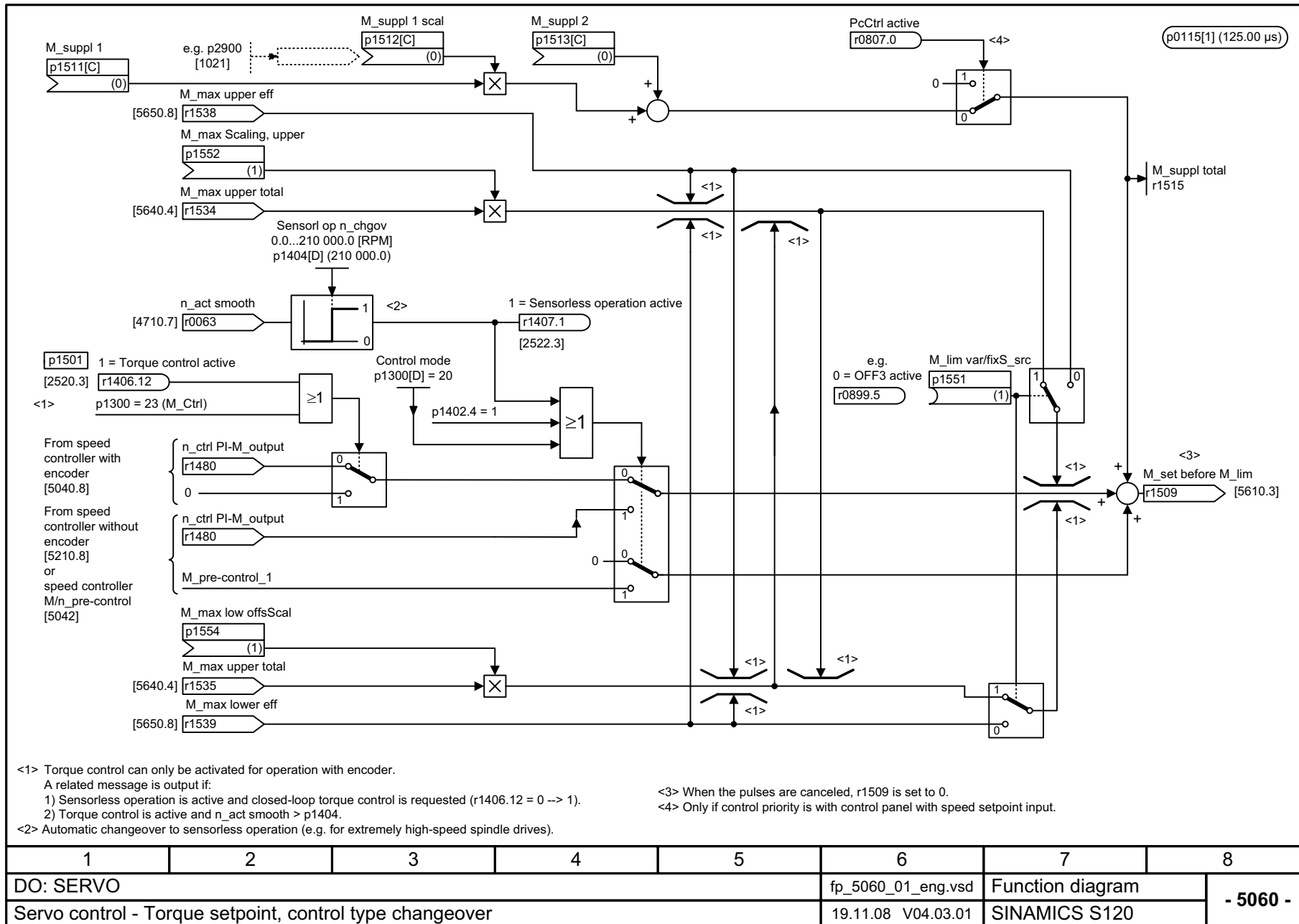
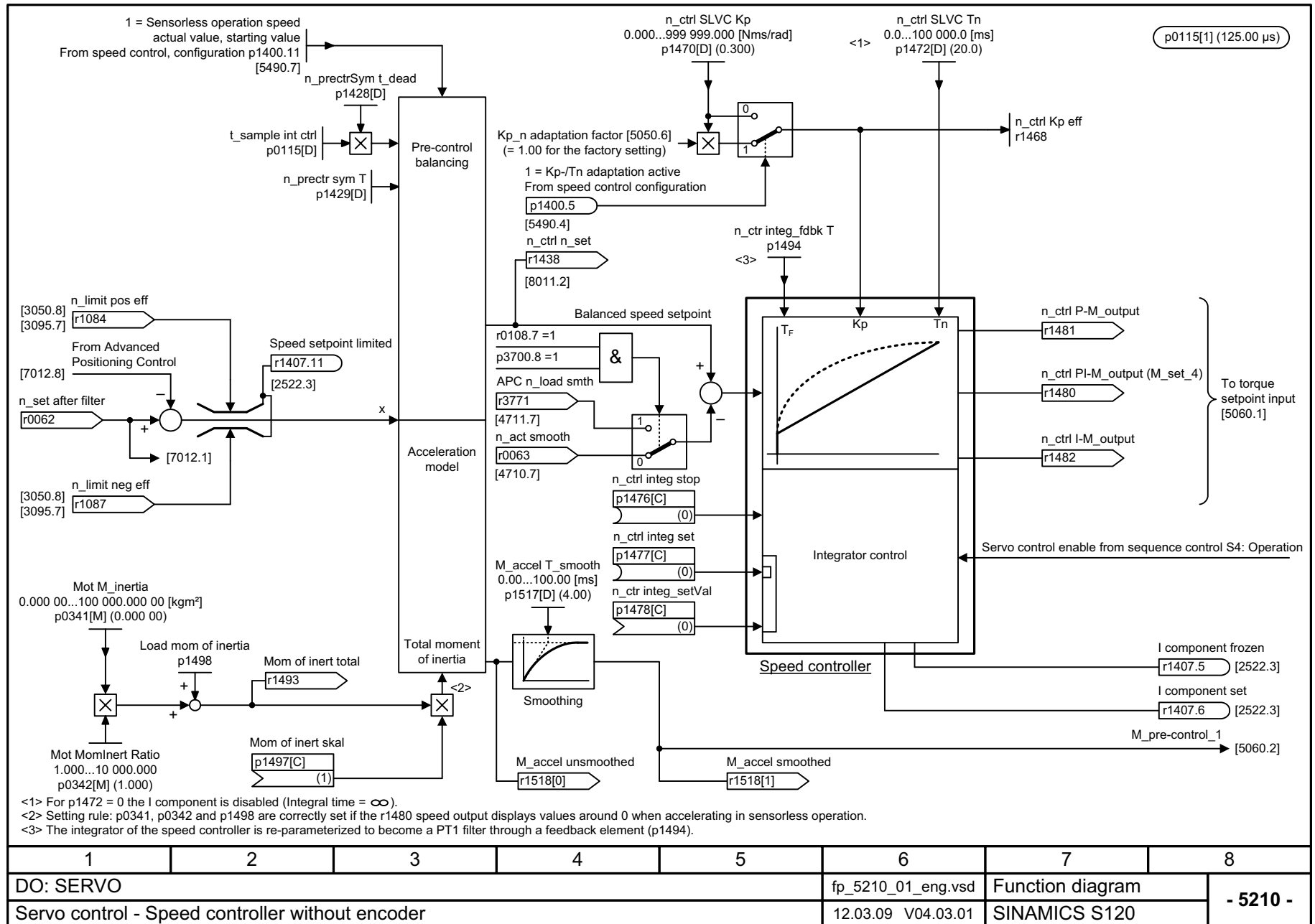


Fig. 2-167 5050 – Kp_n/Tn_n adaptation





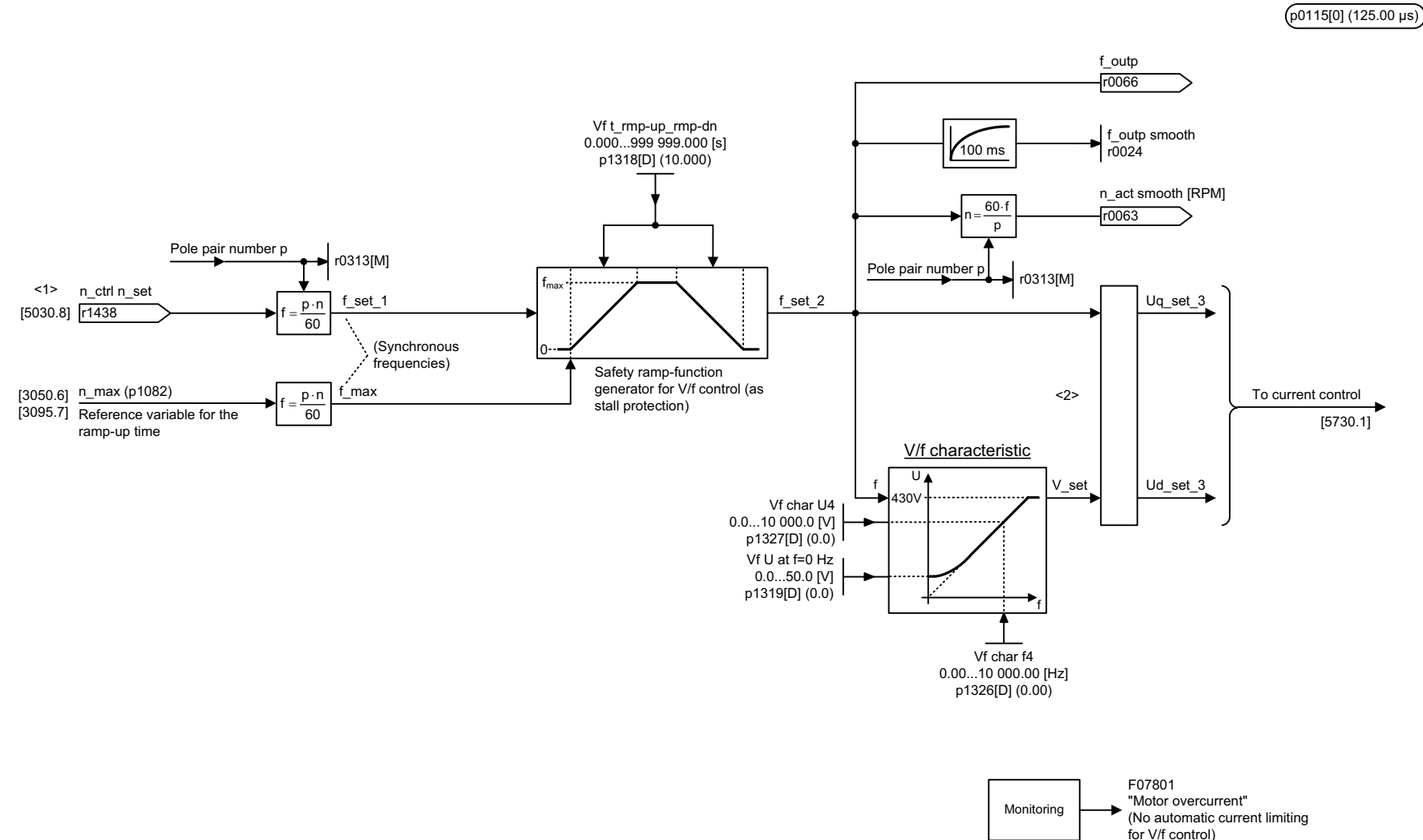
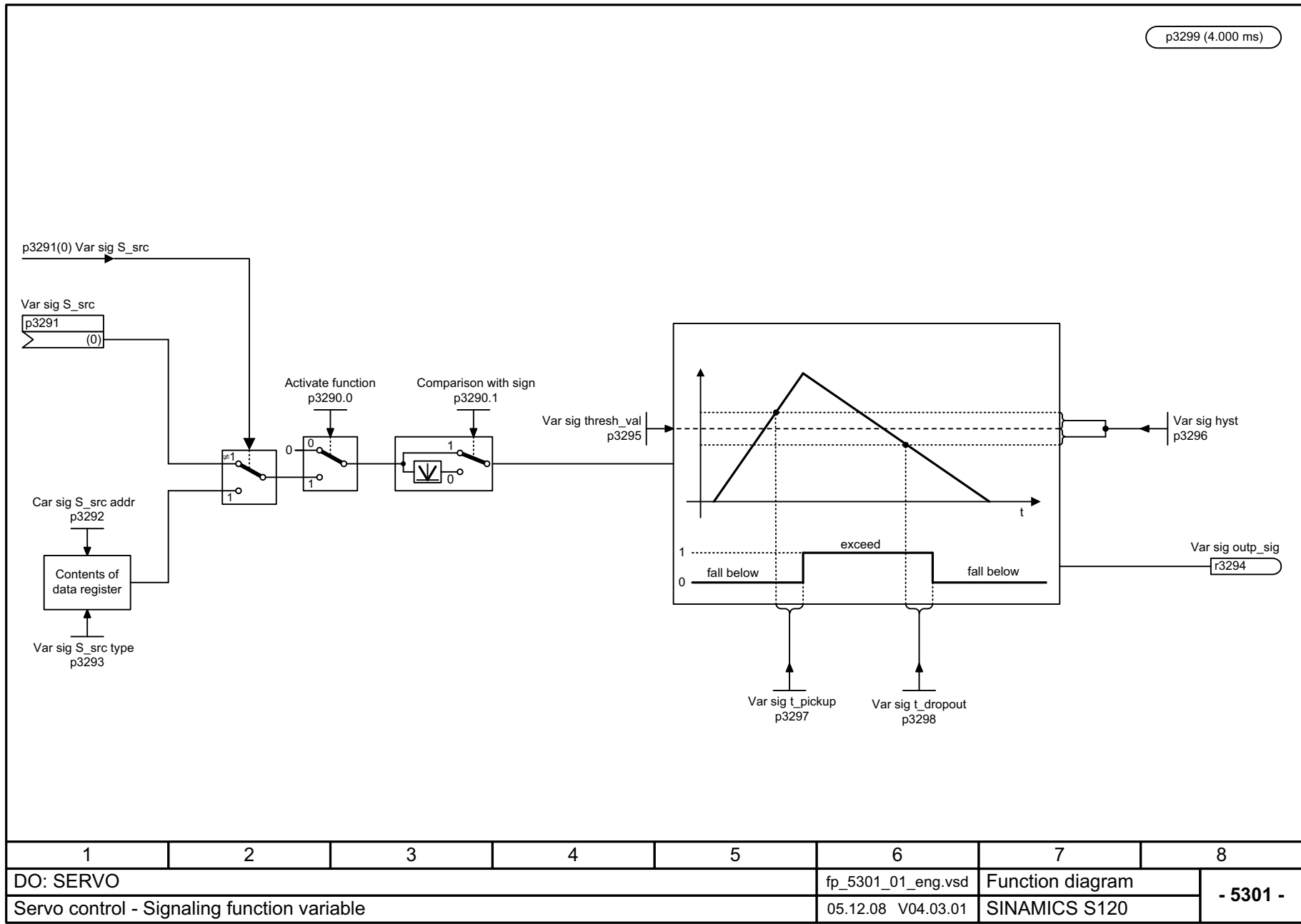


Fig. 2-170



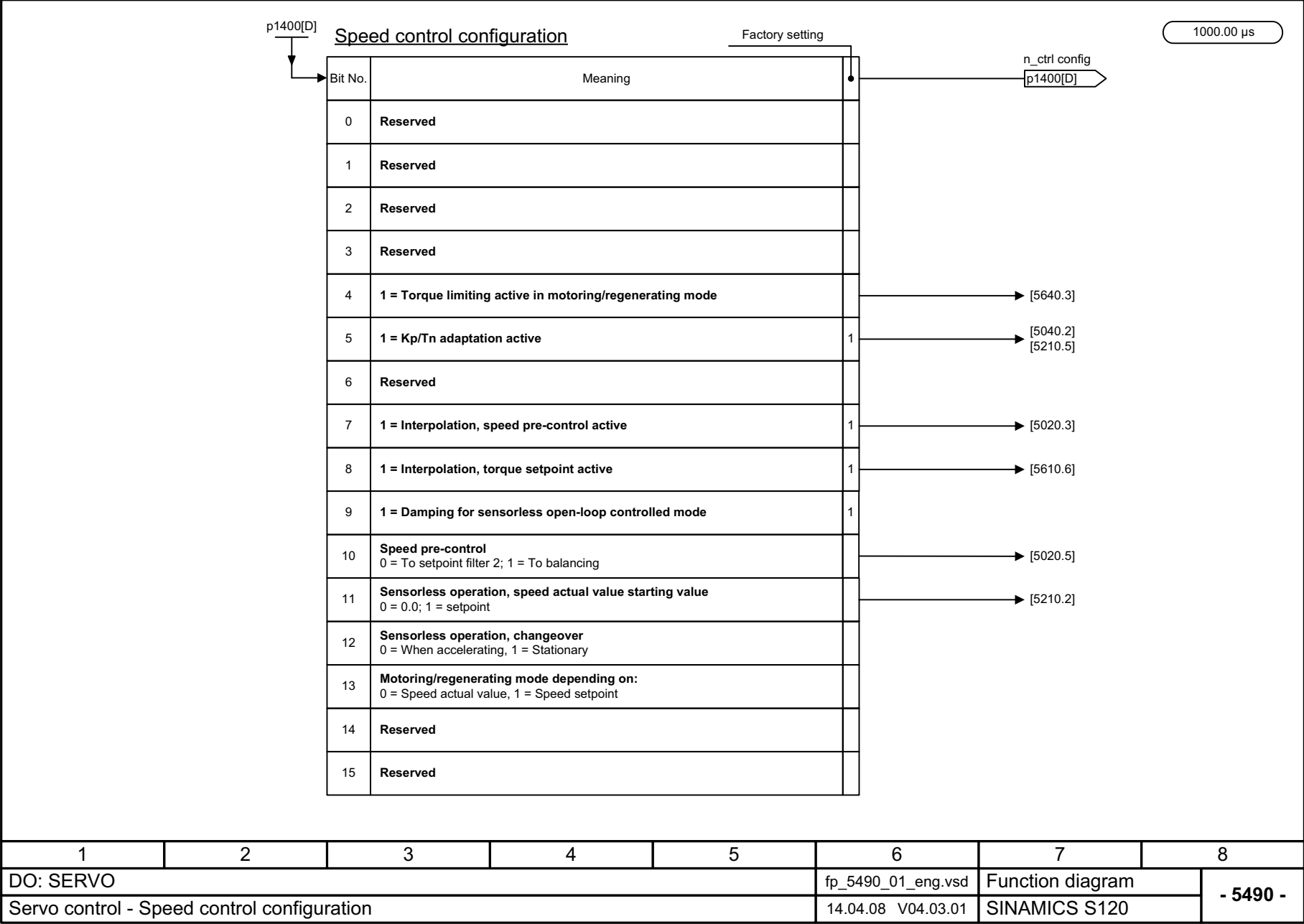
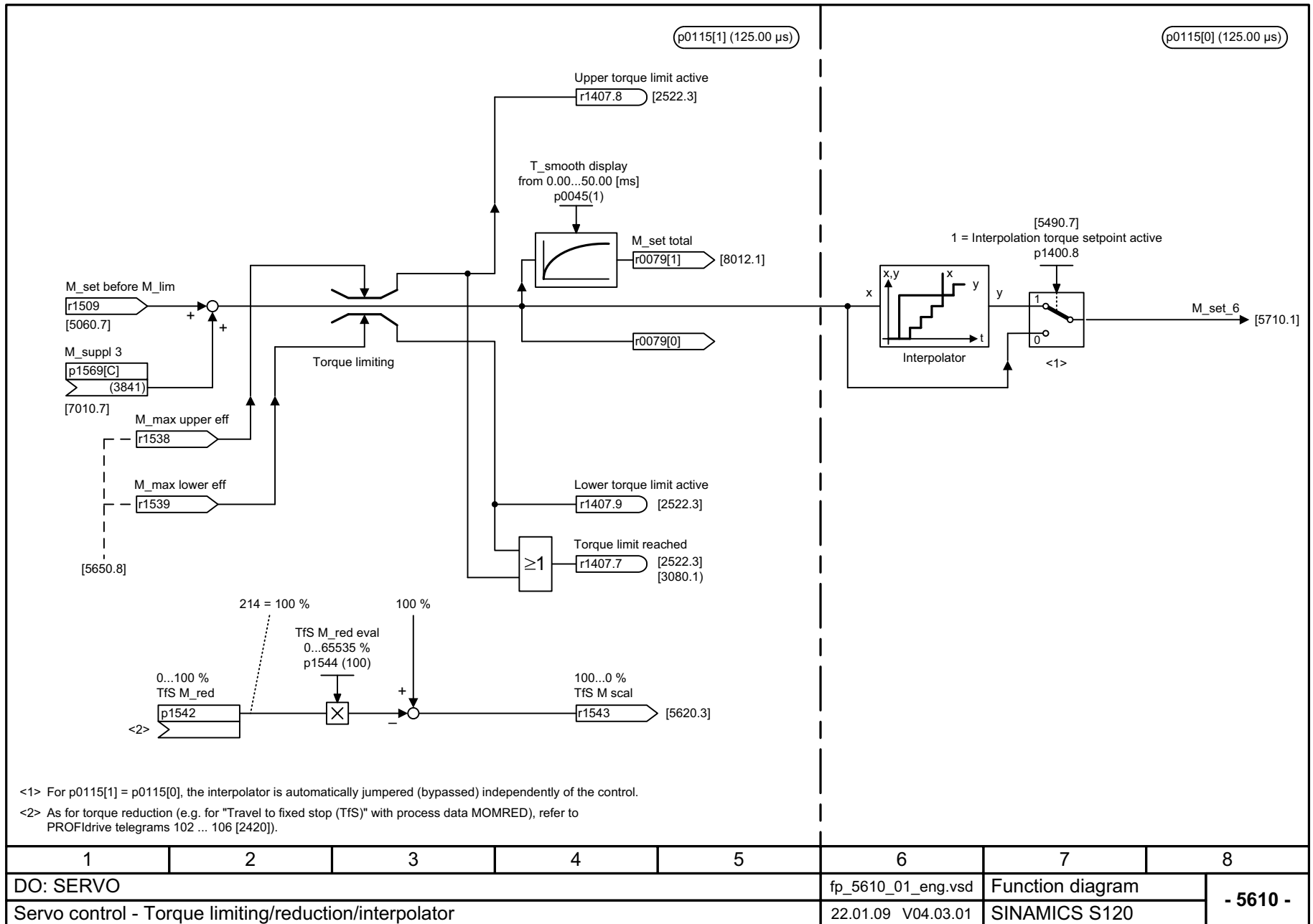
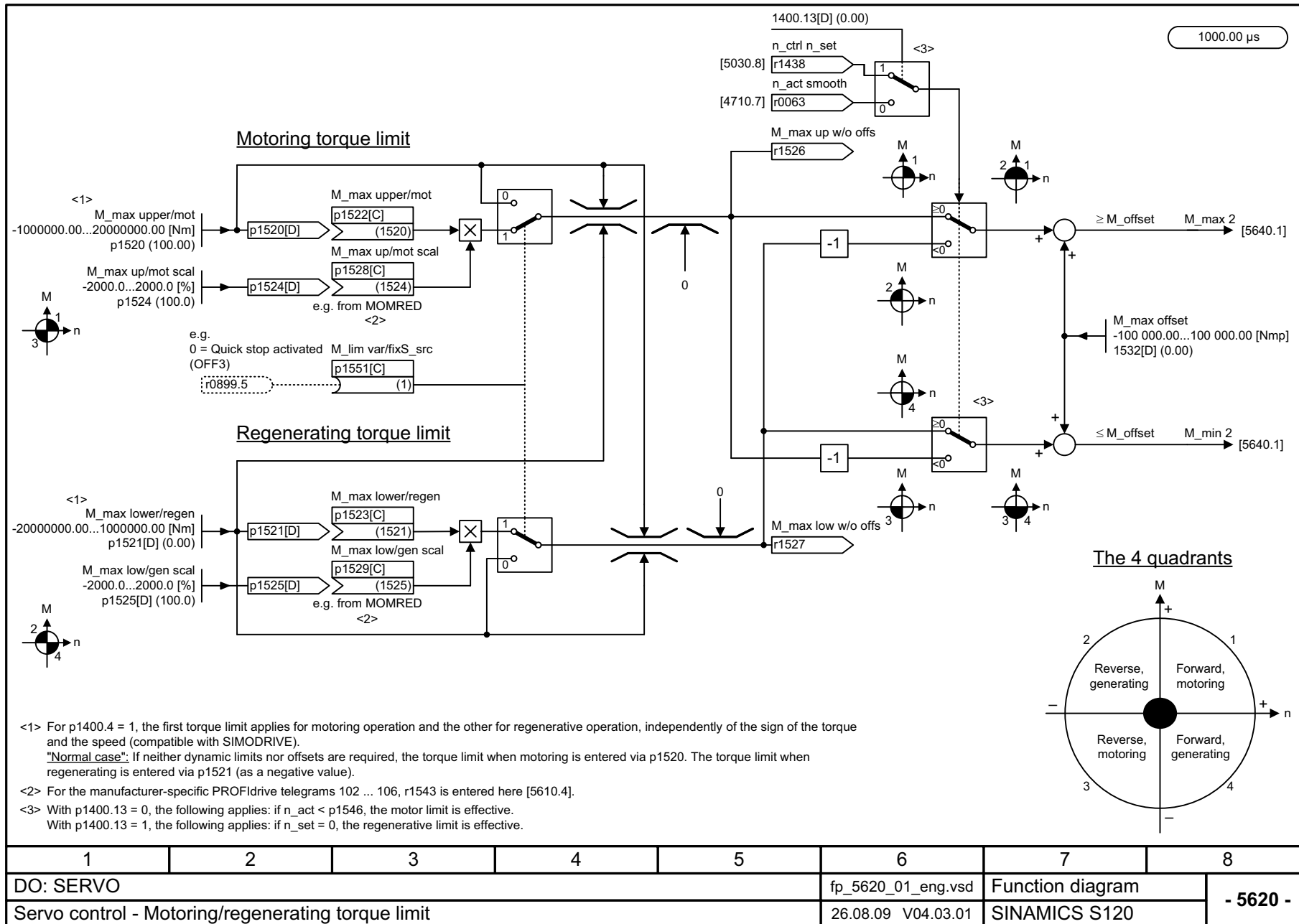


Fig. 2-172 5490 – Speed control configuration





1	2	3	4	5	6	7	8
DO: SERVO					fp_5620_01_eng.vsd	Function diagram	
Servo control - Motoring/regenerating torque limit					26.08.09 V04.03.01	SINAMICS S120	
							- 5620 -

Fig. 2-174 5620 – Motor/generator torque limit

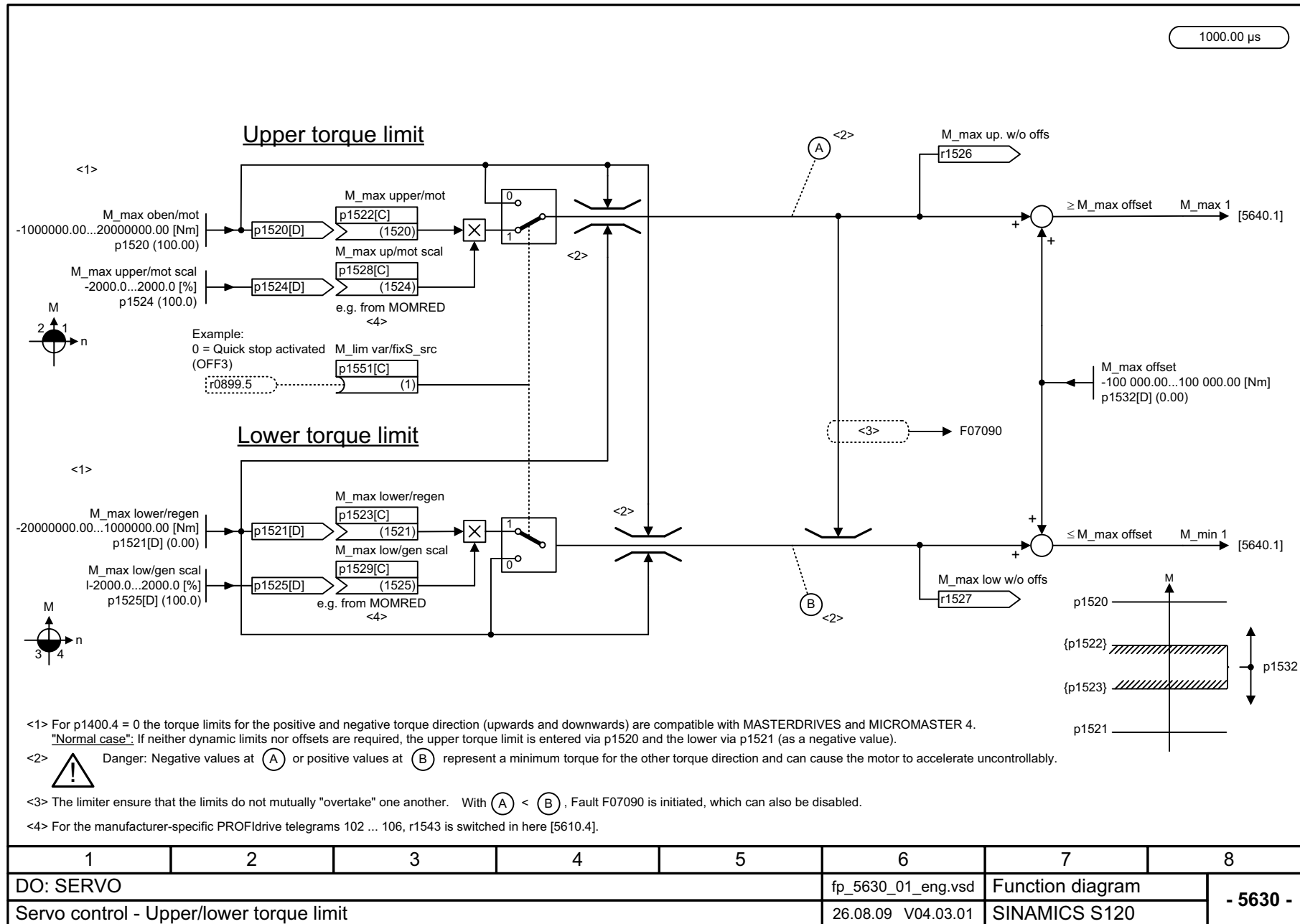


Fig. 2-175 5630 – Upper/lower torque limit

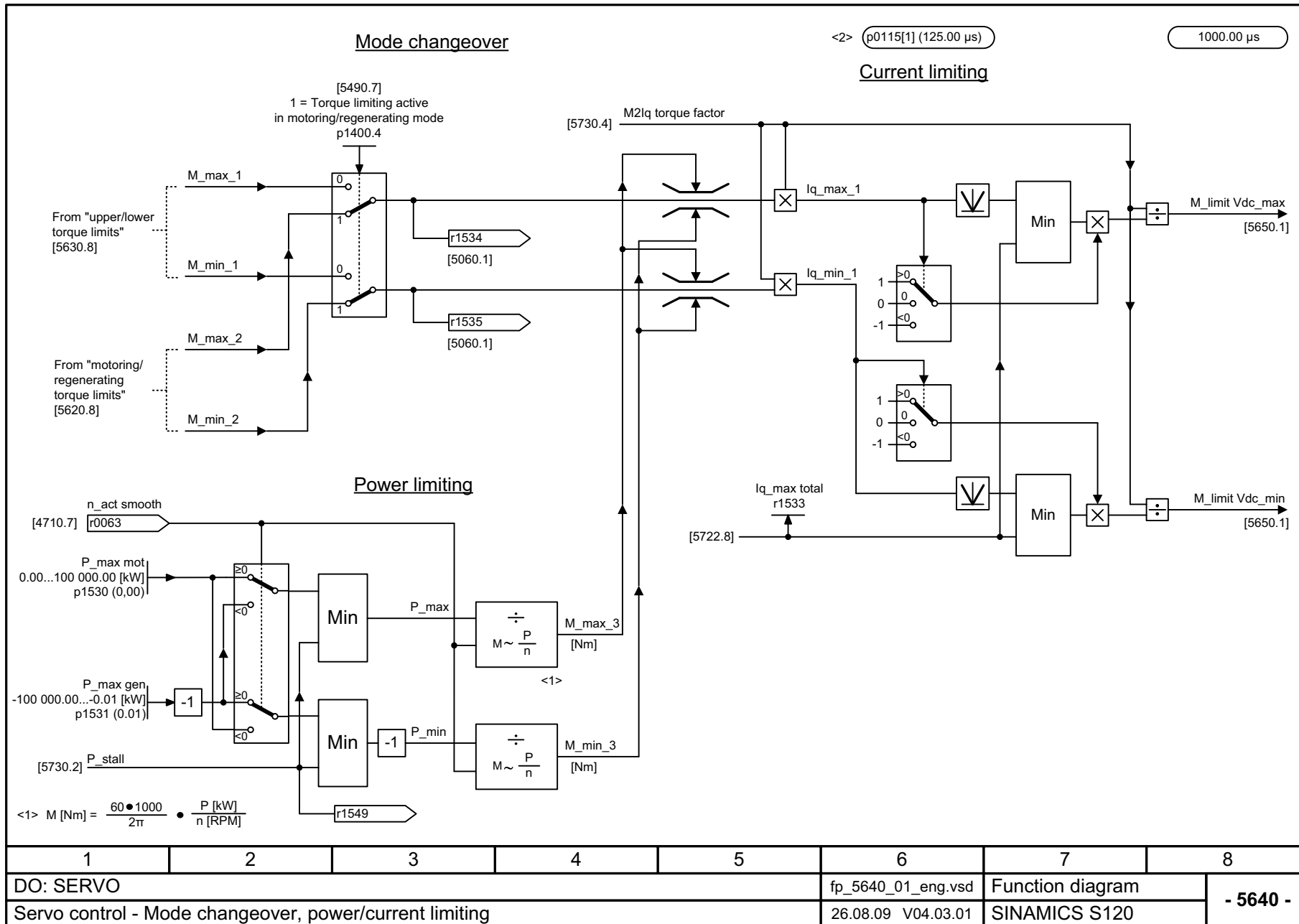
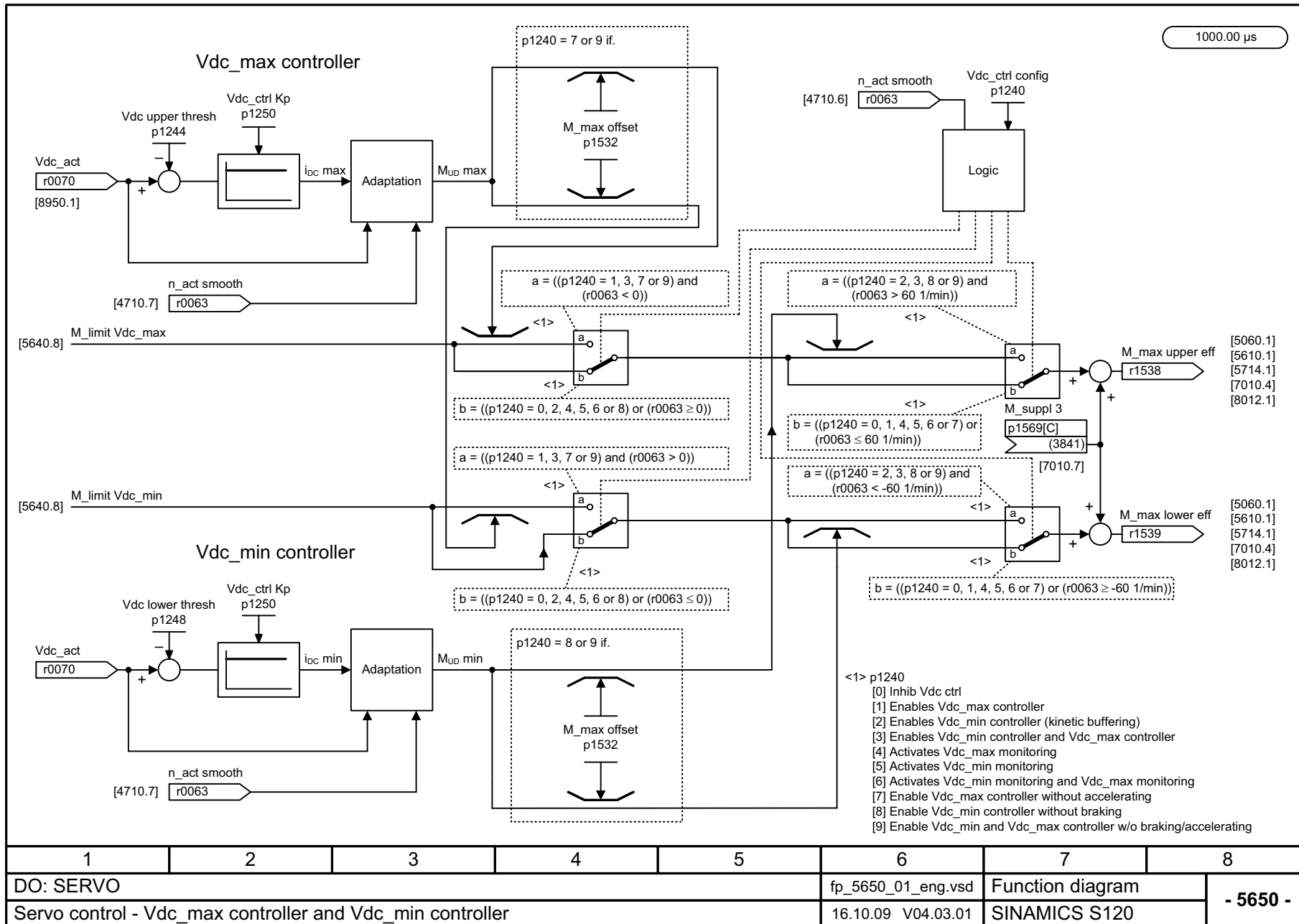
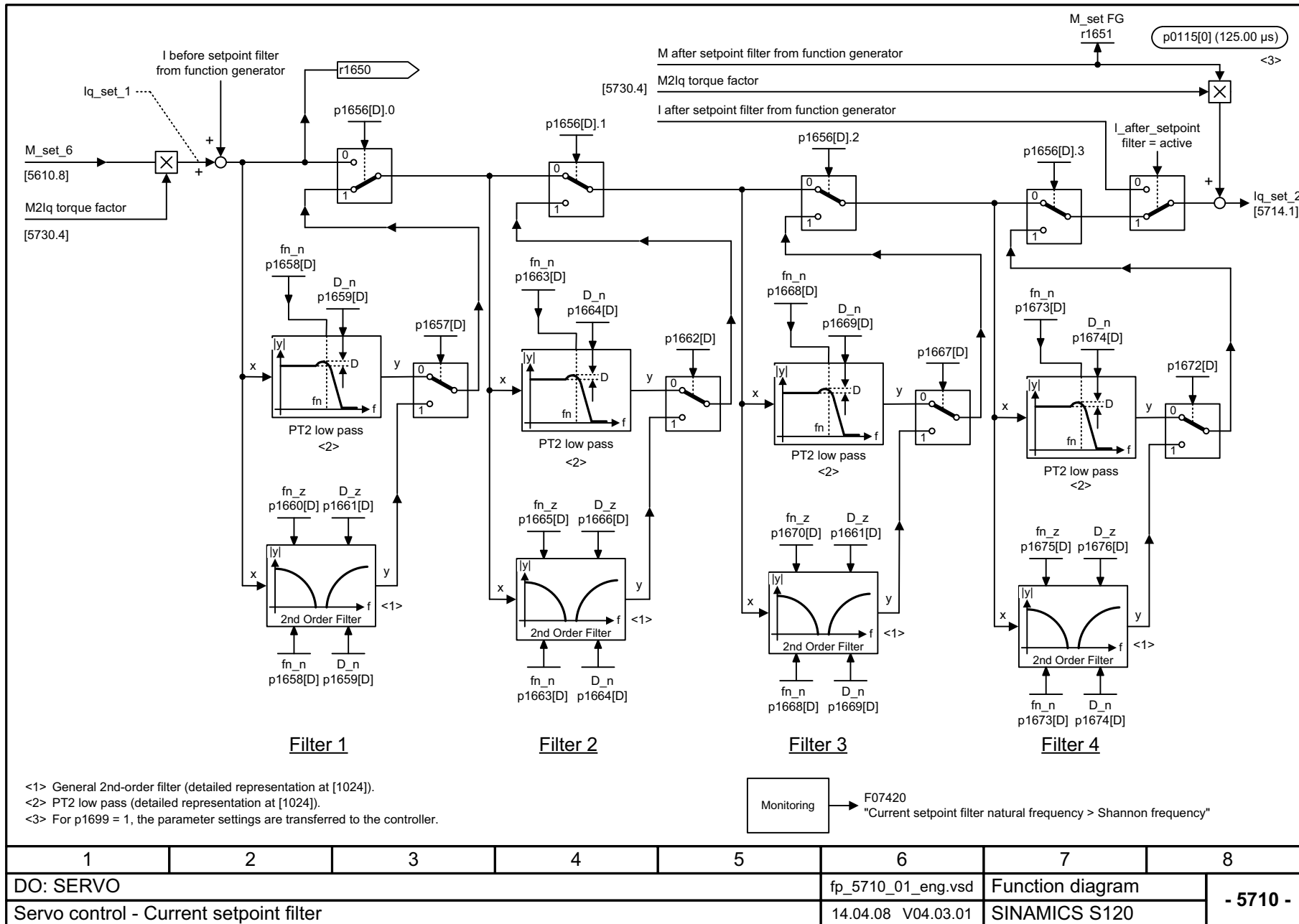


Fig. 2-176 5640 – Mode switchover, power/current limitation





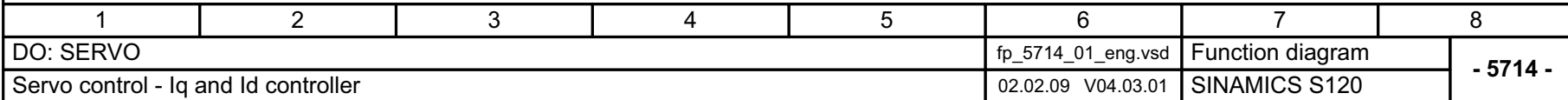
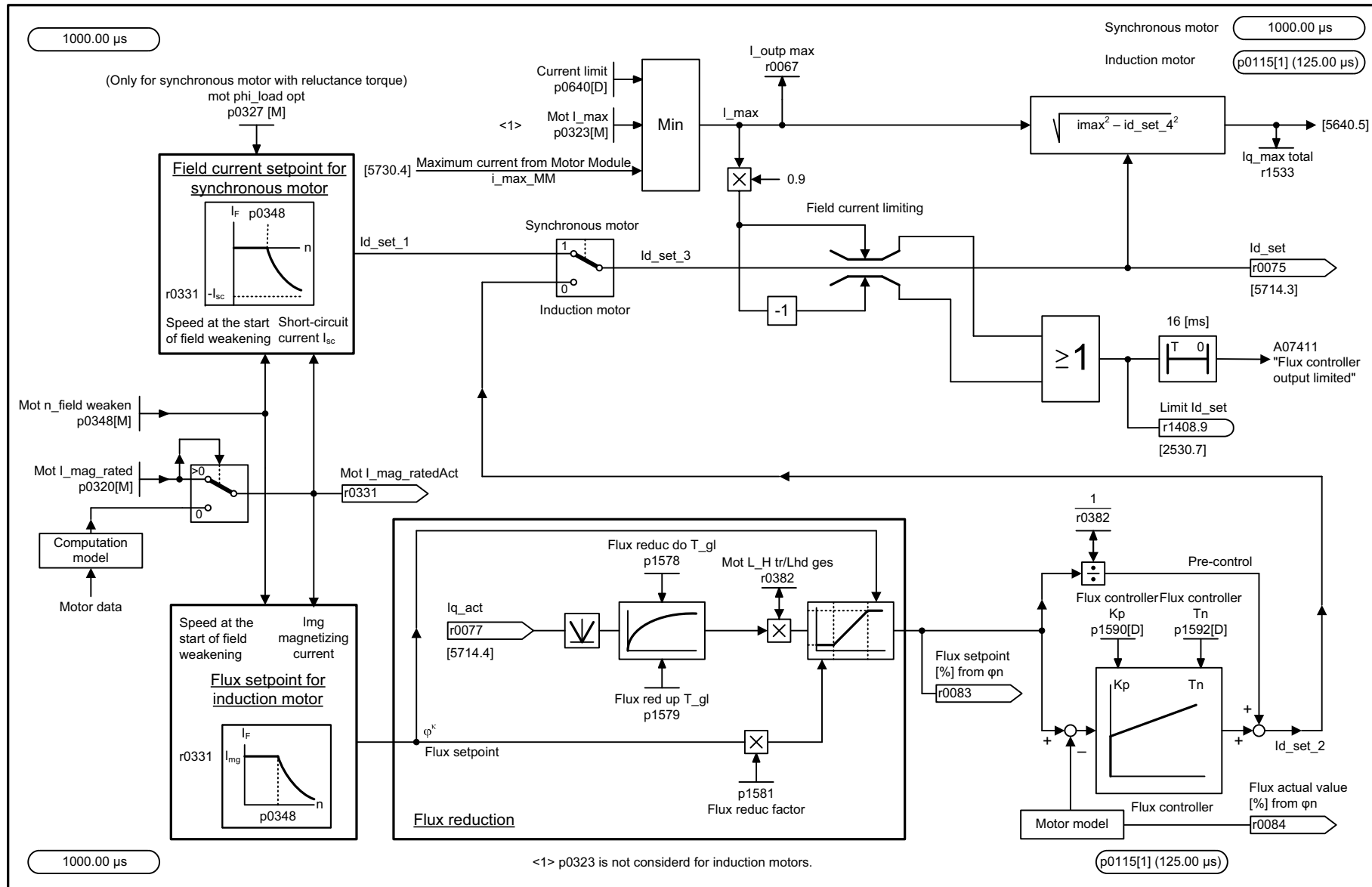


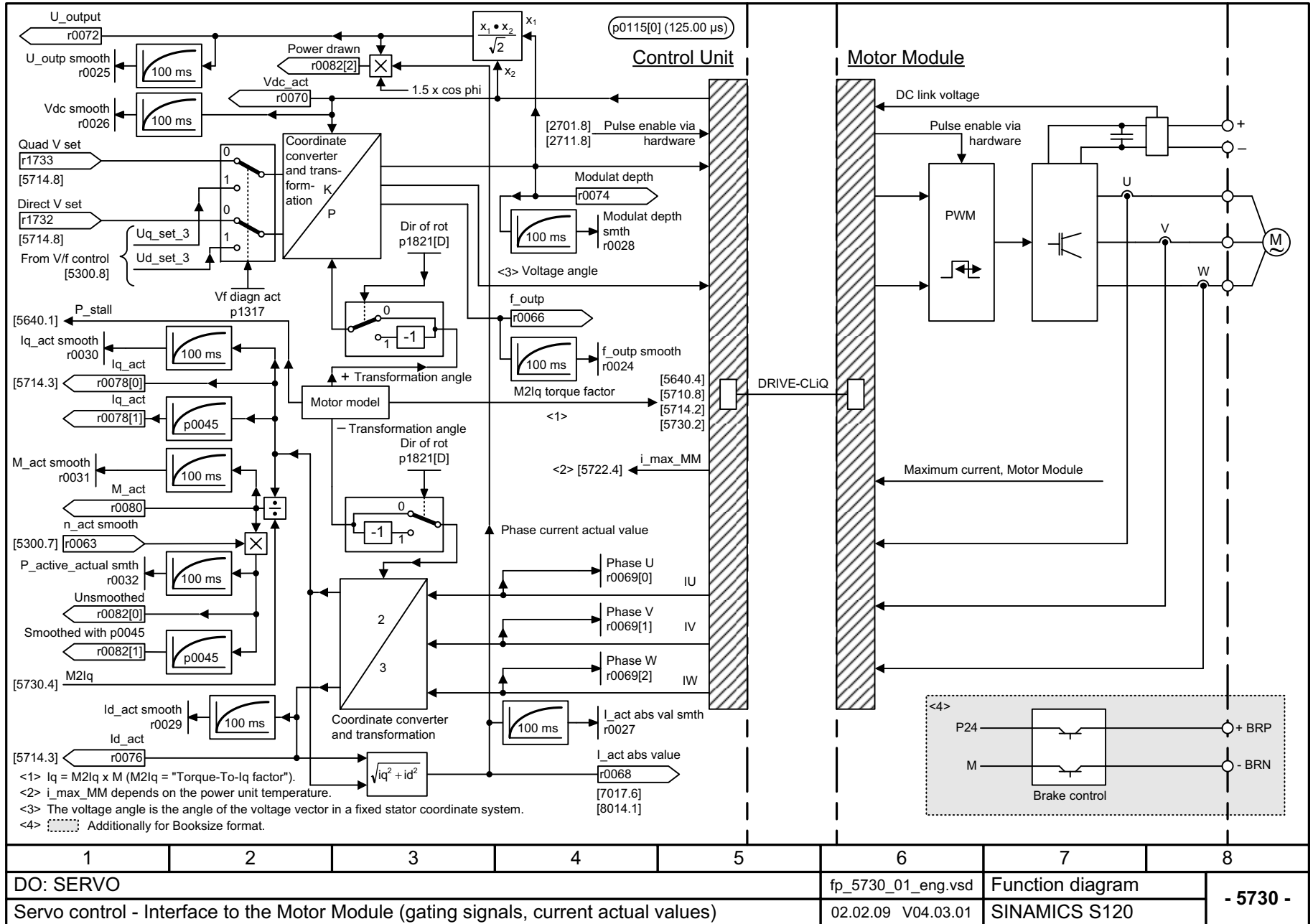
Fig. 2-179



1	2	3	4	5	6	7	8
DO: SERVO					fp_5722_01_eng.vsd	Function diagram	
Servo control - Field current-/flux input, flux reduction, flux controller					26.08.09 V04.03.01	SINAMICS S120	
							- 5722 -

Fig. 2-180 5722 – Field current / flux specification, flux reduction, flux controller

Fig. 2-181 5730 – Interface to the Motor Module (gating signals, actual current values)

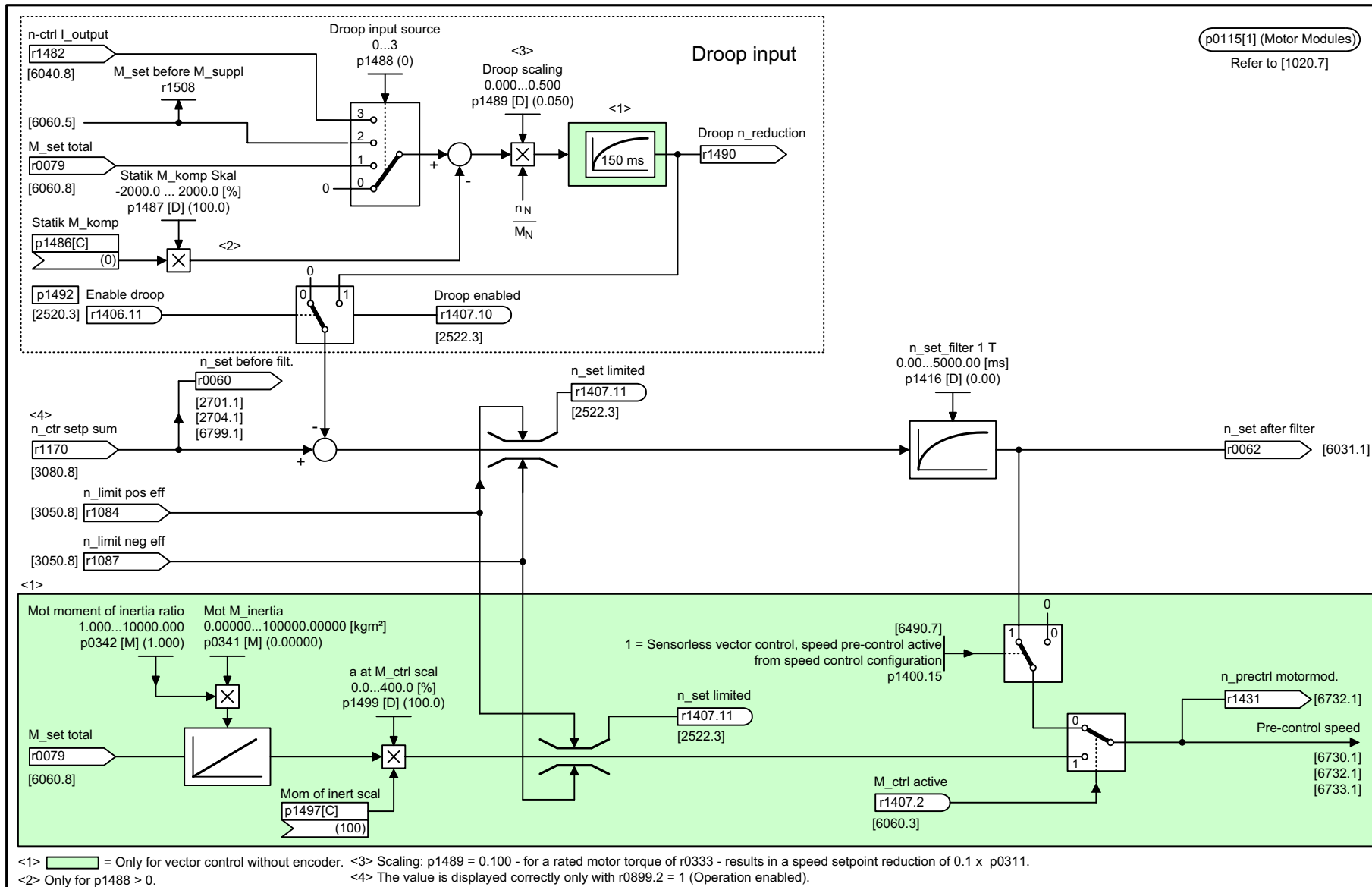


2.18 Vector control

Function diagrams

6030 – Speed setpoint, droop	2-1584
6031 – Pre-control balancing for reference/acceleration model	2-1585
6040 – Speed controller with/without encoder	2-1586
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6731 – Interface to the Motor Module (PEM, p0300 = 2)	2-1608
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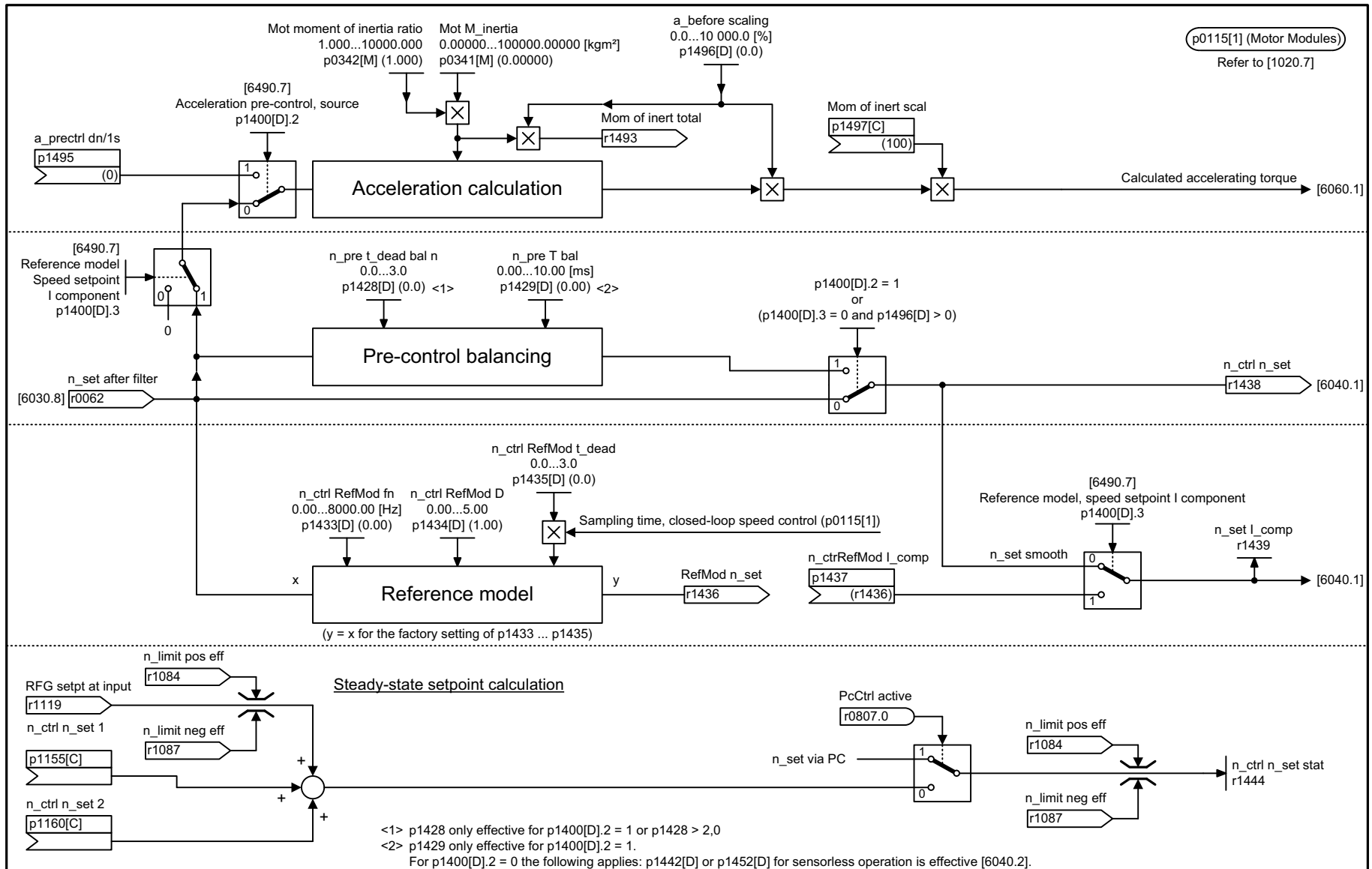
6733 – Motor model selection (FEM, p0300 = 5)	2-1610
6799 – Display signals	2-1611



1	2	3	4	5	6	7	8
DO: VECTOR					fp_6030_54_eng.vsd	Function diagram	
Vector control - Speed setpoint, droop					18.06.09 V04.03.01	S120/S150/G130/G150	

- 6030 -

Fig. 2-183 6031 – Pre-control balancing for reference/acceleration model



DO: VECTOR, VECTORGL, VECTORMV, VECTORSL

Vector control - Pre-control balancing, reference/acceleration model

fp_6031_51_eng.vsd

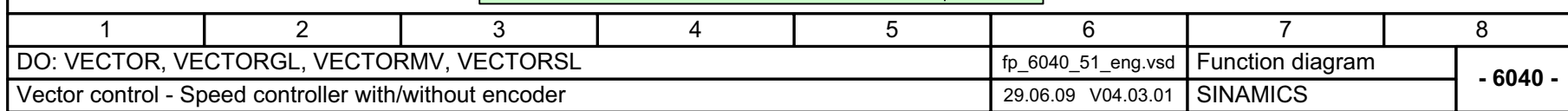
21.08.09 V04.03.01

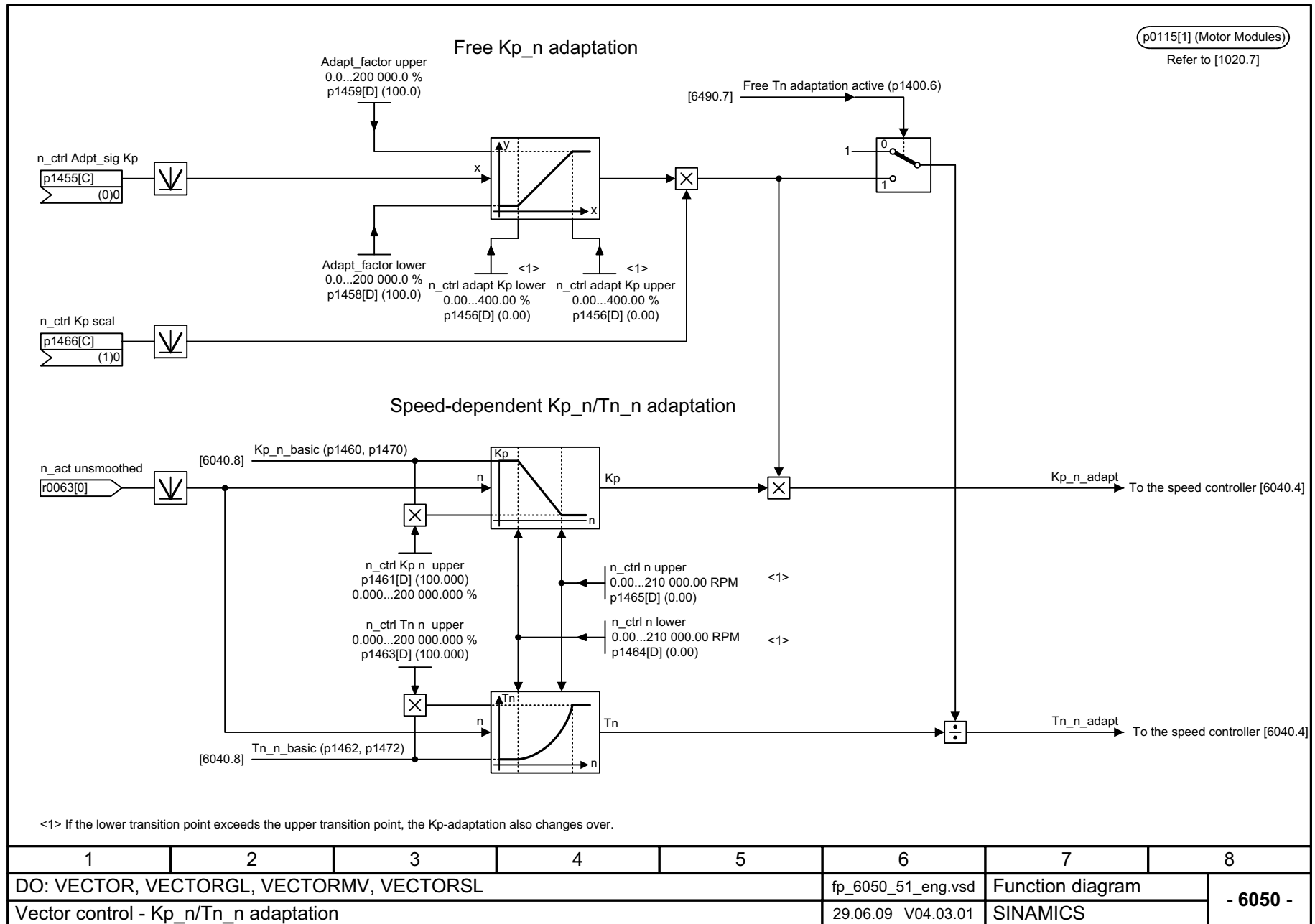
Function diagram

SINAMICS

- 6031 -

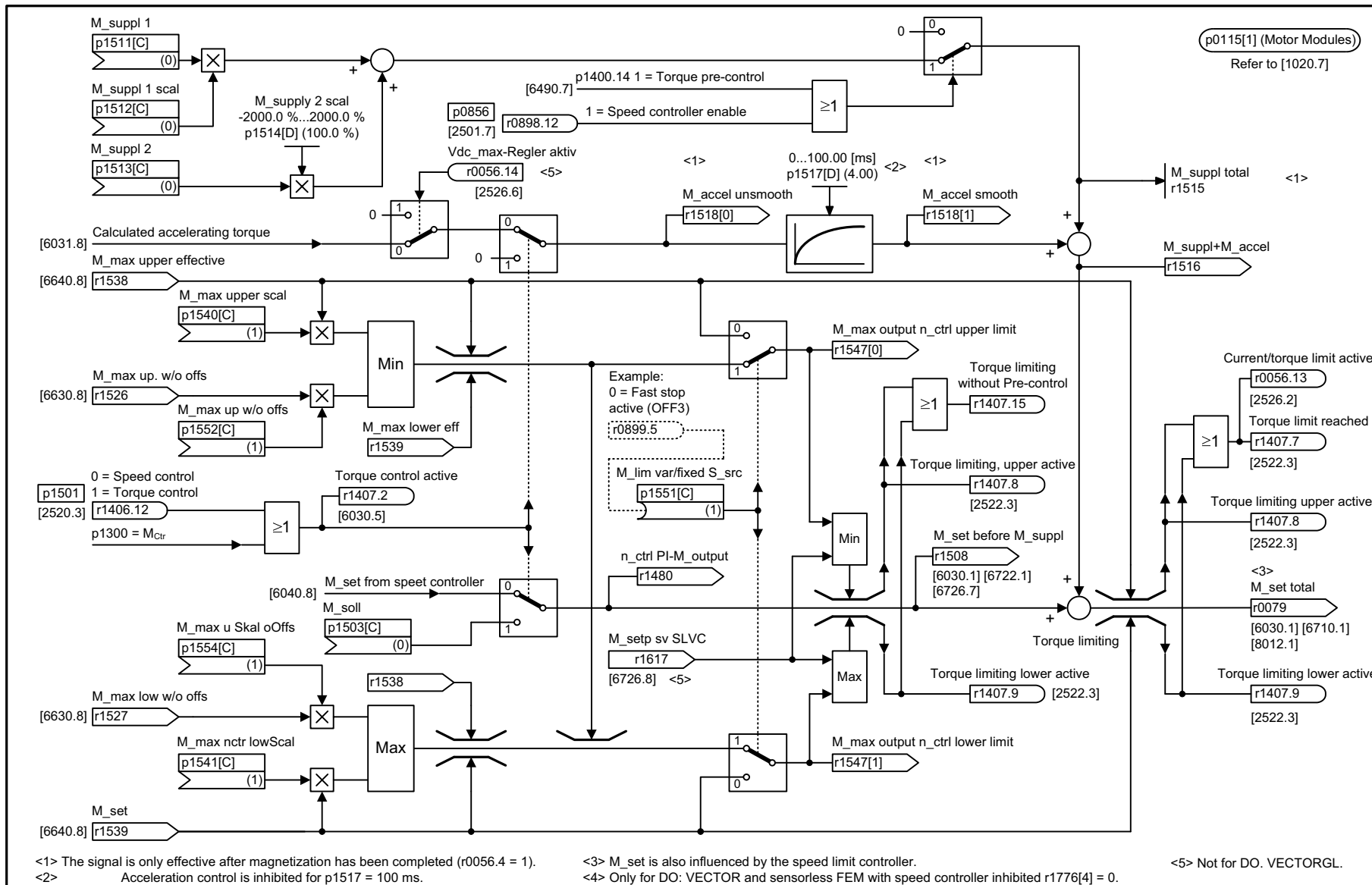
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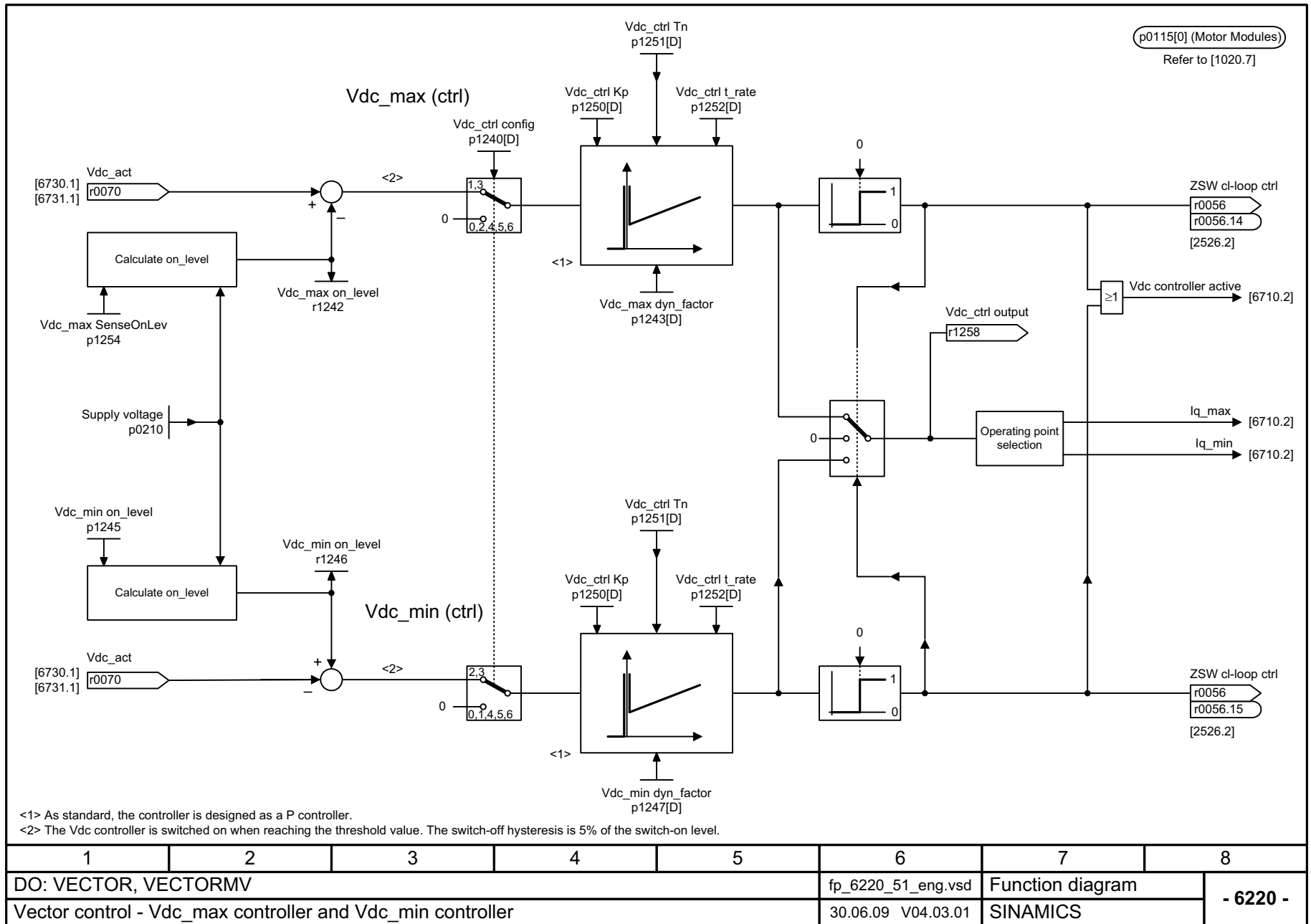
1	2	3	4	5	6	7	8
DO: VECTOR, VECTORGL, VECTORMV, VECTORSL					fp_6050_51_eng.vsd	Function diagram	
Vector control - Kp_n/Tn_n adaptation					29.06.09 V04.03.01	SINAMICS	
							- 6050 -

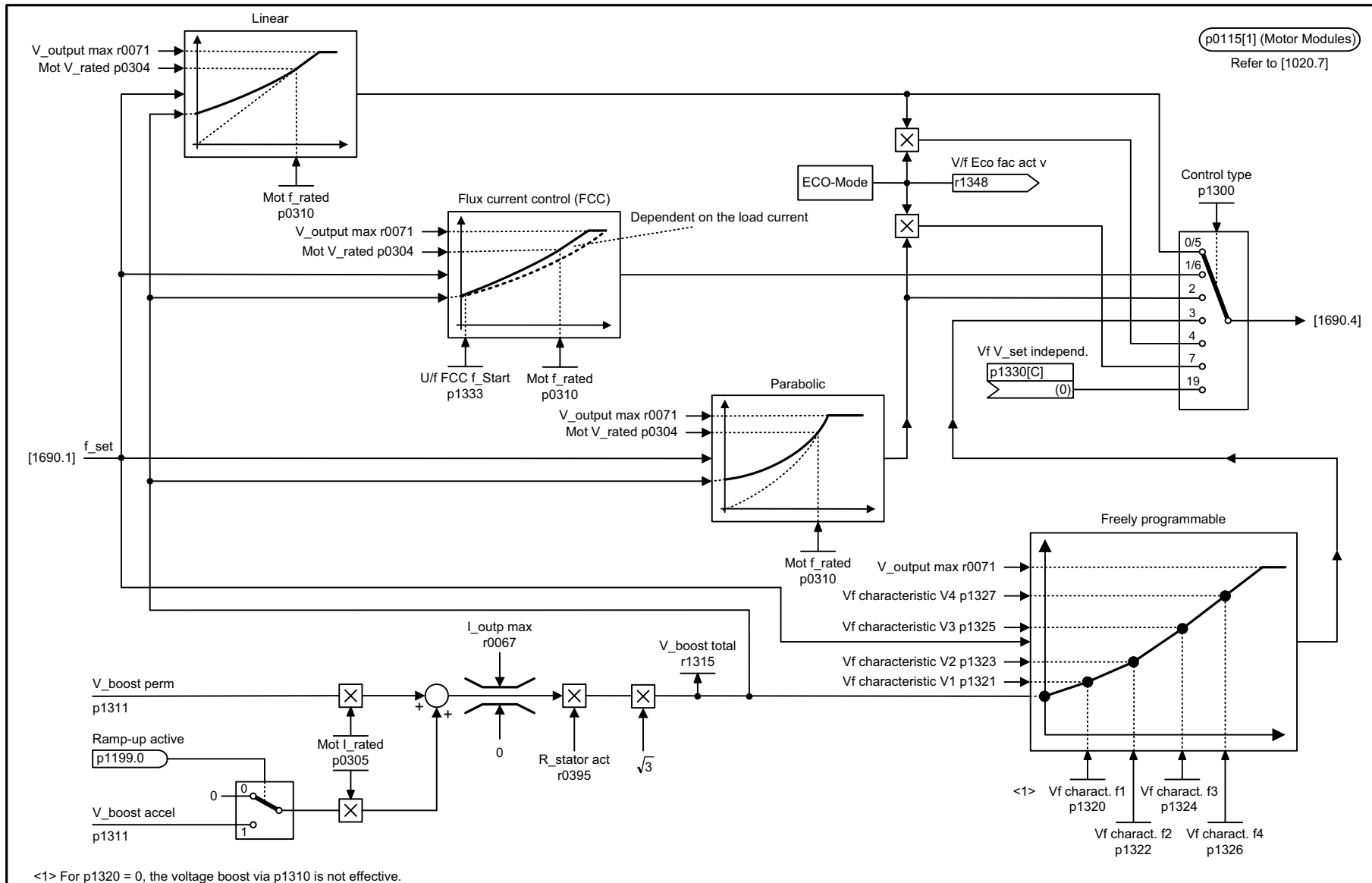
Fig. 2-185 6050 – Kp_n/Tn_n adaptation



1	2	3	4	5	6	7	8	
DO: VECTOR, VECTORGL, VECTORMV, VECTORSL					fp_6060_51_eng.vsd	Function diagram		- 6060 -
Vector control - Torque setpoint					24.11.09 V04.03.01	SINAMICS		

Fig. 2-186 6060 – Torque setpoint





1	2	3	4	5	6	7	8	
DO: VECTOR					fp_6300_54_eng.vsd	Function diagram		- 6300 -
Vector control - V/f characteristic and voltage boost					04.03.09 V04.03.01	S120/S150/G130/G150		

Fig. 2-188 6300 – V/f characteristic and voltage boost

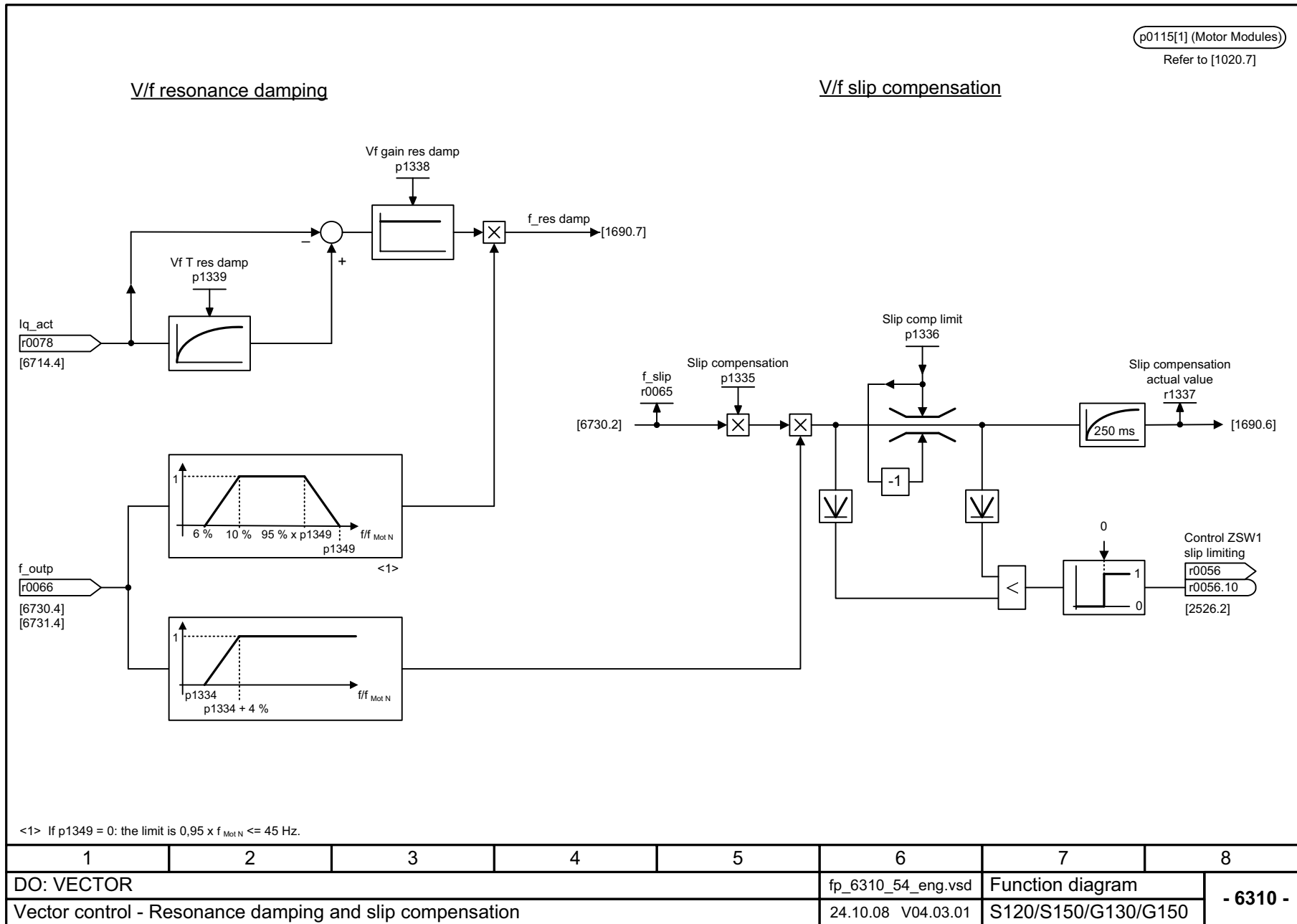
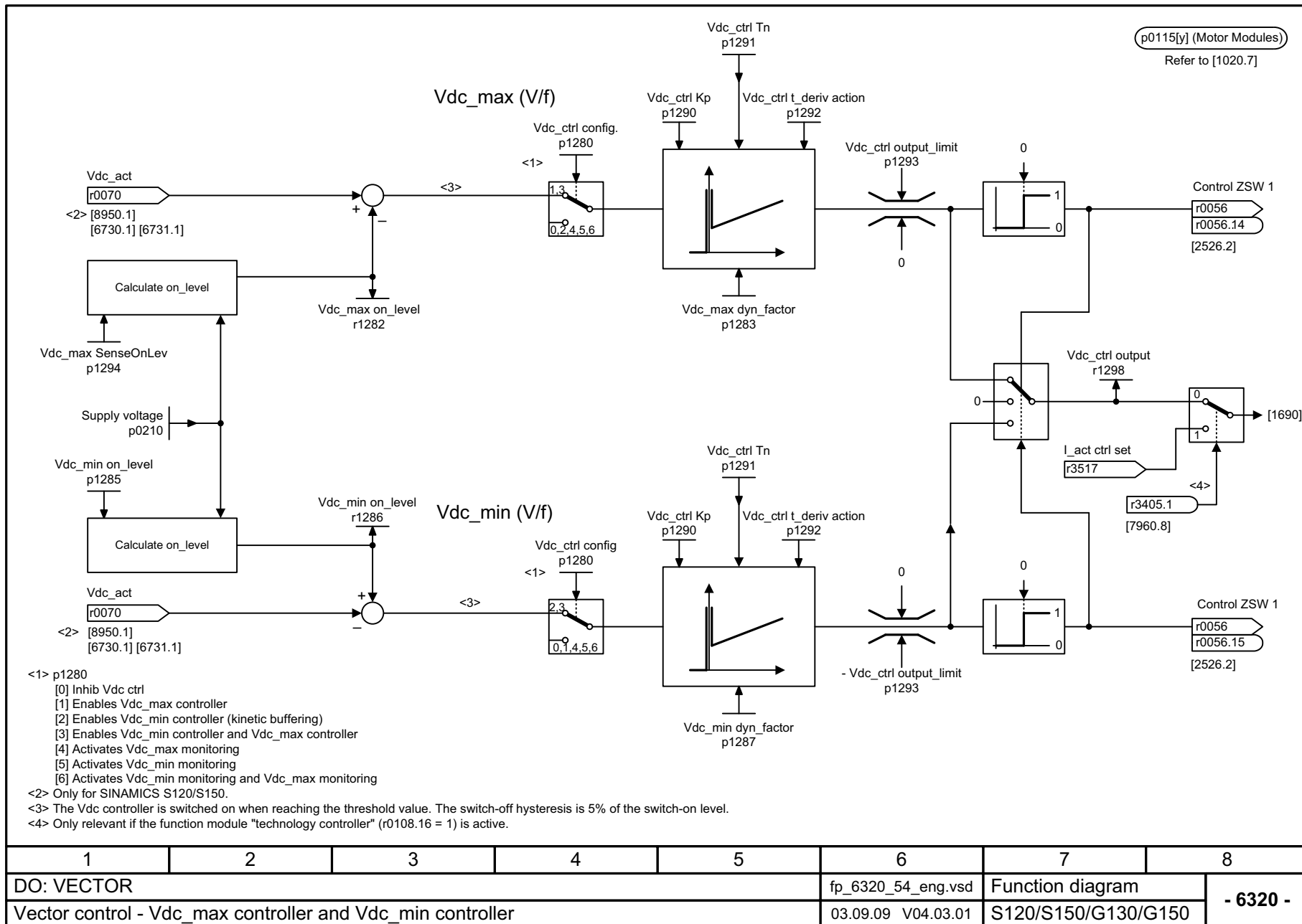


Fig. 2-189 6310 – Resonance damping and slip compensation



n_ctrl config p1400[D] Speed control configuration Factory setting p0115[1] (Motor Modules) Refer to [1020.7]							
	Bit No.	Meaning					
	0	1 = Automatic Kp/Tn adaptation active	1	→	[6040.3]		
	1	1 = Sensorless vector control, freeze I component	0	→	[6040.3]		
	2	1 = Acceleration pre-control, external source (p1495) 0 = Acceleration pre-control, internal source (n_set)	0	→	[6031.2]		
	3	1 = Reference model, speed setpoint I component ON	0	→	[6031.1][6031.7]		
	4	Reserved					
	5	1 = Kp/Tn adaptation active	1	→	[6040.3]		
	6	1 = Free Tn adaptation active	0	→	[6050.6]		
	7	Reserved					
	8	Reserved					
	9	Reserved					
	10	Reserved					
	11	Reserved					
	12	Reserved					
	13	Reserved					
	14	1 = Torque pre-control always active 0 = Torque pre-control for n_ctrl enabled	0	→	[6060.4]		
	15	1 = Sensorless vector control, speed pre-control active	1	→	[6030.5]		
1	2	3	4	5	6	7	8
DO: VECTOR, VECTORGL, VECTORMV, VECTORSL					fp_6490_51_eng.vsd	Function diagram	- 6490 -
Vector control - Speed control configuration					29.06.09 V04.03.01	SINAMICS	

Fig. 2-191 6490 – Speed control configuration

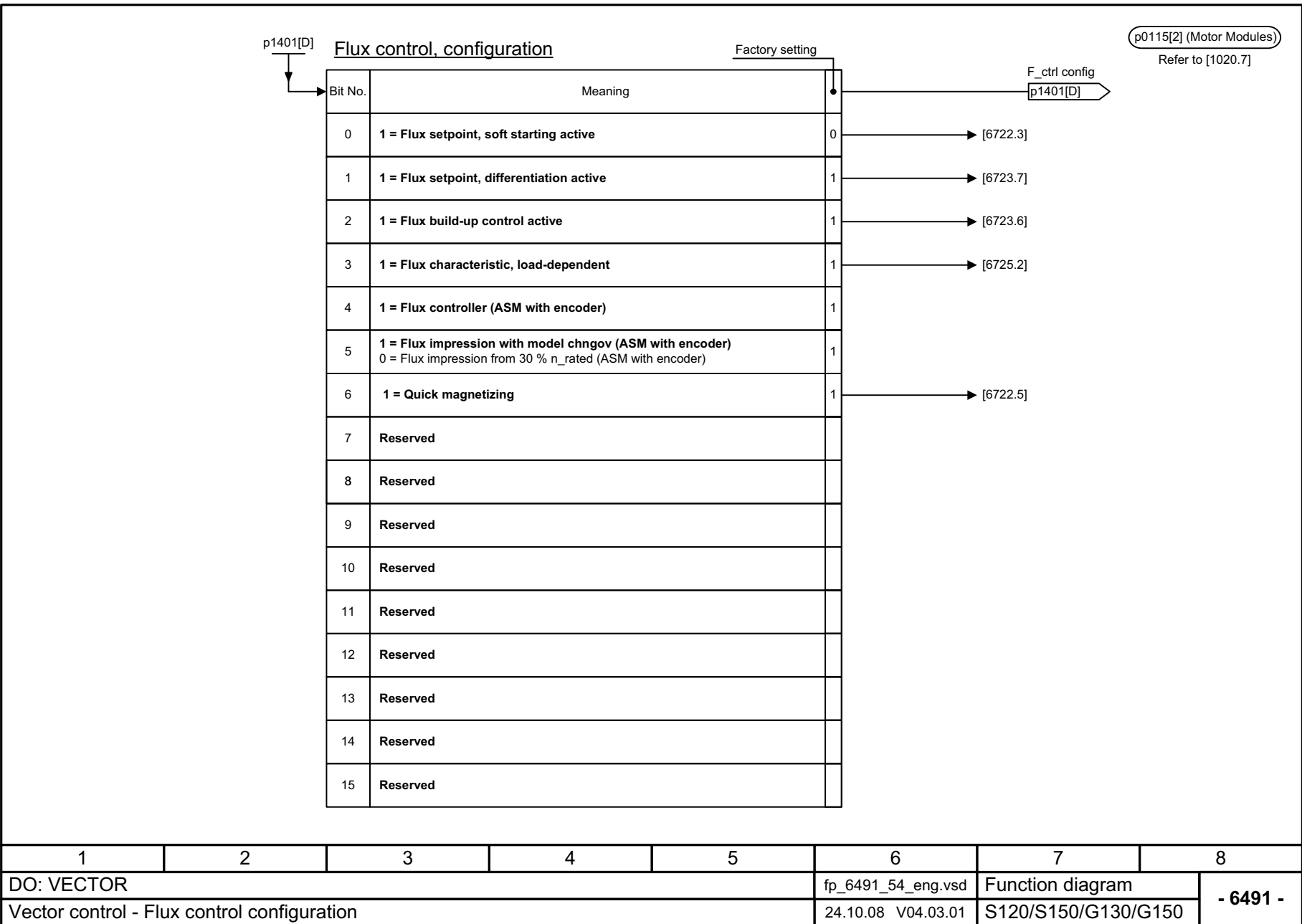
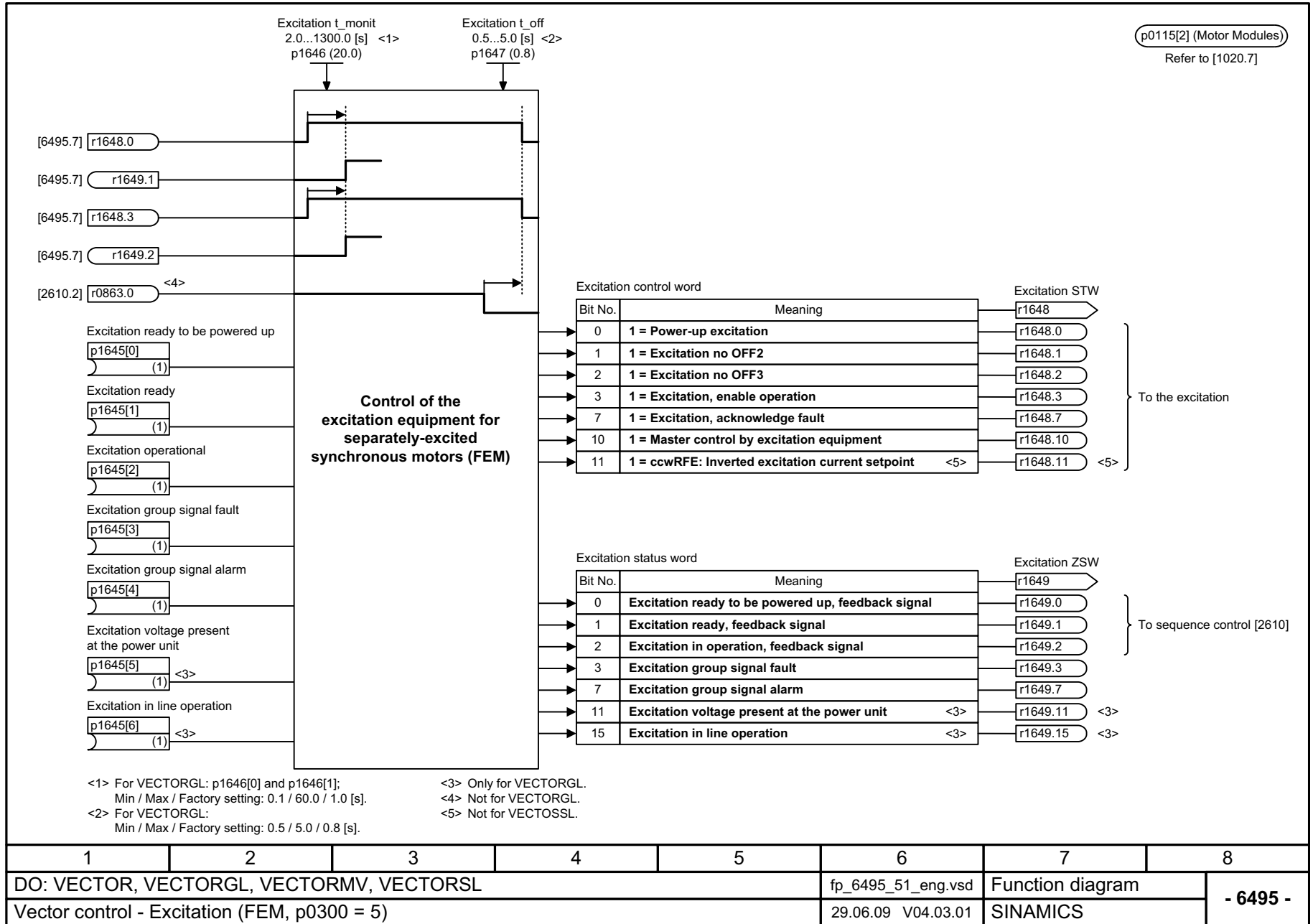


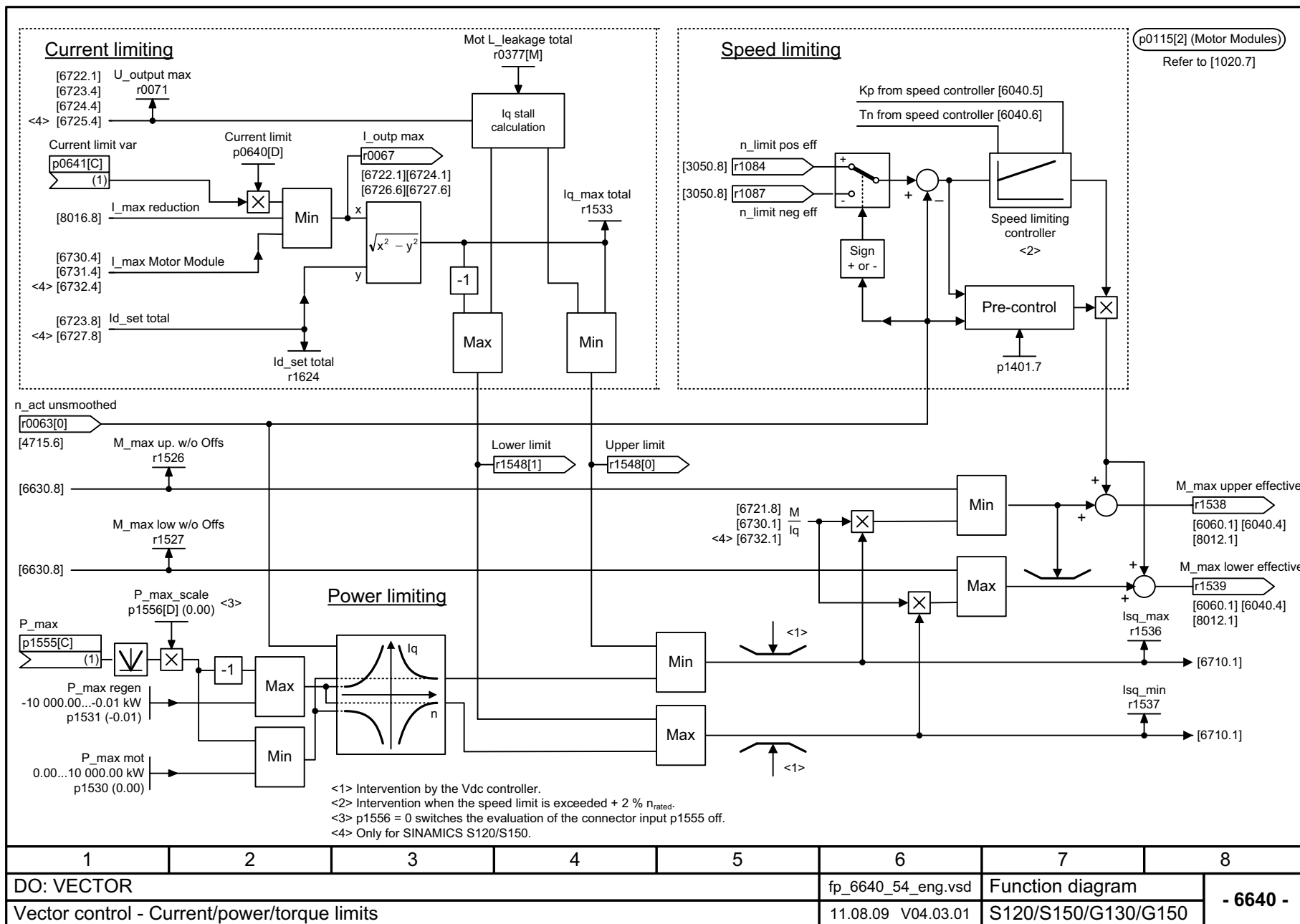
Fig. 2-193 6495 – Excitation (FEM, p0300 = 5)

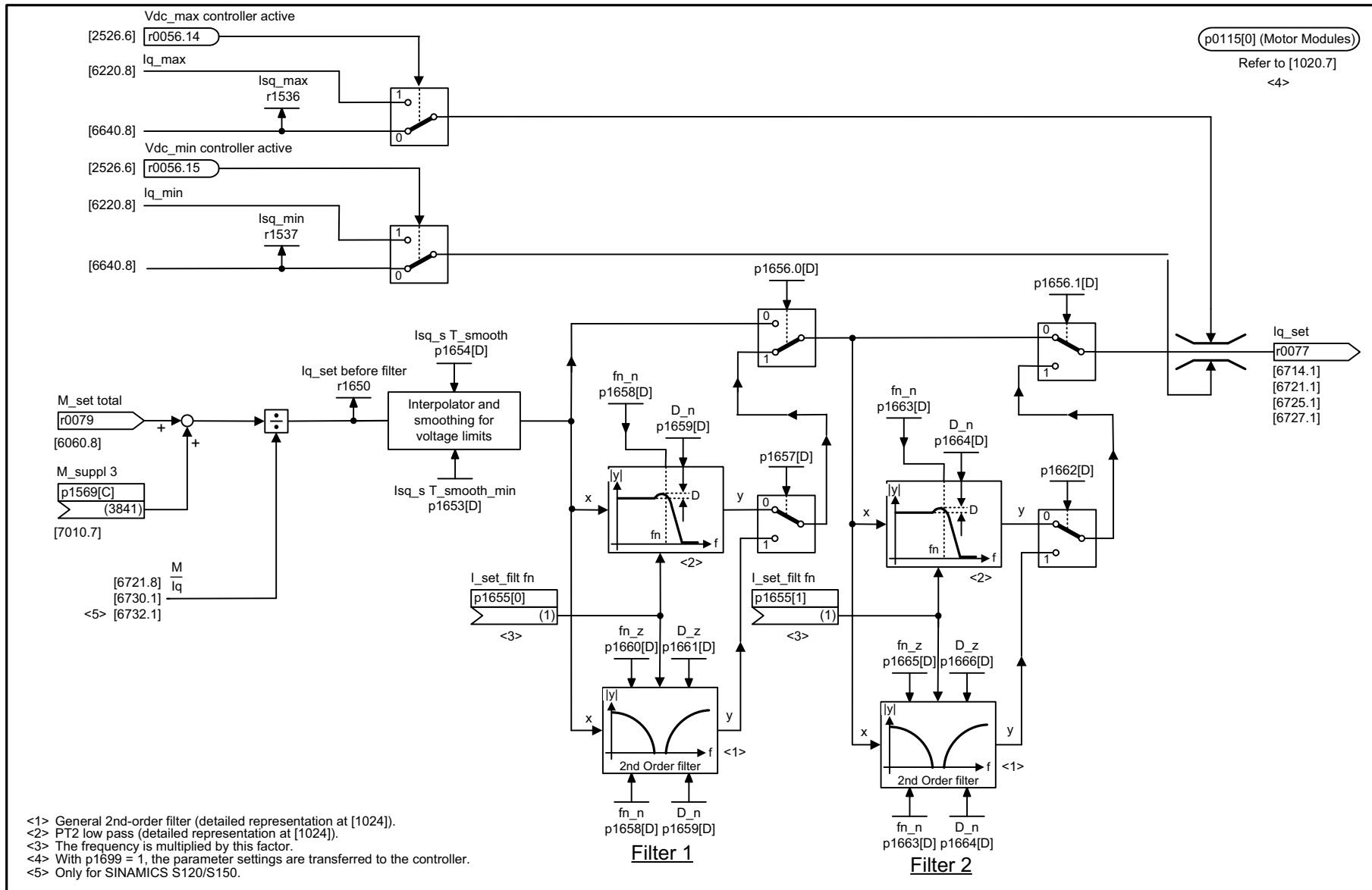


2-1596



Function diagrams
Vector control

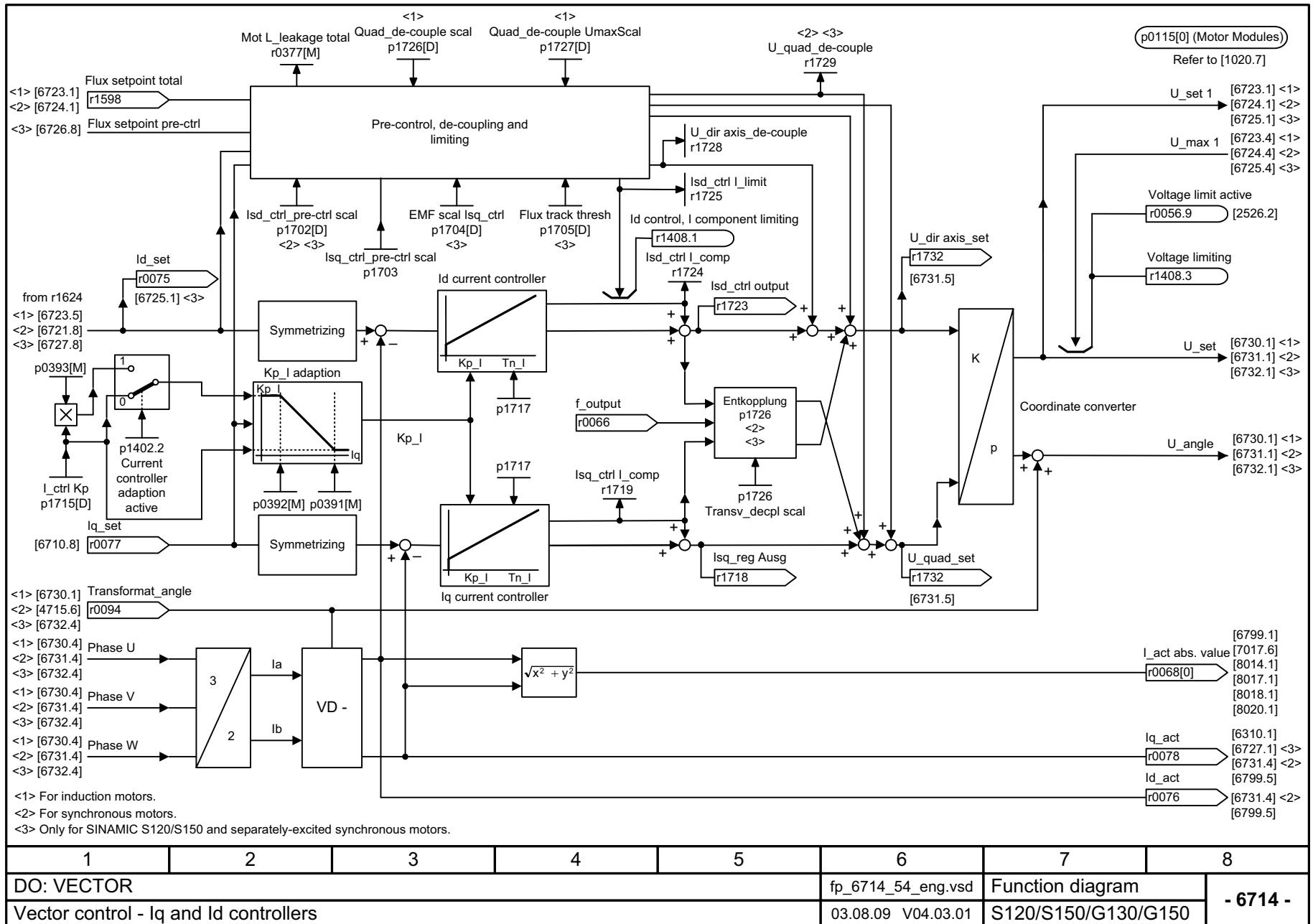


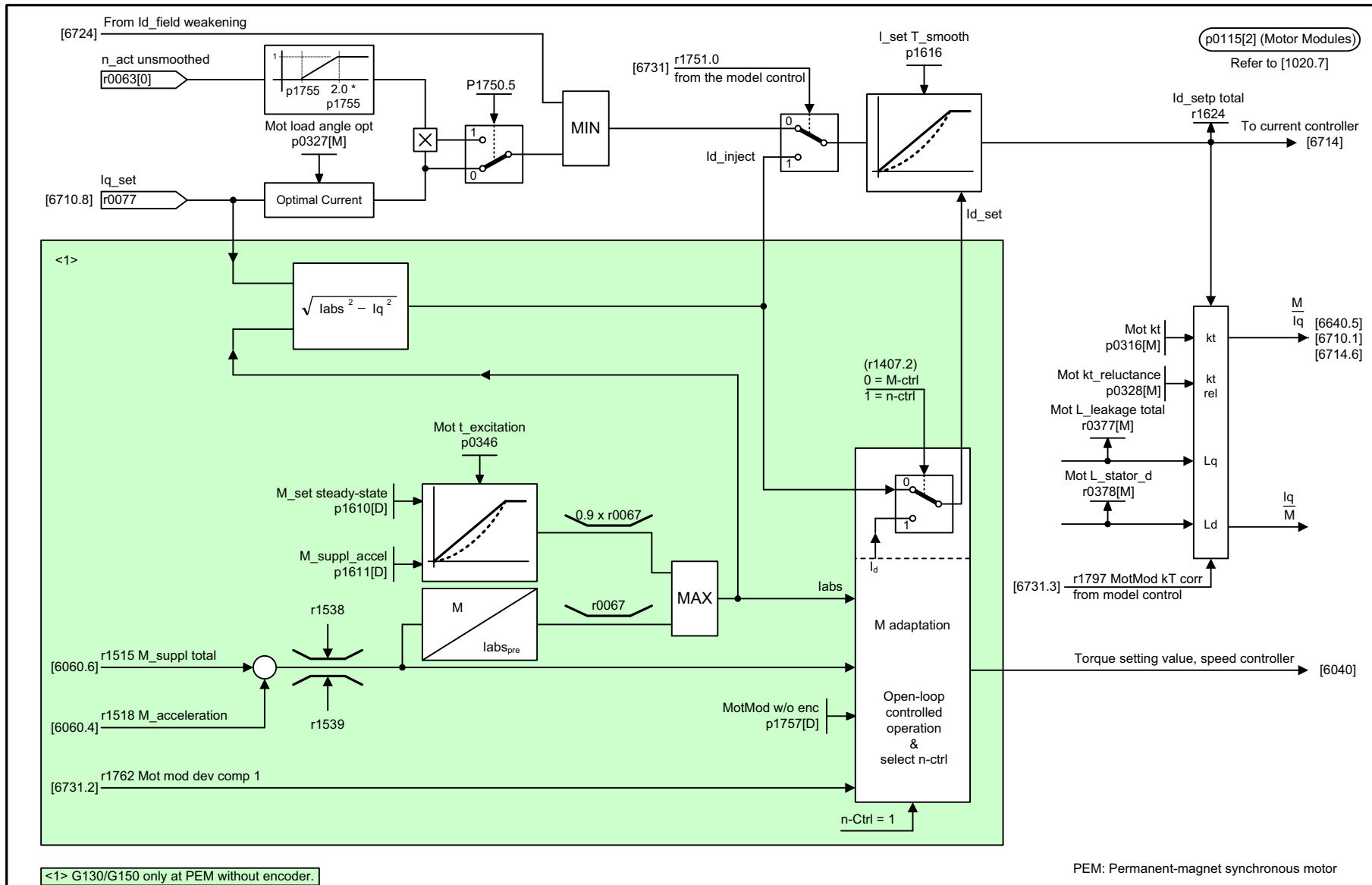


1	2	3	4	5	6	7	8
DO: VECTOR					fp_6710_54_eng.vsd	Function diagram	
Vector control - Current setpoint filter					03.07.09 V04.03.01	S120/S150/G130/G150	
							- 6710 -

Fig. 2-196 6710 – Current setpoint filter

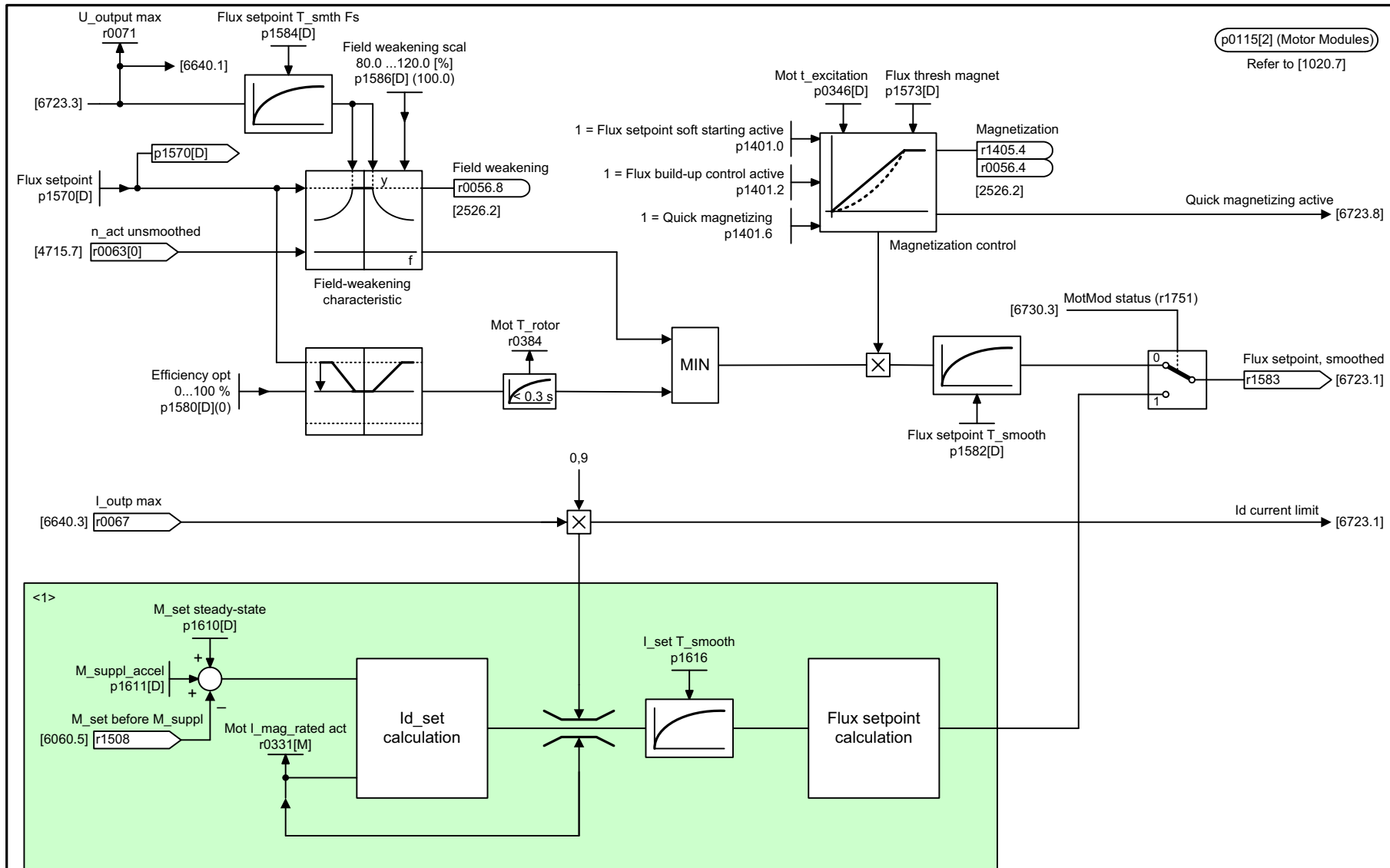
Fig. 2-197 6714 – Iq and Id controller





1	2	3	4	5	6	7	8
DO: VECTOR					fp_6721_54_eng.vsd	Function diagram	
Vector control - Id setpoint (PEM, p0300 = 2)					01.07.09 V04.03.01	S120/S150/G130/G150	
							- 6721 -

Fig. 2-198 6721 – Id setpoint (PEM, p0300 = 2)

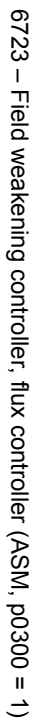


<1> Only for vector control without encoder (SLVC).

ASM: Induction motor

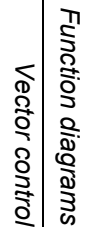
1	2	3	4	5	6	7	8
DO: VECTOR					fp_6722_54_eng.vsd	Function diagram	
Vector control - Field weakening characteristic, Id setpoint (ASM, p0300 = 1)					24.10.08 V04.03.01	S120/S150/G130/G150	
							- 6722 -

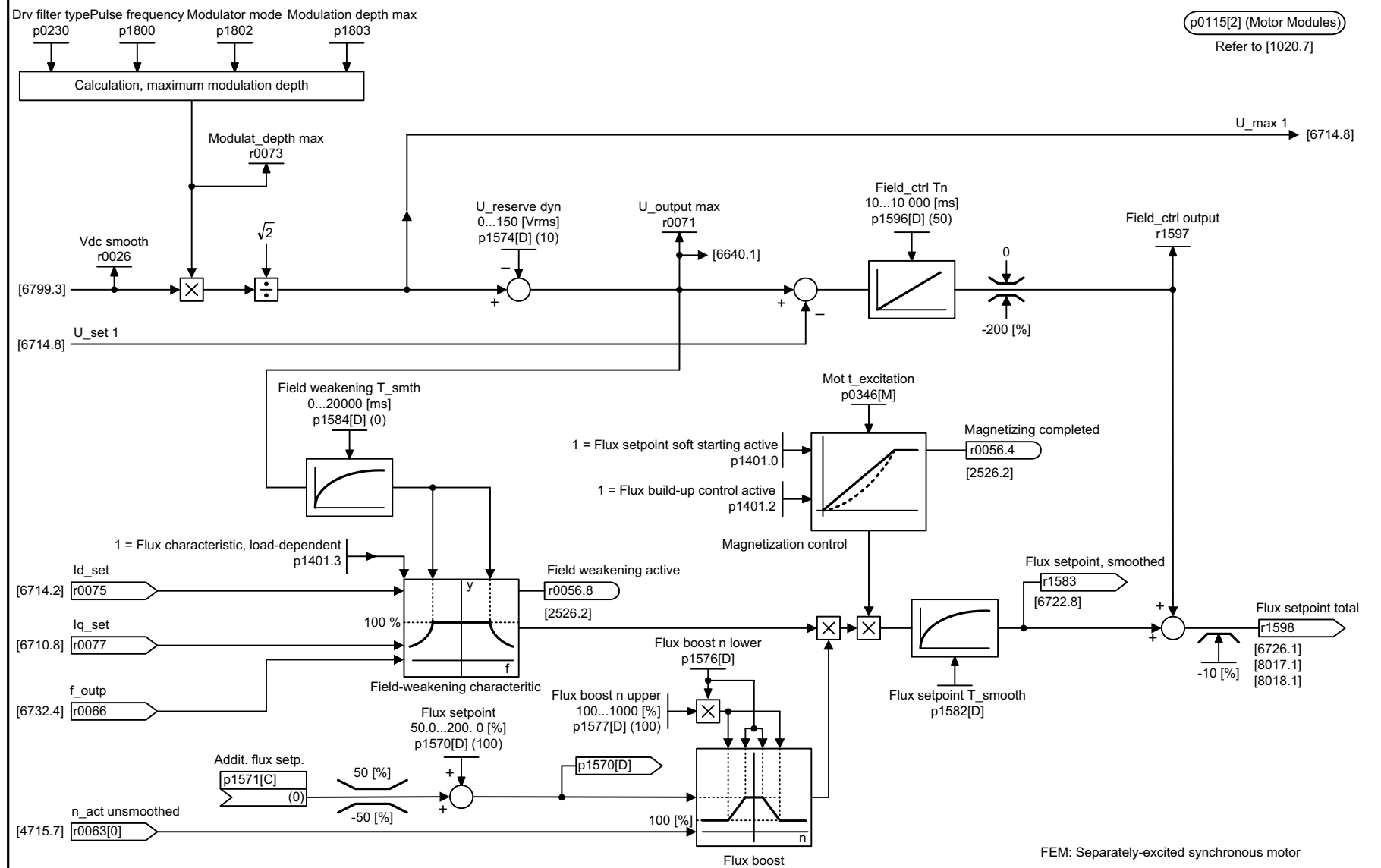
Fig. 2-199 6722 – Field weakening characteristic, Id setpoint (ASM, p0300 = 1)



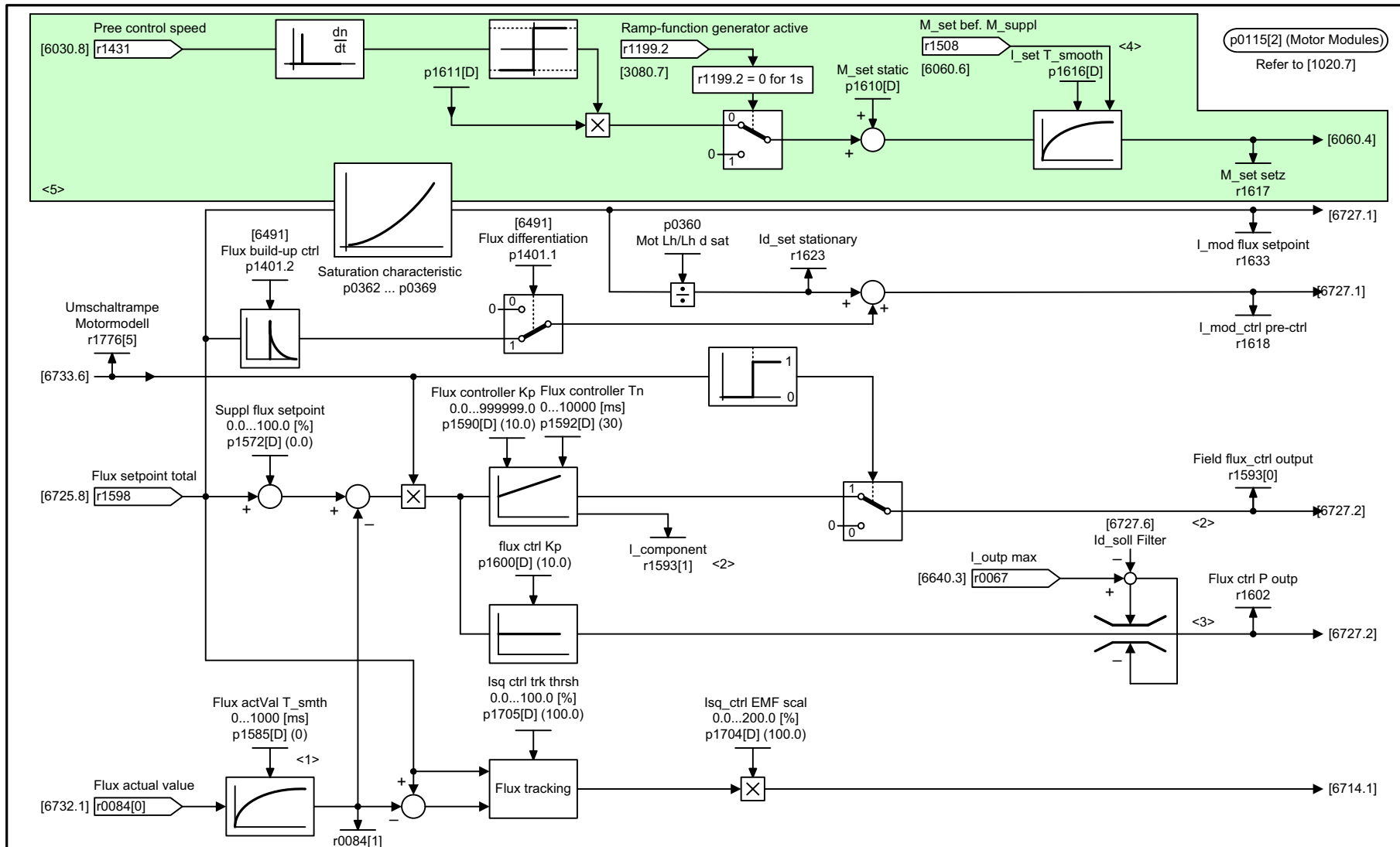
6723 – Field weakening controller, flux controller (ASM, p0300 = 1)

Fig. 2-201





1	2	3	4	5	6	7	8
DO: VECTOR					fp_6725_55_eng.vsd	Function diagram	
Vector control - Flux setpoint, field weakening controller (FEM, p0300 = 5)					02.02.09 V04.03.01	SINAMICS S120/S150	
							- 6725 -



<1> Überbrückt wenn r1776[5] = 0.

<2> Wird eingefroren, wenn die Abweichung im Erregerstrom > p1599 ist.

<3> Wenn <2> zutrifft wird der P- zum PI-Regler (p1592). Dann wird der gesamte Ausgang begrenzt. Bei Begrenzung wird der i-Anteil angehalten.

<4> r1617 = r1508, wenn Drehzahlregler freigegeben (r1776[4] > 0).

<5> Nur bei geberlosem Betrieb.

1	2	3	4	5	6	7	8
DO: VECTOR					fp_6726_55_eng.vsd	Function diagram	
Vector control - Field weakening controller, flux controller (FEM, p0300 = 5)					14.04.08 V04.03.01	SINAMICS S120/S150	
							- 6726 -

Fig. 2-203 6726 – Field weakening controller, flux controller (FEM, p0300 = 5)

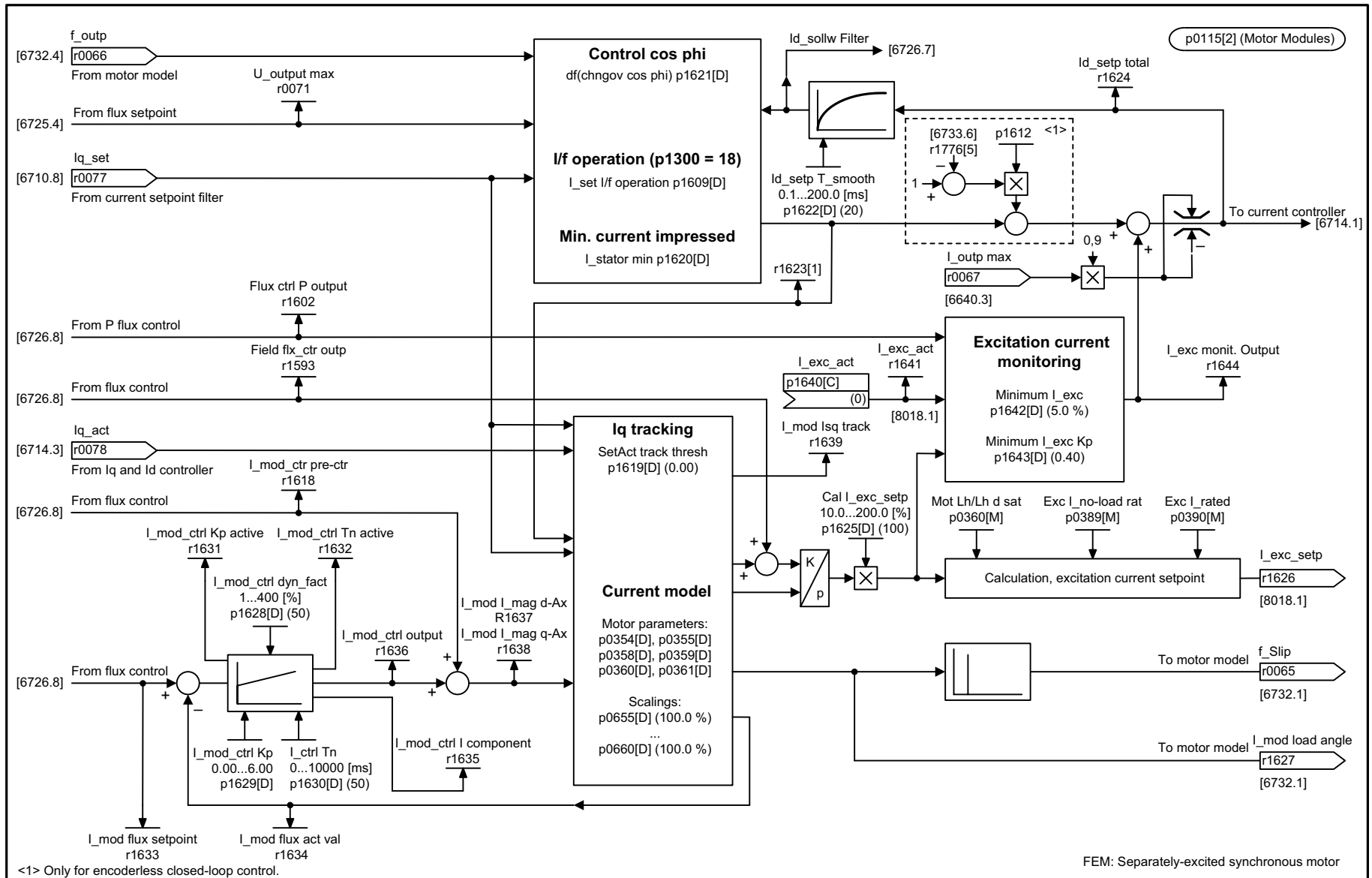
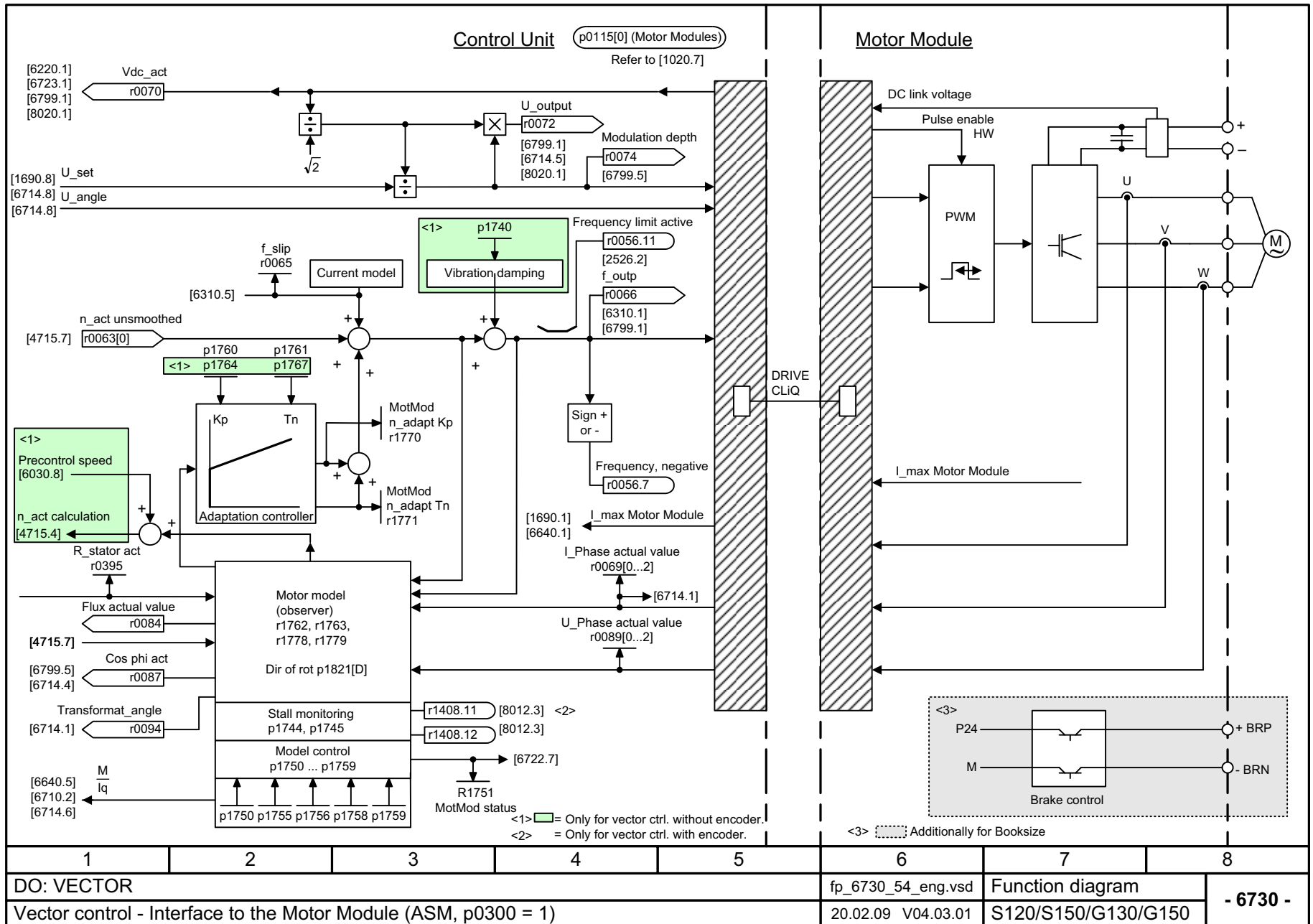
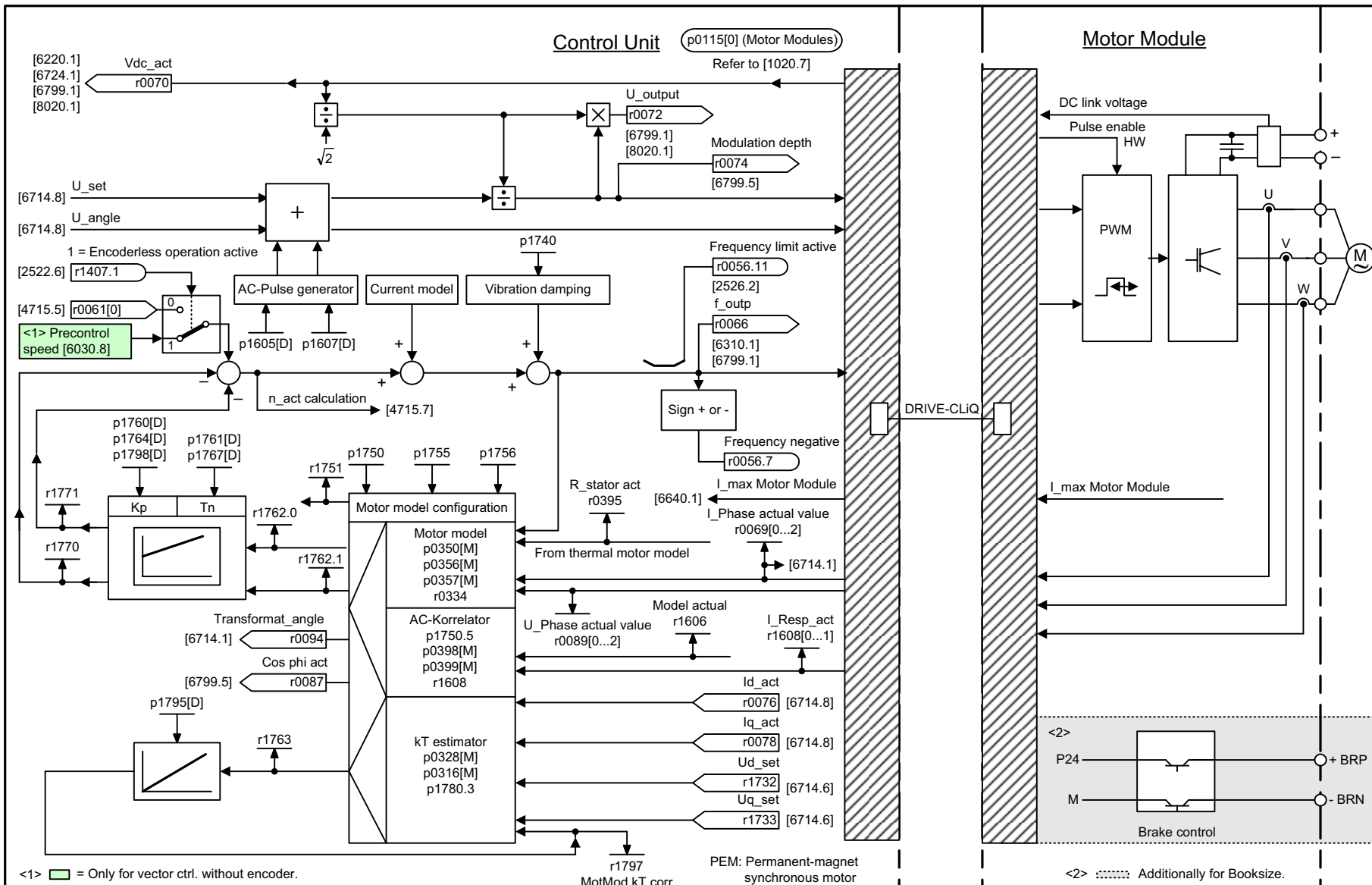


Fig. 2-204 6727 – Current model, excitation current monitoring, control cos phi (FEM, p0300 = 5)

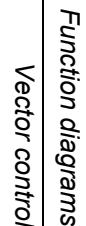
Fig. 2-205 6730 – Interface to Motor Module (ASM, p0300 = 1)

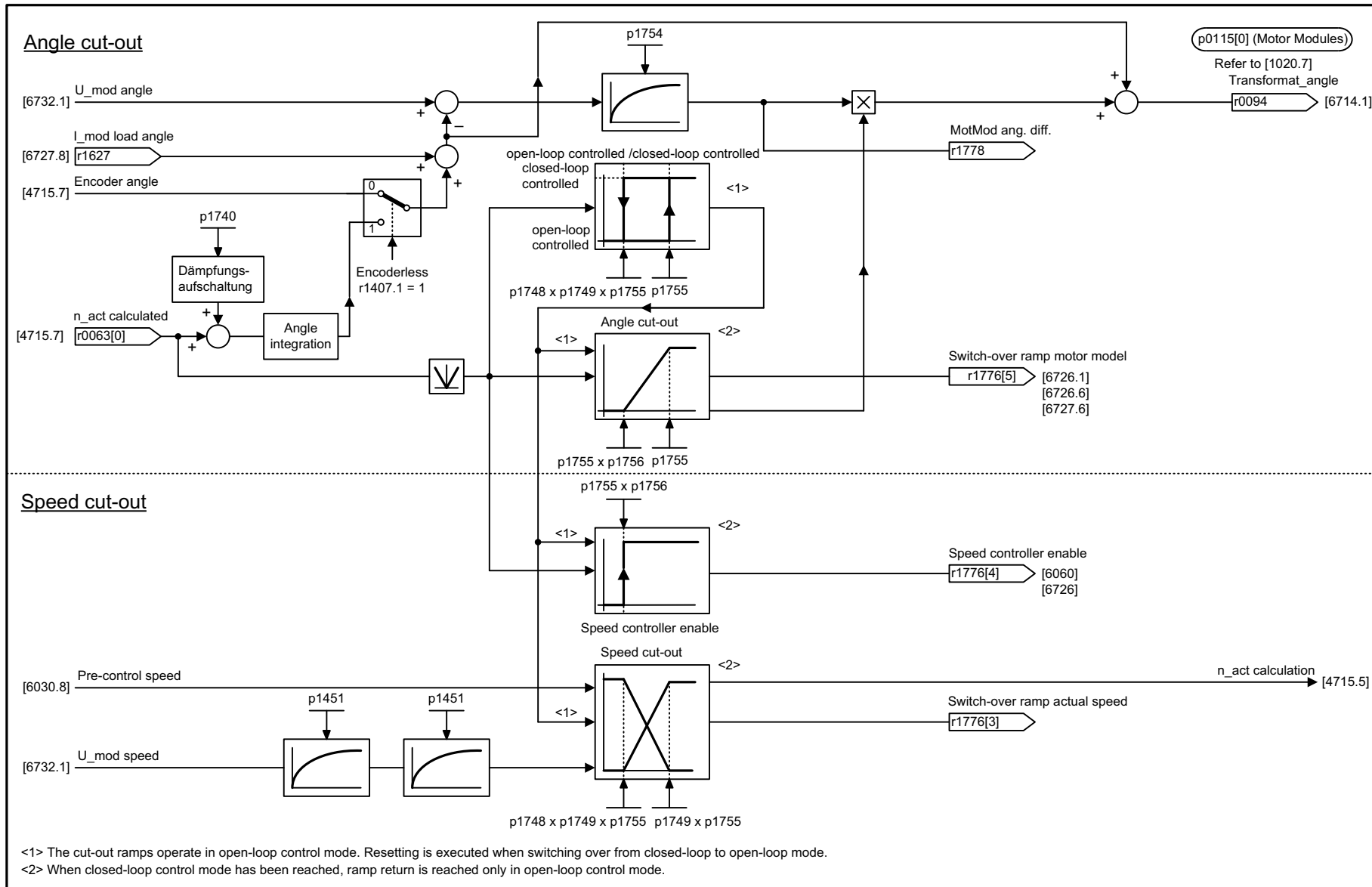




6731 – Interface to the Motor Module (PEM, p0300 = 2)

Fig. 2-207





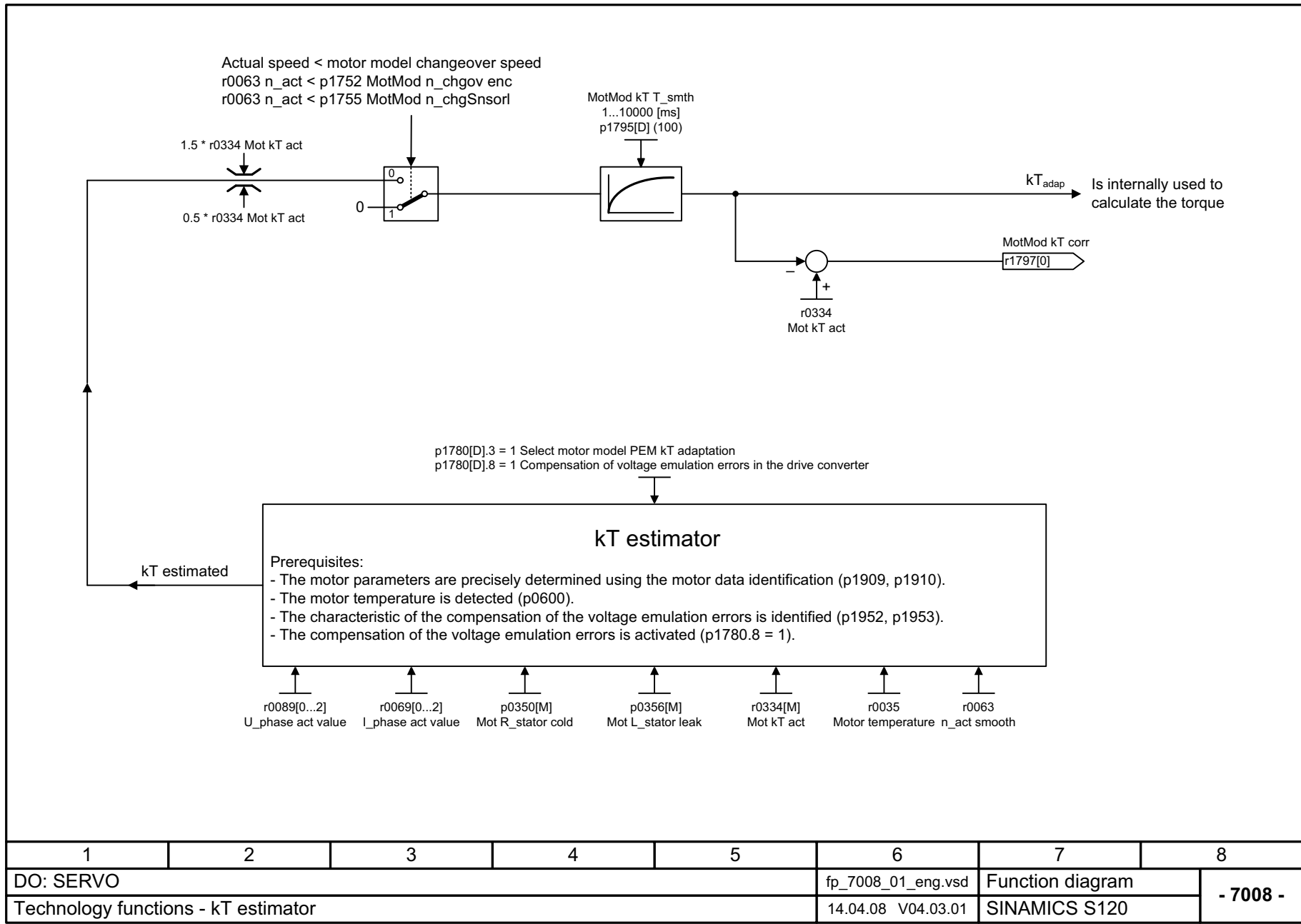
1	2	3	4	5	6	7	8
DO: VECTOR					fp_6733_55_eng.vsd	Function diagram	
Vector control - Motormodell Auswahl (FEM, p0300 = 5)					18.06.09 V04.03.01	SINAMICS S120/S150	
							- 6733 -

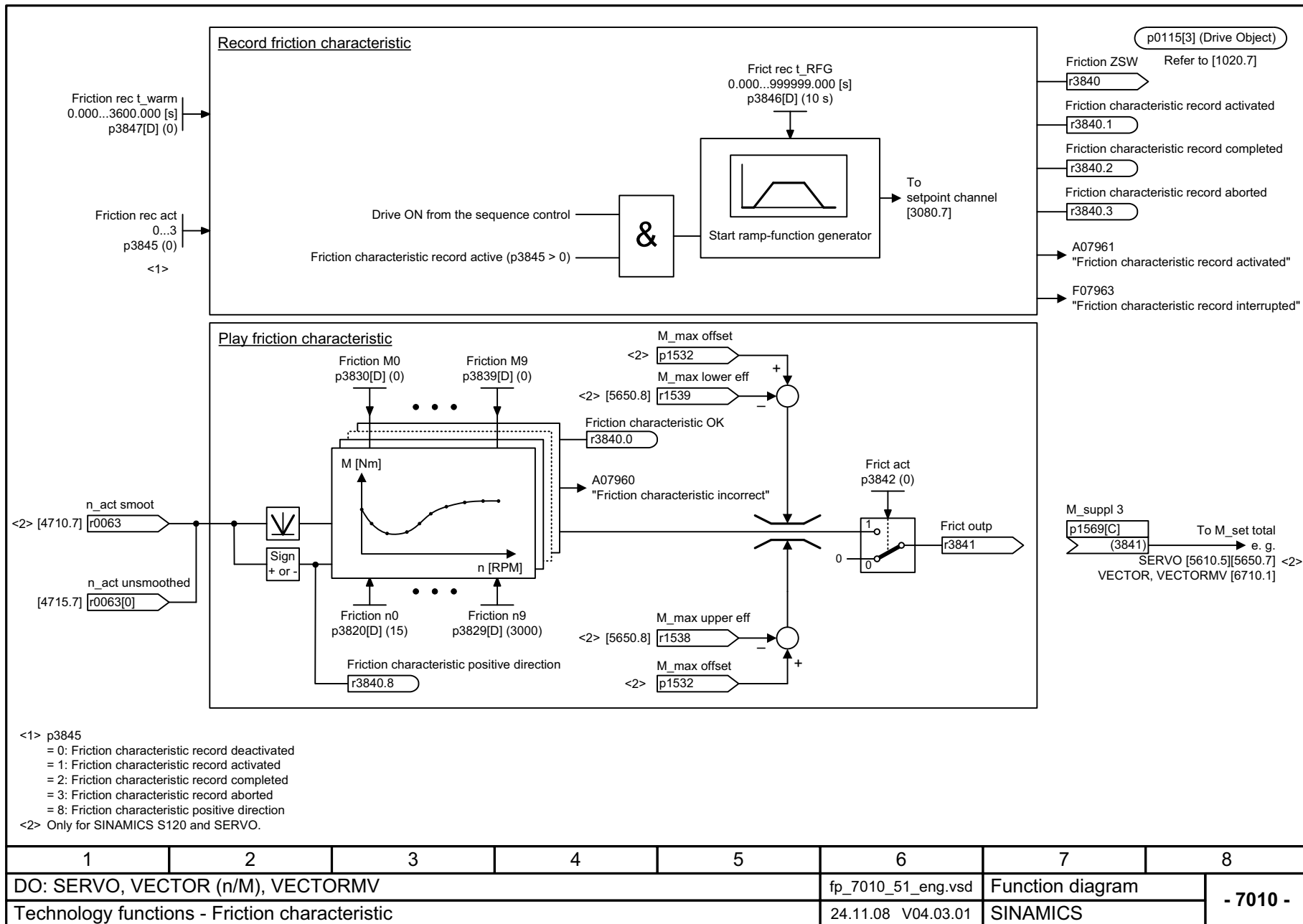
Fig. 2-208 6733 – Motor model selection (FEM, p0300 = 5)

2.19 Technology functions

Function diagrams

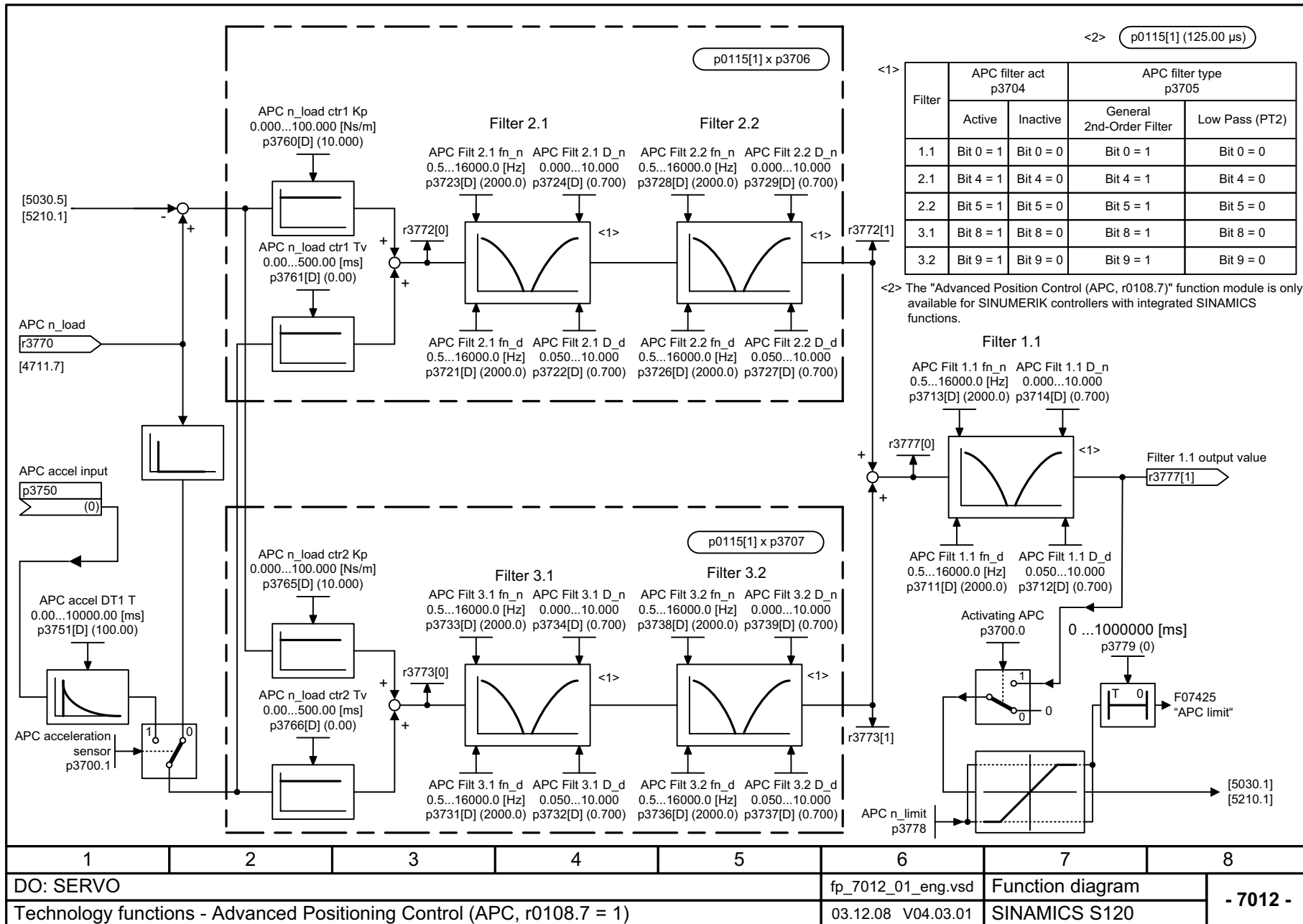
7008 – kT estimator	2-1613
7010 – Friction characteristic	2-1614
7012 – Advanced Positioning Control (APC, r0108.7 = 1)	2-1615
7014 – External armature short-circuit (EASC, p0300 = 2xx or 4xx)	2-1616
7016 – Internal armature short-circuit (IASC, p0300 = 2xx or 4xx)	2-1617
7017 – DC brake (p0300 = 1xx)	2-1618
7020 – Synchronization	2-1619





1	2	3	4	5	6	7	8
DO: SERVO, VECTOR (n/M), VECTORMV					fp_7010_51_eng.vsd	Function diagram	
Technology functions - Friction characteristic					24.11.08 V04.03.01	SINAMICS	

- 7010 -



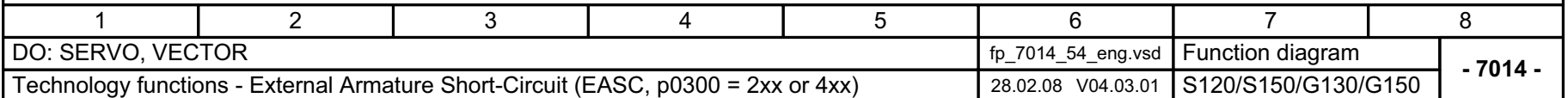


Fig. 2-213

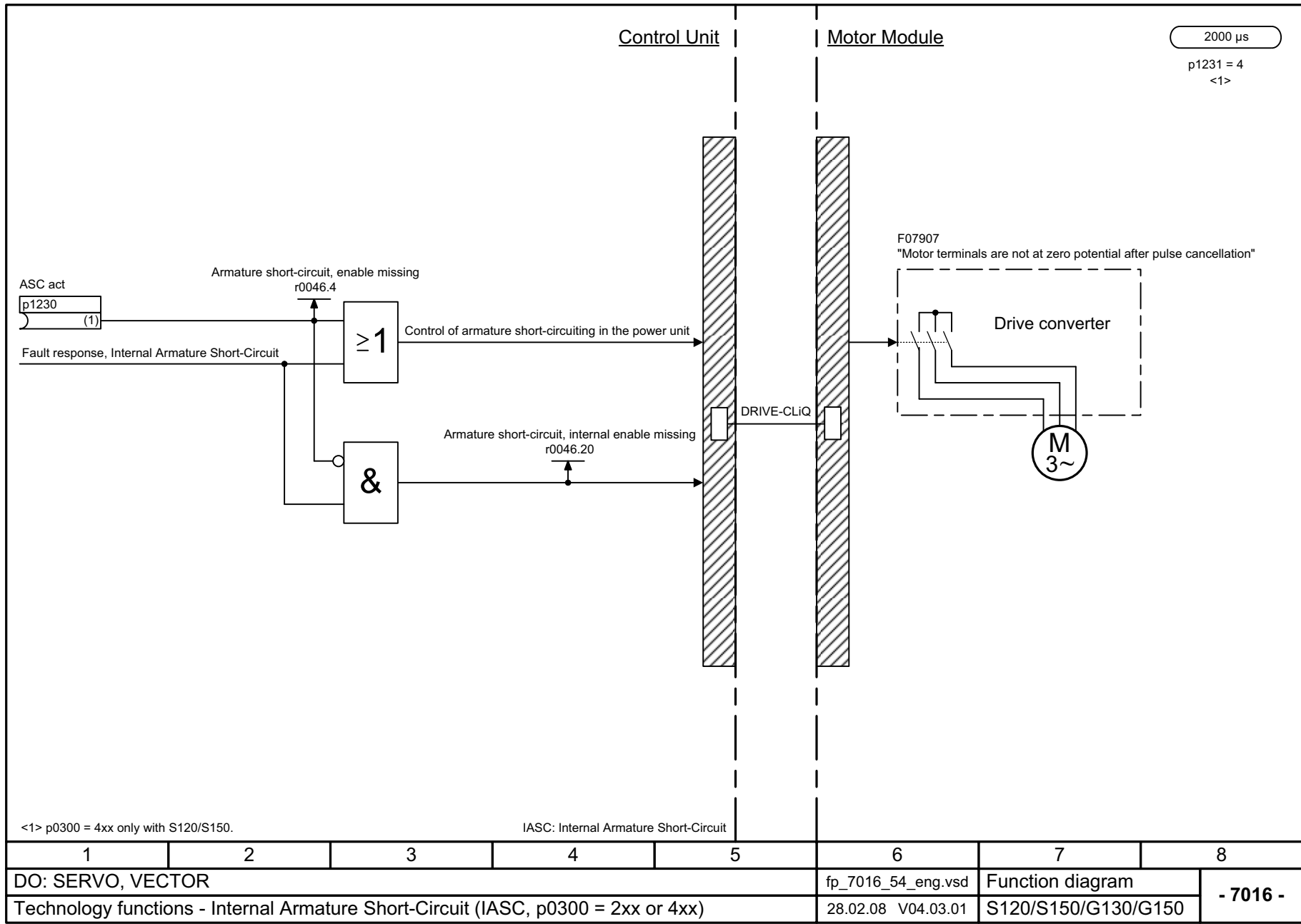
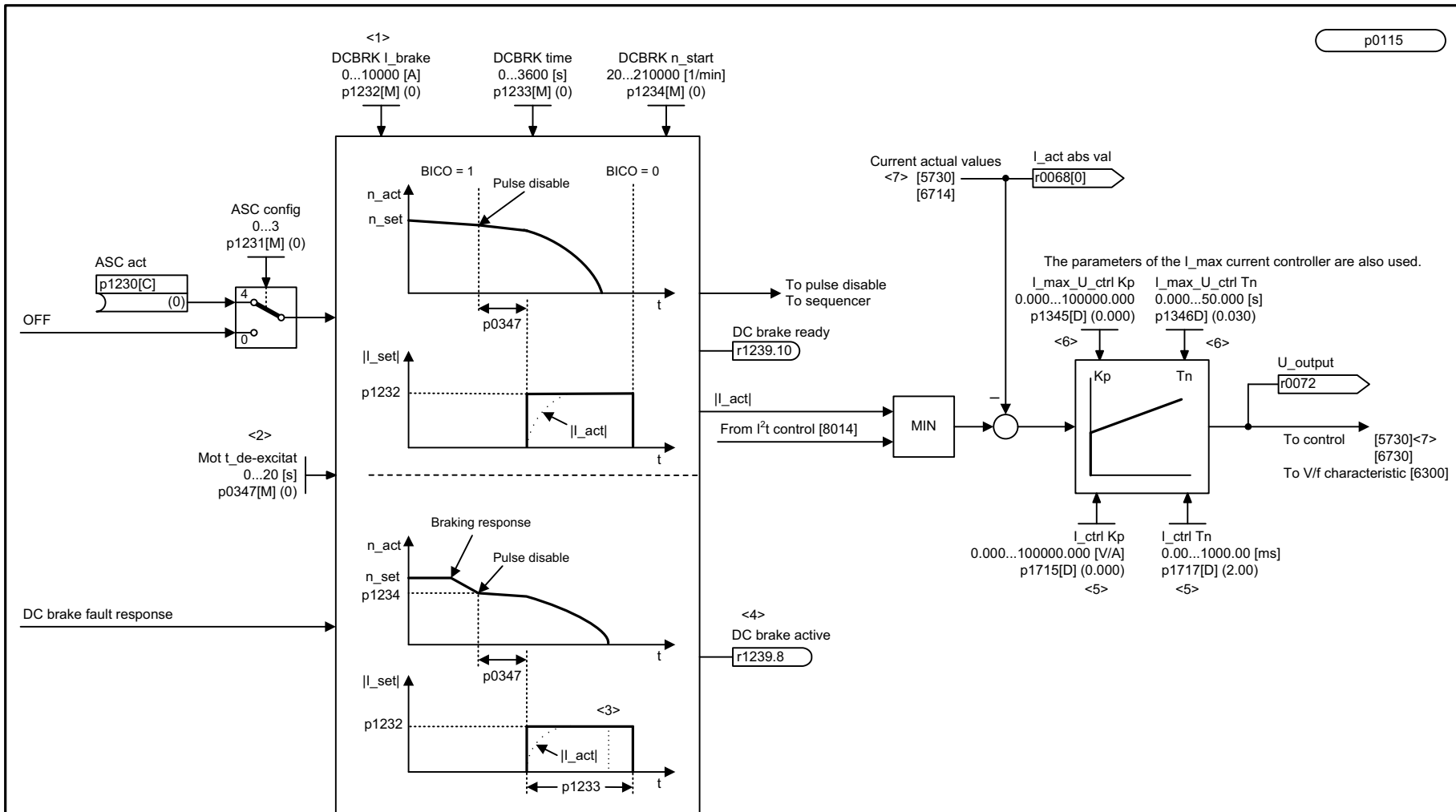


Fig. 2-214 7016 – Internal armature short-circuit (IASC, p0300 = 2xx or 4xx)

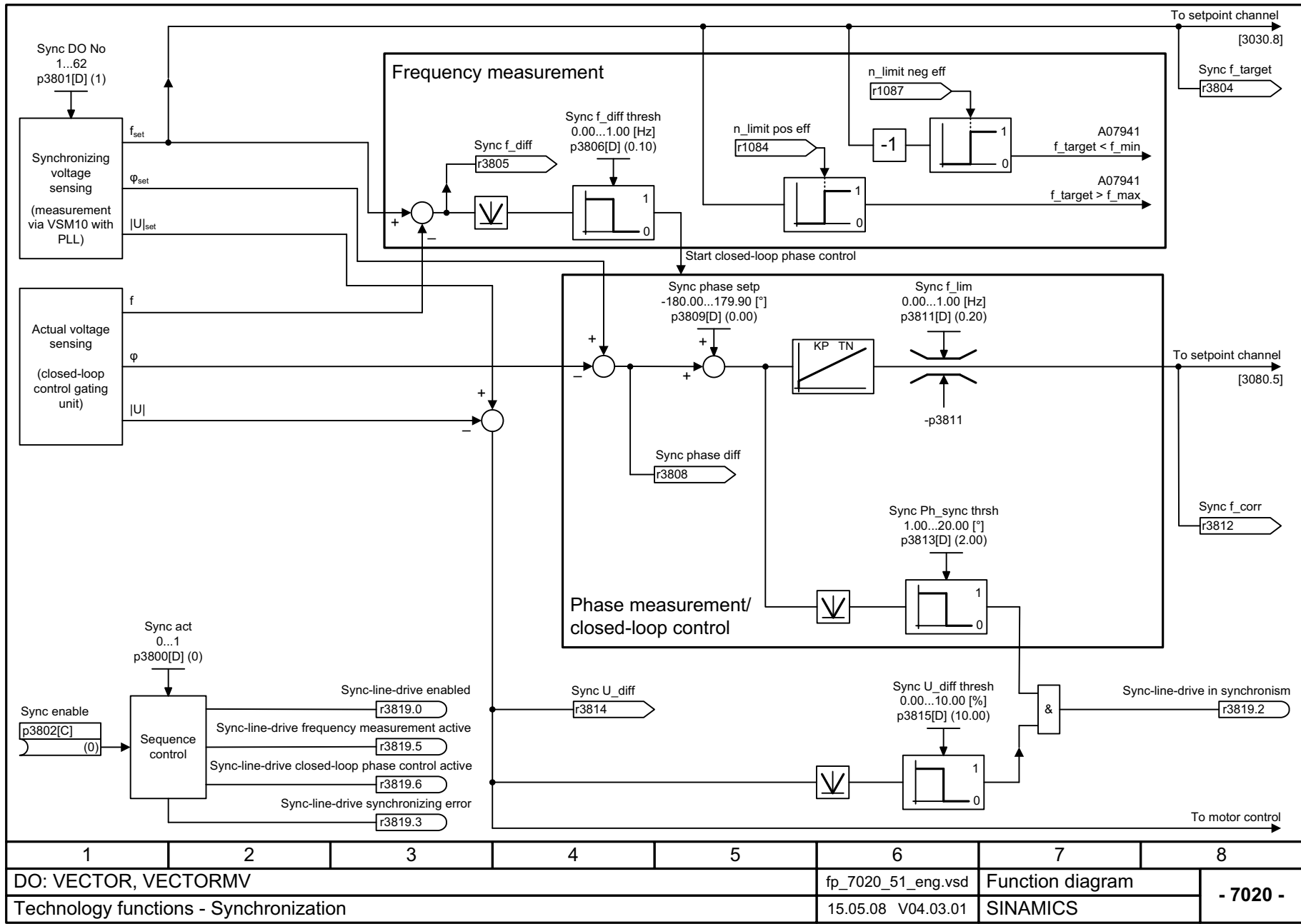


- <1> The DC braking current is determined during automatic calculation (p0340 = 1).
 <2> The de-magnetization time is determined during automatic calculation (p0340 = 1, 3).
 <3> As soon as the standstill threshold (p1226) has been reached, the DC current injection will be aborted prematurely.
 <4> Signal r1239.8 is only set while the DC brake is active.
 <5> Only for SINAMICS S120 and SERVO.
 <6> Only for VECTOR.
 <7> Only for SINAMICS S120.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_7017_54_eng.vsd	Function diagram	
Technology functions - DC brake (p0300 = 1xx)					27.02.08 V04.03.01	S120/S150/G130/G150	

p0115

Fig. 2-215 7017 – DC brake (p0300 = 1xx)



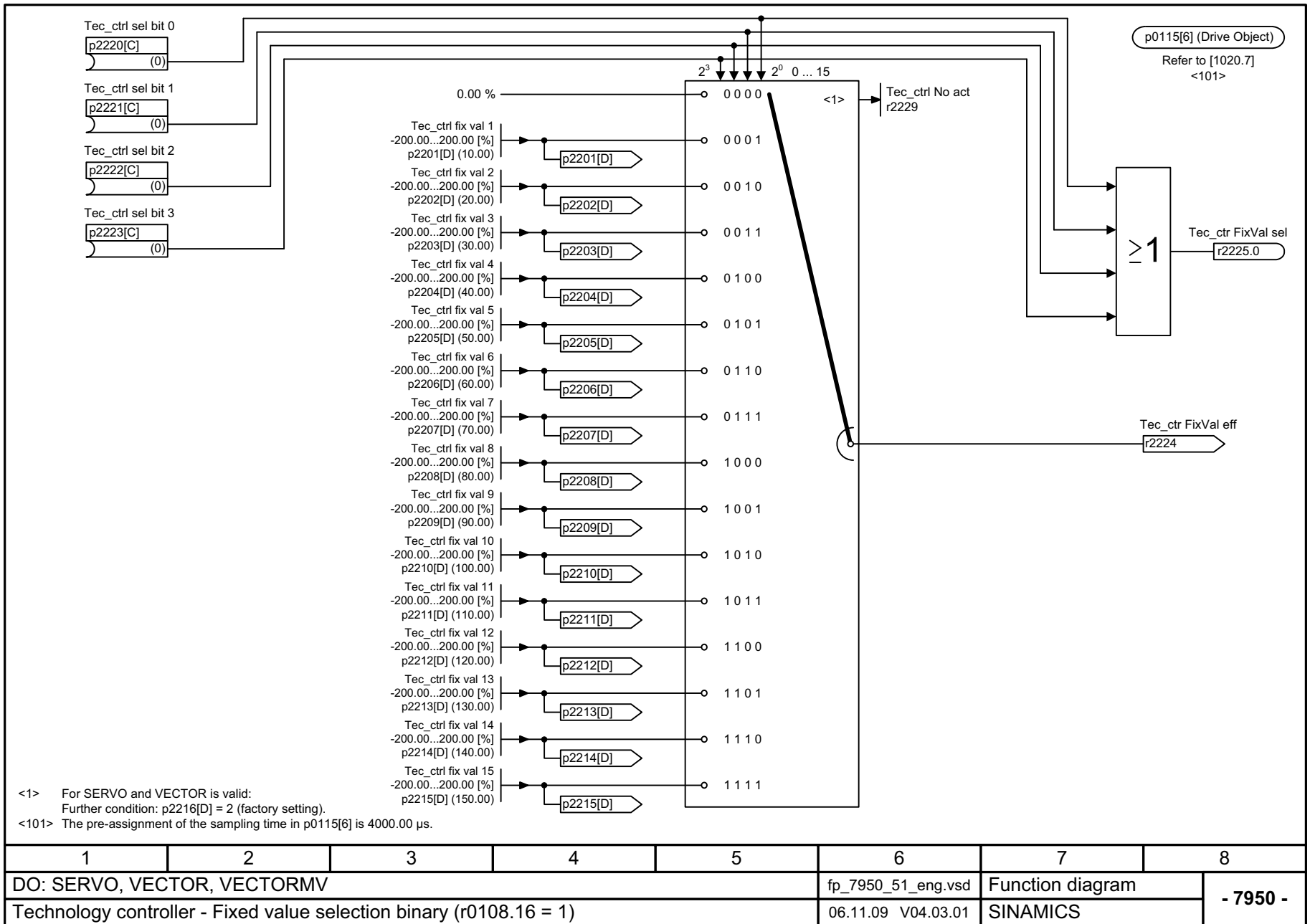
1	2	3	4	5	6	7	8
DO: VECTOR, VECTORMV					fp_7020_51_eng.vsd	Function diagram	
Technology functions - Synchronization					15.05.08 V04.03.01	SINAMICS	

- 7020 -

2.20 Technology controller

Function diagrams

7950 – Fixed values, binary selection (r0108.16 = 1)	2-1621
7951 – Fixed values, direct selection (r0108.16 = 1 and p2216[D] = 1)	2-1622
7954 – Motorized potentiometer (r0108.16 = 1)	2-1623
7958 – Closed-loop control (r0108.16 = 1)	2-1624
7960 – DC-link voltage controller (r0108.16 = 1)	2-1625



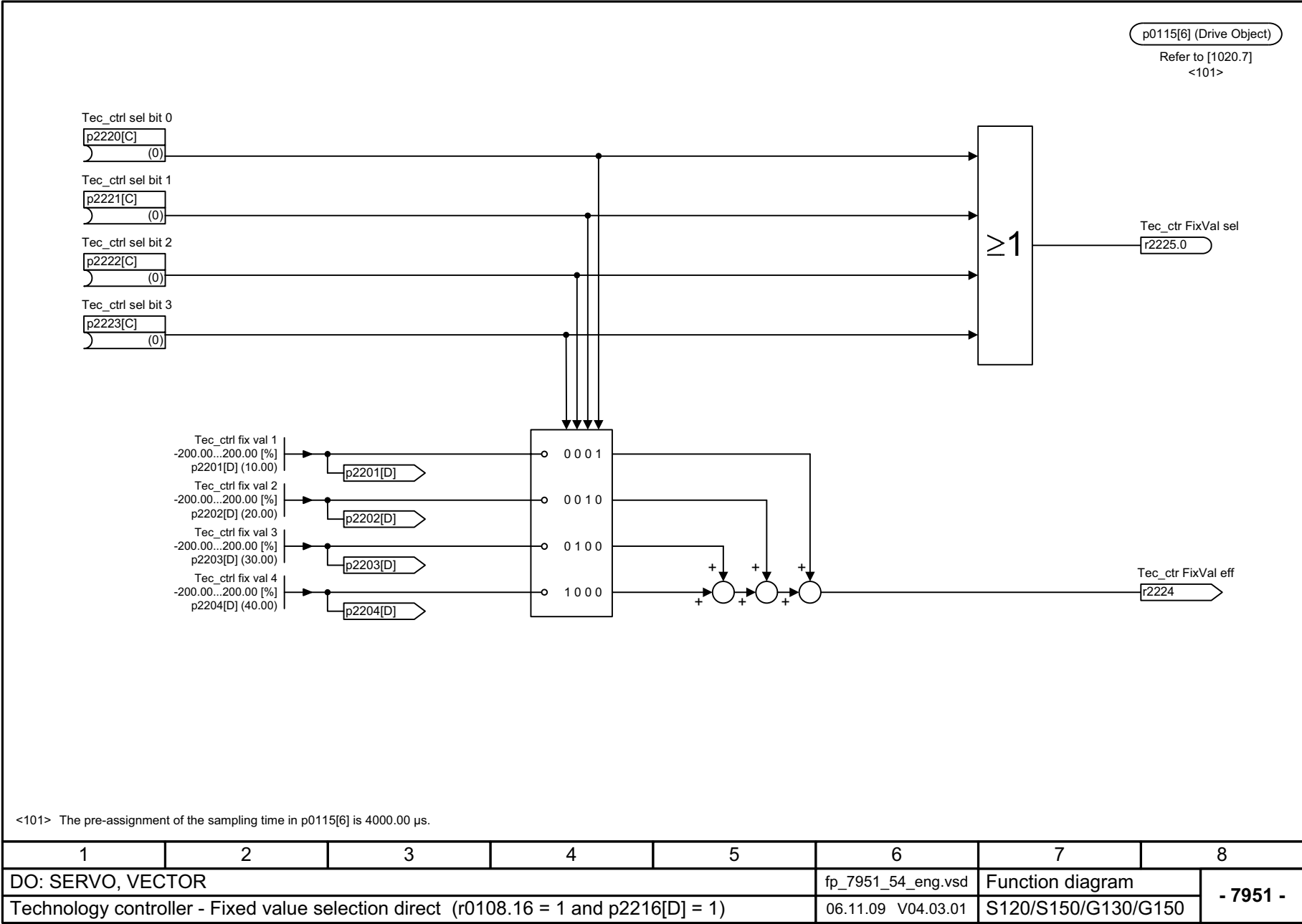
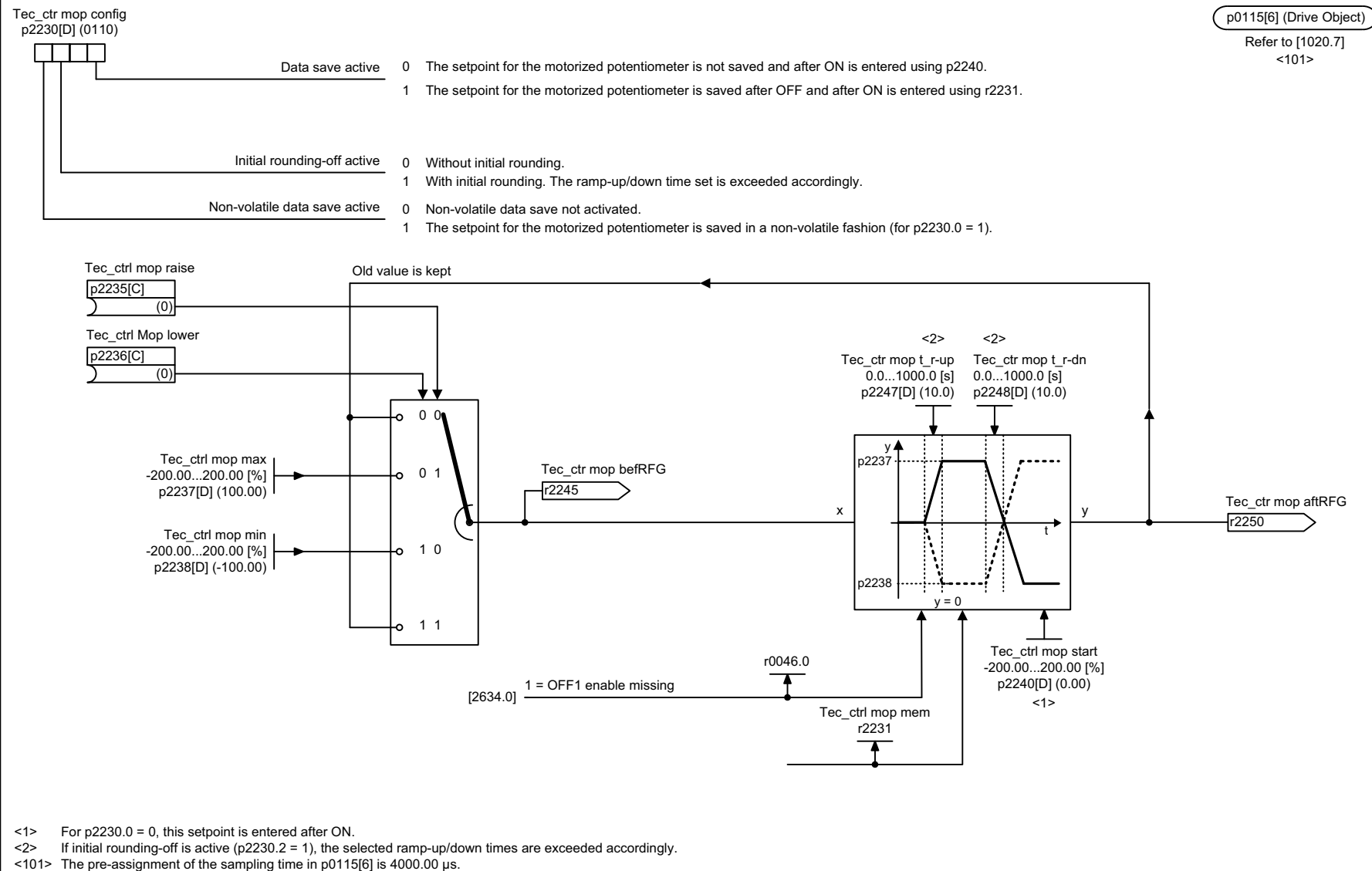
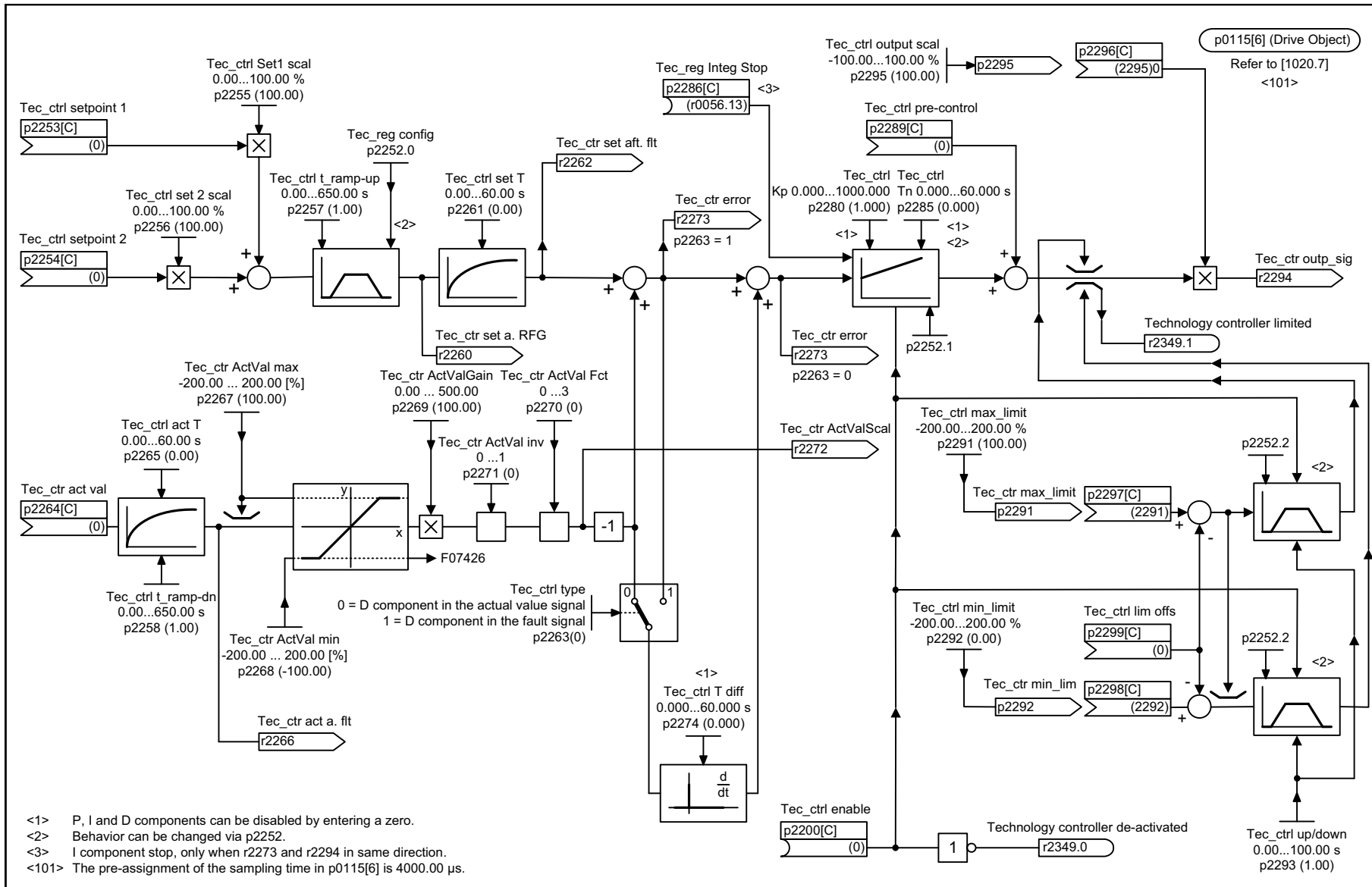


Fig. 2-218 7951 – Fixed values, direct selection (r0108.16 = 1 and p2216[D] = 1)



1	2	3	4	5	6	7	8
DO: SERVO, VECTOR, VECTORMV					fp_7954_51_eng.vsd	Function diagram	
Technology controller - Motorized potentiometer (r0108.16 = 1)					22.10.09 V04.03.01	SINAMICS	
							- 7954 -



1	2	3	4	5	6	7	8
DO: SERVO, VECTOR, VECTORMV					fp_7958_51_eng.vsd	Function diagram	
Technology controller - Closed-loop control (r0108.16 = 1)					10.12.09 V04.03.01	SINAMICS	

Fig. 2-220 7958 – Closed-loop control (r0108.16 = 1)

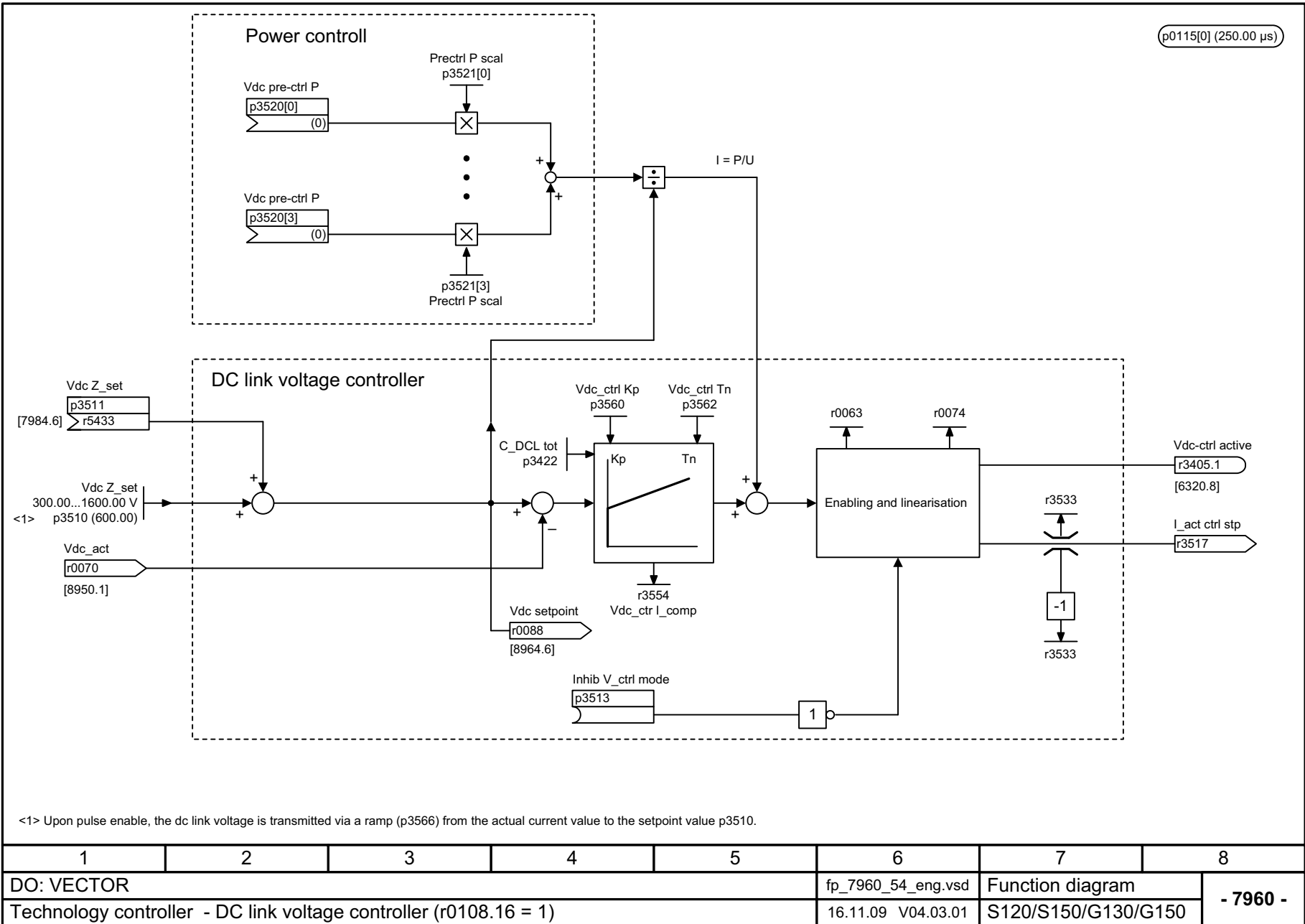


Fig. 2-221 7960 – DC-link voltage controller (r0108.16 = 1)

2.21 Signals and monitoring functions

Function diagrams

8010 – Speed messages 1	2-1627
8011 – Speed messages 2	2-1628
8012 – Torque messages, motor locked/stalled	2-1629
8013 – Load monitoring (r0108.17 = 1)	2-1630
8014 – Thermal monitoring, power unit	2-1631
8016 – Thermal monitoring, motor	2-1632
8017 – Thermal I2t motor model (PEM, p0300 = 2xx)	2-1633
8018 – Separately excited synchronous motor (FEM, p0300 = 5)	2-1634

Fig. 2-222 8010 – Speed messages 1

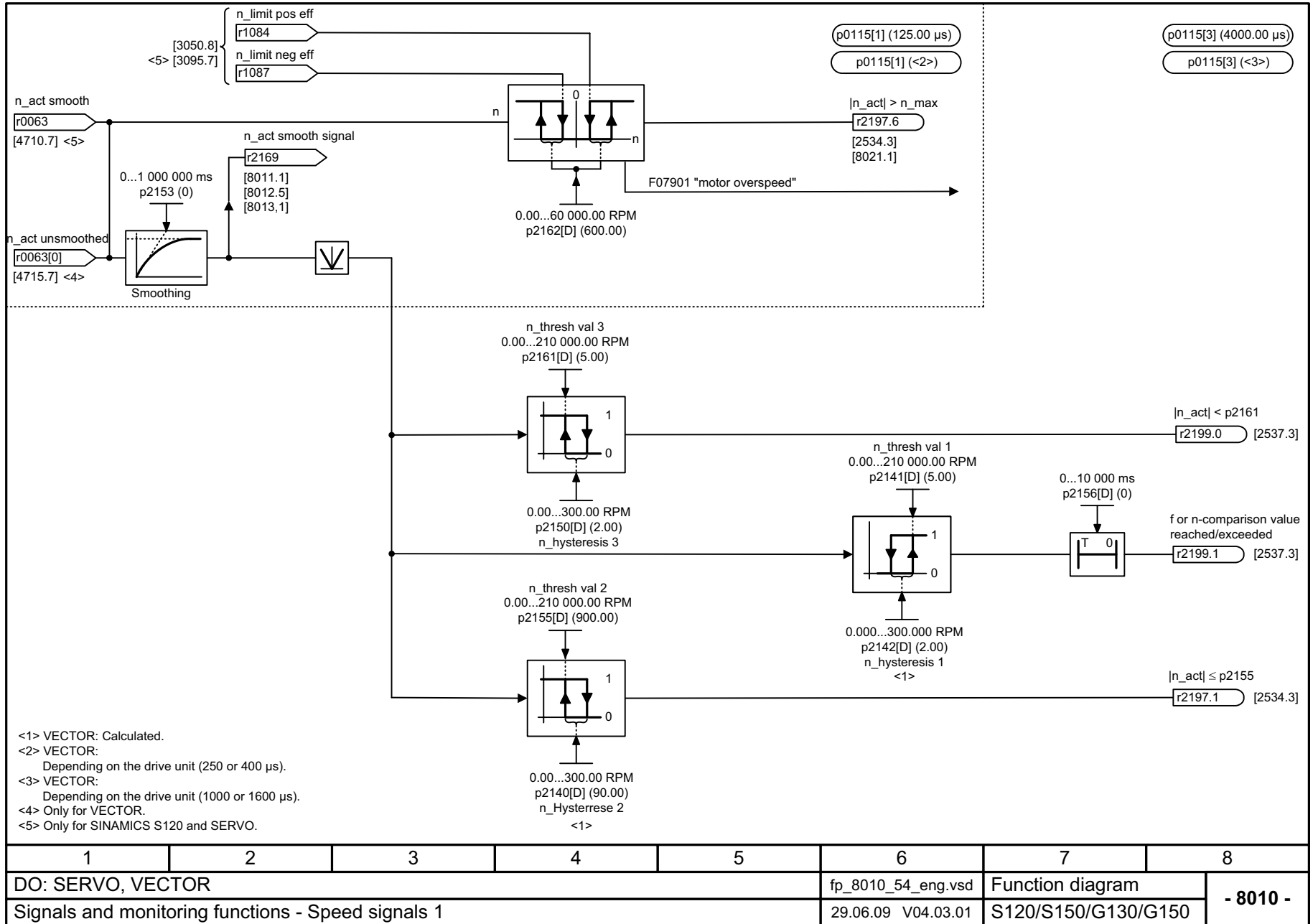
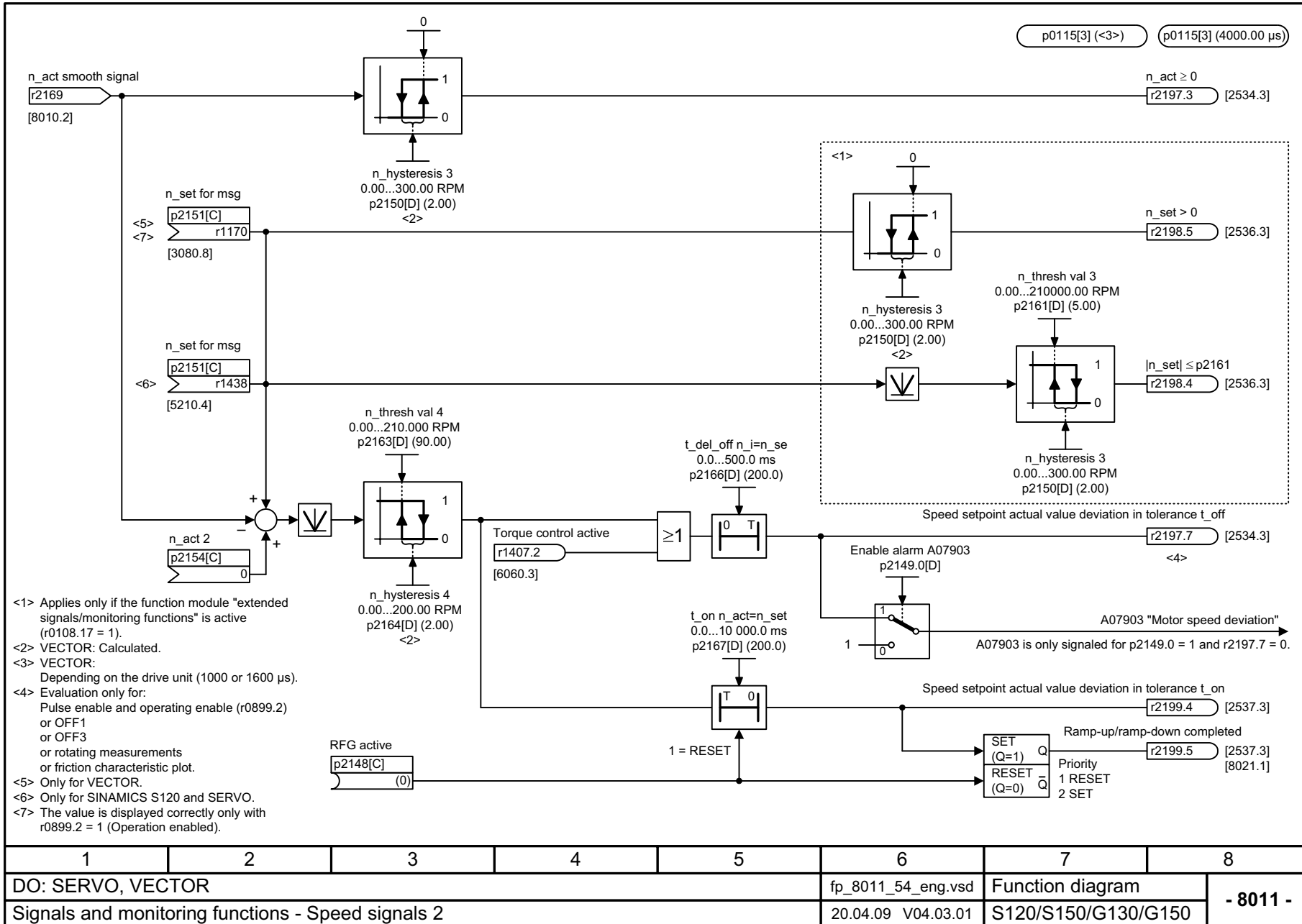
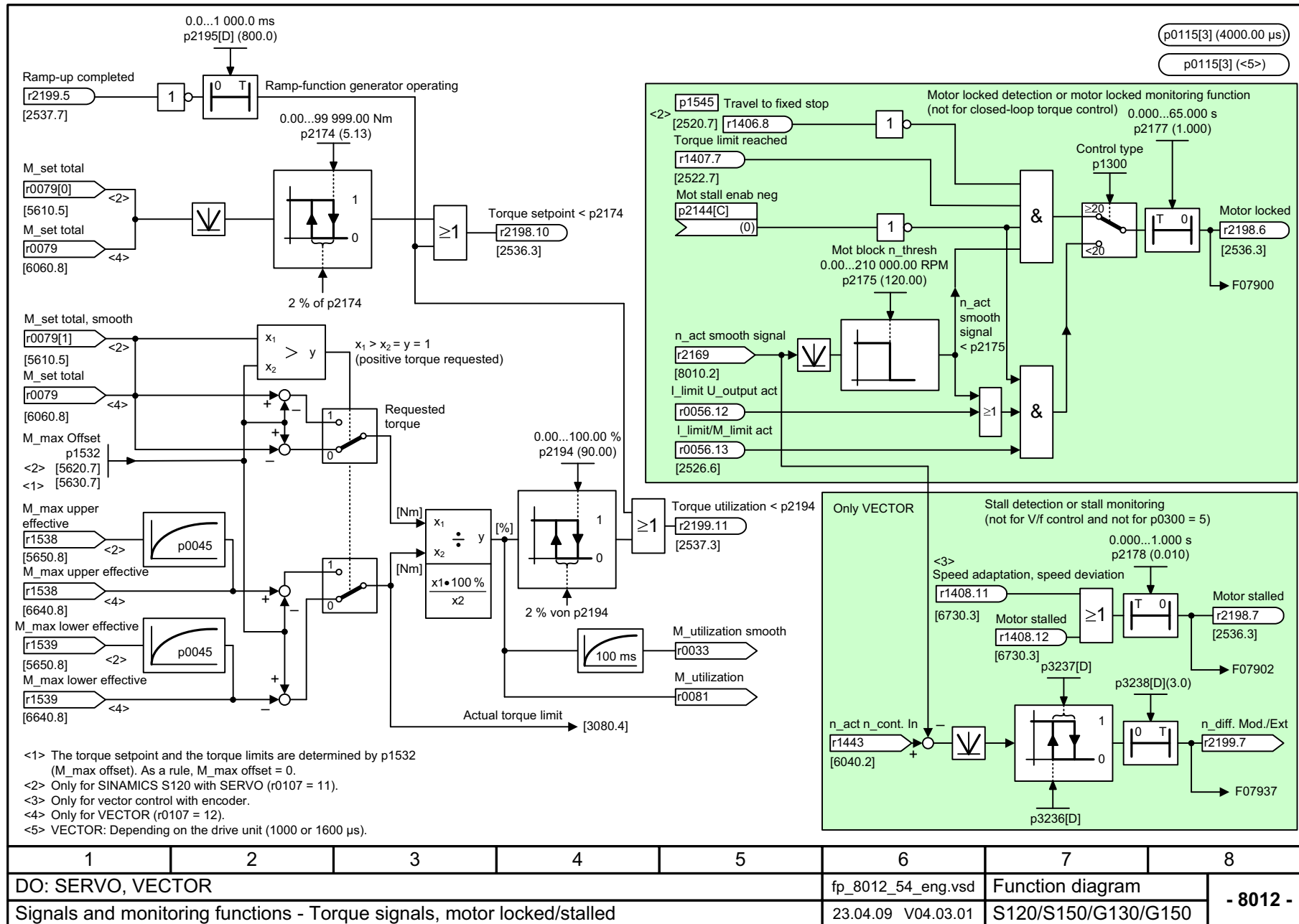
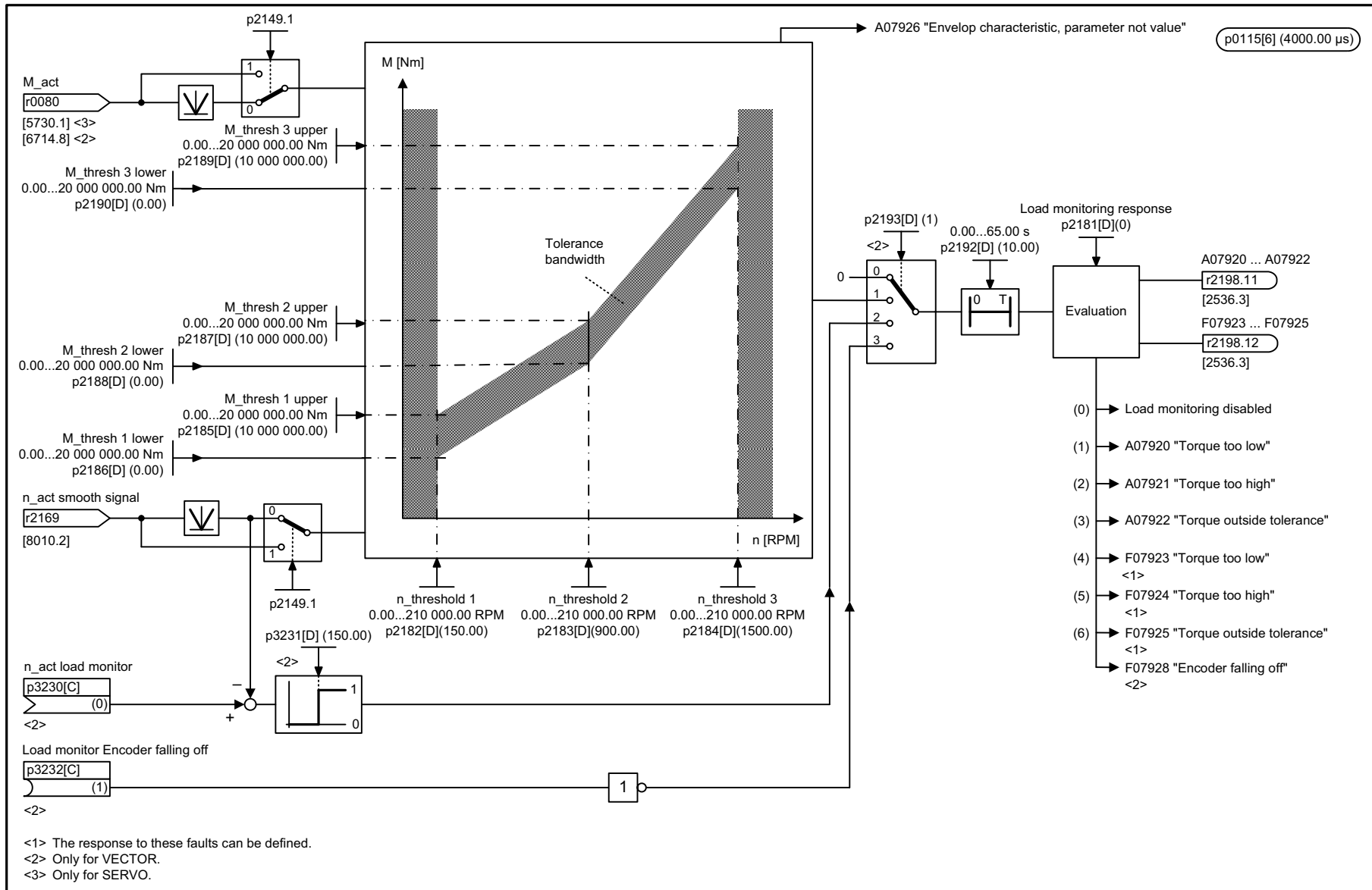


Fig. 2-223

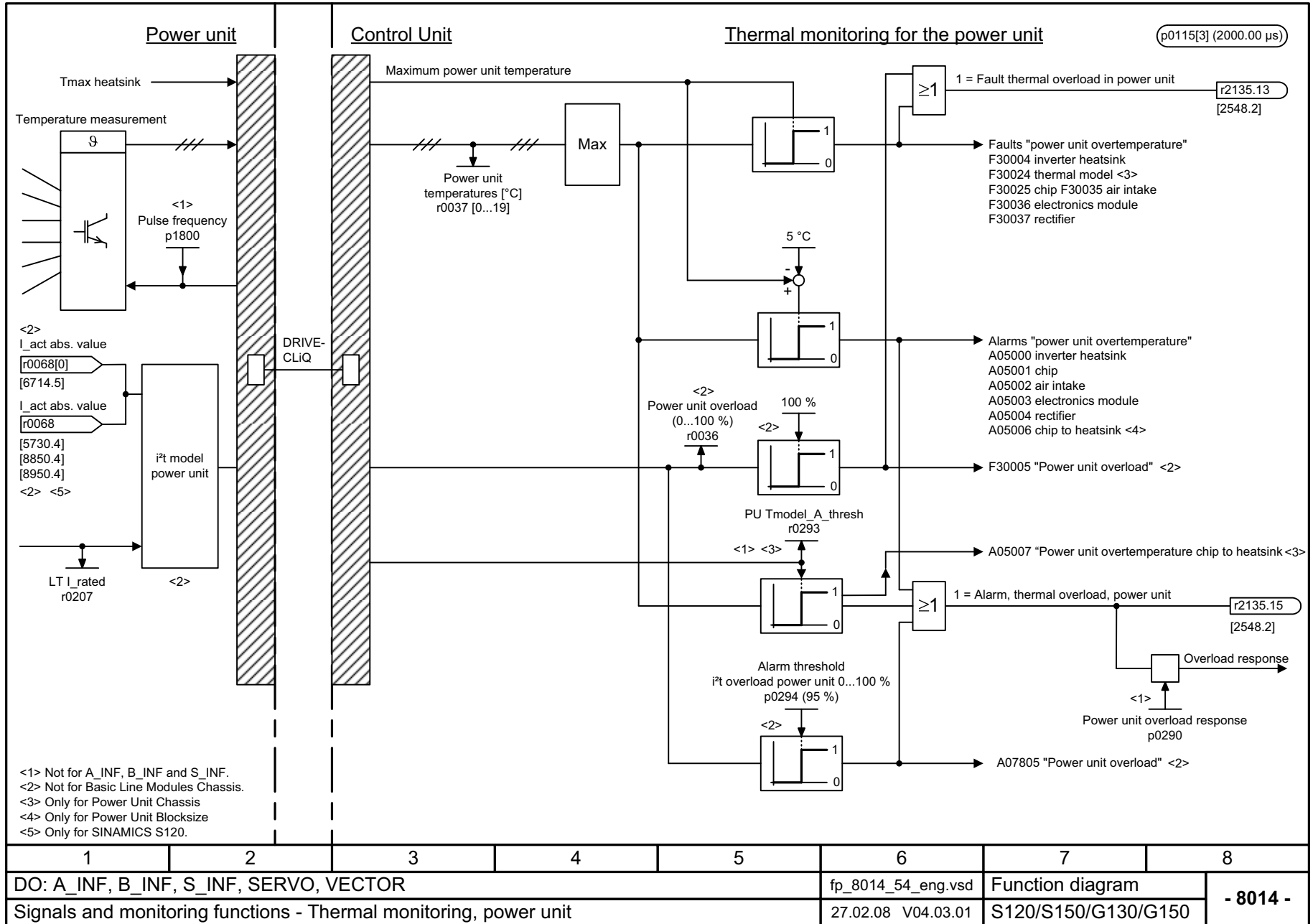


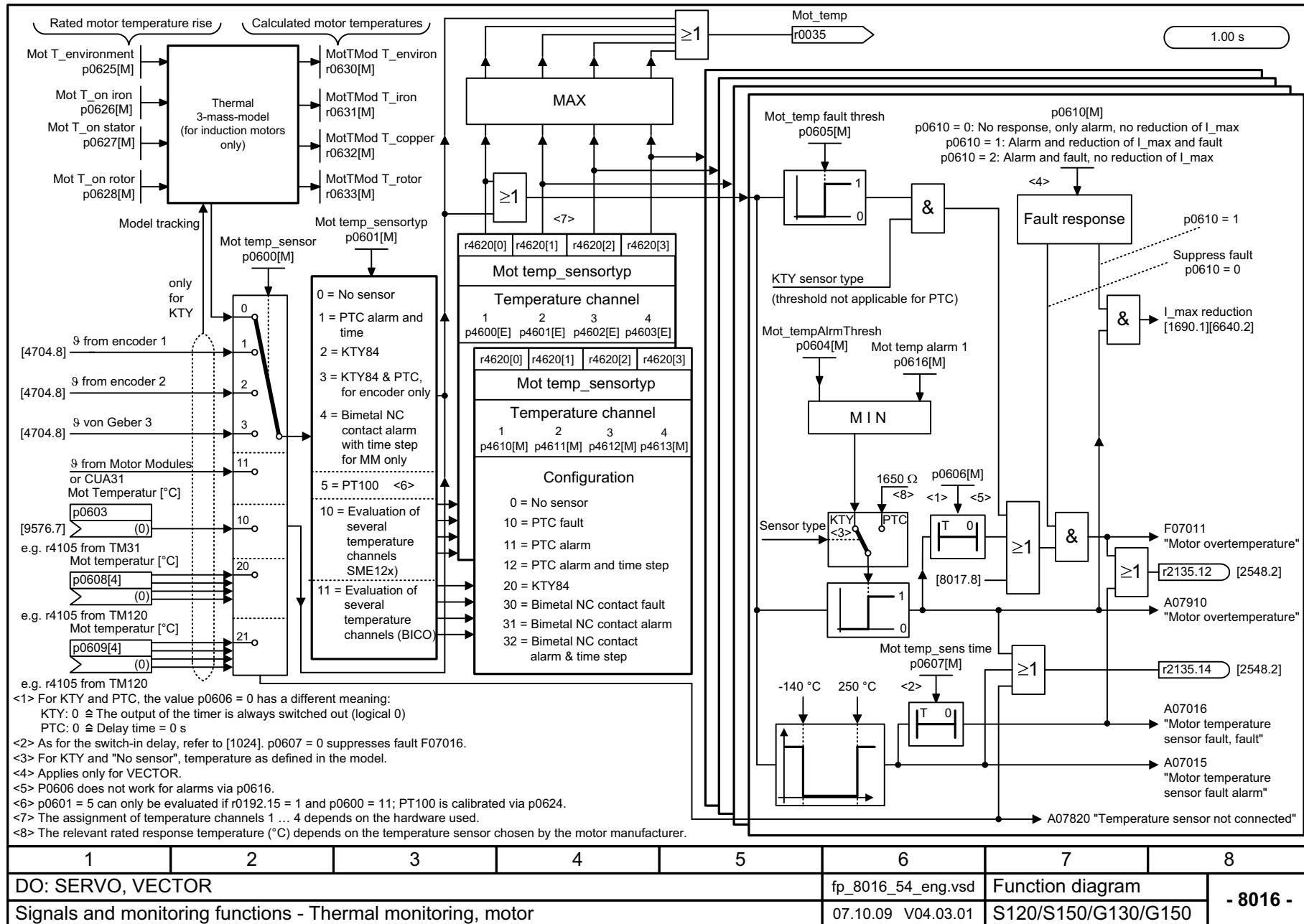




1	2	3	4	5	6	7	8	
DO: SERVO, VECTOR					fp_8013_54_eng.vsd	Function diagram		- 8013 -
Signals and monitoring functions - Load monitoring (r0108.17 = 1)					04.11.09 V04.03.01	S120/S150/G130/G150		

Fig. 2-226 8014 – Thermal monitoring, power unit





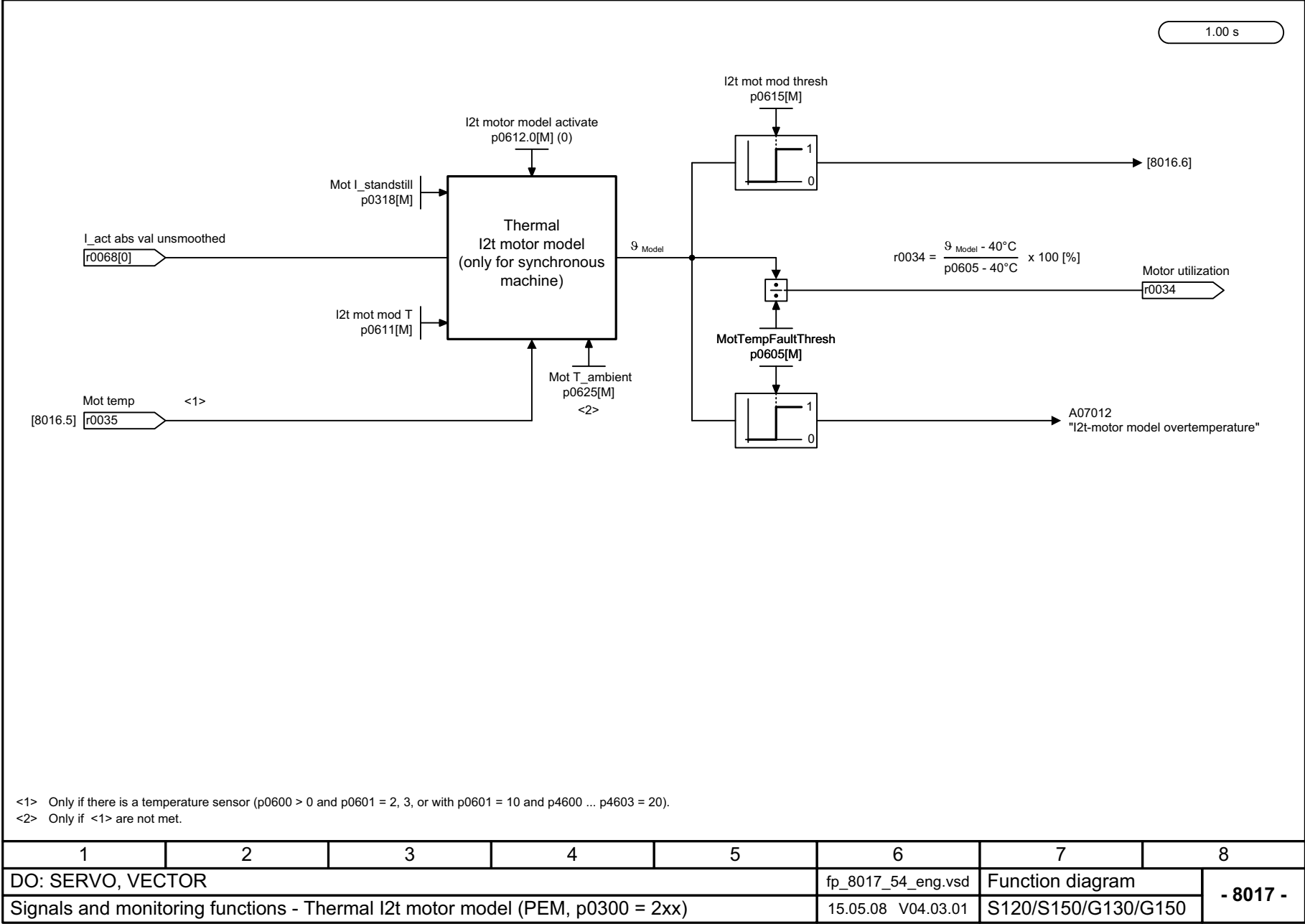


Fig. 2-228 8017 – Thermal I2t motor model (PEM, p0300 = 2xx)

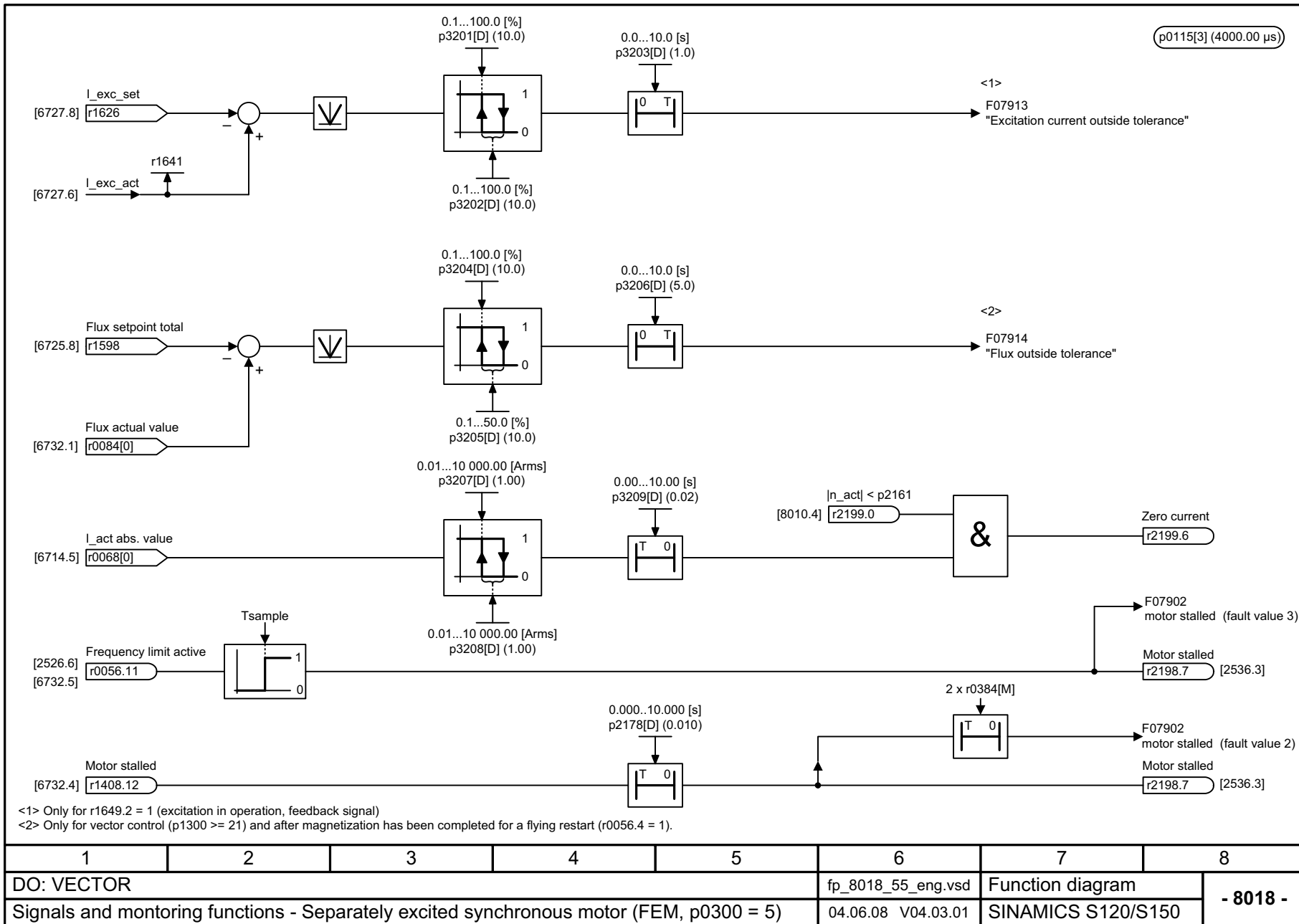
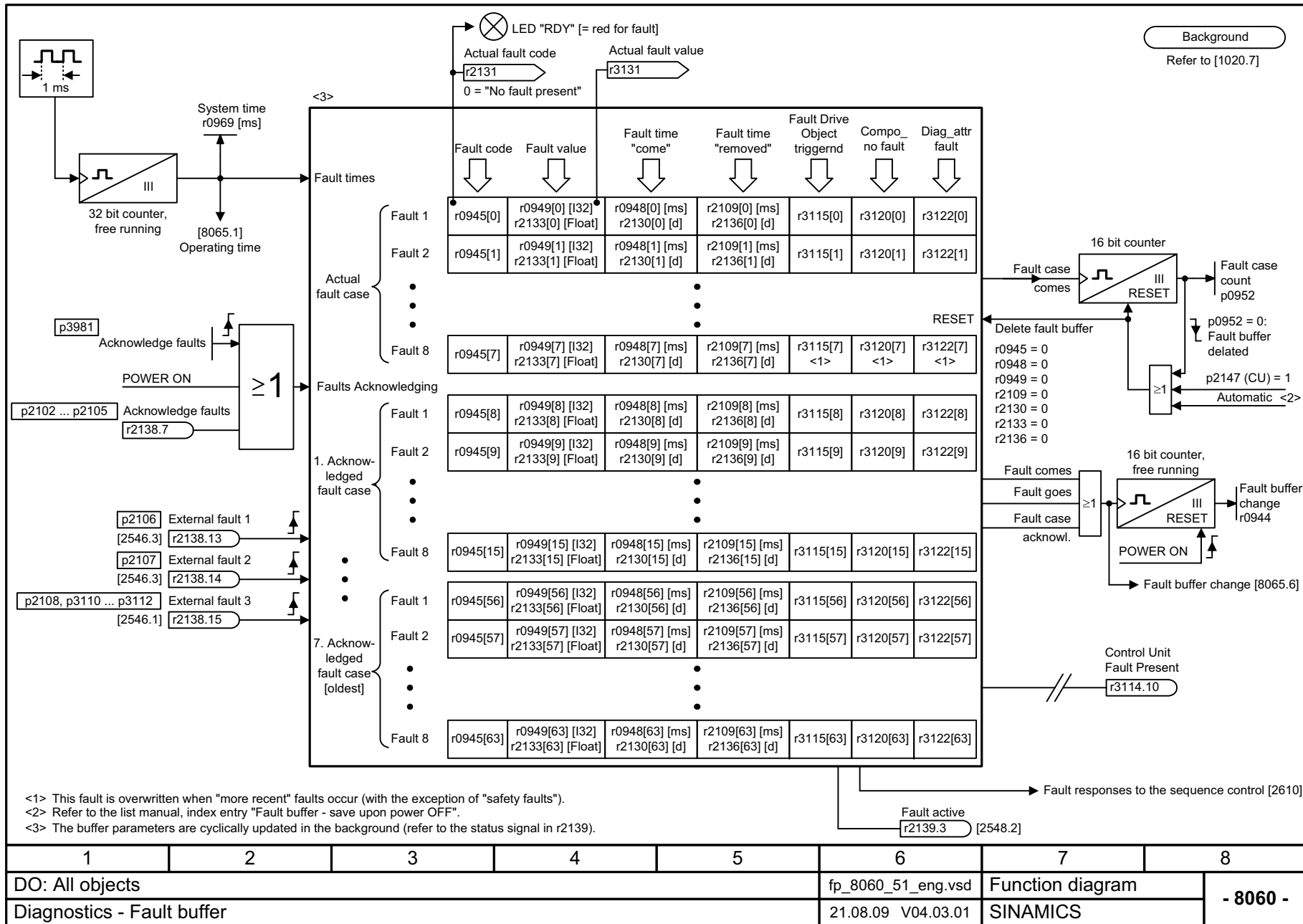


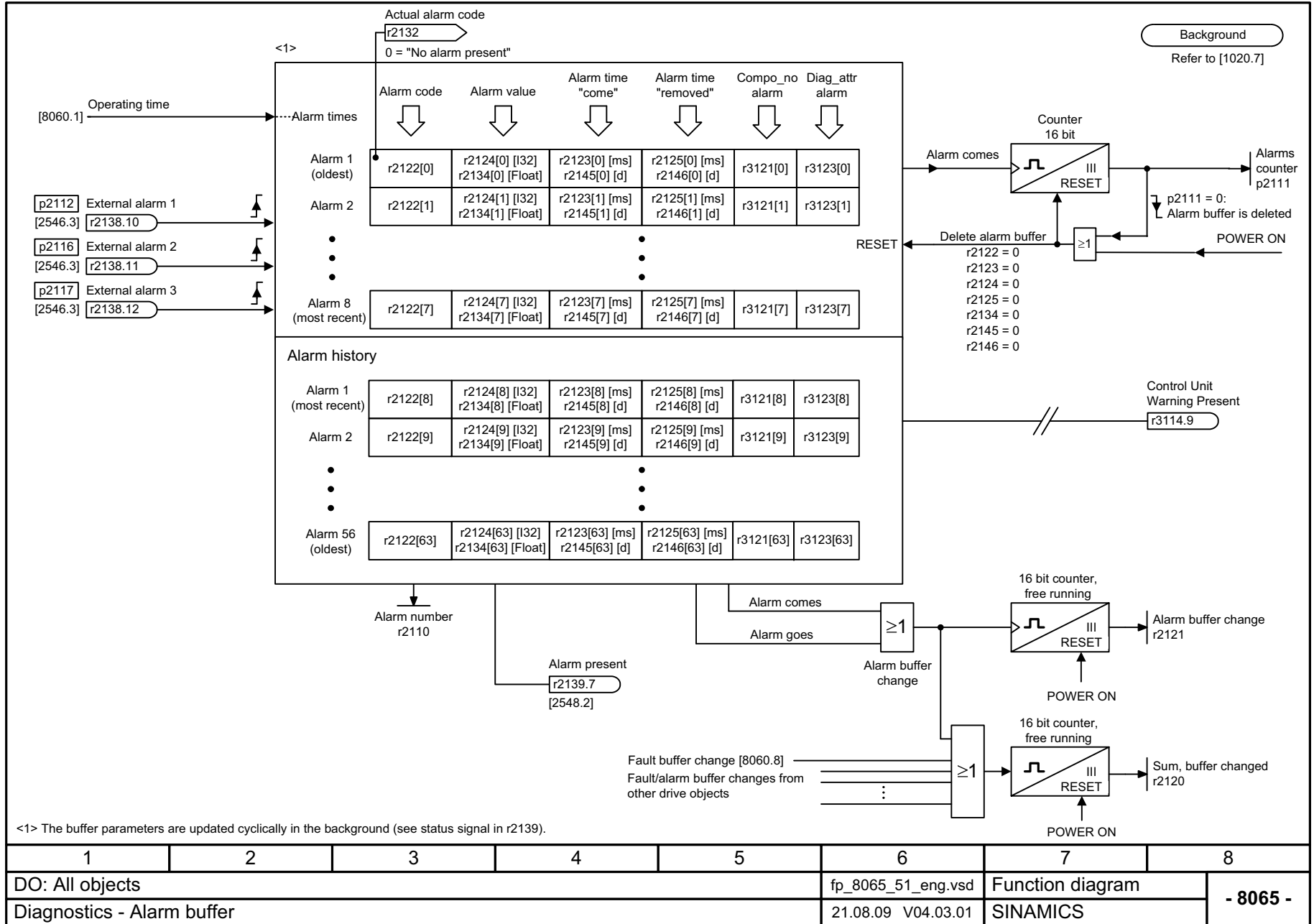
Fig. 2-229 8018 – Separately excited synchronous motor (FEM, p0300 = 5)

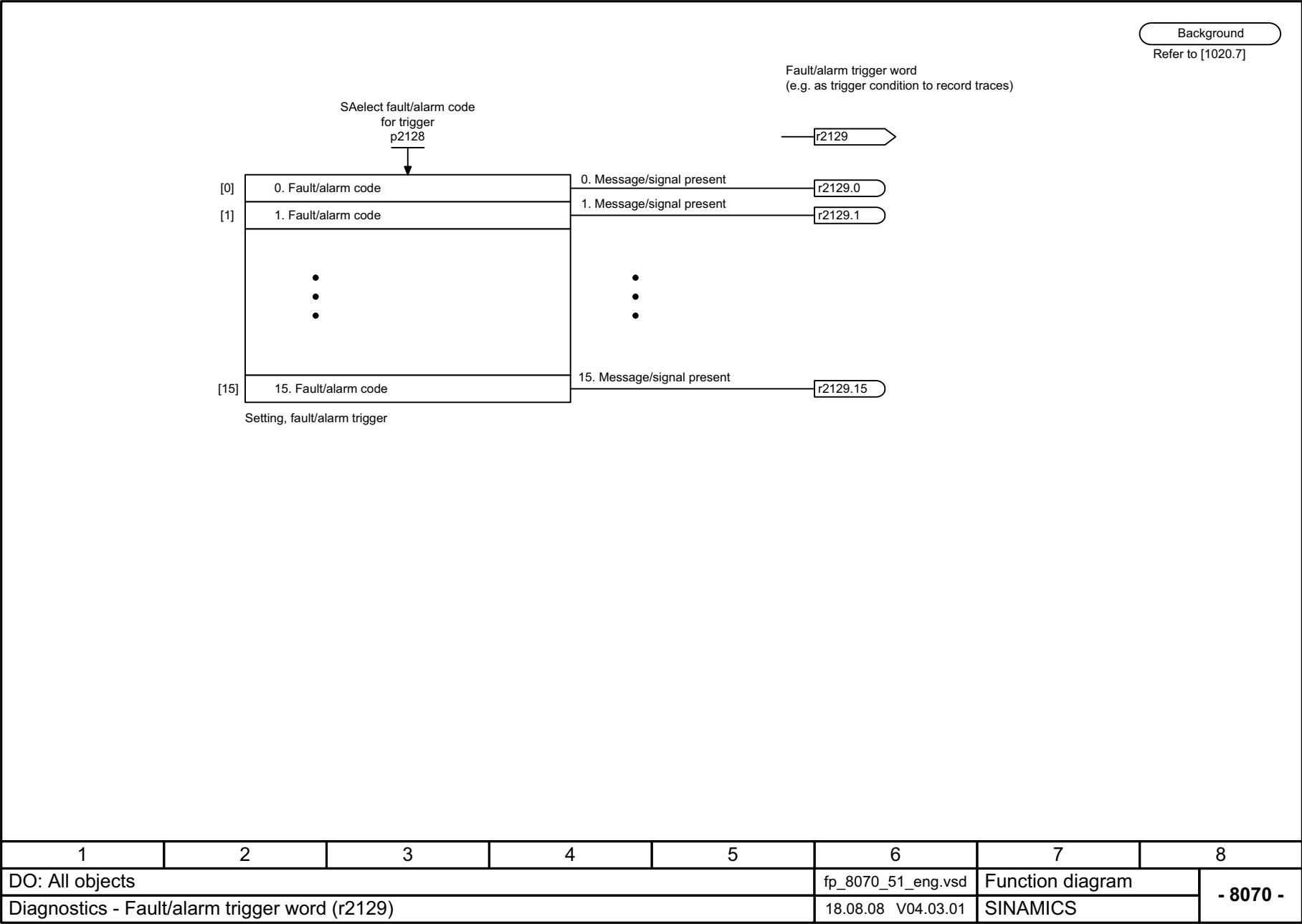
2.22 Diagnostics

Function diagrams

8060 – Fault buffer	2-1636
8065 – Alarm buffer	2-1637
8070 – Fault/alarm trigger word (r2129)	2-1638
8075 – Fault/alarm configuration	2-1639
8134 – Measuring sockets	2-1640

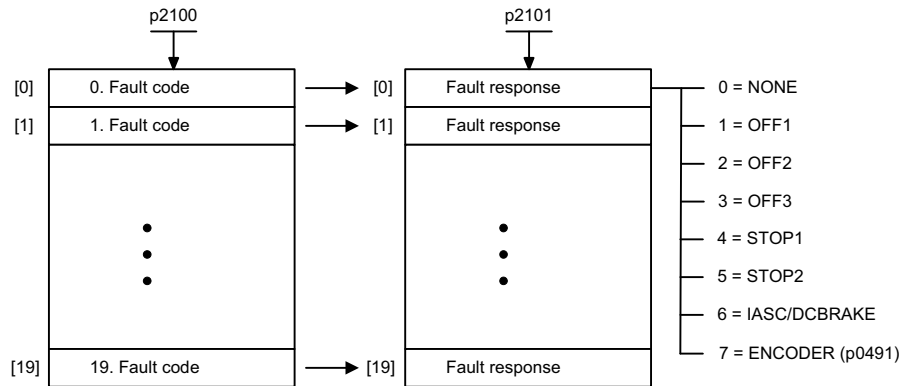




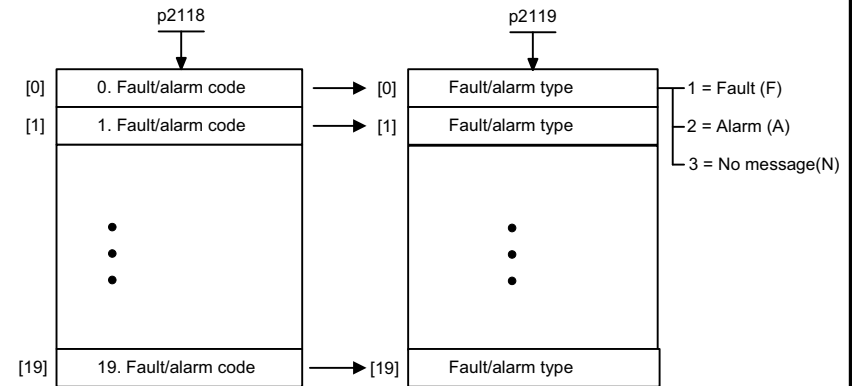


Background
Refer to [1020.7]

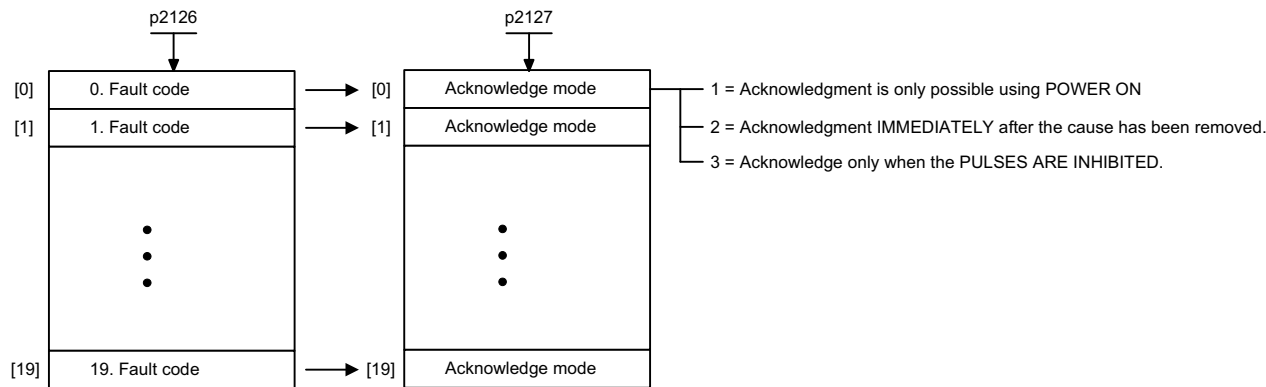
Changing the fault response for maximum 20 faults <1>



Changing the message type - fault <==> alarm for maximum 20 faults/alarms <1>



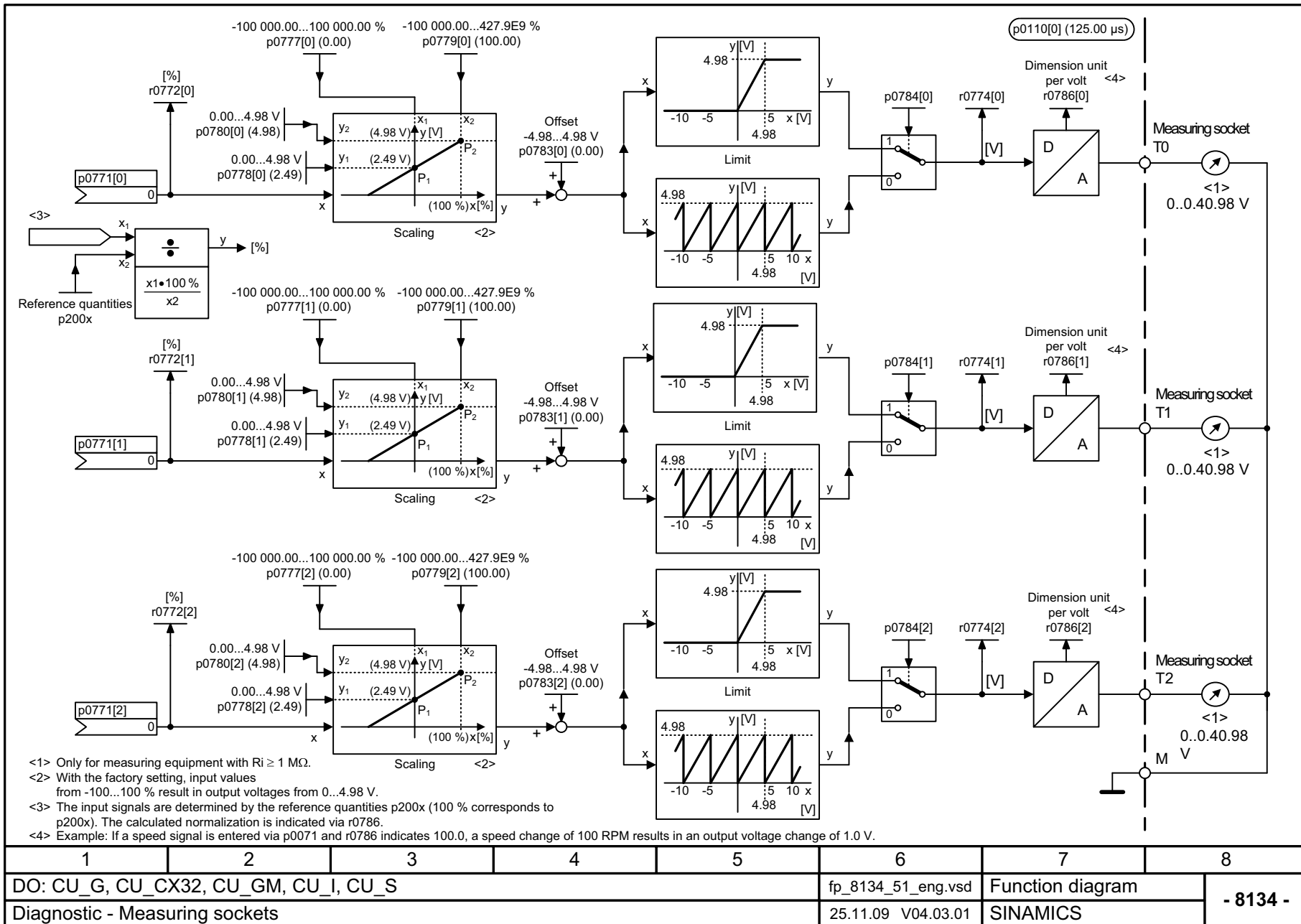
Changing the acknowledge mode for maximum 20 faults <1>



<1> The fault response, acknowledge mode and message type for all faults and alarms are set to meaningful default values in the factory setting. Changes that may be required are only possible in specific value ranges specified by SIEMENS. When the message type is changed, the supplementary information is transferred from fault value r0949 to alarm value r2124 and vice versa.

1	2	3	4	5	6	7	8
DO: All objects					fp_8075_51_eng.vsd	Function diagram	- 8075 -
Diagnostics - Fault/alarm configuration					07.09.09 V04.03.01	SINAMICS	

Fig. 2-233 8075 – Fault/alarm configuration



2.23 Data sets

Function diagrams

8560 – Command data sets (CDS)	2-1642
8565 – Drive data sets (DDS)	2-1643
8570 – Encoder data sets (EDS)	2-1644
8575 – Motor data sets (MDS)	2-1645
8580 – Power unit data sets (PDS)	2-1646

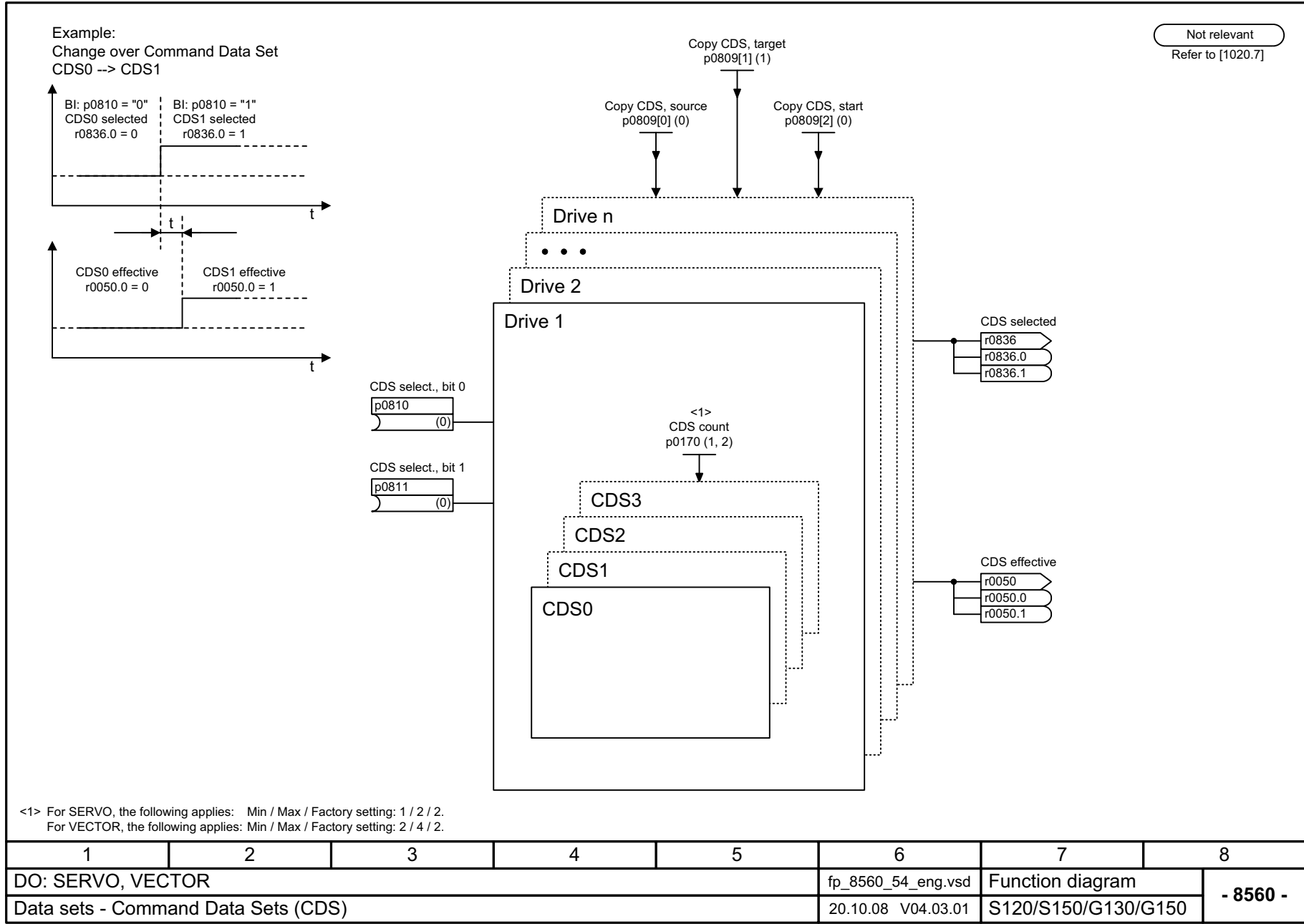
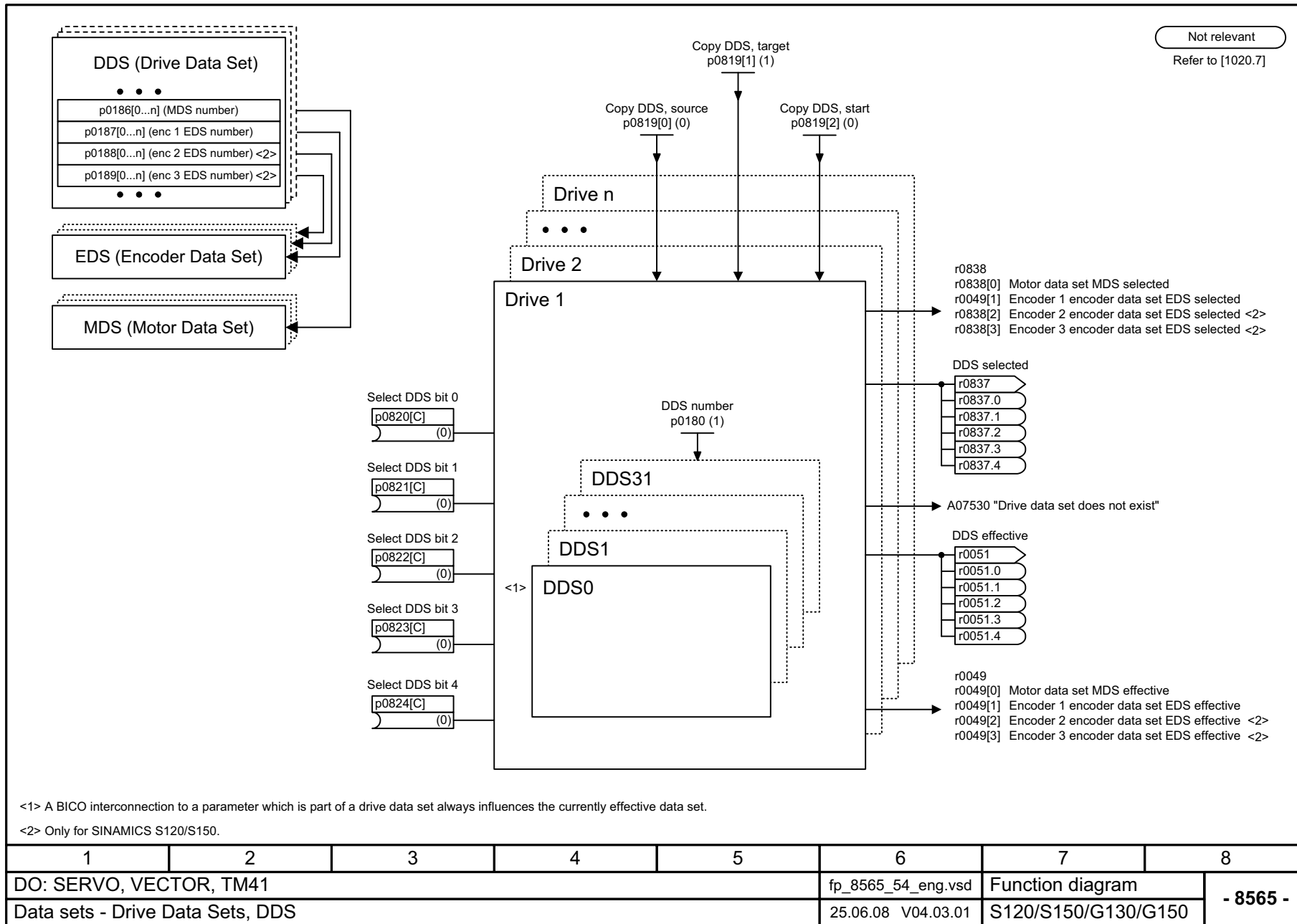


Fig. 2-235 8560 – Command data sets (CDS)



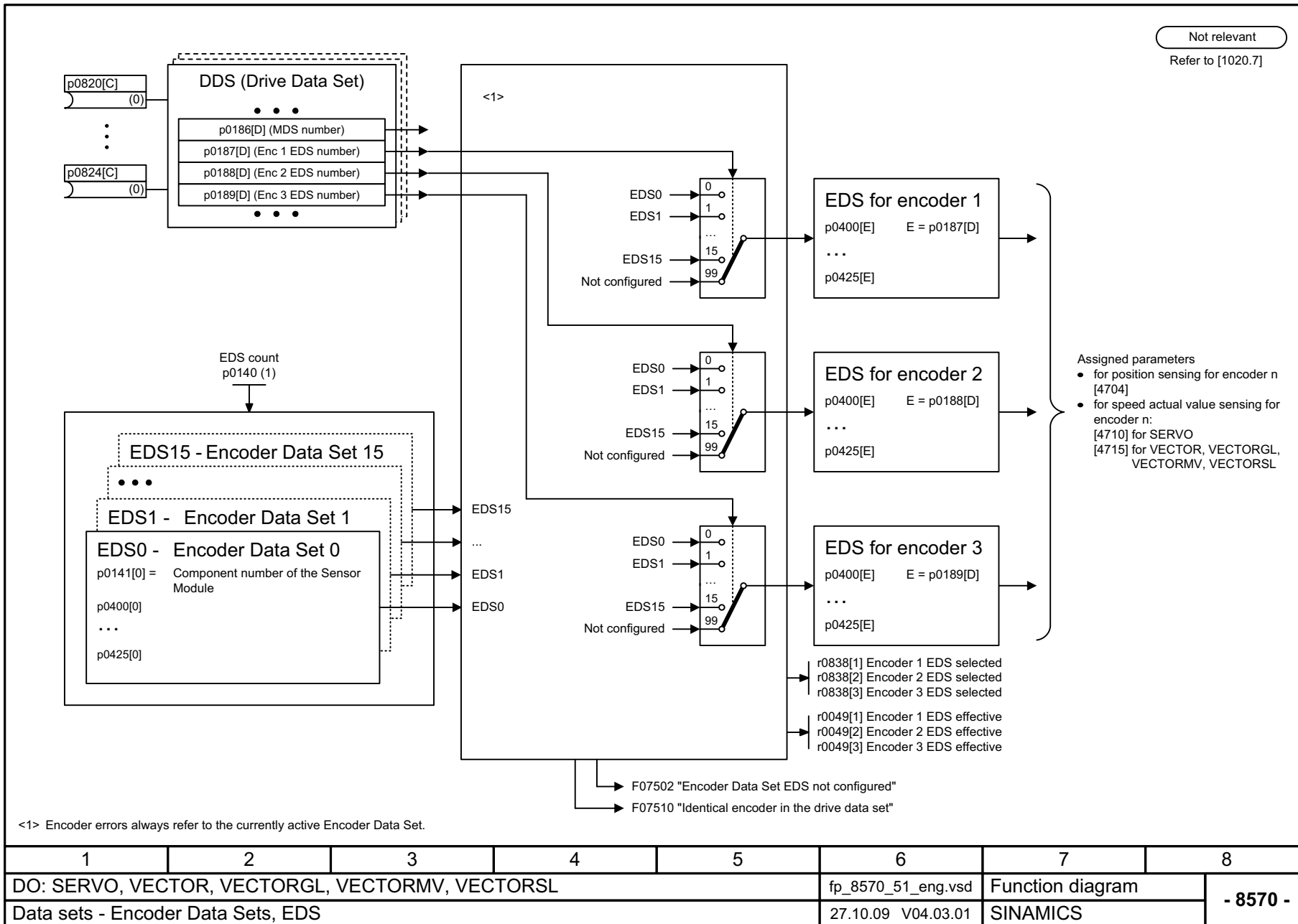
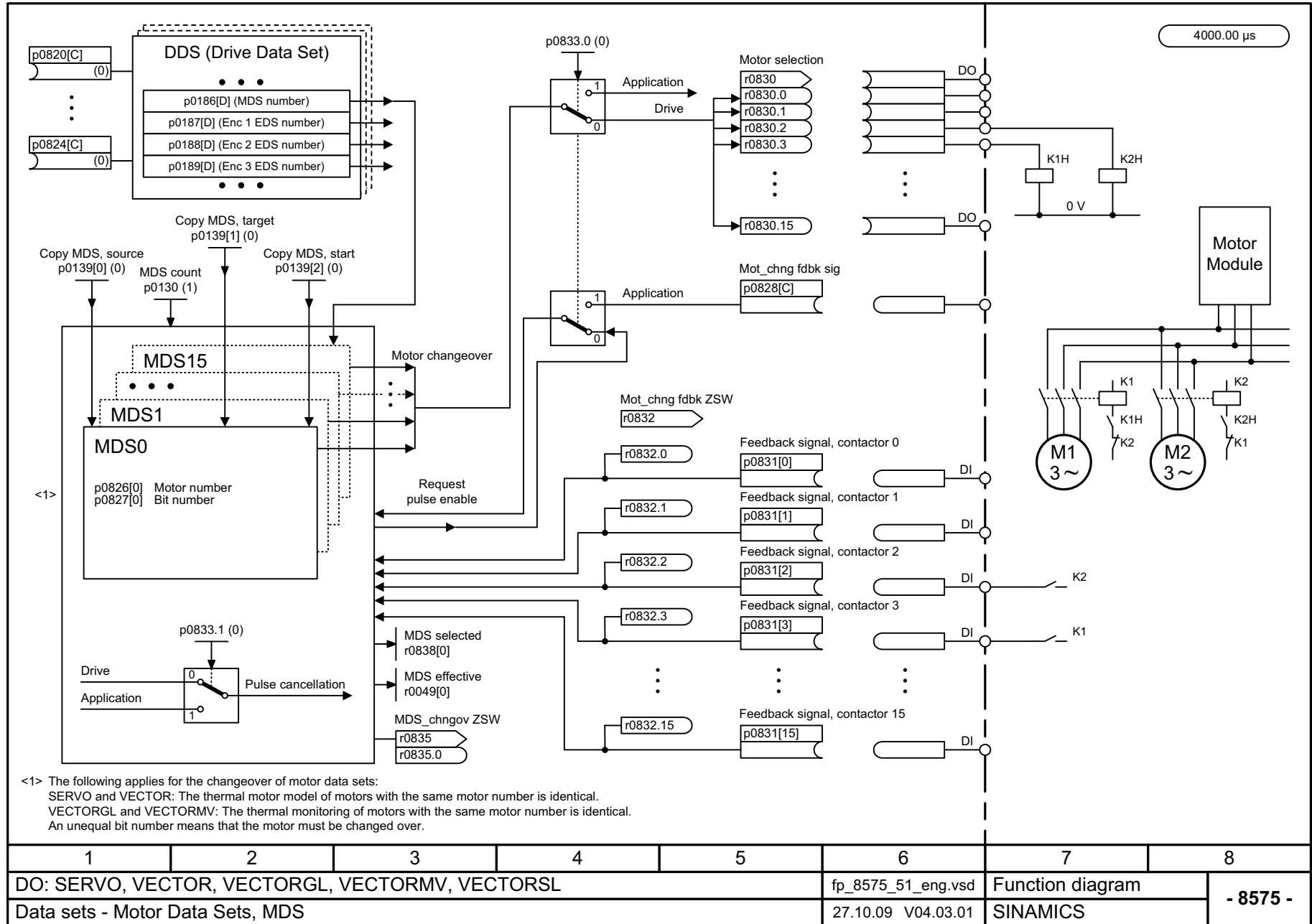


Fig. 2-237 8570 – Encoder data sets (EDS)



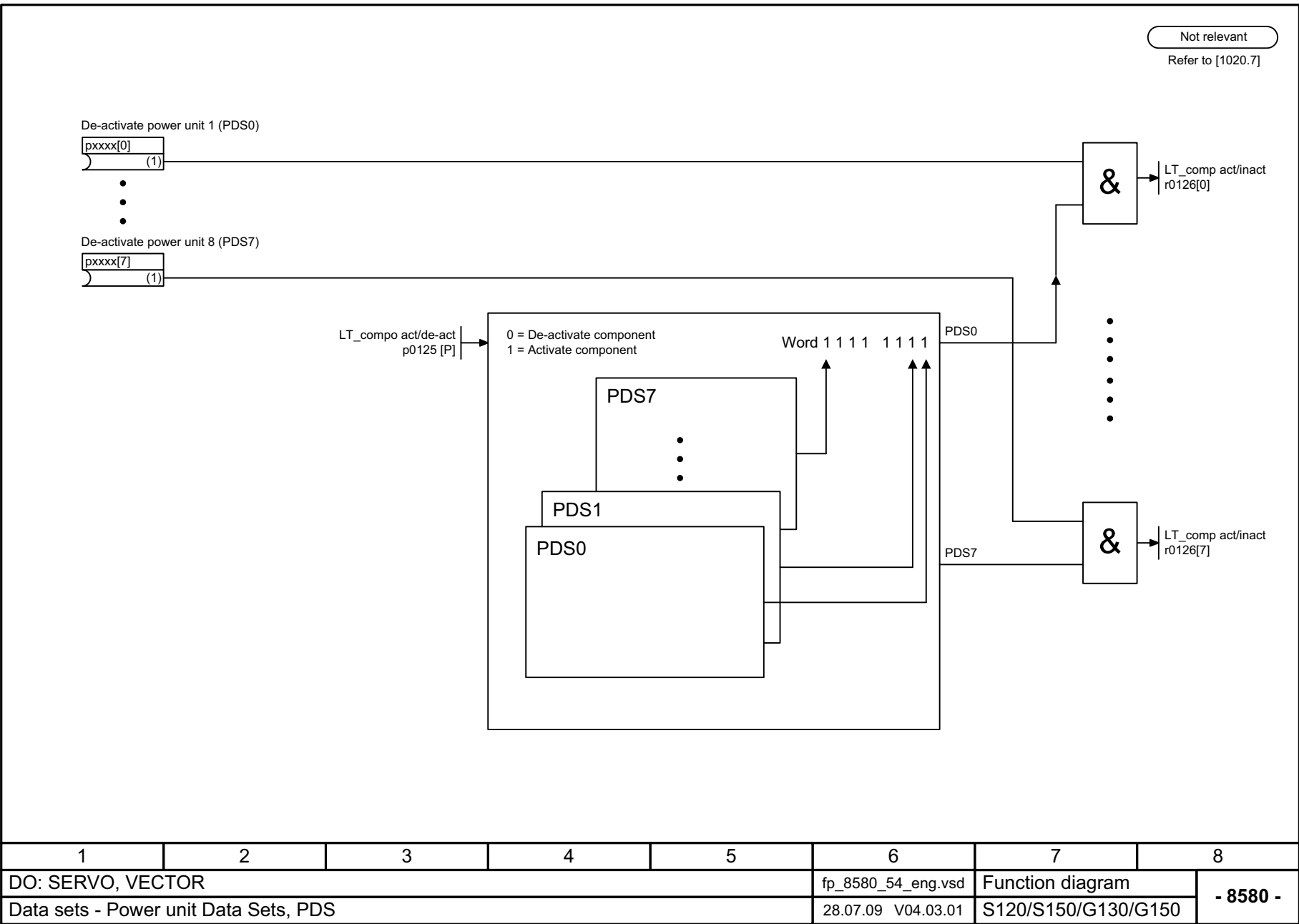


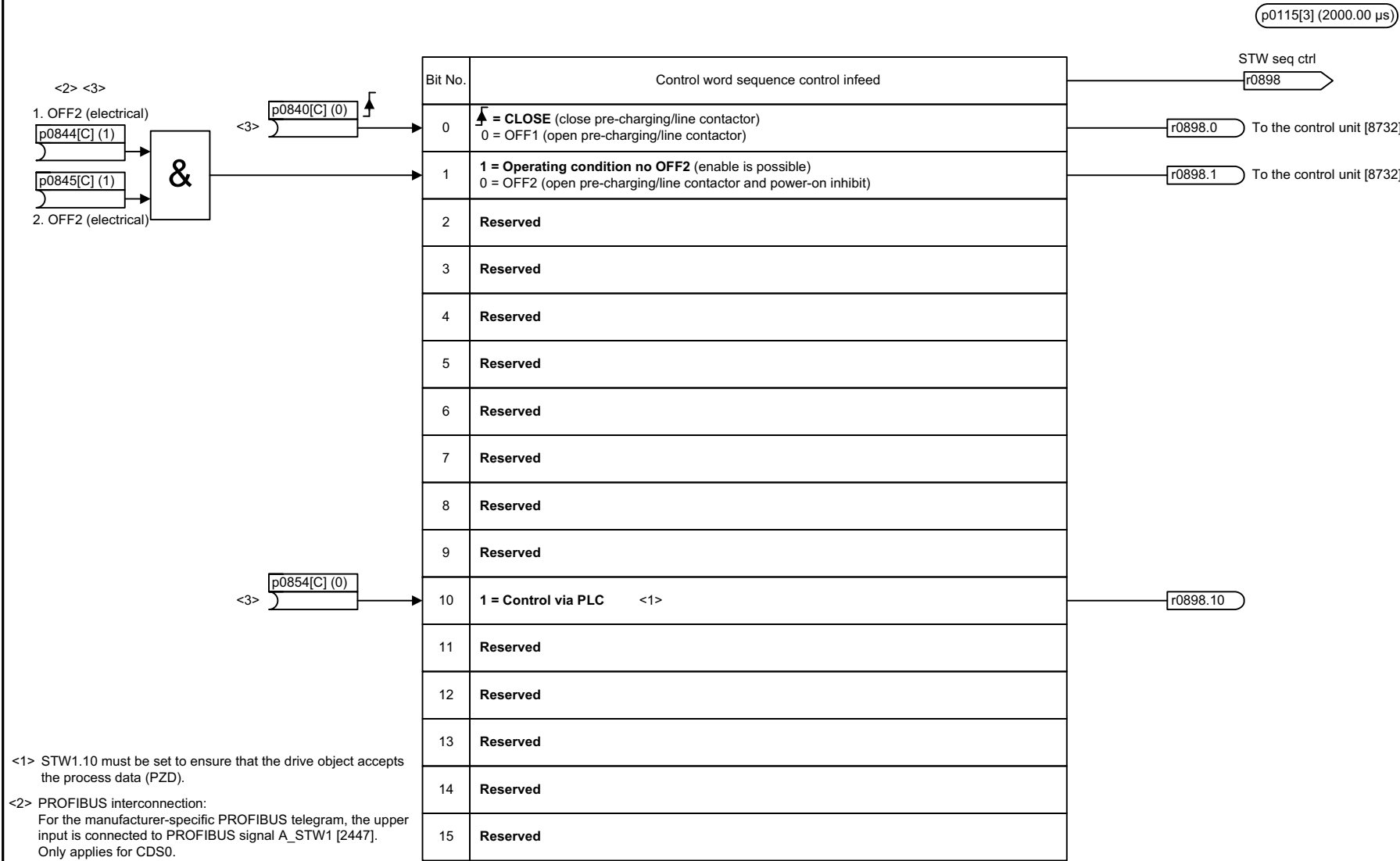
Fig. 2-239 8580 – Power unit data sets (PDS)

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR					fp_8580_54_eng.vsd	Function diagram	
Data sets - Power unit Data Sets, PDS					28.07.09 V04.03.01	S120/S150/G130/G150	
- 8580 -							

2.24 Basic Infeed

Function diagrams

8720 – Control word, sequence control, infeed	2-1648
8726 – Status word, sequence control, infeed	2-1649
8732 – Sequencer	2-1650
8734 – Missing enable signals, line contactor control	2-1651
8750 – Interface to the basic infeed power unit (control signals, actual values)	2-1652
8760 – Signals and monitoring functions (p3400.0 = 0)	2-1653



1	2	3	4	5	6	7	8
DO: B_INF					fp_8720_01_eng.vsd	Function diagram	
Basic Infeed - Control word, sequence control infeed					08.07.08 V04.03.01	SINAMICS S120	
							- 8720 -

Fig. 2-240 8720 – Control word, sequence control, infeed

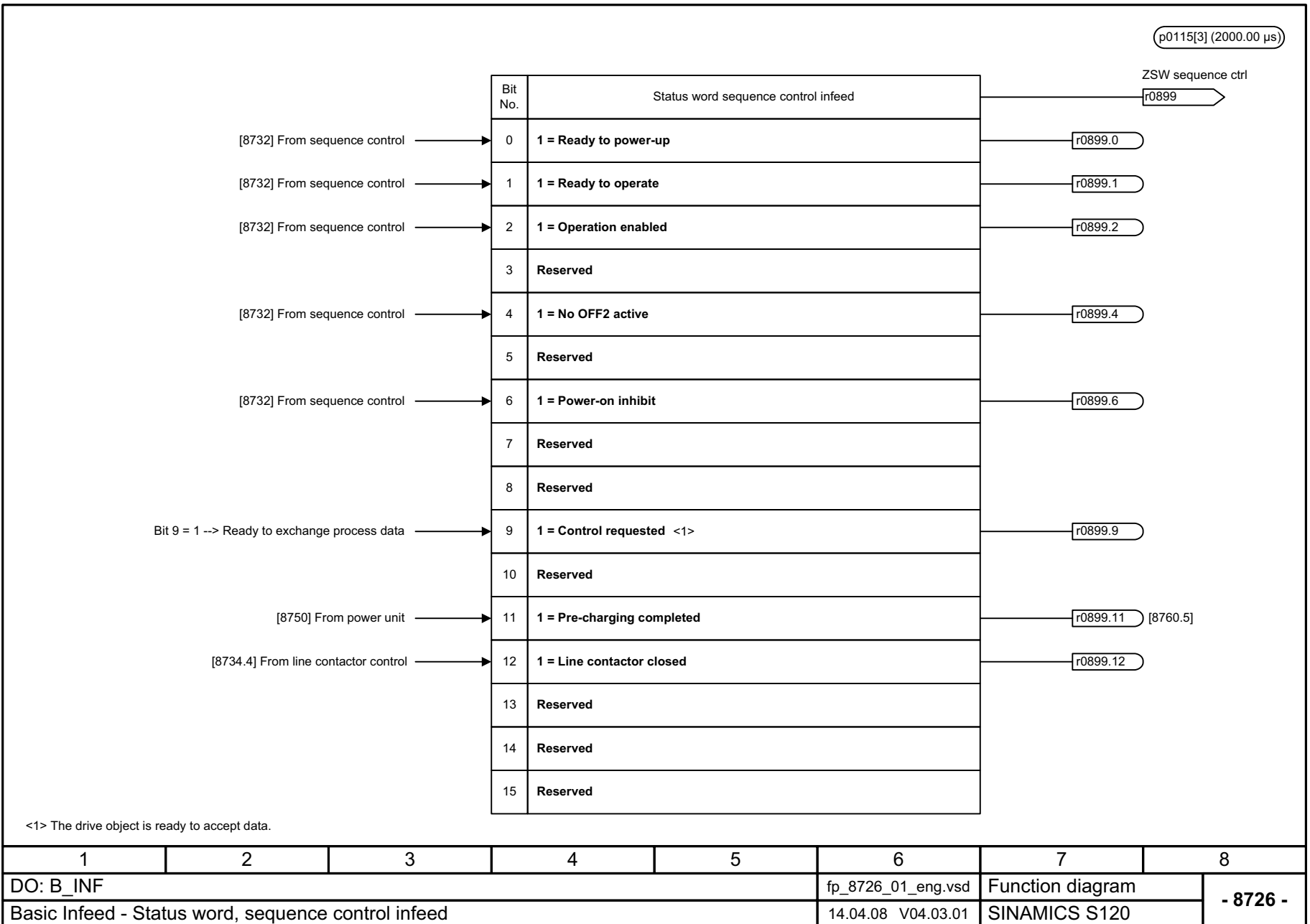


Fig. 2-241 8726 – Status word, sequence control, infeed

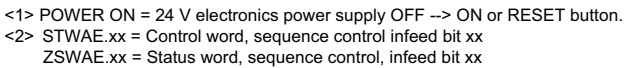


Fig. 2-242

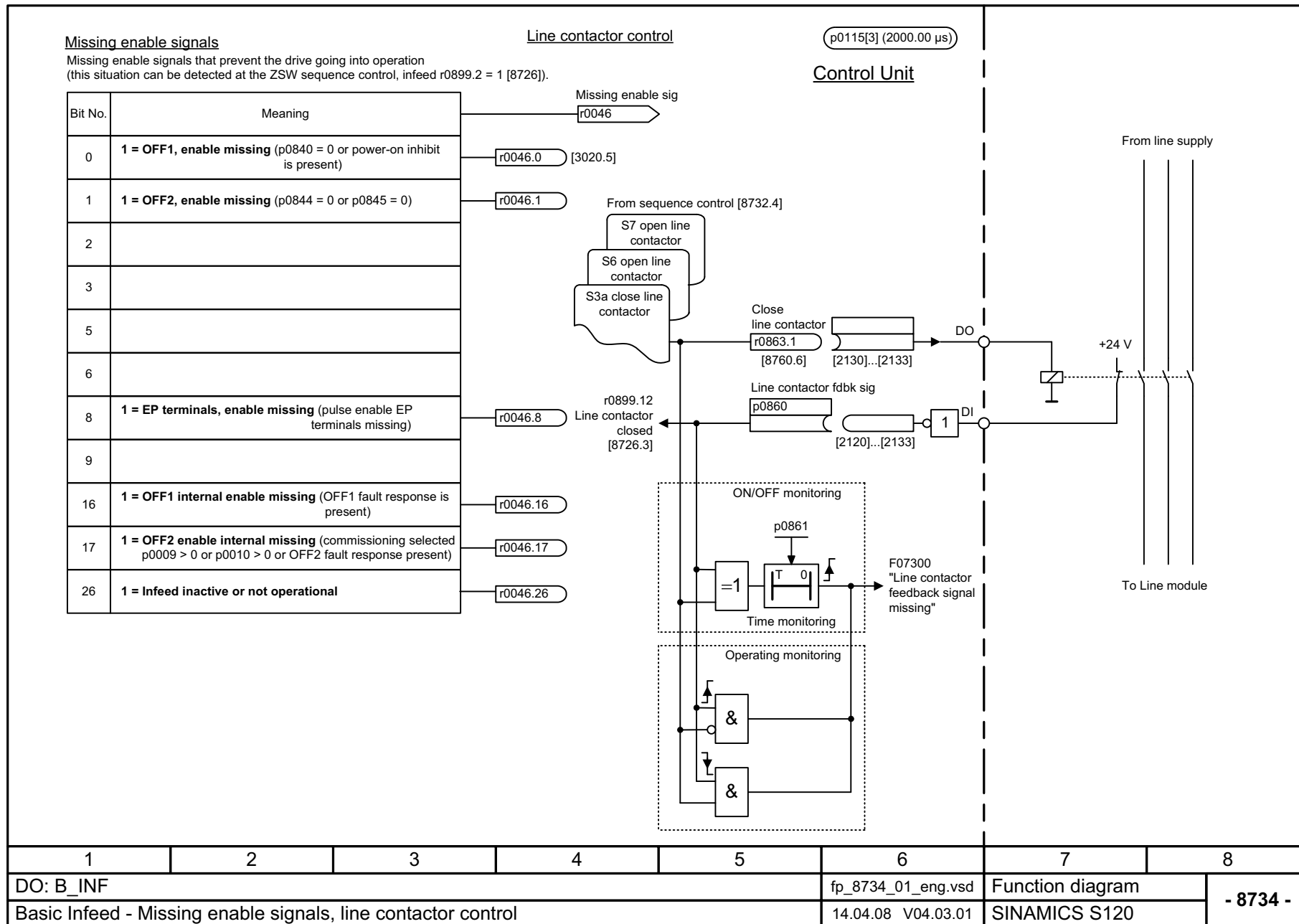
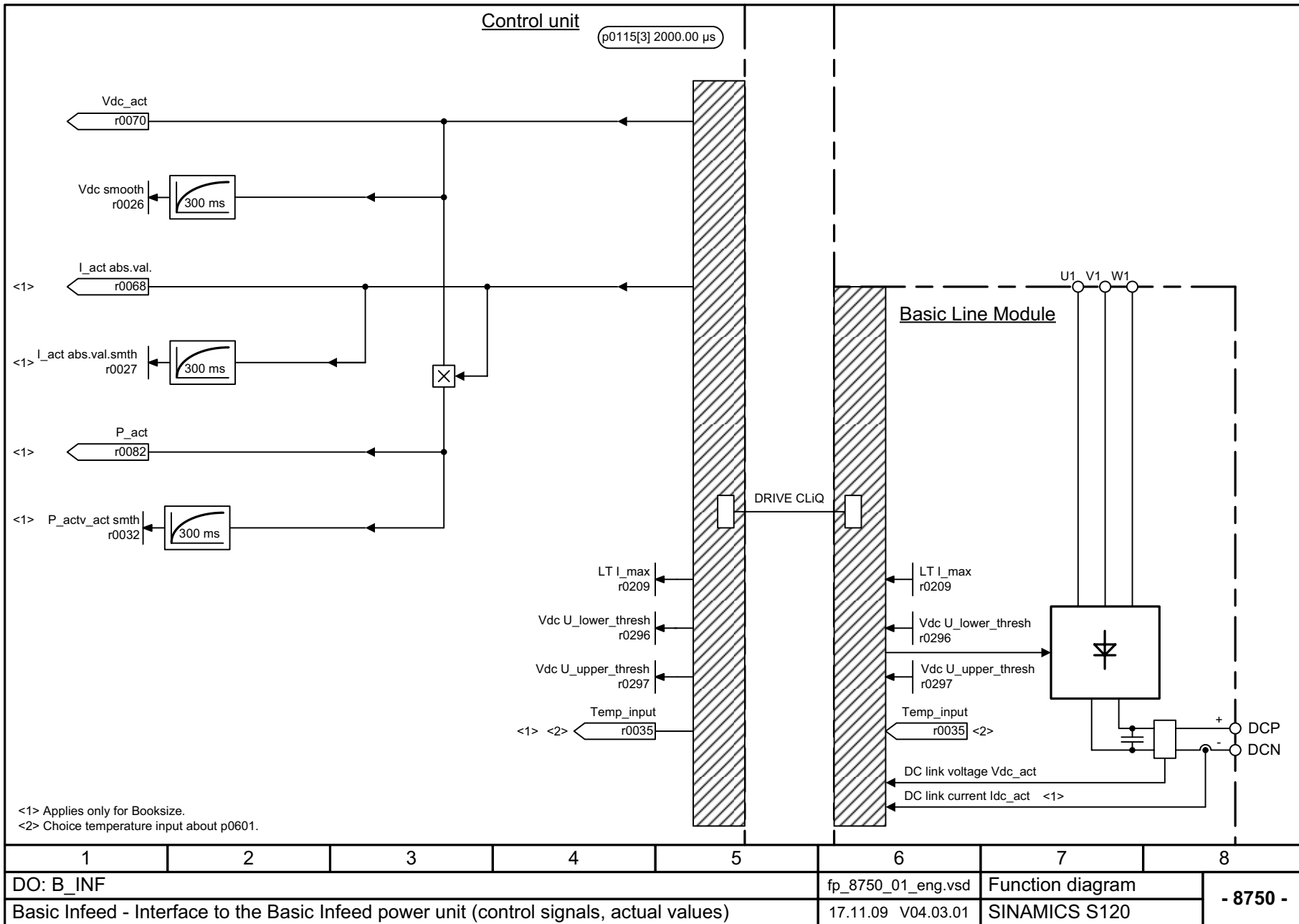
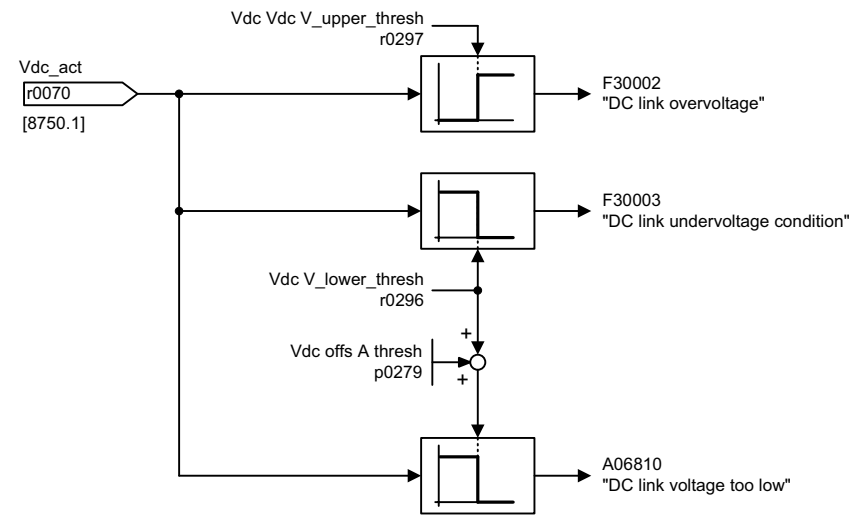


Fig. 2-243 8734 – Missing enable signals, line contactor control

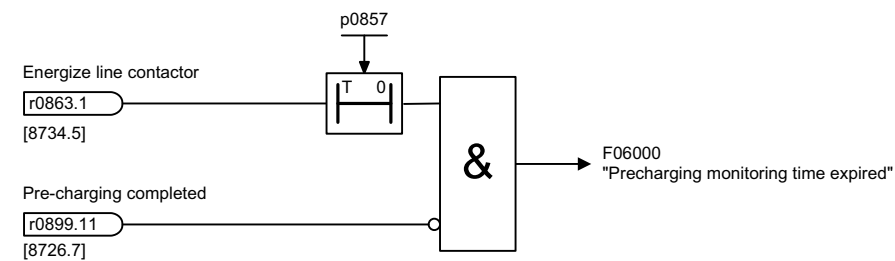


p0115[3] (2000.00 µs)

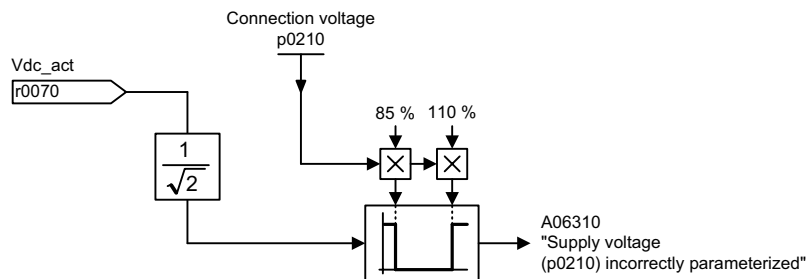
DC link monitoring



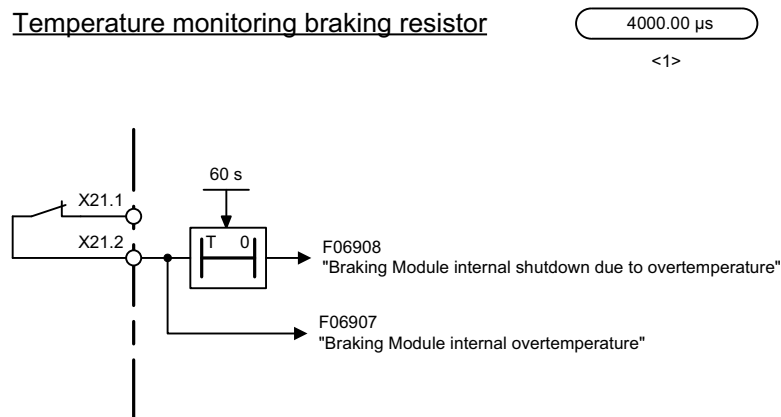
Precharge monitoring for the DC link



Line voltage monitoring when powering-up



Temperature monitoring braking resistor



<1> For B_INF with Braking Module internal only.

1	2	3	4	5	6	7	8
DO: B_INF					fp_8760_01_eng.vsd	Function diagram	
Basic Infeed - Signals and monitoring functions (p3400.0 = 0)					14.07.08 V04.03.01	SINAMICS S120	
							- 8760 -

Fig. 2-245 8760 – Signals and monitoring functions (p3400.0 = 0)

2.25 Smart Infeed

Function diagrams

8820 – Control word, sequence control, infeed	2-1655
8826 – Status word, sequence control, infeed	2-1656
8828 – Status word, infeed	2-1657
8832 – Sequencer	2-1658
8834 – Missing enable signals, line contactor control	2-1659
8850 – Interface for Smart Infeed (control signals, actual values)	2-1660
8860 – Signals and monitoring functions, line supply voltage monitoring	2-1661
8864 – Signals and monitoring functions, line frequency and Vdc monitoring	2-1662

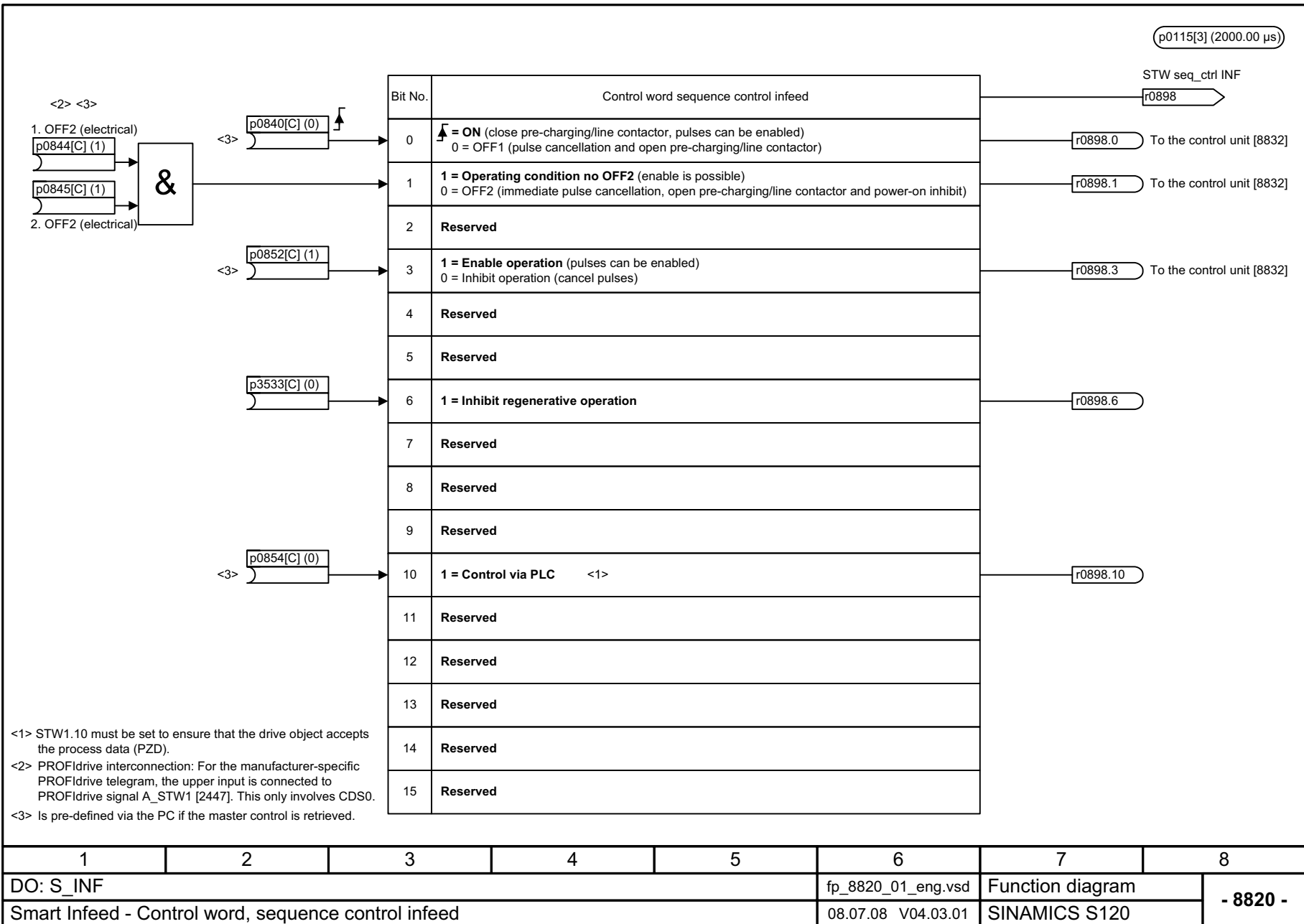


Fig. 2-246 8820 – Control word, sequence control, infeed

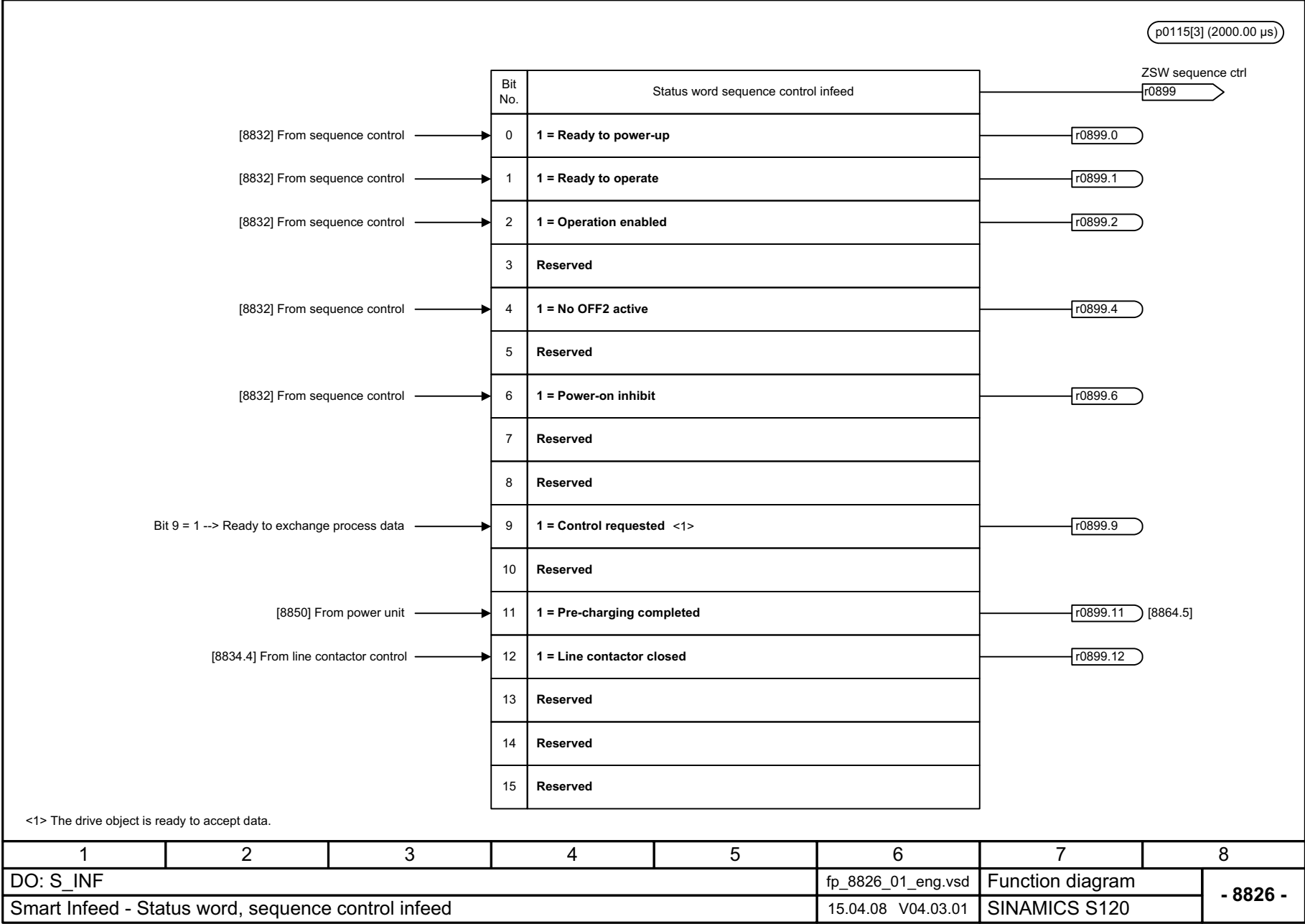


Fig. 2-247 8826 – Status word, sequence control, infeed

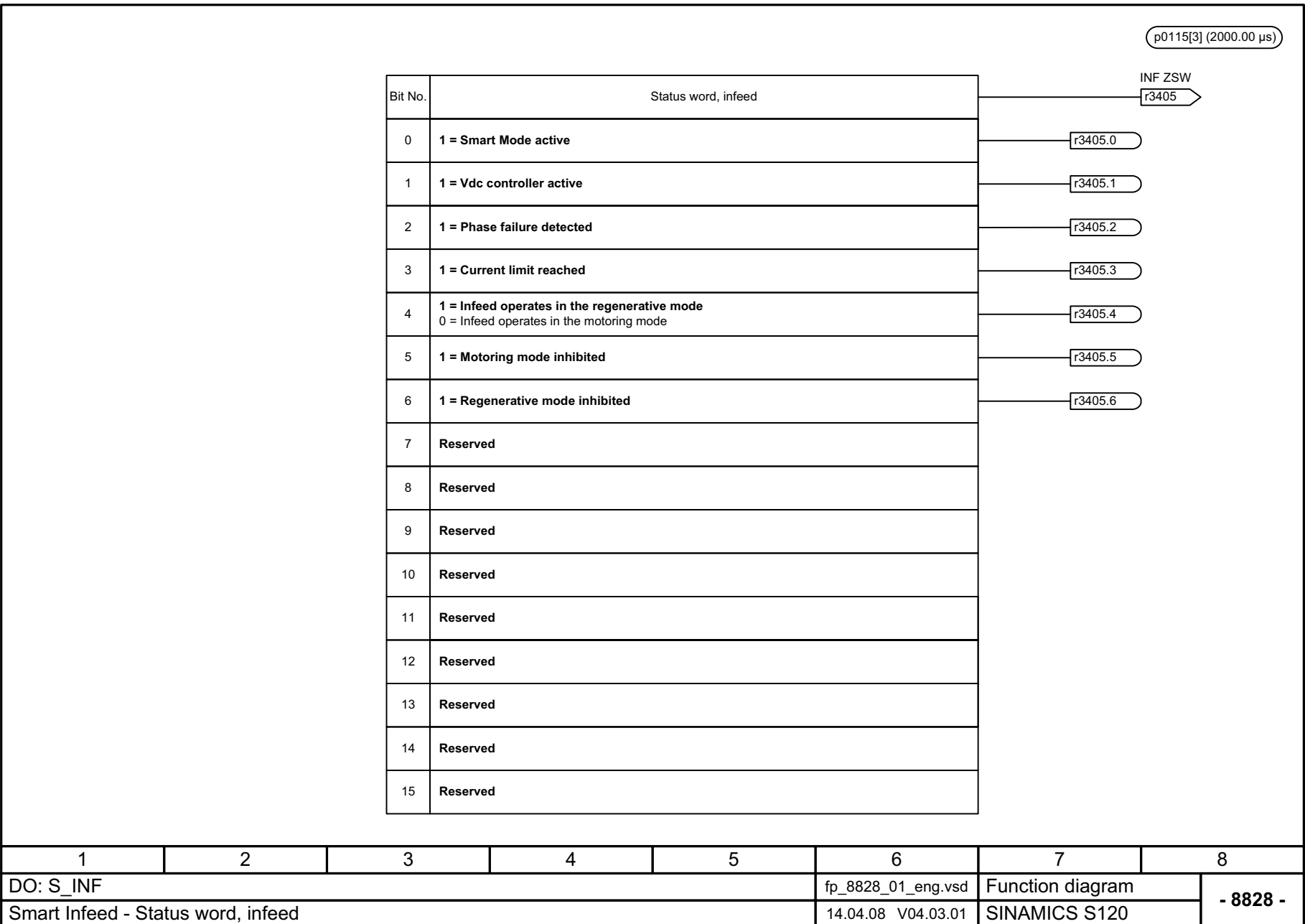
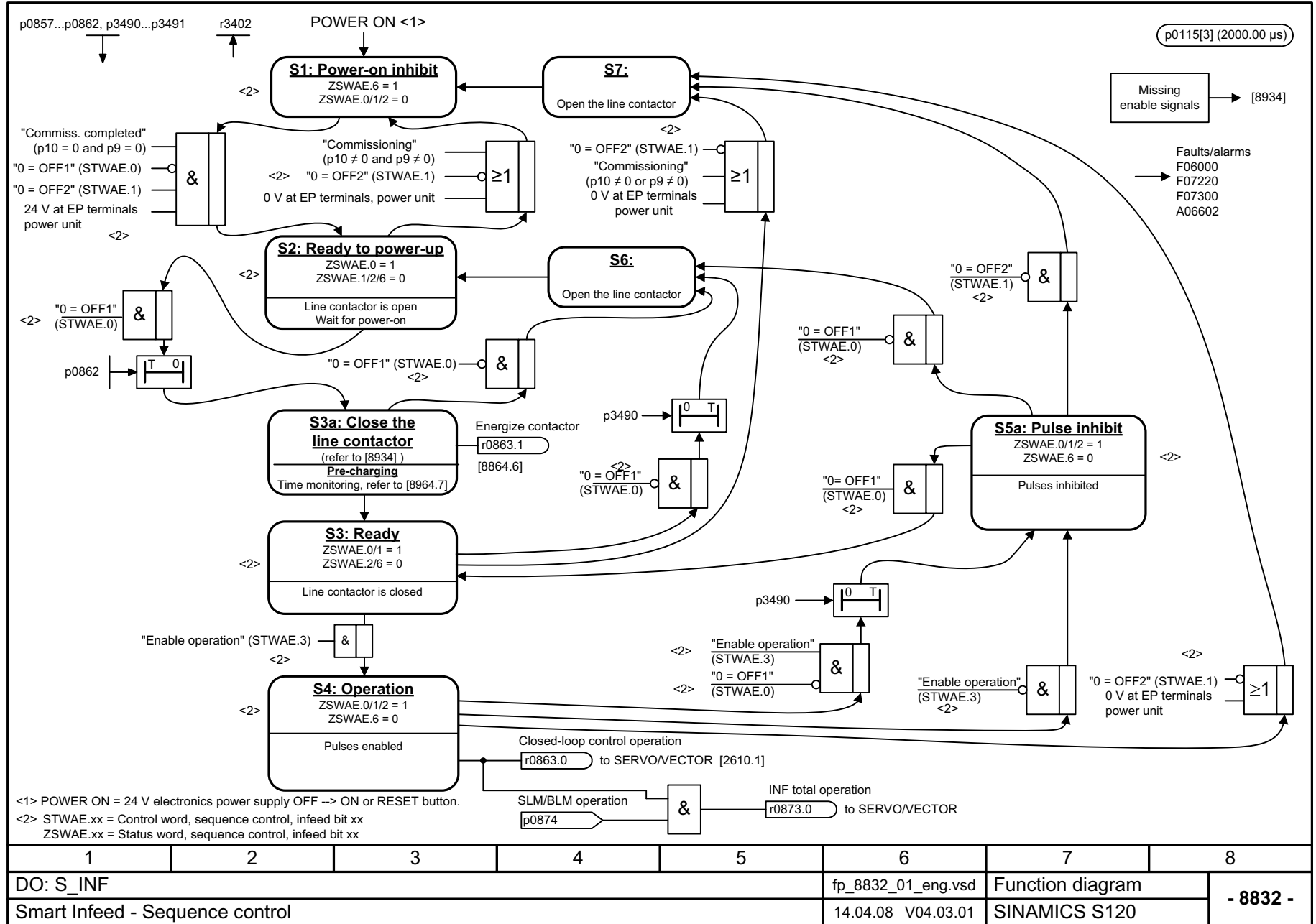


Fig. 2-248 8828 – Status word, infeed

8832 – Sequencer



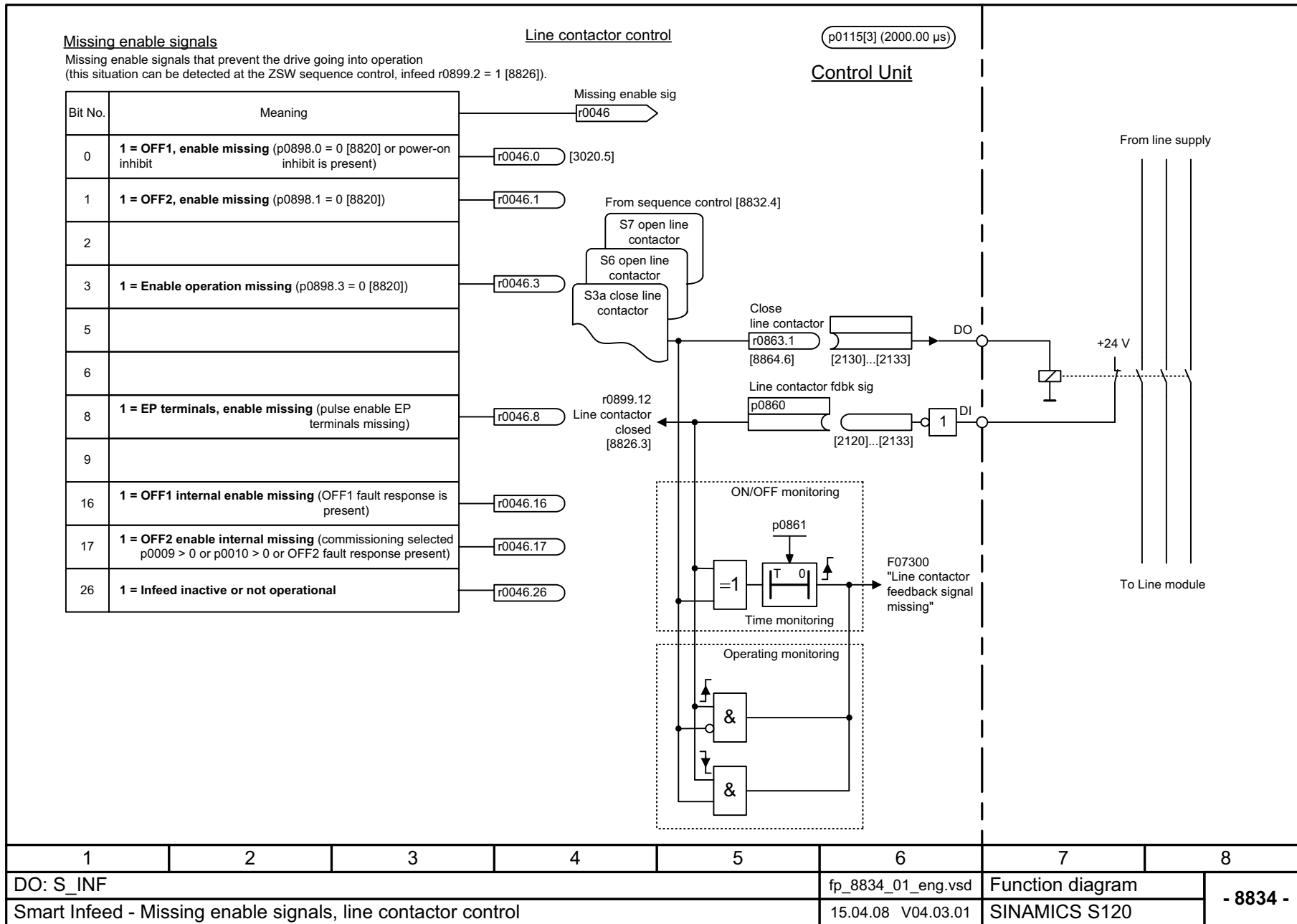
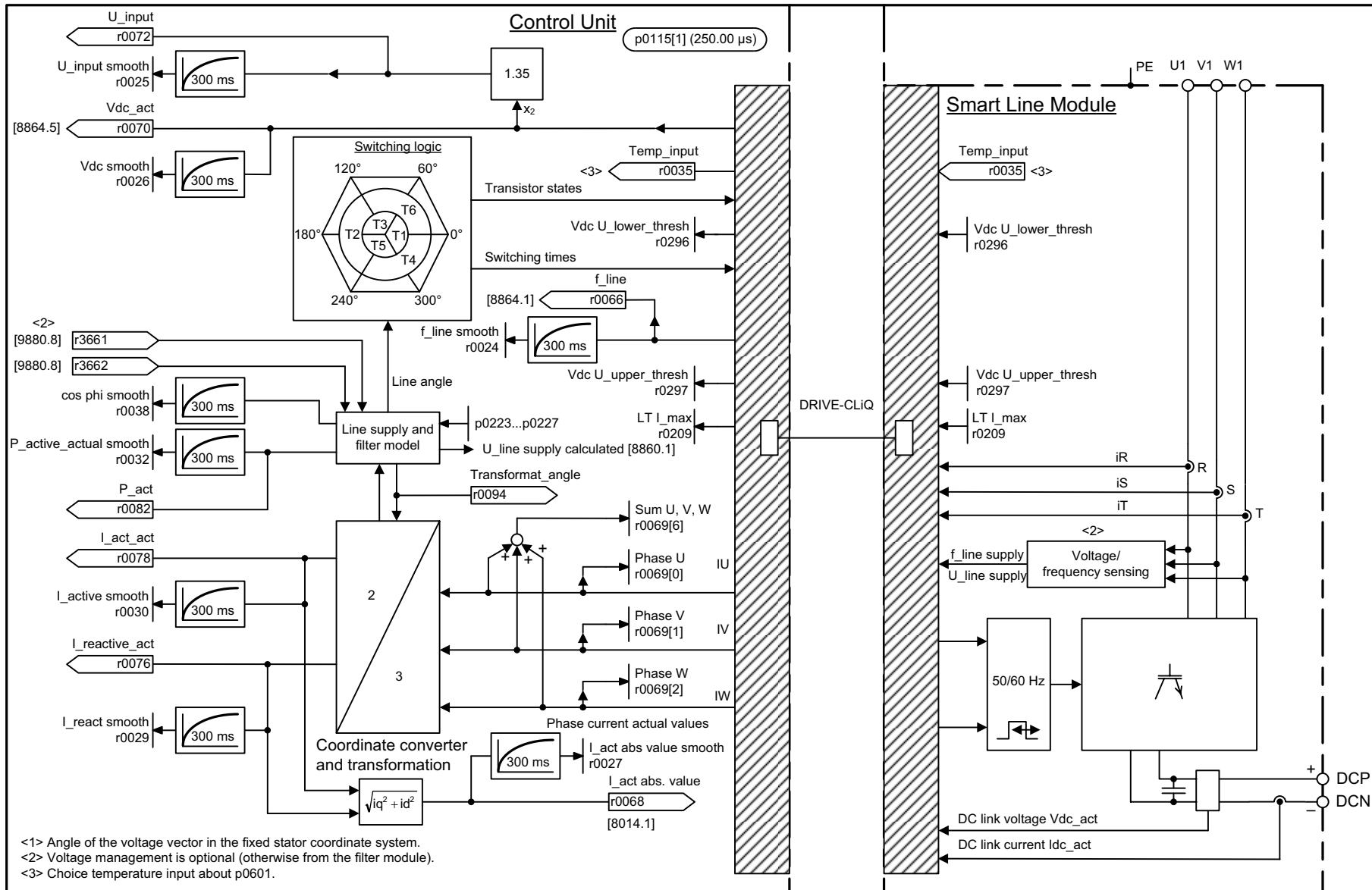


Fig. 2-250 8834 – Missing enable signals, line contactor control



1	2	3	4	5	6	7	8
DO: S_INF					fp_8850_01_eng.vsd	Function diagram	
Smart Infeed - Interface to the Smart Infeed (control signals, actual values)					17.11.09 V04.03.01	SINAMICS S120	

- 8850 -

Fig. 2-251 8850 – Interface for Smart Infeed (control signals, actual values)

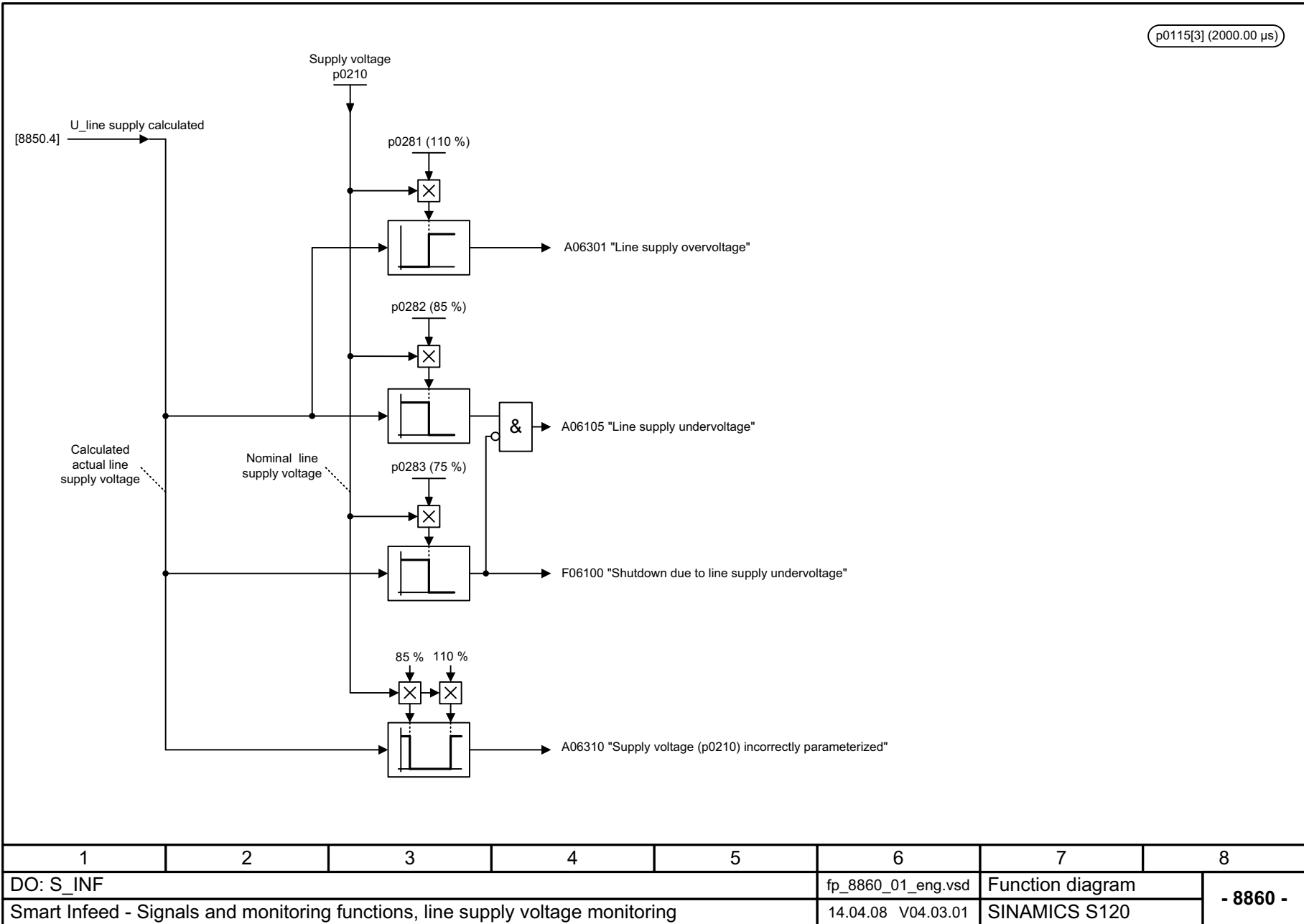
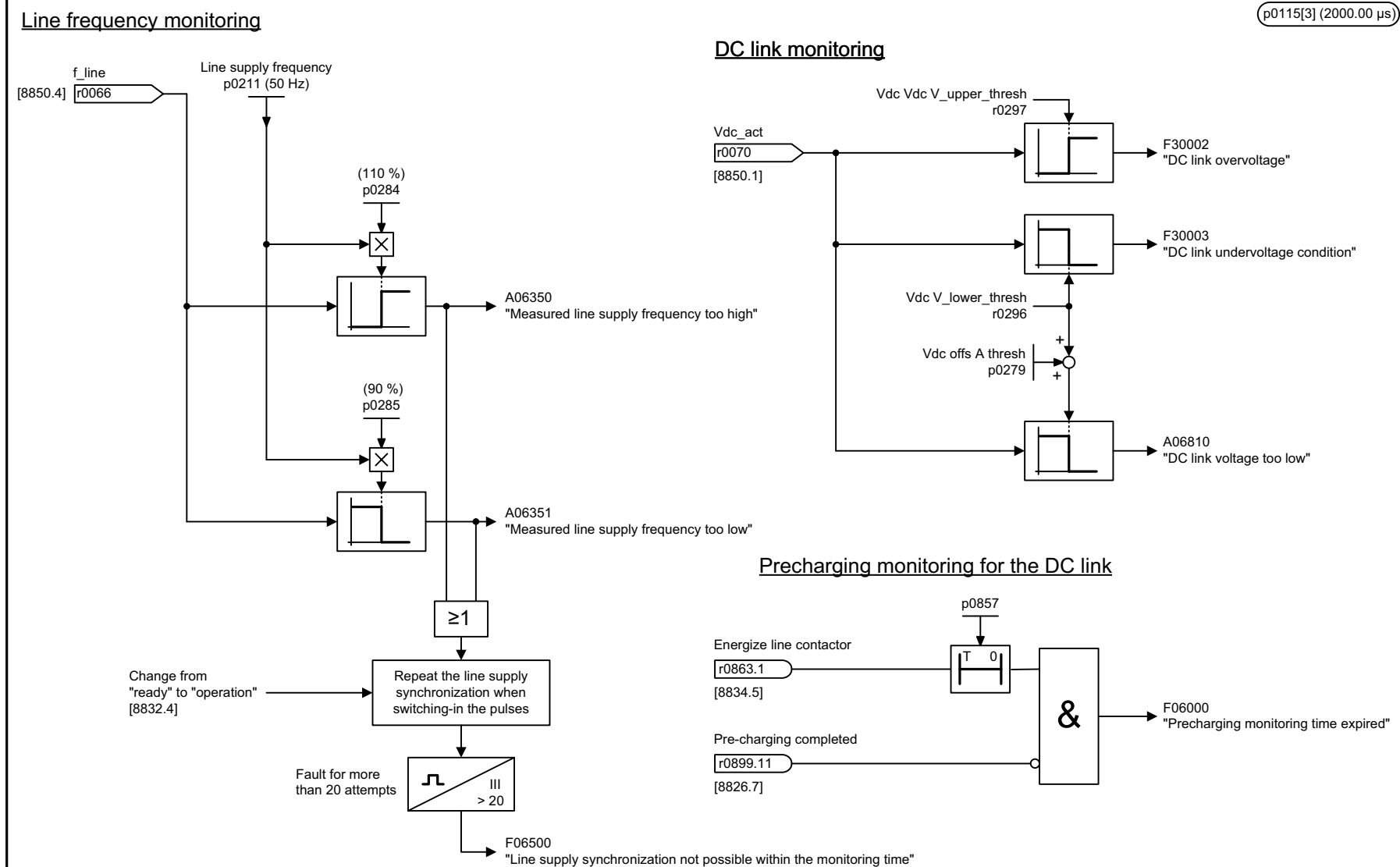


Fig. 2-252 8860 – Signals and monitoring functions, line supply voltage monitoring



1	2	3	4	5	6	7	8
DO: S_INF					fp_8864_01_eng.vsd	Function diagram	
Smart Infeed - Signals and monitoring functions, line frequency and Vdc monitoring					14.07.08 V04.03.01	SINAMICS S120	
- 8864 -							

Fig. 2-253 8864 – Signals and monitoring functions, line frequency and Vdc monitoring

2.26 Active Infeed

Function diagrams

8920 – Control word, sequence control, infeed	2-1664
8926 – Status word, sequence control, infeed	2-1665
8928 – Status word, infeed	2-1666
8932 – Sequencer	2-1667
8934 – Missing enable signals, line contactor control	2-1668
8940 – Modulation depth reserve controller / DC link voltage controller (p3400.0 = 0)	2-1669
8946 – Current pre-control / current controller / gating unit (p3400.0 = 0)	2-1670
8948 – Master/slave (r0108.19 = 1)	2-1671
8950 – Interface for Active Infeed, control signals, actual values (p3400.0 = 0)	2-1672
8960 – Signals and monitoring functions, line supply voltage monitoring (p3400.0 = 0)	2-1673
8964 – Signals and monitoring functions, line frequency/Vdc monitoring (p3400.0 = 0)	2-1674

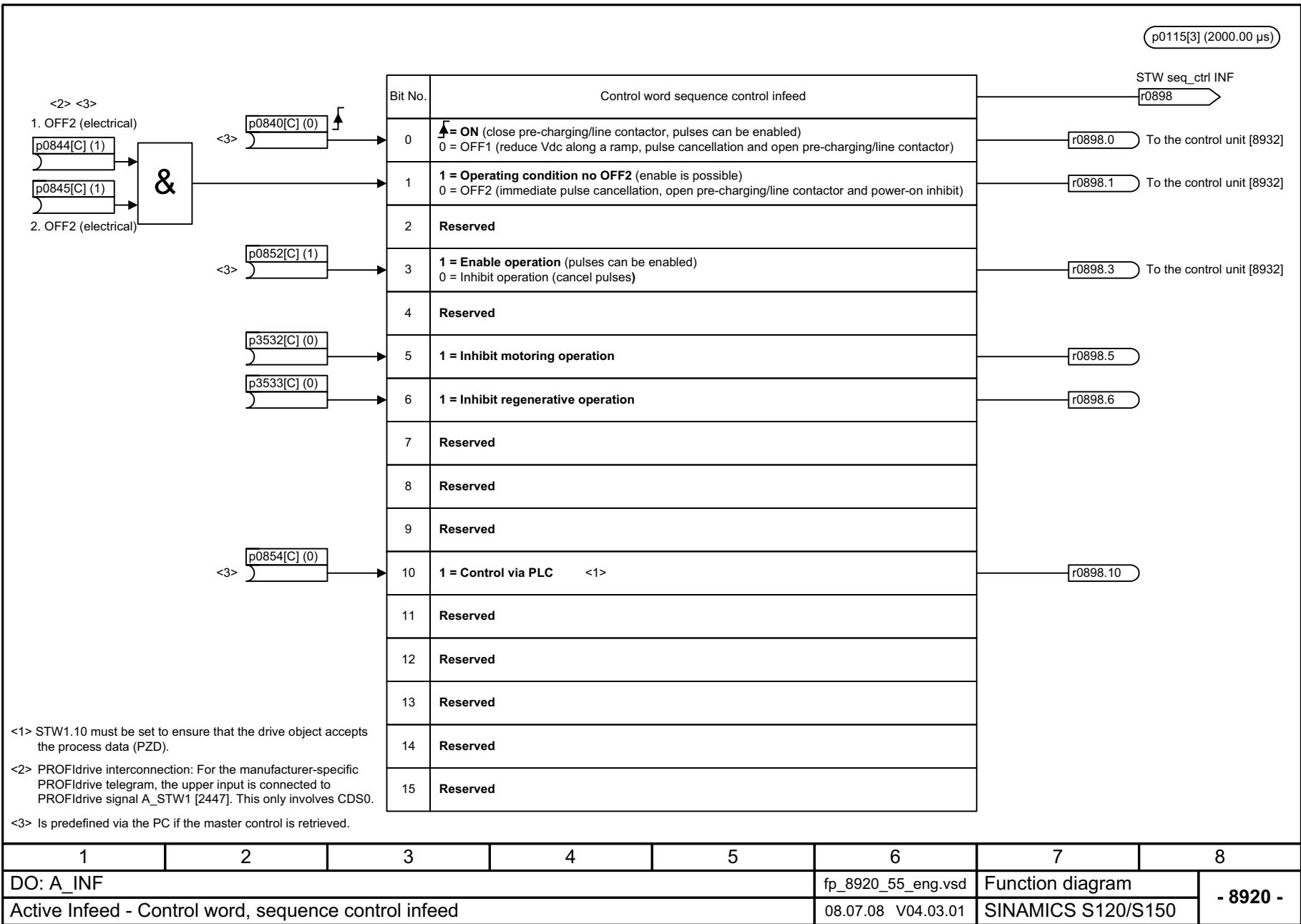


Fig. 2-254 8920 – Control word, sequence control, infeed

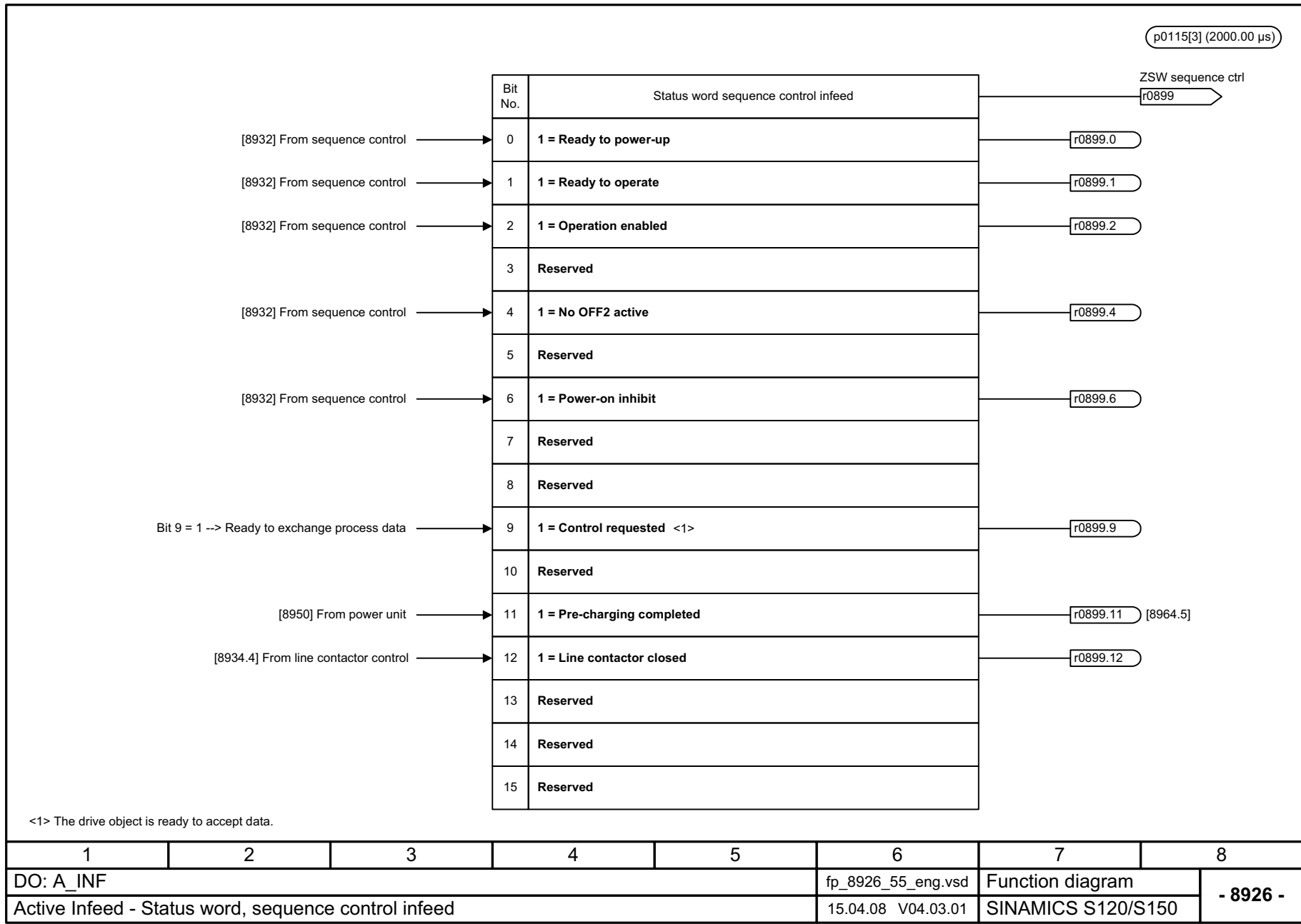


Fig. 2-255 8926 – Status word, sequence control, infeed

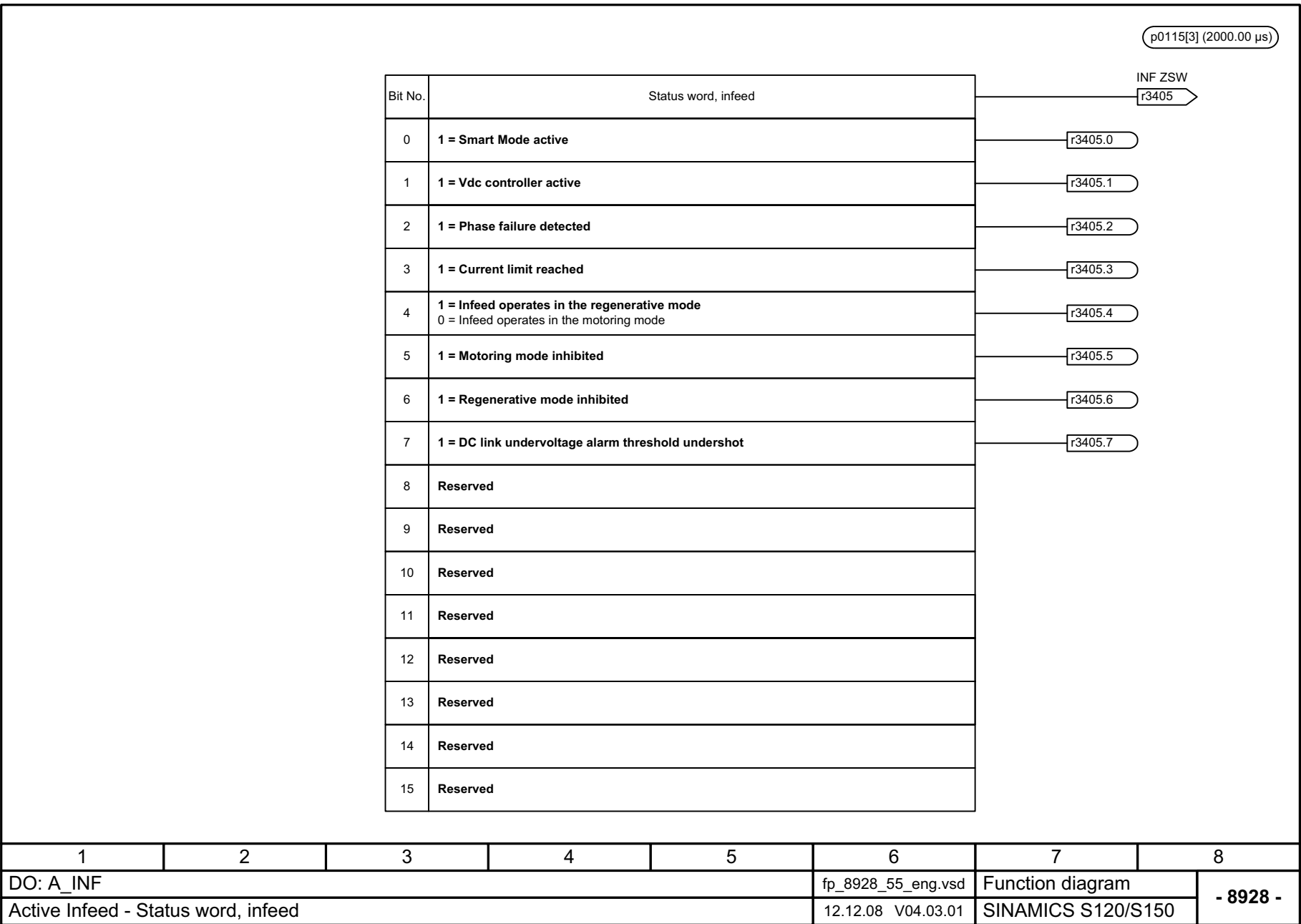
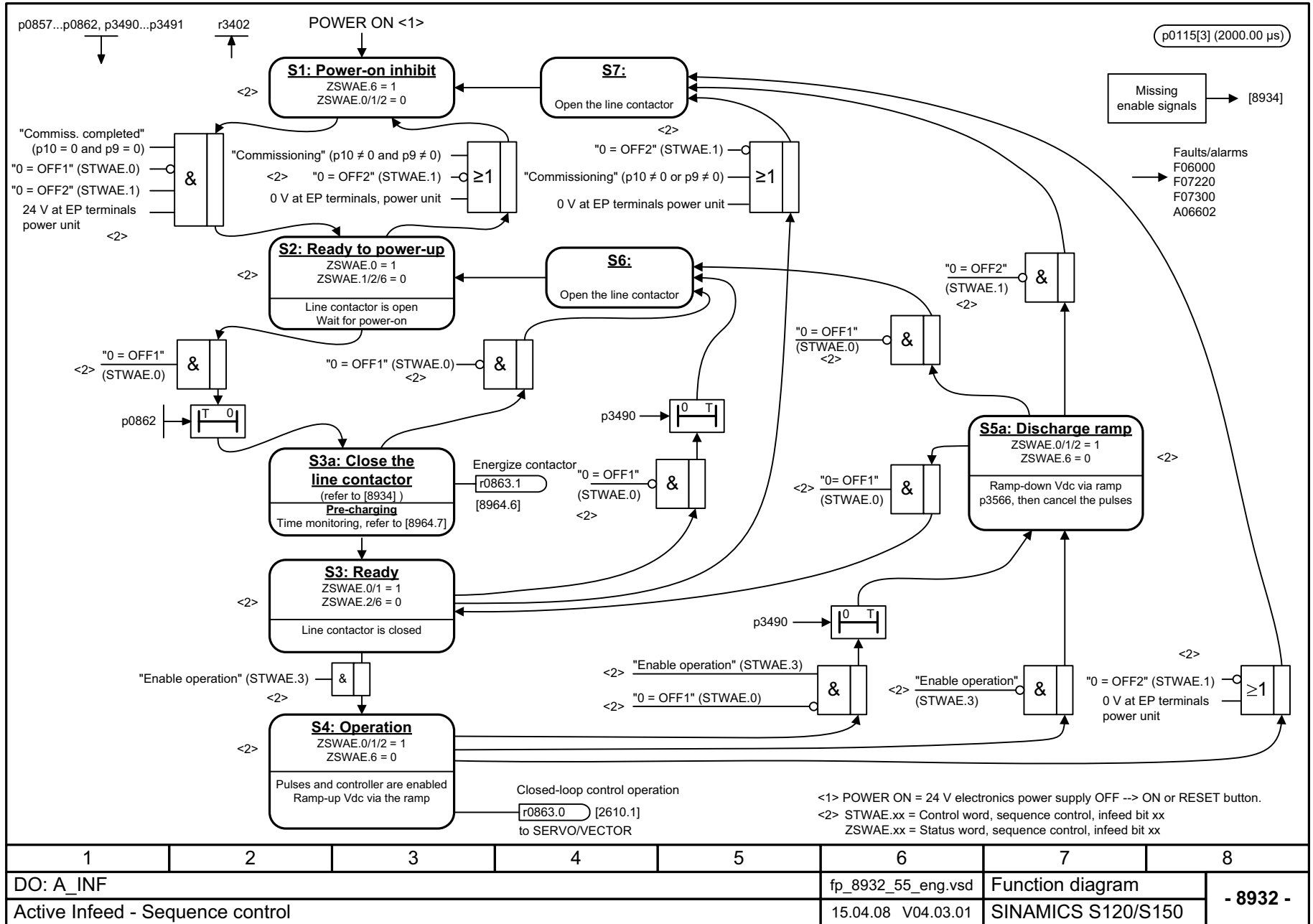
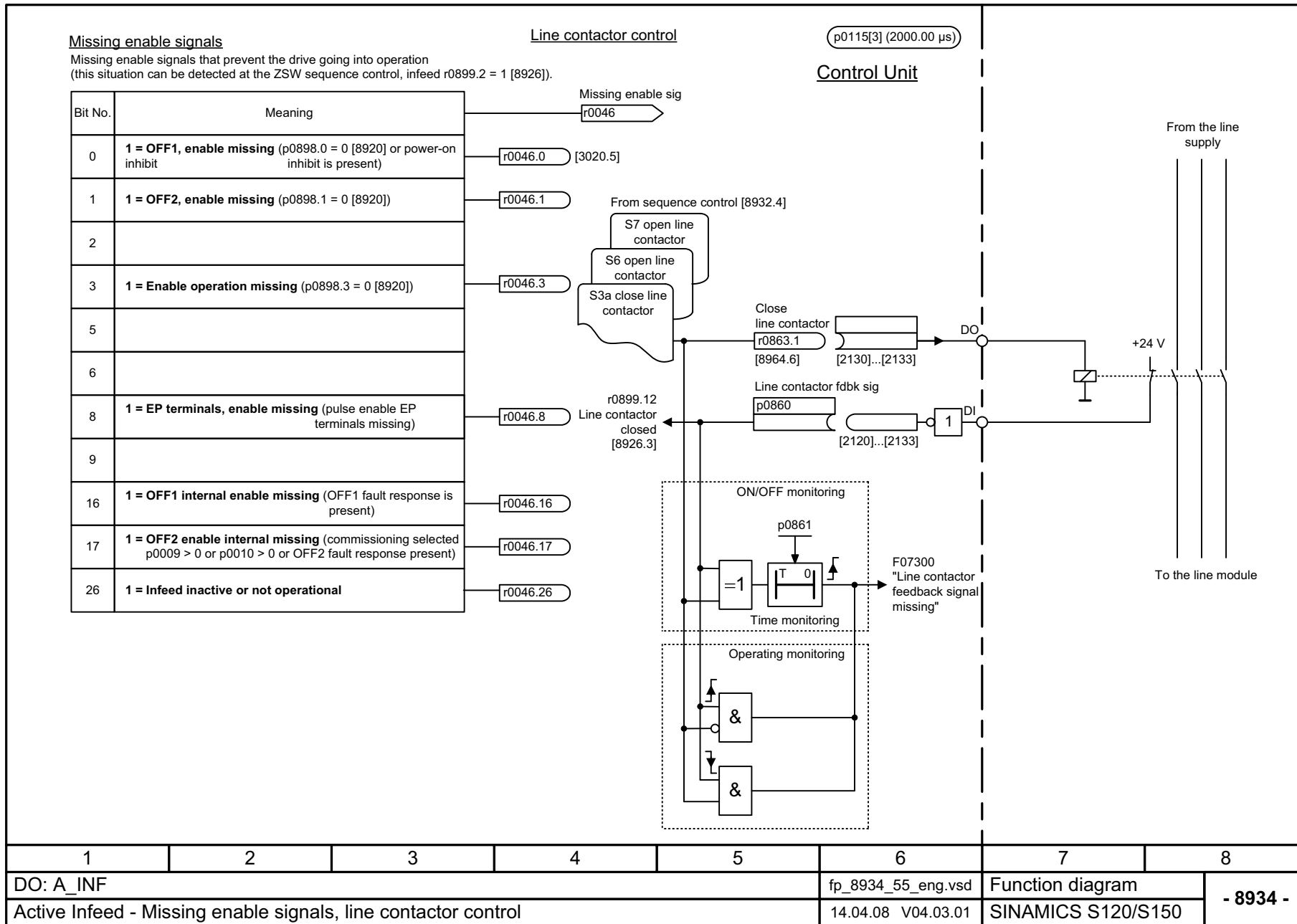


Fig. 2-256 8928 – Status word, infeed

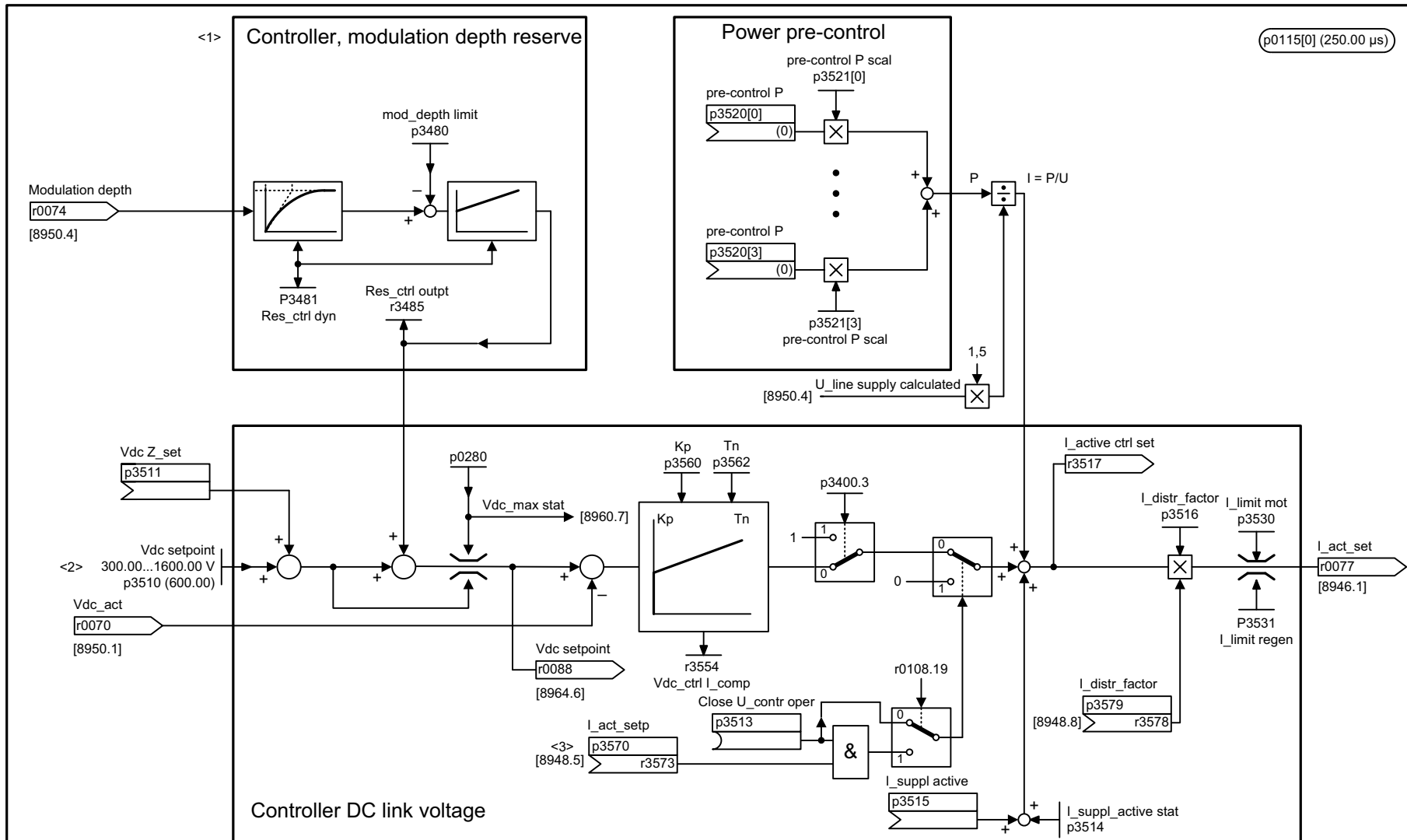
Fig. 2-257 8932 – Sequencer





1	2	3	4	5	6	7	8
DO: A_INF					fp_8934_55_eng.vsd	Function diagram	
Active Infeed - Missing enable signals, line contactor control					14.04.08 V04.03.01	SINAMICS S120/S150	
							- 8934 -

Fig. 2-258 8934 – Missing enable signals, line contactor control



- <1> This controller increases (boosts) the DC link voltage when the modulation depth limit has been reached (p3480).
 <2> When the pulses are enabled, the DC link voltage is ramped (p3566) from the actual value to the setpoint p3510.
 <3> Applies only if the " Master/slave" function module is activated (r0108.19 = 1).

1	2	3	4	5	6	7	8
DO: A_INF					fp_8940_55_eng.vsd	Function diagram	
Active Infeed - Controller modulation depth reserve / controller DC link voltage (p3400.0 = 0)					12.11.09 V04.03.01	SINAMICS S120/S150	
					- 8940 -		

Fig. 2-259 8940 – Modulation depth reserve controller / DC link voltage controller (p3400.0 = 0)

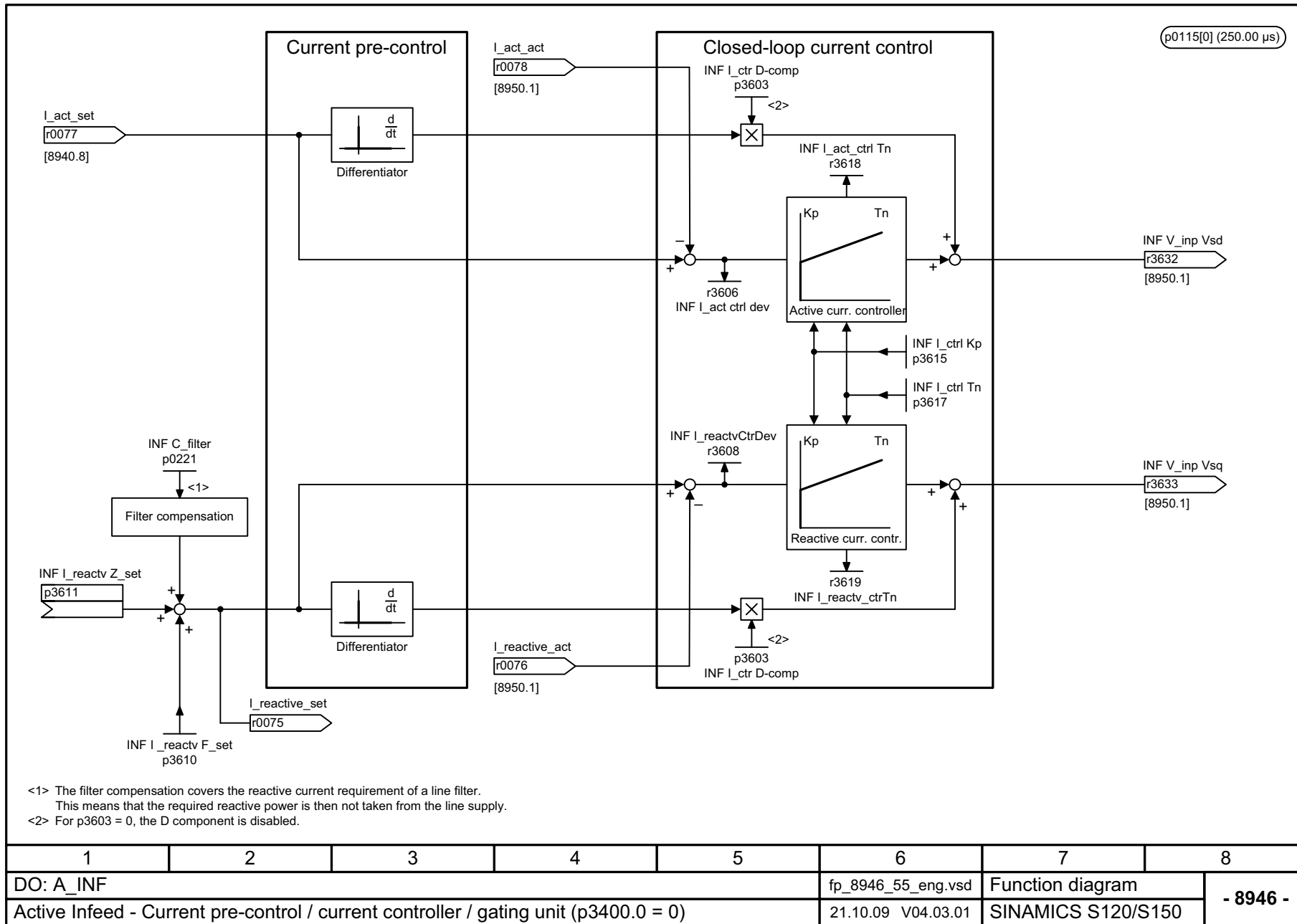


Fig. 2-260 8946 – Current pre-control / current controller / gating unit (p3400.0 = 0)

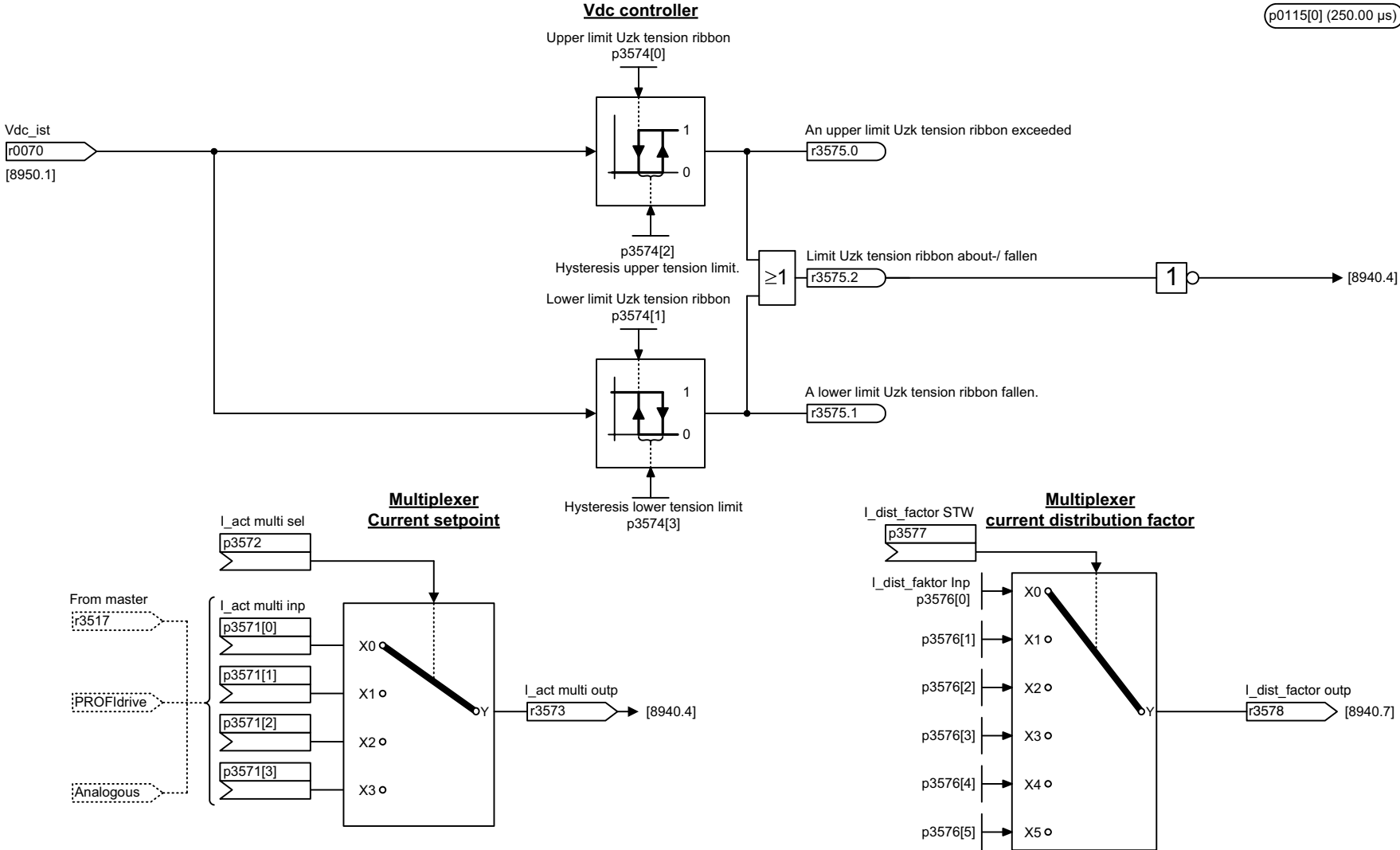


Fig. 2-261

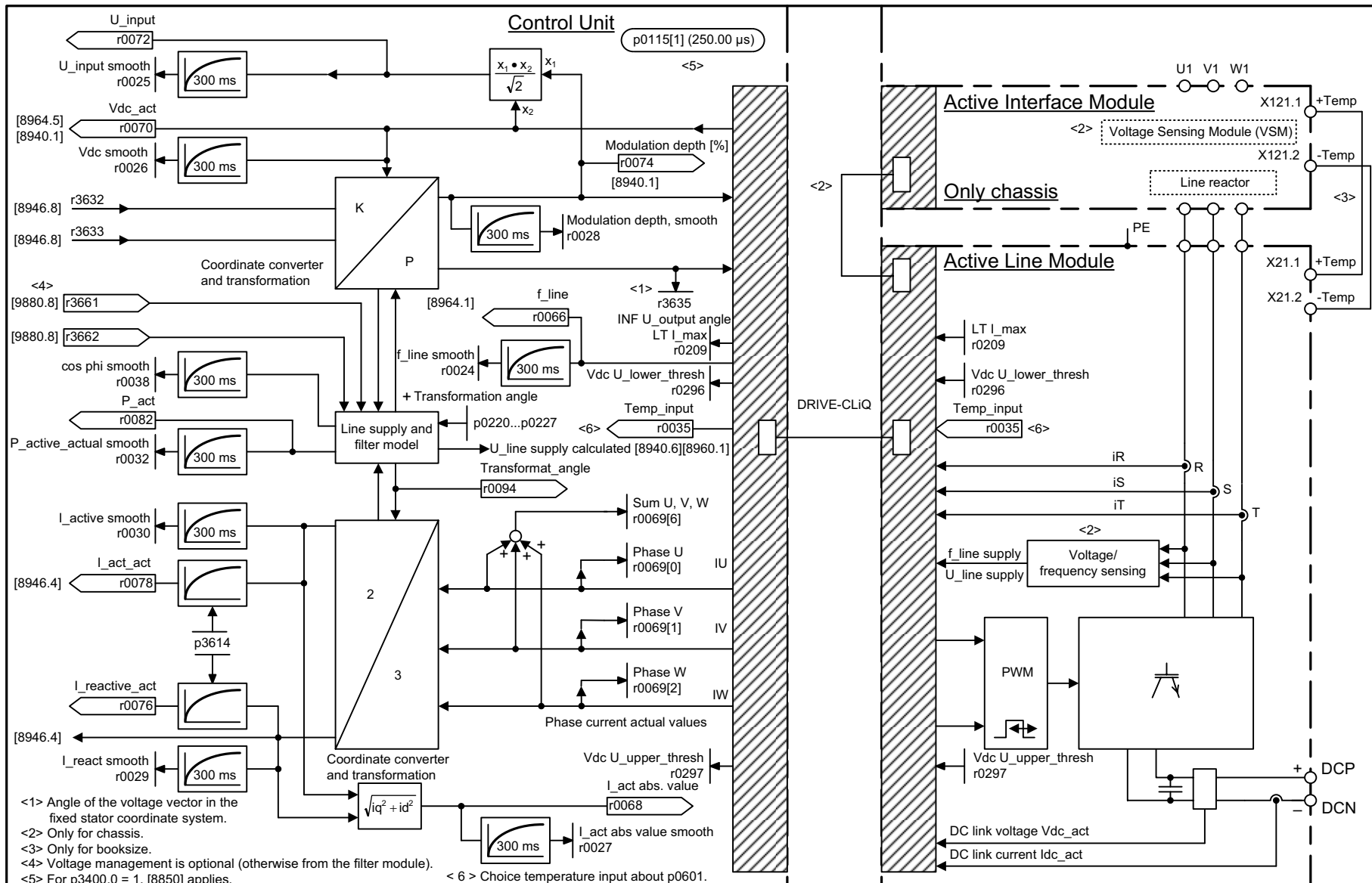
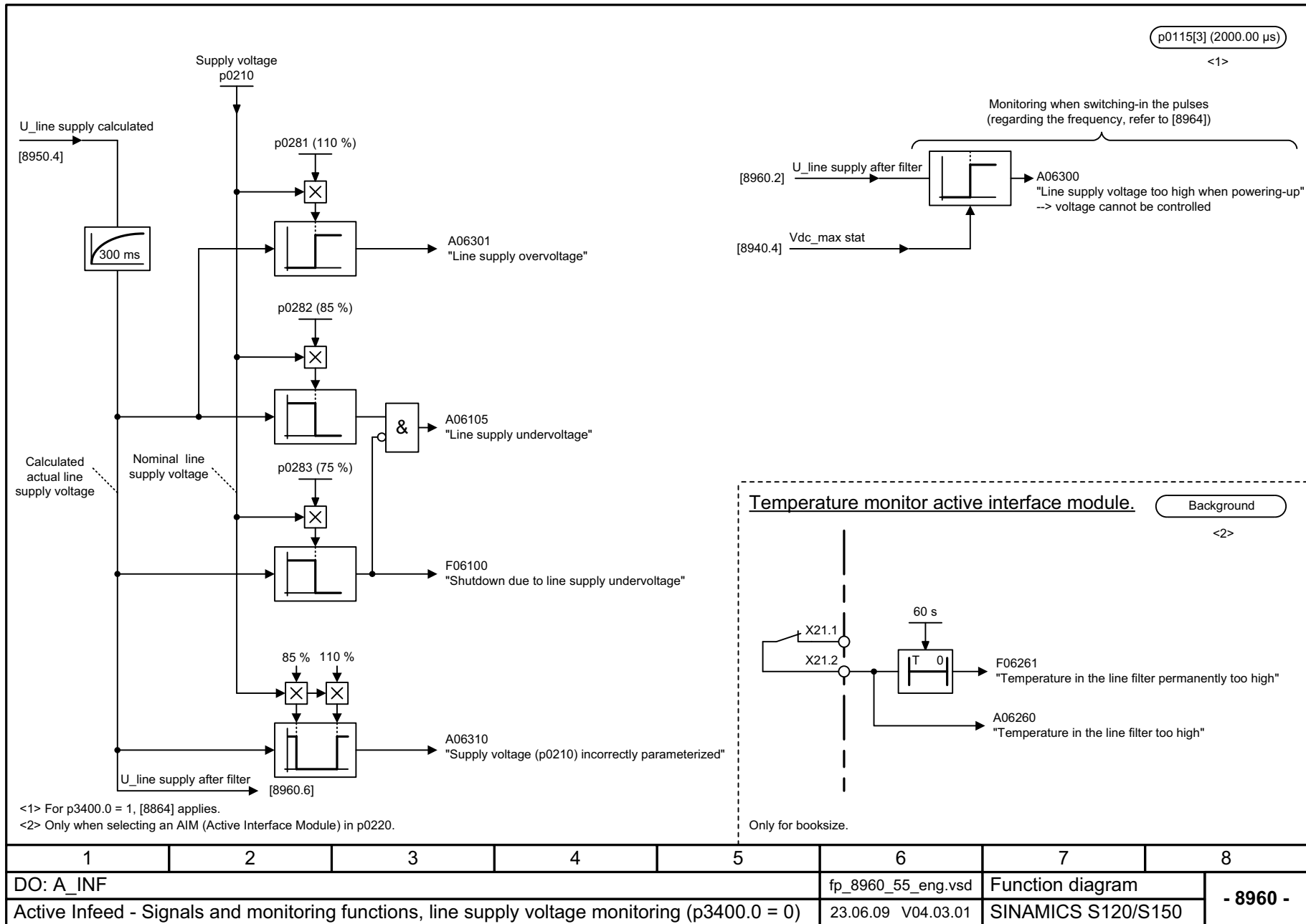


Fig. 2-262 8950 – Interface for Active Infeed, control signals, actual values (p3400.0 = 0)



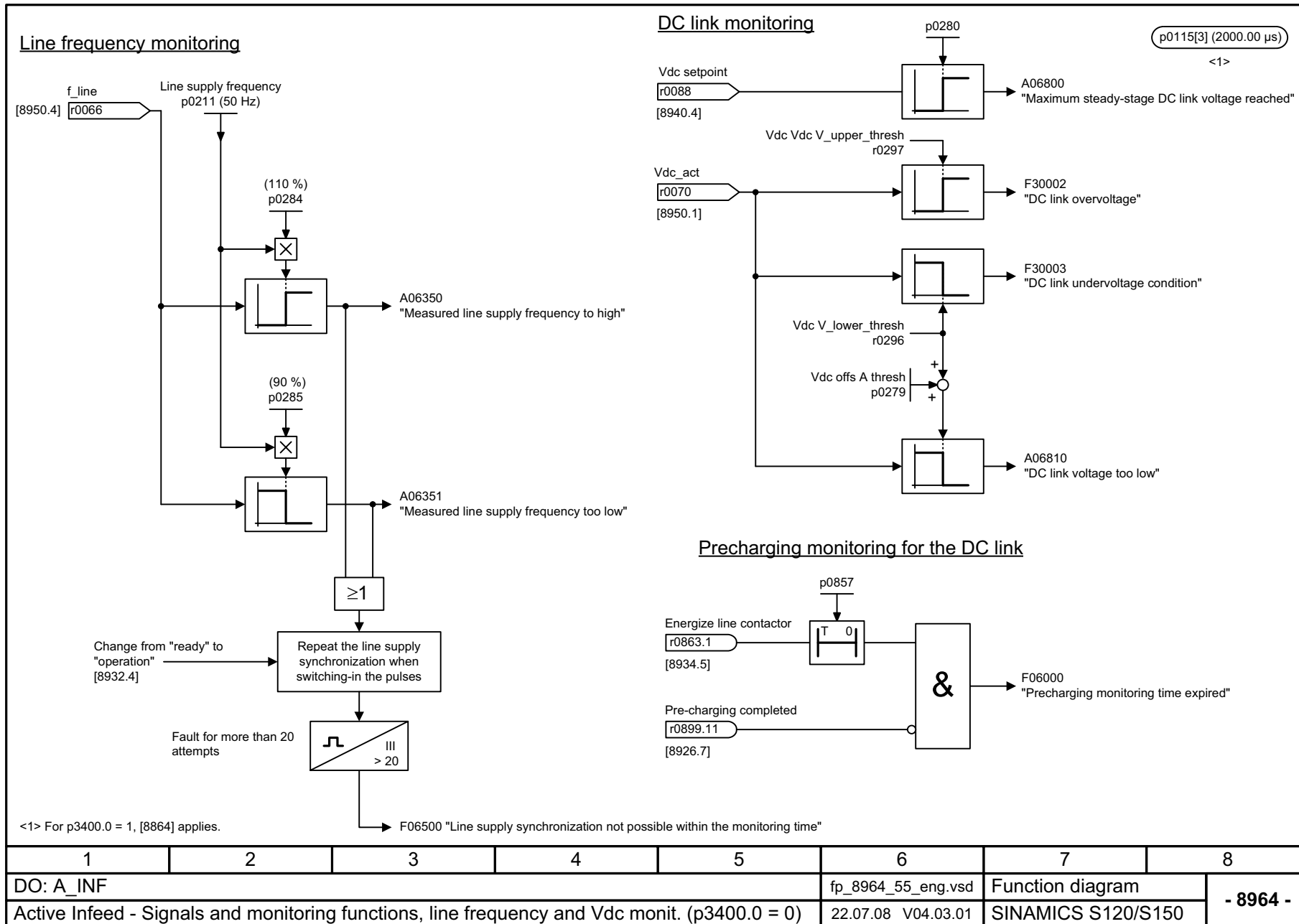


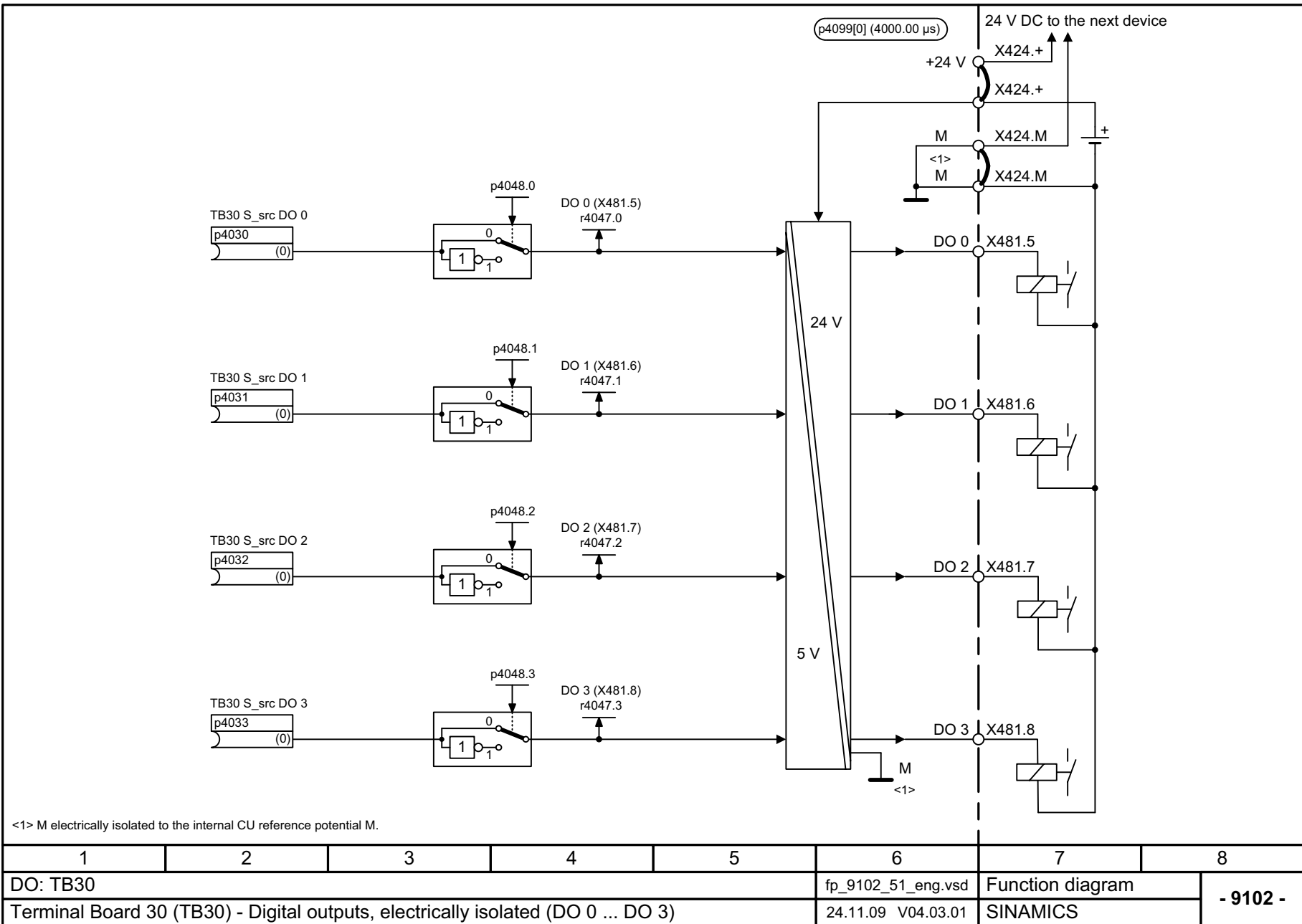
Fig. 2-264 8964 – Signals and monitoring functions, line frequency/Vdc monitoring (p3400.0 = 0)

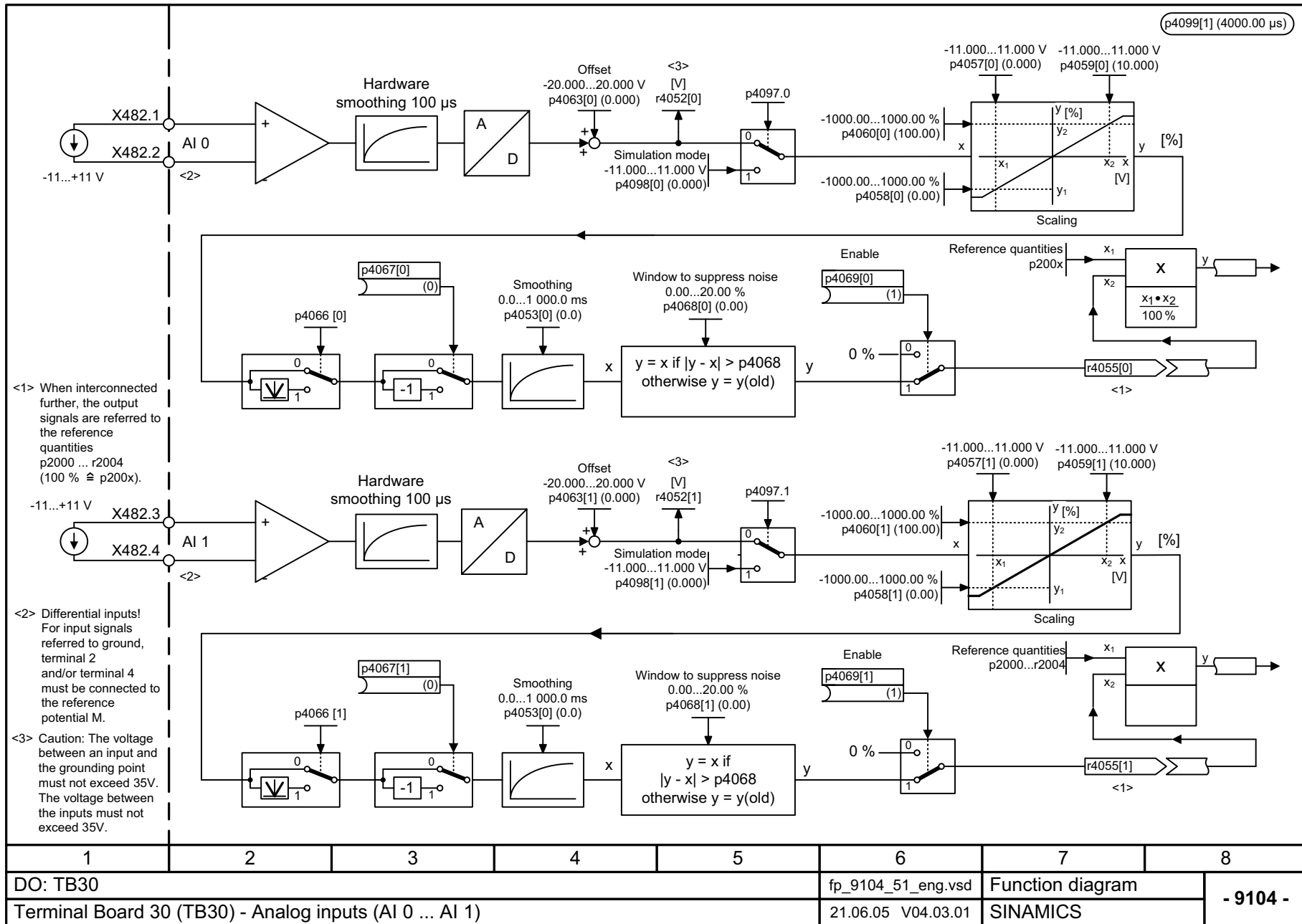
2.27 Terminal Board 30 (TB30)

Function diagrams

9100 – Digital inputs, isolated (DI 0 to DI 3)	2-1676
9102 – Digital outputs, isolated (DO 0 to DO 3)	2-1677
9104 – Analog inputs (AI 0 to AI 1)	2-1678
9106 – Analog outputs (AO 0 to AO 1)	2-1679







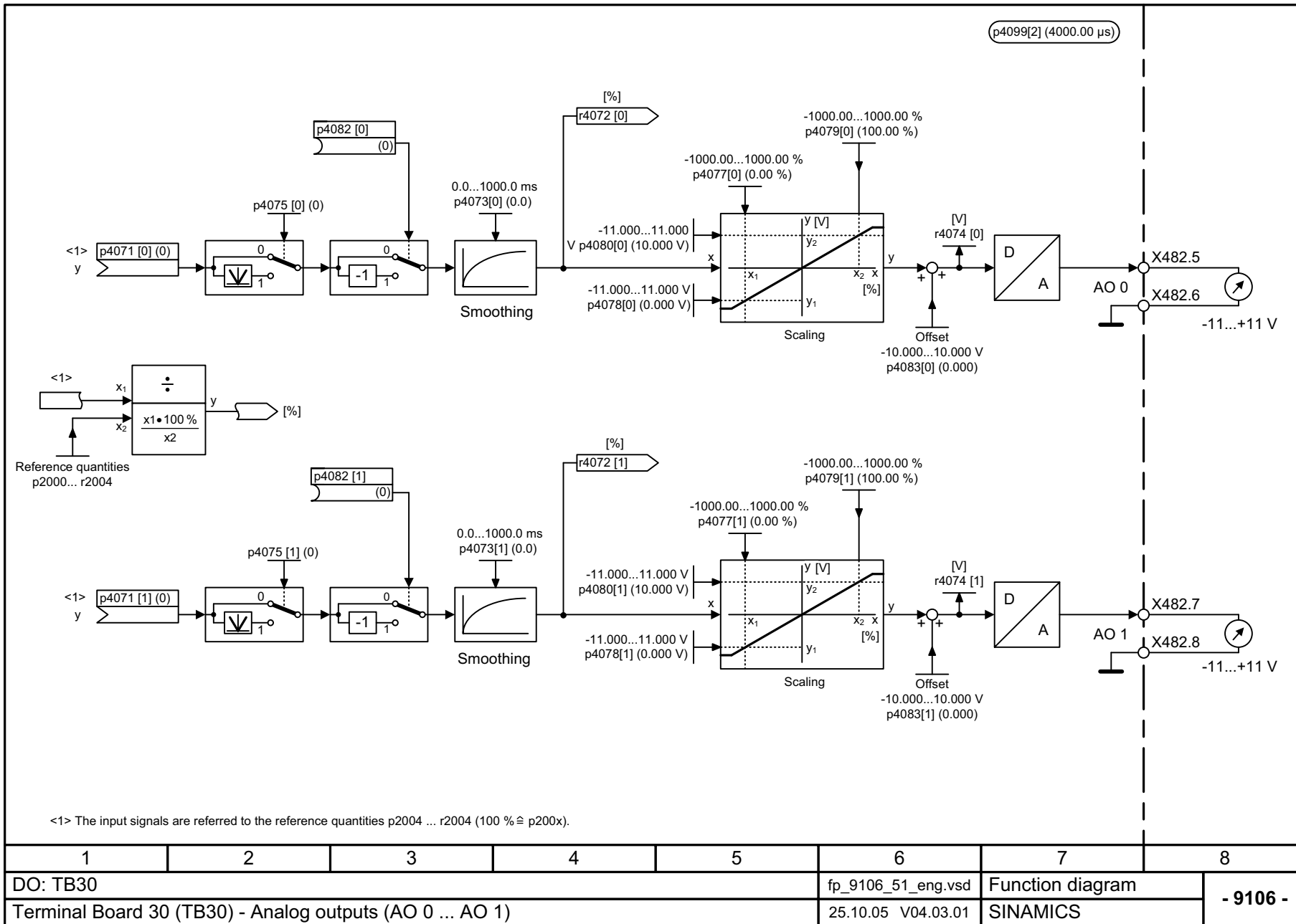
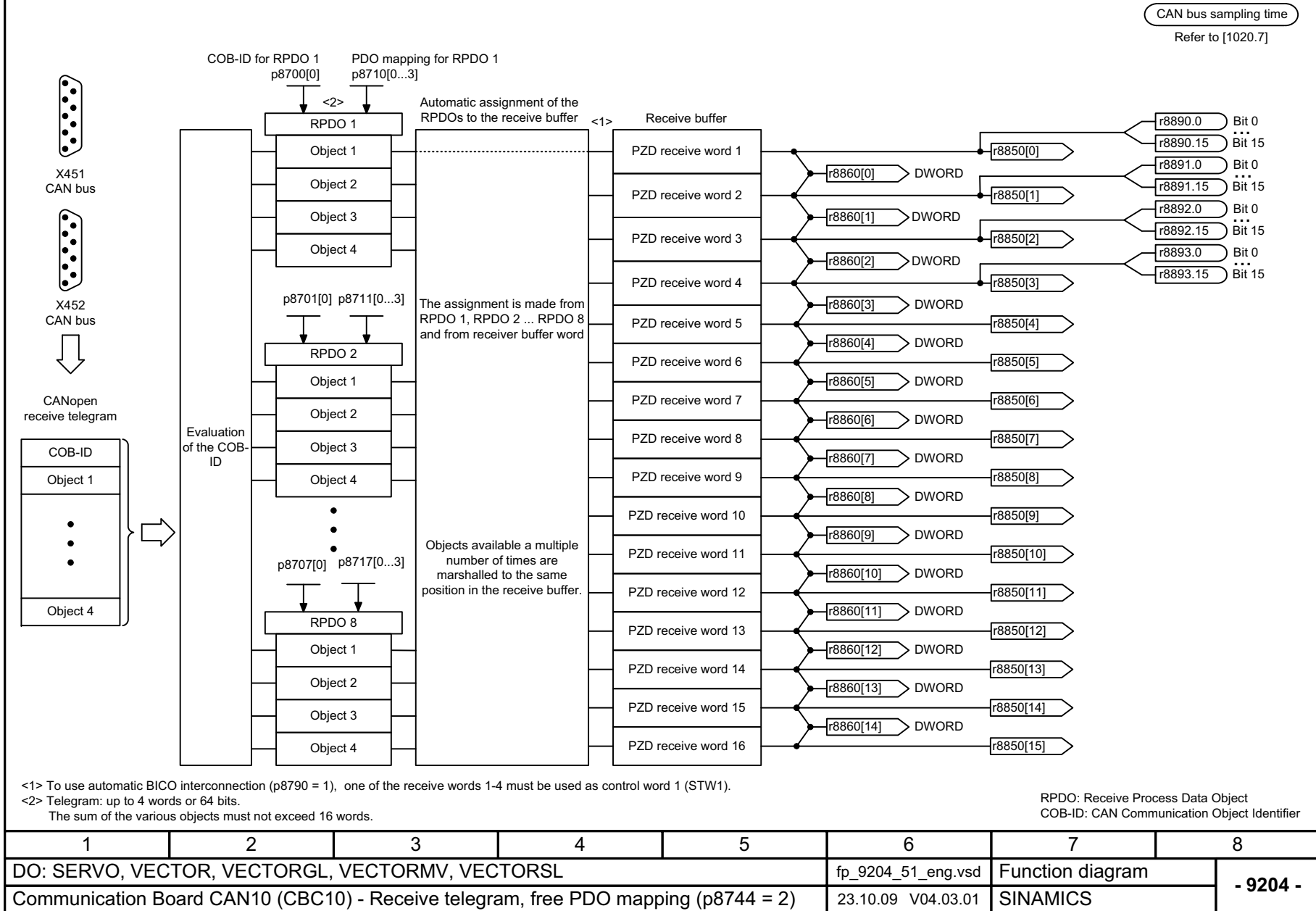


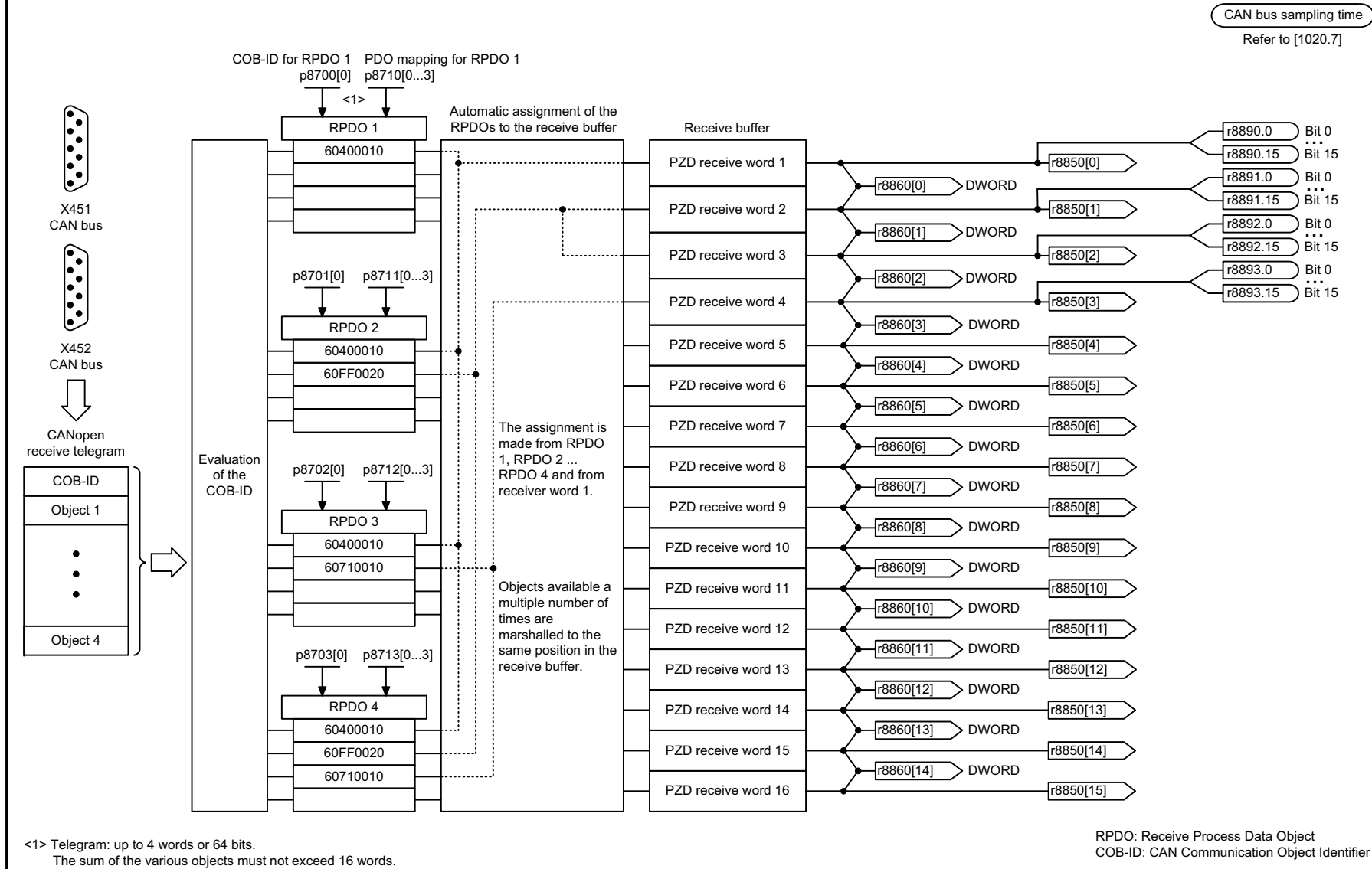
Fig. 2-268 9106 – Analog outputs (AO 0 to AO 1)

2.28 Communication Board CAN10 (CBC10)

Function diagrams

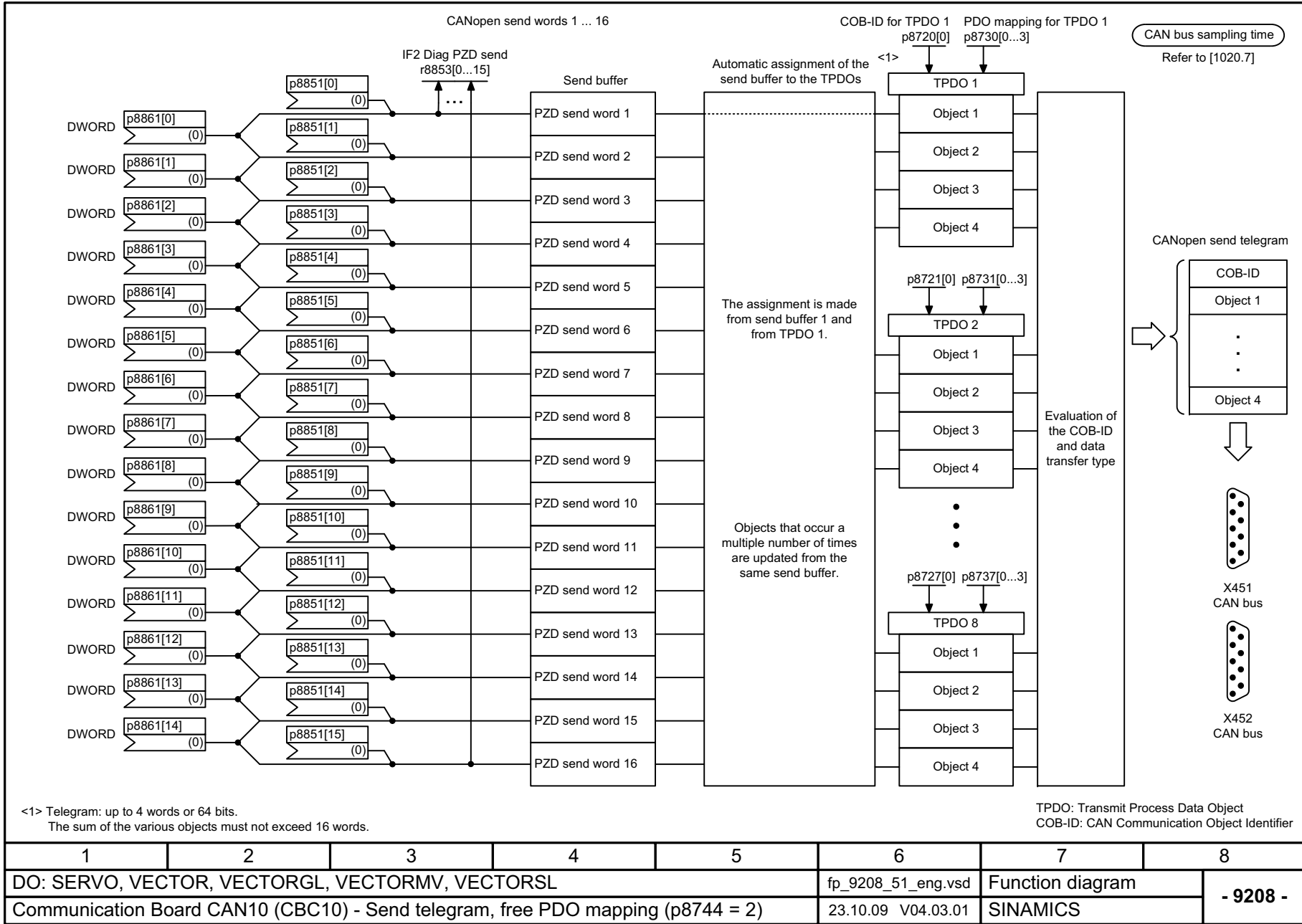
9204 – Incoming telegram, free PDO mapping (p8744 = 2)	2-1681
9206 – Incoming telegram, predefined connection set (p8744 = 1)	2-1682
9208 – Outgoing telegram, free PDO mapping (p8744 = 2)	2-1683
9210 – Outgoing telegram, predefined connection set (p8744 = 1)	2-1684
9220 – Control word CANopen	2-1685
9226 – Status word, CANopen	2-1686





1	2	3	4	5	6	7	8
DO: SERVO, VECTOR, VECTORGL, VECTORMV, VECTORSL					fp_9206_51_eng.vsd	Function diagram	
Communication Board CAN10 (CBC10) - Receive telegram, Predef. Conn. Set (p8744 = 1)					23.10.09 V04.03.01	SINAMICS	
							- 9206 -

Fig. 2-270 9206 – Incoming telegram, predefined connection set (p8744 = 1)



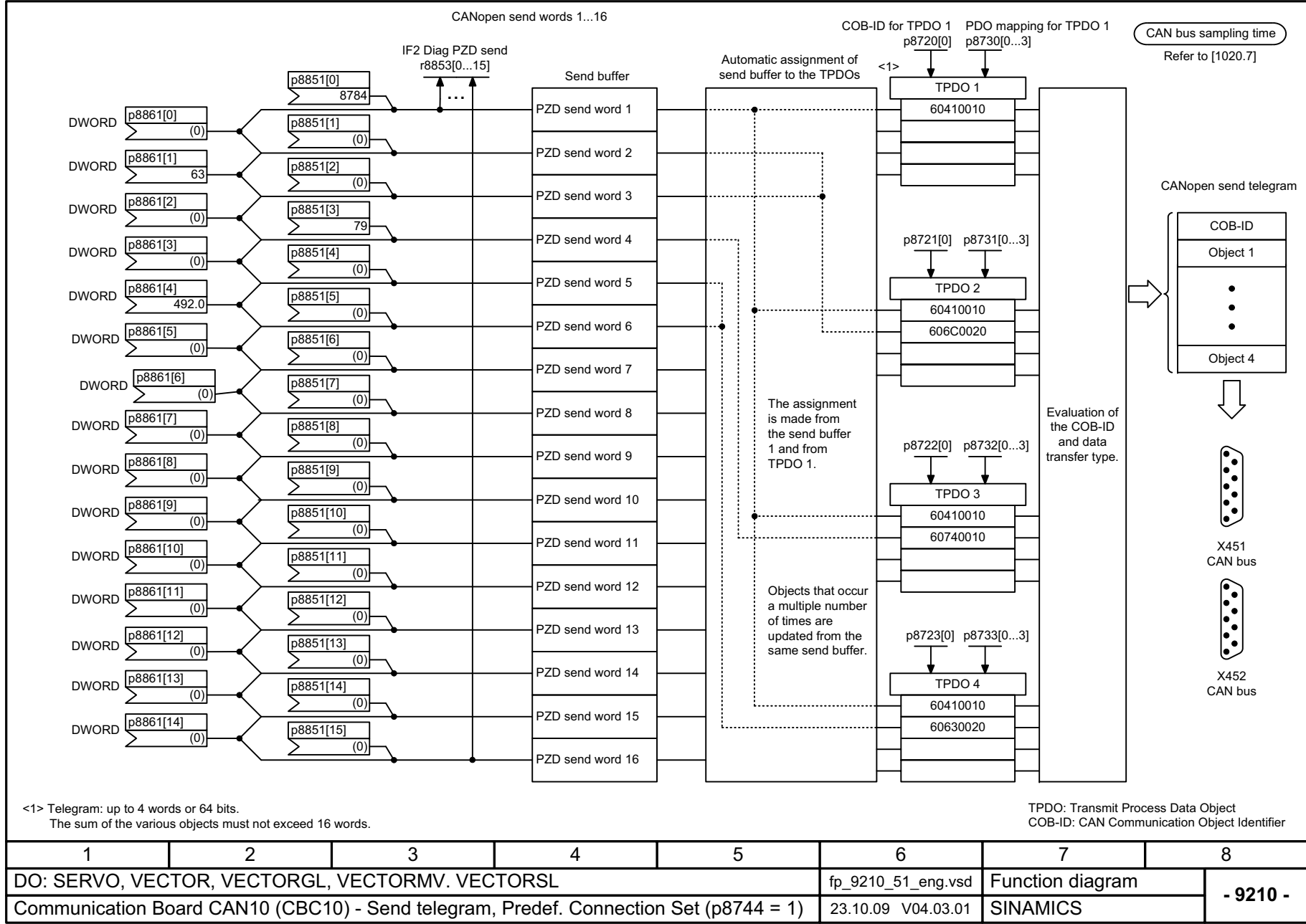


Fig. 2-272 9210 – Outgoing telegram, predefined connection set (p8744 = 1)

CAN bus sampling time
Refer to [1020.7]

Signal targets for control word CANopen					
Signal	Meaning	Interconnection parameters <1>	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW1.0	 = ON (pulses can be enabled) 0 = OFF1 (braking with ramp-function generator, then pulse cancellation and ready-to-power-up)	p0840[0] = r8890.0	[2501.3]	[2610]	-
STW1.1	1 = No coast-down activated (enable possible) 0 = Activate coast-down (immediate pulse cancellation and power-on inhibit)	p0844[0] = r8890.1	[2501.3]	[2610]	-
STW1.2	1 = No fast stop activated (enable possible) 0 = Activate fast stop (braking along an OFF3 ramp p1135, then pulse cancellation and power-on inhibit)	p0848[0] = r8890.2	[2501.3]	[2610]	-
STW1.3	1 = Enable operation (pulses can be enabled) 0 = Inhibit operation (cancel pulses)	p0852[0] = r8890.3	[2501.3]	[2610]	-
STW1.4	Reserved	-	-	-	-
STW1.5	Reserved	-	-	-	-
STW1.6	Reserved	-	-	-	-
STW1.7	 = Acknowledge fault	p2103[0] = r8890.7	[2546.1]	[8060]	-
STW1.8	Reserved	-	-	-	-
STW1.9	Reserved	-	-	-	-
STW1.10	Reserved	-	-	-	-
STW1.11	Can be freely connected	pxxxx[y] = r8890.11			-
STW1.12	Can be freely connected	pxxxx[y] = r8890.12	-	-	-
STW1.13	Can be freely connected	pxxxx[y] = r8890.13	-	-	-
STW1.14	Can be freely connected	pxxxx[y] = r8890.14	-	-	-
STW1.15	Can be freely connected	pxxxx[y] = r8890.15	-	-	-

<1> Depending on the position of the CANopen control word in p8750, the number of the binector to be connected changes.

1	2	3	4	5	6	7	8
DO: SERVO, VECTOR, VECTORGL, VECTORMV, VECTORSLS			fp_9220_51_eng.vsd			Function diagram	- 9220 -
Communication Board CAN10 (CBC10) - Control word, CANopen			04.11.09 V04.03.01			SINAMICS	

Fig. 2-273 9220 – Control word CANopen

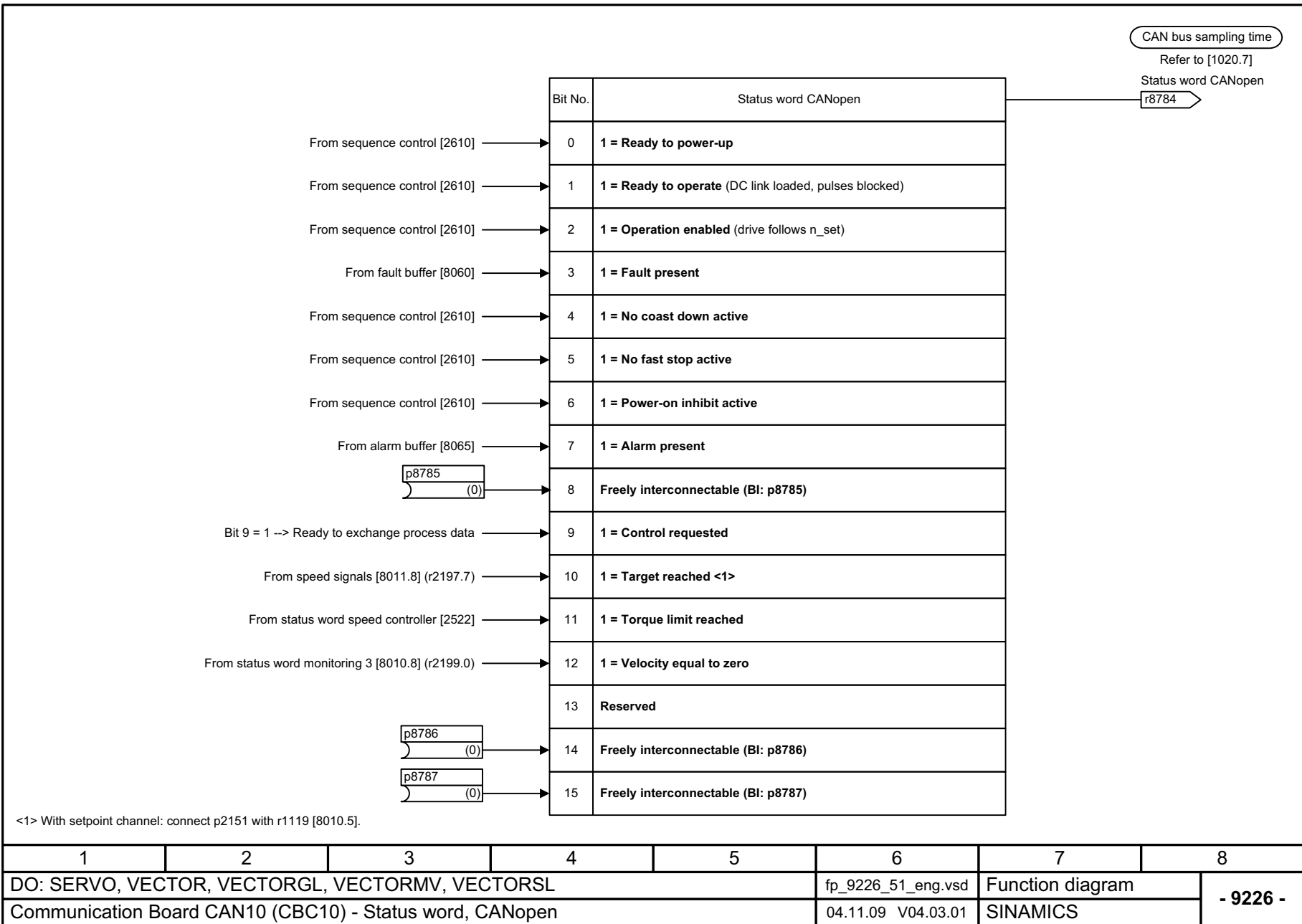


Fig. 2-274 9226 – Status word, CANopen

2.29 Terminal Module 15 for SINAMICS (TM15DI/DO)

Function diagrams

9400 – Digital inputs/outputs, bidirectional (DI/DO 0 to DI/DO 7)	2-1688
9401 – Digital inputs/outputs, bidirectional (DI/DO 8 to DI/DO 15)	2-1689
9402 – Digital inputs/outputs, bidirectional (DI/DO 16 to DI/DO 23)	2-1690

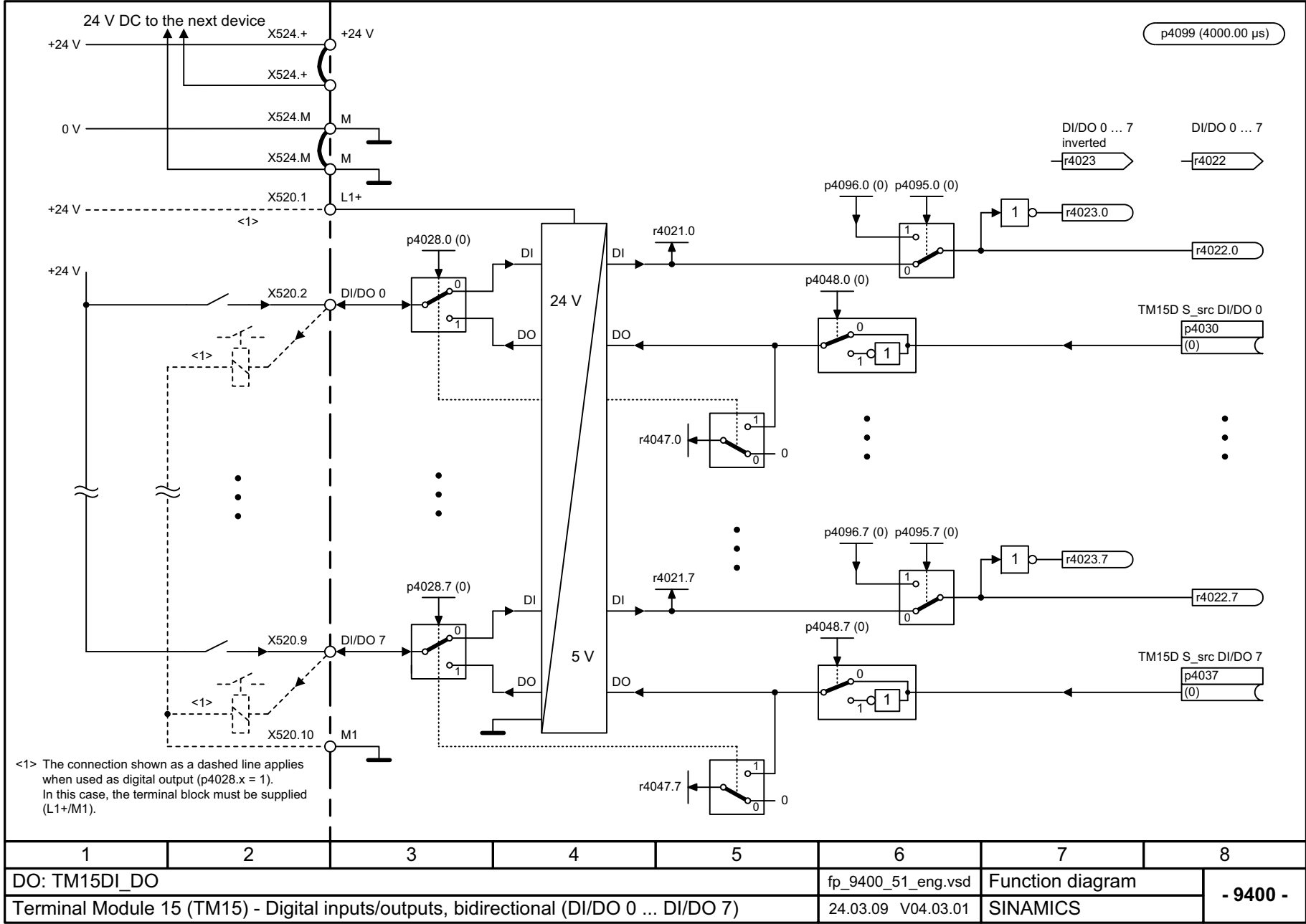


Fig. 2-275 9400 – Digital inputs/outputs, bidirectional (DI/DO 0 to DI/DO 7)

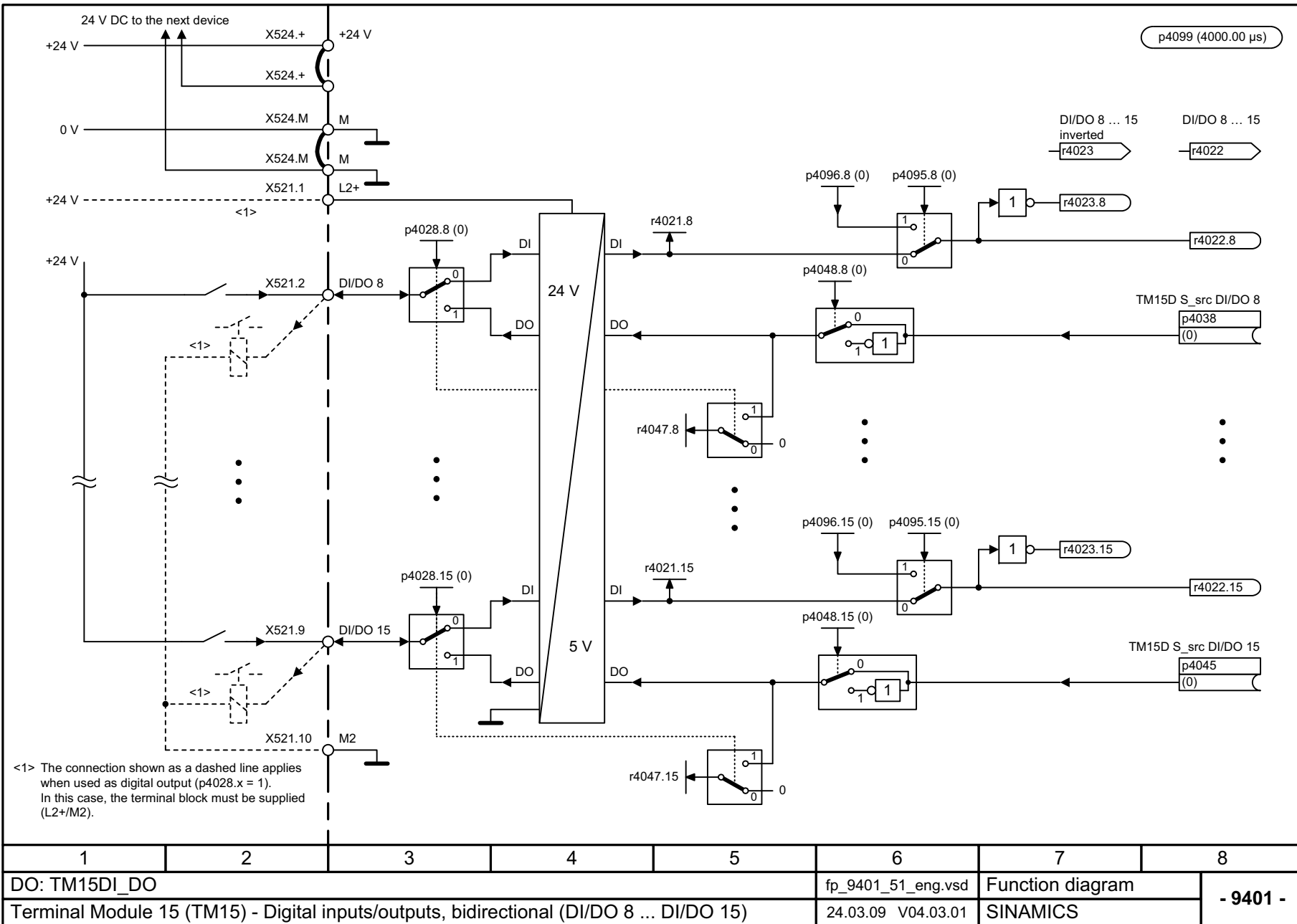


Fig. 2-276 9401 – Digital inputs/outputs, bidirectional (DI/DO 8 to DI/DO 15)



2.30 Terminal Module 31 (TM31)

Function diagrams

9550 – Digital inputs, isolated (DI 0 to DI 3)	2-1692
9552 – Digital inputs, isolated (DI 4 to DI 7)	2-1693
9556 – Digital relay outputs, isolated (DO 0 to DO 1)	2-1694
9560 – Digital inputs/outputs, bidirectional (DI/DO 8 to DI/DO 9)	2-1695
9562 – Digital inputs/outputs, bidirectional (DI/DO 10 to DI/DO 11)	2-1696
9566 – Analog input 0 (AI 0)	2-1697
9568 – Analog input 1 (AI 1)	2-1698
9572 – Analog outputs (AO 0 to AO 1)	2-1699
9576 – Temperature evaluation KTY/PTC	2-1700
9577 – Sensor monitoring KTY/PTC	2-1701

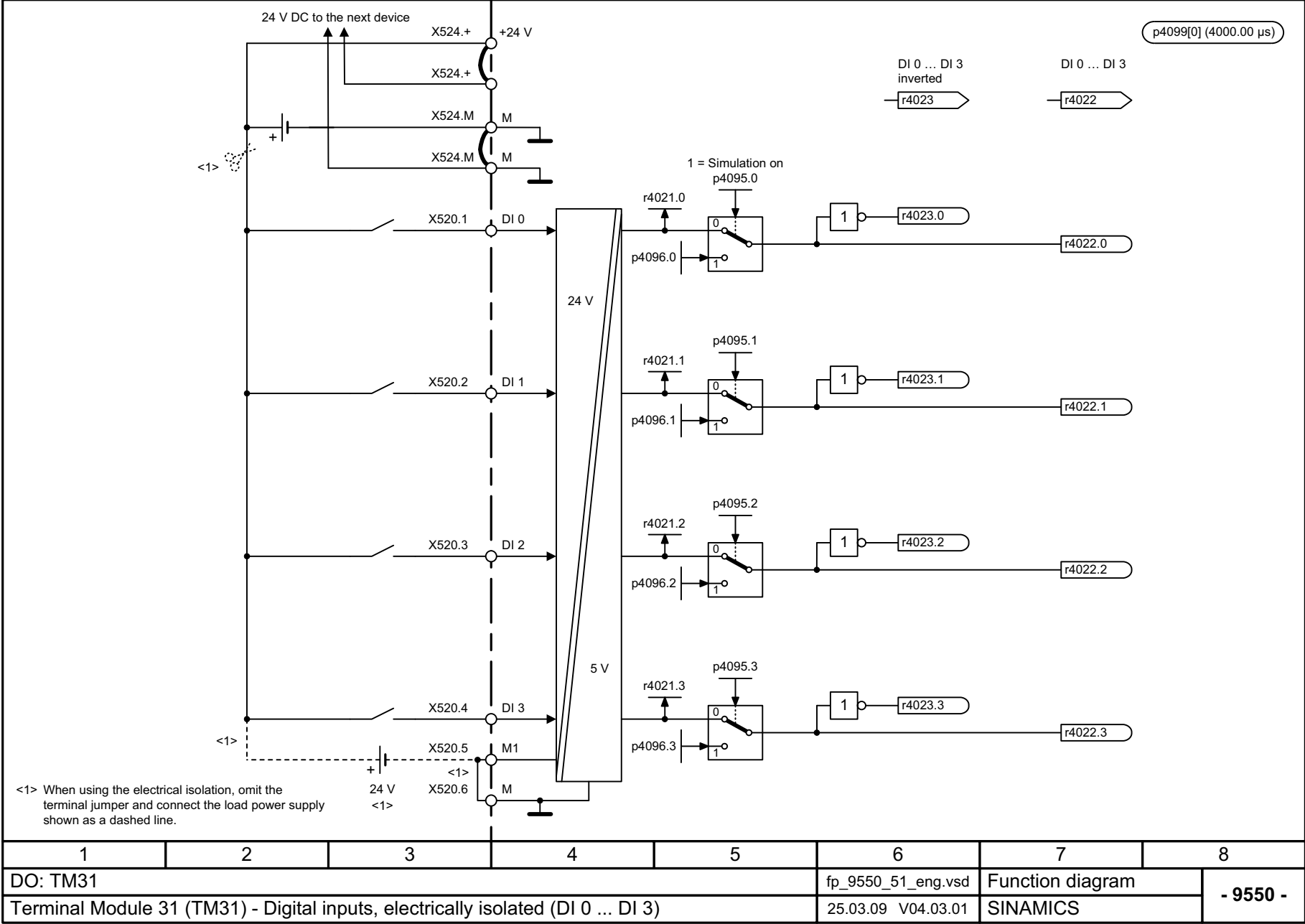


Fig. 2-278 9550 – Digital inputs, isolated (DI 0 to DI 3)

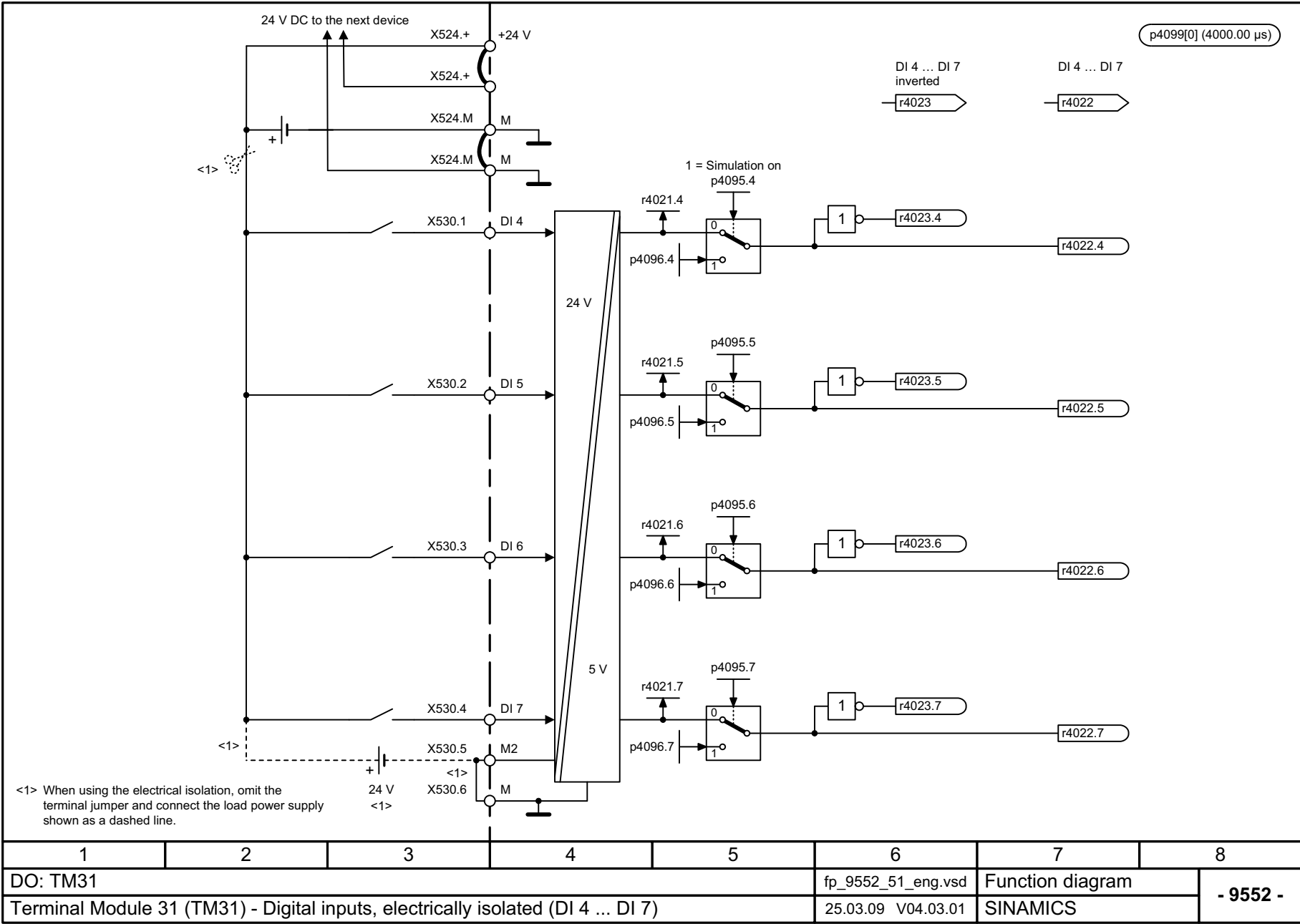


Fig. 2-279 9552 – Digital inputs, isolated (DI 4 to DI 7)

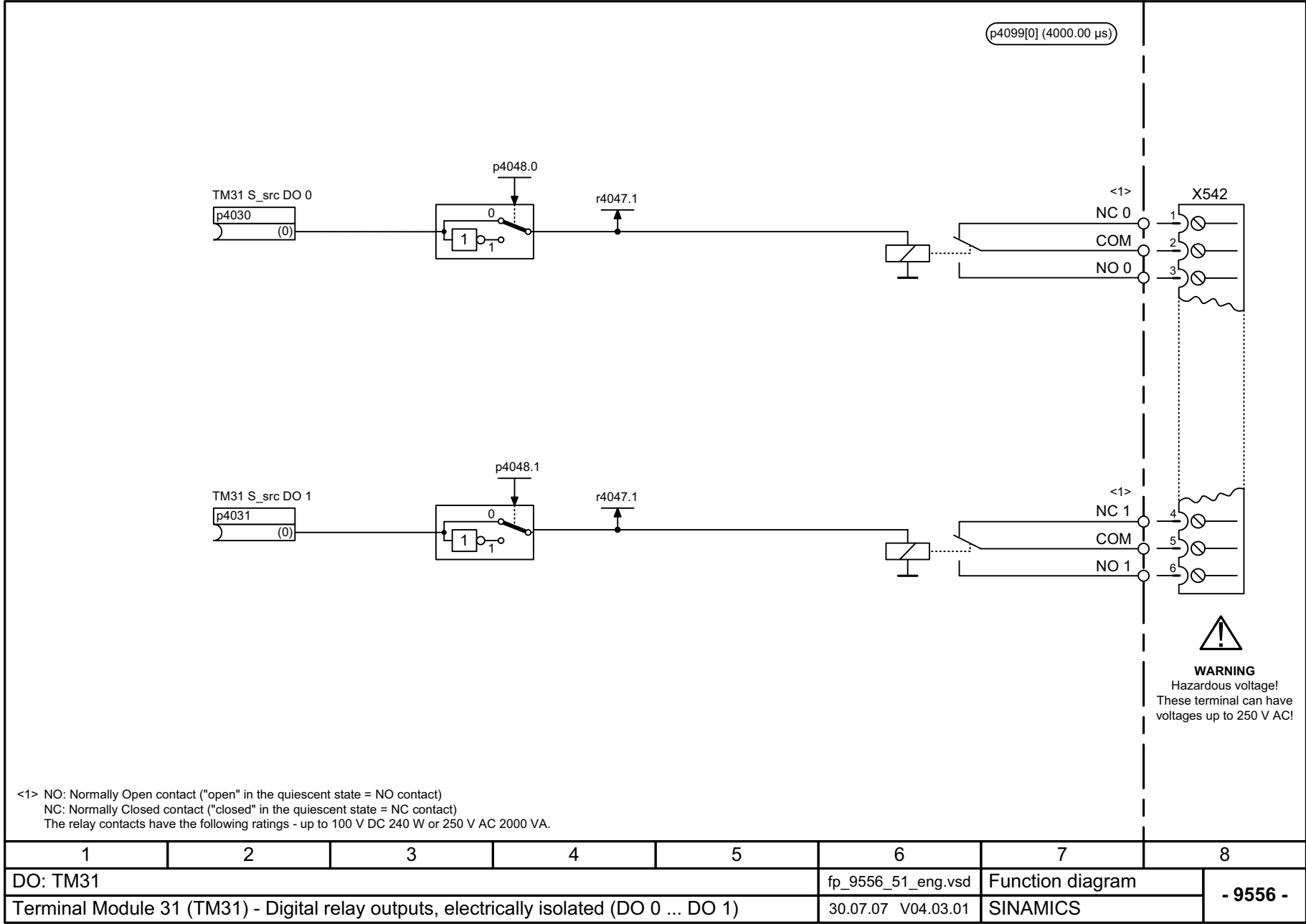
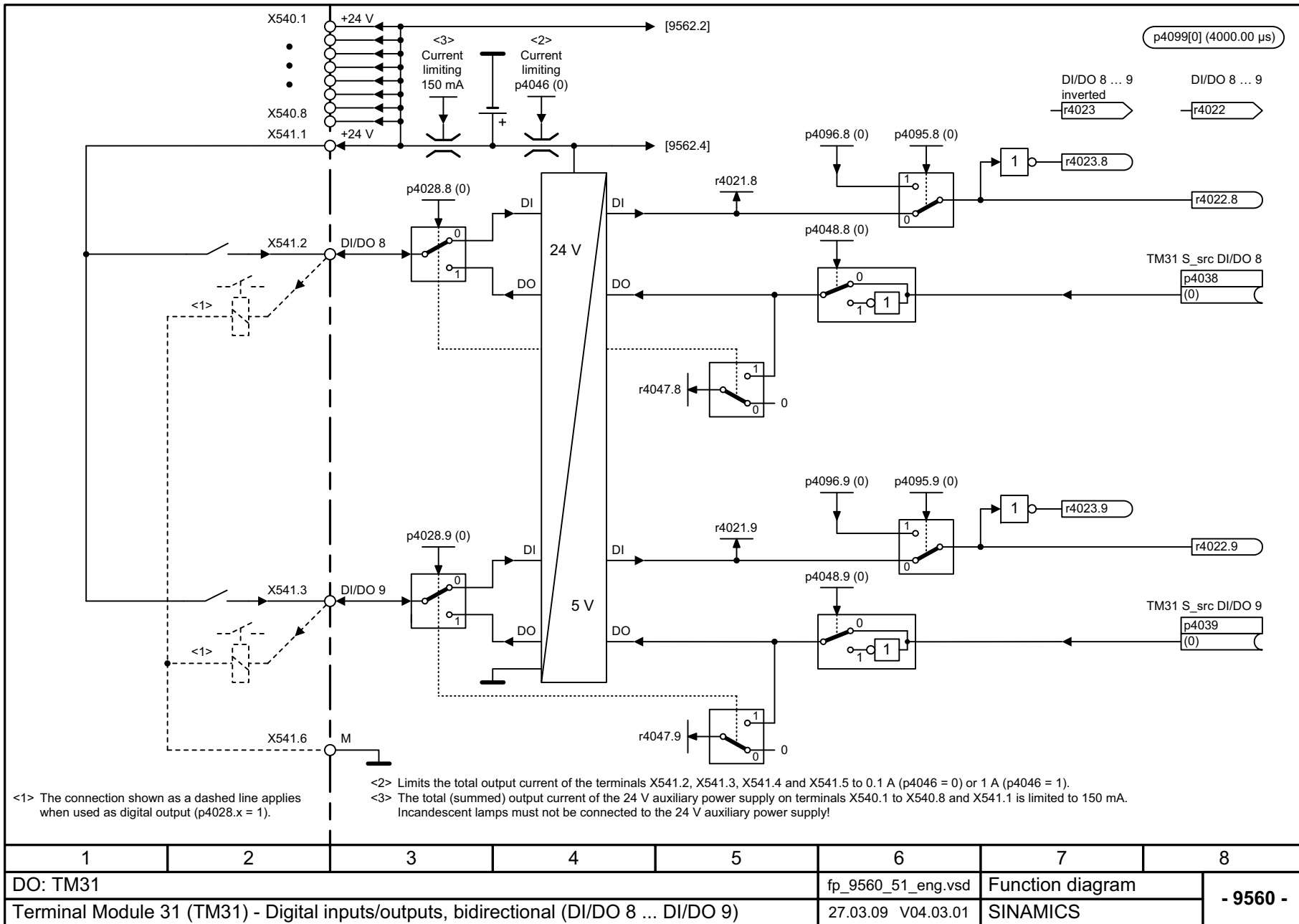


Fig. 2-280 9556 – Digital relay outputs, isolated (DO 0 to DO 1)



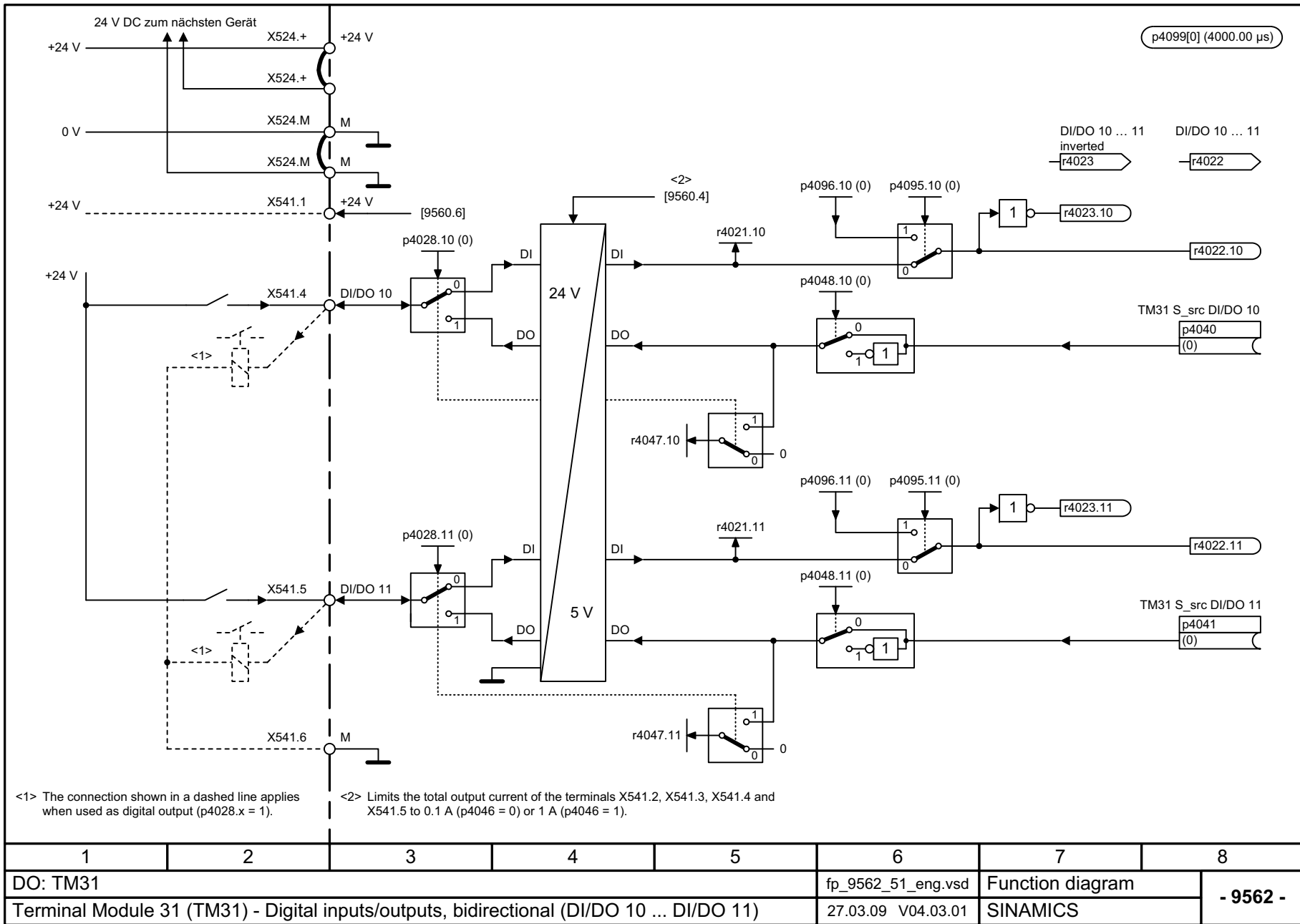
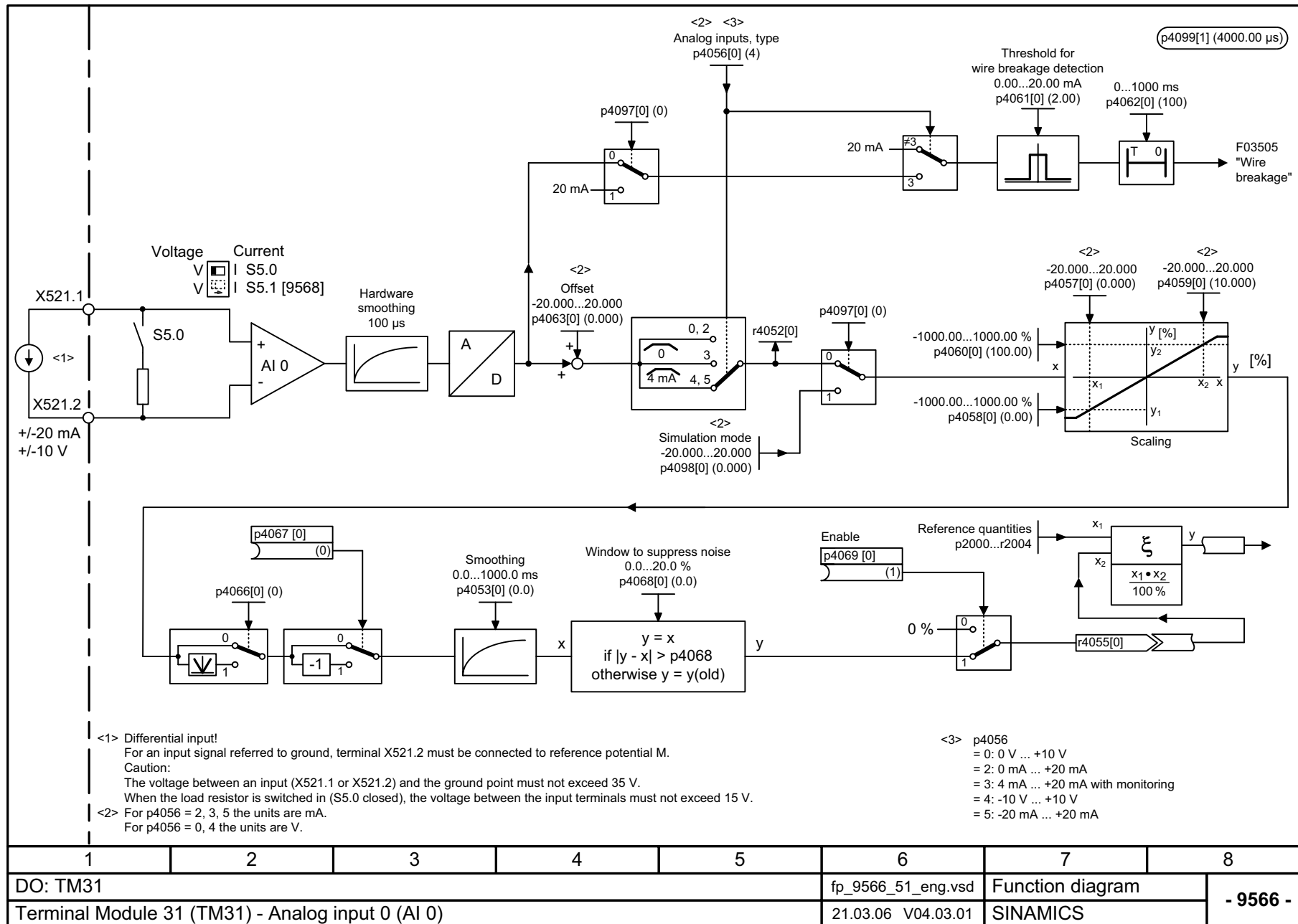
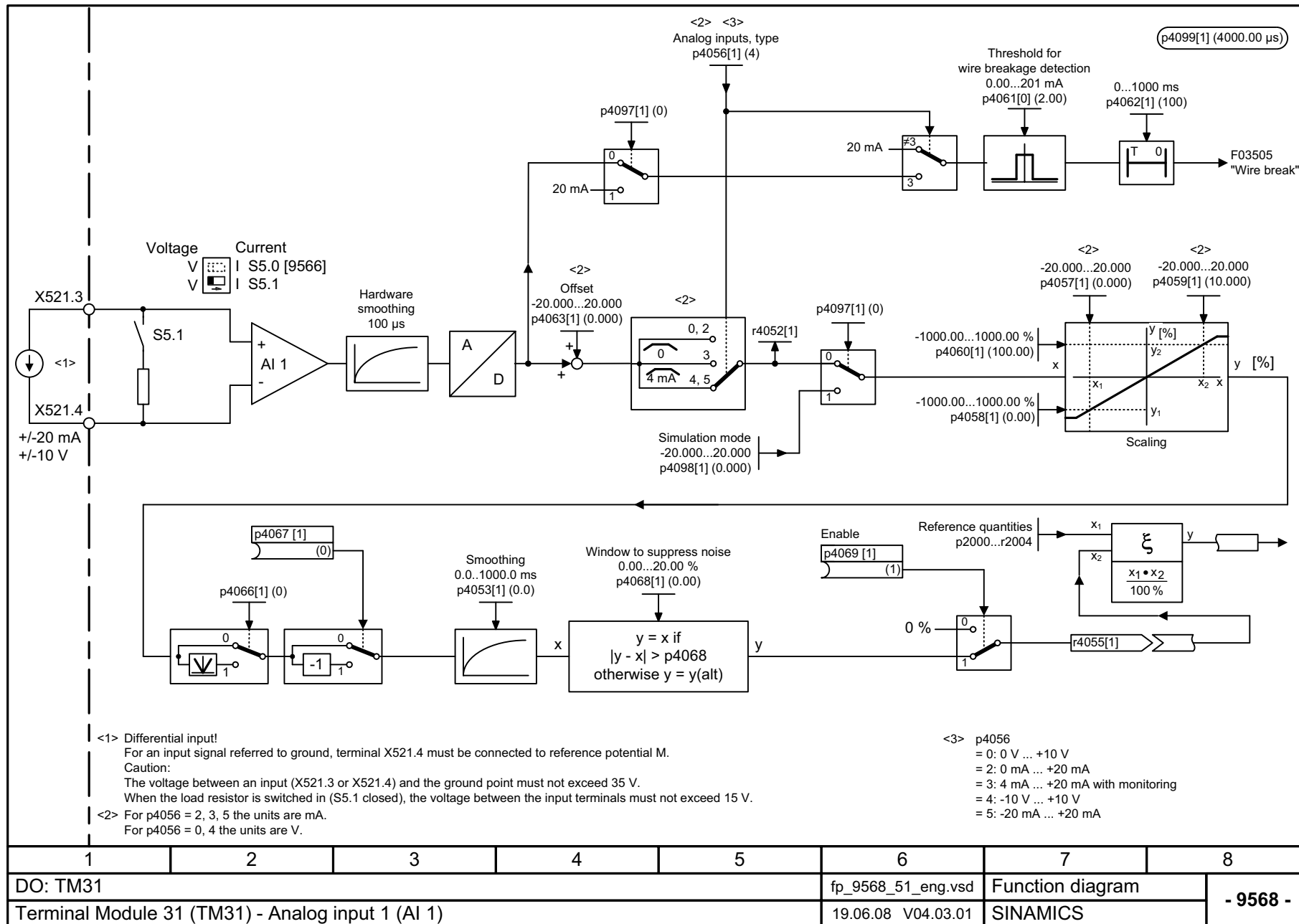
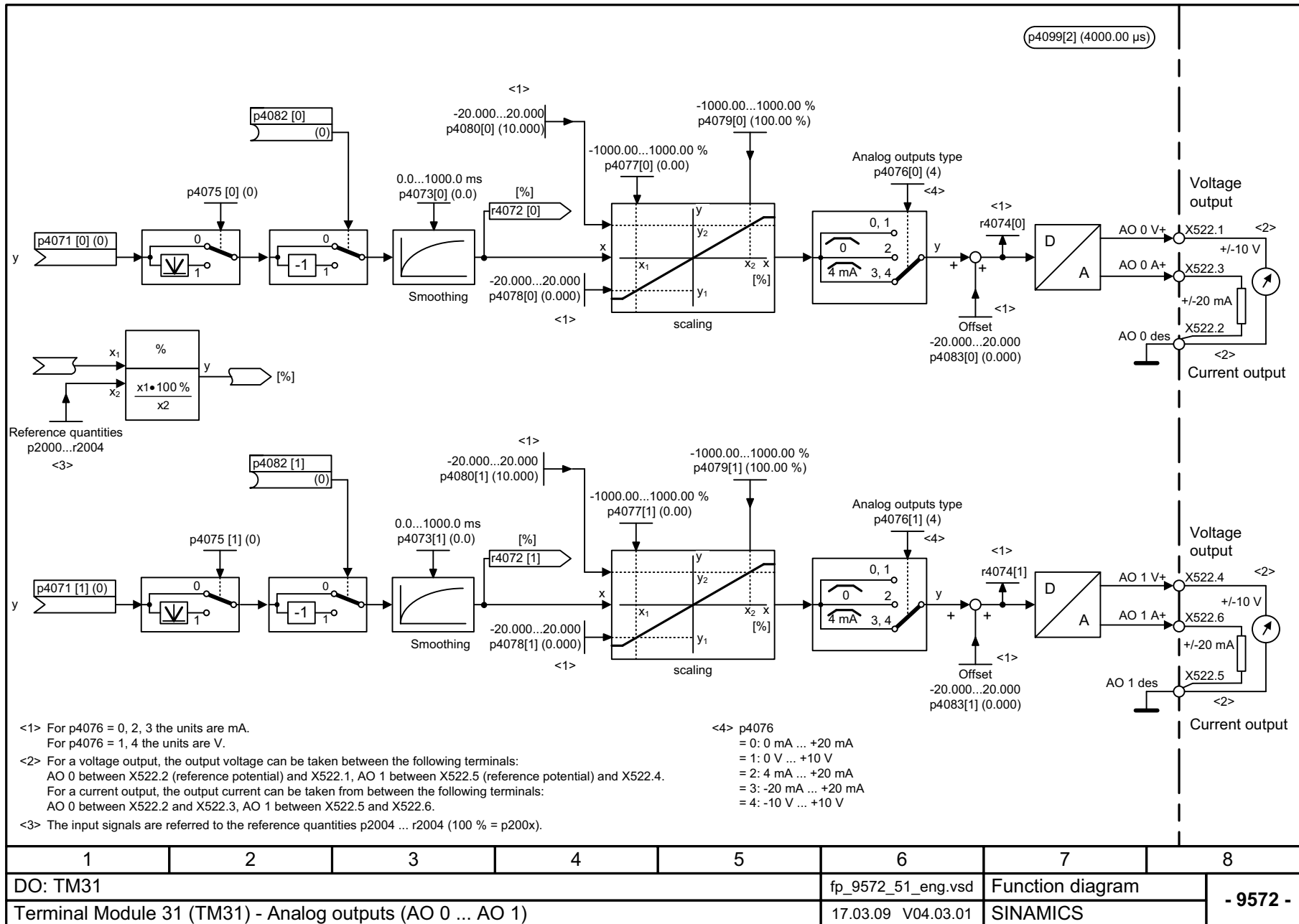


Fig. 2-282 9562 – Digital inputs/outputs, bidirectional (DI/DO 10 to DI/DO 11)







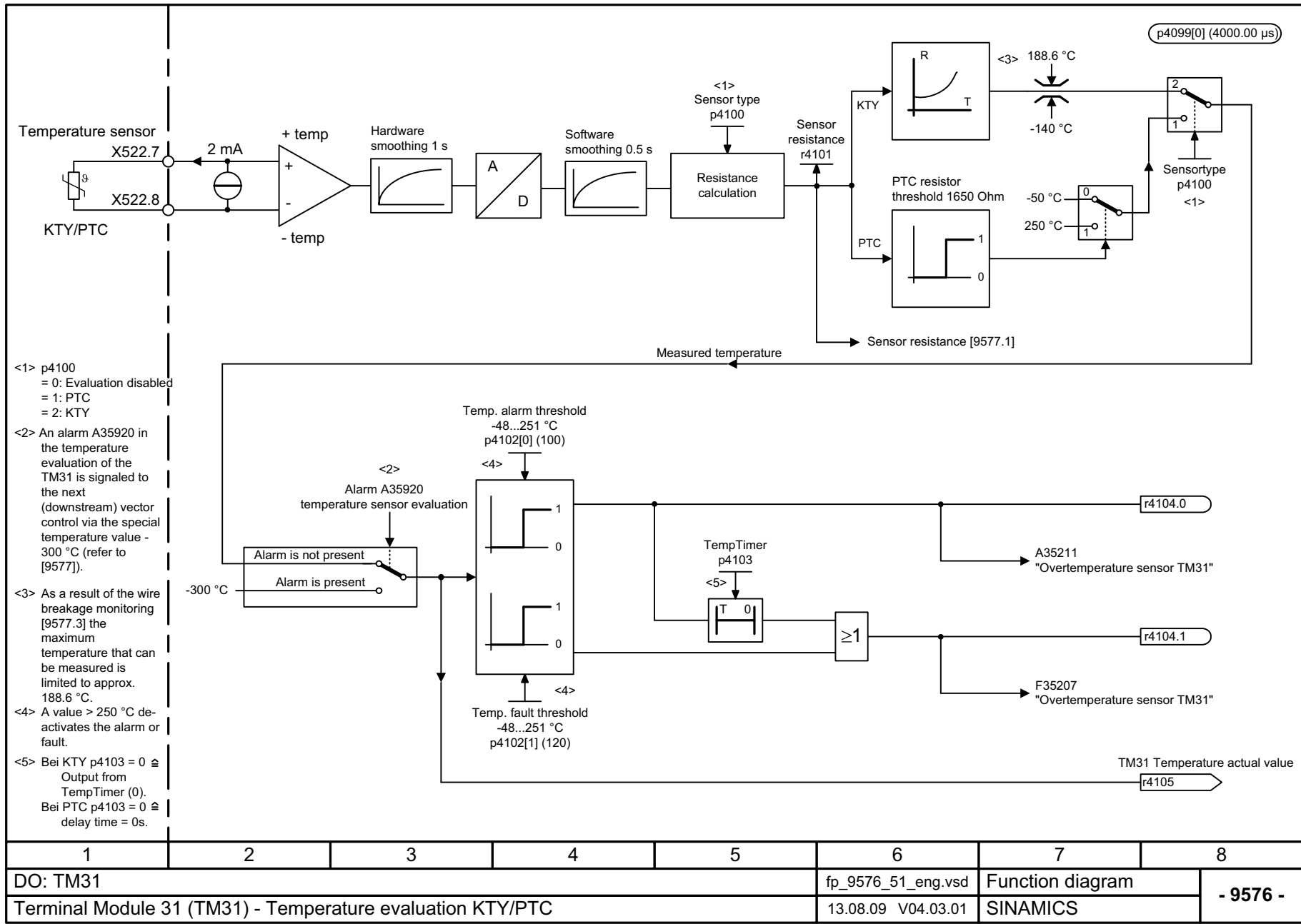
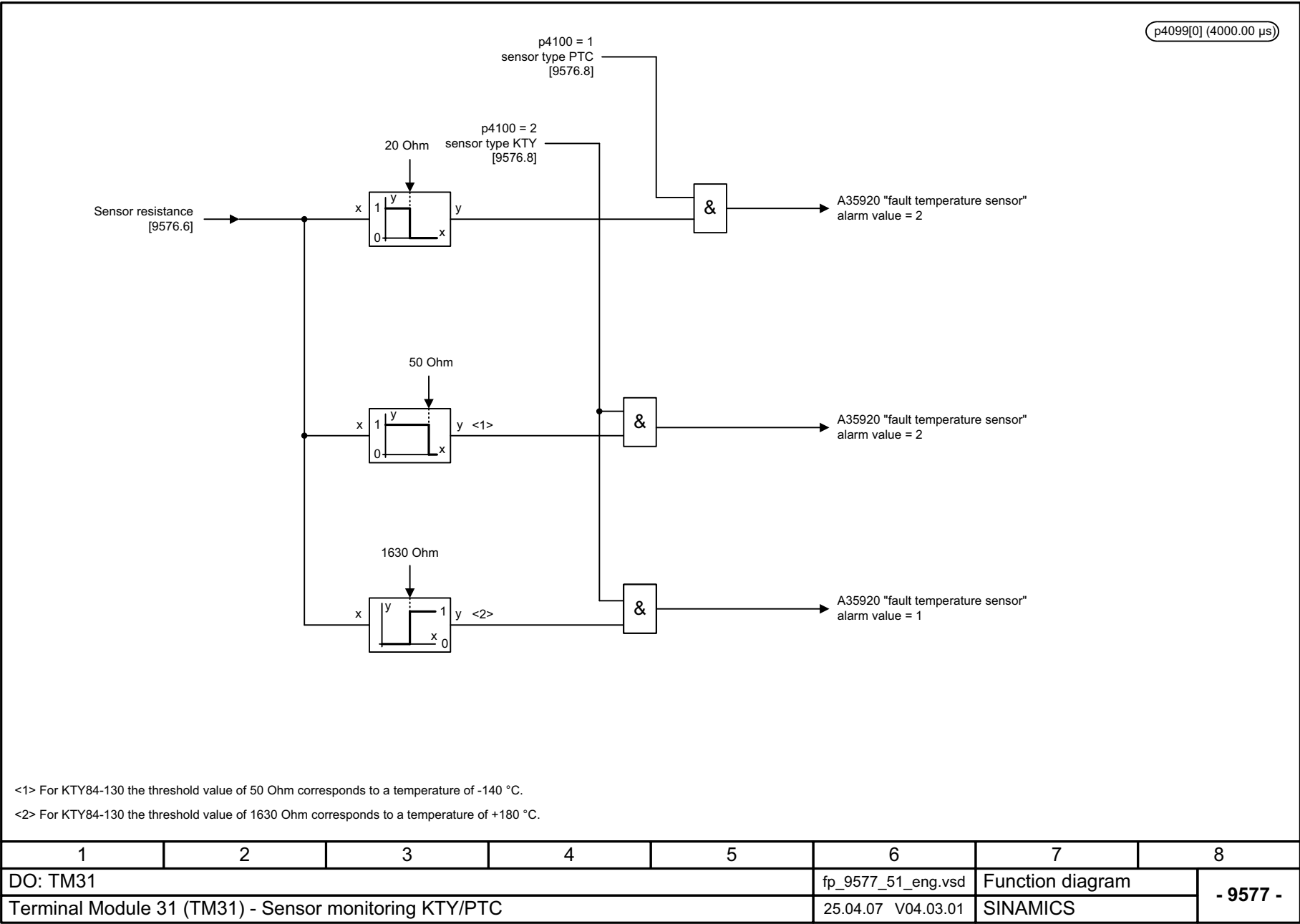


Fig. 2-286 9576 – Temperature evaluation KTY/PTC

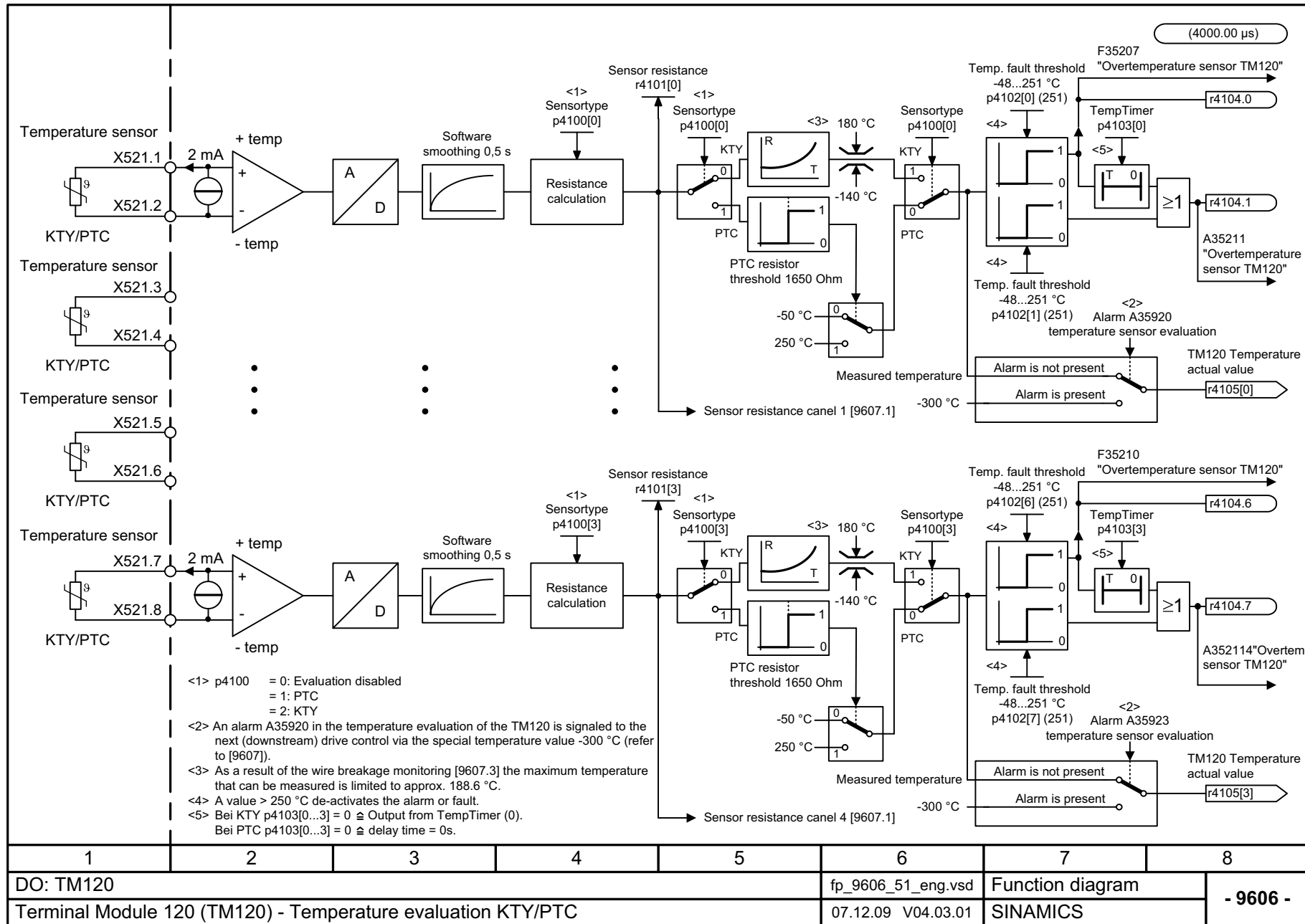
1	2	3	4	5	6	7	8
DO: TM31					fp_9576_51_eng.vsd	Function diagram	
Terminal Module 31 (TM31) - Temperature evaluation KTY/PTC					13.08.09 V04.03.01	SINAMICS	
							- 9576 -



2.31 Terminal Module 120 (TM120)

Function diagrams

9606 – Temperature evaluation KTY/PTC	2-1703
9607 – Sensor monitoring KTY/PTC	2-1704



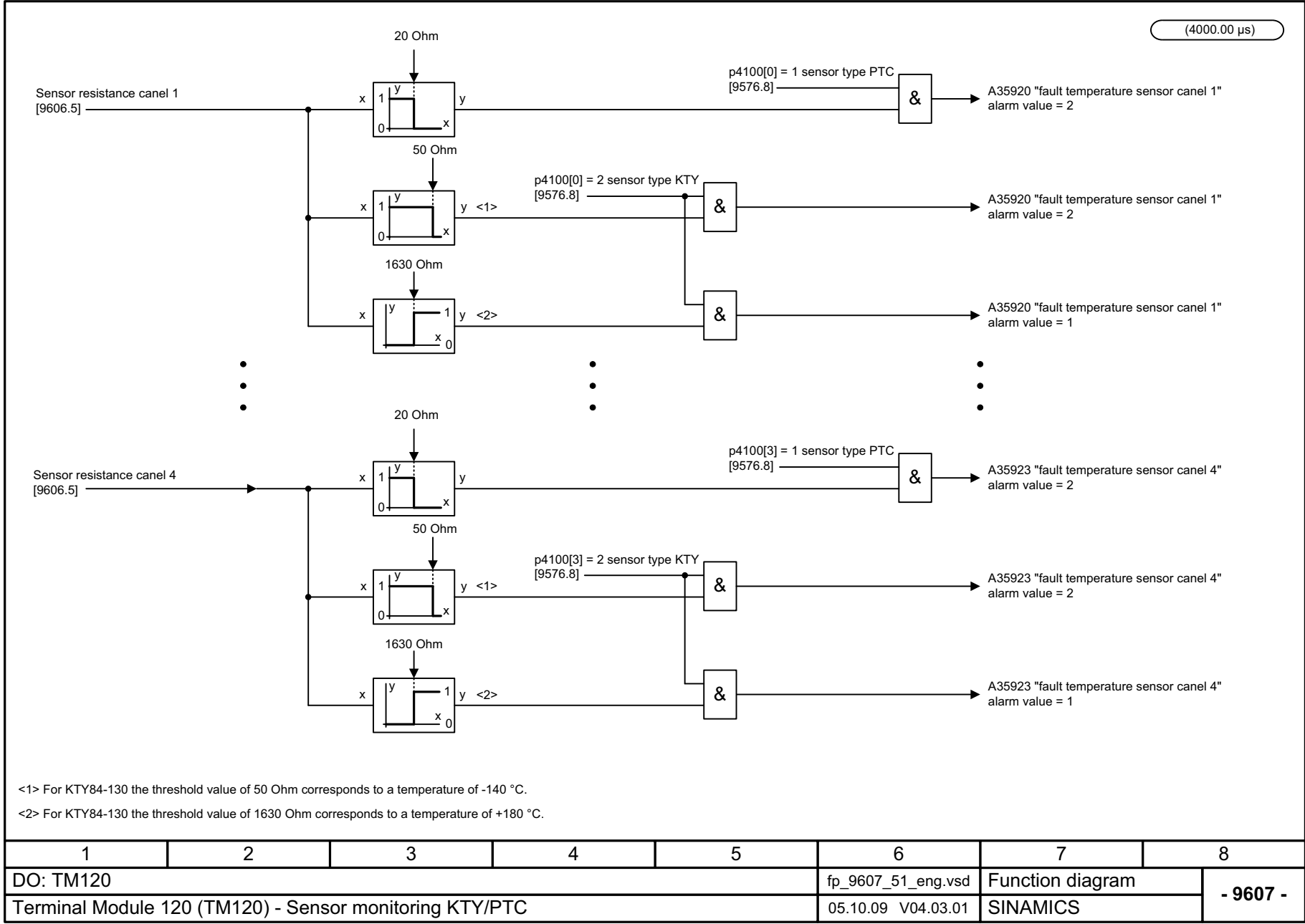
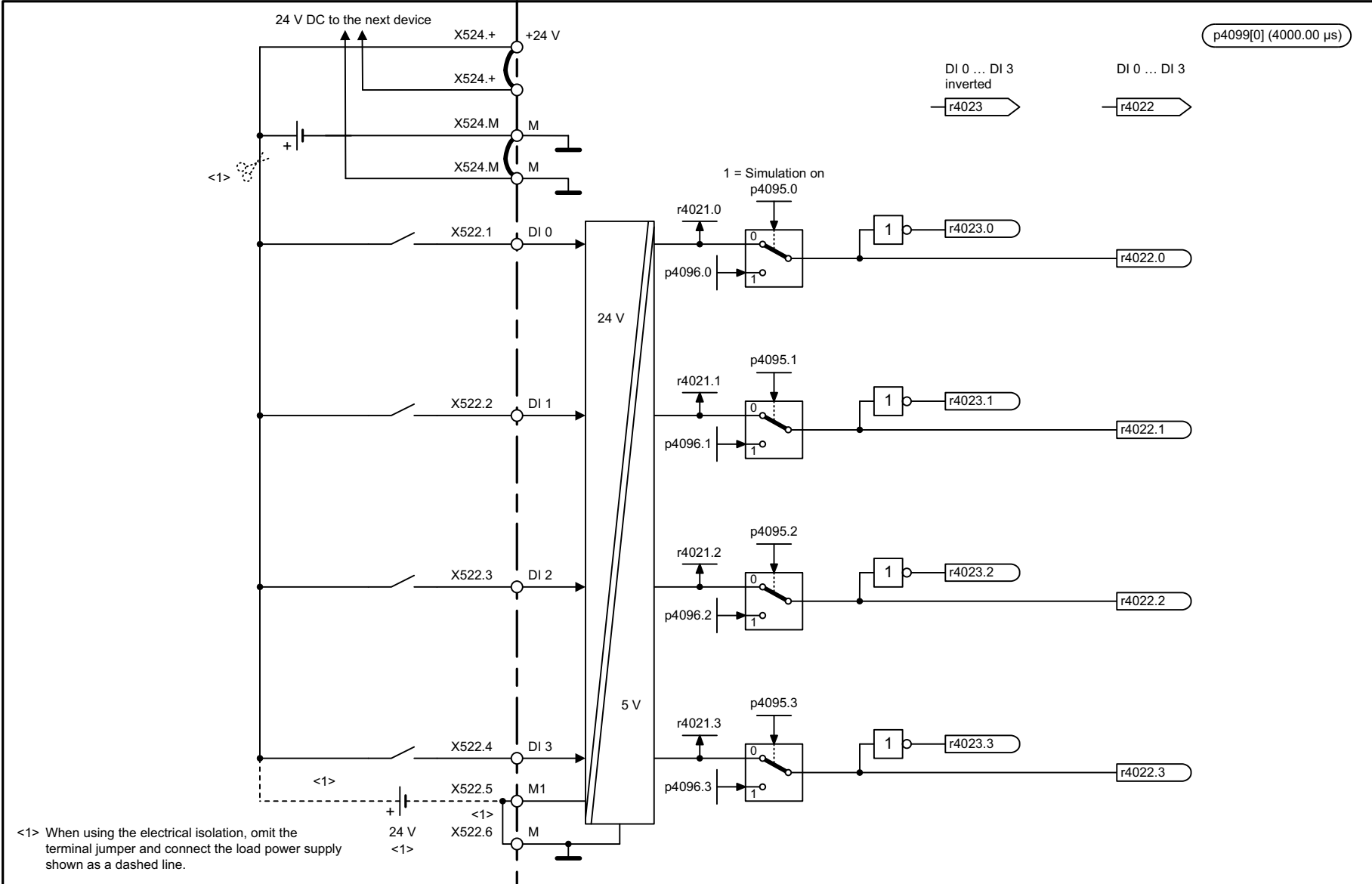


Fig. 2-289 9607 – Sensor monitoring KTY/PTC

2.32 Terminal Module 41 (TM41)

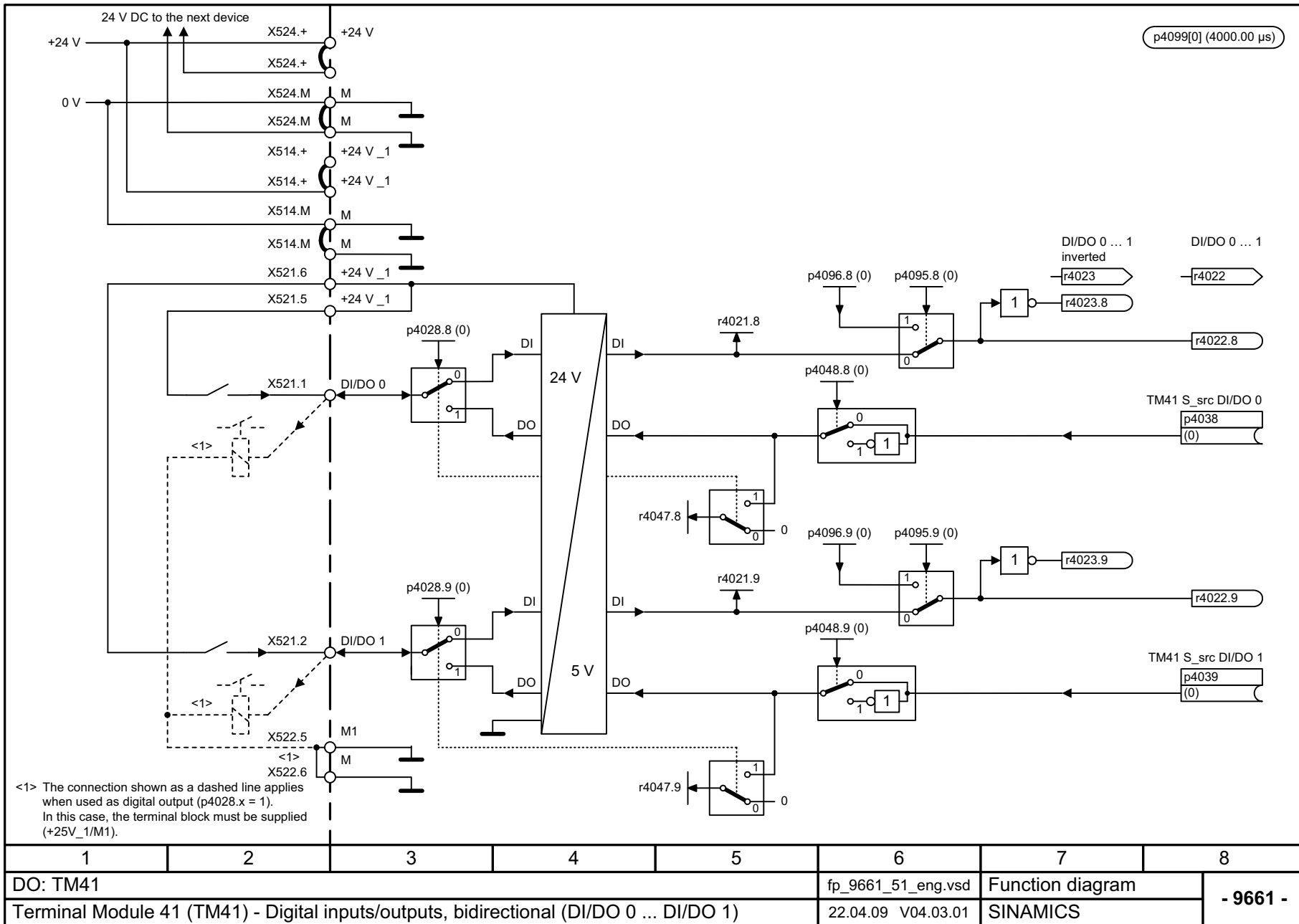
Function diagrams

9660 – Digital inputs, isolated (DI 0 to DI 3)	2-1706
9661 – Digital inputs/outputs, bidirectional (DI/DO 0 to DI/DO 1)	2-1707
9662 – Digital inputs/outputs, bidirectional (DI/DO 2 to DI/DO 3)	2-1708
9663 – Analog input (AI 0)	2-1709
9674 – Incremental encoder emulation (p4400 = 0)	2-1710
9676 – Incremental encoder emulation (p4400 = 1)	2-1711
9677 – STW1 control word interconnection (p0922 = 3)	2-1712
9678 – Control word, sequence control	2-1713
9679 – STW2 control word interconnection (p0922 = 3)	2-1714
9680 – Status word, sequence control	2-1715
9681 – ZSW1 status word interconnection (p0922 = 3)	2-1716
9682 – Sequencer	2-1717
9683 – ZSW2 status word interconnection (p0922 = 3)	2-1718



1	2	3	4	5	6	7	8	
DO: TM41					fp_9660_51_eng.vsd	Function diagram		- 9660 -
Terminal Module 41 (TM41) - Digital inputs, electrically isolated (DI 0 ... DI 3)					25.03.09 V04.03.01	SINAMICS		

Fig. 2-290 9660 – Digital inputs, isolated (DI 0 to DI 3)



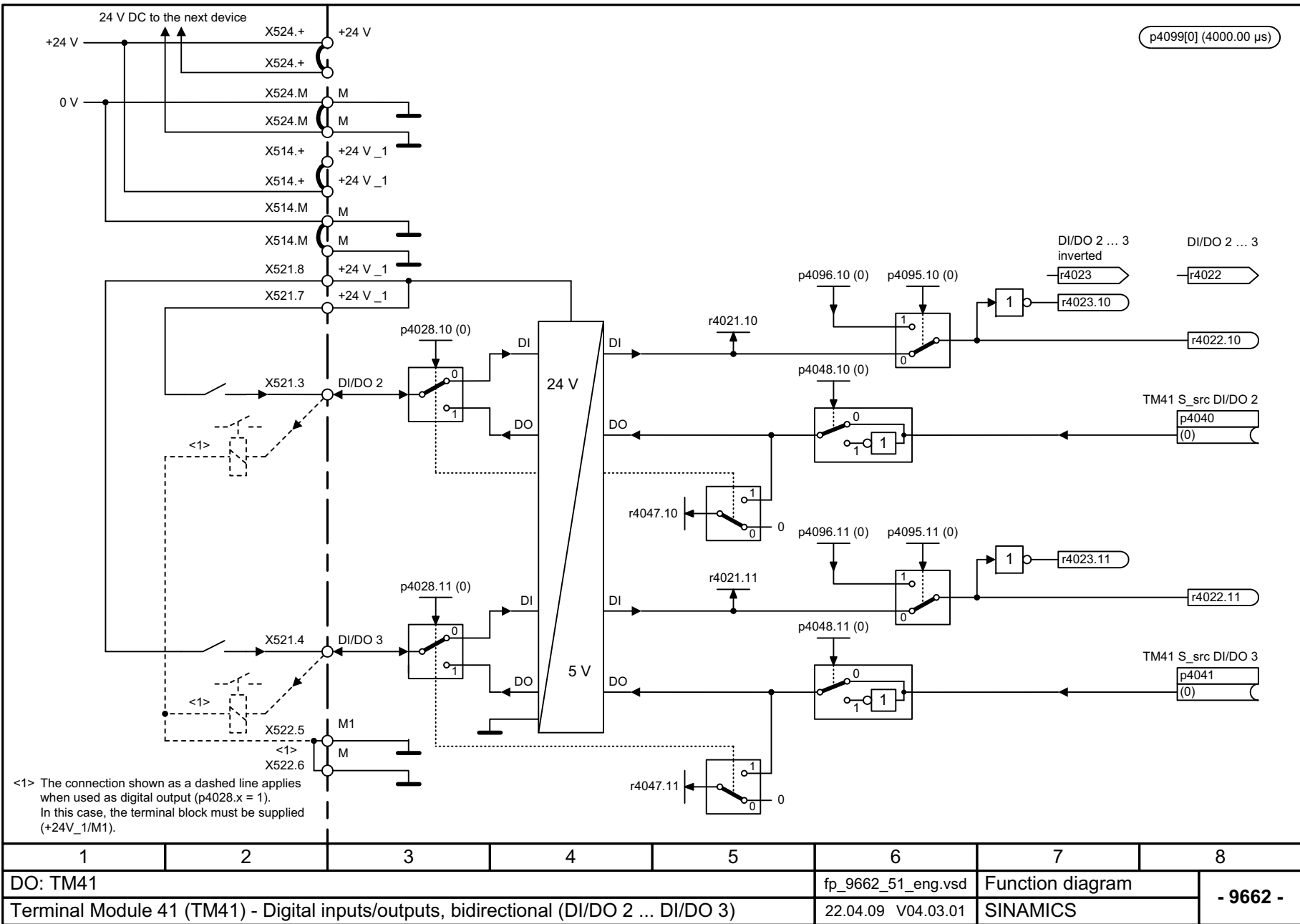
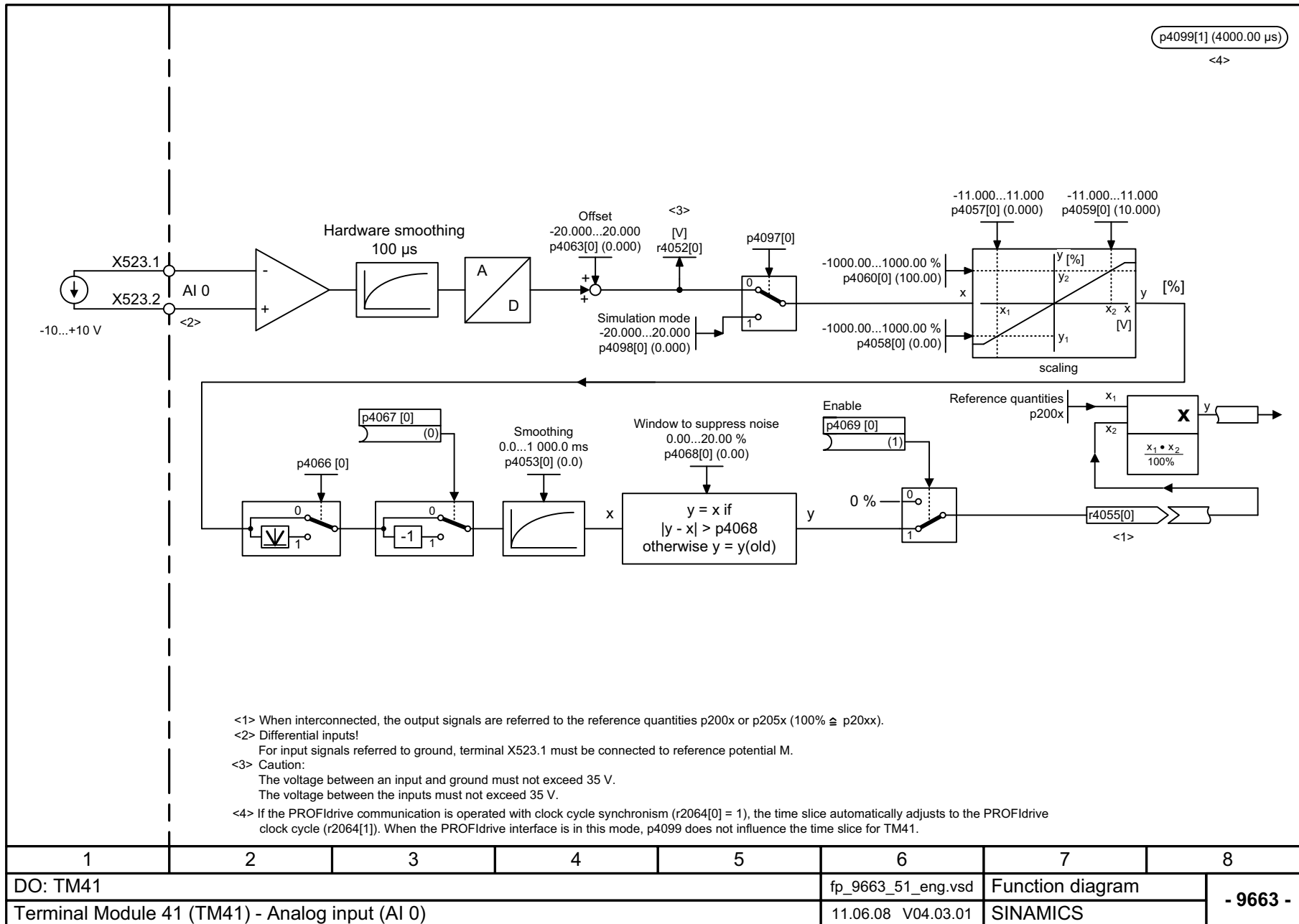


Fig. 2-292 9662 – Digital inputs/outputs, bidirectional (DI/DO 2 to DI/DO 3)



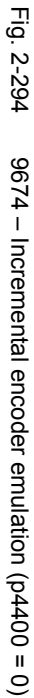


Fig. 2-295



2000.00 µs

Signal targets for STW1 Standard telegram 3 (p0922 = 3)

Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW1.0	 = ON (pulses can be enabled) 0 = OFF1 (braking with ramp-function generator, then pulse cancellation, ready-to-power-up)	p0840 = r2090.0	[9678.3]	[9682]	-
STW1.1	1 = No OFF2 (enable is possible) 0 = OFF2 (immediate pulse cancellation and power-on inhibit)	p0844 = r2090.1	[9678.3]	[9682]	-
STW1.2	1 = No OFF3 (enable possible) 0 = OFF3 (braking with the OFF3 ramp p1135, then pulse cancellation and power-on inhibit)	p0848 = r2090.2	[9678.3]	[9682]	-
STW1.3	1 = Enable operation (pulses can be enabled) 0 = Inhibit operation (cancel pulses)	p0852 = r2090.3	[9678.3]	[9682]	-
STW1.4	1 = Operating condition (the ramp-function generator can be enabled) 0 = Inhibit ramp-function generator (set the ramp-function generator output to zero)	p1140 = r2090.4	[9678.3]	[9682]	-
STW1.5	1 = Enable the ramp-function generator 0 = Stop the ramp-function generator (freeze the ramp-function generator output)	p1141 = r2090.5	[9678.3]	[9682]	-
STW1.6	1 = Enable setpoint 0 = Inhibit setpoint (set the ramp-function generator input to zero)	p1142 = r2090.6	[9678.3]	[9682]	-
STW1.7	 = Acknowledge faults	p2103[0] = r2090.7	[2546.1]	[8060]	-
STW1.8	Reserved	-	-	-	-
STW1.9	Reserved	-	-	-	-
STW1.10	1 = Control via PLC <1>	p0854[0] = r2090.10	[9678.3]	-	-
STW1.11	Reserved	-	-	-	-
STW1.12	Reserved	-	-	-	-
STW1.13	1 = Enable zero marks	p1035 = p2090.13	[9678.3]	-	-
STW1.14	Reserved	-	-	-	-
STW1.15	Reserved	-	-	-	-

<1> The drive object is ready to accept data.

1	2	3	4	5	6	7	8
DO: TM41					fp_9677_51_eng.vsd	Function diagram	- 9677 -
Terminal Module 41 (TM41) - STW1 control word interconnection (p0922 = 3)					20.06.08 V04.03.01	SINAMICS	

Fig. 2-296 9677 – STW1 control word interconnection (p0922 = 3)

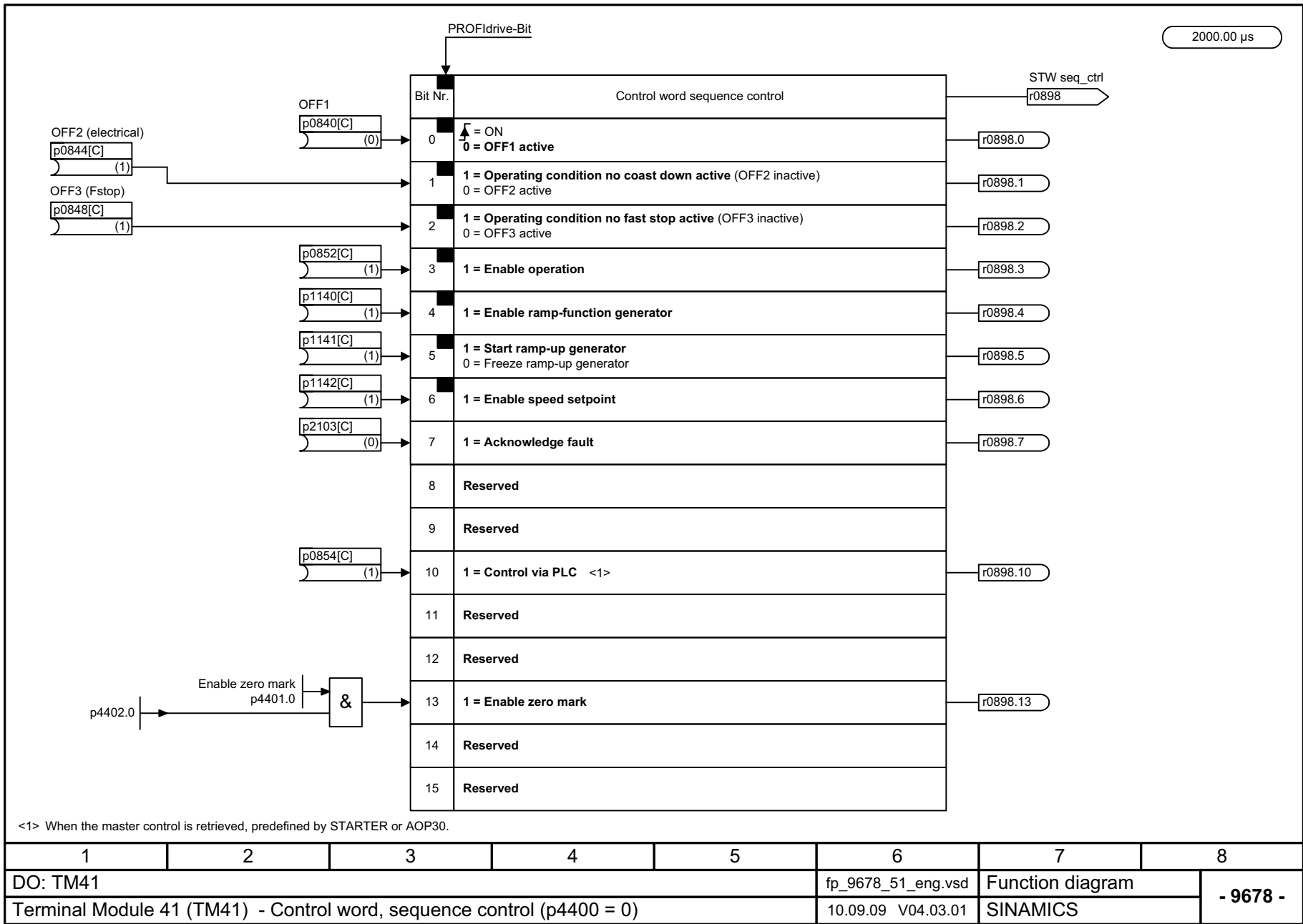


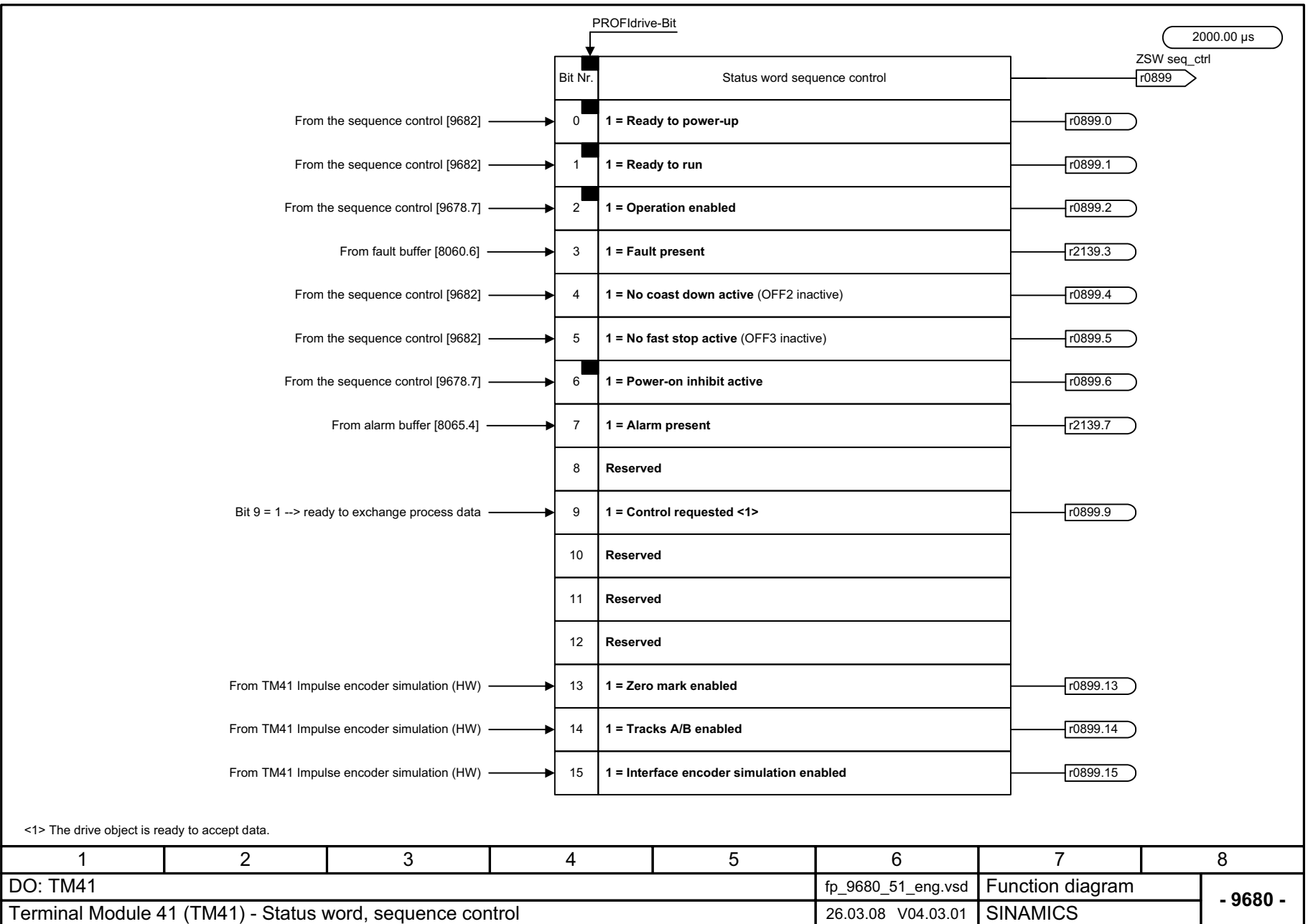
Fig. 2-297 9678 – Control word, sequence control

2000.00 µs

Signal targets for STW2 Standard telegram 3 (p0922 = 3)					
Signal	Meaning	Interconnection parameters	[Function diagram] internal control word	[Function diagram] signal target	Inverted
STW2.0	Drive data set selection DDS, bit 0	p0820[0] = r2093.0	-	-	-
STW2.1	Drive data set selection DDS, bit 1	p0821[0] = r2093.1	-	-	-
STW2.2	Drive data set selection DDS, bit 2	p0822[0] = r2093.2	-	-	-
STW2.3	Drive data set selection DDS, bit 3	p0823[0] = r2093.3	-	-	-
STW2.4	Drive data set selection DDS, bit 4	p0824[0] = r2093.4	-	-	-
STW2.5	Reserved	-	-	-	-
STW2.6	Reserved	-	-	-	-
STW2.7	Reserved	-	-	-	-
STW2.8	Reserved	-	-	-	-
STW2.9	Reserved	-	-	-	-
STW2.10	Reserved	-	-	-	-
STW2.11	Reserved	-	-	-	-
STW2.12	Master sign-of-life, bit 0	p2045 = r2050[3]	-	[2410]	-
STW2.13	Master sign-of-life, bit 1				
STW2.14	Master sign-of-life, bit 2				
STW2.15	Master sign-of-life, bit 3				

1	2	3	4	5	6	7	8
DO: TM41					fp_9679_51_eng.vsd	Function diagram	- 9679 -
Terminal Module 41 (TM41) - STW2 control word interconnection (p0922 = 3)					05.10.07 V04.03.01	SINAMICS	

Fig. 2-298 9679 – STW2 control word interconnection (p0922 = 3)



2000.00 µs

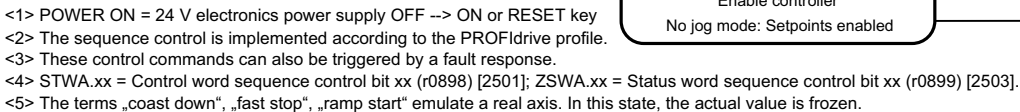
Signal sources for ZSW1 Standard telegram 3 (p0922 = 3)

Signal	Meaning	Interconnection parameters	[Function diagram] Internal status word	[Function diagram] Signal source	Inverted
ZSW1.0	1 = Ready to power up	p2080[0] = r0899.0	[9680.7]	[9682]	-
ZSW1.1	1 = Ready	p2080[1] = r0899.1	[9680.7]	[9682]	-
ZSW1.2	1 = Operation enabled	p2080[2] = r0899.2	[9680.7]	[9682]	-
ZSW1.3	1 = Fault present	p2080[3] = r2139.3	[2548.7]	[8060]	-
ZSW1.4	1 = No coast down active	p2080[4] = r0899.4	[9680.7]	[9682]	-
ZSW1.5	1 = No quick stop active	p2080[5] = r0899.5	[9680.7]	[9682]	-
ZSW1.6	1 = Power-on inhibit active	p2080[6] = r0899.6	[9680.7]	[9682]	-
ZSW1.7	1 = Alarm present	p2080[7] = r2139.7	[2548.7]	[8065]	-
ZSW1.8	Reserved	-	-	-	-
ZSW1.9	1 = Control requested <1>	p2080[9] = r0899.9	[9680.7]	[9680]	-
ZSW1.10	Reserved	-	-	-	-
ZSW1.11	Reserved	-	-	-	-
ZSW1.12	Reserved	-	-	-	-
ZSW1.13	Reserved	-	-	-	-
ZSW1.14	Reserved	-	-	-	-
ZSW1.15	Reserved	-	-	-	-

<1> The drive object is ready accept data.

1	2	3	4	5	6	7	8
DO: TM41			fp_9681_51_eng.vsd			Function diagram	- 9681 -
Terminal Module 41 (TM41) - ZSW1 status word interconnection (p0922 = 3)			26.03.08 V04.03.01			SINAMICS	

Fig. 2-300 9681 – ZSW1 status word interconnection (p0922 = 3)



2000.00 µs

Signal sources for ZSW2 Standard telegram 3 (p0922 = 3)

Signal	Meaning	Interconnection parameters	[Function diagram] Internal status word	[Function diagram] Signal source	Inverted
ZSW2.0	1 = DDS present Bit 0	p2081[0] - r0051.0	-	-	-
ZSW2.1	1 = DDS present Bit 1	p2081[1] - r0051.1	-	-	-
ZSW2.2	1 = DDS present Bit 2	p2081[2] - r0051.2	-	-	-
ZSW2.3	1 = DDS present Bit 3	p2081[3] - r0051.3	-	-	-
ZSW2.4	1 = DDS present Bit 4	p2081[4] - r0051.4	-	-	-
ZSW2.5	1 = Alarm class bit 0	p2081[5] = r2139.11	-	-	-
ZSW2.6	1 = Alarm class bit 1	p2081[6] = r2139.12	-	-	-
ZSW2.7	Reserved	-	-	-	-
ZSW2.8	Reserved	-	-	-	-
ZSW2.9	Reserved	-	-	-	-
ZSW2.10	Reserved	-	-	-	-
ZSW2.11	Reserved	-	-	-	-
ZSW2.12	Slave sign-of-life bit 0	Implicitly interconnected	-	-	-
ZSW2.13	Slave sign-of-life bit 1				
ZSW2.14	Slave sign-of-life bit 2				
ZSW2.15	Slave sign-of-life bit 3				

<1>

<1> These signals are automatically interconnected for clock-cycle synchronous operation.

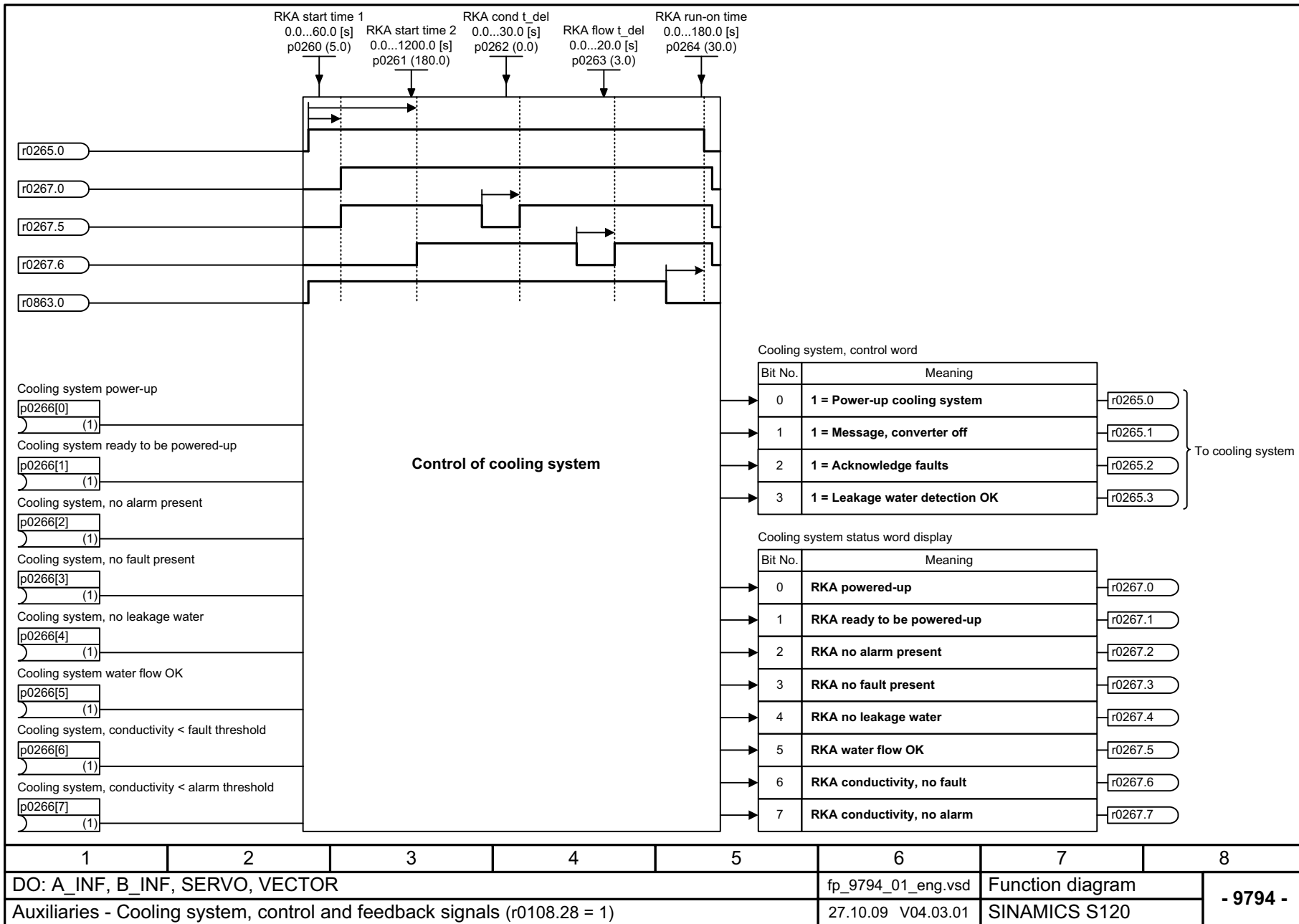
1	2	3	4	5	6	7	8
DO: TM41					fp_9683_51_eng.vsd	Function diagram	- 9683 -
Terminal Module 41 (TM41) - ZSW2 status word interconnection (p0922 = 3)					04.09.09 V04.03.01	SINAMICS	

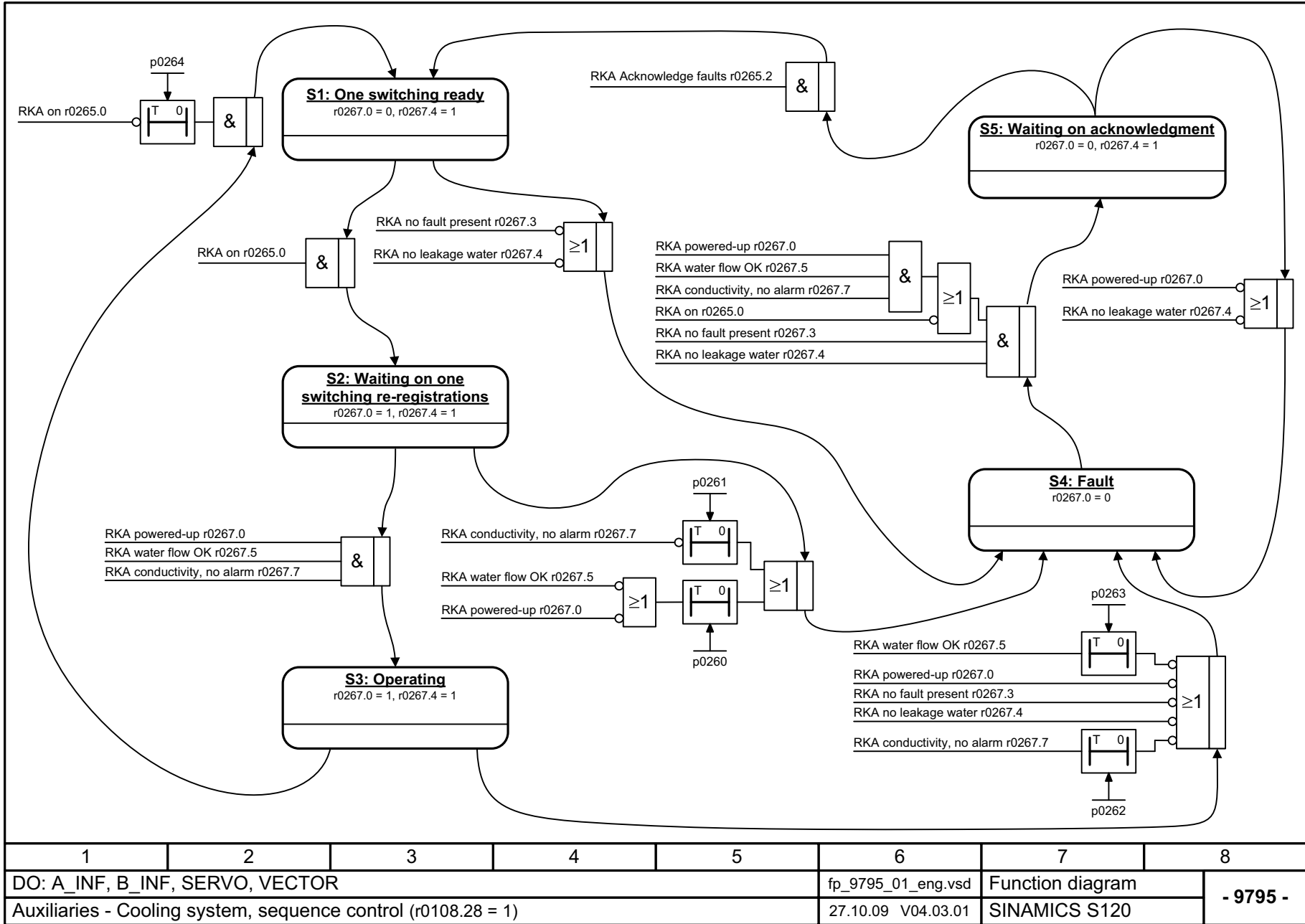
Fig. 2-302 9683 – ZSW2 status word interconnection (p0922 = 3)

2.33 Auxiliary circuits

Function diagrams

9794 – Cooling unit, control and feedback signals (r0108.28 = 1)	2-1720
9795 – Cooling unit, sequence control (r0108.28 = 1)	2-1721



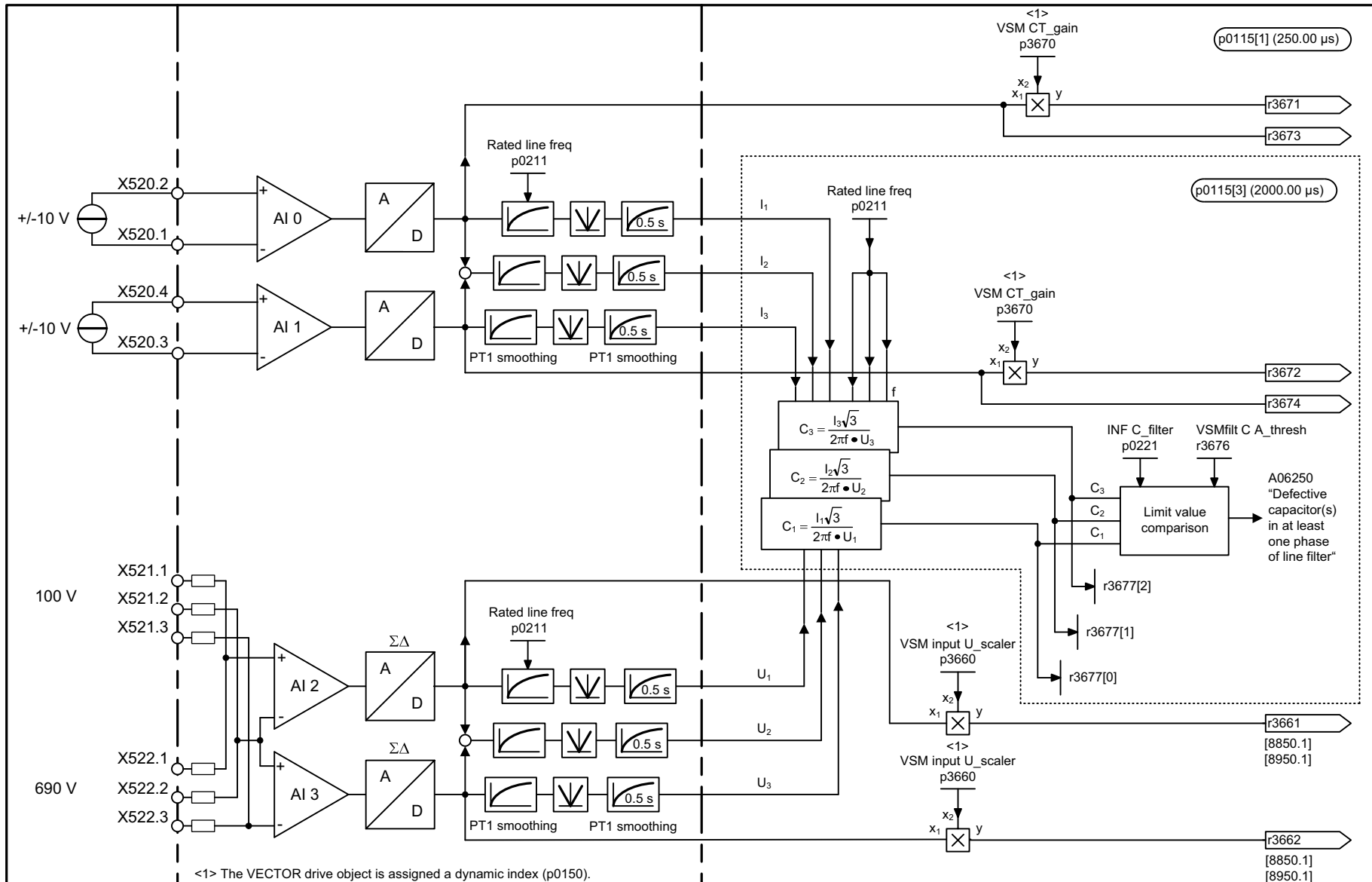


1	2	3	4	5	6	7	8
DO: A_INF, B_INF, SERVO, VECTOR					fp_9795_01_eng.vsd	Function diagram	
Auxiliaries - Cooling system, sequence control (r0108.28 = 1)					27.10.09 V04.03.01	SINAMICS S120	
							- 9795 -

2.34 Voltage Sensing Module (VSM)

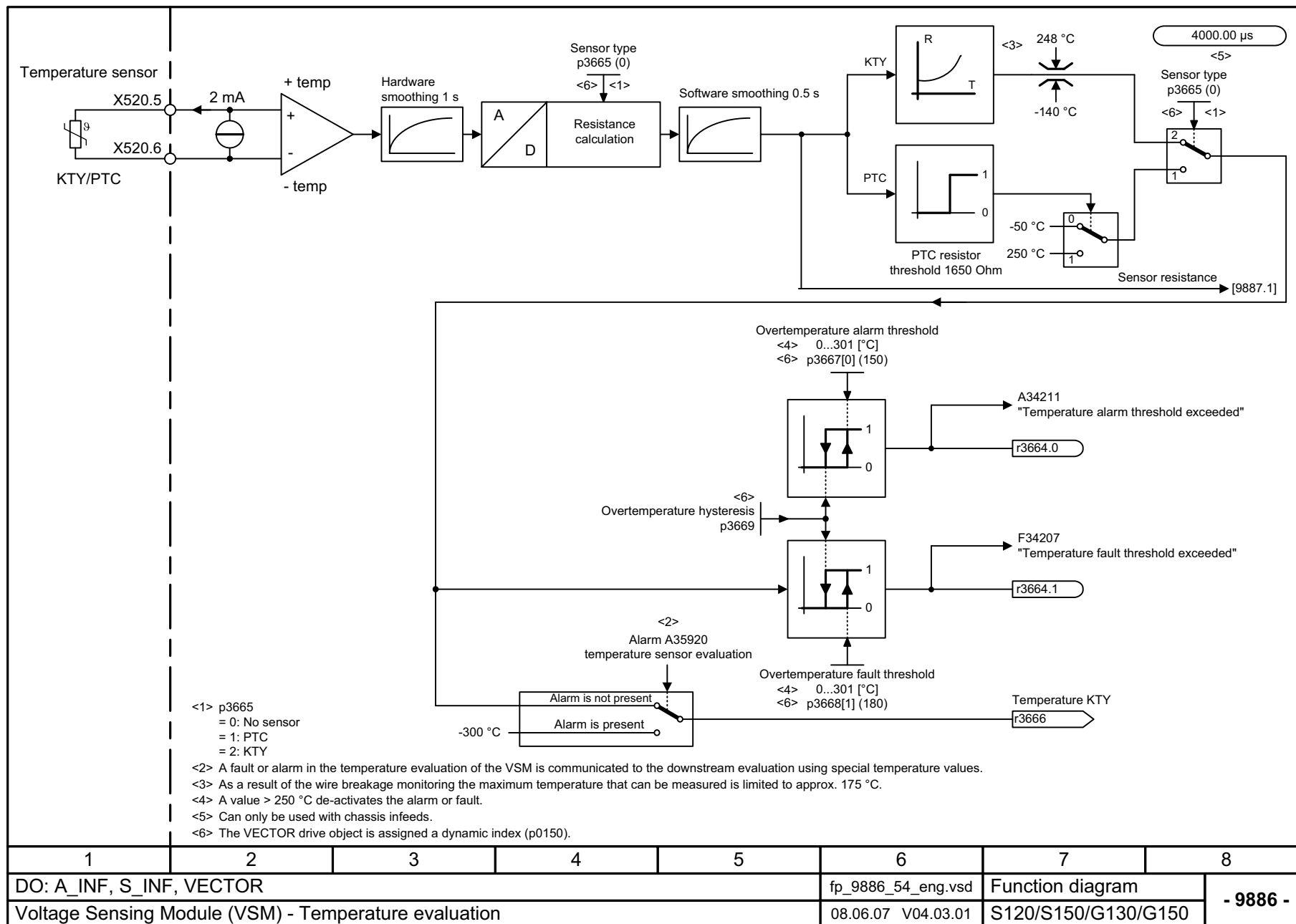
Function diagrams

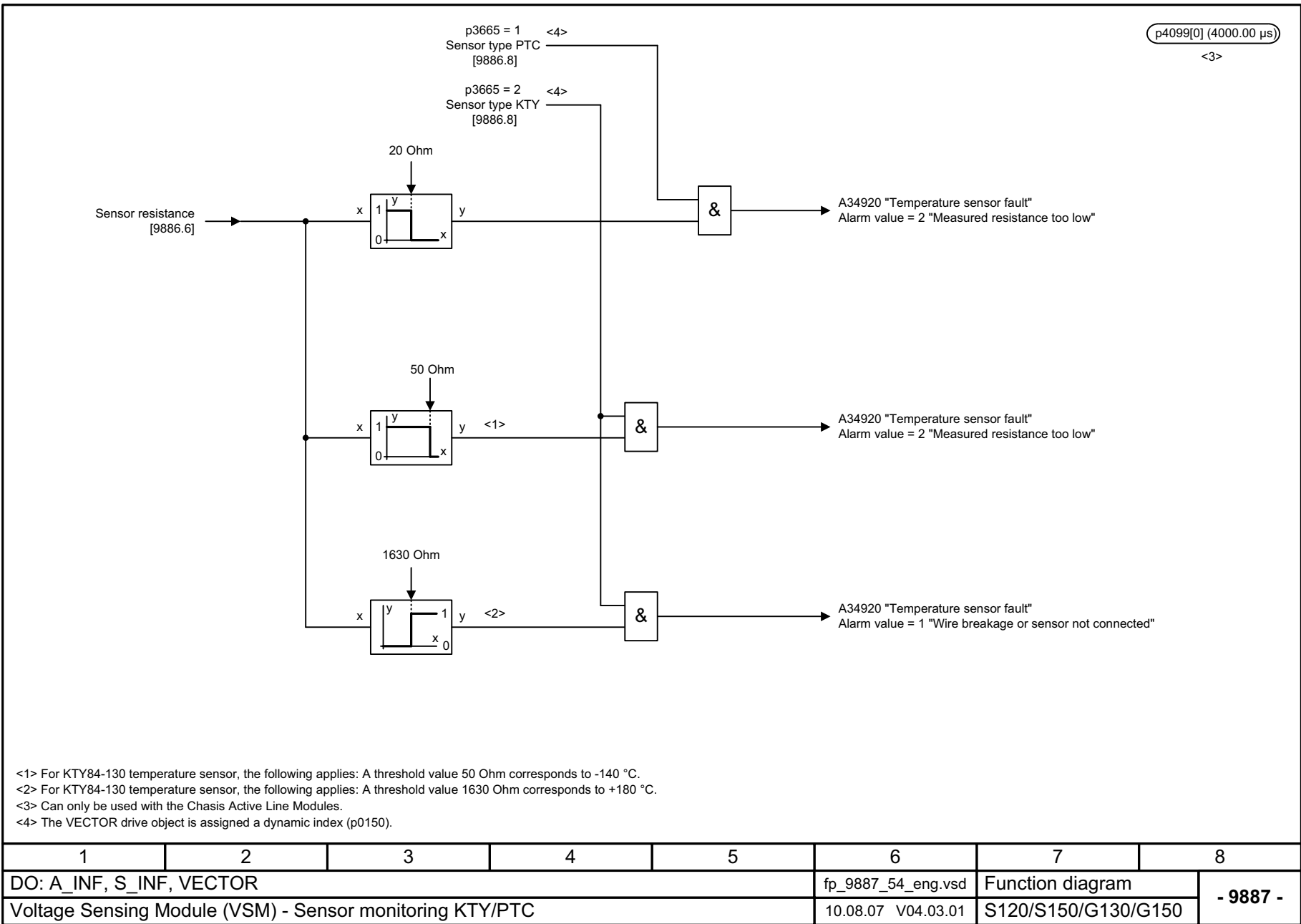
9880 – Analog inputs (AI 0 to AI 3)	2-1723
9886 – Temperature evaluation	2-1724
9887 – Sensor monitoring KTY/PTC	2-1725



1	2	3	4	5	6	7	8
DO: A_INF, S_INF, VECTOR					fp_9880_54_eng.vsd	Function diagram	
Voltage Sensing Module (VSM) - Analog inputs (AI 0 ... AI 3)					26.06.08 V04.03.01	S120/S150/G130/G150	
					- 9880 -		

Fig. 2-305 9880 – Analog inputs (AI 0 to AI 3)





2.35 Basic Operator Panel 20 (BOP20)

Function diagrams

9912 – Control word interconnection

2-1727

PROFIdrive sampling time
Refer to [1020.7]

Interconnection STW BOP (r0019)			<1>
Signal	Meaning	Interconnection parameters	
STW BOP.0	1 = On 0 = OFF (OFF1)	p0840[0] = r0019.0	
STW BOP.1	1 = No coast down 0 = Coast down (OFF2)	p0844[0] = r0019.1	
STW BOP.2	1 = No fast stop 0 = Fast stop (OFF3)	p0848[0] = r0019.2	
STW BOP.3	Reserved	-	
STW BOP.4	Reserved	-	
STW BOP.5	Reserved	-	
STW BOP.6	Reserved	-	
STW BOP.7	↗ = Acknowledge fault	p2102[0] = r0019.7	
STW BOP.8	Reserved	-	
STW BOP.9	Reserved	-	
STW BOP.10	Reserved	-	
STW BOP.11	Reserved	-	
STW BOP.12	Reserved	-	
STW BOP.13	1 = Motorized potentiometer, raise	p1035[0] = r0019.13	
STW BOP.14	1 = Motorized potentiometer, lower	p1036[0] = r0019.14	
STW BOP.15	Reserved	-	

<1> The BICO interconnection represents an example that can be changed by the user.

1	2	3	4	5	6	7	8
DO: CU_G, CU_S					fp_9912_54_eng.vsd	Function diagram	- 9912 -
Basic Operator Panel 20 (BOP20) - Control word interconnection					16.01.07 V04.03.01	S120/S150/G130/G150	

Fig. 2-308 9912 – Control word interconnection

2.36 External brake module

Function diagrams

9951 – Sequencer (r0108.26 = 1)

2-1729

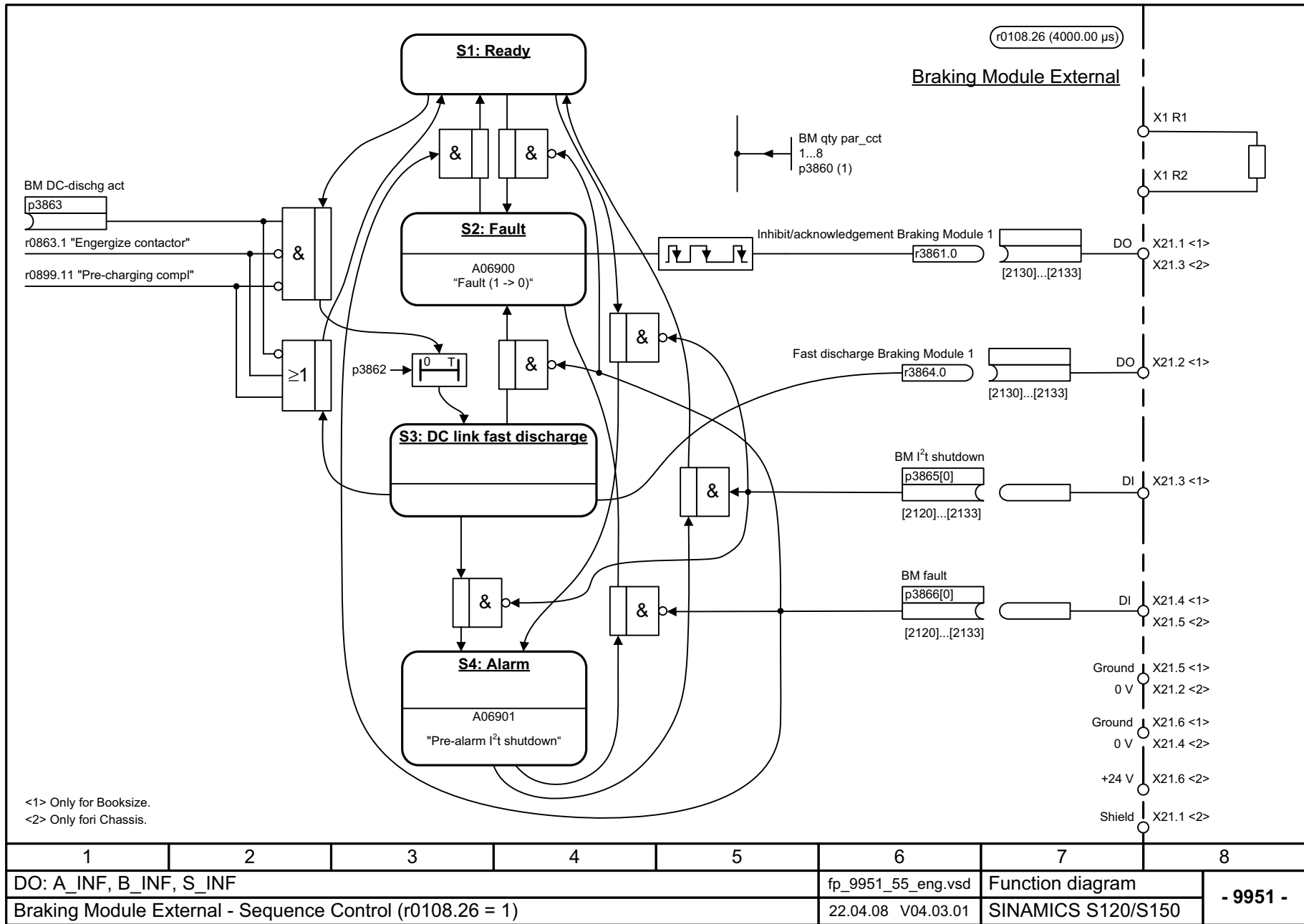


Fig. 2-309 9951 – Sequencer (r0108.26 = 1)

Faults and alarms

3

Contents

3.1	Overview of faults and alarms	3-1732
3.2	List of faults and alarms	3-1742

3.1 Overview of faults and alarms

3.1.1 General information on faults and alarms

Displaying faults and alarms

If a fault occurs, the drive indicates by issuing corresponding fault(s) and/or alarm(s).

The following methods for displaying faults and alarms are available:

- Display via the fault and alarm buffer with PROFIBUS
- Display online via the commissioning software

Differences between faults and alarms

The differences between faults and alarms are as follows:

Table 3-1 Differences between faults and alarms

Type	Description
Faults	What happens when a fault occurs? • The appropriate fault reaction is triggered. • Status signal ZSW1.3 is set. • The fault is entered in the fault buffer. How are faults eliminated? • Remove the original cause of the fault. • Acknowledge the fault.
Alarms	What happens when an alarm occurs? • Status signal ZSW1.7 is set. • The alarm is entered in the alarm buffer. How are alarms eliminated? • Alarms acknowledge themselves. If the cause of the alarm is no longer present, they automatically reset themselves.

Fault reactions

The following fault reactions are defined:

Table 3-2 Fault reactions

List	PROFId- rive	Reaction	Description
NONE	-	None	No reaction when a fault occurs. Note: When the "Basic positioner" function module is activated (r0108.4 = 1), the following applies: When a fault occurs with fault reaction "NONE", an active traversing task is interrupted and the system switches to tracking mode until the fault has been rectified and acknowledged.
OFF1	ON/ OFF	Brake in the ramp-function generator ramp-down followed by pulse disable	Closed-loop speed control (p1300 = 20, 21) • $n_{set} = 0$ is input immediately to brake the drive along the ramp-function generator ramp-down (p1121). • When zero speed is detected, the motor holding brake (if parameters have been assigned for it) is closed (p1215). The pulses are suppressed when the brake closing time (p1217) expires. Zero speed is detected if the actual speed drops below the threshold (p1226) or if the monitoring time (p1227) started when the speed setpoint $\leq$ speed threshold (p1226) has expired. Closed-loop torque control (p1300 = 23) • The following applies to closed-loop torque control mode: Reaction as for OFF2. • When the system switches to closed-loop control with p1501, the following applies: No separate braking reaction. If the actual speed value drops below the speed threshold (p1226) or the timer stage (p1227) has expired, the motor holding brake (if one is being used) is closed. The pulses are suppressed when the brake closing time (p1217) expires.
OFF2	COAST STOP	Internal/external pulse disable	Closed-loop speed and torque control • Instantaneous pulse suppression, the drive "coasts" to a standstill. • The motor holding brake (if one is being used) is closed immediately. • The closing lockout is activated.

Table 3-2 Fault reactions, continued

List	PROFId- rive	Reaction	Description
OFF3	QUICK STOP	Brake along the OFF3 ramp-down followed by pulse disable	Closed-loop speed control (p1300 = 20, 21) • $n_set = 0$ is input immediately to brake the drive along the OFF3 ramp-down (p1135). • When zero speed is detected, the motor holding brake (if parameters have been assigned for it) is closed. The pulses are suppressed when the brake application time (p1217) expires. Zero speed is detected if the actual speed drops below the threshold (p1226) or if the monitoring time (p1227) started when the speed setpoint $\leq$ speed threshold (p1226) has expired. • The closing lockout is activated. Closed-loop torque control (p1300 = 23) • Switchover to speed-controlled operation and other reactions as described for speed-controlled operation.
STOP1	-	-	Under development
STOP2	-	$n_set = 0$	• $n_set = 0$ is input immediately to brake the drive along the OFF3 ramp-down (p1135). • The drive remains in closed-loop speed control mode.
IASC/ DCBRAKE	-	-	• For synchronous motors, the following applies: If a fault occurs with this fault reaction, an internal armature short-circuit is triggered. The conditions for $p1231 = 4$ must be observed. • For induction motors, the following applies: If a fault occurs with this fault reaction, DC braking is triggered. The DC brake must have been put into operation (p1232, p1233, p1234).
ENCODER	-	Internal/external pulse disable (p0491)	The fault reaction ENCODER is applied as a function of the setting in p0491. Factory setting: $p0491 = 0 \rightarrow$ Encoder fault causes OFF2 Notice: When changing p0491, it is imperative that the information in the description of this parameter is carefully observed.

Acknowledging faults

The list of faults and alarms specifies how to acknowledge each fault after the cause has been remedied.

Table 3-3 Acknowledging faults

Acknowledg- ment	Description								
POWER ON	The fault is acknowledged by a POWER ON process (switch drive unit off and on again). Note: If this action has not eliminated the fault cause, the fault is displayed again immediately after power up.								
IMMEDIATE	Faults can be acknowledged on one drive object (Points 1 to 3) or on all drive objects (point 4) as follows: 1 Acknowledge by setting parameter: p3981 = 0 --> 1 2 Acknowledge via binector inputs: <table border="0"> <tr> <td>p2103</td> <td>BI: 1. Acknowledge faults</td> </tr> <tr> <td>p2104</td> <td>BI: 2. Acknowledge faults</td> </tr> <tr> <td>p2105</td> <td>BI: 3. Acknowledge faults</td> </tr> </table> 3 Acknowledge using PROFIBUS control signal: STW1.7 = 0 --> 1 (edge) 4 Acknowledge all faults <table border="0"> <tr> <td>p2102</td> <td>BI: Acknowledge all faults</td> </tr> </table> All of the faults on all of the drive objects of the drive system can be acknowledged using this binector input. Note: • These faults can also be acknowledged by a POWER ON operation. • If this action has not eliminated the fault cause, the fault will continue to be displayed after acknowledgment. • Safety Integrated faults The "Safe standstill" (SH) function must be deselected before these faults are acknowledged.	p2103	BI: 1. Acknowledge faults	p2104	BI: 2. Acknowledge faults	p2105	BI: 3. Acknowledge faults	p2102	BI: Acknowledge all faults
p2103	BI: 1. Acknowledge faults								
p2104	BI: 2. Acknowledge faults								
p2105	BI: 3. Acknowledge faults								
p2102	BI: Acknowledge all faults								
PULSE INHIBIT	The fault can only be acknowledged with a pulse inhibit (r0899.11 = 0). The same options are available for acknowledging as described under acknowledgment with IMMEDIATE.								

Saving the fault buffer when switching power off

The contents of the fault buffer are saved to a non-volatile storage medium when the Control Unit 320 (CU320) is switched off (i.e. the fault buffer history is still available when the unit is switched on again).

Note:

Requirements:

- Firmware version 2.2 or higher.
- Control Unit 320 (CU320) with hardware version C or higher.
The hardware version is shown on the rating plate or can be displayed online with the commissioning software (in Project Navigator under "Drive Unit" --> Configuration --> Version Overview).

If these prerequisites are not met, the contents of the fault buffer are deleted at every POWER ON operation.

The fault buffer of a drive object comprises the following parameters:

- r0945[0 to 63], r0947[0 to 63], r0948[0 to 63], r0949[0 to 63]
- r2109[0 to 63], r2130[0 to 63], r2133[0 to 63], r2136[0 to 63]

The fault buffer contents can be deleted manually as follows:

- Delete fault buffer for all drive objects:
p2147 = 1 --> p2147 = 0 is automatically set after execution.
- Delete fault buffer for a specific drive object:
p0952 = 0 --> The parameter belongs to the specified drive object.

The fault buffer contents are automatically deleted when the following occurs:

- Restore factory setting (p0009 = 30 and p0976 = 1).
- Download with modified structure (e.g. number of drive objects changed)
- Power-up after other parameter values have been loaded (e.g. p0976 = 10)
- Upgrade firmware to later version.

3.1.2 Explanation of the list of faults and alarms

The data in the following example has been chosen at random. The information listed below is the maximum amount of information that a description can contain. Some of the information is optional.

The list of faults and alarms (See Chapter 3.2) is structured as follows:

----- **Start of example** -----

Axxxxx (F, N)	Fault location (optional): Name
Message value:	Component number: %1, cause: %2
Drive object:	List of objects.
Reaction:	NONE
Acknowledgment:	NONE
Cause:	Description of possible causes Fault value (r0949, interpret format): or alarm value (r2124, interpret format): (optional) Information about fault or alarm values (optional)
Remedy:	Description of possible remedies
Reaction to F:	A_INFEED: OFF2 (OFF1, NONE) SERVO: NONE (OFF1, OFF2, OFF3) VECTOR: NONE (OFF1, OFF2, OFF3)
Acknowledgment for F:	IMMEDIATELY (POWER ON)
Reaction to N:	NONE
Acknowledgment:	

Axxxxx	Alarm xxxxx
Axxxxx (F, N)	Alarm xxxxx (message type can be changed to F or N)
Fxxxxx	Fault xxxxx
Fxxxxx (A, N)	Fault xxxxx (report type can be changed to F or N)
Nxxxxx	No message
Nxxxxx (A)	No message (message type can be changed to A)
Cxxxxx	Safety message (separate message buffer)

A message comprises a letter followed by the relevant number.

The meaning of the letters is as follows:

- A means "Alarm".
- F means "Fault".
- N means "No message" or "Internal message".
- C means "Safety message"

The optional parentheses indicate whether the type specified for this message can be changed and which message types can be adjusted via parameters (p2118, p2119).

Information on reaction and acknowledgment is specified independently for a message with an adjustable message type (e.g. reaction to F, acknowledgment for F).

Note:

You can change the default properties of a fault or alarm via parameter assignment.

References: /IH1/ SINAMICS S120 Commissioning Manual
Section "Diagnostics"

The list of faults and alarms (see Chapter 3.2) provides information in relation to the properties of a message that have been set as standard. If the properties of a specific message are changed, the corresponding information may have to be modified in this list.

Fault location (optional): Name

The fault location (optional), the name of the fault or alarm, and the message number are all used to identify the message (e.g. with the commissioning software).

Message value:

The information provided under the message value tells you about the composition of the fault/alarm value.

Example:

Message value: Component number: %1, cause: %2

This fault value or alarm value contains information about the component number and cause. The entries %1 and %2 are placeholders, which are filled appropriately in online operation with the commissioning software.

Drive object:

Each message (fault/alarm) specifies the drive object in which it can be found.

A message can belong to either one, several, or all drive objects.

Reaction: Default fault reaction (adjustable fault reaction)

Specifies the default reaction in the event of a fault.

The optional parentheses indicate whether the default fault reactions can be changed and which fault reactions can be adjusted via parameters (p2100, p2101).

Note:

See Chapter 3.1.1

Acknowledgment: Default acknowledgment (adjustable acknowledgment)

Specifies the default method of acknowledging faults after the cause has been eliminated.

The optional parentheses indicate whether the default acknowledgment can be changed and which acknowledgment can be adjusted via parameters (p2126, p2127).

Note:

See Chapter 3.1.1

Cause:

Description of the possible causes of the fault/alarm. A fault or alarm value can also be specified (optional).

Fault value (r0949, format):

The fault value is entered in the fault buffer in r0949[0 to 63] and specifies additional, more precise information about a fault.

Alarm value (r2124, format):

The alarm value specifies additional, more precise information about an alarm.

The alarm value is entered in the alarm buffer in r2124[0 to 7] and specifies additional, more precise information about an alarm.

Remedy:

Description of the methods available for eliminating the cause of the active fault or alarm



Alarm

In certain cases, servicing and maintenance personnel are responsible for choosing a suitable method to eliminate the cause of faults.

3.1.3 Number ranges of faults and alarms

Note:

The following number ranges represent an overview for all faults and alarms in SINAMICS.

The faults and alarms for the product described in this Parameter Manual are described in detail in Chapter 3.2.

Faults and alarms are organized into the following number ranges:

Table 3-4 Number ranges of faults and alarms

From	To	Range
1000	3999	Control unit
4000	4999	Reserved
5000	5999	Power unit
6000	6899	Infeed
6900	6999	Braking Module
7000	7999	Drive
8000	8999	Option Board
9000	12999	Reserved
13000	13001	Licensing
13002	19999	Reserved
20000	29999	OEM
30000	30999	DRIVE-CLiQ component power unit
31000	31999	DRIVE-CLiQ component encoder 1
32000	32999	DRIVE-CLiQ component encoder 2 Note: Faults that occur are automatically output as an alarm if the encoder is set up as a direct measuring system and does not intervene in the motor control.
33000	33999	DRIVE-CLiQ component encoder 3 Note: Faults that occur are automatically output as an alarm if the encoder is set up as a direct measuring system and does not intervene in the motor control.
34000	34999	Voltage Sensing Module (VSM)
35000	35199	Terminal Module 54F (TM54F)
35200	35999	Terminal Module 31 (TM31)
36000	36999	DRIVE-CLiQ Hub Module

Table 3-4 Number ranges of faults and alarms, continued

From	To	Range
40000	40999	Controller Extension 32 (CX32)
41000	48999	Reserved
49000	49999	SINAMICS GM/SM/GL
50000	50499	Communication Board (COMM BOARD)
50500	59999	OEM Siemens
60000	65535	SINAMICS DC MASTER (DC closed loop control)

3.2 List of faults and alarms

Product: SINAMICS S120/S150, Version: 4301400, Language: eng,
Objects: A_INF, B_INF, CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30,
TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR

F01000	Internal software error
Message value:	%1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	POWER ON
Cause:	An internal software error has occurred. Fault value (r0949, interpret hexadecimal): Only for internal Siemens troubleshooting.
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade firmware to later version. - contact the Hotline. - replace the Control Unit.
F01001	Floating point exception
Message value:	%1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	POWER ON
Cause:	A floating point exception has occurred. The error can be caused by the basic system, DCC, FBLOCKS or OA_applications. Additional debug information can be found in p9999: Index 0: Alarm number Index 1: Program counter at the time the exception occurred Index 2: Cause of floating point exception: To be treated as a hexadecimal: 0x1 - Invalid operation 0x2 - Divide by zero 0x4 - Overflow 0x8 - Underflow 0x10 - Inexact result
Remedy:	- carry out a POWER ON (power off/on) for all components. - check configuration and signals of the free function blocks. - check configuration and signals of the DCC charts. - upgrade firmware to later version. - contact the Hotline.
F01002	Internal software error
Message value:	%1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	An internal software error has occurred. Fault value (r0949, interpret hexadecimal): Only for internal Siemens troubleshooting.
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade firmware to later version. - contact the Hotline.

F01003 Acknowledgement delay when accessing the memory

Message value: %1
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A memory area was accessed that does not return a "READY".
Fault value (r0949, interpret hexadecimal):
Only for internal Siemens troubleshooting.
Remedy: - carry out a POWER ON (power off/on) for all components.
- contact the Hotline.

N01004 (F, A) Internal software error

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: An internal software error has occurred.
Fault value (r0949, hexadecimal):
Only for internal Siemens troubleshooting.
Remedy: - read out diagnostics parameter (r9999).
- contact the Hotline.
See also: r9999 (Software error internal supplementary diagnostics)
Reaction upon F: OFF2
Acknowl. upon F: POWER ON
Reaction upon A: NONE
Acknowl. upon A: NONE

F01005 Firmware download for DRIVE-CLiQ component unsuccessful

Message value: Component number: %1, fault cause: %2
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: It was not possible to download the firmware to a DRIVE-CLiQ component.
Fault value (r0949, interpret hexadecimal):
yyxxxx hex: yy = component number, xxxx = fault cause
xxxx = 000B hex = 11 dec:
DRIVE-CLiQ component has detected a checksum error.
xxxx = 000F hex = 15 dec:
The selected DRIVE-CLiQ component did not accept the contents of the firmware file.
xxxx = 0012 hex = 18 dec:
Firmware version is too old and is not accepted by the component.
xxxx = 0013 hex = 19 dec:
Firmware version is not suitable for the hardware release of the component.
xxxx = 0065 hex = 101 dec:
After several communication attempts, no response from the DRIVE-CLiQ component.
xxxx = 008B hex = 139 dec:
Initially, a new boot loader is loaded (must be repeated after POWER ON).
xxxx = 008C hex = 140 dec:
Firmware file for the DRIVE-CLiQ component not available on the memory card.
xxxx = 008D hex = 141 dec:
An inconsistent length of the firmware file was signaled. The firmware download may have been caused by a loss of connection to the firmware file. This can occur during a project download/reset in the case of a SINAMICS Integrated Control Unit, for example.
xxxx = 008F hex = 143 dec:
Component has not changed to the mode for firmware download. It was not possible to delete the existing firmware.

xxxx = 0090 hex = 144 dec:

When checking the firmware that was downloaded (checksum), the component detected a fault. It is possible that the file on the memory card is defective.

xxxx = 0091 hex = 145 dec:

Checking the loaded firmware (checksum) was not completed by the component in the appropriate time.

xxxx = 009C hex = 156 dec:

Component with the specified component number is not available (p7828).

xxxx = Additional values:

Only for internal Siemens troubleshooting.

Remedy:

- check the selected component number (p7828).
 - check the DRIVE-CLiQ connection.
 - save suitable firmware file for download in the directory "/siemens/sinamics/code/sac/".
 - use a component with a suitable hardware version
 - after POWER ON has been carried out again for the DRIVE-CLiQ component, download the firmware again.
- Depending on p7826, the firmware will be automatically downloaded.

A01006 Firmware update for DRIVE-CLiQ component required

Message value: Component number: %1

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: The firmware of a DRIVE-CLiQ component must be updated as there is no suitable firmware or firmware version in the component for operation with the Control Unit.
Alarm value (r2124, interpret decimal):
Component number of the DRIVE-CLiQ component.

Remedy:

Firmware update using the commissioning software:
The firmware version of all of the components on the "Version overview" page can be read in the Project Navigator under "Configuration" of the associated drive unit and an appropriate firmware update can be carried out.
Firmware update via parameter:
- take the component number from the alarm value and enter into p7828.
- start the firmware download with p7829 = 1.

A01007 POWER ON for DRIVE-CLiQ component required

Message value: Component number: %1

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: A DRIVE-CLiQ component must be powered up again (POWER ON) as, for example, the firmware was updated.
Alarm value (r2124, interpret decimal):
Component number of the DRIVE-CLiQ component.
Note:

For a component number = 1, a POWER ON of the Control Unit is required.

Remedy:

Switch off the power supply of the specified DRIVE-CLiQ component and switch it on again.

A01009 (N) CU: Control module overtemperature

Message value: -

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: The temperature (r0037[0]) of the control module (Control Unit) has exceeded the specified limit value.

Remedy:

- check the air intake for the Control Unit.
 - check the fan for the Control Unit (only for CU310).
- Note:
The alarm automatically disappears after the limit value has been undershot.

Reaction upon N: NONE

Acknowled. upon N: NONE

F01010 Drive type unknown

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: An unknown drive type was found.
Fault value (r0949, interpret decimal):
Drive object number (refer to p0101, p0107).
Remedy:
- Replace Power Module.
- carry out a POWER ON (power off/on) for all components.
- upgrade firmware to later version.
- contact the Hotline.

F01011 (N) Download interrupted

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: The project download was interrupted.
- the project download was prematurely ended by the user or by the commissioning software (e.g. STARTER, SCOUT).
- the communication cable was interrupted (e.g. cable breakage, cable withdrawn).
Note:
The response to an interrupted download is the state "first commissioning".
Remedy:
- check the communication cable.
- download the project again.
- boot from previously saved files (power-down/power-up or p0976).

Reaction upon N: NONE
Acknowl. upon N: NONE

F01012 (N) Project conversion error

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF2 (NONE)
Acknowledge: IMMEDIATELY
Cause: When converting the project of an older firmware version, an error occurred.
Fault value (r0949, interpret decimal):
Parameter number of the parameter causing the error.
For fault value = 600, the following applies:
The temperature evaluation is no longer assigned to the power unit but to the encoder evaluation.
Notice:
Monitoring of the motor temperature is no longer ensured.
Remedy: Check the parameter indicated in the fault value and correctly adjust it accordingly.
Re fault value = 600:
Parameter p0600 must be set to the values 1, 2 or 3 in accordance with the assignment of the internal encoder evaluation to the encoder interface.
Value 1 means: The internal encoder evaluation is assigned to the encoder interface 1 via p0187.
Value 2 means: The internal encoder evaluation is assigned to the encoder interface 2 via p0188.
Value 3 means: The internal encoder evaluation is assigned to the encoder interface 3 via p0189.
- If necessary, the internal encoder evaluation must be assigned to an encoder interface via parameters p0187, p0188 or p0189 accordingly.
- If necessary, upgrade the firmware to a later version.

Reaction upon N: NONE
Acknowl. upon N: NONE

F01015	Internal software error
Message value:	%1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	POWER ON
Cause:	An internal software error has occurred. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade firmware to later version. - contact the Hotline.
A01016 (F)	Firmware changed
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	At least one firmware file in the directory /SIEMENS/SINAMICS/ has been changed without authorization with respect to the version shipped from the factory. No changes are permitted in this directory. Alarm value (r2124, interpret decimal): 0: Checksum of one file is incorrect. 1: File missing. 2: Too many files. 3: Incorrect firmware version. 4: Incorrect checksum of the back-up file. See also: r9925 (Firmware file incorrect)
Remedy:	For the non-volatile memory for the firmware (memory card/device memory), restore the delivery condition. Note: The file involved can be read out using parameter r9925. See also: r9926 (Firmware check status)
Reaction upon F:	OFF2
Acknowl. upon F:	POWER ON
A01017	Component lists changed
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	On the memory card, one file in the directory /SIEMENS/SINAMICS/DATA or /ADDON/SINAMICS/DATA has been illegally changed with respect to that supplied from the factory. No changes are permitted in this directory. Alarm value (r2124, interpret decimal): The problem is indicated in the first digit of the alarm value: 1. File does not exist. 2. Firmware version of the file does not match the software version. 3. The file checksum is incorrect. The second digit of the alarm value indicates in which directory the file is located: 0: Directory /SIEMENS/SINAMICS/DATA/ 1. Directory /ADDON/SINAMICS/DATA/ The third digit of the alarm value indicates the file: 0: File MOTARM.ACX 1: File MOTSRM.ACX 2: File MOTSLM.ACX 3: File ENCDATA.ACX 4: File FILTDATA.ACX 5: File BRKDATA.ACX 6: File DAT_BEAR.ACX 7: File CFG_BEAR.ACX 8: File ENC_GEAR.ACX
Remedy:	For the memory card file involved, restore the status originally supplied from the factory.

F01023	Software timeout (internal)
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	An internal software timeout has occurred. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade firmware to later version. - contact the Hotline.

F01030	Sign-of-life failure for master control
Message value:	-
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: OFF3 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF2, STOP1, STOP2) Vector: OFF3 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF2, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	For active PC master control, no sign-of-life was received within the monitoring time. The master control was returned to the active BICO interconnection.
Remedy:	Set the monitoring time higher at the PC or, if required, completely disable the monitoring function. For the commissioning software, the monitoring time is set as follows: <Drive> -> Commissioning -> Control panel -> Button "Fetch master control" -> A window is displayed to set the monitoring time in milliseconds. Notice: The monitoring time should be set as short as possible. A long monitoring time means a late response when the communication fails!

F01031	Sign-of-life failure for OFF in REMOTE
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: OFF3 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF2, STOP1, STOP2) Vector: OFF3 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF2, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	With the "OFF in REMOTE" mode active, no sign-of-life was received within 3 seconds.
Remedy:	- Check the data cable connection at the serial interface for the Control Unit (CU) and operator panel. - Check the data cable between the Control Unit and operator panel.

F01033	Units changeover: Reference parameter value invalid
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	When changing over the units to the referred representation type, it is not permissible for any of the required reference parameters to be equal to 0.0 Fault value (r0949, parameter): Reference parameter whose value is 0.0. See also: p0349 (System of units, motor equivalent circuit diagram data), p0505 (Selecting the system of units), p0595 (Selecting technological units)
Remedy:	Set the value of the reference parameter to a number different than 0.0. See also: p0304, p0305, p0310, p0596, p2000, p2001, p2002, p2003, r2004

F01034	Units changeover: Calculation parameter values after reference value change unsuccessful
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The change of a reference parameter meant that for an involved parameter the selected value was not able to be recalculated in the per unit representation. The change was rejected and the original parameter value restored. Fault value (r0949, parameter): Parameter whose value was not able to be recalculated. See also: p0304, p0305, p0310, p0596, p2000, p2001, p2002, p2003, r2004
Remedy:	Select the value of the reference parameter such that the parameter involved can be calculated in the per unit representation. See also: p0304, p0305, p0310, p0596, p2000, p2001, p2002, p2003, r2004
A01035 (F)	ACX: Boot from the back-up parameter back-up files
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	When the Control Unit is booted, no complete data set was found from the parameter back-up files. The last time that the parameterization was saved, it was not completely carried out. Instead, a back-up data set or a back-up parameter back-up file is downloaded. Alarm value (r2124, interpret hexadecimal): Only for internal Siemens troubleshooting.
Remedy:	If you have saved the project using the commissioning software, carry out a new download for your project. Save using the function "Copy RAM to ROM" or with p0977 = 1 so that all of the parameter files are again completely written to the non-volatile memory.
Reaction upon F:	Infeed: NONE (OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY
F01036 (A)	ACX: Parameter back-up file missing
Message value:	%1
Drive object:	All objects
Reaction:	Infeed: NONE (OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	When downloading the device parameterization, a parameter back-up file associated with a drive object cannot be found. Neither a PSxxxxyy.ACX, a PSxxxxyy.NEW nor a PSxxxxyy.BAK parameter back-up file exists in the non-volatile memory for this drive object. Fault value (r0949, interpret hexadecimal): Byte 1: yyy in the file name PSxxxxyy.ACX yyy = 000 --> consistency back-up file yyy = 001 ... 062 --> drive object number yyy = 099 --> PROFIBUS parameter back-up file Byte 2, 3, 4: Only for internal Siemens troubleshooting.
Remedy:	If you have saved the project data using the commissioning software, carry out a new download for your project. Save using the function "Copy RAM to ROM" or with p0977 = 1 so that all of the parameter files are again completely written to the non-volatile memory. If you have not saved the project data, then first commissioning of the system has to be carried out again.
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F01037 (A)	ACX: Re-naming the parameter back-up file unsuccessful
Message value:	%1
Drive object:	All objects
Reaction:	Infeed: NONE (OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	Re-naming after saving a parameter back-up file in the non-volatile memory was unsuccessful. One of the parameter back-up files to be re-named had the "read only" attribute. The parameter back-up files are saved in the directory \USER\SINAMICS\DATA. It is possible that the non-volatile memory is defective. Fault value (r0949, interpret hexadecimal): Byte 1: yyy in the file names PSxxxxyy.* or Cxxxxyy.* or CCxxxxyy.* yyy = 000 --> consistency back-up file yyy = 099 --> PROFIBUS parameter back-up file PSxxx099.* Byte 2: xxx in the file name PSxxxxyy.* xxx = 000 --> data save started with p0977 = 1 xxx = 010 --> data save started with p0977 = 10 xxx = 011 --> data save started with p0977 = 11 xxx = 012 --> data save started with p0977 = 12 Byte 4, 3: Only for internal Siemens troubleshooting.
Remedy:	- check whether one of the files to be overwritten has the attribute "read only" and change this file attribute to "writable". Check all of the files (PSxxxxyy.*, CCxxxxyy.*, Cxxxxyy.*) that belong to drive yyy designated in the fault value. - replace the memory card or Control Unit.
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F01038 (A)	ACX: Loading the parameter back-up file unsuccessful
Message value:	%1
Drive object:	All objects
Reaction:	Infeed: NONE (OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	An error has occurred when downloading PSxxxxyy.ACX or PTxxxxyy.ACX files from the non-volatile memory. Fault value (r0949, interpret hexadecimal): Byte 1: yyy in the file name PSxxxxyy.ACX yyy = 000 --> consistency back-up file yyy = 001 ... 062 --> drive object number yyy = 099 --> PROFIBUS parameter back-up file Byte 4, 3, 2: Only for internal Siemens troubleshooting.
Remedy:	- If you have saved the project data using the commissioning software, carry out a new download for your project. Save using the function "Copy RAM to ROM" or with p0977 = 1 so that all of the parameter files are again completely written to the non-volatile memory. - replace the memory card or Control Unit.
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F01039 (A)	ACX: Writing to the parameter back-up file was unsuccessful
Message value:	%1
Drive object:	All objects
Reaction:	Infeed: NONE (OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	Writing to at least one parameter back-up file PSxxxxxyy.*** in the non-volatile memory was unsuccessful. - In the directory /USER/SINAMICS/DATA/ at least one parameter back-up file PSxxxxxyy.*** has the "read only" file attribute and cannot be overwritten. - There is not sufficient free memory space available. - The non-volatile memory is defective and cannot be written to. Fault value (r0949, interpret hexadecimal): dcba hex a = yyy in the file names PSxxxxxyy.*** a = 000 --> consistency back-up file a = 001 ... 062 --> drive object number a = 070 --> FEPROM.BIN a = 080 --> DEL4BOOT.TXT a = 099 --> PROFIBUS parameter back-up file b = xxx in the file names PSxxxxxyy.*** b = 000 --> data save started with p0977 = 1 b = 010 --> data save started with p0977 = 10 b = 011 --> data save started with p0977 = 11 b = 012 --> data save started with p0977 = 12 d, c: Only for internal Siemens troubleshooting.
Remedy:	- check the file attribute of the files (PSxxxxxyy.***, CAxxxxxyy.***, CCxxxxxyy.***) and, if required, change from "read only" to "writeable". - check the free memory space in the non-volatile memory. Approx. 80 kbyte of free memory space is required for every drive object in the system. - replace the memory card or Control Unit.
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F01040	Save parameter settings and carry out a POWER ON
Message value:	-
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	POWER ON
Cause:	A parameter was changed in the drive system which means that it is necessary to save the parameters and re-boot (e.g. p0110).
Remedy:	- save the parameters (p0971/p0977). - carry out a POWER ON (power off/on) for all components.
F01041	Parameter save necessary
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	Defective or missing files were detected on the memory card when booting. Fault value (r0949, interpret decimal): 1: Source file cannot be opened. 2: Source file cannot be read. 3: Target directory cannot be set up. 4: Target file cannot be set up/opened. 5: Target file cannot be written to. Additional values: Only for internal Siemens troubleshooting.

Remedy:

- save the parameters.
- download the project again to the drive unit.
- update the firmware
- if required, replace the Control Unit and/or memory card card.

F01042 Parameter error during project download

Message value: Parameter: %1, Index: %2, fault cause: %3

Drive object: All objects

Reaction: Infeed: OFF2 (NONE, OFF1)
Servo: OFF2 (NONE, OFF1, OFF3)
Vector: OFF2 (NONE, OFF1, OFF3)

Acknowledge: IMMEDIATELY

Cause: An error was detected when downloading a project using the commissioning software (e.g. incorrect parameter value).
For the specified parameter, it was detected that dynamic limits were exceeded that may possibly depend on other parameters.
Fault value (r0949, interpret hexadecimal):
ccbbaaaa hex
aaaa = Parameter
bb = Index
cc = fault cause
0: Parameter number illegal.
1: Parameter value cannot be changed.
2: Lower or upper value limit exceeded.
3: Sub-index incorrect.
4: No array, no sub-index.
5: Data type incorrect.
6: Setting not permitted (only resetting).
7: Descriptive element cannot be changed.
9: Descriptive data not available.
11: No master control.
15: No text array available.
17: Task cannot be executed due to operating state.
20: Illegal value.
21: Response too long.
22: Parameter address illegal.
23: Format illegal.
24: Number of values not consistent.
25: Drive object does not exist.
101: Presently deactivated.
104: Illegal value.
107: Write access not permitted when controller enabled.
108: Unit unknown.
109: Write access only in the commissioning state, encoder (p0010 = 4).
110: Write access only in the commissioning state, motor (p0010 = 3).
111: Write access only in the commissioning state, power unit (p0010 = 2).
112: Write access only in the quick commissioning mode (p0010 = 1).
113: Write access only in the ready mode (p0010 = 0).
114: Write access only in the commissioning state, parameter reset (p0010 = 30).
115: Write access only in the Safety Integrated commissioning state (p0010 = 95).
116: Write access only in the commissioning state, technological application/units (p0010 = 5).
117: Write access only in the commissioning state (p0010 not equal to 0).
118: Write access only in the commissioning state, download (p0010 = 29).
119: Parameter may not be written in download.
120: Write access only in the commissioning state, drive basis configuration (device: p0009 = 3).
121: Write access only in the commissioning state, define drive type (device: p0009 = 2).
122: Write access only in the commissioning state, data set basis configuration (device: p0009 = 4).
123: Write access only in the commissioning state, device configuration (device: p0009 = 1).
124: Write access only in the commissioning state, device download (device: p0009 = 29).
125: Write access only in the commissioning state, device parameter reset (device: p0009 = 30).
126: Write access only in the commissioning state, device ready (device: p0009 = 0).
127: Write access only in the commissioning state, device (device: p0009 not equal to 0).

- 129: Parameter may not be written in download.
- 130: Transfer of the master control is inhibited via BI: p0806.
- 131: Required BICO interconnection not possible because BICO output does not supply floating value
- 132: Free BICO interconnection inhibited via p0922.
- 133: Access method not defined.
- 200: Below the valid values.
- 201: Above the valid values.
- 202: Cannot be accessed from the Basic Operator Panel (BOP).
- 203: Cannot be read from the Basic Operator Panel (BOP).
- 204: Write access not permitted.

Remedy:

- enter the correct value in the specified parameter.
- identify the parameter that restricts the limits of the specified parameter.

F01043 Fatal error at project download

Message value: Fault cause: %1

Drive object: All objects

Reaction: Infeed: OFF2 (OFF1)
Servo: OFF2 (OFF1, OFF3)
Vector: OFF2 (OFF1, OFF3)

Acknowledge: IMMEDIATELY

Cause: A fatal error was detected when downloading a project using the commissioning software.
Fault value (r0949, interpret decimal):

- 1: Device status cannot be changed to Device Download (drive object ON?).
- 2: Incorrect drive object number.
- 3: A drive object that has already been deleted is deleted again.
- 4: Deleting of a drive object that has already been registered for generation.
- 5: Deleting a drive object that does not exist.
- 6: Generating an undeleted drive object that already existed.
- 7: Regenerating a drive object already registered for generation.
- 8: Maximum number of drive objects that can be generated exceeded.
- 9: Error while generating a device drive object.
- 10: Error while generating target topology parameters (p9902 and p9903).
- 11: Error while generating a drive object (global component).
- 12: Error while generating a drive object (drive component).
- 13: Unknown drive object type.
- 14: Drive status cannot be changed to "ready for operation" (p0947 and p0949).
- 15: Drive status cannot be changed to drive download.
- 16: Device status cannot be changed to "ready for operation".
- 17: It is not possible to download the topology. The component wiring should be checked, taking into account the various messages/signals.
- 18: A new download is only possible if the factory settings are restored for the drive unit.
- 19: The slot for the option module has been configured several times (e.g. CAN and COMM BOARD)
- 20: The configuration is inconsistent (e.g. CAN for Control Unit, however no CAN configured for drive objects A_INF, SERVO or VECTOR).

Remedy:

- use the current version of the commissioning software.
- modify the offline project and carry out a new download (e.g. compare the number of drive objects, motor, encoder, power unit in the offline project and at the drive).
- change the drive state (is a drive rotating or is there a message/signal?).
- carefully note any other messages/signals and remove their cause.

F01044 CU: Descriptive data error

Message value: %1

Drive object: All objects

Reaction: OFF2

Acknowledge: POWER ON

Cause: An error was detected when loading the descriptive data saved in the non-volatile memory.

Remedy: Replace the memory card or Control Unit.

A01045	CU: Configuring data invalid
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	An error was detected when evaluating the parameter files PSxxxxxyy.ACX, PTxxxxyy.ACX, CAxxxxyy.ACX, or CCxxxxyy.ACX saved in the non-volatile memory. Alarm value (r2124, interpret hexadecimal): Only for internal Siemens troubleshooting.
Remedy:	Restore the factory setting using (p0976 = 1) and re-load the project to the drive unit. Operation without any restrictions is then possible. After downloading the project, save the parameters in STARTER using "Copy RAM to ROM" or with p0977 = 1. This overwrites the incorrect parameter files in the non-volatile memory.

A01049	CU: It is not possible to write to file
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	It is not possible to write into a write-protected file (PSxxxxxx.acx). The write request was interrupted. Alarm value (r2124, interpret decimal): Drive object number.
Remedy:	Check whether the "write protected" attribute has been set for the files in the non-volatile memory under .../USER/SINAMICS/DATA/... When required, remove write protection and save again (e.g. set p0977 to 1).

F01050	Memory card and device incompatible
Message value:	-
Drive object:	All objects
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (NONE, OFF1, OFF3) Vector: OFF2 (NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The memory card and the device type do not match (e.g. a memory card for SINAMICS S is inserted in SINAMICS G).
Remedy:	- insert the matching memory card. - use the matching Control Unit or power unit.

F01054	CU: System limit exceeded
Message value:	%1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	At least one system overload has been identified. Fault value (r0949, interpret decimal): 1: Computing time load too high (r9976[1]). 5: Peak load too high (r9976[5]). See also: r9976 (System utilization)
Remedy:	Re fault value = 1, 5: - check the sampling times and adjust if necessary (p0115). - deactivate the function module. - deactivate the drive object. - remove the drive object from the target topology.

A01064 (F)	CU: Internal error (CRC)
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	CRC error in the Control Unit program memory
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade firmware to later version. - contact the Hotline.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Servo: NONE (OFF1, OFF2, OFF3, STOP2) Vector: NONE (OFF1, OFF2, OFF3, STOP2)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
A01065	Drive: Fault on non-active encoder
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	One or several non-active encoders indicate an error.
Remedy:	Remove the error for the non-active encoder.
F01068	CU: Data memory, memory overflow
Message value:	%1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A data memory area has been overloaded. Fault value (r0949, interpret binary): Bit 0 = 1: High-speed data memory 1 overloaded Bit 1 = 1: High-speed data memory 2 overloaded Bit 2 = 1: High-speed data memory 3 overloaded Bit 3 = 1: High-speed data memory 4 overloaded
Remedy:	- deactivate the function module. - deactivate the drive object. - remove the drive object from the target topology.
A01099	Tolerance window of time synchronization exited
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The time master exited the selected tolerance window for time synchronization. See also: p3109 (RTC real time synchronization, tolerance window)
Remedy:	Select the re-synchronization interval so that the synchronization deviation between the time master and drive system lies within the tolerance window. See also: r3108 (RTC last synchronization deviation)
A01100	CU: Memory card withdrawn
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The memory card (non-volatile memory) was withdrawn during operation. Notice: It is not permissible for the memory card to be withdrawn or inserted under voltage.

Remedy:

- power down the drive system.
- re-insert the memory card that was withdrawn - this card must match the drive system.
- power up the drive system again.

F01105 (A) CU: Insufficient memory

Message value: %1
Drive object: All objects
Reaction: OFF1
Acknowledge: POWER ON
Cause: Too many functions have been configured on this Control Unit (e.g. too many drives, function modules, data sets, OA applications, blocks, etc).
Fault value (r0949, interpret decimal):
Only for internal Siemens troubleshooting.
Remedy:

- change the configuration on this Control Unit (e.g. fewer drives, function modules, data sets, OA applications, blocks, etc).
- use an additional Control Unit.

Reaction upon A: NONE
Acknowl. upon A: NONE

F01107 CU: Data save in the non-volatile memory unsuccessful

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A data save in the non-volatile memory was not able to be successfully carried out.

- non-volatile memory is defective.
- insufficient space in the non-volatile memory.

Fault value (r0949, interpret decimal):
Only for internal Siemens troubleshooting.
Remedy:

- try to save again.
- replace the memory card or Control Unit.

F01110 CU: More than one SINAMICS G on one Control Unit

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: More than one SINAMICS G type power unit is being operated from the Control Unit.
Fault value (r0949, interpret decimal):
Number of the second drive with a SINAMICS G type power unit.
Remedy: Only one SINAMICS G drive type is permitted.

F01111 CU: Mixed operation of drive units illegal

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: Illegal operation of various drive units on one Control Unit:

- SINAMICS S together with SINAMICS G
- SINAMICS S together with SINAMICS S Value or Combi

Fault value (r0949, interpret decimal):
Number of the first drive object with a different power unit type.
Remedy: Only power units of one particular drive type may be operated with one Control Unit.

F01112 CU: Power unit not permissible

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: The connected power unit cannot be used together with this Control Unit.
 Fault value (r0949, interpret decimal):
 1: Power unit is not supported (e.g. PM240).
 2: DC/AC power unit connected to CU310 not permissible.
Remedy: Replace the power unit that is not permissible by a component that is permissible.

F01120 (A) Terminal initialization has failed

Message value: %1
Drive object: All objects
Reaction: OFF1 (OFF2)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: An internal software error occurred while the terminal functions were being initialized.
 Fault value (r0949, interpret hexadecimal):
 Only for internal Siemens troubleshooting.
Remedy:
 - carry out a POWER ON (power off/on) for all components.
 - upgrade firmware to later version.
 - contact the Hotline.
 - replace the Control Unit.
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F01122 (A) Frequency at the measuring probe input too high

Message value: %1
Drive object: All objects
Reaction: OFF1 (OFF2)
Acknowledge: IMMEDIATELY
Cause: The frequency of the pulses at the measuring probe input is too high.
 Fault value (r0949, interpret decimal):
 1: DI/DO 9 (X122.8)
 2: DI/DO 10 (X122.10)
 4: DI/DO 11 (X122.11)
 8: DI/DO 13 (X132.8)
 16: DI/DO 14 (X132.10)
 32: DI/DO 15 (X132.11)
 64: DI/DO 8 (X122.7)
 128: DI/DO 12 (X132.7)
Remedy: Reduce the frequency of the pulses at the measuring probe input.
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F01150 CU: Number of instances of a drive object type exceeded

Message value: Drive object type: %1, number permitted: %2, current number: %3
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: The maximum permissible number of instances of a drive object type was exceeded.
 Fault value (r0949, interpret hexadecimal):
 Byte 1: Drive object type (p0107).
 Byte 2: Max. permissible number of instances for this drive object type.
 Byte 3: Actual number of instances for this drive object type.

Remedy:

- power down the unit.
- suitably restrict the number of instances of a drive object type by reducing the number of inserted components.
- re-commission the unit.

F01151 CU: Number of drive objects of a category exceeded

Message value: Drive object category: %1, number permitted: %2, current number: %3

Drive object: All objects

Reaction: NONE

Acknowledge: IMMEDIATELY

Cause: The maximum permissible number of drive objects of a category was exceeded.
Fault value (r0949, interpret hexadecimal):
Byte 1: Drive object category.
Byte 2: Max. permissible number for this drive object category.
Byte 3: Actual number for this drive object category.

Remedy:

- power down the unit.
- suitably restrict the number of drive objects of the specified category by reducing the number of inserted components.
- re-commission the unit.

F01200 CU: Time slice management internal software error

Message value: %1

Drive object: All objects

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: A time slice management error has occurred.
It is possible that the sampling times have been inadmissibly set.
Fault value (r0949, interpret hexadecimal):
998: Too many time slices occupied by OA (e.g. DCC)
999: Too many time slices occupied by the basic system
Too many different sampling times may have been set.
Further values for internal Siemens troubleshooting.

Remedy:

- check the sampling time setting (p0112, p0115, p4099).
- contact the Hotline.

F01205 CU: Time slice overflow

Message value: %1

Drive object: All objects

Reaction: OFF2

Acknowledge: POWER ON

Cause: Insufficient processing time is available for the existing topology.
Fault value (r0949, interpret hexadecimal):
Only for internal Siemens troubleshooting.

Remedy:

- reduce the number of drives.
- increase the sampling times.

F01221 CU: Bas clk cyc too low

Message value: %1

Drive object: All objects

Reaction: NONE

Acknowledge: IMMEDIATELY

Cause: The closed-loop control / monitoring cannot maintain the envisaged clock cycle.
The runtime of the closed-loop control/monitoring is too long for the particular clock cycle or the computing time remaining in the system is not sufficient for the closed-loop control/monitoring.
Fault value (r0949, interpret hexadecimal):
Only for internal Siemens troubleshooting.

Remedy: Increase the basic clock cycle of DRIVE-CLiQ communication.
See also: p0112 (Sampling times presetting p0115)

A01223	CU: Sampling time inconsistent
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	When changing a sampling time (p0115[0], p0799 or p4099), inconsistency between the clock cycles has been identified. Alarm value (r2124, interpret decimal): 1: Value, low minimum value. 2: Value, high maximum value. 3: Value not a multiple of 1.25 µs. 4: Value does not match clock-cycle synchronous PROFIBUS operation. 5: Value not a multiple of 125 µs. 6: Value not a multiple of 250 µs. 7: Value not a multiple of 375 µs. 8: Value not a multiple of 400 µs. 10: Special restriction of the drive object violated. 20: On a SERVO with a sampling time of 62.5 µs, more than two drive objects or one drive object of a type other than SERVO have been detected on the same DRIVE-CLiQ line (a maximum of two SERVO type drive objects are permitted). 21: Value can be a multiple of the current controller sampling time of a servo or vector drive in the system (e.g. for TB30, the values of all of the indices should be taken into account). 30: Value less than 31.25 µs. 31: Value less than 62.5 µs. 32: Value less than 125 µs. 40: Nodes have been identified on the DRIVE-CLiQ line whose highest common denominator of the sampling times is less than 125 µs. Further, none of the nodes has a sampling time of less than 125 µs. 41: A chassis unit was identified on the DRIVE-CLiQ line as a node. Further, the highest common denominator of the sampling times of all of the nodes connected to the line is less than 250 µs. 42: An Active Line Module was identified on the DRIVE-CLiQ line as a node. Further, the highest common denominator of the sampling times of all of the nodes connected to the line is less than 125 µs. 43: A Voltage Sensing Module (VSM) was identified on the DRIVE-CLiQ line as a node. Further, the highest common denominator of the sampling times of all of the nodes connected to the line is not equal to the current controller sampling time of the drive object of the VSM. 44: The highest common denominator of the sampling times of all of the components connected to the DRIVE-CLiQ line is not the same for all components of this drive object (e.g. there are components on different DRIVE-CLiQ lines on which different highest common denominators are generated). 52: Nodes have been identified on the DRIVE-CLiQ line whose highest common denominator of the sampling times is less than 31.25 µs. 54: Nodes have been identified on the DRIVE-CLiQ line whose highest common denominator of the sampling times is less than 62.5 µs. 56: Nodes have been identified on the DRIVE-CLiQ line whose highest common denominator of the sampling times is less than 125 µs. 58: Nodes have been identified on the DRIVE-CLiQ line whose highest common denominator of the sampling times is less than 250 µs. 99: Inconsistency of cross drive objects detected. 116: Recommended clock cycle in r0116[0...1]. General note: The topology rules should be noted when connecting up DRIVE-CLiQ (refer to the appropriate product documentation). The parameters of the sampling times can also be changed with automatic calculations. Example for highest common denominator: 125 µs, 125 µs, 62.5 µs --> 62.5 µs
Remedy:	- check the DRIVE-CLiQ cables. - set a valid sampling time. See also: p0115, p0799, p4099

A01224	CU: Pulse frequency inconsistent
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	When changing the minimum pulse frequency (p0113) inconsistency between the pulse frequencies was identified. Alarm value (r2124, interpret decimal): 1: Value, low minimum value. 2: Value, high maximum value. 3: Resulting sampling time is not a multiple of 1.25 µs. 4: Value does not match clock-cycle synchronous PROFIBUS operation. 10: Special restriction of the drive object violated. 99: Inconsistency of cross drive objects detected. 116: Recommended clock cycle in r0116[0...1].
Remedy:	Set a valid pulse frequency. See also: p0113 (Minimum pulse frequency, selection)

F01250	CU: CU-EEPROM incorrect read-only data
Message value:	%1
Drive object:	All objects
Reaction:	NONE (OFF2)
Acknowledge:	POWER ON
Cause:	Error when reading the read-only data of the EEPROM in the Control Unit. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	- carry out a POWER ON. - replace the Control Unit.

A01251	CU: CU-EEPROM incorrect read-write data
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	Error when reading the read-write data of the EEPROM in the Control Unit. Alarm value (r2124, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	For alarm value r2124 < 256, the following applies: - carry out a POWER ON. - replace the Control Unit. For alarm value r2124 >= 256, the following applies: - for the drive object with this alarm, clear the fault memory (p0952 = 0). - as an alternative, clear the fault memory of all drive objects (p2147 = 1). - replace the Control Unit.

F01255	CU: Option Board EEPROM read-only data error
Message value:	%1
Drive object:	All objects
Reaction:	NONE (OFF2)
Acknowledge:	POWER ON
Cause:	Error when reading the read-only data of the EEPROM in the Option Board. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	- carry out a POWER ON. - replace the Control Unit.

A01256	CU: Option Board EEPROM read-write data error
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	Error when reading the read-write data of the EEPROM in the Option Board. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	- carry out a POWER ON. - replace the Control Unit.
F01303	DRIVE-CLiQ component does not support the required function
Message value:	%1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A function requested by the Control Unit is not supported by a DRIVE-CLiQ component. Fault value (r0949, interpret decimal): 1: The component does not support the deactivation. 101: The Motor Module does not support an internal armature short-circuit. 102: The Motor Module does not support the deactivation. 201: The Sensor Module does not support actual value inversion (p0410.0 = 1) when using a Hall sensor (p0404.6 = 1) for the commutation. 202: The Sensor Module does not support parking/unparking. 203: The Sensor Module does not support the deactivation. 204: The firmware of this Terminal Module 15 (TM15) does not support the application TM15DI/DO. 205: The Sensor Module does not support the selected temperature evaluation (r0458). 206: The firmware of this Terminal Modules TM41/TM31/TM15 refers to an old firmware version. It is urgently necessary to upgrade the firmware to ensure disturbance-free operation. 207: The power unit with this hardware version does not support operation with device supply voltages of less than 380 V. 208: The Sensor Module does not support deselection of commutation with zero mark (via p0430.23). 211: The Sensor Module does not support single-track encoders (r0459.10)
Remedy:	Upgrade the firmware of the DRIVE-CLiQ component involved. Re fault value = 205: Check parameter p0600 and p0601 and if required, adapt interpretation. Re fault value = 207: Replace the power unit or if required set the device supply voltage higher (p0210). Re fault value = 208: Check parameter p0430.23 and reset if necessary.
A01304 (F)	Firmware version of DRIVE-CLiQ component is not up-to-date
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The non-volatile memory has a more recent firmware version than the one in the connected DRIVE-CLiQ component. Alarm value (r2124, interpret decimal): Component number of the DRIVE-CLiQ component involved.
Remedy:	Update the firmware (p7828, p7829 and commissioning software).
Reaction upon F:	NONE
Acknowled. upon F:	IMMEDIATELY

F01305	Topology: Component number missing
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The component number from the topology was not parameterized (p0121 (for power unit, refer to p0107), p0131 (for servo/vector drives, refer to p0107), p0141, p0151, p0161). Fault value (r0949, interpret decimal): The fault value includes the particular data set number. Note: The fault also occurs if speed encoders have been configured (p0187 to p0189) but no component numbers exist for them. In this case, the fault value includes the drive data set number plus 100 * encoder number (e.g. 3xx, if a component number was not entered in p0141 for the third encoder (p0189)). See also: p0121, p0131, p0141, p0142, p0151, p0161, p0186, p0187, p0188, p0189
Remedy:	Enter the missing component number or remove the component and restart commissioning. See also: p0121, p0131, p0141, p0142, p0151, p0161, p0186, p0187, p0188, p0189

A01306	Firmware of the DRIVE-CLiQ component being updated
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	Firmware update is active for at least one DRIVE-CLiQ component. Alarm value (r2124, interpret decimal): Component number of the DRIVE-CLiQ component.
Remedy:	None necessary. This alarm automatically disappears after the firmware has been updated.

A01314	Topology: Component must not be present
Message value:	Component number: %1, Component class: %2, Connection number: %3
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	For a component, "deactivate and not present" is set but this component is still in the topology. Alarm value (r2124, interpret hexadecimal): Byte 1: Component number Byte 2: Component class of the component Byte 3: Connection number Note: Component class and connection number are described in F01375.
Remedy:	- remove the corresponding component. - change the setting "deactivate and not present". Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison). See also: p0105, p0125, p0145, p0155

A01315	Drive object not ready for operation
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	For the active drive object involved, at least one activated component is missing. Note: All other active and operational drive objects can be in the "RUN" state.

Remedy: The alarm automatically disappears again with the following actions:

- deactivate the drive object involved (p0105 = 0).
- deactivate the components involved (p0125 = 0, p0145 = 0, p0155 = 0, p0165 = 0).
- re-insert the components involved.

See also: p0105, p0125, p0145, p0155

A01316 Drive object inactive and again ready for operation

Message value: -

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: If, when inserting a component of the target topology, an inactive, non-operational drive object becomes operational again. The associated parameter of the component is, in this case, set to "activate" (p0125, p0145, p0155, p0165).
Note:
 This is the only message that is displayed for a deactivated drive object.

Remedy: The alarm automatically disappears again with the following actions:

- activate the drive object involved (p0105 = 1).
- again withdraw the components involved.

See also: p0105 (Activate/deactivate drive object)

A01317 (N) Deactivated component again present

Message value: -

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: If a component of the target topology for an active drive object is inserted and the associated parameter of the component is set to "deactivate" (p0125, p0145, p0155, p0165).
Note:
 This is the only message that is displayed for a deactivated component.

Remedy: The alarm automatically disappears again with the following actions:

- activate the components involved (p0125 = 1, p0145 = 1, p0155 = 1, p0165 = 1).
- again withdraw the components involved.

See also: p0125 (Activate/deactivate power unit components), p0145, p0155

Reaction upon N: NONE

Acknowl. upon N: NONE

A01318 BICO: Deactivated interconnections present

Message value: %1

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: This alarm is used in the following cases:

- If an inactive/non-operational drive object is active again/ready for operation
- If there are items in the list of BI/CI parameters (r9498[0...29], r9499[0...29])
- If the BICO interconnections saved in the list of BI/CI parameters (r9498[0...29], r9499[0...29]) have actually been changed

Remedy: Reset alarm:

- Set p9496 to 1 or 2

or

- deactivate the drive object again.

A01319	Inserted component not initialized
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Initialization is required for at least one inserted component. This is only possible if the pulses are inhibited for all the drive objects.
Remedy:	Activate pulse inhibit for all drive objects.

A01320	Topology: Drive object number does not exist in configuration
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	A drive object number is missing in p0978 Alarm value (r2124, interpret decimal): Index of p0101 under which the missing drive object number can be determined.
Remedy:	Set p0009 to 1 and change p0978: Rules: - p0978 must include all of the drive object numbers (p0101). - it is not permissible for a drive object number to be repeated. - by entering a 0, the drive objects with PZD are separated from those without PZD. - only 2 partial lists are permitted. After the second 0, all values must be 0. - dummy drive object numbers (255) are only permitted in the first partial list.

A01321	Topology: Drive object number does not exist in configuration
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	p0978 contains a drive object number that does not exist. Alarm value (r2124, interpret decimal): Index of p0978 under which the drive object number can be determined.
Remedy:	Set p0009 to 1 and change p0978: Rules: - p0978 must include all of the drive object numbers (p0101). - it is not permissible for a drive object number to be repeated. - by entering a 0, the drive objects with PZD are separated from those without PZD. - only 2 partial lists are permitted. After the second 0, all values must be 0. - dummy drive object numbers (255) are only permitted in the first partial list.

A01322	Topology: Drive object number present twice in configuration
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	A drive object number is present more than once in p0978. Alarm value (r2124, interpret decimal): Index of p0978 under which the involved drive object number is located.
Remedy:	Set p0009 to 1 and change p0978: Rules: - p0978 must include all of the drive object numbers (p0101). - it is not permissible for a drive object number to be repeated. - by entering a 0, the drive objects with PZD are separated from those without PZD. - only 2 partial lists are permitted. After the second 0, all values must be 0. - dummy drive object numbers (255) are only permitted in the first partial list.

A01323	Topology: More than two partial lists created
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	Partial lists are available more than twice in p0978. After the second 0, all must be 0. Alarm value (r2124, interpret decimal): Index of p0978 under which the illegal value is located.
Remedy:	Set p0009 to 1 and change p0978: Rules: - p0978 must include all of the drive object numbers (p0101). - it is not permissible for a drive object number to be repeated. - by entering a 0, the drive objects with PZD are separated from those without PZD. - only 2 partial lists are permitted. After the second 0, all values must be 0. - dummy drive object numbers (255) are only permitted in the first partial list.
A01324	Topology: Dummy drive object number incorrectly created
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	In p0978, dummy drive object numbers (255) are only permitted in the first partial list. Alarm value (r2124, interpret decimal): Index of p0978 under which the illegal value is located.
Remedy:	Set p0009 to 1 and change p0978: Rules: - p0978 must include all of the drive object numbers (p0101). - it is not permissible for a drive object number to be repeated. - by entering a 0, the drive objects with PZD are separated from those without PZD. - only 2 partial lists are permitted. After the second 0, all values must be 0. - dummy drive object numbers (255) are only permitted in the first partial list.
F01325	Topology: Component Number not present in target topology
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The component configured in a parameter (e.g. p0121, p0131, etc.) is not present in the target topology. Alarm value (r2124, interpret decimal): Configured Component Number that is not present in target topology.
Remedy:	Establish topology and DO configuration consistency.
A01330	Topology: Quick commissioning not possible
Message value:	Fault cause: %1, supplementary information: %2, preliminary component number: %3
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	Unable to carry out a quick commissioning. The existing actual topology does not fulfill the requirements. Alarm value (r2124, interpret hexadecimal): ccccbbaa hex: cccc = preliminary component number, bb = supplementary information, aa = fault cause aa = 01 hex = 1 dec: On one component illegal connections were detected. - bb = 01 hex = 1 dec: For a Motor Module, more than one motor with DRIVE-CLiQ was detected. - bb = 02 hex = 2 dec: For a motor with DRIVE-CLiQ, the DRIVE-CLiQ cable is not connected to a Motor Module.

aa = 02 hex = 2 dec:

The topology contains too many components of a particular type.

- bb = 01 hex = 1 dec: There is more than one master Control Unit.
- bb = 02 hex = 2 dec: There is more than 1 infeed (8 for a parallel circuit configuration).
- bb = 03 hex = 3 dec: There are more than 10 Motor Modules (8 for a parallel circuit configuration).
- bb = 04 hex = 4 dec: There are more than 9 encoders.
- bb = 05 hex = 5 dec: There are more than 8 Terminal Modules.
- bb = 07 hex = 7 dec: Unknown component type
- bb = 08 hex = 8 dec: There are more than 6 drive slaves.
- bb = 09 hex = 9 dec: Connection of a drive slave not permitted.
- bb = 0a hex = 10 dec: There is no drive master.
- bb = 0b hex = 11 dec: There is more than one motor with DRIVE-CLiQ for a parallel circuit.
- cccc: Not used.

aa = 03 hex = 3 dec:

More than 16 components are connected at a DRIVE-CLiQ socket of the Control Unit.

- bb = 0, 1, 2, 3 means e.g. detected at the DRIVE-CLiQ socket X100, X101, X102, X103.
- cccc: Not used.

aa = 04 hex = 4 dec:

The number of components connected one after the other is greater than 125.

- bb: Not used.
- cccc = preliminary component number of the first component and component that resulted in the fault.

aa = 05 hex = 5 dec:

The component is not permissible for SERVO.

- bb = 01 hex = 1 dec: SINAMICS G available.
- bb = 02 hex = 2 dec: Chassis available.
- cccc = preliminary component number of the first component and component that resulted in the fault.

aa = 06 hex = 6 dec:

On one component illegal EEPROM data was detected. These must be corrected before the system continues to boot.

- bb = 01 hex = 1 dec: The Order No. [MLFB] of the power unit that was replaced includes a space retainer. The space retainer (*) must be replaced by a correct character.
- cccc = preliminary component number of the component with illegal EEPROM data.

aa = 07 hex = 7 dec:

The actual topology contains an illegal combination of components.

- bb = 01 hex = 1 dec: Active Line Module (ALM) and Basic Line Module (BLM).
- bb = 02 hex = 2 dec: Active Line Module (ALM) and Smart Line Module (SLM).
- bb = 03 hex = 3 dec: SIMOTION control (e.g. SIMOTION D445) and SINUMERIK component (e.g. NX15).
- bb = 04 hex = 4 dec: SINUMERIK control (e.g. SINUMERIK 730.net) and SIMOTION component (e.g. CX32).
- cccc: Not used.

Note:

Connection type and connection number are described in F01375.

See also: p0097 (Select drive object type), r0098 (Actual device topology), p0099 (Device target topology)

Remedy:

- adapt the output topology to the permissible requirements.
- carry out commissioning using the commissioning software.
- for motors with DRIVE-CLiQ, connect the power and DRIVE-CLiQ cable to the same Motor Module (Single Motor Module: DRIVE-CLiQ at X202, Double Motor Module: DRIVE-CLiQ from motor 1 (X1) to X202, from motor 2 (X2) to X203).

Re aa = 06 hex = 6 dec and bb = 01 hex = 1 dec:

Correct the order number when commissioning using the commissioning software.

See also: p0097 (Select drive object type), r0098 (Actual device topology), p0099 (Device target topology)

A01331

Topology: At least one component not assigned to a drive object

Message value:

Component number: %1

Drive object:

All objects

Reaction:

NONE

Acknowledge:

NONE

Cause:

At least one component is not assigned to a drive object.

- when commissioning, a component was not able to be automatically assigned to a drive object.
- the parameters for the data sets are not correctly set.

Alarm value (r2124, interpret decimal):

Component number of the unassigned component.

Remedy: This component is assigned to a drive object.
Check the parameters for the data sets.
Examples:
- power unit (p0121).
- motor (p0131, p0186).
- encoder interface (p0140, p0141, p0187 ... p0189).
- encoder (p0140, p0142, p0187 ... p0189).
- Terminal Module (p0151).
- option board (p0161).

F01340 Topology: Too many components on one line

Message value: Component number or connection number: %1, fault cause: %2
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: For the selected communications clock cycle, too many DRIVE-CLiQ components are connected to one line of the Control Unit.
Fault value (r0949, interpret hexadecimal):
xyy hex: x = fault cause, yy = component number or connection number.
1yy:
The communications clock cycle of the DRIVE-CLiQ connection on the CU is not sufficient for all read transfers.
2yy:
The communications clock cycle of the DRIVE-CLiQ connection on the CU is not sufficient for all write transfers.
3yy:
Cyclic communication is fully utilized.
4yy:
The DRIVE-CLiQ cycle starts before the earliest end of the application. An additional dead time must be added to the control. Sign-of-life errors can be expected.
5yy:
Internal buffer overflow for net data of a DRIVE-CLiQ connection.
6yy:
Internal buffer overflow for receive data of a DRIVE-CLiQ connection.
7yy:
Internal buffer overflow for send data of a DRIVE-CLiQ connection.
Remedy: Check the DRIVE-CLiQ connection:
Reduce the number of components on the DRIVE-CLiQ line involved and distribute these to other DRIVE-CLiQ connections of the Control Unit. This means that communication is uniformly distributed over several communication lines.
Re fault value = 1yy - 4yy in addition:
- increase the sampling times (p0112, p0115).

F01354 Topology: Actual topology indicates an illegal component

Message value: Fault cause: %1, component number: %2
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The actual topology indicates at least one illegal component.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = cause.
xx = 1: Component at this Control Unit not permissible.
xx = 2: Component in combination with another component not permissible.
Note:
Pulse enable is prevented.
Remedy: Remove the illegal components and restart the system.

F01355	Topology: Actual topology changed
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The device target topology (p0099) does not correspond to the device actual topology (r0098). The fault only occurs if the topology was commissioned using the automatic internal device mechanism and not using the commissioning software. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting. See also: r0098 (Actual device topology), p0099 (Device target topology)
Remedy:	One of the following counter-measures can be selected if no faults have occurred in the topology detection itself: If commissioning was still not completed: - carry out a self-commissioning routine (starting from p0009 = 1). General: Set p0099 to r0098, set p0009 to 0; for existing Motor Modules, this results in servo drives being automatically generated (p0107). Generating servo drives: Set p0097 to 1, set p0009 to 0. Generating vector drives: Set p0097 to 2, set p0009 to 0. Generating vector drives with parallel circuit: Set p0097 to 12, set p0009 to 0. In order to set configurations in p0108, before setting p0009 to 0, it is possible to first set p0009 to 2 and modify p0108. The index corresponds to the drive object (p0107). If commissioning was already completed: - re-establish the original connections and re-connect power to the Control Unit. - restore the factory setting for the complete equipment (all of the drives) and allow automatic self-commissioning again. - change the device parameterization to match the connections (this is only possible using the commissioning software). Notice: Topology changes that result in this fault being generated cannot be accepted by the automatic function in the device, but must be transferred using the commissioning software and parameter download. The automatic function in the device only allows constant topology to be used. Otherwise, when the topology is changed, all of the previous parameter settings are lost and replaced by the factory setting. See also: r0098 (Actual device topology)
F01356	Topology: Defective components in actual topology
Message value:	Fault cause: %1, Component number: %2, Connection number: %3
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The actual topology indicates at least one defective DRIVE-CLiQ component. Fault value (r0949, interpret hexadecimal): zzyyxx hex: zz = connection number of component before the defective component yy = component number of component before the defective component xx = fault cause xx = 1: Component at this Control Unit not permissible. Note: Pulse enable is withdrawn and prevented.
Remedy:	Remove the defective components and restart the system.

F01360	Topology: Actual topology not permissible
Message value:	Fault cause: %1, preliminary component number: %2
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The detected actual topology is not permissible. Fault value (r0949, interpret hexadecimal): ccccbbaa hex: cccc = preliminary component number, aa = fault cause aa = 01 hex = 1 dec: Too many components were detected at the Control Unit. The maximum permissible number of components is 199. aa = 02 hex = 2 dec: The component type of a component is not known. aa = 03 hex = 3 dec: It is illegal to combine ALM and BLM. aa = 04 hex = 4 dec: It is illegal to combine ALM and SLM. aa = 05 hex = 5 dec: It is illegal to combine BLM and SLM. aa = 06 hex = 6 dec: A CX32 was not directly connected to a permitted Control Unit. aa = 07 hex = 7 dec: An NX10 or NX15 was not directly connected to a permitted Control Unit. aa = 08 hex = 8 dec: A component was connected to a Control Unit that is not permitted for this purpose. aa = 0A hex = 10 dec: Too many components of a particular type detected. aa = 0B hex = 11 dec: Too many components of a particular type detected on a single line. Note: The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Re fault cause = 1: Change the configuration. Connect less than 199 components to the Control Unit. Re fault cause = 2: Remove the component with unknown component type. Re fault cause = 3, 4, 5: Establish a valid combination. Re fault cause = 6, 7: Connect the expansion module directly to a permitted Control Unit. Re fault cause = 8: Remove component or use a permissible component. Re fault cause = 10, 11: Reduce the number of components.
A01361	Topology: Actual topology contains SINUMERIK and SIMOTION components
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The detected actual topology contains SINUMERIK and SIMOTION components. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled. Alarm value (r2124, interpret hexadecimal): ddccbbbaa hex: cc = fault cause, bb = component class of the actual topology, aa = component number of the component cc = 01 hex = 1 dec: An NX10 or NX15 was connected to a SIMOTION control. cc = 02 hex = 2 dec: A CX32 was connected to a SINUMERIK control.

Remedy: Re alarm value = 1:
Replace all NX10 or NX15 by a CX32.
Re alarm value = 2:
Replace all CX32 by an NX10 or NX15.

A01362 Topology: Topology rule(s) broken

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: At least one topology rule for the SINAMICS S120 Combi has been broken.
In the event of a fault, the ramping up of the drive system is aborted and closed-loop drive control is not enabled.
Fault value (r0949, interpret decimal):
The fault value indicates which rule has been broken.
1: The S120 Combi may only be wired via DRIVE-CLiQ socket X200 to X1 on the NCU.
2: Only one Terminal Module 54F (TM54F) may be connected to DRIVE-CLiQ socket X2 on the NCU. If a TM54 is being used, it may only be wired via X500 to X2 on the NCU.
3: Only one DRIVE-CLiQ Hub Module (hub) may be connected to DRIVE-CLiQ socket X3 on the NCU. If a hub is being used, it may only be wired via X500 to X3 on the NCU.
4: Only Sensor Modules may be connected to DRIVE-CLiQ sockets X201 through X203 (3-axis) or X204 (4-axis) on the S120 Combi.
5: Only one Single Motor Module or one Double Motor Module may be connected to DRIVE-CLiQ socket X205 (X204 is not available for 3-axis). The module must be wired to the S120 Combi via X200.
6: Only certain Motor Modules may be used for expansion axes.
7: If a Single Motor Module is being used as the first expansion axis, only one more Single Motor Module may be connected (via X200 to X201 on the first Single Motor Module).
8: Only Sensor Modules may be connected to the corresponding DRIVE-CLiQ socket X202 on any Single Motor Modules which may be present.
9: On the second Single Motor Module there must be nothing connected to X201.
10: If a Double Motor Module is used as an expansion axis, only Sensor Modules may be connected to X202 and X203.
11: On the Double Motor Module there must be nothing connected to X201.
12: On the Terminal Module 54F (TM54F) there must be nothing connected to X501.
13: On the DRIVE-CLiQ Hub Module, only Sensor Modules Cabinet (SMC) and Sensor Modules External (SME) may be connected to X501 through X505.
Remedy: Evaluate the fault value and ensure compliance with the corresponding topology rule(s).

F01375 Topology: Actual topology, duplicate connection between two components

Message value: Preliminary component number: %1, component class: %2, connection number: %3
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: When detecting the actual topology, a ring-type connection was detected.
Fault value (r0949, interpret hexadecimal):
ccbbaaaa hex:
cc = connection number
bb = component class
aaaa = preliminary component number of a component included in the ring
Component class:
1: Control Unit.
2: Motor Module.
3: Line Module.
4: Sensor Module (SM).
5: Voltage Sensing Module (VSM).
6: Terminal Module (TM).
7: DRIVE-CLiQ Hub Module.
8: Controller Extension 32 (CX32, NX10, NX15).
49: DRIVE-CLiQ components (non-listed components).
50: Option Slot (e.g. Terminal Board 30).
60: Encoder (e.g. EnDat).
70: Motor with DRIVE-CLiQ.

Remedy:	Component type: Precise designation within a component class (e.g. "SMC20"). Connection number: Consecutive numbers, starting from zero, of the appropriate connection or slot (e.g. DRIVE-CLiQ connection X100 on the Control Unit has the connection number 0). Output the fault value and remove the specified connection. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
F01380	Topology: Actual topology, defective EEPROM
Message value:	Preliminary component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	POWER ON
Cause:	When detecting the actual topology, a component with a defective EEPROM was detected. Fault value (r0949, interpret hexadecimal): bbbbaaaa hex: aaaa = preliminary component number of the defective components
Remedy:	Output the fault value and remove the defected component.
A01381	Topology: Comparison power unit shifted
Message value:	Component number: %1, Component class: %2, Component (target): %3, Connection number: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a power unit in the actual topology that has been shifted with respect to the target topology. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number cc = component number bb = component class aa = component number of the component shifted in the target topology Note: The connection in the actual topology where the shifted component was detected is described in dd, cc and bb. Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - undo the change to the actual topology by changing over the DRIVE-CLiQ cables. - commissioning software: Go online, upload the drive unit, adapt the topology offline and download the modified project. - automatically remove the topology error (p9904). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01382	Topology: Comparison Sensor Module shifted
Message value:	Component number: %1, Component class: %2, Component (target): %3, Connection number: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a Sensor Module in the actual topology that has been shifted with respect to the target topology. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number cc = component number bb = component class aa = component number of the component shifted in the target topology Note: The connection in the actual topology where the shifted component was detected is described in dd, cc and bb. Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - undo the change to the actual topology by changing over the DRIVE-CLiQ cables. - commissioning software: Go online, upload the drive unit, adapt the topology offline and download the modified project. - automatically remove the topology error (p9904). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
A01383	Topology: Comparison Terminal Module shifted
Message value:	Component number: %1, Component class: %2, Component (target): %3, Connection number: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a Terminal Module in the actual topology that has been shifted with respect to the target topology. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number cc = component number bb = component class aa = component number of the component shifted in the target topology Note: The connection in the actual topology where the shifted component was detected is described in dd, cc and bb. Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - undo the change to the actual topology by changing over the DRIVE-CLiQ cables. - commissioning software: Go online, upload the drive unit, adapt the topology offline and download the modified project. - automatically remove the topology error (p9904). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01384	Topology: Comparison DRIVE-CLiQ Hub Module shifted
Message value:	Component number: %1, Component class: %2, Component (target): %3, Connection number: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a DRIVE-CLiQ Hub Module in the actual topology that has been shifted with respect to the target topology. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number cc = component number bb = component class aa = component number of the component shifted in the target topology Note: The connection in the actual topology where the shifted component was detected is described in dd, cc and bb. Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - undo the change to the actual topology by changing over the DRIVE-CLiQ cables. - commissioning software: Go online, upload the drive unit, adapt the topology offline and download the modified project. - automatically remove the topology error (p9904). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
A01385	Topology: Comparison CX32 shifted
Message value:	Component number: %1, Component class: %2, Component (target): %3, Connection number: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a controller extension 32 (CX32) in the actual topology that has been shifted with respect to the target topology. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number cc = component number bb = component class aa = component number of the component shifted in the target topology Note: The connection in the actual topology where the shifted component was detected is described in dd, cc and bb. Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - undo the change to the actual topology by changing over the DRIVE-CLiQ cables. - commissioning software: Go online, upload the drive unit, adapt the topology offline and download the modified project. - automatically remove the topology error (p9904). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01386	Topology: Comparison DRIVE-CLiQ component shifted
Message value:	Component number: %1, Component class: %2, Component (target): %3, Connection number: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a DRIVE-CLiQ component in the actual topology that has been shifted with respect to the target topology. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number cc = component number bb = component class aa = component number of the component shifted in the target topology Note: The connection in the actual topology where the shifted component was detected is described in dd, cc and bb. Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - undo the change to the actual topology by changing over the DRIVE-CLiQ cables. - commissioning software: Go online, upload the drive unit, adapt the topology offline and download the modified project. - automatically remove the topology error (p9904). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
A01387	Topology: Comparison option slot component shifted
Message value:	Component number: %1, Component class: %2, Component (target): %3, Connection number: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a option slot component in the actual topology that has been shifted with respect to the target topology. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number cc = component number bb = component class aa = component number of the component shifted in the target topology Note: The connection in the actual topology where the shifted component was detected is described in dd, cc and bb. Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - undo the change to the actual topology by changing over the DRIVE-CLiQ cables. - commissioning software: Go online, upload the drive unit, adapt the topology offline and download the modified project. - automatically remove the topology error (p9904). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01388	Topology: Comparison EnDat encoder shifted
Message value:	Component number: %1, Component class: %2, Component (target): %3, Connection number: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected an EnDat encoder in the actual topology that has been shifted with respect to the target topology. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number cc = component number bb = component class aa = component number of the component shifted in the target topology Note: The connection in the actual topology where the shifted component was detected is described in dd, cc and bb. Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - undo the change to the actual topology by changing over the DRIVE-CLiQ cables. - commissioning software: Go online, upload the drive unit, adapt the topology offline and download the modified project. - automatically remove the topology error (p9904). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
A01389	Topology: Comparison motor with DRIVE-CLiQ shifted
Message value:	Component number: %1, Component class: %2, Component (target): %3, Connection number: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a motor with DRIVE-CLiQ in the actual topology that has been shifted with respect to the target topology. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number cc = component number bb = component class aa = component number of the component shifted in the target topology Note: The connection in the actual topology where the shifted component was detected is described in dd, cc and bb. Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - undo the change to the actual topology by changing over the DRIVE-CLiQ cables. - commissioning software: Go online, upload the drive unit, adapt the topology offline and download the modified project. - automatically remove the topology error (p9904). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01416	Topology: Comparison additional component in actual topology
Message value:	Component number: %1, Component class: %2, Connection number: %3
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has found a component in the actual topology which is not specified in the target topology. The alarm value includes the component number and connection number of the component with which the additional component is connected. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: cc = connection number bb = component class of the additional component aa = component number Note: - component class and connection number are described in F01375. - components that are connected to this additional component are not operational.
Remedy:	Adapting the topologies: - remove the additional component in the actual topology. - download the target topology that matches the actual topology (commissioning software). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
A01420	Topology: Comparison a component is different
Message value:	Component number: %1, component class target: %2, component class actual: %3, fault cause: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected differences in the actual and target topologies in relation to one component. There are differences in the electronic rating plate. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: aa = component number of the component, bb = component class of the target topology, cc = component class of the actual topology, dd = fault cause dd = 01 hex = 1 dec: Different component type. dd = 02 hex = 2 dec: Different Order No. dd = 03 hex = 3 dec: Different manufacturer. dd = 04 hex = 4 dec: Connection changed over for a multi-component slave (e.g. Double Motor Module), defective EEPROM data in the electronic rating plate, or only part of a multi-component slave set to "deactivate and not present". dd = 05 hex = 5 dec: A CX32 was replaced by an NX10 or NX15. dd = 06 hex = 6 dec: An NX10 or NX15 was replaced by a CX32. Note: Component class and component type are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - check the component soft-wired connections against the hardware configuration of the drive unit in the commissioning software and correct differences. - parameterize the topology comparison of all components (p9906). - parameterize the topology comparison of one components (p9907, p9908). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01421	Topology: Comparison different components
Message value:	Component number: %1, component class target: %2, component class actual: %3, fault cause: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected differences in the actual and target topologies in relation to one component. The component class, the component type or the number of connections differ. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: aa = component number of the component, bb = component class of the target topology, cc = component class of the actual topology, dd = fault cause dd = 01 hex = 1 dec: Different component class. dd = 02 hex = 2 dec: Different component type. dd = 03 hex = 3 dec: Different Order No. dd = 04 hex = 4 dec: Different number of connections. Note: Component class, component type and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Check the component soft-wired connections against the hardware configuration of the drive unit in the commissioning software and correct differences. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
A01425	Topology: Comparison serial number of a component is different
Message value:	Component number: %1, Component class: %2, Differences: %3
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected differences in the actual and target topologies in relation to one component. The serial number is different. Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: cc = number of differences bb = component class aa = component number of the component Note: The component class is described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - change over the actual topology to match the target topology. - download the target topology that matches the actual topology (commissioning software). Re byte cc: cc = 1 --> can be acknowledged using p9904 or p9905. cc > 1 --> can be acknowledged using p9905 and can be deactivated using p9906 or p9907/p9908. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison). See also: p9904 (Topology comparison, acknowledge differences), p9905 (Device specialization), p9906 (Topology comparison, comparison stage of all components), p9907 (Topology comparison, comparison stage of the component number), p9908 (Topology comparison, comparison stage of a component)

A01428	Topo: Comparison connection of a component is different
Message value:	Component number: %1, Component class: %2, Connection number1: %3, Connection number2: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected differences in the actual and target topologies in relation to one component. A component was connected to another connection. The different connections of a component are described in the alarm value: Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number of the target topology cc = connection number of the actual topology bb = component class aa = component number Note: Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - change over the actual topology to match the target topology. - download the target topology that matches the actual topology (commissioning software). - automatically remove the topology error (p9904). Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison). See also: p9904 (Topology comparison, acknowledge differences)
A01429	Topology: Comparison connection is different for more than one component
Message value:	Component number: %1, Component class: %2, Connection number1: %3, Connection number2: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	A topology comparison has found differences between the actual and target topology for several components. A component was connected to another connection. The different connections of a component are described in the alarm value: Alarm value (r2124, interpret hexadecimal): ddccbbaa hex: dd = connection number of the target topology cc = connection number of the actual topology bb = component class aa = component number Note: Component class and connection number are described in F01375. The drive system is no longer booted. In this state, the drive control (closed-loop) cannot be enabled.
Remedy:	Adapting the topologies: - change over the actual topology to match the target topology. - download the target topology that matches the actual topology (commissioning software). Note: In the software, a Double Motor Module behaves just like two separate DRIVE-CLiQ nodes. If a Double Motor Module is re-inserted, this can result in several differences in the actual topology. Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

F01451	Topology: Target topology is invalid
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	An error was detected in the target topology. The target topology is invalid. Fault value (r0949, interpret hexadecimal): ccccbbaa hex: cccc = index error, bb = component number, aa = fault cause aa = 1B hex = 27 dec: Error not specified. aa = 1C hex = 28 dec: Value illegal. aa = 1D hex = 29 dec: Incorrect ID. aa = 1E hex = 30 dec: Incorrect ID length. aa = 1F hex = 31 dec: Too few indices left. aa = 20 hex = 32 dec: component not connected to Control Unit.
Remedy:	Reload the target topology using the commissioning software.
F01470	Topology: Target topology ring-type connection detected
Message value:	Component number: %1, Component class: %2, Connection number: %3
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	A ring-type connection was detected when writing to the target topology. Fault value (r0949, interpret hexadecimal): ddccbbaa hex: cc = connection number bb = component class aa = component number of a component included in the ring Note: Component class and connection number are described in F01375.
Remedy:	Read out the fault value and remove one of the specified connections. Then download the target topology again using the commissioning software. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
F01475	Topology: Target topology duplicate connection between two components
Message value:	Component number: %1, Component class: %2, Connection number1: %3, Connection number2: %4
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	When writing the target topology, a duplicate connection between two components was detected. Fault value (r0949, interpret hexadecimal): ddccbbaa hex: dd = connection number 2 of the duplicate connection cc = connection number 1 of the duplicate connection bb = component class aa = component number of one of the components connected twice Note: Component class and connection number are described in F01375.
Remedy:	Read out the fault value and remove one of the two specified connections. Then download the target topology again using the commissioning software. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01481	Topology: Comparison power unit missing in the actual topology
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a power unit in the target topology that is not available in the actual topology. Alarm value (r2124, interpret decimal): Component number of the additional target components.
Remedy:	- delete the drive belonging to the power unit in the commissioning software project and download the new configuration to the drive unit. - check that the actual topology matches the target topology and if required, change over. - check DRIVE-CLiQ cables for interruption and contact problems. - check the 24 V supply voltage. - check that the power unit is working properly. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01482	Topology: Comparison Sensor Module missing in the actual topology
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a Sensor Module in the target topology that is not available in the actual topology. Alarm value (r2124, interpret decimal): Component number of the additional target components.
Remedy:	- re-configure the drive belonging to the Sensor Module in the commissioning software project (encoder configuration) and download the new configuration to the drive unit. - delete the drive belonging to the Sensor Module in the commissioning software project and download the new configuration to the drive unit. - check that the actual topology matches the target topology and if required, change over. - check DRIVE-CLiQ cables for interruption and contact problems. - check the 24 V supply voltage. - check that the Sensor Module is working properly. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01483	Topology: Comparison Terminal Module missing in the actual topology
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a Terminal Module in the target topology that is not available in the actual topology. Alarm value (r2124, interpret decimal): Component number of the additional target components.
Remedy:	- delete the Terminal Module in the commissioning software project and download the new configuration to the drive unit. - check that the actual topology matches the target topology and if required, change over. - check DRIVE-CLiQ cables for interruption and contact problems. - check the 24 V supply voltage. - check that the Terminal Module is working properly. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01484	Topology: Comparison DRIVE-CLiQ Hub Module missing in the actual topology
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a DRIVE-CLiQ Hub Module in the target topology that does not exist in the actual topology. Alarm value (r2124, interpret decimal): Component number of the additional target components.
Remedy:	- delete the DRIVE-CLiQ Hub Module in the commissioning software project and download the new configuration to the drive unit. - check that the actual topology matches the target topology and if required, change over. - check DRIVE-CLiQ cables for interruption and contact problems. - check the 24 V supply voltage. - test the DRIVE-CLiQ Hub Module to ensure that it functions properly. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
A01485	Topology: Comparison CX32 missing in the actual topology
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a controller extension 32 (CX32) in the target topology that is not available in the actual topology. Alarm value (r2124, interpret decimal): Component number of the additional target components.
Remedy:	- delete the CX32 / NX in the commissioning software project and download the new configuration to the drive unit. - check that the actual topology matches the target topology and if required, change over. - check DRIVE-CLiQ cables for interruption and contact problems. - check the 24 V supply voltage. - check that CX32/NX functions correctly. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
A01486	Topology: Comparison DRIVE-CLiQ components missing in the actual topology
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a DRIVE-CLiQ component in the target topology that is not available in the actual topology. Alarm value (r2124, interpret decimal): Component number of the additional target components.
Remedy:	- delete the drive belonging to this component in the commissioning software project and download the new configuration to the drive unit. - re-configure the drive belonging to this component in the commissioning software project and download the new configuration to the drive unit. - check that the actual topology matches the target topology and if required, change over. - check DRIVE-CLiQ cables for interruption and contact problems. - check the 24 V supply voltage. - check that the component is working properly. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

A01487	Topology: Comparison option slot components missing in the actual topology
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected an option slot module in the target topology that is not available in the actual topology. Alarm value (r2124, interpret decimal): Component number of the additional target components.
Remedy:	- delete the option board in the commissioning software project and download the new configuration to the drive unit. - re-configure the drive unit in the commissioning software project and download the new configuration to the drive unit. - check that the actual topology matches the target topology and if required, change over. - check that the option board is functioning correctly Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
A01488	Topology: Comparison EnDat encoder missing in the actual topology
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected an EnDat encoder in the target topology that is not available in the actual topology. Alarm value (r2124, interpret decimal): Component number of the additional target components.
Remedy:	- re-configure the drive belonging to the encoder in the commissioning software project (encoder configuration) and download the new configuration to the drive unit. - delete the drive belonging to the encoder in the commissioning software project and download the new configuration to the drive unit. - check that the actual topology matches the target topology and if required, change over. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).
A01489	Topology: Comparison motor with DRIVE-CLiQ missing in the actual topology
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The topology comparison has detected a motor with DRIVE-CLiQ in the target topology that is not available in the actual topology. Alarm value (r2124, interpret decimal): Component number of the additional target components.
Remedy:	- re-configure the drive belonging to this motor in the commissioning software project and download the new configuration to the drive unit. - re-configure the drive belonging to this motor in the commissioning software project and download the new configuration to the drive unit. - check that the actual topology matches the target topology and if required, change over. - check DRIVE-CLiQ cables for interruption and contact problems. - check that the motor is working properly. Note: Under "Topology --> Topology view" the commissioning software offers improved diagnostics capability (e.g. set-point/actual value comparison).

F01505 (A)	BICO: Interconnection cannot be established
Message value:	Parameter: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	A PROFIdrive telegram has been set (p0922). An interconnection contained in the telegram was not able to be established. Fault value (r0949, interpret decimal): Parameter receiver that should be changed.
Remedy:	Establish another interconnection.
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F01506 (A)	BICO: No standard telegram
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM31, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The standard telegram in p0922 is not maintained and therefore p0922 is set to 999. Fault value (r0949, interpret decimal): BICO parameter for which the write attempt was unsuccessful.
Remedy:	Again set the required standard telegram (p0922).
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A01507 (F, N)	BICO: Interconnections to inactive objects present
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	There are BICO interconnections as signal sink from a drive object that is either inactive/not operational. The BI/CI parameters involved are listed in r9498. The associated BO/CO parameters are listed in r9499. The list of the BICO interconnections to other drive objects is displayed in r9491 and r9492 of the deactivated drive object. Note: r9498 and r9499 are only written to, if p9495 is not set to 0. Alarm value (r2124, interpret decimal): Number of BICO interconnections found to inactive drive objects.
Remedy:	- set all open BICO interconnections centrally to the factory setting with p9495 = 2. - make the non-operational drive object active/operational again (re-insert or activate components).
Reaction upon F:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2) Vector: OFF2 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A01508	BICO: Interconnections to inactive objects exceeded
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The maximum number of BICO interconnections (signal sinks) when deactivating a drive object was exceeded. When deactivating a drive object, all BICO interconnections (signal sinks) are listed in the following parameters: - r9498[0...29]: List of the BI/CI parameters involved. - r9499[0...29]: List of the associated BO/CO parameters.
Remedy:	The alarm automatically disappears as soon as no BICO interconnection (value = 0) is entered in r9498[29] and r9499[29]. Notice: When reactivating the drive object, all BICO interconnections should be checked and if required, re-established.

F01510	BICO: Signal source is not float type
Message value:	Parameter: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The requested connector output does not have the correct data type. This interconnection is not established. Fault value (r0949, interpret decimal): Parameter number to which an interconnection should be made (connector output).
Remedy:	Interconnect this connector input with a connector output having a float data type.

F01511 (A)	BICO: Interconnection between different scalings
Message value:	Parameter: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The requested interconnection was established. However, a conversion is made between the BICO output and BICO input using the reference values. - the BICO output has different normalized units than the BICO input. - message only for interconnections within a drive object. Example: The BICO output has, as normalized unit, voltage and the BICO input has current. This means that the factor p2002 (contains the reference value for current) / p2001 (contains the reference value for voltage) is calculated between the BICO output and BICO input. Fault value (r0949, interpret decimal): Parameter number of the BICO input (signal sink).
Remedy:	No correction needed.
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F01512	BICO: No scaling available
Message value:	%1
Drive object:	All objects
Reaction:	Infeed: OFF2 (OFF1) Servo: OFF2 Vector: OFF2
Acknowledge:	POWER ON
Cause:	An attempt was made to determine a conversion factor for a scaling that does not exist. Fault value (r0949, interpret decimal): Unit (e.g. corresponding to SPEED) for which an attempt was made to determine a factor.
Remedy:	Apply scaling or check the transfer value.

F01513 (A)	BICO: Spanning DO between different scalings
Message value:	Parameter: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The requested interconnection was established. However, a conversion is made between the BICO output and BICO input using the reference values. An interconnection is made between different drive objects and the BICO output has different normalized units than the BICO input or the normalized units are the same but the reference values are different. Example: The BICO output has, as standard unit, voltage and the BICO input has current; both lie in different drive objects. This means that the factor p2002 (contains the reference value for current) / p2001 (contains the reference value for voltage) is calculated between the BICO output and BICO input. Fault value (r0949, interpret decimal): Parameter number of the BICO input (signal sink).
Remedy:	None necessary.
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A01514 (F)	BICO: Error when writing during a reconnect
Message value:	Parameter: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	During a reconnect operation (e.g. while booting or downloading - but can also occur in normal operation) a parameter was not able to be written to. Example: When writing to a double word BICO input in the second index, the memory areas overlap (e.g. p8861). The parameter is then reset to the factory setting. Alarm value (r2124, interpret decimal): Parameter number of the BICO input (signal sink).
Remedy:	None necessary.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY
F01515 (A)	BICO: Writing to parameter not permitted as the master control is active
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	While changing the number of CDS or when copying from CDS, the master control was active.
Remedy:	None necessary.
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A01590 (F)	Drive: Motor maintenance interval expired
Message value:	Fault cause: %1 bin
Drive object:	A_INF, B_INF, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The selected service/maintenance interval for this motor was reached. Alarm value (r2124, interpret decimal): Motor data set number. See also: p0650 (Actual motor operating hours), p0651 (Motor operating hours maintenance interval)

Remedy: carry out service/maintenance and reset the service/maintenance interval (p0651).
Reaction upon F: NONE
Acknowledged upon F: IMMEDIATELY

F01600 SI CU: STOP A initiated

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The drive-based "Safety Integrated" function in the Control Unit (CU) has detected a fault and initiated a STOP A (pulse suppression via the safety shutdown path of the Control Unit).
- forced checking procedure of the safety shutdown path of the Control Unit unsuccessful.
- subsequent response to fault F01611 (defect in a monitoring channel).
Fault value (r0949, interpret decimal):
0: Stop request from the Motor Module.
1005: Pulses suppressed although STO not selected and there is no internal STOP A present.
1010: Pulses enabled although STO is selected or an internal STOP A is present.
1015: Feedback of the safe pulse suppression for Motor Modules connected in parallel are different.
9999: Subsequent response to fault F01611.
Remedy:
- select Safe Torque Off and deselect again.
- replace the Motor Module involved.
Re fault value = 9999:
- carry out diagnostics for fault F01611.
Note:
CU: Control Unit
MM: Motor Module
SI: Safety Integrated
STO: Safe Torque Off / SH: Safe standstill

F01611 SI CU: Defect in a monitoring channel

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE (OFF1, OFF2, OFF3)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The drive-based "Safety Integrated" function in the Control Unit (CU) has detected a fault in the data cross-check between the CU and Motor Module (MM) and initiated a STOP F.
As a result of this fault, after the parameterized transition has expired (p9658), fault F01600 (SI CU: STOP A initiated) is output.
Fault value (r0949, interpret decimal):
0: Stop request from the Motor Module.
1 ... 999:
Number of the cross-checked data that resulted in this fault. This number is also displayed in r9795.
1: SI monitoring clock cycle (r9780, r9880).
2: SI enable safety functions (p9601, p9801). Crosswise data comparison is only carried out for the supported bits.
3: SI SGE changeover tolerance time (p9650, p9850).
4: SI transition period STOP F to STOP A (p9658, p9858).
5: SI enable Safe Brake Control (p9602, p9802).
6: SI Motion enable, safety-relevant functions (p9501, internal value).
7: SI pulse suppression delay time for Safe Stop 1 (p9652, p9852).
8: SI PROFIsafe address (p9610, p9810).
9: SI debounce time for STO/SBC/SS1 (MM) (p9651, p9851).
11: SI Safe Brake Adapter mode, BICO interconnection (p9621, p9821).
12: SI Safe Brake Adapter relay ON time (p9622[0], p9822[0]).
13: SI Safe Brake Adapter relay OFF time (p9622[1], p9822[1]).
1000: Watchdog timer has expired. Within the time of approx. 5 * p9650 too many switching operations have occurred at terminal EP of the Motor Module, or STO (also as subsequent response) was initiated too frequently via PROFIsafe/TM54F.
1001, 1002: Initialization error, change timer / check timer.

2000: Status of the STO selection on the Control Unit and Motor Module are different.
 2001: Feedback signal for safe pulse suppression on the Control Unit and Motor Module are different.
 2002: Status of the delay timer SS1 on the Control Unit and Motor Module are different.
 2004: Status of the STO selection for modules connected in parallel are different.
 2005: Feedback signal of the safe pulse suppression on the Control Unit and Motor Modules connected in parallel are different.
 6000 ... 6999:
 Error in the PROFIsafe control. The significance of the individual message values is described in safety message C01711 of the Control Unit.

Remedy:

Re fault value = 1 ... 5 and 7 ... 999:
 - check the cross-checked data that resulted in a STOP F.
 - carry out a POWER ON (power off/on) for all components.
 - upgrade the Motor Module software.
 - upgrade the Control Unit software.

Re fault value = 6:
 - carry out a POWER ON (power off/on) for all components.
 - upgrade the Motor Module software.
 - upgrade the Control Unit software.

Re fault value = 1000:
 - check the EP terminal at the Motor Module (contact problems).
 - PROFIsafe: Remove contact problems/faults at the PROFIBUS master/PROFINET controller.
 - check the wiring of the fail-safe inputs at the TM54F (contact problems).

Re fault value = 1001, 1002:
 - carry out a POWER ON (power off/on) for all components.
 - upgrade the Motor Module software.
 - upgrade the Control Unit software.

Re fault value = 2000, 2001, 2002, 2004, 2005:
 - check the tolerance time SGE changeover and if required, increase the value (p9650/p9850, p9652/p9852).
 - check the wiring of the safety-relevant inputs (SGE) (contact problems).
 - check the causes of the STO selection in r9772. When the SMM functions are active (p9501 = 1), STO can also be selected using these functions.
 - replace the Motor Module involved.

Re fault value = 6000 ... 6999:
 Refer to the description of the message values in safety message C01711.

Note:
 CU: Control Unit
 EP: Enable Pulses (pulse enable)
 MM: Motor Module
 SGE: Safety-relevant input
 SI: Safety Integrated
 SMM: Safe Motion Monitoring
 SS1: Safe Stop 1 (corresponds to Stop Category 1 acc. to EN60204)
 STO: Safe Torque Off / SH: Safe standstill

F01612 SI CU: STO inputs for power units connected in parallel different

Message value: Fault cause: %1 bin

Drive object: SERVO, VECTOR

Reaction: NONE (OFF1, OFF2, OFF3)

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive-based "Safety Integrated" function on the Control Unit (CU) has identified different states of the AND'ed STO inputs for power units connected in parallel and has initiated a STOP F.
 As a result of this fault, after the parameterized transition has expired (p9658), fault F01600 (SI CU: STOP A initiated) is output.
 Fault value (r0949, interpret binary):
 Binary image of the digital inputs of the Control Unit that are used as signal source for the function "Safe Torque Off".

Remedy:

- check the tolerance time SGE changeover and if required, increase the value (p9650).
 - check the wiring of the safety-relevant inputs (SGE) (contact problems).

Note:
 CU: Control Unit
 SGE: Safety-relevant input
 SI: Safety Integrated
 STO: Safe Torque Off / SH: Safe standstill

N01620 (F, A) SI CU: Safe Torque Off active

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The "Safe Torque Off" (STO) function has been selected on the Control Unit (CU) using the input terminal and is active.
Note:
This message does not result in a safety stop response.
Remedy: None necessary.
Note:
CU: Control Unit
SI: Safety Integrated
STO: Safe Torque Off / SH: Safe standstill
Reaction upon F: OFF2
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon A: NONE
Acknowl. upon A: NONE

N01621 (F, A) SI CU: Safe Stop 1 active

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The "Safe Stop 1" (SS1) function has been selected on the Control Unit (CU) and is active.
Note:
This message does not result in a safety stop response.
Remedy: None necessary.
Note:
CU: Control Unit
SI: Safety Integrated
SS1: Safe Stop 1 (corresponds to Stop Category 1 acc. to EN60204)
Reaction upon F: OFF3
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon A: NONE
Acknowl. upon A: NONE

F01625 SI CU: Sign-of-life error in safety data

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The drive-based "Safety Integrated" function in the Control Unit (CU) has detected an error in the sign-of-life of the safety data between the CU and Motor Module (MM) and initiated a STOP A.
- there is either a DRIVE-CLiQ communication error or communication has failed.
- a time slice overflow of the safety software has occurred.
Fault value (r0949, interpret decimal):
Only for internal Siemens troubleshooting.

- Remedy:**
- select Safe Torque Off and deselect again.
 - carry out a POWER ON (power off/on) for all components.
 - check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified.
 - deselect all drive functions that are not absolutely necessary.
 - reduce the number of drives.
 - check the electrical cabinet design and cable routing for EMC compliance

Note:

CU: Control Unit

MM: Motor Module

SI: Safety Integrated

F01630 SI CU: Brake control error

Message value: %1

Drive object: SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive-based "Safety Integrated" function in the Control Unit (CU) has detected a brake control error and initiated a STOP A.

Fault value (r0949, interpret decimal):

10, 11:

Fault in "open holding brake" operation.

- Parameter p1278 incorrectly set.

- No brake connected or wire breakage (check whether brake releases for p1278 = 1 and p9602/p9802 = 0 (SBC deactivated)).

- Ground fault in brake cable.

20:

Fault in "brake open" state.

- Short-circuit in brake winding.

30, 31:

Fault in "close holding brake" operation.

- No brake connected or wire breakage (check whether brake releases for p1278 = 1 and p9602/p9802 = 0 (SBC deactivated)).

- Short-circuit in brake winding.

40:

Fault in "brake closed" state.

50:

Fault in the brake control circuit of the Control Unit or communication fault between the Control Unit and Motor Module (brake control).

80:

SafeBrakeAdapter: Fault in the brake control circuit of the Control Unit or communication fault between Control Unit and Motor Module (brake control diagnostics).

Note:

The following causes may apply to fault values:

- motor cable is not shielded correctly.

- defect in control circuit of the Motor Module.

- Remedy:**
- check parameter p1278 (for SBC, only p1278 = 0 is permissible).
 - select Safe Torque Off and deselect again.
 - check the motor holding brake connection.
 - check the function of the motor holding brake.
 - check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified.
 - check that the electrical cabinet design and cable routing are in compliance with EMC regulations (e.g. shield of the motor cable and brake conductors are connected with the shield connecting plate and the motor connectors are tightly screwed to the housing).
 - replace the Motor Module involved.
- Operation with Safe Brake Module or Safe Brake Adapter:
- check the Safe Brake Module or Safe Brake Adapter connection.
 - replace the Safe Brake Module or Safe Brake Adapter.
- Note:**
CU: Control Unit
SBC: Safe Brake Control
SI: Safety Integrated

F01649 SI CU: Internal software error

- Message value:** %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: An internal error in the Safety Integrated software on the Control Unit has occurred.
Note:
This fault results in a STOP A that cannot be acknowledged.
Fault value (r0949, interpret hexadecimal):
Only for internal Siemens troubleshooting.
- Remedy:**
- carry out a POWER ON (power off/on) for all components.
 - re-commission the "Safety Integrated" function and carry out a POWER ON.
 - upgrade the Control Unit software.
 - contact the Hotline.
 - replace the Control Unit.
- Note:**
CU: Control Unit
MM: Motor Module
SI: Safety Integrated

F01650 SI CU: Acceptance test required

- Message value:** %1
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The drive-based "Safety Integrated" function in the Control Unit requires an acceptance test.
Note:
This fault results in a STOP A that can be acknowledged.
Fault value (r0949, interpret decimal):
130: Safety parameters for the Motor Module not available.
1000: Reference and actual checksum on the Control Unit are not identical (booting).
- at least one checksum-checked piece of data is defective.
2000: Reference and actual checksum on the Control Unit are not identical (commissioning mode).
- reference checksum incorrectly entered into the Control Unit (p9799 not equal to r9798).
- when deactivating the safety functions, p9501 or p9503 are not deleted.
2001: Reference and actual checksum on the Motor Module are not identical (commissioning mode).
- reference checksum incorrectly entered into the Motor Module (p9899 not equal to r9898).
- when deactivating the safety functions, p9501 or p9503 are not deleted.
2002: Enable of safety-related functions between the Control Unit and Motor Module differ (p9601 not equal to p9801).
2003: Acceptance test is required as a safety parameter has been changed.
2004: An acceptance test is required because a project with enabled safety-functions has been downloaded.

2005: The Safety LogBook has identified that a functional safety checksum has changed. An acceptance test is required.
2010: Safe Brake Control is enabled differently between the Control Unit and Motor Module (p9602 not equal to p9802).
2020: Error when saving the safety parameters for the Motor Module.
3003: Acceptance test is required as a hardware-related safety parameter has been changed.
3005: The Safety LogBook has identified that a hardware-related safety checksum has changed. An acceptance test is required.
9999: Subsequent response of another safety-related fault that occurred when booting that requires an acceptance test.

Remedy:

Re fault value = 130:
- carry out safety commissioning routine.
Re fault value = 1000:
- again carry out safety commissioning routine.
- replace the memory card or Control Unit.
Re fault value = 2000:
- check the safety parameters in the Control Unit and adapt the reference checksum (p9799).
Re fault value = 2001:
- check the safety parameters in the Motor Module and adapt the reference checksum (p9899).
Re fault value = 2002:
- enable the safety-related functions in the Control Unit and check in the Motor Module (p9601 = p9801).
Re fault value = 2003, 2004, 2005:
- Carry out an acceptance test and generate an acceptance report.
The procedure when carrying out an acceptance test as well as an example of the acceptance report are provided in the documentation for SINAMICS Safety Integrated.
The fault with fault value 3005 can only be acknowledged when the "STO" function is deselected.
Re fault value = 2010:
- check enable of the safety-related brake control in the Control Unit and Motor Module (p9602 = p9802).
Re fault value = 2020:
- again carry out safety commissioning routine.
- replace the memory card or Control Unit.
Re fault value = 3003:
- carry out the function checks for the modified hardware and generate an acceptance report.
The procedure when carrying out an acceptance test as well as an example of the acceptance report are provided in the following literature:
SINAMICS S120 Function Manual Safety Integrated
Re fault value = 3005:
- carry out the function checks for the modified hardware and generate an acceptance report.
The fault with fault value 3005 can only be acknowledged when the "STO" function is deselected.
Re fault value = 9999:
- carry out diagnostics for the other safety-related fault that is present.
Note:
CU: Control Unit
MM: Motor Module
SI: Safety Integrated
STO: Safe Torque Off
See also: p9799 (SI reference checksum SI parameters (Control Unit)), p9899 (SI reference checksum SI parameters (Motor Module))

F01651	SI CU: Synchronization safety time slices unsuccessful
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The "Safety Integrated" function requires a synchronization of the safety time slices between the Control Unit (CU) and Motor Module (MM) and between the Control Unit and the higher-level control. This synchronization routine was unsuccessful. Note: This fault results in a STOP A that cannot be acknowledged. Fault value (r0949, interpret decimal): 150: Fault in the synchronization to the PROFIBUS master. All other values: Only for internal Siemens troubleshooting. See also: p9510 (SI Motion clock-cycle synchronous PROFIBUS master)
Remedy:	Re fault value = 150: - check the setting of p9510 (SI Motion clock-cycle synchronous PROFIBUS master) and if required, correct. General: - carry out a POWER ON (power off/on) for all components. - upgrade the Motor Module software. - upgrade the Control Unit software. - upgrade the software of the higher-level control. Note: CU: Control Unit MM: Motor Module SI: Safety Integrated
F01652	SI CU: Illegal monitoring clock cycle
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	One of the Safety Integrated monitoring clock cycles is not permissible: - the drive-based monitoring clock cycle cannot be maintained due to the communication conditions required in the system. - the monitoring clock cycle for safe motion monitoring functions with the higher-level control is not permissible (p9500). - The sampling time for the current controller (p0112, p0115) cannot be supported. Note: This fault results in a STOP A that cannot be acknowledged. Fault value (r0949, interpret decimal): - for enabled drive-based SI monitoring (p9601/p9801 > 0): Minimum setting for the monitoring clock cycle (in µs). - with the motion monitoring function enabled (p9501 > 0): 100: No matching monitoring clock cycle was able to be found. 101: The monitoring clock cycle is not an integer multiple of the actual value sensing clock cycle. 102: An error has occurred when transferring the DP clock cycle to the Motor Module (MM). 103: An error has occurred when transferring the DP clock cycle to the Sensor Module. 104,105: - four times the current controller sampling time is greater than 1 ms when operating with a non-isochronous PROFIBUS. - four times the current controller sampling time is greater than the DP clock cycle when operating with an isochronous PROFIBUS. - The DP clock cycle is not an integer multiple of the sampling time of the current controller. 106: The monitoring clock cycle does not match the monitoring clock cycle of the TM54F. 107: Four times the current controller sampling time is greater than the actual value sensing clock cycle (p9511) or the actual value sensing clock cycle is not an integer multiple of the current controller sampling time. 108: The parameterized actual value sensing clock cycle cannot be set on this component. 109: If the motion monitoring functions have been parameterized as encoderless (p9506), the actual value sensing clock cycle (p9511) and the current controller clock cycle must be identical. 110: The actual value sensing clock cycle (p9511) for safety with encoder (p9506 = 0) is less than 2 ms for this Control Unit (e.g. CU305).

Remedy:

For enabled drive-based SI monitoring (p9601/p9801 > 0):

- upgrade the Control Unit software.

For enabled motion monitoring function (p9501 > 0):

- correct the monitoring clock cycle (p9500) and carry out POWER ON.

Re fault value = 101:

- actual value sensing clock cycle corresponds to position control clock cycle/DP clock cycle (factory setting).
- for the drive-based motion monitoring functions (p9601/p9801bit 2 = 1) the actual value sensing clock cycle can be directly parameterized in P9511/p9311.

Re fault value = 104, 105:

- set a separate actual value sensing clock cycle in p9511.
- restrict operation to a maximum of two vector drives. For the standard setting in p0112, p0115, the current controller sampling time is automatically reduced to 250 µs. If the standard values were changed, then the current controller sampling time (p0112, p0115) should be appropriately set.
- increase the DP clock cycle for operation with a clock-cycle synchronous PROFIBUS so that there is a multiple clock cycle ratio of at least 4:1 between the DP clock cycle and the current controller sampling time. A clock cycle ratio of at least 8:1 is recommended.
- With firmware version 2.5, please ensure that parameter p9510 is set to 1 in the drive (clock cycle synchronous operation).

Re fault value = 106:

- set the parameters for the monitoring clock cycles the same (p10000 and p9500 / p9300).

Re fault value = 107:

- set an actual value sensing clock cycle in p9511 that matches the current controller clock cycle. A clock cycle ratio of at least 8:1 is recommended.

Re fault value = 108:

- set a suitable actual value sensing clock cycle in p9511.
- if the DP clock cycle is used as the actual value sensing clock cycle for operation with isochronous PROFIBUS (p9511 = 0), then a suitable DP clock cycle must be configured.

A suitable multiple of the DP clock cycle (e.g. 1, 2, 3, 4, 5, 6, 8, 10) must be parameterized on the D410. Otherwise, the clock cycle must be set to less than 8 ms.

Re fault value = 109:

- set the actual value sensing clock cycle in p9511 to the same value as the current controller clock cycle (p115).

Re fault value = 110:

- set the actual value sensing clock cycle in p9511 to 2 ms or higher.

Note:

CU: Control Unit
MM: Motor Module
SI: Safety Integrated

F01653	SI CU: PROFIBUS configuration error
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	There is a PROFIBUS configuration error for using Safety Integrated monitoring functions with a higher-level control (SINUMERIK or F-PLC).
	Note:
	For safety functions that have been enabled, this fault results in a STOP A that cannot be acknowledged.
	Fault value (r0949, interpret decimal):
	200: A safety slot for receive data from the control has not been configured.
	210, 220: The configured safety slot for the receive data from the control has an unknown format.
	230: The configured safety slot for the receive data from the F-PLC has the incorrect length.
	231: The configured safety slot for the receive data from the F-PLC has the incorrect length.
	240: The configured safety slot for the receive data from the SINUMERIK has the incorrect length.
	250: A PROFIsafe slot is configured in the higher-level F control, however PROFIsafe is not enabled in the drive.
	300: A safety slot for the send data to the control has not been configured.
	310, 320: The configured safety slot for the send data to the control has an unknown format.
	330: The configured safety slot for the send data to the F-PLC has the incorrect length.
	331: The configured safety slot for the send data to the F-PLC has the incorrect length.
	340: The configured safety slot for the send data to the SINUMERIK has the incorrect length.

Remedy:

- The following generally applies:
 - check and, if necessary, correct the PROFIBUS configuration of the safety slot on the master side.
 - upgrade the Control Unit software.
- Re fault value = 250:
 - remove the PROFIsafe configuring in the higher-level F control or enable PROFIsafe in the drive.
- Re fault value = 231, 331:
 - configure the PROFIsafe telegram matching the parameterization in the F-PLC.

The following applies for p9501.30 = 1 (FDI via PROFIsafe is enabled): PROFIsafe telegram 900 must be configured.
 The following applies for p9501.30 = 0 (FDI via PROFIsafe not enabled): PROFIsafe telegram 30 must be configured.

F01655 SI CU: Align monitoring functions

Message value: %1

Drive object: SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: An error has occurred when aligning the Safety Integrated monitoring functions on the Control Unit (CU) and Motor Module (MM). Control Unit and Motor Module were not able to determine a common set of supported SI monitoring functions.

- there is either a DRIVE-CLiQ communication error or communication has failed.
- Safety Integrated software releases on the Control Unit and Motor Module are not compatible with one another.

Note:
 This fault results in a STOP A that cannot be acknowledged.
 Fault value (r0949, interpret hexadecimal):
 Only for internal Siemens troubleshooting.

Remedy:

- carry out a POWER ON (power off/on) for all components.
- upgrade the Motor Module software.
- upgrade the Control Unit software.
- check the electrical cabinet design and cable routing for EMC compliance

Note:
 CU: Control Unit
 MM: Motor Module
 SI: Safety Integrated

F01656 SI CU: Motor Module parameter error

Message value: %1

Drive object: SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: When accessing the Safety Integrated parameters for the Motor Module (MM) in the non-volatile memory, an error has occurred.

Note:
 This fault results in a STOP A that can be acknowledged.
 Fault value (r0949, interpret decimal):
 129: Safety parameters for the Motor Module corrupted.
 131: Internal Motor Module software error.
 132: Communication errors when uploading or downloading the safety parameters for the Motor Module.
 255: Internal software error on the Control Unit.

Remedy:

- re-commission the safety functions.
- upgrade the Control Unit software.
- upgrade the Motor Module software.
- replace the memory card or Control Unit.

Re fault value = 132:

- check the electrical cabinet design and cable routing for EMC compliance

Note:
 CU: Control Unit
 MM: Motor Module
 SI: Safety Integrated

F01659	SI CU: Write request for parameter rejected
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The write request for one or several Safety Integrated parameters on the Control Unit (CU) was rejected. Note: This fault does not result in a safety stop response. Fault value (r0949, interpret decimal): 1: The Safety Integrated password is not set. 2: A reset of the drive parameters was selected. However, the Safety Integrated parameters were not reset, as Safety Integrated is presently enabled. 3: The interconnected STO input is in the simulation mode. 10: An attempt was made to enable the STO function although this cannot be supported. 11: An attempt was made to enable the SBC function although this cannot be supported. 12: An attempt was made to enable the SBC function although this cannot be supported for a parallel circuit configuration. 13: An attempt was made to enable the SS1 function although this cannot be supported. 14: An attempt was made to enable the PROFIsafe communication - although this cannot be supported or the version of the PROFIsafe driver used on the CU and MM is different. 15: An attempt was made to enable the motion monitoring functions integrated in the drive although these cannot be supported. 16: An attempt was made to enable the STO function although this cannot be supported when the internal voltage protection (p1231) is enabled. 17: An attempt was made to enable the PROFIsafe function although this cannot be supported for a parallel circuit configuration. 18: An attempt was made to enable the PROFIsafe function for Basic Functions although this cannot be supported. 19: An attempt was made to enable the SBA (Safe Brake Adapter), although this cannot be supported. See also: p0970, p3900, r9771, r9871
Remedy:	Re fault value = 1: - set the Safety Integrated password (p9761). Re fault value = 2: - Inhibit Safety Integrated (p9501, p9601) or reset safety parameters (p0970 = 5), then reset the drive parameters again. Re fault value = 3: - end the simulation mode for the digital input (p0795). Re fault value = 10, 11, 12, 13, 14, 15, 17, 18, 19: - check whether there are faults in the safety function alignment between the Control Unit and the Motor Module involved (F01655, F30655) and if required, carry out diagnostics for the faults involved. - use a Motor Module that supports the required function ("Safe Torque Off", "Safe Brake Control", "PROFIsafe/PROFIsafe V2", "motion monitoring functions integrated in the drive"). - upgrade the Motor Module software. - upgrade the Control Unit software. Re fault value = 16: - inhibit the internal voltage protection (p1231). Note: CU: Control Unit MM: Motor Module SBC: Safe Brake Control SI: Safety Integrated SS1: Safe Stop 1 (corresponds to Stop Category 1 acc. to EN60204) STO: Safe Torque Off / SH: Safe standstill SBA: Safe Brake Adapter See also: p9501 (SI Motion enable safety functions (Control Unit)), p9601 (SI enable, functions integrated in the drive (Control Unit)), p9620 (SI signal source for STO (SH)/SBC/SS1 (Control Unit)), p9761 (SI password input), p9801 (SI enable, functions integrated in the drive (Motor Module))

F01660	SI CU: Safety-related functions not supported
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The Motor Module (MM) does not support the safety-related functions (e.g. the Motor Module version is not the correct one). Safety Integrated cannot be commissioned. Note: This fault does not result in a safety stop response.
Remedy:	- use a Motor Module that supports the safety-related functions. - upgrade the Motor Module software. Note: CU: Control Unit MM: Motor Module SI: Safety Integrated

F01663	SI CU: Copying the SI parameters rejected
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	One of the following values is saved in p9700 or was entered offline: 87 or 208. This is the reason that when booting, an attempt is made to copy SI parameters from the Control Unit to the Motor Module. However, no safety-relevant function has been selected on the Control Unit (p9501 = 0, p9601 = 0). This is the reason that copying is not possible. Note: This fault does not result in a safety stop response. See also: p9700 (SI Motion copy function)
Remedy:	- Set p9700 to 0. - Check p9501 and p9601 and if required, correct. - Restart the copying function by entering the corresponding value into p9700.

F01664	SI CU: No automatic firmware update
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	During booting, the system detected that the "Firmware update automatic" function (p7826 = 1) was not activated. This function must be activated for automatic firmware updates/downgrades to prevent impermissible version combinations when safety functions are enabled. Note: This fault does not result in a safety stop response. See also: p7826 (Firmware update automatic)
Remedy:	For enabled drive-based SI monitoring: 1. Activate the "Firmware update automatic" function (p7826 = 1). 2. Save the parameters (p0977 = 1) and carry out a POWER ON. When deactivating drive-based SI monitoring (p9601 = 0), the fault can be acknowledged after exiting the safety commissioning mode.

F01665	SI CU: System is defective
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Before the drive booted the last time, it detected a defect in the system and carried out an emergency stop. Fault value (r0949, interpret hexadecimal): Only for internal Siemens troubleshooting.

Remedy:

- carry out a POWER ON (power off/on).
- upgrade firmware to later version.
- contact the Hotline.

A01666 (F) SI Motion CU: Steady-state (static) 1 signal at the F-DI for safety-relevant acknowledgement

Message value: -

Drive object: SERVO

Reaction: NONE

Acknowledge: NONE

Cause: A logical 1 signal is present at the F-DI configured in p10006 for more than 10 seconds. A logical 0 signal must be present statically (steady-state) at the F-DI. This avoids unintentional safety-relevant acknowledgement (or the "Internal Event Acknowledge" signal) if a wire breaks or one of the two digital inputs bounces.

Remedy: Set F-DI (see p10006) to logical 0 signal.

Reaction upon F: NONE

Acknowled. upon F: IMMEDIATELY

F01670 SI Motion: Invalid parameterization Sensor Module

Message value: %1

Drive object: SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The parameterization of a Sensor Module used for Safety Integrated is not permissible.

Note:
This fault results in a STOP A that cannot be acknowledged.
Fault value (r0949, interpret decimal):

- 1: No encoder was parameterized for Safety Integrated.
- 2: An encoder was parameterized for Safety Integrated that does not have an A/B track (sine/cosine).
- 3: The encoder data set selected for Safety Integrated is still not valid.
- 4: A communication error with the encoder has occurred.
- 10: For an encoder used for Safety Integrated, not all of the Drive Data Sets (DDS) are assigned to the same Encoder Data Set (EDS) (p0187 ... p0189).

Remedy:

Re fault value = 1, 2:
- use and parameterize an encoder that Safety Integrated supports (encoder with track A/B sine-wave, p0404.4 = 1).

Re fault value = 3:
- check whether the drive or drive commissioning function is active and if required, exit this (p0009 = p00010 = 0), save the parameters (p0971 = 1) and carry out a POWER ON

Re fault value = 4:
- check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Sensor Module involved and if required, carry out a diagnostics routine for the faults identified.

Re fault value = 10:
- align the EDS assignment of all of the encoders used for Safety Integrated (p0187 ... p0189).

Note:
SI: Safety Integrated

F01671 SI Motion: Parameterization encoder error

Message value: %1

Drive object: SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The parameterization of the encoder used by Safety Integrated is different to the parameterization of the standard encoder.

Note:
This fault does not result in a safety stop response.
Fault value (r0949, interpret decimal):
Parameter number of the non-corresponding safety parameter.

Remedy: Align the encoder parameterization between the safety encoder and the standard encoder.

Note:
SI: Safety Integrated

F01672	SI CU: Motor Module software/hardware incompatible
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The existing Motor Module software does not support safe motion monitoring or is not compatible to the software on the Control Unit or there is a communications error between the Control Unit and Motor Module. Note: This fault results in a STOP A that cannot be acknowledged. Fault value (r0949, interpret decimal): 1: The existing Motor Module software does not support the safe motion monitoring function. 2, 3, 6, 8: There is a communications error between the Control Unit and Motor Module. 4, 5, 7: The existing Motor Module software is not compatible to the software on the Control Unit. 9: The current Motor Module software does not support safe encoderless motion monitoring.
Remedy:	- check whether there are faults in the safety function alignment between the Control Unit and the Motor Module involved (F01655, F30655) and if required, carry out the appropriate diagnostics routine for the particular faults. Re fault value = 1: - use a Motor Module that supports safe motion monitoring Re fault value = 2, 3, 6, 8: - check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified. Re fault value = 4, 5, 7: - upgrade the Motor Module software. Re fault value = 9: - upgrade the Motor Module firmware. Note: SI: Safety Integrated

F01673	SI Motion: Sensor Module software/hardware incompatible
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The existing Sensor Module software and/or hardware does not support the safe motion monitoring function with the higher-level control. Note: This fault does not result in a safety stop response. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	- upgrade the Sensor Module software. - use a Sensor Module that supports the safe motion monitoring function. Note: SI: Safety Integrated

F01680	SI Motion CU: Checksum error safety monitoring functions
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The actual checksum calculated by the drive and entered in r9728 via the safety-relevant parameters does not match the reference checksum saved in p9729 at the last machine acceptance. Safety-relevant parameters have been changed or a fault is present. Note: This fault results in a STOP A that can be acknowledged. Fault value (r0949, interpret decimal): 0: Checksum error for SI parameters for motion monitoring. 1: Checksum error for SI parameters for actual values. 2: Checksum error for SI parameters for component assignment.

Remedy:

- check the safety-relevant parameters and if required, correct.
- perform a POWER ON if safety parameters requiring a POWER ON have been modified.
- carry out an acceptance test.

Note:
SI: Safety Integrated

F01681 SI Motion CU: Incorrect parameter value

Message value: Parameter: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The parameter cannot be parameterized with this value.
 Note:
 This fault does not result in a safety stop response.
 Fault value (r0949, interpret decimal):
 Parameter number with the incorrect value.
 The gearbox stage has been set to an excessively high value for the parameter 9522 fault value.

Remedy: Correct the parameter value.
 With hysteresis/filtering enabled (p9501.16=1), the following applies:
 Set parameters p9546/9346 and p9547/9347 acc. to the following rule: p9546 >= 2 * p9547; 9346 >= 2 * p9347.
 The following rule must also be adhered to when actual value synchronization (p9501.3 = 1) is enabled: p9549 <= p9547; p9349 <= p9347.
 Correct the parameter 9522 fault value.
 Also check p9516.0 for fault value 9517.

F01682 SI Motion CU: Monitoring function not supported

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The monitoring function enabled in p9501, p9601 or p9801 is not supported in this firmware version.
 Note:
 This fault results in a STOP A that cannot be acknowledged.
 Fault value (r0949, interpret decimal):
 Monitoring function SLP not supported (p9501.1).
 2: Monitoring function SCA not supported (p9501.7 and p9501.8 ... 15 and p9503).
 3: Monitoring function SLS override not supported (p9501.5).
 4: Monitoring function external ESR activation not supported (p9501.4).
 5: Monitoring function FDI in PROFIsafe not supported (p9501.30).
 6: Enable actual value synchronization not supported (p9501.3).
 9: Monitoring function not supported, enable bit reserved (p9501.2, p9501.17 ... 29, p9501.31, if required p9501.6).
 10: Monitoring functions only supported for a SERVO drive object.
 11: Only drive-based encoderless monitoring functions are supported.
 12: Monitoring functions for ncSI are not supported on a CU305.
 20: Drive-based motion monitoring functions are only supported in conjunction with PROFIsafe (p9501 and p9601.1 ... 2 and p9801.1 ... 2).
 21: PROFIsafe only supported in conjunction with motion monitoring functions in the drive (p9501 and p9601.1 ... 2 and p9801.1 ... 2).

Remedy: Deselect the monitoring function involved (p9501, p9503, p9506, p9601, p9801).
 Note:
 SCA: Safe Cam / SN: Safe software cam
 SI: Safety Integrated
 SLP: Safely-Limited Position / SE: Safe software limit switches
 SLS: Safely-Limited Speed / SG: Safely reduced speed
 See also: p9501 (SI Motion enable safety functions (Control Unit)), p9503 (SI Motion SCA (SN) enable (Control Unit))

F01683 SI Motion CU: SOS/SLS enable missing

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The safety-relevant basic function "SOS/SLS" is not enabled in p9501 although other safety-relevant monitoring functions are enabled.
Note:
This fault does not result in a safety stop response.
Remedy: Enable the function "SOS/SLS" (p9501.0) and carry out a POWER ON.
Note:
SI: Safety Integrated
SLS: Safely-Limited Speed / SG: Safely reduced speed
SOS: Safe Operating Stop / SBH: Safe operating stop
See also: p9501 (SI Motion enable safety functions (Control Unit))

F01684 SI Motion: Safely limited position limit values interchanged

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: For the function "Safely-Limited Position" (SE), a lower value is in p9534 than in p9535.
Note:
This fault does not result in a safety stop response.
Fault value (r0949, interpret decimal):
1: Limit values SLP1 interchanged.
2: Limit values SLP2 interchanged.
Remedy: Correct the limit values in p9534 and p9535 and carry out a POWER ON.
Note:
SI: Safety Integrated
SLP: Safely-Limited Position / SE: Safe software limit switches

F01685 SI Motion CU: Safely-limited speed limit value too high

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The limit value for the function "Safely-Limited Speed" (SLS) is greater than the speed that corresponds to an encoder limit frequency of 500 kHz.
Note:
This fault does not result in a safety stop response.
Fault value (r0949, interpret decimal):
Maximum permissible speed.
Remedy: Correct the limit values for SLS and carry out a POWER ON.
Note:
SI: Safety Integrated
SLS: Safely-Limited Speed / SG: Safely reduced speed
See also: p9531 (SI Motion SLS (SG) limit values (Control Unit))

F01686	SI Motion: Illegal parameterization cam position
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	At least one enabled "Safety Cam" (SCA) is parameterized in p9536 or p9537 too close to the tolerance range around the modulo position. The following conditions must be complied with to assign cams to a cam track: - the cam length of cam x = p9536[x]-p9537[x] must be greater or equal to the cam tolerance + the position tolerance (= p9540 + p9542). This also means that for cams on a cam track, the minus position value must be less than the plus position value. - the distance between 2 cams x and y (minus position value[y] - plus position value[x] = p9537[y] - p9536[x]) on a cam track must be greater than or equal to the cam tolerance + position tolerance (= p9540 + p9542). Note: This fault does not result in a safety stop response. Fault value (r0949, interpret decimal): Number of the "Safe Cam" with an illegal position. See also: p9501 (SI Motion enable safety functions (Control Unit))
Remedy:	Correct the cam position and carry out a POWER ON. Note: SCA: Safe Cam / SN: Safe software cam SI: Safety Integrated See also: p9536 (SI Motion SCA (SN) plus cam position (Control Unit)), p9537 (SI Motion SCA (SN) plus cam position (Control Unit))
F01687	SI Motion: Illegal parameterization modulo value SCA (SN)
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The parameterized modulo value for the "Safe Cam" (SCA) function is not a multiple of 360 000 mDegrees. Note: This fault does not result in a safety stop response.
Remedy:	Correct the modulo value for SCA and carry out a POWER ON. Note: SCA: Safe Cam / SN: Safe software cam SI: Safety Integrated See also: p9505 (SI Motion SCA (SN) modulo value (Control Unit))
F01688	SI Motion CU: Actual value synchronization not permissible
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	- It is not permissible to enable actual value synchronization for a 1-encoder system. - It is not permissible to simultaneously enable actual value synchronization and a monitoring function with absolute reference (SCA/SLP). Note: This fault results in a STOP A that cannot be acknowledged.
Remedy:	- Either select the "actual value synchronization" function or parameterize a 2-encoder system. - Either deselect the function "actual value synchronization" or the monitoring functions with absolute reference (SCA/SLP) and carry out a POWER ON. Note: SCA: Safe Cam / SN: Safe software cam SI: Safety Integrated SLP: Safely-Limited Position / SE: Safe software limit switches See also: p9501 (SI Motion enable safety functions (Control Unit)), p9526 (SI Motion encoder assignment second channel)

C01689 SI Motion: Axis re-configured

Message value: Parameter: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: POWER ON
Cause: The axis configuration was changed (e.g. changeover between linear axis and rotary axis).
Parameter p0108.13 is internally set to the correct value.
Note:
This fault does not result in a safety stop response.
Fault value (r0949, interpret decimal):
Parameter number of parameter that initiated the change.
See also: p9502 (SI Motion axis type (Control Unit))
Remedy: The following should be carried out after the changeover:
- exit the safety commissioning mode (p0010).
- save all parameters (p0977 = 1 or "copy RAM to ROM").
- carry out a POWER ON.
Once the Control Unit has been switched on, safety message F01680 or F30680 indicates that the checksums in r9398[0] and r9728[0] have changed in the drive. The following must, therefore, be carried out:
- activate safety commissioning mode again.
- complete safety commissioning of the drive.
- exit the safety commissioning mode (p0010).
- save all parameters (p0977 = 1 or "copy RAM to ROM").
- carry out a POWER ON.
Note:
For the commissioning software, the units are only consistently displayed after a project upload.

F01690 SI Motion: Data save problem for the NVRAM

Message value: %1
Drive object: All objects
Reaction: Infeed: NONE (OFF1, OFF2)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge: POWER ON
Cause: There is not sufficient memory space in the NVRAM on the drive to save parameters r9781 and r9782 (safety log-book).
Note:
This fault does not result in a safety stop response.
Fault value (r0949, interpret decimal):
0: There is no physical NVRAM available in the drive.
1: There is no longer any free memory space in the NVRAM.
Remedy: Re fault value = 0:
- use a Control Unit NVRAM.
Re fault value = 1:
- deselect functions that are not required and that take up memory space in the NVRAM.
- contact the Hotline.

A01691 (F) SI Motion: Ti and To unsuitable for DP cycle

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The configured times for PROFIBUS communication are not permitted and the DP cycle is used as the actual value acquisition cycle for the safe movement monitoring functions:
Isochronous PROFIBUS: the total of Ti and To is too high for the set DP cycle. The DP cycle should be at least 1 current controller cycle greater than the sum of Ti and To.
Non-isochronous PROFIBUS: the DP cycle must be at least 4 x current controller cycle.

Remedy: Configure Ti and To low so that they are suitable for the DP cycle or increase the DP cycle time.
Alternative when drive-based SI monitoring is enabled (p9601/p9801 > 0):
Use the actual value acquisition cycle p9511/p9311 and, in turn, set independently from DP cycle. The actual value sensing clock cycle must be at least four times the current controller clock cycle. A clock cycle ratio of at least 8:1 is recommended.
See also: p9511 (SI Motion actual value sensing cycle clock (Control Unit))

Reaction upon F: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY (POWER ON)

F01692 SI Motion CU: Parameter value not permitted for encoderless

Message value: Parameter: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The parameter cannot be set to this value if encoderless motion monitoring functions have been selected in p9506.
Note:
This fault does not result in a safety stop response.
Fault value (r0949, interpret decimal):
Parameter number with the incorrect value.
See also: p9501 (SI Motion enable safety functions (Control Unit))

Remedy:
- Correct the parameter specified in the fault value.
- If necessary, deselect encoderless motion monitoring functions (p9506).
See also: p9501 (SI Motion enable safety functions (Control Unit))

A01693 (F) SI Motion CU: Safety parameter setting changed, wam restart/POWER ON required

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Safety parameters have been changed; these will only take effect following a warm restart or POWER ON.
Alarm value (r2124, interpret decimal):
Parameter number of the safety parameter which has changed, necessitating a warm restart or POWER ON.

Remedy:
- carry out a warm restart (p0971 = 3)
- carry out a POWER ON (power off/on) for all components.

Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)

Acknowl. upon F: POWER ON

A01696 (F) SI Motion: Testing of the motion monitoring functions selected when booting

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The test of the motion monitoring functions was already illegally active when booting.
This is the reason that the test is only carried out again after selecting the forced checking procedure parameterized in p9705.
Note:
This message does not result in a safety stop response.
See also: p9705 (SI Motion: Test stop signal source)

Remedy: Deselect the forced checking procedure of the safety motion monitoring functions and then select again.
The signal source for initiation is parameterized in binector input p9705.
Notice:
It is not permissible to use TM54F inputs to start the test stop.
Note:
SI: Safety Integrated
See also: p9705 (SI Motion: Test stop signal source)

Reaction upon F: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY (POWER ON)

A01697 (F) SI Motion: Motion monitoring functions must be tested

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The time set in p9559 for the forced checking procedure of the safety motion monitoring functions has been exceeded. A new test is required.
After next selecting the forced checking procedure parameterized in p9705, the message is withdrawn and the monitoring time is reset.
Note:
This message does not result in a safety stop response.
As the shutdown paths are not automatically checked during booting, an alarm is always issued once booting is complete.
See also: p9559 (SI Motion forced checking procedure timer (Control Unit)), p9705 (SI Motion: Test stop signal source)

Remedy: Carry out the forced checking procedure of the safety motion monitoring functions.
The signal source for initiation is parameterized in binector input p9705.
Notice:
It is not permissible to use TM54F inputs to start the test stop.
Note:
SI: Safety Integrated
See also: p9705 (SI Motion: Test stop signal source)

Reaction upon F: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY (POWER ON)

A01698 (F) SI CU: Commissioning mode active

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The commissioning of the "Safety Integrated" function is selected.
This message is withdrawn after the safety functions have been commissioned.
Note:
This message does not result in a safety stop response.
See also: p0010

Remedy: None necessary.
Note:
CU: Control Unit
SI: Safety Integrated

Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)

Acknowl. upon F: IMMEDIATELY (POWER ON)

A01699 (F)	SI CU: Shutdown path must be tested
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The time set in p9659 for the forced checking procedure of the safety shutdown paths has been exceeded. The safety shutdown paths must be re-tested. After the next time the "STO" function is deselected, the message is withdrawn and the monitoring time is reset. Note: This message does not result in a safety stop response. See also: p9659 (SI forced checking procedure timer)
Remedy:	Select STO and then deselect again. Note: CU: Control Unit SI: Safety Integrated STO: Safe Torque Off / SH: Safe standstill
Reaction upon F:	NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
C01700	SI Motion CU: STOP A initiated
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The drive is stopped via a STOP A (pulses are suppressed via the safety shutdown path of the Control Unit). Possible causes: - stop request from the second monitoring channel - pulses not suppressed after a parameterized time (p9557) after test stop selection. - subsequent response to the message C01706 "SI Motion CU: Safe Acceleration Monitoring limit exceeded". - subsequent response to the message C01714 "SI Motion CU: Safely-Limited Speed exceeded". - subsequent response to the message C01701 "SI Motion CU: STOP B initiated".
Remedy:	- remove the cause of the fault on the second monitoring channel - carry out a diagnostics routine for message C01706. - carry out a diagnostics routine for message C01714. - carry out a diagnostics routine for message C01701. - check the value in p9557 (where available), increase the value if necessary, and carry out a POWER ON - check the shutdown path of the Control Unit (check DRIVE-CLiQ communication if it has been implemented) - replace the Motor Module/Power Module - replace Control Unit. This message can be acknowledged without a POWER ON as follows: - motion monitoring functions integrated in the drive: via Terminal Module 54F (TM54F) or PROFIsafe - motion monitoring functions with SINUMERIK: via the machine control panel in acceptance test mode only Note: SI: Safety Integrated
C01701	SI Motion CU: STOP B initiated
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF3
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The drive is stopped via a STOP B (braking along the OFF3 deceleration ramp). As a result of this fault, after the time parameterized in p9556 has expired, or the speed threshold parameterized in p9560 has been undershot, message C01700 "STOP A initiated" is output. Possible causes: - stop request from the second monitoring channel - subsequent response to the message C01714 "SI Motion: Safely reduced speed exceeded". - subsequent response to the message C01711 "SI Motion: Defect in a monitoring channel".

Remedy:

- remove the cause of the fault on the second monitoring channel
- carry out a diagnostics routine for message C01714.
- carry out a diagnostics routine for message C01711.

This message can be acknowledged without a POWER ON as follows:

- motion monitoring functions integrated in the drive: via Terminal Module 54F (TM54F) or PROFIsafe
- motion monitoring functions with SINUMERIK: via the machine control panel in acceptance test mode only

Note:

SI: Safety Integrated

C01706 SI Motion CU: SBR limit undershot

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: Motion monitoring functions with encoder: SBR - Safe Acceleration Monitoring. After initiating STOP B (SS1) or STOP C (SS2), the speed has exceeded the selected tolerance.
Encoderless motion monitoring function: SBR - Safe Brake Ramp Monitoring. After initiating STOP B (SS1) or SLS changeover to the lower speed stage, the speed has exceeded the selected tolerance.
The drive is shut down by the message C01700 "SI Motion: STOP A initiated".

Remedy: Check the braking behavior and, if necessary, adapt the tolerance for the "Safe Acceleration Monitor" or modify the parameter settings for the "safe brake ramp".
This message can be acknowledged without a POWER ON as follows:

- motion monitoring functions integrated in the drive: via Terminal Module 54F (TM54F) or PROFIsafe
- motion monitoring functions with SINUMERIK: via the machine control panel in acceptance test mode only

Note:

SI: Safety Integrated

See also: p9548 (SI Motion SBR actual velocity tolerance (Control Unit)), p9581 (SI Motion brake ramp reference value (Control Unit)), p9582 (SI Motion brake ramp delay time (Control Unit)), p9583 (SI Motion brake ramp monitoring time (Control Unit))

C01707 SI Motion CU: Tolerance for safe operating stop exceeded

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The actual position has distanced itself further from the target position than the standstill tolerance.
The drive is shut down by the message C01701 "SI Motion: STOP B initiated".

Remedy:

- check whether safety faults are present and if required carry out the appropriate diagnostic routines for the particular faults.
- check whether the standstill tolerance matches the accuracy and control dynamic performance of the axis.
- carry out a POWER ON.

This message can be acknowledged without a POWER ON as follows:

- motion monitoring functions integrated in the drive: via Terminal Module 54F (TM54F) or PROFIsafe
- motion monitoring functions with SINUMERIK: via the machine control panel in acceptance test mode only

Note:

SI: Safety Integrated

SOS: Safe Operating Stop / SBH: Safe operating stop

See also: p9530 (SI Motion standstill tolerance (Control Unit))

C01708 SI Motion CU: STOP C initiated

Message value: -

Drive object: SERVO, VECTOR

Reaction: STOP2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive is stopped via a STOP C (braking along the OFF3 deceleration ramp).
 "Safe Operating Stop" (SOS) is activated after the parameterized timer has expired.
 Possible causes:
 - stop request from the higher-level control.
 - subsequent response to the message C01714 "SI Motion: Safely reduced speed exceeded".
 - subsequent response to the message C01715 "SI Motion: Safe end stop exceeded".
 See also: p9552 (SI Motion transition time STOP C to SOS (SBH) (Control Unit))

Remedy: - remove the cause of the fault at the control.
 - carry out a diagnostics routine for message C01714.
 This message can be acknowledged as follows:
 - motion monitoring functions integrated in the drive: Via Terminal Module 54F (TM54F) or PROFIsafe
 - motion monitoring functions with SINUMERIK: Via the machine control panel
 Note:
 SI: Safety Integrated
 SOS: Safe Operating Stop / SBH: Safe operating stop

C01709 SI Motion CU: STOP D initiated

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive is stopped via a STOP D (braking along the path).
 "Safe Operating Stop" (SOS) is activated after the parameterized timer has expired.
 Possible causes:
 - stop request from the higher-level control.
 - subsequent response to the message C01714 "SI Motion: Safely reduced speed exceeded".
 - subsequent response to the message C01715 "SI Motion: Safe end stop exceeded".
 See also: p9553 (SI Motion transition time STOP D to SOS (SBH) (Control Unit))

Remedy: - remove the cause of the fault at the control.
 - carry out a diagnostics routine for message C01714.
 This message can be acknowledged as follows:
 - motion monitoring functions integrated in the drive: Via Terminal Module 54F (TM54F) or PROFIsafe
 - motion monitoring functions with SINUMERIK: Via the machine control panel
 Note:
 SI: Safety Integrated
 SOS: Safe Operating Stop / SBH: Safe operating stop

C01710 SI Motion CU: STOP E initiated

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive is stopped via a STOP E (retraction motion).
 "Safe Operating Stop" (SOS) is activated after the parameterized timer has expired.
 Possible causes:
 - stop request from the higher-level control.
 - subsequent response to the message C01714 "SI Motion: Safely reduced speed exceeded".
 - subsequent response to the message C01715 "SI Motion: Safe end stop exceeded".
 See also: p9554 (SI Motion transition time STOP E to SOS (SBH) (Control Unit))

Remedy:

- remove the cause of the fault at the control.
- carry out a diagnostics routine for message C01714.

This message can be acknowledged as follows:

- motion monitoring functions integrated in the drive: Via Terminal Module 54F (TM54F) or PROFIsafe
- motion monitoring functions with SINUMERIK: Via the machine control panel

Note:

SI: Safety Integrated
 SOS: Safe Operating Stop / SBH: Safe operating stop

C01711 SI Motion CU: Defect in a monitoring channel

Message value: %1

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: When cross-checking and comparing the two monitoring channels, the drive detected a difference between the input data or results of the monitoring functions and initiated a STOP F. One of the monitoring functions no longer reliably functions - i.e. safe operation is no longer possible.

If at least one monitoring function is active, then after the parameterized timer has expired, the message C01701 "SI Motion: STOP B initiated" is output.

The message value that resulted in a STOP F is displayed in r9725. The described message values involve the data cross-check between the Control Unit and Motor Module. If the drive is operated together with a SINUMERIK, the message values are described in message 27001 of SINUMERIK.

The following message values may also occur in the following cases if the cause that is explicitly mentioned does not apply:

- cycle times not set uniformly (p9500/p9300 and p9511/p9311)
- differently parameterized axis types (p9502/p9302).
- excessively fast cycle times (p9500/p9300, p9511/p9311).
- for message values 3, 44 ... 57, 232, and encoder systems, differently parameterized encoder values (p9516/p9316, p9517/p9317, p9518/p9318, p9520/p9320, p9521/p9321, p9522/p9322, p9526/p9326).
- incorrect synchronization.

Message value (r9749, interpret decimal):

0 to 999: Number of the cross-checked data that resulted in this fault.

Message values that are not subsequently listed are only for internal Siemens troubleshooting.

0: Stop request from the other monitoring channel.

1: Status image of monitoring functions SOS, SLS or SLP (result list 1) (r9710[0], r9710[1]).

2: Status image of monitoring function SCA or n < nx (result list 2) (r9711[0], r9711[1]).

3: The position actual value differential (r9713) between the two monitoring channels is greater than the tolerance in p9542/p9342. When actual value synchronization is enabled (p9501.3/p9301.3), the velocity differential (based on the position actual value) is greater than the tolerance in p9549/p9349.

4: Error when synchronizing the crosswise data comparison between the two channels.

5: Function enable signals (p9501/p9301)

6: Limit value for SLS1 (p9531[0]/p9331[0])

7: Limit value for SLS2 (p9531[1]/p9331[1])

8: Limit value for SLS3 (p9531[2]/p9331[2])

9: Limit value for SLS4 (p9531[3]/p9331[3])

10: Standstill tol. (p9530/p9330)

31: Position tolerance (p9542/p9342) or (p9549/p9349) when actual value synchronization is enabled (p9501.3/p9301.3)

33: Time, velocity changeover (p9551/p9351)

35: Delay time, pulse canc. (p9556/p9356)

36: Checking time, pulse canc. (p9557/p9357)

37: Trans. time, STOP C to SOS (p9552/p9352)

38: Trans. time STOP D to SOS (p9553/p9353)

40: Stop response for SLS (p9561/p9361)

42: Shutdown speed, pulse canc. (p9560/p9360)

43: Memory test, stop response (STOP A).

44 ... 57: General

Possible cause 1 (during commissioning or parameter modification)

The tolerance value for the monitoring function is not the same on the two monitoring channels.

Possible cause 2 (during active operation)

The limit values are based on the current actual value (r9713). If the safe actual values on the two monitoring channels do not match, the limit values, which have been set at a defined interval, will also be different (i.e. corresponding to fault value 3). This can be ascertained by checking the safe actual positions.

44: Position actual value (r9713) + limit value for SLS1 (p9531[0]/p9331[0])

45: Position actual value (r9713) - limit value for SLS1 (p9531[0]/p9331[0])

46: Position actual value (r9713) + limit value for SLS2 (p9531[1]/p9331[1])

47: Position actual value (r9713) - limit value for SLS2 (p9531[1]/p9331[1])

48: Position actual value (r9713) + limit value for SLS3 (p9531[2]/p9331[2])

49: Position actual value (r9713) - limit value for SLS3 (p9531[2]/p9331[2])

50: Position actual value (r9713) + limit value for SLS4 (p9531[3]/p9331[3])

51: Position actual value (r9713) - limit value for SLS4 (p9531[3]/p9331[3])

52: Standstill position + tolerance (p9530/9330)

53: Standstill position - tolerance (p9530/9330)

54: Position actual value (r9713) + limit value nx (p9546/p9346) + tolerance (p9542/p9342)

55: Position actual value (r9713) + limit value nx (p9546/p9346)

56: Position actual value (r9713) - limit value nx (p9546/p9346)

57: Position actual value (r9713) - limit value nx (p9546/p9346) - tolerance (p9542/p9342)

58: Actual stop request.

75: Velocity limit nx (p9546, p9346).

76: Stop response for SLS1 (p9563[0]/p9363[0])

77: Stop response for SLS2 (p9563[1]/p9363[1])

78: Stop response for SLS3 (p9563[2]/p9363[2])

79: Stop response for SLS4 (p9563[3]/p9363[3])

81: Velocity tolerance for SBR (p9548/p9348)

82: SGEs for SLS correction factor.

83: Acceptance test timer (p9558/p9358)

84: Trans. time STOP F (p9555/p9355)

85: Trans. time bus failure (p9580/p9380)

86: ID 1-encoder system (p9526/p9326).

87: Encoder assignment, 2nd channel (p9526/p9326)

89: Encoder limit freq.

230: Filter time constant for n < nx.

231: Hysteresis tolerance for n < nx.

232: Smoothed velocity actual value.

233: Smoothed velocity actual value + limit value nx / safety monitoring clock cycle + hysteresis tolerance.

234: Smoothed velocity actual value + limit value nx / safety monitoring clock cycle.

235: Smoothed velocity actual value - limit value nx / safety monitoring clock cycle.

236: Smoothed velocity actual value - limit value nx / safety monitoring clock cycle - hysteresis tolerance.

237: SGA n < nx.

238: Speed limit value for SBR (p9568/p9368).

239: Acceleration for SBR (p9581/p9381 and p9583/p9383).

240: Inverse value of acceleration for SBR (p9581/p9381 and p9583/p9383).

241: Deceleration time for SBR (p9582/p9382).

242: Encoderless safety (p9506/p9306).

243: Extended alarm acknowledgment (p9507/p9307).

244: Encoderless actual value sensing filter time (p9587/p9387).

245: Encoderless actual value sensing minimum current (p9588/p9388).

246: Voltage tolerance acceleration (p9589/p9389).

1000: Watchdog timer has expired. Too many signal changes have occurred at safety-relevant inputs.

1001: Initialization error of watchdog timer.

1005: Pulses already suppressed for test stop selection.

1011: Acceptance test status between the monitoring channels differ.

1012: Plausibility violation of the actual value from the encoder.

1020: Cyc. communication failure between the monit. cycles.

1021: Cyc. communication failure between the monit. channel and Sensor Module.

1022: Sign-of-life error for DRIVE-CLiQ encoder CU

1023: Error in the effectiveness test in the DRIVE-CLiQ encoder

1032: Sign-of-life error for DRIVE-CLiQ encoder MM

1033: Error checking offset between POS1 and POS2 for DRIVE-CLiQ encoder CU

1034: Error checking offset between POS1 and POS2 for DRIVE-CLiQ encoder MM
 1041: Current absolute value too low (encoderless)
 1042: Current/voltage plausibility error
 1043: Too many acceleration phases
 1044: Actual current values plausibility error.
 5000 ... 5140: PROFIsafe message values.
 5000, 5014, 5023, 5024, 5030 ... 5032, 5042, 5043, 5052, 5053, 5068, 5072, 5073, 5082 ... 5087, 5090, 5091, 5122 ... 5125, 5132 ... 5135, 5140: An internal software error has occurred (only for internal Siemens troubleshooting).
 5012: Error when initializing the PROFIsafe driver.
 5013: The result of the initialization is different for the two controllers.
 5022: Error when evaluating the F parameters. The values of the transferred F parameters do not match the expected values in the PROFIsafe driver.
 5025: The result of the F parameterization is different for the two controllers.
 5026: CRC error for the F parameters. The transferred CRC value of the F parameters does not match the value calculated in the PST.
 5065: A communications error was identified when receiving the PROFIsafe telegram.
 5066: A time monitoring error (timeout) was identified when receiving the PROFIsafe telegram.
 6000 ... 6166: PROFIsafe message values (PROFIsafe driver for PROFIBUS DP V1/V2 and PROFINET).
 6000: An internal software error has occurred (only for internal Siemens troubleshooting).
 6064 ... 6071: Error when evaluating the F parameters. The values of the transferred F parameters do not match the expected values in the PROFIsafe driver.
 6064: Destination address and PROFIsafe address are different (F_Dest_Add).
 6065: Destination address not valid (F_Dest_Add).
 6066: Source address not valid (F_Source_Add).
 6067: Watchdog time not valid (F_WD_Time).
 6068: Incorrect SIL level (F_SIL).
 6069: Incorrect F-CRC length (F_CRC_Length).
 6070: Incorrect F parameter version (F_Par_Version).
 6071: CRC error for the F parameters (CRC1). The transferred CRC value of the F parameters does not match the value calculated in the PROFIsafe driver.
 6072: F parameterization is inconsistent.
 6165: A communications error was identified when receiving the PROFIsafe telegram.
 6166: A time monitoring error (timeout) was identified when receiving the PROFIsafe telegram.
 See also: p9555 (SI Motion transition time STOP F to STOP B (Control Unit)), r9725 (SI Motion, diagnostics STOP F)

Remedy:

The following generally applies:
 The monitoring clock cycles in both channels and the axis types should be checked for equality and the same setting applied if necessary. If the error continues to be identified, increasing the monitoring clock cycles may resolve it.
 Re fault value = 0:
 - no error was identified in this monitoring channel. Note the error message of the other monitoring channel (for MM: C30711).
 Re fault value = 3:
 Commissioning phase:
 Encoder evaluation for own or second channel has been set incorrectly --> Correct the encoder evaluation.
 In operation:
 Check the mechanical design and the encoder signals.
 Re fault value = 4:
 The monitoring clock cycles in both channels should be checked for equality and if required, set the same. In combination with fault value 5 from the other monitoring channel (with MM: C30711), the monitoring clock cycle settings must be increased.
 Re fault value = 1 ... 999:
 - if the fault value is listed under cause: Check the crosswise-compared parameters to which the fault value refers.
 - copy the safety parameters.
 - carry out a POWER ON (power off/on) for all components.
 - upgrade the Motor Module software.
 - upgrade the Control Unit software.
 - correction of the encoder evaluation. The actual values differ as a result of mechanical faults (V belts, travel to a mechanical endstop, wear and window setting that is too narrow, encoder fault, ...).
 Re fault value = 1000:
 - investigate the signal associated with the safety-relevant input (contact problems).
 Re fault value = 1001:
 - carry out a POWER ON (power off/on) for all components.
 - upgrade the Motor Module software.
 - upgrade the Control Unit software.

Re fault value = 1005:

- check the conditions for pulse enable.

Re fault value = 1011:

- for diagnostics, refer to parameter (r9571).

Re fault value = 1012:

- upgrade the Sensor Module software.
- check the encoder parameters in the encoder systems for equality (p9515/p9315, p9519/p9319, p9523/p9323, p9524/p9324, p9525/p9325, p9529/p9329).

Re fault value = 1020, 1021:

- check the communication link.
- increase the monitoring cycle clock settings (p9500, p9511).
- carry out a POWER ON (power off/on) for all components.
- replace the hardware.

Re fault value = 5000, 5014, 5023, 5024, 5030, 5031, 5032, 5042, 5043, 5052, 5053, 5068, 5072, 5073, 5082 ... 5087, 5090, 5091, 5122 ... 5125, 5132 ... 5135, 5140:

- carry out a POWER ON (power off/on) for all components.
- check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified.
- upgrade firmware to later version.
- contact the Hotline.
- replace the Control Unit.

Re fault value = 5012:

- check the setting of the PROFIsafe address of the Control Unit (p9610) and that of the Motor Module (p9810). It is not permissible for the PROFIsafe address to be 0 or FFFF!

Re fault value = 5013, 5025:

- carry out a POWER ON (power off/on) for all components.
- check the setting of the PROFIsafe address of the Control Unit (p9610) and that of the Motor Module (p9810).
- check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified.

Re fault value = 5022:

- check the setting of the values of the F parameters at the PROFIsafe slave (F_SIL, F_CRC_Length, F_Par_Version, F_Source_Add, F_Dest_add, F_WD_Time).

Re fault value = 5026:

- check the settings of the values of the F parameters and the F parameter CRC (CRC1) calculated from these at the PROFIsafe slave and update.

Re fault value = 5065:

- check the configuration and communication at the PROFIsafe slave (cons. No. / CRC).
- check the setting of the value for F parameter F_WD_Time on the PROFIsafe slave and increase if necessary.
- check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified.

Re fault value = 5066:

- check the setting of the value for F parameter F_WD_Time on the PROFIsafe slave and increase if necessary.
- evaluate diagnostic information in the F host.
- check PROFIsafe connection.

Re fault value = 6000:

- carry out a POWER ON (power off/on) for all components.
- check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified.
- increase the monitoring cycle clock settings (p9500, p9511).
- upgrade firmware to later version.
- contact the Hotline.
- replace the Control Unit.

Re fault value = 6064:

- check the setting of the value in the F parameter F_Dest_Add at the PROFIsafe slave.
- check the setting of the PROFIsafe address of the Control Unit (p9610) and that of the Motor Module (p9810).

Re fault value = 6065:

- check the setting of the value in the F parameter F_Dest_Add at the PROFIsafe slave. It is not permissible for the destination address to be either 0 or FFFF!

Re fault value = 6066:

- check the setting of the value in the F parameter F_Source_Add at the PROFIsafe slave. It is not permissible for the source address to be either 0 or FFFF!

Re fault value = 6067:

- check the setting of the value in the F parameter F_WD_Time at the PROFIsafe slave. It is not permissible for the watch time to be 0!

Re fault value = 6068:
- check the setting of the value in the F parameter F_SIL at the PROFIsafe slave. The SIL level must correspond to SIL2!

Re fault value = 6069:
- check the setting of the value in the F parameter F_CRC_Length at the PROFIsafe slave. The setting of the CRC2 length is 2-byte CRC in the V1 mode and 3-byte CRC in the V2 mode!

Re fault value = 6070:
- check the setting of the value in the F parameter F_Par_Version at the PROFIsafe slave. The value for the F parameter version is 0 in the V1 mode and 1 in the V2 mode!

Re fault value = 6071:
- check the settings of the values of the F parameters and the F parameter CRC (CRC1) calculated from these at the PROFIsafe slave and, if required, update.

Re fault value = 6072:
- check the settings of the values for the F parameters and, if required, correct.
The following combinations are permissible for F parameters F_CRC_Length and F_Par_Version:
F_CRC_Length = 2-byte CRC and F_Par_Version = 0
F_CRC_Length = 3-byte CRC and F_Par_Version = 1

Re fault value = 6165:
- check the configuration and communication at the PROFIsafe slave.
- check the setting of the value for F parameter F_WD_Time on the PROFIsafe slave and increase if necessary.
- check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified.

Re fault value = 6166:
- check the configuration and communication at the PROFIsafe slave.
- check the setting of the value for F parameter F_WD_Time on the PROFIsafe slave and increase if necessary.
- evaluate diagnostic information in the F host.
- check PROFIsafe connection.
This message can be acknowledged as follows:
- motion monitoring functions integrated in the drive: Via Terminal Module 54F (TM54F) or PROFIsafe
- motion monitoring functions with SINUMERIK: Via the machine control panel
See also: p9300 (SI Motion monitoring clock cycle (Motor Module)), p9500 (SI Motion monitoring clock cycle (Control Unit))

C01712	SI Motion CU: Defect in F-IO processing
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	When cross checking and comparing the two monitoring channels, the drive detected a difference between parameters or results of the F-IO processing and initiated a STOP F. One of the monitoring functions no longer reliably functions - i.e. safe operation is no longer possible. The safety message C01711 with message value 0 is also displayed due to initiation of STOP F. If at least one monitoring function is active, the safety message C01701 "SI Motion: STOP B initiated" is output after the parameterized timer has expired. Message value (r9749, interpret decimal): Number of the cross-checked data that resulted in this message. 1: SI discrepancy monitoring time inputs (p10002, p10102). 2: SI acknowledgement internal event input terminal (p10006, p10106). 3: SI STO input terminal (p10022, p10122). 4: SI SS1 input terminal (p10023, p10123). 5: SI SS2 input terminal (p10024, p10124). 6: SI SOS input terminal (p10025, p10125). 7: SI SLS input terminal (p10026, p10126). 8: SI SLS_Limit(1) input terminal (p10027, p10127). 9: SI SLS_Limit(2) input terminal (p10028, p10128). 10: SI Safe State signal selection (p10039, p10139). 11 SI F-DI input mode (p10040, p10140). 12: SI F-DO 0 signal sources (p10042, p10142). 13: Different states for static inactive signal sources (p10006, p10022 ... p10028). 14: SI discrepancy monitoring time outputs (p10002, p10102). 15: SI acknowledgment internal event (p10006, p10106). 16: SI test sensor feedback signal test mode selected for test stop (p10046, p10146, p10047, p10147).

17: SI wait time for test stop at DOs (p10001).
 18 ... 25: SI test sensor feedback signal (p10046, p10146, p10047, p10147). Expected state of internal readback signal, generated from the selected test stop mode.
 26 ... 33: SI test sensor feedback signal (p10046, p10146, p10047, p10147). Expected state of external readback signal, generated from the selected test stop mode.
 34 ... 41: SI test sensor feedback signal (p10046, p10146, p10047, p10147). Expected state of second internal readback signal, generated from the selected test stop mode.
 42: Internal data for processing the second internal readback signal, generated from the selected test stop mode (p10047, p10147).
 43: Internal data for processing the internal readback signal, generated from the selected test stop mode (p10047, p10147).
 44: Internal data for processing the external readback signal, generated from the selected test stop mode (p10047, p10147).
 45: Internal data for initialization state of test stop mode, dependent upon test stop parameters.
 46: SI digital inputs debounce time (p10017, p10117)
 47: Selection F-DI for PROFIsafe (p10050, p10150)
 48: Selection F-DI for PROFIsafe (p10050, p10150)

Remedy:

- check parameterization in the parameters involved and correct if required.
- ensure equality by copying the SI data to the second channel and then carry out an acceptance test.
- check monitoring clock cycle in p9500 and p9300 for equality.

Note:

This message can be acknowledged via F-DI or PROFIsafe.

See also: p9300 (SI Motion monitoring clock cycle (Motor Module)), p9500 (SI Motion monitoring clock cycle (Control Unit))

C01714 SI Motion CU: Safely-Limited Speed exceeded

Message value: %1

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive has moved faster than that specified by the velocity limit value (p9531). The drive is stopped as a result of the configured stop response (p9563).

Message value (r9749, interpret decimal):

100: SLS1 exceeded.

200: SLS2 exceeded.

300: SLS3 exceeded.

400: SLS4 exceeded.

1000: Encoder limit frequency exceeded.

Remedy:

- check the traversing/motion program in the control.
- check the limits for "Safely-Limited Speed (SLS) and if required, adapt (p9531).

This message can be acknowledged as follows:

- motion monitoring functions integrated in the drive: Via Terminal Module 54F (TM54F) or PROFIsafe

- motion monitoring functions with SINUMERIK: Via the machine control panel

Note:

SI: Safety Integrated

SLS: Safely-Limited Speed / SG: Safely reduced speed

See also: p9531 (SI Motion SLS (SG) limit values (Control Unit)), p9563 (SI Motion SLS (SG)-specific stop response (Control Unit))

C01745 SI Motion CU: Checking braking torque for the brake test

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: POWER ON (IMMEDIATELY)

Cause: The scaling of the brake torque for the brake test can be changed using parameter p2003.

An acceptance test must be carried out again for the braking test. This determines whether the braking test is still carried out with the correct braking torque.

Remedy:

- carry out a POWER ON (power off/on) for all components.
- repeat the acceptance test for the safe brake test if the brake test is used.

See also: p2003

C01750 SI Motion CU: Hardware fault safety-relevant encoder

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The encoder that is used for the safety-relevant motion monitoring functions signals a hardware fault.
Message value (r9749, interpret decimal):
Encoder status word 1, encoder status word 2 that resulted in the message.
Remedy:
- check the encoder connection.
- replace the encoder.
This message can be acknowledged as follows:
- motion monitoring functions integrated in the drive: Via Terminal Module 54F (TM54F) or PROFIsafe
- motion monitoring functions with SINUMERIK: Via the machine control panel.

C01751 SI Motion CU: Effectivity test error safety-relevant encoder

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The DRIVE-CLiQ encoder for safe motion monitoring signals an error for the effectivity tests.
Message value (r9749, interpret decimal):
Only for internal Siemens troubleshooting.
Remedy:
- check the encoder connection.
- replace the encoder.
This message can be acknowledged as follows:
- motion monitoring functions integrated in the drive: Via Terminal Module 54F (TM54F) or PROFIsafe
- motion monitoring functions with SINUMERIK: Via the machine control panel

C01770 SI Motion CU: Discrepancy error of the fail-safe inputs or outputs

Message value: %1
Drive object: SERVO
Reaction: NONE
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The safety input terminals or output terminals show a different state longer than that parameterized in p10002 (or p10102).
Fault value (r0949, interpret hexadecimal):
yyyyxxxx hex
xxxx:
The safety-relevant input terminals F-DI indicate a discrepancy.
Bit 0: Discrepancy for F-DI 0
Bit 1: Discrepancy for F-DI 1
...
yyyy:
The safety-relevant output terminals F-DO indicate a discrepancy.
Bit 0: Discrepancy for F-DO 0
...
Note:
If several discrepancy errors occur consecutively, then this fault is only signaled for the first error that occurs.

Remedy:

- check the wiring of the F-DI (contact problems).
- carry out safe acknowledgment (p10006 or INTERNAL EVENT ACKNOWLEDGE via PROFIsafe).

Note:
 Discrepancy error of an F-DI can only be completely acknowledged if safe acknowledgement was carried out once the cause of the error was resolved (p10006). As long as safety acknowledgement was not carried out, the corresponding F-DI stays in the safe state.
 F-DI: Failsafe Digital Input
 F-DO: Failsafe Digital Output
 Where switching operations recur cyclically on the FDIs, the discrepancy time must be parameterized as follows:
 td = possible actual discrepancy time (in ms) that can occur with a switching operation. This must correspond to at least 1 SI sampling cycle (see p10000).
 tp = period for a switching operation in ms.
 The following rules must be adhered to:
 $p10002 < (tp/2) - td$ (discrepancy time must be less than half the period minus the actual discrepancy time)
 $p10002 \geq p10000$ (discrepancy time must be no less than P10000)
 $p10002 > td$ (discrepancy time must be greater than the switch discrepancy time which may actually apply)
 Example: If SI sampling cycle is 12 ms and switching frequency is 110 ms, the maximum discrepancy time which can be set is as follows:
 $p10002 \leq 110ms/2 - 12 ms = 43 ms$; this rounds off to $P10002 \leq 36 ms$ (Since the sampling time can only be accepted as a whole SI sampling cycle, the value will need to be rounded up or down to a whole SI sampling time value if it is not an exact multiple of an SI sampling cycle.)

A01772 Test stop fail-safe inputs/outputs active

Message value: -
Drive object: SERVO
Reaction: NONE
Acknowledge: NONE
Cause: The test stop for the fail-safe inputs and/or outputs is currently in progress.
 F1773 is output when a error occurs during the test stop.
Remedy: The alarm disappears automatically after successfully ending or canceling (when a fault condition occurs) the test stop.

F01773 Test stop error

Message value: %1
Drive object: SERVO
Reaction: NONE
Acknowledge: IMMEDIATELY (POWER ON)
Cause: A fault has occurred on the CU side during the test stop for the fail-safe outputs.
 The fault value is output in 0xRRRVWXYZ HEX format.
 The wildcards V,W,X must be interpreted as hexadecimal values and evaluated as binary values.
 Bit0 stands for DO-0, Bit1 for DO-1, Bit2 for DO-2, and Bit3 for DO-4.
 The wildcards Z and Y must be evaluated as hexadecimal values.
 The meaning of the wildcards is as follows:
 R: Reserved
 V: Actual state of the DO channel concerned (see X) on the CU (corresponds to the states read back from the hardware)
 W: Required state of the DO channel concerned (see X)
 X: DO channels concerned which are indicating a fault
 Y:: Reason for the test stop fault
 Z: State of the test stop in which the fault has occurred
 Reason for the test stop fault (Y):
 1=MM side in incorrect test stop state (internal fault)
 2=States of the DO(s) (CU305: readback via DI 22/CU240: readback via DI 2) not as expected
 3=Incorrect timer state on CU side (internal fault)
 4=States of the Diag-DO(s) (CU305: internal readback on the Motor Module channel) not as expected
 5=States of the second Diag-DO(s) (CU305: internal readback on the CU channel) not as expected.
 X and V indicate the DI or Diag-DO state dependent upon the reason for the fault (2, 4 or 5).
 In the event of multiple test stop faults, the first one that occurred is shown.

Test stop state (Z) and associated test actions:

0 to 3 Synchronization phase of test stop between CU and Motor Module no switching operations

4 DO + OFF and DO - OFF

5 Check to see if states are as expected

6 DO + ON and DO - ON

7 Check to see if states are as expected

8 DO + OFF and DO - ON

9 Check to see if states are as expected

10 DO + ON and DO - OFF

11 Check to see if states are as expected

12 DO + OFF and DO - OFF

13 Check to see if states are as expected

14 End of test stop

Diag expected states in table format:

Test stop state: Expectation Mode 1 / Mode 2 / Mode 3 / Mode 4

5 :0/-/-1

7 :0/-/-0

8 :0/-/-0

11 :1/-/-1

13 :0/-/-1

Second Diag expected states in table format:

Test stop state: Expectation Mode 1 / Mode 2 / Mode 3 / Mode 4

5 :-/-/-1

7 :-/-/-0

8 :-/-/-1

11 :-/-/-0

13 :-/-/-1

DI expected states in table format:

Test stop state: Expectation Mode 1 / Mode 2 / Mode 3 / Mode 4

5: -/1/0/-

7: -/0/1/-

8: -/0/0/-

11:-/0/0/-

13:-/1/0/-

Example: Alarm 1773 (CU) is signaled with fault value 0001_0127 and alarm 30773 (MM) is signaled with fault value 0000_0127. This means: In state 7 (Z=7) the state of the external readback signal was not set correctly (Y=2) after DO-0 (X=1) was switched to ON/ON. Fault value 0001_0127 indicates that 0 was expected (W=0) and 1 (V=1) was read back from the hardware. Fault value 0000_0127 on the MM indicates that the states were as expected. In the case of alarm 30773 W and V are always identical; a value of 0 always means that 0 was expected at the readback input but was not present on the other channel (CU).

Remedy:

Check the wiring of the F-DOs and restart the test stop. The fault is withdrawn if the test stop is successfully completed.

In the event of multiple test stop faults, the first one that occurred is shown. Once the test stop has been restarted the next queued test stop fault will be signaled (if there is one).

A01774 Test stop required

Message value: -

Drive object: SERVO

Reaction: NONE

Acknowledge: NONE

Cause:

- after powering up the drive, a test stop has still not been carried out.
- a new test stop is required after commissioning.
- the time to carry out the forced checking procedure (test stop) has expired (p10003).

Remedy: Initiate test stop (BI: p10007).

A01796 (F, N)	SI Motion CU: Wait for communication
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The drive waits for communication to be established with SINUMERIK or TM54F to execute the safety-relevant motion monitoring functions. Note: In this state, the pulses are safely suppressed.
Remedy:	If, after a longer period of time, the message is not automatically withdrawn, the following checks have to be made as appropriate: For communication with SINUMERIK, the following applies: - check any other PROFIBUS messages/signals present and remove their cause. - check that assignment of the axes on the higher-level control to the drives in the drive unit is correct. - check enable signal of the safety-relevant motion monitoring functions for the corresponding axis on the higher-level control and if required, set it. For communication with TM54F, the following applies: - check any other messages/signals present for DRIVE-CLiQ communication with the TM54F and remove their cause. - check the setting of p10010. All the drive objects controlled by the TM54F must be listed. See also: p9601 (SI enable, functions integrated in the drive (Control Unit)), p9801 (SI enable, functions integrated in the drive (Motor Module)), p10010 (SI drive object assignment)
Reaction upon F:	NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
C01798	SI Motion CU: Test stop running
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The test stop is active.
Remedy:	None necessary. The message is withdrawn when the test stop is finished. Note: SI: Safety Integrated
C01799	SI Motion CU: Acceptance test mode active
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The acceptance test mode is active. The POWER ON signals of the safety-relevant motion monitoring functions can be acknowledged during the acceptance test using the RESET button of the higher-level control.
Remedy:	None necessary. The message is withdrawn when exiting the acceptance test mode. Note: SI: Safety Integrated

F01800	DRIVE-CLiQ: Hardware/configuration error
Message value:	%1
Drive object:	All objects
Reaction:	Infeed: NONE (OFF1, OFF2) Servo: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2) Vector: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	A DRIVE-CLiQ connection fault has occurred. Fault value (r0949, interpret decimal): 100 ... 107: Communication via DRIVE-CLiQ socket X100 ... X107 has not been switched to cyclic operation. The cause may be an incorrect structure or a configuration that results in an impossible bus timing. 10: Loss of the DRIVE-CLiQ connection. The cause may be, for example, that the DRIVE-CLiQ cable was withdrawn from the Control Unit or as a result of a short-circuit for motors with DRIVE-CLiQ. This fault can only be acknowledged in cyclic communication. 11: Repeated faults when detecting the connection. This fault can only be acknowledged in cyclic communication. 12: A connection was detected but the node ID exchange mechanism does not function. The reason is probably that the component is defective. This fault can only be acknowledged in cyclic communication.
Remedy:	Re fault value = 100 ... 107: - ensure that the DRIVE-CLiQ components have the same firmware releases. - avoid longer topologies for short current controller clock cycles. Re fault value = 10: - check the DRIVE-CLiQ cables at the Control Unit. - remove any short-circuit for motors with DRIVE-CLiQ. - carry out a POWER ON. Re fault value = 11: - check the electrical cabinet design and cable routing for EMC compliance Re fault value = 12: - replace the component involved.

A01840	SMI: Component found without motor data
Message value:	Component number: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	An SMI/DQI without motor data has been found (e.g. SMI installed as replacement part). Alarm value (r2124, interpret decimal): Component number from target topology.
Remedy:	1. Download the SMI/DQI data (motor/encoder data) from the backup again (p4690, p4691). 2. Carry out a POWER ON (power off/on) for this component. Note: DQI: DRIVE-CLiQ Sensor Integrated SMI: SINAMICS Sensor Module Integrated See also: p4690 (SMI spare part component number), p4691 (Save/download SMI spare part data)

A01900 (F)	PROFIBUS: Configuration telegram error
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	A PROFIBUS master attempts to establish a connection using an incorrect configuring telegram. Alarm value (r2124, interpret decimal): 50: Syntax error. 51: Connection established to more drive objects than configured in the device. The drive objects for process data exchange and their sequence are defined in p0978. 52: Too many PZD data words for output or input to a drive object. The number of possible PZD items in a drive object is determined by the number of indices in r2050/p2051 for PZD IF1, and in r8850/p8851 for PZD IF2. 53: Uneven number of bytes for input or output. 54: Cyclic operation not active. 55: Invalid operating state.
Remedy:	Check the bus configuration on the master and slave sides. Re alarm value = 51: Check the list of the drive objects with process data exchange (p0978). With p0978[x] = 0, all of the following drive objects in the list are excluded from the process data exchange. Re alarm value = 52: Check the number of data words for output and input to a drive object.
Reaction upon F:	NONE (OFF1)
Acknowl. upon F:	IMMEDIATELY
A01901 (F)	PROFIBUS: Parameterizing telegram error
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	A PROFIBUS master attempts to establish a connection using an incorrect parameterizing telegram. Alarm value (r2124, interpret decimal): 1: Incorrect parameterizing bits. 10: Illegal length of an optional parameterizing block. 11: Illegal ID of an optional parameterizing block. 20: Double parameterizing block for clock synchronization. 21: Incorrect parameterizing block for clock synchronization. For more information, see A01902. 22: Incorrect parameterizing bits for clock synchronization. 23: Illegal clock synchronization for PZD interface 2. 30: Double parameterizing block for peer-to-peer data transfer. 31: Incorrect parameterizing block for peer-to-peer data transfer.
Remedy:	Check the bus configuration: - bus addresses - slave configuring
Reaction upon F:	NONE (OFF1)
Acknowl. upon F:	IMMEDIATELY

A01902	IF1: PB/PN clock cycle synchronous operation parameterization not permissible
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	Alarm value (r2124, interpret decimal): 0: Bus cycle time Tdp < 0.5 ms. 1: Bus cycle time Tdp > 32 ms. 2: Bus cycle time Tdp is not an integer multiple of the current controller clock cycle. 3: Instant of the actual value sensing Ti > Bus cycle time Tdp or Ti = 0. 4: Instant of the actual value sensing Ti is not an integer multiple of the current controller clock cycle. 5: Instant of the setpoint acceptance To >= Bus cycle time Tdp or To = 0. 6: Instant of the setpoint acceptance To is not an integer multiple of the current controller clock cycle. 7: Master application cycle time Tmapc is not an integer multiple of the speed controller clock cycle. 8: Bus reserve bus cycle time Tdp - data exchange time Tdx less than two current controller clock cycles. 10: Instant of the setpoint acceptance not To <= data exchange time Tdx + To_min. 11: Master application cycle time Tmapc > 14 or Tmapc = 0. 12: PLL tolerance window Tpll_w > Tpll_w_max. 13: Bus cycle time Tdp is not a multiple of all basic clock cycles p0110[x]. 14: For COMM BOARD with the setting To - 1 = Tdp - Ti, the instant of the setpoint acceptance is not To <= Data exchange time Tdx + 2 * To_min. 15: This configuration is not permitted for Tdp < 1 ms. 16: Instant of the actual value sensing Ti is less than the permitted value (COMM BOARD: Ti >= 2). 17: The setting (To + Ti = Tdp + 2) is not permitted for COMM BOARD.
Remedy:	- adapt the parameterizing telegram. - adapt the current and speed controller clock cycle. Re alarm value = 15: - check the number of specific drive object types in the configuration. Note: IF1: Interface 1 PB: PROFIBUS PN: PROFINET
A01903 (F)	COMM INT: Receive configuration data invalid
Message value:	%1
Drive object:	A_INF, B_INF, CU_CX32, CU_I, CU_LINK, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The drive unit did not accept the receive configuration data. Alarm value (r2124, interpret decimal): Return value of the receive configuration data check. 0: Configuration accepted. 1: Connection established to more drive objects than configured in the device. The drive objects for process data exchange and their sequence are defined in p0978. 2: Too many PZD data words for output or input to a drive object. The number of possible PZD items in a drive object is determined by the number of indices in r2050/p2051. 3: Uneven number of bytes for input or output. 4: Setting data for synchronization not accepted. For more information, see A01902. 5: Cyclic operation not active. 6: Buffer system not accepted. 7: Cyclic channel length too short for this setting. 8: Cyclic channel address not initialized. 9: 3-buffer system not permitted. 10: DRIVE-CLiQ fault. 11: CU-Link fault. 12: CX32 not in cyclic operation.

Remedy:	Check the receive configuration data. Re alarm value = 1: Check the list of the drive objects with process data exchange (p0978). With p0978[x] = 0, all of the following drive objects in the list are excluded from the process data exchange. Re alarm value = 2: Check the number of data words for output and input to a drive object.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY

F01910 (N, A)	PROFIBUS: Setpoint timeout
Message value:	-
Drive object:	All objects
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF3 (IASC/DCBRAKE, NONE, OFF1, OFF2, STOP1, STOP2) Vector: OFF3 (IASC/DCBRAKE, NONE, OFF1, OFF2, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	The receipt of setpoints from the PROFIBUS interface is interrupted because the bus connection is interrupted or the PROFIBUS master is switched off or was set to the STOP state. See also: p2047 (PROFIBUS additional monitoring time)
Remedy:	Restore the bus connection and set the PROFIBUS master to RUN. Slave redundancy: For operation on a Y link, it must be ensured that "DP alarm mode = DPV1" is set in the slave parameterization. See also: p2047 (PROFIBUS additional monitoring time)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F01911	IF1: PB/PN clock cycle synchronous operation clock cycle failure
Message value:	-
Drive object:	All objects
Reaction:	Infeed: OFF1 Servo: OFF1 (OFF3) Vector: OFF1 (OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The global control telegram to synchronize the clock cycles has failed - in cyclic operation - for several DP clock cycles or has violated the time grid specified in the parameterizing telegram over several consecutive DP clock cycles (refer to the bus cycle time, Tdp and Tpllw).
Remedy:	- check the PROFIBUS cables and connectors. - check whether communication was briefly or permanently interrupted. - check the bus and master for utilization level (e.g. bus cycle time Tdp was set too short). Note: IF1: Interface 1 PB: PROFIBUS PN: PROFINET

F01912	IF1: PB/PN clock cycle synchronous operation sign-of-life failure
Message value:	-
Drive object:	All objects
Reaction:	Infeed: OFF1 Servo: OFF1 (OFF3) Vector: OFF1 (OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The maximum permissible number of errors in the master sign-of-life (clock synchronous operation) has been exceeded in cyclic operation.

- Remedy:**
- check the physical bus configuration (terminating resistor, shielding, etc.).
 - correct the interconnection of the master sign-of-life (p2045).
 - check whether the master correctly sends the sign-of-life (e.g. create a trace with STW2.12 ... STW2.15 and trigger signal ZSW1.3).
 - check the permissible telegram failure rate (p0925).
 - check the bus and master for utilization level (e.g. bus cycle time Tdp was set too short).

Note:

IF1: Interface 1

PB: PROFIBUS

PN: PROFINET

F01913 (N, A)	COMM INT: Monitoring time sign-of-life expired
Message value:	-
Drive object:	A_INF, B_INF, CU_CX32, CU_I, CU_LINK, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: OFF1 (NONE, OFF2, OFF3) Vector: OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The monitoring time for the sign-of-life counter has expired. The connection between the drive and the higher-level control (SIMOTION, SINUMERIK) has been interrupted for the following reasons: - the control was reset. - the data transfer to the control was interrupted.
Remedy:	- wait until the control has re-booted. - restore data transfer to the control.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F01914 (N, A)	COMM INT: Monitoring time configuration expired
Message value:	%1
Drive object:	A_INF, B_INF, CU_CX32, CU_I, CU_LINK, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: OFF1 (NONE, OFF2, OFF3) Vector: OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The monitoring time for the configuration has expired. Fault value (r0949, interpret decimal): 0: The transfer time of the send configuration data has been exceeded. 1: The transfer time of the receive configuration data has been exceeded.
Remedy:	- acknowledge faults that are present. - carry out a POWER ON (power off/on) for all components. - upgrade firmware to later version. - contact the Hotline.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F01915	IF1: PB/PN clock cycle synchronous operation sign-of-life failure drive object 1
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	Group display for problems with the sign-of-life of the master (clock-cycle synchronous operation) on the drive object 1 (Control Unit). For central measurements, synchronism with the central master is lost.
Remedy:	Note: IF1: Interface 1 PB: PROFIBUS PN: PROFINET
A01920 (F)	PROFIBUS: Interruption cyclic connection
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The cyclic connection to the PROFIBUS master is interrupted.
Remedy:	Establish the PROFIBUS connection and activate the PROFIBUS master in the cyclic mode.
Reaction upon F:	NONE (OFF1)
Acknowl. upon F:	IMMEDIATELY
A01921 (F)	PROFIBUS: Receive setpoints after To
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	Output data of PROFIBUS master (setpoints) received at the incorrect instant in time within the PROFIBUS clock cycle.
Remedy:	- check bus configuration. - check parameters for clock cycle synchronization (ensure To > Tdx). Note: To: Time of setpoint acceptance Tdx: Data exchange time
Reaction upon F:	NONE (OFF1)
Acknowl. upon F:	IMMEDIATELY
A01930	IF1: PB/PN current controller clock cycle clock cycle synchronous not equal
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The current controller clock cycle of all drives must be set the same for the clock cycle synchronous operation. Alarm value (r2124, interpret decimal): Number of the drive object with different current controller clock cycle.
Remedy:	Set current controller clock cycles to identical values (p0115[0]). Note: IF1: Interface 1 PB: PROFIBUS PN: PROFINET See also: p0115

A01931	IF1: PB/PN speed controller clock cycle clock cycle synchronous not equal
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The speed controller clock cycle of all drives must be set the same for the clock cycle synchronous operation. Alarm value (r2124, interpret decimal): Number of the drive object with the different speed controller clock cycle.
Remedy:	Set the speed controller clock cycles the same (p0115[1]). Note: IF1: Interface 1 PB: PROFIBUS PN: PROFINET See also: p0115

A01932	IF1: PB/PN clock cycle synchronization missing for DSC
Message value:	-
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	There is no clock cycle synchronization and DSC is selected. Note: DSC: Dynamic Servo Control
Remedy:	Set the clock cycle synchronization when configuring the bus.

A01940	IF1: PB/PN clock cycle synchronism not reached
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The bus is in the data exchange state and clock synchronous operation has been selected using the parameterizing telegram. It was not possible to synchronize to the clock cycle specified by the master. - the master does not send a clock synchronous global control telegram although clock synchronous operation was selected when configuring the bus. - the master is using another clock synchronous DP clock cycle than was transferred to the slave in the parameterizing telegram. - at least one drive object has a pulse enable (not controlled from PROFIBUS/PROFINET either).
Remedy:	- check the master application and bus configuration. - check the consistency between the clock cycle input when configuring the slave and clock cycle setting at the master. - check that no drive object has a pulse enable. Only enable the pulses after synchronizing the PROFIBUS/PROFINET drives. Note: IF1: Interface 1 PB: PROFIBUS PN: PROFINET

A01941	IF1: PB/PN clock cycle signal missing when establishing bus communication
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The bus is in the data exchange state and clock synchronous operation has been selected using the parameterizing telegram. The global control telegram for synchronization is not being received.

Remedy: Check the master application and bus configuration.
Note:
 IF1: Interface 1
 PB: PROFIBUS
 PN: PROFINET

A01943 IF1: PB/PN clock cycle signal error when establishing bus communication

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The bus is in the data exchange state and clock synchronous operation has been selected using the parameterizing telegram. The global control telegram for synchronization is being irregularly received.
 - the master is sending an irregular global control telegram.
 - the master is using another clock synchronous DP clock cycle than was transferred to the slave in the parameterizing telegram.
Remedy:
 - check the master application and bus configuration.
 - check the consistency between the clock cycle input when configuring the slave and clock cycle setting at the master.
Note:
 IF1: Interface 1
 PB: PROFIBUS
 PN: PROFINET

A01944 IF1: PB/PN sign-of-life synchronism not reached

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The bus is in the data exchange state and clock synchronous operation has been selected using the parameterizing telegram. Synchronization with the master sign-of-life (STW2.12 ... STW2.15) could not be completed because the sign-of-life is changing differently to how it was configured in the Tmapc time grid.
Remedy:
 - ensure that the master correctly increments the sign-of-life in the master application clock cycle Tmapc.
 - correct the interconnection of the master sign-of-life (p2045).
Note:
 IF1: Interface 1
 PB: PROFIBUS
 PN: PROFINET

A01945 PROFIBUS: Connection to the Publisher failed

Message value: Fault cause: %1 bin
Drive object: A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: For PROFIBUS peer-to-peer data transfer, the connection to at least one Publisher has failed.
 Alarm value (r2124, interpret binary):
 Bit 0 = 1: Publisher with address in r2077[0], connection failed.
 ...
 Bit 15 = 1: Publisher with address in r2077[15], connection failed.
Remedy:
 - check the PROFIBUS cables.
 - carry out a first commissioning of the Publisher that has the failed connection.
 See also: r2077 (PROFIBUS diagnostics peer-to-peer data transfer addresses)

F01946 (A)	PROFIBUS: Connection to the Publisher aborted
Message value:	Fault cause: %1 bin
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: OFF1 (NONE, OFF2, OFF3) Vector: OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	At this drive object, the connection to at least one Publisher for PROFIBUS peer-to-peer data transfer in cyclic operation has been aborted. Alarm value (r2124, interpret binary): Bit 0 = 1: Publisher with address in r2077[0], connection aborted. ... Bit 15 = 1: Publisher with address in r2077[15], connection aborted.
Remedy:	- check the PROFIBUS cables. - check the state of the Publisher that has the aborted connection. See also: r2077 (PROFIBUS diagnostics peer-to-peer data transfer addresses)
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F01950 (N, A)	IF1: PB/PN clock cycle synchronous operation synchronization unsuccessful
Message value:	-
Drive object:	All objects
Reaction:	OFF1 (NONE)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	Synchronization of the internal clock cycle to the global control telegram has failed. The internal clock cycle exhibits an unexpected shift.
Remedy:	Only for internal Siemens troubleshooting. Note: IF1: Interface 1 PB: PROFIBUS PN: PROFINET
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F01951	CU DRIVE-CLiQ: Synchronization application clock cycle missing
Message value:	%1
Drive object:	All objects
Reaction:	OFF2 (NONE)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	If DRIVE-CLiQ components with different application clock cycle are operated on a DRIVE-CLiQ port, this requires synchronization with the Control Unit. This synchronization routine was unsuccessful. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade the software of the DRIVE-CLiQ components. - upgrade the Control Unit software.

F01952	CU DRIVE-CLiQ: Synchronization of component not supported
Message value:	%1
Drive object:	All objects
Reaction:	OFF2 (NONE)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The existing system configuration requires that the connected DRIVE-CLiQ components support the synchronization between the basic clock cycle, DRIVE-CLiQ clock cycle and the application clock cycle. However, not all DRIVE-CLiQ components have this functionality. Fault value (r0949, interpret decimal): Component number of the first faulty DRIVE-CLiQ component.
Remedy:	Upgrade the firmware of the component specified in the fault value. Note: If required, also upgrade additional components in the DRIVE-CLiQ line.
A01953	CU DRIVE-CLiQ: Synchronization not completed
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	After the drive system is powered up, the synchronization between the basic clock cycle, DRIVE-CLiQ clock cycle and application clock cycle was started but was not completed within the selected time tolerance. Alarm value (r2124, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	Carry out a POWER ON (power off/on) for all components. If the error occurs after the drive sampling times were adjusted, and if a Terminal Module 31 (TM31) is being used, the sampling times (p0115, p4099) should be set as integer multiples to the drive clock cycles (p0115).
F01954	CU DRIVE-CLiQ: Synchronization unsuccessful
Message value:	%1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	After the drive system is powered up, the synchronization between the basic clock cycle, DRIVE-CLiQ clock cycle and application clock cycle was started and was not able to be successfully completed. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	1. Ensure perfect functioning of the DRIVE-CLiQ. 2. Initiate a new synchronization, e.g. as follows: - remove the PROFIBUS master and re-insert again. - restart the PROFIBUS master. - power down the Control Unit and power it up again. - press the Control Unit reset button. - reset the parameter and download the saved parameters (p0009 = 30, p0976 = 2).
A01955	CU DRIVE-CLiQ: Synchronization DO not completed
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	After the drive system is powered up, the synchronization between the basic clock cycle, DRIVE-CLiQ clock cycle and application clock cycle was started but was not completed within the selected time tolerance. Alarm value (r2124, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	Carry out a POWER ON (power off/on) for all components of the DO.

A01990 (F)	USS: PZD configuration error
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The configuration of the process data (PZD) for the USS protocol is incorrect. Alarm value (r2124, interpret decimal): 2: PZD amount (p2022) too great for the first drive object (p978[0]). The number of possible PZD items in a drive object is determined by the number of indices in r2050/p2051 for PZD IF1, and in r8850/p8851 for PZD IF2.
Remedy:	Re alarm value = 2: Check the amount of USS PZD (p2022) and the maximum PZD amount (r2050/p2051) for the first drive object (p0978[0]).
Reaction upon F:	NONE (OFF1)
Acknowl. upon F:	IMMEDIATELY
A02000	Function generator: Start not possible
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The function generator has already been started.
Remedy:	Stop the function generator and restart again if necessary. Note: The alarm is reset as follows: - remove the cause of this alarm. - restart the function generator. See also: p4800 (Function generator control)
A02005	Function generator: Drive does not exist
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The drive object specified for connection does not exist. See also: p4815 (Function generator drive number)
Remedy:	Use the existing drive object with the corresponding number. Note: The alarm is reset as follows: - remove the cause of this alarm. - restart the function generator. See also: p4815 (Function generator drive number)
A02006	Function generator: No drive specified for connection
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	No drive specified for connection in p4815. See also: p4815 (Function generator drive number)
Remedy:	At least one drive to be connected must be specified in p4815. Note: The alarm is reset as follows: - remove the cause of this alarm. - restart the function generator. See also: p4815 (Function generator drive number)

A02007	Function generator: Drive not SERVO / VECTOR
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The drive object specified for connection is not a SERVO / VECTOR. See also: p4815 (Function generator drive number)
Remedy:	Use a SERVO / VECTOR drive object with the corresponding number. Note: The alarm is reset as follows: - remove the cause of this alarm. - restart the function generator.
A02008	Function generator: Drive specified a multiple number of times
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The drive object specified for connection is already specified. Alarm value (r2124, interpret decimal): Drive object number of the drive object that is specified a multiple number of times.
Remedy:	Specify a different drive object. Note: The alarm is reset as follows: - remove the cause of this alarm. - restart the function generator.
A02009	Function generator: Illegal mode
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The set operating mode (p1300) of the drive object is not permissible when using the function generator. Alarm value (r2124, interpret decimal): Number of the drive object involved.
Remedy:	Change the operating mode for this drive object to p1300 = 20 (encoderless speed control) or p1300 = 21 (speed control with encoder). Note: The alarm is reset as follows: - remove the cause of this alarm. - restart the function generator.
A02010	Function generator: Speed setpoint from the drive is not zero
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The speed setpoint of a drive selected for connection is greater than the value for the standstill detection set using p1226. Alarm value (r2124, interpret decimal): Number of the drive object involved.
Remedy:	For all of the drives specified for connection, set the speed setpoints to 0. Note: The alarm is reset as follows: - remove the cause of this alarm. - restart the function generator.

A02011	Function generator: The actual drive speed is not zero
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The speed actual value of a drive selected for connection is greater than the value for the standstill detection set using p1226. Alarm value (r2124, interpret decimal): Number of the drive object involved.
Remedy:	Set the relevant drives to zero speed before starting the function generator. Note: The alarm is reset as follows: - remove the cause of this alarm. - restart the function generator.
A02015	Function generator: Drive enable signals missing
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The master control and/or enable signals are missing to connect to the specified drive. Alarm value (r2124, interpret decimal): Number of the drive object involved. See also: p4815 (Function generator drive number)
Remedy:	Fetch the master control to the specified drive object and set all enable signals. Note: The alarm is reset as follows: - remove the cause of this alarm. - restart the function generator.
A02016	Function generator: Magnetizing running
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	Magnetizing has not yet been completed on a drive object specified for connection. Alarm value (r2124, interpret decimal): Number of the drive object involved. See also: p4815 (Function generator drive number)
Remedy:	Wait for magnetizing of the motor (r0056.4). Note: The alarm is reset as follows: - restart the function generator. See also: r0056 (Status word, closed-loop control)
A02020	Function generator: Parameter cannot be changed
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	This parameter setting cannot be changed when the function generator is active (p4800 = 1). See also: p4810, p4812, p4813, p4815, p4820, p4821, p4822, p4823, p4824, p4825, p4826, p4827, p4828, p4829

Remedy:

- stop the function generator before parameterizing (p4800 = 0).
- if required, start the function generator (p4800 = 1).

Note:
The alarm is reset as follows:

- remove the cause of this alarm.
- restart the function generator.

See also: p4800 (Function generator control)

A02025 Function generator: Period too short

Message value: -

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: The value for the period is too short.
See also: p4821 (Function generator period)

Remedy: Check and adapt the value for the period.

Note:
The alarm is reset as follows:

- remove the cause of this alarm.
- restart the function generator.

See also: p4821 (Function generator period)

A02026 Function generator: Pulse width too high

Message value: -

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: The selected pulse width is too high.
The pulse width must be less than the period duration.
See also: p4822 (Function generator pulse width)

Remedy: Reduce pulse width.

Note:
The alarm is reset as follows:

- remove the cause of this alarm.
- restart the function generator.

See also: p4821 (Function generator period), p4822 (Function generator pulse width)

A02030 Function generator: Physical address equals zero

Message value: -

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: The specified physical address is zero.
See also: p4812 (Function generator physical address)

Remedy: Set a physical address with a value other than zero.

Note:
The alarm is reset as follows:

- remove the cause of this alarm.
- restart the function generator.

See also: p4812 (Function generator physical address)

A02040 Function generator: Illegal value for offset

Message value: -

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: The value for the offset is higher than the value for the upper limit or lower than the value for the lower limit.
See also: p4826 (Function generator offset)

Remedy: Adjust the offset value accordingly.
Note:
The alarm is reset as follows:
- remove the cause of this alarm.
- restart the function generator.
See also: p4826 (Function generator offset), p4828 (Function generator lower limit), p4829 (Function generator upper limit)

A02041 Function generator: Illegal value for bandwidth

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The bandwidth referred to the time slice clock cycle of the function generator has either been set too low or too high.
Depending on the time slice clock cycle, the bandwidth is defined as follows:
 $\text{Bandwidth_max} = 1 / (2 * \text{time slice clock cycle})$
 $\text{Bandwidth_min} = \text{Bandwidth_max} / 100000$
Example:
Assumption: p4830 = 125 μs
--> $\text{Bandwidth_max} = 1 / (2 * 125 \mu\text{s}) = 4000 \text{ Hz}$
--> $\text{Bandwidth_min} = 4000 \text{ Hz} / 100000 = 0.04 \text{ Hz}$
Note:
p4823: Function generator bandwidth
p4830: Function generator time slice clock cycle
See also: p4823 (Function generator bandwidth), p4830 (Function generator time slice cycle)
Remedy: Check the value for the bandwidth and adapt accordingly.
Note:
The alarm is reset as follows:
- remove the cause of this alarm.
- restart the function generator.

A02047 Function generator: Time slice clock cycle invalid

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The time slice clock cycle selected does not match any of the existing time slices.
See also: p4830 (Function generator time slice cycle)
Remedy: Enter an existing time slice clock cycle. The existing time slices can be read out via p7901.
Note:
The alarm is reset as follows:
- remove the cause of this alarm.
- restart the function generator.
See also: r7901 (Sampling times)

A02050 Trace: Start not possible

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The trace has already been started.
See also: p4700 (Trace control)
Remedy: Stop the trace and, if necessary, start again.

A02055 Trace: Recording time too short

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The trace duration is too short.
The minimum is twice the value of the trace clock cycle.
See also: p4721 (Trace recording time)
Remedy: Check the selected recording time and, if necessary, adjust.

A02056 Trace: Recording cycle too short

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The selected recording cycle is shorter than the selected basic clock cycle 0 (p0110[0]).
See also: p4720 (Trace recording cycle)
Remedy: Increase the value for the trace cycle.

A02057 Trace: Time slice clock cycle invalid

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The time slice clock cycle selected does not match any of the existing time slices.
See also: p4723 (Time slice cycle for trace)
Remedy: Enter an existing time slice clock cycle. The existing time slices can be read out via p7901.
See also: r7901 (Sampling times)

A02058 Trace: Time slice clock cycle for endless trace not valid

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The selected time slice clock cycle cannot be used for the endless trace
See also: p4723 (Time slice cycle for trace)
Remedy: Enter the clock cycle of an existing time slice with a cycle time ≥ 2 ms for up to 4 recording channels or ≥ 4 ms from 5 recording channels per trace.
The existing time slices can be read out via p7901.
See also: r7901 (Sampling times)

A02059 Trace: Time slice clock cycle for 2 x 8 recording channels not valid

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The selected time slice clock cycle cannot be used for more than 4 recording channels.
See also: p4723 (Time slice cycle for trace)
Remedy: Enter the clock cycle of an existing time slice with a cycle time ≥ 4 ms or reduce the number of recording channels to 4 per trace.
The existing time slices can be read out via p7901.
See also: r7901 (Sampling times)

A02060	Trace: Signal to be traced missing
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	- a signal to be traced was not specified. - the specified signals are not valid. See also: p4730 (Trace record signal 0), p4731 (Trace record signal 1), p4732 (Trace record signal 2), p4733 (Trace record signal 3)
Remedy:	- specify the signal to be traced. - check whether the relevant signal can be traced.
A02061	Trace: Invalid signal
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	- the specified signal does not exist. - the specified signal can no longer be traced (recorded). See also: p4730 (Trace record signal 0), p4731 (Trace record signal 1), p4732 (Trace record signal 2), p4733 (Trace record signal 3)
Remedy:	- specify the signal to be traced. - check whether the relevant signal can be traced.
A02062	Trace: Invalid trigger signal
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	- a trigger signal was not specified. - the specified signal does not exist. - the specified signal is not a fixed-point signal. - the specified signal cannot be used as a trigger signal for the trace. See also: p4711 (Trace trigger signal)
Remedy:	Specify a valid trigger signal.
A02063	Trace: Invalid data type
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The specified data type to select a signal using a physical address is invalid. See also: p4711 (Trace trigger signal), p4730 (Trace record signal 0), p4731 (Trace record signal 1), p4732 (Trace record signal 2), p4733 (Trace record signal 3)
Remedy:	Use a valid data type.
A02070	Trace: Parameter cannot be changed
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The trace parameter settings cannot be changed when the trace is active. See also: p4700, p4710, p4711, p4712, p4713, p4714, p4715, p4716, p4720, p4721, p4722, p4730, p4731, p4732, p4733, p4780, p4781, p4782, p4783, p4789, p4795
Remedy:	- stop the trace before parameterization. - if required, start the trace.

A02075	Trace: Pretrigger time too long
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The selected pretrigger time must be shorter than the trace time. See also: p4721 (Trace recording time), p4722 (Trace trigger delay)
Remedy:	Check the pretrigger time setting and change if necessary.
F02080	Trace: Parameterization deleted due to unit changeover
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The trace parameterization in the drive unit was deleted due to a unit changeover or a change in the reference parameters.
Remedy:	Restart trace.
A02099	Trace: Insufficient Control Unit memory
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The memory space still available on the Control Unit is no longer sufficient for the trace function.
Remedy:	Reduce the memory required, e.g. as follows: - reduce the trace time. - increase the trace clock cycle. - reduce the number of signals to be traced. See also: r4708 (Trace memory space required), r4799 (Trace memory location free)
A02100	CU: Computing dead time current controller too short
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The value in p0118 produces a dead time of one clock cycle because it is prior to setpoint availability. A possible cause could be, for example, that the system characteristics no longer match those parameterized after a component has been replaced. Alarm value (r2134, floating point): The minimum value for p0118 where a dead time no longer occurs.
Remedy:	- set p0118 to a value greater than or equal to the alarm value. - set p0117 to an automatic setting. - check the firmware releases of the components involved. See also: p0117 (Current controller computing dead time mode), p0118 (Current controller computing dead time)
A02150	OA: Application cannot be loaded
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The system was not able to load an OA application. Alarm value (r2124, interpret hexadecimal): Only for internal Siemens troubleshooting.

Remedy:

- carry out a POWER ON (power off/on) for all components.
- upgrade firmware to later version.
- contact the Hotline.

Note:
OA: Open Architecture
See also: r4950, r4955, p4956, r4957

F02151 (A) OA: Internal software error

Message value: %1
Drive object: All objects
Reaction: Infeed: OFF2 (NONE, OFF1)
 Servo: OFF2 (NONE, OFF1, OFF3)
 Vector: OFF2 (NONE, OFF1, OFF3)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: An internal software error has occurred within an OA application.
 Fault value (r0949, interpret hexadecimal):
 Only for internal Siemens troubleshooting.
Remedy: - carry out a POWER ON (power off/on) for all components.
 - upgrade firmware to later version.
 - contact the Hotline.
 - replace the Control Unit.
Note:
OA: Open Architecture
See also: r4950, r4955, p4956, r4957
Reaction upon A: NONE
Acknowl. upon A: NONE

F02152 (A) OA: Insufficient memory

Message value: %1
Drive object: All objects
Reaction: OFF1
Acknowledge: IMMEDIATELY (POWER ON)
Cause: Too many functions have been configured on this Control Unit (e.g. too many drives, function modules, data sets, OA applications, blocks, etc).
 Fault value (r0949, interpret decimal):
 Only for internal Siemens troubleshooting.
Remedy: - change the configuration on this Control Unit (e.g. fewer drives, function modules, data sets, OA applications, blocks, etc).
 - use an additional Control Unit.
Note:
OA: Open Architecture
Reaction upon A: NONE
Acknowl. upon A: NONE

F03000 NVRAM fault on action

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A fault occurred during execution of action p7770 = 1, 2 for the NVRAM data.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex
 yy: Fault cause
 yy = 1:
 The action p7770 = 1 is not supported by this version if Drive Control Chart (DCC) is activated for the drive object concerned.
 yy = 2:
 The data length of the specified application is not the same in the NVRAM and the backup.
 yy = 3:
 The data checksum in p7774 is not correct.
 yy = 4:
 No data available to load.
 xx: Application ID
 See also: p7770 (NVRAM action)
Remedy:

F03001 NVRAM checksum incorrect

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A checksum error occurred when evaluating the non-volatile data (NVRAM) on the Control Unit.
 The NVRAM data affected was deleted.
Remedy: Carry out a POWER ON (power off/on) for all components.

F03500 (A) TM: Initialization

Message value: %1
Drive object: All objects
Reaction: OFF1 (OFF2)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: When initializing the Terminal Modules, the terminals of the Control Unit or the Terminal Board 30, an internal software error has occurred.
 Fault value (r0949, interpret decimal):
 The thousands digit = 1 ... 3:
 The component number (p0151) of the module involved is specified at the units, tens and hundreds digit.
Remedy:
 - power down the power supply for the Control Unit and power it up again.
 - check the DRIVE-CLiQ connection.
 - if required, replace the Terminal Module.
 The Terminal Module should be directly connected to a DRIVE-CLiQ socket of the Control Unit.
 If the fault occurs again, replace the Terminal Module.
 Reaction upon A: NONE
 Acknowl. upon A: NONE

A03501 TM: Sampling time change

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The sampling times of the inputs/outputs were changed.
 This change only becomes valid after the next boot.
Remedy: Carry out a POWER ON.

F03505 (N, A)	TM: Analog input wire breakage
Message value:	%1
Drive object:	A_INF, B_INF, CU_CX32, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL
Reaction:	OFF1 (NONE, OFF2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The input current of the Terminal Module analog input has exceeded the threshold value parameterized in p4061[x]. This fault can only occur if p4056[x] = 3 (4 ... 20 mA with monitoring) is set. Index x = 0: Analog input 0 (X522.1 to .3) Index x = 1: Analog input 1 (X522.4 to .5) Fault value (r0949, interpret decimal): The component number (p0151) of the module involved is specified at the units, tens and hundreds digit. The thousands digit specifies the analog input involved: 0: Analog input 0 (AI 0), 1: Analog input 1 (AI 1). The leading zero of analog input 0 (AI 0) is masked out of the fault value.
Remedy:	Check the connection to the signal source for interruptions. Check the magnitude of the injected current - it is possible that the infed signal is too low. Please note that the input has a load resistance of 250 Ohm. The input current measured by the Terminal Module can be read out from r4052[x].
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F03505 (N, A)	CU: Analog input wire breakage
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF1 (NONE, OFF2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The input current of the Terminal Module analog input has exceeded the threshold value parameterized in p4061[x]. This fault can only occur if p4056[x] = 3 (4 ... 20 mA with monitoring) is set. Index x = 0: Analog input 0 (X522.1 to .3) Index x = 1: Analog input 1 (X522.4 to .5) Fault value (r0949, interpret decimal): The component number (p0151) of the module involved is specified at the units, tens and hundreds digit. The thousands digit specifies the analog input involved: 0: Analog input 0 (AI 0), 1: Analog input 1 (AI 1). The leading zero of analog input 0 (AI 0) is masked out of the fault value.
Remedy:	Check the connection to the signal source for interruptions. Check the magnitude of the injected current - it is possible that the infed signal is too low. Please note that the input has a load resistance of 250 Ohm. The input current measured by the Terminal Module can be read out from r4052[x].
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A03506 (F, N)	24 V power supply missing
Message value:	%1
Drive object:	A_INF, B_INF, CU_I, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The 24 V power supply for the digital outputs (X124) is missing.
Remedy:	Check the terminals for the power supply voltage (X124, L1+, M).
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A03550	TM: Speed setpoint filter natural frequency > Shannon frequency
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The natural filter frequency of the speed setpoint filter (p1417) is greater than the Shannon frequency. The Shannon frequency is calculated according to the following formula: $0.5 / p0115[0]$ See also: p1417
Remedy:	Reduce the natural frequency of the speed setpoint filter (PT2 low pass) (p1417).
F03590 (N, A)	TM: Module not ready
Message value:	%1
Drive object:	All objects
Reaction:	Infeed: OFF2 (NONE) Servo: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2) Vector: OFF2 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The Terminal Module involved does not send a ready signal and no valid cyclic data. Fault value (r0949, interpret decimal): Drive object number of the Terminal Module involved.
Remedy:	- check the 24 V power supply. - check the DRIVE-CLiQ connection. - check whether the sampling time of the drive object involved is not equal to zero (p4099[0]).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A05000 (N)	Power unit: Overtemperature heat sink AC inverter
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The alarm threshold for overtemperature at the inverter heat sink has been reached. The response is set using p0290. If the temperature of the heat sink increases by an additional 5 K, then fault F30004 is initiated.
Remedy:	Check the following: - is the ambient temperature within the defined limit values? - have the load conditions and the load duty cycle been appropriately dimensioned? - has the cooling failed?
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A05001 (N)	Power unit: Chip overtemperature
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Alarm threshold for overtemperature of the power semiconductor in the AC converter has been reached. The response is set using p0290. If the chip temperature increases by an additional 15 K, then fault F30025 is triggered.

Remedy: Check the following:
- is the ambient temperature within the defined limit values?
- have the load conditions and the load duty cycle been appropriately dimensioned?
- has the cooling failed?
- pulse frequency too high?
See also: r0037, p0290 (Power unit overload response)

Reaction upon N: NONE

Acknowl. upon N: NONE

A05002 (N) Power unit: Air intake overtemperature

Message value: -

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The alarm threshold for the air intake overtemperature has been reached. For air-cooled power units, the threshold is 42 °C (hysteresis 2 K). The response is set using p0290.
If the air intake temperature increases by an additional 13 K, then fault F30035 is output.

Remedy: Check the following:
- is the ambient temperature within the defined limit values?
- has the fan failed? Check the direction of rotation.

Reaction upon N: NONE

Acknowl. upon N: NONE

A05003 (N) Power unit: Internal overtemperature

Message value: -

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The alarm threshold for internal overtemperature has been reached.
If the temperature inside the power unit increases by an additional 5 K, then fault F30036 is triggered.

Remedy: Check the following:
- is the ambient temperature within the defined limit values?
- has the fan failed? Check the direction of rotation.

Reaction upon N: NONE

Acknowl. upon N: NONE

A05004 (N) Power unit: Rectifier overtemperature

Message value: -

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The alarm threshold for the overtemperature of the rectifier has been reached. The response is set using p0290.
If the temperature of the rectifier increases by an additional 5 K, then fault F30037 is triggered.

Remedy: Check the following:
- is the ambient temperature within the defined limit values?
- have the load conditions and the load duty cycle been appropriately dimensioned?
- has the fan failed? Check the direction of rotation.
- has a phase of the line supply failed?
- is an arm of the supply (incoming) rectifier defective?

Reaction upon N: NONE

Acknowl. upon N: NONE

A05005	Cooling system: Cooling medium flow rate too low
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Cooling system: Alarm - flow rate has fallen below the alarm value
Remedy:	
A05006 (N)	Power unit: Overtemperature thermal model
Message value:	-
Drive object:	A_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature difference between the chip and heat sink has exceeded the permissible limit value (blocksize power units only). Depending on p0290, an appropriate overload response is initiated. See also: r0037
Remedy:	None necessary. The alarm disappears automatically once the limit value is undershot. Note: If the alarm does not disappear automatically and the temperature continues to rise, this can result in fault F30024. See also: p0290 (Power unit overload response)
Reaction upon N:	NONE
Acknowled. upon N:	NONE
N05007 (A)	Power unit: Overtemperature thermal model (chassis PU)
Message value:	-
Drive object:	A_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature difference between the chip and heat sink has exceeded the permissible limit value (r0293) (chassis power units only). Depending on p0290, an appropriate overload response is initiated. See also: r0037, r0293 (Power unit alarm threshold model temperature)
Remedy:	None necessary. The alarm disappears automatically once the limit value is undershot. See also: p0290 (Power unit overload response)
Reaction upon A:	NONE
Acknowled. upon A:	NONE
F05050	Parallel circuit: Pulse enable in spite of pulse inhibit
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Vector: OFF2 (NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	A power unit signals that the pulses are enabled although the pulses are inhibited. Fault value (r0949, interpret decimal): Number of the power unit involved.
Remedy:	The power unit is defective and must be replaced.

F05051	Parallel circuit: Power unit pulse enable missing
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Vector: OFF2 (NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	For one or several power units, the pulses were not able to be enabled. Fault value (r0949, interpret decimal): Number of the power unit involved.
Remedy:	- acknowledge power unit faults that are still present. - inhibit the pulses of the power unit involved (p7001).

A05052 (F)	Parallel circuit: Illegal current dissymmetry
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The deviation of the individual currents of the power units exceeds the alarm threshold specified in p7010. Alarm value (r2124, interpret decimal): 1: Phase U. 2: Phase V. 3: Phase W.
Remedy:	- inhibit the pulses of the faulted power unit (p7001). - check the connecting cables. Loose contacts can cause current spikes. - the motor reactors are non-symmetrical or faulty and must be replaced. - the CTs must be calibrated or replaced.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Vector: NONE (OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY

A05053 (F)	Parallel circuit: Inadmissible DC link voltage dissymmetry
Message value:	-
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The deviation of the DC link voltage measured values exceeds the alarm threshold specified in p7011.
Remedy:	- inhibit the pulses of the faulted power unit (p7001). - check the DC link connecting cables. - the DC link voltage measurement is incorrect and must be calibrated or renewed.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Vector: NONE (OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY

A05054	Parallel circuit: Power unit deactivated
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	For the drive object involved, fewer power unit components connected in parallel are active than exist in the target topology. Operation is only possible at reduced power (power derating).
Remedy:	Reactivate the deactivated power unit components. See also: p0125 (Activate/deactivate power unit components), p0895 (Activate/deactivate power unit components), p0897 (Parking axis selection)

F05055	Power circuit: Power units with different code numbers
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2 (NONE)
Acknowledge:	IMMEDIATELY
Cause:	The code numbers of the power units do not match. Fault value (r0949, interpret decimal): Parameter in which the first different power unit code number was detected.
Remedy:	For parallel circuit configurations, only power units with identical power unit data may be used.
F05056	Parallel circuit: Power unit EPROM versions differ
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2 (NONE)
Acknowledge:	IMMEDIATELY
Cause:	The EEPROM versions of the power units do not match. Fault value (r0949, interpret decimal): Parameter in which the first different version number was detected.
Remedy:	For parallel circuit configurations, only power units with identical EEPROM versions may be used.
F05057	Parallel circuit: Power unit firmware versions differ
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2 (NONE)
Acknowledge:	IMMEDIATELY
Cause:	The firmware versions of the power units connected in parallel do not match. Fault value (r0949, interpret decimal): Parameter in which the first different version number was detected.
Remedy:	For parallel circuit configurations, only power units with identical firmware versions may be used.
F05058	Parallel circuit: VSM EEPROM versions differ
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The EEPROM versions of the Voltage Sensing Modules (VSM) do not match. Fault value (r0949, interpret decimal): Parameter in which the first different version number was detected.
Remedy:	For parallel circuit configurations, only Voltage Sensing Modules (VSM) with identical EEPROM versions may be used.
F05059	Parallel circuit: VSM firmware versions differ
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The firmware versions of the Voltage Sensing Module (VSM) do not match. Fault value (r0949, interpret decimal): Parameter in which the first different version number was detected.
Remedy:	For parallel circuit configurations, only Voltage Sensing Modules (VSM) with identical firmware versions may be used.

F05060	Parallel circuit: Power unit firmware version does not match
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Firmware from version V02.30.01.00 is required when connecting the power units in parallel.
Remedy:	Update the firmware of the power units (at least V02.30.01.00).
F05061	Infeed, number of VSM
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The number of active Voltage Sensing Modules (VSM) for the drive object infeed with chassis power units is not correct. For A_Infeed, each active power unit must be assigned an active VSM also for a parallel circuit configuration. For S_Infeed, the active drive object, must be assigned at least one active VSM. Fault value (r0949, interpret decimal): Number of VSMs that are currently assigned to the drive object.
Remedy:	Adapts the number of active Voltage Sensing Modules (VSM).
F06000	Infeed: Precharging monitoring time expired
Message value:	-
Drive object:	A_INF, B_INF, S_INF
Reaction:	OFF2 (OFF1)
Acknowledge:	IMMEDIATELY
Cause:	After the line contactor closes the power unit does not signal the READY state within the monitoring time (p0857). The end of the DC link pre-charging was not able to be completed for one of the following reasons: 1) There is no line supply voltage connected. 2) The line contactor/line side switch has not been closed. 3) The line supply voltage is too low. 4) Line supply voltage incorrectly set (p0210). 5) The pre-charging resistors are overheated as there were too many pre-charging operations per time unit. 6) The pre-charging resistors are overheated as the DC link capacitance is too high. 7) The pre-charging resistors are overheated because when there is no "ready for operation" (r0863.0) of the infeed unit, power is taken from the DC link. 8) The pre-charging resistors are overheated as the line contactor was closed during the DC link fast discharge through the Braking Module. 9) The DC link has either a ground fault or a short-circuit. 10) The pre-charging circuit is possibly defective (only for chassis units). See also: p0210 (Drive unit line supply voltage), p0857 (Power unit monitoring time)
Remedy:	In general: - check the line supply voltage at the connecting terminals. - check the line supply voltage setting (p0210). - check the monitoring time and, if required, increase (p0857). - where relevant, observe additional power unit messages/signals (e.g. F30027). - the following applies to booksize units: Wait (approx. 8 min.) until the pre-charging resistors have cooled down. For this purpose, preferably disconnect the infeed unit from the line supply. Re 5): - carefully observe the permissible pre-charging frequency (refer to the appropriate Equipment Manual). Re 6): - check the total capacitance of the DC link and reduce in accordance with the maximum permissible DC-link capacitance if necessary (refer to the appropriate Equipment Manual)

Re 7):

- interconnect the ready-for-operation signal from the infeed unit (r0863.0) in the enable logic of the drives connected to this DC link

Re 8):

- check the connections of the external line contactor. The line contactor must be open during DC-link fast discharge.

Re 9):

- check the DC link for ground faults or short circuits.

F06010 Infeed: Power unit EP 24 V missing in operation

Message value: -

Drive object: A_INF, B_INF, S_INF

Reaction: OFF2 (OFF1)

Acknowledge: IMMEDIATELY (POWER ON)

Cause: In operation, withdraw the pulse enable at terminal EP at the Line Module (X21.3, X21.4).

Remedy: - do not open the Line Side Switch in operation - only when the pulses are inhibited.
- check the wiring of the DP input (X21.3, X21.4) at the Line Module to exclude any poor contacts.

F06050 Infeed: Smart Mode not supported

Message value: -

Drive object: A_INF, S_INF

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The power unit does not support the Smart Mode.

Remedy: - set the suitable sampling time $250 \mu s \leq p0115[0] \leq 400 \mu s$ (e.g. by setting p0112 and p0115 to the factory setting).
- upgrade the power unit software and/or hardware for the Smart Mode. The availability of the Smart Mode function is displayed in r0192.
- for A_INF the following applies: Deactivate the Smart Mode with p3400.0 = 0 and activate the voltage control with p3400.3 = 1. For booksize power units, it must be noted that for a supply voltage $p0210 > 415 V$ only the Smart Mode is possible in the presetting. If DC link voltages above 660 V are permissible in the application, then voltage-controlled operation can be activated with p0280, p0210, p3400 and p3510. The information regarding p0210 should be carefully noted.
See also: r0192 (Power unit firmware properties)

F06052 Infeed: Filter temperature evaluation not supported

Message value: -

Drive object: A_INF, S_INF

Reaction: OFF2 (NONE)

Acknowledge: IMMEDIATELY

Cause: The power unit does not support filter temperature evaluation.
This feature (r0192.11) is required when an Active Interface Module is used as a line filter (p0220 = 41 ... 45).

Remedy: Upgrade the firmware for the power unit to a later version.
See also: r0192 (Power unit firmware properties), p0220 (Infeed line filter type)

F06100 Infeed: Shutdown due to line supply undervoltage condition

Message value: %1

Drive object: A_INF, B_INF, S_INF

Reaction: OFF2 (OFF1)

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The filtered (steady-state) value of the line supply voltage is less than the fault threshold (p0283).
Fault condition: $V_{rms} < p0283 * p0210$
Fault value (r0949, floating point):
Actual steady-state line supply voltage.
See also: p0283 (Line supply undervoltage, shutdown (trip) threshold)

Remedy: - check the line supply.
- check the line supply voltage (p0210).
- check the fault threshold (p0283).

A06105 (F) Infeed: Line supply undervoltage

Message value: %1
Drive object: A_INF, B_INF, S_INF
Reaction: NONE
Acknowledge: NONE
Cause: The filtered (steady-state) value of line supply voltage is lower than the alarm threshold (p0282).
Alarm condition: $V_{rms} < p0282 * p0210$
Alarm value (r2124, floating point):
Actual steady-state line supply voltage.
See also: p0282 (Line supply undervoltage, alarm threshold)
Remedy:
- check the line supply.
- check the line supply voltage (p0210).
- check the alarm threshold (p0282).
Reaction upon F: NONE (OFF1, OFF2)
Acknowl. upon F: IMMEDIATELY (POWER ON)

F06200 Infeed: Failure of one or several line phases

Message value: -
Drive object: A_INF, S_INF
Reaction: OFF2 (OFF1)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: Failure overvoltage in one or several line supply phases.
The fault can be output in two operating states:
1. During the power-on phase of the infeed unit.
The measured line supply angle deviates from the regular characteristic for a 3-phase system - the PLL cannot be synchronized.
The fault occurs immediately after power-up if, when operating with a VSM, the phase assignment L1, L2, L3 at the VSM differs from the phase assignment at the power unit.
2. While the infeed is operational.
After a voltage dip has been detected or an overvoltage (note A06205) in one or several line phases a fault occurred within 100 ms (also refer to other relevant messages). Generally, before fault message F06200 is output, Alarm A06205 occurs at least once, whose warning value can provide information regarding the cause of the line supply fault.
Probable causes of the fault:
- voltage dip on the line side or phase failure or overvoltage lasting longer than 10 ms.
- overload condition on the load side with peak current.
- line reactor missing.
Remedy:
- check the line supply and fuses.
- check the connection and size (rating) of the line reactor.
- check and correct the phase assignment at the VSM (X521 or X522) and at the power unit.
- check the load.
- if failed in operation, carefully note the previous alarm messages A6205 with alarm values.
See also: p3463 (Infeed, line angle change, phase failure detection)

A06205 (F)	Infeed: Voltage dip in at least one line supply phase
Message value:	%1
Drive object:	A_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	Voltage dip or overvoltage in one or several line supply phases has been detected in operation. The pulses are then inhibited for a time of at least 8 ms. The operating signal of the infeed unit in r0863.0 remains and the pulse inhibit due to the phase failure is displayed in r3405.2. Alarm value (r2124, bitwise coded cause of the alarm): Bit 0: Line angle deviation (limit value p3463) due to a line supply fault Bit 2: Active current deviation Bit 3: Line frequency deviation (limit values: 115 % * p0284, 85 % * p0285) Bit 4: Line overvoltage (limit value 120 % * p0281 * p0210) Bit 5: Line undervoltage (limit value 20 % * p0210) Bit 7: Peak current fault Bit 8: Smart Mode without VSM (p3400.5 = 0): Line angle deviation Bit 9: Smart Mode: DC link voltage dip Bit 10: Smart Mode: Line currents not symmetrical
Remedy:	Generally, the following applies when an alarm message is output: - check the line supply and fuses. - check the line supply quality and system fault level. - check the load. Dependent on the alarm value in r2124, the following applies: Bit 0 = 1: Line fault occurred or poor/incorrect controller setting. For poor line quality or frequent line supply change-over operations, when required, limit value p3463 can be increased until the alarm value no longer occurs. Bit 2 = 1: Line fault occurred or poor/incorrect controller setting. - check the controller setting and load. Bit 3 = 1: Line fault occurred. For poor line quality or frequent line changeover operations, when required, limit values p0284 and p0285 can be increased until the alarm value no longer occurs. Bit 4 = 1: Line interrupted or line overvoltage has occurred. Bit 5 = 1: Line interrupted or line undervoltage has occurred. Bit 7 = 1: Peak current trip due to line fault or overload. Check the load. Bit 8 = 1: Line fault occurred. Bit 9 = 1: Line undervoltage or overload. Check the load. Bit 10 = 1: Line supply interrupted in at least one line phase. Check the fuses. See also: r3405, p3463 (Infeed, line angle change, phase failure detection)
Reaction upon F:	NONE (OFF1, OFF2)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
F06207 (N, A)	Infeed: Line currents not symmetrical
Message value:	-
Drive object:	A_INF, S_INF
Reaction:	OFF1 (NONE, OFF2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	Asymmetry of the currents in the line phase too high. The most probable cause is failure of a line phase.
Remedy:	- check the line supply and fuses. - check the connection and size (rating) of the line reactor. - check the previous alarm A06205 and the alarm value.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F06210	Infeed: Summation current too high
Message value:	%1
Drive object:	A_INF, B_INF, S_INF
Reaction:	OFF2 (OFF1)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The smoothed total of the phase currents ($i_1 + i_2 + i_3$) is greater than 4 % of the maximum power unit current (r0209). Possible causes: - the DC link has a ground fault that results in a high summation current (r0069.6). The DC component in the line currents can damage/destroy the power unit, line reactor or line filter! - the zero point calibration of the current measurement was not carried out (p3491, A06602). - defective current measurement in the power unit. Fault value (r0949, floating point): Smoothed total of the phase currents.
Remedy:	- check the DC link for a low-ohmic or high-ohmic ground fault and if present, remove. - increase the monitoring time of the current offset measurement (p3491). - replace the power unit if necessary.

F06211	Infeed: Summation current impermissibly high
Message value:	%1
Drive object:	A_INF
Reaction:	OFF2 (OFF1)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The smoothed sum of the phase currents ($i_1 + i_2 + i_3$) is impermissibly high. The summation current has exceeded the assigned ground fault monitoring threshold (p0287). Possible causes: - there is a ground fault that results in a high summation current (r0069.6). The DC component in the line currents can damage/destroy the power unit, line reactor or line filter! - the zero point calibration of the current measurement was not carried out (p3491, A06602). - defective current measurement in the power unit. Fault value (r0949, floating point): Smoothed total of the phase currents.
Remedy:	- check the line supply for ground faults and remove any that are present. - check the set ground fault monitoring threshold (p0287). - replace the power unit if necessary.

A06215 (F)	Infeed: Summation current too high
Message value:	%1
Drive object:	A_INF, B_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	The smoothed total of the phase currents ($i_1 + i_2 + i_3$) is greater than 3 % of the maximum power unit current (r0209). Possible causes: - the DC link has a ground fault that results in a high summation current (r0069.6). The DC component in the line currents can damage/destroy the power unit, line reactor or line filter! - the zero point calibration of the current measurement was not carried out (p3491, A06602). - defective current measurement in the power unit. Alarm value (r2124, floating point): Smoothed total of the phase currents.
Remedy:	- check the DC link for a low-ohmic or high-ohmic ground fault and if present, remove. - increase the monitoring time of the current offset measurement (p3491). - replace the power unit if necessary.
Reaction upon F:	NONE (OFF1, OFF2)
Acknowl. upon F:	IMMEDIATELY (POWER ON)

A06250 (F, N)	Infeed: Defective capacitor(s) in at least one phase of line filter
Message value:	%1
Drive object:	A_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	A change in the line filter capacitance was detected in at least one line phase. The voltages and phase currents of the line filter, measured using a Voltage Sensing Module (VSM), indicate a deviation of the filter capacitances from the value parameterized in p0221. A change or a defect of the line filter capacitors results in a shift of the resonant frequencies and can result in severe damage to the drive system. Alarm value (r2124, floating point): The calculated present capacitance in μF (rounded-off to an integer number). The 1st decimal point specifies the number of the phase (1, 2, 3) where the capacitance deviates from the specified value.
Remedy:	- check the parameterized value of the filter capacitance (p0221). - check the correct wiring of the Voltage Sensing Module (VSM): Differential voltages u12 and u23 must be present at the 100 V/690 V inputs of the VSM; the phase currents of the line filter must be connected to the 10 V inputs through a current - voltage converter. - check the alarm limits for the permissible filter capacitance deviation (p3676). - check the scaling of the line supply voltage measurement using the VSM (p3660). - check the scaling of the filter current measurement using the VSM (p3670). - check the line filter capacitors and if required, replace the line filter. See also: p0221 (Infeed filter capacitance), p3660 (VSM input line supply voltage, voltage scaler), p3670 (VSM 10 V input CT gain), p3676 (VSM line filter capacitance alarm threshold)
Reaction upon F:	NONE (OFF1, OFF2)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A06260	Infeed: Temperature in the line filter too high
Message value:	-
Drive object:	A_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature monitoring in the line filter has responded. If the temperature remains too high during the complete monitoring time, this results in fault F06261. Note: The temperature monitoring is only available for an Active Interface Module.
Remedy:	- check whether the line filter type set in p0220[0] matches the line filter that is actually connected. Ensure that the line filter specified for the infeed being used is connected or correct the setting of the line filter type in P0220[0]. - temperature monitoring is mandatory for AIM line filters (refer to P0220). Ensure that the line filter temperature switch is correctly and reliably connected to input X21 of the infeed. - reduce the ambient temperature of the line filter. - reduce the load on the infeed and the filter module. - check the magnitude of the line supply voltage. - the internal fan of the filter module is defective. Replace the fan if necessary. - defective temperature switch of the filter module. Replace the filter module if necessary.
F06261	Infeed: Temperature in the line filter permanently too high
Message value:	-
Drive object:	A_INF, S_INF
Reaction:	OFF2 (OFF1)
Acknowledge:	IMMEDIATELY
Cause:	After the temperature monitoring responded, the temperature in the line filter was permanently exceeded. Note: The temperature monitoring is only available for an Active Interface Module.

Remedy:

- check whether the line filter type set in p0220[0] matches the line filter that is actually connected. Ensure that the line filter specified for the infeed being used is connected or correct the setting of the line filter type in P0220[0].
- temperature monitoring is mandatory for AIM line filters (refer to P0220). Ensure that the line filter temperature switch is correctly and reliably connected to input X21 of the infeed.
- reduce the ambient temperature of the line filter.
- reduce the load on the infeed and the filter module.
- check the magnitude of the line supply voltage.
- the internal fan of the filter module is defective. Replace the fan if necessary.
- defective temperature switch of the filter module. Replace the filter module if necessary.

F06262 Infeed: Temperature switch in the line filter open when powering up

Message value: -

Drive object: A_INF, S_INF

Reaction: OFF2 (OFF1)

Acknowledge: IMMEDIATELY

Cause: When powering up the infeed, the temperature in the line filter is too high. Powering up is prevented.

Remedy:

- check whether the line filter type set in p0220[0] matches the line filter that is actually connected. Ensure that the line filter specified for the infeed being used is connected or correct the setting of the line filter type in P0220[0].
- temperature monitoring is mandatory for AIM line filters (refer to P0220). Ensure that the line filter temperature switch is correctly and reliably connected to input X21 of the infeed.
- the filter temperature is too high. Allow the system to cool down.
- the internal fan of the filter module is defective. Replace the fan if necessary.
- defective temperature switch of the filter module. Replace the filter module if necessary.

F06300 Infeed: Line voltage too high at power on

Message value: %1

Drive object: A_INF, S_INF

Reaction: OFF2 (NONE, OFF1)

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The RMS line supply voltage V_{rms} was so high when powering up that controlled operation is not possible without exceeding the permissible maximum voltage in the DC link (p0280).
 Fault condition: $V_{rms} * 1.5 > p0280$.
 Fault value (r0949, floating point):
 Lowest possible controlled DC link voltage for the line supply voltage presently connected.
 See also: p0280 (DC link voltage maximum steady-state)

Remedy:

- check the line supply voltage
- check the maximum DC link voltage and if required, increase (p0280).
- check the line supply voltage and compare with the actual line supply voltage (p0210).
- check whether the power unit is dimensioned for the line supply voltage actually being used.

See also: p0210 (Drive unit line supply voltage), p0280 (DC link voltage maximum steady-state)

A06301 (F) Infeed: Line supply overvoltage

Message value: Line supply voltage: %1

Drive object: A_INF, B_INF, S_INF

Reaction: NONE

Acknowledge: NONE

Cause: The filtered (steady-state) value of the rms line supply voltage V_{rms} is higher than the alarm threshold (p0281).
 Alarm condition: $V_{rms} > p0281 * p0210$.
 Alarm value (r2124, floating point):
 Actual steady-state line supply voltage.
 See also: p0281 (Line supply overvoltage, alarm threshold)

Remedy:

- check the line supply.
- check the line supply voltage (p0210).
- check the alarm threshold (p0281).

See also: p0210 (Drive unit line supply voltage), p0281 (Line supply overvoltage, alarm threshold)

Reaction upon F: NONE (OFF1, OFF2)

Acknowl. upon F: IMMEDIATELY (POWER ON)

F06310 (A)	Infeed: Supply voltage (p0210) incorrectly parameterized
Message value:	Line supply voltage: %1
Drive object:	A_INF, B_INF, S_INF
Reaction:	NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	After pre-charging was completed, the line supply voltage V_{rms} was calculated using the measured DC link voltage. This voltage V_{rms} is not within the tolerance range of the supply voltage. The following applies for the tolerance range: $85 \% * p0210 < V_{rms} < 110 \% * p0210$. Alarm value (r2124, floating point): Line supply voltage V_{rms} present. See also: p0210 (Drive unit line supply voltage)
Remedy:	- check the parameterized supply voltage and if required change (p0210). - check the line supply voltage. See also: p0210 (Drive unit line supply voltage)
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F06310 (A)	Supply voltage (p0210) incorrectly parameterized
Message value:	-
Drive object:	SERVO
Reaction:	NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	For AC/AC drive units, the measured DC voltage lies outside the tolerance range after pre-charging has been completed. The following applies for the tolerance range: $1.16 * p0210 < r0070 < 1.6 * p0210$. The fault can only be acknowledged when the drive is powered down. See also: p0210 (Drive unit line supply voltage)
Remedy:	- check the parameterized supply voltage and if required change (p0210). - check the line supply voltage. See also: p0210 (Drive unit line supply voltage)
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F06310 (A)	Supply voltage (p0210) incorrectly parameterized
Message value:	-
Drive object:	VECTOR
Reaction:	NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	For AC/AC drive units, the measured DC voltage lies outside the tolerance range after pre-charging has been completed: $1.16 * p0210 < r0070 < 1.6 * p0210$. The fault can only be acknowledged when the drive is powered down. See also: p0210 (Drive unit line supply voltage)
Remedy:	- check the parameterized supply voltage and if required change (p0210). - check the line supply voltage. See also: p0210 (Drive unit line supply voltage)
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F06311 Infeed: Supply voltage (p0210) incorrect

Message value: Line supply voltage: %1
Drive object: A_INF, B_INF, S_INF
Reaction: OFF2 (OFF1)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The line voltage nominal value indicated in p0210 is outside the nominal voltage range of the power unit.
After pre-charging was completed, the current line supply voltage Vrms was calculated using the measured DC link voltage. This voltage Vrms does not lie within the extended tolerance range of the supply voltage set in p0210.
The following applies for the extended tolerance range: $75 \% * p0210 < Vrms < 120 \% * p0210$
Alarm value (r2124, floating point):
Line supply voltage Vrms present.
See also: p0210 (Drive unit line supply voltage)
Remedy: - check the parameterized supply voltage and if required change (p0210).
- check the line supply voltage.
See also: p0210 (Drive unit line supply voltage)

F06320 Master/slave: 4-channel multiplexer control not valid

Message value: %1
Drive object: A_INF
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: Values 0, 1, 2, and 3 are valid to control the 4-channel multiplexer via connector input 3572.
In this case, an invalid value was identified. The control remains effective with the previous value.
Fault value (r0949, interpret decimal):
Invalid value to control the multiplexer.
See also: p3572 (Master/slave active current setpoint, multiplexer selection)
Remedy: - check the interconnection to control the multiplexer (CI: p3572).
- check the signal source signal value of the BICO interconnection.
See also: p3572 (Master/slave active current setpoint, multiplexer selection)

F06321 Master/slave: 6-channel multiplexer control not valid

Message value: %1
Drive object: A_INF
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: Values 0, 1, 2, 3, 4 and 5 are valid to control the 6-channel multiplexer via CI: 3577. In this case, an invalid value was identified. The control remains effective with the previous value.
Fault value (r0949, interpret decimal):
Invalid value to control the multiplexer.
See also: p3577 (Master/slave current distribution factor, multiplexer selection)
Remedy: - check the interconnection to control the multiplexer (CI: p3577).
- check the signal source signal value of the BICO interconnection.
See also: p3577 (Master/slave current distribution factor, multiplexer selection)

A06350 (F)	Infeed: Measured line frequency too high
Message value:	Line frequency: %1
Drive object:	A_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	The actual line frequency f_{line} is higher than the parameterized alarm threshold ($f_{line} > p0211 * p0284$). The alarm can be output in two operating states: 1. During the power-on phase of the infeed unit. Consequence: Synchronization of the infeed to the line supply is interrupted and is restarted. 2. While the infeed is operational. Consequence: The infeed remains in the operating (run) state and alarm A6350 is output. This signifies a critical operational fault. Alarm value (r2124, floating point): Actual line frequency determined. See also: p0284 (Line supply frequency exceeded, alarm threshold)
Remedy:	- check the parameterized line frequency and if required change (p0211). - check the alarm threshold (p0284). - check the line supply. - check the line supply quality. See also: p0211 (Rated line freq), p0284 (Line supply frequency exceeded, alarm threshold)
Reaction upon F:	NONE (OFF1, OFF2)
Acknowled. upon F:	IMMEDIATELY (POWER ON)
A06351 (F)	Infeed: Measured line frequency too low
Message value:	Line frequency: %1
Drive object:	A_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	The actual line frequency f_{line} is lower than the parameterized alarm threshold ($f_{line} < p0211 * p0285$). The alarm can be output in two operating states: 1. During the power-on phase of the infeed unit. Consequence: Synchronization of the infeed to the line supply is interrupted and is restarted. 2. While the infeed is operational. Consequence: The infeed remains in the operating (run) state and alarm A06351 is output. This signifies a critical operational fault. Alarm value (r2124, floating point): Actual line frequency determined. See also: p0285 (Line supply frequency undershot, alarm threshold)
Remedy:	- check the parameterized line frequency and if required change (p0211). - check the alarm threshold (p0285). - check the line supply. - check the line supply quality. See also: p0211 (Rated line freq), p0285 (Line supply frequency undershot, alarm threshold)
Reaction upon F:	NONE (OFF1, OFF2)
Acknowled. upon F:	IMMEDIATELY (POWER ON)
A06400	Infeed: Line supply data identification selected/active
Message value:	-
Drive object:	A_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	The line supply data identification is selected and active. The line inductance and the DC link capacitance are measured at the next pulse enable. See also: p3410 (Infeed identification method)
Remedy:	No remedial action required.

A06401	Infeed: Transformer data identification/test mode selected/active
Message value:	%1
Drive object:	A_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	A transformer data identification type or a transformer test mode has been selected or is active. Alarm value (r2124, interpret decimal): 11: Identification type 1 selected for transformer data (automatic determination of the magnetizing inductance). 12: Identification type 2 selected for transformer data (automatic determination of transformer phase shift and gain correction). 13: Identification type 3 selected for transformer data (determination of total leakage inductance of transformer during line data identification). 101: Test mode 1 selected. 102: Test mode 2 selected. See also: p5480 (Transformer magnetization mode)
Remedy:	No remedial action required. The alarm automatically disappears after identification has stopped.

F06500	Infeed: Line synchronization not possible
Message value:	-
Drive object:	A_INF, S_INF
Reaction:	OFF2 (OFF1)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The line synchronization is not possible within the monitoring time. The infeed was re-synchronized to the line supply because it was interrupted due to a line frequency that was determined to be either too low or too high. After 20 attempts, synchronization - and therefore also the power-on operation - were interrupted.
Remedy:	- check the parameterized line frequency and if required change (p0211). - check the setting of the threshold values (p0284, p0285). - check the line supply. - when a Voltage Sensing Module (VSM) is used: check the line supply to the VSM terminals X521 or X522. - check the line supply quality. See also: p0211 (Rated line freq), p0284 (Line supply frequency exceeded, alarm threshold), p0285 (Line supply frequency undershot, alarm threshold)

A06502 (F, N)	Infeed: Unable to achieve line synchronization in transformer magnetization
Message value:	-
Drive object:	A_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	Line synchronization is not possible within the monitoring time (p5481[2]).
Remedy:	- check the setting of the threshold value (p5485) - check the setting of the maximum time (p5481[2]) - check the line supply quality. See also: p5481 (Transformer magnetization ramp-up time/bounce time/timeout), p5485 (Transf magnetization voltage threshold)
Reaction upon F:	NONE (OFF1, OFF2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A06601 (F)	Infeed: Current offset measurement interrupted
Message value:	%1
Drive object:	A_INF, B_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	Defective current measurement or a DC current is present during the offset measurement. Alarm value (r2124, interpret decimal): 1: Excessively high phase current has occurred during the current offset calibration. 2: The measured current offset is greater than the 3% of the maximum permissible converter current (e.g. due to a ground fault in the DC link).
Remedy:	Re alarm value = 1: - possible counter-measure if there is no line contactor: Power up an adequately long time before OFF1 = 1. Re alarm value = 2: - defective current measurement or a DC current is present during the offset measurement. - check the DC link for a ground fault.
Reaction upon F:	NONE (OFF1, OFF2)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
A06602 (F)	Infeed: Current offset measurement not possible
Message value:	-
Drive object:	A_INF, B_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	After an OFF1 = 1 no valid current offset measurement was able to be made within the monitoring time (p3491) before closing the line contactor. The current offset is set to 0. See also: p3491 (Infeed I-offset measurement monitoring time)
Remedy:	- check the DC link for a ground fault. A ground fault can destroy parts and components! - Check the monitoring time setting and if required increase (p3491). At least 100 ms is required for a valid measurement (p3491 > 100 ms). Notice: If there is no valid measurement, then under certain circumstances the quality of the DC link control will be reduced. See also: p3491 (Infeed I-offset measurement monitoring time)
Reaction upon F:	NONE (OFF1, OFF2)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
F06700 (A)	Infeed: Switch line contactor for load condition
Message value:	-
Drive object:	A_INF, B_INF, S_INF
Reaction:	NONE (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	For an ON command, the infeed line contactor should be switched under load.
Remedy:	- do not load the DC link if the infeed has not issued an operating signal (r0863.0 = 1). - after the infeed has been powered down, all power units connected to the DC link should be powered down. To realize this, the operating signal of the infeed (r0863.0) must be suitably interconnected.
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A06800 (F) Infeed: Maximum steady-state DC link voltage reached

Message value: -

Drive object: A_INF, S_INF

Reaction: NONE

Acknowledge: NONE

Cause: The DC link voltage setpoint has reached the maximum steady-state voltage parameterized in p0280. The DC link voltage is increased by the modulation depth reserve controller for the following reasons:

- modulation depth reserve is too low (p3480).
- line supply voltage is too high.
- supply voltage (p0210) parameterized to be too low.
- excessively high setpoint for the reactive line current.

Remedy:

- check the line supply voltage setting (p0210).
- check the line supply for an overvoltage condition.
- reduce the modulation depth reserve (p3480).
- reduce the reactive current setpoint.

See also: p0210 (Drive unit line supply voltage), p0280 (DC link voltage maximum steady-state), p3480 (Infeed modulation depth limit)

Reaction upon F: NONE (OFF1, OFF2)

Acknowl. upon F: IMMEDIATELY (POWER ON)

A06810 (F) Infeed: DC link voltage alarm threshold

Message value: -

Drive object: A_INF, B_INF, S_INF

Reaction: NONE

Acknowledge: NONE

Cause: In operation, the DC link voltage has dropped to below the alarm threshold. The alarm threshold is obtained from the sum of p0279 and r0296. Possible causes include:

- line supply voltage dip or another line supply fault.
- overload of the infeed.
- for ALM: Incorrect controller parameterization.

See also: p0279 (DC link voltage offset alarm threshold), r0296 (DC link voltage undervoltage threshold)

Remedy:

- check the line voltage and line supply quality.
- reduce the power drawn, avoid step-like load changes
- for ALM: Adapt the controller parameterization, e.g. using an automatic line supply identification (p3410 = 4, 5).

Reaction upon F: NONE (OFF1, OFF2)

Acknowl. upon F: IMMEDIATELY (POWER ON)

F06850 Infeed: Short-circuit prevailing for too long

Message value: -

Drive object: A_INF

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The maximum permissible duration (p5458[1]) for the short-circuit has been exceeded. The short-circuit could not be cleared within this time.

Remedy:

- check the minimum time parameter setting (p5458[1]).
- check the line supply and fuses.

See also: p5458 (Current hysteresis controller minimum time operating state)

F06855	Infeed: Threshold value violation in line filter
Message value:	%1
Drive object:	A_INF
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A threshold set in p3678 was exceeded in the line filter for a time set in p3679. Fault value: 0: Violation of voltage threshold value 1: Violation of current threshold value See also: p3678 (Filter monitoring threshold values), p3679 (Filter monitoring minimum times)
Remedy:	
A06860	Infeed: Function module activation not possible
Message value:	-
Drive object:	A_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	Function module activation is not possible. The power unit used does not have a "gating unit with current limitation control" (p0192.19). The affected function module is identified in fault code r0949 (the value of r0949 corresponds to the bit of parameter p0108). r0949=7 "Fault ride-through" function module r0949=12 "Line droop control" function module See also: r0192 (Power unit firmware properties), p5401 (Line droop control activation)
Remedy:	- check whether the power unit used has a "gating unit with current limitation control" (p0192.19). - if the power unit used does not have a "gating unit with current limitation control" replace it with a power unit which does have a "gating unit with current limitation control".
A06900 (F)	Braking Module: Fault (1 -> 0)
Message value:	%1
Drive object:	A_INF, B_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	The Braking Module signals a fault (1 -> 0) via X21.4 ("booksize" format) or terminal X21.5 ("chassis" format). This signal is interconnected via binector input BI: p3866[0...7]. See also: p3866 (Braking Module fault)
Remedy:	- reduce the number of braking operations. - check binector input BI: p3866[0...7] and the wiring from terminal X21.4 ("booksize" format) or terminal X21.5 ("chassis" format).
Reaction upon F:	NONE (OFF2)
Acknowl. upon F:	IMMEDIATELY
A06901	Braking Module: Pre-alarm I2t shutdown
Message value:	%1
Drive object:	A_INF, B_INF, S_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	The Braking Module signals "Pre-alarm I2t shutdown" via terminal X21.3. This signal is interconnected via binector input p3865[0...7]. Note: The pre-alarm I2t shutdown is only possible for "booksize" formats. This function is not supported for "chassis" formats.
Remedy:	- reduce the number of braking operations. - check binector input BI: p3865[0...7] and the wiring from terminal X21.3 of the particular Braking Module.

A06904 (N)	Braking Module internal is inhibited
Message value:	%1
Drive object:	B_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	The internal Braking Module was inhibited via the binector input BI: p3680 = 1 signal. In the inhibited state, energy cannot be dissipated using the braking resistor. See also: p3680 (Braking Module internal inhibit)
Remedy:	Release the internal Braking Module (BI: p3680 = 0 signal).
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A06905	Braking Module internal I2t shutdown alarm
Message value:	%1
Drive object:	B_INF
Reaction:	NONE
Acknowledge:	NONE
Cause:	The internal Braking Module outputs an alarm due to the high I2t value. 80% of the maximum switch-on duration of the braking resistor has been reached. Note: This message is also displayed via BO: p3685. See also: r3685 (Digital Braking Module: Pre-alarm I2t shutdown)
Remedy:	Reduce the number of braking operations.

F06906 (A)	Braking Module internal fault
Message value:	%1
Drive object:	B_INF
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The internal Braking Module outputs a fault due to overcurrent or an excessively high I2t value and is therefore inhibited. Note: This message is also displayed via BO: p3686. Fault value (r0949, interpret bitwise binary): Bit 0 = 1: I2t exceeded Bit 1 = 1: overcurrent See also: r3686 (Digital Braking Module Fault)
Remedy:	Reduce the number of braking operations.
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F06907	Braking Module internal overtemperature
Message value:	-
Drive object:	B_INF
Reaction:	OFF2 (NONE, OFF1)
Acknowledge:	IMMEDIATELY
Cause:	The temperature sensor connected to the braking resistor signals an overtemperature. The Braking Module is still active. If the overtemperature continues for an additional 60s, the Braking Module is shut down (F6908). See also: r3687 (Digital Braking Module pre-alarm overtemperature)
Remedy:	- reduce the temperature at the sensor. - check the temperature sensor connection.

F06908	Braking Module internal shutdown due to overtemperature
Message value:	-
Drive object:	B_INF
Reaction:	OFF2 (OFF1)
Acknowledge:	IMMEDIATELY
Cause:	Shutdown of the Braking Module due to overtemperature at the temperature sensor of the braking resistor for more than 60s. See also: r3688 (Digital Braking Module fault overtemperature)
Remedy:	- reduce the temperature at the sensor. - check the temperature sensor connection.
F06909	Braking Module internal Vce fault
Message value:	%1
Drive object:	B_INF
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Trip due to Vce fault. Collector emitter voltage dip (Vce) See also: r3689 (Digital Braking Module Vce fault)
Remedy:	- Power ON - replace the unit.
F07011	Drive: Motor overtemperature
Message value:	%1
Drive object:	SERVO
Reaction:	OFF2 (NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	KTY: The motor temperature has exceeded the fault threshold (p0605) or the timer (p0606) after the alarm threshold was exceeded (p0604) has expired. The response parameterized in p0610 becomes active. PTC: The response threshold of 1650 Ohm was exceeded and the timer (p0606) has expired. The response parameterized in p0610 becomes active. Possible causes: - motor is overloaded. - motor ambient temperature too high. - wire breakage or sensor not connected. Fault value (r0949, interpret decimal): With SME (p0601 = 10), TM120 (p0601 = 11) selected, the number of the sensor channel leading to the message. When the I2t motor model is activated (p0612 bit[0] = yes, p0611 > 0), fault value= 200 refers to the fact that the fault has been triggered by the I2t motor model. See also: p0604 (Motor temperature alarm threshold), p0605 (Motor temperature fault threshold), p0606 (Motor temperature timer), p0610 (Motor overtemperature response)
Remedy:	- reduce the motor load. - check the ambient temperature. - check the wiring and sensor connector. See also: p0604 (Motor temperature alarm threshold), p0605 (Motor temperature fault threshold), p0606 (Motor temperature timer)

F07011	Drive: Motor overtemperature
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF2 (NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	KTY or no sensor: The measured motor temperature or model temperature has exceeded the fault threshold (p0605) or the timer (p0606) has elapsed following the alarm threshold (p0604) being exceeded. The response parameterized in p0610 becomes active. PTC or bimetallic NC contact: The response threshold of 1650 Ohm was exceeded or the NC contact opened and the timer (p0606) has expired. The response parameterized in p0610 becomes active. Possible causes: - motor is overloaded. - motor ambient temperature too high. - wire breakage or sensor not connected. Fault value (r0949, interpret decimal): With SME (p0601 = 10), TM120 (p0601 = 11) selected, the number of the sensor channel leading to the message. When the I2t motor model is activated (p0612 bit[0] = yes, p0611 > 0), fault value= 200 refers to the fact that the fault has been triggered by the I2t motor model. See also: p0604 (Motor temperature alarm threshold), p0605 (Motor temperature fault threshold), p0606 (Motor temperature timer), p0610 (Motor overtemperature response)
Remedy:	- reduce the motor load. - check the ambient temperature. - check the wiring and sensor connector. See also: p0604 (Motor temperature alarm threshold), p0605 (Motor temperature fault threshold), p0606 (Motor temperature timer)
A07012 (N)	Drive: I2t motor model overtemperature
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The thermal I2t motor model (for synchronous motors) identified that the temperature alarm threshold was exceeded. See also: r0034 (Motor utilization), p0605 (Motor temperature fault threshold), p0611 (I2t motor model thermal time constant)
Remedy:	- check the motor load and if required, reduce. - check the motor ambient temperature. - check the thermal time constant p0611. - check the overtemperature fault threshold p0605 (= alarm threshold for the I2t motor model, see p0612)
Reaction upon N:	NONE
Acknowled. upon N:	NONE
A07015	Drive: Motor temperature sensor alarm
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	An error was detected when evaluating the temperature sensor set in p0600 and p0601. With the fault, the time in p0607 is started. If the fault is still present after this time has expired, then fault F07016 is output; however, at the earliest, 0.2 s after alarm A07015. Possible causes: - wire breakage or sensor not connected (KTY: R > 1630 Ohm). - measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm). Alarm value (r2124, interpret decimal): With SME (p0601 = 10), TM120 (p0601 = 11) selected, the number of the sensor channel leading to the message.
Remedy:	- make sure that the sensor is connected correctly. - check the parameterization (p0600, p0601). See also: r0035, p0600, p0601, p0607

F07016	Drive: Motor temperature sensor fault
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	An error was detected when evaluating the temperature sensor set in p0600 and p0601. Possible causes: - wire breakage or sensor not connected (KTY: R > 1630 Ohm). - measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm). Note: If alarm A07015 is present, the time in p0607 is started. If the fault is still present after this time has expired, then fault F07016 is output; however, at the earliest, 0.2 s after alarm A07015. Fault value (r0949, interpret decimal): With SME (p0601 = 10), TM120 (p0601 = 11) selected, the number of the sensor channel leading to the message. See also: p0607 (Temperature sensor fault timer)
Remedy:	- make sure that the sensor is connected correctly. - check the parameterization (p0600, p0601). - induction motors: Deactivate temperature sensor fault (p0607 = 0). See also: r0035, p0600, p0601, p0607
F07080	Drive: Incorrect control parameter
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The closed-loop control parameters have been parameterized incorrectly (e.g. p0356 = L_spread = 0). Fault value (r0949, interpret decimal): The fault value includes the parameter number involved. The following parameter numbers only occur as fault values for vector drives: p0310, for synchronous motors: p0341, p0344, p0350, p0357 The following parameter numbers do not occur as fault values for synchronous motors: p0354, p0358, p0360 See also: p0310, p0311, p0341, p0344, p0350, p0354, p0356, p0357, p0358, p0360, p0400, p0404, p0408, p0640, p1082, p1300
Remedy:	Modify the parameter indicated in the fault value (r0949) (e.g. p0640 = current limit > 0). See also: p0311, p0341, p0344, p0350, p0354, p0356, p0358, p0360, p0400, p0404, p0408, p0640, p1082
F07082	Macro: Execution not possible
Message value:	Fault cause: %1, supplementary information: %2, preliminary parameter number: %3
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The macro cannot be executed. Fault value (r0949, interpret hexadecimal): ccccbaa hex: cccc = preliminary parameter number, bb = supplementary information, aa = fault cause Fault causes for the trigger parameter itself: -19: Called file is not valid for the trigger parameter. -20: Called file is not valid for parameter 15. -21: Called file is not valid for parameter 700. -22: Called file is not valid for parameter 1000. -23: Called file is not valid for parameter 1500. -24: Data type of a TAG is incorrect (e.g.: Index, number or bit is not U16). Fault causes for the parameters to be set: -25: Error level has an undefined value. -26: Mode has an undefined value. -27: A value was entered as string in the tag value that is not "DEFAULT". -31: Entered drive object type unknown. -32: A device was not able to be found for the determined drive object number.

- 34: A trigger parameter was recursively called.
- 35: It is not permissible to write to the parameter via macro.
- 36: Check, writing to a parameter unsuccessful, parameter can only be read, not available, incorrect data type, value range or assignment incorrect.
- 37: Source parameter for a BICO interconnection was not able to be determined.
- 38: An index was set for a non-indexed (or CDS-dependent) parameter.
- 39: No index was set for an indexed parameter.
- 41: A bit operation is only permissible for parameters with the parameter format DISPLAY_BIN.
- 42: A value not equal to 0 or 1 was set for a BitOperation.
- 43: Reading the parameter to be changed by the BitOperation was unsuccessful.
- 51: Factory setting for DEVICE may only be executed on the DEVICE.
- 61: The setting of a value was unsuccessful.

Remedy:

- check the parameter involved.
- check the macro file and BICO interconnection.

See also: p0015, p0700, p1000, p1500

F07083 Macro: ACX file not found

Message value: Parameter: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: The ACX file (macro) to be executed was not able to be found in the appropriate directory.
Fault value (r0949, interpret decimal):
Parameter number with which the execution was started.
See also: p0015, p0700, p1000, p1500
Remedy: - check whether the file is saved in the appropriate directory on the memory card.
Example:
If p0015 is set to 1501, then the selected ACX file must be located in the following directory:
... /PMACROS/DEVICE/P15/PM001501.ACX

F07084 Macro: Condition for WaitUntil not fulfilled

Message value: Parameter: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: The WaitUntil condition set in the macro was not fulfilled in a certain number of attempts.
Fault value (r0949, interpret decimal):
Parameter number for which the condition was set.
Remedy: Check and correct the conditions for the WaitUntil loop.

F07085 Drive: Open-loop/closed-loop control parameters changed

Message value: Parameter: %1
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY (POWER ON)
Cause: Open-loop/closed-loop control parameters have had to be changed because:
1) They have exceeded dynamic limits because of other parameters, or
2) They cannot be used due to the hardware detected not having certain properties.
Fault value (r0949, interpret decimal):
The fault value includes the modified parameter number.
340: The motor and control parameters were automatically calculated (p0340 = 1), because the vector control was subsequently activated as configuration (r0108.2).
See also: p0640, p1082, p1300, p1800
Remedy: It is not necessary to change the parameters as they have already been correctly limited.

F07086	Units changeover: Parameter limit violation due to reference value change
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	A reference parameter was changed in the system. This resulted in the fact that for the parameters involved, the selected value was not able to be written in the per unit representation (cause: e.g. the steady-state minimum/maximum limit or that defined in the application was violated). The values of the parameters were set to the corresponding violated minimum/maximum limit or to the factory setting. Fault value (r0949, parameter): Diagnostics parameter r9450 to display the parameters that were not able to be recalculated. See also: p0304, p0305, p0310, p0596, p2000, p2001, p2002, p2003, r2004
Remedy:	Check the adapted parameter value and if required correct. See also: r9450 (Reference value change parameter with unsuccessful calculation)
F07087	Drive: Encoderless operation not possible for the selected pulse frequency
Message value:	Parameter: %1
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	Encoderless operation is not possible for the selected pulse frequency (p1800). Encoderless operation is activated under the following conditions: - the changeover speed for encoderless operation (p1404) is less than the maximum speed (p0322). - a control type with encoderless operation has been selected (p1300). - encoder faults of the motor encoder result in a fault response with encoderless operation (p0491). See also: p0491, p1300, p1404, p1800
Remedy:	Increase the pulse frequency (p1800). Note: In encoderless operation, the pulse frequency must be at least as high as half the current controller clock cycle (1/p0115[0]).
F07088	Units changeover: Parameter limit violation due to units changeover
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	A changeover of units was initiated. Possible causes for the violation of a parameter limit are: - when rounding off a parameter corresponding to its decimal places, the steady-state minimum or maximum limit was violated. - inaccuracies for the data type "Floating Point". In these cases, when the minimum limit is violated then the parameter value is rounded up and when the maximum limited is violated the parameter value is rounded down. Fault value (r0949, interpret decimal): Diagnostics parameter r9451 to display all parameters whose value had to be adapted. See also: p0100 (IEC/NEMA mot stds), p0349 (System of units, motor equivalent circuit diagram data), p0505 (Selecting the system of units), p0595 (Selecting technological units)
Remedy:	Check the adapted parameter values and if required correct. See also: r9451 (Units changeover adapted parameters)

A07089	Changing over units: Function module activation is blocked because the units have been changed over
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	An attempt was made to activate a function module. This is not permissible if the units have already been changed over. See also: p0100 (IEC/NEMA mot stds), p0349 (System of units, motor equivalent circuit diagram data), p0505 (Selecting the system of units)
Remedy:	Restore units that have been changed over to the factory setting.

F07090	Drive: Upper torque limit less than the lower torque limit
Message value:	-
Drive object:	SERVO
Reaction:	OFF2 (NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The upper torque limit is lower than the lower torque limit.
Remedy:	P1 must be >= P2 if parameter P1 is connected to p1522 and parameter P2 to p1523.

F07100	Drive: Sampling times cannot be reset
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	When resetting drive parameter (p0976) sampling times cannot be reset using p0111, p0112, p0115. Fault value (r0949, interpret decimal): Parameter whose setting prevents the sampling times being reset. See also: r0110 (Basic sampling times)
Remedy:	- continue to work with the set sampling times. - before resetting the drive parameters, set the basic clock cycle p0110[0] to the original value. See also: r0110 (Basic sampling times)

F07110	Drive: Sampling times and basic clock cycle do not match
Message value:	Parameter: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The parameterized sampling times do not match the basic clock cycle. Fault value (r0949, interpret decimal): The fault value specifies the parameter involved. See also: r0110, r0111, p0115
Remedy:	Enter the current controller sampling times so that they are identical to the basic clock cycle, e.g. by selecting p0112. Note which basic clock cycle is selected in p0111. The sampling times in p0115 can only be changed manually in the sampling times presetting "Expert" (p0112). See also: r0110, r0111, p0112, p0115

A07200	Drive: Master control ON/OFF1 command present
Message value:	-
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The ON/OFF1 command is not 0, either via binector input p0840 (current CDS) or in control word bit 0 via the master control.
Remedy:	Both the signal at binector input p0840 (current CDS) and bit 0 via the master control must be 0.

F07220 (N, A)	Drive: Master control by PLC missing
Message value:	-
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: OFF1 (NONE, OFF2, OFF3, STOP1, STOP2) Vector: OFF1 (NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	The "master control by PLC" signal was missing in operation. - interconnection of the binector input for "master control by PLC" is incorrect (p0854). - the higher-level control has withdrawn the "master control by PLC" signal. - data transfer via the fieldbus (master/drive) was interrupted.
Remedy:	- check the interconnection of the binector input for "master control by PLC" (p0854). - check the "master control by PLC" signal and, if required, switch in. - check the data transfer via the fieldbus (master/drive). Note: If the drive should continue to operate after withdrawing "master control by PLC" then fault response must be parameterized to NONE or the message type should be parameterized as alarm.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F07300 (A)	Drive: Line contactor feedback signal missing
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2 (NONE)
Acknowledge:	IMMEDIATELY
Cause:	- the line contactor was not able to be closed within the time in p0861. - the line contactor was not able to be opened within the time in p0861. - the line contactor dropped out during operation - the line contactor has closed although the drive converter is powered down.
Remedy:	- check the setting of p0860. - check the feedback circuit from the line contactor. - increase the monitoring time in p0861. See also: p0860 (Line cont. fdbk sig), p0861 (Line contactor monitoring time)
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F07311	Bypass motor switch
Message value:	Fault cause: %1 bin
Drive object:	VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Fault value: Bit field BYPASS_CONTACTOR_ERROR_STATE Bit 1 BYPASS_CONTACTOR_ERR_FBK_ON_MISSING Switch "Closed" feedback signal missing Bit 2 BYPASS_CONTACTOR_ERR_FBK_OFF_MISSING Switch "Open" feedback signal missing Bit 3 BYPASS_CONTACTOR_ERR_TOO_SLOW Switch feedback signal too slow: After switching, the system waits for the positive feedback signal. If the feedback signal is received later than the specified time, then a fault trip (shutdown) is issued.

Bit 6
 BYPASS_CONTACTOR_ERR_BYPASS_INCONSISTENCY
 Drive switch feedback signal is not consistent with the bypass state:
 When powering up or for STAGING, the drive switch is closed.
 See also: p1260 (Bypass configuration), r1261 (Bypass control/status word), p1266 (Bypass, control command), p1267 (Bypass changeover source configuration), p1269 (Bypass switch feedback signal), p1274 (Bypass switch monitoring time)

Remedy:

- check the transfer of the feedback signals.
- check the switch.

F07312 Bypass LSS:

Message value: Fault cause: %1 bin
Drive object: VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: Fault value: Bit field BYPASS_CONTACTOR_ERROR_STATE
 Bit 1
 BYPASS_CONTACTOR_ERR_FBK_ON_MISSING
 Switch "Closed" feedback signal missing
 Bit 2
 BYPASS_CONTACTOR_ERR_FBK_OFF_MISSING
 Switch "Open" feedback signal missing
 Bit 3
 BYPASS_CONTACTOR_ERR_TOO_SLOW
 Switch feedback signal too slow:
 After switching, the system waits for the positive feedback signal. If the feedback signal is received later than the specified time, then a fault trip (shutdown) is issued.
 Bit 6
 BYPASS_CONTACTOR_ERR_BYPASS_INCONSISTENCY
 Line Side Switch feedback signal is not consistent with the bypass state:
 When powering up or for STAGING, the Line Side Switch is closed without this having been requested from the bypass.
 See also: p1260 (Bypass configuration), r1261 (Bypass control/status word), p1266 (Bypass, control command), p1267 (Bypass changeover source configuration), p1269 (Bypass switch feedback signal), p1274 (Bypass switch monitoring time)

Remedy:

- check the transfer of the feedback signals.
- check the switch.

F07320 Drive: Automatic restart interrupted

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: - The specified number of restart attempts (p1211) has been completely used up because within the monitoring time (p1213) the faults were not able to be acknowledged. The number of restart attempts (p1211) is decremented at each new start attempt.
 - there is no active ON command.
 - the monitoring time for the power unit has expired (p0857).
 - when exiting commissioning or at the end of the motor identification routine or the speed controller optimization, the drive unit is not automatically powered up again.
 Fault value (r0949, interpret hexadecimal):
 Only for internal Siemens troubleshooting.

Remedy:

- increase the number of restart attempts (p1211). The actual number of starting attempts is displayed in r1214.
- increase the delay time in p1212 and/or the monitoring time in p1213.
- issue an ON command (p0840).
- either increase or disable the monitoring time of the power unit (p0857).

F07320	Drive: Automatic restart interrupted
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	- The specified number of restart attempts (p1211) has been completely used up because within the monitoring time (p1213) the faults were not able to be acknowledged. The number of restart attempts (p1211) is decremented at each new start attempt. - there is no active ON command. - the monitoring time for the power unit has expired (p0857). - when exiting commissioning or at the end of the motor identification routine or the speed controller optimization, the drive unit is not automatically powered up again. Fault value (r0949, interpret hexadecimal): Only for internal Siemens troubleshooting.
Remedy:	- increase the number of restart attempts (p1211). The actual number of starting attempts is displayed in r1214. - increase the delay time in p1212 and/or the monitoring time in p1213. - issue an ON command (p0840). - either increase or disable the monitoring time of the power unit (p0857). - Reduce the waiting time for resetting the fault counter p1213[1] so that fewer faults are registered in the time interval.
A07321	Drive: Automatic restart active
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The automatic restart (AR) is active. When the line supply returns and/or the causes of the existing faults are removed the drive is automatically restarted. The pulses are enabled and the motor starts to rotate.
Remedy:	- the automatic restart (AR) should, if required, be inhibited (p1210 = 0). - an automatic restart can be directly interrupted by withdrawing the power-on command (BI: p0840).
A07329 (N)	Drive: kT estimator, kT(iq) characteristic or voltage compensation does not function
Message value:	%1
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	A function of the function module "extended torque control" (r0108.1) was activated - however the (complete) function is not available. Fault value (r0949, interpret decimal): 1 ... 3: The kT estimator is active (p1780.3 = 1) without a functioning compensation of the voltage emulation error in the drive converter. This means that the accuracy is severely restricted. 1: The drive converter voltage emulation error "final value" is 0 (p1952). 2: The drive converter voltage emulation error "current offset" is 0 (p1953). 3: The compensation of the voltage emulation error is disabled (p1780.8 = 0). 4: The kT estimator (p1780.3 = 1), the kT(iq) characteristic (p1780.9 = 1) or the compensation of the voltage emulation error (p1780.8 = 1) was activated without activating the function module "extended torque control" (when the function module is activated, the following must apply: r0108.1 = 1).
Remedy:	Re fault value = 1, 2: - carry out an identification of the voltage emulation error in the drive converter (p1909.14 = 1, p1910 = 1). - set the parameter to compensate the voltage emulation error in the drive converter (p1952, p1953). Re fault value = 3: - enable the compensation of the voltage emulation error in the drive converter (p1780.8 = 1). Re fault value = 4: - activate the function module "extended torque control" (r0108.1 = 1) or deactivate the corresponding functions (p1780.3 = 0, p1780.8 = 0, p1780.9 = 0).
Reaction upon N:	NONE
Acknowl. upon N:	NONE

F07330	Flying restart: Measured search current too low
Message value:	-
Drive object:	VECTOR
Reaction:	OFF2 (NONE, OFF1)
Acknowledge:	IMMEDIATELY
Cause:	During a flying restart, it was identified that the search current reached is too low. It is possible that the motor is not connected.
Remedy:	Check the motor feeder cables.

F07331	FlyRestart: Not supported
Message value:	-
Drive object:	VECTOR
Reaction:	OFF2 (NONE, OFF1)
Acknowledge:	IMMEDIATELY
Cause:	It is not possible to power up with the motor rotating (no flying restart). In the following cases, the "flying restart" function is not supported: Permanent-magnet and separately-excited synchronous motors (PEM, FEM): Operation with V/f characteristic. Permanent-magnet synchronous motor (PEM): Encoderless operation without a Voltage Sensing Module (VSM) being connected.
Remedy:	- deactivate the "flying restart" function (p1200 = 0). - change the open-loop/closed-loop control mode (p1300). - connect a Voltage Sensing Module (VSM) (voltage measurement).

A07350 (F)	Drive: Measuring probe parameterized to a digital output
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	The measuring probe is connected to a bi-directional digital input/output and the terminal is set as output. Alarm value (r2124, interpret decimal): 8: DI/DO 8 (X122.9/X132.1) 9: DI/DO 9 (X122.10/X132.2) 10: DI/DO 10 (X122.12/X132.3) 11: DI/DO 11 (X122.13/X132.4) 12: DI/DO 12 (X132.9) 13: DI/DO 13 (X132.10) 14: DI/DO 14 (X132.12) 15: DI/DO 15 (X132.13) To the terminal designation: The first designation is valid for CU320, the second for CU305.
Remedy:	- set the terminal as input (p0728). - deselect the measuring probe (p0488, p0489, p0580).
Reaction upon F:	OFF1
Acknowl. upon F:	IMMEDIATELY

A07400 (N)	Drive: DC link voltage maximum controller active
Message value:	-
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	The DC link voltage controller has been activated because of the upper switch-in threshold (r1244). A system deviation can occur between the setpoint and actual speed. See also: r0056 (Status word, closed-loop control), p1240 (Vdc controller or Vdc monitoring configuration)

Remedy: None necessary.
This alarm automatically disappears after the upper threshold has been distinctly undershot.
Otherwise, apply the following measures:
- use a Braking Module or regenerative feedback unit.
- increase the ramp-down times (p1121, p1135).
- shut down the Vdc_max controller (p1240 = 0).

Reaction upon N: NONE
Acknowl. upon N: NONE

A07400 (N) Drive: DC link voltage maximum controller active

Message value: -
Drive object: VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The DC link voltage controller has been activated because the upper switch-in threshold has been exceeded (r1242, r1282).
The ramp-down times are automatically increased in order to maintain the DC link voltage (r0070) within the permissible limits. There is a system deviation between the setpoint and actual speeds.
When the DC link voltage controller is switched out (disabled), this is the reason that the ramp-function generator output is set to the speed actual value.
See also: p1240 (Vdc controller or Vdc monitoring configuration), p1280 (Vdc controller or Vdc monitoring configuration (V/f))

Remedy: If the controller is not to intervene:
- increase the ramp-down times.
- shut down the Vdc_max controller (p1280 = 0).
If the ramp-down times are not to be changed:
- use a chopper or regenerative feedback unit.

Reaction upon N: NONE
Acknowl. upon N: NONE

A07401 (N) Drive: DC link voltage maximum controller deactivated

Message value: -
Drive object: VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The Vdc_max controller can no longer maintain the DC link voltage (r0070) below the limit value (r1242, r1282) and was therefore switched out (disabled).
- the line supply voltage is permanently higher than specified for the power unit.
- the motor is permanently in the regenerative mode as a result of a load that is driving the motor.

Remedy: - check whether the input voltage is within the permissible range.
- check whether the load duty cycle and load limits are within the permissible limits.

Reaction upon N: NONE
Acknowl. upon N: NONE

A07402 (N) Drive: DC link voltage minimum controller active

Message value: -
Drive object: SERVO
Reaction: NONE
Acknowledge: NONE
Cause: The DC link voltage controller has been activated due to the lower switch-in threshold (r1248).
A system deviation can occur between the setpoint and actual speed.
A possible cause can be e.g. that the line supply has failed.
See also: r0056 (Status word, closed-loop control), p1240 (Vdc controller or Vdc monitoring configuration), p1248 (DC link voltage threshold lower)

Remedy: None necessary.
This alarm automatically disappears after the lower threshold has been distinctly exceeded.
Otherwise, apply the following measures:
- check the line supply and infeed.
- increase the ramp-up times (p1120).
- shut down the Vdc_min controller (p1240 = 0).

Reaction upon N: NONE

Acknowl. upon N: NONE

A07402 (N) Drive: DC link voltage minimum controller active

Message value: -

Drive object: VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The DC link voltage controller has been activated as the lower switch-in threshold has been undershot (r1246, r1286).
The kinetic energy of the motor is used to buffer the DC link. The drive is therefore braked.
See also: r0056 (Status word, closed-loop control), p1240 (Vdc controller or Vdc monitoring configuration), p1280 (Vdc controller or Vdc monitoring configuration (V/f))

Remedy: The alarm disappears when power supply returns.

Reaction upon N: NONE

Acknowl. upon N: NONE

F07403 (N, A) Drive: Lower DC link voltage threshold reached

Message value: -

Drive object: SERVO

Reaction: OFF1 (NONE, OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: The DC link voltage monitoring is active (p1240 = 5, 6) and the lower DC link voltage threshold (p1248) was reached in the "Operation" state.

Remedy: - check the line supply voltage.
- check the infeed module
- reduce the lower DC link threshold (p1248).
- switch out (disable) the DC link voltage monitoring (p1240 = 0).

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F07403 (N, A) Drive: Lower DC link voltage threshold reached

Message value: -

Drive object: VECTOR

Reaction: OFF1 (NONE, OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: The DC link voltage monitoring is active (p1240 , p1280 = 5, 6) and the lower DC link voltage threshold (r1246, r1286) was reached in the "Operation" state.

Remedy: - check the line supply voltage.
- check the infeed module
- adapt the device supply voltage (p0210) or the switch-on level (p1245, p1285).
- disable the DC link voltage monitoring (p1240, p1280 = 0).

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F07404	Drive: Upper DC link voltage threshold reached
Message value:	-
Drive object:	SERVO
Reaction:	OFF2 (NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The DC link voltage monitoring is active (p1240 = 4, 6) and the upper DC link voltage threshold (p1244) was reached in the "Operation" state.
Remedy:	- check the line supply voltage. - check the infeed module or the Braking Module. - increase the upper DC link voltage threshold (p1244). - switch out (disable) the DC link voltage monitoring (p1240 = 0).
F07404	Drive: Upper DC link voltage threshold reached
Message value:	-
Drive object:	VECTOR
Reaction:	OFF2 (NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The DC link voltage monitoring is active (p1240 , p1280 = 4, 6) and the upper DC link voltage threshold (r1242, r1282) was reached in the "Operation" state.
Remedy:	- check the line supply voltage. - check the infeed module - adapt the device supply voltage (p0210). - disable the DC link voltage monitoring (p1240, p1280 = 0).
F07405 (N, A)	Drive: Kinetic buffering minimum speed not reached
Message value:	-
Drive object:	VECTOR
Reaction:	OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	During kinetic buffering the speed fell below minimum speed (p1257 or p1297 for vector drives with V/f control) and the line supply did not return.
Remedy:	Check the speed threshold for the Vdc_min controller (kinetic buffering) (p1257, p1297). See also: p1257 (Vdc_min controller speed threshold)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F07406 (N, A)	Drive: Kinetic buffering maximum time exceeded
Message value:	-
Drive object:	VECTOR
Reaction:	OFF3 (IASC/DCBRAKE, NONE, OFF1, OFF2, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	The maximum buffer time (p1255 and p1295 for vector drives with V/f control) has been exceeded without the line supply having returned.
Remedy:	Check the time threshold for Vdc-min controller (kinetic buffering) (p1255, p1295). See also: p1255 (Vdc_min controller time threshold)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A07409	Drive: V/f control, current limiting controller active
Message value:	%1
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The current limiting controller of the V/f control was activated because the current limit was exceeded.
Remedy:	The alarm is automatically withdrawn when increasing the current limit (p0640), reducing the load or using a slower up ramp for the setpoint (reference) speed.

F07410	Drive: Current controller output limited
Message value:	-
Drive object:	SERVO
Reaction:	OFF2 (NONE, OFF1)
Acknowledge:	IMMEDIATELY
Cause:	The condition " $I_{act} = 0$ and $U_{q_set_1}$ longer than 16 ms at its limit" is present and can be caused by the following: - motor not connected or motor contactor open. - no DC link voltage present. - Motor Module defective.
Remedy:	- connect the motor or check the motor contactor. - check the DC link voltage (r0070). - check the Motor Module.

F07410	Drive: Current controller output limited
Message value:	-
Drive object:	VECTOR
Reaction:	OFF2 (NONE, OFF1)
Acknowledge:	IMMEDIATELY
Cause:	The condition " $I_{act} = 0$ and $U_{q_set_1}$ longer than 16 ms at its limit" is present and can be caused by the following: - motor not connected or motor contactor open. - motor data and motor configuration (star-delta) do not match. - no DC link voltage present. - power unit defective. - the "flying restart" function is not activated.
Remedy:	- connect the motor or check the motor contactor. - check the motor parameterization and the connection configuration (star-delta). - check the DC link voltage (r0070). - check the power unit. - activate the "flying restart" function (p1200).

F07411	Drive: Flux controller output limited
Message value:	-
Drive object:	SERVO
Reaction:	Servo: OFF2 (NONE, OFF1) Vector: OFF2
Acknowledge:	IMMEDIATELY
Cause:	The specified flux setpoint cannot be reached although 90% of the maximum current has been specified. - incorrect motor data. - motor data and motor configuration (star-delta) do not match. - the current limit has been set too low for the motor. - induction motor (encoderless, open-loop controlled) in I2t limiting. - the Motor Module is too small.
Remedy:	- correct the motor data. - check the motor configuration. - correct the current limits (p0640, p0323). - reduce the induction motor load. - if required, use a larger Motor Module.

F07411	Drive: Flux setpoint not reached when building up excitation
Message value:	-
Drive object:	VECTOR
Reaction:	Servo: OFF2 (NONE, OFF1) Vector: OFF2
Acknowledge:	IMMEDIATELY
Cause:	When quick magnetizing is configured (p1401.6 = 1) the specified flux setpoint is not reached although 90% of the maximum current is specified. - incorrect motor data. - motor data and motor configuration (star-delta) do not match. - the current limit has been set too low for the motor. - induction motor (encoderless, open-loop controlled) in I2t limiting. - the Motor Module is too small. - the magnetizing time p0346 is too short.
Remedy:	- correct the motor data. Perform motor data identification and rotating measurement. - check the motor configuration. - correct the current limits (p0640). - reduce the induction motor load. - if required, use a larger Motor Module. - check motor supply cable. - check power unit. - increase p0346.
F07412	Drive: Commutation angle incorrect (motor model)
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	ENCODER (NONE, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	An incorrect commutation angle was detected that can result in a positive coupling in the speed controller. Possible causes: - the motor encoder is incorrectly adjusted with respect to the magnet position. - the motor encoder is damaged. - the angular commutation offset is incorrectly set (p0431). - data to calculate the motor model has been incorrectly set (p0356 (motor-stator leakage inductance) and/or p0350 (motor-stator resistance) and/or p0352 (cable resistance)). - the changeover speed for the motor model is too low (p1752). The monitoring function only becomes effective above the changeover speed. - pole position identification might have calculated an incorrect value when activated (p1982 = 1). - the motor encoder speed signal is faulted. - the control loop is instable due to incorrect parameterization. Fault value (r0949, interpret decimal): SERVO: 0: The comparison of the pole position angle from the encoder and the motor model resulted in an excessively high value (> 80 ° electrical). 1: - VECTOR: 0: The comparison of the pole position angle from the encoder and the motor model resulted in an excessively high value (> 45 ° electrical). 1: The change in the speed signal from the motor encoder has changed by > p0492 within a current controller clock cycle.

Remedy:

- if the encoder mounting was changed - re-adjust the encoder.
- replace the defective motor encoder.
- correctly set the angular commutation offset (p0431).
- correctly set the motor stator resistance, cable resistance and motor-stator leakage inductance (p0350, p0352, p0356).
- increase the changeover speed for the motor model (p1752). The monitoring is completely deactivated for p1752 > p1082 (maximum speed).
- with pole position identification activated (p1982 = 1) check the procedure for pole position identification (p1980) and force a new pole position identification procedure by means of deselection followed by selection (p1982 = 0 -> 1).

Note:
For High Dynamic Motors (1FK7xxx-7xxx), for applications with a higher current, if necessary, the monitoring should be disabled.

F07413 Drive: Commutation angle incorrect (pole position identification)

Message value: -

Drive object: SERVO, VECTOR

Reaction: ENCODER (NONE, OFF2)

Acknowledge: IMMEDIATELY

Cause: An incorrect commutation angle was detected that can result in a positive coupling in the speed controller.

- within the pole position identification routine (p1982 = 2):
A difference of > 45° electrical to the encoder angle was determined.
- for VECTOR, within the encoder adjustment (p1990 = 2):
A difference of > 6° electrical to the encoder angle was determined.

Remedy:

- correctly set the angular commutation offset (p0431).
- re-adjust the motor encoder after the encoder has been replaced.
- replace the defective motor encoder.
- check the pole position identification routine. If the pole position identification routine is not suitable for this motor type, then disable the plausibility check (p1982 = 0).

F07414 (N, A) Drive: Encoder serial number changed

Message value: -

Drive object: SERVO

Reaction: ENCODER (NONE, OFF2)

Acknowledge: IMMEDIATELY

Cause: The serial number of the motor encoder of a synchronous motor has changed. The change was only checked for encoders with serial number (e.g. EnDat encoders) and build-in motors (e.g. p0300 = 401) or third-party motors (p0300 = 2).

Cause 1:
The encoder was replaced.

Cause 2:
A third-party, build-in or linear motor was re-commissioned.

Cause 3:
The motor with integrated and adjusted encoder was replaced.

Cause 4:
The firmware was updated to a version that checks the encoder serial number.

Remedy: Re causes 1, 2:
Carry out an automatic adjustment using the pole position identification routine. Acknowledge the fault. Initiate the pole position identification routine with p1990 = 1. Then check that the pole position identification routine is correctly executed.
SERVO:
If a pole position identification technique is selected in p1980, and if p0301 does not contain a motor type with an encoder adjusted in the factory, then p1990 is automatically activated.
or
Set the adjustment via p0431. In this case, the new serial number is automatically accepted.
or
Mechanically adjust the encoder. Accept the new serial number with p0440 = 1.
Re causes 3, 4:
Accept the new serial number with p0440 = 1.

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

N07415 (F) Drive: Angular commutation offset transfer running

Message value: -
Drive object: SERVO
Reaction: OFF2
Acknowledge: NONE
Cause: The angular commutation offset was automatically determined using p1990 = 1.
This fault causes the pulses to be suppressed - this is necessary to transfer the angular commutation offset to p0431.
See also: p1990 (Encoder adjustment, determine angular commutation offset)

Remedy: The fault can be acknowledged without any additional measures.

Reaction upon F: OFF2
Acknowl. upon F: IMMEDIATELY

A07416 Drive: Flux controller configuration

Message value: Parameter: %1, Index: %2, fault cause: %3
Drive object: VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The configuration of the flux control (p1401) is contradictory.
Alarm value (r2124, interpret hexadecimal):
ccbbaaaa hex
aaaa = Parameter
bb = Index
cc = fault cause
cc = 01 hex = 1 dec:
Quick magnetizing (p1401.6) for soft start (p1401.0).
cc = 02 hex = 2 dec:
Quick magnetizing (p1401.6) for flux build-up control (p1401.2).
cc = 03 hex = 3 dec:
Quick magnetizing (p1401.6) for Rs identification after restart (p0621 = 2).

Remedy: Re fault cause = 1:
- Shut down soft start (p1401.0 = 0).
- Shut down quick magnetizing (p1401.6 = 0).
Re fault cause = 2:
- De-energize flux build-up control (p1401.2 = 0).
- Shut down quick magnetizing (p1401.6 = 0).
Re fault cause = 3:
- Re-parameterize Rs identification (p0621 = 0, 1)
- Shut down quick magnetizing (p1401.6 = 0).

F07417	Drive: Pulse technique not plausible (motor model)
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF2 (NONE)
Acknowledge:	IMMEDIATELY
Cause:	The evaluation of the test pulse response found incorrect values. Fault value (r0949, interpret decimal): Fault value 0: An impermissible pulse technique configuration was detected during ramp-up. Possible causes: - The pulse technique was initially selected during system ramp-up (p1750.5=1) but the power unit component does not support the current oversampling required (see r192.23). As a consequence, p1750.0 was deselected automatically. Fault value 10: The pulse response is repeatedly implausible. Possible causes: - Incorrect configuration of the power unit component - The power unit component is faulty. Fault value 20: At the preset pulse amplitude the measured pulse response is much higher than the expected value. Possible causes: - The orientation loop has been affected by major vibration phenomena. - The motor is short-circuited for high frequencies (output filter). - The motor is damaged.
Remedy:	Re fault value 0: Once the pulse technique has been deselected automatically (p1750.5=0), there are two possible options: - acknowledge the fault and save parameter p1750.5=0 => field-oriented control mode to standstill is forfeited and replaced by transition to open-loop control at low speeds. - replace the power unit firmware (at least V4.30) => field-oriented control mode to standstill is available. Re fault value 10: With active selection of the pulse technique (p1750.5) - power both the CU and the power unit component down and then back up again, or - perform a manual warm restart (see P0009, P0976). If this does not solve the problem: Replace the power unit component. Re fault value 20: - control parameters might have been adjusted (factory setting, commissioning). - filters must not be connected between motor and converter/inverter. - check the motor.
F07420	Drive: Current setpoint filter natural frequency > Shannon frequency
Message value:	%1
Drive object:	SERVO
Reaction:	NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	One of the filter natural frequencies is greater than the Shannon frequency. The Shannon frequency is calculated according to the following formula: $0.5 / p0115[0]$ Fault value (r0949, interpret hexadecimal): Bit 0: Filter 1 (p1658, p1660) Bit 1: Filter 2 (p1663, p1665) Bit 2: Filter 3 (p1668, p1670) Bit 3: Filter 4 (p1673, p1675) Bit 8 ... 15: Data set number (starting from zero).
Remedy:	- reduce the numerator or denominator natural frequency of the current setpoint filter involved. - reduce the current controller sampling time (p0115[0]). - switch out the filter involved (p1656).

F07421	Drive: Speed setpoint filter natural frequency > Shannon frequency
Message value:	%1
Drive object:	SERVO
Reaction:	NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	One of the filter natural frequencies is greater than the Shannon frequency. The Shannon frequency is calculated according to the following formula: $0.5 / p0115[1]$ Fault value (r0949, interpret hexadecimal): Bit 0: Filter 1 (p1417, p1419) Bit 1: Filter 2 (p1423, p1425) Bit 8 ... 15: Data set number (starting from zero)
Remedy:	- reduce the numerator or denominator natural frequency of the speed setpoint filter involved. - reduce the speed controller sampling time (p0115[1]). - switch out the filter involved (p1414).
F07422	Drive: Reference model natural frequency > Shannon frequency
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The natural filter frequency of the PT2 element for the reference model (p1433) is greater than the Shannon frequency. The Shannon frequency is calculated according to the following formula: $0.5 / p0115[1]$
Remedy:	- reduce the natural frequency of PT2 element for reference model (p1433). - reduce the speed controller sampling time (p0115[1]).
F07423	Drive: APC filter natural frequency > Shannon frequency
Message value:	%1
Drive object:	SERVO
Reaction:	NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	One of the filter natural frequencies is greater than the Shannon frequency. The Shannon frequency is calculated according to the following formula: $0.5 / (p0115[1] * x)$ Fault value (r0949, interpret hexadecimal): Bit 0: Filter 1.1 (p3711, p3713), $x = 1$ Bit 4: Filter 2.1 (p3721, p3723), $x = p3706$ Bit 5: Filter 2.2 (p3726, p3728), $x = p3706$ Bit 8: Filter 3.1 (p3731, p3733), $x = p3707$ Bit 9: Filter 3.2 (p3736, p3738), $x = p3707$ Bit 16 ... 32: Data set number (starting from zero)
Remedy:	- reduce the numerator or denominator natural frequency of the filter involved. - reduce the speed controller sampling time (p0115[1]) or the sub-sampling (p3706, p3707). - switch out the filter involved (p3704).
A07424	Drive: Operating condition for APC not valid
Message value:	Fault cause: %1 bin
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	The APC function (Advanced Positioning Control) has identified an invalid operating condition. Alarm value (r2124, interpret hexadecimal): Bit 0 = 1: APC is operating without encoder Bit 1 = 1: The load measuring system for APC, selected using p3701, has a fault. The APC function is disabled. Bit 2 = 1: The load measuring system for APC, selected using p3701, has a fault. The pulse de-coupling is disabled, i.e. the speed of the motor measuring system is used as speed for the closed-loop motor speed control.

Remedy: Re bit 0:
Only use the APC function in operation with an encoder.
Re bit 1, 2:
Check the load measuring system.

F07425 Drive: APC monitoring time for speed limit expired

Message value: -
Drive object: SERVO
Reaction: OFF1
Acknowledge: IMMEDIATELY
Cause: The limit value (p3778) for the speed/velocity was exceeded for a time longer than that set in the monitoring time (p3779).
Note:
APC: Advanced Positioning Control
Remedy: - check the measured value.
- check the limit value and monitoring time (p3778, p3779).

F07426 (A) Technology controller actual value limited

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The actual signal of the technology controller has reached its limit.
- Fault value = 1: upper limit reached
- Fault value = 2: lower limit reached
Remedy: - Adapt the limits in P2267, P2268 to the signal level.
- Check scaling of the actual signal.
See also: p2264 (Technology controller actual value), p2267 (Technology controller upper limit actual value), p2268 (Technology controller lower limit actual value)
Reaction upon A: NONE
Acknowl. upon A: NONE

F07429 Drive: DSC without encoder not possible

Message value: -
Drive object: SERVO
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The function DSC (Dynamic Servo Control) was activated although there is no encoder.
See also: p1191 (DSC position controller gain KPC)
Remedy: If there is no encoder and connector input p1191 (DSC position controller gain) is interconnected, then connector input p1191 must have a 0 signal.

F07430 Drive: Changeover to open-loop torque controlled operation not possible

Message value: -
Drive object: SERVO
Reaction: OFF2 (NONE, OFF1, OFF3)
Acknowledge: IMMEDIATELY
Cause: For encoderless operation, the converter cannot change over to closed-loop torque-controlled operation (BI: p1501).
Remedy: Do not attempt to cover over to closed-loop torque-controlled operation.

F07431	Drive: Changeover to encoderless operation not possible
Message value:	-
Drive object:	SERVO
Reaction:	OFF2 (OFF1)
Acknowledge:	IMMEDIATELY
Cause:	For closed-loop torque control, the converter cannot change over to encoderless operation (p1404).
Remedy:	Do not attempt to change over to encoderless operation.
F07432	Drive: Synchronous motor without overvoltage protection
Message value:	%1
Drive object:	SERVO
Reaction:	OFF2 (OFF1)
Acknowledge:	IMMEDIATELY
Cause:	Under voltage conditions, a synchronous motor can generate an overvoltage condition that can destroy the drive system. Fault value (r0949, interpret hexadecimal): Associated Drive Data Set (DDS).
Remedy:	Overvoltage protection can be implemented in the following ways: - limit the maximum speed (p1082) without any additional protection. The maximum speed without protection is calculated as follows: Rotary motors: $p1082 \text{ [rpm]} \leq 11.695 * p0297/p0316 \text{ [Nm/A]}$ Linear motors: $p1082 \text{ [m/min]} \leq 73.484 * p0297/p0316 \text{ [N/A]}$ - use a voltage protection module (VPM) in conjunction with the function "Safe Torque Off" (p9601, p9801). When a fault condition exists, the VPM short-circuits the motors. During the short-circuit, the pulses must be suppressed - this means that the terminals for the function "Safe Torque Off" must be connected to the VPM. When using a VPM, p0643 must be set to 1. - activating the internal voltage protection (IVP) with p1231 = 3. See also: p0643 (Overvoltage protection for synchronous motors), p1231 (Armature short-circuit / DC brake configuration)
F07433	Drive: Closed-loop control with encoder is not possible as the encoder has not been unparked
Message value:	%1
Drive object:	SERVO
Reaction:	NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The changeover to closed-loop control with encoder is not possible as the encoder has not been unparked.
Remedy:	- check whether the encoder firmware supports the "parking" function (r0481.6 = 1). - upgrade the firmware. Note: For long-stator motors (p3870.0 = 1), the following applies: The encoder must have completed the unparking procedure (r3875.0 = 1) before a changeover can be made to closed-loop control with encoder. The encoder is unparked with a 0/1 edge at BI: p3876 and remains unparked until a 0 signal is again present.
F07434	Drive: It is not possible to change the direction of rotation with the pulses enabled
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A drive data set was selected - with the pulses enabled - which has a different parameterized direction of rotation (p1821). It is only possible to change the motor direction of rotation using p1821 when the pulses are inhibited.
Remedy:	- change over the drive data set with the pulses inhibited. - ensure that the changeover to a drive data set does not result in the motor direction of rotation being changed (i.e. for these drive data sets, the same value must be in p1821). See also: p1821

F07435 (N)	Drive: Setting the ramp-function generator for encoderless vector control
Message value:	Parameter: %1
Drive object:	VECTOR
Reaction:	OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	During operation with encoderless vector control (r1407.1) the ramp-function generator was stopped (p1141) or bypassed (p1122). An internal setting command of the ramp-function generator output caused the set setpoint speed to be frozen or was not able to be realized.
Remedy:	- deactivate the holding command for the ramp-function generator (p1141). - do not bypass the ramp-function generator (p1122). - suppress the fault (p2101, p2119). This is necessary if the ramp-function generator is held using jogging and the speed setpoint is simultaneously inhibited (r0898.6). Note: For encoderless vector control it is not practical to read-in the main setpoint of the speed control via p1155 or p1160 (p0922). In this case, the main setpoint should be injected before the ramp-function generator (p1070). The reason for this is that the ramp-function generator output is automatically set when transitioning from closed-loop speed controlled into open-loop speed controlled operation.
Reaction upon N:	NONE
Acknowled. upon N:	NONE
A07440	EPOS: Jerk time is limited
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The calculation of the jerk time $Tr = \text{MAX}(p2572, p2573) / p2574$ resulted in an excessively high value so that the jerk time is internally limited to 1000 ms.
	Note: The alarm is also output if jerk limiting is not active.
Remedy:	- increase the jerk limiting (p2574). - reduce maximum acceleration or maximum deceleration (p2572, p2573). See also: p2572 (EPOS maximum acceleration), p2573 (EPOS maximum deceleration), p2574 (EPOS jerk limiting)
A07441	LR: Save the position offset of the absolute encoder adjustment
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The status of the absolute encoder adjustment has changed. In order to permanently save the determined position offset (p2525) it must be saved in a non-volatile fashion (p0971, p0977).
Remedy:	None necessary. This alarm automatically disappears after the offset has been saved. See also: p2507 (LR absolute encoder adjustment status), p2525 (LR encoder adjustment, offset)
F07442 (A)	LR: Multiturn does not match the modulo range
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The ratio between the multiturn resolution and the modulo range (p2576) is not an integer number. This results in the adjustment being set back, as the position actual value cannot be reproduced after power-off/power-on.

Remedy: Make the ratio between the multiturn resolution and the modulo range an integer number.

The ratio v is calculated as follows:

1. Motor encoder without position tracking:

$$v = (p0421 * p2506 * p0433 * p2505) / (p0432 * p2504 * p2576)$$

2. Motor encoder with position tracking for the measuring gear:

$$v = (p0412 * p2506 * p2505) / (p2504 * p2576)$$

3. Motor encoder with position tracking for the load gear:

$$v = (p2721 * p2506 * p0433) / (p0432 * p2576)$$

4. Motor encoder with position tracking for the load and measuring gear:

$$v = (p2721 * p2506) / p2576$$

5. Direct encoder without position tracking:

$$v = (p0421 * p2506 * p0433) / (p0432 * p2576)$$

6. Direct encoder with position tracking for the measuring gear:

$$v = (p0412 * p2506) / p2576$$

Note:

With position tracking, it is recommended that p0412 and p2721 are changed

See also: p0412, p0432, p0433, p2504, p2505, p2506, p2576, p2721

Reaction upon A: NONE

Acknowl. upon A: NONE

F07443 (A) LR: Reference point coordinate not in the permissible range

Message value: %1

Drive object: SERVO, VECTOR

Reaction: OFF1 (OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: The reference point coordinate received when adjusting the encoder via connector input p2599 lies outside the half of the encoder range and cannot be set as actual axis position.

Fault value (r0949, interpret decimal):

Maximum permissible value for the reference point coordinate.

Remedy: Set the reference point coordinate to a lower value than specified in the fault value.

See also: p2598 (EPOS reference point coordinate, signal source), p2599 (EPOS reference point coordinate value)

Reaction upon A: NONE

Acknowl. upon A: NONE

F07446 (A) Load gear: Position tracking cannot be reset

Message value: %1

Drive object: SERVO, VECTOR

Reaction: OFF1 (OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: The position tracking cannot be reset.

Remedy: Reset the position tracking as follows:

- select encoder commissioning (p0010 = 4).

- reset position tracking, position (p2720.2 = 1).

- deselect encoder commissioning (p0010 = 0).

The fault should then be acknowledged and, if necessary, the absolute encoder adjusted (p2507).

Reaction upon A: NONE

Acknowl. upon A: NONE

F07447	Load gear: Position tracking, maximum actual value exceeded
Message value:	Component number: %1, encoder data set: %2, drive data set: %3
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	When the position tracking of the load gear is configured, the drive/encoder (motor encoder) identifies a maximum possible absolute position actual value (r2723) that can no longer be represented within 32 bits. Maximum value: $p0408 * 2^{p0419}$ Fault value (r0949, interpret hexadecimal): ccbbaa hex aa = encoder data set bb = component number cc = drive data set See also: p0408 (Rotary encoder pulse No.), p0419 (Fine resolution absolute value Gx_XIST2 (in bits)), p2721 (Load gear, rotary absolute gearbox, revolutions, virtual)
Remedy:	- reduce the fine resolution (p0419). - reduce the multiturn resolution (p2721). See also: p0419 (Fine resolution absolute value Gx_XIST2 (in bits)), p2721 (Load gear, rotary absolute gearbox, revolutions, virtual)
F07448 (A)	Load gear: Position tracking, linear axis has exceeded the maximum range
Message value:	-
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	For a configured linear axis/no modulo axis, the currently effective motor encoder (encoder 1) has exceeded the maximum possible traversing range. For the configured linear axis, the maximum traversing range is defined to be $64x (+/- 32x)$ of p0421. It should be read in p2721 and interpreted as the number of load revolutions. Note: Only the motor encoder in the currently effective drive data set is monitored here. The actual effective drive data set is displayed in $x = r0051$ and the corresponding motor encoder is specified in in p0187[x].
Remedy:	The fault should be resolved as follows: - select encoder commissioning (p0010 = 4). - reset position tracking, position (p2720.2 = 1). - deselect encoder commissioning (p0010 = 0). The fault should then be acknowledged and the absolute encoder adjusted.
Reaction upon A:	NONE
Acknowled. upon A:	NONE
F07449 (A)	Load gear: Position tracking, actual position outside tolerance window
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	When powered down, the currently effective motor encoder was moved through a distance greater than was parameterized in the tolerance window. It is possible that there is no longer any reference between the mechanical system and encoder. Note: Only the motor encoder in the currently effective drive data set is monitored here. The actual effective drive data set is displayed in $x = r0051$ and the corresponding motor encoder is specified in in p0187[x]. Fault value (r0949, interpret decimal): Deviation (difference) to the last encoder position in increments of the absolute value after the measuring gear - if one is being used. The sign designates the traversing direction. Note: The deviation (difference) found is also displayed in r2724. See also: p2722 (Load gear, position tracking tolerance window), r2724 (Load gear position difference)

Remedy: Reset the position tracking as follows:
 - select encoder commissioning (p0010 = 4).
 - reset position tracking, position (p2720.2 = 1).
 - deselect encoder commissioning (p0010 = 0).
 The fault should then be acknowledged and, if necessary, the absolute encoder adjusted (p2507).
 See also: p0010, p2507

Reaction upon A: NONE
 Acknowl. upon A: NONE

F07450 (A) LR: Standstill monitoring has responded

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: After the standstill monitoring time (p2543) expired, the drive left the standstill window (p2542).
 - position actual value inversion incorrectly set (p0410).
 - standstill window set too small (p2542).
 - standstill monitoring time set too low (p2543).
 - position loop gain too low (p2538).
 - position loop gain too high (instability/oscillation, p2538).
 - mechanical overload.
 - check the connecting cable, motor/drive converter (phase missing, interchange).
 - when selecting motor identification, select tracking mode (BI: p2655[0] = 1 signal).
 - when selecting function generator, select tracking mode (BI: p2655[0] = 1 signal) and deactivate position control (BI:p2550 = 0 signal).
Remedy: Check the causes and resolve.
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F07451 (A) LR: Position monitoring has responded

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: When the position monitoring time (p2545) expired, the drive had still not reached the positioning window (p2544).
 - positioning window parameterized too small (p2544).
 - position monitoring time parameterized too short (p2545).
 - position loop gain too low (p2538).
 - position loop gain too high (instability/oscillation, p2538).
 - drive mechanically locked.
Remedy: Check the causes and resolve.
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F07452 (A) LR: Following error too high

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The difference between the position setpoint position actual value (following error dynamic model, r2563) is greater than the tolerance (p2546).
 - the drive torque or accelerating capacity exceeded.
 - position measuring system fault.
 - position control sense incorrect.
 - mechanical system locked.
 - excessively high traversing velocity or excessively high position reference value (setpoint) differences

Remedy: Check the causes and resolve.
Reaction upon A: NONE
Acknowled. upon A: NONE

F07453 **LR: Position actual value preprocessing error**
Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: An error has occurred during the position actual value preprocessing.
Remedy: Check the encoder for the position actual value preprocessing.
See also: p2502 (LR encoder assignment)

A07454 **LR: Position actual value preprocessing does not have a valid encoder**
Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: One of the following problems has occurred with the position actual value preprocessing:
- an encoder is not assigned for the position actual value preprocessing (p2502 = 0).
- an encoder is assigned, but no encoder data set (p0187 = 99 or p0188 = 99 or p0189 = 99).
- an encoder and an encoder data set have been assigned, however, the encoder data set does not contain any encoder data (p0400 = 0) or invalid data (e.g. p0408 = 0).
Remedy: Check the drive data sets, encoder data sets and encoder assignment.
See also: p0187 (Encoder 1 encoder data set number), p0188 (Encoder 2 encoder data set number), p0189 (Encoder 3 encoder data set number), p0400 (Encoder type selection), p2502 (LR encoder assignment)

A07455 **EPOS: Maximum velocity limited**
Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The maximum velocity (p2571) is too high to correctly calculate the modulo correction.
Within the sampling time for positioning (p0115[5]), with the maximum velocity, a maximum of the half modulo length must be moved through. p2571 was limited to this value.
Remedy: - reduce the maximum velocity (p2571).
- increase the sampling time for positioning (p0115[5]).

A07456 **EPOS: Setpoint velocity limited**
Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The actual setpoint velocity is greater than the parameterized maximum velocity (p2571) and is therefore limited.
Remedy: - check the entered setpoint velocity.
- reduce the velocity override (CI: p2646).
- increase the maximum velocity (p2571).
- check the signal source for the externally limited velocity (CI: p2594).

A07457	EPOS: Combination of input signals illegal
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	An illegal combination of input signals that are simultaneously set was identified. Alarm value (r2124, interpret decimal): 0: Jog 1 and jog 2 (p2589, p2590). 1: Jog 1 or jog 2 and direct setpoint input/MDI (p2589, p2590, p2647). 2: Jog 1 or jog 2 and start referencing (p2589, p2590, p2595). 3: Jog 1 or jog 2 and activate traversing task (p2589, p2590, p2631). 4: Direct setpoint input/MDI and starting referencing (p2647, p2595). 5: Direct setpoint input/MDI and activate traversing task (p2647, p2631). 6: Start referencing and activate traversing task (p2595, p2631).
Remedy:	Check the appropriate input signals and correct.
F07458	EPOS: Reference cam not found
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	After starting the search for reference, the axis moved through the maximum permissible distance to search for the reference cam without actually finding the reference cam.
Remedy:	- check the "reference cam" binector input (BI: p2612). - check the maximum permissible distance to the reference cam (p2606). - if axis does not have any reference cam, then set p2607 to 0. See also: p2606 (EPOS search for reference, reference cam, maximum distance), p2607 (EPOS search for reference, reference cam present), p2612 (EPOS search for reference, reference cam)
F07459	EPOS: No zero mark
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	After leaving the reference cam, the axis has traversed the maximum permissible distance between the reference cam and zero mark without finding the zero mark.
Remedy:	- check the encoder regarding the zero mark - check the maximum permissible distance between the reference cam and zero mark (p2609). - use an external encoder zero mark (equivalent zero mark) (p0495). See also: p0495 (Equivalent zero mark, input terminal), p2609 (EPOS search for reference, max. distance ref. cam and zero mark)
F07460	EPOS: End of reference cam not found
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	During the search for reference, when the axis reached the zero mark it also reached the end of the traversing range without detecting an edge at the binector input "reference cam" (BI: p2612). Maximum traversing range: -2147483648 [LU] ... -2147483647 [LU]
Remedy:	- check the "reference cam" binector input (BI: p2612). - repeat the search for reference. See also: p2612 (EPOS search for reference, reference cam)

A07461	EPOS: Reference point not set
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When starting a traversing block/direct setpoint input, a reference point is not set (r2684.11 = 0).
Remedy:	Reference the system (search for reference, flying referencing, set reference point).

A07462	EPOS: Selected traversing block number does not exist
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A traversing block selected via BI: p2625 to BI: p2630 was started via BI: p2631 = 0/1 edge "Activate traversing task". - the number of the started traversing block is not contained in p2616[0...n]. - the started traversing block is suppressed. Alarm value (r2124, interpret decimal): Number of the selected traversing block that is also not available.
Remedy:	- correct the traversing program. - select an available traversing block number.

A07463 (F)	EPOS: External block change not requested in the traversing block
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	For a traversing block with the block change enable CONTINUE_EXTERNAL_ALARM, the external block change was not requested. Alarm value (r2124, interpret decimal): Number of the traversing block.
Remedy:	Resolve the reason as to why the edge is missing at binector input (BI: p2632).
Reaction upon F:	OFF1
Acknowl. upon F:	IMMEDIATELY

F07464	EPOS: Traversing block is inconsistent
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The traversing block does not contain valid information. Alarm value (r2124, interpret decimal): Number of the traversing block with invalid information.
Remedy:	Check the traversing block and where relevant, take into consideration alarms that are present.

A07465	EPOS: Traversing block does not have a subsequent block
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	There is no subsequent block in the traversing block. Alarm value (r2124, interpret decimal): Number of the traversing block with the missing subsequent block.
Remedy:	- parameterize this traversing block with the block change enable END. - parameterize additional traversing blocks with a higher block number and for the last block, using the block change enable END.

A07466	EPOS: Traversing block number assigned a multiple number of times
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The same traversing block number was assigned a multiple number of times. Alarm value (r2124, interpret decimal): Number of the traversing block that was assigned a multiple number of times.
Remedy:	Correct the traversing blocks.
A07467	EPOS: Traversing block has illegal task parameters
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The task parameter in the traversing block contains an illegal value. Alarm value (r2124, interpret decimal): Number of the traversing block with an illegal task parameter.
Remedy:	Correct the task parameter in the traversing block.
A07468	EPOS: Traversing block jump destination does not exist
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In a traversing block, a jump was programmed to a non-existent block. Alarm value (r2124, interpret decimal): Number of the traversing block with a jump destination that does not exist.
Remedy:	- correct the traversing block. - add the missing traversing block.
A07469	EPOS: Traversing block < target position < software limit switch minus
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the traversing block the specified absolute target position lies outside the range limited by the software limit switch minus. Alarm value (r2124, interpret decimal): Number of the traversing block with illegal target position.
Remedy:	- correct the traversing block. - change software limit switch minus (CI: p2578, p2580).
A07470	EPOS: Traversing block > target position > software limit switch plus
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the traversing block the specified absolute target position lies outside the range limited by the software limit switch plus. Alarm value (r2124, interpret decimal): Number of the traversing block with illegal target position.
Remedy:	- correct the traversing block. - change software limit switch plus (CI: p2579, p2581).

A07471	EPOS: Traversing block target position outside the modulo range
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the traversing block the target position lies outside the modulo range. Alarm value (r2124, interpret decimal): Number of the traversing block with illegal target position.
Remedy:	- in the traversing block, correct the target position. - change the modulo range (p2576).

A07472	EPOS: Traversing block ABS_POS/ABS_NEG not possible
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the traversing block the positioning mode ABS_POS or ABS_NEG were parameterized with the modulo correction not activated. Alarm value (r2124, interpret decimal): Number of the traversing block with the illegal positioning mode.
Remedy:	Correct the traversing block.

A07473 (F)	EPOS: Beginning of traversing range reached
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When traversing, the axis has moved to the traversing range limit.
Remedy:	Move away in the positive direction.
Reaction upon F:	OFF1 (OFF2, OFF3)
Acknowled. upon F:	IMMEDIATELY

A07474 (F)	EPOS: End of traversing range reached
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When traversing, the axis has moved to the traversing range limit.
Remedy:	Move away in the negative direction.
Reaction upon F:	OFF1 (OFF2, OFF3)
Acknowled. upon F:	IMMEDIATELY

F07475 (A)	EPOS: Target position < start of traversing range
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The target position for relative traversing lies outside the traversing range.
Remedy:	Correct the target position.
Reaction upon A:	NONE
Acknowled. upon A:	NONE

F07476 (A)	EPOS: Target position > end of the traversing range
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The target position for relative traversing lies outside the traversing range.
Remedy:	Correct the target position.
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A07477 (F)	EPOS: Target position < software limit switch minus
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the actual traversing operation, the target position is less than the software limit switch minus.
Remedy:	- correct the target position. - change software limit switch minus (CI: p2578, p2580). See also: p2578 (EPOS software limit switch minus signal source), p2580 (EPOS software limit switch minus), p2582 (EPOS software limit switch activation)
Reaction upon F:	OFF1 (OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY
A07478 (F)	EPOS: Target position > software limit switch plus
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the actual traversing operation, the target position is greater than the software limit switch plus.
Remedy:	- correct the target position. - change software limit switch plus (CI: p2579, p2581). See also: p2579 (EPOS software limit switch plus signal source), p2581 (EPOS software limit switch plus), p2582 (EPOS software limit switch activation)
Reaction upon F:	OFF1 (OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY
A07479	EPOS: Software limit switch minus reached
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The axis is at the position of the software limit switch minus. An active traversing block was interrupted.
Remedy:	- correct the target position. - change software limit switch minus (CI: p2578, p2580). See also: p2578 (EPOS software limit switch minus signal source), p2580 (EPOS software limit switch minus), p2582 (EPOS software limit switch activation)
A07480	EPOS: Software limit switch plus reached
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The axis is at the position of the software limit switch plus. An active traversing block was interrupted.

Remedy:

- correct the target position.
- change software limit switch plus (CI: p2579, p2581).

See also: p2579 (EPOS software limit switch plus signal source), p2581 (EPOS software limit switch plus), p2582 (EPOS software limit switch activation)

F07481 (A) EPOS: Axis position < software limit switch minus

Message value: -

Drive object: SERVO, VECTOR

Reaction: OFF1 (OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: The actual position of the axis is less than the position of the software limit switch minus.

Remedy:

- correct the target position.
- change software limit switch minus (CI: p2578, p2580).

See also: p2578 (EPOS software limit switch minus signal source), p2580 (EPOS software limit switch minus), p2582 (EPOS software limit switch activation)

Reaction upon A: NONE

Acknowled. upon A: NONE

F07482 (A) EPOS: Axis position > software limit switch plus

Message value: -

Drive object: SERVO, VECTOR

Reaction: OFF1 (OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: The actual position of the axis is greater than the position of the software limit switch plus.

Remedy:

- correct the target position.
- change software limit switch plus (CI: p2579, p2581).

See also: p2579 (EPOS software limit switch plus signal source), p2581 (EPOS software limit switch plus), p2582 (EPOS software limit switch activation)

Reaction upon A: NONE

Acknowled. upon A: NONE

A07483 EPOS: Travel to fixed stop clamping torque not reached

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The fixed stop in the traversing block was reached without the clamping torque/clamping force having been achieved.

Remedy:

- Check the maximum torque-generating current (r1533).
- check the torque limits (p1520, p1521).
- check the power limits (p1530, p1531).
- check the BICO interconnections of the torque limits (p1522, p1523, p1528, p1529).

F07484 EPOS: Fixed stop outside the monitoring window

Message value: -

Drive object: SERVO, VECTOR

Reaction: OFF3 (OFF1, OFF2)

Acknowledge: IMMEDIATELY

Cause: In the "fixed stop reached" state, the axis has moved outside the defined monitoring window (p2635).

Remedy:

- check the monitoring window (p2635).
- check the mechanical system.

F07485 (A)	EPOS: Fixed stop not reached
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	In a traversing block with the task FIXED STOP, the end position was reached without detecting a fixed stop.
Remedy:	- check the traversing block and locate the target position further into the workpiece. - check the "fixed stop reached" control signal (p2637). - if required, reduce the maximum following error window to detect the fixed stop (p2634).
Reaction upon A:	NONE
Acknowled. upon A:	NONE
A07486	EPOS: Intermediate stop missing
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the modes "traversing blocks" or "direct setpoint input/MDI" at the start of motion, the binector input "no intermediate stop/intermediate stop" (BI: p2640) did not have a 1 signal.
Remedy:	Connect a 1 signal to the binector input "no intermediate stop/intermediate stop" (BI: p2640) and re-start motion. See also: p2640 (EPOS intermediate stop (0 signal))
A07487	EPOS: Reject traversing task missing
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the modes "traversing blocks" or "direct setpoint input/MDI" at the start of motion, the binector input "do not reject traversing task/reject traversing task" (BI: p2641) does not have a 1 signal.
Remedy:	Connect a 1 signal to the binector input "do not reject traversing task/reject traversing task" (BI: p2641) and restart motion. See also: p2641 (EPOS reject traversing task (0 signal))
F07488	EPOS: Relative positioning not possible
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	In the mode "direct setpoint input/MDI", for continuous transfer (p2649 = 1) relative positioning was selected (BI: p2648 = 0 signal).
Remedy:	Check the control.
A07489	EPOS: Reference point correction outside the window
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	For the function "flying referencing" the difference between the measured position at the measuring probe and the reference point coordinate lies outside the parameterized window.
Remedy:	- check the mechanical system. - check the parameterization of the window (p2602).

F07490	EPOS: Enable signal withdrawn while traversing
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	- for a standard assignment, another fault may have occurred as a result of withdrawing the enable signals. - the drive is in the "switching on inhibited" state (for a standard assignment).
Remedy:	- set the enable signals or check the cause of the fault that first occurred and then result (for a standard assignment). - check the assignment to enable the basic positioning function.

F07491 (A)	EPOS: STOP cam minus reached
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF3
Acknowledge:	IMMEDIATELY
Cause:	A zero signal was detected at binector input BI: p2569, i.e. the STOP cam minus was reached. For a positive traversing direction, the STOP cam minus was reached - i.e. the wiring of the STOP cam is incorrect. See also: p2569 (EPOS STOP cam minus)
Remedy:	- leave the STOP cam minus in the positive traversing direction and return the axis to the valid traversing range. - check the wiring of the STOP cam.
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F07492 (A)	EPOS: STOP cam plus reached
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF3
Acknowledge:	IMMEDIATELY
Cause:	A zero signal was detected at binector input BI: p2570, i.e. the STOP cam plus was reached. For a negative traversing direction, the STOP cam plus was reached - i.e. the wiring of the STOP cam is incorrect. See also: p2570 (EPOS STOP cam plus)
Remedy:	- leave the STOP cam plus in the negative traversing direction and return the axis to the valid traversing range. - check the wiring of the STOP cam.
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F07493	LR: Overflow of the value range for position actual value
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The value range (-2147483648 ... 2147483647) for the position actual value representation was exceeded. When the overflow occurs, the "referenced" or "adjustment absolute measuring system" status is reset. Fault value (r0949, interpret decimal): 1: The position actual value (r2521) has exceeded the value range. 2: The encoder position actual value Gn_XIST2 (r0483) or the absolute value after the load gear (r2723) has exceeded the value range. 3: The maximum encoder value times the factor to convert the absolute position (r0483 and/or r2723) from increments to length units (LU) has exceeded the value range for displaying the position actual value.

Remedy: If required, reduce the traversing range or position resolution (p2506).
 Increase the fine resolution of absolute position actual value (p0419).
 Reference to fault value = 3:
 If the value for the maximum possible absolute position (LU) is greater than 4294967296, then it is not possible to make an adjustment due to an overflow.
 For rotary encoders, the maximum possible absolute position (LU) is calculated as follows:
 1. Motor encoder without position tracking
 $p2506 * p0433 * p2505 / (p0432 * p2504)$
 $p2506 * p0433 * p2505 * p0421 / (p0432 * p2504)$ for multiturn encoders
 2. Motor encoder with position tracking for measuring gear:
 $p2506 * p0412 * p2505 / p2504$
 3. Motor encoder with position tracking for load gear
 $p2506 * p2721 * p0433 / p0432$
 4. Motor encoder with position tracking for load and measuring gear
 $p2506 * p2721$
 5. Direct encoder without position tracking
 $p2506 * p0433 / p0432$
 $p2506 * p0433 * p0421 / p0432$ for multiturn encoders
 6. Direct encoder with position tracking for measuring gear
 $p2506 * p0412$

F07494 LR: Drive Data Set changeover in operation

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A drive data set changeover (DDS changeover) when the mechanical relationships change (p2503 .. 2506), the direction of rotation (p1821) or the encoder assignment (p2502) were requested during operation.
Remedy: To changeover the drive data set, initially, exit the "operation" mode.

A07495 (F) LR: Reference function interrupted

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An activated reference function (reference mark search or measuring probe evaluation) was interrupted.
 - an encoder fault has occurred (Gn_ZSW.15 = 1).
 - position actual value was set during an activated reference function.
 - simultaneously activate reference mark search and measuring probe evaluation (BI: p2508 and BI: p2509 = 1 signal).
 - activated reference function (reference mark search or measuring probe evaluation) was deactivated (BI: p2508 and BI: p2509 = 0 signal).
Remedy:
 - check the causes and resolve.
 - reset the control (BI: p2508 and BI: p2509 = 0 signal) and activate the requested function.
 Reaction upon F: OFF1 (OFF2, OFF3)
 Acknowled. upon F: IMMEDIATELY

A07496 EPOS: Enable not possible

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: It is not possible to enable the basic positioner because at least one signal is missing.
 Alarm value (r2124, interpret decimal):
 1: EPOS enable missing (BI: p2656).
 2: Position actual value, valid feedback signal missing (BI: p2658).
 See also: p2656 (EPOS enable basic positioner), p2658 (EPOS pos. actual value valid, feedback signal)
Remedy: Check the appropriate binector inputs and signals.

A07497 LR: Position setting value activated

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The position actual value is set to the value received via CI: p2515 while BI: p2514 = 1 signal. A possible system deviation cannot be corrected.
Remedy: None necessary.
The alarm automatically disappears with BI: p2514 = 0 signal.

A07498 (F) LR: Measuring probe evaluation not possible

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When evaluating the measuring probe, an error occurred.
Alarm value (r2124, interpret decimal):
6: The input terminal for the measuring probe is not set.
4098: Error when initializing the measuring probe.
4100: The measuring pulse frequency is too high.
> 50000: The measuring clock cycle is not a multiple integer of the position controller clock cycle.
Remedy: Deactivate the measuring probe evaluation (BI: p2509 = 0 signal).
Re alarm value = 6:
Set the input terminal for the measuring probe (p0488, p0489 or p2517, p2518).
Re alarm value = 4098:
Check the Control Unit hardware.
Re alarm value = 4100:
Reduce the frequency of the measuring pulses at the measuring probe.
Re alarm value > 50000:
Set the clock cycle ratio of the measuring clock cycle to the position controller clock cycle to an integer multiple.
To do this, the currently effective measuring clock cycle can be determined from the alarm value as follows:
 $T_{\text{meas}} [125 \mu\text{s}] = \text{alarm value} - 50000$
With PROFIBUS, the measuring clock cycle corresponds to the PROFIBUS clock cycle (r2064[1]).
Without PROFIBUS, the measuring clock cycle is an internal cycle time that cannot be influenced.
Reaction upon F: OFF1
Acknowled. upon F: IMMEDIATELY

F07499 (A) EPOS: Reversing cam approached with the incorrect traversing direction

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF3
Acknowledge: IMMEDIATELY
Cause: The reversing cam MINUS was approached in the positive traversing direction or the reversing cam PLUS was approached in the negative traversing direction.
See also: p2613 (EPOS search for reference reversing cam minus), p2614 (EPOS search for reference reversing cam plus)
Remedy: - check the wiring of the reversing cam (BI: p2613, BI: p2614).
- check the traversing direction to approach the reversing cam.
Reaction upon A: NONE
Acknowled. upon A: NONE

F07500	Drive: Power unit data set PDS not configured
Message value:	Drive data set: %1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	Only for controlled line supply infeed/regenerative feedback units: The power unit data set was not configured - this means that a data set number was not entered into the drive data set. Fault value (r0949, interpret decimal): Drive data set number of p0185.
Remedy:	The index of the power unit data set associated with the drive data set should be entered into p0185.
F07501	Drive: Motor Data Set MDS not configured
Message value:	Drive data set: %1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Only for power units: The motor data set was not configured - this means that a data set number was not entered into the associated drive data set. Fault value (r0949, interpret decimal): The fault value includes the drive data set number of p0186.
Remedy:	The index of the motor data set associated with the drive data set should be entered into p0186. See also: p0186 (Motor Data Sets (MDS) number)
F07502	Drive: Encoder Data Set EDS not configured
Message value:	Drive data set: %1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Only for power units: The encoder data set was not configured - this means that a data set number was not entered into the associated drive data set. Fault value (r0949, interpret decimal): The fault value includes the drive data set number of p0187, p0188 and p0189. The fault value is increased by 100 * encoder number (e.g. for p0189: Fault value 3xx with xx = data set number).
Remedy:	The index of the encoder data set associated with the drive data set should be entered into p0187 (1st encoder), p0188 (2nd encoder) and p0189 (3rd encoder).
A07504	Drive: Motor data set is not assigned to a drive data set
Message value:	%1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A motor data set is not assigned to a drive object. All of the existing motor data sets in the drive data sets must be assigned using the MDS number (p0186[0...n]). There must be at least as many drive data sets as motor data sets. Alarm value (r2124, interpret decimal): Number of the motor data set that has not been assigned.
Remedy:	In the drive data sets, assign the non-assigned motor data set using the MDS number (p0186[0...n]). - check whether all of the motor data sets are assigned to drive data sets. - if required, delete superfluous motor data sets. - if required, set up new drive data sets and assign to the corresponding motor data sets. See also: p0186 (Motor Data Sets (MDS) number)

F07509	Drive: Component number missing
Message value:	%1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A Drive Data Set (DDS) is assigned to a Motor Data Set (MDS) or Encoder Data Set (EDS) that does not have a component number. Alarm value (r2124, interpret decimal): nnmmmmxyyy nn: Number of the MDS/EDS. mmm: Parameter number of the missing component number. xx: Number of the DDS that is assigned to the MDS/EDS. yyy: Parameter number that references the MDS/EDS. Example: p0186[7] = 5: DDS 7 is assigned MDS 5. p0131[5] = 0: There is no component number set in MDS 5. Alarm value = 0513107186
Remedy:	In the drive data sets, no longer assign MDS/EDS using p0186, p0187, p0188, p0189 or set a valid component number. See also: p0131, p0141, p0142, p0186, p0187, p0188, p0189
F07510	Drive: Identical encoder in the drive data set
Message value:	%1
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	More than one encoder with identical component number is assigned to a single drive data set. In one drive data set, it is not permissible that identical encoders are operated together. Fault value (r0949, interpret decimal): 1000 * first identical encoder + 100 * second identical encoder + drive data set. Example: Fault value = 1203 means: In drive data set 3, the first (p0187[3]) and second encoder (p0188[3]) are identical.
Remedy:	Assign the drive data set to different encoders. See also: p0141, p0187 (Encoder 1 encoder data set number), p0188 (Encoder 2 encoder data set number), p0189 (Encoder 3 encoder data set number)
F07511	Drive: Encoder used a multiple number of times
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	Each encoder may only be assigned to one drive and within a drive must - in each drive data set - either always be encoder 1, always encoder 2 or always encoder 3. This unique assignment has been violated. Fault value (r0949, interpret decimal): The two parameters in coded form, that refer to the same component number. First parameter: Index: First and second decimal place (99 for EDS, not assigned DDS) Parameter number: Third decimal place (1 for p0187, 2 for p0188, 3 for p0189, 4 for EDS not assigned DDS) Drive number: Fourth and fifth decimal place Second parameter: Index: Sixth and seventh decimal place (99 for EDS, not assigned DDS) Parameter number: Eighth decimal place (1 for p0187, 2 for p0188, 3 for p0189, 4 for EDS, not assigned DDS) Drive number: Ninth and tenth decimal place See also: p0141
Remedy:	Correct the double use of a component number using the two parameters coded in the fault value.

F07512	Drive: Encoder data set changeover cannot be parameterized
Message value:	%1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Using p0141, a changeover of the encoder data set is prepared that is illegal. In this firmware release, an encoder data set changeover is only permitted for the components in the actual topology. Alarm value (r2124, interpret decimal): Incorrect EDS data set number. See also: p0187 (Encoder 1 encoder data set number), p0188 (Encoder 2 encoder data set number), p0189 (Encoder 3 encoder data set number)
Remedy:	Every encoder data set must be assigned its own dedicated DRIVE-CLiQ socket. The component numbers of the encoder interfaces (p0141) must have different values within a drive object. The following must apply: p0141[0] not equal to p0141[1] not equal to ... not equal to p0141[n]
A07514 (N)	Drive: Data structure does not correspond to the interface module
Message value:	-
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The interface mode "SIMODRIVE 611 universal" was set (p2038 = 1) and the data structure does not correspond to this mode. For the data structure, the following rule must be complied with. Within the group of 8 drive data sets, the assignment to the motor data set must be set the same: p0186[0] = p0186[1] = ... = p0186[7] p0186[8] = p0186[9] = ... = p0186[15] p0186[16] = p0186[17] = ... = p0186[23] p0186[24] = p0186[25] = ... = p0186[31] See also: p0180 (Number of Drive Data Sets (DDS)), p0186 (Motor Data Sets (MDS) number), p2038 (PROFIdrive STW/ZSW interface mode)
Remedy:	- structure the data according to the rules of the "SIMODRIVE 611 universal" interface mode. - check the interface mode (p2038).
Reaction upon N:	NONE
Acknowled. upon N:	NONE
F07515	Drive: Power unit and motor incorrectly connected
Message value:	%1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A power unit (via PDS) was assigned to a motor (via MDS) in a drive data set that is not connected in the target topology. Alarm value (r2124, interpret decimal): Number of the incorrectly parameterized drive data set.
Remedy:	- assign the drive data set to a combination of motor and power unit permitted by the target topology. - adapt the target topology. See also: p0121 (Power unit component number), p0131 (Motor component number), p0186 (Motor Data Sets (MDS) number)

F07516	Drive: Re-commission the data set
Message value:	%1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The assignment between the drive data set and motor data set (p0186) or between the drive data set and the encoder data set was modified (p0187). This is the reason that the drive data set must re-commissioned. Fault value (r0949, interpret decimal): Drive data set to be re-commissioned.
Remedy:	Commission the drive data set specified in the fault value (r0949).

F07517	Drive: Encoder data set changeover incorrectly parameterized
Message value:	%1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	An MDS cannot have different motor encoders in two different DDS. The following parameterization therefore results results in an error: p0186[0] = 0, p0187[0] = 0 p0186[0] = 0, p0187[0] = 1 Alarm value (r2124, interpret decimal): The lower 16 bits indicate the first DDS and the upper 16 bits indicate the second DDS.
Remedy:	If you wish to operate a motor once with one motor encoder and then another time with the other motor encoder, then you must set up two different MDSs, in which the motor data are the same. Example: p0186[0] = 0, p0187[0] = 0 p0186[0] = 1, p0187[0] = 1

F07518	Drive: Motor data set changeover incorrectly parameterized
Message value:	%1
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The system has identified that two motor data sets were incorrectly parameterized. Parameter r0313 (calculated from p0314, p0310, p0311), r0315 and p1982 may only have different values if the motor data sets are assigned different motors. p0827 is used to assign the motors and/contactors. It is not possible to toggle between motor data sets. Alarm value (r2124, interpret hexadecimal): xxxxyyyy: xxxx: First DDS with assigned MDS, yyyy: Second DDS with assigned MDS
Remedy:	Correct the parameterization of the motor data sets.

A07519	Drive: Motor changeover incorrectly parameterized
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	With the setting p0833.0 = 1, a motor changeover via the application is selected. This is the reason that p0827 must have different values in the appropriate motor data set. Alarm value (r2124, interpret hexadecimal): xxxxyyyy: xxxx: First MDS, yyyy: Second MDS
Remedy:	- parameterize the appropriate motor data sets differently (p0827). - select the setting p0833.0 = 0 (motor changeover via the drive).

A07520	Drive: Motor cannot be changed over
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The motor cannot be changed over. Alarm value (r2124, interpret decimal): 1: The contactor for the motor that is presently active cannot be opened, because for a synchronous motor, the speed (r0063) is greater than the speed at the start of field weakening (p3048). As long as r0063 > p0348, the current in the motor does not decay in spite of the pulses being suppressed. 2: The "contactor opened" feedback signal was not detected within 1 s. 3: The "contactor closed" feedback signal was not detected within 1 s.
Remedy:	Re alarm value = 1: Set the speed lower than the speed at the start of field weakening (r0063 < p0348). Re alarm value = 2, 3: Check the feedback signals of the contactor involved.
A07530	Drive: Drive Data Set DDS not present
Message value:	-
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The selected drive data set is not available (p0837 > p0180). The drive data set was not changed over. See also: p0180, p0820, p0821, p0822, p0823, p0824, r0837
Remedy:	- select the existing drive data set. - set up additional drive data sets.
A07541	Drive: Data set changeover not possible
Message value:	-
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The selected drive data set changeover and the assigned motor changeover are not possible and are not carried out. For synchronous motors, the motor contactor may only be switched for actual speeds less than the speed at the start of field weakening (r0063 < p0348). See also: r0063, p0348
Remedy:	Reduce the speed below the speed at the start of field weakening.
A07550 (F, N)	Drive: Not possible to reset encoder parameters
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When carrying out a factory setting (e.g. using p0970 = 1), it was not possible to reset the encoder parameters. The encoder parameters are directly read out of the encoder via DRIVE-CLiQ. Alarm value (r2124, interpret decimal): Component number of the encoder involved.
Remedy:	- repeat the operation. - check the DRIVE-CLiQ connection.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE

F07551	Drive encoder: No commutation angle information
Message value:	Fault cause: %1, drive data set: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF2 (IASC/DCBRAKE)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The commutation angle information is missing. This means that synchronous motors cannot be controlled (closed-loop control) Fault value (r0949, interpret decimal): yyyyxxxx dec: yyyy = fault cause, xxxx = drive data set yyyy = 1 dec: The motor encoder used does not supply an absolute commutation angle. yyyy = 2 dec: The selected ratio of the measuring gear does not match the motor pole pair number.
Remedy:	Re fault cause = 1: - check the encoder parameterization (p0404). - use an encoder with track C/D, EnDat interface of Hall sensors. - use an encoder with sinusoidal A/B track for which the motor pole pair number (r0313) is an integer multiple of the encoder pulse number (p0408). - activate the pole position identification routine (p1982 = 1). Re fault cause = 2: - the quotient of the pole pair number divided by the ratio of the measuring gear must be an integer number: (p0314 * p0433) / p0432. Note: For operation with track C/D, this quotient must be less than 8. See also: p0402 (Gearbox type selection), p0404 (Encoder configuration effective), p0432 (Gearbox factor, encoder revolutions), p0433 (Gearbox factor, motor/load revolutions)
F07552 (A)	Drive encoder: Encoder configuration not supported
Message value:	Fault cause: %1, component number: %2, encoder data set: %3
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The requested encoder configuration is not supported. Only bits may be requested in p0404 that are signaled as being supported by the encoder evaluation in r0456. Fault value (r0949, interpret decimal): Low word low byte: Encoder data set number Low word high byte: Component number High word: The encoder evaluation does not support a function selected in p0404. 1: sin/cos encoder with absolute track (this is supported by SME25). 3: Squarewave encoder (this is supported by SMC30). 4: sin/cos encoder (this is supported by SMC20, SMI20, SME20, SME25). 12: sin/cos encoder with reference mark (this is supported by SME20). 15: Commutation with zero mark for separately-excited synchronous motors with VECTORMV. 23: Resolver (this is supported by SMC10, SMI10). 65535: Other function (compare r0456 and p0404). See also: p0404 (Encoder configuration effective), r0456 (Encoder configuration supported)
Remedy:	- check the encoder parameterization (p0400, p0404). - use the matching encoder evaluation (r0456).
Reaction upon A:	NONE
Acknowled. upon A:	NONE

F07553 (A)	Drive encoder: Sensor Module configuration not supported
Message value:	Encoder data set: %1, first incorrect bit: %2, incorrect parameter: %3
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The Sensor Module does not support the requested configuration. For incorrect p0430 (cc = 0), the following applies: - In p0430 (requested functions), at least 1 bit was set that is not set in r0458 (supported functions) (exception: Bit 19, 28, 29, 30, 31). - p1982 > 0 (pole position identification requested), but r0458.16 = 0 (pole position identification not supported). For incorrect p0437 (cc = 1), the following applies: - In p0437 (requested functions), at least 1 bit was set that is not set in r0459 (supported functions). Fault value (r0949, interpret hexadecimal): ddccbbaa hex aa: encoder data set number bb: first incorrect bit cc: incorrect parameter cc = 0: incorrect parameter is p0430 cc = 1: incorrect parameter is p0437 cc = 2: incorrect parameter is r0459 dd: reserved (always 0)
Remedy:	- check the encoder parameterization (p0430, p0437). - check the pole position identification routine (p1982). - use the matching encoder evaluation (r0458, r0459). See also: p0430, p0437, r0458, r0459, p1982
Reaction upon A:	NONE
Acknowled. upon A:	NONE
F07555 (A)	Drive encoder: Configuration position tracking
Message value:	Component number: %1, encoder data set: %2, drive data set: %3, fault cause: %4
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The configuration of the position tracking is not supported. Position tracking can only be activated for absolute encoders. For linear axes, it is not possible to simultaneously activate the position tracking for load and measuring gears. Fault value (r0949, interpret hexadecimal): ddccbbaa hex aa = encoder data set bb = component number cc = drive data set dd = fault cause dd = 00 hex = 0 dez An absolute encoder is not being used. dd = 01 hex = 1 dez Position tracking cannot be activated because the memory of the internal NVRAM is not sufficient or a Control Unit does not have an NVRAM. dd = 02 hex = 2 dez For a linear axis, the position tracking was activated for the load and measuring gear. dd = 03 hex = 3 dez Position tracking cannot be activated because position tracking with another gear ratio, axis type or tolerance window has already been detected for this encoder data set. dd = 04 hex = 4 dez A linear encoder is being used. See also: p0404 (Encoder configuration effective), p0411 (Measuring gear, configuration)

Remedy:

- use an absolute encoder.
- if necessary, deselect the position tracking (p0411 for the measuring gear, p2720 for the load gear).
- use a Control Unit with sufficient NVRAM.
- Only activate position tracking of the load gear in the same encoder data set if the gear ratio (p2504, p2505), axis type (p2720.1) and tolerance window (p2722) are also the same.

Reaction upon A: NONE

Acknowled. upon A: NONE

F07556 Measuring gear: Position tracking, maximum actual value exceeded

Message value: Component number: %1, encoder data set: %2

Drive object: ENCODER, SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY

Cause: When the position tracking of the measuring gear is configured, the drive/encoder identifies a maximum possible absolute position actual value (r0483) that cannot be represented within 32 bits.
Maximum value: $p0408 * p0412 * 2^{p0419}$
Fault value (r0949, interpret decimal):
Low word low byte: Encoder data set number
Low word high byte: Component number
See also: p0408 (Rotary encoder pulse No.), p0412 (Measuring gear, absolute encoder, rotary, revolutions, virtual), p0419 (Fine resolution absolute value Gx_XIST2 (in bits))

Remedy:

- reduce the fine resolution (p0419).
- reduce the multiturn resolution (p0412).

See also: p0412 (Measuring gear, absolute encoder, rotary, revolutions, virtual), p0419 (Fine resolution absolute value Gx_XIST2 (in bits))

A07557 (F) Encoder 1: Reference point coordinate not in the permissible range

Message value: %1

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The reference point coordinate received when adjusting the encoder via connector input CI:p2599 lies outside the half of the encoder range and cannot be set as actual axis position. The maximum permissible value is displayed in the supplementary information.

Remedy: Set the reference point coordinate less than the value from the supplementary information.
See also: p2598 (EPOS reference point coordinate, signal source)

Reaction upon F: OFF1 (OFF2, OFF3)

Acknowled. upon F: IMMEDIATELY

A07558 (F) Encoder 2: Reference point coordinate not in the permissible range

Message value: %1

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The reference point coordinate received when adjusting the encoder via connector input CI:p2599 lies outside the half of the encoder range and cannot be set as actual axis position. The maximum permissible value is displayed in the supplementary information.

Remedy: Set the reference point coordinate less than the value from the supplementary information.
See also: p2598 (EPOS reference point coordinate, signal source)

Reaction upon F: OFF1 (OFF2, OFF3)

Acknowled. upon F: IMMEDIATELY

A07559 (F)	Encoder 3: Reference point coordinate not in the permissible range
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The reference point coordinate received when adjusting the encoder via connector input CI:p2599 lies outside the half of the encoder range and cannot be set as actual axis position. The maximum permissible value is displayed in the supplementary information.
Remedy:	Set the reference point coordinate less than the value from the supplementary information. See also: p2598 (EPOS reference point coordinate, signal source)
Reaction upon F:	OFF1 (OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY
F07560	Drive encoder: Number of pulses is not to the power of two
Message value:	Encoder data set: %1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	For rotary absolute encoders, the pulse number in p0408 must be to the power of two. Fault value (r0949, interpret decimal): The fault value includes the encoder data set number involved.
Remedy:	- check the parameterization (p0408, p0404.1, r0458.5). - upgrade the Sensor Module firmware if necessary
F07561	Drive encoder: Number of multiturn pulses is not to the power of two
Message value:	Encoder data set: %1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The multiturn resolution in p0421 must be to the power of two. Fault value (r0949, interpret decimal): The fault value includes the encoder data set number involved.
Remedy:	- check the parameterization (p0421, p0404.1, r0458.5). - upgrade the Sensor Module firmware if necessary
F07562 (A)	Drive, encoder: Position tracking, incremental encoder not possible
Message value:	Fault cause: %1, component number: %2, encoder data set: %3
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The requested position tracking for incremental encoders is not supported. Fault value (r0949, interpret hexadecimal): ccccbbaa hex aa = encoder Data Set number bb = component number cccc = fault cause cccc = 00 hex = 0 dec The encoder type does not support the "Position tracking incremental encoder" function. cccc = 01 hex = 1 dec Position tracking cannot be activated because the memory of the internal NVRAM is not sufficient or a Control Unit does not have an NVRAM. cccc = 04 hex = 4 dec A linear encoder is used that does not support the position tracking function. See also: p0404 (Encoder configuration effective), p0411 (Measuring gear, configuration), r0456 (Encoder configuration supported)

Remedy:

- check the encoder parameterization (p0400, p0404).
- use a Control Unit with sufficient NVRAM.
- if required, deselect position tracking for the incremental encoder (p0411.3 = 0).

Reaction upon A: NONE
Acknowl. upon A: NONE

A07565 (F, N) Drive: Encoder error in PROFIdrive encoder interface 1

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An encoder error was signaled for encoder 1 via the PROFIdrive encoder interface (G1_ZSW.15).
Alarm value (r2124, interpret decimal):
Error code from G1_XIST2, refer to the description regarding r0483.
Note:
This alarm is only output if p0480[0] is not equal to zero.
Remedy: Acknowledge the encoder error using the encoder control word (G1_STW.15 = 1).
Reaction upon F: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A07566 (F, N) Drive: Encoder error in PROFIdrive encoder interface 2

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An encoder error was signaled for encoder 2 via the PROFIdrive encoder interface (G2_ZSW.15).
Alarm value (r2124, interpret decimal):
Error code from G2_XIST2, refer to the description regarding r0483.
Note:
This alarm is only output if p0480[1] is not equal to zero.
Remedy: Acknowledge the encoder error using the encoder control word (G2_STW.15 = 1).
Reaction upon F: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A07567 (F, N) Drive: Encoder error in PROFIdrive encoder interface 3

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An encoder error was signaled for encoder 3 via the PROFIdrive encoder interface (G3_ZSW.15).
Alarm value (r2124, interpret decimal):
Error code from G3_XIST2, refer to the description regarding r0483.
Note:
This alarm is only output if p0480[2] is not equal to zero.
Remedy: Acknowledge the encoder error using the encoder control word (G3_STW.15 = 1).
Reaction upon F: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A07569 (F)	Encoder could not be identified
Message value:	-
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	During encoder identification (waiting) with p0400 = 10100, the encoder could not be identified. Either the wrong encoder has been installed or no encoder has been installed, the wrong encoder cable has been connected or no encoder cable has been connected to the Sensor Module, or the DRIVE-CLiQ component has not been connected to DRIVE-CLiQ. Note: Encoder identification must be supported by the encoder and is possible in the following cases: Encoder with EnDat interface, motor with DRIVE-CLiQ.
Remedy:	- check and, if necessary, connect the encoder and/or encoder cable. - check and, if necessary, establish the DRIVE-CLiQ connection. - in the case of encoders that cannot be identified (e.g. encoders without EnDat interface), the correct encoder type must be entered in p0400.
Reaction upon F:	NONE (OFF1, OFF2, OFF3)
Acknowled. upon F:	IMMEDIATELY
F07575	Drive: Motor encoder not ready
Message value:	-
Drive object:	A_INF, B_INF, ENCODER, S_INF, SERVO, TM41, VECTOR
Reaction:	Infeed: OFF2 Servo: OFF2 (ENCODER) Vector: OFF2 (ENCODER)
Acknowledge:	IMMEDIATELY
Cause:	The motor encoder signals that it is not ready. - initialization of encoder 1 (motor encoder) was unsuccessful. - the function "parking encoder" is active (encoder control word G1_STW.14 = 1). - the encoder interface (Sensor Module) is deactivated (p0145). - the Sensor Module is defective.
Remedy:	Evaluate other queued faults via encoder 1.
A07576	Drive: Encoderless operation due to a fault active
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Encoderless operation is active due to a fault (r1407.13). The required response when an encoder fault occurs is parameterized in p0491. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	
A07577 (F)	Encoder 1: Measuring probe evaluation not possible
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When evaluating the measuring probe, an error occurred. Alarm value (r2124, interpret decimal): 6: The input terminal for the measuring probe is not set. 4098: Error when initializing the measuring probe. 4100: The measuring pulse frequency is too high. 4200: The PROFIBUS clock cycle is not a multiple of integer of the position controller clock cycle.

Remedy: Deactivate the measuring probe evaluation (BI: p2509 = 0 signal).
Re alarm value = 6:
Set the input terminal for the measuring probe (p0488, p0489 or p2517, p2518).
Re alarm value = 4098:
Check the Control Unit hardware.
Re alarm value = 4100:
Reduce the frequency of the measuring pulses at the measuring probe.
Re alarm value = 4200:
Set the clock cycle ratio between the PROFIBUS clock cycle and the position controller clock cycle to an integer multiple.
Reaction upon F: OFF1
Acknowled. upon F: IMMEDIATELY

A07578 (F) Encoder 2: Measuring probe evaluation not possible

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When evaluating the measuring probe, an error occurred.
Alarm value (r2124, interpret decimal):
6: The input terminal for the measuring probe is not set.
4098: Error when initializing the measuring probe.
4100: The measuring pulse frequency is too high.
4200: The PROFIBUS clock cycle is not a multiple of integer of the position controller clock cycle.
Remedy: Deactivate the measuring probe evaluation (BI: p2509 = 0 signal).
Re alarm value = 6:
Set the input terminal for the measuring probe (p0488, p0489 or p2517, p2518).
Re alarm value = 4098:
Check the Control Unit hardware.
Re alarm value = 4100:
Reduce the frequency of the measuring pulses at the measuring probe.
Re alarm value = 4200:
Set the clock cycle ratio between the PROFIBUS clock cycle and the position controller clock cycle to an integer multiple.
Reaction upon F: OFF1
Acknowled. upon F: IMMEDIATELY

A07579 (F) Encoder 3: Measuring probe evaluation not possible

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When evaluating the measuring probe, an error occurred.
Alarm value (r2124, interpret decimal):
6: The input terminal for the measuring probe is not set.
4098: Error when initializing the measuring probe.
4100: The measuring pulse frequency is too high.
4200: The PROFIBUS clock cycle is not a multiple of integer of the position controller clock cycle.
Remedy: Deactivate the measuring probe evaluation (BI: p2509 = 0 signal).
Re alarm value = 6:
Set the input terminal for the measuring probe (p0488, p0489 or p2517, p2518).
Re alarm value = 4098:
Check the Control Unit hardware.
Re alarm value = 4100:
Reduce the frequency of the measuring pulses at the measuring probe.
Re alarm value = 4200:
Set the clock cycle ratio between the PROFIBUS clock cycle and the position controller clock cycle to an integer multiple.
Reaction upon F: OFF1
Acknowled. upon F: IMMEDIATELY

A07580 (F, N) Drive: No Sensor Module with matching component number

Message value: Encoder data set: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: A Sensor Module with the component number specified in p0141 was not found.
 Alarm value (r2124, interpret decimal):
 Encoder data set involved (index of p0141).
Remedy: Correct parameter p0141.
 Reaction upon F: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
 Acknowl. upon F: IMMEDIATELY (POWER ON)
 Reaction upon N: NONE
 Acknowl. upon N: NONE

A07581 (F) Encoder 1: Position actual value preprocessing error

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An error has occurred during the position actual value preprocessing.
Remedy: Check the encoder for the position actual value preprocessing.
 See also: p2502 (LR encoder assignment)
 Reaction upon F: OFF1 (OFF2, OFF3)
 Acknowl. upon F: IMMEDIATELY

A07582 (F) Encoder 2: Position actual value preprocessing error

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An error has occurred during the position actual value preprocessing.
Remedy: Check the encoder for the position actual value preprocessing.
 See also: p2502 (LR encoder assignment)
 Reaction upon F: OFF1 (OFF2, OFF3)
 Acknowl. upon F: IMMEDIATELY

A07583 (F) Encoder 3: Position actual value preprocessing error

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An error has occurred during the position actual value preprocessing.
Remedy: Check the encoder for the position actual value preprocessing.
 See also: p2502 (LR encoder assignment)
 Reaction upon F: OFF1 (OFF2, OFF3)
 Acknowl. upon F: IMMEDIATELY

A07584 Encoder 1: Position setting value activated

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The position actual value is set to the value received via CI: p2515 while BI: p2514 = 1 signal. A possible system deviation cannot be corrected.
Remedy: None necessary.
The alarm automatically disappears with BI: p2514 = 0 signal.

A07585 Encoder 2: Position setting value activated

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The position actual value is set to the value received via CI: p2515 while BI: p2514 = 1 signal. A possible system deviation cannot be corrected.
Remedy: None necessary.
The alarm automatically disappears with BI: p2514 = 0 signal.

A07586 Encoder 3: Position setting value activated

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The position actual value is set to the value received via CI: p2515 while BI: p2514 = 1 signal. A possible system deviation cannot be corrected.
Remedy: None necessary.
The alarm automatically disappears with BI: p2514 = 0 signal.

A07587 Encoder 1: Position actual value preprocessing does not have a valid encoder

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The following problem has occurred during the position actual value preprocessing.
- an encoder data set has been assigned, however, the encoder data set does not contain any encoder data (p0400 = 0) or invalid data (e.g. p0408 = 0).
Remedy: Check the drive data sets, encoder data sets.
See also: p0187 (Encoder 1 encoder data set number), p0188 (Encoder 2 encoder data set number), p0189 (Encoder 3 encoder data set number), p0400 (Encoder type selection), p2502 (LR encoder assignment)

A07588 Encoder 2: Position actual value preprocessing does not have a valid encoder

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The following problem has occurred during the position actual value preprocessing.
- an encoder data set has been assigned, however, the encoder data set does not contain any encoder data (p0400 = 0) or invalid data (e.g. p0408 = 0).
Remedy: Check the drive data sets, encoder data sets.
See also: p0187 (Encoder 1 encoder data set number), p0188 (Encoder 2 encoder data set number), p0189 (Encoder 3 encoder data set number), p0400 (Encoder type selection), p2502 (LR encoder assignment)

A07589	Encoder 3: Position actual value preprocessing does not have a valid encoder
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The following problem has occurred during the position actual value preprocessing. - an encoder data set has been assigned, however, the encoder data set does not contain any encoder data (p0400 = 0) or invalid data (e.g. p0408 = 0).
Remedy:	Check the drive data sets, encoder data sets. See also: p0187 (Encoder 1 encoder data set number), p0188 (Encoder 2 encoder data set number), p0189 (Encoder 3 encoder data set number), p0400 (Encoder type selection), p2502 (LR encoder assignment)
A07590 (F)	Encoder 1: Drive Data Set changeover in operation
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A Drive Data Set changeover (DDS) with a change of the mechanical relationships and the encoder assignment (p2502) was requested in operation.
Remedy:	To changeover the drive data set, initially, exit the "operation" mode.
Reaction upon F:	OFF1 (OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY
A07591 (F)	Encoder 2: Drive Data Set changeover in operation
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A Drive Data Set changeover (DDS) with a change of the mechanical relationships and the encoder assignment (p2502) was requested in operation.
Remedy:	To changeover the drive data set, initially, exit the "operation" mode.
Reaction upon F:	OFF1 (OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY
A07592 (F)	Encoder 3: Drive Data Set changeover in operation
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A Drive Data Set changeover (DDS) with a change of the mechanical relationships and the encoder assignment (p2502) was requested in operation.
Remedy:	To changeover the drive data set, initially, exit the "operation" mode.
Reaction upon F:	OFF1 (OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY

A07593 (F)	Encoder 1: Value range for position actual value exceeded
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The value range (-2147483648 ... 2147483647) for the position actual value representation was exceeded. When the overflow occurs, the "referenced" or "adjustment absolute measuring system" status is reset. Fault value (r0949, interpret decimal): 1: The position actual value (r2521) has exceeded the value range. 2: The encoder position actual value Gn_XIST2 (r0483) or the absolute value after the load gear (r2723) has exceeded the value range. 3: The maximum encoder value times the factor to convert the absolute position (r0483 and/or r2723) from increments to length units (LU) has exceeded the value range for displaying the position actual value.
Remedy:	If required, reduce the traversing range or position resolution. Re alarm value = 3: Reducing the position resolution and conversion factor: - reduce the length unit (LU) per load revolution for rotary encoders (p2506). - increase the fine resolution of absolute position actual values (p0419).
Reaction upon F:	OFF1 (OFF2, OFF3)
Acknowled. upon F:	IMMEDIATELY

A07594 (F)	Encoder 2: Value range for position actual value exceeded
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The value range (-2147483648 ... 2147483647) for the position actual value representation was exceeded. When the overflow occurs, the "referenced" or "adjustment absolute measuring system" status is reset. Fault value (r0949, interpret decimal): 1: The position actual value (r2521) has exceeded the value range. 2: The encoder position actual value Gn_XIST2 (r0483) or the absolute value after the load gear (r2723) has exceeded the value range. 3: The maximum encoder value times the factor to convert the absolute position (r0483 and/or r2723) from increments to length units (LU) has exceeded the value range for displaying the position actual value.
Remedy:	If required, reduce the traversing range or position resolution. Re alarm value = 3: Reducing the position resolution and conversion factor: - reduce the length unit (LU) per load revolution for rotary encoders (p2506). - increase the fine resolution of absolute position actual values (p0419).
Reaction upon F:	OFF1 (OFF2, OFF3)
Acknowled. upon F:	IMMEDIATELY

A07595 (F)	Encoder 3: Value range for position actual value exceeded
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The value range (-2147483648 ... 2147483647) for the position actual value representation was exceeded. When the overflow occurs, the "referenced" or "adjustment absolute measuring system" status is reset. Fault value (r0949, interpret decimal): 1: The position actual value (r2521) has exceeded the value range. 2: The encoder position actual value Gn_XIST2 (r0483) or the absolute value after the load gear (r2723) has exceeded the value range. 3: The maximum encoder value times the factor to convert the absolute position (r0483 and/or r2723) from increments to length units (LU) has exceeded the value range for displaying the position actual value.

Remedy: If required, reduce the traversing range or position resolution.
Re alarm value = 3:
Reducing the position resolution and conversion factor:
- reduce the length unit (LU) per load revolution for rotary encoders (p2506).
- increase the fine resolution of absolute position actual values (p0419).

Reaction upon F: OFF1 (OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY

A07596 (F) Encoder 1: Reference function interrupted

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An activated reference function (reference mark search or measuring probe evaluation) was interrupted.
- an encoder fault has occurred (Gn_ZSW.15 = 1).
- position actual value was set during an activated reference function.
- simultaneously activate reference mark search and measuring probe evaluation (BI: p2508 and BI: p2509 = 1 signal).
- activated reference function (reference mark search or measuring probe evaluation) was deactivated (BI: p2508 and BI: p2509 = 0 signal).

Remedy: - check the causes and resolve.
- reset the control (BI: p2508 and BI: p2509 = 0 signal) and activate the requested function.

Reaction upon F: OFF1 (OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY

A07597 (F) Encoder 2: Reference function interrupted

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An activated reference function (reference mark search or measuring probe evaluation) was interrupted.
- an encoder fault has occurred (Gn_ZSW.15 = 1).
- position actual value was set during an activated reference function.
- simultaneously activate reference mark search and measuring probe evaluation (BI: p2508 and BI: p2509 = 1 signal).
- activated reference function (reference mark search or measuring probe evaluation) was deactivated (BI: p2508 and BI: p2509 = 0 signal).

Remedy: - check the causes and resolve.
- reset the control (BI: p2508 and BI: p2509 = 0 signal) and activate the requested function.

Reaction upon F: OFF1 (OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY

A07598 (F) Encoder 3: Reference function interrupted

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An activated reference function (reference mark search or measuring probe evaluation) was interrupted.
- an encoder fault has occurred (Gn_ZSW.15 = 1).
- position actual value was set during an activated reference function.
- simultaneously activate reference mark search and measuring probe evaluation (BI: p2508 and BI: p2509 = 1 signal).
- activated reference function (reference mark search or measuring probe evaluation) was deactivated (BI: p2508 and BI: p2509 = 0 signal).

Remedy: - check the causes and resolve.
- reset the control (BI: p2508 and BI: p2509 = 0 signal) and activate the requested function.

Reaction upon F: OFF1 (OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY

F07599 (A) Encoder 1: Adjustment not possible

Message value: Drive data set: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The maximum encoder value times the factor to convert the absolute position (r0483 and/or r2723) from increments to length units (LU) has exceeded the value range (-2147483648 ... 2147483647) for displaying the position actual value.
Remedy: If the value for the maximum possible absolute position (LU) is greater than 4294967296, then it is not possible to make an adjustment due to an overflow.
For rotary encoders, the maximum possible absolute position (LU) is calculated as follows:
1. Motor encoder without position tracking:
 $p2506 * p0433 * p2505 / (p0432 * p2504)$
 $p2506 * p0433 * p2505 * p0421 / (p0432 * p2504)$ for multiturn encoders
2. Motor encoder with position tracking for measuring gear:
 $p2506 * p0412 * p2505 / p2504$
3. Motor encoder with position tracking for load gear:
 $p2506 * p2721 * p0433 / p0432$
4. Motor encoder with position tracking for load and measuring gear:
 $p2506 * p2721$
5. Direct encoder without position tracking:
 $p2506 * p0433 / p0432$
 $p2506 * p0433 * p0421 / p0432$ for multiturn encoders
6. Direct encoder with position tracking for measuring gear:
 $p2506 * p0412$
Reaction upon A: NONE
Acknowl. upon A: NONE

F07600 (A) Encoder 2: Adjustment not possible

Message value: Drive data set: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The maximum encoder value times the factor to convert the absolute position (r0483 and/or r2723) from increments to length units (LU) has exceeded the value range (-2147483648 ... 2147483647) for displaying the position actual value.
Remedy: If the value for the maximum possible absolute position (LU) is greater than 4294967296, then it is not possible to make an adjustment due to an overflow.
For rotary encoders, the maximum possible absolute position (LU) is calculated as follows:
1. Motor encoder without position tracking:
 $p2506 * p0433 * p2505 / (p0432 * p2504)$
 $p2506 * p0433 * p2505 * p0421 / (p0432 * p2504)$ for multiturn encoders
2. Motor encoder with position tracking for measuring gear:
 $p2506 * p0412 * p2505 / p2504$
3. Motor encoder with position tracking for load gear:
 $p2506 * p2721 * p0433 / p0432$
4. Motor encoder with position tracking for load and measuring gear:
 $p2506 * p2721$
5. Direct encoder without position tracking:
 $p2506 * p0433 / p0432$
 $p2506 * p0433 * p0421 / p0432$ for multiturn encoders
6. Direct encoder with position tracking for measuring gear:
 $p2506 * p0412$
Reaction upon A: NONE
Acknowl. upon A: NONE

F07601 (A)	Encoder 3: Adjustment not possible
Message value:	Drive data set: %1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The maximum encoder value times the factor to convert the absolute position (r0483 and/or r2723) from increments to length units (LU) has exceeded the value range (-2147483648 ... 2147483647) for displaying the position actual value.
Remedy:	If the value for the maximum possible absolute position (LU) is greater than 4294967296, then it is not possible to make an adjustment due to an overflow. For rotary encoders, the maximum possible absolute position (LU) is calculated as follows: 1. Motor encoder without position tracking: $p2506 * p0433 * p2505 / (p0432 * p2504)$ $p2506 * p0433 * p2505 * p0421 / (p0432 * p2504)$ for multiturn encoders 2. Motor encoder with position tracking for measuring gear: $p2506 * p0412 * p2505 / p2504$ 3. Motor encoder with position tracking for load gear: $p2506 * p2721 * p0433 / p0432$ 4. Motor encoder with position tracking for load and measuring gear: $p2506 * p2721$ 5. Direct encoder without position tracking: $p2506 * p0433 / p0432$ $p2506 * p0433 * p0421 / p0432$ for multiturn encoders 6. Direct encoder with position tracking for measuring gear: $p2506 * p0412$
Reaction upon A:	NONE
Acknowled. upon A:	NONE
F07800	Drive: No power unit present
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The power unit parameters cannot be read or no parameters are stored in the power unit. Connection between the Control Unit and the Motor Module was interrupted or is defective. Note: This fault also occurs if an incorrect topology was selected in the commissioning software and this parameterization is then downloaded to the Control Unit. See also: r0200 (Power unit code number actual)
Remedy:	- connect the data line to power unit and restart the Control Unit (POWER ON). - check or replace the Control Unit. - check the cable between the Control Unit and Motor Module. - after correcting the topology, the parameters must be again downloaded using the commissioning software.
F07801	Drive: Motor overcurrent
Message value:	-
Drive object:	SERVO
Reaction:	OFF2 (NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The permissible motor limit current was exceeded. - effective current limit set too low. - current controller not correctly set. - motor was braked with an excessively high stall torque correction factor. - V/f operation: Up ramp was set too short or the load is too high. - V/f operation: Short-circuit in the motor cable or ground fault. - V/f operation: Motor current does not match the current of Motor Module. Note: Synchronous motor: Limit current= $1.3 * p0323$ Induction motor: Limit current= $1.3 * r0209$

- Remedy:**
- check the current limits (p0323, p0640).
 - check the current controller (p1715, p1717).
 - reduce the stall torque correction factor (p0326).
 - increase the up ramp (p1318) or reduce the load.
 - check the motor and motor cables for short-circuit and ground fault.
 - check the Motor Module and motor combination.

F07801 Drive: Motor overcurrent

- Message value:** -
- Drive object:** VECTOR
- Reaction:** OFF2 (NONE, OFF1, OFF3)
- Acknowledge:** IMMEDIATELY
- Cause:** The permissible motor limit current was exceeded.
- effective current limit set too low.
 - current controller not correctly set.
 - V/f operation: Up ramp was set too short or the load is too high.
 - V/f operation: Short-circuit in the motor cable or ground fault.
 - V/f operation: Motor current does not match current of power unit.
 - Switch to rotating motor without flying restart function (p1200).
- Note:**
Limit current = $2 * \text{Minimum}(p0640, 4 * p0305) \geq 2 * p0305$
- Remedy:**
- check the current limits (p0640).
 - vector control: Check the current controller (p1715, p1717).
 - V/f control: Check the current limiting controller (p1340 ... p1346).
 - increase the up ramp (p1120) or reduce the load.
 - check the motor and motor cables for short-circuit and ground fault.
 - check the motor for the star-delta configuration and rating plate parameterization.
 - check the power unit and motor combination.
 - Choose "flying restart" function (p1200) if switched to rotating motor.

F07802 Drive: Infeed or power unit not ready

- Message value:** -
- Drive object:** SERVO, VECTOR
- Reaction:** OFF2 (NONE)
- Acknowledge:** IMMEDIATELY
- Cause:** After an internal power-on command, the infeed or drive does not signal ready.
- monitoring time is too short.
 - DC link voltage is not present.
 - associated infeed or drive of the signaling component is defective.
 - supply voltage incorrectly set.
- Remedy:**
- increase the monitoring time (p0857).
 - ensure that there is a DC link voltage. Check the DC link busbar. Enable the infeed.
 - replace the associated infeed or drive of the signaling component.
 - check the line supply voltage setting (p0210).
- See also: p0857 (Power unit monitoring time)

A07805 (N) Infeed: Power unit overload I2t

- Message value:** -
- Drive object:** A_INF, B_INF, S_INF
- Reaction:** NONE
- Acknowledge:** NONE
- Cause:** Alarm threshold for I2t overload (p0294) of the power unit exceeded.
- Remedy:**
- reduce the continuous load.
 - adapt the load duty cycle.
- Reaction upon N: NONE
- Acknowled. upon N: NONE

A07805 (N)	Drive: Power unit overload I2t
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Alarm threshold for I2t overload (p0294) of the power unit exceeded. The response parameterized in p0290 becomes active. See also: p0290 (Power unit overload response)
Remedy:	- reduce the continuous load. - adapt the load duty cycle. - check the assignment of the rated currents of the motor and Motor Module.
Reaction upon N:	NONE
Acknowled. upon N:	NONE
F07810	Drive: Power unit EEPROM without rated data
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	No rated data are stored in the power unit EEPROM. See also: p0205 (Power unit application), r0206 (Rated power unit power), r0207 (Rated power unit current), r0208 (Rated power unit line supply voltage), r0209 (Power unit, maximum current)
Remedy:	Replace the power unit or inform Siemens Customer Service.
F07815	Drive: Power unit has been changed
Message value:	Parameter: %1
Drive object:	A_INF, B_INF, S_INF
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The code number of the actual power unit does not match the saved number. The fault only occurs if the comparator in p9906 or p9908 is not at f 2 (low) or 3 (minimum). Fault value (r0949, interpret decimal): Number of the incorrect parameter. See also: r0200 (Power unit code number actual), p0201 (Power unit code number)
Remedy:	Connect the original power unit and power up the Control Unit again (POWER ON) or set p0201 to r0200 and exit commissioning with p0010 = 0. For infeeds, the following applies: Line reactors or line filters must be used that are specified for the new power unit. A line supply and DC link identification routine (p3410 = 5) must then be carried out. It is not possible to change the power unit without re-commissioning the system if the type of infeed (A_Infeed, B_Infeed, S_Infeed), the type of construction/design (booksize, chassis) or the voltage class differ between the old and new power units. For inverters, the following applies: If the new power unit is accepted, then if required, the current limit p0640 can be reduced by a lower maximum current of the power unit (r0209) (torque limits stay the same). If not only the power unit is changed, but also the motor, then the motor must be re-commissioned (e.g. using p0010 = 1). This is also necessary if motor data is still to be downloaded via DRIVE-CLiQ. See also: r0200 (Power unit code number actual)

F07815	Drive: Power unit has been changed
Message value:	Parameter: %1
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The code number of the actual power unit does not match the saved number. The fault only occurs if the comparator in p9906 or p9908 is not at f 2 (low) or 3 (minimum). Fault value (r0949, interpret decimal): Number of the incorrect parameter. See also: r0200 (Power unit code number actual), p0201 (Power unit code number)
Remedy:	Connect the original power unit and power up the Control Unit again (POWER ON) or set p0201 to r0200 and exit commissioning with p0010 = 0. For infeeds, the following applies: Line reactors or line filters must be used that are specified for the new power unit. A line supply and DC link identification routine (p3410 = 5) must then be carried out. It is not possible to change the power unit without re-commissioning the system if the type of infeed (A_Infeed, B_Infeed, S_Infeed), the type of construction/design (booksize, chassis) or the voltage class differ between the old and new power units. For inverters, the following applies: If the new power unit is accepted, then if required, the current limit p0640 can be reduced by a lower maximum current of the power unit (r0209) (torque limits stay the same). If not only the power unit is changed, but also the motor, then the motor must be re-commissioned (e.g. using p0010 = 1). This is also necessary if motor data is still to be downloaded via DRIVE-CLiQ. If the comparison stage in p9906 is set to 2, 3, then commissioning can be exited (p0010 = 0) and the fault acknowledged. See also: r0200 (Power unit code number actual)
F07815	Drive: Power unit has been changed
Message value:	Parameter: %1
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The code number of the actual power unit does not match the saved number. The fault only occurs if the comparator in p9906 or p9908 is not at f 2 (low) or 3 (minimum). Fault value (r0949, interpret decimal): Number of the incorrect parameter. See also: r0200 (Power unit code number actual), p0201 (Power unit code number)
Remedy:	Connect the original power unit and power up the Control Unit again (POWER ON) or set p0201 to r0200 and exit commissioning with p0010 = 0. If the new power unit is accepted, then if required, the current limit p0640 can be reduced by a lower maximum current of the power unit (r0209) (torque limits stay the same). If not only the power unit is changed, but also the motor, then the motor must be re-commissioned (e.g. using p0010 = 1). This is also necessary if motor data is still to be downloaded via DRIVE-CLiQ. If the comparison stage in p9906 is set to 2, 3, then commissioning can be exited (p0010 = 0) and the fault acknowledged. See also: r0200 (Power unit code number actual)
A07820	Drive: Temperature sensor not connected
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature sensor for motor temperature monitoring, specified in p0600, is not available. - 10 (SME) is set in p0601, but a value other than 1, 2, or 3 is set in p0600 - 11 (SME) is set in p0601, but a value other than 20 or 21 is set in p0600 - 11 (BICO) is set in p0601, and a sensor is set in p4610 - p4613, but the associated BICO input (p0608, p0609) is not interconnected - module with sensor evaluation not present or has been removed in the meantime. - temperature sensor via Motor Module, not for CU310.

Remedy:

- connect the module with temperature sensor.
- set the available temperature sensor (p0600, p0601).
- set p4610 - p4613 to no sensor (0), or interconnect p0608 and p0609 with an external sensor value (only if BICO (11) is set in p0601)

See also: p0600 (Motor temperature sensor for monitoring), p0601

A07825 (N) Drive: Simulation mode activated

Message value: -

Drive object: VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The simulation mode is activated.
The drive can only be powered up if the DC link voltage is less than 40 V.

Remedy: The alarm automatically disappears if simulation mode is deactivated with p1272 = 0.

Reaction upon N: NONE

Acknowl. upon N: NONE

F07826 Drive: Simulation mode with DC link voltage too high

Message value: -

Drive object: VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY

Cause: The simulation mode is activated and the DC link voltage is greater than the permissible value of 40 V.

Remedy:

- switch out (disable) simulation mode (p1272 = 0) and acknowledge the fault.
- reduce the input voltage in order to reach a DC link voltage below 40 V.

F07840 Drive: Infeed operation missing

Message value: -

Drive object: SERVO, VECTOR

Reaction: OFF2 (NONE)

Acknowledge: IMMEDIATELY

Cause: The signal "infeed operation" is not present although the enable signals for the drive have been present for longer than the parameterized monitoring time (p0857).

- infeed not operational.
- interconnection of the binector input for the ready signal is either incorrect or missing (p0864).
- infeed is presently carrying out a line supply identification routine.

Remedy:

- bring the infeed into an operational state.
- check the interconnection of the binector input for the signal "infeed operation" (p0864).
- increase the monitoring time (p0857).
- wait until the infeed has completed the line supply identification routine.

See also: p0857 (Power unit monitoring time), p0864 (Infeed operation)

F07841 (A) Drive: Infeed operation withdrawn

Message value: -

Drive object: SERVO, VECTOR

Reaction: OFF2 (NONE, OFF1, OFF3)

Acknowledge: IMMEDIATELY

Cause: The signal "infeed operation" was withdrawn in operation.

- interconnection of the binector input for the signal "infeed operation" is either incorrect or missing (p0864).
- the enable signals of the infeed were disabled.
- due to a fault, the infeed withdraws the signal "infeed operation".

Remedy:

- check the interconnection of the binector input for the "infeed operation" signal (p0864).
- check the enable signals of the infeed and if required, enable.
- remove and acknowledge an infeed fault.

Note:
If this drive is intended to back up the DC link regeneratively, then the fault response must be parameterized for NONE, OFF1 or OFF3. so that the drive can continue to operate even after the infeed fails.

Reaction upon A: NONE
Acknowl. upon A: NONE

A07850 (F) External alarm 1

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The BICO signal for "external alarm 1" was triggered.
The condition for this external alarm is fulfilled.
See also: p2112 (External alarm 1)
Remedy: Eliminate the causes of this alarm.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY (POWER ON)

A07851 (F) External alarm 2

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The BICO signal for "external alarm 2" was triggered.
The condition for this external alarm is fulfilled.
See also: p2116 (External alarm 2)
Remedy: Eliminate the causes of this alarm.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY (POWER ON)

A07852 (F) External alarm 3

Message value: -
Drive object: All objects
Reaction: NONE
Acknowledge: NONE
Cause: The BICO signal for "external alarm 3" was triggered.
The condition for this external alarm is fulfilled.
See also: p2117 (External alarm 3)
Remedy: Eliminate the causes of this alarm.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY (POWER ON)

F07860 (A)	External fault 1
Message value:	-
Drive object:	All objects
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2) Vector: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The BICO signal "external fault 1" was triggered. See also: p2106 (External fault 1)
Remedy:	Eliminate the causes of this fault.
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F07861 (A)	External fault 2
Message value:	-
Drive object:	All objects
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2) Vector: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The BICO signal "external fault 2" was triggered. See also: p2107 (External fault 2)
Remedy:	Eliminate the causes of this fault.
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F07862 (A)	External fault 3
Message value:	-
Drive object:	All objects
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2) Vector: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The BICO signal "external fault 3" was triggered. See also: p2108, p3111, p3112
Remedy:	Eliminate the causes of this fault.
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F07890	Internal voltage protection/internal armature short-circuit with Safe Torque Off active
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The internal armature short-circuit (p1231 = 4) is not possible as Safe Torque Off (STO) is enabled. The pulses cannot be enabled.
Remedy:	Switch out the internal armature short-circuit (p1231=0) or deactivate Safe Torque Off (p9501 = p9561 = 0). Note: STO: Safe Torque Off / SH: Safe standstill

F07900 (N, A)	Drive: Motor locked/speed controller at its limit
Message value:	-
Drive object:	SERVO
Reaction:	OFF2 (NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	Motor has been operating at the torque limit longer than the time specified in p2177 and below the speed threshold set in p2175. This signal can also be initiated if the speed actual value is oscillating and the speed controller output repeatedly goes to its limit. See also: p2175, p2177 (Motor locked delay time)
Remedy:	- check that the motor can rotate freely. - check the torque limit: For a positive direction of rotation r1538, for a negative direction of rotation r1539. - check the parameter, message "Motor locked" and if required, correct (p2175, p2177). - check the inversion of the actual value (p0410). - check the motor encoder connection. - check the encoder pulse number (p0408). - for SERVO with encoderless operation and motors with low power ratings (< 300 W), increase the pulse frequency (p1800). - after deselecting the "Basic positioner" (EPOS) function mode, check the motoring (p1528) and regenerative (p1529) torque limit and modify again.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F07900 (N, A)	Drive: Motor blocked
Message value:	-
Drive object:	VECTOR
Reaction:	OFF2 (NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	Motor has been operating at the torque limit longer than the time specified in p2177 and below the speed threshold set in p2175. This signal can also be initiated if the speed actual value is oscillating and the speed controller output repeatedly goes to its limit. If the simulation mode is enabled (p1272 = 1) and the closed-loop control with speed encoder activated (p1300 = 21), then the inhibit signal is generated if the encoder signal is not received from a motor that is driven with the torque setpoint of the closed-loop control. See also: p2175, p2177 (Motor locked delay time)
Remedy:	- check that the motor can rotate freely. - check the torque limit: For a positive direction of rotation r1538, for a negative direction of rotation r1539. - check the parameter, message "Motor locked" and if required, correct (p2175, p2177). - check the inversion of the actual value (p0410). - check the motor encoder connection. - check the encoder pulse number (p0408). - for SERVO with encoderless operation and motors with low power ratings (< 300 W), increase the pulse frequency (p1800). - after deselecting the "Basic positioner" (EPOS) function mode, check the motoring (p1528) and regenerative (p1529) torque limit and modify again. - in the simulation mode and operation with speed encoder, the power unit to which the motor is connected must be powered up and must be supplied with the torque setpoint of the simulated closed-loop control. Otherwise, change over to encoderless control (see p1300).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F07901 Drive: Motor overspeed

Message value: -

Drive object: SERVO

Reaction: OFF2 (IASC/DCBRAKE)

Acknowledge: IMMEDIATELY

Cause: The maximum permissible speed was either positively or negatively exceeded.
The maximum permissible positive speed is formed as follows: Minimum (p1082, Cl: p1085) + p2162
The maximum permissible negative speed is formed as follows: Maximum (-p1082, Cl: 1088) - p2162

Remedy: The following applies for a positive direction of rotation:
- check r1084 and if required, correct p1082, Cl:p1085 and p2162.
The following applies for a negative direction of rotation:
- check r1087 and if required, correct p1082, Cl:p1088 and p2162.

F07901 Drive: Motor overspeed

Message value: -

Drive object: VECTOR

Reaction: OFF2 (IASC/DCBRAKE)

Acknowledge: IMMEDIATELY

Cause: The maximum permissible speed was either positively or negatively exceeded.
The maximum permissible positive speed is formed as follows: Minimum (p1082, Cl: p1085) + p2162
The maximum permissible negative speed is formed as follows: Maximum (-p1082, Cl: 1088) - p2162

Remedy: The following applies for a positive direction of rotation:
- check r1084 and if required, correct p1082, Cl:p1085 and p2162.
The following applies for a negative direction of rotation:
- check r1087 and if required, correct p1082, Cl:p1088 and p2162.
Activate pre-control of the speed limiting controller (p1401 bit 7 = 1).
Increase the hysteresis for the overspeed signal p2162. This upper limit is dependent upon the maximum motor speed p0322 and the maximum speed p1082 of the setpoint channel.

F07902 (N, A) Drive: Motor stalled

Message value: %1

Drive object: SERVO

Reaction: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)

Acknowledge: IMMEDIATELY

Cause: For a vector drive the system has identified that the motor has stall for a time longer than is set in p2178.
Fault value (r0949, interpret decimal):
1: Stall detection using r1408.11 (p1744 or p0492).
2: Stall detection using r1408.12 (p1745).
3: Stall detection using r0056.11 (only for separately excited synchronous motors).
See also: p1744 (Motor model speed threshold stall detection), p2178 (Motor stalled delay time)

Remedy: For closed-loop speed and torque control with speed encoder, the following applies:
- check the speed signal (interrupted cable, polarity, pulse number, broken encoder shaft).
- check the speed encoder, if another speed encoder was selected using the data set changeover. This must be connected to the same motor that is controlled for the data set changeover.
If there is no fault, then the fault tolerance (p1744 and p0492) can be increased.
For closed-loop speed and torque control without speed encoder, the following applies:
- check whether the drive in the open-loop controlled mode (r1750.0) stalls under load. If yes, then increase the current setpoint using p1610.
- check whether the drive stalls due to the load if the speed setpoint is still zero. If yes, then increase the current setpoint using p1610.
- if the motor excitation (magnetizing) time (r0346) was significantly reduced, then it should be increased again.
- check the current limits (p0640, r0067). If the current limits are too low, then the drive cannot be magnetized.
- check the current controller (p1715, p1717) and the speed adaptation controller (p1764, p1767). If the dynamic response was significantly reduced, then this should be increased again.
- check the speed encoder, if another speed encoder was selected using the data set changeover. This must be connected to the motor that is controlled for the data set changeover.
If there is no fault, then the fault tolerance (p1745) or the delay time (p2178) can be increased.

For separately-excited synchronous motors (closed-loop control with speed encoder), the following applies:

- check the speed signal (interrupted cable, polarity, pulse number).
- ensure the correct motor parameterization (rating plate and equivalent circuit diagram parameters).
- check the excitation equipment and the interface to the closed-loop control.
- encoder the highest possible dynamic response of the closed-loop excitation current control.
- check the speed control for any tendency to oscillate and if resonance effects occur, use a bandstop filter.
- do not exceed the maximum speed (p2162).

If there is no fault, then the delay time can be increased (p2178).

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F07902 (N, A) Drive: Motor stalled

Message value: %1

Drive object: VECTOR

Reaction: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)

Acknowledge: IMMEDIATELY

Cause: For a vector drive the system has identified that the motor has stall for a time longer than is set in p2178.

Fault value (r0949, interpret decimal):

1: Stall detection using r1408.11 (p1744 or p0492).

2: Stall detection using r1408.12 (p1745).

3: Stall detection using r0056.11 (only for separately excited synchronous motors).

See also: p1744 (Motor model speed threshold stall detection), p2178 (Motor stalled delay time)

Remedy: It should always be carefully ensured that the motor data identification (p1910) as well as the rotating measurement (p1960) were carried out (also refer to r3925). For synchronous motors with encoder, the encoder must have been adjusted (p1990).

For closed-loop speed and torque control with speed encoder, the following applies:

- check the speed signal (interrupted cable, polarity, pulse number, broken encoder shaft).

- check the speed encoder, if another speed encoder was selected using the data set changeover. This must be connected to the same motor that is controlled for the data set changeover.

If there is no fault, then the fault tolerance (p1744 and p0492) can be increased.

If the stalled motor should take place in the range of the monitor model and for speeds of less than 30 % of the rated motor speed, then a change can be made directly from the current model into the flux impression (p1401.5 = 1). We therefore recommend that the time-controlled model change is switched in (p1750.4 = 1) or the model changeover limits are significantly increased (p1752 > 0.35 * p0311; p1753 = 5 %).

For closed-loop speed and torque control without speed encoder, the following applies:

- Check whether the drive stalls solely due to the load in controlled mode (r1750.0) or when the speed setpoint is still zero. If so, increase the current setpoint via p1610 or set p1750 bit 2 = 1 (encoderless vector control to standstill for passive loads).

- If the motor excitation time (p0346) was significantly reduced and the drive stalls when it is switched on and run immediately, p0346 should be increased again or quick magnetizing selected (p1401).

- Check the current limits (p0640, r0067, r0289). If the current limits are too low, then the drive cannot be magnetized.

- check the current controller (p1715, p1717) and the speed adaptation controller (p1764, p1767). If the dynamic response was significantly reduced, then this should be increased again.

- check the speed encoder, if another speed encoder was selected using the data set changeover. This must be connected to the motor that is controlled for the data set changeover.

If there is no fault, then the fault tolerance (p1745) or the delay time (p2178) can be increased.

The following generally apply for closed-loop and torque control:

- Check whether the motor cables are disconnected.

- If the fault occurs with fault value 2 when the motor accelerates very quickly to the field weakening range, the deviation between the flux setpoint and actual value can be reduced and, in turn, the fault message prevented, by reducing p1596.

For separately-excited synchronous motors (closed-loop control with speed encoder), the following applies:

- check the speed signal (interrupted cable, polarity, pulse number).

- ensure the correct motor parameterization (rating plate and equivalent circuit diagram parameters).

- check the excitation equipment and the interface to the closed-loop control.

- encoder the highest possible dynamic response of the closed-loop excitation current control.

- check the speed control for any tendency to oscillate and if resonance effects occur, use a bandstop filter.

- do not exceed the maximum speed (p2162).

If there is no fault, then the delay time can be increased (p2178).

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

A07903 Drive: Motor speed deviation

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The absolute value of the speed difference from the two setpoints (p2151, p2154) and the speed actual value (r2169) exceeds the tolerance threshold (p2163) longer than tolerated (p2164, p2166).
 The alarm is only enabled for p2149.0 = 1.
 Possible causes could be:
 - the load torque is greater than the torque setpoint.
 - when accelerating, the torque/current/power limit is reached. If the limits are not sufficient, then it is possible that the drive has been dimensioned too small.
 - for closed-loop torque control, the speed setpoint does not track the speed actual value.
 - for active Vdc controller.
 For V/f control, the overload condition is detected as the I_{max} controller is active.
 See also: p2149 (Monitoring configuration)
Remedy:
 - increase p2163 and/or p2166.
 - increase the torque/current/power limits.
 - for closed-loop torque control: The speed setpoint should track the speed actual value.
 - deactivate alarm with p2149.0 = 0.

A07904 (N) External armature short-circuit: Contactor feedback signal "Closed" missing

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When closing, the contactor feedback signal (p1235) did not issue the signal "Closed" (r1239.1 = 1) within the monitoring time (p1236).
Remedy:
 - check that the contactor feedback signal is correctly connected (p1235).
 - check the logic of the contactor feedback signal (r1239.1 = 1: "Closed", r1239.1 = 0: "Open").
 - increase the monitoring time (p1236).
 - if required, set the external armature short-circuit without contactor feedback signal (p1231=2).
 Reaction upon N: NONE
 Acknowl. upon N: NONE

F07905 (N, A) External armature short-circuit: Contactor feedback signal "Open" missing

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2 (NONE)
Acknowledge: IMMEDIATELY
Cause: When opening, the contactor feedback signal (p1235) did not issue the signal "Open" (r1239.1 = 0) within the monitoring time (p1236).
Remedy:
 - check that the contactor feedback signal is correctly connected (p1235).
 - check the logic of the contactor feedback signal (r1239.1 = 1: "Closed", r1239.1 = 0: "Open").
 - increase the monitoring time (p1236).
 - if required, set the external armature short-circuit without contactor feedback signal (p1231=2).
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F07906	Armature short-circuit / internal voltage protection: Parameterization error
Message value:	Fault cause: %1, motor data set: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The armature short-circuit is incorrectly parameterized. Fault value (r0949, interpret decimal): Low word: Motor data set number High word: Cause: 1: A permanent-magnet synchronous motor has not been selected. 2: No induction motor selected. 101: External armature short-circuit: Output (r1239.0) not connected up. 102: External armature short-circuit with contactor feedback signal: No feedback signal connected (BI:p1235). 103: External armature short-circuit without contactor feedback signal: Delay time when opening (p1237) is 0. 201: Internal voltage protection: The maximum output current of the Motor Module (r0209) is less than 1.8 * motor short-circuit current (r0331). 202: Internal voltage protection: A Motor Module in booksize or chassis format is not being used. 203: Internal voltage protection: The motor short-circuit current (p0320) is greater than the maximum motor current (p0323). 204: Internal voltage protection: The activation (p1231 = 4) is not given for all motor data sets with synchronous motors (p0300 = 2xx, 4xx).
Remedy:	Re cause 1: - an armature short-circuit / voltage protection is only permissible for permanent-magnetic synchronous motors. The highest position of the motor type in p0300 must either be 2 or 4. Re cause 101: - the contactor for the external armature short-circuit configuration should be controlled using output signal r1239.0. The signal can, e.g. be connected to an output terminal BI: p0738. Before this fault can be acknowledged, p1231 must be set again. Re cause 102: - if the external armature short-circuit with contactor feedback signal (p1231 = 1) is selected, this feedback signal must be connected to an input terminal (e.g. r722.x) and then connected to BI: p1235. - alternatively, the external armature short-circuit without contactor feedback signal (p1231 = 2) can be selected. Re cause 103: - if the external armature short-circuit without contactor feedback signal (p1231 = 2) is selected, then a delay time must be parameterized in p1237. This time must always be greater than the actual contactor opening time, as otherwise the Motor Module would be short-circuited! Re cause 201: - a Motor Module with a higher maximum current or a motor with a lower short-circuit current must be used. The maximum Motor Module current must be higher than 1.8 * short-circuit current of the motor. Re cause 202: - for internal voltage protection, use a Motor Module in booksize or chassis format. Re cause 203: - for internal voltage protection, only use short-circuit proof motors. Re cause 204: - The internal voltage protection must either be activated for all motor data sets with synchronous motors (p0300 = 2xx, 4xx) (p1231 = 3) or it must be deactivated for all motor data sets (p1231 not equal to 3). This therefore ensures that the protection cannot be accidentally withdrawn as a result of a data set changeover. The fault can only be acknowledged if this condition is fulfilled.

F07907	Internal armature short-circuit: Motor terminals are not at zero potential after pulse suppression
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The function "Internal voltage protection" (p1231 = 3) was activated. The following must be observed: - when the internal voltage protection is active, after pulse suppression, all of the motor terminals are at half of the DC link voltage (without an internal voltage protection, the motor terminals are at zero potential)! - it is only permissible to use motors that are short-circuit proof (p0320 < p0323). - the Motor Module must be able to continually conduct 180% short-circuit current (r0331) of the motor (r0289). - the internal voltage protection cannot be interrupted due to a fault response. If an overcurrent condition occurs during the active, internal voltage protection, then this can destroy the Motor Module and/or the motor. - if the Motor Module does not support the autonomous, internal voltage protection (r0192.10 = 0), in order to ensure safe, reliable functioning when the line supply fails, an external 24 V power supply (UPS) must be used for the components. - if the Motor Module does support the autonomous, internal voltage protection (r0192.10 = 1), in order to ensure safe, reliable functioning when the line supply fails, the 24 V power supply for the components must be provided through a Control Supply Module. - if the internal voltage protection is active, it is not permissible that the motor is driven by the load for a longer period of time (e.g. as a result of loads that move the motor or another coupled motor).
Remedy:	None necessary. This a note for the user.
A07908	Internal armature short-circuit active
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The Motor Module signals that the motor is short-circuited through the power semiconductors (r1239.5 = 1). The pulses cannot be enabled. The internal armature short-circuit is selected (p1231 = 4):
Remedy:	For synchronous motors, the armature short-circuit braking is activated if a 1 signal is present via binector input p1230. See also: p1230 (Armature short-circuit / DC brake activation), p1231 (Armature short-circuit / DC brake configuration)
F07909	Internal voltage protection: Deactivation only effective after POWER ON
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	POWER ON
Cause:	The deactivation of the internal voltage protection (p1231 not equal to 3) only becomes effective after POWER ON. The status signal r1239.6 = 1 indicates that the internal voltage protection is ready.
Remedy:	None necessary. This a note for the user.

A07910 (N)	Drive: Motor overtemperature
Message value:	%1
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	KTY: The motor temperature has exceeded the fault threshold (p0604 or p0616). VECTOR: The response parameterized in p0610 becomes active. PTC: The response threshold of 1650 Ohm was exceeded. Alarm value (r2124, interpret decimal): SME not selected in p0601: 1: No output current reduction. 2: Output current reduction active. With SME (p0601 = 10), TM120 (p0601 = 11) selected: Number of the sensor channel leading to the message. See also: p0604 (Motor temperature alarm threshold), p0610 (Motor overtemperature response)
Remedy:	- check the motor load. - check the motor ambient temperature. - check KTY84.
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A07910 (N)	Drive: Motor overtemperature
Message value:	%1
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	KTY or no sensor: The measured motor temperature or temperature of the thermal model has exceeded the alarm threshold (p0604 or p0616). The response parameterized in p0610 becomes active. PTC or bimetallic NC contact: The response threshold of 1650 Ohm was exceeded or the NC contact opened. Alarm value (r2124, interpret decimal): SME not selected in p0601: 1: No output current reduction. 2: Output current reduction active. SME or TM120 selected in p0601 (p0601 = 10, 11): The number specifies the sensor channel that resulted in the alarm being output. See also: p0604 (Motor temperature alarm threshold), p0610 (Motor overtemperature response)
Remedy:	- check the motor load. - check the motor ambient temperature. - check KTY84.
Reaction upon N:	NONE
Acknowl. upon N:	NONE

F07913	Excitation current outside the tolerance range
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The difference between the excitation current actual value and setpoint has exceeded the tolerance: $\text{abs}(r1641 - r1626) > p3201 + p3202$ The cause of this fault is again reset for $\text{abs}(r1641 - r1626) < p3201$.
Remedy:	- check the parameterization (p1640, p3201, p3202). - check the interfaces to the excitation equipment (r1626, p1640). - check the excitation equipment.

F07914	Flux out of tolerance
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The difference between the flux actual value and setpoint has exceeded the tolerance: $\text{abs}(r0084 - r1598) > p3204 + p3205$ The cause of this fault is again reset for $\text{abs}(r0084 - r1598) < p3204$. The fault is only issued after the delay time in p3206 has expired.
Remedy:	- check the parameterization (p3204, p3205). - check the interfaces to the excitation equipment (r1626, p1640). - check the excitation equipment. - check the flux control (p1592, p1592, p1597). - check the control for oscillation and take the appropriate counter measures (e.g. optimize the speed control loop, parameterize a bandstop filter).
A07918 (N)	Three-phase setpoint generator operation selected/active
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Only for separately excited synchronous motors (p0300 = 5): The actual open-loop/closed-loop control mode is I/f control (open-loop) with a fixed current (p1300 = 18). The speed is entered via the setpoint channel and the current setpoint is given by the minimum current (p1620). It must be ensured that in this mode, the control dynamic performance is very limited. This is the reason that longer ramp-up times should be set for the setpoint speed than for normal operation. See also: p1620 (Stator current, minimum)
Remedy:	Select another open-loop/closed-loop control mode See also: p1300 (Open-loop/closed-loop control operating mode)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A07920	Drive: Torque/speed too low
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The torque deviates from the torque/speed envelope characteristic (too low). See also: p2181 (Load monitoring response)
Remedy:	- check the connection between the motor and load. - adapt the parameterization corresponding to the load.
A07921	Drive: Torque/speed too high
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The torque deviates from the torque/speed envelope characteristic (too high).
Remedy:	Check the connection between the motor and load. Adapt the parameterization according to the load.

A07922 Drive: Torque/speed out of tolerance

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The torque deviates from the torque/speed envelope characteristic.
Remedy: Check the connection between the motor and load.
Adapt the parameterization according to the load.

F07923 Drive: Torque/speed too low

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The torque deviates from the torque/speed envelope characteristic (too low).
Remedy: Check the connection between the motor and load.
Adapt the parameterization according to the load.

F07924 Drive: Torque/speed too high

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The torque deviates from the torque/speed envelope characteristic (too high).
Remedy: Check the connection between the motor and load.
Adapt the parameterization according to the load.

F07925 Drive: Torque/speed out of tolerance

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The torque deviates from the torque/speed envelope characteristic.
Remedy: Check the connection between the motor and load.
Adapt the parameterization according to the load.

A07926 Drive: Envelope curve, parameter invalid

Message value: Parameter: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Invalid parameter values were entered for the envelope characteristic of the load monitoring.
The following rules apply for the speed thresholds:
p2182 < p2183 < p2184
The following rules apply for the torque thresholds:
p2185 > p2186
p2187 > p2188
p2189 > p2190
Alarm value (r2124, interpret decimal):
Number of the parameter with the invalid value.
Remedy: Set the parameters for the load monitoring according to the applicable rules or deactivate load monitoring (p2181 = 0; p2193 = 0).

A07927	DC brake active
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The motor is braked with a DC current, the DC current brake is active. 1) An alarm with response DCBRAKE is active. The motor is braked with the braking current set in p1232 for the duration set in in p1233. If the standstill threshold p1226 is undershot, then braking is prematurely canceled. 2) DC braking has been activated at binector input p1230 with the DC brake set (p1230 = 4). Braking current p1232 is injected until this binector input becomes inactive.
Remedy:	None necessary.
F07928	Internal voltage protection initiated
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The Motor Module signals that the motor is short-circuited through the power semiconductors (r1239.5 = 1). The pulses cannot be enabled. The internal voltage protection is selected (p1231 = 3).
Remedy:	If the Motor Module supports the autonomous internal voltage protection (r0192.10 = 1), then the Motor Module automatically decides - using the DC link voltage - as to whether the armature short-circuit should be activated. The armature short-circuit is activated and response OFF2 is initiated if the DC link voltage exceeds 800 V. If the DC link voltage falls below 450 V, then the armature short-circuit is withdrawn. If the motor is still in a critical speed range, the armature short-circuit is reactivated once the DC link voltage exceeds the threshold of 800 V. If the autonomous (independent) internal voltage protection is active (r1239.5 = 1) and the line supply returns (450 V < DC link voltage < 800 V), the armature short-circuit is withdrawn after 3 minute.
F07930	Drive: Brake control error
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The Control Unit has detected a brake control error. Fault value (r0949, interpret decimal): 10, 11: Fault in "open holding brake" operation. - No brake connected or wire breakage (check whether brake releases for p1278 = 1). - Ground fault in brake cable. 20: Fault in "brake open" state. - Short-circuit in brake winding. 30, 31: Fault in "close holding brake" operation. - No brake connected or wire breakage (check whether brake releases for p1278 = 1). - Short-circuit in brake winding. 40: Fault in "brake closed" state. 50: Fault in the brake control circuit of the Control Unit or communication fault between the Control Unit and Motor Module (brake control). Note: The following causes may apply to fault values: - motor cable is not shielded correctly. - defect in control circuit of the Motor Module. See also: p1278 (Brake control, diagnostics evaluation)

Remedy:

- check the motor holding brake connection.
- check the function of the motor holding brake.
- check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified.
- check that the electrical cabinet design and cable routing are in compliance with EMC regulations (e.g. shield of the motor cable and brake conductors are connected with the shield connecting plate and the motor connectors are tightly screwed to the housing).
- replace the Motor Module involved.

Operation with Safe Brake Module:

- check the Safe Brake Modules connection.
- replace the Safe Brake Module.

See also: p1215 (Motor holding brake configuration), p1278 (Brake control, diagnostics evaluation)

A07931 (F, N) Brake does not open

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: This alarm is output for r1229.4 = 1.
See also: p1216 (Motor holding brake, opening time), r1229 (Motor holding brake status word)

Remedy:

- check the functionality of the motor holding brake.
- check the feedback signal (p1223).

Reaction upon F: NONE (OFF1, OFF2, OFF3)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

A07932 Brake does not close

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: This alarm is output for r1229.5 = 1.
For r1229.5 = 1, OFF/OFF3 are suppressed to prevent the drive accelerating by a load that drives the motor - whereby OFF2 remains effective.
See also: p1217 (Motor holding brake closing time), r1229 (Motor holding brake status word)

Remedy:

- check the functionality of the motor holding brake.
- check the feedback signal (p1222).

F07934 (N) Drive: S120 Combi motor holding brake configuration

Message value: %1

Drive object: All objects

Reaction: Infeed: OFF2 (NONE, OFF1)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)

Acknowledge: IMMEDIATELY (POWER ON)

Cause: A connected motor holding brake has been detected with an S120 Combi unit. However, this motor holding brake has not been assigned to just one Combi feed drive and, therefore, brake control is not configured (correctly).
Fault value (r0949, interpret decimal):
0: No motor holding brake assigned (p1215 = 0 or 3 on all S120 Combi feed drives)
1: Multiple holding brakes assigned (p1215 = 1 or 2 on more than one S120 Combi feed drive)
MHB: Motor holding brake.

Remedy: Check whether the motor holding brake has been assigned to one S120 Combi feed drive exclusively (p1215 = 1 or 2).
The fault will only be withdrawn once the motor holding brake has been assigned to just one of the S120 Combi feed drives (p1215 = 1 or 2 for this one drive). From this point, the motor holding brake will be controlled by this drive.
See also: p1215 (Motor holding brake configuration)

Reaction upon N: NONE

Acknowl. upon N: NONE

F07935 (N) Drv: Motor holding brake detected

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE (OFF1, OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: A motor holding brake was detected where the brake control has not been configured (p1215 = 0). The brake control configuration was then set to "motor holding brake the same as sequence control" (p1215 = 1).

Remedy: None necessary.
See also: p1215 (Motor holding brake configuration)

Reaction upon N: NONE

Acknowl. upon N: NONE

F07937 (N) Drive: Speed deviation between motor model and external speed

Message value: -

Drive object: VECTOR

Reaction: OFF2 (NONE, OFF1, OFF3)

Acknowledge: IMMEDIATELY

Cause: The absolute value of the speed difference from the two actual values (r2169, r1443) exceeds the tolerance threshold (p3236) for longer than the time set in p3238.

Possible causes could be:

- External encoder signal wired incorrectly (p1440)
 - Speed encoder for external encoder signal faulty
 - Encoder signal's polarity or gain incorrect
 - Smoothing time constant for model speed for monitoring p2157 too high
 - Smoothing time constant or threshold values for monitoring too low (p3236, p3238)
- See also: p2149 (Monitoring configuration)

Remedy: - Check that the external speed (see p1440, r1443) matches the motor speed.
- Check the polarity of the external speed (r1443).
- Check connector connection p1440.

Reaction upon N: NONE

Acknowl. upon N: NONE

F07940 Sync-line-drive: Synchronization error

Message value: -

Drive object: VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY

Cause: After synchronization has been completed, the phase difference (r3808) is greater than the threshold value, phase synchronism (p3813).

OFF1 or OFF3 response, while the closed-loop phase control is active (r3819.6 = 1) or synchronism reached (r3819.2 = 1).

Enable signal withdrawn (p3802 = 0), while the closed-loop phase control was active (r3819.6 = 1).

Remedy: If required increase the threshold value phase synchronism (p3813) for synchronizing the line supply to the drive.
Before OFF1 or OFF3, complete synchronizing (r3819.0 = 0).
Before withdrawing the enable signal (p3802 = 0), reach synchronism (r3819.2 = 1).
See also: p3813 (Sync-line-drive phase synchronism threshold value)

A07941	Sync-line-drive: Target frequency not permissible
Message value:	Parameter: %1
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The target frequency is outside the permissible value range. Alarm value (r2124, interpret decimal): 1084: Target frequency greater than the positive speed limit, $f_{\text{sync}} > f_{\text{max}}$ (r1084). 1087: Target frequency less than the negative speed limit, $f_{\text{sync}} < f_{\text{min}}$ (r1087).
Remedy:	Fulfill the conditions for the target frequency for line-drive synchronization. See also: r1084, r1087

A07942	Sync-line-drive: Setpoint frequency is completely different than the target frequency
Message value:	-
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	There is a considerable difference between the setpoint frequency and the target frequency ($f_{\text{set}} \neq f_{\text{target}}$). The deviation that can be tolerated is set in p3806.
Remedy:	The alarm automatically disappears after the difference that can be tolerated between the setpoint and target frequencies (p3806) is reached. See also: p3806 (Sync-line-drive frequency difference threshold value)

A07943	Sync-line-drive: Synchronization not permitted
Message value:	Parameter: %1
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Synchronization is not permitted. Alarm value (r2124, interpret decimal): 1300: The control mode (p1300) has not been set to encoderless closed-loop speed control or V/f characteristic. 1910: Motor data identification activated. 1960: Speed controller optimization activated. 1990: Encoder adjustment activated. 3801: Voltage Sensing Module (VSM) not found. 3845: Friction characteristic record activated.
Remedy:	Fulfill the conditions for the line-drive synchronization. Re alarm value = 1300: Set the control mode (p1300) to encoderless closed-loop speed control (p1300 = 20) or V/f characteristic (p1300 = 0 ... 19). Re alarm value = 1910: Exit the motor data identification routine (p1910). Re alarm value = 1960: Exit the speed controller optimization routine (p1960). Re alarm value = 1990: Exit the encoder adjustment (p1990). Re alarm value = 3801: Connect-up a Voltage Sensing Module (VSM) and when connecting to an adjacent drive object ensure that the drive objects have the same basis clock cycle. Re alarm value = 3845: Exit the friction characteristic record (p3845).

F07950 (A)	Drive: Incorrect motor parameter
Message value:	Parameter: %1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	- the motor parameters were incorrectly entered while commissioning (e.g. p0300 = 0, no motor) - The braking resistor (p6811) has still not been parameterized - commissioning cannot be completed. Fault value (r0949, interpret decimal): The parameter number involved. See also: p0300, p0301, p0304, p0305, p0307, p0310, p0311, p0314, p0315, p0316, p0320, p0322, p0323
Remedy:	Compare the motor data with the rating plate data and if required, correct. See also: p0300, p0301, p0304, p0305, p0307, p0310, p0311, p0314, p0316, p0320, p0322, p0323
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F07955	Drive: Motor has been changed
Message value:	Parameter: %1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The code number of the actual motor with DRIVE-CLiQ does not match the saved number. Fault value (r0949, interpret decimal): Number of the incorrect parameter. See also: p0301 (Motor code number selection), r0302 (Motor code number of motor with DRIVE-CLiQ)
Remedy:	Connect the original motor, power up the Control Unit again (POWER ON) and exit the quick commissioning by setting p0010 to 0. Or set p0300 = 10000 (load the motor parameter with DRIVE-CLiQ) and re-commission. Quick commissioning (p0010 = 1) is automatically exited with p3900 > 0. If quick commissioning was exited by setting p0010 to 0, then an automatic controller calculation (p0340 = 1) is not carried out.
F07956 (A)	Drive: Motor code does not match the list (catalog) motor
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The motor code of the connected motor with DRIVE-CLiQ does not match the possible list motor types (see selection in p0300). The connected motor with DRIVE-CLiQ might not be supported by this firmware version. Fault value (r0949, interpret decimal): Motor code of the connected motor with DRIVE-CLiQ. Note: The first three digits of the motor code generally correspond to the list motor type.
Remedy:	Use a motor with DRIVE-CLiQ and the matching motor code.
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A07960 Drive: Incorrect friction characteristic

Message value: Parameter: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The friction characteristic is incorrect.
Alarm value (r2124, interpret decimal):
1538:
The friction torque is greater than the maximum from the upper effective torque limit (p1538) and zero. This is the reason that the output of the friction characteristic (r3841) is limited to this value.
1539:
The friction torque is less than the minimum from the lower effective torque limit (p1539) and zero. This is the reason that the output of the friction characteristic (r3841) is limited to this value.
3820 ... 3829:
Incorrect parameter number. The speeds entered in the parameters for the friction characteristic do not correspond to the following condition:
 $0.0 < p3820 < p3821 < \dots < p3829 \leq p0322$ or $p1082$, if $p0322 = 0$
Therefore the output of the friction characteristic (r3841) is set to zero.
3830 ... 3839:
Incorrect parameter number. The torques entered in the parameters for the friction characteristic do not correspond to the following condition:
 $0 \leq p3830, p3831 \dots p3839 \leq p0333$
Therefore the output of the friction characteristic (r3841) is set to zero.
See also: r3840 (Friction characteristic, status word)
Remedy: Fulfill the conditions for the friction characteristic.
Re alarm value = 1538:
Check the upper effective torque limit (e.g. in the field weakening range).
Re alarm value = 1539:
Check the lower effective torque limit (e.g. in the field weakening range).
Re alarm value = 3820 ... 3839:
Fulfill the conditions to set the parameters of the friction characteristic.
If the motor data (e.g. the maximum speed p0322) are changed during commissioning ($p0010 = 1, 3$), then the technological limits and threshold values, dependent on this, must be recalculated by selecting $p0340 = 5$.

A07961 Drive: Friction characteristic record activated

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The automatic friction characteristic record is activated.
The friction characteristic is recorded at the next power-on command.
Remedy: None necessary.
The alarm disappears automatically after the friction characteristic record has been successfully completed or the record is deactivated ($p3845 = 0$).

F07963	Drive: Friction characteristic record interrupted
Message value:	Parameter: %1
Drive object:	SERVO, VECTOR
Reaction:	OFF1
Acknowledge:	IMMEDIATELY
Cause:	The conditions to record the friction characteristic are not fulfilled. Fault value (r0949, interpret decimal): 0046: Missing enable signals (r0046). 1082: The highest speed value to be approached (p3829) is greater than the maximum speed (p1082). 1084: The highest speed value to be approached (p3829) is greater than the maximum speed (r1084, p1083, p1085). 1087: The highest speed value to be approached (p3829) is greater than the maximum speed (r1087, p1086, p1088). 1110: Friction characteristic record, negative direction selected (p3845) and negative direction inhibited (p1110). 1111: Friction characteristic record, positive direction selected (p3845) and positive direction inhibited (p1111). 1198: Friction characteristic record selected (p3845 > 0) and negative (p1110) and positive directions (p1111) inhibited (r1198). 1300: The control mode (p1300) has not been set to closed-loop speed control. 1755: For encoderless closed-loop control (p1300 = 20), the lowest speed value to be approached (p3820) is less than or equal to the changeover speed, open-loop controlled operation (p1755). 1910: Motor data identification activated. 1960: Speed controller optimization activated. 3820 ... 3829: Speed (p382x) cannot be approached. 3840: Friction characteristic incorrect. 3845: Friction characteristic record deselected.
Remedy:	Fulfill the conditions to record the friction characteristic. Re fault value = 0046: Establish missing enable signals. Re fault value = 1082, 1084, 1087: Select the highest speed value to be approached (p3829) less than or equal to the maximum speed (p1082, r1084, r1087). Recalculate the speed points along the friction characteristic (p0340 = 5). Re fault value = 1110: Select the friction characteristic record, positive direction (p3845). Re fault value = 1111: Select the friction characteristic record, negative direction (p3845). Re fault value = 1198: Enable the permitted direction (p1110, p1111, r1198). Re fault value = 1300: Set the control mode (p1300) on the closed-loop speed control (p1300 = 20, 21). Re fault value = 1755: For encoderless closed-loop speed control (p1300 = 20) select the lowest speed value to be approached (p3820) greater than the changeover speed of open-loop controlled operation (p1755). Recalculate the speed points along the friction characteristic (p0340 = 5). Re fault value = 1910: Exit the motor data identification routine (p1910). Re fault value = 1960: Exist the speed controller optimization routine (p1960). Re fault value 3820 ... 3829: - check the load at speed p382x. - check the speed signal (r0063) for oscillation at speed p382x. Check the settings of the speed controller if applicable. Re fault value = 3840: Make the friction characteristic error-free (p3820 to p3829, p3830 to p3839, p3840). Re fault value = 3845: Activate the friction characteristic record (p3845).

A07965 (N)	Drive: Save required
Message value:	-
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	The angular commutation offset (p0431) was re-defined and has still not been saved. In order to permanently accept the new value, it must be saved in a non-volatile fashion (p0971, p0977). See also: p0431 (Angular commutation offset), p1990 (Encoder adjustment, determine angular commutation offset)
Remedy:	None necessary. This alarm automatically disappears after the data has been saved. See also: p0971 (Save drive object parameters), p0977 (Save all parameters)
Reaction upon N:	NONE
Acknowled. upon N:	NONE

F07966	Drive: Check the commutation angle
Message value:	-
Drive object:	SERVO
Reaction:	OFF2 (NONE)
Acknowledge:	IMMEDIATELY
Cause:	The speed actual value was inverted and the associated angular commutation offset is not equal to zero and is therefore possibly incorrect.
Remedy:	Angular commutation offset after the actual value inversion or determine it again (p1990=1).

F07967	Drive: Automatic encoder adjustment incorrect
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF2 (NONE, OFF1)
Acknowledge:	IMMEDIATELY
Cause:	A fault has occurred during the automatic encoder adjustment or the pole position identification. Only for internal Siemens troubleshooting.
Remedy:	Carry out a POWER ON.

F07968	Drive: Lq-Ld measurement incorrect
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A fault has occurred during the Lq-Ld measurement. Fault value (r0949, interpret decimal): 10: Stage 1: The ratio between the measured current and zero current is too low. 12: Stage 1: The maximum current was exceeded. 15: Second harmonic too low. 16: Drive converter too small for the measuring technique. 17: Abort due to pulse inhibit.
Remedy:	Re fault value = 10: Check whether the motor is correctly connected. Replace the power unit involved. Deactivate technique (p1909). Re fault value = 12: Check whether motor data have been correctly entered. Deactivate technique (p1909). Re fault value = 16: Deactivate technique (p1909). Re fault value = 17: Repeat technique.

F07969	Drive: Incorrect pole position identification
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A fault has occurred during the pole position identification routine. Fault value (r0949, interpret decimal): 1: Current controller limited 2: Motor shaft locked. 4: Encoder speed signal not plausible. 10: Stage 1: The ratio between the measured current and zero current is too low. 11: Stage 2: The ratio between the measured current and zero current is too low. 12: Stage 1: The maximum current was exceeded. 13: Stage 2: The maximum current was exceeded. 14: Current difference to determine the +d axis too low. 15: Second harmonic too low. 16: Drive converter too small for the measuring technique. 17: Abort due to pulse inhibit. 18: First harmonic too low. 20: Pole position identification requested with the motor shaft rotating and activated flying restart function.
Remedy:	Re fault value = 1: Check whether the motor is correctly connected. Check whether motor data have been correctly entered. Replace the Motor Module involved. Re fault value = 2: Open the motor holding brake (p1215 = 2) and bring the motor into a no-load condition. Re fault value = 4: Check whether the encoder pulse number (p0408) and gearbox factor (p0432, p0433) are correct. Check whether the motor pole pair number is correct (p0314). Re fault value = 10: When selecting p1980 = 4: Increase the value for p0325. When selecting p1980 = 1: Increase the value for p0329. Check whether the motor is correctly connected. Replace the Motor Module involved. Re fault value = 11: Increase the value for p0329. Check whether the motor is correctly connected. Replace the Motor Module involved. Re fault value = 12: When selecting p1980 = 4: Reduce the value for p0325. When selecting p1980 = 1: Reduce the value for p0329. Check whether motor data have been correctly entered. Re fault value = 13: Reduce the value for p0329. Check whether motor data have been correctly entered. Re fault value = 14: Increase the value for p0329. Re fault value = 15: Increase the value for p0325. Motor not sufficiently anisotropic, change the technique (p1980 = 1 or 10). Re fault value = 16: Deactivate technique (p1982). Re fault value = 17: Repeat technique. Re fault value = 18: Increase the value for p0329. Saturation not sufficient, change the technique (p1980 = 10). Re fault value = 20: Before carrying out a pole position identification routine ensure that the motor shaft is absolutely stationary (zero speed).

F07970	Drive: Automatic encoder adjustment incorrect
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF2 (NONE)
Acknowledge:	IMMEDIATELY
Cause:	A fault has occurred during the automatic encoder adjustment. Fault value (r0949, decimal): 1: Current controller limited 2: Motor shaft locked. 4: Encoder speed signal not plausible. 10: Stage 1: The ratio between the measured current and zero current is too low. 11: Stage 2: The ratio between the measured current and zero current is too low. 12: Stage 1: The maximum current was exceeded. 13: Stage 2: The maximum current was exceeded. 14: Current difference to determine the +d axis too low. 15: Second harmonic too low. 16: Drive converter too small for the measuring technique. 17: Abort due to pulse inhibit.
Remedy:	Re fault value = 1: Check whether the motor is correctly connected. Check whether motor data have been correctly entered. Replace the power unit involved. Re fault value = 2: Open the motor holding brake (p1215 = 2) and bring the motor into a no-load condition. Re fault value = 4: Check whether the speed actual value inversion is correct (p0410.0). Check whether the motor is correctly connected. Check whether the encoder pulse number (p0408) and gearbox factor (p0432, p0433) are correct. Check whether the motor pole pair number is correct (p0314). Re fault value = 10: Increase the value for p0325. Check whether the motor is correctly connected. Replace the power unit involved. Re fault value = 11: Increase the value for p0329. Check whether the motor is correctly connected. Replace the power unit involved. Re fault value = 12: Reduce the value for p0325. Check whether motor data have been correctly entered. Re fault value = 13: Reduce the value for p0329. Check whether motor data have been correctly entered. Re fault value = 14: Increase the value for p0329. Re fault value = 15: Increase the value for p0325. Re fault value = 16: Deactivate technique (p1982). Re fault value = 17: Repeat technique.

A07971 (N)	Drive: Angular commutation offset determination activated
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The automatic determination of the angular commutation offset (encoder adjustment) is activated (p1990 = 1). The automatic determination is carried out at the next power-on command. For SERVO and fault F07414 present, the following applies: The determination of the angular commutation offset is automatically activated (p1990 = 1), if a pole position identification technique is set in p1980. See also: p1990 (Encoder adjustment, determine angular commutation offset)
Remedy:	None necessary. The alarm automatically disappears after determination or for the setting p1990 = 0.
Reaction upon N:	NONE
Acknowled. upon N:	NONE
A07980	Drive: Rotating measurement activated
Message value:	-
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	The rotating measurement is activated. For the rotating measurement, the motor can accelerate up to the maximum speed and with maximum torque. Only the parameterized current limit (p0640) and the maximum speed (p1082) are effective. The behavior of the motor can be influenced using the direction inhibit (p1959.14, p1959.15) and the ramp-up/ramp-down time (p1958). The rotating measurement is carried out at the next power-on command. See also: p1960
Remedy:	None necessary. The alarm automatically disappears after the rotating measurement has been successfully completed or for the setting p1960 = 0. If a POWER ON or a warm restart is performed with motor data identification selected, the motor data identification request will be lost. If motor data identification is required, it will need to be selected again manually following ramp-up.
A07980	Drive: Rotating measurement activated
Message value:	-
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The rotating measurement (automatic speed controller optimization) is activated. The rotating measurement is carried out at the next power-on command. See also: p1960
Remedy:	None necessary. The alarm disappears automatically after the speed controller optimization has been successfully completed or for the setting p1900 = 0.
A07981	Drive: Enable signals for the rotating measurement missing
Message value:	-
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The rotating measurement cannot be started due to missing enable signals.
Remedy:	- acknowledge faults that are present. - establish missing enable signals. See also: r0002, r0046

F07982 Drive: Rotating measurement encoder test

Message value:	%1
Drive object:	VECTOR
Reaction:	OFF1 (NONE, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A fault has occurred during the encoder test. Fault value (r0949, interpret decimal): 1: The speed did not reach a steady-state condition. 2: The speed setpoint was not able to be approached as the minimum limiting is active. 3: The speed setpoint was not able to be approached as the suppression (skip) bandwidth is active. 4: The speed setpoint was not able to be approached as the maximum limiting is active. 5: The encoder does not supply a signal. 6: Incorrect polarity. 7: Incorrect pulse number. 8: Noise in the encoder signal or speed controller unstable. 9: Voltage Sensing Module (VSM) incorrectly connected.
Remedy:	Re fault value = 1: - check the motor parameters. - carry out a motor data identification routine (p1910). - if required, reduce the dynamic factor (p1967 < 25 %). Re fault value = 2: - adapt the speed setpoint (p1965) or adapt the minimum limit (p1080). Re fault value = 3: - adapt the speed setpoint (p1965) or suppression (skip) bandwidths (p1091 ... p1094, p1101). Re fault value = 4: - adapt the speed setpoint (p1965) or maximum limit (p1082, p1083 and p1086). Re fault value = 5: - check the encoder connection. If required, replace the encoder. Re fault value = 6: - check the connection assignment of the encoder cable. Adapt the polarity (p0410). Re fault value = 7: - adapt the pulse number (p0408). Re fault value = 8: - check the encoder connection and encoder cable. It is possible that there is a problem associated with the ground connection. - reduce the dynamic response of the speed controller (p1460, p1462 and p1470, p1472). Re fault value = 9: - check the connections of the Voltage Sensing Module (VSM). Note: The encoder test can be switched out (disabled) using p1959.0. See also: p1959

F07983 Drive: Rotating measurement saturation characteristic

Message value:	%1
Drive object:	VECTOR
Reaction:	OFF1 (NONE, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A fault has occurred while determining the saturation characteristic. Fault value (r0949, interpret decimal): 1: The speed did not reach a steady-state condition. 2: The rotor flux did not reach a steady-state condition. 3: The adaptation circuit did not reach a steady-state condition. 4: The adaptation circuit was not enabled. 5: Field weakening active. 6: The speed setpoint was not able to be approached as the minimum limiting is active. 7: The speed setpoint was not able to be approached as the suppression (skip) bandwidth is active. 8: The speed setpoint was not able to be approached as the maximum limiting is active. 9: Several values of the determined saturation characteristic are not plausible. 10: Saturation characteristic could not be sensibly determined because load torque too high.

Remedy:

Re fault value = 1:

- the total drive moment of inertia is far higher than that of the motor (p0341, p0342).

Deselect rotating measurement (p1960), enter the moment of inertia p0342, recalculate the speed controller p0340 = 4 and repeat the measurement.

Re fault value = 1 ... 2:

- increase the measuring speed (p1961) and repeat the measurement.

Re fault value = 1 ... 4:

- check the motor parameters (rating plate data). After the change: Calculate p0340 = 3.
- check the moment of inertia (p0341, p0342). After the change: Calculate p0340 = 3.
- carry out a motor data identification routine (p1910).
- if required, reduce the dynamic factor (p1967 < 25 %).

Re fault value = 5:

- the speed setpoint (p1961) is too high. Reduce the speed.

Re fault value = 6:

- adapt the speed setpoint (p1961) or minimum limiting (p1080).

Re fault value = 7:

- adapt the speed setpoint (p1961) or suppression (skip) bandwidths (p1091 ... p1094, p1101).

Re fault value = 8:

- adapt the speed setpoint (p1961) or maximum limit (p1082, p1083 and p1086).

Re fault value = 9, 10:

- the measurement was carried out at an operating point where the load torque is too high. Select a more suitable operating point, either by changing the speed setpoint (p1961) or by reducing the load torque. The load torque may not be varied while making measurements.

Note:

The saturation characteristic identification routine can be disabled using p1959.1.

See also: p1959

F07984	Drive: Speed controller optimization, moment of inertia
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF1 (NONE, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A fault has occurred while identifying the moment of inertia. Fault value (r0949, interpret decimal): 1: The speed did not reach a steady-state condition. 2: The speed setpoint was not able to be approached as the minimum limiting is active. 3: The speed setpoint was not able to be approached as the suppression (skip) bandwidth is active. 4: The speed setpoint was not able to be approached as the maximum limiting is active. 5: It is not possible to increase the speed by 10% as the minimum limiting is active. 6: It is not possible to increase the speed by 10% as the suppression (skip) bandwidth is active. 7: It is not possible to increase the speed by 10% as the maximum limiting is active. 8: The torque difference after the speed setpoint step is too low in order to be able to still reliably identify the moment of inertia. 9: Too few data to be able to reliably identify the moment of inertia. 10: After the setpoint step, the speed either changed too little or in the incorrect direction. 11: The identified moment of inertia is not plausible.

Remedy:

Re fault value = 1:

- check the motor parameters (rating plate data). After the change: Calculate p0340 = 3.
- check the moment of inertia (p0341, p0342). After the change: Calculate p0340 = 3.
- carry out a motor data identification routine (p1910).
- if required, reduce the dynamic factor (p1967 < 25 %).

Re fault value = 2, 5:

- adapt the speed setpoint (p1965) or adapt the minimum limit (p1080).

Re fault value = 3, 6:

- adapt the speed setpoint (p1965) or suppression (skip) bandwidths (p1091 ... p1094, p1101).

Re fault value = 4, 7:

- adapt the speed setpoint (p1965) or maximum limit (p1082, p1083 and p1086).

Re fault value = 8:

- the total drive moment of inertia is far higher than that of the motor (refer to p0341, p0342). Deselect rotating measurement (p1960), enter the moment of inertia p0342, recalculate the speed controller p0340 = 4 and repeat the measurement.

Re fault value = 9:

- check the moment of inertia (p0341, p0342). After the change, recalculate (p0340 = 3 or 4).

Re fault value = 10:

- check the moment of inertia (p0341, p0342). After the change: Calculate p0340 = 3.

Note:

The moment of inertia identification routine can be disabled using p1959.2.

See also: p1959

F07985 Drive: Speed controller optimization (oscillation test)

Message value: %1

Drive object: VECTOR

Reaction: OFF1 (NONE, OFF2)

Acknowledge: IMMEDIATELY

Cause:

A fault has occurred during the vibration test.

Fault value (r0949, interpret decimal):

- 1: The speed did not reach a steady-state condition.
- 2: The speed setpoint was not able to be approached as the minimum limiting is active.
- 3: The speed setpoint was not able to be approached as the suppression (skip) bandwidth is active.
- 4: The speed setpoint was not able to be approached as the maximum limiting is active.
- 5: Torque limits too low for a torque step.
- 6: No suitable speed controller setting was found.

Remedy:

Re fault value = 1:

- check the motor parameters (rating plate data). After the change: Calculate p0340 = 3.
- check the moment of inertia (p0341, p0342). After the change: Calculate p0340 = 3.
- carry out a motor data identification routine (p1910).
- if required, reduce the dynamic factor (p1967 < 25 %).

Re fault value = 2:

- adapt the speed setpoint (p1965) or adapt the minimum limit (p1080).

Re fault value = 3:

- adapt the speed setpoint (p1965) or suppression (skip) bandwidths (p1091 ... p1094, p1101).

Re fault value = 4:

- adapt the speed setpoint (p1965) or maximum limit (p1082, p1083 and p1086).

Re fault value = 5:

- increase the torque limits (e.g. p1520, p1521).

Re fault value = 6:

- reduce the dynamic factor (p1967).
- disable the vibration test (p1959.4 = 0) and repeat the rotating measurement.

See also: p1959

F07986	Drive: Rotating measurement ramp-function generator
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF1 (NONE, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	During the rotating measurements, problems with the ramp-function generator occurred. Fault value (r0949, interpret decimal): 1: The positive and negative directions are inhibited.
Remedy:	Re fault value = 1: Enable the direction (p1110 or p1111).
A07987	Drive: Rotating measurement, no encoder available
Message value:	%1
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	No encoder available. The rotating measurement was carried out without encoder.
Remedy:	Connect encoder or select p1960 = 1, 3.
F07988	Drive: Rotating measurement, no configuration selected
Message value:	-
Drive object:	VECTOR
Reaction:	OFF2 (NONE, OFF1)
Acknowledge:	IMMEDIATELY
Cause:	When configuring the rotating measurement (p1959), no function was selected.
Remedy:	Select at least one function for automatic optimization of the speed controller (p1959). See also: p1959
F07989	Drive: Rotating measurement leakage inductance (q-axis)
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF1 (NONE, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	An error has occurred while measuring the dynamic leakage inductance. Fault value (r0949, interpret decimal): 1: The speed did not reach a steady-state condition. 2: The speed setpoint was not able to be approached as the minimum limiting is active. 3: The speed setpoint was not able to be approached as the suppression (skip) bandwidth is active. 4: The speed setpoint was not able to be approached as the maximum limiting is active. 5: The 100% flux setpoint was not reached. 6: No Lq measurement possible because field weakening is active. 7: Speed actual value exceeds the maximum speed p1082 or 75% of the rated motor speed. 8: Speed actual value is below 2 % of the rated motor speed.

Remedy:

Re fault value = 1:

- check the motor parameters.
- carry out a motor data identification routine (p1910).
- if required, reduce the dynamic factor (p1967 < 25 %).

Re fault value = 2:

- adapt the speed setpoint (p1965) or adapt the minimum limit (p1080).

Re fault value = 3:

- adapt the speed setpoint (p1965) or suppression (skip) bandwidths (p1091 ... p1094, p1101).

Re fault value = 4:

- adapt the speed setpoint (p1965) or maximum limit (p1082, p1083 and p1086).

Re fault value = 5:

- flux setpoint p1570 = 100 % and current setpoint p1610 = 0 % kept during the Lq measurement.

Re fault value = 6:

- reduce the regenerative load so that the drive does not reach field weakening when accelerating.
- reduce p1965 so that the q leakage inductance is recorded at lower speeds.

Re fault value = 7:

- increase p1082 if this is technically permissible.
- reduce p1965 so that the q leakage inductance is recorded at lower speeds.

Re fault value = 8:

- reduce the load when motoring so that the drive is not braked.
- increase p1965 so that the measurement may be taken at higher speeds.

Note:

The measurement of the q leakage inductance can be disabled using p1959.5. If only p1959.5 is set, then only this measurement is carried out if p1960 is set to 1, 2 and the drive is powered up.

See also: p1959

F07990 Drive: Incorrect motor data identification

Message value: %1

Drive object: SERVO

Reaction: OFF2 (NONE, OFF1)

Acknowledge: IMMEDIATELY

Cause: A fault has occurred during the identification routine.

Fault value (r0949, interpret decimal):

- 1: Current limit value reached.
- 2: Identified stator resistance lies outside the expected range 0.1 ... 100% of Zn.
- 3: Identified rotor resistance lies outside the expected range 0.1 ... 100% of Zn.
- 4: Identified stator reactance lies outside the expected range 50 ... 500% of Zn.
- 5: Identified magnetizing reactance lies outside the expected range 50 ... 500% of Zn.
- 6: Identified rotor time constant lies outside the expected range 10 ms ... 5 s.
- 7: Identified total leakage reactance lies outside the expected range 4 ... 50% of Zn.
- 8: Identified stator leakage reactance lies outside the expected range 2 ... 50% of Zn.
- 9: Identified rotor leakage reactance lies outside the expected range 2 ... 50% of Zn.
- 10: Data set changeover during motor data identification.
- 11: Motor shaft rotates.
- 20: Identified threshold voltage of the semiconductor devices lies outside the expected range 0 ... 10 V.
- 30: Current controller in voltage limiting.
- 40: At least one identification contains errors. The identified parameters are not saved to prevent inconsistencies.
- 50: With the selected current controller sampling rate, the pulse frequency cannot be implemented.

Note:

Percentage values are referred to the rated motor impedance:

$$Z_n = V_{mot,nom} / \sqrt{3} / I_{mot,nom}$$

- 101: Voltage amplitude even at 30% maximum current amplitude is too low to measure the inductance.
- 102, 104: Voltage limiting while measuring the inductance.
- 103: Maximum frequency exceeded during the rotating inductance measurement.
- 110: Motor not finely synchronized before the rotating measurement.
- 111: The zero mark is not received within 2 revolutions.
- 112: Fine synchronization is not realized within 8 seconds after the zero mark has been passed.
- 113: The power, torque or current limit is zero.
- 115: V/f control is active.
- 120: Error when evaluating the magnetizing inductance.
- 125: Cable resistance greater than the total resistance.
- 126: Series inductance greater than the total leakage inductance.

- 127: Identified leakage inductance negative.
- 128: Identified stator resistance negative.
- 129: Identified rotor resistance negative.
- 130: Drive data set changeover during the motor data identification routine.
- 140: The setpoint channel inhibits both directions.
- 160: Accelerating when determining kT, moment of inertia or reluctance torque too short or the accelerating time is too long.
- 173: Internal problem.
- 180: Identification speed (maximum speed, rated speed, $0.9 \cdot p0348$) less than p1755.
- 190: Speed setpoint not equal to zero.
- 191: An actual speed of zero is not reached.
- 192: Speed setpoint not reached.
- 193: Inadmissible motion of the motor when identifying the voltage emulation error.
- 194: Supplementary torque (r1515) not equal to zero.
- 195: Closed-loop torque control active.
- 200, 201: Not possible to identify the voltage emulation error characteristic of the drive converter (p1952, p1953).

Remedy:

- Re fault value = 0:
 - check whether the motor is correctly connected. Observe configuration (star-delta).
- Re fault value = 1 ... 40:
 - check whether motor data have been correctly entered in p0300, p0304 ... p0311.
 - is there an appropriate relationship between the motor power rating and that of the Motor Module? The ratio of the Motor Module to the rated motor current should not be less than 0.5 and not be greater than 4.
 - check configuration (star-delta).
- Re fault value = 2:
 - for parallel circuits, check the motor winding system in p7003. If, for power units connected in parallel, a motor is specified with a single-winding system (p7003 = 0), although a multi-winding system is being used, then a large proportion of the stator resistance is interpreted as feeder cable resistance and entered in p0352.
- Re fault value = 4, 7:
 - check whether inductances are correctly entered in p0233 and p0353.
 - check whether motor has been correctly connected (star-delta).
- Re fault value = 50:
 - reduce the current controller sampling rate.
- Re fault value = 101:
 - increase current limit (p0640) or torque limit (p1520, p1521).
 - check current controller gain (p1715).
 - reduce current controller sampling time (p0115).
- It may be impossible to completely identify the L characteristic, as required current amplitude is too high.
 - suppress meas. (p1909, p1959).
- Re fault value = 102, 104:
 - reduce current limit (p0640).
 - check current controller P gain.
 - suppress meas. (p1909, p1959).
- Re fault value = 103:
 - increase external moment of inertia (if possible).
 - reduce current controller sampling time (p0115).
 - suppress meas. (p1909, p1959).
- Re fault value = 110:
 - before rotating measurement, traverse motor over zero mark.
- Re fault value = 111:
 - it is possible that encoder does not have zero mark. Correct setting in p0404.15.
 - encoder pulse number was incorrectly entered. Correct setting in p0408.
 - if zero mark signal is defective, replace encoder.
- Re fault value = 112:
 - upgrade encoder software.
- Re fault value = 113:
 - check the limits (p0640, p1520, p1521, p1530, p1531), correct the zero values.
- Re fault value = 115:
 - deselect V/f control (p1317 = 0).
- Re fault value = 120:
 - check current controller P gain (p1715) and if required, reduce.
 - increase the pulse frequency (p1800).
- Re fault value = 125:
 - reduce cable resistance (p0352).

Re fault value = 126:

- reduce series inductance (p0353).

Re fault = 127, 128, 129:

- it is possible that current controller is oscillating. Reduce p1715 before next measurement.

Re fault value = 130:

- do not initiate a drive data set changeover during motor ident. routine.

Re fault value = 140:

- before the measurement, enable at least one direction (p1110 = 0 or p1111 = 0 or p1959.14 = 1 or p1959.15 = 1).

Re fault value = 160:

- extend accelerating time when determining kT, moment of inertia and reluctance torque, e.g. by increasing max. speed (p1082), increasing moment of inertia or reducing max. current (p0640).
- in encoderless operation with load moment of inertia, parameterize the load moment of inertia (p1498).
- reduce the ramp-up time (p1958).
- increase speed controller P-gain (p1460).
- suppress meas. (p1959).

Re fault value = 173:

-

Re fault value = 180:

- increase max. speed (p1082).
- reduce p1755.
- suppress meas. (p1909, p1959).

Re fault value = 190:

- set speed setpoint to zero.

Re fault value = 191:

- do not start motor data ident. routine while motor is still rotating.

Re fault value = 192:

- check closed-loop speed control (motor rotor may be locked or closed-loop speed control is not functioning).
- for p1215 = 1, 3 (brake the same as the sequence control) check the control sense (p0410.0).
- ensure that enable signals are present during measurement.
- remove any pulling loads from motor.
- increase max. current (p0640).
- reduce max. speed (p1082).
- suppress meas. (p1959).

Re fault value = 193:

- the motor has moved through more than 5 ° electrical (r0093). Lock motor rotor at one of these pole position angles (r0093): 90 °, 210 ° or 330 ° (+/-5 °) and then start identification.

Re fault value = 194:

- switch out all supplementary torques (e.g. CI: p1511).
- for hanging/suspended axes: Lock motor rotor at one of these pole position angles (r0093): 90 °, 210 ° or 330 ° (+/- 1 °) and then start identification.

Re fault value = 195:

- deselect closed-loop torque control (p1300 = 21 or 20, or set the signal source in p1501 to a 0 signal).

Re fault value = 200, 201:

- set pulse frequency to 0.5 * current controller frequency (e.g. 4 kHz for a current controller clock cycle of 125 us).
- reduce cable length between Motor Module and motor.
- read-out measured values (r1950, r1951) and therefore determine suitable values for p1952, p1953 according to your own estimation.

F07990	Drive: Incorrect motor data identification
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF2 (NONE, OFF1)
Acknowledge:	IMMEDIATELY
Cause:	A fault has occurred during the identification routine. Fault value (r0949, interpret decimal): 1: Current limit value reached. 2: Identified stator resistance lies outside the expected range 0.1 ... 100% of Zn. 3: Identified rotor resistance lies outside the expected range 0.1 ... 100% of Zn. Separately excited synchronous motors: damping resistance outside 1.0 ... 15 % of Zn. 4: Identified stator reactance lies outside the expected range 50 ... 900 % of Zn. Separately excited synchronous motors: stator reactance outside 20 ... 500 % of Zn. 5: Identified magnetizing reactance lies outside the expected range 50 ... 900 % of Zn. Separately excited synchronous motors: magnetizing reactance outside 20 ... 500 % of Zn. 6: Identified rotor time constant lies outside the expected range 10 ms ... 5 s. Separately excited synchronous motors: damping time constant outside 5 ms ... 1 s. 7: Identified total leakage reactance lies outside the expected range 4 ... 100 % of Zn. 8: Identified stator leakage reactance lies outside the expected range 2 ... 50% of Zn. Separately excited synchronous motors: stator leakage reactance outside 2 ... 40 % of Zn. 9: Identified rotor leakage reactance lies outside the expected range 2 ... 50% of Zn. Separately excited synchronous motors: damping leakage reactance outside 1.5 ... 20 % of Zn. 10: Motor has been incorrectly connected. 11: Motor shaft rotates. 20: Identified threshold voltage of the semiconductor devices lies outside the expected range 0 ... 10 V. 30: Current controller in voltage limiting. 40: At least one identification contains errors. The identified parameters are not saved to prevent inconsistencies. Note: Percentage values are referred to the rated motor impedance: $Z_n = V_{mot,nom} / \sqrt{3} / I_{mot,nom}$
Remedy:	Re fault value = 0: - check whether the motor is correctly connected. Observe configuration (star-delta). Re fault value = 1 ... 40: - check whether motor data have been correctly entered in p0300, p0304 ... p0311. - is there an appropriate relationship between the motor power rating and that of the Motor Module? The ratio of the Motor Module to the rated motor current should not be less than 0.5 and not be greater than 4. - check configuration (star-delta). Also for fault value = 11: - deactivate vibration monitoring (p1909 bit 7 = 1). Re fault value = 2: - for parallel circuits, check the motor winding system in p7003. If, for power units connected in parallel, a motor is specified with a single-winding system (p7003 = 0), although a multi-winding system is being used, then a large proportion of the stator resistance is interpreted as feeder cable resistance and entered in p0352. Re fault value = 4, 7: - check whether inductances are correctly entered in p0233 and p0353. - check whether motor has been correctly connected (star-delta).
A07991 (N)	Drive: Motor data identification activated
Message value:	-
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	The motor data ident. routine is activated. The motor data identification routine is carried out at the next power-on command. See also: p1910, p1960

Remedy: None necessary.
The alarm automatically disappears after the motor data identification routine has been successfully completed or for the setting p1910 = 0 or p1960 = 0.
If a POWER ON or a warm restart is performed with motor data identification selected, the motor data identification request will be lost. If motor data identification is required, it will need to be selected again manually following ramp-up.

Reaction upon N: NONE
Acknowl. upon N: NONE

A07991 (N) Drive: Motor data identification activated

Message value: -
Drive object: VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The motor data ident. routine is activated.
The motor data identification routine is carried out at the next power-on command.
If rotating measurement is selected (see p1900, p1960), it will not be possible to save the parameter assignment. Once motor data identification has been completed or deactivated, the option to save the parameter assignment will be made available again.
See also: p1910

Remedy: None necessary.
The alarm automatically disappears after the motor data identification routine has been successfully completed or for the setting p1900 = 0.

Reaction upon N: NONE
Acknowl. upon N: NONE

F07993 Drive: Incorrect direction of rotation of the field or encoder actual value inversion

Message value: -
Drive object: SERVO
Reaction: OFF2 (NONE)
Acknowledge: IMMEDIATELY
Cause: Either the direction of the rotating field or the encoder actual value has an incorrect sign. The motor data identification automatically changed the actual value inversion (p0410) in order to correct the control sense. This can result in a direction of rotation change. To acknowledge this fault, the correctness of the direction of rotation must first be acknowledged with p1910 = -2.

Remedy: Check the direction of rotation, also for the position controller, if one is being used.
If the direction of rotation is correct, the following applies:
No additional measures are required (except p1910 = -2 and acknowledge fault).
If the direction of rotation is incorrect, the following applies:
To change the direction of rotation, two phases must be interchanged and the motor identification routine must be repeated.

A07994 (N) Drive: motor data identification not performed

Message value: -
Drive object: VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Motor data identification has not yet been performed with the actual data set.
The alarm is only triggered if the selection of the data set changes (see r0051) and vector control is parameterized (p1300 >= 20) in the new data set selected and motor data identification has not yet been performed (see r3925).

Remedy: - Perform motor data identification (see p1900).
- Return data set.
- Parameterize V/f control (p1300 < 20).

Reaction upon N: NONE
Acknowl. upon N: NONE

F07995	Drive: Pole position identification not successful
Message value:	%1
Drive object:	SERVO
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The pole position identification routine was unsuccessful. Fault value (r0949, interpret decimal): 1: No current is established. 2: The starting current is not zero. 3: The selected max. distance was exceeded (p1981). 4x: The measuring signal does not permit a clear evaluation. 5: The max. current was exceeded during the measurement. 6: The current measurement must be re-calibrated. 7x: The Sensor Module does not support the pole position identification routine. 70 ... 79: Only for internal Siemens troubleshooting. 8: The pole position identification routine current required is greater than the max. current. 9: The set pole position identification routine current is zero. 10: Data set changeover during the pole position identification. 11: The encoder adjustment to determine the commutation angle (p1990 = 1) and the encoder without zero mark is not finely synchronized or does not have any valid data. 100: Motion-based pole position identification, 1st and 2nd measurement different. Motor locked or current (p1993) too low. 101: Motion-based position position identification, insufficient motion, motor locked or current (p1993) too low. 102: Motion-based pole position identification, brake is being used and is closed. The motion-based position position identification in conjunction with the brake is not permitted. 103: Motion-based pole position identification without encoder. 104: Motion-based pole position identification, speed actual value not zero after stabilizing time. Note: x = 0 ... 9
Remedy:	Re fault value = 1: Check the motor connection and DC link voltage. For the following parameters, set practical values that are not zero (p0325, p0329). Re fault value = 3: Increase the max. distance (p1981). Reduce the currents for the pole position identification routine (p0325, p0329). Stop the motor in order to carry out the pole position identification routine. Re fault value = 40 ... 49: Increase the currents for the pole position identification routine (p0325, p0329). Stop the motor in order to carry out the pole position identification routine. Select another technique for pole position identification routine (p1980). Use another motor, absolute encoder or Hall sensors. Re fault value = 5: Reduce the currents for the pole position identification routine (p0325, p0329). Re fault value = 6: Re-calibrate the Motor Module. Re fault value = 7x: Upgrade the software in the Sensor Module. Re fault value = 8: Reduce the currents for the pole position identification routine (p0329, p0325, p1993). The power unit cannot provide the necessary pole position identification routine current (p0209 < p0329, p0325, p1993), replace the power unit by a power unit with a higher max. current. Re fault value = 9: Enter a value not equal to zero in the pole position identification routine current (p0329, p0325, p1993). Re fault value = 10: Do not initiate a data set changeover during the pole position identification. Re fault value = 11: - for incremental encoders without commutation with zero mark (p0404.15 = 0), it does not make sense to adjust the encoder to determine the commutation angle (p1990 = 1). In this case, the function should be deselected (p1990 = 0) or, for an encoder with suitable zero mark, commutation with zero mark should be selected (p0404.15 = 1). - for absolute encoders, only adjust the encoder to determine the commutation angle (p1990 = 1) if the encoder supplies commutation information and is finely synchronized (p1992.8 = 1 and p1992.10 = 1). The encoder is possibly parked, deactivated (p0145), not ready to operate or signals a fault condition. - deselect the encoder adjustment to determine the commutation angle (set p1990 to 0).

Re fault value = 100, 101:
Check and ensure that the motor is free to move.
Increase the current for motion-based pole position identification (p1993).
Re fault value = 102:
If the motor is to be operated with a brake: Select a different technique to identify the pole position (p1980).
If the motor can be operated without a brake: Open the brake (p1215 = 2).
Re fault value = 103:
The motion-based pole position identification can only be carried out using an encoder. Connect an encoder or select another technique for pole position identification routine (p1980).
Re fault value = 104:
Pole position identification, increase the smoothing time, motion-based (p1997).
Pole position identification, increase the rise time, motion-based (p1994).
Pole position identification, check the gain, motion-based (p1995).
Pole position identification, check the integral time, motion-based (p1996).
For motor encoders with track A/B sq-wave (p0404.3 = 1) and flank time measurement (p0430.20 = 0), the integral time must be disabled (p1996 = 0).

F07996	Drive: Pole position identification routine not carried out
Message value:	-
Drive object:	SERVO
Reaction:	ENCODER (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	In operation, the operating mode that requires a pole position identification was changed over, which is not possible in this state: - the drive was changed over, flying, from encoderless operation to operation with encoder without having previously carried out a pole position identification for the encoder. p1404 is then at a value between zero and the max. speed and the pulses in the speed range above p1404 were enabled without a pole position ident. routine having been previously carried out in operation with encoder. - in operation, an EDS changeover was made to an encoder where it is necessary to carry out a pole position identification. However, this has still not been carried out (p1982 = 1 or 2 and p1992.7 = 0).
Remedy:	- for a flying changeover between operation with and without encoder with pole position identification after POWER ON or commissioning (p0010 not equal to zero) enable the pulses once at zero speed. This means that the pole position identification routine is carried out and the result is available for operation. - carry out the EDS changeover with the pulses inhibited, or, before the changeover, carry out a pole position identification using this data set.
A07998	Drive: Motor data identification active on another drive
Message value:	%1
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	The motor data identification is activated on the drive object specified in the fault value and interlocks the other drive objects so they cannot be powered up. Fault value (r0949, interpret decimal): Drive object with the active motor data identification. See also: p1910, p1960
Remedy:	- wait for the complete execution of the motor data identification of the drive object designated in the fault value. - deselect the motor data identification for the drive object designated in the fault value (p1910 = 0 or p1960 = 0).
A07999	Drive: Motor data identification cannot be activated
Message value:	%1
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	Closed-loop control is enabled on a SERVO drive object type. To select motor data identification, pulses must be suppressed for all SERVO drive objects. Fault value (r0949, interpret decimal): Drive object with enabled closed-loop control.
Remedy:	Withdraw the pulse enable on all drives and reactivate the motor data identification.

F08000 (N, A) TB: +/-15 V power supply faulted

Message value: %1
Drive object: All objects
Reaction: Infeed: NONE (OFF1, OFF2)
 Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
 Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: Terminal Board 30 detects an incorrect internal power supply voltage.
 Fault value (r0949, interpret decimal):
 0: Error when testing the monitoring circuit.
 1: Fault in normal operation.
Remedy: - replace Terminal Board 30.
 - replace Control Unit.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F08010 (N, A) TB: Analog-digital converter

Message value: -
Drive object: All objects
Reaction: Infeed: NONE (OFF1, OFF2)
 Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
 Vector: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The analog/digital converter on Terminal Board 30 has not supplied any converted data.
Remedy: - check the power supply.
 - replace Terminal Board 30.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F08500 (A) COMM BOARD: Monitoring time configuration expired

Message value: %1
Drive object: A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: Infeed: OFF1 (OFF2)
 Servo: OFF1 (OFF2, OFF3)
 Vector: OFF1 (OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The monitoring time for the configuration has expired.
 Fault value (r0949, interpret decimal):
 0: The transfer time of the send configuration data has been exceeded.
 1: The transfer time of the receive configuration data has been exceeded.
Remedy: Check communication line.
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F08501 (N, A) COMM BOARD: Monitoring time process data expired

Message value: -

Drive object: A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR

Reaction: Infeed: OFF1 (OFF2)
Servo: OFF3 (IASC/DCBRAKE, NONE, OFF1, OFF2, STOP1, STOP2)
Vector: OFF3 (IASC/DCBRAKE, NONE, OFF1, OFF2, STOP1, STOP2)

Acknowledge: IMMEDIATELY

Cause: The set monitoring time expired while transferring process data via COMM BOARD.
See also: p8840 (COMM BOARD monitoring time)

Remedy: - check communications link.
- check the set monitoring time if the error persists.
See also: p8840 (COMM BOARD monitoring time)

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F08502 (A) COMM BOARD: Monitoring time sign-of-life expired

Message value: -

Drive object: A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR

Reaction: Infeed: OFF1 (OFF2)
Servo: OFF1 (OFF2, OFF3)
Vector: OFF1 (OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: The monitoring time for the sign-of-life counter has expired.
The connection to the COMM BOARD was interrupted.

Remedy: - check communications link.
- check COMM BOARD.

Reaction upon A: NONE
Acknowl. upon A: NONE

A08504 (F) COMM BOARD: Internal cyclic data transfer error

Message value: %1

Drive object: A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The cyclic actual and/or setpoint values were not transferred within the specified times.
Alarm value (r2124, interpret decimal):
Only for internal Siemens troubleshooting.

Remedy: Check the parameterizing telegram (Ti, To, Tdp, etc.).

Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)

Acknowl. upon F: IMMEDIATELY

F08510 (A)	COMM BOARD: Send configuration data invalid
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	Infeed: OFF1 (OFF2) Servo: OFF1 (OFF2, OFF3) Vector: OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	COMM BOARD did not accept the send-configuration data. Fault value (r0949, interpret decimal): Return value of the send-configuration data check.
Remedy:	Check the send configuration data.
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A08511 (F)	COMM BOARD: Receive configuration data invalid
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The drive unit did not accept the receive configuration data. Alarm value (r2124, interpret decimal): Return value of the receive configuration data check. 0: Configuration accepted. 1: Connection established to more drive objects than configured in the device. The drive objects for process data exchange and their sequence are defined in p0978. 2: Too many PZD data words for output or input to a drive object. The number of possible PZD items in a drive object is determined by the number of indices in r2050/p2051 for PZD IF1, and in r8850/p8851 for PZD IF2. 3: Uneven number of bytes for input or output. 4: Setting data for synchronization not accepted. For more information, see A01902. 5: Cyclic operation not active. 6: Buffer system not accepted. 7: Cyclic channel length too short for this setting. 8: Cyclic channel address not initialized. 9: 3-buffer system not permitted. 23: Illegal clock synchronization for PZD interface 2.
Remedy:	Check the receive configuration data. Re alarm value = 1: Check the list of the drive objects with process data exchange (p0978). With p0978[x] = 0, all of the following drive objects in the list are excluded from the process data exchange. Re alarm value = 2: Check the number of data words for output and input to a drive object.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY
A08520 (F)	COMM BOARD: Non-cyclic channel error
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The memory or the buffer status of the non-cyclic channel has an error. Alarm value (r2124, interpret decimal): 0: Error in the buffer status. 1: Error in the memory.

Remedy: Check communication line.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY

A08526 (F) COMM BOARD: No cyclic connection

Message value: -
Drive object: A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: There is no cyclic connection to the control.
Remedy: Establish the cyclic connection and activate the control with cyclic operation.
For PROFINET, check the parameters "Name of Station" and "IP of Station" (r61000, r61001).
Reaction upon F: NONE (OFF1)
Acknowl. upon F: IMMEDIATELY

A08530 (F) COMM BOARD: Message channel error

Message value: %1
Drive object: A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The memory or the buffer status of the message channel has an error.
Alarm value (r2124, interpret decimal):
0: Error in the buffer status.
1: Error in the memory.
Remedy: Check communication line.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY

A08550 PZD Interface Hardware assignment error

Message value: %1
Drive object: A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The assignment of the hardware to the PZD interface has been incorrectly parameterized.
Alarm value (r2124, interpret decimal):
Only one of the two indices is not equal to 99 (automatic).
2: Both PZD interfaces are assigned to the same hardware.
3: Assigned COMM BOARD missing.
4: CBC10 is assigned to interface 1.
See also: p8839 (PZD interface hardware assignment)
Remedy: Correct the parameterization (p8839).

A08560	IE: Syntax error in configuration file
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A syntax error has been detected in the ASCII configuration file for the Industrial Ethernet interface (X127). The saved configuration file has not been loaded. Note: IE: Industrial Ethernet
Remedy:	- Check the interface configuration (p8900 and following), correct if necessary, and activate (p8905 = 1). - Save the parameters for interface configuration (e.g. p8905 = 2) or - Reconfigure the station via the "Edit Ethernet node" screen form (e.g. with STARTER commissioning software). See also: p8905 (IE Interface configuration)
A08561	IE: Consistency error affecting adjustable parameters
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A consistency error was detected when activating the configuration (p8905 = 1) for the Industrial Ethernet interface (X127). The currently set configuration has not been activated. Possible causes: - IP address, subnet mask or default gateway is not correct - IP address or station name used twice in the network - station name contains invalid characters, etc. Note: IE: Industrial Ethernet See also: p8900 (IE Name of Station), p8901 (IE IP Address of Station), p8902 (IE Default Gateway of Station), p8903 (IE Subnet Mask of Station)
Remedy:	- Check the required interface configuration (p8900 and following), correct if necessary, and activate (p8905 = 1). or - Reconfigure the station via the "Edit Ethernet node" screen form (e.g. with STARTER commissioning software). See also: p8905 (IE Interface configuration)
A08564	CBE20: Syntax error in configuration file
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A syntax error has been detected in the ASCII configuration file for the Communication Board Ethernet 20 (CBE20). The saved configuration file has not been loaded.
Remedy:	- Check the CBE20 configuration (p8940 and following), correct if necessary, and activate (p8945 = 1). Note: The configuration is not applied until the next POWER ON! - reconfigure the CBE20 (e.g. using the STARTER commissioning software) See also: p8945 (CBE20 interface configuration)

A08565	CBE20: Consistency error affecting adjustable parameters
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A consistency error was detected when activating the configuration (p8945 = 1) for the Communication Board Ethernet 20 (CBE20). The currently set configuration has not been activated. Possible causes: - IP address, subnet mask or default gateway is not correct - IP address or station name used twice in the network - station name contains invalid characters, etc. See also: p8940 (CBE20 Name of Station), p8941 (CBE20 IP Address of Station), p8942 (CBE20 Default Gateway of Station), p8943 (CBE20 Subnet Mask of Station), p8944 (CBE20 DHCP Mode)
Remedy:	Check the required interface configuration (p8940 and following), correct if necessary, and activate (p8945 = 1). See also: p8945 (CBE20 interface configuration)
F08700 (A)	CAN: Communications error
Message value:	%1
Drive object:	CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	Infeed: NONE Servo: OFF3 (NONE, OFF1, OFF2) Vector: OFF3 (NONE, OFF1, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A CAN communications error has occurred. Fault value (r0949, interpret decimal): 1: The error counter for the send telegrams has exceeded the BUS OFF value 255. The bus disables the CAN controller. - bus cable short circuit. - incorrect baud rate. - incorrect bit timing. 2: The master no longer interrogated the CAN node status longer than for its "life time". The "life time" is obtained from the "guard time" (p8604[0]) multiplied by the "life time factor" (p8604[1]). - bus cable interrupted. - bus cable not connected. - incorrect baud rate. - incorrect bit timing. - master fault. Note: The fault response can be set as required using p8641. See also: p8604 (CAN node guarding), p8641 (CAN Abort Connection Option Code)
Remedy:	- check the bus cable - check the baud rate (p8622). - check the bit timing (p8623). - check the master. The CAN controller must be manually restarted with p8608 = 1 after the cause of the fault has been resolved! See also: p8608 (CAN Clear Bus Off Error), p8622 (CAN baud rate), p8623 (CAN Bit Timing selection)
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F08700 (A)	CAN: Communications error
Message value:	%1
Drive object:	A_INF, B_INF, S_INF
Reaction:	Infeed: NONE Servo: OFF3 (NONE, OFF1, OFF2) Vector: OFF3 (NONE, OFF1, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A CAN communications error has occurred. Fault value (r0949, interpret decimal): 1: The error counter for the send telegrams has exceeded the BUS OFF value 255. The bus disables the CAN controller. - bus cable interrupted. - bus cable not connected. - incorrect baud rate. - incorrect bit timing. 2: The master no longer interrogated the CAN node status longer than for its "life time". The "life time" is obtained from the "guard time" (p8604[0]) multiplied by the "life time factor" (p8604[1]). - bus cable interrupted. - bus cable not connected. - incorrect baud rate. - incorrect bit timing. - master fault. See also: p8604 (CAN node guarding), p8641 (CAN Abort Connection Option Code)
Remedy:	- check the bus cable - check the baud rate (p8622). - check the bit timing (p8623). - check the master. The CAN controller must be manually restarted with p8608 = 1 after the cause of the fault has been resolved! See also: p8608 (CAN Clear Bus Off Error), p8622 (CAN baud rate), p8623 (CAN Bit Timing selection)
Reaction upon A:	NONE
Acknowled. upon A:	NONE
F08701	CAN: NMT state change
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	Infeed: OFF2 Servo: OFF3 Vector: OFF3
Acknowledge:	IMMEDIATELY
Cause:	A CANopen NMT state transition from "operational" to "pre-operational" or after "stopped". Fault value (r0949, interpret decimal): 1: CANopen NMT state transition from "operational" to "pre-operational". 2: CANopen NMT state transition from "operational" to "stopped". Note: In the NMT state "pre-operational", process data cannot be transferred and in the NMT state "stopped", no process data and no service data can be transferred.
Remedy:	None necessary. Acknowledge the fault and continue operation.
A08751	CAN: Telegram loss
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The CAN controller has lost a receive message (telegram).
Remedy:	Reduce the cycle times of the receive messages.

A08752	CAN: Error counter for error passive exceeded
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The error counter for the send or receive telegrams has exceeded the value 127.
Remedy:	- check the bus cable - set a higher baud rate (p8622). - check the bit timing and if required optimize (p8623). See also: p8622 (CAN baud rate), p8623 (CAN Bit Timing selection)
A08753	CAN: Message buffer overflow
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A message buffer overflow. Alarm value (r2124, interpret decimal): 1: Non-cyclic send buffer (SDO response buffer) overflow. 2: Non-cyclic receive buffer (SDO receive buffer) overflow. 3: Cyclic send buffer (PDO send buffer) overflow.
Remedy:	Check the bus cable. Set a higher baud rate (p8622). Check the bit timing and if required optimize (p8623). Alarm value = 2: - reduce the cycle times of the SDO receive messages. - SDO request from master only after SDO feedback for previous SDO request. See also: p8622 (CAN baud rate), p8623 (CAN Bit Timing selection)
A08754	CAN: Incorrect communications mode
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the "operational" mode, an attempt was made to change parameters p8700 ... p8737.
Remedy:	Change to the "pre-operational" or "stopped" mode.
A08755	CAN: Object cannot be mapped
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The CANopen object is not provided for the Process Data Object (PDO) Mapping.
Remedy:	Use a CANopen object intended for the PDO mapping or enter 0. The following objects can be mapped in the Receive Process Data Object (RPDO) or Transmit Process Data Object (TPDO): - RPDO: 6040 hex, 6060 hex, 60FF hex, 6071 hex; 5800 hex - 580F hex; 5820 hex - 5827 hex - TPDO: 6041 hex, 6061 hex, 6063 hex, 6069 hex, 606B hex, 606C hex, 6074 hex; 5810 hex - 581F hex; 5830 hex - 5837 hex Only sub-index 0 of the specified objects can be mapped. Note: As long as A08755 is present, the COB-ID cannot be set to valid.

A08756	CAN: Number of mapped bytes exceeded
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The number of bytes of the mapped objects exceeds the telegram size for net data. A max. of 8 bytes is permissible.
Remedy:	Map fewer objects or objects with a smaller data type. See also: p8710, p8711, p8712, p8713, p8714, p8715, p8716, p8717, p8730, p8731, p8732, p8733, p8734, p8735, p8736, p8737
A08757	CAN: Set COB-ID invalid
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	For online operation, the appropriate COB-ID must be set invalid before mapping. Example: Mapping for RPDO 1 should be changed (p8710[0]). --> set p8700[0] = C00006E0 hex (invalid COB-ID) --> set p8710[0] as required. --> p8700[0] enter a valid COB-ID
Remedy:	Set the COB-ID to invalid.
A08758	CAN: Number of PDO channels too low
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The number of PDO channels in p8740 has either been set to 0 or too low. See also: p8740 (CAN channel distribution)
Remedy:	The number of channels set in p8740 must be greater than or equal to the number of PDOs. There are 2 possibilities: Increase the number of channels in p8740 and confirm the selection using p8741. Reduce the number of PDOs by setting the COB-ID to invalid. See also: p8740 (CAN channel distribution), p8741 (CAN PDO configuration acknowledgement)
A08759	CAN: PDO COB-ID already available
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	An existing PDO COB-ID was allocated.
Remedy:	Select another PDO COB-ID.

A13000	License not adequate
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	- for the drive unit, the options that require a license are being used but the licenses are not sufficient. - an error occurred when checking the existing licenses. Alarm value (r2124, interpret decimal): 0: The existing license is not sufficient. 1: An adequate license was not able to be determined as the memory card with the required licensing data was withdrawn in operation. 2: An adequate license was not able to be determined, as an error occurred when reading-out the required licensing data from the memory card. 3: An adequate license was not able to be determined as there is a checksum error in the license key. 4: An internal error occurred when checking the license.
Remedy:	Re alarm value = 0: Additional licenses are required and these must be activated (p9920, p9921). Re alarm value = 1: With the system powered down, re-insert the memory card that matches the system. Re alarm value = 2: Enter and activate the license key (p9920, p9921). Re alarm value = 3: Compare the license key (p9920) entered with the license key on the certificate of license. Re-enter the license key and activate (p9920, p9921). Re alarm value = 4: - carry out a POWER ON. - upgrade firmware to later version. - contact the Hotline.

A13001	Error in license checksum
Message value:	-
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When checking the checksum of the license key, an error was detected.
Remedy:	Compare the license key (p9920) entered with the license key on the certificate of license. Re-enter the license key and activate (p9920, p9921).

F13009	Licensing OA application not licensed
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1
Acknowledge:	IMMEDIATELY
Cause:	At least one OA application which is under license does not have a license. Code for unlicensed OA applications corresponds to p4956 or p4955.
Remedy:	- Enter and activate the license key which licenses OA applications under license (p9920, p9921). - Deactivate unlicensed OA applications (p4956).

F13010	Licensing function module not licensed
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1
Acknowledge:	IMMEDIATELY
Cause:	At least one function module which is under license does not have a license. Fault value (r0949, interpret hexadecimal): Unlicensed function modules are coded bit by bit. The code corresponds to that used in r0108.
Remedy:	- Deactivate unlicensed function modules (p108). - Enter and activate the license key which licenses function modules under license (p9920, p9921).
F30001	Power unit: Overcurrent
Message value:	Fault cause: %1 bin
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The power unit has detected an overcurrent condition. - closed-loop control is incorrectly parameterized. - motor has a short-circuit or fault to ground (frame). - V/f operation: Up ramp set too low. - V/f operation: Rated motor current is significantly greater than that of the Motor Module. - infeed: High discharge and post-charging current for voltage dip. - infeed: High post-charging currents for overload when motoring and DC link voltage dip. - infeed: Short-circuit currents at power-up due to the missing line reactor. - power cables are not correctly connected. - power cables exceed the maximum permissible length. - power unit defective. - line phase interrupted. Additional causes for a parallel switching device (r0108.15 = 1): - a power unit has tripped (powered down) due to a ground fault. - the closed-loop circulating current control is either too slow or has been set too fast. Fault value (r0949, interpret bitwise binary): Bit 0: Phase U. Bit 1: Phase V. Bit 2: Phase W. Bit 3: Overcurrent in the DC link. Note: Fault value = 0 means that the phase with overcurrent is not recognized (e.g. for blocksize device).
Remedy:	- check the motor data - if required, carry out commissioning. - check the motor circuit configuration (star-delta) - V/f operation: Increase up ramp. - V/f operation: Check the assignment of the rated currents of the motor and Motor Module. - infeed: Check the line supply quality. - infeed: Reduce the motor load. - infeed: Correct connection of the line reactor. - check the power cable connections. - check the power cables for short-circuit or ground fault. - check the length of the power cables. - replace power unit. - check the line supply phases. For a parallel switching device (r0108.15 = 1) the following additionally applies: - check the ground fault monitoring thresholds (p0287). - check the setting of the closed-loop circulating current control (p7036, p7037).

F30002 Power unit: DC link voltage, overvoltage

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The power unit has detected overvoltage in the DC link.
- motor regenerates too much energy.
- device connection voltage too high.
- when operating with a Voltage Sensing Module (VSM), the phase assignment L1, L2, L3 at the VSM differs from the phase assignment at the power unit.
- line phase interrupted.
Fault value (r0949, interpret decimal):
DC link voltage [1 bit = 100 mV].
For SINAMICS GM/SM, the following applies:
Fault value (r0949, interpret decimal):
32: Overvoltage in the negative partial DC link (VdcN).
64: Overvoltage in the positive partial DC link (VdcP).
96: Overvoltage in both partial DC links.
Remedy:
- increase the ramp-down time
- activate the DC link voltage controller
- use a brake resistor or Active Line Module
- increase the current limit of the infeed or use a larger module (for the Active Line Module)
- check the device supply voltage
- check and correct the phase assignment at the VSM and at the power unit
- check the line supply phases.
See also: p0210 (Drive unit line supply voltage), p1240 (Vdc controller or Vdc monitoring configuration)

F30002 Power unit: DC link voltage, overvoltage

Message value: %1
Drive object: VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The power unit has detected overvoltage in the DC link.
- motor regenerates too much energy.
- device connection voltage too high.
- when operating with a Voltage Sensing Module (VSM), the phase assignment L1, L2, L3 at the VSM differs from the phase assignment at the power unit.
- line phase interrupted.
Fault value (r0949, interpret decimal):
DC link voltage [1 bit = 100 mV].
For SINAMICS GM/SM, the following applies:
Fault value (r0949, interpret decimal):
32: Overvoltage in the negative partial DC link (VdcN).
64: Overvoltage in the positive partial DC link (VdcP).
96: Overvoltage in both partial DC links.
Remedy:
- increase the ramp-down time
- activate the DC link voltage controller
- use a brake resistor or Active Line Module
- increase the current limit of the infeed or use a larger module (for the Active Line Module)
- check the device supply voltage
- check and correct the phase assignment at the VSM and at the power unit
- check the line supply phases.
- set the rounding times (p1130, p1136). This is particularly recommended in V/f operation to relieve the DC link voltage controller with rapid ramp-down times of the ramp-function generator.
See also: p0210 (Drive unit line supply voltage), p1240 (Vdc controller or Vdc monitoring configuration)

F30003 Power unit: DC link voltage, undervoltage

Message value: -

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY

Cause: The power unit has detected an undervoltage condition in the DC link.
- line supply failure
- line supply voltage below the permissible value.
- line supply infeed failed or interrupted.
- line phase interrupted.
Note:
The monitoring threshold for the DC link undervoltage is the minimum of the following values:
- for a calculation, refer to p0210.

Remedy:
- check the line supply voltage
- check the line supply infeed and observe the fault messages relating to it (if there are any)
- check the line supply phases.
Note:
The ready signal for the infeed (r0863) must be connected to the associated drive inputs (p0864).
See also: p0210 (Drive unit line supply voltage)

F30004 Power unit: Overtemperature heat sink AC inverter

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY

Cause: The temperature of the power unit heat sink has exceeded the permissible limit value.
- insufficient cooling, fan failure.
- overload.
- ambient temperature too high.
- pulse frequency too high.
Fault value (r0949):
Temperature [1 bit = 0.01 °C].

Remedy:
- check whether the fan is running.
- check the fan elements.
- check whether the ambient temperature is in the permissible range.
- check the motor load.
- reduce the pulse frequency if this is higher than the rated pulse frequency.
Notice:
This fault can only be acknowledged after this alarm threshold for alarm A05000 has been undershot.
See also: p1800 (Pulse frequency setpoint)

F30005 Power unit: Overload I2t

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO

Reaction: OFF2

Acknowledge: IMMEDIATELY

Cause: The power unit was overloaded (r0036 = 100 %).
- the permissible rated power unit current was exceeded for an inadmissibly long time.
- the permissible load duty cycle was not maintained.
Fault value (r0949, interpret decimal):
I2t [100 % = 16384].

Remedy:
- reduce the continuous load.
- adapt the load duty cycle.
- check the motor and power unit rated currents.
See also: r0036 (Power unit overload I2t), r0206 (Rated power unit power), p0307 (Rated motor power)

F30005	Power unit: Overload I2t
Message value:	%1
Drive object:	VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The power unit was overloaded (r0036 = 100 %). - the permissible rated power unit current was exceeded for an inadmissibly long time. - the permissible load duty cycle was not maintained. Fault value (r0949, interpret decimal): I2t [100 % = 16384].
Remedy:	- reduce the continuous load. - adapt the load duty cycle. - check the motor and power unit rated currents. - increase p0294 See also: r0036 (Power unit overload I2t), r0206 (Rated power unit power), p0307 (Rated motor power)

F30006	Power unit: Thyristor Control Board
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The Thyristor Control Board (TCB) of the Basic Line Module signals a fault. - there is no line supply voltage. - the line contactor is not closed. - the line supply voltage is too low. - line supply frequency outside the permissible range (45 ... 66 Hz). - there is a DC link short-circuit. - there is a DC link short-circuit (during the pre-charging phase). - voltage supply for the Thyristor Control Board outside the nominal range (5 ... 18 V) and line voltage >30 V. - there is an internal fault in the Thyristor Control Board.
Remedy:	The faults must be saved in the Thyristor Control Board and must be acknowledged. To do this, the supply voltage of the Thyristor Control Board must be switched out for at least 10 s! - check the line supply voltage - check or energize the line contactor. - check the monitoring time and, if required, increase (p0857). - if required, observe additional power unit messages/signals. - check the DC link regarding short-circuit or ground fault. - evaluate diagnostic LEDs for the Thyristor Control Board.

F30008	Power unit: Sign-of-life error cyclic data
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	Infeed: NONE (OFF1, OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The power unit has detected that the cyclic setpoint telegrams of the Control Unit have not been updated on time. The number of consecutive sign-of-life errors has exceeded the fault threshold (p7789).
Remedy:	- check the electrical cabinet design and cable routing for EMC compliance - for projects with the VECTOR drive object, check whether p0117 = 6 has been set on the Control Unit. - increase the fault threshold (p7789). See also: p0117 (Current controller computing dead time mode)

A30010 (F)	Power unit: Sign-of-life error cyclic data
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A DRIVE-CLiQ communication error has occurred between the Control Unit and the power unit involved. The cyclic setpoint telegrams of the Control Unit were not received on time by the power unit for at least one clock cycle.
Remedy:	- check the electrical cabinet design and cable routing for EMC compliance
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
F30011	Power unit: Line phase failure in main circuit
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2 (OFF1)
Acknowledge:	IMMEDIATELY
Cause:	A line phase failure was detected at the power unit. - the fuse of a phase of a main circuit has ruptured. - the DC link voltage ripple has exceeded the permissible limit value. Note: The cause may also be a phase failure in the motor feeder cable.
Remedy:	- check the main circuit fuses. - check the motor feeder cables.
F30012	Power unit: Temperature sensor heat sink wire breakage
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	The connection to one of the heat sink temperature sensors in the power unit is interrupted. Fault value (r0949, interpret hexadecimal): Bit 0: Module slot (electronics slot) Bit 1: Air intake Bit 2: Inverter 1 Bit 3: Inverter 2 Bit 4: Inverter 3 Bit 5: Inverter 4 Bit 6: Inverter 5 Bit 7: Inverter 6 Bit 8: Rectifier 1 Bit 9: Rectifier 2
Remedy:	Contact the manufacturer.

F30013	Power unit: Temperature sensor heat sink short-circuit
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	The heat sink temperature sensor in the power unit is short-circuited. Fault value (r0949, interpret hexadecimal): Bit 0: Module slot (electronics slot) Bit 1: Air intake Bit 2: Inverter 1 Bit 3: Inverter 2 Bit 4: Inverter 3 Bit 5: Inverter 4 Bit 6: Inverter 5 Bit 7: Inverter 6 Bit 8: Rectifier 1 Bit 9: Rectifier 2
Remedy:	Contact the manufacturer.

F30015	Power unit: Phase failure motor feeder cable
Message value:	-
Drive object:	SERVO
Reaction:	OFF2 (NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A phase failure was detected at the power unit output (motor feeder cable). Chassis power units do not feature phase failure monitoring.
Remedy:	Check the motor feeder cables.

F30015	Power unit: Phase failure motor feeder cable
Message value:	-
Drive object:	VECTOR
Reaction:	OFF2 (NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A phase failure was detected at the power unit output (motor feeder cable). Chassis power units do not feature phase failure monitoring. Note: The fault can also be signaled if the drive stalled in V/f control. In this particular case, a phase failure does not necessarily have to be present, but a current of 0 was measured in one phase due to current dissymmetry.
Remedy:	1. Check the motor feeder cables. 2. If the drive has stalled in V/f control, the acceleration or deceleration time (p1120) should be increased.

A30016 (N)	Power unit: Load supply switched out
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The DC link voltage is too low. Alarm value (r2124, interpret decimal): DC link voltage in [V].
Remedy:	- switch on load supply. - check the line supply if necessary.
Reaction upon N:	NONE
Acknowled. upon N:	NONE

F30017	Power unit: Hardware current limit has responded too often
Message value:	Fault cause: %1 bin
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The hardware current limitation in the relevant phase (see A30031, A30032, A30033) has responded too often. The number of times the limit has been exceeded depends on the design and type of power unit. For infeed units, the following applies: - closed-loop control is incorrectly parameterized. - load on the infeed is too high. - Voltage Sensing Module incorrectly connected. - line reactor missing or the incorrect type. - power unit defective. The following applies to Motor Modules: - closed-loop control is incorrectly parameterized. - fault in the motor or in the power cables. - the power cables exceed the maximum permissible length. - motor load too high - power unit defective. Fault value (r0949, interpret binary): Bit 0: Phase U Bit 1: Phase V Bit 2: Phase W
Remedy:	For infeed units, the following applies: - check the controller settings and reset and identify the controller if necessary (p0340 = 2, p3410 = 5) - reduce the load and increase the DC-link capacitance or use a higher-rating infeed if necessary - check the connection of the optional Voltage Sensing Module - check the connection and technical data of the line reactor - check the power cables for short-circuit or ground fault. - replace power unit. The following applies to Motor Modules: - check the motor data. - check the motor circuit configuration (star-delta). - check the motor load. - check the power cable connections. - check the power cables for short-circuit or ground fault. - check the length of the power cables. - replace power unit.
F30020	Power unit: Configuration not supported
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A configuration is requested that is not supported by the power unit. Fault value (r0949, interpret hexadecimal): 0: Autonomous operation is requested but is not supported.
Remedy:	Re fault value = 0: If internal voltage protection is active (p1231 = 3), deactivate it if necessary. See also: p1231 (Armature short-circuit / DC brake configuration)

F30021 Power unit: Ground fault

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: Power unit has detected a ground fault.
- ground fault in the power cables.
- winding fault or ground fault at the motor.
- CT defective.
Additional cause for CU310/CUA31:
- when the brake is applied, this causes the hardware DC current monitoring to respond.
Additional cause for parallel switching devices (r0108.15 = 1):
- the closed-loop circulating current control is either too slow or has been set too fast.
Fault value (r0949, interpret decimal):
Absolute value, total current amplitude [20479 = r0209 * 1.4142].
Remedy: - check the power cable connections.
- check the motor.
- check the CT.
The following applies additionally for CU310/CUA31:
- check the cables and contacts of the brake connection (a wire is possibly broken).
For parallel switching devices (r0108.15 = 1) the following additionally applies:
- check the ground fault monitoring thresholds (p0287).
- check the setting of the closed-loop circulating current control (p7036, p7037).
See also: p0287 (Ground fault monitoring thresholds)

F30022 Power unit: Monitoring V_{ce}

Message value: Fault cause: %1 bin
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: POWER ON
Cause: In the power unit, the monitoring of the collector-emitter voltage (V_{ce}) of the semiconductor has responded.
Possible causes:
- fiber-optic cable interrupted.
- power supply of the IGBT gating module missing.
- short-circuit at the power unit output.
- defective semiconductor in the power unit.
Fault value (r0949, interpret binary):
Bit 0: Short-circuit in phase U
Bit 1: Short circuit in phase V
Bit 2: Short-circuit in phase W
Bit 3: Light transmitter enable defective
Bit 4: V_{ce} group fault signal interrupted
See also: r0949 (Fault value)
Remedy: - check the fiber-optic cable and if required, replace.
- check the power supply of the IGBT gating module (24 V).
- check the power cable connections.
- select the defective semiconductor and replace.

A30023	Power unit: Overtemperature thermal model alarm
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature difference between the heat sink and chip has exceeded the permissible limit value. - the permissible load duty cycle was not maintained. - insufficient cooling, fan failure. - overload. - ambient temperature too high. - pulse frequency too high. See also: r0037
Remedy:	- adapt the load duty cycle. - check whether the fan is running. - check the fan elements. - check whether the ambient temperature is in the permissible range. - check the motor load. - reduce the pulse frequency if this is higher than the rated pulse frequency.

A30023	Power unit: Overtemperature thermal model alarm
Message value:	-
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature difference between the heat sink and chip has exceeded the permissible limit value. - the permissible load duty cycle was not maintained. - insufficient cooling, fan failure. - overload. - ambient temperature too high. - pulse frequency too high. See also: r0037
Remedy:	- adapt the load duty cycle. - check whether the fan is running. - check the fan elements. - check whether the ambient temperature is in the permissible range. - check the motor load. - reduce the pulse frequency if this is higher than the rated pulse frequency. - if DC brake is active: reduce braking current (see p1232).

F30024	Power unit: Overtemperature thermal model
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The temperature difference between the heat sink and chip has exceeded the permissible limit value. - the permissible load duty cycle was not maintained. - insufficient cooling, fan failure. - overload. - ambient temperature too high. - pulse frequency too high. See also: r0037
Remedy:	- adapt the load duty cycle. - check whether the fan is running. - check the fan elements. - check whether the ambient temperature is in the permissible range. - check the motor load. - reduce the pulse frequency if this is higher than the rated pulse frequency.

F30024 Power unit: Overtemperature thermal model

Message value: -
Drive object: VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The temperature difference between the heat sink and chip has exceeded the permissible limit value.
- the permissible load duty cycle was not maintained.
- insufficient cooling, fan failure.
- overload.
- ambient temperature too high.
- pulse frequency too high.
See also: r0037
Remedy: - adapt the load duty cycle.
- check whether the fan is running.
- check the fan elements.
- check whether the ambient temperature is in the permissible range.
- check the motor load.
- reduce the pulse frequency if this is higher than the rated pulse frequency.
- if DC brake is active: reduce braking current (see p1232).

F30025 Power unit: Chip overtemperature

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: Chip temperature of the semiconductor has exceeded the permissible limit value.
- the permissible load duty cycle was not maintained.
- insufficient cooling, fan failure.
- overload.
- ambient temperature too high.
- pulse frequency too high.
Fault value (r0949):
Temperature difference between the heat sink and chip [1 Bit = 0.01 °C].
Remedy: - adapt the load duty cycle.
- check whether the fan is running.
- check the fan elements.
- check whether the ambient temperature is in the permissible range.
- check the motor load.
- reduce the pulse frequency if this is higher than the rated pulse frequency.
Notice:
This fault can only be acknowledged after this alarm threshold for alarm A05001 has been undershot.
See also: r0037

F30027	Power unit: Precharging DC link time monitoring
Message value:	Enable signals: %1, Status: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The power unit DC link was not able to be pre-charged within the expected time. 1) There is no line supply voltage connected. 2) The line contactor/line side switch has not been closed. 3) The line supply voltage is too low. 4) Line supply voltage incorrectly set (p0210). 5) The pre-charging resistors are overheated as there were too many pre-charging operations per time unit. 6) The pre-charging resistors are overheated as the DC link capacitance is too high. 7) The pre-charging resistors are overheated because when there is no "ready for operation" (r0863.0) of the infeed unit, power is taken from the DC link. 8) The pre-charging resistors are overheated as the line contactor was closed during the DC link fast discharge through the Braking Module. 9) The DC link has either a ground fault or a short-circuit. 10) The pre-charging circuit is possibly defective (only for chassis units). Fault value (r0949, interpret binary): Missing internal enable signals, power unit (lower 16 bit): (Inverted bit-coded representation FFFF hex -> all internal enable signals available) Bit 0: Power supply of the IGBT gating shut down Bit 1: Reserved Bit 2: Reserved Bit 3: Ground fault detected Bit 4: Peak current intervention Bit 5: I2t exceeded Bit 6: Thermal model overtemperature calculated Bit 7: (heat sink, gating module, power unit) overtemperature measured Bit 8: Reserved Bit 9: Overvoltage detected Bit 10: Power unit has completed pre-charging, ready for pulse enable Bit 11: STO terminal missing Bit 12: Overcurrent detected Bit 13: Armature short-circuit active Bit 14: DRIVE-CLiQ fault active Bit 15: Uce fault detected, transistor de-saturated due to overcurrent/short-circuit Status, power unit (upper 16 bit, hexadecimal number): 0: Fault status (wait for OFF and fault acknowledgement) 1: Restart inhibit (wait for OFF) 2: Overvoltage condition detected -> change into the fault state 3: Undervoltage condition detected -> change into the fault state 4: Wait for bypass contactor to open -> change into the fault state 5: Wait for bypass contactor to open -> change into restart inhibit 6: Commissioning 7: Ready for pre-charging 8: Pre-charging started, DC link voltage lower than the minimum switch-on voltage 9: Pre-charging, DC link voltage end of pre-charging still not detected 10: Wait for the end of the de-bounce time of the main contactor after pre-charging has been completed 11: Pre-charging completed, ready for pulse enable 12: It was detected that the STO terminal was energized at the power unit See also: p0210 (Drive unit line supply voltage)

Remedy:

In general:

- check the line supply voltage at the input terminals.
- check the line supply voltage setting (p0210).
- the following applies to booksize units: Wait (approx. 8 min.) until the pre-charging resistors have cooled down. For this purpose, preferably disconnect the infeed unit from the line supply.

Re 5):

- carefully observe the permissible pre-charging frequency (refer to the appropriate Equipment Manual).

Re 6):

- check the total capacitance of the DC link and reduce in accordance with the maximum permissible DC-link capacitance if necessary (refer to the appropriate Equipment Manual)

Re 7):

- interconnect the ready-for-operation signal from the infeed unit (r0863.0) in the enable logic of the drives connected to this DC link

Re 8):

- check the connections of the external line contactor. The line contactor must be open during DC-link fast discharge.

Re 9):

- check the DC link for ground faults or short circuits.

See also: p0210 (Drive unit line supply voltage)

A30031 Power unit: Hardware current limiting, phase U

Message value: -

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: Hardware current limit for phase U responded. The pulsing in this phase is inhibited for one pulse period.

- closed-loop control is incorrectly parameterized.
- fault in the motor or in the power cables.
- the power cables exceed the maximum permissible length.
- motor load too high
- power unit defective.

Note:
Alarm A30031 is always output if, for a Power Module, the hardware current limiting of phase U, V or W responds.

Remedy:

- check the motor data.
- check the motor circuit configuration (star-delta)
- check the motor load.
- check the power cable connections.
- check the power cables for short-circuit or ground fault.
- check the length of the power cables.

A30032 Power unit: Hardware current limiting, phase V

Message value: -

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: Hardware current limit for phase V responded. The pulsing in this phase is inhibited for one pulse period.

- closed-loop control is incorrectly parameterized.
- fault in the motor or in the power cables.
- the power cables exceed the maximum permissible length.
- motor load too high
- power unit defective.

Note:
Alarm A30031 is always output if, for a Power Module, the hardware current limiting of phase U, V or W responds.

Remedy:

- check the motor data.
- check the motor circuit configuration (star-delta)
- check the motor load.
- check the power cable connections.
- check the power cables for short-circuit or ground fault.
- check the length of the power cables.

A30033	Power unit: Hardware current limiting, phase W
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Hardware current limit for phase W responded. The pulsing in this phase is inhibited for one pulse period. - closed-loop control is incorrectly parameterized. - fault in the motor or in the power cables. - the power cables exceed the maximum permissible length. - motor load too high - power unit defective. Note: Alarm A30031 is always output if, for a Power Module, the hardware current limiting of phase U, V or W responds.
Remedy:	- check the motor data. - check the motor circuit configuration (star-delta) - check the motor load. - check the power cable connections. - check the power cables for short-circuit or ground fault. - check the length of the power cables.
A30034	Power unit: Internal overtemperature
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The alarm threshold for internal overtemperature has been reached. If the temperature inside the unit continues to increase, fault F30036 may be triggered. - ambient temperature might be too high. - insufficient cooling, fan failure. Fault value (r0949, interpret binary): Bit 0 = 1: Control electronics range. Bit 1 = 1: Power electronics range.
Remedy:	- check the ambient temperature. - check the fan for the inside of the unit.
F30035	Power unit: Air intake overtemperature
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	The air intake in the power unit has exceeded the permissible temperature limit. For air-cooled power units, the temperature limit is at 55 °C. - ambient temperature too high. - insufficient cooling, fan failure. Fault value (r0949, interpret decimal): Temperature [1 bit = 0.01 °C].
Remedy:	- check whether the fan is running. - check the fan elements. - check whether the ambient temperature is in the permissible range. Notice: This fault can only be acknowledged after this alarm threshold for alarm A05002 has been undershot.

F30036 Power unit: Internal overtemperature

Message value: %1
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: For chassis power units, the following applies:
The temperature inside the drive converter has exceeded the permissible temperature limit.
- insufficient cooling, fan failure.
- overload.
- ambient temperature too high.
Fault value (r0949, interpret binary):
Bit 0 = 1: Control electronics range.
Bit 1 = 1: Power electronics range.
Remedy: - check whether the fan is running.
- check the fan elements.
- check whether the ambient temperature is in the permissible range.
Notice:
This fault can only be acknowledged once the permissible temperature limit minus 5 K has been undershot.

F30037 Power unit: Rectifier overtemperature

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The temperature in the rectifier of the power unit has exceeded the permissible temperature limit.
- insufficient cooling, fan failure.
- overload.
- ambient temperature too high.
- line supply phase failure.
Fault value (r0949, interpret decimal):
Temperature [1 bit = 0.01 °C].
Remedy: - check whether the fan is running.
- check the fan elements.
- check whether the ambient temperature is in the permissible range.
- check the motor load.
- check the line supply phases.
Notice:
This fault can only be acknowledged after this alarm threshold for alarm A05004 has been undershot.

A30038 Power unit: Capacitor fan monitoring

Message value: %1
Drive object: B_INF
Reaction: NONE
Acknowledge: NONE
Cause: The capacitor fan signals a fault.
Remedy: Replace the capacitor fan in the power unit.

F30039 Power unit: Failure capacitor fan

Message value: %1
Drive object: B_INF
Reaction: OFF1
Acknowledge: IMMEDIATELY
Cause: The capacitor fan has failed.
Remedy: Replace the capacitor fan in the power unit.

F30040	Power unit: Undervolt 24 V
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	Failure of the 24 V power supply for the power unit. - The undervoltage threshold was undershot for longer than 3 ms. Fault value (r0949, interpret decimal): 24 V voltage [1 bit = 0.1 V].
Remedy:	- check the 24 V DC voltage supply to power unit. - carry out a POWER ON (power off/on) for the component.
A30041 (F)	Power unit: Undervoltage 24 V alarm
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	24 V power supply fault for the power unit. - the 16 V threshold was undershot.. Fault value (r0949, interpret decimal): 24 V voltage [1 bit = 0.1 V].
Remedy:	- check the 24 V DC voltage supply to power unit. - carry out a POWER ON (power off/on) for the component.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
A30042	Power unit: Fan operating time reached or exceeded
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The maximum operating time of the fan in the power unit is set in p0252. This message indicates the following: Fault value (r0949, interpret decimal): 0: The maximum fan operating time is 500 hours. 1: The maximum fan operating time has been exceeded.
Remedy:	Replace the fan in the power unit and reset the operating hours counter to 0 (p0251 = 0). See also: p0251 (Operating hours counter power unit fan), p0252 (Maximum operating time power unit fan)
F30043	Power unit: Overvolt 24 V
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	POWER ON
Cause:	The following applies for CU31x: Overvoltage of the 24 V power supply for the power unit. - the 31.5 V threshold was exceeded for more than 3 ms. Fault value (r0949): 24 V voltage [1 bit = 0.1 V].
Remedy:	Check the 24 V DC voltage supply to the power unit.

A30044 (F) Power unit: Overvoltage 24 V alarm

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The following applies for CU31x:
24 V power supply fault for the power unit.
- the 32.0 V threshold was exceeded.
Fault value (r0949):
24 V voltage [1 bit = 0.1 V].
Remedy: Check the 24 V DC voltage supply to the power unit.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY (POWER ON)

F30045 Power unit: Supply undervoltage

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: Power supply fault in the power unit.
- The voltage monitor signals an undervoltage fault on the module.
The following applies for CU31x:
- the voltage monitoring on the DAC board signals an undervoltage fault on the module.
Remedy: - check the 24 V DC voltage supply to power unit.
- carry out a POWER ON (power off/on) for the component.
- replace the module if necessary.

A30046 (F) Power unit: Undervoltage, alarm

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Before the last restart, a problem occurred at the power unit power supply.
- the voltage monitor in the internal FPGA of the PSA signals an undervoltage fault on the module.
Fault value (r0949):
Register value of the voltage fault register.
Remedy: - check the 24 V DC voltage supply to power unit.
- carry out a POWER ON (power off/on) for the component.
- replace the module if necessary.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY (POWER ON)

F30047 Cooling system: Cooling medium flow rate too low

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: Cooling system: Fault - flow rate has fallen below the fault value
Remedy:

A30048	Power unit: External fan faulty
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The feedback signal from the external fan indicates a fault. - fan faulty, blocked. - feedback signal inaccurate.
Remedy:	- check the external fan and replace if necessary. - if you are using an external fan with feedback, check its wiring (X12.2 or X13.2). Note: If you are using an external fan without feedback, check that the feedback terminal wiring on the power unit is connected to ground and make this connection if necessary (X12.1/2 or X13.1/2).
A30049	Power unit: Internal fan faulty
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The internal fan has failed.
Remedy:	Check the internal fan and replace if necessary.
F30050	Power unit: 24 V supply overvoltage
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	POWER ON
Cause:	The voltage monitor signals an overvoltage fault on the module.
Remedy:	- check the 24 V power supply. - replace the module if necessary.
F30052	EEPROM data error
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	POWER ON
Cause:	EEPROM data error of the power unit module. Fault value (r0949, interpret hexadecimal): 0: The EEPROM data read in from the power unit module is inconsistent. 1: EEPROM data is not compatible to the firmware of the power unit application. Additional values: Only for internal Siemens troubleshooting.
Remedy:	Re fault value = 0: Replace the power unit module or update the EEPROM data. Re fault value = 1: The following applies for CU31x and CUA31: Update the firmware \SIEMENS\SINAMICS\CODE\SAC\cu31xi.ufw (cua31.ufw)
F30053	FPGA data faulty
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	POWER ON
Cause:	Power unit module FPGA data error.
Remedy:	Replace the power unit module or update the FPGA data.

A30054 (F)	Power unit: 24 V braking undervoltage
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	If the brake needs to be released during brake control, the P24 voltage must be at least 21.6 V (24 V - 10%) so that the brake is sure to release. If the voltage undershoots the 21.6 V threshold when the brake is released, the alarm is triggered; it goes away again as soon as the voltage rises above a threshold value of 22 V. When the alarm is triggered, the voltage value is shown in the fault value in increments of 0.1 V (e.g. 21.2 V = 212).
Remedy:	Make sure that the 24 V voltage is stable and does not fluctuate downwards by more than 10%.
Reaction upon F:	NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY

F30060 (A)	Pre-charge contactor monitoring
Message value:	Fault cause: %1 bin
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2 (NONE, OFF1)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	At the end of the monitoring time (p0255[0]), the actual state of the pre-charge contactor does not match the desired state. Bit 0: Monitoring time exceeded. Bit 1: Contactor opened during operation. Bit 2: Contactor closed in OFF state.
Remedy:	- check the monitoring time setting p0255[0]. - check the contactor wiring and activation. - replace the contactor.
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F30061 (A)	Bridging contactor monitoring
Message value:	Fault cause: %1 bin
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2 (NONE, OFF1)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	At the end of the monitoring time (p0255[1]), the actual state of the bridging contactor does not match the desired state. Bit 0: Monitoring time exceeded. Bit 1: Contactor opened during operation. Bit 2: Contactor closed in OFF state.
Remedy:	- check the monitoring time setting p0255[1]. - check the contactor wiring and activation. - replace the contactor.
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F30062 (N, A)	Bypass contactor has been opened under current
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The bypass contactor for the infeed has been damaged due to (multiple) opening under current. Possible causes: - Scheduled opening under load can be necessary to protect the converter line-up in the event of a ground fault in high-frequency spindles. - Operating the contactor under load can result from incorrect operation of the infeed; for example, if motoring power is drawn from the DC link although no operating enable is present for the infeed.

Remedy: Damaged infeeds must be replaced, to prevent serious damage to the entire converter line-up.
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F30070 Cycle requested by the power unit module not supported

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A cycle is requested that is not supported by the power unit.
 Fault value (r0949, interpret hexadecimal):
 0: The current control cycle is not supported.
 1: The DRIVE-CLiQ cycle is not supported.
 2: Internal timing problem (clearance between RX and TX instants too low).
 3: Internal timing problem (TX instant too early).
Remedy: The power unit only supports the following cycles:
 62.5 µs, 125 µs, 250 µs and 500 µs
 Re fault value = 0:
 Set a permitted current control cycle.
 Re fault value = 1:
 Set a permitted DRIVE-CLiQ cycle.
 Re fault value = 2, 3:
 Contact the manufacturer (you may have an incompatible firmware version).

F30071 No new actual values received from the power unit module

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The number of actual value telegrams from the power unit module that have failed has exceeded the permissible number.
Remedy: Check the interface (adjustment and locking) to the power unit module.

F30072 Setpoints are no longer being transferred to the power unit

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The following applies for CU31x and CUA31:
 More than one setpoint telegram was not able to be transferred to the power unit module.
Remedy: The following applies for CU31x and CUA31:
 Check the interface (adjustment and locking) to the power unit module.

A30073 (N) Actual value/setpoint preprocessing no longer synchronous

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Communication with the power unit module is no longer in synchronism with the current control cycle.
Remedy: Wait until synchronization is re-established.
Reaction upon N: NONE
Acknowl. upon N: NONE

F30074	Communications error to the power unit module
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	Communication is not possible with the power unit module via the plug connection.
Remedy:	The following applies for CU31x and CUA31: Either replace the Control Unit or the power unit. You must check which of the two components needs to be replaced by replacing one component and then the other (ensuring that the component which is not being tested in each case is fully functioning). If these are not available, then both components must be returned.
F30080	Power unit: Current increasing too quickly
Message value:	Fault cause: %1 bin
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The power unit has detected an excessive rate of rise in the overvoltage range. - closed-loop control is incorrectly parameterized. - motor has a short-circuit or fault to ground (frame). - V/f operation: Up ramp set too low. - V/f operation: Rated motor current is significantly greater than that of the Motor Module. - infeed: High discharge and post-charging current for voltage dip. - infeed: High post-charging currents for overload when motoring and DC link voltage dip. - infeed: Short-circuit currents at power-up due to the missing line reactor. - power cables are not correctly connected. - power cables exceed the maximum permissible length. - power unit defective. Additional causes for a parallel switching device (r0108.15 = 1): - a power unit has tripped (powered down) due to a ground fault. - the closed-loop circulating current control is either too slow or has been set too fast. Fault value (r0949, interpret bitwise binary): Bit 0: Phase U. Bit 1: Phase V. Bit 2: Phase W.
Remedy:	- check the motor data - if required, carry out commissioning. - check the motor circuit configuration (star-delta) - V/f operation: Increase up ramp. - V/f operation: Check the assignment of the rated currents of the motor and Motor Module. - infeed: Check the line supply quality. - infeed: Reduce the motor load. - infeed: Correct connection of the line reactor. - check the power cable connections. - check the power cables for short-circuit or ground fault. - check the length of the power cables. - replace power unit. For a parallel switching device (r0108.15 = 1) the following additionally applies: - check the ground fault monitoring thresholds (p0287). - check the setting of the closed-loop circulating current control (p7036, p7037).

F30081	Power unit: Switching operations too frequent
Message value:	Fault cause: %1 bin
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The power unit has executed too many switching operations for current limitation. - closed-loop control is incorrectly parameterized. - motor has a short-circuit or fault to ground (frame). - V/f operation: Up ramp set too low. - V/f operation: Rated motor current is significantly greater than that of the Motor Module. - infeed: High discharge and post-charging current for voltage dip. - infeed: High post-charging currents for overload when motoring and DC link voltage dip. - infeed: Short-circuit currents at power-up due to the missing line reactor. - power cables are not correctly connected. - power cables exceed the maximum permissible length. - power unit defective. Additional causes for a parallel switching device (r0108.15 = 1): - a power unit has tripped (powered down) due to a ground fault. - the closed-loop circulating current control is either too slow or has been set too fast. Fault value (r0949, interpret bitwise binary): Bit 0: Phase U. Bit 1: Phase V. Bit 2: Phase W.
Remedy:	- check the motor data - if required, carry out commissioning. - check the motor circuit configuration (star-delta) - V/f operation: Increase up ramp. - V/f operation: Check the assignment of the rated currents of the motor and Motor Module. - infeed: Check the line supply quality. - infeed: Reduce the motor load. - infeed: Correct connection of the line reactor. - check the power cable connections. - check the power cables for short-circuit or ground fault. - check the length of the power cables. - replace power unit. For a parallel switching device (r0108.15 = 1) the following additionally applies: - check the ground fault monitoring thresholds (p0287). - check the setting of the closed-loop circulating current control (p7036, p7037).
F30105	PU: Actual value sensing fault
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	At least one incorrect actual value channel was detected on the Power Stack Adapter (PSA). The incorrect actual value channels are displayed in the following diagnostic parameters.
Remedy:	Evaluate the diagnostic parameters. If the actual value channel is incorrect, check the components and if required, replace.
A30502	Power unit: DC link voltage, overvoltage
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The power unit has detected overvoltage in the DC link on a pulse inhibit. - device connection voltage too high. - line reactor incorrectly dimensioned. Fault value (r0949, interpret decimal): DC link voltage [1 bit = 100 mV]. See also: r0070 (Actual DC link voltage)

Remedy:

- check the device supply voltage (p0210).
- check the dimensioning of the line reactor.

See also: p0210 (Drive unit line supply voltage)

F30600 SI MM: STOP A initiated

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive-based "Safety Integrated" function in the Motor Module (MM) has detected a fault and initiated STOP A (pulse suppression via the safety shutdown path of the Motor Module).

- forced checking procedure of the safety shutdown path of the Motor Module unsuccessful.
- subsequent response to fault F30611 (defect in a monitoring channel).

Fault value (r0949, interpret decimal):

0: Stop request from the Control Unit.

1005: Pulses suppressed although STO not selected and there is no internal STOP A present.

1010: Pulses enabled although STO is selected or an internal STOP A is present.

1020: Internal software error in the "Internal voltage protection" function. The "internal voltage protection" function is withdrawn. A STOP A that cannot be acknowledged is initiated.

9999: Subsequent response to fault F30611.

Remedy:

- select Safe Torque Off and deselect again.
- replace the Motor Module involved.

Re fault value = 1020:

- carry out a POWER ON (power off/on) for all components.
- upgrade the Motor Module software.
- replace the Motor Module.

Re fault value = 9999:

- carry out diagnostics for fault F30611.

Note:

CU: Control Unit

MM: Motor Module

SI: Safety Integrated

STO: Safe Torque Off / SH: Safe standstill

F30611 SI MM: Defect in a monitoring channel

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: Infeed: NONE (OFF1, OFF2)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive-based "Safety Integrated" function in the Motor Module (MM) has detected a fault in the data cross-check between the Control Unit (CU) and MM and initiated a STOP F.

As a result of this fault, after the parameterized transition has expired (p9858), fault F30600 is output (SI MM: STOP A initiated).

Fault value (r0949, interpret decimal):

0: Stop request from the Control Unit.

1 ... 999:

Number of the cross-checked data that resulted in this fault. This number is also displayed in r9895.

- 1: SI monitoring clock cycle (r9780, r9880).
- 2: SI enable safety functions (p9601, p9801). Crosswise data comparison is only carried out for the supported bits.
- 3: SI SGE changeover tolerance time (p9650, p9850).
- 4: SI transition period STOP F to STOP A (p9658, p9858).
- 5: SI enable Safe Brake Control (p9602, p9802).
- 6: SI Motion enable, safety-relevant functions (p9501, internal value).
- 7: SI pulse suppression delay time for Safe Stop 1 (p9652, p9852).
- 8: SI PROFIsafe address (p9610, p9810).
- 9: SI debounce time for STO/SBC/SS1 (MM) (p9651, p9851).
- 11: SI Safe Brake Adapter mode, BICO interconnection (p9621, p9821).
- 12: SI Safe Brake Adapter relay ON time (p9622[0], p9822[0]).
- 13: SI Safe Brake Adapter relay OFF time (p9622[1], p9822[1]).

1000: Watchdog timer has expired. Within the time of approx. 5 * p9850 too many switching operations have occurred at the safety-related inputs of the Control Unit, or STO (also as subsequent response) was initiated too frequently via PROFIsafe/TM54F.

1001, 1002: Initialization error, change timer / check timer.

2000: Status of the STO selection on the Control Unit and Motor Module are different.

2001: Feedback signal for safe pulse suppression on the Control Unit and Motor Module are different.

2002: Status of the delay timer SS1 on the Control Unit and Motor Module are different.

6000 ... 6999:

Error in the PROFIsafe control. The significance of the individual message values is described in safety message C01711 of the Control Unit.

Remedy:

Re fault value = 1 ... 5 and 7 ... 999:

- check the cross-checked data that resulted in a STOP F.
- carry out a POWER ON (power off/on) for all components.
- upgrade the Motor Module software.
- upgrade the Control Unit software.

Re fault value = 6:

- carry out a POWER ON (power off/on) for all components.
- upgrade the Motor Module software.
- upgrade the Control Unit software.

Re fault value = 1000:

- check the wiring of the safety-relevant inputs (SGE) on the Control Unit (contact problems).
- PROFIsafe: Remove contact problems/faults at the PROFIBUS master/PROFINET controller. - check the wiring of the fail-safe inputs at the TM54F (contact problems).

Re fault value = 1001, 1002:

- carry out a POWER ON (power off/on) for all components.
- upgrade the Motor Module software.
- upgrade the Control Unit software.

Re fault value = 2000, 2001, 2002:

- check the tolerance time SGE changeover and if required, increase the value (p9650/p9850, p9652/p9852).
- check the wiring of the safety-relevant inputs (SGE) (contact problems).
- check the cause of the STO selection in r9772. When the SMM functions are active (p9501 = 1), STO can also be selected using these functions.
- replace the Motor Module involved.

Re fault value = 6000 ... 6999:

Refer to the description of the message values in safety message C01711.

Note:

CU: Control Unit

MM: Motor Module

SGE: Safety-relevant input

SI: Safety Integrated

SMM: Safe Motion Monitoring

SS1: Safe Stop 1 (corresponds to Stop Category 1 acc. to EN60204)

STO: Safe Torque Off / SH: Safe standstill

N30620 (F, A) SI MM: Safe Torque Off active

Message value: -

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The "Safe Torque Off" function was selected on the Motor Module (MM) via the input terminal and is active.

Note:

This message does not result in a safety stop response.

Remedy: None necessary.

Note:

MM: Motor Module

SI: Safety Integrated

STO: Safe Torque Off / SH: Safe standstill

Reaction upon F: OFF2

Acknowl. upon F: IMMEDIATELY (POWER ON)

Reaction upon A: NONE

Acknowl. upon A: NONE

N30621 (F, A)	SI MM: Safe Stop 1 active
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The "Safe Stop 1" function (SS1) was selected on the Motor Module (MM) and is active. Note: This message does not result in a safety stop response.
Remedy:	None necessary. Note: MM: Motor Module SI: Safety Integrated SS1: Safe Stop 1 (corresponds to Stop Category 1 acc. to EN60204)
Reaction upon F:	Infeed: OFF2 Servo: OFF3 Vector: OFF3
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F30625	SI MM: Sign-of-life error in safety data
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The drive-based "Safety Integrated" function on the Motor Module (MM) has detected an error in the sign-of-life of the safety data between the Control Unit (CU) and MM and initiated a STOP A. - there is either a DRIVE-CLiQ communication error or communication has failed. - a time slice overflow of the safety software has occurred. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting.
Remedy:	- select Safe Torque Off and deselect again. - carry out a POWER ON (power off/on) for all components. - check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified. - deselect all drive functions that are not absolutely necessary. - reduce the number of drives. - check the electrical cabinet design and cable routing for EMC compliance Note: CU: Control Unit MM: Motor Module SI: Safety Integrated
F30630	SI MM: Brake control error
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The drive-based "Safety Integrated" function on the Motor Module (MM) has detected a brake control error and initiated a STOP A. Fault value (r0949, interpret decimal): 10: Fault in "open holding brake" operation. - Parameter p1278 incorrectly set. - No brake connected or wire breakage (check whether brake releases for p1278 = 1 and p9602/p9802 = 0 (SBC deactivated)). - Ground fault in brake cable.

List of faults and alarms

30:

Fault in "close holding brake" operation.

- No brake connected or wire breakage (check whether brake releases for p1278 = 1 and p9602/p9802 = 0 (SBC deactivated)).

- Short-circuit in brake winding.

40:

Fault in "brake closed" state.

60, 70:

Fault in the brake control circuit of the Control Unit or communication fault between the Control Unit and Motor Module (brake control).

81: SafeBrakeAdapter: Fault in "brake closed" state.

82: SafeBrakeAdapter: Fault in "open brake" state.

83: SafeBrakeAdapter: Fault in "close brake" state.

84,85:

SafeBrakeAdapter: Fault in the brake control circuit of the Control Unit or communication fault between Control Unit and Motor Module (brake control).

Note:

The following causes may apply to fault values:

- motor cable is not shielded correctly.

- defect in control circuit of the Motor Module.

Remedy:

- check parameter p1278 (for SBC, only p1278 = 0 is permissible).

- select Safe Torque Off and deselect again.

- check the motor holding brake connection.

- check the function of the motor holding brake.

- check whether there is a DRIVE-CLiQ communication error between the Control Unit and the Motor Module involved and, if required, carry out a diagnostics routine for the faults identified.

- check that the electrical cabinet design and cable routing are in compliance with EMC regulations (e.g. shield of the motor cable and brake conductors are connected with the shield connecting plate and the motor connectors are tightly screwed to the housing).

- replace the Motor Module involved.

Operation with Safe Brake Module or Safe Brake Adapter:

- check the Safe Brake Module or Safe Brake Adapter connection.

- Replace the Safe Brake Module or Safe Brake Adapter.

Note:

MM: Motor Module

SBC: Safe Brake Control

SI: Safety Integrated

F30640

SI MM: Fault in the shutdown path of the second channel

Message value:

%1

Drive object:

A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction:

OFF2

Acknowledge:

IMMEDIATELY (POWER ON)

Cause:

The Motor Module has detected a communications error with the higher-level control or the TM54F to transfer the safety-relevant information.

Note:

This fault results in a STOP A that can be acknowledged.

Fault value (r0949, interpret decimal):

Only for internal Siemens troubleshooting.

Remedy:

For the higher-level control, the following applies:

- check the PROFIsafe address in the higher-level control and Motor Modules and if required, align.
- save all parameters (p0977 = 1).
- carry out a POWER ON (power off/on) for all components.

For TM54F, carry out the following steps:

- start the copy function for the node identifier (p9700 = 1D hex).
- acknowledge hardware CRC (p9701 = EC hex).
- save all parameters (p0977 = 1).
- carry out a POWER ON (power off/on) for all components.

The following generally applies:

- upgrade the Motor Module software.

Note:

MM: Motor Module
SI: Safety Integrated
See also: p9810 (SI PROFIsafe address (Motor Module))

F30649 SI MM: Internal software error

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: An internal error in the Safety Integrated software on the Motor Module has occurred.

Note:

This fault results in a STOP A that cannot be acknowledged.

Fault value (r0949, interpret hexadecimal):

Only for internal Siemens troubleshooting.

Remedy:

- carry out a POWER ON (power off/on) for all components.
- re-commission the Safety Integrated function and carry out a POWER ON.
- upgrade the Motor Module software.
- contact the Hotline.
- replace the Motor Module.

Note:

MM: Motor Module
SI: Safety Integrated

F30650 SI MM: Acceptance test required

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The "Safety Integrated" function on the Motor Module requires an acceptance test.

Note:

This fault results in a STOP A that can be acknowledged.

Fault value (r0949, interpret decimal):

130: Safety parameters for the Motor Module not available.

1000: Reference and actual checksum in the Motor Module are not identical (booting).

- at least one checksum-checked piece of data is defective.

2000: Reference and actual checksum on the Motor Module are not identical (commissioning mode).

- reference checksum incorrectly entered into the Motor Module (p9899 not equal to r9898).

2003: Acceptance test is required as a safety parameter has been changed.

2005: The safety logbook has identified that the safety checksums have changed. An acceptance test is required.

3003: Acceptance test is required as a hardware-related safety parameter has been changed.

9999: Subsequent response of another safety-related fault that occurred when booting that requires an acceptance test.

Remedy:

Re fault value = 130:
 - carry out safety commissioning routine.

Re fault value = 1000:
 - again carry out safety commissioning routine.
 - replace the memory card or Control Unit.

Re fault value = 2000:
 - check the safety parameters in the Motor Module and adapt the reference checksum (p9899).

Re fault value = 2003, 2005:
 - Carry out an acceptance test and generate an acceptance report.
 The procedure when carrying out an acceptance test as well as an example of the acceptance report are provided in the following literature:
 SINAMICS S120 Function Manual Safety Integrated

Re fault value = 3003:
 - carry out the function checks for the modified hardware and generate an acceptance report.
 The procedure when carrying out an acceptance test as well as an example of the acceptance report are provided in the following literature:
 SINAMICS S120 Function Manual Safety Integrated

Re fault value = 9999:
 - carry out diagnostics for the other safety-related fault that is present.

Note:
 MM: Motor Module
 SI: Safety Integrated
 See also: p9799 (SI reference checksum SI parameters (Control Unit)), p9899 (SI reference checksum SI parameters (Motor Module))

F30651 SI MM: Synchronization with Control Unit unsuccessful

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive-based "Safety Integrated" function is requesting synchronization of the safety time slices on the Control Unit and Motor Module. This synchronization routine was unsuccessful.
 Note:
 This fault results in a STOP A that cannot be acknowledged.
 Fault value (r0949, interpret decimal):
 Only for internal Siemens troubleshooting.

Remedy:
 - carry out a POWER ON (power off/on) for all components.
 - upgrade the Motor Module software.
 - upgrade the Control Unit software.
 Note:
 MM: Motor Module
 SI: Safety Integrated

F30652 SI MM: Illegal monitoring clock cycle

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The Safety Integrated monitoring clock cycle cannot be maintained due to the communication conditions requested in the system.
 Note:
 This fault results in a STOP A that cannot be acknowledged.
 Fault value (r0949, interpret decimal):
 Only for internal Siemens troubleshooting.

Remedy:
 Upgrade the Motor Module software.
 Note:
 MM: Motor Module
 SI: Safety Integrated

F30655	SI MM: Align monitoring functions
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	An error has occurred when aligning the Safety Integrated monitoring functions on the Control Unit (CU) and Motor Module (MM). Control Unit and Motor Module were not able to determine a common set of supported SI monitoring functions. - there is either a DRIVE-CLiQ communication error or communication has failed. - Safety Integrated software releases on the Control Unit and Motor Module are not compatible with one another. Note: This fault results in a STOP A that cannot be acknowledged. Fault value (r0949, interpret hexadecimal): Only for internal Siemens troubleshooting.
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade the Motor Module software. - upgrade the Control Unit software. - check the electrical cabinet design and cable routing for EMC compliance Note: CU: Control Unit MM: Motor Module SI: Safety Integrated
F30656	SI MM: Motor Module parameter error
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	When accessing the Safety Integrated parameters for the Motor Module (MM) in the non-volatile memory, an error has occurred. Note: This fault results in a STOP A that can be acknowledged. Fault value (r0949, interpret decimal): 129: Safety parameters for the Motor Module corrupted. 131: Internal software error on the Control Unit. 255: Internal Motor Module software error.
Remedy:	- re-commission the safety functions. - upgrade the Control Unit software. - upgrade the Motor Module software. - replace the memory card or Control Unit. Note: MM: Motor Module SI: Safety Integrated

F30659	SI MM: Write request for parameter rejected
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The write request for one or several Safety Integrated parameters on the Motor Module (MM) was rejected. Note: This fault does not result in a safety stop response. Fault value (r0949, interpret decimal): 10: An attempt was made to enable the STO function although this cannot be supported. 11: An attempt was made to enable the SBC function although this cannot be supported. 13: An attempt was made to enable the SS1 function although this cannot be supported. 14: An attempt was made to enable the safe motion monitoring function with the higher-level control, although this cannot be supported. 15: An attempt was made to enable the motion monitoring functions integrated in the drive although these cannot be supported. 16: An attempt was made to enable the PROFIsafe communication - although this cannot be supported or the version of the PROFIsafe driver used on the CU and MM is different. 18: An attempt was made to enable the PROFIsafe function for Basic Functions although this cannot be supported. See also: r9771 (SI common functions (Control Unit)), r9871 (SI common functions (Motor Module))
Remedy:	Re fault value = 10, 11, 13, 14, 15, 16, 18: - check whether there are faults in the safety function alignment between the Control Unit and the Motor Module involved (F01655, F30655) and if required, carry out diagnostics for the faults involved. - use a Motor Module that supports the required function ("Safe Torque Off", "Safe Brake Control", "PROFIsafe/PROFIsafe V2", "motion monitoring functions integrated in the drive"). - upgrade the Motor Module software. - upgrade the Control Unit software. Note: CU: Control Unit MM: Motor Module SBC: Safe Brake Control SI: Safety Integrated SS1: Safe Stop 1 (corresponds to Stop Category 1 acc. to EN60204) STO: Safe Torque Off / SH: Safe standstill
F30662	Error in internal communications
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF2
Acknowledge:	POWER ON
Cause:	A module-internal communication error has occurred. Fault value (r0949, interpret hexadecimal): Only for internal Siemens troubleshooting.
Remedy:	- carry out a POWER ON (power off/on). - upgrade firmware to later version. - contact the Hotline.
F30664	Error while booting
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF2
Acknowledge:	POWER ON
Cause:	An error has occurred during booting. Fault value (r0949, interpret hexadecimal): Only for internal Siemens troubleshooting.

Remedy:

- carry out a POWER ON (power off/on).
- upgrade firmware to later version.
- contact the Hotline.

F30665 SI MM: System is defective

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The drive has detected a defect in the system and performed an emergency shutdown.
 Fault value (r0949, interpret hexadecimal):
 Only for internal Siemens troubleshooting.
Remedy: - carry out a POWER ON (power off/on).
 - upgrade firmware to later version.
 - contact the Hotline.

A30666 (F) SI Motion MM: Steady-state (static) 1 signal at the F-DI for safety-relevant acknowledgement

Message value: -
Drive object: SERVO
Reaction: NONE
Acknowledge: NONE
Cause: A logical 1 signal is present at the F-DI parameterized in p10106 for more than 10 seconds. A logical 0 signal must be present statically (steady-state) at the F-DI. This avoids unintentional safety-relevant acknowledgement (or the "Internal Event Acknowledge" signal) if a wire breaks or one of the two digital inputs bounces.
Remedy: Set F-DI (see p10106) to a logical 0 signal.
Reaction upon F: NONE
Acknowl. upon F: IMMEDIATELY

F30672 SI CU: Control Unit software incompatible

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The existing Control Unit software does not support the safe drive-based motion monitoring function.
 Note:
 This fault results in a STOP A that cannot be acknowledged.
 Fault value (r0949, interpret decimal):
 Only for internal Siemens troubleshooting.
Remedy: - check whether there are faults in the safety function alignment between the Control Unit and the Motor Module involved (F01655, F30655) and if required, carry out diagnostics for the faults involved.
 - use a Control Unit that supports the safe motion monitoring function.
 - upgrade the Control Unit software.
 Note:
 SI: Safety Integrated

F30680	SI Motion MM: Checksum error safety monitoring functions
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The actual checksum calculated by the Motor Module and entered in r9398 over the safety-relevant parameters does not match the reference checksum saved in p9399 at the last machine acceptance. Safety-relevant parameters have been changed or a fault is present. Note: This fault results in a STOP A that can be acknowledged. Fault value (r0949, interpret decimal): 0: Checksum error for SI parameters for motion monitoring. 1: Checksum error for SI parameters for component assignment.
Remedy:	- check the safety-relevant parameters and if required, correct. - set the reference checksum to the actual checksum. - perform a POWER ON if safety parameters requiring a POWER ON have been modified. - carry out an acceptance test. Note: SI: Safety Integrated
F30681	SI Motion MM: Incorrect parameter value
Message value:	Parameter: %1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The parameter value cannot be parameterized with this value. Note: This message does not result in a safety stop response. Fault value (r0949, interpret decimal): Parameter number with the incorrect value.
Remedy:	Correct the parameter value. If the encoder parameters (p9526/p9326) have different values, start the copy function for SI parameters on the drive (p9700 = 57 hex). Also check p9316.0 for fault value 9317.
F30682	SI Motion MM: Monitoring function not supported
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The monitoring function enabled in p9301, p9501, p9601 or p9801 is not supported in this firmware version. Note: This message does not result in a safety stop response. Fault value (r0949, interpret decimal): 1: Monitoring function SLP not supported (p9301.1). 2: Monitoring function SCA not supported (p9301.7 and p9301.8 ... 15). 3: Monitoring function SLS override not supported (p9301.5). 4: Monitoring function external ESR activation not supported (p9301.4). 5: Monitoring function FDI in PROFI-safe not supported (p9301.30). 6: Enable actual value synchronization not supported (p9301.3). 9: Monitoring function not supported, enable bit reserved (p9301.2, p9301.17 ... 29, p9301.31, if required p9301.6). 30: The firmware version of the Motor Module is older than the version of the Control Unit.
Remedy:	- deselect the monitoring function involved (p9301, p9301, p9303, p9601, p9801). - Upgrade the Motor Module firmware. See also: p9301 (SI Motion enable safety functions (Motor Module)), p9501 (SI Motion enable safety functions (Control Unit)), p9503 (SI Motion SCA (SN) enable (Control Unit)), p9601 (SI enable, functions integrated in the drive (Control Unit)), p9801 (SI enable, functions integrated in the drive (Motor Module))

F30683 SI Motion MM: SOS/SLS enable missing

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The safety-relevant basic function "SOS/SLS" is not enabled in p9301 although other safety-relevant monitoring functions are enabled.
Note:
This message does not result in a safety stop response.
Remedy: Enable the function "SOS/SLS" (p9301.0).
Note:
SI: Safety Integrated
SLS: Safely-Limited Speed / SG: Safely reduced speed
SOS: Safe Operating Stop / SBH: Safe operating stop
See also: p9301 (SI Motion enable safety functions (Motor Module))

F30685 SI Motion MM: Safely-Limited Speed limit value too high

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The limit value for the function "Safely-Limited Speed" (SLS) is greater than the speed that corresponds to an encoder limit frequency of 500 kHz.
Note:
This message does not result in a safety stop response.
Fault value (r0949, interpret decimal):
Maximum permissible speed.
Remedy: Correct the limit values for SLS and carry out a POWER ON.
Note:
SI: Safety Integrated
SLS: Safely-Limited Speed / SG: Safely reduced speed
See also: p9331 (SI Motion SLS limit values (Motor Module))

F30688 SI Motion MM: Actual value synchronization not permissible

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY (POWER ON)
Cause:
- It is not permissible to enable actual value synchronization for a 1-encoder system.
- It is not permissible to simultaneously enable actual value synchronization and a monitoring function with absolute reference (SCA/SLP).
Note:
This fault results in a STOP A that cannot be acknowledged.
Remedy:
- Either select the "actual value synchronization" function or parameterize a 2-encoder system.
- Either deselect the function "actual value synchronization" or the monitoring functions with absolute reference (SCA/SLP) and carry out a POWER ON.
Note:
SCA: Safe Cam / SN: Safe software cam
SI: Safety Integrated
SLP: Safely-Limited Position / SE: Safe software limit switches
See also: p9501 (SI Motion enable safety functions (Control Unit)), p9526 (SI Motion encoder assignment second channel)

F30692	SI Motion MM: Incorrect parameter value encoderless
Message value:	Parameter: %1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The parameter cannot be set to this value if encoderless motion monitoring functions have been parameterized in p9306. Note: This message does not result in a safety stop response. Fault value (r0949, interpret decimal): Parameter number with the incorrect value. See also: p9301 (SI Motion enable safety functions (Motor Module))
Remedy:	Correct the parameter value or deselect encoderless motion monitoring functions. See also: p9301 (SI Motion enable safety functions (Motor Module)), p9501 (SI Motion enable safety functions (Control Unit))
A30693 (F)	SI MM: Safety parameter settings changed, warm restart/POWER ON required
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Safety parameters have been changed; these will only take effect following a warm restart or POWER ON. Alarm value (r2124, interpret decimal): Parameter number of the safety parameter which has changed, necessitating a warm restart or POWER ON.
Remedy:	- carry out a warm restart (p0971 = 3) - carry out a POWER ON (power off/on) for all components.
Reaction upon F:	NONE (OFF1, OFF2, OFF3)
Acknowled. upon F:	POWER ON
C30700	SI Motion MM: STOP A initiated
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The drive is stopped via a STOP A (pulses are suppressed via the safety shutdown path of the Control Unit). Possible causes: - stop request from the Control Unit. - pulses not suppressed after a parameterized time (p9357) after test stop selection. - subsequent response to the message C30706 "SI Motion MM: Safe Acceleration Monitoring, limit exceeded". - subsequent response to the message C30714 "SI Motion MM: Safely-Limited Speed exceeded". - subsequent response to the message C30701 "SI Motion MM: STOP B initiated".
Remedy:	- remove the cause to the fault on the Control Unit. - check the value in p9357, if required, increase the value. - check the shutdown path of the Control Unit (check DRIVE-CLiQ communication). - carry out a diagnostics routine for message C30706. - carry out a diagnostics routine for message C30714. - carry out a diagnostics routine for message C30701. - replace the Motor Module/Power Module - replace Control Unit. This message can only be acknowledged via the Terminal Module 54F (TM54F) or PROFIsafe. Note: SI: Safety Integrated

C30701 SI Motion MM: STOP B initiated

Message value: -

Drive object: SERVO, VECTOR

Reaction: OFF3

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive is stopped via a STOP B (braking along the OFF3 ramp).
As a result of this fault, after the time parameterized in p9356 has expired, or the speed threshold parameterized in p9360 has been undershot, message C30700 "SI Motion MM: STOP A initiated" is output.
Possible causes:
- stop request from the Control Unit.
- subsequent response to the message C30714 "SI Motion MM: Safely limited speed exceeded".
- subsequent response to the message C30711 "SI Motion MM: Defect in a monitoring channel".

Remedy: - remove the cause of the fault on the Control Unit
- carry out a diagnostics routine for message C01714.
- carry out a diagnostics routine for message C01711.
This message can only be acknowledged via the Terminal Module 54F (TM54F) or PROFIsafe.
Note:
SI: Safety Integrated

C30706 SI Motion MM: SBR limit exceeded

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: Motion monitoring functions with encoder: SBR - Safe Acceleration Monitoring. After initiating STOP B (SS1) or STOP C (SS2), the speed has exceeded the selected tolerance.
Encoderless motion monitoring function: SBR - Safe Brake Ramp Monitoring. After initiating STOP B (SS1) or SLS changeover to the lower speed stage, the speed has exceeded the selected tolerance.
The drive is shut down by the message C30700 "SI Motion MM: STOP A initiated".

Remedy: Check the braking behavior and, if necessary, adapt the tolerance for the "Safe Acceleration Monitor" or modify the parameter settings for the "safe brake ramp".
This message can only be acknowledged via the Terminal Module 54F (TM54F) or PROFIsafe.
Note:
SI: Safety Integrated
See also: p9548 (SI Motion SBR actual velocity tolerance (Control Unit))

C30707 SI Motion MM: Tolerance for safe operating stop exceeded

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The actual position has distanced itself further from the target position than the standstill tolerance.
The drive is shut down by the message C30701 "SI Motion MM: STOP B initiated".

Remedy: - check whether safety faults are present and if required carry out the appropriate diagnostic routines for the particular faults.
- check whether the standstill tolerance matches the accuracy and control dynamic performance of the axis.
This message can only be acknowledged via the Terminal Module 54F (TM54F) or PROFIsafe.
Note:
SI: Safety Integrated
SOS: Safe Operating Stop / SBH: Safe operating stop
See also: p9530 (SI Motion standstill tolerance (Control Unit))

C30708 SI Motion MM: STOP C initiated

Message value: -

Drive object: SERVO, VECTOR

Reaction: STOP2

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive is stopped via a STOP C (braking along the OFF3 ramp).
 "Safe Operating Stop" (SOS) is activated after the parameterized timer has expired.
 Possible causes:
 - stop request from the higher-level control.
 - subsequent response to the message C30714 "SI Motion MM: Safely limited speed exceeded".
 See also: p9552 (SI Motion transition time STOP C to SOS (SBH) (Control Unit))

Remedy: - remove the cause of the fault at the control.
 - carry out a diagnostics routine for message C30714.
 This message can only be acknowledged via the Terminal Module 54F (TM54F) or PROFIsafe.

Note:
 SI: Safety Integrated
 SOS: Safe Operating Stop / SBH: Safe operating stop

C30709 SI Motion MM: STOP D initiated

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The drive is stopped via a STOP D (braking along the path).
 "Safe Operating Stop" (SOS) is activated after the parameterized timer has expired.
 Possible causes:
 - stop request from the Control Unit.
 - subsequent response to the message C30714 "SI Motion: Safely limited speed exceeded".
 See also: p9353 (SI Motion transition time STOP D to SOS (Motor Module)), p9553 (SI Motion transition time STOP D to SOS (SBH) (Control Unit))

Remedy: - remove the cause of the fault at the control.
 - carry out a diagnostics routine for message C30714.
 This message can only be acknowledged via the Terminal Module 54F (TM54F) or PROFIsafe.

Note:
 SI: Safety Integrated
 SOS: Safe Operating Stop / SBH: Safe operating stop

C30711 SI MM MM: Defect in a monitoring channel

Message value: %1

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: When cross-checking and comparing the two monitoring channels, the drive detected a difference between the input data or results of the monitoring functions and initiated a STOP F. One of the monitoring functions no longer reliably functions - i.e. safe operation is no longer possible.
 If at least one monitoring function is active, then after the parameterized timer has expired, the message C30701 "SI Motion: STOP B initiated" is output. The message is output with message value 1031 when the Sensor Module hardware is replaced.
 The following message values may also occur in the following cases if the cause that is explicitly mentioned does not apply:
 - differently parameterized cycle times (p9500/p9300, p9511/p9311).
 - differently parameterized axis types (p9502/p9302).
 - excessively fast cycle times (p9500/p9300, p9511/p9311).
 - incorrect synchronization.
 Message value (r9749, interpret decimal):
 0 ... 999: Number of the cross-checked data that resulted in this message. Refer to safety message C01711 for a description of the individual data.
 The significance of the individual message values is described in safety message C01711 of the Control Unit.
 1000: Watchdog timer has expired. Too many signal changes have occurred at safety-relevant inputs.

1001: Initialization error of watchdog timer.
 1005: Pulses already suppressed for test stop selection.
 1011: Acceptance test status between the monitoring channels differ.
 1012: Plausibility violation of the actual value from the encoder.
 1020: Cyc. communication failure between the monit. cycles.
 1021: Cyc. communication failure between the monit. channel and Sensor Module.
 1023: Error in the effectiveness test in the DRIVE-CLiQ encoder
 1030: Encoder fault detected from another monitoring channel.
 1031: Data transfer error between the monitoring channel and the Sensor Module (p9526/p9326).
 1040: Pulses suppressed with active encoderless monitoring functions.
 1041: Current absolute value too low (encoderless)
 1042: Current/voltage plausibility error
 1043: Too many acceleration phases
 1044: Actual current values plausibility error.
 5000 ... 5140: PROFIsafe message values.
 The significance of the individual message values is described in safety message C01711 of the Control Unit.
 6000 ... 6166: PROFIsafe message values (PROFIsafe driver for PROFIBUS DP V1/V2 and PROFINET).
 The significance of the individual message values is described in safety message C01711 of the Control Unit.
 See also: p9555 (SI Motion transition time STOP F to STOP B (Control Unit)), r9725 (SI Motion, diagnostics STOP F)

Remedy:

Re message value = 1030:
 - check the encoder connection.
 - if required, replace the encoder.

Re message value = 1031:
 When replacing a Sensor Module, carry out the following steps:
 - start the copy function for the node identifier on the drive (p9700 = 1D hex).
 - acknowledge the hardware CRC on the drive (p9701 = EC hex).
 - save all parameters (p0977 = 1).
 - carry out a POWER ON (power off/on) for all components.
 The following always applies:
 - check the encoder connection.
 - if required, replace the encoder.

Re message value = 1040:
 - deselect encoderless monitoring functions, select and deselect STO.
 - if monitoring function is active, issue SLS pulse enable within 5 s of deselecting STO.

Re other message values:
 - The significance of the individual message values is described in safety message C01711 of the Control Unit.

Note:
 This message can only be acknowledged via the Terminal Module 54F (TM54F) or PROFIsafe.
 See also: p9300 (SI Motion monitoring clock cycle (Motor Module)), p9500 (SI Motion monitoring clock cycle (Control Unit))

C30712 SI Motion MM: Defect in F-IO processing

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY (POWER ON)
Cause: When cross checking and comparing the two monitoring channels, the drive detected a difference between parameters or results of the F-IO processing and initiated a STOP F. One of the monitoring functions no longer reliably functions - i.e. safe operation is no longer possible.
 The safety message C30711 with message value 0 is also displayed due to initiation of STOP F.
 If at least one monitoring function is active, the safety message C30701 "SI Motion: STOP B initiated" is output after the parameterized timer has expired.
 Message value (r9749, interpret decimal):
 Number of the cross-checked data that resulted in this message.
 Refer to the description of the message values in safety message C01712.

Remedy:

- check parameterization in the parameters involved and correct if required.
 - ensure equality by copying the SI data to the second channel and then carry out an acceptance test.
 - check monitoring clock cycle in p9500 and p9300 for equality.

Note:
 This message can be acknowledged via F-DI or PROFIsafe.
 See also: p9300 (SI Motion monitoring clock cycle (Motor Module)), p9500 (SI Motion monitoring clock cycle (Control Unit))

C30714	SI Motion MM: Safely-Limited Speed exceeded
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The drive had moved faster than that specified by the velocity limit value (p9331). The drive is stopped as a result of the configured stop response (p9363). Message value (r9749, interpret decimal): 100: SLS1 exceeded. 200: SLS2 exceeded. 300: SLS3 exceeded. 400: SLS4 exceeded. 1000: Encoder limit frequency exceeded.
Remedy:	- check the traversing/motion program in the control. - check the limits for "Safely-Limited Speed" (SLS) and if required, adapt (p9331). This message can only be acknowledged via the Terminal Module 54F (TM54F) or PROFIsafe. Note: SI: Safety Integrated SLS: Safely-Limited Speed / SG: Safely reduced speed See also: p9331 (SI Motion SLS limit values (Motor Module)), p9363 (SI Motion SLS stop response (Motor Module))
C30770	SI Motion MM: Discrepancy error affecting the fail-safe inputs/outputs
Message value:	%1
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The safety input terminals or output terminals show a different state longer than that parameterized in p10002 (or p10102). Fault value (r0949, interpret hexadecimal): yyyyxxxx hex xxxx: The safety-relevant input terminals F-DI indicate a discrepancy. Bit 0: Discrepancy for F-DI 0 Bit 1: Discrepancy for F-DI 1 ... yyyy: The safety-relevant output terminals F-DO indicate a discrepancy. Bit 0: Discrepancy for F-DO 0 ... Note: If several discrepancy errors occur consecutively, then this fault is only signaled for the first error that occurs.

Remedy:

- check the wiring of the F-DI (contact problems).
- carry out safe acknowledgment (p10006 or INTERNAL EVENT ACKNOWLEDGE via PROFIsafe).

Note:
 Discrepancy error of an F-DI can only be completely acknowledged if safe acknowledgement was carried out once the cause of the error was resolved (p10106). As long as safety acknowledgement was not carried out, the corresponding F-DI stays in the safe state.
 F-DI: Failsafe Digital Input
 F-DO: Failsafe Digital Output
 Where switching operations recur cyclically on the FDIs, the discrepancy time must be parameterized as follows:
 td = possible actual discrepancy time (in ms) that can occur with a switching operation. This must correspond to at least 1 SI sampling cycle (see p10000).
 tp = period for a switching operation in ms.
 The following rules must be adhered to:
 $p10002 < (tp/2) - td$ (discrepancy time must be less than half the period minus the actual discrepancy time)
 $p10002 \geq p10000$ (discrepancy time must be no less than P10000)
 $p10002 > td$ (discrepancy time must be greater than the switch discrepancy time which may actually apply)
 Example: If SI sampling cycle is 12 ms and switching frequency is 110 ms, the maximum discrepancy time which can be set is as follows:
 $p10002 \leq 110ms/2 - 12 ms = 43 ms$; this rounds off to $P10002 \leq 36 ms$ (Since the sampling time can only be accepted as a whole SI sampling cycle, the value will need to be rounded up or down to a whole SI sampling time value if it is not an exact multiple of an SI sampling cycle.)

A30772 Test stop fail-safe inputs/outputs active

Message value: -
Drive object: SERVO
Reaction: NONE
Acknowledge: NONE
Cause: The test stop for the fail-safe inputs and/or outputs is currently in progress.
 F1773 is output when an error occurs during the test stop.
Remedy: The alarm disappears automatically after successfully ending or canceling (when a fault condition occurs) the test stop.

F30773 Motor Module test stop faulty

Message value: %1
Drive object: SERVO
Reaction: NONE
Acknowledge: IMMEDIATELY (POWER ON)
Cause: A fault has occurred on the MM side during the test stop for the fail-safe outputs.
 The fault value is output in RRRVWXYZ hex format.
 The wildcards V, W, X must be interpreted as hexadecimal values and evaluated as binary values.
 Bit 0 stands for DO 0, bit 1 for DO 1, bit 2 for DO 2, and bit 3 for DO 4.
 The wildcards Z and Y must be evaluated as hexadecimal values.
 The meaning of the wildcards is as follows:
 R: Reserved
 V: State of the affected DO channel (see X) on the MM (corresponds to the expected state in the case of faulty test stop state)
 W: Required state of the affected DO channel (see X)
 X: DO channels concerned which are indicating a fault
 Y: Reason for the test stop fault
 Z: State of the test stop in which the fault has occurred
 Reason for the test stop fault (Y):
 1 = CU side in incorrect test stop state (internal fault)
 2 = States of the DO(s) (CU305: readback via DI22/CU240, readback via DI2) not as expected
 3 = Incorrect timer state on MM side (internal fault)
 4 = States of the Diag-DO(s) (CU305: internal readback on the Motor Module channel) not as expected
 5 = States of the second Diag-DO(s) (CU305: internal readback on the CU channel) not as expected.
 X and V indicate the DI or Diag-DO state dependent upon the reason for the fault (2 or 4).
 In the event of multiple test stop faults, the first one that occurred is shown.

Test stop state (Z) and associated test actions:

0 to 3 Synchronization phase of test stop between CU and Motor Module no switching operations

4 DO + OFF and DO - OFF

5 Check to see if states are as expected

6 DO + ON and DO - ON

7 Check to see if states are as expected

8 DO + OFF and DO - ON

9 Check to see if states are as expected

10 DO + ON and DO - OFF

11 Check to see if states are as expected

12 DO + OFF and DO - OFF

13 Check to see if states are as expected

14 End of test stop

Diag expected states in table format:

Test stop state: Expectation Mode 1 / Mode 2 / Mode 3 / Mode 4

5 :0/-/-/1

7 :0/-/-/0

8 :0/-/-/0

11 :1/-/-/1

13 :0/-/-/1

Second Diag expected states in table format:

Test stop state: Expectation Mode 1 / Mode 2 / Mode 3 / Mode 4

5 :-/-/-/1

7 :-/-/-/0

8 :-/-/-/1

11 :-/-/-/0

13 :-/-/-/1

DI expected states in table format:

Test stop state: Expectation Mode 1 / Mode 2 / Mode 3 / Mode 4

5: -/1/0/-

7: -/0/1/-

8: -/0/0/-

11:-/0/0/-

13:-/1/0/-

Example:

Alarm 1773 (CU) is signaled with fault value 0001_0127 and alarm 30773 (MM) is signaled with fault value 0000_0127. This means: In state 7 (Z=7) the state of the external readback signal was not set correctly (Y=2) after DO-0 (X=1) was switched to ON/ON. Fault value 0001_0127 indicates that 0 was expected (W=0) and 1 (V=1) was read back from the hardware. Fault value 0000_0127 on the MM indicates that the states were as expected. In the case of alarm 30773 W and V are always identical; a value of 0 always means that 0 was expected at the readback input but was not present on the other channel (CU).

Remedy:

Check the wiring of the F-DOs and restart the test stop. The fault is withdrawn once the test stop is successfully completed.

In the event of multiple test stop faults, the first one that occurred is shown. Once the test stop has been restarted the next queued test stop fault will be signaled (if there is one).

C30798

SI Motion MM: Test stop running

Message value:

-

Drive object:

SERVO, VECTOR

Reaction:

NONE

Acknowledge:

IMMEDIATELY (POWER ON)

Cause:

The test stop is active.

Remedy:

None necessary.

The message is withdrawn when the test stop is finished.

Note:

SI: Safety Integrated

C30799	SI Motion MM: Acceptance test mode active
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The acceptance test mode is active. The POWER ON signals of the safety-relevant motion monitoring functions can be acknowledged during the acceptance test using the acknowledgement functions of the higher-level control.
Remedy:	None necessary. The message is withdrawn when exiting the acceptance test mode. Note: SI: Safety Integrated

N30800 (F)	Power unit: Group signal
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	NONE
Cause:	The power unit has detected at least one fault.
Remedy:	Evaluates other actual messages.
Reaction upon F:	OFF2
Acknowl. upon F:	IMMEDIATELY

F30801	Power unit DRIVE-CLiQ: Sign-of-life missing
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Control Unit to the power unit concerned. The computing time load might be too high. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0A hex: The sign-of-life bit in the receive telegram is not set.
Remedy:	- check the electrical cabinet design and cable routing for EMC compliance - remove DRIVE-CLiQ components that are not required. - deselect functions that are not required. - if required, increase the sampling times (p0112, p0115). - replace the component involved.

F30802	Power unit: Time slice overflow
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Time slide overflow.
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade firmware to later version. - contact the Hotline.

A30804 (F)	Power unit: CRC
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	CRC error actuator
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade firmware to later version. - contact the Hotline.
Reaction upon F:	Infeed: OFF2 (OFF1) Servo: OFF2 (OFF1, OFF3) Vector: OFF2 (OFF1, OFF3)
Acknowl. upon F:	IMMEDIATELY
F30805	Power unit: EPROM checksum error
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Internal parameter data is corrupted. Fault value (r0949, interpret hexadecimal): 01: EEPROM access error. 02: Too many blocks in the EEPROM.
Remedy:	Replace the module.
F30809	Power unit: Switching information not valid
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	For 3P gating unit: The last switching status word in the setpoint telegram is identified by the end ID. Such an end ID was not found.
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade firmware to later version. - contact the Hotline.
A30810 (F)	Power unit: Watchdog timer
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When booting it was detected that the cause of the previous reset was an SAC watchdog timer overflow.
Remedy:	- carry out a POWER ON (power off/on) for all components. - upgrade firmware to later version. - contact the Hotline.
Reaction upon F:	NONE (OFF2)
Acknowl. upon F:	IMMEDIATELY

F30820 Power unit DRIVE-CLiQ: Telegram error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Control Unit to the power unit concerned.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 01 hex:
CRC error.
xx = 02 hex:
Telegram is shorter than specified in the length byte or in the receive list.
xx = 03 hex:
Telegram is longer than specified in the length byte or in the receive list.
xx = 04 hex:
The length of the receive telegram does not match the receive list.
xx = 05 hex:
The type of the receive telegram does not match the receive list.
xx = 06 hex:
The address of the component in the telegram and in the receive list do not match.
xx = 07 hex:
A SYNC telegram is expected - but the received telegram is not a SYNC telegram.
xx = 08 hex:
No SYNC telegram is expected - but the received telegram is one.
xx = 09 hex:
The error bit in the receive telegram is set.
xx = 10 hex:
The receive telegram is too early.
Remedy:
- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F30835 Power unit DRIVE-CLiQ: Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Control Unit to the power unit concerned. The nodes do not send and receive in synchronism.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 21 hex:
The cyclic telegram has not been received.
xx = 22 hex:
Timeout in the telegram receive list.
xx = 40 hex:
Timeout in the telegram send list.
Remedy:
- carry out a POWER ON.
- replace the component involved.
See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F30836	Power unit DRIVE-CLiQ: Send error for DRIVE-CLiQ data
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Control Unit to the power unit concerned. Data were not able to be sent. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 41 hex: Telegram type does not match send list.
Remedy:	Carry out a POWER ON.
F30837	Power unit DRIVE-CLiQ: Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component concerned. Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error.
Remedy:	- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). - check the electrical cabinet design and cable routing for EMC compliance - if required, use another DRIVE-CLiQ socket (p9904). - replace the component involved.
F30845	Power unit DRIVE-CLiQ: Cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Control Unit to the power unit concerned. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex: Synchronization error during alternating cyclic data transfer.
Remedy:	Carry out a POWER ON. See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
F30850	Power unit: Internal software error
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: OFF1 (NONE, OFF2, OFF3) Vector: OFF1 (NONE, OFF2, OFF3)
Acknowledge:	POWER ON
Cause:	An internal software error has occurred in the power unit. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting.

Remedy:

- replace power unit.
- if required, upgrade the firmware in the power unit.
- contact the Hotline.

F30851 Power unit DRIVE-CLiQ (CU): Sign-of-life missing

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
Servo: OFF2 (NONE, OFF1, OFF3)
Vector: OFF2 (NONE, OFF1, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the power unit to the Control Unit involved.
The DRIVE-CLiQ component did not set the sign-of-life to the Control Unit.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 0A hex = 10 dec:
The sign-of-life bit in the receive telegram is not set.
Remedy: Upgrade the firmware of the component involved.

A30853 Power unit: Sign-of-life error cyclic data

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The power unit has detected that the cyclic setpoint telegrams of the Control Unit have not been updated on time. At least two sign-of-life errors have occurred within the window set in p7788.
Remedy: - check the electrical cabinet design and cable routing for EMC compliance
- reduce the size of the window (p7788) for monitoring.

F30860 Power unit DRIVE-CLiQ (CU): Telegram error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the power unit to the Control Unit involved.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 11 hex = 17 dec:
CRC error and the receive telegram is too early.
xx = 01 hex = 01 dec:
Checksum error (CRC error).
xx = 12 hex = 18 dec:
The telegram is shorter than that specified in the length byte or in the receive list and the receive telegram is too early.
xx = 02 hex = 02 dec:
Telegram is shorter than specified in the length byte or in the receive list.
xx = 13 hex = 19 dec:
The telegram is longer than that specified in the length byte or in the receive list and the receive telegram is too early.
xx = 03 hex = 03 dec:
Telegram is longer than specified in the length byte or in the receive list.
xx = 14 hex = 20 dec:
The length of the receive telegram does not match the receive list and the receive telegram is too early.
xx = 04 hex = 04 dec:
The length of the receive telegram does not match the receive list.
xx = 15 hex = 21 dec:
The type of the receive telegram does not match the receive list and the receive telegram is too early.
xx = 05 hex = 05 dec:
The type of the receive telegram does not match the receive list.

xx = 16 hex = 22 dec:
The address of the power unit in the telegram and in the receive list does not match and the receive telegram is too early.
xx = 06 hex = 06 dec:
The address of the power unit in the telegram and in the receive list do not match.
xx = 19 hex = 25 dec:
The error bit in the receive telegram is set and the receive telegram is too early.
xx = 09 hex = 09 dec:
The error bit in the receive telegram is set.
xx = 10 hex = 16 dec:
The receive telegram is too early.

Remedy:

- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F30885 CU DRIVE-CLiQ (CU): Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the power unit to the Control Unit involved.
The nodes do not send and receive in synchronism.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 1A hex = 26 dec:
Sign-of-life bit in the receive telegram not set and the receive telegram is too early.
xx = 21 hex = 33 dec:
The cyclic telegram has not been received.
xx = 22 hex = 34 dec:
Timeout in the telegram receive list.
xx = 40 hex = 64 dec:
Timeout in the telegram send list.
xx = 62 hex = 98 dec:
Error at the transition to cyclic operation.

Remedy:

- check the power supply voltage of the component involved.
- carry out a POWER ON.
- replace the component involved.

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F30886 PU DRIVE-CLiQ (CU): Error when sending DRIVE-CLiQ data

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the power unit to the Control Unit involved.
Data were not able to be sent.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 41 hex:
Telegram type does not match send list.

Remedy:

Carry out a POWER ON.

F30887	Power unit DRIVE-CLiQ (CU): Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component (power unit) involved. Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error. xx = 60 hex: Response received too late during runtime measurement. xx = 61 hex: Time taken to exchange characteristic data too long.
Remedy:	- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). - check the electrical cabinet design and cable routing for EMC compliance - if required, use another DRIVE-CLiQ socket (p9904). - replace the component involved.

F30895	PU DRIVE-CLiQ (CU): Alternating cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2) Vector: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the power unit to the Control Unit involved. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex: Synchronization error during alternating cyclic data transfer.
Remedy:	Carry out a POWER ON. See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F30896	Power unit DRIVE-CLiQ (CU): Inconsistent component properties
Message value:	Component number: %1
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2) Vector: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	The properties of the DRIVE-CLiQ component (power unit), specified by the fault value, have changed in an incompatible fashion with respect to the properties when booted. One cause can be, e.g. that a DRIVE-CLiQ cable or DRIVE-CLiQ component has been replaced. Fault value (r0949, interpret decimal): Component number.
Remedy:	- carry out a POWER ON. - when a component is replaced, the same component type and if possible the same firmware version should be used. - when a cable is replaced, only cables whose length is the same as or as close as possible to the length of the original cables should be used (ensure compliance with the maximum cable length).

F30897	DRIVE-CLiQ (CU) power unit: no communication with component
Message value:	Preliminary component number: %1
Drive object:	VECTOR
Reaction:	OFF2 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	Communication with the DRIVE-CLiQ component (power unit) specified by the fault value is not possible. One cause may, for example, be that a DRIVE-CLiQ cable has been withdrawn. Fault value (r0949, interpret decimal): Component ID.
Remedy:	- check the DRIVE-CLiQ connections. - carry out a POWER ON.
F30899 (N, A)	Power unit: Unknown fault
Message value:	New message: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	Infeed: NONE (OFF1, OFF2) Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2) Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	A fault occurred on the power unit that cannot be interpreted by the Control Unit firmware. This can occur if the firmware on this component is more recent than the firmware on the Control Unit. Fault value (r0949, interpret decimal): Fault number. Note: If required, the significance of this new fault can be read about in a more recent description of the Control Unit.
Remedy:	- replace the firmware on the power unit by an older firmware version (r0128). - upgrade the firmware on the Control Unit (r0018).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F30903	Power unit: I2C bus error occurred
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	Infeed: NONE (OFF1, OFF2) Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2) Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	Communications error with an EEPROM or A/D converter. Fault value (r0949, interpret hexadecimal): 80000000 hex: - internal software error. 00000001 hex ... 0000FFFF hex: - module fault.
Remedy:	Re fault value = 80000000 hex: - upgrade firmware to later version. Re fault value = 00000001 hex ... 0000FFFF hex: - replace the module.

F30907 Power unit: FPGA configuration unsuccessful

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO
Reaction: Infeed: OFF2 (NONE, OFF1)
Servo: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge: IMMEDIATELY
Cause: During initialization within the power unit, an internal software error has occurred.
Remedy: - if required, upgrade the firmware in the power unit.
- replace power unit.
- contact the Hotline.

A30920 (F) Power unit: Temperature sensor fault

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When evaluating the temperature sensor, an error occurred.
Alarm value (r2124, interpret decimal):
1: Wire breakage or sensor not connected (KTY: R > 1630 Ohm, PT100: R > 375 Ohm).
2: Measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm, PT100: R < 30 Ohm).
Remedy: - make sure that the sensor is connected correctly.
- replace the sensor.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY

A30999 (F, N) Power unit: Unknown alarm

Message value: New message: %1
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An alarm occurred on the power unit that cannot be interpreted by the Control Unit firmware.
This can occur if the firmware on this component is more recent than the firmware on the Control Unit.
Alarm value (r2124, interpret decimal):
Alarm number.
Note:
If required, the significance of this new alarm can be read about in a more recent description of the Control Unit.
Remedy: - replace the firmware on the power unit by an older firmware version (r0128).
- upgrade the firmware on the Control Unit (r0018).
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

F31100 (N, A)	Encoder 1: Zero mark distance error
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	The measured zero mark distance does not correspond to the parameterized zero mark distance. For distance-coded encoders, the zero mark distance is determined from zero marks detected pairs. This means that if a zero mark is missing, depending on the pair generation, this cannot result in a fault and also has no effect in the system. The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder). Fault value (r0949, interpret decimal): Last measured zero mark distance in increments (4 increments = 1 encoder pulse). The sign designates the direction of motion when detecting the zero mark distance. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - check the encoder type (encoder with equidistant zero marks). - adapt the parameter for the distance between zero marks (p0424, p0425). - if message output above speed threshold, reduce filter time if necessary (p0438). - replace the encoder or encoder cable
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31101 (N, A)	Encoder 1: Zero marked failed
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	The 1.5 x parameterized zero mark distance was exceeded. The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder). Fault value (r0949, interpret decimal): Number of increments after POWER ON or since the last zero mark that was detected (4 increments = 1 encoder pulse). See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - check the encoder type (encoder with equidistant zero marks). - adapt the parameter for the clearance between zero marks (p0425). - if message output above speed threshold, reduce filter time if necessary (p0438). - when p0437.1 is active, check p4686. - replace the encoder or encoder cable
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F31103 (N, A)	Encoder 1: Amplitude error, track R
Message value:	R track: %1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The amplitude of the reference track signal (track R) does not lie within the tolerance bandwidth for encoder 1. The fault can be initiated when the unipolar voltage level is exceeded (RP/RN) or if the differential amplitude is under-shot. Fault value (r0949, interpret hexadecimal): xxxx hex: xxxx = Signal level, track R (16 bits with sign). The response thresholds of the unipolar signal levels of the encoder are between < 1400 mV and > 3500 mV. The response threshold for the differential signal level of the encoder is < -1600 mV. A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec. Note: The analog value of the amplitude error is not measured at the same time with the hardware fault output by the Sensor Module. The fault value can only be represented between -32767dec and 32767dec (-770 mV ... 770 mV). The signal level is not evaluated unless the following conditions are satisfied: - Sensor Module properties available (r0459.31 = 1). - Monitoring active (p0437.31 = 1). See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- check the speed range; frequency characteristic (amplitude characteristic) of the measuring equipment might not be sufficient for the speed range - check that the encoder cables and shielding are routed in compliance with EMC. - check the plug connections and contacts of the encoder cable. - check whether the zero mark is connected and the signal cables RP and RN have been connected correctly - replace the encoder cable. - if the coding disk is soiled or the lighting aged, replace the encoder.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31110 (N, A)	Encoder 1: Serial communications error
Message value:	Fault cause: %1 bin
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	Serial communication protocol transfer error between the encoder and evaluation module. Fault value (r0949, interpret binary): Bit 0: Alarm bit in the position protocol. Bit 1: Incorrect quiescent level on the data line. Bit 2: Encoder does not respond (does not supply a start bit within 50 ms). Bit 3: CRC error: The checksum in the protocol from the encoder does not match the data. Bit 4: Encoder acknowledgement error: The encoder incorrectly understood the task (request) or cannot execute it. Bit 5: Internal error in the serial driver: An illegal mode command was requested. Bit 6: Timeout when cyclically reading. Bit 8: Protocol is too long (e.g. > 64 bits). Bit 9: Receive buffer overflow. Bit 10: Frame error when reading twice. Bit 11: Parity error. Bit 12: Data line signal level error during the monoflop time. Bit 13: Data line incorrect.

Remedy:	Re fault value, bit 0 = 1: - Enc defect F31111 may provide additional details. Re fault value, bit 1 = 1: - Incorrect encoder type / replace the encoder or encoder cable. Re fault value, bit 2 = 1: - Incorrect encoder type / replace the encoder or encoder cable. Re fault value, bit 3 = 1: - EMC / connect the cable shield, replace the encoder or encoder cable. Re fault value, bit 4 = 1: - EMC / connect the cable shield, replace the encoder or encoder cable, replace the Sensor Module. Re fault value, bit 5 = 1: - EMC / connect the cable shield, replace the encoder or encoder cable, replace the Sensor Module. Re fault value, bit 6 = 1: - Update Sensor Module firmware. Re fault value, bit 8 = 1: - Check parameterization (p0429.2). Re fault value, bit 9 = 1: - EMC / connect the cable shield, replace the encoder or encoder cable, replace the Sensor Module. Re fault value, bit 10 = 1: - Check parameterization (p0429.2, p0449). Re fault value, bit 11 = 1: - Check parameterization (p0436). Re fault value, bit 12 = 1: - Check parameterization (p0429.6). Re fault value, bit 13 = 1: - Check data line.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F31111 (N, A)	Encoder 1: Absolute encoder EnDat, internal fault/error
Message value:	Fault cause: %1 bin
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The EnDat encoder fault word supplies fault bits that have been set. Fault value (r0949, interpret binary): Bit 0: Lighting system failed. Bit 1: Signal amplitude too low. Bit 2: Position value incorrect. Bit 3: Encoder power supply overvoltage condition. Bit 4: Encoder power supply undervoltage condition. Bit 5: Encoder power supply overcurrent condition. Bit 6: The battery must be changed. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	Re fault value, bit 0 = 1: Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor. Re fault value, bit 1 = 1: Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor. Re fault value, bit 2 = 1: Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor.

Re fault value, bit 3 = 1:
 5 V power supply voltage fault.
 When using an SMC: Check the plug-in cable between the encoder and SMC or replace the SMC.
 When a motor encoder with a direct DRIVE-CLiQ connection is used: Replace the motor.
 Re fault value, bit 4 = 1:
 5 V power supply voltage fault.
 When using an SMC: Check the plug-in cable between the encoder and SMC or replace the SMC.
 When using a motor with DRIVE-CLiQ: Replace the motor.
 Re fault value, bit 5 = 1:
 Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor.
 Re fault value, bit 6 = 1:
 The battery must be changed (only for encoders with battery back-up).

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F31112 (N, A) Encoder 1: Error bit set in the serial protocol

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: PULSE INHIBIT
Cause: The encoder sends a set error bit via the serial protocol.
 Fault value (r0949, interpret binary):
 Bit 0: Fault bit in the position protocol.
Remedy: For fault value, bit 0 = 1:
 In the case of an EnDat encoder, F31111 may provide further details.

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F31115 (N, A) Encoder 1: Amplitude error track A or B ($A^2 + B^2$)

Message value: A track: %1, B-track: %2
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: PULSE INHIBIT
Cause: The amplitude (root of $A^2 + B^2$) for encoder 1 exceeds the permissible tolerance.
 Fault value (r0949, interpret hexadecimal):
 yyyxxxx hex:
 yyyy = Signal level, track B (16 bits with sign).
 xxxx = Signal level, track A (16 bits with sign).
 The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %).
 The response thresholds are < 230 mV (observe the frequency response of the encoder) and > 750 mV.
 A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
 Note for sensors modules for resolvers (e. g. SMC10):
 The nominal signal level is at 2900 mV (2.0 Vrms). The response thresholds are < 1070 mV and > 3582 mV.
 A signal level of 2900 mV peak value corresponds to the numerical value 6666 hex = 26214 dec.
 Note:
 The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.
 See also: p0491 (Motor encoder fault response ENCODER)

Remedy:

- check that the encoder cables and shielding are routed in compliance with EMC.
- check the plug connections
- replace the encoder or encoder cable
- check the Sensor Module (e.g. contacts).

The following applies to measuring systems without their own bearing system:

- adjust the scanning head and check the bearing system of the measuring wheel.

The following applies for measuring systems with their own bearing system:

- ensure that the encoder housing is not subject to any axial force.

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F31116 (N, A) Encoder 1: Amplitude error monitoring track A + B

Message value: A track: %1, B-track: %2
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: IMMEDIATELY
Cause: The amplitude of the rectified encoder signals A and B and the amplitude from the roots of $A^2 + B^2$ for encoder 1 are not within the tolerance bandwidth.
 Fault value (r0949, interpret hexadecimal):
 yyyyxxxx hex:
 yyyy = Signal level, track B (16 bits with sign).
 xxxx = Signal level, track A (16 bits with sign).
 The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %).
 The response thresholds are < 176 mV (observe the frequency response of the encoder) and > 955 mV.
 A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
Note:
 The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.
 See also: p0491 (Motor encoder fault response ENCODER)

Remedy:

- check that the encoder cables and shielding are routed in compliance with EMC.
- check the plug connections
- replace the encoder or encoder cable
- check the Sensor Module (e.g. contacts).

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F31117 (N, A) Encoder 1: Inversion error signals A and B and R

Message value: Fault cause: %1 bin
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: IMMEDIATELY
Cause: For a square-wave encoder (bipolar, double ended) the A* and B* and R* signals are not inverted with respect to signals A and B and R.
 Fault value (r0949, interpret binary):
 Bits 0 - 15: Only for internal Siemens troubleshooting.
 Bit 16: Error track A.
 Bit 17: Error track B.
 Bit 18: Error track R.
Note:
 For SMC30 (order no.. 6SL3055-0AA00-5CA0 and 6SL3055-0AA00-5CA1 only), CUA32, and CU310, the following applies:
 A square-wave encoder without track R is used and track monitoring (p0405.2 = 1) is activated.
 See also: p0491 (Motor encoder fault response ENCODER)

Remedy: Check the encoder/cable: Does the encoder provide signals and the associated inverted signals?
Note:
For SMC30 (order no. 6SL3055-0AA00-5CA0 and 6SL3055-0AA00-5CA1 only), the following applies:
- check the setting of p0405 (p0405.2 = 1 is only possible if the encoder is connected at X520).
For a square-wave encoder without track R, the following jumpers must be set for the connection at X520 (SMC30) or X23 (CUA32, CU310):
- pin 10 (reference signal R) <--> pin 7 (encoder power supply, ground)
- pin 11 (reference signal R inverted) <--> pin 4 (encoder power supply)

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F31118 (N, A) Encoder 1: Speed difference outside the tolerance range

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: PULSE INHIBIT
Cause: For an HTL/TTL encoder, the speed difference has exceeded the value in p0492 over several sampling cycles. The change to the averaged speed actual value - if applicable - is monitored in the current controller sampling time. Encoder 1 is used as motor encoder and can be effective has fault response to change over to encoderless operation.
Fault value (r0949, interpret decimal):
Only for internal Siemens troubleshooting.
See also: p0491, p0492

Remedy: - check the tachometer feeder cable for interruptions.
- check the grounding of the tachometer shielding.
- if required, increase the maximum speed difference per sampling cycle (p0492).

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F31120 (N, A) Encoder 1: Power supply voltage fault

Message value: Fault cause: %1 bin
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: PULSE INHIBIT
Cause: Encoder 1 power supply voltage fault.
Note:
If the encoder cables 6FX2002-2EQ00-.... and 6FX2002-2CH00-.... are interchanged, this can result in the encoder being destroyed because the pins of the operating voltage are reversed.
Fault value (r0949, interpret binary):
Bit 0: Undervoltage condition on the sense line.
Bit 1: Overcurrent condition for the encoder power supply.
Bit 2: Overcurrent condition for encoder power supply on cable resolver excitation negative.
Bit 3: Overcurrent condition for encoder power supply on cable resolver excitation positive.
See also: p0491 (Motor encoder fault response ENCODER)

Remedy:

For fault value, bit 0 = 1:

- correct encoder cable connected?
- check the plug connections of the encoder cable.
- SMC30: Check the parameterization (p0404.22).

For fault value, bit 1 = 1:

- correct encoder cable connected?
- replace the encoder or encoder cable

For fault value, bit 2 = 1:

- correct encoder cable connected?
- replace the encoder or encoder cable

For fault value, bit 3 = 1:

- correct encoder cable connected?
- replace the encoder or encoder cable

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F31121 (N, A) Encoder 1: Coarse position error

Message value: -

Drive object: ENCODER, SERVO, VECTOR

Reaction: ENCODER (NONE)

Acknowledge: PULSE INHIBIT

Cause: For the actual value sensing, an error was detected on the module. As a result of this error, it must be assumed that the actual value sensing supplies an incorrect coarse position.
 See also: p0491 (Motor encoder fault response ENCODER)

Remedy: Replace the motor with DRIVE-CLiQ or the appropriate Sensor Module.

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F31122 Encoder 1: Internal power supply voltage faulty

Message value: %1

Drive object: ENCODER, SERVO, VECTOR

Reaction: ENCODER

Acknowledge: IMMEDIATELY

Cause: Fault in internal reference voltage of ASICs for encoder 1.
 Fault value (r0949, interpret decimal):
 1: Reference voltage error.
 2: Internal undervoltage.
 3: Internal overvoltage.

Remedy: Replace the motor with DRIVE-CLiQ or the appropriate Sensor Module.

F31123 (N, A) Encoder 1: Signal level A/B unipolar outside tolerance

Message value: Fault cause: %1 bin
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: PULSE INHIBIT
Cause: The unipolar level (AP/AN or BP/BN) for encoder 1 is outside the permissible tolerance.
Fault value (r0949, interpret binary):
Bit 0 = 1: Either AP or AN outside the tolerance.
Bit 16 = 1: Either BP or BN outside the tolerance.
The unipolar nominal signal level of the encoder must lie in the range 2500 mV +/- 500 mV.
The response thresholds are < 1700 mV and > 3300 mV.
Note:
The signal level is not evaluated unless the following conditions are satisfied:
- Sensor Module properties available (r0459.31 = 1).
- Monitoring active (p0437.31 = 1).
See also: p0491 (Motor encoder fault response ENCODER)
Remedy:
- make sure that the encoder cables and shielding are installed in an EMC-compliant manner.
- check the plug connections and contacts of the encoder cable.
- check the short-circuit of a signal cable with mass or the operating voltage.
- replace the encoder cable.

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F31125 (N, A) Encoder 1: Amplitude error track A or B overcontrolled

Message value: A track: %1, B-track: %2
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: PULSE INHIBIT
Cause: The amplitude of track A or B for encoder 1 exceeds the permissible tolerance band.
Fault value (r0949, interpret hexadecimal):
yyyyxxxx hex:
yyyy = Signal level, track B (16 bits with sign).
xxxx = Signal level, track A (16 bits with sign).
The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %).
The response threshold is > 750 mV. This fault also occurs if the A/D converter is overcontrolled.
A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
Note for sensors modules for resolvers (e. g. SMC10):
The nominal signal level is at 2900 mV (2.0 Vrms). The response threshold is > 3582 mV.
A signal level of 2900 mV peak value corresponds to the numerical value 6666 hex = 26214 dec.
Note:
The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.
See also: p0491 (Motor encoder fault response ENCODER)
Remedy:
- check that the encoder cables and shielding are routed in compliance with EMC.
- replace the encoder or encoder cable

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F31126 (N, A)	Encoder 1: Amplitude AB too high
Message value:	Amplitude: %1, Angle: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The amplitude (root of $A^2 + B^2$ or $A + B$) for encoder 1 exceeds the permissible tolerance. Fault value (r0949, interpret hexadecimal): yyyyxxxx hex: yyyy = Angle xxxx = Amplitude, i.e. root from $A^2 + B^2$ (16 bits without sign) The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %). The response threshold for $(A + B)$ is > 1120 mV or the root of $(A^2 + B^2) > 955$ mV. A signal level of 500 mV peak value corresponds to the numerical value of 299A hex = 10650 dec. The angle 0 ... FFFF hex corresponds to 0 ... 360 degrees of the fine position. Zero degrees is at the negative zero crossover of track B. Note: The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- check that the encoder cables and shielding are routed in compliance with EMC. - replace the encoder or encoder cable
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31129 (N, A)	Encoder 1: Position difference, hall sensor/track C/D and A/B too large
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The error for track C/D is greater than $\pm 15^\circ$ mechanical or $\pm 60^\circ$ electrical or the error for the Hall signals is greater than $\pm 60^\circ$ electrical. One period of track C/D corresponds to 360° mechanical. One period of the Hall signal corresponds to 360° electrical. The monitoring responds if, for example, Hall sensors are connected as equivalent for the C/D tracks with the incorrect rotational sense or supply values that are not accurate enough. After the fine synchronization using one reference mark or 2 reference marks for distance-coded encoders, this fault is no longer initiated, but instead, Alarm A31429. Fault value (r0949, interpret decimal): For track C/D, the following applies: Measured deviation as mechanical angle (16 bits with sign, 182 dec corresponds to 1°). For Hall signals, the following applies: Measured deviation as electrical angle (16 bits with sign, 182 dec corresponds to 1°). See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- track C or D not connected. - correct the direction of rotation of the Hall sensor possibly connected as equivalent for track C/D. - check that the encoder cables are routed in compliance with EMC. - check the adjustment of the Hall sensor.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F31130 (N, A)	Encoder 1: Zero mark and position error from the coarse synchronization
Message value:	Angular deviation, electrical: %1, angle, mechanical: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	After initializing the pole position using track C/D, Hall signals or pole position identification routine, the zero mark was detected outside the permissible range. For distance-coded encoders, the test is carried out after passing 2 zero marks. Fine synchronization was not carried out. When initializing via track C/D (p0404) then it is checked whether the zero mark occurs in an angular range of +/-18 ° mechanical. When initializing via Hall sensors (p0404) or pole position identification (p1982) it is checked whether the zero mark occurs in an angular range of +/-60 ° electrical. Fault value (r0949, interpret hexadecimal): yyyyxxxx hex yyyy: Determined mechanical zero mark position (can only be used for track C/D). xxxx: Deviation of the zero mark from the expected position as electrical angle. Scaling: 32768 dec = 180 ° See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- Check p0431 and, if necessary, correct (trigger via p1990 = 1 if necessary). - check that the encoder cables are routed in compliance with EMC. - check the plug connections - if the Hall sensor is used as an equivalent for track C/D, check the connection. - Check the connection of track C or D. - replace the encoder or encoder cable
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F31131 (N, A)	Encoder 1: Deviation, position incremental/absolute too large
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	Absolute encoder: When cyclically reading the absolute position, an excessively high difference to the incremental position was detected. The absolute position that was read is rejected. Limit value for the deviation: - EnDat encoder: Is supplied from the encoder and is a minimum of 2 quadrants (e.g. EQI 1325 > 2 quadrants, EQN 1325 > 50 quadrants). - other encoders: 15 pulses = 60 quadrants. Incremental encoder: When the zero pulse is passed, a deviation in the incremental position was detected. For equidistant zero marks, the following applies: - The first zero mark passed supplies the reference point for all subsequent checks. The other zero marks must have n times the distance referred to the first zero mark. For distance-coded zero marks, the following applies: - the first zero mark pair supplies the reference point for all subsequent checks. The other zero mark pairs must have the expected distance to the first zero mark pair. Fault value (r0949, interpret decimal): Deviation in quadrants (1 pulse = 4 quadrants). See also: p0491 (Motor encoder fault response ENCODER)

- Remedy:**
- check that the encoder cables are routed in compliance with EMC.
 - check the plug connections
 - replace the encoder or encoder cable
 - check whether the coding disk is dirty or there are strong ambient magnetic fields.
 - adapt the parameter for the clearance between zero marks (p0425).
 - if message output above speed threshold, reduce filter time if necessary (p0438).

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F31135 Encoder 1: Fault when determining the position

Message value: Fault cause: %1 bin

Drive object: ENCODER, SERVO, VECTOR

Reaction: ENCODER (IASC/DCBRAKE, NONE)

Acknowledge: PULSE INHIBIT

Cause: The DRIVE-CLiQ encoder supplies status information via bits in an internal status/fault word. Some of these bits cause this fault to be triggered. Other bits are status displays. The status/fault word is displayed in the fault value.

Fault value (r0949, interpret binary):

Bit 0: F1 (safety status display)

Bit 1: F2 (safety status display)

Bit 2: Lighting (reserved)

Bit 3: Signal amplitude (reserved)

Bit 4: Position value (reserved)

Bit 5: Overvoltage (reserved)

Bit 6: Undervoltage (reserved)

Bit 7: Overcurrent (reserved)

Bit 8: Battery (reserved)

Bit 16: Lighting (--> F3x135, x = 1, 2, 3)

Bit 17: Signal amplitude (--> F3x135, x = 1, 2, 3)

Bit 18: Singleturn position 1 (--> F3x135, x = 1, 2, 3)

Bit 19: Overvoltage (--> F3x135, x = 1, 2, 3)

Bit 20: Undervoltage (--> F3x135, x = 1, 2, 3)

Bit 21: Overcurrent (--> F3x135, x = 1, 2, 3)

Bit 22: Temperature exceeded (--> F3x405, x = 1, 2, 3)

Bit 23: Singleturn position 2 (safety status display)

Bit 24: Singleturn system (--> F3x135, x = 1, 2, 3)

Bit 25: Singleturn power down (--> F3x135, x = 1, 2, 3)

Bit 26: Multiturn position 1 (--> F3x136, x = 1, 2, 3)

Bit 27: Multiturn position 2 (--> F3x136, x = 1, 2, 3)

Bit 28: Multiturn system (--> F3x136, x = 1, 2, 3)

Bit 29: Multiturn power down (--> F3x136, x = 1, 2, 3)

Bit 30: Multiturn overflow/underflow (--> F3x136, x = 1, 2, 3)

Bit 31: Multiturn battery (reserved)

Remedy: Replace DRIVE-CLiQ encoder.

F31136	Encoder 1: Error when determining multiturn information
Message value:	Fault cause: %1 bin
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The DRIVE-CLiQ encoder supplies status information via bits in an internal status/fault word. Some of these bits cause this fault to be triggered. Other bits are status displays. The status/fault word is displayed in the fault value. Fault value (r0949, interpret binary): Bit 0: F1 (safety status display) Bit 1: F2 (safety status display) Bit 2: Lighting (reserved) Bit 3: Signal amplitude (reserved) Bit 4: Position value (reserved) Bit 5: Overvoltage (reserved) Bit 6: Undervoltage (reserved) Bit 7: Overcurrent (reserved) Bit 8: Battery (reserved) Bit 16: Lighting (--> F3x135, x = 1, 2, 3) Bit 17: Signal amplitude (--> F3x135, x = 1, 2, 3) Bit 18: Singleturn position 1 (--> F3x135, x = 1, 2, 3) Bit 19: Overvoltage (--> F3x135, x = 1, 2, 3) Bit 20: Undervoltage (--> F3x135, x = 1, 2, 3) Bit 21: Overcurrent (--> F3x135, x = 1, 2, 3) Bit 22: Temperature exceeded (--> F3x405, x = 1, 2, 3) Bit 23: Singleturn position 2 (safety status display) Bit 24: Singleturn system (--> F3x135, x = 1, 2, 3) Bit 25: Singleturn power down (--> F3x135, x = 1, 2, 3) Bit 26: Multiturn position 1 (--> F3x136, x = 1, 2, 3) Bit 27: Multiturn position 2 (--> F3x136, x = 1, 2, 3) Bit 28: Multiturn system (--> F3x136, x = 1, 2, 3) Bit 29: Multiturn power down (--> F3x136, x = 1, 2, 3) Bit 30: Multiturn overflow/underflow (--> F3x136, x = 1, 2, 3) Bit 31: Multiturn battery (reserved)
Remedy:	Replace DRIVE-CLiQ encoder.

F31137	Encoder 1: Internal error when determining the position
Message value:	Fault cause: %1 bin
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The DRIVE-CLiQ encoder fault word supplies fault bits that have been set. Fault value (r0949, interpret binary): Only for internal SIEMENS use.
Remedy:	Replace encoder

F31138	Encoder 1: Internal error when determining multiturn information
Message value:	Fault cause: %1 bin
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The DRIVE-CLiQ encoder fault word supplies fault bits that have been set. Fault value (r0949, interpret binary): Only for internal SIEMENS use.
Remedy:	Replace encoder

F31150 (N, A)	Encoder 1: Initialization error
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	Encoder functionality selected in p0404 is not operating correctly. Fault value (r0949, interpret hexadecimal): The fault value is a bit field. Every set bit indicates functionality that is faulted. The bit assignment corresponds to that of p0404 (e.g. bit 5 set: Error track C/D). See also: p0404 (Encoder configuration effective), p0491 (Motor encoder fault response ENCODER)
Remedy:	- Check that p0404 is correctly set. - check the encoder type used (incremental/absolute value) and for SMCxx, the encoder cable. - if relevant, note additional fault messages that describe the fault in detail.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31151 (N, A)	Encoder 1: Encoder speed for initialization AB too high
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	The encoder speed is too high during while initializing the sensor.
Remedy:	Reduce the speed of the encoder accordingly during initialization. If necessary, deactivate monitoring (p0437.29). See also: p0437 (Sensor Module configuration extended)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31160 (N, A)	Encoder 1: Analog sensor channel A failed
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The input voltage of the analog sensor is outside the permissible limits. Fault value (r0949, interpret decimal): 1: Input voltage outside detectable measuring range. 2: Input voltage outside measuring range set in p4673. 3: The absolute value of the input voltage has exceeded the range limit (p4676).
Remedy:	Re fault value = 1: - check the output voltage of the analog sensor. Re fault value = 2: - check the voltage setting for each encoder period (p4673). Re fault value = 3: - check the range limit setting and increase it if necessary (p4676).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F31161 (N, A) Encoder 1: Analog sensor channel B failed

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: PULSE INHIBIT
Cause: The input voltage of the analog sensor is outside the permissible limits.
Fault value (r0949, interpret decimal):
1: Input voltage outside detectable measuring range.
2: Input voltage outside measuring range set in p4675.
3: The absolute value of the input voltage has exceeded the range limit (p4676).
Remedy: Re fault value = 1:
- check the output voltage of the analog sensor.
Re fault value = 2:
- check the voltage setting for each encoder period (p4675).
Re fault value = 3:
- check the range limit setting and increase it if necessary (p4676).
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

A31400 (F, N) Encoder 1: Alarm threshold zero mark distance error

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The measured zero mark distance does not correspond to the parameterized zero mark distance.
For distance-coded encoders, the zero mark distance is determined from zero marks detected pairs. This means that if a zero mark is missing, depending on the pair generation, this cannot result in a fault and also has no effect in the system.
The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder).
Alarm value (r2124, interpret decimal):
Last measured zero mark distance in increments (4 increments = 1 encoder pulse).
The sign designates the direction of motion when detecting the zero mark distance.
Remedy: - check that the encoder cables are routed in compliance with EMC.
- check the plug connections
- check the encoder type (encoder with equidistant zero marks).
- adapt the parameter for the distance between zero marks (p0424, p0425).
- replace the encoder or encoder cable
Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A31401 (F, N) Encoder 1: Alarm threshold zero marked failed

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The 1.5 x parameterized zero mark distance was exceeded.
The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder).
Alarm value (r2124, interpret decimal):
Number of increments after POWER ON or since the last zero mark that was detected (4 increments = 1 encoder pulse).

Remedy:

- check that the encoder cables are routed in compliance with EMC.
- check the plug connections
- check the encoder type (encoder with equidistant zero marks).
- adapt the parameter for the clearance between zero marks (p0425).
- replace the encoder or encoder cable

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

F31405 (N, A) Encoder 1: Temperature in the encoder evaluation inadmissible

Message value: %1

Drive object: ENCODER, SERVO, VECTOR

Reaction: ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowledge: IMMEDIATELY (POWER ON)

Cause: The encoder evaluation for a motor with DRIVE-CLiQ has detected an inadmissible temperature.
The fault threshold is 125 ° C.
Alarm value (r2124, interpret decimal):
Measured board/module temperature in 0.1 °C.

Remedy: Reduce the ambient temperature for the DRIVE-CLiQ connection of the motor.

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

A31407 (F, N) Encoder 1: Function limit reached

Message value: %1

Drive object: ENCODER, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The encoder has reached one of its function limits. A service is recommended.
Alarm value (r2124, interpret decimal):
1 : Incremental signals
3 : Absolute track
4 : Code connection

Remedy: Perform service. Replace the encoder if necessary.
Note:
The current functional reserve of an encoder can be displayed via r4651.
See also: p4650 (Encoder functional reserve component number), r4651 (Encoder functional reserve)

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

A31410 (F, N)	Encoder 1: Serial communications
Message value:	Fault cause: %1 bin
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Serial communication protocol transfer error between the encoder and evaluation module. Alarm value (r2124, interpret binary): Bit 0: Alarm bit in the position protocol. Bit 1: Incorrect quiescent level on the data line. Bit 2: Encoder does not respond (does not supply a start bit within 50 ms). Bit 3: CRC error: The checksum in the protocol from the encoder does not match the data. Bit 4: Encoder acknowledgement error: The encoder incorrectly understood the task (request) or cannot execute it. Bit 5: Internal error in the serial driver: An illegal mode command was requested. Bit 6: Timeout when cyclically reading. Bit 8: Protocol is too long (e.g. > 64 bits). Bit 9: Receive buffer overflow. Bit 10: Frame error when reading twice. Bit 11: Parity error. Bit 12: Data line signal level error during the monoflop time.
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A31411 (F, N)	Encoder 1: EnDat encoder signals alarms
Message value:	Fault cause: %1 bin
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The error word of the EnDat encoder has alarm bits that have been set. Alarm value (r2124, interpret binary): Bit 0: Frequency exceeded (speed too high). Bit 1: Temperature exceeded. Bit 2: Control reserve, lighting system exceeded. Bit 3: Battery discharged. Bit 4: Reference point passed. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	Replace encoder.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A31412 (F, N)	Encoder 1: Error bit set in the serial protocol
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The encoder sends a set error bit via the serial protocol. Alarm value (r2124, interpret binary): Bit 0: Fault bit in the position protocol. Bit 1: Alarm bit in the position protocol.

Remedy:	- carry out a POWER ON (power off/on) for all components. - check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A31414 (F, N) Encoder 1: Amplitude error track C or D ($C^2 + D^2$)

Message value:	C track: %1, D track: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The amplitude ($C^2 + D^2$) of track C or D of the encoder or from the Hall signals, is not within the tolerance bandwidth. Alarm value (r2124, interpret hexadecimal): yyyyyxxx hex: yyyy = Signal level, track D (16 bits with sign). xxxx = Signal level, track C (16 bits with sign). The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %). The response thresholds are < 230 mV (observe the frequency response of the encoder) and > 750 mV. A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec. Note: If the amplitude is not within the tolerance bandwidth, then it cannot be used to initialize the start position.
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - check the Sensor Module (e.g. contacts). - check the Hall sensor box
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

N31415 (F, A) Encoder 1: Amplitude alarm track A or B ($A^2 + B^2$)

Message value:	Amplitude: %1, Angle: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The amplitude (root of $A^2 + B^2$) for encoder 1 exceeds the permissible tolerance. Alarm value (r2124, interpret hexadecimal): yyyyyxxx hex: yyyy = Angle xxxx = Amplitude, i.e. root from $A^2 + B^2$ (16 bits without sign) The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %). The response threshold is < 300 mV (observe the frequency response of the encoder). A signal level of 500 mV peak value corresponds to the numerical value 299A hex = 10650 dec. The angle 0 ... FFFF hex corresponds to 0 ... 360 degrees of the fine position. Zero degrees is at the negative zero crossover of track B. Note for sensors modules for resolvers (e. g. SMC10): The nominal signal level is at 2900 mV (2.0 Vrms). The response threshold is < 1414 mV (1.0 Vrms). A signal level of 2900 mV peak value corresponds to the numerical value 3333 hex = 13107 dec. Note: The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module. See also: p0491 (Motor encoder fault response ENCODER)

Remedy:

- check the speed range, frequency characteristic (amplitude characteristic) of the measuring equipment is not sufficient for the speed range.
- check that the encoder cables and shielding are routed in compliance with EMC.
- check the plug connections
- replace the encoder or encoder cable
- check the Sensor Module (e.g. contacts).
- if the coding disk is soiled or the lighting aged, replace the encoder.

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon A: NONE

Acknowl. upon A: NONE

A31418 (F, N) Encoder 1: Speed difference per sampling rate exceeded

Message value: %1

Drive object: ENCODER, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: For an HTL/TTL encoder, the speed difference between two sampling cycles has exceeded the value in p0492. The change to the averaged speed actual value - if applicable - is monitored in the current controller sampling time. Alarm value (r2124, interpret decimal): Only for internal Siemens troubleshooting. See also: p0492

Remedy:

- check the tachometer feeder cable for interruptions.
- check the grounding of the tachometer shielding.
- if required, increase the setting of p0492.

Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

A31419 (F, N) Encoder 1: Track A or B outside the tolerance range

Message value: %1

Drive object: ENCODER, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The amplitude, phase or offset correction for track A or B is at the limit. Amplitude error correction: Amplitude B / Amplitude A = 0.78 ... 1.27
Phase: <84 degrees or >96 degrees
SMC20: Offset correction: +/-140 mV
SMC10: Offset correction: +/-650 mV
Alarm value (r2124, interpret hexadecimal):
xxxx1: Minimum of the offset correction, track B
xxxx2: Maximum of the offset correction, track B
xxx1x: Minimum of the offset correction, track A
xxx2x: Maximum of the offset correction, track A
xx1xx: Minimum of the amplitude correction, track B/A
xx2xx: Maximum of the amplitude correction, track B/A
x1xxx: Minimum of the phase error correction
x2xxx: Maximum of the phase error correction
1xxxx: Minimum of the cubic correction
2xxxx: Maximum of the cubic correction
See also: p0491 (Motor encoder fault response ENCODER)

Remedy:

- check mechanical mounting tolerances for encoders without their own bearings (e.g. toothed-wheel encoders).
- check the plug connections (also the transition resistance).
- check the encoder signals.
- replace the encoder or encoder cable

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

A31421 (F, N) Encoder 1: Coarse position error

Message value: %1

Drive object: ENCODER, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: For the actual value sensing, an error was detected. As a result of this error, it must be assumed that the actual value sensing supplies an incorrect coarse position.

Alarm value (r2124, interpret decimal):

3: The absolute position of the serial protocol and track A/B differ by half an encoder pulse. The absolute position must have its zero position in the quadrants in which both tracks are negative. In the case of a fault, the position can be incorrect by one encoder pulse.

Remedy: Re alarm value = 3:

- for a standard encoder with cable, if required, contact the manufacturer.
- correct the assignment of the tracks to the position value that is serially transferred. To do this, the two tracks must be connected, inverted, at the Sensor Module (interchange A with A* and B with B*) or, for a programmable encoder, check the zero offset of the position.

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

A31422 (F, N) Encoder 1: Pulses per revolution square-wave encoder outside tolerance bandwidth

Message value: %1

Drive object: ENCODER, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The measured zero mark distance does not correspond to the parameterized zero mark distance. This alarm is triggered with active square-wave encoder PPR correction and re-parameterized fault 31131 if the accumulator contains larger values than p4683 or p4684.

The zero mark distance for zero mark monitoring is set in p0425 (rotary encoder).

Alarm value (r2124, interpret decimal):

accumulated differential pulses in encoder pulses.

See also: p0491 (Motor encoder fault response ENCODER)

Remedy:

- check that the encoder cables are routed in compliance with EMC.
- check the plug connections
- check the encoder type (encoder with equidistant zero marks).
- adapt the parameter for the distance between zero marks (p0424, p0425).
- replace the encoder or encoder cable

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

A31429 (F, N)	Encoder 1: Position difference, hall sensor/track C/D and A/B too large
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The error for track C/D is greater than +/-15 ° mechanical or +/-60 ° electrical or the error for the Hall signals is greater than +/-60 ° electrical. One period of track C/D corresponds to 360 ° mechanical. One period of the Hall signal corresponds to 360 ° electrical. The monitoring responds if, for example, Hall sensors are connected as equivalent for the C/D tracks with the incorrect rotational sense or supply values that are not accurate enough. Alarm value (r2124, interpret decimal): For track C/D, the following applies: Measured deviation as mechanical angle (16 bits with sign, 182 dec corresponds to 1 °). For Hall signals, the following applies: Measured deviation as electrical angle (16 bits with sign, 182 dec corresponds to 1 °). See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- track C or D not connected. - correct the direction of rotation of the Hall sensor possibly connected as equivalent for track C/D. - check that the encoder cables are routed in compliance with EMC. - check the adjustment of the Hall sensor.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A31431 (F, N)	Encoder 1: Deviation, position incremental/absolute too large
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When the zero pulse is passed, a deviation in the incremental position was detected. For equidistant zero marks, the following applies: - The first zero mark passed supplies the reference point for all subsequent checks. The other zero marks must have n times the distance referred to the first zero mark. For distance-coded zero marks, the following applies: - the first zero mark pair supplies the reference point for all subsequent checks. The other zero mark pairs must have the expected distance to the first zero mark pair. Alarm value (r2124, interpret decimal): Deviation in quadrants (1 pulse = 4 quadrants). See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - Clean coding disk or remove strong magnetic fields.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A31432 (F, N)	Encoder 1: Rotor position adaptation corrects deviation
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	For track A/B, pulses have been lost or too many have been counted. These pulses are presently being corrected. Alarm value (r2124, interpret decimal): Last measured deviation of the zero mark distance in increments (4 increments = 1 encoder pulse). The sign designates the direction of motion when detecting the zero mark distance.

Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - check encoder limit frequency. - adapt the parameter for the distance between zero marks (p0424, p0425).
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A31442 (F, N) Encoder 1: Battery voltage pre-alarm

Message value:	-
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- replace battery.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A31443 (F, N) Encoder 1: Unipolar CD signal level outside specification

Message value:	Fault cause: %1 bin
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The unipolar level (CP/CN or DP/DN) for encoder 1 is outside the permissible tolerance. Alarm value (r2124, interpret binary): Bit 0 = 1: Either CP or CN outside the tolerance. Bit 16 = 1: Either DP or DN outside the tolerance. The unipolar nominal signal level of the encoder must lie in the range 2500 mV +/- 500 mV. The response thresholds are < 1700 mV and > 3300 mV. Note: The signal level is not evaluated unless the following conditions are satisfied: - Sensor Module properties available (r0459.31 = 1). - Monitoring active (p0437.31 = 1). See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- check that the encoder cables and shielding are routed in compliance with EMC. - check the plug connections and contacts of the encoder cable. - are the C/D tracks connected correctly (have the signal lines CP and CN or DP and DN been interchanged)? - replace the encoder cable.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A31460 (N) Encoder 1: Analog sensor channel A failed

Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The input voltage of the analog sensor is outside the permissible limits. Alarm value (r2124, interpret decimal): 1: Input voltage outside detectable measuring range. 2: Input voltage outside measuring range set in p4673. 3: The absolute value of the input voltage has exceeded the range limit (p4676).

Remedy: Re alarm value = 1:
- check the output voltage of the analog sensor.
Re alarm value = 2:
- check the voltage setting for each encoder period (p4673).
Re alarm value = 3:
- check the range limit setting and increase it if necessary (p4676).

Reaction upon N: NONE

Acknowled. upon N: NONE

A31461 (N) Encoder 1: Analog sensor channel B failed

Message value: %1

Drive object: ENCODER, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The input voltage of the analog sensor is outside the permissible limits.
Alarm value (r2124, interpret decimal):
1: Input voltage outside detectable measuring range.
2: Input voltage outside measuring range set in p4675.
3: The absolute value of the input voltage has exceeded the range limit (p4676).

Remedy: Re alarm value = 1:
- check the output voltage of the analog sensor.
Re alarm value = 2:
- check the voltage setting for each encoder period (p4675).
Re alarm value = 3:
- check the range limit setting and increase it if necessary (p4676).

Reaction upon N: NONE

Acknowled. upon N: NONE

A31462 (N) Encoder 1: Analog sensor, no channel active

Message value: %1

Drive object: ENCODER, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: Channel A and B are not activated for the analog sensor.

Remedy: - activate channel A and/or channel B (p4670).
- check the encoder configuration (p0404.17).
See also: p4670 (Analog sensor configuration)

Reaction upon N: NONE

Acknowled. upon N: NONE

A31470 (F, N) Encoder 1: Soiling detected

Message value: -

Drive object: ENCODER, SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: In the case of the alternative encoder system interface on the Sensor Module Cabinet 30 (SMC30), encoder soiling is signaled via a 0 signal at terminal X521.7.

Remedy: - check the plug connections
- replace the encoder or encoder cable

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowled. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowled. upon N: NONE

F31500 (N, A)	Encoder 1: Position tracking traversing range exceeded
Message value:	-
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	For a configured linear axis without modulo correction, the drive/encoder has exceeded the maximum possible traversing range. The value should be read in p0412 and interpreted as the number of motor revolutions. For p0411.0 = 1, the maximum traversing range for the configured linear axis is defined to be 64x (+/- 32x) of p0421. For p0411.3 = 1, the maximum traversing range for the configured linear axis is preset (default value) to the highest possible value and is +/-p0412/2 (rounded off to complete revolutions). The highest possible value depends on the pulse number (p0408) and the fine resolution (p0419).
Remedy:	The fault should be resolved as follows: - select encoder commissioning (p0010 = 4). - reset the position tracking as follows (p0411.2 = 1). - deselect encoder commissioning (p0010 = 0). The fault should then be acknowledged and the absolute encoder adjusted.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31501 (N, A)	Encoder 1: Position tracking encoder position outside tolerance window
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	When powered down, the drive/encoder was moved through a distance greater than was parameterized in the tolerance window. It is possible that there is no longer any reference between the mechanical system and encoder. Fault value (r0949, decimal): Deviation (difference) to the last encoder position in increments of the absolute value. The sign designates the traversing direction. Note: The deviation (difference) found is also displayed in r0477. See also: p0413 (Measuring gear, position tracking tolerance window), r0477 (Measuring gear, position difference)
Remedy:	Reset the position tracking as follows: - select encoder commissioning (p0010 = 4). - reset the position tracking as follows (p0411.2 = 1). - deselect encoder commissioning (p0010 = 0). The fault should then be acknowledged and, if necessary, the absolute encoder adjusted (p2507). See also: p0010, p2507
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31502 (N, A)	Encoder 1: Encoder with measuring gear, without valid signals
Message value:	-
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The encoder with measuring gear no longer provides any valid signals.
Remedy:	It must be ensured that all of the encoders, with mounted measuring gear, provide valid actual values in operation.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F31503 (N, A) Encoder 1: Position tracking cannot be reset

Message value: -
Drive object: ENCODER, SERVO, VECTOR
Reaction: OFF1 (NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The position tracking for the measuring gear cannot be reset.
Remedy: The fault should be resolved as follows:
- select encoder commissioning (p0010 = 4).
- reset the position tracking as follows (p0411.2 = 1).
- deselect encoder commissioning (p0010 = 0).
The fault should then be acknowledged and the absolute encoder adjusted.

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

A31700 Encoder 1: Effectivity test does not supply the expected value

Message value: Fault cause: %1 bin
Drive object: ENCODER, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The DRIVE-CLiQ encoder fault word supplies fault bits that have been set.
Fault value (r0949, interpret binary):
Bit x = 1: Effectivity test x unsuccessful.

Remedy:

N31800 (F) Encoder 1: Group signal

Message value: -
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: NONE
Cause: The motor encoder has detected at least one fault.
See also: p0491 (Motor encoder fault response ENCODER)
Remedy: Evaluates other actual messages.
Reaction upon F: ENCODER (IASC/DCBRAKE, NONE)
Acknowl. upon F: IMMEDIATELY

F31801 (N, A) Encoder 1 DRIVE-CLiQ: Sign-of-life missing

Message value: Component number: %1, fault cause: %2
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 0A hex:
The sign-of-life bit in the receive telegram is not set.
See also: p0491 (Motor encoder fault response ENCODER)
Remedy: - check the electrical cabinet design and cable routing for EMC compliance
- replace the component involved.
See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F31802 (N, A) Encoder 1: Time slice overflow

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: IMMEDIATELY
Cause: Time slice overflow, encoder 1.
 Fault value (r0949, interpret decimal):
 9: Time slice overflow of the fast (current controller clock cycle) time slice.
 10: Time slice overflow of the average time slice.
 12: Time slice overflow of the slow time slice.
 999: Timeout when waiting for SYNO, e.g. unexpected return to non-cyclic operation.
 See also: p0491 (Motor encoder fault response ENCODER)
Remedy: Reduce the current controller frequency.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F31804 (N, A) Encoder 1: Checksum error

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: IMMEDIATELY
Cause: A checksum error has occurred when reading-out the program memory on the Sensor Module.
 Fault value (r0949, interpret hexadecimal):
 yyyxxxx hex
 yyyy: Memory area involved.
 xxxx: Difference between the checksum at POWER ON and the actual checksum.
 See also: p0491 (Motor encoder fault response ENCODER)
Remedy: - check whether the permissible ambient temperature for the component is maintained.
 - replace the Sensor Module.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F31805 (N, A) Encoder 1: EPROM checksum error

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: IMMEDIATELY
Cause: Internal parameter data is corrupted.
 Fault value (r0949, interpret hexadecimal):
 01: EEPROM access error.
 02: Too many blocks in the EEPROM.
 See also: p0491 (Motor encoder fault response ENCODER)
Remedy: Replace the module.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F31806 (N, A)	Encoder 1: Initialization error
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The encoder was not successfully initialized. Fault value (r0949, interpret hexadecimal): Bit 0,1: Encoder initialization with the motor rotating has failed (deviation involving coarse and fine position in encoder pulse/4) Bit 2: Mid-voltage matching for track A unsuccessful. Bit 3: Mid-voltage matching for track B unsuccessful. Bit 4: Mid-voltage matching for acceleration input (under development) unsuccessful. Bit 5: Mid-voltage matching for track safety A unsuccessful. Bit 6: Mid-voltage matching for track safety B unsuccessful. Bit 7: Mid-voltage matching for track C unsuccessful. Bit 8: Mid-voltage matching for track D unsuccessful. Bit 9: Mid-voltage matching for track R unsuccessful. Bit 10: The difference in mid-voltages between A and B is too great (> 0.5 V) Bit 11: The difference in mid-voltages between C and D is too great (> 0.5 V) Bit 12: The difference in mid-voltages between safety A and safety B is too great (> 0.5 V) Bit 13: The difference in mid-voltages between A and safety B is too great (> 0.5 V) Bit 14: The difference in mid-voltages between B and safety A is too great (> 0.5 V) Bit 15: The standard deviation of the calculated mid-voltages is too great (> 0.3 V) Bit 16: Internal fault - fault reading a register (CAFE) Bit 17: Internal fault - fault writing a register (CAFE) Bit 18: Internal fault: No mid-voltage matching available Bit 19: Internal error - ADC access error. Bit 20: Internal error - no zero crossover found. Bit 0,1 : Up to 6SL3055-0AA00-5*A0 Bits 2..20 : 6SL3055-0AA00-5*A1 and higher See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	Acknowledge the fault. If the fault cannot be acknowledged: Bits 2 - 9: Check encoder power supply, Bits 2 -14: Check the corresponding cable, Bit 15 with no other bits: Check track R, check settings in p404.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A31811 (F, N)	Encoder 1: Encoder serial number changed
Message value:	-
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The serial number of the motor encoder of a synchronous motor has changed. The change was only checked for encoders with serial number (e.g. EnDat encoders) and build-in motors (e.g. p0300 = 401) or third-party motors (p0300 = 2). Cause 1: The encoder was replaced. Cause 2: A third-party, build-in or linear motor was re-commissioned. Cause 3: The motor with integrated and adjusted encoder was replaced. Cause 4: The firmware was updated to a version that checks the encoder serial number.

Note:

With closed-loop position control, the serial number is accepted when starting the adjustment (p2507 = 2).

When the encoder is adjusted (p2507 = 3), the serial number is checked for changes and if required, the adjustment is reset (p2507 = 1).

Proceed as follows to hide serial number monitoring:

- set the following serial numbers for the corresponding Encoder Data Set: p0441= FF, p0442 = 0, p0443 = 0, p0444 = 0, p0445 = 0.

- parameterize F07414 as message type N (p2100, p2101).

See also: p0491 (Motor encoder fault response ENCODER)

Remedy:

Re causes 1, 2:

Carry out an automatic adjustment using the pole position identification routine. Acknowledge the fault. Initiate the pole position identification routine with p1990 = 1. Then check that the pole position identification routine is correctly executed.

SERVO:

If a pole position identification technique is selected in p1980, and if p0301 does not contain a motor type with an encoder adjusted in the factory, then p1990 is automatically activated.

or

Set the adjustment via p0431. In this case, the new serial number is automatically accepted.

or

Mechanically adjust the encoder. Accept the new serial number with p0440 = 1.

Re causes 3, 4:

Accept the new serial number with p0440 = 1.

Reaction upon F: NONE (ENCODER, OFF2)

Acknowled. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowled. upon N: NONE

F31812 (N, A) Encoder 1: Requested cycle or RX-/TX timing not supported

Message value: %1

Drive object: ENCODER, SERVO, VECTOR

Reaction: OFF2

Acknowledge: IMMEDIATELY

Cause: A cycle requested from the Control Unit or RX/TX timing is not supported.

Alarm value (r2124, interpret decimal):

0: Application cycle is not supported.

1: DQ cycle is not supported.

2: Distance between RX and TX instants in time too low.

3: TX instant in time too early.

Remedy:

Reaction upon N: NONE

Acknowled. upon N: NONE

Reaction upon A: NONE

Acknowled. upon A: NONE

F31813 Encoder 1: Hardware logic unit failed

Message value: Fault cause: %1 bin

Drive object: ENCODER, SERVO, VECTOR

Reaction: ENCODER (IASC/DCBRAKE, NONE)

Acknowledge: PULSE INHIBIT

Cause: The DRIVE-CLiQ encoder fault word supplies fault bits that have been set.

Fault value (r0949, interpret binary):

Bit 0: ALU watchdog has responded.

Bit 1: ALU has detected a sign-of-life error.

Remedy: Replace encoder

F31820 (N, A) Encoder 1 DRIVE-CLiQ: Telegram error

Message value:	Component number: %1, fault cause: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 01 hex: CRC error. xx = 02 hex: Telegram is shorter than specified in the length byte or in the receive list. xx = 03 hex: Telegram is longer than specified in the length byte or in the receive list. xx = 04 hex: The length of the receive telegram does not match the receive list. xx = 05 hex: The type of the receive telegram does not match the receive list. xx = 06 hex: The address of the component in the telegram and in the receive list do not match. xx = 07 hex: A SYNC telegram is expected - but the received telegram is not a SYNC telegram. xx = 08 hex: No SYNC telegram is expected - but the received telegram is one. xx = 09 hex: The error bit in the receive telegram is set. xx = 10 hex: The receive telegram is too early. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- carry out a POWER ON. - check the electrical cabinet design and cable routing for EMC compliance - check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F31835 (N, A) Encoder 1 DRIVE-CLiQ: Cyclic data transfer error

Message value:	Component number: %1, fault cause: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved. The nodes do not send and receive in synchronism. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 21 hex: The cyclic telegram has not been received. xx = 22 hex: Timeout in the telegram receive list. xx = 40 hex: Timeout in the telegram send list. See also: p0491 (Motor encoder fault response ENCODER)

Remedy:

- carry out a POWER ON.
- replace the component involved.

See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F31836 (N, A) Encoder 1 DRIVE-CLiQ: Send error for DRIVE-CLiQ data

Message value: Component number: %1, fault cause: %2

Drive object: ENCODER, SERVO, VECTOR

Reaction: ENCODER (IASC/DCBRAKE, NONE)

Acknowledge: IMMEDIATELY

Cause: A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved. Data were not able to be sent.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 41 hex:
 Telegram type does not match send list.
 See also: p0491 (Motor encoder fault response ENCODER)

Remedy: Carry out a POWER ON.

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F31837 (N, A) Encoder 1 DRIVE-CLiQ: Component fault

Message value: Component number: %1, fault cause: %2

Drive object: ENCODER, SERVO, VECTOR

Reaction: ENCODER (IASC/DCBRAKE, NONE)

Acknowledge: IMMEDIATELY

Cause: Fault detected on the DRIVE-CLiQ component concerned. Faulty hardware cannot be excluded.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 20 hex:
 Error in the telegram header.
 xx = 23 hex:
 Receive error: The telegram buffer memory contains an error.
 xx = 42 hex:
 Send error: The telegram buffer memory contains an error.
 xx = 43 hex:
 Send error: The telegram buffer memory contains an error.
 See also: p0491 (Motor encoder fault response ENCODER)

Remedy:

- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
- check the electrical cabinet design and cable routing for EMC compliance
- if required, use another DRIVE-CLiQ socket (p9904).
- replace the component involved.

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F31845 (N, A)	Encoder 1 DRIVE-CLiQ: Cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex: Synchronization error during alternating cyclic data transfer. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	Carry out a POWER ON. See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31850 (N, A)	Encoder 1: Encoder evaluation, internal software error
Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	POWER ON
Cause:	An internal software error has occurred in the Sensor Module of encoder 1. Fault value (r0949, interpret decimal): 1: Background time slice is blocked. 2: Checksum over the code memory is not OK. 10000: OEM memory of the EnDat encoder contains data that cannot be interpreted. 11000 ... 11499: Descriptive data from EEPROM incorrect. 11500 ... 11899: Calibration data from EEPROM incorrect. 11900 ... 11999: Configuration data from EEPROM incorrect. 16000: DRIVE-CLiQ encoder initialization application error. 16001: DRIVE-CLiQ encoder initialization ALU error. 16002: DRIVE-CLiQ encoder HISI / SISI initialization error. 16003: DRIVE-CLiQ encoder safety initialization error. 16004: DRIVE-CLiQ encoder internal system error. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- replace the Sensor Module. - if required, upgrade the firmware in the Sensor Module. - contact the Hotline.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31851 (N, A)	Encoder 1 DRIVE-CLiQ (CU): Sign-of-life missing
Message value:	Component number: %1, fault cause: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 1) involved to the Control Unit. The DRIVE-CLiQ component did not set the sign-of-life to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0A hex = 10 dec: The sign-of-life bit in the receive telegram is not set.

Remedy: Upgrade the firmware of the component involved.
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F31860 (N, A) Encoder 1 DRIVE-CLiQ (CU): Telegram error

Message value: Component number: %1, fault cause: %2
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 1) involved to the Control Unit.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 11 hex = 17 dec:
 CRC error and the receive telegram is too early.
 xx = 01 hex = 01 dec:
 Checksum error (CRC error).
 xx = 12 hex = 18 dec:
 The telegram is shorter than that specified in the length byte or in the receive list and the receive telegram is too early.
 xx = 02 hex = 02 dec:
 Telegram is shorter than specified in the length byte or in the receive list.
 xx = 13 hex = 19 dec:
 The telegram is longer than that specified in the length byte or in the receive list and the receive telegram is too early.
 xx = 03 hex = 03 dec:
 Telegram is longer than specified in the length byte or in the receive list.
 xx = 14 hex = 20 dec:
 The length of the receive telegram does not match the receive list and the receive telegram is too early.
 xx = 04 hex = 04 dec:
 The length of the receive telegram does not match the receive list.
 xx = 15 hex = 21 dec:
 The type of the receive telegram does not match the receive list and the receive telegram is too early.
 xx = 05 hex = 05 dec:
 The type of the receive telegram does not match the receive list.
 xx = 16 hex = 22 dec:
 The address of the power unit in the telegram and in the receive list does not match and the receive telegram is too early.
 xx = 06 hex = 06 dec:
 The address of the power unit in the telegram and in the receive list do not match.
 xx = 19 hex = 25 dec:
 The error bit in the receive telegram is set and the receive telegram is too early.
 xx = 09 hex = 09 dec:
 The error bit in the receive telegram is set.
 xx = 10 hex = 16 dec:
 The receive telegram is too early.

Remedy:
 - carry out a POWER ON.
 - check the electrical cabinet design and cable routing for EMC compliance
 - check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
 See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F31885 (N, A)	Encoder 1 DRIVE-CLiQ (CU): Cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 1) involved to the Control Unit. The nodes do not send and receive in synchronism. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 1A hex = 26 dec: Sign-of-life bit in the receive telegram not set and the receive telegram is too early. xx = 21 hex = 33 dec: The cyclic telegram has not been received. xx = 22 hex = 34 dec: Timeout in the telegram receive list. xx = 40 hex = 64 dec: Timeout in the telegram send list. xx = 62 hex = 98 dec: Error at the transition to cyclic operation.
Remedy:	- check the power supply voltage of the component involved. - carry out a POWER ON. - replace the component involved. See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F31886 (N, A)	Encoder 1 DRIVE-CLiQ (CU): Error when sending DRIVE-CLiQ data
Message value:	Component number: %1, fault cause: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 1) involved to the Control Unit. Data were not able to be sent. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 41 hex: Telegram type does not match send list.
Remedy:	- carry out a POWER ON. - check whether the firmware version of the encoder (r0148) matches the firmware version of Control Unit (r0018).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F31887 (N, A)	Encoder 1 DRIVE-CLiQ (CU): Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component involved (Sensor Module for encoder 1). Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error. xx = 60 hex: Response received too late during runtime measurement. xx = 61 hex: Time taken to exchange characteristic data too long.
Remedy:	- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). - check the electrical cabinet design and cable routing for EMC compliance - if required, use another DRIVE-CLiQ socket (p9904). - replace the component involved.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31895 (N, A)	Encoder 1 DRIVE-CLiQ (CU): Alternating cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 1) involved to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex: Synchronization error during alternating cyclic data transfer.
Remedy:	Carry out a POWER ON. See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F31896 (N, A)	Encoder 1 DRIVE-CLiQ (CU): Inconsistent component properties
Message value:	Component number: %1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	OFF2 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	The properties of the DRIVE-CLiQ component (Sensor Module for encoder 1), specified by the fault value, have changed in an incompatible fashion with respect to the properties when booted. One cause can be, e.g. that a DRIVE-CLiQ cable or DRIVE-CLiQ component has been replaced. Fault value (r0949, interpret decimal): Component number.

Remedy:

- carry out a POWER ON.
- when a component is replaced, the same component type and if possible the same firmware version should be used.
- when a cable is replaced, only cables whose length is the same as or as close as possible to the length of the original cables should be used (ensure compliance with the maximum cable length).

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F31899 (N, A) Encoder 1: Unknown fault

Message value: New message: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: A fault occurred on the Sensor Module for encoder 1 that cannot be interpreted by the Control Unit firmware. This can occur if the firmware on this component is more recent than the firmware on the Control Unit.
Fault value (r0949, interpret decimal):
Fault number.
Note:
If required, the significance of this new fault can be read about in a more recent description of the Control Unit.
See also: p0491 (Motor encoder fault response ENCODER)

Remedy:

- replace the firmware on the Sensor Module by an older firmware version (r0148).
- upgrade the firmware on the Control Unit (r0018).

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

A31902 (F, N) Encoder 1: SPI-BUS error occurred

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Error when operating the internal SPI bus.
Fault value (r0949, interpret hexadecimal):
Only for internal Siemens troubleshooting.

Remedy:

- replace the Sensor Module.
- if required, upgrade the firmware in the Sensor Module.
- contact the Hotline.

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A31903 (F, N) Encoder 1: I2C-BUS error occurred

Message value: %1
Drive object: ENCODER, SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Error when operating the internal I2C bus.
Fault value (r0949, interpret hexadecimal):
Only for internal Siemens troubleshooting.

Remedy:	- replace the Sensor Module. - if required, upgrade the firmware in the Sensor Module. - contact the Hotline.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

F31905 (N, A) Encoder 1: Parameterization error

Message value:	Parameter: %1, supplementary information: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	A parameter of encoder 1 was detected as being incorrect. It is possible that the parameterized encoder type does not match the connected encoder. The parameter involved can be determined as follows: - determine the parameter number using the fault value (r0949). - determine the parameter index (p0187). Fault value (r0949, interpret decimal): yyyyxxxx dec: yyyy = supplementary information, xxxx = parameter Supplementary information = 0: No information available. Supplementary information = 1: The component does not support HTL level (p0405.1 = 0) combined with track monitoring A/B <> -A/B (p0405.2 = 1). Supplementary information = 2: A code number for an identified encoder has been entered into p0400, however, no identification was carried out. Please start a new encoder identification. Supplementary information = 3: A code number for an identified encoder has been entered into p0400, however, no identification was carried out. Please select a listed encoder in p0400 with a code number < 10000. Supplementary information = 4: This component does not support SSI encoders (p0404.9 = 1) without track A/B. Supplementary information = 5: For SQW encoder, value in p4686 greater than in p0425. Supplementary information = 6: DRIVE-CLiQ encoder cannot be used with this firmware version. Supplementary information = 7: For the SQW encoder, the Xact1 correction (p0437.2) is only permitted with equidistant zero marks. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- check whether the connected encoder type matches the encoder that has been parameterized. - correct the parameter specified by the fault value (r0949) and p0187. - re parameter number 314: Check the pole pair number and measuring gear ratio. The quotient of the "pole pair number" divided by the "measuring gear ratio" must be less than or equal to 1000 ((r0313 * p0433) / p0432 <= 1000).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A31915 (F, N) Encoder 1: Configuration error

Message value:	%1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The configuration for encoder 1 is incorrect. Fault value (r0949, interpret decimal): 1: Re-parameterization between fault/alarm is not permissible. 419: When the fine resolution Gx_ACT2 is configured, the encoder identifies a maximum possible absolute position actual value (r0483) that can no longer be represented within 32 bits.

Remedy: 1: No re-parameterization between fault/alarm.
419: Reduce the fine resolution (p0419).
Reaction upon F: NONE (ENCODER, IASC/DCBRAKE)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

F31916 (N, A) Encoder 1: Parameterization fault
Message value: Parameter: %1, supplementary information: %2
Drive object: SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge: IMMEDIATELY
Cause: A parameter of encoder 1 was detected as being incorrect.
It is possible that the parameterized encoder type does not match the connected encoder.
The parameter involved can be determined as follows:
- determine the parameter number using the fault value (r0949).
- determine the parameter index (p0187).
Fault value (r0949, interpret decimal):
Parameter number.
Note:
This fault is only output for encoders where r0404.10 = 1 or r0404.11 = 1. It corresponds to A31905 with encoders where r0404.10 = 0 and r0404.11 = 0.
See also: p0491 (Motor encoder fault response ENCODER)
Remedy: - check whether the connected encoder type matches the encoder that has been parameterized.
- correct the parameter specified by the fault value (r0949) and p0187.
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F31916 (N, A) Encoder 1: Parameterization fault
Message value: Parameter: %1, supplementary information: %2
Drive object: ENCODER
Reaction: ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge: IMMEDIATELY
Cause: A parameter of encoder 1 was detected as being incorrect.
In the case of the ENCODER drive object, the selected encoder type (rotary/linear) might not match the function module setting (r0108.12).
The parameter involved can be determined as follows:
- determine the parameter number using the fault value (r0949).
- determine the parameter index (p0187).
Fault value (r0949, interpret decimal):
Parameter number.
See also: p0491 (Motor encoder fault response ENCODER)
Remedy: - check whether the connected encoder type matches the encoder that has been parameterized.
- correct the parameter specified by the fault value (r0949) and p0187.
- if a linear encoder has been selected in parameter p0400/p0404, the "linear encoder" function module has to be activated (r0108.12 = 1)
- if a rotary encoder has been selected in parameter p0400/p0404, the "linear encoder" function module should not be activated (r0108.12 = 0)
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

A31920 (F, N)	Encoder 1: Temperature sensor fault
Message value:	Fault cause: %1, channel number: %2
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When evaluating the temperature sensor, an error occurred. Alarm value (r2124, interpret decimal): Low word low byte: Cause: 1: Wire breakage or sensor not connected (KTY: R > 1630 Ohm). 2: Measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm). Additional values: Only for internal Siemens troubleshooting. Low word high byte: Channel number. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- check that the encoder cable is the correct type and is correctly connected. - check the temperature sensor selection in p0600 to p0603. - replace the Sensor Module (hardware defect or incorrect calibration data).
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A31999 (F, N)	Encoder 1: Unknown alarm
Message value:	New message: %1
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A alarm has occurred on the Sensor Module for encoder 1 that cannot be interpreted by the Control Unit firmware. This can occur if the firmware on this component is more recent than the firmware on the Control Unit. Alarm value (r2124, interpret decimal): Alarm number. Note: If required, the significance of this new alarm can be read about in a more recent description of the Control Unit. See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- replace the firmware on the Sensor Module by an older firmware version (r0148). - upgrade the firmware on the Control Unit (r0018).
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
F32100 (N, A)	Encoder 2: Zero mark distance error
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	The measured zero mark distance does not correspond to the parameterized zero mark distance. For distance-coded encoders, the zero mark distance is determined from zero marks detected pairs. This means that if a zero mark is missing, depending on the pair generation, this cannot result in a fault and also has no effect in the system. The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder). Fault value (r0949, interpret decimal): Last measured zero mark distance in increments (4 increments = 1 encoder pulse). The sign designates the direction of motion when detecting the zero mark distance.

Remedy:

- check that the encoder cables are routed in compliance with EMC.
- check the plug connections
- check the encoder type (encoder with equidistant zero marks).
- adapt the parameter for the distance between zero marks (p0424, p0425).
- if message output above speed threshold, reduce filter time if necessary (p0438).
- replace the encoder or encoder cable

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F32101 (N, A) Encoder 2: Zero marked failed

Message value: %1

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)

Acknowledge: PULSE INHIBIT

Cause: The 1.5 x parameterized zero mark distance was exceeded.
The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder).
Fault value (r0949, interpret decimal):
Number of increments after POWER ON or since the last zero mark that was detected (4 increments = 1 encoder pulse).

Remedy:

- check that the encoder cables are routed in compliance with EMC.
- check the plug connections
- check the encoder type (encoder with equidistant zero marks).
- adapt the parameter for the clearance between zero marks (p0425).
- if message output above speed threshold, reduce filter time if necessary (p0438).
- when p0437.1 is active, check p4686.
- replace the encoder or encoder cable

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F32103 (N, A) Encoder 2: Amplitude error, track R

Message value: R track: %1

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: The amplitude of the reference track signal (track R) does not lie within the tolerance bandwidth for encoder 2.
The fault can be initiated when the unipolar voltage level is exceeded (RP/RN) or if the differential amplitude is under-shot.
Fault value (r0949, interpret hexadecimal):
xxxx hex:
xxxx = Signal level, track R (16 bits with sign).
The response thresholds of the unipolar signal levels of the encoder are between < 1400 mV and > 3500 mV.
The response threshold for the differential signal level of the encoder is < -1600 mV.
A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
Note:
The analog value of the amplitude error is not measured at the same time with the hardware fault output by the Sensor Module.
The fault value can only be represented between -32767dec and 32767dec (-770 mV ... 770 mV).
The signal level is not evaluated unless the following conditions are satisfied:
- Sensor Module properties available (r0459.31 = 1).
- Monitoring active (p0437.31 = 1).

- Remedy:**
- check the speed range; frequency characteristic (amplitude characteristic) of the measuring equipment might not be sufficient for the speed range
 - check that the encoder cables and shielding are routed in compliance with EMC.
 - check the plug connections and contacts of the encoder cable.
 - check whether the zero mark is connected and the signal cables RP and RN have been connected correctly
 - replace the encoder cable.
 - if the coding disk is soiled or the lighting aged, replace the encoder.

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F32110 (N, A) Encoder 2: Serial communications error

Message value: Fault cause: %1 bin

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)

Acknowledge: PULSE INHIBIT

Cause: Serial communication protocol transfer error between the encoder and evaluation module.

Fault value (r0949, interpret binary):

Bit 0: Alarm bit in the position protocol.

Bit 1: Incorrect quiescent level on the data line.

Bit 2: Encoder does not respond (does not supply a start bit within 50 ms).

Bit 3: CRC error: The checksum in the protocol from the encoder does not match the data.

Bit 4: Encoder acknowledgement error: The encoder incorrectly understood the task (request) or cannot execute it.

Bit 5: Internal error in the serial driver: An illegal mode command was requested.

Bit 6: Timeout when cyclically reading.

Bit 8: Protocol is too long (e.g. > 64 bits).

Bit 9: Receive buffer overflow.

Bit 10: Frame error when reading twice.

Bit 11: Parity error.

Bit 12: Data line signal level error during the monoflop time.

Bit 13: Data line incorrect.

- Remedy:**
- Re fault value, bit 0 = 1:
- Enc defect F31111 may provide additional details.

Re fault value, bit 1 = 1:

- Incorrect encoder type / replace the encoder or encoder cable.

Re fault value, bit 2 = 1:

- Incorrect encoder type / replace the encoder or encoder cable.

Re fault value, bit 3 = 1:

- EMC / connect the cable shield, replace the encoder or encoder cable.

Re fault value, bit 4 = 1:

- EMC / connect the cable shield, replace the encoder or encoder cable, replace the Sensor Module.

Re fault value, bit 5 = 1:

- EMC / connect the cable shield, replace the encoder or encoder cable, replace the Sensor Module.

Re fault value, bit 6 = 1:

- Update Sensor Module firmware.

Re fault value, bit 8 = 1:

- Check parameterization (p0429.2).

Re fault value, bit 9 = 1:

- EMC / connect the cable shield, replace the encoder or encoder cable, replace the Sensor Module.

Re fault value, bit 10 = 1:

- Check parameterization (p0429.2, p0449).

Re fault value, bit 11 = 1:

- Check parameterization (p0436).

Re fault value, bit 12 = 1:

- Check parameterization (p0429.6).

Re fault value, bit 13 = 1:

- Check data line.

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F32111 (N, A) Encoder 2: Absolute encoder EnDat, internal fault/error

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: PULSE INHIBIT
Cause: The EnDat encoder fault word supplies fault bits that have been set.
 Fault value (r0949, interpret binary):
 Bit 0: Lighting system failed.
 Bit 1: Signal amplitude too low.
 Bit 2: Position value incorrect.
 Bit 3: Encoder power supply overvoltage condition.
 Bit 4: Encoder power supply undervoltage condition.
 Bit 5: Encoder power supply overcurrent condition.
 Bit 6: The battery must be changed.
Remedy: Re fault value, bit 0 = 1:
 Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor.
 Re fault value, bit 1 = 1:
 Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor.
 Re fault value, bit 2 = 1:
 Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor.
 Re fault value, bit 3 = 1:
 5 V power supply voltage fault.
 When using an SMC: Check the plug-in cable between the encoder and SMC or replace the SMC.
 When a motor encoder with a direct DRIVE-CLiQ connection is used: Replace the motor.
 Re fault value, bit 4 = 1:
 5 V power supply voltage fault.
 When using an SMC: Check the plug-in cable between the encoder and SMC or replace the SMC.
 When using a motor with DRIVE-CLiQ: Replace the motor.
 Re fault value, bit 5 = 1:
 Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor.
 Re fault value, bit 6 = 1:
 The battery must be changed (only for encoders with battery back-up).

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F32112 (N, A) Encoder 2: Error bit set in the serial protocol

Message value: %1
Drive object: SERVO
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: PULSE INHIBIT
Cause: The encoder sends a set error bit via the serial protocol.
 Fault value (r0949, interpret binary):
 Bit 0: Fault bit in the position protocol.

Remedy: For fault value, bit 0 = 1:
In the case of an EnDat encoder, F31111 may provide further details.

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F32115 (N, A) Encoder 2: Amplitude error track A or B ($A^2 + B^2$)

Message value: A track: %1, B-track: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: PULSE INHIBIT
Cause: The amplitude (root of $A^2 + B^2$) for encoder 2 exceeds the permissible tolerance.
 Fault value (r0949, interpret hexadecimal):
 yyyxxxxx hex:
 yyyy = Signal level, track B (16 bits with sign).
 xxxx = Signal level, track A (16 bits with sign).
 The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %).
 The response thresholds are < 230 mV (observe the frequency response of the encoder) and > 750 mV.
 A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
 Note for sensors modules for resolvers (e. g. SMC10):
 The nominal signal level is at 2900 mV (2.0 Vrms). The response thresholds are < 1070 mV and > 3582 mV.
 A signal level of 2900 mV peak value corresponds to the numerical value 6666 hex = 26214 dec.
 Note:
 The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.

Remedy: - check that the encoder cables and shielding are routed in compliance with EMC.
 - check the plug connections
 - replace the encoder or encoder cable
 - check the Sensor Module (e.g. contacts).
 The following applies to measuring systems without their own bearing system:
 - adjust the scanning head and check the bearing system of the measuring wheel.
 The following applies for measuring systems with their own bearing system:
 - ensure that the encoder housing is not subject to any axial force.

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F32116 (N, A) Encoder 2: Amplitude error monitoring track A + B

Message value: Amplitude: %1, Angle: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The amplitude of the rectified encoder signals A and B and the amplitude from the roots of $A^2 + B^2$ for encoder 2 are not within the tolerance bandwidth.
 Fault value (r0949, interpret hexadecimal):
 yyyxxxxx hex:
 yyyy = Signal level, track B (16 bits with sign).
 xxxx = Signal level, track A (16 bits with sign).
 The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %).
 The response thresholds are < 176 mV (observe the frequency response of the encoder) and > 955 mV.
 A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
 Note:
 The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.

Remedy:

- check that the encoder cables and shielding are routed in compliance with EMC.
- check the plug connections
- replace the encoder or encoder cable
- check the Sensor Module (e.g. contacts).

Reaction upon N: NONE

Acknowled. upon N: NONE

Reaction upon A: NONE

Acknowled. upon A: NONE

F32117 (N, A) Encoder 2: Inversion error signals A and B and R

Message value: Fault cause: %1 bin

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: For a square-wave encoder (bipolar, double ended) the A* and B* and R* signals are not inverted with respect to signals A and B and R.

Fault value (r0949, interpret binary):

Bits 0 - 15: Only for internal Siemens troubleshooting.

Bit 16: Error track A.

Bit 17: Error track B.

Bit 18: Error track R.

Note:

For SMC30 (order no.. 6SL3055-0AA00-5CA0 and 6SL3055-0AA00-5CA1 only), CUA32, and CU310, the following applies:

A square-wave encoder without track R is used and track monitoring (p0405.2 = 1) is activated.

Remedy: Check the encoder/cable: Does the encoder provide signals and the associated inverted signals?

Note:

For SMC30 (order no. 6SL3055-0AA00-5CA0 and 6SL3055-0AA00-5CA1 only), the following applies:

- check the setting of p0405 (p0405.2 = 1 is only possible if the encoder is connected at X520).

For a square-wave encoder without track R, the following jumpers must be set for the connection at X520 (SMC30) or X23 (CUA32, CU310):

- pin 10 (reference signal R) <--> pin 7 (encoder power supply, ground)

- pin 11 (reference signal R inverted) <--> pin 4 (encoder power supply)

Reaction upon N: NONE

Acknowled. upon N: NONE

Reaction upon A: NONE

Acknowled. upon A: NONE

F32118 (N, A) Encoder 2: Speed difference outside the tolerance range

Message value: %1

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)

Acknowledge: PULSE INHIBIT

Cause: For an HTL/TTL encoder, the speed difference has exceeded the value in p0492 over several sampling cycles.

The change to the averaged speed actual value - if applicable - is monitored in the current controller sampling time.

Fault value (r0949, interpret decimal):

Only for internal Siemens troubleshooting.

See also: p0492

Remedy:

- check the tachometer feeder cable for interruptions.
- check the grounding of the tachometer shielding.
- if required, increase the maximum speed difference per sampling cycle (p0492).

Reaction upon N: NONE

Acknowled. upon N: NONE

Reaction upon A: NONE

Acknowled. upon A: NONE

F32120 (N, A)	Encoder 2: Power supply voltage fault
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	Encoder 2 power supply voltage fault. Note: If the encoder cables 6FX2002-2EQ00-.... and 6FX2002-2CH00-.... are interchanged, this can result in the encoder being destroyed because the pins of the operating voltage are reversed. Fault value (r0949, interpret binary): Bit 0: Undervoltage condition on the sense line. Bit 1: Overcurrent condition for the encoder power supply. Bit 2: Overcurrent condition for encoder power supply on cable resolver excitation negative. Bit 3: Overcurrent condition for encoder power supply on cable resolver excitation positive.
Remedy:	For fault value, bit 0 = 1: - correct encoder cable connected? - check the plug connections of the encoder cable. - SMC30: Check the parameterization (p0404.22). For fault value, bit 1 = 1: - correct encoder cable connected? - replace the encoder or encoder cable For fault value, bit 2 = 1: - correct encoder cable connected? - replace the encoder or encoder cable For fault value, bit 3 = 1: - correct encoder cable connected? - replace the encoder or encoder cable
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F32121 (N, A)	Encoder 2: Coarse position error
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	For the actual value sensing, an error was detected on the module. As a result of this error, it must be assumed that the actual value sensing supplies an incorrect coarse position.
Remedy:	Replace the motor with DRIVE-CLiQ or the appropriate Sensor Module.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F32122	Encoder 2: Internal power supply voltage faulty
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	ENCODER
Acknowledge:	IMMEDIATELY
Cause:	Fault in internal reference voltage of ASICs for encoder 2. Fault value (r0949, interpret decimal): 1: Reference voltage error. 2: Internal undervoltage. 3: Internal overvoltage.
Remedy:	Replace the motor with DRIVE-CLiQ or the appropriate Sensor Module.

F32123 (N, A) Encoder 2: Signal level A/B unipolar outside tolerance

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The unipolar level (AP/AN or BP/BN) for encoder 2 is outside the permissible tolerance.
Fault value (r0949, interpret binary):
Bit 0 = 1: Either AP or AN outside the tolerance.
Bit 16 = 1: Either BP or BN outside the tolerance.
The unipolar nominal signal level of the encoder must lie in the range 2500 mV +/- 500 mV.
The response thresholds are < 1700 mV and > 3300 mV.
Note:
The signal level is not evaluated unless the following conditions are satisfied:
- Sensor Module properties available (r0459.31 = 1).
- Monitoring active (p0437.31 = 1).
Remedy:
- make sure that the encoder cables and shielding are installed in an EMC-compliant manner.
- check the plug connections and contacts of the encoder cable.
- check the short-circuit of a signal cable with mass or the operating voltage.
- replace the encoder cable.
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F32125 (N, A) Encoder 2: Amplitude error track A or B overcontrolled

Message value: A track: %1, B-track: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge: PULSE INHIBIT
Cause: The amplitude of track A or B for encoder 2 exceeds the permissible tolerance band.
Fault value (r0949, interpret hexadecimal):
yyyyxxxx hex:
yyyy = Signal level, track B (16 bits with sign).
xxxx = Signal level, track A (16 bits with sign).
The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %).
The response threshold is > 750 mV. This fault also occurs if the A/D converter is overcontrolled.
A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
Note for sensors modules for resolvers (e. g. SMC10):
The nominal signal level is at 2900 mV (2.0 Vrms). The response threshold is > 3582 mV.
A signal level of 2900 mV peak value corresponds to the numerical value 6666 hex = 26214 dec.
Note:
The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.
Remedy:
- check that the encoder cables and shielding are routed in compliance with EMC.
- replace the encoder or encoder cable
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F32126 (N, A)	Encoder 2: Amplitude AB too high
Message value:	Amplitude: %1, Angle: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	The amplitude (root of $A^2 + B^2$ or $A + B$) for encoder 2 exceeds the permissible tolerance. Fault value (r0949, interpret hexadecimal): yyyyxxxx hex: yyyy = Angle xxxx = Amplitude, i.e. root from $A^2 + B^2$ (16 bits without sign) The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %). The response threshold for $(A + B)$ is > 1120 mV or the root of $(A^2 + B^2) > 955$ mV. A signal level of 500 mV peak value corresponds to the numerical value of 299A hex = 10650 dec. The angle 0 ... FFFF hex corresponds to 0 ... 360 degrees of the fine position. Zero degrees is at the negative zero crossover of track B. Note: The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.
Remedy:	- check that the encoder cables and shielding are routed in compliance with EMC. - replace the encoder or encoder cable
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F32129 (N, A)	Encoder 2: Position difference, hall sensor/track C/D and A/B too large
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	The error for track C/D is greater than $\pm 15^\circ$ mechanical or $\pm 60^\circ$ electrical or the error for the Hall signals is greater than $\pm 60^\circ$ electrical. One period of track C/D corresponds to 360° mechanical. One period of the Hall signal corresponds to 360° electrical. The monitoring responds if, for example, Hall sensors are connected as equivalent for the C/D tracks with the incorrect rotational sense or supply values that are not accurate enough. After the fine synchronization using one reference mark or 2 reference marks for distance-coded encoders, this fault is no longer initiated, but instead, Alarm A32429. Fault value (r0949, interpret decimal): For track C/D, the following applies: Measured deviation as mechanical angle (16 bits with sign, 182 dec corresponds to 1°). For Hall signals, the following applies: Measured deviation as electrical angle (16 bits with sign, 182 dec corresponds to 1°).
Remedy:	- track C or D not connected. - correct the direction of rotation of the Hall sensor possibly connected as equivalent for track C/D. - check that the encoder cables are routed in compliance with EMC. - check the adjustment of the Hall sensor.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F32130 (N, A)	Encoder 2: Zero mark and position error from the coarse synchronization
Message value:	Angular deviation, electrical: %1, angle, mechanical: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	After initializing the pole position using track C/D, Hall signals or pole position identification routine, the zero mark was detected outside the permissible range. For distance-coded encoders, the test is carried out after passing 2 zero marks. Fine synchronization was not carried out. When initializing via track C/D (p0404) then it is checked whether the zero mark occurs in an angular range of +/-18 ° mechanical. When initializing via Hall sensors (p0404) or pole position identification (p1982) it is checked whether the zero mark occurs in an angular range of +/-60 ° electrical. Fault value (r0949, interpret hexadecimal): yyyyxxxx hex yyyy: Determined mechanical zero mark position (can only be used for track C/D). xxxx: Deviation of the zero mark from the expected position as electrical angle. Scaling: 32768 dec = 180 °
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - if the Hall sensor is used as an equivalent for track C/D, check the connection. - Check the connection of track C or D. - replace the encoder or encoder cable
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F32131 (N, A)	Encoder 2: Deviation, position incremental/absolute too large
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	Absolute encoder: When cyclically reading the absolute position, an excessively high difference to the incremental position was detected. The absolute position that was read is rejected. Limit value for the deviation: - EnDat encoder: Is supplied from the encoder and is a minimum of 2 quadrants (e.g. EQI 1325 > 2 quadrants, EQN 1325 > 50 quadrants). - other encoders: 15 pulses = 60 quadrants. Incremental encoder: When the zero pulse is passed, a deviation in the incremental position was detected. For equidistant zero marks, the following applies: - The first zero mark passed supplies the reference point for all subsequent checks. The other zero marks must have n times the distance referred to the first zero mark. For distance-coded zero marks, the following applies: - the first zero mark pair supplies the reference point for all subsequent checks. The other zero mark pairs must have the expected distance to the first zero mark pair. Fault value (r0949, interpret decimal): Deviation in quadrants (1 pulse = 4 quadrants).
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - check whether the coding disk is dirty or there are strong ambient magnetic fields. - adapt the parameter for the clearance between zero marks (p0425). - if message output above speed threshold, reduce filter time if necessary (p0438).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F32135	Encoder 2: Fault when determining the position
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	The DRIVE-CLiQ encoder supplies status information via bits in an internal status/fault word. Some of these bits cause this fault to be triggered. Other bits are status displays. The status/fault word is displayed in the fault value. Fault value (r0949, interpret binary): Bit 0: F1 (safety status display) Bit 1: F2 (safety status display) Bit 2: Lighting (reserved) Bit 3: Signal amplitude (reserved) Bit 4: Position value (reserved) Bit 5: Overvoltage (reserved) Bit 6: Undervoltage (reserved) Bit 7: Overcurrent (reserved) Bit 8: Battery (reserved) Bit 16: Lighting (--> F3x135, x = 1, 2, 3) Bit 17: Signal amplitude (--> F3x135, x = 1, 2, 3) Bit 18: Singleturn position 1 (--> F3x135, x = 1, 2, 3) Bit 19: Overvoltage (--> F3x135, x = 1, 2, 3) Bit 20: Undervoltage (--> F3x135, x = 1, 2, 3) Bit 21: Overcurrent (--> F3x135, x = 1, 2, 3) Bit 22: Temperature exceeded (--> F3x405, x = 1, 2, 3) Bit 23: Singleturn position 2 (safety status display) Bit 24: Singleturn system (--> F3x135, x = 1, 2, 3) Bit 25: Singleturn power down (--> F3x135, x = 1, 2, 3) Bit 26: Multiturn position 1 (--> F3x136, x = 1, 2, 3) Bit 27: Multiturn position 2 (--> F3x136, x = 1, 2, 3) Bit 28: Multiturn system (--> F3x136, x = 1, 2, 3) Bit 29: Multiturn power down (--> F3x136, x = 1, 2, 3) Bit 30: Multiturn overflow/underflow (--> F3x136, x = 1, 2, 3) Bit 31: Multiturn battery (reserved)
Remedy:	Replace DRIVE-CLiQ encoder.
F32136	Encoder 2: Error when determining multiturn information
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	The DRIVE-CLiQ encoder supplies status information via bits in an internal status/fault word. Some of these bits cause this fault to be triggered. Other bits are status displays. The status/fault word is displayed in the fault value. Fault value (r0949, interpret binary): Bit 0: F1 (safety status display) Bit 1: F2 (safety status display) Bit 2: Lighting (reserved) Bit 3: Signal amplitude (reserved) Bit 4: Position value (reserved) Bit 5: Overvoltage (reserved) Bit 6: Undervoltage (reserved) Bit 7: Overcurrent (reserved) Bit 8: Battery (reserved) Bit 16: Lighting (--> F3x135, x = 1, 2, 3) Bit 17: Signal amplitude (--> F3x135, x = 1, 2, 3) Bit 18: Singleturn position 1 (--> F3x135, x = 1, 2, 3) Bit 19: Overvoltage (--> F3x135, x = 1, 2, 3) Bit 20: Undervoltage (--> F3x135, x = 1, 2, 3) Bit 21: Overcurrent (--> F3x135, x = 1, 2, 3) Bit 22: Temperature exceeded (--> F3x405, x = 1, 2, 3)

Bit 23: Singleturn position 2 (safety status display)
 Bit 24: Singleturn system (--> F3x135, x = 1, 2, 3)
 Bit 25: Singleturn power down (--> F3x135, x = 1, 2, 3)
 Bit 26: Multiturn position 1 (--> F3x136, x = 1, 2, 3)
 Bit 27: Multiturn position 2 (--> F3x136, x = 1, 2, 3)
 Bit 28: Multiturn system (--> F3x136, x = 1, 2, 3)
 Bit 29: Multiturn power down (--> F3x136, x = 1, 2, 3)
 Bit 30: Multiturn overflow/underflow (--> F3x136, x = 1, 2, 3)
 Bit 31: Multiturn battery (reserved)

Remedy: Replace DRIVE-CLiQ encoder.

F32137 Encoder 2: Internal error when determining the position

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: PULSE INHIBIT
Cause: The DRIVE-CLiQ encoder fault word supplies fault bits that have been set.
 Fault value (r0949, interpret binary):
 Only for internal SIEMENS use.
Remedy: Replace encoder

F32138 Encoder 2: Internal error when determining multiturn information

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: PULSE INHIBIT
Cause: The DRIVE-CLiQ encoder fault word supplies fault bits that have been set.
 Fault value (r0949, interpret binary):
 Only for internal SIEMENS use.
Remedy: Replace encoder

F32150 (N, A) Encoder 2: Initialization error

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge: PULSE INHIBIT
Cause: Encoder functionality selected in p0404 is not operating correctly.
 Fault value (r0949, interpret hexadecimal):
 The fault value is a bit field. Every set bit indicates functionality that is faulted.
 The bit assignment corresponds to that of p0404 (e.g. bit 5 set: Error track C/D).
Remedy:
 - Check that p0404 is correctly set.
 - check the encoder type used (incremental/absolute value) and for SMCxx, the encoder cable.
 - if relevant, note additional fault messages that describe the fault in detail.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F32151 (N, A)	Encoder 2: Encoder speed for initialization AB too high
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	The encoder speed is too high during while initializing the sensor.
Remedy:	Reduce the speed of the encoder accordingly during initialization. If necessary, deactivate monitoring (p0437.29). See also: p0437 (Sensor Module configuration extended)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F32160 (N, A)	Encoder 2: Analog sensor channel A failed
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The input voltage of the analog sensor is outside the permissible limits. Fault value (r0949, interpret decimal): 1: Input voltage outside detectable measuring range. 2: Input voltage outside measuring range set in p4673. 3: The absolute value of the input voltage has exceeded the range limit (p4676).
Remedy:	Re fault value = 1: - check the output voltage of the analog sensor. Re fault value = 2: - check the voltage setting for each encoder period (p4673). Re fault value = 3: - check the range limit setting and increase it if necessary (p4676).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F32161 (N, A)	Encoder 2: Analog sensor channel B failed
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE)
Acknowledge:	PULSE INHIBIT
Cause:	The input voltage of the analog sensor is outside the permissible limits. Fault value (r0949, interpret decimal): 1: Input voltage outside detectable measuring range. 2: Input voltage outside measuring range set in p4675. 3: The absolute value of the input voltage has exceeded the range limit (p4676).
Remedy:	Re fault value = 1: - check the output voltage of the analog sensor. Re fault value = 2: - check the voltage setting for each encoder period (p4675). Re fault value = 3: - check the range limit setting and increase it if necessary (p4676).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A32400 (F, N)	Encoder 2: Alarm threshold zero mark distance error
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The measured zero mark distance does not correspond to the parameterized zero mark distance. For distance-coded encoders, the zero mark distance is determined from zero marks detected pairs. This means that if a zero mark is missing, depending on the pair generation, this cannot result in a fault and also has no effect in the system. The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder). Alarm value (r2124, interpret decimal): Last measured zero mark distance in increments (4 increments = 1 encoder pulse). The sign designates the direction of motion when detecting the zero mark distance.
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - check the encoder type (encoder with equidistant zero marks). - adapt the parameter for the distance between zero marks (p0424, p0425). - replace the encoder or encoder cable
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A32401 (F, N)	Encoder 2: Alarm threshold zero marked failed
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The 1.5 x parameterized zero mark distance was exceeded. The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder). Alarm value (r2124, interpret decimal): Number of increments after POWER ON or since the last zero mark that was detected (4 increments = 1 encoder pulse).
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - check the encoder type (encoder with equidistant zero marks). - adapt the parameter for the clearance between zero marks (p0425). - replace the encoder or encoder cable
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

F32405 (N, A)	Encoder 2: Temperature in the encoder evaluation inadmissible
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The encoder evaluation for a motor with DRIVE-CLiQ has detected an inadmissible temperature. The fault threshold is 125 ° C. Alarm value (r2124, interpret decimal): Measured board/module temperature in 0.1 °C.
Remedy:	Reduce the ambient temperature for the DRIVE-CLiQ connection of the motor.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A32407 (F, N)	Encoder 2: Function limit reached
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The encoder has reached one of its function limits. A service is recommended. Alarm value (r2124, interpret decimal): 1 : Incremental signals 3 : Absolute track 4 : Code connection
Remedy:	Perform service. Replace the encoder if necessary. Note: The current functional reserve of an encoder can be displayed via r4651. See also: p4650 (Encoder functional reserve component number), r4651 (Encoder functional reserve)
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A32410 (F, N)	Encoder 2: Serial communications
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Serial communication protocol transfer error between the encoder and evaluation module. Alarm value (r2124, interpret binary): Bit 0: Alarm bit in the position protocol. Bit 1: Incorrect quiescent level on the data line. Bit 2: Encoder does not respond (does not supply a start bit within 50 ms). Bit 3: CRC error: The checksum in the protocol from the encoder does not match the data. Bit 4: Encoder acknowledgement error: The encoder incorrectly understood the task (request) or cannot execute it. Bit 5: Internal error in the serial driver: An illegal mode command was requested. Bit 6: Timeout when cyclically reading. Bit 8: Protocol is too long (e.g. > 64 bits). Bit 9: Receive buffer overflow. Bit 10: Frame error when reading twice. Bit 11: Parity error. Bit 12: Data line signal level error during the monoflop time.
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A32411 (F, N)	Encoder 2: EnDat encoder signals alarms
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The error word of the EnDat encoder has alarm bits that have been set. Alarm value (r2124, interpret binary): Bit 0: Frequency exceeded (speed too high). Bit 1: Temperature exceeded. Bit 2: Control reserve, lighting system exceeded. Bit 3: Battery discharged. Bit 4: Reference point passed.

Remedy: Replace encoder.
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A32412 (F, N) Encoder 2: Error bit set in the serial protocol

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The encoder sends a set error bit via the serial protocol.
 Alarm value (r2124, interpret binary):
 Bit 0: Fault bit in the position protocol.
 Bit 1: Alarm bit in the position protocol.
Remedy:
 - carry out a POWER ON (power off/on) for all components.
 - check that the encoder cables are routed in compliance with EMC.
 - check the plug connections
 - replace the encoder.
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A32414 (F, N) Encoder 2: Amplitude error track C or D ($C^2 + D^2$)

Message value: C track: %1, D track: %2
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The amplitude ($C^2 + D^2$) of track C or D of the encoder or from the Hall signals, is not within the tolerance bandwidth.
 Alarm value (r2124, interpret hexadecimal):
 yyyyxxxx hex:
 yyyy = Signal level, track D (16 bits with sign).
 xxxx = Signal level, track C (16 bits with sign).
 The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %).
 The response thresholds are < 230 mV (observe the frequency response of the encoder) and > 750 mV.
 A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
Note:
 If the amplitude is not within the tolerance bandwidth, then it cannot be used to initialize the start position.
Remedy:
 - check that the encoder cables are routed in compliance with EMC.
 - check the plug connections
 - replace the encoder or encoder cable
 - check the Sensor Module (e.g. contacts).
 - check the Hall sensor box
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

N32415 (F, A)	Encoder 2: Amplitude alarm track A or B ($A^2 + B^2$)
Message value:	Amplitude: %1, Angle: %2
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The amplitude (root of $A^2 + B^2$) for encoder 2 exceeds the permissible tolerance. Alarm value (r2124, interpret hexadecimal): yyyyxxxx hex: yyyy = Angle xxxx = Amplitude, i.e. root from $A^2 + B^2$ (16 bits without sign) The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %). The response threshold is < 300 mV (observe the frequency response of the encoder). A signal level of 500 mV peak value corresponds to the numerical value 299A hex = 10650 dec. The angle 0 ... FFFF hex corresponds to 0 ... 360 degrees of the fine position. Zero degrees is at the negative zero crossover of track B. Note for sensors modules for resolvers (e. g. SMC10): The nominal signal level is at 2900 mV (2.0 Vrms). The response threshold is < 1414 mV (1.0 Vrms). A signal level of 2900 mV peak value corresponds to the numerical value 3333 hex = 13107 dec. Note: The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.
Remedy:	- check the speed range, frequency characteristic (amplitude characteristic) of the measuring equipment is not sufficient for the speed range. - check that the encoder cables and shielding are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - check the Sensor Module (e.g. contacts). - if the coding disk is soiled or the lighting aged, replace the encoder.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A32418 (F, N)	Encoder 2: Speed difference per sampling rate exceeded
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	For an HTL/TTL encoder, the speed difference between two sampling cycles has exceeded the value in p0492. The change to the averaged speed actual value - if applicable - is monitored in the current controller sampling time. Alarm value (r2124, interpret decimal): Only for internal Siemens troubleshooting. See also: p0492
Remedy:	- check the tachometer feeder cable for interruptions. - check the grounding of the tachometer shielding. - if required, increase the setting of p0492.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A32419 (F, N) Encoder 2: Track A or B outside the tolerance range

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The amplitude, phase or offset correction for track A or B is at the limit.
Amplitude error correction: Amplitude B / Amplitude A = 0.78 ... 1.27
Phase: <84 degrees or >96 degrees
SMC20: Offset correction: +/-140 mV
SMC10: Offset correction: +/-650 mV
Alarm value (r2124, interpret hexadecimal):
xxxx1: Minimum of the offset correction, track B
xxxx2: Maximum of the offset correction, track B
xxx1x: Minimum of the offset correction, track A
xxx2x: Maximum of the offset correction, track A
xx1xx: Minimum of the amplitude correction, track B/A
xx2xx: Maximum of the amplitude correction, track B/A
x1xxx: Minimum of the phase error correction
x2xxx: Maximum of the phase error correction
1xxxx: Minimum of the cubic correction
2xxxx: Maximum of the cubic correction
Remedy:
- check mechanical mounting tolerances for encoders without their own bearings (e.g. toothed-wheel encoders).
- check the plug connections (also the transition resistance).
- check the encoder signals.
- replace the encoder or encoder cable
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A32421 (F, N) Encoder 2: Coarse position error

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: For the actual value sensing, an error was detected. As a result of this error, it must be assumed that the actual value sensing supplies an incorrect coarse position.
Alarm value (r2124, interpret decimal):
3: The absolute position of the serial protocol and track A/B differ by half an encoder pulse. The absolute position must have its zero position in the quadrants in which both tracks are negative. In the case of a fault, the position can be incorrect by one encoder pulse.
Remedy:
Re alarm value = 3:
- for a standard encoder with cable, if required, contact the manufacturer.
- correct the assignment of the tracks to the position value that is serially transferred. To do this, the two tracks must be connected, inverted, at the Sensor Module (interchange A with A* and B with B*) or, for a programmable encoder, check the zero offset of the position.
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A32422 (F, N)	Encoder 2: Pulses per revolution square-wave encoder outside tolerance bandwidth
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The measured zero mark distance does not correspond to the parameterized zero mark distance. This alarm is triggered with active square-wave encoder PPR correction and re-parameterized fault 31131 if the accumulator contains larger values than p4683 or p4684. The zero mark distance for zero mark monitoring is set in p0425 (rotary encoder). Alarm value (r2124, interpret decimal): accumulated differential pulses in encoder pulses.
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - check the encoder type (encoder with equidistant zero marks). - adapt the parameter for the distance between zero marks (p0424, p0425). - replace the encoder or encoder cable
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A32429 (F, N)	Encoder 2: Position difference, hall sensor/track C/D and A/B too large
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The error for track C/D is greater than +/-15 ° mechanical or +/-60 ° electrical or the error for the Hall signals is greater than +/-60 ° electrical. One period of track C/D corresponds to 360 ° mechanical. One period of the Hall signal corresponds to 360 ° electrical. The monitoring responds if, for example, Hall sensors are connected as equivalent for the C/D tracks with the incorrect rotational sense or supply values that are not accurate enough. Alarm value (r2124, interpret decimal): For track C/D, the following applies: Measured deviation as mechanical angle (16 bits with sign, 182 dec corresponds to 1 °). For Hall signals, the following applies: Measured deviation as electrical angle (16 bits with sign, 182 dec corresponds to 1 °).
Remedy:	- track C or D not connected. - correct the direction of rotation of the Hall sensor possibly connected as equivalent for track C/D. - check that the encoder cables are routed in compliance with EMC. - check the adjustment of the Hall sensor.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A32431 (F, N)	Encoder 2: Deviation, position incremental/absolute too large
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When the zero pulse is passed, a deviation in the incremental position was detected. For equidistant zero marks, the following applies: - The first zero mark passed supplies the reference point for all subsequent checks. The other zero marks must have n times the distance referred to the first zero mark. For distance-coded zero marks, the following applies: - the first zero mark pair supplies the reference point for all subsequent checks. The other zero mark pairs must have the expected distance to the first zero mark pair. Alarm value (r2124, interpret decimal): Deviation in quadrants (1 pulse = 4 quadrants).
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - Clean coding disk or remove strong magnetic fields.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A32432 (F, N)	Encoder 2: Rotor position adaptation corrects deviation
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	For track A/B, pulses have been lost or too many have been counted. These pulses are presently being corrected. Alarm value (r2124, interpret decimal): Last measured deviation of the zero mark distance in increments (4 increments = 1 encoder pulse). The sign designates the direction of motion when detecting the zero mark distance.
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - check encoder limit frequency. - adapt the parameter for the distance between zero marks (p0424, p0425).
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A32442 (F, N)	Encoder 2: Battery voltage pre-alarm
Message value:	-
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- replace battery.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A32443 (F, N)	Encoder 2: Unipolar CD signal level outside specification
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The unipolar level (CP/CN or DP/DN) for encoder 2 is outside the permissible tolerance. Alarm value (r2124, interpret binary): Bit 0 = 1: Either CP or CN outside the tolerance. Bit 16 = 1: Either DP or DN outside the tolerance. The unipolar nominal signal level of the encoder must lie in the range 2500 mV +/- 500 mV. The response thresholds are < 1700 mV and > 3300 mV. Note: The signal level is not evaluated unless the following conditions are satisfied: - Sensor Module properties available (r0459.31 = 1). - Monitoring active (p0437.31 = 1).
Remedy:	- check that the encoder cables and shielding are routed in compliance with EMC. - check the plug connections and contacts of the encoder cable. - are the C/D tracks connected correctly (have the signal lines CP and CN or DP and DN been interchanged)? - replace the encoder cable.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A32460 (N)	Encoder 2: Analog sensor channel A failed
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The input voltage of the analog sensor is outside the permissible limits. Alarm value (r2124, interpret decimal): 1: Input voltage outside detectable measuring range. 2: Input voltage outside measuring range set in p4673. 3: The absolute value of the input voltage has exceeded the range limit (p4676).
Remedy:	Re alarm value = 1: - check the output voltage of the analog sensor. Re alarm value = 2: - check the voltage setting for each encoder period (p4673). Re alarm value = 3: - check the range limit setting and increase it if necessary (p4676).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A32461 (N)	Encoder 2: Analog sensor channel B failed
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The input voltage of the analog sensor is outside the permissible limits. Alarm value (r2124, interpret decimal): 1: Input voltage outside detectable measuring range. 2: Input voltage outside measuring range set in p4675. 3: The absolute value of the input voltage has exceeded the range limit (p4676).

Remedy: Re alarm value = 1:
- check the output voltage of the analog sensor.
Re alarm value = 2:
- check the voltage setting for each encoder period (p4675).
Re alarm value = 3:
- check the range limit setting and increase it if necessary (p4676).
Reaction upon N: NONE
Acknowl. upon N: NONE

A32462 (N) Encoder 2: Analog sensor, no channel active

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Channel A and B are not activated for the analog sensor.
Remedy: - activate channel A and/or channel B (p4670).
- check the encoder configuration (p0404.17).
See also: p4670 (Analog sensor configuration)
Reaction upon N: NONE
Acknowl. upon N: NONE

A32470 (F, N) Encoder 2: Soiling detected

Message value: -
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: In the case of the alternative encoder system interface on the Sensor Module Cabinet 30 (SMC30), encoder soiling is signaled via a 0 signal at terminal X521.7.
Remedy: - check the plug connections
- replace the encoder or encoder cable
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

F32500 (N, A) Encoder 2: Position tracking traversing range exceeded

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: For a configured linear axis without modulo correction, the drive/encoder has exceeded the maximum possible traversing range. The value should be read in p0412 and interpreted as the number of motor revolutions.
For p0411.0 = 1, the maximum traversing range for the configured linear axis is defined to be 64x (+/- 32x) of p0421.
For p0411.3 = 1, the maximum traversing range for the configured linear axis is preset (default value) to the highest possible value and is +/-p0412/2 (rounded off to complete revolutions). The highest possible value depends on the pulse number (p0408) and the fine resolution (p0419).
Remedy: The fault should be resolved as follows:
- select encoder commissioning (p0010 = 4).
- reset the position tracking as follows (p0411.2 = 1).
- deselect encoder commissioning (p0010 = 0).
The fault should then be acknowledged and the absolute encoder adjusted.
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F32501 (N, A)	Encoder 2: Position tracking encoder position outside tolerance window
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	When powered down, the drive/encoder was moved through a distance greater than was parameterized in the tolerance window. It is possible that there is no longer any reference between the mechanical system and encoder. Fault value (r0949, decimal): Deviation (difference) to the last encoder position in increments of the absolute value. The sign designates the traversing direction. Note: The deviation (difference) found is also displayed in r0477. See also: p0413 (Measuring gear, position tracking tolerance window), r0477 (Measuring gear, position difference)
Remedy:	Reset the position tracking as follows: - select encoder commissioning (p0010 = 4). - reset the position tracking as follows (p0411.2 = 1). - deselect encoder commissioning (p0010 = 0). The fault should then be acknowledged and, if necessary, the absolute encoder adjusted (p2507). See also: p0010, p2507
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F32502 (N, A)	Encoder 2: Encoder with measuring gear, without valid signals
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The encoder with measuring gear no longer provides any valid signals.
Remedy:	It must be ensured that all of the encoders, with mounted measuring gear, provide valid actual values in operation.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F32503 (N, A)	Encoder 2: Position tracking cannot be reset
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The position tracking for the measuring gear cannot be reset.
Remedy:	The fault should be resolved as follows: - select encoder commissioning (p0010 = 4). - reset the position tracking as follows (p0411.2 = 1). - deselect encoder commissioning (p0010 = 0). The fault should then be acknowledged and the absolute encoder adjusted.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A32700 Encoder 2: Effectivity test does not supply the expected value

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The DRIVE-CLiQ encoder fault word supplies fault bits that have been set.
Fault value (r0949, interpret binary):
Bit x = 1: Effectivity test x unsuccessful.
Remedy:

N32800 (F) Encoder 2: Group signal

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: NONE
Cause: The motor encoder has detected at least one fault.
Remedy: Evaluates other actual messages.
Reaction upon F: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY

F32801 (N, A) Encoder 2 DRIVE-CLiQ: Sign-of-life missing

Message value: Component number: %1, fault cause: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 0A hex:
The sign-of-life bit in the receive telegram is not set.
Remedy: - check the electrical cabinet design and cable routing for EMC compliance
- replace the component involved.
See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F32802 (N, A) Encoder 2: Time slice overflow

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: Time slice overflow, encoder 2.
Fault value (r0949, interpret decimal):
9: Time slice overflow of the fast (current controller clock cycle) time slice.
10: Time slice overflow of the average time slice.
12: Time slice overflow of the slow time slice.
999: Timeout when waiting for SYNO, e.g. unexpected return to non-cyclic operation.
Remedy: Reduce the current controller frequency.
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F32804 (N, A)	Encoder 2: Checksum error
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A checksum error has occurred when reading-out the program memory on the Sensor Module. Fault value (r0949, interpret hexadecimal): yyyyxxxx hex yyyy: Memory area involved. xxxx: Difference between the checksum at POWER ON and the actual checksum.
Remedy:	- check whether the permissible ambient temperature for the component is maintained. - replace the Sensor Module.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F32805 (N, A)	Encoder 2: EPROM checksum error
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	Internal parameter data is corrupted. Fault value (r0949, interpret hexadecimal): 01: EEPROM access error. 02: Too many blocks in the EEPROM.
Remedy:	Replace the module.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F32806 (N, A)	Encoder 2: Initialization error
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	The encoder was not successfully initialized. Fault value (r0949, interpret hexadecimal): Bit 0,1: Encoder initialization with the motor rotating has failed (deviation involving coarse and fine position in encoder pulse/4) Bit 2: Mid-voltage matching for track A unsuccessful. Bit 3: Mid-voltage matching for track B unsuccessful. Bit 4: Mid-voltage matching for acceleration input (under development) unsuccessful. Bit 5: Mid-voltage matching for track safety A unsuccessful. Bit 6: Mid-voltage matching for track safety B unsuccessful. Bit 7: Mid-voltage matching for track C unsuccessful. Bit 8: Mid-voltage matching for track D unsuccessful. Bit 9: Mid-voltage matching for track R unsuccessful. Bit 10: The difference in mid-voltages between A and B is too great (> 0.5 V) Bit 11: The difference in mid-voltages between C and D is too great (> 0.5 V) Bit 12: The difference in mid-voltages between safety A and safety B is too great (> 0.5 V) Bit 13: The difference in mid-voltages between A and safety B is too great (> 0.5 V) Bit 14: The difference in mid-voltages between B and safety A is too great (> 0.5 V) Bit 15: The standard deviation of the calculated mid-voltages is too great (> 0.3 V) Bit 16: Internal fault - fault reading a register (CAFE) Bit 17: Internal fault - fault writing a register (CAFE)

Bit 18: Internal fault: No mid-voltage matching available
Bit 19: Internal error - ADC access error.
Bit 20: Internal error - no zero crossover found.
Bit 0,1 : Up to 6SL3055-0AA00-5*A0
Bits 2..20 : 6SL3055-0AA00-5*A1 and higher

Remedy: Acknowledge the fault.
If the fault cannot be acknowledged:
Bits 2 - 9: Check encoder power supply,
Bits 2 -14: Check the corresponding cable,
Bit 15 with no other bits: Check track R, check settings in p404.

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F32811 (N, A) Encoder 2: Encoder serial number changed

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The encoder serial number has changed. The change is only checked for encoders with serial number (e.g. EnDat encoders).
Cause: The encoder was replaced.
Note: With closed-loop position control, the serial number is accepted when starting the adjustment (p2507 = 2).
When the encoder is adjusted (p2507 = 3), the serial number is checked for changes and if required, the adjustment is reset (p2507 = 1).
Proceed as follows to hide serial number monitoring:
- set the following serial numbers for the corresponding Encoder Data Set: p0441= FF, p0442 = 0, p0442 = 0, p0444 = 0, p0445 = 0.

Remedy: Mechanically adjust the encoder. Accept the new serial number with p0440 = 1.

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F32812 (N, A) Encoder 2: Requested cycle or RX-/TX timing not supported

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A cycle requested from the Control Unit or RX/TX timing is not supported.
Alarm value (r2124, interpret decimal):
0: Application cycle is not supported.
1: DQ cycle is not supported.
2: Distance between RX and TX instants in time too low.
3: TX instant in time too early.

Remedy:

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F32813 Encoder 2: Hardware logic unit failed

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: PULSE INHIBIT
Cause: The DRIVE-CLiQ encoder fault word supplies fault bits that have been set.
 Fault value (r0949, interpret binary):
 Bit 0: ALU watchdog has responded.
 Bit 1: ALU has detected a sign-of-life error.
Remedy: Replace encoder

F32820 (N, A) Encoder 2 DRIVE-CLiQ: Telegram error

Message value: Component number: %1, fault cause: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 01 hex:
 CRC error.
 xx = 02 hex:
 Telegram is shorter than specified in the length byte or in the receive list.
 xx = 03 hex:
 Telegram is longer than specified in the length byte or in the receive list.
 xx = 04 hex:
 The length of the receive telegram does not match the receive list.
 xx = 05 hex:
 The type of the receive telegram does not match the receive list.
 xx = 06 hex:
 The address of the component in the telegram and in the receive list do not match.
 xx = 07 hex:
 A SYNC telegram is expected - but the received telegram is not a SYNC telegram.
 xx = 08 hex:
 No SYNC telegram is expected - but the received telegram is one.
 xx = 09 hex:
 The error bit in the receive telegram is set.
 xx = 10 hex:
 The receive telegram is too early.
Remedy:
 - carry out a POWER ON.
 - check the electrical cabinet design and cable routing for EMC compliance
 - check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
 See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F32835 (N, A)	Encoder 2 DRIVE-CLiQ: Cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved. The nodes do not send and receive in synchronism. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 21 hex: The cyclic telegram has not been received. xx = 22 hex: Timeout in the telegram receive list. xx = 40 hex: Timeout in the telegram send list.
Remedy:	- carry out a POWER ON. - replace the component involved. See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F32836 (N, A)	Encoder 2 DRIVE-CLiQ: Send error for DRIVE-CLiQ data
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved. Data were not able to be sent. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 41 hex: Telegram type does not match send list.
Remedy:	Carry out a POWER ON.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F32837 (N, A)	Encoder 2 DRIVE-CLiQ: Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component concerned. Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error.

Remedy:

- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
- check the electrical cabinet design and cable routing for EMC compliance
- if required, use another DRIVE-CLiQ socket (p9904).
- replace the component involved.

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F32845 (N, A) Encoder 2 DRIVE-CLiQ: Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 0B hex:
 Synchronization error during alternating cyclic data transfer.

Remedy: Carry out a POWER ON.
 See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F32850 (N, A) Encoder 2: Encoder evaluation, internal software error

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: POWER ON
Cause: An internal software error has occurred in the Sensor Module of encoder 2.
 Fault value (r0949, interpret decimal):
 1: Background time slice is blocked.
 2: Checksum over the code memory is not OK.
 10000: OEM memory of the EnDat encoder contains data that cannot be interpreted.
 11000 ... 11499: Descriptive data from EEPROM incorrect.
 11500 ... 11899: Calibration data from EEPROM incorrect.
 11900 ... 11999: Configuration data from EEPROM incorrect.
 16000: DRIVE-CLiQ encoder initialization application error.
 16001: DRIVE-CLiQ encoder initialization ALU error.
 16002: DRIVE-CLiQ encoder HISI / SISI initialization error.
 16003: DRIVE-CLiQ encoder safety initialization error.
 16004: DRIVE-CLiQ encoder internal system error.

Remedy:

- replace the Sensor Module.
- if required, upgrade the firmware in the Sensor Module.
- contact the Hotline.

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F32851 (N, A)	Encoder 2 DRIVE-CLiQ (CU): Sign-of-life missing
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 2) involved to the Control Unit. The DRIVE-CLiQ component did not set the sign-of-life to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0A hex = 10 dec: The sign-of-life bit in the receive telegram is not set.
Remedy:	Upgrade the firmware of the component involved.
Reaction upon N:	NONE
Acknowled. upon N:	NONE
Reaction upon A:	NONE
Acknowled. upon A:	NONE

F32860 (N, A)	Encoder 2 DRIVE-CLiQ (CU): Telegram error
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 2) involved to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 11 hex = 17 dec: CRC error and the receive telegram is too early. xx = 01 hex = 01 dec: Checksum error (CRC error). xx = 12 hex = 18 dec: The telegram is shorter than that specified in the length byte or in the receive list and the receive telegram is too early. xx = 02 hex = 02 dec: Telegram is shorter than specified in the length byte or in the receive list. xx = 13 hex = 19 dec: The telegram is longer than that specified in the length byte or in the receive list and the receive telegram is too early. xx = 03 hex = 03 dec: Telegram is longer than specified in the length byte or in the receive list. xx = 14 hex = 20 dec: The length of the receive telegram does not match the receive list and the receive telegram is too early. xx = 04 hex = 04 dec: The length of the receive telegram does not match the receive list. xx = 15 hex = 21 dec: The type of the receive telegram does not match the receive list and the receive telegram is too early. xx = 05 hex = 05 dec: The type of the receive telegram does not match the receive list. xx = 16 hex = 22 dec: The address of the power unit in the telegram and in the receive list does not match and the receive telegram is too early. xx = 06 hex = 06 dec: The address of the power unit in the telegram and in the receive list do not match. xx = 19 hex = 25 dec: The error bit in the receive telegram is set and the receive telegram is too early. xx = 09 hex = 09 dec: The error bit in the receive telegram is set. xx = 10 hex = 16 dec: The receive telegram is too early.

Remedy:

- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

Reaction upon N: NONE

Acknowled. upon N: NONE

Reaction upon A: NONE

Acknowled. upon A: NONE

F32885 (N, A) Encoder 2 DRIVE-CLiQ (CU): Cyclic data transfer error

Message value: Component number: %1, fault cause: %2

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 2) involved to the Control Unit. The nodes do not send and receive in synchronism.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 1A hex = 26 dec:
 Sign-of-life bit in the receive telegram not set and the receive telegram is too early.
 xx = 21 hex = 33 dec:
 The cyclic telegram has not been received.
 xx = 22 hex = 34 dec:
 Timeout in the telegram receive list.
 xx = 40 hex = 64 dec:
 Timeout in the telegram send list.
 xx = 62 hex = 98 dec:
 Error at the transition to cyclic operation.

Remedy:

- check the power supply voltage of the component involved.
- carry out a POWER ON.
- replace the component involved.

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

Reaction upon N: NONE

Acknowled. upon N: NONE

Reaction upon A: NONE

Acknowled. upon A: NONE

F32886 (N, A) Encoder 2 DRIVE-CLiQ (CU): Error when sending DRIVE-CLiQ data

Message value: Component number: %1, fault cause: %2

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 2) involved to the Control Unit. Data were not able to be sent.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 41 hex:
 Telegram type does not match send list.

Remedy: Carry out a POWER ON.

Reaction upon N: NONE

Acknowled. upon N: NONE

Reaction upon A: NONE

Acknowled. upon A: NONE

F32887 (N, A)	Encoder 2 DRIVE-CLiQ (CU): Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component involved (Sensor Module for encoder 2). Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error. xx = 60 hex: Response received too late during runtime measurement. xx = 61 hex: Time taken to exchange characteristic data too long.
Remedy:	- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). - check the electrical cabinet design and cable routing for EMC compliance - if required, use another DRIVE-CLiQ socket (p9904). - replace the component involved.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F32895 (N, A)	Encoder 2 DRIVE-CLiQ (CU): Alternating cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 2) involved to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex: Synchronization error during alternating cyclic data transfer.
Remedy:	Carry out a POWER ON. See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F32896 (N, A)	Encoder 2 DRIVE-CLiQ (CU): Inconsistent component properties
Message value:	Component number: %1
Drive object:	SERVO, VECTOR
Reaction:	OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	The properties of the DRIVE-CLiQ component (Sensor Module for encoder 2), specified by the fault value, have changed in an incompatible fashion with respect to the properties when booted. One cause can be, e.g. that a DRIVE-CLiQ cable or DRIVE-CLiQ component has been replaced. Fault value (r0949, interpret decimal): Component number.

Remedy:	- carry out a POWER ON. - when a component is replaced, the same component type and if possible the same firmware version should be used. - when a cable is replaced, only cables whose length is the same as or as close as possible to the length of the original cables should be used (ensure compliance with the maximum cable length).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F32899 (N, A) Encoder 2: Unknown fault

Message value:	New message: %1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	A fault occurred on the Sensor Module for encoder 2 that cannot be interpreted by the Control Unit firmware. This can occur if the firmware on this component is more recent than the firmware on the Control Unit. Fault value (r0949, interpret decimal): Fault number. Note: If required, the significance of this new fault can be read about in a more recent description of the Control Unit.
Remedy:	- replace the firmware on the Sensor Module by an older firmware version (r0148). - upgrade the firmware on the Control Unit (r0018).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A32902 (F, N) Encoder 2: SPI-BUS error occurred

Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Error when operating the internal SPI bus. Fault value (r0949, interpret hexadecimal): Only for internal Siemens troubleshooting.
Remedy:	- replace the Sensor Module. - if required, upgrade the firmware in the Sensor Module. - contact the Hotline.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A32903 (F, N) Encoder 2: I2C-BUS error occurred

Message value:	%1
Drive object:	SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	Error when operating the internal I2C bus. Fault value (r0949, interpret hexadecimal): Only for internal Siemens troubleshooting.

Remedy:

- replace the Sensor Module.
- if required, upgrade the firmware in the Sensor Module.
- contact the Hotline.

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

F32905 (N, A) Encoder 2: Parameterization error

Message value: Parameter: %1, supplementary information: %2

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)

Acknowledge: IMMEDIATELY

Cause:

A parameter of encoder 2 was detected as being incorrect.
It is possible that the parameterized encoder type does not match the connected encoder.
The parameter involved can be determined as follows:

- determine the parameter number using the fault value (r0949).
- determine the parameter index (p0187).

Fault value (r0949, interpret decimal):
yyyyxxxx dec: yyyy = supplementary information, xxxx = parameter

Supplementary information = 0:
No information available.

Supplementary information = 1:
The component does not support HTL level (p0405.1 = 0) combined with track monitoring A/B <> -A/B (p0405.2 = 1).

Supplementary information = 2:
A code number for an identified encoder has been entered into p0400, however, no identification was carried out.
Please start a new encoder identification.

Supplementary information = 3:
A code number for an identified encoder has been entered into p0400, however, no identification was carried out.
Please select a listed encoder in p0400 with a code number < 10000.

Supplementary information = 4:
This component does not support SSI encoders (p0404.9 = 1) without track A/B.

Supplementary information = 5:
For SQW encoder, value in p4686 greater than in p0425.

Supplementary information = 6:
DRIVE-CLiQ encoder cannot be used with this firmware version.

Supplementary information = 7:
For the SQW encoder, the Xact1 correction (p0437.2) is only permitted with equidistant zero marks.

Remedy:

- check whether the connected encoder type matches the encoder that has been parameterized.
- correct the parameter specified by the fault value (r0949) and p0187.
- re parameter number 314: Check the pole pair number and measuring gear ratio. The quotient of the "pole pair number" divided by the "measuring gear ratio" must be less than or equal to 1000 ((r0313 * p0433) / p0432 <= 1000).

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

A32915 (F, N) Encoder 2: Configuration error

Message value: %1

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause:

The configuration for encoder 2 is incorrect.
Fault value (r0949, interpret decimal):

- 1: Re-parameterization between fault/alarm is not permissible.
- 419: When the fine resolution Gx_ACT2 is configured, the encoder identifies a maximum possible absolute position actual value (r0483) that can no longer be represented within 32 bits.

Remedy: 1: No re-parameterization between fault/alarm.
419: Reduce the fine resolution (p0419).

Reaction upon F: NONE (IASC/DCBRAKE)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

F32916 (N, A) Encoder 2: Parameterization fault

Message value: Parameter: %1, supplementary information: %2

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)

Acknowledge: IMMEDIATELY

Cause: A parameter of encoder 2 was detected as being incorrect.
It is possible that the parameterized encoder type does not match the connected encoder.
The parameter involved can be determined as follows:
- determine the parameter number using the fault value (r0949).
- determine the parameter index (p0187).
Fault value (r0949, interpret decimal):
Parameter number.
Note:
This fault is only output for encoders where r0404.10 = 1 or r0404.11 = 1. It corresponds to A32905 with encoders where r0404.10 = 0 and r0404.11 = 0.

Remedy: - check whether the connected encoder type matches the encoder that has been parameterized.
- correct the parameter specified by the fault value (r0949) and p0187.

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

A32920 (F, N) Encoder 2: Temperature sensor fault

Message value: Fault cause: %1, channel number: %2

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: When evaluating the temperature sensor, an error occurred.
Alarm value (r2124, interpret decimal):
Low word low byte: Cause:
1: Wire breakage or sensor not connected (KTY: R > 1630 Ohm).
2: Measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm).
Additional values:
Only for internal Siemens troubleshooting.
Low word high byte: Channel number.

Remedy: - check that the encoder cable is the correct type and is correctly connected.
- check the temperature sensor selection in p0600 to p0603.
- replace the Sensor Module (hardware defect or incorrect calibration data).

Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

A32999 (F, N) Encoder 2: Unknown alarm

Message value: New message: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: A alarm has occurred on the Sensor Module for encoder 2 that cannot be interpreted by the Control Unit firmware. This can occur if the firmware on this component is more recent than the firmware on the Control Unit.
Alarm value (r2124, interpret decimal):
Alarm number.
Note:
If required, the significance of this new alarm can be read about in a more recent description of the Control Unit.
Remedy:
- replace the firmware on the Sensor Module by an older firmware version (r0148).
- upgrade the firmware on the Control Unit (r0018).
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

F33100 (N, A) Encoder 3: Zero mark distance error

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge: PULSE INHIBIT
Cause: The measured zero mark distance does not correspond to the parameterized zero mark distance.
For distance-coded encoders, the zero mark distance is determined from zero marks detected pairs. This means that if a zero mark is missing, depending on the pair generation, this cannot result in a fault and also has no effect in the system.
The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder).
Fault value (r0949, interpret decimal):
Last measured zero mark distance in increments (4 increments = 1 encoder pulse).
The sign designates the direction of motion when detecting the zero mark distance.
Remedy:
- check that the encoder cables are routed in compliance with EMC.
- check the plug connections
- check the encoder type (encoder with equidistant zero marks).
- adapt the parameter for the distance between zero marks (p0424, p0425).
- if message output above speed threshold, reduce filter time if necessary (p0438).
- replace the encoder or encoder cable
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F33101 (N, A) Encoder 3: Zero marked failed

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge: PULSE INHIBIT
Cause: The 1.5 x parameterized zero mark distance was exceeded.
The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder).
Fault value (r0949, interpret decimal):
Number of increments after POWER ON or since the last zero mark that was detected (4 increments = 1 encoder pulse).

- Remedy:**
- check that the encoder cables are routed in compliance with EMC.
 - check the plug connections
 - check the encoder type (encoder with equidistant zero marks).
 - adapt the parameter for the clearance between zero marks (p0425).
 - if message output above speed threshold, reduce filter time if necessary (p0438).
 - when p0437.1 is active, check p4686.
 - replace the encoder or encoder cable

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F33103 (N, A) Encoder 3: Amplitude error, track R

- Message value:** R track: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: The amplitude of the reference track signal (track R) does not lie within the tolerance bandwidth for encoder 3. The fault can be initiated when the unipolar voltage level is exceeded (RP/RN) or if the differential amplitude is under-shot.
 Fault value (r0949, interpret hexadecimal):
 xxxx hex:
 xxxx = Signal level, track R (16 bits with sign).
 The response thresholds of the unipolar signal levels of the encoder are between < 1400 mV and > 3500 mV.
 The response threshold for the differential signal level of the encoder is < -1600 mV.
 A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
Note:
 The analog value of the amplitude error is not measured at the same time with the hardware fault output by the Sensor Module.
 The fault value can only be represented between -32767dec and 32767dec (-770 mV ... 770 mV).
 The signal level is not evaluated unless the following conditions are satisfied:
 - Sensor Module properties available (r0459.31 = 1).
 - Monitoring active (p0437.31 = 1).
Remedy:
- check the speed range; frequency characteristic (amplitude characteristic) of the measuring equipment might not be sufficient for the speed range
 - check that the encoder cables and shielding are routed in compliance with EMC.
 - check the plug connections and contacts of the encoder cable.
 - check whether the zero mark is connected and the signal cables RP and RN have been connected correctly
 - replace the encoder cable.
 - if the coding disk is soiled or the lighting aged, replace the encoder.
- Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F33110 (N, A) Encoder 3: Serial communications error

Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	Serial communication protocol transfer error between the encoder and evaluation module. Fault value (r0949, interpret binary): Bit 0: Alarm bit in the position protocol. Bit 1: Incorrect quiescent level on the data line. Bit 2: Encoder does not respond (does not supply a start bit within 50 ms). Bit 3: CRC error: The checksum in the protocol from the encoder does not match the data. Bit 4: Encoder acknowledgement error: The encoder incorrectly understood the task (request) or cannot execute it. Bit 5: Internal error in the serial driver: An illegal mode command was requested. Bit 6: Timeout when cyclically reading. Bit 8: Protocol is too long (e.g. > 64 bits). Bit 9: Receive buffer overflow. Bit 10: Frame error when reading twice. Bit 11: Parity error. Bit 12: Data line signal level error during the monoflop time. Bit 13: Data line incorrect.
Remedy:	Re fault value, bit 0 = 1: - Enc defect F31111 may provide additional details. Re fault value, bit 1 = 1: - Incorrect encoder type / replace the encoder or encoder cable. Re fault value, bit 2 = 1: - Incorrect encoder type / replace the encoder or encoder cable. Re fault value, bit 3 = 1: - EMC / connect the cable shield, replace the encoder or encoder cable. Re fault value, bit 4 = 1: - EMC / connect the cable shield, replace the encoder or encoder cable, replace the Sensor Module. Re fault value, bit 5 = 1: - EMC / connect the cable shield, replace the encoder or encoder cable, replace the Sensor Module. Re fault value, bit 6 = 1: - Update Sensor Module firmware. Re fault value, bit 8 = 1: - Check parameterization (p0429.2). Re fault value, bit 9 = 1: - EMC / connect the cable shield, replace the encoder or encoder cable, replace the Sensor Module. Re fault value, bit 10 = 1: - Check parameterization (p0429.2, p0449). Re fault value, bit 11 = 1: - Check parameterization (p0436). Re fault value, bit 12 = 1: - Check parameterization (p0429.6). Re fault value, bit 13 = 1: - Check data line.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33111 (N, A)	Encoder 3: Absolute encoder EnDat, internal fault/error
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	The EnDat encoder fault word supplies fault bits that have been set. Fault value (r0949, interpret binary): Bit 0: Lighting system failed. Bit 1: Signal amplitude too low. Bit 2: Position value incorrect. Bit 3: Encoder power supply overvoltage condition. Bit 4: Encoder power supply undervoltage condition. Bit 5: Encoder power supply overcurrent condition. Bit 6: The battery must be changed.
Remedy:	Re fault value, bit 0 = 1: Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor. Re fault value, bit 1 = 1: Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor. Re fault value, bit 2 = 1: Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor. Re fault value, bit 3 = 1: 5 V power supply voltage fault. When using an SMC: Check the plug-in cable between the encoder and SMC or replace the SMC. When a motor encoder with a direct DRIVE-CLiQ connection is used: Replace the motor. Re fault value, bit 4 = 1: 5 V power supply voltage fault. When using an SMC: Check the plug-in cable between the encoder and SMC or replace the SMC. When using a motor with DRIVE-CLiQ: Replace the motor. Re fault value, bit 5 = 1: Encoder is defective. Replace the encoder, where the motor encoder has a direct DRIVE-CLiQ socket: Replace the motor. Re fault value, bit 6 = 1: The battery must be changed (only for encoders with battery back-up).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F33112 (N, A)	Encoder 3: Error bit set in the serial protocol
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	The encoder sends a set error bit via the serial protocol. Fault value (r0949, interpret binary): Bit 0: Fault bit in the position protocol.
Remedy:	For fault value, bit 0 = 1: In the case of an EnDat encoder, F31111 may provide further details.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33115 (N, A)	Encoder 3: Amplitude error track A or B ($A^2 + B^2$)
Message value:	A track: %1, B-track: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	The amplitude (root of $A^2 + B^2$) for encoder 3 exceeds the permissible tolerance. Fault value (r0949, interpret hexadecimal): yyyyxxxx hex: yyyy = Signal level, track B (16 bits with sign). xxxx = Signal level, track A (16 bits with sign). The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %). The response thresholds are < 230 mV (observe the frequency response of the encoder) and > 750 mV. A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec. Note for sensors modules for resolvers (e. g. SMC10): The nominal signal level is at 2900 mV (2.0 Vrms). The response thresholds are < 1070 mV and > 3582 mV. A signal level of 2900 mV peak value corresponds to the numerical value 6666 hex = 26214 dec. Note: The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.
Remedy:	- check that the encoder cables and shielding are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - check the Sensor Module (e.g. contacts). The following applies to measuring systems without their own bearing system: - adjust the scanning head and check the bearing system of the measuring wheel. The following applies for measuring systems with their own bearing system: - ensure that the encoder housing is not subject to any axial force.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33116 (N, A)	Encoder 3: Amplitude error monitoring track A + B
Message value:	Amplitude: %1, Angle: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The amplitude of the rectified encoder signals A and B and the amplitude from the roots of $A^2 + B^2$ for encoder 3 are not within the tolerance bandwidth. Fault value (r0949, interpret hexadecimal): yyyyxxxx hex: yyyy = Signal level, track B (16 bits with sign). xxxx = Signal level, track A (16 bits with sign). The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %). The response thresholds are < 176 mV (observe the frequency response of the encoder) and > 955 mV. A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec. Note: The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.
Remedy:	- check that the encoder cables and shielding are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - check the Sensor Module (e.g. contacts).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33117 (N, A)	Encoder 3: Inversion error signals A and B and R
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	For a square-wave encoder (bipolar, double ended) the A* and B* and R* signals are not inverted with respect to signals A and B and R. Fault value (r0949, interpret binary): Bits 0 - 15: Only for internal Siemens troubleshooting. Bit 16: Error track A. Bit 17: Error track B. Bit 18: Error track R. Note: For SMC30 (order no.. 6SL3055-0AA00-5CA0 and 6SL3055-0AA00-5CA1 only), CUA32, and CU310, the following applies: A square-wave encoder without track R is used and track monitoring (p0405.2 = 1) is activated.
Remedy:	Check the encoder/cable: Does the encoder provide signals and the associated inverted signals? Note: For SMC30 (order no. 6SL3055-0AA00-5CA0 and 6SL3055-0AA00-5CA1 only), the following applies: - check the setting of p0405 (p0405.2 = 1 is only possible if the encoder is connected at X520). For a square-wave encoder without track R, the following jumpers must be set for the connection at X520 (SMC30) or X23 (CUA32, CU310): - pin 10 (reference signal R) <--> pin 7 (encoder power supply, ground) - pin 11 (reference signal R inverted) <--> pin 4 (encoder power supply)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F33118 (N, A)	Encoder 3: Speed difference outside the tolerance range
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	For an HTL/TTL encoder, the speed difference has exceeded the value in p0492 over several sampling cycles. The change to the averaged speed actual value - if applicable - is monitored in the current controller sampling time. Fault value (r0949, interpret decimal): Only for internal Siemens troubleshooting. See also: p0492
Remedy:	- check the tachometer feeder cable for interruptions. - check the grounding of the tachometer shielding. - if required, increase the maximum speed difference per sampling cycle (p0492).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33120 (N, A)	Encoder 3: Power supply voltage fault
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	Encoder 3 power supply voltage fault. Note: If the encoder cables 6FX2002-2EQ00-.... and 6FX2002-2CH00-.... are interchanged, this can result in the encoder being destroyed because the pins of the operating voltage are reversed. Fault value (r0949, interpret binary): Bit 0: Undervoltage condition on the sense line. Bit 1: Overcurrent condition for the encoder power supply. Bit 2: Overcurrent condition for encoder power supply on cable resolver excitation negative. Bit 3: Overcurrent condition for encoder power supply on cable resolver excitation positive.
Remedy:	For fault value, bit 0 = 1: - correct encoder cable connected? - check the plug connections of the encoder cable. - SMC30: Check the parameterization (p0404.22). For fault value, bit 1 = 1: - correct encoder cable connected? - replace the encoder or encoder cable For fault value, bit 2 = 1: - correct encoder cable connected? - replace the encoder or encoder cable For fault value, bit 3 = 1: - correct encoder cable connected? - replace the encoder or encoder cable
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33121 (N, A)	Encoder 3: Coarse position error
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	For the actual value sensing, an error was detected on the module. As a result of this error, it must be assumed that the actual value sensing supplies an incorrect coarse position.
Remedy:	Replace the motor with DRIVE-CLiQ or the appropriate Sensor Module.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33122	Encoder 3: Internal power supply voltage faulty
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	ENCODER
Acknowledge:	IMMEDIATELY
Cause:	Fault in internal reference voltage of ASICs for encoder 3. Fault value (r0949, interpret decimal): 1: Reference voltage error. 2: Internal undervoltage. 3: Internal overvoltage.
Remedy:	Replace the motor with DRIVE-CLiQ or the appropriate Sensor Module.

F33123 (N, A) Encoder 3: Signal level A/B unipolar outside tolerance

Message value: Fault cause: %1 bin

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: The unipolar level (AP/AN or BP/BN) for encoder 3 is outside the permissible tolerance.
 Fault value (r0949, interpret binary):
 Bit 0 = 1: Either AP or AN outside the tolerance.
 Bit 16 = 1: Either BP or BN outside the tolerance.
 The unipolar nominal signal level of the encoder must lie in the range 2500 mV +/- 500 mV.
 The response thresholds are < 1700 mV and > 3300 mV.
 Note:
 The signal level is not evaluated unless the following conditions are satisfied:
 - Sensor Module properties available (r0459.31 = 1).
 - Monitoring active (p0437.31 = 1).

Remedy: - make sure that the encoder cables and shielding are installed in an EMC-compliant manner.
 - check the plug connections and contacts of the encoder cable.
 - check the short-circuit of a signal cable with mass or the operating voltage.
 - replace the encoder cable.

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F33125 (N, A) Encoder 3: Amplitude error track A or B overcontrolled

Message value: A track: %1, B-track: %2

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)

Acknowledge: PULSE INHIBIT

Cause: The amplitude of track A or B for encoder 3 exceeds the permissible tolerance band.
 Fault value (r0949, interpret hexadecimal):
 yyyxxxx hex:
 yyyy = Signal level, track B (16 bits with sign).
 xxxx = Signal level, track A (16 bits with sign).
 The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %).
 The response threshold is > 750 mV. This fault also occurs if the A/D converter is overcontrolled.
 A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
 Note for sensors modules for resolvers (e. g. SMC10):
 The nominal signal level is at 2900 mV (2.0 Vrms). The response threshold is > 3582 mV.
 A signal level of 2900 mV peak value corresponds to the numerical value 6666 hex = 26214 dec.
 Note:
 The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.

Remedy: - check that the encoder cables and shielding are routed in compliance with EMC.
 - replace the encoder or encoder cable

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F33126 (N, A)	Encoder 3: Amplitude AB too high
Message value:	Amplitude: %1, Angle: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	The amplitude (root of $A^2 + B^2$ or $A + B$) for encoder 3 exceeds the permissible tolerance. Fault value (r0949, interpret hexadecimal): yyyyxxxx hex: yyyy = Angle xxxx = Amplitude, i.e. root from $A^2 + B^2$ (16 bits without sign) The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %). The response threshold for ($A + B$) is > 1120 mV or the root of $(A^2 + B^2) > 955$ mV. A signal level of 500 mV peak value corresponds to the numerical value of 299A hex = 10650 dec. The angle 0 ... FFFF hex corresponds to 0 ... 360 degrees of the fine position. Zero degrees is at the negative zero crossover of track B. Note: The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.
Remedy:	- check that the encoder cables and shielding are routed in compliance with EMC. - replace the encoder or encoder cable
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33129 (N, A)	Encoder 3: Position difference, hall sensor/track C/D and A/B too large
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	The error for track C/D is greater than $\pm 15^\circ$ mechanical or $\pm 60^\circ$ electrical or the error for the Hall signals is greater than $\pm 60^\circ$ electrical. One period of track C/D corresponds to 360° mechanical. One period of the Hall signal corresponds to 360° electrical. The monitoring responds if, for example, Hall sensors are connected as equivalent for the C/D tracks with the incorrect rotational sense or supply values that are not accurate enough. After the fine synchronization using one reference mark or 2 reference marks for distance-coded encoders, this fault is no longer initiated, but instead, Alarm A33429. Fault value (r0949, interpret decimal): For track C/D, the following applies: Measured deviation as mechanical angle (16 bits with sign, 182 dec corresponds to 1°). For Hall signals, the following applies: Measured deviation as electrical angle (16 bits with sign, 182 dec corresponds to 1°).
Remedy:	- track C or D not connected. - correct the direction of rotation of the Hall sensor possibly connected as equivalent for track C/D. - check that the encoder cables are routed in compliance with EMC. - check the adjustment of the Hall sensor.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33130 (N, A)	Encoder 3: Zero mark and position error from the coarse synchronization
Message value:	Angular deviation, electrical: %1, angle, mechanical: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	After initializing the pole position using track C/D, Hall signals or pole position identification routine, the zero mark was detected outside the permissible range. For distance-coded encoders, the test is carried out after passing 2 zero marks. Fine synchronization was not carried out. When initializing via track C/D (p0404) then it is checked whether the zero mark occurs in an angular range of +/-18 ° mechanical. When initializing via Hall sensors (p0404) or pole position identification (p1982) it is checked whether the zero mark occurs in an angular range of +/-60 ° electrical. Fault value (r0949, interpret hexadecimal): yyyyxxxx hex yyyy: Determined mechanical zero mark position (can only be used for track C/D). xxxx: Deviation of the zero mark from the expected position as electrical angle. Scaling: 32768 dec = 180 °
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - if the Hall sensor is used as an equivalent for track C/D, check the connection. - Check the connection of track C or D. - replace the encoder or encoder cable
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F33131 (N, A)	Encoder 3: Deviation, position incremental/absolute too large
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	PULSE INHIBIT
Cause:	Absolute encoder: When cyclically reading the absolute position, an excessively high difference to the incremental position was detected. The absolute position that was read is rejected. Limit value for the deviation: - EnDat encoder: Is supplied from the encoder and is a minimum of 2 quadrants (e.g. EQ1 1325 > 2 quadrants, EQN 1325 > 50 quadrants). - other encoders: 15 pulses = 60 quadrants. Incremental encoder: When the zero pulse is passed, a deviation in the incremental position was detected. For equidistant zero marks, the following applies: - The first zero mark passed supplies the reference point for all subsequent checks. The other zero marks must have n times the distance referred to the first zero mark. For distance-coded zero marks, the following applies: - the first zero mark pair supplies the reference point for all subsequent checks. The other zero mark pairs must have the expected distance to the first zero mark pair. Fault value (r0949, interpret decimal): Deviation in quadrants (1 pulse = 4 quadrants).
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - check whether the coding disk is dirty or there are strong ambient magnetic fields. - adapt the parameter for the clearance between zero marks (p0425). - if message output above speed threshold, reduce filter time if necessary (p0438).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33135	Encoder 3: Fault when determining the position
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	The DRIVE-CLiQ encoder supplies status information via bits in an internal status/fault word. Some of these bits cause this fault to be triggered. Other bits are status displays. The status/fault word is displayed in the fault value. Fault value (r0949, interpret binary): Bit 0: F1 (safety status display) Bit 1: F2 (safety status display) Bit 2: Lighting (reserved) Bit 3: Signal amplitude (reserved) Bit 4: Position value (reserved) Bit 5: Overvoltage (reserved) Bit 6: Undervoltage (reserved) Bit 7: Overcurrent (reserved) Bit 8: Battery (reserved) Bit 16: Lighting (--> F3x135, x = 1, 2, 3) Bit 17: Signal amplitude (--> F3x135, x = 1, 2, 3) Bit 18: Singleturn position 1 (--> F3x135, x = 1, 2, 3) Bit 19: Overvoltage (--> F3x135, x = 1, 2, 3) Bit 20: Undervoltage (--> F3x135, x = 1, 2, 3) Bit 21: Overcurrent (--> F3x135, x = 1, 2, 3) Bit 22: Temperature exceeded (--> F3x405, x = 1, 2, 3) Bit 23: Singleturn position 2 (safety status display) Bit 24: Singleturn system (--> F3x135, x = 1, 2, 3) Bit 25: Singleturn power down (--> F3x135, x = 1, 2, 3) Bit 26: Multiturn position 1 (--> F3x136, x = 1, 2, 3) Bit 27: Multiturn position 2 (--> F3x136, x = 1, 2, 3) Bit 28: Multiturn system (--> F3x136, x = 1, 2, 3) Bit 29: Multiturn power down (--> F3x136, x = 1, 2, 3) Bit 30: Multiturn overflow/underflow (--> F3x136, x = 1, 2, 3) Bit 31: Multiturn battery (reserved)
Remedy:	Replace DRIVE-CLiQ encoder.

F33136	Encoder 3: Error when determining multiturn information
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	PULSE INHIBIT
Cause:	The DRIVE-CLiQ encoder supplies status information via bits in an internal status/fault word. Some of these bits cause this fault to be triggered. Other bits are status displays. The status/fault word is displayed in the fault value. Fault value (r0949, interpret binary): Bit 0: F1 (safety status display) Bit 1: F2 (safety status display) Bit 2: Lighting (reserved) Bit 3: Signal amplitude (reserved) Bit 4: Position value (reserved) Bit 5: Overvoltage (reserved) Bit 6: Undervoltage (reserved) Bit 7: Overcurrent (reserved) Bit 8: Battery (reserved) Bit 16: Lighting (--> F3x135, x = 1, 2, 3) Bit 17: Signal amplitude (--> F3x135, x = 1, 2, 3) Bit 18: Singleturn position 1 (--> F3x135, x = 1, 2, 3) Bit 19: Overvoltage (--> F3x135, x = 1, 2, 3) Bit 20: Undervoltage (--> F3x135, x = 1, 2, 3) Bit 21: Overcurrent (--> F3x135, x = 1, 2, 3) Bit 22: Temperature exceeded (--> F3x405, x = 1, 2, 3)

Bit 23: Singleturn position 2 (safety status display)
 Bit 24: Singleturn system (--> F3x135, x = 1, 2, 3)
 Bit 25: Singleturn power down (--> F3x135, x = 1, 2, 3)
 Bit 26: Multiturn position 1 (--> F3x136, x = 1, 2, 3)
 Bit 27: Multiturn position 2 (--> F3x136, x = 1, 2, 3)
 Bit 28: Multiturn system (--> F3x136, x = 1, 2, 3)
 Bit 29: Multiturn power down (--> F3x136, x = 1, 2, 3)
 Bit 30: Multiturn overflow/underflow (--> F3x136, x = 1, 2, 3)
 Bit 31: Multiturn battery (reserved)

Remedy: Replace DRIVE-CLiQ encoder.

F33137 Encoder 3: Internal error when determining the position

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: PULSE INHIBIT
Cause: The DRIVE-CLiQ encoder fault word supplies fault bits that have been set.
 Fault value (r0949, interpret binary):
 Only for internal SIEMENS use.
Remedy: Replace encoder

F33138 Encoder 3: Internal error when determining multiturn information

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: PULSE INHIBIT
Cause: The DRIVE-CLiQ encoder fault word supplies fault bits that have been set.
 Fault value (r0949, interpret binary):
 Only for internal SIEMENS use.
Remedy: Replace encoder

F33150 (N, A) Encoder 3: Initialization error

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge: PULSE INHIBIT
Cause: Encoder functionality selected in p0404 is not operating correctly.
 Fault value (r0949, interpret hexadecimal):
 The fault value is a bit field. Every set bit indicates functionality that is faulted.
 The bit assignment corresponds to that of p0404 (e.g. bit 5 set: Error track C/D).
Remedy:
 - Check that p0404 is correctly set.
 - check the encoder type used (incremental/absolute value) and for SMCxx, the encoder cable.
 - if relevant, note additional fault messages that describe the fault in detail.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F33151 (N, A) Encoder 3: Encoder speed for initialization AB too high

Message value: %1
Drive object: SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge: PULSE INHIBIT
Cause: The encoder speed is too high during while initializing the sensor.

Remedy: Reduce the speed of the encoder accordingly during initialization.
If necessary, deactivate monitoring (p0437.29).
See also: p0437 (Sensor Module configuration extended)

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F33160 (N, A) Encoder 3: Analog sensor channel A failed

Message value: %1
Drive object: SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: PULSE INHIBIT
Cause: The input voltage of the analog sensor is outside the permissible limits.
Fault value (r0949, interpret decimal):
1: Input voltage outside detectable measuring range.
2: Input voltage outside measuring range set in p4673.
3: The absolute value of the input voltage has exceeded the range limit (p4676).

Remedy: Re fault value = 1:
- check the output voltage of the analog sensor.
Re fault value = 2:
- check the voltage setting for each encoder period (p4673).
Re fault value = 3:
- check the range limit setting and increase it if necessary (p4676).

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F33161 (N, A) Encoder 3: Analog sensor channel B failed

Message value: %1
Drive object: SERVO, VECTOR
Reaction: ENCODER (IASC/DCBRAKE, NONE)
Acknowledge: PULSE INHIBIT
Cause: The input voltage of the analog sensor is outside the permissible limits.
Fault value (r0949, interpret decimal):
1: Input voltage outside detectable measuring range.
2: Input voltage outside measuring range set in p4675.
3: The absolute value of the input voltage has exceeded the range limit (p4676).

Remedy: Re fault value = 1:
- check the output voltage of the analog sensor.
Re fault value = 2:
- check the voltage setting for each encoder period (p4675).
Re fault value = 3:
- check the range limit setting and increase it if necessary (p4676).

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

A33400 (F, N)	Encoder 3: Alarm threshold zero mark distance error
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The measured zero mark distance does not correspond to the parameterized zero mark distance. For distance-coded encoders, the zero mark distance is determined from zero marks detected pairs. This means that if a zero mark is missing, depending on the pair generation, this cannot result in a fault and also has no effect in the system. The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder). Alarm value (r2124, interpret decimal): Last measured zero mark distance in increments (4 increments = 1 encoder pulse). The sign designates the direction of motion when detecting the zero mark distance.
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - check the encoder type (encoder with equidistant zero marks). - adapt the parameter for the distance between zero marks (p0424, p0425). - replace the encoder or encoder cable
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A33401 (F, N)	Encoder 3: Alarm threshold zero marked failed
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The 1.5 x parameterized zero mark distance was exceeded. The zero mark distance for the zero mark monitoring is set in p0425 (rotary encoder) or p0424 (linear encoder). Alarm value (r2124, interpret decimal): Number of increments after POWER ON or since the last zero mark that was detected (4 increments = 1 encoder pulse).
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - check the encoder type (encoder with equidistant zero marks). - adapt the parameter for the clearance between zero marks (p0425). - replace the encoder or encoder cable
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
F33405 (N, A)	Encoder 3: Temperature in the encoder evaluation inadmissible
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	ENCODER (IASC/DCBRAKE, NONE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The encoder evaluation for a motor with DRIVE-CLiQ has detected an inadmissible temperature. The fault threshold is 125 ° C. Alarm value (r2124, interpret decimal): Measured board/module temperature in 0.1 °C.
Remedy:	Reduce the ambient temperature for the DRIVE-CLiQ connection of the motor.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A33407 (F, N)	Encoder 3: Function limit reached
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The encoder has reached one of its function limits. A service is recommended. Alarm value (r2124, interpret decimal): 1 : Incremental signals 3 : Absolute track 4 : Code connection
Remedy:	Perform service. Replace the encoder if necessary. Note: The current functional reserve of an encoder can be displayed via r4651. See also: p4650 (Encoder functional reserve component number), r4651 (Encoder functional reserve)
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A33410 (F, N)	Encoder 3: Serial communications
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Serial communication protocol transfer error between the encoder and evaluation module. Alarm value (r2124, interpret binary): Bit 0: Alarm bit in the position protocol. Bit 1: Incorrect quiescent level on the data line. Bit 2: Encoder does not respond (does not supply a start bit within 50 ms). Bit 3: CRC error: The checksum in the protocol from the encoder does not match the data. Bit 4: Encoder acknowledgement error: The encoder incorrectly understood the task (request) or cannot execute it. Bit 5: Internal error in the serial driver: An illegal mode command was requested. Bit 6: Timeout when cyclically reading. Bit 8: Protocol is too long (e.g. > 64 bits). Bit 9: Receive buffer overflow. Bit 10: Frame error when reading twice. Bit 11: Parity error. Bit 12: Data line signal level error during the monoflop time.
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A33411 (F, N)	Encoder 3: EnDat encoder signals alarms
Message value:	Fault cause: %1 bin
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The error word of the EnDat encoder has alarm bits that have been set. Alarm value (r2124, interpret binary): Bit 0: Frequency exceeded (speed too high). Bit 1: Temperature exceeded. Bit 2: Control reserve, lighting system exceeded. Bit 3: Battery discharged. Bit 4: Reference point passed.

Remedy: Replace encoder.
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A33412 (F, N) Encoder 3: Error bit set in the serial protocol

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The encoder sends a set error bit via the serial protocol.
 Alarm value (r2124, interpret binary):
 Bit 0: Fault bit in the position protocol.
 Bit 1: Alarm bit in the position protocol.
Remedy:
 - carry out a POWER ON (power off/on) for all components.
 - check that the encoder cables are routed in compliance with EMC.
 - check the plug connections
 - replace the encoder.
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A33414 (F, N) Encoder 3: Amplitude error track C or D ($C^2 + D^2$)

Message value: C track: %1, D track: %2
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The amplitude ($C^2 + D^2$) of track C or D of the encoder or from the Hall signals, is not within the tolerance bandwidth.
 Alarm value (r2124, interpret hexadecimal):
 yyyyxxxx hex:
 yyyy = Signal level, track D (16 bits with sign).
 xxxx = Signal level, track C (16 bits with sign).
 The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %).
 The response thresholds are < 230 mV (observe the frequency response of the encoder) and > 750 mV.
 A signal level of 500 mV peak value corresponds to the numerical value 5333 hex = 21299 dec.
Note:
 If the amplitude is not within the tolerance bandwidth, then it cannot be used to initialize the start position.
Remedy:
 - check that the encoder cables are routed in compliance with EMC.
 - check the plug connections
 - replace the encoder or encoder cable
 - check the Sensor Module (e.g. contacts).
 - check the Hall sensor box
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

N33415 (F, A)	Encoder 3: Amplitude alarm track A or B ($A^2 + B^2$)
Message value:	Amplitude: %1, Angle: %2
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The amplitude (root of $A^2 + B^2$) for encoder 3 exceeds the permissible tolerance. Alarm value (r2124, interpret hexadecimal): yyyyxxxx hex: yyyy = Angle xxxx = Amplitude, i.e. root from $A^2 + B^2$ (16 bits without sign) The nominal signal level of the encoder must lie in the range 375 mV to 600 mV (500 mV -25/+20 %). The response threshold is < 300 mV (observe the frequency response of the encoder). A signal level of 500 mV peak value corresponds to the numerical value 299A hex = 10650 dec. The angle 0 ... FFFF hex corresponds to 0 ... 360 degrees of the fine position. Zero degrees is at the negative zero crossover of track B. Note for sensors modules for resolvers (e. g. SMC10): The nominal signal level is at 2900 mV (2.0 Vrms). The response threshold is < 1414 mV (1.0 Vrms). A signal level of 2900 mV peak value corresponds to the numerical value 3333 hex = 13107 dec. Note: The analog values of the amplitude error are not measured at the same time with the hardware fault output by the Sensor Module.
Remedy:	- check the speed range, frequency characteristic (amplitude characteristic) of the measuring equipment is not sufficient for the speed range. - check that the encoder cables and shielding are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - check the Sensor Module (e.g. contacts). - if the coding disk is soiled or the lighting aged, replace the encoder.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A33418 (F, N)	Encoder 3: Speed difference per sampling rate exceeded
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	For an HTL/TTL encoder, the speed difference between two sampling cycles has exceeded the value in p0492. The change to the averaged speed actual value - if applicable - is monitored in the current controller sampling time. Alarm value (r2124, interpret decimal): Only for internal Siemens troubleshooting. See also: p0492
Remedy:	- check the tachometer feeder cable for interruptions. - check the grounding of the tachometer shielding. - if required, increase the setting of p0492.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A33419 (F, N)	Encoder 3: Track A or B outside the tolerance range
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The amplitude, phase or offset correction for track A or B is at the limit. Amplitude error correction: Amplitude B / Amplitude A = 0.78 ... 1.27 Phase: <84 degrees or >96 degrees SMC20: Offset correction: +/-140 mV SMC10: Offset correction: +/-650 mV Alarm value (r2124, interpret hexadecimal): xxxx1: Minimum of the offset correction, track B xxxx2: Maximum of the offset correction, track B xxx1x: Minimum of the offset correction, track A xxx2x: Maximum of the offset correction, track A xx1xx: Minimum of the amplitude correction, track B/A xx2xx: Maximum of the amplitude correction, track B/A x1xxx: Minimum of the phase error correction x2xxx: Maximum of the phase error correction 1xxxx: Minimum of the cubic correction 2xxxx: Maximum of the cubic correction
Remedy:	- check mechanical mounting tolerances for encoders without their own bearings (e.g. toothed-wheel encoders). - check the plug connections (also the transition resistance). - check the encoder signals. - replace the encoder or encoder cable
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A33421 (F, N)	Encoder 3: Coarse position error
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	For the actual value sensing, an error was detected. As a result of this error, it must be assumed that the actual value sensing supplies an incorrect coarse position. Alarm value (r2124, interpret decimal): 3: The absolute position of the serial protocol and track A/B differ by half an encoder pulse. The absolute position must have its zero position in the quadrants in which both tracks are negative. In the case of a fault, the position can be incorrect by one encoder pulse.
Remedy:	Re alarm value = 3: - for a standard encoder with cable, if required, contact the manufacturer. - correct the assignment of the tracks to the position value that is serially transferred. To do this, the two tracks must be connected, inverted, at the Sensor Module (interchange A with A* and B with B*) or, for a programmable encoder, check the zero offset of the position.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A33422 (F, N)	Encoder 3: Pulses per revolution square-wave encoder outside tolerance bandwidth
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The measured zero mark distance does not correspond to the parameterized zero mark distance. This alarm is triggered with active square-wave encoder PPR correction and re-parameterized fault 31131 if the accumulator contains larger values than p4683 or p4684. The zero mark distance for zero mark monitoring is set in p0425 (rotary encoder). Alarm value (r2124, interpret decimal): accumulated differential pulses in encoder pulses.
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - check the encoder type (encoder with equidistant zero marks). - adapt the parameter for the distance between zero marks (p0424, p0425). - replace the encoder or encoder cable
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A33429 (F, N)	Encoder 3: Position difference, hall sensor/track C/D and A/B too large
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The error for track C/D is greater than +/-15 ° mechanical or +/-60 ° electrical or the error for the Hall signals is greater than +/-60 ° electrical. One period of track C/D corresponds to 360 ° mechanical. One period of the Hall signal corresponds to 360 ° electrical. The monitoring responds if, for example, Hall sensors are connected as equivalent for the C/D tracks with the incorrect rotational sense or supply values that are not accurate enough. Alarm value (r2124, interpret decimal): For track C/D, the following applies: Measured deviation as mechanical angle (16 bits with sign, 182 dec corresponds to 1 °). For Hall signals, the following applies: Measured deviation as electrical angle (16 bits with sign, 182 dec corresponds to 1 °).
Remedy:	- track C or D not connected. - correct the direction of rotation of the Hall sensor possibly connected as equivalent for track C/D. - check that the encoder cables are routed in compliance with EMC. - check the adjustment of the Hall sensor.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A33431 (F, N)	Encoder 3: Deviation, position incremental/absolute too large
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When the zero pulse is passed, a deviation in the incremental position was detected. For equidistant zero marks, the following applies: - The first zero mark passed supplies the reference point for all subsequent checks. The other zero marks must have n times the distance referred to the first zero mark. For distance-coded zero marks, the following applies: - the first zero mark pair supplies the reference point for all subsequent checks. The other zero mark pairs must have the expected distance to the first zero mark pair. Alarm value (r2124, interpret decimal): Deviation in quadrants (1 pulse = 4 quadrants).
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - Clean coding disk or remove strong magnetic fields.
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A33432 (F, N)	Encoder 3: Rotor position adaptation corrects deviation
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	For track A/B, pulses have been lost or too many have been counted. These pulses are presently being corrected. Alarm value (r2124, interpret decimal): Last measured deviation of the zero mark distance in increments (4 increments = 1 encoder pulse). The sign designates the direction of motion when detecting the zero mark distance.
Remedy:	- check that the encoder cables are routed in compliance with EMC. - check the plug connections - replace the encoder or encoder cable - check encoder limit frequency. - adapt the parameter for the distance between zero marks (p0424, p0425).
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A33442 (F, N)	Encoder 3: Battery voltage pre-alarm
Message value:	-
Drive object:	ENCODER, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	See also: p0491 (Motor encoder fault response ENCODER)
Remedy:	- replace battery.
Reaction upon F:	NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A33443 (F, N) Encoder 3: Unipolar CD signal level outside specification

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The unipolar level (CP/CN or DP/DN) for encoder 3 is outside the permissible tolerance.
Alarm value (r2124, interpret binary):
Bit 0 = 1: Either CP or CN outside the tolerance.
Bit 16 = 1: Either DP or DN outside the tolerance.
The unipolar nominal signal level of the encoder must lie in the range 2500 mV +/- 500 mV.
The response thresholds are < 1700 mV and > 3300 mV.
Note:
The signal level is not evaluated unless the following conditions are satisfied:
- Sensor Module properties available (r0459.31 = 1).
- Monitoring active (p0437.31 = 1).
Remedy:
- check that the encoder cables and shielding are routed in compliance with EMC.
- check the plug connections and contacts of the encoder cable.
- are the C/D tracks connected correctly (have the signal lines CP and CN or DP and DN been interchanged)?
- replace the encoder cable.
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A33460 (N) Encoder 3: Analog sensor channel A failed

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The input voltage of the analog sensor is outside the permissible limits.
Alarm value (r2124, interpret decimal):
1: Input voltage outside detectable measuring range.
2: Input voltage outside measuring range set in p4673.
3: The absolute value of the input voltage has exceeded the range limit (p4676).
Remedy:
Re alarm value = 1:
- check the output voltage of the analog sensor.
Re alarm value = 2:
- check the voltage setting for each encoder period (p4673).
Re alarm value = 3:
- check the range limit setting and increase it if necessary (p4676).
Reaction upon N: NONE
Acknowl. upon N: NONE

A33461 (N) Encoder 3: Analog sensor channel B failed

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The input voltage of the analog sensor is outside the permissible limits.
Alarm value (r2124, interpret decimal):
1: Input voltage outside detectable measuring range.
2: Input voltage outside measuring range set in p4675.
3: The absolute value of the input voltage has exceeded the range limit (p4676).

Remedy:
 Re alarm value = 1:
 - check the output voltage of the analog sensor.
 Re alarm value = 2:
 - check the voltage setting for each encoder period (p4675).
 Re alarm value = 3:
 - check the range limit setting and increase it if necessary (p4676).

Reaction upon N: NONE

Acknowl. upon N: NONE

A33462 (N) Encoder 3: Analog sensor, no channel active

Message value: %1

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: Channel A and B are not activated for the analog sensor.

Remedy:
 - activate channel A and/or channel B (p4670).
 - check the encoder configuration (p0404.17).
 See also: p4670 (Analog sensor configuration)

Reaction upon N: NONE

Acknowl. upon N: NONE

A33470 (F, N) Encoder 3: Soiling detected

Message value: -

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: In the case of the alternative encoder system interface on the Sensor Module Cabinet 30 (SMC30), encoder soiling is signaled via a 0 signal at terminal X521.7.

Remedy:
 - check the plug connections
 - replace the encoder or encoder cable

Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

F33500 (N, A) Encoder 3: Position tracking traversing range exceeded

Message value: -

Drive object: SERVO, VECTOR

Reaction: OFF1 (NONE, OFF2, OFF3)

Acknowledge: IMMEDIATELY

Cause: For a configured linear axis without modulo correction, the drive/encoder has exceeded the maximum possible traversing range. The value should be read in p0412 and interpreted as the number of motor revolutions.
 For p0411.0 = 1, the maximum traversing range for the configured linear axis is defined to be 64x (+/- 32x) of p0421.
 For p0411.3 = 1, the maximum traversing range for the configured linear axis is preset (default value) to the highest possible value and is +/-p0412/2 (rounded off to complete revolutions). The highest possible value depends on the pulse number (p0408) and the fine resolution (p0419).

Remedy: The fault should be resolved as follows:
 - select encoder commissioning (p0010 = 4).
 - reset the position tracking as follows (p0411.2 = 1).
 - deselect encoder commissioning (p0010 = 0).
 The fault should then be acknowledged and the absolute encoder adjusted.

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F33501 (N, A)	Encoder 3: Position tracking encoder position outside tolerance window
Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	When powered down, the drive/encoder was moved through a distance greater than was parameterized in the tolerance window. It is possible that there is no longer any reference between the mechanical system and encoder. Fault value (r0949, decimal): Deviation (difference) to the last encoder position in increments of the absolute value. The sign designates the traversing direction. Note: The deviation (difference) found is also displayed in r0477. See also: p0413 (Measuring gear, position tracking tolerance window), r0477 (Measuring gear, position difference)
Remedy:	Reset the position tracking as follows: - select encoder commissioning (p0010 = 4). - reset the position tracking as follows (p0411.2 = 1). - deselect encoder commissioning (p0010 = 0). The fault should then be acknowledged and, if necessary, the absolute encoder adjusted (p2507). See also: p0010, p2507
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33502 (N, A)	Encoder 3: Encoder with measuring gear, without valid signals
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The encoder with measuring gear no longer provides any valid signals.
Remedy:	It must be ensured that all of the encoders, with mounted measuring gear, provide valid actual values in operation.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33503 (N, A)	Encoder 3: Position tracking cannot be reset
Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The position tracking for the measuring gear cannot be reset.
Remedy:	The fault should be resolved as follows: - select encoder commissioning (p0010 = 4). - reset the position tracking as follows (p0411.2 = 1). - deselect encoder commissioning (p0010 = 0). The fault should then be acknowledged and the absolute encoder adjusted.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A33700 Encoder 3: Effectivity test does not supply the expected value

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The DRIVE-CLiQ encoder fault word supplies fault bits that have been set.
 Fault value (r0949, interpret binary):
 Bit x = 1: Effectivity test x unsuccessful.

Remedy:

N33800 (F) Encoder 3: Group signal

Message value: -
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: NONE
Cause: The motor encoder has detected at least one fault.
Remedy: Evaluates other actual messages.
 Reaction upon F: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
 Acknowl. upon F: IMMEDIATELY

F33801 (N, A) Encoder 3 DRIVE-CLiQ: Sign-of-life missing

Message value: Component number: %1, fault cause: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 0A hex:
 The sign-of-life bit in the receive telegram is not set.
Remedy: - check the electrical cabinet design and cable routing for EMC compliance
 - replace the component involved.
 See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F33802 (N, A) Encoder 3: Time slice overflow

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: Time slice overflow, encoder 3.
 Fault value (r0949, interpret decimal):
 9: Time slice overflow of the fast (current controller clock cycle) time slice.
 10: Time slice overflow of the average time slice.
 12: Time slice overflow of the slow time slice.
 999: Timeout when waiting for SYNO, e.g. unexpected return to non-cyclic operation.
Remedy: Reduce the current controller frequency.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F33804 (N, A) Encoder 3: Checksum error

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A checksum error has occurred when reading-out the program memory on the Sensor Module.
Fault value (r0949, interpret hexadecimal):
yyyyxxxx hex
yyyy: Memory area involved.
xxxx: Difference between the checksum at POWER ON and the actual checksum.
Remedy:
- check whether the permissible ambient temperature for the component is maintained.
- replace the Sensor Module.

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F33805 (N, A) Encoder 3: EPROM checksum error

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: Internal parameter data is corrupted.
Fault value (r0949, interpret hexadecimal):
01: EEPROM access error.
02: Too many blocks in the EEPROM.
Remedy: Replace the module.

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F33806 (N, A) Encoder 3: Initialization error

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: PULSE INHIBIT
Cause: The encoder was not successfully initialized.
Fault value (r0949, interpret hexadecimal):
Bit 0,1: Encoder initialization with the motor rotating has failed (deviation involving coarse and fine position in encoder pulse/4)
Bit 2: Mid-voltage matching for track A unsuccessful.
Bit 3: Mid-voltage matching for track B unsuccessful.
Bit 4: Mid-voltage matching for acceleration input (under development) unsuccessful.
Bit 5: Mid-voltage matching for track safety A unsuccessful.
Bit 6: Mid-voltage matching for track safety B unsuccessful.
Bit 7: Mid-voltage matching for track C unsuccessful.
Bit 8: Mid-voltage matching for track D unsuccessful.
Bit 9: Mid-voltage matching for track R unsuccessful.
Bit 10: The difference in mid-voltages between A and B is too great (> 0.5 V)
Bit 11: The difference in mid-voltages between C and D is too great (> 0.5 V)
Bit 12: The difference in mid-voltages between safety A and safety B is too great (> 0.5 V)
Bit 13: The difference in mid-voltages between A and safety B is too great (> 0.5 V)
Bit 14: The difference in mid-voltages between B and safety A is too great (> 0.5 V)
Bit 15: The standard deviation of the calculated mid-voltages is too great (> 0.3 V)

	Bit 16: Internal fault - fault reading a register (CAFE)
	Bit 17: Internal fault - fault writing a register (CAFE)
	Bit 18: Internal fault: No mid-voltage matching available
	Bit 19: Internal error - ADC access error.
	Bit 20: Internal error - no zero crossover found.
	Bit 0,1 : Up to 6SL3055-0AA00-5*A0
	Bits 2..20 : 6SL3055-0AA00-5*A1 and higher
Remedy:	Acknowledge the fault. If the fault cannot be acknowledged: Bits 2 - 9: Check encoder power supply, Bits 2 -14: Check the corresponding cable, Bit 15 with no other bits: Check track R, check settings in p404.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33811 (N, A) Encoder 3: Encoder serial number changed

Message value:	-
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	The encoder serial number has changed. The change is only checked for encoders with serial number (e.g. EnDat encoders). Cause: The encoder was replaced. Note: With closed-loop position control, the serial number is accepted when starting the adjustment (p2507 = 2). When the encoder is adjusted (p2507 = 3), the serial number is checked for changes and if required, the adjustment is reset (p2507 = 1). Proceed as follows to hide serial number monitoring: - set the following serial numbers for the corresponding Encoder Data Set: p0441= FF, p0442 = 0, p0442 = 0, p0444 = 0, p0445 = 0.
Remedy:	Mechanically adjust the encoder. Accept the new serial number with p0440 = 1.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33812 (N, A) Encoder 3: Requested cycle or RX-/TX timing not supported

Message value:	%1
Drive object:	SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A cycle requested from the Control Unit or RX/TX timing is not supported. Alarm value (r2124, interpret decimal): 0: Application cycle is not supported. 1: DQ cycle is not supported. 2: Distance between RX and TX instants in time too low. 3: TX instant in time too early.
Remedy:	
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F33813 Encoder 3: Hardware logic unit failed

Message value: Fault cause: %1 bin
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: PULSE INHIBIT
Cause: The DRIVE-CLiQ encoder fault word supplies fault bits that have been set.
Fault value (r0949, interpret binary):
Bit 0: ALU watchdog has responded.
Bit 1: ALU has detected a sign-of-life error.
Remedy: Replace encoder

F33820 (N, A) Encoder 3 DRIVE-CLiQ: Telegram error

Message value: Component number: %1, fault cause: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 01 hex:
CRC error.
xx = 02 hex:
Telegram is shorter than specified in the length byte or in the receive list.
xx = 03 hex:
Telegram is longer than specified in the length byte or in the receive list.
xx = 04 hex:
The length of the receive telegram does not match the receive list.
xx = 05 hex:
The type of the receive telegram does not match the receive list.
xx = 06 hex:
The address of the component in the telegram and in the receive list do not match.
xx = 07 hex:
A SYNC telegram is expected - but the received telegram is not a SYNC telegram.
xx = 08 hex:
No SYNC telegram is expected - but the received telegram is one.
xx = 09 hex:
The error bit in the receive telegram is set.
xx = 10 hex:
The receive telegram is too early.
Remedy:
- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F33835 (N, A)	Encoder 3 DRIVE-CLiQ: Cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved. The nodes do not send and receive in synchronism. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 21 hex: The cyclic telegram has not been received. xx = 22 hex: Timeout in the telegram receive list. xx = 40 hex: Timeout in the telegram send list.
Remedy:	- carry out a POWER ON. - replace the component involved. See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F33836 (N, A)	Encoder 3 DRIVE-CLiQ: Send error for DRIVE-CLiQ data
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved. Data were not able to be sent. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 41 hex: Telegram type does not match send list.
Remedy:	Carry out a POWER ON.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F33837 (N, A)	Encoder 3 DRIVE-CLiQ: Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component concerned. Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error.

Remedy:

- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
- check the electrical cabinet design and cable routing for EMC compliance
- if required, use another DRIVE-CLiQ socket (p9904).
- replace the component involved.

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F33845 (N, A) Encoder 3 DRIVE-CLiQ: Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Control Unit to the encoder involved.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 0B hex:
Synchronization error during alternating cyclic data transfer.

Remedy: Carry out a POWER ON.
See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F33850 (N, A) Encoder 3: Encoder evaluation, internal software error

Message value: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: POWER ON
Cause: An internal software error has occurred in the Sensor Module of encoder 3.
Fault value (r0949, interpret decimal):
1: Background time slice is blocked.
2: Checksum over the code memory is not OK.
10000: OEM memory of the EnDat encoder contains data that cannot be interpreted.
11000 ... 11499: Descriptive data from EEPROM incorrect.
11500 ... 11899: Calibration data from EEPROM incorrect.
11900 ... 11999: Configuration data from EEPROM incorrect.
16000: DRIVE-CLiQ encoder initialization application error.
16001: DRIVE-CLiQ encoder initialization ALU error.
16002: DRIVE-CLiQ encoder HISI / SISI initialization error.
16003: DRIVE-CLiQ encoder safety initialization error.
16004: DRIVE-CLiQ encoder internal system error.

Remedy:

- replace the Sensor Module.
- if required, upgrade the firmware in the Sensor Module.
- contact the Hotline.

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F33851 (N, A)	Encoder 3 DRIVE-CLiQ (CU): Sign-of-life missing
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 3) involved to the Control Unit. The DRIVE-CLiQ component did not set the sign-of-life to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0A hex = 10 dec: The sign-of-life bit in the receive telegram is not set.
Remedy:	Upgrade the firmware of the component involved.
Reaction upon N:	NONE
Acknowled. upon N:	NONE
Reaction upon A:	NONE
Acknowled. upon A:	NONE
F33860 (N, A)	Encoder 3 DRIVE-CLiQ (CU): Telegram error
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 3) involved to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 11 hex = 17 dec: CRC error and the receive telegram is too early. xx = 01 hex = 01 dec: Checksum error (CRC error). xx = 12 hex = 18 dec: The telegram is shorter than that specified in the length byte or in the receive list and the receive telegram is too early. xx = 02 hex = 02 dec: Telegram is shorter than specified in the length byte or in the receive list. xx = 13 hex = 19 dec: The telegram is longer than that specified in the length byte or in the receive list and the receive telegram is too early. xx = 03 hex = 03 dec: Telegram is longer than specified in the length byte or in the receive list. xx = 14 hex = 20 dec: The length of the receive telegram does not match the receive list and the receive telegram is too early. xx = 04 hex = 04 dec: The length of the receive telegram does not match the receive list. xx = 15 hex = 21 dec: The type of the receive telegram does not match the receive list and the receive telegram is too early. xx = 05 hex = 05 dec: The type of the receive telegram does not match the receive list. xx = 16 hex = 22 dec: The address of the power unit in the telegram and in the receive list does not match and the receive telegram is too early. xx = 06 hex = 06 dec: The address of the power unit in the telegram and in the receive list do not match. xx = 19 hex = 25 dec: The error bit in the receive telegram is set and the receive telegram is too early. xx = 09 hex = 09 dec: The error bit in the receive telegram is set. xx = 10 hex = 16 dec: The receive telegram is too early.

Remedy:

- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F33885 (N, A) Encoder 3 DRIVE-CLiQ (CU): Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 3) involved to the Control Unit.
 The nodes do not send and receive in synchronism.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 1A hex = 26 dec:
 Sign-of-life bit in the receive telegram not set and the receive telegram is too early.
 xx = 21 hex = 33 dec:
 The cyclic telegram has not been received.
 xx = 22 hex = 34 dec:
 Timeout in the telegram receive list.
 xx = 40 hex = 64 dec:
 Timeout in the telegram send list.
 xx = 62 hex = 98 dec:
 Error at the transition to cyclic operation.

Remedy:

- check the power supply voltage of the component involved.
- carry out a POWER ON.
- replace the component involved.

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F33886 (N, A) Encoder 3 DRIVE-CLiQ (CU): Error when sending DRIVE-CLiQ data

Message value: Component number: %1, fault cause: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 3) involved to the Control Unit.
 Data were not able to be sent.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 41 hex:
 Telegram type does not match send list.

Remedy: Carry out a POWER ON.

Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F33887 (N, A)	Encoder 3 DRIVE-CLiQ (CU): Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component involved (Sensor Module for encoder 3). Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error. xx = 60 hex: Response received too late during runtime measurement. xx = 61 hex: Time taken to exchange characteristic data too long.
Remedy:	- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). - check the electrical cabinet design and cable routing for EMC compliance - if required, use another DRIVE-CLiQ socket (p9904). - replace the component involved.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F33895 (N, A)	Encoder 3 DRIVE-CLiQ (CU): Alternating cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	SERVO, VECTOR
Reaction:	OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Sensor Module (encoder 3) involved to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex: Synchronization error during alternating cyclic data transfer.
Remedy:	Carry out a POWER ON. See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F33896 (N, A)	Encoder 3 DRIVE-CLiQ (CU): Inconsistent component properties
Message value:	Component number: %1
Drive object:	SERVO, VECTOR
Reaction:	OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	The properties of the DRIVE-CLiQ component (Sensor Module for encoder 3), specified by the fault value, have changed in an incompatible fashion with respect to the properties when booted. One cause can be, e.g. that a DRIVE-CLiQ cable or DRIVE-CLiQ component has been replaced. Fault value (r0949, interpret decimal): Component number.

Remedy:

- carry out a POWER ON.
- when a component is replaced, the same component type and if possible the same firmware version should be used.
- when a cable is replaced, only cables whose length is the same as or as close as possible to the length of the original cables should be used (ensure compliance with the maximum cable length).

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

F33899 (N, A) Encoder 3: Unknown fault

Message value: New message: %1
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: A fault occurred on the Sensor Module for encoder 3 that cannot be interpreted by the Control Unit firmware. This can occur if the firmware on this component is more recent than the firmware on the Control Unit.
Fault value (r0949, interpret decimal):
Fault number.
Note:
If required, the significance of this new fault can be read about in a more recent description of the Control Unit.

Remedy:

- replace the firmware on the Sensor Module by an older firmware version (r0148).
- upgrade the firmware on the Control Unit (r0018).

Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

A33902 (F, N) Encoder 3: SPI-BUS error occurred

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Error when operating the internal SPI bus.
Fault value (r0949, interpret hexadecimal):
Only for internal Siemens troubleshooting.

Remedy:

- replace the Sensor Module.
- if required, upgrade the firmware in the Sensor Module.
- contact the Hotline.

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A33903 (F, N) Encoder 3: I2C-BUS error occurred

Message value: %1
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Error when operating the internal I2C bus.
Fault value (r0949, interpret hexadecimal):
Only for internal Siemens troubleshooting.

Remedy:

- replace the Sensor Module.
- if required, upgrade the firmware in the Sensor Module.
- contact the Hotline.

Reaction upon F: NONE (ENCODER, IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)

Acknowl. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowl. upon N: NONE

F33905 (N, A) Encoder 3: Parameterization error

Message value: Parameter: %1, supplementary information: %2

Drive object: SERVO, VECTOR

Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)

Acknowledge: IMMEDIATELY

Cause:

A parameter of encoder 3 was detected as being incorrect.
 It is possible that the parameterized encoder type does not match the connected encoder.
 The parameter involved can be determined as follows:

- determine the parameter number using the fault value (r0949).
- determine the parameter index (p0187).

Fault value (r0949, interpret decimal):
 yyyyxxxx dec: yyyy = supplementary information, xxxx = parameter

Supplementary information = 0:
 No information available.

Supplementary information = 1:
 The component does not support HTL level (p0405.1 = 0) combined with track monitoring A/B <> -A/B (p0405.2 = 1).

Supplementary information = 2:
 A code number for an identified encoder has been entered into p0400, however, no identification was carried out.
 Please start a new encoder identification.

Supplementary information = 3:
 A code number for an identified encoder has been entered into p0400, however, no identification was carried out.
 Please select a listed encoder in p0400 with a code number < 10000.

Supplementary information = 4:
 This component does not support SSI encoders (p0404.9 = 1) without track A/B.

Supplementary information = 5:
 For SQW encoder, value in p4686 greater than in p0425.

Supplementary information = 6:
 DRIVE-CLiQ encoder cannot be used with this firmware version.

Supplementary information = 7:
 For the SQW encoder, the Xact1 correction (p0437.2) is only permitted with equidistant zero marks.

Remedy:

- check whether the connected encoder type matches the encoder that has been parameterized.
- correct the parameter specified by the fault value (r0949) and p0187.
- re parameter number 314: Check the pole pair number and measuring gear ratio. The quotient of the "pole pair number" divided by the "measuring gear ratio" must be less than or equal to 1000 ((r0313 * p0433) / p0432 <= 1000).

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

A33915 (F, N) Encoder 3: Configuration error

Message value: %1

Drive object: SERVO, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause:

The configuration for encoder 3 is incorrect.

Fault value (r0949, interpret decimal):

- 1: Re-parameterization between fault/alarm is not permissible.
- 419: When the fine resolution Gx_ACT2 is configured, the encoder identifies a maximum possible absolute position actual value (r0483) that can no longer be represented within 32 bits.

Remedy: 1: No re-parameterization between fault/alarm.
419: Reduce the fine resolution (p0419).
Reaction upon F: NONE (IASC/DCBRAKE)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

F33916 (N, A) Encoder 3: Parameterization fault

Message value: Parameter: %1, supplementary information: %2
Drive object: SERVO, VECTOR
Reaction: OFF1 (IASC/DCBRAKE, NONE, OFF2, OFF3, STOP1, STOP2)
Acknowledge: IMMEDIATELY
Cause: A parameter of encoder 3 was detected as being incorrect.
It is possible that the parameterized encoder type does not match the connected encoder.
The parameter involved can be determined as follows:
- determine the parameter number using the fault value (r0949).
- determine the parameter index (p0187).
Fault value (r0949, interpret decimal):
Parameter number.
Note:
This fault is only output for encoders where r0404.10 = 1 or r0404.11 = 1. It corresponds to A33905 with encoders where r0404.10 = 0 and r0404.11 = 0.
Remedy: - check whether the connected encoder type matches the encoder that has been parameterized.
- correct the parameter specified by the fault value (r0949) and p0187.
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

A33920 (F, N) Encoder 3: Temperature sensor fault

Message value: Fault cause: %1, channel number: %2
Drive object: SERVO, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When evaluating the temperature sensor, an error occurred.
Alarm value (r2124, interpret decimal):
Low word low byte: Cause:
1: Wire breakage or sensor not connected (KTY: R > 1630 Ohm).
2: Measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm).
Additional values:
Only for internal Siemens troubleshooting.
Low word high byte: Channel number.
Remedy: - check that the encoder cable is the correct type and is correctly connected.
- check the temperature sensor selection in p0600 to p0603.
- replace the Sensor Module (hardware defect or incorrect calibration data).
Reaction upon F: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY
Reaction upon N: NONE
Acknowl. upon N: NONE

A33999 (F, N)	Encoder 3: Unknown alarm
Message value:	New message: %1
Drive object:	SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A alarm has occurred on the Sensor Module for encoder 3 that cannot be interpreted by the Control Unit firmware. This can occur if the firmware on this component is more recent than the firmware on the Control Unit. Alarm value (r2124, interpret decimal): Alarm number. Note: If required, the significance of this new alarm can be read about in a more recent description of the Control Unit.
Remedy:	- replace the firmware on the Sensor Module by an older firmware version (r0148). - upgrade the firmware on the Control Unit (r0018).
Reaction upon F:	NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
F34207 (N, A)	VSM: Temperature fault threshold exceeded
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Vector: NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The temperature (r3666) measured using the Voltage Sensing Module (VSM) has exceeded the threshold value (p3668). Note: This fault can only be initiated if the temperature evaluation was activated (p3665 = 2 for a KTY sensor or p3665 = 1 for a PTC sensor). Fault value (r0949, interpret decimal): yyxxxx dec: yy: Component number of the component which detected the fault.
Remedy:	- check the fan. - reduce the power.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A34211 (F, N)	VSM: Temperature alarm threshold exceeded
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature (r3666) measured using the Voltage Sensing Module (VSM) has exceeded the threshold value (p3667). Alarm value (r2124, interpret decimal): The hundred-thousands and ten-thousands position specifies the component number of the VSM which detected the fault.
Remedy:	- check the fan. - reduce the power.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Vector: NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE

N34800 (F) VSM: Group signal

Message value: -
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge: NONE
Cause: The Voltage Sensing Module (VSM) has detected at least one fault.
Remedy: Evaluates other actual messages.
Reaction upon F: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY

F34801 VSM DRIVE-CLiQ: Sign-of-life missing

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, S_INF
Reaction: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the Voltage Sensing Module (VSM).
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 0A hex:
The sign-of-life bit in the receive telegram is not set.
Remedy: - check the DRIVE-CLiQ connection.
- replace the Voltage Sensing Module (VSM).

F34801 VSM DRIVE-CLiQ: Sign-of-life missing

Message value: Component number: %1, fault cause: %2
Drive object: VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the Voltage Sensing Module (VSM).
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 0A hex:
The sign-of-life bit in the receive telegram is not set.
Remedy: - check the DRIVE-CLiQ connection.
- replace the Terminal Module.

F34802 VSM: Time slice overflow

Message value: -
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: Time slice overflow on the Voltage Sensing Module.
Remedy: Replace the Voltage Sensing Module.

F34803 VSM: Memory test

Message value: -
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: An error has occurred during the memory test on the Voltage Sensing Module.
Remedy: - check whether the permissible ambient temperature for the Voltage Sensing Module is being maintained.
- replace the Voltage Sensing Module.

F34804 VSM: CRC

Message value: %1
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: A checksum error has occurred when reading-out the program memory on the Voltage Sensing Module (VSM).
Remedy: - check whether the permissible ambient temperature for the component is maintained.
- replace the Voltage Sensing Module.

F34805 VSM: EPROM checksum error

Message value: %1
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: Internal parameter data is corrupted.
Fault value (r0949, interpret hexadecimal):
01: EEPROM access error.
02: Too many blocks in the EEPROM.
Remedy: - check whether the permissible ambient temperature for the component is maintained.
- replace the Voltage Sensing Module (VSM).

F34806 VSM: Initialization

Message value: -
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowledge: IMMEDIATELY
Cause: For the Voltage Sensing Module (VSM), a fault has occurred while initializing.
Remedy: Replace the Voltage Sensing Module.

A34807 (F, N) VSM: Sequence control time monitoring

Message value: -
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Error, timeout in the sequence control on the Voltage Sensing Module (VSM).
Remedy: Replace the Voltage Sensing Module.
Reaction upon F: NONE
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

F34820 VSM DRIVE-CLiQ: Telegram error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the Voltage Sensing Module.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 01 hex:
CRC error.
xx = 02 hex:
Telegram is shorter than specified in the length byte or in the receive list.
xx = 03 hex:
Telegram is longer than specified in the length byte or in the receive list.
xx = 04 hex:
The length of the receive telegram does not match the receive list.
xx = 05 hex:
The type of the receive telegram does not match the receive list.
xx = 06 hex:
The address of the component in the telegram and in the receive list do not match.
xx = 07 hex:
A SYNC telegram is expected - but the received telegram is not a SYNC telegram.
xx = 08 hex:
No SYNC telegram is expected - but the received telegram is one.
xx = 09 hex:
The error bit in the receive telegram is set.
xx = 10 hex:
The receive telegram is too early.
Remedy: - carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F34835 VSM DRIVE-CLiQ: Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
Vector: NONE (OFF1, OFF2)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the Voltage Sensing Module. The nodes do not send and receive in synchronism.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 21 hex:
The cyclic telegram has not been received.
xx = 22 hex:
Timeout in the telegram receive list.
xx = 40 hex:
Timeout in the telegram send list.
Remedy: - carry out a POWER ON.
- replace the component involved.

F34836	VSM DRIVE-CLiQ: Send error for DRIVE-CLiQ data
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Vector: NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communications error has occurred from the Control Unit to the Voltage Sensing Module. Data were not able to be sent. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 41 hex: Telegram type does not match send list.
Remedy:	Carry out a POWER ON.
F34837	VSM DRIVE-CLiQ: Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Vector: NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component concerned. Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error.
Remedy:	- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). - check the electrical cabinet design and cable routing for EMC compliance - if required, use another DRIVE-CLiQ socket (p9904). - replace the component involved.
F34845	VSM DRIVE-CLiQ: Cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Vector: NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communications error has occurred from the Control Unit to the Voltage Sensing Module (VSM). Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex: Synchronization error during alternating cyclic data transfer.
Remedy:	Carry out a POWER ON. See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F34850	VSM: Internal software error
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Vector: OFF1 (NONE, OFF2, OFF3)
Acknowledge:	POWER ON
Cause:	An internal software error in the Voltage Sensing Module (VSM) has occurred. Fault value (r0949, interpret decimal): 1: Background time slice is blocked. 2: Checksum over the code memory is not OK.
Remedy:	- replace the Voltage Sensing Module (VSM). - if required, upgrade the firmware in the Voltage Sensing Module. - contact the Hotline.

F34851	VSM DRIVE-CLiQ (CU): Sign-of-life missing
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: NONE (OFF1, OFF2) Vector: NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communications error has occurred from the Voltage Sensing Module (VSM) to the Control Unit. The DRIVE-CLiQ component did not set the sign-of-life to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0A hex = 10 dec: The sign-of-life bit in the receive telegram is not set.
Remedy:	Upgrade the firmware of the component involved.

F34860	VSM DRIVE-CLiQ (CU): Telegram error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: NONE (OFF1, OFF2) Vector: NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communications error has occurred from the Voltage Sensing Module (VSM) to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 11 hex = 17 dec: CRC error and the receive telegram is too early. xx = 01 hex = 01 dec: Checksum error (CRC error). xx = 12 hex = 18 dec: The telegram is shorter than that specified in the length byte or in the receive list and the receive telegram is too early. xx = 02 hex = 02 dec: Telegram is shorter than specified in the length byte or in the receive list. xx = 13 hex = 19 dec: The telegram is longer than that specified in the length byte or in the receive list and the receive telegram is too early. xx = 03 hex = 03 dec: Telegram is longer than specified in the length byte or in the receive list. xx = 14 hex = 20 dec: The length of the receive telegram does not match the receive list and the receive telegram is too early. xx = 04 hex = 04 dec: The length of the receive telegram does not match the receive list. xx = 15 hex = 21 dec: The type of the receive telegram does not match the receive list and the receive telegram is too early. xx = 05 hex = 05 dec: The type of the receive telegram does not match the receive list.

xx = 16 hex = 22 dec:
The address of the power unit in the telegram and in the receive list does not match and the receive telegram is too early.
xx = 06 hex = 06 dec:
The address of the power unit in the telegram and in the receive list do not match.
xx = 19 hex = 25 dec:
The error bit in the receive telegram is set and the receive telegram is too early.
xx = 09 hex = 09 dec:
The error bit in the receive telegram is set.
xx = 10 hex = 16 dec:
The receive telegram is too early.

Remedy:

- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F34885 VSM DRIVE-CLiQ (CU): Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
 Servo: NONE (OFF1, OFF2)
 Vector: NONE (OFF1, OFF2)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the Voltage Sensing Module (VSM) to the Control Unit.
 The nodes do not send and receive in synchronism.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 1A hex = 26 dec:
 Sign-of-life bit in the receive telegram not set and the receive telegram is too early.
 xx = 21 hex = 33 dec:
 The cyclic telegram has not been received.
 xx = 22 hex = 34 dec:
 Timeout in the telegram receive list.
 xx = 40 hex = 64 dec:
 Timeout in the telegram send list.
 xx = 62 hex = 98 dec:
 Error at the transition to cyclic operation.

Remedy:

- check the power supply voltage of the component involved.
- carry out a POWER ON.
- replace the component involved.

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F34886 VSM DRIVE-CLiQ (CU): Error when sending DRIVE-CLiQ data

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: Infeed: OFF2 (NONE, OFF1)
 Servo: NONE (OFF1, OFF2)
 Vector: NONE (OFF1, OFF2)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the Voltage Sensing Module (VSM) to the Control Unit.
 Data were not able to be sent.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 41 hex:
 Telegram type does not match send list.

Remedy:

- Carry out a POWER ON.

F34887	VSM DRIVE-CLiQ (CU): Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: NONE (OFF1, OFF2) Vector: NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component (Voltage Sensing Module) involved. Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error. xx = 60 hex: Response received too late during runtime measurement. xx = 61 hex: Time taken to exchange characteristic data too long.
Remedy:	- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). - check the electrical cabinet design and cable routing for EMC compliance - if required, use another DRIVE-CLiQ socket (p9904). - replace the component involved.

F34895	VSM DRIVE-CLiQ (CU): Alternating cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: NONE (OFF1, OFF2) Vector: NONE (OFF1, OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communications error has occurred from the Voltage Sensing Module (VSM) to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex: Synchronization error during alternating cyclic data transfer.
Remedy:	Carry out a POWER ON. See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F34896	VSM DRIVE-CLiQ (CU): Inconsistent component properties
Message value:	Component number: %1
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2) Vector: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	The properties of the DRIVE-CLiQ component (Voltage Sensing Module), specified by the fault value, have changed in an incompatible fashion with respect to the properties when booted. One cause can be, e.g. that a DRIVE-CLiQ cable or DRIVE-CLiQ component has been replaced. Fault value (r0949, interpret decimal): Component number.

Remedy:

- carry out a POWER ON.
- when a component is replaced, the same component type and if possible the same firmware version should be used.
- when a cable is replaced, only cables whose length is the same as or as close as possible to the length of the original cables should be used (ensure compliance with the maximum cable length).

F34897 VSM DRIVE-CLiQ (CU): no communication with component

Message value: Preliminary component number: %1

Drive object: VECTOR

Reaction: OFF2 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)

Acknowledge: IMMEDIATELY (POWER ON)

Cause: Communication with the DRIVE-CLiQ component (Voltage Sensing Module) specified by the fault value is not possible.
One cause may, for example, be that a DRIVE-CLiQ cable has been withdrawn.
Fault value (r0949, interpret decimal):
Component ID.

Remedy:

- check the DRIVE-CLiQ connections.
- carry out a POWER ON.

F34899 (N, A) VSM: Unknown fault

Message value: New message: %1

Drive object: A_INF, B_INF, S_INF, VECTOR

Reaction: Infeed: NONE (OFF1, OFF2)
Vector: NONE (OFF1, OFF2, OFF3)

Acknowledge: IMMEDIATELY (POWER ON)

Cause: A fault occurred on the Voltage Sensing Module that cannot be interpreted by the Control Unit firmware. This can occur if the firmware on this component is more recent than the firmware on the Control Unit.
Fault value (r0949, interpret decimal):
Fault number.
Note:
If required, the significance of this new fault can be read about in a more recent description of the Control Unit.

Remedy:

- replace the firmware on the Voltage Sensing Module by an older firmware version (r0158).
- upgrade the firmware on the Control Unit (r0018).

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

A34903 (F, N) VSM: I2C bus error occurred

Message value: -

Drive object: A_INF, B_INF, S_INF

Reaction: NONE

Acknowledge: NONE

Cause: An error has occurred when accessing the module-internal I2C bus.

Remedy: Replace Voltage Sensing Module (VSM).

Reaction upon F: Infeed: NONE (OFF1, OFF2)
Vector: NONE

Acknowl. upon F: IMMEDIATELY (POWER ON)

Reaction upon N: NONE

Acknowl. upon N: NONE

A34903 (F, N) VSM: I2C bus error occurred

Message value: -

Drive object: VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: An error has occurred when accessing the module-internal I2C bus.

Remedy: Replace the Terminal Module.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Vector: NONE
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

A34904 (F, N) VSM: EEPROM

Message value: -
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An error has occurred accessing the non-volatile memory on the Terminal Module.
Remedy: Replace Voltage Sensing Module (VSM).
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Vector: NONE
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

A34905 (F, N) VSM: Parameter access

Message value: -
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The Control Unit attempted to write an illegal parameter value to the Voltage Sensing Module (VSM).
Remedy: - check whether the firmware version of the VSM (r0158) matches the firmware version of Control Unit (r0018).
- if required, replace the Voltage Sensing Module.
Note:
The firmware versions that match each other are in the readme.txt file on the memory card.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Vector: NONE
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

A34920 (F, N) VSM: Temperature sensor fault

Message value: %1
Drive object: A_INF, B_INF, S_INF, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When evaluating the temperature sensor, an error occurred.
Alarm value (r2124, interpret decimal):
1: Wire breakage or sensor not connected (KTY: R > 1630 Ohm).
2: Measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm).
Remedy: - make sure that the sensor is connected correctly.
- replace the sensor.
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Vector: NONE
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

A34999 (F, N)	VSM: Unknown alarm
Message value:	New message: %1
Drive object:	A_INF, B_INF, S_INF, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A fault occurred on the Voltage Sensing Module (VSM) an alarm has occurred that cannot be interpreted by the Control Unit firmware. This can occur if the firmware on this component is more recent than the firmware on the Control Unit. Alarm value (r2124, interpret decimal): Alarm number. Note: If required, the significance of this new alarm can be read about in a more recent description of the Control Unit.
Remedy:	- replace the firmware on the Voltage Sensing Module by an older firmware version (r0148). - upgrade the firmware on the Control Unit (r0018).
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
F35000	TM54F: Sampling time invalid
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	POWER ON
Cause:	The set sampling time is invalid. - not a multiple integer of the DP clock cycle. Fault value (r0949, floating point): Recommended valid sampling time.
Remedy:	Adapt the sampling time (e.g. set the recommended valid sampling time). See also: p10000 (SI sampling time)
F35001	TM54F: Parameter value invalid
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The entered value is invalid. Fault value (r0949, interpret decimal): Parameter number with the invalid value.
Remedy:	Correct the parameter value.
F35002	TM54F: Commissioning not possible
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The commissioning mode setting was rejected because for at least one drive belonging to the TM54F, the pulses had not been suppressed. Fault value (r0949, interpret decimal): Drive object number of the first drive found without pulse suppression.
Remedy:	Cancel the pulses for the drive specified in the fault value.

F35003	TM54F: Acknowledgement on the Control Unit is required
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	A fault on the Terminal Module 54F (TM54) was acknowledged using the safe acknowledgement (P10006). An additional acknowledgement is also required at the Control Unit.
Remedy:	Acknowledge the fault at the Control Unit.

F35011	TM54F: Drive object number assignment illegal
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	A drive object number was assigned twice. Each drive object number can be assigned only once.
Remedy:	Correct the assignment of the drive object numbers. See also: p10010 (SI drive object assignment)

A35012	TM54F: Test stop active
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The test stop for the Terminal Module 54F (TM54F) is presently being executed. F35013 is output when a error occurs during the test stop.
Remedy:	The alarm disappears automatically after successfully ending or canceling (when a fault condition occurs) the test stop.

F35013	TM54F: Test stop error
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	An error was detected when carrying out the test stop on the TM54F. As fault response fail-safe input terminal signals are transferred to the motion monitoring functions. Fault value 0xaaaabbcc hex (r0949, interpret hexadecimal): aaaa: Specifies the DOs or F-DIs (dependent on test step cc) for which the expected state was not assumed. The number is bit-coded (bit 0 = F-DI 0 or F-DO 0; bit 3 = F-DI 3 or F-DO 3). bb: Describes the cause of the fault 0x01 = Internal error (error state on the opposite side). 0x02 = Fault during comparison of switching signals: FDIs or DIs 0x03 = Internal error (delay time in the new state has still not expired). 0x04 = Fault during comparison of switching signals: DiagDOs cc: Describes the test step of the test stop in which the fault occurred

The information below is displayed in the following format:

Test stop step slave : (test actions)(test actions) | Step Master : (test actions)(test actions)

| Description

List of test stop steps:

0x00:(L1+OFF)(L2+ON) | 0x0A:(=)(=) | Synchronization / Switching step

0x0A:(L1+OFF)(L2+ON) | 0x15:(=)(=) | Wait step

0x15:(L1+OFF)(L2+OFF) | 0x20:(=) (=) | FDIs 0..4 Test for 0 V / Switching step

0x20:(L1+OFF)(L2+OFF) | 0x2B:(=) (=) | Wait step

0x2B:(L1+ON)(L2+ON) | 0x36:(=) (=) | FDIs 5..9 Test for 0 V / Switching step

0x36:(DO OFF)() | 0x41:(DO OFF)() | Wait step / Switching step

0x41:(DO OFF)() | 0x4C:(DO OFF)() | Wait step

0x4C:(DO ON)() | 0x57:(DO ON)() | Test DiagDOs or DiagDI / Switching step

0x57:(DO ON)() | 0x62:(DO ON)() | Wait step

0x62:(DO OFF)() | 0x6D:(DO ON)() | Test DiagDOs or DiagDI / Switching step

0x6D:(DO OFF)() | 0x78:(DO ON)() | Wait step

0x78:(DO ON)() | 0x83:(DO OFF)() | Test DiagDOs or DiagDI / Switching step

0x83:(DO ON)() | 0x8E:(DO OFF)() | Wait step

0x8E:(DO OFF)() | 0x99:(DO OFF)() | Test DiagDOs or DiagDI / Switching step

0x99:(DO OFF)() | 0xA4:(DO OFF)() | Wait step

0xA4:(DO OFF)() | 0xAF:(DO OFF)() | Test DiagDOs or DiagDI

0xAF:(DO original state)() | 0xC5:(DO original state)() | Switching step

0xC5: End of test

The following expected states are tested in the test steps when testing the FDOs:

The information below is displayed in the following format:

Test step (SL MA): Expected DiagDO mode 1 | Expected DI20..23 mode 2 | Expected DI20..23 mode 3

(0x4C 0x57): DiagDO=0V | DI=24V | DI=24V

(0x62 0x6D): DiagDO=0V | DI=0V | DI=0V

(0x78 0x83): DiagDO=0V | DI=0V | DI=24V

(0x8E 0x99): DiagDO=24V | DI=0V | DI=24V

(0xA4 0xAF): DiagDO=0V | DI=24V | DI=24V

Example:

If an error with fault causes (bb) 0x02 or 0x04 occurs in a test stop step, the test action for the fault took place in the previous test stop step. The expected states are tested in the next step.

Master signals fault value 0001_04AF and slave signals fault value 0001_04A4.

aaaa = 1 i.e. FDO 0 affected

bb = 04h i.e. testing of DiagFDO has failed

cc = The expected states were tested in test stop step AF on the master and A4 on the slave. Expected state DiagDO=0V is being tested in the table; in other words, the Diag DO was at 0 V instead of the expected 24 V. The associated test action took place in the previous step (0x99 DO OFF 0xA4 DO OFF). Both DOs were switched to OFF.

Remedy: Check the wiring of the F-DIs and F-DOs and restart the test stop. The fault is withdrawn if the test stop is successfully completed.

Fault values 0xC0000000 / 0xDDDDDDDD / 0xEEEEEEEE : These fault values are triggered together with alarm 35152. If they appear, you should check all test stop parameters. You should also check whether the firmware version of the TM54F matches the Control Unit's software version.

You also need to check P10001, P10017, P10046, and P10047.

A35014 TM54F: Test stop required

Message value: -

Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause:

- after powering up the drive, a test stop has still not been carried out.
- a new test stop is required after commissioning.
- the time to carry out the forced checking procedure (test stop) has expired (p10003).

Remedy: Initiate test stop (BI: p10007).

A35015 TM54F: Communication with drive not established

Message value: Fault cause: %1 bin
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Cyclic communication of one or several drives with the Terminal Module 54F (TM54F) is not active.
Fault value (r0949, interpret binary):
Bit 0 = 1: No communication with drive 1.
...
Bit 5 = 1: No communication with drive 6.
For fault value = 0, the following applies:
The number of drive objects specified in p10010 is not equal to the number of drives that have drive-based motion monitoring functions that have been enabled.
The drive object number for drive n is set in p10010[n-1].
When this fault is present, none of the drives that have drive-based motion monitoring functions operating with TM54F, are enabled.
Remedy: For all drive objects specified in p10010, check whether the drive-based motion monitoring functions with TM54F are enabled (p9601).

A35016 TM54F: Net data communication with drive not established

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The cyclic net data communication within the Terminal Module 54F (TM54F) is still not active.
This message is output after the TM54F master and TM54F slave have booted and is automatically withdrawn as soon as communications have been established.
If a drive does not communicate with the TM54F, then none of the drives parameterized in p10010 are enabled.
Remedy: When replacing a Motor Module, carry out the following steps:
- start the copy function for the node identifier on the TM54F (p9700 = 1D hex).
- acknowledge the hardware CRC on the TM54F (p9701 = EC hex).
- save all parameters (p0977 = 1).
- carry out a POWER ON (power off/on) for all components.
The following always applies:
- for all drive objects specified in p10010, check whether the drive-based motion monitoring functions with TM54F are enabled (p9601).
- check whether fault F35150 is present and if required, remove the cause of the fault.
See also: r10055 (SI TM54F communication status drive-specific)

F35040 TM54F: 24 V undervoltage

Message value: Fault cause: %1 bin
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: For the 24 V power supply for the Terminal Module 54F (TM54F) an undervoltage condition was detected.
As fault response fail-safe input terminal signals are transferred to the motion monitoring functions.
Fault value (r0949, interpret binary):
Bit 0 = 1: Power supply undervoltage at connection X524.
Bit 1 = 1: Power supply undervoltage at connection X514.
Remedy: - check the 24 V DC power supply for the TM54F.
- carry out safe acknowledgement (p10006).

F35043	TM54F: 24 V overvoltage
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	For the 24 V power supply for the Terminal Module 54F (TM54F) an overvoltage condition was detected. As fault response fail-safe input terminal signals are transferred to the motion monitoring functions.
Remedy:	- check the 24 V DC power supply for the TM54F. - carry out safe acknowledgement (p10006).
F35051	TM54F: Defect in a monitoring channel
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The Terminal Module 54F (TM54F) has identified an error in the data cross check between the two control channels. As fault response fail-safe input terminal signals are transferred to the motion monitoring functions. Fault value (r0949, interpret hexadecimal): aaaabbcc hex aaaa: A value greater than zero indicates an internal software error. bb: Data to be cross-checked that resulted in the error. bb = 00 hex: p10000 bb = 01 hex: p10001 bb = 02 hex: p10002 bb = 03 hex: p10006 bb = 04 hex: p10008 bb = 05 hex: p10010 bb = 06 hex: p10011 bb = 07 hex: p10020 bb = 08 hex: p10021 bb = 09 hex: p10022 bb = 0A hex: p10023 bb = 0B hex: p10024 bb = 0C hex: p10025 bb = 0D hex: p10026 bb = 0E hex: p10027 bb = 0F hex: p10028 bb = 10 hex: p10036 bb = 11 hex: p10037 bb = 12 hex: p10038 bb = 13 hex: p10039 bb = 14 hex: p10040 bb = 15 hex: p10041 bb = 16 hex: p10042 bb = 17 hex: p10043 bb = 18 hex: p10044 bb = 19 hex: p10045 bb = 1A hex: p10046 bb = 1B hex: Test stop internal p10041 bb = 1C hex: Test stop internal p10046 bb = 1D hex: - 1F hex test stop internal p10017, p10002, p10000 bb = 20 hex: - 2A hex test stop internal p10040 bb = 2B hex: Test stop initialization bb = 2C hex: Input/output calculation initialization bb = 2D hex: - 45 hex internal data for output calculation, P10042 - P10045 bb = 46 hex - 63 hex data for SGE calculation of drive group 1 bb = 64 hex - 81 hex data for SGE calculation of drive group 2 bb = 82 hex - 9F hex data for SGE calculation of drive group 3

bb = A0 hex - BD hex data for SGE calculation of drive group 4
 bb = BE hex debounce time of FDI inputs, P10017
 bb = BF hex debounce time of DI inputs, P10017
 bb = C0 hex debounce time of diag inputs, P10017
 cc: Index of the data to be cross-checked that resulted in the error.

Remedy:

Carry out the following steps on the TM54F:
 - activate the safety commissioning mode (p0010 = 95).
 - start the copy function for SI parameters (p9700 = 57 hex).
 - acknowledge complete data change (p9701 = AC hex).
 - exit the safety commissioning mode (p0010 = 0).
 - save all parameters (p0977 = 1).
 - carry out safe acknowledgement (p10006).
 For an internal software error (aaaa greater than zero):
 - upgrade the software on the TM54F.
 - contact the Hotline.
 - replace the TM54F.

F35052 (A) TM54F: Internal hardware error

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY (POWER ON)
Cause: An internal software/hardware error has been detected on the Terminal Module 54F (TM54F).
 Fault value (r0949, interpret decimal):
 Only for internal Siemens troubleshooting.
Remedy:
 - check the electrical cabinet design and cable routing for EMC compliance
 - upgrade TM54F firmware to more recent version.
 - contact the Hotline.
 - replace the TM54F.
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F35053 TM54F: Temperature fault threshold exceeded

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: The temperature measured using the temperature sensing on the TM54F has exceeded the threshold value to initiate this fault.
 As fault response fail-safe input terminal signals are transferred to the motion monitoring functions.
 Fault value (r0949, interpret decimal):
 Only for internal Siemens troubleshooting.
Remedy:
 - allow the TM54F to cool down.
 - carry out safe acknowledgement (p10006).

A35054 TM54F: Temperature alarm threshold exceeded

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The temperature measured using the temperature sensing on the TM54F has exceeded the threshold value to initiate this alarm.
Remedy:
 - allow the TM54F to cool down.
 - carry out safe acknowledgement (p10006).

A35075 (F)	TM54F: Error during internal communication
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	An internal communications error has occurred in the Terminal Module 54F (TM54F). This alarm can also occur if the TM54F exists and no safety function has yet been parameterized. Alarm value (r2124, interpret decimal): Only for internal Siemens diagnostics.
Remedy:	For internal communication errors: - check the electrical cabinet design and cable routing for EMC compliance - upgrade the software on the TM54F. - contact the Hotline. - replace the TM54F. If TM54F exists and no safety function has yet been parameterized: - None necessary. The alarm disappears automatically after a safety function has been parameterized.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
A35080 (F)	TM54F: Checksum error safety parameters
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The calculated checksum entered in r10004 over the safety-relevant parameters does not match the reference checksum saved in p10005 at the last machine acceptance. Fault value (r0949, interpret decimal): 1: Checksum error for functional SI parameters. 2: Checksum error for SI parameters for component assignment.
Remedy:	- check the safety-relevant parameters and if required, correct. - set the reference checksum to the actual checksum. - acknowledge that hardware was replaced - carry out a POWER ON. - carry out an acceptance test.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
A35081 (F)	TM54F: Static (steady state) 1 signal at the F-DI for safety-relevant acknowledgement
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A logical 1 signal is present at the F-DI configured in p10006 for more than 10 seconds. A logical 0 signal must be present statically (steady-state) at the F-DI. This avoids unintentional safety-relevant acknowledgement (or the "Internal Event Acknowledge" signal) if a wire breaks or one of the two digital inputs bounces.
Remedy:	Set F-DI (see p10006) to logical 0 signal.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY

F35150 TM54F: Communication error

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY (POWER ON)
Cause: A communication error between the TM54F master and Control Unit or between the TM54F slave and the Motor Module was detected.
Fault value (r0949, interpret hexadecimal):
Only for internal Siemens troubleshooting.
Remedy: When replacing a Motor Module, carry out the following steps:
- start the copy function for the node identifier on the TM54F (p9700 = 1D hex).
- acknowledge the hardware CRC on the TM54F (p9701 = EC hex).
- save all parameters (p0977 = 1).
- carry out a POWER ON (power off/on) for all components.
The following always applies:
- check the electrical cabinet design and cable routing for EMC compliance
- upgrade the software on the TM54F.
- contact the Hotline.
- replace the TM54F.

F35151 TM54F: Discrepancy error

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: The safety input terminals or output terminals show a different state longer than that parameterized in p10002.
Fault value (r0949, interpret hexadecimal):
yyyyxxxx hex
xxxx:
The safety-relevant input terminals F-DI indicate a discrepancy.
Bit 0: Discrepancy for F-DI 0
...
Bit 9: Discrepancy for F-DI 9
yyyy:
The safety-relevant output terminals F-DO indicate a discrepancy.
Bit 0: Discrepancy for F-DO 0
...
Bit 3: Discrepancy for F-DO 3
Note:
If several discrepancy errors occur consecutively, then this fault is only signaled for the first error that occurs.
The following possibilities exist of diagnosing all of the discrepancy errors:
- in the commissioning software, evaluate the input states and output states of the TM54F. All discrepancy errors are displayed here.
- compare parameters p10051 and p10052 from the TM54F master and TM54F slave for discrepancy.

Remedy: Check the wiring of the F-DI and F-DO (contact problems).
Note:
 A discrepancy of the F-DO also occurs (in this special case, in conjunction with fault F35150 for the TM54F slave), if, after replacing a Motor Module, it was forgotten to acknowledge this.
 When replacing a Motor Module, carry out the following steps:
 - start the copy function for the node identifier on the TM54F (p9700 = 1D hex).
 - acknowledge the hardware CRC on the TM54F (p9701 = EC hex).
 - save all parameters (p0977 = 1).
 - carry out a POWER ON (power off/on) for all components.
 Discrepancy errors in the F-DIs can only be completely acknowledged if, after the cause of the error was resolved, safe acknowledgement was carried out (see p10006). As long as safety acknowledgement was not carried out, the corresponding F-DI stays in the safe state.
 F-DI: Failsafe Digital Input
 F-DO: Failsafe Digital Output
 Where switching operations recur cyclically on the FDIs, the discrepancy time must be parameterized as follows:
 td = possible actual discrepancy time (in ms) that can occur with a switching operation. This must correspond to at least 1 SI sampling cycle (see p10000).
 tp = period for a switching operation in ms.
 The following rules must be adhered to:
 $p10002 < (tp/2) - td$ (discrepancy time must be less than half the period minus the actual discrepancy time)
 $p10002 > = p10000$ (discrepancy time must be no less than P10000)
 $p10002 > td$ (discrepancy time must be greater than the switch discrepancy time which may actually apply)
 Example: If SI sampling cycle is 12 ms and switching frequency is 110 ms, the maximum discrepancy time which can be set is as follows:
 $p10002 < = 110ms/2 - 12ms = 43ms$; this rounds off to $P10002 < = 36 ms$.
 (Since the sampling time can only be accepted as a whole SI sampling cycle, the value will need to be rounded up or down to a whole SI sampling time value if it is not an exact multiple of an SI sampling cycle.)

F35152 TM54F: Internal software error
Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: An internal software error has occurred in the Terminal Module 54F (TM54F).
 The fail-safe digital inputs and digital outputs (F-DI, F-DO) on the TM54F have been set to the safe state.
Note:
 F-DI: Failsafe Digital Input
 F-DO: Failsafe Digital Output
Remedy: Check that the firmware version of the TM54F matches the Control Unit's firmware version.
 The automatic firmware update must be activated in the project.
Note:
 This signal will also appear, for example, in conjunction with signal 35013. In this case you should check all the parameters for the test stop on the TM54F (p10001, p10003, p10007, p10041, p10046, p10047).

A35200 (F, N) TM: Calibration data
Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An error was detected in the calibration data of the Terminal Module.
 Alarm value (r2124, interpret decimal):
 The hundred-thousands and ten-thousands position specifies the component ID of the Terminal Module which detected the fault.
 The thousands location specifies whether the analog input 0 (=0) or analog output 1 (= 1) is involved.
 The hundreds location specifies the fault type:
 No calibration data available.
 1: Offset too high (> 100 mV).
 The tens and ones location specifies the number of the input involved.

Remedy: Power down the unit and power up again.
If the fault is still present, replace the module/board.

Reaction upon F: NONE

Acknowl. upon F: IMMEDIATELY (POWER ON)

Reaction upon N: NONE

Acknowl. upon N: NONE

F35207 (N, A) TM: Temperature fault/alarm threshold channel 1 exceeded

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR

Reaction: Infeed: OFF2 (NONE, OFF1)
Servo: OFF2 (NONE, OFF1, OFF3)
Vector: OFF2 (NONE, OFF1, OFF3)

Acknowledge: IMMEDIATELY (POWER ON)

Cause: For the temperature evaluation via the Terminal Module (TM), at least one of the following conditions to initiate this fault is fulfilled:
- alarm threshold has been exceeded longer than that set in the timer stage (p4102[0], p4103).
or
- fault threshold exceeded (p4102[1]).
Note:
For PTC, the following applies:
- if r4101 > 1650 ohms, the temperature r4105 = 250 °C
- if r4101 <= 1650 ohms, the temperature r4105 = -50 °C
The measured temperature is displayed in r4105.
This fault can only be initiated if temperature evaluation was activated (p4100 = 2 for KTY sensor or p4100 = 1 for PTC sensor or p4100 = 4 for bimetal).
Notice:
This fault only causes the drive to shut down if there is at least one BICO interconnection between the drive and the Terminal Module.
Fault value (r0949, interpret decimal):
yyxxxx dec:
yy: Component number of the component which detected the fault.
xxxx: Temperature value multiplied by 10 at the point of initiation

Remedy: - reduce ambient temperature of temperature sensor to below p4102[1] hysteresis.
- if required, set the fault response to NONE (p2100, p2101).

Reaction upon N: NONE

Acknowl. upon N: NONE

Reaction upon A: NONE

Acknowl. upon A: NONE

F35208 (N, A)	TM: Temperature fault/alarm threshold channel 2 exceeded
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM41, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (NONE, OFF1, OFF3) Vector: OFF2 (NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	For the temperature evaluation via the Terminal Module 120 (TM120), one condition to initiate this fault is fulfilled. The conditions to initiate this fault are as follows: - alarm threshold has been exceeded longer than that set in the timer stage (p4102[2], p4103). or - fault threshold exceeded (p4102[3]). Note: The measured temperature is displayed in r4105. This fault can only be initiated if temperature evaluation was activated (p4100 = 2 for KTY sensor or p4100 = 1 for PTC sensor or p4100 = 4 for bimetal). Notice: This fault only causes the drive to shut down if there is at least one BICO interconnection between the drive and the TM. Fault value (r0949, interpret decimal): yyxxxx dec: yy: Component number of the component where the fault occurred. xxxx: Temperature value multiplied by 10 at the point of initiation
Remedy:	- reduce ambient temperature of temperature sensor to below p4102[3] hysteresis. - if required, set the fault response to NONE (p2100, p2101).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F35209 (N, A)	TM: Temperature fault/alarm threshold channel 3 exceeded
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM41, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (NONE, OFF1, OFF3) Vector: OFF2 (NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	For the temperature evaluation via the Terminal Module 120 (TM120), one condition to initiate this fault is fulfilled. The conditions to initiate this fault are as follows: - alarm threshold has been exceeded longer than that set in the timer stage (p4102[4], p4103). or - fault threshold exceeded (p4102[5]). Note: The measured temperature is displayed in r4105. This fault can only be initiated if temperature evaluation was activated (p4100 = 2 for KTY sensor or p4100 = 1 for PTC sensor or p4100 = 4 for bimetal). Notice: This fault only causes the drive to shut down if there is at least one BICO interconnection between the drive and the TM. Fault value (r0949, interpret decimal): yyxxxx dec: yy: Component number of the component where the fault occurred. xxxx: Temperature value multiplied by 10 at the point of initiation
Remedy:	- reduce ambient temperature of temperature sensor to below p4102[5] hysteresis. - if required, set the fault response to NONE (p2100, p2101).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

F35210 (N, A)	TM: Temperature fault/alarm threshold channel 4 exceeded
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM41, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (NONE, OFF1, OFF3) Vector: OFF2 (NONE, OFF1, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	For the temperature evaluation via the Terminal Module 120 (TM120), one condition to initiate this fault is fulfilled. The conditions to initiate this fault are as follows: - alarm threshold has been exceeded longer than that set in the timer stage (p4102[6], p4103). or - fault threshold exceeded (p4102[7]). Note: The measured temperature is displayed in r4105. This fault can only be initiated if temperature evaluation was activated (p4100 = 2 for KTY sensor or p4100 = 1 for PTC sensor or p4100 = 4 for bimetal). Notice: This fault only causes the drive to shut down if there is at least one BICO interconnection between the drive and the TM. Fault value (r0949, interpret decimal): yyxxx dec: yy: Component number of the component where the fault occurred. xxx: Temperature value multiplied by 10 at the point of initiation
Remedy:	- reduce ambient temperature of temperature sensor to below p4102[7] hysteresis. - if required, set the fault response to NONE (p2100, p2101).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A35211 (F, N)	TM: Temperature alarm threshold channel 1 exceeded
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature measured using the temperature sensing of the Terminal Module (TM) (r4105) has exceeded the threshold value to initiate this alarm (p4102[0]). Alarm value (r2124, interpret decimal): The hundred-thousands and ten-thousands position specifies the component number of the component which detected the fault. For PTC, the following applies: - if r4101 > 1650 ohms, the temperature r4105 = 250 °C - if r4101 <= 1650 ohms, the temperature r4105 = -50 °C Alarm value (r2124, interpret decimal): yyxxx dec: yy: Component number of the component where the fault occurred. xxx: Temperature value multiplied by 10 at the point of initiation
Remedy:	Cool temperature sensor to below p4102[0] hysteresis.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A35212 (F, N)	TM: Temperature alarm threshold channel 2 exceeded
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature measured using the temperature sensing of the Terminal Module 120 (TM120) (r4105) has exceeded the threshold value to initiate this alarm (p4102[2]). Alarm value (r2124, interpret decimal): yyxxxx dec: yy: Component number of the component where the fault occurred. xxxx: Temperature value multiplied by 10 at the point of initiation
Remedy:	Cool temperature sensor to below p4102[2] hysteresis.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A35213 (F, N)	TM: Temperature alarm threshold channel 3 exceeded
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature measured using the temperature sensing of the Terminal Module 120 (TM120) (r4105) has exceeded the threshold value to initiate this alarm (p4102[4]). Alarm value (r2124, interpret decimal): yyxxxx dec: yy: Component number of the component where the fault occurred. xxxx: Temperature value multiplied by 10 at the point of initiation
Remedy:	Cool temperature sensor to below p4102[4] hysteresis.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A35214 (F, N)	TM: Temperature alarm threshold channel 4 exceeded
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The temperature measured using the temperature sensing of the Terminal Module 120 (TM120) (r4105) has exceeded the threshold value to initiate this alarm (p4102[6]). Alarm value (r2124, interpret decimal): yyxxxx dec: yy: Component number of the component where the fault occurred. xxxx: Temperature value multiplied by 10 at the point of initiation
Remedy:	Cool temperature sensor to below p4102[6] hysteresis.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE

F35220 (N, A)	TM: Frequency limit reached for signal output
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: OFF1 (NONE, OFF2, OFF3) Vector: OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The signals output from the Terminal Module 41 (TM41) for tracks A/B have reached the limit frequency. The output signals are no longer in synchronism with the specified setpoint. Note: If with SIMOTION the TM41 has been configured as the technology project, this fault is also output in response to short-circuited A/B signals in X520.
Remedy:	SIMOTION (p4400 = 0) operating mode: - enter a lower speed setpoint (p1155). - reduce the encoder pulse number (p0408). - check track A/B for short-circuits. SINAMICS (p4400 = 1) operating mode: - the fine resolution of TM41 in p0418 does not match that of the connector input that was interconnected at P4420 - the encoder position actual value r0479 interconnected at connector input p4420 has an excessively high actual speed
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
F35221 (N, A)	TM: Setpoint - actual value deviation, outside the tolerance range
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: OFF1 (NONE, OFF2, OFF3) Vector: OFF1 (NONE, OFF2, OFF3)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	The deviation between the setpoint and the output signals (track A/B) exceeds the tolerance of +/-3 %. The deviation between the internal and external measured value is too high.
Remedy:	- reduce the basic clock cycle (p0110, p0111). - replace the module.
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE
A35222 (F, N)	TM: Encoder pulse number not permissible
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The encoder pulse number entered does not match the permissible pulse number from a hardware perspective. Fault value (r0949, interpret decimal): 1: Encoder pulse number is too high. 2: Encoder pulse number is too low. 4: Encoder pulse number is less than the zero mark offset (p4426).

Remedy:

- enter the encoder pulse number in the permissible range (p0408).
- if necessary, replace TM41 SAC with TM41 DAC.

Note:

TM41 SAC: order no. = 6SL3055-0AA00-3PA0

TM41 DAC: order no. = 6SL3055-0AA00-3PA1

The following applies for TM41 SAC:

- minimum/maximum value for p0408: 1000/8192

The following applies for TM41 DAC:

- minimum/maximum value for p0408: 1000/16384

See also: p0408 (Rotary encoder pulse No.)

Reaction upon F: Infeed: OFF1 (NONE, OFF2)
 Servo: OFF1 (NONE, OFF2, OFF3)
 Vector: OFF1 (NONE, OFF2, OFF3)

Acknowl. upon F: IMMEDIATELY (POWER ON)

Reaction upon N: NONE

Acknowl. upon N: NONE

A35223 (F, N) TM: Zero mark offset not permissible

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The entered zero mark offset is not permissible.
 Fault value (r0949, interpret decimal):
 1: Zero mark offset is too high.
 See also: p4426 (Incremental encoder emulation, pulses for zero mark)

Remedy: Enter the zero mark offset in the permissible range (p4426).

Reaction upon F: Infeed: OFF1 (NONE, OFF2)
 Servo: OFF1 (NONE, OFF2, OFF3)
 Vector: OFF1 (NONE, OFF2, OFF3)

Acknowl. upon F: IMMEDIATELY (POWER ON)

Reaction upon N: NONE

Acknowl. upon N: NONE

A35224 (N) TM: Zero mark synchronization interrupted

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, TM41, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The zero mark synchronization with the encoder to be emulated was interrupted.
 Alarm value (r2124, interpret decimal):
 0: The encoder is not in the ready state (e.g. encoder parked)
 1: An absolute encoder was connected.
 2: The encoder r0479[0...2] interconnected with CI: p4420 is already communicating with another TM41 (precisely one TM41 can be interconnected with a specific r0479[0...2]).
 3: The BICO interconnection to Terminal Module 41 (TM41) was removed (CI: p4420 = 0 signal).
 4: The encoder connected with CI: p4420 has carried out an EDS changeover or has been re-parameterized (this operation is not supported, set p4420 to 0 and interconnect again).
 5: The maximum number of revolutions of the encoder was exceeded.
 6: Encoder in an invalid state.
 7: Encoder in an invalid state.
 8: Encoder in an invalid state (the encoder is not parameterized or the interconnected signal source is not in the cyclic state).

Remedy: None necessary.
- if the encoder changes into the ready state, then a synchronization operation that was previously interrupted is carried out again.
- if the synchronization was interrupted due to the maximum permissible synchronization duration, then a new synchronization is not carried out.
- for an absolute encoder, no synchronization is carried out, the zero mark is always output at the zero revolution of the TM41.

Reaction upon N: NONE
Acknowl. upon N: NONE

A35225 TM: Zero mark synchronization held - encoder not in the ready state

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The zero mark synchronization with the encoder to be emulated was held.
The encoder is not in the "ready" state.
Remedy: Bring the encoder into the "ready" state.

A35226 TM: Tracks A/B are deactivated

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The output of tracks A/B of the Terminal Module 41 (TM41) has been held (frozen).
The encoder emulation of the TM41 hardware is enabled (this is necessary so that no TRI state of the A/B tracks occurs). The hardware receives a setpoint of zero so that no motion occurs at the A/B tracks.
Reasons for this alarm:
- CI: p4420 was not interconnected (in this case, the encoder emulation of the hardware is deactivated)
- the encoder is not in the "ready" state (parking encoder or non-parameterized encoder data set).
- for TM41 there is an additional fault.
Remedy: - establish an interconnection from CI: p4420.
- bring the encoder into the "ready" state.
- remove any TM41 faults.

A35227 EDS changeover/encoder data set change not supported

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: Terminal Module 41 (TM41) does not support these particular application cases:
- the encoder interconnected via connector input p4420 has carried out an EDS changeover
- the encoder connected to the TM41 was re-parameterized such that the position actual value of the encoder required complete re-interpretation. This is the case with, for example, a change to p410 or p1821 (motor direction of rotation reversal) or a change to the fine resolution. This may amount to a sudden change in the position actual value, which cannot be output on the TM41.
Terminal Module 41 (TM41) does not support these particular application cases.
See also: p4420 (TM41 incremental encoder emulation position actual value)
Remedy: Set connector input p4420 = 0 and re-wire.

F35228	TM: Sampling time p4099[3] invalid
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: NONE Vector: NONE
Acknowledge:	IMMEDIATELY
Cause:	The value of the cycle time for the incremental encoder emulation, specified in p4099[3] does not correspond to a valid value. The system has already changed p4099[3] to a valid value. The parameters of the TM41 in question must be saved on the memory card and a POWER ON carried out. If necessary, the sampling time can be checked again the next time the system is switched on, taking into account any other TM41s located on the same DRIVE-CLiQ line. The following rules must be observed when setting p4099[3]: - if several TM41 are located on a DRIVE-CLiQ line, the same sampling time in p4099[3] must be set for all components. - the sampling time of a TM41 in SINAMICS mode (p4400) must correspond to that of the emulated encoder. The encoder sampling time is normally the same as the parameter value p0115[0] of the drive object used to interconnect the TM41 via connector input p4420. - it is not possible to operate two TM41s on one line if they emulate encoders with different cycles. - this fault cannot be resolved by removing the BICO connection (p4420 == 0). With subsequent reconnection
Remedy:	This fault can be temporarily resolved by removing the BICO connection (p4420 == 0), so that it is possible to continue working without alarms. This renders the TM41 inactive. Following subsequent BICO reconnection, the sampling time is checked again and the alarm output if necessary.
F35229	TM time slice deactivated
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: NONE Vector: NONE
Acknowledge:	IMMEDIATELY
Cause:	The required value of a cycle time in p4099[0...2] is invalid. The corresponding time slice was not activated. Alarm value (r2124, interpret decimal): 0: Digital input/outputs (p4099[0]) 1: Analog inputs (p4099[1]) 3: Encoder emulation (p4099[3]). 4: Encoder emulation speed setpoint (p4099[3]). 5: Encoder emulation speed setpoint (p4099[3]). 6: Internal sequence control of the TM41 (internal error)
Remedy:	The sampling time p4099[0] may not be zero. Change the sampling time corresponding to the error code.
F35230	HW problem with the TM module
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15DI_DO, TM31, TM41, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: NONE Vector: NONE
Acknowledge:	POWER ON
Cause:	The terminal module used has signaled an internal error. Signals of this module may not be evaluated and are potentially incorrect.
Remedy:	The module must be replaced if no other alarms that refer to a communications error are present in the system.

A35231	TM: Master control by PLC missing
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The "master control by PLC" signal was missing in operation. - interconnection of the binector input for "master control by PLC" is incorrect (p0854). - the higher-level control has withdrawn the "master control by PLC" signal. - data transfer via the fieldbus (master/drive) was interrupted. Note: This alarm is only decisive in the "SIMOTION" operating mode (p4400 = 0). In the "SINAMICS" operating mode, the setpoints at p4420 are evaluated independent of binector input p0854.
Remedy:	- check the interconnection of the binector input for "master control by PLC" (p0854). - check the "master control by PLC" signal and, if required, switch in. - check the data transfer via the fieldbus (master/drive). - check the setting of parameter p2037.

A35232	TM41: Zero mark no longer synchronous : Power ON required
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	SINAMICS (p4400 = 1) operating mode: When parameterizing a Terminal Module 41 (TM41) or when operating a TM41 Module, an operating state was reached which required a POWER ON. These include: - changing the encoder pulse number (p0408). - changing the fine resolution (p0418). - withdrawing the DRIVE-CLiQ cable without first deactivating TM41 via p0105. If this alarm was output, then the zero mark of the TM41 can no longer be output in synchronism to that of the encoder interconnected at p4420. SIMOTION (p4400 = 0) operating mode: A previously set zero mark position (p4426) no longer matches encoder position (r0479) due to the change in the pulse number (p0408).
Remedy:	The incremental position at output X520 of TM41 can still be evaluated independent of the zero mark. A POWER ON must be carried out if the TM41 zero mark is evaluated.

F35233	DRIVE-CLiQ component does not support function
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM31, TM41, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A function requested by the Control Unit is not supported by a DRIVE-CLiQ component. Fault value (r0949, interpret decimal): 1: Terminal Module 31 does not support the function "Timer for temperature evaluation" (X522.7/8, p4103 > 0.000).
Remedy:	Re fault value = 1: - Deactivate timer for temperature evaluation (X522.7/8) (p4103 = 0.000). - Use Terminal Module 31 and the relevant firmware version to enable the "Timer for temperature evaluation" function (Order No. 6SL3055-0AA00-3AA1, firmware version 2.6 and higher). See also: p4103

N35800 (F)	TM: Group signal
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2) Vector: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	NONE
Cause:	The Terminal Module has detected at least one fault.
Remedy:	Evaluates other actual messages.
Reaction upon F:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2) Vector: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowl. upon F:	IMMEDIATELY
A35801 (F, N)	TM DRIVE-CLiQ: Sign-of-life missing
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A DRIVE-CLiQ communication error has occurred from the Terminal Module to the encoder involved. Alarm value (r2124, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0A hex: The sign-of-life bit in the receive telegram is not set.
Remedy:	- check the DRIVE-CLiQ connection. - replace the component involved. See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A35802 (F, N)	TM: Time slice overflow
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Time slice overflow on Terminal Module.
Remedy:	Replace the Terminal Module.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A35803 (F, N)	TM: Memory test
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	An error has occurred during the memory test on the Terminal Module.

Remedy: - check whether the permissible ambient temperature for the Terminal Module is being maintained.
- replace the Terminal Module.

Reaction upon F: NONE

Acknowl. upon F: IMMEDIATELY (POWER ON)

Reaction upon N: NONE

Acknowl. upon N: NONE

A35804 (F, N) TM: CRC

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: A checksum error has occurred when reading-out the program memory on the Terminal Module.
Fault value (r0949, interpret hexadecimal):
Difference between the checksum at POWER ON and the actual checksum.

Remedy: - check whether the permissible ambient temperature for the component is maintained.
- replace the Terminal Module.

Reaction upon F: NONE

Acknowl. upon F: IMMEDIATELY (POWER ON)

Reaction upon N: NONE

Acknowl. upon N: NONE

A35805 (F, N) TM: EPROM checksum error

Message value: %1

Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: Internal parameter data is corrupted.
Alarm value (r2124, interpret hexadecimal):
01: EEPROM access error.
02: Too many blocks in the EEPROM.

Remedy: - check whether the permissible ambient temperature for the component is maintained.
- replace the Terminal Module 31 (TM31).

Reaction upon F: NONE

Acknowl. upon F: IMMEDIATELY (POWER ON)

Reaction upon N: NONE

Acknowl. upon N: NONE

A35807 (F, N) TM: Sequence control time monitoring

Message value: -

Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: Error, timeout, sequence control on the Terminal Module.

Remedy: Replace the Terminal Module.

Reaction upon F: NONE

Acknowl. upon F: IMMEDIATELY (POWER ON)

Reaction upon N: NONE

Acknowl. upon N: NONE

F35820	TM DRIVE-CLiQ: Telegram error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Terminal Module to the encoder involved. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 01 hex: CRC error. xx = 02 hex: Telegram is shorter than specified in the length byte or in the receive list. xx = 03 hex: Telegram is longer than specified in the length byte or in the receive list. xx = 04 hex: The length of the receive telegram does not match the receive list. xx = 05 hex: The type of the receive telegram does not match the receive list. xx = 06 hex: The address of the component in the telegram and in the receive list do not match. xx = 07 hex: A SYNC telegram is expected - but the received telegram is not a SYNC telegram. xx = 08 hex: No SYNC telegram is expected - but the received telegram is one. xx = 09 hex: The error bit in the receive telegram is set. xx = 10 hex: The receive telegram is too early.
Remedy:	- carry out a POWER ON. - check the electrical cabinet design and cable routing for EMC compliance - check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
F35835	TM DRIVE-CLiQ: Cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Terminal Module to the encoder involved. The nodes do not send and receive in synchronism. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 21 hex: The cyclic telegram has not been received. xx = 22 hex: Timeout in the telegram receive list. xx = 40 hex: Timeout in the telegram send list.
Remedy:	- carry out a POWER ON. - replace the component involved. See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F35836	TM DRIVE-CLiQ: Send error for DRIVE-CLiQ data
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Terminal Module to the encoder involved. Data were not able to be sent. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 41 hex: Telegram type does not match send list.
Remedy:	Carry out a POWER ON.

F35837	PTM DRIVE-CLiQ: Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component concerned. Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error.
Remedy:	- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). - check the electrical cabinet design and cable routing for EMC compliance - if required, use another DRIVE-CLiQ socket (p9904). - replace the component involved.

F35845	TM DRIVE-CLiQ: Cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Terminal Module (TM) to the encoder involved. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex: Synchronization error during alternating cyclic data transfer.
Remedy:	Carry out a POWER ON. See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F35850	TM: Internal software error
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	Infeed: OFF1 (NONE, OFF2) Servo: OFF1 (NONE, OFF2, OFF3) Vector: OFF1 (NONE, OFF2, OFF3)
Acknowledge:	POWER ON
Cause:	An internal software error in the Terminal Module (TM) has occurred. Fault value (r0949, interpret decimal): 1: Background time slice is blocked. 2: Checksum over the code memory is not OK.
Remedy:	- replace the Terminal Module (TM). - if required, upgrade the firmware in the Terminal Module. - contact the Hotline.
F35851	TM DRIVE-CLiQ (CU): Sign-of-life missing
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Terminal Module (TM) to the Control Unit involved. The DRIVE-CLiQ component did not set the sign-of-life to the Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0A hex = 10 dec: The sign-of-life bit in the receive telegram is not set.
Remedy:	Upgrade the firmware of the component involved.
F35860	TM DRIVE-CLiQ (CU): Telegram error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Terminal Module (TM) to the Control Unit involved. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 11 hex = 17 dec: CRC error and the receive telegram is too early. xx = 01 hex = 01 dec: Checksum error (CRC error). xx = 12 hex = 18 dec: The telegram is shorter than that specified in the length byte or in the receive list and the receive telegram is too early. xx = 02 hex = 02 dec: Telegram is shorter than specified in the length byte or in the receive list. xx = 13 hex = 19 dec: The telegram is longer than that specified in the length byte or in the receive list and the receive telegram is too early. xx = 03 hex = 03 dec: Telegram is longer than specified in the length byte or in the receive list. xx = 14 hex = 20 dec: The length of the receive telegram does not match the receive list and the receive telegram is too early. xx = 04 hex = 04 dec: The length of the receive telegram does not match the receive list. xx = 15 hex = 21 dec: The type of the receive telegram does not match the receive list and the receive telegram is too early. xx = 05 hex = 05 dec: The type of the receive telegram does not match the receive list.

xx = 16 hex = 22 dec:
The address of the power unit in the telegram and in the receive list does not match and the receive telegram is too early.
xx = 06 hex = 06 dec:
The address of the power unit in the telegram and in the receive list do not match.
xx = 19 hex = 25 dec:
The error bit in the receive telegram is set and the receive telegram is too early.
xx = 09 hex = 09 dec:
The error bit in the receive telegram is set.
xx = 10 hex = 16 dec:
The receive telegram is too early.

Remedy:

- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F35885 TM DRIVE-CLiQ (CU): Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: OFF1 (OFF2)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Terminal Module (TM) to the Control Unit involved.
The nodes do not send and receive in synchronism.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 1A hex = 26 dec:
Sign-of-life bit in the receive telegram not set and the receive telegram is too early.
xx = 21 hex = 33 dec:
The cyclic telegram has not been received.
xx = 22 hex = 34 dec:
Timeout in the telegram receive list.
xx = 40 hex = 64 dec:
Timeout in the telegram send list.
xx = 62 hex = 98 dec:
Error at the transition to cyclic operation.

Remedy:

- check the power supply voltage of the component involved.
- carry out a POWER ON.
- replace the component involved.

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F35886 TM DRIVE-CLiQ (CU): Error when sending DRIVE-CLiQ data

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: OFF1 (OFF2)
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communication error has occurred from the Terminal Module (TM) to the Control Unit involved.
Data were not able to be sent.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 41 hex:
Telegram type does not match send list.

Remedy:

Carry out a POWER ON.

F35887	TM DRIVE-CLiQ (CU): Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component (Terminal Module) involved. Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex: Error in the telegram header. xx = 23 hex: Receive error: The telegram buffer memory contains an error. xx = 42 hex: Send error: The telegram buffer memory contains an error. xx = 43 hex: Send error: The telegram buffer memory contains an error. xx = 60 hex: Response received too late during runtime measurement. xx = 61 hex: Time taken to exchange characteristic data too long.
Remedy:	- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). - check the electrical cabinet design and cable routing for EMC compliance - if required, use another DRIVE-CLiQ socket (p9904). - replace the component involved.
F35895	TM DRIVE-CLiQ (CU): Alternating cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	OFF1 (OFF2)
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communication error has occurred from the Terminal Module (TM) to the Control Unit involved. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex: Synchronization error during alternating cyclic data transfer.
Remedy:	Carry out a POWER ON. See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)
F35896	TM DRIVE-CLiQ (CU): Inconsistent component properties
Message value:	Component number: %1
Drive object:	A_INF, B_INF, CU_LINK, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	Infeed: OFF2 (NONE, OFF1) Servo: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2) Vector: OFF2 (IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY
Cause:	The properties of the DRIVE-CLiQ component (Terminal Module), specified by the fault value, have changed in an incompatible fashion with respect to the properties when booted. One cause can be, e.g. that a DRIVE-CLiQ cable or DRIVE-CLiQ component has been replaced. Fault value (r0949, interpret decimal): Component number.
Remedy:	- carry out a POWER ON. - when a component is replaced, the same component type and if possible the same firmware version should be used. - when a cable is replaced, only cables whose length is the same as or as close as possible to the length of the original cables should be used (ensure compliance with the maximum cable length).

F35897	TM DRIVE-CLiQ (CU): no communication with component
Message value:	Preliminary component number: %1
Drive object:	VECTOR
Reaction:	OFF2 (ENCODER, IASC/DCBRAKE, NONE, OFF1, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	Communications with the DRIVE-CLiQ component (Terminal Module) specified by the fault value is not possible. One cause may, for example, be that a DRIVE-CLiQ cable has been withdrawn. Fault value (r0949, interpret decimal): Component ID.
Remedy:	- check the DRIVE-CLiQ connections. - carry out a POWER ON.

F35899 (N, A)	TM: Unknown fault
Message value:	New message: %1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	Infeed: NONE (OFF1, OFF2) Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2) Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge:	IMMEDIATELY (POWER ON)
Cause:	A fault has occurred on the Terminal Module that cannot be interpreted by the Control Unit firmware. This can occur if the firmware on this component is more recent than the firmware on the Control Unit. Fault value (r0949, interpret decimal): Fault number. Note: If required, the significance of this new fault can be read about in a more recent description of the Control Unit.
Remedy:	- replace the firmware on the Terminal Module by an older firmware version (r0158). - upgrade the firmware on the Control Unit (r0018).
Reaction upon N:	NONE
Acknowl. upon N:	NONE
Reaction upon A:	NONE
Acknowl. upon A:	NONE

A35903 (F, N)	TM: I2C bus error occurred
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	An error has occurred while accessing the internal I2C bus of the Terminal Module.
Remedy:	Replace the Terminal Module.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A35904 (F, N)	TM: EEPROM
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	An error has occurred accessing the non-volatile memory on the Terminal Module.
Remedy:	Replace the Terminal Module.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A35905 (F, N)	TM: Parameter access
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The Control Unit attempted to write an illegal parameter value to the Terminal Module.
Remedy:	- check whether the firmware version of the Terminal Module (r0158) matches the firmware version of Control Unit (r0018). - if required, replace the Terminal Module.
	Note: The firmware versions that match each other are in the readme.txt file on the memory card.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A35906 (F, N)	TM: 24 V power supply missing
Message value:	%1
Drive object:	A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The 24 V power supply for the digital outputs is missing. Alarm value (r2124, interpret hexadecimal): 01: TM17 24 V power supply for DI/DO 0 ... 7 missing. 02: TM17 24 V power supply for DI/DO 8 ... 15 missing. 04: TM15 24 V power supply for DI/DO 0 ... 7 (X520) missing. 08: TM15 24 V power supply for DI/DO 8 ... 15 (X521) missing. 10: TM15 24 V power supply for DI/DO 16 ... 23 (X522) missing. 20: TM41 24 V power supply for DI/DO 0 ... 3 missing.
Remedy:	Check the terminals for the power supply voltage (L1+, L2+, L3+, M).
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE

A35907 (F, N) TM: Hardware initialization error

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The Terminal Module was not successfully initialized.
Alarm value (r2124, interpret hexadecimal):
01: TM17 or TM41 - incorrect configuration request.
02: TM17 or TM41 - programming not successful.
04: TM17 or TM41 - invalid time stamp
Remedy: Carry out a POWER ON.
Reaction upon F: NONE
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

A35910 (F, N) TM: Module overtemperature

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The temperature in the module has exceeded the highest permissible limit.
Remedy: - reduce the ambient temperature.
- replace the Terminal Module.
Reaction upon F: NONE
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

A35911 (F, N) TM: Clock synchronous operation sign-of-life missing

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The maximum permissible number of errors in the master sign-of-life (clock synchronous operation) has been exceeded in cyclic operation.
When the alarm is output, the module outputs are reset up to the next synchronization.
Remedy: - check the physical bus configuration (terminating resistor, shielding, etc.).
- check the interconnection of the master sign-of-life (r4201 via p0915).
- check whether the master correctly sends the sign-of-life (e.g. set up a trace with r4201.12 ... r4201.15 and trigger signal r4301.9).
- check the bus and master for utilization level (e.g. bus cycle time Tdp was set too short).
Reaction upon F: NONE
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

A35920 (F, N) TM: Error temperature sensor channel 1

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When evaluating the temperature sensor, an error occurred.
 Alarm value (r2124, interpret decimal):
 1: Wire breakage or sensor not connected (KTY: R > 1630 Ohm).
 2: Measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm).
Remedy: - make sure that the sensor is connected correctly.
 - replace the sensor.
 Reaction upon F: NONE
 Acknowl. upon F: IMMEDIATELY (POWER ON)
 Reaction upon N: NONE
 Acknowl. upon N: NONE

A35921 (F, N) TM: Error temperature sensor channel 2

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When evaluating the temperature sensor, an error occurred.
 Alarm value (r2124, interpret decimal):
 1: Wire breakage or sensor not connected (KTY: R > 1630 Ohm).
 2: Measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm).
Remedy: - make sure that the sensor is connected correctly.
 - replace the sensor.
 Reaction upon F: NONE
 Acknowl. upon F: IMMEDIATELY (POWER ON)
 Reaction upon N: NONE
 Acknowl. upon N: NONE

A35922 (F, N) TM: Error temperature sensor channel 3

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When evaluating the temperature sensor, an error occurred.
 Alarm value (r2124, interpret decimal):
 1: Wire breakage or sensor not connected (KTY: R > 1630 Ohm).
 2: Measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm).
Remedy: - make sure that the sensor is connected correctly.
 - replace the sensor.
 Reaction upon F: NONE
 Acknowl. upon F: IMMEDIATELY (POWER ON)
 Reaction upon N: NONE
 Acknowl. upon N: NONE

A35923 (F, N) TM: Error temperature sensor channel 4

Message value: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: When evaluating the temperature sensor, an error occurred.
 Alarm value (r2124, interpret decimal):
 1: Wire breakage or sensor not connected (KTY: R > 1630 Ohm).
 2: Measured resistance too low (PTC: R < 20 Ohm, KTY: R < 50 Ohm).
Remedy:
 - make sure that the sensor is connected correctly.
 - replace the sensor.
 Reaction upon F: NONE
 Acknowl. upon F: IMMEDIATELY (POWER ON)
 Reaction upon N: NONE
 Acknowl. upon N: NONE

A35999 (F, N) TM: Unknown alarm

Message value: New message: %1
Drive object: A_INF, B_INF, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An alarm has occurred on the Terminal Module that cannot be interpreted by the Control Unit firmware.
 This can occur if the firmware on this component is more recent than the firmware on the Control Unit.
 Alarm value (r2124, interpret decimal):
 Alarm number.
 Note:
 If required, the significance of this new alarm can be read about in a more recent description of the Control Unit.
Remedy:
 - replace the firmware on the Terminal Module by an older firmware version (r0158).
 - upgrade the firmware on the Control Unit (r0018).
 Reaction upon F: Infeed: NONE (OFF1, OFF2)
 Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
 Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
 Acknowl. upon F: IMMEDIATELY (POWER ON)
 Reaction upon N: NONE
 Acknowl. upon N: NONE

F36207 (N, A) Hub: Overtemperature component

Message value: %1
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: NONE (OFF1, OFF2)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The temperature on the DRIVE-CLiQ Hub Module has exceeded the fault threshold.
 Fault value (r0949, interpret decimal):
 Actual temperature in 0.1 °C resolution.
Remedy:
 - Check ambient temperature at component installation location.
 - replace the component involved.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

A36211 (F, N) Hub: Overtemperature alarm component

Message value: %1
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The temperature on the DRIVE-CLiQ Hub Module has exceeded the alarm threshold.
 Alarm value (r2124, interpret decimal):
 Actual temperature in 0.1 °C resolution.
Remedy:
 - Check ambient temperature at component installation location.
 - replace the component involved.
 Reaction upon F: NONE
 Acknowl. upon F: IMMEDIATELY
 Reaction upon N: NONE
 Acknowl. upon N: NONE

F36214 (N, A) Hub: overvoltage fault 24 V supply

Message value: %1
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: NONE (OFF1, OFF2)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The 24 V power supply on the DRIVE-CLiQ Hub Module has exceeded the fault threshold.
 Fault value (r0949, interpret decimal):
 Actual operating voltage in 0.1 °C resolution
Remedy:
 - check the supply voltage of the component involved.
 - replace the component involved.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

F36216 (N, A) Hub: undervoltage fault 24 V supply

Message value: %1
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: NONE (OFF1, OFF2)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: The 24 V power supply on the DRIVE-CLiQ Hub Module has undershot the fault threshold.
 Fault value (r0949, interpret decimal):
 Actual operating voltage in 0.1 °C resolution
Remedy:
 - check the supply voltage of the component involved.
 - replace the component involved.
 Reaction upon N: NONE
 Acknowl. upon N: NONE
 Reaction upon A: NONE
 Acknowl. upon A: NONE

A36217 (N) Hub: undervoltage alarm 24 V supply

Message value: %1
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The 24 V power supply on the DRIVE-CLiQ Hub Module has undershot the alarm threshold.
 Alarm value (r2124, interpret decimal):
 Actual operating voltage in 0.1 °C resolution

Remedy: - check the supply voltage of the component involved.
 - replace the component involved.

Reaction upon N: NONE

Acknowled. upon N: NONE

N36800 (F) Hub: Group signal

Message value: -

Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: The DRIVE-CLiQ Hub Module has detected at least one fault.

Remedy: Evaluates other actual messages.

Reaction upon F: NONE

Acknowled. upon F: IMMEDIATELY

A36801 (F, N) Hub DRIVE-CLiQ: Sign-of-life missing

Message value: Component number: %1, fault cause: %2

Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR

Reaction: NONE

Acknowledge: NONE

Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the DRIVE-CLiQ Hub Module involved.
Alarm value (r2124, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 0A hex = 10 dec:
The sign-of-life bit in the receive telegram is not set.

Remedy: - check the DRIVE-CLiQ connection.
 - replace the component involved.
See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

Reaction upon F: NONE

Acknowled. upon F: IMMEDIATELY

Reaction upon N: NONE

Acknowled. upon N: NONE

F36802 (N, A) Hub: Time slice overflow

Message value: %1

Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR

Reaction: Infeed: OFF2 (NONE)
 Servo: NONE
 Vector: NONE

Acknowledge: IMMEDIATELY (POWER ON)

Cause: A time slice overflow has occurred on the DRIVE-CLiQ Hub Module.
Fault value (r0949, interpret decimal):
xx: Time slice number xx

Remedy: - reduce the current controller frequency.
 - carry out a POWER ON (power off/on) for all components.
 - upgrade firmware to later version.
 - contact the Hotline.

Reaction upon N: NONE

Acknowled. upon N: NONE

Reaction upon A: NONE

Acknowled. upon A: NONE

A36804 (F, N)	Hub: Checksum error
Message value:	%1
Drive object:	A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	A checksum error has occurred when reading out the program memory on the DRIVE-CLiQ Hub Module. Alarm value (r2124, interpret hexadecimal): Difference between the checksum at POWER ON and the actual checksum.
Remedy:	- check whether the permissible ambient temperature for the component is maintained. - replace the DRIVE-CLiQ Hub Module.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
A36805 (F, N)	Hub: EEPROM checksum incorrect
Message value:	%1
Drive object:	A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The internal parameter data on the DRIVE-CLiQ Hub Module is incorrect. Alarm value (r2124, interpret hexadecimal): 01: EEPROM access error. 02: Too many blocks in the EEPROM.
Remedy:	- check whether the permissible ambient temperature for the component is maintained. - replace the DRIVE-CLiQ Hub Module.
Reaction upon F:	NONE
Acknowl. upon F:	IMMEDIATELY (POWER ON)
Reaction upon N:	NONE
Acknowl. upon N:	NONE
F36820	Hub DRIVE-CLiQ: Telegram error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communications error has occurred from the Control Unit to the DRIVE-CLiQ Hub Module involved. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 01 hex = 1 dec: Checksum error (CRC error). xx = 02 hex = 2 dec: Telegram is shorter than specified in the length byte or in the receive list. xx = 03 hex = 3 dec: Telegram is longer than specified in the length byte or in the receive list. xx = 04 hex = 4 dec: The length of the receive telegram does not match the receive list. xx = 05 hex = 5 dec: The type of the receive telegram does not match the receive list. xx = 06 hex = 6 dec: The address of the component in the telegram and in the receive list do not match. xx = 07 hex = 7 dec: A SYNC telegram is expected - but the received telegram is not a SYNC telegram. xx = 08 hex = 8 dec: No SYNC telegram is expected - but the received telegram is one.

xx = 09 hex = 9 dec:
The error bit in the receive telegram is set.
xx = 10 hex = 16 dec:
The receive telegram is too early.

Remedy:

- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).

See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F36835 Hub DRIVE-CLiQ: Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the DRIVE-CLiQ Hub Module involved.
The nodes do not send and receive in synchronism.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 21 hex = 33 dec:
The cyclic telegram has not been received.
xx = 22 hex = 34 dec:
Timeout in the telegram receive list.
xx = 40 hex = 64 dec:
Timeout in the telegram send list.

Remedy:

- carry out a POWER ON.
- replace the component involved.

See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F36836 Hub DRIVE-CLiQ: Send error for DRIVE-CLiQ data

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the DRIVE-CLiQ Hub Module involved.
Data were not able to be sent.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 41 hex = 65 dec:
Telegram type does not match send list.

Remedy: Carry out a POWER ON.

F36837 Hub DRIVE-CLiQ: Component fault

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: Fault detected on the DRIVE-CLiQ component concerned. Faulty hardware cannot be excluded.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 20 hex = 32 dec:
Error in the telegram header.
xx = 23 hex = 35 dec:
Receive error: The telegram buffer memory contains an error.
xx = 42 hex = 66 dec:
Send error: The telegram buffer memory contains an error.
xx = 43 hex = 67 dec:
Send error: The telegram buffer memory contains an error.

Remedy:

- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
- check the electrical cabinet design and cable routing for EMC compliance
- if required, use another DRIVE-CLiQ socket (p9904).
- replace the component involved.

F36845 Hub DRIVE-CLiQ: Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the DRIVE-CLiQ Hub Module involved.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 0B hex = 11 dec:
 Synchronization error during alternating cyclic data transfer.
Remedy: Carry out a POWER ON.
 See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F36851 Hub DRIVE-CLiQ (CU): Sign-of-life missing

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, HUB, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: DRIVE-CLiQ communication error from DRIVE-CLiQ Hub Module in question to Control Unit.
 The DRIVE-CLiQ component did not set the sign-of-life to the Control Unit.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 0A hex = 10 dec:
 The sign-of-life bit in the receive telegram is not set.
Remedy: Upgrade the firmware of the component involved.

F36860 Hub DRIVE-CLiQ (CU): Telegram error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, HUB, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: DRIVE-CLiQ communication error from DRIVE-CLiQ Hub Module in question to Control Unit.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 11 hex = 17 dec:
 Checksum error (CRC error) and the receive telegram is too early.
 xx = 01 hex = 01 dec:
 Checksum error (CRC error).
 xx = 12 hex = 18 dec:
 The telegram is shorter than that specified in the length byte or in the receive list and the receive telegram is too early.
 xx = 02 hex = 2 dec:
 Telegram is shorter than specified in the length byte or in the receive list.
 xx = 13 hex = 19 dec:
 The telegram is longer than that specified in the length byte or in the receive list and the receive telegram is too early.
 xx = 03 hex = 3 dec:
 Telegram is longer than specified in the length byte or in the receive list.
 xx = 14 hex = 20 dec:
 The length of the receive telegram does not match the receive list and the receive telegram is too early.
 xx = 04 hex = 4 dec:
 The length of the receive telegram does not match the receive list.
 xx = 15 hex = 21 dec:
 The type of the receive telegram does not match the receive list and the receive telegram is too early.
 xx = 05 hex = 5 dec:
 The type of the receive telegram does not match the receive list.

xx = 16 hex = 22 dec:
The address of the power unit in the telegram and in the receive list does not match and the receive telegram is too early.
xx = 06 hex = 6 dec:
The address of the power unit in the telegram and in the receive list do not match.
xx = 19 hex = 25 dec:
The error bit in the receive telegram is set and the receive telegram is too early.
xx = 09 hex = 9 dec:
The error bit in the receive telegram is set.
xx = 10 hex = 16 dec:
The receive telegram is too early.

Remedy:

- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).

F36885 Hub DRIVE-CLiQ (CU): Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, HUB, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: DRIVE-CLiQ communication error from DRIVE-CLiQ Hub Module in question to Control Unit.
The nodes do not send and receive in synchronism.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 1A hex = 26 dec:
Sign-of-life bit in the receive telegram not set and the receive telegram is too early.
xx = 21 hex = 33 dec:
The cyclic telegram has not been received.
xx = 22 hex = 34 dec:
Timeout in the telegram receive list.
xx = 40 hex = 64 dec:
Timeout in the telegram send list.
xx = 62 hex = 98 dec:
Error at the transition to cyclic operation.

Remedy:

- check the supply voltage of the component involved.
- carry out a POWER ON.
- replace the component involved.

F36886 Hub DRIVE-CLiQ (CU): Error when sending DRIVE-CLiQ data

Message value: Component number: %1, fault cause: %2
Drive object: A_INF, B_INF, CU_LINK, HUB, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: DRIVE-CLiQ communication error from DRIVE-CLiQ Hub Module in question to Control Unit.
Data were not able to be sent.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 41 hex = 65 dec:
Telegram type does not match send list.

Remedy:

- Carry out a POWER ON.

F36887	Hub DRIVE-CLiQ (CU): Component fault
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, HUB, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	Fault detected on the DRIVE-CLiQ component (DRIVE-CLiQ Hub Module) involved. Faulty hardware cannot be excluded. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 20 hex = 32 dec: Error in the telegram header. xx = 23 hex = 35 dec: Receive error: The telegram buffer memory contains an error. xx = 42 hex = 66 dec: Send error: The telegram buffer memory contains an error. xx = 43 hex = 67 dec: Send error: The telegram buffer memory contains an error. xx = 60 hex = 96 dec: Response received too late during runtime measurement. xx = 61 hex = 97 dec: Time taken to exchange characteristic data too long.
Remedy:	- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...). - check the electrical cabinet design and cable routing for EMC compliance - if required, use another DRIVE-CLiQ socket (p9904). - replace the component involved.
F36895	Hub DRIVE-CLiQ (CU): Alternating cyclic data transfer error
Message value:	Component number: %1, fault cause: %2
Drive object:	A_INF, B_INF, CU_LINK, HUB, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	DRIVE-CLiQ communication error from DRIVE-CLiQ Hub Module in question to Control Unit. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0B hex = 11 dec: Synchronization error during alternating cyclic data transfer.
Remedy:	Carry out a POWER ON. See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)
F36896	Hub DRIVE-CLiQ (CU): Inconsistent component properties
Message value:	Component number: %1
Drive object:	A_INF, B_INF, CU_LINK, HUB, S_INF, SERVO, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, VECTOR
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The properties of the DRIVE-CLiQ component (DRIVE-CLiQ Hub Module), specified by the fault value, have changed in an incompatible fashion with respect to the properties when booted. One cause can be, e.g. that a DRIVE-CLiQ cable or DRIVE-CLiQ component has been replaced. Fault value (r0949, interpret decimal): Component number.
Remedy:	- carry out a POWER ON. - when a component is replaced, the same component type and if possible the same firmware version should be used. - when a cable is replaced, only cables whose length is the same as or as close as possible to the length of the original cables should be used (ensure compliance with the maximum cable length).

F36899 (N, A) Hub: Unknown fault

Message value: New message: %1
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: Infeed: NONE (OFF1, OFF2)
Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowledge: IMMEDIATELY (POWER ON)
Cause: A fault occurred on the DRIVE-CLiQ Hub Module that cannot be interpreted by the Control Unit firmware.
This can occur if the firmware on this component is more recent than the firmware on the Control Unit.
Fault value (r0949, interpret decimal):
Fault number.
Note:
If required, the significance of this new fault can be read about in a more recent description of the Control Unit.
Remedy:
- replace the firmware on the DRIVE-CLiQ Hub Module with older firmware (r0158).
- upgrade the firmware on the Control Unit (r0018).
Reaction upon N: NONE
Acknowl. upon N: NONE
Reaction upon A: NONE
Acknowl. upon A: NONE

A36999 (F, N) Hub: Unknown alarm

Message value: New message: %1
Drive object: A_INF, B_INF, HUB, S_INF, SERVO, TM41, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: An alarm occurred on the DRIVE-CLiQ Hub Module that cannot be interpreted by the Control Unit firmware.
This can occur if the firmware on this component is more recent than the firmware on the Control Unit.
Alarm value (r2124, interpret decimal):
Alarm number.
Note:
If required, the significance of this new alarm can be read about in a more recent description of the Control Unit.
Remedy:
- replace the firmware on the DRIVE-CLiQ Hub Module with older firmware (r0158).
- upgrade the firmware on the Control Unit (r0018).
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Vector: NONE (IASC/DCBRAKE, OFF1, OFF2, OFF3, STOP1, STOP2)
Acknowl. upon F: IMMEDIATELY (POWER ON)
Reaction upon N: NONE
Acknowl. upon N: NONE

F40000 Fault at DRIVE-CLiQ socket X100

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A fault has occurred at the drive object at the DRIVE-CLiQ socket X100.
Fault value (r0949, interpret decimal):
First fault that has occurred for this drive object.
Remedy: Evaluate the fault buffer of the specified object.

F40001 Fault at DRIVE-CLiQ socket X101

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A fault has occurred at the drive object at the DRIVE-CLiQ socket X101.
 Fault value (r0949, interpret decimal):
 First fault that has occurred for this drive object.
Remedy: Evaluate the fault buffer of the specified object.

F40002 Fault at DRIVE-CLiQ socket X102

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A fault has occurred at the drive object at the DRIVE-CLiQ socket X102.
 Fault value (r0949, interpret decimal):
 First fault that has occurred for this drive object.
Remedy: Evaluate the fault buffer of the specified object.

F40003 Fault at DRIVE-CLiQ socket X103

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A fault has occurred at the drive object at the DRIVE-CLiQ socket X103.
 Fault value (r0949, interpret decimal):
 First fault that has occurred for this drive object.
Remedy: Evaluate the fault buffer of the specified object.

F40004 Fault at DRIVE-CLiQ socket X104

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A fault has occurred at the drive object at the DRIVE-CLiQ socket X104.
 Fault value (r0949, interpret decimal):
 First fault that has occurred for this drive object.
Remedy: Evaluate the fault buffer of the specified object.

F40005 Fault at DRIVE-CLiQ socket X105

Message value: %1
Drive object: All objects
Reaction: NONE
Acknowledge: IMMEDIATELY
Cause: A fault has occurred at the drive object at the DRIVE-CLiQ socket X105.
 Fault value (r0949, interpret decimal):
 First fault that has occurred for this drive object.
Remedy: Evaluate the fault buffer of the specified object.

A40100 Alarm at DRIVE-CLiQ socket X100

Message value: %1

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: An alarm has occurred at the drive object at the DRIVE-CLiQ socket X100.
Alarm value (r2124, interpret decimal):
First alarm that has occurred for this drive object.

Remedy: Evaluate the alarm buffer of the specified object.

A40101 Alarm at DRIVE-CLiQ socket X101

Message value: %1

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: An alarm has occurred at the drive object at the DRIVE-CLiQ socket X101.
Alarm value (r2124, interpret decimal):
First alarm that has occurred for this drive object.

Remedy: Evaluate the alarm buffer of the specified object.

A40102 Alarm at DRIVE-CLiQ socket X102

Message value: %1

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: An alarm has occurred at the drive object at the DRIVE-CLiQ socket X102.
Alarm value (r2124, interpret decimal):
First alarm that has occurred for this drive object.

Remedy: Evaluate the alarm buffer of the specified object.

A40103 Alarm at DRIVE-CLiQ socket X103

Message value: %1

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: An alarm has occurred at the drive object at the DRIVE-CLiQ socket X103.
Alarm value (r2124, interpret decimal):
First alarm that has occurred for this drive object.

Remedy: Evaluate the alarm buffer of the specified object.

A40104 Alarm at DRIVE-CLiQ socket X104

Message value: %1

Drive object: All objects

Reaction: NONE

Acknowledge: NONE

Cause: An alarm has occurred at the drive object at the DRIVE-CLiQ socket X104.
Alarm value (r2124, interpret decimal):
First alarm that has occurred for this drive object.

Remedy: Evaluate the alarm buffer of the specified object.

A40105	Alarm at DRIVE-CLiQ socket X105
Message value:	%1
Drive object:	All objects
Reaction:	NONE
Acknowledge:	NONE
Cause:	An alarm has occurred at the drive object at the DRIVE-CLiQ socket X105. Alarm value (r2124, interpret decimal): First alarm that has occurred for this drive object.
Remedy:	Evaluate the alarm buffer of the specified object.
F40799	CX32: Configured transfer end time exceeded
Message value:	-
Drive object:	All objects
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The configured transfer end time when transferring the cyclic actual values was exceeded.
Remedy:	- carry out a POWER ON (power off/on) for all components. - contact the Hotline.
F40801	CX32 DRIVE-CLiQ: Sign-of-life missing
Message value:	Component number: %1, fault cause: %2
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communications error has occurred from the Control Unit to the controller extension involved. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 0A hex: The sign-of-life bit in the receive telegram is not set.
Remedy:	- carry out a POWER ON. - replace the component involved. See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)
F40820	CX32 DRIVE-CLiQ: Telegram error
Message value:	Component number: %1, fault cause: %2
Drive object:	All objects
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	A DRIVE-CLiQ communications error has occurred from the Control Unit to the controller extension involved. Fault value (r0949, interpret hexadecimal): yyxx hex: yy = component number, xx = fault cause xx = 01 hex: CRC error. xx = 02 hex: Telegram is shorter than specified in the length byte or in the receive list. xx = 03 hex: Telegram is longer than specified in the length byte or in the receive list. xx = 04 hex: The length of the receive telegram does not match the receive list. xx = 05 hex: The type of the receive telegram does not match the receive list. xx = 06 hex: The address of the component in the telegram and in the receive list do not match. xx = 07 hex: A SYNC telegram is expected - but the received telegram is not a SYNC telegram. xx = 08 hex: No SYNC telegram is expected - but the received telegram is one.

xx = 09 hex:
The error bit in the receive telegram is set.
xx = 10 hex:
The receive telegram is too early.

Remedy:

- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).

See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F40835 CX32 DRIVE-CLiQ: Cyclic data transfer error

Message value: Component number: %1, fault cause: %2

Drive object: All objects

Reaction: OFF2

Acknowledge: IMMEDIATELY

Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the controller extension involved. The nodes do not send and receive in synchronism.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 21 hex:
The cyclic telegram has not been received.
xx = 22 hex:
Timeout in the telegram receive list.
xx = 40 hex:
Timeout in the telegram send list.

Remedy:

- carry out a POWER ON.
- replace the component involved.

See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F40836 CX32 DRIVE-CLiQ: Send error for DRIVE-CLiQ data

Message value: Component number: %1, fault cause: %2

Drive object: All objects

Reaction: OFF2

Acknowledge: IMMEDIATELY

Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the controller extension involved. Data were not able to be sent.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 41 hex:
Telegram type does not match send list.

Remedy: Carry out a POWER ON.

F40837 CX32 DRIVE-CLiQ: Component fault

Message value: Component number: %1, fault cause: %2

Drive object: All objects

Reaction: OFF2

Acknowledge: IMMEDIATELY

Cause: Fault detected on the DRIVE-CLiQ component concerned. Faulty hardware cannot be excluded.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 20 hex:
Error in the telegram header.
xx = 23 hex:
Receive error: The telegram buffer memory contains an error.
xx = 42 hex:
Send error: The telegram buffer memory contains an error.
xx = 43 hex:
Send error: The telegram buffer memory contains an error.

Remedy:

- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
- check the electrical cabinet design and cable routing for EMC compliance
- if required, use another DRIVE-CLiQ socket (p9904).
- replace the component involved.

F40845 CX32 DRIVE-CLiQ: Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the Control Unit to the controller extension involved.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 0B hex:
 Synchronization error during alternating cyclic data transfer.
Remedy: Carry out a POWER ON.
 See also: p9916 (DRIVE-CLiQ data transfer error shutdown threshold slave)

F40851 CX32 DRIVE-CLiQ (CU): Sign-of-life missing

Message value: Component number: %1, fault cause: %2
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the controller extension involved to the Control Unit.
 The DRIVE-CLiQ component did not set the sign-of-life to the Control Unit.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 0A hex:
 The sign-of-life bit in the receive telegram is not set.
Remedy: Upgrade the firmware of the component involved.

F40860 CX32 DRIVE-CLiQ (CU): Telegram error

Message value: Component number: %1, fault cause: %2
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the controller extension involved to the Control Unit.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 11 hex:
 CRC error and the receive telegram is too early.
 xx = 01 hex:
 CRC error.
 xx = 12 hex:
 The telegram is shorter than that specified in the length byte or in the receive list and the receive telegram is too early.
 xx = 02 hex:
 Telegram is shorter than specified in the length byte or in the receive list.
 xx = 13 hex:
 The telegram is longer than that specified in the length byte or in the receive list and the receive telegram is too early.
 xx = 03 hex:
 Telegram is longer than specified in the length byte or in the receive list.
 xx = 14 hex:
 The length of the receive telegram does not match the receive list and the receive telegram is too early.
 xx = 04 hex:
 The length of the receive telegram does not match the receive list.
 xx = 15 hex:
 The type of the receive telegram does not match the receive list and the receive telegram is too early.
 xx = 05 hex:
 The type of the receive telegram does not match the receive list.

xx = 16 hex:
The address of the power unit in the telegram and in the receive list does not match and the receive telegram is too early.
xx = 06 hex:
The address of the power unit in the telegram and in the receive list do not match.
xx = 19 hex:
The error bit in the receive telegram is set and the receive telegram is too early.
xx = 09 hex:
The error bit in the receive telegram is set.
xx = 10 hex:
The receive telegram is too early.

Remedy:

- carry out a POWER ON.
- check the electrical cabinet design and cable routing for EMC compliance
- check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F40885 CX32 DRIVE-CLiQ (CU): Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the controller extension involved to the Control Unit.
The nodes do not send and receive in synchronism.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 1A hex:
Sign-of-life bit in the receive telegram not set and the receive telegram is too early.
xx = 21 hex:
The cyclic telegram has not been received.
xx = 22 hex:
Timeout in the telegram receive list.
xx = 40 hex:
Timeout in the telegram send list.
xx = 62 hex:
Error at the transition to cyclic operation.

Remedy:

- check the power supply voltage of the component involved.
- carry out a POWER ON.
- replace the component involved.

See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F40886 CX32 DRIVE-CLiQ (CU): Error when sending DRIVE-CLiQ data

Message value: Component number: %1, fault cause: %2
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the controller extension involved to the Control Unit.
Data were not able to be sent.
Fault value (r0949, interpret hexadecimal):
yyxx hex: yy = component number, xx = fault cause
xx = 41 hex:
Telegram type does not match send list.

Remedy: Carry out a POWER ON.

F40887 CX32 DRIVE-CLiQ (CU): Component fault

Message value: Component number: %1, fault cause: %2
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: Fault detected on the DRIVE-CLiQ component concerned. Faulty hardware cannot be excluded.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 20 hex:
 Error in the telegram header.
 xx = 23 hex:
 Receive error: The telegram buffer memory contains an error.
 xx = 42 hex:
 Send error: The telegram buffer memory contains an error.
 xx = 43 hex:
 Send error: The telegram buffer memory contains an error.
 xx = 60 hex:
 Response received too late during runtime measurement.
 xx = 61 hex:
 Time taken to exchange characteristic data too long.
Remedy:
 - check the DRIVE-CLiQ wiring (interrupted cable, contacts, ...).
 - check the electrical cabinet design and cable routing for EMC compliance
 - if required, use another DRIVE-CLiQ socket (p9904).
 - replace the component involved.

F40895 CX32 DRIVE-CLiQ (CU): Cyclic data transfer error

Message value: Component number: %1, fault cause: %2
Drive object: All objects
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: A DRIVE-CLiQ communications error has occurred from the controller extension involved to the Control Unit.
 Fault value (r0949, interpret hexadecimal):
 yyxx hex: yy = component number, xx = fault cause
 xx = 0B hex:
 Synchronization error during alternating cyclic data transfer.
Remedy:
 Carry out a POWER ON.
 See also: p9915 (DRIVE-CLiQ data transfer error shutdown threshold master)

F49150 Cooling system: Fault occurred

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The cooling system signals a general fault.
Remedy:
 - check the wiring between the cooling system and the input terminal (Terminal Module).
 - check the external control device for the cooling system.
 See also: p0266 (Cooling system, feedback signals, signal source)

F49151 Cooling system: Conductivity has exceeded the fault threshold

Message value: -
Drive object: A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The conductivity of the cooling liquid has exceeded the selected fault threshold (p0269[2]).
 See also: p0261 (Cooling system, starting time 2), p0262 (Cooling system, fault conductivity delay time), p0266 (Cooling system, feedback signals, signal source)
Remedy:
 Check the device to de-ionize the cooling liquid.

F49152	Cooling system: ON command feedback signal missing
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The feedback signal of the ON command of the cooling system is missing. - after the ON command, the feedback signal has not been received within the selected starting time (p0260). - the feedback signal has failed in operation. See also: p0260 (Cooling system, starting time 1), r0267 (Cooling system status word)
Remedy:	- check the wiring between the cooling system and the input terminal (Terminal Module). - check the external control device for the cooling system.

F49153	Cooling system: Liquid flow too low
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The drive converter cooling system signals that the cooling liquid flow is too low. - after the ON command, the feedback signal has not been received within the selected starting time (p0260). - in operation, the feedback signal has failed for longer than the permitted failure time (p0263). See also: p0260 (Cooling system, starting time 1), p0263 (Cooling system fault liquid flow, delay time), r0267 (Cooling system status word)
Remedy:	- check the wiring between the cooling system and the input terminal (Terminal Module). - check the external control device for the cooling system.

F49154 (A)	Cooling system: Liquid leak is present
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The liquid leakage monitoring function has responded. Caution: If this fault is re-parameterized as an alarm then using other monitoring functions it must be ensured that when cooling water is lost, the drive is powered down! See also: r0267 (Cooling system status word)
Remedy:	- check the cooling system for leaks in the cooling circuit. - check the wiring of the input terminal (Terminal Module) used to monitor leaking fluid.
Reaction upon A:	NONE
Acknowled. upon A:	NONE

F49155	Cooling system: Power Stack Adapter, firmware version too old
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	POWER ON
Cause:	The firmware version in the Power Stack Adapter (PSA) is too old and does not support the liquid cooling.
Remedy:	Upgrade the firmware. Check EEPROM data.

F49156	Cooling system: Cooling liquid temperature has exceeded the fault threshold
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The cooling liquid intake temperature has exceeded the permanently set fault threshold.
Remedy:	Check the cooling system and the ambient conditions.

A49170	Cooling system: Alarm has occurred
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The cooling system signals a general alarm.
Remedy:	- check the wiring between the cooling system and the input terminal (Terminal Module). - check the external control device for the cooling system.
A49171	Cooling system: Conductivity has exceeded the alarm threshold
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO
Reaction:	NONE
Acknowledge:	NONE
Cause:	The conductivity of the cooling liquid has exceeded the selected alarm threshold (p0269[1]). See also: p0261 (Cooling system, starting time 2), p0262 (Cooling system, fault conductivity delay time), p0266 (Cooling system, feedback signals, signal source)
Remedy:	Check the device to de-ionize the cooling liquid.
A49171	Cooling system: Conductivity has exceeded the alarm threshold
Message value:	-
Drive object:	VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	Conductivity monitoring is set for the cooling liquid (r0267.7, from p0266[7]). See also: p0261 (Cooling system, starting time 2), p0262 (Cooling system, fault conductivity delay time), p0266 (Cooling system, feedback signals, signal source), r0267 (Cooling system status word)
Remedy:	Check the device to de-ionize the cooling liquid.
A49172	Cooling system: Conductivity actual value is not valid
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	When monitoring the conductivity of the cooling liquid, there is a fault in the wiring or in the sensor.
Remedy:	- check the wiring between the cooling system and the Power Stack Adapter (PSA). - check the function of the sensor to measure the conductivity.
A49173	Cooling system: Cooling liquid temperature has exceeded the alarm threshold
Message value:	-
Drive object:	A_INF, B_INF, S_INF, SERVO, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	The cooling liquid intake temperature has exceeded the specified alarm threshold.
Remedy:	Check the cooling system and the ambient conditions.

F49200 Excitation group signal fault

Message value: %1
Drive object: VECTOR
Reaction: OFF2
Acknowledge: IMMEDIATELY
Cause: The excitation sequence control signals a fault.
Fault value (r0949, interpret hexadecimal):
Bit 0:
When powered down or when powering down the excitation, the signal "excitation ready to power up feedback signal" was not received within the monitoring time.
Bit 1:
After an ON command, the signal "excitation ready feedback signal" was not received within the monitoring time.
Bit 2:
After the pulses were enabled, the signal "excitation operational feedback signal" was not received within the monitoring time.
Bit 3:
The "excitation group signal fault" signal is present.
Remedy: - check the excitation.
- check commands, feedback signals and BICO interconnections.

A49201 (F) Excitation group signal alarm

Message value: -
Drive object: VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: The "excitation group signal alarm" signal is present.
Remedy: Check the excitation equipment.
Reaction upon F: NONE
Acknowl. upon F: IMMEDIATELY

A50002 (F) COMM BOARD: Alarm 2

Message value: %1
Drive object: A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction: NONE
Acknowledge: NONE
Cause: In the case of CBE20 SINAMICS Link:
A specific telegram word (send) is being used twice.
Alarm value (r2124, interpret decimal):
Telegram word used twice
See also: p8871 (SINAMICS Link send telegram word PZD)
Remedy: In the case of CBE20 SINAMICS Link:
Correct the parameter assignment.
See also: p8871 (SINAMICS Link send telegram word PZD)
Reaction upon F: Infeed: NONE (OFF1, OFF2)
Servo: NONE (OFF1, OFF2, OFF3)
Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F: IMMEDIATELY

A50003 (F)	COMM BOARD: Alarm 3
Message value:	Info. 1: %1, info. 2: %2
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the case of CBE20 SINAMICS Link: A specific telegram word (receive) is being used twice. Alarm value (r2124, interpret hexadecimal): yyyyxxxx hex: yyyy = info. 1, xxxx = info. 2 Info. 1 (interpret decimal) = Address of sender Info. 2 (interpret decimal) = Receive telegram word See also: p8870 (SINAMICS Link receive telegram word PZD), p8872 (SINAMICS Link address receive PZD)
Remedy:	In the case of CBE20 SINAMICS Link: Correct the parameter assignment.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY
A50004 (F)	COMM BOARD: Alarm 4
Message value:	Info. 1: %1, info. 2: %2
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the case of CBE20 SINAMICS Link: Telegram word (receive) and address of sender inconsistent. Both values have to be either equal to zero or not equal to zero. Alarm value (r2124, interpret hexadecimal): yyyyxxxx hex: yyyy = info. 1, xxxx = info. 2 Info. 1 (interpret decimal) = Drive object number from p8870, p8872 Info. 2 (interpret decimal) = Index from p8870, p8872 See also: p8870 (SINAMICS Link receive telegram word PZD), p8872 (SINAMICS Link address receive PZD)
Remedy:	In the case of CBE20 SINAMICS Link: Correct the parameter assignment.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY
A50005 (F)	COMM BOARD: Alarm 5
Message value:	%1
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the case of CBE20 SINAMICS Link: Sender not found on SINAMICS Link. Alarm value (r2124, interpret decimal): Address of sender that cannot be located See also: p8872 (SINAMICS Link address receive PZD)
Remedy:	In the case of CBE20 SINAMICS Link: Check the connection to the sender.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY

A50006 (F)	COMM BOARD: Alarm 6
Message value:	Info. 1: %1, info. 2: %2
Drive object:	A_INF, B_INF, CU_LINK, CU_S, CU_S_CU310DP, CU_S_CU310PN, CU_S_S150, ENCODER, HUB, S_INF, SERVO, TB30, TM120, TM15, TM15DI_DO, TM17, TM31, TM41, TM54F_MA, TM54F_SL, VECTOR
Reaction:	NONE
Acknowledge:	NONE
Cause:	In the case of CBE20 SINAMICS Link: The parameter assignment indicates that the sender and the receiver are one and the same. This is not permitted. Alarm value (r2124, interpret hexadecimal): yyyyxxxx hex: yyyy = info. 1, xxxx = info. 2 Info. 1 (interpret decimal) = Drive object number from p8872 Info. 2 (interpret decimal) = Index from p8872 See also: p8836 (SINAMICS Link address), p8872 (SINAMICS Link address receive PZD)
Remedy:	In the case of CBE20 SINAMICS Link: Correct the parameter assignment. All p8872[index] must be set to a value not equal to p8836.
Reaction upon F:	Infeed: NONE (OFF1, OFF2) Servo: NONE (OFF1, OFF2, OFF3) Vector: NONE (OFF1, OFF2, OFF3)
Acknowl. upon F:	IMMEDIATELY

Appendix

A

Contents

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A.1 ASCII table (excerpt)

The following table includes the decimal and hexadecimal notation of selected ASCII characters.

Table A-1 ASCII table (excerpt)

Character	Decimal	Hexadecimal	Character	Decimal	Hexadecimal
Space	32	20	H	72	48
-	45	2D	I	73	49
0	48	30	J	74	4A
1	49	31	K	75	4B
2	50	32	L	76	4C
3	51	33	M	77	4D
4	52	34	N	78	4E
5	53	35	O	79	4F
6	54	36	P	80	50
7	55	37	Q	81	51
8	56	38	R	82	52
9	57	39	S	83	53
A	65	41	T	84	54
B	66	42	U	85	55
C	67	43	V	86	56
D	68	44	W	87	57
E	69	45	X	88	58
F	70	46	Y	89	59
G	71	47	Z	90	5A

A.2 List for motor code/encoder code

A.2.1 Motor code

Induction motors (Version: 4301700)

Table A-2 Motor code for induction motors

Order number	Motor type (p0300)	Motor code (p0301)
1PH2092-4WG4x-xxxx	102	10201
1PH2093-6WF4x-xxxx	102	10202
1PH2095-6WF4x-xxxx	102	10203
1PH2096-4WG4x-xxxx	102	10204
1PH2113-6WF4x-xxxx	102	10205
1PH2115-6WF4x-xxxx	102	10206
1PH2117-6WF4x-xxxx	102	10207
1PH2118-6WF4x-xxxx	102	10208
1PH2123-4WF4x-xxxx	102	10209
1PH2127-4WF4x-xxxx	102	10210
1PH2128-4WF4x-xxxx	102	10211
1PH2143-4WF4x-xxxx	102	10212
1PH2147-4WF4x-xxxx	102	10213
1PH2182-6WC4x-xxxx	102	10214
1PH2184-6WP4x-xxxx	102	10215
1PH2186-6WB4x-xxxx	102	10216
1PH2188-6WB4x-xxxx	102	10217
1PH2254-6WB4x-xxxx	102	10218
1PH2256-6WB4x-xxxx	102	10219
1PH4103-4NF2x-xxxx	104	10401
1PH4103-4xF5x-xxxx	104	10421
1PH4105-4NF2x-xxxx	104	10403
1PH4105-4xF5x-xxxx	104	10422
1PH4107-4NF2x-xxxx	104	10405
1PH4107-4xF5x-xxxx	104	10423
1PH4133-4NF2x-xxxx	104	10407
1PH4133-4xF5x-xxxx	104	10424
1PH4135-4NF2x-xxxx	104	10409
1PH4135-4xF5x-xxxx	104	10425

Table A-2 Motor code for induction motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1PH4137-4NF2x-xxxx	104	10411
1PH4137-4xF5x-xxxx	104	10426
1PH4138-4NF2x-xxxx	104	10413
1PH4163-4NF2x-xxxx	104	10416
1PH4163-4xF5x-xxxx	104	10427
1PH4163-xxF2x(L37)	104	10431
1PH4167-4NF2x-xxxx	104	10418
1PH4167-4xF5x-xxxx	104	10428
1PH4168-4NF2x-xxxx	104	10420
1PH4168-4xF5x-xxxx	104	10429
1PH7101-xxFxx-xLxx	107	12701
1PH7101-xxFxx-xxxx	107	10701
1PH7103-xxDxx-xLxx	107	12702
1PH7103-xxDxx-xxxx	107	10702
1PH7103-xxFxx-xLxx	107	12703
1PH7103-xxFxx-xxxx	107	10703
1PH7103-xxGxx-xLxx	107	12704
1PH7103-xxGxx-xxxx	107	10704
1PH7105-xxFxx-xLxx	107	12705
1PH7105-xxFxx-xxxx	107	10705
1PH7107-xxDxx-xLxx	107	12706
1PH7107-xxDxx-xxxx	107	10706
1PH7107-xxFxx-xLxx	107	12707
1PH7107-xxFxx-xxxx	107	10707
1PH7107-xxGxx-xLxx	107	12708
1PH7107-xxGxx-xxxx	107	10708
1PH7131-xxFxx-xLxx	107	12709
1PH7131-xxFxx-xxxx	107	10709
1PH7133-xxDxx-xLxx	107	12710
1PH7133-xxDxx-xxxx	107	10710
1PH7133-xxFxx-xLxx	107	12711
1PH7133-xxFxx-xxxx	107	10711
1PH7133-xxGxx-xLxx	107	12712
1PH7133-xxGxx-xxxx	107	10712
1PH7135-xxFxx-xLxx	107	12713

Table A-2 Motor code for induction motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1PH7135-xxFxx-xxxx	107	10713
1PH7137-xxBxx-xxxx	107	10766
1PH7137-xxDxx-xLxx	107	12714
1PH7137-xxDxx-xxxx	107	10714
1PH7137-xxFxx-xLxx	107	12715
1PH7137-xxFxx-xxxx	107	10715
1PH7137-xxGxx-xLxx	107	12716
1PH7137-xxGxx-xxxx	107	10716
1PH7163-xxBxx-xLxx	107	12717
1PH7163-xxBxx-xxxx	107	10717
1PH7163-xxDxx-xLxx	107	12718
1PH7163-xxDxx-xxxx	107	10718
1PH7163-xxFxx-xLxx	107	12719
1PH7163-xxFxx-xxxx	107	10719
1PH7163-xxGxx-xLxx	107	12720
1PH7163-xxGxx-xxxx	107	10720
1PH7167-xxBxx-xLxx	107	12721
1PH7167-xxBxx-xxxx	107	10721
1PH7167-xxDxx-xLxx	107	12722
1PH7167-xxDxx-xxxx	107	10722
1PH7167-xxFxx-xLxx	107	12723
1PH7167-xxFxx-xxxx	107	10723
1PH7167-xxGxx-xLxx	107	12724
1PH7167-xxGxx-xxxx	107	10724
1PH7184-xxBxx-xxxx	107	10725
1PH7184-xxDxx-xxxx	107	10735
1PH7184-xxExx-xxxx	107	10727
1PH7184-xxFxx-xxxx	107	10736
1PH7184-xxLxx-xxxx	107	10737
1PH7184-xxTxx-xxxx	107	10726
1PH7186-xxBxx-xxxx	107	10770
1PH7186-xxDxx-xxxx	107	10734
1PH7186-xxExx-xxxx	107	10730
1PH7186-xxFxx-xxxx	107	10768
1PH7186-xxLxx-xxxx	107	10769

Table A-2 Motor code for induction motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1PH7186-xxTxx-xxxx	107	10729
1PH7224-xxBxx-xxxx	107	10743
1PH7224-xxCxx-xxxx	107	10731
1PH7224-xxDxx-xxxx	107	10738
1PH7224-xxFxx-xxxx	107	10732
1PH7224-xxLxx-xxxx	107	10744
1PH7224-xxUxx-xxxx	107	10745
1PH7226-xxBxx-xxxx	107	10746
1PH7226-xxDxx-xxxx	107	10747
1PH7226-xxFxx-xxxx	107	10739
1PH7226-xxLxx-xxxx	107	10748
1PH7228-xxBxx-xxxx	107	10749
1PH7228-xxDxx-xxxx	107	10750
1PH7228-xxFxx-xxxx	107	10741
1PH7228-xxLxx-xxxx	107	10751
1PH7284-xxBxx-xxxx	107	10752
1PH7284-xxCxx-xxxx	107	10753
1PH7284-xxDxx-xxxx	107	10754
1PH7284-xxFxx-xxxx	107	10755
1PH7286-xxBxx-xxxx	107	10756
1PH7286-xxCxx-xxxx	107	10757
1PH7286-xxDxx-xxxx	107	10758
1PH7286-xxFxx-xxxx	107	10759
1PH7288-xxBxx-xxxx	107	10760
1PH7288-xxCxx-xxxx	107	10761
1PH7288-xxDxx-xxxx	107	10762
1PH7288-xxFxx-xxxx	107	10763
1PH8083-1xF0x-xxxx	108	10801
1PH8083-1xF1x-xxxx	108	10864
1PH8083-1xF2x-xxxx	108	10865
1PH8083-1xG0x-xxxx	108	10866
1PH8083-1xG1x-xxxx	108	10867
1PH8083-1xG2x-xxxx	108	10868
1PH8083-1xM0x-xxxx	108	10869
1PH8083-1xM1x-xxxx	108	10870

Table A-2 Motor code for induction motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1PH8083-1xN0x-xxxx	108	10879
1PH8083-1xN1x-xxxx	108	10880
1PH8083-1xN2x-xxxx	108	10881
1PH8087-1xF0x-xxxx	108	10871
1PH8087-1xF1x-xxxx	108	10872
1PH8087-1xF2x-xxxx	108	10873
1PH8087-1xG0x-xxxx	108	10874
1PH8087-1xG1x-xxxx	108	10875
1PH8087-1xG2x-xxxx	108	10876
1PH8087-1xM0x-xxxx	108	10877
1PH8087-1xM1x-xxxx	108	10878
1PH8087-1xN0x-xxxx	108	10882
1PH8087-1xN1x-xxxx	108	10883
1PH8087-1xN2x-xxxx	108	10884
1PH8101-1xF0x-xxxx	108	10885
1PH8101-1xF1x-xxxx	108	10886
1PH8101-1xF2x-xxxx	108	10887
1PH8101-1xG2x-xxxx	108	10888
1PH8101-1xS0x-xxxx	108	10889
1PH8101-1xS0x-xxxx	108	10890
1PH8101-1xS1x-xxxx	108	10891
1PH8101-1xS1x-xxxx	108	10892
1PH8103-1xD0x-xxxx	108	10893
1PH8103-1xD1x-xxxx	108	10894
1PH8103-1xF0x-xxxx	108	10895
1PH8103-1xF1x-xxxx	108	10896
1PH8103-1xF2x-xxxx	108	10897
1PH8103-1xG0x-xxxx	108	10898
1PH8103-1xG1x-xxxx	108	10899
1PH8103-1xG2x-xxxx	108	11800
1PH8103-1xM0x-xxxx	108	11820
1PH8103-1xM1x-xxxx	108	11821
1PH8103-1xM2x-xxxx	108	11822
1PH8105-1xF0x-xxxx	108	11801
1PH8105-1xF1x-xxxx	108	11802

Table A-2 Motor code for induction motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1PH8105-1xF2x-xxxx	108	11803
1PH8105-1xG2x-xxxx	108	11804
1PH8105-1xM2x-xxxx	108	11823
1PH8105-1xS0x-xxxx	108	11805
1PH8105-1xS0x-xxxx	108	11806
1PH8105-1xS1x-xxxx	108	11807
1PH8105-1xS1x-xxxx	108	11808
1PH8107-1xD0x-xxxx	108	11809
1PH8107-1xD1x-xxxx	108	11810
1PH8107-1xF0x-xxxx	108	11811
1PH8107-1xF1x-xxxx	108	11812
1PH8107-1xF2x-xxxx	108	11813
1PH8107-1xG0x-xxxx	108	11814
1PH8107-1xG1x-xxxx	108	11815
1PH8107-1xM0x-xxxx	108	11824
1PH8107-1xM1x-xxxx	108	11825
1PH8107-1xM2x-xxxx	108	11826
1PH8107-1xS0x-xxxx	108	11816
1PH8107-1xS0x-xxxx	108	11817
1PH8107-1xS1x-xxxx	108	11818
1PH8107-1xS1x-xxxx	108	11819
1PH8131-1xF0x-xxxx	108	10803
1PH8131-1xF1x-xxxx	108	10804
1PH8131-1xF2x-xxxx	108	10805
1PH8131-1xG2x-xxxx	108	10806
1PH8131-1xS0x-xxxx	108	10807
1PH8131-1xS0x-xxxx	108	10808
1PH8131-1xS1x-xxxx	108	10809
1PH8131-1xS1x-xxxx	108	10810
1PH8133-1xD0x-xxxx	108	10811
1PH8133-1xD1x-xxxx	108	10812
1PH8133-1xF0x-xxxx	108	10813
1PH8133-1xF1x-xxxx	108	10814
1PH8133-1xF2x-xxxx	108	10815
1PH8133-1xG0x-xxxx	108	10816

Table A-2 Motor code for induction motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1PH8133-1xG1x-xxxx	108	10817
1PH8133-1xG2x-xxxx	108	10818
1PH8135-1xF0x-xxxx	108	10819
1PH8135-1xF1x-xxxx	108	10820
1PH8135-1xF2x-xxxx	108	10821
1PH8135-1xG2x-xxxx	108	10822
1PH8135-1xS0x-xxxx	108	10823
1PH8135-1xS0x-xxxx	108	10824
1PH8135-1xS1x-xxxx	108	10825
1PH8135-1xS1x-xxxx	108	10826
1PH8137-1xD0x-xxxx	108	10827
1PH8137-1xD1x-xxxx	108	10828
1PH8137-1xS0x-xxxx	108	10834
1PH8137-1xS0x-xxxx	108	10835
1PH8137-1xS1x-xxxx	108	10836
1PH8137-1xS1x-xxxx	108	10837
1PH8184-1xB2x-xxxx	108	10839
1PH8184-1xC2x-xxxx	108	10840
1PH8184-1xD2x-xxxx	108	10841
1PH8184-1xF2x-xxxx	108	10842
1PH8184-1xL2x-xxxx	108	10843
1PH8186-1xB2x-xxxx	108	10844
1PH8186-1xC2x-xxxx	108	10845
1PH8186-1xD2x-xxxx	108	10846
1PH8186-1xF2x-xxxx	108	10847
1PH8186-1xL2x-xxxx	108	10848
1PH8224-1xB2x-xxxx	108	10849
1PH8224-1xC2x-xxxx	108	10850
1PH8224-1xD2x-xxxx	108	10851
1PH8224-1xF2x-xxxx	108	10852
1PH8224-1xL2x-xxxx	108	10853
1PH8226-1xB2x-xxxx	108	10854
1PH8226-1xC2x-xxxx	108	10855
1PH8226-1xD2x-xxxx	108	10856
1PH8226-1xF2x-xxxx	108	10857

Table A-2 Motor code for induction motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1PH8226-1xL2x-xxxx	108	10858
1PH8228-1xB2x-xxxx	108	10859
1PH8228-1xC2x-xxxx	108	10860
1PH8228-1xD2x-xxxx	108	10861
1PH8228-1xF2x-xxxx	108	10862
1PH8228-1xL2x-xxxx	108	10863
1PH8284-1xB2x-xxxx	108	11827
1PH8284-1xC2x-xxxx	108	11828
1PH8284-1xD2x-xxxx	108	11829
1PH8284-1xF2x-xxxx	108	11830
1PH8284-1xH2x-xxxx	108	11831
1PH8286-1xB2x-xxxx	108	11835
1PH8286-1xC2x-xxxx	108	11836
1PH8286-1xD2x-xxxx	108	11837
1PH8286-1xF2x-xxxx	108	11838
1PH8288-1xB2x-xxxx	108	11843
1PH8288-1xC2x-xxxx	108	11844
1PH8288-1xD2x-xxxx	108	11845
1PH8288-1xF2x-xxxx	108	11846
1PL6184-xxBxx-xxxx	166	16600
1PL6184-xxCxx-xxxx	166	16631
1PL6184-xxDxx-xxxx	166	16601
1PL6184-xxFxx-xxxx	166	16602
1PL6184-xxLxx-xxxx	166	16603
1PL6186-xxBxx-xxxx	166	16604
1PL6186-xxDxx-xxxx	166	16605
1PL6186-xxFxx-xxxx	166	16606
1PL6186-xxLxx-xxxx	166	16630
1PL6224-xxBxx-xxxx	166	16608
1PL6224-xxDxx-xxxx	166	16609
1PL6224-xxFxx-xxxx	166	16610
1PL6224-xxLxx-xxxx	166	16611
1PL6226-xxBxx-xxxx	166	16612
1PL6226-xxDxx-xxxx	166	16614
1PL6226-xxFxx-xxxx	166	16615

Table A-2 Motor code for induction motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1PL6226-xxLxx-xxxx	166	16616
1PL6228-xxBxx-xxxx	166	16617
1PL6228-xxDxx-xxxx	166	16618
1PL6228-xxFxx-xxxx	166	16619
1PL6228-xxLxx-xxxx	166	16620
1PL6284-xxCxx-xxxx	166	16621
1PL6284-xxDxx-xxxx	166	16622
1PL6284-xxFxx-xxxx	166	16623
1PL6286-xxCxx-xxxx	166	16624
1PL6286-xxDxx-xxxx	166	16626
1PL6286-xxFxx-xxxx	166	16625
1PL6288-xxCxx-xxxx	166	16627
1PL6288-xxDxx-xxxx	166	16628
1PL6288-xxFxx-xxxx	166	16629
1PM4101-xxF8x(L37)	134	14409
1PM4101-xxF8x(L37)	134	14410
1PM4101-xxF8x-xxxx	134	14401
1PM4101-xxF8x-xxxx	134	14402
1PM4101-xxW2x(L37)	134	13409
1PM4101-xxW2x-xxxx	134	13401
1PM4105-xxF8x(L37)	134	14411
1PM4105-xxF8x(L37)	134	14412
1PM4105-xxF8x-xxxx	134	14403
1PM4105-xxF8x-xxxx	134	14404
1PM4105-xxW2x(L37)	134	13411
1PM4105-xxW2x-xxxx	134	13403
1PM4133-xxF8x(L37)	134	14414
1PM4133-xxF8x(L37)	134	14413
1PM4133-xxF8x-xxxx	134	14405
1PM4133-xxF8x-xxxx	134	14406
1PM4133-xxW2x(L37)	134	13413
1PM4133-xxW2x-xxxx	134	13405
1PM4137-xxF8x(L37)	134	14415
1PM4137-xxF8x(L37)	134	14416
1PM4137-xxF8x-xxxx	134	14407

Table A-2 Motor code for induction motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1PM4137-xxF8x-xxxx	134	14408
1PM4137-xxW2x(L37)	134	13415
1PM4137-xxW2x-xxxx	134	13407
1PM6101-xxF8x(L37)	136	14615
1PM6101-xxF8x(L37)	136	14616
1PM6101-xxF8x-xxxx	136	14601
1PM6101-xxF8x-xxxx	136	14602
1PM6105-xxF8x(L37)	136	14617
1PM6105-xxF8x(L37)	136	14618
1PM6105-xxF8x-xxxx	136	14603
1PM6105-xxF8x-xxxx	136	14604
1PM6107-xxF8x-xxxx	136	14619
1PM6107-xxF8x-xxxx	136	14620
1PM6133-xxF8x(L37)	136	14621
1PM6133-xxF8x(L37)	136	14622
1PM6133-xxF8x-xxxx	136	14605
1PM6133-xxF8x-xxxx	136	14606
1PM6137-xxF8x(L37)	136	14623
1PM6137-xxF8x(L37)	136	14624
1PM6137-xxF8x-xxxx	136	14607
1PM6137-xxF8x-xxxx	136	14608
1PM6138-xxF8x(L37)	136	14626
1PM6138-xxF8x(L37)	136	14625
1PM6138-xxF8x-xxxx	136	14609
1PM6138-xxF8x-xxxx	136	14610
2SP1253-8xAxx-0xxx	191	19102
2SP1253-8xAxx-0xxx	191	19101
2SP1255-8xAxx-0xxx	191	19103
2SP1255-8xAxx-0xxx	191	19104

Synchronous motors (Version: 4301700)

Table A-3 Motor code for synchronous motors

Order number	Motor type (p0300)	Motor code (p0301)
1FE1051-4WL11-xxxx	261	26103
1FE1051-4WL51-xxxx	261	26104
1FE1051-4WN11-xxxx	261	26105
1FE1051-6WK10-xxxx	261	26106
1FE1051-6WN00-xxxx	261	26107
1FE1051-6WN10-xxxx	261	26108
1FE1051-6WN20-xxxx	261	26109
1FE1051-6WN30-xxxx	261	26110
1FE1052-4WK11-xxxx	261	26111
1FE1052-4WN11-xxxx	261	26112
1FE1052-4WN51-xxxx	261	26113
1FE1052-6LK00-xxxx	261	26114
1FE1052-6WK10-xxxx	261	26115
1FE1052-6WN00-xxxx	261	26116
1FE1052-6WN10-xxxx	261	26117
1FE1052-6WY10-xxxx	261	26118
1FE1053-4WN11-xxxx	261	26119
1FE1054-6LR00-xxxx	261	26120
1FE1054-6WQ10-xxxx	261	26122
1FE1054-6WR10-xxxx	261	26287
1FE1055-6LU00-xxxx	261	26123
1FE1055-6LX00-xxxx	261	26124
1FE1061-6LW00-xxxx	261	26125
1FE1061-6WV10-xxxx	261	26284
1FE1061-6WY10-xxxx	261	26126
1FE1064-6LQ00-xxxx	261	26127
1FE1064-6WN11-xxxx	261	26128
1FE1072-4WH11-xxxx	261	26129
1FE1072-4WL11-xxxx	261	26130
1FE1072-4WN01-xxxx	261	26131
1FE1072-4WN11-xxxx	261	26132
1FE1072-4WN31-xxxx	261	26133
1FE1073-4WL11-xxxx	261	26289
1FE1073-4WN01-xxxx	261	26134

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FE1073-4WN11-xxxx	261	26135
1FE1073-4WR01-xxxx	261	26136
1FE1073-4WT11-xxxx	261	26137
1FE1073-4WT31-xxxx	261	26138
1FE1074-4WM11-xxxx	261	26139
1FE1074-4WN11-xxxx	261	26140
1FE1074-4WN51-xxxx	261	26141
1FE1082-4WN01-xxxx	261	26142
1FE1082-4WN11-xxxx	261	26143
1FE1082-4WN51-xxxx	261	26144
1FE1082-4WP11-xxxx	261	26145
1FE1082-4WR11-xxxx	261	26146
1FE1082-4WR31-xxxx	261	26147
1FE1082-6WE11-xxxx	261	26285
1FE1082-6WP10-xxxx	261	26148
1FE1082-6WQ11-xxxx	261	26149
1FE1082-6WS10-xxxx	261	26150
1FE1082-6WS30-xxxx	261	26151
1FE1082-6WW11-xxxx	261	26152
1FE1083-4WN01-xxxx	261	26153
1FE1083-4WN11-xxxx	261	26154
1FE1084-4WN11-xxxx	261	26155
1FE1084-4WN31-xxxx	261	26156
1FE1084-4WP11-xxxx	261	26157
1FE1084-4WQ11-xxxx	261	26158
1FE1084-4WQ51-xxxx	261	26159
1FE1084-4WT11-xxxx	261	26160
1FE1084-4WT51-xxxx	261	26161
1FE1084-6LN00-xxxx	261	26162
1FE1084-6WN11-xxxx	261	26163
1FE1084-6WR11-xxxx	261	26164
1FE1084-6WX11-xxxx	261	26165
1FE1085-4WN11-xxxx	261	26166
1FE1085-4WQ11-xxxx	261	26167
1FE1085-4WT11-xxxx	261	26168

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FE1091-6WN10-xxxx	261	26169
1FE1091-6WN30-xxxx	261	26170
1FE1091-6WS10-xxxx	261	26171
1FE1092-4WV11-xxxx	261	26172
1FE1092-6WN00-xxxx	261	26173
1FE1092-6WN10-xxxx	261	26174
1FE1092-6WN30-xxxx	261	26175
1FE1092-6WR11-xxxx	261	26176
1FE1093-4WF01-xxxx	261	26177
1FE1093-4WH11-xxxx	261	26178
1FE1093-4WK01-xxxx	261	26179
1FE1093-4WM11-xxxx	261	26180
1FE1093-4WN01-xxxx	261	26181
1FE1093-4WN10-xxxx	261	26182
1FE1093-4WN11-xxxx	261	26183
1FE1093-6WN10-xxxx	261	26184
1FE1093-6WS10-xxxx	261	26185
1FE1093-6WS30-xxxx	261	26186
1FE1093-6WV01-xxxx	261	26286
1FE1093-6WV11-xxxx	261	26187
1FE1093-6WV31-xxxx	261	26188
1FE1093-7LN00-xxxx	261	26189
1FE1094-4LW01-xxxx	261	26190
1FE1094-4WK11-xxxx	261	26191
1FE1094-4WL11-xxxx	261	26192
1FE1094-4WS11-xxxx	261	26193
1FE1094-4WU11-xxxx	261	26243
1FE1095-4WN11-xxxx	261	26194
1FE1095-6LT01-xxxx	261	26195
1FE1095-6WU11-xxxx	261	26290
1FE1096-4WK10-xxxx	261	26196
1FE1096-4WN11-xxxx	261	26197
1FE1103-4WN01-xxxx	261	26245
1FE1103-4WN11-xxxx	261	26198
1FE1103-4WN31-xxxx	261	26199

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FE1103-4WQ01-xxxx	261	26200
1FE1103-4WQ11-xxxx	261	26201
1FE1103-4WT01-xxxx	261	26202
1FE1103-4WT11-xxxx	261	26203
1FE1103-4WU01-xxxx	261	26204
1FE1104-4WN11-xxxx	261	26205
1FE1105-4WN01-xxxx	261	26206
1FE1105-4WN11-xxxx	261	26207
1FE1105-4WQ01-xxxx	261	26208
1FE1105-4WQ11-xxxx	261	26209
1FE1106-4WN11-xxxx	261	26210
1FE1106-4WS11-xxxx	261	26211
1FE1106-4WY11-xxxx	261	26212
1FE1112-6LW01-xxxx	261	26213
1FE1113-6LU01-xxxx	261	26214
1FE1114-6LU11-xxxx	261	26215
1FE1114-6WR11-xxxx	261	26216
1FE1114-6WR31-xxxx	261	26217
1FE1114-6WT10-xxxx	261	26218
1FE1114-6WT11-xxxx	261	26219
1FE1114-6WT31-xxxx	261	26220
1FE1114-6WT51-xxxx	261	26221
1FE1114-6WW11-xxxx	261	26222
1FE1114-6WW31-xxxx	261	26223
1FE1116-6LS01-xxxx	261	26224
1FE1116-6LT01-xxxx	261	26225
1FE1116-6WR11-xxxx	261	26226
1FE1116-6WT11-xxxx	261	26227
1FE1116-6WW11-xxxx	261	26242
1FE1116-6WY11-xxxx	261	26228
1FE1124-4WN11-xxxx	261	26229
1FE1125-4WN11-xxxx	261	26230
1FE1125-4WP11-xxxx	261	26231
1FE1126-4WN11-xxxx	261	26232
1FE1126-4WP11-xxxx	261	26233

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FE1126-4WQ11-xxxx	261	26234
1FE1144-8WT10-xxxx	261	26244
1FE1145-8WN11-xxxx	261	26235
1FE1145-8WS11-xxxx	261	26237
1FE1147-8WN11-xxxx	261	26238
1FE1147-8WQ11-xxxx	261	26239
1FE1147-8WQ31-xxxx	261	26240
1FE1147-8WS11-xxxx	261	26241
1FK6032-6AK7x-xxxx	236	23601
1FK6033-7AK7x-xxxx	236	23602
1FK6040-6AK7x-xxxx	236	23603
1FK6042-6AF7x-xxxx	236	23604
1FK6043-7AH7x-xxxx	236	23605
1FK6043-7AK7x-xxxx	236	23606
1FK6044-7AF7x-xxxx	236	23607
1FK6044-7AH7x-xxxx	236	23608
1FK6060-6AF7x-xxxx	236	23609
1FK6061-7AF7x-xxxx	236	23610
1FK6061-7AH7x-xxxx	236	23611
1FK6063-6AF7x-xxxx	236	23612
1FK6064-7AF7x-xxxx	236	23613
1FK6064-7AH7x-xxxx	236	23614
1FK6080-6AF7x-xxxx	236	23615
1FK6082-7AF7x-xxxx	236	23616
1FK6083-6AF7x-xxxx	236	23617
1FK6085-7AF7x-xxxx	236	23618
1FK6100-8AF7x-xxxx	236	23619
1FK6101-8AF7x-xxxx	236	23620
1FK6103-8AF7x-xxxx	236	23621
1FK7011-xAK2x-xxxx	237	23738
1FK7011-xAK7x-xxxx	237	23747
1FK7015-xAK2x-xxxx	237	23739
1FK7015-xAK7x-xxxx	237	23748
1FK7022-xAK2x-xxxx	237	23733
1FK7022-xAK7x-xxxx	237	23726

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FK7024-xAK7x-xxxx	237	23753
1FK7032-xAF2x-xxxx	237	23742
1FK7032-xAK7x-xxxx	237	23727
1FK7033-xAF2x-xxxx	237	23741
1FK7033-xAK7x-xxxx	237	23701
1FK7034-xAF2x-xxxx	237	23740
1FK7034-xAK7x-xxxx	237	23732
1FK7040-xAK7x-xxxx	237	23702
1FK7042-xAC7x-xxxx	237	23749
1FK7042-xAF2x-xxxx	237	23735
1FK7042-xAF7x-xxxx	237	23703
1FK7042-xAK7x-xxxx	237	23704
1FK7042-xBK7x-xxxx	237	23765
1FK7043-xAF2x-xxxx	237	23743
1FK7043-xAH7x-xxxx	237	23705
1FK7043-xAK7x-xxxx	237	23706
1FK7044-xAF7x-xxxx	237	23707
1FK7044-xAH7x-xxxx	237	23708
1FK7060-xAF7x-xxxx	237	23709
1FK7060-xAH7x-xxxx	237	23710
1FK7060-xBF7x-xxxx	237	23766
1FK7061-xAF7x-xxxx	237	23711
1FK7061-xAH7x-xxxx	237	23712
1FK7062-xBF7x-xxxx	237	23750
1FK7063-xAF7x-xxxx	237	23713
1FK7063-xAH7x-xxxx	237	23714
1FK7064-xAF7x-xxxx	237	23715
1FK7064-xAH7x-xxxx	237	23716
1FK7080-xAF7x-xxxx	237	23717
1FK7080-xAH7x-xxxx	237	23718
1FK7081-xBF7x-xxxx	237	23751
1FK7082-xAF7x-xxxx	237	23719
1FK7083-xAF7x-xxxx	237	23720
1FK7083-xAH7x-xxxx	237	23721
1FK7084-xBC7x-xxxx	237	23752

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FK7084-xBF7x-xxxx	237	23772
1FK7085-xAF7x-xxxx	237	23722
1FK7086-xAA7x-xxxx	237	23737
1FK7086-xAC7x-xxxx	237	23744
1FK7086-xAF7x-xxxx	237	23731
1FK7086-xSF7x-xxxx	237	23730
1FK7100-xAF7x-xxxx	237	23723
1FK7101-xAC7x-xxxx	237	23745
1FK7101-xAF7x-xxxx	237	23724
1FK7103-xAC7x-xxxx	237	23746
1FK7103-xAF7x-xxxx	237	23725
1FK7105-xAC7x-xxxx	237	23728
1FK7105-xAF7x-xxxx	237	23729
1FS6074-xAC7x-xxxx	276	27601
1FS6074-xAF7x-xxxx	276	27602
1FS6074-xAH7x-xxxx	276	27603
1FS6074-xAK7x-xxxx	276	27604
1FS6096-xAC7x-xxxx	276	27605
1FS6096-xAF7x-xxxx	276	27606
1FS6096-xAH7x-xxxx	276	27607
1FS6115-xAB7x-xxxx	276	27608
1FS6115-xAC7x-xxxx	276	27609
1FS6115-xAF7x-xxxx	276	27610
1FS6134-xAB7x-xxxx	276	27611
1FS6134-xAC7x-xxxx	276	27612
1FS6134-xAF7x-xxxx	276	27613
1FT6021-6AK7x-xxxx	206	20601
1FT6024-6AK7x-xxxx	206	20602
1FT6031-xAK7x-xxxx	206	20603
1FT6034-xAK7x-xxxx	206	20604
1FT6041-xAF7x-xxxx	206	20605
1FT6041-xAK7x-xxxx	206	20606
1FT6044-xAF7x-xxxx	206	20607
1FT6044-xAK7x-xxxx	206	20608
1FT6061-xAC7x-xxxx	206	20609

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FT6061-xAF7x-xxxx	206	20610
1FT6061-xAH7x-xxxx	206	20611
1FT6061-xAK7x-xxxx	206	20612
1FT6062-xAC7x-xxxx	206	20613
1FT6062-xAF7x-xxxx	206	20614
1FT6062-xAH7x-xxxx	206	20615
1FT6062-xAK7x-xxxx	206	20616
1FT6062-xWF7x-xxxx	206	22601
1FT6062-xWH7x-xxxx	206	22602
1FT6062-xWK7x-xxxx	206	22603
1FT6064-xAC7x-xxxx	206	20617
1FT6064-xAF7x-xxxx	206	20618
1FT6064-xAH7x-xxxx	206	20619
1FT6064-xAK7x-xxxx	206	20620
1FT6064-xWF7x-xxxx	206	22604
1FT6064-xWH7x-xxxx	206	22605
1FT6064-xWK7x-xxxx	206	22606
1FT6081-xAC7x-xxxx	206	20621
1FT6081-xAF7x-xxxx	206	20622
1FT6081-xAH7x-xxxx	206	20623
1FT6081-xAK7x-xxxx	206	20624
1FT6082-xAC7x-xxxx	206	20625
1FT6082-xAF7x-xxxx	206	20626
1FT6082-xAH7x-xxxx	206	20627
1FT6082-xAK7x-xxxx	206	20628
1FT6082-xWH7x-xxxx	206	22630
1FT6084-xAC7x-xxxx	206	20629
1FT6084-xAF7x-xxxx	206	20630
1FT6084-xAH7x-xxxx	206	20631
1FT6084-xAK7x-xxxx	206	20632
1FT6084-xSF7x-xxxx	206	21601
1FT6084-xSH7x-xxxx	206	21602
1FT6084-xSK7x-xxxx	206	21603
1FT6084-xWF7x-xxxx	206	22607
1FT6084-xWH7x-xxxx	206	22608

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FT6084-xWK7x-xxxx	206	22609
1FT6086-xAC7x-xxxx	206	20633
1FT6086-xAF7x-xxxx	206	20634
1FT6086-xAH7x-xxxx	206	20635
1FT6086-xSF7x-xxxx	206	21604
1FT6086-xSG7x-xxxx	206	21626
1FT6086-xSH7x-xxxx	206	21605
1FT6086-xSK7x-xxxx	206	21606
1FT6086-xWF7x-xxxx	206	22610
1FT6086-xWH7x-xxxx	206	22611
1FT6086-xWK7x-xxxx	206	22612
1FT6102-xAB7x-xxxx	206	20636
1FT6102-xAC7x-xxxx	206	20637
1FT6102-xAF7x-xxxx	206	20638
1FT6102-xAH7x-xxxx	206	20639
1FT6105-xAB7x-xxxx	206	20640
1FT6105-xAC7x-xxxx	206	20641
1FT6105-xAF7x-xxxx	206	20642
1FT6105-xSB7x-xxxx	206	21607
1FT6105-xSC7x-xxxx	206	21608
1FT6105-xSF7x-xxxx	206	21609
1FT6105-xSH7x-xxxx	206	21610
1FT6105-xWC7x-xxxx	206	22613
1FT6105-xWF7x-xxxx	206	22614
1FT6108-xAB7x-xxxx	206	20643
1FT6108-xAC7x-xxxx	206	20644
1FT6108-xAF7x-xxxx	206	20645
1FT6108-xSB7x-xxxx	206	21611
1FT6108-xSC7x-xxxx	206	21612
1FT6108-xSF7x-xxxx	206	21613
1FT6108-xWB7x-xxxx	206	22615
1FT6108-xWC7x-xxxx	206	22616
1FT6108-xWF7x-xxxx	206	22617
1FT6132-xAB7x-xxxx	206	20646
1FT6132-xAC7x-xxxx	206	20647

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FT6132-xAF7x-xxxx	206	20648
1FT6132-xSB7x-xxxx	206	21614
1FT6132-xSC7x-xxxx	206	21615
1FT6132-xSF7x-xxxx	206	21616
1FT6132-xWB7x-xxxx	206	22618
1FT6132-xWD7x-xxxx	206	22619
1FT6134-xAB7x-xxxx	206	20649
1FT6134-xAC7x-xxxx	206	20650
1FT6134-xSB7x-xxxx	206	21617
1FT6134-xSC7x-xxxx	206	21618
1FT6134-xSF7x-xxxx	206	21619
1FT6134-xWB7x-xxxx	206	22620
1FT6134-xWD7x-xxxx	206	22621
1FT6136-xAB7x-xxxx	206	20651
1FT6136-xAC7x-xxxx	206	20652
1FT6136-xSB7x-xxxx	206	21620
1FT6136-xSC7x-xxxx	206	21621
1FT6136-xSF7x-xxxx	206	21622
1FT6136-xWB7x-xxxx	206	22622
1FT6136-xWD7x-xxxx	206	22623
1FT6138-xWB7x-xxxx	206	22624
1FT6138-xWD7x-xxxx	206	22625
1FT6163-xSB7x-xxxx	206	21623
1FT6163-xSD7x-xxxx	206	21624
1FT6163-xWB7x-xxxx	206	22626
1FT6163-xWD7x-xxxx	206	22627
1FT6168-xSB7x-xxxx	206	21625
1FT6168-xWB7x-xxxx	206	22628
1FT7034-xAK7x-xxxx	207	20740
1FT7036-xAK7x-xxxx	207	20741
1FT7042-xAF7x-xxxx	207	20701
1FT7042-xAK7x-xxxx	207	20702
1FT7044-xAF7x-xxxx	207	20703
1FT7044-xAK7x-xxxx	207	20704
1FT7046-xAF7x-xxxx	207	20705

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FT7046-xAH7x-xxxx	207	20732
1FT7046-xAK7x-xxxx	207	20706
1FT7062-xAF7x-xxxx	207	20716
1FT7062-xAK7x-xxxx	207	20717
1FT7062-xWF7x-xxxx	207	20745
1FT7062-xWK7x-xxxx	207	20746
1FT7064-xAF7x-xxxx	207	20720
1FT7064-xAK7x-xxxx	207	20721
1FT7064-xWF7x-xxxx	207	20747
1FT7064-xWK7x-xxxx	207	20748
1FT7065-xSF7x-xxxx	207	20781
1FT7065-xSH7x-xxxx	207	20782
1FT7065-xWF7x-xxxx	207	20770
1FT7065-xWH7x-xxxx	207	20771
1FT7066-xAF7x-xxxx	207	20722
1FT7066-xAH7x-xxxx	207	20733
1FT7066-xWF7x-xxxx	207	20749
1FT7066-xWH7x-xxxx	207	20750
1FT7067-xSF7x-xxxx	207	20783
1FT7067-xSH7x-xxxx	207	20784
1FT7067-xWF7x-xxxx	207	20772
1FT7067-xWH7x-xxxx	207	20773
1FT7068-xAF7x-xxxx	207	20725
1FT7068-xWF7x-xxxx	207	20751
1FT7082-xAC7x-xxxx	207	20734
1FT7082-xAF7x-xxxx	207	20709
1FT7082-xAH7x-xxxx	207	20707
1FT7082-xWC7x-xxxx	207	20752
1FT7082-xWF7x-xxxx	207	20753
1FT7082-xWH7x-xxxx	207	20754
1FT7084-xAC7x-xxxx	207	20735
1FT7084-xAF7x-xxxx	207	20711
1FT7084-xAH7x-xxxx	207	20712
1FT7084-xSC7x-xxxx	207	20789
1FT7084-xSF7x-xxxx	207	20790

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FT7084-xSH7x-xxxx	207	20791
1FT7084-xWC7x-xxxx	207	20755
1FT7084-xWF7x-xxxx	207	20756
1FT7084-xWH7x-xxxx	207	20757
1FT7085-xSF7x-xxxx	207	20774
1FT7085-xSH7x-xxxx	207	20775
1FT7085-xWF7x-xxxx	207	20776
1FT7085-xWH7x-xxxx	207	20777
1FT7086-xAC7x-xxxx	207	20736
1FT7086-xAF7x-xxxx	207	20714
1FT7086-xAH7x-xxxx	207	20715
1FT7086-xSC7x-xxxx	207	20792
1FT7086-xSF7x-xxxx	207	20793
1FT7086-xSH7x-xxxx	207	20794
1FT7086-xWC7x-xxxx	207	20758
1FT7086-xWF7x-xxxx	207	20759
1FT7086-xWH7x-xxxx	207	20760
1FT7087-xSF7x-xxxx	207	20778
1FT7087-xSH7x-xxxx	207	20779
1FT7087-xWF7x-xxxx	207	20769
1FT7087-xWH7x-xxxx	207	20780
1FT7102-xAB7x-xxxx	207	20726
1FT7102-xAC7x-xxxx	207	20737
1FT7102-xAF7x-xxxx	207	20727
1FT7102-xWB7x-xxxx	207	20761
1FT7102-xWC7x-xxxx	207	20762
1FT7102-xWF7x-xxxx	207	20763
1FT7105-xAB7x-xxxx	207	20728
1FT7105-xAC7x-xxxx	207	20738
1FT7105-xAF7x-xxxx	207	20729
1FT7105-xSC7x-xxxx	207	20785
1FT7105-xSF7x-xxxx	207	20786
1FT7105-xWB7x-xxxx	207	20744
1FT7105-xWC7x-xxxx	207	20764
1FT7105-xWF7x-xxxx	207	20765

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FT7108-xAB7x-xxxx	207	20730
1FT7108-xAC7x-xxxx	207	20739
1FT7108-xAF7x-xxxx	207	20731
1FT7108-xSC7x-xxxx	207	20787
1FT7108-xSF7x-xxxx	207	20788
1FT7108-xWB7x-xxxx	207	20742
1FT7108-xWC7x-xxxx	207	20766
1FT7108-xWF7x-xxxx	207	20767
1FW3150-1xH7x-xxxx	283	28301
1FW3150-1xL7x-xxxx	283	28302
1FW3150-1xP7x-xxxx	283	28303
1FW3152-1xH7x-xxxx	283	28304
1FW3152-1xL7x-xxxx	283	28305
1FW3152-1xP7x-xxxx	283	28306
1FW3154-1xH7x-xxxx	283	28307
1FW3154-1xL7x-xxxx	283	28308
1FW3154-1xP7x-xxxx	283	28309
1FW3155-1xH7x-xxxx	283	28310
1FW3155-1xL7x-xxxx	283	28311
1FW3155-1xP7x-xxxx	283	28312
1FW3156-1xH7x-xxxx	283	28313
1FW3156-1xL7x-xxxx	283	28314
1FW3156-1xP7x-xxxx	283	28315
1FW3201-1xE7x-xxxx	283	28316
1FW3201-1xH7x-xxxx	283	28317
1FW3201-1xL7x-xxxx	283	28318
1FW3202-1xE7x-xxxx	283	28319
1FW3202-1xH7x-xxxx	283	28320
1FW3202-1xL7x-xxxx	283	28321
1FW3203-1xE7x-xxxx	283	28322
1FW3203-1xH7x-xxxx	283	28323
1FW3203-1xL7x-xxxx	283	28324
1FW3204-1xE7x-xxxx	283	28325
1FW3204-1xH7x-xxxx	283	28326
1FW3204-1xL7x-xxxx	283	28327

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FW3206-1xE7x-xxxx	283	28328
1FW3206-1xH7x-xxxx	283	28329
1FW3206-1xL7x-xxxx	283	28330
1FW3208-1xE7x-xxxx	283	28331
1FW3208-1xH7x-xxxx	283	28332
1FW3208-1xL7x-xxxx	283	28333
1FW3281-1xE7x-xxxx	283	28334
1FW3281-1xG7x-xxxx	283	28335
1FW3281-2xE7x-xxxx	283	29301
1FW3281-2xG7x-xxxx	283	29302
1FW3281-3xJ7x-xxxx	283	29303
1FW3281-3xM7x-xxxx	283	29304
1FW3283-1xE7x-xxxx	283	28336
1FW3283-1xG7x-xxxx	283	28337
1FW3283-2xE7x-xxxx	283	29305
1FW3283-2xG7x-xxxx	283	29306
1FW3283-3xJ7x-xxxx	283	29307
1FW3283-3xM7x-xxxx	283	29308
1FW3285-1xE7x-xxxx	283	28338
1FW3285-1xG7x-xxxx	283	28339
1FW3285-2xE7x-xxxx	283	29309
1FW3285-2xG7x-xxxx	283	29310
1FW3285-3xJ7x-xxxx	283	29311
1FW3285-3xM7x-xxxx	283	29312
1FW3287-2xE7x-xxxx	283	29313
1FW3287-2xG7x-xxxx	283	29314
1FW3287-3xJ7x-xxxx	283	29315
1FW3287-3xM7x-xxxx	283	29316
1FW3288-1xE7x-xxxx	283	28340
1FW3288-1xG7x-xxxx	283	28341
1FW6090-0xx05-0Fxx	286	28601
1FW6090-0xx05-0Kxx	286	28602
1FW6090-0xx07-0Kxx	286	28603
1FW6090-0xx07-1Jxx	286	28604
1FW6090-0xx10-0Kxx	286	28605

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FW6090-0xx10-1Jxx	286	28606
1FW6090-0xx15-1Jxx	286	28607
1FW6090-0xx15-2Jxx	286	28608
1FW6130-0xx05-0Kxx	286	28620
1FW6130-0xx05-1Jxx	286	28621
1FW6130-0xx07-0Kxx	286	28622
1FW6130-0xx07-1Jxx	286	28623
1FW6130-0xx10-1Jxx	286	28624
1FW6130-0xx10-2Jxx	286	28625
1FW6130-0xx15-1Jxx	286	28626
1FW6130-0xx15-2Jxx	286	28627
1FW6150-0xx05-1Jxx	286	28642
1FW6150-0xx05-4Fxx	286	28643
1FW6150-0xx07-2Jxx	286	28644
1FW6150-0xx07-4Fxx	286	28645
1FW6150-0xx10-2Jxx	286	28646
1FW6150-0xx10-4Fxx	286	28647
1FW6150-0xx15-2Jxx	286	28648
1FW6150-0xx15-4Fxx	286	28649
1FW6160-0xx05-1Jxx	286	28628
1FW6160-0xx05-2Jxx	286	28629
1FW6160-0xx07-1Jxx	286	28630
1FW6160-0xx07-2Jxx	286	28631
1FW6160-0xx10-1Jxx	286	28632
1FW6160-0xx10-2Jxx	286	28633
1FW6160-0xx15-2Jxx	286	28634
1FW6160-0xx15-5Gxx	286	28635
1FW6160-xxx05-5Gxx	286	28658
1FW6160-xxx07-5Gxx	286	28659
1FW6160-xxx07-8Fxx	286	28660
1FW6160-xxx10-2Pxx	286	28661
1FW6160-xxx10-5Gxx	286	28662
1FW6160-xxx10-8Fxx	286	28663
1FW6160-xxx15-0Wxx	286	28664
1FW6160-xxx15-2Pxx	286	28665

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FW6160-xxx15-8Fxx	286	28666
1FW6160-xxx20-0Wxx	286	28667
1FW6160-xxx20-2Pxx	286	28668
1FW6160-xxx20-5Gxx	286	28669
1FW6160-xxx20-8Fxx	286	28670
1FW6190-0xx05-1Jxx	286	28636
1FW6190-0xx05-2Jxx	286	28637
1FW6190-0xx07-1Jxx	286	28638
1FW6190-0xx07-2Jxx	286	28639
1FW6190-0xx10-1Jxx	286	28640
1FW6190-0xx10-2Jxx	286	28641
1FW6190-0xx15-2Jxx	286	28609
1FW6190-0xx15-5Gxx	286	28610
1FW6190-xxx05-5Gxx	286	28671
1FW6190-xxx07-5Gxx	286	28672
1FW6190-xxx07-8Fxx	286	28673
1FW6190-xxx10-2Pxx	286	28674
1FW6190-xxx10-5Gxx	286	28675
1FW6190-xxx10-8Fxx	286	28676
1FW6190-xxx15-0Wxx	286	28677
1FW6190-xxx15-2Pxx	286	28678
1FW6190-xxx15-8Fxx	286	28679
1FW6190-xxx20-0Wxx	286	28680
1FW6190-xxx20-2Pxx	286	28681
1FW6190-xxx20-5Gxx	286	28682
1FW6190-xxx20-8Fxx	286	28683
1FW6230-0xx05-1Jxx	286	28611
1FW6230-0xx05-2Jxx	286	28612
1FW6230-0xx07-1Jxxx	286	28613
1FW6230-0xx07-2Jxx	286	28614
1FW6230-0xx10-2Jxx	286	28615
1FW6230-0xx10-5Gxx	286	28616
1FW6230-0xx15-4Cxx	286	28617
1FW6230-0xx15-5Gxx	286	28618
1FW6230-xxx05-5Gxx	286	28684

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FW6230-xxx07-5Gxx	286	28685
1FW6230-xxx07-8Fxx	286	28686
1FW6230-xxx10-2Pxx	286	28687
1FW6230-xxx10-8Fxx	286	28688
1FW6230-xxx15-0Wxx	286	28689
1FW6230-xxx15-2Pxx	286	28690
1FW6230-xxx15-8Fxx	286	28691
1FW6230-xxx20-0Wxx	286	28692
1FW6230-xxx20-2Pxx	286	28693
1FW6230-xxx20-5Gxx	286	28694
1FW6230-xxx20-8Fxx	286	28695
1FW6290-0xx15-7Axx	286	28619
1FW6290-xxx07-0Lxx	286	28696
1FW6290-xxx07-2Pxx	286	28697
1FW6290-xxx07-5Gxx	286	28698
1FW6290-xxx11-0Lxx	286	28699
1FW6290-xxx11-2Pxx	286	29600
1FW6290-xxx11-7Axx	286	29601
1FW6290-xxx15-0Lxx	286	29602
1FW6290-xxx15-2Pxx	286	29603
1FW6290-xxx20-0Lxx	286	29604
1FW6290-xxx20-2Pxx	286	29605
1LE400x-1ABxx-xxxx	204	20401
1LE400x-1BBxx-xxxx	204	20402
1PH8131-2xF0x-xxxx	200	20001
1PH8131-2xF1x-xxxx	200	20002
1PH8131-2xF2x-xxxx	200	20003
1PH8131-2xL0x-xxxx	200	20004
1PH8131-2xL1x-xxxx	200	20005
1PH8131-2xL2x-xxxx	200	20006
1PH8133-2xF0x-xxxx	200	20007
1PH8133-2xF1x-xxxx	200	20008
1PH8133-2xF2x-xxxx	200	20009
1PH8133-2xG2x-xxxx	200	20010
1PH8133-2xL0x-xxxx	200	20011

Table A-3 Motor code for synchronous motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1PH8133-2xL1x-xxxx	200	20012
1PH8135-2xF0x-xxxx	200	20013
1PH8135-2xF1x-xxxx	200	20014
1PH8135-2xF2x-xxxx	200	20015
1PH8135-2xG0x-xxxx	200	20016
1PH8135-2xG1x-xxxx	200	20017
1PH8135-2xG2x-xxxx	200	20018
1PH8137-2xF0x-xxxx	200	20019
1PH8137-2xF1x-xxxx	200	20020
1PH8137-2xF2x-xxxx	200	20021
1PH8137-2xG2x-xxxx	200	20022
1PH8137-2xL0x-xxxx	200	20023
1PH8137-2xL1x-xxxx	200	20024
1PH8137-2xM0x-xxxx	200	20025
1PH8137-2xM1x-xxxx	200	20026
1PH8138-2xF2x-xxxx	200	20027
1PH8138-2xG2x-xxxx	200	20028
1PH8184-2xC2x-xxxx	200	20029
1PH8184-2xD2x-xxxx	200	20030
1PH8184-2xF2x-xxxx	200	20031
1PH8184-2xL2x-xxxx	200	20032
1PH8186-2xC2x-xxxx	200	20033
1PH8186-2xD2x-xxxx	200	20034
1PH8186-2xF2x-xxxx	200	20035
1PH8186-2xL2x-xxxx	200	20036
2SP1202-1HAxx-xxxx	291	29101
2SP1202-1HBxx-xxxx	291	29102
2SP1204-1HAxx-xxxx	291	29103
2SP1204-1HBxx-xxxx	291	29104
2SP1253-1xAxx-xxxx	291	29105
2SP1253-1xBxx-xxxx	291	29106
2SP1255-1xAxx-xxxx	291	29107
2SP1255-1xBxx-xxxx	291	29108

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Table A-4 Motor code for linear motors

Order number	Motor type (p0300)	Motor code (p0301)
1FN1072-3xF7x-xxxx	401	40131
1FN1076-3xF7x-xxxx	401	40132
1FN1122-5xC7x-xxxx	401	40103
1FN1122-5xF7x-xxxx	401	40121
1FN1124-5xC7x-xxxx	401	40101
1FN1124-5xF7x-xxxx	401	40123
1FN1126-5xC7x-xxxx	401	40104
1FN1126-5xF7x-xxxx	401	40122
1FN1184-5xC7x-xxxx	401	40102
1FN1184-5xF7x-xxxx	401	40124
1FN1186-5xC7x-xxxx	401	40105
1FN1186-5xF7x-xxxx	401	40125
1FN1244-5xC7x-xxxx	401	40106
1FN1244-5xF7x-xxxx	401	40126
1FN1246-5xC7x-xxxx	401	40107
1FN1246-5xF7x-xxxx	401	40127
1FN3050-1KD0x-xxxx	403	41329
1FN3050-1ND0x-xxxx	403	41301
1FN3050-2KC4x-xxxx	403	41328
1FN3050-2NB8x-xxxx	403	41302
1FN3050-2WC0x-xxxx	403	40349
1FN3100-1KC5x-xxxx	403	41331
1FN3100-1NC0x-xxxx	403	41303
1FN3100-1WC0x-xxxx	403	40341
1FN3100-2KC5x-xxxx	403	41325
1FN3100-2NC8x-xxxx	403	41304
1FN3100-2WC0x-xxxx	403	40302
1FN3100-2WE0x-xxxx	403	40303
1FN3100-3KC5x-xxxx	403	41326
1FN3100-3NC0x-xxxx	403	41305
1FN3100-3WC0x-xxxx	403	40342
1FN3100-3WE0x-xxxx	403	40304
1FN3100-4NC8x-xxxx	403	41306
1FN3100-4WC0x-xxxx	403	40305

Table A-4 Motor code for linear motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FN3100-4WE0x-xxxx	403	40306
1FN3100-5WC0x-xxxx	403	40307
1FN3150-1KC7x-xxxx	403	41324
1FN3150-1NC2x-xxxx	403	41307
1FN3150-1WC0x-xxxx	403	40308
1FN3150-1WE0x-xxxx	403	40309
1FN3150-2KC7x-xxxx	403	41327
1FN3150-2NB8x-xxxx	403	41308
1FN3150-2WC0x-xxxx	403	40310
1FN3150-3KC7x-xxxx	403	41330
1FN3150-3NC7x-xxxx	403	41309
1FN3150-3WC0x-xxxx	403	40311
1FN3150-4NB8x-xxxx	403	41310
1FN3150-4WC0x-xxxx	403	40312
1FN3150-5WC0x-xxxx	403	40313
1FN3300-1NC1x-xxxx	403	41311
1FN3300-1WC0x-xxxx	403	40343
1FN3300-2NC1x-xxxx	403	41312
1FN3300-2WB0x-xxxx	403	40314
1FN3300-2WC0x-xxxx	403	40315
1FN3300-2WG0x-xxxx	403	40316
1FN3300-3NC4x-xxxx	403	41313
1FN3300-3WC0x-xxxx	403	40317
1FN3300-3WG0x-xxxx	403	40318
1FN3300-4NB8x-xxxx	403	41314
1FN3300-4WB0x-xxxx	403	40319
1FN3300-4WC0x-xxxx	403	40320
1FN3450-2NC5x-xxxx	403	41315
1FN3450-2WA5x-xxxx	403	40344
1FN3450-2WC0x-xxxx	403	40321
1FN3450-2WE0x-xxxx	403	40322
1FN3450-3NC5x-xxxx	403	41316
1FN3450-3WA5x-xxxx	403	40345
1FN3450-3WB0x-xxxx	403	40323
1FN3450-3WB5x-xxxx	403	40324

Table A-4 Motor code for linear motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FN3450-3WC0x-xxxx	403	40325
1FN3450-3WE0x-xxxx	403	40326
1FN3450-4NB8x-xxxx	403	41317
1FN3450-4WB0x-xxxx	403	40327
1FN3450-4WB5x-xxxx	403	40328
1FN3450-4WC0x-xxxx	403	40329
1FN3450-4WE0x-xxxx	403	40330
1FN3600-2NB8x-xxxx	403	41318
1FN3600-2WA5x-xxxx	403	40346
1FN3600-3NB8x-xxxx	403	41319
1FN3600-3WB0x-xxxx	403	40331
1FN3600-3WC0x-xxxx	403	40332
1FN3600-4NB8x-xxxx	403	41320
1FN3600-4WA3x-xxxx	403	40347
1FN3600-4WB0x-xxxx	403	40333
1FN3600-4WB5x-xxxx	403	40334
1FN3600-4WC0x-xxxx	403	40335
1FN3900-2NB2x-xxxx	403	41321
1FN3900-2WB0x-xxxx	403	40336
1FN3900-2WC0x-xxxx	403	40337
1FN3900-3NB2x-xxxx	403	41322
1FN3900-3WB0x-xxxx	403	40348
1FN3900-4NB2x-xxxx	403	41323
1FN3900-4WB0x-xxxx	403	40338
1FN3900-4WB5x-xxxx	403	40339
1FN3900-4WC0x-xxxx	403	40340
1FN6003-xLC57-xxxx	406	40601
1FN6003-xLC84-xxxx	406	40602
1FN6003-xLE38-xxxx	406	40603
1FN6003-xLE88-xxxx	406	40604
1FN6003-xLG24-xxxx	406	40605
1FN6003-xLG61-xxxx	406	40606
1FN6003-xLJ17-xxxx	406	40607
1FN6003-xLJ44-xxxx	406	40608
1FN6003-xLL12-xxxx	406	40609

Table A-4 Motor code for linear motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FN6003-xLL35-xxxx	406	40610
1FN6003-xLN10-xxxx	406	40611
1FN6003-xLN28-xxxx	406	40612
1FN6003-xWC57-xxxx	406	40615
1FN6003-xWC84-xxxx	406	40616
1FN6003-xWE38-xxxx	406	40617
1FN6003-xWE88-xxxx	406	40618
1FN6003-xWG24-xxxx	406	40619
1FN6003-xWG61-xxxx	406	40620
1FN6003-xWJ17-xxxx	406	40621
1FN6003-xWJ44-xxxx	406	40622
1FN6003-xWL12-xxxx	406	40623
1FN6003-xWL35-xxxx	406	40624
1FN6003-xWN10-xxxx	406	40625
1FN6003-xWN28-xxxx	406	40626
1FN6007-xLC31-xxxx	406	40629
1FN6007-xLC46-xxxx	406	40630
1FN6007-xLE20-xxxx	406	40631
1FN6007-xLE53-xxxx	406	40632
1FN6007-xLG12-xxxx	406	40633
1FN6007-xLG33-xxxx	406	40634
1FN6007-xLJ08-xxxx	406	40635
1FN6007-xLJ24-xxxx	406	40636
1FN6007-xLL05-xxxx	406	40637
1FN6007-xLL18-xxxx	406	40638
1FN6007-xLN15-xxxx	406	40639
1FN6007-xLN32-xxxx	406	40640
1FN6007-xWC31-xxxx	406	40643
1FN6007-xWC46-xxxx	406	40644
1FN6007-xWE20-xxxx	406	40645
1FN6007-xWE53-xxxx	406	40646
1FN6007-xWG12-xxxx	406	40647
1FN6007-xWG33-xxxx	406	40648
1FN6007-xWJ08-xxxx	406	40649
1FN6007-xWJ24-xxxx	406	40650

Table A-4 Motor code for linear motors, continued

Order number	Motor type (p0300)	Motor code (p0301)
1FN6007-xWL05-xxxx	406	40651
1FN6007-xWL18-xxxx	406	40652
1FN6007-xWN15-xxxx	406	40653
1FN6007-xWN32-xxxx	406	40654
1FN6008-xLC17-xxxx	406	40657
1FN6008-xLC37-xxxx	406	40658
1FN6008-xLE16-xxxx	406	40659
1FN6008-xLE34-xxxx	406	40660
1FN6008-xLG16-xxxx	406	40661
1FN6008-xLG33-xxxx	406	40662
1FN6016-xLC18-xxxx	406	40663
1FN6016-xLC30-xxxx	406	40664
1FN6016-xLE17-xxxx	406	40665
1FN6016-xLE27-xxxx	406	40666
1FN6016-xLG16-xxxx	406	40667
1FN6016-xLG26-xxxx	406	40668
1FN6024-xLC12-xxxx	406	40669
1FN6024-xLC20-xxxx	406	40670
1FN6024-xLE11-xxxx	406	40671
1FN6024-xLE18-xxxx	406	40672
1FN6024-xLG10-xxxx	406	40673
1FN6024-xLG17-xxxx	406	40674

A.2.2 Encoder code

1FK6 encoders

Table A-5 Encoder code for 1FK6 encoders

Order number	Encoder code (p0400)	Remark
1FK6xxx-xxxxx-xAxx	2001	-
1FK6xxx-xxxxx-xExx	2051	-
1FK6xxx-xxxxx-xGxx	2052	-
1FK6xxx-xxxxx-xHxx	2053	-
1FK6xxx-xxxxx-xJxx	2054	-
1FK6xxx-xxxxx-xSxx	1002 1003 1004	4p (2-speed) 6p (3-speed) 8p (4-speed) The pole number of the resolver corresponds to the pole number of the motor (see catalog).
1FK6xxx-xxxxx-xTxx	1001	-

1FK7 encoders

Table A-6 Encoder code for 1FK7 encoders

Order number	Encoder code (p0400)	Remark
1FK7xxx-xxxxx-xAxx	2001	-
1FK7xxx-xxxxx-xExx	2051	-
1FK7xxx-xxxxx-xGxx	2052	-
1FK7xxx-xxxxx-xHxx	2053	-
1FK7xxx-xxxxx-xJxx	2054	-
1FK7xxx-xxxxx-xSxx	1002 1003 1004	4p (2-speed) 6p (3-speed) 8p (4-speed) The pole number of the resolver corresponds to the pole number of the motor (see catalog).
1FK7xxx-xxxxx-xTxx	1001	-

1FS6 encoders

Table A-7 Encoder code for 1FS6 encoders

Order number	Encoder code (p0400)	Remark
1FS6xxx-xxxxx-xAxx	2001	-
1FS6xxx-xxxxx-xExx	2051	-

1FT6 encoders

Table A-8 Encoder code for 1FT6 encoders

Order number	Encoder code (p0400)	Remark
1FT6xxx-xxxxx-xAxx	2001	-
1FT6xxx-xxxxx-xExx	2051	-
1FT6xxx-xxxxx-xHxx	2053	-
1FT6xxx-4xxxx-xSxx	1002	4p (2-speed)
1FT6xxx-6xxxx-xSxx	1003	6p (3-speed)
1FT6xxx-8xxxx-xSxx	1004	8p (4-speed)
1FT6xxx-xxxxx-xTxx	1001	-

1FW3 encoders

Table A-9 Encoder code for 1FW3 encoders

Order number	Encoder code (p0400)	Remark
1FW3xxx-xAxxx-xxxx	2001	-
1FW3xxx-xExxx-xxxx	2051	-
1FW3xxx-xGxxx-xxxx	2052	-
1FW3xxx-xSxxx-xxxx	1003	-

1PH4 encoders

Table A-10 Encoder code for 1PH4 encoders

Order number	Encoder code (p0400)	Remark
1PH4xxx-xNxxx-xxxx	2002	-

1PH7 encoders

Table A-11 Encoder code for 1PH7 encoders

Order number	Encoder code (p0400)	Remark
1PH7xxx-xExxx-xxxx	2051	-
1PH7xxx-xHxxx-xxxx	3002	-
1PH7xxx-xJxxx-xxxx	3003	-
1PH7xxx-xMxxx-xxxx	2001	-
1PH7xxx-xNxxx-xxxx	2002	-
1PH7xxx-xRxxx-xxxx	1001	-

1PM4 encoders

Table A-12 Encoder code for 1PM4 encoders

Order number	Encoder code (p0400)	Remark
1PM4xxx-xGxxx-xxxx	2002	-
1PM4xxx-xLxxx-xxxx	2003	-

1PM6 encoders

Table A-13 Encoder code for 1PM6 encoders

Order number	Encoder code (p0400)	Remark
1PM6xxx-xGxxx-xxxx	2002	-
1PM6xxx-xLxxx-xxxx	2003	-

2SP1 encoders

Table A-14 Encoder code for 2SP1 encoders

Order number	Encoder code (p0400)	Remark
2SP1xxx-xHxxx-xxxx	2003	-

List of abbreviations

B

Note:

The following list of abbreviations includes all abbreviations and their meanings used in the entire SINAMICS user documentation.

Abbreviation	Source of abbreviation	Meaning
A		
A...	Alarm	Alarm
AC	Alternating Current	Alternating current
ADC	Analog Digital Converter	A-D converter
AI	Analog Input	Analog input
AIM	Active Interface Module	Active interface module
ALM	Active Line Module	Active Line Module
AO	Analog Output	Analog output
AOP	Advanced Operator Panel	Advanced operator panel
APC	Advanced Positioning Control	Advanced Positioning Control
AR	Automatic Restart	Automatic restart
ASC	Armature Short-Circuit	Armature short-circuit
ASCII	American Standard Code for Information Interchange	American Standard Code for Information Interchange
ASM	Induction motor	Induction motor
B		
BB	Operating condition	Operating condition
BERO	-	Contact-free proximity switch
BI	Binector Input	Binector input
BIA	Berufsgenossenschaftliches Institut für Arbeitssicherheit	Germany's Institute for Occupational Safety and Health
BICO	Binector Connector Technology	Binector connector technology
BLM	Basic Line Module	Basic Line Module
BO	Binector Output	Binector output
BOP	Basic Operator Panel	Basic Operator Panel
C		
C	Capacitance	Capacitance
C...	-	Safety message
CAN	Controller Area Network	Serial bus system

Abbreviation	Source of abbreviation	Meaning
CBC	Communication Board CAN	Communication Board CAN
CD	Compact disc	Compact disc
CDS	Command Data Set	Command data set
CF Card	CompactFlash Card	CompactFlash memory card
CI	Connector Input	Connector input
CLC	Clearance Control	Clearance control
CNC	Computerized Numerical Control	Computer-supported numerical control
CO	Connector Output	Connector output
CO/BO	Connector Output/Binector Output	Connector output/binector output
COB ID	CAN Object Identification	CAN object identification
COM	Common contact of a change-over relay	Center contact on a changeover contact
COMM	Commissioning	Commissioning
CP	Communications Processor	Communications processor
CPU	Central Processing Unit	Central processing unit
CRC	Cyclic Redundancy Check	Cyclic redundancy check
CSM	Control Supply Module	Control Supply Module
CU	Control Unit	Control unit
CUA	Control Unit Adapter	Control Unit Adapter
CUD	Control Unit DC MASTER	Control Unit DC MASTER
D		
DAC	Digital Analog Converter	D-A converter
DC	Direct Current	Direct current
DCB	Drive Control Block	Drive Control Block
DCC	Drive Control Chart	Drive control chart
DCC	Data Cross Check	Crosswise data comparison
DCN	Direct Current Negative	Direct current negative
DCP	Direct Current Positive	Direct current positive
DDS	Drive Data Set	Drive data set
DI	Digital Input	Digital input
DI/DO	Digital Input/Digital Output	Digital input/digital output, bidirectional
DMC	DRIVE-CLiQ Hub Module Cabinet	DRIVE-CLiQ Hub Module Cabinet
DME	DRIVE-CLiQ Hub Module External	DRIVE-CLiQ Hub Module External
DO	Digital Output	Digital output
DO	Drive Object	Drive object
DP	Decentralized Peripherals	Distributed I/Os
DPRAM	Dual-Port Random Access Memory	Memory with dual access
DRAM	Dynamic Random Access Memory	Dynamic memory
DRIVE-CLiQ	Drive Component Link with IQ	Drive Component Link with IQ
DSC	Dynamic Servo Control	Dynamic Servo Control
E		
EASC	External Armature Short-Circuit	External armature short-circuit

Abbreviation	Source of abbreviation	Meaning
EDS	Encoder Data Set	Encoder data set
EGB	Electrostatically Sensitive Devices (ESD)	Electrostatically Sensitive Devices (ESD)
ELCB	Earth Leakage Circuit Breaker	Residual current operated circuit breaker
ELP	Earth Leakage Protection	Ground-fault monitoring
EMC	Electromagnetic Compatibility	Electromagnetic compatibility
EMF	Electromagnetic Force	Electromagnetic force
EMK	Electromagnetic force	Electromagnetic force
EMV	Electromagnetic compatibility	Electromagnetic compatibility
EN	Europäische Norm (European standard)	Europäische Norm (European standard)
EnDat	Encoder Data Interface	Encoder interface
EP	Enable Pulses	Pulse enable
EPOS	Basic positioner	Basic positioner
ES	Engineering System	Engineering system
ESB	Equivalent circuit diagram	Equivalent circuit diagram
ESD	Electrostatically Sensitive Devices	Electrostatically Sensitive Devices (ESD)
ESR	Extended Stop and Retract	Extended stop and retraction
F		
F...	Fault	Fault
FAQs	Frequently Asked Questions	Frequently asked questions
FBL	Free Blocks	Free function blocks
FCC	Function Control Chart	Function Control Chart
FCC	Flux Current Control	Flux current control
FD	Function Block Diagram	Function block diagram
F-DI	Failsafe Digital Input	Failsafe digital input
F-DO	Failsafe Digital Output	Failsafe digital output
FEM	Separately excited synchronous motor	Separately excited synchronous motor
FEPROM	Flash EPROM	Non-volatile read/write memory
FG	Function Generator	Function generator
FI	-	Fault current
FOC	Fiber-Optic Cable	Fiber-optic cable
FP	Function block diagram	Function block diagram
FPGA	Field Programmable Gate Array	Field Programmable Gate Array
FW	Firmware	Firmware
G		
GB	Gigabyte	Gigabyte
GC	Global Control	Global Control Telegram (broadcast telegram)
GND	Ground	Reference potential for all signal and operating voltages, usually defined as 0 V (also referred to as G)
GSD	GSD file	Master device file: Describes the characteristics of a PROFIBUS slave

Abbreviation	Source of abbreviation	Meaning
GSV	Gate Supply Voltage	Gate Supply Voltage
GUID	Globally Unique Identifier	Globally unique identifier
H		
HF	High Frequency	High frequency
HFD	High-frequency reactor	High-frequency reactor
HLG	Ramp-function generator	Ramp-function generator
HMI	Human Machine Interface	Human-machine interface
HTL	High-Threshold Logic	Logic with a high fault threshold
HW	Hardware	Hardware
I		
i. V.	Under development	Under development: This feature is not currently available.
I/O	Input/Output	Input/output
I2C	Inter-Integrated Circuit	Internal serial data bus
IASC	Internal Armature Short-Circuit	Internal armature short-circuit
IBN	Commissioning	Commissioning
ID	Identifier	Identification
IE	Industrial Ethernet	Industrial Ethernet
IEC	International Electrotechnical Commission	International electrotechnical standard
IF	Interface	Interface
IGBT	Insulated Gate Bipolar Transistor	Insulated gate bipolar transistor
IGCT	Integrated Gate-Controlled Thyristor	Semiconductor circuit breaker with integrated control electrode
IL	Pulse cancellation (Impulslöschung)	Pulse cancellation
IP	Internet Protocol	Internet Protocol
IPO	Interpolator	Interpolator
IT	Isolé Terré	Isolated three-phase supply network
IVP	Internal Voltage Protection	Internal voltage protection
J		
JOG	Jogging	Jogging
K		
KDV	Crosswise data comparison	Crosswise data comparison
KIP	Kinetic buffering	Kinetic buffering
Kp	-	Proportional gain
KTY	-	Special temperature sensor
L		
L	-	Formula symbol for inductance
LED	Light Emitting Diode	Light Emitting Diode
LIN	Linear motor	Linear motor
LR	Position controller	Position controller
LSB	Least Significant Bit	Least significant bit
LSC	Line-Side Converter	Line-side converter

Abbreviation	Source of abbreviation	Meaning
LSS	Line-side switch	Line-side switch
LU	Length Unit	Length unit
LWL	Fiber-optic cable	Fiber-optic cable
M		
M	-	Formula symbol for torque
M	Ground	Reference potential for all signal and operating voltages, usually defined as 0 V (also referred to as GND)
MB	Megabyte	Megabyte
MCC	Motion Control Chart	Motion Control Chart
MDS	Motor Data Set	Motor data set
MLFB	Machine-Readable Product Code	Machine-Readable Product Code
MMC	Man-Machine Communication	Man-machine communication
MMC	Micro Memory Card	Micro memory card
MSB	Most Significant Bit	Most significant bit
MSC	Motor-Side Converter	Motor-side converter
MSCY_C1	Master Slave Cycle Class 1	Cyclic communication between master (Class 1) and slave
MSR	Motor-side converter	Motor-side converter
MT	Probe	Probe
N		
N. C.	Not Connected	Not connected
N...	No Report	No message or internal message
NAMUR	Standardization association for measurement and control in chemical industries	Standardization association for measurement and control in chemical industries
NC	Normally Closed (contact)	NC contact
NC	Numerical Control	Numerical control
NEMA	National Electrical Manufacturers Association	Standardization body in the USA
NM	Zero mark	Zero mark
NO	Normally Open (contact)	N/O contact
NSR	Line-side converter	Line-side converter
NVRAM	Non-Volatile Random Access Memory	Non-volatile read/write memory
O		
OA	Open Architecture	Open Architecture
OC	Operating Condition	Operating condition
OEM	Original Equipment Manufacturer	Original Equipment Manufacturer
OLP	Optical Link Plug	Fiber-optic bus connector
OMI	Option Module Interface	Option module interface
P		
p...	-	Adjustable parameters
PB	PROFIBUS	PROFIBUS

Abbreviation	Source of abbreviation	Meaning
PcCtrl	PC Control	Control for master
PD	PROFIdrive	PROFIdrive
PDS	Power Unit Data Set	Power unit data set
PE	Protective Earth	Protective earth
PELV	Protective Extra Low Voltage	Protective extra low voltage
PEM	Permanent-magnet synchronous motor	Permanent-magnet synchronous motor
PG	Programming device	Programming device
PI	Proportional Integral	Proportional integral
PID	Proportional Integral Differential	Proportional integral differential
PLC	Programmable Logic Controller	Programmable logic controller
PLL	Phase-locked loop	Phase-locked loop
PN	PROFINET	PROFINET
PNO	PROFIBUS user organization	PROFIBUS user organization
PPI	Point-to-Point Interface	Point-to-point interface
PRBS	Pseudo Random Binary Signal	White noise
PROFIBUS	Process Field Bus	Serial data bus
PS	Power Supply	Power supply
PSA	Power Stack Adapter	Power Stack Adapter
PTC	Positive Temperature Coefficient	Positive temperature coefficient
PTP	Point-To-Point	Point-to-Point
PWM	Pulse Width Modulation	Pulse width modulation
PZD	Process data	Process data
R		
r...	-	Display parameters (read-only)
RAM	Random Access Memory	Read/write memory
RCCB	Residual Current Circuit Breaker	Ground-fault circuit interrupter
RCD	Residual Current Device	Ground-fault circuit interrupter
RCM	Residual Current Monitor	Residual current monitor
RFG	Ramp-Function Generator	Ramp-function generator
RJ45	Registered Jack 45	Term for an 8-pin socket system for data transmission with shielded or non-shielded multi-wire copper cables
RKA	Cooling unit	Cooling unit
RO	Read Only	Read only
RPDO	Receive Process Data Object	Receive process data object

Abbreviation	Source of abbreviation	Meaning
RS232	Recommended Standard 232	Interface standard for cable-connected serial data transmission between a sender and receiver (also known under EIA232)
RS485	Recommended Standard 485	Interface standard for a cable-connected differential, parallel, and/or serial bus system (data transmission between a number of senders and receivers, also known as EIA485)
RTC	Real Time Clock	Real time clock
RZA	Space vector approximation	Space vector approximation
S		
S1	-	Continuous operation
S3	-	Intermittent operation
SBC	Safe Brake Control	Safe brake control
SBH	Safe operating stop	Safe operating stop
SBR	-	Safe acceleration monitoring
SCA	Safe Cam	Safe cam
SD Card	Secure Digital Card	Secure digital memory card
SE	Safe software limit switch	Safe software limit switch
SG	Safely-limited speed	Safely-limited speed
SGA	Safety-related output	Safety-related output
SGE	Safety-related input	Safety-related input
SH	Safe standstill	Safe standstill
SI	Safety Integrated	Safety Integrated
SIL	Safety Integrity Level	Safety Integrity Level
SLM	Smart Line Module	Smart Line Module
SLP	Safely-Limited Position	Safely-limited position
SLS	Safely-Limited Speed	Safely-limited speed
SLVC	Sensorless Vector Control	Vector control without encoder
SM	Sensor Module	Sensor Module
SMC	Sensor Module Cabinet	Sensor Module Cabinet
SME	Sensor Module External	Sensor Module External
SN	Safe software cam	Safe software cam
SOS	Safe Operating Stop	Safe operating stop
SP	Service Pack	Service pack
SPC	Setpoint Channel	Setpoint channel
SPI	Serial Peripheral Interface	Serial interface for connecting peripherals
SPS	Programmable logic controller	Programmable logic controller
SS1	Safe Stop 1	Safe stop 1 (monitored for time and ramping up)
SS2	Safe Stop 2	Safe stop 2
SSI	Synchronous Serial Interface	Synchronous serial interface

Abbreviation	Source of abbreviation	Meaning
SSM	Safe Speed Monitor	Safe feedback for speed monitoring ($n < n_x$)
SSP	SINAMICS Support Package	SINAMICS support package
STO	Safe Torque Off	Safe torque off
STW	Control word	Control word
T		
TB	Terminal Board	Terminal board
TIA	Totally Integrated Automation	Totally Integrated Automation
TM	Terminal Module	Terminal module
TN	Terre Neutre	Grounded three-phase supply network
Tn	-	Integral time
TPDO	Transmit Process Data Object	Transmit process data object
TT	Terre Terre	Grounded three-phase supply network
TTL	Transistor-Transistor Logic	Transistor-transistor logic
Tv	-	Rate time
U		
UL	Underwriters Laboratories Inc.	Underwriters Laboratories Inc.
UPS	Uninterruptible Power Supply	Uninterruptible power supply
USV	Uninterruptible power supply	Uninterruptible power supply
UTC	Universal Time Coordinated	Universal Time Coordinated
V		
VC	Vector Control	Vector control
Vdc	-	DC-link voltage
VdcN	-	Partial DC-link voltage, negative
VdcP	-	Partial DC-link voltage, positive
VDE	Verband Deutscher Elektrotechniker	Association of German Electrical Engineers
VDI	Verein Deutscher Ingenieure	Association of German Engineers
VPM	Voltage Protection Module	Voltage Protection Module
Vpp	Volt peak to peak	Volt peak to peak
VSM	Voltage Sensing Module	Voltage sensing module
W		
WEA	Automatic restart	Automatic restart
WZM	Machine tool	Machine tool
X		
XML	Extensible Markup Language	Standard language for web publishing and document management
Z		
ZK	DC link	DC link
ZM	Zero Mark	Zero mark
ZSW	Status word	Status word

List of References

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Haid-und-Neu-Straße 7, D-76131 Karlsruhe
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- /PDP/ PROFIBUS Installation Guidelines**
Installation Guideline for PROFIBUS-FMS/DP
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Documentation for Safety Equipment

Note

For more information on technical documentation in and around the topic of "Safety Integrated", please follow the link below:

<http://www.siemens.de/safety>

The following list contains some of the safety-related documentation available.

/LV1/	Low Voltage Switchgear SIRIUS-SENTRON-SIVACON		
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/SICD/	Safety Integrated		
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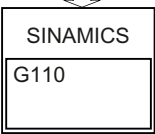
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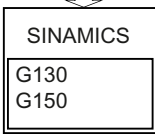
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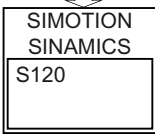
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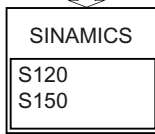
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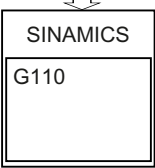


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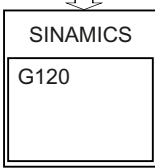


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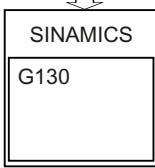
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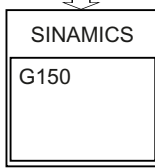
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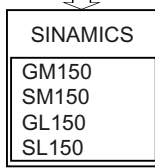
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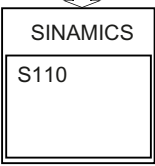


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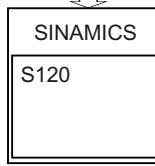


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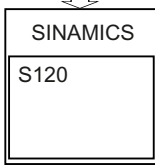
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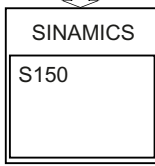
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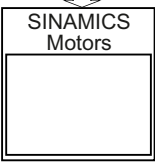


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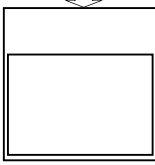


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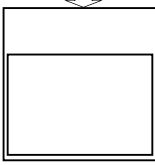
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